

SUNDAY, JULY 25

Council Symposium

TRACK 1

International Council Symposium: AAPM Global Engagement; Planning for the Future

SU-A-TRACK 1-00

International Council Symposium: AAPM Global Engagement; Planning for the Future

Moderator: Ana Maria Marques da Silva, Pontifícia Universidade Católica do RS, Porto Alegre, RS

SU-A-TRACK 1-01

S. Avery, University of Pennsylvania:

Global Medical Physics Education and Training

SU-A-TRACK 1-02

S. Parker, Wake Forest Baptist Health High Point Medical Center:

Global Need Assessment

SU-A-TRACK 1-03

J. Palta, Virginia Commonwealth University:

Global Collaborations

SU-A-TRACK 1-04

R. Jeraj, University of Wisconsin:

Global Research and Scientific Innovation

SU-A-TRACK 1-05

G. Kim, University of California, San Diego:

Global Data and Information Exchange

SU-A-TRACK 1-06

S. Wadi-Ramahi, University of Pittsburgh Medical Center:

Global Clinical Education and Training

SU-A-TRACK 1-07

Q&A

Non communicable diseases (NCD) cause one in every six deaths worldwide, more than major communicable diseases (AIDS, tuberculosis, and malaria) combined. Cancer is the most dominant NCD. The global incidence of cancer is increasing rapidly in countries with low and medium human development indices (HDI). It is anticipated that by 2022, up to 70% (14 million) of the 20 million annual new cancers are expected to occur in these countries. Multimodality imaging for the diagnosis and radiotherapy for treatment will be required for these patients to cure localized disease and to effectively palliate symptoms. The imaging and radiotherapy equipment manufacturers anticipate that the growth of their new business is going to occur in low to middle income countries (LMICs) with a projection of over 200% growth in equipment sales in the next 5 years. Correspondingly, the global pool of radiation oncologists and radiologists needs to increase by 100%, medical physicists and dosimetrists by 300%, and technologists by over 300%. Unfortunately, this personnel shortage is exacerbated by the fact that many of the LMICs do not have the resources and infrastructure to train and educate the growing numbers of professionals needed in the safe and effective use of diagnostic and therapeutic technologies.

The professional organizations such as AAPM, ASTRO, RSNA, and ESTRO; international agencies such as IAEA and UICC; NGOs such as ICEC, Rayos Contra Cancer, Radiating Hope, and RADAID International; and vendors are doing their best to mitigate these challenges by offering their own solutions. NGOs have tried to fill the coordination gap, but there has been limited success without full institutional backing. Thus, there remains little or no coordination of these efforts at the present time while the financial commitment, especially by radiotherapy vendors and the IAEA is substantial.

AAPM has made a strategic decision to coordinate all international activities under the umbrella of International Council. The committee structure of IC will support efforts in education, clinical training, research, and a broader scope of collaboration with other global stakeholders that has largely been unexplored. It will provide opportunities to grow its base globally and improve domestically and it will clearly demonstrate AAPM's global leadership in the field of medical physics.

Learning Objectives:

1. Understand the need for global collaboration.
2. Understand the scope of activities of categorical committees of International Council.
3. Understand the need for coordination and harmonization of projects with other global stakeholders.

Students and Trainees

TRACK 2

2021 Annual Student Meeting: Beyond the Clinic - Medical Physics in Industry

SU-AB-TRACK 2-00

2021 Annual Student Meeting: Beyond the Clinic - Medical Physics in Industry

Moderators: Ahtesham Ullah Khan, University of Wisconsin - ADCL, Madison, WI, and Reed Kolany, University of Wisconsin - ADCL, Madison, WI

SU-AB-TRACK 2-01

Introduction

SU-AB-TRACK 2-02

P. Naine, Elekta, Inc.:

Medical Physicists in Industry: Present and Future

SU-AB-TRACK 2-03

M. Bellon, ViewRay Incorporated:
Transitioning from a Clinical to an Industry Career

SU-AB-TRACK 2-04

C. Lin, OnLume Inc:
Beginning a Novel Career as a Medical Physicist

SU-AB-TRACK 2-05

AAPM Initiatives to Promote Industry Careers

SU-AB-TRACK 2-06

Questions/Discussions/Interactive Exercise

The 2021 Annual Student Meeting aims to discuss the role of medical physicists in the industrial workplace and the necessary skills required to succeed in such an environment. The majority of the medical physics graduate students and trainees are unaware of the industrial internship opportunities available to them and lack guidance in terms of what elective coursework to take to successfully obtain an industry job post-graduation. Furthermore, many mid-career medical physicists struggle to successfully transition from a clinical/academic job to an industry job in case they desire to switch careers.

As part of this meeting, the three speakers will share their experiences as industry physicists, including the journey to where they are now. As a recent graduate, Dr. Christie Lin will connect with current graduate students and recent grads, discussing her recent transition from graduate school to the industry and offer advice for those interested in a similar career path. For mid-career clinical physicists who are considering switching careers to the industry, Maria Bellon will be speaking about her transition to ViewRay after many years as a clinical physicist. Additionally, an established physicist who has worked their entire career in industry, Paul Naine, will speak about his experiences and the future of medical physics in the industry.

These presentations will be followed by an interactive discussion and Q&A session with the speakers, facilitated by student moderators. Join us in this session to learn about life as an industry physicist and learn about the many paths taken by those currently in the industry.

Learning Objectives:

1. Learn about a day in the life of an industry physicist
2. Understand the didactic and non-didactic requirements to be successful in an industry job
3. Get insights into the future of the medical physics industry

SAM Imaging Scientific Symposium**TRACK 3****SIIM-AAPM Joint Symposium on Machine Intelligence in Medical Imaging****SU-AB-TRACK 3-00**

SIIM-AAPM Joint Symposium on Machine Intelligence in Medical Imaging

Moderators: Paul Nagy, Johns Hopkins University, Baltimore, MD and Jeffrey Siewerdsen, Johns Hopkins University, Baltimore, MD

SU-AB-TRACK 3-01

Introduction

SU-AB-TRACK 3-02

B. Landman, Vanderbilt University Medical Center:
Challenges and Opportunities in Validation of AI for Medical Imaging

SU-AB-TRACK 3-03

K. Brock, UT MD Anderson Cancer Center:
Harnessing AI to Enable Advanced Image-Guidance in Interventional Procedures

SU-AB-TRACK 3-04

J. Fessler, University of Michigan:
Joint Optimization of Learning-Based Image Reconstruction and K-Space Trajectories for MRI

SU-AB-TRACK 3-05

K. Andriole, Brigham & Women's Hospital, Harvard Medical School:
Standards-Based Integration of AI Tools into Clinical Practice

SU-AB-TRACK 3-06

Q&A

This symposium features 4 invited presentations on the development and application of machine learning approaches, infrastructure, and standards in medical imaging, including experts in health informatics representing the Society for Imaging Informatics in Medicine (SIIM) and image acquisition and reconstruction from the AAPM.

(1) Dr. Bennett Landman will discuss “Challenges and Opportunities in Validation of AI for Medical Imaging.” Artificial Intelligence (AI) is promising to transform how we capture, interpret, and understand medical images from both basic science and clinical perspectives. However, it is becoming increasingly apparent that biases in study populations, site-factors, and imaging designs can have unintended, and sometimes catastrophic, impacts on AI performance. Hence, AI validation is rapidly becoming an essential field of research.

(2) Dr. Kristy Brock will present on “Harnessing AI to Enable Advanced Image Guidance in Interventional Procedures.” AI has the potential to be both an enabling and transformational tool in image guided interventions. The ability to obtain automated, accurate, and consistent segmentation of tumors and normal tissues can enable advanced image analysis, targeting, and treatment evaluation. In addition, AI can transform the way that clinicians can predict the outcomes of treatment, allowing real-time optimization to drive improved outcomes.

(3) Dr. Jeff Fessler will address the topic of “Joint Optimization of Learning-Based Image Reconstruction and K-Space Trajectories for MRI.” Machine learning approaches to medical image reconstruction are of considerable recent interest, especially supervised approaches that use a corpus of training data. Accelerated MRI scans, where fewer k-space points than image voxels are acquired, is a natural setting for such reconstruction methods. Recently, machine learning methods for optimizing the k-space sampling have also had growing interest. This talk will summarize recent work where we jointly optimize non-Cartesian

k-space sampling, heeding physical constraints like gradient slew rate, and a learning-based image reconstruction method.

(4) Dr. Kathy Andriole will discuss “Standards-Based Integration of AI Tools into Clinical Practice.” The goal of many machine learning (ML) models is to provide clinical decision support for imaging examination interpretation. But ML can be applied at various points in the medical imaging chain, including at examination ordering, imaging modality protocoling, image creation and quality assessment, study diagnosis, reporting, and billing. Such applications may improve workflow efficiency and reduce errors and cost. Integrating ML tools into the clinical arena is challenging. Topics covered in this presentation include the integration standards required for ML inference implementation at various points throughout the radiological workflow. Illustrative clinical examples will be shown and the role of the medical physicist will be discussed.

Together, these presentations highlight the potential for interaction between experts in health informatics and medical physics to bring machine learning methods to clinical benefit in medical imaging. The joint symposium will reciprocate at the 2021 SIIM Conference on Machine Intelligence in Medical Imaging (CMIMI).

Learning objectives include understanding advances in: (1) infrastructure to enable collaboration on machine learning methods; (2) machine learning methods to enable advanced image-guided interventional approaches; (3) learning-based optimization for MR image reconstruction; and (4) standards for integrating AI in clinical practice.

SAM Multi-Disciplinary Scientific Symposium

TRACK 4

Solid State PET in Radiology and Radiation Oncology: from SiPM to AI in the Clinic

SU-A-TRACK 4-00

Solid State PET in Radiology and Radiation Oncology: From SiPM to AI in the Clinic

Moderator: Jun Zhang, The Ohio State University, Columbus, OH

SU-A-TRACK 4-01

J. Zhang, The Ohio State University:

Technologies and AI Features of SiPM PET Instrumentation

SU-A-TRACK 4-02

S. Bowen, University of Washington, School of Medicine:

Clinical Considerations of SiPM PET in Radiology and Radiation Oncology

SU-A-TRACK 4-03

Q&A & Panel Discussion

With the disruptive technology leap of solid state SiPM PET detectors, improvements in PET instrumentation and technologies are happening at a staggering pace. At the same time, rapid development of computational technologies has made artificial intelligence (AI) more feasible and increasingly important.

This symposium aims to describe current clinical status and future expectations of utilizing SiPM PET in radiology and radiation oncology patient care, as well as innovative technologies and novel AI features of emerging SiPM PET instrumentation focusing on how digital PET is better than analog PET. Physicists, physicians and technologists wishing to better understand novel PET systems and technologies are expected to attend this symposium.

Learning Objectives:

1. Understand and learn current challenges of clinical care as well as expectations of future improvement using SiPM PET technologies in both radiology and radiation oncology.
2. Learn physics of SiPM PET technologies and specify emerging SiPM PET instrumentations and their novel AI features.
3. Identify and learn the advantages of digital PET than analog PET from imaging acquisition to clinical application.

Topics:

1. Clinical Considerations of SiPM PET in Radiology and Radiation Oncology: Motivating Applications and Needs.
2. Technologies and AI Features of SiPM PET Instrumentation: What is available and how is digital PET better than analog PET?

SAM Therapy Scientific Symposium

TRACK 5

Affordable Cancer Care for All

SU-AB-TRACK 5-00

Affordable Cancer Care for All

Moderator: Laurence Edward Court, University of Texas MD Anderson Cancer Center, Houston, TX

SU-AB-TRACK 5-01

Introduction

SU-AB-TRACK 5-02

C. Fumiko, Memorial Sloan Kettering Cancer Center:

The Real Impact of Cost on Patients

SU-AB-TRACK 5-03

M. Weil, Sirius Medicine:

Low-Cost X-Ray Treatment System

SU-AB-TRACK 5-04

M. Salehpour, M.D. Anderson Cancer Center:

Focused X-Ray Beams for Low-Cost Radiotherapy

SU-AB-TRACK 5-05

T. Ige, National Hospital:

A Rethink of the Linear Accelerator for Resource-Limited Environments

SU-AB-TRACK 5-06

M. Dosahjh, Cern:

A Rethink of the Linear Accelerator for Resource-Limited Environments

SU-AB-TRACK 5-07

P. Halvorsen, Beth Israel - Lahey Health:

Factors Affecting Resource Utilization in Radiation Oncology

SU-AB-TRACK 5-08

T. Netherton, MD Anderson Cancer Center:

Fully Automated Treatment Planning

SU-AB-TRACK 5-017

Q&A

In the USA, cancer patients are at very high risk for bankruptcy. Outside the USA, 90% of patients in low-income countries lack access to radiotherapy. Resolving this situation requires an understanding of what patients experience, and significant, out-of-the-box efforts to reduce costs and improve the robustness and efficiency of our current clinical equipment, workflows, and software tools. What role can medical physicists take in this important drive?

In this session we will first describe how radiotherapy costs are affecting our patients, followed by descriptions of physics-led efforts in reducing costs and improving access to care for patients across the world. We will describe projects to develop new treatment devices that will provide robust and low-cost solutions worldwide, and also improvements in process and workflow that will improve efficiency, thus reducing overall costs.

Learning Objectives:

1. Understand some of the challenges that patients can have in finding and receiving radiation therapy
2. Understand physics-related efforts in treatment design to improve treatment availability of radiotherapy across the world
3. Understand physics-related efforts in process efficiency to reduce costs of radiotherapy across the world

NCI, Varian, CPRIT, Wellcome Trust

Therapy Scientific Session

TRACK 6

Artificial Intelligence in Treatment Planning and Delivery

SU-A-TRACK 6-01

Curbing the Errors from Automated Tools Based on An FMEA of an AI-Based Treatment Planning System

K Nealon^{1,2*}, P Balter^{1,2}, R Douglas², D Fullen², S Hernandez^{1,2}, P Nitsch², A Olanrewaju², M Soliman², L Court^{1,2}, (1) UT MD Anderson UTHealth Graduate School of Biomedical Sciences, Houston, TX (2) The University of Texas MD Anderson Cancer Center, Houston, TX

Purpose: To prospectively assess the patient risk associated with the clinical use of a web-based, automated contouring and treatment planning tool. **Methods:** A Failure Modes and Effects Analysis (FMEA) was performed to quantify risk prior to clinical deployment of the Radiation Planning Assistant (RPA), an automated contouring and treatment planning tool. Potential errors were identified by a multidisciplinary team, consisting of representatives from physics, radiation therapy, dosimetry, physician, RPA development and quality improvement teams. Each failure mode was individually discussed and scored by all participants, over the course of twenty 90-minute meetings. A 1-10 scale for severity, occurrence and detectability was used, following TG-100 recommendations. **Results:** Of the identified failure modes, 56.6% (164/290) were common in manual radiotherapy processes, leaving 126 unique to RPA assisted workflows. The mean RPN of these errors was 56.3, with a maximum of 486. The top ten failure modes were caused by automation bias, a tendency of users to rely too heavily on results from automated systems, operator and software errors. An action threshold of 125 was set, and corrective actions were taken to the 21 failure modes above that limit. As a result, changes were made to the RPA including modifications to the user interface to remove unnecessary input and better training to highlight the importance of review of the automated output. After these changes, the FMEA was rescored and the mean and maximum RPN decreased to 33.7 and 288, respectively. **Conclusion:** The majority of high-risk failure modes were caused by automation bias or operator error, emphasizing the need to limit unnecessary user input while maximizing user review of output. As a result of this analysis, changes were made to the RPA software and RPN scores subsequently decreased, showing that the FMEA approach is an effective way to reduce risk for automated planning tools.

Partial funding for this work was provided by Varian Medical Systems

SU-A-TRACK 6-02

Generalizability Study of a Fluence Map Prediction Network

L Ma*, M Chen, X Gu, W Lu, University of Texas Southwestern Medical Center, Dallas, TX

Purpose: A deep learning-based fluence map prediction network (FMPN) was developed for predicting fluence maps from desired dose distribution. The network was trained with clinical head and neck (HN) VMAT plans. This work is to investigate the performance of FMPN for various clinical scenarios apart from the training data. **Methods:** The FMPN which maps projections of dose to fluence maps was trained only with clinical HN full-arc VMAT plans. To study the generalizability of FMPN, we designed three tests, each of which has a feature different from the training scenario. First, we test the trained network on a different treatment site, prostate. Second, we test the network on new delivery modalities, with three partial-arc VMAT plans and one IMRT plans. Third, we test the network using plans with different degrees of modulation (DOM), to see how well it predicts fluence maps for doses in a wide spectrum of DOM. The generalizability was quantified by comparing the dose of FMPN-predicted fluence maps to the ground truth dose (the desired dose, input of FMPN) using 3D Gamma passing rate with 3mm/3% criteria. **Results:** In the first test, FMPN achieved a prediction accuracy in prostate site (99.06±1.75%) as high as in training site (HN, 98.06±2.64%). In the second test, performance was excellent on partial-arc VMAT (99%-89%) and degraded on IMRT (87% with 5mm/5% criteria), which was reasonable because IMRT is too far away from training data (full-arc VMAT) from data distribution point of view. In the third test, FMPN can predict fluence map for plans with various DOM/quality (95%-97%) and accuracy doesn't vary with DOM. **Conclusion:** The FMPN trained on HN full-arc VMAT plans could generalize well to other treatment site, delivery modality and various DOM, showing great potential for clinical use.

R01 CA235723, R01 CA218402

SU-A-TRACK 6-03

Inverse Treatment Planning Using a Virtual Treatment Planner (VTP) for Prostate Cancer Treated with Intensity Modulated Radiation Therapy
D Sprouts^{1*}, C Shen², Y Chi¹, (1) University of Texas at Arlington, Arlington, Tx (2) University of Texas Southwestern Medical Center, Dallas, TX

Purpose: Although commercial treatment planning systems (TPSs) can automatically solve the optimization problem for treatment planning, human planners need to define and adjust the planning objectives/constraints to obtain clinically acceptable plans. Such a process is labor-intensive and time-consuming. In this work, we show an end-to-end study to train a deep reinforcement learning (DRL) based virtual treatment planner (VTP) that can behave like a human to operate the commercial TPS for high-quality planning. **Methods:** We considered prostate cancer IMRT treatment planning as a testbed. The VTP takes DVH of a plan as input and predicts the optimal adjustment action to improve plan quality. The training of VTP followed the state-of-the-art Q-learning framework. Experience replay was implemented with epsilon-greedy search to explore the impacts of taking different actions on a large number of automatically generated plans, from which an optimal policy can be learned. Since the major computational efforts in training were spent to perform plan optimization repeatedly, we implemented a GPU-based Eclipse-equivalent TPS using PYCUDA to improve the optimization efficiency. Upon the completion of training, the established VTP was deployed to plan for a set of testing patient cases. Similar to a human planner, VTP keeps adjusting the planning objectives/constraints in the TPS to improve plan quality until the plan is acceptable or achieves maximum adjusting steps. The generated plans were evaluated using the ProKnow scoring system. **Results:** VTP was successfully trained, validated, and tested with 10, 2, and 50 cases, respectively. With PYCUDA, the training efficiency was improved by 3 folds. The mean plan score ($\pm$ std.) of the 50 test cases was improved from 6.18 ± 1.75 to 8.11 ± 1.27 by VTP, with 9 being the maximal score-value. **Conclusion:** The proposed DRL-based VTP was able to operate the Eclipse-equivalent TPS to generate high-quality plans for prostate cancer IMRT.

SU-A-TRACK 6-04

Predicting Daily Record of Machine Performance Check Using Multi-Long Short-Term Memory Neural Networks

M Ma*, J Dai, National Cancer Center/National Clinical Research Center for Cancer/Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, China, Beijing, CN

Purpose: The test of machine performance check (MPC) monitors the status of Varian truebeam machine. The cumulative record of MPC tests shows the trend of machine performance and provides warning message for preventive actions. To precisely predict the future records of MPC tests, a generalized model using multi-long short-term memory (multi-LSTM) was introduced. **Methods:** A total of 1044 sets of daily MPC test data were collected. The 80% of data were used to train multi-LSTM model, while the remaining 20% of data were used to test the accuracy of the model. To enrich the data, cubic interpolation is used in the model. Furthermore, a greedy coordinate descent method was employed to find the optimal hyper-parameters of the model. The accuracy of the model was quantified by the mean absolute error (MAE), root-mean-square error (RMSE), and coefficient of determination (R^2). **Results:** The hyper-parameters are determined as 15 for the length of time lags, Adam for the optimizers, 0.001 for the learning rates, 150 for the number of epochs, 50 for the number of hidden units, and 32 for the batch sizes. The majority of MPC test records was predicted with an RMSE less than or equal to 0.005. Exceptions are the jaw collimation, collimation rotation offset, beam output change, and beam uniformity. The similar result was observed in MAE metric. The result of all MPC tests records with R^2 was higher than 85.00%. **Conclusion:** The multi-LSTM model can accurately predict all MPC test record. Predicting the future performance records of MPC tests will foresee possible machine failure, which allows early action of machine maintenance and reduces unscheduled machine downtime.

the National Natural Science Foundation of China (Grant No. 11875320)

SU-A-TRACK 6-05

Towards Automatic Metastasis-Directed Therapy Planning in a Three-Dimensional Beam Model Using Reinforcement Learning

W Hrinivich^{1*}, M Deek¹, R Phillips², H Li¹, P Tran¹, J Lee¹, (1) Johns Hopkins University, Baltimore, MD, (2) Mayo Clinic, Rochester MN

Purpose: Automating metastasis-directed therapy (MDT) planning may improve plan consistency between users and centers, but variability in characteristics of metastasis lesions hinders existing supervised learning approaches due to training data requirements. The purpose of this study is to investigate the feasibility of reinforcement learning (RL)-based machine parameter optimization for automatic MDT planning for oligometastatic prostate cancer (OMPC), which eliminates dependence on prior plans for training and directly optimizes linear accelerator parameters. **Methods:** We implemented a three-dimensional (3D) deep-Q RL volumetric modulated arc therapy optimization algorithm for MDT as follows. Dose was computed in a 3D grid using matRad. The dose grid, target and organ contours were re-sampled to extract axial slices aligned with individual pairs of leaves in the multi-leaf collimator (MLC) model. These slices were used as input in a 2D deep-Q network, which predicted slice-specific leaf position and dose rate updates. Slice-specific dose rate updates were recombined for each control point using a voting strategy. RL was conducted using four OMPC patients to maximize target coverage and minimize bowel dose, and applied to one independent test case. Resultant dose was compared to simple conformal arcs to quantify improvement over un-optimized plans. **Results:** Training was conducted for 4,800 iterations taking 3 days. RL MDT maintained mean $\pm$ SD target coverage of $95.1 \pm 7.9\%$ in the training cohort and 100.0% in the test patient. Compared to the conformal arcs, RL MDT reduced maximum and mean bowel dose by $20.0 \pm 16.3\%$ and $38.0 \pm 18.6\%$ in the training cohort, and 1.5% and 37.6% in the test patient, demonstrating a preliminary trend toward improved dosimetry. **Conclusion:** These preliminary results suggest that RL MDT optimization is feasible in a 3D beam model despite variable targets and limited training time. Ongoing work involves increasing training cohort size and training time to enable high-quality auto-planning for MDT.

This work was supported by an AAPM Seed Research Funding Grant.

SU-A-TRACK 6-06

Towards Interpretable Intelligent Automatic Treatment Planning in Radiotherapy: Understanding the Decision-Making Behaviors of a Hierarchical Deep Reinforcement Learning Based Virtual Treatment Planner Network

C Shen*, L Chen, X Jia, (1) University of Texas Southwestern Medical Center, Dallas, TX

Purpose: Intelligent automatic treatment planning (IATP) in radiotherapy builds a virtual treatment planner network (VTPN) that automatically operates a treatment planning system (TPS) to generate high-quality plans by adjusting dose constraints in TPS like a human. It is important to interpret VTPN's decision-making behaviors to gain confidence for clinical applications. This study proposed a VTPN of a hierarchical architecture (HieVTPN) motivated by the decision-making process of human planners, and performed numerical studies to analyze and understand its behaviors in planning process. **Methods:** HieVTPN consists of three networks, i.e. Structure-Net, Parameter-Net, and Action-Net. In each step controlling the TPS to improve a plan, HieVTPN sequentially determines the anatomical structure to adjust dose constraints, the specific constraint of the selected structure, and the detailed adjustment action, respectively. A hierarchical deep reinforcement learning approach was developed to train HieVTPN to operate an in-house developed TPS to generate plans for prostate cancer intensity modulated radiotherapy. We analyzed the decision-making behaviors of the established HieVTPN on anatomical structure, parameter, and adjustment action levels, respectively. Scenarios triggering distinct decisions were characterized to understand the rationale of the decision-making behaviors, which were then compared with human intuitions. **Results:** HieVTPN were successfully trained to operate the in-house developed TPS to generate high-quality plans for all 59 testing patient cases, achieving an average plan quality score of $8.62 (\pm 0.83)$, with 9 being the maximal score. We found that the decision-making

behaviors of HieVTPN were understandable. By observing a plan, HieVTPN tended to pick the anatomical structure that needs the most attention, and adjusted the appropriate parameter corresponding to the selected structure towards the direction to improve the plan. Such decision-making behaviors generally agreed with the human experience. **Conclusion:** Our study demonstrated the agreement between the decision-making behaviors of HieVTPN in planning process with that of human based on intuitions.

SU-A-TRACK 6-07

Treatment Parameters That Impact IROC SRS Phantom Performance Evaluated Using AI

H Mehrens*, T Nguyen, S Edward, S Hartzell, M Glenn, D Branco, N Hernandez, P Alvarez, A Molineu, P Taylor, S Kry, UT MD Anderson Cancer Center, Houston, TX

Purpose: To analyze institutional performance irradiating IROC Houston's SRS head phantom. **Methods:** Irradiation results of 1072 SRS head phantoms between 2012-2020 were abstracted. Univariate analysis and random forest models were used to associate irradiation conditions with phantom results. Phantom results included pass/fail classification, average TLD ratio, and percent of pixels passing gamma. The following categories were evaluated in terms of how they predicted these outcomes: irradiation year, TPS algorithm, machine model, energy, and field size. Specifically, algorithm was categorized per TG-329, collimators were divided into three categories: cones, high-definition (MLC Leaf Width < 0.5 cm), and low-definition (MLC Leaf Width > 0.5 cm), and field size was calculated from the average extent of the measured 50% isodose line. **Results:** Pearson chi-squared test showed that algorithm ($p < 0.001$) and collimator ($p = 0.24$) were statistically significant predictors of pass/fail. Per Tukey's post-hoc test, pencil beam algorithm (AAA: $p = 0.01$, GBBS: $p = 0.15$, Monte Carlo: $p = 0.001$, Measured: $p < 0.001$, and Superposition/Convolution: $p = 0.015$) and cone collimator (high-definition: $p = 0.035$, low-definition: $p = 0.055$) were both more likely to be associated with failing phantom results. Our random forest models were able to achieve an overall prediction accuracy of $90.9 \pm 0.3\%$. For overall pass/fail, TLD result, and percent of pixels passing, the field size was the most important predictor, with poorer results associated with smaller fields. The next three most important variables were irradiation year, machine, and algorithm; however, the specific order of importance varied. **Conclusion:** Field size is the most important factor in determining the outcome of the phantom. Specifically, smaller fields (≤ 3 cm), which encapsulates the majority of cone collimators (72.7%) used, tends to produce more failures. Pencil beam algorithms were inferior to accurately predicting dose as compared to all other algorithms.

This work received funding from the NIH/NCI grants #CA180803 and #CA214526.

SAM Therapy Educational Course

TRACK 7

Clinical Implementation of MR-IGRT

SU-AB-TRACK 7-00

Clinical Implementation of MR-IGRT

Moderator: Anil Sethi, Loyola University Medical Center, Maywood, IL

SU-AB-TRACK 7-01

B. Raaymakers, University Medical Center Utrecht:

Tools of the Trade: MR Linacs & Image Optimization for MRgART

SU-AB-TRACK 7-02

P. Yadav, School of Medicine and Public Health UW-Madison:

State of the ART 1: Low Field MRgART

SU-AB-TRACK 7-03

D. Hyer, University of Iowa:

State of the ART 2: High Field MRgART

SU-AB-TRACK 7-04

A. Sethi, Loyola University Medical Center:

Proof Is in the Plan: Pros and Cons of MRgART

During the course of treatment, a patient treatment plan may be compromised by anatomical changes resulting from weight loss, soft-tissue deformation, organ filling/emptying, peristalsis etc. This often requires plan adjustment or full-re-planning (adaptive radiation therapy). MR Guided Adaptive RT (MRgART) is a novel treatment paradigm to address dose variations due to above changes. Therapeutic ratio for many treatment sites can be improved with MRgART, but at the added expense of time investment with the patient on the table. The adaptive workflow introduces several new steps to the treatment process, such as, deformable image registration, contouring, plan optimization, QA and tumor tracking. Clinical implementation of MRgART will be discussed with a focus on resource utilization. We will also propose guidelines for clinical workflow enhancements in imaging, treatment planning and delivery. Importance of comprehensive system QA (end-to-end test) will be discussed along with QA thresholds and consistency checks for each step in the process chain. Finally, interesting clinical cases will be presented to emphasize efficient planning techniques for daily plan adaptation and improved clinical workflow.

Learning Objectives:

1. Present fundamentals of MR imaging in the IGRT environment while highlighting features unique to hybrid MR linacs.
2. Share clinical experiences of performing MRgART on low & high Tesla MR linacs.
3. Present clinical examples that emphasize smart plan adaptation and associated benefits.

Research Funding Support/Speaker's Bureau, ViewRay, Inc.

Speaker's Bureau, Standard Imaging, Inc.

SAM Educational Topic-Specific Guided Tours-Therapy

TRACK 8

Patient-Specific QA Devices/Software

SU-AB-TRACK 8-00

Patient-Specific QA Devices/Software

Moderators: Sharareh Koufigar, Piedmont Hospital, Atlanta, GA and Kevinraj Sukumar, Piedmont Hospital, Atlanta, GA

SU-AB-TRACK 8-01

S. Koufigar, Piedmont Hospital:
Patient-Specific QA Devices/Software

SU-AB-TRACK 8-02

K. Sukumar, Piedmont Hospital :
Patient-Specific QA Devices/Software

SU-AB-TRACK 8-03

J. Hamilton, Sun Nuclear Corporation:
Improving Patient Outcomes through Automation of Error Detection During Treatment

SU-AB-TRACK 8-04

M. Napolitano, Standard Imaging, Inc.:
Patient Specific QA Throughout Treatment

SU-AB-TRACK 8-05

A. Nagle, Iba Dosimetry GmbH:
Monte Carlo Based Patient QA Efficiency and Accuracy Through Automation and Prediction

SU-AB-TRACK 8-06

N. Brand, PTW North America Corporation:
Fast, Independent, True 3D: SRS Patient QA for SIMT Plans with OCTAVIUS 4D 1600 SRS

SU-AB-TRACK 8-07**Q&A**

Patient specific quality assurance is conducted to validate the accurate deliverability of a treatment plan. This QA procedure was designed to improve the quality and safety of radiation treatments particularly those utilizing inverse planning techniques and/or hypofractionation. Numerous products have been developed for this task. This guided tour will begin with an introductory presentation from the moderators on patient specific QA including necessity and the broad array of available techniques. Vendors will then showcase their latest developments and products in patient specific QA followed by a question-and-answer session.

Learning Objectives:

1. Describe the rationale for conducting patient specific QA including formal guidance and recommendations.
2. Review the inherent limitations associated with patient specific QA testing.
3. Explore the pros and cons of the varying techniques currently available.

Council Symposium**TRACK 1*****Joint Council Symposium: Creativity in Medical Physics*****SU-B-TRACK 1-00**

Joint Council Symposium: Creativity in Medical Physics
Moderator: Brent Parker, The University of Texas M.D. Anderson Cancer Center, Galveston, TX

SU-B-TRACK 1-01**Introduction****SU-B-TRACK 1-02**

P. Basran, Cornell University:
Virtual Fellowships: A Novel Approach to Undergraduate Research Training and Mentorship

SU-B-TRACK 1-03

M. Giger, University of Chicago:
Extending AAPM Leadership to Advance Data Science in the Age of COVID-19

SU-B-TRACK 1-04

B. Parker, The University of Texas M.D. Anderson Cancer Center:
Sailing the High C's

SU-B-TRACK 1-05**Q&A**

While science is frequently considered a rigid discipline, recent times have highlighted the need for flexibility and adaptability in all areas of medical physics. Though COVID-19 forced our profession to creatively respond to external factors, opportunities exist for us to implement this type of creativity in our normal work environments on a regular basis. In this symposium, the speakers will discuss experiences with and opportunities for implementing creative change.

Parminder Basran, PhD – "Virtual Fellowships: A Novel Approach to Undergraduate Research Training and Mentorship" (Education Council)

For the summer of 2020, COVID-19 required the AAPM Summer Undergraduate Fellowship and Outreach Program to dramatically pivot from a conventional in-person undergraduate training experience to a virtual experience. Virtualizing fellowships forces a conscientious definition and refining of roles and responsibilities for the mentor, trainee, hosting institution and sponsors. Apart from the research experience itself, fellowship experiences include exposure to the clinical/research workplace, communicating with supervisors, researchers, non-physicists, residents, graduate students and fellow undergraduates, access to labs, equipment, and software, and the institution itself, by way of 'on-boarding' and such. While there are clear disadvantages of eliminating an in-person research experience, the successful virtualization of fellowships demonstrates many facets of the fellowship experience can indeed be provided online. This suggests virtual fellowships could become a novel vehicle for outreach to students currently unable to participate in this fellowship experience.

Maryellen Giger, PhD – "Extending AAPM Leadership to Advance Data Science in the Age of COVID-19" (Science Council)

MIDRC represents an NIBIB-funded partnership spearheaded by the medical imaging community and aimed at building data repositories to fuel COVID-19 machine intelligence research, coupled with optimal standardization, curation, and compliance with ethical responsibilities to honor patients' privacy. This partnership will lead to development and implementation of new diagnostics, including machine learning algorithms, that will empower population-wide

preventive and management strategies for COVID-19. This leadership collaboration among the ACR, RSNA, and AAPM is based on each organization's unique and complementary expertise within the medical imaging community, with AAPM bringing its expertise in physical image quality/harmonization and tailored distribution/metrology standards/evaluation metrics.

Brent Parker, PhD – “Sailing the High C's” (Professional Council)

The concept of creativity tends to be associated with the research and educational areas of medical physics, with the professional realm being thought of as being unimaginative. However, the COVID-19 experience forced many of us to creatively adapt to a changed work environment without even realizing the creativity that adaptation required. As Professional Council discussed the theme of this symposium and how it applies to professional issues, we identified areas where medical physicists have creative opportunities. As the proverb says, “Necessity is the mother of invention”, or in this case, creativity. As we look to the future of our profession, professional creativity will be necessary to demonstrate our value to our employers, our colleagues, and our patients.

Multi-Disciplinary Scientific Symposium

TRACK 4

WMIS/AAPM Joint Symposium: Novel and Theranostic PET Agents

SU-B-TRACK 4-00

WMIS/AAPM Joint Symposium: Novel and Theranostic PET Agents

Moderators: John Hazle, UT MD Anderson Cancer Center, Houston, TX and Yuan-Chuan Tai, Washington University, Saint Louis, MO

SU-B-TRACK 4-01

T. Hope, UCSF:

Theranostics and PET/MR

SU-B-TRACK 4-02

S. Lapi, University of Alabama at Birmingham:

Novel Theranostics

SU-B-TRACK 4-03

N. Vasdev, University of Toronto:

Developing Novel PET Tracers for Brain Imaging

Theranostics, or the combination of both therapeutics and diagnostics for image guided therapies, is an emerging area of interest toward personalized and highly targeted approach. This joint symposium between the World Molecular Imaging Society (WMIS) and AAPM will highlight the current status and latest technologies in the growing field of theranostics. Dr. Hope will provide an overview of theranostics in PET/MR. Dr. Lapi will describe novel areas of interest in theranostics. Finally, Dr. Vasdev will discuss some current efforts in developing novel PET tracers in the brain. When taken together, these three invited speakers will summarize the state of the art knowledge in theranostics, review the current challenges and potential solutions, and share their perspectives of future directions for research and clinical implementation.

Learning Objectives:

1. Understand the principles of theranostics, the different approaches, and their advantages and drawbacks
2. Learn about emerging areas of clinical and research theranostics implementation
3. Understand the potential of using theranostics for target therapy.

Multi-Disciplinary Scientific Session

TRACK 6

Imaging in Treatment Planning

SU-B-TRACK 6-01

A Fully Automated Approach for Labeling, Contouring, and Treatment Planning of Vertebral Bodies, Including Automatic Verification to Prevent Errors

T Netherton*, C Nguyen, C Cardenas, C Chung, A Klopp, L Colbert, D Rhee, C Peterson, R Howell, P Balter, L Court, MD Anderson Cancer Center, Houston, TX

Purpose: To develop a fully-automated framework that labels vertebral levels, contours vertebral bodies, detects labeling and contouring errors, and generates treatment plans within 10 minutes. Additionally, to ensure that diagnostic and simulation CT-based auto-planning is safe, efficient, and flags potential mistreatments. **Methods:** For labeling vertebral levels(C1-L5), a methodology was developed using two CNNs trained with independent patient populations(n1=897, n2=290). For contouring, UNet++ was trained, validated, and tested using diagnostic and simulation CT scans(n=220). For detection of contouring and labeling errors, 9 overlap, distance, and intensity metric features were calculated using 752 auto-contours from 56 CBCT, PET-CT, diagnostic-CT, and simulation-CT scans. Using these features as inputs, hyper-parameter optimization refined 9 machine-learning algorithms to detect errors. For planning, treatment plans were automatically generated based on auto-contour projections. To validate the entire framework, 60 treatment plans were generated on a new cohort of diagnostic and simulation scans varying in scan length, imaging protocol, presence of surgical implants, and metastatic burden. Clinical acceptability of the plans was evaluated by three radiation oncologists using a 5-point scale. **Results:** Automatic labeling was accurate for 94% of scans and demonstrated mean localization error <2.2mm. Mean Dice-Similarity-Coefficient/95th-percentile Hausdorff distance was 85.0%/1.8mm(cervical), 90.3%/1.8mm(thoracic), and 93.7%/1.6mm(lumbar). A random forest model detected contouring and labeling errors across various CT scan types with precision-recall-AUC=0.82. Within the validation cohort, the random forest model flagged all contouring and labeling errors (11/11) within prescribed treatment regions, including errors from patient plans (6/6) with atypical anatomy (e.g. T13, L6). For patients with typical anatomy, radiation oncologists scored 98% of simulation CT- and 92% of diagnostic CT-based plans as clinically acceptable or needing minor edits. End-to-end treatment planning time was 7.0±2.0 minutes. **Conclusion:** We demonstrated that this novel, end-to-end, fully automated process to label, contour, and plan was efficient, effective, and safe across various CT scan types.

I am a member of the Radiation Treatment Planning Assistant team at UT MD Anderson

Funding for this work was provided by the Larry Deaven PhD Fellowship in Biomedical Sciences

SU-B-TRACK 6-02

AI Agent Competing with Human in Planning Challenge: A Human-Expert Level Virtual Treatment Planner Network for Prostate Cancer Stereotactic Body Radiation Therapy (SBRT)

C Shen^{1*}, C Men², Y Gao¹, L Chen¹, X Jia¹, (1) University of Texas Southwestern Medical Center, Dallas, TX, (2) Elekta, Newbury Park, CA

Purpose: Virtual treatment planner network (VTPN) has been previously developed to operate treatment-planning systems (TPSs) and adjust planning parameters to generate a plan in lieu of human planners. Previous applications were limited to relatively simple planning tasks on unrealistic in-house TPSs. This study reports the development of VTPN that is able to operate a realistic TPS for a real clinical task of prostate cancer SBRT treatment planning. We evaluate it objectively against human performance in the case in the planning challenge of ProKnow 2016 AAMD/RSS Plan Study. **Methods:** Due to technical challenge of directly interacting with a commercial TPS, we built a realistic TPS equivalent to a commercial TPS in terms of inverse planning. It uses the actual dose engine in the real TPS, and solves the identical optimization problem defined by the user-adjustable dose-volume constraints. A hierarchical VTPN was constructed to perform multi-level decision-making in planning process to adjust constraints to improve plan quality. We trained VTPN via end-to-end deep reinforcement learning with human experience as guidance. The established VTPN was applied to the case in the planning challenge of ProKnow 2016 AAMD/RSS Plan Study to generate a 7-field intensity-modulated radiotherapy plan. The resulting plan quality was compared against top plans in the challenge generated by expert human planners. **Results:** VTPN was successfully trained to operate the TPS, achieving an average plan score of 139.8 on validation cases. When applying the trained VTPN to plan for the case used in the planning challenge, it achieved a plan score of 140.5, ranked as the third highest score among all the 31 plans submitted for competition. **Conclusion:** VTPN can spontaneously learn planning strategies and handle complex prostate cancer SBRT treatment planning. This is the first time that VTPN achieves performance similar to expert human planners.

SU-B-TRACK 6-03

Evaluating the Clinical Acceptability of Deep Learning Contours in Prostate Radiotherapy

J Duan*, M Bernard, B Willows, L Downes, W Mourad, W St Clair, X Feng, Q Chen, University of Kentucky, Lexington, KY

Purpose: To comprehensively evaluate whether contours generated from Deep learning (DL) model can be safely used clinically in prostate radiotherapy. **Methods:** A 3D U-Net based model was trained on 84 patients and an additional 23 patients with intact prostate were randomly selected for testing. Structures that were used in clinical treatment were further review and modified by an experienced radiation oncologist (RO) as ground-truth (GT). Contours generated by the DL model (AI) were compared with GT contours using dice similarity coefficient (DSC), 95% Hausdorff distance (HD95), and mean surface distance (MSD). To quantify interobserver variability, prostate was recontoured by additional two expert ROs in every case. GT and AI contours were evaluated by the 4th RO in a double-blind study. Finally, Varian RapidPlan was used to create plans using AI or GT contours. Then both plans were evaluated on GT contours to see if the use of AI contour would produce a comparable plan. **Results:** AI demonstrates good geometrical accuracy on prostate with mean DSC, HD95, and MSD of 0.83 ± 0.05 , 0.60 ± 0.18 cm, and 0.21 ± 0.07 cm respectively. It shows no significant differences between the contours generated from three ROs ($p > 0.05$). Organ at Risk (OARs) also showed good agreement with the GT. In the double-blind study, 95.7% of AI contours were scored as either "Great" (34.8%) or "Acceptable without changes" (60.9%). Totally 69.6% (16 of 23) AI contours were considered equal or better than their counterparts. No significant differences were found in OAR dose from the use of AI instead of GT contours except for bladder and Seminal Vesicle ($P < 0.05$). However, all plans generated with AI contours satisfy the RTOG-0815 dose constraints even when evaluated using GT contours. **Conclusion:** The investigated 3D U-Net model provided reasonable contours that can be used directly in clinics.

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SU-B-TRACK 6-04

Managing Tumor Changes During Fractionated Radiotherapy: Evaluation on a Deep-Learning-Based Predictive Model and a Re-Planning Strategy

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Purpose: We propose a treatment planning framework that accounts for weekly lung tumor shrinkage using CBCT images with a deep learning-based model. **Methods:** Sixteen patients with non-small cell lung cancer were reviewed with one planning CT and six weekly CBCTs each. A convolutional long short-term memory-based model was applied to predict the weekly deformation of the primary tumor based on the spatial and temporal features extracted from previous weekly CBCTs and target contours. Starting from Week 3, the tumor contour at Week N was predicted by the model based on the input from all the prior weeks (1, 2, ..., N-1), and was evaluated against the manually contoured tumor using dice coefficient (DSC) and average surface distance (ASD). Information about the predicted tumor was then used to re-optimize the treatment plan every week. The objectives were set to maximize the dose coverage in the target region while minimizing the toxicity to the surrounding OARs. Dosimetric evaluation of the target and organs at risk was performed on four cases, whereby we compared the conventional plan (ignoring shrinkage) to the shrinkage-based plan. **Results:** Average DSC and ASD between the predicted tumor and the actual tumor for Week 3, 4, 5, 6 were (0.81, 0.80, 0.76, 0.78) and (1.72, 1.64, 2.04, 1.97) mm, respectively, which are significantly superior to the scores of (0.70, 0.68, 0.66, and 0.63) and (2.81, 3.22, 3.69, and 3.74) mm between the actual and original volume. While target coverage metrics were maintained for the re-optimized plans, lung mean dose dropped by 2.32, 2.38, 0.36 and 1.25 Gy for four sample cases when compared to the original plan. **Conclusion:** We developed a deep-learning-based model for tumor shrinkage prediction. The proposed framework produces reasonable tumor contours and adaption can maintain target coverage while reducing the mean lung dose.

SU-B-TRACK 6-05

Mapping Motion-Induced Dose Variations on Bladder Surfaces: A Novel Surface-Based Dose Modeling for Accurate Dose Assessment in Fractionated Prostate Cancer Radiotherapy

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Purpose: To introduce a novel surface-based dose mapping method for improved quantitative bladder dosimetric assessment in prostate cancer (PC) radiotherapy. **Methods:** Based on the planning and daily pre- and post-fraction MRIs of 12 PC patients, bladder surface models (SM) were generated on manually delineated contours and regionally aligned via surface-based constrained harmonic registration. Subsequently, bladder surface dose models (SDM) were created using face-wise dose sampling. This was followed by localized surface mapping of bladder intrafractional and interfractional dose variation on fractional SDMs. Discrepancies between the received dose, accumulated from daily SDMs, and the planned dose were then assessed on the corresponding SDMs. Complementary to the surface dose mapping, dose-surface-histograms (DSHs) based comparisons were also performed. **Results:** The intra-fraction motion substantially altered dose to mid bladder body, but not the bladder surface areas distal to or contiguous with the target. The overall intra-fraction comparisons, after multiple comparison corrections, are significant (< 0.0001) for both the SDMs and DSHs. A similar pattern of dose variations was also detected by inter-fraction comparisons. With surface registration to the common SM, the cumulative bladder dose significantly differs from the planned dose. The discrepancy is evident in the mid-posterior range that corresponds to a mid-to-high dose region. The received DSH significantly differs from the planned

DSH after permutation correction ($p=0.0122$), and the overall surface-based comparison after multiple comparison correction revealed a trend of difference ($p=0.0800$). **Conclusion:** We developed a novel surface-based intra- and inter-dose mapping framework, which was applied to a unique MR dataset for planning and guidance. The framework identified the pattern of inter-fraction dose changes, localized significant intra-fraction dose variations, and quantified planned vs. received dose differences on the bladder surface. The result indicates the importance of adopting the motion-integrated bladder SDM for bladder dose management.

SU-B-TRACK 6-06

Towards Daily Online Adaptive SBRT for Pancreatic Cancer: Dosimetric Analysis of Daily Adaptive Re-Planning Using Re-Optimization and Dose Escalation Techniques

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Purpose: To evaluate the benefit of daily adaptive re-planning for pancreas SBRT by using either daily plan re-optimization or fractional dose-escalation to attain maximal tumor coverage. **Methods:** Eight pancreatic cancer patients, originally treated with SBRT (36–40 Gy, 5 fractions) using daily CT-on-rails image guidance and breath-hold motion management, were re-planned on the daily treatment CT-on-rails images using 2 different techniques: (i) all patients had each of their 5 fractional plans re-optimized to preserve clinical target coverage and meet OAR constraints; and (ii) all patients were re-planned on each fraction to receive dose-escalated SBRT, using an iterative strategy that consisted of a dose-escalated PTV (DE-PTV), as well as simultaneous integrated boosts to both the tumor-vessel-interface (TVI) and gross-tumor-volume (GTV), while meeting organ-at-risk (OAR) constraints. Both the original and dose-escalated treatment plans, using the original planning CT, were recalculated on each treatment's CT-on-rails image and accumulated dose was calculated. Additionally, dose was accumulated using the 5 daily re-plans, for both strategies, and then compared to the clinically delivered and the dose-escalated treatment plans. **Results:** When compared to the accumulated dose from the original clinical plan, daily re-optimization resulted in an average 7.1% /6.7% increase in D100% of the GTV/TVI ($p=0.17/0.12$). Compared to the accumulated dose from the dose-escalated treatment plan, the daily adaptive dose-escalation approach had an average 10.7%/21.2%/20.4% increase in D100% for the GTV/TVI/DE-PTV ($p=0.20/0.02*/0.03*$), respectively. The daily adaptive dose-escalated approach was able to deliver an average 22.4%/32.8% increase to the GTV/TVI ($p=0.007/0.0001*$), when compared to the delivered dose of the original clinical plan. Without daily adaptation, dose-escalated plans would violate the D1.0cc<35Gy OAR constraint in 5/1/1 patients for the duodenum/stomach/small bowel, according to accumulated dose. **Conclusion:** Daily re-optimization of pancreas SBRT treatments leads to superior plan quality. Daily adaptive re-planning can maximize dose escalation, without violating OAR constraints.

SU-B-TRACK 6-07

Online Adaptive Radiotherapy: Assessment of Planning Technique and On-Table Adaptive Plan Quality

K Mittauer^{1,2*}, J Bryant¹, A Gomez¹, T Romaguera^{1,2}, D Alvarez^{1,2}, J McCulloch^{1,2}, R Herrera¹, D Doty¹, M Mehta^{1,2}, A Gutierrez^{1,2}, M Chuong^{1,2}, (1) Miami Cancer Institute, Baptist Health South Florida, Miami, FL, (2) Florida International University, Herbert Wertheim College of Medicine, Miami, FL

Purpose: Stereotactic body radiation therapy (SBRT) planning based on a static dataset of the simulation day anatomy is historically created over at least a week. We hypothesize that the quality of daily on-table adaptive plans is similar to that of the baseline plan when delivering stereotactic magnetic resonance image guided adaptive radiation therapy (SMART) for pancreas cancer (PCa) on a 0.35T MR Linac. **Methods:** Fifty ($n=50$) PCa patients received 50 Gy/5-fraction SMART. Baseline planning included: 3 mm gastrointestinal (GI) PRV, 50 Gy/100% and 33 Gy/66% optimization targets based on GI PRV, high-dose and low-dose conformity rings, and a contracted GTV for ablative 120%-140% region. For each adaptive fraction, GI anatomy was re-contoured, followed by propagation of planning structures and optimization. If conformity, coverage, or organ at risk (OAR) goals were not achieved, optimization was repeated until satisfactory. Plan quality was evaluated for target coverage (TC=PTV V100%/PTV vol), PTV D90% and D80%, homogeneity index (HI=PTV D2%/D98%), high-dose conformity (PITV), low-dose conformity (D2cm), and gradient (R50%). **Results:** OAR constraints were met on all baseline ($n=50$) and adaptive ($n=250$) plans. Mean $\pm$ SD PTV coverages for baseline and adaptive plans were 41.7 $\pm$ 6.3 Gy and 40.5 $\pm$ 6.3 Gy (D90%), 45.6 $\pm$ 5.6 Gy and 44.4 $\pm$ 5.5 Gy (D80%), and 76 $\pm$ 20% and 70 $\pm$ 20% (TC), respectively, all not statistically significant (NSS). Mean homogeneity and conformity indices were 0.84, 0.76 (PITV), 3.37, 3.43 (R50%), 1.54, 1.57 (HI), and 58%, 60% (D2cm), respectively, all NSS. Median number of adaptive re-optimizations was 2. **Conclusion:** For a site with GI interfraction motion causing daily changes in the spatial position of the dosimetric gradient, similar target coverage quality was observed between baseline and adaptive plans due to robustness of GI PRV technique subdividing the target. Planning rings enabled robustness in conformity. This work supports the feasibility towards utilizing a patient-independent template baseline plan for daily adaptation.

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Imaging Interactive ePoster Discussion

TRACK 1

Imaging: New and Emerging Technologies

SU-lePD-TRACK 1-01

Electroacoustic Tomography for Monitoring Electroporation in Real Time

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Purpose: Electroporation is a new technique that uses nanosecond to microsecond duration high-voltage electric pulses to create temporary pores in cell membranes or irreversibly destroys target cells for cancer treatment. The versatility of electroporation makes the technique ideal for direct drug delivery and cancer treatment. Electroacoustic tomography (EAT) is a novel imaging modality for electroporation treatment monitoring, that utilizes the existing energy deposited by high-voltage electrical fields created in the electroporation process. **Methods:** In EAT, ultrasound waves are generated by the deposited electric potential energy and captured by different configurations of ultrasound transducers. After being captured, the ultrasound information is used to reconstruct the geometric information of the electric field distribution in the target medium. In this work, we demonstrate the feasibility of electroacoustic tomography (EAT) by creating a 2D reconstruction of an electric field in the water. A custom high-voltage electric pulse generating device was made for delivering nanosecond duration pulses in water through two electrodes. A 500Khz ultrasound transducer was placed near the electrodes and rotated to capture the acoustic information from 180 positions. The corresponding 2D reconstruction of the electric field in water was completed by using filtered back-projection algorithm. **Results:** A 2-dimensional reconstruction of the electric field in water was obtained using the EAT system. The electric field and the electrodes were clearly visualized. **Conclusion:** The successful visualization of electric fields in water demonstrated the feasibility of EAT model for electroporation therapy. As a real-time capable and 3D capable novel imaging modality, EAT has the potential to greatly enhance the precision of electroporation treatment in the clinic.

SU-lePD-TRACK 1-02

Liver Delineation Using the Weighted A-Star Pathfinding Algorithm and Deep Learning on Ultrasound B-Mode Images for Cirrhosis Assessment Using Liver Biopsy as the 'Gold Standard'

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Purpose: To evaluate the diagnostic performance of a deep learning scheme on liver cirrhosis detection. Input data were segmented left liver lobes from ultrasound (US) B-Mode images delineated by a segmentation algorithm based on modified Weighted A-Star algorithm, on patients with Chronic Liver Disease (CLD) using liver biopsy (LB) as 'Gold Standard'. **Methods:** 69 consecutive CLD diagnosed patients (14 F0-F3 and 55 F4) underwent a regular abdomen US B-Mode and LB examination using the Metavir scale (F0-F4). A B-Mode image containing the left liver lobe was acquired and saved by an expert Radiologist performing the US examination. Then, a sophisticated segmentation algorithm based on 2D continuous wavelet transform (2D-CWT) and Weighted A-Star pathfinding was employed to delineate the liver surface boundaries from each patient's US image. Subsequently, a binary image was produced representing the left liver's lobe mask. A radiologist's delineation on the same images was used as 'Gold Standard' for segmentation. The segmentation results were evaluated by the Jaccard's Coefficient Index. Consequently, all mask images were fed to the GoogLeNet pre-trained deep learning scheme using LB fibrosis staging (F0-F4) as labels for fine tuning. Data was separated in two classes according to their steatosis grade: F0-F3 (Non-cirrhotic – Class 1), and F4 (Cirrhotic Class – Class 2). Mask images were separated in training (70%) and testing (30%) sets, and testing accuracies were calculated. The random image separation and GoogLeNet fine tuning processes were repeated 30 times. **Results:** The segmentation algorithm achieved a mean Jaccard Coefficient performance of 0.92 using the radiologist's delineation as reference. The deep learning scheme's mean accuracy reached 97% for the training data and 90% for the test data respectively. **Conclusion:** The results indicate that this scheme can be used as a supplementary diagnostic tool for cirrhosis detection on regular US B-Mode abdomen examination.

SU-lePD-TRACK 1-03

Quantitative Image Guidance In Vivo: Dual-Energy CT for Thermochemical Ablation

E Thompson*, M Jacobsen, R Layman, E Cressman, UT MD Anderson Cancer Center, Houston, TX

Purpose: Thermochemical ablation (TCA) is a minimally invasive therapy under development for hepatocellular carcinoma. TCA utilizes acid/base chemistry delivered simultaneously directly into the tumor to induce local ablation when administered. When delivered via a mixing catheter, acid (AcOH) and base (NaOH) react to completion at the catheter tip before delivery into tumor, producing the acetate salt, water, and releasing heat ($\Delta > 50^{\circ}\text{C}$) in sufficient quantities to induce lethal osmotic and thermal stress in tumor cells. However, these two reagents are not distinguishable from tissues, which makes monitoring delivery of TCA difficult. We address this issue by developing CsOH as a novel theranostic component of TCA that is detectable with CT. **Methods:** VX2 tumor fragments (0.3mL) from carrier animals were inoculated into the flank of New Zealand white rabbits and allowed to grow for 10 days until tumor diameter reached 1-2cm. Catheters were placed under ultrasound guidance and TCA was delivered as 5M AcOH and 5M NaOH doped with 250mM CsOH. Concentration selection was informed by previous studies. A split-filter DECT system (Definition Edge, Siemens Healthineers) was used to acquire images pre- and post-TCA treatment with the following parameters: 120kVp, Au/Sn filter, and CTDIvol=1.39mGy. 20mm line profiles were drawn on 70keV monoenergetic images (Syngo.via) through the ablation center (ImageJ, NIH). Cesium concentration was quantified using the HU and CsOH concentration relationship ($R^2=0.999$) determined in previous studies. **Results:** The resulting line profile has a maximum CT number of 692HU (220mM) near the center of the injection site and decreases to approximately 40HU (12mM) at the periphery of the ablated region, demonstrating higher cesium concentration near the center, decreased concentration at the periphery, and none beyond the ablation zone. **Conclusion:** DECT is a viable technique for quantitative monitoring to monitor tumor ablation when CsOH is incorporated as a theranostic agent in image-guided TCA interventions.

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SU-lePD-TRACK 1-04

Recurrence Prediction for Head and Neck Squamous Cell Cancer Patients Using Local Binary Pattern-Based Dosimetrics

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Purpose: Our aim was to predict the recurrence after radiation therapy by using local binary pattern (LBP)-based dosimetrics for head and neck squamous cell cancer (HNSCC) patients. **Methods:** A total of 131 HNSCC patients has recurrence/non-recurrence information after radiation therapy were collected from HNSCC collection in The Cancer Imaging Archive. The cases were split into a training (80%) and a test (20%) dataset. The numbers of recurrence and non-recurrence were 39 (training: 29, test: 10) and 92 (training: 73, test: 19), respectively. Recurrence versus non-recurrence cases were imbalanced. Therefore, the recurrence group in the training dataset was balanced via the synthetic minority over-sampling technique method. A total of 327 dosimetrics features enrolled cold spot volume, histogram features, and texture features (gray-level co-occurrence matrix, gray-level run length matrix, gray-level size zone matrix, neighborhood gray tone matrix) were extracted from the original dose distribution (ODD) and LBP on gross tumor volume, clinical target volume, and planning target volume. The Cox-net algorithm was employed for feature selection and dosimetrics signature building in training dataset. The ODD and LBP models using support vector machine for the recurrence prediction were constructed with dosimetrics features extracted from ODD and LBP. The prediction performances were evaluated in training and test datasets by the accuracy and the area under the receiver operating characteristic curve (AUC). **Results:** The accuracy and AUC in the training dataset were 92% and 0.95, respectively, for the LBP model and 77% and 0.81, respectively, for the ODD model. In the test dataset, the LBP model demonstrated the higher performance (accuracy: 79%, AUC: 0.72) than the ODD model (accuracy: 48%, AUC: 0.71). **Conclusion:** The LBP-based dosimetrics (LBP model) could be feasible for accurately prediction of recurrence after radiation therapy in HNSCC patients.

SU-lePD-TRACK 1-05

Functional MRI-Based Radiomics for the Diagnosis of Major Depressive Disorder

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Purpose: We developed a functional magnetic resonance imaging (fMRI)-based radiomics model for the diagnosis of major depressive disorder (MDD). **Methods:** A total of 146 participants (84 healthy controls, 62 MDD patients) were randomly divided into a training set ($n = 110$) and a validation set ($n = 36$) using stratified sampling. The resting-state fMRI images were preprocessed, including co-registration, correction, normalization and segmentation, using public toolbox DPABI in MATLAB, resulting in three amplitude of low-frequency fluctuation (ALFF) images, three fractional ALFF (fALFF) images, and four regional homogeneity (ReHo) images for each participant. We extracted 1118 radiomics features from each preprocessed images using an open-source python package Pyradiomics, including first-order statistical descriptors, high-order texture features, wavelet features, and Laplacian of Gaussian (LoG) features. The Shapiro-wilk test was used to calculate the normality of data. The differences between healthy controls and MDD patients in normally distributed features were calculated using two-sample t-test, otherwise, two-sided Wilcoxon rank sum test. All statistical analyses were performed by using SPSS software. The features with $P < 0.05$ were enrolled into least absolute shrinkage and selection operator (LASSO) regression model for further feature selection. The most valuable

features selected by LASSO were used for a support vector machine (SVM) model. **Results:** Among the ten models that were built based on different fMRI images processing methods, three models performed best in classifying healthy controls and MDD patients, including ALFF, mfALFF (i.e., each voxel's fALFF was divided by the global mean fALFF), and ReHo models. The ALFF model outperformed mfALFF and ReHo (accuracy: 86% and 83%, area under curve [AUC]: 0.88 and 0.88, sensitivity: 75% and 75%, specificity: 92% and 87.5%) models with an accuracy of 100% (AUC: 1). **Conclusion:** This study demonstrates that the fMRI-based radiomics model can provide favorable predictive efficacy for the diagnosis of MDD.

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SU-lePD-TRACK 1-06

Uniqueness of Radiomic Features in NSCLC: Beyond Reproducibility/Repeatability

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Purpose: The uniqueness of radiomic features, combined with their reproducibility, determines the success of radiomic studies. In general, radiomic features extracted from a defined region of interest (ROI) is hypothesized unique to the ROI. This study is to investigate if the hypothesis is valid. **Methods:** Two cohorts of NSCLC patients were retrospectively retrieved from GE and Siemens CT scanners. The lung nodule (defined region) was delineated manually and radiomic features were extracted using IBEX. The same ROI was then translated randomly to four other tissue regions of the same image set: adipose, heart, lung beyond nodule, and muscle. The same features were extracted from the four tissue regions for comparison. Coefficient of Variation (CV) was calculated to test variation of features within different ROIs. The Concordance Correlation Coefficient (CCC) between radiomic features of the defined region (lung nodule) and a given tissue region were calculated to determine feature uniqueness. Identifying features as non-unique follows this process: 1) A feature is flagged within a tissue region when the CCC > 0.85 for ≥ 50% of patients; then 2) a feature is identified as non-unique when ≥ 2 tissue regions are flagged for that feature. **Results:** We analyze 14 patients from GE and 18 patients from Siemens. Of 123 features, 18 radiomic features in GE and 16 features in Siemens are identified as non-unique, with 11 features overlapping. The overlap features are features included in the GradientOrientHistogram, and InverseDiffMomentNorm and InverseDiffNorm within the 3D GrayLevelCooccurrence (GLC). The remaining non-unique features for GE are SumAverage from the 3DGLC and features in the Intensity Histogram Gauss Fit. While for Siemens the features are Energy and Entropy for both 2D/3DGLC and MaxProbability for 3DGLC. **Conclusion:** Approximately 10% or more radiomic features are not unique to a defined region. Radiomic feature uniqueness needs to be considered in future radiomics studies.

SU-lePD-TRACK 1-07

Investigation of Bias-Voltage Dependent Operational Characteristics of An Energy-Resolving Pixelated Cadmium Telluride Detector System Within An Experimental Benchtop X-Ray Fluorescence Imaging System

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Purpose: To study the bias-voltage dependent operational characteristics of an energy-resolving pixelated cadmium telluride (CdTe) detector system coupled with a parallel-hole collimator within an experimental benchtop x-ray fluorescence (XRF) imaging system. **Methods:** An experimental XRF CT (XFCT) system was revamped with a commercially available energy-resolving x-ray detector system, known as High-Energy X-ray Imaging Technology (HEXITEC), consisting of 1-mm-thick CdTe sensor with 80 × 80 pixels on a 250-μm pitch. HEXITEC was coupled with a 2mm-diameter aperture parallel-hole collimator. A cone-beam was produced from an x-ray tube operated at 125kVp and 24mA. This cone-beam beam was filtered with 1.8mm Tin and used to irradiate a phantom containing 200-μL of 0.1wt.% gold nanoparticle (GNP) solution placed at 15-cm from the x-ray source. The detector-collimator system was placed at 10cm from the phantom, orthogonal to the excitation beam direction. The Compton/XRF photon spectra from the GNP solution were measured for 60s at different bias-voltages ranging from -100 to -600V. The x-ray spectra from all the detector pixels corresponding to the 2-mm collimator aperture were added to obtain the final spectra. The net XRF signal was extracted using the deconvolution method developed in-house. The gold Kα1 XRF peak at 68.8keV was fitted with Gaussian distribution to obtain the energy resolution (FWHM). The net XRF signals and energy resolution at 68.8keV were used to determine the optimal bias-voltage setting for the benchtop XFCT system. **Results:** As the bias-voltage increased from -100 to -500V, the net gold XRF signal, mean per-pixel energy resolution, and the homogeneity of energy resolution across the collimated 7 × 7 detector pixel area were improved by ~44, ~41, and ~66%, respectively. **Conclusion:** This work investigated the bias-voltage dependent operational characteristics of HEXITEC consisting of 1-mm-thick CdTe sensor, and determined the optimal bias-voltage under typical operating conditions of benchtop XRF imaging/XFCT systems.

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Imaging Interactive ePoster Discussion TRACK 2

Imaging: CT Radiomics and Clinical Applications

SU-lePD-TRACK 2-01

Coronary Computed Tomography Angiography Combined with Coronary Heart Disease Risk Stratification Prediction for Asymptomatic Type 2 Diabetes Mellitus Patients Screening

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Purpose: The clinical significance of coronary computed tomography angiography (CCTA) in asymptomatic type 2 diabetes mellitus (T2DM) patients was not clear. We performed a retrospective study to evaluate a method of coronary heart disease (CHD) screening in asymptomatic T2DM patients by the combination of CCTA and CHD risk stratification prediction. **Methods:** Data of 141 T2DM patients (58±8 years, 57% males) without known symptoms suggestive of CHD and underwent CCTA were retrospectively analyzed. Patients were classified into three groups based on UK Prospective Diabetes Study (UKPDS) CHD risk stratification prediction. Other 76 healthy screening persons underwent CCTA successively were chosen as control group. Difference of CCTA scores among groups were compared using Post Hoc Multiple comparisons. **Results:** Segment involvement score in intermediate risk group (4.6±3.2) and high risk group (5.3±3.3) was significantly higher than control group (2.9±2.9). This indicates that stenosis was more widely distributed in intermediate and high risk groups. Segment stenosis score in intermediate risk group (7.2±7.0) and high risk group (9.2±8.0) was significantly higher than control group (3.5±4.4). Stenosis modulus in intermediate risk group (1.2±0.6) and high risk group (1.5±0.6) was also significantly higher than control group (0.8±0.5). This means stenosis was more severe in intermediate and high risk groups. Percentage of severe proximal plaque in intermediate risk group (14.5%) and high risk group (28.6%) was significantly higher than control group (3.9%), which had been known to be associated with a poor prognosis. **Conclusion:** In the intermediate-high CHD risk groups, the distribution of coronary artery stenosis was more extensive and more severe. Combined with risk stratification prediction, CCTA may be used for screening CHD in intermediate-high risk asymptomatic T2DM patients.

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SU-lePD-TRACK 2-02

Prediction of EGFR Mutations, Subtypes, and Uncommon Mutations in Lung Adenocarcinoma Based On Machine Learning

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Purpose: Molecularly targeted drugs (MTD) have been widely used for patients who have non-small cell lung cancer (NSCLC). The epidermal growth factor receptor (EGFR) tyrosine kinase inhibitor is one of MTD and its therapy has a longer progression free survival (PFS) than conventional chemotherapy. The purpose of this study is to predict EGFR mutation status using machine learning and CT images of NSCLC. **Methods:** This study was consisted of 172 patients with NSCLC who had undergone biopsy or surgical specimen. All NSCLCs were pathologically confirmed adenocarcinoma and EGFR mutation status (whether EGFR mutations or wild-type). Then, EGFR mutations were confirmed common or uncommon mutations. Furthermore, common mutations were confirmed subtypes whether 19del or L858R or others. Patients data were divided into training and test data. Total 1048 radiomic features were extracted, then radiomic features and 4 clinical features were used to construct the three prediction models (EGFR mutations vs. wild-type, common vs. uncommon mutations and 19del vs. L858R). Lasso was used to select features and two machine learning algorithms (support vector machine (SVM) and logistic regression (LR)) with five-fold cross-validation were applied. The prediction performance of each model was evaluated in terms of the AUCs in training and test data. **Results:** For the training data of EGFR mutations vs. wild-type, AUC of SVM and LR were 0.85 and 0.84. For the test data, AUCs were 0.78 and 0.77. For the training data of common vs. uncommon mutations, AUC of SVM and LR were 0.84 and 0.84. For the test data, AUCs were 0.82 and 0.79. For the training data of 19del vs. L858R, AUC of SVM and LR were 0.82 and 0.84. For the test data, AUCs were 0.71 and 0.67. **Conclusion:** The constructed models were relatively high performance. These may be useful in selecting appropriate MTD.

SU-lePD-TRACK 2-03

A Deep Learning-Based Radiomics Approach to Identify Patient with Early Tumor Regression Utilizing Planning CT Images for Adaptive Radiotherapy

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Purpose: We developed a deep learning-based radiomics (DLR) approach to identify early tumor regression in head and neck (H&N) cancer patients based only on planning computed tomography (CT) images for adaptive radiotherapy. **Methods:** Our study used CT images of 62 patients with H&N cancer who received radiation therapy. We calculated the regression rate of gross tumor volume primaries (GTVp) and nodal GTV (GTVn) per treatment day by subtracting the volume of boost CT from the volume of initial CT, and then dividing the value obtained by the number of treatment days. Subsequently, we used the median regression rate as a threshold and assigned two labels (change and non-change) for GTVp and GTVn to all patients. We extracted one axial image for each maximum GTVp and GTVn volume from the initial CT image, which was then loaded into the pre-trained deep learning (DL) model via transfer learning. Eventually, we succeeded in extracting several deep features from the last fully connected layer. The robust, non-redundant, and predictive deep features were then selected using four-feature selection metrics. The selected deep features were used to predict change and non-change of GTVp and GTVn in five machine learning (ML). A predictive performance was then evaluated by 0.632+ area under the curve (AUC) bootstrap. **Results:** Darknet DL and the Random forest (RF) ML model for predicting GTVp regression had the highest AUC values (0.89 [0.885–0.894, 95% confidence interval (CI)]), whereas values achieved from radiomics and clinical factors were 0.723 and 0.66, respectively. In addition, Densenet DL and RF ML model for predicting GTVn regression achieved the highest AUC values (0.86 [0.856–0.866, 95% CI]), whereas values achieved by radiomics and clinical factors were 0.723 and 0.66, respectively. **Conclusion:** Our DLR approach to identify patients with early tumor regression could outperform existing radiomics features and clinical factors.

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SU-lePD-TRACK 2-04

Classification of Head and Neck Contours Using Extremely Randomized Trees

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Purpose: To train a classification model that provides a class-probability estimate for labeling delineated Head and Neck (HN) organs at risk (OARs). **Methods:** A knowledge-based quality control system composed of an extremely-randomized-trees classifier (ERT) model trained using segmentations from 234 publicly available CT Head and Neck image-sets were used to solve an Organ-at-Risk (OAR) classification problem. The feature vector for each OAR comprises shape, image intensity, location, and orientation features. The dataset was augmented using SMOTE. 5-fold stratified cross-validation was used to obtain learning curves for ERT k-nearest neighbors vote (KNN) and Dummy Classifiers (DC) for a random classification baseline. Each model was evaluated using a stratified 5-fold iterator with different randomization in three repetitions. ERT output was further derived from the probability of an OAR being an HN organ. A probability threshold was selected that maximizes the detection recall, consequently minimizing the fraction of false negatives. The dataset contains 28 OAR delineations. The model was optimized to classify 12 common HN structures, and the remaining ones were kept as hold-out OARs. **Results:** The cross-validated recall was DC 0.09±0.02, KNN 0.98±0.01, and ERT 0.994±0.005. The training curves for ERT showed convergence after a sample size of around 500 OARs, showing excellent recall scores in both train and validation samples as a low probability of overfitting. The mean classification probability for in-sample and hold-out OARs were respectively 0.97±0.07 and 0.41±0.08. A minimum classification probability of 0.77 was obtained using grid-search, resulting in a global 0.96 recall score **Conclusion:** ERT classifies HN OARs with high recall and provides superior results compared to KNN and DC. ERT reduces the likelihood of ROI delineation mislabeling and might detect aberrant delineations. It enables mass ROI name standardization with low miss-classification errors for data-mining of heterogeneous historical priors.

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SU-lePD-TRACK 2-05

Evaluation of Computer-Aided Nodule Assessment and Risk Yield (CANARY) in Korean Patients for Prediction of Invasiveness of Early Lung Adenocarcinoma

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Purpose: Discriminating the invasiveness of early lung adenocarcinomas clinically important, and several institutions have developed their own approaches

by using computed tomography images of patients. The purpose of this study is to evaluate Computer-Aided Analysis of Risk Yield (CANARY), a validated virtual biopsy and risk-stratification machine-learning tool for lung adenocarcinomas, in a Korean patient population. **Methods:** A total of 380 lesions were reviewed from 360 patients who underwent lung resection between January 2018 and July 2020. The lesions were classified as "indolent" (atypical adenomatous hyperplasia, adenocarcinomas in situ, or minimally invasive adenocarcinoma) or "invasive" (invasive adenocarcinoma) by following the Score Indicative of Lung Cancer Aggression (SILA) from the CANARY analysis results, and compared with the pathology report. **Results:** The mean age of the patients was 63 ± 9.8 years, and 219 (60.9%) were female. Each of the area under the curve (AUC) from the entire cohort and a dataset upsampled for an even distribution of two cases were 0.814 and 0.809, while the CANARY developer group announced the AUC of 0.912 with the cut-off SILA of 0.338. At the optimal threshold from our data, which was SILA of 0.229, CANARY found 47 out of 65 (72.3%) indolent cases and 242 out of 315 (76.8%) invasive cases. **Conclusion:** The AUCs of CANARY in Korean patients were considered excellent even though there was a meaningful difference in the optimal thresholds from the developer group. We found that adjusting the cut-off threshold is needed to use CANARY in other countries or races.

SU-lePD-TRACK 2-06

Development of Machine Learning Based Algorithm for Prediction of Invasiveness of Early-Lung Adenocarcinoma by Using Chest Computed Tomography

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Purpose: To locate and classify lung adenocarcinoma lesions, which are also called ground-glass nodules (GGN) from computed tomography (CT) images, are clinically helpful in determining the treatment plans and the extent of resection before surgery to lung cancer patients with high mortality rates. Herein we report a machine learning-based auto-detection and classification model of early lung adenocarcinoma by using chest CT images. **Methods:** An image dataset of 190 chest CT scan datasets from the same number of patients who received the pulmonary resection surgery was used to train and validation of the model. To detect and the GGNs and distinguish them into two large groups: "non-invasive" (including atypical adenomatous hyperplasia, adenocarcinoma in situ, and minimally invasive adenocarcinoma) or "invasive" (invasive adenocarcinoma), the two-step combination model was used in this study. First, Fully Convolutional DenseNet was applied to train the GGN segmentation model from the CT images. Then the true positive cases were categorized into one of two groups by using radiomics, a method of comprehensively classifying the features of tumors by quantifying large amounts of image features. **Results:** The mean age of the patients was 62 ± 9.6 years, and 122 (64.2%) were female. The segmentation model showed a dice score of 0.65, a true positive rate of 0.88 (36 out of 41), and averaged 1.5 false positives per case. Among 36 GGNs that the segmentation model detected, the area under the curve of the classification model scored 0.90. **Conclusion:** Our preliminary machine learning-based two-step GGN segmentation and classification model required management of false-positive cases but showed excellent performance on the classification for true positive cases. Therefore, our model could be expected to be of great help in clinical use if the false positives are reduced.

SU-lePD-TRACK 2-07

Multi-Scale Regional Fusion Based Interactive 3D Tumor Segmentation in Lung Cancer

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Purpose: Accurate lung tumor segmentation in computed tomography (CT) images is of great importance in diagnosis and target identification in lung cancer treatment. Most current segmentation methods require a large scale labeled data to train, while it is always difficult and time consuming to acquire labeled data. We aim to develop a new multi-scale regional fusion based interactive method for three-dimension (3D) tumor segmentation (MFIS). **Methods:** Since no data is needed to train the model, seven patients collected from Peking University Cancer Hospital (Beijing, China) are used for testing. The manually contoured tumor by an experienced radiologist is used as ground truth. MFIS consists of three steps: (1) normalization. To reduce the potential influence of image acquisition protocols on tumor segmentation, CT images are normalized through min-max normalization. (2) 3D tumor block extraction through interactive information. The interactive markers such as a simple line are drawing on 2D images to indicate the tumor region. (3) Multi-scale regional fusion. Three phases are included in this step, they are multi-scale super-voxel pre-segmentation to divide the 3D image into a number of super-voxel regions in first phase. Then Super-voxel region merging is performed to merge the region with maximal similarity as the tumor. Finally, multi-scale average fusion is performed to improve segmentation accuracy through capturing both local and global contextual information. Dice coefficient (Dice), true positive rate (TPR) and false positive rate (FPR) are used for evaluation. The ideal segmentation is higher Dice and TPR, while lower FPR. **Results:** The quantitative evaluation results for MFIS on Dice, TPR and FPR are 0.8821, 0.0052, and 0.8943, respectively. **Conclusion:** A new multi-scale regional fusion based interactive method for three-dimension (3D) tumor segmentation (MFIS) in lung cancer was developed. The experimental results demonstrated that MFIS can obtain better segmentation performance in a more efficient way.

Multi-Disciplinary Interactive ePoster Discussion

TRACK 3

Multi-Disciplinary: Segmentation I

SU-lePD-TRACK 3-01

Quantitative and Qualitative Evaluation of Translating an Adult Auto-Contouring Algorithm to Pediatric Patients

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Purpose: To perform a quantitative and qualitative assessment of using an adult auto-contouring algorithm to generate pediatric normal tissue contours. **Methods:** A clinically deployed, adult head-and-neck (HN) auto-contouring model was originally trained to delineate 16 normal tissue structures on 3,495 retrospective, adult HN patients. This work tested the model's ability to segment the same 16 structures on 143 retrospective pediatric CSI CT scans. Dice similarity coefficient (DSC) and Hausdorff distance (HD) were computed for each of the 16 HN structures. The DSC and HD were compared to the metrics achieved on the adult data set, using an unpaired t-test for each structure ($p < 0.05$ as statistically significant). To assess the clinical acceptability of the contours, we conducted a physician review using 20 patients and a scoring system of 1=no edits, 2=minor edits, and 3=major edits. The clinical acceptability of the pediatric contours was compared to what was achieved on the adult contours. **Results:** The average $DSC \pm 1\sigma / HD \pm 1\sigma$ (cm) achieved on the brain, brainstem, eyes(avg), lenses(avg), optic nerves(avg), parotids(avg), cochleae(avg), spinal cord, esophagus and optic chiasm, were $0.98 \pm 0.01 / 0.97 \pm 1.51$, $0.80 \pm 0.08 / 0.97 \pm 0.76$, $0.87 \pm 0.53 / 0.46 \pm 4.4$, $0.65 \pm 0.12 / 0.51 \pm 4.10$, $0.61 \pm 0.17 / 0.37 \pm 4.97$, $0.60 \pm 0.23 / 1.17 \pm 8.10$, $0.53 \pm 0.17 / 0.48 \pm 0.24$, $0.67 \pm 0.15 / 7.40 \pm 7.10$, $0.57 \pm 0.13 / 3.31 \pm 1.40$, and $0.34 \pm 0.16 / 0.89 \pm 5.40$, respectively. The difference between the adult and pediatric DSC values was not statistically significant. The difference between the HD of the esophagus was statistically significant ($p = 0.0001$). Physician review scored 8/16 pediatric and 14/16 adult contours as clinical acceptable (no or minor edits), respectively. Clinically unacceptable pediatric structures included the lenses (5%), optic nerves (5%), optic chiasm (42%), parotids (55%), mandible (5%), and esophagus (11%). The number of structures requiring major edits for patients <2 and >2 years old was 6 and 1, respectively.

Conclusion: Our results suggest that apart from very young patients (<2 years old), adult auto-contouring algorithms can automatically generate clinically acceptable contours on pediatric patients for a majority of HN structures.

SU-lePD-TRACK 3-02

Multi-Organ Segmentation on An Inference-Combined Dataset

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Purpose: To evaluate the performance of a single multi-organ segmentation by a convolutional neural network trained on a combination of two publicly available datasets of CT images. **Methods:** 88 CT images with segmentations of the liver, left kidney, spleen, gallbladder, esophagus, stomach, pancreas, and duodenum and 83 segmentations of the liver, bladder, and left and right kidneys are prepared from two different publicly available datasets. Five segmentations from the first set and eight from the second are excluded from training and used for validation. A DenseVNet is trained on each dataset, and inference performed on the other to generate segmentations of the missing organs. A combined dataset of 121 segmentations (including validation) is used to train a network to segment all 10 abdominopelvic organs. Dice similarity coefficients are used to evaluate accuracy, with a T-test used to identify significantly different accuracies. **Results:** For the liver and left kidney, differences in the average dice coefficients between segmentations are largely explained by differences in ground-truth segmentations provided by different observers. For organs inferred for part of the combined dataset, average dice scores on the validation set tend to decrease when trained on the combined dataset. However, for the spleen, esophagus, and stomach the difference is negligible, indicated by a very large p-value (>0.5) from a T-test on the dice coefficients. For other organs these p-values range from 0.005 to 0.16. **Conclusion:** By combining two datasets, the robustness of a multi-organ segmentation network can be improved by providing a more varied set of images and segmentations produced by more human observers. For more easily segmented organs, the accuracy of a multi-organ network may be preserved even when these organs are inferred for part of the dataset. For smaller, more difficult organs, using several dedicated networks may be necessary to achieve best results.

SU-lePD-TRACK 3-03

Segmentation of a Highly Deformable Organ Using a Machine-Assisted Interpolation Algorithm

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Purpose: Accurate and robust auto-segmentation of highly deformable organs, e.g. stomach, remains an outstanding problem due to these organs' frequent and large anatomical variances. We propose a machine-assisted interpolation (MAI) that uses prior information, in the form of sparse manual delineations, to auto-segment the stomach in low field magnetic resonance images (MRI). **Methods:** Images from 96 patients undergoing 0.35T MRI-guided abdominal radiotherapy were collected. For each patient volume, the manual delineation of the stomach was extracted from every 8th slice. These manually drawn contours were first interpolated to obtain a rough segmentation mask of the stomach. A 64x64 pixel patch-based convolutional neural network (CNN) was first used to localize the position of the organ's boundary on each slice within a 5-pixel wide road. This boundary prediction was then used as an anatomical constraint for a second CNN, which output the final stomach segmentation. Both networks used a Dense-UNet architecture and were trained, validated, and tested on 79, 7, and 10 patients respectively. Algorithm performance was compared against linear interpolation and de novo auto-segmentation using a Dense-UNet trained on the same dataset. **Results:** MAI resulted in a mean 50th Hausdorff Distance (50HD) of 1.6 ± 1.1 mm and mean Dice Similarity Coefficient (DSC) of 0.91 ± 0.07 with 6% of slices with DSC<0.8. De novo auto-segmentation resulted in 50HD of 4.7 ± 11.8 mm and DSC of 0.80 ± 0.24 with 22% of slices having DSC<0.8. Linear interpolation resulted in 50HD of 2.7 ± 2.3 mm and DSC of 0.86 ± 0.12 with 26% of slices having DSC<0.8. Differences in 50HD, DSC, and fraction of slices with DSC<0.8 were all statistically significant between MAI and comparison methods ($p<0.001$). **Conclusion:** The proposed machine-assisted interpolation algorithm significantly outperformed de novo segmentation and linear interpolation in terms of accuracy and robustness for stomach segmentation.

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SU-lePD-TRACK 3-04

A Deep Learning Approach for Contour Interpolation

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Purpose: Contour interpolation is an important tool to expedite manual segmentation of anatomical structures by allowing users to contour on discontinuous slices and fill-in the gaps. The conventional interpolation tool in clinical TPS is based on distance map interpolation and has difficulties in generating satisfactory results when the existing manual contours differ dramatically. This is especially apparent near the superior and inferior borders of organs, and for the gastrointestinal structures such as bowel. In this study, we developed a deep learning model to provide improved contour interpolation robustness and accuracy for these historically difficult cases. **Methods:** The deep contour interpolator was developed and trained using selected 5882 datasets containing images of multiple organs (lung, liver, esophagus, heart). Preprocessing steps include cropping, image resizing and intensity normalizing. A 3D UNet model was utilized and its softmax layer and segmentation layer were replaced by a regression layer. The input size is $128 \times 128 \times 8$ in 2 channels. The first channel is the top and bottom slices of a 5-slice patch and three middle slices duplicated. The second channel is the related masks, empty mask for middle slices. The outputs are the predicted organ masks for the 3 middle slices. **Results:** The dice coefficients and DTA (distance-to-agreement) values were computed to evaluate the performance. For 1117 independent testing cases, the dice score was 0.90 ± 0.13 for the difficult cases, comparing to 0.85 ± 0.17 by the conventional method. The max and 95th percentile DTA values, were 7.6 ± 13.2 and 4.5 ± 11 mm, comparing to 8.7 ± 14.8 and 5.2 ± 5.2 mm by the conventional method. A t-test was conducted that confirmed that our model statistically significantly out-performed the conventional method ($p<0.05$). **Conclusion:** These results suggest that the deep interpolator may produce more robust and accurate interpolation results than the conventional contour interpolation method for historically difficult cases.

SU-lePD-TRACK 3-05

Automated CTV Contouring for Prostate and Nodes Utilizing Deep Learning Contouring

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Purpose: Applications of automation are increasingly expanding into different aspects of cancer management in radiation therapy. At the forefront is the contouring of organs at risk utilizing artificial intelligence. Few studies, however, have utilized artificial intelligence for target segmentation in the pelvic region. We implement a first of its kind deep learning contouring (DLC) for the prostate and nodes clinical target volume (CTV). This study investigates the similarity of manual delineated CTV and planning target volumes (PTV), done by radiation oncologists, with ones completed using autosegmentation with a DLC model. **Methods:** DLC model training used one hundred patient's contours of the prostate and nodes. The model was validated by generated CTV contours on 30 patients and expanding by 0.5cm to get PTV. Fifteen of those patients had physician CTV contours and subsequent expansion for PTVEX, and fifteen had direct PTV (PTVDIR) volumes by the physician. Evaluation of the geometric similarity using the Dice Similarity Coefficient (DSC) and average distance (AD), a modification of the Hausdorff Distance, allows for contour comparison. **Results:** Average values and standard deviation for the DSC generated

on the CTV, PTVEX, and PTVDir are 0.68 ± 0.08 , 0.77 ± 0.05 , and 0.75 ± 0.05 . AD yielded 5.5 ± 1.1 mm, 6.0 ± 1.2 mm, and 6.5 ± 1.2 mm. Typically, deviations occurred at the superior aspect of the nodal volume and on the prostate's inferior and posterior, attributed to known inconsistencies in the training volume. A subjective observation by physicians yields that, in general, the contours are beneficial and should yield time savings. **Conclusion:** We demonstrate the automating DLC for the prostate and nodes is possible. Although not statistically significant, we find an increased correlation between PTV volumes when physician contours the CTV over PTV. Further curating of training data should yield increased effectiveness in CTV contouring.

Research performed in collaboration with Mirada Medical

SU-lePD-TRACK 3-06

Patient-Specific Anatomic Context and Shape Prior (PACS)-Aware Recurrent Network for Cone-Beam CT Lung Cancer Segmentation

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Purpose: To train a AI auto-segmentation of lung tumors on weekly cone-beam (CBCT) using planning CT (pCT) and its clinical contour. **Methods:** We developed a novel Patient-specific Anatomic Context and Shape prior information or PACS-aware 3D recurrent registration-segmentation deep network for segmenting lung tumors from CBCT. Using an end-to-end trained network, the anatomic context from higher contrast planning CT and shape (delineated contours) priors are incorporated as additional input channels into a 3D recurrent CBCT segmentation network through progressively deformed sequence of images and contours, produced by the recurrent registration network. Our approach learns to progressively deform images and handles large deformations between pCT and a weekly CBCT without requiring deformation vector field (DVF) for training. The segmentation network losses are optimized using CBCT expert segmentations. We trained our model using 98,000 pCT and CBCT image patches of size $192 \times 192 \times 48$ extracted from 315 CBCT scans from 55 patients in a cross-validation setting. Independent testing was performed on 54 CBCT scans obtained from 10 patients. Performance comparisons was done against multiple state-of-the-art methods using Dice similarity coefficient (DSC), surface DSC and Hausdorff distance at 95th percentile (HD95) metrics. **Results:** Our method produced significantly ($p < 0.001$) more accurate tumor segmentations (DSC of 0.84 ± 0.08 , surface DSC of 0.98 ± 0.04 , and Hausdorff distance at 95th percentile of 3.33 ± 2.02 mm) than both deformable image registration and deep learning voxelmorph methods on independent test set. **Conclusion:** We introduced a patient-specific anatomic context and shape prior aware multi-modal recurrent registration-segmentation network for segmenting tumors on treatment CBCTs. Our approach significantly outperformed multiple methods and achieved promising accuracies on noisy and artifact-abundant cone-beam images.

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SU-lePD-TRACK 3-07

Deep Learning-Based Intracranial Vessel Wall Segmentation: Good DSC with Biased NWI?

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Purpose: Vessel wall (VW) contouring is important in cardiovascular disease diagnosis. Deep learning methods have been applied to MR VW images to delineate the lumen and the VW, with good common segmentation objectives, in Dice similarity coefficients (DSC). On the other hand, to support quantitative clinical judgement, further measures need to be assessed and characterized. This study provides an in-depth investigation on the behavior of a popular clinical metric – the normalized wall index (NWI), which is defined by the VW area divided by the whole vessel area, on the cross-sectional slices. **Methods:** We investigated the combination of (a) different network structures, including 2D UNet, 2.5D UNet, and 2.5D UNet++, all with good DSC performance 0.9163 ± 0.0522 , 0.7452 ± 0.1046 ; 0.9080 ± 0.0641 , 0.7521 ± 0.1006 ; and 0.9116 ± 0.0723 , 0.7758 ± 0.0957 for the lumen and the VW, respectively; (b) two objective functions: DSC, and DSC plus Hausdorff distance loss; and (c) different training sample reweighting schemes, including random, uniform, importance sampling, and matched distribution between training and testing to establish performance upper-bound. We examined closely the impact on NWI prediction and its variation along the vessel, to generate hypothesis for the cause of such bias and guide further quantitative radiomic study. **Results:** The results showed persistent systematic bias and large variation on the testing NWI, regardless of network structure, objective function, or training set sampling scheme. The NWI bias ranged from 0.0592 ± 0.0237 to 0.0975 ± 0.0425 . **Conclusion:** NWI may not be sufficient as the sole quantitative basis for plaque burden judgement. Its ratio form could be more sensitive to mislabeling. We hypothesized (1) it may be beneficial to optimize the clinical endpoints directly as the segmentation objective; (2) the seemingly small variation in contour labeling in training labels could have a large impact on derived quantitative omics features and their associated confidence. A better scheme to homogenize manual labels is required.

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Therapy Interactive ePoster Discussion

TRACK 5

Therapy: AI in Radiation Therapy

SU-lePD-TRACK 5-01

Knowledge-Based HDR Catheter Position for Prostate Brachytherapy

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Purpose: Current high-dose-rate (HDR) prostate brachytherapy catheter placement is based on physicians' experience. It may lead to large plan quality variations among physicians, and increased placement randomness for individual physician. In this study, we proposed a learning-based method to predict HDR needle position for prostate cancer patients. **Methods:** The proposed learning-based framework consists of deformable registration network (Reg-Net), multi-atlas ranking and catheter regression. Binary masks of the organs-at-risks are first transformed into distance maps which describe the distance of each local voxel to the organ surfaces. For a new patient, Reg-Net deformably registers the distance maps from multi-atlas set to match this patient's distance map and then bring catheter maps from multi-atlas to this patient. Based on several similarity criterions, we select the top-ranked atlas' deformed catheter positions as predicted catheter position for this patient. Finally, catheter regression is used to refine the final catheter positions. A retrospective study on 90 patients with a five-fold cross-validation was used to evaluate the proposed method's feasibility. Plans were optimized based on both the clinical and predicted catheter pattern. **Results:** For all patients, on average, both clinical plan dose and new plan dose meet the common dose constraints when prostate dose coverage is kept at $V100=95\%$. The plans from predicted catheter pattern have slightly higher hotspot in terms of $V150$ by 5.0% and $V200$ by 2.9% on average. For bladder $V75$, rectum $V75$ and urethra $V125$, the average difference is close to zero, and the range of most patients is within ± 1 cc. **Conclusion:** We developed a novel learning-based method in predicting catheter locations for HDR prostate brachytherapy. It has great clinical potential since it can provide catheter location estimation prior to catheter placement, which could reduce the dependence on physicians' experience in catheter implantation and improve the quality of

prostate HDR treatment plans.

SU-lePD-TRACK 5-02

Accurate Automatic Rectum Segmentation for High-Dose Rate Brachytherapy of Cervical Cancer Treatment Planning

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Purpose: Despite advances in deep-learning (DL) based automatic organ segmentation, automatic rectum segmentation is still challenging in high-dose-rate brachytherapy (HDRBT) of cervical cancer treatment planning, because of a lack of well-defined rectosigmoid boundary. DL models often cannot accurately determine this boundary, causing large errors when evaluating D2cc, as the superior part of rectum falls in the high dose region and 2cc volume is sensitive to segmentation results. This study develops a method to address this problem. **Methods:** We first performed an initial segmentation of rectum using a 3D U-Net that mapped a CT image to a rectum volume. We then used the superior-most slice of the initial rectum volume as initial condition to segment the rectum/sigmoid structure in a slice-by-slice fashion for 5cm along the superior direction to ensure rectosigmoid junction was included in the segmented volume. As such, a 2.5D U-Net was established to predict the sigmoid on a given slice based on the connectivity between previous adjacent CT slice and segmentation. We finally calculated the centroid of segmented volume in each slice and determined the centerline of the structure. The final rectum structure was determined by keeping the part inferior to the rectosigmoid junction defined as the place where the structure centerline turned for an angle over a threshold value. **Results:** We evaluated our method and a simple 3D U-Net-based rectum segmentation method against ground truth drawn by physicians. The error in superior rectum boundary position was 5.0 ± 5.0 mm in our method, while the error of 3D U-net was 15.7 ± 12.1 mm. When using the segmented structure for D2cc calculation, the error of our method and 3D U-net was $4.8\% \pm 3.9\%$ and $9.8\% \pm 10.2\%$. **Conclusion:** The proposed method overcame the challenge of conventional DL-based rectum segmentation, which allowed accurate computations of D2cc in HDRBT of cervical cancer treatment planning.

SU-lePD-TRACK 5-03

Influence of Loss Functions on Accuracy of Deep Learning Based Dose Distribution Prediction for Intensity Modulated Proton Radiation Therapy

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Purpose: This study investigated the influence of different loss functions on dose prediction accuracy for intensity modulated proton radiation therapy (IMPT). **Methods:** We proposed a dual-pyramid-network (DPN) with late-fusion-network (LFN) to generate high-quality pseudo-Monte Carlo (MC) proton plans using low-quality pencil-beam (PB)-based dose plans and corresponding CT as inputs. The proposed deep-learning network architecture includes three subnetworks: CT-only, PB dose-distribution only and late fusion subnetwork. The CT only and PB dose-distribution only subnetworks are based on U-Net architecture and extract features from the CT and PB dose distribution of each patient. Individual features are then combined in the LFN. The subnetworks were trained through three loss functions (L1-norm, L0.5-norm, and combined L1+L0.5), respectively, with constant hyperparameters in three separate models. Five-fold cross-validation experiments were performed on 40 prostate cancer patients. Predicted pseudo-MC dose distributions were compared with the target MC dose via gamma analysis under 1%/1mm, 2%/2mm and 3%/3mm acceptance criteria. Obtained dose-volume-histograms (DVHs) from each trained model were evaluated against the high-quality MC DVHs. **Results:** Dose map comparison results show that L1-norm loss function performs well in low dose region, whereas L0.5-norm function performs well in high dose regions. The combined loss function achieved the best dose distribution prediction accuracy. L1norm, L0.5norm, and combined (L1+L0.5) loss functions achieved $30.8 \pm 20.3\%$, $71.9 \pm 17.3\%$, and $90.1 \pm 5.8\%$ under 1%/1mm acceptance criteria and $56.2 \pm 26.2\%$, $92.5 \pm 5.6\%$ and $95.1 \pm 0.8\%$ under 2%/2mm acceptance criteria averaged over 40 patients from cross validation. The DVHs obtained from combined loss model had the closer agreement with the ground-truth MC DVHs compared to model with individual losses. **Conclusion:** We quantitatively investigated dose prediction accuracy based on three different loss functions to generate high-quality pseudo-MC proton plans via the proposed LFN-DPN. Overall results identify the combined (L1+L0.5) loss function to be the best candidate for deep-learning-based dose prediction of IMPT and highlights the importance of choosing an appropriate loss function.

SU-lePD-TRACK 5-04

Application of An Artificial Neural Network to Predict Equivalent Uniform Dose and Tumor Control Probability in Radiation Therapy of Lung Cancer Using Treatment Plan Data

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Purpose: To apply an artificial neural network (ANN) in order to predict equivalent uniform dose and tumor control probability in external beam radiation therapy of lung cancer treatment plans. **Methods:** Retroactive lung cancer treatment plans of 100 patients were chosen for this study. The tumor control probability (TCP) of tumor target and equivalent uniform dose (EUD) were calculated for all plans using Niemierko's EUD based linear-quadratic model. The expected outputs for ANN were calculated EUD and TCP. The inputs for ANN were treatment modality, location of tumor, prescribed dose, fractions, planning target volume (PTV), the maximum dose to the tumor, and mean doses to the OARs. All numeric data were normalized in the range 0 to 1, while non-numeric inputs were categorized into binary numbers (0 or 1). Our ANN is based on a Scaled Conjugate gradient algorithm with two hidden layers having 15-10 nodes with 11 inputs and 2 outputs. The data were employed 70% for training and 30% for validation and testing. **Results:** We found EUD and TCP for the target volume with an overall regression value of 0.97. The Mean absolute difference for predicted and expected values for EUD and TCP are found 0.0510 ± 0.0704 and 0.0439 ± 0.0708 respectively. In some cases, we encountered a large error (~ 0.14), where TCP were 50% or lower in palliative treatment case. **Conclusion:** These results indicate that ANN can be employed to predict TCP and EUD. The outcome of this research can be further improved by studying a large cohort of patient's treatment plans.

SU-lePD-TRACK 5-05

Improved Dosimetric Performance of VMAT Total Marrow Irradiation Using Knowledge-Based Planning Approach

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Purpose: We present updates on knowledge-based planning (KBP) driven linac-based total marrow irradiation (TMI). A KBP model was developed to handle multiple-isocenter volumetric modulated arc therapy (VMAT) covering a large target volume from head to mid femur. **Methods:** Fifty-one patients previously treated in our clinic were selected for the model training, while 12 patients from another clinic were used for validation. All plans used a 3-isocenter, 13-arc VMAT technique to cover sub-target volumes of head and neck (HN), chest, and pelvis. Both the HN and pelvis plans used the chest plan as the base dose to reduce hot spots around the field junctions. This leads to a wide range in dose-volume histogram (DVH) plots of target volumes. To address this, re-optimized plans without the base-dose plan were added to the library to train the KBP model. The performance of KBP-TMI model was then tested using a cohort of 11 clinical plans that did not use KBP. Conformity index and homogeneity index were evaluated using V95%/planning target volume (PTV) and D0.03cc/Rx dose, respectively. **Results:** KBP achieved our clinical goals (Rx dose = D95%) within 2-3 iterative optimizations, which used to take 4-6 days of effort without

KBP. Both KBP and clinical plans had mean dose values in the range of 108%-112% of the prescribed dose for PTVs, 22%-30% for lenses and oral cavity, and 41%-64% for brain, heart, lungs, bowels, liver, kidneys, eyes. Statistically significant reductions with KBP were observed in the mean dose values to brain, lungs, eyes and lenses. KBP substantially improved conformity index (1.5 ± 0.1 vs 1.8 ± 0.1) and homogeneity index (1.25 ± 0.02 vs 1.33 ± 0.03). **Conclusion:** Our study demonstrates that VMAT-TMI powered by KBP improves dosimetric performance with uniform quality. Moreover, it improves planning time considerably to allow its wide-spread clinical implementation.

SU-lePD-TRACK 5-06

Implementation of a Conditional Generative Adversarial Network for Automatic Prediction of 3D Dose Distributions in Prostate Cancer

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Purpose: The treatment planning process is time-consuming due to the complexity of iterative optimizations, this bottleneck can be resolved through deep learning dose prediction. However, this process excludes the mechanical parameters necessary for a deliverable plan. We aim to generate a streamlined VMAT planning process for prostate cancer using the Eclipse photon optimizer (PO) to generate a deliverable plan from an automatically predicted 3-dimensional (3D) radiation dose distribution. **Methods:** A conditional generative adversarial network, trained with 127 patients, was used to predict normalized 3D radiation dose distributions for 5 prostate cancer patients treated to 79.2Gy in 44 fractions, which were then imported into the treatment planning system. The PO was used to define the optimization parameters based off the predicted dose volume histogram. One optimization was performed using three consistent optimization objectives at predefined points for each OAR (rectum, bladder, penile bulb and right/left femoral heads) and PTV. Key dosimetric parameters were compared between the original planned, predicted and deliverable doses. 3D gamma analysis using plan delivery logs was performed using 3%3mm criteria. **Results:** Dosimetric comparison demonstrated that the deliverable doses had improved PTV coverage and comparable dose to OARs to the predicted and planned dose. The PTV coverage (D95%) on average increased from 96.0% to 100.6% between predicted and deliverable dose. The D50%(Gy) and V20Gy(%) for the bladder and rectum on average increased from 27.6Gy and 57.0% to 28.6Gy and 57.4% and 37.3Gy and 78.7% and 39.4Gy and 79.5% respectively. The right and left femoral D5%(Gy) decreased from 29.9Gy to 29.6Gy and 34.0Gy and 33.9Gy, respectively. Mean dose to the penile bulb increased from 26.2Gy to 29.1Gy. The average passing rate for 3D gamma analysis was 99.6%. **Conclusion:** With deep learning predicted doses, we can generate a deliverable plan with radiation dose distributions comparable to the predicted dose.

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SU-lePD-TRACK 5-07

Improving Target Detection for Real-Time Markless Lung Tumor Tracking Using Deep Learning

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Purpose: To investigate the feasibility of using deep learning to improve markless lung tumor detection on x-ray radiograph images during real-time motion tracking and correction (RMTC). **Methods:** A convolutional neural network (CNN), designed with Matlab (Mathworks, R2020a), was trained, validated and tested with 194 radiograph images acquired at different gantry angles and couch positions from 12 lung cancer patients treated with RMTC during helical tomotherapy. The dataset was divided into training (70%), validation (15%) and test (15%) subsets. The ground truth lung tumor contours on the radiographs were obtained by transferring the contours from DRR with manual editing. All radiographs were normalized based on the intensities of voxels enclosed in a region of interest (ROI) corresponding to the tumor motion ranges on the DRR. Performance of the trained and validated CNN was measured by comparing the predicted target contours of the test data subset with the ground truth using Dice similarity coefficient (DSC) and the distance between the centers from the predicted and the ground truth contours, (dCNN-GT), which was also compared to the distance between the detected center in the RMTC system and the ground truth center (dRMTC-GT). **Results:** The training and validation achieved sufficient accuracy (99.68%). The average DSC was 0.76 ± 0.24 . The average dCNN-GT was 1.48 ± 1.05 mm, slightly smaller than the average of dRMTC-GT (1.69 ± 1.00 mm). On average, the time needed for a prediction was 119 ± 29 ms with an Intel Core™ i7-6700 CPU@3.40 GHz PC. **Conclusion:** The CNN model can quickly predict target center on x-ray radiograph with accuracy equal or slightly better than the current system detection. With further development using large datasets, the deep learning may be used to improve target prediction for real-time markless lung tumor tracking.

Therapy Interactive ePoster Discussion TRACK 6

Therapy: Brachytherapy Dosimetry

SU-lePD-TRACK 6-01

Monte Carlo Characterization of High Atomic Number Inorganic Scintillators for in Vivo Dosimetry in Ir-192 Brachytherapy

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Purpose: To use Monte Carlo (MC) simulations for the characterization of several existing high atomic number (Z) inorganic scintillator dosimeters designed for in vivo dosimetry of ¹⁹²Ir brachytherapy treatments. **Methods:** An MC code system PENELOPE-2014 was used to determine the absorbed-dose energy dependence of ZnSe:O (Z(eff)= 32) and CsI:Tl (Z(eff)= 54) relative to water (D(det)/D(wat)) and the influence of scatter conditions which change between detector calibration, experimental characterization, and in vivo measurements. Additionally, the change in photon spectrum due to the pelvic bones and its impact on detector response was evaluated to mimic treatments in the pelvic area. Detector design influence was also investigated. **Results:** ZnSe:O and CsI:Tl have a large absorbed-dose energy dependence relative to water (up to four and ten times, respectively). Its change over the radial distance range relevant for in vivo dosimetry was comparable for the two materials despite a large difference in their Z(eff). The D(det)/D(wat) dependence on phantom size was negligible up to 2 cm distance but increased to 14 % at 5 cm distance from the source. Neither detector design nor pelvic bones had influence on the detector response. Volume averaging was few percent. **Conclusion:** Most of the experimental studies investigating high-Z detectors use water phantoms that provide full scatter which differ from the in vivo measurement conditions. Our work shows that the change in ZnSe:O and CsI:Tl detector response is not negligible and shall be evaluated to avoid erroneous absorbed dose to water determination. Similar conclusions would apply for other high-Z detector systems. Additionally, although the D(det)/D(wat) dependence is large for high-Z inorganic scintillators, it can be well-characterized and agrees with the experimental data for ZnSe:O. Without an MC study, such conclusions could not be drawn since the experimentally determined detector response could be affected by its scintillation properties.

SU-lePD-TRACK 6-02

Evaluating Dosimetric Accuracy of EBT3 Radiochromic Film Calibration Using 6 MV Photon Beam in HDR Ir-192 Dose Measurement

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Purpose: To evaluate the accuracy of Ir-192 film dosimetry using the calibration curve from 6 MV photon beams, and design a novel method for direct film calibration using the Ir-192 source. **Methods:** EBT3 films were calibrated with the 6 MV photon beams on one Truebeam and the Ir-192 source on the GammaMed unit of our institution. We irradiated 18 film pieces (0 to 10 Gy) for each radiation source. All films were scanned 1-day post-irradiation on an EPSON EXPRESSION 10000XL scanner with "Portrait" placement using 48 bits, 150 dpi, and no color correction. Pixel values and calibration curves were acquired from RIT V6.8 using all 3 channels. The irradiated films using the Ir-192 source were reloaded in RIT and a different ROI was chosen to test the accuracy of both calibrations. Moreover, we irradiated an independent test case with the Varian surface applicator (1.5 cm cone) and performed composite analysis between dose maps generated by these two calibrations. **Results:** It was found the red channel has the best performance in point dose measurement (on average, 6 MV: 3.3% from the expected value, Ir-192: 0.3%), comparing to green (6 MV: 3.91%, Ir-192: 0.65%) and blue (6 MV: 4.43%, Ir-192: 2.15%) channels. The discrepancy was less significant (~1.82%) in the high dose region (300 cGy~1000 cGy) for red channels as well. With red channel, the composite analysis showed 83.15% passing rate for 1mm&1%, 89.23% for 1mm&2%, and 98.71% for 1mm&3%. The number of points failed only the percent dose difference (%Diff) criteria is found to be 0 when %Diff > 4%. **Conclusion:** This preliminary study showed a higher error for Ir-192 film dosimetry if films are calibrated using 6 MV photon beams. We suggest the films should be calibrated directly using the cylinder applicator and Ir-192 source for more accurate film dosimetry.

SU-lePD-TRACK 6-03

Evaluating the Fidelity of Post-Plan Vs. Pre-Plan Dosimetry for 3D-Printed Brachytherapy Surface Molds

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Purpose: To investigate dosimetry of pre-plan vs. post-plan dosimetry for patients treated with 3D-printed brachytherapy surface applicators. **Methods:** Patients treated in our clinic with surface brachytherapy undergo iterative pre-planning prior to the creation of a 3D-printed mold. The pre-plans are used to determine mold designs that result in the desired coverage and limited hot spots for each patient's anatomy and treatment site. Pre-plans for treated patients were compared to post-plans obtained with the printed mold on the patient anatomy to determine the extent to which pre-plans predict the final plans as evaluated with the actual treatment CT-scan. **Results:** To-date, five patients have been treated with 3D-printed molds for surface brachytherapy including three noses, one three-site forehead, and one cheek. For all patients, pre- vs. post-plan D90% and D95% were within 2.5% (average differences: 0.6% and 0.7% for D90% and D95%, respectively), where post-plans had higher coverage compared to pre-plans for all but one case. The post-plan V150% were all within 2% (average 0.6%) of the pre-plan, with most post-plans being slightly hotter than the pre-plan likely due to improved coverage in the post-plan. The average change in CTV minimum, mean, D90%, D95%, D100%, V100%, V110%, V120%, V130%, V140%, and V150% parameters were all less than $\pm 1.5\%$, while average maximum point CTV difference was within 4% over all cases. **Conclusion:** 3D-printed brachytherapy surface molds created for clinical use exhibit very good agreement between planned and final dosimetry on the patient. The pre-planning process for 3D-printed applicators allows for modifications to depth of catheters, intra-catheter spacing, and catheter orientation to achieve the desired dosimetry for surface brachytherapy. This process allows for dosimetric goals to drive the mold design, as opposed to mold design governing the achievable dosimetry which is often the case when using handmade or commercially available flap molds.

SU-lePD-TRACK 6-04

A Comprehensive Study to Design A Novel Direction Modulated Brachytherapy (DMBT) Vaginal Cylinder (VC) Using GEANT4 Monte Carlo Simulation Code

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Purpose: To design a new DMBT VC applicator to address the lack of dose coverage at the VC apex due to anisotropy effect and to provide directional modulation for coverage of lesions that cannot be covered using the common VC applicators at the very critical region of the vaginal cuff. **Methods:** The new DMBT-VC was modeled in GEANT4 MC code, as single lumen and made of PPSU plastic. The central part of that including the lumen was considered to be a detachable 8 mm diameter tandem rod made of PEEK plastic. First, we extracted a standard dose distribution from BrachyVision TPS of the VC applicator with PEEK tandem. Second, to provide directional intensity modulation, the PEEK tandem was replaced with MR-compatible tungsten alloy or alternatively iridium tandems, with similar dimensions to optimize the coverage at the apex utilizing directional radiation beam generated. Two widely used 192Ir sources, Varian's GammaMedPlus and VS2000 were simulated. Further, we studied the applicator functionality for improving coverage for lesions that are located beyond the typical 5mm isodose prescription line from the surface using both DMBT tandems. **Results:** The analysis showed that the new applicator can improve the lack of coverage at the apex due to anisotropy which were about 2mm and 4mm for GammaMedPlus and VS2000, respectively, using all DMBT tandems. In addition, the DMBT applicator was able to successfully modulate the prescription isodose line by up to 13 mm and 18mm using tungsten and iridium tandems, respectively, beyond the VC surface at vaginal apex without overdosing the peripheral sides through inverse optimization. **Conclusion:** Results showed that the novel DMBT VC can remedy the anisotropy effect and provide coverage for deep-seated lesions at vaginal cuff. This is of major clinical value as the vaginal apex is a frequent site of recurrence in patients.

SU-lePD-TRACK 6-05

Predictive Modeling for Rapid Online Evaluation of Plan Quality During Intraoperative Gynecological Brachytherapy

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Purpose: The current standard of care for locally advanced gynecological cancers is external beam radiotherapy (EBRT) with brachytherapy boost to the high risk clinical target volume (HRCTV). During intraoperative brachytherapy treatment planning, the patient is under anesthesia, therefore an acceptable plan must be achieved within a limited timeframe. The purpose of this work is to develop a model that can predict the plan quality, allowing the planner to evaluate if an acceptable plan has been achieved. **Methods:** Using an internal database of patients treated with gynecological brachytherapy, two models were developed using MATLAB (Mathworks, R2018b); (1) a distance-based model and (2) an iterative computational model. The distance model evaluates the achieved HRCTV D90/OAR D2cc as a function of quality metric OAR distance to HRCTV. The iterative model evaluates the achieved HRCTV D90/OAR D2cc as a function of the quality metric predicted HRCTV D90/OAR D2cc. The iterative model further accounts for implant quality, represented by activated dwell positions. Regression analysis was performed on both models. **Results:** The distance based model is less precise in its ability to predict achieved HRCTV D90/OAR D2cc ($R^2=0.83$) in comparison to the iterative model ($R^2=0.97$) indicating the importance in accounting for implant quality in the iterative model. Assuming an EBRT dose of 45Gy in 25 fractions, the iterative model can consistently predict the combined dose from brachytherapy and EBRT within $\pm 7\%$, with 95% confidence. **Conclusion:** The iterative model can be used during brachytherapy treatment planning to rapidly determine whether a new intraoperative treatment plan is consistent with the plan quality that has been achieved previously within the center. Future work will include demonstrating the utility of this model in a clinical setting by evaluating its robustness over a range of patient anatomy and implant quality.

SU-lePD-TRACK 6-06

Monte Carlo Simulation of the Permanent Implant - GammaTile

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Purpose: GammaTile (GT Medical Technologies, Tempe AZ) is a permanent brain implant used to treat certain brain tumors. A GammaTile treatment program has been established at our institution. Post-op treatment planning is performed clinically with a TG-43 based treatment planning system (TPS). The TPS calculates dose to water and doesn't account for heterogeneities. The purpose of this work was to investigate differences in dose deposited due to heterogeneities, in particular to examine dose to the skull and skin following GammaTile implantation utilizing Monte Carlo simulation. **Methods:** Each GammaTile contains four CS-1, Rev2 131Cs seeds (IsoRay Medical Inc., Richland, WA) embedded in DuraGen (Integra Lifesciences, Plainsboro, NJ) collagen tile measuring $2.0 \times 2.0 \times 0.4 \text{ cm}^3$. Each seed has an air kerma strength of 3.5U at the date of implant. MCNP Monte Carlo simulation (MCNP5) was used to simulate the dose distribution around a single GammaTile implanted permanently. MC simulated doses were obtained both in water and in a heterogeneous medium, including brain, skull bone (thickness=0.7 cm), skin (thickness=0.5 cm), and air. For comparison, the total dose deposited was also calculated using BrachyVision (Varian Medical Systems, Palo Alto, CA). **Results:** Initial simulated dose distributions in water were within 7% agreement with comparable dose distributions generated by BrachyVision. When bone was located 1 cm from a single GammaTile, the total dose simulated to the bone and skin were 28.3 Gy and 2.74 Gy, respectively. Doses for bone and skin at the same distance calculated with the treatment planning system were 10.4 Gy and 4.96 Gy, respectively. MC simulated values are 2-3 times the values calculated using TG-43. **Conclusion:** Considerable dose to the skull bone is possible and even likely from GammaTile therapy. Initial simulations included a single GammaTile. Future simulations will include more representative of a clinical arrangement containing multiple GammaTile units.

SU-lePD-TRACK 6-07

Correction Factors for a Planar Windowless Extrapolation Chamber Measuring Surface Absorbed Dose to Water from An Unsealed Thin-Film Alpha-Emitting Radionuclide

A Khan*, W Culberson, L DeWerd, Department of Medical Physics, School of Medicine and Public Health, University of Wisconsin-Madison, Madison, WI

Purpose: To establish a primary standard of absorbed-dose-to-water for targeted alpha therapy radionuclides by measuring surface absorbed-dose-to-water from a point-like Po-210 radionuclide using a planar, windowless extrapolation chamber. The measured ionization current at each airgap must be corrected to account for side-losses of charged particles at finite air-gaps ($k(\text{div})$), the finite radius of the Po-210 source ($k(\text{point})$), the difference in backscatter between the medium surrounding the air-cavity and water ($k(\text{backscatter})$), and δ -ray disequilibrium due to the absence of build-up material ($k(\delta)$). In this study, we determine the magnitude of these correction factors using Monte Carlo (MC) simulation. **Methods:** GEANT4 was used to model the extrapolation chamber and circular, 3 mm diameter Po-210 source on a silver substrate along with a cylindrical air-cavity with the radius determined by COMSOL Multiphysics™ electrostatic simulations. The Po-210 source was simulated as a surface source and the absorbed dose to air was calculated for 100-250 μm airgaps. $k(\text{div})$ was determined by extrapolating the calculated absorbed dose at zero airgap, $k(\text{point})$ was determined by simulating a point source, and $k(\text{backscatter})$ was determined by setting the material surrounding the cavity as water. The $k(\delta)$ correction factor was calculated by considering local absorption of δ -rays generated in the cavity. **Results:** The radius of the air-cavity was calculated to be 2.02 mm according to the electrostatic simulations. The $k(\text{backscatter})$ and $k(\delta)$ correction factors were found to be in the 0.882-0.911 and 1.020-1.041 ranges, respectively, and their deviation from unity was observed to decrease with increasing airgap. In contrast, $k(\text{point})$ and $k(\text{div})$ were observed to linearly increase with increasing airgap with magnitudes in the 1.051-1.062 and 1.243-1.591 ranges, respectively. $k(\text{div})$ was determined to be the largest correction factor. **Conclusion:** MC-based correction factors were calculated for an extrapolation chamber measuring surface absorbed-dose-to-water from an alpha-emitting radionuclide.

Council Symposium TRACK 1

Education Council Symposium: Lessons Learned During the COVID-19 Pandemic, Innovative Approaches to Didactic and Clinical Training

SU-C-TRACK 1-00

Education Council Symposium: Lessons Learned During the COVID-19 Pandemic, Innovative Approaches to Didactic and Clinical Training

Moderator: Jay Burmeister, Wayne State University, Karmanos Cancer Center, Detroit, MI

SU-C-TRACK 1-01

Introduction

SU-C-TRACK 1-02

I. Barreto, University of Florida College of Medicine:

It Only Took a Pandemic: Engaging Online Students with Active Learning Strategies

SU-C-TRACK 1-03

C. Watchman, Memorial Sloan Kettering Cancer Center:

Using Digital Training Tools to Address Resident Training During a Pandemic

SU-C-TRACK 1-04

V. Montemayor, Germantown Academy:

Silver Linings: Tools I Will Take with Me When I Leave Virtual Teaching Behind

SU-C-TRACK 1-05

J. Burmeister, Wayne State University, Karmanos Cancer Center:

Post-Pandemic Education and Training: Will We Do More Than Just Endure?

SU-C-TRACK 1-06

Q&A

On March 11, 2020 the World Health Organization officially declared the novel coronavirus (COVID-19) a pandemic. In light of the subsequent restrictions and recommendations in educational and clinical settings, many educators were required to make radical changes to their courses mid-semester in an attempt to transition from face-to-face to online instruction. In the clinic, residency program directors and faculty grappled with ways to provide “equivalent” hands-on experiences remotely. Academic instructors had to quickly learn how to use several new platforms with varying levels of assistance from institutions to provide video-conferencing, assignment submissions, and virtual exam proctoring. What quickly became apparent was that didactic and clinical training techniques used for face-to-face interactions could not be directly virtualized. During the symposium, four educators will discuss their experience, the challenges they encountered, and the positive lessons they learned in order to improve instructional strategies moving forward.

Learning Objectives:

After attending this Symposium, the participant will:

1. Understand how to overcome challenges of online learning by applying active learning strategies that spark curiosity.
2. Learn how to apply opportunities to humanize courses to foster a learning community that encourages learners to participate.
3. Identify what digital education resources are currently available online.

4. Learn about the usefulness and drawbacks of commercial tools and applications in teaching and mentoring when coupled with the use of a tablet.
5. Understand how eLearning course modules may be used to supplement/replace some elements of residency training.
6. Identify issues that can inhibit student and resident learning in the virtual environment and techniques to manage these issues.
7. Consider how student and resident evaluation can be performed effectively in a virtual environment.
8. Recognize opportunities to improve our pedagogical strategies resulting from the forced transition to the virtual environment.

SAM Imaging Educational Course

TRACK 2

Fluoroscopy Dose Management

SU-CD-TRACK 2-00

Fluoroscopy Dose Management

Moderator: Ryan Fisher, MetroHealth, Cleveland, OH

SU-CD-TRACK 2-01

R. Fisher, MetroHealth:

MPPG #12: Fluoroscopy Dose Management

SU-CD-TRACK 2-02

M. Mahesh, Johns Hopkins University:

First CPT Code for Diagnostic Medical Physics: What to Know and How to Use This Code

SU-CD-TRACK 2-03

A. Jones, UT MD Anderson Cancer Center:

ACR Dose Index Registry for Fluoroscopy (DIR-Fluoro)

SU-CD-TRACK 2-04

Audience Q & A

Ryan Fisher - MPPG #12: Fluoroscopy Dose Management

Medical physics practice guideline # 12, "Fluoroscopy Dose Management" aims to outline the role of the diagnostic qualified medical physicist in practical patient dose management for fluoroscopically-guided interventional (FGI) procedures. This role includes helping facilities set up policies related to dose management, including pre-procedure patient consent, intra-procedure dose index level notification, and post-procedure follow up for potential tissue reactions. Additionally, the document includes suggestions for methods of complying with various state regulatory requirements and Joint Commission standards related to fluoroscopy, including tracking radiation doses, setting dose index thresholds, and analyzing dose index data, and includes discussion of the challenges posed by these requirements. This presentation will cover the content of the new practice guideline, as well as potential changes to the Joint Commission fluoroscopy radiation overdose sentinel event.

Learning Objectives:

1. Understand recommendations for physicists' responsibilities related to fluoroscopically-guided interventions outlined in the forthcoming medical physics practice guideline.
2. Understand changes related to the Joint Commission's prolonged fluoroscopy sentinel event standard.

Mahadevappa Mahesh - First CPT Code for Diagnostic Medical Physics: What to Know and How to Use This Code

The very first CPT code (76145 – Medical physics dose evaluation for radiation exposure that exceeds institutional review threshold for diagnostic medical physics) was approved and became active from January 1st, 2021. The presentation includes explanation of the code and its origin along with discussion on when it could be used. The presentation will include sample cases with step-by-step guide on how to evaluate and complete the report. There will be discussion on how to go about setting the code in the clinic.

Learning Objectives:

1. To learn what is CPT code 76145
2. To examine when the code could be used and what should be in the report.
3. To become familiar with the threshold values that warrants the use of this code

A. Kyle Jones- ACR Dose Index Registry for Fluoroscopy (DIR-Fluoro)

The American College of Radiology (ACR) Fluoroscopy Dose Index Registry (DIR) is the latest addition to the National Radiology Data Registry (NRDR). During the pilot phase of the registry, fluoroscopy dose index data was collected from nine pilot sites over a period of 22 months from March 1, 2018 through December 31, 2019. Ultimately, dose index data from almost 40,000 unique clinical procedures was collected, reviewed, and analyzed. These procedures were mapped to one of 127 ACR Common IDs which allowed for comparison of dose indices for similar procedures among pilot sites. Broader roll-up categories were also developed, which allowed for comparison of procedures across multiple ACR Common IDs and provided sites with less granular clinical facility procedure names a pathway to analyze their submitted data and produce reasonable comparisons to the pilot group as a whole.

Learning Objectives:

1. Outline the process for sites to participate in the ACR Fluoroscopy DIR.
2. Review the dose index data collected during the pilot phase of the ACR Fluoroscopy DIR.
3. Discuss strategies for using DIR data and comparisons for quality improvement.

Content reviewer for fluorosafety.com online fluoroscopy training

SAM Imaging Scientific Symposium

TRACK 3

Medical Imaging and Data Resource Center: Imaging in Covid

SU-CD-TRACK 3-00

Medical Imaging and Data Resource Center: Imaging in Covid

Moderator: Maryellen Giger, University of Chicago, Chicago, IL

SU-CD-TRACK 3-01

M. Giger, University of Chicago:

Overview of MIDRC and COVID-19 AI Research Projects

SU-CD-TRACK 3-02

P. Kinahan, University of Washington:

Image Quality and Harmonization in MIDRC

SU-CD-TRACK 3-03

M. McNitt-Gray, David Geffen School of Medicine at UCLA:

Rigorous Validation, Appropriate Metrics, and Task-Based Distribution of Data for Evaluating COVID-19 AI

SU-CD-TRACK 3-04

K. Myers, Office of Science and Engineering Laboratories, FDA:

Rigorous Validation, Appropriate Metrics, and Task-Based Distribution of Data for Evaluating COVID-19 AI

SU-CD-TRACK 3-05

S. Armato, The University of Chicago:

Grand Challenges Through MIDRC

SU-CD-TRACK 3-06

Q&A & Panel Discussion

The COVID-19 pandemic presents a critical public health crisis. Essential biomedical research and development is needed to urgently address surveillance and early detection of COVID-19 resurgence; differential diagnosis of COVID-19 patients; prognosis; and prediction and monitoring of response for use in patient management. Since the start of the pandemic, medical imaging has played key roles in these patient monitoring and assessment tasks. In response to this need, leaders from the RSNA, ACR, and AAPM along with NIBIB have jointly developed the Medical Imaging and Data Resource Center (MIDRC) for rapid, flexible, and large collection of well-curated imaging studies to facilitate development of and externally validate machine intelligence (AI) algorithms for the various tasks described above. MIDRC is a multi-group, two-year NIBIB-funded project hosted at the University of Chicago and coordinated among 23 institutions including two intake portals operated by RSNA-RICORD and ACR-CIRR that integrate quality control and data harmonization. A public access, exploration, and analysis portal and commons is being developed on the Gen3 data ecosystem. Twelve collaborative research projects exist to expedite translation of machine learning and AI from scientific findings and technical resources to public dissemination and clinical benefit. In this symposium, we will review the status of MIDRC, the role of AAPM's Data Science Committee, and the important role that medical physicists have in MIDRC for quality assessment and data harmonization, machine intelligence assessment through metrology and task-based evaluations, and the conduct of grand challenges.

Learning Objectives:

1. Learn about MIDRC and the AAPM's leadership and development roles in it.
2. Understand the need for large, well-curated medical imaging datasets to facilitate development of AI and to externally validate such algorithms in order to enable translation to clinical use.
3. Understand the essential medical physics research needed to address the role of medical imaging for monitoring, diagnosis, and assessment of the COVID-19 patient.

Funding for MIDRC is from NIBIB COVID-19 Contract 75N92020D00021

SAM Multi-Disciplinary Scientific Symposium

TRACK 4

Novel Imaging Technologies for Motion Management in Radiotherapy

SU-C-TRACK 4-00

Novel Imaging Technologies for Motion Management in Radiotherapy

Moderator: Tinsu Pan, UT MD Anderson Cancer Center, Houston, TX

SU-C-TRACK 4-01

W. Yang, Keck School of Medicine of USC:

4D-MRI, Motion and Beyond

SU-C-TRACK 4-02

L. Ren, University of Maryland:

4D-CBCT: Where Are We and What's Next?

SU-C-TRACK 4-03

T. Pan, UT MD Anderson Cancer Center:

Data Driven Gated PET/CT

Respiratory motion is an important topic in radiotherapy. In this presentation, we will review the new developments of 4D MRI, 4D CBCT and 4D PET/CT, and their applications to radiotherapy.

Learning Objectives:

- 1: New development of 4D MRI;
- 2: New development of 4D CBCT;
- 3: New development of 4D PET/CT.

NIH 1R21CA222749-01A1 and R03EB030280, and a ROSI grant from M.D. Anderson Cancer Center, Division of Radiation Oncology. This research was conducted at the M.D. Anderson Cancer Center for Advanced Biomedical Imaging in-part with equipment support from General Electric Healthcare. TP is consultant for Bracco Medical Systems, LLC.

SAM Therapy Scientific Symposium

TRACK 5

Advances in Automatic Segmentation, Treatment Planning and QA

SU-CD-TRACK 5-00

Advances in Automatic Segmentation, Treatment Planning and QA

Moderator: Qing-Rong Jackie Wu, Duke University Medical Center, Durham, NC

SU-CD-TRACK 5-01

H. Veeraraghavan, Memorial Sloan Kettering Cancer Center:

Deep Learning Automated Segmentation for Radiation Therapy Planning

SU-CD-TRACK 5-02

Q. Wu, Duke University Medical Center:

Automation in Treatment Planning Process

SU-CD-TRACK 5-03

E. Ford, University of Washington:

Advances in Quality Assurance Through Automation

SU-CD-TRACK 5-04

S. Jiang, UT Southwestern Medical Center:

Thoughts on The Clinical Implementation of AI Tools in Treatment Planning Process

Recent advances in segmentation, planning, and QA are largely enabled by innovative applications of artificial intelligence (AI) techniques. AI refers to methods that improve and automate challenging human tasks by systematically capturing and applying relevant knowledge in these tasks.

Over the past decades, a number of AI approaches have been developed to address different needs of system intelligence in treatment planning, ranging from search strategies to knowledge representation to machine learning. In particular, deep learning techniques have found tremendous success in the segmentation of normal anatomical structures as well as treatment target volumes; and also have shown great promise in accurately predicting dose distributions and fluence maps resulting in completely automated algorithms for generating treatment plans with clinically viable qualities. Robust systems that integrate multiple AI and machine learning algorithms have also been developed to automate entire clinical workflows in radiation therapy. Moreover, quality assurance has also benefited significantly from recent advances. Data-driven intelligent algorithms are making the QA process more efficient and effective.

This session presents recent developments of AI algorithms and other novel approaches to improving segmentation, planning, and QA in the treatment planning process; and discusses the potential changes of clinical workflow these technologies may bring.

Learning Objectives:

1. Understand current AI techniques and algorithms that have been developed for segmentation, planning and quality assurance
2. Understand the issues related to developing AI tools in these targeted applications
3. Understand potential challenges in implementing and integrating these techniques in clinical workflow and preparations for adoption

NIH R01CA201212

Varian Medical System Master Research Agreement

Imaging Scientific Session

TRACK 6

Nuclear Medicine and PET

SU-C-TRACK 6-01

A Robust Method to Quantitate Tracer Activity in Organs and Lesions with Clinical SPECT/CT Systems

J James*, J Bian, J Halama, Loyola University Medical Center, Maywood, IL

Purpose: It is essential that a robust method be used to quantitate activity from SPECT/CT for organ based radiation dosimetry. Aim of this study is to validate the accuracy of a universal method to perform quantitative measurements across vendors and differing reconstruction methods. **Methods:** An IEC Image Quality phantom cover was 3D printed to hold 1-2-3-4-5-6 cm diameter hollow spheres with a minimum separation of 2.5 cm between surfaces. The spheres and the phantom chamber had ^{99m}Tc sphere/background activity concentrations of no background, 8:1, 4:1 and 2:1. SPECT/CT acquisitions were performed on the Philips-BrightView and GE-Discovery-670 using clinical acquisition protocols. Quantitative SPECT reconstructions for: 1) BrightView was performed using SUV-SPECT on Hermes and 2) GE using (a) Q.Metrix on GE-Xeleris, and (b) SUV-SPECT on Hermes. Image reconstructions were iterative with CT attenuation-correction, scatter-correction and resolution-recovery. The total activity in the spheres was measured by placing volumes-of-interest (VOI) of diameter: 1) matching the sphere diameter (VOI₁), 2) expanded by 2 cm (VOI₂), and 3) expanded by 3 cm (VOI₃). VOI₂ includes spill-out of sphere tracer that is observed outside the sphere boundary. VOI₃ is to estimate background tracer activity concentration just outside VOI₂ and is needed to apply a background correction for VOI₂. The measured/true activity fractions (TF(spheres)) for the 6 spheres and the background concentration (TF(BKG)) were computed. **Results:** Overall the TF's were within 10% (GE) and 15% (BrightView) of the truth, except for scans with 2:1 sphere/background that has significant artifacts and non-visualization of all spheres that lead to measurement inaccuracies. Across all measurements, BrightView with Hermes reconstruction was 0.89 ± 0.05 , GE-Discovery-670 with Q.Metrix was 0.97 ± 0.03 and with Hermes was 0.95 ± 0.04 . **Conclusion:** Utilizing an expanded VOI to capture all sphere activity in the measurement together with a background correction is a robust method for performing quantitative analysis of SPECT/CT images.

SU-C-TRACK 6-02

Harmonization of Four PET Scanners Through the Use of a Uniformity Phantom and a Sphere Simulation Tool

D Gomez Cardona¹*, J Browne², L Stecker³, R Pooley³, (1) Gundersen Health System, La Crosse, WI, (2) Mayo Clinic, Rochester, MN, (3) Mayo Clinic, Jacksonville, FL

Purpose: Quantitative positron emission tomography (PET) with 18F-fluorodeoxyglucose (18F-FDG) can provide important information for staging cancer and monitoring therapy response. However, to reduce quantitative bias and variability, hospitals should standardize their imaging procedures and harmonize

the SUV response of their fleet of PET scanners. The aim of this study was to harmonize the response of four PET scanners in our practice through the use of a uniformity phantom and a sphere simulation tool which predicts optimal reconstruction filter settings[1]. The predicted reconstruction settings were validated experimentally using the NEMA NU2-2007 phantom. **Methods:** Four PET systems, GE Discovery MI, GE Signa PETMR, Siemens Biograph 40, and Siemens Biograph Vision were assessed. The total activity of 18F-FDG in the uniformity and NEMA phantoms was 70MBq; the NEMA phantom spheres-background ratio was 20:2kBq/ml. The uniformity and NEMA phantoms were scanned for 20 min/bed position (1-3 bed positions) and 2 min/bed position (1 -2 bed positions), respectively. Two clinical reconstruction settings were evaluated for each scanner to identify baseline settings. The optimal post-filter settings were predicted using the uniformity phantom and the sphere simulation tool. These settings were applied to the NEMA phantom data to validate the sphere simulation tool. For each sphere in the NEMA phantom, the maximum voxel value and an adaptive 50% isocontour mean value were calculated[2]. **Results:** The recovery coefficient (RC) curves obtained for the NEMA and uniformity phantom show close agreement (Figure 1a and 1b). Figure 1b also displays a harmonized RC curve after specific filter settings were applied. Figure 2a and 2b display pre- and post-harmonization results of the NEMA phantom following the application of the predicted filter settings. **Conclusion:** Scanner harmonization can be achieved using vendor specific uniformity phantoms and the sphere simulation tool rather than the more expensive and complex NEMA phantom.

SU-C-TRACK 6-03

BEST IN PHYSICS (IMAGING): Image Quality and Lesion Detectability in New DDG-PET/CT Utilizing 50% of PET Data

MA Thomas*, T Pan, UT MD Anderson Cancer Center, Houston, TX

Purpose: Prospective data-driven gated (DDG) PET requires advance selection of patients who may benefit from gating, and leaves DDG-PET/CT registration to chance. Retrospective DDG of both PET and CT can reduce CT radiation dose, shorten scan time, improve PET/CT registration, and correct tumor motion on all patients. The first DDG-PET/CT (DDG-PET and DDG-CT) has been prototyped for patient care in a network of seven PET/CT scanners. Here, the impacts of misregistration and motion correction on image quality and lesion detectability were assessed with the new DDG-PET/CT technique. **Methods:** A low-dose cine CT was acquired in misregistered regions to enable both average CT (ACT) and DDG-CT for attenuation-correction of static and DDG-PET, respectively, yielding baseline PET/CT, PET/ACT and DDG-PET/CT. For DDG-PET, end-expiration data was derived from 50% of PET data at 30% from end-inspiration. For DDG-CT, end-expiration CT data was extracted from cine CT by lung CT number and body contour. SUVmax, SUVmean, SNR, and CNR were compared for 121 lesions from 21 consecutive patients. **Results:** Improved registration with ACT did not change SNR. DDG-PET/CT showed decreased background and lesion SNR with median $\pm\sigma$ changes of $-24\pm10\%$ ($p<0.0001$), equivalent to using $\sim 57\%$ of total PET counts. Changes from baseline in SUVmax and CNR were $21\pm41\%$ ($p<0.0001$) and $6\pm76\%$ ($p=0.27$) for PET/ACT, and $68\pm65\%$ ($p<0.0001$) and $37\pm141\%$ ($p<0.0001$) for DDG-PET/CT. Lesions with ambiguous detectability at baseline (CNR <10) showed the most significant CNR increases in DDG-PET/CT, with nearly all lesions reaching CNR >5 . **Conclusion:** Using 50% of PET data in DDG-PET leads to the expected increase in image noise. However, the increase in SUV due to improved registration and motion correction in DDG-PET/CT is more substantial than the increased image noise. Therefore, lesion detectability is still largely improved with DDG, especially for lesions with uncertain detectability (CNR <10) at baseline.

This research was supported in part by NIH grants 1R21CA222749-01A and R03EB030280, and a ROSI grant from UT MD Anderson Cancer Center. It was conducted at the MD Anderson Center for Advanced Biomedical Imaging in-part with equipment support from GE Healthcare. T.P. was a consultant for Bracco Medical Systems, LLC.

SU-C-TRACK 6-04

Longitudinal PET/CT Imaging of ^{64}Cu for Radiopharmaceutical Therapy Dosimetry

M Merrick*, L Dunnwald, A Tiwari, J Sunderland, S Graves, University of Iowa, Iowa City, IA

Purpose: Radiopharmaceuticals for diagnostic and therapeutic purposes have rapidly emerged in recent years. With recent FDA approval of ^{64}Cu -DOTATATE ($t(1/2):12.7\text{h}$; $\beta^+ :17.6\%$), there has been excitement over the possibility of using this imaging agent to predict the dosimetry of ^{177}Lu -DOTATATE ($t(1/2):6.6\text{d}$) or ^{67}Cu -analogs ($t(1/2):2.6\text{d}$). Pre-treatment PET/CT imaging is considered standard practice, so the possibilities of pre-treatment planning and quantitative assessment of potential benefit from therapy are highly attractive. The purpose of this work is to establish the qualitative and quantitative limitations of late-timepoint ^{64}Cu PET imaging. **Methods:** An anthropomorphic chest phantom (SNMMI-CTN) containing spherical 'tumors' was filled with ^{64}Cu -DTPA. Initial tumor-to-background ratio was 1.07:9.61 $\mu\text{Ci/mL}$, for a total fill activity of 11.02mCi. PET/CT images were acquired (GE Discovery MI PET/CT;10min/bed position;2-bed-positions/scan) and reconstructed (VPFX-4i16s) every 12hours for 7days and then every 24hours for 3days. Reconstructed images were assessed qualitatively to determine tumor detectability as a function of tumor size and time. Quantitative activity recovery-coefficients (RC-SUVmean) were calculated as a function of sphere-size and decay-time. Additionally, a radioactive half-life was calculated from activity measurements over-time to assess system linearity over a range of activity levels. **Results:** All lesions (7–37mm) were visualized 4days (~ 8 half-lives) after injection. Spheres with diameter range 7-17mm were gradually lost to background noise during days 5–7, with only the 22, 28, and 37mm spheres being visualized on day 8. Quantitative analysis of measured radioactivity vs. time indicated excellent PET response linearity even under low-count-density conditions. A ^{64}Cu half-life of $12.79\pm0.03\text{hours}$ was measured, which is $\sim 0.7\%$ higher than accepted value. RC-curves demonstrated an increase in variability at late time-points, however, generally the measured RC-curve shape was maintained out to 6days post-injection. **Conclusion:** Copper-64 appears to be an excellent diagnostic pair for theranostic applications. This study has confirmed that late-timepoint imaging following administration is possible, thus demonstrating promise for pre-treatment radiopharmaceutical dosimetry.

SU-C-TRACK 6-05

NCINM2.0: Organ Dose Calculator for Patients Undergoing Nuclear Medicine Procedures - Implementation of the ICRP Reference Phantoms and Biokinetic Models

D Villoing, C Lee*, National Cancer Institute, Rockville, MD

Purpose: To implement two major updates into the previously-published organ dose calculation program, NCI dosimetry system for Nuclear Medicine (NCINM): (1) a library of S values calculated using the International Commission on Radiological Protection (ICRP) reference pediatric and adult voxel phantoms and (2) the ICRP biokinetic models for selected radiopharmaceuticals. **Methods:** We calculated a library of photon and electron SAFs for a total of 68 source organs and 55 target organs by using the ICRP reference newborn, 1-, 5-, 10-, 15-year-old, and adult male and female voxel phantoms. We then derived S values from the SAFs for a total of 299 radionuclides frequently used in nuclear medicine procedures using the energy spectra from ICRP Publication 107. We also extracted the biokinetic models for twelve radiopharmaceuticals from ICRP Publication 80 and 106. The S values and biokinetic models were implemented into the NCINM program. We compared our organ doses for 18F-FDG with those from ICRP Publication 106. **Results:** We found that the organ doses from ICRP Publication 106 are overall smaller than those from NCINM. The breast dose from the ICRP 106 is about 20% of that of NCINM for the 1-year-old child phantoms. As an exception, the urinary bladder wall dose of ICRP is up to 2.5-fold greater than that of NCINM. **Conclusion:** We implemented a new library of S values based on the ICRP reference pediatric and adult phantoms and the biokinetic models of twelve radiopharmaceuticals into NCINM. The upgraded NCINM should be useful for nuclear medicine physicists to readily estimate organ doses in major nuclear medicine procedures based on the international reference anatomical and biokinetic models.

SU-C-TRACK 6-06

Organ Doses Following Intra-Hepatic-Arterial Administration of 99mTc-MAA in Planning Dosimetry for 90Y-SIRT

SC Kappadath*, BP Lopez, A Mahvash, UT MD Anderson Cancer Center, Houston, TX

Purpose: Expected organ doses from the intra-hepatic-arterial (IHA) administration of 99mTc-macro-aggregated-albumin (MAA) for lung-shunt-fraction (LSF) estimation prior to 90Y-SIRT or radioembolization are now requested by the FDA for new 90Y-radioembolization trials. Unfortunately to date, no such values have been published. Traditionally, organ doses from any radionuclide administrations are estimated through MIRD-based dosimetry following a laborious prospective study to establish radiotracer biodistribution through serial imaging. Here, we instead relate the possible IHA 99mTc-MAA biodistributions, considering the worst-case LSF and 99mTc-MAA disassociation values, to the published biodistributions and organ doses following intravenous 99mTc-MAA and intravenous 99mTc-pertechnetate administrations to estimate organ doses following IHA 99mTc-MAA. **Methods:** Three potential IHA 99mTc-MAA biodistributions were considered. Case1) all 99mTc-MAA is sequestered in the liver (0% LSF, 0% disassociation), and organ doses are estimated with OLINDA1.1 with a liver source organ of 99mTc-MAA (8.67MBq-h/MBq residence time). Case2) all 99mTc-MAA exits the liver into venous circulation (100% LSF, 0% disassociation), and organ doses follow intravenous 99mTc-MAA ICRP-80 values. Case3) all 99mTc-MAA disassociates and exits the liver into venous circulation (100% LSF, 100% disassociation), and organ doses follow intravenous 99mTc-pertechnetate ICRP-80 values. We report worst-possible individual organ doses (highest values across Cases) and the expected organ doses for typical LSF and disassociation (weighted linear combinations of Cases). **Results:** The liver (98.6 mGy/GBq), lung (66.0 mGy/GBq), and colon (42.0 mGy/GBq) received the largest doses in Case1, Case2, and Case3, respectively. Following 0.185 GBq (5 mCi) IHA 99mTc-MAA, the liver and lung doses were 18.2 and 12.2 mGy (worst-case), 15.8 and 2.3 mGy (typical 10% LSF, nominal 5% disassociation), and 11.7 and 3.3 mGy (high 20% LSF, high 20% disassociation). **Conclusion:** Our novel approach uses published intravenous 99mTc-MAA and 99mTc-pertechnetate dosimetry in conjunction with OLINDA1.1 liver dosimetry to provide previously unknown and unpublished worst-case and typical-case organ doses following IHA 99mTc-MAA.

SU-C-TRACK 6-07

Uncertainty Propagation Methods for Personalized, Image-Based 131Iodine SPECT/CT Radiopharmaceutical Therapy Dosimetry

D Adam*, I Marsh, J Grudzinski, T Bradshaw, P Hill, B Bednarz, University of Wisconsin, Madison, WI

Purpose: The assessment of uncertainty in personalized, image-based, radiopharmaceutical therapy (RPT) dosimetry is complicated and is commonly ignored in clinical routines because it is not as developed as it is for external beam radiotherapy. As such, this work investigates the uncertainties associated with ¹³¹I SPECT/CT based Jaszczak phantom studies and provides the groundwork for the estimation of dosimetric uncertainties for patient-specific RPT treatments. **Methods:** ¹³¹I SPECT/CT cylindrical Jaszczak phantom studies were conducted. The first phantom study (5.64L with 105 kBq/ml ¹³¹I) established a calibration factor for the SPECT camera and the second phantom study included hot-sphere inserts (257 kBq/ml ¹³¹I with 10:1 insert to background ratio) with volumes of 0.5-16ml to establish recovery coefficient (RC) partial volume corrections. GE Xeleris 4.0 was used to reconstruct images with CT-based attenuation, scatter, and resolution corrections enabled. Segmentations were manually performed on the CT image. The uncertainties in the volume, RC, mean counts, total sensitivity, and cumulative uncertainty were then calculated. **Results:** In general, the uncertainty decreased as the size of the structure increased. The largest uncertainties were the mean counts and RC which are dependent on the volume determination and the RC fitting function. The range of uncertainties for volume (17.3% to 54.7%), RC (22.32% to 95.71%), mean counts (13.2% to 106.4%), total sensitivity (7.47%), and cumulative activity (25.6% to 113%) were calculated. **Conclusion:** The uncertainties for ¹³¹I SPECT/CT RPT can be larger than 100% for small structures and thus require the utmost consideration in the administration of RPT treatments. Patient-specific RPT dosimetry becomes much more valuable for clinical translation if the associated uncertainties are also measured and reported for the physician's reference.

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SAM Therapy Educational Course

TRACK 7

Bottleneck Issues in the Clinic Implementation of Monte Carlo Method in Proton Therapy

SU-C-TRACK 7-00

Bottleneck Issues in the Clinic Implementation of Monte Carlo Method in Proton Therapy

Moderators: Liyong Lin, Emory Proton Therapy Center, Atlanta, GA and Ying Xiao, University of Pennsylvania, Philadelphia, PA

SU-C-TRACK 7-01

L. Lin, Emory Proton Therapy Center:

Bottleneck Issues in Proton Monte Carlo Clinical Implementation and Companion Imaging Techniques – Report of NRG Survey Results

SU-C-TRACK 7-02

J. Schuemann, Physics Division, Department of Radiation Oncology, Massachusetts General Hospital and Harvard Medical School, Boston, Massachusetts, USA:

Proton Monte Carlo Platforms – Treatment Planning and Latest Developments

SU-C-TRACK 7-03

S. Leng, Mayo Clinic:

Dual Energy Computed Tomography, Metal Artifact Reduction and Photon Counting Detectors – Latest Development and Potential for Monte Carlo Dose Calculation

Monte Carlo (MC) dose calculation has recently been implemented in primary commercial treatment planning systems (TPS) as well as various in-house platforms for proton therapy. MC promises smaller dose uncertainties, particularly in patients with high heterogeneities within the treatment field (e.g. bone/lung) or large surgical implants. Newly developed dual-energy computed tomography (DECT) and metal artifact reduction (MAR) techniques complement MC capabilities. However, there are no reports yet on how proton therapy centers implement these new technologies. Therefore, an NCI-sponsored NRG survey was distributed to operating USA proton centers in 2019 to determine the feasibility for including MC and companion techniques in cooperative group clinical trials. The use of MC for primary dose calculation and optimization is found to be prevalent and, therefore, likely feasible for clinical trials, where MC is encouraged for patients without implant materials. For patients with metal implants, there is consensus to use MAR and override tissues surrounding metals. However, there is no consensus about DECT regarding virtual mono-energetic images or the extraction of material information for implants and human tissues. The development and standardization of these advanced technologies are strongly encouraged for vendors and clinical physicists alike. In this joint imaging therapy symposium, we will review the latest developments of MC and imaging methods to facilitate their implementations in clinical trials with updated consensus practice recommendations and to mitigate the identified bottleneck issues. (max 250 words)

Learning Objectives:

1. Report of NRG survey results of MC implementation and companion imaging techniques, including current status and bottleneck issues (Liyong Lin)
2. Report on the latest development of MC in commercial TPS and in-house platforms (Jan Schuemann)
3. Report on the latest development of DECT and MAR techniques (Shuai Leng)

Vendor Showcase**TRACK 8****LAP Laser, LLC****SU-SHOWCASE-TRACK 8-1-01****RadCalc with 3D Monte Carlo**

C Bohorquez¹*, (1) Lifeline Software Inc, Mount Pleasant, SC

Council Symposium**TRACK 1*****Professional Council Symposium: Professional Skills for Future Medical Physicists: A Case Study Example*****SU-D-TRACK 1-00****Professional Council Symposium: Professional Skills for Future Medical Physicists: A Case Study Example**

Moderator: Brent Parker, The University of Texas M.D. Anderson Cancer Center, Galveston, TX

SU-D-TRACK 1-01**Introduction****SU-D-TRACK 1-02****R. Fahrig, Siemens Healthcare GmbH:**

A View of Clinical Trials from Industry

SU-D-TRACK 1-03**W. Newhauser, Louisiana State University:**

A View of Clinical Trials from Education

SU-D-TRACK 1-04**R. Nishikawa, University of Pittsburgh:**

A View of Clinical Trials from Academia

SU-D-TRACK 1-05**Q&A**

The session will present a case study on the processes involved in conducting a clinical trial and the skill sets needed for various steps in the process. Perspectives will be provided for research, industry, and educational points of view. While Medical Physicists can take on the role of 'clinical studies manager', specific training and certification already exists for that role and is not the focus of this discussion. The case study will be used as an example of the need for medical physicists to develop new skill to meet the future needs of the profession (e.g., data anonymization and deidentification, financial understanding, bringing innovation closer to marketable commercial product, educational needs to prepare future physicists in these areas).

Rebecca Fahrig, PhD

A View of Clinical Trials from Industry

Medical Physicists have the opportunity to play several different roles in Industry, including supporting research, pre-development, development, and product definition and productization. One key role is to support the creation of clinical evidence that proves the benefit of medical physics technology (imaging and therapeutic) to the Healthcare Community. Clinical studies literacy helps to ensure that the evaluation and clinical implementation of innovative and complex technology is supported by Medical Physicists, and that Medical Physicists continue to be seen as crucial to successful transition of a technology from concept to clinic.

Understanding the pre-development process

-What makes a prototype 'close to' product?

oWhen is a prototype ready for clinical evaluation?

-Different regulatory environments for Medical Product testing and release : EU, USA, China, etc.

Bringing the technical details to the table in the planning phase

-Evaluating Risks and Benefits – our sweet spot

-AI and Software as a Product : clinical data access and management

Wayne Newhauser, PhD

A View of Clinical Trials from Education

There are opportunities for many medical physics faculty, students and residents to learn about and participate in clinical trials. Example areas of interest include study design, quality assurance, dosimetry intercomparisons, treatment planning protocols, IRB protocols, data analysis, dose reconstructions, compliance, accreditation, scientific communications, to name a few. Based on the example case presented, we will compare the list of competencies needed against the typical graduate curricula (e.g., as specified by the AAPM Report 197), e.g., scientific methods, radiation physics, dosimetry, medical research ethics, protected health information, etc. Graduated levels of skills in related scientific disciplines will be considered, including statistics, epidemiology, radiation biology, where requirements may vary from familiarity with basics concepts to proficiency to perform certain tasks. Lastly, we will identify selected additional resources available to build skills of relevance to participating in clinical trials.

Robert Nishikawa, PhD

A View of Clinical Trials from Academia

Medical Physicists play a pivotal role in establishing the safety and efficacy of a new technology. Safety and efficacy are the sine qua non for approval of a

medical device to be used clinically, as defined by the Food and Drug Administration (FDA). The FDA often requires a clinical trial to establish safety and efficacy, and these trials are almost always conducted by academia. Medical Physicists can either lead the study or support the study in collaboration with other scientists. In either role, it is necessary or at least beneficial to understand the different components in conducting the trial. In this presentation, I will discuss the different components of a clinical trial, with emphasis on data anonymization and deidentification; and the IRB process.

Multi-Disciplinary Scientific Session

TRACK 4

Motion Assessment and Management

SU-D-TRACK 4-01

A Subject-Specific 5D Cardiac and Respiratory Motion Model Reconstructed from Free-Breathing 2D Cine MRIs

X Wu*, y Zhang, H An, H Gach, S Goddu, H Li, D Yang, Washington University in St. Louis, St. Louis, MO

Purpose: To treat tachycardia disease using the noninvasive cardiac radiation ablation, an accurate patient-specific heart motion model is required for treatment planning and motion management. In this study, we developed a procedure to generate a subject-specific 5D (3D+cardiac+respiration) heart motion model from free-breathing continuous 2D CINE MRI movies without additional ECG and respiratory signals. **Methods:** Seventeen 2D CINE MRI movies of a healthy volunteer was acquired continuously in the sagittal view at 17 positions at 5mm intervals, covering the entire heart. Each CINE movie was acquired for 60 seconds (8f/s, 100x100x480, 1.5x1.5x5mm3). In each movie frame. The cardiac signal was extracted from the heart area, and the respiratory signal was extracted from the average image intensity of the selected box in the diaphragm. Each signal was fitted to a combination of short cine curves. The final cardiac and respiratory phase values were assigned based on the fitting results. 3D volumes of selected cardiac and respiratory phase pairs were constructed from the 2D frames at all sagittal positions based on the assigned values and the temporal and spatial continuity. The deformation fields among the constructed 3D volumes of the selected cardiac and respiratory phases were computed using a group-wise deformable registration method. A PCA analysis was applied on the computed deformation fields to create the final 5D motion models. **Results:** The proposed procedure was applied for 2D cine MRIs of a healthy volunteer. The 5D motion model was composed successfully and 3D heart volumes at any cardiac and respiratory phases can be constructed using this model. **Conclusion:** We developed a procedure to generate a 5D subject-specific heart motion model from continuous 2D CINE MRI movies. It will be useful in managing cardiac and respiratory motions in radiation therapy for cardiac diseases.

The project described was partially supported by the Agency for Healthcare Research and Quality (AHRQ) grant number R01-HS022888, National Institute of Biomedical Imaging and Bioengineering (NIBIB) grant R03-EB028427 and National Heart, Lung, and Blood Institute (NHLBI) grant R01-HL148210.

SU-D-TRACK 4-02

Automated Predictive Models for Lung ART

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Purpose: Mid-treatment anatomic changes for patients undergoing radiation therapy for lung cancer can alter the accuracy of delivered vs. planned dose. Changes visible on the daily conebeam CT (CBCT) can be classified into five types: tumor regression, tumor alignment, nodal target alignment via two surrogates (carina and thoracic vertebrae) and lung density. To maintain the intended treatment quality, adaptive radiotherapy (ART) modifies the treatment plan to account for each type of change. However, the evaluation on when to adapt is completed manually, resulting in a subjective and inconsistent application. To address these limitations, five predictive ART models were created to identify when to implement ART. **Methods:** Models were trained utilizing 1158 CBCT datasets from 43 patients treated 2010–2018. The “ground-truth” need for ART was evaluated on each CBCT by two trained clinical staff, with ART indicated when anatomic deviations exceeded predefined tolerances across three consecutive days. The rate and magnitude of anatomic changes were quantified using a subset of 368 features derived from image registration similarity criteria applied between the simulation CT and on-treatment CBCT. These quantifying criteria, along with the “ground-truth” ART evaluation, were input data for predictive logistic regression models, with one model for each type of anatomic changes. The models were evaluated with a 10-fold cross-validation, focusing on the specificity, sensitivity, and first identified fraction compared to “ground-truth”. **Results:** The models identified ART with a range of sensitivity (78.9%–100%) and specificity (93.5%–100%) across five types of anatomic changes. Model accuracy to predict the specific fraction for ART ranged from -3.58 ± 5.35 to 0.14 ± 7.54 fractions, with negative numbers indicating ART prediction occurring before the “ground-truth”. **Conclusion:** A method for automated ART predictive modeling in lung cancer patients was developed and provided a high degree of accuracy to predict the need for ART for each individual fraction.

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SU-D-TRACK 4-03

Biomechanical Simulation of Lung Motion for Tumor Position Estimation During Radiation Therapy

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Purpose: To address challenges inherent in single-cycle respiratory-correlated CT (4DCT), several surrogate-based motion models (SMMs) have been proposed. These models utilize the a-priori correspondence between the surrogate and internal anatomy observed during the 4D-simulation. Commonly occurring changes in this correlation and extrapolation beyond their training dataset can result in model breakdown. To address such issues, we present a hybrid finite-element (FE) model with transient boundary conditions (BCs) to estimate tumor position during radiotherapy delivery. **Methods:** 4DCT data, VisionRT surfaces (VRT), and ~30–50s lateral and anterior-posterior fluoroscopic projections (FLs) were collected from five lung cancer patients. A previously validated volumetric SMM that integrated a few FLs to construct and update the correlation between VRT and internal motion was used to generate transient deformations for the lung-surface (as BCs). The exhale lung and tumor topology were imported into ABAQUS multi-physics software and a FE lung-tumor model was constructed. GTV position estimation was evaluated by comparing estimated centroid and Hausdorff distance with respect to the corresponding 4DCT phase. Transient BCs were used, and lung-tumor motion was simulated during the treatment session. **Results:** 95th percentile absolute difference between FE-estimated and 4DCT-delineated GTV centroid in the SI/AP/LR directions were 2.7/2.4/1 mm respectively. Submillimetre median errors were observed for all patients. For two patients, large errors (8mm and 5 mm) were present in peak-inspiration due to GTV-delineation inaccuracies from 4DCT binning artifacts. GTV excursions outside of the ITV were observed in transient simulations, but GTV motion was mainly within the PTV. For one FL-acquisition, baseline-shifts were present, with GTV outside of the PTV more than 50% of its time. **Conclusion:** Our findings indicate that the tumour can move outside the irradiated volume. Clinical implementation and application of our model can lead to improvements in more precise RT delivery and target margin accuracy.

SU-D-TRACK 4-04

Motion Management During Breath-Hold SBRT Liver Treatments: External Optical Monitoring Is Not Adequate – First Results of the TROG 17.03 Liver Ablative Radiotherapy with KIM (LARK)

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Purpose: During liver SABR treatment, breath-hold at end exhale or deep inhale is often used to manage target motion. Typically, breath-holds are monitored with a surface monitoring technique such as the Varian RPM. In the LARK trial, we deployed a real-time IGRT system to monitor the internal target motion. Here, we investigate the differences between external and internal monitoring for breath-hold SABR. **Methods:** Seven patients with colorectal metastases in the liver were treated at end-exhale (5 patients) or deep-inhale (2 patients) breath-hold on a Varian TrueBeam (dual-arc VMAT, 6FFF). A margin of 5mm in all directions was applied to the GTV to create the PTV. Three gold fiducial markers (3mm×1mm diameter) were implanted around the patient's tumours for real-time IGRT tracking surrogate. During treatment, the patient internal motion was monitored with an in-house 3D real-time IGRT method utilizing intrafraction kilovoltage imaging at 10Hz. External surface position monitoring (Varian RPM) was also used to measure the patient's breath-hold. Couch shifts were performed following kV pair imaging if the internal motion exceeded 3mm for two consecutive breath-holds. An isocenter shift method was used to reconstruct the delivered motion-included dose with real-time IGRT and with external monitoring alone. **Results:** With real-time IGRT, the 95% confidence interval of the target motion was [-2.3, 2.5]mm, [-3.0 3.0]mm, [-2.7 3.0]mm in LR, SI and AP. With external monitoring, the 95% confidence interval of the target motion would be [-2.4, 2.9]mm, [-3.7, 5.4]mm, [-3.6 3.3]mm in LR, SI and AP. Dosimetrically, with real-time IGRT, the PTV D95 was within 2.7% of the planned value for all patients. With external monitoring alone, the PTV D95 would be underdosed by 4.5±3.8%. **Conclusion:** Real-time IGRT is more accurate for target motion monitoring during breath-hold liver SABR treatments. External monitoring alone resulted in target underdose due to undetected internal motion.

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SU-D-TRACK 4-05

Reconstruction of DRR-Like KV-DR Using CycleGAN-Based Image Synthesis for Intra- and Extracranial SRT/SRS

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Purpose: A kV digital radiograph (kV-DR) using ExacTrac® X-ray source (BrainLAB AG, Munich, Germany) undergoes degradation of image quality due to radiotherapy accessories equipped for patient immobilization. We proposed a deep learning model based on cycleGAN image synthesis to improve the image quality of ExacTrac kV-DR. **Methods:** A total of 2574 paired kV-DRs and digitally reconstructed radiograph (DRR) were obtained from stereotactic radiotherapy (SRT/SRS) treatments for patients with brain or spine cancer. The DRR-like synthetic kV-DRs were generated from kV-DRs directly obtaining from ExacTrac using a deep learning-based image synthesis network. 2166 and 234 of kV-DR and DRR pairs were used for training for testing, and 92 and 82 pairs were used for testing from brain and spine cases, respectively. The images were normalized into [0, 1], and then the order of the paired data cases was shuffled to prevent the overfitting in network training. The mean absolute error (MAE), the root-mean-square error (RMSE), and the peak signal-to-noise ratio (PSNR) were used to quantify the image quality of the synthetic kV-DR. **Results:** The synthetic kV-DR (MAE=0.26±0.08, RMSE=0.33±0.09, and PSNR=9.83±0.33 dB) showed a better image quality compared to ExacTrac kV-DRs (MAE=0.34±0.14, RMSE=0.39±0.14, and PSNR=8.62±3.07 dB). The average MAE, RMSE, and PSNR values were 0.21±0.06, 0.29±0.07, and 11.06±8.45 dB for the brain cases and 0.31±0.07, 0.39±0.08, and 9.83±2.42 dB for spine cases, respectively. **Conclusion:** We successfully suppressed the artifacts in the kV-DRs originated from radiotherapy accessories and generated a DRR-like synthetic kV-DRs from ExacTrac based on deep learning network. The proposed model can improve the tumor localization accuracy for SRS/SRT radiotherapy.

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SU-D-TRACK 4-06

Respiratory Adaptive Computed Tomography (REACT) to Reduce Artifacts in Helical 4DCT Scans

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Purpose: 4DCT imaging plays an important role in current 4D radiotherapy treatment planning. Conventional helical 4DCT scans are prone to image artifacts caused by breathing irregularity and the difficulty in satisfying the data sufficiency condition. To reduce image artifacts in helical 4DCT scans, we have developed a prospective couch velocity adaptation and gating protocol, known as REACT that allows an acquisition in a phase-specific displacement gating window and adapts the pitch based on the patient's breathing. **Methods:** Conventional helical 4DCT scans and prospective respiratory gated 4DCT acquisition (REACT helical) were simulated for 10 patients from the Virginia Commonwealth University tumour motion database using the REACT software. The REACT acquisition in helical mode was simulated both with a constant pitch and a variable pitch value which was updated in response to real-time changes in the patient's breathing rate. An initial training period was used to learn the patient's breathing characteristics and then to determine potential beam gating. The performance of the scans was measured using trace RMSE during imaging (a surrogate for breathing irregularity and image quality) and beam on time (a surrogate for imaging dose). **Results:** As compared to a conventional helical 4DCT, the average breathing irregularity (RMSE) during imaging was reduced by 36.84% and 12.06% for the constant pitch REACT acquisition and the variable pitch REACT acquisition respectively. The average beam on time increased by 49.23% for the variable pitch REACT acquisition as compared to a constant pitch REACT acquisition. **Conclusion:** The results demonstrate the potential of using the REACT method with an adaptive pitch to reduce image artifacts in helical 4DCT scans. While the gating protocol allows for reduction in breathing irregularity related artifacts, varying pitch allows to close the data sufficiency gap.

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SU-D-TRACK 4-07

Towards Closed-Loop Control Head Motion Correction with Soft Actuators in MRI-LINAC Systems

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Purpose: To realize real-time automatic patient head motion correction during radiation therapy with non-magnetic and non-radiation attenuating actuators. Contrary to current methods, this (i) removes need for frames or thermoplastic masks; (ii) corrects the complex intra/inter-fractional geometric uncertainties; (iii) and is compatible with standalone MRIs or combined MRI-Linac systems. **Methods:** Following our recent investigation, we now present the real-time deformation framework of a soft actuator within the erstwhile-fabricated mechanism and describe the sensing and closed-loop pneumatic control system. We clamp fabricated actuators using 3D-printed PLA holders and fit the planar actuator with a pipe fitting connector. Compressed air at low pressure (~3-10 psi)

is supplied through a firm polyurethane air tubing into a pressure transducer connection, whose outlet conveys the airflow into a proportional solenoid valve, which in turn leads to the pipe fitting connection of the soft. We construct a standard current source electronic regulator that varies the air flow into the solenoid valve. Finally, a flex sensor, meshed laterally along the center of the actuator during curing measures deformation. **Results:** Effective closed-loop deformation of the actuator chamber is now realized as a result of the designed mechanism. By measuring the deformation of the actuator with an integrated flex sensor, we can proportionally adjust the airflow into the solenoid valves thereby allowing accurate regulation of the head motion. **Conclusion:** This work shows the potential of real-time head motion correction with non-magnetic and a radiation-transparent soft actuator, arranged in systematic fashion around a patient's head. This mechanism is useful in emerging hybrid MRI-Linear accelerators. Its inherent actuation medium (air) eliminates the risk of electronic/metallic parts that are unsuitable for the MRI's magnetic fields and assures safe human-robot interaction.

Therapy Scientific Session

TRACK 6

Patient Safety and Quality Improvement

SU-D-TRACK 6-01

A Systematic Approach to Balancing Radiotherapy Risk Management and Infection Control During a Pandemic

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Purpose: To develop a systematic risk management method that uses failure modes and effects analysis (FMEA) to design mitigation efforts to control pandemic infection spread, while ensuring safe delivery of quality radiotherapy. **Methods:** We define Infection containment failure mode (ICFM) to be any scenario where transmission of the infection occurs within the clinic. A two-stage approach has been developed for application of FMEA to radiotherapy workflow during a pandemic. In stage 1, the FMEA is focused on ICFMs, where risk priorities are evaluated based on available data and modeling of the pandemic infection. An infection risk priority number (IRPN) has been defined as a function of time, distance and protective gear. Subsequently, infection containment mitigation plans are integrated into the radiotherapy workflow. In stage 2, a traditional FMEA will be conducted on the adjusted workflow with a focus on radiotherapy failure mode (RFM) prevention. Furthermore, additional RFM mitigation plans will be introduced in the workflow, thereby iteratively returning to stage 1 for reevaluation of additional ICFMs. **Results:** The COVID-19 pandemic has been used to illustrate the two-stage systematic approach to risk management developed in this work. Selected parts of the radiotherapy process have been used to demonstrate examples of ICFM identification and failure mode mitigation. For a pre-pandemic CT-simulation workflow, 14 ICFMs were identified. For each ICFM, the IRPN was calculated based on average time, distance, and degree of protective gear with values ranging from 3 - 100. Various infection control measures were subsequently added to the CT-simulation workflow thereby reducing the IRPN scores. The modified workflow was then evaluated for RFMs and subsequently additional mitigation plans were introduced. **Conclusion:** The framework developed in this work offers a tool for radiotherapy clinics to systematically assess risks, adjust workflows, and allocate resources in response to the challenges during a pandemic.

SU-D-TRACK 6-02

Application of Machinery FMEA (MFMEA) for the Risk Assessment of Machine Issues Using An In-House Electronic Reporting System

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Purpose: Despite the capability to perform remote monitoring and preventive maintenance for radiotherapy machines, treatments are often interrupted by unforeseen machine issues. In contrast to the process-based FMEA for clinical workflow analysis (i.e. AAPM TG100), a machinery FMEA (MFMEA) was devised for the machine issues and treatment interruptions recorded in an in-house electronic reporting system. The impact of these events on treatment schedule, patient safety, and operators' competency in resolving these issues is assessed. **Methods:** Reported machine issues were classified under five sub-systems: beam-generating system, beam-limiting, kV-imaging, control software, and patient couch. MFMEA was implemented with four modules to analyze the impact of a component failure in each of the sub-systems. MFMEA modules include (1) functional analysis – identifies components of each sub-system (e.g. MLC for beam-limiting); (2) failure analysis – describes functional failures (e.g. stuck MLC), impacts (e.g. treatment terminations) and causes (e.g. faulty MLC-motor); (3) risk analysis – describes the ease of detecting the failure; and (4) risk management – risk prioritization and preventive measures. Scoring for occurrence, severity, and detectability for each component failure were guided by data such as the frequency of error reported, duration of downtime, and detection rate through routine QA and from a remote monitoring software. **Results:** Based on a total record of 215 issues reported over one year on one machine, 40% (kV-imaging), 29% (beam-limiting), 16% (beam-generating), 8% (control software), and 7% (patient couch) failures correspond to 20 component failures generated. The top three component failures (with their risk-priority-number) are – CBCT (424), couch positioner (214), and MLC (176). **Conclusion:** The risks encountered with machine issues and interruptions have been analyzed for the first time using an MFMEA model with data from an electronic reporting system. Trainings and preventive measures for improving treatment efficiency are identified based on the risk model

SU-D-TRACK 6-03

Automatic Detection of Vertebral Body Misalignment Errors in Cone-Beam CT Guided Radiotherapy Using a Convolutional Neural Network

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Purpose: Vertebral misalignments during cone-beam computed tomography (CBCT)-guided radiotherapy can lead to wrong-site treatment. We propose an automatic error detection algorithm that uses a three-branch convolutional neural network (CNN) to detect vertebral body misalignments using planning computed tomography (CT) images and setup CBCT images. **Methods:** Planning CTs and setup CBCTs were collected from 160 patients undergoing radiotherapy treatment in the thoracic or abdominal region using Varian TrueBeam and NovalisTx treatment machines. To simulate vertebral body misalignment errors, the setup and planning images were misaligned by one vertebral body and for each patient, one superior and one inferior misalignment were generated. The clinically applied registration was used to derive true-negative (no error) data. For each of the aligned and misaligned 3D image pairs, 2D slice pairs were extracted in each anatomical plane about an isocenter within the vertebral column. The 3 slice pairs obtained were then used as inputs to a three-branch CNN which returned a probability of vertebral misalignment. Each of the 3 branches was used to extract features from one of the orthogonal image planes, which were then joined in a single densely-connected layer. The CNN was trained on 140 patients and tested on 20 patients. The algorithm performance was quantified using a receiver operating characteristic (ROC) analysis. Due to the rarity of vertebral body misalignments in the clinic, a minimum threshold value yielding a specificity of at least 95% was selected. **Results:** The ROC analysis resulted in an area under curve of 0.985 with a 95% confidence interval of [0.978 0.992]. The sensitivity was found to be 90% for a fixed specificity value of 95%. **Conclusion:** Our CNN-based algorithm can accurately identify off-by-one vertebral body misalignments in planning CT and setup CBCT registrations, leading towards the possibility of an on-line error prevention system.

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SU-D-TRACK 6-04

Multi-Institutional Validation of a Convolutional Neural Network-Based Approach to the Detection of Vertebral Body Misalignments in Planar X-Ray Setup Images

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Purpose: Alignment to the incorrect vertebral body remains a rare but serious patient safety risk in image-guided radiotherapy (IGRT). Our group has previously presented a prototype convolutional neural network (CNN)-based approach for automatic detection of these errors in a planar stereoscopic x-ray IGRT system. Here we expand on our previous work to examine the robustness of our model when applied to multi-institutional data. **Methods:** A single-institutional model was trained using x-ray and digitally reconstructed radiograph (DRR) image pairs from 87 thoracic spine radiotherapy patients treated at a single institution using a stereoscopic x-ray IGRT system. Training data used simulated off-by-one vertebral body errors created by shifting DRRs by one vertebral body in both directions along the spinal column using a semi-automated method. 70 patient datasets were used to train the CNN model and 17 used for testing the detection accuracy. This trained model was then applied to 167 patient datasets collected from 3 external institutions to evaluate model robustness. Subsequently, the CNN was re-trained using data from all four institutions. Of this combined dataset, 204 patients were used for re-training and 49 reserved for testing. For all datasets, all treatment fractions were used in each of the patients. **Results:** When the single-institution CNN was used to classify unseen image pairs from collaborating institutions, the resulting receiver operating characteristic area under the curve (AUC) was 0.94. However, with the specificity fixed at 99% the corresponding sensitivity was 32.9%. For the model re-trained and tested on multi-institutional data, an AUC of 0.99 was obtained, and a sensitivity of 91.9% when specificity was fixed at 99%. **Conclusion:** We have demonstrated a CNN-based vertebral body misalignment model that is robust to multi-institutional image data. The incorporation of cross-institutional data into the training dataset significantly improves final classification accuracy.

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SU-D-TRACK 6-05

Radiotherapy Application of Causal Analysis Based On Systems Theory (CAST)

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Purpose: Radiation therapy (RT) accidents are commonly investigated with Root Cause Analysis (RCA) based on chain-of-events models. This work aims to analyze an RT incident using a technique based on a systems- and control-theory model and to describe early impressions of this approach. **Methods:** Causal Analysis based on Systems Theory (CAST) was applied to an RT incident recreated as a publicly-available video (i.treatsafely.org, #118350879) involving a pediatric patient that received a suboptimal treatment. We performed the five parts of the CAST methodology: 1) gather basic information about the accident, 2) create a safety control structure, 3) analyze the relevant components, 4) identify systemic factors, and 5) create a safety improvement program. These steps were performed by an experienced clinical physicist who has extensive experience with RCA and an expert in systems safety engineering and CAST. A qualitative comparison between CAST and RCA was completed. **Results:** A diverse set of contextual factors contributed to the incident. Including the mental model flaws of the frontline staff, these contextual factors revealed additional contributions by technology providers, management, as well as regulatory and advisory bodies. Four systemic factors (e.g., communication and coordination, economics, etc.) were also involved in the incident. Multiple intervention ideas were generated covering technology, practice change, and management to prevent incident recurrence. Overall, CAST enabled a robust analysis of the accident. For the non-safety expert, the CAST analysis forced incident analysis from multiple levels which is different than the conventional RCA approach that primarily targets frontline personnel. However, the CAST methodology is more complicated to use than RCA for the non-safety expert. **Conclusion:** While CAST appears to be a more robust methodology than RCA, achieving its potential requires prior knowledge about the procedure. Also, some terminology modifications can be beneficial to make it more easily comprehensible when introducing it to clinicians.

SU-D-TRACK 6-06

The Utility of Medical Physics Proton Peer Review for Clinical Trials

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Purpose: To highlight the value of IROC's on-site proton audits and key findings. **Methods:** IROC has performed 38 on-site measurement-based peer reviews of proton centers participating in NCI-funded clinical trials. The reviews covered beam calibration, lateral and depth measurements, mechanical checks, treatment planning, and quality assurance (QA) practices. The CT Number to Relative Linear Stopping Power (RLSP) conversion was tested with a phantom with known RLSP values, and by comparing the institution's curve to other proton centers. **Results:** Institutions received an average of 3 [1, 8] recommendations for practice improvements. The number of deficiencies did not decrease over time, highlighting the continued need for this type of peer review. The most common deficiencies were for QA (97% of centers), CT Number to RLSP conversion (50%), and treatment planning (45%). 32% of institutions failed at least one lateral beam profile measurement (<90% of pixels passing 3%/3mm, 10% threshold), despite passing internal QA measurements. These failures occurred for several different plan configurations (large, small, shallow, and deep targets), and at different depths in the beam path (proximal to target, central, and distal). CT Number to RLSP conversion curves showed deviations at low, mid, and high CT Numbers. This test highlights areas of inconsistency between proton centers, with many centers falling outside 2σ of the mean curve of their peers. All deficiencies from the peer review were discussed with the institutions, and many implemented dosimetric, treatment planning, and practice changes to improve the accuracy of their system and consistency with other institutions. **Conclusion:** This peer review program has been integral in confirming comparability and consistency across proton centers for clinical trials, minimizing deviations for outcomes data.

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SU-D-TRACK 6-07

Development of a Text-Mining Based Methodology for the Identification of Trending Topics in a Radiation Oncology Incident Learning Database

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Purpose: Incident-learning is a powerful tool necessary for continuous improvement of quality and safety within radiation oncology. However, as the percentage of captured incidents increases, so does the difficulty of identifying important trends/topics meriting the highest attention of the internal safety committee. This study utilized automated text-mining procedures to identify important topics for review. **Methods:** Data was exported to CSV for 133 incident reports from our institutional RO-ILS database. Narratives were extracted, and data was imported to Orange, an open-source data-mining tool, for processing and analysis. Text was converted to lower case before using regexp for tokenization (breaking into single-word pieces). Punctuation and unimportant words were filtered using a list of stopwords (words such as "the"). Finally, tokens were used to create a range of N-gram phrases (phrases with N number of words) which were counted and regularized to relative weighted "importance" before visualization using word clouds. Phrases with largest weightings were further investigated to determine if this was indicative of a topic that needed additional action. **Results:** The N-gram phrase "v-sim" was observed to receive a high

weighting. This represented a previously unidentified issue and, upon further investigation, was recognized as being indicative of both the number of events identified at the patient's v-sim appointment, as well as events that occurred due to shortened sim-to-treatment timelines. "Setup instructions" also received a high weighting, and was identified as being indicative of a previously known trend caused by incorrect or unclear setup instructions. This was encouraging, as it validated the potential of the tool. Other notable findings included "iso-center", which was tagged in many events with various topics. This was ascertained to be indicative of the large number of error pathways that can result in isocenter related problems. **Conclusion:** Preliminary use of text-mining tools shows promise for identifying relevant safety-related topics for review.

SAM Therapy Educational Course

TRACK 7

Current Status of Proton Thoracic Radiotherapy

SU-D-TRACK 7-00

Current Status of Proton Thoracic Radiotherapy

Moderator: Stella Flampouri, Emory University, Atlanta, GA

SU-D-TRACK 7-01

H. Li, Johns Hopkins Medicine:

Treatment Planning and Delivery for Proton Thoracic Therapy – An US Center's Experience

SU-D-TRACK 7-02

S. Both, University Medical Center Groningen:

Treatment Planning and Delivery for Proton Thoracic Therapy – A European Center's Experience

SU-D-TRACK 7-03

S. Flampouri, Emory University:

Parallels and Intersections in Thoracic Proton Therapy

SU-D-TRACK 7-04

Q&A

Thoracic proton therapy is challenging because of high tissue heterogeneity and volatility. The effects of uncertainties on proton range are accentuated in the thorax due to the respiratory motion and time structure of the proton pencil beam, making implementation of new developments more intricate than for other sites. This session will focus on how technological advancements in a. Plan optimization and evaluation and b. Adaptive planning and delivered dose accumulation, are implemented for thoracic targets. The latest in treatment planning systems such as Monte Carlo-based 4-dimensional robust optimization and estimation of biological weighted dose will be presented along with their benefits, disadvantages, and inadequacies in and around the lungs. Robustness evaluation methods based on dose calculation of error and anatomy scenarios recently replaced PTV evaluations in proton therapy and will also be introduced. In addition, estimations of delivered dose will be explored in the session. Methods of reporting dose distributions that include delivery log files, daily IGRT and patient monitoring data, and repeated scan images will be presented. Remaining issues in reporting fractional and cumulative doses that include the effects of uncertainties, errors, physiological motions, and anatomical deformations as well as plan adaptations will be reviewed. Practice differences between photons, older and new technology protons and their repercussions will be discussed. The Dutch Model Based perspective from thoracic patient selection to planning techniques and treatment evaluation methods will be presented.

Learning Objectives:

1. Be updated on the latest treatment planning techniques applied in proton thoracic therapy
2. Learn about dose evaluation both planned and delivered
3. Explore differences between European and North American practices in thoracic proton therapy

Vendor Showcase

TRACK 8

LAP Laser, LLC

SU-SHOWCASE-TRACK 8-2-1

THALES - Motorized Water Phantom for MRIdian & Halcyon-New Developments & User Experiences

T Mertens^{1*}, (1) LAP GmbH, Luneburg, DE

Imaging Interactive ePoster Discussion

TRACK 1

Imaging: CBCT Image Processing, Image Analysis, and Applications

SU-lePD-TRACK 1-01

CBCT Manual Vs. Semi-Automatic Single and Multi-Slice Low Contrast Resolution Analysis

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Purpose: To compare methods for quantifying low contrast resolution for monthly CBCT QA using manual and semi-automatic software analysis. **Methods:** The Phantom Laboratory's Catphan 504 CBCT images containing sub-slice and supra-slice low contrast disks were evaluated for twelve months. The number of supra-slice disks visible out of a maximum of twenty-seven was recorded after visual inspection of the images. Using Radiological Imaging Technology's (RIT) Radia Complete Imaging QA software analysis, a central slice in the CT515 module was selected and analyzed for each month. Selecting the same central slice as the middle, an average of seven slices was taken and evaluated using the same software. The number of disks detected using each **Method:** manual, single slice, and multi-slice, was then compared using Bland Altman analysis, and paired two-sample t-test. **Results:** In all three **Methods:** manual vs. RIT single slice, manual vs. RIT seven slice averaging, and RIT single slice vs. RIT seven slice averaging, the differences were statistically significant within a significance interval of $\alpha=0.05$. The degree of statistically significant difference was lowest between the manual vs. RIT single slice ($t=-3.70$, $p=0.0018$), increasing in the RIT single slice vs. RIT seven slice averaging ($t=-3.94$, $p=0.0012$), and reaching a maximum in the manual vs. RIT seven slice averaging ($t=-12.50$, $p=3.83E-8$). This maximum difference coincides with higher low contrast disk detectability with the RIT seven slice averaging method, and lower disk detectability with manual user observation. Bland Altman analysis shows an increasing bias going from manual to seven-slice averaging, and generally wide

limits of agreement. **Conclusion:** Semi-automatic multi-slice averaging greatly improves supra-slice low contrast disk detectability in the CT515 module. The method can provide users with a reliable mean to accurately assess monthly CBCT low contrast module performance.

SU-lePD-TRACK 1-02

Evaluation of Megavoltage Cone-Beam Computed Tomography Doses in Thoracic Radiation Therapy

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Purpose: Daily megavoltage cone-beam computed tomography (MVCBCT) imaging verification ensures the accuracy of treatment delivery but raises a cumulative dose to normal tissue. This study aimed to investigate imaging doses and cancer risk attributable to megavoltage cone-beam computed tomography undergoing thoracic radiation therapy. **Methods:** A total of 206 patient datasets (age range: 2-60 years; chest circumference range: 40–109 cm) were retrospectively assessed with chest circumference. An Eclipse clinical treatment planning system was used to calculate imaging dose distributions of MVCBCT scans using 10×10 cm² field size, 200° arc length, and low dose (5MU) protocol. Optically stimulated luminescence dosimeters (OSLDs) inserted in three anthropomorphic phantoms (equivalent to a 5-year, 10-year old child, and adult male) were then scanned using this scanning protocol by a Halcyon MV-CBCT. We used regression analysis of more than 200 MVCBCT doses to develop an empirical model of chest circumference, and cancer risk was calculated using the BEIR VII. **Results:** The maximum difference between the TPS calculated dose and the phantom validated dose was less than 4.9%. The organs at risk (OAR) mean doses decreased monotonically with increasing chest circumference. The median doses to the heart, spinal cord, lungs were 4.3, 3.1, and 2.9 cGy/scan. **Conclusion:** Results showed a good agreement between experimental measurements (with OSLDs) and TPS simulations for three anthropomorphic phantoms. Imaging doses and cancer risk are highly correlated with chest circumference in image-guided thoracic radiation therapy. Under the same CBCT imaging protocols, pediatric patients received even more than twice of imaging dose compared to adults due to their smaller body sizes.

SU-lePD-TRACK 1-03

Efficient Visualisation of Changes in CBCT During Radiotherapy to Detect Lung Infections

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Purpose: Cancer patients are vulnerable to COVID-19 and other infections. Daily CBCT is common for image-guided radiotherapy. To protect patients and staff lung infections must be detected timely, which is possible on daily thoracic CBCT. However, it is time consuming to inspect all slices of a thoracic CBCT, especially for longitudinal changes. Here we develop and optimise a tool for rapid detection lung of infections by visual comparison of daily images projected in 2D. **Methods:** All daily thoracic CBCTs are processed overnight by a pipeline containing registration, cropping to lungs, filtration, and maximum intensity projection. Parameters were optimised visually using 150 cases. A visual timeline of the treatment is presented in a report and animation. Images are reviewed each morning by a radiographer who flags issues to a clinical oncologist to determine if further action is required. The tool was retrospectively evaluated in 285 patients treated between January and June 2020 and prospectively implemented in a clinical workflow and departmental quality system in July 2020. **Results:** Rigid registration outperformed deformable registration, the latter being affected by lung lesions. Optimal cropping was the planning CT lungs - 3mm, balancing sensitivity for peripheral lesions and rejection of pleural effusion changes. Blurring ($\sigma=1\text{cm}$) in AP direction suppressed healthy lung structures while maintaining infection contrast. Report generation took ~60s per case, and inspection ~12s. To date, over 400 patients were prospectively evaluated and 4 asymptomatic patients were diagnosed with COVID-19 based on CBCT imaging changes. A further patient was identified as having a non COVID-19 lung infection on CBCT. **Conclusion:** We developed and validated a near real-time tool to identify lung density changes on CBCT indicative of COVID-19 or other lung infections. CBCT imaging provides a unique opportunity to study temporal progression of lung infections and protect patients and staff.

SU-lePD-TRACK 1-04

Cone-Beam Computed Tomography (CBCT) to Computed Tomography (CT) Image Translation with Perceptual Loss to Eliminate Cone-Beam Artifacts

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Purpose: Cone-beam computed tomography (CBCT) is widely used for image-guided radiation therapy (IGRT). The application of CBCT can be extended to adaptive radiation therapy (ART) for dose accumulation and re-planning, which are currently limited in accuracy due to degraded image quality from cone-beam artifacts. Thus, this work presents an advanced deep learning (DL) algorithm for generating synthetic CT from CBCT by incorporating perceptual loss. **Methods:** Patient cohort for the DL model consisted of 43 scans with paired CBCT and CT images. The dataset was split into 35 patients for train-set, 3 for valid-set, and 5 for test-set. We utilized a two-dimensional FC-DenseNet as the DL architecture to eliminate the cone-beam artifacts of CBCT. CT slices were deformable registered and resampled to CBCT using MIM software and MATLAB. With L1-loss only, we found that details in bone structure disappeared in synthetic CT generation. To enhance the translating performance, the perceptual loss besides L1-loss was newly applied. Data augmentations (flip, rotation, blur) were conducted to increase the data size for training. The similarity metric to assess the trained networks employed structural similarity index (SSIM) and mean absolute error (MAE). **Results:** 5 patients with paired CBCT/CT images were randomly selected to validate the trained networks. Two networks with L1-loss and (perceptual loss+L1-loss) resulted in similar MAEs (0.050 and 0.044). The L1-loss based network may well produce slightly lower MAE as the L1-loss emphasizes compressing MAE. Contrarily, in SSIM, our proposed network with additional perceptual loss outperformed that with L1-loss only (0.756 to 0.824). It implies that the perceptual loss substantially contributed to improving the similarity of synthetic CT produced from CBCT to real CT images. **Conclusion:** Our DL model with perceptual loss was demonstrated to promote the performance of CBCT to CT image translation, which can strengthen the usefulness of CBCT in radiotherapy.

SU-lePD-TRACK 1-05

Identification of Modes of Tumour Regression During Radiotherapy

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Purpose: Safety of adapting lung cancer radiotherapy (RT) to tumour shrinkage depends on whether microscopic disease is left behind - the question of elastic versus non-elastic regression. The purpose of this project is to quantitatively distinguish these modes of tumour regression in non-small cell lung cancer (NSCLC) patients. **Methods:** First, simulations of pure elastic and non-elastic regression were created. To quantify tumour regression, in both real and simulated cone-beam CTs (CBCTs), registration of all CBCT to day one CBCT was performed using NiftyReg algorithm. Data from 150 locally advanced NSCLC patients receiving the same prescription was used. From simulations, a methodology based on the determinant of the Jacobian (local volume change) was developed. The gross tumour volume (GTV) was triangulated and, at each facet, ten Jacobian values were sampled along lines perpendicular to the surface. Jacobian values were combined to classify tumour changes spatially and in time. Finally, modes of tumour change were compared between tumours of different histology and TNM stage. **Results:** In simulated data, our approach can distinguish elastic and non-elastic tumour regression combining Jacobian values 1.2cm

inside and outside the GTV boundary. Values near the boundary were unreliable. Patients showed a mix of elastic and non-elastic behaviour, but no real patient showed pure elastic regression. 88%(N=125) of patients showed stable disease in the first two weeks of RT. At treatment completion, 32%(N=46) of the patients showed tumour regression, predominately non-elastic. Slight differences in the mode of tumour change were found with histology and TNM stage. **Conclusion:** This work found regional differences in mode of lung tumours change on CBCTs. Patients with pure elastic regression, however, were not observed in our data. Our results suggest that safe adaptive RT for lung cancer patients requires careful consideration to avoid underdosing of microscopic tumour residuals after non-elastic regression.

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SU-lePD-TRACK 1-06

ROI Based Deep Learning Enhancement for CBCT Based Radiomics Analysis

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Purpose: To investigate the impact of CBCT image quality and ROI-based deep learning image enhancement on radiomics analysis. **Methods:** CBCT image quality is affected by scattering, beam hardening and noise, which directly impact its accuracy for radiomics analysis. This study focuses on investigating the accuracy of CBCT-based radiomics analysis and the efficacy of using deep learning to improve the accuracy. Around 900 CBCT projections were simulated from each patient's CT images using Monte Carlo method to contain primary signal, scatter signal and noise. The projections were then fed into the Varian Itool package to mimic clinical CBCT reconstruction. A Pix2Pix Generative-Adversarial-Network was used to improve the CBCT image quality based on CT images as ground truth. The deep learning model was trained for both enhancing the whole body and the ROI around the tumor using 11 lung patients and tested using 3 patients. Radiomics analysis was performed for both CBCT and enhanced CBCT and compared to ground-truth CT to evaluate their accuracy. **Results:** The CBCT image quality was significantly improved after the AI-based enhancement. 18 First order, 22 GLCM, 16 GLRLM, 16 GLSZM, 14 GLDM, and 688 wavelet features were computed for the CT, original CBCT, whole body-based enhanced CBCT, and ROI-enhanced CBCT. Under best scenario, the first 86 features improvement was 83.4%, 18.9%, 13.3%, -11.9% and 17.6% for each radiomics feature zone based on ROI image augmentation, and improved 88.9%, 21.7%, 15.9%, -12.2%, 18.2% based on whole-body image augmentation. The sensitivity of features against image quality was consistent for all 3 patients. **Conclusion:** Image quality is vital for establishing accurate CBCT based radiomics analysis, and deep learning can effectively improve the radiomic feature accuracy, especially for low-order features. The superior performance for low-order features is likely due to their consistency with the L1-loss function used in deep learning training.

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SU-lePD-TRACK 1-07

Patient-Specific Deep Learning Model for Enhancing 4D-CBCT Image for Radiomic Analysis

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Purpose: 4D-CBCT provides phase-resolved images valuable for radiomics analysis for outcome prediction throughout the treatment course. However, 4D-CBCT suffers from streak artifacts caused by under-sampling, which severely degrades its accuracy for deriving radiomic features. Previously we developed group-patient-trained deep learning methods to enhance the 4D-CBCT quality for radiomics analysis, which was not optimized for individual patients. In this study, we developed a patient-specific model to further improve the robustness and accuracy of 4D-CBCT based radiomics analysis for individual patients. **Methods:** The patient-specific model was trained using intra-patient data. Specifically, patient planning 4D-CT was augmented through image translation, rotation and deformation to generate 305 CT volumes from 10 volumes to simulate possible patient positions during the image acquisition. 120 projections were simulated from 4D-CT per phase and were used to reconstruct 4D-CBCT using FDK back-projection algorithm. Breathing motion traces were simulated using a regular breathing curve containing 120 breathing cycles. The patient-specific model was trained using these 305 sets of patient-specific 4D-CT and 4D-CBCT data to enhance the 4D-CBCT to match with 4D-CT. For model testing, 4D-CBCT were simulated from a separate 4D-CT scan acquired from the same patient, and were then enhanced by the patient-specific model. Radiomics features were then extracted from the testing 4D-CT, 4D-CBCT, and enhanced 4D-CBCT for comparison. **Results:** Compared with a group-based model, patient-specific training improved the accuracy of radiomic features, especially for the first order features. First order improved 75.97% and the total radiomics improved by 31.62% from the group-based model to patient specific model. For example, first-order median feature improved 82.39% and texture non-uniformity-GLDM improved 88.09%. **Conclusion:** This study demonstrated that the patient-specific model is more effective than group-based model in improving the accuracy of the 4D-CBCT radiomic features, which can consequently improve the precision for outcome prediction in radiotherapy.

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Imaging Interactive ePoster Discussion TRACK 2

Imaging: Mammography and Tomosynthesis

SU-lePD-TRACK 2-01

Theoretical Investigation of the Signal and Spatial Resolution Performance of HgI₂ Converters Incorporating Pillar-Supported Frisch Grid Structures for Digital Breast Tomosynthesis

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Purpose: The theoretical signal and spatial resolution performance of particle-in-binder mercuric iodide (PIB HgI₂) x-ray converters incorporating pillar-supported Frisch grid structures are reported. The grid serves to reduce hole signal contribution, and thus image lag, under digital breast tomosynthesis (DBT) irradiation conditions. **Methods:** Each converter employs a top, continuous electrode and 100 μ m pitch, bottom pixelated electrodes. Finite element analysis was used to calculate electric field and weighting potential maps in the converter. A trajectory tracking algorithm used these maps to follow the transit of electrons and holes through the converter and determine induced signal. The signal performance of the converter along with the modulation transfer function (MTF) were investigated as a function of various grid parameters in order to identify grid designs that optimize performance. **Results:** Examination of MTF and signal performance as a function of grid pitch (the center-to-center distance between two neighboring grid wires), RGRID (the ratio of grid wire width to grid pitch) and projection image number in a DBT scan, indicates the following. (1) Higher RGRID and smaller grid pitch provide better suppression of hole signal, but diminish total signal due to narrower gaps between grid wires. (2) Grid pitches considerably smaller than the pixel pitch are more effective for

maintaining MTF close to the ideal of a sinc function corresponding to the pixel aperture – irrespective of RGRID. (3) For a total DBT scan exposure of 15 mR, charge accumulation on the insulator pillars has insignificant effect upon performance – other than a slight improvement of total signal across consecutive images for high RGRID. **Conclusion:** PIB HgI₂ converters incorporating a pillar-supported Frisch grid at a grid pitch of 20 μ m and RGRID of 35% suppress up to ~76.5% of hole signal, while preserving total signal and MTF compared to a detector without a grid.

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SU-lePD-TRACK 2-02

Spectral Simulation for a Dual-Layer Detector for Dual-Energy Contrast-Enhanced Breast Imaging

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Purpose: Dual-energy (DE) contrast-enhanced (CE) breast imaging requires registered low energy (LE) and high energy (HE) images for DE subtraction and is susceptible to patient motion. The motion artifacts could be more severe for CE digital breast tomosynthesis. Dual-layer (DL) detectors allow acquisition of LE and HE images in one x-ray exposure which minimizes patient motion. We aim to investigate a DL detector design for DE CE breast imaging and evaluate the impact of the thickness of x-ray filter and detector layer on the iodine detectability. **Methods:** The DL detector consists of a top layer (TL) of amorphous selenium to generate the LE images and a bottom layer (BL) of columnar cesium iodine for the HE images with a glass substrate in between. The x-ray spectrum was simulated using tungsten target and rhodium (Rh) filter at 49 kVp with intensity that leads to a total radiation dose of 1.5 mGy to a 4-cm breast with 50% density. We calculated the x-ray spectrum for photons interacted in the TL and the BL for different Rh filter and TL thicknesses. The image signal was determined based on the x-ray energy deposition in the TL and BL and included quantum noise. We calculated the signal-difference-to-noise-ratio (SDNR) for 1 mg/cm² iodine in the DE-subtracted image and evaluated the tube-current time product (mAs) for different Rh filter thicknesses. **Results:** Increasing the Rh filter thickness improves the spectral separation in the TL and BL and increases the iodine SDNR. Decreasing the TL thickness reduces the HE photon interactions in the TL and further increases the iodine SDNR. Rh filter with 200 μ m thickness achieves the highest iodine SDNR with practical mAs. **Conclusion:** The proposed dual-layer detector is feasible for DE CE breast imaging to minimize patient motion.

SU-lePD-TRACK 2-03

Breast Thickness Map Estimation and Its Associated Correction in DBT Imaging

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Purpose: Breast compression is still required during the digital breast tomosynthesis (DBT) image acquisition. Because of the rounded shape of a breast, it cannot be constricted into a single uniform thickness particularly near the periphery. Current segmentation-based thickness correction methods are error-prone and do not necessarily represent the x-ray physics. **Methods:** We propose a convolutional neural network (CNN) based breast thickness correction method. Not only a normalized log projection image but also a Euclidean distance map were prepared as network inputs. A distance map is generated by calculating the Euclidean distance of each internal pixel of the projection image to the closest point of the skin-line. We utilized patch-based learning to reduce computational costs whilst maintaining the image resolution of DBT projections. By providing distance map along with a log projection image, one can preserve the positional information of the patched image. We introduce a concept of the thickness map, which is a projection of a hypothetical binary volume mask that distinguishes inside and outside of a breast. We define a thickness corrected image by dividing the projection image with the corresponding thickness map in a pixel-wise manner. Thickness corrected images are the targets of the network training. Dataset for training and testing was constructed with help of the VICTRE phantom and MC-GPU simulation. **Results:** In the simulation study, the proposed network successfully corrected the breast thickness nonuniformity. A low root-mean-square deviation (NRMSE) down to 1.976% and high structural similarity (SSIM) up to 99.997% were obtained quantitatively. **Conclusion:** We have developed a CNN-based breast thickness correction method. By offering a distance map along with the projection, the proposed network successfully performed the breast thickness correction. Our method may benefit several applications including DBT image quality enhancement, scatter correction, density estimation, and multimodal imaging such as DBT plus diffuse optical tomography.

SU-lePD-TRACK 2-04

Monte-Carlo Simulation of the Detective Quantum Efficiency of Cadmium Telluride X-Ray Detectors for Photon-Counting Mammography

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Purpose: Photon counting offers the possibility of single-shot spectroscopic imaging and electronic noise rejection, but charge sharing will degrade image quality. We used Monte Carlo (MC) methods to simulate the detective quantum efficiency (DQE) of cadmium-telluride (CdTe) photon-counting x-ray detectors for breast imaging and compared results against published DQEs for energy-integrating systems that use amorphous selenium (a-Se) detectors. **Methods:** We calibrated and validated a MC model of x-ray interaction and detection in CdTe photon-counting x-ray detectors against an XCounter Thor detector. Our model accounted for charge sharing, electronic noise and anticoincidence logic. We then optimized the detector thickness for a 28kVp Mo/Mo spectrum hardened by 3.9cm of lucite as a function of electronic noise level and pixel size with and without inter-pixel anticoincidence logic. The figure of merit used for optimization was the integral of the DQE. The optimized DQE was compared to published values of the DQE of energy-integrating a-Se systems for 85 μ m pixels. **Results:** Without anti-coincidence logic, the optimal DQE(0) was 0.72 at the optimal detector thickness of 300 μ m, which is comparable to the DQE of a-Se systems in clinical use. Anti-coincidence logic reduced the photon-counting DQE(0) by 14% due to an associated increase in electronic noise. Reducing the electronic noise level to ~1.5 keV in combination with anti-coincidence logic resulted in a DQE(0) of 0.84. **Conclusion:** Increasing the DQE of photon-counting CdTe detectors relative to that of energy-integrating a-Se detectors will require the use of anti-coincidence logic to suppress charge sharing, and electronic noise levels to less than ~1.5keV, which is approximately half of that of the system used in this work. Future work will focus on contrast-enhanced spectral mammography.

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SU-lePD-TRACK 2-05

Assessment of Case-Based Repeatability of AI Using Mammographic Radiomics of Breast Lesions

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Purpose: Assessment of case-based repeatability of artificial intelligence algorithms can provide insight into potential clinical use over a range of cases. Previous studies have demonstrated that variation in training cases impacts repeatability. This study investigated impact of classifier on case-based classification repeatability using human-engineered radiomic features extracted from full-field digital mammography (FFDM) images of breast lesions, for three different classifiers: linear discriminant analysis, support vector machine, and random forest. **Methods:** A retrospective, HIPAA/IRB compliant collection of 268 unique lesions (138 malignant, 130 benign) was used. There were 516 FFDM images; most lesions (144) were imaged in two or three views. Lesions were automatically

segmented; 29 radiomic features were extracted for each image. Classifier training/testing was conducted using a 0.632 bootstrap/1000 iterations, with cancer prevalence maintained in each bootstrap sample. The same bootstrap samples were used for the classifiers. Classifier outputs were averaged from different views of the same lesion to obtain case-based output (CBO). Performance in the task of distinguishing malignant from benign lesions was evaluated by the 0.632+ bootstrap-corrected area under the receiver operating characteristic curve (AUC). The bootstrap samples were used to calculate confidence intervals for AUC and the difference in AUC in pair-wise comparisons between classifiers. Difference in AUC was significant if the 98.33% confidence interval (CI, adjusted from 95% for multiple comparisons) failed to include zero. Case-based repeatability by classifier was characterized by repeatability profiles, i.e., histograms of median 95% CI of CBO as function of median CBO across all cases from the bootstrap samples. **Results:** Classifier performance failed to demonstrate statistically significant differences, but classifier output was more repeatable when the random forest classifier was used. **Conclusion:** This suggests that even if classifiers appear to perform similarly in terms of AUC, classifier choice can impact repeatability and should be considered when developing computer-aided diagnosis systems.

KD receives royalties from Hologic, Inc.

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SU-lePD-TRACK 2-06

Deep Learning Method for Volumetric Segmentation of Dense Tissue in Tomosynthesis Using Computer Simulations

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Purpose: The use of deep-learning systems in clinical care requires extensive evaluations to develop effective, safe, and acceptable artificial intelligence applications. This study aims to evaluate and optimize nnU-Net models to estimate volumetric breast density (VBD) in digital breast tomosynthesis (DBT) images. **Methods:** A virtual clinical trial (VCT) framework was employed to simulate 2,010 anatomical breast models, with distribution of dense compartments varying from 0.1% to 50%. DBT projections were simulated using the acquisition geometry of a clinical tomosynthesis system. The projections were reconstructed using 0.1mm, 0.2mm, and 0.5mm slice spacing. Further, we developed and trained three nnU-Net models using the 51 center-most slices of the DBT reconstructions along with the respective ground-truth tissue labels. For each model, 75% and 25% of ground-truth labels and DBT images were used for training and test, respectively. The effect of the reconstructed slice spacing in the nnU-net models was evaluated using sensitivity (TPR), Jaccard, and Dice metrics. **Results:** The accuracy of the nnU-Net varies significantly with smaller reconstructed slice spacing. Images reconstructed with 0.1mm slice spacing resulted in a performance improvement that represents a mean Jaccard increase of up to 41%, 32%, 31%, and 37% (P<0.001), for phantoms simulated using 10%, 20%, 30%, and 50% of dense compartments, respectively. Changes in performance are also noted in the mean Dice and mean TPR metrics. **Conclusion:** Computer simulations can optimize DBT acquisition parameters to improve VBD segmentation. Results from simulations with ground truth provide evidence for a better understanding of deep learning outcomes. The nnU-Net performance has been shown to improve the VBD segmentation in DBT using reconstructions with smaller slice spacing. Ultimately, we believe that a plethora of VCT-based strategies will be developed to optimize VBD segmentation.

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Owner, Daimroc Imaging, LLC

Member, RTT Scientific Advisory Board.

SU-lePD-TRACK 2-07

A Reader Model to Compute Detectability of Lesions with Varying Size in X-Ray Images of the Breast

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Purpose: Model observers have traditionally been applied to simple tasks where the lesion location is known exactly (LKE) or the lesion morphology is known exactly (SKE). However, these conditions are far from a clinical scenario. Here, we propose an observer model to detect masses of different size. This model can be used in virtual clinical trials comparing Digital Mammography (DM) and Digital Breast Tomosynthesis (DBT) images. **Methods:** We generated irregular spiculated masses of seven different sizes. Using the VICTRE pipeline, we generated 2270 DM and DBT samples for each size. Average glandular dose was consistent with the clinical dose levels and with the dose reported in the VICTRE project. We trained a Channelized Hotelling Observer (CHO) template per size by isolating the mass from the rest of the tissues. With target-only images, we trained multiple CHO models for different channel sizes and computed the performance using full sample sizes containing all the breast tissues in target-present and target-absent images. Finally, we analyzed the AUC for both modalities at different lesion sizes for 30 reader models. **Results:** The best channel set combination for DBT and DM show a consistent increase in detectability with DBT always higher than DM. Both modalities seem to track in terms of the benefits of having larger signals in the AUC values. As lesion size increases, the size of the channels should increase accordingly. We compared the AUC for a wide range of channel sizes and found that DBT images are more sensitive to channel sizes than DM. **Conclusion:** As the size of breast lesions increases, we found a monotonic increase in AUC for both modalities. The DBT modality shows the best AUC performance for all sizes for the best corresponding channel combination.

Multi-Disciplinary Interactive ePoster Discussion

TRACK 3

Multi-Disciplinary: Image Guidance: Cone-Beam CT

SU-lePD-TRACK 3-01

Cone Beam CT (CBCT) Based Synthetic CT Generation Using Deep Learning Methods for Dose Calculation of Nasopharyngeal Carcinoma Radiotherapy

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Purpose: Cone beam CT (CBCT) for the purpose of setup alignment inhibits its further use in adaptive radiotherapy due to artifacts and inaccurate Hounsfield

units (HU). We compared three deep learning methods to generate high quality synthetic CT (sCT) images from CBCT and planning CT (pCT) for dose calculation of nasopharyngeal carcinoma (NPC) radiotherapy. **Methods:** 169 NPC patients with a total of 20926 slices of CBCT and pCT images were included. The CycleGAN, Pix2pix and U-Net models were used to generate the sCT images. The mean absolute error (MAE), root mean squared error (RMSE), peak signal to noise ratio (PSNR), and structural similarity index (SSIM) were used to quantify the accuracy of the proposed models in a testing cohort of 34 patients. Radiation dose were calculated on pCT and sCT by three deep learning models following the same protocol. Dose distributions were evaluated for 4 patients by comparing the dose-volume-histogram (DVH) and 2D gamma index analysis. **Results:** The average MAE and RMSE values between sCT by three models and pCT reduced by 15.4 HU and 26.8 HU at least, while the mean PSNR and SSIM metrics between sCT by different models and pCT added by 10.6 and 0.05 at most, respectively. There were only slight differences for DVH of selected contours between different plans. The passing rates of gamma index analysis for 4 patients under 3mm/3% 3mm/2%, 2mm/3% and 2mm/2% criteria were all higher than 95%. All the sCT had achieved better evaluation metrics than those of original CBCT, while the performance of CycleGAN model was proved to be best among three methods. **Conclusion:** The dosimetric agreement confirmed the HU accuracy and consistent anatomical structures of sCT by deep learning methods. The results demonstrated that the sCT could be readily incorporated for further adaptive radiotherapy dose calculation in clinical practice.

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SU-lePD-TRACK 3-02

Deep Learning Based Synthetic CT Generation with Multi-Sequence MRI for MRI-Only Radiotherapy

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Purpose: Synthetic CT(sCT) generation with MR image plays an essential role in MR-only radiotherapy, which can be used for density estimation and dose calculation in treatment planning. Currently, the reliability of sCT generation methods depends on two aspects: the MR-CT registration errors, which can be eliminated by CycleGAN with unpaired training data, and contract divergence in different sequences which could affect the nonlinear mapping of MR-CT. This study is to investigate the effect of different sequence MRI by evaluating the image quality of sCTs. **Methods:** T1, T2, T1C-MR images and planning CT(pCT) images of 151 patients were acquired on the same day. 126 cases were randomly selected for training. Three CycleGAN-based models were respectively trained with each sequence MRI. 25 cases with non-rigid registration were selected as the testing set. The mean error(ME) and mean absolute error(MAE) between plan CT and sCT of each model were used to measure the CT value accuracy, which was calculated within soft tissue, bone, and whole body, respectively. **Results:** The MEs calculated between pCT and T1, T2, T1C-based sCT images, were (154.92±87.93) vs. (173.27±97.62) vs. (262.78±91.74) within soft tissue, (1363.99±294.94) vs. (337.21±50.4) vs. (661.63±168.43) within bone, (22.12±29.47) vs. (17.05±10.51) vs. (35.28±20.75) within the whole body. The MAEs were (186.41±61.75) vs. (197.33±58.62) vs. (293.56±72.72) within soft tissue, (186.41±61.75) vs. (197.33±58.62) vs. (293.56±72.72) within bone, (31.28±28.1) vs. (27.64±10.41) vs. (42.6±23.82) within the whole body. The quantitative analysis indicated that T1C based generated model achieves higher accuracy than the other models in H&N radiotherapy. **Conclusion:** The T1C MRI can provide sufficient anatomy information and tissue contract in synthetic CT prediction of head and neck region, which offers a substantial potential for density estimation and dose calculation in radiotherapy. The generation method and evaluation will investigate synthetic CT prediction with more complex tissue in various body parts.

SU-lePD-TRACK 3-03

Image Synthesis for Planning and Target Tracking of MR-Based Stereotactic Radiation Therapy

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Purpose: To identify the model training parameters that result in the synthetic CTs with the best fiducial visualization and normal tissue accuracy for MR-only stereotactic radiation therapy to the pelvis. **Methods:** We previously developed a machine learning model to create synthetic CTs from the paired MRs. The model is applied for 11 CT/MR (conventional T2) image sets used for previous Cyberknife (Accuray, Inc) radiosurgery treatments. The MR images were pre-processed to mimic the appearance of fiducial-enhancing images. Model performance was evaluated on three factors, fiducial contrast ("realistic" vs manually enhanced), patch sizes and numbers of image pairs in the training. Four observers identified the fiducial centers in the synthetic and real CTs and associated DRRs to evaluate the fiducial reconstruction accuracy. The CT and DRR images were also qualitatively compared using mean squared error (MSE) and peak-signal-to-noise-ratio (PSNR). The same treatment plan was calculated on the real and synthetic CT. **Results:** We found that the manually enhanced images, largest patch size (128x128 pixels), and 9 image pairs in the training set gave the best clinically applicable synthetic CTs and the highest contrast for the reconstructed fiducials. The fiducial appearance in the synthetic CTs, as well as the DRRs generated from them had similar contrast and visualization to the real CT. There was no significant difference found for the fiducial localization for the CTs and DRRs ($p>0.05$). The mean MSE between the true and synthetic CTs was 2.77%±1.31% and the mean PSNR was 32.45±4.61. The MSE and PSNR for DRRs were 1.2±1.5% and 22.8±5.7, respectively. 3D gamma analysis on the dose distributions calculated on the real and synthetic CTs both pass 100% (2%/2mm)**Conclusion:** Our machine learning based method generates synthetic CTs for accurate treatment planning and DRRs with sufficient image contrast for target tracking in prostate stereotactic radiotherapy.

SU-lePD-TRACK 3-04

Implementation and Validation of Non-Uniform Body Region Specific 3D Age Scaling Factors (ASF) to Scale Whole-Body Computed Tomography (CT) Anatomy to Any Arbitrary Age

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Purpose: To scale whole-body computed tomography (CT) based anatomy non-uniformly in 3D to provide age-specific phantoms for retrospective dosimetry in late-effects studies. This is necessary because the University of Florida/National Cancer Institute (UF/NCI) computational phantom series (the gold-standard pediatric phantoms) are limited to discrete ages which could result in uncertainty in dose reconstruction when discrete-aged phantoms are used instead of age-specific phantoms. **Methods:** Fourteen UF/NCI phantoms (52 organs in each) were downsampled from their nearest older discrete age (1, 5, 10, 15, and 30 years old) using our in-house body-region-specific age scaling factors which accounts for non-uniform growth of head, neck, trunk, and legs in 3D. To validate the scaling, percent change in height, Dice Similarity Coefficients (DSC), mean distance to agreement (MDA), and Euclidean distance (ED) were calculated between the original and scaled anatomy for whole-body, brain, heart, liver, pancreas, and kidneys. Typical 6MV Wilms' tumor plans were constructed on a downsampled 3.9-year-old (median age of Wilms' tumor cohort from the Childhood Cancer Survivor Study) and 5-year-old male phantoms to quantify differences in V10, mean dose, D1, D50, and D95 for in-/near-beam organs. **Results:** For whole-body, brain, and heart, the DSC, MDA, and ED were ≥ 0.61 , ≤ 1.23 cm, and ≤ 2.90 cm, respectively. For liver, pancreas, and kidneys, the DSCs fell below 0.50 with MDA and ED up to 2.29cm and 4.09cm, respectively. Lower agreements were attributed to higher variation in organs' shape across the age range. The heights of phantoms were within 5% of reference heights. In the dose study, a maximum V10 difference of 0.5% was obtained for the pancreas (near-beam organ). Absolute dose differences in organ-to-organ dose comparisons were within 20cGy for each metric. **Conclusion:** CT based anatomy can be scaled to any arbitrary age with accurate anatomical shape, size, and

position and accurate dose reconstruction for late-effect studies.

SU-lePD-TRACK 3-05

Self-Attention Generative Adversarial Network for Synthetic CT Generation from CBCT in Head and Neck Radiotherapy

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Purpose: Cone Beam CT(CBCT) is commonly used for accurate patient positioning and anatomic changes tracking during the IGRT, which performs poorly on organ segmentation and dose calculation in adaptive radiotherapy(ART) because of the strong artifacts and Hounsfield unit(HU) inaccuracy. To overcome the CBCT limitation, we proposed a self-attention Generative Adversarial Network(GAN) to generate Synthetic CT(sCT) from CBCT, improving the CT number accuracy while keeping the same anatomy. **Methods:** 38 cases H&N patients(30 for training, 8 for testing) with planning-CT(pCT) and CBCT in the first treatment fraction were included. We proposed a self-attention condition GAN generate CBCT-based sCT, with non-local block for image artifacts denoising and feature representation enhancement. To demonstrate the model performance, a Res-Unet generation model was trained for the comparison. The mean absolute error(MAE) and peak signal-noise ratio(pSNR) between the pCTs and sCTs were calculated for evaluating synthesis performance. **Results:** The MAEs were (79.52±13.54) HU vs. (77.48±6.6) HU (P=0.0015) within the whole body, (97.29±12.52) HU vs. (56.06±6.28) HU (P=0.0005) within soft tissue for the U-net vs. the proposed method. The pSNRs were(33.24±0.83) vs. (34.06±0.69) (P=0.0494)within the whole body, (35.96±1.18) vs. (37.44±0.77) (P=0.0098) within soft tissue for the U-net vs. the proposed method. Significant differences in MAE and pSNR were observed in the whole body and soft tissue. **Conclusion:** Qualitative and quantitative evaluation indicates that the proposed method can synthesize sCT with accurate CT numbers and best texture information, especially in soft tissue and the body. This potentially enables advanced CBCT applications for ART.

SU-lePD-TRACK 3-06

Synthetic Contrast MR Image Generation Using Deep Learning

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Purpose: Gadolinium-based dye is often used as the contrast agent during MR imaging examinations. In addition to extra cost, labor and time, contrast-enhanced MR imaging can cause patient discomfort and side effects. We propose a deep learning-based method to synthesize contrast-enhanced MR from non-contrast MR images. **Methods:** A cohort of 274 brain cancer patients from a public dataset, the Brain Tumor Segmentation Challenge (BRATS2015) was used. They have both non-contrast and contrast-enhanced MR images. A self-attention cycle generative adversarial network (cycleGAN) was used to learn the mapping from non-contrast images to contrast MR images. Attention U-Net was used as a generator of CycleGAN to force the trained model to extract information that can represent the specific difference between non-contrast MRI and contrast MRI, namely, the tissue contrast difference. The proposed method was trained on 224 patients and evaluated on the rest 50 patients. **Results:** The discrepancy between the ground truth contrast MR and synthetic contrast MR images were compared. The mean absolute error (NMAE) was 0.035±0.004. The peak signal-to-noise ratio (PSNR) was 32.4±2.6. And structural similarity index (SSIM) was 0.97±0.06. **Conclusion:** It is shown that the proposed method has great potential in accurately generating contrast MR, and therefore bypassing the contrast administration step during MR scan, which can save labor, time, cost and mitigate patient discomfort.

SU-lePD-TRACK 3-07

Artificial Intelligence (AI) Based Missing Tissue Generated Synthetic CT Preparation for Planning in Limited Field of View (LFOV) MR-Only Simulation

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Purpose: MR-only simulations often suffer from limited field of view (LFOV), resulting in incomplete body outlines. A novel approach of generating such missing tissue using machine learning and optical imaging has been conceived. **Methods:** The method consists of 3 steps. 1) Patient body outline in interest is obtained using an optical imaging method. 2) An MR image set in limited FOV (MR-in-LFOV) is converted to a synthetic CT image set in limited FOV (syn-CT-in-LFOV). This task can be accomplished using any one or combination of existing methods such as an intensity based approach, an atlas based approach and a machine learning based approach. 3) The syn-CT-in-LFOV is expanded to a synthetic CT image set in full FOV (syn-CT-in-FFOV) using a machine learning based missing body generation algorithm. The main goal of step 3 (i.e., generation of missing part of image) has not been proven and we focused on its realization using regular CT images in this study. For proof of the concept, we built a model using 10,005 training images taken from the cancer imaging archive (TCIA) and performed tests with other data sets not used in training. All training and evaluation were performed on a 64-bit Windows 10 Enterprise workstation. Computing hardware included an Intel Xeon W3520 quad-core CPU, 32 GB RAM, and two Nvidia GeForce GTX 1080 Ti graphic cards with a total 7168 cores and 22GB GPU RAM. **Results:** The completion network was trained for 250 iterations and the entire training procedure took roughly 5 days. The model provided missing tissues in tested cases. **Conclusion:** It is demonstrated that the approach introduced can realistically complete CT images despite large occluded areas. This approach is also expected to reduce geometrical distortion that is dominant at periphery in typical MR images

Expanding Horizons

TRACK 4

Expanding Horizons

SU-lePD-TRACK 4-00

Expanding Horizons

Moderators: William Donahue, Yale-New Haven Hospital, New Haven, CT and Suman Shrestha, University of Texas MD Anderson Cancer Center, Houston, TX

SU-lePD-TRACK 4-01

P. Jermain, Massachusetts General Hospital:

AACR 2021 Annual Meeting

SU-lePD-TRACK 4-02

J. Zhang, Emory University:

2020 Conference on Computer Vision and Pattern Recognition (CVPR)

SU-lePD-TRACK 4-03

A. Imran, Stanford University:

International Conference on Pattern Recognition (ICPR) 2020

Awardees from the Expanding Horizons Travel Grant program will discuss their experiences attending conferences that are not specifically geared toward traditional medical physics, focusing on broad scientific impact and opportunities for medical physicists in various fields. Virtual meeting attendees will have the ability to view the Expanding Horizons ePosters and learn about the hot topics and emerging research areas presented at these conferences, understand the relevance to future medical physics research, and submit questions to awardees during an interactive ePoster session. Students interested in applying in the 2021 Round Two application cycle are highly encouraged to participate!

Therapy Interactive ePoster Discussion

TRACK 5

Therapy: Brachytherapy treatment and QA

SU-lePD-TRACK 5-01

A 3D-Printed Prostate Phantom for Quality Assurance in HDR Brachytherapy

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Purpose: To design and build an anthropomorphic prostate phantom for dosimetric quality assurance testing in HDR brachytherapy. **Methods:** The prostate model was developed from 3D ultrasound images of a fully implanted prostate previously treated with HDR brachytherapy at our institution. Images and clinical contours were imported into Slicer3D to create the design elements of the phantom. Space was included in the model to allow for insertion of clinical brachytherapy needles and a variety of dosimeters. The model was printed on a Lulzbot 3D printer at 100% infill using 2.85 mm PLA filament. A CT of the printed phantom was acquired, and a test brachytherapy plan created in BrachyVision and then delivered using a GammaMedIX afterloader. Dosimetry measurements were taken in triplicate using an A1SL ion chamber and corrected for temperature and pressure. **Results:** The phantom consists of a 7.2 cm (L/R) x 6 cm (A/P) x 7.5 cm (S/I) block which encompasses four removable prostate quadrants and 12 needle tracks. It is bisected to include a central coronal plane film; the prostate is further split at midline for a sagittal plane film. There are 4 locations to insert optically stimulated luminescence dosimeters (OSLDs). There are 2 locations to insert an A1SL ion chamber, roughly corresponding to the positions of the urethra and rectum. The nominal total treatment time for the test plan was 315.5 seconds for a 10 Ci source. At the urethral and rectal positions, the average total charge over triplicate measurements was 28.0 ± 0.2 nC and 18.3 ± 0.1 nC, respectively; the percent difference with repeat measurements was <2% and <2.5%, respectively. **Conclusion:** 3D printing an anthropomorphic prostate phantom for HDR brachytherapy quality assurance was feasible using an inexpensive desktop printer. Initial ion chamber constancy measurements have reasonable agreement. Future work will include self-calibration as well as OSLD and film dosimetry.

SU-lePD-TRACK 5-02

Development of Reference Plan-Based Verification System for Intracavitary Brachytherapy for Uterine Cervical Cancer

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Purpose: For high-dose-rate intracavitary brachytherapy for uterine cervical cancer, all procedures including CT imaging, treatment planning, verification, and irradiation are performed within a few hours. Although verification is important for safe treatment, patient-specific quality assurance by secondary check software or measurement are not common. In this study, we developed a system which automatically check treatment plans. **Methods:** Reference treatment plans were generated for all patterns of Tandem & Ovoid applicators constructed in air. A software was developed to compare the clinical plans with the reference plan and automatically check the dwell positions, applicator length and offset, weights of dwell time, applicator geometry, and the coordinates of the reference points (Point A). The total dwell time of the reference plan was corrected for source intensity and prescribed dose to predict the total dwell time of the patient plans. Three-dimensional dose distribution was also calculated using the AAPM Task group 43 update 1 equation, and V100%, D90% of target, and D2cc of critical organs were evaluated. 517 plans of 155 cases were retrospectively analyzed. **Results:** Calculation time for each plan was 10.4 ± 3.3 s (range 3.3 - 21.1 s). Differences between the predicted and clinical total dwell time were $3.8\% \pm 5.4\%$ (range -13.1% - 19.1%) and $-0.4\% \pm 11.7\%$ (range -28.9% - 36.5%) for point A prescription and optimized plans, respectively. The ratio of total dwell time to the predicted value showed quadratic correlation with the ratio of V100 of the patient plans to that of reference plans, with an error of $0.0\% \pm 0.5\%$ (range -1.5% - 3.0%). The software successfully detected errors of the plans with deliberate mistakes. **Conclusion:** The software developed in this study enabled quick evaluation of the accuracy and validity of the treatment plans even for the optimized plans.

SU-lePD-TRACK 5-03

Development of a Dynamic-Shield Applicator Enabling Intensity-Modulated Brachytherapy for the Treatment of Rectal Cancer

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Purpose: Rotationally-symmetric dose distributions provided by brachytherapy sources deliver a high dose to the tumor, but often with poor target conformity resulting in dose spillage to the organs at risk (OAR). By incorporating dynamically-rotating metallic shields, intensity-modulated brachytherapy (IMBT) opens the possibility to deliver more conformal dose distributions. The goal of this study was to develop a novel MRI-compatible dynamic-shield IMBT applicator for the treatment of rectal cancer and compare the results with conventional high-dose-rate brachytherapy (HDR-BT) with and without static shielding. **Methods:** For dynamic-shield IMBT, a single-grooved, MRI-compatible, 15 mm diameter tungsten shield with a 180° emission window was designed. A cylindrical intracavitary applicator (length 28 cm, diameter 2 cm) (Elekta Brachytherapy, Veenendaal, The Netherlands) with a central lumen that fits an 8 mm diameter tungsten rod for static shielding was also modeled. For both applicator models, four rectal cancer patients were retrospectively planned using an in-house Monte Carlo based treatment planning system called RapidBrachyMCTPS and a generic Ir-192 source. The plans were optimized using a fast mixed-integer optimization method. The prescribed dose was 3 fractions of 10 Gy. The resolution of the scoring grid was $3 \times 3 \times 3$ mm³ and 10⁶ decay events were simulated. **Results:** For the same target coverage, the contralateral healthy rectal wall D2cc, rectum D50 and pelvis D50 decreased on average by 38.93%, 18.82% and 18.75% when using dynamic-shield IMBT compared with HDR-BT with the static shield and by 53.04%, 68.35% and 36.62% compared with HDR-BT without the static shield. However, the inferior and superior dose spill regions D2cc were 11.14% and 9.57% greater compared with static-shielding HDR-BT. The treatment times were below 30 minutes for all cases. **Conclusion:** The designed dynamically-rotating IMBT applicator for the treatment of rectal cancer shields the OAR more effectively with the possibility to increase the delivered dose to the tumor.

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SU-lePD-TRACK 5-04

Commissioning and QA of 3D-Printed Molds for Surface Brachytherapy

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Purpose: To describe key commissioning and quality assurance methods for 3D-printed brachytherapy surface applicators. **Methods:** Prior to clinical implementation of 3D-printed surface brachytherapy molds, a comprehensive commissioning plan was developed, including tests to determine channel tunnel radius, deliverable radius of curvature, and HU thresholds for accurate body contour creation. QA procedures for patient-specific molds were also established. **Results:** Tunnel radius and curvature must be determined to ensure deliverability. Based on catheters and afterloader used, the proper tunnel radius and curvature radius were determined by iterative testing. For 4.7Fr catheters (~1.55mm radius) a tunnel radius of 1.8 mm was too narrow, while a 2mm tunnel radius was compatible. The radius of curvature recommended by the vendor (14 mm), plus larger and smaller radii, were tested for deliverability with the afterloader (Varisource iX, Varian). Plans with radii of curvature <14mm were undeliverable. Body contour directly affects mold fit and must be optimized. HU settings were iteratively adjusted and used to create the body contour for test prints on an anthropomorphic phantom. A window/level of -380HU to +300HU (with hand editing of the surface at this W/L setting) produced molds that fit the phantom with minimal air gaps. This method was utilized on several clinical cases, producing molds with excellent fit (<3mm gap). QA of 3D-printed molds is essential to ensure accurate plan creation and delivery. QA included visual inspection, deliverability of test plans with catheters fixed in place, post-processing of rough patches, and patient CTs with the mold to evaluate both fit and mold integrity. **Conclusion:** This report describes commissioning and QA necessary for clinical implementation of 3D-printed surface brachytherapy applicators. These techniques have allowed clinical implementation of deliverable, highly customizable, and well-fitting molds for treatment of complex surface geometries with brachytherapy where commercial surface flaps or handmade molds are often problematic.

SU-lePD-TRACK 5-05

Integrating Rotatable Tandem Applicator to Conventional Ovoid Applicator for Complete Framework of Intensity Modulated Brachytherapy (IMBT) for Cervical Cancer

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Purpose: A new rotatable tandem applicator for intensity modulated brachytherapy (IMBT) in intracavitary brachytherapy(ICR) was recently proposed. This work presents an integrated framework with rotatable tandem applicator and ovoid applicator by modifying dose deposition matrix and plan optimization algorithm. **Methods:** The rotatable, single-channel tandem applicator enables for generating asymmetric dose distribution by rotating Ir-192 source in selected transverse direction. The previous work focused on plan optimization with tandem applicator only that constrains the number of source dwell points and activated rotational angles for enhancing delivery efficiency, while excluding ovoid applicator. To comprehend ovoid applicator in inverse planning, this work computed Monte-Carlo (MC) simulated dose distributions for 10 available dwell points belonging to ovoid applicator. Combining the dose information from tandem and ovoid applicators constructs dose deposition matrix for inverse planning. The optimization algorithm constrained the dwell points in tandem with group-wise in L2, 1/2-norm and element-wise sparsity in approximated L0-norm to minimize number of activated angles and entire elements, while constraining the dwell points by element-wise sparsity only in ovoid applicator. The complete framework is validated by three clinical cases, compared against conventional ICR, and IMBT with tandem applicator. **Results:** Integrated IMBT framework increased the number of dwell point by comprehending ovoid applicator, while it did not affect the number of dwell points and activated angles assigned to tandem applicator. Although IMBT with tandem enabling fluence-modulation spared critical organs relative to conventional ICR, insufficient geometric target coverage led to greater target volume exposed to high dose of radiation. IMBT with both tandem and ovoid applicators enhanced target volume homogeneity by optimizing dwell points in ovoid applicator, while maintaining enhanced dose sparing to critical organs. **Conclusion:** IMBT with integrated framework was achieved by new dose deposition matrix and modified plan optimization algorithm, which yield superior dose sparing and target volume homogeneity simultaneously.

SU-lePD-TRACK 5-06

Dual-Source Optimization for HDR Prostate Brachytherapy

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Purpose: High dose rate (HDR) brachytherapy is used to treat prostate cancer in mono-, combination-, or salvage settings. Iridium-192 is the most common source, but Ytterbium-169 has also been considered for its lower energy and potential to spare nearby sensitive organs, e.g., the urethra. However, a planning method to optimize the utilization of both energy sources for efficient plan delivery has not been presented. In this study, we develop a framework that jointly optimizes Ir-192 and Yb-169 dwell times for urethra sparing with a total treatment time penalty. **Methods:** The planning problem was formulated as a group sparsity optimization problem to minimize the number of channels used by both sources. The optimization problem was solved using the fast-iterative shrinkage-thresholding algorithm. The HDR source dose distributions were modeled following the AAPM TG-43 formalism for 10 Ci Varian Varisource (VS2000 ¹⁹²Ir) and 10 Ci Model M42 ¹⁶⁹Yb HDR sources. Three treatment plans were generated to deliver a 15 Gy/fx prescription dose for a retrospective patient with 19 channels using Ir-192 only, Yb-169 only, and dual-source. **Results:** The three plans resulted in similar dose distribution to the PTV, rectum, and bladder. The minimum dose in the most irradiated 0.1 cm³ of the urethra were 15.13, 16.50, and 15.00 Gy for the Ir-192, Yb-169, and dual-source plans, respectively. The group sparsity term effectively reduced the total number of channels used for the dual-energy approach. The total dwell times were 15.41, 49.43, and 26.70 minutes for the Ir-192, Yb-169, and dual-source plans, respectively. **Conclusion:** Dual-energy brachytherapy combines the penetration of high energy and rapid drop-off of low energy Gamma rays for greater flexibility in plan dosimetry. We developed a novel framework to optimize the delivery efficiency for dual-source treatment. Further improvement in plan dosimetry and OAR sparing may require optimization of catheter locations.

SU-lePD-TRACK 5-07

Proton Arc Therapy as An Alternative to Syed Interstitial High Dose Rate (HDR) Brachytherapy for Treating a Gynecological Cancer Patient with Lateral Disease Extension

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Purpose: To investigate if single energy proton arc therapy (SEPAT) plans generate improved dose to organs at risk (OARs) compared to the Syed interstitial high dose rate brachytherapy (Syed HDR) for a patient with gynecological cancer with lateral disease extension. **Methods:** SEPAT delivers proton arc beams by modulating energy at the beam nozzle while maintaining the same incident energy at the gantry during rotation. An index case, high-risk CTV (HRCTV) measuring 76 cc with the lateral dimension of 5.1 cm treated with 14 interstitial needles and 6-cm long tandem in the uterus was selected. Full arc SEPAT plan was designed using 125-157 MeV modulated 157 MeV incident proton beams. **Results:** Target coverage of the SEPAT plan was comparable to the Syed HDR plan: 100% of the HRCTV received 90 % of the prescribed dose (V90) for SEPAT, compared to V90 of 90% for Syed HDR. SEPAT robust plan (± 3mm, ±2.5%) generated lower target coverage, 93% dose for V90, which remains to be higher than Syed HDR. D2cc of bladder and rectum were considerably lower for SEPAT plan with 90% (SEPAT) vs 121% (Syed HDR) and 55% (SEPAT) vs 69% (Syed HDR) of prescribed dose, respectively. Other OAR doses such as small bowel and large bowel were also lower with the SEPAT plan than Syed HDR. The SEPAT robust plan demonstrated 98% and 70% of D2cc of bladder and rectum, respectively. **Conclusion:** In a case with locally advanced gynecological cancer with lateral disease extension, SEPAT generated a dosimetrically superior plan in comparison to the Syed HDR brachytherapy with interstitial needles. SEPAT may provide a non-invasive alternate to HDR brachytherapy with superior dosimetric profile. Further study is needed with more cases to confirm the result with statistical confidence.

Therapy Interactive ePoster Discussion

TRACK 6

Therapy: Electron Treatment

SU-lePD-TRACK 6-01

Monte Carlo (MC) Simulation of Total Skin Electron Therapy (TSET) with TOPAS and EGSnrc

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Purpose: Cherenkov imaging is applied to TSET as a non-invasive and real-time remote monitoring technique. Monte Carlo (MC) simulations of TSET with TOPAS are capable of generating and propagating Cherenkov photons as well as transporting radiation. This study compares doses that are simulated using both TOPAS and EGSnrc, and are benchmarked with data in order to validate the accuracy of simulations of realistic incidents in a real clinical treatment geometry. **Methods:** The complete physical processes of all particles from a clinical electron beam interacting with different materials of phantoms are simulated by using TOPAS. A treatment head model, patient phantom model, and TSET clinical treatment setting including geometry, chemical compositions, and optical properties of materials, especially of human tissues, are all studied and simulated with TOPAS. The 6 MeV electron beam phase space files scored before incidence on the JAWS are provided by Varian. Realistic clinical incident beams used in TSET including the setup geometry are simulated with both TOPAS and EGSnrc. The results are compared and benchmarked with experiments. **Results:** MC simulations of TSET are validated by comparing with measurements, such as PDD in water at 100 cm SSD, PDD in solid water at 500 cm SSD with a 3 mm beam spoiler, and vertical profiles at 500 cm SSD in a PVC board with the spoiler in single and dual fields. The calculated dose distributions from TOPAS agree with those from EGSnrc. Simulations with TOPAS also show the relationship between dose distributions and distributions of Cherenkov photons emitted from phantom surfaces. **Conclusion:** The agreement between TOPAS, EGSnrc, and data establishes the accuracy of the simulated incident beams and dose distributions. By using the relationship between dose distribution and Cherenkov photon distribution we are able to monitor the patient skin dose during the radiation delivery of TSET.

SU-lePD-TRACK 6-02

Initial Assessment of the Suitability of 3D PRESAGE Phantom for Relative Dose Measurements in a FLASH Electron Beam

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Purpose: To assess the suitability of the 3D PRESAGE phantom for relative dose measurements in a high dose per pulse FLASH electron beam. **Methods:** Irradiations were performed at the Radiological Research Accelerator Facility (RARAF) using a Varian iX linear accelerator, modified to deliver a high dose per pulse 9 MeV electron beam. By varying the pulse frequency and SSD, PRESAGE phantoms were irradiated at four dose rates: 6 Gy/min, 40 Gy/s, 86 Gy/s, and ~900 Gy/s. At each position, except for the exit window (~900 Gy/s), the dose rate was determined using an advanced Markus chamber. A dose of 5-7.1 Gy was delivered to each phantom. The dose to the phantom was then evaluated using an in-house optical CT scanner. Due to the expected uncertainty in measuring absolute dosimetry at such a high pulse rate, we compared the percentage depth dose curves for this study. The resultant PDD curves were compared with each other and to the PDD curve measured using EBT3 Gafchromic film which has been shown to have no dose rate dependence. **Results:** PDD curves measured for the four PRESAGE phantoms were very similar over dose rates ranging from 6 Gy/min used in conventional radiation therapy to an ultra-high FLASH dose rate of about 900 Gy/s. No dose rate dependency was seen. Comparisons to EBT3 Gafchromic film were similar but some differences were observed. Discrepancies among between PDD are possibly due to difficulties in setup and alignment. **Conclusion:** PDD curves measured in PRESAGE 3D phantoms for a wide range of dose rates were comparable to those obtained using Gafchromic film, indicating the PRESAGE may be useful for relative dose measurements in FLASH electron beams. Further studies should be performed to investigate differences observed between film and 3D phantom and to investigate PRESAGE for absolute dosimetry in FLASH.

SU-lePD-TRACK 6-03

The LIAC 12 MeV Intraoperative Electron Radiotherapy Unit: Bi-Directional Assessment of Field Symmetry and Independent Evaluation of the SWL-LIAC Simulation Software

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Purpose: The LIAC intraoperative electron radiotherapy (IOERT) accelerator is commissioned for clinical use mostly based on the manufacturer's SWL-LIAC simulation software, which substantially reduces the number of difficult-to-set-up measurements needed. Moreover, in clinical IOERT practice, the appropriate beam energy is chosen based on the target tissue thickness using the beam data modeled by this software. We aimed to independently validate the accuracy of SWL-LIAC's calculated outputs, namely, percentage depth-doses (PDDs), off-axis dose profiles and relative output factors (ROFs), against experimental data. **Methods:** For all the available energies (6, 8, 10, 12 MeV), PDD measurements were made for all sizes of the flat-ended applicators (3 to 10 cm) and several beveled applicators. Profiles and ROFs were measured for the more frequently used applicators in clinical practice and the reference 10 cm diameter applicator. Meticulous measurements were made using a p-type electron diode in a water phantom, aided by inhouse custom-made positioning tools and confirmed using radiochromic film dosimetry. The measured and calculated scans were compared using multi-criteria one-dimensional gamma analysis. **Results:** Eighty-five percent of the calculated ROFs showed a $\leq 2\%$ agreement with the measurements. The average gamma passing rate for PDDs was 100% using a 2mm/1% (global dose difference) criterion. Repeated measurements showed that the measured in-plane profiles were somewhat tilted, which was confirmed by the manufacturer. The average gamma passing rate of all profiles was 95.5% (2mm/3%); Without considering the in-plane profiles, this value was 97.6%. Most of the profile points that failed the gamma criteria were in the peripheral region of the inner field and the shoulder regions of profiles. **Conclusion:** The accuracy of SWL-LIAC calculations can be deemed acceptable for most clinical applications. However, assuming isotropic off-axis profiles is unwarranted and measuring profiles in both directions for all energies at installation and acceptance testing is indicated.

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Radiation-oncology Department of Shiraz University of Medical Sciences is a Reference Hospital for PTW-Freiburg.

SU-lePD-TRACK 6-04

Customize Electron Energy Beams on TrueBeam Linear Accelerators

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Purpose: To facilitate optimal planning, we implemented two new electron energy beams (7- and 11-MeV) on two Varian TrueBeam linacs. **Methods:** To better meet clinical needs we worked with the vendor to create two additional customized electron energies without hardware modifications. For each of these

beams we set the bending magnet current and then optimized other beam specific parameters to achieve depths of 50% ionization (I50) of 2.9-cm for 7-MeV and 4.2-cm for the 11-MeV beams with the 15×15cm² cone. This was done using an Ionization Chamber Profiler (ICP, Sun Nuclear) with double-wedge-phantom (DW). Beams were steered and balanced to optimize symmetry using the ICP. After all parameters were set with the vendor, a full commissioning was done including measuring beam profiles, PDDs, output factors (OF) at standard and extended SSDs. The measured data was compared between two linacs and also against the calculated values from our TPS model (RayStation) following MPPG 5.a. **Results:** The values of I50 initially determined with ICP/DW agree to those from PDD scanned in-water phantom within 0.2-mm for 7- and 11-MeV on both linacs. Comparison of the beam characteristics from two linacs indicated that, flatness and symmetry were in agreement within 0.4%, point-by-point differences of PDD are within 0.01±0.3% for 7- and 0.01±0.3% for 11-MeV beams. The OF ratios between the two linacs were 1.000±0.007 for 7- and 1.004±0.007 for 11-MeV beams. The TPS calculations and measurements indicated that the point-to-point ratios (low-gradient regions) were 0.2±0.8% and Distance-To-Agreements (high-gradient regions) were 0.8±0.6mm, all meet the MPPG5.a criteria. **Conclusion:** We were able to add in two new beam energies with no hardware modifications. The tuning of the new beams was facilitated by the ICP/DW system allowing us to have the procedures done in a few hours and achieve highly consistent results across two linacs.

SU-lePD-TRACK 6-05

A TPS Independent Automated Software to Detect Electron Applicator-Body Collisions

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Purpose: Electron applicator collisions with the patient are difficult to discern in most treatment planning systems (TPS) which do not display a model of the applicator. Collisions discovered during a pre-treatment plan check or worse, during the first treatment must be re-planned, causing additional work and costly patient treatment delays. We developed a TPS independent program that accurately predicts collisions and determines the proximity of the body and bolus to the applicator using DICOM files exported from any TPS. **Methods:** External dimensions and position of all electron applicators relative to isocenter for a Varian linac were physically measured. A MATLAB script was developed that creates and renders 3D representations of the applicator, body contour, and bolus contour extracted from the plan and structure DICOM files. The script calculates either the overlap or minimum distance between the applicator and body+bolus contours. The script was retrospectively tested on 9 electron plans (n=3 shoulder, n=1 scalp, n=5 breast) from 8 patients planned in either Eclipse (n=4) or RayStation (n=4), and prospectively utilized during an initial plan check. **Results:** Retrospectively, the software correctly identified 2 plans for one patient where the applicator intersected with the body, which was discovered during the first day of treatment. For the remaining 7 plans, the shortest distance between the applicator and body/bolus contour were calculated and rendered, and accurately predicted no collisions (n=6, dist.>3cm; n=1 dist.<3cm). Prospectively, the software was utilized during an initial breast-boost plan check and predicted an overlap at the armpit preventing a patient delay. This predicted collision was confirmed by the therapists by simulating the original boost-setup before treating with a corrected plan. **Conclusion:** The developed software provides quantitative measurements and accurate visualizations of the proximity of the electron applicator to the external body and bolus contours for any TPS.

SU-lePD-TRACK 6-06

Verification of a Cost-Effective and Cerrobend-Free Workflow for Customized 3D-Printing Based Skin Collimation

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Purpose: Secondary skin collimation is the optimal methodology for reducing electron scatter and preventing irradiation of non-involved tissue. However, the process of molds is cumbersome and commercial solutions are cost-prohibitive. We investigate the feasible clinical use of a 3D printed shaper filled with tungsten balls for skin collimation. **Methods:** Using a 3D model a 3D printing based skin collimator (3D-SC) was produced. The thickness of the 3D-SC was set at 8 mm and filled with 2 mm tungsten balls (17.5 g/cm³). Measurements were performed with 6 MeV in solid water at 0.5 cm depth with GafchromicTM EBT3 films. The film was irradiated with 300 MU. The irradiated film was scanned and analyzed with an OmniPro I^mRT (IBA Dosimetry). A clinical case is used to demonstrate the ease of implementation. **Results:** The penumbra (90% to 20%) value measured was 0.175 mm for the 3D-SC. Compared to this, the conventional Cerrobend cut-out created 0.72 mm penumbra. The total print time of the 3D-SC was less than 5 hours. Once the CT scan was acquired, our clinical case was modeled in our 3D planning software within 30 minutes. **Conclusion:** Skin collimation's efficacy is confirmed with the 3D-SC. This process is efficient (e.g., less labor-intensive than molding), cost-effective, and the shielding (tungsten balls) is reusable. Future work will verify this methodology with additional patient cases involving large curvature and irregularity.

SU-lePD-TRACK 6-07

Feasibility Study of Using Custom 3D-Printed Bismuth Shields in Electron Beam Treatment

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Purpose: Low-energy electrons (6-9 MeV) are often used to irradiate tumors on patient's skin. External shields are used to minimize the radiation exposure to non-treatment areas. Comparing to lead shield, 3D printed shield can be made more conformal to the patient's skin surrounding the lesion, which reduces the radiation to the outside region and increases patient's comfort. This study evaluates the capability of bismuth-based 3D-shield as an alternative to lead in attenuating the electrons. **Methods:** Four rectangular (12x12cm) shields of varying thicknesses (one 1mm, two 2mm, and one 5mm) were 3D-printed using bismuth-based filament, which can be combined to create thickness from 1mm to 10mm. Each shield was printed solid with a 100% infill on a MatterHackers Pulse XE 3D printer with an enclosure using GMASS bismuth ABS filament. Transmission for the shield of different thickness from electrons with energies of 6, 9, 12, 16, and 20 MeV was measured using an ion chamber and a 10x10 cm² cone. The depth of measurement was 1.4cm for 6E and 2.5cm for other energies. SSD was 100cm. **Results:** The transmission of the 3D-shield were within 5% for 6E and 9E beam with ≤ 10mm thickness. The transmission was 3.57%, 1.81%, 1.28%, and 1.12% for 6E beam with shield of 7 to 10mm, respectively. For the 9E beam, the acceptable thickness was 10mm with the transmission of 4.77%. **Discussion and Conclusion:** To use 3D-printed bismuth radiation shields as an alternative to lead sheets is promising. For 6E beam, from 7mm above, the dose can be reduced to less than 5%. The transmission for 10mm shield is within a reasonable range for 9E beam. For the other energies, the transmission was over 5% with 10mm bismuth shield. However, tungsten can be used to reduce the thickness and will be studied in the future.

SAM Professional Symposium

TRACK 1

Establishing Efficient and Effective Systems for Policy and Procedures

SU-E-TRACK 1-00

Establishing Efficient and Effective Systems for Policy and Procedures

Moderator: Mohammad Bakhtiari, William E. Kahlert Regional Cancer Center, Westminster, MD

SU-E-TRACK 1-01**P. Halvorsen, Beth Israel - Lahey Health:****Policies and Procedures: Why Bother, and Why Should the Physicist Play a Key Role?****SU-E-TRACK 1-02****M. Bakhtiari, William E. Kahlert Regional Cancer Center:****Policies and Procedures: Developing Policies and Procedures That Work for Your Clinic****SU-E-TRACK 1-03****K. Moore, UC San Diego:****Policies and Procedures: Practical Strategies for Effective Implementation and Maintenance****SU-E-TRACK 1-04****Audience Q&A**

The phrase “Policies and Procedures” is viewed by many as a dry, administrative function without significant value to clinical practice. For many years, the primary focus of the AAPM’s task group activities was centered on equipment, with numerous recommendations for commissioning and ongoing quality control and quality assurance related to equipment safety and performance. Recent years have seen an increased recognition of the importance of process toward ensuring safety and consistent quality. The multi-society Safety is no Accident publication, the broad adoption of structured incident learning, and the publication of the Task Group 100 report have emphasized the risks of poorly defined workflows. This presentation will summarize what we have learned from recent initiatives and how policies and procedures fit into a comprehensive quality management system, with emphasis on the importance of accessible policies and procedures to guide clinical workflows.

While we have learned that policies and procedures provide value to clinical operations, where do they really fit in and how do they influence our actions? Developing effective policies and procedures is much more than just simply meeting an administrative checkbox. The culture of both the larger organization and in the immediate environment will influence design. Design of policies and procedures will clearly affect the effectiveness, so a suggested hierarchical structure of policy-procedure-work instruction-records will be presented. This presentation will look at where organizational management and task and technology factors may be organized to set team members to succeed rather than fail in the clinical process (notice, interpret, decide how to address a problem, develop an effective plan, or act on a developing problem).

The purpose of Policy and Procedure documents is to be used, practical considerations of how to craft them to be effectively integrated into normal clinical workflows must be considered. We will discuss examples of effective and ineffective Policy and Procedure documents, focusing on clarity (or lack thereof) in writing and presentation. Nearly as important as writing documents effectively are the processes for accessibility, version control, traceable approvals, and maintenance. We will discuss strategies for efficiently satisfying these criteria, with a focus on accessibility in a highly digitized, modern quality management system.

Learning Objectives:

1. Importance of policies and procedures in a practice. Importance of following P&Ps in preventing incidents (data from RO-ILS will be presented). Get familiar with the differences between policies, procedures, and processes. Examples of “good” and “not so good” P&Ps.
2. Incorporating P&Ps into a quality improvement program (TG-100, FMEA and incident learning). The quality metrics for evaluating the P&Ps.
3. Standardization for Writing and editing P&Ps (formatting templates, outlines, terminologies, etc)
4. Document management systems; technical aspects of incorporating P&Ps into daily practice. Easy and quick access to P&Ps from different applications, platforms, and devices. Task management tools for easily assigning tasks and monitoring the performance of the P&Ps.

SAM Imaging Educational Course**TRACK 2*****Non-anger Gamma Cameras: Physics, Quality Control and Evaluation*****SU-EF-TRACK 2-00****Non-Anger Gamma Cameras: Physics, Quality Control and Evaluation**

Moderator: William Erwin, UT MD Anderson Cancer Center, Houston, TX

SU-EF-TRACK 2-01**P. Slomka, Cedars-Sinai Medical Center:****Dedicated Myocardial Perfusion Imagers****SU-EF-TRACK 2-02****L. Zimmermann, University Hospitals Cleveland Medical Center:****Large Field-Of-View Pixelated Gamma Cameras****SU-EF-TRACK 2-03****M. O'Connor, Mayo Clinic:****New and Innovative CZT-Based Gamma Cameras****SU-EF-TRACK 2-04****Audience Q & A**

A number of non-Anger planar and SPECT gamma cameras (pixelated, solid state, multiple small field-of-view), including those dedicated to myocardial perfusion and molecular breast imaging, are now commercially available, and their clinical use is becoming common. As a consequence, the likelihood that a nuclear and diagnostic medical physicist will encounter these non-traditional cameras in clinical practice is increasing. The underlying physics, performance characteristics, calibration and quality control, and annual evaluation of these cameras differ substantially from those of the conventional Anger camera. Secondly, due to their unique construction and purpose, the collection of conventional phantom data for accreditation present several problems that are not trivial to overcome. The purpose of this session is to provide attendees with information related to the underlying physics, calibrations, quality control and performance characteristics of various classes of non-Anger gamma cameras; as well as, appropriate accreditation tests.

Dedicated myocardial perfusion imagers (Piotr Slomka, PhD)

This lecture will describe solid state detector performance, collimator design and characteristics, dedicated quality control, and new types of artifacts and evaluation in the clinical practice, associated with dedicated cardiac non-Anger SPECT cameras.

Large field-of-view pixelated gamma cameras (Leonard W. Zimmermann, MS)

This lecture will focus on non-Anger large field-of-view gamma camera detector and collimator characteristics, system calibration and commissioning, and annual performance evaluation.

New and innovative CZT-based gamma cameras (Michael K. O'Connor, PhD)

This lecture will describe molecular breast imaging (MBI) and next-generation Ring gantry CZT-based systems, as well as acceptance testing and ACR accreditation associated with both of those classes of non-Anger gamma cameras.

Learning Objectives:

1. Understand the underlying physical principles of the various types of non-Anger gamma cameras
2. Become familiar with non-Anger gamma camera calibrations and quality control
3. Learn about non-Anger gamma camera performance characteristics and how to evaluate them to satisfy accreditation groups

SAM Imaging Scientific Symposium

TRACK 3

Imaging Markers of Breast Cancer Risk

SU-E-TRACK 3-00

Imaging Markers of Breast Cancer Risk

Moderator: Carrie Hruska, Mayo Clinic, Rochester, MN

SU-E-TRACK 3-01

C. Vachon, Mayo Clinic:

Breast Cancer Risk Markers from Mammography

SU-E-TRACK 3-02

E. Morris, University of California, Davis:

Emerging AI Breast Imaging Applications

SU-E-TRACK 3-03

Q&A & Panel Discussion

Although the primary goal of breast imaging modalities is to detect and diagnose breast cancers, mammography and advanced imaging techniques such as breast MRI may also provide useful information about breast cancer risk. A wealth of evidence has identified imaging markers that, when used in conjunction with other patient information, can aid in predicting whether breast cancer is currently present or more likely to develop in the future. In this session, we will hear about the latest research tools and commercial products to assess breast density on 2D mammography and tomosynthesis. We will learn about research of radiomic features that can be extracted from mammography to inform clinical risk prediction. We will discuss new risk markers identified on MRI and other contrast-based imaging that can provide additional risk information beyond density. Finally, we will discuss the role of AI tools for identification and application of imaging markers associated with breast cancer risk. Together, the work discussed here has high potential to impact clinical practice by identifying women who may benefit from tailored breast screening or risk-reduction strategies.

Learning Objectives:

1. Discuss the latest research tools and commercial software for breast density assessment and extraction of radiomic features with mammographic techniques.
2. Recognize new risk markers from breast imaging and their potential contribution to risk prediction.
3. Consider the role of AI tools in breast imaging for risk assessment.

Licensing agreement with CMR Naviscan

Research agreement with Siemens

SAM Multi-Disciplinary Scientific Symposium

TRACK 4

State-of-the-art in MRI-guided Radiotherapy

SU-EF-TRACK 4-00

State-Of-The-Art in MRI-Guided Radiotherapy

Moderator: Carri Glide-Hurst, University of Wisconsin, Madison, WI

SU-EF-TRACK 4-01

C. Glide-Hurst, University of Wisconsin:

Introduction to the State-Of-The-Art in MRI-Guided Radiotherapy

SU-EF-TRACK 4-02

P. Liu, The University of Sydney:

Next-Generation Target Tracking Using MRI Guidance

SU-EF-TRACK 4-03

D. Ma, Case Western Reserve:

MRI Fingerprinting

SU-EF-TRACK 4-04

N. Mickevicius, University of Wisconsin:

Challenges and Solutions for Quantitative Imaging On Low-Field MR-Guided RT Systems

SU-EF-TRACK 4-05

M. Kim, University of Michigan:

Toward Patient-Specific Radiotherapy: Biologically Informed Tumor Targeting in Glioblastoma

SU-EF-TRACK 4-06

M. Boss, American College of Radiology:
Standards to Reduce Variability in Quantitative MRI

SU-EF-TRACK 4-07

Panel Discussion

Magnetic resonance imaging (MRI) is playing an increasingly large role in the treatment of cancer with radiation therapy. With applications ranging from biologically informed treatment planning to in-room motion management, MRI is a versatile tool with far-reaching advantages. This session covers several cutting-edge MRI methods and their application to radiation therapy planning and delivery. The first talk will cover the use of real-time MR guidance for tumor tracking for multiple targets in the presence of patient motion. Quantitative MRI metrics are rapidly being adopted as imaging biomarker surrogates for tumors and organs at risk. One approach, magnetic resonance fingerprinting, will be described as a state-of-the-art method for rapid quantitative MRI. Next, a presentation will be given on overcoming the challenges of quantitative imaging on low field MRI-guided radiation therapy systems. To ensure rigor and reproducibility in quantitative MRI, a presentation on standards to reduce variability will be given. Finally, a talk on the novel use of MRI for patient specific radiation therapy treatment planning for high-grade brain tumors in a clinical trial will be presented by a physician.

Learning Objectives:

1. Describe real-time multi-target tracking on an MRI-guided radiation therapy system.
2. Understand the concept of magnetic resonance fingerprinting for rapid quantitative MRI.
3. Describe the challenges of quantitative imaging on low field strength MRI scanners.
4. Describe standards to reduce variability in quantitative imaging.
5. Introduce methods for patient-specific treatment of glioblastoma tumors in a clinical trial.

Therapy Scientific Session

TRACK 5

Treatment Planning: Dose Calculation and Prediction

SU-E-TRACK 5-01

A Consensus Report from the Global Harmonisation Group On Clinical Trials: Report Dose-To-Medium

S Kry^{1*}, J Lye², C Clark³, N Andratschke⁴, A Dimitriadis⁵, D Followill⁶, R Howell⁷, M Hussein⁸, M Ishikawa⁹, S Kito¹⁰, T Kron¹¹, J Lee¹², J Michalski¹³, A Monti¹⁴, N Reynaert¹⁵, P Taylor¹⁶, K Venables¹⁷, Y Xiao¹⁸, J Lehmann¹⁹, (1) MD Anderson Cancer Ctr., Houston, TX, (2) Australian Radiation Protection and Nuclear Safety Agency, Yallambie, VIC, AU, (3) UCLH, London, SRY, GB, (4) University Hospital Zurich And University of Zurich, (5) (6) UT MD Anderson Cancer Center, Houston, TX, (7) UT MD Anderson Cancer Center, Houston, TX, (8) National Physics Laboratory, (9) Hokkaido University, Sapporo, JP, (10) Jcog, (11) Peter MacCallum Cancer Instit., Melbourne, AU, (12) Nih Radiotherapy Trials Quality Assurance Group, (13) Washington University, Saint Louis, MO, (14) European Organisation For Research And Treatment of Cancer, (15) University of Gent, Gent, BE, (16) UT MD Anderson Cancer Center, Houston, TX, (17) Nih Radiotherapy Trials Quality Assurance Group, (18) University of Pennsylvania, Philadelphia, PA, (19) Newcastle Mater Hospital, Hunter Region Mail C, AU

Purpose: To promote consistency in global clinical trials by recommending a uniform framework as it relates to radiation transport and dose calculation in water versus in medium. **Methods:** The Global Quality Assurance of Radiation Therapy Clinical Trials Harmonisation Group (GHG; www.rtgaharmonization.org) is comprised of 5 primary member organizations: IROC, EORTC, RTTQA, JCOG, and TROG and several observer organizations: IAEA, RDS, ADCS and NPL. The GHG compared the differences between dose to water in water (Dw,w), dose to water in medium (Dw,m), and dose to medium in medium (Dm,m). This was done based on a review of historical frameworks, existing literature and standards, clinical issues in the context of clinical trials, and the trajectory of radiation dose calculations. Based on these factors, recommendations were developed. **Results:** No framework was found to be ideal or perfect given the history, complexity, and current status of radiation therapy. Nevertheless, based on the evidence available, the GHG established a recommendation preferring dose to medium in medium (Dm,m). If an institution's planning system can only calculate dose to water in water (Dw,w), this is acceptable. **Conclusion:** Dose to medium in medium (Dm,m) is the preferred dose calculation and reporting framework. Use of this common framework will provide global standardization for clinical trial data reporting and interpretation of trial results.

SU-E-TRACK 5-02

A GPU-Accelerated Monte Carlo Engine for Calculation of MLC-Collimated Electron Fields

E Brost*, H Wan Chan Tseung, J Antolak, Department of Radiation Oncology, Mayo Clinic, Rochester, MN

Purpose: To develop a GPU-accelerated Monte Carlo (MC) engine incorporating the Varian TrueBeam linac head geometry for rapid calculation of electron fields collimated using the photon multi-leaf collimator (MLC). **Methods:** A CUDA framework was created for the following: (1) transport of electrons and photons through the linac head geometry, taking into account multiple scattering, Bremsstrahlung, Moller, Compton, and pair production interactions; (2) electron and photon propagation through the CT geometry, taking into account all aforementioned interactions plus the photoelectric effect; and (3) secondary particle cascades through the linac head and within the CT geometry. The linac head collimating geometry was modelled according the specifications provided by the vendor, who also provided phase space files above the collimator jaws. The MC was benchmarked against EGSnrc by simulating individual interactions with simple geometries and pencil beam dose calculations in water phantoms. Lastly, MC-calculated dose distributions for MLC-collimated electron fields were compared to measurements in a water phantom. **Results:** Pencil beam dose distributions within a water phantom are in good agreement with DOSXYZnrc. Angular and spatial distributions for multiple scattering and secondary particle production are in good agreement with EGSnrc. Dose profiles for MLC-collimated 6-20 MeV electron beams had differences of <5% for 5x5 cm² and 10x10 cm² field sizes as compared to measurement. The computation time on a single NVIDIA GTX TITAN X card is 15 minutes for 10⁸ electron histories, which is approximately 100 times faster than published results in a similar geometry using a single-CPU core. **Conclusion:** The GPU-based MC can quickly calculate dose for electron fields collimated using the conventional photon MLC. Dose profiles in homogeneous medium agree well with DOSXYZnrc and measured data. The fast calculation times will allow for rapid calculation of electron fields for mixed photon and electron particle therapy.

Dr. Eric Brost was funded for his work by the Mayo Clinic School of Graduate Medical Education. Dr. John Antolak is funded, in-part, by Varian Medical Systems, CA, USA.

SU-E-TRACK 5-03

Correlation of Treatment Time to Target Volume for GammaPod Treatments: A Simple Second Calculation"

L Chen*, S Becker, S. McAvoy, E. Nichols and M Guerrero, University of Maryland School of Medicine, Baltimore, MD

Purpose: The GammaPod is a novel device for stereotactic breast treatments that employs 25 rotating Co-60 sources while the patient is continuously translated in three axes to deliver a highly conformal dose to the target. There is no commercial software available for independent second calculations. The purpose of this study is to determine an efficient way to estimate GammaPod treatment times based on target volume and use it as a second calculation for patient-specific QA. **Methods:** Data was extracted from the plan reports of 65 breast cancer patient treatments. Similar to the Curie-seconds concept in brachytherapy, we considered dose-rate*time/(prescribed dose) as a function of target volumes. Using a MATLAB script, we generated linear (with 95% confidence interval) and quadratic fits and tested the resulting equations in an additional set of 18 patients. **Results:** We found a strong correlation between the dose-rate*time/(prescribed dose) and a patient's target volume for both the linear and quadratic models. The linear fit was selected for use and using the polyval function in MATLAB, a 95% confidence interval (CI) graph was created to depict the accuracy of the prediction for treatment times. Testing the model in 18 additional patients with target volumes ranging from 29cc to 193cc yielded treatment times from 10 to 25 minutes that in all cases were within the predicted CI. The average absolute difference between the predicted and actual treatment times was 1.2 minutes (range 0-4 minutes). The average percent difference was 7.3% (range 0-20%). **Conclusion:** This work has resulted in a viable independent calculation for GammaPod treatment times. After multiple tests, the spreadsheets are ready for clinical use to predict and verify the accuracy of breast cancer treatment times.

SU-E-TRACK 5-04

Introducing Kernel Truncation and Sparsity into Dose Calculations for IMRT Optimization

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Purpose: The adoption of AI and Machine Learning techniques is greatly improving planning efficiency. Modeling these algorithms can involve numerous iterations in which the dose calculations can become a bottleneck. The reliance on commercial treatment planning systems introduces unnecessary overhead due to the constant flow of data. Here we present and investigate the effect of kernel truncation and sparsity on the dose and fluence calculation for the purposes of designing a dose engine for AI/ML algorithms. **Methods:** The basis for this algorithm relies on beamlet and voxel discrimination in which numerous beamlets and voxels are pruned. This results in a constant time reduction in computation time and storage. The resulting deposition matrix is now highly sparse and can be treated and manipulated as such. We compared our dose calculation against calculations in both a water phantom and patient anatomy geometry to the AAA algorithm in Eclipse, the standard clinical dose engine. Our fluence optimization algorithm was tested on the TG-119 prostate phantom. Our system was developed using the Julia programming language. **Results:** The dose calculation algorithm had errors of less than 1% for field sizes up to 30x30 cm². Comparing dose-volume objectives the optimizer produced a distribution that differed from the constraints on average by 0.1% for the PTV, 2.9% for the rectum, and 2.85% for the bladder. For beamlet resolutions common in multi-beam treatments, the calculation had a performance around 17 times faster than a full dense calculation on the same hardware. **Conclusion:** The sparsity and truncation greatly improved computational efficiency while maintaining accuracies needed to be introduced in AI/ML applications. This improvement will greatly aid in the development cycle for AI/ML applications in radiation therapy.

SU-E-TRACK 5-05

Polar Coordinate Based Pencil Beam Dose Calculation Algorithm for VMAT

W Tseng¹*, B Lu, University of Florida, Gainesville, FL

Purpose: To propose a pencil beam (PB) dose calculation algorithm for volumetric modulated arc therapy (VMAT) that can reduce dose discrepancies due to angular under-sampling issue and improve the computational efficiency. **Methods:** 6 MV 1x1 mm² pencil beams were simulated and calculated on ArcCheck phantom under EGSnrc environment. Monte Carlo generated PB kernels were collected in Polar system for each bixel in a 40x40 cm² grid. These kernels were fine-tuned based on a squared field dose from a commercial treatment planning system (TPS, Pinnacle³) and then fitted using multiple Gaussians. Fluence for VMAT arc was calculated using detectors' eye view with leaf end, tongue and groove, and transmission factors corrected. Dose in Polar system for an arc can be calculated by a summation of convolutions of the corresponding PB fitting kernels and fluence for each bixel using fast Fourier transforms. Dose was converted to Cartesian system and compared with the dose from TPS using gamma analysis. Ten VMAT arcs were studied to evaluate the robustness of the algorithm. **Results:** The average gamma passing rate between the calculated and TPS dose was 96%(2%/2mm) and 98%(3%/3mm). With a 2x2x2 mm³ calculation grid, the proposed algorithm can provide a comparable computational efficiency (~150s) as TPS for common arcs and a shorter computational time for arcs with large fields using a PC equipped with a quad-core CPU. **Conclusion:** We proposed a Polar coordinate based pencil beam algorithm that can be implemented for VMAT dose calculation. It improves the computational efficiency of VMAT dose calculation since the convolution is applied to Polar system, whereas the convolution in the traditional discretized algorithm needs to be done for each discretized apertures with small angular separations. This algorithm can also reduce dose discrepancies due to angular under-sampling issue, which improves the accuracy of VMAT dose calculation.

SU-E-TRACK 5-06

BEST IN PHYSICS (THERAPY): Clinical Implementation of AI-Based 3D Dose Prediction Model as Physician Directive Support Tool to Streamline Head and Neck Planning Process

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Purpose: There exist some inherent shortcomings with the current treatment planning practice—e.g. deliverable dose to OARs is unknown a priori. In addition, the heterogeneity of clinical practices can lead to variation in the achievability of the planning goals and the resulted plan quality. The goal of this study is to incorporate an AI-based dose predictor into the treatment planning pipeline to develop a hybrid approach of physician intelligence and artificial intelligence that improves the planning efficiency, uniformity, and quality. **Methods:** We developed a 3D AI-based dose predictor based on a hierarchically densely connected U-net model. The model is trained separately on definitive and post-operative plans. We integrated the model with Eclipse treatment planning system via Application Programming Interface (API) to allow seamless 3D dose and the uncertainty estimation in TPS upon the completion of contour. In phase one implementation, we tested the API and the expert model on 60 retrospective cases treated by 3 different physicians to build the confidence level of prediction quality for different practice styles. In phase two implementation, all four head and neck physicians use the dose predictor as a decision support tool to construct their planning directives for 60 patients prospectively. The physician directives and the achieved plans for phase one and two were compared to quantify the efficacy of AI-guided physician directive workflow. **Results:** Our analysis shows increased practice uniformity among the physicians and improved achievability and plan quality of the plans upon application of AI. A dose difference of 3Gy was considered clinically significant. The frequency that resulted plan achieved within 3Gy from the physician directives has increased from only 47.5% to 86.7% upon implementing AI. **Conclusion:** AI guided physician decision support improved the uniformity of practice, directive achievability, and the final plan metrics in a significant percentage of patients.

SU-E-TRACK 5-07

Prediction of Organ-At-Risk Doses Using An Artificial Intelligence Algorithm: Clinical Validation and Estimated Benefit to Treatment Planning for Lung SBRT

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Purpose: To validate the clinical performance of an artificial intelligence (AI) algorithm to predict pre-treatment organ-at-risk (OAR) doses for lung SBRT

using only the planning CT, target and OAR contours. We also assessed the treatment planning benefit of having access to these dose predictions. **Methods:** AI models were trained for dose prediction using 213 patients treated with lung SBRT, using a gradient-boosted regression tree algorithm with regularized feature selection. Twenty patients not included in the training set were used to test OAR dose prediction performance, ten with 5x10Gy and ten with 3x18Gy. We also performed blinded re-planning with access to the OAR dose predictions but without access to clinically delivered plans, with target coverage set to equal that of the delivered plans. Differences between predicted and delivered doses were assessed by root-mean square deviation (RMSD) and OAR doses across all three comparisons were evaluated using one-way ANOVA tests. **Results:** Prediction performance was good with ANOVA tests showing no significant difference between prediction, delivered and re-planned OAR doses (All $p > 0.30$). The RMSD was 2.9Gy, 5.3Gy, 4.3Gy and 1.7Gy for max dose to spinal cord, esophagus, heart and trachea for 5x10Gy and respectively 1.7Gy, 1.9Gy, 2.9Gy and 2.6Gy for 3x18Gy. Three patients with doses close to the acceptable 52.5Gy max for great vessels had excellent dose prediction with average 3.4Gy difference. In blinded re-planning, lower OAR doses than clinically delivered were achieved for 6/10, 3/10 and 6/10 patients for spinal cord, heart and esophagus at 3x18Gy, and respectively 5/10, 5/10 and 7/10 patients for 5x10Gy. **Conclusion:** Pre-treatment prediction of OAR doses showed good agreement with delivered lung SBRT plans and access to these predictions led to lower OAR doses for several patients when re-planned. This prediction will allow upfront discussions of achievable dose fractionation and important OAR and target dose trade-offs.

Leslie Shulte, Damon Pappas and Colin Carpenter are employed by SIRIS Medical

Multi-Disciplinary Scientific Session

TRACK 6

Data Science, Radiomics, and Computing

SU-E-TRACK 6-01

A Multi-Modality Radiomics-Based Model for Recurrence Risk Stratification in Non-Small Cell Lung Cancer

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Purpose: To develop a multi-modal model integrating quantitative imaging features from the tumor and non-tumor regions of interest, descriptive features, and clinical data to improve risk-stratification of patients with non-small cell lung cancer (NSCLC) over cancer staging. **Methods:** A dataset of 135 patients with early-stage NSCLC who had surgery as their primary form of treatment was analyzed. The tumor and peri-tumoral volumes on both CT and PET were segmented, while the bone marrow (L3-L5 vertebral bodies) was segmented on PET. The Quantitative Image Feature Engine was used to compute shape, size, first-order, and texture radiomic features from these volumes. Combined with the clinical and descriptive features, feature selection was performed on the 1030 features to select the top features to predict time to recurrence in the training cohort ($n=94$). A Cox model was built in the training cohort and evaluated on the testing cohort ($n=41$). Model performance was assessed using the concordance index and compared to a baseline clinical model with cancer staging. **Results:** A total of 14 features were selected as the top performing features. Twelve of the top features were texture features with the remaining two features being cancer stage and patient age. The model achieved a concordance of 0.84 and 0.81 in the training and testing cohorts, respectively. The locked model outperformed that of a gold standard clinical model consisting of stage-only, with concordances of 0.69 and 0.68 in the training and testing cohorts, respectively. **Conclusion:** Our model outperformed the stage-only model for NSCLC recurrence prediction and risk stratification. These results demonstrate that radiomics has the potential to augment staging information in a clinical setting. These non-invasive software tools can be implemented into the clinic and will allow physicians to more accurately identify patients who are at higher risk of treatment failure.

Natural Sciences and Engineering Research Council of Canada (NSERC)

SU-E-TRACK 6-02

Computer-Assisted Diagnosis of Hepatic Portal Hypertension: A Novel, Attention-Guided Deep Learning Framework Based On CT Imaging and Laboratory Data Integration

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Purpose: To develop a deep learning framework that integrates CT imaging with laboratory data to predict the hepatic venous pressure gradient (HVPG) of patients at risk of portal hypertension (PHTN). **Methods:** We retrospectively identified 198 patients with CT imaging, laboratory results, and HVPG measurements. PHTN was defined as HVPG ≥ 5 mmHg. Laboratory results included albumin, platelet count, and APRI, all of which are associated with the pathogenesis of PHTN. We designed a novel attention-guided deep learning approach that targets the left hepatic lobe surface nodularity, which has been linked to HVPG. The liver was segmented using a self-adapting nnU-Net model, from which the liver tip and center-of-mass were detected as anatomic landmarks. The surface of the left hepatic lobe was then detected using a rotationally-invariant bounding box technique to partition the liver relative to its tip and center-of-mass. Surface nodularity variation in the superior-inferior direction was captured by incorporating adjacent CT slices. The resulting 3-channel image of the left hepatic lobe's surface served as input to a ResNet-152 deep learning architecture trained to predict PHTN. To incorporate laboratory data, positional encoding was used to transcribe raw lab results into a high-dimensional feature-space, which was concatenated to the fully connected layer of the ResNet-152 model prior to PHTN prediction. Performance of the joint-model (imaging+laboratory) was compared to both the imaging-only and laboratory-only models. **Results:** The joint-model consisting of both imaging data and laboratory data achieved an AUC of 0.74 and an accuracy of 0.71, resulting in an increase in model performance relative to both the imaging-only model (AUC=0.60; accuracy=0.62) and the laboratory-only model (AUC=0.58; accuracy=0.60). **Conclusion:** A novel deep learning paradigm was developed to predict PHTN based on the integration of CT imaging and laboratory results. Our results suggest that CT imaging and laboratory data provide complementary diagnostic information.

SU-E-TRACK 6-03

JACK KROHMER EARLY-CAREER INVESTIGATOR COMPETITION WINNER: Multi-Group Multi-Block Data Integration for Harmonizing 18F-FDG-PET/CT Radiomics Associated with Circulating Tumor Cells and Predicting Recurrence-Free Survival Across Independent Lung Cancer Radiotherapy Studies

S Lee*, G Kao, S Feigenberg, Y Fan, Y Xiao, University of Pennsylvania, Philadelphia, PA

Purpose: To combine independent studies with a multi-group multi-block (MGMB) integrative method for harmonizing ¹⁸F-FDG-PET/CT radiomics associated with circulating tumor cells (CTCs), and evaluate the impact on predicting recurrence-free survival (RFS) after radiotherapy in non-small cell lung cancer (NSCLC). **Methods:** The present study investigated two independent studies, consisting of 56/50 patients with early-stage/locally-advanced NSCLC (ES/LA-NSCLC) treated with stereotactic body radiotherapy (SBRT)/definitive concurrent chemoradiotherapy (CCRT), with pretreatment ¹⁸F-FDG-PET/CT imaging and CTC assessment via a telomerase-based assay. Each study included 1,548/1,562 PET/CT radiomic features extracted from gross tumor volume plus 5 baseline clinical parameters (CPs). MGMB data integration was performed to integrate four (CT/PET/CP/CTC)-block feature sets and combine both studies (ES-NSCLC SBRT and LA-NSCLC CCRT) by using sparse MGMB generalized partial least squares (PLS) regression with unsupervised feature

selection (UFS). Algorithm for RFS data stacking was implemented for RFS analysis as a classification problem. Block PLS-discriminant analysis (PLS-DA) was performed to integrate the four-block feature sets and predict locoregional/distant RFS after radiotherapy, where one study group served as a training set while the other as a testing set. The same analysis was conducted via single-group multi-block (SGMB) data integration that naively concatenated both studies while integrating the four-block feature sets by using sparse MB generalized PLS regression with UFS. Predictive performance of the block PLS-DA on the testing set for locoregional/distant RFS was compared using Harrell's c-index between MGMB and SGMB integrative approaches. **Results:** For predicting locoregional/distant RFS, the MGMB vs. SGMB integration obtained c-indices of 0.653/0.609 vs. 0.547/0.530 for the pre-SBRT ES-NSCLC study and c-indices of 0.736/0.758 vs. 0.730/0.882 for the pre-CCRT LA-NSCLC study, respectively. **Conclusion:** The MGMB integration of ^{18}F -FDG-PET/CT radiomics and CTC improved predictive performance through exploring potential correlation between studies, indicating that the MGMB integration may reduce prediction bias for RFS when combining independent NSCLC radiotherapy studies.

This project was supported by NCI grants, U24CA180803(IROC) and U10CA180868(NRG). Dr. Kao is a co-founder and has equity in Liquid Biotech USA, Inc., a University of Pennsylvania PCI-developed company through the UPStart program.

SU-E-TRACK 6-04

Multi-Institutional Data Analysis of Radiomic Signature Set to Predict Overall Survival in Glioblastoma Patients

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Purpose: This study aims to expand upon published literature in radiomics and deep learning to: 1) Apply deep learning to segment GBM on MRI, generating consistent ROI for radiomics. 2) Effectively employ harmonization to find a radiomic signature set (RSS) that competitively predicts GBM patient overall survival on independent datasets. 3) Integrate RSS with clinical/genetic patient information to perform a complete prognostic assessment. **Methods:** 180 MR datasets including T1, T2, T1CE, and Flair were obtained through BraTS, IVYGAP, and our institution (140/18/22) for training/ independent validation. Preprocessing included resampling, registration, skull-stripping, N4 bias-correction, and normalization. Whole Tumor, Tumor Core and Enhanced Tumor were delineated on the multi-modality MRI by physicians in the training dataset. Contours used for validation datasets were generated by 2D U-Net employing synthetic MR (synMR) data augmentation by GAN. Features were extracted by CAPTK, a radiomic toolbox, and feature harmonization included ComBat with quantile normalization. Feature selection was done by LASSO and a 12-feature signature set was constructed. Predictive power of RSS was assessed by calculating the area under the curve of a receiver operating characteristic curve (AUC-ROC) created by Support Vector Machine (SVM). RSS robustness was assessed using independent validation datasets. Complete prognostic assessment was performed by comparing RSS against known clinical/genetic prognostic features (Age, Karnofsky Performance Scale, IDH1, MGMT, EGFR3) using a sub-dataset containing 59 of 180 patients, based on availability. **Results:** U-Net performance and RSS prognostic ability was assessed using independent auto-contoured datasets; AUC for IVYGAP and Institutional datasets are 0.93 and 0.96, respectively. Using the sub-dataset, RSS outperformed known clinical/genetic features; AUC of 0.57. Including RSS with clinical/genetic features improved AUC to 0.83. **Conclusion:** U-Net performance shows ability to fully automate RSS workflow. RSS shows encouraging preliminary results as it shows strong predictive power on independent datasets and outperforms/aids commonly used clinical/genetic features.

SU-E-TRACK 6-05

Multi-Class Classification Based On Multi-Loss Strategy and Auxiliary Deep Learning Network with Applications in Medical Imaging

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Purpose: Multi-class classification has wide applications in medical imaging fields. However, the performance decreases due to the issues of inter-class data similarity and data imbalance. Previously proposed deep-learning methods commonly trained the network through the minimization of single loss function to learn discriminative features for classification, which might increase the risks of overfitting. Therefore, a multi-class classification strategy is proposed to address the issues and improve the classification performance. **Methods:** The proposed framework backbone includes a feature-extraction network (Feat-Net) and a multi-class classifier. Two auxiliary networks--reconstruction network (Recon-Net) and discriminator-network (Disc-Net)-- are proposed to support the training of Feat-Net and extract class-differentiable features to improve classification performance. Inspired by the design of GAN, Recon-Net aims to recover the images to be similar to the input images using Feat-Net-extracted features, while Disc-Net aims to distinguish the reconstructed and original images. Three networks are trained iteratively by updating Disc-Net only and updating Feat-Net and Recon-Net jointly through the minimization of pre-defined multiple loss functions. In this way, the Feat-Net can extract more informative features stably from the input images compared to traditional methods. Two case studies, the 3-class COVID-19 classification with X-ray images and the two-class stratification of risks of head-and-neck cancer treatment failures with PET images, were employed to evaluate the framework performance. **Results:** The framework achieved superior performances compared to the baseline method with 4.6% and 2.5% improvement on F1-score on two case studies, respectively. Also, it was observed that the effectiveness of the proposed framework depends on the depth of Feat-Net and the size of input images. **Conclusion:** This study indicates that the auxiliary networks and multi-loss strategy can improve the performance of multi-class classification. The modularized design allows flexibly tailored architectures fitting for varied applications. Extending the framework to other complex tasks would be investigated in the future.

SU-E-TRACK 6-06

A Radiomics-Boosted Deep Learning Model for COVID-19 and Non-COVID-19 Pneumonia Detection Using Chest X-Ray Image

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Purpose: To develop a deep-learning model that integrates radiomics analysis for enhanced performance of COVID-19 and Non-COVID-19 pneumonia detection using chest X-ray image **Methods:** Based on a pre-trained VGG-16 architecture, the deep-learning design consists of a 5-Dense layer neural network with varying sizes, while the last three convolutional layers were set as free parameters for training. In radiomics analysis, a 2D sliding kernel was implemented to map the impulse response of radiomic features throughout the entire X-ray image; thus, each feature is rendered as a 2D map in the same dimension as X-ray image. Two deep-learning models were trained: in the 1st model, X-ray image was the sole input; in the 2nd model, X-ray image and 2 radiomic feature maps(RFM) selected by the saliency map analysis of the 1st model were stacked as the input. Both models were developed using 812 chest X-ray images with 262/288/262 COVID-19/Non-COVID-19 pneumonia/healthy cases, and 649/163 cases were assigned as training-validation/independent test sets. For each model design, 25 versions were trained with random assignments of training/validation cases following 7:1 ratio in training-validation set. Sensitivity, specificity, accuracy, and ROC curve results from both model designs were analyzed. **Results:** In 1st model using X-ray as the sole input, the 1)sensitivity, 2)specificity, 3)accuracy, and 4)ROC Area-Under-the-Curve (AUC) of COVID-19/Non-COVID-19 pneumonia detection were 1)0.91±0.08/0.78±0.07, 2) 0.93±0.04/0.94±0.05, 3)0.92±0.03/0.89±0.02, and 4)0.96±0.03/0.92±0.03. In the 2nd model design, two RFMs, Entropy and Short-Run-Emphasize, were selected with their highest cross-correlations with the saliency maps of the 1st model. The corresponding results demonstrated significant improvements($p<0.05$) of disease detection: 1)0.95±0.03/0.85±0.04, 2)0.97±0.02/0.96±0.02, 3)0.97±0.02/0.93±0.02, and 4)0.99±0.01/0.97±0.02. The reduced variations suggested a superior robustness of 2nd model design. **Conclusion:** The inclusion of radiomic analysis in deep-learning design improved the performance and robustness of COVID-19/Non-COVID-19 pneumonia detection, which holds great potential for clinical applications in the COVID-19 pandemic.

SU-E-TRACK 6-07

Spatial Reconstruction of Statistically Significant Radiomics Signatures Using 3D Wavelet Decomposition in Tumors of Oropharyngeal Cancer
H Bagher-Ebadian*, F Siddiqui, A Ghanem, S Zhu, M Lu, B Movsas, I Chetty, Henry Ford Health System, Detroit, MI

Purpose: We applied 3D-wavelet-decomposition to spatially reconstruct statistically significant radiomics-features within images of the tumor for patients with oropharyngeal-squamous-cell-carcinomas(OPSCC's), for characterization of HPV. **Methods:** One-hundred-twenty-eight OPSCC-patients with known HPV-status (60-HPV+ and 68-HPV-, confirmed by immunohistochemistry-P16-protein-testing) were retrospectively studied. Three-dimensional-wavelet-decomposition analysis was performed on the contrast-enhanced CT-images of patient gross-tumor-volumes to decompose the tumor's intensity and spatial-frequency information into 3D-decomposition levels with series of orthonormal high-pass and low-pass wavelet-coefficients (WCs). Log-Energy-Entropy of the WCs for all the decomposition levels were calculated as radiomic-features. A Least-Absolute-Shrinkage-and-Selection-Operation (Lasso) technique combined with a Generalized-Linear-Model (Lasso-GLM) was applied on these features to identify and rank the optimal feature subsets with most representative information for prediction of HPV-status. Lasso-GLM classifier was constructed and validated using random-permutation-sampling of the features. Average Area-Under-Receiver-Operating-Characteristic (AUC), and Positive and Negative Predictive-values (PPV/NPV) were computed to estimate the generalization-error and prediction performance of the classifier. Ultimately, the optimal radiomic-features selected by the Lasso-analysis were used to select the significant wavelet-sub-bands and their respective WCs to perform inverse-wavelet-transform to reconstruct the tumor zones containing the highest information towards prediction of HPV-status. **Results:** Four entropy-based features from two decomposition-levels, including two low-frequency (LLL3/LHL3) and two high-frequency components (HHL1/HLL1) were found to be statistically significant discriminators between HPV-negative and HPV-positive cohorts respectively. The wavelet-based-Lasso-GLM classifier's prediction performance was: AUC/PPV/NPV=0.798/0.745/0.826. Results imply that tumor zones with smaller-components (higher-details/spatial-frequencies: components-sizes<~2mm) contain important radiomic information for characterization of HPV+ patients while tumor zones with larger-components (low-spatial-frequencies: components-sizes>~8mm) contain information for classification of HPV-cases. This study also confirms that compared to the central-zones of tumors, peritumoral-regions contain more information for characterization of the HPV-status. **Conclusion:** Results suggest that radiomics can be used to discriminate spatial-regions of importance toward the contribution of HPV. By associating this information with tumor pathology, one can potentially link radiomics to underlying biological mechanisms.

This work was supported in part by a grant from Varian Medical Systems (Palo Alto, CA)

SAM Therapy Educational Course

TRACK 7

Radiation Biology for Radiation Therapy Physicists

SU-EF-TRACK 7-00

Radiation Biology for Radiation Therapy Physicists

Moderators: Caridad Borrás, Washington, DC and Colin Orton, Wayne State University, The Villages, FL

SU-EF-TRACK 7-01

C. Borrás:

Introduction to Radiation Biology

SU-EF-TRACK 7-02

C. Orton, Wayne State University:

Radiobiological Basis of Conventional and Hypofractionated Radiotherapy

SU-EF-TRACK 7-03

K. Parodi, Ludwig-Maximilians-Universität München:

Radiobiological Aspects of Proton and Heavy Ion Radiotherapy

SU-EF-TRACK 7-04

J. Jin, University Hospitals, Case Medical Center:

Potential Radiobiological Mechanisms for FLASH Radiotherapy

SU-EF-TRACK 7-05

J. Deasy, Memorial Sloan Kettering Cancer Center:

Minimizing Secondary Adverse Effects of Radiation Therapy: QUANTEC and Newer Results

SU-EF-TRACK 7-06

Q&A

The course will start with an introduction to radiation biology. The mechanisms involved in biological effects from ionizing radiation at the cellular and tissue levels and the radiation dosimetry terms used in radiation biology and in radiation protection will be reviewed. Concepts such as Linear Energy Transfer (LET) and Relative Biological Effectiveness (RBE) will be defined. Epidemiological data on cancer induction and tissue injuries, including radiation effects on children and the developing embryo, will be described. Increases in the risk of second cancers following radiation therapy, as well as data on radiation induced tissue reactions, such as cardiovascular effects, for follow up times up to 20-40 years, published by ICRP, NCRP and BEIR Committees, will be examined. The latest risk estimates per unit dose for stochastic effects and the threshold doses for tissue injuries will be presented and their potential impact on radiotherapy treatments assessed.

Fractionation of radiotherapy treatments will be discussed. Topics presented will include: Why has radiotherapy been fractionated (repair of late-reacting normal tissue cells, the L-Q model and the narrow "Window of Opportunity"), and what is the effect of overall treatment time (repopulation of cancer cells and representation in the L-Q model)? Also presented will be the radiobiological basis of clinical trials of various fractionation schemes such as hyperfractionation, accelerated fractionation, accelerated hyperfractionation, and hypofractionation, and the reason for all the recent interest in hypofractionation (highly conformal radiotherapy widens the Window of Opportunity).

Proton and light ion beams are rapidly emerging external-beam radiotherapy modalities. These combine the advantages of ballistic selectivity with distinguished biological properties emerging from their elevated ionization density which, especially for heavier ions, can result in a selectively enhanced biological effectiveness in the tumor region. However, this introduces a new level of complexity to be properly accounted for in treatment planning. This presentation will review the basic radiobiological principles, quantities, and models for clinical application of protons and light ions, along with highlighting the still open questions, emerging new trends, and instrumentation requirements for addressing them.

The physics and biology of ultra-fast FLASH radiation will be discussed at two levels: the cell and body levels. At the cell level, a recombination theory during

radiolysis will be discussed. FLASH radiation increases the recombination of intermediate products that leads to reduction of free radicals and hence reduces DNA damage. A simulation and an experimental study will be presented to support this theory. At the body level, FLASH radiation can “freeze” motion and blood flow. A simulation study demonstrated that FLASH radiation dramatically reduces the radiation killing of immune cells in blood, and thus potentially reduces radiation-induced lymphopenia, which may lead to improve tumor control and normal tissue sparing.

Outcome-related mathematical models seek to capture the known trends in dose-volume-tolerance for normal tissues (normal tissue complication probability, or NTCP), or dose-volume-eradication for tumors (tumor control probability, or TCP). We will briefly review the underlying radiobiological and mathematical characteristics of a few predictive models for normal tissues and tumors and point to their uses to understand radiobiological mechanisms, design protocols, and, eventually, guide personalized plan optimization.

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The course will end with a general discussion.

Learning Objectives:

1. To distinguish between stochastic effects and tissue reactions and their impact on radiotherapy treatments
2. To realize how the linear-quadratic equations model the effects of radiobiological parameters in radiotherapy
3. To understand the radiobiological rationale of proton and heavier ion therapy, along with remaining challenges and opportunities
4. To learn outcome-related mathematical models for normal tissues and tumors that eventually will guide personalized plan optimization
5. To understand the FLASH mechanism and its potential reduction of normal tissue toxicity

SAM Educational Topic-Specific Guided Tours-Therapy

TRACK 8

Patient-Specific QA Devices/Software

SU-EF-TRACK 8-00

Patient-Specific QA Devices/Software

Moderators: Vimal Desai, UW Health, Madison, WI and Sara Thrower, UT MD Anderson Cancer Center, Houston, TX

SU-EF-TRACK 8-01

V. Desai, UW Health:

Patient-Specific QA Devices/Software

SU-EF-TRACK 8-02

S. Thrower, UT MD Anderson Cancer Center:

Patient-Specific QA Devices/Software

SU-EF-TRACK 8-03

Y. Lee-Bartlett, Elekta:

Patient QA – A Solution for Every Clinic

SU-EF-TRACK 8-04

P. Tobe, Varian Medical Systems:

Varian QA Vision and SRS End-to-end Phantom

SU-EF-TRACK 8-05

J. Kunz, University of Utah, Huntsman Cancer Hospital:

ScandiDos Delta4 Discover Transmission Detector: A New Paradigm for Patient Safety and Quality Assurance

SU-EF-TRACK 8-06

Q&A

Patient-specific quality assurance (PSQA) is an integral aspect of modern radiotherapy. As the complexity of radiotherapy treatment plans has grown over the years, so too has the need to verify that various treatment delivery systems are accurately delivering each patient-specific treatment. This guided tour will begin with an introductory presentation from the moderators covering the basics of PSQA including its rationale and various types of equipment that have been used historically to measure PSQA. Vendors will then showcase their most recent hardware and software solutions for measuring, analyzing, and recording PSQA data followed by a question-and-answer session.

Learning Objectives:

1. Describe the rationale of patient-specific QA
2. Review various classes of equipment that have been used to measure PSQA and understand their advantages and technical limitations
3. Understand the extent of what PSQA is checking for and how it fits into the larger radiotherapy QA process

Professional Symposium

TRACK 1

Professional Economics Updates

SU-F-TRACK 1-00

Professional Economics Updates

Moderators: Jonas Fontenot, Mary Bird Perkins Cancer Center, Baton Rouge, LA and Michele Ferenci, Penn State Cancer Institute, Hershey, PA

SU-F-TRACK 1-01

J. Fontenot, Mary Bird Perkins Cancer Center:
Professional Economics Updates

SU-F-TRACK 1-02

W. Fuss, Health Policy Solutions:
Professional Economics Updates

SU-F-TRACK 1-03

Audience Q&A

This presentation will describe reimbursement changes implemented in 2021 and proposed changes for 2022, including the radiation oncology alternative payment model. A review of changes that impact radiation oncology and radiology will be provided. A discussion of the Hospital Outpatient Prospective Payment System (HOPPS), the Medicare Physician Fee Schedule (MPFS), and the ambulatory surgery settings will be discussed. Lastly, an overview of the resources available to membership and tips for interacting with the professional economics committee will be provided.

Learning Objectives:

1. Basic understanding of payment models impacting medical physics clinical services.
2. Basic understanding of the 2022 HOPPS and MPFS proposed rules, including the RO-APM.
3. Basic understanding of economics and policy resources that are available from the AAPM and how to interact with the professional economics committee.

Imaging Scientific Session

TRACK 3

Breast Imaging

SU-F-TRACK 3-01

Anisotropies in Super-Resolution in 3D Magnification Mammography Using a Next-Generation Tomosynthesis System

R Acciavatti*, T Vent, C Choi, P Noel, A Maidment, Hospital of University of Pennsylvania, Philadelphia, PA

Purpose: In breast imaging, microcalcifications detected in a screening exam can be characterized with better detail in magnification mammography. In this exam, the breast is elevated closer to the x-ray source, and a 2D image is acquired. Unlike clinical systems, the next-generation tomosynthesis (NGT) system built in our lab is capable of 3D magnification mammography. We are also designing the system to generate super-resolution (SR) reconstructions. In our earlier work modeling the tomosynthesis acquisition in screening mammography, we found that there are anisotropies in SR at various positions in the image. This study analyzes how the anisotropies in SR vary with the use of magnification. **Methods:** A high-frequency sinusoidal test object was analyzed with theoretical modeling. The input frequency was oriented in the direction of x-ray source motion. Image quality in the reconstruction was quantified using the Fourier transform. The amplitude of low-frequency information was measured relative to the input frequency; this ratio should be as small as possible to achieve SR. We used these calculations to identify anisotropies in SR at various magnifications. For experimental validation of the anisotropies, a bar pattern phantom was analyzed with the NGT system. **Results:** In screening mode, there are anisotropies in SR in the direction perpendicular to the breast support; i.e., in various slices of the reconstruction. With theoretical modeling, we demonstrated that the spacing between anisotropies is reduced with increasing magnification; however, the relative amplitude of low-frequency (aliasing) signal at the anisotropies is not as pronounced as in screening mode. The anisotropies were validated experimentally with the NGT system. **Conclusion:** SR can be achieved in tomosynthesis reconstructions. Using the conventional acquisition geometry, anisotropies in SR exist in both screening and magnification tomosynthesis. Our future work will use this model of SR to optimize the acquisition geometry for magnification tomosynthesis.

A.M. is the spouse of a shareholder of Real Time Tomography (RTT). P.N. receives support from Philips Healthcare and is consultant to Stryker. Support was provided by grants W81XWH-18-1-0082 (DoD), IRSA 1016451 (Burroughs Wellcome), R01CA196528 (NIH), and IIR13264610 (Komen). Equipment support was provided by Analogic Inc., Barco NV, and RTT.

SU-F-TRACK 3-02

Histological Validation of Anatomical Imaging for Breast Modeling

A Sen*, C Kingsley, A Kulp, M McCulloch, B Willis, K Brewer Savannah, A Contreras, M Bordes, T Huynh, N Fowlkes, G Reece, K Brock, UT MD Anderson Cancer Center, Houston, TX

Purpose: Develop a framework to register in vivo and ex vivo images with histology using a novel cryo-microtome for validation of anatomical structures to improve tissue modeling and imaging signals. **Methods:** Breast tissue obtained from a mastectomy patient and a formalin-fixed entire breast including chest wall from a cadaver were imaged using magnetic resonance (MR) imaging (3D T1 and 3D T2 Dixon, fat/water-saturated, in/out-of-phase). Specimens were frozen and embedded in optical cutting temperature (OCT) compound. The OCT block was placed in a cryo-microtome mounted with an overhead camera (Xerra, Emit Imaging). Slices of 33µm and 50µm were successively shaved off the block for the tissue and cadaver specimens, respectively. After each shaving, the blockface was photographed. At select tissue sites (connective/adipose, muscle, skin, fibroglandular), 20µm sections were transferred onto cryotape for manual H&E staining and histological analysis. A 3D blockface image was automatically reconstructed from the photographs by aligning fiducial markers embedded into the OCT block. The 3D MR, blockface images, and histology images were rigidly registered. Target registration errors (TRE) were computed based on corresponding points marked at fibroglandular intersections (5 for tissue, 10 for cadaver). The overall MR-histology registration was used to compare the MR intensities at tissue extraction sites with a one-way ANOVA. **Results:** The MRI-blockface TREs were 0.34±0.11mm and 0.73±25mm for the tissue and cadaver datasets, respectively (both less than respective MR slice thicknesses of 0.35 and 1.00mm). The blockface-histology registration showed alignment of anatomical structures and tissue boundaries. The MR intensities at the four tissue sites showed statistically significant differences (P<0.01). Each pair of tissues except skin-connective/adipose were also significantly different (P<0.01). **Conclusion:** Fine sectioning in a highly controlled imaging/sectioning environment enabled accurate registration between the MR image and histology and differentiation of tissue types in the MR image.

Research reported was supported in part by resources of the Image Guided Cancer Therapy Research Program from The University of Texas MD Anderson Cancer Center and the Helen Black Image-Guided Fund.

SU-F-TRACK 3-03

Direct-Indirect Dual-Layer Detector for Contrast Enhanced Digital Breast Tomosynthesis: Feasibility Using Image Simulation

X Duan*, H Huang, A Howansky, J Stavro, A Lubinsky, A Goldan, W Zhao, Stony Brook Medicine, Stony Brook, NY

Purpose: Contrast enhanced digital mammography and tomosynthesis (CEDM and CEDBT) use weighted subtraction of images acquired at energies below and above the k-edge of iodine to enhance the conspicuity of malignant lesions. Traditionally, low energy (LE) and high energy (HE) exposures are taken separately (dual-shot), which allows better energy separation and breast tissue cancellation. However, it is more prone to patient motion artifact. Dual-layer detector allows CEDM with a single shot, eliminating motion between LE and HE images. In this work, we propose the combination of direct and indirect flat-panel imagers (FPI) and an x-ray filter to improve spectral separation of dual-layer detectors for CEDM and CEDBT. **Methods:** The system consists of a k-edge filter (200 micron Rh) and a dual-layer detector, with a direct conversion (200 micron a-Se) FPI in the front (LE) and an indirect CsI (400 micron) FPI in the back (HE). We used the VICTRE tool to conduct image simulation of a digital breast phantom (half uniform and half with breast tissue structure), which was inserted with spherical iodinated objects with different sizes and concentrations. The LE, HE and subtracted images were compared for three different conditions: dual-shot, dual-layer (both indirect detectors) and dual-layer (with direct-indirect-filter combination). **Results:** Dual-shot offers excellent tissue separation and high iodine SDNR if no patient motion is involved. Iodinated objects have reasonable SDNR values with the k-edge filter and a-Se/CsI detector combination, even with the imperfect energy separation. If the top a-Se FPI is replaced with an a-Se based photon counting detector (PCD) SWAD, the highest SDNR can be obtained. **Conclusion:** The combination of direct-indirect FPI dual-layer detector and k-edge filter can potentially produce good breast tissue cancellation and SDNR that approaches that of dual-shot technique. The results are substantially better than dual-layer detectors that uses the same x-ray conversion material.

SU-F-TRACK 3-04

Image Quality Evaluation of a Compressed-Sensing Algorithm for Image Reconstruction in Dedicated Cone-Beam Breast CT Using Offset-Detector
S Vedantham*, H Tseng, A Karellas, University of Arizona, Tucson, AZ

Purpose: We have developed a new version of a dedicated, prone, cone-beam breast computed tomography (CBBCT). This system uses a CMOS detector in offset-detector configuration for imaging large breasts, while maintaining an optimal air-gap between the breast and the detector for scatter rejection. For this geometry, image quality achieved with a compressed-sensing (CS) algorithm was compared to Feldkamp-Davis-Kress algorithm with Wang's weights (FDK-Wang). **Methods:** 30 clinical breast CT datasets with calcifications acquired on a CBBCT system (300 views in 360-degrees) with a large-area (40x30-cm), centered, a flat-panel detector with 1024x768 pixel matrix, were selected. The sinograms were truncated to 768x768 and 640x768 -pixels along the fan-angle to emulate offset-detectors. A modified and GPU-enabled version of a CS-based algorithm, referred to as FIRST, and the FDK algorithm with a weighting window (FDK-Wang), were used to reconstruct these truncated sinograms. FDK reconstructions from complete sinograms served as reference. Signal-to-noise ratio (SDNR), bias, variance, root-mean-squared-error (RMSE), and the full-width at half-maximum (FWHM) of calcified lesions in two orthogonal directions were quantified. **Results:** The FWHM of calcifications did not differ ($P>0.262$) among reconstruction algorithms and detector formats, implying comparable spatial resolution. For a chosen offset-detector, the FIRST algorithm outperformed the FDK-Wang in terms of RMSE, bias, variance, and SDNR. For FIRST reconstructions with 768x768 (640x768) pixels, the median RMSE (cm-2), bias (cm-1), variance (cm-2) and SDNR were 6.45E-6 (7.63E-6), 7.17E-4 (9.26E-4), 3.37E-5 (4.29E-5), and 8.5 (6.54), indicating better performance with the 768x768 pixel matrix. The variance and SDNR from FIRST reconstructions with 768x768 pixels outperformed the reference FDK reconstructions from complete sinograms. The reconstruction time varied with breast size (25-45 minutes). **Conclusion:** This study indicates the feasibility of using CS-based, FIRST algorithm to reconstruct sinograms from offset-detectors. Our results indicate that FIRST reconstructions from 768x768 pixel matrix achieved the best performance.

SU-F-TRACK 3-05

Two-Dimensional Breast Tomosynthesis with a Stationary Array of Thermionic Cathodes for Higher Resolution Images and Fast Acquisition

J Boone^{1*}, A Sisniega², J Siewersden³, P Schwoebel⁴, (1) UC Davis Medical Center, Sacramento, CA, (2) Johns Hopkins University, Baltimore, MD, (3) Johns Hopkins University, Department of Biomedical Engineering, Baltimore, MD, (4) SRI International, Albuquerque, NM

Purpose: To develop a two-dimensional (2D) stationary thermionic x-ray source array for breast tomosynthesis, in order to dramatically improve image quality and reduce acquisition time. **Methods:** A Multiple X-ray source Array (MXA) was developed with 11 focal spots distributed along a ~28 cm long rotating anode shaft which supports eleven ~10 cm diameter anode disks. Individual thermionic (hot) grid-pulsed cathode assemblies supply pulsed electron beams which strike the end of each disk. All x-ray sources are within the same vacuum enclosure and are individually addressable using TTL control electronics. Coupled with a flat panel detector, this 11-source array of x-ray sources was used to collect tomosynthesis data using an orange as a test object. By adding a second row of cathodes on the other side of the anode disks, two rows of 11 x-ray focal spots will produce a 2D array (2 x 11) of sources for breast tomosynthesis. A breast CT data set was used to simulate 22 projection images with the 2D source distribution, and a model-based iterative reconstruction algorithm with spatially invariant regularization was used to reconstruct tomosynthesis images. **Results:** The 11-source x-ray tube demonstrated rapid acquisition, robust performance at 35 kV, and small focal spots (~0.3 mm each). Computer simulation of the proposed 2D array tomosynthesis system with 22 source positions shows a marked improvement for in-plane sharpness and thinner slices. Acquisition time using a 20 FPS detector would be 1.1 sec. **Conclusion:** All clinical breast tomosynthesis systems use a one-dimensional arc with acquisition times ranging from 3.7 to 25 s. Due to constraints on acquisition time, moving sources are not practical for 2D geometry tomosynthesis. The proposed stationary hot-cathode source array provides rapid acquisition (1.1 s for 22 sources) for low motion artifact tomosynthesis images with much better slice separation and in-plane object delineation.

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SU-F-TRACK 3-06

Validation of Automated MTF Measurement for Digital X-Ray Imaging Quality Control

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Purpose: To validate an automated MTF-measurement procedure for achieving accurate and objective assessment of digital x-ray system spatial resolution in routine quality control surveys. **Methods:** The automated procedure (autoMTF) was evaluated for a full-field digital mammography system, a Senographe Pristina (GE Healthcare, IL), wherein the manufacturer prescribes sub-system MTF sampling at defined object frequencies. The autoMTF procedure utilizes a detection algorithm to (i) locate the line-pair phantom (ii) place elliptical ROIs in prescribed locations, and (iii) output tabulated ROIs statistics from which MTF samples can be calculated. Phantom images were acquired for both targets (Mo, Rh), both focal spots (small SFS, large LFS), two bar patterns, and phantom directions (width, length). Manual and autoMTF measurements were performed by placing four ROIs in each image: at the "space", "bar", and two line-pair sections (2.09 and 3.93 lp/mm for LFS, 5 and 8 lp/mm for SFS). Images were copied to external storage media (CD or USB drive) for autoMTF processing. Performance in terms of procedure time and accuracy was evaluated by three participating medical physicists. **Results:** The autoMTF completed the resolution testing, including storage media manipulation, in an average of 4 min. compared to manual measurements in 15-30 min. Test outcome (pass/fail) was equivalent for both procedures. Physicist measurements exhibited larger percent deviations (2-24%) compared to autoMTF, as well as more inter-variability, for the most challenging ROIs, i.e., higher spatial frequencies. **Conclusion:** This work produced a tool to facilitate oftentimes tedious and lengthy spatial resolution evaluations that are prone to error and inter-observer variability. The autoMTF algorithm is versatile and can be ported to similar spatial

resolution testing in other modalities such as digital radiography and fluoroscopy. Future investigations will utilize the autoMTF tool to quantitatively assess physicist subjectivity in performing line-pair phantom tasks.

SU-F-TRACK 3-07

Automating Tumor Functional Diameter Measurement with Molecular Breast Imaging for Treatment Response Assessment

BP Lopez*, GM Rauch, SC Kappadath, UT MD Anderson Cancer Center, Houston, TX

Purpose: Recent small-cohort studies have investigated using Molecular Breast Imaging (MBI) with ^{99m}Tc -sestamibi to assess tumor response following neoadjuvant chemotherapy (NAC). We present a robust automated methodology to estimate true functional uptake diameters in MBI capable of accurate size-based response assessment. **Methods:** Over 1,000 unique simulated images, sampling clinically-relevant tumor diameters ($d=5\text{--}25\text{mm}$), tumor depths ($z=1\text{--}11\text{cm}$), tumor/background volumetric-concentration ratios ($\text{TBR}=10,20,40$), breast thicknesses ($b=1\text{--}12\text{cm}$), were generated using our validated Geant4-based simulation of GE Discovery NM750b MBI to design and test a diameter estimation methodology. The methodology's accuracy/precision was additionally tested on simulated images of 78 ellipsoid tumors and phantom images of 3 spheres ($d=9.89, 15.43, 24.82\text{mm}$, $\text{TBR}=10,20$, $z=2,3\text{cm}$) within a $b=9\text{cm}$ container. **Results:** Our methodology calculates uptake diameter as $1.17 \cdot \text{FWHM} + 1.64$, where FWHM is automatically interpolated from a background-corrected profile through the tumor centroid (orientation user-specified) in a 3×3 median filtered single-detector image. The resulting absolute error [$\text{mean}=\text{accuracy}, 1\sigma=\text{precision}$] for $d \geq 15\text{mm}$ spherical tumors was $[0.1\text{mm}, 0.9\text{mm}]$, with accuracy differing $<0.6\text{mm}$ across tumor depths, TBRs, and breast thicknesses. Thus, we are able to accurately measure tumor uptake diameters $\geq 15\text{mm}$ and identify differences $\geq 3\text{mm}$ (corresponding to $\geq 20\%$ change after intervention) with $>99\%$ confidence, independently of most common biologic (e.g., uptake distribution) and imaging (e.g., view and compression) conditions during acquisition. These values are clinically relevant, as ultrasound size analysis in 28 patients revealed that 1) pre-NAC all tumors had diameters $\geq 15\text{mm}$ and 2) mid-NAC 100% of partial responders (per RECIST1.1) and 89% of patients had changes $\geq 3\text{mm}$. Methodology performance remained constant at $[0.1\text{mm}, 0.7\text{mm}]$ with $d \geq 15\text{mm}$ ellipsoid tumors but decreased to $[1.9\text{mm}, 2.3\text{mm}]$ with all $d=10\text{mm}$ simulated tumors. Performance in the phantom images was $[0.0\text{mm}, 1.1\text{mm}]$, $[2.2\text{mm}, 1.7\text{mm}]$, and $[5.5\text{mm}, 5.0\text{mm}]$ for 24.82mm , 15.43mm , and 9.89mm spheres, respectively. **Conclusion:** Our automated methodology accurately estimates functional uptake diameters $\geq 15\text{mm}$. MBI-based response assessment in 70+ patients undergoing NAC is underway.

Research supported in part by GE Healthcare

Therapy Scientific Session

TRACK 5

Treatment Planning: Planning Automation and Assessment

SU-F-TRACK 5-01

A Hierarchical Optimization Framework for Gamma Knife Treatment Planning

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Purpose: Gamma Knife (GK) is a widely used stereotactic radiosurgery system for brain lesion treatment. Its treatment planning, however, may require solving a large-scale optimization to determine many variables, including an unknown number of isocenter positions, collimators, and intensities. We propose a hierarchical optimization framework to address this large-scale problem. **Methods:** The framework is built by three critical components: kernel-decomposition-based (KD) dose model; multi-resolution shot initialization and dose grid with shot migration capability; and hierarchical collimator optimization. The KD dose is a shift-invariant kernel scaled by the normalized mean TMR. Both shot initialization and dose grid take resolutions of 1, 2, and 4 mm on the 1-mm-margin, 2-mm-margin, and the remaining interiors of the region of interest (ROI), respectively. For shot initialization, the ROI is the target; while for dose grid, the ROIs are the 4-mm dilations of the planning structures, including the target and organs-at-risk. Hierarchical collimator optimization starts with cones for shot locations and migrations followed by sector durations. Though not limited to, the optimization is primarily based on linear programming for its simplicity and effectiveness in sparse solution. We tested this framework using 12 AVM clinical cases retrospectively. **Results:** The KD dose had Gamma Index (2%/2 mm) passing rates of 99% compared with Monte Carlo calculation for the studied cases and linear complexity that calculated shot dose in sub seconds. The sampling strategy further reduced the complexity compared to using the whole ROI without losing its representation. Hierarchical collimator optimization sped up the convergence and completed in few seconds. Compared to the clinical plans, the optimized plans by this framework on average had improved coverage (96% $>100\%$), selectivity (62% $>78\%$), homogeneity (1.99 >1.62), gradient index (5.42 >3.78), and radiation time (59min $>55\text{min}$). **Conclusion:** The proposed framework addresses multiple fundamental aspects in GK optimization for efficient and quality treatment planning.

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SU-F-TRACK 5-02

Assessing the Practicality of Using a Single KBP Model for Multiple Linac Vendors

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Purpose: To evaluate the effectiveness of a commercial knowledge-based planning (KBP) tool by applying a single KBP model to three treatment machines with different beam and MLC characteristics (Varian C-arm, Varian Halcyon and Elekta Versa HD). **Methods:** A RapidPlan model for head-neck VMAT treatments was developed and finetuned for a Varian 2100 linac. For fifty retrospective head-neck patients, we generated three sets of VMAT plans using an automated treatment planning system (Radiation Planning Assistant). Each plan used a different treatment machine: Varian 2100, Elekta Versa HD, and Varian Halcyon. The Varian 2100 plans were reviewed by radiation oncologists in a separate study. For each patient, we calculated the dosimetric differences (Versa or Halcyon minus Varian 2100) between each plan. We also calculated the percentage of plans that meet established clinical dosimetric recommendations for normal and planning structures. **Results:** The mean differences from our Varian 2100 plan for brainstem (D_{max}), contralateral parotid (D_{mean}), ipsilateral parotid (D_{mean}) and spinal cord (D_{max}) were $0.76 \pm 1.57\text{Gy}$, $0.46 \pm 0.59\text{Gy}$, $0.53 \pm 0.44\text{Gy}$, $0.80 \pm 1.1\text{Gy}$, respectively for the Versa plans; and $0.46 \pm 3.46\text{Gy}$, $-0.45 \pm 0.59\text{Gy}$, $-0.34 \pm 0.49\text{Gy}$, $-0.14 \pm 1.24\text{Gy}$, respectively for the Halcyon plans. The mean difference in high dose PTV coverage were less than 3% for prescription dose ($V100\%$) and hotspots ($V105\%$ and $V110\%$). 100% of plans for all machines met the clinical dosimetric recommendations for brainstem ($D_{\text{max}} < 54\text{Gy}$), spinal cord ($D_{\text{max}} < 45\text{Gy}$) and high dose PTV ($V95\% > 95\%$). For contralateral parotid, 41 Varian 2100 plans met recommendation ($D_{\text{mean}} < 26\text{Gy}$), compared to 41 Versa and 40 Halcyon plans. For ipsilateral parotid, 23 Varian 2100 plans met recommendation, compared to 20 Versa and 25 Halcyon plans. **Conclusion:** We successfully applied a single RapidPlan model to generate comparable plans across three different treatment machines with different beam and MLC characteristics. This supports the use of head-neck RapidPlan models in centralized planning workflows that support clinics with different linac models/vendors.

SU-F-TRACK 5-03

Landmark-Based Field Aperture Generation for Automatic Whole-Brain Radiation Planning

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Purpose: To develop and compare two landmark-based approaches for automatic whole-brain treatment planning that can be configured to suit local clinical practices. **Methods:** Field apertures for both approaches used landmarks based on organ contours seen in beams-eye-view digitally reconstructed radiographs (DRRs), with the relationship between the landmarks and field borders configurable to suit local clinical practice. For Approach 1, a previously trained deep learning model was used to create contours on CT scans, and these contours were projected onto DRRs. For Approach 2, organ contours were directly segmented on DRRs using a deep learning model trained/validated/tested using 724/182/224 patients. Field apertures were then created using the DRR contours from both approaches and compared with manually outlined field apertures using Hausdorff Distance (HD) and Mean Surface Distance (MSD). Additionally, quality assurance for the contours on DRRs from Approach 1 and 2 were compared using Dice-Similarity-Coefficient (DSC). In total, eight landmark-based rules consisting of anterior, carinal, posterior, caudal, and anterior-caudal bounds were applied, and five setting options included selecting vertebral bodies, anterior-caudal bound shapes, and extent of brain expansion and skin flash were provided for creating field aperture variations. **Results:** The average DSC of organ contours segmented on DRRs regarding brain, eyes, lens, C1, and C2 were 0.97 ± 0.01 , 0.88 ± 0.05 , 0.55 ± 0.18 , 0.89 ± 0.04 , and 0.87 ± 0.07 , respectively. The performance of the generated field apertures from a single configuration in terms of the average HD and MSD for Approach 1 resulted in 16.43 ± 6.95 and 11.63 ± 7.26 mm, respectively, and 16.93 ± 6.87 and 12.18 ± 6.92 mm for Approach 2. **Conclusion:** We have developed and tested two landmark-based methods to define whole-brain field apertures. Both gave similar results compared to the manually outlined field apertures. The flexible configuration for creating field aperture variations also provided flexibility in adapting this traditional treatment approach to local clinical practices.

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SU-F-TRACK 5-04

Toward Semi-Automatic Biologically Effective Dose Treatment Planning for Gamma Knife Radiosurgery

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Purpose: Gamma Knife (GK) radiosurgery treatments exhibit considerable variations in treatment times, influencing the delivered radiation's biological effectiveness. We investigate the feasibility of generating semi-automated treatment plans based on the biologically effective dose (BED) to tackle this issue. We thus create high-quality iso-effective treatment plans through the optimisation of per iso-centre beam-on times and delivery sequence. **Methods:** Using a BED model considering bi-exponential repair during treatment delivery leads to a non-convex optimisation problem, and optimising the shot order poses a combinatorial problem. Therefore, we implemented two optimisation algorithms. One accurate but slow, based on mixed-integer linear programming (MILP) using a purposely developed convex under-estimator for the BED. The other utilising two established local methods (L-BFGS-B and an adapted 2-opt) alternatively, significantly faster but more prone to local minima issues. We evaluated the post-optimisation results to determine if the more rapid approaches could produce results close to those from the more demanding MILP approach. The different approaches were executed on 14 cases of Vestibular Schwannoma, using the iso-centre configurations from the original physical dose treatment plans and using a 60 min treatment as a BED reference. **Results:** In terms of the final objective function values, the faster optimisations came within 1.2% (0.02-2.08%) of the convex MILP approach while requiring significantly less computation time (minutes vs hours-days). The issue of reduced biological effectiveness with increasing treatment times, which was evident in the initial BED95 values, was resolved by the BED optimisations. Overall, the Paddick Conformity Index (PCI) for the BED distribution was improved, both in average and range, from initially 0.81 (0.58-0.9) to 0.88 (0.78-0.93) with negligible differences between the optimisation approaches. **Conclusion:** We demonstrate the feasibility of efficient BED-based treatment planning to ensure iso-effective treatments across varying treatment times and highlight the prospect of further improvements in treatment plan quality.

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SU-F-TRACK 5-05

Knowledge-Based Cardiac Sparing Treatment Planning for Lung Radiotherapy

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Purpose: High radiation doses to the heart are correlated with poor overall survival in radiotherapy patients with stage III non-small cell lung cancer (NSCLC). Cardiac sparing is an optimization goal of all lung treatment plans, but optimization based on the whole heart has limitations. Optimization of treatment plans using cardiac substructures can reduce cardiac dose, but is not feasible in most clinics due to the extra time needed for planning. This study builds a knowledge-based cardiac sparing planning (KBCSP) model for cardiac dose reduction using a set of treatment plans for patients with stage III NSCLC. **Methods:** Cardiac substructures were delineated for 33 patients and treatment plans were re-optimized to reduce dose to substructures and the heart. A KBCSP model was developed from these plans using target volumes and organs-at-risk (OARs) including the heart but excluding substructures. To test the model's efficacy for cardiac dose reduction, a second KBP model was trained from the same 33 patients, using the original clinical treatment plans. Both models were evaluated in a cohort of 30 separate stage III NSCLC patients. **Results:** Both models produced acceptable target coverage, dose uniformity, and dose to OARs. Compared to the KBP-based plans, KBCSP-based plans showed significant reductions in mean dose to the esophagus and lungs while performing similar or better in all evaluated heart dose metrics. Compared to the clinical KBP model, the KBCSP model showed reduced ($p < 0.05$) heart mean and maximum doses, as well as volumes receiving 60 Gy, 50 Gy, and 30 Gy. **Conclusion:** The KBCSP model successfully captured the cardiac-sparing expertise and improved plan quality for patients in a validation cohort. This KBCSP model offers an automated approach to reduced cardiac dose without explicit delineation of cardiac substructures, which may reduce heart toxicity and improve overall survival in radiotherapy patients with NSCLC.

SU-F-TRACK 5-06

Deep Learning-Based Dosimetric Plan Quality Assessment for Prostate Radiotherapy

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Purpose: This study developed a novel dosimetric plan quality evaluation method for prostate radiotherapy using a multivariate deep neural network (DNN). **Methods:** First, 60 prostate CT images with segmented contours of planning target volume (PTV), clinical target volume (CTV) and organs at risk (OARs)

were collected. These were divided into 48 patients for training of a DNN model, 6 for validation, and 6 for testing, respectively. Second, intensity-modulated radiation therapy (IMRT) dose distributions were calculated for each patient from seven equally spaced photon beams. Dose-volume histograms (DVHs) of PTV, CTV and OARs and homogeneity index (HI), conformity number (CN), dose coverage index (DCI) and gradient index (GI) of PTV were calculated for each dose distribution. Third, reference quality of the dose distributions was manually assessed by a treatment planner who had access to volumetric CT images, dose distributions and DVHs. Moreover, an in-house plan quality metric (PQM) technique was applied to calculate analytical dosimetric plan quality values. Finally, a multivariate DNN model was generated to predict PQM or the manual plan quality by using HI, CN, DCI, GI and DVHs as inputs. For evaluation, the DNN-calculated qualities of test data were compared with actual plan quality values. **Results:** The DNN model estimates PQM and the manual quality values accurately. Averaged relative absolute error and its standard deviation over all test data were less than 4.00% for both PQM and the manual plan quality values. Computation time for calculating one quality value is 0.03 seconds with a single GPU. **Conclusion:** A novel dosimetric plan quality assessment method was established using a multivariate DNN. This technique can predict, as a “virtual” planner, plan qualities calculated from multiple assessment methods with high accuracy. This method will be useful for DNN-assisted treatment planning to improve treatment plans and reduce inter-planner quality variation.

SU-F-TRACK 5-07

Development of a Framework to Assess Robustness of Dynamic Trajectory Treatment Plans

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Purpose: To develop a flexible Monte Carlo (MC) based robustness calculation and evaluation framework to assess treatment plan robustness of dynamic trajectory radiotherapy (DTRT, extending VMAT by dynamic table and collimator rotation) treatment plans, by exploring the impact of uncertainties in patient setup and mechanical accuracy of machine components on the dose distribution. Additionally, a novel robustness index (RI) is defined, with the aim to condense the dosimetric robustness against multiple error scenarios (ESs) into one number. **Methods:** An in-house MC based dose calculation framework is extended to assess robustness of treatment plans. This extension allows to automatically start dose calculations for user-defined ESs. Each ES can be a combination of patient setup uncertainties and of the following machine components: Angles of gantry, collimator, table-yaw, table-pitch and table-roll and positions of jaws, MLC banks and single MLC leaves. The framework allows the simultaneous evaluation of the ESs on a GUI according to user-specified dosimetric criteria such as D2%, mean dose, gamma analysis or dose difference to the nominal dose distribution (no error). The RI is defined by the fraction of ESs passing the previously defined DVH/Gamma criteria. The application of the framework is demonstrated on a DTRT plan for a nasopharynx case. **Results:** The DTRT plan is evaluated for combinations of systematic errors in gantry, collimator and table angle between -4° and +4°. Robustness criteria of >90% Gamma-passing-rate (2%/2mm evaluated on CTV) lead to a RI 64.1%; and <1 Gy deterioration of D2% and D98% (of CTV) to RI 99.7%. **Conclusion:** The extended framework was successfully implemented and demonstrated on a DTRT plan. Effects of uncertainties in patient setup and mechanical accuracy of machine components on the dose distribution can be evaluated in a user-friendly GUI to assess treatment plan robustness. This work was partially supported by Varian Medical Systems.

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Multi-Disciplinary Scientific Session TRACK 6

Real-time Tracking and Adaptive Radiation Therapy

SU-F-TRACK 6-01

Automatic Stent Recognition Using Deep Neural Network for Quantitative Intra-Fractional Motion Monitoring in Pancreatic Cancer Radiotherapy

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Purpose: Stent has been often used as an internal surrogate to monitor intra-fractional tumor motion during pancreatic cancer radiotherapy. Based on the stent contours generated from planning CT images, current intrafraction motion review (IMR) system on Varian TrueBeam only provide a tool to verify the stent motion but lack of quantitative information visually. The purpose of this study is to develop an automatic stent recognition method for quantitative intra-fractional tumor motion monitoring in pancreas cancer treatment. **Methods:** 466 IMR images from 11 pancreatic cancer patients were retrospectively selected in this study, with the manual contour of the stent on each image served as the ground truth. An objective attention modeling mechanism was integrated into the U-net framework to deal with the optimization difficulties when training a deep network with 2D kV IMR images and limited training data. For supervision, we used the binary cross-entropy loss combined with a Dice loss for training. The deep neural network was trained to capture more contextual information to predict binary stent masks. 68% of 466 images were randomly selected for training (85%, 269 images) and validation (15%, 47 images), while the rest (150 images) were used for testing. **Results:** Our stent segmentation results were compared with the manually segmented contours. The trained model achieved a detection rate of 93%. The mean stent Dice similarity coefficient (DSC), 95% Hausdorff distance (HD95), 2D mean surface distance (MSD), and residual mean square distance (RMSD) were 0.91 ± 0.08 , 1.12 ± 1.84 mm, 0.26 ± 0.43 mm, and 0.38 ± 0.59 mm, respectively. **Conclusion:** We developed a novel deep learning-based approach to segment the stent from kV IMR images automatically, demonstrated its clinical feasibility, and validated its accuracy compared to manual segmentation. The proposed technique could be a useful tool for quantitative intra-fractional motion monitoring in pancreatic cancer radiotherapy.

SU-F-TRACK 6-02

Dual-Energy Real-Time Stereoscopic X-Ray Image Guidance for Markerless Lung Tumor Tracking

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Purpose: The purpose of this work is to develop real-time markerless tracking of lung tumors during patient free-breathing, guided by stereoscopic dual energy (DE) x-ray imaging. **Methods:** Incremental and motorized motion stages were designed and used to simulate lung tumor respiratory motion within an anthropomorphic phantom. A set of tumor motion experiments were performed using spherical tumor models with diameters ranging from 6.4 to 25.4 mm. A series of stereoscopic high (140 kVp) and low (60 kVp) energy x-ray images of the phantom were acquired with the moving tumor insert using the ExacTrac stereoscopic imaging system (Brainlab AG, Germany). The high and low energy images were combined by weighted log subtraction to create stereoscopic DE images. Conventional stereoscopic single energy (SE) (120 kVp) images were also acquired for comparison. Digitally reconstructed radiographs (DRRs) were calculated from tumor-only volumes within CT images of the phantom, and tumor templates were cropped from these DRRs. Normalized template-image cross correlation and stereoscopic triangulation were used to calculate the 3D tumor positions from stereoscopic image pairs, which were compared to a ground truth from known tumor locations and the accuracy, precision and success rate of localization were calculated. **Results:** The success rate of tumor localization using DE imaging was consistently higher than that of the SE technique for both incremental and dynamic motion experiments with most dramatic improvement for the smaller tumors (<20mm). The accuracy of tumor localization using the DE technique was generally sub-millimeter (range 0.53 - 1.10 mm). **Conclusion:** DE stereoscopic imaging outperforms the SE technique to localize lung tumors in real-time particularly for smaller tumors. With some alterations to existing stereoscopic image guidance systems, markerless lung tumor tracking would be feasible.

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SU-F-TRACK 6-03

Implementation of High-Quality Motion-Compensated Simultaneous Algebraic Reconstruction Technique (mc-SART) Cone-Beam CT (CBCT) Imaging Using the 5D Model in a Prospective Patient Study

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Purpose: Clinically used pre-treatment imaging methods for external-beam radiotherapy of lung cancer such as 3D/4D-based CBCT imaging can result in low quality images. Here, we implement a novel 5D-model-based motion-compensated simultaneous algebraic reconstruction technique (mc-SART) method to create a patient- and treatment-day-specific volumetric image and motion model from free-breathing cone-beam projections and respiratory surrogate measurements. **Methods:** Four lung cancer patients receiving external beam radiotherapy were prospectively enrolled in this study. Patient-specific breathing traces were recorded using an in-house respiratory monitoring system. Raw CBCT projections were saved after a single fraction for each patient. CBCT projections were sorted into 8 bins based on breathing amplitude and flow. The mc-SART framework involves creating an initial reference image and motion model, and then iteratively improving them using a modified version of SART. The initial reference image was defined using a reconstructed SART image from end exhale/zero flow projections. The motion model was created using deformable image registration between the reconstructed bins and least squares fitting to the 5D model parameters. The initial reference image was deformed to a particular amplitude and flow and a modified version of SART was applied to include motion compensation. The model output is a volumetric reference image and a motion model that can be used to generate volumetric images at any other time point from breathing traces. **Results:** Signal-to-noise-ratio (SNR) measurements were performed in soft tissue and lung regions for FDK-reconstructed, 4D-SART and mc-SART reference images. The soft tissue and lung SNR measurements for mc-SART images was 160% and 28% greater than FDK-reconstructed image and, 26% and 43% greater than 4D-SART reconstructed images. **Conclusion:** We have developed a method to create a high-quality motion-compensated reference image and motion model with real patient data. This framework opens up the possibility of creating model-based volumetric images during treatment delivery.

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SU-F-TRACK 6-04

One-Second into the Future: A Deep Learning Method to Predict 3D Lung Cancer Target Motion to Account for Adaptation Latency

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Purpose: To develop an intrafraction 3D motion prediction algorithm to compensate for latency of up to one second between motion observation and adaptation in real-time IGRT systems. The AAPM TG264 recommended an end-to-end latency of less than 0.5 seconds for MLC tracking for real-time adaptation of intrafraction motion. This is hard to achieve when current real-time imaging systems has 200-500 ms between images (MRI linac, Varian and Elekta intrafraction fluoroscopy), hence the need for motion prediction. **Methods:** A motion prediction system based on a Long-Short-Term-Memory (LSTM) network topology for signal forecasting was developed. A separate network was trained for each motion axes of Left-Right (LR), Superior-Inferior (SI) and Anterior-Posterior (AP). Intrafraction motion acquired with Calypso from 7 lung cancer was used. Each patient received either 4 for 5 fractions, resulting in 29 fractions and 15 hours of recording in total. The network was trained with two different training configurations: with 2-minutes and 4-minutes of motion for each patient. The LSTM weights were updated when a new target position was observed. The prediction interval values addressed were 0.5s and 1s. The neural network's prediction was compared against the actual motion. **Results:** With 2-minutes training, the prediction error was 0.04 ± 1.13 mm, 0.02 ± 1.18 mm and -0.05 ± 1.28 mm for 0.5s prediction and -0.05 ± 1.12 , 0.04 ± 1.27 mm and -0.05 ± 1.30 mm for 1s prediction in LR, SI and AP, respectively. The error reduced when the training data increases to 4 minutes for both 0.5s and 1s prediction. The root-mean-square-error was <1 mm for all motion axes. The range of motion in the tested patient database was $[-8.3, 11.3]$ mm, $[-34.8, 7.7]$ mm and $[-16.1, 9.3]$ mm in LR, SI and AP. **Conclusion:** A deep learning prediction algorithm was developed and tested for 3D lung tumour motion. Up to 1 second forecasting was achieved with sub-mm accuracy, enabling compensation of latency in motion adaptation systems in real-time.

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SU-F-TRACK 6-05

BEST IN PHYSICS (MULTI-DISCIPLINARY): Real-Time Dose-Optimized Multi-Target MLC Tracking for Locally Advanced Prostate Cancer

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Purpose: The accuracy of radiotherapy for patients with locally advanced cancer is compromised by independent motion of multiple tumors. Multi-target MLC tracking has been developed to adapt to multiple targets in real time. However, geometrically adapting the MLC aperture to target motion by directly translating the planned aperture results in sub-optimal delivered dose. It was hypothesized that optimizing multi-target MLC tracking based on accumulated dose would reduce dosimetric errors compared to geometric-based multi-target tracking for locally advanced prostate cancer. **Methods:** A dose-optimized multi-target tracking algorithm that adapts the MLC aperture to correct for dosimetric error was developed for moving prostate and static lymph node targets. The dose-optimized algorithm accumulated the planned dose to the prostate and lymph node PTVs during treatment and shifted the prostate dose volume in the direction of motion. The adapted MLC apertures were calculated to minimize the difference between the planned and delivered doses accumulated up to that point. This algorithm was evaluated by simulating three locally advanced prostate cancer VMAT plans and three prostate motion traces with a relative interfraction displacement. The performance of dose-optimized multi-target tracking was compared to geometric-based multi-target tracking and no tracking using a 3D gamma comparison between delivered and planned doses with a 2%/2 mm pass criterion. **Results:** The mean gamma-failure rate was lowest for dose-optimized tracking, with $11.2 \pm 3.7\%$ for the prostate and $0.6 \pm 0.4\%$ for the nodes. The gamma-failure rate for geometric-based tracking was $24.4 \pm 7.0\%$ for the prostate and $0.8 \pm 0.8\%$ for the nodes. When no tracking was used, the gamma-failure rate was $34.0 \pm 15.6\%$ for the prostate and $21.7 \pm 1.8\%$ for the nodes. **Conclusion:** This study developed a dose-optimized multi-target tracking method that can correct for dosimetric deviations during treatment for locally advanced prostate cancer. Dose-optimized tracking reduced error compared to geometric-based tracking and no tracking for both the prostate and nodes.

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SU-F-TRACK 6-06

Synthesizing Real-Time In-Treatment 4D Images Based on Optical Surface Signals and Pre-Treatment Images: A Proof-Of-Concept Study

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Purpose: To propose a proof-of-concept solution to real-time virtual visualization of target-in-motion based on in-treatment optical surface signals and pre-treatment images. A deep learning model was developed and validated to map body surface displacement to internal anatomy deformation. **Methods:** Body contours of 4DCT were segmented to simulate optical surface. Free-form deformable-image-registration (DIR) algorithm with isotropic variation regularization was used to register the end-of-exhalation (EOE) phase to other phases. Dimension of deformation vector field was reduced to the first two principal components (PC). A simple neural network composed of convolutional and fully connection layers was trained to predict the two PC scores of each phase from surface displacement with respect to EOE surface. The instant deformation field was reconstructed and used to warp EOE images to obtain real-time CT images. The approach was tested on three schemes: (1) Different respirations of 4D-XCAT digital phantom mimicking pre-treatment (training) and in-treatment motions respectively. (2-3) 4DCT images with annotated landmarks (5 patients) and with physician-delineated tumor contours (5 patients) from public DIR-Lab and TCIA 4D-Lung dataset respectively. The EOE-to-EOI (end-of-inhalation) pairs were excluded in training but used for validation. **Results:** Due to the relatively low complexity of respiration motion, the highest validation accuracy of tumor centroid trajectory ($0.04\text{mm} \pm 0.02\text{mm}$) was observed on XCAT phantom. For the 300 landmarks annotated on EOI images of DIR-Lab, the mean displacements between their predicted and reference positions were 1.05mm, 0.88mm, 1.18mm, 1.72mm and 1.53mm for the 5 patients respectively. The DICE-coefficients between predicted and reference tumor contours at EOI phase were 0.92, 0.84, 0.93, 0.85 and 0.71 respectively for the 5 patients from TCIA 4D-Lung dataset. **Conclusion:** The preliminary experiment displayed close relationship between surface geometry and internal anatomy, demonstrating the potential of using deep learning model to synthesize real-time 4D images out of in-treatment dose-free surface signals.

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SU-F-TRACK 6-07

Evaluating the Clinical Impact and Accuracy of Real-Time KV Imaging in Liver SBRT

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Purpose: In liver SBRT, high dose gradients and tumor motion make it challenging to accurately deliver dose. Real-time tumor tracking using on-board kV imaging can address intrafraction positional errors online. The purpose of this study was to evaluate the impact of real-time kV imaging on the delivery and accuracy of liver SBRT in a retrospective patient cohort. **Methods:** 33 patients treated with 41 courses of liver SBRT on a conventional linac were retrospectively analyzed. Patients were treated in 3-5 fractions with 6-18 Gy/fx, and kV images were acquired every 20° at the start of and during arc-based treatment to visualize the location of implanted fiducial markers as surrogate for tumor position. Corrections were made if markers were >3 mm from the expected reference position. Treatment interruptions and corrections were recorded from each fraction and the average interruption rate was calculated for each patient. Statistical analysis was performed using multiway ANOVA to evaluate interruption rate as a function of gating, compression, number of implanted fiducial markers, treatment time, and PTV volume. The elapsed time of pauses and shifts was recorded as well as the magnitude of shifts. **Results:** Treatment interruptions occurred at a rate of 0.69 per fraction. Gating threshold adjustment and patient re-alignment shifts occurred in 61% and 39% of interruptions, respectively. The interruption rate was significantly higher in patients with longer treatment times ($p < 0.01$). Average time per interruption was 1.9 ± 1.6 minutes. No significant difference was observed in the elapsed time for gating adjustment and re-alignment shifts. Median shifts for patient re-alignment were 1.4 mm (AP), 4.0 mm (SI), and 1.2 mm (LR). The median radial (3D) shift was 5.7 mm. **Conclusion:** Real-time imaging can correct for significant intrafraction positional errors with relatively minimal clinical impact and ensures accurate dose delivery in liver SBRT.

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Imaging Interactive ePoster Discussion TRACK 1

Imaging: AI in Imaging

SU-lePD-TRACK 1-01

Convolutional Neural Network Based Fully Automatic Lesion Localization in Rectal Cancer

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Purpose: To implement deep learning methodology involved in multi-modalities MRI imaging, to fully automatically localize and segment rectal tumors. **Methods:** A total of 54 patients with stage T3 and T4 colorectal cancer were studied. The imaging protocol included T2w, T1w DCE and DWI. All images were reviewed by a single experienced oncologist specialized in rectal MRI. Tumors were outlined as a region of interest (ROI) on the third-phase of DCE. As input to the neural network architecture, we choose image patches with size of 21x21 pixels which locate at the same position on one slice from all frames. Using these patches, a 7 layer Convolutional Neural Network(CNN) was established to test whether the middle of the patch belong to the ROI. The output can be interpreted as the probability that the central pixel in the input image patch is associated with a tumor. Five-fold cross-validation was used to evaluate the performance of the CNN classifier. For the training process, a large number of batches were employed and extracted from different areas inside and outside ROIs. For the validation set, all of the image pixels were tested in the trained network architecture to determine whether each pixel belongs to the ROI areas. **Results:** To evaluate the performance of the segmentation, the Dice Similarity Coefficient (DSC) was used to compare the results of the proposed algorithm and the ROI outlined by reader. Among all 54 patients, the mean of DSC is 0.76. **Conclusion:** Based on the results, CNN can be utilized to develop a rectal cancer segmentation method. This can provide helpful information for rectal cancer patients to assess their risk more accurately to choose an optimal strategy when designing radiation therapy planning.

SU-lePD-TRACK 1-02

Optimized Image Segmentation with An Interpretation Method

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Purpose: A common challenge of deep-learning-based interactive segmentation methods is the need of retraining the original network after user's clicks. The retraining process is computationally expensive, and the labels that the user provides can be easily down-weighted during retraining, leading to inaccurate segmentation results. In this study, we propose a novel interactive method, instead of performing optimization over the weights of a neural network, to optimize the activation in the intermediate layer. Meanwhile, we introduce the gradient-based network interpretation method to guide our optimization. **Methods:** We use two types of operations to optimize the activation, namely Activation Replacement and Activation Reparameterization. We adopt the gradient-based interpretation method to find the spatial regions and feature channels in the activation that are associated with the target category. We replace the activation value in these spatial regions with the feature value corresponding to the category (called Activation Replacement). Besides, we introduce channel-wise scaling and bias to these associated feature channels as auxiliary parameters. The auxiliary parameters are optimized by the backpropagating refinement scheme (called Activation Reparameterization). Finally, we obtain the segmentation mask of the target object by updating the activation and performing the forward inference process again. The performance is evaluated over 120 MRI sequences for liver, kidney, and spleen segmentation. **Results:** We use the deep object selection (DOS) algorithm as the baseline, which is a conventional deep-learning-based method. The performance of the interactive segmentation algorithms is evaluated by the Number of Clicks when the segmentation mask achieves 90% IOU over the ground truth (NoC@90). The average NoC@90 scores are 3.20 and 5.60 for our proposed method and the DOS respectively. **Conclusion:** We propose an interactive segmentation algorithm by performing optimization over the activation in the intermediate layer. Experimental results suggest satisfactory performance achieved using our method, compared with the baseline.

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SU-lePD-TRACK 1-03

Self-Supervised Deep Learning in the Assessment of Enteral Feeding Tube Positioning Based on A Small Dataset

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Purpose: Enteral nutrition (EN) through feeding tubes serves as the primary method of nutritional supplementation for critically ill patients unable to feed themselves. The position of the Nasoenteric feeding tubes is routinely confirmed by plain radiographs after insertion and before the commencement of tube feeds. This work uses a self-supervised training strategy to improve the DCNN performance on an extremely small dataset. **Methods:** A self-supervised pre-training strategy called Comparing to Learn (C2L) was used as the backbone of our EN position DCNN model, which included a batch-level mix-up method for data augmentation and a novel teacher-student network for feature learning. The plain abdominal radiographs of 100 patients were retrospectively retrieved and used in this study. The images were resized to 256x256 and equally split into five folds for five-fold cross-validation. Different windows of pixel values (0-4095) were tested. Nine DCNN architecture were investigated. In total, over 100 models were trained and evaluated. Each model was trained five times. The accuracy and area under the receiver operating characteristic (ROC) curve (AUC) on the test data were used to assess the models. The ground-truth classification for enteric feeding tube positions was performed by a board-certified abdominal radiologist. **Results:** The preliminary results show that, by including a self-supervised pre-training strategy, DL provides an encouraging solution in the binary classification of the appropriate tube positioning vs. wrong positioning based on the extremely small dataset with an accuracy of 0.76 (95% CI 0.75-0.77) and AUC of 0.73 (95% CI 0.72-0.74), which is also an over 7% improvement of accuracy and over 5% improvement of AUC compared with the ImageNet pre-trained weights. Windowing setting (0-2750) helps improve the performance of DL models. **Conclusion:** Self-supervised DL based on a small dataset demonstrates promise in the assessment of appropriate vs inappropriate EN feeding tube placements.

SU-lePD-TRACK 1-04

Wavelet U-Net: Incorporating Wavelet Transform Into U-Net for Liver Segmentation

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Purpose: To incorporate the concept of wavelet transform into the U-net for the organ segmentation. The segmentation of organs in CT images is a crucial part of the overall treatment planning process in radiation oncology. While many automation packages, both atlas- and deep learning-based, are becoming available recently, current clinical practice for both organ-at-risks (OARs) and targets still relies almost entirely on human inputs. Here we incorporated the wavelet's multi-level spatially encoding characteristics into the U-net, and examined the clinical performance of the resultant automated segmentation system. **Methods:** A total of 50 liver cases are anonymized for this study. The liver contours are manually delineated in each case. Forty (40) cases are used for training, and ten (10) are reserved for validation. Dual-tree complex wavelet transform(DTCWT) are used in the deep learning U-net model. DTCWT provides multiresolution and sparse representation properties into the structure of an image. The deep convolution model provides the hierarchical structure to extract information from different feature levels. Similar to traditional U-Nets, we attempt to preserve information flow and details by interchanging maxpooling layers with discrete wavelet transformation and upsampling layers with inverse wavelet transformations. **Results:** The segmentation mask of the liver is generated by feeding CT images into the wavelet U-Net. Evaluation of the wavelet U-Net image segmentation model is done with dice and pixel-wise accuracy. The model achieved a dice score of 0.896 and the pixel-wise accuracy of 0.83% in the validation dataset. This model is found to have achieved comparable results as currently available commercial packages. Further adjustments of the model parameters are in progress and can potentially improve its performance. **Conclusion:** This Wavelet U-Net can currently be used as an automated quality assurance tool for commercial packages. Further development is in progress to extend its capability to segmenting liver targets.

SU-lePD-TRACK 1-05

Multi-Frequency-Based CXR Data Normalization for Deep-Neural-Network Classifier

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Purpose: While deep-neural-network (DNN) classifiers have arisen as promising tools showing state-of-the-art performances for detecting abnormalities in chest x-ray (CXR), one of the challenging problems that hampers the networks' performance is the data heterogeneity originated from a variety of sources such as devices, scan conditions, and image processing algorithms. We investigate the effect of a multi-frequency-based data normalization technique for the DNN-based diagnosis. **Methods:** The reference data set consists of hundreds of normal and abnormal CXR images collected from a single hospital. Each CXR image has been decomposed into various frequency bands by use of the Gaussian-Laplacian pyramids technique. We calculated five levels of Laplacian pyramids. Then, five images at the Laplacian pyramids and the highest level image at the Gaussian pyramids were used for data normalization. We calculated the standard deviation value of each frequency band image. The average value of the Gaussian pyramid image was calculated as well. After that, we set the reference values at each frequency band by averaging calculated values. Each image in the reference database has been normalized iteratively to meet a convergence condition. We trained a DNN classifier with the normalized reference dataset. In the testing phase, we collected another CXR database that has different frequency characteristics. We normalized the test dataset in the same way. **Results:** The inference of the non-reference dataset using the trained network combined with the data normalization resulted in an average precision score about 20% higher than that of the other trained network without data normalization. **Conclusion:** We have developed a multiscale data normalization algorithm to increase the performance of the deep-neural-network classifier. Our study is believed to play an important role in developing many DNN CXR classifiers.

SU-lePD-TRACK 1-06

Autosegmentation of Videofluoroscopic Swallow Studies Using U-Net for Improved Analysis of Radiation Induced Dysphagia in Head and Neck Cancer Patients

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Purpose: Radiation induced dysphagia is a common complication from head and neck cancer (HNC) treatment. Dysphagic patients will experience malnutrition and a large increased risk for aspiration pneumonia, leading to future hospitalizations. Current dysphagia diagnosis and treatment requires extensive manual analysis of a videofluoroscopic swallow study (VFS). Current methods are time consuming and solely qualitative. This work develops an auto segmentation algorithm for assisted analysis of VFS and further implementation of automated quantitative measures. **Methods:** 40 VFS were collected from a varying severity dysphagic cohort. Shielded edges of the images and non-swallowing related motion were cropped. Images were standardized and resized to 256 x 256 pixels. Hyoid, bolus, and C2 – C4 vertebrae contours were done on unprocessed images and underwent the same cropping and resizing. A 2D modified U – Net structure was used. The network was trained using 80% of available image slices (1386,4490, 2881) for the hyoid, bolus, and C2-C4 separately. A non-decaying learning rate and Adam optimizer in TensorFlow was used. **Results:** Dice coefficients on the remaining validation images were 0.87, 0.92, and 0.95 for the hyoid, bolus, and C2-C4, respectively. **Conclusion:** Resulting dice coefficients reflected accurate contours. The C2-C4 contours were most similar given the lack of motion and contrast in that region. With the Hyoid and C2-C4, edge pixels were most uniformly missed around the contour. The bulk of bolus contour was accurate, but small portions (residue) were missed in some cases. Identification of residue is debated with manual analysis leading to required further work for accurate bolus contouring.

UW - Madison Fall Research Competition

SU-lePD-TRACK 1-07

Super-Resolution CT Image Via Convolution Neural Network with An Observer Loss Function

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Purpose: To enhance the CT image resolution, we proposed a convolutional neural network (CNN) based super-resolution (SR) technique. To train SR network, we proposed a perceptual loss function that uses the classifier learned by CT images as a feature extractor (i.e., observer loss). **Methods:** We used patient data authorized by Mayo Clinics for "the 2016 NIH-AAPM-Mayo Clinic Low Dose CT Grand Challenge". We used normal dose CT images of 10 patients. We acquired projection data using Siddon's algorithm in fan-beam CT geometry and applied 1x2 detector binning. High-resolution (HR) images and low-resolution (LR) images were generated by performing back-projection on original sinogram and binned sinogram, respectively. We implemented 13-layer CNN for CT classifier. The classifier was trained to classify lesion-present and lesion-absent. We randomly extracted 64x64 patches from the HR images and added Gaussian signal with various intensity, size, and location. After training, we extracted features from the 10-th convolution layer to calculate perceptual loss. For the SR network, we adapted RED-CNN based architecture, and bicubic interpolation was added at the beginning. We trained SR network to minimize the loss functions (i.e., mean-squared-error (MSE), ImageNet pretrained VGG, and proposed observer loss). We compared the results with different loss functions using peak signal-to-noise ratio (PSNR), structural similarity (SSIM), and visual information fidelity (VIF). **Results:** The quantitative evaluation results are best when using MSE loss but blurring occurs in the result images. On the other hand, when training the SR network using perceptual loss functions, structural details are preserved. In particular, the SR network with the observer loss represents more detailed textures than the network with VGG loss. **Conclusion:** In this work, we proposed observer loss to train SR network for CT images. SR network with proposed observer loss could preserve structural details and textures better than that with other loss functions.

Multi-Disciplinary Interactive ePoster Discussion

TRACK 3

Multi-Disciplinary: Imaging for Treatment Planning

SU-lePD-TRACK 3-01

A Realistic DRR Rendering Method Based on Cycle-GAN

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Purpose: In recent study, deep learning network trained with paired data was utilized to reduce the intensity inconsistency between DRR and x-ray projection. But for lung cancer patients, it is hard to obtain adequate paired data for network training due to respiration. In this work, we developed and validated a DRR rendering method based on Cycle Generative Adversarial Networks (Cycle-GAN), which could improve the similarity between DRR and x-ray projection using unpaired training data. **Methods:** In the proposed method, a Cycle-GAN was trained to generate synthetic x-ray projection from DRR, which was used to resemble real x-ray projection in tumor localization algorithms. Since respiratory motion made it hard to obtain paired data of DRR and x-ray projection, the Cycle-GAN was trained with unpaired data of DRR and x-ray projection from multiple patients. Wasserstein distance and gradient penalty terms were added to the loss function to improve the performance of the network. To apply to the new patient, the synthetic x-ray projection can be generated with DRR fed into the trained network. **Results:** The proposed method was evaluated with four patients' data. Three patients' data was utilized to train the Cycle-GAN, and the last one was used to test the network. The x-ray projections at angle ranging from 0° to 90° were sorted into different phases and DRR of corresponded CT was computed as the input to the network. For this experiment, the relative error between synthetic x-ray projection and realistic ones ranged from 0.04 to 0.11 with an average of 0.087, while the relative error between raw DRR and real x-ray projection ranged from 0.09 to 0.32 with an average of 0.163. **Conclusion:** We proposed a Cycle-GAN based DRR rendering method, which could greatly reduce the intensity difference between DRR and x-ray projection using unpaired training data.

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SU-lePD-TRACK 3-02

Angiogram-Free Intracranial Vessel Localization for Plaque Assessment

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Purpose: Inspection of vessel morphology is important in assessing cardiovascular risks. Magnetic resonance (MR) vessel wall imaging (VWI) is routinely used with a companion MR angiogram (MRA) that localizes the vessel segments of interest. However, the acquisition of MRA prolongs scanning time, extra cost in clinical workflow, and is also subject to slight displacement with respect to the VWI. This study aims to investigate the feasibility to infer the vessel location directly from VWI. **Methods:** The (VWI, MRA) pair of images from 30 patients were included; and were randomly split into 25 cases for training doubled as atlas, and 5 cases for testing. To regulate vessel topology while preserving the target's specific appearance, such as plaque presentation, a novel atlas-based pipeline with deep correction was proposed. For each target VWI, all atlas VWI were first rigidly registered and the one with the highest mutual information was chosen to proceed to be deformable, generating an initial deformation vector field (DVF). An unsupervised error correction network was further introduced to improve the DVF, particularly in alignment and conformality to vessel defined via training MRA. The end-to-end inference applied the refined DVF to the reference MRA for final vessel extraction and localization. **Results:** The 90% Hausdorff distance on the vessel segment of interest is around 10mm, which is satisfactory for localization purposes. Close inspection shows that major vessels such as the internal carotid artery (ICA) agree well with the benchmark MRA, while vertebral arteries and basilar arteries showed inferior local match. **Conclusion:** Our method showed significant improvement, particularly in vessel structure integrity compared to the commonly used deep segmentation or deep synthesis. We are working on regulating the error correction network to better account for DVF heterogeneity and focal vessel performance.

SU-lePD-TRACK 3-03

Intelligent CT Reconstruction with Simultaneous Learning of Reconstruction Policy and Image Quality Assessment Using Quality-Guided Deep Reinforcement Learning

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Purpose: Although iterative CT reconstruction methods can generate high-quality images by solving an optimization problem, it faces two challenges: to adjust regularization parameters to produce desired results and to define a proper image quality metric to guide regularization parameter adjustment. This study aims to employ a quality-guided deep reinforcement learning (QgDRL) framework to solve the two problems simultaneously. **Methods:** A parameter-tuning policy network (PTPN) and a quality assessment network (QAN) are constructed and jointly trained under a QgDRL framework. PTPN is trained to learn how to adjust regularization parameters for input image patches to improve reconstructed image quality via an end-to-end reinforcement learning process using an epsilon-greedy algorithm with experience replay. Intermediate results generated in the reconstruction process are stored to train QAN, which is trained to differentiate high-quality true CT image patches and low-quality generated intermediate image patches, as measured by a probability score for the input patch to be of high quality. The probability in turn serves as a reward function to guide the reinforcement learning process of PTPN training. We implemented the system in a problem of iterative CT reconstruction with pixel-wise total-variation regularization. **Results:** The effectiveness of QAN is demonstrated by being able to assign higher probability scores to patches of visually better quality. The trained PTPN can automatically decide regularization parameter adjustments in the reconstruction process to improve the resulting image quality. Relative errors of images with respect to the ground truth images reconstructed using a fixed regularization parameter, using the best PTPN-adjusted parameters, and using the best manually adjusted parameters are 19.6%, 9.1%, 9.5%, respectively. **Conclusion:** The QgDRL framework can simultaneously and spontaneously learn the policy on regularization parameter adjustment and image quality assessment in the iterative CT reconstruction problem.

SU-lePD-TRACK 3-04

Ortho2D: An Accurate, Memory-Efficient Framework for 3D Vertebrae Labeling

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Purpose: Automatic labeling of vertebrae in CT is a valuable capability for more quantitative diagnostic analysis, interventional planning, and an independent check against wrong-level surgery or radiosurgery. We propose a novel neural network approach (Ortho2D) that operates on 2D orthogonal slices of a 3D image to aggregate, localize, and classify features of interest, thereby avoid memory bottlenecks associated with fully 3D networks and enabling operation on higher resolution datasets. **Methods:** Ortho2D uses two independent Faster R-CNN networks to label vertebrae in sagittal and coronal CT slices. The centroids of 2D detection bounding boxes are clustered in 3D to detect vertebrae centroids, identify the anatomical region (cervical, thoracic, lumbar, or sacral), and identify the individual vertebral levels therein. A post-process sorting method incorporates the confidence in network output to refine classifications and reduce outliers. Ortho2D was evaluated on a publicly available dataset containing 302 normal and pathological spine CT images. The labeling accuracy and memory requirements were assessed in comparison to other methods recently reported in literature. The approach was extended to higher resolution CT data to investigate potential improvements in accuracy. **Results:** Ortho2D achieved vertebrae detection accuracy of 97.1%, region identification accuracy of 94.3%, and individual vertebral level identification accuracy of 91.0%. Ortho2D met or exceeded the performance of previously reported 2D and 3D labeling methods and reduced memory consumption by a factor of ~50 (at 1 mm voxel size) compared to a 3D U-Net. Extension of Ortho2D to higher resolution datasets than normally afforded by fully 3D networks demonstrated a 15% boost in labeling accuracy by virtue of improved memory efficiency and the ability to operate on high-resolution data. **Conclusion:** The Ortho2D method achieved vertebrae labeling performance comparable to other recently reported methods with significant reduction in memory consumption, permitting further performance boosts via application to high-resolution CT.

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SU-lePD-TRACK 3-05

Self-Supervised Learning Based CT Imaging with High Through-Plane Resolution

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Purpose: CT images for radiation therapy planning are usually acquired in thick slice to reduce imaging dose, especially for pediatric patients. However, this practice may degrade the quality of dose calculation and target delineation. In this work, a self-supervised deep learning workflow is proposed to synthesize high through-plane (sagittal and coronal planes) resolution CT images without requiring ground truth images for training, as patient images of thin slices are often not available for many tumor sites. **Methods:** The proposed workflow was designed to facilitate the convolutional neural networks to learn the mapping from low resolution (LR) to high resolution (HR) images in the in-plane (axial) direction. In the workflow, the through-plane LR CT images were sliced along the sagittal and coronal planes and fed through two parallel data flow paths. During the inference step, the through-plane HR images were generated by feeding the respective LR sagittal and coronal images to the trained networks. **Results:** The qualitative results show the capability of the proposed method for generating high through-plane resolution with data of 75 head and neck and 20 lung cancer patients' CT images. The image quality metrics can be all enhanced with the proposed method by from ~7% for structural similarity index measurement (SSIM) to ~130% for edge keeping index (EKI). All the improvements of the measure metrics are confirmed to be statistically significant with paired two-sample t-test analysis ($p=0.014$ for SSIM and $p<0.001$ for EKI). **Conclusion:** The proposed deep learning method for CT image generation with high through-plane resolution is self-supervised, which means it does not rely on ground truth

for network training. In addition, the assumption that the in-plane HR information can supervise the through-plane HR generation is confirmed and anticipated to potentially inspire more researches on this topic to further improve the through-plane resolution of medical images.

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SU-lePD-TRACK 3-06

Squamous Cell Tumor Characterization in Head and Neck Cancer Patients Using Dual-Energy Computed Tomography

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Purpose: To characterize contrast agent concentrations in head and neck squamous cell cancers (SCC) as a function of tumor subsite using contrast enhanced dual-energy computed tomography (CE-DECT) pre-treatment scans. **Methods:** Patients with larynx, hypopharynx and oropharynx SCC that underwent treatment planning with CE-DECT of the neck were studied. Primary and nodal gross tumor volumes (GTVp, GTVn) were contoured and their CE-DECT data were analysed. A two-material decomposition method was performed to extract iodine concentration (IC) maps. The IC distributions were grouped according to GTV (GTVp, GTVn) and tumor subsite (larynx-hypopharynx, oropharynx). The oropharynx and larynx-hypopharynx IC distributions were compared for both GTVp and GTVn subgroups. Both a Student's t-test and a Welch t-test were realized to compare the GTVp and GTVn IC distribution's average 90th percentile, mean, variance, skewness and kurtosis. A Welch t-test was used when the variances of the sample means were unequal. **Results:** 65 patients of which 57 had a GTVp and 30 a GTVn were analysed. 35%/65% of GTVp and 73%/27% of GTVn were in the oropharynx/larynx-hypopharynx subsites. There was a significant difference in the average 90th percentile, mean, variance, skewness and kurtosis of GTVp ICs between the larynx-hypopharynx ($p_0=9.97$, $M=2.00$, $VAR=8.80$, $SKEW=2.15$, $KURT=3.35$) and oropharynx ($p_0=7.85$, $M=1.53$, $VAR=7.78$, $SKEW=2.58$, $KURT=8.20$) subgroups: $p_{p_0}=0.0002$, $p_M=0.007$, $p_{VAR}=0.001$, $p_{SKEW}=0.004$, $p_{KURT}=0.002$. There was no significant difference in the average 90th percentile, mean, variance, skewness and kurtosis of GTVn ICs between the larynx-hypopharynx and oropharynx subgroups. These preliminary results suggest that IC maps have a better clinical resolving power for GTVp than for GTVn. **Conclusion:** This work implies that IC maps could be used as a biomarker to gain further insight into head and neck tumors. This information may be useful when investigating the prediction of locoregional recurrence using tumor IC distributions.

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SU-lePD-TRACK 3-07

Using the Planning CT Scan for Pixel-To-Pixel Corrections of Cherenkov Intensity for Dose Imaging

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Purpose: The Cherenkov light emitted during radiotherapy provides a useful measure of the dose delivered to the patient's superficial tissues, yet variability between patients and treated sites contributes a substantial amount of nonlinearity to the Cherenkov light-to-dose relationship. This work developed a means of correcting for patient tissue properties and highly-attenuating features (localized vasculature, areola etc.) in breast cancer patients using the planning CT. A secondary application to correct for intra-fractional changes from erythema was examined. **Methods:** The average CT# was projected on a spatial mapping of the patient and co-registered to the Cherenkov image. Both 6MV/10MV beams were used in clinical treatment, and Cherenkov cameras mounted on the ceiling recorded emission images from all fractions of $n = 13$ patients. An updated correction model was generated to map the CT/Cherenkov light relationship, which was used to introduce point-to-point corrections of each Cherenkov image. The coefficient of variation (COV), σ/μ was used as a metric of success. **Results:** Variation diminished from 30.3% to 18.8% in 6MV beams, and in 10 MV beams was reduced from 30.5% to 12.6%. The linear regression for uncorrected Cherenkov intensities with respect to dose for 6 MV beams became stronger, from $R^2 = 0.59$ uncorrected, to a corrected value of $R^2 = 0.81$, and in 10 MV beams, uncorrected $R^2 = 0.72$ was corrected to $R^2 = 0.92$, gaining substantial linearity even in highly attenuating regions. It was also shown that the background image, acquired in tandem with the Cherenkov image, could be used to correct for signal loss from erythema. **Conclusion:** A substantial increase in light-to-dose linearity was introduced by calibrating for attenuation from surface-rendered CT scan image data. This was successful for both inter-patient tissue differences and challenging highly attenuating features. Lastly, a new means of correcting for inter-fractional optical changes was examined.

Authors Prof. Brian Pogue, Prof. M Jermyn, Prof. P. Bruza and Dr. Lesley Jarvis M.D. have competing interests with DoseOptics LLC, a company that manufactures Cherenkov cameras used to monitor radiation therapy.

Therapy Interactive ePoster Discussion

TRACK 5

Therapy: FLASH Techniques

SU-lePD-TRACK 5-01

Simultaneous Dose and Dose Rate Optimization with Transmission Beams and Bragg Peaks for Proton FLASH Therapy

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Purpose: With the radiation at ultra-high dose rates, the FLASH therapy (FLASH-RT) can substantially reduce normal tissue toxicities while maintaining tumor response. By far the only commercially available devices that can deliver ultra-high dose rates required for general-purpose clinical FLASH-RT are isochronous cyclotron based proton systems, for which high-energy transmission beams (TB) are needed to achieve ultra-high dose rates for deep-seated tumor targets. However, planning with TB alone does not fully leverage the degrees of freedom for dose shaping as traditional intensity modulated proton therapy (IMPT) via the Bragg peaks (BP) of multi-energy proton beams at the tumor targets. This work aims to develop a simultaneous dose and dose rate optimization (SDDRO) method with the joint use of TB and BP, namely SDDRO-Joint. **Methods:** Different from IMPT that only optimizes the dose distribution, SDDRO enables the dose rate optimization for the purpose of FLASH-RT, in addition to the dose optimization. During SDDRO-Joint, while TB primarily cover the tumor boundary in order to achieve ultra-high dose rate coverage of organs-at-risk (OAR) surrounding tumor targets, BP are placed inside tumor targets to further improve the target dose conformity and spare the normal-tissue dose, i.e., the FLASH sparing by TB and the dose sparing by BP. **Results:** SDDRO-Joint (both TB and BP) was validated in comparison with IMPT-BP (BP only), IMPT-TB (TB only), and SDDRO-TB (TB only). IMPT-BP serves as the gold standard for the dose distribution. SDDRO-TB serves as the gold standard for the dose rate distribution. The results suggest that SDDRO-Joint substantially increased the target dose conformity from SDDRO-TB, while preserving the dose rate coverage. Noticeably SDDRO-Joint had the best conformal index values, e.g., higher than IMPT-BP as well. **Conclusion:** A new treatment planning method SDDRO-Joint is developed to simultaneously optimize dose and

dose rate distribution for FLASH-RT.

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SU-lePD-TRACK 5-02

Differential Dose-Rate Effects on Pancreatic Cell Survival from Proton Therapy: FLASH Versus Conventional Dose Rate

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Purpose: To compare the cell survival rates following proton irradiation at ultra-high FLASH vs. conventional dose rates. **Methods:** A synchrotron-based horizontal proton beamline (87 MeV) at our institution has been commissioned to deliver dose in either FLASH or non-FLASH mode. The dose rate within each beam delivery cycle was calculated based on the measured dose and its pulse width (~100 ms). A high-precision and high-efficiency experimental platform was developed by using a laser positioning system and a robotic arm. MIA PaCa-2 pancreatic cancer cells and HPNE normal pancreatic epithelial cells were seeded in a 96-well plate, then irradiated at the entrance region of the pristine Bragg dose curve. In the step-and-shoot irradiation mode, each well can have its specific dose (2, 4, 6 or 8 Gy in a single fraction) and dose rate (100 or 0.2 Gy/s). Post irradiation, the cells were re-plated into 6-well plates for colony formation. **Results:** For MIA PaCa-2 cells, from 0 to 6 Gy, there was no significant difference in cell surviving fraction (SF) between FLASH and conventional dose rates. However, at 8 Gy, there was a sharp increase of cell survival using the FLASH (SF = 0.02) compared with the non-FLASH (SF = 0.002). For the HPNE human pancreatic ductal epithelial cells, cell survival from the FLASH group is higher than that of the non-FLASH group at most of the dose levels. **Conclusion:** Our preliminary experimental data have shown that for pancreatic tumor cells, a dose threshold may exist for the differential effects in cell killing between FLASH and non-FLASH protons. However, for normal pancreatic cells, proton FLASH could increase their radio-resistance and protect cell survival. More experiments with additional conditions such as various oxygen concentrations are needed to further explore the clinical potential of proton FLASH in improving the therapeutic index.

SU-lePD-TRACK 5-03

Feasibility of SOBP Proton FLASH Beam Delivery Using a Gantry-Mounted Clinical Synchrocyclotron

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Purpose: To demonstrate the feasibility of delivering a spread-out Bragg peak (SOBP) proton beam in ultra-high dose rate (FLASH) using a proton therapy synchrocyclotron as a major step toward implementing an experimental setup for preclinical studies. **Methods:** A Mevion HYPERSCAN synchrocyclotron was adjusted for the experiments. Protons with 230 MeV energy were delivered in pulses with temporal width ranging from 12.5 μ s to 24 μ s spanning from conventional to ultra-high dose rates. A boron carbide absorber and a slab of PMMA with arrays of holes, as the range modulator, were placed in the beam path for creating an SOBP profile. The radiation field was defined by a brass aperture with 11 mm diameter. The number of protons per pulse at various dose rates was determined by a Faraday cup. The response of two cylindrical (IBA CC04 and CC13) and two plane-parallel (IBA PPC05 and PTW Advanced Markus) ionization chambers were characterized. The dose rate was measured using the PPC05 chamber. The integral depth dose (IDD) was measured with a PTW Bragg Peak ionization chamber. The lateral beam profile was measured with EBT-XD radiochromic film. Monte Carlo simulation was performed in TOPAS as the secondary check for the measurements. **Results:** Faraday cup measurement showed 39.9 pC protons per pulse at 24 μ s pulse width. The measured IDD and beam profiles agreed with the simulation. The cylindrical chambers showed very high ion recombination and deemed not suitable for absolute dosimetry at ultra-high dose rates. The average dose rate measured using the PPC05 chamber showed 126 Gy/s at 1 cm depth in the SOBP beam. The beam range and modulation were 2.4 and 2 cm, respectively. **Conclusion:** It is feasible to deliver protons in an SOBP pattern with >100 Gy/s average dose rate in an ~1 cm³ volume using a clinical synchrocyclotron.

Townsend Zwart, Miles Wagner, and Tucker Evans are employees of Mevion Medical Systems.

SU-lePD-TRACK 5-04

Commissioning a Fixed Field FLASH Pencil Beam Scanning Proton Therapy System in Support of the First In-Human Clinical Trial

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Purpose: Ultra-high dose rate radiotherapy, or FLASH radiotherapy (FRT), delivers therapeutic doses at dose rates exceeding 40Gy/s. With small modifications, some modern proton therapy systems are capable of treating deep-seated tumors with FRT. Pre-clinical and translational data shows potential for normal tissue sparing at ultra-high dose rates with no reduction in tumor control. FAST-01, the first in-human proton therapy FLASH clinical trial, began accruing patients in late 2020. **Methods:** For FAST-01, the pencil beam scanning proton therapy system delivers pre-defined FLASH treatment plans using a 250MeV transmission beam at a minimum of 40Gy/s. Commissioning and characterization of the delivery system was performed using, primarily, ion chamber and film. A treatment planning beam model was validated and end-to-end tests performed. The shielding was evaluated for the FRT trial doses and dose rates. A quality assurance program was adapted to include FLASH deliveries. **Results:** The parallel plate ion chamber was validated against a graphite calorimeter, with agreement within 1.0%. The radiochromic film dose-response curve was calibrated in the FLASH fields, cross-calibrated to the parallel plate ion chamber. A treatment planning beam model was created and validated for transmission FLASH fields. The machine trend analysis shows absolute dose variation within 1% and dose rate variation within 5% and always above 40Gy/s threshold. End-to-end tests and shielding evaluations all passed. Patient QA results meet a 90% passing rate at gamma criteria of 3%/3mm. Initial FLASH clinical treatments are performed with little modification to standard clinical workflow. In vivo dosimetry and log file analysis further confirms the accuracy and constancy of the FRT treatment delivery. **Conclusion:** While following a standard commissioning methodology while also adhering to a quality assurance program that includes special consideration with respect to dose rate, the proton therapy system is capable of safe and efficient FLASH radiotherapy.

Funding/support from Varian Medical Systems

SU-lePD-TRACK 5-05

Development and Validation of a FLASH Pencil Beam Scanning Beam Model for Treatment Planning and Verification in First In-Human Trial

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Purpose: State-of-art clinical proton accelerators can readily deliver scanning proton beam at ultra-high dose rate. While conventional PBS beam models were validated for dose accuracy in an SOBP, the purpose of this work is to develop and validate a dose calculation model in support of treatment planning and

verification under single-layer transmission mode. **Methods:** FAST-01, the first in-human clinical trial for proton FLASH, adopted a library of pre-defined 250MeV rectangular plans to cover a variety of field size, ranging from 7.5x7.5cm to 7.5x20cm. The required beam data, including depth dose, in-air spot and absolute dose, are acquired, processed and calculated to create a beam model in Varian Eclipse. Prior to validations, reference dosimetry was established using a calorimeter, and absolute dose was cross-validated using various detectors at both conventional and FLASH dose rate. The beam model was validated for the pre-defined library of plans delivered in the gantry in FLASH mode. Ion chamber measurements at different locations using a parallel plate chamber were compared to the calculation. Radiochromic film was also deployed to validate 2D dose distribution using gamma analysis in mock and real patients. **Results:** The measurement results were compared against dose calculations from FLASH beam model in Eclipse. For the plan library, the overall dose deviation was $0.2\% \pm 1.3\%$ at 1cm and 5cm depths. So far, FAST-01 has accrued four patients. For patient QA, the absolute doses were measured at 5cm using PTW Advanced Markus chamber. Cross-calibrated Gafchromic EBT3 film was used to acquire 2D dose distribution for gamma analysis. For these patients, the average dose difference was $1.8\% \pm 0.3\%$, and 2D gamma index all passed 90% at 3mm/3%. **Conclusion:** The validation results have demonstrated that the beam model is an accurate and indispensable tool for transmission flash planning and dose verification in support of trial.

The research is supported by Varian Medical System

SU-lePD-TRACK 5-06

Online Control of Micro-Beam Synchrotron Radiation Therapy Using Micro-Stripped Diamond Detectors

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Purpose: A diamond-based detector is being developed to solve the limitation of online monitoring in the case of Microbeam Radiation Therapy (MRT), a synchrotron-based technique using a micro-beam matrix of low energy photons (100 keV range) at very high dose rates (kGy/s range). **Methods:** To demonstrate the feasibility of using diamond materials for portal online monitoring in MRT, bulk diamond-based detectors (4.5x4.5 mm² area) have been tested under a single 50x800 μm^2 micro-beam with the same characteristics as MRT photons beam). Five repetitions of 2 s measurements were performed under a 4000 Gy/s photon flux to test the stability and repeatability of our detectors under those extreme conditions, and compare the quantitative response to simulations. As our detector will be used in a portal mode, measurements has been completed using absorbers of PMMA (0 to 47 cm thickness) to study the response of our detector as a function of the dose rate between 1 and 4000 Gy/s. **Results:** The study of our detector response exhibits a maximum difference within a 2 s pulse better than 0.2 % and a maximum difference on pulse mean response of 0.1 %. The reproduction of the measurement after approximately 500 kGy irradiation also shows middle term stability better than 0.3 %. **Conclusion:** This study shows the suitability of the diamond-based detector for the MRT portal beam monitoring. Linearity over a wide range of dose rates, allows us to measure micro-beam-irradiated and non-irradiated zones at the same time. These results lead us to proceed to the next stage. A multi-stripped diamond detector, read out with multi-channel analog charge digitizers, is being tested to perform each micro-beam and inter-beam flux measurement simultaneously. The first results will be presented.

SU-lePD-TRACK 5-07

Novel Plastic Scintillator for Online Dosimetry in Electron FLASH-RT

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Purpose: The accurate delivery of electrons at FLASH-RT dose rates in radiobiological experiments require new dosimeters that are capable of accurately measuring the radiation dose delivered at >0.55 Gy per pulse (>100 Gy/s) in real-time. We investigate the use of the novel Hyperscint RP100 plastic dosimeter as a direct pulse counter. **Methods:** A Varian 21EX clinac was converted to deliver 16 MeV electrons at FLASH-RT (100 Gy/s) dose rates. The linearity and reproducibility of the plastic scintillator at FLASH-RT dose rates was established by comparing with a pinpoint ionization chamber placed under 10 cm of solid water, where it acts as a relative dosimeter due to intense recombination effects. The plastic scintillator detector and to EBT-XD Gafchromic radiochromic film. were calibrated at conventional dose rates (1000 MU/min) and then irradiated at FLASH-RT Dose rate over a range of 3-20 Gy. **Results:** The plastic dosimeter response tracked with relative ion chamber measurements within $\leq 1\%$ over a wide range of delivered doses and dose rates, showing excellent response linearity and reproducibility. Plastic dosimeter dose measurements agreed with Gafchromic EBT-XD film measurements within $\pm 4-6\%$. Most importantly, the plastic scintillator system was able to detect the number of pulses delivered by the linac directly, allowing for instant verification of proper delivery of FLASH-RT beams. **Conclusion:** The plastic dosimeter was able to accurately measure the delivered radiation absorbed dose under characterization and biological experimental conditions, with a higher degree of reliability than conventional radiochromic film. Furthermore, it was shown to directly and accurately measure the number of pulses delivered in real time. This shows potential for use as a real-time in-vivo dosimeter during biological experiments, as well as potential clinical applications.

Francois Theriault-Proulx is the founder and CEO of Medscint, Inc. No financial support from Medscint was provided for this study.

Therapy Interactive ePoster Discussion

TRACK 6

Therapy: MR-guided RT Dosimetry and QA

SU-lePD-TRACK 6-01

Real-Time Reduction in Isocenter Shifts During MR-IGRT Gantry Rotation

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Purpose: To quantify imaging isocenter shifts due to fluctuations in the primary magnetic field (B_0) as a result of gantry rotation in a commercial 0.35T magnetic resonance image guided radiotherapy (MR-IGRT) system and to correct those shifts within the imaging sequence in real time. **Methods:** A hard pulse navigator (FA:90, dwell time = 8 μs , complex points = 64) was added to a sagittal 2D cartesian spoiled gradient echo (GRE) sequence (TR/TE: 20/3.7ms, FA: 20, rBW: 260 Hz/pixel) following each image on a commercial 0.35T MR-IGRT system. The gantry was rotated from 33° to 153° during imaging and measurements were acquired using a spherical 24 cm DSV phantom (T1/T2: 330/260 ms) using the system body coil. The resonant frequency offset was calculated by comparing the phase data from the first (reference) hard pulse navigator with each subsequent navigator. The calculated frequency offset was then fed back into the sequence from the image reconstruction computer as a real-time feedback object. This data was then used to modify the imaging pulse and receiver frequencies for the next image. Phantom images were acquired with and without the correction. Shimming was done at 33° for both image sets. Images were time aligned and the center of the phantom was calculated for each image using MATLAB. The phantom shifts in the frequency encoding direction were then compared. **Results:** RMS isocenter shifts in the frequency encoding direction were reduced using the resonance frequency correction (RMS = 2.70mm/1.14mm). Peak displacement in the frequency encoding direction was additionally reduced from 5.7mm without real time correction to 2.5mm with correction. **Conclusion:** This work provides a method for real-time reduction of imaging isocenter shifts due to B_0 fluctuations during gantry rotation in GRE

sequences. More precise gantry rotation data could yield further reductions.

SU-lePD-TRACK 6-02

Experimental Verification of Scanning Magnet with Distributed Winding Coils for Particle Beam Therapy

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Purpose: Size of scanning system in particle beam therapy can be reduced by using a combined function scanning magnet compared to separated scanning magnets. We propose a combined function scanning magnet with distributed winding coils (DW-SCM). Its homogeneity of the dipole magnetic field and the correction method of the magnetic field are experimentally verified. **Methods:** DW-SCM can generate the rotating dipole magnetic field by the three-phase sinusoidal current. By changing balance of the three-phase current, the good field region (radius) of field homogeneity within $\pm 0.5\%$ can be expanded. We have made a prototype of DW-SCM with 30 mm bore diameter which is one-third the actual size of the actual one. It consists of 12 magnetic poles and 3 coils of hollow conductors. The magnetic field distribution in the scanning direction with respect to the central magnetic field has been measured to evaluate the good field region with and without the correction. **Results:** The measured magnetic field distribution of the prototype DW-SCM within the good field region is in agreement with the simulation results calculated by Poisson Superfish code by 0.01%. The good field region after applying the correction was 15 mm which is 50% of the bore diameter. **Conclusion:** We have made the prototype of DW-SCM and evaluated the good field region. After applying the magnetic field correction, the good field region reaches 50% of the bore diameter.

SU-lePD-TRACK 6-03

Response of Parallel-Plate Ionization Chambers and a Diode Detector on the Buildup Dose Measurement in Inline Magnetic Fields

N Anami*

Purpose: To investigate the response of parallel-plate ionization chambers and a diode detector on the buildup dose measurement in inline magnetic fields using Monte Carlo (MC) simulation. **Methods:** The ROOS, NACP-02 and MARKUS parallel-plate chambers and a PTW-60018 diode detector were modeled using the EGSnrc/cavity code. A 6 MV photon beam from a Varian Novalis Tx head was irradiated with a $10 \times 10 \text{ cm}^2$ field at a source-to-surface distance of 100 cm. The magnetic field strengths of 0, 0.35 and 1.0 T were applied parallel to the beam axis. The dose to water, $D(w)$, and the dose in sensitive volume of detector, D_{det} , were calculated at 0.1–1.5 cm depth in a water phantom. The overall perturbation correction factor, $P(Q,B)$, for the detectors was calculated with/without magnetic fields according to the Spencer–Attix cavity theory. The average restricted mass collision stopping-power ratio of water to air was calculated by the EGSnrc/SPRRZ code. In all dose calculations, photon and electron cutoff energies were set to 0.01 MeV and 0.521 MeV, respectively. The statistical uncertainty of the calculated dose was less than 0.1% in one standard deviation. **Results:** The dose to water at a 0.1 cm depth in inline magnetic fields was higher up to 37.4% than that of $B=0 \text{ T}$. The depth of the dose maximum in all inline magnetic fields was shifted 0.7 cm toward the surface compared to that of $B=0 \text{ T}$. The $P(Q,B)$ values for PTW-60018 were 0.814–0.889 at 1.0 T. Similarly, the $P(Q,B)$ values for ROOS, NACP-02, and MARKUS chambers they were 0.983–1.015, 0.960–1.022, 0.959–1.027 at 1.0 T, respectively, and larger than those of PTW 60018. **Conclusion:** The response of parallel-plate ionization chambers and a diode detector is affected by focusing electrons in the buildup dose measurement under the inline magnetic fields.

SU-lePD-TRACK 6-04

A Monte Carlo Characterization of Positional Accuracy of a Double-Focused and Double-Stack Multileaf Collimator On a 0.35 T MR-Linac

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Purpose: Characterization of a multileaf collimator (MLC) quality assurance (QA) technique for the positional accuracy of a double-focused and double-stacked MLC on a 0.35 T MR-linac using an ionization chamber array (ICA) is presented. As the experimental evaluation of the proposed QA technique may suffer from imperfect positioning of the ICA and gantry sag, Monte Carlo (MC) methods were employed. **Methods:** A previously benchmarked GEANT4 MC model of a 0.35 T MR-linac was used. Measured and simulated beam profiles for $10 \times 10 \text{ cm}^2$ and $24 \times 24 \text{ cm}^2$ field sizes were compared using gamma analysis for further model validation. Simulations were performed for the X1 and X2 banks of the MLC. The leaves of the bank under investigation occluded half of the inline detector array, creating a half-beam block, while the other bank created a 6.64 cm field size. Positional errors ranging from 0.5–2.5 mm along the crossline direction were introduced in 0.5 mm increments to correlate detector response with the MLC positional offset. **Results:** The measured and simulated beam profiles agreed with a 100% pass rate using a gamma criterion of 2%/2mm. A linear relationship was found between the introduced MLC offset and detector response with the coefficient of determination for the X1 and X2 banks being 0.9987 and 0.9991, respectively. The change in detector response with MLC offset was 22.7%/mm and 17.4%/mm for the X1 and X2 banks, respectively, where this difference in sensitivity was determined to be statistically significant ($p < 0.001$). **Conclusion:** The proposed QA method was found to be sensitive to MLC positional offset and has potential to be useful for routine machine QA. Furthermore, differences in sensitivity to MLC offset were found for each MLC bank, which could be the result of the 0.35 T magnetic field influencing the secondary electron trajectories.

SU-lePD-TRACK 6-05

Mechanical Wind-Up Motion Phantom for MR-Linac Daily Quality Assurance

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Purpose: Visualizing organ motion in real-time is a primary advantage of combined MRI and linear accelerator treatment systems. However, the cine sequences used for motion visualization often trade image quality and spatial distortion for temporal resolution. Here we present a simple MR-safe phantom for evaluating daily consistency of such sequences. **Methods:** We designed the components of the phantom in AutoDesk Fusion 360TM. To avoid eddy currents, we utilized plastic for all moving components of the phantom. We made design choices for ease of 3D printing and simple laser cut production. We used glass bearings to reduce friction in the system, and leaded plastic (density of $\sim 4 \text{ g/cc}$) for counterweights and energy storage. **Results:** The phantom is 25cm in height and width, and 15cm in depth. A weighted tray is wound to the top of the phantom, and the stored energy is released via an escapement mechanism attached to an external oscillating arm. Attached to the oscillating arm is a 3cm diameter water-filled ball. The period of oscillation is adjusted via a slidable counterweight. The motion phantom produces an adjustable periodic oscillation in a frequency range of roughly 0.4 to 2Hz. This corresponds to a run time of between 15 to 3 minutes. This range allows enough time to setup the phantom and run a test of the cine sequences. **Conclusion:** It is possible to construct a simple, lightweight, MR-safe mechanical phantom that oscillates with a long enough period to provide a useful motion test for cine MR sequences. The phantom can be built with a combination of 3D printed components, laser-cut plastic, and inexpensive plastic/glass bearings. STL files for all pieces will be uploaded to facilitate others wishing to instantiate the phantom.

SU-lePD-TRACK 6-06

Evaluation of the Dosimetric Accuracy and Angular Dependency of the Novel SRSSMapCheck Design in the Magnetic Field

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Purpose: SRS MapCheck (SRS-MC) developed by the SunNuclear Corporation is a commercial device used for small field patient-specific dosimetry. However, functionality in the presence of a magnetic field for the purposes of MRI-guided radiation therapy QA comes with certain challenges and requires novel engineering solutions, as well as rigorous testing to confirm the improved performance of the new design. **Methods:** Angular dependency of the central axis ion chamber reading was investigated for 2.5x2.5 cm and 5x5 cm fields. An additional set of measurements were taken at both the isocenter and at a 2.5 cm lateral offset for a SRS-MC detector plane parallel and perpendicular to the couch on the MRidian ViewRay Linac, which has constant background magnetic field of 0.35T. Control sets of measurements were taken on a Varian Edge Linac. Measurements were acquired for a traditional SRS-MC design and a proprietary novel engineering design. Several plans of varying complexity were planned in the ViewRay TPS and then measured on the ViewRay Linac, with acquired measurements being compared to the pre-planned and evaluated based on γ -analysis. Absolute dose calibration was performed with readings acquired with A1SL Extradin IC and PTW diamond detector in a solid water phantom. **Results:** Angular dependency observed in the traditional design of the SRS-MC was rectified with the new design. CAX readings matched were similar with and without the magnetic field. A γ -rate passing rate of 97.8% (3%/3mm) was achieved for all plans in relative dose mode. Absolute dose mode also reveals good agreement when properly calibrated for the dose at the StereoPhan center. **Conclusion:** Novel design of the MRI-compatible SRS-MC addresses previously discovered deficiencies and irregularities in the dose measurement introduced by the presence of the magnetic field and ensures the suitability of the SRS-MC as a tool for patient-specific QA in small field dosimetry.

SU-lePD-TRACK 6-07

Evaluation of MRI-Guided Linear Accelerator Based Stereotactic Radiosurgery for Brain Metastasis

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Purpose: Assess the feasibility of an MRI-guided linac system in treating brain metastasis by comparing the plan quality to a frameless automated linac-based SRS system for treatment of brain metastasis. **Methods:** Ten patients with a single metastatic lesion were retrospectively planned using a frameless automated linac-based SRS planning system with HDMLC (2.5mm leaf-width) and an MRI-guided linac system with double-focused double-stacked multileaf collimator (4.15mm leaf-width). Lesion sizes ranged from 0.3cc to 27.1cc, with an average of 5.1cc. Energy for all plans was 6MV FFF and normalization set to cover 99.5% of the target volume with the prescription dose. The HDMLC plans utilized VMAT technique with 4-5 non-coplanar arcs at 1200 MU/min while the MRI-guided linac system used step-and-shoot technique with multiple coplanar beams at 600 MU/min. Plan quality was assessed by the following metrics: gradient index (GI), RTOG conformity index (CI(R)(T)(O)(G)), brain volume receiving 12Gy or more (V12Gy) and maximum dose (D(m)(a)(x)) for brainstem, optic chiasm, and optic nerves. Two-sided Wilcoxon sign-rank test (statistical significance at $p < 0.05$) was used to test differences between metrics. **Results:** All plans met OAR constraints. D(m)(a)(x) to the brainstem, optic chiasm and nerves were not significantly different between the plans. HDMLC-linac plans were statistically better in terms of GI ($p = 0.00$), CI(R)(T)(O)(G) ($p = 0.00$) and V12 ($p = 0.01$). **Conclusion:** Our study demonstrates that MRI-guided linac system can provide comparable organs-at-risk sparing to the HDMLC-linac for treatment of single metastatic lesions, although the HDMLC-linac plans had better GI, CI(R)(T)(O)(G), and V12. The GI for the MRI-guided plans improved with increasing target volumes, achieving acceptable values (< 4.0) for targets greater than 3cc, suggesting it could be used for SRS treatment of single metastasis targets larger than this size.

SAM Professional Symposium

TRACK 1

An Introduction to LGBTQIA+ Issues in Radiation Oncology from the Medical Physicist's Perspective

MO-A-TRACK 1-00

An Introduction to LGBTQIA+ Issues in Radiation Oncology from the Medical Physicist's Perspective

Moderators: Victoria Ainsworth, E. Harwich, MA and K. Sunshine Osterman, NYU Langone Health, New York, NY

MO-A-TRACK 1-01

J. Vadas, Beaumont Health:

Current State of LGBTQIA+ Healthcare (Jessica, She/Her)

MO-A-TRACK 1-02

T. Roth, Washington University School of Medicine:

Intersection of Race and LGBTQIA+ (Toni, She/Her)

MO-A-TRACK 1-03

V. Ainsworth:

Two Sides of the Same Coin: Looking at Inclusion from Patient Facing and Trainee Directions (Victoria, She/They and Sunshine, She/Her)

MO-A-TRACK 1-04

K. Osterman, NYU Langone Health:

Two Sides of the Same Coin: Looking at Inclusion from Patient Facing and Trainee Directions (Victoria, She/They and Sunshine, She/Her)

MO-A-TRACK 1-05

Audience Q&A

The relationship between the LGBTQIA+ community and the healthcare industry has historically been laden with fear and distrust. This persists today with access to care being denied or limited through policies including religious exemptions, discriminatory insurance requirements, and legislation targeting the LGBTQIA+ community. In addition, a lack of high-quality training for medical practitioners and supporting staff often results in a climate that fails to make LGBTQIA+ patients feel safe when they are at their most vulnerable. These barriers to care result in an inability and/or unwillingness to receive routine healthcare screenings, lower the likelihood of completing treatment, and ultimately lead to poorer outcomes among this population. The LGBTQIA+ community is multifaceted and varied; one's socioeconomic status and racial identity are key factors in dictating the level of severity of these health disparities.

While the challenges to reducing inequality can seem formidable, there are ways in which we, in the Medical Physics community, can begin laying the foundation for compassionate care for our LGBTQIA+ patients and provide support for our LGBTQIA+ colleagues. In this session, we aim to provide a brief historical background on this community's relationship to healthcare, tools that can be utilized to enhance both the patient and colleague experience, and examples pertaining to ongoing departmental efforts dedicated to promoting and championing inclusivity.

Learning Objectives:

1. Identify the historical and current healthcare disparities facing the LGBTQIA+ community.
2. Critically explore the intersection of race and LGBTQIA+ identities and its implications.
3. Identify and discuss steps being taken by clinics to enhance the comfort and safety of our LGBTQIA+ patients.
4. Demonstrate ways in which Members can ensure a welcoming and safe environment for LGBTQIA+ trainees and colleagues.

SAM Imaging Educational Course

TRACK 2

MR Protocol Basics: Brain, Spine and Prostate

MO-A-TRACK 2-00

MR Protocol Basics: Brain, Spine and Prostate

Moderators: Ho-Ling Anthony Liu, UT MD Anderson Cancer Center, Houston, TX and Trevor Andrews, Washington University at St Louis, Saint Louis, MO

MO-A-TRACK 2-01

P. Hou, UT MD Anderson Cancer Center:

Brain MRI Protocols

MO-A-TRACK 2-02

T. Andrews, Washington University at St Louis:

Spine MRI Protocols

MO-A-TRACK 2-03

Q. Yuan, UT Southwestern Medical Center:

Prostate MRI Protocols

MO-A-TRACK 2-04

Audience Q & A

Speaker 1 (Ping Hou, PhD) – Brain MRI protocols

Magnetic resonance imaging (MRI) is a painless, noninvasive procedure that produces highly detailed images. Unlike X-rays or CT scans that use radiation to generate an image, MRI relies on hydrogen in the body to create images using a strong static magnetic field, RF waves, and spatially varying gradients. MRI brain scan can detect numerous diseases such as brain tumors and injuries, strokes, seizures, and developmental abnormalities as well as monitor treatment responses. Brain protocols share some common sequences which are tailored differently for unique diseases. This lecture will focus on the protocol of basic brain study, the physics of each pulse sequence, and image parameter optimizations. We will discuss several basic image artifacts and techniques to minimize them as well as introduce some new emerging techniques.

Learning Objectives:

1. To understand composition of common clinical MR brain protocols and their indications.
2. To understand pulse sequence selection, optimization and some basic artifacts.
3. To be able to implement some simple MR protocol for brain study.

Speaker 2 (Trevor J. Andrews, PhD) – Spine MRI protocols

MRI of spine is one of the most commonly clinical MRI studies. It is a powerful test for detecting anatomic abnormalities and assessing severity of pathology. This lecture will discuss several common indications for MRI of the cervical, thoracic, and lumbar spine. In addition several specific scanning techniques will be discussed as well as typical parameter choices which address these indications. Finally, we will address some of the key quality and safety issues physicists can address for MRI of the spine.

Learning Objectives:

1. To be able to list and discuss basic features of several common indications for spine MRI.
2. To be able to list and discuss the most common clinical MRI techniques used for the spine.
3. To be able to list and discuss the key image quality and MR safety issues for MRI of the spine.

Speaker 3 (Qing Yuan, PhD) – Prostate MRI protocols

Clinical prostate MRI protocols have evolved from robust anatomic T2-weighted (T2W) imaging to multiparametric MRI (mpMRI) with additional functional information from diffusion-weighted imaging (DWI) and dynamic contrast-enhanced (DCE) imaging in the management of patients with confirmed or suspected prostate cancer. Prostate Imaging Reporting and Data System (PI-RADS) adopted by medical communities not only serves as a guideline for standardized interpretation and reporting of prostate mpMRI data, but also defines explicitly minimal acceptable imaging acquisition parameters for prostate MRI protocols. This lecture will provide details of implementation and optimization of pulse sequences for prostate mpMRI protocols, and discuss emerging techniques for MRI-guided in-bore prostate biopsy and therapeutic interventions.

Learning Objectives:

1. To understand the pulse sequences in common clinical prostate MRI protocols.
2. To be able to implement and optimize pulse sequences tailored for specific clinical indications.
3. To be familiar with emerging MR techniques in prostate MRI applications.

SAM Imaging Educational Course

TRACK 3

Biophotonics

MO-A-TRACK 3-00

Biophotonics

Moderator: Brian Pogue, Thayer Engineering, Hanover, NH

MO-A-TRACK 3-01

X. Li, John Hopkins:

Microscopic Biophotonics In-Vivo

MO-A-TRACK 3-02

B. Pogue, Thayer Engineering:

Macroscopic Biophotonics In-Vivo

MO-A-TRACK 3-03

Audience Q & A

Microscopic Biophotonics - One major objective of microscopic biophotonics technologies is to enable in vivo, in situ, and noninvasive or minimally invasive optical histology. This presentation will focus on the recent developments of two representative microscopic biophotonics technologies: optical coherence tomography and multiphoton endomicroscopy. These technologies are capable of imaging tissue microanatomy at a resolution from a few microns to the submicron level in real time and without the need for tissue removal, staining or processing. The basic physics principles, engineering challenges and solutions will be discussed. The performance of these imaging technologies for in vivo animal models, ex vivo human tissues, and translational in vivo human subject imaging will be presented.

Macroscopic Biophotonics – The major objective of macroscopic biophotonic technologies is to enable human physiological or pathophysiological study or clinical assessment. The presentation will focus on both research based and clinical tools for imaging tissue, guiding procedures, and measuring tissue function. Examples will be illustrated from endoscopy, surgery and radiation therapy, where innovations in procedure guidance are changing how delivery of care is achieved. More basic instruments that are portable, mobile or even implantable, make use of low cost innovations in light sources and detectors. The economic footprint of macroscopic optics is perhaps the largest single technology sector in medicine today.

Learning Objectives:

1. To learn about the recent developments and translational potential of microscopic biophotonics imaging technologies.
2. To learn about existing and emerging tools in surgical guidance, endoscopy, and radiation therapy as examples of medium to high cost macroscopic optical systems.

SAM Multi-Disciplinary Scientific Symposium

TRACK 4

Frontiers in AI and Its Applications in Medical Physics

MO-AB-TRACK 4-00

Frontiers in AI and Its Applications in Medical Physics

Moderator: Lei Xing, Stanford University, Stanford, CA

MO-AB-TRACK 4-01

R. Zeng, Food & Drug Administration:
Growing Significance of Deep Learning in Modern Medical Imaging

MO-AB-TRACK 4-02

T. Purdie, The Princess Margaret Cancer Centre - UHN:
Emerging Data-Driven Approaches for Treatment Planning

MO-AB-TRACK 4-03

N. Wen, Henry Ford Health System:
Unconventional Neural Networks and Their Applications in Cancer Detection and Decision-Making

MO-AB-TRACK 4-04

L. Xing, Stanford University:
Pitfalls of Current Data-Driven Solutions and Pathways Toward XAI for Medical Physics

Artificial intelligence driven by deep learning have attracted much attention in the several years and is being increasing adopted in medical physics for various applications. The enormous success of deep learning stems from its unique capabilities of extracting essential features from big data and then making inferences. However, the data-driven process has many potential flaws, such as the demand for a large amount of annotated data and lack of interpretability. The purpose of this symposium is to summarize recent advances in deep learning techniques, in particular, unsupervised and semi-supervised deep learning, graphical neural networks, and technical tools that make AI more interpretable and trustworthy. Specific applications of the new generation of deep learning techniques in imaging, treatment planning, dose delivery, and clinical decision-making will be discussed in detail.

Learning Objectives:

1. Highlighting recent advances in deep learning techniques, such as unsupervised learning and interpretable AI.
2. Presenting new applications of these deep learning algorithms in quantitative imaging, image analysis and treatment planning.
3. Providing a comprehensive overview of the roles of the new generation of neural networks in imaging, treatment planning and therapeutic delivery.

PI of Master Research Agreement (MRA) with Varian Medical Systems
Adviser for More Health Inc., Luca Medical Systems, and Huiyihuiyong Med Co.

SAM Therapy Scientific Symposium**TRACK 5*****New Developments in Flash RT*****MO-AB-TRACK 5-00**

New Developments in Flash RT
Moderator: Magdalena Bazalova-Carter, University of Victoria, Victoria, BC

MO-AB-TRACK 5-01

S. Krishnan, MD Anderson Cancer Center:
FLASH RT – Tempering Optimism with Caution

MO-AB-TRACK 5-02

P. Montay-gruel, UC Irvine Health:
Pre-Clinical Investigations on Neurocognitive Sparing with FLASH

MO-AB-TRACK 5-03

M. Kim, University of Pennsylvania:
First Canine Clinical Trial with FLASH RT

MO-AB-TRACK 5-04

C. Desrosiers, University Hospital:
FLASH Using VHEE

MO-AB-TRACK 5-05

A. Mascia, University of Cincinnati Medical Center:
First Human Clinical Trial on Treating Bone Metastases with FLASH

MO-AB-TRACK 5-06**Q&A & Panel Discussion**

Abstract: Ultrahigh dose-rate (FLASH) radiotherapy is on the forefront of radiotherapy with the potential to 'revolutionize cancer treatment'. There is a great interest as well as skepticism about this novel radiotherapy technique. The goal of this symposium is not to re-iterate the most known results to date, but to present the newest developments in terms of preclinical and clinical trials as well as technology developments.

The session will be opened by Sunil Krishnan, MD by discussing clinical challenges of FLASH in his talk entitled FLASH RT – the devil is in the details. Dr. Krishnan will speak about the conflicting data and focus more on the negative results to date. For balance, he will acknowledge that there are compelling positive results. Then he will discuss the recent results from Dresden, Australia, MDACC and MGH. He will wrap up by saying that it is critical to democratize FLASH and make it readily accessible to all so that more groups can perform benchmarking studies with appropriate controls and implore all groups to publish both positive and negative data and minimize publication bias.

Pierre Montay-Gruel, PhD, a radiobiologist, will present on the Benefits of hypo-fractionated FLASH-RT on the brain function. The advantages of single doses of FLASH-RT on the preservation of neurocognitive functions along with brain homeostasis has been intensively described. Recently, his group showed that FLASH-RT delivered with hypo-fractionated regimen was able to delay murine glioblastoma growth. Investigations of the impact of such hypo-fractionated doses to the normal brain are currently being performed, on the neurocognitive aspect but also on the preservation of the blood brain barrier and brain homeostasis.

Michelle Kim, PhD will present a talk entitled Implementation and Design of a Proton FLASH System for the First Canine Clinical Trial. Dr. Kim will talk about the initial experiences with our dedicated proton research room and talk about the efforts on the design and commissioning of the proton FLASH capabilities of our system. Additionally, she will discuss the current ongoing canine clinical trial for osteosarcoma.

Colleen Des Rossiers, PhD will present a talk entitled VHEE – rationale for therapy and FLASH potential. In her presentation, she will summarize the work on very high-energy electrons (VHEE), a candidate modality for clinical FLASH to large deep-seated tumors. The utility of very high energy electrons in the energy range from 150 MeV – 250 MeV (VHEE) in radiation therapy has been explored by many researchers in the last twenty years, but currently there are no commercially available medical accelerators producing electrons in this energy range. The FLASH effect has been most demonstrated with electron beams and data supporting the FLASH effect with photon beams is sparse. The dosimetric benefits of VHEE as compared with photons will be presented with the potential for FLASH therapy using VHEE. Additionally, the current state of the VHEE medical accelerator will be presented.

Anthony Mascia, PhD will conclude the session by presenting the technical and perhaps some clinical challenges of the first FLASH clinical trial with proton beams. Dr. Mascia's presentation will focus mostly on preparation, commissioning and delivery of FAST-01, the first FLASH clinical trial. The narrative is to present the pre-existing scientific literature then show some Cincinnati data that supported the clinical trial. Dr. Mascia will describe their animal irradiation systems that prepared them for commissioning a clinical gantry for a clinical trial. Most of the presentation will focus on the FAST-01: study design, technical challenge, planning, dosimetry, measurements, procedures and summary of data thus far available.

Learning Objectives:

1. To learn about the clinical barriers for translation of FLASH
2. To learn about the most current understanding of FLASH radiobiology
3. To learn about the technology developments potentially enabling clinical FLASH
4. To learn about the ongoing canine and clinical FLASH trials

Multi-Disciplinary Scientific Session

TRACK 6

Image-Guided Surgery and Interventions

MO-A-TRACK 6-01

Cardiac Radioablation: MRI-Guided Tracking of Cardiac-Induced Target Motion

S Lydiard^{1*}, B Pontre², N Hindley¹, B Lowe³, G Sasso⁴, P Keall¹, (1) ACRF Image X Institute, University of Sydney, Sydney, AU, (2) Department of Anatomy and Medical Imaging, University of Auckland, Auckland, NZ (3) Green Lane Cardiovascular Service, Auckland City Hospital, Auckland, NZ (4) Cancer & Blood, Radiation Oncology, Auckland City Hospital, Auckland, NZ

Purpose: Cardiac radioablation (CR) for atrial fibrillation (AF) presents one of the most challenging radiotherapy tracking scenarios: multiple small targets undergo rapid differentional cardio-respiratory motion and are near organs-at-risk. MR-guidance offers a solution for real-time target tracking. This work aims to develop and investigate MRI-guided tracking for cardiac-induced motion of CR targets. **Methods:** A tracking algorithm utilizing template-matching concepts, image manipulation optimized for target visualization and population-based target motion assumptions was developed. Tracking performance was characterized using breath-hold MRI images from six healthy and six AF participants. Multi-slice, ECG-gated, 2D cines were interpolated into 3D images for 18-30 cardiac phases, creating 4D datasets. Ground-truth targets were contoured on every cardiac phase. Retrospective tracking was performed on a single transverse slice of a 4D dataset. 3D geometric and volumetric tracking errors were calculated, defined as the difference between the ground-truth and tracked target centroids and volumes respectively. Single slice, transverse, ungated cines (50-60 images) were acquired at 4Hz in three healthy and three AF participants and used to characterize retrospective tracking performance on images more comparable to MRI-Linac acquisitions. Each image was visually assessed for critical tracking errors, defined as the tracked target overlapping an organ-at-risk. **Results:** The average 3D geometric tracking error was 1.6±1mm and 1.3±1mm for healthy and AF participants respectively. 90% of 3D geometric tracking errors were less than 3mm and 71% were less than 2mm. The average volume difference for all evaluated images and targets was 0.27±3cc and 0.02±2cc in healthy and AF participants respectively. 8.0±5% and 11±10% of the ungated images in healthy and AF participants respectively had critical 2D tracking errors. **Conclusion:** The feasibility of MRI-guided AF CR cardiac-induced target motion tracking has been demonstrated. The promising results warrant further investigation as a possible means to improve CR for AF treatment efficacy.

Funding: This project was funded by the Auckland Academic Health Alliance Collaboration Fund.

Disclosure: The authors would like to thank MIM Maestro for use of a research license .

MO-A-TRACK 6-02

Deep Learning for Near Real-Time Image-Guided Focal Ablation

B Anderson*, B Rigaud, Y Lin, E Lin, G Cazoulat, E Koay, A Jones, B Odisio, K Brock, UT MD Anderson Cancer Center, Houston, TX

Purpose: Harness deep-learning based segmentation and registration capabilities combined with biomechanical model-based registration to train an outcome prediction algorithm for liver focal ablation outcomes. **Methods:** 129 patients with 204 lesions who underwent ablation therapy were retrospectively evaluated, including follow-up status of the ablated lesion. Liver, and disease segmentations were defined on pre-treatment and post-treatment CT images using a clinically validated AI-based algorithm. Lesions were separated based on patient MRN into a training/validation cohort, with 29 locally progressing and 24 non-progressing sites with > 6 month follow-up in the validation cohort. Image features from the pre-treatment CT were extracted via convolutions and fed into fully-connected layers for outcome prediction. Tuned hyper-parameters included dense layers, dense connections, convolutional layers, dropout, transition blocks, number of filters, unique loss functions (cosine/categorical cross entropy), and the images presented. We trained and evaluated the model using both rigid and deformable image registration (DIR), including liver segmentations, and/or disease segmentations based on validation area under the curve (AUC). **Results:** Qualitative assessment of our AI-based segmentation models showed immediate clinical usability or < 10 seconds editing in 96% (48/50) of liver, and 100% (24/24) of disease segmentations. AUC values were maximized when presenting the model with DIR images and the disease contour (AUC=0.75), and when presenting the model with rigid images, liver, and disease contours (AUC=0.74). **Conclusion:** Our fully automated outcome prediction model can provide the clinician with a likelihood of future tumor progression based on intra-procedural imaging in a clinically actionable timeframe (< 1 minute), providing feedback on ablation efficacy.

Funding was provided by the Society of Interventional Radiology Allied Scientist Grant, and the Dr. John J Kopchick Fellowship.

MO-A-TRACK 6-03

Head, Shoulders, Knees, and Toes: Towards Full Body Upright Cone-Beam CT Imaging

T Reynolds*, O Dillon, R O'Brien, ACRF Image X Institute, University of Sydney, Sydney, NSW, AU

Purpose: Limb realignment surgery is increasingly guided by preoperative 3D imaging to develop precise patient specific treatments. Current standard 3D imaging systems such as CT scanners require the patient be imaged supine, complicating surgical decisions attempting to improve limb performance under load. Here, we investigate the possibility of conducting weight bearing 3D imaging on a standard robotic cone-beam CT system, imaging the ankle, knee and hip in a single scan. **Methods:** A Siemens ARTIS pheno robotic cone-beam CT system (Siemens Healthcare) was used to image a synthetic ankle, knee and hip phantom (Sawbones) standing upright with a height of 110cm, mimicking stationary weight bearing conditions. The ARTIS pheno C-arm was manually manoeuvred away from the table

and positioned with the source and detector parallel to the floor. In this position, the C-arm was able to rotate around the phantom through an arc of 135°. The phantom was imaged (7.5 fps, 90 kV) from ankle to hip via a series of eight 135° arcs separated by approximately 17cm upward axial translations. The 3D images were reconstructed using a total variation regularised reconstruction. **Results:** The multi-arc acquisition enabled a total axial coverage of 100 cm, capturing the ankle, knee and hip joints in a single scan. The scan took ~120 seconds to complete, acquiring 1357 projections. The gantry movements were manually controlled via the pilot-joystick, limiting the gantry rotation velocity to 10°/s, elongating the scan time. **Conclusion:** This is the first-time upright 3D imaging of the ankle, knee and hip in a single scan has been implemented on a standard robotic angiography system. It represents the first step towards full body upright 3D imaging on a standard system, as well as 3D imaging of lower body limbs moving (i.e. bending and walking) for high quality preoperative planning and postoperative assessment.

This research was supported by G201166 IPA2 with Siemens Healthineers; ACRF grant G175269; Ricky O'Brien Fellowship CI NSW Fellowship G195559, NHMRC Project Grant G193048; Tess Reynolds USyd Postdoctoral Fellowship Scheme G200793.

MO-A-TRACK 6-04

Mechatronics-Assisted MRI-Guided Focal Laser Ablation of the Prostate: A Phantom Study

E Knull^{1,2}, C Park^{1,2*}, J Bax², D Tessier², A Fenster^{1,2}, (1) Western University, London, ON, CA, (2) Robarts Research Institute, London, ON, CA

Purpose: Magnetic resonance imaging (MRI)-guided focal laser ablation (FLA) is an emerging, minimally invasive treatment for prostate cancer, which involves the therapy and monitoring of localized regions to preserve critical structures and healthy tissues. Accurate needle delivery is pivotal to the therapeutic success of FLA. We present a mechatronic-assisted MRI-guided needle delivery system for prostate FLA allowing for in-bore needle positioning. This work aims to validate its free-space and in-bore needle positioning accuracy, and to demonstrate its utility in prostate phantoms. **Methods:** A compact MRI-conditional mechatronic system was developed with non-ferromagnetic components with actuation controlled by piezoelectric motors and optical encoders. It enables four degrees-of-freedom for transperineal positioning and an adaptable needle-guide. The needle-guide is remotely actuated, and needles are manually inserted for FLA to retain clinician expertise during the workflow. Needle positioning accuracy over the range-of-motion was assessed via free-space and in-bore MRI experiments (GE Discovery MR750 3T MRI) to virtual targets and the needle tip error was quantified. Prostate phantom experiments were performed to a 10.0cm needle insertion depth and needle tip error and needle trajectory error were quantified. Needle tracks were manually segmented on gradient recalled-echo (GRE) 3T-MRI images. **Results:** Free-space and in-bore needle positioning resulted in a mean needle tip error (SD) of 0.80±0.36mm (N=40) and 2.11±1.05mm (N=25), respectively. Preliminary phantom studies show a mean needle tip error and needle trajectory error of 2.61±0.62mm and 0.32±0.62° (N=10), respectively. This error enables an accurate FLA ablation radius of 3.02mm, within a 95% confidence interval and under the intended MRI acquisition parameters. **Conclusion:** We demonstrated the utility of a mechatronics-assisted system for MRI-guided prostate FLA. Preliminary prostate phantom studies show potential for accurate and precise MRI-guided needle delivery to localized focal zones, while maintaining manual needle insertion to improve safety with haptic feedback and optimizing MRI-guided FLA workflow.

This work was supported by the Ontario Institute for Cancer Research (OICR) Imaging Program, the Natural Sciences and Engineering Research Council of Canada (NSERC), and the Canadian Institutes of Health Research (CIHR).

MO-A-TRACK 6-05

An Investigation of Microwave Ablation Equipment Interference with Wireless Patient Monitoring

C Favazza*, A Ferrero, J Felmlee, Mayo Clinic, Rochester, MN

Purpose: To determine the extent of microwave ablation equipment interference with wireless patient monitoring in neighboring clinical spaces and estimate a shielding goal to allow simultaneous operation. **Methods:** Two Covidien Emprint microwave ablation units were clinically oriented in a space for a future procedure room, adjacent to an anesthesia induction bay where anesthetized patient vitals are wirelessly monitored with a Philips Expression MR400. One antenna per generator was inserted into an anthropomorphic water tank phantom at locations that represented worst-case clinical scenario for leakage microwave energy. Patient wireless monitoring devices were operated in the induction bay and interference was assessed with microwave generators set to different power levels. Ambient RF power (2.4–2.5 GHz range) was measured in the induction bay using a Tektronix RSA306B USB Spectrum Analyzer. The following measurements were performed 10 times over a 3-week span: (1)background, (2)MR400 communication, (3)full microwave antenna power (I(max)), (4)microwave antenna power level that preceded loss of vital trace (I(Edge)), and (5)power level that did not cause significant interference (I₀) with MR400. To assess the effect of room shielding, microwave units were operated inside an MRI scan room while wireless patient monitoring devices were operated in the adjacent induction bay. **Results:** Experiments demonstrated that microwave interference could result in complete loss of patient vitals data. Repeated RF power measurements showed: (1) significant overlap between MR400 communication and leakage microwave at I(max), (2) partial overlap at I(Edge) (3) and negligible overlap at I₀. From these results, a shielding design goal of 16dB was calculated. Lastly, MRI scan room shielding attenuated leakage microwave to power levels at or below background and prevented interference with MR400 in the adjacent room. **Conclusion:** Leakage radiation from microwave ablation devices can catastrophically interfere with wireless patient monitoring in nearby rooms. Proper room shielding can prevent such interference.

MO-A-TRACK 6-06

Micro-CT Enables Margin Assessment of Full Tissue Surface in Breast Conserving Surgery Without Impeding Clinical Workflow

B Maloney^{1*}, R Zuurbier², R Diflorio-alexander², S Streeter¹, W Wells², R Barth², K Paulsen^{1,2}, B Pogue^{1,2}, (1) Dartmouth College, Hanover, NH, (2) Dartmouth Hitchcock Medical Center

Purpose: Breast conserving surgery has a high positive margin rate leading to 15% of patients requiring a second surgery. Clinically accepted tools such as standard specimen x-ray have severe limitations. This study aims to show that micro-CT can improve malignancy detection across the entire surface of the resected specimen without significantly changing clinical workflow. **Methods:** Specimens from 33 patients undergoing breast conserving surgery were imaged as a part of a clinical trial at the time of surgery. Micro-CT scans were prospectively analyzed by two radiologists to determine margin status and tumor-to-edge distances for all six anatomic margins as well as the primary tumor diameter with a time-constraint of five minutes to mimic clinical conditions. **Results:** Micro-CT imaging of breast cancer specimens was done in 3-4 minutes and scans were analyzed in under five minutes. Sensitivity, specificity, and accuracy compared to pathology were 28.6-57.1%, 94.2%-90.1%, and 91.9-88.9% respectively. This can be compared to standard specimen x-ray imaging on the same data which gives values of 29%, 98%, and 95%. The primary tumor diameter obtained from the micro-CT scans compared to measurements in Pathology showed a Pearson Correlation Coefficient of 0.61 or 0.43 depending on the radiologist. **Conclusion:** This is the first study comparing micro-CT scans taken at the time of surgery to standard-of-care intraoperative tools. This study was designed to mimic clinical conditions, with imaging and analysis taking less than 10 minutes. This is comparable to the timing of current tools. The results are comparable to standard 2D specimen x-ray but with the ability to determine margin status across the entire tissue surface. There is inter radiologist variation in the analysis of these micro-CT scans. This could be due to a lack of training data for this novel modality.

MO-A-TRACK 6-07

Evaluation of Geometric Distortion in 7T MRI for Frameless Surgical Procedures

K Kirby*, K Welker, E Koons, A Fagan, Mayo Clinic, Rochester, MN

Purpose: To investigate the feasibility of using 7T MRI for image guidance during neurosurgical procedures that utilize skin-adhesive fiducials. These techniques require high spatial accuracy, which in MRI is limited by distortions due to inhomogeneity of the static field, nonlinearity of the applied gradients, susceptibility of tissues, and chemical shift artifacts. These distortion artifacts can affect targeting accuracy and are more pronounced in 7T MRI than at lower field strengths. **Methods:** A custom head-shaped phantom and 3 research subjects were imaged at 3T (Siemens Prisma) and 7T (Siemens Magnetom Terra). 10 skin-adhesive fiducials were placed on the phantom surface and on the exposed skin surface of the subjects, according to standard scanning procedures. Imaging sequences (MPRAGE, T2_SPACE, T2_SPACE_FLAIR) were optimized at 7T to reduce distortions, and all were corrected for gradient nonlinearity using the vendor distortion correction algorithm. The ground truth (GT) positions of the phantom fiducials were established via CT imaging (Siemens Somatom Force). Rigid registration of all image volumes was performed. Residual geometric distortions were quantified in phantoms via comparison with GT locations. Differences in fiducial positions in subjects between 3T and 7T were measured. **Results:** The phantom skin fiducial location error was 1.9 ± 0.6 mm for 7T MPRAGE, 1.3 ± 0.6 mm for T2_SPACE, and 1.6 ± 0.5 mm for T2_SPACE_FLAIR. Average subject fiducial location differences (3T compared to 7T) were 1.9 ± 0.1 mm for MPRAGE, 2.1 ± 0.1 mm for T2_SPACE, and 2.9 ± 0.5 mm for T2_SPACE_FLAIR. **Conclusion:** The data presented herein demonstrate that 7T-optimized imaging sequences can be used for neurosurgical planning applications where errors of up to 2 mm are acceptable, using standard skin-fiducial positioning procedures.

Education Session

TRACK 7

Innovation in Medical Physics Education

MO-AB-TRACK 7-01

A Novel Formalism for Incorporating Professionalism, Interpersonal and Communication Skills into Medical Physics Residency Training

I Vergalasova*, N Yue, K Nie, Rutgers Cancer Institute of New Jersey, New Brunswick, NJ

Purpose: The evolution of value-based healthcare and personalized medicine together with the rise of Medical Physics 3.0 has placed great emphasis on the importance of cultivating our skillset beyond science and analytics, to include professionalism, interpersonal and communication skills. Traditionally, these soft skills have been acquired indirectly via on-the-job training. However, it has become strikingly evident that formalized training of this skillset is a necessity for the success of medical physicists. Thus, a novel framework for incorporating formalized training of these soft skills into medical physics residency programs is proposed. **Methods:** The Accreditation Council for Graduate Medical Education (ACGME) provides a template of 5 milestone levels that can be applied to each of their required six core competencies, two of which are “Professionalism” and “Interpersonal and Communication Skills.” Given the success of the ACGME in training resident physicians over the course of decades, we decided to apply their milestone level template by customizing it to fit within each of the required medical physics residency clinical rotations. The formalism presents a framework for both the training that should be provided to the resident, as well as how to evaluate whether the resident has successfully mastered this training. **Results:** Five milestone levels were developed within each clinical rotation, as per CAMPEP standards. Along with each milestone template are guidelines for baseline assessment, formalized training and continuous evaluation for rotation leader(s) to follow their trainee’s progress attaining each of the milestones. Depending on the length of the rotation, a timeline was recommended for achieving each of the levels within the scope of that rotation. **Conclusion:** We believe the proposed framework is universal enough to be adapted into CAMPEP-accredited residencies nationwide. Although a significant time investment for both training and evaluation is required, the cultivation of successful medical physicists is rewarding beyond measure.

MO-AB-TRACK 7-02

Career Preparation for Graduate Students: Preliminary Evaluation of An Introductory Course on the Profession of Medical Physics

S Leon*, B Schwarz, L Rill, M Arreola, University of Florida, Gainesville, FL

Purpose: Medical physics graduate students often receive an excellent education in academic topics but lack knowledge of what to expect after graduate studies. Students may be unfamiliar with career options available to them; the role of regulatory, accreditation, and certifying bodies; the residency application process; ethical and professional behavior; medical errors; and the role of professional societies. A one-credit course was created to address these topics. **Methods:** A new graduate course, featuring didactic lectures and practical, interactive assignments, was developed to address medical physics career awareness. Students learned about different subspecialties and work environments by holding and presenting interviews with practicing medical physicists across the country; they prepared for residency by writing a CV, personal statement, and participating in mock residency interviews with various program faculty; they engaged in small group discussions to debate real-world ethical issues; they performed a root-cause analysis; and they planned their own “career path”. Improvement in student career awareness and preparation was assessed by course evaluations and pre- and post-class surveys. **Results:** The course has been offered for two years and taken by 20 students. Course evaluations were completed by 15 students, with the educational value of the course rated as 4.3/5 (3.71 in year one, 4.75 in year two). In the second year, a pre-class and post-class survey was instituted to gauge the impact of the course on individual topics. Students reported increased awareness of career options, regulatory agencies, board certification, ethical and professional behavior, and professional societies. They reported a particularly strong increase in feeling prepared for the residency application process, with an initial score of 4.2/10 and a final score of 8.3/10. **Conclusion:** Medical physics students can benefit from a one-credit course dedicated to career preparation. Comments by students in course evaluations were overwhelmingly positive.

MO-AB-TRACK 7-03

Feasibility of Virtual Reality (VR) with Open-Source Software for Interactive Medical Physics Education

S Yoon*, T Li, University of Pennsylvania, Philadelphia, PA

Purpose: To investigate capabilities of an open-source-based VR tool for interactive multi-volume data presentation and medical physics education on a commercially available computer. **Methods:** VR headset and controller system (Samsung Odyssey+) was set-up with 3D-Slicer (v4.10.2) on a PC with modern graphics card (NVIDIA RTX 2080). Six VR scenes were constructed within 3D Slicer based on anonymized radiation therapy (RT) cases, including lung stereotactic body radiotherapy and head and neck proton therapy. Position and 6D motion data from headset and controllers were inputted into 3D-Slicer’s volume renderer to visualize multi-volumetric information in real-time. User could navigate freely within the VR scene using natural head and body movement. Stability and frame-rate (frames-per-second, or fps) of VR displays were qualitatively tested under various stressing conditions (e.g. multiple volume visualization, real-time re-slicing), and subjective experience such as motion sickness was tested on volunteers. **Results:** VR cases with less than 4 volumes simultaneously rendered in real-time had minimal perceptible loss in VR display fps~60. Performance suffered minimally with 4D-CT visualization involving 8 breathing phases (fps>50). Interaction with a single volumetric image decreased performance slightly while interaction with two simultaneously decreased performance significantly (fps<50 vs. <30), especially when real-time volume re-slicing was involved (fps<20). Performance increased significantly when images were rendered only on VR headset and not on the computer (50 vs. 35fps). Turning off shading also improved volume quality and performance substantially, while minimally impacting data visualization. Overall volunteers reported minimal motion sickness, except when frame-rate dropped below 20fps. **Conclusion:** VR scenes based on radiotherapy patient cases were created using open-source software. The ability to navigate and visualize complex 3D multi-volume information with intuitive body movements makes this a valuable tool for medical physics education. Various frame-rate optimization strategies can be employed to improved usability and reduce motion sickness of VR on this platform.

MO-AB-TRACK 7-04

Valued Discourse in Oral Examinations for Medical Physicists

A Cetnar^{1*}, (1) The Ohio State University - James Cancer Hospital, Columbus, OH

Purpose: Oral exams are commonly used as formative assessment for graduate programs and medical specialty certification and claim to assess problem-solving, probe depth of knowledge, recognize safe and competent clinicians, and assess professionalism and ethics. This study explores discourse between examiners and candidates during oral examinations to understand how these exchanges during the exam are related to the outcomes of the candidates in mock oral examinations.

Methods: Mock oral examinations were studied to examine discourse between examiners and candidates. Video recordings of the sessions (N = 53) were collected and transcribed. A mixed-methods design was used to inform a theory of questioning, justification, and the determination of acceptable responses. A grounded theory approach was used to analyze the discourse during these oral board examinations to understand how answers are assessed so that candidates can be more successful in how they present their responses during the examination. The scores of the exam questions, overall score, and examiner comments were provided correlation with the recordings. **Results:** After reviewing the data, examiners overwhelmingly asked questions involving requests for additional details or facts (74%). Candidates spoke more during the examination than the examiners with the average of $65\% \pm 13\%$ of the words being spoken by the candidate and $35\% \pm 13\%$ of the words from the examiner. **Conclusion:** Oral examinations can be a useful tool for evaluation if conducted with suitable assessment conditions. The questions should be well organized and examiners should receive appropriate training in asking valuable questions during the examination to be of greatest benefit to the candidate.

MO-AB-TRACK 7-05

Cooperative Education in a Medical Physics Masters Graduate Degree Program

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Purpose: Graduate medical physics education is often geared towards three career paths: Academic medical physics through post-doctoral research, clinical medical physics via board certification, or some weighted combination of the two. There are, however, many jobs where medical physics expertise may be useful. We developed a cooperative education or "co-op" course within our graduate program to connect employers to second year master's students. The purpose of the co-op was threefold: First, provide additional professional experience in our field; second, expose students to career pathways previously not considered; third, provide a means to financial support during graduate study. **Methods:** The co-op course was first offered in summer 2019. Students were eligible to participate after completing one year of study with a minimum cumulative GPA of 3.0. Three local companies agreed to offer a co-op: A medical physics consulting group, a radiation oncology software company, and an intellectual property law firm. Employers were required to sign a statement of understanding regarding educational expectations and minimum hourly wage. Student were hired directly by employers. Students and employers were asked to complete performance evaluations twice a semester to ensure a positive experience for both participants. Two graduates and two current students have participated in the co-op. **Results:** Survey results from participating students, alumni, and employers have been overwhelmingly positive with students saying they would recommend the co-op to other students and employers saying that having a co-op student was very positive for their company. One firm hired their co-op student full-time after graduation. **Conclusion:** Cooperative education is an innovative program that, in addition to providing much-needed funding to graduate students, yields invaluable professional experience to students that may illuminate career opportunities beyond traditional academic and clinical medical physics. We plan to expand the co-op program to include more local firms in the near future.

MO-AB-TRACK 7-06

Enhancing Brachytherapy Education with Immersive Virtual Reality (VR) Video Technology: Best Practice and Preliminary Outcome

T Li*, N Shah, E Hubley, S Anamalayil, N Taunk, University of Pennsylvania, Philadelphia, PA

Purpose: To develop and evaluate an education platform for brachytherapy using immersive video technologies. **Methods:** Immersive VR video technology uses a single 360-degree camera to record the entire 4-Pi space around the camera. When played back on supported devices, viewers can freely look around using their head movement when viewing on a head-mounted display or mouse when on computers. In this study, we compared cameras with different resolutions and features such as high-dynamic range videos. Different content delivery platforms were also compared. Finally, a gynecological brachytherapy immersive video simulation was delivered using VR headset to 14 medical residents. Enhancement of training outcomes was evaluated based on pre- and post-simulation surveys. **Results:** Consumer-grade VR cameras are preferred over professional VR cameras because of their smaller size and ease of use, which reduces interference with operating-room workflow and the chance of operator-related errors during video acquisition. If surgical light is used during procedure, it is important to use cameras capable of recording high-dynamic-range video. A minimal of 5.7k (5760x2880 pixels) resolution is required for brachytherapy immersive simulation due to the need to visualize small instruments such as applicator/needles, and ultrasound images/videos. 8k (7680x3840 pixels) resolution enhanced the visibility of small instruments and ultrasound but is not currently supported on mobile devices or headsets. VR brachytherapy training is best delivered using VR headset because it creates an immersive experience and reduce distractions. Applying VR simulation to medical residents increased their self-assessed scores in confidence, applicator assembling skills, insertion skills, and comfort with independent procedure from 1.5 to above 3 ($p < 0.001$). **Conclusion:** For creating VR video for brachytherapy education, a small camera with $\geq 5.7k$ resolution and high-dynamic-range video should be used. Preliminary results demonstrated substantially enhanced training outcome when VR simulation is used with high degrees of usability, engagement, and enjoyability.

This study is partially supported by McCabe Foundation and American Brachytherapy Society

Vendor Showcase

TRACK 8

Sun Nuclear

MO-SHOWCASE-TRACK 8-3-01

What's New from Sun Nuclear

J Hamilton^{1*}, (1) Sun Nuclear Corporation, Indianapolis, IN

SAM Professional Symposium

TRACK 1

Who's Culpable? Moving Beyond Blame to Create a Just Culture and Avoid Burnout

MO-B-TRACK 1-00

Who's Culpable? Moving Beyond Blame to Create a Just Culture and Avoid Burnout

Moderators: Jacqueline Faight, St. Jude Children's Research Hospital, Memphis, TN and Erica Kinsey, SCL Health St. Mary's Medical Center, Grand Junction, CO

MO-B-TRACK 1-01

J. Johnson, Kelsey-Seybold Clinic:

Areas of Worklife: Fairness and the Role of a Just Culture in a Healthy Work Environment

MO-B-TRACK 1-02

D. Schofield, AdventHealth Orlando:

The Toll of the Job: Burnout and the Medical Physicist

MO-B-TRACK 1-03

Audience Q&A

When clinics first began to care about patient incidents, punitive action was often taken against the healthcare worker. When safety experts recognized the environment's role, some clinics shifted to a blame free culture. We currently know however, that the Goldilocks "just right" clinical environment is one with a just culture. An organization with a just culture uses systems methodology when investigating an error or incident and evaluates the role of both the involved healthcare worker as well as the organization. Instead of focusing solely on the "sharp end", or the employee who made the mistake, a just culture looks to identify systematic factors that led to the error. The organization is responsible for a work environment that is fair to its staff while still endorsing personal accountability. Often, an organization with a genuine just culture environment also has a healthy work environment.

Burnout has recently been recognized by the World Health Organization as an occupational phenomenon that can manifest itself in a number of different ways. One of the current theories suggests that the probability of burnout increases as the mismatch between the individual's own value system and that of their organization diverges. One example of this potential mismatch is an employee's expectation of a just and fair work environment while the organization operates from a punitive stance. Based on data from a recent survey on burnout in medical physicists, the prevalence of burnout in our profession, and correlates to organizational features will be presented.

This session will provide an overview of just culture and explore ways to encourage a fair work environment in your workplace. An overview of burnout, its relationship with your work environment, and the potential implications of burnout will also be discussed.

Learning Objectives:

1. Understand just culture and ways to encourage a fair environment in your workplace.
2. Review the signs and potential causes of burnout.
3. Explore the prevalence of burnout in our profession and understand the potential implications of burnout on yourself and others.

SAM Imaging Educational Course

TRACK 2

Session in Memory of Edward "Ed" F. Jackson

MO-B-TRACK 2-00

Session in Memory of Edward "Ed" F. Jackson

Moderator: Kristy Brock, UT MD Anderson Cancer Center, Houston, TX

MO-B-TRACK 2-01

J. Hazle, UT MD Anderson Cancer Center:

Introduction

MO-B-TRACK 2-02

R. Golden, UW School of Medicine and Public Health:

Introduction

MO-B-TRACK 2-03

R. Stafford, UT MD Anderson Cancer Center:

The Expanding Role of the Physicist in MRI

MO-B-TRACK 2-04

P. Kinahan, University of Washington:

Quantitative imaging: The Translation from Research Tool to Clinical Practice

MO-B-TRACK 2-05

J. Prisciandaro, University of Michigan:

Innovations in Medical Physics Education

MO-B-TRACK 2-06

Q&A

Dr. Edward F Jackson was a gentleman and a scholar. Ed's passing left a hole in our world and our hearts. His contributions to the field of medical physics, the science of magnetic resonance imaging (MRI) and quantitative imaging, MD Anderson Cancer Center, the University of Wisconsin, the AAPM and education were all impactful. In this symposium, we will attempt to convey the impact Ed had on these areas and on so many colleagues in medical physics around the world.

The landscape of MRI has changed substantially over the last couple of decades. MRI is an extremely versatile, non-ionizing modality with multiple soft-tissue contrast mechanisms for visualizing and characterizing anatomy, physiological function and metabolism. Constant technological advances are resulting in higher-performing and more reliable systems increasing the footprint of MRI in diagnostic and oncologic imaging. Additionally, MRI has also experienced rapid growth outside traditional diagnostic radiology departments and into hybrid interventional, intraoperative and radiation oncology suites. With these clinical and technical shifts, the role of the physicist in MRI has expanded to better meet the quality and safety needs of patients and practitioners for a rich variety of protocols and environments.

During the past few decades, remarkable advances in medical imaging technology have made it possible to obtain many anatomic, functional, metabolic, and physiological measurements from clinical images. With appropriate calibration, most of these imaging technologies can provide quantifiable information on properties of the materials from which the imaging signal emanated. We can combine quantitative imaging with the concept of objective biomarkers to define quantitative imaging biomarkers (QIBs), giving us objective measurements from clinical medical images. We will discuss the development and application of QIBs in clinical practice and clinical trials.

The experiences we have encountered over the last year as a result of COVID-19 have taught us the importance of adaptability. As educators, we were required to make radical changes in our approach to didactics and clinical training. However, adaptability is a hallmark of our continually evolving field. In this presentation will we discuss innovative educational approaches, as well as ways to educate, train, and prepare students and trainees for the future of our profession.

Learning Objectives:

1. Describe recent advances of MRI for diagnostic and oncologic imaging
2. Understand the development and application of QIBs in clinical practice and clinical trials
3. Describe methods to educate, train, and prepare students and trainees for the future of our profession

SAM Imaging Scientific Symposium

TRACK 3

Biophotonics

MO-B-TRACK 3-00

Biophotonics

Moderator: Brian Pogue, Thayer Engineering, Hanover, NH

MO-B-TRACK 3-01

T. Zhu, University of Pennsylvania:

Biophotonics in Radiation and Photodynamic Therapy

MO-B-TRACK 3-02

P. La Riviere:

Computational Imaging: from X-ray Tomography to Biophotonics

MO-B-TRACK 3-03

B. Shabestari, NIH NIBIB:

Biophotonics at the NIBIB

MO-B-TRACK 3-04

Q&A

Biophotonics in Radiation and Photodynamic Therapy - Therapeutic delivery can be enhanced by the synergy between non-ionizing and ionizing photons, where light-radiation interactions can inform about dose or amplify the clinical therapeutic value. In radiation therapy (RT), emerging trends include optical surface imaging, 3D Gel and water tank imaging, Scintillator detectors, and utilization of the interaction by scintillation or Cherenkov light to serve as molecular intermediates for imaging. In photodynamic therapy (PDT), emerging trends include direct measurements of light fluence, photosensitizer (PS) concentration, tissue oxygen concentration, blood flow, and their combinations leading to reactive oxygen species explicit dosimetry.

Computational Imaging: from X-ray Tomography to Biophotonics - Computational solution of inverse problems plays a critical role in a wide range of biomedical imaging modalities, ranging from human scale computed tomography to microscopic fluorescence tomography. In this overview, we will review common themes found across all these scales and give examples of algorithmic cross-fertilization between modalities. We will draw on examples from computed tomography, light-sheet microscopy, and hybrid x-ray/optical modalities.

Biophotonics Research Programs and Funding Opportunities at NIBIB - The NIBIB mission is to improve health by leading the development and accelerating the application of biomedical technologies. The Institute is committed to integrating the physical and engineering sciences with the life sciences to advance basic biomedical research and medical care. This presentation will provide an overview of the scientific programs and funding opportunities supported by NIBIB, highlighting the program priorities with the areas of particular importance to the field of Optical Imaging and Spectroscopy. Selected research awards with their potential clinical impacts and the National Centers for Biomedical Imaging and Bioengineering (NCBIB) in biophotonic will be highlighted.

Learning Objectives:

1. To learn common examples of applications of biophotonics in Radiotherapy and Photodynamic therapy.
2. To learn about the role of computational imaging and inverse problems in biomedical imaging and biophotonics.
3. To learn about existing funding opportunities and priorities at the National Institute of Biomedical Imaging and Bioengineering.

Multi-Disciplinary Scientific Session

TRACK 6

Quantitative Imaging

MO-B-TRACK 6-01

A Quantitative Analysis of Lung Elastography Performance Using Large-Deformation CT Scans Acquired at Residual Volume (RV) and Total Lung Capacity (TLC)

B Stiehl^{1*}, M Lauria¹, L Naumann¹, K Singhrao¹, I Barjaktarevic², J Goldin², M McNitt-Gray³, D Low¹, A Santhanam¹, (1) Department of Radiation Oncology, University of California, Los Angeles, Los Angeles, CA, (2) Division of Pulmonary and Critical Care, University of California, Los Angeles, Los Angeles, CA, (3) Department of Radiology, University of California, Los Angeles, Los Angeles, CA

Purpose: Lung elastography during normal breathing has proven to be a useful tool in characterizing regional lung function and has been shown to be an image-based biomarker for COPD patients. However, residual volume (RV) and total lung capacity (TLC) scans representative of deep breathing are often acquired for COPD patients. In this study, we systematically analyze the feasibility of performing lung elastography using large deformation breath-hold CT images acquired at RV and TLC. **Methods:** A quantitative analysis of biomechanical model performance for elasticity estimation was performed using 10 patient datasets with CT scans acquired at RV and TLC, retrospectively. Deformable image registration (DIR) was performed between RV and TLC image pairs with a published registration technique (pTVreg) to obtain ground-truth deformation vector fields (DVF). For each DVF, a well-validated inverse elasticity approach, which employs a lung biomechanical model, was performed to calculate elasticity estimations for each voxel. For validation of elastography performance, the percentage of voxels converging successfully in accordance with the ground-truth DVFs was calculated with convergence criteria of 5% and 10% of maximum observed deformation. Jacobian distributions were calculated for ground-truth DVF and displacement vectors produced by the biomechanical model during elasticity estimation and evaluated for convergence. **Results:** The mean convergence between ground-truth DVF values and displacement vectors output from the biomechanical model within

5% and 10% of the maximum deformation value were measured at 67.31% and 87.62%, respectively, with mean displacement error of 2.21 mm. Jacobian convergence within 0.3 (-0.3 mm³) of the ground-truth registration value was observed to be 80.15%. **Conclusion:** A high level of convergence demonstrates the feasibility of performing lung elastography using RV and TLC CT images. The ability to acquire elasticity estimation distributions for large deformation images allows for characterization of regional biomechanical phenomena observed in patients with lung diseases such as COPD.

MO-B-TRACK 6-02

Localized GBM Recurrence Prediction on Post-Operative Multi-Modal MRIs Through Stem Cell Niches Proximity Estimation Coupled SVM

Y Lao^{1*}, V Yu², A Pham³, T Wang³, D Ruan¹, E Chang³, K Sheng¹, W Yang³, (1) UCLA School of Medicine, Los Angeles, CA, (2) Memorial Sloan Kettering Cancer Center, New York, NY, (3) University of Southern California, Los Angeles, CA

Purpose: To provide early and localized GBM recurrence prediction, we introduce a novel post-surgery multi-parametric MR based support vector machine (SVM) method coupling with stem cell niches (SCN) proximity estimation. **Methods:** This study utilized post-surgery MRI scans ~2 months before clinically diagnosed recurrence from 58 GBM patients. The main prediction pipeline consists of a proximity-based calculation step, to identify regions with high risks of recurrence, and an SVM classifier training step, to further leverage voxel-wise prediction in the high-risk regions (HRR). The HRRs were estimated using the weighted sum of inverse distances to two possible origins of recurrence - SCN (PS) and tumor cavity (PT). Subsequently, multi-parametric voxels (from T1, T1 GD, FLAIR, T2, ADC MR) within the HRR were grouped into the recurrent subregions (warped from the time of clinical diagnosis) and non-recurrent subregions, and feed into a radial basis function SVM classifier. 49 patients were used for training and validation, and 9 patients were used for testing. Besides SVMPE, an SVM classifier using tumor margin as input high-risk regions (SVMTM) was also trained for comparison. **Results:** On pre-recurrence MRIs from 9 testing patients, the SCN proximity based HRRs achieved a maximized inclusion of recurrence, with a recall of 0.97. The fine-tuned voxel-wise prediction from the SVMPE classifier reached a recall of 0.84, a precision of 0.74, and an F1-score of 0.77. In contrast, performance from SVMTM was consistently lower (recall 0.66, precision 0.59, and F1 0.60). **Conclusion:** We introduced a novel voxel-wise early prediction method for brain GBM recurrence based on clinical follow-up MR scans using an SCN proximity estimation coupled SVM framework-SVMPE. The result shows promise in localizing subclinical traces of recurrence 2-month ahead of clinical diagnosis and may be used to guide early retreatment to a smaller target volume and individualized prescription.

MO-B-TRACK 6-03

Predicting PD-L1 Expression Level in Non-Small Cell Lung Cancer on Computed tomography Using Machine Learning

T Shiinoki^{*}, K Fujimoto, Y Kawazoe, Y Yuasa, M Kajima, Y Manabe, T Hirano, K Matsunaga, H Tanaka, Yamaguchi University

Purpose: Immune checkpoint therapy has been the research hotspot in clinical practice since new cancer therapy was discovered. Immune inhibitors which can hinder the combination between programmed death 1 (PD-1) and programmed death ligand 1 (PD-L1) may extend patients' progression-free survival or overall survival with lower side effects. The purpose of this study is to construct the model to predict the PD-L1 expression level in Non-small cell lung cancer (NSCLC) using CT images and machine learning. **Methods:** 203 patients who underwent CT scan were enrolled in this study. PD-L1 expression level was tested through the immunohistochemistry (IHC). Patients' data were divided into two parts (training and test data) by random selection during feature selection procedure. For each Tumor, Tumor-5 mm and Ring structures, 1130 radiomics features were calculated. Then, these features and 7 clinical features (age, sex, stage, histology, location, smoking, CEA) were used to construct the prediction model. Feature selections with Null importance and Lasso algorithms were performed. The prediction models were constructed using three machine learning algorithms (LightGBM, SVM: Support vector machine and LR: Logistic regression) with five-fold cross validation. For training and test data, AUCs were calculated to evaluate the accuracy of prediction model for each algorithm. **Results:** For training data of PD-L1 $\geq$ 50%, prediction models for LightGBM, SVM and LR resulted in an AUC of 0.95, 0.50, and 0.71, respectively. For test data, AUC were 0.79, 0.63, and 0.67, respectively. For training data of PD-L1 $\geq$ 1%, those resulted in AUC of 0.98, 0.69, and 0.68, respectively. For test data of PD-L1 $\geq$ 1%, AUC were 0.76, 0.59, and 0.58, respectively. **Conclusion:** The radiomic-based predictive approach may predict PD-L1 expression status in NSCLC patients relatively accurate. It may be helpful in guiding immunotherapy in clinical practice and deserves further analysis.

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MO-B-TRACK 6-04

Principal Component Analysis of Quantitative Computed Tomography Features and Visual Emphysema Scores: Association with Lung Function Decline

M Koo^{1*}, W Tan², J Hogg², J Bourbeau³, C Hague², J Leipsic², and M Kirby^{1,2} for the CanCOLD Collaborative Research Group and The Canadian Respiratory Research Network, (1) Ryerson University, Toronto, ON, CA, (2) Centre for Heart Lung Innovation, St. Paul's Hospital, Vancouver, BC, CA, (3) McGill University Health Centre Research Institute, McGill University, Montreal, QC, CA

Purpose: Quantitative computed tomography (QCT) provides fully-automated measurements reflecting disease features in chronic obstructive pulmonary disease (COPD), yet visual scoring of emphysema remains the clinical standard and outperforms QCT for prediction of longitudinal outcomes, such as mortality. However, previous studies largely focus on only single QCT features. Our objective is to determine if multiple QCT features in combination are associated with COPD disease progression, independent of visual scoring. **Methods:** Participants with and without COPD from the Canadian-Cohort-Obstructive-Lung-Disease (CanCOLD) study were evaluated. CT images were visually scored by a radiologist for emphysema severity. 14 QCT features were extracted (VIDA software), including: low-attenuation-area<950 HU (LAA₉₅₀), wall-area-percent, total-airway-count, low-attenuation-cluster slope, disease-probability-map measures of emphysema and small airway disease, and vessel-volume. Principal component analysis (PCA) was performed for dimension reduction. Multiple linear regression models for annualized change in forced-expiratory-volume-in-one-second (Δ FEV₁) were constructed with linear combinations of features from each component along with visual emphysema score (VES), adjusted by confounding variables. **Results:** We evaluated 588 participants; never-smoker: n=145, ever-smoker: n=161; mild COPD: n=171; moderate-severe COPD: n=111. Three PCA components were extracted that explained 68% of the variance in Δ FEV₁; these components reflected airway, emphysema/vessels, and emphysema clustering/air trapping measurements. In a multiple linear regression model for Δ FEV₁, VES was significant (standardized estimate (SE)=-0.098, p=0.04). In a model for Δ FEV₁ with LAA₉₅₀ added, VES remained significant (SE=-0.094, p=0.04) but LAA₉₅₀ was not. In a model with the three PCA components added, the emphysema clustering/air trapping component was significant (SE=-0.233, p<0.0001), while VES and all other components were not. **Conclusion:** Linear combinations of QCT features increased the relative importance of QCT compared to visual scoring in terms of the association with FEV₁ change. These findings will motivate future studies to develop and investigate QCT measurements in COPD.

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MO-B-TRACK 6-05

Quantitative Relaxometry for Ultra-Hypofractionated MR-Guided Radiotherapy to the Prostate and DIL: A Feasibility Study

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Purpose: Treatment of the prostate with an integrated boost to the dominant intra-prostatic lesion (DIL) using ultra-hypofractionated radiotherapy has been

implemented in a high-field MR-Linac. We study the feasibility of using quantitative relaxometry for response assessment during the course of this treatment regimen. **Methods:** Quantitative maps of the longitudinal (T1) and transverse (T2) relaxation times were acquired on the Elekta Unity MR-Linac using the variable flip-angle (VFA) and multi-echo spin-echo (MESE) method. Imaging parameters for the VFA method were: TR/TE=20/2.3ms, FA={4°,22°}, FOV=320x320x200mm, reconstructed voxel=0.7x0.7x3mm, acquisition time=1.5mins. For MESE: TR/TE/ΔTE=4000/22/11ms, 8 echoes, FOV=320x320x100mm, reconstructed voxel=0.5x0.5x3mm, acquisition time=5.5mins. The bias and uncertainty in the quantitative maps were first determined using a NIST-traceable phantom with known relaxation values. T1/T2 was measured at each fraction in eight patients undergoing ultra-hypofractionated therapy to the prostate with an integrated boost to the DIL. The prescription was 800cGy/900cGy x 5 to the prostate/DIL, respectively. The VFA, MESE, online-planning images, structure set, and dose were imported in CERR (Computational Environment for Radiotherapy Research) where the images from each relaxometry acquisition were rigidly registered to the online-planning dataset. T1 and T2 values were analyzed within the GTV and CTV as a function of fraction number. **Results:** In the NIST phantom, the measured bias and coefficient-of-variation for T1=[100ms–2600ms] was [-1.1%–12.3%] and [2.8%–9.6%] respectively. For T2=[30ms–1200ms], the measured bias and coefficient-of-variation was [-13.4%–5.1%] and [3.1%–16.5%] respectively. Across all patients and fractions, T1(CTV)=1775±102ms, T1(DIL)=1525±142ms, T2(CTV)=124±19ms, T2(DIL)=104±11ms. T1 and T2 were significantly longer in the CTV. Linear regression analysis showed that in the DIL there is a significant increase in T1 and a significant decrease in T2. This trend was also observed in the CTV but was not significant. **Conclusion:** Quantitative relaxometry may provide useful and complementary guidance as an imaging biomarker for therapy response assessment.

MO-B-TRACK 6-06

BEST IN PHYSICS (MULTI-DISCIPLINARY): Treatment Response Prediction Using Extracellular Volume Fraction Derived from Dual-Source DECT for Chemoradiation Therapy of Pancreatic Cancer

G Noid^{1*}, G Godfrey¹, E Paulson¹, W Hall¹, B Erickson¹, J Shah², X Li¹, (1) Medical College of Wisconsin, Milwaukee, WI, (2) Siemens Healthineers, Durham, NC

Purpose: Dual-source dual-energy CT (DECT) data can be used to calculate the extracellular volume fraction (ECVf) in tumors, which has been correlated with treatment outcome. In this work, we seek to find a correlation between ECVf and change in Carbohydrate Antigen (CA) 19-9 as a surrogate for treatment response for chemoradiation therapy (CRT) of pancreatic cancer. **Methods:** DECT data at the late arterial contrast phase acquired in the standard simulation on a dual-source DECT simulator for 22 pancreatic cancer patients, along with their CA19-9 and hematocrit data, were analyzed. Each patient underwent CRT with a prescription of 50.4Gy in 28 fractions. The patients were chosen based on the presence of a solid tumor in the pancreas which could be clearly delineated. A region of interest (ROI) was placed in the tumor and in the aorta. From the ratio of the Iodine density calculated from the DECT in the ROI and the hematocrit taken at the time of simulation, the ECVf was calculated. The ECVf was then compared to the change in CA19-9 before and after the CRT. Distant metastases as the cause of CA 19-9 elevation were ruled out on subsequent restaging studies prior to surgery. The DECT derived iodine ratio was validated using a phantom study. **Results:** The average hematocrit, ECVf, and change in CA19-9 during the treatment for the 22 patients was 35.6±5.4%, 7.3±4.9% and -4.6±21.8 respectively. We found a linear correlation between the ECV fraction and the change in CA19-9 with an R² of 0.7: ΔCA19-9 = 3.63×ECVf – 31.1. The correlation was statistically significant (P=0.006). The DECT derived iodine concentration agreed with the phantom measurements (R²=1.0). **Conclusion:** The pre-treatment ECV fraction obtained based on Iodine maps from dual-source DECT may be used to predict treatment response for adjuvant chemoradiation therapy of pancreatic cancer.

Research partially supported by Siemens

MO-B-TRACK 6-07

Use of Short Duration Dynamic [¹⁸F]DCFPyL PET and CT Perfusion Imaging to Localize Dominant Intraprostatic Lesions in Prostate Cancer: Validation Against Digital Histopathology

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Purpose: Identifying the dominant prostate cancer nodule (dominant intraprostatic lesion, DIL) may enable more accurate diagnosis and therapy through more precise targeting of biopsy, radiotherapy, and focal ablative therapies. Our goal is to validate the performance of [¹⁸F]DCFPyL positron-emission tomography (PET) and CT Perfusion (CTP) for localizing DIL against digital histopathology images. **Methods:** Multi-modality image sets: in-vivo T2-weighted (T2w)-magnetic resonance imaging (MRI), 22-min dynamic [¹⁸F]DCFPyL PET/CT, 2-hr post-injection PET/MR, CTP were acquired in patients prior to radical prostatectomy. The explanted gland with implanted fiducial markers was imaged with T2w-MRI. All PET, CT, and MR images were co-registered to the pathologist-annotated whole-mount mid-gland prostatectomy histology using fiducial markers and anatomical landmarks. Regions of interest encompassing DIL and non-DIL tissue were drawn on the digital histopathology images and superimposed on PET and CTP parametric maps. Logistic regression with backward elimination of parameters was used to select the most sensitive parameter set to distinguish DIL from non-DIL voxels. Leave-one-patient-out cross-validation was performed to determine diagnostic performance. **Results:** 49,256 voxels from [¹⁸F]DCFPyL PET and CTP parametric maps of 15 patients were analyzed. 2-hr post-injection [¹⁸F]DCFPyL SUV (SUV_{late}) and a model combining K_i (net uptake rate constant) and k₄ (dissociation rate constant) of [¹⁸F]DCFPyL achieved the most accurate performance distinguishing DIL from non-DIL voxel. Both detection models achieved an AUC of 0.90 and an error rate of <10% from leave-one-patient-out cross-validation. Compared to digital histopathology, the detected DILs had a mean dice similarity coefficient of 0.76 for the K_i and k₄ model and 0.73 for SUV_{late}. **Conclusion:** We have validated using co-registered digital histopathology images that parameters from kinetic analysis of 22-min dynamic [¹⁸F]DCFPyL PET can accurately localize DIL in prostate cancer for targeting of biopsy, radiotherapy, and focal ablative therapies. Short duration dynamic [¹⁸F]DCFPyL PET was not inferior to SUV_{late} in this diagnostic task.

Author Ting-Yim Lee licenses the CT Perfusion software to GE Healthcare.
All other authors have nothing to disclose.

Vendor Showcase

TRACK 8

RaySearch Laboratories AB

MO-SHOWCASE-TRACK 8-4-01

Data Driven Oncology in RayStation

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Imaging Interactive ePoster Discussion

TRACK 1

Imaging: Rad/Fluoro Detectors, Acquisition, and Novel Systems

MO-lePD-TRACK 1-01

Evaluation of a Prototype Ratio 29:1 Grid for Large Patient Cardiovascular Angiography Imaging

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Purpose: This work assessed the performance of a high ratio (r29) 80 line cm⁻¹ grid under adult patient cardiovascular angiography imaging condition. Non-grid and the OEM r15 80 line cm⁻¹ grid were used as reference conditions. **Methods:** Plastic WaterTM phantom thickness in the range 20-44 cm was used to attenuate the primary beam and create scatter. X-ray source to detector (SID) distance 107 and 120 cm was used to match the 105 and 118 cm focus distances of the grids. Fields of view (FOV) 20 and 25 cm were tested. The X-ray technique was fixed to ensure equal phantom-specific dose for all conditions. Primary and scatter transmission (Tp and Ts) of the grids and non-grid scatter to primary ratio (SPR) was measured using a lead beam stop method. Object quantum SNR improvement factor (Ksnr) theory was expanded to consider variable geometric magnification of a hypothetical test object located at the X-ray system isocenter and used to assess the relative SNR across experimental conditions. **Results:** For equivalent imaging conditions, Tp of the r15 and r29 grids was 0.75 and 0.69 and Ts was 0.34 and 0.18, respectively. SPR was in the range 0.27-4.1; increased by 30-39% for FOV 25 versus 20 cm; and increased exponentially with phantom thickness. Ksnr increased approximately linearly with increasing phantom thickness. Ksnr of the 29 grid at SID 120 cm versus the r15 grid at SID 107 cm was in the range 1.0-1.17 for phantom thicknesses 20-44 cm. Predicted radiation dose reduction with preserved image quality ranged from 0-26%. **Conclusion:** Compared to the r15 grid, the very low scatter transmission properties of the r29 grid is expected to improve the relationship between image quality and radiation dose for adult patients. The predicted benefit of the r29 grid increased with increasing phantom thickness.

This work was supported by a grant from Siemens Healthineers.

MO-lePD-TRACK 1-02

Clinical Performance of An AI Assisted Dose Reduction System for Fluoroscopic Guided Interventional Procedures

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Purpose: To evaluate the clinical performance of a dose reduction system utilizing AI guidance and a shuttered attenuator system to reduce radiation doses, to both patients and operators, during fluoroscopically guided interventional procedures. **Methods:** Two interventional fluoroscopy systems were retrofit with a shuttered attenuator system that intermittently attenuates the x-ray field outside of the primary focus area of the interventionalist. An integrated AI system adjusts the primary focus area based on interpretation of the tools and actions of the interventionalist to maintain high image quality in a dynamic, primary region of interest (ROI). Evaluation of the clinical performance of the system included characterization of entrance exposure, exposure from scattered radiation, high contrast resolution, low contrast resolution, and tracking performance of the AI system. **Results:** Dosimetry measurements demonstrate that the dose reductions of both DAP and exposure measurements scale in proportion to the relative areas of the attenuated and primary focus areas. However, the dose reduction algorithm is not immediately effective upon initiating an exposure. Fluoroscopists who utilize multiple short exposures are unlikely to realize the dose savings afforded by the system. High contrast spatial resolution in the peripheral region is reduced slightly compared to that in the primary ROI, but maintain spatial resolution of 2.36 lp per mm. CNR and contrast detail object detectability are more significantly affected by the increased noise present in the area outside of the primary ROI. The temporal response of the AI system is on the order of 1 s and can result in image discontinuities between the regions inside and outside of the primary ROI due to different refresh rates. **Conclusion:** The AI system generally performs as expected although subtle features of the system performance may require adjustments to operator technique to fully realize the dose reduction benefits of the system.

MO-lePD-TRACK 1-03

A Novel Integrating Proton Radiography System with a Monolithic Scintillator Detector

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Purpose: In this work, we demonstrate the system design and preliminary images obtained using our novel integrating proton radiography system consisting of a monolithic volume of plastic scintillator and an optical camera. **Methods:** The system consists of a monolithic plastic scintillator volume (20 x 20 x 20 cm³) capable of capturing a wide range of proton beam energy depositions. The scintillator captures the slowed and scattered proton beam after it exits from the phantom, producing a scintillation light distribution. A charge-coupled device (CCD) placed along the beam's-eye-view direction captures an integral of the scintillation light along the beam axis to generate the radiograph. The imaging system was tested for uniformity, stability, linearity, and accuracy in measuring water-equivalent thickness (WET). A custom spatial and contrast resolution phantom was radiographed for qualitative assessment. **Results:** A coefficient of variation (CV) was calculated over 10 radiographs collected by the system and the resulting system stability was determined to be 0.37%. The system's ability to image the dose uniformly over a radiograph was evaluated by calculating the CV over a 5 x 5 cm area and was found to be 2.6%. The system response was found to be highly linear ($R^2 = 1$) over a range of two orders of magnitude in deposited dose (1 – 92 mGy). The WET was calculated for a Gammex solid water phantom insert with a relative accuracy of 0.25%. To improve contrast for the resolution phantom, radiograph image thresholding was applied by rejecting low pixel intensities (< 30% of the peak pixel intensity in the image). **Conclusion:** The preliminary images and studies conducted on the monolithic scintillator system design demonstrate its excellent linearity, uniformity, stability, and WET calculation accuracy. Our system therefore has the potential to be adopted clinically for proton radiography.

MO-lePD-TRACK 1-04

A Simulation Study of Quasi-Monochromatic X-Ray Source Based on Reflection Thin-Film Target

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Purpose: Monochromatic X-ray sources are effective in contrast imaging, especially when their energy matches the contrast's K-edge. In this study, we evaluated the performance of the quasi-monochromatic X-ray source based on reflective thin-film Cerium oxide (CeO₂) target deposited on a diamond substrate. **Methods:** A thin-film target produces isotropic discrete k-lines and forward-distributed anisotropic white Bremsstrahlung x-rays. In the reflective direction, a quasi-monochromatic spectrum can be obtained when the Bremsstrahlung production is largely suppressed. Monte Carlo simulations were performed using Geant4 software to calculate the spectrum from the thin-film target of various thickness bombarded by electrons with energies ranged from 100-450keV. The absorbed dose in a water block was also simulated in Geant4 for each incident spectrum. The signal-to-noise ratio (SNR) and contrast to noise ratio (CNR) of a 4.5 cm water with iodine contrast were calculated based on the predicted spectrum and with that of conventional mammography x-ray sources under the same dose conditions. **Results:** Monte Carlo simulation shows that the Bremsstrahlung's production by the thin-film target decreases with electron energy and increases with the thin-film thickness. Comparing with conventional mammography sources, the same SNR and CNR of iodine images can be achieved by the quasi-monochromatic source with Cerium target but with 2 to 3 times lower x-ray exposure. **Conclusion:** The x-ray source with a CeO₂ thin-film target can produce a quasi-monochromatic spectrum that matches the K-edge of iodine to enhance the contrast. It may reduce x-ray exposure and increase CNR of iodine contrast in mammography.

MO-lePD-TRACK 1-05

Estimating Bone Densities Using a Cadmium Zinc Telluride Photon Counting Detector

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Purpose: The purpose of this work was to develop a technique to measure bone densities using a cadmium zinc telluride (CZT) photon counting detector (PCD). **Methods:** Seven bone phantoms were prepared by mixing a suitable amount of calcium perchlorate with water to achieve densities ranging between 1-1.45 g/cm³, which were pipetted into 600 μ L microcentrifuge tubes. A 10 cm diameter cylindrical polyethylene phantom with 18 cavities was used as a soft tissue equivalent material. Each bone tube was inserted in the same cavity and the phantom was imaged with a 120 kVp x-ray beam and CZT PCD while the remaining 17 cavities were filled with water. Energy thresholds of 38, 42, 68, 72, 98 and 102 keV were chosen to create 5 energy bins. Planar x-ray images were acquired at a tube current of 1 mA and the acquisition time was 25 ms. **Results:** A linear calibration curve was produced by plotting the ratio of the total number of counts in the bone phantom region of interest between two different energy bins as a function of bone density. Between all energy bins, the ratio of bins that resulted in the highest sensitivity was between bins 1 (38-42 keV) and 5 (98-102 keV), which resulted in a slope of 0.17 and an adjusted R² value of 0.976. Five additional bone phantoms with unknown densities were imaged and their densities were estimated from the calibration curve where the mean absolute error was 3.8% and 6.6% for the 40/70 and 40/100 keV ratios, respectively. **Conclusion:** It was demonstrated that measuring bone densities was possible by using a single image acquired with a PCD, a new technique that does not rely on dual-exposure imaging or a dual-layer detector. With further optimizations, future clinical applications to measure bone densities in vivo could be achieved.

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MO-lePD-TRACK 1-06

Implementation of Detailed Monte Carlo Simulation for Semiconductor Detectors Using the PENELOPE Code

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Purpose: The modeling of radiographic detection can provide insight into the optimization strategies of the imaging system. This work aims to implement a detailed Monte Carlo (MC) simulation of DR digital detectors (semiconductor) considering pairs electron-hole creation and diffusion. **Methods:** The modeling was divided into two steps: radiation interaction and charge dispersion based on the penEasy Imaging code. The transport of photons and electrons is simulated using the PENELOPE code, with 50 eV as their cut-off energies. In each electron interaction, the absorbed energy was converted into electron-hole pairs, considering the Poisson distribution, modulated by the Fano factor of the sensor material. The electron-hole pair creation energy (W) depends on the absorbed energy and the electric field (F). The point of charge detection was sampled considering a two-dimensional Gaussian distribution, where the standard deviation was obtained from the Einstein equation for electron diffusion. The pulse-height spectrum (PHS) was obtained from the simulations for monoenergetic pencil beams with energy between 10 and 100 keV, considering a detector composed of amorphous selenium, 150 μ m thick, with 139 μ m pixel, and (F) equals to 3, 10, and 30 V/ μ m. The Swank factor was calculated from the PHS moments. **Results:** For energies lower than the K-edge the PHS consists of a single spectral peak. However, for energies above the K-edge, an additional peak is observed due to fluorescent photons. Moreover, the number of electron-hole pairs increases with F, resulting in a decrease in W. The Swank factor also increases with F, with the maximum difference of 2% between 3 and 30 V/ μ m cases. The Swank factor is up to 50% lower than the calculated without charge dispersion. **Conclusion:** The simulation presented includes the electron-hole pair dispersion in the detector providing a more detailed detector modeling and, consequently, insight into the imaging system's fundamental limitations.

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MO-lePD-TRACK 1-07

Comparison of Ethiodol Contrast Media with Iodinated Microspheres for Use in 1000 Fps High-Speed Angiography (HSA) to Determine Blood Flow Details and Blood Velocity

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Purpose: To compare the performance of Ethiodol, a non-water-soluble iodinated contrast agent, with iodinated microspheres in obtaining blood velocity distributions in order to achieve accurate blood flow evaluations important for diagnosis and outcome predictions in vascular disease. **Methods:** A 3D-printed patient-specific basilar bifurcation aneurysm phantom was placed in a constant flow loop, with a flow rate of 315 mL/min as measured at the phantom inlet. For each contrast media, 1500 images were acquired at 70 kV, 100 mA, and 1000 fps using the Actaeon photon counting detector (PCD), which has 100 micron pixel pitch. The respective contrast media were injected using an automated syringe pump through a 4 Fr catheter at a fixed rate of 60 mL/min. 1 mL of Ethiodol and 0.5 mL of 700-900 micrometer diameter iodine-coated microspheres were injected. The Ethiodol stream breaks up into droplets of similar size to that of the microspheres upon injection. X-ray Particle Image Velocimetry (XPIV) was performed using an automated particle tracking algorithm to obtain velocity distributions from the 1000 fps HSA sequences, which were then compared with each other. Contrast-to-noise ratios (CNR) of the two methods were evaluated for 10 different samples of tracked microspheres and 10 different samples of tracked Ethiodol droplets. **Results:** Similar velocity distributions were obtained around the aneurysm sac ROI for the Ethiodol and microspheres. The average CNR was 4.96 for the Ethiodol and 2.78 CNR for the iodinated microspheres. **Conclusion:** This work indicates that Ethiodol XPIV produces comparable velocity information to iodinated-microsphere XPIV for in-vitro applications. The Ethiodol droplets also produce a higher CNR than the iodinated microspheres, indicating a smaller radiation dose could be used for its particle tracking capability.

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Imaging Interactive ePoster Discussion

TRACK 2

Imaging: Dosimetry in CT/CBCT

MO-lePD-TRACK 2-01

Dedicated Cone-Beam Breast CT with Offset-Detector: Estimates of Mean Glandular Dose (MGD) with Heterogeneous Tissue Distribution and Real Breast Shapes from Clinical Datasets

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Purpose: We are developing a high-resolution CMOS detector-based cone-beam breast CT (CBBCT) system with offset-detector geometry. The purpose is to estimate the mean glandular dose (MGD) for offset-detector geometry using clinical datasets that provide for heterogeneous tissue distribution and real shape and compare it to the reference geometry using a centered detector. **Methods:** 30 clinical breast CT datasets acquired on a CBBCT system with a 40x30-cm centered (no offset) detector (1024x768 pixels) were reconstructed using Feldkamp-Davis-Kress (FDK) algorithm. The images were segmented into skin, adipose, and fibroglandular tissues. The segmented breast models were imported to a Monte Carlo simulation toolkit (GATE 8.0) and were assigned Hammerstein's elemental compositions. Three imaging geometries were considered: (1) 40x30-cm centered detector corresponding to acquisition geometry as reference; (2) 30x30-cm detector with 5cm offset; and (3) 25x30-cm detector with 7.5cm offset. MC simulations were conducted with 49kV (1.39mm HVL), 0.3x0.3-mm focal spot, and with 10E6 photons. The normalized glandular dose coefficient (DgNCT) was obtained for each geometry and breast. MGD was obtained as the product of DgNCT and the measured air kerma at the axis-of-rotation. **Results:** The coefficient of variation was less than 0.7%. For the 30 clinical breast CT datasets, the mean $\pm$ SD and the median (inter-quartile range) of the MGD (mGy) were 9.63 $\pm$ 2.84 and 8.99 (7.68–11.31) for the 40x30-cm reference geometry, 8.19 $\pm$ 2.62 and 7.70 (6.74–8.86) for

the 30x30-cm detector with 5cm offset, and 5.35 ± 1.39 and 5.15 (4.22–5.94) for the 25x30-cm detector with 7.5cm offset. The median computational time for MC simulations was approximately 40 minutes for each case. **Conclusion:** This study demonstrates the potential for dose reduction using offset-detector geometry with heterogeneous tissue distribution and real breast shapes. The estimated dose reduction for the 30x30-cm detector with 5cm offset and the 25x30-cm detector with 7.5cm offset were 14.8% and 44.4%, respectively.

MO-lePD-TRACK 2-02

Feasibility of Using Size-Specific Dose Estimate (SSDE) to Estimate Effective Dose of Individual Patients

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Purpose: The aim of this study is to investigate the relationship between size-specific dose estimate (SSDE) and a derivative form of effective dose called size-specific effective dose (SED) for individual patients. **Methods:** Values of SED resulting from CT scans of six scan regions over the trunk were estimated for 193 adult phantoms using NCICT software. The library covers a wide range of adult patient sizes of different heights (150 – 190 cm) and weights (40 – 125 kg). The regions studied were: (1) chest, (2) abdomen, (3) pelvis, (4) chest-abdomen, (5) abdomen-pelvis, and (6) the whole trunk (CAP). SED values were assessed for 120 kV and a pitch of 1 using standard scan lengths similar to those used in the clinic. A MATLAB code developed in-house was employed to calculate the patient size in terms of water-equivalent diameter for the same phantom set used in NCICT software, which allowed values for SSDE to be calculated for CT scans investigated. **Results:** Good correlations between SED and SSDE values were observed for all scan regions studied, except the pelvis. The variations for the regions were all within $\pm 8\%$, but they reached 16% for pelvis scans due to differences in colon doses. These occurred because sections of the transverse colon lay inside the scan range for some phantoms and outside for others depending on the colon position for the individual. **Conclusion:** The strong correlations between SED and SSDE values suggest that coefficients of the best-fit for the correlations could be derived, from which risk of an individual patient could be assessed based on SSDE of a given scan taking into account the patient size and the gender. This may provide a simple and straightforward approach to assess SED in a manner similar to using size-based k-factors linked to dose-length product.

MO-lePD-TRACK 2-03

Fetal Dose in Body CT: Z-Axis Dose Profile Based Calculation Compared to Monte Carlo Simulation

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Purpose: In CT exams of pregnant women, fetus (or embryo) may be encompassed by, partly inside, or completely outside scan range. This study is to compare two fetal dose evaluation methods (Monte Carlo simulation, z-axis dose profile based direct calculation) for abdominopelvic, abdominal, and chest exams. **Methods:** This work considered three whole-body models of a pregnant patient (3, 6, 9 gestational months; 62, 67, 73 kg) undergoing chest, abdominal, and abdominopelvic exams on Siemens Somatom Definition CT with fixed mA, 120 kV, 19.2-mm beam width, helical pitch at 1, body bowtie filter, and CTDIvol of 15.6 mGy. Maternal water equivalent diameter (WED) at the fetus centroid was 30.7, 35.3, and 36.3 cm, respectively. A previously validated Monte Carlo simulation program (Geant4) was used to evaluate fetal dose, where helical scan was simulated with 360 projections and 360 patient table translations per gantry rotation, each translation by pitch $\times$ (collimator width)/360. In parallel, another previously reported method was used to evaluate z-axis dose profile of a water phantom, with inputs of scan range, CTDIvol and WED. Fetal average dose was calculated over the fetal volume in the cranial-caudal direction, and was compared with that of simulation. **Results:** Fetal dose from Geant4 simulation was 18.7, 18.3 and 17.7 mGy (3, 6, 9 months) in abdominopelvic exam; 0.69, 1.88 and 3.2 mGy (3, 6, 9 months) in abdominal exam; 0.10, 0.24 and 0.38 mGy (3, 6, 9 months) in chest exam. The results of profile-based evaluation were consistent with those of simulation, where their difference in magnitude was $(8.8 \pm 5.9)\%$ (maximum, 17.1%) for three exams. **Conclusion:** The dose profile based fetal dose calculation is easier to implement than Monte Carlo simulation and is suited for body CT of pregnant women at all gestational ages.

MO-lePD-TRACK 2-04

Implementation of Tube Current Modulation for Pediatric and Adult CT Patients with Various Body Sizes into NCICT2.0

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Purpose: To implement a prospective tube current modulation (TCM) feature into an organ dose calculation program for CT patients with various body sizes, NCI dosimetry system for CT (NCICT) 2.0, by using a library of body size-dependent computational human phantoms. **Methods:** We created a comprehensive library of TCM profiles for a total of 351 pediatric and adult body size-dependent computational human phantoms. We adopted a generic ray-tracing algorithm (Li et al. PMB 2014) to create TCM profiles: calculation of the attenuation at a given angle within a CT fan beam geometry for the energy spectrum of a reference CT scanner (Siemens Sensation 16). We averaged angular modulation in a single rotation to derive surrogate longitudinal modulation profiles with 1-cm intervals in each phantom. The resulting TCM profiles were implemented into the NCICT2.0 program. We compared organ doses for different scan protocols between fixed tube current and TCM scans. **Results:** A comprehensive library of TCM profiles were created from a total of 351 pediatric and adult computational phantoms with various body sizes and implemented into NCICT2.0. When the TCM strength was set 0.5, dose reduction from TCM was clearly observed in low to normal BMI phantoms but organ doses under TCM were greater than those under fixed current for high BMI phantoms. Lung dose for the 60-kg adult male phantom with the height of 175 cm reduced by 51% under TCM but that for the 100-kg phantom at the same height increased by 12% under TCM compared to fixed current scans. **Conclusion:** We established a library of generic TCM profiles from a library of body size-dependent computational phantoms and implemented the feature into NCICT2.0. The upgraded program should be useful to estimate organ doses for CT patients undergoing TCM-based CT scans.

NCICT licensed to PACS Health, LLC.

MO-lePD-TRACK 2-05

Direct Measurement of CTDIw On Helical CT Scans

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Purpose: In present day clinical practice the majority of CT exams employ helical scan protocols. Helical protocol CTDIvol is routinely determined by measuring CTDIw for axial scans. On some CT scanners axial scans with the same detector configuration employed on helical protocols are not available, and CTDIw cannot be measured directly with the standard CTDI phantom. Presented is a method to directly measure CTDIw on helical scan protocols. **Methods:** CTDIw is measured on helical scans by centering the CTDI phantom in the CT gantry as with axial CTDI measurements. The body CTDI phantom was modified by the addition of brackets to the sides of the phantom. The brackets support the phantom on the CT gantry shroud with the tabletop below. The phantom remains stationary during helical scans, and surface and axis doses measured. The measured doses are divided by the number of gantry rotations to determine values for one rotation, and CTDIw calculated. CTDIw for body and head helical scan protocols were measured and CTDIvol determined on a recently installed Siemens Somatom Drive CT Scanner that did not have axial scan protocols with the same detector configurations. CTDIw adult body measurements were made for helical and axial scans on a GE Discovery CT Scanner employing the same detector configuration and technical factors. **Results:** The body and head CTDIvol measured on the Siemens Scanner were 13.45 and 48.93 mGy. The corresponding displayed values were 12.80 and 47.78 mGy; 5% and 2.4% lower. The GE helical scan and axial body CTDIw measurements were 10.90 and 10.94 mGy. **Conclusion:** From the good agreement of axial and helical results for the GE scanner, the suspended phantom method is

viable and allows one to directly measure CTDI_w with helical scans. The results for the Siemens scanner show how the method can be applied.

MO-lePD-TRACK 2-06

Scanner-Independent Coefficients to Estimate Organ Doses of CT Pediatric Patients Based On Water-Equivalent Diameter

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Purpose: This investigation aimed to study development of size-specific and scanner-independent dose conversion coefficients (DCCs) to assess organ doses of pediatric patients resulting from CT scans in the trunk based on the patient size, that was measured by water equivalent diameter (D_w). **Methods:** Organ doses of a wide range of pediatric patient sizes ranging from 1-15 years old were assessed using NCICT software based on Monte Carlo simulations. More than 150 pediatric phantoms with weights and heights in the range of 10–125 kg and 0.85-1.85 m, respectively, were involved. Organ doses were estimated using a 120 kV spectrum for CT scans performed in the body over six different areas, namely: chest, abdomen, pelvis, chest-abdomen, abdomen-pelvis, and all the trunk (CAP). In order to establish scanner-independent DCCs, organ doses of each patient were normalized with respect to the corresponding CTDI_{vol}. The patient size that was reported in terms of D_w was calculated using the same phantom set with a MATLAB software developed for this purpose using an equivalent spectrum. D_w was assessed for each phantom over the six areas studied. **Results:** The correlations between D_w values and DCCs of organs that were fully or largely inside the scan areas were good. Values of R² were more than 0.90 for the majority of organs. This would allow establishing size-specific and scanner-independent DCCs that could be utilized to assess doses of pediatric patients resulting from any CT scanner. **Conclusion:** The DCCs only depend on the patient size in terms of D_w, which takes attenuation of the scan area into account, and CTDI_{vol} that is given in the CT dose report. Most of DCCs published in the literature are based on patient sizes that depend on the patient dimensions, which are expected to lead to some uncertainties, especially for the chest area.

MO-lePD-TRACK 2-07

Variability of Water-Equivalent Diameter (D_w) with Tube Potential of CT Scans

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Purpose: The aim of this work was to study variability of the patient size measure, which is reported by water-equivalent diameter (D_w) with the applied tube potential for given CT scans for pediatric and adult patients. **Methods:** A large library of 363 phantoms (168 pediatric and 195 adult) were used. The library was designed to represent characteristic of the whole patient population. It covers a wide range of ages from newborn baby to adult with various weights (10 – 125 kg) and heights (85 – 190 cm) for male and female phantoms. Values of D_w were assessed for each phantom using a MATLAB developed in house for this purpose. The values were calculated over six regions of the trunk that represent CT protocols applied in the clinic. The regions studied were chest, abdomen, pelvis, chest-abdomen, abdomen-pelvis, and the trunk (CAP). Five equivalent monoenergetic beams, which represented tube potentials of 80 – 140 kVp, were used to calculate D_w values. **Results:** The differences between D_w values assessed for the various spectra were found to be insignificant. For all phantoms, the differences were within ±2% over the scan regions, with the majority of differences being less than 1%. The differences were not be affected by size or gender of the patient. **Conclusion:** D_w is the metric that has been recommended by the AAPM to replace dimension-based metrics to estimate size-specific dose estimate (SSDE) for an individual patient resulting from a given CT. Although D_w values were affected by the area attenuation, results of this study showed that impact of tube potential on D_w of the trunk regions was minimal. Since, assessment of D_w requires advanced computational tools, the results suggest that energy-independent D_w values could be established and linked to effective diameter, which can be simply measured for an individual patient.

Multi-Disciplinary Interactive ePoster Discussion

TRACK 3

Multi-Disciplinary: MRI in Radiotherapy

MO-lePD-TRACK 3-01

Comparison of CT-Guided and MR-Guided Adaptive Radiotherapy for Intramuscular Metastasis

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Purpose: An intramuscular tumor is not visible on CT but delineable on MR scans. We compared dosimetric differences between CT-guided radiotherapy (CTgRT) and MR-guided radiotherapy (MRgRT) to demonstrate the advantages of treating such tumor with MRgRT. **Methods:** A patient with a renal cell carcinoma gluteal intramuscular oligometastasis was treated with a 1.5T MR-Linac using daily adaptive MRgRT. The prescription was 60 Gy to GTV and 45 Gy to PTV delivered in 15 fractions. Fractional delivered dose of MRgRT was accumulated to the simulation MR scan by aligning the GTV on daily treatment MR scans to the simulation MR scan using rigid registration. In parallel, we developed an in-house cycle-consistent generative adversarial network (cycleGAN) model to generate synthetic CT for each daily MR scan. The synthetic CT was used to simulate daily CBCT alignment using rigid registration based on nearby bony landmarks because tumor is invisible in CT. The fractional dose based on CTgRT was accumulated to the simulation MR scan for comparison with MRgRT. **Results:** MRgRT D95% dose accumulation was 60.1 Gy and 47.5 Gy with average dose of 61.1 Gy and 55.0 Gy to GTV and PTV respectively. CTgRT D95% dose accumulation was 37.3 Gy and 24.4 Gy with average dose of 44.8 Gy and 38.8 Gy to GTV and PTV respectively. CTgRT dose accumulation resulted in >35% and >45% under-dose compared to the planned GTV and PTV D95% dose respectively. Qualitatively, the low-dose regions in CTgRT were comparable to MRgRT, however the high-dose regions were significantly different. The GTV was partially missed in 10/15 fractions and completely missed in 5/15 fractions to receive full planned dose in CTgRT simulated treatments. **Conclusion:** Treating intramuscular tumors using CTgRT can significantly under-dose the tumor without excessively generous margin. MRgRT is ideal for cases where tumor is not visible on CT.

MO-lePD-TRACK 3-02

Correction of Gantry Induced Geometric Image Distortion on MRgRT at Various Gantry Angles with Deformation Vector Field

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Purpose: Geometric accuracy is vital for delivering high quality MR-guided radiotherapy (MRgRT). MR images acquired on 0.35T MR-linac systems are distorted when acquired at different gantry angles, requiring repositioning of the gantry for acquiring new volumetric images. This work investigates correction of distorted images acquired at different gantry positions using deformation vector fields (DVF) from reference images. **Methods:** Two phantoms were imaged on a 0.35T MR-linac using a clinical 3D TRUFI sequence. The phantoms included a cylindrical phantom, part of the QUASARTM MRID3D geometric distortion analysis system, and a QUASARTM 3D grid phantom. Acquisitions were performed at 30° increments, from 0° to 330°, with TR: 3ms, TE: 1ms, flip angle: 60°, FoV: 450x450x360mm³, imaging matrix: 300x300x240. Acquisitions were performed with the onboard Siemens distortion correction function turned on and off (DstOn and DstOff). The MRID3D phantom images were imported into the distortion analysis software to produce DVFs for each gantry angle with DstOn and DstOff. These DVFs were then applied to the 3D grid phantom images in the proprietary System Distortion Resampler software from ModusQA. Resampled images were compared to the original and resampled G0 distortion corrected images visually and with the structural similarity index measure (SSIM). **Results:** Significant

distortion was present in 3D grid phantom images with DstOff, and was partially corrected with DstOn. After resampling with DVFs the phantom appeared visually identical to the reference image. SSIM increased from a mean ($\pm$ SD) of 0.26 ± 0.001 and 0.06 ± 0.001 before resampling to 0.70 ± 0.004 and 0.69 ± 0.005 after resampling for DstOn and DstOff, respectively. **Conclusion:** The resampling software was successfully able to apply DVFs from one phantom to a second to correct for geometric distortion artifacts generated by imaging at different gantry angles on an MR-linac system. This gives the possibility of applying DVFs to patient images for accurate reconstruction during MRgRT.

MO-lePD-TRACK 3-03

Generative Adversarial Networks for Synthetic CT Generation from MR Scans with Truncated Anatomy

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Purpose: MR scans in MR-guided radiotherapy can be partially truncated due to limited field of view, which could significantly affect dose calculation accuracy for MR-based online adaptive planning. We developed a cycle consistent generative adversarial networks (cycleGAN) based approach to create synthetic CT (sCT) images compensating the missing anatomy from the truncated MR scans. **Methods:** CT scans and T1-weighted MR scans with complete anatomy of 24 head and neck patients were used for this study. The MR scans were manually cropped 10-15 mm off at the posterior region of head to simulate truncated MR scans. Six patients were randomly chosen for testing and the rest were used for training. We proposed a modified cycleGAN using a residual-Unet as the generator to create sCT from MR. Truncated/complete MR scan pairs and CT scans were used to train the cycleGAN. The sCT pairs were generated for both truncated and complete MR scans. The mean absolute difference (MAD) of Hounsfield units (HU) was used to quantify the sCT performance. To evaluate the sCT in the truncated region, body contours were generated for each sCT pair. In the truncated region, the structural similarity index (SSIM), mean surface distance (MSD), and Dice similarity of body contours were calculated between sCT pairs to quantify the sCT performance in compensating the truncated anatomy. **Results:** The mean MAD were 83.2 ± 9.3 HU and 81.6 ± 9.2 HU for the sCT generated from truncated and complete MRI, respectively. In the truncated region, the mean SSIM was 0.86 ± 0.02 , the MSD was 1.91 ± 0.52 mm, and the Dice of body contour was 0.92 ± 0.02 , indicating a good performance in compensating the truncated anatomy. **Conclusion:** We developed a novel cycleGAN model that can create synthetic CT with complete anatomy from truncated MR scans, which could potentially benefit online MR-guided adaptive treatment planning.

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MO-lePD-TRACK 3-04

3D Deep Convolutional Neural Network for Mapping MRI to Synthetic MVCT for Radiotherapy Planning

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Purpose: To synthesize MVCT (sMVCT) datasets from MRI using a 3D deep CNN. This is a novel solution for MRI-only treatment planning where MRI can be utilized for anatomical delineation and MRI-generated sMVCTs for more accurate electron density mapping versus synthetic conventional kVCTs. **Methods:** MVCT (3.5MV), kVCT (120kV), and T1-weighted MRIs for 120 head and neck cancer patients were retrospectively acquired and co-registered. MRI intensity values were normalized to standardize datasets acquired on different scanners. A deep neural network based on a fully-convolutional 3D residual U-Net architecture was implemented to map MRI intensity to sMVCT HU. Model inputs were volumetric patches generated from 100 paired MRI and MVCT datasets. The U-Net was initialized with random parameters and trained on a mean absolute error (MAE) objective function. Model accuracy was evaluated on 20 withheld test exams. For comparison, this method was applied to generate conventional synthetic kVCTs (skVCTs). sCTs were compared to respective actual CTs using MAE. To demonstrate clinical proof-of-concept, IMRT plans were generated on a MVCT then calculated with fixed plan parameters onto the corresponding sMVCT. Dose distributions were compared using gamma (3%/3mm local dose threshold) criteria. **Results:** The model produced MAE of 118.4 ± 17.3 and 127.7 ± 33.3 HU for MVCT and kVCT datasets, respectively. Overall, there was good agreement between sCTs and CTs. Distinguishing bone from air posed the greatest challenges. The retrospective dataset introduced additional deviations due to sinus filling or tumor growth/shrinkage between scans, differences in external contours due to variability in patient positioning, or when immobilization devices were absent from diagnostic MRIs. Dose-volume-histograms were in close agreement between sMVCT and MVCT datasets and the gamma value between dose distributions was 100%. **Conclusion:** sMVCT datasets can be generated from T1-weighted MRI using a 3D DCNN with dose calculation on a sample sMVCT in excellent agreement with the MVCT.

MO-lePD-TRACK 3-05

Diffusion Profiling Via a Histogram Approach in Locally Advanced Cervical Cancer Treated with Chemoradiotherapy: Challenges and Opportunities for Dynamic Targeting with MRI

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Purpose: There is need for imaging biomarkers to identify radiation resistant tumors and guide radiotherapy of cervical cancer. As such, MR-guided adaptive radiotherapy has the potential to provide improvements to the therapeutic ratio. Parameters derived from diffusion-weighted MRI (DW-MRI) are appealing quantitative biomarkers because DW-MRI does not require exogenous contrast agents, has relatively short scan time, and is ubiquitous across systems owing to its role in diagnosis. Important histologic properties that correlate with apparent diffusion coefficient (ADC) include tumor proliferation index, tumor grade, presence of necrosis, and post-therapy tumor cell apoptosis. The purpose of this study was to evaluate cervical cancer ADC histograms prior to and following concurrent chemoradiotherapy (CCRT). **Methods:** Five cervical cancer patients who underwent a 4-week CCRT regimen with MRI prior to and following external beam radiotherapy (EBRT) were retrospectively analyzed. Histogram features were calculated from volumes of interest (VOIs) contoured by a radiologist on ADC maps. Tumor volume, surface area, and descriptive statistics for the ADC voxel data, including mean, median, standard deviation, quantiles, skewness, and kurtosis were compared using a paired t-test. **Results:** Post-EBRT scans were acquired an average of 41.4 days after baseline DWI. Tumor volume decreased by an average of 58.2 ± 55.2 cm³ ($p = 0.04$) and surface area decreased by 9823 ± 8011 mm² ($p = 0.03$) from baseline. Mean ADC was 1184.8 ± 169.5 10⁻⁶ mm²/s and 1342.0 ± 172.2 10⁻⁶ mm²/s ($p = 0.12$) pre-CCRT and post-CCRT, respectively. Skewness and kurtosis were 0.82 ± 0.16 and 4.56 ± 1.36 pre-CCRT and 0.50 ± 0.51 and 3.68 ± 0.38 post-CCRT ($p = 0.11$ and $p = 0.10$). **Conclusion:** Chemoradiotherapy resulted in increases in mean tumor ADC of locally advanced cervical carcinomas, with statistically significant decreases in tumor volume and surface area in this cohort. Future work will evaluate ADC of viable tumor subvolumes over time and correlate histogram features with treatment response.

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MO-lePD-TRACK 3-06

Evaluation of Registration Accuracy for Online MR-Guided Adaptive Planning

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Purpose: To quantify the accuracy of rigid and deformable image registration (DIR) by using an in-house -developed phantom for online MR-guided adaptive planning with a Unity MR-Linac. **Methods:** We developed a phantom to evaluate the registration accuracy between modalities (MR-CT) and within MR (MR-MR). This acrylic phantom consists of a rotational base and 3 inserts of different shapes (circle, oval, and irregular), which were filled with oil, agarose gel, and Styrofoam jelly to simulate fat, prostate, and lung tissues. Deformations and location changes were simulated by changing the inserts. One CT scan and 48 MR scans (T1-weighted short [2-min] and long [6-min], and T2-weighted [2-min and 6-min] scans) were imported into the Monaco treatment planning system for rigid and DIR tests. Contours of all inserts were manually drawn on each image for evaluation. After DIR, deformed contours were compared with manual contours in terms of Dice similarity coefficient (DSC), Hausdorff distance (HD), and mean distance to agreement (MDA). **Results:** Rigid registration accuracy was within 2 mm and 2 degree for all scans (CT-MR, MR-MR). For MR-MR, T1 images had better DIR accuracy than T2 images: DSC T1-T1=0.93, HDT1-T1=0.69 cm, and MDA T1-T1=0.12 cm (vs T2: DSC T2-T2=0.86, HDT2-T2=0.12 cm and MDA T2-T2=0.42 cm). For MR-CT, T1 was also more accurate than T2 (T1: DSC T1-CT=0.82, HDT1-CT=1.56 cm and MDA T1-CT=0.32 cm, compared with T2: DSC T2-CT=0.75, HDT2-CT=1.62 cm and MDA T2-CT=0.47 cm). No difference was found in DIR accuracy between the T1 short and long scans. **Conclusion:** Different MR sequences can greatly affect DIR accuracy. However, low-quality T1 scans [2-min] can be used without sacrificing DIR accuracy. This phantom allows end-to-end testing for online MR-based adaptive planning.

MO-lePD-TRACK 3-07

Real-Time Motion Tracking Based on Orthogonal Cine MRI for Abdominal Tumors On 1.5T MR-Linac

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Purpose: This work investigates the feasibility of real-time motion tracking (RMT) for abdominal targets using a template registration-based algorithm between real-time 2D-cine and 3D-MRI on MR-Linac. **Methods:** The RMT was evaluated using 9 sets of daily mid-position images (MidP) derived from an in-house 4D-MRI and balanced T2/T1-weighted bFFE cine images acquired with free breathing on a 1.5T MR-Linac for 8 patients with abdominal tumors. The cine images were acquired with a temporal resolution of 0.2 s for a total duration of about 10 minutes interleaved between coronal and sagittal orientations. A 2D-template was created in both orientations using data acquired over the first 20 s, followed by a rigid registration to the 3D-volume using a mask defined on the MidP. Motion tracking of the target was performed by registering the rest of the 2D-cine series with the corresponding template with a known relationship to the 3D-volume. Manually delineated contours on sampled cine frames were used as the ground truth. Tracking accuracy was defined as the standard deviation of the error (SDE) comparing the center-of-mass (CM) of the ground truth and the tracked motion. **Results:** The tracking algorithm was qualitatively validated by visual assessment of 9 abdomen sets for the 8 patients. It was quantitatively validated on 4/7 Pancreatic and 2/2 Renal Fossa cases. The mean SDEs (and ranges), were 1.08 mm (0.31-1.9 mm), 1.07 mm (0.33-1.68 mm), and 0.76 mm (0.37-1.24 mm) in superior-inferior, left-right, and anterior-posterior directions, respectively. For the three remaining pancreas cases, performing tracking with ROIs including a margin encompassing the target demonstrated the robustness of the algorithm to implant-induced image artefacts in two cases and suboptimal imaging plane selection. **Conclusion:** We have demonstrated the feasibility of accurate free-breathing motion tracking using the template registration for abdominal targets based on real-time orthogonal cine on MR-Linac.

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Therapy Interactive ePoster Discussion

TRACK 5

Therapy: External Beam: Automatic Treatment Planning

MO-lePD-TRACK 5-01

Deep Learning Based Multiple-Arc VMAT MLC Sequence Prediction in Prostate Cancer

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Purpose: To predict sequential multileaf collimator (MLC) apertures and dose for multiple-arc volumetric modulated arc therapy (VMAT), we introduce a deep learning (DL) based model utilizing the beam eye's views (BEVs) of the given dose distributions in prostate cancer patients. **Methods:** Clinical dose distributions and VMAT plans for 70 prostate cancer patients were solicited. All VMAT plans consist of 4 half-arcs, each consisting of 98 control points. A 6-channel 7-level U-NET was used as the main network. The input was the 6 stacked 2D dose distribution slices in the BEV of each gantry angle, spanning from the isocenter to the source with 5mm intervals. The output was the 2 corresponding MLC apertures in each gantry angle, with the associated dose being the intensity. The dataset was divided into 80% for training, 10% for validation, and 10% for testing. 200 epochs were trained using an aggregated loss combining mean squared error and dice coefficient (DC), and the Adam optimizer. For comparison, networks using 4 and 8 input slices per gantry angle were also trained with the same hyperparameters. **Results:** On 7 test subjects, the predictions from the 6-channel model (6s-BEV) achieved an averaged MLC apertures DC of 0.73 (± 0.04) and a step-wise dose error (DE) of -4.89% ($\pm 7.98\%$), normalized to each arc. In contrast, models using fewer (4s) or more (8s) BEV slices as inputs resulted in compromised performance: 0.69 (± 0.04) DC and -10.96% ($\pm 8.35\%$) DE for 4s-BEV, and 0.71 (± 0.04) DC and -9.06% ($\pm 8.67\%$) DE for 8s-BEV. **Conclusion:** We successfully implemented a U-net for the prediction of multiple-arc prostate VMAT plans. The trained model predicted MLC apertures and the associated dose without involving the time-consuming planning process. The proposed DL model may thus facilitate online applications that benefit from fast planning.

MO-lePD-TRACK 5-02

Capturing Head-And-Neck Planning Trends with a Knowledge-Based Tradeoff Model

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Purpose: Since the clinical implementation of inverse planning, treatment planning knowledge has developed substantially, yet the knowledge shift is not well-understood. This study's purpose is to analyze the progression of the clinical planning knowledge over time and investigate if knowledge-based planning (KBP) models built using historical data can represent the current planning practice. **Methods:** To evaluate how clinical knowledge has progressed, we retrieved 304 anonymized head-and-neck treatment plans and binned them into four cohorts based on treatment date: 2011-2012, 2013-2014, 2015-2016, and 2017. Planning parameters and anatomic features were extracted from the treatment planning system via scripting. The first three cohorts were used for model training and the 2017 dataset was reserved for evaluation. A recently-proposed knowledge-based tradeoff model was used to quantify planning tradeoff considerations. Tradeoff directions projected on the 2017 dataset were recorded. Model prediction errors after taking tradeoff considerations were also evaluated. Additional plan information, including optimization constraints and optimization structures were analyzed. **Results:** During the four time periods, the numbers of beams, structures with optimization constraints, and optimization constraints have not changed significantly ($p>0.05$). However, the standard deviations of these parameters have all decreased, indicating the planning practice has become increasingly standardized. Using the latest 2015-2016 dataset for model training reduces validation error by 26.0% for parotid, 21.9% for oral cavity, and 25.0% for pharynx, compared with models trained using cases from previous periods. However, the tradeoff decisions made clinically remain consistent: parotid and oral cavity have been involved in most tradeoff considerations. **Conclusion:** We have demonstrated that the planning trends can be captured by a knowledge-based tradeoff model. Our results suggest that clinical planning knowledge changes over time and care must be taken when using prior treatment plans to guide new planning efforts.

MO-lePD-TRACK 5-03

A Deep-Learning-Based Dual-Arc VMAT Plan Generation from Patient Anatomy for Prostate Simultaneous Integrated Boost (SIB) Cases

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Purpose: To develop a deep-learning algorithm to automatically generate dual-arc VMAT plans using a patient's 2D integrated CT attenuation coefficients for prostate SIB cases. **Methods:** A customized U-net were designed with 4-resolution-step analysis path and a 4-resolution-step synthesis path. The inputs are 2D integrated CT attenuation coefficients and associated structure contours at the gantry position projection. The output prediction of intensity maps generates two sets of MLC sequences for two VMAT arcs with the same collimator angle. The results were then sent to a commercial TPS for plan finalization. This study involves 140 prostate patients who received simultaneously-integrated-boost (SIB) treatment (PTV(58.8)/PTV(70) in 28fx). A total of 129 one-arc VMAT plans (library plans) were used for network training and validation. For the 11 test patients, the predicted intensity maps are divided into two parts from the isocenter. An empirical elliptical Gaussian distribution was multiplied with the predicted intensity maps. These intensity maps were then divided into upper and lower parts to calculate MLC positions. Key dosimetry metrics in the resulting dual-arc VMAT plans (DL plans) were compared against the one-arc AVP-DSP plans in network training. **Results:** After dose normalization (PTV(70) V70Gy=95%), some of 12 DL test case plans met institutional clinic guidelines of dose distribution for OAR and normal tissue sparing. The tested plans showed a high consistency. In an example DL plan, OAR parameters were comparable to one-arc library plan (Bladder V70Gy=5.6cc, V40Gy=15.5%; rectum V70Gy=3.9cc, V40Gy=35.9%). Single-arc: (Bladder (V70Gy=6.8cc, V40Gy=19.4%; rectum (V70Gy=2.8cc, V40Gy=26.3%). 3D max dose results in DL plans (D1cc=114.6% were lower than the one-arc library plan (D1cc=118.9%). On average, DL plan generation takes 30s/case in contrast to the current clinical practice. **Conclusion:** Compared to one-arc library plans, the proposed dual-arc VMAT plan generation algorithm has potentials to achieve higher modulation and lower maximum dose.

MO-lePD-TRACK 5-04

A Feasibility Study of Automated Inverse Planning Using Hyperparameter Tuning

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Purpose: Inverse planning in current practice requires treatment planners to modify multiple parameters in the objective function to produce clinically acceptable plans. Due to the manual steps in this process, the quality of plans could widely vary depending on planning time available and treatment planner's skills. The purpose of this study is to automate the inverse planning process to maintain plan quality with reduced planning time. **Methods:** The parameters for tuning were the limit dose parameters of each organ-at-risk (OAR) objective. For given parameters, the treatment plan optimization was done in Raystation using the scripting interface to obtain the dose distributions deliverable. We investigated random and Bayesian searches with various forms of utility functions to achieve clinical goals. We also varied the range of dose parameters in the search space. We applied the framework developed to 5 patients who received SBRT to their peripherally located lung tumors with distinct clinical concerns. We constrained the dose to OARs according to RTOG0915. We normalized all plans to have PTV D95 equal the prescription dose and compared the OAR dose metrics in the clinically used plans with those in the automatically generated plans. **Results:** The number of iterations required to produce the satisfactory plan quality was 100 and the average planning time was 2.2 hours. The OAR doses achieved by the automatically generated plans were lower than the clinical plans by up to 76.8%. When the OAR doses were larger than the clinical plans, they were still less than the clinical goals by 3.6-98.9% indicating they were clinically acceptable. **Conclusion:** We successfully devised an automated inverse planning framework using hyperparameter tuning. We demonstrated that the plan quality could be improved, and the average treatment planning time (no planner intervention required) was 2.2 hours for SBRT lung cases.

MO-lePD-TRACK 5-05

Plan N Check: An Open-Source Tool for Automatic Head-And-Neck VMAT Planning in Varian Eclipse™

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Purpose: To demonstrate the efficacy of a novel and open source tool, "Plan n Check," for automated creation of head-and-neck Volumetric Modulated Arc Therapy (VMAT) radiotherapy plans in Varian Eclipse™. **Methods:** The Eclipse Scripting Application Programming Interface was used to develop a binary script which runs within Eclipse™, reading and operating on patient data. Plan n Check creates dual concentric arcs centred on the primary target volume with a 60° collimator angle difference aligned with the patient's superior-inferior axis. Collimator jaws are automatically adjusted to adequately encompass target volumes. Standard institutional dose constraints are automatically applied to relevant structures which can be verified and overridden by the planner if needed. Optimization structures are automatically created where regions-of-interest overlap planning target volumes, excluding higher-priority tissues (e.g., nervous system). To fine-tune optimization objectives, N preliminary optimization cycle iterations followed by automatic constraint-checking and updating of objective penalties take place before proceeding with full VMAT optimization for 4 additional optimization cycles. **Results:** Plan n Check created treatment plans conforming to institutional best-practices and standards-of-care. Plans optimized with N = 1 and N = 2 met all institutional constraints, while plans optimized without preliminary iterations resulted in constraint violations for all but one patient (median: 2 violations per plan). **Conclusion:** Plan n Check is an open source tool for automating VMAT treatment planning in Varian Eclipse™. Use of preliminary optimization cycle iterations improves clinical constraint satisfaction without requiring any additional effort from the planner. Source code is freely available online.

MO-lePD-TRACK 5-06

A Fully-Automated Field-In-Field Algorithm Translatable to Multiple Disease Sites

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Purpose: To develop an adaptive automated field-in-field technique compatible with various beam arrangements and treatment sites. **Methods:** We created an algorithm that automates the clinical workflow for creating field-in-field plans independent of treatment planning system. The algorithm automatically identifies a hotspot, creates a subfield, calculates dose, and optimizes beam weight without user intervention. This process is repeated until the hotspot is reduced. Parameters used in the algorithm are configurable and include the definition of hotspot, the target volume, the maximum number of subfields, the minimum MU per field, and the optimization solver, and so on. A sub-field is determined based on hotspot area. The beam weights are optimized based on user-configured physical constraints for DVH coverage and least-squared cost functions. The final plans were evaluated based on target volume coverage and the remaining hotspot dose. The algorithm was tested on 20 patients each for whole brain (2-field), rectal (3-field), and cervix (4-field-box) with the clinically delivered primary fields as inputs. **Results:** For whole brain plans, all plans achieved V98.5% with hotspot less than 108% of prescription dose. For rectal plans, all plans achieved V95% with hotspot less than 110% of prescription dose. For cervix plans, all plans achieved V97% with hotspot less than 109% of prescription dose. On average, the algorithm reduced the hotspots of the plans from 114% to 107%, 130% to 107%, and 111% to 107% for whole brain, rectal, and cervix, respectively. The average time to complete a plan is 8.5 mins, 22.2 mins, and 6.5 mins for whole brain, rectal and cervix plans, respectively. **Conclusion:** We automated the clinical workflow for generating FIF to reduce hotspots in 3D conformal plans. This algorithm successfully produced FIF plans for beam arrangements and multiple disease sites and can improve treatment planning efficiency when incorporated into automatic planning workflow.

MO-lePD-TRACK 5-07

Robust Lung SBRT Knowledge-Based Planning with Missing Data Management

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Purpose: Lung SBRT is a complex site and treatment technique. A knowledge-based planning (KBP) technique is harder to implement because not all organs-at-risk in the thorax are delineated based on the location of the tumor and its size, which often results in a heterogeneous plan database. We developed a novel technique to predict dosimetric parameters for the lung SBRT effective even in the presence of missing data. **Methods:** A retrospective study was made of 399 lung SBRT patients (219 training, 30 validation and 50 test) treated with VMAT. Prescribed dose was 48 or 52 Gy in 4 fractions and 50 Gy in 5 fractions. Bayesian Stochastic Frontier Analysis (BSFA) with Markov Chain Monte Carlo (MCMC) algorithm were used to design a knowledge-based technique to predict achievable dose sparing to organs-at-risk. 95% Highest Density Interval (HDI) is also calculated within the MCMC framework and give us credible interval of dose prediction given a patient specific anatomy. Missing data for non-delineated OARs were estimated following a truncated normal distribution specific to each OAR. **Results:** In total, 16 dosimetric parameters were predicted for the spinal cord PRV, esophagus, heart, main bronchus, great vessels and chest wall. Mean difference between the observed and predicted values for the test group ranges between 1.5 (1.9) Gy for the D0.35cc of the spinal cord PRV and 4.9 (5.3) Gy for the D0.035cc of the main bronchus. Investigation of the missing data model shows that it is robust in our predictive model. **Conclusion:** BSFA for lung SBRT with missing data management shows convincing results in predicting dosimetric parameters for lung SBRT.

Therapy Interactive ePoster Discussion

TRACK 6

Therapy: Outcome Modeling and Assessment I

MO-lePD-TRACK 6-01

Patient-Reported Outcomes+ (PRO+) for Remote symptom Management, Featuring Automated Triage Pathways

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Purpose: Despite the promising results, symptom monitoring is not widely implemented in the clinic. This is largely due to the additional staffing that is needed to triage the increased patients' symptom reporting. We herein propose a PRO+ platform for remote symptom management, where the plus sign represents automated triage pathways. This includes AI and evidence-based triage chatbot and severity classification algorithm, to reduce manual triage needs. **Methods:** 22 most common symptoms were included in the triage system. For each symptom, triage pathways were built in chatbot format. Three metrics were used for ongoing evaluation during development. 1) Patient engagement rate. Volunteer patients used the system to report symptoms. If a chatbot conversation was initiated and finished, it was counted as a successful engagement. 2) Chatbot triage efficiency. Nurses reviewed the reported symptom details and marked the situations that enough information was collected to provide patients clinical advice. Corresponding percentage was calculated. 3) Classification accuracy. The classification algorithm results were compared with manual results by nurses, and accuracy was reported. **Results:** The PRO+ platform was built. Patients can select any symptom to report. Once a symptom is selected, the chatbot will carry a conversation with patient to collect necessary information for triage. The chat bot conversation is adapted real-time to the reported symptoms and conversation. 150 volunteer patients reported symptoms, and the engagement rate was 91%. 80% of the time the chatbot had collected enough information for nurses to provide clinical advice. The severity classification discrepancy was 95%. **Conclusion:** A PRO+ platform was built for symptom management. It has automated triage pathways that can potentially allow clinics to provide remote symptom monitoring with reduced manual triage needs. More study is needed to examine the chatbot questions, severity classification performance, and staffing model required before wide adoption in the clinic.

MO-lePD-TRACK 6-02

Using in Vivo EPID Measurements to Model Acute Gastrointestinal Toxicity

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Purpose: Currently for abdominal treatments, estimates of bowel toxicity are based on pre-treatment dose-volume histogram data. However, the actual dose the bowel receives can depend on intrafraction anatomy changes, such as weight loss and changes in bowel filling. We propose a method to model bowel toxicities, incorporating on-treatment patient information using in vivo transit electronic portal imaging device (EPID) images. **Methods:** For 63 patients treated to the lower thorax, abdomen, or pelvis on the Varian Halcyon, chart review was performed to measure incidences of toxicity. Twenty patients presented with acute gastrointestinal (GI) toxicity. For each treatment plan, the absolute volume dose-volume histogram of the bowel was exported and analyzed. The Halcyon collects in vivo EPID images for all treatments, which were downloaded and analyzed with an automated framework. Over the course of treatment, EPID measurements quantifying changes in radiation transmission in vivo were summed after every fraction. Logistic models were used to test correlations between acute GI toxicity and bowel dosimetric parameters as well as in vivo EPID measurements. **Results:** The incidence of toxicity versus the volume of 40 Gy fit with a logistic function was superior to an average model ($p < 0.0001$) and agrees with previously published models. The incidence of toxicity versus the sum of in vivo transmission measurements showed marginal significance after 15 fractions ($p=0.10$) and a significance of $p < 0.05$ is seen after 20 fractions when compared to an average model. **Conclusion:** This study agrees with other models that reducing bowel dose results in lower acute GI toxicity. Additionally, the study supports the hypothesis that in vivo EPID measurements can be used to monitor and model acute GI toxicity. From this study, it is unknown if the change in the patient's anatomy is a source of toxicity or just correlated.

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MO-lePD-TRACK 6-03

Impact of Heart Irradiation on Survival (OS) in Two Patient Cohorts: Locally Advanced Non-Small Cell Lung Cancer (NSCLC) Patients (pts) Treated Conventionally and Early Lung Cancer Patients Treated with Lung SBRT

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Purpose: Recent evidence suggests that heart irradiation decreases OS in patients treated with radiation therapy (RT) for locally advanced (LA) NSCLC. Less is known about cardiac toxicity pts who receive SBRT to the lung. **Methods:** 134 pts treated for LA-NSCLC with conventionally fractionated RT and 197 pts treated for early stage cancer with lung SBRT at Mayo Clinic Arizona were included in this study. Each group was analyzed separately. We used a multivariate Cox model Akaike Information Criterion (AIC) to select patient specific covariates predictive for OS. We digitally subdivided each heart into 64 segments and used a multivariate Cox model with patient specific covariates V%_D for each segment to determine irradiated segments predictive for OS. We used a clustering algorithm to determine irradiated heart regions predictive for OS. **Results:** 134 patients were treated for LA-NSCLC (median dose: 62Gy, 1.8-2.0 Gy/fx) and 197 pts were treated with SBRT (48Gy in 4fx and 50Gy in 5fx). High doses delivered to the right-superior region of the heart were predictive for decreased OS in LA-NSCLC patients. The best predictive index was V%_55Gy (HR = 1.04/1%, $p<0.01$). Low doses to the inferior region of the heart were predictive for decreased OS in SBRT patients. The most predictive index was V%_2Gy (2Gy equivalent dose, HR = 1.01/1%, $p=0.01$). The LA-NSCLC model applied to SBRT patients did not predict a reduction of

OS because of significantly lower doses to the heart in SBRT treatments. **Conclusion:** Heart irradiation is associated with reduced OS in LA-NSCLC patients and in patients treated with lung SBRT. A comparison of the two patient cohorts suggests that different etiologies may be responsible for reduced OS in each.

MO-lePD-TRACK 6-04

Predictive Factors of Biochemical Failure Free Survival for Prostate Cancer Patients After External Beam Radiation Therapy

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Purpose: To identify external beam radiation therapy (EBRT)-related and diagnostic factors that predict biochemical failure-free survival (BFFS) for prostate cancer patients. **Methods:** Patients diagnosed with localized prostate adenocarcinoma (N0, M0) between January 2005 and December 2016 and treated with curative EBRT using a Varian linear accelerator at one of the four cancer centers in Alberta, Canada, were retrospectively included. Biochemical failure-free survival (Astro-Phoenix definition) was calculated using Kaplan-Meier estimator. Patient demographic and diagnostic factors, as well as EBRT treatment planning and delivery parameters, were collected. Cox proportional hazard model with LASSO regularization and random survival forest were used to identify the predictive factors. **Results:** This study included a cohort of 2827 patients with a median follow-up of 6.4 years (IQR: 4.0 - 9.1 years). The BFFS for this patient cohort was 85% at 5 years (CI: 83.5%-86.4%). The statistically significant factors in the Cox model with LASSO regularization were: Gleason score, prostate-specific antigen (PSA) at diagnosis, clinical T stage (increased risk of biochemical failure with worse diagnosis), PTV V98% (lower risk with better coverage), and irradiation technique (2-phase pelvic irradiation had a lower risk than seminal vesicles having lower dose than prostate). The five most predictive variables in random survival forest were, in order of decreasing importance: Gleason score, PSA at diagnosis, PTV V98%, clinical T stage, and CTV D99%. Variables including dose calculation algorithm, bladder volume, hip implant, PTV D99%, hormone treatment, PTV volume, age at diagnosis, image-guidance type, RT modality, number of malignancies, equivalent PTV margin, and fiducial marker usage were not predictive in either method. **Conclusion:** Using a large retrospective cohort of patients from multiple centers, we identified that dosimetric parameters including PTV V98% and CTV D99% were predictive of BFFS in addition to diagnostic factors including Gleason score, PSA at diagnosis, and clinical T stage.

This study is supported by Alberta Innovates Graduate Studentship in Health Solution.

MO-lePD-TRACK 6-05

Survival Prediction of Locally Advanced Non-Small Cell Lung Cancer Via Machine Learning Techniques Based On Multi-Region Radiomics Features and Clinical Risk Factors

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Purpose: To develop and validate a nomogram for predicting overall survival(OS) in patients with locally advanced non-small cell lung cancer(NSCLC) using multi-region radiomics analysis that analyze CT image in the image-based segmentation and the dosimetric-based segmentation. **Methods:** From 2008 to 2018, 77 patients with locally advanced non-small cell lung cancer were included. A total of 66196 features were extracted from pretreatment computed tomography (CT) images. Rad-score was generated using Cox proportional hazards regression. The Clinical model was established by the use of clinical predictors which was selected by multivariable Cox regression model. The Nomogram model, combined with Rad-score and clinical predictors significantly associated with OS, compared with Rad-score and the Clinical model. The prediction model was evaluated by under the receiver operating characteristics curve, concordance index (C-index), calibration. Kaplan-Meier survival curve was calculated for OS comparing high-risk and low-risk cohorts based on nomogram scores. **Results:** Thirteen features were selected to construct Rad-score, significantly associated with OS. Univariable and Multivariate analysis showed that Rad-score and Histologic type were independent predictors of OS, with p values of <0.001 and 0.01, respectively. Nomogram model showed best performance (C-index: 0.87 [95%CI: 0.81-0.93], AUC: 0.88) compared with the Clinical model (C-index:0.76 [95%CI: 0.64-0.88], AUC: 0.63) and Rad-score (C-index:0.84 [95%CI: 0.77-0.91], AUC: 0.88). Additionally, participants could be classified into low- and high-risk groups by the nomogram. The calibration plots also showed good consistency between the prediction and the observation. **Conclusion:** A nomogram model, incorporate Rad-score and clinical predictors, is a useful tool for predicting OS in NSCLC patients which might facilitate clinical decision-making process.

MO-lePD-TRACK 6-06

Prediction of Therapeutic Outcomes of Patients with Non-Small Cell Lung Cancer for Stereotactic Ablative Radiotherapy Based on a Differential Model

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Purpose: For unresectable non-small cell lung cancer (NSCLC) patients at a Stage 1 or 2, stereotactic ablative radiotherapy (SART) is the first treatment choice alongside surgery. A five-year survival rate for patients received SART is reportedly about 70%, whose patients may have been suitable for SART. Therefore, the evaluation approaches to choose suitable patients for SART could be needed. We aimed to predict radiotherapeutic outcomes of patients with NSCLC from computed tomography (CT) scans using a differential model. **Methods:** Five NSCLC patients receiving SART that had planning CT and follow-up CT images [total scans: 23 (4.6 scan/patient)] were employed for extraction of GTVs and calculation of tumor cell numbers. The number of tumor cells at a time (CT scan) for each tumor, except CT images with radiation pneumonitis, was calculated by multiplying the density of tumor cells with the tumor volume obtained from the CT images. The tumor growth trajectories with the time-variant number of tumor cells were predicted by a Gompertz model with a linear-quadratic model under the assumption that each tumor cell could be resistant or sensitive to radiation. Three parameters in the models (degrees of radiation resistance and sensitivity, and the ratio of two types of cells) were optimized by a Levenberg-Marquardt method. The proposed method was evaluated with the mean absolute percentage error (MAPE) between predicted and reference numbers of tumor cells. **Results:** The MAPE ranged from 6.09 to 83.10%, and the average MAPE was 29.3%. **Conclusion:** This study showed the potential of predicting the therapeutic effect after SART.

MO-lePD-TRACK 6-07

Dosimetric Predictors of Hematological Toxicity in Cervical Cancer Patients Using the Lyman Normal Tissue Complication Probability Model

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Purpose: To identify dosimetric parameters associated with acute hematological toxicity(HT) and analyze HT in cervical cancer patients receiving helical tomotherapy(Tomo) and fix-field intensity-modulated radiation therapy(ff-IMRT) with concurrent chemo-radiotherapy using the Lyman-Kutcher-Burman (LKB) Normal tissue complication probability(NTCP) model. **Methods:** Data were collected from 269 cervical cancer patients who received Tomo or ff-IMRT from 2015-2018. Pelvis bone marrow (PBM) was contoured from the superior to the inferior boundary of the planning target volume (PTV). The cumulated dose of PBM was

evaluated by integrating two plans for each patient on the MIMTM system. The parameters (TD50(1), m, n) of the LKB model predicting $\geq$ grade 2HT (RTOG grading criteria) were determined using maximum likelihood analyses. Univariate and multivariate logistic regressions were enlisted to find the correlation between dose-volumetric parameters and the clinical factors of HT. **Results:** 202 (75%) patients experienced $\geq$ grade 2HT. The fitted LKB model parameters were: TD50(1)=30.23Gy, m=0.46, n=0.05. Per univariate analysis, NTCP (p=0.023), maximal PBM dose (p=0.064), V6-11Gy (p<0.100), and V46-55Gy (p<0.100) were associated with $\geq$ grade 2HT. NTCP (p=0.032; AUC=0.606), maximal PBM dose $\geq$ 54.5Gy (p=0.024, AUC=0.601), and V47 $\geq$ 11.5% (p=0.036; AUC=0.596) were significant predictive factors of $\geq$ grade 2HT from the multivariate logistic regression analysis. **Conclusion:** The volume of the PBM of patients treated with concurrent chemo-radiotherapy subjected to both low and high-dose irradiation was associated with HT. NTCP, maximal PBM dose $\geq$ 54.5Gy, and V47Gy $\geq$ 11.5% were strong predictors of toxicity. Hence, avoiding radiation hot spots on PBM could reduce the incidence of severe HT.

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Vendor Showcase

TRACK 8

Siemens Healthineers

MO-SHOWCASE-TRACK 8-5-01

First Impressions with Direct Organs

A Whitley^{1*}, M Skowronski^{1*}, (1) Montgomery Cancer Center, Montgomery, AL

President's Symposium

TRACK 1

President's Symposium: The Importance of Creativity in Science

MO-CD-TRACK 1-00

The Importance of Creativity in Science

Moderator: James Dobbins, Duke University Medical Center, Durham, NC

MO-CD-TRACK 1-01

Welcome and Introduction

MO-CD-TRACK 1-02

RSNA President, Remarks

MO-CD-TRACK 1-03

ASTRO President, Remarks

MO-CD-TRACK 1-04

J. Dobbins, Duke University Medical Center:

AAPM President, Introduction The Importance of Creativity in Science: A Brief History....

MO-CD-TRACK 1-05

B. Tromberg, NIBIB:

Physics and Bioengineering: Converging Disciplines with Creativity and Impact

MO-CD-TRACK 1-06

Panel Discussion

MO-CD-TRACK 1-07

T. Mackie, University of Wisconsin:

Panel Member

MO-CD-TRACK 1-08

M. Giger, University of Chicago:

Panel Member

MO-CD-TRACK 1-09

D. Jaffray, The University of Texas MD Anderson Cancer Center:

Panel Member

MO-CD-TRACK 1-010

N. Ramanujam, Duke University:

Panel Member

MO-CD-TRACK 1-011

R. Jeraj, University of Wisconsin:

Panel Member

MO-CD-TRACK 1-012

C. Mistretta, University of Wisconsin:

Panel Member

MO-CD-TRACK 1-013

Q&A | Closing Comments

Over the past few decades, the field of medicine has undergone tremendous changes as new paradigms have altered the way in which we understand health and disease, and have accelerated the pace of innovation from laboratory to clinical practice. The speed with which the COVID-19 vaccine has been developed is a good example of modern medicine's increasingly molecular, prospective, and data driven nature. We can anticipate even greater advances over the next 10-20 years that will make medicine of the future very different than that practiced for the past century.

Where does physics fit into this evolving paradigm of contemporary medicine? How is our role in medicine likely to be different in the next few decades? What can and should we be doing to prepare for this change – and more importantly – to ensure that we contribute the greatest societal impact possible going forward? To answer these questions, we need to look at what drives innovation in science and medicine. In short: an essential element is creativity.

The suggestion that creativity is an essential element of our future as physicists may at first seem surprising, since we often associate creativity with scholars in the arts and humanities rather than the natural sciences. But science is indeed extraordinarily creative. The part of science that involves methodical collection of data and careful analysis may not seem creative at first glance – but the leaps in understanding that occur with new theories and approaches that have transformed how we understand the world inherently involve the highest levels of creativity. Creativity at its core is the ability to see what has not been seen, to imagine what has not been imagined, and to bring that into tangible expression. Key advances in science have involved exactly those kinds of creative leaps. And our ability to envision how we apply physics in new ways in medicine going forward will involve exactly that type of creativity.

We will begin our Symposium by looking at a brief history of creativity in the sciences in order to understand how essential it is to cultivate this important human endeavor. Our keynote speaker, Bruce Tromberg, PhD, Director of the National Institute of Biomedical Imaging and Bioengineering (NIBIB) at NIH will then describe areas in which he sees the greatest need for creativity in the application of physics and engineering to medicine today. He will describe some of the key areas that NIH views as essential drivers of medical advances and where creativity is needed in our discipline of medical physics. We will conclude our Symposium with a panel discussion among several of our distinguished colleagues who have embodied the best of creativity in their careers to create new areas of biomedical science, and will explore how they cultivated creativity in their work. We will also ask them to address those things that we should be doing as a community to ensure that we promote the creativity that will be needed to create our future.

Where exactly will medical physics go in the coming decades? We can anticipate that it might involve new emphases on big data, machine learning, and molecular diagnostics and therapeutics, as examples, and will involve increasing collaboration with our colleagues in other disciplines. While we may not know the exact form this future will take, we can be assured that it will take creativity to get us there. It is our hope that you will leave this Symposium inspired to bring your creative spark to this important effort.

Learning Objectives:

1. Understand how creativity has played an essential role in the history of science and medicine.
2. Understand areas in which creativity is needed today to create the future of physics in medicine.
3. Understand how individual innovators have cultivated creativity in their own careers and what needs to be done to ensure that we as a field lay the groundwork for creativity that will build our future.

Imaging Interactive ePoster Discussion

TRACK 1

Imaging: Rad/Fluoro Image Quality and Imaging Protocols

MO-lePD-TRACK 1-01

Using Exposure Index to Develop Thickness-Based Techniques for the Skull Views of a Pediatric Skeletal Survey

X Li^{1*}, K Hulme¹, E Park¹, (1) Imaging Institute, Cleveland Clinic, Cleveland, OH

Purpose: Skeletal survey is an important tool in evaluating suspected physical abuse in preverbal children (0-3 years). The need to demonstrate subtle injuries demands high contrast and spatial resolution. This is especially true for the imaging of infant skulls. The goal of this study was to develop radiographic techniques based on head thickness and the target exposure indices (EI_T) needed for clinical diagnosis. **Methods:** The study included skull radiographs of 30 patients (aged 0-32 months) who underwent pediatric skeletal surveys using one type of digital system. All images were taken table-top using manual techniques without grid. From these images, head thicknesses were measured. Exposure techniques and EI were also extracted. Most images were taken at 60 kV, with the rest at 65 kV. For the latter group, an equivalent mAs needed at 60 kV was calculated following the 15% rule. Because EI scales linearly with mAs, a new mAs was then calculated for each image to provide a hypothetical EI_T of 800 (mAs_800). Regression analysis was performed to relate mAs_800 to head thickness. From this relationship, the median head thickness of each age group was used to calculate mAs_800, which was then scaled to obtain the mAs needed for a clinically desired EI_T. **Results:** The natural logarithm of mAs_800 increased linearly with head thickness ($R^2=0.38-0.59$). For the same head thickness, a higher mAs was needed in the lateral direction. Evaluation of clinical image quality by experienced pediatric radiologists showed that EI_T of 2000 was desirable for newborns who have small and underdeveloped skulls, whereas EI_T of 1000 was adequate for toddlers. The relationship between mAs_800 and head thickness allowed a new technique chart to be developed to provide such age-dependent EI_T. **Conclusion:** The regression relationship between mAs_EI_T and anatomical thickness may be exploited to develop manual technique charts for children.

MO-lePD-TRACK 1-02

Development of a Size-Based Technique Chart for Portable Chest Images to Reduce Variability in Image Quality

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Purpose: To use the patient height and weight, which are accessible from most Radiology Information Systems, to create a radiographic technique chart for portable chest images. By providing technologists with size-based imaging techniques, it is expected that diagnostic quality images may be captured for a variety of patient sizes without exposing the patient to excess radiation. **Methods:** Technique charts for portable chest images were derived from image data, over a period of three months (170 exposures). Patients were modeled as water-equivalent cylinders, having a diameter determined from their weight and assumed density of 1.00g/cm³, and length equal to their height. Technique charts were created through a two-step process: first calculating soft-tissue attenuation coefficients for x-ray spectra ranging from 50-120kVp using SPEKTR spectrum generating code, then estimating the patients' tissue-equivalent attenuation thickness based off the image technique. The kVp and mAs were then chosen to achieve the desired detector exposure for a range of patient heights and weights. **Results:** The relationship between the patients' tissue-equivalent attenuation thickness and the square root of their weight over height was linear, having a slope of 0.63 ± 0.03 , intercept of 0.59 ± 0.51 cm and R^2 of 0.701. The technique chart greatly improved the variability in EI values, reducing the range from 360 ± 200 to 300 ± 80 . The target EI for this technique chart was 300. Further, 90% of images (64 of 71) captured after implementing the technique chart had a DI between -2 and +2. The variability in EI values was significantly less in the proposed technique chart ($p < 0.001$) than the standard technique chart. **Conclusion:** Constructing a size-specific technique chart proved useful in maintaining a consistent image quality, as defined by the EI over a wide range of patient sizes. A size-based technique chart will guide technologists in adapting standard techniques to accommodate a wide range of patient sizes.

MO-lePD-TRACK 1-03

Evaluation of Exposure for High-Speed Angiography (HSA) with the 1000 Fps Aries Image Detector

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Purpose: The Aries x-ray photon-counting detector (Direct Conversion Inc.) can image at 1000 fps and is being investigated to observe details of vascular blood flow. This study considers the input-exposure requirements of the imager and the radiation dose implications for clinical use. **Methods:** The Aries image receptor has 100-micron pixels and a 5x7.5 cm FOV. Image frames were captured using one millisecond acquisition times, an RQA5 phantom placed in the beam and technique parameters set to 70 kVp and 100 mA. A 6-cc PTW ionization chamber was used to measure exposure at the detector input. An idealized ICA saccular-aneurysm model was placed over the phantom and images of contrast flow taken at 1000 fps were evaluated. The image signal-to-noise ratio and the vessel contrast-to-noise ratio were calculated from the counts per pixel. The integrated input detector exposure was determined over the transit time of a contrast bolus through the vasculature of interest. **Results:** The RQA5 phantom images had a detector input-exposure-per-frame (EPF) of 38 μ R, with a signal-to-noise ratio (SNR) of 7.2 and contrast-to-noise ratio (CNR) for the aneurysm of 2.4. At this dose level, flow details could be clearly seen in images of the aneurysm model. Contrast-bolus transit times are typically under 200 ms, requiring less than 200 1-ms frames. The integrated detector exposure for the RQA5 phantom run would correspond to about 7.5 seconds of DSA (300 μ R/frame) at 3fps. **Conclusion:** The Aries imager has been shown to give high temporal and spatial resolution, providing visualization of detailed blood-flow patterns in vascular pathology previously unseen. Estimated exposure for HSA runs appear to be comparable to present DSA runs.

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MO-lePD-TRACK 1-04

Adaptive Dual-Energy X-Ray Imaging with Pre-Calibrated Patient-Specific Weighting Factors

I Romadanov¹, M Sattarivand^{2*}, (1) Nova Scotia Health Authority, Halifax, NS, CA, (2) Nova Scotia Health Authority, Halifax, NS, CA

Purpose: In this work we developed a novel adaptive dual-energy (aDE) algorithm, which accounts for differences in beam attenuation within imaging plane allowing for enhanced material cancellation and noise removal compared to conventional log-subtraction dual-energy (DE) algorithm. **Methods:** Pre-calibrated weighting factors were obtained with a step phantom, which consists of overlapping slabs of solid water and bone with different thicknesses. Bone and solid water thickness were varied in [0-6]cm and [0-30]cm range respectively. Thicknesses varied in orthogonal directions forming 49 regions with overlapping materials of different thicknesses. Optimal material cancellation weighting factors for each region was determined by varying weighting factors to achieve zero contrast-to-noise (CNR) between regions with overlapping materials and corresponding target material (bone or soft tissue). The optimal weighting factors were fitted to average intensities of each region in low (LE) and high energy (HE) images. Image noise was suppressed with anti-correlated noise reduction (ACNR) algorithm. ACNR weighting factors were determined by maximizing signal-to-noise-ratio of each overlapping region, and then were fitted with corresponding LE and HE region intensities. To test the DE algorithm, HE and LE images of Rando phantom were fed as inputs to the optimal fitting functions to create patient-specific weighting factors maps. Soft tissue and bone only DE images of Rando phantom were obtained with different DE algorithms and CNR for several selected regions of interest within the Rando phantom were compared for different algorithms. **Results:** CNR values for aDE algorithm were higher compared to simple log-subtraction algorithm. Implementation of aACNR algorithm results in further reduction of image noise and improvements in CNR. **Conclusion:** A novel DE algorithm was developed, which has improved material cancellation and noise reduction compared to the conventional DE technique. This method can be easily implemented in a clinical environment for real-time DE image generation.

MO-lePD-TRACK 1-05

Theoretical Investigation of X-Ray Spectral Shape Impact on Dual-Energy Image Quality: Effects of Dose Allocation, Total Dose, and Energy Ratio

I Romadanov, M Sattarivand*, Nova Scotia Health Authority, Halifax, NSCA

Purpose: The aim of this study was to develop a theoretical framework for the investigation of the impact of x-ray spectral shapes and dependent parameters (dose allocation, total dose, energy ratio) on the resulting dual-energy (DE) image quality. **Methods:** Spectral shape incident on the object was described with an analytical fitting function, partially depicting the physical mechanisms behind the polychromatic x-ray beam generation such as Bremsstrahlung radiation and its attenuation. This function was written in terms of the physical parameters, such as x-ray tube potential Emax (kVp), parameter α (related to tube mAs), and parameter β (related to photoelectric absorption coefficient). Simple phantom was developed to evaluate image quality and this model was used in theoretical calculations. The contrast to noise ratio (CNR) between a tumor region (behind bone) and its background was used as a figure of merit to quantify image quality. Analytical CNR was derived and parametrically investigated in terms of spectral shape parameters, dose allocation, and total dose. Experimental verification of the derived equation as a function of dose allocation was performed. Various dose allocations for different energy ratios were achieved by varying beam filtrations and mAs with constant total dose. **Results:** Equation for DE CNR as a function of spectral parameters, total dose, and dose allocation was derived. It was parametrically investigated and experimentally validated. It was shown that the contrast is defined by material properties only, while noise depends on relationship between spectra. Expression for dose allocation corresponding to maximum tumor CNR function was derived. **Conclusion:** The derived analytical CNR function allows for investigation of the spectral parameters and their impact on the image contrast and noise. It was shown that dose allocation corresponding to maximum CNR has strong dependency on material composition of the imaged object and cannot be reduced to one near-optimal number.

MO-lePD-TRACK 1-06

A Fully Automated Algorithm for Accurately Analyzing Star Shot Images from Linear Accelerators

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Purpose: To design a robust and fully automated image processing algorithm for accurately analyzing star shot (also known as 'spoke shot') images, as recommended by the AAPM TG-142 guidelines. The proposed algorithm was specifically designed to eliminate external inputs (i.e., 100% automated) with a focus on generalizability for maintaining increased accuracy across a multitude of highly variable beam characteristics and image qualities, while also providing a significant time-save in the clinic. **Methods:** A fully automated ad hoc algorithm was designed for accurately and reliably analyzing star shot images without the need for external inputs. Performance of the algorithm was tested across a set of 29 star shot images. The images were comprised of a total of 165 beams. Analysis was performed on both a per-image and a per-beam basis. Generalizability was demonstrated with an array of gantry, collimator, and couch star shot configurations typically found in the clinical setting. In addition, because beam properties are likely to vary within the clinical settings, whether due to machine restraints, unique procedures, or preferences of the physicist themselves, the algorithm was designed to be invariant to beam widths, number of beams, and relative beam doses. **Results:** The proposed algorithm correctly analyzed all beams in 28 of the 29 star shot images (96.6%). Additionally, of the 165 beams, 163 (98.8%) were correctly identified. The single image that failed served as a test case for the resolution of the image processing algorithm as it could not resolve two beams with a large proportion of overlapping dose. **Conclusion:** This work demonstrated a robust solution for fully automating the star shot image analysis process with high accuracy and reliability, while also saving valuable time. Such automation provides a vastly improved workflow for medical physicists to perform necessary QA tasks more efficiently.

MO-lePD-TRACK 1-07

Impact of Flexible Noise Control (FNC) Processing Parameters on Image Quality: A Chest Phantom Study

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Purpose: To investigate the impact of flexible noise control (FNC) processing (Fujifilm, Tokyo, Japan) on digital radiography chest imaging. **Methods:** An anthropomorphic chest phantom was imaged using a clinical chest protocol with 85 kV and five radiation dose levels (0.7, 1.0, 1.4, 1.8, and 2.2 mAs) at 40'' source-

to-image distance with scatter removal image processing applied. All images were processed without and with FNC. Selected options for the four processing parameters of FNC included filter control FFC (C and F), frequency balance FNB (A, C, E, and G), density control FNT (A, B, and C), and strength control FNE (0.1, 0.5, and 0.9), resulting in 72 combinations. Subtractions were performed between images without and with FNC from the same setting to assess noise and grayscale changes. Noise-only images were generated by subtracting two identical acquisitions and two regions-of-interest (ROIs), one over lung field and the other over liver area, were drawn to quantify the noise level for each setting. Texture analysis was performed in the liver ROI using the gray-level co-occurrence matrix (GLCM). Edge-preserving capability was evaluated using a line profile through a lung/rib interface. **Results:** Use of FNC resulted in 15-70% noise reduction depending on the dose levels, parameter choices, and ROI selections. Options in FNB and FNE parameters did not show performance difference. Option F for FFC and A for FNT demonstrated the highest noise reduction compared to other options. Information measure of correlation from GLCM demonstrated the largest difference among the parameter combinations, with CGC the closest to baseline and FGA/B the most different. The appearance of edge interfaces was also affected, with FGB and CGB providing better edge-preserving capabilities. **Conclusion:** The impact of FNC on chest imaging was demonstrated using a phantom. Noise suppression and edge-preserving capability depend on radiation dose level and FNC parameter choice.

Imaging Interactive ePoster Discussion

TRACK 2

Imaging: MRI Motion, Safety, Phantoms and more

MO-lePD-TRACK 2-01

Development of 3D Rosette K-Space Trajectory in Ultra-Short Echo Time MRI Applications

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Purpose: Ultra-short echo time (UTE) MRI is the promising imaging modality for the direct measurement of tissues with short transverse relaxations (T₂), e.g., myelin in the white matter, globus pallidus, sodium imaging, and lung imaging. The most common k-space patterns for UTE applications are the 2D and 3D radial trajectories. The purpose of this study is to develop 3D rosette k-space pattern for UTE applications. Since rosette k-space pattern shares the advantages of radial acquisition, and it is capable of combining compressed sensing for acceleration, rosette trajectory may be a better choice for UTE MRI. **Methods:** The equations and figures of 3D rosette trajectory are shown in the supporting document. The performance of 3D rosette and 3D radial k-space patterns with different sampling strategy was compared based on the reconstructed image quality from a simulated phantom, in terms of the signal-to-noise ratio (SNR) and accuracy. In addition, the 3D UTE sequence using rosette trajectory was simulated on the JEMRIS software (<http://www.jemris.org/>). The parameters were: FOV=240mm, matrix size=120x120x120, hard RF pulse for excitation with duration of 0.1ms, 15 degree flip angle, TE=0.05ms, and TR=5ms. **Results:** The SNR of 3D rosette k-space trajectory is larger than the SNR of 3D radial, regardless of sampling pattern (11.86 > 9.10 with full k-space acquisition, 12.39 > 7.85 with undersampling pattern of 25% spokes). In addition, 3D rosette k-space provided more accurate reconstructed images than 3D radial. The simulation results of 3D UTE sequence with rosette trajectory were shown in the supporting document. The reconstructed images captured most of the properties in original images. **Conclusion:** The performance of 3D rosette k-space trajectory is generally better than 3D radial, in terms of SNR and accuracy. In addition, the feasibility of using rosette trajectory in UTE MRI has been demonstrated by simulation.

MO-lePD-TRACK 2-02

Characterization of Diffusion Tensor Parameters in Normal-Appearing White Matter

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Purpose: To quantify the fractional anisotropy (FA), the apparent diffusion coefficient (ADC) and the radial diffusivity (RD) in normal-appearing white matter (NAWM) surrounding focal brain lesions in patients with multiple sclerosis (MS) and to establish a change gradient of these 3 factors over the distance to the focal lesion. Then, by comparing the gradient between patients with several stages of MS, evaluate the possibility of predicting new focal lesions. **Methods:** A diffusion tensor (DT) post processing was completed to magnetic resonance images (MRI) of MS patients. The FA, ADC and RD values in focal lesions were obtained using the software toolkit. Then, a color map in the images was constructed to better visualize the lesion limits, also 2 regions of interest (ROI) of different sizes were defined surrounding the focal lesions. Once these 3 variables were measured in the lesions and in every ROI, an analysis of change gradient was determined along the distance to the focal lesion. This was done in 7 different patients. **Results:** In an early stage of the disease (relapsing-remitting MS), the patients showed a significant decrease in ADC and RD values as the ROI moved away from the lesion, being bigger in the lesions than in the ROIs. In contrast, the FA had its greatest increase in this early disease stage as the ROI was further away from the lesion. New lesions were shown in cases where there was a greater change gradient. **Conclusion:** This work helps to provide an accurate prognosis of the disease and the possibility to analyze the use of early treatment to prevent the development of clinically definite MS.

MO-lePD-TRACK 2-03

Effects of MLC Motion on B₀ in 0.35 T MRI-Linac

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Purpose: Currently, commercial MRI-Linacs only turn on the beam when the gantry and MLCs are stationary. However, treatment times could potentially be reduced by allowing the beam to be on during MLC motion (e.g., Sliding Window IMRT or VMAT). The purpose of this study was to quantify the effects of MLC motion on B₀ for a 0.35 T MRI-Linac (ViewRay, Mountain View) and determine their effect on image quality. **Methods:** B₀ center frequencies and field homogeneities were measured in a 24 cm diameter spherical volume (DSV) water phantom doped with 5 mM NiSO₄ (T₁ = 330 ms at 0.35 T) during MLC sweeps. Free induction decays (FIDs) were acquired every 0.29 s while the gantry was static at gantry angles 0-330° in increments of 30°. The spectra were calculated from the FIDs. The peak offset and full width at half maximum (FWHM) were calculated after fitting the spectra to a Lorentzian function. Measurements were conducted without gradient shimming. MRIs were also acquired during MLC sweeps using the 2D cine radial (8 fps) and Cartesian (4 fps) balanced steady state free precession (bSSFP) acquisitions normally used during treatment. MRIs were inspected for presence of artifacts. **Results:** Variations in the mean center frequency and FWHM between gantry angles are within expected ranges. B₀ and field homogeneity standard deviations from MLC motion were <2.0 and <1.3 Hz, respectively. No conspicuous image artifacts were observed during the MLC sweeps. **Conclusion:** MLC motion had a minimal effect on image quality and B₀. Therefore, dynamic beam shaping using MLCs could potentially be performed without an impact to image quality and target tracking.

ViewRay provides financial support and speaker fees to Washington University in St Louis and their faculty (unrelated to this abstract). Washington University has a Master Research Agreement with ViewRay.

MO-lePD-TRACK 2-04

Exploration of the Predictive Value of Delta Radiomics Texture Features Extracted from Low-Field Magnetic Resonance Images in Pancreas Stereotactic Body Radiotherapy Patients

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Purpose: Magnetic resonance-guided radiotherapy (MRgRT) provides radiation oncology departments a modality that acquires high contrast images ideal for anatomical visualization for target localization, motion tracking, and gating. The purpose of this work is to explore the predictive potential of delta radiomics texture features (i.e., changes in radiomic texture features over time) extracted from daily set up images of pancreas patients treated with MRgRT. **Methods:** The gross tumor volume (GTV) was delineated on daily set up images of 27 pancreas patients treated with 33-50 Gy in 5 fractions. Gray level intensities for each GTV were quantized to 64 levels using the histogram equalization method and 39 second order texture features were extracted. Two sets of delta radiomics features were calculated: absolute change (DF(ABS)) and normalized change (DF(NORM)). DFABS is the absolute difference between fraction 1 and fractions 2-5. DF(NORM) features are DF(ABS) normalized by fraction 1 feature values. Delta radiomics features were ranked by predictive importance using the Gini Index obtained during training of a random forest (RF) model. The RF models constructed using DF(ABS) and DF(NORM) from fraction 1 and 2 produced in the highest area under (AUC) the receiver operating characteristic curve and selected the same two DF(ABS) and DF(NORM) features as most important. Gray level co-occurrence matrix-based sum average and gray level size zone matrix-based large zone low gray level emphasis were used to construct RF models for evaluation of predictive performance using leave-one-out (LOO) analysis to obtain AUC. **Results:** The RF using DF(ABS) features achieved an AUC, with a 95% confidence interval, of 0.77(0.58–0.96) and the RF with DF(NORM) achieved an AUC = 0.78(0.59–0.97). **Conclusion:** Delta radiomics texture features calculated from fraction 1 and 2 set up images during SBRT may provide valuable prediction of treatment response with time to modify the treatment and improve outcomes.

MO-lePD-TRACK 2-05

Improved Myelin Water Imaging Using 3D GRASE and Orthogonal Matching Pursuit Reconstruction

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Purpose: Noninvasive MRI-based quantification of myelin provides valuable information for the study of neurological and neuropsychiatric disorders. Multi-echo spin echo MRI data are analyzed using multi-exponential T2 relaxation to assess myelin by computing the fraction of myelin water (usually assumed as 10<T2<40 ms) to the total signal. This approach for myelin water fraction (MWF) is challenged by the fast decay of the myelin water signal and the need for high resolution and signal-to-noise ratio. Three dimensional GRASE imaging accelerates image acquisition for whole-brain application. However, data fitting is sensitive to noise due to the ill-conditioned nature of the problem, which is conventionally addressed using regularized non-negative least-squares (NNLS) analysis. Here we investigate the application of orthogonal matching pursuit for improving the quality of MWF maps. **Methods:** 3D GRASE data were acquired from one patient with multiple sclerosis in four different session in a study of oral acetazolamide over a period of 5.5 month using a 3.0 T Philips MRI system with a multi-channel head coil. Acquisition parameters were TR/TE/ΔTE = 747/7/7 ms; number of echoes, 32. Non-negative least squares reconstruction was used as reference. Orthogonal matching pursuit was implemented with nonnegativity constraint (NNOMP). The number of components (sparsity) was empirically optimized and four components was found to yield the best results. **Results:** The NNOMP-reconstructed MWF maps consistently showed improved visual quality in all scans. Figure 1 shows a comparison between NNLS and NNOMP in one scan. White matter MWF values calculated from NNOMP were approximately 10-15% higher than those obtained from NNLS. **Conclusion:** NNOMP produces MWF maps with improved quality. The improved performance is attributed to the regularization imposed by the sparsity constraints in NNOMP. No spatial regularization was imposed on either of the techniques. Analysis on a larger sample is underway to assess NNOMP both qualitatively and quantitatively.

MO-lePD-TRACK 2-06

Development of a Clinical Protocol to Rapidly Determine the Potential Impact of Radiofrequency Induced Heating in Patients with Abandoned Epicardial Leads During MRI

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Purpose: In the 2017 Heart Rhythm Society consensus statements, scanning of MR nonconditional CIEDs in the presence of epicardial leads were classified as contraindications. While there are numerous studies on transvenous leads, there is insufficient data to comment on the safety of patients with epicardial leads during MRI. This study seeks to clarify safety concerns in patients with epicardial leads owing to the sensitivity of its applications. **Methods:** a tissue-equivalent phantom was designed according to the ASTM standard for measurement of radiofrequency (RF) induced heating on or near passive implants. The polycarbonate phantom and was filled with a polyacrylic acid (PAA) gel-saline that has similar electrical and thermal properties with the human body tissue. Forms fabricated to hold the lead and fluoroptic temperature probes permits easy localization and configuration (lead length and shape) within the gel. An initial thermometric study was performed on a 3T Siemens Tim Trio MRI system with routine clinical pulse sequences on a straight 58.4cm bipolar epicardial lead positioned in the phantom at a depth of 5cm below the surface. **Results:** Preliminary data shows no significant heating above the background in presence of the lead for all pulse sequences examined. **Conclusion:** The design of the phantom, lead, and temperature probe holder permits a rapid determination of the potential heating that may occur during an MRI study given the specific presentation of the lead within the patient. The system is relatively easy to setup and use within the clinic and can provide additional information to better evaluate safety concerns with border-line cases where the potential benefit of MRI may outweigh the risks. Further investigation on the variation of induced RF heating with lead type, insulation, lengths, and presentation within the body are underway and will be discussed.

MO-lePD-TRACK 2-07

Characterization of a Tissue-Equivalent Phantom for Thermometric Studies in MRI

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Purpose: To measure the complex permittivity and thermal properties of a gel-saline phantom designed for thermometric studies in a 3T MRI scanner. **Methods:** A gel-saline phantom was designed according to the ASTM International standard for measurement of radiofrequency (RF) induced heating on or near passive implants during magnetic resonance imaging. The dielectric constant and electrical conductivity of the phantom was obtained using an open-ended coaxial probe permittivity measurement method that relies on changes in the complex reflection coefficient of an optimized probe due to the material under test. A simple implementation of this method was achieved with a basic RF electronic circuit consisting of a signal generator, power splitter, circulator, and oscilloscope, as opposed to the widely used Vector Network Analyzers (VNA). The non-ideal behavior of the coaxial probe due to spurious reflections and ohmic losses were accounted for by a scattering matrix that relates the measured reflection coefficient to the reflection coefficient at the probe tip. The probe was calibrated with three standard terminations including a short circuit, an open circuit, and a saline solution with a similar permittivity to that of the phantom. Assessment of the thermal conductivity and specific heat capacity of the phantom is underway and will be discussed. **Results:** A dielectric constant of 82.1 ± 1.4 and electrical conductivity of 0.47 ± 0.04 S/m was measured at a frequency of 127.8 MHz. This agrees with the recommended values of the ASTM standard of 80 ± 20 and $0.47 \pm 10\%$ S/m, respectively, for the phantom. **Conclusion:** A novel equivalent circuit for open-ended probe permittivity measurement has been developed and verified in the electrical characterization of a tissue-equivalent phantom. The complex permittivity of the phantom agrees with recommended values and indicates its usability for safety assessment in MRI.

Multi-Disciplinary Interactive ePoster Discussion

TRACK 3

Multi-Disciplinary: Image Guidance: Cone-beam CT

MO-lePD-TRACK 3-01

A Hybrid Approach for Correcting CBCT by Exploiting Individual Strength of Deep Learning and Deformable Registration

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Purpose: To propose a novel hybrid approach, which utilizes prior planning CT (pCT) with deep learning, for correcting daily pretreatment CBCT to facilitate online verification of proton range. **Methods:** Previously proposed methods for correcting CBCT using CycleGAN are often limited in suppressing image artifacts, while maintaining a high geometric fidelity. In contrast, classic methods of deforming pCT onto CBCT provide more accurate intensities but suffers from geometric uncertainties. In the proposed method, high and low spatial-frequency components in each CBCT and pCT image are separated via Fourier transform. Then, CycleGAN and deformation are respectively used to correct geometry-weighted (high frequency) and intensity-weighted (low frequency) components which are subsequently combined, thereby each process deals with only the component weighted toward its strength. The CycleGAN was trained with 50 high-frequency image pairs of abdominopelvic CBCT/CT. A fast deformation algorithm followed by a threshold-based internal air correction was employed to process pCT. Additional datasets of 4 and 7 CBCT/pCT image pairs were used respectively to tune meta-parameters and evaluate the mean absolute error (MAE) performance of the proposed method against the same-day CT. **Results:** Correcting a CBCT dataset took less than a minute including the CycleGAN and deformation processes. The proposed method produced more accurate intensities than using CycleGAN alone (MAE, 37 ± 4 vs. 42 ± 4 ; $P<0.001$, paired t-test) and significantly reduced artifacts. When comparing with the deformation-alone method, the improved accuracy was pronounced in bony structures (defined by HU > 200; MAE, 73 ± 12 vs. 87 ± 19 ; $P=0.045$, paired t-test) reflecting robustness against misregistered bones in the deformed pCT. **Conclusion:** In the proposed hybrid approach, CycleGAN and deformable registration complemented each other to perform better than either individual method. The improvement in bony structure fidelity has implications for estimating proton range which is sensitive to geometric errors in those structures.

MO-lePD-TRACK 3-02

CBCT Air Artifact Reduction Using a Simulation-Based Image Translation Model

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Purpose: Cone-beam computed tomography (CBCT) has long been an established standard for patient setup. With the growing focus on adaptive radiotherapy (ART) and the rise of purpose-built ART platforms, the importance of CBCT-based guidance in facilitating adaptive workflows is only rising. A primary challenge in this reliance on CBCT is the persistent issue of image quality in scans that are marred by artifacts, especially streak artifacts caused by the motion of intestinal gas during CBCT acquisition. We present preliminary results of a simulated reconstruction image translation model dedicated to artifact reduction for CBCT-guided abdominal ART. **Methods:** The image-to-image translation approach employed here relies on a training data set consisting of paired CT images and simulated reconstructions derived from these images ($n = 35$ patients, 20 pairs per patient). To simulate gas motion-induced artifacts in a reconstructed image, deformations of the air containing regions in the source CT image were introduced during projection acquisition every 36 projections over a total of 360 projections. The trained model, which maps from an input reconstruction to a synthetic CT output, was tested on unseen simulated reconstructions as well clinical CBCT data. **Results:** When applied to unseen simulated reconstructions, the trained model achieves a notable reduction of streak artifacts and accurate HU values in the synthetic CT outputs. Preliminary tests on out-of-domain clinical CBCT data demonstrate a general smoothing effect that similarly acts to reduce the presence of artifacts. **Conclusion:** Results achieved with the simulation-enabled image translation model explored here demonstrate the promise of a fast, image-to-image translation model in the CBCT artifact reduction task, which is a topic of special relevance to an abdominal ART workflow.

MO-lePD-TRACK 3-03

Deformable Image Registration Using Spatial Transformation-Based Network for CBCT-Guided Radiotherapy

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Purpose: Longitudinal CBCTs from image guided radiation therapy reveal volumetric changes in patient anatomy that may warrant plan changes or be predictive of outcomes. However, quantitative assessments are challenged by poor CBCT image quality, especially in the abdomen, and by the need for fast and accurate deformable registration to make real-time clinical decisions. To address these challenges, we propose a deep learning-based CBCT-CBCT deformable registration method. **Methods:** The proposed deformable registration workflow consists of training and inference stages that share the same feed-forward path through a spatial transformation-based network (STN). The STN consists of a global generative adversarial network (GlobalGAN) and a local GAN (LocalGAN), which capture the overall motion and the local motion respectively. The network was trained by minimizing the losses combining image similarity with the deformable vector fields (DVF) regularization without the supervision of ground truth DVFs. In the inference stage, the global DVF and patches of local DVF were predicted by the trained GlobalGAN and LocalGAN. The local DVF patches were subsequently fused to generate a local whole-image DVF, which is summed with the global DVF to obtain the final DVF. The proposed method was investigated using 100 abdominal CBCTs from 20 radiotherapy patients. Each patient was treated in five fractions over 1 to 2 weeks. **Results:** The resulting images show that the deformably registered CBCT images are visually similar to the target CBCT image. Additionally, quantitative assessment shows that the target registration error (TRE) across all patients and fractions is 2.91 ± 1.16 mm after registration with the proposed method. An accurate CBCT-CBCT deformable registration took less than 3 seconds. **Conclusion:** We developed a deep learning-based CBCT-CBCT registration method and investigated its feasibility and performance in fractionated radiotherapy. This registration tool could provide fast and straightforward image registration for CBCT-guided radiotherapy.

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MO-lePD-TRACK 3-04

High-Quality CBCT Reconstruction from Sparse and Limited-Size Projections Using a Joint Deformable Registration and Reconstruction Network (Reg-ReconNet)

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Purpose: To develop a joint deformable registration and reconstruction network (Reg-ReconNet) to reconstruct high-quality, field-of-view (FOV)-extended CBCT images from sparse and limited-size projections. **Methods:** Reg-ReconNet leverages prior CT/CBCT information to reduce the number of cone-beam projections needed for on-board CBCT reconstruction, and to extend the effective FOV. It starts with an unsupervised 2D-to-3D deformable registration network (2D3D-RegNet), which deforms prior images to estimate on-board CBCT images by matching digitally-reconstructed-radiographs (DRRs) of the deformed prior images to acquired on-board cone-beam projections. The 2D3D-RegNet was followed with a deep cascaded CBCT reconstruction network, which features recurrent inputs of cone-beam projections at each cascade level, repeated data fidelity enforcement layers composed of embedded forward/backward projection operators, and task-specific convolutions and residual connections to correct aliasing, streaking and truncation artifacts from reconstruction. We used 14 patient cases from a TCIA lung library to train and evaluate Reg-ReconNet, each of which has multiple 4D-CT image sets acquired throughout the radiotherapy treatments. For each patient, we extracted one phase image from the first 4D-CT set as prior information, and simulated cone-beam projections from all intra-fractional/inter-fractional 4D-CTs for training/testing (10/4 patients, corresponding to 400/200 image volumes). The reconstructed images were compared against the 'ground-truth' original 4D-CTs using

the relative-error (RE) metric. **Results:** Using 5 projections, the average($\pm$ s.d.) REs of reconstructed CBCTs by the Feldkamp-Davis-Kress (FDK) algorithm, iterative algebraic-reconstruction-technique with total-variation regularization (ART-TV), 2D3D-RegNet (without following ReconNet), ReconNet (without preceding 2D3D-RegNet), and Reg-ReconNet were $198.4\pm 37.8\%$, $44.2\pm 3.8\%$, $20.3\pm 5.1\%$, $17.8\pm 6.8\%$, and $16.9\pm 3.8\%$, respectively. Corresponding results were $112.8\pm 21.8\%$, $27.6\pm 1.8\%$, $18.1\pm 4.8\%$, $11.5\pm 4.3\%$, and $10.4\pm 2.6\%$ for 20 projections, and $78.9\pm 14.3\%$, $19.7\pm 3.2\%$, $16.8\pm 4.4\%$, $9.1\pm 3.2\%$ and $7.6\pm 1.8\%$ for 60 projections. **Conclusion:** Reg-ReconNet incorporates prior information into new CBCT reconstruction through joint deformable registration and image reconstruction. It can reconstruct high-quality images from sparse projections and extend the effective FOV by merging and updating prior information.

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MO-lePD-TRACK 3-05

Physics-Based Motion Artifact Simulation in CT/CBCT Images Using 4DCT Phases

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Purpose: To realistically mimic motion and scatter artifacts in lung CT and cone-beam CT (CBCT) using 4DCT phases and physics-based simulation. In our recent paper [1], we presented a new data augmentation pipeline for physics-based simulation of baseline CBCT scatter artifacts on planning CT to allow utilization of more accurate contouring. Here, we extend our pipeline to include motion artifacts. **Methods:** 600 projections equivalent to a 60 second (10 projections/second) 4DCT scan were used for generating motion plus scatter/noise artifact induced synthetic CBCTs. First scatter artifacts/noises were extracted from week1 CBCTs using power-law adaptive histogram equalization method and induced to each 10 phases of 4DCT. Four projections/phase were repeated 15 times to reconstruct a 60 second scan. Hence, we reconstructed [4 projections/phase x 10 phases] x 15, equivalent to 600 projections for a 60 sec scan. Eventually, corresponding motion plus scatter artifact induced synthetic CBCT was reconstructed using CBCT reconstruction parameters using iterative Ordered-Subset Simultaneous Algebraic Reconstruction Technique. The geometric parameters used for reconstruction were taken directly from the clinical practice for CBCT reconstruction. Synthetic CBCTs were evaluated against their ground-truth CBCT using histogram comparison and structure similarity index (SSIM) & root mean square error (RMSE). **Results:** The histogram of motion-artifact simulated data via our pipeline aligned very well with histogram against their ground-truth CTs and achieved RMSE and SSIM of 0.04 and 0.83, respectively. **Conclusion:** We show that we can realistically simulate motion artifacts using our pipeline for CT as well as CBCT images using 4DCT phases. The augmented motion and scatter-artifact simulated data will allow to segment OAR/GTV structures across CT and CBCT images using the same 3D deep learning segmentation model we presented in [1].Ref:[1] Alam SR, et al. Generalizable cone beam CT esophagus segmentation using physics-based data augmentation. Physics in Medicine & Biology, 2021

MO-lePD-TRACK 3-06

Synthetic CT Generation from Contrast Enhanced Planning CT and Low-Quality Treatment CBCT Using a Novel Dual Convolutional Neural Network

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Purpose: The study aims to produce quantitative cone-beam computed tomography (CBCT) by learning HU features from planning CT while maintaining the structural integrity of on-treatment CBCT using a dual convolutional neural network (CNN) architecture. **Methods:** The study included paired CT and CBCT images from 49 patients (33 training, 8 validation, and 8 testing). A primal-dual 3D CNN architecture was built based on a loop, which contains a primal network and a dual network. For the primal network, CBCT was treated as initial synthetic CBCT (S-CBCT) to obtain the synthetic CT (S-CT), aiming to capture the HU information from planning CT. In the dual network, the S-CT were treated as input data to generate S-CBCT using another CNN, aiming to capture the geometry information from CBCT. The energy function was computed by a combination of the pixel intensity loss of S-CT with CT, and the pixel+texture loss of S-CBCT with CBCT. Thus, the full objective was: $E_T = aE_{CT}(SCT-CT) + bE_{CBCT}(SCBCT-CBCT)$, where a and b are weightings to control the contribution of each network. Mean absolute error (MAE), peak signal-to-noise ratio (PSNR), normalized cross-correlation (NCC), and structural similarity index (SSIM) were used to evaluate the model performance. **Results:** The dual CNN model performed better when the total loss was close to the loss between S-CT and CT using unequal weightings ($a=0.9$ and $b=0.1$). The visual inspection of S-CT showed improved image quality. All parameters tested were improved from CBCT to S-CT (equal weightings) and S-CT (unequal weightings). The results for the 4 abovementioned images are: MAE (78.88, 65.45, and 59.97); NCC (0.89, 0.92 and 0.95); PSNR (24.28, 26.30 and 28.65); SSIM (0.58, 0.67, and 0.75), respectively. **Conclusion:** In this study, a primal-dual CNN was built to generate synthetic CT with more accurate HU, reduced noise, and organ structure integrity for the treatment day.

MO-lePD-TRACK 3-07

Thoracic CBCT-Based Synthetic CT for Lung Stereotactic Body Radiation Therapy

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Purpose: Cone beam CT (CBCT) has been widely implemented to guide the daily delivery of lung radiotherapy including Stereotactic Body Radiation Therapy (SBRT), however, clinical applications that depend on quantitative CBCT are limited by poor tissue contrast, image artifacts, and instability of Hounsfield Unit (HU) values. The purpose of this study is to develop a deep learning-based approach to improve CBCT image quality for quantitative analysis during radiotherapy. **Methods:** A deep learning-based model, which integrates histogram matching and perceptual supervision into a cycle-consistent adversarial network (Cycle-GAN), was trained to learn the mapping between paired thoracic CBCT and planning CT images. Cycle-GAN was introduced due to its ability to learn a mapping between CBCT images and paired planning CT images. Perceptual supervision was adopted to suppress interface blurring. Histogram matching was introduced into Cycle-GAN to force the synthetic CT (sCT)'s intensity distribution to be close to that of planning CT. The proposed deep learning-based algorithm was evaluated using 80 thoracic breath-hold CBCT images from 20 lung SBRT patients, each with 3-5 CBCTs. Planning CT images were used as the ground truth for evaluating the sCTs derived from co-registered CBCTs. **Results:** The mean absolute error (MAE), peak-signal-to-noise-ratio (PSNR) and normalized cross-correlation (NCC) indexes calculated between the sCTs and planning CTs were 63.2 HU, 30.2 dB, and 0.96, respectively. The proposed sCT method outperforms other current deep-learning methods and can generate a quantitatively accurate sCT in seconds. **Conclusion:** We developed a new CBCT correction method to generate a quantitatively accurate thoracic sCT from a CBCT and demonstrated that this method could significantly improve CBCT image quality. This method has the potential to enhance the accuracy of CBCT-guided radiotherapy, in particular for online adaptive applications that depend on accurate, real-time HU values for dose calculations.

Multi-Disciplinary Interactive ePoster Discussion

TRACK 4

Multi-Disciplinary: Data Science/Radiomics

MO-lePD-TRACK 4-01

Evaluation of Cone-Beam Computed Tomography-Based Radiomic Features Reproducibility: A Phantom Study

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Purpose: To explore the most influential factors on the reproducibility of cone-beam computed tomography (CBCT) radiomics features. **Methods:** Three sphere materials (acrylic, rubber, and cork: 30 mm ϕ) were placed at three different locations (upper-central, lower-central, lower-peripheral) in an anthropomorphic chest phantom. The CBCT images were acquired using Halcyon (125 kV, 168 mAs, CTDI(vol): 3.36 mGy) and TrueBeam (125 kV, 225 mAs, CTDI(vol): 3.46 mGy; Half-fan). The images were reconstructed using following three algorithms: (i) Feldkamp-Davis-Kress (FDK) algorithm in Halcyon (FDK(Halcyon)), (ii) FDK algorithm in TrueBeam (FDK(TrueBeam)), and (iii) iterative CBCT (iCBCT) algorithm with medium noise suppression level in TrueBeam (iCBCT(TrueBeam)). The targets were manually delineated in Eclipse treatment planning system version 15.6 by three medical physicists under the lung window setting. A total of 93 radiomics features, including 18 first order and 75 texture features, were extracted from the inside region of interest for each material using PyRadiomics (version 2.2.0). The effects of the inter-observer, target location and reconstruction algorithm variability on radiomics features were evaluated using coefficient of variation (CV). Features with CV value less than 5% were considered highly reproducible. **Results:** For inter-observer and inter-target location comparison, the median percentage of reproducible features in FDK(Halcyon), FDK(TrueBeam) and iCBCT(TrueBeam) were 51.6% (range: 38.7-62.4), 51.6% (range: 43.0-76.3) and 55.9% (range: 45.2-69.9), and 35.5% (range: 28.0-48.4), 33.3% (range: 26.9-40.9) and 33.3% (range: 24.7-38.7), respectively. For inter-reconstruction algorithm comparison, the median percentage of reproducible features were 20.4% (range: 11.8-29.0). **Conclusion:** Inter-reconstruction algorithm variability may have a greater influence on the reproducibility of CBCT-based radiomic features than the inter-observer and target location variability. Even with the same FDK algorithm, reconstruction device differences have a large effect on the radiomics features reproducibility. It is recommended that the reconstruction algorithm should be unified for CBCT-based radiomics analysis.

MO-lePD-TRACK 4-02

Graph Theory-Based Radiomics Features: Application of Tumor Network Structures On CT-Based Radiomics for Prognostic Prediction

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Purpose: We developed graph theory-based features and tested their feasibility for predicting prognosis in non-small cell lung cancer (NSCLC) using computed tomography (CT)-based radiomics. **Methods:** We used CT images and gross tumor volume contours of 304 patients treated with radiotherapy for NSCLC at our hospital. Patients were divided into training (n = 213) and test (n = 91) datasets. Graphs were constructed with pixels as vertices and the difference in CT values between pixels as weights given to the edges. Thresholds were set for the weights, and edges lower than each threshold were removed. Several graphs were constructed with different thresholds, and features based on graph theory concepts, such as 1 vertex feature, 51 edge features, 20 degree features, 40 minimum spanning tree features, and 40 Laplacian matrix features, were extracted. Next, we developed graph theory histogram-based features for each of these features with 45 gradients and 15 y-axis intercepts of approximate curve features, 1 peak feature, 1 kurtosis feature, and 1 skewness feature. Then, we extracted 215 graph theory-based features. Univariate cox regression analysis was used to assess each feature's prognostic power and the C-index was calculated to evaluate predictive performance. The least absolute shrinkage and selection operator cox regression model and the Kaplan-Meier method were used to assess the performance of the prognostic prediction, and results were evaluated by comparing C-index of 107 standard features (14 shape features, 18 statistical features, and 75 texture features). **Results:** C-indexes for graph theory-based features in the training and test datasets were 0.624 and 0.667, respectively, whereas those for standard features were 0.642 and 0.625, respectively, showing that, in test dataset, graph theory-based features had higher C-index than standard features. **Conclusion:** Graph theory-based features may be useful for improving the performance of radiomics-based prognostic prediction.

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MO-lePD-TRACK 4-03

Explainable AI Model for COVID-19 Diagnosis Through Joint Deep Learning and Radiomics

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Purpose: Deep learning based computer-aided diagnosis is a promising and efficient tool for the early detection of the Coronavirus Disease 2019 (COVID-19). However, the "black box" characteristic makes it difficult for clinicians to comprehend. This study aims to develop an explainable automatic COVID-19 detection system by combining deep learning classification and radiomics feature analysis. **Methods:** A dataset of 4671 Chest X-rays from multiple public datasets was used in this study. A deep learning model was developed by inserting deformable convolutional layer into the last denseblock in DenseNet121. During training, data augmentation was applied to training data to prevent overfitting. The proposed deformable convolutional neural network predicted the category of the input Chest X-ray and output a series of visualization results with Grad-CAM++ method. A mask for the region of interest (ROI) was generated via element-wise multiplication between lung region mask and visualized attention mask. Pyradiomics was then used to extract the features of the ROI. Correlation between each feature and category was calculated by using statistical methods. Ten features were selected and evaluated to describe the variance between categories based on statistical scores and random forest modeling. **Results:** The proposed network obtained an overall test accuracy of 95.0%. The recall and precision of COVID-19 were both above 0.95. Meanwhile, saliency maps with bounding boxes were provided for interpretability. In radiomics feature analysis, all selected features had a p-value less than 0.01. The feature based classifier achieved 79.6% on testing set, which proved the validity and consistency of using selected features to represent different categories. **Conclusion:** We have developed an explainable deep learning based COVID-19 Chest X-ray diagnosis method. To the best of the author's knowledge, our method is the first of its kind to combine class activation map and radiomics features to achieve the interpretability of convolutional neural network.

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MO-lePD-TRACK 4-04

Treatment Response Prediction for MRI-Guided Adaptive Radiation Therapy of Pancreatic Cancer Using Multiscale Wavelet-Based Delta-Radiomics

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Purpose: To investigate whether delta radiomics (change of radiomic features) based on daily MRI acquired during MR-guided adaptive radiation therapy (MRgART) of pancreatic cancer can predict treatment response. **Methods:** Daily motion-average MRI sets derived from 4D-MRI acquired on a 1.5T MR-Linac during MRgART for 18 pancreatic cancer patients undergoing chemoradiation therapy (CRT) with 30-35 Gy in 5-fractions were acquired. CA19-9 response, pathological response for the cases treated with pre-operative CRT, and metastasis status were correlated with radiomic data. Wavelet decompositions were utilized to overcome challenges with inter- and intra-patient MRI intensity variations and to provide a robust way to extract radiomics data. On each daily MRI set, 438 wavelet-based delta-radiomics features (WDRF) were calculated from multiscale features extracted from different decomposition levels. Good or bad response groups were defined based on either pathological treatment response or distant metastasis free progression, and 50% reduction in CA19-9. Spearman correlations were applied to rule out redundant information. T-test, regression models, and linear mixed effect model were used to identify independent WDRFs with significant changes that correlate to response. Bayesian classifier with a leave-one-out cross validation method was used to determine WDRF combination with highest outcome predictions. The performance of the model to predict response was measured using AUC of ROC curve. **Results:** Spearman correlation showed that 67 features from different wavelet decomposition were not highly correlated and can be used for the analysis. 30 WDRFs showed trends correlating to response, with 8 WDRFs passing t-test and linear mixed effect (p<0.05) and demonstrated significant changes by 3rd or 4th fraction during treatment. Best performing model was a 3-feature combination that can predict response with an AUC of 0.94. **Conclusion:** Machine learning identified wavelet delta-radiomics features from daily MRI in MRgART that can predict treatment response during chemoradiation therapy for pancreatic cancer.

MO-lePD-TRACK 4-05

Exploratory Unsupervised Structure-Learning Based Radiomics Approach for Brain Metastases Treatment Response Modeling of Stereotactic Radiosurgery

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Purpose: Stereotactic radiosurgery (SRS) has been shown to reduce neurocognitive decline for patients with multiple (>4) brain metastases (BMs). Early, accurate and noninvasive treatment response diagnosis is essential for SRS management to assist subsequent therapy decision making. In this study, we develop a novel unsupervised structure learning-based radiomics approach for BMs treatment response modeling. **Methods:** The study cohort includes 14 BMs SRS patients (307 BMs) with planning MRIs and dose distribution available. BMs segmentations are generated via our En-DeepMedic segmentation platform. For each BMs, 791 features are collected from MRI and dose distribution including clinical parameters, geometry features, first-order and texture features, using 7 types of ROI masks with different dilation margins. An unsupervised structure learning algorithm integrating both global and local information of the input is implemented. After optimization, a tree graph with skeleton and clusters on branches is learned to represent the intrinsic structure of the dataset in a reduced dimension, where similar subjects are clustered together and the main trajectory of the graph represents the inherent data structure. In addition, the algorithm can also perform clustering task based on input features. **Results:** This unsupervised algorithm can discover a stable latent tree structure suggesting a linear progression trajectory starting from no size-change BMs towards size-changed BMs. The main skeleton diverges into three sub-branches in size-changed group, suggesting that possible responses subtypes can be further distinguished in that group. The algorithm can also cluster size-changed and no-change BMs into different subsets with F-measure score and Rand index as 0.70 and 0.65 respectively without any supervision. Specifically, no size-change group is clustered with only 6 misclassification. **Conclusion:** This proposed unsupervised structure learning-based radiomics approach can reveal a tree structure with clusters associated with BMs treatment responses. This indicates that BMs treatment responses are separable only utilizing treatment images and dose.

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MO-lePD-TRACK 4-06

CT-Based Deep Learning Radiomics for Predicting Chemoradiation Treatment Response in Locally Advanced Rectal Cancer

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Purpose: Many studies have used the conventional handcrafted radiomic features to predict neoadjuvant chemoradiation treatment (nCRT) response in patients with locally advanced rectal cancer (LARC). The goal of this study is to investigate deep learning (DL)-based features for nCRT response prediction based on the planning CT. **Methods:** 86 LARC patients treated at a single institution were retrospectively enrolled. All patients received preoperative nCRT, with a prescription dose of 50.4Gy in 28 fractions, before total mesorectal excision. Patients were split into a responder group (n=45) and a non-responder group (n=41) based on pathological tumor regression grades. Each patient had a planning CT along with the manually delineated gross tumor volume. Handcrafted features and DL-based features were extracted from the planning CT using PyRadiomics and a pre-trained convolutional neural network (VGG19), respectively. Two logistic regression models with LASSO regularization were constructed using handcrafted and DL-based features, respectively. The model performance was evaluated with the stratified 5-fold cross-validation using receiver operating characteristic (ROC) curves. **Results:** The model constructed using handcrafted features achieved an area under the ROC curve (AUC) of 0.63±0.13, while the one constructed using DL-based features yielded an AUC of 0.76±0.04. **Conclusion:** Deep learning-based features extracted from planning CT achieved more accurate nCRT response prediction for LARC patients, as compared to handcrafted features. Deep learning-based features selected by the LASSO regularization are potential biomarkers for outcome prediction and personalized therapy.

MO-lePD-TRACK 4-07

Can Unified Data Improve the Performance of Radiomics-Based Prognostic Prediction in Lung Cancer Patients?

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Purpose: We investigated the radiomics-based survival prediction of non-small-cell lung cancer (NSCLC) patients by focusing on subgroups with identical characteristics. **Methods:** 304 NSCLC (Stages I-IV) patients treated with radiotherapy in our hospital were used. We extracted 107 radiomic features (14 shape features, 18 first-order statistical features, and 75 textural features) from the GTV drawn on the free-breathing treatment-planning CT image. Three feature selection methods [test-retest and multiple-segmentation (FS1), Pearson's correlation analysis (FS2), and a method that combined FS1 and FS2 (FS3)] were used to clarify how they affect survival prediction performance. The radiomic features selected by each selection method were combined with clinical features to form the explanatory variables in this analysis. A least absolute shrinkage and selection operator cox regression was used for subgroup analysis including each histological subtype and each T stage, where the most accurate feature selection method was adapted. Five-fold cross-validation was used to ensure the reliability of the model, and the predictive performance was evaluated with the Concordance-index (C-index) and Kaplan-Meier methods. **Results:** In all data, the C-index were 0.64 (FS1), 0.65 (FS2), and 0.64 (FS3) for the training dataset and 0.62 (FS1), 0.63 (FS2), and 0.62 (FS3) for the test dataset. The subgroup analysis indicated that prediction models based on specific histological subtypes and T stages had higher C-index than prediction model based on all data, especially for model based on T4 (training dataset, 0.72; test dataset, 0.70). Moreover, prediction models based on each T stage in adenocarcinoma (ADC) had a higher C-index than prediction model of ADC. **Conclusion:** Our results showed that feature selection methods moderately impacted the survival prediction performance. In addition, prediction models based on specific histological subtypes and T stages may improve the prediction performance. These results may prove useful for determining the optimal radiomics-based predication model.

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Therapy Interactive ePoster Discussion

TRACK 5

Therapy: Outcome Modeling and Assessment II

MO-lePD-TRACK 5-01

A Functional Dosimetry-Based Prediction Model for Radiation Pneumonitis

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Purpose: Ventilation image (VI) reveals the presence of functional heterogeneity within lungs. This study adopted the omics method to mine the heterogeneous information to better predict the incidence of radiation pneumonitis (RP). **Methods:** VI was derived from the 4-dimensional computed tomography (4DCT). The functional dose (FD) distribution was obtained by weighting OD with VI. The omics features were extracted, and then selected by the minimum redundancy maximum

relevance (mRMR) method. The prediction models were constructed with the logistic regression (LR), support vector machine (SVM), k-nearest neighbors (KNN), and random forest (RF) classifiers. Cross validation, bootstrap and nested sampling and methods were adopted for model training and classifier parameter hyper-tuning. The prediction models based on the omics features extracted from OD and the dosimetric factors (V5, V20 and mean lung dose, DF) based model were constructed for comparison. **Results:** 217 thoracic cancer patients treated with radiotherapy were used to train and validate the prediction model. The 4DCT-based VI showed the heterogeneous pulmonary function of the lungs. 19 first-order, 27 gray level (GL) co-occurrence matrix (GLCM), 16 GL run length matrix (GLRLM) and 16 GL size zone (GLSZM) features were extracted from the ipsilateral, contralateral and total lungs, separately. In total, 234 (78×3) features were extracted for each patient. The FD-omics and OD-omics models achieved optimal performance when 5 features were selected using the LR classifier. And the area under curve (AUC) were 0.755 (0.753-0.757), 0.716 (0.713-0.718), respectively. In comparison, the AUC of the DF model was 0.680 (0.675-0.684). **Conclusion:** The FD-omics model outperformed the OD-omics and DF modes for RP prediction due to the use of more data points and the data-mining in greater depth. It also indicates that the heterogeneity of pulmonary distribution revealed by VI helps to better model the dose response of radiation.

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MO-lePD-TRACK 5-02

Prediction of Glioblastoma Patient's Survival After Radiation Therapy with Random Survival Forest Model

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Purpose: To predict the outcome of radiation therapy (RT) in patients with glioblastoma from parameters before and after RT and to identify predictive variables using a machine learning. **Methods:** We assessed 336 glioblastoma patients treated between 2006 and 2017. A random survival forest was trained to predict survival from baseline values of total 16 parameters including tumor, demographic, immunologic, and treatment factors. The feature importance was determined using the amounts of increased predictive power by adding each parameter. **Results:** The trained model achieved concordance index (c-index) of 0.628 in cross-validation, comparable to the one (0.629) in the similar published study with cox proportional hazard regression. The importance analysis identified methylation of the MGMT gene promoter, the extent of resection, and sub-ventricular zone involvement as the most important variables (permutation importance = 0.082, 0.034, and 0.021 respectively), followed by gross target volume (0.018), age (0.008), and acute severe lymphopenia (0.004, lymphocyte counts of < 500 μ L). More number of variables did not necessarily guarantee the predictive power of the trained model. The model achieved the maximal concordance index (0.658) by including the most important 3 or 4 parameters. **Conclusion:** Random survival forests were an explainable machine-learning approach for prediction of an entire patient's survival curve. Only three or four baseline parameters were necessary to achieve comparable predictive power of the trained model. Those parameters were the methylation of MGMT gene promoter, extent of resection, sub-ventricular zone involvement, or gross target volume, for the glioblastoma patients.

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MO-lePD-TRACK 5-03

Moving Beyond Dose: The Role of HPV Status in Predicting Normal Tissue Toxicity for Head & Neck Cancer Patients

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Purpose: Planning objectives used in radiotherapy plan optimization are driven by population-based studies, however patient-to-patient variation in tumor and normal tissue response is well established. Various patient characteristics have been explored as potential biomarkers to personalize treatment planning for this variance. The objective of this study was to evaluate if HPV status is a significant, independent factor affecting normal tissue toxicities for head-and-neck cancer patients treated with radiotherapy. **Methods:** HPV status, radiation treatment doses to organs-at-risk, and reported early and late dysphagia were retrospectively obtained for 101 patients treated at our clinic. Mean doses were extracted for the entire pharynx, superior/middle/inferior pharyngeal constrictor muscles (PCM), and cricopharyngeus. Early and late dysphagia grades were collected on the 0-5 CTCAE scale and binarized to 0-2 versus 3-5. T-tests were used to evaluate relationships between outcome and dose, and chi-squared tests to evaluate associations between outcome and HPV. Multivariate logistic regression models were built to predict outcome as a function of HPV status, sex, age, and dose metrics. P-values < 0.05 were considered significant. **Results:** Significant dose metrics for early dysphagia included mean dose to the Pharynx (p=0.032), middle PGM (p=0.020), inferior PGM (p=0.015), and cricopharyngeus (p=0.022) with respective increased dose differences of 7.0, 7.1, 12.1, and 11.4 Gy. No dose metrics were significantly associated with late dysphagia; however, HPV-negative status was significantly associated with late dysphagia (p=0.028). HPV-negative status was not significantly associated with early dysphagia. In logistic models, results were consistent: all four organ mean doses remained significant covariates for early dysphagia while HPV was the only significant covariate for late dysphagia. **Conclusion:** HPV status was a significant predictive covariate for late dysphagia in our cohort. While further investigation is warranted to evaluate the root cause of this association, these results may be useful for assessing an individual patient's toxicity risk and personalizing their treatment or follow-up.

X Ray discloses a lab services agreement with Varian. P Sanghvi discloses past relationships with Varian and VisionRT.

MO-lePD-TRACK 5-04

Analysis of Time Dependency of NTCP Modeling and Dose-Volume Correlations of Patient Reported Xerostomia After HPV Associated Oropharyngeal Cancer

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Purpose: This study aims at analyzing the time dependency of dose-volume correlations with patient reported (PRO) xerostomia after radiotherapy and the corresponding normal tissue complication probability (NTCP) modeling. **Methods:** 151 patients with favorable risk, HPV-associated oropharyngeal squamous cell carcinoma were treated with radiotherapy. A dose of 60Gy delivered with IMRT was prescribed to the patients. Xerostomia was assessed based on the PRO-CTCAE score and it was defined as a ≥ 2 point increase from baseline at 6, 12, 18 and 24 months post-RT. The dose volume histograms (DVH) of the contralateral parotid and contralateral glands (parotid and submandibular glands combined) of each patient were exported and correlated with the follow-up data of xerostomia. The clinical data was fitted by the Relative Seriality (RS) NTCP model. **Results:** Xerostomia was observed in 59%, 38%, 29% and 31% of the patients at 6, 12, 18 and 24 months, respectively. The AUC values of the correlations ranged between 0.59-0.66. For the contralateral parotid, the values of the D50, γ and s parameters of the RS model were 24.4Gy, 0.46 and 0.01 at 12 months and 31.5Gy, 0.34 and 0.01 at 24 months post-RT. Statistically significant Odds Ratios (ORs) of 3.1 (1.2-7.9) and 2.9 (1.3-6.6) were found for the contralateral parotid for biological doses < 23Gy and 21Gy at 12 and 24 months post-RT, respectively. Similarly, for the combined contralateral glands ORs of 2.3 (1.1-5.0) and 2.8 (1.1-6.8) were found for biological doses < 27Gy and 31Gy at 12 and 24 months post-RT, respectively. **Conclusion:** The dose-response curve of PRO xerostomia at 6, 12, 18 and 24 months post-RT could be determined by the RS NTCP model. Keeping the biological dose to the contralateral parotid < 21Gy or to the combined contralateral glands < 27Gy was found to significantly reduce the risk for patient reported xerostomia.

MO-lePD-TRACK 5-05

Identifying Correlations Between Radiation Dose and Blood Counts Drops After Radiotherapy of Prostate Cancer

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Purpose: The purpose of this study is to determine dose-response relations regarding the endpoint of hematocrit level drop after pelvic irradiation in patients undergoing external beam radiotherapy for prostate carcinomas. **Methods:** A retrospective analysis was conducted on 221 patients treated between 2014 and 2016 at a single institution for prostate cancer. Patients received either 78Gy to primary prostate or 70Gy post-prostatectomy. All patients had complete blood counts collected at baseline (pre-RT) and 3 months post-RT, which were analyzed in conjunction with dose volume histograms corresponding to eight different pelvic structures. **Results:** Pelvic structures that upon irradiation correlated with significant decreases in hematocrit were the following: the bone volume of os-coxae bilaterally superior to the acetabulum (OCUB), the bone volume of the entire os-coxae bilaterally (OCTB), and the bone volume of the whole pelvis (BVWP). The structure showing the highest correlation was OCUB with a maximum AUC of 0.74. For V20Gy < 30% the odds ratio was 9.8 with 95% confidence interval (CI) of 2.9-32.9. For mean dose to OCUB, an AUC of 0.73 was observed where the dose threshold was 23Gy and the odds ratio was 2.7 and 95%CI 1.3-5.6. For OCTB, AUC was 0.69 with a V20Gy < 50% showing an odds ratio of 9.3 (95%CI 2.7-31.8). For BVWP, AUC was 0.71 with a V20Gy < 40% demonstrating an odds ratio of 7.5 (95%CI 2.2-25.4). Further, mean dose < 24Gy had an odds ratio of 2.6 (95%CI 1.1-6.1) showing an AUC of 0.72. **Conclusion:** These findings confirm the association of damage to pelvic bone marrow with blood counts drop. The doses to OCUB, OCTB and BVWP were found to correlate with a significant drop of hematocrit. A threshold of V20Gy < 30% or mean dose < 23Gy to OCUB may reduce almost ten-fold the risk for this endpoint.

MO-lePD-TRACK 5-06

Association Between Dosiomics-Based Index with Radiation Pneumonitis Induced by Radiation Therapy of Breast Cancers

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Purpose: To investigate the feasibility of predicting the occurrence of radiation pneumonia (RP) in patients after radiation therapy (RT) of breast cancer using dosiomics index. **Methods:** A pilot retrospective study of 42 breast cancer patients was performed under an institutional IRB protocol. 11 of 42 patients were observed with RP complications based on CT scans 3-5 months post RT. We hypothesize that the dosiomics features can better characterize the occurrence of RP than the traditional volume-based dosimetric metrics (such as V5, V20, and mean dose). To test, we first evaluated the potential dosimetric deviations due to inter-fractional mispositioning and intra-fractional patient respiratory motion using daily cone-beam CT (CBCT). The actual dose distribution within the lung was re-calculated by incorporating the CBCT images taken before each fractional treatment. 91 dosiomics features were extracted from both the whole lung dose and the high dose region (>20 Gy), respectively. T-test was performed to find the optimal features as potential indexes for RP. The comparison of p-value between the dosiomics features and the traditional dosimetric indexes was also conducted to evaluate the overall performance. **Results:** Dosiomics feature GLCOM-based Sum Variance extracted from whole lung dose presents the significant statistical difference between 11 RP patients and rest 31 patients (p-value = 0.0487), and feature GLCOM-based Cluster Shade extracted from high dose region gives lower p-value (0.0316). While the traditional dose constraints are insufficient to characterize the dose distribution differences between two groups of patients (p-value for mean dose = 0.5447, p-value for V5 = 0.7926, and p-value for V20 = 0.4058). **Conclusion:** This pilot work verified the feasibility of evaluating the occurrence of RP in patients after breast cancer RT with dosiomics features. The dosiomics-based indexes, feature GLCOM-based Sum Variance and Cluster Shade, showed the potential to replace the traditional dosimetric indexes in predicting RP.

MO-lePD-TRACK 5-07

Principal Component Analysis of Dose Clustering Patterns for Parotid Toxicity Modeling in Head and Neck Radiation Therapy

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Purpose: To develop a predictive model using principal component analysis (PCA) and logistic regression (LR) for parotid complication based on 3D dose clustering in head-and-neck cancer irradiation. **Methods:** Spatial dose distribution inside the parotid gland was incorporated into the percolation-based cluster model for 155 patients receiving intensity modulated radiation therapy from three institutions. Radiation induced complication was classified into two categories for patients who never developed xerostomia and those who developed xerostomia with grade 1 and above, respectively. For each patient, forty bins of the normalized largest size of dose clusters were formed with cutoff dose ranging from 11-50 Gy in 1 Gy step size. PCA was applied to these 40-column variables from all enrolled patients. The principal components (PCs) were determined and subsequently input into a multinomial logistic regression model for quantitative analysis of toxicity correlation. The returned fitting parameters were examined to establish the correlation between the principal components and parotid gland toxicity. In addition, three cluster connectivity options representing cluster formation were evaluated for the association of optimal morphological clustering patterns with the gland toxicity. **Results:** PCA analysis revealed that first three PCs are the most important ones with the first PC explaining over 91% for all three connectivity variances. The scatter plots between the dominating PCs show distinct clustering patterns among the two groups of patients. These first three PCs were input into the LR model and the p values (<0.05) were found for the first and the third principal components. No significant difference was observed among three cluster connectivity choices. **Conclusion:** The PCA-LR model can reveal correlations between the percolation-based cluster model and parotid toxicity. Thus, it can be a valuable tool to augment normal tissue complication modeling and facilitate treatment planning decision making in the head and neck radiation therapy.

Therapy Interactive ePoster Discussion

TRACK 6

Therapy: External Beam: Segmentation, Dose Calculation and Treatment Planning

MO-lePD-TRACK 6-01

Application of Random Forest Algorithm for Dose Reconstruction Based on Classification of Clean-Up Workers of the ChNPP Accident

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Purpose: As a result of the accident at Chernobyl Nuclear Power Plant a heavy contamination of the environment occurred. Thousands of workers took part in cleanup work, and only few percent of them have sufficient dosimetry. Thus, methods are required for dose reconstruction for large groups of liquidators based on a limited set of data. **Methods:** A dosimetry method was developed based on random forest algorithm. It was applied to a cohort of 11105 liquidators from Clinical and epidemiological registry of Ukraine. Neural network was trained (70%) and tested (30%) on database of 1807 liquidators, whose doses were reconstructed according to detailed questionnaires using the Realistic Analytical Dose Reconstruction with Uncertainty Estimation (RADRUE) method. The developed algorithm was based on the date of the mission and duration, occupation, place of work and residence, affiliation to the organization. Dose reconstruction for study cohort was made in 2 steps: reconstruction of the specific group (contingent) of the liquidator and then dose assessment within the contingent. Rules for division into continents described in RADRUE method. **Results:** Liquidators from study cohort were divided into 9 contingents. The contingents were also divided by year from 1986 to

1990. Based on RADRUE database the dose distribution for each contingent any year was calculated. The developed method determines with an accuracy of 91.5% whether the liquidator was a military or civilian. For other contingents, the correctness of the method was from 73.7% to 99.8%. **Conclusion:** Developed algorithm applicable for contingent and dose range reconstruction for liquidator if the training dataset contains all possible variants from the studied dataset. Results of the study and the cohort will be used as a resource to investigate novel hypotheses about the radiation-related risk of cancer and benign outcomes by combining historical dose records, epidemiological data and medical records.

MO-lePD-TRACK 6-02

An Experimental Study of Intrabeam IORT Dosimetry Accuracy

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Purpose: The INTRABEAM (IB) System has been reported to have a large dosimetric difference between the vendor provided and user measured data. This study is aimed to verify the dosimetric accuracy by performing ionization chamber measurements of absorbed dose in a water phantom and comparing with the annual calibration data for past 10 years. **Methods:** Ionization chamber measurements were performed with a PTW 34013 parallel-plate chamber at source to detector distances of 2 to 30 mm with and without a spherical applicator. The experimental output dose rates were derived by using the formula recommended by the manufacturer. The source calibration data, taken from the annual certificate files provided by manufacturer from 2011 to 2020, were reviewed and compared with the experimental results. **Results:** It is found that the source data format was changed in 2017, from one dataset to two distinct datasets: one is Calibration V4.0, and other is TARGIT. The dose rate fluctuation from 2011 to 2016 at the same distance was within $\pm 12\%$, from 2017 to 2020 was within 5%. However, the dose rates of 2017 certificate file increased up to 210% and 58% respectively for Calibration V4.0 and Target datasets comparing to the previous years. Experimental data with and without a spherical applicator taken in 2017 and after were found to still agree with 2011 certificate data within $\pm 5\%$ uncertainty, but a difference up to 200% was seen comparing with the Zeiss 2017 source data. **Conclusion:** Comparing with our experimental data, the source output was found to remain almost the same from 2011 to 2020, therefore TARGIT trial dosimetry is still accurate. The large dosimetric difference is attributed to the lack of updated Zeiss QA water phantom conversion factor. We provided a new conversion factor to account for the change of calibration standard in Zeiss dosimetry laboratory.

MO-lePD-TRACK 6-03

Liver Segmentation Workflow Development for Pre-Treatment Hepatic Y90 Resin Microspheres Therapy Using a Commercially Available Software

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Purpose: Liver segmentation for the purpose of pre-treatment planning of hepatic Y90 resin microspheres therapy is not a standardized practice. To that end, a semi-automatic, customizable workflow has been developed within MIM Sureplan® that enables standardization within a clinical practice. **Methods:** The clinical workflow at Miami Cancer Institute (MCI) for pre-treatment planning of hepatic Y90 resin microspheres therapy was re-created in the MIM SurePlan platform. Briefly, the original steps were performed manually, utilizing a segmentation software where the liver was drawn on the non-contrast CT of the MAA mapping SPECT/CT. Early and late arterial phase CBCTs for left and right hepatic lobes were then registered to the non-contrast CT. The early phase CBCTs, which determine perfusion paths, were used to delineate left/right lobes, while late phase CBCTs were used for GTV delineation. For select cases, a recently acquired diagnostic contrast CT or MRI may be used for GTV delineation. After these five image registrations, the RT/LT lobe and RT/LT GTV volumes were segmented and their volumes used to calculate the prescribed activity of Sirspheres® for each hepatic lobe. This manual process was designed into a streamlined, semi-automated workflow. **Results:** The MIM workflow was designed to standardize the MCI pre-Y90 treatment process. It is robust enough to accommodate several scenarios to best serve the unique needs of the patient, including pauses, stop gaps between major processes, and editing points, as well as flexibility as to which image studies to use for segmentation. An auto-generated volume statistics table is ultimately rendered for activity calculation. **Conclusion:** A clinically robust, semi-automated workflow process for pre-treatment of hepatic Y90 resin microspheres therapy has been created in MIM SurePlan. Streamlining the workflow should achieve time-savings and standardization in clinical practice. Consideration should be given to this workflow as a means for standardizing the pre-treatment planning method.

Dr. Chuong reports receiving honoraria from Sirtex. Dr. Gandhi is a consultant and proctor for Sirtex.

MO-lePD-TRACK 6-04

Demonstration of An Automatic Segmentation Method for Evaluating Cardiac Structure Doses in Support of Research on Cardiovascular Morbidity Subsequent to Breast Radiotherapy

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Purpose: To develop an automatic segmentation algorithm to contour heart substructures on a planning CT for estimating doses in support of research on cardiovascular morbidity subsequent to breast radiotherapy. **Methods:** We collected DICOM-RT data for 100 breast cancer patients enrolled in the Radiotherapy Comparative Effectiveness clinical trial who were treated using photons. For each CT, a cardiology team provided a manual segmentation of the whole heart (WH), left/right atria, left/right ventricles and left anterior descending artery (LAD). A cardiac atlas library was developed for the automatic segmentation using 30 patients and the remaining 70 patients (40 left-side, 30 right-side) were used for performance testing. The input to the algorithm was the manual contouring of the WH which is drawn at time of treatment planning. The automatic segmentation consists of a selection of a most similar heart from the atlas library followed by a structure-guided B-spline transformation. The automatically contoured results were then compared with manual delineations to evaluate geometric and dosimetric similarity. **Results:** The median dice similarity coefficient (DSC) for the four chambers ranged from 66% to 84% and the average surface distance (ASD) was ~4 mm. Performance for the LAD was worse with a mean DSC of 6% and median ASD of ~6 mm which can be explained by the difficulty segmenting such a small structure. The doses calculated using the manual and automatic delineations showed excellent correlation within expected variability of manual segmentation. For left-sided (right-sided) treatments, the mean dose difference for the chambers and LAD was <0.7 Gy (<0.4 Gy) and 1.8 Gy (0.4 Gy), respectively. **Conclusion:** Cardiac structure doses based on our automatic segmentation agree with that of manual segmentation. When applied to large patient datasets our automatic segmentation method should facilitate the development of better prescriptive criteria for mitigating cardiovascular morbidity following breast radiotherapy.

MO-lePD-TRACK 6-05

Improving Workflow Efficiency and Safety for RefleXion X1 Treatment Planning Process Via Eclipse API Scripting

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Purpose: The RefleXion™ X1 system is the first biology-guided radiotherapy machine that can target cancer in real-time using on-board PET imaging in combination with a 6MV FFF beam. The current version of the system is not yet integrated with any record and verify system, has limited contouring, dose evaluation, and isocenter placement tools. To improve efficiency and safety of the treatment planning process, application programming interface (API) scripting was utilized to overcome these limitations. **Methods:** Eclipse, the primary treatment planning system (TPS) at our institution, is used to perform all contouring and plan dose

evaluation for the RefleXion treatment plans. Therefore, two scripts were developed using the Eclipse API scripting framework. The first script, RefleXion Assistant, focuses on preparing structure sets for import and planning in the RefleXion TPS. Such preparation includes checking all structures are within the body contour, inserting a special couch contour, and verifying the proposed isocenter placement in outside the collision zone. The second script, Automated Plan Check (APC), generates the shift note and focuses on physics plan checks and plan quality evaluation of the treatment plans. **Results:** RefleXion Assistant can check several planning failure-modes in the RefleXion TPS within 30sec, e.g., if a structure is outside the body, the RefleXion TPS will throw an error preventing optimization and requiring the planner to identify the offending structure, fix it, export/import the new structure, and adjust the plan setup. Example items checked with the APC include high-severity failure-modes, i.e., physician's prescription dose/fractionation in Aria match the RefleXion plan, shift note accuracy, cardiac device policy compliance, etc. Plan quality is evaluated by computing DVH metrics and comparing them to institutional/clinical trials planning constraints. **Conclusion:** Two API scripts were developed and implemented to prevent failure-modes with the RefleXion TPS, resulting in significant improvements in planning efficiency and safety.

MO-lePD-TRACK 6-06

Optimal Combination of Spatial and Temporal Fractionation for the Treatment of Multiple Brain Metastases

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Purpose: Stereotactic radiosurgery (SRS) is a standard treatment modality for intracranial metastases. While small lesions are preferably treated in a single fraction, large metastases must be temporally fractionated to reduce dose toxicities in normal tissues. In this work, an algorithm for determining the best trade-off between single-fraction and hypofractionated SRS for the treatment of multiple brain metastases is developed and demonstrated. **Methods:** Non-coplanar IMRT plans are created to obtain spatial dose distributions for the treatment of each metastasis. We then assume that each metastasis is either treated to the full dose in one of the fractions (spatial fractionation) or is fractionated uniformly over all fractions (temporal fractionation). The optimal fractionation scheme is obtained by solving a binary quadratic optimization problem, which, for a fixed tumor BED, minimizes an equally weighted sum of the mean cumulative biologically effective doses (BED α/β) to the whole brain and to a 5 mm-thick rim structure around the metastases. The algorithm is demonstrated for a clinical case consisting of twenty-nine brain metastases of varying sizes to be treated with a BED10Gy of 45 Gy. **Results:** A four-fraction SRS plan is computed with the proposed fractionation optimization algorithm and compared to two SRS plans treating all metastases with one and four uniform fractions, respectively. The optimal fractionation scheme suggests the temporal fractionation of large metastases and the spatial fractionation of smaller lesions, leading to mean BED2Gy of 14.90 Gy and 74.96 Gy in the healthy brain and rim structure, respectively. Single-fraction and hypofractionated SRS plans achieve mean BED2Gy in the healthy brain and rim structure of 20.18 Gy and 90.59 Gy, and 18.48 Gy and 72.17 Gy, respectively. **Conclusion:** The proposed algorithm allows to optimally combine spatial and temporal fractionation for different lesions, potentially outperforming current state-of-the-art uniform fractionation schemes for the treatment of multiple intracranial metastases.

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MO-lePD-TRACK 6-07

Effective Organs-At-Risk Dose Sparing in Volumetric Modulated Arc Therapy Using a Half-Beam Technique in Whole Pelvic Irradiation

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Purpose: To effectively spare organs-at-risk (OAR) doses using multi-leaf collimator (MLC)'s optimal segments, volumetric modulated arc therapy (VMAT) using a half-beam technique (HF) was devised for whole pelvic radiation therapy (WPRT). The potential dosimetric benefits of VMAT-HF were suggested for clinical application with statistical analysis. **Methods:** Fifteen eligible patients diagnosed with anal, vaginal, and cervical cancer were selected. Dosimetric benefits of VMAT-HF, presented with dose conformity and dose-volume parameters in terms of modulation complex score, were compared to those of VMAT optimized using a fully opened field (FF). The left or right side of FF was alternatively opened in VMAT-HF by using a pair of arcs rotating clockwise and counterclockwise. Consequent normal tissue complication probability (NTCP) by reducing the irradiated volumes and delivered doses for OAR were evaluated as well as dose-volume parameters. **Results:** Compared to VMAT-FF, VMAT-HF showed superior conformal dose distribution, where the 75% isodose line compactly surrounded the separated regional lymph nodes. Dose conformity was comparable between VMAT-HF and VMAT-FF; however, VMAT-HF achieved the conformal dose distribution using 60%–70% less intensity modulation complexity than that used by VMAT-FF. The small intestine and colon showed a noticeable reduction in the irradiated volumes of up to 35% and 15%, respectively, at an intermediate dose of 20–45 Gy. The small intestine showed statistically significant dose sparing at the volumes that received a dose every 5 Gy from 15 to 45 Gy. Such a dose reduction for the small intestine and colon in VMAT-HF presented significant NTCP reduction than that in VMAT-FF. **Conclusion:** VMAT-HF provided more optimal composition of MLC segments to deliver conformal doses without excessive intensity modulation. It achieved distinguishable dose sparing for gastrointestinal OAR in WPRT without sacrificing the target dose conformity. VMAT-HF can be usefully used in reducing OAR toxicities associated with WPRT.

Vendor Showcase

TRACK 8

VisionRT

MO-SHOWCASE-TRACK 8-6-01

The Value Proposition of SGRT for Patient Care

T Pawlicki^{1*}, (1) UC San Diego, La Jolla, CA

SAM Professional Symposium

TRACK 1

Formalizing Professionalism Education: Shared Experiences with Medical Physics Trainees

MO-E-TRACK 1-00

Formalizing Professionalism Education: Shared Experiences with Medical Physics Trainees

Moderators: Irina Vergalasova, Rutgers, Cancer Institute of New Jersey, New Brunswick, NJ and Hania Al-Hallaq, The University of Chicago, Chicago, IL

MO-E-TRACK 1-01

I. Vergalasova, Rutgers, Cancer Institute of New Jersey:

What Is Formalized Professionalism Training?

MO-E-TRACK 1-02

J. Wilson, Duke University Health System:

From Day 1 Through Graduation: Introducing and Instilling a Professional Mindset in Graduate Students for the Medical Physics Career Ahead

MO-E-TRACK 1-03

M. Price, Vanderbilt University Medical Center:

Flattening the Learning Curve: Concise Coursework to Prepare Medical Physics Residents for Leadership

MO-E-TRACK 1-04

J. Hoisak, UC San Diego:

Developing Professionalism Competencies in Therapy Physics Residents: An Experiential Learning and Systems-Based Approach

MO-E-TRACK 1-05

Audience Q&A

Professionalism education is one of the six main categories required of CAMPEP-accredited programs. Often, professionalism is taught in an ad hoc manner or in the course of on-the-job training. Several graduate programs and residency programs have recently formalized professionalism education. This session will begin with an introduction of professionalism and how it's taught in other allied health fields. Next, graduate and residency programs will describe their professionalism curricula and motivation for selecting topics, which vary widely from business acumen to communicating with patients. They will describe at least one way in which formalizing professionalism training has benefited their trainees and one way in which it has directly benefited the educational program. The session will conclude by discussing how professionalism training can be formalized and unified across Medical Physics.

Learning Objectives:

1. Present concrete examples of residency professionalism training and education currently incorporated by several medical physics programs
2. Discuss the advantages to trainees of formalized professionalism education
3. Discuss the advantages to programs of formalized professionalism education

SAM Imaging Educational Course

TRACK 2

Testing the Misfits: Tips and Tricks for Testing Your Least Favorite Diagnostic Imaging Equipment

MO-EF-TRACK 2-00

Testing the Misfits: Tips and Tricks for Testing Your Least Favorite Diagnostic Imaging Equipment

Moderators: Rebecca Marsh, University of Colorado School of Medicine, Aurora, CO and Jeff Frimeth, JF Medical Physics Inc., Sudbury, ON

MO-EF-TRACK 2-01

N. Bevins, Henry Ford Health System:

Considerations for Evaluating Color Displays

MO-EF-TRACK 2-02

J. Gray, Diquad, LLC:

National Council on Radiation Protection and Measurements (NCRP), Radiation Protection in Dentistry and Oral & Maxillofacial Imaging, NCRP Report 177 (2019) Summary

MO-EF-TRACK 2-03

M. Supanich, Rush:

What Physicists Need to Know About the Medtronic O-arm

MO-EF-TRACK 2-04

Z. Lu, University of Chicago:

Considerations for Evaluating Ultrasound Units

MO-EF-TRACK 2-05

R. Marsh, University of Colorado School of Medicine:

Encounters with Non-standard Fluoroscopy Systems

MO-EF-TRACK 2-06

J. Frimeth, JF Medical Physics Inc.:

Troubleshooting DXA QC issues

MO-EF-TRACK 2-07

Audience Q & A

Equipment testing is a key component of the life of a clinical medical physicist. For some imaging systems, this process is straightforward and well-established. Some systems, however, have unique characteristics that require adaptation of traditional testing methods. This session will feature active clinical medical physicists, each of whom will reveal how to tackle testing some of these "misfit" systems that fall outside of the norm among imaging systems. The discussion will include what QC evaluations one should consider, how these may be adapted for "non-standard" systems, and how to approach QC for technically advanced equipment that does not yet have clearly-defined, accepted testing methods. Speakers will specifically address breast ultrasound, color displays, digital dental units, O-arms, mini c-arms, and DXA units.

Learning Objectives:

1. Obtain practical tips for successfully evaluating less common imaging systems.
2. Understand some of the unique characteristics of these less common systems.
3. Understand how to incorporate adapted/new QC practices when evaluating technologically advanced equipment.

SAM Imaging Educational Course

TRACK 3

Accelerated Imaging in MR

MO-E-TRACK 3-00

Accelerated Imaging in MR

Moderator: Courtney Morrison, Henry Ford Health System, Detroit, MI

MO-E-TRACK 3-01

S. Ragunathan, Barrow Neurological Institute:

Accelerated Imaging Strategies for Combined Spin and Gradient Echo (SAGE) Acquisitions

MO-E-TRACK 3-02

S. Kurugol, Boston Children's Hospital:

Compressed Sensing and Deep Learning in DCE-MRI

MO-E-TRACK 3-03

Audience Q & A

Temporal resolution and spatial resolution present competing demands in MRI. Several new techniques have been developed in recent years to accelerate MR image acquisition, some of which have been introduced commercially and others still in the research realm. This session will review some of these techniques. The first talk will provide an overview of the combined spin and gradient echo imaging (SAGE) technique and introduce spatial and temporal acceleration strategies for application in dynamic susceptibility contrast MRI and fMRI. This talk will also highlight the advantages of a multi-echo acquisition approach over the current clinical implementations. The second talk will discuss dynamic contrast-enhanced MRI reconstruction of undersampled k-space data using compressed sensing and deep learning.

Learning Objectives:

1. Understand the basic implementation the SAGE pulse sequence
2. Understand the advantages of multi-echo acquisition vs. conventional EPI acquisition for DSC-MRI and fMRI
3. Understand the requirements of DCE-MRI in clinical practice
4. Discuss DCE-MRI reconstruction from undersampled k-space data using compressed sensing and deep learning methods

Early-Career Investigator Symposium

TRACK 4

John R. Cameron Early-Career Investigator Symposium

MO-EF-TRACK 4-01

Design and Performance of the ARIEL X-Ray FLASH Irradiation Platform at TRIUMF

N Esplen^{1,*}, L Egoriti², B Paley³, T Planche³, C Hoeft³, A Gottberg³, M Bazalova-Carter¹, (1) University of Victoria, Victoria, BC (2) University of British Columbia, Vancouver, BC (3) TRIUMF, Vancouver, BC

Purpose: To optimize, build and commission a novel Bremsstrahlung target and megavoltage irradiation platform for delivering ultrahigh dose-rate radiation to small-animals in an exploration of X-ray FLASH radiotherapy at TRIUMF. **Methods:** The design of a tantalum-aluminum (Ta-Al) explosion-bonded conversion target has been optimized using Monte Carlo (MC) and Finite Element Analysis (FEA) simulation methods. The water-cooled Ta-Al target facilitates conversion of a 10MeV, 1kW electron beam at the ARIEL e-linac into an ultrahigh dose-rate ($>40\text{Gy/s}$) x-ray source with $<1\text{s}$ pulse-lengths and 0.05%-100% Duty Factor (FLASH mode) while preserving routine beam-dump capabilities in a lower power steady-state. Dose rates in water-phantoms were calculated in slab and CAD geometries using EGSnrc and TOPAS MC codes, respectively, to inform the design of the Ta-target, Al-flange and W-shield thicknesses and geometry. Thermo-mechanical FEA simulations in ANSYS subsequently informed the stress-strain conditions and fatigue life of the target assembly under prescribed conditions resulting in the re-design, and thus revised dose-rate capabilities, of the final prototype. Erosion tests and commissioning will be followed by dosimetric characterization ahead of FLASH treatment planning in small-animal models. **Results:** Simulated water-phantom irradiations demonstrate that surface dose-rates of 128Gy/s may be achieved for a $1\times 1\text{cm}^2$ field size and 7.5cm source-to-surface distance using the treatment-beam configuration ($E=10\text{MeV}$, $2\sigma=5\text{mm}$, $P=1\text{kW}$). Dose rates $>40\text{Gy/s}$ are maintained to a depth of 5cm in water. Modular collimation and SSD flexibility allows for $>200\text{Gy/s}$ to be achieved at larger field sizes. Assembly temperatures are maintained below the Ta, Al and cooling-water thresholds of 2000, 300 and 100°C , respectively, while the Al strain behavior remains everywhere elastic. **Conclusion:** The ARIEL x-ray FLASH experimental platform achieves ultra-high dose-rates within the thermo-mechanical constraints of the system. Target cooling, mechanical robustness and failure mitigation have culminated in a design intended for FLASH and steady-state applications with installation and commissioning planned for May 2021.

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MO-EF-TRACK 4-02

Electron FLASH in Clinical Setting: LINAC Conversion, Commissioning and Treatment Planning

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Purpose: Clinical use of ultra-high dose-rate (UHDR, $>40\text{Gy/s}$) electron FLASH radiotherapy (eFLASH-RT), a modality demonstrating greater normal tissue sparing than conventional radiotherapy with equivalent tumor control, is constrained by the availability of FLASH systems that include minimally modified clinical settings, well-defined beam characteristics and FLASH-enabled treatment planning system (TPS). In this study, we present a potential solution with rigorous commissioning of a modified common LINAC and implementation of an electron FLASH beam model in a widely-adopted TPS. **Methods:** A Varian Clinac 2100C/D was converted to deliver UHDR electron beams by withdrawing the target and scattering foil while running in 10MV x-ray mode. Beam characteristics and stability were quantified by film, Cherenkov and scintillation imaging. The eFLASH beam model, established in Geant4 and implemented in Varian Eclipse TPS, was validated by comparing calculations with film measurements. eFLASH plans were generated for representative canine and human patient cases to demonstrate the planning and delivery capability in minimally modified clinical settings involving complex geometries and anatomical inhomogeneities. **Results:** The surface average dose rate at the isocenter was $>230\text{Gy/s}$ for all measured fields with adequate long-term stability (deviations of output $<10\%$, symmetry/flatness $<2\%$, spatial shift and shape $<2\text{mm}$). The Geant4 beam model and its Eclipse implementation were validated to clinical accuracy (absolute average error was $<1.5\%$ for lateral profiles, and $<2\%$ for percent-depth-dose profiles). An eFLASH-RT plan was generated and accurately delivered to a canine patient with a malignant melanoma tumor in the posterior oral cavity. By utilizing routine accessories, oblique gantry angle and couch kick, a human eFLASH-RT plan comparable to a conventional electron plan was achieved. **Conclusion:** Treatment planning and accurate delivery of eFLASH-RT were demonstrated feasible in minimally modified radiation oncology clinical settings. The presented solution and its open-source beam model are readily transferable to facilitate the clinical translation of eFLASH-RT.

MO-EF-TRACK 4-03

Fast 4D-CBCT Reconstruction Using a Novel Feature-Compensated Deformable Convolutional Network (FeaCo-DCN)

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Purpose: 4D-CBCT is valuable for localizing targets affected by respiratory motion in radiotherapy. However, it is limited by long scanning time, high imaging dose, and degraded image quality due to the under-sampling in each respiratory phase. Compressed sensing-based methods suffer from severe blurriness, while motion-compensated methods are limited by their slow speed and compensation errors. In this study, we proposed a novel feature-compensated deformable convolutional network (FeaCo-DCN) to realize the fast 4D-CBCT with high image quality. **Methods:** The proposed FeaCo-DCN consists of (1) encoding networks which extract features from each phase of 4D-CBCT reconstructed by the FDK method, (2) deformable convolutional networks which align features of other phases to those of the target phase, and (3) a decoding network which combines and decodes features from all phases to yield high-quality images of the target phase. Different phases of 4D-CBCT are reconstructed in parallel using FeaCo-DCN. We trained and tested FeaCo-DCN on 9 fast (1-minute) 4D-CBCT scans in the AAPM SPARE Challenge using the leave-one-out strategy. Results were compared to the top-ranking methods in the Challenge in both image quality and tumor localization accuracy. **Results:** FeaCo-DCN generated high-quality 4D-CBCT while reducing about 90% of the conventional scanning time. Images reconstructed by FeaCo-DCN showed accurate and clear structures and achieved 3D tumor localization accuracy within 2.5 mm, considerably outperforming the compressed sensing-based and motion-compensated algorithms. Additionally, FeaCo-DCN took only about 2.5 seconds to reconstruct images of one respiratory phase. **Conclusion:** FeaCo-DCN showed effectiveness in realizing fast and high-quality 4D-CBCT while reducing about 90% of the acquisition time. The proposed feature-compensated reconstruction can be applicable for other 4D imaging modalities beyond 4D-CBCT, such as 4D-MRI and 4D-PET. FeaCo-DCN can become a valuable tool for 4D imaging for motion management in radiotherapy.

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MO-EF-TRACK 4-04

Imaging Based Prediction of Proliferative Foci as a Target for Surgical Intervention Across Glioma Grades

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Purpose: True proliferation of gliomas can only be measured by biopsy and is prone to sampling error due to tumor heterogeneity. We estimate tumor proliferation using MR imaging then demonstrate the ability for predictions to inform prognosis and guide surgical treatment. **Methods:** 52 image-guided(sub-millimeter accuracy) biopsy samples from 23 patients were used to train a random forest model to estimate proliferative index(%Ki-67 expression). Model inputs were routine MRI sequences: T1w (pre- and post-contrast), T2w, and FLAIR. This model was applied to a glioma patient cohort from our institution containing 207 grade II, 246 grade III, and 728 grade IV tumors. MR images for these patients were co-registered and normalized using normal tissue intensities. The random forest was applied voxel-wise through the automatically segmented tumor volume to estimate the pointwise proliferative index. We then calculated the maximum estimated proliferative index with at least 0.1cc volume in each tumor. For 811 cases with postoperative imaging available we also calculated the maximum postoperative proliferative index within unresected tumor. Maximum estimated proliferation was correlated with survival using univariate and multivariate(including mental status, age, grade) Cox models. **Results:** The root-mean-square-error for predicting Ki-67 with the random forest was 5.4%. A threshold of 11.6% on maximum preoperative proliferative index showed strong survival differences between patients with high and low proliferation tumors: univariate hazard ratio of 2.18(log-rank $p < 0.05$), multivariate hazard ratio 1.38($p < 1E-4$). Furthermore, reduction in highly proliferative disease postoperatively was associated with improved survival. Patients with high maximum postoperative proliferation had significantly worse survival($p < 0.05$) than patients where highly proliferative disease was removed. **Conclusion:** MR imaging non-invasively predicts pointwise proliferative activity. Estimated proliferation also predicts survival in glioma patients independently of clinical factors and tumor grade. Changes after surgery are prognostic, supporting the use of estimated proliferation to both guide and assess surgical intervention.

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MO-EF-TRACK 4-05

Intentional Deep Overfit Learning (IDOL): A Novel Deep Learning Strategy for Adaptive Radiation Therapy

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Purpose: Applications of deep learning (DL) are promising to realize an effective adaptive radiotherapy (ART) workflow. Despite the promise demonstrated by DL approaches in several critical ART tasks, there remain unsolved challenges to achieve satisfactory generalizability for a trained model in a clinical setting. Foremost among these is the difficulty of collecting a task-specific training dataset with high quality and consistent annotations. In this study, we propose a tailored DL framework for patient-specific performance that leverages the behavior of a model intentionally overfit to a patient's available prior information – an approach we term Intentional Deep Overfit Learning (IDOL). **Methods:** Implementing the IDOL framework in any task in radiotherapy consists of two training stages: 1) training a generalized model with a diverse set of n patients, just as in the conventional DL approach, and 2) intentionally overfitting this general model to the patient of interest ($n+1$) using perturbations and augmentations of the available task- and patient-specific prior information to establish a personalized IDOL model. The IDOL framework itself is task-agnostic and is thus widely applicable to many components of the ART workflow, three of which we explore here. **Results:** In the replanning CT auto-contouring task, the Dice similarity coefficient for the parotid glands improves from 0.865 with the general model to 0.939 by adopting the IDOL model. In the case of MRI super-resolution, the mean absolute error (MAE) is improved by 28% using the IDOL framework over the conventional model. Finally, in the synthetic CT reconstruction task, the MAE is reduced from 68 to 22 HU by utilizing the IDOL framework. **Conclusion:** In this study, we demonstrate the wide applicability of the IDOL framework to common ART tasks. We expect the IDOL framework to be especially useful in creating personally tailored models in situations with limited availability of training data.

MO-EF-TRACK 4-06

Joint Optimization of Subsampling Pattern and Reconstruction of DCE MRI for Pharmacokinetic Parameter Estimation

J Zou^{*}, J Balter, Y Cao, University of Michigan, Ann Arbor, MI

Purpose: To optimize a deep neural network (NN) that incorporates k-space subsampling pattern selection and pharmacokinetic (PK) parameter estimation for reconstruction of dynamic contrast-enhanced (DCE) MRI. **Methods:** 2D Cartesian phase-encoding subsampling patterns in a 3D Cartesian acquisition are modeled as independent and identically distributed samples from a multivariate Bernoulli distribution which is jointly learned within a convolutional recurrent reconstruction NN. A PK parameter estimation layer which produces PK parameters from the extended Tofts (eTofts) model using linear least-squares fitting follows the reconstruction network. The PK parameters of 25 real DCE-MRI scans were used to generate simulated dynamic image time series using the eTofts model which were then used for network training and testing. Gaussian noise with CNR=20 was added to the simulated data. 15 scans were used for training, 5 for validation, and

5 for testing. Loss functions investigated included combinations of 1) image reconstruction L_2 loss and 2) PK parameter estimation L_2 loss. An undersampling acceleration rate of 9 was used in all experiments. PK parameter estimation accuracy was evaluated by root mean squared error (RMSE). Performance was compared to a dictionary learning-based (DL) method. **Results:** The combined image reconstruction and PK parameter loss yielded the lowest RMSE for all 3 eTofts model parameters, followed by the image reconstruction loss alone, and finally the DL method which demonstrated the largest deviation from ground truth. The combined loss reduced the RMSEs by 38.5%, 54.5%, and 41.0% for K^{trans} , $k(ep)$, and $v(p)$, respectively, compared with the DL method, with corresponding improved SSIM of reconstructed images (.98 vs .69). **Conclusion:** A NN that jointly optimizes image subsampling patterns and reconstruction yields images and quantitative model parameters from undersampled MR scans that outperform DL techniques. Different subsampling pattern parameterization strategies and application on real DCE data will be investigated in the future.

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MO-EF-TRACK 4-07

Latent Space Arc Therapy Optimization

N Bice^{1*}, N Kirby¹, D Nguyen², C Kabat¹, P Myers¹, N Papanikolaou¹, M Fakhreddine¹, (1) UT Health San Antonio MD Anderson Cancer Center, San Antonio, Texas, (2) UT Southwestern Medical Center, Dallas, TX.

Purpose: Volumetric modulated arc therapy (VMAT) planning is a challenging problem in high-dimensional, non-convex optimization. Current arc therapy optimization algorithms favor solutions near the initialization point and are slower than necessary due to plan overparameterization. In this work, we reduce the effective dimension of arc therapy plans with unsupervised deep learning and navigate the learned low-dimensional coordinates during VMAT optimization. **Methods:** We collected a dataset of 1,874 clinically delivered VMAT arcs from historical treatments documented in ElektaVersaHD log files. Multileaf collimator (MLC) leaf positions are represented at 80 evenly-spaced control points for 80 leaves on 2 banks (12,800-dimensional parameterization) over the duration of the arc. A family of 2D convolutional variational autoencoders (VAE) is used to reduce the dimension of the arc data (embedding dimensions $d \in \{32, 64, 128\}$). For comparison with linear machine learning dimensionality reduction, principal component analysis (PCA) is used. Trained models are used to search for held-out arcs with a leaf-position-based objective function and simulated annealing. For comparison with a traditional routine, stochastic global direct aperture optimization with geometry-based segment initialization is implemented. **Results:** VAEs can compress arcs into 32, 64, and 128 dimensions and reconstruct them with median absolute leaf position errors of 1.59, 1.43, and 1.58 mm, respectively. PCA with the same embedding dimensions returns median errors of 7.99, 6.83, and 5.55 mm. When allowed the same number of maximum iterations, we find that the latent space algorithms converge in significantly fewer iterations compared to traditional methods. Given the same optimization problem, latent space optimization approaches a solution significantly faster (2.5 minutes) compared to conventional DAO (7.5+ minutes). **Conclusion:** Learned latent spaces provide a setting for efficient VMAT optimization. While global DAO can reach lower objective scores given enough time, optimization in learned latent spaces can achieve similar accuracy much faster.

MO-EF-TRACK 4-08

MR Multitasking-Based Dynamic Imaging for Cerebrovascular Evaluation (MT-DICE): Development and Feasibility Study on Brain Cancer

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Purpose: Dynamic contrast enhanced MRI (DCE-MRI) and dynamic susceptibility contrast MRI (DSC-MRI) provide permeability- and perfusion-related parameters, respectively, and are evolving as increasingly common modalities for evaluating a variety of brain cancers. However, acquiring both in one exam requires two separate scans and multiple contrast injections. In this work, we developed an MR MultiTasking-based Dynamic Imaging for Cerebrovascular Evaluation (MT-DICE) technique for simultaneous high-temporal-resolution, distortion-free DCE-MRI and leakage-insensitive DSC-MRI in one scan with a single-dose injection. **Methods:** MT-DICE employs a 3D Cartesian acquisition with periodic non-selective saturation recovery (SR) preparations followed by multi-echo FLASH readouts. Image reconstruction is performed based on MR multitasking framework with an SR time dimension and an echo time dimension for dynamic T_1/T_2^* quantification, respectively, and a contrast phase dimension for contrast dynamics imaging. DCE- and leakage-insensitive DSC-MRI metrics are computed by tracer kinetic modeling with tracer concentrations derived directly from the dynamic T_1/T_2^* values. The technique was tested on 8 healthy subjects and 5 patients with known brain tumors at 3-T with a 7.6-minute scan ($1.5 \times 1.5 \times 4.0 \text{ mm}^3$ spatial resolution, 1.2s temporal resolution). Inversion-recovery TSE and multi-echo GRE were performed in healthy subjects to provide T_1/T_2^* references, respectively. Three healthy subjects were returned for a second session for reproducibility assessment. **Results:** MT-DICE T_1/T_2^* values showed excellent quantitative agreement with references with all intraclass correlation coefficients (ICC) ≥ 0.85 . The permeability and perfusion parameters demonstrated good inter-session reproducibility, with the ICCs of v_p , K^{trans} , CBV, and CBF at 0.822, 0.694, 0.872, and 0.857, respectively. Derived values of the kinetic parameters in all healthy subject and patient cases were within the literature range. **Conclusion:** MT-DICE simultaneously quantifies permeability and leakage-insensitive perfusion parameters in a single 7.6-minute scan with a single-dose injection. Such a spatially matched multiparametric evaluation would potentially improve the performance of tumor characterization and treatment response monitoring.

MO-EF-TRACK 4-09

Non-Circular Orbits on a Clinical Robotic C-Arm for Reducing Metal Artifacts in Orthopedic Interventions

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Purpose: Metal artifacts pose a challenging problem to cone-beam CT (CBCT) in image-guided orthopedic interventions. Highly attenuating metal implants effectively create a missing-data problem and this incomplete data yields reconstruction artifacts. Previous work has demonstrated that non-circular orbits can improve data completeness and potentially avoid missing data. In this work, we present the first implementation of continuous non-circular orbit acquisitions on a clinical robotic C-arm for reduction of metal artifacts. **Methods:** A Siemens ARTIS pheno robotic C-arm with a Siemens Test Automation Control System (TACS) was used to image a 3D-printed (copper/PLA blend) cervical spine phantom implanted with four stainless steel pedicle screws using two non-circular orbits: a sawtooth orbit and a double-circle orbit. The sawtooth orbit rotated 360° continuously in the LAO/RAO direction tilted between $+25^\circ$ and -25° in two cycles in the CRAN/CAUD direction. For the double-circle orbit, two 360° LAO/RAO rotations were conducted with fixed 30° and -30° CRAN/CAUD tilt. To accommodate the irregular projection data, model-based iterative reconstruction was applied using a penalized-likelihood approach. Comparisons between orbits including a conventional circular scan are presented. Additionally, we analyze the effect of precision of system geometry on image quality. **Results:** Both non-circular orbits significantly reduce the artifacts surrounding the implants, permitting better visualization of pedicle screws and the surrounding anatomy. In extreme cases where two screws coincide in the same axial plane, the anatomical structures are highly obscured in the conventional scan, while the non-circular orbits preserve anatomical details of the vertebrae, surrounding soft tissue, and threads on the screws. Moreover, by further refining the machine recorded geometry, we reduce the noise level by 40.67%. **Conclusion:** This is the first continuous acquisition of non-circular orbits on a clinical imaging system. In an emulated cervical spine scan, we demonstrate the ability of non-circular orbits to dramatically reduce metal artifacts.

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MO-EF-TRACK 4-10

Tumor Dosimetry and Radiobiological Modeling of Combined Radiopharmaceutical Therapy and External Beam Radiotherapy for Recurrent Head and Neck Cancer

D Adam*, I Marsh, J Grudzinski, T Bradshaw, P Hill, S Cho, C Hullett, A Burr, P Harari, B Bednarz, University of Wisconsin, Madison, WI

Purpose: Radiopharmaceutical therapy (RPT) in combination with external beam radiotherapy (EBRT) shows promise as a method to offer disease control and mitigate potential high-grade toxicities associated with re-treatment for patients with recurrent head and neck cancer. This work establishes a patient-specific dosimetry framework that uses radiobiological models to combine the dosimetry from RPT and EBRT and applies it to a first-of-a-kind clinical trial where patients are treated with a ^{131}I -based RPT compound (CLR 131) and reduced fractionation EBRT. **Methods:** Serial SPECT/CT patient scans performed at approximately 24-, 48-, 72-, and 168-hours post-injection of 15.6 mCi/m² CLR 131 were used as input to a GEANT4 based RPT dosimetry platform named RAPID. Phantom-based recovery coefficients were applied to physician delineated tumor volumes to account for partial volume effects. Separately, EBRT treatment plans were created using standard of care (SOC) clinical practices (prescribed doses of 60-66 Gy in 2 Gy fractions). For treatment, the total number of EBRT fractions delivered was reduced accordingly to achieve the SOC total dose when including the CLR 131 RPT mean absorbed dose to the PTV. Radiobiological models (BED, EUD, EQD2) were used to compare the two radiation modalities. **Results:** The RPT mean absorbed dose to the PTV reduced EBRT treatments by 3-4 fractions. The RPT BED to the PTV was nearly equivalent to the absorbed dose yet the RPT EUD indicated dose heterogeneity in two patients. The EBRT EUD indicated the absorbed dose was largely homogeneous and the EBRT BED was greater than the EBRT absorbed dose. **Conclusion:** RPT proffers a method to reduce the EBRT dose from current SOC treatments. Incorporating RPT dose during the optimization of EBRT plans could further reduce normal tissue toxicities. The framework employed here is extendible to other permutations of EBRT and RPT treatments.

This work is supported by NIH SPORE CA196513-01. BB and JG are co-founders of Voximetry, Inc., a nuclear medicine dosimetry company in Madison, WI.

SAM Therapy Educational Course

TRACK 5

Overview of MRI Implementation in HDR Brachytherapy for Treatment of Gynecologic and Prostate Cancer

MO-E-TRACK 5-00

Overview of MRI Implementation in HDR Brachytherapy for Treatment of Gynecologic and Prostate Cancer

Moderator: Mahta McKee, Wake Forest University School of Medicine, Winston Salem, NC

MO-E-TRACK 5-01

M. McKee, Wake Forest University School of Medicine:

Use of MR in Gynecological Brachytherapy

MO-E-TRACK 5-02

F. Mourtada, Christiana Care Hospital:

Nuts and Bolts to Establish Your MRI-HDR Program

MO-E-TRACK 5-03

G. Cohen, Memorial Sloan-Kettering Cancer Center:

Prostate HDR and Utilization of MR

The ability of magnetic resonance imaging (MRI) to create distinct images of the body's soft tissues and the tumors that arise amid them is helping Brachytherapy Radiation Oncologists to precisely shape doses to cervical and prostate tumors, while at the same time avoiding exposure to critical healthy organs and tissues. Conforming brachytherapy doses closely to the tumor's shape and position is important in order to apply the highest possible dose to the tumor while limiting the dose to critical structures, such as the bladder rectum, and urethra thereby decreasing the likelihood of treatment side effects.

To achieve the above, comprehensive quality assurance and safety considerations need to be put in place prior to clinical use of MRI is Radiation Oncology imaging and treatment planning.

There are three approaches to incorporating MRI into the brachytherapy workflow. They are: MR-informed MR-guided, and MR-based brachytherapy. Workflows for MR based HDR brachytherapy will be discussed, however, many of these processes apply to MR-guided techniques as well. Safety consideration will be reviewed and their implications to applicator, and workflow selection.

Implementation of MR-based HDR gynecologic and prostate brachytherapy will be discussed, including development of an imaging protocol, applicator commissioning and reconstruction, simulation techniques, contouring, treatment planning and dose adaptation.

Learning Objectives:

1. To understand the advantages of having MR based brachytherapy
2. To understand the safety considerations when using MR based brachytherapy
3. Clinical considerations when utilizing MR for GYN and prostate brachytherapy

SAM Therapy Educational Course

TRACK 6

Perspectives on Imaging Dose in IGRT

MO-E-TRACK 6-00

Perspectives on Imaging Dose in IGRT

Moderators: George Ding, Vanderbilt University, Nashville, TN and Grace Tang, Memorial Sloan-Kettering Cancer Center, New York, NY

MO-E-TRACK 6-01

P. Alaei, University of Minnesota:

Advances in Image Guidance with Emphasis on X Ray-Based Modalities

MO-E-TRACK 6-02

G. Ding, Vanderbilt University:

Weighing the Risks and Benefits of Imaging Guidance with Ionizing Radiation

MO-E-TRACK 6-03

Q&A Panel

Image guided radiation therapy (IGRT) has become the standard of care in radiotherapy. Modern day IGRT, first introduced in late 1990s, has evolved over the past two decades by introduction of new techniques, often involving multiple modalities. During a daily treatment fraction, the patient may be imaged more than once in order to position the patient and to confirm repositioning if necessary. Image guidance is also employed for intrafraction motion management and adaptive radiotherapy. These imaging procedures, if x ray based, expose additional radiation dose to healthy tissues outside radiotherapy targets. Therefore, understanding the magnitude of this extra dose and the risks associated with these doses are important. Both AAPM TG-75 and TG-180 reports were developed to address the concerns and to provide guidelines and strategies for minimizing imaging dose while improving treatment delivery.

This lecture reviews advances in image guidance during the radiotherapy procedures with emphasis on x ray-based modalities. The evolution of IGRT and currently available IGRT tools and techniques will be reviewed with emphasis on x ray-based ones. Newer specialty radiation delivery units like proton therapy often employ customizable IGRT devices that may need attention, especially with regard to radiation dose. These devices will also be reviewed during this session. The session also provides an overview of the AAPM TG-180 report as well as the recommendations of the report on managing image guidance doses.

The session further discusses techniques for reducing the imaging dose and explains the recommended threshold beyond which imaging dose should be considered in the treatment planning process. It addresses the techniques and methods that can be used in clinical practice to estimate imaging dose resulting from different kV and MV imaging procedures. It explains two recommended methods of estimating patient imaging dose during treatment planning. The lecture also presents a perspective view of radiation dose to healthy tissues resulting from the sources such as leakage, out-of-field dose, etc. and from image-guidance procedures. While the commonly adopted radiation protection safety philosophy of As Low As Reasonably Achievable (ALARA) is applicable to imaging guidance dose, minimizing imaging doses by employing reasonable methods should not compromise target localization. Therefore, balancing ALARA principles with the requirement for effective target localization requires that imaging dose be managed based on the consideration of weighing risks and benefits to the patient.

Learning Objectives:

1. Learn the latest advances in IGRT modalities and techniques
2. Learn the applications of IGRT in specialty delivery units
3. Learn the magnitude of typical organ doses resulting from different imaging procedures and modalities
4. Learn available techniques to reduce the imaging dose in daily clinical practice and how to account for kV and MV imaging dose.
5. Develop a perspective view of radiation to sensitive organs between imaging dose and unintended dose from therapeutic beams.
6. Understand weighing the risks and benefits of imaging guidance for effective target localization.

SAM Professional Symposium

TRACK 7

Increasing the Contribution of Radiology and Imaging Physics to Global Health Initiatives

MO-E-TRACK 7-00

Increasing the Contribution of Radiology and Imaging Physics to Global Health Initiatives

Moderators: Ashley Rubinstein, Henry Ford Health System, Detroit, MI and Joseph Weygand, Henry Ford Health System, Willow Grove, PA

MO-E-TRACK 7-01

E. Medel Baez:

A Therapy Physicist's Perspective: Developing a Radiology Quality Assurance Program in An Underserved Region

MO-E-TRACK 7-02

C. Morrison, Henry Ford Health System:

An Imaging Physicist's Perspective: Volunteering in Underserved Regions

MO-E-TRACK 7-03

D. Mollura, Rad-aid International:

RAD-AID International's Initiatives to Improve Access to Radiological Care

MO-E-TRACK 7-04

C. Borrás:

The Role of Intergovernmental Organizations in Promoting Radiology to Improve Global Health

MO-E-TRACK 7-05

Audience Q&A

AAPM and its individual members have been engaged in global outreach efforts for some time. AAPM's imaging physics community is involved in projects with several intergovernmental and non-governmental international organizations (e.g., the IAEA and RAD-AID). However, some imaging physicists - especially those early in their career - may find it difficult to break into global health work. Even once actively involved, they may have difficulty making meaningful, effective contributions. In some regions, physicists' work is hindered by a lack of regulations, enforcement of regulations, and resources needed to comply with regulations. Often, the work of individual physicists is not part of a concerted, continuous effort and would benefit from more organized partnerships. With the formation of the International Council, AAPM's global activities have been centralized to allow for more efficient use of resources and greater collaboration between individual efforts. To foster participation at a global level, this session will highlight AAPM's current contributions to global health initiatives in radiology and will discuss challenges and areas of growth. We will demonstrate the need for radiology and imaging physics services in regions with limited resources, provide examples of successful partnerships, and illustrate opportunities for continued growth.

Learning Objectives:

1. Learn about challenges faced in regions with limited access to radiology and imaging physics services

2. Increase awareness of opportunities for imaging physicists to contribute to global health initiatives
3. Learn about creating successful, effective partnerships with institutions in low-resource regions

Vendor Showcase

TRACK 8

MIM Software, Inc.

MO-SHOWCASE-TRACK 8-7-01

Practical Applications of MIM SureCalc® MonteCarlo for Adaptive Assessment and Plan Second Check

J Kuhn^{1*}, (1) MIM Software, Inc.

History Symposium

TRACK 1

History Symposium: Exploring the Foundation of Our Medical Physics Profession

MO-F-TRACK 1-00

History Symposium: Exploring the Foundation of Our Medical Physics Profession

Moderators: Slavik Tabakov, King's College London - KSCMD, London, and Geoffrey Ibbott, The American Board of Radiology, Tucson, AZ

MO-F-TRACK 1-01

Introduction and History Project Overview

MO-F-TRACK 1-02

P. Sprawls, Emory University and Sprawls Educational Foundation:

The Physics Foundation and Evolutionary Technical Developments of Computed Tomography, the Quest for Enhanced Visibility

MO-F-TRACK 1-03

J. Van Dyk, Western University:

The Atom Bomb That Saves Lives™: The Role of Cobalt-60 in Cancer Treatment Around the World

MO-F-TRACK 1-04

J. Battista:

Harold Johns: Combining Innovation and Education in Establishing the Role of Medical Physicists in Radiation Oncology

MO-F-TRACK 1-05

Concluding Discussion

Physics is the fundamental science of many modern diagnostic and therapeutic medical procedures. For over 135 years physical principles have been applied in the development of technologies and methods to enhance the practice of clinical medicine with two major goals, more effective diagnostic imaging procedures and applying radiation for the treatment of diseases, especially cancer.

This continues to be a dynamic process with the evolution of technology throughout society and especially the impact of digital science, technology, and applications. Our current medical physics activities and clinical applications are built on the foundation of the research and developments by many of our medical physics pioneers that are our history and heritage.

The International Organization for Medical Physics (IOMP) in collaboration with the AAPM has an extensive project for documenting and publishing elements of this history, often by physicists who were active participants in the developments during the recent half-century. These are published at: <http://www.mpjournal.org/history.aspx> and selected ones presented at the AAPM History Symposium.

Learning Objectives:

1. Follow the events and contributions of individuals in the development and evolution of computed tomography (CT).
2. Explore the Role of Cobalt-60 in Cancer Treatment Around the World.
3. Appreciate the contributions of Harold Johns to the medical physics profession and practice.

SAM Imaging Educational Course

TRACK 3

MRI Homogeneity Testing

MO-F-TRACK 3-00

MRI Homogeneity Testing

Moderator: Joseph Och, Geisinger Medical Center, Danville, PA

MO-F-TRACK 3-01

J. Och, Geisinger Medical Center:

Resources for Homogeneity Testing of MRI Systems

MO-F-TRACK 3-02

C. Brunquell, University of Washington:

Vendor-Specific Approaches to Main Magnetic Field Homogeneity Testing

MO-F-TRACK 3-03

Audience Q & A

MRI Homogeneity Testing

In MRI, main magnetic field (B0) homogeneity is an important factor in performance. Annual testing of B0 homogeneity is required by the ACR accreditation program and Joint Commission standards. However, many physicists are uncertain as to how to conduct this test and interpret the results, and each vendor has their

own accepted methods for testing B0 homogeneity and corresponding tolerance levels.

This session will explain the main methods of homogeneity testing and provide specific instructions on how to perform these tests on individual vendor platforms. Advice on interpreting the results will also be discussed. Resources to assist the physicist with homogeneity testing will be identified.

Learning Objectives:

1. Learn vendor-specific methods for testing homogeneity.
2. Understand how to set performance criteria.
3. Identify resources to assist the physicist with homogeneity testing.

SAM Therapy Educational Course

TRACK 5

Report of TG 182: Electronic Intracavitary Brachytherapy Quality Management Based on Risk Analysis

MO-F-TRACK 5-00

Report of TG 182: Electronic Intracavitary Brachytherapy Quality Management Based on Risk Analysis

Moderator: Bruce Thomadsen, University of Wisconsin, Madison, WI

MO-F-TRACK 5-01

B. Thomadsen, University of Wisconsin:

Electronic Brachytherapy and Quality Management Development

MO-F-TRACK 5-02

M. Rivard, Rhode Island Hospital / Brown University:

The Xofig Axxent System and Sample Design of Quality Management

MO-F-TRACK 5-03

S. Pillai, Oregon Health & Science Univ:

The Zeiss INTRABEAM System and Sample Design of Quality Management

MO-F-TRACK 5-04

Q&A

Purpose: To provide guidance on quality management for electronic brachytherapy.

Materials and Methods: The task group used the risk-assessment approach of Task Group 100 of the American Association of Physicists in Medicine. Because the quality management program for a device is intimately tied to the procedure in which it is used, the task group first designed quality interventions for intracavitary brachytherapy for both commercial electronic brachytherapy units in the setting of accelerated partial-breast irradiation. To demonstrate the methodology to extend an existing risk-analysis for a different application, the task group modified the analysis for the case of post-hysterectomy, vaginal cuff irradiation for one of the devices.

Results: The analysis illustrated how the TG-100 methodology can lead to interventions to reduce risks and improve quality for each unit and procedure addressed.

Conclusion: This presentation on this report provides a model to guide facilities establishing a quality management program for electronic brachytherapy.

Learning Objectives:

1. To become familiar with the contents of the report of TG 182.
2. To see how the TG-100 approach can be applied to the design of QM for an electronic brachytherapy unit.
3. To understand why the design of a QM program is customized to a clinic's procedure.

SAM Imaging Educational Course

TRACK 6

Providing Value Beyond Accreditation and Compliance Testing: Imaging Physics

MO-F-TRACK 6-00

Providing Value Beyond Accreditation and Compliance Testing: Imaging Physics

Moderator: Timothy Szczykutowicz, University Wisconsin-Madison, Madison, WI

MO-F-TRACK 6-01

C. Brunnquell, University of Washington:

Physics Pitch to Radiology Administrator: 1

MO-F-TRACK 6-02

C. Bush, Touchstone Medical Imaging:

Radiology Administrator's Perspective: 1

MO-F-TRACK 6-03

S. Rose, University of Texas at Houston Medical School:

Physics Pitch to Radiology Administrator: 2

MO-F-TRACK 6-04

A. Bush:

Radiology Administrator's Perspective: 2

MO-F-TRACK 6-05

Audience Q & A

Radiology administrators are not likely eager to pay for physicists to engage in activities outside of accreditation and regulatory compliance roles. Physicists, however, have a skill set that is aptly suited to tackle problems outside of these traditional roles. This symposium will teach the physicist how to bring additional value to a

radiology department. Our speakers are split 50/50 between physicists and administrators. Topics will include non-traditional physics tasks like: building scanner/indication based scheduling template rules, contrast administration/timing/uniformity across scanners and sites, scan speed optimization in CT for pediatrics to mitigate motion and sedation, monitoring exam timeliness in the setting of rapid imaging needs, and monitoring protocol selection and dose consistency across an enterprise. Examples will be presented in a manner which does not require on-site physics support. Therefore, our examples should resonate with the consulting physicist or the off-site academic physicist in addition to onsite physics support.

Learning Objectives:

1. Learn how to quantify potential time and contrast agent savings and provide opportunities for increased efficiency to support physics time spent outside of traditional roles.
2. Walk away with references and examples of non-traditional physics projects you can implement.

TPS consultant to GE Healthcare, Cybermed.ai, IMALOGIX, ischemaview, AstoCT, AiDoc. TPS on advisory board GE and imalogix.

SAM Professional Symposium

TRACK 7

Virtual Global Mentoring: Now More Than Ever

MO-F-TRACK 7-00

Virtual Global Mentoring: Now More Than Ever

Moderators: Jacob Van Dyk, Western University, London, ON and Robert Jeraj, University of Wisconsin, Madison, WI

MO-F-TRACK 7-01

J. Van Dyk, Western University:

Introduction and Overview on Virtual Mentoring

MO-F-TRACK 7-02

L. Hirschhorn, Northwestern University:

Mentoring Across Oceans: Experiences, Benefits and Pitfalls

MO-F-TRACK 7-03

B. Goldberg, Northwestern University:

Mentoring with Social and Emotional Intelligence – The Role of Personal Narrative

MO-F-TRACK 7-04

R. Jeraj, University of Wisconsin:

Mentoring Activities Within the AAPM

MO-F-TRACK 7-05

Discussion

The importance of mentorship within health care settings is well recognized and serves as a key component of training to build clinical, teaching and research knowledge, and to advance interpersonal, communication and leadership skills. Usually, this involves an experienced, well-regarded, empathetic person (the mentor) who guides another (usually less experienced and possibly younger) individual (the mentee) in the development of their clinical, teaching, and research skills, as well as their personal and professional growth. Formalized mentoring has been promoted in the academic world for many years to advance an individual's career development and improve teaching and research. Virtual mentoring is a more recent phenomenon enabled with expansion of broadband internet, together with a variety of tools that allow easier communication across the world. Virtual mentoring has the further potential of providing professional training and education support to under-resourced regions. Virtual mentoring activities were significantly escalated in the last year with the COVID-19 pandemic and the additional emphasis on remote instruction.

Both the mentor and mentee benefit from the mentoring process. A challenge for the medical physicist who is mentoring someone from an under-resourced region is to truly understand the local circumstances of the mentee's environment including the institutional, societal, economic, cultural, political, clinical, and technological perspectives. Additional challenges relate to time-zone differences, insufficient internet bandwidth, and limitations in time and availability. A further challenge relates to mentoring associated with practical hands-on activities in a clinical environment, especially in the context of medical physics residency programs.

There is evidence that mentorship works the best when the mentor and mentee have explicitly shared their values, interests, expectations, and goals. During the past year, research and practice in remote mentoring have further highlighted the need to reassess, re-evaluate, and reconnect intentionally between mentor and mentee. We will demonstrate mentoring approaches that extend best practices of shared understanding and trust building to remote mentoring.

This session will provide: (1) an overview of global mentoring, in healthcare and academic contexts, and at multiple levels (e.g., graduate student, medical physics resident, early career medical physicist, mid-career medical physicist, academic medical physicist), (2) a discussion of the foundational frameworks of successful mentoring methods, with an emphasis on virtual mentoring, (3) a discussion of factors influencing the success of virtual mentoring in medical physics, and (4) a summary of AAPM mentoring activities, particularly those aimed at trainees and early careers.

Learning Objectives:

1. Describe an overview of global mentoring in the clinical and research settings and the benefits that it provides.
2. Define the characteristics of successful approaches to virtual mentoring.
3. Explore and self-reflect on virtual mentoring methods, i.e., techniques and technologies and identify strategies to apply to their practice.
4. Describe AAPM mentoring activities.

License agreement with Modus Medical Devices, Inc. of London, Ontario, Canada

Vendor Showcase

TRACK 8

IBA Dosimetry

MO-SHOWCASE-TRACK 8-8-01

Patient QA Efficiency and Accuracy Through High-Resolution Measurements, Automation and Prediction

I Fotina^{1*}

Imaging Interactive ePoster Discussion TRACK 1

Imaging: Multimodality Imaging Topics

MO-lePD-TRACK 1-01

Multi-Scale Residual Network for Tumor Segmentation in Pancreatic Cancer

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Purpose: Accurate segmenting tumor is of great importance on effective diagnosis and treatment for pancreatic cancer. We aim to develop a new multi-scale residual network (MSRN) for accurate and automatic tumor segmentation. **Methods:** Totally 71 positron emission tomography (PET) and 71 MR images are used in this study. In each scan, the tumor was contoured by an experienced radiologist and used as ground truth. Due to the heterogeneity of pancreatic components and unclear tumor boundaries, achieving segmentation of pancreatic tumors may present challenges. Accurately and automatically segment the tumor can greatly help the diagnosis and treatment of pancreatic tumors. The model is designed based on multi-modal feature fusion network. In order to fuse information in the multi-modal image, the feature maps of PET and MRI were extracted through two parallel multi-scale networks respectively and were fused. And then the squeezing-and -excitation block (SE-block) was used to automatically improve the valid features in the fused feature map and suppress the invalid features. The registered training images are input into the multi-modal feature fusion network to train the segmentation model, and the lesion features are extracted through the multi-scale residual network. The testing phase includes two steps: (1) tumor segmentation through trained multi-modal fusion network; (2) the distance regularized level set evolution (DRLSE) method is applied to post-processing. The segmentation results are evaluated through dice similarity coefficient (DSC), sensitivity (SEN) and specificity (SPE). **Results:** Five-fold cross validation was performed in this study. The average results on quantitative analysis for DSC, SEN, and SPE are 0.742, 0.771, and 0.993, respectively. **Conclusion:** A new multi-scale residual network (MSRD) was developed for tumor segmentation in pancreatic cancer in this study. The experimental results demonstrated that MSRD can obtain promising segmentation results.

MO-lePD-TRACK 1-02

Quantitative Evaluation of Intra- and Inter-Modal Deformable Image Registration of CT and MR Images for Heart, Left-Ventricle and Thoracic Aorta

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Purpose: The first step in correlating local radiation dose and regional cardiac magnetic resonance (CMR) findings is mapping CT-derived radiation doses onto CMR via image registration. This is the first study to evaluate the accuracy of intra- and inter-modality deformable image registration (DIR) for heart, left ventricle (LV), and thoracic aorta (TA), using paired longitudinal CT and pre/post contrast MR imaging. **Methods:** 0% (INH) and 50% (EXH) breathing-phases from 4DCT, and inspiratory/expiratory pre-/post-contrast MR image sets were acquired from five lung cancer patients twice during radiotherapy treatment. DIR was performed for the following comparisons: Different time points - (C1) intramodal CT, (C2) intramodal MR, (C3) intermodal CT/MRI; Same time point - (C4) intermodal CT/MR, (C5) intramodal MR pre/post contrast. Registration performance was assessed using HD, MDA, and DICE for the original and deformed contours. **Results:** Mean MDA ranged 1-2 mm, except for the two intermodality comparisons of the LV (2.6mm for both). Mean HD ranged 8-17 mm. Mean DICE ranged 0.85-0.95. Overall, all metrics performed better intramodality, with/without contrast. Contour comparisons via HD and MDA were least favorable for LV, and similar for TA and heart. DICE was best in heart, with LV and TA performing similarly. The following scenarios had statistically significant differences ($p < 0.05$): HD: (C2), (C4), (C5) for LV_vs._TA; MDA: (C1), (C2), (C4), (C5) LV_vs._heart (C2)-(C5) LV_vs._TA; DICE: (C1)-(C5) for TA_vs._heart and LV_vs._heart. When the various modalities were compared, the differences were found significant mainly between (C1) vs. (C4), and (C2) vs. (C4), for almost all structures of interest. **Conclusion:** Though intramodal registration remains slightly superior, both intramodal and intermodal DIR are within TG 132 recommended range for the structures of interest, suggesting that quantitative regional analyses of CT and MR images from patients undergoing radiotherapy may be spatially correlated without significant error following registration, even across modalities.

MO-lePD-TRACK 1-03

Spatial Transformation Network-Based MRI-CBCT Deformable Registration for CBCT-Guided Prostate Radiotherapy

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Purpose: Radiotherapeutic dose escalation to dominant intraprostatic lesions (DIL) in prostate cancer could potentially improve tumor control. This study aims to propose an unsupervised method to accurately register multiparametric MRI to CBCT images for improved DIL delineation, treatment planning, and dose monitoring in CBCT-guided prostate radiotherapy. **Methods:** Our proposed spatial transformation network (STN) consists of two consecutive networks: zoomed-out and zoomed-in feature networks. Both networks have adversarial learning integrated to further force the realism of derived deformation vector field (DVF). The network begins with MRI and CBCT as inputs to the zoomed-out network to predict an overall approximation of DVF, which is transferred to spatial transformer to generate zoomed-out version of deformed MRI. For subsequent zoomed-in network training, 3D patches are extracted from zoomed-out version of deformed MRI. Zoomed-in network learns smaller changes within the extracted patches to generate more local DVF, which is transferred to spatial transformer for generating deformed patch. We performed leave-one-out cross-validation experiment on 30 pelvic cancer patients to evaluate the proposed registration method. The target-registration-error (TRE) as the Euclidean distance of the identified landmarks was used to quantify the accuracy of the proposed MRI-CBCT registration. **Results:** Overall results of this work showed feasibility of our proposed network architecture to accurately register multimodal images with good spatial alignment of different structures. An average TRE of 2.9mm (range 2.1-3.7mm) was achieved after registration for all patients, compared with 5.7mm (range 4.9-8.3mm) before registration. The proposed network can generate a final DVF in a single forward prediction, which is faster than the conventional iterative registration algorithms. **Conclusion:** We developed a new framework to accurately register the prostate on MRI to CBCT images for external beam radiotherapy. The proposed method could be used to aid DIL delineation on CBCT, treatment planning, dose escalation to DIL, and dose monitoring.

MO-lePD-TRACK 1-04

To Design and Construct Customized Multimodal Imaging Phantoms by 3D Printing

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Purpose: To develop a method for constructing customizable, multimodal quality control (QC) imaging phantoms based on 3D printing technology **Methods:** Four

phantoms were designed and constructed through 3D-printing technology using three unique printing materials. Physical parameters of the 3D-printed materials were evaluated, including density, shore hardness, porosity, deformation temperature, computed tomography (CT) number, absorption coefficient, and printing accuracy. Imaging performance of the phantoms was studied using MRI, CT, PET/MR, and PET/CT, and compared with conventional/commercial phantoms. Imaging assessments included high contrast resolution, low contrast resolution, uniformity, deformation, SNR, slice accuracy/slice thickness, location accuracy/laser alignment, CT number, relaxation time, and registration. **Results:** All three printing materials have a shore hardness of 90. The physical densities of these materials are 1.15 g/cm³, 0.76 g/cm³ and 1.27 g/cm³, respectively. The porosities are 9.09%, 6.81%, and 18.56%, respectively. The threshold temperature of deformation for the three materials is greater than 105°C, which is higher than that of PMMA and silica gel. Imaging scans of the constructed phantoms for single modality scanners (MRI and CT) and dual-modality scanners (PET/MRI and PET/CT) were compared with those of the commercial phantoms. The standard deviation of the HU value uniformity test was less than 3 HU for CT scans. **Conclusion:** 3D-printed medical imaging phantoms allow for rapid, customized phantom fabrication for clinical situations across single and dual modality imaging platforms. Further imaging parameter analysis is underway to provide more quantitative evaluation of the proposed phantoms.

MO-lePD-TRACK 1-05

Effects of Permanent Magnet Shapes Used in Halbach Cylinders On FFL-MPI

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Purpose: Selection field, which contains a field-free line (FFL), is used in Magnetic Particle Imaging (MPI) to allow spatial selection of a specific region to achieve spatial encoding along a line. Since permanent magnets can generate a static magnetic field, there is no need to consider the issue of power consumption. Studies show that the shape and performance of permanent magnets affect the quantity of the magnetic field generated. In this study, FFL was formed by using dipole Halbach cylinders and the effects of magnet shapes on FFL performance were investigated. **Methods:** Dipole Halbach rings consist of six different shapes of permanent magnets (square, hexagon, octagon, dodecagon, hexadecagon, circle) were used for generating FFL. NdFeB (neodymium-iron-boron) was chosen as the magnet material. For each model, magnetic flux density and gradient field strength were simulated using COMSOL Multiphysics 5.4. **Results:** Magnetic flux density and gradient field strength increase on the y-axis from the rectangular magnet shape to the cylindrical shape. Gradient values obtained from the permanent magnets (square, hexagon, octagon, dodecagon, hexadecagon, circle) are 3.6 T/m, 4 T/m, 4.4 T/m, 4.6 T/m, 4.7 T/m, 4.8 T/m, respectively. The homogeneity of the gradient for each design was above 95% and did not change according to the shapes of the magnets used. Also, the gradient values were stable along 35 mm. **Conclusion:** The outcomes show that Halbach magnet systems formed with different magnet shapes have a significant impact on FFL performance. Polygons were used to improve the square-shaped permanent magnets. Gradient has been increased by 25% using the polygonal magnet shapes. Future studies will be continued to increase the stability and bore size of the system.

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No conflict of interest.

Multi-Disciplinary Interactive ePoster Discussion

TRACK 3

Multi-Disciplinary: MRI in Radiation Therapy

MO-lePD-TRACK 3-01

Autosegmentation On Low-Resolution T2-Weighted MRI of Head and Neck Cancers for Off-Line Dose Reconstruction in MR-Linac Adapt-To-Position Workflow

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Purpose: To compare autosegmentation methods on low-resolution T2-weighted on-board setup MRIs from a 1.5T MR-linac for off-line reconstruction of delivered dose. **Methods:** 7 organs-at-risk (OARs) (parotids, submandibulars, mandible, cord, brainstem) were contoured by 7 observers each in 43 images. Ground truth (GT) contours were generated using STAPLE. 20 autosegmentation methods were evaluated in ADMIRE: 1-9) atlas-based autosegmentation using a population atlas library (PAL) of 5/10/15 patients with STAPLE, patch fusion (PF), random forest (RF) for label fusion; 10-19) autosegmentation using images from a patient's 1-4 prior fractions (individualized patient prior (IPP)) using STAPLE/PF/RF; 20) deep learning (DL) (3D ResUNet trained on 43 GT structure sets plus 45 contoured by one observer). Autosegmentation methods were evaluated on 5 images using Dice similarity coefficient, mean surface distance, Hausdorff distance, Jaccard Index against GT. Execution time was measured for each method. Inter-observer variability was quantified using pairwise comparison. For each metric and OAR, performance was compared to inter-observer variability using Dunn's test with control. Methods were compared pairwise using Steel-Dwass test for each metric pooled across OARs. Further dosimetric analysis was performed on contours from DL (fastest), IPP_RF_4fractions (best performance), and PAL_STAPLE_5atlases (worst performance): Delivered doses from clinical plans were recalculated on setup images with each structure set. Percent differences in Dmean and Dmax were measured between GT and each method. **Results:** DL and IPP methods performed best, all significantly outperforming inter-observer variability and with no significant difference between methods in pairwise comparison. Most PAL methods were not significantly different from inter-observer variability or each other. DL was the fastest and PAL methods the slowest. Execution time increased with number of prior fractions/atlas. Dosimetric differences were minimal (median<6%) for DL and IPP_RF_4fractions but greater for PAL_STAPLE_5atlases (median<16%). **Conclusion:** Autosegmentation using DL or IPP is superior to PAL considering both geometric and dosimetric criteria.

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MO-lePD-TRACK 3-02

Dosimetric Benefits of MR Guided Adaptive Radiation Therapy

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Purpose: To quantify dosimetric gains achieved with MR guided adaptive radiation therapy (MRgART) for pancreas cancer patients treated with stereotactic body radiotherapy (SBRT). **Methods:** This IRB approved study evaluated SBRT plans of 15 pancreas cancer patients treated with MRgART on a 0.35T MR linac in our department between 2018 and 2020. Prescription dose was 40Gy to PTV and 50Gy to GTV in 5 fractions with 6MV FFF X-rays. For each fraction, a reference treatment plan was re-computed on the daily MR image for dose predictions. If OARs exceeded dose tolerance, the treatment plan was re-optimized and adapted.

The decision to adapt was driven by an iso-toxicity approach: $V36 \leq 0.5\text{cc}$ for stomach, duodenum, small and large bowels (SB, LB). We looked at adapted plan specifications, reasons for adaptation, and dosimetric gains with adaptation for each OAR (Dmax and V36Gy) and PTV coverage (V40Gy) by comparing predicted vs. re-optimized plans. **Results:** Based on a review of 72 treated fractions (42 adapted), duodenum was the most common reason for plan adaptation (38.1% fractions) owing to its close proximity to pancreas, followed by stomach (31%), SB(26.2%), and LB(9.5%). For 14.3% of fractions, PTV coverage was the decisive factor. For adapted fractions, average improvements in V36Gy (cc) for stomach, duodenum, SB, and LB were (mean $\pm$ s.d.) 1.7 ± 1.2 , 2.3 ± 1.2 , 1.1 ± 0.7 , and 4.0 ± 3.2 cc respectively. The corresponding dose reductions in Dmax for Stomach, Duodenum, SB, LB were $18.1 \pm 7.0\%$, $16.7 \pm 10.3\%$, $15.0 \pm 10.8\%$, and $37.0 \pm 37.8\%$ respectively. For target coverage driven MRgART, V40Gy for PTV improved on average by $5.0 \pm 4.7\%$. **Conclusion:** Intra-abdominal OARs undergo significant motion due to soft-tissue deformation, patient weight loss, respiration, and peristalsis. MRgART can be effective in mitigating the dose impact of these anatomical variations.

MO-lePD-TRACK 3-03

Improving DWI Spatial Resolution with Respiratory Motion Modeling on An Interleaved BSSFP-DWI Sequence

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Purpose: Diffusion-weighted imaging (DWI) may serve as a biomarker for lung and pancreatic tumors undergoing magnetic resonance imaging (MRI)-guided radiotherapy (MRgRT). However, obtaining DWI with sufficient signal-to-noise ratio (SNR) and resolution for biomarker use is challenging for these tumors due to substantial respiratory motion. Furthermore, direct co-registration and summation of DWI frames of moving tumors is difficult due to lack of anatomic information. Here we demonstrate a motion deblurring technique based on interleaving high-resolution, high SNR balanced steady-state free precession (bSSFP) images with low-resolution, low SNR DWI images. **Methods:** Studies on a dynamic thorax phantom and one volunteer were performed on a 0.35T MRgRT machine. The subjects experienced a sequence of bSSFP and DWI multislice scans, which repeated 100 times over approximately 17 minutes in order to adequately capture respiratory motion. A respiratory bellows surrogate was processed in LabVIEW and served as the motion surrogate. The bSSFP images and respiratory bellows were used to extract a motion model which related tissue motion to the bellows trace. The model coefficients were resampled from the bSSFP grid (2mm x 2mm spacing) to the DWI grid (2.73mm x 2.73mm spacing). The warped coefficients, together with bellows surrogate values corresponding to the DWI images, were used as estimated DWI deformation vector fields (DVF). Signal averaging across all DWI images was compared to averaging across the model-registered DWI images. Resolution enhancements were evaluated visually and quantified by fitting a sigmoid function to line profiles across intensity edges. **Results:** Model-estimated DVF accuracy on the bSSFP phantom images was subpixel for nearly every image. Line profile sigmoid widths for the phantom were 8.3 and 3.6 pixels for non-registered and phantom-registered images. Visually, motion blurring was reduced. **Conclusion:** Our work demonstrates the feasibility of motion de-blurring of DWI images using a motion model built from interleaved bSSFP images.

MO-lePD-TRACK 3-04

Optimizing Site-Specific Diffusion Weighted Imaging in an MR-Linac

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Purpose: Diffusion weighted imaging (DWI) is sensitive to changes in tumor microstructure and can be used to non-invasively monitor radiotherapy treatment response. Despite being clinically available in MR-scanners, DWI can be further optimized in MR-guided radiotherapy systems. Maximum b-value is an important sequence parameter that controls the amount of diffusion weighting; high diffusion weighting leads to noise and less accurate apparent diffusion coefficient (ADC) estimates. MR-Linac users follow the recommended maximum b-value of 500 s/mm², however, higher maximum b-values are routinely used for DWI in sites such as the brain. The goal of this work was to optimize DWI in an MR-Linac using a standardized phantom. **Methods:** DWI sequences for brain, prostate, and abdomen were optimized with respect to maximum b-value and number of signal averages (NEX). DWI scans of the NIST-QIBA phantom (High Precision Devices, Boulder, USA) were conducted in the Elekta Unity MR-Linac's (Elekta AB, Stockholm, Sweden) Philips Marlin 1.5T MR scanner. ADCs were estimated by fitting with offline reconstruction (MATLAB). Maximum b-value and NEX were varied and differences between measured and manufacturer-specified ADC were reported. The impact of varying NEX on scan duration was also quantified. **Results:** For the brain sequence, maximum b-value achieving accurate ADC was 3,000 s/mm² ($\pm 4.0\%$ of standard across all diffusion vials). For all three sites, fitting with a subset of b-values below 400 s/mm² reduced ADC accuracy ($\pm 17.6\%$, 22.3% , 24.1% for brain, prostate and abdomen, respectively). Reducing NEX from clinical protocol to the scanner's minimum resulted in underestimation of ADC by $14.3\% (\pm 1.7\%)$, $18.4\% (\pm 5.1\%)$, and $17.5\% (\pm 4.3\%)$ for the brain, prostate, and abdomen sequences, respectively but reduced average scan duration from 5.8 to 1.1 minutes. **Conclusion:** Site-specific DWI sequences were optimized for ADC accuracy with respect to maximum b-value and scan time using a standardized phantom. Future work involves repeating this methodology in vivo.

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MO-lePD-TRACK 3-05

Reduced Field-Of-View Diffusion Weighted MRI On An MR-Guided Radiotherapy System

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Purpose: Longitudinal on-board diffusion MRI (DWI) using MR-guided radiotherapy (MRgRT) system has the potential to monitor localized treatment response throughout treatment course. Clinically, DWI is acquired together with an anatomical MRI with a large field-of-view (FOV), while the DWI focuses primarily on the tumor within a smaller FOV. Accordingly, reduced FOV (rFOV) DWI has been introduced to remove artifacts from motion, flow and other signal interferences from the surrounding regions. In this study, a rFOV DW-single-shot echo planar imaging (DW-ssEPI) using saturation bands was implemented on an MRgRT system and its performance was evaluated with phantom studies. **Methods:** Full FOV DW-ssEPI (fFOV-EPI) and rFOV DW-ssEPI (rFOV-EPI) were evaluated on a 0.35T MRgRT system using a diffusion phantom and a spatial integrity phantom. Both scans used the same resolution ($2.7 \times 2.7 \times 6\text{mm}^3$), TE (120ms), and TR (3200ms). The FOV of fFOV-EPI and rFOV-EPI was $350 \times 350\text{mm}^2$ and $225 \times 225\text{mm}^2$, respectively. Saturation bands were placed immediately outside of the rFOV in the phase encoding direction to minimize aliasing artifacts. Signal-to-noise ratio (SNR) compensation was introduced for the rFOV-EPI to equalize SNR between the two scans. As a result, fFOV-EPI and rFOV-EPI used 6 and 9 averages for a scan time of 1:46 and 2:34min, respectively. Apparent diffusion coefficient (ADC) measurements were evaluated relative to the reference values by the manufacturer. Spatial integrity was evaluated using an open-source toolbox to quantify spatial distortion. **Results:** Both fFOV-EPI and rFOV-EPI provided accurate ADC quantification on the diffusion phantom ($p < 0.05$). rFOV-EPI had similarly low levels of distortion as fFOV-EPI at 50, 75, and 100mm radius of the spatial integrity phantom. **Conclusion:** rFOV-EPI is a feasible method to generate rFOV DWI and is comparable to conventional fFOV-EPI on phantom scans. Future work will focus on patients and volunteers to test the efficacy of rFOV-EPI for image quality and ADC quantification improvements.

Dr. Yang has received honoraria and consulting fees from ViewRay. Dr. Hu has received consulting fees and research funding from ViewRay.

MO-lePD-TRACK 3-06

Dosimetry Impact of Gating Threshold Selection in Cine-MR Image Guided Breath-Hold Pancreatic Cancer Radiotherapy

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Purpose: To investigate the dosimetry effects of gating threshold selection in cine-MRI guided breath-hold pancreatic cancer radiotherapy. **Methods:** Seventeen pancreatic cancer patients underwent MRI guided breath-hold stereotactic body radiation therapy. The sagittal cine-MRI was used to monitor tumor motion and gate radiation beam. Gross tumor volume (GTV) was set as the tracking target, and a margin of 3-5 mm was used as gating boundary. Beam-off event would not be triggered until the target-out percentage exceeded a predefined gating threshold. We re-generated VMAT plans using 33Gy prescription. We calculated the GTV displacement in each MRI frame that satisfied the gating threshold, shifted the isocenter by the displacement, and then added up all iso-shift plans to create the motion plan. The coverage for GTV and PTV and dose to nearby organs at risk (OARs) were compared between motion plans and original plan. **Results:** The GTV V33Gy was 89.34%±12.58% in the original plan, and 88.53%±12.27%, 88.39%±12.16%, 88.30%±11.91% and 88.26%±11.82% for motion plans generated with 0%, 1%, 3% and 5% gating threshold. ($p>0.05$ for all motion plans). The PTV V33Gy was 97.63%±1.33% in the original plan, and 92.87%±3.16%, 92.32%±3.27%, 91.64%±3.38% and 91.27%±3.34% for motion plans generated with 0%, 1%, 3% and 5% gating threshold. ($p<0.05$ for all motion plans). The duodenum V12.5Gy ($p<0.05$) and V18Gy in the motion plan increased significantly when gating threshold was greater than 2% ($p<0.05$). No significant difference was found for other OARs. No significant difference was found for all parameters between motion plans. **Conclusion:** The dose parameters deteriorate with increasing gating threshold but rendered no difference among thresholds 0-5%. But the efficiency gradually improves with increasing thresholds. A threshold of 3-5% was deemed a reasonable selection considering the tradeoff between effectiveness and efficiency.

MO-lePD-TRACK 3-07

Optimization of the imaging Plane for Real-Time Tumor Tracking in MR-Guided Radiotherapy of Moving Targets

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Purpose: To provide a patient-specific optimization of the 2D cine-imaging plane in MR-guided radiotherapy with respiratory gating, based on pre-treatment 4D information. **Methods:** The sampling directions in MR imaging can be selected arbitrarily, thus leading to different imaging planes in 2D cine-MR. Our method performs a multi-threaded grid search of candidate planes for cine-MR imaging, whose orientation can be adjusted at user-defined angular steps and total range. The optimization maximizes the peak-to-peak motion amplitude over the tumor trajectory, as depicted in the treatment planning 4D-CT. The GTV contours delineated on the 4D-CT motion phases are processed to create the corresponding ITV. Candidate planes are centered in the center of mass of the ITV, whose cross-sectional area is assumed to be proportional to the motion amplitude sampled with cine-MR. The imaging plane orientation is defined by rotation angles about the principal direction of the ITV, which provide the grid search initialization. The algorithm has been tested on artificial images (ellipsoids and cuboids), where the ground truth motion trace allows determining the optimal plane unambiguously. Finally, the algorithm has been applied to 16 clinical 4D-CT datasets including lung, liver and pancreas tumors. **Results:** The algorithm checked up to 4275 planes, with a computational time proportional to the angular range in the grid search procedure. Optimal planes in the artificial images were returned as expected. In clinical cases, motion in the optimal plane was larger (10.3 mm on average, range [4.1, 17.4] mm) compared to the clinically used sagittal orientation (8.5 mm on average, range [2.3, 13.8] mm). **Conclusion:** An algorithm for the optimization of the cine-MR imaging plane based on pre-treatment 4D-CT images has been implemented and tested. The optimal plane consistently provides a larger motion amplitude compared to the sagittal plane, thus effectively minimizing effects due to out of plane motion.

The Department of Radiation Oncology of the University Hospital of LMU Munich has research agreements with Brainlab, Elekta and ViewRay

Multi-Disciplinary Interactive ePoster Discussion

TRACK 4

Multi-Disciplinary: Deformable Image Registration

MO-lePD-TRACK 4-01

Automatic Liver Tumor Localization Using a Combined Deep Learning and Biomechanical Model (DL-Bio)

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Purpose: Liver tumor localization is challenging for X-ray-based image-guided radiotherapy, due to the low tumor contrast against neighboring normal tissues. We developed a liver tumor localization technique through combining 2D-3D deformation, deep-learning, and biomechanical-modeling (DL-Bio) to automatically localize liver tumors through solved deformation-vector-fields (DVF). **Methods:** DL-Bio solves DVFs between planning CT images and new on-board cone-beam projections for tumor contour propagation and localization. It initializes the DVFs through 2D-3D deformation, which solves DVFs by intensity-matching digitally-reconstructed-radiographs of the deformed prior image to on-board cone-beam projections. DL-Bio further applies a deep-learning model to fine-tune the 2D-3D DVFs at the liver boundary, especially for the caudal liver side where lacks sufficient intensity contrast for 2D-3D deformation. The deep-learning model corrects the DVFs at the caudal liver boundary, using a liver surface ring structure, 2D-3D DVFs at the high-contrast cranial liver boundary, and liver boundary motion correlations learnt through model training. Final intra-liver DVFs, including those around the tumor region, were derived with a biomechanical model using the deep-learning-optimized liver boundary DVFs as the boundary condition for finite element analysis. We trained and validated the deep-learning model of DL-Bio using 24 liver patients, and evaluated the combined DL-Bio framework using an independent set of 10 patients. The tumor localization accuracy of DL-Bio was evaluated using the tumor DICE coefficient and center-of-mass-error (COME). **Results:** On a total of 90 evaluation images and tumor contours, the average(±s.d.) liver tumor COMEs of the 2D-3D deformation (2D-3D), 2D-3D deformation with biomechanical-modeling (2D-3D-Bio), and DL-Bio were 4.7±1.9 mm, 2.9±1.0 mm, and 1.7±0.35 mm, respectively. The corresponding average(±s.d.) DICE coefficients were 0.60±0.120, 0.71±0.072, and 0.78±0.028, respectively. **Conclusion:** The deep-learning model of DL-Bio successfully captures the motion correlations at the liver boundaries for DVF correction, and generates more accurate boundary conditions for biomechanical modeling to improve liver tumor localization accuracy.

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MO-lePD-TRACK 4-02

Deformable MRI-CT Image Registration Using Self-Similarity Network for Head and Neck Cancer Radiotherapy

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Purpose: To propose an unsupervised deep learning-based image registration method for MRI-CT head neck (HN) image registration using image self-similarity descriptors. **Methods:** The MRI and CT images were first separately processed to extract self-similarity descriptors for MRI and CT, respectively. The self-similarity descriptors were obtained by constructing normalized local intensity differences, which was used as local structural descriptors. The self-similarity descriptor for MRI and CT could effectively represent the local image features using the same gray scale, which enables traditional similarity metrics to be used for similarity calculation. An unsupervised network takes the concatenated similarity descriptors and the original images as input and directly predict the deformation vector field (DVF). The network was trained by minimizing the normalized cross correlation loss between the deformed MRI and fixed CT and an additional DVF regularization loss. The network was trained using 20 patients' datasets and evaluated on a separate 10 patients' datasets. **Results:** Our results show that the image alignment at the shoulder region and the image boundaries has greatly improved after registration. Landmarks located at the center of the humeral head and tip of the axis vertebra dens were manually selected to calculate the landmark distance errors. The average landmark distance errors were reduced from 6.13 to 2.92 mm at the center of humeral head, and from 3.98 to 2.62 mm at the tip of the axis dens. **Conclusion:** We have developed a novel unsupervised multimodal image registration method for MRI-CT HN image registration. Taking the combined MRI and CT image and their respective self-similarity descriptors as input, the network performs direct DVF prediction to deformably align the MRI HN images to the CT images for facilitating HN cancer treatment planning.

MO-lePD-TRACK 4-03

Investigation of the Sensitivity of Ventilation Accuracy to the Deformable Image Registration Regularization Parameter

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Purpose: To investigate the sensitivity of producing accurate ventilation information from deformable image registration (DIR) to the regularization parameter, λ , by comparing DIR-derived ventilation with Galligas PET images for a range of values of λ . **Methods:** Paired inhale/exhale breath hold CT scans and Galligas PET ventilation images were obtained from 8 patients. The exhale CT was deformably registered to the inhale CT for a range of λ from 0.05 to 1.5. Two previously described methods were used to produce ventilation maps: first the Hounsfield unit change between the deformed inhale and exhale image to infer lung density change, second the Jacobian determinant of the deformation vector field to infer air volume change. Three assessment metrics were used (1) Spearman correlation between the CT ventilation maps and the Galligas PET images, (2) percentage of negative values in the Jacobian determinant (indicating singularities in the motion field, see AAPM report TG132) and (3) mean square error between the exhale and deformed inhale CT images. **Results:** Spearman correlations between CT ventilation maps and the Galligas PET images generally increased as λ increased from 0.05 to around 0.75, with the gains stabilizing as λ was increased further. Image match accuracy measured by mean square error increases with smaller λ as expected. However, $\lambda < 0.5$ created motion models implying up to 6.3% of the lung contracted during inhale, and derived ventilation maps using the Jacobian determinant had Spearman correlations as low as 0.20, implying non-physiological deformation. The dependence of the Spearman correlation on λ was more marked for the Jacobian determinant method than the Hounsfield unit method. **Conclusion:** Physiological ventilation information derived from deformable image registration using two ventilation algorithms was benchmarked against Galligas PET as the regularization parameter was varied. The Hounsfield unit algorithm proved robust across the range of λ tested.

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MO-lePD-TRACK 4-04

Learning-Based Multiscale Self-Similarity Descriptors for Non-Rigid MRI-CT Liver Image Registration

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Purpose: To propose a multi-scale image self-similarity descriptor for non-rigid MRI-CT liver image registration using deep learning-based network. **Methods:** The lack of multimodal image similarity metric is one of the challenges in multimodal image registration. We propose to use multiscale image self-similarity descriptors together with deep learning networks for MRI-CT liver image registration. To capture both the global and local descriptors, a multiscale deep learning network was used. The original MRI and CT images were down-sampled by factors of 8, 4, 2 in a pyramid network architecture. CT and MRI image self-similarity descriptors were calculated by normalizing the image intensity differences in the superior-inferior, anterior-posterior and lateral directions. The normalization of the image similarity descriptor enables the calculation of cross-modality image similarity measure since they share the same gray scale levels for MRI and CT. Normalized cross correlation was maximized between the fixed CT and the deformed MRI using their respective multi-scale image similarity descriptors. We evaluated the proposed network on 40 patients' datasets using five-fold cross validation. The trained network was able to predict the deformation that aligns the MRI to CT in a single shot. **Results:** For evaluation purpose, the MRI liver contours were deformed using the predicted deformation to obtain the liver contours of the deformed MRI. For all patients, the average dice similarity coefficient of liver has increased from 0.89±0.04 before registration to 0.95±0.03 after registration. The alignment of hepatic vessels and liver boundaries have both improved after the proposed non-rigid image registration. **Conclusion:** We have proposed a novel multiscale image self-similarity descriptor for MRI-CT liver image registration. Integrated into an unsupervised registration network, the proposed image similarity descriptors enable direct alignment of the MRI to CT images with high accuracy. This image tool may have a great potential on facilitating the liver radiotherapy treatment planning.

MO-lePD-TRACK 4-05

Training an Unsupervised Deformable Image Registration Model with Thoracic Fast, Low-Dose Helical CT Acquisitions

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Purpose: Monitoring respiratory motion is critical while patients are undergoing thoracic radiotherapy. Low-dose, fast helical CT scans can capture the variability in respiratory motion while patients are free breathing. These scans have been used previously to fit a model (5DCT) to derive CT volumes from a baseline CT volume, as well as the tidal volume and air flow derived from an external surrogate marker. Here, we leverage these data to apply an unsupervised deep-learning based deformable image registration algorithm (VoxelMorph) to build towards a motion management model. **Methods:** We obtained data from 32 retrospective patients (27 for training, 5 for validation) each scanned with 25 fast, low-dose helical CT acquisitions. A VoxelMorph model was trained at 2-mm resolution using mean square error for image similarity loss with regularization. A random pair of images within each patient served as input, training randomly across patients for a total of 25,000 pairs. The generated deformation vector fields from the model were used to warp each structure to a selected reference volume for each patient. Dice scores between the warped structure and an autosegmentation were computed and also compared to a state-of-the-art algorithm, deeds. **Results:** Median Dice scores were higher than that of deeds on the 5 independent test sets for 19 of 20 structures (stomach was lower). Median Dice scores ranged from 76.2% in the duodenum to >95% in the right and left lungs. On inference, deformations could be generated on the 5DCT VoxelMorph model in less than 1 second on a GPU, whereas deformations for deeds took approximately 10 minutes on CPUs. **Conclusion:** Fast, helical free breathing scanning provides a rich dataset to train deep-learning based image deformation models. Together, this framework is being developed towards developing a rapid motion model to help predict respiratory motion in real-time for radiotherapy.

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MO-lePD-TRACK 4-06

Using Automatic Segmentation to Improve Deformable Image Registration

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Purpose: Deformable image registration (DIR) aims to find a geometric transformation between corresponding image data and brings them into a common coordinate frame. DIR accuracy can be improved using a hybrid image similarity metric which matches both image intensities and the location of known structures. This work evaluates the application of the point-to-distance map (PD) hybrid similarity metric for DIR when the segmented structures have been created using a deep learning segmentation method. **Methods:** The approach was evaluated using a publicly available head and neck dataset. Brainstem, mandible, larynx, and parotid glands were segmented automatically using a 3D UNet model. Pairs of images with segmented structures were registered using a hybrid similarity metric, which penalizes the distance between structure boundaries. Cubic B-splines were used for the registration transform and sum of squared difference for the intensity similarity. **Results:** Accuracy of the deformable registration was quantified using the Dice Similarity Coefficient (DSC) and 95% Hausdorff Distance (95% HD), with and without the PD metric. Results demonstrate much improved overlap between the fixed and warped contours once the PD metric is applied. For example, with a grid spacing of 40 mm, mean DSC for the larynx increases from 0.72 to 0.8 with the PD metric. The computational overhead associated with adding the PD metric is minimal. **Conclusion:** A PD hybrid similarity metric was found to improve DIR accuracy when the segmented structures were created by deep learning. This means that a significant improvement in terms of registration accuracy can be achieved without resorting to manual contouring.

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MO-lePD-TRACK 4-07

Using Neural Networks to Extend Cropped Medical Images for Deformable Registration Among Images with Differing Scan Extents

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Purpose: To improve deformable image registration (DIR) among images with different scan extents, we train a neural network that fills the missing volume of head-and-neck CT's. **Methods:** We trained a generative adversarial network on a dataset of 409 head-and-neck CT's to predict anatomy in the missing volume using the cropped 3D images as input. The network used a U-Net generator with VGG deep feature losses. To test the DIR improvement, we deformably registered combinations of a randomly cropped, full, and predicted full volume on 53 unseen test volumes to a single cropped, full, and predicted full target. We tested our method's robustness by performing the same registration combinations with progressively increasing the amount of cropping. Registration performance was measured using 95% Hausdorff distance across 16 contours. **Results:** We successfully trained a network to recover large missing volumes of incomplete images. Registration with predicted full images was not significantly different from registration using original full images. The average contour matching error for full image registration was 9.9mm, while our method was 11.6mm, 12.1mm, and 13.6mm for predicted-to-full, full-to-predicted, and predicted-to-predicted registrations, respectively. Registration of the cropped images had errors of 31.7mm and higher. Plotting the registered image contour error as a function of the initial error shows that our method is robust to registration difficulty. Predicted-to-full registration was statistically independent of cropping extent up to 18.7cm superiorly cropped. Predicted-to-predicted registration was robust, with a -0.04mm change in average contour error for every millimeter of cropping. **Conclusion:** Differences in scan extent is a major contributor to inaccuracies in deformable image registration. We address this by training a network to complete cropped 3D images. Predicted full images can be registered to the same accuracy as full images and are robust to differences in crop extent.

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Therapy Interactive ePoster Discussion

TRACK 5

Therapy: External Beam: treatment planning Optimization

MO-lePD-TRACK 5-01

Neuro-Fuzzy Guided Objective Function Parameter Optimization of Inverse Treatment Planning

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Purpose: IMRT is a complex, multi-criteria issue, which contains various non-linear relationships. The tuning for the weighting factors in the objective function for an IMRT plan is driven by humans to achieve an ideal dose distribution. This is done manually in a tedious and time-consuming process based on trial and error and user experience. This work aims to develop a neuro-fuzzy guided system to automate the tuning process. **Methods:** A machine learning technique, based on an adaptive network fuzzy inference system (ANFIS), was used. It integrates both fuzzy inference systems (FIS) and neural network (NN) principles. FIS logic theory has been widely used for IMRT optimization. There are criticisms that most studies use pre-defined simple linear membership functions and fuzzy rules, which are derived from experts' knowledge. However, IMRT contains various non-linear relationships. This study explores a new approach using ANFIS to tune parameters of the FIS model to increase the exactness of the AI model. The capability and the accuracy of FIS and ANFIS were analyzed and compared against each other. **Results:** The algorithm's performance was examined using the C-Shape TG119 IMRT phantom. The experimental results showed that ANFIS is an effective way to build an optimal FIS, and analysis of ANFIS showed improvement on both DVH and dose distribution. Using ANFIS, the OAR volume encompassed by characteristic percentages of isodose was reduced by 15%. The ANFIS guided inverse planning system is capable of finding the optimal combination of weighting factors for different structures, finding an optimal balance between the target and the critical organ dose by finding an optimal set of weighting factors. **Conclusion:** It is feasible to automatically tune the weighting factors for an IMRT plan under the guidance of an optimized FIS without human intervention other than providing the treatment plan parameters and set of constraints.

MO-lePD-TRACK 5-02

Solving the Non-Convex Volumetric Modulated Arc Therapy (VMAT) Problem Via a Sequence of Convex Optimization Problems

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Purpose: Volumetric modulated arc therapy (VMAT) optimization problem is a complex and highly non-convex problem and existing algorithms suffer from local optimality. We propose a novel sequential convex programming (SCP) algorithm that directly optimizes aperture shapes and promotes delivery efficiency and global optimality. **Methods:** We propose an algorithm for VMAT that conducts an effective global search by solving a sequence of convex local approximations. The algorithm starts with optimizing the aperture weights of 72 evenly distributed beams using the beam's eye view (BEV) of the target from each direction. Thereafter, a convex approximation problem is solved to optimize the leaf positions and the aperture weights within a search space where the leaves can move in and out within a pre-defined step-size. There are both local and global search strategies in the algorithm to ensure a high-quality plan. The complexity of the apertures is constrained, to exclude highly irregular aperture shapes, and is also penalized in the objective function to increase shape regularity. The algorithm is tested on paraspinal, prostate and oligometastasis sites. The VMAT plans are compared against ideal 72-beam IMRT plans. Also, the performance of aperture regularization strategies is analyzed.

Results: The DVHs of the optimized VMAT plans are similar to 72-beam IMRT plans. Due to the convexity of the resultant sub-problems, the algorithm is computationally tractable, converging in less than an hour on average on a regular Windows workstation. Significant reductions in aperture complexity at a reasonable cost of plan quality are observed after adding constraints and the aperture complexity regularization term in the objective function. **Conclusion:** We developed a computationally efficient algorithm for VMAT which finds a near-global optimal solution by alternating large (global) and small (local) search steps. Aperture complexity is effectively constrained (hard constraint) and penalized (objective function) to produce efficient delivery plans.

MO-lePD-TRACK 5-03

Characterization and Optimization of GRID Treatment Planning

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Purpose: Spatially fractionated radiation therapy or GRID is known to be effective to treat bulky mass in breast and head & neck tumors. Biological mechanism includes abscopal effect, escalated damage to endothelial cells and immune system activation. Difference in radiation damage to healthy tissue and tumor cell due to the heterogeneous dose distribution in GRID treatment can be evaluated using therapeutic ratio (TR) defined by the surviving fraction (SF) in normal tissue with GRID dose distribution to the equivalent uniform dose (EUD) with which SF of tumor is maintained. In this study, the efficacy of GRID dose distribution is characterized as a function of collimator design, beam energy, depth of tumor, dose fraction size and tumor type. **Methods:** Dose distribution in water phantom was acquired in the treatment planning system, Pinnacle, using three different GRID collimators of 10mm, 12mm, and 14mm with separation of 21mm as well as fluence pattern of 10mm by 10mm segments with 17mm separation using MLCs. Three different beam energies of 6MV, 10MV and 15MV were used with the GRID field size of 10cm x 10cm. Dose was prescribed at depth of 5cm. Dose volume histogram (DVH) in the phantom was collected in every 1cm depth. Three different tumor type (radio-resistance, semi-resistance, and radiosensitive) were considered. **Results:** The maximum TR was found at the depth of dose maximum in each beam energy for all collimator design. For the same prescription dose of 15Gy, TR at depth of 5cm increased by 12%-21% and 13-26% from 6MV-to-10MV and 6MV-to-15MV, respectively. EUD for tumor also increased by 22%-32% and 24%-33%. To maintain the same EUD of 5.5Gy for tumor, the low energy photon of 6MV required 27%-59% and 28%-65% more prescription dose compared to 10MV and 15MV, respectively. TR increased with the ratio of collimator opening in the order of 10mm, 12mm, 14mm and MLCs. **Conclusion:** The TR and EUD of tumor is characterized to optimize the efficacy of GRID planning.

MO-lePD-TRACK 5-04

Development of Plan Complexity Mitigation Algorithm for VMAT

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Purpose: The purpose of this study was to develop an algorithm to mitigate plan complexity in volumetric-modulated arc therapy (VMAT) and to evaluate its usefulness. **Methods:** Twenty patients who underwent VMAT (including seven Dynamic WaveArc plans) between September 2015 and December 2020 were enrolled in this study. Clinical sites included the brain (n=6), head and neck (n=2), lungs (n=3), esophagus (n=3), whole pelvis (n=3), and prostate (n=3). Patients with less than 85% gamma passing rate (GPR) of 3%/2mm were arbitrarily selected. Using the GPR prediction model we previously developed, we identified the leaf positions responsible for the decrease in GPR. Using a limited nonlinear algorithm, the multi-leaf collimator (MLC) positions were optimized within the mechanical limitations. The dose prescription for the target was re-normalized, and the monitor unit (MU) was modified. The Modulated Complexity Score (MCS), average aperture area (AA: mm²), and MU/Gy were evaluated to determine the difference in plan complexity between the optimized and original plans. The calculated doses were evaluated for the target and surrounding organs. Additionally, the ArcCHECK cylindrical diode array was used to measure the dose, and GPR (5%/1 mm) and GPR (3%/2 mm) were evaluated to determine the difference between the optimized and original plans. Differences were presented as means $\pm$ standard deviations. **Results:** The differences in plan complexity parameters between the optimized and original plans were $4.3 \pm 2.7 \times 10^{-3}$ for MCS, 67.6 ± 113.3 mm² for AA, and -0.3 ± 1.3 MU/Gy, respectively. For the calculated dose, dose volume histograms were within 1% for the target and surrounding organs. The difference of GPR between optimized and original plans were $0.9 \pm 1.1\%$ point (%p) for GPR (5%/1 mm) and $1.0 \pm 1.8\%$ p for GPR (3%/2 mm). The GPRs were slightly improved with the optimized plans. **Conclusion:** The developed algorithm could mitigate plan complexity and improve GPR with minor leaf position modifications for VMAT.

MO-lePD-TRACK 5-05

IMRT Treatment Planning Study for the First Clinical Biology-Guided Radiotherapy System

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Purpose: The first clinical biology-guided radiotherapy (BgRT) system – RefleXion X1 - is installed and being commissioned for clinical use at our institution. RefleXion X1 is a 6MV-FFF linac mounted on a ring gantry rotating at 60 rpm while the couch advances in 2.1mm increments. The modulation is achieved via 50 firing positions with 64 binary MLCs (6.25mm at 85cm SAD) with either 1cm or 2cm jaws. This study was conducted to compare the treatment plan quality for IMRT cases without PET-guidance. **Methods:** Nineteen patients previously treated with VMAT on C-arm linacs were selected for this retrospective study including 8 prostate cancer patients (80Gy/40fx), 6 lung cancer patients (66Gy/30fx) and 5 post-surgical parotid cancer patients (60Gy–66Gy/30fx). For each VMAT plan, a corresponding plan was generated on the RefleXion treatment planning system (TPS) using our institutional planning constraints. All clinically relevant metrics in this study, including plan PTV D95%, PTV D1%, Conformity Index (CI), and organs at risk constraints were analyzed and compared between the VMAT and RefleXion plans using paired t-tests. **Results:** Clinically acceptable plans were obtained with both techniques. For the prostate, lung, and head and neck sites, no statistically significant difference was observed in PTV D95%, PTV D1%, or CI between the VMAT and RefleXion treatment plans. Due to inherent OAR dose penalties in the RefleXion optimizer, dose reduction to certain critical structures was observed in RefleXion plans: brainstem Dmax (-4.8Gy, p<0.008), contralateral cochlea Dmean (-4.8Gy, p<0.01), lips Dmax (-3.5Gy, p<0.007) for head and neck cases; esophagus Dmean (-2.3Gy, p<0.039), spinal cord Dmax (-3.6Gy, p<0.001) for lung cases; femurs left and right Dmax (-4.8Gy, p<0.003 and -5.0Gy, p<0.01) and rectum V75% (-1.6%, p<0.01) for prostate cases. **Conclusion:** The RefleXion TPS provided comparable plan quality to VMAT plans with slightly superior OAR sparing.

MO-lePD-TRACK 5-06

NYU Clinical Experience with Total Body Irradiation: From 3D Laterals to Image-Guided VMAT

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Purpose: Patients receiving myeloablative total body irradiation (TBI) doses ≥ 12 Gy are at risk of developing interstitial pneumonitis. At our institution, TBI transitioned from extended distance opposed Laterals to image-guided VMAT, in an effort to improve coverage while sparing lungs and kidneys. This work presents a dosimetric comparison between 3D Laterals and VMAT. **Methods:** Nine patients were treated with VMAT as part of an ongoing phase II single-arm clinical trial. VMAT patients were CT-simulated supine, with thermoplastic masks for head/neck, chest/abdomen/pelvis and feet. VMAT planning (12Gy (n=6) or 13.2Gy (n=3) in 8-BID fractions) utilizes 6MV multi-isocentric arcs and AP/PA beams to treat the upper and lower body, respectively. Ten 3D Lateral patients were CT-simulated supine with arms positioned/immobilized for lung shielding, with rice compensation between legs/feet. Laterals plan (12Gy in 8-BID fractions) uses 15MV, beam

spoiler, head compensation, and subfields to maintain coverage and mean-lungs dose <10.5Gy. Target (Body-5mm, extending 3mm into lungs and kidneys for VMAT; Body-2cm for Laterals) coverage was evaluated at V100%, and D98% (percentage of Rx). Absolute dose to lungs and kidneys were reported. **Results:** Median Target V100% and D98% for VMAT was 93.2% (Range: 95.6% to 92.1%) and 90.2% (94.3% to 88.3%), whereas for Laterals V100% and D98% was 57.3% (66.5% to 46.3%) and 80.6% (75.5% to 84%). The median Lung mean dose was 7.6Gy (7.3Gy to 7.9Gy) for VMAT. The median mean dose to kidney was 10.4Gy (10.1Gy to 10.7Gy) for VMAT, and 12.5Gy (11.9Gy to 13.5Gy) for Laterals. **Conclusion:** We have established a VMAT-TBI program for patients requiring myeloablative irradiation. Improvement in target coverage is demonstrated by V100% and D98%, while reducing the mean dose to the lungs significantly from 10.5Gy to 8Gy.

MO-lePD-TRACK 5-07

Using the Pareto Optimal Projection Search (POPS) Algorithm for Fully Automated Treatment Planning

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Purpose: Radiation therapy treatment planning has traditionally been a time-consuming process with potentially high inter-planner variability. The Pareto optimal projection search (POPS) algorithm, which produces Pareto optimal and clinically acceptable plans, provides a fully automated framework for treatment planning without requiring any active planning. **Methods:** The POPS algorithm searches the feasibility boundary for desirable plans, as defined by a scoring function. Overall, POPS has two main components—gradient-free search in the decision variable space and projection of decision variables to the Pareto front using the bisection method. As treatment plan evaluation is highly nuanced with many candidate scoring functions available in literature, we evaluate the performance of the POPS automated plans under the following two conditions: 1) when POPS performs plan scoring using a normalized equivalent uniform dose (NEUD) metric and 2) when POPS performs plan scoring using a scoring function (SF) based on dose-volume histogram (DVH) limits. **Results:** Treatment plans were generated automatically using the POPS algorithm for a dataset of 21 prostate cases collected as part of clinical workflow. Dose conformity, dose homogeneity, and sparing of organs-at-risk are assessed using a Wilcoxon signed-rank test. In general, plans generated using the POPS algorithm were of high quality. Plans scored using the NEUD metric had improved conformity and comparable homogeneity as compared to plans scored using the SF score. Sparing of organs-at-risk for POPS generated plans was also comparable when using either of the scoring functions. **Conclusion:** The POPS algorithm allows for fully automated treatment planning and produces plans that are Pareto optimal and clinically acceptable. We anticipate that POPS will substantially improve treatment planning workflow.

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Therapy Interactive ePoster Discussion

TRACK 6

Therapy: Novel Dosimeters

MO-lePD-TRACK 6-01

Quantifying of the Afterglow Effect from a Chromium-Doped Alumina Scintillator Following Proton Irradiation

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Purpose: The purpose of this work was to investigate the afterglow effect, which refers to continued light emittance post-irradiation, of a chromium-doped alumina scintillation screen (Chromox) when irradiated with a proton beam. To our knowledge, the afterglow effect has not previously been quantified in the literature. **Methods:** Chromox is a desirable scintillation material for visualization of proton beam profiles due to its high light yield and radiation hardness. The afterglow effect was studied using a clinical IBA Proteus Plus delivery system to irradiate a 1 mm thick 10x10 cm² Chromox scintillation screen. A uniform planar dose (7x7 cm²) with a single energy (150 MeV and 9125 total monitor units) was delivered to the screen. Following the irradiation, the afterglow images were acquired every 3 minutes for 18 minutes. **Results:** The afterglow was still detectable after 10 minutes using an IR-filtered camera with the exposure time set to 7 seconds. The afterglow half-life was calculated to be approximately 110 seconds. **Conclusion:** While Chromox appears to be very suitable for static beam diagnostics, the afterglow will be present if consecutive, varied images are acquired. Ideally, use of a Chromox scintillation screen should be limited to applications that require a high light output for single static exposures. In some applications, the afterglow could be viewed as an advantage as imaging can be completed post exposure rather than in real time.

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MO-lePD-TRACK 6-02

Investigations of a Novel Hyperscint™ Plastic Scintillator Detector and Hyperspectral Analysis Approach in a 74 MeV Proton Beam

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Purpose: A novel plastic scintillator and spectral analysis approach (Hyperscint™) was investigated in a 74 MeV proton beam for stem effect, dose and dose rate response, and suitability to provide a Cerenkov-free reference spectrum for use in MV x-ray dose calibrations. **Methods:** Irradiations were performed in a passively scattered 74 MeV horizontal proton beam line with a 2.5 cm circular field aperture. The plastic scintillator detector (PSD) was 1 mm in diameter and 1 mm in length, coupled to the spectrometer with PMMA optical fibre. The probe was positioned at the centre of a 2 cm spread out Bragg peak in polystyrene phantom 7 cm distal to the proton nozzle. Doses from 200 cGy to 5000 cGy were delivered using the nominal 6 nA beam current, followed by comparison with 1 nA, 2 nA, and 10 nA beam currents. The optical emission spectrum in the proton beam was compared with that acquired in a 150 kVp ortho-voltage x-ray beam, to assess suitability for use as a Cerenkov-free reference spectrum. Stem effect was further investigated in the proton beam by irradiating the fibre cable alone. **Results:** The 74 MeV proton beam and 150 kVp x-ray beam produced indistinguishable Cerenkov-free optical emission spectra in the PSD. Integrated area under the spectral curve was linear with dose and independent of dose rate. Optical fibre emission was observed to increase with length of cable irradiated at 0.1 % of the PSD signal per cm of irradiated cable. **Conclusion:** The Hyperscint™ dosimetry system shows potential for dose measurement in a 74 MeV proton beam with negligible stem effect. The Cerenkov free spectrum may be used to facilitate calibration of the device in MV x-ray beams to improve Cerenkov removal and performance in small field dosimetry.

MO-lePD-TRACK 6-03

Development of a Novel Hybrid Scintillation-Cerenkov Detector for Simultaneous Dose and Irradiation Angle Measurements

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Purpose: This study introduces a novel hybrid detector capable of simultaneous dose and direct irradiation angle measurements based on Cerenkov angular dependency. **Methods:** A Cerenkov probe composed of a 10-mm long sensitive volume of clear PMMA fiber separated by an absorptive filter from a 1-mm diameter

transport fiber was built. Filtered and raw Cerenkov signals respectively from the sensitive volume and the transport fiber were collected using the Hyperscint RP-200 scintillation dosimetry platform (Medscint, Quebec, QC). An independent scintillation probe composed of 10-mm BCF12 scintillating fiber (Saint-Gobain, Hiram, OH) was also used for simultaneous dose measurements. Dose calibration of each detector signal was accomplished with repeated irradiations of 200 cGy (6 and 18 MV) and the total signal was deconvolved using a hyperspectral approach. Using a solid-water phantom, measurements at fixed incident angles covering a wide range of doses, dose rates as well as output factors were realized. A cylindrical phantom was then used to obtain an angular calibration curve for fixed dose irradiations and perform irradiation angle measurements. **Results:** For fixed incident angle, signal characterization of the Cerenkov detector displays a linear dose-light relationship for the whole range of doses and dose rates tested. The sensitive volume signal shows a discrepancy with the predicted dose that reaches 1.1 % for the lowest dose rate. Output factors were accurately measured within ± 0.8 % for field size up to 25×25 cm². Irradiation angle measurements using various doses exhibit an absolute mean error of 1.86° and 1.02° at 6 and 18 MV, respectively. **Conclusion:** The ability to measure both irradiation angle and dose from a clinical photon beam using novel hybrid scintillation-Cerenkov detector was shown. These results offer promising perspectives for future applications in external beam radiotherapy, especially for stereotactic radiosurgery (SRS) and magnetic resonance image-guided radiotherapy (MR-IGRT).

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MO-lePD-TRACK 6-04

Output Constancy of the W2 Scintillator in Flattening Filter Free Fields in a Medical Linac

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Purpose: Ion chambers are prone to saturation effects at the dose rates delivered by modern flattening filter free (FFF) beams. The purpose of this work is to characterize the dose linearity and dose rate constancy of an Exradin W2 scintillator in a FFF beam as a potential alternative to ion chambers for FFF beam dosimetry. **Methods:** An Exradin W2 scintillator detector was placed at reference conditions (100cm SSD, 10cmx10cm field, depth of dmax). All measurements were taken in a 10MV FFF beam calibrated to deliver 1cGy/MU at reference conditions according to AAPM-TG51 protocol. Cerenkov light ratio corrections were computed according to the manufacturer's instructions. The W2 output was calibrated by defining the reading from 100MU at the reference point as 100cGy. To test the linearity of the response with delivered dose, 200, 400, 600, and 800cGy were delivered at dose rates of 800 and 2400MU/min. To test the detector's constancy with dose rate, 100MU was delivered at nominal dose rates of 400, 800, 1200, 1600, 2000, and 2400MU/min. The measurements were repeated at 90cm and 80cm SSD, where the inverse-square adjusted dose rates are higher than the nominal rates, reaching a maximum of 3750cGy/min. The dose rate constancy was quantified by evaluating the standard error (std/expected value) of the measurements at each dose rate. **Results:** At 800 and 2400 MU/min, the difference between measured and delivered dose was <0.15%; and the slope of the linear fit of the data was 0.99 and 1.00, respectively. The standard error across dose rates were 0.04%, 0.04%, and 0.06% for measurements at 100cm, 90cm, and 80cm SSD, respectively. **Conclusion:** The W2 produces consistent readings up to 3750cGy/min without showing any saturation effects, and therefore may be a better choice than an ion chamber for dosimetry of FFF beams.

The authors would like to thank Standard Imaging for their guidance and for providing the W2 detector so that we could perform this study.

MO-lePD-TRACK 6-05

Geometry Optimization and Optical Sectioning for In-Water Cherenkov Dosimetry of Large and Small Photon Beams

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Purpose: Cherenkov emission (CE)-based dosimetry is an emerging technique that is especially promising for small-field measurements. Here, we present Monte Carlo (MC) simulations and relative experimental validation of photon-beam CE, optimize detection angle geometry, and simulate optical sectioning in a small field with added noise. **Methods:** Additional photon beam experimental validation is provided for a previously-published MC code. Both large and small fields are measured at 90° to beam in water with lenses focusing into a fiber+CCD. Small field measurements are 1D with field size varied only transverse to the optical axis (10x10 to 0.5x10 cm²), pending detector development. Optimization of polar angle (relative to beam) and aperture is performed for a 10 MV photon beam MC model, on percent-depth dose-CE (PDD-PDC) root-mean-squared error and Gamma index, with consideration of convenience. In addition, we simulate 1D (along optical axis) optical sectioning of a 5x5-cm² small-field Matlab model, assuming diffraction-limited optics and using the theoretical PSF with added Poisson noise. Based on these findings and the literature, the achievable detection system specifications and potential costs are laid out. **Results:** Measurements and simulations were in agreement within uncertainty (<1% for 10x10 cm², <7% for 0.5x10 cm²). Optimal detection included multiple angles, but a sufficiently accurate single-angle compromise was 57.5° $\pm$ 5.0° [uncertainty (k=1): 1.2%/0.3mm in build-up and 0.2% in drop-off, with <1mm upstream shift in build-up and <0.5% correction in drop-off]. The optical sectioning test case and accompanying analysis revealed that shot-noise limited detection should be possible and reasonably priced, with expected system quantum efficiency >90%, $\pm 2.5^\circ$ polar angle resolution, short acquisition times (~seconds), sensitive volume of 0.0013 mm³ (5 μ m along optical axis), and traceable calibration uncertainty of 0.5%. **Conclusion:** These findings indicate that, especially for small fields, optical sectioning may be an attractive (viz. accurate and affordable) dosimetry alternative, unique to CE.

MO-lePD-TRACK 6-06

Effects of Shipping and Handling On a DNA-Based Detector

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Purpose: Our lab has developed a DNA-based detector that uses plasmid DNA as a biosensor of DNA damage from ionizing radiation. We aimed to quantify the effects of shipping and handling on the stability of DNA detector readings to assess radiobiological damage. **Methods:** DNA detectors were constructed by pipetting plasmid DNA, pBR322 (Thermo Fisher Scientific, Inc.), into nine 6.28 mm³ cylindrical wells drilled into biocompatible near tissue equivalent material (polycarbonate). Each well contained 3 μ l of DNA solution with a concentration of 12 ng/ μ l for a total of 27 μ l per detector. Each detector was sealed using aluminum foil seals. One control and two detectors to be irradiated to 20 and 30 Gy, respectively, using the XStrahl 300 orthovoltage unit, were shipped on ice in an insulated box via FedEx airmail from San Diego State University (SDSU), San Diego, USA, to Odette Cancer Centre Sunnybrook, Toronto, Canada following biomaterials shipment guidelines. A stationary control was kept at -20 °C at SDSU. The detectors were shipped back to SDSU via FedEx airmail. Agarose gel electrophoresis was used to separate the plasmid DNA open circular, linear, and supercoiled isoform bands corresponding to single strand breaks (SSBs), double strand breaks (DSBs), and no strand breaks, respectively. The gel band intensities were imaged to quantify the relative SSBs and DSBs. **Results:** Total shipment distance and time was 8300 km in ~3.6 days. The shipped controls resulted in mean increases of 18.3 $\pm$ 3.0% and 1.64 $\pm$ 0.30% in the SSBs and DSBs, respectively. Adjusted for these changes, the 20 and 30 Gy irradiated samples showed DNA DSB damage of 4.73 $\pm$ 1.10% and 8.26 $\pm$ 0.67%, respectively. **Conclusion:** Increases in SSBs and DSBs occurred with shipping and handling of the DNA. However, relative damage between the shipped and stationary control can be used to quantify DNA damage in the irradiated detectors.

MO-lePD-TRACK 6-07

Evaluation of a New High-Resolution Digital Detector Array for SRS and SBRT Quality Assurance

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Purpose: The myQA SRS device is a recently developed 2D digital detector array dedicated for patient specific quality assurance (QA) and commissioning of stereotactic radiosurgery (SRS) and stereotactic body radiation therapy (SBRT). A preliminary investigation was performed to evaluate the performance of this high-resolution device for clinical use. **Methods:** The myQA SRS device came with the SRS phantom and three different inserts for film and ion chamber measurements. It provided an active area of 120x140mm² and exploited complementary metal-oxide-semiconductor (CMOS) technology. Five clinically treated SRS/SBRT plans were randomly selected and evaluated. All treatment plans were originally planned in RayStation (V8.0A) treatment planning system (TPS) with VMAT using 6FFF. The QA plans were then generated on scanned CT images of the myQA SRS device within the SRS phantom and delivered on a Varian Truebeam linac equipped with HD MLCs. The same QA plans were also measured with EBT3 radiochromic film using the film insert for absolute dosimetry comparison. Gamma index analysis was performed with 1mm distance to agreement (DTA) and 3% absolute global dose difference (DD) using myQA Patient software. **Results:** The large active area allowed myQA SRS to measure dose to multiple targets in one measurement. The CMOS technology provided a spatial resolution of 0.4mm. With 1mm DTA and 3% DD criteria and 10% threshold, the average gamma passing rate of TPS compared to myQA SRS and EBT3 film was 99.5% and 99.1%, respectively. The average gamma passing rate of myQA SRS measurement compared to EBT3 film was 96.0%. **Conclusion:** The myQA SRS device was evaluated for patient specific QA with both SRS and SBRT treatment plans. Gamma analysis demonstrated that myQA SRS provided accurate dose measurements for small field. The large active area and high resolution improved the VMAT QA efficiency for single isocenter multiple targets SRS plans.

This study is supported by IBA Dosimetry GmbH

SAM Professional Symposium**TRACK 1****AAPM/IAEA Joint Professional Symposium: Ethics in Practice****TU-A-TRACK 1-00****AAPM/IAEA Joint Professional Symposium: Ethics in Practice**

Moderators: Deborah van der Merwe, International Atomic Energy Agency, Florida Hills, GP and Per Halvorsen, Beth Israel - Lahey Health, Burlington, MA

TU-A-TRACK 1-01**Introduction****TU-A-TRACK 1-02****G. White, Medical Physics Services:****Ethics in Medical Physics: The AAPM Perspective****TU-A-TRACK 1-03****G. Loreti, International Atomic Energy Agency:****Ethics in Medical Physics: The IAEA Perspective****TU-A-TRACK 1-04****C. Skourou, Beaumont Hospital:****Ethics in Translation****TU-A-TRACK 1-05****Discussion and Audience Q&A**

The AAPM has developed a code of ethics to guide professional conduct, as well as a published curriculum to educate medical physics residents in ethics. This symposium seeks to promote a conversation in the field of ethics as it pertains to global medical physics practice and outreach. The IAEA has identified a lack of awareness of the importance of ethics and its potential to strengthen professional behavior, e.g. the need for competencies to encourage medical physics professionals to question and raise ethical dilemmas in daily practice. Wider recognition of medical physics as a health profession and more exposure to ethics in medical physics education, are key to addressing challenges in ethical behavior.

Learning Objectives:

1. Realize the extent to which ethical principles can be applied to practice, procedures and tools
2. Explain the importance of awareness of ethics in global activities and cooperation
3. Explore what the AAPM and IAEA can do to promote education in ethics for medical physics worldwide by sharing experiences and supporting international initiatives

Imaging Scientific Session**TRACK 2****Radiography and Fluoroscopy****TU-A-TRACK 2-01****BEST IN PHYSICS (IMAGING): 3D Assessment of Complex Spine Deformity Using Linear Tomosynthesis and Statistical Shape Model Registration**

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Purpose: Spine deformities (e.g. pediatric scoliosis and kyphosis) are conventionally evaluated using anteroposterior (AP) and lateral (LAT) long-view radiography or slot-scans. However, the evaluation of vertebral axial rotation is not possible in two-view planar imaging. We investigate the feasibility of comprehensive 3D assessment of spine deformity using linear unidirectional tomosynthesis (AP only) augmented with 3D-2D registration of a vertebral statistical shape model (SSM). **Methods:** We consider an imaging protocol involving a longitudinal linear scan (1cm translation/view) of the x-ray source and flat-panel detector (varied collimation: 5x40 cm to 20x40 cm, 0.3 mm pixel) to acquire AP projections of the entire spine (180 - 60 views depending on collimation). Such scans can be implemented using a variety of robotic x-ray systems. To obtain 3D pose of an individual vertebra, its SSM was 3D-2D registered to the tomosynthesis projections. Feasibility studies utilized a previously validated high-fidelity x-ray system simulation framework. A digital pediatric anthropomorphic phantom with the capability to introduce spine deformations was developed. The accuracy of the 3D-2D SSM registration was tested on 120 instances with randomized initialization errors (± 10 mm in translations and $\pm 10^\circ$ in rotations, 60 instances each for normal and deformed spines). Radiation dose was varied 1 - 5 mAs/view. **Results:** The 20x40 cm collimation (60 views covering 60 cm length) provided sufficient depth information to accurately estimate 3D vertebral pose. For both normal and deformed spine, coronal plane translations and rotations were measured with less than ± 1 mm and $\pm 2^\circ$ errors. Vertebral axial rotation errors were $< \pm 3^\circ$; AP translation errors were within ± 5 mm. The registration was robust down to 2 mAs/view exposure, corresponding to ~ 0.11 mGy mean dose to the spine (estimated by Monte Carlo simulation). **Conclusion:** 3D-2D SSM registration using AP linear tomosynthesis achieves reliable 3D quantification of lumbar spine deformity with modest radiation dose.

Work supported by academic-industry collaboration with Siemens Healthineers, XP Division

TU-A-TRACK 2-02**A Framework for Calculating 2D Wall Shear Stress (WSS) In-Vitro Using 1000 Fps High-Speed Angiography (HSA)**

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Purpose: There is evidence that wall shear stress (WSS), or the frictional force of flowing blood on the vessel wall, is involved in the pathogenesis of aneurysms and vessel stenoses. The objective of this investigation was to create a framework for calculation of 2D WSS using velocity distributions extracted from high-speed angiography (HSA) sequences. The impact of different parameters on WSS estimation was explored. **Methods:** A 3D-printed patient-specific basilar-aneurysm model was imaged under constant-flow conditions using 1000-fps HSA. The average/maximum velocity values were determined pixelwise over the flow sequence to create velocity-mapping distributions. At each coordinate position along the vessel wall a normal vector was defined, along which the X/Y-components of velocity vectors were indexed with increasing distance from the wall. It was assumed that there is no flow through the vessel wall, i.e. the velocity at the wall was forced to zero. The shear rates were obtained from the spatial gradients of fitted curves, or the rate of change of tangential-velocity values along the direction of the normal. The effect

of tangential-velocity curve fitting parameters was investigated. **Results:** The proposed framework for calculating 2D WSS allows for manipulation of parameters such as the number and spacing of points used along the normal vector. Finer resolution of data points allows for truer estimation of velocity nearest the wall, the tradeoff being increased noise. Smoothing splines produced the best fit for tangential-velocity curves. The highest WSS values were found along the aneurysm dome, as well as the bifurcation region between the outflow vessels and aneurysm sac where the flow impinges on the wall. **Conclusion:** A method to obtain average and peak WSS per pixel from in-vitro HSA sequences was described. The application of this method to HSA sequences has the potential to provide clinically useful information during a vascular intervention.

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TU-A-TRACK 2-03

Development of a Hybrid Direct-Indirect Active Matrix Flat Panel Imager for Digital Radiography

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Purpose: Recent work has proposed “hybrid” direct-indirect active matrix flat panel imagers (AMFPIs) for x-ray imaging. Hybrid AMFPIs comprise an amorphous selenium (a-Se) direct AMFPI which is optically coupled to a turbid scintillator, such that a-Se acts as both an x-ray and optical sensor. This work investigates how beam quality and x-ray converter parameters influence the sensitivity, spatial resolution, noise characteristics, and detective quantum efficiency (DQE) of Hybrid AMFPIs. **Methods:** A prototype Hybrid AMFPI was constructed by modifying a commercial direct AMFPI (AXS-Screen+v2, 85 μ m pitch). Its active matrix was fixed upright to permit imaging in a back-irradiation (BI) geometry, wherein x-rays first expose the detector through the pixel array. Gd₂O₂S:Tb (GOS) scintillators were coupled to the a-Se to compose different Hybrid AMFPIs. Their x-ray sensitivity, presampling modulation transfer function (MTF), quantum-noise-limited noise power spectrum (NPS), and DQE were measured at RQA5 and RQA9 beam qualities using conventional methods. Phantom imaging experiments were performed to evaluate differences in their imaging characteristics. **Results:** Hybrid configurations had greater sensitivity than the direct AMFPI alone; the largest improvement was observed at RQA9 with the thickest GOS due to the increased x-ray quantum efficiency provided by scintillator. Hybrid MTFs were between the direct AMFPI and scintillator MTFs across all configurations and beam qualities. Hybrid NPS featured correlated and white noise components, and an overall noise reduction at low spatial frequencies compared to the direct AMFPI NPS. Hybrid DQEs were accordingly enhanced at low frequencies without penalty at higher frequencies. Phantom images demonstrated more favorable tradeoffs in spatial- and low-contrast resolution with Hybrid AMFPIs than with their constituent direct and indirect imaging components. **Conclusion:** Hybrid AMFPIs leverage the favorable x-ray imaging characteristics of direct and indirect AMFPIs, which may improve tradeoffs between spatial- and low-contrast resolution in digital radiography.

TU-A-TRACK 2-04

Dual-Energy Reduces Severity of Motion Artifacts in Free-Breathing 2D Pulmonary Angiography

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Purpose: To quantify the severity of motion artifacts in 2D pulmonary angiography using conventional digital subtraction angiography (DSA) and dual-energy (DE) subtraction angiography in a free-breathing animal model. **Methods:** Real-time dual-energy pulmonary angiography was performed in three free-breathing porcine subjects on an interventional C-arm system equipped with 30 Hz fast-kV switching capability. Tissue-subtracted dual-energy images were generated by weighted log subtraction of low and high kV image pairs. The relative size and intensity of respiratory-induced motion artifacts over a single respiratory cycle were quantified and compared between dual-energy and conventional DSA images. Severity of motion artifacts was measured from image profiles placed across the diaphragm boundary. Deformable image registration was implemented to correct inter-frame respiratory motion in the 1/30 s period between low and high kV images. Additionally, dual-energy images were generated after application of a machine-learning based denoising algorithm. Dual-energy images with one or both algorithms applied were subjected to the same artifact analysis and compared with regular dual-energy. **Results:** For dual-energy acquisitions, the areas of diaphragm motion artifact profiles were reduced by 93.2% to 96.8% compared to DSA. Application of deformable registration further reduced artifact areas (98.1% to 98.7%). Artifact profile intensity was reduced by 32.2% to 74.6% using dual-energy, and by 80.3% to 90.3% for registered dual-energy. Noise reduction in dual-energy images across all porcine subjects ranged from 58.5% to 67.4%. **Conclusion:** Motion artifacts in dual-energy and conventional DSA on an interventional C-arm have been evaluated in an animal model of free-breathing pulmonary angiography. Dual-energy imaging with 30 Hz kV switching enables a large reduction in diaphragm motion artifacts seen in DSA images. Application of a deformable registration algorithm further reduced residual inter-frame motion artifacts seen in dual-energy subtraction images.

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TU-A-TRACK 2-05

In-Vitro 1000 Fps High-Speed Angiographic (HSA) Evaluation of the Hemodynamics in the Ascending Aorta Resulting from the Graft of a Left Ventricular Assist Device (LVAD)

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Purpose: Implantable left ventricular assist devices (LVAD) can restore blood circulation for patients with end-stage heart failure. However, significant complications such as stroke may occur due to the unfavorable hemodynamic environments produced by suboptimal implantation configuration. High-speed angiography (HSA) was performed at 1000 fps to resolve previously unseen vascular flow dynamics in-vitro for one LVAD outflow graft configuration. **Methods:** Flow detail was evaluated within a 3D-printed aortic vascular model. The imaging ROI focused on the anastomosis of a 10-mm diameter LVAD graft with the wall of the ascending aorta, at an insertion angle of 90 degrees with respect to the centerline of the aorta. The model was connected to a constant flow loop, with a system flow rate set to 5 L/min, as measured at the graft. The aortic valve was assumed to remain closed, so all flow into the aorta originated from the LVAD graft mimicking complete LVAD support. Iodine-based contrast media was slowly injected with an automated syringe injector and imaged at 1000 fps with the new 5cm x7.5cm FOV CdTe-based Aries photon-counting detector. **Results:** Examination of the image sequences illustrates detailed flow dynamic patterns made available through use of the high-frame-rate-enabled, larger-FOV detector. The iodine contrast-labeled fluid can be seen exiting the LVAD graft as a high-velocity jet, which directly impinges on the aortic wall opposite the graft. The 90-degree configuration causes a strong recirculation pattern to form inferior to the anastomosis zone. **Conclusion:** This investigation presents a novel HSA application, where the severe hemodynamic conditions produced by the high velocity jet from a 90-degree LVAD graft configuration are directly visualized in-vitro. This initial study sets the groundwork for further investigation into other LVAD graft configurations, where better alignment of the outflow graft anastomoses with the aortic axis may produce more favorable flow patterns in the aorta.

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TU-A-TRACK 2-06

Multiparameter Dense Neural Network (DNN) to Estimate Patient Eye-Lens Dose During Neuro-Interventional Procedures

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Purpose: Fluoroscopically-guided neuro-interventional procedures can result in high patient radiation dose to the eye lens with associated deterministic risks such as cataract formation. Deep learning (DL) models were investigated to determine if the patient lens dose for given exposure conditions could be estimated in real-time hence better controlled. **Methods:** Monte-Carlo simulations were done using NCICT computational phantoms to create a dataset of lens dose values for the right and left eyes to train the DL models. The dataset was obtained for a range of geometric and exposure parameters used in neuro-interventional procedures including entrance-field size, kVp, filter type, gantry angulation, patient head size, and patient x, y, z head position relative to the beam isocenter. The dose for each combination of parameters was expressed as lens dose per entrance air kerma (mGy/Gy). Data was split into training, validation and testing sets. Stacked models and median algorithms were implemented to create more robust models. The parameters to be input into the DNN during a procedure are obtained from our dose-tracking system (DTS) which is interfaced to the imaging system. Model performance was evaluated using the mean-absolute-percentage error (MAPE). **Results:** Using the median and stacked algorithms both led to a MAPE of less than 10% for prediction of the testing set. The lens dose prediction time for a single-exposure projection was 8 ms and 6 ms for the median and stacked algorithms, respectively. **Conclusion:** A DNN method is able to accurately predict the patient eye lens dose based on the geometric and exposure parameters that would be used in a fluoroscopic neuro-interventional procedure. This work shows that using a DNN is a viable option to be implemented in the DTS to predict patient eye-lens dose in real time and eliminate the need to have a large database of pre-calculated factors.

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TU-A-TRACK 2-07

Single-Shot Quantitative X-Ray Imaging Using Primary Modulation and Dual-Layer Detector: A Simulation Study

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Purpose: Conventional x-ray imaging provides limited quantitative information due to scatter, beam hardening, and overlaying tissues. A single-shot quantitative x-ray imaging (SSQI) method proposed in this work quantifies material-specific densities in x-ray imaging by combining the use of a primary modulator (PM) and a dual-layer (DL) detector. **Methods:** The PM features a checkerboard arrangement of alternating unattenuated and partially attenuated squares. Conceptually, the PM performs scatter correction for the DL images, while the DL images remove beam hardening from the PM. The DL detector further generates low- and high-energy images with superior registration, which are used to create material-specific images. In this work, a simulation study was performed for chest x-ray to test the feasibility of SSQI. We first simulated the x-ray images of a digital phantom (Lungman, Kyoto Kagaku) by incorporating a 120 kV spectrum, PM, and DL detector, and including scatter. The DL images contained four measurements that were obtained behind the hole (unattenuated) and fill (partially attenuated) regions of the PM of each layer. Using the low-frequency property of scatter and a pre-calibrated material decomposition (MD), four unknowns (i.e., two scatter images and two material-specific images) were jointly recovered by directly solving four equations given by the four measurements. **Results:** SSQI accurately performed MD to separate soft tissue from bone. Its performance improved with a smaller PM pitch and reduced focal spot blur. A pitch of 457 μ m and source-to-modulator distance of 36.3 cm can be an optimal combination for practical use, reducing the RMSE by 80% and 96% for PMMA and copper MD images, respectively, as compared to without scatter correction. **Conclusion:** We demonstrated the feasibility of SSQI to perform MD that is robust against scatter. The simplicity of SSQI may enable its widespread adoption beyond x-ray imaging to real-time image guidance and cone-beam CT.

SAM Imaging Scientific Symposium

TRACK 3

Recent Advances in X-ray CT - from Ultra-High-Resolution Imaging to AI Reconstruction

TU-AB-TRACK 3-00

Recent Advances in X-Ray CT - From Ultra-High-Resolution Imaging to AI Reconstruction

Moderator: Wojciech Zbijewski, Johns Hopkins University, Baltimore, MD

TU-AB-TRACK 3-01

J. Boone, UC Davis Medical Center:

High-Resolution Whole-Body CT: The Next Generation of CT?

TU-AB-TRACK 3-02

T. Schmidt, Marquette University:

Patient-Specific CT Dose Estimation

TU-AB-TRACK 3-03

E. Hoffman, University of Iowa:

CT and Dual-Energy CT in Assessment of Pulmonary Function

TU-AB-TRACK 3-04

S. Ravishankar, Michigan State University:

Machine Learning for Low-Dose CT Reconstruction

TU-AB-TRACK 3-05

Q&A

Despite its long history and established clinical utility, X-ray Computed Tomography (CT) remains an area of active research and innovation. In the recent years, significant progress has been made in scanner hardware, reconstruction algorithms, and dose reduction techniques. The technical development is paralleled by continuous expansion of diagnostic applications of CT. In this session, we will present an overview of some of these important advances, including (1) high-resolution CT systems, (2) personalized CT dosimetry, (3) new clinical capabilities in assessment of lung disease and (4) learning-based reconstruction algorithms.

One of the areas of major current progress in CT hardware has been in improving the spatial resolution of whole-body CT. The first talk will present comprehensive measurements of technical and clinical performance of one of the recently introduced high-resolution CT scanners. The system has six user-selectable focal spots, and three different detector configurations for acquisition. The MTF measurements in both the in-plane (x, y) and longitudinal (z) axes demonstrate a true doubling in spatial resolution. Clinical applications which benefit from the resolution of this scanner include thoracic, small vessel, pediatric, and musculoskeletal imaging. Advances in computational and deep learning technologies are enabling rapid, patient-specific CT organ dose estimation. Patient-specific organ dose estimation uses the CT images from the patient's scan to develop a personalized phantom model and to accurately model patient-specific aspects of the CT scan that affect dose,

such as tube current modulation and patient positioning. Patient-specific organ dose estimation is typically performed through two steps: (1) estimating a dose distribution map and (2) determining organ boundaries that are then applied to the dose map. Monte Carlo, Linear Boltzmann Transport Equation (LBTE) solver, and deep learning methods have been developed for rapid dose map estimation. Deep learning methods have also been applied in organ segmentation. The second talk will present an overview of patient-specific CT organ dose estimation methods and potential applications. The development and validation of a specific tool, using a rapid LBTE solver and deep learning segmentation methods, will be described.

The third talk will focus on recent advances in clinical application of CT. With careful attention to image protocol details, CT and Dual Energy CT now provides for a detailed map of parenchyma, vessels and airways along with a characterization of pathophysiology. Image matching between multiple lung volumes yields regional metrics related to lung mechanics and provides a marker of underlying functional small airways disease. Dual Energy-based material decomposition of contrast enhanced images provide a measure of regional perfused blood volume which serves as a surrogate for regional perfusion. With regional Jacobians from image matching providing regional measures of ventilation and DECT-PBV providing regional measures of perfusion, one can now assess the matching of ventilation and perfusion which is the primary function of the lung, required for gas exchange. The application of these CT metrics to the understanding of the etiology of smoking-associated lung disease will be discussed.

The session will conclude with a presentation of the emerging learning-based methods that exploit datasets of regular-dose and/or low-dose CT scans for improved image reconstruction. The talk will first review an approach based on model-based image reconstruction (MBIR), where the sparsity-based regularizer parameters are learned in an unsupervised manner from a dataset of high-quality CT images. A union of sparsifying transforms is pre-learned to cluster CT image patches into multiple groups, with a specific transform well-matched to each group. When incorporated as a data-driven regularizer in MBIR, the learned transforms provide much better image reconstruction than conventional filtered backprojection and non-adaptive regularization methods especially at low X-ray doses. Next, an extension of this approach to a unified supervised-unsupervised (SUPER) learning scheme will be discussed. This scheme combines classical MBIR optimization and unsupervised transform learning together with recent supervised deep learning in a common framework. Multiple interpretations of this unified framework will be discussed and shown to provide much better image reconstructions than the constituent models with limited training data.

Learning Objectives:

1. Understand the performance characteristics of high-resolution whole-body CT
2. Understand the computer modeling techniques that enable personalized CT dose estimation
3. Understand the advanced CT techniques used in assessment of lung function
4. Understand the unsupervised and supervised-unsupervised approaches for learning-based CT image reconstruction.

WZ receives research support from Siemens Healthineers

SAM Multi-Disciplinary Scientific Symposium

TRACK 4

Functional Image-guided Targeting and Avoidance in Radiation Therapy

TU-AB-TRACK 4-00

Functional Image-Guided Targeting and Avoidance in Radiation Therapy

Moderators: Amit Sawant, University of Maryland School of Medicine, Baltimore, MD and Martha Matuszak, University of Michigan, Ann Arbor, MI

TU-AB-TRACK 4-01

Y. Vinogradskiy, Thomas Jefferson University:

Regional Functional Avoidance in Lung Radiotherapy Using Ventilation Mapping

TU-AB-TRACK 4-02

M. Matuszak, University of Michigan:

Functional Avoidance in Liver RT Using DCE MRI

TU-AB-TRACK 4-03

B. Han, Stanford University School of Medicine:

Initial Experience on a Real-Time Biologically-Guided Radiotherapy System

TU-AB-TRACK 4-04

A. Sawant, University of Maryland School of Medicine:

Combining Virtual Bronchoscopy and Ventilation Mapping for Preserving Post-SAbR Respiratory Function

TU-AB-TRACK 4-05

Q&A & Panel Discussion

Function-guided radiation therapy using various imaging modalities such as CT, MRI, PET, and SPECT, may play a role in reducing toxicity as well as improving local control by targeting resistant disease and enabling treatment planning that can use a normal tissue regional function map in order to minimize the probability of radiation-induced toxicity or maximize the probability of post-treatment function. This session is focused on efforts to develop and clinically translate various functional image-guidance technologies across a variety of disease sites and disease stages.

In the first talk, we will discuss functional avoidance lung radiotherapy, which uses 4DCT to map regional lung ventilation. These maps are used to generate radiotherapy plans that reduce doses to functional regions of the lung. We will present early findings from a phase II, two-center, study that evaluated 4DCT-ventilation functional avoidance radiotherapy. The talk will report on the primary trial outcome of radiation pneumonitis. Secondary outcomes reported include non-pneumonitis thoracic clinical toxicity and functional avoidance dosimetric results.

In the second talk, we will highlight recent work on function guided treatment planning using multiple imaging modalities in liver, lung, and head and neck. We will discuss in detail the clinical workflow and preliminary results from recent, prospective clinical trials on response-based adaptive therapy using FDG-PET and V/Q SPECT in advanced stage lung cancer and gadoteric acid MRI for liver SBRT

In the third talk, we will discuss the first clinical experience with a novel biology-guided radiotherapy (BgRT) system equipped with PET imaging, Reflexion X1. We will discuss beam QA and commissioning procedures following the TG-148 protocol. We will report on performance characterization of the PET subsystem following the NEMA NU-2 2018 standard. We will also report on the imaging performance in CT mode and on the dosimetric and geometric accuracy of the system. We will report on the design of an early-phase clinical trial to quantify and evaluate the FDG dose for daily PET guidance, expected to start in April 2021.

In the fourth talk, we will discuss a novel lung functional avoidance strategy that accounts for serial (central and peripheral airways) and parallel (parenchyma)

components of lung function. We will discuss a virtual bronchoscopy-based technique to spatially map the airway tree and connect the air supply pathways with functional sub-lobe lung regions. We will discuss the development of a radiation response model for individual airway segments and the incorporation this model with NTCP models for lung parenchyma to create functional avoidance plans that minimize post-radiotherapy ventilation loss in lung cancer patients.

In the last part of the symposium, we will have a Q&A session for discussions between session organizers, speakers, and the audience.

Learning Objectives:

1. Understand the motivation and the basic concepts of functional guidance in radiotherapy
2. Keep up to date on the latest functionally-guided RT clinical trials
3. Understand the clinical requirements and technical characterization of a novel biologically-guided radiotherapy system
4. Understand the current challenges in functional-image-based guidance and novel strategies to address these challenges

Amit Sawant has funding support from the National Cancer Institute and from Varian Medical Systems

SAM Therapy Scientific Symposium

TRACK 5

Radiation as a Biological Rather than a Physical Tool to Combat Cancer

TU-AB-TRACK 5-00

Radiation as a Biological Rather Than a Physical Tool to Combat Cancer

Moderator: Gabriel Sawakuchi, UT MD Anderson Cancer Center, Houston, TX

TU-AB-TRACK 5-01

Introduction

TU-AB-TRACK 5-02

A. McNamara, Physics Division, Department of Radiation Oncology, Massachusetts General Hospital and Harvard Medical School, Boston, Massachusetts, USA:

Cellular Targets of Proton Radiation-Induced Bone Fractures

TU-AB-TRACK 5-03

J. Eley, Vanderbilt University School of Medicine:

Minibeam Therapy with Protons and Heavy Ions to Treat Brain Cancers

TU-AB-TRACK 5-04

S. Bright, UT MD Anderson Cancer Center:

A Physics Approach to Combine Radiotherapy with Molecularly Targeted Therapies to Preferentially Radiosensitize Tumors But Not Normal Tissues

TU-AB-TRACK 5-05

W. Ngwa, Brigham and Women's Hospital:

Radio-Immunotherapy Dose-Painting

TU-AB-TRACK 5-06

Q&A & Panel Discussion

Advances in radiation therapy (RT) have been driven mainly by technologic developments that enable precise delivery of ablative radiation doses to tumors while sparing adjacent critical structures. Substantial advances in RT technology, particularly in image-guided RT, over the past decade have paved the way for escalating radiation doses to levels that were historically inconceivable. However, dose escalation is subject to physical limits, and it cannot combat radioresistance in many tumors. The current "brute force" approach, in which delivering higher radiation doses to tumors creates more DNA damage and thus kills more tumor cells, neglects the vast knowledge accrued on the biology and immunology of tumors and normal tissues in response to radiation. This session highlights new advances with a focus on the biological rather than the physical aspects of radiation. Topics covered include mechanisms of proton-induced bone fractures; minibeam therapy using protons and heavy ions; combinations of radiation and DNA-repair inhibitors for radiosensitization and for antitumor immunity; and radio-immunotherapy dose-painting. The hope is that this session will encourage, motivate, inspire, and guide physicists to assume leading roles, rather than secondary ones, with regard to biological research applied to radiotherapy.

Learning Objectives:

1. Understand and discuss opportunities for physicists in biological research applied to radiotherapy.
2. Highlight how original physics ideas can help to create biological hypotheses.
3. Encourage, motivate, inspire, and guide physicists to take a leading role rather than a secondary one on applying biological research to radiotherapy.

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Therapy Scientific Session

TRACK 6

Outcome Modeling and Assessment

TU-A-TRACK 6-01

Compact Neural Network to Predict Radiation-Induced Lymphocyte Depletion

Y Kim^{1*}, I Chamseddine², A Saraf³, Y Cho⁴, F Keane⁵, W Sung⁶, H Yoon⁷, H Paganetti⁸, J Kim⁹, S Cho¹⁰, C Grassberger¹¹, (1) KAIST, Daejeon, , KR, (2) Harvard Medical School/Massachusetts General Hospital, Quincy, MA, (3) Massachusetts General Hospital, , (4) Yonsei University College of Medicine, Seoul, , KR, (5) Massachusetts General Hospital, , (6) The Catholic University of Korea, Seoul, , KR, (7) Yonsei University College of Medicine, Seoul, , KR, (8) Mass General Hospital and Harvard Medical School, Boston, MA, (9) Yonsei University College of Medicine, Seoul, , KR, (10) KAIST, Daejeon, , KR, (11) Massachusetts General Hospital, Cambridge, MA

Purpose: Radiation-induced lymphopenia (RIL) is known to be associated with survival after radiotherapy (RT). We investigate the correlation of dose distributions with RIL in lung and liver cancer patients and develop a compact neural network to predict RIL in this study. **Methods:** We investigated dose distributions in 84 lung

cancer patients and 63 liver cancer patients treated at two institutions, and investigated RIL directly after RT and the lymphocyte recovery 3 months later. Dose distributions were deformably registered to a common target coordinate system for spatial normalization, and non-parametric correlation tests were used to study local effects of dose to specific regions. A compact multi-path neural network was designed that integrates clinical patient data in fully connected layers with the dose distribution in convolutional layers for predicting the post-RT absolute lymphocyte count of lung cancer patients. We used relatively a small (less than 3) number of convolutional layers, a limited number of kernels, and extensive max-pooling for a light-weight network. Training and validation were performed separately on data from different institutions. **Results:** Clear correlation of RIL with dose was observed around large vessels - pulmonary vessels and heart in lung cancer patients and large hepatic vessels in liver cancer patients. When focusing on lymphocyte recovery, these correlations shifted to bone marrow. Keeping the neural network structure compact enabled convergence of the network in both training ($n=63$) and validation sets ($n=21$), and the network predicted the post-RT lymphocyte count (average $0.9/\mu\text{l}$, range $[0.1-2.2]/\mu\text{l}$) with a mean square error of $0.06-0.08$. **Conclusion:** Dose to large vessels is strongly correlated with RIL directly after RT in liver and lung cancer patients, while dose to bone marrow correlates with the recovery from RIL. The developed neural network integrates 3D dose information directly with clinical data and successfully predicts RIL.

This study is supported by NCI (grant nr: R21 CA241918) and Korean Health Industry Development Institute (grant number:HI19C1332)

TU-A-TRACK 6-02

From Person-Specific to Population-Based Colorectal Models: An Age-Scalable Computational Anatomical Colorectal Model for Radiation Dosimetry in Late Effects Studies of Childhood Cancer Survivors

C Owens^{1,2*}, B Rigaud¹, E Ludmir¹, A Gupta^{1,2}, S Shrestha^{1,2}, A Paulino¹, C Peterson^{1,2}, S Kry^{1,2}, S Smith¹, K Brock^{1,2}, R Howell^{1,2}, (1) The University of Texas MD Anderson Cancer Center, Houston, TX, (2) The University of Texas Graduate School of Biomedical Sciences at Houston, Houston, TX

Purpose: To develop, validate and integrate into an age-scalable computational phantom an “average” colorectal model that incorporates anatomical variations of pediatric patients. **Methods:** Whole-body non-contrast CT scans of fifty-five male pediatric patients (age range: 2.1-21.6years) were retrospectively selected. Manual colorectal contours were reviewed and approved by two radiation oncologists. Forty-five and ten patients were used for training and testing, respectively. All contours were spatially normalized using median colorectal length and registered to an anatomical template using the constrained symmetric thin-plate spline robust point matching (sTPS-RPM) method. Deformed contours were used to generate a principal component analysis (PCA)-based statistical shape model (SSM) of the colorectum to extract the dominant deformation modes of the population. The sTPS-RPM inter-patient registration was evaluated using Dice similarity coefficient (DSC), distance-to-agreement (DTA), and Hausdorff distance (HD). To validate the generalizability of the PCA model to represent unseen shapes, we computed the DSC between the original and PCA reconstructed contours for ten test patients. The population PCA-based colorectal model was integrated into our phantom, which can be scaled to any age (infant-to-adult). Additionally, the length and mass of the model was matched to ICRP reference data within 5% across the age range. **Results:** The sTPS-RPM inter-patient registration mean(min-max) DSC, DTA and HD was $0.96(0.93-0.98)$, $1.5\text{mm}(1.2-2.0)$, and $6.0\text{mm}(3.2-19.7)$, respectively. On the training set, the PCA-based SSM was able to describe 95% of inter-patient colorectal variance with 11 of the 45 dominant principal components. Using the PCA-based SSM on the unseen 10 test patients, the mean(min-max) DSC between the validation and reconstructed colorectal contours was $0.80(0.76-0.84)$. **Conclusion:** We developed, validated and integrated a population PCA-based SSM of the colorectum into our phantom. The model will be used to reconstruct colorectal doses for a study of subsequent colorectal cancers in childhood cancer survivors treated in the pre-CT era of RT.

TU-A-TRACK 6-03

MRI-Based Multi-Region Delta-Radiomics for Prediction of Recurrence After Chemoradiotherapy in Locally Advanced Cervical Cancer

M Kawamura^{1*}, D Kawahara², I Nishibuchi², A Saito², T Yoshida³, Y Nagata^{2,4}, (1) Graduate School of Biomedical & Health Sciences, Hiroshima University, Hiroshima, JP, (2) Department of Radiation Oncology, Institute of Biomedical & Health Sciences, Hiroshima University, Hiroshima, JP, (3) School of Medicine, Hiroshima University, Hiroshima, JP, (4) Hiroshima High Precision Radiotherapy Cancer Center, Hiroshima, JP

Purpose: The aim of this study is to predict the recurrence of locally advanced cervical cancer after radiotherapy from delta-radiomics features using magnetic resonance imaging (MRI). **Methods:** This study included 61 patients with locally advanced cervical cancer treated with chemoradiotherapy between 2003 and 2017. Region of interest(ROI) were created with the uterus as the clinical target volume(CTV). In addition to the ROIs that expanded and shrunk from CTV were created, the inner and shell regions were analyzed. The inner region was defined as the region that the voxels excluded the tumor boundaries. The shell region was defined as the region that the voxels around the tumor boundaries were analyzed. Delta-radiomics feature was defined as the difference in radiomics feature before and at 4 weeks after radiation therapy begins. A total of 9394 features were extracted from T1-weighted images(T1WI) and T2-weighted images(T2WI). Using the features selected by the method of least absolute shrinkage and selection operator (LASSO) regression, two single sequence radiomics models and mixture of the two sequence were constructed using a neural network with 5-fold cross-validation. The performance of the prediction models was assessed in terms of sensitivity, specificity, accuracy and area under the receiver operator characteristic curve (AUC). **Results:** From Lasso regression, 4 radiomics features from T1WI and 8 radiomics features from T2WI and 8 radiomics features from combination of T1WI and T2WI. The accuracy, specificity, sensitivity, and AUC of the prediction model for the dataset were 70.0%, 87.5%, 41.2%, and 0.77 with T1WI, 85.6%, 91.9%, 71.4%, and 0.90 with T2WI, 82.2%, 87.8%, 75.6%, and 0.91 with the combined. **Conclusion:** The current study showed that multi-region delta-radiomics features could be an important factor to predict recurrence of cervical cancer.

TU-A-TRACK 6-04

Predicting Osteoradionecrosis from Head and Neck Radiotherapy Using a Residual convolutional Neural Network

B Reber^{*}, B Anderson, A Mohamed, L Van Dijk, B Rigaud, M McCulloch, Y He, M Woodland, C Fuller, S Lai, K Brock, UT MD Anderson Cancer Center, Houston, TX

Purpose: Osteoradionecrosis (ORN) is a detrimental radiation therapy (RT)-induced toxicity for patients after treatment of head and neck cancer. The ability to predict ORN before radiation treatment is performed could enable further optimization of the RT plan or stratification to advanced techniques (e.g. MR-LINAC, adaptive RT, dose de-escalation). A residual convolutional neural network is developed to predict ORN occurrence to aid in this endeavor. **Methods:** A residual convolutional neural network was constructed using a hyperparameter search to find the model with optimum performance, including 2D and 3D architectures. Model input consisted of 3D computed tomography (CT) images, dose distributions, and mandible contours with size (32, 128, 128). Binary outcome of ORN is used for model output. The 148 subject data was split into 111 cases (40 ORN+) training and 37 cases (11 ORN+) validation. Positive cases were oversampled to ensure a match between ORN+ and ORN- in both datasets. Accuracy, sensitivity, specificity, and area under the receiver operator characteristic curve (AUC) are used for model evaluation. Results were compared with logistic regression models using a 0.8/0.2 train/test split and 10-fold cross-validation. **Results:** The best logistic regression models had 0.68 AUC using only the dose distribution (V60) and 0.70 AUC using the dose distribution and clinical variables ($N_{\text{stage}}+V65$). The best deep learning model had 64 starting filters, 5 down-sampling operations and 1 residual block in-between down-sampling operations. The model was a 3D architecture using CT, dose distribution, and mandible contour and had 0.79 accuracy, 0.85 recall, 0.76 precision, 0.83 AUC. 3D deep learning architectures outperformed 2D architectures (0.53 AUC). **Conclusion:** The model predicts ORN with higher performance metrics compared to logistic regression models. Further evaluation on multi-institutional test data is needed. The model could be implemented during the treatment planning stages to aid in patient decision making and treatment stratifications.

TU-A-TRACK 6-05

Relationship Between Coronary Artery Radiation Doses and Risk of Late Coronary Artery Disease: A Report from the Childhood Cancer Survivor Study (CCSS)

S Shrestha^{1*}, J Bates², Q Liu³, S Smith¹, D Mulrooney⁴, A Gupta¹, C Owens¹, W Leisenring⁵, T Gibson⁶, E Chow⁵, K Oeffinger⁷, L Robison⁴, G Armstrong⁴, L Constine⁸, B Hoppe⁹, C Lee⁶, L Court¹, S Kry¹, Y Yasui⁴, R Howell¹, (1) University of Texas MD Anderson Cancer Center, Houston, TX, (2) Emory University School of Medicine, Atlanta, GA, (3) University Of Alberta, Alberta, CA, (4) St. Jude Children's Research Hospital, Memphis, TN, (5) Fred Hutchinson Cancer Research Center, Seattle, WA, (6) National Cancer Institute, Rockville, MD, (7) Duke University, Durham, NC, (8) University of Rochester Medical Center, Rochester, NY, (9) Mayo Clinic, Jacksonville, FL

Purpose: Among childhood cancer survivors, heart dose from radiotherapy (RT) is an established risk factor for coronary artery disease (CAD). Since contemporary RT can be optimized for individual substructures, coronary artery-specific dose-response relationships could refine cardiac dose constraints beyond whole heart metrics. We investigated associations between coronary artery doses and risk of CAD. **Methods:** We estimated 35-year cumulative incidence and incidence rate ratios (RR) of CTCAE grade 3-5 CAD for 25,481 five-year cancer survivors in the CCSS diagnosed at median age 6.1 years (0-20 years) between 1970-1999 at 32 institutions. For the 12,228 individuals treated with RT, fields were reconstructed on an age-scaled computational phantom with a detailed heart model with four coronary arteries. Mean doses to the left main (LMA), left anterior descending (LAD), left circumflex (LCX), and right coronary (RCA) arteries and whole heart were calculated. Associations of heart and artery doses with CAD were evaluated with piecewise exponential models adjusted for demographic, treatment-related characteristics, and cumulative anthracycline dose. **Results:** For all arteries and whole heart, mean doses ≥ 10 Gy were associated with significantly increased CAD risk. After a mean dose of 10- <20 Gy to the LMA, LAD, LCX, and RCA respectively, 35-year cumulative incidences were 4.7% (95%CI:2.7-6.7), 10.7% (95%CI:7.7-13.7), 6.8% (95%CI:4.8-8.7), and 12.6% (95%CI:10.3-15.0); RR values were 2.5-5.3 times higher than for survivors unexposed to RT. Lower doses (5- <10 Gy) to the LAD (RR 1.9, 95%CI:1.6-4.1, $p<0.001$) and RCA (RR 2.6, 95%CI:1.6-4.1, $p<0.001$) were also associated with significantly increased risk. **Conclusion:** This study expands our understanding of RT-related CAD beyond whole heart doses and establishes relationships between dose to specific coronary arteries and CAD. We uniquely identified that LAD and RCA doses as low as 5 Gy were associated with increased risk. These data can guide contemporary pediatric RT to prioritize coronary artery dose constraints.

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TU-A-TRACK 6-06

Seed Spots Analysis to Characterize Linear-Energy-Transfer (LET) Effect in the Adverse Event Regions of Head and Neck Cancer Patients Treated by Intensity-Modulated Proton Therapy (IMPT)

Y Yang^{1*}, S Patel¹, J Bridhikitti², W Wong¹, M Halyard¹, L McGee¹, J Rwigema¹, S Schild¹, S Vora¹, T Liu³, M Bues¹, M Fatyga¹, R Foote², W Liu¹, (1) Department of Radiation Oncology, Mayo Clinic Arizona, Phoenix, Arizona, USA (2) Department of Radiation Oncology, Mayo Clinic Rochester, Rochester, Minnesota, USA (3) Department of Computer Science, the University of Georgia, Athens, GA

Purpose: High linear energy transfer (LET) radiotherapy may cause unexpected adverse events (AEs) in head and neck (H&N) cancer patients treated by intensity-modulated proton therapy (IMPT). In this exploratory study, we propose to investigate the effects of LET upon AEs in IMPT using dose-linear-energy-transfer volume histograms (DLVHs) and seed spots analysis. **Methods:** We included 14 H&N patients with unanticipated CTCAEv4.0 grade ≥ 3 AEs for analysis. The AE regions were contoured and the corresponding DLVHs of the AE regions were generated. DLVH was constructed with physical dose (Gy) and LET (keV/ μ m) as independent variables. The normalized volume of the structure was contoured as iso-volume lines in the dose-LET plane. All voxels in the structure were mapped into the dose-LET plane as dots to see how the dots with similar dose and LET were clustered. For seed spot analysis, we selected voxels at the top edge of the DLVH plots as critical voxels. Individually clustered critical voxels that are geometrically apart were considered as independent seed spots. Median dose/LET from seed spots were extracted. Bivariate-linear-regression models were established for all AEs. The dose and LET product (xBD) volume constraint of osteoradionecrosis was obtained using the receiver operating characteristic curve of 4 osteoradionecrosis patients and 15 additional independent control patients. **Results:** Dose played a dominant role for in-field AEs such as hemorrhage and ulceration, while LET played an important role in out-of-field AEs such as osteoradionecrosis and mucositis. Intercept-free linear models between the reciprocal of dose and LET were found for different AEs. The xBD volume constraint of osteoradionecrosis was derived, $V(xBD \geq 275.18 \text{ Gy-keV}/\mu\text{m}) < 0.061\text{cc}$, with an area under curve of 0.85. **Conclusion:** A potentially important LET-enhancing effect to induce AEs in H&N cancer treated with IMPT was observed. Voxel-based DLVH and seed spots analysis are powerful tools to study AEs in IMPT.

This research was supported by Arizona Biomedical Research Commission Investigator Award, the Lawrence W. and Marilyn W. Matteson Fund for Cancer Research, and the Kemper Marley Foundation.

TU-A-TRACK 6-07

Survival Prediction Models for Patients with Malignant Pleural Mesothelioma After Adjuvant Radiotherapy

Z Wang^{1*}, V Li², X Qi³, (1) UCLA School of Medicine, Los Angeles, CA, (2) USC Dornsife College of Letters, Arts and Sciences, Los Angeles, CA, (3) UCLA School of Medicine, Los Angeles, CA

Purpose: Complex target shape and unique location pose big challenges of plan quality and achievable dosimetric parameters and subsequent treatment outcome for Malignant pleural mesothelioma (MPM) on individual basis. Correlations between target characteristics, planning dosimetric parameters and overall survival were explored using machine learning methods. **Methods:** A retrospective cohort of 59 MPM patients, receiving 45 Gy/25 fractions on a Tomotherapy unit, were pooled and analyzed. The achieved dosimetric parameters for target and critical OARs were analyzed for 23 left-sided mesothelioma (LSM) and 36 right-sided mesothelioma patients (RSM) respectively. LASSO function $1/2 \|Ax - b\|_2^2 + \gamma \|x\|_1$ was adopted for selecting the most relevant predictors with dosimetric endpoints and/or overall patient survival. Machine learning methods including Multiple Layer Perceptron (MLP) were applied to construct predictive models for MPM response after a pleurectomy/decortication. The model performance was evaluated using the Area Under the ROC Curve (AUC) with 3-fold cross validation. **Results:** No differences were observed between planning target volume (PTV) coverage 94.2 $\pm$ 1.2% (LSM) vs 93.4 $\pm$ 2.4% (RSM). The achieved dosimetric endpoints were found significantly different ($p<0.05$) for OARs between the LSM and RSM: ipsilateral lung V20 (91.2 $\pm$ 6.7 vs. 85.3 $\pm$ 6.5%), ipsilateral mean lung doses (39.1 $\pm$ 3.8 vs 36.9 $\pm$ 2.5 Gy), heart mean doses (25.2 $\pm$ 3.6 vs 18.8 $\pm$ 3.7 Gy), liver mean doses (10.6 $\pm$ 2.7 vs. 23.4 $\pm$ 4.2 Gy), respectively. There was no significant difference in total lung V20-PTV mean dose 13.4 $\pm$ 2.0 vs. 14.4 $\pm$ 1.5 Gy nor expected lung pneumonitis for LSM and RSM. The MLP reached an AUC of 0.71 in predicting overall survival from combined characteristics and planning dosimetric parameters. The identified predictors that are most associated with overall survival were age, PTV volume, ipsilateral lung volume and V20Gy, and ipsilateral lung-PTV V20. **Conclusion:** Significantly dosimetric differences of OARs were observed for left- and right-side MPM. The predictive models can assist in achieving patient-specific optimal plan quantity and treatment outcome via personalized treatment.

SAM Therapy Educational Course

TRACK 7

HyTEC SBRT Dose Constraints

TU-A-TRACK 7-00

HyTEC SBRT Dose Constraints

Moderator: Jimm Grimm, Geisinger Health System, Wilkes Barre, PA

TU-A-TRACK 7-01

J. Grimm, Geisinger Health System:

HyTEC - The Project and the Product

TU-A-TRACK 7-02

E. Yorke, Memorial Sloan-Kettering Cancer Center:

HyTEC - The Project and the Product

TU-A-TRACK 7-03

A. Mahadevan, Geisinger Medical Center:

Pancreas TCP as a Case Study of HyTEC Process

TU-A-TRACK 7-04

A. Jackson, Memorial Sloan-Kettering Cancer Center:

Physicist/Modeler's Perspective on HyTEC

TU-A-TRACK 7-05

L. Marks, University of North Carolina at Chapel Hill:

Physician Perspective on HyTEC - the Clinical Needs, Opportunities and Limits

HyTEC Overview: The HyTEC (High Dose per Fraction, Hypofractionated Treatment Effects in the Clinic) review is an effort to summarize the currently-available stereotactic body radiation therapy (SBRT) dose / volume / outcome data to update and refine the normal tissue tolerance guidelines provided by the classic “Emami” paper (IJROBP May 1991) and the recent QUANTEC publications (IJROBP March 2010), for radiosurgery throughout the body in 1 to 5 fractions. HyTEC was formed as the SBRT Working Group (WGSBRT) within the American Association of Physicists in Medicine (AAPM) under the Biological Effects Subcommittee (BESC) and consists of more than 100 authors who are physicians, physicists, radiobiologists, and biomathematicians. The effort has resulted in 7 articles addressing NTCP, 9 articles addressing TCP, and several vision and editorial articles, that will constitute a special issue of the Red Journal (May or June 2021) with free access for AAPM members. This session will provide an updated summary of the overall effort, with an emphasis on general principles and findings of the project. The recently completed Pancreas TCP paper is an example of the methods and results.

HyTEC_ the project and the product: The 16 HyTEC organ/tumor-specific papers are analyses of selected, peer-reviewed publications of sbrt treatments spanning the entire body: Cranial, Head and Neck, Thoracic, Abdominal, Pelvic, and Spinal. Results are presented in a format similar to that of QUANTEC. For each paper, a self-assembled group of experts (physicians, physicists, radiobiologists and biomathematicians) systematically triaged the literature to find papers that included sufficient data for a quantitative synthesis, preferably (but not always) in the form of a sigmoidal curve -a mathematical model-usually requiring a radiobiological dose correction to account for fractionation. Additionally, each paper summarizes such site-specific questions as volume definition and ambiguities in outcome determination. From the models, readers can determine dosimetric metrics and uncertainties associated with literature-based NTCP or TCP outcomes and can compare them with their own experience. As in QUANTEC, the HyTEC introductory article summarizes consensus dose and dose-volume metrics for the major SBRT treatment types, risk organs and tumors in tabular form, such that readers can evaluate and compare them with those used in their own clinics.

Pancreas TCP: Pancreatic cancer is the fourth leading cause of cancer death in the United States, though it is only 3% of all new cancers. Five-year overall survival (OS) is less than 25%. Treatments often combine chemotherapy, surgery and radiation therapy with a major radiation therapy goal of converting locally advanced and borderline resectable pancreatic cancer (LAPC and BRPC) to resectable. SBRT is a desirable form of radiation treatment because short treatment schedules interfere minimally with chemotherapy. However, treatment planning and delivery are complicated by respiratory motion and inter- and intra-fraction normal organ variability and proximity of the tumor to stomach and bowel limits safe prescription doses.

A PubMed search initiated in 6/2019 resulted in 414 articles of which only 32 met the inclusion criteria of providing the cumulative prescription, number of fractions, number of patients, Kaplan-Meier (KM) or individual patient data on OS or Local Control (LC), at least 75% of patients without prior pancreatic radiation and at least 90% without distant disease. These articles provided 48 data points; all but three studies used between 1 and 5 fractions. Study sizes ranged from 12 to 91 patients. There was considerable variability among these studies including the starting points for effect duration (e.g. time from diagnosis vs time from start of radiation), different applications of respiratory motion control and image guidance, the definition of local failure. Detailed dose distributions were rarely provided. For study comparisons, prescription doses were converted to equivalent dose in 3-fractions using the Linear-Quadratic model with $\alpha/\beta=10$ Gy (3fxED). Given the variability, modeling LC beyond a year was not attempted.

Resectability was the strongest factor affecting outcome; average 1 year LC was over 90% for patients with R0 resection. For unresected patients, 1 year LC was <70% for 3fxED<24 Gy, approximately 77% for 3fxED approximately 28.2 Gy and approximately 86% for 3fxED of 36 Gy. The dose response from studies in which at least 80% of patients were unresected throughout the study was significant; a logistic model was constructed: 1-year LC=100%/(1+(D50/D)⁴ g50) where D50=17.6 Gy (95% CI 8.8-21.5) and g50=0.64 (95% CI 0.27-1.02).

It is desirable for future studies to agree on starting time for KM analysis, to better stratify patients by extent of resectability, provide more detail about motion management and image guidance and provide dosimetric information beyond the prescription dose.

Physicist/Modeler's perspective: The HyTEC project has used published literature to investigate and synthesize normal tissue complication probabilities in 7 tissues and local control in 9 tumor types resulting from hypo-fractionated and single fraction radiotherapy/SBRT, resulting in models of treatment tolerances and outcomes. In this talk we will discuss the successes and limitations of this effort, in terms of the kinds of information that could be gleaned from the published data, and the subsequent uncertainties in the outcomes models, tolerances, and rates of local control. The reporting standards required to overcome the limitations that have hampered HyTEC will be summarized. Finally, we will highlight the opportunities for modelling treatment outcome that may result from SBRT treatments.

Physician's perspective: Caring for patients with cancer is a privilege. We are fortunate to be able to offer patients and their families the opportunity of cure from an often-lethal disease, as well as to relieve much pain and suffering. At the same time, with responsibility comes a level of burden. Strikingly an appropriate balance between attaining local control/cure vs. minimizing the risk of serious morbidity can be difficult. While we have “class solutions” for common clinical situations, there are often the patient-specific issues (e.g. anatomy, comorbid conditions) that can make treatment planning quite challenging. In our quest to best serve our patients, we try to make data driven objective decisions. However, sorting through the literature can be a daunting task- especially for busy clinicians. Thus, expert reviews that provide readily accessible, clinically-useful information can be most helpful. The dose/volume/outcome data provided by the prior QUANTEC, the current HyTEC, and similar initiatives (e.g. PENTEC; Pediatric Normal Tissue Effects in the Clinic), provide this type of helpful information, and we believe that these efforts have helped, and will continue to help, a large number of patients. At the same time, physicians/planners need to be cognizant of the limitations inherent in these systematic/ pooled reviews. For example, there are inherent inaccuracies in the pooling of data from different centers (e.g. related to differences in image

segmentation, treatment techniques, outcome definitions; reporting biases; patient heterogeneities; and mathematical models of uncertain accuracy). In addition, our estimates of local control and toxicity risk may appear overly optimistic; e.g. in some settings local recurrence may be difficult to establish with certainty by non-invasive imaging methods, and/or complications may not be accurately recognized, and/or there is a bias towards the publication of favorable results. Further, a wide-spread problem in much of our literature is the use of actuarial techniques to compute estimates of local control, with the censoring patients at the time of death. The accuracy of actuarial techniques requires that censoring events should be independent of the endpoint under consideration. Since the pace of disease beyond the treated site (that can cause the censoring event of death) and the pace of regrowth of treated site (that obviously impacts local recurrence) are likely related, actuarial estimates may not be accurate and may overstate the local control. Nevertheless, despite these limitations, we believe that the data provided by the HyTEC reviews are useful. These issues, as well as an “Idealized Future State” to yield better data (e.g. leveraging advances in electronic health records; standards for contouring, structure naming, outcome reporting; and patient empowerment) will be discussed.

Learning Objectives:

1. Understand AAPM's role in the HyTEC effort
2. Understand some ways in which a clinical physicist or dosimetrist can use the HyTEC 'organ papers'
3. Understand the efficacy of current SBRT techniques for treatment of pancreatic cancer
4. Understand causes of successes and failures of outcome modeling for SBRT
5. Understand future opportunities in building on reported clinical experience to improve SBRT

JG: patent for DVH evaluator; grants from Accuray, Novocure

AM: Varian and Accuray Advisory Board

EY, AJ: NCI P30CA008748

Vendor Showcase

TRACK 8

Radformation

TU-SHOWCASE-TRACK 8-9-1

Focus on Your Strengths, Not Your Weaknesses: Automation for the Clinical Physicist

K Sysock1*, (1) Radformation, New York, NY

SAM Professional Symposium

TRACK 1

Ethics Unwrapped: Ethics Case Studies

TU-B-TRACK 1-00

Ethics Unwrapped: Ethics Case Studies

Moderator: Nina Bahar, St. Peter's Health Partners, Albany, NY

TU-B-TRACK 1-01

N. Bahar, St. Peter's Health Partners:

General Introduction to AAPM Code of Ethics

TU-B-TRACK 1-02

A. Riegel, Northwell Health:

Case Study: Conflict of Interest

TU-B-TRACK 1-03

Breakout Session

TU-B-TRACK 1-04

Discussion, Reflection, and Audience Q&A

TU-B-TRACK 1-05

Wrap Up

The Ethics Case Study Working Group will present a case study concerning conflict of interest in a professional medical physics environment. Specifically, the case study focuses on a young physicist whose medical physics practice was recently acquired by a larger firm that also has an equipment manufacturing arm. The case study explores the ethical concerns of purchasing equipment when one potential vendor is financially linked to the purchaser. After a brief introduction to the AAPM Code of Ethics, presenters will describe the case study and participants will break into smaller discussion groups led by working group members to navigate through the ethical decision making process and discuss potential consequences of their decision. Smaller groups will rejoin the larger group for open discussion and potential guidance from the work group concerning the AAPM Code of Ethics.

Learning Objectives for this session include:

1. To become familiar with the AAPM code of Ethics, specifically the sections relating to conflict of interest
2. To identify the key issues in a case scenario as applies to the AAPM code of Ethics
3. To facilitate the discussion of theories and ideas regarding the case study scenarios and extract meaning from the scenario presented

Imaging Scientific Session

TRACK 2

Multimodality, Optical, and Emerging Imaging Technologies

TU-B-TRACK 2-01

Quantitative and Sensitive Diagnosis of Early Tumor Metastases Using X-Ray and Bioluminescence Tomography

J Chen*, N Zhao, Y Yang, University of Science and Technology of China, Hefei, Anhui, P.R. China

Purpose: To develop an imaging technique for sensitive and accurate detection of early tumor metastases in vivo by using dual-modality x-ray and bioluminescence tomography. **Methods:** Cone beam X-ray computed tomography (CBCT) and bioluminescence tomography (BLT) have been integrated on the image guided small animal arc radiation treatment platform. In this study, its capability of imaging early-phase tumor metastases was challenged with a metastatic castrated-resistant prostate cancer mouse model. Luciferase-expressing cancer cells were injected through left ventricles of immune-deficient mice (n=4). The tumor cells were allowed for seeding and forming metastasis for 4 weeks before imaging. Cone beam computed tomography was first conducted to provide anatomy information and three dimensional mesh for BLT reconstruction. After that bioluminescence projections were captured every 90° rotation, then collected data were put into the CBCT-assisted BLT reconstruction framework. BLT imaging performance including tumor location and size was assessed by postmortem histology. **Results:** All tumors observed in histology section were also found in BLT imaging results in the same location, and all tumors were at millimeter scale as measured in histology section. Two tumors as small as ~1mm were simultaneously detected in each of two mice. One tumor was detected in each of the other two mice. Histology results confirmed that the onboard CBCT/BLT system was able to detect millimeter scale tumor metastases in living animal with sub-organ level accuracy. **Conclusion:** The onboard CBCT/BLT system is able to precisely locate, contour and quantify the early tumor metastases in living animal at millimeter scale, which provides an exciting imaging tool for the investigation of tumor metastasis mechanisms.

TU-B-TRACK 2-02

Liquid Immunogenic Fiducial Eluter (LIFE) Gel for Image-Guided Radiotherapy

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Purpose: To design Liquid Immunogenic Fiducial Eluter (LIFE) gel using natural biodegradable polymers: chitosan and sodium alginate, as a nano-Smart Radiotherapy Biomaterial (n-SRB) that gelifies within the calcium rich tumor micro-environment. This study shows that the LIFE gel loaded with nanoparticles and immunoadjuvants can provide Magnetic Resonance Imaging (MRI) / Computed Tomography (CT) contrast and cause regression of both local and distant unirradiated tumors. **Methods:** Two formulations of the LIFE gel were created to investigate prolonged release of nanoparticles, sodium alginate polymer crosslinked with calcium, and Chitosan-Alginate polymer crosslinked with calcium. In vitro release for these polymer formulations was characterized via fluorescence from FITC dye using a spectrophotometer. In-vitro release was studied in PBS PH 7.4 solution. MR and CT imaging of LIFE gel formulations were conducted to further characterize the release and establish that signal contrast can be obtained from the nanoparticles over time in vitro and also within C57BL6/Jac mice pancreatic tumor. Therapeutic benefit was also assessed in animals with two cervical tumors on each animal where only one was treated with the LIFE gel in combination with radiotherapy at 6Gy compared to control cohorts. **Results:** Results showed burst release from both gel formulations as well as approximately 15% higher release from alginate gel compared to that from chitosan-alginate polymer at day-5. Nanoparticle contrast decreased over time from MR and CT images highlighting prolonged retention of nanoparticles within the chitosan-alginate gel versus the alginate gel in-vitro and within mice tumor microenvironment. Cohorts of mice treated with the LIFE gel loaded with anti-CD40 showed regression of both treated and untreated tumors. **Conclusion:** This work demonstrates the potential of LIFE gel for image-guided radiotherapy, with major potential to provide MR or CT imaging contrast while also boosting therapeutic outcomes.

TU-B-TRACK 2-03

Quantification of Tumor Location and Growth for Orthotopic Pancreatic Cancer Model Using Bioluminescence Tomography-Guided System

Z Deng^{1*}, X Xu¹, H Dehghani², J Reyes³, J Wong³, P Tran³, K Wang¹, (1) University of Texas Southwestern Medical Center, Dallas, TX, (2) University of Birmingham, Birmingham, UK, (3) Johns Hopkins University, Baltimore, MD

Purpose: The developing radio-immunotherapy has cast radiation therapy(RT) into a pivotal role for pancreatic cancer. There is a need to study pancreatic cancer in pre-clinical setting to advance our understanding of RT. The challenge to study abdominal malignancy in vivo is lack of proper imaging guidance. Bioluminescence with its strong imaging contrast has been widely used to monitor tumor growth. We innovated bioluminescence tomography(BLT) to guide irradiation for orthotopic pancreatic ductal adenocarcinoma(PDAC), and monitor its growth and treatment response. **Methods:** To establish orthotopic PDAC model, we implanted a fraction of BxPc3-Red-Fluc tumor into nude mouse pancreas. For BLT, mice were subject to multi-spectral multi-projection bioluminescence imaging(BLI), followed by small animal radiation research platform(SARRP) CBCT imaging. CBCT image was acquired to generate anatomical mesh for BLT reconstruction. We inserted a titanium wire to tumor center as a marker to validate BLT localization accuracy. We will also quantify the accuracy of BLT-reconstructed volume with MRI and ex vivo method. Volumetric bioluminescent power(BP), related to tumor viability, will also be assessed in comparing with surface BLI and ex vivo method, to determine if it is a proper metric for tumor monitoring and assessment. We will further irradiate orthotopic PDAC guided by BLT and demonstrate the application of BLT to longitudinally assess the change of PDAC volume/power in response to treatment. **Results:** Our initial result shows BLT can retrieve the orthotopic PDAC location within 1.5mm accuracy at depth from 3-5mm, common depth observed in pancreatic study. The BP is observed linearly by 12-fold along with the tumor growth 4 weeks after tumor implanted. With BLT, we can reconstruct the 3D bioluminescent tumor volume using multiple-projection imaging. **Conclusion:** The quantitative BLT system offers unique opportunities for researchers to study important pancreatic tumor in orthotopic setting to quantify its location, growth and response to therapeutic intervention.

The research was supported by National Institutes of Health/National Cancer Institute (R21CA223403, R37CA230341, and R01CA240811), and Cancer Prevention and Research Institute of Texas (RR200042). J. Wong receives royalty payment from a licensing agreement between Xstrahl Ltd. and Johns Hopkins University.

TU-B-TRACK 2-04

Preliminary Investigation of a Novel 3D Dosimetry System Consisting of ClearView 3D Radiochromic Dosimeters with Telecentric Optical Computed Tomography Scanning

S Hoopingarner^{1*}, K Penev², M Oldham¹, (1) Duke University Medical Center, Durham, NC, (2) Modus Medical Devices, London, Ontario, CA

Purpose: To investigate and characterize a novel 3D dosimetry system consisting of ClearView radiochromic gel dosimeters and telecentric optical CT imaging. **Methods:** ClearView radiochromic dosimeters (Modus Medical Devices) are gellan gum based radiochromic dosimeters containing a water-soluble tetrazolium salt which reduces into an insoluble formazan dye (with associated color change) under ionizing radiation. Initial spectrophotometric studies investigated linearity of dose response on small volumes of ClearView in optical cuvettes. Simple single beam benchmark radiation therapy treatments were then delivered to 10 and 15-cm diameter ClearView dosimeters. The treatments were modeled with a commissioned Eclipse treatment planning system. Dosimeters were scanned submerged in a refractive index (RI) matching fluid (ca. 10% propylene glycol) by optical CT both pre- and post-irradiation (within 24h) and 3D reconstructions of the change in linear-optical-attenuation was determined using in-house software. Percent depth-dose, cross plane and in-plane profiles, and 3D gamma analysis were performed and compared to the commissioned Eclipse dose, which served as ground truth. **Results:** Linearity of dose response was confirmed in the cuvette study with excellent agreement ($R^2 \geq 0.9986$) at two wavelengths (520-and 632 nm) at 3 post-irradiation time points: 21 hours, 6 and 10 days. Dosimeter reconstructions were performed at 1mm³ resolution in full 3D. Relative dose profiles in both 15-and 10 cm dosimeters show good agreement in comparison to Eclipse dose calculations, with root mean square errors (RMSE) 0.00107-0.006649, and $R^2 \geq 0.9808$. Relative 3D gamma analysis performed at 3%3mm range from 82.91% - 99.99% agreement. Regions of known artifacts were excluded from analysis (jar walls and neck), but some remain unaccounted for (e.g., ring and cupping artifacts). **Conclusion:** This work presents the first use of a telecentric optical-CT scanner with ClearView. The system shows substantive promise for a new comprehensive 3D dosimetry system.

None. ClearView dosimeters provided by Modus Medical Devices.

TU-B-TRACK 2-05

Evolution and Development of Micro-CT and Optical Imaging for Breast Conserving Surgery

B Maloney^{1*}, S Streeter¹, M Gesner², J Kempner², K Paulsen¹, B Pogue¹, (1) Dartmouth College, Hanover, NH, (2) PerkinElmer

Purpose: A need exists for better intraoperative tools in breast conserving surgery, where there is a high re-excision rate due to malignancy missed using conventional surgical tools. One potential solution is combined micro-CT and optical imaging which have shown promise in clinical studies. The micro-CT rapidly scans the macroscale morphology and interior information of the specimen, while the optical imaging provides surface-weighted scatter contrast from fibrous tissues. The purpose of this study is to highlight the evolution of this technology towards utility as a surgical tool. **Methods:** A multimodal micro-CT and optical system imaged breast conserving surgery specimens at the time of surgery in a prospective clinical trial. In addition, a new multimodal system was developed for clinical translation. The system was built specifically to meet clinical time and cost constraints, as well as to enable ease of system control and image acquisition, reconstruction, and visualization by clinical staff. **Results:** In the prospective clinical trial, image acquisition and reconstruction of full volume micro-CT and surface optical imaging of two margins was completed in under ten minutes without impeding clinical workflow from n=46 patients, to date. The new multimodal system was developed with a focus on speed and ease of clinical use. CT imaging and reconstruction times were significantly reduced in the new system. Software tools were developed such that the system could be operated with minimal training. **Conclusion:** Micro-CT and optical imaging is feasible in the breast conserving surgery clinical workflow and has shown promise for use as a clinical tool to reduce the high positive margin rate in breast conserving surgery. This study represents the next steps toward clinical utility.

TU-B-TRACK 2-06

Use of Computer-Controlled Automatic Injector to Perform 1000-Fps High Speed Angiography During Neurointervention Procedures

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Purpose: It is well accepted that details of blood flow can influence the formation and progression of vessel pathologies such as aneurysms and stenoses. High Speed Angiography (HSA) at 1000 fps can provide detailed flow pattern images that were previously unavailable during neuro-interventional procedures. To derive such flow information, it is important to have the ability to control the rate and volume of contrast injection. In this work we present injection techniques performed by using a computer-controlled syringe injector and the resulting flow pattern. **Methods:** A 3D-printed neuro-vascular model was connected in a flow loop with a 40% glycerol and 60% water solution used as the circulating fluid. An Aries single photon counting detector (Direct Conversion Inc.) with a pixel size of 100 μ m and FOV of 7.5cmx5cm was used to acquire HSA or digital subtraction HSA sequences at 1000 fps. A KD Scientific Legato 110 syringe injector was used to inject 100% iodine-contrast agent in the flow loop. The flow patterns were evaluated for both pulsed injections (rate: 100 ml/min; injection duration: 80ms; time between injections: 120ms; total number of injections: 15; duration of image acquisition: 1.5s) and continuous injections (rate 100 ml/min; total injection duration : 2s; duration of image acquisition: 1.5s). **Results:** Pulsing the injections results in consistent formation of glob-like features when the contrast comes out of the catheter. These structures can be used by tracking algorithms to generate velocity profiles in the vasculature and subsequent wall-shear-stress determination. However, a continuous injection fills up the vessel and does not result in any trackable features that can be used for velocity determination. **Conclusion:** A properly controlled contrast injection technique that creates trackable contrast media discontinuities (globs) is necessary to be able to visualize details in the flow patterns when performing HSA at 1000- fps speeds.

TU-B-TRACK 2-07

Development and Testing of a Quality Assurance Phantom for Microwave Imaging

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Purpose: The goal of this project is to develop a quality assurance phantom that can be used to characterize the image quality and provide quantitative measures of linearity, uniformity, spatial and contrast resolutions of different microwave imaging systems. **Methods:** Different phantom materials were used for testing the microwave response that include water, solid water phantoms, and phantoms with tissue-equivalent materials such as lung, breast, adipose, muscle, liver and bone. The microwave image quality parameters including linearity, uniformity, spatial and contrast resolutions were quantified using the previous phantom materials. The microwave projections were acquired using and an in-house developed microwave imaging system that uses a transmitter with 2.85cm wave length, frequency 10.525GHz, and a receiver with variable amplification 1X-10X (PASCO scientific 10101 Foothills Blvd. Roseville, CA). Microwave images were reconstructed using a back-projection algorithm considering corrections of interference and polarization effects of the microwave signal. **Results:** The microwave response was not linear as a function of density of the different tissue equivalent materials. The minimal microwave response was linear with the water volume in imaging view. The uniformity of the microwave response was within 3% over a uniform region that was made from a large solid water phantom. Spatial resolution was tested by changes in the position (2mm) of the minimal microwave intensity in regions that included small water objects that were shifted across the microwave field of view. The contrast resolution was tested with objects that included variable content of water where the microwave intensity decreased almost linearly with increasing water volume in the fields of view. **Conclusion:** A quality assurance phantom for microwave imaging is an essential tool to quantify the different image quality parameters, provides quality control tool for testing and maintaining the image quality of microwave imaging systems over time and enable the performance of quantitative imaging.

Multi-Disciplinary Scientific Session

TRACK 6

Imaging for Particle Therapy

TU-B-TRACK 6-01

Dose Tracking Using CBCT During Head and Neck Proton PBS Radiotherapy: A Feasibility Study

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Purpose: Compared to photons, protons are more sensitive to organ motion and anatomy change. Verification CTs (VFCT) are often needed to evaluate the anatomy change and dose coverage during the proton treatment course. This study investigates the feasibility of using CycleGAN deep learning algorithm to generate synthetic CT (CycleGAN-sCT) from Proteus®ONE daily CBCT and use it to track the daily dose and trigger the verification CT for adaptive proton therapy. **Methods:** Among 29 head and neck patients, 39 same-day CBCT-VFCT pairs from Proteus®ONE imaging system and Philips Big Bore CT were retrospectively collected. VFCT to CBCT rigid registrations were first performed to align and standardize the size of the images. They were then fed into the deep learning training algorithm as ground truth and input, respectively. sCT generated from deformable image registration (DIR-sCT) was used for comparison. The dose was calculated with the in-house commissioned MCsquare calculation engine. 3D gamma analyses were performed to evaluate the dose accuracy calculated on CycleGAN-sCT and DIR-sCT. **Results:** Compared to CBCT, the preliminary results from two patients showed that both CycleGAN-sCT and DIR-sCT have noticeable improvement. The dose calculation gamma passing rate (TPS dose as reference) is increased from ~80% to ~90% for 3%/3mm, comparable to VFCT. DIR-sCT sometimes failed to deal with detailed

information, while CycleGAN-sCT showed better robustness, though the 3D dose gamma passing rates from both methods are comparable to VFCT. In terms of efficiency, generate 3D CycleGAN-sCT takes a couple of seconds, and DIR-sCT takes a couple of minutes. **Conclusion:** While more cases are still needed to consolidate this study, preliminary results show that the CycleGAN-sCT generated from daily CBCT for proton radiation therapy has comparable image quality as VFCT. CycleGAN-sCT can be used to track the dose and trigger the verification for adaptive proton therapy.

This work is partially supported by a master research grant from Ion Beam Applications.

TU-B-TRACK 6-02

Model-Based Reconstruction for Proton CT Detected with Multiple-Layer Ionization Chamber

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Purpose: Proton Computed Tomography (pCT), reconstructed from the integral depth dose (IDD) detected by a multiple-layer ionization chamber (MLIC), suffers from inferior image resolution and suboptimal accuracy in proton stopping power determination due to the large spot size. Although a previous study showed significant improvement with beamline collimation, it would require modification of beamline and also cause extra ambient neutron doses. Therefore, this study proposes a novel and universal model-based reconstruction for pencil beam scanning (PBS) machines with various spot sizes. **Methods:** The IDD of an uncollimated proton beam detected by MLIC was modeled as a weighted sum of percentage depth doses of elemental infinitesimal beamlets evenly spaced by 1mm. The weights followed the source distribution. Squared L2 norm between calculated and measured IDD were minimized at each gantry angle by correcting the water equivalent path length (WEPL) of all beamlets. Two iterations of correction with an intermediate smoothing were performed as a strategy to escape local minima. The corrected WEPLs were filtered back-projected to the final stopping power ratio (SPR) map. The reconstruction method was verified on a digital cylindrical water-based phantom with tissue-equivalent inserts and an ICRP adult phantom. SPR accuracy, spatial resolution, and imaging dose were evaluated. **Results:** The mean absolute deviation of SPR in ROIs was 0.66%, 0.33%, and 0.48% for the cylindrical phantom, head, and lung region of the adult phantom, respectively. MTF(10%) derived from radial edge spread function of a cortical insert was 5.30cm^{-1} . The mean imaging dose was 1.71cGy and 1.95cGy at the head and lung region of the adult phantom, accordingly. **Conclusion:** A novel model-based reconstruction method has been developed for pCT that can be utilized by all commercial PBS machines without complicated detector tracking individual protons. Excellent SPR accuracy, improved spatial resolution, and low imaging dose were achieved without additional collimation.

TU-B-TRACK 6-03

Neural Network Based Event Classification to Improve Compton Imaging for Proton Beam Range Verification

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Purpose: To determine if Compton camera (CC) images of prompt gamma (PG) emission that occurs during proton radiotherapy (RT) can be improved by using a neural network (NN) to identify 1) "True" and "False" PG events, and 2) the correct interaction order for "True" PG events recorded by the CC. **Methods:** The Monte Carlo plus Detector Effects (MCDE) model was used to produce realistic PG interactions that occur in a CC during clinical proton RT beam delivery. The PG data was used to train a fully connected NN to identify True (due to a single PG) and False (due to multiple PGs) PG double scatter (DS) and triple scatter (TS) events. The NN predicted type and interaction order for the events were compared to the known values (from the MCDE model) to determine the accuracy of the NN predictions. Images were reconstructed using the raw MCDE data and a dataset with the NN identified False DS/TS events removed and using the NN predicted True DS/TS interaction order. **Results:** The fully trained NN was able to predict the PG event type (True/False) and interaction order with > 80% accuracy. The proton beam range could easily be identified on the PG images reconstructed using NN processed data, even for beams delivered at the highest clinical dose rates. **Conclusion:** NN processing of CC data can significantly improve the acquired images of PG emission occurring during the delivery of clinical proton RT beams. These results show that further studies of NNs for this purpose are warranted.

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TU-B-TRACK 6-04

Performance Evaluation of An Optimization Method for Fluence-Modulated Proton CT with Dose and Variance Objectives

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Purpose: To optimize dynamically modulated fluence fields for proton computed tomography (pCT) resulting in desired spatial distributions of imaging dose and image variance, with an application for adaptive particle therapy treatment planning based on frequent low-dose and high-quality imaging. **Methods:** We present an optimization algorithm for fluence-modulated pCT (FMpCT), which divides the image into a region-of-interest (ROI), in which high image quality is required and a non-ROI part for which imaging dose should be reduced. The algorithm makes use of concepts of treatment plan optimization to find modulated imaging fluences that minimize imaging dose outside of the ROI and achieve a target image variance inside the ROI using a joint cost function. Imaging doses to susceptible organs can be further reduced. Using Monte Carlo simulation, FMpCT plans were optimized for three pediatric patients with tumors in the head region for which robust proton treatment plans were used to define the ROI. Imaging doses were scored, and the accuracy of low-dose images was evaluated by re-calculating the ground truth proton plans on simulated FMpCT images. **Results:** Using the optimization, the imaging dose could be reduced by 74% outside of the ROI on average over all patients, bringing it down to only 0.3mGy per tomography. Imaging doses to susceptible structures were further reduced with up to 87% dose saving. The quality of FMpCT scans for treatment planning was maintained with treatment dose passing rates for a 1% criterion on the difference to a uniform fluence pCT scan above 98% and range passing rates for a 1mm criterion above 97%. **Conclusion:** A fluence modulation optimization algorithm for FMpCT was evaluated for its ability to save imaging dose while maintaining sufficient quality for treatment planning using three pediatric patients. Imaging doses of only 0.3mGy could pave the way towards daily adaptive particle therapy.

This work was supported by the German Research Foundation (DFG) project #388731804 and the DFG's Cluster of Excellence Munich-Centre for Advanced Photonics (MAP) and by the Bavaria-California Technology Center (BaCaTeC). Additional funding from the Zusatzfinanzierung 2019 fuer DFG-Sachbeihilfen from the Forschungsdekanat of the Faculty of Medicine of the LMU Munich.

TU-B-TRACK 6-05

Scatter Correction of 4D Cone Beam Computed Tomography for Time-Resolved Proton Dose Calculation: Porcine Lung Phantom Validation

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Purpose: A respiratory-phase-specific CBCT scatter correction algorithm, using a 4D virtual CT (4DvCT) as prior, was developed for time-resolved proton therapy dose calculations for lung tumors. The time-resolved dose calculation accuracy achievable with the resulting 4DCBCTcor, as well as with the 4DvCT, was evaluated using a porcine lung phantom allowing reproducible 4DCBCT and 4DCT scanning. **Methods:** 4DCBCT projections of a periodically breathing ex-vivo porcine lung phantom with 4 injected artificial lesions, as well as 4DCTs with two different breathing patterns (planning and day-of-treatment), were acquired. The average planning CT was rigidly registered onto the day-of-treatment 3DCBCT. Subsequently, CT mid-position images were computed using deformable image registration (DIR). Exploiting the deformation vector fields, obtained from the previous step, a 4DCBCT was reconstructed using MA-ROOSTER, followed by mid-position image generation. Applying DIR on the two mid-position images yielded a mid-position vCT, which was propagated to 4D with the inverted vector fields from the 4DCBCT mid-positioning. All DIRs were performed using a diffeomorphic Morphons algorithm. DRRs of each of the 4DvCT phases, assumed to be scatter-free, were individually exploited to scatter correct the corresponding CBCT projections. The corrected projection set was again reconstructed with the MA-ROOSTER yielding the 4DCBCTcor. The 4DvCT and 4DCBCTcor methods were evaluated by comparing dose calculations for lesion-specific robust proton plans vs. day-of-treatment reference 4DCT for individual phases and accumulated doses. **Results:** Accumulated dose DVH differences of the 4 plans for D(2%), D(98%), D(mean,ITV) and D(mean,lung), were 1.7%, 1.3%, 0.9% and 1.3% for 4DvCT-4DCT and 2.0%, 1.6%, 1.0% and 1.2% for 4DCBCTcor-4DCT. The 2%/2mm gamma pass-rate using a 10% dose threshold was always above 93.2% (4DvCT) and 94.2% (4DCBCTcor). **Conclusion:** The qualitative and quantitative agreement between the different modalities suggests that 4DCBCTcor provides sufficient image quality to perform accurate time-resolved proton dose calculation for the used porcine lung phantom.

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TU-B-TRACK 6-06

Synthetic Dual-Energy CT for MRI-Based Proton Therapy Treatment Planning Using Label-GAN

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Purpose: MRI-only treatment planning is highly desirable in current proton radiation therapy workflow due to its appealing advantages such as bypassing MR-CT co-registration, avoiding x-ray CT exposure dose and reduced medical cost. However, MRI alone cannot provide stopping power ratio (SPR) information for dose calculations. Given that dual energy CT (DECT) can estimate SPR with higher accuracy than conventional single energy CT (SECT), we propose a deep learning-based method in this study to generate synthetic DECT (sDECT) from MRI to calculate SPR. **Methods:** Since the contrast difference between high-energy and low-energy CT is important, and in order to accurately model this difference, we propose a novel label generative adversarial network (label-GAN)-based model which can not only discriminate the realism of sDECT but also differentiate high-energy CT (HECT) and low-energy CT (LECT) from DECT. A cohort of 57 head-and-neck cancer patients with DECT and MRI pairs were used to validate the performance of the proposed framework. The results of sDECT and its derived SPR maps were compared with clinical DECT and the corresponding SPR, respectively. **Results:** The mean absolute error (MAE) for synthetic LECT and HECT were 79.98±18.11 HU and 80.15±16.27 HU, respectively. The corresponding SPR maps generated from sDECT showed a normalized mean absolute error (NMAE) as 5.22%±1.23%. By comparing with the traditional Cycle GANs, our proposed method significantly improves the accuracy of sDECT. **Conclusion:** We developed a deep learning-based method to generate sDECT from MRI for proton radiotherapy. The results indicate that on our dataset, the synthetic DECT image from MRI is close to planning DECT, and thus shows promising potential for generating SPR maps for proton therapy. The image similarity and SPR agreement between sDECT and DECT warrants further study and development of an MRI-only workflow for MRI-based proton radiotherapy.

TU-B-TRACK 6-07

Beam Path Length from Isocenter to Skin On Cone Beam CT Images as An Adaptive Planning Indicator of Head and Neck Patients Undergoing Proton Therapy

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Purpose: This study tested the beam path length (BPL) of daily cone beam CT (CBCT) images as an adaptive planning indicator of head-and-neck (HN) patients treated with proton beam therapy (PBT). **Methods:** In this IRB approved study, we retrospectively analyzed 70 HN patients who were treated with PBT. Out of 70 patients, 30 of them underwent adaptive planning on their quality assurance CT scans (QACTs) due to different reasons, e.g. weight loss (10 patients), anatomical changes (11 patients), tumor shrinkage (8 patients) and loss of target coverage (1 patient). Each beam was ray traced from the isocenter to the skin surface along the central axis using the daily CBCT images. The patterns of change in BPL of CBCT images have been compared to those of the adaptive planning using weekly QACT. **Results:** In our study, we used a threshold of BPL change of 3 mm, beyond which an adaptive plan must be needed. Out of 119 beams for all 30 the adaptively planned patients, 58 (48.7 %) of them had BPL changes more than 3 mm. Similarly, out of 128 beams for the 40 non-adaptive planned patients, only 3 (2.3 %) beams had BPL changes of more than 3 mm. The maximum change in BPL for 29 out of 30 adaptive planned patients were larger than 3 mm, this led to a sensitivity of 96.7 % for BPL change as an indicator of adaptive planning. 37 out of 40 non-adaptive planned patients had BPL changes less than 3 mm, led to a specificity of 92.5 %. **Conclusion:** The beam path length (BPL) change from isocenter to skin surface along the beam central axis on CBCT images can be used as an adaptive plan indicator for proton beam therapy (PBT) head-and-neck patients with 96.7 % sensitivity and 92.5 % specificity.

SAM Therapy Educational Course

TRACK 7

Adaptive SBRT for Liver and Pancreas Patients

TU-B-TRACK 7-00

Adaptive SBRT for Liver and Pancreas Patients

Moderator: Rachael Martin, UT MD Anderson Cancer Center, Houston, TX

TU-B-TRACK 7-01

R. Martin, UT MD Anderson Cancer Center:

Intro: Challenges in Pancreas and Liver Patients Receiving SBRT

TU-B-TRACK 7-02

J. Niedzielski, UT MD Anderson Cancer Center:

Importance of Daily Adaption for the Management of Liver and Pancreas Patients Receiving SBRT

TU-B-TRACK 7-03

K. Mittauer, Miami Cancer Institute, Baptist Health South Florida:

MR Guidance and Online Adaptation of Liver and Pancreas Patients

TU-B-TRACK 7-04

Q&A

Liver and pancreas cancers are intrinsically radioresistant, and thus require large radiation doses for tumor control. However, several radiosensitive gastrointestinal (GI) organs at risk (OARs) are located near tumors in these locations, including the duodenum, stomach, and bowel. This complicates hypofractionated and stereotactic ablative radiation treatment (SBRT) of the liver and pancreas, because the tolerance doses for these OARs must be maintained to avoid life-threatening toxicities. Maintaining GI tolerances can greatly reduce the amount of ablative dose to the tumor, and hence may limit the SBRT effectiveness in terms of local control and overall survival. Moreover, respiratory motion and daily anatomic variations from intra- and inter-fractional motion impose additional therapeutic challenges. This is especially true for pancreatic cancer, as tumors often lie next to the duodenum, requiring sharp dosimetric gradients between the target and this OAR. Obtaining sharp dosimetric gradients is also complicated by daily anatomic variations from transient GI gas filling and peristalsis; these variations can affect daily dose distributions and warrant dynamic changes in the dosimetric gradient position to maintain GI OAR tolerances. Thus, consideration of these variations in the treatment planning and delivery workflow is essential. Image guidance using cone-beam CT may not provide adequate soft-tissue visualization of OARs adjacent to the tumor in the abdomen. Even advanced image guidance methods involving CT-on-rails or MR-guided radiotherapy (MRgRT) may not provide optimal treatments if an online adaptation approach is not used. Therefore, selecting patients for SBRT based on optimal tumor position with respect to OARs is essential to ensure safe delivery of SBRT and avoid exceeding GI tolerances, but this limits the population that can be treated with this ablative approach. Implementing a daily online adaptive workflow may allow additional pancreatic and liver patients to receive SBRT and provide the optimal therapeutic intervention. In this session, we will highlight the challenges with using SBRT for liver and pancreas cancers and present potential alternatives using online adaptation with both CT-on-rails and MRgRT approaches that may allow safe dose escalation beyond what can be achieved with current methods.

Learning Objectives:

1. Highlight the importance of daily volumetric imaging with soft-tissue contrast in the treatment of liver and pancreas tumors.
2. Highlight the effect of sharp dose gradients and daily variations in dose distributions to OARs
3. Present strategies using online adaptation to minimize the compounded dosimetric effects from high-dose gradients and daily variations in gastrointestinal anatomy.

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Vendor Showcase

TRACK 8

Sun Nuclear

TU-SHOWCASE-TRACK 8-10-1

What's New from Sun Nuclear

J Holzmann^{1*}, (1) Sun Nuclear Corp., Waunakee, WI

Imaging Interactive ePoster Discussion

TRACK 1

Imaging: Phantoms and Simulations in Radiography and CT/CBCT

TU-IePD-TRACK 1-01

A Quantitative Evaluation of Computational Paediatric Phantoms for Radiotherapy Applications

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Purpose: Computational phantoms have important applications in paediatric radiotherapy, in quality assurance of diagnostic/therapeutic protocols, and in reconstructing historical radiation doses. Detailed age-specific computational anatomical models are available, developed from average and/or healthy individuals, which may not be representative of cancer patients due to pathology and/or treatment effects. This study investigated the capability of existing phantoms in representing the paediatric radiotherapy population. **Methods:** Computational models evaluated were the International Commission on Radiological Protection (ICRP) paediatric reference computational phantoms (n=8, median age 8y, range: 1–15y) and the default 4D extended cardiac torso (XCAT) (n=75, median age 9y, range: 1–18y). Five key organs (kidneys, lungs, spleen, liver and brain) were automatically segmented on the virtual phantoms similar to clinical organ at risk segmentation protocols. Anatomical similarity was assessed in terms of organ length and mass. These quantities were measured on the phantoms and on a clinical radiotherapy dataset, consisting of planning CT images/contours from craniospinal irradiation patients (n=68, median age 7, range: 2–16y). We also compared clinical measures with published literature on healthy children (9 publications, median age 8y, range 1–16y). **Results:** For each dataset (phantom, published and clinical data) we performed a linear fit of the mass and length across the ages. Differences between clinical data and virtual phantoms/published data were calculated as the average relative difference between the linear fits for integer ages across the clinical data. For the phantoms, differences across all organs ranged from 1–22% for lengths and 4–35% for masses. For published data these were 5–23% and 7–39%, respectively. The smallest and largest differences were found for the liver and spleen, respectively. **Conclusion:** Quantitative anatomical differences were described between phantoms, literature, and routine radiotherapy data. Our findings will help selecting and tailoring the phantoms most representative of this population.

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TU-IePD-TRACK 1-02

Anatomically Realistic Simulation of Random Human Airway Trees

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Purpose: To develop a tool for generating high resolution, realistic and random human airway trees that abide by physiological and anatomical constraints. This tool is meant to facilitate training and validation of deep learning airway segmentation algorithms for surgical planning and pulmonary disease diagnosis. **Methods:** Airway tree generation begins with selecting terminal airway segments (endpoints) within the lung by randomly sampling a given oxygen demand map. The tree is progressively generated by calculating a connection cost between every existing branch within the tree and an endpoint. A new branch is then created from the endpoint to the lowest cost branch by forming a new bifurcation. The cost function is designed to optimize the physiological principle of minimum work (Murray's law and volume-minimization) under anatomical constraints (main airway branch to each lobe). The algorithm was adjusted to assure no airway intersections. Airways were simulated down to .4mm, equivalent to the average minimum human airway diameter. Realism was evaluated by comparing length to diameter ratios (LDR), homothety ratios (HR), and branching angles with literature data for human subjects. Simulation randomness was evaluated through a dissimilarity metric (1-Dice Coefficient) measured between airway trees. **Results:** For 40 synthetic airway trees, the LDR, HR, and branching angles [mean(std)] were found to be 2.9(1.3), 0.72(0.2) and 76.3(13) degrees, respectively. Corresponding literature values for human subjects were 2.9(2.0), 0.71(0.1) and 76.1(46) degrees. The airway tree randomness as characterized with the dissimilarity metric was 0.73(.01). **Conclusion:** A method for the simulation of realistic airway trees with LDR, HR, and branching angles close to human subjects has been developed. This algorithm can be paired with any existing digital phantoms for realistic background and could allow the training of airway segmentation algorithms with large datasets and known ground truth.

TU-IePD-TRACK 1-03

Development and Characterization of a Dynamic, Anatomically Realistic, Abdominal Phantom for Simulating Complex Deformable Motion in Cone-Beam CT

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Purpose: Cone-beam CT (CBCT) provides 3D guidance for interventional radiology and radiotherapy, but long acquisition time results in patient motion. Motion compensation methods require realistic models to replicate the complex interplay between image features and motion in CBCT. We report a phantom that presents realistic anatomical deformable motion via simple linear actuation. **Methods:** The phantom incorporates numerous elements within a plastic cylindrical container (23 x 24 cm): ultrasound gel interstitium; synthetic liver (Syndaver Inc.); kidneys (SuperFlab, Bebig); aorta (3 cm PMMA cylinder); spine (4 cm Teflon tube and Delrin insert); and fat fascia (vegetable shortening - Crisco, P&G). 11 Teflon spheres (3 - 9.5 mm) and two cylindrical vessels were inserted into the liver to simulate contrast-enhanced lesions and vasculature. Key to deformable motion are two rubber apertures – one at the diaphragm connected to a linear actuator for SUP-INF motion and another to allow AP motion. CBCT images were acquired on a C-arm (body protocol, Cios Spin 3D, Siemens) with 4 - 16 mm amplitude of the linear actuator. CBCT images were registered to the static reference using Elastix (mutual information, 14³ knots) to assess the imparted deformation. **Results:** X-ray attenuation of the simulated structures was accurate: 30 HU liver, 60 HU kidneys, 660 HU cortical (300 HU inner) bone, and -160 HU fat. Linear actuation yielded a rich variety of deformable motion of structures: ~30% AP in mid-anterior regions, while posterior regions remained nearly static. The magnitude and complexity of the motion field was controllable via the actuation amplitude – e.g., 0.1 - 3.8 mm and 0.3 - 12.5 mm, for 4 mm and 16 mm actuation. **Conclusion:** A realistic abdominal phantom for deformable motion was designed and tested. Linear motion applied to the diaphragm generated a rich deformable motion field, enabling testing of CBCT motion compensation techniques.

Academic-industry collaboration with Siemens Healthineers.

TU-IePD-TRACK 1-04

Smoothed Delineation Uncertainties Contouring (SDUC) Model

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Purpose: Contouring is one of the largest sources of variability in radiotherapy. The purpose of this work is to develop an algorithm to inject known levels of error to clinically accepted contours, giving realistic contours with varying levels of variability. This is an important first step to investigating the dosimetric impact of contouring variability, and to developing approaches to automatically verify contouring quality. **Methods:** We have developed a smoothed-delineation-uncertainties-contouring (SDUC) model to automatically generate alternative, realistic contours. The SDUC model transforms contours (deformation, contraction and/or expansion) as a function of image contrast, based on the expectation that contouring variability is largest when image contrast is low (i.e. when structure edges are less clear). The magnitude of the delineation uncertainty is scaled by a voxel-specific displacement vector in 3 dimensions by applying a random base uncertainty, scaled by the local contrast. Finally, 3D smoothing is applied to ensure the final contours appear realistic. **Results:** For our evaluation, we use the algorithm for auto-contouring of 3 organs (the prostate, bladder, and rectum) on different variation parameters C and on 4 different patients. From the evaluation, we find that there is a decreasing trend in Dice Score Coefficient (DSC) as the variation rate C increases. Hence, for variability rate C of 5, 50, 100, 200, 300, the DSC are 0.917±0.04, 0.899±0.05, 0.861±0.06, 0.859±0.06, and 0.832±0.07, respectively. **Conclusion:** We have developed an algorithm that can simulate the effect of variability in contouring, generating variable but realistic contours.

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TU-IePD-TRACK 1-05

Some Observations on the Basis Material Paradigm

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Purpose: To describe a computational approach to the basis material paradigm and describe some of its applications. **Methods:** The basis material paradigm asserts that the linear attenuation coefficient of an arbitrary material can be accurately represented by a linear combination of the linear attenuation coefficients of two dissimilar materials. The combined thicknesses of the basis materials do not, in general, match the thickness of the material being characterized. Adding a third material resolves that issue. Two equations in three coefficients, plus an equation for the volume constraint, complete the theoretical system. Attenuation coefficient data were generated by the program XCOM from NIST. Data for elements from 1 to 83 are tabulated at energies from 1.0 to 150.0 keV in steps of 0.5 keV. Total and partial (coherent scatter, incoherent (Compton) scatter, and photoelectric effect) coefficients are provided. Also included in the package is a library of materials that have been suggested as phantom components, as well as data on tissue composition from ICRU Report 44. **Results:** Solutions generally show good agreement in total and partial coefficient ratios (better than 1%) for total and Compton coefficients. Deviations for coherent scattering and photoelectric effect can be several percent. Pairs of solutions for some tissues have been found where photoelectric deviations are in opposite directions. Combinations of these can produce agreement of a few tenths of a percent from 20 to 150 keV. Electron densities calculated using the two-component and the three-component versions of the basis material paradigm match the values for the tissue in question to within a few hundredths of a percent. **Conclusion:** The method described produces accurate results and is

easy to use. Often the only input needed consists of the names of the material being studied and the suggested simulation components.

TU-IePD-TRACK 1-06

Task-Based Assessment of Pediatric Fluoroscopy by Use of Vesico-Ureteral Reflux Grading

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Purpose: To propose a methodology for task-based evaluation of pediatric fluoroscopy systems analogous to the clinical diagnostic task of grading the severity of vesicoureteral reflux (VUR) in neonates. **Methods:** Four depictions of VUR corresponding to grades 2-5 were created based on published references and embedded into selected regions from 20 neonate radiographs. Synthetic patches were simulated to match the noise-power spectrum of the selected regions, with mean and standard deviation matched to each radiograph region. A reader study program was written in MATLAB to display a simulated background image with an embedded task and a random displacement and rotation of the task of up to 15 pixels and 20-degrees, respectively. The reader was then prompted to assign the task a grade, from 2-5. Three observers performed a total of 15 reader studies using the simulated patches and five different contrast levels, determined as number of printing passes. Accuracy of grade identification for each contrast level and sensitivity of each class were calculated, with standard errors obtained using a bootstrapping technique. **Results:** Accuracy increased as a function of contrast level with values of 0.45 ± 0.054 , 0.68 ± 0.040 , 0.70 ± 0.020 , 0.76 ± 0.035 , and 0.95 ± 0.0094 for 10, 12, 13, 15, and 17 passes, respectively. In this study guessing would yield an accuracy of 0.25. Grades four and five were more easily confounded yielding sensitivity values of 0.68 ± 0.040 and 0.66 ± 0.044 , respectively. Grades two and three were easier to identify, and yielded sensitivity values of 0.79 ± 0.040 and 0.78 ± 0.037 . **Conclusion:** The proposed methodology was used to simulate a diagnostic task and to establish the feasibility of using VUR grading as a surrogate task for quantitative, task-based assessment of the image quality of pediatric fluoroscopy. Future work will refine the reader study to improve reproducibility, and will extend this methodology to image quality assessment of real fluoroscopy systems.

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TU-IePD-TRACK 1-07

Understanding Variability in Quantitative Imaging and Radiomics: The Use of Different Phantoms in CT to Explore the Effects of Acquisition and Reconstruction Parameters

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Purpose: Quantitative Imaging and Radiomic Features (QIFs) have been shown to be affected by the CT Imaging acquisition and reconstruction parameters used. The purpose of this study was to use a series of phantoms of increasing complexity to understand the effects of acquisition and reconstruction parameters on QIF variability. **Methods:** Image data was used from three different phantoms: (a) publicly available image data from the Credence Cartridge Radiomics (CCR) phantom; (b) publicly available image data from FDA anthropomorphic phantom with simulated lesions and (c) a water phantom. Each phantom was scanned and reconstructed at a variety of acquisition (e.g. different mAs) and reconstruction (slice thickness and kernel) conditions. For each phantom and each condition, a volume of interest (VOI) was created. For the anthro phantom, the VOI was the segmented boundary of the phantom lesion. For each VOI, three QIFs were computed – the mean HU, the standard deviation and the Gray Level Co-occurrence Matrix measure Entropy (GLCM-Entropy). QIFs were compared for pairs of phantoms (water vs. CCR; water vs. Anthro) across conditions. **Results:** The mean HU for the water and CCR phantom was very constant across conditions; however the mean HU of the anthro phantom lesion varied with slice thickness due to volume averaging effects. The standard deviation values varied substantially in almost all phantoms and conditions. Interestingly, the standard deviation of the CCR phantom was constant across acquisition (mAs) values. This was due to the heterogeneity of the material contributing more to the standard deviation than the acquisition parameters. The GLCM-Entropy values did not display consistent patterns. **Conclusion:** The use of these phantoms allowed us to explore sources of QIF variability due to acquisition and reconstruction parameters. They also allowed us to demonstrate the effects that volume averaging and material heterogeneity have on QIFs as well.

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Imaging Interactive ePoster Discussion

TRACK 2

Imaging: Rad/Fluoro Dosimetry and Safety

TU-IePD-TRACK 2-01

Distribution of Peak Skin Doses from Fluoroscopic Procedures Exceeding 5 Gy Reference Air Kerma

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Purpose: The reference point air kerma ($K_{a,r}$) is a readily-available dose quantity provided by most modern fluoroscopes that can provide guidance for patient dose management actions following fluoroscopically-guided procedures. This study investigates the patient peak skin dose (PSD) distribution for procedures exceeding the 5 Gy $K_{a,r}$ which is the recommended substantial radiation dose level (SRDL) by NCRP 168 for potential patient follow-up. **Methods:** Using DICOM radiation dose structured report data, PSD was calculated for 125 fluoroscopically-guided interventions (FGIs) exceeding a single procedure $K_{a,r}$ of 5 Gy using an internally-validated PSD calculation method. Additionally, for each procedure, the ratio of PSD to $K_{a,r}$ was calculated and the results pooled into one of three clinical categories: cardiac (N=67), body-interventional (N=24), and neuro-interventional (N=34). The $K_{a,r}$ distribution and PSD distribution were analyzed for all procedures in aggregate while the PSD to $K_{a,r}$ ratios were analyzed by the clinical procedure type. **Results:** The $K_{a,r}$ distribution resulted in a mean value of 6.7 Gy with a range of 5.1–18.6 Gy. The PSD distribution resulted in a mean value of 5.1 Gy with a range of 1.6–11.0 Gy. For the PSD to $K_{a,r}$ ratio differentiated by clinical categories, cardiac FGIs resulted in a mean ratio of 0.84 with a range of 0.31–1.72, body-interventional FGIs resulted in a mean ratio of 0.74 with a range of 0.43–1.41, and neuro-interventional FGIs resulted in a mean ratio of 0.53 with a range of 0.37–0.88. **Conclusion:** PSD to $K_{a,r}$ ratios yielded substantially different results with respect to the clinical categories which illustrates the possible need of a $K_{a,r}$ SRDL specific to the clinical service. Additionally, using a static $K_{a,r}$ SRDL for any clinical service could over or under predict the need for patient follow-up indicating it may be better to establish SRDLs based on PSD.

TU-IePD-TRACK 2-02

Enhancements of the Scattered Radiation Display System Training Module

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Purpose: The real-time scattered-radiation display system (SDS) has been expanded in functionality to facilitate interactive resident and staff-member training in a virtual-reality world. In this work we report on improvements to the training system, which enhance the information provided to the trainee. **Methods:** The SDS has

been previously expanded to include a virtual reality (VR) training module (TM). Unity3D was utilized for designing a virtual interventional room consisting of a C-Arm, patient, and patient table. When the software is initiated, a main menu is opened to select either the dose-management module (DM) or the TM. The TM has two options. In the first, the trainee can view the scatter distributions for previous fluoroscopically-guided-interventional (FGI) procedures using machine-parameter log files stored by the DM. In addition, we utilize a time-of-flight depth-sensing camera along with a custom-built convolutional neural network and NiTE 2 for identifying and tracking staff-members' locations during procedures which are also recorded in log files for post-procedure evaluation. The second TM mode allows the trainee to control parameters of an interactive virtual procedure and visualize the room-scatter-distribution display. **Results:** Within the TM, three distinct enhancements have been made: 1) the ability to select staff-locational log files for playback allowing trainees to observe doses for each of the individuals involved with a procedure; previously, only machine-parameter log files were available; 2) a dose probe to numerically analyze the scatter distribution; 3) fluoroscopic image display, which gives the trainee a frame of reference to the clinical procedure and the ability to correlate the scatter distributions with clinical technique and conduct of the procedure. **Conclusion:** We present enhancements to the SDS TM, which will further improve the training experience for staff and residents. A software system capable of VR training and real-time intraprocedural feedback will facilitate improved staff awareness and radiation-dose management during fluoroscopic-interventional procedures.

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TU-IePD-TRACK 2-03

Fluoroscopic Guided Interventional Procedures, Patient and Related Staff Radiation Doses Monitoring in Mississippi

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Purpose: Interventional Radiology (IR) procedures using fluoroscopic guiding are widely implemented in healthcare systems for the diagnostic and/or treatment of diseases. Interventionists are from different specialists including surgeons, cardiologists, and radiologists. Based on complications, patient weight and habits, interventionist experience, and exposure parameters setting, the patient and staff dose will vary. In this study, the details of patients and staff dose were monitored, evaluated and the area of dose optimization was proposed in the largest healthcare system in Mississippi. **Methods:** Patient and staff doses in past 3 years, for different procedures and 15376 patients/procedures were monitored and evaluated in four distinct IR locations: Vascular Surgeons in the operating room (VSOR), Cardiologists in adult and pediatric Cath-labs, Neuro-surgeons, and IR radiologists in old and new IR divisions. Fluoro and acquisition (high-quality image) times, Dose Area Products (DAP), Cumulative KERMA at reference points (RP), and the number of events were evaluated for different procedures and IR physicians. Patient effective dose and Staff dose in vascular surgery OR were evaluated. **Results:** The highest and the lowest values were for VSOR and Pediatric Cath-lab, respectively. The average patient effective dose was 41.36 mSv, while the average monthly Staff deep, Lenses and shallow dose equivalents (DDE, LDE, and SDE) were 1.44, 4.84, and 4.81 mSv, respectively in VSOR. Recognizable patterns were seen for different staff dose metrics and patient doses delivered by known physicians. **Conclusion:** The in-house patient dose monitoring system is in place, the threshold dose has decreased from 8 Gy to 5 Gy in institutional radiation safety policy, moreover, the awareness of Interventionists was increased and the trend of the patient dose is decreasing which depicted in supporting figures. Setting the default protocols to low levels exposure parameters for both Fluoro and Acquisition were implemented as well as some practical actions in dose reduction.

TU-IePD-TRACK 2-04

Fluoroscopic Imaging Data Analysis: A Practical Study to Guide Protocol Optimization and Operator Training

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Purpose: Fluoroscopy, 2D acquisition (2D-ACQ), and 3D acquisition (3D-ACQ) are fluoroscope imaging modes with substantially different radiation dose rates and image quality that may be used during fluoroscopically-guided interventions (FGIs). The purpose of this study was to analyze clinical procedure data to better understand imaging mode use frequency and its contribution to the reference point air kerma ($K_{a,r}$) to potentially guide protocol optimization and focused educational opportunities. **Methods:** 48,454 DICOM radiation structured dose reports originating from 59 interventional fluoroscopes in calendar year 2020 were analyzed in Excel. The data were parsed into four clinical groups performing FGIs, cardiac catheterization, cardiac electrophysiology, body-interventional, and neuro-interventional. The average percentage of irradiation events and average percentage of $K_{a,r}$ per study from each imaging mode were calculated for each clinical service. **Results:** Cardiac catheterization procedures resulted in fluoroscopy [85% of irradiation events, 65.8% of $K_{a,r}$], 2D-ACQ [15% of irradiation events, 34.2% of $K_{a,r}$], and 3D-ACQ [0% of irradiation events, 0% of $K_{a,r}$] from 19,439 procedures. Cardiac electrophysiology procedures resulted in fluoroscopy [97.6% of irradiation events, 85% of $K_{a,r}$], 2D-ACQ [2.49% of irradiation events, 15% of $K_{a,r}$], and 3D-ACQ [0% of irradiation events, 0% of $K_{a,r}$] from 6,125 procedures. Body-interventional procedures resulted in fluoroscopy [91.6% of irradiation events, 46.4% of $K_{a,r}$], 2D-ACQ [8.1% of irradiation events, 44.5% of $K_{a,r}$], and 3D-ACQ [0.3% of irradiation events, 9.1% of $K_{a,r}$] from 20,198 procedures. Neuro-interventional procedures resulted in fluoroscopy [83.3% of irradiation events, 48.4% of $K_{a,r}$], 2D-ACQ [16.3% of irradiation events, 48.6% of $K_{a,r}$], and 3D-ACQ [0.4% of irradiation events, 3% of $K_{a,r}$] from 2,692 procedures. **Conclusion:** Institutions that perform FGI procedures may benefit from similarly analyzing their procedure data to better understand how the various imaging modes are used by different clinical services and use that data to guide protocol optimization and/or educational opportunities.

TU-IePD-TRACK 2-05

Convenient Calculation of Radiation Workloads for Cardiac Angiography Using Dose Area Product

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Purpose: The dose area product (DAP) may be a more reliable and convenient metric than mA-min for radiation shielding design. DAP is more readily available and better correlated with x-ray scatter than mA-min. We investigated the use of DAP as an alternative metric of radiation workload for modern cardiac angiography suites. **Methods:** Irradiation event data was collected from radiation dose structured reports (RDSRs) for 11 cardiac angiography suites across 4 hospitals for 3 weeks in 2019. Workload distributions as a function of kVp were calculated for DAP and mA-min. DAP was consistently provided in the RDSR but mA-min required calculation of the product of tube current (mA) and exposure time using DICOM fields in the RDSR that were populated inconsistently among vendors. To enable comparison of DAP workloads with mA-min workloads (from NCRP Report 147), a DAP-to-mA-min conversion factor was estimated. **Results:** The average DAP/patient workload was 490 dGy-cm²/patient. The DAP-to-mA-min conversion factor was 0.31. Of the 10 suites with data from ≥20 patients, this conversion factor predicted the total mA-min with a mean absolute error of 21% (error range: -45% to 58%). The average workload was slightly lower than that of NCRP-147 (150 mA-min/patient vs. 160 mA-min/patient, respectively) and had a higher average tube potential (93 kVp vs. 88 kVp, respectively). Notably, the workload at high kVp (115 – 120 kVp) was approximately five times greater than in NCRP-147. **Conclusion:** The radiation workload for modern cardiac angiography suites was readily calculated using DAP. DAP and mA-min workload distributions were compared to estimate a DAP-to-mA-min conversion factor. This conversion factor may help facilitate future comparisons between DAP workloads and mA-min workloads (from NCRP-147). Future work will investigate secondary air kerma rates for this DAP distribution which may be useful for shielding calculations.

TU-IePD-TRACK 2-06

Evaluating the Kerma Correction Factor of Modern Fluoroscopy Systems

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Purpose: A model for the kerma correction factor (KCF), used for improving peak skin dose (PSD) calculation of fluoroscopy guided procedures, was presented by DeLorenzo and Goode (DG-KCF). We propose to validate the author's KCF model, and extend its application to other fluoroscopy systems. **Methods:** Under fluoroscopic service mode, table attenuated/scatter included air-kerma rates and free-in-air kerma rates were measured, using a 0.6cc thimble ionization chamber (IC) (10X6-0.6, Radcal®). The measurements were acquired with a Siemens Artis Zee under different tube voltages [60–120kVp], pre-patient copper filtration [0–0.9mm] and X-ray field sizes [9×9, 12×12, 14.5×14.5cm²], defined at an clinical patient skin location. The KCF is computed from the ratio of the two types of kerma rates. Other analytical nonlinear fits of the KCF are then generated and compared against DG-KCF. **Results:** Preliminary KCF values range between 0.79 and 1.26 for all parameters. For different kVp, the KCF directly correlates with the maximum energy output of the X-ray tube for most circumstances, except for high filtration. The change in the KCF with tube voltage was the highest in the absence of copper filtration and 14.5×14.5cm² field size and lowest in the presence of 0.9mm copper filtration and 9×9cm² field size. The increase in copper filtration increased the KCF by 1.13, 1.25 and 1.36 for the respective field sizes of 9×9cm², 12×12cm², 14.5×14.5cm². The fitted nonlinear model demonstrates a strong correlation to the recorded KCF with a square residual norm of 0.051 and maximum percent difference of 7%. **Conclusion:** Validation of the DG-KCF model was performed. Preliminary results suggest good agreement to measured data. Further validation of the model will be extended to other fluoroscopy systems and parameters. Use of the KCF model with the radiation dose structured report data will improve the accuracy of PSD calculation.

TU-lePD-TRACK 2-07

A Low Hanging Fruit for Children: Anti-Scatter Grid Removal for Significant Radiation Dose Reduction in Pediatric Cerebral Angiography

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Purpose: To achieve significant dose reduction by the removal of the anti-scatter grid during cerebral angiography, using a bi-plane interventional fluoroscopy system in a tertiary pediatric hospital. **Methods:** Cerebral angiography is a fluoroscopic X-ray examination that requires significant spatial resolution in order to see small blood vessels in the cerebral vasculature. Typical cerebral angiogram is performed using bi-plane interventional fluoroscopy to see the blood vessels of interest from two angles. Modern interventional fluoroscopy systems are equipped with removable anti-scatter grids that improve image quality at the expense of higher radiation to the patient. However, we hypothesized that the use of anti-scatter grid would not drastically improve the diagnostic quality of the imaging in pediatric cerebral angiography. The anti-scatter grid was removed from the lateral plane of a bi-plane interventional fluoroscopy used for cerebral angiograms, which were thereafter performed by an experienced interventional neuroradiologist who had previously used to use the anti-scatter grid for the same procedures. Dose-Area-Product (DAP) and Reference Point Air Kerma (K) from both fluoroscopic and acquisition procedures along with fluoroscopic time of the lateral fluoroscopic examinations were compared for 54 cerebral angiograms performed with the anti-scatter grid in place and 48 procedures without the grid. **Results:** The interventional neuroradiologist observed no noticeable changes in diagnostic image quality and in fluoroscopic time between the procedures with and without the anti-scatter grid among all 102 procedures. The median reduction of DAP was 47% and 33%, and the median reduction of K was 40% and 29% for patients younger and older than 10 years old, respectively. **Conclusion:** The removal of the anti-scatter grid from the lateral plane during cerebral angiography is feasible, and results in significant reduction of radiation exposure to pediatric patients and clinical staff without sacrificing the overall diagnostic performance of this complex procedure.

Multi-Disciplinary Interactive ePoster Discussion

TRACK 3

Multi-Disciplinary: Innovative Technologies

TU-lePD-TRACK 3-01

Deep Learning-Based Real-Time Volumetric Imaging Using Single CBCT Projections

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Purpose: Due to the inter- and intra-fraction variations of respiratory motion, real-time volumetric imaging is highly desirable during lung SBRT delivery. Real-time volumetric imaging could improve active motion management and better account for motion interplay effects in the reconstruction of delivered dose. We recently developed a novel deep learning-based method to derive 3D CT images from a single 2D projection image and validated its efficacy via CT simulation studies. The purpose of this study is to investigate clinical feasibility using actual patient CBCT data and associated 2D projection images from lung SBRT treatments. **Methods:** The training and the inference steps for a new case follow a feed-forward path in our proposed TransNet network. TransNet is trained by several supervision mechanisms that optimize the learnable parameters to derive 2D-to-3D transformations. A single 2D projection image is input to generate a synthetic 3D CT. The training dataset is comprised of phase-binned 3D CT images and their corresponding 2D projection images from patient 4D CT studies. A 3D CT of the same respiratory phase as a 2D projection image is used as the ground truth to supervise TransNet. During inference, the trained TransNet extracts features from a patient's 2D CBCT projection image and derives a synthetic 3D CBCT. Our method was validated using clinical images from the treatments of 45 lung SBRT patients. **Results:** The mean absolute error (MAE), peak-signal-to-noise-ratio (PSNR) and normalized cross-correlation (NCC) indexes between the synthetic CBCT and original CBCT images were 134.8 ± 30.8 HU, 18.8 ± 1.3 dB, and 0.93 ± 0.02 , respectively. **Conclusion:** We have evaluated the effectiveness of our recently developed deep learning-based method to derive instantaneous synthetic CBCT images from single 2D projection images using real patient data. This study shows the potential to achieve real-time volumetric imaging during SBRT delivery to ensure treatment accuracy.

TU-lePD-TRACK 3-02

Regional Dose-Responses of Liver Functions Derived from Dynamic Gadoxetic Acid-Enhanced (DGAE) MRI of Patients with Hepatocellular Carcinoma (HCC) Treated with SBRT

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Purpose: To investigate dose-responses of regional liver functions from Gadoxetic Acid uptake rates (k1) mapped by DGAE-MRI and to relate them to baseline global and regional liver functions. **Methods:** 25 patients with HCC were enrolled in an IRB-approved adaptive SBRT trial. The patients had DGAE-MRI of liver pre-RT, and one-month after 3 or 5 fx of SBRT on a 3T scanner to quantify k1 maps. The accumulated doses up to the post scan were converted into EQD2 ($\square\square\square=2.5$) and co-registered to pre/post-RT k1 maps. Percentage changes in k1 from pre- to post-RT were analyzed in dose bins of every 5-Gy and 5 pre-RT k1 bins. Dose-response patterns of the whole group and individuals were analyzed by regression analysis. Responses patterns were further clustered into two groups using k-means and tested the group differences with ANOVA. Correlations between percentage changes in k1 and pre-, post-RT or changes in Child-Pugh scores were analyzed using correlations. **Results:** The averaged percentage change in k1 of the whole group showed a linear relationship with 0.44%/Gy with an adjusted R²=0.99 and a significant greater response at the highest pre-RT k1 value (P<0.0001). Two groups were clustered based on the dose-response patterns at low doses, with one group showing a linear dose-response to all doses and another having a non-linear but over-response to low doses and a linear response for the doses > 80 Gy. Responses of the two groups were significantly different (P<0.000002). The response patterns were not significantly correlated with pre, post-RT or changes in Child-Pugh scores. **Conclusion:** Different dose-response subgroups (over-sensitive to low doses vs not) were identified based on changes in GA uptake rates from pre- to post-SBRT in the patients with HCC. This could have an impact on how to spare liver functional regions in personalized SBRT for HCC.

TU-lePD-TRACK 3-03

Automatic Segmentation of Regions of Interest in Radiotherapy of Lower Abdominal Tumors with Lifelong Learning

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Purpose: A main challenge that Convolutional neural networks (CNNs) face in segmentation is catastrophic forgetting. In this study, we used a published lifelong learning method to learn segmentation of regions of interest (ROIs) for lower abdominal tumors continuously without forgetting previous tasks. **Methods:** The cohort included three abdominal tumors, 600 patients of which had rectal cancer, 300 patients had cervical cancer, and 125 patients had prostate cancer. The tasks included segmentation of the three clinical target volume (cervical_CTV, prostate_CTV, and rectal_CTV), bladder, left femoral head, right femoral head, and rectum. A CNN with 103 convolutional layers was used as the basic segmentation network for a single task. The proposed lifelong learning network adopted dilation adapter to learn seven segmentation tasks one by one. Only the newly added dilation adapter (seven layers) was fine-tuning for incoming new task, whereas all the other learned layers were frozen. For comparison, seven single models for each task were trained separately. **Results:** The proposed lifelong learning can achieve better or comparable segmentation accuracy, with dice similarity coefficients of 0.84 ± 0.02 for cervical_CTV, 0.80 ± 0.02 vs. 0.79 ± 0.03 for prostate_CTV, 0.87 ± 0.02 vs. 0.87 ± 0.03 for rectal_CTV, 0.94 ± 0.02 vs. 0.94 ± 0.02 for bladder, 0.89 ± 0.03 vs. 0.89 ± 0.03 for femoral heads, and 0.80 ± 0.03 vs. 0.80 ± 0.05 for rectum. It can yield good performance with less training data. At the same time, it greatly improved the modeling efficiency, which reduced the number of parameters, the size of model, and training time by $\sim 75.5\%$, $\sim 71.5\%$, and $\sim 25.3\%$, respectively. **Conclusion:** The proposed method preserved the knowledge of previous tasks while learning a new one by using dilation adapter. It could yield comparable performance with much less training data, model parameters, and training time.

TU-lePD-TRACK 3-04

Using Artificial Intelligence to Automatically Track the Response of Brain Metastases to Stereotactic Radiosurgery

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Purpose: To automatically track control and progression of brain metastases (BrM) following stereotactic radiosurgery (SRS); and to provide a platform for studying BrM management with SRS and systemic therapies. **Methods:** 299 patients with 527 BrM of diameter 10+ mm were retrospectively identified for this study. They were imaged with T1 MR post-Gd prior to SRS treatment for BrM, then imaged again at least once within 1 year of SRS. The target SRS dose varied between 18 and 30 Gy in 1-5 fractions. Median time interval between SRS and first follow-up was 61 days, with interquartile range of [48, 75] days. A 3D convolutional neural network was developed in-house to segment BrM gross tumor volumes (GTVs) on T1 MR post-Gd images with 100% lesion sensitivity above 10 mm diameter, and average Hausdorff distances of 2 mm. Pre- and post-treatment images were rigidly registered using an automatic multi-step process that considers the whole head, brain, and lateral ventricles. Then, the deep learning model was applied to automatically contour BrM GTVs on all images. Based on changes in longest diameter post-SRS, BrM were classified on follow-up imaging as: size increased ($+20\%$ change or more), size stable, size decreased (-30% change or more), and unappreciable. **Results:** The registration method was robust, with error of approximately 1 mm. Relative size changes of 527 BrM were assessed on first available follow-up imaging after SRS, with 53 lesions increased in size, 83 stable, 193 decreased, and 84 unappreciable. Changes in 208 BrM (126 patients) were also analyzed after 180 days post-SRS, with 26 lesions increased in size, 58 stable, 59 decreased, and 65 unappreciable. **Conclusion:** This translational study demonstrates a proof-of-concept for automatic longitudinal tracking of BrM. Future work can build on this to study BrM dose response and interactions between systemic therapies and SRS.

TU-lePD-TRACK 3-05

Novel-View X-Ray Projection Synthesis Through Geometry-Integrated Deep Learning

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Purpose: X-ray imaging is a widely used approach to view the internal structure of a subject for clinical diagnosis, image-guided interventions and decision-making. Here, we for the first time investigate the novel-view projection synthesis: obtain the X-ray projection image at a novel view angle from a given projection image at a specific view angle to alleviate the need for actual projection measurement. **Methods:** We propose a deep learning-based geometry-integrated projection synthesis model to generate novel-view X-ray projections through feature disentanglement and geometry transformation. The proposed deep learning model extracts geometry and texture features from a source-view projection, and conducts geometry transformation on the geometry features to accommodate the change of view angle. Then, X-ray projection in target view is synthesized from the transformed geometry and shared texture features via an image generator. We validate feasibility of the proposed approach by experimenting on the one-to-one and multi-to-multi X-ray projection synthesis using lung imaging cases across various patients. **Results:** We deploy the trained model on the held-out testing set for novel-view projection synthesis. The generated results are compared with ground truth qualitatively and quantitatively. For AP-to-lateral and lateral-to-AP view synthesis, the average NRMSE / SSIM / PSNR values over all testing data are $0.185 / 0.880 / 22.646$ and $0.173 / 0.911 / 23.908$, respectively. Promising results are also obtained for multi-to-multi view synthesis. By visualizing the generated projections, we observe the proposed model can generate images closely to the targets despite the various anatomic structures of different patients, indicating the potential of the proposed model for X-ray projection synthesis. **Conclusion:** We present a robust model combining the deep learning and geometry transformation. The generated X-ray projections reveal the internal anatomy distribution from new viewpoints, which potentially provides a new paradigm for volumetric imaging with substantially reduced efforts in data acquisition.

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TU-lePD-TRACK 3-06

Liver 4D Blood Flow Model for Dose Calculation to Circulating Blood and Lymphocytes

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Purpose: To develop a novel 4D dynamic liver blood flow model, capable of accurate dose estimation to circulating blood during fractionated radiotherapy. **Methods:** Adult male and female liver phantoms with detailed vascular trees including the hepatic arterial, portal venous and hepatic venous trees were developed. A discrete time Markov Chain approach was applied to determine the spatiotemporal distribution of 10^5 blood particles (BP) in the human body, based on reference cardiac output and blood volume values. For BPs entering the liver, an explicit Monte Carlo simulation was implemented to track the propagation of individual BPs through the vascular trees and time-dependent radiation fields. The model was evaluated for photon (VMAT, IMRT) and proton (passive SOBP and active PBS) treatments and the impact of total delivery time on circulating blood was investigated and quantified using mean dose(μ dose), V0Gy, V0.5Gy. **Results:** The model tracks over 10^4 BPs in the liver simultaneously along 1996 distinct vascular pathways, accumulating dose from time-dependent radiation fields with a 0.1s time resolution. The simulations of the four modalities estimate μ dose(VMAT) (2.44Gy) $>\mu$ dose(IMRT) (2.42Gy) $>\mu$ dose(SOBP) (2.20Gy) $>\mu$ dose(PBS) (2.08Gy) after 15 fractions, related to the differences in mean liver dose. However, V0Gy and V0.5Gy are linked to the total delivery time, where V0Gy is lowest for VMAT (37%, 60s delivery), and highest for PBS (81%, 255s delivery). V0.5Gy shows the opposite trend, highest for VMAT (15%) and lowest for PBS (0.6%), demonstrating the tradeoff between low dose to large fractions of circulating blood and high dose to a small fraction depending on delivery speed. **Conclusion:** The 4D dynamic liver blood flow model enables realistic dose estimation to circulating blood during radiotherapy. The model allows us to study the impact of treatment modalities and delivery time on the dose to circulating blood, which will provide insights into radiation-induced lymphopenia and guide radiotherapy planning.

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TU-lePD-TRACK 3-07

Development of a Deep Learning Model That Can Create Multiple Auto-Plans Prioritizing Different Clinical Goals to Facilitate Decision Making in Complex Pancreatic VMAT with Dose Painting

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Purpose: To develop a deep learning (DL) model that can create multiple auto-plans prioritizing different clinical goals to support decision making in complex pancreatic VMAT with dose painting. **Methods:** A DL model was trained with 109 VMAT plans in RayStation 10B. The pancreatic tumor and regional nodes receive 50.4 Gy, with the involved vessels painted to 58.8 Gy. The model used three different sets of dose prediction and mimicking rules to generate auto-plans prioritizing PTV coverage, OAR dose and the balance between the two. The model was validated and tested on ten and 20 new patients, respectively. Each auto-plan was compared to the clinical plan using 12 clinical goals, with the p value calculated by a two-tailed t-test. The auto-plan most similar to the clinical plan was selected as the favorable plan. **Results:** On average, all three auto-plans offered some universal advantages over the clinical plan: slightly higher CTV coverage, significant reduction in stomach mean dose, kidney V18 and V10, and V50 for GI organs excluding PTV5040. For PTV coverage, uniformity in low-dose PTV and GI maximum dose, the OAR auto-plans were the most similar to the clinical plans, whereas the PTV auto-plans were the least. The PTV, balance and OAR auto-plans were picked 20%, 40% and 40% of the time as the favorable plan, respectively. Collectively, the favorable auto-plans offered similar performance to the clinical plans on seven goals and significant improvement on five. 30% of the favorable auto-plans were directly acceptable. 20% and 40% required minor and moderate processing, respectively. 10% exhibited large difference, but due to significant relaxation of some OAR constraints in the clinical plan. **Conclusion:** For patients with standard OAR constraints, the DL model could always produce an auto-plan very similar to the clinical plan, which was acceptable directly or after minor or moderate processing.

Multi-Disciplinary Interactive ePoster Discussion TRACK 4

Multi-Disciplinary: Research Tools and Databases

TU-lePD-TRACK 4-01

A Clinically Implemented Watchdog Program to Streamline Adaptive LINAC-Based Head and Neck Radiotherapy

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Purpose: We have developed and implemented an adaptive radiotherapy program, enabled by a watchdog system that detects and alerts the clinical team of anatomical changes that may require plan adaptation. In this study we report our initial clinical experience. **Methods:** Nineteen patients with cancer of the oropharynx and hypopharynx were studied. Patients that received definitive radiotherapy with no resection were included. The patients were selected at the time of planning and were scheduled on an in-house software program called AWARE (Automated Watchdog for Adaptive Radiotherapy Environment). The AWARE software automatically identifies weekly CBCTs for the scheduled patients, deforms tracked structures to the new anatomy, obtains volumetric information of these structures and alerts the adaptive team of significant changes via a daily morning email. The changes are then reviewed for accuracy and discussed with the treating physician who decides whether the plan needs to be adapted. For this initial study the primary and nodal GTVs, and the parotids were tracked. **Results:** The AWARE program was successfully deployed and used to monitor 19 patients with weekly CBCT. The primary GTV decreased on average by approximately 10% the first 3 weeks, while remaining largely stable weeks 4-7. The nodal GTV showed the largest average reduction (15%) after one week of treatment and continued to see reduction until the final week. We observed an average parotid volume change of -24% [-46% - +8%]. Of the 19 patients that were tracked, 5 received an adaptive plan. In the majority of adapted cases, adaptive plans resulted in significant reductions (>10%) in dose to critical OARs. **Conclusion:** We successfully developed and deployed watchdog software that automatically monitors anatomical changes on CBCT and alerts the clinical team. The developed software can enable and streamline personalized treatments by finding the optimum time point for adapting treatment.

TU-lePD-TRACK 4-02

A Federated Database Design for Pediatric Outcomes Modelling

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Purpose: To model acute and long-term treatment outcomes, a well-labeled, organized database that pulls from multiple data sources is needed. When focused on a comparatively rare patient population in pediatrics, making use of all data available is essential. We set out to design a database and application capable of handling modern and historical data generated by our department. **Methods:** A DICOM RT PACS system was established using the big data toolbox provided ProKnow DS. Dose accumulation for multi-phase and adapted treatments was established using a treatment planning system (TPS) agnostic, deformable image registration workflow that transferred and accumulated all doses on a single CT simulation image. Renaming rules were applied to structures to bring historical data in line with TG-263 recommendations. The ProKnow API was used to link with SQL queries against the MOSAIQ database that updated and organized patients based on protocol and trial enrollment. The API was then used to link DICOM RT and dosimetry data with a custom, in-house database developed using PostgreSQL. Importing scripts were developed to pull in clinical data from active and legacy institutional research databases. **Results:** Dosimetry data for a total of 918 patients and 958 courses of treatment were aggregated from three different TPSs. Clinical research outcomes were loaded from 3 unique databases. Data were organized at an institutional level and into 10 unique subgroups corresponding to institutional protocols and clinical trial enrollment. A responsive web application was built which includes dashboards to interact with and easily export clinical data collected through trial and protocol databases. **Conclusion:** Well-formatted data and insightful visualizations are essential in the move towards big data style analytics. Our early success has seeded the results of several studies investigating clinical outcomes, dosimetric predictors, and physics-based planning studies.

TU-lePD-TRACK 4-03

Integrated Structure Name Mapping with CNN

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Purpose: Physicians give widely varying customized names to radiotherapy structures present in the prostate and lungs. The standardization of the physician-given names of the Organs at Risk (OAR), Target Organs (PTV), and other Organs inside the area of interest is a significant problem towards radiotherapy quality assurance. Prior work in this area either considered image or textual data but not both. We designed and evaluated an integrated model that considers both types of data compiled from the radiotherapy centers administered by VHA and the Department of Radiation Oncology at VCU. **Methods:** The combined VCU and VHA radiation oncology data has 16,290 prostate and 13,999 lung structures. Within each structure set name, all digits and English alphabets were converted into lower-case. The data was stratified and split into 70:30 training and testing sets. BioBERT was used to tokenize the text and the embedding layer was based on our corpus. Since the textual

data performed better than the feature-reduced image data, the textual features were assigned higher importance. Then two parallel CNNs were designed for the textual dataset and another CNN for the feature-reduced image data (with 1000 features) and all three outputs are concatenated together and fed into two dense layers of the CNN to standardize structure names. **Results:** Macro-averaged precision, recall, and F1-score for prostate were 0.886, 0.94, 0.911, and that for lung were 0.896, 0.878, 0.915, respectively. **Conclusion:** CNN-based integration of image and text data provides higher F1-scores than on individual datasets delivering state-of-the-art performance for structure name standardization. The majority of structures in real clinical datasets are not true OARs or planning targets adding class imbalance challenges leading to lower overall accuracy despite improved F1-scores. Addressing the large number of non-OAR and non-PTV structures is important for future research.

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TU-lePD-TRACK 4-04

NCI Imaging Data Commons

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Purpose: The National Cancer Institute's (NCI) Cancer Research Data Commons (CRDC) aims to establish a cloud-based data science infrastructure. The Imaging Data Commons (IDC) is a component of CRDC to enable access and exploration of imaging data, and to support integrated analyses with non-imaging data available in other components of CRDC, such as genomics and proteomics repositories. IDC builds on the strengths of established projects such as The Cancer Imaging Archive (TCIA) to collect and share FAIR (Findable Accessible Interoperable Reusable) imaging data. **Methods:** IDC uses a combination of commercially available Google Cloud Platform (GCP) and open source components. While the initial focus is to support clinical radiology and radiotherapy data, IDC aims to support brightfield microscopy, multi-channel immunofluorescence and other imaging modalities. Equally important is the ability to support image-derived data, such as annotations of regions of interest. IDC relies on the DICOM standard to harmonize both imaging and image-derived data. IDC data is public and contains no Protected Health Information (PHI). As CDRC grows, imaging datasets will be cross-linked to genomic, proteomic, and clinical data about the subjects. **Results:** The IDC pilot, released in October 2020, focused on radiology data from The Cancer Genome Atlas (TCGA) project. The IDC portal is available at <https://portal.imagingdatacommons.cancer.gov>, and integrates a customized web viewer for visualization of images and image annotations (specifically, DICOM Segmentation and Radiotherapy Structure Set, including multiplanar reformatting). **Conclusion:** The IDC pilot which is available to the cancer research community explores the promise of cloud-hosted public imaging collections co-located with the compute resources and a growing number of tools to support data analysis. Production release of IDC is planned for Fall 2021, and will include all of the public TCIA collections, in particular those shared by radiotherapy studies and clinical trials.

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TU-lePD-TRACK 4-05

PIXNAT: An XNAT Imaging Archive for Collaborative Radiotherapy Research

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Purpose: We introduce our research imaging archive, the "Predictive Informatics XNAT" ("PIXNAT"). The platform is built on the well-known XNAT platform, which we work to enable with the CERR computational platform, enhanced with custom pipelines to assist with conversion between DICOM and analysis-friendly data formats. PIXNAT-housed data is secure and accessible to external researchers for research and processing. **Methods:** The PIXNAT server is a RedHat Enterprise Linux instance running XNAT v1.7.5. Pipeline processes securely run on institutional HPC cluster with multi-GPU for rapid deep network processing. PIXNAT is approved by institutional information security for housing and sharing research data. All imaging data is fully anonymized using CTP and according to 18-point institutional guidelines. PIXNAT is extended with OHIF-XNAT and ROI datatype plugins. CERR computational routines are implemented as PIXNAT pipelines to provide access to a variety of image analysis tools. Deep learning segmentation pipelines from CERR work directly from PIXNAT imaging data and produce RTSTRUCT-based segmentations to be imported and overlaid in the viewer. CERR texture map analysis is also implemented on the system, the resulting maps are converted to DICOM and imported into the session list for review with the integrated OHIF viewer. We have enabled a launch pipeline for JupyterLab notebooks which can be pre-loaded with images from PIXNAT to interactively run Python and CERR routines and visualize in-line results. **Results:** PIXNAT currently houses 24 projects, 2243 subjects, and 3067 imaging sessions. It has been used to collaborate with 5 different institutions. **Conclusion:** We have demonstrated a close integration between CERR and XNAT, resulting in a useful inter-institutional collaboration paradigm. We have furnished PIXNAT as a destination for medical physics research, open source image processing and data sharing.

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TU-lePD-TRACK 4-06

The National Biomedical Imaging Archive

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Purpose: To develop an open-source software application that allows for the storage, search, and downloading of DICOM images to support data sharing, enabling imaging informatics applications and data mining for cancer research. **Methods:** The system utilizes a Java server that provides REST services to independent applications such as the Angular-based graphical user interfaces. The system also includes a Solr server that indexes all tags in the DICOM headers and allows rapid retrieval based on any of the information in the tags. The applications are loosely coupled, enabling independent upgrades, speeding the delivery of features to users. **Results:** The National Biomedical Imaging Archive (NBIA) provides web-based and programmatic access to DICOM images by using role-based security. NBIA consists of a search client, data administration tools, user administration tools, and the NBIA Data Retriever, which supports downloading of images to the user's local computer. NBIA enables several medical imaging archives, including The Cancer Imaging Archive (TCIA) of the National Cancer Institute (NCI), which currently hosts over 27.5 million images and empowers downloads of over 100 TB per month. NBIA incorporates the Clinical Trial Processor software from RSNA and supports customized de-identification of images. NBIA can integrate with other applications through a set of APIs that can facilitate data query and data transfer between multiple repositories, such as between the NCI image repositories, TCIA, and Imaging Data Commons, a cloud-based resource. The NBIA code is available on GitHub. Technical documentation including an API guide can be found on the NCI's NBIA wiki page. **Conclusion:** NBIA is an open-source software application that enables the creation of secure and searchable archives of medical images in DICOM format. NBIA is a constantly evolving system that takes advantage of new technologies and approaches to meet the needs of the cancer imaging community.

TU-lePD-TRACK 4-07

Radiation Therapy Ontology: Report of AAPM Working Group

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Purpose: To build an ontology that represents the domain of radiation oncology in a manner that makes it useful for medical and scientific progress. **Methods:** A working group was established by the AAPM Big Data Subcommittee to organize and lead this effort. The working group has met on a biweekly basis since November 2018. The ontology's design follows the guidelines of the OBO Foundry whose mission is "to develop a family of interoperable ontologies that are both logically well-formed and scientifically accurate". To this end, a foundational ontology (Basic Formal Ontology) was chosen to provide the guiding principles and the basic classes and relations. Additional principles include: (a) to reuse whenever possible entities already described in other OBO ontologies to avoid duplication and confusion; (b) to embrace "realism" in the definition of entities and relations to encourage its use in clinical practice and research, and (c) to coordinate our efforts with other organizations working on standards-related projects. On a practical level, the github platform provides the infrastructure, development is accomplished using the Web Ontology Language (OWL), and the ontology software Protégé is used to develop and view the ontology. **Results:** A domain ontology has been constructed. In line with the goal of coordinating with related efforts, mapping to the ASTRO minimum data elements has been accomplished. The work has also spawned the development of an applied ontology with more specific goals. **Conclusion:** A robust fundamental ontology can be a key element in the development of data and concept-driven research in biomedicine. It provides a link to a number of practical taxonomies. Ontologies have proven to be useful in machine learning efforts in medicine. In the future, the ontology will as a basis for building applied ontologies and coordination with standards-forming organizations.

Therapy Interactive ePoster Discussion

TRACK 5

Therapy: Particle Therapy Treatment Planning

TU-lePD-TRACK 5-01

The First Modeling of the Spot-Scanning Proton Arc (SPArc) Delivery Sequence and Investigating Its Efficiency Improvement in the Clinical Proton Treatment Workflow

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Purpose: To quantitatively model a precise spot-scanning proton arc (SPArc) delivery sequence and assess its efficiency improvement in the routine proton clinical operation. **Methods:** The SPARC delivery sequence model (DSMSPARC) includes two kinds of parameters: (1) mechanical parameters (the maximum gantry velocity, acceleration, and deceleration speed), (2) irradiation parameters (tolerance window and buffer, spot scanning speed, energy layer switching time, and burst switching time). An independent gantry inclinometer was used to measure mechanical parameters. Log files were used to derive the irradiation parameters through a series of SPARC test plans. The in-house DSMSPARC was established by fitting both mechanical and irradiation parameters. Eight SPARC plans from different disease sites (brain, HN, lung, and liver cancer) were used to validate the model's accuracy. To quantitatively assess the treatment efficiency improvement compared to the clinical IMPT, a random clinical operation date of our proton center (total 21 cases on Jan 6th 2021) was selected, and SPARC plans were generated for all the cases. The DSMSPARC was used to simulate the SPARC treatment delivery sequence and compared to the clinical IMPT treatment logfiles. **Results:** The relative difference of treatment time between log files and DSMSPARC's prediction was $6.1\% \pm 3.9\%$ on average, and the gantry angle vs. delivery time showed a good agreement between the DSMSPARC and log file. Additionally, the SPARC plan could effectively save two hours out of 10 hours of clinical operation by simplifying the treatment workflow for a single room proton therapy center. The average treatment delivery time (including gantry rotation and irradiation) per patient was reduced to 226 ± 149 s using SPARC compared to 665 ± 407 s using IMPT ($p < 0.01$). **Conclusion:** This is the first modeling of the SPARC delivery sequence, which paves the roadmap for implementing the delivery speed and time into the SPARC optimization algorithm. Additionally, SPARC can offer a superior delivery efficiency to improve clinical treatment throughput, compared to IMPT.

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TU-lePD-TRACK 5-02

4D Robust Optimization in Small Spot Intensity-Modulated Proton Therapy (IMPT) for Distal Esophageal Carcinoma

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Purpose: To compare the dosimetric performances of small-spot 3D and 4D robustly optimized intensity-modulated proton (IMPT) plans in the presence of uncertainties and interplay effect simultaneously for distal esophageal carcinoma. **Methods:** Thirteen (13) patients were selected and re-planned with small-spot ($\sigma \sim 2-6$ mm) 3D and 4D robust optimization in IMPT respectively. The internal clinical target volumes (CTV_{high}, CTV_{low}) were used in 3D robust optimization. Different CTVs (CTV_{high}4d, CTV_{low}4d) were generated by subtracting an inner margin of the motion amplitudes in 3 cardinal directions from the internal CTVs and used in 4D robust optimization. All patients were prescribed the same dose to CTVs (50 Gy[RBE] for CTV_{high}/CTV_{high}4d and 45 Gy[RBE] for CTV_{low}/CTV_{low}4d). Dose-volume-histogram (DVH) indices were calculated to assess plan quality. Comprehensive plan robustness evaluations that consisted of 300 perturbed scenarios (10 different motion patterns to consider irregular motion and 30 different uncertainties scenarios combined, both were randomly sampled from the corresponding Gaussian distributions) were done to quantify robustness to uncertainties and interplay effect simultaneously. Wilcoxon signed-rank test was used for statistical analysis. **Results:** Compared to 3D robustly optimized plans, 4D robustly optimized plans had statistically improved target coverage and better sparing of lungs and heart (Dmean: $P = 0.001$; V30Gy[RBE]: $P = 0.001$). 4D robustly optimized plans had better heart protection (Dmean: $P = 0.010$; V30Gy[RBE]: $P = 0.017$; V40Gy[RBE]: $P = 0.010$) when uncertainties and interplay effect were considered. **Conclusion:** Even with small spots in IMPT, 4D robust optimization outperformed 3D robust optimization in terms of normal tissues protection and robustness to uncertainties and interplay effect simultaneously. Our findings support the use of 4D robust optimization to treat distal esophageal carcinoma with small spots in IMPT.

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TU-lePD-TRACK 5-03

A Monte Carlo Study on Fetal Dose from Craniospinal Irradiation with Proton Pencil Beam Scanning Treatment

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Purpose: To evaluate fetal dose from proton pencil beam scanning (PBS) craniospinal irradiation (CSI) during pregnancy by conducting Monte Carlo (MC) dose calculations coupled with eight gestational-age dependent pregnant female phantoms (8, 10, 15, 20, 25, 30, 35, and 38 weeks). **Methods:** The pregnant phantoms were converted into DICOM-RT data (CT images and structures) and imported into a treatment planning system (TPS) (Eclipse v15.6) commissioned for an IBA PBS nozzle. From the TPS, a proton PBS CSI plan (prescribed dose: 36 Gy) was created on the phantoms. MC simulation of the proton PBS CSI to estimate fetal dose was conducted using the TOPAS MC code, for which MC beam properties at the nozzle exit (spot size, spot divergence, mean energy, and energy spread) as a function of the nominal energy were matched to the IBA PBS nozzle beam measurement data. We calculated organ/tissue-averaged absorbed doses for 28 fetal organs and tissues. For contextual purposes, the fetal organ/tissue doses from the treatment planning CT scan of the mother's head and abdomen were also estimated by using the National Cancer Institute dosimetry system for CT (NCICT, Version 3) considering a low-dose CT protocol (CTDIvol: 8.97 mGy). **Results:** The majority of the fetal organ/tissue doses from the proton PBS CSI onto the eight pregnant phantoms fell within a range from 3 to 6 mGy. The fetal doses from the proton PBS CSI were generally lower than those from the planning CT scan, which fell within a range from 7 to 13 mGy. **Conclusion:** The organ/tissue doses produced from this comprehensive MC study provide one of the first clinically informative predictions on the magnitude of fetal dose during proton PBS CSI during pregnancy because most proton TPSs cannot estimate out-of-field fetal dose.

TU-lePD-TRACK 5-04

A Novel Component Method to Delineate Surgical Spine Implants for Proton Monte Carlo Dose Calculation

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Purpose: Spine implants has been shown to decrease local control for spinal sarcoma and chordoma patients because of artifact-induced dose calculation inaccuracies. Artificial intelligence (AI)-based methods have shown advantages for metal artifact reduction (MAR) by assuming that prior knowledge of implants is available. We propose a novel component method to accurately delineate implants from CT images to train AI-based models that potentially can increase image accuracy for proton Monte Carlo dose calculation. **Methods:** We propose a novel method to identify implant components (tulip, screw & rod), utilizing various levels of prior knowledge such as CT scanned implant components and implant volumes and screw lengths from medical records. The tulips from two major vendors are manufactured with different materials such as titanium and Chromalloy. With the use of extended Hounsfield units (HU), and a fine resolution reconstruction algorithm, the implant component method was developed and demonstrated for various levels of prior knowledge both in a phantom and in vivo for one patient. **Results:** Implant systems from two major vendors were analyzed. It was found that mischaracterization of 8 typical implants in a patient can cause under coverage of a clinical target volume (CTV) around 20 cm³, consistent with excessive proton range uncertainties of 10 mm reported in the literature. Using our method, tulip, the most important implant component, can be characterized with proper material and volume as well as other components. Intended CTV coverage can be maintained within tolerance. **Conclusion:** Our component method provides accurate implant characterization for proton dosimetry and can potentially enhance the accuracy of AI-based MAR models. The current study primarily focuses on the implant characterization for lumbar spine. Future work will include cervical spine and dental implants for head-and-neck patients where tighter margins are required due to proximity of organs-at-risk.

TU-lePD-TRACK 5-05

Adaptive Spot Placement Method for Pencil Beam Scanning Proton Therapy

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Purpose: Although pencil beam scanning (PBS) proton radiotherapy (RT) can provide flexible proton spot placement for intensity modulated proton therapy (IMPT), currently the spot placement is mostly based on a fixed Cartesian grid (FG), without fully accounting for the geometry of tumor targets. This work will develop an adaptive Cartesian grid (AG) spot placement method, which will be highly adaptive to the geometry of tumor targets. **Methods:** Given the simplicity and universality of the Cartesian grid to place spots for tumor targets that can be arbitrarily shaped, AG is based on the Cartesian grid. However, compared with FG, AG places (1) a relatively fine grid of spots at the boundary of tumor targets to account for the geometry of tumor targets and treatment uncertainties (setup and range uncertainty) for improving dose conformality, and (2) a relatively coarse grid of spots in the interior of tumor targets to reduce the number of spots and increase the spot weights for improving delivery efficiency. **Results:** AG was validated using representative clinical cases in comparison with FG, with clinically used DVH planning objectives, beam angles, and uncertainty level. The same plan objectives, the same optimization algorithm, and the same plan normalization were used for the fair comparison of AG and FG. The preliminary results demonstrate that AG provided better plan quality than FG in both target coverage and OAR sparing, with up to 50-60% reduction in number of spots. **Conclusion:** We have developed an AG spot placement method with adaptive Cartesian grid that is fully compatible with existing treatment planning convention and systems. Compared with the conventional FG on a fixed Cartesian grid, AG can improve both lateral dose falloff (for better plan quality) and spot utility (for better delivery efficiency).

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TU-lePD-TRACK 5-06

Robustness in Biological Dose-Guided IMPT Planning for Rectal Cancer Using a Monte Carlo-Based Optimizer

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Purpose: Intensity modulated proton beam therapy (IMPT) affords exquisite geometric control of dose deposition in targets and enables improved sparing of normal tissues; however, critical, dose-limiting structures may prevent the delivery of curative radiation doses. We previously demonstrated the feasibility of using an in-house Monte Carlo-based inverse planning optimizer to capitalize on a convergence of Bragg peaks to enhance the linear energy transfer (LET) in a given volume. The resulting increase in biological dose (BD) achievable may provide superior anti-tumor effects for treating otherwise highly radio-resistant tumors. This planning study systematically evaluates various parameters in the optimizer in an effort to escalate high-LET within targets while limiting deposition of high-LET within organs-at-risk (OAR). We hypothesize modest LET escalation in rectal plan GTVs may be leveraged to enhance BD while meeting OAR constraints. **Methods:** Standard and LET-enhanced IMPT plans are created for 10 resectable (T2-T3) rectal cancer patients. Patients are planned with Eclipse and Monte Carlo-based optimization to the same prescription dose, including two target volumes (CTV 2500cGy, GTV 2800cGy) and two equally-weighted posterior-oblique fields. Multivariate generalized linear models are used to assess the influence of planning and optimization parameters on LET-enhancement, physical, and biological dose. Relevant metrics include CTV/GTV coverage reproducibility with changes in internal organ motion, rectal gas filling, patient positioning, and tissue density

inhomogeneities. **Results:** Dose-volume histogram (DVH)-analyses reveal LET-guided plans achieved comparable CTV physical dose homogeneity and OAR sparing compared to standard IMPT plans. LET-guided plans also achieved significant LET and BD enhancement in GTVs. However, robustness analyses of plans cast onto verification CTs indicate LET-enhancement in GTV may inversely correlate with plan Robustness for select optimization parameters. **Conclusion:** LET-guided IMPT planning for rectal cancer GTV biological dose escalation is feasible. Although tradeoffs exist between optimization parameters, LET-enhanced rectal treatments may allow for enhanced therapeutic windows.

TU-lePD-TRACK 5-07

Toward Online Adaptive Proton Therapy of Sinonasal Cancer: The Dosimetric Accuracy of Utilizing Deep- Learning Synthetic CT from CBCT for Daily Treatment Dose Verification

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Purpose: Proton beam is sensitive to anatomy variation, especially for sinonasal cancer patients due to daily nasal cavity fillings variation and tumor shrinkage. With online CBCT increasingly available and daily acquired in the proton treatment room, we hypothesized that daily proton dose can be accurately reconstructed from CBCT using a deep-learning model for sinonasal cancer patients. **Methods:** 35 pairs of pre-treatment CT and first-fraction CBCT were selected from patients treated with proton therapy in our institution for training. Deformable-registration(DIR) was performed to eliminate potential anatomy variation between the simulation and first treatment. A Generative-Adversarial-Networks(GAN) model, including one generator with Residual-Unet architecture, and one discriminator using a multi-layer convolutional neural net(CNN), were trained from scratch to predict CT from CBCT, with random 3D B-spline deformation as augmentation. Five additional sinonasal patients with same-day re-plan CT and CBCT were selected for dosimetric evaluation. The re-plan CTs were deformed to the same-day CBCT to eliminate any neck flexion variation and served as the ground-truth(GT-CT). The synthetic CTs(Syn-CT) were compared to the GT-CTs by Mean-Absolute-Error(MAE). The clinical plans(3-5beams) were re-calculated on both Syn-CT and GT-CT using a commercial Monte-Carlo algorithm with 0.5% uncertainty. Clinically relevant DVH parameters and γ -passing rates were utilized to quantify the dosimetric discrepancy. **Results:** The prediction takes less than 10s per patient. The MAE of Syn-CT was 53.58 ± 5.3 HU. Compared to the reference doses on GT-CT, the average 1%/1mm, 2%/2mm, 3%/3mm γ -passing rates of the doses on Syn-CT were 82.6%, 97.5%, and 99.0%. The mean D₉₉ of CTV, D₁ of (brain-stem, cord, optical nerves, chiasm), and D_{mean} of parotids on Syn-CT are 99.9%, 101.7%, 102.5%, 104.5%, 98.9%, and 100.2% of those on GT-CT, respectively. **Conclusion:** The Synthetic CTs predicted by the GAN-model achieved clinically acceptable dosimetric accuracy as evaluated by the same-day re-plan CT. The method could be utilized to monitor daily treatment dose online and trigger plan adaptation for patients with undesired dosimetric variations.

Therapy Interactive ePoster Discussion

TRACK 6

Therapy: Patient Safety and Quality Improvement

TU-lePD-TRACK 6-01

Development of a Multi-Layer Quality Assurance System for Adaptive Radiotherapy

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Purpose: Dose accumulation is a key component in adaptive radiotherapy (ART). The purpose of this study is to develop a multi-layer quality assurance system to verify dose accumulation performed in an ART program. **Methods:** The development involves the following steps: converting HU and dose in daily and planning CTs to mass and energy, generating a pseudo-inverse map ϕ -1 for each displacement vector field ϕ generated by the ART program, and using the map ϕ -1 and a mass-and-energy congruent mapping (EMCM) method to reconstruct dose on the planning CT. To generate ϕ -1, the source voxels of ϕ were divided into three categories: empty, unique and clustered voxels, and the center-of-mass and tetrahedron-based continuous mapping methods were used to generate the inverse displacements for the clustered and empty source voxels, respectively. The HU-mass density tables used in clinic were adopted for the dose-to-energy conversions. An inverse consistency error (ICE) metric, energy conservation and Gamma index were taken as criteria to evaluate the displacements, energy and dose reconstructed by the ART program. Five registration cases were evaluated: three performed from daily MVCTs to planning CTs using PreciseART and two performed between the end-inhale and end-exhale of 4DCTs using MIM software. **Results:** The results showed that the empty, unique, and clustered voxels are 45.6%, 10.2% and 44.2% on average for the three daily image registrations, and 20.7%, 65.9% and 13.4% for the 4DCT registrations. Their average ICEs are 1.05 ± 0.92 mm and 1.26 ± 1.22 mm, respectively. After dose reconstruction, the average energy in PTV changed by 0.4 ± 0.13 % for EMCM and 3.22 ± 0.58 % for PreciseART. Compared with the results of EMCM, the dose and energy reconstructed by PreciseART show the 5%/5 mm gamma passing rates of 96.9 ± 2.1 % and 90.6 ± 3.3 %, respectively. **Conclusion:** A multi-layer quality assurance system has been developed to verify deformable dose accumulations performed in ART.

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TU-lePD-TRACK 6-02

Development of a Quantitative Method to Evaluate Organ Segmentation for Enhanced Error Detection

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Purpose: Accurate segmentation of organs at risk (OARs) for treatment planning is crucial for the delivery of high-quality care and the prevention of adverse events in radiation therapy. A method to rapidly and quantitatively evaluate the geometric characteristics of an organ against the expected geometry is proposed as a technique for detecting segmentation errors. **Methods:** Relationships between organ volume, patient gender, and geometric characteristics including length, width, depth, mean area, and complexity were modeled as a least-squares linear fit for 30 common organs based on metrics gathered from 1,331 patients. Complexity is defined in this context as the ratio of the organ's volume (V_{org}) to that of the bounding-box. For each individual organ, the perpendicular distance from the linear fit with slope M and intercept B was multiplied by a weighting-factor (ω) to generate a sub-score (μ) for each characteristic: $\mu_{\text{area}} = \omega_{\text{area}} * |M_{\text{area}} * V_{\text{org}} - \text{AREA}_{\text{org}} - B_{\text{area}}| * (M_{\text{area}}^2 + 1)^{-2}$. The final score (S_{org}) was calculated as the sum of individual sub-scores and a sub-score based on the organ's volume and the mean organ volume (V_{av}) and standard deviation (σ_v) determined from an independent dataset: $S_{\text{org}} = \mu_{\text{area}} + \mu_{\text{depth}} + \mu_{\text{width}} + \mu_{\text{length}} + \omega_v * |V_{\text{org}} - V_{\text{av}}| * \sigma_v^{-1}$. Individual weighting-factors were iteratively adjusted using a subset of the data to optimally assign higher scores to those organs with contouring errors, with a value of $S_{\text{org}} > 1$ indicating a suspected error. **Results:** Of the 10,846 organs in the dataset not used in algorithm training, 746 erroneous contours were correctly identified, 7 erroneous contours were not identified, 2,071 contours were incorrectly identified as erroneous, and the remaining 8,022 were correctly identified as error-free, giving an overall sensitivity of 99.1% and specificity of 79.5%. **Conclusion:** A method of quantitatively evaluating OAR contours against demographic data has been developed with the goal of identifying contouring errors. As a classifier, the proposed structure score demonstrates excellent sensitivity, and reasonable specificity in detecting errors in OAR delineation.

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GDH reports personal fees from Varian Medical Systems outside the scope of the present work.

MCS reports personal consulting fees and honoraria with Varian Medical Systems outside of the submitted work.

TU-lePD-TRACK 6-03

A Convolutional Neural Network-Based Retrospective Search for Previously Unreported Treatment Errors in a Planar X-Ray Setup Image Database
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Purpose: The radiation oncology clinical workflow involves numerous patient safety checks; however, rare but serious treatment errors do still occur and may go unreported. Here we present a semi-automated process, based on a convolutional neural network (CNN) misalignment detection algorithm, to retrospectively search a setup imaging database for near miss events and treatment errors not previously reported. **Methods:** A CNN-based error detection model was developed using 57,036 digitally reconstructed radiograph (DRR)/ x-ray image pairs obtained from the records of patients treated at our institution from 2014-2017 using a stereoscopic onboard imaging guidance system. The model was trained to detect alignment errors between the x-ray and the DRR using simulated 1 cm translational errors. The model was applied to previously unseen data from 2018 in order to flag potential errors. Flagged events were cross-referenced with record-and-verify systems, incident learning systems, and external image-guidance systems to determine whether they represented treatment errors, near misses, or false positives. **Results:** When the CNN was used to classify DRR/x-ray image pairs from 2018, a total of 1,233 image pairs (out of 27,278 total) were identified as misaligned. 136 image pairs were manually reviewed. Cross-referencing led to the following categorizations: treatment error, near miss, alignment mismatch corrected with repeat imaging, alignment mismatch corrected by imaging with a different system, fiducial marker match with bony anatomy mismatch, image acquisition failure, and algorithmic false positives. **Conclusion:** Manual retrospective audits of image guided radiotherapy databases could lead to detection of previously unreported treatment errors, but are infeasible due to the sheer size of such databases. Previously unreported errors and near misses can provide key insights to improve the safety of the radiotherapy workflow. To our knowledge this report is the first use of an automated misalignment detection algorithm to pre-select potential errors, thus making systematic comprehensive retrospective review feasible.

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TU-lePD-TRACK 6-04

Retrospective Evaluation of Incident Learning Data in the Six Months Prior to and During COVID-19
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Purpose: The COVID-19 pandemic caused many changes in our clinic, including variable patient volume, a shift towards hypofractionation, and changes in on-site staff availability. This work sought to retrospectively evaluate incident learning data before and after our state declared a public health emergency due to COVID-19. **Methods:** Our clinic has been operating an electronic incident reporting system since January 2017. All staff members can voluntarily report patient safety incidents and other clinical issues. The reports submitted during the six months pre- and post-March 13th, 2020, were reviewed and analyzed for trends that could be associated with clinical changes due to COVID-19. The reports (n=475) were categorized with respect to event classification (incident, near miss, etc.), whether a dosimetric change was required, anatomical site, the process step where incidents occurred and were discovered, the role of the reporter, and the type of intervention. Within each category, the number and distribution of reports before and during COVID-19 were compared to identify changes over time. **Results:** The overall number of reports decreased by 19% (262 to 213) in the 6 months following March 13th. The event classifications that had the greatest decrease were “near miss” (-36%) and “operational/process improvement” (-19%). Of all team members, incident reporting by physicists remained the closest to baseline (-7%). While the overall number of reports submitted decreased, the number that involved a plan revision increased by 18%. The proportion of reports addressed with a quality improvement intervention increased during COVID-19 from 45% to 63%. In direct response to the pandemic, a change in the operational model for physician coverage was implemented. **Conclusion:** Patterns of incident reporting changed during COVID-19, with fewer reports being submitted for low-severity issues. This change may be attributable to staff effort being redirected to respond to COVID-19-associated clinical changes.

TU-lePD-TRACK 6-05

Always-On Cherenkov Imaging for Incident Learning in Radiation Therapy

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Purpose: Cherenkov imaging has been shown to be capable of displaying certain treatment irregularities that might otherwise go unnoticed with standard QA processes. In this study we report on a clinical QA/QI study investigating the use of always-on treatment delivery monitoring with Cherenkov imaging in a radiotherapy clinic. **Methods:** Cherenkov imaging systems (BeamSite, DoseOptics LLC, Lebanon NH) were installed in two linac bunkers in our clinic, which provided continuous imaging of Cherenkov and background images of all treatments. For this project, a custom data transfer and central storage infrastructure was employed to efficiently handle the collection and review of the current 3500-fraction dataset. Images were reviewed post-treatment by a team clinical staff consisting of therapists, physicists, and physicians, and any irregularities were interpreted and analyzed using standard clinical tools and procedures. **Results:** Offline review of 200 patient treatments uncovered 3 independent delivery events, all related to patient positioning. One case consisted of unfavorable chin positioning during an AP-PA thoracic spine treatment which was corrected mid-fraction. The remaining two cases involved arm/hand placement which resulted in unintended exposure; one was found to have persisted for 7/10 fractions resulting in an estimated 3 Gy of exit dose to the arm and no negative impact on the treatment area. Independently, during this same period, 12 events were reported in the hospital-based incident reporting system, all deemed to have no detectable harm to the patient. **Conclusion:** Clinical review of Cherenkov images identified non-ideal patient positioning events during treatment which were not identified in the existing QA program. This system has the potential to improve radiation oncology incident learning through the increased visibility of clinical events during patient treatment. Future work will focus on data-driven automatic incident detection with improved sensitivity and development of appropriate action procedures to improve efficacy and efficiency.

Hardware and software support for this project was provided by DoseOptics LLC, a company manufacturing Cherenkov imaging cameras. B. Pogue is the President and M. Jermyn and P. Bruza are employees of DoseOptics LLC. D. Alexander reports receiving consulting fees from DoseOptics LLC.

TU-lePD-TRACK 6-06

Risk Assessment for Custom Software in the Clinic

E Covington*, R Cardan, R Popple, UAB University of Alabama, Birmingham, Birmingham, AL

Purpose: To present a method for assessing risk of custom clinical software using an approach based on failure modes and effects analysis (FMEA). **Methods:** We created a novel model for assessing risk for custom clinical software analogous to FMEA. Using this risk model, we integrated a literature review and institutional experience to form a practical guide for risk mitigation. This model was refined by application in a clinic with a high volume of clinical software development. **Results:** We identified three parameters, population (P), intent (I), and complexity (C), analogous to the FMEA parameters of occurrence, severity, and detectability, respectively. Population classifies the scale of the software tool and is a direct measure of the percentage of the clinic's population that the tool will impact. Intent refers to the classification of the software and how it's used in clinical decision making. Complexity is a measure of how difficult it is to find an error by an independent reviewer. Analogous to FMEA risk priority number, multiplying these parameters gives the software risk number (SRN). **Conclusion:** Software risk number serves as a quantitative metric for the posed risk if the code does not perform as anticipated or if the design is flawed. After SRN is evaluated, clinics can rank tools by

highest SRN and intent to determine the most hazardous tools. Tools with the highest SRN and those used for direct clinical decision making can be allotted the appropriate resources during development, commissioning, and routine quality assurance. This model has been integrated into our clinical software development practice.

TU-IePD-TRACK 6-07

How Do More Relaxed Treatment Couch Tolerance Tables Lead to More Consistent Patient Positions? A Statistical Approach to Help Reduce Alert Fatigue
C Hui*, M Egan, Arizona Oncology Associates, Phoenix, AZ

Purpose: To show the benefits of deriving meaningful tolerance tables from data, and how the proper number of warnings could improve setup consistency. **Methods:** One year of historical patient positions were collected from the record and verify database. A set of new tolerance tables for different anatomical sites was created based on twice the standard deviation of the corresponding patient positions. In most scenarios, the new tolerance tables were more relaxed than the historical ones, and they were subsequently implemented for clinical use. To investigate the impacts of the new tolerance tables, three months of patient positions controlled by either the historical or new tolerance tables were studied and compared. **Results:** For photon beam therapy, the couch override frequencies for patient positions were 6.7% lateral, 11.9% longitudinal, and 2.8% vertical using the historical tolerance tables. This was compared to 2.1% lateral, 1.8% longitudinal, and 0.4% vertical for the new tables. The standard deviations in patient positions were 1.13 cm lateral, 2.43 cm longitudinal, and 0.29 cm vertical for the former, versus 1.07 cm lateral, 2.04 cm longitudinal, and 0.29 cm vertical for the latter. In general, using the new tolerance tables resulted in smaller lateral and longitudinal standard deviations ($p < 0.01$). As for electron therapy, the override frequencies were 4.4% lateral, 8.5% longitudinal, and 6.1% vertical using the historical tolerance tables, and 5.9% lateral, 3.0% longitudinal, and 2.6% vertical using the new tables. The standard deviations in patient positions were 1.75 cm lateral, 4.00 cm longitudinal, and 0.74 cm vertical for the former, versus 1.17 cm lateral, 2.23 cm longitudinal, and 0.51 cm vertical for the latter. The differences in all three dimensions were significant ($p < 0.01$). **Conclusion:** Applying statistically derived tolerance tables, even with more relaxed tolerances, can significantly reduce the variation in patient positions.

Students and Trainees

TRACK 1

MedPhys Slam

TU-CD-TRACK 1-00

MedPhys Slam

Moderators: Mary Gronberg, UT MD Anderson Cancer Center, Houston, TX and Soleil Hernandez, MD Anderson Cancer Center, Houston, TX

TU-CD-TRACK 1-01

M. Gronberg, UT MD Anderson Cancer Center:

Host

TU-CD-TRACK 1-02

S. Hernandez, MD Anderson Cancer Center:

Host

MedPhys Slam is a research communication competition for student, resident, and junior members of the AAPM in which participants prepare a three-minute presentation aimed at sharing the significance of their science to the general public in a compelling and coherent manner. Researchers spend their days investigating every minutia of their topics, but to engage investors, employers, and the general public, the importance of their research must be understood by more than just the scientific community. It's all in the art of communication, and this is what MedPhys Slam aims to foster in the medical physics community.

The preliminary competitions took place at the local chapter meetings throughout the year, and those winners will participate in the final competition at the AAPM 2019 Annual Meeting. The rules are simple: participants have three minutes to present their research using a maximum of three PowerPoint slides. Participants will be judged on two equally weighted categories, comprehension/content and communication/engagement. Not only will this be of interest to those within medical physics, it can be expanded as an AAPM media initiative aimed at communicating what we do to the public.

Learning Objectives:

1. Develop the ability to communicate research objectives and results to a general audience.
2. Develop confidence and conciseness in presentation skills.
3. Learn to incorporate creativity into science.

SAM Imaging Educational Course

TRACK 2

Nuclear Medicine Dosimetry in Imaging and Therapy

TU-CD-TRACK 2-00

Nuclear Medicine Dosimetry in Imaging and Therapy

Moderator: Sara St. James, UCSF, San Francisco, CA

TU-CD-TRACK 2-01

W. Bolch, University of Florida:

Dosimetry for Nuclear Medicine Imaging

TU-CD-TRACK 2-02

C. Lawhn Heath, University Of California San Francisco:

Clinical Perspective On Therapeutic Nuclear Medicine

TU-CD-TRACK 2-03

W. Erwin, UT MD Anderson Cancer Center:

Practical Dosimetry for Therapeutic Nuclear Medicine, MIBG

TU-CD-TRACK 2-04

S. St. James, UCSF:

Practical Dosimetry for Therapeutic Nuclear Medicine, ^{177}Lu Based Therapies

TU-CD-TRACK 2-05

Audience Q & A

In Nuclear Medicine imaging and therapy applications, molecularly targeted radionuclides may be used to either provide medical images or to deliver a therapeutic dose of radiation. Across this wide range of applications, there are different methods of estimating the organ and tumor doses received by the patient. In this session, dose calculations for nuclear medicine imaging applications will be discussed along with the need for accurate dose calculations in the therapeutic setting. The session will conclude with a discussion of the current methodologies for dose calculations for therapeutic applications including 131I-MIBG, 177Lu-DOTATATE and 177Lu-PSMA.

Learning Objectives:

1. Understanding the MIRD schema for assessment of radiation dose at multiple anatomic levels.
2. Understanding fixed activity versus dosimetry driven treatment planning
3. To become familiar with the current methodology for I131-MIBG dosimetry based treatment planning.
4. To become familiar with the requirements for patient specific dosimetry for 177-Lu based therapies.

Imaging Scientific Session

TRACK 3

4D CT/CBCT and Sparse Acquisitions

TU-C-TRACK 3-01

A Deep Learning Technique for Projection Interpolation in CBCT Reconstruction

K Lu^{1*}, Z Zhang¹, L Ren², F Yin², (1) Duke University, Durham, NC, (2) Duke University Medical Center, Durham, NC

Purpose: Some previous studies used DL techniques for sinogram interpolation prior to CT reconstruction to reduce the imaging dose. However, few studies have applied DL techniques on CBCT projection interpolation for cone-beam CT (CBCT) reconstruction. This study develops a novel DL technique that performs CBCT projection interpolation before the reconstruction step to improve CBCT image quality, reduce artifacts caused by sparse sampling, as well as reduce the imaging dose. **Methods:** Due to the limitation of GPU memory, the proposed technique re-slice the stack of projection data along the axial direction. The processed data are fed into the custom-designed network slice by slice. As the network iterates through each slice in the input stack of sparsely sampled projections, the outputs of the network are stacked together to form the stack of fully sampled projections. These projections are then used to reconstruct CBCT with the Feldkamp, Davis, and Kress (FDK) algorithm. The proposed technique is tested for CBCT reconstruction using sparsely sampled data. We compared the CBCT images reconstructed from projections interpolated with the conventional bilinear interpolation method and the proposed technique to demonstrate its feasibility and advantages. The quantitative evaluation metrics used are peak-signal-to-noise ratio (PSNR), structural similarity (SSIM) index, and root-mean-square error (RMSE). **Results:** Average PSNR, SSIM, and RMSE values are 41.552, 0.988, and 0.008 for CBCT images reconstructed from projections interpolated with the proposed DL technique; 37.774, 0.975, and 0.013 for bilinear interpolation method; 39.692, 0.978, and 0.010 for images reconstructed without projection interpolation. The proposed technique shows an overall improvement in PSNR, SSIM, and RMSE. **Conclusion:** The proposed DL interpolation technique can perform accurate CBCT projection interpolation that reduces artifacts for CBCT reconstructed from sparsely sampled projections as well as the imaging dose.

This work is partially supported by Duke University Chancellor Scholarship and is partially supported by NIH 1R01EB028324-01 and R01CA184173.

TU-C-TRACK 3-02

BEST IN PHYSICS (IMAGING): A GAN-Based Technique for Synthesizing Realistic Respiratory Motion in the 4D-XCAT Phantoms

Y Chang*, Z Jiang, P Segars, Z Zhang, F Yin, L Ren, Duke University Medical Center, Cary, NC

Purpose: Synthesize realistic and controllable respiratory motions in the 4D-XCAT phantoms by developing a GAN-based deep learning technique. **Methods:** A motion generation model was developed using bicycle-GAN with a novel 4D generator. The model input includes the end-of-inhale (EOI) phase of 4D-CTs and a Gaussian perturbation, and it generates inter-phase deformable-vector-fields (DVF), which are compounded and applied to the EOI image to generate 4D images. A separate machine-learning respiratory motion amplitude control model was built to predict the perturbation needed to generate a specific motion amplitude. The models were trained and validated using 71 patient 4D-CT data and tested using 6 XCAT to generate 4D-XCAT phantoms with realistic respiration. The simulated 4D images were evaluated by Dice coefficients calculated for the lungs. The generated DVFs were evaluated by deformation energy, the correlations of DVFs and derived ventilation maps with those from the reference images. Results were compared to those from the original 4D-XCAT to demonstrate the improvement of motion realism. **Results:** For the simulated 4D-CTs, the lung volume deviation was within 5.8%, and the Dice coefficients of the lungs reached at least 0.95. The cross-correlation of DVFs achieved $0.89 \pm 0.10 / 0.86 \pm 0.12 / 0.95 \pm 0.04$ along the x/y/z directions in the testing group. The cross-correlation of ventilation maps derived achieved $0.80 \pm 0.05 / 0.67 \pm 0.09 / 0.68 \pm 0.13$, and the Spearman's correlation was $0.70 \pm 0.05 / 0.60 \pm 0.09 / 0.53 \pm 0.01$, respectively, in the training/validation/testing groups. The generated 4D-XCAT phantoms presented similar deformation energy (0.36 ± 0.07) as patient data (training: 0.33 ± 0.17 ; validation: 0.22 ± 0.14 ; testing: 0.36 ± 0.14), compared with 0.07 ± 0.01 in the original 4D-XCAT. The Dice coefficient between the original and generated 4D-XCAT phantoms of similar breathing amplitudes was 0.95. The motion amplitude control error was less than 0.5 mm. **Conclusion:** The results demonstrate the efficacy of synthesizing realistic controllable respiratory motion in the 4D-XCAT phantom using the proposed method. This crucial development enhances the value of XCAT phantoms for various 4D imaging and therapy studies.

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TU-C-TRACK 3-03

A Method for 4D Cone-Beam CT in Under 20 Seconds with New Generation Linacs

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Purpose: The Halcyon linear accelerator (Varian Medical Systems, Palo Alto, CA) has an integrated x-ray imaging system capable of performing cone-beam CT (CBCT) acquisitions four times faster than conventional systems, including a free-breathing fast acquisition protocol for 3DCBCT in which 491 projections are acquired in 16.6 seconds. A conventional 4DCBCT acquires 1320 projections over 240 seconds to provide enough information for conventional reconstructions to produce images of sufficient quality across the respiratory cycle. This study presents a method by which the fast <20 second acquisition can produce 4DCBCT images and quantify image quality relative to conventional 4DCBCT. **Methods:** We used two 4DCT scans of the same patient from the Cancer Imaging Archive 4D-Lung dataset, defining the first as a "planning" 4DCT and the second as a "ground-truth" for 4DCBCT simulation and image quality quantification. The open-source Reconstruction Tool Kit (RTK) was used to simulate conventional and rapid CBCT acquisitions over a 200-degree arc by generating 1320 and 491 forward projections respectively, assuming 13-15 breaths per minute. Both acquisitions were reconstructed using the 4D Feldkamp-Davis-Kress (4DFDK) algorithm. The rapid

acquisition was also reconstructed using the Motion Compensated FDK (MCFDK) algorithm, in which data is backprojected along curved paths to account for motion estimated from the planning 4DCT. Image quality was quantified in terms of contrast-to-noise ratio (CNR), root-mean-square error to ground-truth (RMSE-GT) and structural similarity index to ground-truth (SSIM-GT). **Results:** The phase averaged CNR, RMSE-GT and SSIM-GT for the conventional 4DFDK were 2.695, 78.93 and 0.415 respectively. The phase averaged CNR, RMSE-GT and SSIM-GT for the rapid 4DFDK and MCFDK reconstructions were 2.006, 292.8, 0.072 and 2.864, 134.7, 0.205. **Conclusion:** The data acquired in the <20 second protocols offered on new generation linacs can deliver 4DCBCT images of quality comparable to 4-minute protocols but only when motion compensated reconstruction is implemented.

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TU-C-TRACK 3-04

Dual-Encoder Convolutional Neural Network (DeCNN) for Average Image-Constrained 4D-CBCT Reconstruction

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Purpose: 4D-CBCT is valuable for 4D-target verification in radiotherapy. However, due to the scanning time and imaging dose constraints, projections acquired for each respiratory phase are intrinsically under-sampled, leading to streak artifacts in FDK reconstruction or blurriness in iterative reconstruction using compressed sensing (CS). Introducing the average 4D-image constraint to the CS-based reconstruction, such as prior-image-constrained CS (PICCS), can improve the edge sharpness of the static structures. However, PICCS can lead to motion artifacts in the moving regions. In this study, we proposed a dual-encoder convolutional neural network (DeCNN) for fast and high-quality average image-constrained 4D-CBCT reconstruction. **Methods:** The proposed DeCNN has two parallel encoders to extract features from the average images and the under-sampled target phase images, respectively. The two encoders have the same architecture but do not share weights. Average image features and under-sampled image features are extracted at multi-scale levels and are then concatenated and fed into the decoder to reconstruct the high-quality images of the target phase. DeCNN was trained on 28 lung 4D-CBCT and tested on 12 lung 4D-CBCT data. Its results were both qualitatively and quantitatively evaluated using RMSE, PSNR, and SSIM, and compared against other state-of-the-art methods. **Results:** 4D-CBCT reconstructed by DeCNN showed: (1) qualitatively, clear and accurate edges for both stable and moving structures; (2) quantitatively, low RMSE, high PSNR, and high SSIM compared to the ground-truth images; and (3) superior quality to those reconstructed by other methods, including back-projection, CS total-variation, PICCS, and the single-encoder CNN. DeCNN took only about 1.8 seconds to reconstruct images of one respiratory phase. **Conclusion:** The proposed dual-encoder architecture demonstrated effectiveness in improving the 4D-CBCT image quality with average image constraint, thus improving 4D-CBCT's clinical utility for the moving target localization.

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TU-C-TRACK 3-05

Faster Safer Clearer 4DCBCT: Results of the ADAPT Clinical Trial

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Purpose: Conventional pre-treatment thoracic 4DCBCT scans take 1320 projections over 240 seconds. The ADAPT clinical trial (NCT04070586) uses first-in-world implementations of motion compensated acquisition and reconstruction to reduce scans to just 200 projections over 20 breaths (40 to 160 seconds). We present results of the first 20 of 30 trial patients and investigate our motion compensated acquisition and reconstruction relative to conventional 4DCBCT. **Methods:** The ADAPT trial enrolled 30 lung cancer radiation therapy patients for additional imaging in their first 2 treatment fractions. For each fraction, they receive a conventional 4DCBCT, followed by a 200 projection motion compensated 4DCBCT. In motion compensated acquisition, the imaging hardware is adapted to patient respiration in real time to ensure even data spread across the respiratory cycle. In motion compensated reconstruction, the Motion Compensated McKinnon Bates (MCMKB) algorithm estimates patient anatomy and motion so that data acquired at every phase contributes to reconstruction of each phase. Acquisition performance was quantified by scan dose and time, as well as Root-Mean-Square-Deviation (RMSD) between our real time adaptive gantry control against perfectly even breathing and data spread. Reconstruction performance was quantified with Contrast-to-Noise-Ratio (CNR), Structural SIMilarity Index (SSIM) to conventional and Tissue Interface Sharpness (TIS) at the diaphragm and tumour boundaries. **Results:** Relative to conventional 4DCBCT, motion compensated 4DCBCT delivered 85% less dose, was on average 64% faster, had RMSD of 1.8° from ideal, images had 35% better CNR, 99% structural similarity, 47% better diaphragm TIS and 56% better tumour TIS. **Conclusion:** The ADAPT clinical trial provides real world evidence that motion compensated acquisition and reconstruction can be implemented on standard linacs to perform thoracic 4DCBCT with 85% less dose, 64% faster and with improved image quality.

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TU-C-TRACK 3-06

Rapid Low Dose 4DCBCT

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Purpose: Prior to thoracic radiotherapy treatments, patients are commonly aligned using conventional 4DCBCT scans (1320 projections over 4 minutes). We determine the viability of rapid low dose 4DCBCT scans by simulating scans as low as 60 projections over 25 seconds, using data from 20 patients of the ADAPT clinical trial (NCT04070586). **Methods:** The ADAPT clinical trial performs adaptive 4DCBCT scans by adapting gantry rotational speed and projection acquisition in accordance to real-time patient breathing rate changes, this produces 600 and 200 projection scans in a quarter of conventional scan time. In this study, these adaptive 4DCBCT scans were down sampled and simulated into even faster and lower imaging dose 310, 160, 110 and 60 projection. Reconstructed image quality was quantified using Structural Similarity Index (SSIM), Signal-to-Noise-Ratio (SNR), Contrast-to-Noise-Ratio (CNR), Tissue Interface Sharpness-Diaphragm (TIS-D) and Tissue Interface Sharpness-Tumour (TIS-T). Conventional 4DCBCT scans were used as a comparator. **Results:** The simulated 310, 160, 110 and 60 projection adaptive acquisitions deliver approximately (76%, 87%, 91%, 95%) reduction in imaging dose via reduction in total number of projections acquired and (48%, 73%, 82%, 90%) shorter scan time on average compared to conventional 4DCBCT scans, respectively. The 310, 160 and 110 projection adaptive acquisitions from all patients matched or improved compared to the conventional acquisition across all image quality metrics. The minimum number of projections to maintain the image quality metrics of conventional scans (SSIM=1±0, SNR=11.9±11.7, CNR=9.5±3.1, TIS-D=0.59±0.3 and TIS-T=0.41±0.1) was the 110 projection adaptive scan (SSIM=0.98±0.01, SNR=18.5±8.9, CNR=10.2±3.5, TIS-D=0.72±0.4, TIS-T=0.57±0.7). **Conclusion:** 110 projection adaptive acquisition results show image quality can be maintained comparable to conventional acquisition whilst reducing scan time by 95% and dose by 90%.

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development fellowship, and PK is supported by an Australian NHMRC Senior Principal Research Fellowship.

TU-C-TRACK 3-07

Reference-Free, Learning-Based Rigid Motion Compensation in Cone-Beam CT for Interventional Neuroradiology

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Purpose: Cone-beam CT (CBCT) provides guidance in interventional neuroradiology, but limited acquisition speed (4 – 30 s) results in motion artifacts. Autofocus motion compensation (MoCo) was successfully applied in head CBCT but autofocus metrics do not guarantee preservation of anatomical structures. Similarity metrics (e.g., visual information fidelity, VIF) provide quantification of image quality and structural similarity to a reference. However, motion-free, matched, references are seldom available. We propose DL-VIF, a learned reference-free autofocus metric that preserves anatomy. **Methods:** DL-VIF was estimated with a deep CNN implementing a cascade of 3D residual blocks acting on downsampled head CBCTs (128³ voxels). The network was trained on simulated motion-free and motion-contaminated CBCT pairs. Random rigid motion trajectories (0-8mm amplitude) were applied to 38 head MDCT volumes (12100 instances). DL-VIF was integrated into an autofocus framework and assessed on 20 CBCTs generated analogously to the training data. Further validation was obtained on 5 CBCTs of a cadaver head acquired on a robotic C-arm (Artis Zeego, Siemens Healthineers) with motion induced by shifting and rotating the specimen. The performance of DL-VIF was compared to a conventional autofocus metric (gradient entropy). **Results:** DL-VIF showed good agreement with the reference (slope = 0.980, R² = 0.996). DL-VIF MoCo showed reduction of motion artifacts in simulated data, with 0.099 improvement in SSIM, compared to 0.094 for conventional autofocus. In the cadaver study, DL-VIF outperformed conventional autofocus, providing robust compensation in cases of moderate and severe motion, where conventional autofocus failed. An overall 6x larger increase in SSIM and 7x reduction in SSIM standard deviation was observed, restoring visualization of cranial anatomy. **Conclusion:** The learned DL-VIF provided reliable estimation of structural similarity without a reference image and yielded consistent reduction of motion artifacts with anatomically realistic solutions. DL-VIF offers an important step in robust autofocus MoCo methods for CBCT.

Academic-industry collaboration with Siemens Healthineers.

Multi-Disciplinary Scientific Session

TRACK 4

Biologically- and Functionally-Guided Radiation Therapy

TU-C-TRACK 4-01

Evaluation of CT-Derived Lung Perfusion with Deep Learning in Radiotherapy Patients

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Purpose: Deep learning-based CT perfusion mapping (DL-CTPM) has been proposed to provide regional function information for functional lung avoidance radiotherapy (FLART). This study aims to evaluate the DL-CTPM synthesized perfusion images in radiotherapy patients for FLART. **Methods:** SPECT/CT perfusion scans of 19 lung cancer patients were retrospectively collected from Henan Anti-cancer Hospital. All patients underwent radiation therapy afterward. The left/right lungs were separated to augment the dataset to 38 (20 for transfer training; 18 for testing). These images were processed by a deep learning-based framework to extract features from 3D-CT images and synthesize CT-derived perfusion (CTDP). Transfer learning strategies of fine-tuning and fixed-encoder were used to reuse information learned from a previously built dataset of different types of lung diseases. The Spearman's correlation coefficient (R) and structural similarity index measure (SSIM) between the CTDP and the corresponding SPECT were computed to assess the statistical and perceptual image similarities, respectively. To assess the function-wise concordance, the Dice similarity coefficient (DSC) was computed to determine the similarity of the low/high functional lung volumes. The dosimetry evaluation is undergoing. **Results:** The fine-tune transfer learning model has the best performance in terms of the voxel-wise agreement. The evaluation shows a high voxel value correlation (0.8111±0.1090) and high structural similarity (0.8094±0.0517) between the CTDP images and SPECT images. The evaluation of the function-wise concordance obtains an average DSC value of 0.7869±0.0634 for high-functional lungs, ranging from 0.6460 to 0.8984, and 0.7806±0.0727 for low-functional lungs, ranging from 0.6743 to 0.8902. Of the testing cases, 94% have high functional lung larger than DSC > 0.7, and 83% for low functional lung. **Conclusion:** The CT-derived lung perfusion using the DL-CTPM method showed a high agreement with the SPECT perfusion for lung cancer patients. This method holds great promise to provide lung function images for image-guided FLART.

This work is supported by the Health and Medical Research Fund (HMRF 07183266); the General Research Fund (GRF 151022/19M).

TU-C-TRACK 4-02

Functional Dose Sparing of the Parotid Glands: How Sparing the Parotid Ducts Yields Improvements in Patient Reported Xerostomia

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Purpose: To assess efficacy of prospectively sparing parotid ducts during treatment planning. **Methods:** 38 patients prospectively underwent MRI sialography-guided parotid ductal sparing (PDS) and had their patient reported xerostomia (PRX) evaluated at 6 and 12 months. 28/38 patients received de-escalated chemoradiotherapy. A parotid duct atlas was created, evaluated and used to generate parotid duct contours on a baseline cohort who did not receive PDS but were on the same de-escalation trials. PRX, defined as moderate to very severe symptoms, in the de-escalated PDS patients was compared to the aforementioned baseline cohort. Nested multivariate models including mean dose to the contralateral submandibular and parotid glands with and without mean parotid ductal doses were generated to determine if ductal dose improved model fit for PRX. **Results:** The atlas-generated parotid duct contours had a mean distance-to-agreement of 3.6mm compared to the MRI sialography imaged contours and an average absolute dose difference of 3.5Gy. Atlas based contours found the mean dose to the contralateral and ipsilateral ducts was lower in patients receiving PDS by an average of 11.4Gy and 17.2Gy, respectively. PRX was lower in patients receiving PDS at 6 months post-CRT (46% vs 67%, p=0.03), and trended lower at 12 months post-CRT (32% vs 49%, p=0.11). The nested models were compared and the addition of mean parotid ductal dose significantly improved model fit at 6 and 12 months post CRT (p=0.030 and 0.047, respectively). **Conclusion:** Atlas-based contours of parotid ducts demonstrated a substantial reduction of parotid ductal dose is possible with prospective sparing. This dose reduction yielded an approximately 20% absolute reduction in rates of PRX at 6 and 12 months. The significant improvement of model fit upon including mean parotid ductal dose suggests that the dose received by the parotid ducts relates to PRX even when considering traditional dosimetric predictors.

TU-C-TRACK 4-03

BEST IN PHYSICS (MULTI-DISCIPLINARY): Lung Treatment Planning with Parametric Response Mapping Function Guidance

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Purpose: Advancements in functional lung imaging have improved the pretreatment understanding of a patient's pulmonary function and have enabled the development of function-guided radiotherapy (RT) planning. Parametric response mapping (PRM) is a voxel-wise image analysis technique used for CT-based functional imaging that classifies lung abnormality phenotypes and has previously shown utility in the assessment of pulmonary complication risk in a diagnostic setting. The purpose of this work was to implement PRM guidance in a clinical RT planning workflow and examine the potential clinical impact. **Methods:** PRM analysis was performed for 18 lung cancer patients using paired pretreatment inspiration/expiration CT scans to classify each voxel as normal, emphysema, small airway disease (SAD), or parenchymal disease (PD). The non-normal PRM data were converted to topological maps and high-density regions of each classification were contoured as avoidance structures for VMAT treatment planning. Treatment plans were optimized to maintain clinical organ-at-risk dose limits to the Lungs-GTV, spinal cord, heart, esophagus, and brachial plexus contours. Dosimetric comparisons were performed between non-function-guided plans and PRM-guided plans which included additional low priority goals reducing the V20Gy of PRM avoidance contours. **Results:** The PRM-guided workflow was successfully implemented and the inclusion of PRM guidance resulted in statistically significant ($p < 0.05$) improvements to the V20Gy of the PRM avoidance contours. The average reduction of the V20Gy(%) of all PRM avoidance contours was 5.4%. The inclusion of PRM guidance did not significantly ($p > 0.05$) increase relevant dose metrics for OARs or decrease PTV coverage but did result in increased plan complexity. **Conclusion:** PRM guidance in RT treatment planning was found to be effective at redistributing dose away from avoidance structures in the lung while maintaining OAR dose limits and PTV coverage constraints. The presented workflow provides a framework for the potential clinical implementation of PRM-guided treatment planning.

Craig Galban has a financial interest in Imbio, LLC, which has licensed PRM from the University of Michigan. This work was funded by NCI grant P01-CA059872, and NHLBI grants R01-HL139690 and R01-HL150023.

TU-C-TRACK 4-04

Predicting Lung Ventilation Loss Caused by Local and Distant Radiation-Induced Damage

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Purpose: This work challenges the conventional assumption of “locoregional radiation damage” to lung parenchyma, implicit in current functional lung avoidance (FLA) radiotherapy (RT). Here, we investigate the impact of including “distant damage” by accounting for the radiation-induced injury to airways, branching serial structures distant from the point where the damage is observed. Consequently, we propose a method to predict post-RT ventilation loss that considers not only the locoregional parenchymal damage but also the airflow disruption caused by injury to airways. **Methods:** Breath-hold CTs (BHCTs) and simulation 4DCTs were acquired from seven lung RT patients. Airways were auto-segmented (12 generations, ~235 airways/patient) on the BHCTs using a virtual bronchoscopy software. 4DCT-based ventilation maps were used to estimate pre-RT ventilation. Conventional IMRT plans were recreated to obtain dose distribution. Post-RT ventilation loss was estimated studying three contributions: (i) Using the pre-RT ventilation and a normal tissue complication probability model we estimated loss due to parenchymal damage ($\Delta V_{\text{loss(P)}}$). (ii) Connecting each terminal airway with a lung sub-lobe volume (SLV), we estimated loss due to airway damage ($\Delta V_{\text{loss(A)}}$) by calculating the contribution of pre-RT ventilation in not-ventilated SLVs. A SLV was considered not ventilated if only one of the upstream airways was collapsed post-RT, using our probability of airway collapse model. (iii) Finally, the loss due to both effects ($\Delta V_{\text{loss(P+A)}}$) was calculated by subtracting from the total pre-RT ventilation the contribution of the post-RT ventilation (due to parenchymal damage) in the post-RT ventilated SLVs. **Results:** Preliminary results showed that average $\Delta V_{\text{loss(P)}}$ was 13%, $\Delta V_{\text{loss(A)}}$ was 22%, and the combined effect, $\Delta V_{\text{loss(P+A)}}$, was 25%. **Conclusion:** These results reveal that airway damage is not only important but may be a more critical factor than locoregional parenchymal damage. Ignoring this distant effect and focusing primarily on parenchymal damage can lead to spurious choices in current FLA strategies.

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TU-C-TRACK 4-05

Simultaneous Whole-Lung Ventilation-Perfusion Imaging with Volumetric CT in Non-Small Cell Lung Cancer: A Feasibility Study

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Purpose: To generate a comprehensive map of regional lung function including both ventilation and perfusion in a single dynamic contrast-enhanced CT scan using a volumetric CT scanner. **Methods:** Two patients with non-small cell lung cancer were imaged on a GE Revolution volumetric CT scanner before and 6 weeks after radiotherapy with standard fractionation. Images were acquired at a single couch position using the following parameters: cine mode, 0.28s/revolution, 100kV, 100mA, and 160mm axial field-of-view for 45-50s. Iodine-based contrast was injected 6s after the start of the acquisition. All images in the series were non-rigidly registered in Elastix using a b-splines transformation and a normalized mutual information metric. The lung volume was segmented on the registered images using a semi-automated density-based method. Blood flow maps were generated using GE's prototype CT Perfusion software. End-inhale and end-exhale images were identified from the pre-contrast images and were used to calculate specific ventilation maps using an existing density-based method. Ventilation (V) and perfusion (Q) maps were normalized to the 99th percentile of the distribution to facilitate direct comparison, and V/Q maps were calculated as a ratio of the normalized maps. **Results:** After radiotherapy, ventilation in the treated lung decreased for both patients and blood flow decreased in one patient but increased in the second. Tumour blood flow was heterogeneous in one patient, and uniformly low in the second. In both patients, regions of the lung that appear to have normal function based on ventilation or perfusion imaging alone are shown to have V/Q mismatch. **Conclusion:** These results demonstrate the feasibility of generating both ventilation and perfusion imaging using volumetric CT in a single imaging protocol performed in under 1 minute. This method may facilitate the clinical translation of functional lung imaging in radiation treatment planning.

TU-C-TRACK 4-06

Ventilation Derived from Clinical 4D-CBCT Using a Deep Learning-Based Model: First Comparison with Technegas SPECT Ventilation

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Purpose: Ventilation derived from 4D cone-beam computed tomography (CBCT) may predict specific radiation sensitivity, but the current algorithms for measuring ventilation highly depend on deformable image registration (DIR). With single-photon emission-computed tomography ventilation (SPECT-V) acting as a gold standard, this study proposes new deep learning (DL) model-independent of DIR for deriving ventilation from 4D-CBCT and investigates whether the accuracy could be improved in comparison with DIR dependent methods. **Methods:** The study consists of 4D-CBCT and 99mTc-Technegas SPECT/CT scans of 28 esophagus or lung cancer patients. For each patient, the scans were acquired before the first radiotherapy treatment and rigidly registered. Using these data, ventilation images from 4D-CBCT (CBCTVI) were derived by using a deep-learning-based model with two forms of the dataset [(a) ten phases and (b) two phases of peak-exhalation and peak-inhalation] as input. A Sevenfold cross-validation procedure was used to evaluate the performance. For comparative evaluation, density-change- and Jacobian-based (HU and JAC) methodologies are also used to calculate CBCTVIs for each patient. The SPECT-V and derived CBCTVIs were segmented into high, medium, and low functional lung (HFL, MFL, and LFL) regions. The spatial overlap of corresponding HFL, MFL, and LFL for SPECT-V against CBCTVIs was evaluated by using the dice similarity coefficient (DSC) and analyzed with a one-factor ANOVA model among different methods. **Results:** The averaged DSC values were

0.34, 0.34, and 0.59/0.58 for the CTVI(HU), CTVI(JAC), and CTVI(DL(a)/DL(b)), respectively. These results showed the DL method yielded the highest similarity with SPECT-V with the prominently significant difference. **Conclusion:** This study proposed a DL model for deriving CBCTVI and performed a validation against SPECT-V. The results demonstrated that DL method increased the accuracy in comparison to HU and JAC methods, and can be a stepping stone to extract dynamic changes in the respiration patterns of patients during radiotherapy.

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TU-C-TRACK 4-07

Protons Combined with PARP Inhibition Sensitize Cancer Cells and Activate Immune Signaling More Than Photons

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Purpose: Due to their higher ionization density compared to photons, protons induce more clustered DNA lesions including clustered double strand breaks (DSBs), clustered single strand breaks (SSBs) and clustered base damages (BDs). Clustered SSBs and BDs can be converted into DSBs during attempted repair, increasing the number of DSBs. If not properly repaired, DSBs can generate micronuclei (MN), which are displaced chromosomes or chromosome fragments enclosed by the nuclear membrane and are known precursors of immune activation. Here we show that protons+PARP inhibitor (PARPi) sensitize cancer cells more than photons+PARPi, and that this combination might also induce antitumor immune signaling. **Methods:** We obtained clonogenic cell survival for H460, H1299, PANC-1 and Panc 10.05 cell lines to quantify the effect of a PARPi on radiosensitization. We used immunohistochemistry in H1299, PANC-1, HCC1937 (BRCA mut) and HCC1937 (BRCA complemented) cell lines to measure MN and cGAS-positive MN (MN-cGAS+) numbers to quantify antitumor immune signaling at 24 and 72 h post-radiation. Cells were treated with 6 MV x-rays at 10 cm water equivalent depth or 100 MeV unmodulated protons at 4.3 water equivalent depth (9.9 keV/ μ m dose-weighted LET in water) alone or with a PARPi (0.1 or 1 μ M, Olaparib). **Results:** PARPi radiosensitizes multiple cell lines to both photons and protons. Relative to photons alone, protons+PARPi provided sensitizations of 1.70 ± 0.04 (H460), 1.94 ± 0.06 (H1299), 1.93 ± 0.05 (PANC-1) and 1.70 ± 0.09 (Panc 10.05). PARPi resulted in a modest increase (5%) or a significant decrease (-10%) in RBE. MN and MN-cGAS+ were increase by both photons and protons. However, protons+PARPi further amplified MN-cGAS+ compared to all other treatment combinations. **Conclusion:** Protons+PARPi increase cell death to levels that are only achievable with carbon ions. Protons+PARPi also amplify precursors associated with antitumor immune signaling. Additional investigation is warranted to evaluate if these effects are seen in vivo.

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Therapy Scientific Symposium

TRACK 5

Radiomics and Outcome Prediction

TU-CD-TRACK 5-00

Radiomics and Outcome Prediction

Moderator: Fang-Fang Yin, Duke University, Chapel Hill, NC

TU-CD-TRACK 5-01

H. Aerts, Dana-Farber/Brigham Women's Cancer Center:

Artificial Intelligence in Cancer Imaging

TU-CD-TRACK 5-02

Y. Xiao, University of Pennsylvania:

Standardizing Radiomics Applications for Clinical Trials

TU-CD-TRACK 5-03

F. Yin, Duke University:

Impact of Image Quality on Radiomics Applications

TU-CD-TRACK 5-04

E. Deutsch, Universite Paris-saclay:

State-Of-The-Art Multi-Omics

Rapid development in machine learning (ML) and radiomics in the last decade has stimulated substantial interest in precision medicine as well as research activities in applying radiomics and artificial intelligence (AI) for radiation therapy treatment assessment and outcome prediction. At the same time, the radiation oncology community is eager to understand its efficacy and limitation in order to narrow down the gap between scientific research and clinical application and prepare for emerging field evolution. This symposium aims to provide fundamental knowledge of radiomics and artificial-intelligence (AI) development in precision medicine, including open-source informatics. The impact of imaging parameters and image processing techniques on radiomics/ML as well as some general and specific guidelines recently recommended by a number of medical professional societies will be discussed to improve the robustness of using medical images for the outcome research. As an extension of radiomic application in multimodality cancer treatment, multiple omics applications in cancer treatment assessment and outcome prediction will also be introduced.

Learning Objectives:

1. To learn the role of radiologic AI with other -omics data for precision medicine and open-source informatics developments
2. To learn the current recommendations for incorporating radiomics in oncology clinical studies and the impact of imaging and image processing on radiomics applications
3. To learn the methodology of incorporating biological data into radiomics for the assessment of tumor heterogeneity

Therapy Scientific Session

TRACK 6

Radiobiology

TU-C-TRACK 6-01

A DNA Repair Inhibitor Targeting ATR Modulates Immune Responses to Radiotherapy

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Purpose: Radiotherapy (RT) has been shown to induce antitumor immune responses whereby the host immune system is stimulated to recognize tumor and activate mechanisms to clear it. RT-induced antitumor immunity is rare but once achieved it offers robust and long-lasting antitumor immunity. Because RT-induced antitumor immunity is thought to be mediated through DNA damage we set out to establish if inhibitors of DNA damage repair proteins, particularly targeting ataxia telangiectasia and Rad3-related protein (ATR), can induce greater immune responses than RT alone. **Methods:** We used a triple negative breast cancer murine model, 4T1, on the hind leg of BALB/c mice. Animals were treated with vehicle or an ATR inhibitor (ATRi) (75 mg/kg, AZD6738) 2 h prior to each radiation fraction (3x6 Gy using 6 MV x-rays) or sham exposure. Tumors were harvested 7 days after the last fraction to isolate sub-populations of immune cells (CD8⁺ T-cells, CD4⁺ T-cells, T-regs, CD8⁺ PD-1⁺ T-cells, type 1 and 2 macrophages and classical dendritic cells type 1 and 2) within the tumor. Tumor growth delay and survival data were also acquired. **Results:** RT delayed tumor growth and modestly increased survival. RT+ATRi enhanced these effects significantly. RT alone was able to increase the percentage of macrophages present in the tumor (12.9%) compared to vehicle alone (4.6%). RT+ATRi also increased macrophage numbers (24.8%) but also significantly increased CD4⁺ (9.3% vs vehicle 3.2%) and CD8⁺ (5.6% vs vehicle 2.6%) T-cells. Dendritic cells were also increased after RT+ATRi. **Conclusion:** RT alone can alter the tumor microenvironment by increasing macrophage recruitment. This effect can be increased with RT+ATRi. Furthermore, RT+ATRi significantly increased the number T-cells (CD4⁺ and CD8⁺) and non-significantly, dendritic cells. This suggests differential signaling at 7 days between RT and RT+ATRi. Further analysis is required to identify pro- and anti-tumor subpopulations within these sub-types.

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TU-C-TRACK 6-02

DNA Repair Inhibitors Combined with Protons Increase Cell Radiosensitivity in Vitro

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Purpose: To compare the competing radiosensitizing effects of DNA-repair inhibitors and proton LET in vitro. **Methods:** We treated H460 and H1299 non-small cell lung cancer and PANC-1 and Panc 10.05 pancreatic adenocarcinoma cells with varying concentrations (0.1-10 μ M) of DNA repair inhibitors (DRIs) targeting the DNA repair proteins Rad51 (BO2), DNA-PKcs (NU7441), ATM (KU5593), ATR (ADZ6738) and PARP (AZD2281), or with DMSO vehicle as a control. We irradiated the cells with 6 MV x-rays or 9.9 keV/ μ m protons (dose-weighted LET in water) and performed clonogenic assays to characterize their survival. To parameterize cell radiosensitivity, we estimated the dose required to achieve 10% survival, D10%. To quantify drug sensitization at a given radiation quality, Q, we calculated the sensitization enhancement ratio, SER(Q)=D10%(DMSO,Q)/D10%(drug,Q). To quantify the proton LET effect, we calculated the RBE for each drug combination as: RBE=D10%(drug,x-rays)/D10%(drug,protons). To quantify the combined sensitizing effect of protons and drugs, we calculated the total radiosensitivity enhancement, TRE=D10%(DMSO,x-rays)/D10%(drug,protons). **Results:** The SERs varied greatly across drug combinations and cell lines, ranging from 1.00 $\pm$ 0.02 (H460, 5 μ M BO2, x-rays) to 3.23 $\pm$ 0.13 (H1299, 1 μ M ADZ6738, x-rays). Between radiation qualities, the SERs were positively correlated ($r=0.8807$), suggesting that DRIs that are effective for x-rays are similarly effective for protons. However, the RBE values were negatively correlated with SER(x-ray) ($r=-0.6160$), suggesting that the greater the DRI-induced sensitization, the less benefit there will be from the proton LET effect. Nevertheless, the TREs from combining DRIs with protons were generally larger than SER(x-ray) values (30 $\pm$ 3% on average), and were positively correlated with SER(x-ray) ($r=0.8911$), suggesting that even when DRIs reduce proton RBE, there is still a net radiosensitivity gain when combining them with protons. **Conclusion:** Although protons+DRI may reduce proton RBE, the DRI-induced sensitization outweighs the RBE loss, resulting in a substantial net sensitization beyond that achieved with DRI+photons.

This research was supported in part by RP170040, 1R21CA252411-01, Emerson Collective, and MD Anderson Cancer Center. Dr. Sawakuchi has research agreements with Alpha Tau Medical and Artios Pharma. Dr. Shaitelman has a research agreement with Exact Sciences.

TU-C-TRACK 6-03

Features Of TOPAS-nBio – Release 1.0

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Purpose: The goal of the TOPAS-nBio project is to provide intuitive nanometer scale Monte Carlo (MC) simulations for radiobiology experiments that do not require programming expertise. Here we present new functionalities of the TOPAS-nBio system available in the first full release (v1.0). **Methods:** Since the open-source beta-release of TOPAS-nBio in 2019, the framework offers to connect energy deposition within irradiated cells (physics) via molecular reactions (chemistry) to cell kill/repair (biology) at the level of sub-cellular targets such as DNA. To facilitate the setup of simulations we further developed a Graphical User (GUI) Interface. TOPAS-nBio is an extension to TOPAS and layered on top of the Geant4/Geant4-DNA MC toolkit. The new release was built for TOPAS release 3.6 (based on Geant4.10.6.p3) and will be compatible with all future releases of TOPAS. **Results:** New geometries that were developed include two new methods to fill the cell nucleus: a whole nucleus DNA model using chromatin fibers and a fractal walk filling pattern, as well as a filling of the nucleus using spheres to represent topologically associated domains (TADs) of DNA based on the Hi-C technique. New scoring options offer direct scoring of single and double strand breaks (SSB and DSB) as well as output the DNA damage in the SDD (Standard for DNA Damage) scoring format. Chemistry was improved to provide better agreement with experimental data of G-values, both for step-by-step chemistry and a newly included independent reaction time method, which offers faster chemistry simulations. Biological outcome can now be simulated directly within TOPAS-nBio using the DaMaRiS biological effect model, or via the SDD which offers a convenient interface to other models such as MEDRAS. **Conclusion:** The new features of TOPAS-nBio v1.0, offer improved modeling from initial DNA damage to cell outcome, or from energy depositions to G-value propagation over time is possible.

TU-C-TRACK 6-04

Investigating the Impact of Sustainable Delivery of Immunoadjuvant From Smart Radiotherapy Biomaterial in Pancreatic Cancer

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Purpose: To investigate the impact of sustained delivery of an immunoadjuvant drug, anti-CD40 monoclonal antibody (Ab_CD40), compared to its single direct in-situ administration in preclinical studies for determining optimal delivery of treatment for metastatic pancreatic tumors. **Methods:** Smart radiotherapy biomaterials (SRBs) were designed similar to currently used gold fiducials, but with smart polymers that can provide imaging contrast and containing similar amount of anti-CD40 that lead tumor regression in animals. C57BL6/Tac mice were subcutaneously injected with KPC cells on one flank to study: a) the release of a fluorescent monoclonal antibody (anti-CD40-AF750) for up to 12-days and b) abscopal effect and mice survival post treatment. A small animal radiation research platform (SARRP) was used to deliver 5Gy at the tumor site followed by either a) free intratumoral injection of Ab_CD40 and b) SRB_Ab_CD40 treatment. Tumor volume size, survival and Immune cell infiltrations in tumors were assessed and results compared for the free direct injection of Ab_CD40 versus the SRB_Ab_CD40 that affords sustained delivery. **Results:** Results showed sustained presence of the anti-CD40 in the tumors treated with SRB_Ab_CD40. The results further showed significant increase in abscopal response rates for animals treated with SRB_Ab_CD40 ($p < 0.001$) compared to animals treated with direct injection. Moreover, significant increase in survival was also observed for the SRB_Ab_CD40 cohort ($p < 0.01$) compared to direct free injection. Significantly more CD8+ T-cells ($p < 0.001$) and CD4+ T-cells ($p < 0.01$) were observed to infiltrate the SRB_CD40_5Gy tumor compared to the cohort that received free injection of Ab_CD40_5Gy. **Conclusion:** Sustained delivery of anti-CD40 with smart radiotherapy biomaterials was more effective in boosting abscopal response rates compared to direct intratumoral injection. This establishes additional potential advantage of using smart sustained delivery of immunotherapy to boost the immune-mediated abscopal effect to engender robust local and metastatic tumor kill.

TU-C-TRACK 6-05

Nanoparticle-Mediated Assessment of ROS and Radiosensitization of Brain Cancer Cells for Improved Radiotherapy Outcomes

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Purpose: Although glioblastoma is the most common primary brain tumor in adults, it remains one of the least treatable. The median survival is only 15 months following current standard of care which includes combinations of surgery, radiotherapy, and chemotherapy. This dismal outcome is partly due to the high radio-resistance of glioblastoma. The purpose of this work is to develop strategies that will ultimately enhance radiotherapy outcomes for glioblastoma through radiosensitization. **Methods:** Having recently developed and published a novel assay using fluorescence intensity modulation of PEGylated (biocompatible) CdSe/ZnS quantum dots (QDs) to assess reactive oxygen species (ROS) generation during chemotherapy and radiotherapy for cancer cells, we have added other nanoparticles to this assay and are applying it for concurrent measurement of ROS and radiosensitization. Using a Faxitron Cell Irradiator, we irradiate (5 Gy, 10 Gy, 20 Gy, 50 Gy) glioblastoma cancer cell lines (T98G and U87 cells) treated with QDs and carbon quantum dots (CQD) and measure both their QD/CQD fluorescence intensity and their migration. We measure and quantify the migration using a commercially available Electric Cell Impedance Sensor (ECIS). **Results:** U87 and T98G cells attach and migrate significantly ($p < 0.0001$) more than non-irradiated cells in the first 20 hours post-irradiation with 20 Gy. Our results show QD intensity reduction due to ROS production as expected, but intensity enhancement for CQD. Furthermore, relative peak fluorescence intensity ratio calculations and average intensity comparisons show highly significant ($p < 0.001$) enhancement of ROS for 5 Gy and 20 Gy irradiation. **Conclusion:** We have developed a nanoparticle-based assay for in vitro concurrent measurement of ROS and radiosensitization of cancer cells. This suggests using PEGylated QDs, CQDs and similar conjugates to improve radiotherapeutic outcomes for glioblastoma and other highly radio-resistant tumors.

TU-C-TRACK 6-06

Potential Anti-Kras Activity OffFlavonoid Derivative of Cannabis Is Enhanced by a Single Fraction of Low Dose of Radiation Therapy

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Purpose: Usually, 2 out of 3 patients of pancreatic cancer have Kras mutation causing fast progression of the disease. Prior research shows that, flavonoid derived from Cannabis sativa (FBL-03G) demonstrates significant therapy potential in the treatment of murine pancreatic adenocarcinoma, including potential for radiosensitization and treatment of metastatic tumors through immunomodulation. Here we demonstrate the potential for FBL-03 in targeting the Kras-gene expression in a murine pancreatic cancer model. **Methods:** FBL-03G (Mw = 368.38 g/mol) was delivered intratumorally loaded in smart radiotherapy biomaterial (SRB) for sustained action. In house SRBs were developed with Poly (lactic-co-glycolic) acid (PLGA) polymer. KPC cells were derived from a LSL-Kras;p53+/floxex,Pdx-cre mouse used for generating orthotopic pancreatic tumors in C57BL/6 background mice. Tumors were treated with SRB-FL-03 or 6 Gy of radiation or in combination. Post-vivo tumors were collected on day 10 post treatment and tumor weight and intratumor immune cell population were analyzed. To further evaluate the role of these treatment models in Kras-gene expression, western blot analysis was performed on same treated tumor tissue using Kras antibody. **Results:** We observed significant reduction of tumor weight with the treatment of SRB-FBL-03G and in combination treatment with RT+SRB-FBL-03G. Radiation was shown to further enhance the antitumor activity of SRB-FBL-03G. Further investigation with flowcytometry also demonstrates increase infiltration of T helper cells (CD45+CD3+CD4+) in both SRB-FL-03 ($p < 0.01$) and RT+SRB-FL-03 ($p < 0.001$) treated groups. Similarly, most important cytotoxic T cells (CD45+CD3+CD8+) was also increased in SRB-FBL-03G ($p < 0.01$), and RT+SRB-FBL-03G (0.001) treated groups, confirming the antitumor immune response with these treatments. Western Blot analysis of the treated tumor tissue also confirmed the suppression of Kras-gene expression in both SRB-FBL-03G ($p < 0.05$) and RT+SRB-FBL-03G ($p < 0.01$) treated groups. **Conclusion:** Significant suppression of Kras expression was observed along with immune-modulation and reduction of tumor growth in the FBL-03G treatment, which was further enhanced by radiation.

There is no conflict of interest.

TU-C-TRACK 6-07

BEST IN PHYSICS (THERAPY): Radio-Immunotherapydose-Painting with Survival Boost Using Checkpoint Inhibitors

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Purpose: Abscopal responses during radiotherapy are rare, especially for immunologically cold tumors like Castration resistant prostate cancer. Here we investigate a new approach called radio-immunotherapy dose-painting (RAID), which specifically targets tumor sub-volumes with radiotherapy and in-situ delivered immunoadjuvant to boost abscopal responses. The study is conducted in metastatic castration-resistant prostate cancer animal model with and without check-point inhibitors. **Methods:** Syngeneic murine model of castration resistant prostate adenocarcinoma (CRPA) was generated in both flanks of wild C57/BL6 background mice using androgen deprived TRAMP-C1 cell lines. The palpable sized tumor of one of the flanks was treated as four different randomized cohorts: control with no treatment, direct treatment with 5 Gy of radiation, intra-tumor treatment with Immunogenic Biomaterial (IBM) loaded with AntiCD40, and in combination. Tumor growth was measured on both sides. The highest responded treatment method (RT+IBM) was rechallenged by adding checkpoint inhibitors, AntiPD1 and CTLA4,

for survival study. Checkpoint inhibitors were given in systemic doses for 3 consequent administrations. Image Guided Radiotherapy (IGRT) targeting tumor and tumor sub-volumes was administered with the small animal radiation research platform. **Results:** We demonstrate that the RAID approach substantially boosts abscopal responses for prostate cancer. Adding checkpoint inhibitors with 5 Gy of IGRT+IBM, further increases the survival duration, where the combination treatment with (RT+IBM) demonstrates enhancement of abscopal effect ($p<0.001$) compares with only RT, and only IBM. Adding AntiPD-1 ($p<0.001$) and CTLA4 ($p<0.01$) with the combination of RT+IBM, show significant increase in the survival duration. whereas, AntiPD-1 shows the longest duration of survival (>250 days), compared to others (<50 days) for at least one third of the treated mice. **Conclusion:** RAID with IBM represents a promising new approach to significantly boost abscopal response rates for immunologically cold tumors like prostate cancer. Moreover, Checkpoint inhibitors like AntiPD-1 can further boost treatment outcomes.

SAM Therapy Educational Course

TRACK 7

State of the Art (Non-MRI) Motion Management for External Radiotherapy

TU-CD-TRACK 7-00

State of the Art (Non-MRI) Motion Management for External Radiotherapy

Moderator: Jennifer Smilowitz, University of Wisconsin, Madison, WI

TU-CD-TRACK 7-01

T. McCaw, University of Wisconsin-Madison:

Application of Optical and Infrared Imaging Systems for Respiratory Motion Management

TU-CD-TRACK 7-02

E. Colvill, Paul Scherrer Institute:

MLC Tracking for Motion Management

TU-CD-TRACK 7-03

G. Mageras, Memorial Sloan-Kettering Cancer Center:

Planar Radiographic Imaging for Prostate Fiducial Tracking

TU-CD-TRACK 7-04

X. Li, Medical College of Wisconsin:

Motion Tracking with MLC and Jaw During Tomotherapy Delivery

Motion management solutions for external beam radiation therapy can potentially enable a smaller volume of normal tissue irradiated and target dose escalation. In recent years, many optical, infrared, kV and MV radiographic imaging techniques have emerged clinically, and continued to be developed, to manage the effect of intra-fraction motion. Imaging and positional data are also often coupled with tracking systems using MLC and jaw to manage respiratory and non-respiratory motion during radiation therapy. The applications and procedural considerations for select imaging technology for respiratory motion management will be presented. Using implanted fiducials, prostate motion can be monitored with planar radiographic during radiation treatment delivery. Moreover, MLC and jaw tracking approaches are now available different linear accelerator systems. An educational overview of these systems, their clinical implementation and associated quality assurance considerations will be presented.

Learning Objectives:

1. Review methods to account for respiratory motion using optical and infrared imaging systems.
2. Present treatment planning and delivery requirements for implementing (and performing QA on) MLC tracking systems.
3. Present methods of intra-fraction monitoring for prostate motion with planar radiographic imaging of implanted fiducial markers.
4. Understand implementation and clinical experience of motion tracking using jaws and MLC during helical tomotherapy delivery.

SAM Partners in Solutions Imaging

TRACK 8

Imaging: Artificial Intelligence in Medical Imaging

TU-CD-TRACK 8-00

Imaging: Artificial Intelligence in Medical Imaging

Moderators: Colin Schaeffer, University of Florida, Gainesville, FL and Stephanie Leon, University of Florida, Gainesville, FL

TU-CD-TRACK 8-01

C. Schaeffer, University of Florida:

Artificial Intelligence in Medical Imaging

TU-CD-TRACK 8-02

S. Leon, University of Florida:

Artificial Intelligence in Medical Imaging

TU-CD-TRACK 8-03

N. Mistry, Siemens Healthineers GmbH:

Artificial Intelligence in Medical Imaging – Progresses in Radiation Oncology

TU-CD-TRACK 8-04

J. Vaishnav, Canon Medical Systems, USA:

Deep Learning in Medical Imaging: From Image Generation to Workflow Automation

TU-CD-TRACK 8-05

R. Bammer, Rapidai:

Embracing Artificial Intelligence and Machine Learning in Radiology to Streamline Emergent Care

TU-CD-TRACK 8-06

A. Dresner, Philips Healthcare:

Next Generation Synthetic CT for MR-Only Brain Radiotherapy Planning

TU-CD-TRACK 8-07

Q&A

In the last few years, artificial intelligence (AI) has emerged as powerful tool in the field of diagnostic medical imaging with applications in areas such as image processing, image reconstruction, and image interpretation. AI is a multidisciplinary field that takes ideas from computer science and neuroscience to try to replicate human intelligence in machines with the idea of solving complex problems typically requiring human intelligence. Machine learning (ML) is a subset of AI where the algorithm is designed to learn from data without explicit programming. A popular ML algorithm is the artificial neural network (ANN) which was designed to mimic the neural networking in the human brain. Deep learning (DL) is a further subset of ML involving very deep neural network models with many layers. When applied to medical images, a popular variant of the ANN is the convolutional neural network (CNN) which can learn image patterns and features from training images. These learned features can then be used to solve a wide array of problems such as classification or segmentation.

Current and potential AI applications in diagnostic radiology may include improving image quality, reducing radiation dose, decreasing MRI imaging time, enhancing technical protocols, standardizing the detection and characterization of image findings, and improving workflow. With an ever-increasing amount of data, the potential for AI in the field of diagnostic medical imaging is greater than ever.

Learning Objectives:

1. Understand the basics concepts of Artificial Intelligence (AI) and it's relation with Machine Learning (ML) and Deep Learning (DL)
2. Understand primary components of DL architectures and neural network.
3. Understand selected AI applications in diagnostic radiology.

References:

1. Rik R. Ranschaert, Sergey Morozov, Paul R. Algra, "Artificial Intelligence in Medical Imaging, Springer, 2019.
2. Ahmed Hosny, Chintan Parmar, John Quackenbush, et al, "Artificial Intelligence in Radiology", Nat Rev Cancer, 2018 August; 18(8): 500-510. Doi: 10.1038/s41568-018-0016-5.

Imaging Scientific Session

TRACK 3

AI in CT and CBCT: Image Enhancement and Synthesis

TU-D-TRACK 3-01

Convolutional Neural Network Based Metal Artifact Correction for Sparse View Dental CT Imaging

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Purpose: We aim to reduce complex artifacts due to both metal implants and sparse view sampling in dental CT images by utilizing prior images via convolutional neural network (CNN). **Methods:** To generate sparse view dental CT images in the presence of metal implants, we inserted two circular-shaped gold implants within a region of the teeth and acquired projection data by simulating polychromatic X-ray CT scans with one-fourth of full view data set. CT images were then reconstructed with the filtered backprojection algorithm. Since the image quality is severely degraded by metal implants, we initially reduced the metal artifacts using linear metal artifact reduction (LMAR) algorithm. To reduce the streak artifacts, we trained CNN in the image domain, which is denoted by LMARCNN. Then, by using a thresholding, the prior image was acquired from the results of LMARCNN. By utilizing prior images, normalized metal artifacts reduction (NMAR) method is applied for the sparse view sampled sinogram data. Then, the residual streak artifacts are reduced again by CNN in the image domain, which is denoted by NMARCNN. For both CNNs, we employed a U-net architecture to ensure a large receptive field to capture the globally distributed streak artifacts. **Results:** It is observed that NMARCNN can reduce the residual errors of NMAR and shows the best qualitative and quantitative quality of images through ROIs. The quantitative evaluation of mean squared error (MSE) and structural similarity (SSIM) also confirm our observations. Since the remaining streak artifacts in LMAR were reduced in LMARCNN, clear prior images can be acquired. **Conclusion:** In this work, we proposed CNN-based NMAR approach for sparse view dental CT image reconstruction. Our results show that the combination of metal artifacts and streak artifacts due to sparse view sampling can be reduced effectively by the proposed method.

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TU-D-TRACK 3-02

Low-Dose CT Image Enhancement Through a Texture Transformer

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Purpose: To improve the spatial resolution and to suppress the noise of the low-dose computed tomography (CT) images, we adapted a Texture Transformer network for image Super-Resolution (TTSR) method. **Methods:** The TTSR is a reference-based image super-resolution method. We adapted it to translate the low-dose (low-resolution and noisy) CT images to the full-dose (high-resolution and clean) CT images. The low-dose CT images and full-dose images are severed as queries and keys in a transformer, respectively. The image translation is optimized through deep neural network (DNN) texture extraction, relevance embedding, and attention-based texture transfer and synthesis to enable joint feature learning between low-dose and full-dose images. 4D XCAT phantom program based on 18 patients' data was used to generate both benchmark full-dose CT images (512x512 in-plane resolution) and low-dose images (20% dose and 128x128 in-plane resolution). The data from 17 patients were used as training data and the rest one was used as test data. The reference images in TTSR were randomly selected from the training set, which were not the same as the test images. The peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) were used as the quantitative metrics. For comparison, we used cubic spline interpolation and generative adversarial network (GAN) with cycle consistency (GAN-CIRCLE). **Results:** The performance of different methods (mean $\pm$ std) are PSNR = 31.57 $\pm$ 0.89dB, and SSIM = 0.74 $\pm$ 0.02 for cubic spline interpolation, PSNR = 34.85 $\pm$ 0.32dB and SSIM = 0.88 $\pm$ 0.02 for GAN-CIRCLE, and PSNR = 37.23 $\pm$ 0.42dB and SSIM=0.84 $\pm$ 0.02 for TTSR. TTSR also recovers more details in enhanced low-dose CT images than the other two methods, although some artifacts are noticeable. **Conclusion:** TTSR based on texture transformer and attention mechanism is effective to improve the spatial resolution and to suppress the noise of low-dose CT images in general. Further work is needed to minimize the artifacts.

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TU-D-TRACK 3-03

Multi-Resolution Residual Deep Neural Network for Generating Synthetic CT Images with High HU Accuracy and Structural Fidelity

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Purpose: Cone-beam computed tomography (CBCT) is frequently used for accurate image guided radiation therapy (IGRT). However, the poor CBCT image quality prevents its further clinical use. Thus, it is important to improve the HU accuracy and structure preservation of CBCT images. **Methods:** In this study, we proposed a novel method to generate synthetic CT (sCT) images from CBCT images. A multi-resolution residual deep neural network (RDNN) was adopted for image

regression from CBCT images to planning CT (pCT) images. At the coarse level, RDNN was first trained with a large amount of lower resolution images, which can make the network focus on coarse information and prevent overfitting problems. More fine information was obtained gradually by fine-tuning the coarse model using fewer number of higher resolution images. Our model was optimized by using aligned pCT and CBCT image pairs of a particular body region of 153 prostate cancer patients treated in our hospital (120 for training, 33 for testing). **Results:** The averaged accuracy of mean absolute error (MAE) between CBCT and pCT on testing data was 121.61 HU, while the MAE between the sCT and pCT images was only 27.62 HU. And the averaged structural similarity index measure between sCT and CBCT was 48.89% higher than that of CBCT and pCT. Our proposed multi-resolution RDNN also provides an overall better image quality in terms of HU accuracy and structural fidelity than U-Net and RDNN trained only using high-resolution CBCT and pCT image pairs. **Conclusion:** The sCT images generated by our proposed multi-resolution RDNN have higher HU accuracy and structural fidelity, which may promote the further applications of CBCT images in the clinic for structure segmentation, dose calculation and adaptive radiotherapy planning.

TU-D-TRACK 3-04

Renal Stone Quantification in Contrast Enhanced CT Using Convolutional Neural Network Assisted Dual-Energy Virtual Non-Contrast Imaging

H Gong*, A Ferrero, J Marsh, N Huber, J Fletcher, C McCollough, S Leng, Mayo Clinic, Rochester, MN

Purpose: Virtual non-contrast (VNC) technique has been proposed to detect renal stones in contrast-enhanced dual-energy CT (DECT) but has demonstrated limitations of reduced stone volume and removal of small stones. We aim to improve stone detection and quantification accuracy by developing a novel convolutional neural network (CNN) assisted VNC technique. **Methods:** A dual-task CNN with customized loss functions was developed to synergistically integrate material-classification and -decomposition. The classification provided a probability weighting scheme to material decomposition, to minimize the ambiguity between calcium and iodinated soft-tissue. Multi-energy CT phantoms were scanned on a dual source DECT (80/Sn150 kV) and image patches (50x50 pixels) of insert materials (e.g. hydroxyapatite, iodine, and iodinated-blood mixture) were used to train the CNN. Random-shaped numerical inserts with additional densities were simulated to augment training dataset. Iodine component was subtracted from original CT images to generate VNC images. Human stone specimens (n=35; 20 calcium-based/15 non-calcium-based) embedded in an abdomen-sized water phantom were scanned with and without iodine contrast (20mg/cc). CT number and volume per stone was measured using in-house stone quantification software and compared to commercial VNC (Syngo.VIA, Siemens). Mean-absolute-percent-error (MAPE) was calculated using true non-contrast (TNC) images as ground truth and Wilcoxon signed rank test (5% significance level) was used to gauge stone volume discrepancy between VNC and TNC images. **Results:** CT number of stones from commercial VNC were significantly reduced (MAPE: 32.9%), resulting in significantly-underestimated stone volume (bias: -26.4%, MAPE 26.5%, p<0.01). Stones were much brighter in CNN-assisted VNC than those in commercial VNC, and density plot showed CT numbers recovered to those of the original CT images. The stone volume was also more accurate in CNN-assisted VNC, with not significant difference from that of TNC (bias: -2.1%, MAPE 10.6%, p=0.568). **Conclusion:** The CNN-assisted VNC improves the quantification of renal stones in contrast-enhanced DECT.

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TU-D-TRACK 3-05

Synthesize 3D Realistic CT Textures and Anatomy in the XCAT Phantom Using Generative Adversarial Network (GAN)

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Purpose: Generate 3D realistic textures and anatomy in the thorax region of extended cardiac torso (XCAT) phantom. **Methods:** We proposed to generate realistic anatomical textures and structures in the XCAT phantom using a conditional generative adversarial network (CGAN) and 3D U-net. 300 and 100 3D CT images were used for training and validation, respectively. Organ maps were generated from patient CT images with uniform intensities in each organ to mimic XCAT images. The model was trained to simulated realistic CT textures and structures in organ maps to match with real CT images. To capture the fine details and guarantee the continuity of lung vessels, we used multiple generators to generate lung vessels and other body parts separately before integrating them together. The results from the training and validation group were evaluated by measuring the SSIM, L1, and L2 metric between the real CT images and simulation images generated from organ maps. After training, XCAT phantoms were input into the model to generate textured XCAT phantoms. The result was compared to our prior work using the 2D GAN model. **Results:** Compared to the 2D GAN model, the 3D model achieved higher SSIM and PSNR and lower L1 and L2 losses in the validation group. In the generated XCAT phantoms, the model generated similar contrast and anatomical textures as real CT images. The generated phantom also showed 3D continuity by visual examination from different views, especially for lung vessels. Additionally, we found the strategy of using separate generators for lung vessels and other parts greatly enhanced the realism of the generated images. **Conclusion:** The study demonstrated the feasibility of generating realistic 3D anatomical textures and structures in XCAT phantoms using the proposed method. This development dramatically enhances the phantom's value for various clinical applications, such as testing and optimizing novel imaging techniques.

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TU-D-TRACK 3-06

Task-Based Loss Function for Convolutional Neural Network-Based CT Denoising

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Purpose: To develop a task-based loss function in a deep convolutional neural network (CNN) to reduce noise and maintain spatial details simultaneously. **Methods:** A U-Net CNN was trained with simulated-quarter-dose head CT images reconstructed with smooth and sharp kernels and using routine-dose images of each kernel as targets. A baseline model was trained using mean-squared-error (MSE) loss and image patches from 100 scans; 5 additional scans were used for validation and 5 more for testing. To better retain sharpness in the bone and reduce noise in the brain, a loss function with two task-based loss terms was introduced. The first term enforced similarity with the sharp kernel target images in the HU range outside of the brain (<0 or >80) while penalizing sharp kernel noise within the brain HU range using total variation loss. The second term enforced similarity with the smooth kernel target images within the brain HU range. Results from the task-based model were compared against those of the baseline model using region of interest (ROI) measurements of standard deviation to assess noise reduction and line profile measurements to assess spatial resolution preservation. **Results:** The CNN model trained using the task-based loss function achieved a 51.6% and 70.5% greater noise reduction when applied to routine and low-dose CT images compared to the MSE loss model while simultaneously maintaining image sharpness in the bone comparable to the sharp kernel input. **Conclusion:** A task-based loss function that specified where each input kernel should be emphasized, outperformed the same model trained under equal conditions but using MSE loss alone. Superior performance was noted in terms of noise reduction and sharpness preservation.

TU-D-TRACK 3-07

Evaluation of Image-Domain Training Frameworks for Deep Learning Based Denoising and Deconvolution

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Purpose: Spatially matched full-dose/low-dose (FD/LD) pairs are often used as training input and target for supervised learning. Training data is commonly simulated

using projection noise insertion; however, inability of many institutions to access CT projection data motivates development of widely accessible image domain methods. In this work, we evaluated two image domain training frameworks for CNN-based dual-energy denoising and deconvolution applied to ex-vivo kidney stone specimens. **Methods:** We compared two image domain training frameworks, a Generalizable Noise and Artifact Reduction Network (GARNET) and an isotropic colored noise insertion technique (Iso-NI), with an existing reference technique using projection-domain noise insertion (PD-NI). GARNET training data was synthesized using the following procedure: noise-only images were extracted from phantom scans, re-scaled to the desired dose, and added back to FD images to mimic LD images for training. Iso-NI training data was synthesized by backprojecting filtered Gaussian random noise into an image, which was then blurred to match the noise power spectrum of the LD images. Patches derived from FD/LD pairs were used to train a separate CNN for each technique, and denoised/deconvolved kidney stone images were inferred. Inferred images were input into a dual-energy stone composition algorithm to evaluate the performance in a clinical task. **Results:** GARNET and the PD-NI reference technique were found to have similar noise power spectra, whereas Iso-NI inferred images were slightly noisier. All methods resulted in comparable, excellent performance for kidney stone composition quantification (9 out of 11 mixed stones properly identified by all methods vs 0/11 in the original data). **Conclusion:** Inferred images using both GARNET and Iso-NI trained CNNs were shown to significantly reduce noise and improve stone quantification relative to the original images. These methods provide a suitable substitute to projection domain noise insertion when access to proprietary projection data is unavailable.

This work is supported by NIH grant R01 EB028591.

Multi-Disciplinary Scientific Symposium

TRACK 4

The Anne and Donald Herbert Distinguished Lectureship in Modern Statistical Modeling

TU-D-TRACK 4-00

The Anne and Donald Herbert Distinguished Lectureship in Modern Statistical Modeling

Moderator: Ioannis Sechopoulos, Radboud University Medical Centre, Nijmegen, The Netherlands

TU-D-TRACK 4-01

Introduction

TU-D-TRACK 4-02

N. Obuchowski, Cleveland Clinic Foundation:

Importance of Quantitative Imaging Biomarkers' Technical Performance Characteristics in Designing and Analyzing Clinical Trials

TU-D-TRACK 4-03

Q&A

Importance of Quantitative Imaging Biomarkers' Technical Performance Characteristics in Designing and Analyzing Clinical Trials

Quantitative imaging biomarkers (QIBs) are being used increasingly to diagnose disease, predict patients' outcomes, and monitor and adapt treatment. In clinical trials, QIBs are used to identify patients eligible for trials and often serve as non-invasive endpoints, providing results earlier than traditional patient outcomes. Yet biomarker measurements are inherently biased and imprecise. When their bias and imprecision are overlooked, measurements can be mis-interpreted, studies can be underpowered, and treatment effects can be over- or under-estimated.

In this presentation I first review the key technical performance characteristics of a QIB, namely precision, bias, linearity, and the regression slope against true values. I then examine four scenarios using QIBs in clinical practice and in trials. I illustrate how the technical performance characteristics of a QIB affect the interpretation of pulmonary nodule volume doubling time in ruling out malignancy, the sample size of a trial investigating ultrasound elastography to diagnose stage F4 liver cirrhosis, the eligibility of Parkinson's disease subjects for a trial of a new intervention, and the treatment effect of the rate of brain atrophy in a randomized clinical trial of progressive multiple sclerosis. Methods for correcting for the QIB measurement error are presented for each scenario.

Dr. Obuchowski is the statistician for the Quantitative Imaging Biomarker Alliance, through a contract between RSNA and the Cleveland Clinic

Therapy Scientific Session

TRACK 6

Radiobiology and Preclinical Systems

TU-D-TRACK 6-01

ATR Inhibition Suppresses DNA Damage Signaling and Sensitizes Cancer Cells to Photons and Protons

D Martinus*, S Bright, D Flint, B Turner, M Manandhar, M Ben Kacem, S Shaitelman, G Sawakuchi, UT MD Anderson Cancer Center, Houston, TX

Purpose: ATR is a DNA damage response protein essential to double strand break (DSB) repair and cell cycle checkpoint regulation. Here we investigate the influence an ATR inhibitor (ATRi) has on DSB repair and radiosensitization in response to photons and protons. **Methods:** We exposed H1299 cancer cells to 6 MeV x-rays and 9.9 keV/μm protons (dose-weighted LET) with and without an ATRi (0.1-5 μM, AZD6738). We then performed immunofluorescent staining of 53BP1, a DNA repair protein involved in double strand break repair and fixed cells at 2, 4, and 24 h timepoints post-irradiation. Clonogenic cell survival assay was also performed. **Results:** ATRi radiosensitized to both photons and protons and increased the RBE of protons. At 4 h post-irradiation, increasing ATRi concentration suppressed the number of 53BP1 foci. Cells exposed to 5 Gy of photons alone had an average of 28.55±0.17 foci/nucleus while the photons + ATRi (1 μM) had a significantly lower average of 21.7±0.2 foci/nucleus at 4 h post-irradiation. However, at 24 h post-irradiation, increasing ATRi concentration resulted in a significant increase in 53BP1 foci (p<0.0001). Photons alone had an average of 10.61±0.11 foci/nucleus while the photons + ATRi (1 μM) had a significant increase at 16.43±0.15 foci/nucleus. Additionally, we observed a threshold on the ATRi concentration in the 24 h timepoint. Below a concentration of 0.5 μM, ATRi had very little effect on 53BP1 foci number for all timepoints. This suggests that ATR inhibition has an effective time window relating to the stage of DNA repair. **Conclusion:** ATR inhibition sensitizes cancer cells to both photons and protons. The sensitization is due in part to the role of ATR in DSB repair as the number of persistent 53BP1 foci at 24 h post-irradiation was much higher relative to radiation alone.

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TU-D-TRACK 6-02

Modeling DNA Structure with Different Chromatin Compaction in TOPAS-nBio Using Hi-C Representations

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Purpose: TOPAS-nBio is an open source MC application for nanodosimetry developed to advance our understanding of radiobiology effects at the (sub-)cellular scale. Several approaches were introduced to calculate DNA damages induced by ionizing radiation using TOPAS-nBio. In the TOPAS-nBio, a human fibroblast nucleus model was provided that includes chromatin territories with the same DNA density across the nucleus (6.08 Gbp DNA and DNA density of 14.4 Mbp/um³). However, the DNA density varies for each cell, chromatin territory and cell cycle, consisting of higher and lower DNA density (heterochromatin and euchromatin, respectively). The aim of this study is to design a realistic DNA geometry containing both heterochromatin and euchromatin. **Methods:** In order to design a realistic DNA geometry containing both heterochromatin and euchromatin, we use a previously developed method of filling a nucleus with spheres representing topologically associated domains (TADs) using the Hi-C technique. Hi-C data consist of several thousand TADs that classify the chromosome territories (separated by color). Each TAD (sphere) was filled with voxels using three different voxel sizes (12, 24, and 48 nm) depending on the TAD radius to efficiently model DNA packing inside the TADs. **Results:** Various geometries of DNA packing were placed inside the TADs to construct heterochromatin and euchromatin regions. Depending on the voxel size, the number of nucleosomes inside the voxel was adjusted differently. A maximum of one nucleosome was packed inside the 12 nm voxel, 8 nucleosomes were packed inside the 24 nm and 16 nucleosomes were packed inside the 48 nm voxel. The nucleosome packed inside the TADs was organically linked to each other to construct the heterochromatin and euchromatin regions. **Conclusion:** The modeling of DNA structure with different chromatin compaction organized in TADs based on Hi-C data will allow studies of DNA damage considering the heterochromatin and euchromatin regions.

TU-D-TRACK 6-03

Mechanistic Modelling of Oxygen Enhancement Ratio with DNA Damage by gMicroMC

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Purpose: Oxygen plays an important role determining the radiobiological effects of ionizing radiation. It is understood that oxygen can fix damages to DNA, hence enhancing strand breaks. In this work, we used a GPU-based Monte Carlo (MC) simulation package to mechanistically model the oxygen fixation hypothesis and compute oxygen enhancement ratio (OER). **Methods:** We implemented two kernel functions in gMicroMC to model the OER. The first one simulated the water radiolysis process in the presence of DNA and recorded the initial DNA damage. Chemical reactions involving oxygen was included in the simulation. The second one was a probabilistic model to capture the oxygen fixation effect on DNA damage sites. As cell survival fraction is mainly dominated by the number of DNA double strand break (DSB), we assume that the same number of DSB results in the same biological effect, OER was computed as the ratio of doses reaching the same level of DSB with and without having oxygen. We searched in the parameter space of reaction rate between damage site and oxygen, probability of repairing a damage site with/without oxygen fixation to match OER with published experimental results of photon irradiation. We applied the developed model to proton irradiation cases. **Results:** Parameters was found to reaction rate 4.75×10^7 L/(mol·s), probability of repairing with and without oxygen 0.72 and 0.92, respectively. The computed OER as a function of oxygen concentration matched with published experimental results of photon irradiation with a mean relative error 3.18%. The model gave OER of 3.04 for photon case and 2.93 for proton case at oxygen partial pressure of 15.2 mmHg. **Conclusion:** We modelled OER in a mechanistic approach using microscopic MC simulations. The mechanistic modeling approach is expected to be important for understanding oxygen impacts in different scenarios, such as FLASH with multi pulses.

TU-D-TRACK 6-04

Radiodynamic Therapy for Treatment of Lung Cancer: An In-Vivo Study

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Purpose: Radiodynamic therapy (RDT) uses Cerenkov light induced by photon beams to treat deep-seated tumors that photodynamic therapy cannot perform due to the low penetration depth of laser beams. The purpose of this study was to investigate the therapeutic effect of RDT with 15MV photon beams combined with 5-aminolevulinic acid (5-ALA) using a mouse model. **Methods:** A subcutaneous murine KP1 small-cell lung cancer model using male C57 black mice was used. The tumors (n=18) were randomly assigned into four groups, including control (untreated, n=4), radiation treatment (RT) only (n=4), 5-ALA only (n=4), and RDT (n=6), to observe individual and synergistic effects of 5-ALA and radiation treatment. For mice treated with radiation, 4 Gy in a single fraction was delivered to the tumors using 15MV photons on a Varian Clinac iX accelerator. For the RDT group, 5-ALA was injected at a dosage of 100mg/kg by tail-vein 4 hours prior to RT. Tumor volumes were measured using a 1.5T MR scanner on the treatment day and weekly post-treatment. Tumor growth for each tumor was quantified as the ratio of tumor volume one and two weeks after treatment to the tumor volume on the treatment day. **Results:** RDT achieved a decrease in tumor volume by $27.5 \pm 11.4\%$ and $26.9 \pm 11.4\%$ at one- and two-week post-treatment, respectively, compared to the tumor volume on the treatment day. The tumor growth in the RDT group was delayed by $73.1 \pm 25.4\%$, $82.6 \pm 28.9\%$, and $81.2 \pm 19.4\%$ two weeks after treatment compared to 5-ALA only ($P < 0.05$), RT only ($P < 0.01$), and control group ($P < 0.01$), respectively. There were no significant differences in tumor growth between 5-ALA only, RT only, and the control group. **Conclusion:** Radiation treatment with 5-ALA resulted in a significant tumor growth control compared to the other groups. The preliminary (hypothesis-generating) results advocate a further investigation into larger sample sizes.

TU-D-TRACK 6-05

Second Generation Silica-Based Bismuth Gadolinium Nanoparticles for Tumor Dose Enhancement and Real-Time MR-Tracking

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Purpose: The purpose of this study is to develop a new generation of theranostic silica-based bismuth-gadolinium nanoparticles for local tumor dose enhancement and adaptive treatment planning using a clinical Viewray MRIdian MRI-Linac System. **Methods:** First generation silica-based gadolinium nanoparticles (AGuIX) were modified at different ratios to exchange gadolinium-DOTA complexes with bismuth-DOTA complexes. A new generation of AGuIX-Bi nanoparticles were developed with three ratios of Gd/Bi (70/30, 50/50, and 30/70). Nanoparticles were characterized via ICP-MS, HPLC-UV, DLS, and T1 relaxometry. In vitro toxicity analysis and clonogenic assays were conducted to confirm radiosensitization of nanoparticles in a human non-small cell lung carcinoma (NSCLC) cell line. In vivo studies in agar phantoms and NSCLC mouse models were conducted to evaluate changes in MR contrast of AGuIX-Bi nanoparticles compared to AGuIX in a clinical 3T Siemens MR scanner and a 0.35T MRIdian MRI-Linac. **Results:** Each AGuIX-Bi nanoparticle formulation was characterized with final ratios of 70/30, 50/50, and 30/70 of Gd/Bi with an approximate size of 4.5-5.1 nm and 93-96% purity. As the ratio of Gd relative to Bi increased, T1 shortening was observed, confirming an enhancement of MR contrast. Even nanoparticles with the lowest ratio of Gd (30/70 Gd/Bi) showed strong T1 shortening, thus retaining improved contrast. In vitro MTT studies confirmed relatively low toxicity of all nanoparticles up to 3 g/L in a human NSCLC A549 cell line. Clonogenic assays identified a sensitizer enhancement ratio (SER = 50%) of 1.28 for 30/70 AGuIX-Bi nanoparticles compared to 1.125 for AGuIX (gadolinium only). In vivo MR imaging of subcutaneous

NSCLCs further confirmed retention of strong signal enhancement of 30/70 AGuIX-Bi nanoparticles compared to AGuIX in a TrueFISP and T1-weighted MR-Linac scan and a T1-weighted 3T MR scan. **Conclusion:** This study develops a new generation of theranostic nanoparticles for real-time MR-guidance and localized tumor dose amplification.

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TU-D-TRACK 6-06

Development of a Novel System for Preclinical Small Animal MR-Guided Radiation Experiments

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Purpose: Small Animal Radiation Research (SARR) serves as a critical counterpart to human radiation therapy (RT). Current SARR platforms use cone-beam CT for image-guidance, which results in poor soft-tissue contrast, additional radiation dose, and lack of real-time imaging. Whereas magnetic resonance-guided RT (MRgRT) has become available for human RT, SARR platforms currently lack on-board MRI capabilities. This study proposes a novel MR-guided SARR platform and reports our on-going progress. **Methods:** We propose the use of an inhomogeneous unilateral low-field MR design to ease the integration of MR to a SmART irradiator (Precision X-Ray Inc.). Targeting a 4cm-diameter field of view (FOV), we designed a magnet by optimizing the arrangement of neodymium magnets in a Halbach cylinder using a genetic algorithm. The decaying magnetic field along the axial direction is naturally used for spatial encoding. We constructed the magnet using 3D-printed holders to mount the magnets accurately. We designed and constructed gradient coils with a 4cm gap to allow the beam to pass unobstructed. We designed a 4cm gapped radiofrequency (RF) coil with three resonant circuits to span a wide bandwidth needed to cover the FOV. The inhomogeneous magnetic field allows acquisition of images in a set of curved surfaces, which are resampled to form a volumetric image in Cartesian coordinates. We simulated the MR imaging formation process. **Results:** Optimizing magnet orientations was able to increase field strength by 47%, resulting in a 78mT field strength at a distance of 2cm from the surface of the magnet assembly. The measured field strength agreed with our calculations within 3%. The gradient coils achieved linearity through a gapped distance of 4cm. RF coil is under construction. **Conclusion:** We proposed a novel approach for MR-guided SARR and are in the process of constructing the system.

TU-D-TRACK 6-07

Improving the Efficiency of Small Animal IMRT with Submillimeter Dose Deposition Kernels and Total Variation Regularization

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Purpose: There is growing interest in the use of modern 3D printing technology to implement intensity-modulated radiation therapy (IMRT) on the preclinical scale which is analogous to clinical IMRT. However, current 3D-printed IMRT methods suffer from long delivery times, excess filament usage, and complex modulation patterns. In this work, a comprehensive practical implementation of IMRT using 3D-printed beam compensators on the Xstrahl Small Animal Radiation Research Platform (SARRP) is presented, with an emphasis on accurate submillimeter dose calculation and the use of total variation regularization (TVR) to facilitate 3D printing. **Methods:** Custom micrometer-scale dose deposition kernels were developed based on actual beam commissioning data. TVR-IMRT, a technique designed to minimize the intensity difference between neighboring beamlets, was used to optimize the beamlet intensity map, which was then converted to corresponding compensator thicknesses in copper-doped PLA filament. IMRT and TVR-IMRT plans using five beams were generated to treat a mouse heart while sparing lung tissue. The individual field doses and composite dose were delivered to film and compared to the corresponding planned doses using gamma analysis. **Results:** The use of micrometer-scale dose kernels led to better agreement between the planned and measured doses, especially in low-dose regions. TVR-IMRT reduced the total variation of both the beamlet intensities and compensator thicknesses by 45-55%. The total mass of compensator material consumed and radiation beam-on time were both reduced by 15-25%, while DVH curves remained comparable. Gamma analysis passing rate with 3%/0.5mm criterion was 93.6% for IMRT and 97.3% for TVR-IMRT. **Conclusion:** TVR can be applied to small animal IMRT in order to produce fluence maps and subsequent 3D-printed compensator patterns with less total variation, facilitating 3D printing and reducing the amount of filament required. The TVR-IMRT plan required less beam-on time while maintaining the dose conformity when compared to a traditional IMRT plan.

Imaging Interactive ePoster Discussion

TRACK 1

Imaging: CT Artifacts, Reconstruction, and Imaging Performance Assessment

TU-IePD-TRACK 1-01

Development of An Automated Analysis Software for Daily CT QA

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Purpose: Positron emission tomography/computed tomography (PET/CT) imaging is used for a wide range of applications from diagnosis, to staging and radiotherapy treatment planning. Daily quality assurance (QA) testing is an essential duty of both physicists and technologists in order to verify that the x-ray CT images produced by PET/CT scanners meet performance standards. As part of an effort to bring scanners into compliance with standards in an efficient manner, a software package was developed to automate the analysis of x-ray CT images of the GE VCT QA phantom. This work aimed at developing an automatic solution that would analyze the results of the daily CT QA scans. Our tool reduces the burden on technologists and contributes to a more robust QA procedure. **Methods:** The software was developed using Python (v3.8.2) to aid in the daily CT QA procedure for GE DMI and D690 PET/CT scanners. The developed software automates each analysis step by automatically placing ROIs and performing calculations for the following tests: laser alignment, contrast scale, spatial resolution, noise and uniformity, slice thickness, and low contrast detectability. **Results:** The software has been implemented clinically in three centres. The software can automatically log data through an API connection. All analysis tests conducted using the automatic software agreed with the manual analysis and results have been within manufacturer specification. For example, the differences in measured image noise ($\Delta SD = 0.16\text{HU}$) and slice thickness ($\Delta x = 0.01\text{mm}$) were not statistically significant ($p > 0.05$). **Conclusion:** The automated analysis software developed in this work requires fewer manual hours from the technologists and allows the physicists overseeing the QA program to easily audit the performance of scanners located in several centres from a centralized location. This software is also compatible with standalone CT systems that use the GE VCT QA phantom for daily tests.

TU-IePD-TRACK 1-02

CARE-CT: CARDiac and RESpiratory Adaptive Computed Tomography

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Purpose: To safely plan high dose radiotherapy treatments for thoracic cancers and novel treatments such as non-invasive cardiac radioablation, 4DCT requires the capacity to account for multiple motion sources. We have developed a novel acquisition method, CARE-CT (CARDiac and RESpiratory adaptive Computed Tomography), to gate for dual motion and quantified the reduction in image artifacts. **Methods:** CARE-CT and conventional cine 4DCT imaging were simulated on

a deformable digital phantom (XCAT) for nineteen patient-measured ECG and respiratory traces. Our software monitored motion in real-time, triggering a simulated CT acquisition during combined cardiac and respiratory phases with 0.36s gantry rotation and 2cm/1s couch translation. Six and ten respiratory phases during cardiac diastole (60-80% through cardiac cycle) were acquired for each scan. Image artifacts were quantified and compared for 13 thoracic substructures using the normalized cross correlation between axial slices (Δ NCC). The 13 substructure volumes were compared to an average XCAT ground truth and compared between protocols. Total scan and beam-on times were recorded. **Results:** Compared to conventional 4DCT, the pooled mean reduction in the Δ NCC was 19% and 20% for the 6 and 10 bin CARE-CT scans respectively, with the largest reduction of 37% for the left ventricular wall. The mean variation in total structure volumes compared to the ground truth was reduced from 3% (-19% to 59%) for conventional 4DCT to 0% (-1% to 2%) for CARE-CT. The mean total scan times were increased from 1.8 $\pm$ 0.4min for conventional 4DCT to 3.5 $\pm$ 0.9min and 7.5 $\pm$ 2.2min for 6 and 10 bin scans, respectively. The total beam on time was reduced from 98 $\pm$ 28s for conventional 4DCT to 27 $\pm$ 0.2s and 45 $\pm$ 0.6s for 6 and 10 bin scans, respectively. **Conclusion:** CARE-CT reduced image artifacts in thoracic substructures while reducing beam-on time as compared. This study provides a crucial first step towards dual-motion management for central tumours.

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TU-lePD-TRACK 1-03

A Spatiotemporal Denoising Method for Low-Dose Cardiac CT Images

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Purpose: To develop a spatiotemporal denoising method using an unpaired video-to-video translation model, which can improve the image quality of low-dose cardiac CT images by exploiting both spatial and temporal correlations of dynamic image frames. **Methods:** The new spatiotemporal denoising model is built upon the generative adversarial network with cycle consistency (CycleGAN), which is an unpaired image-to-image translation model. In order to utilize the temporal dynamics of image frames, a recurrent loss of image frames in both domain (low-dose and full-dose CT images) is added to enforce the temporal correlation between image time frames. This is particularly useful for denoising of low-dose dynamic CT images, such as multi-phase cardiac CT angiography, where consecutive cardiac gates exhibit strong temporal dependencies. The 4D XCAT phantom program based on 18 patients' data is used to generate both benchmark full-dose and low-dose (80% reduction) cardiac CT images (8 gates for each patient). 17 of 18 patients are used as training data and the rest one is used as test data. To qualitatively evaluate the model performance, peak signal noise ratio (PSNR) and structural similarity index (SSIM) are used. **Results:** The spatiotemporal denoising method achieves better quantitative performance (PSNR: 38.35 $\pm$ 1.46 dB and SSIM: 0.9300 $\pm$ 0.0112) than the low-dose images (PSNR: 31.56 $\pm$ 1.74 dB and SSIM: 0.8436 $\pm$ 0.0284) and CycleGAN (PSNR: 36.85 $\pm$ 1.29 dB and SSIM: 0.9240 $\pm$ 0.0122). It also significantly improves the low-dose CT image quality compared to CycleGAN. **Conclusion:** We proposed a new unpaired deep-learning based denoising method using spatial cycle consistent and temporal recurrent consistency for low-dose CT image sequences. The simulation study demonstrated the superior performance of the proposed method. The future study will investigate its effectiveness on real patient data.

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TU-lePD-TRACK 1-04

Automated Assessment of Image Sharpness Degradation in Iterative CT Reconstruction Techniques: Vessel-Sharpness in Contrast-Enhanced Liver

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Purpose: Iterative CT reconstruction techniques were often accompanied with the degradation of lesion sharpness. In this study, we developed a quantitative vessel sharpness measurement techniques and compared them with the previous slope estimation method in contrast-enhance liver CT. **Methods:** After IRB approval, a total of 120 CT images were acquired. The CT images were scanned with four scanners of GE Discovery CT750 HD (GE), Ingenuity CT (Philips), Definition Flash (Siemens), and Aquilion One (Canon). For a given image, the structure coherence feature was firstly calculated which is consisted of an edginess feature to represent the likelihood of a pixel being located on an anatomical structure and the randomness of the pixel orientation to signify the absence of an anatomical structure. The outline of enhanced vessel was drawn, and then was dilated with structural element of 7 pixels. An average of structure coherence feature for this region was defined as edge sharpness. To demonstrate the efficacy of suggested algorithm, the median slope value of the least squares regression line fit was obtained in the same edge profile. The correlation between two sharpness was analyzed. Fully automated sharpness measurement contains segmentation of candidate liver mask, adopting regions with structure coherence feature greater than the 70th percentile, and defining the representative sharpness as an average of structure coherence features on the segmented edges. The mean sharpness in FBP and iterative reconstruction image was compared. **Results:** For 120 FBP images, the Pearson correlation coefficient between manual and automated sharpness was 0.844. Compared to the FBP images, the mean sharpness was degraded in each vendor's iterative reconstruction algorithm by 17.8%, 16.3%, 9.8%, and 26.0% for GE, Philips, Siemens, and Canon, respectively. **Conclusion:** Degradation of image sharpness in iterative CT reconstruction techniques could be quantified automatically by using our proposed vessel sharpness measurement technique.

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TU-lePD-TRACK 1-05

Quantitative Evaluation of Noise Texture of Deep-Learning-Based CT Reconstruction with An Anthropomorphic Phantom

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Purpose: Using an anthropomorphic phantom, quantitatively study noise texture of CT images reconstructed with deep-learning image reconstruction (DLIR, True Fidelity, GE) and compare them with filtered back-projection (FBP) and iterative reconstruction (ASiR-V). **Methods:** Axial CT scans of abdomen region of an anthropomorphic phantom (Kyoto Kagaku CTU-41) were acquired from a GE Revolution scanner with four CT DIvol levels ranging from 2.5 mGy (25% of TCM determined mAs) to 46.94 mGy (scanner max mAs). CT slices were reconstructed using all three algorithms, 1.25 mm and 5 mm slice thicknesses, five strength levels of ASiR-V (0% for FBP, 10%, 40%, 80%, and 100%) and three levels of DLIR (low, medium, and high). The 2D noise power spectrum (NPS) was estimated through Fourier analysis of difference images by subtracting two matching images from identical scans. Visual comparison of noise texture was performed between recon algorithms and dose levels via 1D NPS plots. Noise variance (0D NPS) as a quantitative metric was used to study the dependence of noise on algorithm, dose level, and slice thickness. **Results:** Image variance had a strong ($R^2 > 0.99$) power relationship with dose level for all algorithms, with the power around -0.65. Image variance was reduced by 66% from 1.25 mm to 5 mm slice thickness, by 8% for each 10% of ASiR-V strength increment and by 24% for each DLIR level increment from FBP. The shape of NPS from ASiR-V gradually biased towards lower spatial frequency and deviated away from that of FBP. Our results show a deterioration in spatial resolution with increasing ASiR-V strength. NPS from DLIR had very close shapes to that of FBP, for all three strength levels. **Conclusion:** Noise texture with DLIR reconstruction is closer to FBP than ASiR-V. DLIR, as compared to ASiR-V, preserves spatial resolution while reducing imaging noise.

TU-IePD-TRACK 1-06

Assessing the Severity of Metal Artifacts and Detectability of Lesions in CT Metal Artifact Reduction Images

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Purpose: Metal Artifact Reduction (MAR) algorithms are becoming increasingly utilized to improve image quality and lesion detectability around prostheses. Performance assessment of these algorithms in clinical studies is complex due to ethical issues and lack of reference standards to confirm the presence of lesions. The goal of this study was to develop a quantitative method to assess the severity of streak artifacts and detectability of lesions in CT phantom images with and without MAR via change in image noise and HUs around metal inserts. **Methods:** A semi-anthropomorphic phantom (40 HU) manufactured with spherical lesions of various attenuations (20, 25 and 30 HU) embedded around arthroplasty inserts, cobalt-chromium sphere attached to a titanium rod, was scanned at 140 kVp, 300 mAs and 0.6-mm collimation. A control material (120 HU), simulating bone, was used instead of metal inserts. Images were reconstructed using FBP and MAR with identical parameters. Axial images were retrospectively analyzed with a post-processing software where multiple circular ROIs, near and away from the inserts were drawn and applied consistently to each series. **Results:** The degree of streak artifact (noise ratio to the control within corresponding ROIs) in MAR images were graded as significantly less ($p < 0.001$, Wilcoxon signed rank test) than for the FBP images at both near and away from the inserts. There was less CT number change (exceeding 20 HU compared to control) in MAR images than FBP (30% and 94% of the ROIs respectively) around the cobalt-chromium insert while this change was lower around the Titanium rod (25% for FBP and none for MAR images). These changes made two of the lesions undetectable in FBP images while all were distinguishable in MAR. **Conclusion:** MAR achieved more accurate attenuation values and less streak artifact than FBP, providing better detection of lesions near metal inserts in a phantom model.

TU-IePD-TRACK 1-07

"Focused" CT: A Reconstruction Algorithm Permitting Radiation Dose Reduction for Region-Of-Interest Viewing

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Purpose: CT scans often image an anatomical region of interest (ROI), such as the spine. Traditional reconstruction algorithms require roughly-equal signal levels throughout the axial plane of the body, a limitation which has stymied any attempt to reduce radiation dose to off-ROI tissues not of clinical interest. We propose a modified filtered back-projection (FBP) algorithm, Focused CT (FCT), that permits accurate reconstruction of a limited scan-FOV with a 90% or greater reduction of radiation exposure outside the clinical ROI. **Methods:** FCT splits the Fourier transform into low-frequency and high-frequency components. The high-frequency component uses low-noise data from within the ROI (acquired at a typical clinical dose). The low-frequency component uses high-noise data from the entire axial plane, with the off-ROI data acquired at 10% of the typical dose. To simulate a single CT exam acquired at different doses, a cadaver was scanned twice, with CTDIvol values of 17 mGy and 1.7 mGy. Matlab was used to create several hybrid slices combining ROI data from the 17-mGy acquisition with off-ROI data from the 1.7-mGy acquisition. Projection data for these combined images were simulated, then re-reconstructed using the FCT algorithm. The resulting HU values were compared to those from the 17-mGy image alone: quantitatively by use of the mean-square error, and visually by creation of a difference image. **Results:** ROI images generated with the 1.7-mGy off-ROI dose appeared identical to those using the 17-mGy dose throughout the axial plane. The mean-square error was negligible (0.0041). The difference image showed no perceptible structures within the ROI. **Conclusion:** The FCT algorithm permits accurate reconstruction of an ROI, with off-ROI dose reduction of at least 90%. The development of a mathematical basis for "focused" reconstruction may lead to the engineering of a physical mechanism to reduce radiation dose to off-ROI tissues.

Multi-Disciplinary Interactive ePoster Discussion TRACK 3

Multi-Disciplinary: Adaptive Radiation Therapy

TU-IePD-TRACK 3-01

A Method to Select Optimal Candidates for Adaptive Radiation Therapy (ART) in Anal Cancer

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Purpose: Adaptive Radiation Therapy (ART) is a treatment technique that may allow for reducing radiation dose to normal structures, such as bowel. However, accessibility to this treatment modality is limited because it is resource intensive. Here, we propose a method to select anal cancer patients that will benefit most from ART in terms of bowel sparing, allowing more efficient allocation of resources. **Methods:** Retrospective treatment data from 4 patients (5400cGy/30 fractions) were evaluated. Daily adapted CTs were created in Velocity v4.1 based on daily CBCTs. The structure sets from the planning CTs were deformed onto each adapted CT and ultimately corrected and revised by two qualified radiation oncologists. The original treatment plan was recalculated on each fraction's adapted CT to establish the true dose received during treatment. Dose statistics (Mean, Max, Min) and DVHs were extracted and analyzed. **Results:** The ratio (V4500cGyRatio) of the initial plan bowel V4500cGy to the plan-specific constraint was used to stratify patients into three groups (V4500cGyRatio < 0.65, 0.65 < V4500cGyRatio < 0.85, V4500cGyRatio > 0.85). Every fraction was classified as pass/fail based on the true daily bowel V4500cGy meeting constraint. Daily bowel V4500cGy significantly ($p < 0.05$) correlates with daily bladder volume. A scatter plot of the daily/planned bladder volume ratio vs. the V4500cGyRatio of the initial plan for every fraction shows which combinations of these lead to passing or failing the plan-specific constraint. For inter-fraction bladder volume variability at our clinic, V4500cGyRatio < 0.65 needs little adaptation (2/30 fractions), 0.65 < V4500cGyRatio < 0.85 needs daily adaptation depending on daily bladder volume (10/30 fractions), and V4500cGyRatio > 0.85 requires daily adaptation (29/30). **Conclusion:** Patient plan V4500cGyRatio can help determine which anal cancer patients will benefit most from ART in terms of bowel sparing. Decreasing inter-fraction variability of bladder volume may reduce the number of patients requiring ART.

TU-IePD-TRACK 3-02

A Prediction Model for Dosimetric-Based Lung Adaptive Radiotherapy

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Purpose: To quantify anatomical changes of lung cancer patients and evaluate the resulting dosimetric effect according to on-treatment cone-beam computed tomography (CBCT) scans. To develop a dosimetric-based lung adaptive radiotherapy (ART) decision tool based on features of patients' anatomy and anatomical changes. **Methods:** Eleven non-small cell lung cancer patients were enrolled in this study. A planning CT (pCT) and weekly CBCTs of each patient were collected. A virtual CT (vCT) was generated by deforming the pCT to each CBCT and the dose distribution was recalculated on the vCT (DvCT). The replanning status of a CBCT was labeled as positive if the DvCT was clinically unacceptable or could be potentially improved according to two sets of dosimetric criteria, otherwise labeled as negative. The region of anatomical changes (region of interest, ROI) between each CBCT and the pCT was extracted. Twenty-four features from the ROI and the pCT anatomy were extracted for each pCT-CBCT pair. A model based on nonlinear supporting vector machine was built for replanning status prediction. A nested cross-validation (CV) scheme was used for model evaluation, where the recursive feature elimination method was applied for feature selection. **Results:** Thirty-Five CBCT-pCT pairs were identified of having a ROI, among which only 7 CBCTs had positive replanning status. Six features were selected for model fitting and the top 6 most frequently selected features were from both feature categories. A high prediction accuracy of 0.84 was achieved and the area under the mean curve was

0.82. A balanced sensitivity and specificity of 0.70 could be obtained by lowering the probability threshold. **Conclusion:** Our preliminary results demonstrated that the proposed model has the potential to identify the replanning need for lung cancer patients from CBCTs. Further study will be done to alleviate the imbalanced class problem and refine the model performance.

TU-IePD-TRACK 3-03

Analysis of the Benefits of Daily Online Adaptive Radiotherapy Via Ethos for Prostate Cancer

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Purpose: to evaluate the Varian Ethos online-adaptive platform for prostate cancer including 1.auto-segmentation accuracy for CTV and OARs, 2.achievable dosimetric improvements, and 3.minimal margin necessary with daily adaptation. **Methods:** 25 retrospective prostate patients were re-planned using an Ethos emulator, 12-field IMRT, and 2Gy fractions. CTVs included intact prostate and 1cm proximal seminal vesicles. Our clinical asymmetric margin was used: 3mm posterior, 5mm left/right/anterior, and 7mm superior/inferior. Adapted and non-adapted dose distributions for 10 fractions per patient were calculated using their on-treatment CBCTs. No corrections to the auto-segmentation were performed prior to adaptive plan optimization, however all contours were reviewed/corrected for dosimetric evaluation. The auto-segmentation was evaluated by comparing DVH metrics from auto-segmented versus corrected contours. Dosimetric improvements were measured by comparing DVH metrics from adapted and non-adapted doses to corrected contours. Minimum safe CTV-to-PTV margins were determined by replanning each fraction with a clinically-validated RapidPlan™ knowledge-based planning routine and symmetric 2-5mm margins. Then CTV-Dmin with reduced margins and daily adaptation was compared to standard margin without adaptation. **Results:** 70% of CTV edits (88% Bladder, 90% Rectum), were less than 10% of the total volumes and resulted in small average changes of: 0.13%±2.9% CTV-D95%, 0.43%±3.3% Bladder-V90%, and 0.12%±5.1% Rectum-V90%. Without adaptation, CTV-Dmin was greater than 95% of prescribed dose for only 45% of fractions; adaptation with clinical margin increased this to 83%. For non-adapted fractions where the OAR-V90% exceeded its clinical threshold, adaptation reduced it on average by 0.6%±4.0%(Bladder) and 2.0%±5.5%(Rectum). For these cases, using 2mm margin preserved an improvement in tumor coverage(65% fractions CTV-Dmin>100%), and further improved the OAR-V90 metrics by 3.5%±6.0%(bladder) and 11.5%±4.7%(rectum). **Conclusion:** we established Ethos's online adaptive workflow without pre-treatment segmentation edits can substantially improve prostate CTV coverage and reduce OAR dose. This approach minimizes the impact on the clinical workflow and dosimetrically benefits patients.

Dr. Ray has a lab services agreement with Varian Medical Systems. Dr. Moore reports income for personal consulting and speaker's honoraria from Varian Medical Systems. This work was supported in part by the Agency for Healthcare Research and Quality (AHRQ R01HS025440)

TU-IePD-TRACK 3-04

Elimination of the Set-Up Margin Using Daily Adaptive Radiotherapy for Head and Neck Radiotherapy: A Feasibility Study

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Purpose: Involved Nodal Radiotherapy using AI-based Radiomics (INRT-AIR) for head-and-neck cancer eliminates the elective neck treatment altogether and focus treatment on the nodes themselves. This pilot dosimetric study leverages adaptive therapy to further reduce PTV margins and assess resulting dosimetric gain. **Methods:** A total of 10 patients treated with INRT-AIR were recruited and replanned with Ethos treatment-planning-system. Ethos plans were created with planning-target-volume (PTV) margin 5mm and 1mm (for intrafractional-motion), respectively. The plans were virtually delivered in the Ethos emulator using the CBCT acquired in the patient's prior treatments. The 5mm plans were delivered with image-guidance-radiation-therapy (IGRT) to mimic the current delivery while the 1mm plans were delivered with daily adaptive therapy. The initial plan quality was compared to quantify the dosimetric improvements. The adaptive plan quality was compared with the initial plan to evaluate the robustness of the 1mm INRT-AIR plan against patient's daily anatomy change. Independent fractions of delivered dose of the two cohorts were compared to evaluate the efficacy of 1mm adaptive therapy over the conventional 5mm IGRT. The gross-tumor-target (GTV) and clinical-target-volume (CTV) coverages along with the organs-at-risk (OAR) dose were the denominators. A dose difference of 300cGy was considered clinically significant. **Results:** Among the high impact OARs, the 1mmINRT-AIR plans exhibited clinically significant improvements in contralateral-submandibular-gland (Avg.1775cGy), constrictor muscle (Avg.457cGy), contralateral-parotid-gland (Avg.363cGy), and mandible(Avg.335cGy). Despite the adaptive workflow is fully automatic and user has very minimum control of adaptive plan optimization, over 83% frequency that 1mm adaptive plan recovers the initial plan quality. While 5mmIGRT delivery degraded both the target coverage and OAR sparing, 1mm adaptive delivery holds the promise of dosimetric benefit from margin reduction. **Conclusion:** We demonstrated the significant dosimetric benefit of dramatic PTV margin reduction made viable using daily-adaptive-radiotherapy, even in the context of INRT-AIR planning that has improved OAR dosimetry.

TU-IePD-TRACK 3-05

Initial Experience of CBCT-Based Adaptive Radiotherapy with Ethos for Postoperative Prostate SBRT

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Purpose: To investigate whether Cone-Beam Computed Tomography (CBCT)-based adaptive planning improved dosimetry for patients receiving postoperative prostate stereotactic body radiotherapy (SBRT) over 5 fractions. **Methods:** A retrospective cohort of 5 patients with prostate cancer receiving adjuvant or salvage radiotherapy to the prostatic fossa were included. With the Ethos emulator environment, initial treatment plans were created using baseline CT simulation images and prescribed to 32.50Gy in 5 fractions with a 3mm planned target volume (PTV) expansion (initial plan). Subsequently, 25 fractions were delivered by two separate **Methods:** (1) adaptation to the CBCT of the day (adapted plan) and (2) dose recalculation using the initial plan on the same CBCT (scheduled plan). Dosimetry for targets and organs at risk were recorded. Independent fractions were compared between the initial, adapted, and scheduled plans with paired-samples t-tests. **Results:** Rectal dosimetry improved with CBCT-based adaptation (V25Gy 10.7 + 3.4 cc), when compared to both the initial (16.4 + 7.2 cc, p=0.001) and scheduled (16.4 + 8.3 cc, p<0.001) fractions, which appeared primarily secondary to changes in rectal distension from gas or stool. Bladder wall dosimetry slightly improved with adaptation (V25Gy 32.7 + 6.4 cc) when compared to the initial (35.5 + 33.6 cc, p=0.002) and scheduled (33.6 + 6.4 cc, p=0.024) fractions. PTV coverage was acceptable for all fractions: adapted (D95% = 100%), initial (100%), and scheduled (98.5% + 1%). Further, this adaptive planning process on Ethos is largely automated, taking an average of 12.9 + 1.7 minutes to generate a new adaptive plan from the patient's daily CBCT, of which only 5 to 6.5 minutes were spent manually editing contours. **Conclusion:** Overall, CBCT-based adaptation with Ethos provided superior plans with better rectal sparing whilst maintaining excellent target coverage. Ethos offers a fast and effective method to generate adaptive plans for postoperative prostate SBRT.

TU-IePD-TRACK 3-06

Use of Deep Learning Segmentation and Biomechanical Models to Improve Dose Accumulation Accuracy in GI Structures

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Purpose: Dose accumulation in luminal gastrointestinal (GI) structures during radiotherapy (RT) for abdominal cancers can potentially improve toxicity prediction

models. The purpose of this study is to assess the benefit of deep learning (AI) and biomechanical model-based deformable image registration (DIR) to improve DIR-based dose accumulation accuracy. **Methods:** Dose accumulation was conducted on 75 patients with primary and metastatic liver cancer treated with external beam RT guided by daily CT-on-rails (CTOR), using a “Merged” DIR approach with biomechanical model-based DIR (FEM) inside the liver and intensity-based DIR outside. On a sub-cohort, FEM with boundary conditions on the liver and combined stomach and duodenum (GI_ROI) was applied (FEM+GI). The GI_ROI was auto-contoured on CTOR using a 3D UNet model trained on another 102-patient cohort. Contours were compared to physician-drawn contours on 8 CTOR, using Dice similarity coefficient (DSC), mean distance to agreement (DTA), Hausdorff distance (HD), and planned dose metrics. Intra-observer variability was evaluated. DSC was evaluated on the stomach and duodenum propagated after application of the Merged DIR approach. Accumulated doses based on Merged and FEM+GI were converted to EQD2 ($\alpha/\beta=3.5$) and compared. **Results:** Average(SD) DSC, DTA, and HD between AI and physician-drawn GI_ROI were 0.9(0.1), 0.2(0.1)cm, and 2.4(2.2)cm; intra-observer: 0.9(0.0), 0.1(0.0)cm, and 2.3(0.8)cm, $p>0.1$. Average D50(SD) and D1(SD) differences between AI and physician-drawn contours were 0.6(1.1)Gy and 0.5(0.2)Gy; intra-observer: 0.2(0.3)Gy and 0.3(0.3)Gy, $p>0.1$. Average(SD) DSC of the stomach and duodenum mapped via Merged were 0.8(0.1) and 0.55(0.2), respectively, compared to manual contours. Average(SD) accumulated D1 (EQD2) using Merged and FEM+GI were 26.2(13.8) and 25.6(13.7)Gy for stomach, and 20.3(18.0) and 19.6(18.1) for duodenum, respectively. These distributions were significantly different ($p<0.05$), with differences ranging 0.7-3.9Gy. **Conclusion:** The use of AI-generated segmentation to drive biomechanical model-based DIR led to significant differences in the accumulated dose estimations which may improve toxicity prediction models.

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TU-lePD-TRACK 3-07

CBCT-Based, Physicist-Driven Daily Adaptive Protocol for Locally Advanced Cervical Cancer: Improved Target Coverage and Reduced Margins in a 20-Minute Treatment Slot

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Purpose: To evaluate a physicist-driven, CBCT-based daily adaptive protocol in locally advanced cervical cancer for multiple endpoints: on-couch time, physicist contouring accuracy of daily CTVs, and target coverage with reduced PTV margins versus non-adapted standard-of-care. **Methods:** CT scans of 13 retrospective cervical cancer patients with original CTV_SIM were re-planned on the Varian Ethos platform using 12-field IMRT and 3mm symmetric CTV-to-PTV margins. Each plan was adapted on five CBCT scans to simulate Ethos’ auto-segmentation and auto-planning workflow. Required contouring edits were made by one of three physicists, mimicking daily adaptation without real-time physician review. Time spent editing OARs and CTVs was recorded, as was total treatment time (CBCT acquisition to treatment start, inclusive of auto-segmentation, contour review/edits, replanning, and evaluation). All physicist-adapted targets, CTV_PHYS, were reviewed by an experienced gynecologic radiation oncologist for accuracy and, when necessary, corrected to establish ground truth, CTV_GT. The adaptive coverage frequency was compared to non-adaptive standard-of-care treatment by assessing whether daily CTV_GT was contained inside PTV_PHYS=CTV_PHYS+3mm and/or PTV_SOC=CTV_SIM+(5mm-nodal, 10mm-vagina/parametrium, 15mm-uterus/cervix). **Results:** The average time spent editing OAR/CTV contours was 3.8(range:1.0-15) and 7.1(range:2.5-12.5) minutes, respectively. The average total time was 16.7(range:7.0-29.0) minutes. 54/65 sessions were completed <20 min; only 1/65 required >23 min. Physician reviews of CTV_PHYS resulted in: 47/65 requiring no further editing (CTV_GT=CTV_PHYS), 14/65 requiring target size reduction (CTV_GT<CTV_PHYS), and 4/65 requiring target size expansion (CTV_GT>CTV_PHYS). CTV_GT extended outside the non-adapted PTV_SOC in 53/65(81.5%) fractions, while the daily CTV_GT extended outside of PTV_PHYS in only 4/65(6.2%) fractions. **Conclusion:** Our results demonstrate that CBCT-driven, daily adaptation of cervical cancer without real-time physician review is feasible in a typical time slot (<20 minutes). With daily adaptation, even substantially reduced PTV margins resulted in markedly increased CTV coverage frequency. These margin reductions would also improve normal tissue sparing, providing further motivation for daily adaptation.

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Multi-Disciplinary Interactive ePoster Discussion

TRACK 4

Multi-Disciplinary: Segmentation II

TU-lePD-TRACK 4-01

A Deep Reinforcement Learning Based Pipeline for Prostate Segmentation on MRI with Low Variance Performance

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Purpose: To effectively address the scarce annotation problem, we proposed a Deep Reinforcement Learning (DRL) based pipeline approach for image segmentation. Unlike supervised machine learning, DRL is a self-learning algorithm to learn through the interactions with the given environment, and adjust the action based on observed state and received reward. We aim to develop a DRL-based approach for prostate gland segmentation on MRI by self-learning, and introduce a novel post processing approach based on the Kalman filter to enhance the segmentation accuracy in 3D volumes. **Methods:** We considered the segmentation as a painting task and trained a DRL agent to draw the structure’s boundary in a sequential decision-making approach. The pipeline has two phases: (a) a backbone encoder capturing the global information and providing the start point for the agent; (b) a DRL based agent to delineate boundary on 2D MRI by interacting with the environment. The segmented slices for each patient were then aggregated and post-processed using a Kalman filter to rectify the contour results in the 3D space. The proposed model was trained with SPIE-AAPM-NCI PROSTATEx-2 Challenge (SPIE) dataset, and evaluated on both SPIE and an in-house dataset. Dice similarity coefficient (DSC) and 95% Hausdorff distance (HD) were used as the evaluation metrics. **Results:** The proposed method achieved a DSC of 87.88%±3.0%, 87.89%±1.7% on the validation and test based on the 2D slices, and 85.20%±10.3% and 85.27%±10.4% on the 3D volumes using the SPIE data. The results were 84.18%±4.3% and 81.72%±11.7 on the 3D and 2D respectively using the in house data. The 3D 95% HD was 4.08±0.94 mm and 5.28±2.21 mm on the validation and test, 6.49±2.48 mm on the in-house dataset. **Conclusion:** The DRL agent provides promising prediction results on organ segmentation and the Kalman filter-based post segmentation shows its effectiveness in alleviating the segmentation variance.

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TU-lePD-TRACK 4-02

Automatic Segmentation of Low-Contrast Head and Neck Tumors for Palliative Radiotherapy

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Purpose: To automatically contour gross tumor volumes (GTVs) for palliative head-and-neck (HN) radiotherapy at multiple treatment sites from single-modality, low-contrast scans. **Methods:** 110 palliative and 234 curative treatment plans (CT scans and physician-approved GTV contours) for nine HN treatment sites were retrospectively obtained. Mean target volume was 88.9 (σ 123.3) cc, and both palliative and curative cases were evenly split 80%/20% into training-validation and test sets, respectively. First, state-of-the-art CNNs (e.g. U-Net) were evaluated. Then, a novel, coarse-to-fine cascading 3D Attention U-Net was developed to segment the GTVs. Model performance was evaluated by comparing Dice similarity coefficient (DSC) and mean surface distance (MSD) to measure target coverage and structure similarity of predicted and clinical GTVs. **Results:** Over the entire test dataset (curative & palliative cases), the best scores achieved with the conventional U-Net architecture were median DSC and median MSD of 0.65 (σ 0.22) and 2.08 (σ 7.65) mm, respectively. For the cascading 3D Attention U-Net, these scores changed to 0.67 (σ 0.23) and 2.26 (σ 4.27) mm; thus, this approach improved both median DSC and the extent of MSD outliers. Segmentation of large, high-visibility tumors performed better than smaller or low-visibility tumors. Across treatment sites, base of tongue, mandible, and thyroid median DSC scores were 0.70 or greater, with mandible achieving the highest median DSC (0.74). Additionally, median DSC scores for glossopharyngeal sulcus, head and neck, thyroid, tonsil, and other were 0.60 or greater. **Conclusion:** We developed a novel 3D segmentation network to find GTV delineations for palliative HN treatments. To our knowledge, this is the first work developed for multiple HN treatment sites.

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TU-lePD-TRACK 4-03

Evaluation of Deep Learning-Based Automatic Segmentation of the Pancreas

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Purpose: Investigate the performance of multiple deep learning (DL) models for the automatic segmentation of pancreas on contrast enhanced CTs. **Methods:** A total of 595 CTs of pancreas with various characteristics were gathered from our institution and publicly available datasets. The population was separated into 425, 60, and 110 CTs for the training, validation and withheld test datasets. Both the training and validation datasets were composed of cyst, pancreatic and extra pancreatic tumor, pancreas from our institution (n=204) and the MICCAI Medical Segmentation Decathlon (MSD) (n=281). The withheld test dataset was composed of healthy pancreas from the NIH-82 (n=80) and SYNAPSE MICCAI (n=30) challenges. Four DL models were investigated: attention Unet (attUnet), 2D DeepLabV3+, 3D patch-based BasicUnet from MONAI, and 2-step Unet from RaySearch Laboratories (RSLab). The segmentation resulting from the majority vote (MV) of all DL models was also evaluated. The DL models' performance compared to manual segmentation was reported using the Dice similarity coefficient (DSC), distance to agreement (DTA), Hausdorff distances (100thHD and 95thHD). The intra- and inter-observer variability was reported for 20 pancreas segmentations. **Results:** The median (min-max) DSC between manual and automatic pancreas segmentations on the test dataset were 0.78 (0.16-0.88), 0.80 (0.44-0.89), 0.80 (0.00-0.88), 0.82 (0.34-0.88), 0.82 (0.59-0.89), for the attUnet, DeepLabV3+, RSLab, MONAI, and MV models. Majority vote was able to reduce the number of segmentations with a DSC <0.7 to 5.0% and 7.3% for the validation and test datasets. Using the MONAI as reference, the disagreement between the MONAI and DLV3 models was able to detect 72% of the failing cases (DSC<0.7). The median (min-max) intra- and inter-observer delineation DSC were 0.83 (0.62-0.92) and 0.87 (0.78-0.91). **Conclusion:** Multiple DL model's majority vote improved the performance similarly to observer variability. DL model agreement could be used to automatically detect cases where DL fails.

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TU-lePD-TRACK 4-04

Improving Deep Learning Auto-Segmentation Using an Adaptive Spatial Resolution Approach

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Purpose: Deep learning-based auto-segmentation (DLAS), although has been shown to substantially improve organ delineation efficiency, performs generally poor for organs with small/narrow size (e.g., inner ears) or complex shape (e.g., bowels). To overcome this problem, we introduce an adaptive spatial resolution approach (ASRA) for DLAS. **Methods:** The framework of ASRA uses a fine spatial resolution in a subregion that contains small/narrow and/or complex anatomy while using default or coarse resolutions in other regions to balance the accuracy and efficiency of DLAS. We implemented ASRA in 3DResUnet and trained the DLAS models for selected organs in head and neck (HN) based on CT by using a fine spatial resolution of $1 \times 1 \times 2 \text{ mm}^3$ inside the subregions, and $2 \times 2 \times 2 \text{ mm}^3$ outside it. To overcome a common issue that a portion of a large organ (e.g., small bowel) far away from the target may not delineated or may be missing in the training dataset, we extended existing contours via 8-fold repetition of last contoured slice using 3D elastic transformation, thus stimulating a larger field-of-view. DLAS models were trained with 47 datasets for HN and 58 for abdomen. Performance of the obtained models with and without ASRA was compared. **Results:** DLAS accuracy was substantially improved with the use of ASRA compared to that without ASRA, with maximum improvement observed for small/narrow organs, e.g., eye lens and optic nerves with dice similarity coefficient (DSC) of 0.74 – 0.8 vs. 0.25 – 0.45. For complex structures, e.g., small and large bowels, DSC/mean distance to agreement (MDA) of 0.84/3.8 mm and 0.87/2.37 mm vs 0.71/7.9 mm and 0.7/14 mm, were observed with 3D extended volume respectively. **Conclusion:** The proposed adaptive spatial resolution approach can substantially improve accuracy of deep learning-based auto-segmentation, particularly for organs with small/narrow sizes or complex shapes.

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TU-lePD-TRACK 4-05

On the Application of a Variational Autoencoder (VAE) and Transfer Learning to Account for Inter-Observer Uncertainties in Automatic Prostate Gland Segmentation

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Purpose: Inter-observer variation is an important issue for automated segmentation in the clinic. To account for inter-observer variability in delineating the prostate gland, we employed a deep neural network architecture, comprised of a U-net and a variational autoencoder (VAE). VAE encodes inter-observer variability and produces multiple segmentations for each image CT slice robust to the inter-user uncertainty. To our knowledge, this is the first such application of a VAE. **Methods:** The source dataset contained 300 patient CT images in which the prostate was delineated by a single physician. The target dataset used for incorporating inter-observer variation was comprised of 10 patient CT datasets, where the prostate for each patient was segmented by 5 independent physicians. Data augmentation utilizing random rotation (<5degrees), cropping and horizontal flipping was applied to each dataset to increase the sample size by factor of 100. Probabilistic hierarchical U-net with VAE (PHiSeg) was pre-trained using the source dataset for 30 epochs. Model weights were then transferred to fine tune PHiSeg using the target dataset for 100 epochs via transfer learning. Ten iterations using random permutation sampling (with training/validation/test ratio of 6:1:3) of the augmented

target dataset were performed. **Results:** Average results of models trained with the target dataset only versus source followed by target datasets (transfer learning) were as follows: Dice score=0.76+0.03 vs. 0.80+0.02 ($p=0.001$); Hausdorff distance (mm)=11.48+2.28 vs. 10.18+1.35 ($p=0.019$); normalized-cross-correlation=0.52+0.10 vs. 0.62+0.06 ($p=0.006$); generalized energy distance=0.33+0.09 vs. 0.26+0.06 ($p=0.002$). All metrics were shown to improve significantly using the proposed VAE and transfer learning approach. **Conclusion:** A VAE combined with a hierarchical U-net demonstrated promise toward accounting for inter-observer variability in automatic prostate segmentation. Multiple segmentations for each CT slice enable users to determine clinical tradeoffs in selecting the “best fitting” contour, which offers advantages over a standard U-net, where output is limited to just one contour.

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TU-IePD-TRACK 4-06

Deploying Deep Learning-Based Image Segmentation Models Via CERR

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Purpose: Despite the availability of several general-purpose platforms for deep learning-based segmentation (DLS), we lack similarly comprehensive platforms with functionality specific to medical images. We present an open-source extension to the Computational Environment for Radiological Research (CERR) for deploying DLS models that leverages CERR's existing functionality for radiological data import, transformation, management, and visualization. **Methods:** CERR's deep learning pipeline utilizes Java Script Object Notation (JSON) format configuration files to specify input and output data formats; methods and parameters for transforming images, populating channels, and post-processing segmentations. Several pre-processing operations specific to the medical imaging domain are supported, including automated cropping around patient outline, 2D/3D resampling, 2D/2.5D/3D and multimodal channels, and orientation transformations (transverse, sagittal, coronal). CPU and GPU implementations of pre-trained DLS models are packaged using Singularity containers (Linux) and Conda environment archives (Windows, macOS) for distribution. The returned segmentation labels are imported to CERR for further processing and export to DICOM format. Various post-processing operations are supported to minimize manual editing, including filtering to retain connected component(s) by size or within a user-defined region of interest. Custom pre/post-processing functions and dynamic label-to-structure mapping (where the number output labels is variable) are supported through the use of flexible JSON-format configuration files. This pipeline has additionally been made compatible with GNU Octave for license-free use. **Results:** CERR's DLS pipeline, along with a JupyterLab notebook demonstrating its usage, are distributed as open-source GPL-copyrighted software at <https://www.github.com/cerr/CERR>. It has been successfully used in conjunction with MIM workflows to deploy DLS models developed in-house for segmenting organs-at-risk on head & neck CT images and T2-weighted MR prostate images in the clinic. **Conclusion:** We developed a comprehensive, open-source pipeline for deploying deep learning-based medical image segmentation models. In addition to its clinical applications, this pipeline facilitates reproducible and consistent segmentation across institutions for research purposes.

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TU-IePD-TRACK 4-07

Saliency-Guided Deep Learning Network for Automatic Gross Target Volume Delineation in Partial Breast Irradiation

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Purpose: To accommodate an efficient scan-plan-treat clinical workflow in partial breast irradiation (PBI) such as GammaPod, fast, accurate and automated target delineation is desired. In this study, we develop a saliency-based deep learning segmentation (SDL-Seg) algorithm for automatic gross tumor volume (GTV) delineation in post-op breast irradiation. **Methods:** Our approach incorporates saliency information into a U-Net model for target delineation. The visual saliency maps were generated using surgical clips as markers identified on CT images, and a distance-transformation coupled with a Gaussian filter were adopted to convert markers' locations to probability maps. The CT images and the corresponding probability maps form a two-channel input for the segmentation network. The dataset used for model training, validation and testing (19:5:5) is comprised of 145 prone CT images from 29 post-operative breast cancer patients, who had surgical clips and received 5-fraction PBI regimen on GammaPod. We used the Dice similarity coefficient (DSC), 95 percentile Hausdorff distance (HD95) and average symmetric surface distance (ASD) to assess segmentation results using SDL-Seg and compared it with those generated with basic U-Net. **Results:** Our model achieves mean (standard deviation) of 76.4(2.7) %, 6.76(1.83)mm, and 1.9(0.66)mm for DSC, HD95, and ASD respectively outperforming basic U-Net by 13.8%, 1.63mm, and 0.9mm. For all 5 testing cases, the saliency-guided U-Net showed increased DSC compared to basic U-Net. The average DSC for each case using SDL-Seg are 74.3%±4.8%, 79.35±3.7%, 75.6±1.7%, 73.6%±4.4% and 79.3%±1.5%, while using basic U-Net the outcomes are: 70.8.6±2.1%, 57.8%±21.1%, 63.2%±4.3%, 51.9%±9.5% and 69.2±3.7%. **Conclusion:** We developed a deep learning model integrating visual saliency information into the network for GTV delineation. Results demonstrate that SDL-Seg outperforms basic U-Net and is a promising approach for improving the efficiency and accuracy of the on-line treatment planning procedure of PBI, such as GammaPod.

Therapy Interactive ePoster Discussion

TRACK 5

Therapy: Particle Therapy Verification

TU-IePD-TRACK 5-01

A Feasibility Study of Proton Beam Dosimetry Using Cerenkov Radiation

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Purpose: To study the relationship between Cerenkov photon emission and absorbed dose in water, soft tissue, and cortical bone for a primary proton beam. A proportionality constant between the number of Cerenkov photons and the absorbed dose is produced, which will aid in future dosimetry work. **Methods:** A proportionality constant is introduced to quantify the amount of Cerenkov radiation emission per Gray of absorbed dose. MCNP 6.2 is then used to obtain said proportionality constant for a typical proton therapy spread-out Bragg peak. A correlation between proton energy and Cerenkov emission is also obtained for a water, soft tissue, and cortical bone phantom for proton energies of 10 – 150 MeV. The central axis depth dose curve for a solid water phantom was also produced and compared to the central axis Cerenkov emission profile to ascertain the feasibility of using Cerenkov output as an estimator of the Bragg peak location. **Results:** The cortical bone emitted the most Cerenkov photons, followed by water and then soft tissue. The cortical bone produced the most Cerenkov photons due to its larger Sellmeier coefficients, and therefore larger refractive index. The differences between the water and soft tissue output is likely due to the larger electron density for water. The proportionality constant between Cerenkov photon emission and absorbed dose is 6.99×10^8 photons/Gray. The Cerenkov output was found to be a poor predictor of the Bragg peak location. **Conclusion:** Body parenchyma emits Cerenkov photons in proportion with increasing proton energy. A proportionality constant, which relates the quantity of Cerenkov photon emission to absorbed dose, was determined for a primary proton beam. A polyenergetic proton beam was used to incorporate some energy dependence into the parameter. The Cerenkov light output was found to be an inadequate predictor of the Bragg peak location.

TU-lePD-TRACK 5-02

Characteristics Evaluation of Contact Lens in Vivo Ocular Dosimeter for Proton Therapy

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Purpose: The aims of this study were to investigate the characteristics of contact lens in vivo ocular dosimeter (CLOD) which directly measured the in vivo dose delivered by treatment beam in ocular tumor for proton therapy. **Methods:** The proton beam irradiation was carried out using IBA cyclotron PROTEUS 235 at National Cancer Center in Korea. CLOD was composed of biocompatible materials such as LiPCDA, Tartrazine and Sorta clear 40. It was analyzed using an EPSON Expression 10000XL scanner in reflective mode. The analysis of CLODs was used to RIT software (Radiological Imaging Technology, Inc., Colorado Springs, CO, USA). For characteristics evaluation, we had investigated the dose response for RGB channels, linearity, energy & dose-rate dependence, angular dependence and sensitivity. **Results:** For RGB channels, the red channel is the most sensitive one among the RGB channels. The results of linearity showed that the CLOD is linearly related to the irradiated dose in the range from 0 to 15 Gy. The energy response obtained for the 122.9, 159.9, 206.9 MeV/u were 0.2%, 1.6% and 1.6% compared to the 147.3 MeV/u. In the range from 1 to 5 Gy/min, the dose rate responses differ by up to 1.8% compared to the 3 Gy/min. The angular dependence normalized to the response at 0 obtained for different angle was also observed to be <2.0% for all angle. **Conclusion:** The CLOD has been proven to be an in vivo dosimeter which verify the delivered dose compared to the planned dose for proton therapy. The evaluation of dose characteristics in open field condition shows that the CLOD can be applied to an in vivo dosimeter for ocular tumor in proton therapy.

This work was supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT) (No. NRF-2019R1F1A1063078).

TU-lePD-TRACK 5-03

Validation of a Commercial Treatment Planning System for Small-Field Patient-Specific Apertures in Pencil Beam Scanning Proton Therapy

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Purpose: Patient-specific apertures are increasingly being studied and developed for pencil beam scanning proton therapy. RayStationTM, a commercial treatment planning system, is capable of simulating apertures and has previously been validated for general use, however validation is needed for small fields with apertures. This work aims to validate the modelling of apertures with small openings in RayStationTM by comparison to in-water point-dose and film measurements and by cross-validation in patient geometries with MCsquare, an open source Monte Carlo dose calculation engine. **Methods:** The original MCsquare does not support apertures. We enhanced MCsquare to support apertures for cross-validating RayStationTM. Apertures with openings of 1,2,3,4, and 5 cm have been manufactured and used in the irradiation of proton beams in a water phantom. Two setups were used in the irradiation, one with a range shifter and the other without. The doses were measured in the phantom using point-dose detectors and films and were cross-validated by the enhanced MCsquare. We further cross-validated MCsquare and RayStationTM for 10 small-field treatment plans in patient geometries. **Results:** Comparing to the absolute point-dose in-water measurements, MCsquare differed by $1.8\% \pm 3.6\%$ while RayStationTM differed by $1.2\% \pm 1.0\%$. Compared to the in-water film measurements, MCsquare and RayStationTM both performed well with an average 2D Gamma passing-rate of 99.7% and 99.4% (3%/3mm) respectively. A t-test by comparing the film measurements' agreement to the calculation results between MCsquare and RayStationTM suggests that the dose distributions calculated by MCsquare and RayStationTM are statistically indistinguishable. Directly comparing the dose calculations between MCsquare and RayStationTM over 10 patients resulted in 3D Gamma passing-rates of 98.5% (3%/3mm) and 94.1% (2%/2mm) respectively, where the average target volume size was 8.3 cc. **Conclusion:** RayStationTM and the enhanced MCsquare were validated for small field dose calculation with patient-specific apertures in pencil beam scanning proton therapy.

This research was supported by Arizona Biomedical Research Commission Investigator Award, the Lawrence W. and Marilyn W. Matteson Fund for Cancer Research, and the Kemper Marley Foundation.

TU-lePD-TRACK 5-04

Proton Modulation Technique But Not Acceleration Method Is Associated with Variations in Dose-Averaged Linear Energy Transfer

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Purpose: Although the association between proton linear energy transfer (LET) and biologic effectiveness is well established in vitro, this association is unclear in patients. Analyzing this association in clinical data is further complicated by confounding factors such as treatment delivery variations. We previously assessed the effect of proton modulation techniques on dose-weighted LET (LET(D)), but the contribution of different acceleration methods to LET(D) variability remained unknown. Therefore, the purpose of this study was to determine the extent to which proton acceleration methods affect LET(D). **Methods:** We used TOPAS Monte Carlo transport code to model the protons exiting medical cyclotrons and synchrotrons. We calculated the dose and LET(D) from a single proton beam incident on a virtual water phantom with three treatment delivery conditions: passively scattered (PS) protons from a cyclotron, pencil-beam-scanned (PBS) protons from a cyclotron, and PBS protons from a synchrotron. Treatment plans for all conditions matched the absorbed dose distributions. **Results:** Comparing the spatial LET(D) distributions revealed no significant difference between cyclotron- and synchrotron-based treatments with the same modulation technique in (IQR: [2.70, 4.20] keV/μm and [2.65, 4.12] keV/μm, respectively) or near (IQR: [11.89, 14.91] keV/μm and [11.55, 13.94] keV/μm, respectively) the treatment volume. LET(D) significantly differed between the modulation techniques but was unaffected by the acceleration method (average LET(D) ratio (PBS/PS): cyclotron 2.29 +/- 0.44; synchrotron 2.18 +/- 0.39). Furthermore, the spectral fluence incident on the water phantom was characterized by the modulation technique, with minimal variation with acceleration method. **Conclusion:** LET(D) is unaffected by different proton acceleration methods but is affected by modulation technique. These results suggest that multicenter studies aiming to elucidate the biologic effectiveness of proton therapy in patients can safely integrate data from synchrotron- and cyclotron-based systems but should exercise caution when considering data from different modulation techniques.

TU-lePD-TRACK 5-05

Investigate the Delivery Efficiency and Accuracy of Spot-Scanning Proton Arc (SPArc) in Comparison with Conventional IMPT, Automatic Step-And-Shoot Multifield IMPT

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Purpose: To quantitatively investigate the treatment delivery efficiency improvement using Spot-Scanning Proton Arc (SPArc) in comparison with automated step and shoot multi-field IMPT (IMPTauto) treatment and clinical IMPT treatment. **Methods:** Total 12 cases from the three disease sites such as head neck, lung and brain cancer were retrospectively selected. SPARC and IMPTauto plans were generated with the same plan objectives and robustness parameters like clinical IMPT. SPARC utilized a partial arc with 2.5 degree of sampling frequency and IMPTauto used beam angles 40 degree apart covering the partial arc trajectory. A dynamic arc controller is designed to connect the velocity between control points during SPARC and IMPTauto treatment delivery with the following machine configurations (maximal gantry rotation speed 5 degree/s; The maximal acceleration and deceleration are 0.6 degree/s² and 0.48 degree/s²; ±1 degree delivery tolerance window). Total treatment delivery time of each case was compared among SPARC, IMPTauto simulation and clinical treatment log files. 3D gamma index was used to evaluate the treatment delivery accuracy between SPARC and IMPTauto from the simulated dose reconstruction. **Results:** The SPARC and IMPTauto delivery accuracy is clinical acceptable with ±1 degree delivery tolerance window. The average 3D gamma index comparison showed 97.85± 2.77% passing rate for SPARC and 99.07±

0.73% for IMPTauto. Excluding the time spent on the initial patient setup and IGRT workflow, the SPARC could reduce the average treatment time to 175 ± 74 s compared to the clinical IMPT (641 ± 428 s) and IMPTauto (595 ± 254 s). In comparison with clinical IMPT, SPARC simplified the clinical treatment workflow by eliminating the time for each beam request, preparation, transfer time from OIS to delivery system and manually gantry rotation between each treatment field. In comparison with IMPTauto, SPARC effectively reduced the energy layers and spot number. **Conclusion:** SPARC technique could significantly simplify and improve the proton treatment workflow.

The senior author Xuanfeng Ding reports personal fee from the IBA speaker Bureau.

TU-lePD-TRACK 5-06

Hausdorff Distance Between Plan and Delivery Logs for Daily Treatment Quality Assurance

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Purpose: Intensity-modulated proton therapy (IMPT) provides better treatment quality than conventional radiotherapy at the cost of increased complexity. Parameters that could cause catastrophic consequences include energy, spot location, spot weight (MU), and leaf positions for some proton units. This work proposes a novel method for quality assurance (QA) of daily IMPT treatment records. **Methods:** This method was tested on a synchrotron unit equipped with adaptive apertures for shaper penumbra with layer-by-layer and spot-by-spot collimation. Four types of parameters were extracted from daily delivery logs and arranged into a 35-D Hausdorff space (HS), including 18 range shifter plates for energy layer, 14 leaves (7 pairs) for the adaptive aperture, 2 coordinates for spot position, and weight. A unitless weighted Hausdorff Distance (HD) was calculated between the planned and delivered HS for all pulses in a field. The weight of each dimension in the HS is based on the severity of any deviation. An automated report tool was developed to verify treatment records for all patients daily. **Results:** The tool reported deviations in delivery with HD larger than 5. Among the 42 fractions tagged by the tool in the last two months, 36 were not reported by our R&V system. Those errors have been analyzed, and none caused significant dosimetric concern. The analysis provides valuable information identifying the spot index, energy layer, and range shift plates involved in any fault to support repair and preventative maintenance. **Conclusion:** Our machine log file analysis can catch errors originating from data transfer problems, machine delivery issues, and other human mistakes. The dosimetric impact of these errors can be recomputed based on machine delivery conditions. The historic failure trends can assist in maintaining system operation at high efficiency.

TU-lePD-TRACK 5-07

A 3D Printed Phantom for Range Verification of Low-Energy Proton Fields

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Purpose: Range uncertainties have been identified as a major source of error in the delivery of proton radiation therapy. In recent years, tolerances for ranges in clinical proton beams have been proposed and are typically measured using either a scanning chamber in water or a dedicated multi-layer ion chamber. For small fields with range on the order of 3 cm, such as the ones used to treat ocular melanoma, these methods are not practical. To address this unmet need we designed and tested a phantom that can be used to quickly measure the range in these challenging conditions. **Methods:** 3D printed plastics were first evaluated to determine their water equivalent thickness (WET) in a 67.5 MeV proton beam. Once the WET was known, two phantoms were designed. The main component of the phantoms is water with the secondary component being the printed plastic. Both phantoms had a base of 2.5 cm, with either a cone of height 0.7 cm and base diameter of 2.3 cm, or a reverse cone of the same height and diameter. The phantoms were irradiated with a pristine 67.5 MeV proton beam with gafchromic film placed behind the phantom. The resulting 2 irradiations resulted in either a cold circle in the film center (cone) or a cold annulus (reverse cone). The diameter of the circles was related to the range of the measured proton beam. **Results:** The range measurements were less than 1 mm different from the measured range performed using a water column. **Conclusion:** The method presented here is robust and accurate and will be used in an updated Quality Assurance program for small fields with range less than 3 cm. Additional measurements will be performed to determine the sensitivity of this phantom to detect differences of ± 1 mm in range.

Therapy Interactive ePoster Discussion

TRACK 6

Therapy: Pre-Clinical

TU-lePD-TRACK 6-01

Imputation of Truncated Tumor Growth Data Via Bayesian Mixed Modelling

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Purpose: Dropout occurs when an experimental unit, on which one is taking serial measurements, becomes unavailable for further evaluation prior to the planned end of follow-up. A common instance of dropout is the removal of mice from tumor xenograft experiments, which occurs when the animals either naturally die or undergo sacrifice for morbidity. The unobserved tumor volumes from the lost animals are not, strictly speaking, missing, because the animals are no longer alive. Nevertheless, one approach to analyzing such studies is to treat these truncated data values as missing observations and apply techniques from the statistical modeling of dropout in longitudinal studies. We introduce a novel method for imputing the lost tumor volume values, as well as an R package that implements the method. **Methods:** Based on data of measured tumor volumes of mice at multiple prescribed times, we modeled the series of log tumor volumes for each mouse as a set of observations from a multivariate Gaussian distribution. We estimated this model using a Bayesian approach, and sampling values from the posterior distribution of the mean and variance. Using these, for each mouse with dropout we impute the counterfactual tumor volumes (i.e., sample from its predictive distribution), conditional on its observed sequence. We have implemented the method in an R package. **Results:** Plots of the imputed values reveal consistency with the trend of the observed data. Simulation revealed that the method captures true values reliably. Our R package, *tgmix*, is available on Github. **Conclusion:** We have created a computational tool which allows users to impute counterfactual values in longitudinal datasets with dropout. Further directions may include computational optimization of efficiency in implementation, more extensive tests of the method via simulation, and modification of the method to account for different types of dropout, potentially including nonignorable (biased) dropout.

TU-lePD-TRACK 6-02

Amplification of Reactive Oxygen Species Generation Under Gamma-Ray Irradiation for Gold Nanoparticle Mediated Radiation Therapy

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Purpose: An approach of selective enhancement of the tumoricidal radiation effect with gold nanoparticles (AuNPs) have been successfully explored experimentally, both in-vivo and in-vitro. Significant amplification of reactive oxygen species (ROS) generation under irradiation in the presence of AuNPs is proposed as the main mechanism responsible for the biological effect. We have a two-fold purpose in this study: 1) finding a sensor suitable for quantitative characterization of ROS

species in both aqueous media and cell cultures; 2) establishing a reliable protocol for fluorescence measurements with the sensor. **Methods:** Radiolytic products such as hydroxyl radicals and hydrogen peroxide, generated under high-energy photon irradiation can be quantitatively characterized by fluorescence spectroscopy, providing a reliable sensor is identified. 2',7'-dichlorodihydrofluorescein diacetate (H2DCFDA) dissolved in DMSO (dimethyl sulfoxide) was used as a ROS sensor in experiments comparing radiolysis in pure water and 0.005mg/ml concentration AuNP solution (particle diameter 30nm, citrate coating). Samples in glass capillaries were irradiated under Ir-192 HDR treatment source (average energy ~380keV) to dose levels up to 100Gy. Fluorescence signal measurements were performed on irradiated samples within 1 hour after HDR treatment. **Results:** The fluorescence intensity signals were normalized to those of the 0Gy, used as control in each experiment. The signal intensity increased with the dose for both aqueous and AuNP solutions, demonstrating AuNP-mediated ROS enhancement above 30% for higher doses. The sensor was found suitable for aqueous media only after cleavage in cell culture. The concentration of AuNPs used in the study leads to negligible physical dose enhancement. The same sensor was used with cells, demonstrating strong fluorescence signal after irradiation. **Conclusion:** An experiment procedure for quantitative characterization of ROS species in both aqueous media and cell cultures using H2DCFDA was established and validated. Fluorescence signal enhancement is attributed to amplification in ROS production in presence of AuNPs.

TU-IePD-TRACK 6-03

Monte Carlo Simulation of Enhanced Free Radical Generation Due to Presence of Gold Nanoparticle Under High Energy Photon Irradiation

K Xie*, A Ray, D Shvydka, University of Toledo Health Science Campus, Toledo, OH

Purpose: Quantitative experimental characterization of free radical generation under high-energy photon irradiation, demonstrate enhancements in presence of gold nanoparticle (GNP). A concomitant free radical quenching effect leads to decreasing enhancement if nanoparticles are not precipitated immediately after irradiation. The two opposing effects are extremely difficult to decouple in experiment. In this study, Monte Carlo method is used to simulate those reaction processes in order to estimate the true amounts of free radicals generated in water radiolysis. **Methods:** An in-house developed Monte Carlo model is based on a series of interaction probabilities, of which some are fixed based on established reaction parameters, some are set to adjustable parameters varied until a modeled outcome approaches that acquired through experiments. Statistics of different pathways of free radicals are tallied and analyzed as the results. **Results:** Simulations for both pure water and GNP solutions were conducted to fit the measured data. Analysis of simulated data shows total amount of free radical generated with 30 nm diameter GNP in presence at 0.0005 mg/ml concentration is 265%±2% of that of pure water at all dose levels. About 37% of the free radicals are quenched by GNP. Excluding quenched, the ratio of released free radical is 159%±2%. The ratio of detected free radical amounts of two groups are dose dependent, similar to experimental result. **Conclusion:** With Monte Carlo simulations we can successfully estimate the actual total amount of free radicals generated with the presence of GNP, including the fraction quenched, which is not easily measurable through experiments. The total free radical generation enhancement due to GNP compared to pure water is much higher than can be detected. The decreasing enhancements vs dose is due to faster decline of sensor molecule concentration in the presence of GNPs.

TU-IePD-TRACK 6-04

Development of a Treatment Planning System for Preclinical Small Animal Irradiation with Intensity-Modulated Radiation Delivery Capability

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Purpose: Preclinical research on small animals pivots the discovery of novel radiotherapy techniques. Intensity-modulated radiation delivery (IMRD) – a gold-standard in clinical settings – still finds its way into small animal research. To circumvent this issue, we are developing a new generation irradiator that entangles IMRD via an automated rectangular collimator. This work elaborates on the ongoing development of the custom small animal irradiator with emphasis on the treatment planning system development. **Methods:** The developed in-house treatment planning system enables to accomplish the following tasks: load CT images, segment CT geometry into structures, define irradiation setup (isocenter, beam angles, etc.), simulate distribution dose for the forward or inverse radiation delivery using Monte Carlo (MC) simulations. Of note, the modulation of intensities is achieved via apertures of rectangular shapes by a motorized collimator. The x-ray focal-spot intensity map impacts the dose distribution; hence, we measured the x-ray projections of a tungsten ball bearing with the imaging detector available on our irradiator and determined the focal-spot intensity map via an optimization approach. The x-ray focal spot was modeled in our in-house GPU-based MC dose-calculation engine. For IMRD optimization, the problem was solved using the Direct Aperture Optimization algorithm. **Results:** The measured x-ray focal spot size was 1.62 × 1.50 mm² with an inhomogeneous intensity map. IMRD performance was assessed in comparison with the conformal radiation delivery that utilizes a fixed-size rectangular collimator. The evaluation was performed on two phantoms with the bony structure inside the PTV and a nearby C-shape organ. IMRD enabled reduction of the maximum dose in PTV from ~340% to ~230% when normalizing to PTV D95%, and organ D50% from ~117% to ~63%. **Conclusion:** We developed a treatment planning system for IMRD to support preclinical small animal radiation research. Integration with hardware is currently in progress.

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TU-IePD-TRACK 6-05

Hesperetin Enhances Radiotherapy of Human Lung Cancer Cells

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Purpose: Lung cancer is the leading cause of cancer death in the United States and new/improved treatment options are needed. Recent studies show that flavonoids like Hesperetin can bind strongly to the ACE2 receptors abundant on lung cancer cells. We hypothesize that hesperetin, a naturally occurring major flavonoid, inhibits the progression of lung tumor cells and amplifies damage to tumors during radiotherapy. **Methods:** In-vitro survival studies were performed in A549 human lung cancer cell line (adenocarcinoma human alveolar basal epithelial cells) to compare cells treated with hesperetin, with and without radiotherapy vs. controls. Cells were treated with 5 to 100 ug/ml of hesperetin, with and without radiation treatment. Radiation doses of 2, 4, 6, and 8 Gy were delivered via a Small Animal Radiation Research Platform (SARRP). The cells were treated with 0, 25, 50, 75, and 100 ug/ml of Hesperetin concentrations. The survival, toxicity, and cell viability were analyzed using clonogenic death assays. **Results:** Our data show that radiation enhances the effects of hesperetin treatment on human lung cancer cells A549 by reducing the colony-forming ability when cells are seeded in 6-well plates with 400 cells in each well. Irradiation with 2Gy photons demonstrated complete inhibition of colony growth when combined with at least 25 ug/ml of hesperetin treatment. Substantial colony-forming ability was observed via treatment of up to 50 ug/ml concentration hesperetin solution. However, a single treatment of Hesperetin of 75 ug/ml concentration or higher was enough to kill all plated cancer cells in 7 days. **Conclusion:** Radiation demonstrates enhancement of the anti-tumor effect of Hesperetin. The results justify additional studies in-vivo, including delivery with Hesperetin-loaded nanoparticles ('drones') to target lung cancer during radiotherapy.

TU-IePD-TRACK 6-06

Monte Carlo Dose Calculations for Radiobiology Studies

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Purpose: To provide small-animal percentage depth-dose (PDD) data as a function of source-to-skin distance (SSD), field size (FS) and animal size (AS) in order to

translate experimentally-measured TG-319 skin dose to the dose at target depth. **Methods:** The x-ray tube designs for two biological irradiators, the RT250 and the PXi XRAD-320, were simulated using the BEAMnrc Monte Carlo (MC) code to generate 200 and 320kVp photon beams, respectively. The 320 kV beam was simulated with two filtrations: a soft F1 aluminium filter and a hard F2 filter made of aluminium, tin and copper. Beams were collimated into circular fields of 0.5–10cm in diameter at SSDs of 10–60cm. MC dose calculations in 1–5cm diameter homogeneous small-animal phantoms as well as in heterogeneous phantoms with 3-mm diameter cylindrical bone inserts were performed using DOSXYZnrc. The depth doses in three simulated test-cases were estimated by applying SSD, FS and AS correction factors to a reference case (40cm SSD, 1cm FS and 5cm AS) and comparing the results with the simulated doses to assess the accuracy of dose estimation. **Results:** The depth doses for three test-cases calculated at 200, 320 F1 and 320 F2 kVp agreed well with simulated doses, yielding dose differences of 1.4, 0.1 and 1.0%, respectively. Dose enhancement on the bone-tissue proximal interface ranged from 1.2 to 2.5 times the dose in the absence of bone for rib and 1.3 to 3.7 times for cortical bone, whilst the dose drop-off at 1cm depth behind the bone ranged from 1% to 6% for rib and 3% to 12% for cortical bone. **Conclusion:** The developed dose estimation method can be used to translate TG-319 measured skin dose to dose at any depth.

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TU-IePD-TRACK 6-07

Radiotherapy Enhanced by Cherenkov Photo-Activation in a Novel Rat Brain Slice Model

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Purpose: RECA (Radiotherapy Enhanced by Cherenkov photo-Activation) is a novel treatment with potential to add an anti-cancer immunogenic component through Cherenkov activation of a photo-chemotherapeutic (psoralen). Here we investigate RECA in a novel tissue representative in-vitro model consisting of 4T1 murine cancer cells grown on thin slices of viable rat-brain tissue. **Methods:** Six, 12-well-plates, each containing 1cm of agarose supporting a 400 μ m thick coronal slice of viable rat brain tissue were created. Each plate corresponded to an arm of an experiment incorporating the psoralen derivative 4'-aminomethyl trioxsalen (AMT): MV control, MV+AMT, no-irradiation control, and AMT-alone control. 20,000 4T1 cells expressing both mCherry-flourescent and firefly luciferase-luminescent reporter proteins were plated on each rat brain slice hemisphere. The MV arms received ~4 Gy from a 15 MV Varian linear accelerator beam. Images were taken of the plates each day for 5 days, including filtered for mCherry signal. An optimized CellProfiler pipeline quantified cell colony survival and growth. A luciferase assay was done on day 5 as another measure of relative viability. **Results:** The brain slice model was assessed daily, and found viable for supporting 4T1 cell colony growth for up to 5 days. The optimized CellProfiler colony counting pipeline was validated in simulations, and found to have excellent performance, also enabling uncertainty estimation. Integrated intensity analysis of mCherry signal revealed a significant decrease in tumor proliferation by day 5 between the MV control (5.65 ± 0.78 -fold growth) and MV+AMT (3.49 ± 0.52 -fold growth) arms. This result is consistent with psoralen activation in the MV+AMT arm. The growth observed in the Dark control arm was consistent with the 13.6 ± 1.5 hour doubling time for 4T1 cells. **Conclusion:** The early results from ongoing experiments are consistent with psoralen activation and cell kill in RECA treated cells (MV+AMT arm). Further work will be presented to fully quantify the effect.

SAM Professional Symposium

TRACK 1

Overview of Equity, Diversity and Inclusion (EDI) Efforts in AAPM

TU-E-TRACK 1-00

Overview of Equity, Diversity and Inclusion (EDI) Efforts in AAPM

Moderators: Laura Cervino, Memorial Sloan Kettering Cancer Center, New York, NY and Stephen Avery, University of Pennsylvania, Philadelphia, PA

TU-E-TRACK 1-01

B. Gittens, University of Arkansas:

What Is EDI and How Can It Benefit AAPM?

TU-E-TRACK 1-02

J. Pollard-Larkin, University of Texas MD Anderson Cancer Center:

Overview of EDI Efforts in AAPM

TU-E-TRACK 1-03

Audience Q&A and Feedback

Medical Physics is a Physics specialty with documented inequities with regards to gender and race. Currently, data is being collected to better understand the proportion of underrepresented scientists in the field and their experiences in the workplace and within the AAPM. To that end, the AAPM has created the Ad Hoc Committee on Equity, Diversity and Inclusion (AHCDI) to address equity, diversity and inclusion (EDI) goals and to make recommendations to improve the experience of diverse members in the AAPM. This session will explain what EDI is, will discuss the EDI goals of AAPM, and will highlight the achievements and data collection and metrics organized by AHCDI, the AAPM Diversity and Inclusion subcommittee (WMRSC) and the Women's Professional Subcommittee (WPSC). It is important for AAPM members to understand the purpose of the data collected in the 2021 AAPM EDI Climate Survey and through other means as baseline data, and the benefit of tracking EDI metrics for our membership. This workshop will establish a sense of transparency regarding AAPM EDI goals and provide opportunity for feedback from membership to leadership, AHCDI, WMRSC and WPSC, in order to further improve AAPM EDI efforts.

Learning Objectives:

1. Define EDI terms
2. List current and upcoming EDI efforts
3. Describe recently collected EDI data
4. Provide feedback to leadership and involved committees related to EDI efforts

SAM Imaging Educational Course

TRACK 2

Breast Imaging 1: Current State of the Art

TU-EF-TRACK 2-00

Breast Imaging 1: Current State of the Art

Moderator: Ingrid Reiser, University of Chicago, Chicago, IL

TU-EF-TRACK 2-01

R. Zuley, University of Pittsburgh Medical Center:
Clinical Experience: Contrast Enhanced Mammography

TU-EF-TRACK 2-02

R. Mann, Radboud University Nijmegen Medical Center:
Novel Approaches to Screening for Breast Cancer

TU-EF-TRACK 2-03

K. Little, Ohio State Univ:
Contrast-Enhanced Mammography: Physics and QC Testing

TU-EF-TRACK 2-04

C. Hruska, Mayo Clinic:
Molecular Breast Imaging (MBI): Physics and QC Testing

TU-EF-TRACK 2-05

T. Hall, University of WI-Madison/ADCL:
Novel Ultrasound for Breast Imaging

TU-EF-TRACK 2-06

Audience Q & A

Novel breast imaging technologies are increasingly becoming part of routine clinical practice, including contrast-enhanced mammography, molecular breast imaging, and emerging breast ultrasound applications. The goal of this education symposium is to prepare and familiarize the medical physicist with these new modalities. For each modality, clinician and physicist presenters team up to discuss clinical applications and utility, as well as basic physics principles and quality-control procedures. This will allow the medical physicist to gain a thorough understanding of the clinical use, potential pitfalls, and will help them establish and implement a quality control program.

Learning Objectives:

1. To understand clinical utility of novel breast imaging technologies
2. To become familiar with the physics principles of each modality
3. To be able to implement quality control procedures for each modality

SAM Imaging Scientific Symposium

TRACK 3

Quantitative SPECT for Radionuclide and External Beam Treatment Planning

TU-E-TRACK 3-00

Quantitative SPECT for Radionuclide and External Beam Treatment Planning
Moderator: Srinivas Cheenu Kappadath, UT MD Anderson Cancer Center, Houston, TX

TU-E-TRACK 3-01

J. Dickson, University College London:
Evolution of SPECT From Qualitative to Quantitative Imaging

TU-E-TRACK 3-02

S. Kappadath, UT MD Anderson Cancer Center:
SPECT in Radionuclide and External Beam Treatment Planning

TU-E-TRACK 3-03

Q&A & Panel Discussion

Quantitative SPECT is an emerging technology that incorporates standardized uptake values (SUV) and/or activity concentration information to conventional SPECT/CT imaging. In this session we will report on the technical developments that have enabled quantitative SPECT and provide context for understanding its capabilities and limitations. The current role of SPECT/CT for radionuclide treatment planning and external beam treatment planning will be presented through clinical examples with a focus on value and limitations.

Learning Objectives:

1. To learn about the technical developments in terms of acquisition and processing that enabled the evolution of SPECT from qualitative to quantitative imaging
2. To learn about the application of SPECT in radionuclide and external-beam treatment planning

JD has received research grants from GE Healthcare and served as consultant for GE Healthcare. SCK has received research grants from GE Healthcare, Boston Scientific, ABK Biomedical and has served as a consultant for Sirtex Medical, Boston Scientific, ABK Biomedical.

Science Council Session

TRACK 4

Science Council Session: Innovative Technologies to Advance Diagnosis and Treatment

TU-EF-TRACK 4-01

An Imaging Based Biological Framework for Personalizing Radiation Therapy of Lung Cancer Using Interpretable Bayesian Networks and NTCP-Based Treatment Planning

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Purpose: To develop a biological framework for adapting radiation therapy (RT) treatment plans of lung cancer patients by integrating functional imaging, machine learning, and convex optimization. **Methods:** Seventy-four non-small cell lung cancer (NSCLC) patients treated with passive scattered proton (PSP, n=28) and intensity modulated RT (IMRT, n=46) were retrospectively analyzed. Baseline and mid-treatment [18F]-fluorodeoxyglucose (FDG) positron emission tomography (PET) imaging information was integrated with clinicopathological/dosimetric information to estimate the patient-specific response. A novel Bayesian Network (BN)

was trained and validated for predicting grade2+ radiation pneumonitis (RP) at three months post-RT. BN predictions were used to estimate an individualized normal tissue toxicity probability (NTCP) function for each patient. The updated NTCP functions were then integrated in the RT inverse optimization and a convex optimization algorithm was developed to solve the resulting problem. The framework was tested on 20 patients (PSP, n=10; IMRT, n=10) to optimally adapt the dose distribution according to mid-treatment biological information. **Results:** BN model was able to predict the RP risk with an AUC of 0.82 (95% CI=[0.72-0.91]). The treatment plans for 17 patients (85%) were adapted to keep the updated RP risk below 20% (9 in PST and 8 in IMRT groups). Treatment plans for patients whose predicted RP risk exceed the predefined threshold were de-escalated (n=14, average target dose reduction = 5.4Gy). Patients with predicted NTCP lower than 20% saw an increase in average tumor dose after RT adaptation (n=6, average target dose increase = 4.1Gy). Overall, 17% decrease in predicted RP risk was achieved using the proposed methodology (SD=7.4%). No difference between PSP and IMRT groups were observed in terms of RP risk prediction or plan adaptation. **Conclusion:** Mid-treatment adaptation could significantly lower the risk of RP without compromising tumor control. The proposed methodology could help the design of personalized clinical trials for NSCLC patients.

TU-EF-TRACK 4-02

BEST IN PHYSICS (MULTI-DISCIPLINARY): Focused KV X-Rays for Radiotherapy and Imaging of Small Targets

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Purpose: Radiotherapy machines deliver intrinsically divergent x-ray radiation, requiring collimation to a diseased region. The penumbra and power efficiency limit the minimal size of the field to ~5 millimeters for conventional photon and charged particle beams. Therefore, many medical conditions that may be responsive to local radiation cannot be effectively treated because of their small size and susceptibility to intolerable collateral damages to surrounding critical organs. To overcome the fundamental issues of collimated radiation, we are developing a focused kV technique based on novel usage of polycapillary x-ray lenses, which can focus x-ray beams to <0.2 mm in diameter with a dose peak well below the skin region. **Methods:** Unlike diffraction lenses, polycapillary optics are broadband, based on total external reflection. We simulated the optics using Monte Carlo (MC)-based ray tracing. Multi-arc scanning was simulated to deliver highly conformal radiation treatment to 3 types of small targets of a few millimeters size. Because the lens output focal spot is smaller than the x-ray tube focal spot, we also used the lens output focus as a virtual x-ray source to acquire phantom and rat head x-ray and CBCT images (60 kVp) to demonstrate improved resolution. **Results:** The simulated beam dosimetric characteristics were verified by EBT3 film measurements within 10%. MC simulated treatments of choroidal melanoma and age-related macular degeneration demonstrated more conformal target coverage and substantially reduced critical structure doses compared to standard treatments. Simulations also demonstrated irradiating 1-2 mm neurofunctional regions in rat heads with sharp penumbras. CBCT of a thin wire showed 33% reduction of the FWHM of a reconstructed wire with vs without lens. CBCT of a rat head could distinguish ~0.3mm fine bony structure with lens, but not without. **Conclusion:** Focused kV x-ray technique offers the capability of treating mm size targets and improved x-ray imaging resolution.

TU-EF-TRACK 4-03

Biologically Guided Deep Learning for Post-Radiation PET Image Outcome Prediction: A Feasibility Study of Oropharyngeal Cancer Application

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Purpose: To develop a method of biologically guided deep learning for post-radiation ¹⁸FDG-PET image outcome prediction based on pre-radiation images and radiotherapy dose information. **Methods:** Based on the classic reaction-diffusion mechanism, a novel biological model was proposed using a partial differential equation that incorporates spatial radiation dose distribution as a patient-specific treatment information variable. A 7-layer encoder-decoder based convolutional neural network (CNN) was designed and trained to learn the proposed biological model. As such, the model could generate post-radiation ¹⁸FDG-PET image outcome predictions with possible time-series transition from pre-radiotherapy image states to post-radiotherapy states. The proposed method was developed using 64 oropharyngeal patients with paired ¹⁸FDG-PET studies before and after 20Gy delivery (2Gy/fx) by IMRT. In a two-branch deep learning execution, the proposed CNN learns specific terms in the biological model from paired ¹⁸FDG-PET images and spatial dose distribution as in one branch, and the biological model generates post-20Gy ¹⁸FDG-PET image prediction in the other branch. As in 2D execution, 718/233/230 axial slices from 38/13/13 patients were used for training/validation/independent test. The prediction image results in test cases were compared with the ground-truth results quantitatively. **Results:** The proposed method successfully generated post-20Gy ¹⁸FDG-PET image outcome predictions and the associated time-series evolutions with breakdown illustrations of biological model components. SUV mean values in ¹⁸FDG high-uptake regions of predicted images (2.45±0.25) were similar to ground-truth results (2.51±0.33). In 2D-based Gamma analysis, the median/mean Gamma Index (<1) passing rate of test images was 96.5%/92.8% using 5%/5mm criterion; such result was improved to 99.9%/99.6% when 10%/10mm was adopted. **Conclusion:** The developed biologically guided deep learning method achieved post-20Gy ¹⁸FDG-PET image outcome predictions in good agreement with ground-truth results. With break-down biological modelling components, the outcome image predictions could be used in adaptive radiotherapy decision-making to optimize personalized plans for the best outcome in future.

TU-EF-TRACK 4-04

Computing Dose to Circulating Blood Cells Using Whole-Body Blood Flow Simulations

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Purpose: To develop a time-dependent computational method to estimate dose to circulating blood from radiation fields for any treatment site. **Methods:** Two independent dynamic models were implemented: one describing the spatiotemporal distribution of blood particles (BPs) in organs and a second describing the time-dependent radiation field delivery. A whole-body blood flow network based on blood volumes and flow rates from ICRP Publication 89 was simulated to produce the spatiotemporal distribution of BPs in organs across the entire body using a discrete-time Markov process. Constant or time-varying transition probabilities were applied to the Markov process and their impact on transition time was investigated. The impact of treatment time and different anatomical sites was investigated using a liver and brain cancer patient. The impact on the blood dose-volume histograms (bDVHs) was assessed using bDVH metrics (V0Gy, V0.05Gy, D2%). **Results:** Simulations revealed different dose levels to the circulating blood for brain compared to liver even for similar field sizes due to the different blood flow properties of the two organs. For the constant probability model, the volume of blood receiving any dose (V0Gy) after a single radiation fraction increases from 1.2% for 1s delivery time to 25.8% for 120s for brain and from 10% (1s) to 55.8% (120s) for liver, respectively. With time-dependent probability, V0Gy increases from 1.2% to 20.9% for brain and from 10% to 48.7% for liver, respectively. Changes in V0.05Gy show a similar pattern, while D2% decreases with longer delivery time. Unexpectedly, delivering the dose using a realistic time structure for every field only changes the bDVH negligibly compared to delivering the entire dose distribution over the treatment time. **Conclusion:** This publicly available framework estimates patient-specific dose to circulating blood cells based on organ DVHs and enables studying the impact of different treatment plans, dose rates, and fractionation schemes.

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TU-EF-TRACK 4-05

Deep Learning-Based Framework for the Assessment of Radiation Dermatitis in Nasopharyngeal Carcinoma (NPC) Patients

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Purpose: Radiation dermatitis (RD) is a common, unpleasant side-effect of patients receiving radiotherapy. Its severity is graded manually through visual inspection in clinical practice, which is resource-demanding and leads to inter-rater variation. To overcome this difficulty, this study aims to develop an end-to-end automatic RD grading framework based on deep learning (DL) techniques. **Methods:** A dataset of 1120 photographs of the head and neck region was built for NPC patients undergoing radiotherapy. These photographs were graded by qualified assessors based on Radiation Therapy Oncology Group (RTOG) guidance. We further classified RD Grade 0&1 as mild cases and RD Grade 2 and above as severe cases. An end-to-end RD grading framework was developed by combining DL-based segmentation and RD severity classification models. Models were trained separately for the front and side view. First, the neck region was segmented from the camera-captured photographs via U-Net. Secondly, a convolutional neural network (CNN) classifier (VGG16-BN with pretrained) was applied to the segmented region for RD severity classification. The performance of the proposed DL methods was qualitatively and quantitatively evaluated for both segmentation and classification parts. **Results:** Preliminary results of segmentation showed that the averaged Dice Similarity Coefficients (DSCs) were 91.2% and 90.8% for front and side view, respectively. For RD severity classification, the overall accuracy of 111 test photographs was 93%. Our method accurately classified 94% (82/87) of the mild cases and 92% (22/24) of the severe cases. Mild cases were predicted with precision of 0.98, recall of 0.94 and F1-score of 0.96; severe cases with precision of 0.81, recall of 0.92 and F1-score of 0.86. **Conclusion:** We demonstrated a DL-based end-to-end framework for automatic grading of RD severity in NPC patients. This method holds great potential to facilitate the monitoring and management of RD in NPC patients and improve patients' quality of life.

TU-EF-TRACK 4-06

Patient-Specific Deep Learning-Based Self-High-Resolution for MR Imaging

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Purpose: High-resolution (HR) magnetic resonance imaging (MRI) is desirable in many clinical applications, but there is a trade-off between image resolution and acquisition time. HR generation methods may yield fine details from fast acquisitions; however, most current methods attempt to solve this ill-posed problem using external atlases to derive the transformation from low-resolution (LR) to HR images. In this study, we propose a deep-learning-based self-HR method, which instead generates HR images directly from a patient's thick-cut MRI. **Methods:** We propose a new deep-learning-based, patient-specific method using a self-HR framework. Two LR-to-HR deep-learning models are pre-trained on HR in-plane images from a training dataset and then are fine-tuned on each test patient's in-plane images. To generate HR through-plane MR images for a test patient, the LR sagittal and coronal images are input into the trained model, and the resulting HR MRIs are fused together to obtain a HR 3D MRI. This technique was validated across 50 clinical datasets of multimodal MRIs: Flair, T1-weighted contrast and T2-weighted images. The native HR images were used as the ground truth for evaluating resolution recovery from down-sampled images. **Results:** The normalized mean absolute error (NMAE) and peak-signal-to-noise-ratio (PSNR) were calculated between the generated and original HR images. The LR data was obtained via down-sampling the MRI in z-axis. The down-sampling factor was set to 3. Overall, the mean NMAE and PSNR were 0.019±0.005 and 32.1±1.6 dB for Flair, 0.018±0.005 and 32.4±1.5 dB for T1c, and 0.015±0.003 and 33.7±1.1 dB for T2, which demonstrated the accuracy of the proposed method. **Conclusion:** We have proposed a patient-specific deep learning-based self-HR method to derive high through-plane resolution MR images and validated its feasibility. This proof-of-concept study shows potential to improve image resolution without any high through-plane resolution MR data and its self-HR capability could have broad clinical applications.

TU-EF-TRACK 4-07

Spatiotemporal Dose Characterization of An Electron FLASH Beam from a LINAC Using Radioluminescence and Cherenkov Imaging

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Purpose: Spatiotemporal pulse to pulse beam profile for an ultra-high dose rate (UHDR, >40Gy/s) electron FLASH (eFLASH) irradiator was characterized via Cherenkov emission and radioluminescence imaging using iCMOS cameras. **Methods:** Cameras time-gated to the delivered clinical LINAC pulses, imaged radioluminescence and Cherenkov emission signals incited by single pulses of 10MeV eFLASH irradiation at the treatment room isocenter. The two cameras imaged the entrance Cherenkov emission from a solid water phantom surface or scintillation from a screen on top of the phantom and the projected depth profile from a tank filled with water or quinine sulfate solution. A 450nm peak bandpass filter was considered for improving imaged depth profiles. The optical results were compared to lateral profiles measured by Gafchromic film at varying depths including the surface. **Results:** The per pulse beam output, from Cherenkov imaging, agreed with the photomultiplier tube (PMT) Cherenkov output to within 3%, after the first ~5-7 ramp-up pulses. In comparison to film measurements, Cherenkov emission and scintillation images were dose rate independent to >300Gy/s and linear with dose ($R^2=0.987, 0.995$, respectively). The surface dose profiles from film agreed more with scintillation than Cherenkov emission imaging (3%/3mm gamma pass rate of 98.9% and 88.8% respectively). A 450nm bandpass filter improved the quinine sulfate-based water images of the projected depth optical profiles to match projected film dose within 5%. **Conclusion:** The cameras resolved dose profiles of the UHDR beam with a single pulse (60Hz) temporal and 1mm spatial resolution. The agreement of scintillation screen imaging with film measured dose suggests it can accurately confirm the consistency of the beam's parameters for quality assurance. The cross validation of Cherenkov images with PMT temporal profile of each pulse suggest it can monitor beam output but required its optical filtering to acquire accurate dose via quinine sulfate-based water imaging.

TU-EF-TRACK 4-08

Thermoacoustic Range Verification During Pencil Beam Delivery of a Clinical Plan to An Abdominal Imaging Phantom

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Purpose: The purpose of this phantom study is to demonstrate that thermoacoustic range verification could be performed clinically. Thermoacoustic emissions generated in an anatomical multimodality imaging phantom during delivery of a clinical plan are compared to simulated emissions to estimate range shifts compared to the treatment plan. **Methods:** A single-field 12-layer proton pencil beam scanning (PBS) treatment plan created in Pinnacle prescribing 6 Gy/fraction was delivered by a superconducting synchrocyclotron to a triple modality (CT, MRI, and US) abdominal imaging phantom. Data was acquired by four acoustic receivers rigidly affixed to a linear ultrasound array. Acoustic receivers (transducer + amplifier) were tuned to this application provided 15-25 dB amplification relative to 1 mV/Pa over 10-100 kHz. Receivers 1-2 were located distal to the treatment volume, whereas 3-4 were lateral. Receivers' room coordinates were computed relative to the ultrasound image plane after co-registration to the planning CT volume. For each prescribed beamlet, a Monte Carlo simulation of the energy density provided initial pressure from which simulated emissions were computed. To overcome the diffraction limit, range estimates were computed by comparing measured to simulated emissions. **Results:** Shifts were small for high-dose beamlets that stopped in soft tissue. Signals acquired by channels 1-2 yielded shifts of -0.2 ± 0.7 mm relative to Monte Carlo simulations for high dose spots (~40 cGy) in the second layer. Additionally, for beam energy ≥ 125 MeV, thermoacoustic emissions qualitatively tracked lateral motion of pristine beams in a layered gelatin phantom, and time shifts induced by changing phantom layers were self-consistent within

nanoseconds. **Conclusion:** Acoustic receivers tuned to spectra of thermoacoustic emissions may enable range verification during proton therapy. Improving receive sensitivity by another 20 dB could enable range verification during delivery of a standard 2 Gy fraction.

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Dr. Patch founded Acoustic Range Estimates and holds US patent #10,758,127.

Mr. Nguyen was employed by A.R.E.

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TU-EF-TRACK 4-09

Thoracic 4D Cone-Beam CT Registration Incorporating a Fan-Beam CT-Derived Feasible Motion Descriptor

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Purpose: Cone-beam CTs (CBCTs) are readily available for on-table acquisition in companion with treatment and have lower imaging dose compared to fan-beam CTs (FBCTs). Deformable registration between phases in a 4D-CBCT can potentially provide on-table 4D volumetric information of target location and motion. However, CBCT registration is very challenging due to strong artifacts and the lack of structural details. In this study, we aim to develop an enabling approach for thoracic 4D CBCT registration. **Methods:** We propose a novel approach to incorporate a learned implicit regularizer as a feasibility prior of respiratory motion, into the training of an unsupervised image registration network to improve registration accuracy and robustness to noise and artifacts. Specifically, a feasibility descriptor from a set of deformation vector fields (DVs) is first generated from FBCTs. Subsequently, this FBCT-derived feasibility descriptor was used as a spatially variant regularizer on DVF Jacobian during the unsupervised training for 4D-CBCT registration. In doing so, the higher-quality, higher-confidence information from FBCT is transferred into the much challenging problem of CBCT registration, without explicit FB-CB synthesis. The method was evaluated using manually identified landmarks on five real CBCTs and automatically detected landmarks on nine simulated CBCTs. **Results:** The method presented good robustness to noise and artifacts and generated physically more feasible DVFs. The target registration errors on the real and simulated data were (1.63 ± 0.98) and (2.16 ± 1.91) mm, respectively, significantly better than the classic bending energy regularization in both the conventional method in SimpleElastix and the unsupervised network. The average registration time was 0.04 s. **Conclusion:** The proposed method imposes a strong and robust motion feasibility prior and enhances registration networks' resistance against undesired image qualities. It demonstrates the strength of a learned and implicit feasibility prior, in comparison to the pre-defined regularization energies.

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TU-EF-TRACK 4-10

Toward Magnetic Resonance Fingerprinting for Low-Field MR-Guided Radiation Therapy

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Purpose: Magnetic resonance fingerprinting (MRF) unlocks incredible potential for efficient multi-parametric quantitative MRI. When coupled with MR-guided radiation therapy, MRF will push new boundaries for online functional plan adaptation. This work establishes the feasibility of MRF on a 0.35T MR-linac and introduces a novel deep learning-based method to improve quantitative parameter precision from clinically feasible MRF scan times. **Methods:** MRF was benchmarked on a 0.35T MR-linac in the NIST/ISMRM System quantitative phantom. Gold standard multi-inversion recovery and multi-echo spin echo scans were acquired to derive reference quantitative T_1 and T_2 maps. A fully sampled MRF was designed and acquired in ~20 minutes. MRF-derived T_1 and T_2 times were compared to gold standard via linear regression and mean percent error. To assess the impact of accelerating MRF, the fully sampled dataset was undersampled to three radial k-space spokes/frame (28 seconds). Two acceleration approaches were tested: (1) uniformly spaced spoke angles with a linear reconstruction, and (2) spoke angles determined by a neural network with a deep learning regularized reconstruction. T_1 and T_2 times from both acceleration approaches were compared to fully sampled MRF using paired t-tests. **Results:** Fully encoded MRF at 0.35T yielded precise T_1 and T_2 estimates relative to gold standard (mean percent errors of 4.65% and 8.24%, respectively) and correlation coefficients >0.996 . Acceleration approach (1) produced T_1 and T_2 times different from those measured with the full 21-minute MRF scan ($p=0.009$ and $p=0.013$, respectively). The deep learning acceleration approach produced similar estimates for T_1 and T_2 ($p=0.121$ and $p=0.363$, respectively). **Conclusion:** These encouraging results suggest that MR fingerprinting is feasible at 0.35T and, when coupled with deep learning-assisted acceleration, accurate MRF can be performed efficiently. Future work will optimize MRF for in vivo quantitative imaging, thereby opening the door for subvolume tumor targeting and treatment response assessment.

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Therapy Scientific Session

TRACK 5

FLASH: Radiobiology

TU-E-TRACK 5-01

Abdominal Flash Proton Irradiation Does Not Benefit the Intestine

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Purpose: Abdominal Flash irradiation has been reported to spare the intestinal tissue, while some other reports suggested no gastrointestinal benefit was gained. In the present study, we plan to investigate the potential effect of Flash proton irradiation on the intestine. **Methods:** A homogenous elliptical field of 1.2×1.6 cm² was obtained through the established Flash proton platform to cover 60~80% of the whole abdomen. The MRI scanning was conducted before irradiation and the mice were fixed on a designed holder to receive a single dose of 13.8 Gy, 16 Gy, 16.2 Gy, or 17.5 Gy. The dose rate of Flash irradiation was ~120 Gy/s and the one for the conventional dose rate (CDR) group was ~0.36 Gy/s. The Ki67 immunofluorescent staining of the intestine tissue was performed at 7 days and 14 days after irradiation. For the late effect, the mice were kept for as long as 210 days for the weight detection. **Results:** Higher survival fraction of the mice was observed after 16 Gy conventional dose rate proton irradiation. There was no significant survival difference between the Flash and CDR groups after 17.5 Gy exposure. The Ki67 staining showed that the ratio of the proliferating cells in the crypts recovered at 7 days in both 13.8 Gy Flash and CDR groups. Compared with the 16.2 Gy CDR group, the long-term weight monitoring indicated more severe weight loss in the Flash group with the corresponding dose. **Conclusion:** No benefit of Flash proton irradiation was gained after partial abdominal exposure. There should be some undiscovered physical and biological factors that impact the Flash effect on the intestine if it does exist.

This work was supported by the Damon Runyon-Rachleff Innovation Award (DRR 57-19).

TU-E-TRACK 5-02

Analysis of Proton Inter-Track Interactions at Ultra-High Dose Rates

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Purpose: Laser-accelerated ultra-high dose rate proton therapy offers dose rates on the order of 10^9 Gy/sec. Such dose rates reduce delivery to nanosecond timescales, equivalent to the lifetime of chemical species within a biological medium. This means reactive chemical species from individual tracks could interact and recombine, and tracks can no longer be considered independently. This work investigated the temporal evolution of chemical species for a range of proton LETs to quantify inter-track interactions. **Methods:** The TOPAS-nBio Monte Carlo toolkit was used to find the radial dimensions of proton tracks from 1 MeV to 100 MeV between 1 ps to 1 μ s timepoints, defined as the radius which contained 80% of the physical and chemical interactions. From this, the likelihood of interaction was calculated using the geometric probability of incident track areas overlapping as a function of dose, time and LET. **Results:** Track radius ranged from 50 to 254 nm, increasing with diffusion time after irradiation, and falling with LETs above 2 keV/ μ m. Physical inter-track effects were rare at clinical doses, with a probability of 32% for 2 Gy of 100 MeV protons, falling to 0.4% for 1 MeV protons due to reduced fluence and track area. Radiochemical overlap probabilities at 1 μ s were larger due to diffusion, with over 90% overlap probability predicted for 80 and 100 MeV protons. However, when considering the reduced lifetime of chemical species in cells due to biological scavenging, the likelihood of inter-track effects remains small, with a maximum probability of 29% for a shorter 1 ns diffusion time. **Conclusion:** The likelihood of track overlaps has been evaluated in a pure water and scavenging setup, considering both the physical and chemical stages of irradiation. These observations suggest that inter-track interactions play a negligible role at clinically relevant doses, even at ultra-high dose rates.

TU-E-TRACK 5-03

BEST IN PHYSICS (THERAPY): FLASH Sparing of Normal Tissue in a Proton Spread Out Bragg Peak

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Purpose: To determine the biological effects of FLASH proton radiotherapy (F-PRT) using the spread out Bragg peak (SOBP) compared to standard dose rate proton radiotherapy (S-PRT) in a whole abdomen mouse model. **Methods:** An IBA Proteus Plus C230 Cyclotron was used to deliver 230 MeV proton beams (range 32 g/cm²) to a fixed beam line in a dedicated research room. An SOBP was generated with a custom made ridge filter placed between the first and second scatterer in the beam. Range shifting material was placed prior to the final collimator to irradiate mouse whole abdomen with the SOBP. To investigate the acute radiation damage, 8-10 week old C57BL/6J mice were randomly assigned to 15 Gy whole abdomen irradiation into one of four groups: (1) SOBP F-PRT, (2) SOBP S-PRT, (3) entrance F-PRT, (4) entrance S-PRT. Mice were injected with EdU at 3.5 days after irradiation and jejunum segments were harvested and preserved for further staining. Upon staining, EdU-positive proliferating cells and regenerated intestinal crypts were quantified. **Results:** The generated SOBP had a modulation (width) of 2.5 cm from the proximal to distal 90%. Mice were irradiated with FLASH dose rates of 101.3 ± 2.8 Gy/s and standard dose rates of 0.94 ± 0.02 Gy/s with a SOBP. Using the entrance region of the proton beam, dose rates were 115.4 ± 7.4 Gy/s and 1.1 ± 0.01 Gy/s for F-PRT and S-PRT, respectively. Film irradiation showed uniform dose delivery across the whole abdomen throughout the mouse body width. FLASH preserved a significantly higher percentage of regenerated crypts compared to standard dose rate radiation using both the entrance and SOBP. **Conclusion:** SOBP F-PRT beams were delivered to healthy mouse whole abdomen. FLASH-irradiated mice exhibited normal tissue sparing when compared to S-PRT. SOBP for proton FLASH can lead to better dose conformity around targets and increased normal tissue sparing.

TU-E-TRACK 5-04

Measuring Oxygen Concentration at Ultra-Fast Rates During Proton FLASH Delivery Using Phosphorescence Quenching of Soluble Oxyphor Probes

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Purpose: The mechanism underlying the tissue sparing effect of FLASH radiation therapy is currently unknown, but the depletion of tissue oxygen at FLASH dose rates has been proposed as a potential candidate. This hypothesis has only been supported by modeling studies, primarily due to the difficulty of measuring oxygen concentration on millisecond timescales during irradiation. Here we use the phosphorescence quenching method to monitor oxygen concentration during proton delivery at conventional and FLASH dose rates. **Methods:** Oxygen concentration was measured at rates of up to 3kHz using a soluble phosphorescent oxygen probe and a phosphorometer. Sealed glass vials, containing the experimental solution and probe, were placed in a proton beam and irradiated at conventional (conv) (~1Gy/s) and FLASH (~100Gy/s) dose rates. The g-value of oxygen depletion was measured at multiple dose rates and oxygen concentrations. Experiments were performed using three solutions containing 1, 2, and 5% albumin and a fourth designed to emulate the intracellular environment, containing glucose, glycerol, and glutathione (Buffer 3G). Oxygen concentration was continuously monitored before, during, and after irradiation. **Results:** We found that FLASH dose rates resulted in lower g-values for oxygen depletion when compared to conventional dose rates by ~15% (range: 7.2-22.5%) for all solutions tested. E.g., for a 30Gy dose to Buffer 3G, the g-value was 16% lower for FLASH than for conventional ($g_{conv} = 0.72 \pm 0.02 \mu\text{M}/\text{Gy}$, $g_{FLASH} = 0.63 \pm 0.02 \mu\text{M}/\text{Gy}$) (all values listed are mean $\pm$ S.E.M.). The g-value was higher for Buffer 3G than for the albumin solutions ($g_{conv} = 0.72 \pm 0.02$ and $0.58 \pm 0.03 \mu\text{M}/\text{Gy}$ respectively, $g_{FLASH} = 0.63 \pm 0.02$ and $0.54 \pm 0.01 \mu\text{M}/\text{Gy}$ respectively). **Conclusion:** We have demonstrated that ultra-fast measurements of oxygen concentration can be performed during actual proton irradiation. Furthermore, we found that the g-value of oxygen depletion at FLASH dose rates is lower than that of conventional dose rates by ~15%.

TU-E-TRACK 5-05

Modeling of Cellular Response After FLASH Irradiation: A Quantitative Analysis Based On the Radiolytic Oxygen Depletion Hypothesis

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Purpose: Recent studies have suggested that ultra-high dose rate (FLASH) irradiation can spare normal tissues from radiotoxicity, while efficiently controlling the tumor, and this is known as the "FLASH effect". This study performed theoretical analyses to quantitatively investigate the impact of radiolytic oxygen depletion (ROD) on the cellular responses after FLASH irradiation. **Methods:** Monte Carlo simulation was used to model the ROD process, determine the DNA damage, and calculate the amount of oxygen depleted (LROD) during FLASH exposure. A mathematical model was applied for the analysis of oxygen tension (pO₂) distribution in human tissues and the recovery of pO₂ after FLASH irradiation. DNA damage and cell survival fractions after FLASH irradiation were calculated, and the impact of initial cellular pO₂, FLASH pulse fractionation, pulse interval, and radiation quality of the source particles on ROD and the subsequent cellular responses were systematically evaluated. **Results:** The simulated LROD range was 0.220–0.248 mmHg/Gy when pO₂ ranged from 7.5–160 mmHg. The calculated DNA damage and cell survival fractions show that the radioprotective effect is more evident in cells with a lower pO₂. Different irradiation setups alter the cellular responses by modifying the pO₂. Single pulse delivery or multi-pulse delivery with pulse intervals lower than 10–50 ms resulted in fewer DNA damages and higher cell survival fractions. Source particles with a low radiation quality have a higher capacity to deplete oxygen, and thus, lead to a more conspicuous radioprotective effect. **Conclusion:** Here, a systematic analysis of the cellular response following FLASH irradiation was performed to provided suggestions for future FLASH sparing effect study. Single pulse delivery or multi-pulse delivery with short pulse intervals are suggested for FLASH irradiation to avoid oxygen tension recovery during pulse intervals. Source particles with low radiation quality are preferred for their conspicuous radioprotective effects.

TU-E-TRACK 5-06

TOPAS-NBio Simulation of Radiation Chemistry Following FLASH Irradiation Including Reactions of Biological Importance

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Purpose: Ultra-high dose rate irradiation has demonstrated to improve sparing of normal tissues while maintaining an equivalent tumor response (the FLASH effect). Although the biological mechanism of the FLASH effect is still not understood, oxygen depletion seems to be one of the plausible hypotheses. However, current mechanistic studies rely on pure liquid water calculations. We implemented two models of water radiolysis with chemical reactions of oxygen and molecules of biological importance, which might help evaluate the FLASH effect in tissue in a more reliable way. **Methods:** We extended TOPAS-nBio to allow delivering pulses of radiation and reactions with oxygen in two different scenarios: including reactions between free radicals and DNA (Model 1); and additional reactions with RNA, proteins, amino acids and free nucleotides (Model 2). Those molecules were implemented as scavengers with reaction rates and concentrations obtained from the literature. Our simulations were carried out using microsecond-width pulses of 1 MeV electrons uniformly irradiating a water phantom cube of 1 μm side. The oxygen depletion without replenishing was compared for a conventional dose rate (0.29Gy/s) and FLASH irradiation conditions (... Gy/s) for 21% oxygen concentration. **Results:** At the end of the electron pulse, and for all irradiation settings, both models predict that FLASH irradiation depletes 2.7 times more oxygen than conventional irradiation for the same absolute dose. This is caused by the intertrack and interpulse reactions between reactive species. The oxygen consumption varied by 16.7% between the two Models, which results from introducing competing scavengers that react with free radicals. **Conclusion:** The proposed models under pulsed irradiation allowed us to quantify the time course of oxygen depletion under more realistic biological conditions than pure liquid water. Further work is performed under even more realistic conditions mimicking the cell environment and will be validated by experimental studies.

TU-E-TRACK 5-07

Modeling Radiation-Induced Oxygen Depletion and Its Effects on Normal Tissue Complications in FLASH-RT

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Purpose: Ultra-high dose rate irradiation (FLASH) has emerged as a promising technique that may increase the therapeutic ratio of radiation therapy. A leading hypothesis is that the reduction of normal tissue toxicity is mediated by transient radioprotective hypoxia. A computational model is developed to simulate heterogeneous oxygen levels in the mouse brain during FLASH and investigate the effect of transient oxygen depletion on normal tissue complication probability. **Methods:** A 3D model of mouse brain vasculature was constructed using a publicly available two-photon fluorescence microscopy dataset. A dynamic oxygen distribution was modeled by considering oxygen diffusion, metabolism and radiolytic depletion. During FLASH, the oxygen enhancement ratio (OER) was incorporated into the linear-quadratic (LQ) model to obtain cell killing rate and compute an effective dose distribution considering oxygen depletion and heterogeneous distribution. **Results:** The effective dose is calculated for a FLASH pulse of 2-50Gy and mean oxygen levels 0.1-35mmHg in extra-vascular space. For a fixed physical dose, dose modifying factors (DMFs) first decrease and then increase with increasing oxygen tension. The oxygen tension at which DMF is minimized (maximal FLASH effects) shifts toward higher oxygen levels with increasing dose. For a fixed initial oxygen level, FLASH effects increase and then saturate with increasing physical dose. In the range 12-22Gy, V12 (a predictor of brain necrosis) yields a higher value for extreme cases of 0.1 and 35mmHg and a lower value for 6, 9, and 15mmHg. **Conclusion:** The proposed computation model is a useful tool to analyze various factors, i.e., alpha, beta (in LQ), metabolism, oxygen tension, OER, physical dose affecting the FLASH effects. The tissue oxygen heterogeneity may play an important role in mediating the FLASH effect. The same concept can be easily applied to the tumor and eventually help provide insight into determining the optimal treatment parameters for the FLASH RT.

Multi-Disciplinary Scientific Session

TRACK 6

AI-Based Auto-Segmentation and Auto-Contouring - I

TU-E-TRACK 6-01

Automatic Segmentation of High-Risk CTV for Tandem and Ovoid Brachytherapy Patients Using a Dual-Path Convolutional Neural Network

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Purpose: Pre-implant MRI is the gold standard on image-guided brachytherapy for cervical cancer. However, HDR planning is typically done on post-implant CT-based high-risk CTV (HR-CTV) because transferring of pre-implant MR-based GTV to post-implant planning CT is difficult due to large anatomical changes. The goal of this study is to train a dual-path convolutional neural network (CNN) for automatic segmentation of HR-CTV on post-implant planning CT. **Methods:** Pre-implant T2-weighted MR images and post-implant CT for 55 (48 for training and 7 for testing) patients were retrospectively solicited. MR was aligned to the corresponding CT using rigid registration. HR-CTV and GTV were manually contoured on CT and MR by an experienced radiation oncologist. All images were then resampled to a spatial resolution of 0.5x0.5x1.25 mm. A 3D CNN architecture was constructed with two encoding paths based on CT (no mask) and MR (masked by GTV). Each path used the same 3x3x3 kernel to capture the features from CT and MR. As a comparison, single path CNN with only CT input was trained. Voxel-based dice similarity coefficient (DSC), sensitivity, precision, and Hausdorff distance (HD) were used to evaluate model performance. Cross-validation was performed to assess model stability. **Results:** The dual-path CNN model achieved average DSC of 0.76 (0.70-0.81), Sensitivity of 0.82 (0.79-0.86), and Precision of 0.85 (0.83-0.90). The single-path CNN with only CT input achieved DSC of 0.65 (0.55-0.76), Sensitivity of 0.76 (0.68-0.80), and Precision of 0.78 (0.75-0.83). The area under the curve (AUC) values were 0.83 (0.82-0.84) and 0.76 (0.75-0.77) for dual-path and single-path models, respectively. The HD for the two models were 7.6 mm (4.2-10.5 mm) and 8.3 mm (5.1-12.5 mm). **Conclusion:** A CNN model with two encoding paths from pre-implant MR and post-implant CT was successfully developed for automatic segmentation of HR-CTVs for tandem-and-ovoid brachytherapy patients.

TU-E-TRACK 6-02

Cardiac Substructure Segmentation Using a Mask-Scoring Attention Convolutional Neural Network

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Purpose: Radiation-induced heart disease is a post-treatment toxicity which directly impacts overall survival for many thoracic cancer patients. Dose to specific cardiac substructures can better predict post-treatment toxicity than whole heart dose. However, delineation of substructures during treatment planning is not feasible in most clinics. This study proposes a deep-learning-based framework for simultaneous, automatic generation of substructure contours which can be used to better understand the relationship between radiation dose and post-treatment toxicity. **Methods:** The proposed mask-scoring regional convolutional neural network (MS-R-CNN) with attention gate consists of five subnetworks: backbone, regional proposal network (RPN), R-CNN, mask network with an attention gate, and mask-scoring network. Relying on CT alone, the backbone extracts multi-scale feature maps. The RPN utilizes these maps to detect the volume-of-interest (VOI) around all substructures, and the final three subnetworks extract structural information from these VOIs. An attention gate enforces a deep attention strategy which highlights informative features for accurate substructure segmentation. A retrospective study with 55 patients undergoing radiotherapy for lung cancer has been conducted to evaluate the proposed segmentation method. The whole heart and 15 cardiac substructures have quantitatively been accessed based on Dice score coefficients (DSC) and mean surface distances (MSD). **Results:** The proposed method achieved average DSC for the whole heart, chambers, great vessels, coronary arteries, the valves

of the heart of 0.96, 0.94, 0.93, 0.66, and 0.77 respectively. MSD between substructures segmented by the proposed method and the physicians' manual contours were 1-5 mm for all substructures. After network training, all substructures and the whole heart can be segmented on new datasets in less than five seconds. **Conclusion:** A deep-learning network was developed for automatic delineation of cardiac substructures based on CT alone. The tool has high potential for investigating cardiac substructure dose and treatment toxicities for improved cardiac sparing treatment planning.

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TU-E-TRACK 6-03

GAN-Driven Anomaly Detection for Active Learning in Medical Imaging Segmentation

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Purpose: One barrier to the clinical deployment of deep learning-based image segmentation models is the existence of “anomalous” cases where the segmentation model may fail, despite external validation. We developed methods that enable the detection of images on which the segmentation models are likely to fail, thereby aiding in clinical deployment of these models. **Methods:** We designed an anomaly detector that approximates the performance of a liver segmentation model on CT images by determining whether the images have a high likelihood of falling within the model's training distribution. We consider an image to be within the training distribution if our anomaly detection model can reconstruct the image. We use backpropagation to map input images to a latent space and a GAN to reconstruct the images. We used a StyleGAN2 network, due to its cutting-edge performance on high-resolution images. We will score reconstructed images using a residual loss. If this score is over a specified threshold, the image will be classified as anomalous. We trained the detection model on 96 CT scans and have 50 unseen CT scans available for anomaly detection. We use standard image similarity metrics for evaluation. **Results:** We successfully trained the StyleGAN2 network on slices extracted from non-contrast enhanced abdominal CT images. The network achieved a Fréchet-Inception score of 31.102. It also attained a Wasserstein distance of 1.388, a KL-divergence of 0.004, and a mutual information score of 0.241 between the pixel distributions of 120 real and 120 generated images. After successful reconstruction of images, a radiologist classified real and generated images as such with equivalent accuracy (60%). **Conclusion:** We defined an anomaly detection paradigm that determines when a segmentation model is likely to fail on an image. This paradigm can aid the clinical deployment of segmentation models, diversify medical imaging datasets, and prioritize images for labeling.

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TU-E-TRACK 6-04

Multi-Organ Auto-Segmentation of Abdominal Structures from Contrast-Enhanced and Non-Contrast-Enhanced CT Images

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Purpose: To develop a deep-learning-based tool for accurate and robust upper abdominal organs-at-risk(OAR) autosegmentation suitable for treatment planning and adaptive replanning. **Methods:** Forty pancreas SBRT patients with contrast-enhanced breath-hold planning CT images were selected. All OARs (duodenum, small bowel, large bowel, stomach, liver, spleen, left kidney, right kidney and spinal cord) were edited or recontoured under physician supervision to ensure accuracy and consistency. We used the self-adapting nnUNet framework to customize a 3D U-Net for our data to predict all organs concurrently. We tested our model using five-fold-cross-validation and calculated Dice-similarity-coefficients(DSC) and 95th-percentile Hausdorff(HD95) distances. A radiation oncologist assessed the performance of the tool on 8 randomly selected patients. This assessment was then repeated on non-contrast-enhanced daily CT images of the same patients. **Results:** The mean dice-similarity-coefficient/95th-percentile Hausdorff distance between the automatic segmentation and the ground truth on contrast-enhanced CT images were 0.88±0.07/8.94±10.29mm for small bowel, 0.89±0.08/12.46±15.09mm for large bowel, 0.80±0.08/12.29±8.10mm for duodenum, 0.91±0.04/5.98±4.59mm for stomach, 0.96±0.02/5.43±9.93mm for liver, 0.97±0.01/2.28±2.09mm for spleen, 0.95±0.05/2.78±2.02mm for left kidney and 0.96±0.02/2.46±1.42mm for right kidney. For clinical performance evaluation, all contours on contrast-enhanced planning CT images and non-contrast-enhanced daily CT images were clinically acceptable. The radiation oncologist suggested minor edits for the following OARs: large bowel (1 patient), small bowel (1 patient), liver (1 patient), and duodenum (7 patients). All other OARs were rated as “use-as-is”. For non-contrast-enhanced daily CT images, large bowel (2 patients), small bowel (2 patients), liver (1 patient), stomach (1 patient), and duodenum (7 patients) required minor edits. All other OARs were rated as “use-as-is”. **Conclusion:** We developed an auto-contouring tool for upper abdominal OAR segmentation for both contrast-enhanced and non-contrast-enhanced CT images. Our tool achieved a high-level of clinical acceptability on both types of images, which can provide accurate and consistent contours for treatment planning and adaptive replanning.

TU-E-TRACK 6-05

Rapid Organ-At-Risk Delineation in Pancreatic CBCT for CBCT-Guided Adaptive Radiotherapy

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Purpose: Rapid re-planning is a fundamental requirement in on-line adaptive radiotherapy. The delineation of organs-at-risk (OARs) is rather time-consuming and labor-intensive in the plan adaptation process. This study aims to develop a deep learning-based multi-organ segmentation to facilitate efficient and accurate re-planning for CBCT-guided adaptive pancreatic radiotherapy. **Methods:** Considering the relatively low quality of daily CBCT acquired in the treatment machine, two-stage-in-one model was designed. In the first stage, taking CBCT as input, a cycle-consistent generative adversarial network is used to generate synthetic CT images with improved image quality. Then, in the second stage, the synthetic CT is fed into a mask-scoring regional convolutional neural network (MS-R-CNN) which is constructed by five subnetworks including a backbone, a regional proposal network (RPN), a R-CNN, a mask network, and a mask scoring network for the detection of the positions and shapes of multiple organs simultaneously. The model was trained and tested on a cohort of 40 patients. Metrics including the Dice similarity coefficient (DSC), 95th percentile Hausdorff distance (HD95), mean surface distance (MSD), and residual mean square distance (RMS) were computed to quantitatively assess the proposed method. **Results:** A mean DSC of 0.92 (0.89 - 0.97), mean HD95 of 2.90 mm (1.63 mm - 4.19 mm), mean MSD of 0.89 mm (0.61 mm - 1.36 mm), and mean RMS of 1.43 mm (0.90 mm - 2.10 mm) were achieved by our proposed method across eight abdominal OARs including duodenum, large bowel, small bowel, left and right kidneys, liver, spinal cord and stomach. The contours of eight OARs can be obtained on the order of seconds once the model was trained. **Conclusion:** We have proposed a novel deep learning-based multi-organ segmentation algorithm for CBCT-guided adaptive pancreatic radiotherapy. It could be implemented in the setting of pancreatic adaptive radiotherapy to rapidly contour OARs.

TU-E-TRACK 6-06

Simultaneous Segmentation of Target and Organ at Risk in Thoracic Kilovoltage Images

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Purpose: The ability to simultaneously localize the treatment target and local organs-at-risks (OARs) can improve SABR treatment efficacy and reduce side effects by ensuring the target dose and avoiding irradiating the OARs. We present a fiducial-less Multi-Target-Segmentation (MTS) approach that utilizes the standard-equipped kV-imager of a conventional linac. **Methods:** MTS used a conditional generative adversarial network to simultaneously segment the tumor and the heart in kV-images. MTS was trained on the DRRs of the planning CT for each patient, such that it can be deployed for intra-fractional target tracking. A patient-specific training dataset consisted of 36000 DRRs, created by forward-projecting each of ten planning 4D-CT phases and segmented target volumes equidistantly over a full imaging rotation. The labels of multiple targets were stacked as separate image channels. Data was augmented during training by randomly cropping the field-of-view within ± 3 mm. MTS was trained individually for three lung SABR patients and tested on kV-images of three treatment fractions per patient. The ground truth target positions were determined through manual alignment of the forward-projected target volumes onto the kV-projections. The geometric accuracy was determined by comparing the centroid of the segmented targets on each image to the ground truths. Additionally, the dice-similarity-coefficient (DSC) was calculated to evaluate MTS. **Results:** MTS had a mean tumor localization accuracy of -0.4 ± 2.6 mm and 0.7 ± 1.7 mm in the lateral/ anterior-posterior and superior-inferior directions of the images, respectively. The mean heart localization accuracy was 1.9 ± 1.6 mm and 1.4 ± 1.1 mm in the lateral/anterior-posterior and superior-inferior directions of the images, respectively. The mean DSC was 0.72 ± 0.12 for the tumor and 0.88 ± 0.04 for the heart, respectively. **Conclusion:** Simultaneous target and OAR localization without fiducials is feasible and can achieve clinically relevant accuracy for clinical deployment. MTS may widen the therapeutic index of SBRT by minimizing thoracic toxicity for targets in close proximity of critical OARs.

TU-E-TRACK 6-07

Towards Artificial Intelligence and Clinician Integrated Systems (AICIS): Interactive Contour Revision with Deep Boundary Net

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Purpose: Accurate organ contours are required to ensure the quality of a radiotherapy treatment plan. However, contours generated by current auto-segmentation tools are not perfect, will never be, and may require significant revisions which can be tedious and time consuming. This work proposes an artificial intelligence and clinician integrated systems (AICIS) that can interact and assist clinicians to quickly revise the contours. **Methods:** The pipeline starts with a convolutional neural network (CNN)-based auto-segmentation model, which generates an initial contour. After review by the clinician, if it is not acceptable, the clinician will click on the boundary that exhibits large errors. This boundary click will be converted into a Gaussian point and then fed into the network to guide an interactive model revising the contour. This interaction process can be iterated multiple times. The input of our interactive model consists of three channels: the CT image, the current segmentation result, and the Gaussian point. The training dataset is built upon the existing dataset for auto-segmentation model training by simulating clinician's click. In this work, we reasonably assume the clinician prefers to click on the boundary point that exhibit large errors which can be quantified by Hausdorff distance (HD). We used three different head-and-neck (HN) datasets to train and test our model. For quantification, we measure the percentage of acceptable contours that meet the criterion $HD95 < 2.5$ mm and the number of clicks required to reach this criterion. **Results:** On average, 42.6% of the auto-generated contours were acceptable. With the aid of our AICIS system, the percentage of acceptable contours were improved to 92.3%. The mean number of clicks required to reach our criterion was 1.57. **Conclusion:** We developed an artificial intelligence and clinician integrated systems that can substantially improve the clinician's contour revision efficiency.

SAM Therapy Scientific Symposium

TRACK 7

Joint AAPM-EFOMP Symposium: Non-conventional Treatment in Radiation Therapy

TU-EF-TRACK 7-00

Joint AAPM-EFOMP Symposium: Non-Conventional Treatment in Radiation Therapy

Moderator: Yolanda Prezado, Institut Curie, Paris

TU-EF-TRACK 7-01

R. Timmerman, The University of Texas Southwestern Medical Ctr:

PULSAR: The Adaptation of SabR Toward Personalized Medicine

TU-EF-TRACK 7-02

Y. Prezado, Institut Curie:

Overview of Spatially Fractionated Radiation Therapy: From Photons to Charged Particles

TU-EF-TRACK 7-03

S. Chang, University of North Carolina, School of Medicine:

Understanding the Correlation Between Dosimetric Parameters and Treatment Responses for Spatially Fractionated Radiation Therapy

TU-EF-TRACK 7-04

K. Sheng, UCLA School of Medicine:

Radiotherapy Beyond 4pi

TU-EF-TRACK 7-05

Q&A

Speaker 1: Prof. Robert Timmerman, UT Southwestern

Presentation Title: "PULSAR: The adaptation of SabR toward personalized medicine."

Radiotherapy dogma, based on conventional daily dosing, insists that treatments be completed without interruption. In fact, randomized trials conducted in the 70's and 80's with a "split" inserted into a course of conventional daily dose treatment resulted in improved toxicity (as predicted), but poorer tumor control (tumor control penalty) leading to abandonment. The implementation of Stereotactic Ablative Radiotherapy (SABR) successfully challenged another radiotherapy dogma condemning hypo-fractionation by showing higher rates of tumor control WITHOUT increasing late toxicity. Building on this reality, we and others have hypothesized that potential benefits of a "split course" might be re-examined with SABR where more potent dosing could avoid the tumor control penalty while providing other opportunities to exploit for improving therapy. New radiotherapy platforms with sophisticated on-board imaging and software are capable of rapid re-planning enabling an "adaptive" approach where change triggers a modified treatment. The benefit of adaptive radiotherapy using conventional small dose daily fractionation is marginal because little changes. So here we introduce a new treatment paradigm called Personalized Ultra-fractionated Stereotactic Adaptive Radiotherapy (PULSAR). PULSAR uses independently planned, potent SABR treatments called "pulses" to trigger change in the tumor, in its microenvironment and in the patient as a whole. Pulses are separated by ample time (weeks to months) hoping for ample changes to occur. Multiple interrogations proceed between pulses to guide the next pulse which is independently planned (adapted). Interrogations might include imaging (anatomical and functional), blood/urine tests, assays, history and physical exam, toxicity assessments and even repeat biopsy. With time, these tests are likely to uniquely change for each patient facilitating adaptations personalizing the therapy. Much work needs to be done to guide the adaptations as this is much more than just adapting to position and shape. Treatments may be altogether changed combining different drugs with different pulses depending on the evolution of change, perhaps even abandoning the original treatment altogether. PULSAR defines

both opportunities and complicated assessments. Artificial intelligence tools will likely be helpful in making PULSAR a revolutionary new treatment paradigm

Learning Objectives:

1. Learn about the concept and the advantages of Personalized Ultra-fractionated Stereotactic Adaptive Radiotherapy (PULSAR) for adaptive radiotherapy
2. Learn about potential changes in the tumor and its microenvironment triggered by PULSAR and how to use them to optimize the treatments.

Speaker 2: Dr. Yolanda Prezado, Institut Curie (France), EFOMP's representative

Presentation Title: "Overview of Spatially Fractionated Radiation Therapy: from photons to charged particles."

Despite remarkable advancements, the dose tolerances of normal tissues continue to be the main limitation in radiation therapy (RT). One possible solution could be to employ distinct dose delivery methods, activating different biological processes from those ones in standard RT. Along this line, the strong spatial modulation used in techniques such as GRID, Lattice therapy or minibeam radiation therapy (MBRT), has already demonstrated a significant reduction of normal tissue toxicity and an important increase of the therapeutic index for some (radio-resistant) tumors. While spatially fractionated radiation therapy (SFRT) has mainly used photon beams, the alliance of SFRT and the benefits of charged particles for therapy has recently started. This presentation will provide a general overview on SFRT and how it is evolving from the use of photons to charged particles.

Learning Objectives:

1. Learn about spatial fractionation of the dose in radiation therapy and its advantages
2. Learn about radiobiology in SFRT

Speaker 3: Prof. Sha Chang, University of North Carolina

Presentation Title: "Understanding the correlation between dosimetric parameters and treatment responses for spatially fractionated radiation therapy"

A solid understanding of correlation between dosimetry and treatment response is crucial for radiation therapy for cancer. We have gained such an understanding for conventional seamless radiation therapy but it may not apply for spatially-fractionated radiation therapy (SFRT).

SFRT is a nonconventional radiation therapy where dose distribution is highly fractionated in space. Decades of clinical and preclinical data have shown that SFRT has promising potentials as a very high therapeutic ratio radiation therapy, especially when used as a neoadjuvant therapy and for cancers that cannot be adequately managed by conventional therapies. These cancers include large late stages cancers, in- or near-field reoccurrences, refractory cancers, and cancer within or near highly radiosensitive critical organs.

Two major obstacles for broad SFRT clinical application are lack of understanding of working mechanism and of the correlation between complex SFRT dosimetric parameters and treatment responses. The key questions to be answered include what is the most proper dosimetric parameter to be used for SFRT treatment prescription and what is an "apple to apple" comparison between a SFRT treatment and the "gold standard" conventional radiation therapy for a given endpoint. In this presentation we intend to shed lights on these and other important questions by reviewing and analyzing available data and discuss directions for future research.

Learning Objectives:

1. Recognize that understanding the correlation between dosimetric parameters and treatment response is crucial to advance SFRT.
2. Understand that preclinical SFRT studies indicate that peak dose may not be closely correlated with treatment response.
3. Learn methods to conduct research on SFRT dosimetry vs. treatment response.

Speaker 4: Prof. Ke Sheng, University of California,

Presentation Title: "Radiotherapy beyond 4pi"

Despite the potential for significantly improve the dose conformity, OAR sparing, and target dose escalation, gaps exist to prevent 4 π radiotherapy from being widely used in the clinic. The gaps include the complexity of the optimization, the cumbersomeness of the C-arm gantry system to achieve the non-coplanar angles, and the long time needed for treatment delivery. The presentation will discuss the technical roadmaps to overcome these challenges. Artificial intelligence can be used to streamline treatment planning in two aspects. First, the plan dose can be individually predicted based on previously treated patients. Second, Artificial intelligence methods can be used to simplify dose calculation and optimization or circumvent certain steps of the optimization to speed up the calculation. A new robotic hardware platform is expected to effectively and efficiently access the non-coplanar angles without rotating the patient couch. The reduction in the form factor afforded by cutting-edge linear accelerator affords additional beam angles that were unavailable previously. The new many-isocenter optimization methods blur the traditional definitions of field of view and intensity modulation resolution. Ultimately, the combination of hardware and software innovations will move 4 π closer to clinical reality.

Learning Objectives:

1. Learn the challenges that exist in 4 π radiotherapy and the hardware development to overcome the challenges.
2. Learn the optimization methods that allow the same system to treat both small targets with high intensity modulation resolution in SRS and large tumors with conventional fractionation.

SAM Partners in Solutions Therapy

TRACK 8

Therapy: Adaptive Therapy AI Software Delivery Systems

TU-EF-TRACK 8-00

Therapy: Adaptive Therapy AI Software Delivery Systems

Moderators: Leigh Conroy, Princess Margaret Cancer Centre, Toronto, ON and Deborah Schofield, AdventHealth Orlando, Orlando, FL

TU-EF-TRACK 8-01

L. Conroy, Princess Margaret Cancer Centre:

Adaptive Therapy AI Software Delivery Systems

TU-EF-TRACK 8-02

A. Yock, Vanderbilt University Medical Center:

Ethos™ Therapy, An Adaptive Intelligence™ Solution

TU-EF-TRACK 8-03

D. Casey, Accuray Inc.:

Real-Time Delivery Adaptation with Synchrony®, and Adaptive Therapy with Automated Dose Trending on the Radixact® System

TU-EF-TRACK 8-04

D. Ly, Elekta, Inc.:

Adaptive Planning with Monaco TPS for Unity

TU-EF-TRACK 8-05

Q&A

A treatment plan captures a single moment in time and does not account for changes in anatomy over the course of treatment. Changes in weight or variations in internal anatomy, for example, can result in displacement of the target volume and OARs relative to the planned dose. The process of adapting a plan can be labor-intensive and time-consuming for both physicians and staff. Artificial intelligence (AI) has the potential to improve the accuracy, precision, efficiency, and overall quality of radiation therapy for patients by automating adaptive radiotherapy workflows such as target segmentation, treatment planning, quality assurance and, in some cases, radiotherapy delivery. Vendors will present their AI adaptive solutions and describe how their products can improve the workflow and empower the clinician and staff to make informed decisions related to patient cancer treatments.

Keywords: Artificial intelligence, automated contouring, adaptive, treatment planning, treatment delivery

Learning Objectives:

1. Review the rationale behind adaptive planning.
2. Explore the different means of adapting a plan including off-line, on-line, and real-time adaptive planning.
3. Explain how AI can help streamline the adaptive treatment process.

Professional Symposium

TRACK 1

New Member Symposium

TU-F-TRACK 1-00

New Member Symposium

Moderators: Kathryn Mittauer, Miami Cancer Institute, Baptist Health South Florida, Miami, FL and Shelby Grzetic, Advocate Christ Medical Center, Oak Lawn, IL

TU-F-TRACK 1-01

Introduction

TU-F-TRACK 1-02

J. Dobbins, Duke University Medical Center:

President's Remarks: Volunteer History of AAPM / How You Can Become Involved

TU-F-TRACK 1-03

F. Fahey, Children's Hospital:

Administrative Council

TU-F-TRACK 1-04

J. Prisciandaro, University of Michigan:

Education Council

TU-F-TRACK 1-05

J. Palta, Virginia Commonwealth University:

International Council

TU-F-TRACK 1-06

B. Parker, The University of Texas M.D. Anderson Cancer Center:

Professional Council

TU-F-TRACK 1-07

J. Seuntjens, McGill University:

Science Council

TU-F-TRACK 1-08

Moderated Panel Discussion / Drawing for Attendees

The New Professionals Subcommittee (NPSC) offers a New Member Symposium (NMS) at each Summer Meeting in order to provide information about how to become involved with AAPM and how to deal with issues that students, trainees, and anyone new to the AAPM organization are faced with. This summer we invite you to come with your questions about how to search for a job, become a team member, and how to become a volunteer for AAPM.

AAPM President James Dobbins will discuss leadership of the AAPM, and a panel of experts will discuss topics relevant to AAPM's volunteer organization: Frederic Fahey, Chair of Administrative Council, Joann Prisciandaro, Chair of Education Council, Jatinder Palta, Chair of International Council, Brent Parker, Chair Professional Council Chair, Jan Seuntjens, Chair of Science Council. A question and answer will follow to discuss your interests, concerns, and ideas.

Learning Objectives:

1. Gain familiarity of the leadership structure of the AAPM
2. Explore how to become involved in committees in AAPM and in AAPM leadership
3. Gain understanding of new initiatives and future directions in each AAPM council

Imaging Scientific Session

TRACK 3

New Technologies and Image Reconstruction in CT and CBCT

TU-F-TRACK 3-01

Estimating Ischemic Core Volume with Four Dynamic Images in CT Perfusion

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Purpose: Simplifying CT perfusion (CTP) protocols may encourage its use at less-experienced centers to guide patient selection for thrombectomy in acute ischemic stroke. In this simulation study, we demonstrate that the cerebral blood flow (CBF) map can be calculated with four images equivalent to standard of care non-contrast CT (NCCT) and multiphase CT angiography (mCTA) to measure the ischemic core volume required for patient selection. **Methods:** Twenty-eight consecutive patients with acute ischemic stroke and CTP at admission were included in this study. The standard CTP protocol consisted of 22 dynamic images acquired over 60 s. NCCT and mCTA were simulated by manually selecting the four dynamic CTP images that corresponded to the pre-contrast baseline (NCCT), peak arterial enhancement, and two delayed images at 8 and 16 s after the peak (mCTA). Early or late timing of the peak and of the delayed images was also simulated by shifting the selection of contrast-enhanced images up to three time points (<9 s) before or after the true peak. CBF maps were generated by a model-based deconvolution algorithm from which ischemic core was identified by $\text{CBF} < 30\%$ of the contralateral hemisphere. Bland-Altman analysis and the intraclass correlation coefficient (ICC) assessed agreement and reliability, respectively, of measured core volumes compared to those from a standard CTP study. **Results:** With correct timing of the arterial peak, the mean difference (MD; limits of agreement) and ICC were 3.7 (−4.9 to 12.4) ml and 0.94 (0.72 to 0.98), respectively. MD and ICC were similarly good when the peak was timed early by up to 6 s or late by 3 s ($\text{MD} < 5.9$ ml; $\text{ICC} > 0.86$) but poor with earlier (≈ 9 s) or later timings (> 3 s). **Conclusion:** Core volumes measured with only four images equivalent to NCCT and mCTA were consistent to those from a standard CTP study.

K.J. Chung is supported by an NSERC Canada Graduate Scholarship - Master's award. T.Y. Lee licenses CT Perfusion software to GE Healthcare.

TU-F-TRACK 3-02

JACK FOWLER EARLY-CAREER INVESTIGATOR COMPETITION WINNER: Imaging from the Cervical to the Lumbar Spine with a Continuous Multi-Turn Reverse Helical 3D Cone-Beam CT Scan

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Purpose: The current axial coverage of intraoperative 3D cone-beam CT imaging is limited by detector length. For neurosurgical interventions, for example, this prevents the alignment of the entire spine to be assessed from a single 3D scan. Here, we investigate the feasibility of realising whole spine 3D cone-beam CT imaging for the first time within the interventional suite. **Methods:** A Siemens ARTIS pheno robotic angiography imaging system in conjunction with a Siemens Test Automation Control System (TACS) was used to image a 3D printed spine phantom (Sawbones) containing cervical, thoracic, and lumbar vertebra with a multi-turn reverse helical scan. The TACS was used to control both the C-arm rotation and table movement. The C-arm was rotated continuously through a total of 1350° (400° clockwise, 400° anticlockwise, 400° clockwise and 150° anticlockwise) around the table at $20^\circ/\text{s}$, acquiring projections at a fixed frame rate of 10 f/s, while the table was translated backward from the C-arm at a rate of 10 mm/s. The 3D images were reconstructed using penalized-likelihood reconstructions. To compare the axial coverage with current 3D imaging capacity, a conventional circular scan (4 second, 248 projections) was acquired of the thoracic spine. Image quality was examined visually and via the Structural Similarity Index Metric (SSIM). **Results:** The multi-turn reverse helical scan enabled a total axial coverage of 76 cm, capturing all 24 vertebrae. Comparatively, the conventional scan enabled 17 cm of axial coverage, encapsulating 6 thoracic vertebrae. The multi-turn reverse helical scan took 80 seconds to complete, acquiring 790 projections. Structural similarity of the multi-turn to the conventional scan from T-4 to T-9 was 97.83%. **Conclusion:** This is the first time a continuous multi-turn reverse helical scan has been implemented on a clinical robotic angiography system, allowing the entire spine to be imaged in a single scan.

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TU-F-TRACK 3-03

Intraoperative Cone-Beam CT Image Quality and Dose with a Slot Collimator On the O-Arm

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Purpose: An actuated slot-beam collimator was implemented on an intraoperative cone-beam CT system (Medtronic O-armTM), hypothesizing benefits to dose and soft-tissue contrast resolution for the slot-beam geometry. This study investigates the dose, scatter-to-primary, and contrast-to-noise characteristics for slot- and cone-beam configurations for scenarios that may benefit from improved soft-tissue image quality. **Methods:** The collimator features three narrow slots that can be automatically moved into position during scanning, each exposing a detector area of 40 cm width x 2.5 cm longitudinal. Slot- and cone-beam configurations were evaluated with and without a focused linear anti-scatter grid (12:1 GR). Dose, scatter, and soft-tissue contrast resolution were evaluated in phantom studies for head and body protocols (~ 745 projections over 360° , $0.388 \times 0.776 \text{ mm}^2$ pixels, 80-120 kV, 75-940 mAs) with reconstruction performed via 3D filtered backprojection. Quantitative evaluation included scatter-to-primary ratio (SPR), contrast-to-noise ratio (CNR), modulation transfer function (MTF), and absorbed dose. **Results:** The slot-beam configuration reduced SPR by $\sim 6\times$ compared to a cone-beam, and the grid reduced SPR by $\sim 3\times$ compared to no-grid. As anticipated, the slot-beam with a grid yielded the lowest SPR (< 0.1 and < 0.3 for the head and body, respectively) and increased soft-tissue CNR by 2-2.5x compared to cone-beam at equivalent dose. MTF was unaffected among all configurations. The dose for the slot-beam configuration was $\sim 1/5$ that for the cone-beam due to reduction of scatter dose, ranging 0.013-0.037 mGy/mAs for the head and 0.006-0.017 mGy/mAs for the body. **Conclusion:** Implementation of an actuated slot-beam collimator demonstrated strong improvements in image quality and dose that may benefit certain interventional imaging scenarios – e.g., high-quality soft-tissue visualization within a region of interest to assess the quality of surgical product or as a check against hemorrhage. It also provides a potential step toward intraoperative helical scanning on such systems.

The research was supported by academic-industry partnership with Medtronic Inc.

TU-F-TRACK 3-04

Ordered Subsets PWLS Reconstruction of Multi-Layer Detector MV-CBCT Data Using TIGRE

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Purpose: Recently, clinical use of MV cone beam CT (CBCT) for radiotherapy guidance has increased. However, even in systems capable of lower MV energies, image quality suffers from detection efficiency limitations in traditional x-ray detectors. By combining a high DQE multi-layer detector prototype with dose-efficient, iterative Penalized Weighted Least Squares (PWLS) reconstruction, improved MV-CBCT image quality is pursued. **Methods:** The proposed imager consists of 4 detector layers, each with a 0.506 mm GOS scintillator and a 0.7 mm photodiode sub-layer. Copper blocking layers are omitted to improve DQE at lower (2.5 MV) energies. The iterations of PWLS are accelerated by using the Ordered Subsets Momentum-Based (OSMOM) technique to enhance convergence rate, and the public-domain Tomographic Iterative GPU-based Reconstruction (TIGRE) library to alleviate forward/back projection time. Refinements to TIGRE were made that allowed it to communicate input/output as MATLAB gpuArray objects. This enabled all iterative computations to be executed within MATLAB on the GPU, eliminating overhead from CPU-to-GPU transfers. The overall technique is evaluated in terms of noise and contrast-to-noise (CNR) in scans of the Catphan 604, and qualitatively in a pelvis phantom. **Results:** The proposed technique demonstrated 3 times lower noise as a function of spatial resolution compared to a conventional MV detector architecture and reconstruction algorithm (FDK). Reconstructed noise levels closely approached that of the Varian Edge LINAC's clinical kV-CBCT Spotlight protocol at matched dose (CTDI=12.5 mGy). This resulted in CNR about half that of kV-CBCT in a soft tissue (LDPE) region of interest. On the GTX 1080 Ti graphics card, reconstruction time for the pelvis was about 30 sec. using a 400x400x170 voxel grid and 200x320x400 projections. **Conclusion:** Combining a multi-layer imager with PWLS reconstruction greatly diminishes the image quality gap between MV-CBCT and standard kV-CBCT. Moreover, short reconstruction times are implementable with MATLAB and non-proprietary GPU software.

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TU-F-TRACK 3-05

Reconstructing Dual KV Data Using the COSSCIR "One-Step" Direct Inversion Algorithm

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Purpose: This study applies the one-step inversion material decomposition algorithm, cOSSCIR, to Dual kV spectral CT and validates our approach on digital and experimental phantom data. **Methods:** The cOSSCIR algorithm is a one-step inversion approach in which the basis images are directly optimized from the spectral projection measurements. cOSSCIR demonstrated reduced artifacts caused by beam hardening and photon starvation for photon-counting CT. The cOSSCIR approach does not require spectral measurements to be registered and thus may provide additional benefits for clinical Dual kV CT systems. By modifying the spectral response and accounting for the detector gain, an integrating detector model was developed and tested on a simple digital phantom. A subsequent inverse crime study was used to validate the approach. A preliminary experimental study was also performed using a physical phantom composed of cylindrical rods of PMMA, Polystyrene, and Teflon. Projection data was acquired at 80 and 130 kV over 500 equally spaced view angles. The system consists of a micro-focal x-ray source (Hamamatsu 9181-02) and a high resolution flat panel detector (Varian PaxScan 2520 DX). The Aluminum and PMMA basis images were then reconstructed using cOSSCIR. For each basis image, the mean value in each material ROI was compared with their predicted values. **Results:** Convergence to the digital phantom suggests that our integrating detector model is consistent with the mathematical framework developed for cOSSCIR. Furthermore, the experimental results demonstrate good recovery of the materials present within the physical phantom. The percent error in the PMMA basis image was 1.9% 3.3%, 4.3% in the polystyrene, PMMA, and Teflon ROIs. Similarly, in the Aluminum basis image the percent error was 7.5%, and 38.4% in the Polystyrene and Teflon ROIs. **Conclusion:** By incorporating an integrating detector model into the cOSSCIR framework we demonstrate direct inversion material decomposition for Dual kV data.

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TU-F-TRACK 3-06

Weight-Bearing C-Arm Cone-Beam CT Using a Dynamic Range Reducer (DyRaR): Initial Prototype and Phantom Studies

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Purpose: Cone-beam CT (CBCT) with automatic exposure control saturates the flat-panel detector in peripheral regions, leading to inferior image quality and excess skin dose. For example, in weight-bearing CBCT of knees, a traditional fixed bowtie filter cannot fully address this challenge. Thus, we constructed a prototype dynamic range reducer (DyRaR) which comprises four independently controlled brass wedges on each lateral side. The DyRaR is pre-programmed to conform to the patient-specific knee shape and continuously attenuate the beam beyond the knee outline as the gantry rotates. In this work, we demonstrate the DyRaR for weight-bearing CBCT using a knee phantom. **Methods:** A knee phantom was first scanned using the weight-bearing CBCT trajectory to establish baseline image quality and to design phantom-specific DyRaR wedge positions. A second scan of the knee phantom was then acquired with the DyRaR mounted to the collimator of the C-arm. Finally, an air scan with just the DyRaR was acquired to correct the previous scan. Note that direct subtraction of the two scans is not ideal due to differences in beam hardening and scattering. Thus, we segmented the projections of knee+DyRaR scan into knee only, DyRaR only, and overlap regions, and used a 2nd-order polynomial to correct different regions. **Results:** The DyRaR substantially attenuated the beam and prevented overexposure in air or near the knee outline by precisely following the designed trajectory. With proper projection correction, we could reconstruct CBCT images of the knee phantom with minor artifacts around the knee boundaries, although further image processing is needed to fully correct beam hardening and residual artifacts. **Conclusion:** We have designed a prototype DyRaR and evaluated its application on a C-arm system for weight-bearing CBCT of knees. This study demonstrates the potential to design patient-specific DyRaR trajectories toward achieving optimal CBCT image quality.

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TU-F-TRACK 3-07

Simultaneous Analytical Image Reconstruction and Material Decomposition Denoising in Single-Scan Dual-Energy Cone-Beam CT Using Dual Energy Vectorization and Joint Bilateral Filtering

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Purpose: Dual-energy cone-beam CT (DE-CBCT) with a flat-panel detector (FPD) has promising applications in various clinical tasks. Rotating-filter based DE-CBCT (RF-DECT) technique could acquire dual-energy projections within single circular rotation, enabling shorter scan time and reduced patient dose compared with two-scan method. Here we propose a new method for RF-DECT imaging which achieves analytical image reconstruction and material decomposition denoising

simultaneously. **Methods:** RF-CBCT employs a rotatable filter to produce low and high energy projections alternately. CT images were first separately reconstructed using filtered back-projection (FBP) method on the down-sampled projection set of each energy, then we compactly represented the reconstructed low and high energy images in a vectorized form, which was further characterized as an amplitude image and an argument image. Based on the analysis of noise dependence and image complementarity, a joint bilateral filter using amplitude image as a guide was then designed for edge-preserved smoothing of argument image. Finally, reconstructed images with reduced streaks were recovered from the amplitude image and the smoothed argument image, decomposed noise could be effectively suppressed simultaneously. **Results:** On Catphan®600 phantom, the proposed method reduced streaking artifacts while preserving spatial resolution in the reconstructed images. Compared with typical two-scan results, proposed method reduced noise by a factor of 21.2, 38.0, 2.5 for soft tissue, bone tissue and electron density image, respectively, and achieved electron density accuracy within 1.5% for each material. On anthropomorphic head phantom, the proposed method successfully recovered the intricate brain structures in both high energy and electron density images. **Conclusion:** Using down-sampled projections in single-scan DE-CBCT, proposed method achieves simultaneous analytical image reconstruction and material decomposition denoising. This method demonstrated superior performance on spatial resolution enhancement, decomposition noise reduction and electron density accuracy, allowing better material differentiation and dose calculation, while using only half radiation dose.

Therapy Scientific Session

TRACK 5

FLASH: Treatment Planning, Delivery, and Verification

TU-F-TRACK 5-01

Simulating Spatial-Temporal Dose Rate Deposition on Tissue Sparing for Scanned Pencil-Beam Vs Double-Scattered Protons

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Purpose: To develop a proton dose calculation framework simulating both spatial distribution and time structure of ultra-high dose rate deliveries, enabling comparison of scanned pencil-beam (PBS) vs double-scattered (DS) sequences on local tissue sparing. **Methods:** Physical proton dose was implemented based on the validated superposition model in Eclipse, with a novel grid interpolation framework enabling rapid computation of each time-step. Oxygen depletion and tissue sparing was modeled across the calculation grid as exponential depletion and recovery proportional to concentration gradient, each time-step coupled to the Oxygen-Enhancement-Ratio k-curve, yielding effective dose. Dose equivalent PBS and DS plans were created with various time structures to explore their effective dose which depends on oxygen concentration at each moment of energy deposition. Beam currents were extrapolated to FLASH dose rates. **Results:** DS deliveries consistently yielded greater tissue sparing proximal to the spread out Bragg peak (SOBP). Slow time between spots and layers inherent to PBS deliveries allowed oxygen to recover to nearly physiologic levels, whereas DS modulation irradiates the entire field near simultaneously, for widespread oxygen depletion even at a fraction of the PBS instantaneous dose rate. DS offered nearly twice the differential sparing of regions proximal to the SOBP, where dose to organs-at-risk are of concern. PBS can be improved by minimizing slow time and number or dose repainting. **Conclusion:** The spatial-temporal structure of high dose rate deliveries yields insight into differential tissue sparing, as groundwork for proton treatment planning while considering FLASH effects. PBS scanning trajectories can be explored which maximize differential sparing in regions of interest. Radiochemistry and tissue sparing models can be refined with further experimental evidence towards a planning paradigm considering dose rate modulated tissue sparing.

TU-F-TRACK 5-02

Feasibility of 3D Printed Ridge Filter to Enable SBRT FLASH Therapy Using Scanning Proton Beam

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Purpose: Due to long proton energy switching time, most FLASH therapy is limited to single energy and to spare normal tissue without conformity of treatment target along the depth direction. We investigate if a 3D ridge filter can enable dose conformity of stereotactic radiation therapy (SBRT) target in FLASH treatment using scanning proton beams. **Methods:** Varian ProBEAM's FLASH energy of 250 MeV was degraded to 120 MeV to treat simulated SBRT spherical targets of 30 mm and 50 mm diameters at depths of ~80 mm of a water phantom. We develop a method to optimize the height and location of the PMMA pins of the ridge filter so that the spread-out Bragg peaks at each location can correspond to that of the sphere targets. The ridge filter pins must meet the 0.1-mm lateral and height resolutions of a 3D printer. Geant4 was used to simulate radiation transport. The spot weights were also optimized to achieve lateral target conformity. **Results:** The simulation results showed that it is feasible to use 3D printed ridge filters to achieve location-dependent spread-out Bragg peaks for SBRT spherical targets for one fixed energy. Delivering 10 Gy dose in such spot maps is estimated to be below 250 ms, achieving a Flash dose rate of 40 Gy/s. The validation of timing and simulated dosimetric distributions will be performed since Flash mode has been recently delivered at Emory. **Conclusion:** According to Geant4 simulation, it is feasible to use a 3D ridge filter to enable proton SBRT FLASH therapy up to 50 mm diameter sphere target.

TU-F-TRACK 5-03

Need for Beam Collimation Design for Dose-Rate Fall-Off in Proton Flash Treatment Using Pencil Beam Scanning

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Purpose: Flash irradiation requires delivering a spatial high dose-rate that is consistently above the Flash dose-rate threshold (>40Gy/sec). This requirement is not fully met for Gaussian beam distributions such as proton pencil beam scanning as the lateral fall-off of dose (or dose-rate) profile can be below the Flash threshold. In this study, we evaluate the use of proton beam collimation to reduce the non-Flash dose-rate of proton pencil beam. **Methods:** The profile of a proton pencil beam captured using a plastic scintillator with and without 8 mm pinhole collimation using a lead aperture. The measurements were performed using 245 MeV proton beam at 5, 15 and 37 cm depth in water, similarly, for 150 MeV beam at of 5 and 15 cm depths. The data analysis focused on (i) Comparison of the spatial Intensity distribution of the collimated and non-collimated beams, (ii) Determining the adequate collimation ratio, defined as the ratio of beam spot-size to diameter of the pinhole aperture, and (iii) Determining the proportion of non-flash dose delivered by the collimated beam. **Results:** The collimation of 245 MeV (with the spot size of $\sigma=3.62$ mm) reduces the non-Flash dose by to 6%, 9% and 15% of the delivered dose at the respective depths of 5, 15 and 37 cm in comparison to 26% from non-collimated beam. The collimation of 150 MeV beam ($\sigma=4.47$ mm) is satisfactory as it limits the non-Flash dose to less than 4.3% for all depths comparing to 25% for non-collimated beam. **Conclusion:** The collimation of Gaussian proton beam can reduce the Flash beam dose-rate fall-off as long as the collimation factor is higher than 0.5.

TU-F-TRACK 5-04

Prompt Gamma Timing as a Real-Time Relative Dose Rate Monitor of FLASH Proton Delivery

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Purpose: To investigate the feasibility of real-time proton FLASH relative dose rate monitoring via prompt gamma ray (PG) timing. PG event measurements are used for secondary relative verification of absolute dose rate delivery and for visualization into the temporal dose rate characteristics of proton FLASH deliveries up to 138 Gy/s. PG counts are processed to provide insight into an optimized PG detection system performance. **Methods:** A PG detection and measurement system consisting of a NaI(Tl) scintillator and nuclear electronic pulse processing system are used along with transmission ionization chamber absolute dose measurements. Varying dose rates were delivered to measure PG counts and count rates to investigate PG count and proton FLASH dose rate linearity, time structures, and delivery stability. **Results:** PG counts and count rates are found to increase linearly with delivered dose and dose rate. Upon beam termination, PG counts exhibit an exponential decay due to scintillation crystal afterglow with a half-life of 2.495 ± 1.35 ms, which corresponds to the theoretical minimum response time in which a real-time dose rate monitoring system would be able to terminate beam delivery upon recognition of $\geq 7.0\%$ dose rate variation per single delivery. With no detector collimation, FLASH dose rates cause detection system paralyzation, leading to PG count saturation; after detector dead time and paralyzation correction calculations, true unsaturated PG count rates for dose rates up to 138 Gy/s were determined. Variations in absolute dose rate measurements provide a $>6.59\%$ threshold variation for which PG counts could provide substantial dose rate variation detectability. **Conclusion:** PG counting systems can be designed to perform real-time relative proton dose rate monitoring for <36 Gy/s deliveries, and with corrections accounting for PG count saturation, real-time relative monitoring may be feasible for FLASH dose rates; allowing for verification of critical research and clinical setting delivery parameters.

TU-F-TRACK 5-05

Pulse Resolved Beam Characterization and Feedback for FLASH-RT Using Radioluminescent Dosimeters

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Purpose: Beam monitoring for FLASH-RT is challenging due to ultra-high dose rate conditions. In this study, dose-rate independent radioluminescent based techniques are presented for pulse-resolved beam characterization and real-time feedback to a FLASH enabled Linac. **Methods:** A modified clinical linac delivered a radially symmetric 10 MeV electron FLASH beam (~ 300 Gy/s, 1 Gy/Pulse) to isocenter. Lateral projected 2D dose distributions for each linac pulse were imaged in a fluorophore-doped water tank using a time-gated camera. An inverse Abel transform reconstruction provided 3D images for on-axis depth dose values. The central axis depth dose values were compared against film data. For pulse-by-pulse feedback, a scintillating point dosimeter was coupled to a gated integrator and a field programmable gate array (FPGA)-based control system. **Results:** The camera images were reconstructed at a spatial resolution of 0.5 mm. The D_{max} , R_{50} , and R_p measured with film and camera based method agreed to within 1 mm for a 1.5 cm circular beam and the beam with jaws wide open (40×40 cm²). Central axis cross beam profiles for both beams agreed with film data with $>95\%$ passing rate (2%/2mm gamma criteria). Beam energy measurements based on the R_{50} criteria using optical measurements revealed a stable beam energy between adjacent pulses. However, a ramp-up period was observed where the dose per pulse increased for the first few pulses and then stabilized. The dose-read out scheme for individual pulses using the gated integrator and the FPGA based system was characterized. **Conclusion:** Radioluminescence based techniques were presented as a high spatio-temporal resolution technique for online monitoring of FLASH beams. The presence of a ramp-up requires that feedback to the Linac should be based on dose accumulation. Therefore, the gated integrator and the FPGA based control system can enable accurate feedback for FLASH.

TU-F-TRACK 5-06

Absolute Dosimetry of Ultra-High Dose Rate Pulsed Electron Beams Using a Graphite Probe Calorimeter

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Purpose: To evaluate and verify the dosimetric response of a clinically-deployable probe-format graphite calorimeter when operated in ultra-high dose rate (or 'FLASH') pulsed electron beams. **Methods:** The ion chamber-sized graphite calorimeter, termed Aerrow, was operated in quasi-adiabatic mode. In this mode, the absorbed dose deposited in the graphite core can be obtained by calculating the product of the specific heat capacity and the radiation-induced temperature rise. The measurements were carried out at the Metrological Electron Accelerator Facility of the German National Metrology Institute (PTB). Ultra-high dose rate pulses were delivered with a 20 MeV electron beam. The dose-per-pulse was varied in the range of 0.5 – 5.5 Gy. The detector response measurement was repeated 5 times for each pulse, with pulse deliveries separated by 90 seconds. The pulse charge was monitored non-destructively with a previously validated beam current transformer (ICT). Dose conversion factors from graphite to water were calculated using the EGSnrc Monte Carlo code system. The heat transfer correction factor was assessed through finite element analysis using COMSOL Multiphysics. **Results:** The response of the calorimeter was found to be independent of dose-per-pulse in the investigated range of pulse charges. The absorbed dose values measured by Aerrow had an average absolute deviation of 0.11 % and a maximum absolute deviation of 0.42 % from the fit on the ICT signals (which have uncertainties of 0.05 %). The response uncertainty corresponding to a coverage factor of $k = 2$ was 0.2 % (on average) for the higher dose rates (≥ 4.5 Gy/pulse) and reached a maximum of 1 % at the lowest dose rate. **Conclusion:** This feasibility study demonstrates that Aerrow can produce accurate and reproducible dose measurements in the ultra-high dose rate regime of pulsed electron beams, making it a suitable candidate for absolute dosimetry of clinical FLASH radiotherapy beams.

This work was supported financially by the CREATE Medical Physics Research Training Network grant of the Natural Sciences and Engineering Research Council (NSERC) (#432290) and NSERC operating grant (#RGPIN 2019-06746). Absorbed dose calorimetry research at McGill University is in part supported by Sun Nuclear Corporation.

TU-F-TRACK 5-07

A Pyroelectric Calorimeter with Electrometer Readout for the Absolute Dosimetry of FLASH Radiation: Proof of Principle

J Renaud^{*}, B Muir, National Research Council, Ottawa, ON

Purpose: Given the extremely short delivery times of FLASH beams, real-time dosimetry is essential. The poor performance of ion chambers in high dose-per-pulse beams is well known and the alternative proposed to date is absorbed dose calorimetry. However, to-date, such devices are complex to operate and require instrumentation for readout not normally possessed by clinics. This work describes the development and initial characterization of a pyroelectric calorimeter capable of FLASH dosimetry measurements that can be read out with only an electrometer. **Methods:** A square pyroelectric crystal (lead-zirconate-titanate; PZT) with a side length of 2.8 mm was mounted in a cylindrical Delrin holder. The holder reproducibly compresses the crystal between two electrodes. The pyroelectric crystal is a current source with a signal magnitude that is directly proportional to a change in temperature, hence a calorimeter. As an initial test, a calibrated thermistor was affixed to the crystal and the assembly was embedded in a copper block. The temperature of the copper was then modulated by exposing it to the heat of a halogen lamp, simulating the effect of a FLASH beam. Pyroelectric readout was performed with an electrometer (Keithley 6517A) in current mode, while thermistor resistance was measured with an 8.5-digit multimeter (HP 3458A). **Results:** A set of 6000 data points were acquired spanning rates of temperature change from ± 8 mK/s, or 3.5 Gy/s in terms of equivalent dose rate. Pyroelectric current varied linearly with the time derivative of the thermistor temperature. The proportionality constant, which is the detector's calibration coefficient, was measured to be $2.300(3)$ nC/K, a value consistent with the literature on PZT. **Conclusion:** This proof of principle demonstrates the feasibility of operating a pyroelectric calorimeter with an electrometer for the purpose of absolute determination of absorbed dose rates greater than a few Gy/s, relevant for FLASH dosimetry.

Multi-Disciplinary Scientific Session

TRACK 6

AI-Based Auto-Segmentation and Auto-Contouring - II

TU-F-TRACK 6-01

3D Dense U-Net for Fully Automated Multi-Organ Segmentation in Female Pelvic Magnetic Resonance Imaging

F Zabihollahy*, E Schmidt, A Viswanathan, J Lee, Department of Radiation Oncology and Molecular Radiation Sciences, Johns Hopkins University, Baltimore, MD

Purpose: Brachytherapy (BT) combined with external beam radiotherapy (RT) is the current standard of care for cervical cancer patients. Accurate segmentation of organs at risk (OAR) is a prerequisite step for RT planning. In this study, we propose a two-step automatic multi-organ segmentation method using convolutional neural network (CNN) in female pelvic magnetic resonance (MR) images. **Methods:** Our segmentation model consists of two cascaded steps; 1) organ localization and 2) fine segmentation of individual organs. The first step employs a 3D U-Net to localize each organ from which organ-specific regions of interest (ROIs) are produced. The original MR volume is then cropped, based on the ROIs, yielding organ-specific volumes for the next step. For fine segmentation, modified 3D Dense Block (BD) is embedded into a 3D U-Net-like architecture to simultaneously capture intra- and inter-slice information while facilitating information flow in the network. A total of 221 3D T2-weighted MR volumes from 135 patients, were used for this study. Training images were augmented using shift and rotation operations and the networks were trained for 200 epochs using a total of 804 MR volumes. **Results:** The proposed method was applied to segment the bladder, rectum, and sigmoid. Compared to the expert radiation oncologists' manual segmentations, average dice similarity coefficients (DSC, mean $\pm$ standard deviation) for 20 test cases are 0.93 ± 0.04 (bladder), 0.87 ± 0.03 (rectum), and 0.78 ± 0.09 (sigmoid). Our methods was also compared to the standard 3D U-Net and 3D Dense U-Net and showed superior performance ($p < 0.05$). Using this newly created model, the segmentation takes 7-8 seconds per each organ. **Conclusion:** We proposed a novel fully automated multi-organ segmentation algorithm for female pelvic MR images. Our experimental results demonstrate that the developed method is accurate, fast, and reproducible, potentially reducing the burden of tedious manual image contouring and thus improving efficiency in RT planning.

TU-F-TRACK 6-02

Automatic Segmentation and Physician Review of Intact and Post-Operative Prostate Radiotherapy Contours

M El Basha^{1,2*}, D Fuentes^{1,2}, J Pollard-Larkin^{1,2}, F Poenisch², Z Yu^{1,2}, S Frank², C Tang², A Aggarwal³, L Court^{1,2}, C Cardenas^{1,2}, (1) UT MD Anderson UT Health Graduate School of Biomedical Sciences, Houston, TX, (2) University of Texas MD Anderson Cancer Center, Houston, TX, (3) Guy's and St. Thomas' NHS Foundation Trust, London, UK

Purpose: To develop and clinically review a deep learning segmentation algorithm that segments intact and post-operative targets (prostate, prostate bed, seminal vesicles (SV), and SV fossa) and five normal structures for prostate cancer treatment planning. **Methods:** A modified two-stage 3D UNet model was developed using retrospective clinical CT data from 670 patients (8:1:1 train, validation, test split). Stage one inputs a downsampled CT volume and outputs a localized coarse region of interest (ROI) containing multiclass segmentations. The ROI segmentations from stage one are then processed using k-means clustering to identify centroids for 3D patches that are cropped from the full-resolution CT image to generate final segmentation predictions. The 3D patches are then combined by averaging pixel-wise prediction probabilities when there is overlap between 3D patches to generate individual ROI segmentations. The resulting tissue contours were then evaluated quantitatively (Dice, mean surface distance) and qualitatively (radiation oncologist review). **Results:** Segmentations of 9 planning structures were generated with the following average Dice/mean surface distances: prostate $0.79 \pm 0.11/2.9 \pm 2.0$ mm, prostate bed $0.75 \pm 0.05/3.1 \pm 0.6$ mm, SV $0.6 \pm 0.17/3.7 \pm 4.3$ mm, SV-Fossa $0.49 \pm 0.18/4.2 \pm 3.1$ mm, rectum $0.87 \pm 0.05/2.1 \pm 1.0$ mm, bladder $0.96 \pm 0.01/0.8 \pm 0.1$ mm, penile bulb $0.74 \pm 0.11/1.6 \pm 0.8$ mm, femoral heads $0.92 \pm 0.03/1.6 \pm 1.0$ mm, sigmoid $0.40 \pm 0.25/12.6 \pm 13.5$ mm. Two radiation oncologists each reviewed at least two intact and two post-operative patient datasets. The preliminary qualitative review of contours showed the majority of bladder, penile bulb, and femoral head contours were acceptable without edits; the rectum, prostate bed, and SV contours required minor edits; and the sigmoid, SV fossa, and prostate contours required either minor or major edits. **Conclusion:** We developed and evaluated an auto-segmentation algorithm for intact and post-operative prostate radiotherapy. Additional qualitative review is necessary to determine clinical acceptability of the automated contours.

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TU-F-TRACK 6-03

Autosegmentation Anomaly Detection by Clustering of Shapes

D Sawkey*, Varian Medical Systems, Toronto, CA

Purpose: As machine learning (ML) systems become more and more widely used, it is critical to build mechanisms for them to report the confidence level of predictions. For organ-at-risk (OAR) autosegmentation, one measure of confidence is whether the shape determined at prediction time is similar to shapes used to train the model. Here, we develop an ML method to characterise the distribution of OAR shapes and use it to find erroneous training data and anomalous shapes in simulated prediction data. **Methods:** We built a 3D Resnet-style classifier with 13 layers using TensorFlow(TM) 2.3.1 and Keras, and trained it to classify 11 head and neck OAR from a publicly-available dataset [1]. 20 of each OAR were used. The trained model was used in inference mode to classify 878 OAR from the same dataset. The shape distribution was determined by reducing dimensionality of weights in intermediate layers of the classifier to 2, using principal component analysis and t-SNE. Outliers were detected by visual inspection or calculating distances of data points from cluster center-of-mass. t-SNE variables (perplexity) and classifier layer were varied. [1] Wee, L., & Dekker, A. (2019). Data from Head-Neck-Radiomics-HN1 [Data set]. The Cancer Imaging Archive. <https://doi.org/10.7937/tcia.2019.8kap372n> **Results:** Examples of anomalous shapes were detected include mislabelling (both cochlea labelled 'left cochlea' in training data); unusual anatomy (oral cavity of patient without teeth); patient positioning differences (brain of a patient with reduced neck extension); contouring styles (volume of cochlea, inferior limit of spine contour). One OAR (oral cavity) showed two distinct clusters indicating variation in contouring styles. **Conclusion:** Clustering of OAR shapes is useful for cleaning ML training data and flagging potentially incorrect predictions. This may be used to help autosegmentation programs determine prediction confidence levels.

TU-F-TRACK 6-04

Auto-Segmentation of Organs-At-Risk in Head and Neck CT Images with Dual Shape Guided Network

S Wang*, T Yanagihara, B Chera, C Shen, P Yap, J Lian, UNC at Chapel Hill, Chapel Hill, NC

Purpose: The accurate segmentation of organs-at-risk (OARs) in CT images is a critical step for radiation therapy of head and neck cancer patients. However, manual delineation for numerous OARs in this region is time-consuming and challenging due to low soft-tissue contrast. We propose a novel dual shape guided network (DSGnet) to delineate nine important OARs in CT images automatically. **Methods:** To deal with the large shape variation and unclear boundary of OARs in head and neck CT images, we represent the organ shape using an organ-specific unilateral inverse-distance map (UIDM) and guide the segmentation task from two different

perspectives: direct shape guidance by following the segmentation prediction and across shape guidance by sharing the segmentation feature. In direct shape guidance, the segmentation prediction is supervised by the true label mask and the true UIDM, implemented through an encoder-decoder mapping from the label space to the distance space. In the across shape guidance, UIDM is used to facilitate the segmentation by optimizing the shared feature maps. Our method was applied to 699 images. The Dice Similarity Coefficient (DSC) and Average Surface Distance (ASD) of the segmentation result were compared with another seven published start-of-art deep learning methods. The time spent in revising automatically generated contours by our algorithm was compared with the time for the complete manual contouring. **Results:** Our model showed the overall DSC value of 0.842 and ASD value of 1.204 mm across nine OARs. Compared with competing methods, our method achieved the best DSC for all nine OARs and the best ASD for eight OARs. Manual delineation time is saved 64% when aided by the contours created from our algorithm. **Conclusion:** We have developed a novel dual shape guided fully convolutional network for the accurate segmentation of organs-at-risk in head and neck CT images.

SW and JL are in part supported by NIH 1R01CA206100.

TU-F-TRACK 6-05

Improving Cone-Beam CT Based Organ Segmentation with Attention and Knowledge Transfer

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Purpose: It is desirable to utilize Cone-beam CT (CBCT) for on-table patient anatomy monitoring to perform daily dose accumulation and even plan adaptation. However, CBCT suffers from low image quality, resulting in difficulty to perform segmentation, either manually or automatically; and are usually associated with a high level of uncertainty and inconsistency. In this study, we aim to improve CBCT based segmentation by proposing a deep learning approach with common domain embedding to transfer planning quality image inference and contour confidence to CBCT-based segmentation. **Methods:** A YOLO detector trained from multiple views was first applied to localize the region of segmentation interests. Using 2.5D UNet as the basic segmentation structure, two approaches injected FBCT into the segmentation module were explored: (1) a brute-force approach where training data contained images and the corresponding contours from both modalities, entering the network randomly; (2) a two-stream model with partially coupled encoder and early inference, where CBCT went through one stream, and a randomized subset of CBCT and FBCT went through another stream. The proposed approach was applied to the prostate region, where rectum and bladder were segmented from the background. **Results:** The two-stream model enhanced the segmentation accuracy of the rectum/bladder to DSC $0.68\pm0.09/0.92\pm0.02$ for testing subject1, and $0.79\pm0.04/0.87\pm0.01$ for testing subject2; and MSD $3.67\pm1.23/1.77\pm0.51$ mm and $3.24\pm1.62/3.75\pm0.64$ mm; compared to the baseline UNet trained with CBCT which had DSC $0.66\pm0.07/0.90\pm0.03$ and $0.76\pm0.04/0.77\pm0.03$; $3.47\pm0.58/2.20\pm0.88$ mm and MSD $2.80\pm0.54/5.43\pm0.81$ mm; and with CBCT+FBCT of DSC $0.67\pm0.08/0.91\pm0.02$ and $0.78\pm0.04/0.86\pm0.01$; MSD $3.51\pm0.85/1.96\pm0.64$ mm and $3.59\pm0.97/4.55\pm0.64$ mm, respectively. **Conclusion:** The proposed YOLO + two-stream model effectively enhanced the segmentation performance in CBCT by regulating its features extraction and early inference with FBCT. We will adjust the coupling to allow for modality-specific feature variation and also regulate the output contour. Moreover, we are actively investigating approaches to minimize uncertainty associated with the CBCT label themselves.

TU-F-TRACK 6-06

Measuring the Clinical Impact of the Introduction of a Novel Auto-Contouring Workflow for 0.35T MRI-Guided Pelvic Radiotherapy

Y Abdulkadir*, D Luximon, E Morris, P Chow, J Lamb, University of California, Los Angeles, Los Angeles, CA

Purpose: Deep neural nets have revolutionized the science of auto-segmentation and present great promise for treatment planning automation. However, little data exists regarding clinical implementation. We evaluated the performance and clinical implementation of a novel deep learning-based auto-contouring workflow for 0.35T magnetic resonance imaging (MRI)-guided pelvic radiotherapy. **Methods:** An auto-contouring model was developed using a UNet-derived architecture for the femoral heads, bladder, and rectum. Training data was taken from 75 patients treated on an 0.35T MRI-guided radiotherapy machine. Post-processing steps were developed in order to optimize usability of the resulting contours. The model was tested against 20 retrospective cases outside the training set, and subsequently was clinically implemented by interfacing it to the software used for clinical contouring in our department. Usability was evaluated on the first 10 consecutive clinical cases by computing 2D slice-by-slice Dice similarity coefficient (2D-DSC) and the fraction of slices that were used un-modified by planners. Final contours were retrospectively reviewed by an experienced treatment planner and clinical significance of deviations was graded as negligible, low, moderate and high probability of leading to actionable dosimetric variations. **Results:** Average 2D-DSC for the retrospective test data were 92.8 ± 4.9 , 91.8 ± 8.2 , 91.6 ± 13.3 , and 87.2 ± 11.8 for the right and left femoral heads, bladder, and rectum respectively. Post-implementation, average 2D-DSC were 100.0 ± 0.4 , 99.3 ± 4.3 , 94.9 ± 10.5 , 94.5 ± 14.4 , respectively. For each organ, 99.1, 96.3, 53.7, and 64.7 percent of slices were used unmodified by the planner. In retrospective review of contours used for planning, a total of 6 deviating slices in 2 patients were graded as low potential clinical significance. No deviations were graded as moderate or high clinical significance. **Conclusion:** We have presented an analysis of the clinical implantation of a novel auto-contouring workflow. Research-oriented performance metrics correlate imperfectly with clinical utility. Substantial workflow savings were obtained, and little automation bias was observed.

TU-F-TRACK 6-07

Weaving Attention U-Net for Head and Neck Multi-Organ CT Segmentation

Z Zhang*, T Zhao, H Gay, W Zhang, B Sun, Washington University in St. Louis, St. Louis, MO

Purpose: Manual contouring used in the current treatment planning process is tedious, time-consuming, and can produce inconsistent results. We develop a novel and efficient CNN-based deep learning method for automated multi-organ segmentation on head and neck (HN) CT images. **Methods:** The new method uses a densely weaving attention U-net (WAU-net) structure as the backbone and implements a novel boosting-style training strategy. WAU-net follows the basic U-net structure and uses attention fusion blocks for the decoder path. It evolves from shallow structures to more complex ones while forcing itself to exploit deeper features by down-weighting the regions correctly contoured by simpler networks. Under the boosting-style training strategy, the network evolves gradually, stressing previously wrongly labeled pixels to achieve finer segmentation outcomes. We trained our new method to contour 10 OARs within a single forward propagation. We performed 10-fold cross-validation with 90 cases and tested the performance on the rest 25 cases. The accuracy of modeled contours was evaluated with the Dice similarity coefficient (DSC), average Hausdorff distance (AHD), and average surface Hausdorff distance (ASHD) using clinical contours as a reference to the ground truth. The accuracy of the model was compared to other popular deep learning methods. **Results:** Our method achieved superior or comparable results to several existing deep learning approaches. In addition, our new model used substantially fewer model parameters and less computation cost. The model also achieved state-of-the-art performance on the testing dataset. The average DSC on the testing data sets were as following: brainstem (0.88), chiasm (0.71), mandible (0.95), spinal cord (0.89), left optic nerve (0.74), right optic nerve (0.75), left sub-mandibular (0.81), right sub-mandibular (0.80), left & right parotid (0.85) and right parotid (0.83). **Conclusion:** WAU-net provides state-of-the-arts accurate contouring of multiple OARs, which can potentially facilitate the routine OARs delineation process for HN cancer.

This work is funded by Varian Medical Systems.

SAM Professional Symposium**TRACK 1*****Building and Mentoring Diverse Teams to Achieve Equity, Diversity, Inclusion and Greater Success*****WE-A-TRACK 1-00****Building and Mentoring Diverse Teams to Achieve Equity, Diversity, Inclusion and Greater Success**

Moderators: Jeremy Hoisak, UC San Diego, Chula Vista, CA and Ghada Aldosary, Carleton University, Ottawa, ON

WE-A-TRACK 1-01**Introduction****WE-A-TRACK 1-02**

T. Juang, UC San Diego:

Strategies for Equity, Diversity, and Inclusion in the Medical Physics Residency Recruitment Process

WE-A-TRACK 1-03

A. Johnson, Kaiser Permanente:

Strategies for Mentoring and Retaining Underrepresented Minorities in Medical Physics Practice Environments

WE-A-TRACK 1-04

M. Siker, Medical College of Wisconsin:

Representation Matters: Our Responsibility in Cultivating Inclusion and Belonging in Radiation Oncology

WE-A-TRACK 1-05**Panel Discussion and Audience Q&A**

There has been a focus on generating awareness and cultivating equity, diversity, and inclusion (EDI) in the medical physics community, with significant efforts made by the AAPM and COMP at the last joint scientific meeting. However, there is still a need to exchange readily adoptable methods to improve EDI in recruitment, education, residency training, and professional practice. In this session, we will review how mentorship and intentional professional development can improve EDI in medical physics.

First, we will discuss how mentorship plays an important role in retaining underrepresented minorities (URM) in medical physics. We will provide strategies to develop holistic developmental relationships with URM mentees. Attendees will also be able to recognize unintentionally harmful and exclusionary practices that may discourage an URM's training, professional engagement, and career advancement.

Second, medical physicists involved in education, training, and management will learn about the concept of adverse impact and how adverse impact assessments can be used while reviewing screening criteria and procedures for residency applications. We will introduce structured interviews and their value in improving reliability, validity, and fairness, and examine how structure can be added to the recruitment interview process by implementing content-related components (e.g., standardized questions, behavioral or situational interviewing) and evaluation-related components (e.g., rating scales, interviewer training).

Then, we will highlight EDI initiatives adopted by our colleagues in radiation oncology, discuss the importance of diversity in the radiation oncology workforce, and identify effective strategies to improve EDI. This discussion will return to why diversity matters in both medical physics and in the general scope of healthcare.

Finally, the audience will engage with the speakers in a panel discussion and have the opportunity to exchange their own experiences with successful EDI strategies.

At the end of this session, participants will:

1. Learn about methods for fostering EDI in education, training, and professional practice.
2. Become familiar with mentorship strategies to advance developmental relationships with underrepresented minorities in medical physics.
3. Understand how adverse impact assessments and structured interviews can be applied as valuable tools for improving reliability, validity, and fairness in the recruitment process.
4. Gain insight on why diversity matters in healthcare, and learn about strategies used to improve diversity in the current radiation oncology workforce.

SAM Imaging Educational Course**TRACK 2*****Principles and Applications of Multienergy CT: Report of AAPM Task Group 291*****WE-AB-TRACK 2-00****Principles and Applications of Multienergy CT: Report of AAPM Task Group 291**

Moderator: Cynthia McCollough, Mayo Clinic, Rochester, MN

WE-AB-TRACK 2-01

C. McCollough, Mayo Clinic:

Introduction & Clinical Motivation

WE-AB-TRACK 2-02

T. Flohr, Siemens Healthcare GmbH:

Physical Principles of Multi-Energy CT

WE-AB-TRACK 2-03

X. Duan, UT Southwestern Medical Center:

Technical Implementations of Multi-Energy CT

WE-AB-TRACK 2-04

C. McCollough, Mayo Clinic:

Clinical Applications & Dosimetric Considerations

The motivation for dual- or multi-energy CT (DECT or MECT) is that in x-ray computed tomography (CT), materials with different elemental compositions can have identical CT number values, depending on the mass density of each material and the energy of the detected x-ray beam. Differentiating and classifying different tissue types and contrast agents can thus be extremely challenging. In MECT, one or more additional attenuation measurements are obtained at a second, third or more energy. This allows the differentiation of at least two materials. Commercial DECT systems (only two energy measurements) are now available either using sequential acquisitions of low- and high-tube potential scans, fast tube-potential switching, beam filtration combined with spiral scanning, dual-source, or dual-layer detector approaches. The use of energy-resolving, photon-counting detectors is now being evaluated on research systems. Irrespective of the technological approach to data acquisition, all commercial MECT systems today provide dual-energy data. Material decomposition algorithms are then used to identify specific materials according to their effective atomic number and/or to quantitate mass density. These algorithms are applied to either projection or image data. Since 2006, a number of clinical applications have been developed for commercial release, including those that automatically 1) remove the signal from bony anatomy and/or calcified plaque; 2) create iodine concentration maps from contrast-enhanced CT data and/or quantify absolute iodine concentration; 3) create virtual non-enhanced images from contrast-enhanced scans; 4) identify perfused blood volume in lung parenchyma or the myocardium; and 5) characterize materials according to their elemental compositions, which can allow in vivo differentiation between uric-acid and non-uric-acid urinary stones or uric acid (gout) or non-uric-acid (calcium pyrophosphate) deposits in articulating joints and surrounding tissues. AAPM Report 291 provides an authoritative, open-access, peer-reviewed review of the topic of MECT for both the imaging and therapy communities. It describes the fundamental principles of MECT in an intuitive manner and with mathematical rigor, current and emerging manufacturer implementations, and current and emerging clinical applications.

Learning Objectives:

1. Understand the motivation for and fundamental principles of multi-energy CT
2. Appreciate the range of technical implementations of multi-energy CT
3. Recognize strengths and weaknesses of different technical implementation strategies
4. Identify clinical CT applications made possible by multi-energy CT

Cynthia McCollough receives research support from Siemens Healthcare, which is not directly related to this work.

SAM Imaging Scientific Symposium TRACK 3

Robotic X-ray Systems: Emerging Platform for Improved Radiography, 3D Imaging, and Interventional Radiology

WE-AB-TRACK 3-00

Robotic X-Ray Systems: Emerging Platform for Improved Radiography, 3D Imaging, and Interventional Radiology

Moderator: Tess Reynolds, University of Sydney, Eveleigh

WE-AB-TRACK 3-01

T. Reynolds, University of Sydney:

Adaptive Imaging - Syncing with Patients for Safer and Smarter 3D Imaging

WE-AB-TRACK 3-02

G. Gang, Johns Hopkins University:

Task Driven/Metal Artefact Reduction Orbits

WE-AB-TRACK 3-03

F. Noo, University of Utah:

Beyond the Conventional Circular Trajectory in Cone-Beam CT Imaging

WE-AB-TRACK 3-04

L. Ritschl, Siemens Healthineers:

Advanced Imaging using Multitom Rax

The vast majority of medical Cone Beam Computed Tomography (CBCT) imaging occurs with the gantry rotating at a constant velocity in a circular trajectory, independent of the specific task and patient. Advancements in robotic X-ray systems have paved the way for variable velocity and non-circular exotic acquisition orbits to be investigated in clinical settings. The clinical drivers for such acquisitions include enabling patient-specific imaging, reducing overall imaging dose and providing the ability to tailor the image acquisition for a specific clinical task. Imaging patients with variable velocity and non-circular exotic orbits has applications in fields ranging from radiation therapy and ablation, to neuro- and orthopaedic surgery through to interventional radiology and beyond.

Learning Objectives:

1. To understand the clinical drivers for adaptive and exotic acquisition orbits on robotic X-ray systems
2. To learn the current exotic acquisition orbits being implemented
3. To examine the future directions of clinical imaging with robotic X-ray systems

SAM Multi-Disciplinary Scientific Symposium TRACK 4

AI in Image Guided Radiation Therapy

WE-AB-TRACK 4-00

AI in Image Guided Radiation Therapy

Moderator: Xiaofeng Yang, Emory University, Atlanta, GA

WE-AB-TRACK 4-01

G. Chen, University of Wisconsin:

Deep Learning in CT Image Reconstruction

WE-AB-TRACK 4-02

L. Ren, University of Maryland:

AI Based Image Enhancement: Advances and Clinical Implications

WE-AB-TRACK 4-03

**X. Yang, Emory University:
Deep Learning in Image Registration**

WE-AB-TRACK 4-04

**D. Nguyen, University of Technology Sydney:
AI for Intrafraction Motion Monitoring: Are We There Yet?**

The progress of artificial intelligence (AI) technologies has recently been exponential and shown to be both transformative and disruptive in many fields, which has impacted many aspects of human life. These AI technologies are now being researched, developed, and applied to healthcare, including radiation therapy. Imaging has become essential not only for the detection and monitoring of disease but also for improving the outcome of radiotherapy. Image-guided radiation therapy (IGRT) uses imaging during radiotherapy to improve the precision and accuracy of treatment delivery. Treatment machines are equipped with imaging technology to image patient anatomy before and during treatment. By comparing these images to the reference images taken during simulation, the patient's position and/or the radiation beams may be adjusted to more precisely deliver the radiation dose to the target. AI-based analytical and computational tools have been developed to enhance the role of quantitative imaging in IGRT and further improve the accuracy and precision of radiotherapy.

In the session, we will focus on the potential impact of AI on four important aspects of IGRT: image reconstruction, image augmentation, image registration, and intrafraction motion monitoring. This session will identify challenges and opportunities of AI, especially deep learning in IGRT, explore strengths and limitations of different AI models in IGRT, outline future trends of AI in IGRT, and highlight specific clinical applications.

Learning Objectives:

1. Learn the challenges in current IGRT and the technologies available or being developed for IGRT.
2. Learn how AI can be utilized to improve current IGRT practice.
3. Identify the rationale, significance, quality, and safety issues of AI in IGRT.
4. Understand the challenges, opportunities, strengths, and limitations of AI in IGRT.

Multi-Disciplinary Scientific Symposium

TRACK 5

Grand Challenges: Deep Learning Sparse-View CT and DBTex

WE-A-TRACK 5-00

Grand Challenges: Deep Learning Sparse-View CT and DBTex
Moderator: Samuel Armato, The University of Chicago, Chicago, IL

WE-A-TRACK 5-01

**X. Pan, University of Chicago:
DL-Sparse-View CT Challenge**

WE-A-TRACK 5-02

**E. Sidky, University of Chicago:
DL-Sparse-View CT Challenge**

WE-A-TRACK 5-03

DL-Sparse-View CT Challenge, Group 1

WE-A-TRACK 5-04

DL-Sparse-View CT Challenge, Group 2

WE-A-TRACK 5-05

**M. Mazurowski, Duke Univ:
DBTex Challenge**

WE-A-TRACK 5-06

DBTex Challenge, Group 1

WE-A-TRACK 5-07

DBTex Challenge, Group 2

Advancement of imaging research requires years of effort from dedicated research groups around the world. Such groups, working independently, typically suffer from limited local resources in terms of patient data and access to the “ground truth” required to properly train and test their algorithms. When these research groups report their methods in the literature, it is difficult for the research community to compare the relative merits of different approaches, since performance can depend on factors such as database composition, image quality, “truth” definition, and performance evaluation metric. Grand challenges allow for a direct comparison of different algorithms designed for a specific task, with all algorithms following the same set of rules, operating on a common set of data, and being evaluated with a uniform performance assessment approach. Comparisons among the methods of participating groups can help identify approaches that are the most promising for a specific task. The AAPM Working Group on Grand Challenges (WGGC) was created to promote the conduct of grand challenges in medical imaging by (1) developing recommendations for hosting computational challenges through AAPM activities designed to assess or improve the use of medical imaging in diagnostic and/or therapeutic applications, (2) vetting proposals from groups that wish to host a challenge in conjunction with the Annual Meeting, (3) facilitating the execution of these challenges, and (4) generating ideas for challenges to advance the field. Two challenges sponsored by the WGGC were conducted in the months leading up to the 2021 AAPM Annual Meeting and provided a unique opportunity for participants to compare their algorithms with those of other groups in a structured, direct way. Overviews of the Deep Learning for Inverse Problems: Sparse-View Computed Tomography Image Reconstruction (DL-Sparse-View CT) Challenge and the SPIE-AAPM-NCI DAIR Digital Breast Tomosynthesis Lesion Detection (DBTex) Challenge along with the DAIR Digital Breast Tomosynthesis Lesion Detection Challenge, Part 2 (DBTex2) will be presented along with presentations from the top-performing groups from the DL-Sparse-View and DBTex2 Challenges.

Learning Objectives:

1. To understand the role of grand challenges and public image datasets in medical imaging research.
2. To learn about the methods used by the top-performing participants in the DL-Sparse-View CT Challenge.
3. To learn about the methods used by the top-performing participants in the DBTex2 Challenge.

The DL-Sparse-View CT Challenge was supported by NIH grant numbers R01 EB026282 and R01 EB023968 and by a Grayson-Jockey Club research grant. The DBTex2 Challenge was supported by NIH grant number R01 EB021360.

Therapy Scientific Session TRACK 6 Particle Therapy: Treatment Planning and Radiobiology

WE-A-TRACK 6-01

Analytical Modeling of the Dynamic Collimation System Using Dual Trimmer Planes

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Purpose: Previous implementations of the Dynamic Collimation System (DCS) in the Astroid Treatment Planning System (TPS) modeled the trimmers as a single infinitesimally thin rectangular aperture. While this method is appropriate for a single brass aperture, it is insufficient to model DCS-based pencil beam scanning (PBS), which has two trimmer banks at different distances from the isocenter. This work introduces an analytical model that utilizes dual trimmer planes to enable more accurate dose calculation, with and without a range shifter present upstream of the trimmers. **Methods:** The lateral dose distribution from a single proton beamlet was modeled as the weighted summation of sub-beamlets (bixels). The interaction of protons with the apertures was accounted for by modifying bixel weights (fluences). Weights were modified according to the overlap of apertures with each sub-beamlet and were dependent on beam energy, range shifter interaction, and aperture location. Our model introduces multiple trimmer planes 6.5 cm and 10.65 cm from isocenter in the extended nozzle position and 19.65 cm and 23.80 cm from isocenter in the retracted nozzle position. The analytical dose calculation model was validated against a commissioned Monte Carlo model in TOPAS. **Results:** Without a range shifter present, 2D gamma passage rates at 1.0%/1.0mm were 100% between analytical and Monte Carlo for all energies considered (75 MeV, 100 MeV, and 150 MeV) and all trimmer positions relative to the central beamlet axis (1 mm, 5 mm) for both the single and dual trimmer plane models. With the range shifter present, the 2D gamma pass rates for all energies considered (80 MeV, 125 MeV, and 170 MeV) and all trimmer positions were >82% for the single plane model and >92% for the dual plane model. **Conclusion:** Using dual trimmer planes is necessary to accurately calculate dose for DCS plans using analytical dose algorithms.

Research reported in this abstract was supported by the National Cancer Institute of the National Institutes of Health under award number R37CA226518. Hyer, Flynn, and Hill are co-inventors on a patent that has been licensed to IBA.

WE-A-TRACK 6-02

Dose-Averaged LET in Proton Therapy for Patients with MRI Abnormalities and Its Role in Normal Tissue Complication Probability

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Purpose: To elucidate the role of LET in the appearance of abnormalities in MRI after undergoing proton radiotherapy. Results helped feed a normal tissue complication probability (NTCP) explicitly including LET as a predictor variable. **Methods:** We calculated dose-averaged LET (LET_d) in 26 meningioma patients that developed abnormalities in MRI studies following proton radiotherapy treatments. An analytical method (MicroCalc) based on the computation of the spectral fluence of primary protons, secondary protons and alpha particles was employed. MRI were registered to planning CT and LET_d values across voxels belonging abnormalities were scored. We divided dose (D) and LET_d ranges in bins and a priori probability of abnormality P(D,LET_d) for each bin was obtained as the fraction of voxels belonging abnormalities and belonging normal areas. This probability is modeled as a function of dose and the linear-quadratic (LQ) parameters $\alpha(LET_d) = \alpha_0 + \alpha_1 LET_d$ and $\beta = \beta_0$, which, in turn, depend on LET_d in this model. Data from patients were used to estimate these dependencies under the model. Finally, NTCP was modeled assuming independence between voxels. **Results:** For (D, LET_d) pair-values with a priori probability higher than 50% of abnormality, LET_d was found to increase roughly linearly as dose decreased, being above 8 keV/μm at doses below 1 Gy and about 3 keV/μm at a dose of 60 Gy. This points towards an enhancement of low dose effectiveness because of high LET_d. LQ parameters were estimated as $\alpha_0 = 0.011 \pm 0.02 \text{ Gy}^{-1}$, $\alpha_1 = 0.033 \pm 0.002 \text{ Gy}^{-1} \mu\text{m keV}^{-1}$ and $\beta_0 = 0.0015 \pm 0.0004 \text{ Gy}^{-2}$. **Conclusion:** We illustrated a practical method to infer the potential role that LET may play in the induction of normal tissue complications and on derived parameter estimates of an LET-dependent NTCP model. However, a larger patient cohort including patients without abnormalities are required to further validate these results.

A. Bertolet, R. Abolfath and A. Carabe-Fernandez were supported by Varian Medical Systems for this work

WE-A-TRACK 6-03

Robust Optimization of Proton Therapy Treatments Using a Variable Range Uncertainty

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Purpose: In proton therapy (PT), a range uncertainty of 2.6% is typically assumed when planning robust treatments with Monte Carlo dose engines. This number is based on a dominant contribution from tissues I-values. However, it was recently shown that expressing tissues as a mixture of water and “dry” material during the CT calibration process allowed for a significant reduction of this uncertainty. We thus propose an adapted framework for PT robust optimization. First, we move towards a spot-specific range uncertainty (SSRU) determination. Second, we use the water-based formalism to reduce range uncertainties and potentially better spare OAR. The methodology is applied to the treatment planning of a head and neck (H&N) primary tumor. **Methods:** For each spot, a ray-tracing method was used to propagate I-values uncertainties and obtain the corresponding range uncertainties. These were then combined with other sources of range uncertainties. Two plans of the same patient were optimized: the first one with the classical 2.6% range uncertainty, the second one with the SSRU. These plans were compared in terms of OAR mean dose reduction. Robustness evaluations were also performed, using the SSRU for both plans in order to simulate errors as realistically as possible. **Results:** The SSRU plan was found to have a very good robustness level at a 90% confidence interval while sparing OAR better than the classical plan. Mean dose reductions of 2.20 Gy and 4.27 Gy were respectively observed in the oral cavity and superior pharyngeal muscle for a similar target coverage. The classical plan showed an unnecessary robustness level, even when considering 100% of the simulated scenarios. **Conclusion:** Promising results of the SSRU framework were observed on a simplified H&N case. We believe the methodology could benefit even more for a full H&N treatment where more OAR will be close to the target volume.

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K. Souris is funded by the Walloon Region (MECHATECH/BIOWIN, Grant No. 8090).

J. Lee is a senior research associate with the F.R.S-FNRS.

WE-A-TRACK 6-04

The Effect of Treatment Planning Approach on Head-And-Neck IMPT for Patients with Implanted Dental Fixtures

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Purpose: Proton dose perturbations from dental fixtures (e.g. amalgams, crowns, and implants) are unverified for clinical treatment plans. The effect of several treatment planning approaches for patients with installed dental fixtures is quantified. Four scenarios were measured and compared: (1) standard-operating procedure (SOP) treatment geometry (two anterior oblique and one posterior beam) with no fixtures in phantom; (2) SOP plan for patient with installed fixtures (e.g. spot delivery through fixtures prohibited); (3) SOP no-fixture plan delivered to phantom with installed hardware; (4) installed fixtures overridden and plan reoptimized without limitations for spot positions/delivery. **Methods:** An anthropomorphic head phantom with removable maxilla, mandible, and tongue inserts was simulated and clinically representative treatment plans were generated, delivering 54 Gy dose to a base-of-tongue (BoT) clinical target volume (CTV). An approximately 1.5 mm depth central groove occlusal amalgam and a PFM crown were installed on separate maxilla inserts and tongue inserts were 3D printed accommodating either a microdiamond diode (PTW 60019; Fricberg, Germany) or several axial or sagittal oriented pieces of EBT3 film (Ashland: New Jersey, USA). **Results:** For the amalgam, case (3) delivered to a SFO BoT treatment plan resulted in a dose shadow parallel with the field delivering through the fixture. The dose shadow corresponded to an approximately 12% decrease in dose directly behind the amalgam. Dosimetric losses were local, dropping to a 1% deviation from control within 1 cm of maximum in the direction of the affected beam. Further, the microdiamond diode, which was not set up directly in line with the amalgam, measured effectively no change in delivered dose. **Conclusion:** For SFO BoT plans, amalgams do perturb the beam, resulting in a shadowing effect, reducing the dose directly behind the amalgam in the direction of the affected beam. Losses were observed to be largely local to the implanted fixture.

WE-A-TRACK 6-05

The Relative Variability in Radiosensitivity Between Cell Lines Is the Same for Carbon Ions, Protons and X-Rays

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Purpose: To demonstrate that the large variations in radiosensitivity between cell lines is the same relative size for higher LET radiation as for x-rays. **Methods:** We compiled clonogenic survival data we measured with data from the literature for human cancer cell lines (n=38) exposed to x-rays and protons (9.9 keV/μm) or carbon (C) ions (13.3-77.1 keV/μm), parameterizing each cell's radiosensitivity as the dose required to achieve 10% survival, D10%. For each radiation quality, we quantified the relative variability in radiosensitivity between cell lines using the coefficient of variation, COV(D10%). For each particle radiation quality, we compared the COV(D10%) to that for the same subset of cells exposed to x-rays via two-tailed z-tests. We further stratified the data into three histological subsets (brain, pancreatic and non-small cell lung cancer) based on tumor type, including only tumors for which no distinction is made in clinical dose prescription. Then, within each histological category, we compared the COV(D10%) values between radiation qualities. **Results:** Even for high LET C-ions, there were remarkable variations in radiosensitivity between cell lines, with COV(D10%) values of 28±6% at 77.1 keV/μm compared to 34±7% for x-rays. Consequently, we could not detect significant differences in the COV(D10%) values for x-rays and any particle radiation quality. The same was true even after we stratified the data into histologic subsets. Indeed, within histologic subsets, we still found remarkable differences in radiosensitivity at high LETs, with brain cancer cells varying 43±18% to 77.1 keV/μm C-ions, non-small cell lung cancer cells varying 24±7% to 60.5 keV/μm C-ions and pancreatic cancer cells varying 16±4% to 60.5 keV/μm C-ions, compared to 37±15%, 34±10% and 15±4%, to x-rays, respectively. **Conclusion:** Even for cell lines of the same histology, there are remarkable variations in intrinsic radiosensitivity that do not differ significantly between photon, proton or C-ion radiation.

This research was supported in part by RP170040, 1R21CA252411-01, Emerson Collective, and MD Anderson Cancer Center. Dr. Sawakuchi has research agreements with Alpha Tau Medical and Artios Pharma. Dr. Shaitelman has a research agreement with Exact Sciences.

WE-A-TRACK 6-06

A Beamlet-Based Optimization Algorithm for Arc Proton Therapy Planning

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Purpose: Arc Proton Therapy (ArcPT) is an emerging technique that has the possibility of improving both the outcome and the efficiency of the treatment. This modality constrains the number of energy switches as it primarily determines the treatment delivery time. Therefore, the challenge lies in selecting as few energy layers as possible for each beam direction. In this study, we designed a beamlet-based algorithm for the optimization of ArcPT plans that distributes a number of energy layers similar to conventional therapy over more gantry angles. **Methods:** The ArcPT optimization problem is formulated with an objective function defined conventionally as the quantitative evaluation criterion for the delivered dose distribution combined with new objectives respecting constraints on the number of energy layers used over the arc. The proposed method was tested on a lung tumor case and evaluated according to these criteria: objective function value, layer sparsity and DVH metrics. **Results:** An ArcPT plan with 98.4% of the beams delivering only one energy layer per control point was successfully obtained with good dosimetry results, some of which are even better than the ones obtained with a conventional IMPT plan. Actually, the target D95 reached 96.5% of the prescription in ArcPT versus 95.6% in IMPT. Although the mean doses in the heart (0.4 Gy) and right lung (6.1 Gy) obtained with ArcPT slightly exceed those with IMPT (0.3 Gy and 5.6 Gy for the heart and right lung respectively), they still achieve the clinical goals. **Conclusion:** The algorithm proposed for ArcPT treatment plan optimization has successfully solved the energy layer selection problem by generating acceptable plans with good dosimetric results. The definition of our objective function will be completed with a sequencing energy term that favors small consecutive energy decrements to further reduce irradiation time.

Funded by the Walloon Region. Research agreement with Ion Beam Applications SA (IBA).

WE-A-TRACK 6-07

The Combined Effects of Protons and An Inhibitor of ATR Increase Micronuclei Formation and Potentiate Anti-Tumor Immunity

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Purpose: High linear energy transfer (LET) protons are more densely ionizing than photons and low LET protons. Therefore, high LET protons create more clustered DNA damage and double strand breaks (DSBs). The double stranded DNA fragments generated in the form of micronuclei are the key activators of the cGAS-STING pathway, thereby potentiating anti-tumor immunity. The cGAS-STING pathway results in downstream activation of pro-inflammatory cytokines and signals that can lead to destruction of tumor cells. Here we show that the combination of high LET protons with an inhibitor of ATR (ATRi), a key protein in DNA damage response, amplify the production of precursors to antitumor immune activation, specifically cGAS-positive micronuclei (MN-cGAS+), more so than photons+ATRi. **Methods:** H1299 and PANC-1 cell lines were treated with 6 MV x-rays and 9.9 keV/μm (dose-weighted LET in water) protons alone or with an ATRi (1 μM, AZD6738). We then assessed MN and MN-cGAS+ 24 and 72 h after radiation+ATRi. The Micronuclei were assessed as the proportion of micronuclei or cGAS+ relative to the quantity of nuclei counted for each treatment tested. This was done using a custom auto-counting macro generated using ImageJ. **Results:** ATRi radiosensitized both

cell lines to photons and protons with sensitizations relative to photons alone of $1.126 \pm 0.049 / 1.003 \pm 0.025$ (H1299/PANC-1) (photons+ATRi) and $1.547 \pm 0.029 / 1.448 \pm 0.043$ (H1299/PANC-1) (protons+ATRi). At 72 h and 5 Gy for H1299 cells, protons+ATRi (0.5326 ± 0.015 MN/nucleus and 0.2426 ± 0.044 MN-cGAS+/nucleus) amplified MN and MN-cGAS+ compared to photons alone (0.3184 ± 0.033 MN/nucleus and 0.1716 ± 0.042 MN-cGAS+/nucleus), protons alone (0.4529 ± 0.007 MN/nucleus and 0.2444 ± 0.018 MN-cGAS+/nucleus) and photons+ATRi. Although of lower magnitude, the same trends were observed at 24 h/5 Gy and 72 h/2 Gy. Results for PANC-1 were also similar. **Conclusion:** In addition to radiosensitization from combining protons+ATRi, high LET protons+ATRi amplify MN-cGAS+, which has been reported to be a precursor of antitumor immune signaling.

This research was supported in part by RP170040, 1R21CA252411-01, Emerson Collective, and MD Anderson Cancer Center. Dr. Sawakuchi has research agreements with Alpha Tau Medical and Artios Pharma. Dr. Shaitelman has a research agreement with Exact Sciences.

Ultrasound Symposium

TRACK 7

Ultrasound QA/QC Workshop

WE-A-TRACK 7-00

Ultrasound QA/QC Workshop

Moderator: Zheng Feng Lu, University of Chicago, Chicago, IL

WE-A-TRACK 7-01

Z. Long, Mayo Clinic:

ACR Required QC Tests

WE-A-TRACK 7-02

M. Holland, Indiana University- School of Medicine:

Automated Whole-Breast US QC

WE-A-TRACK 7-03

J. Zagzebski, University of Wisconsin:

Doppler QC

WE-A-TRACK 7-04

I. Rosado-Mendez, University of Wisconsin-Madison:

Quantitative Ultrasound and Shear Wave Elastography Evaluation

WE-A-TRACK 7-05

S. Dekker, Cablon Medical:

Automated Ultrasound QC Methods

WE-A-TRACK 7-06

Q&A

The workshop will include five presentations covering a range of current topics in diagnostic ultrasound QA/QC. “Annual Survey” requirements will be presented, outlining steps and pitfalls of ACR-required ultrasound system tests. These include image uniformity and artifact surveys, geometric accuracy and system sensitivity tests, and display monitor evaluations. QC phantoms will be described for testing general purpose ultrasound scanners. This will be followed by descriptions of specialized phantoms and procedures for performance assessment of automated whole-breast screening ultrasound units. In addition, an overview of Doppler system testing using flow phantoms will be provided. Efforts to assess shear wave elastography and an initiative to estimate “pulse-echo quantitative ultrasound” parameters, including attenuation and backscatter coefficients and speed of sound, by the Quantitative Imaging Biomarkers Alliance (QIBA) and the American Institute of Ultrasound in Medicine (AIUM), will be introduced. Lastly, innovative software programs for automated QC will be described.

The targeted attendees are medical physicists and other medical professionals in diagnostic imaging and radiation oncology with interest in diagnostic ultrasound QC.

Learning Objectives:

1. To gain familiarity with common elements of a QA/QC program for diagnostic ultrasound scanners, including automated whole-breast screening ultrasound systems. QC tools, phantoms and automated software programs will be described.
2. To provide an overview of important phantom characteristics for evaluating Doppler ultrasound equipment and to outline Doppler system tests that can be performed using flow phantoms.
3. To introduce the AIUM/QIBA efforts in assessing shear wave elastography and the emerging pulse-echo quantitative ultrasound features.

Vendor Showcase

TRACK 8

Sun Nuclear

WE-SHOWCASE-TRACK 8-11-1

What's New from Sun Nuclear

J Tracy^{1*}, (1) Sun Nuclear Corporation, Brookville, KS

SAM Professional Symposium

TRACK 1

MPLA Career Development: Secrets to Your Success

WE-B-TRACK 1-00

MPLA Career Development: Secrets to Your Success

Moderators: Jatinder Saini, SCCA Proton Therapy Center, Seattle, WA and Paige Taylor, UT MD Anderson Cancer Center, Houston, TX

WE-B-TRACK 1-01

I. Barreto, University of Florida College of Medicine:

Your Technical Skills Won't Get You There: Identifying Vital Skills to Thrive in Your Career

WE-B-TRACK 1-02

J. Johnson, Kelsey-Seybold Clinic:

Leaders Making #Goals Happen: Taking Your Discovery Journey

WE-B-TRACK 1-03

A. Schreuder, Leo Cancer Care:

Leadership Lessons: Key Factors Affecting Your Career Advancement (Panel Discussion)

WE-B-TRACK 1-04

K. Kanai, University of Washington:

Leadership Lessons: Key Factors Affecting Your Career Advancement (Panel Discussion)

WE-B-TRACK 1-05

M. Howard, Parkridge Medical Ctr:

Leadership Lessons: Key Factors Affecting Your Career Advancement (Panel Discussion)

WE-B-TRACK 1-06

E. Angel, Canon Medical Systems, USA:

Leadership Lessons: Key Factors Affecting Your Career Advancement (Panel Discussion)

WE-B-TRACK 1-07

Audience Q&A

Looking for that next career step? Do you have what it takes to be considered? Lots of factors can influence your advancement, ranging from expertise, problem-solving ability, and experience. Yet sometimes these factors that look-good-on-paper are not enough. Leadership at the next level means making a personal impact in addition to getting results.

In this session, MPLA will present its recommended leadership curriculum for where to efficiently put your effort for further developing your leadership skills based on your career stage. These core leadership skills should be transferable regardless of your work environment. MPLA will also present career planning steps to use in designing your path forward best suited to you. Additionally, the session will have a panel of medical physicist leaders discuss the need for passion in what you do, for creative as well as strategic thinking, and for the ability to make a strong case. The future of medicine is rapidly changing, so how can you be best equipped to play a role in defining and driving medical physics in your future?

Learning Objectives:

1. Apply the MPLA curriculum matrix of skills versus career stage
2. Value the need for medical physicists to foster both creative thinking and strategic thinking
3. Adapt lessons from a panel of medical physics leaders with their experiences of factors that affected their career advancement

SAM Therapy Educational Course**TRACK 5*****TG-302 Surface Image Guided Radiotherapy (SGRT): Prevalence and Clinical Applications*****WE-B-TRACK 5-00**

TG-302 Surface Image Guided Radiotherapy (SGRT): Prevalence and Clinical Applications

Moderator: Hania Al-Hallaq, The University of Chicago, Chicago, IL

WE-B-TRACK 5-01

H. Al-Hallaq, The University of Chicago:

SGRT for Breast Radiotherapy Including DIBH Techniques

WE-B-TRACK 5-02

T. Pawlicki, UC San Diego:

SGRT-Based Cranial SRS and Risk Mitigation Strategies

WE-B-TRACK 5-03

L. Padilla, Virginia Commonwealth University:

Current Prevalence and Uses of SGRT in the US and Future Applications

WE-B-TRACK 5-04

L. Cervino, Memorial Sloan Kettering Cancer Center:

Considerations for Commissioning and QA of SGRT Systems

Surface imaging is becoming a prevalent technology for many radiotherapy treatments, with its clinical applications expanding. The AAPM formed the TG-302 to provide recommendations on the common SGRT clinical applications and discuss its implementation. In addition, TG-302 surveyed AAPM members in 2018 on the prevalence and clinical uses of SGRT. This session will review the survey results to describe the current state of clinical use, including high-dose treatments such as stereotactic body radiotherapy (SBRT). Considerations for specific commissioning and quality assurance tests for SGRT systems will be reviewed. The recommendations of TG-302 regarding use of SGRT for breast radiotherapy and frameless brain SRS treatments will be presented, and some of the potential pitfalls that could occur during clinical application will be summarized.

Learning Objectives:

1. Summarize results of survey regarding the prevalence of SGRT
2. Learn how to commission and QA SGRT systems
3. Discuss the use of SGRT for breast deep-inspiration breath hold treatments including pitfalls
4. Discuss the use of SGRT for frameless brain SRS treatments including risk mitigation

Todd Pawlicki receives royalties from CRC Press as Editor of the book entitled Surface Guided Radiation Therapy.

Therapy Scientific Session

TRACK 6

Particle Therapy: Treatment Delivery and Verification

WE-B-TRACK 6-01

Electrostatic Focusing Lens System for Low MeV-Ion Microprobe: A Simulation and Optimization Study

H Arya*, V Chirayath, Y Chi, University of Texas at Arlington, Arlington, TX

Purpose: Low MeV ion microprobe, that focuses an ion beam into submicron size at the cellular matrix plane, plays an important role in advancing fundamental mechanism study in particle-therapy. The dominant state-of-the-art beam focusing systems are electromagnetic quadrupole lens based, which can be hard to duplicate or of high cost. In this work, we investigated the focusing capability of the electrostatic quadrupole (EQ) lens system with a compact, easy-to-adopt, and cost-effective design goal. **Methods:** We have performed the investigation with three simulation tools: GICOSY (low accuracy but high efficiency) for raw optimization, SIMION 8.1 (high accuracy but low efficiency) for fine tuning, and WinTRAX for aberration studies. We optimized the lens system with considering the lens numbers, lengths, intervals, applying voltages, and aperture-lens distance (DAL) under a given input beam condition. We compared the performance of the optimized system to that of the state-of-art EQ sextuplet (EQS) system used in the RARAF center of Columbia University and our previously optimized EQ quadruplet (EQQ) system, in the aspects of system length (Ls), working distance (Dw), demagnification factor (Df) and aberrations. **Results:** An EQ triplet (EQT) lens system was optimized with superior performance than EQS and EQQ lens for a 3 MeV input proton beam with beam diameter of 30 μ m and angular divergence of 0.0067mrad. Specifically, comparing to Columbia EQS, Ls for EQT is $\sim$ 2 folds shorter (1.9 vs. 3.887 m), Dw is $\sim$ 1.4 times longer (170.2 vs. 126 mm) and Df is $\sim$ 2 times larger (198 vs. 94). Using the 'Q' value to comprehensively evaluate the focusing strength and aberration, EQT gains a Q of 30.4, while that for EQS and EQQ are 9.4 and 5.0, respectively. **Conclusion:** The results support our goal of compact, stable and high performance EQ design, providing promise to advance relevant radiobiological studies.

WE-B-TRACK 6-02

The Potential of Online-Adaptive Head and Neck Proton Therapy to Reduce Normal Tissue Complication Probability

A Lalonde*, M Bobic, B Winey, J Verburg, G Sharp, H Paganetti, Massachusetts General Hospital / Harvard Medical School, Boston, MA

Purpose: To evaluate the impact of daily online adaptation on normal tissue complication probability (NTCP) in the context of head and neck intensity modulated proton therapy (IMPT). **Methods:** IMPT treatment plans are created without setup margins for 10 retrospective head and neck patients with daily CBCTs (median=33). The plans are adapted at each fraction using an in-house developed online adaptation (OA) workflow based on GPU accelerated Monte Carlo dose calculation and a deep learning-based CBCT scatter correction algorithm. OA is performed by re-adjusting the intensity of a subset of highly weighted spots without changing their energy or position. For comparison, a second plan is created for each patient using robust optimization (RO) with a 3 mm set-up uncertainty on which no adaptation is performed during delivery. For both scenarios, dose accumulation is performed using deformable image registration between each daily CBCT and the planning CT. NTCP models for xerostomia, aspiration, dysphagia, oral mucositis, liquid and solid swallowing are applied to the accumulated dose distributions in order to quantify the toxicity risk associated with each treatment technique. **Results:** Both methods achieved similar target coverage over the course of treatment. NTCP calculated on the cumulative dose distributions showed a reduction in toxicity risk for aspiration, dysphagia, liquid and solid swallowing in all patients using OA over RO. The most noticeable differences were for aspiration, with median NTCP values of 28.4% and 17.1% for RO and OA respectively. Xerostomia and oral mucositis NTCP were also reduced over the whole patient cohort using OA over RO, with mean reductions of 4.7 and 8.3 percentage point respectively. **Conclusion:** This work demonstrates the potential of online adaptive proton therapy to expand the therapeutic window of head and neck IMPT by reducing the risk of long-term side effects while ensuring adequate target coverage.

WE-B-TRACK 6-03

Design and Initial Testing of An Isocenter Alignment Prototype for the Proton Dynamic Collimation System

T Geoghegan^{1*}, K Patwardhan¹, B Smith¹, R Flynn¹, J Yu², A Gutierrez², D Hyer¹, (1) University of Iowa Hospital and Clinics, Iowa City, IA, (2) Miami Cancer Institute, Baptist Health South Florida, Miami, FL

Purpose: The purpose of this work is to design and validate an isocenter alignment system for the Dynamic Collimation System (DCS), a proton collimator comprised of four nickel trimmers that move in synchrony with a scanned proton beam to sharpen the lateral penumbra. Accurate alignment of the DCS with the radiation isocenter is necessary as the lateral dose distribution is sensitive to the spatial relationship between the proton beam and the collimator. **Methods:** An alignment system was designed to be affixed to the base of the DCS and consists of a chromium-doped alumina scintillating screen (Chromox), a mirror, and a Raspberry Pi camera system. The camera system, which is mounted outside of the primary beam, views the scintillator via the mirror which is tilted at 45°. Each axis is aligned to radiation isocenter by irradiating a nickel trimmer with a uniform, 150 MeV scanned proton field (7x7 cm²) while acquiring a long exposure image of the Chromox screen. From the acquired images, the center of the trimmer is compared with the center of the radiation beam to determine the offset between the radiation and mechanical isocenter of the DCS. This calculated offset is passed along to the DCS control system such that the mechanical DCS coordinate system can be re-calibrated to match the radiation isocenter. This process is then repeated for each axis. **Results:** The alignment system was able to calculate the radiation isocenter offset within 1.1 mm as compared to a beams-eye-view x-ray image. **Conclusion:** Initial results indicate that the camera system is capable of aligning the trimmers to the center of a scanned proton field. Automation and further refinement of this system could result in an efficient QA alignment platform for the DCS

Research reported in this abstract was supported by the National Cancer Institute of the National Institutes of Health under award number R37CA226518. Hyer and Flynn are co-inventors on a patent that has been licensed to IBA.

WE-B-TRACK 6-04

Fast MCsquare-Based Independent Dose Verification Platform for Pencil Beam Scanning Proton Therapy

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Purpose: Secondary dose verification has not been introduced to proton therapy as a routine practice. The purpose of this study is to commission MCsquare (a multi-cores CPU-based dose calculation engine), integrate it into RayStation treatment plan system (TPS) to create a dedicated platform for fast, independent pencil beam scanning (PBS) dose verification. **Methods:** A MCsquare-based independent dose verification platform (MC2InRS) was developed to realize automatic dose re-calculation for clinical use, including data preparation, dose calculation, 2D/3D gamma analysis. MCsquare was commissioned based on in-air lateral dose profiles, integrated depth dose, and the absolute dose of different beam energies for Proteus@ONE. It was then integrated into RayStation using Python to build the independent dose verification platform - MC2InRS. MC2InRS was validated with measurement data using various targets and depths in a water phantom. This study also investigated fifteen clinical cases to demonstrate the feasibility and effectiveness of MC2InRS platform in clinic practice. **Results:** Between simulation and

measurement, the distal range differences at 80% (R80) and 20% (R20) dose levels for each energy were below 0.05 mm, and 0.1 mm, respectively, and the absolute dose differences were below 0.5%. 29 out of 36 QA planes reached a 100% gamma passing rate (GPR) for 2%/2mm criteria, and a minimum of 98.3% gamma was obtained in water phantom between simulation and measurement. For the 15 clinical cases investigated, the average 2D GPR (2%/2mm) was 95.4%, 99.3% for MCsquare vs. measurement, MCsquare vs. TPS, respectively. The average 3D GPR (2%/2mm) was 98.9%, 95.3% for MCsquare vs. TPS in water, and computed tomography (CT), respectively. **Conclusion:** A fast, independent dose verification platform – MC2InRS, has been developed to perform dose verification with high accuracy and efficiency for Pencil Beam Scanning (PBS). The feasibility of using MC2InRS in routine clinical practice has also been demonstrated.

WE-B-TRACK 6-05

BEST IN PHYSICS (THERAPY): New System for Fast and Independent Proton Beam Energy Measurement

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Purpose: A new system to measure the energy of clinical proton beams was developed. It is based on the precise measurement of the protons' time of flight (TOF) and a self-calibration procedure. Results from preliminary validation tests will be presented. **Methods:** A telescope was built using two Ultra-Fast Silicon Detectors segmented in 2.2 mm² strips and mounted on dedicated 8-channels front-end readout boards. A high-precision positioning system allows the TOF measurement at different configurations. Two linear stages are used for the transversal alignment of the second sensor while a third, 1 m long, linear stage moves one sensor to change the distance from 30 to 95 cm along the beam direction (axes Z). This displacement is measured with 10 µm of accuracy using an absolute optical encoder. The 16 signals from strips were acquired with a 5Gs/s digitizer and processed with a MATLAB application. To validate the self-calibration procedure, measurements were performed with a clinical proton beam at 7 different energies (69, 72, 104, 142, 183, 223, and 227 MeV), 4 distances (40, 60, 80 and 95 cm) and 12 milliseconds of data from 30 seconds of acquisition time. **Results:** The linear interpolations of the averaged TOFs as a function of the distance shown deviations of a few hundred keV for most of the energies and all the distances, corresponding to less than 1 mm range difference. The consistent energy values obtained for the four distances reflect the good calibration of the system. **Conclusion:** A new system able to measure independently the proton beam energy in a few seconds has been developed. This new technology, fast and precise, would radically improve the commissioning operations and could introduce new methods for beam control and validation in particle therapy.

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WE-B-TRACK 6-06

Simultaneous Proton Dose and LET Measurements Using Dual Storage Phosphors

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Purpose: To investigate the feasibility of using two optical storage phosphor dosimeters of different proton numbers (Zeff) for simultaneous proton dose and linear-energy-transfer (LET) measurements. **Methods:** A high Zeff commercial BaFBr:Eu²⁺ storage phosphor detector was investigated and compared with KCl:Eu²⁺, a low Zeff (Zeff=18) storage phosphor made in-house. BaFBr:Eu²⁺'s proton stopping-power-ratio (SPR) was calculated using the Bethe-Bloch theory with elemental data obtained via energy dispersive x-ray spectroscopy (EDS) analysis. Then, its optical characteristics were investigated using excitation and emission spectral analysis. Finally, its dosimetric response under varying proton irradiation conditions was measured under a reference proton SOBP geometry (Range: 15.0 g/cm², Modulation: 10.0 g/cm²) generated using a Mevion S250 proton therapy unit. The LET environment varied with depth in water and their values were calculated using analytical methods. The dosimetric responses of BaFBr:Eu²⁺ under varying irradiation conditions including dose linearity, dose-rate dependence, dynamic range, temporal profiles, radiation hardness and LET dependence were measured via photostimulated luminescence (PSL) intensity measurements. **Results:** EDS analysis revealed its exact elemental composition to be BaFBr_{0.85}IO_{1.5}:Eu²⁺ (Zeff=49) and subsequent analytical calculations showed a good, monotonic signal separation by virtue of the proton energy dependence of its proton SPR. Its PSL excitation and emission spectrum overlapped with KCl:Eu²⁺'s showing good cross-compatibility. In addition, it had good proton dose linearity, dose-rate independence and high reusability. It showed a strong LET-dependent PSL under-response across an LET spectrum from 0.64 keV/µm to 11.1 keV/µm that could be sufficiently explained using stopping-power-ratio variations. **Conclusion:** We have demonstrated that BaFBr_{0.85}IO_{1.5}:Eu²⁺ and KCl:Eu²⁺ constitute a pair of compatible storage phosphor dosimeters having desirable optical and dosimetric properties and having the ability to measure absolute proton doses (using PSL intensities from the near LET-independent KCl:Eu²⁺ dosimeters) and LET values (via the relative PSL under-response of the high Zeff BaFBr_{0.85}IO_{1.5}:Eu²⁺ dosimeters) simultaneously.

WE-B-TRACK 6-07

Development of An Imaging System for Proton Pencil Beam Spot Profile Determination

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Purpose: To develop an independent dosimetric imaging system consisting of an optical scanner and reusable storage phosphor dosimeters that can obtain two-dimensional (2D) dosimetric images of proton spots at sub-millimeter resolution. **Methods:** An in-house scanner was constructed using stepper motors that can achieve translations to sub-millimeter precision. BaFBr:Eu²⁺ storage phosphor detectors were then irradiated using proton pencil beams at maximum energy of 227.1 MeV. During optical readout, the detector panels were mounted vertically and actively translated relative to a static, focused stimulating laser light produced by a 594nm helium-neon laser. The intensities of the photostimulated luminescence (PSL) produced will correlate to the proton dose delivered at specific locations of the latent dosimetric image. Proton spots were delivered while varying the proton doses delivered from 1-150 monitor units (MUs), where 1MU corresponded to 6cGy peak proton dose. Typical dose delivery errors such as lateral displacements and beam divergence were simulated too. **Results:** The combined system managed to attain a sub-millimeter precision and close agreement with 1MU proton dose images collected with a commercial scintillator. Dosimeters showed good linearity and no saturation of up to 150MU proton dose. A signal-to-background ratios of at least 2400:1 was attainable at 150MU demonstrating the ability for proton halo measurements down to at least 0.04% of peak proton doses. The system was sensitive enough to detect ±1mm lateral displacements inline and crossline and ±0.30mm divergence in the Gaussian profile. **Conclusion:** We have developed an independent dosimetric platform that is able to perform profile and dosimetric checks of proton pencil beams at sub-millimeter precision and at high dosimetric accuracy. It was sensitive enough to detect beam delivery errors including lateral shifts and beam divergence. The demonstrated high dynamic range shows potential utility for versatile measurement applications such as proton beam halo measurements.

Ultrasound Symposium

TRACK 7

Ultrasound Image Guidance for Interventions

WE-B-TRACK 7-00

Ultrasound Image Guidance for Interventions

Moderator: Davide Fontanarosa, Queensland University of Technology, Brisbane

WE-B-TRACK 7-01

E. Harris, Institute of Cancer Research and Royal Marsden Hospital:
Frustrations in Ultrasound Guided Radiotherapy: Automated Solutions

WE-B-TRACK 7-02

S. Camps, EBAMed:

Ultrasound-Guided Cardiac Radio-Ablation

WE-B-TRACK 7-03

M. Antico, Queensland University of Technology:

Platform for Ultrasound-Guided Autonomous Minimally Invasive Surgery: Application to the Knee

WE-B-TRACK 7-04

Q&A

Speaker #1

Frustrations in Ultrasound-guided Radiotherapy: Automated Solutions

Dr. Emma Harris

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Ultrasound can image soft tissues with high resolution, in real-time and also offers volumetric imaging. These attributes make it ideal to guide radiation therapy. Ultrasound is also recognised as a user dependent modality, with the quality of the acquired images and the eventual interpretation of those images highly dependent on the skill and experience of the user. Clinical staff in the radiation therapy department often have little or zero experience with ultrasound. This has posed a significant barrier to the implementation of ultrasound in guidance in radiotherapy. The additional training and time required to use ultrasound, adds significant pressure to an already highly-pressured clinical environment and often under-resourced departments. On a practical level, ultrasound does not compare well with tools such as cone beam CT, now effectively single push button acquisition and complemented with a suite of image processing tools that automate image registration. It is eminently possible to improve the "user friendliness" of ultrasound. Alternative hardware which is more suited to the radiation therapy clinic, to improve speed of set up and provide a more seamless integration with radiotherapy workflow can be employed. In diagnostic ultrasound setting, new software tools are being developed to allow unskilled users to acquire images, and automatic segmentation of tissues is being addressed. This talk will review these new innovations and explore how a new generation of ultrasound guided radiation therapy can be successfully translated into the clinic.

Learning Objectives:

1. Understand the practical barriers to clinical implementation of ultrasound guided radiotherapy.
2. Knowledge of innovations in hardware that could be applied to ultrasound to improve integration of ultrasound with the radiotherapy workflow.
3. Knowledge of state-of-the-art machine learning approaches that are being applied to ultrasound to improve the ability of users to acquire good quality images and interpret images.

Speaker #2

Ultrasound guidance during radioablation treatment of cardiac arrhythmias

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Heart arrhythmias are disruptions in the normal heartbeat, which can result in too fast, too slow or irregular heartbeats. As the occurrence of these arrhythmias is strongly linked to heart attacks and strokes, this can be a life threatening condition. Nowadays, catheter ablations are the main curative treatment option for atrial fibrillation and ventricular tachycardia. These procedures require hospitalization of the patient, can be lengthy and risky and they have shown varying outcomes depending on e.g. the arrhythmia type and skills of the treating physician. For this reason, there has been a growing interest for treating cardiac arrhythmia patients non-invasively and potentially more effectively by means of radiation.

To safely and effectively perform radioablation of the heart, several motion components which can affect the shape and position of the heart and therefore also the shape and position of the treatment target should be taken into account. One could think of motion resulting from the heartbeat of the patient, as well as of inter-fraction motion and intra-fraction drift of the heart. This talk will focus on how ultrasound imaging can be used to monitor these motion components in order to provide the clinicians with real-time information on the motion of the heart during radioablation treatment of cardiac arrhythmias.

Learning Objectives:

1. To learn more about the use of radiation for the treatment of cardiac arrhythmias.
2. To understand how ultrasound imaging can be used for motion monitoring during cardiac radioablation treatments.

Speaker #3

Platform for ultrasound-guided autonomous minimally invasive surgery: application to the knee

Dr. Maria Antico

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Ultrasound imaging can be employed to improve automation levels in minimally invasive surgery and eventually enable full automation by providing real-time situational awareness to medical robots via intra-operative, real-time, volumetric mapping of the surgical site.

This talk describes the development of a prototype automated image guidance system, combining high-refresh-rate, volumetric ultrasound (US) imaging (or 4D US) and advanced deep learning strategies for an autonomous robotic platform, currently being investigated for knee surgery. The feasibility of using 4D US for guidance in knee minimally invasive surgery was first proved through cadaver and volunteer studies. The workflow to enable automatic interpretation of 4D US imaging for robotic guidance was then developed and implemented. The essential steps include automatic image quality assessment, tissue segmentation and tracking (in combination with the assessment of uncertainties) and surgical tool tracking. The results obtained show US imaging's potential for quantitative autonomous tasks and for the creation of autonomous, intelligent robotic surgical systems that will hopefully soon make surgery more sustainable and improve people's quality of life.

Learning Objectives:

1. US imaging combined with deep learning strategies can provide a solution for real-time volumetric guidance for autonomous systems.

2. The algorithms developed and implemented can be extended to US images of other body districts.
3. The workflow elucidated provides solutions to the issues currently limiting the application of US imaging to quantitative autonomous tasks.

Vendor Showcase TRACK 8 MIM Software, Inc.

WE-SHOWCASE-TRACK 8-12-1

Overcoming Peer Review Challenges with MIM Harmony

K Ostergren^{1*}, (1) Mim Software, Cleveland, OH

Imaging Interactive ePoster Discussion

TRACK 1

Imaging: CBCT Acquisition and Reconstruction

WE-lePD-TRACK 1-01

An X-Ray Fluence Modulation Apparatus for Cone-Beam CT

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Purpose: To develop and evaluate the performance an apparatus of fluence modulation and scatter shielding for cone-beam CT, and the corresponding method of image acquisition. **Methods:** The apparatus is composed of two assemblies, referred to herein as the dynamic Fluence Modulation (FM) and Scatter Shield (SS), that work in synchrony to form a narrow beam sweeping the entire field of view of the imaging system in each projection. The positioning of the FM with respect to the focal spot leads a change in the flux of x-rays incident on a patient – providing the means to optimize the dose depending on the size, shape and positioning of the organ of interest. Feasibility of this apparatus was verified through a series of Monte Carlo (MC) simulations, followed by incorporating the FM-SS assembly into an existing cone-beam CT system. A series of UHMW cylindrical phantoms were utilized to measure the detected signal level, scatter build-up, and the dose, in presence and absence of the FM-SS apparatus. **Results:** Implementation of FM-SS results in near scatter-free projections. The scatter-to-primary ratio dropped from 90% in the absence of the FM-SS to below 15% in its presence. The recorded dose, tallied throughout a phantom with a diameter similar to that of an average-size breast (14 cm), was dropped 32%. The variabilities in the recorded raw signal, resulting from the differences in the attenuation path length through a cylinder, was almost entirely resolved (coefficient of variation dropping from 850% to 20%). **Conclusion:** Utilization of the FM-SS apparatus and technique was demonstrated via simulations and physical implementation to result in: significant reductions in dose, reductions in scatter uptake in projections, and a reduction in the required dynamic range of the imager. This novel solution addresses the limitations of bow-tie filters, and high-scatter projections in cone-beam CT geometry.

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WE-lePD-TRACK 1-02

The Evaluation of the Impact of Rotational Corrections on Translational Setup Errors

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Purpose: To compare the translational and rotational patients setup errors of different body regions with six-dimensional (6D) matching using CBCT, and therefore to quantitatively evaluate the impact of rotational corrections on translational setup errors. **Methods:** Daily online treatment 6D setup errors data from 302 patients for 4078 fractions treated by tomotherapy using Varian VitalBeam were obtained. Systematic and random errors were calculated and statistical methods were then applied. In addition, a phantom experiment was conducted. Also, 736 fractions from 40 patients received pelvic treatment were reevaluated offline by matching the CBCT scans to the planning CT scans with and without rotational corrections. With translational corrections only, unsuccessful registrations were classified and successful registrations were quantitatively analyzed. **Results:** Larger rotational setup errors were observed in the pelvis and breast regions. The changes in the anterior-posterior(y axis) and superior-inferior(z axis) directions were caused by uncorrected pitch. The changes in the superior-inferior (z axis) and lateral-medial (x axis) directions were caused by uncorrected yaw. The single uncorrected roll had little effect on the translational errors. The single uncorrected yaw had the greatest influenced on the translational errors. A shift change of nearly 3mm was caused by uncorrected 3° of yaw. Registration on bony tissue of 736 fractions from forty patients was successful for all offline 6D matches, 43.75% of which were unsuccessful with translational correction only. However, the rules of combinations in these rotational angles were summarized with the unsuccessful registration. The changes of translational shifts >1mm in 27.05% and >2mm in 3.38% of the rest successful registration were recorded. The changes of maximal translational shifts were 3.3mm. **Conclusion:** Translational setup errors have been affected by rotational corrections. Rotational corrections significantly improve the accuracy and efficiency of patient positioning before irradiation. It is of great significance for rotational corrections on high-precision radiotherapy.

WE-lePD-TRACK 1-03

Online Bayesian Geometric Calibration of CBCT Systems

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Purpose: C-arm x-ray imaging systems have found widespread clinical use, for example taking 2D projections to guide interventional surgeries. Accurate 3D reconstruction from 2D images requires precise calibration of geometric parameters. Geometric calibration is typically offline, which restricts 3D imaging to a narrow library of acquisition arcs on systems with highly reproducible geometry. We present a real time online geometric calibration framework utilising a treatment couch attachment applicable to patient adaptive scans and portable systems, and show results obtained with a commercial c-arm. **Methods:** We derived a Bayesian estimate of geometric parameters from observed markers. We designed a couch attachment that presents required markers during online imaging. A CIRS Dynamic Cardiac Phantom was imaged on a Siemens ARTIS pheno c-arm system with the couch attachment placed. First an automatic 3D acquisition with “gold standard” offline calibration was performed, then a manually controlled irreproducible acquisition. The first acquisition was reconstructed with uncalibrated, offline and online calibration geometry, the second with uncalibrated and online calibration as offline not applicable. The difference in parameter sets was quantified with Root-Mean-Square Deviation (RMSD) and the 5 3D images were inspected for calibration artefacts. **Results:** The RMSD between offline calibrated and uncalibrated geometry was 0.37° to 0.44° and 1.0mm to 3.7mm while RMSD between offline calibrated and online Bayesian calibrated was 0.0013° to 0.0017° and 0.4mm to 0.9mm. The manual acquisition uncalibrated geometry had the most artefacts, then inbuilt acquisition uncalibrated, then automatic and manual acquisition online calibration with minimal artefacts completely absent in the automatic acquisition offline calibration reconstruction. **Conclusion:** The online Bayesian calibration approach is able to

recover geometric parameters and produce reconstructions to nearly the quality of “gold standard” offline calibration, with the benefit of being applicable to adaptive acquisitions and portable systems.

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WE-lePD-TRACK 1-04

Dictionary-Learning Iterative Reconstruction for Sparse Data, Slot-Scanning, Breast CT

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Purpose: Breast CT improves the visualization of breast anatomy over conventional 2D mammography and has been widely investigated for the diagnosis and treatment of breast cancer. Slot-scanning mammography has potential advantages with the use of high-resolution linear detectors and reduced x-ray scatter; however, integrating such technology into a CT acquisition introduces acquisition time challenges. In this work, we explore an implementation of slot-scanning CT which uses sparsely sampled projection data. We propose a dictionary-learning (DL) iteration reconstruction method to handle the sparse data and to facilitate fast breast CT scans using the slot-scanning approach. **Methods:** We implemented a dictionary-learning penalized-weighted least squares (DL-PWLS) algorithm for breast CT. A global redundant dictionary with 16x16 voxel atoms was generated to emulate breast anatomical details. In each iteration, the image was extracted into patches that were estimated by a sparse combination of dictionary atoms. The DL-PWLS algorithm penalizes the difference between the reconstructed image and its representation assembled using the dictionary. The developed algorithm was tested using complete 360-view (49 kV, 243 mAs) projection data and 10% of data with 1D θ -sparsity (36 views) and 2D R, θ -sparsity (72 view but only 50% of the radial bins with 180° phase shifts between neighboring views). **Results:** The DL-PWLS algorithm demonstrates enhanced noise reduction and low-contrast tissue classification over traditional filtered backprojection (FBP) and PWLS algorithms. For sparsely sampled data, visualizations of microcalcifications and soft tissues as measured by structural similarity (SSIM) are further improved using the DL-PWLS reconstruction with 10% 2D R, θ -sparsity (SSIM = 0.77 and 0.84 for microcalcifications and low-contrast tissue). **Conclusion:** A DL iterative reconstruction algorithm is developed for breast CT. The DL-PWLS reconstruction in combination with doubly sparse data is expected to substantially shorten the breast CT scan time. The effective noise reduction also shows potential capability for patient dose reduction.

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WE-lePD-TRACK 1-05

Towards Megavoltage Cone Beam CT (CBCT) with Image Quality of Kilovoltage CBCT

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Purpose: Firstly, to develop fastCAT, a cone beam CT (CBCT) simulation platform capable of modeling the physics, geometries, and phantoms of commercial linacs with high computational efficiency. Secondly, the platform was used to conduct a simulation study investigating novel MV imaging beam and detector combinations. **Methods:** FastCAT combines GPU ray-tracing with Monte Carlo (MC) methods for improved simulation efficiency. FastCAT requires photon spectra and detector optical spread functions (OSFs) as input. 2.5 and 6 MV photon spectra were modelled in EGSnrc using carbon, aluminum, and tungsten targets. kV x-ray tube spectra were simulated analytically. OSFs were simulated in Topas for cadmium tungstate (CWO), gadolinium oxysulfide (GOS), and cesium iodide (CsI) detectors. FastCAT contrast and noise magnitude were validated using a Truebeam linac and a Catphan 504 phantom for 100 kVp and 6 MV CBCTs, respectively. The simulation study calculated CNRs for all detector/beam combinations using a contrast phantom containing rib and spongiosa bone, lung, and adipose tissue inserts. Detector modulation transfer function (MTF) for each beam/detector combination was calculated using an angled slit phantom. **Results:** Truebeam contrast and noise magnitude matched all fastCAT values within the 99% confidence interval. FastCAT simulation times were 51s and 77s for the kV and MV CBCT, respectively, significantly shorter than the estimated 1-year long full MC simulations. The simulation study showed CNRs for the carbon 2.5 MV/CWO combination were more than 1.5 times greater than the next best 120 kVp/CsI combination, while MTF 50 values were only 1.2% lower for the carbon 2.5 MV/CWO combination. **Conclusion:** Our finding makes a compelling case that clinical CBCT imaging could be greatly improved using carbon target beams with a CWO detector.

WE-lePD-TRACK 1-06

Evaluation of U-Net++ On Low Dose CT Enhancement

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Purpose: Deep learning has been widely applied in traditional medical imaging tasks like segmentation and registration. Some fundamental CNN based deep learning methods have shown great potential in low dose CT enhancement. This study applied U-Net++ model to enhance CT images with low dose and compared the performance of U-net++ and symmetric residual convolution neural network (SR-CNN) quantitatively. **Methods:** 11 patient CT images were chosen as the ground truth in the training process. Under-sampled projections were simulated from the ground truth volumes with a uniform distribution. CBCT was then reconstructed from the under-sampled projections using the ASD-POCS TV algorithm with 40 iterations and was treated as the input of the models. The U-net++ model was improved based on U-net model by connecting the decoders, reserving better dense feature along skip connections. (shown in the figure 1). Deep supervision (DS) were used to make a combined loss between each upper node and the ground truth to enhance the image feature preserving capacity. U-net and SR-CNN were used as standard model for comparison. MSE loss and structure similarity (SSIM) loss were used in different attempts. The generated images were compared quantitatively using SSIM and peak signal-to-noise ratio (PSNR). **Results:** Both models succeeded to improve the quality of the low dose CT images. The U-net++ model trained with MSE loss had best average PSNR of 41.4 on the test dataset and average SSIM of 0.9993 in terms of the whole images. RS-CNN model trained using SSIM loss had the best average SSIM of 0.99996 and the average PSNR of 38.6. **Conclusion:** Deep learning method showed its potential dealing with the high dose caused by modern CT technique. Different CNN model could influence the quality of the generated image on different evaluation criterions.

WE-lePD-TRACK 1-07

A Geometry-Guided Deep Learning Technique for CBCT Reconstruction

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Purpose: Despite the success of using deep learning for CT reconstruction, deep-learning-based cone-beam CT (CBCT) reconstruction remains challenging due to GPU memory limitations. This study is to develop a novel deep-learning (DL) technique to perform CBCT reconstruction with significantly reduced computer memory requirement and demonstrate its feasibility through CBCT reconstruction with sparsely sampled projection data. **Methods:** The novel geometry-guided deep-learning (GDL) technique contains a GDL reconstruction module and a DL post-processing module. GDL replaces the traditional single fully connected layer with many small fully connected layers in the network architecture based on the projection geometry. GDL reconstruction module learns and performs projection-to-image domain transformation. The DL post-processing module further improves image quality after reconstruction. We demonstrated the feasibility and advantages of the model by comparing ground truth CT volumes with CBCT images reconstructed from simulated digitally-reconstructed-radiographs (DRRs) using 1) GDL reconstruction module only, 2) GDL reconstruction module with deep-learning post-processing module, 3) Feldkamp, Davis, and Kress (FDK) only, and 4) FDK

with deep-learning post-processing module. The differences are quantified by peak-signal-to-noise ratio (PSNR), structural similarity (SSIM) index, and root-mean-square error (RMSE). **Results:** Average values for PSNR, SSIM, and RMSE of reconstructed images are 14.092, 0.857, and 0.198 for FDK only, 18.086, 0.894, and 0.125 for GDL only, 23.015, 0.901, and 0.071 for FDK with post-processing, and 25.313, 0.930, and 0.054 for GDL with post-processing. Reconstruction time for all reconstruction methods is comparable (around seconds). The estimated memory requirement using GDL is reduced for more than 10,000-fold compared to DL methods using large fully connected layers for CBCT reconstruction. **Conclusion:** The GDL technique was demonstrated to be able to perform fast and accurate CBCT image reconstruction from sparsely sampled data with significantly reduced memory requirement compared to other existing networks.

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Imaging Interactive ePoster Discussion

TRACK 2

Imaging: Nuclear Medicine, PET

WE-lePD-TRACK 2-01

Patient-Specific Estimates of Absorbed and Effective Dose in Pediatric 18FDG PET/CT Imaging Studies

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Purpose: To estimate patient-specific organ dose in children undergoing ¹⁸FDG-PET/CT scans and evaluate the correlation between CT size specific dose estimate (SSDE) and effective dose. **Methods:** Organ and whole body doses were calculated on 181 scans in 100 pediatric patients (females/males: 47/53; 11.4±5.2 YO) who underwent ¹⁸FDG-PET/CT scans on a Siemens mCT Biograph-64 scanner. PET/CT scans consisted of 75 diagnostic (Dx) partial body CT, 36 whole body (WB) low-dose attenuation correction (AC) CT, and 74 partial body AC-CT. The CT scan parameters were similar between Dx and AC scans except for Quality Reference mAs (QRM) which was 150 for Dx-CT and 20/35 for AC-CT. Tube current modulation (TCM) was applied to the CT acquisitions and modeled in the dose estimation. The UF/NCI phantom library was used to calculate CT and PET patient-specific organ dose. The FDG administered activity was based on the North American Consensus Guidelines. In addition, CT SSDE values were calculated based on the AAPM TG-220 formulation and the correlation between CT effective dose and SSDE was investigated. **Results:** The average PET effective dose was 7.63±2.06mSv. The mean WB-AC, WB-Dx-CT effective doses were 1.50±0.70mSv and 5.97±1.71mSv respectively. The mean total PET+CT effective dose for WB-AC-CT and WB-Dx-CT exams were 6.56±1.06mSv (range: 4.33-9.13mSv) and 11.11±2.29mSv (range: 7.61-16.78mSv) respectively. The four organs which received the highest PET dose were: bladder, heart, brain and liver; and, for the highest WB-AC-CTs, gonads, heart, bladder and small intestine. The estimated CT effective doses showed high correlation with SSDE ($R^2=0.9$) for all examination types (excluding neck); the correlation increased with the CT scan length, demonstrating almost perfect correlation for WB-AC scans ($R^2=0.99$). **Conclusion:** This work provides a reference library of organ dose for a wide range of age/weight of pediatric ¹⁸FDG-PET/CT patients. Also, a linear relationship between CT effective dose and SSDE was determined.

WE-lePD-TRACK 2-02

Learning Feature Fusion for Multi-Modality PET-CT Segmentation with An Interpretation Method

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Purpose: The performance of a multi-modality PET-CT segmentation method heavily relies on the quality of the used information fusion strategy. We utilize the interpretation methods as tools to quantitatively measure the importance of information from each modality. We propose a novel feature fusion strategy according to the measured importance of information and design an iterative algorithm to automatically learn how to fuse information from different modalities. **Methods:** The co-segmentation framework consists of three stages. The first stage contains two parallel branches, and each is responsible for high-dimensional feature extraction either from PET or CT using an encoder-decoder style network. The second stage consists of an interpretable module, in which the perturbation-based interpretation method is used to measure the importance of different features. According to the measured importance of features, we obtain an effectively weighted features fusion strategy. The third stage is composed of cascaded convolutional blocks, which follow the interpretable module and are used to re-extract the semantic features and output the final segmentation mask. The algorithm is trained and tested on a PET-CT dataset with 84 patients, in which 60 are selected randomly for training, while the remaining 24 are used for test. **Results:** We use W-Net as the baseline, which is a classic co-segmentation method. Compared to the metrics (DSC=0.82, VE=0.36, CE=0.17) of the baseline, our method (DSC=0.85, VE=0.30, CE=0.15) achieves a superior performance. Meanwhile, we decompose our framework to investigate the contribution of the learned feature fusion strategy in the interpretable module. The backbone with the interpretable module performs much better than without the interpretable module. **Conclusion:** We propose a co-segmentation framework for PET-CT segmentation, and introduce the perturbation-based interpretation method to learn how to integrate features from multi-modality effectively. The proposed network with the learned feature fusion strategy achieved superior performance compared to the baseline.

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WE-lePD-TRACK 2-03

Independent Verification of a Ge-68/Ga-68 Well Counter Accuracy Source Activity Mis-Calibration

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Purpose: Annual efficiencies of our PET hot lab well counters using a new ⁶⁸Ge/⁶⁸Ga calibration rod source in 2020, were substantially lower than historical averages. An independent verification as to whether or not the stated activity of the new source was correct, was performed. **Methods:** A positron detection lab setup was employed for coincidence counting of the source. Two Harshaw 3" diam. x 3" thick NaI(Tl) detectors faced each other, separated by 18.6 cm and connected to Ortec pre-amplifiers (PAs). PA signals were input into Ortec amplifier and single-channel analyzers (SCAs) windowed around 511 keV. The two SCA logic pulses were input to an Ortec universal coincidence (UC) unit, and UC output logic pulses were then fed into an Ortec timer and counter. ²²Na coincidence efficiency (cpm/dpm) was determined using two 3-mm calibrated button sources centered between the detectors, and converted to that for ⁶⁸Ga based on positron yield. The 2020 and a 2015 source were then coincidence counted (cpm) and the results were converted to dpm. **Results:** The estimated dpm of the 2015 source was 7.6% lower than the decayed stated activity, but that for the 2020 source was 64.4% lower. A second new source sent by the manufacturer yielded an identical result (64.4% lower). All three sources were returned for investigation, and the manufacturer discovered an error in their source activity calibration method. A new source calibrated with a corrected method was sent, and the efficiencies of two well counters using that source (0.60 cpm/dpm) were within the range of those for the prior four years using the 2015 source (0.58 - 0.64). **Conclusion:** A coincidence counting method of measuring ⁶⁸Ge/⁶⁸Ga source activity, in addition to the well counter efficiencies based on that source, resulted in discovery and correction of an error in the manufacturer's activity calibration procedure.

WE-lePD-TRACK 2-04

Monte Carlo Study on PET Imaging Used as An Image-Guidance Tool in Very High-Energy Electron Therapy

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Purpose: Monte Carlo (MC) study on the use of PET imaging as an image-guidance tool in Very High-Energy Electron (VHEE) therapy. **Methods:** Using the MC code TOPAS, the simulations were aiming to assess the number of positron emitters created along the beam path when irradiating DICOM images for abdomen. While no specific spot was targeted, the 250 MeV VHEE beam was going through a part of a lung and the heart. The main positron emitters created in biological tissue were scored, namely oxygen-15, carbon-11, nitrogen-13, phosphorous-30 and potassium-38 and the corresponding activity was then calculated. **Results:** The total number of positron emitters produced by the VHEE beam was averaging 110 emitters.Gy⁻¹.mm⁻³ and can reach a maximum of 330 emitters.Gy⁻¹.mm⁻³. The interface between tissue and air within the lungs can be observed as there is no creation of positron emitters in air. Moreover, the relationship between positron emitter production and material density allows PET imaging to discern structures with strong density differences such as air/tissue or tissue/bone. Finally, an in-beam PET system is required to prevent significant signal losses during acquisition due to the short half-life of oxygen-15 (2.03 min) which is the most abundant emitters. The overall production of positron emitters is lower with VHEE therapy than with proton therapy because it occurs indirectly by photonuclear reactions, whose cross-sections are small. However, the resulting total activity remains sufficient for detection with an average of 0.6 Bq.Gy⁻¹.mm⁻³ and a maximum of 1.8 Bq.Gy⁻¹.mm⁻³. **Conclusion:** The potential use of PET imaging as an image-guidance tool in VHEE therapy has been demonstrated as long as an in-beam PET system is used.

WE-lePD-TRACK 2-05

Predicting Breast Cancer Hormone Receptor Status with PET/MR Images Using Dual-Input Deep Convolutional Neural Network (DCNN)

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Purpose: The use of endocrine agents in breast cancer therapy depends on the response to estrogen hormone receptor. Tissue sampling and surrogate genetic testing are the gold standard methodology to figure out the hormone receptor (HR) status which are the basis of genetic expression of cancer cells. However, it is sub-optimal to personalized treatment plan due to its time consuming process. A simple deep convolutional neural network (DCNN) approach with dual-image (PET and MR) input has been proposed to classify the estrogen receptor (ER) status (positive/negative). The purpose of the study is to develop an automatic pipeline to predict prognostic factors. **Methods:** Dynamic contrast enhancement MR sequences and 18F-FDG PET imaging were performed using a synchronized PET/MR scanner. A total 212 patients were patients were retrospectively collected who had received immunohistochemistry (IHC) genetic testing after tissue sampling. We annotated the breast tumors by drawing binary masks on PET images and then warped them into the corresponding MR images for each patient. Image preprocessing includes MR and PET slice thickness and field-of-view matching. A total 2,565 image patches were independently trained with three different neural networks to predict the binary classes of ER positive and ER negative status. We used cross-validation scheme to avoid possible unseen coincidence when shuffling the dataset during training. **Results:** The resulted accuracies from three different DCNNs of single MR input, single PET input, and dual PET/MR input were 54%, 63%, and 76%, respectively, which demonstrates that combining the features of both PET and MR images can predict the ER status well, rather than solely training the images with one image type. **Conclusion:** Image-based prediction of hormone receptor status without IHC genetic testing would demonstrate that the extracted PET and MR image features from DCNN may include prognostic factors, which is a potential for radiogenetic imaging biomarker.

Multi-Disciplinary Interactive ePoster Discussion

TRACK 3

Multi-Disciplinary: Biologically and Functionally-Guided Radiation Therapy

WE-lePD-TRACK 3-01

Change in ADC as An Indicator of Radionecrosis

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Purpose: To investigate if Apparent Diffusion Coefficient (ADC) can identify radionecrosis (RN) which is a serious potential side-effect to stereotactic radiosurgery (SRS) and can be mistaken for tumour progression on imaging. There is an urgent need to develop tools that distinguish between necrosis, progression, and immunotherapy-induced pseudoprogression. **Methods:** A retrospective study of 74 patients treated for melanoma with ipilimumab (prescription: 4x infusion every 3 weeks) and longitudinal Diffusion-weighted, T1-with-contrast, and FLAIR MRI on a range of clinical systems is being undertaken. The preliminary cohort consists of 27 patients. 5 were excluded as diagnosis to treatment was >1 year. 8 patients had diffusion-weighted-imaging prior to and after SRS. The SRS to ipilimumab time was +43-170 days (mean 93). SRS prescriptions ranged from 15-21Gy/1-3 fractions according to published protocols. 7 patients were imaged <20 days prior to SRS, 1 at 118 days, and all at times from 20-200 (mean: 109) days post-SRS. Lesions were contoured by an experienced radiation oncologist. Additional ROIs were derived by contracting originals by 2mm in all directions, thus separating the tumour core from the peritumoral rim. The mean ADC value for each ROI prior to radiosurgery, was compared to its corresponding mean value after to obtain a change in ADC value for each ROI. RN and non-RN groups, identified by an experienced oncologist through examination of clinical reports, were compared with a t-test. **Results:** The change in ADC observed in core (peritumoral rim excluded) ROIs from prior to post treatment was significantly different between the RN and non-RN groups (p=0.047). This exaggerated a similar, not significant trend, observed when the whole tumour ROIs were examined (p=0.140). **Conclusion:** Increasing ADC in the inner-core may be predictive of RN in this study. Observed differences between the complete ROI and contraction ROIs should be considered in future investigations.

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WE-lePD-TRACK 3-02

Characterization of Lobe-Wise Ventilation and Elasticity in Patients with Chronic Obstructive Pulmonary Disease

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Purpose: Chronic Obstructive Pulmonary Disease (COPD) is a heterogeneous disease but is currently measured holistically by spirometry. This project aims to begin to describe the heterogeneous nature of COPD through ventilation and elasticity by characterizing these parameters for individual lobes. **Methods:** Eight patients with varying degrees of COPD and two patients without COPD were scanned 25 times using a free-breathing fast helical CT protocol. The 5DCT motion model was used to simulate end-exhalation and end-inhalation images from the 25 images and recorded breathing amplitudes. The DVFs between these pseudo-images were used to guide a biomechanical model to estimate elasticity and to calculate the Jacobian, which represents ventilation. The Pulmonary Toolkit was used to segment the lobes semi-automatically. The resulting lobe masks were used to analyze the Jacobian and elasticity distributions in each lobe, separately. A Ventilation Heterogeneity Index (VHI) was defined as the ratio of the minimum lobe Jacobian divided by the mean of the other lobe Jacobians. The relative volume of low-elasticity values (1-3 kPa), an elasticity range that is reflective of diseased lung tissue, was determined for each lobe. An Elasticity Heterogeneity Index (EHI) was

also defined to describe the ratio of the lobe with the maximum relative volume of low elasticity to the mean values of the other lobes. **Results:** Patients with severe COPD had a mean VHI of 0.00230 while patients with mild or no COPD had a mean VHI of 0.104. The EHI for patients with severe COPD was 1.69, denoting further heterogeneity than the 3.13 calculated for mild/non-COPD patients. **Conclusion:** We have measured a marked increase in heterogeneity in tissue elasticity and ventilation in patients with severe COPD. This work is one step towards personalizing treatment for patients with COPD to better spare the healthy tissue and maintain lung functionality.

WE-lePD-TRACK 3-03

Machine Learning Based Prediction of Big Inter-Fractional Changes in Lung Tumor Motion During Radiotherapy Course Using Robust Feature Selection

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Purpose: During lung cancer radiotherapy, tumor motion amplitude may change due to various reasons. If motion amplitude varies greater than a clinical threshold (3 mm was used in this study), adaptive radiotherapy may be needed. In this work, we proposed an ensemble machine learning model to identify potential patients for adaptive radiotherapy after 4DCT simulation. **Methods:** We employed 5 conventional classification models (logistic regression, decision tree, support vector machine, random forest, and XGBoost) as base predictors. Then the results were used as inputs for a two-layer neural network to get a final prediction. For each first-level predictor, the inputs are based on 17 features (clinical factors and imaging features from planning CT) from 140 patients. To understand the correlation between each feature and the prediction task, we proposed a robust feature ranking method. We compared the ensemble learning model with each individual base predictor. To evaluate the model robustness, the error and area under receiver operating characteristic (ROC) curve were derived based on 1000-times 5-fold cross validation. **Results:** The feature ranking results showed that the initial tumor motion along the superior-inferior (SI) direction best correlates with the big tumor motion change. By sequentially removing the least important features, the prediction performance of base predictors was consistently improved. Furthermore, the proposed ensemble learning model effectively improved the best predictive accuracies of a single base predictor (Random Forest or Logistic Regression) from 0.78 to 0.83, and the best F1 score of XGBoost from 0.50 to 0.60. **Conclusion:** The proposed ensemble learning model can effectively detect a large change in tumor motion, which helps predict patients that may need adaptive radiotherapy. Experimental results demonstrated superior predictive performance over 5 base predictors. By involving a feature ranking strategy, the prediction accuracy of both base predictors and the overall prediction model are improved.

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WE-lePD-TRACK 3-04

Powering Predictive Treatment Planning by a Seq2seq Deep Learning Predictor

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Purpose: To develop a novel deep learning algorithm, Seq2Seq, for predicting weekly anatomical changes of locally advanced lung tumor and esophagus during definitive radiotherapy, incorporate the potential tumor shrinkage into a predictive treatment planning paradigm for adaptive radiotherapy, and improve the therapeutic ratio. **Methods:** Seq2Seq starts with the tumor and esophagus observed on the planning CT to predict their geometric evolutions during radiotherapy on a weekly basis (week 1 to 6), and subsequently updates the predictions with new snapshots acquired via weekly CBCTs. Seq2Seq is composed of an encoder to parse the changes of images, and a decoder to reconstruct the prediction; both equipped with convolutional long-short term memory for sequence analysis. The longitudinal dataset (planning CT plus 6 consecutive weekly CBCTs) included 55 lung cancer patients, in which we used 11-fold cross-validation to comprehensively evaluate predictions for all patients. Dice coefficient and average Hausdorff distance (d_{hd}) between the predicted and actual weekly contours were calculated for accuracy evaluation. Simulating a workflow of weekly adaptation, six plans were optimized according to each weekly prediction, and made ready for weekly deployment to mitigate the clinical burden of online replanning. **Results:** Seq2Seq tracks structural changes well: DICE between predicted and actual weekly tumor and esophagus were $(0.82 \pm 0.10, 0.80 \pm 0.14, 0.79 \pm 0.13, 0.76 \pm 0.12, 0.74 \pm 0.13, 0.70 \pm 0.16)$, and $(0.72 \pm 0.16, 0.73 \pm 0.11, 0.75 \pm 0.08, 0.74 \pm 0.09, 0.72 \pm 0.14, 0.71 \pm 0.14)$, respectively, while d_{hd} were within 2mm. Evaluating dose to the actual weekly tumor and esophagus, a 5.3Gy reduction in esophagus mean dose while maintaining 60Gy tumor coverage was achieved with the predictive weekly plans, compared to the plan optimized using the initial tumor and esophagus alone, primarily due to noticeable tumor shrinkage during radiotherapy. **Conclusion:** It is feasible to predict the longitudinal changes of tumor and esophagus with the Seq2Seq network, which could lead to improving the efficiency and effectiveness of lung adaptive radiotherapy.

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WE-lePD-TRACK 3-05

Positron Emission Tomography (PET) Characterization for Biology-Guided Radiotherapy (BgRT)

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Purpose: To characterize the performance of the PET subsystem of the first Reflexion X1 machine installation using the NEMA NU-2 2018 standard. **Methods:** The X1 machine consists of two symmetrically opposing 90° arcs of PET detectors incorporated into the architecture of a ring-gantry linear accelerator rotating up to 60 RPM to aim radiotherapy beams using PET emissions of the tumors. Each PET arc has 32 detectors arranged in 2 rings with 90cm inner diameter and 5cm axial extent. Each detector is made up of 12x6 Lutetium-based crystals with a dimension of 4.14x4.14x20mm³. Sensitivity assessed using 2M coincidence events collected for each position. Spatial resolution tests utilized six point-sources arranged horizontally 1, 10 and 20cm away from the center axis, three in and three off the isoplane, measured simultaneously for 10 mins. Count loss assessed using 20 frames were scanned, 2M coincidence events each, with a 15 min break between scans. Image quality assessed using a phantom filled with FDG (4:1 sphere-to-background ratio). **Results:** X1 PET Subsystem sensitivity: 182.4 and 160.9 cps/MBq at the center and off-center positions, respectively. Spatial resolution: average FWHM values of 4.3, 5.1, and 6.7mm for the point sources at 1, 10, and 20cm off center, respectively. For count loss, max NECR: 2625.5 cps, max true coincidence rate: 5557.4 cps, and scatter fraction: 39.8%. Image-quality contrast values: 29.6%, 64.9%, 66.5%, 81.8%, 81.2%, and background variability: 14.8%, 12.4%, 10.3%, 8.8%, 8.3%, for the 13, 17, 22, 28, 37 mm sphere sizes, respectively. **Conclusion:** When operating in an imaging mode, spatial resolution and image contrast were equivalent to those of typical diagnostic systems, while sensitivity and count rate were lower due to design of the system which requires a far smaller detector area than a typical diagnostic system. Clinical efficacy when used in BgRT remains to be validated.

Zhiqiang Hu, Manoj Narayanan, Peter Olcott, Matthew Bieniosek, Thomas Laurence and Shervin M. Shirvani are employees of Reflexion Medical, Inc. There is a master research agreement between Reflexion Medical and Stanford University.

WE-lePD-TRACK 3-06

Comparison of 3D Deep Convolutional Neural Networks and Training Strategies for Ventilated Lung Segmentation Using Multi-Nuclear Hyperpolarized Gas MRI

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Purpose: Deep learning (DL) has shown competency in numerous image segmentation tasks, including delineating ventilated lung volumes from hyperpolarized gas MRI. We previously demonstrated the utility of a VNet convolutional neural network (CNN), trained on a combination of ^3He and ^{129}Xe scans, in producing segmentations that outperform conventional methods. Here, we compare the performance of several 3D CNNs with varying training strategies for segmenting ventilated lungs on a large multi-nuclear hyperpolarized gas MRI dataset comprising healthy subjects and patients with a range of pulmonary pathologies. **Methods:** The dataset contained 743 volumetric hyperpolarized gas MRI scans, with either ^3He (248 scans) or ^{129}Xe (495 scans) and corresponding expert segmentations, from 326 healthy subjects and patients with pulmonary pathologies. Five experiments were performed to train each CNN: (1,2) the model was trained on either ^{129}Xe or ^3He images; (3,4) transfer learning was applied to the pre-trained models in (1,2) to fine-tune the network for the opposite gas images; (5) the model was trained on combined ^3He and ^{129}Xe data. Experiments were performed on the UNet and VNet architectures with cross-entropy loss and PRELU activation functions. Each trained model was evaluated on a combined testing dataset of ^3He and ^{129}Xe images ($n=75$). Segmentation accuracy was evaluated via Dice similarity coefficients (DSCs), average boundary Hausdorff distances (Avg-HDs) and volume comparisons using Pearson correlation and Bland-Altman analysis. **Results:** The combined UNet model exhibited statistically significant improvements over the combined VNet model using DSC and Avg-HD metrics. Using Pearson correlation and Bland-Altman analysis, DL segmentation volume was highly correlated with the expert segmentations and exhibited minimal bias. **Conclusion:** We compared the performance of several 3D CNNs and training strategies for segmenting ventilated lungs on a large multi-nuclear hyperpolarized gas MRI dataset. The UNet CNN trained on both ^3He and ^{129}Xe data was the highest performing model.

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WE-lePD-TRACK 3-07

First Beam QA and Commissioning Report of a Novel Biology-Guided Radiotherapy (BgRT) System

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Purpose: Reflexion X1, the first biology-guided radiotherapy (BgRT) system equipped with PET imaging, was installed at our institution. This study reports the QA and commissioning results of the X1 system following the TG-148 protocol. **Methods:** The X1 linac produces 6MV flattening filter free beams with a dose rate of 850cGy/min. The rotational treatment fields are modulated by a binary multi-leaf collimator (MLC) with 64 leaves with 6.25mm leaf width at 85cm SAD and IEC y-jaws switching between 1cm and 2cm. The mechanical tests were performed to verify the centering, alignment, and divergence of the MLC and y-jaws. The beam parameters were commissioned using a 3D water phantom and a diode detector. Beam quality, transverse and longitudinal profiles of each slice width were measured and compared with the beam model. The localization accuracies of the laser and kVCT to MV beam isocenter were checked using a ball cube phantom. **Results:** The source sweep test showed the y-jaw misalignment was 0.049mm. The y-jaw divergence was 0.36mm and the twist was 0.03°. The longitudinal field center variation along the x-axis was less than 0.03mm. The normalized MLC tongue-and-groove film result showed the x-axis out-of-focus value was 0.66%. The MLC alignment test showed an offset of 0.57mm and a twist of 0.15°. Comparing the measurement and TPS modeling, the PDD10 differences were 0.2% and -0.3%, and the mean transverse profile differences were 0.55% and 0.73% for the 40x2cm² and 40x1cm² fields, respectively. The FWHM differences of the longitudinal profiles were within 0.5mm. The localization accuracies of the laser and kVCT to MV isocenters were within 1mm. **Conclusion:** This evaluation represents the first QA and commissioning evaluation of the clinical BgRT system as a future reference.

Multi-Disciplinary Interactive ePoster Discussion TRACK 4

Multi-Disciplinary: Imaging for Particle Therapy

WE-lePD-TRACK 4-01

IGRT-Based On-Line Dose Estimation in Proton Therapy

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Purpose: In proton therapy, dose deposition is sensitive to both geometry and tissue density along the beam-path. This work investigates the feasibility of rapid daily dose estimation to guide patient setup. **Methods:** A reference CBCT (refCBCT) is a synthetic CBCT produced once by deforming first day CBCT to the planning CT, corrected by histogram matching to the planning CT. Same correction is applied to daily CBCT (iCBCT) for the shape of HU-to-RPSP. Software was developed to compare iCBCTs to refCBCTs and adjust proton dose based on corrected HU differences along each beam-path. Daily total dose was estimated by summing individual beam doses. 10 patients receiving bilateral neck radiation for oropharyngeal cancer were used for validation. Validation was performed by comparing estimated fractional dose to our off-line standard Monte Carlo calculation using a synthetic CT acquired by deforming the closest in time CT to iCBCT. **Results:** After acquiring iCBCT and its registration to the planning CT, dose estimation for 5-field bilateral HN plans took < 1min. Estimated doses for 4 fractions x 10 patients when compared with the standard dose calculations produced Gamma-Index (3mm/3%) with average pass-rates 97.4%, $\sigma = 0.8\%$. 95th percentile of voxel-by-voxel dose difference was <5% of the daily prescription. There was no statistically significant difference on how the method performed on the second and last treatment fractions. **Conclusion:** Using density-based dose deformation along each proton beam, a rapid and accurate fractional dose estimate for oropharyngeal radiation was obtained which could be used for real-time guidance on patient setup.

WE-lePD-TRACK 4-02

A Scatter-Rejection Method for Proton Computed Tomography

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Purpose: Proton computed tomography (pCT) facilitates direct determination of the electron density distribution within a patient, enabling accurate proton therapy treatment planning. However, pCT has never been implemented clinically due to its low spatial resolution, largely an effect of multiple Coulomb scattering. Currently, this effect is accounted for by extrapolating each proton's most-likely-path and algebraically reconstructing the image. This method is an approximation and requires significant computing power. As an alternative, we have developed a simpler method in which detected protons scattered by an angle greater than a cutoff value θ_c are rejected, and standard cone-beam reconstruction algorithms are employed to generate images. **Methods:** Using GEANT4, a 250 MeV proton cone-beam was placed at a distance of 100 cm from the target object. Two silicon strip detectors were modeled to determine each proton's entry and exit positions with respect to the object for calculating scatter angle. Protons with scatter angle exceeding θ_c were rejected. A QA phantom was designed to determine reconstruction accuracy and spatial resolution, and an anthropomorphic head phantom (cropped from a full-body extended cardiac-torso, XCAT phantom) was used to evaluate dose. Scans were performed in 1° steps with 2×10^8 primary protons generated at each step. Images were reconstructed for $\theta_c = 0.1^\circ$, 0.5° , 1.0° , and unrestricted. **Results:** Spatial resolution and reconstruction accuracy generally improved with decreasing θ_c . At $\theta_c = 0.1^\circ$, resolution was 8 lp/cm for bone-water and 9 lp/cm for air-water. Reconstruction accuracy for bone and water was within 1% of the expected electron density. The absorbed dose in the head phantom was 43.85 mGy, which is below the median dose for head X-ray CT. **Conclusion:** Proton CT images exceeding clinical standards for spatial resolution, reconstruction accuracy, noise, and dose are achievable using a method in which detected protons scattered by an angle greater than some cutoff value are rejected.

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WE-lePD-TRACK 4-03

Improving the Performance of a Proton Computed Tomography Scanner with An Empirical Artifact Correction Method

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Purpose: To improve the accuracy of a prototype proton computed tomography (pCT) scanner in imaging of the stopping power relative to water (RSP) used for particle therapy treatment planning, by reducing image artifacts using an empirical artifact correction method. **Methods:** Inaccurate measurements of the water-equivalent pathlengths (WEPLs; RSP line integrals) degrade the performance of a prototype pCT scanner. An x-ray CT beam hardening correction was adapted to the requirements of pCT and used to calculate a correction function mapping distorted WEPLs to their corrected values. The function was optimized exploiting the linearity of the filtered backprojection operator and based on an experimental scan of a custom-built elliptical correction phantom with known RSP. The proposed method is simple and robust since it does not require knowledge of the correction phantom's geometry. It was applied in subsequent scans of a water phantom, a sensitometry phantom and a pediatric head phantom. **Results:** Application of the optimized correction function reduced an offset in the RSP maps and removed ring-shaped artifacts mainly stemming from protons stopping close to interfaces between segments of the energy detector. The scan of the water phantom was considerably more homogeneous. The mean absolute percentage RSP error reduced from 0.87% to 0.46%, which is an improvement by 47%. Artifacts visible in the head phantom were largely obviated. **Conclusion:** With the proposed method, the amplitude of artifacts of a prototype pCT scanner could be almost halved. This is to date the best RSP accuracy reported for a pCT scanner. There may also be an improvement compared to state-of-the-art dual-energy x-ray CT, for which errors of 0.7% were reported. Using pCT together with the empirical artifact correction method may help to reduce range calculation uncertainties in image-guided particle therapy.

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WE-lePD-TRACK 4-04

Assessment of Clinical to High Energy Proton Radiography Viability Using TOPAS

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Purpose: Proton radiography could provide instantaneous imaging for proton therapy, reducing position error and facilitating a more accurate dose calculation. Using a magnetic lens system, it has proven efficacious at high proton energies, but yields mixed results at lower energies. Here, a proton radiography system based on that at the Los Alamos Neutron Science Center (LANSCE) is modeled using the TOol for Particle Simulation (TOPAS). Clinical to high-energy protons irradiated head, rat, and mouse phantoms in simulation to determine the energy at which proton radiography becomes feasible. **Methods:** A proton flux of 108 particles traverses a magnetic lens system forming a Fourier plane (collimation point) and imaging plane downstream. Phantoms are simulated as stacked cylinders of tissue-equivalent material with varying diameters atop a 'plus sign' resolution object of bone-equivalent material; parameters changed per phantom to replicate approximate human, rat, and mouse anatomical dimensions. **Results:** Percent transmission was assessed through each bone and tissue combination; approximately 40% transmission was the threshold for visibility. While the bone was visible for all energies at the smallest tissue depth, it only became visible at the greatest depth starting at 530 MeV. Resolution across the bone edge was assessed at all energies and depths for the head phantom. Bone is visible with all energies at 4 cm thickness; at 8 cm, a sub-mm resolution image is obtained starting at 330 MeV, with resolution decreasing at higher energies due to edge effects; for 12 cm, at least 530 MeV is required for sub-mm spatial resolution; and 630 MeV required for 16 cm of tissue. **Conclusion:** Clinical 230 MeV protons are sufficient only for radiography of shallow tissues or preclinical applications. At least 630 MeV protons would be required to radiograph bone under 16 cm of tissue and image the full depth of an adult patient's head.

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WE-lePD-TRACK 4-05

Histogram Analysis Method of Water Equivalent Path Length to Evaluate Dose Variation Due to Anatomical Change in Spot-Scanning Proton Beam Therapy

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Purpose: Anatomical changes in a patient during radiation therapy can cause imprecise delivery of the irradiation dose, resulting in an increased dose to normal tissue and a reduced dose to tumor tissue. We have developed a method that uses the deviation in the water equivalent length (Δ WEL) along the spot beam path to quickly check for anatomical changes and then estimate the resulting dose differences (Δ Dose) from the planned dose distribution. It is designed for use in spot-scanning proton beam therapy (SSPT). **Methods:** To determine whether the WEL can be used to predict dose changes due to respiration, we investigated the correlation between the Δ WEL obtained with the proposed method and the Δ Dose obtained from freely available 4D-computed tomography (CT) data. Three Δ WEL analysis methods were evaluated: 1) Δ WEL map, 2) Δ WEL spot, and 3) monitor-unit (MU)-weighted Δ WEL spot. Treatment plans were prepared for a simulated target implanted into a chest defined by a 4D CT dataset provided by DIR-Lab. The Δ WELs between the planned CT paths and the comparison target CT paths were calculated spot by spot using DICOM-Radiation Therapy software. Linear regression fits between the Δ WEL histogram (median and 95% width) and Δ Dose (root mean square deviation (RMSD) and 3% dose pass rate) were calculated for the three analysis methods. **Results:** The combination of RMSD and Δ WEL showed a relatively high correlation expressed as the coefficient of determination ($0.3631 < R^2 < 0.8894$) while the combination of the 3% dose pass rate and Δ WEL showed a low correlation ($0.0002 < R^2 < 0.3428$). **Conclusion:** These results show that the Δ WEL spot histogram can provide information for predicting the dose deviation to the target and for judging whether to continue or terminate the treatment course.

Fujii and Hirayama are employees of Hitachi, Ltd.

WE-lePD-TRACK 4-06

A Novel Reconstruction Method for 3D Cone Beam CT Images in Proton Therapy Using RTK

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Purpose: CBCT reconstruction research requires access to different algorithms. Offline reconstruction for the Varian ProBeam CBCT system has not been reported. We aim to: 1) develop a novel offline reconstruction method for ProBeam CBCT and 2) simulate ultra-low-dose CBCT images. **Methods:** Straightforward reconstruction of ProBeam projection images with RTK results in significant artefacts. Our reconstruction pipeline includes processing on projection images as well as the reconstructed CBCT. 10 air calibration images are used to correct for bowtie filter, source, and detector shifts during gantry rotation (flood field flex). Beam hardening by the bowtie filter is corrected using an empirically-derived attenuation space map. RTK version 1.4 is used for filtered backprojection and slice averaging is then applied. The pipeline was validated by comparison to clinically-reconstructed CBCT images. To test the simulation of ultra-low-dose images, a Catphan 503 phantom was scanned at 9 mAs exposures (35-811 mAs/projection, 100 kV, SDD 370 cm). The signal-noise relationship was measured from phantom projection data to inform the amount of non-uniform Gaussian noise to be added to simulate low dose scans. Ultra-low-dose CBCTs at 30% and 10% exposure were simulated using datasets from paediatric patients treated on ProBeam. **Results:** The exposure and hardening corrections completely eliminate artefacts. The processed reconstructed images show excellent visual agreement with clinically reconstructed CBCT images. SNR measured in simulated phantom projection data shows good agreement with that in acquired images. Simulated low-exposure scans of paediatric patients are noisier but maintain good visibility of bone and some soft tissue, even at 10% exposure. **Conclusion:** We successfully implemented offline reconstruction for Varian Probeam CBCT images. Correcting for flood field image flex is essential in large-geometry imaging systems. Our methods allow optimisation of CBCT acquisition and reconstruction, especially relevant for paediatric treatments, and can be easily applied to other scan types e.g. 4DCBCT.

WE-lePD-TRACK 4-07

Performance of Dose Restoration in Head and Neck Cancer for Adaptive IMPT

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Purpose: Proton plans are very sensitive to patient anatomical changes due to the nature of proton interaction with tissue. Initial plans tend to distort throughout the treatment fractions due to anatomical changes occurring during treatment. Dose restoration (DR), a fully-automated adaptive strategy which aims to keep the dose distribution constant in every fraction, has proven powerful in lung and pelvic locations. The objective of this work was to test the performance of DR in head and neck (HN) patients. **Methods:** Ten clinically planned HN patients underwent between 4 and 6 repeated-CTs (rCTs) during treatment. Robust DR using the same robust optimization parameters as in planning phase (setup-error=3mm, range-error=3%) was performed for all rCTs (n=50). Isodose contours generated from the initial dose and patient-specific minimum and maximum objectives were used in the DR workflow to re-optimize the plan and reproduce the initial dose in every rCT. Robust evaluations (1mm, 3%) of non-adapted and of restored plans were compared for each rCT. Differences to clinical plan on the planning CT scan in DVH-metrics (nominal and worst-case values) were analysed for non-adapted and for restored plans. **Results:** An improvement in DVH-metrics and robustness was seen for the restored plans compared to non-adapted plans. For non-adapted plans, 28% (14/50) of rCTs did not fulfill the CTV coverage criteria (i.e. D98% > 95% Dprescription (Dp)) and therefore needed offline adaptation. By implementing DR, the median of the D98% over the 50 rCTs was increased by 0.61 Gy and 0.17 Gy in the primary tumour CTV (Dp=70Gy) and the nodal target CTV (Dp=54.25Gy), respectively. Because of large anatomical changes and/or inaccurate patient repositioning, 14% (7/50) of rCTs still needed offline adaptation to ensure an optimal treatment. **Conclusion:** DR has the potential to reduce offline adaptation rate by 50%. However, a minority of rCTs (14%) still need to be visually inspected and fully-automation might not be possible for these HN patients.

Therapy Interactive ePoster Discussion

TRACK 5

Therapy: Particle Therapy: Biological Effect and Modeling

WE-lePD-TRACK 5-01

Modelling Toxicity Risk and Uncertainty in Breast Radiotherapy with Bayesian Networks

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Purpose: To predict post-treatment normal tissue toxicity in patients undergoing accelerated partial breast irradiation (APBI) therapy and to quantitatively identify which diagnostic, anatomical, and dosimetric features are contributing to these outcomes. **Methods:** A retrospective study of APBI treatments was performed using 32 diagnostic, anatomical, and dosimetric features pertaining to various phases of the patient's treatment journey. These features were used to inform and construct a Bayesian network (BN) based on both statistical analysis of feature distributions and relative clinical importance. The target feature for prediction was defined as a measurable worsening of telangiectasia, or subcutaneous tissue induration, or fibrosis when compared against the observed baseline. Parameter learning for the network was performed using data from the 299 patients included in the ACCEL trial. Predictive performance of the BN was measured and compared against a variety of conventional machine learning (ML) approaches. **Results:** Cross validated performance of the BN for predicting toxicity was consistently higher across all binary classification metrics when compared against ML techniques including decision trees, naive Bayes, ensemble trees, and support vector machines. The measured BN receiver operating characteristic area under the curve (ROC-AUC) was 0.938±0.028 against the best ML result of 0.910±0.032 using 5-fold cross validation across 100 trials. The volume of the clinical target volume and baseline fibrosis measurements were found to have the highest mutual dependence with normal tissue toxicity in the network, representing the strongest contribution to patient outcomes. **Conclusion:** The BN outperformed conventional ML techniques in predicting tissue toxicity outcomes and provided deeper insight into which features are contributing to these outcomes. With additional data and further development of the network topology, stronger predictive performance is possible. BNs are a powerful tool for many predictive and diagnostic clinical tasks, particularly for applications with limited data availability or completeness.

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WE-lePD-TRACK 5-02

Multi-Domain Statistical Modeling of Treatment Tolerance in Patients with Gastric and Esophageal Adenocarcinoma

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Purpose: To investigate associations between demographic, clinical, and CT-based estimates of muscle and fat mass with treatment tolerance in patients with gastric or esophageal cancer. **Methods:** We retrospectively identified 142 patients with gastroesophageal adenocarcinoma treated with neoadjuvant chemotherapy +/- radiation. Clinical endpoints were defined as: treatment breaks in neoadjuvant therapy; reduction in neoadjuvant therapy dose; admission to the Emergency Department (ED) / inpatient encounters; and neoadjuvant treatment completion. Demographic and clinical features included age, gender, race, ethnicity, BMI, smoking status, pack years, TNM Stage, and treatment site. A deep learning model using the UNET architecture was developed to segment visceral fat (VF), subcutaneous (SF), and muscle on CT images from T12-L5. These values were used to derive: average transverse area of VF, SF, and muscle; ratio of VF/SF area; ratio of muscle area to total fat (VF+SF) area; skeletal muscle index (muscle area/[patient height]²). Univariate statistical associations between each clinical endpoint and the demographic, clinical, and imaging features were analyzed using T-tests or Chi-squared tests, as appropriate, with Bonferroni correction. Multivariate machine learning models were then developed for each clinical endpoint, where Regularized Logistic Regression, Support Vector Machine (SVM), and Random Forest (RF)

formalisms were compared. Model training and hyperparameter tuning was achieved with 5-fold cross-validation, and model generalization was evaluated based on a stratified Monte Carlo sub-sampling technique. **Results:** On univariate analysis, age was associated with breaks in neoadjuvant therapy ($p=0.047$), and VF/SF was associated with ED admissions ($p=0.023$). On multivariate analysis, SVM modeling achieved the best performance regarding breaks in neoadjuvant therapy ($AUC=0.64$) and neoadjuvant treatment completion ($AUC=0.64$). In contrast, RF modeling achieved the best performance for ED admissions ($AUC=0.64$) and dose reductions ($AUC=0.54$). **Conclusion:** Our preliminary analysis suggests that demographic, clinical, and imaging features may be associated with adverse outcome in patients with gastric and/or esophageal cancer.

WE-lePD-TRACK 5-03

Influence of Sub-Lethal Damage Repair On Biological Effectiveness of Proton with the Consideration of Dose Delivery Time Structure in Scanning Proton Therapy

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Purpose: To evaluate how the sub-lethal damage repair influences on the biological effectiveness of proton in prolonged scanning irradiation. Previous studies revealed the reduction of biological effectiveness caused by the repair effect in passively-scattered irradiation where the dose rate is independent of position over target volume. However, the effect in scanning irradiation was not clarified because of the different dose delivery time structure. **Methods:** Spread-out Bragg-peak (SOBP) irradiation was simulated in water phantom with one- and two-field plans. Three scenarios (instantaneous, passively-scattered and layered-scan) were considered with corresponding models of dose delivery time structures for each irradiation technique. The repair effect was evaluated with the biological dose (dose $\times$ relative biological effectiveness (RBE)) obtained from linear quadratic model combined with theory of dual radiation action. McNamara model was used to include the LET dependence of RBE. In addition, the simulation using clinical plans with one and two fields (post-operative seminoma and prostate tumor, respectively) were carried out. **Results:** Dose delivery time structure which differs for each scenario was found to be an important factor for evaluating the repair effect. In the passively-scattered scenario, relative dose decrease against the instantaneous scenario was about 3.3% over SOBP with 19 min of total irradiation time. On the other hand, layered-scan scenario resulted in the variable dose decrease with 3.5% and 0.5% at the proximal and distal side of SOBP, respectively. The variability of dose decrease was mainly caused by the spatial dependence of dose delivery time structure in scanning irradiation. The similar results were obtained from the clinical plans indicating the potential distortion of biological dose distribution in layered-scan scenario. **Conclusion:** Consideration of the dose delivery time structure in scanning proton irradiation is important for the evaluation of the repair effect.

WE-lePD-TRACK 5-04

A Mechanistic Model for the Calculation of Oxygen Dependent DNA Damage After Radiation Exposure

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Purpose: Cellular responses at different oxygen concentrations (OCs) continues to be of significant research interest in cancer radiobiology. Therefore, this study aimed to develop a mechanistic model for the analysis of cellular responses at various OCs. **Methods:** A DNA damage model (the different cell oxygen level DNA damage [DICOLDD] model) that examines the oxygen effect was developed based on the oxygen fixation hypothesis, which states that dissolved oxygen can alter the reaction kinetics of DNA-derived radicals generated by ionizing radiation. Radiation induced DNA-derived radicals were simulated using the Monte Carlo method, and the decay of DNA-derived radicals was described using differential equations. The DICOLDD model was fitted to previously published experimental data obtained under different irradiation configurations. Furthermore, the DICOLDD model was validated by calculating yields of DNA double strand breaks (DSBs) after exposure to 137Cs as well as cell survival fractions (SFs) after exposure to different particles. **Results:** Generally, DSB yields calculated after exposure to 137Cs at different OCs correspond to statistical uncertainties of previous experimental results. Calculated cell SFs exposed to photons, protons, and alpha particles at different OCs concur with those obtained in previous studies. Moreover, the DICOLDD model was applied for the calculation of DNA damage yields after irradiation with 0.5–50 MeV protons. Our results demonstrated that the variation in DSB yields was less than 10% when the cellular OC decreased from 21% to 5%. Additionally, DSB yields changed drastically when OC dropped below 1%. **Conclusion:** We developed a DNA damage model to evaluate the oxygen effect and evidenced that it provides a mechanistic description of the reaction kinetics of DNA-derived radicals induced by ionizing radiation. Therefore, the DICOLDD model could be a powerful tool for the analysis of cellular responses at different OCs after exposure to different types of radiation.

WE-lePD-TRACK 5-05

Contribution of Nuclear Fragmentation to Dose and Relative Biological Effectiveness (RBE) Across Common RBE Models in Carbon Radiotherapy

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Purpose: Variable RBE in carbon radiotherapy may be calculated using several models, including the microdosimetric kinetic model (MKM), stochastic MKM (SMKM), and Local Effect Model I (LEM), which have not been thoroughly compared. This work compares how models handle the nuclear fragmentation of carbon beams to provide insight to where model differences arise. **Methods:** Geant4 Monte Carlo was used to simulate various monoenergetic and SOBP carbon beams incident on a water phantom. From these, input parameters for each RBE model (microdosimetric spectra, double strand break yield, kinetic energy spectra, dose fragment contributions) were calculated for each contributing fragment of the carbon beam (namely H, He, Li, Be, B, and secondary C). The spectra for each fragment was used to calculate the linear (α) and quadratic (β) portions of each RBE model, which were combined with α and β values of reference radiation and physical dose to calculate RBE. The contribution of RBE by each fragment was then compared across models. **Results:** Calculations found that secondary fragment contributions could exceed 20% of total physical dose. When calculated using identical beam parameters, RBE varied greatly in magnitude across models and was typically lowest using MKM. When compared across fragments, RBE typically decreased with atomic number when $Z < 3$ and increased when $Z \geq 3$ for RBE(MKM) and RBE(SMKM). Using LEM, RBE increased with Z until $Z=6$, where RBE dropped sharply. Trends of RBE by fragment varied by region (entrance vs. SOBP) for microdosimetric models only. **Conclusion:** This study demonstrated that secondary fragments could contribute notably to physical dose, indicating that fragmentation is an important factor in treatment delivery. Similar trends were seen in fluctuations of RBE by atomic number for microdosimetric models, which differed from those of RBE(LEM). Further insight into model differences may drive future cell studies and increase consistency in model use.

WE-lePD-TRACK 5-06

Robust Alternating RBE- and LET-Weighted Beam Orientation and Fluence Map Optimization for Intensity Modulated Proton Therapy

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Purpose: Empirical relative biological effectiveness (RBE) models and dose averaged LET surrogates have been separately used to estimate the biological dose in proton therapy. Neither method adequately captures the factors influencing RBE values for treatment planning. Here we incorporate McNamara RBE (RBE(M)) and LET in a beam orientation (BOO) and fluence map optimization (FMO) framework with alternating objective functions. **Methods:** Based on our previously developed robust sensitivity-regularized and heterogeneity weighted BOO (SHBOO) framework, we formulated the optimization problem using two different methods. The

first method optimized RBE(M)-weighted dose, updating RBE(M) values with each round of optimization until they converged (RBEwFMO). The second alternated between RBE(M)-weighted and LET-weighted doses, until the RBE values converged (RBELETwFMO). We compare their performance with a physical dose optimizer assuming RBE(1.1) (PHYSFMO). Three head and neck patients were planned with the three techniques and compared on dosimetry and robustness. **Results:** Compared to PHYSFMO, CTV homogeneity and Dmax were improved by an average of 0.02 and 1.4 GyRBE for RBELETwFMO, and 0.03 and 2.3 GyRBE for RBEwFMO. D98% was improved by RBEwFMO by 0.6 GyRBE. Slight decrease in mean and maximum RBE dose was seen in the larynx, esophagus, and constrictors with RBEwFMO and RBELETwFMO. Regarding robustness, worst [Dmax, V95%, D95%, D98%] for CTVs improved by [4.2%, 31.9%, 5.5%, 4.7%] with RBEwFMO and [2.4%, 33.2%, 6.7%, 5.4%] with RBELETwFMO under range and setup uncertainties. For OARs, the worst [Dmax, Dmean] improved by an average of [3.1, 0.8] GyRBE for RBEwFMO and [5.3, 2.0] GyRBE for RBELETwFMO under setup uncertainties. **Conclusion:** The three optimization techniques deliver comparable physical doses for the head and neck cases. Beside modest OAR RBE dose sparing improvement, CTV coverage and overall robustness were substantially improved with RBEwFMO and RBELETwFMO, showing potential benefit for directly incorporating McNamara RBE in proton treatment planning.

WE-lePD-TRACK 5-07

Improving the Uniqueness of Numerically Extracted Proton Pencil-Beam Energy Spectrum from Integral Depth Dose

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Purpose: Proton pencil-beam energy spectrum is an essential parameter for calculations of dose and linear energy transfer(LET). We extract energy spectrum from measured integral depth dose(IDD) by decomposing the measured IDD into IDD of mono-energetic protons. The decomposition is unique if the mono-energetic IDD are linearly independent, but this condition may not be satisfied especially at low energy when the proton range < the step size that is used in generating the mono-energetic IDD. In this work, we examined the condition of linear independence and improved the uniqueness. **Methods:** Spectra with single and double peaked Gaussian distribution were used to simulate multi-energetic proton beams. The IDD of the beam was synthesized from mono-energetic IDD with its energy spectrum. The energy interval of mono-energetic IDD is 0.05 MeV so that the calculation error of LET will not be larger than 0.5 keV/μm. Optimal decomposition algorithms were performed on the synthesized IDD. The uniqueness of the extracted energy spectrum was examined by the difference of the extracted energy spectrum from the original. To improve the uniqueness and meanwhile save the memory and computation, we used multiple resolution mono-energetic IDD. The step sizes of the IDD of mono-energetic pencil beams are 1 and 0.1 mm. **Results:** The difference of the extracted and original Gaussian energy spectrum peaked at 75 and 80 MeV was <1%. The difference increased as the energy decreased. With 1 mm step size of IDD, we obtained well extracted Gaussian spectrum peaked at and larger than 30 MeV, and with 0.1 mm step size of IDD, peaked at and larger than 20 MeV. **Conclusion:** High resolution mono-energetic IDD are necessary to extract low energy spectrum from measured IDD. Use of energy interval larger than 0.05 MeV in the mono-energetic IDD will be investigated to optimize the calculated dose and LET.

Therapy Interactive ePoster Discussion

TRACK 6

Therapy: Quality assurance

WE-lePD-TRACK 6-01

Design and Testing of a Novel Phantom for Camera Calibration of Skin Tones with SGRT

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Purpose: Surface guided imaging relies on exposure to a patient's surface and anatomical topographic information to accurately track patient positioning for setup and to manage motion during treatment. Camera sensitivity parameters permit adjustments according to different reflections on the skin surface. In this study we attempt to quantify Integration time (μs) and Signal Amplification, or Gain (%) values with the CRAD Catalyst HD for four different skin tones. **Methods:** Four ceramic skulls (Fun Express) were painted using four different shades of a matte mousse makeup foundation (Rimmel). We chose the darkest and lightest shades and two shades in between to simulate a variety of skin tones. A head cushion was molded to provide reproducible setup and positioning. Phantom was setup with and without the head rest and automatic camera sensitivity parameters were evaluated at multiple gantry angles (180°, 225°, 135°, and 360°). **Results:** The average integration time generated by the automatic settings for no headrest in order from Lightest to Darkest tones were 2685μs, 2700μs, 4008μs, and 6608μs and the addition of a headrest decreased these values to 2562μs, 2686μs, 2007μs and 2109μs. The average Gain generated by the auto settings for no headrest in order from Lightest to Darkest tones were 167%, 363%, 314%, 321% and the addition of a headrest changed these values to 178%, 326%, 89%, 50%. **Conclusion:** It is important to quantify camera sensitivity parameters for a variety of skin tones so that clinicians can rely on surface imaging for accurate positioning of patients at treatment time. Use of makeup foundation is a novel method for evaluating maximum imaging signal for a variety of skin tones. We demonstrated that darker skin tones required much greater integration time values than lighter skin tones to maximize surface reflection.

WE-lePD-TRACK 6-02

EPID-Based Detection of Anatomical Changes During Radiotherapy Treatment for Head-And-Neck and Prostate Cancer Patients

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Purpose: To quantify and analyze the effectiveness of an EPID-based cine transmission dosimetry to detect gross anatomical errors of cancer patients during radiotherapy treatment. **Methods:** Cine EPID patient transmission images were simulated for 360° 6MV VMAT fields from a TrueBeam linac with an aS1024 EPID imager system for 10 head-and-neck and 19 prostate patients. Simulations were performed to determine the ability to distinguish in-tolerance (1-3 mm/1-3 degree) from out-of-tolerance (10-20 mm/15-20 degree) miss-alignments, and in-tolerance inter-fraction patient pose variations from large out-of-tolerance anatomical changes or a different patient. Predicted EPID image frames mimicked the ~10 Hz EPID image acquisition, with an EPID frame for each 100 msec delivery interval. An image distance metric was used to classify the in-tolerance and out-of-tolerance EPID frame images. The beam error classification was made by checking the failed-frame rate with respect to a threshold. The error detection system was tested with patient datasets which have 8-13 CTs acquired on different days throughout the treatment course. **Results:** Results showed good performance for all classification types of errors. The areas under ROC curves (AUCs) are 0.88 for the miss-alignment scenario and 0.91 for the incorrect patient scenario. The system readily detects substantial anatomical changes between treatment days due to rectal gas. **Conclusion:** This simulation study indicates that this gross error detection system is a promising tool for detecting patient pose errors with EPID dosimetry in real-time mode. Such a system could detect errors that were previously undetectable.

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WE-lePD-TRACK 6-03

Modeling the Geometric Relationship Between Targets and OARs for Patient-Specific Plan Quality Evaluation in Head and Neck Radiotherapy

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Purpose: Head and neck (HN) radiotherapy presents highly variable treatment geometries, which make optimal patient-specific organ at risk (OAR) dose objectives difficult to define. We sought to develop models of expected HN OAR doses based on geometric relationships between OARs and planning target volumes (PTV) to facilitate quantitative patient-specific plan quality evaluation. **Methods:** Fifty-five treatment plans (VMAT or IMRT) from 41 patients were retrospectively analyzed. Primary and boost plans were assessed independently. For parotids, submandibular glands, and oral cavity, fractional volumes within 1, 5, 10, 20, 30, 40, and 50mm of the nearest PTV surface (including all dose levels) were computed for each plan. Linear, quadratic, and cubic polynomial fits were then generated to relate each distance metric with mean OAR dose (normalized to the lowest prescription dose in each plan). Optimal distance metrics were then selected for each OAR to minimize the mean absolute error (MAE) of the fit. The optimal polynomial order was set to 1 (linear) unless higher order models significantly improved MAE ($P < 0.05$). **Results:** The optimal parotid model ($N=103$ parotid contours including left and right) was a linear fit to the 10mm distance metric, which modelled mean parotid dose with MAE=4.2% (95% confidence interval, 95CI=0.2-10.6%). For oral cavity ($N=55$), the optimal model was a linear fit to the 20mm distance metric (MAE=6.9%, 95CI=1.1-15.1%) and for the submandibular glands ($N=106$), it was a quadratic fit to the 5mm distance metric (MAE=8.6%, 95CI=0.4-23.0%). MAE values did not change substantially ($<0.4\%$) upon five-fold cross-validation. Target coverage was consistent across plans (D95% range: 98-103%, V95% range: 97-100%). **Conclusion:** Simple models based on OAR-PTV distance metrics can produce accurate models of mean OAR doses in HN treatment plans. These models can be easily implemented in treatment planning systems to objectively assess plan quality against cases with similar treatment geometries.

WE-lePD-TRACK 6-04

An Uncollided Energy Fluence Convolution Method for EPID Image Prediction

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Purpose: To determine the dosimetric accuracy of EPID images computed based upon Varian Acuros predicted exit fluence. **Methods:** AcurosPD is an application prototype that calculates the uncollided energy fluence transmitted through a patient image-set, and incident upon on the portal imager panel. In this work, we developed a method that convolves this output to estimate the recorded dose image on the EPID panel. The method uses Monte Carlo generated energy deposition kernels to account for the primary and scattered components of the EPID response separately via convolution with the corresponding AcurosPD calculated transmitted energy fluence. Different mono-energetic dose kernels were calculated to account for the EPID's energy-dependent responses to primary photons and for photons that penetrate phantoms of different thicknesses (1, 5, and 10 cm). The fluence output of AcurosPD is been structured into 19 energy groups ranging from 0.01 to 6 MeV. The convolved results were validated by comparing with the open field measurement from 6 MV flattened beams at 150 cm source to detector distance. **Results:** The results show good agreement for most of the field sizes and phantom thicknesses with the relative differences within 5%. For the 2x2 field size, the energy spectrum reweighting is needed to achieve acceptable agreement between prediction and measurement. **Conclusion:** The EPID image prediction can be performed by the Acuros energy fluence calculation together with the proposed convolution method. The inclusion of thickness-dependent scatter kernels improves the prediction accuracy.

This work is supported by Varian Medical Systems.

WE-lePD-TRACK 6-05

An Automatic, Uncertainty Model-Guided, Target-Generating Algorithm Improves Target Volumes in Lung SBRT

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Purpose: This work presents the improvement in lung SBRT internal target volumes generated with an automatic algorithm that used voxelized position uncertainty models derived from individual 4DCT simulation scans. **Methods:** For 19 previously treated lung SBRT targets, deformable image registration was applied to all respiratory phases of 4DCT simulation scans. From the resulting deformation vector fields, voxelized models of gross tumor volume motion were created and fed into an in-house, target-generating algorithm to produce a new internal target volume, the aITV. The efficacy of the aITV was evaluated by comparing its volume and the coverage it provided to each gross tumor volume voxel with those of the clinically-utilized ITV. In addition, the magnitude of an isotropic margin that could be added to the aITV and yield a volume equivalent to the clinical ITV was determined. **Results:** The target-generating algorithm was used to create aITVs that covered at least 95% of the tumor voxels at least 95% of the time according to the voxelized motion model. The aITV increased the coverage of the 5th-percentile-most-covered voxel 50.2% (range -4.9% to 95%) ($p < 0.001$) with a volume that was 29.6% less than the clinical ITV (range 11.4% to 57.3%) ($p < 0.001$). Furthermore, an extra 1.8 mm isotropic margin (range 0.5 to 3.2 mm) was able to be added to the aITV and still not yield a volume greater than the clinical ITV. **Conclusion:** An automatic, uncertainty model-guided, target-generating algorithm was used to create lung SBRT internal target volumes that, when compared to the clinically-utilized ITV, provided both a significant increase in tumor voxel coverage and a significant decrease in volume. The greater efficacy of this target has the potential to increase the therapeutic ratio for lung SBRT treatments, and the algorithm can be extended to other radiotherapy treatment sites exhibiting positional uncertainties.

WE-lePD-TRACK 6-06

Miniaturized Scintillator Dosimeter for Small Field Dosimetry

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Purpose: A miniaturized inorganic scintillator detector (MSD) is demonstrated in the analysis of the small static photon fields used in external radiotherapy **Methods:** The detector is constituted by a 0.25 millimeter diameter and 0.48 millimeter long inorganic scintillating cell efficiently coupled to a narrow 125 micron diameter silica optical fiber using a photonic interface. Response of our MSD is tested under a 6MV photon beam of a linac for various field sizes (from 1x1cm² to 4x4cm²). The MSD measurements are compared to those obtained with two high-resolution reference detectors commonly used in small field dosimetry: a micro-diamond detector and a silicon diode. The spurious Cerenkov signal is rejected with a spectral filtering **Results:** The MSD shows a linear response regarding the dose, repeatability within 0.1% and radial directional dependence of 0.36% (standard deviations). Beam profile 5cm depth with the MSD and the micro-diamond detector shows a mismatch in the measurement of the full widths at 80% and 50% of the maximum which does not exceed 0.25mm. The deviation of the percentage depth dose between the MSD and micro-diamond detector remains below 2.3% within the first fifteen centimeters of the decay region for field sizes of 1x1cm², 2x2cm² and 3x3cm² (0.76% between the silicon diode and the micro-diamond in the same field range). The 2D dose mapping of a 0.6x0.6cm² photon field evidences the strong 3D character of the radiation-matter interaction in small photon field regime. Our detector is a right compromise between high resolution, compactness, flexibility and ease of use. The MSD overcomes problem of volume averaging, stem effects, and despite its water non-equivalence it is expected to minimize electron fluence perturbation due to its extreme compactness. **Conclusion:** MSD has the potential to become a valuable dose verification tool in small-field radiation therapy, in Brachytherapy, FLASH-radiotherapy and microbeam radiation therapy.

WE-lePD-TRACK 6-07

Small-Field Dosimetry in Single-Isocenter Stereotactic Irradiation for Brain Metastases Using a Detector-Specific Field Output Correction Factor

T Ono^{1*}, M Nakamura², M Uto¹, T Mizowaki¹, (1) Department of Radiation Oncology and Image-applied Therapy, Graduate School of Medicine, Kyoto University, Kyoto, Japan, (2) Division of Medical Physics, Department of Information Technology and Medical Engineering, Human Health Sciences, Kyoto University Graduate School of Medicine, Kyoto, Japan

Purpose: In small-field dosimetry such as stereotactic irradiation for brain metastases, the measurement dose may include large uncertainties. Recently, IAEA-TRS483 has reported detector-dependent correction factors for small irradiation fields that are expected to reduce the uncertainty of the measured dose in small irradiation fields. This study aimed to evaluate the measured doses considering the correction factors in single-isocenter stereotactic irradiation for brain metastases. **Methods:** A total of 14 sites: 9 cases of single brain metastasis and 2 cases of multiple brain metastases (including 2 sites and 3 sites) were included in this study. Single-isocenter stereotactic irradiation plans were created with a dynamic conformal arc using XVMC (calculation grid size: 1 mm) in Elements Multiple Brain Mets SRS (Brainlab). The dose was measured with a water-equivalent phantom using a cc04 chamber (IBA). The multi-leaf collimator (MLC) field size of each target was calculated for each control point (CP) using Python. Each correction factor was calculated from the MLC field size referred to in the IAEA-TRS483 report. Finally, the plan correction factor was calculated from the average correction factor weighted by the monitor unit for each CP. The measured doses with and without the correction factor were compared with the values calculated by the XVMC, and the differences were evaluated as the mean value $\pm$ standard deviation. **Results:** The difference between the measured and calculated doses was $-3.2\% \pm 1.5\%$ (range: -5.3 – 0.6%) without correction and $-1.8\% \pm 1.2\%$ (range: -3.6 – 1.0%) with correction, indicating that the difference between the measured and calculated doses decreased with correction. **Conclusion:** The measured doses considering the correction factors were evaluated in single-isocenter stereotactic irradiation for brain metastases. A more reliable evaluation of the dose in patients is expected by considering the output correction that takes into account the detector characteristics in small-field dosimetry.

Vendor Showcase

TRACK 8

Elekta

WE-SHOWCASE-TRACK 8-13-1

Elekta Unity - Designed for Today and the Future

K Brown^{1*}, (1) Elekta Oncology Systems LTD, Crawley, WSX, GB

Professional Symposium

TRACK 1

Personal Finance for Medical Physicists

WE-C-TRACK 1-00

Personal Finance for Medical Physicists

Moderators: Mahadevappa Mahesh, Johns Hopkins University, Baltimore, MD and Julianne Pollard-Larkin, University of Texas MD Anderson Cancer Center, Houston, TX

WE-C-TRACK 1-01

M. Mahesh, Johns Hopkins University:

Personal Financing 101

WE-C-TRACK 1-02

J. Harper, Capstone Financial:

Holistic Financial Planning

WE-C-TRACK 1-03

Q&A

In this session, there will be two presentations which focusses on personal financing for medical physicists. Each presentation will focus on different aspects of personal financing, including savings and investments and financial planning for securing financial freedom.

1. To learn the fundamentals of personal financing
2. To appreciate risks and benefits of different savings and investment vehicles
3. To become familiar with the various tools available to assist in modeling savings/investments

SAM Imaging Educational Course

TRACK 2

ACR Activities and Updates

WE-CD-TRACK 2-00

ACR Activities and Updates

Moderator: Dustin Gress, American College of Radiology, Reston, VA

WE-CD-TRACK 2-01

D. Gress, American College of Radiology:

Update From HQ

WE-CD-TRACK 2-02

W. Geiser, UT MD Anderson Cancer Center:

SBB QC Manual in Development

WE-CD-TRACK 2-03

B. Harkness, Henry Ford Hospital System:

NM-PET QC Manual in Development

WE-CD-TRACK 2-04

D. Reeve, :

MRAP Medium Phantom

WE-CD-TRACK 2-05

G. Clarke, UT Health San Antonio:

MRAP Medium Phantom

WE-CD-TRACK 2-06

E. Samei, Duke University:

In Pursuit of Meaningful and Efficient Automatic Quantification of Clinical CT Exams

WE-CD-TRACK 2-07

Audience Q & A

The American College of Radiology (ACR) quality control manuals have represented standardized programs for quality and safety for many years. The most recent stereotactic breast biopsy QC manual is from 1999; this session will include an introductory vision for an updated stereotactic and/or x-ray image-guided breast biopsy QC manual that is now in development. This session will also include an introductory vision for QC manuals for nuclear medicine and PET, which are also now in progress. Summer 2021 will also see introduction of a new MR Accreditation Program (MRAP) phantom in a medium size. This medium phantom will accommodate clinical head coils that are too small for the large phantom, necessitating extra handling of MR coils in order to conduct required MRAP QC. This session will include key updates to the MRAP related to the new medium phantom. This session will conclude with a discussion of a new ACR project focused on the pursuit of meaningful and efficient automatic quantification of clinical CT exams.

Learning Objectives:

1. Understand the vision for the ACR stereotactic breast biopsy QC manual revision in progress;
2. Understand the vision for the ACR NM and PET QC manuals in development;
3. Learn requirements for using the new medium phantom in the ACR MR Accreditation Program;
4. Learn key elements of ACR's project to identify and develop processes for automatically quantifying and reporting indices of clinical image quality in CT.

Therapy Scientific Session

TRACK 3

Quality Assurance

WE-C-TRACK 3-01

Clinical Implementation of Fully-Automated Comprehensive Quality Assurance Tool Using In-Vivo EPID Image Analysis for Varian Halcyon

A Chalise*, C Bojcheko, University of California San Diego, La Jolla, CA

Purpose: An automated, comprehensive in-vivo system to detect changes in patient anatomy and machine output was developed using a novel analysis of in-vivo EPID images for every fraction of treatment on a Varian Halcyon. The in-vivo approach was used to identify errors that go undetected by routine quality assurance (QA) to compliment daily machine performance check (MPC), with minimal additional physicist workload. **Methods:** Images for all fractions treated on Halcyon were automatically downloaded and analyzed at the end of the treatment day. For image analysis, we select a high-dose region that has been shown to be a predictive metric for changes in the PTV mean dose (PTV_μ). Flags are raised for: (Type-A) patients whose PTV_μ exceeds 10%, to protect against large errors, and (Type-B) patients with three consecutive fractions higher than 3%, to protect against systematic trends. A detailed report for flagged patients is e-mailed to a physicist for further investigation. Furthermore, daily averages and standard deviations are uploaded to Total QA, our QA portal, in addition to MPC that are signed-off by physicist daily, ensuring comprehensive QA for Halcyon. To guide clinical implementation, beginning on 12-15-2020, a retrospective study was conducted for treatments from Nov-2018 till Dec-2020, grouping errors by treatment site. **Results:** From retrospective data of 1033 patients, no Type-A errors were found and only 31 patients (3 %) had a Type-B error. Most of these deviations for Type-B were due to Head/Neck weight loss. For 3 months of use, no flags have been raised by the system that required immediate intervention, indicating the system is running nominally. **Conclusion:** This system protects against errors that can occur in-vivo to provide a more comprehensive QA. This fully automated framework can easily be implemented in other centers with a Halcyon and only requires a standard desktop computer and analysis scripts.

Supported by grant funding from Varian Medical Systems.

WE-C-TRACK 3-02

Establishment of a Standard High-Energy X-Ray Beam for Replacement of Cs-137

E King*, L DeWerd, University of Wisconsin-Madison, Madison, WI

Purpose: To establish a standard high-energy x-ray beam to act as a surrogate for Cs-137. **Methods:** To achieve an x-ray beam with a high average energy, a combination filter of tin, copper and aluminum was chosen. A Varex NDI-451Be x-ray irradiator, with a maximum energy of 500 kV, was used for developing this beam. A beam model of the irradiator was created with the EGSnrc user code BEAMnrc, which was benchmarked with percent depth dose (PDD) and half-value layer (HVL) measurements. The beam model allowed the spectrum of the standard x-ray beam to be simulated to iterate the filtration. The filter thicknesses were optimized to increase the average energy of the spectrum while allowing the beam intensity to stay above an acceptable level. An air kerma rate that is higher than our highest energy, highest filtered NIST-matched x-ray beam was chosen as the beam intensity metric. **Results:** The standard x-ray beam that was created has a thickness of 5.36 mm of tin, 1.28 mm of copper, and 2.01 mm of aluminum. It creates a spectrum with an average energy of 265 keV and a HVL of 6.64 mm of copper. The air kerma rate at 1 m is 2.735 mGy/s which is above the air kerma rate of our highest filtered NIST-matched beam indicating an acceptable beam intensity. **Conclusion:** This work establishes a standard high-energy, more highly filtered x-ray beam than what is currently available. As clinics and research institutions are phasing out Cs-137 irradiators in favor of x-ray irradiators, a standard high-energy x-ray beam for calibration use becomes increasingly more critical.

WE-C-TRACK 3-03

Evaluation of MRI-Only Treatment Planning (MRTP) Quality for Gynecologic HDR Brachytherapy Using PETRA Sequence

CY Lee*, E Kaza, TC Harris, DA O'Farrell, MT King, RA Cormack, I Buzurovic, Brigham and Women's Hospital, Dana-Farber Cancer Institute, Harvard Medical School, Boston, MA

Purpose: Interstitial HDR brachytherapy uses MRI for the clinical target volumes (CTV) and the organs at risk (OARs) delineation for treatment planning. CT is utilized for needle localization, as opposed to MRI, which often does not provide sufficient contrast to visualize needles. An optimized 'Pointwise Encoding Time Reduction with Radial Acquisition' (PETRA) MRI sequence is a promising candidate for implementing an MRI-only treatment planning (MRTP). The feasibility of administering MRTP in gynecological patients was evaluated against the clinical treatment plan (CTP). **Methods:** Five gynecological cancer patients with interstitial needle implants were treated with the CTP generated using the standard CT/MRI-based method employing 3D SPACE images. To generate a plan using MRTP approach, an optimized 3D axial PETRA MRI (TR/TE 3.3/0.07) was acquired on a 3T MAGNETOM Vida (Siemens Healthcare, Erlangen, Germany). The MRI-only needle digitization was performed on Oncentra Brachy (Elekta Brachytherapy, Netherlands) Treatment Planning System using both PETRA and SPACE images for better visualization of the needles through tissues and the interstitial template, respectively. The MRI-based plan was generated by activating and assigning the same dwell times as in the CTP. The quality of the MRI-based plan was evaluated against the CTP by comparing the DVH and EQD2 metrics for the CTV and the

OARs (rectum, bladder and sigmoid). D90 and D2cc were used for the CTV and OARs, respectively. **Results:** In these five patients, the maximum relative differences in the DVH metrics were 5%, 11%, 12%, and 5% for the CTV, rectum, bladder and sigmoid, respectively. These differences correspond to the maximum relative EQD2 metric difference of 6% for all CTVs and OARs. **Conclusion:** The optimized PETRA shows the potential to make MRTP feasible in this pilot cohort of gynecological cancer patients. A larger cohort of patients is required to confirm these preliminary observations.

WE-C-TRACK 3-04

Monte Carlo Dose Calculations for Patient-Specific QA Based On Machine Log Files

H Parenica^{1*}, C Kabat¹, P Papaconstadopoulos², N Kirby¹, P Myers¹, T Wagner¹, N Papanikolaou¹, S Stathakis¹, (1) UT Health San Antonio MD Anderson Cancer Center, San Antonio, Texas, (2) The Netherlands Cancer Institute, Amsterdam, Netherlands

Purpose: Two-dimensional detector arrays are commonly used to verify IMRT dose delivery. An area of interest is the use of machine log files for machine and patient-specific quality assurance. We aim to develop an application that uses a true Monte Carlo model to calculate dose based on machine parameters extracted from log files from patient plan delivery. **Methods:** A model of the Elekta VersaHD linear accelerator was developed using the EGSnrc codes. This model was validated to be within 1%/1 mm accuracy of measured PDDs and profiles. A tool was created in MATLAB that extracts information from machine log files from delivered beams and “injects” it into the patient’s DICOM RP file, replacing the initial parameters used to calculate the “planned” dose. Plans were created from log files from 10 patients’ IMRT QA deliveries. These plans were converted to the inputs required by BEAMnrc/DOSXYZnrc using a MATLAB script developed in-house. Doses were calculated using Source 21 in DOSXYZnrc. Results were compared to measurements made with the PTW Octavius phantom and evaluated in Verisoft with a gamma criterion of 3%/2mm. **Results:** The 3D gamma passing rates for log file-based plans calculated in DOSXYZnrc ranged from 95.7% to 99.9%, and all statistical uncertainties were below 3%. In 9 out of 10 patients, doses calculated in DOSXYZ showed better agreement with measured values than the doses calculated in Pinnacle. **Conclusion:** Using plans with parameters from log files, we can obtain a more accurate representation of the dose that was delivered to the patient and determine the differences from the dose that was initially planned. The “DICOM Injection” tool can provide greater insight on the impact of machine errors in delivered plan quality. When used with Monte Carlo, users have the potential to see an accurate representation of the dose delivered to the patient.

The project described was supported by the National Center for Advancing Translational Sciences, National Institutes of Health, through Grant TL1 TR002647. This project was also supported by the Cancer Prevention and Research Institute of Texas (CPRIT) Research Training Award (RP170345).

WE-C-TRACK 3-05

Multi-Institution Plan Complexity Characteristics Based on IROC Phantoms

V Desai^{1*}, S Kry², (1) UW Health, Madison, WI, (2) MD Anderson Cancer Ctr., Houston, TX

Purpose: To evaluate how much treatment plan variability exists across institutions developing plans to satisfy common objectives. **Methods:** IROC phantoms provide consistent anatomy and plan objectives, but the institution is allowed to develop a plan based on their treatment techniques. Four standard IROC phantoms (head-and-neck, prostate, lung, and spine) were evaluated based on 24 complexity metrics for more than 1700 plans developed across the radiotherapy community. Statistical tests were used to assess the spread within, and differences between, the complexity of phantom plans. These metrics were also evaluated on a single institution database containing 627 clinical patient plans over the same 4 anatomical sites. A k-means clustering algorithm was applied to the principal components of the 24 complexity metrics to determine whether any distinct plan clusters existed. **Results:** For the 4 IROC phantoms, median complexity results were distinct from one another: the H&N phantom and spine phantom were most complex, the prostate and lung least complex, consistent with the complexity of the phantom anatomy. However, there was also a large spread of results for each phantom in terms of solution complexity. These two trends were also seen for the clinical plans. The k-means algorithm demonstrated that complexity metrics successfully stratified the IROC phantom data into two groups, namely prostate/lung and H&N/spine. However, these clusters demonstrated low inter-cluster variance and high intracluster variance. Additionally, step-and-shoot and sliding window plans were quite distinct in terms of complexity, while VMAT plans were not. **Conclusion:** Consistency in planning across the community is relatively low as substantial plan variability was seen for common plan objectives on identical treatment geometries. In addition, plan complexity metrics were not uniquely descriptive of a particular class of plans, even for a single phantom geometry, suggesting they may have limited utility as plan descriptors for QA.

WE-C-TRACK 3-06

Predicting Ion Chamber Disagreements for Patient-Specific Quality Assurance of VMAT Plans Using Machine Learning

P Wall^{*}, M Lafreniere, E Hirata, O Morin, G Valdes, A Witztum, University of California San Francisco, San Francisco, CA

Purpose: To develop and validate a machine learning model for predicting ion chamber disagreements used in patient-specific quality assurance (QA) of VMAT plans at a multi-site institution. **Methods:** A dataset of 579 VMAT treatment plans and associated QA measurements was retrospectively collected at our institution. 30 classes of complexity features representing different failure modes (totaling 293 raw features) were extracted from each DICOM RT-Plan file using in-house software. QA outcomes, represented by the percent difference between planned dose and ion chamber measured dose, were extracted from a RedCAP database used in our clinic. An XGBoost model was trained (10-fold cross-validation) to predict ion chamber measurement disagreements using the complexity features as input. A grid search over a range of values was used to optimize maximum tree depth and learning rate hyperparameters, after which the tuned model was tested on a previously unseen dataset. **Results:** The beam model, edge metric, monitor unit factor, small aperture score at 25 mm, and modulation complexity score were found to be the five most important features for predicting ion chamber disagreements by total gain. The initial model was able to predict all but 24 of the 579 testing samples within $\pm 3\%$ across the 10 testing folds. The average training and testing mean absolute error (MAE) were $0.73 \pm 0.14\%$ and $1.10 \pm 0.17\%$. The 24 samples predicted outside of 3% error were remeasured after which the model was retrained and reoptimized. All testing samples were subsequently predicted within 3% error and the testing MAE was improved to $1.01 \pm 0.08\%$. **Conclusion:** This is the first predictive model developed for ion chamber disagreements in patient-specific QA and was shown to be accurate within 3%. The model has been implemented in our clinical workflow and prospectively validated on 147 plans with MAE < 1%.

A.W. and G.V. report an ownership stake in Foretell Med, LLC, which is developing machine learning models in medicine

WE-C-TRACK 3-07

Sensitivity Evaluation of Patient Specific Quality Assurance - Independent Assessment of Quality

J Lehmann^{1,2,3*}, M Hussein⁴, S Siva⁵, A Moore^{3,6}, T Standen¹, B Subramanian⁷, P Greer^{1,3}, CH Clark^{4,8,9}, (1) Calvary Mater Newcastle Hospital, Newcastle, AU (2) University of Sydney, Sydney, AU (3) University of Newcastle, Newcastle, AU (4) National Physical Laboratory, Teddington, UK (5) Peter MacCallum Cancer Centre, Melbourne, AU (6) TROG Cancer Research, Newcastle, AU (7) GenesisCare, Bundoora, AU (8) University College London Hospital, London, UK (9) National Radiotherapy Trials Quality Assurance Group, London, UK

Purpose: The quality of radiation delivery has been shown to impact clinical outcomes significantly. External dosimetry audits are widely implemented as additional independent tools to ensure safe delivery of treatments, however have found problems despite facilities passing the audit plans with their own patient specific quality assurance (PSQA). This work presents a new concept for dosimetry audits which assesses the sensitivity of a facility’s clinical PSQA to detect relevant errors. **Methods:** Treatment delivery errors which produce clinically significant changes to treatment plans, yet are within or close to the specifications of the treatment

delivery system, were designed and introduced into Spine SBRT treatment plans from participating facilities. Each facility submitted one plan using a provided data set and planning guidelines. They received twelve versions of the plan back, on which they then performed their standard clinical PSQA. Identical errors were used for all facilities and the impact of the errors in the plans was assessed individually. **Results:** Seventeen facilities from Australia, the UK and the US participated in the study, using six different radiation treatment planning systems and the two major radiation treatment delivery systems. Of the twelve versions of a participant's plan, between two and eight passed the local PSQA. Six of the error plans (from four facilities) which had caused a >10% increase in radiation dose to an organ at risk still passed the PSQA. If the threshold was reduced to >5% this rose to 18 plans (seven facilities). **Conclusion:** Going beyond the current approach to test a facility's ability to deliver a selected test case, this novel audit methodology could improve the precision of institutional PSQA, thus improving quality in every facility, impacting many patients. The developed system is administered fully remotely, which provides advantages in cost, environmental impact and logistics.

This work has been supported by a Hunter Cancer Research Alliance (HCRA) Radiation Oncology, Cancer Imaging & Technology (ROCIT) Flagship Program grant (Newcastle, Australia).

Ultrasound Symposium

TRACK 4

Carson-Zagzebski Distinguished Lectureship on Medical Ultrasound

WE-C-TRACK 4-00

Carson-Zagzebski Distinguished Lectureship on Medical Ultrasound

Moderators: Tian Liu, Emory University, Atlanta, GA and Chris John Diederich, UC San Francisco, San Francisco, CA

WE-C-TRACK 4-01

Introduction

WE-C-TRACK 4-02

E. Konofagou, Columbia University:

Ultrasound-Based Neurotherapeutics and Imaging of Cardiovascular Disease and Cancer

WE-C-TRACK 4-03

Q&A

Therapeutic ultrasound is older than ultrasound imaging and yet has not translated to the clinic until only recently. Focused ultrasound (FUS) has been shown to facilitate drug delivery through the opening of the blood-brain barrier (BBB). Protein and gene delivery will be shown in the neuroprotection and neurorestoration of the nigrostriatal pathway through the combined use of FUS and AAV-GDNF in an early-stage Parkinson's disease (PD) mouse model. Our group has demonstrated that only the striatum that received a combination of AAV-GDNF and FUS exhibited protection against a subsequent MPTP insult while the mice with already existing and stable MPTP lesions exhibited signs of both behavioral and morphological amelioration following AAV-GDNF and FUS with up to 76% restoration, especially in the neuronal terminals. A second study will also be presented for the treatment of Alzheimer's disease (AD) in a mouse model expressing both amyloid and tau. In this case, FUS targets the hippocampus and both amyloid and tau are shown to be reduced with behavioral working memory also improved in the FUS-treated group only. FUS has thus the capability of overcoming certain limitations of pharmacological agents that are incapable of permeating the BBB at clinically relevant dosages; thereby opening new avenues for disease-modifying strategies in undertreated neurodegenerative disorders such as PD and AD.

FUS can also affect brain cognition, either with but also without opening the BBB and merely exerting mechanical and/or thermal effects on the brain or peripheral nerves and driving neuronal modulation. Examples of lateralized motor and sensory responses will be shown in mice and improvement in motor and cognitive performance in non-human primates (NHP) will be shown as well as pain suppression in human subjects by targeting the median nerve.

Finally, FUS is used for ablation of tumors in the clinic and in this lecture, an elastographic technique developed by our group, Harmonic Motion Imaging, will be demonstrated in its capability of identifying the type, targeting and treating breast and pancreatic tumors while also monitoring and assessing the procedure before and after chemotherapy based on the change in the underlying tumor stiffness as a result of the treatment.

Learning Objectives:

1. To understand the mechanisms of inducing neurorestoration through protein and gene delivery by opening the blood-brain barrier.
2. To understand the mechanisms of neuromodulation using therapeutic ultrasound in the brain and peripheral nerves.
3. To understand the imaging capabilities of monitoring elasticity changes in tumors to assess responses to chemotherapy and ultrasound ablation.

SAM Therapy Scientific Symposium

TRACK 5

Advances in Proton Therapy

WE-CD-TRACK 5-00

Advances in Proton Therapy

Moderators: Thomas Bortfeld, Massachusetts General Hospital, Boston, MA and Wei Liu, Mayo Clinic Arizona, Phoenix, AZ

WE-CD-TRACK 5-01

Introduction

WE-CD-TRACK 5-02

W. Liu, Mayo Clinic Arizona:

LET/RBE in Proton Planning

WE-CD-TRACK 5-03

A. Knopf, University of Groningen:

4D Adaptive Proton Therapy

WE-CD-TRACK 5-04

J. Langendijk, University of Groningen:

Patient Selection Models

WE-CD-TRACK 5-05

J. Unkelbach, University Hospital Zurich:
Combining Proton-Photon Treatments

WE-CD-TRACK 5-06

S. Yan, Massachusetts General Hospital:
Low-Cost Proton Therapy

WE-CD-TRACK 5-07

Q&A & Panel Discussion

Proton therapy (PT), especially pencil beam scanned proton therapy (PBS-PT), has been increasingly used to treat cancer patients. PT has a physical advantage, and thus in some cases a demonstrated clinical advantage over standard radiation therapy with photons. Despite the physics advantage, PT poses many challenges. Therefore, while at least 15% of all patients who receive radiation are expected to benefit from proton therapy, less than 1% receive this form of treatment. In this symposium we will present recent advances in treatment planning, modeling, and delivery technology that may broaden the accessibility of proton therapy from a “luxury” treatment option for relatively few selected patients to all patients who are expected to benefit from it.

In the first talk, we will discuss the unanticipated adverse events in PT induced by its high linear energy transfer (LET) at the end of proton ranges via a newly developed tool, dose-LET volume histogram (DLVH). The need of understanding LET effect for PT from patient outcomes combining both dose and LET for evaluation and treatment planning will be discussed.

In the second talk, we will discuss the 4D adaptive PT to account for the intra-fractional anatomical changes in PT to treat mobile tumors. 4D adaptive proton therapy will enable an automated seamless workflow by integrating imaging, treatment planning, and treatment quality assurance into a real-time adaptive PT treatment loop.

In the third talk, we will discuss the model-based selection of those patients who are expected to benefit from proton therapy, which can be considered an alternative evidence-based method for randomized studies. This approach has been implemented for patient selection in the Netherlands. Positive experiences as well as challenges will be reported.

In the fourth talk, we will cover recent advances in combined photon and proton optimization. The role of combined treatments for maximizing the benefit of limited proton resources for a population of patients will be analyzed. Situations in which individual patients benefit from a combination of protons and photons will be discussed.

In the fifth talk, we will discuss technological advances that have recently enabled the installation of proton therapy in spaces designed for conventional radiotherapy and future developments to the level of standard radiation therapy. This will dramatically increase the availability of proton therapy to more patients and reduce the disability adjusted life year.

In the last part of the symposium, we will have a Q&A session for some further in-depth discussion between session organizers, speakers, and audience.

Learning Objectives:

1. Describe advanced concepts of integrating the variable linear energy transfer (LET) in the adverse events study.
2. Identify the most important factors requiring treatment adaptation.
3. Describe the motivation behind model-based patient selection.
4. Name two advantages of combining proton and photon therapy.
5. What are the main drivers of proton therapy cost?

Multi-Disciplinary Scientific Session

TRACK 6

MRI for Adaptive Treatment Planning and Delivery

WE-C-TRACK 6-01

A Physics-Aware Deep Neural Network for Real Time Volumetric MRI in Support of MR-Guided Radiotherapy

L Liu^{1*}, L Shen¹, A Johansson², J Balter³, Y Cao³, L Xing¹, (1) Stanford University, Palo Alto, CA, (2) Uppsala University, (3) University of Michigan, Ann Arbor, MI

Purpose: To investigate the feasibility of real time volumetric MRI using a physics-aware deep neural network, in support of MRI-based 3D motion tracking for MR-guided radiotherapy. **Methods:** A deep neural network was proposed to generate volumetric MRI from orthogonal Cine MRI, which can be acquired on MR-Linac systems real time. The network consisted of a 2D generation module that synthesized oblique slices from two orthogonal slices and a 3D refinement module to generate final high quality volumetric images. A physics module that utilized prior knowledge of imaging physics was used to bridge 2D and 3D domains. The proposed method was evaluated for 6 patients with intrahepatic carcinoma. Each patient received two golden-angle radial MRI examinations that were more than 1 month apart, where one examination was used for network training and another for evaluation. Time series of 100 volumetric images with a sampling rate of 340 ms were reconstructed via retrospective binning and served as the ground truth. Orthogonal Cine MRI were sampled from volumetric MRI and served as network inputs. A reference MRI with gross tumor volume (GTV) contours was deformed to match the time series. Deformed GTV centroid positions and contours were compared between ground truth and network prediction. **Results:** Across the 6 patients evaluated, the averaged distance between predicted and ground truth GTV centroids was less than 1 mm in all 3 directions (anterior-posterior, inferior-superior and lateral). The averaged 95-percentile Hausdorff distances between predicted and ground truth GTV contours was 4.6 mm, which was similar to the cross-plane imaging resolution (4 mm). **Conclusion:** Physics-aware deep neural network enabled real time volumetric MRI with sufficient accuracy for 3D motion tracking. The proposed technique is robust to longitudinal patient changes and has the potential of reducing treatment margins and improving treatment delivery precision on a MR-Linac system.

This work was supported by NIH R01 EB016079.

WE-C-TRACK 6-02

BEST IN PHYSICS (MULTI-DISCIPLINARY): Multi-Task MR (MT-MR) for Free-Breathing Simulation in Abdominal Radiotherapy Treatment Planning

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Purpose: MR simulation in the abdomen region is challenging because of respiratory motion. Multiple scans with different contrast weightings are ideally needed for better delineation of the tumor target and organs-at-risk (OARs). However, conventional diagnostic MR sequences often use variable motion compensation strategies (e.g. respiratory gated or breath holds) and are susceptible to inter-scan misalignment. Emerging motion resolved volumetric MR (i.e. 4D-MR) techniques are limited to a single contrast weighting. In this study, we present a free-breathing MR sequence, namely multi-task MR (MT-MR), that produces multi-contrast, motion-resolved volumetric images for abdominal radiotherapy simulation. **Methods:** Our technique builds upon the recently proposed MR Multitasking imaging framework that uses low-rank tensor modeling and exploits the correlation between images of different motion states and contrast weightings to achieve accelerated acquisitions. During each repetition time, a combination of saturation recovery pulses, idle gaps and T2 preparation pulses allows T1, proton density and T2 contrast weightings to be successively generated. MR data are collected using FLASH readouts with a Cartesian spiral-in sampling pattern, which allows for denser coverage of k-space center region and reduced eddy-current effect. This sequence was validated on an abdomen digital MR phantom as well as 7 healthy volunteers at 3.0T with a 1.6SI x 1.6LR x 3.2AP mm³ resolution. Additionally, we acquired 2D real-time MR images as a motion reference in 4 of the 7 volunteers. **Results:** The digital simulation confirmed the image accuracy of our MR technique. In vivo scans demonstrated our capabilities in producing motion-resolved, volumetric multi-contrast abdomen MR images from a free breathing scan under 10 mins. Differences between the respiratory motion range resolved by MT and the 2D real-time MR image reference are within 0.65mm. **Conclusion:** MT-MR is a multi-contrast, motion resolved MR simulation sequence that may provide versatile, useful information for abdominal radiotherapy treatment planning.

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WE-C-TRACK 6-03

Real-Time MRI Motion Estimation Through an Unsupervised K-Space-Driven Deformable Registration Network (KS-RegNet)

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Purpose: Real-time MRI is highly desired for intra-treatment target localization to guide radiotherapy. However, imaging acquisition constraints limit the amount of K-space data that can be acquired for real-time MRI. We developed an unsupervised, K-space-driven, end-to-end deformable registration network (KS-RegNet) for real-time MRI motion estimation from fully-sampled priors and under-sampled K-space data. **Methods:** KS-RegNet solves deformation-vector-fields (DVF) to deform prior MRIs (for instance, a fully-sampled setup MRI) to real-time new MRIs, through direct data fidelity matching in K-space. The network takes in complex-valued prior MRIs and real-time under-sampled K-space data, and outputs DVFs for real-time motion tracking. Without access to high-quality 'gold-standard' real-time MRIs, the loss function directly measures similarity between K-space transformation of the DVF-morphed prior MRI and the real-time new K-space signal. Inverse-consistency is incorporated into the network design to reduce the DVF solution space and improve its accuracy. A dynamic cardiac MR set with raw K-space data from 16 patients were used to train and evaluate KS-RegNet (12/1/3 for training/validation/testing). Radial sampling masks were generated to down-sample the K-space to 10% - 30% of full sampling. The real-time MRIs estimated by KS-RegNet, respect to each under-sampling mask, were compared against the fully-sampled gold-standard images via metrics including mean-squared-error (MSE) and relative-error (RE). **Results:** Under 10% sampling, KS-RegNet generated dynamic MRIs with average MSE and RE of 0.95e-4 and 16.2%, as compared to 1.79e-4 and 21.3% for prior MRIs (without deformable registration), and 7.63e-4 and 47.6% for non-uniform fast-Fourier-transform (NUFFT) images. The corresponding results were 0.83e-4/15.0% and 2.86e-4/28.9% for KS-RegNet and NUFFT by 20% sampling, and 0.76e-4/14.3% and 1.41e-4/20.3% by 30% sampling, respectively. KS-RegNet solved each DVF within 200 ms, meeting clinical demands for real-time imaging. **Conclusion:** KS-RegNet allows motion estimation from substantially under-sampled K-space acquisition, paving the way for real-time MRI motion tracking and treatment adaptation.

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WE-C-TRACK 6-04

The Optimal Imaging Plane Determination for Motion Monitoring in MRI Guided Radiation Therapy

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Purpose: To develop a framework that can identify optimal imaging plane to monitor tumor motion in MRI guided radiation therapy using combined motion trajectory and plan dose information. **Methods:** A total of 10 patients with 6 lung and 4 liver patients underwent IMRT or SBRT were retrospectively analyzed. The target volumes were identified from the maximum inhale and exhale breathing cycles. Voxel-based motion trajectories were obtained by deforming images between these two respiratory phases. The voxelized dose gradient was also calculated between the 50%-100% prescription dose regions. A heatmap was constructed by calculating the products between these two gradient vectors, with higher values representing the motion directions more parallel to the dose gradient directions. The heatmap identified regions that could result more clinical-relevant plan failure thus close motion monitoring for those regions would be needed. Principle component analysis (PCA) was further conducted to identify the primary imaging plane that can capture major clinical-relevant motions. The tumor motion extracted from the optimal imaging angle was then compared to that monitored with the standard image plane used clinically. **Results:** From the heat maps generated from all the cases, there was no specific direction or region consistently showing high values. The PCA-determined imaging planes were also patient-specific. Our proposed method was able to capture larger in-plane motion with an average of 2.8 mm(range [1.6-6.8 mm]) compared to 2.1 mm([1.2-5.7mm]) as seen in clinical used image plane. The through-plane motion was also reduced from 2.4mm [0.8-6.6mm] to 1.7mm [0.3-4.2 mm]. **Conclusion:** The proposed method used the respiratory tumor motion trajectory combined with plan dose to determine the optimal imaging angle before the treatment. Results showed the proposed method had overall better motion tracking performance when compared against the imaging planes used in routine clinic.

WE-C-TRACK 6-05

Time-Resolved Estimated Synthetic CTs Based on Orthogonal Cine MRI for Low-Field MR-Linac Treatments

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Purpose: To develop and experimentally validate a method to create continuous time-resolved estimated synthetic CTs (tresCTs) based on 3DCT and orthogonal cine MRI data for MR-guided radiotherapy. **Methods:** A breathing porcine lung phantom was consecutively scanned at a CT scanner and a 0.35T MR-Linac. A mid-exhale breath-hold 3DCT was deformably registered to a mid-exhale 3DMRI. The 3DMRI was deformably registered to orthogonal cine MRI series (sagittal/coronal orientation) at 7.3Hz, intersecting different injected tumor-mimicking gelatin nodules. The time-resolved deformation vector fields were extrapolated to 3D and applied to the deformed 3DCT to create 82s-long tresCTs at 3.65Hz. A 4DCT scan served as ground truth (GT). Ten tresCTs were created for ten tracked nodules with different motion patterns in two lungs. Each intersected nodule, expanded by a 5mm margin, was used as PTV for stereotactic step-and-shoot IMRT treatment planning (8x7.5Gy) on the GT-4DCT mid-exhale image to mimic gated MR-Linac plan generation. Each plan was recalculated on all GT-4DCT phases and one tresCT image per breathing phase, randomly sampled from different breathing cycles. Phase-dependent DVH parameter and dose differences were quantified. **Results:** Median nodule motion amplitudes were 8mm (range 3-16mm), with median centroid position deviations between GT and tresCTs of 2.3mm. For all ten

nodules and breathing phases (100 dose comparisons), the median absolute differences of D98%, D50% and D2% were 2.1%, 0.8% and 0.5% for the PTV. The dose difference analysis with a 2% prescription dose acceptance level (10% dose threshold) yielded a median pass rate of 97.4%. Dose deviations in the vicinity of the GTVs were observed, where phase-dependent lung density variations were not accurately modeled. **Conclusion:** Dose differences between GT and tresCTs were at a clinically acceptable level. Hence, tresCTs could be used for time-resolved dose accumulation to guide inter- and intra-fractional treatment adaptation in the future.

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WE-C-TRACK 6-06

Treatment Window Determination for Cardiac Synchronized VMAT

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Purpose: The use of SABR for treatment of ventricular tachycardia is negatively affected by target motion. The effect of motion can be reduced by treating only during the quiescent interval of the cardiac cycle using a technique such as cardiac synchronized VMAT (CSVMAT). In this work, cardiac magnetic resonance (CMR) cine images for an illustrative patient with impaired left ventricle (LV) function were analyzed to track LV contour point motion and determine an optimum treatment window. **Methods:** CMR data included breath-hold scans for 2 short-axis slices and 2-, 3-, and 4-chamber view long-axis slices. Image data and myocardial tissue masks for 30 cardiac cycle phases were converted to endocardial and epicardial contours. Coherent point drift, a non-rigid point set registration algorithm, was used to assign contour point correspondences across all cardiac phases. The motion profile of the target was then determined according to the displacement of the associated contour points. From this temporal motion profile, a CSVMAT or gated treatment window can then be designed to minimize target displacement. **Results:** Minimum, mean, and maximum displacements for each segment were displayed in a 17-segment bullseye plot for either the short-axis or long-axis direction. Displacement due to cardiac motion was generally greater towards the basal end of the LV compared to the apex. Choosing basal segment 6 as the target, an optimal beam delivery window of 200 ms (for a 1 s cardiac cycle) reduces the maximum target displacement during delivery from 18.4 mm to 4.2 mm. For mid-cavity segment 12, maximum displacement is reduced from 7.0 mm to 1.2 mm. **Conclusion:** A method of estimating LV wall displacement due to cardiac motion was developed. This method may be used to create optimal treatment margin guidelines for cardiac radioablation targeting specific areas of the LV.

This work was supported by a Vancouver Coastal Health Research Institute (VCHRI) Innovation & Translational Research Award. Authors Steven Thomas, Kirpal Kohli, and Justin Poon have filed a patent application for CSVMAT.

WE-C-TRACK 6-07

Auto-Detection of Structural Similarity Index Map from Daily MRI as An Indicator for MR-Guided Online Adaptation Replanning

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Purpose: MRI-guided adaptive radiation therapy(MRgART), particularly daily online adaptive replanning (OLAR), can substantially improve radiation therapy delivery, however, can be labor-intensive and time-consuming. Currently, whether to perform OLAR for a treatment fraction is determined subjectively. In this work, we develop a machine learning classifier based on structural similarity index map (SSIM) to quickly, automatically, and objectively determine whether OLAR is necessary for a daily MRI set. **Methods:** The classifier was developed based on a total of 70 daily MRI sets acquired on a 1.5T MR-Linac during MRgART of 14 pancreatic cancer patients each treated with 5 fractions. For each daily MRI set, OLAR and reposition plans were created and the superior plan for the daily fraction was determined per clinical dose-volume criteria. SSIM was extracted for each daily MRI set with respect to its reference (e.g., planning) MRI in six regions enclosed by: 50%, 80%, 100%, 50–80%, 50–100%, and 80–100% isodose surfaces, respectively. A series of histogram features, i.e., mean, median, moment (2nd order), skewness's, kurtosis, and full width at half maximum (FWHM), along with a new parameter MMF ratio=Mean/(Moment*FWHM)*10⁵, were extracted from each SSIM. A linear regression, t-SNE, and 5-fold cross-validation ensemble classifier was determined to predict if OLAR is necessary for a daily MRI set. **Results:** Higher significant difference was observed for SSIM features extracted from the 50–100% isodose surface regions based on initial linear regression analysis ($p = 0.007$) and t-SNE projection between OLAR vs. reposition. The developed ensemble classifier in term of MMF ratio can predict 90% reposition and 95.9% OLAR with five-fold cross validation. The prediction accuracy was 94.2% and AUC was 0.97. A simplified decision tree with six-nodes was created. **Conclusion:** An ensemble-based classifier using SSIM was developed, allowing to automatically and objectively determine if OLAR is necessary for a treatment fraction during MRgART.

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SAM Therapy Educational Course TRACK 7

Artificial Intelligence in the Clinic - Is It Going to Help or Hurt Our Productivity?

WE-CD-TRACK 7-00

Artificial Intelligence in the Clinic - Is It Going to Help or Hurt Our Productivity?

Moderator: Eduard Schreibmann, Emory University, Atlanta, GA

WE-CD-TRACK 7-01

C. Cardenas, UT MD Anderson Cancer Center:

Adding AI-Based Automated Contouring and Treatment Planning to Every Medical Physicist's Toolbox

WE-CD-TRACK 7-02

Y. Yang, Stanford University Cancer Center:

Clinical Implementation of An Auto-Planning and Plan Check Tool

WE-CD-TRACK 7-03

A. Apte, Memorial Sloan-Kettering Cancer Center:

From Data to Segmentation Models: AI for Everybody

WE-CD-TRACK 7-04

K. Moore, UC San Diego:

Widespread Clinical Implementation of Data-Driven Automated Treatment Planning

WE-CD-TRACK 7-05

D. Ruan, UCLA School of Medicine:
The Two Sides of the AI Coin

This session is intended for clinicians who are asking themselves how artificial intelligence may change their clinical tasks. Recent research has shown that artificial intelligence algorithms can solve tasks with an accuracy comparable to a human interpreter. Similarly, scripting has been shown to automate some repetitive tasks. Most clinicians would question if the promises of research abstracts will hold in day-to-day clinical work where many practical considerations are essential in addition to the pure accuracy of automation algorithms. The session will explore how AI algorithms can be integrated with the clinical workflow by using scripting to enable them in commercial software, with the ultimate question of if AI research can indeed become a tool that keeps up to its promise to improve productivity.

Another function of this session is to explain in layman terms what AI and scripting can and can't do, and how we see the field changed by automation in the upcoming future. It is not intended as a collection of research or clinical topics; instead, an example of how AI and clinical duties can be bridged by scripts to include automation algorithms in our clinical lives, for tasks where current research can replace human thinking. The session's objectives are balanced on both the advantages and pitfalls of automation through AI, presented to a clinically minded audience.

Each speaker will give an overview of an AI application, covering integration with the clinical software and describing their experience in adapting these changes to daily clinical hurdles. It should not focus on mathematics but on the experience with implementing these in the clinic. A panel discussion will be held to answer questions.

Learning Objectives:

1. Present planning tasks that can be automated by current research to a clinical audience
2. Describe the pros and cons of automation from the end-user's perspective
3. Detail the hurdles encountered when implementing AI algorithms on real clinical datasets

Vendor Showcase

TRACK 8

Radiological Imaging Technology, Inc.

WE-SHOWCASE-TRACK 8-14-1

The Clinical Stereotactic Environment: Revealing a Paradoxical Inverse Correlation Between Individual Maximum 3D Deviations and the Overall Winston-Lutz 3D Displacement

D Ritt*, N Battaglia*, Radiological Imaging Technology, Inc., Colorado Springs, CO

Therapy Scientific Session

TRACK 1

Quality Control in Treatment Planning and Delivery

WE-D-TRACK 1-01

Towards Contour Quality Assurance of Cardiac Structures with Automatic Segmentation

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Purpose: This study aims to determine the clinical readiness of two auto-segmentation approaches for cardiac structures contouring quality assurance. **Methods:** Five experienced radiation oncologists delineated the pericardium, atria, and ventricles of 469 RTOG0617 CT scans using a consistent guideline. We developed atlas-based automatic contouring (ABAC) library using 100 randomly sampled CT scans. The CT scans of 200 patients (including the 100 scans for ABAC) were trained on an in-house developed convolutional neural network (CNN) model. The remaining patients were used to assess the performance of the two models, based on the Dice similarity coefficient (DSC), Hausdorff distances (HD), and mean distance-to-agreement (MDA). For dosimetric comparison, the mean dose (Dmean), and the maximum dose (Dmax), were computed for all the cardiac structures. Relative dose-volume factors, including V5 - V60, were analyzed to study the effect of dose heterogeneity. Spearman correlation enabled us to evaluate the relationships between geometric and dosimetric metrics. **Results:** Compared with ABAC, the geometric metrics (DSC, HD, and MDA) of CNN for the cardiac structures depicted a better alignment with the physician-drawn contours. Auto-segmentation with CNN also attained relatively lower Dmean, Dmax, and V5 to V60 to the cardiac structures. However, significant mean and maximum dose differences between the two auto-segmentation methods in the atria and ventricles highlighted a non-negligible disparity between the two auto-segmentation approaches. Analysis of the geometric and dosimetric metrics showed no clear relationship with the automatic contours due to dose heterogeneity. **Conclusion:** Although auto-segmentation shows promise to reduce cardiac structures' contouring inconsistencies, the heterogeneous distribution of radiation doses within the heart segments negates the use of geometric metrics as suitable surrogates for dosimetric parameters. Thus, auto-segmentation would continue to require a thorough validation of a qualified expert to ensure its safe clinical use.

WE-D-TRACK 1-02

Automated ICC Tool for a Novel Planning Incident Learning Environment

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Purpose: Initial chart checks (ICCs) are integral to ensuring safe and accurate delivery of radiation. Over a third of treatment plans at our three-site institution are submitted for ICC within a day of treatment, creating a large and urgent workload. We developed a tool to automate a large proportion of the ICC by connecting our TPS (Raystation) and record and verify system (Mosaik), with the goal of reducing the likelihood of errors reaching the patient and introducing automated reporting in a new planning incident learning environment (PILE). **Methods:** Plan parameters from the TPS and Mosaik database were extracted and compared using in-house code. Automated checks were written to match those documented in the ICC list, with the output presented through a graphical user interface (GUI) in the TPS and through a web interface (WI). The TPS GUI can be used by dosimetrists prior to plan approval, and the WI can be used by physicists to complete the ICC. **Results:** A tool was developed to automate a portion of ICCs. Using this, the average time to complete an ICC was reduced by approximately 50%. Additionally, each time the tool is used, an entry in the PILE is created, enabling the tracking and remedying of repeating issues. A retrospective analysis of plans in Mosaik identified minor (non-reportable) errors in plans that would have been flagged by this tool. **Conclusion:** A chart check tool was developed and integrated into the workflow to automate ICCs where possible, simultaneously reducing the likelihood of errors reaching the patient and reducing the physicists' time burden. The output of each ICC is stored in the PILE database, allowing trends in behaviors to be identified and providing a framework for how each ICC can provide information to further improve our

WE-D-TRACK 1-03

Plan Quality Assessment of the Leksell Gamma Knife Lightning Dose Optimizer

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Purpose: To compare plan quality metrics of manual versus Elekta Lightning™-based inversely optimized radiosurgery plans for Gamma Knife. **Methods:** Clinically treated plans for 100 consecutive patients (115 lesions; 67 metastatic and 48 benign) were replanned with the Lightning™ dose optimizer (based on a convex linear programming formulation). Plans were generated to match or exceed the following metrics in order of importance: Target Coverage, Paddick Conformity Index (PCI), delivery treatment time, and Gradient Index. Plan quality metrics in addition to delivery parameters such as spatial distribution of shots, collimator sizes and optimization times were investigated. **Results:** Overall, 76% (87/115) of Lightning plans showed a statistically significant improvement in plan quality. As compared to manual plans, Lightning plans showed significant improvements of 2.0% in the median PCI ($p < 0.01$), 4.7% in median GI ($p < 0.01$) and an increase in median number of shots by 60.3% ($p < 0.01$). Target coverage and delivery treatment time differences were not statistically significant. Regarding plan specifics, Lightning plans showed a significant increase in use of 16mm ($p < 0.01$) and blocked shots ($p < 0.01$) while also showing a significant decrease in 8mm ($p < 0.01$) and 4mm ($p < 0.01$) shots when compared to manual plans. Use of multiple shots per coordinate was substantially more frequent in Lightning plans. The Lightning optimizer did fail to produce clinically acceptable results in 4/115 (3.5%) cases. Median optimization time for Lightning plans was 4.0 min (Range: 1.0 – 15.0 min). **Conclusion:** Lightning offers a significant reduction in efforts for treatment planning while achieving comparable plan quality of an expert planner. Lightning plans employ non-intuitive increased use of blocked sectors and shot-in-shot technique to achieve high quality plans. A handful of Lightning plans severely failed in matching clinical plans and warrants further investigation.

WE-D-TRACK 1-04

Power Transformations on Multivariate Non-Normal Radiotherapy Plan Quality Measures

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Purpose: To investigate the application of power transformations on skewed plan quality measures used by multivariate quality control charts in radiotherapy. **Methods:** We compute multivariate risk-adjusted control charts for treatment plan quality control, which employ the resulting residuals of a risk-adjustment model that uses a set of organ-at-risk (OAR) dose-volume histogram (DVH) points as proxies for plan quality. The risk-adjustment model adjusts the DVH points for a set of patient- and treatment-specific risk factors. Given an ordinary least squares (OLS) regression-based risk-adjustment model, an assumption is that the data is normally distributed, which is typically not the case for DVH data. Thus, we employ the Box-Cox transformation that is a power transformation to reduce the impact of skewness in the DVH points. The control charts are evaluated on 69 head-and-neck cases with the brainstem serving as the primary OAR. **Results:** In the multivariate control chart with Box-Cox transformation, two patients (43 and 49) are signaled as out-of-control (OC), while the chart without Box-Cox transformation signals three patients (24, 43, and 49) as OC. The DVHs show that patient 49 is a high-quality plan, while patients 24 and 43 are poor quality plans with high brainstem dose. Replanning led to a sizable improvement in patient 43's brainstem dose while patient 24 could not be improved without sacrificing tumor dose criteria. The risk-adjustment model and the control chart fails to account for the skewed DVH data without the Box-Cox transformation, leading to a false signal for patient 24. **Conclusion:** A power transformation is necessary for non-normal DVH data for statistical modeling and quality control. The proposed Box-Cox transformation can scale and adjust the skewness of DVH data, leading to an improved performance of control charts for quality control in radiotherapy.

WE-D-TRACK 1-05

Artificial Intelligence for Dose-Volume Histogram Based Clinical Decision-Making Support System in Radiation Therapy Plans for Brain Tumors

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Purpose: To create and investigate a novel, clinical decision-support system using artificial intelligence (A.I.). **Methods:** The A.I. model was developed based on 79 radiotherapy plans of brain tumor patients that were prescribed a total dose of 60 Gy delivered in 30 fractions (2 Gy per fraction) with volumetric-modulated arc therapy (VMAT). Structures considered for analysis included planning target volume (PTV), brainstem, cochleae, and optic chiasm. Data for model training consisted of five dosimetric and five geometric features extracted from radiation therapy plans. The model aimed to classify the target variable that included class-0 corresponding to plans for which the PTV treatment planning objective was met and class-1 that was associated with plans for which the PTV objective was not met due to the priority trade-off to meet one or more organs-at-risk constraints. One model was selected for further investigation amongst those evaluated: Support-Vector Machine, Elastic-Net, Logistic-Regression, and Random-Forest using double-nested cross-validation and an area under the curve (AUC) metric. The model predictions were explained both globally and locally with Shapely additive explanation (SHAP) interaction values. **Results:** The highest-performing model was Logistic Regression achieving an accuracy of 93.8±4.1% and AUC of 0.98±0.02 on the testing data. The SHAP analysis indicated that the $\Delta D99\%$ metric for PTV had the greatest influence on the model predictions and was approximately 3.5 times more significant than the second most important feature – $\Delta D1\%$ for optic chiasm. The least important feature was $\Delta DMAX$ for the left and right cochleae. **Conclusion:** The trained model achieved satisfactory accuracy and can be used by medical physicists in a data-driven quality assurance program as well as by radiation oncologists to support their decision-making process in terms of treatment plan approval and potential plan modifications. Model explanation analysis showed that the model relies on clinically valid logic when making predictions.

WE-D-TRACK 1-06

Feasibility of Time-Resolved DVH Based Treatment Delivery Verification of VMAT with EPID

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Purpose: EPID presents a potentially efficient and comprehensive option for verification of VMAT with no setup time. However dosimetric verification of VMAT with EPID is usually performed using integrated EPID images therefore the time-resolved dose delivery is not verified. Moreover gamma based analysis is difficult to interpret for the clinical significance of any errors. This work investigates the feasibility of performing time-resolved dose-volume histogram (DVH) based analysis of VMAT with EPID. **Methods:** Feasibility was assessed with a 2-arc (200 degrees rotation) VMAT lung SBRT plan modelled as 40 subarcs of 10 degrees. Dose in flat phantom for each subarc at 10 cm depth, gantry zero, was calculated with Eclipse V15.6.05. Image frames at 13 frames per second were collected during in-air delivery by a framegrabber computer with in-house acquisition software and interpolated to form 10 degree subarc images. These were then converted to dose at 10 cm depth in flat phantom using an in-house EPID to dose conversion model and dose ratio (DR) matrices to Eclipse dose generated. These DR matrices were backprojected through the original patient CT dose using ray-tracing to give a measured-weighted dose. Dose volume histogram comparisons were then performed. **Results:** The 2D dose maps (n=40) were evaluated with traditional planar gamma evaluation methods with gamma pass rates (mean±SD) of 99.9±1.1% at 2%,2 mm criteria. The DVH curves generated from measured-weighted dose were virtually indistinguishable with the original planned DVH. **Conclusion:** This study successfully demonstrated feasibility of time-resolved DVH based verification of VMAT with EPID. Using EPID subarc measurements during VMAT the DVHs are

reproduced.

WE-D-TRACK 1-07

6-DoF Robotic Motion Phantom: An Open-Source Project for Quality Assurance of Real-Time Image-Guided Radiotherapy Technologies

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Purpose: The geometric and dosimetric impact of tumor translation and rotation, or 6-degree-of-freedom (6-DoF) motion, in radiation therapy is becoming increasingly important. Since there is currently no commercial 6-DoF solution that could reproduce patient-measured motion, we designed, built and programmed a robot-based system for 6-DoF Image-guided Radiation Therapy (IGRT) research and Quality Assurance (QA). To enable advancement of intrafraction motion management in our community, we have made the complete package open-source. The purpose of this work is to describe the applications of the 6-DoF system, the research results and the open-source repository. **Methods:** The 6-DoF system consists of a six-axis robotic arm (Universal Robot, UR3), an acrylic phantom, a custom-base plate to mount the robotic phantom onto the treatment couch and a software application which controls the robotic phantom to reproduce the patient-measured tumor motion. The integrated system is capable of accurately mimicking patient-measured tumor traces to provide precise 6-DoF QA for real-time IGRT approaches. The repository contains everything required to assemble a similar 6-DoF system: source code, CAD files for the phantom and hardware tools, robot specifications, bill of materials, documentation, and user-guide to operate the system, and, step-by-step QA procedure. **Results:** The 6-DoF system has been used successfully for several applications including geometric QA tests for the real-time tumor tracking technology, Kilovoltage Intrafraction Monitoring (KIM) in two multi-center clinical trials, for prostate and liver cancer patients and evaluation of the dosimetric impact of prostate rotational motion using films. **Conclusion:** The development of this system has enabled us to perform 6-DoF QA processes for KIM and it has great potential to perform QA for other real-time radiotherapy systems. Full details of the system and its application have been shared on GitHub for the benefit of the medical physics community: <https://github.com/ACRF-Image-X-Institute/6-DoF-Robotic-Motion-Phantom>.

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Imaging Scientific Session

TRACK 3

Dose and Imaging Performance Assessment in CT and CBCT

WE-D-TRACK 3-01

A Method to Estimate a Fan-Beam CT NPS Using Two Basis Functions with a Limited Number of Noise Realizations

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Purpose: Traditionally, noise power spectrum (NPS) is estimated by ensemble averaging of multiple realizations of noise-only images. However, with a limited number of images, the estimation accuracy of NPS becomes lower. Recent works proposed NPS estimation methods using radial 1D-NPS as its basis, but these methods couldn't reflect the non-stationary noise property of a fan-beam CT image. In this work, we proposed a more accurate NPS estimation method that reflects the non-stationary noise property of a fan-beam CT image. **Methods:** Each angular spoke of the 2D-NPS can be expressed as a sum of two basis functions with different cutoff frequencies determined from complimentary projection magnification factors. Therefore, the proposed method uses two basis functions by reflecting view angle dependent variations of cutoff frequencies in each angular spoke of the 2D-NPS. To examine the non-stationary property of NPS in a fan-beam CT image, NPS was estimated at 3 different local regions centered at (-100mm, 0), (0, 0) and (0, -100mm). The weighting factors of the two basis functions were estimated by minimizing the mean squared error (MSE) from the realized NPS image calculated from 10 noise realizations. We compared the estimation accuracy of NPS calculated from 10 noise realizations, previous method, and proposed method. Analytical NPS was used as a reference. **Results:** The NPS estimated by the proposed method showed high similarity to analytical NPS, while the NPS estimated by the previous method couldn't reflect the non-stationary noise property of a fan-beam CT image. **Conclusion:** In this work, we proposed a more accurate method using two basis functions to estimate the NPS which reflects the non-stationary noise property of a fan-beam CT image. The NPS estimated by the proposed method showed higher similarity to analytical NPS compared to the NPS estimated by the previous method.

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WE-D-TRACK 3-02

An Assessment of Uncertainties in Pediatric CT Organ Dosimetry When Scan Parameters and Individual Body Morphometry Are Unknown

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Purpose: In many epidemiology studies of patient CT exposures, details regarding patient-specific body morphometry and/or exam-specific technique factors are typically absent, and thus various assumptions are employed during organ dose reconstruction. The purpose of this study is to highlight the importance of these parameter sets in estimating effective and organ doses from a CT scan. **Methods:** The University of Florida/National Cancer Institute (UF/NCI) computational phantom library (12 reference phantoms and 351 non-reference phantoms) were used to compute organ doses within a population of 2,815 American pediatric patients undergoing CT imaging from 1998 to 2013 in which both patient height/weight and exam-specific scan lengths were available. Next, patient morphometry and scan length data were incrementally removed and replaced with various estimates to assess uncertainties in CT dosimetry when these data are unavailable. **Results:** Using the effective diameter of the patient to estimate height and weight resulted in up to an 82% overestimation of the effective dose. Matching to a reference phantom resulted in up to a 54% overestimation of effective dose. Estimating height based on patient age and weight based on effective diameter resulted in effective dose being underestimated by a maximum of 8% and overestimated by a maximum of 10%. Estimating height and weight based on patient age resulted in effective dose being underestimated by a maximum of 25% and overestimated by a maximum of 12%. Using a default scan length in place of a patient-specific scan length resulted in up to a >200% overestimate or a 51% underestimate of the effective dose. **Conclusion:** The differences in organ and effective doses found in this study highlight the need for patient- and exam-specific information when estimating organ and effective doses from CT imaging in pediatric patient populations.

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WE-D-TRACK 3-03

Diagnostic Reference Levels (DRLs) and Achievable Doses (ADs) for the 11 Most Commonly Performed Pediatric CT Examinations in the United States (USA) as a Function of Patient Age and Size Using 1.5 Million Examinations in the American College of Radiology (ACR) CT Dose Index Registry

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Purpose: To develop diagnostic reference levels (DRLs) and achievable doses (ADs) for the 11 most commonly performed pediatric CT examinations in the United States (USA) as a function of patient age and size using the American College of Radiology (ACR) CT Dose Index Registry. **Methods:** Data on the 11 most commonly performed pediatric CT examinations acquired between 2016 – 2020 from 1637 facilities were analyzed. For head, neck and extremity examinations, doses were analyzed based on patient age; for body examinations, doses were analyzed based on both patient age and effective diameter. Data from 1,562,016 examinations provided medians (AD) as well as 75th percentiles (DRL) for CTDIvol, DLP and SSDE. Results were compared with DRLs from other countries. **Results:** Of all facilities analyzed, 65.2% of the facilities were community hospitals, 16.8% were freestanding centers, 9.5% were academic facilities and 3.5% were dedicated children's hospitals. Head CT was the most frequently performed examination in the study (56.1%) followed by the abdomen/pelvis CT (28.1%). ADs and DRLs (for both CTDIvol and DLP) for all examinations generally increased as a function of age and patient effective diameter. For head without contrast examinations, the CTDIvol AD and DRL ranged from 19-46 mGy and 23-55 mGy, respectively increasing with age. For abdomen/pelvis without contrast examinations, the CTDIvol ADs and DRLs ranged from 2.1-24 mGy and 2.6-28 mGy, respectively with increasing effective diameter. Although international pediatric DRL methodology and results vary, our results fall within the ranges reported across other countries. **Conclusion:** DRLs and ADs as a function of patient age and effective diameter were developed from a wide range of USA practice environments for the 11 most common CT pediatric examinations using data from over 1.5 million pediatric examinations. These determinations can serve as practical benchmarks for accountability and for improvement in pediatric CT practice.

WE-D-TRACK 3-04

Fast Monte Carlo Simulation of Non-Isotropic X-Ray Source for CT Dose Calculation

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Purpose: To accurately and efficiently estimate individualized patient dose in CT scans with Monte Carlo (MC) simulations. **Methods:** A GPU-based MC tool with the ability to model a non-isotropic source was developed from MC-GPU v1.3. Unlike the isotropic source in the original MC-GPU, the anode heel effect and bowtie filtration are realized in this tool. The heel effect leads to photon intensity variation for azimuthal angles that can be modeled as a probability function, while the bowtie model is based on analytic attenuation. The bowtie is represented by two basis materials so that with inputs of material attenuation coefficients and thickness combinations at different fan angles, photon transmission and filtered spectra for each fan angle can be calculated for a given source spectrum. Using a precomputed lookup table of the bowtie, the photon's direction and energy can be efficiently randomly selected using the alias method. An air scan was used to validate the simulation tool by comparing against the analytically calculated ground truth image. For individualized patient dose maps, we generated spatial maps of both material type and mass density based on the CT numbers as the inputs for the MC tool along with necessary scanner-specific parameters. **Results:** The simulation was performed on a workstation with two Nvidia Titan RTX GPUs. The average processing speed was $\sim 3.5e7$ photons/sec (original MC-GPU with isotropic source: $4.7e7$ photons/sec). In the air scan validation, the average relative error in a spectral test was 0.86% for 25-80 keV and the average relative errors in energy-integrated detection were 1.69% and 0.92% for bowtie filtration and heel effect, respectively. **Conclusion:** In this work, a near real-time MC dose tool is introduced which has the potential to provide realistic and individualized dose map from non-isotropic sources. Ongoing work will include tube current modulation and automatic organ segmentation.

This work was supported by GE Healthcare. Sen Wang received Zijing Scholarship from Tsinghua University.

WE-D-TRACK 3-05

Longitudinal QC Testing of CBCT and Helical MDCT Systems with a Unified Image Quality and Dose Phantom

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Purpose: The first clinical use of a phantom for unified quality control testing of both cone-beam CT (CBCT) and helical MDCT is reported, with automated analysis and structured reporting of quantitative imaging performance and dose in longitudinal studies spanning 6 clinical systems (3 CBCT + 3 MDCT) for 6 months. **Methods:** The phantom (CorgiTM, The Phantom Lab) was used in monthly QC of 3 CBCT systems – Synergy XVI (Elekta), O-arm (Medtronic), and Artis Zee (Siemens) – and 3 MDCT systems – Edge (Siemens), Precision (Canon), and Aquilion One (Canon). Automated software provided one-click analysis and structured reporting of voxel value linearity, contrast-to-noise ratio (CNR), image uniformity, spatial resolution (MTF in axial and non-axial planes), noise, noise-power spectrum (NPS), and cone-beam artifact (signal modulation about two parallel disks). For each system, image quality tests were performed for standard Head (with 13.5 cm diameter phantom) and Body protocols (phantom within a 32 cm annulus). **Results:** The results demonstrate a unified system for more rigorous, quantitative, objective QC across a broad spectrum of CBCT and MDCT systems in clinical applications ranging from image-guided radiotherapy, image-guided surgery, interventional radiology to diagnostic radiology and cardiology. Quantitative tests revealed factors not conventionally evaluated and had positive impact on QC of each system – example, quantitative analysis of spatial resolution (MTF, rather than subjective evaluation and line-pair tests that do not probe the limits of spatial resolution on modern CBCT or MDCT systems), the extent to which spatial resolution is isotropic (MTF in axial and non-axial planes), and NPS analysis that was valuable in identifying shortfalls in detector and system geometry calibration. **Conclusion:** The phantom, test procedures, and automated analysis of quantitative QC metrics provide a valuable basis for more standardized, rigorous, and quantitative QC in CBCT and with helical MDCT.

Boone JM & Siewerdsen JH have licensing agreement with The Phantom Lab (Greenwich NY) for the phantom used in the work.

Uneri A has a consulting agreement with The Phantom Lab (Greenwich NY) for software development.

Mahesh M and Fernandez M do not have any COI related to this study.

WE-D-TRACK 3-06

A Scanner-Specific Simulation Platform for Virtual Imaging Trials in Deep Si-Based Photon-Counting CT

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Purpose: Deep Si-based photon-counting CT (Si-PCCT), a candidate emerging CT detector technology, is expected to provide multiple benefits by virtue of fast charge collection times, reduced K-fluorescence crosstalk, and ease of manufacturing. To translate Si-PCCT to clinical use, its performance needs to be comprehensively evaluated and optimized in a task-specific manner. Often performed through large-scale imaging studies on human subjects, such evaluations are impractical for Si-PCCT due to the lack of access to clinical prototype systems. This limitation can be overcome by using computational anthropomorphic phantoms and imaging simulators, in a process known as virtual imaging trials (VITs). To enable such VITs, we present a scanner-specific simulation platform to simulate realistic clinical images for Si-PCCT systems. **Methods:** A hybrid simulation platform for VITs (DukeSim) was extended to model the scanner-specific geometry

and components of a PCCT prototype, including models incorporating energy-dependent response and crosstalk characteristics for a deep Si-based detector. To simulate images, a potential pilot system was emulated to image a high-resolution (0.1 mm) virtual patient with a lung lesion. The obtained projections with energy-binned photon counts were processed to introduce Poisson noise, summed across energy bins, and corrected for beam hardening before reconstruction. To demonstrate the utility of this platform, the reconstructed images were utilized to qualitatively and quantitatively compare the spatial resolution performance of PCCT to conventional energy-integrating CT (ECT), including their relative performance for estimation of morphological radiomics features. **Results:** The platform successfully simulated realistic images for prototype Si-PCCT designs. The evaluation of spatial resolution revealed superior performance of Si-PCCT over ECT both qualitatively and quantitatively, with a dramatic improvement of accuracy (30.7% mean) in the estimation of morphological radiomics features. **Conclusion:** The developed simulation platform simulated realistic images for a pilot Si-PCCT system, enabling the qualitative and quantitative evaluations integral to performing clinical VITs.

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WE-D-TRACK 3-07

Accurate and Efficient Measurement of Channelized Hotelling Observer-Based Low-Contrast Detectability on the ACR Accreditation Phantom

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Purpose: To optimize channel filters and many other parameters in channelized Hotelling observer (CHO) so that the low-contrast detectability can be accurately and efficiently estimated for the widely used American College of Radiology (ACR) accreditation phantom. **Methods:** A large number of repeated scans is typically required in CHO calculation to provide accurate and precise estimate of index of detectability (d'). The main goal of optimization is to minimize the number of repeated scans and make it practically feasible. To provide a reference standard for optimization, an ACR phantom was repeatedly scanned 100 times at three dose levels (24, 12 and 6 mGy). Images were reconstructed with two kernels (FBP Br44 and IR Br44-3). d' as a function of number of repeated scans was determined for different channel filter options, different number of low-contrast objects, different number of background regions of interest (ROIs), and different number of slices per each object. The bias of d' was quantified using the d' calculated from all 100 repeated scans as reference. The variation of d' at each condition was estimated using a resampling method combining bootstrapping and random sampling among all 100 repeated scans. **Results:** Optimized parameters in CHO calculation were determined: 9 non-overlapping slices per object, 4 objects per low-contrast object size, 2 background ROIs per object, and 4 channel filters. The bias and standard deviation were estimated at different numbers of repeated scans using these parameters. If only one single scan was used in the CHO calculation, the bias of d' was within 6%. However, the uncertainty remains high (16-21%). In order to limit the uncertainty to be 10%, 2-3 repeated scans were still required at routine dose levels. **Conclusion:** With optimized parameter settings in CHO, efficient and accurate measurement of low-contrast detectability on the commonly used ACR phantom becomes feasible.

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Ultrasound Symposium

TRACK 4

Mechanical Tissue Disruption with Focused Ultrasound

WE-D-TRACK 4-00

Mechanical Tissue Disruption with Focused Ultrasound

Moderators: Allison Payne, University of Utah, Salt Lake City, UT and Kenneth Bader, University of Chicago, Chicago, IL

WE-D-TRACK 4-01

T. Khokhlova, University of Washington:

Boiling Histotripsy for Ablation of Intrabdominal Targets

WE-D-TRACK 4-02

A. Maxwell, University of Washington:

Cavitation Control in Burst Wave Lithotripsy of Kidney Stones: A Delicate Balance

WE-D-TRACK 4-03

K. Bader, University of Chicago:

The Development of Metrics to Gauge Treatment Outcomes from Mechanical Ablation with Focused Ultrasound

WE-D-TRACK 4-04

Q&A

Ablation is among the primary tools for interventional procedures. Focused ultrasound is gaining traction for noninvasive ablation in oncological and urological applications. Early stage focused ultrasound technologies relied on heating to achieve ablation, but more recent developments indicate ultrasound-induced mechanical effects can also be utilized to achieve the desired clinical outcomes. Acoustic cavitation is the most prominent mechanical effect induced by ultrasound for therapeutic purposes. Histotripsy is a focused ultrasound technology that generates cavitation spontaneously to disintegrate soft tissues. Histotripsy has some advantages over its thermal counterparts, including its ability to ablate large and highly vascularized and large tumors with specialized sources. Cavitation also plays a role for breaking down kidney stones. Burst wave lithotripsy (BWL) is an experimental treatment that uses sinusoidal focused ultrasound pulses to noninvasively fragment urinary stones into small pieces that can naturally pass through the urinary tract. Cavitation induced by the BWL pulse produces stress at the surface of the stone that contribute to erosion of the material, but also can lead to complications from off target damage. The noninvasive application of each of these therapies requires imaging peri-treatment guidance and monitoring, and post-treatment assessment of efficacy. The choice of imaging modality depends on the intended therapeutic mechanism of action.

This session will provide an overview of the mechanical effects of ultrasound as a therapeutic modality. Focused ultrasound instrumentation will be discussed for oncological and urological applications. Standard and novel methods for image guidance of therapeutic ultrasound will be introduced, including the relevant imaging metrics to gauge therapeutic efficacy.

Learning Objectives:

1. Understand the origin of ultrasound-based therapeutic mechanical mechanisms
2. Understand how cavitation impacts stone fragmentation and tissue injury for burst wave lithotripsy, and the role of cavitation in ablation of soft tissues with histotripsy

3. Recognize the origin of cavitation-generated acoustic emissions, and how the emissions are processed to monitor cavitation-based therapies

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Multi-Disciplinary Scientific Session

TRACK 6

Autosegmentation and Image Analysis in MRI-guided Radiation Therapy

WE-D-TRACK 6-01

Auto-Correction of Inaccurate Auto-Segmentation for MRI-Guided Adaptive Radiotherapy

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Purpose: Fast and accurate auto-segmentation on daily image is essential for MR-guided adaptive radiotherapy (MRgART). However, the state-of-the-art auto-segmentation based on deep learning (e.g., deep convolutional neural networks, DCNN) has a limited success particularly for complex structures, e.g., bowels. To address this issue, we develop a method to quickly and automatically correct for unacceptable auto-segmented contours for MRgART. **Methods:** An improved active contour model (ACM) based on level set was implemented to utilize the probability maps from standard DCNN models to set per-pixel parameters and initialize contour evolution. The ACM method was tested on the DCNN auto-segmented bowel contours (initial contours) for 46 abdominal T2w-MRI sets. All MRI images were pre-processed with bias correction and contrast enhancement. For each image slice, bowel region was divided into multiple subregions with each including a bowel loop. The ACM contour correction was performed on the initial contour in each subregion. The correction performance was measured by comparing the Dice Similarity Coefficients (DSC) for the corrected contours with those for the initial contours. Manually delineated contours were used as ground truth. **Results:** The DSC for the ACM-corrected contours was increased compared to those for the initial contours. For bowel contours with initial DSC<0.5, the mean DSC was increased from 0.307 to 0.574 after correction, with 16% of the corrected contours reaching DSC≥0.8. For contours with DSC=0.5–0.8, the mean DSC was increased from 0.694 to 0.773 after correction, with DSC≥0.8 in 47% of cases. The execution time of the correction on a subregion was less than 10 seconds. **Conclusion:** A fully-automated contour correction method based on ACM was performed to quickly correct for inaccurate deep learning auto-segmentation and its effectiveness was demonstrated for complex anatomy, e.g., bowels. The method can be integrated into the segmentation step to improve and accelerate the process of MRgART.

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WE-D-TRACK 6-02

Multi-Parametric Ultra-Quality 4D-MRI Synthesis Using VoxelMorph-Based Deformable Image Registration

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Purpose: To develop a novel multi-parametric ultra-quality 4D-MRI technique through deep learning synthesis using a VoxelMorph-based deformable image registration (DIR) model. **Methods:** Twenty-seven liver tumor patients undergoing radiotherapy were collected from Beijing Cancer Hospital with IRB approval. Each patient received a T1w 4D-MRI scan using TWIST volumetric interpolated breath-hold examination (TWIST-VIBE) sequence. Patients also received multi-parametric 3D MR scans, including T1w, T2w, and diffusion-weighted MR imaging (DWI) with b-value of 50 and 800. 4D-MRI frames at the same respiratory phases with 3D MR scans were selected, and a VoxelMorph-based model was trained to obtain the deformations between selected frames and other frames. The deformations were then applied to those 3D MRIs to synthesize multi-parametric ultra-quality 4D-MRIs at corresponding respiratory phases. Contrast-to-noise ratio and full-width-half-maximum (FWHM) of the profile along lung-liver interface were calculated on different MRIs to quantitatively evaluate image quality, and tumor motion trajectories were measured to determine motion errors and subsequently the motion accuracy of synthetic 4D-MRIs. **Results:** CNR of the original T1w 4D-MRI was 6.62±3.85, and it increased to 10.54±5.64, 12.83±7.66, 25.36±14.26, and 30.42±14.02 in the synthetic ultra-quality T1w, T2w 4D-MRI, 4D-DWI (b=50), and 4D-DWI (b=800), respectively. FWHM of the liver-lung profile reduced from 6.7±2.9 mm to 4.7±2.4 mm, 3.1±1.3 mm, 6.4±4.5 mm, and 4.5±1.4 mm in the synthetic ultra-quality T1w and T2w 4D-MRI, 4D-DWI (b=50), and 4D-DWI (b=800), respectively. In addition, relative tumor motion errors were much smaller than the voxel sizes of the 3D-MRIs. **Conclusion:** This study demonstrated the feasibility of a novel multi-parametric ultra-quality 4D-MRI with deep learning-based DIR. This method overcomes many limitations of current 4D-MRI techniques and holds great promises for clinical applications.

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WE-D-TRACK 6-03

Radial MR Fingerprinting for Accelerated High Resolution Quantitative Prostate MRI in Radiation Therapy

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Purpose: magnetic resonance fingerprinting (MRF) provides simultaneous T1/T2 quantitative maps, allowing for longitudinal assessment of treatment response and disease progression. We examine the feasibility of a golden-angle radial MR fingerprinting for radiotherapy applications in prostate with a dedicated MRI simulator. **Methods:** A 2D steady-state-free-precession(SSFP) MRF sequence with inversion preparation was used with golden angle radial trajectory readout. All scans were performed on a 3T Philips scanner equipped with a flat tabletop for MRI simulation. MR fingerprinting imaging parameters are as follows: in plane resolution 1.1x1.1 mm², slice thickness=5mm, [echo time(TE),repetition time(TR)] = [8.1ms,4ms], variable flip angle=0-60 degrees. The acquisition schedule contains 500 time points, and 3 golden-angle-radial-spokes per time point. The radial MRF sequence was tested with the NIST imaging phantom and the measured T1/T2 values were validated against the ground truth values. Prostate MRF images of a healthy volunteer subject were acquired. T1/T2 maps were calculated by dot product matching to the simulated MRF dictionary. Standard relaxation mapping methods, namely Variable-flip-angle(VFA) for T1 and Multi-single-echo-spin-echo(MSE) for T2, were performed for comparison. The prostate gland(P), peripheral zone(PZ) and transition zone(TZ) regions of interest were delineated, and the statistics and distribution of the quantitative mapping values were analyzed. **Results:** T1/T2 quantitative maps were obtained covering the whole prostate gland in 3 minutes. The generated parameter maps on the NIST Phantom were in good agreement with expected T1/T2 values. All values from standard method or MRF were in agreement with published literature. The MRF-T1 average values for [P,PZ,TZ]=[1960ms,2171ms,1858ms]. Compared with VFA, MRF-T1 values were lower but achieve significantly lower standard deviations. The MRF T2 mean value for [P,PZ,TZ]=[52.7ms,68.4ms,47.4ms]. MSE-T2 resulted in higher average values and superior image quality at the cost of longer scan time. **Conclusion:** feasibility of prostate radial MRF in the RT MRI simulation has been demonstrated.

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WE-D-TRACK 6-04

Synthetic Magnetic Resonance Imaging from Computed Tomography Simulation Scans Using Deep-Learning Models for High-Dose-Rate Prostate Brachytherapy

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Purpose: For high dose rate prostate (HDR) brachytherapy at our institution, both CT and MRI are acquired to identify catheters and delineate the prostate, respectively. We propose to build a deep-learning model to generate synthetic magnetic resonance imaging (sMRI) with enhanced soft-tissue contrast from computed tomography (CT) scans. sMRI would assist physicians to accurately delineate the prostate without needing to acquire additional planning MRI. **Methods:** 58 paired post-implant CT and T2-weighted MRI sets acquired on the same day for HDR prostate patients were retrospectively curated to train and validate the conditional Generative Adversarial Networks (cGAN) algorithm Pix2pix to generate sMRI. CT and MRI images were fused using a mutual information rigid registration algorithm in the Eclipse treatment planning system. The CT images for each patient were resampled to match to the dimensions of the MRI scan. Pix2pix was then trained to generate sMRI from the CT images using the MRI images as the ground truth. The performance of the algorithm was quantitatively evaluated using the metrics of mean absolute error (MAE), mean squared error (MSE) and cosine similarity (CS) over 400 images. MAE and MSE are 0 and CS is 1 when sMRI and MRI images are identical. **Results:** sMRI images with enhanced soft-tissue contrast and catheter visualization were generated using Pix2pix for HDR prostate patients. The following performance metrics were obtained: MAE = 0.062±0.008, MSE = 0.008±0.002, CS = 0.987±0.006. Qualitatively, sMRI closely resembled the patient MRI in both soft tissues and bony anatomies in the region of near the prostate. **Conclusion:** It is feasible to generate sMRI from CT scans clearly visualizing soft tissues, the boundary of the prostate, and catheters inside the prostate. To our knowledge, this is the first time both the boundary of the prostate and catheters have been visualized on sMRI.

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WE-D-TRACK 6-05

Synthetic MRI-Aided Delineation of Organs at Risk in Head-And-Neck Radiotherapy

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Purpose: Eliminating the labor-intensive, time-consuming, and observer-dependent manual delineation of organs-at-risk (OARs) is highly desired in radiotherapy treatment planning. This study aimed to tackle the challenges remaining in achieving reliable expert performance among current auto-delineation algorithms through developing a novel synthetic MRI-aided method. **Methods:** MRI provides superior soft-tissue contrast compared to CT. By taking the complementary contrasts provided by CT and MRI into account, the accuracy of OAR delineation can be improved. In the study, synthetic MR images were firstly synthesized from CT images utilizing a cycle-consistent generative adversarial network. A novel model namely, mask scoring regional convolutional neural network, was implemented to obtaining delineation of organs through extracting and combining the features of CT and synthetic MRI. The performance of the proposed method was quantitatively assessed using both in-house and public datasets containing CT scans from head-and-neck (HN) cancer patients. Metrics including Dice similarity coefficient (DSC), 95th percentile Hausdorff distance (HD95), mean surface distance (MSD), and residual mean square distance (RMS) were calculated to evaluate the outperformance of our proposed method against current state-of-the-art algorithms. **Results:** A mean DSC of 0.77, mean HD95 of 2.90 mm, mean MSD of 0.89 mm, mean RMS of 1.44 mm were achieved by our proposed method across all of 18 OARs in our in-house dataset which contains images from 70 patients, which outperforms the current state-of-the-art algorithms by 6%, 16%, 25%, and 36%, respectively. A mean DSC of 0.86 was achieved for all 9 OARs on public dataset which has 48 patients, which is 6% better than the competing methods. **Conclusion:** In this study, a synthetic MRI-aided deep learning framework for automated delineation of OARs in HN CT has been developed. The proposed method will have a high impact in rapid OAR contouring in routine HN cancer radiotherapy treatment planning.

WE-D-TRACK 6-06

Learn from What You Drew - a Deep Learning Assisted Fast Semi-Automatic Segmentation for Complex Anatomy On MRI

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Purpose: Existing auto-segmentation has limited success for complex organs (e.g., bowels). Consequently, time-consuming slice-by-slice review and editing is often required. To address this issue, we propose a deep learning based, fast, semi-automatic contouring method that predicts a contour on a slice using the user-approved contour from the previous slice. **Methods:** To demonstrate the idea, the deep learning method was trained and tested for stomach and small bowel contours on T2-weighted MRIs acquired from 48 patients with abdominal tumors. All the MRIs were pre-processed (e.g., bias corrected, normalized, and cropped) and carefully delineated (ground truth). The network backbone used was UNet with 3-channel input, including an image slice (to be segmented), and the image and contour of previous slice. Data augmentation by randomized 3D rotation, scaling and different image pair sampling using 1, 2, or 3 slices intervals was used in the model training. Patient-based leave-one-out-cross-validation was employed to evaluate the model performance. Quality of the predicted contour was evaluated quantitatively using Dice similarity coefficient (DSC) and qualitatively by a human rater with scores of 1, 2 and 3 indicating acceptable, minor editing, and major editing, respectively. **Results:** The average 2D-DSC of the predicted contours for each patient was in the range of (0.82-0.96) / (0.74-0.93) with a median value of 0.90/0.88 for stomach/small bowel contours. The percentage of the predicated contours with scores of 1 to 3 was 82.4 ± 10.5%, 14.6 ± 8.9% and 3.1 ± 4.8% for stomach and 72.7 ± 19.0%, 23.4 ± 15.2% and 3.9 ± 5.7% for small bowel. **Conclusion:** The proposed method can robustly predict a contour on an MRI slice based on the contour from previous slice for complex structures (e.g. stomach and small bowel). It can be integrated into existing workflows, facilitating fast and accurate segmentation, which is particularly important for MR-guided adaptive radiation therapy.

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WE-D-TRACK 6-07

A Prior Knowledge Guided Deep Learning Model for GTV Auto-Segmentation On 4DMRI in MR-Guided Adaptive Radiotherapy of Pancreatic Cancer

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Purpose: Fast and consistent GTV segmentation in non-contrast enhanced daily 4DMRI is challenging for MR-guided Online Adaptive Radiation Therapy (MRgOART) for abdominal tumors. In this work, we develop a prior knowledge guided deep learning model specifically for auto-segmentation of pancreatic GTV on daily 4DMRI for MRgOART. **Methods:** Seventy-three pairs of verification and daily mid-position MRI sets acquired from 12 pancreatic cancer patients using 4D Vane sequence on a 1.5T MR-Linac during MRgOART were used to train and test the deep learning model with patient-based leave-one-out cross-validation. The model used the verification MRI and GTV mask (obtained before the first fraction) as a guidance for the daily MRI. A region of interest (ROI) was selected as the minimum bounding box of the verification GTV contour plus a margin of 15 mm to account for inter-fractional variations. Model input size was 128×128×3 with 3 channels corresponding to verification MRI, verification GTV mask, and daily MRI. Data augmentation methods such as translation, rotation, flip, guidance-target inversion, and rigid registration error introduction were applied for robust model training. Model performance was evaluated using dice similarity coefficient (DSC). **Results:** With the small size of the 4D Vane MRI dataset, the proposed model achieved an average DSC of 0.66±0.13 for all the cases studied. While for the 8 cases

with GTV >5 cc, the average DSC was 0.71, comparable to the inter-observer variation of DSC = 0.71 estimated in a previous study. The proposed model took less than 1 second to create the GTV contour on a daily MRI set. **Conclusion:** We developed a prior knowledge guided deep learning model to segment the pancreatic GTV in the daily non-contrast enhanced MRI using a composite image-mask input. The proposed model achieved clinically relevant DSC for cases with GTV >5 cc.

Imaging Interactive ePoster Discussion

TRACK 1

Imaging: CBCT Artifacts

WE-lePD-TRACK 1-01

Deep Learning Model for Scatter Correction in Dedicated Breast CT

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Purpose: To develop a deep learning (DL) model for scatter correction of dedicated breast CT (BCT) projection images. **Methods:** 118 digital breast phantoms were obtained by automatic segmentation of as many BCT patient images and augmented via random translation of the breast in the image field of view to a total of 470 samples. The samples were used to simulate the primary and scatter projections of a clinical BCT system using Monte Carlo (MC) simulations. Twelve projections were simulated for each sample, one every 30° covering a full 360° acquisition. The projections simulated from 450 samples were used to train the DL model to predict the scatter from the input primary+scatter projections. The model was trained end-to-end by unifying information from four orthogonal projections, to account for breast size and position. It was then tested on the remaining samples, and on acquired projections of a physical breast phantom. The scatter-corrected projections of this phantom were reconstructed, and then compared to the uncorrected reconstruction and a reconstruction from MC-corrected projections. **Results:** Inside the breast, the mean percentage error (min, max) between the MC-simulated and the DL-predicted scatter was -1% (-10%, 4%) and the mean standard deviation of the error within a projection was 6% (2%, 13%). In a homogeneous region of the physical phantom reconstruction, the mean ΔHU between ROIs close to the perimeter and the center were 86, 24, -64 for the uncorrected, MC-corrected, and DL-corrected images, respectively. **Conclusion:** The proposed DL model showed promising results for scatter correction in BCT images. Moreover, its performance allows for almost real-time processing, correcting 300 projections in 17 seconds on a standard PC. Future work will involve the evaluation of different DL architectures and learning algorithms to further improve the precision of the method, and additional evaluation with physical phantoms and patient images.

WE-lePD-TRACK 1-02

Scatter Correlation and Correction for Cone-Beam CT Using Dual-Layer Flat Panel Detector

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Purpose: Scatter remains a big challenge and dominating factor of affecting dual-energy CT performance. The goal of this work is to study the correlation and correction of the x rays scattering between the layers for Cone-Beam CT using a prototype dual-layer flat-panel detector. **Methods:** A physical model about the relationship of the scatter distribution between layers is proposed, it may be described by a function related to the average scatter energy. Since the average scatter energy and the projection value are both related to the size of the phantom, we approximately assume the correlation is a function of the low and high projections. The dual layer detector consists of a top and bottom CsI scintillator with the same thickness of 550 μm, no extra filter in-between the two layers, the active area of 43.2*43.2 cm², and the pixel size of 150 μm. Experiments were performed using 120 kV X-ray voltage, 2*2 detector binning mode, and a multi-energy CT phantom (Gammex, Model 1472). The projections of the phantom in short, long axis, and the separate head part were acquired using cone beam and fan beam respectively. The difference between cone beam and fan beam transmission images is considered to be the contribution from scatter. **Results:** Three groups of the experiments show that scatter distribution from top (Is1) and bottom (Is2) layers satisfy the relation $Is2 = \alpha Is1$, where α follows our physics model. The correlation coefficients of the bottom distribution and the normalized top distribution are 99.3%, 99.7%, 93.4%, respectively. **Conclusion:** A scatter correlation model is proposed to describe the relationship of the scatter between two layers. The preliminary experimental results are basically consistent with the expectation, further validation of the scatter correlation model and scatter correction are underway with MC simulations and real experiments.

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WE-lePD-TRACK 1-03

Correcting Cone-Beam CT for Patient-Generated Scatter Using Bayesian Monte Carlo Extrapolation

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Purpose: Cone-beam computed tomography (CBCT) is used in image-guided radiotherapy to ensure correct patient positioning prior to treatment delivery. However, patient-generated x-ray scatter degrades CBCT image quality. Monte Carlo (MC) simulations offer the most physically accurate means to predict scatter signals, which can improve CBCT images via subtraction in projection space. Unfortunately, MC simulations can be too computationally costly for clinical application. We present a novel Bayesian Monte Carlo extrapolation (BMCE) algorithm that uses Bayesian statistics to mitigate the statistical noise present in MC-estimated scatter signals. The BMCE scatter correction method is then used to correct a 200-projection CBCT scan of a lung cancer patient. **Methods:** The BMCE method uses MC simulations to estimate the scatter component of each CBCT projection by transporting a 120 kVp x-ray beam through the planning 4DCT volume. Statistical noise present in the MC-simulated scatter signals was then corrected using an iterative Bayesian extrapolation postprocessing algorithm. Finally, the Bayesian-extrapolated scatter signals were subtracted from CBCT projections and images were reconstructed in 3D and 4D using the Feldkamp-Davis-Kress (FDK) algorithm. The quality of the CBCT images corrected using our BMCE algorithm was quantified in terms of contrast-to-noise ratio (CNR) and Hounsfield unit (HU) integrity. Image quality was then compared to images reconstructed without any scatter correction. **Results:** The BMCE scatter correction algorithm increased CNR for 3D and 4D reconstructions by factors of 3.4 and 3.8 respectively, and decreased the root mean square error (RMSE) between the HU-calibrated 3DCBCT reconstruction and the planning 3DCT by 30% relative to the 3DCBCT image reconstructed without any scatter correction. **Conclusion:** We demonstrate that Bayesian extrapolation of scatter signals estimated using Monte Carlo simulations with low numbers of primary x-rays can render otherwise noisy scatter distributions suitable for implementation in a CBCT scatter correction algorithm.

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WE-lePD-TRACK 1-04

Gas Bubble Motion Artifact Reduction Through Simultaneous Motion Estimation and Image Reconstruction

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Purpose: Motion of gas bubbles in abdominal region can produce significant artifacts during on-board CBCT scanning, which adversely affects the imaging quality

and limits the process of CBCT-based adaptive planning and image-guided radiotherapy. In this study, we aim to test the effectiveness of the simultaneous motion estimation and image reconstruction technique (SMRIE) for improving CBCT image quality and HU accuracy for abdominal scan. **Methods:** Two 3D abdominal CT images for a same patient acquired with 3 mins time interval were used in this study and the deformation vector field (DVF) between these two scans were calculated. Based on the calculated DVF, we simulated n ($n=0, 2, 4, 6, 8$) different DVFs through adding random motion vectors, and the corresponding n different CT images were generated. A $n+2$ phases 360° CBCT projection was then simulated through combining the projection images from the $n+2$ CT images, where projections from every continuous $360^\circ/(n+2)$ are from the same CT. Then we applied SMEIR algorithm onto the generated multi-phase CBCT projection, the motions of gas bubbles between these phases were iteratively predicted through inverse-consistent 2D-3D registration following motion compensated simultaneous algebraic reconstruction technique (SART), and the motion induced artifacts were corrected in the meanwhile. **Results:** When using SART for CBCT reconstruction, gas bubble motion artifacts in the simulated CBCT image showed the same pattern to those in the real abdominal CBCT scan. By introducing SMRIE technique, the boundary of structures and HU inaccuracy of gas bubble region are well corrected in the reconstructed CBCT image compared with SART reconstruction result. **Conclusion:** We introduced SMRIE technique for gas bubble motion artifact reduction, the improved image quality in the result demonstrated that SMRIE technique is promising for on-board CBCT gas bubble motion artifact reduction.

WE-lePD-TRACK 1-05

Scatter Correction for Cone Beam CT Using Half-Beam Blocker and Local-Filtration Scatter Estimation

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Purpose: We propose a half beam blocker-based scatter correction method combined with local filtration algorithm to remove the scatter-induced artifacts in single-scan cone-beam CT(CBCT) imaging. **Methods:** A head phantom and an electron density phantom were scanned in our in-house developed CBCT platform. A half beam blocker composed of lead strips was inserted between the x-ray source and the phantom, and used to block about half of the beam area. The shaded regions behind lead strips were considered as scatter signal. The scatter signal was smoothened with a threshold-based local filtration algorithm. Finally, the scatter free projections were produced by subtracting the estimated scatter from the unblocked half projections, and were used for volumetric reconstruction with FDK algorithm. The scatter corrected CBCT images were compared with fan beam CT and those without scatter correction. **Results:** Compared with the CBCT without correction, the error of CT number was reduced from more than 200 HU to less than 7 HU for the electron density phantom, and from more than 148 HU to less than 14 HU for the head phantom. The spatial uniformity was improved by a factor of 3. **Conclusion:** The proposed scatter correction method obtained accurate CT numbers and improved image spatial uniformity, without additional effort of another scan. This method has promising application in CBCT guided radiotherapy for better adaptive radiation dose calculation.

WE-lePD-TRACK 1-06

Reducing Scan Time and Dose: Motion Compensated Reconstruction for Adaptive 4DCBCT

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Purpose: Prior to thoracic radiotherapy treatments, patients are commonly aligned using conventional 4DCBCT scans (1320 projections over 4 minutes). The ADAPT clinical trial performs adaptive 4DCBCT scans by modulating gantry rotation speed and projection acquisition in response to real-time changes in the patients breathing rate with as low as 200 projections. This study determines reconstruction algorithms that best complement adaptive 4DCBCT acquisitions for reducing imaging dose and scantime whilst maintaining or improving image quality compared to conventional 4DCBCT acquisitions from 20 ADAPT clinical trial patients (NCT04070586). **Methods:** Adaptive acquisitions from 20 patients treated for lung cancer as part of the ADAPT clinical trial were used in this study. Each patient dataset consisted of a planning 4DCT and 3 4DCBCT scans acquired on two treatment fractions (conventional 1320, adaptive 600 and adaptive 200 projections scans were acquired). All scans were reconstructed using Feldkamp-Davis-Kress (FDK) as a comparator, and the adaptive scans are reconstructed using McKinnon-Bates (MKB), Motion Compensated MKB (MCMKB) and motion compensated FDK (MCFDK). Reconstructed image quality was quantified using Structural Similarity Index (SSIM), Signal-to-Noise-Ratio (SNR), Contrast-to-Noise-Ratio (CNR), Tissue Interface Sharpness at the diaphragm (TIS-D) and tumor (TIS-T). **Results:** 200 and 600 projection adaptive 4DCBCT acquisition delivered 85% and 55% less imaging dose and 62% and 4% shorter scan times, respectively. Motion compensated reconstruction maintained or improved image quality metrics comparable to conventional reconstructions. The 200 projection adaptive scan with MCFDK (SNR=21.0±9.9, CNR=13.0±4.9, TIS-D=0.8±0.9, TIS-T=0.8±0.6) and MCMKB (SNR=22.1±10.7, CNR=14.1±5.3, TIS-D=1.3±1.1, TIS-T=0.9±0.3) reconstructions improved image quality metrics compared to conventional scans (SNR=11.9±4.2, CNR=9.5±3.1, TIS-D=0.58±0.3, TIS-T=0.4±0.1). **Conclusion:** Combining adaptive acquisition with motion compensation allows significant reductions in 4DCBCT scan times to be realized in the clinic on a standard linac.

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WE-lePD-TRACK 1-07

A Generative Adversarial Network (GAN)-Based Technique for Synthesizing On-Board Imaging Artifacts in the Extended Cardiac-Torso (XCAT) Phantoms

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Purpose: Simulating on-board imaging in digital phantoms is valuable for evaluating and optimizing of on-board treatment techniques like adaptive therapy. This study aims to synthesize CBCT imaging artifacts for different acquisition conditions in XCAT phantoms by developing a GAN-based deep-learning technique. **Methods:** An artifact simulation model is developed using CT images and image acquisition conditions as inputs, and is trained to generate CBCT images with the conditions-specific imaging artifacts. Then, XCAT phantoms are input to the model with user-desired image acquisition conditions to simulate CBCT artifacts in XCAT. The model was trained in two steps: (1) simulate cone and under-sampling artifacts for different projection numbers (450, 150, or 100); (2) simulate scatter and beam-hardening artifacts. Cone-beam projections of the CT volumes were generated by ray-tracing and used to reconstruct CBCTs using FDK-backprojection, which were used for the output for step 1 and input for step 2. The second-step model outputs corresponding CBCTs reconstructed using Monte-Carlo (MC) projections. We incorporated 15 patients and 5 XCAT phantoms for model training, validation, and testing. We evaluated the model performance with peak-signal-to-noise-ratio (PSNR), structural-similarity (SSIM), cross-correlation and NPS (noise power spectrum) correlation coefficient (NCC). **Results:** For the under-sampling and cone artifacts simulation, the PSNR reached 38.58/37.24/36.92, and the SSIM reached 0.987/0.978/0.966 for the testing patient with 450/225/100 projections, respectively. The cross-correlation and NCC achieved at least 0.99 for all projection numbers. Qualitatively, the model successfully simulated the streak artifact due to under-sampling and cone artifact in the XCAT phantom. For scatter simulation, the PSNR reached 28.05, and SSIM reached 0.904 for the testing patient. The cross-correlation was 0.927, and NCC was 0.984. **Conclusion:** The results demonstrated the feasibility of synthesizing CBCT artifacts in the 4D-XCAT phantom using the proposed method. This crucial development enhances the value of XCAT phantoms for evaluating and optimizing various on-board imaging and treatment delivery techniques.

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Imaging Interactive ePoster Discussion

TRACK 2

Imaging: MRI Image Analysis, Biomarkers, and Radiomics

WE-IePD-TRACK 2-01

High Spatial Resolution Simultaneous Quantitative T1rho and T2 Mapping at 3T

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Purpose: Quantitative T1ρ and T2 MR mapping are promising MRI techniques for biochemical evaluation for soft-tissue, particularly articular cartilages, brain white matter, and tumors. Both techniques suffer from long scan duration. We present a novel approach in which T1ρ and T2 MRI are acquired and reconstructed simultaneously with quadrupled spatial resolution and nearly halved scan time compared to traditional approaches. **Methods:** All experiments were performed using a 3T MR scanner (Ingenia, Philips) equipped with a 1Tx/16Rx knee coil. A 3D Magnetization-Prepared Angle-Modulated Partitioned k-Space Spoiled Gradient Echo Snapshots (3D MAPSS) sequence was implemented for high resolution 3D T1ρ and T2 mapping. To improve sensitivity, reduce artifacts, and reduce total scan duration, T1ρ preparation with adiabatic RF pulses, synovial fluid suppression, unpaired RF phase cycling, optimized spin-lock time or TE sampling schemes, and combined T1ρ and T2 imaging and map generation were implemented in a novel sequence termed “FastMAPSS”. The single combined sequence had high spatial resolution of 0.3×0.5×2 mm³ with 5:55min scan duration, while the standard T1ρ and T2 sequences had 0.44×0.77×4 mm³ acquired within 5 min each. **Results:** SNR efficiency was greatly improved, and artifacts were maximally suppressed in the FastMAPSS sequence when integrating T1ρ and T2 mapping in one sequence and employing synergistically complementary curve fitting in the same equation for both maps using the complex MRI data. The combined T1ρ and T2 acquisition also avoided the need to co-register T1ρ and T2 images. As a result, high-resolution MRI (0.3×0.5×2 mm³) with laminar specificity on a 3T clinical scanner was achieved by simultaneous T1ρ and T2 mapping of the whole-knee cartilage in less than 6 minutes. **Conclusion:** The combined T1ρ and T2 FastMAPSS MRI sequence allowed quadrupled spatial resolution with nearly halved scan time compared to the traditional approaches.

WE-IePD-TRACK 2-02

Evaluating and Improving the Bloch Fitting Method for the Analysis of AcidoCEST MRI

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Purpose: The Bloch-McConnell equations can be used to fit CEST spectra to estimate the proton chemical exchange rates (k_{ex}) and pH of a sample. This study aims to test the improvement of the fitting algorithm when the fitting parameters are fixed with T₁, T₂, B₁ and B₀ experimental values. **Methods:** Two-hundred iopamidol samples were prepared with 5 concentrations (5-50 mM), 5 T₁ values (0.4-3.9 sec) and 8 pH levels (6.25-7.30, equally spaced). Thirty-six CEST scans were performed on each sample using a FISP acquisition sequence and a cross-combination of six saturation powers (0.5-6 μT) and six saturation times (0.5-6 sec) with a Bruker 7T preclinical scanner. The set of 36 CEST scans was repeated at five temperatures (31-43 °C). In addition, T₁, T₂, B₁ and B₀ maps were also acquired. Image analysis and Bloch-fitting was performed with MATLAB. **Results:** The changes in the iopamidol amide contrasts in Z-spectra with CEST agent concentrations, T₁ values, sample pH values, saturation powers and saturation times match well with previous studies. Our Bloch fitting algorithm can detect the increase of amide k_{ex} with increasing sample pH and temperatures. Reducing the number of parameters fitted in the Bloch-fitting process by fixing the parameters with experimentally measured values does not significantly affect the fitting results of mid-to-high pH (>6.55) iopamidol samples, but can increase the sensitivity to the pH changes of low pH (<6.55) samples. Including T₁, T₂ and B₁ experimental information also significantly reduces the computation time. **Conclusion:** Our Bloch-fitting algorithm is capable of detecting the changes of k_{ex} of iopamidol protons at different pH levels. The inclusion of experimental information in the Bloch-fitting process improves the accuracy and precision of the fitting result and shortens the fitting time.

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WE-IePD-TRACK 2-03

3D Brain Tumor Segmentation Through Multimodal Weighted Network in MRI

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Purpose: Accurate brain tumor segmentation in Magnetic resonance imaging (MRI) images plays an important role in image-guided surgery and treatment planning, while unreliable may cause mislead surgery. We aim to develop a new multimodal weighted network (MW-Net) for three-dimension (3D) brain tumor segmentation. **Methods:** Totally 285 patients including 210 high grade and 75 low grade gliomas from Brats 2018 dataset are used in this study. The registration is performed on four available modalities such as T1, T1Gd, T2 and FLAIR on each case. In MW-Net, 3D-U-Net which is developed based on U-net are used as base model, which consists of convolutional, pooling and upsampling layers in 3D way. The activation function is ReLU, and the loss function is cross entropy. Adam function is used to tune the model parameters. Then four modalities with the corresponding weights fed into the shared 3D-U-Net model. A backpropagation algorithm is employed to train MW-Net. In testing stage, the 3D brain image is directly fed into trained MW-Net and the 3D segmentation results can be obtained. Dice coefficient (Dice), Positive Predictive Value (PPV) are sensitivity (SEN) are used as evaluation criterion. The better segmentation results can be obtained with higher Dice, SEN and lower PPV. **Results:** Five-fold cross validation is performed in this study. The evaluation results for MW-Net on Dice, PPV and SEN are 0.8824, 0.8823, and 0.8963, respectively, while 3D-U-Net are 0.8524, 0.8995, and 0.8475. **Conclusion:** A new multimodal weighted network (MW-Net) for three-dimension (3D) brain tumor segmentation was developed. The experimental results demonstrated that MW-Net can obtain better segmentation performance compared with available methods.

WE-IePD-TRACK 2-04

MRI Super-Resolution Via Real-World-Like Training Set Preparation

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Purpose: Current super-resolution (SR) methods suffer from poor performance on real-world MR images due to the simple translation model (e.g., Gaussian blur) used when preparing low-resolution (LR) images for training. We propose a deep learning-based (DL-based) method that generates more realistic LR images and use them to train our SR network to achieve better performance on real-world LR MR images. **Methods:** A two-step workflow was used: (a) dataset preparation via the proposed method. (b) SR network training. In the first step, a Generative Adversarial Network (GAN) was constructed to simulate the downsampling process. The generator of the proposed GAN is comprised of six convolutional layers without non-linear activation layer to simulate the low-pass filtration. The receptive field of the discriminator is 7-by-7. During the adversarial training, the generator output fake LR MR images that mimicked the real LR MR images on patch-level. Matched LR and HR MRI pairs were not needed since the receptive field of discriminator is small. An Enhanced Deep Super-Resolution (EDSR) network was used for SR.

A total of 2624 axial HR and LR images from 20 patients scanned with TWIST-VIBE sequence were used for training. Two experiments were conducted using different methods to prepare LR MR images: proposed GAN and conventional Gaussian filtration. Real LR images in this dataset were used to evaluate the SR network. **Results:** Severe artifacts were observed with the SR network that was trained with Gaussian blurred LR images. Better resolution enhancement was observed with the SR network that was trained with the data prepared by the proposed adversarial training method. This visual difference was verified by SSIM and PSNR. The SSIM improved by 0.111 ± 0.016 and the PSNR improved by 2.76 ± 0.98 dB. **Conclusion:** The proposed method outperformed the Gaussian blur method by demonstrating better resolution enhancement and fewer artifacts on real-world MRI.

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WE-lePD-TRACK 2-05

Towards a Probabilistic Neurochemical Atlas of the Human Cerebellum Using ZOOM MRSI

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Purpose: Magnetic Resonance Spectroscopic Imaging (MRSI) combines the principles of spectroscopy and anatomical imaging to create images mapping relative metabolite concentrations within each pixel. However, traditional MRSI sequences lack the necessary resolving power needed to distinguish functional subregions of the cerebellum (such as white and grey matter boundaries). This work's purpose is to create a probabilistic neurochemical model of the cerebellum using ZOOM MRSI, a novel sequence developed by our group which obtains 2.5mm in-plane resolutions in 9 minutes and 36 seconds. **Methods:** Six healthy volunteers gave informed consent to be subjects for this experiment, which involved in-vivo scans in the cerebellum for two and the posterior cingulate for the other four. The scans were performed on a Siemens Prisma 3 Tesla (Siemens, Germany) scanner with a 64-channel receive only head coil. ZOOM MRSI was performed on each subject via metabolite-cycled semi-LASER localization (TE = 32ms, TR = 2s). Localizations were placed on the central cerebellar slice and posterior cingulate slice within the transverse plane. MATLAB (R2016b, MathWorks, Natick, MA) was used as the reconstruction software, with LCModel (Version 6.3-1M) being utilized for metabolite quantification. Normalized metabolite levels were calculated using the water-scaled signal intensity of each metabolite. Lastly, test/retest reliability will be evaluated with Intraclass Coefficient (ICC), Percent Volume Change (PVC), and Pearson Correlation Coefficient (PCC). **Results:** Coronal, sagittal, and axial images of the Montreal Neurologist Institute-152 template were fitted with ZOOM MRSI images and found to have significant overlap. Additionally, normalized maps for Glutamate+Glutamine for two healthy subjects (for two different scans) were also developed. **Conclusion:** ZOOM MRSI advantages over conventional MRSI methods is demonstrated through its reliable quantification for tNAA, tCr, tCho, Glx, and myo- ins in a voxel of isotropic size equivalent to 62.5uL. This is due to the sequence's improved resolution and spectral quality.

WE-lePD-TRACK 2-06

Classification of Tumor Progression and Radiation Necrosis Following Gamma Knife for Brain Metastases Using Radiomic Features and Clinical Variables

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Purpose: To develop a classification tool to differentiate between tumor progression and necrosis following Gamma Knife treatment for brain metastases based on both radiomic features and clinical variables. **Methods:** Under IRB approval, this analysis includes a retrospective cohort of 154 patients treated with Gamma Knife for brain metastases who had contrast-enhanced T1-weighted MR images acquired using 3-D spoiled gradient echo (SPGR) sequences and surgical resection of the suspected lesion. Lesions were segmented using semi-automated methods, and tumor/necrosis was determined pathologically (100% necrosis vs. tumor present). 3-D radiomic features ($3 \times 3 \times 3$ -mm spatial normalization, 64-bin intensity discretization) were extracted using PyRadiomics and clinical variables via chart review. Z-scores were used to normalize feature values. Feature reduction was performed using principal component analysis (components represented 95% explained variance). Fifteen classification models and ensembles were trained, and performance was assessed using 5-fold cross-validation. **Results:** 100 radiomic features were selected a priori and extracted. Ten clinical variables were implemented, including gender, race, age at diagnosis, time from Gamma Knife treatment, histology, time between primary and metastasis diagnoses, metastasis location, Gamma Knife dose, and Gamma Knife treatment volume. A RUSBoosted trees ensemble yielded a classification accuracy of 66.2% (AUC = 0.71). 12 out of 17 necrosis cases were correctly predicted, and 90 out of 137 tumor progression cases were correctly predicted. **Conclusion:** These results are promising for the development of a reliable classification tool for tumor progression and radiation necrosis using data that is readily clinically available. Such a tool could have significant value by assisting with clinical decision making. Further investigation of classifiers and parameter optimization may yield higher performance.

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WE-lePD-TRACK 2-07

Prediction of pH Using AcidoCEST MRI and Machine Learning

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Purpose: To test and evaluate the performance of a machine learning (ML) algorithm in predicting pH from CEST spectra of iopamidol. **Methods:** Two-hundred iopamidol samples were prepared with 5 concentrations (5-50 mM), 5 T₁ values (0.4-3.9 sec) and 8 pH levels (6.25-7.30, equally spaced). Thirty-six CEST scans were performed on each sample using a FISP acquisition sequence and a cross-combination of six saturation powers (0.5-6 μ T) and six saturation times (0.5-6 sec) with a Bruker 7T preclinical scanner. The set of 36 CEST scans was repeated at five temperatures. For each CEST scan, 81 images were collected at different saturation frequencies. In addition, T₁, T₂, B₁ and B₀ maps were also acquired. All 36,000 CEST spectra from the 200 samples and the T₁, T₂, B₁ and B₀ data for each phantom sample were used to the train and validate the ML algorithm. **Results:** ML is able to produce reliable pH prediction from iopamidol CEST spectra. As expected, the accuracy of pH prediction increases as the increase of saturation powers and times. However, to our surprise, the ML algorithm can achieve a 95% accuracy even for spectra acquired with 0.5 μ T power and 0.5 sec time, which is not easily achievable by other analysis methods. This advantage of ML likely arises from that no assumptions need to be made about what parts of the spectra are useful. We also demonstrated that the pH prediction accuracy can be retained when the number of saturation frequencies is decreased from 81 to 27, therefore eliminating the need for data acquisition for redundant frequencies. **Conclusion:** Employing a ML algorithm is an effective method in predicting pH based on iopamidol spectra across a wide range of pH and saturation parameters. The accuracy of pH prediction can be retained when the saturation frequency list is truncated.

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Multi-Disciplinary Interactive ePoster Discussion TRACK 3

Multi-Disciplinary: MR-guided Adaptive Radiation Therapy

WE-lePD-TRACK 3-01

A Framework of Utilizing Biological Models for Iso-Toxic Adaptive Replanning in MR-Guided Radiotherapy

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Purpose: MR-guided online adaptive radiotherapy (MRgOART) allows iso-toxic treatments, i.e., adjusting fraction target dose to take advantage of daily anatomic and/or biologic changes while maintaining the same doses to organs at risk (OAR). This work describes a framework for iso-toxic planning in MRgOART using biophysical models. **Methods:** This framework includes (1) converting fractional adaptive plan optimized based on the daily anatomy to biologically effective dose (BED) distribution, (2) calculating composite BED dose distribution for total or a portion of the fractions delivered, (3) converting the composite BED plan to the physical dose distribution with the original fractionation, and (4) calculating tumor control probability (TCP) and normal tissue complication probability (NTCP) using the dose volume histograms and the tumor/organ specific model parameters derived from published dose-response data. Data collected from selected pancreatic cancer cases treated with SBRT using MRgOART were used to demonstrate the utility of the proposed framework. The fractional target dose was rescaled such that the same OAR maximum doses (Dmax), or doses to 0.03 or 1.0 cm³ (D0.03cc or D1cc), were maintained. Comparison was made with respect to the treatment planning goals (TPGs) and the sum daily dose (SDD) of D0.03cc & D1cc and the ratio of SDD to TPGs was calculated. **Results:** The proposed framework allowed to calculate BEDs for increased/decreased fraction doses based on daily anatomy change (e.g., OAR moved away from the target), and to estimate fraction doses for the remaining fractions considering total BED. For example, the ratio of SDD to TPGs were < 0.65 in one case, which would allow to increase fraction dose in subsequent fractions or to deliver an extra fraction. **Conclusion:** A framework for iso-toxic adaptive planning is developed to account for biological effect of different fraction doses and to optimize the therapeutic ratio in MRgOART.

WE-lePD-TRACK 3-02

Changes in Prostate Volume and Dosimetry with Stereotactic MR-Guided Adaptive Radiotherapy

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Purpose: MR-guided adaptive radiotherapy enables visualization of the prostate and surrounding anatomy at each treatment fraction and ability to adapt the treatment plan to account for interfraction variations in anatomy. This study investigates changes in prostate volume throughout treatment and its impact dosimetrically. **Methods:** Prostate cancer patients were treated with MR-guided adaptive SBRT (MRIdian, Viewray Inc.) at our institution between 06-2020 and 02-2021 to 36.25Gy in 5 fractions. The “original” plan was re-optimized at each treatment based on daily anatomy to improve target coverage and meet organ-at-risk (OAR) metrics. A “predicted” plan was generated at each fraction by re-calculating the original plan on the day-of anatomy, estimating what would have been delivered without adaptive capabilities. Dose to the PTV (V36.25) and OARs (rectum, bladder, urethra) were assessed. Differences between simulation and treatment fractions for prostate volume and PTV V36.25 were calculated. **Results:** A cohort of 13 patients treated in 65 fractions was analyzed. The median prostate volume was 29.3cc (range: 16.2-104.6cc) at time of simulation. Increases in prostate volume were observed for 51/65 fractions. At the first treatment, the prostate volume was similar to simulation (median difference of -0.2cc (range: -1.5-0.49cc)). However, the median volume increased from simulation at successive fractions (fractions 2, 3, 4 and 5) by 1.1cc (range: 0.18-6.8cc), 1.2cc (range: 0.02-6.52cc), 1.0cc (range: -0.81-8.2cc) and 0.8cc (range: -1.79-7.71cc), respectively. As prostate volume increased, the predicted plan PTV V36.25 decreased by a median of -2.2% on the first fraction, and -10%, -8.9%, -6.6%, -8.4% on successive fractions. Adaptive planning improved PTV V36.25 on 54/65 fractions; remaining fractions required sacrificing PTV coverage to meet OAR metrics. **Conclusion:** MR-guided prostate SBRT revealed a consistent increase in prostate volume after first fraction of treatment, which can impact PTV coverage. Adaptive re-optimization can significantly improve PTV coverage while minimizing toxicity to OARs.

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WE-lePD-TRACK 3-03

Deep Learning-Based Automatic Plan Prediction for MR-Guided Adaptive Radiotherapy

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Purpose: Fast and automated plan generation is desirable for MR-guided adaptive radiation therapy (MRgART). Recently, significant progress has been made in predicting an optimal dose distribution using deep learning approaches. As a next step for automatic planning, we investigate the feasibility of predicting a deliverable plan for a target dose distribution based on deep learning. **Methods:** A conditional generative adversarial network (cGAN) was trained to predict the MLC leaf sequence corresponding to a target dose distribution. The training data were 50 ground truth dose distributions and corresponding IMRT plans that were optimized during MRgART in a treatment planning system (TPS) for 10 pancreatic cancer patients (each with 5 fractions). The model input was the dose distribution from each individual beam and the output was the predicted corresponding field segments with specific shape and weight. The predicted plan was then generated and transferred back to TPS to calculate the corresponding dose. Patient-based leave-one-out-cross-validation was employed and the model performance was evaluated by calculating the gamma passing rate of the ground truth dose vs. the dose from the predicted plan. **Results:** The average gamma passing rate (95%, 3mm/3%) among 10 test cases was 88%. With relaxed passing criteria (95%, 5mm/5%), the average passing rate was 96%. In general, we observed 95% of the prescription dose to PTV achieved with a slight increase of dose to OARs. Complete plans were predicted in <15 seconds using a GTX 1660TI GPU. **Conclusion:** We have developed a deep learning method to predict IMRT plans rapidly and automatically for pancreatic cancer. With further developments using large datasets, the method may be implemented to accelerate MRgART process or to facilitate real-time MRgART.

This work is supported by Manteia Technologies.

WE-lePD-TRACK 3-04

Dose Accumulation Comparison Between Adapt-To-Shape and Adapt-To-Position for MRL Adaptive SBRT Prostate Patients

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Purpose: To use dose accumulation to directly compare two strategies for online adaptation on an MR-Linac (MRL): the adapt-to-position (ATP) isocentric shift approach versus the full re-optimization adapt-to-shape (ATS) approach for prostate SBRT treatment. **Methods:** This study analyzed 12 prostate SBRT patients treated at our centre with 30 Gy in 5 fractions on a 1.5 T MRL using 9-field IMRT with MR-based reference planning. The ATP workflow adapts the beams following translation-only rigid registration of the prostate. The ATS workflow generates a new treatment plan to match the daily anatomy using deformable image registration (DIR) and manual online contour editing by a Radiation Oncologist. Treatment plans on the localization MR images were created for all fractions using both ATP (simulated) and ATS (clinical) workflows. All plans were imported into a commercial treatment planning system to perform DIRs and accumulate dose on the reference MR images. Clinical goal DVH metrics for accumulated ATP and ATS dose distributions were compared with Student's t-tests. **Results:** Mean ($\pm$ STD) CTV D98 was not significantly different between ATP (3132 ± 129 cGy) and ATS (3180 ± 90 cGy), and satisfied the > 2850 cGy clinical goal for 11 of 12 patients for ATP and all patients for ATS. DVH goals for femurs (D5 < 1200 cGy), rectum (D1cm3 < 3000 cGy, D20 < 2000 cGy, D50 < 1000 cGy) and bladder (D5cm3 < 3000 cGy) were not significantly different between ATP and ATS. Organ-at-risk clinical goals were not met in 6 patients (12/72 goals) for ATP and 2 patients (3/72 goals) for ATS. **Conclusion:** Dose accumulation demonstrates that CTV coverage and OAR sparing are statistically equivalent for ATP and ATS workflows. The

ATP workflow may be preferred for MR-guided adaptive treatments to eliminate need for online recontouring and thus reduce the overall treatment time.

WE-lePD-TRACK 3-05

Feasibility of Dynamic Contrast Enhanced MR Perfusion for Glioblastoma On a Low-Field MRI-Linac System

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Purpose: The purpose of this study is to evaluate the feasibility of measuring dynamic contrast enhanced (DCE) perfusion in glioblastoma on an MRI-Linac at 0.35T. **Methods:** Images were acquired on a glioblastoma patient prior to first radiotherapy treatment on a 0.35T MRI-linac system using 15-channel surface array coils. A 3D gradient recalled echo sequence was used to acquire continuous 45-volumes with 18 axial slices of the brain during gadoteridol infusion. Each volume and total acquisition time were 9.75s and 7min19s, respectively. The tumor lesion was segmented on post-contrast enhancement T1-weighted images. An extended volume of interest (VOI) was created by expanding the margins of the tumor contour by 8mm to incorporate peritumoral voxels to the analysis. Constrained Non-Negative Matrix Factorization (cNMF) method was applied to the DCE curves in the VOI. The cNMF was set to seek for 3 solutions with the condition that the weights of a given pattern should be more than 40% in a given voxel to label it accordingly. Finally, the signal-vs-time curves, from voxels in a given solution were averaged and analyzed using Tofts' pharmacokinetic model. **Results:** The average VOI signal intensity late in the DCE time series was 21% higher relative to the pre-contrast signal. The three temporal curves identified by cNMF and their spatial distribution were consistent with the patterns of: (VAS)-blood vessels, characterized by rapid contrast wash-in and wash-out; (TC1)-well perfused tumor tissue, contrast fast wash-in and gradual wash-out and (TC2)-possibly hypoxic/necrotic tumor tissue, characterized by slow and continuous contrast uptake. The transfer constant K_{trans} for patterns TC1 and TC2 were 0.82 and 0.47 min⁻¹, respectively. **Conclusion:** To the best of our knowledge, this is the first report of DCE perfusion on a 0.35T MRI-linac system. This technique can be used to evaluate early tumor perfusion response for adaptive radiotherapy of glioblastoma.

WE-lePD-TRACK 3-06

Longitudinal Evaluation of Plan Quality Metrics During the Course of Pancreatic Online Adaptive MRI-Guided Radiotherapy

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Purpose: To evaluate plan quality longitudinally over the course of pancreatic MRI-guided adaptive stereotactic body radiation therapy (SBRT). **Methods:** A script was developed to retrospectively extract SBRT plan quality metrics from dose volume histogram files generated during MRI-guided adaptive radiotherapy treatments on a ViewRay MRIdian Linac. Plan quality metrics, including conformity index (CI), ratio of the 50% isodose to the 100% prescription isodose (R50), and ratio of the 36% isodose to the 100% isodose (R36), were computed. At our institution, a planning target volume optimization (PTVopt) structure is generated for each adaptive case by subtracting away an organ-at-risk+5mm planning risk volume from the planning target volume. The CI was computed relative to the PTVopt. These plan quality metrics were computed for the initial plan prepared prior to treatment, as well as for each adapted plan. A paired t-test was used to compare initial plan quality to all subsequent adapted fraction metrics. A Cronbach's alpha was calculated for each of the 3 dosimetric indices to determine consistency within adaptive fractions across all patients. **Results:** Plan quality metrics were evaluated retrospectively for 42 pancreas patients. Initial results indicate that plan quality was stable throughout the course of radiotherapy. The average CI across all patients ranged from 1.03 to 1.05 throughout the course of treatment. The difference between the initial and all subsequent adapted CI values were not statistically significant. All dosimetric indices had Cronbach's alpha ≥ 0.96 , indicating high overall consistency across all adapted fractions. However, some differences between initial and adapted R50 and R36 values were statistically significant. **Conclusion:** Initial results indicate that plan conformity was consistent throughout adaptive MRI-guided radiotherapy. However, lower isodose falloff varied more throughout the course of therapy. Work is underway to investigate other plan quality metrics and organ-at-risk sparing.

WE-lePD-TRACK 3-07

Super Resolution for Improving 2D Cine-MRI Resolution During MR-Guided Radiotherapy of Pancreatic Cancer Patients

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Purpose: Spatiotemporal resolution of MR images is limited by hardware and acquisition parameters, especially at low magnetic fields where long imaging times are required to improve SNR and feature identification. These factors are vital to target tracking and beam gating during MR-guided radiotherapy (MRgRT). Super resolution (SR) reconstruction with deep-learning frameworks provides the ability to generate SR images from low resolution (LR) images. This study applies the SR framework to 2D cine MRI acquired on an MRgRT system during pancreatic cancer treatment. **Methods:** Real time 4 frames-per-second (FPS) cine MR images were collected retrospectively for fifty pancreatic cancer patients treated on the ViewRay MR-linac system. A previously trained super resolution framework was used to generate SR images from the original 3.5x3.5mm² pixel images, with a new resolution of 0.9x0.9mm². SR images were compared to the original LR using inherent metrics including: NIQE, PIQE, and BRISQUE. Inherent image metrics score from 0 to 100, 0 indicates an ideal image. SR images were also compared to paired high resolution (HR) 3D images using SSIM and anterior wall intensity gradient as a surrogate for edge sharpness. **Results:** SR images had statistically significant improvements for inherent metrics, compared to the LR images. SR images scored a mean ($\pm$ SD) of 3.17 $\pm$ 0.18 for NIQE, 34.58 $\pm$ 3.38 for PIQE, and 29.65 $\pm$ 2.98 for BRISQUE. LR images scored a mean ($\pm$ SD) of 7.40 $\pm$ 0.69 for NIQE, and 69.46 $\pm$ 1.52 for PIQE, and 42.48 $\pm$ 0.98 for BRISQUE. SR images also had greater SSIM values and sharper line profiles compared to LR images with SSIM of 0.633 $\pm$ 0.063 and 0.587 $\pm$ 0.067, and gradients of 17.71 $\pm$ 3.17 and 14.39 $\pm$ 3.51, for SR and LR images, respectively. **Conclusion:** Super resolution images provided improved image quality and target identification without introducing artifacts or anatomical inconsistencies. This leads the way for real time SR image generation during MRgRT for target tracking and gating.

Multi-Disciplinary Interactive ePoster Discussion

TRACK 4

Multi-Disciplinary: Image-Guided Treatment Response Assessment

WE-lePD-TRACK 4-01

Forecasting Absolute Lymphocyte Count Depletion During and After Radiation Therapy Using Deep Learning

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Purpose: Numerous studies have shown that severe radiation-induced lymphopenia (RIL - depletion of absolute lymphocyte count (ALC)) adversely impacts patient survival after radiation therapy (RT) of most solid tumors. In this study, we aimed to identify pre-RT patient characteristics and dosimetric features that could predict radiation-induced lymphocyte depletion in esophageal cancer patients during and after the course of RT to identify patients who may benefit from treatment plan reoptimization, use of a different modality, or a pharmacological intervention and ultimately improve outcomes. **Methods:** Data of 860 esophageal cancer patients who received concurrent chemoradiotherapy were used for this study. All patients were treated with protons or photons radiation modalities with a dose of 50.4 Gy.

Pre-, weekly during RT and post-RT ALCs, clinical characteristics and dosimetric parameters were extracted from data bases. A hybrid deep learning model using deep neural network and long short-term memory (LSTM) network was applied to predict ALC depletion trend using baseline value of ALCs measured before or during early stages of RT course. **Results:** Several important prediction metrics were employed to evaluate the performance of the proposed model compared to other commonly used prediction methods. The results showed that the proposed model achieved predictions with the mean squared error (MSE) of 0.046 for 258 patients in test set. Inclusion of ALC after the first week reduced the MSE to 0.014. Moreover, this model outperformed off-the-shelf prediction methods by at least 30% reduction in MSE of weekly ALC predictions. **Conclusion:** The proposed model performed well in predicting radiation-induced lymphocyte depletion. The ALC prediction would enable physician to identify patients at high risk of severe radiation-induced lymphopenia and potentially benefit from modified treatment approaches. Furthermore, if measured ALC were to be available after the first few fractions, it could be used for improved confidence in risk prediction.

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WE-lePD-TRACK 4-02

Imaging Biomarkers for Radiation-Induced Airway Changes in a Swine Model

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Purpose: To investigate changes in imaging biomarkers as precursors to atelectasis post-RT in a swine model and develop a dose-response relationship for bronchial stenosis. **Methods:** A cohort of three swine were analyzed. The swine underwent a research course of 60 Gy in 5 fractions delivered to a targeted airway in the left inferior lung. The right lung was used as control as it was left unirradiated. Airway segmentation was performed on the pre- and three months post- radiation therapy (RT) maximum inhale phase of the four dimensional (4D) computed tomography (CT) scans. Segmentation repeatability was assessed by calculating the percent relative standard deviation of the inner cross-sectional area (A_i) for each airway. A decrease in A_i was assumed to be a surrogate of bronchial stenosis. **Results:** Repeatability analysis showed good repeatability for both timepoints ($r > 0.99$, small deviations $< 5\%$). Airways ($n=30$) in the right (unirradiated) lung for all swine showed no significant changes in A_i post-RT compared to pre-RT. Airways ($n=28$) in the left (irradiated) lung of all swine were found to have a decreased A_i post-RT compared to pre-RT, which was significantly ($p < 0.001$) correlated (Adjusted $R^2 = 0.95$) with dose, and independent of pre-RT inner radius. Additionally, while the square root of wall area (i.e. thickness) decreased significantly ($p = 0.0001$) with dose, the wall area percent increased significantly ($p = 0.04$) with dose. **Conclusion:** This work shows there is a significant correlation between radiation dose and bronchial stenosis as well as wall area percent. Airway toxicities beyond the proximal bronchial tree have mostly been ignored in treatment planning. These results support the hypothesis that airway dose should be considered during treatment planning in order to preserve functional lung and reduce lung toxicities. Further work should be performed in order to verify these effects in humans and the consequences on pulmonary function.

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WE-lePD-TRACK 4-03

Overall Survival Prediction Via an Interpretable Machine Learning Model for Patients with Oropharyngeal Cancer

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Purpose: Treatment outcome prediction is first step towards plan adaptation and personalized treatment. We aimed to explore and use interpretable machine learning (ML) methods, with a hope of adding more prognosis value, to predict overall survival of patients with Oropharyngeal Cancer (OPC). **Methods:** 200 patients with OPC from TCIA database were pooled and analyzed. 70 cases were used for training, 30 cases were used for evaluation, the remaining cases were used as an independent validation cohort. Survival prediction model was created using Support vector machine (SVM), based on imaging features of gross tumor volume (GTV) extracted from planning CT using Pyradiomics. Patient characteristics, including ECOG_Performance, smoking/drinking status, et al were also considered as potential predictors to construct the prediction models. The shapley value (SV) of each feature was calculated according to the alliance game theory to determine its importance in the prediction model. A positive SV corresponds to the prediction result of the instance is positive, while a negative SV means a negative contribution to the prediction. The final prediction model performance was evaluated using the area-under-ROC-curve (AUC). **Results:** The prediction model achieved AUC=0.83 combining both feature categories. Additionally, the correlations between both features categories and survival model can be captured by the SV from the interpretable ML model with interpretable graphs. The top five features that were most correlated with overall prediction were ECOG_Performance, smoking_PY, original_shape_Maximum2DDiameterRow, original_gldm_DependenceNonUniformityNormalized, wavelet-HLH_glszm_SizeZoneNonUniformityNormalized, etc. **Conclusion:** We demonstrated the predictive value of combined patient characteristics and imaging features for OPC patients. The mostly relevant features to patient survival and their contribution to the predictive model were determined via interpretable machine learning models, which hold great promises of addition prognostic value compared to blackbox radiomics analysis. The interpretable ML could greatly improve the application of ML model in medical and clinical applications.

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WE-lePD-TRACK 4-04

Predicting Pathological Complete Response Through Automatic Belief Rule Base in Gastric Cancer

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Purpose: Neoadjuvant chemotherapy (NAC) has been proved as an effective way to improve survival time in gastric cancer treatment. Accurate predicting pathological complete response (pCR) status after NAC plays an important role in making the personalized treatment plan. We aim to develop a new automated belief rule base (AutoBRB) model to predict pCR after NAC in gastric cancer. **Methods:** Totally 55 patients' clinical data (Peking Cancer Hospital, Beijing, China) are used in this study. Since belief rule base (BRB) can integrate clinical domain knowledge and be interpretable, it is used in this study. However, it is always hard to initialize the model parameters. As such, a new automatic BRB (AutoBRB) is developed in this study. In AutoBRB, the referential values, adopted to construct the rule base, are initialized through the information gain ratio, and the corresponding belief degrees are initialized through a new grid strategy. To obtain optimized model, a differential evolutionary algorithm is employed to train the model parameters, and an adaptive searching strategy is adopted to set the confidence threshold. Furthermore, the final output is inferred through an evidential reasoning approach where the belief degrees are obtained and used for making the decision. In the experiments, accuracy, sensitivity, specificity, and area under the curve (AUC) are used as the evaluation measures. **Results:** The mean and standard deviation value of AutoBRB on the accuracy, sensitivity, specificity, and AUC are 0.8962 ± 0.0562 , 0.8530 ± 0.1161 , 0.9394 ± 0.0683 , and 0.9488 ± 0.0345 , respectively, while BRB is 0.7659 ± 0.0560 , 0.9046 ± 0.0756 , 0.6273 ± 0.1081 , 0.9179 ± 0.0450 . **Conclusion:** A new AutoBRB model for predicting pCR after NAC in gastric cancer was developed in this study. The experimental results demonstrated that AutoBRB is superior to BRB and other available methods.

WE-lePD-TRACK 4-05

Prediction of Cardiotoxicity for Breast Cancer Patients Using Light Gradient Boosting Machine

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Purpose: To predict the increased risk of severe cardiotoxicities in breast cancer patients receiving chemoradiotherapy could be challenging due to the variations of current methods used to evaluate the cardiotoxicity symptoms, individual susceptibility and early markers of injury. In this study, we developed, for the first time to our knowledge, a Light Gradient Boosting Machine (LightGBM)-enabled predictive model to integrate patients' chart from electronic medical records (EMRs) for early prediction of severe cardiotoxicities. **Methods:** A total of 179 breast cancer patients were included. The patients were randomly partitioned to the training set and the validation set for LightGBM predictive model development and validation. The training features extracted from each patient include age, cancer stage, tumor size, tumor location, medical history, chemotherapy drugs, targeted therapy drugs, hormone therapy drugs, distant metastases, surgery, radiation therapy position, radiotherapy dose, electrocardiograph (ECG) signal score and left ventricular ejection fraction (LVEF) value before chemoradiotherapy. The utility of the LightGBM model constructed in predicting severe cardiotoxicities was evaluated by ROC analysis in the validation set. **Results:** The AUC value of the LightGBM predictive model achieved in the validation set was 0.82. The sensitivity, specificity, F1 score, precision, and overall accuracy of LightGBM model in the validation set were 78.6%, 81.8%, 81.5%, 84.6% and 80.0%, respectively. The feature important analysis showed that age, the LVEF value before chemoradiotherapy, cancer position, targeted therapy, tumor stage and hormone therapy were the most valuable risk predicting factors. **Conclusion:** The LightGBM framework proposed herein affords a means to use EMR data to individualize the prediction of severe cardiotoxicities at point of care of patients with breast cancer receiving chemoradiotherapy, which can facilitate the identification of patients for whom early intervention are warranted before the therapy, thus potentially improving the utility of chemoradiotherapy for breast cancer from a precision treatment perspective.

WE-lePD-TRACK 4-06

Ultrasound Histogram Assessment of Radiation-Induced Breast Toxicity in Breast Cancer Radiotherapy: A Longitudinal Study of Acute Toxicity

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Purpose: To develop an ultrasound histogram technique capable of evaluating radiation induced acute toxicity in patients receiving radiotherapy (RT) for early-stage breast cancer. **Methods:** Thirty-one patients who underwent conservative breast surgery followed by hypofractionated RT were enrolled in the IRB-approved study. RT involves 15 fractions of whole breast irradiation with simultaneous integrated boost. Each participant received longitudinal breast ultrasound at four timepoints: prior to RT, last day of RT, 3 weeks and 10 weeks post RT. Ultrasound scans were performed using a clinical ultrasound scanner with a 12-MHz linear probe on the irradiated breast and contralateral breast at 5 locations (3, 6, 9, 12 o'clock and tumor bed). The scans on the contralateral breast served as controls. RT-induced breast toxicity was graded in 3 levels (11 patients rated as mild, 10 as moderate, and 10 as severe) based in ultrasound B-mode images. Normalized echo-intensity histograms were computed for the regions of interest including the skin and subcutaneous tissue. The histogram difference (Bhattacharyya distance) between irradiated and contralateral breasts was calculated to capture radiation-induced tissue changes. **Results:** Bhattacharyya distance increased from mild to severe toxicity in all locations. Bhattacharyya distance also increased from mild to severe toxicity at various timepoints (Mild: 0.28 ± 0.06 Pre-RT, 0.32 ± 0.10 Last-RT, 0.32 ± 0.09 at 3-week post-RT, 0.31 ± 0.13 at 10-week post RT; Moderate: $.39 \pm 0.12$ Pre-RT, 0.37 ± 0.17 Last-RT, 0.38 ± 0.15 3-week post-RT, 0.4 ± 0.14 10-week post-RT; Severe: 0.42 ± 0.11 Pre-RT, 0.47 ± 0.1 Last-RT, 0.42 ± 0.14 3-week post-RT, 0.50 ± 0.14 10-week post-RT). **Conclusion:** Radiation-induced breast toxicity remains a major obstacle to effective RT. This study has shown that ultrasound histogram analysis is feasible and may be developed into a low-cost imaging method for radiation-induced tissue toxicity monitoring and assessment.

WE-lePD-TRACK 4-07

Unique Changes on Functional Magnetic Resonance Imaging Maps Following D2 Receptor and Cholinergic Specific Knock-Out of TorsinA

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Purpose: The objective of this study was to compare how loss of torsinA function in D2R and cholinergic cells within the striatum impairs motor deficits, sensory evoked blood oxygen level dependent (BOLD) functional magnetic resonance imaging (fMRI), sensory-evoked functional connectivity, and tissue microstructure using in vivo high field magnetic resonance imaging. **Methods:** We used two genetic mouse models characterized by conditional knock-out of torsinA in either D2R-expressing striatal neurons or striatal cholinergic interneurons for motor behavior assessment (accelerated Rotarod and Beam-walking). Furthermore, we acquired in vivo sensory-evoked fMRI and in vivo diffusion magnetic resonance imaging (dMRI) to assess tissue microstructural changes across the whole brain, as this technique has shown promise in studying neurodegeneration. **Results:** The findings demonstrate that knock-out of torsinA from d2R-expressing striatal cells and striatal cholinergic interneurons have profound effects on sensory-evoked BOLD fMRI, sensory-evoked functional connectivity, and motor deficits: (1) The within-group effects on the β -coefficients. In all groups, BOLD activation was significantly increased within the sensorimotor cortex, cingulate, retrosplenial area, parietal cortex, striatum, globus pallidus, thalamus, cerebellar cortex, and brainstem. (2) The seed-based correlation of functional connectivity with the striatum during sensory-evoked fMRI. Dyt1 d2KO and Ch2KO mice showed abnormal functional connectivity in sensory-evoked functional MRI compared to their control littermates. (3) Correlation between fMRI and behavior deficits, with the region of significance between-group difference found for the sensory-evoked BOLD activation and sensory-evoked functional connectivity. **Conclusion:** The study determines how systemic level changes in brain function, connectivity, and microstructure are influenced by knock-out of torsinA using high field 11T fMRI and dMRI. The findings demonstrate that knock-out of torsinA from d2R-expressing striatal cells has profound effects on sensory-evoked BOLD fMRI, sensory-evoked functional connectivity, which have high correlation with the subjects' motor deficits.

Therapy Interactive ePoster Discussion

TRACK 5

Therapy: Radiation dosimetry

WE-lePD-TRACK 5-01

Development and Validation of a TOPAS Monte Carlo Model of a Dynamic Collimation System for Pencil Beam Scanning Proton Therapy

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Purpose: To develop and experimentally validate a TOPAS Monte Carlo model of the Dynamic Collimation System (DCS) and the IBA dedicated nozzle (DN). The DCS is an energy layer-specific collimator specifically designed for low energy pencil beam scanning proton therapy and consists of four nickel trimmer blades that translate and rotate as a function of beam position located downstream from a polyethylene range shifter. **Methods:** Uncollimated integral depth dose (IDD) curves,

in-air spot profiles, and absolute dose measurements were used to commission a source model of the IBA DN system in TOPAS. Collimated pristine and range shifted IDD curves and in-air and in-phantom 2D spot profiles were acquired to benchmark the model's performance for a variety of collimator configurations. The water equivalent thickness (WET) of the range shifter was determined by the shift in the depth of the 80% dose point distal to the Bragg peak between the range shifted and pristine uncollimated beams. **Results:** Uncollimated IDDs 1D gamma pass rates were 99.4% and 91.2% at the 1%/1mm and 0.5%/0.5mm evaluation criteria, respectively. For the pristine collimated beams, the average 1%/1mm gamma pass rates across all collimator configurations investigated were 99.8% for IDDs and 97.6% and 95.2% for in-air and in-phantom lateral profiles. All range shifted IDDs passed at 100% while in-air and in-phantom lateral profiles had average pass rates of 99.1% and 99.8%, respectively. The measured and simulated WETs of the polyethylene range shifter were determined to be 40.9 and 41.0mm, respectively. **Conclusion:** A TOPAS Monte Carlo model of the DCS and IBA DN system was developed and validated against uncollimated and collimated IDD and lateral profile measurements. As the DCS moves towards clinical implementation, this model will be used to assess the accuracy of analytical algorithms used in commercial treatment planning systems and help guide future dosimetric investigations.

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WE-lePD-TRACK 5-02

3D Dosimetry of Lung and Cardiac Shielding for Total Body Irradiation

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Purpose: Despite technological improvements in treatment planning, dosimetry for total body irradiation (TBI) remains relatively crude. This work employs a unique 3D TBI beam model to inform how different shielding parameters may affect patient dose distributions. **Methods:** An 18 MV clinical beam model was commissioned in RayStation to accommodate classical AP/PA TBI at an extended distance using spoilers and compensators. The low-energy photon spectrum and electron contamination were modified to ensure optimal agreement between the TPS-computed and measured tissue phantom ratios (TPR). Validation measurements were performed using solid water in the standing TBI setup. TPS calculations of the newly-validated model were cross-checked with hand calculations for clinical cases: 3 CT scans of patients without gross disease were planned for the standing treatment position with a total dose of 1200 cGy in 6 fractions. Lung shielding was simulated in RayStation with varying margins (0-2 cm) as well as cases for which the cardiac silhouette was blocked or unblocked. Each case was examined for changes in mean heart and lung dose, lung V10Gy and V12Gy, and point doses behind the blocked tissues. **Results:** The modeled TBI beam agreed with measured TPRs within 2% at 1 cm depth and 0.4% beyond a depth of 3 cm. The average difference in calculated MU between the 3D TBI model and hand calculations for the three patients was 1.7%, confirming agreement with our current clinical practice. Increasing the lung block margins systematically increased the average lung dose by 10%, whereas V10Gy increased nearly two-fold, from 31% to 60%. The mean heart dose increased by 7.5% when the cardiac silhouette is unblocked, while the cardiac V12Gy increased by 25%. **Conclusion:** 3D TBI planning for extended distance is feasible and can provide dosimetric information that traditional planning methods lack, especially regarding lung and heart doses.

WE-lePD-TRACK 5-03

RayStation Photon Beam Modeling Challenges for Small Targets Treated with VMAT and Dynamic Conformal Arcs

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Purpose: A TrueBeamSTx 6FFF beam model geared toward treating small targets with VMAT and dynamic conformal arcs (DCA) was created. However, a revision was necessitated due to IMRT QA failures. The purpose of this abstract is to describe the needed changes and required validation, as well as to discuss the challenges encountered in creating an adequate model for two distinct treatment techniques. **Methods:** Since its clinical release, our beam model was predominately used for small VMAT targets (2-4 cm in diameter). However, IMRT QA failures, where the calculated dose was higher than measured, were occurring. Of 70 total plans, 15 (21.4%) plans exhibited median dose differences of >3.0%, with an average median dose difference of -2.3%. To improve absolute dose agreement, several MLC parameters were adjusted. Subsequent validation work included film, ion chamber, and Delta4 measurements for representative DCA and VMAT plans. **Results:** Post-revision, profile agreement in the low-dose shoulders for simple DCA plans worsened based on film measurements. Original DTA values were 0.1-0.2 mm and the average 5%/0.5mm gamma passing rate was 95.5% for a range of target sizes. Corresponding values for the revised model were 0.1-0.3 mm and 90.9%. For VMAT deliveries, the absolute dose agreement within targets improved, with the average discrepancy decreasing from -1.7% to -0.6% for TG-119 type validation plans. Post-revision, the average median dose difference for IMRT QA was -0.5% (59 plans), with all passing clinical criteria. **Conclusion:** Our experiences highlight the challenges with creating a single beam model that is accurate for a range of target sizes and treatment techniques. For small targets, a compromise between DCA profile accuracy and VMAT absolute dose accuracy was needed. Ultimately, a model revision was performed to optimize the model based on actual clinical utilization.

WE-lePD-TRACK 5-04

Sensitivity of IMRT and VMAT Dose Calculation to Changes in MLC Parameters

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Purpose: To evaluate the sensitivity of IMRT/VMAT dose calculation to changes in dosimetric leaf gap(DLG) and MLC-transmission for a suite of test plans. **Methods:** A set of 5 test shapes were used to generate IMRT and VMAT plans for dose calculation sensitivity analysis. Three test shapes were selected from the TG-119 report: CShape, H&N, and Prostate. Additional test shapes were used as well: H&N_2, which contains an SIB volume, and Pelvis, which mimics prostate with pelvic lymph node irradiation. IMRT and VMAT plans were generated in Eclipse using AAAv15.6 algorithm for both 6 MV and 10 MV. Each plan was calculated on phantom image set for each DLG and MLC-transmission setting evaluated (3 settings were used for each energy). The sensitivity for each parameter was taken to be the difference in point dose relative to the baseline dose calculation. The sensitivity was also compared to the modulation factor (MU per Gy delivered for each plan). **Results:** For 6MV, the mean dose sensitivity (normalized to 1 mm change in DLG) was 2.4% (VMAT) and 4.1% (IMRT). For 10 MV, DLG sensitivity was 2.0% (VMAT) and 5.0% (IMRT). For 6 MV, the mean dose sensitivity (normalized to 1% absolute change in MLC-transmission) was 1.4% (VMAT) and 3.1% (IMRT). For 10 MV, MLC-transmission sensitivity was 1.3% (VMAT) and 3.4% (IMRT). The dose sensitivity (for both DLG and MLC-transmission) was largest for the plans with largest modulation factor-CShape and Pelvis IMRT plans. **Conclusion:** For both 6 and 10 MV, IMRT planning exhibited higher sensitivity to changes in MLC parameters – DLG and MLC-transmission. Further, the plan modulation factor (defined as the MU/Gy) appears to be correlated with the plan sensitivity to changes in MLC parameters. The sensitivity analysis in this study is a potential reference for physicists when tweaking MLC parameters during machine commissioning.

WE-lePD-TRACK 5-05

The Lifetime of a Monitor Unit Ion Chamber

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Purpose: This study shares our clinical experience of the lifetime of Varian Kapton sealed ion chambers as a retrospective review. The data provided shows the average lifetimes of the ion chambers in our institution for nine clinical linear accelerators, so other physicists can be informed of the average lifetime and prepare

for replacement. **Methods:** The lifetime of ion chambers (N=11) has been analyzed using daily quality assurance, monthly quality assurance, delivered treatment field data, and machine applied gain values pulled from configuration files saved after each output adjustment. We investigate the length of time before replacement, beam hours, and filament hours captured during routine preventative maintenance inspections. The estimated clinical monitor units (MU) delivered, cumulative MU after ion chamber replacement, and output adjustments are also analyzed. Data for all clinical linear accelerators was exported and analyzed using a combination of Microsoft Excel and custom Python programs. **Results:** The average number of MU delivered per machine before ion chamber replacement was 13.9 Million (STDEV 4.1 Million). The average amount of time before the first ion chamber replacement was 1,731 days (STDEV 567) which is approximately 4.75 years. The average increase in the applied gain to the ion chamber before replacement was 8.7% (STDEV 3.4%) with a maximum increase over the life of the ion chamber of 13.1%. **Conclusion:** The average lifetime of an ion chamber is approximately 14 million clinically delivered MU. We have found replaced chambers have greater initial changes in output than would be anticipated based on current literature for output changes, and we would recommend this be closely monitored after replacement. We did not observe a plateau in the output changes over time as suggested by other reports.

WE-lePD-TRACK 5-06

Dosimetric Characterization of Orthopaedic Carbon Fiber Implants for Photon and Proton Radiotherapy

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Purpose: Impending/actual pathologic femoral fractures secondary to metastatic bone disease, lymphoma, or multiple myeloma often require intramedullary fixation followed by radiotherapy. Traditionally, titanium-alloy (TA) rods are used but recently, carbon fiber (CF) has emerged as an alternative. Due to its low atomic number, there is reduced CT-imaging artifact and dose perturbation when planning postoperative radiation. Herein, we characterize the dosimetric properties of CF implants compared to TA for proton and photon radiotherapy and develop recommendations for treatment planning. **Methods:** TA and CF samples were acquired from an implant manufacturer as flat pieces 5cm x 5cm and 1.1cm thickness. CT scans were acquired in a solid water phantom with the implants in a custom-made Lucite holder. Material characteristics were evaluated on CT scans with and without metal artifact reduction (MAR). Relative stopping power (RSP) was determined from the range pull-back of each sample in a 20cm range proton beam. Photon transmission measurements were made in a solid water phantom and compared to the RayStation planning system. Photon plans for patients with implants were evaluated for dose perturbation. **Results:** CF caused no visible CT artifacts and MAR was not necessary for Hounsfield Unit determination (median 364 HU) or contouring, while TA (median 3071 HU) caused substantial artifacts, which were improved, but not eliminated by MAR. The proton water-equivalent-thickness and RSP was measured as 35.4mm and 3.204 for TA and 15.7mm and 1.414 for CF. For 6 MV photons, the measured transmission was 89.3% for TA and 98% for CF. CF RSP calculation and transmission from CT HU showed a physical density overestimate compared to measurements. **Conclusion:** With a density similar to bone, CF implants cause no imaging artifacts and minimal dose perturbation compared to TA. Overestimation of RSP and transmission from CT caused slight, but acceptable dose uncertainty (<1.5mm proton range).

Dr Lozano-Calderon has been a paid speaker for CarboFix Orthopedics Ltd, Amgen, and Daiichi Sankyo.

WE-lePD-TRACK 5-07

Validation of Monte Carlo Simulated Tc-99m Absorbed-Dose-To-Water Inside a Custom SPECT/CT Phantom Using Active and Passive Dosimeters: A Novel Approach to Benchmarking Monte Carlo Methods for Targeted Radionuclide Therapy Dosimetry

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Purpose: Targeted radionuclide therapy (TRT) is a growing field within the realm of radiation therapy despite its shortcomings in dosimetry. This work aims to introduce a novel method for benchmarking TRT Monte Carlo simulations using a custom-made SPECT/CT phantom and active and passive dosimeters. **Methods:** A cylindrical, acrylic SPECT/CT phantom of diameter 8" and length 8" capable of housing an IBA EFD diode, SNC600c farmer type ion chamber, and TLD-100 microcubes was designed and built for the purpose of assessing internal absorbed-dose-to-water at various points within a solution of ^{99m}Tc. The phantom is equipped with removable inserts that allow for numerous detector configurations. All detectors were calibrated using a 250 kVp NIST traceable x-ray beam in either Virtual Water or a water tank. Three experiments were conducted with exposure times ranging from 11 to 21 hours with starting activities of approximately 350 mCi. Automation of the electrometer allowed for data collection over the extended acquisition time. Measurement data was appropriately corrected for pressure and temperature fluctuations, or energy dependence, and subsequently compared to Monte Carlo simulations using the egs_chamber user code in EGSnrc 2018. **Results:** In general, both the diode and ion chamber measurements agreed with the predicted Monte Carlo simulated doses within k=1 calculated uncertainty values. TLD measurements generally produced an under response relative to the predicted doses. **Conclusion:** Our results show that measurements can be obtained with generally good agreement of Monte Carlo simulations for the active dosimeters, and within at least 20% for the passive dosimeter investigated. Future work will incorporate beta emitting radionuclides and modified attachments to the current phantom to assess the feasibility of measuring dose from these commonly used TRT agents.

Therapy Interactive ePoster Discussion TRACK 6

Therapy: Radiobiology and modeling

WE-lePD-TRACK 6-01

Incorporating Organ Deformation in Biological Modeling and Patient Reported Outcome Study for Permanent Prostate Brachytherapy

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Purpose: Permanent prostate brachytherapy has inherent intraoperative organ deformation due to the inflatable trans-rectal ultrasound probe cover. Since the majority of the dose is delivered with reduced prostate deformation, the dosimetry approved at the time of implant may not accurately represent the dose delivered to the target and organs at risk. We aimed to evaluate the biological effect of the prostate deformation and its correlation with patient-reported outcome. **Methods:** We prospectively acquired ultrasound images of the prostate pre- and post-probe cover inflation for 27 patients undergoing Iodine-125 seed implant. Coordinates of implanted seeds from the approved clinical plan were transferred to the undeformed prostate to simulate the actual dosimetry using a machine learning-based deformable image registration. DVHs of both plans were reduced to Tumor Control Probability (TCP) and Normal Tissue Complication Probability (NTCP) with assumed generalized equivalent uniform dose distribution. The change in patient-reported rectal and urinary symptoms six months post procedure were correlated with NTCP using area under the curve (AUC) and odds ratio (OR). **Results:** From the deformed clinical plan to the undeformed plan, the average TCP decreased from 98.9% to 94.2% and the average NTCP changed from 0.4% and 26.7% to 18.0% and 17.3% for the urethra and rectum, respectively (p<0.05). For the diarrhea symptom NTCP>20%, the undeformed plan showed AUC=0.75, OR=8.9 (1.3-58.8) while the deformed plan showed AUC=0.56 and OR=1.4 (0.2-9.0). For the urinary control NTCP>15%, the undeformed plan showed AUC=0.70, OR=6.9 (0.6-78.0) while the deformed plan showed AUC=0.51 and no reported OR since all NTCP were below 6.6%. **Conclusion:** With the consideration of organ deformation, TCP decreased while urethra NTCP showed worse sparing and rectal NTCP showed better sparing. The NTCP of the undeformed plan shows a strong correction with patient-reported diarrhea and urinary control, while the original clinical plan does not.

WE-lePD-TRACK 6-02

Prediction of Radiation Pneumonitis for NSCLC Using Crop Convolutional Neural Network Based Model

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Purpose: To predict grade 2 radiation pneumonitis (RP) in locally advanced non-small cell lung cancer (NSCLC) patients with high accuracy by using convolutional-neural network (CNN) model. **Methods:** Three CNN models were created: the whole body region, normal lung (nLung) region, and nLung region overlapping the region over 20 Gy (nLung $\cap$ 20Gy) were fed into 3D CNN model. Models were evaluated on 54 patients and were independently tested on 23 newly treated patients. The risk of RP was categorized into two groups (Group I: more than RP grade 2 and Group II: within RP grade1). The model was optimized using Adam. Fivefold cross-validation was applied to verify its predictive performance. The precision, accuracy, and sensitivity by generating confusion matrices and the areas under the receiver operating characteristic curve (AUC) for each model were evaluated. **Results:** The accuracy, specificity, sensitivity, and AUC were 53.9%, 80.0%, 25.5%, and 0.58, respectively for whole body method and 60.0%, 81.7%, 36.4%, and 0.64, respectively for nLung method. For the nLung $\cap$ 20Gy method, the accuracy, specificity, sensitivity, and AUC were improved to 75.7%, 80.0%, 70.9%, and 0.84, respectively. The segmented input data can be learned effectively and useful for CNN model. **Conclusion:** The CNN model which the input image was segmented and considered dose distribution can help to predict grade 2 RP for NSCLC after definitive radiotherapy.

WE-lePD-TRACK 6-03

Development of Radioimmunotherapy for Brain Tumors Using In-Vitro Assays

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Purpose: In the last decade, anticancer agents that act as immune-checkpoint inhibitors have been approved and deployed successfully in treating some cancers such as melanoma, leading to the emergence of immunotherapy as a possible fourth pillar of cancer treatment in addition to surgery, radiotherapy and chemotherapy. For primary brain tumors such as glioblastoma, which show high radioresistance as well as high chemoresistance, there is an urgent need to develop synergistic strategies combining immunotherapy with traditional modalities. The purpose of our work is to develop new combination modalities for glioblastoma which currently has a dismal 5-10% 5-year survival rate. **Methods:** Having recently used a commercially available Electric Cell Impedance Sensor (ECIS) to quantify the migration of various cancer cell lines following chemotherapy and following radiotherapy (2 Gy, 10 Gy and 20 Gy) using a cell irradiator, Faxitron CellRad, we now focus on quantifying cell migration following combinations of radiotherapy with temozolomide (TMZ), and other approved immunotherapeutic agents. **Results:** Our preliminary results show that irradiated T98G cells (Glioblastoma, brain cancer cells) attach and migrate significantly more ($p < 0.0001$) than non-irradiated cells in the first 20-40 hours post irradiation. Irradiated cells (5 Gy and 20 Gy) show markedly different proliferative and migratory patterns in the course of 7 days following irradiation (Figures 1 and 2 in supplement). Along with clonogenic assays, therapeutic windows suggested by these results will be explored and presented. **Conclusion:** Our preliminary results suggest that real time cell migration assays can be used to explore therapeutic windows for optimal combination of immunotherapy and radiotherapy especially in the case of highly radioresistant and chemoresistant cancers such as glioblastoma.

WE-lePD-TRACK 6-04

Characterization of the Response of 9L Orthotopic Brain Tumors to 3D Conformal Radiation Therapy Using Contrast Enhanced MRI

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Purpose: Our long-term goal is to evaluate early predictors of tumor radiation response. As a first step, the response of 9L orthotopic brain tumors were measured following MRI guided planning of stereotactic radiotherapy (SRS) in order to determine the dose to cure 50%, TCD50. **Methods:** 118 Fischer-344 rats were implanted with 105 rat 9L cells in 10 μ l using males and females. Tumors either grew untreated or were treated with 20 or 25 Gy to the 95% isodose as follows. Around 8 days post-implantation, rats were imaged with contrast enhanced MRI (MRI-CE) and irradiated using a Small Animal Radiation Research Platform (SARRP) from Xstrahl Inc. (Suwanee, GA) operating at 220 kV and 13 mA with an effective energy of ~ 70 keV and dose rate of ~ 2.5 Gy per minute. Rat orientations were assessed using cone-beam CT; tumor locations were co-registered using the MRI-CE images. Treatment planning using MuriPlan software from Xstrahl Inc. allowed delivery of four non-coplanar arcs with an identical isocenter. Rats were monitored daily for physical and behavioral abnormalities. Study endpoints were animal survival at 200 days with no weight loss exceeding 20% of maximum animal weight. **Results:** The TCD50 was established from Kaplan-Meier survival analysis as between 20 and 25 Gy. Animals with large tumors assessed from MRI-CE on the day of treatment did not survive long, as expected. For example, tumors > 150 mm³ correlated inversely with survival for animals treated with 25 Gy (Wilcoxon Rank Sum test). Cox Proportions-Hazards modeling did not suggest an effect of sex with the caveat of the wide confidence intervals. **Conclusion:** The radiation response of orthotopic 9L tumors was characterized using clinically relevant treatment techniques. One future direction is to evaluate the predictive value of pre-treatment vascular measures (blood flow, vascular permeability, etc.) on radiation response.

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WE-lePD-TRACK 6-05

Comparison of Dose Escalation Strategies for Head and Neck Cancer Patients Using a Radiobiological Model of Reoxygenation

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Purpose: Hypoxia is a big challenge for tumor control in head and neck cancers. With advanced imaging techniques, individual patient hypoxic regions can be identified. Several clinical trials perform dose-escalation based on pre-treatment hypoxia, typically boosting hypoxic tumor areas. However, longitudinal imaging studies showed that the location of hypoxic cells changes throughout treatment. The purpose of this work is to quantify the potential loss of biological effect due to changes in hypoxia during treatment in dose-escalation studies based on initial hypoxia only and compare this strategy to one based on patient imaging at weeks 0 and 2 of treatment. **Methods:** A previously published reoxygenation model based on LQ survival formulas with reoxygenation parameter Δ is extended to allow for dose-escalation with dose $d(H)$ per fraction for initial hypoxic cells. The new model allows for some hypoxic cells to miss the escalated dose depending on parameter "m" ($m=0$ no missed cells, $m=1$ all cells missed). Using published radiobiological parameters and hypoxia reduction factor (HRF) of 1.5-2.5, we calculate the EQD2 when hypoxia is present for 70Gy delivered in 35 fractions with $d(H)(2.2-2.6Gy)$ to the hypoxic region. **Results:** For an HRF of 1.5, the biological effect with escalated dose per fraction up to 2.6Gy cannot sufficiently compensate hypoxia if there is no reoxygenation ($\Delta=0$). As Δ is increased to 0.2 and beyond (20% or more hypoxic cells re-oxygenate after each fraction), the EQD2 becomes comparable to the non-hypoxic values. Based on previous results, the dose-escalation of the whole tumor after 20Gy based on hypoxia imaging at 2 weeks yields comparable or higher EQD2 values for more modest escalated doses to the whole tumor. **Conclusion:** The new model allows to evaluate and compare strategies for dose-escalation to compensate for hypoxia during treatment and quantifies limitations of dose-escalation based on initial hypoxia.

WE-lePD-TRACK 6-06

Towards the Biological Meaning of the Site in Microdosimetry: Relation to DNA Compactness

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Purpose: To study the biological meaning of site in microdosimetry and DNA direct damage at different degrees of DNA compactness under alpha particle irradiation. **Methods:** To associate the concept of site in microdosimetry with a biological endpoint, we performed TOPAS-nBio simulations of direct damage caused by alpha particles to two DNA arrangements in a water sphere of 500 nm radius: (i) straight linear plasmids of DNA, i.e., double helix arranged in parallel, with axial separations of 2.8 nm, 4.2 nm, and 5.6 nm; and (ii) supercoiled plasmids of DNA (pBR322) available in TOPAS-nBio. DNA geometries were uniformly irradiated by an isotropic source of alpha particles with initial energies between 1 and 15 MeV/u. Scored double strand breaks (DSBs) were defined as damage (i.e., energy imparted above 17.5 eV) in two base pairs separated by less than 10 base pairs. Yield of DSBs per Gy per giga base pair (Gbp) were obtained and reported as a function of the corresponding dose-mean lineal energy (y_D) and dose-mean specific energy ($z_D(r)$) for sites of variable size. **Results:** Yields of DSBs in the linear plasmids, regardless of the separation between them, showed the same linear relation with y_D . Yields of DSBs for supercoiled plasmids were greater than for linear plasmids but also exhibited an approximately linear relation. Yields as a function of $z_D(r)$ from both linear and supercoiled configurations were aligned for radii equal to 385 nm and 293 nm, respectively. **Conclusion:** Compacted DNA have neighbor base pairs closer than uncoiled DNA, which makes a track more likely to damage two of them. This associates with smaller volumes in which the pattern of energy deposited is relevant, i.e., smaller sites. Therefore, prediction of direct DNA induction from microdosimetry can be separated into two components: y_D (radiation quality-dependent) and site radius (DNA geometry-dependent).

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WE-lePD-TRACK 6-07

Extracting Organ-Specific Radiobiological Model Parameters from Clinical Data for Radiation Therapy Planning of Head and Neck Cancers

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Purpose: Different radiation therapy (RT) strategies, e.g., conventional fractionation, hypofractionation, stereotactic body radiotherapy (SBRT), adaptive RT, and re-irradiation are often used to treat head and neck cancers. Combining and/or comparing these strategies require to calculate biological effective dose (BED). The purpose of this study is to develop a practical process to extract organ-specific radiobiologic model parameters that may be used for BED calculations in individualized RT planning for head and neck cancer. **Methods:** The dose volume constraint data for 12 organs at risk (OAR) corresponding to no grade ≥ 3 toxicity for SBRT, hypofractionated and conventional fractionation treatments were compiled from various sources such as TG101, QUANTEC and clinical trial database. These data cover 10 fractionation schemes. A linear-quadratic-linear (LQ-L) model was used to extract model parameters (e.g., α/β , γ/α and d_t) from fitting the Iso-BED curves of those dose volume constraints for the OARs using an in-house python code. The obtained model parameters were integrated into a practical workflow in MIM software. Their practicality was tested for treatment planning of re-irradiation. **Results:** The LQ-L model fitted the data well as indicated by smooth transition curves from SBRT to conventional fractionation data. The organ-specific model parameters were obtained, for example, $\alpha/\beta=2.57\text{Gy}$, $\gamma/\alpha=9.07\text{Gy}$ and $d_t=7.65\text{Gy}$, and $\alpha/\beta=2.34\text{Gy}$, $\gamma/\alpha=9.67\text{Gy}$ and $d_t=6.80\text{Gy}$ for spinal cord and brainstem, respectively. The composite dose plans for retreatment of head and neck cancers with considering the obtained model parameters show up to 10% difference in physical dose constraints from traditional EQD2 method with $\alpha/\beta=3.0\text{Gy}$. **Conclusion:** A practical process was developed to obtain organ-specific radiobiological model parameters from clinical data and was shown to generate more clinically relevant model parameters for calculating BED distributions compared to those extracted from cell line survival data. The practicality of the obtained parameters was demonstrated in creating composite plans for retreatment.

Vendor Showcase

TRACK 8

Reflexion Medical

WE-SHOWCASE-TRACK 8-15-1

First Clinical Experience: Overview of Commissioning the Reflexion BgRT System (Linac, kV system and Treatment Planning)

J Crawford*, B Han*, M Surucu*, N Kovalchuk*, Stanford University School of Medicine, Stanford, CA

Professional Proffered

TRACK 1

Medical Physics Practice Changes in Response to the Covid-19 Pandemic

WE-E-TRACK 1-01

Microsoft Teams® as An Operational Tool for Evaluation of Machine Performance and QA Tasks

M Gopalakrishnan*, S Kucuker Dogan, B Zawislak, T Teo, I Das, Northwestern Memorial Hospital, Northwestern University Feinberg School of Medicine, Chicago, IL

Purpose: Maintaining machine performance and service database in an academic department with multiple machines, different locations and various vendors is a challenging task. Microsoft Teams®, used for virtual meeting is explored for tracking machine status and performance. **Methods:** Microsoft Teams®, a collaborative workspace hub for conversations, document sharing, assigning and completing tasks is utilized for machine related communications in the department. An equipment dashboard was created for each machine to assign tasks and alert physicist for necessary services. Physicist completes and add notes for the task which would then trigger an email to members alerting the status of the service, maintenance and QA. A timeline for each of the above tasks is maintained for assessing workflow in the system. As part of machine performance QA, this data is reviewed during the weekly physics meetings where all the machine related QA issues are discussed and remedial actions evaluated. **Results:** Department maintains 5 Linacs, 2 CT scanners, GK ICON and host of other peripheral equipment at two locations with mixed vendors and different vintage. Teams® dashboard provides every information that is entered to the entire department and have reduced the bottleneck. It provides instantaneous information for the service issues, report and summarize the incidence of each machine in real-time. This information is vital for communication and for future planning. Machine related issues for different components including downtime, maintenance reports, service calls, physics support are visible and tracked. Visibility, communications, and overall management of equipment issues has dramatically improved after utilizing Microsoft Teams®. **Conclusion:** Microsoft Teams® has been successfully implemented as a platform to track QA and service issues without investing huge amount of money or writing a code. It is concluded that Teams can be used in any department for providing essential communication and exchange of information among staff.

WE-E-TRACK 1-02

Successful VMAT Commissioning in Low- and Middle-Income Country (LMIC) in the Era of COVID-19 Using Remote Connectivity

S Wadi-Ramahi^{1*}, F Waqqad^{2,3}, A Alsharif³, (1) University of Pittsburgh Medical Center, Hillman Cancer Center, Pittsburgh, PA, (2) Radiation Oncology Center, King Abdullah University Hospital, Irbid, Jo, (3) Radiation Oncology Dept, Afia Nuclear Medicine and Radiation Oncology Center, Amman, Jo

Purpose: Outlining a systematic process to successfully commission VMAT in Jordan in a time of restricted travel and increased virtual connectivity. **Methods:** Process started on 9/2019 with a 2-days physical visit to initiate trust and discuss scope of work. Observing clinical workflow was intended for a week-long visit to follow. After COVID19, VPN access to both center's Mosaic and TPS to evaluate treatment setup, reproducibility and reviewing IMRT cases was given. Due to time difference, there was no interruption in clinical workflow when the systems were accessed. Extensive conference calling and email communication to discuss findings. Confidential communication with administration is important. Remote commissioning began with agreeing on a detailed time-table with clear assigned responsibilities. Remote supervision through VPN, videoconferencing, group chats and emails. **Results:** Workflow baseline unveiled opportunities for improvements that were accepted and implemented. TPS re-acceptance (TG119 and TecDoc1583) was done by local physicists. Various QAs were implemented using TG142. Linac VMAT acceptance testing was done followed by virtual training by vendor. Lastly, 20 different disease site VMAT plans were done, using actual patient scans for different disease sites, for QA purposes resulting in recommending 3%/3mm for gamma analysis. Clinical protocols written for dose objectives and QA methodology. All work was remotely followed and supervised. **Conclusion:** COVID-19 has taught us a very important lesson: Professionals from high-income countries can effectively help clinics in LMICs through virtual connectivity. This process can be adapted to after-COVID to reduce cost. Other lessons learned: 1. A short visit is a must to meet the team, 2. Written communication should be clear. Lapses in communication may greatly hinder the project, 3. Always start with measuring the clinical-workflow baseline. Implementing change to improve and assessing local staff's effective participation is critical, 4. Agree on a clear timetable with assigned responsibilities.

WE-E-TRACK 1-03

Remote MRI Physics During a Pandemic: Challenges and Opportunities

K Vigen*, L Henze Bancroft, J Garrett, F Korosec, J Holmes, University of Wisconsin-Madison, Madison, WI

Purpose: Prior to the COVID-19 pandemic, our MRI physics group often provided in-person clinical MRI exam physics support, including monitoring exams of patients with certain MR Conditional and unlabeled implants; managing artifacts; suggesting protocol modifications; and providing basic MRI system troubleshooting. This work will describe the transition to an almost entirely virtual MRI physics model during the pandemic, including example cases. **Methods:** To minimize risk to patients and staff, our MRI physics team adopted a virtual support model immediately after the pandemic started. Remote support tools included remote PACS viewers, secure access to the MRI host computer, and a prototype remote MRI console viewer installed on a single hospital MRI scanner. **Results:** Prior to the pandemic, 3.5 cases per week required in-person MRI physics support. The majority of exams requiring physics support were shifted to the MRI system with the remote MRI console viewer. Following the initial decrease in all MRI exams at the onset of the pandemic onset, MRI volumes at our institution recovered to pre-pandemic levels; our virtual model has been able to support nearly all cases requiring MRI physicist presence: 5 exams in the first 4 weeks of the pandemic shutdown, up to 5.6 cases per week by late summer, normalizing to 3.8 cases per week since that time. MRI physicists can work remotely, with onsite after-hours visits for QC, pulse sequence and protocol development, and basic troubleshooting. **Conclusion:** We have shifted the majority of in-person clinical MRI physics activities to a virtual model. This helped our institution navigate initial and subsequent phases of the pandemic by enabling remote work and minimizing facility occupancy, while maintaining a high level of MRI physics services. We anticipate continuing to use these tools for efficient MRI physics support across our network spanning several imaging facilities in the community.

Our department received research support from GE Healthcare.

WE-E-TRACK 1-04

Did the COVID-19 Pandemic Disproportionally Affect Patients Receiving Radiation Therapy at a Public Safety-Net Hospital Compared with Patients Treated at a Private Hospital in the Same Metropolitan Area?

J Cui*, J Cui, W Yang, A Zhuang, O Ragab, D Hong, S Bian, E Chang, R Jennelle, Keck School of Medicine of USC, Los Angeles, CA

Purpose: To compare the number of radiotherapy treatments delivered to patients at two departments at an epicenter of the COVID-19 pandemic. One is a public safety-net hospital serving a socioeconomically disadvantaged and under-insured population. The other is a private hospital serving patients with private insurance. **Methods:** On March 11, 2020, the World Health Organization declared COVID-19 a global pandemic. Considering the response to COVID-19, the number of External Beam Radiation Therapy (EBRT) and Gynecologic High-Dose-Rate (HDR) brachytherapy treatments delivered between May 2020 and Feb 2021 (during pandemic) was compared to treatments delivered between May 2019 to Feb 2020 (pre pandemic). We also reported differences in patient access to telemedicine, COVID-19 screening, and clinical trials at the two centers. **Results:** At the safety-net hospital, there were 8291 EBRT treatments pre-pandemic and 7475 during the pandemic, which reflects a 10% decrease. At the private hospital, the numbers were 10662 and 8231 respectively, a 23% decrease. At the safety-net hospital, the HDR treatments were 45, 32, 26 for cylinder, interstitial, tandem-and-ovoid respectively pre pandemic and 38, 14, 56 during the pandemic. The total number of HDR treatments were 103 and 108 respectively. The decrease in interstitial therapy was balanced by an increase in hybrid tandem-and-ovoid with needles when needed. For the private hospital, Gyn HDR increased from 14 to 32, likely due to detrimental impact of treatment delay on Gyn cancers. During the pandemic, clinical trials were administratively paused at the safety-net hospital to allow for reassignment of resources directed to the care of critically ill COVID-19 patients. The private center enrolled 9 patients during the pandemic, same number as pre-pandemic. **Conclusion:** Our analysis demonstrates that the COVID-19 pandemic did not disproportionately affect patients' accessibility to RT at a safety-net hospital versus a private hospital. However, access to clinical trials was severely impacted.

WE-E-TRACK 1-05

Medical Physics During COVID-19: The North American Perspective | Audience Q&A

B Parker^{1*}, (1) The University of Texas M.D. Anderson Cancer Center, Galveston, TX

Medical physics was not immune to the impact of the COVID-19 pandemic, and the profession required creative methods to adapt to working in this new age. This presentation first will introduce a recent publication on the global medical physics response to the pandemic (Medical Physics During the COVID-19 Pandemic: Global Perspectives in Clinical Practice, Education, and Research) and then focus on the North American response in the areas of clinical practice, education, research, and regulatory and accreditation. It will conclude with lessons learned and opportunities identified from the COVID-19 experience.

Learning Objectives:

1. Understand the impact the pandemic had on various aspects of the profession
2. Review the North American medical physics response for various aspects of the profession
3. Identify lessons learned and opportunities to improve future responses for similar emergent situations

SAM Imaging Educational Course

TRACK 2

Updates on CT Dosimetry

WE-EF-TRACK 2-00

Updates on CT Dosimetry

Moderator: Mark Supanich, Rush, Chicago, IL

WE-EF-TRACK 2-01

A. Hernandez, University of California-Davis:

SSDE: Coming Soon to A Scanner Near You! Updates on Head Coefficients (AAPM Report 293) and IEC Codification

WE-EF-TRACK 2-02

M. Bostani, University of California, Los Angeles:

How Looooong Can You Go? Dose Measurements in a Long Phantom (Review of AAPM Report 200)

WE-EF-TRACK 2-03

I. Barreto, University of Florida College of Medicine:

You Spin Me Right Round: Approaches to Measuring CT Dose in Helical Modes

WE-EF-TRACK 2-04

M. Supanich, Rush:

Free Falling: Measurements of Dose in Air

WE-EF-TRACK 2-05

Audience Q & A

Updates on CT Dosimetry Abstract

Accurate measurement and assessment of CT Dose indices is vital for equipment constancy and acceptance testing, estimating patient absorbed doses, and comparison of imaging protocols across scanners. In this educational session, updates on several key topics in CT dosimetry will be discussed. Recent scientific and regulatory developments in CT dosimetry offer new indices to better approach patient dose and new approaches to measure dose outside of the “traditional” 4 cm or less beam width scenario.

The first talk will be focused on Size Specific Dose Estimate (SSDE) updates, including how SSDE may be calculated for examinations focused on the head as described by AAPM Report 293. In addition, prospects for scanner implementations of SSDE following approval of a new IEC standard will be discussed. The 2nd presentation will review the findings of AAPM Report 200 and highlight options for measuring dose indices in long phantoms to ensure scatter tails for wide beam geometries are captured. Further discussion of dose index measurement under non-conventional scenarios will follow in the third talk, which will focus on approaches to measure CT dose from helical acquisitions. Finally, approaches to measuring CT dose in air will be discussed, including approaches described by manufacturers in technical manuals when measurements in phantoms alone are not sufficient to assess CTDIvol.

Learning Objectives:

1. Understand how SSDE is calculated in head and body and how it will be implemented on scanners in the future.
2. Understand how CT dose may be measured in a long phantom as described by AAPM report 200
3. Understand how CT dose may be measured without a CTDI phantom or when axial acquisition modes are not available.

SAM Ultrasound Symposium

TRACK 3

Frontiers of Ultrasound Imaging

WE-E-TRACK 3-00

Frontiers of Ultrasound Imaging

Moderators: Ivan Rosado-Mendez, University of Wisconsin-Madison, Madison, WI and Marie Muller, North Carolina State University, Raleigh, NC

WE-E-TRACK 3-01

G. Cloutier, Université De Montreal:

Unifying Concepts of Spectral and Statistical Backscatter, and In Vivo Application

WE-E-TRACK 3-02

J. Bamber, Institute of Cancer Research and Royal Marsden Hospital:

Pressure-Wave Elastography (Sound-Speed Measurement and Imaging) and Its Biomedical Applications

WE-E-TRACK 3-03

M. Muller, North Carolina State University:

Quantitative Lung Ultrasound, Beyond Artifact Imaging

WE-E-TRACK 3-04

Q&A

Ultrasound imaging is one of the most used imaging modalities due to its portability, relative low cost and use of non-ionizing radiation. Despite these advantages, variability in image acquisition and interpretation, as well as the complexity of the image that result from the many interactions of ultrasound with tissue, and a lack of compensation for system dependent factors, have limited a more extended use. Additionally, conventional ultrasound methods are not well suited to imaging organs such as the lung, because of large amounts of scattering by the air-filled alveoli. This session will discuss technological developments that are expanding the frontiers of ultrasound imaging, enhancing it with quantitative measuring features. The goal of this expansion is to overcome some of these limitations by offering clinicians tools to quantify physical characteristics of tissue such as its scattering properties (reflectivity) and sound speed (related to compressibility), which can be used as biomarkers or surrogates of tissue healthy or disease status. Invited speakers will provide the physical basis of these quantitative ultrasound techniques, and present recent applications in liver, breast, and other tissues. The challenges of extending this approach to the lung, where new sources of contrast must be established so that new biomarkers of lung diseases can be developed, will also be discussed. Proffered talks will expand on these applications by presenting the state of the art of ultrasound imaging research.

After this session, attendees will be able to:

1. Describe the physical principles of novel scattering and sound speed-based quantitative ultrasound techniques.
2. Discuss recent applications of these techniques in liver, breast, lung, and other tissues.
3. Describe how the proposed techniques can be used to solve specific clinical needs.
4. Identify the main challenges of incorporating these novel quantitative ultrasound techniques in their clinical practice

Targeted attendees are scientists and medical physicists and other medical professionals with interests in the development of quantitative imaging, particularly quantitative ultrasound, and its clinical translation.

Cloutier: Canadian Institutes of Health Research, NSERC, Cancer Research Society, Research funding from Siemens Healthcare.

Bamber: Engineering and Physical Sciences Research Council, Cancer Research UK, Research Council of Norway, among others. In-kind funding from iThera GmbH and Delphinus, lecture fees from Samsung Healthcare.

Muller: NIH, Department of Defense

Therapy Educational Course

TRACK 4

In Memoriam of John "Jack" Cunningham

WE-EF-TRACK 4-00

In Memoriam of John "Jack" Cunningham

Moderators: Jacob Van Dyk, Western University, London, ON and Jerry Battista, Western University, London, ON

WE-EF-TRACK 4-01

J. Van Dyk, Western University:

Jack Cunningham and Treatment Planning From the 1960s Through the 2000s

WE-EF-TRACK 4-02

J. Battista, :

Conceptual Overview of Megavoltage Photon Dose Calculation Algorithms

WE-EF-TRACK 4-03

D. Rogers, Carleton Univ:

Monte Carlo Modelling for Radiation Dosimetry and Treatment Planning

WE-EF-TRACK 4-04

A. Ahnesjö, Uppsala University:

Dose and Beyond in Treatment Planning Calculations

WE-EF-TRACK 4-05

T. Mackie, University of Wisconsin:

The Future of Treatment Planning

WE-EF-TRACK 4-06

Discussion

This symposium will celebrate the life and accomplishments of John Robert (Jack) Cunningham O.C., B.Eng., M.Sc., Ph.D., FCCPM, FAAPM, FCOMP, one of the great pioneers in Medical Physics. His breadth and depth of Medical Physics experience and knowledge have resulted in multiple, wide-ranging contributions extending from operating a radon plant in the 1940s, to evaluating range-energy relationships for electrons from a betatron in the early 1950s, to the development of radiation therapy apparatus including the design and construction of cobalt-60 machines in the 1950s and 60s, to generating radiation-related basic quantities for dose calibration protocols in the 1970s and 80s. He co-authored *The Physics of Radiology* along with Harold Johns, "the Johns and Cunningham book," generally considered as the "bible" in Medical Physics. However, his greatest and most impactful contributions started in the 1960s with his early development of dose calculation algorithms for use with computerized, clinical, radiation treatment planning programs. The algorithms used calculations of primary dose summed with differential scatter doses using quantities known as scatter-air ratios (SARs). His programs were made commercially available beginning in the 1970s evolving through various commercial enterprises, starting with Atomic Energy of Canada Limited (AECL) and moving through to Theratronics, MDS Nordion, and Nucletron. The principles and concepts were implemented in various other commercial systems as well.

In view of his significant contributions to Medical Physics and the resulting impact on many patients around the world, Jack has received multiple awards, including the AAPM Coolidge award in 1988, the inaugural COMP Gold Medal in 2006 and the Order of Canada, the top award given by the Canadian government for those "who exemplify citizenship and whose contributions enrich the lives of their contemporaries". A more detailed review of Jack's career can be found in *Medical Physics*: <https://doi.org/10.1002/mp.14050>.

In honor of Jack Cunningham and out of recognition of his intense desire to develop elegant physics solutions to clinical problems, this symposium will focus on radiation treatment planning-related algorithms by considering the past, evaluating the present and projecting into the future. The speakers are all acquaintances of Jack and have interacted with him at a variety of levels.

Learning Objectives

1. Learn about the contributions Jack Cunningham made to the field of Medical Physics.
2. Learn about the dose calculation algorithms developed by Jack Cunningham and their implementation in commercial treatment planning systems.
3. Learn about the state-of-the-art treatment planning and dose determination algorithms by considering the past, evaluating the present and projecting into the future.

License agreement with Modus Medical Devices, Inc. of London, Ontario, Canada

SAM Therapy Scientific Symposium

TRACK 5

Cardiac Radioablation: How Medical Physicists Can Shape the Future?

WE-E-TRACK 5-00

Cardiac Radioablation: How Medical Physicists Can Shape the Future?

Moderator: Paul Keall, University of Sydney, Camperdown, AU

WE-E-TRACK 5-01

M. De Riva Silva, Leiden University Medical Center:

Cardiac Radioablation Research and Development Needs at the Juncture of Electrophysiology and Radiation Therapy

WE-E-TRACK 5-02

S. Lydiard, University of Sydney:

Research Opportunities from Academia: A Summary of the State-Of-The-Art Cardiac Radioablation Medical Physics Research and Future Opportunities

WE-E-TRACK 5-03

A. Deisher, Mayo Clinic:

Open Questions in the Radiobiology of Cardiac Radioablation and the Promise of Particle Therapy

WE-E-TRACK 5-04

Q&A

The stellar clinical results of the early clinical applications of cardiac radioablation¹⁻³ mirror the enthusiasm for the early applications of lung cancer SABR that has now demonstrated efficacy compared with surgery and conventional radiotherapy in phase III trials.⁴⁻⁵ Lung SABR is now a mainstream treatment. In order for cardiac radioablation to navigate the path from early adopters to the mainstream, the enthusiasm for the broader implementation must be accompanied by multidisciplinary education, clinical evidence and practice guidelines to maximize patient safety and procedure efficacy. There are many opportunities for imaging and therapy medical physicists to contribute to the development of new and better technology to image, target and monitor treatment response. This symposium will cover:

- Cardiac radioablation R&D needs at the juncture of electrophysiology and radiation therapy
- Research opportunities from academia: A summary of the state-of-the-art cardiac radioablation medical physics R&D opportunities, including cardiac imaging and gating
- Open questions in the radiobiology of cardiac radioablation and the promise of particle therapy
- The challenges and opportunities to contribute to improved cardiac radioablation patient treatments and outcomes

Learning Objectives:

1. To understand electrophysiology and radiobiology relevant to a medical physicist planning to implement cardiac radioablation
2. To learn the current technology approaches implemented for cardiac radioablation
3. To appreciate the future opportunities for medical physicists to define cardiac radioablation practice

References

1. Cuculich. New England Journal of Medicine 2017.
2. Robinson. Circulation 2019
3. van der Ree. Heart Rhythm 2020
4. Chang. Lancet Oncology 2015
5. Ball. Lancet Oncology 2019

PK's conflicts are updated and publicly disclosed at: <https://image-x.sydney.edu.au/home/disclosures/>

Therapy Scientific Session

TRACK 6

Brachytherapy

WE-E-TRACK 6-01

3D Printed Patient-Specific HDR Brachytherapy Applicator for Penile Cancer Plesiotherapy

H Lee*, B Hryckushko, Z Xiong, M Folkert, N Desai, A Garant, T Chiu, UT Southwestern Medical Center, Dallas, TX

Purpose: To develop a custom 3D printed HDR plesiotherapy applicator for localized squamous cell carcinoma on the glans penis. **Methods:** A rigid shell-like applicator having eight channels for standard plastic catheter insertion was designed such that it could slide onto the penis. A spiral pattern was designed at the distal end to accommodate the spherical curvature of the glans while achieving a large radius of curvature for smooth source wire transfer. This applicator was printed on a SLA 3D printer to achieve a highly smoothed finish for standard plastic catheter insertion. Before putting on the applicator, a custom-made silicone bolus/sleeve (4mm-thickness) was placed to provide space from the source to reduce the surface dose and for patient comfort. Tests were conducted to ensure the standard plastic channel diameter and radius of curvature were appropriate for guide tube insertion and source delivery. **Results:** With the bolus and applicator placed, a patient with a rcT1a penile cancer was imaged on a CT scanner with wires placed in the channels for treatment planning digitization. For a penis of diameter ~30mm, a 36mm applicator with a 26mm silicone bolus provided sufficient immobilization with minimal air gaps. Brachytherapy planning was performed with a prescription dose of 40 Gy in 10 fractions at the surface, and 35 Gy at 5mm depth, bidaily. Dose coverage to the targeted glans was 94%(D90). During the course of treatment, a highly reliable and reproducible setup (5-10 minutes setup time) was made possible. 6-month follow-up monitoring showed no significant changes in urinary and sexual function. **Conclusion:** A 3D printed patient specific applicator with a silicone bolus was shown effective in providing a reproducible setup and dose delivery for HDR brachytherapy plesiotherapy.

WE-E-TRACK 6-02

A Fast and Accurate Analytical Dose Calculation Method for Seed-Loaded Stent

T Xiong^{1*}, B Liu^{2,3}, J Lu⁴, S Li⁴, X Bai^{1,2}, F Zhou^{1,2}, Q Wu⁵, (1) Department of Physics, Beihang University, Beijing, CN (2) Image Processing Center, Beihang University, Beijing, CN (3) Beijing Advanced Innovation Center for Biomedical Engineering, Beihang University, Beijing, CN (4) Center of Interventional Radiology & Vascular Surgery, Department of Radiology, Zhongda Hospital, Medical School, Southeast University, Nanjing, CN (5) Duke University Medical Center, Durham, NC

Purpose: The safety and clinical efficacy of 125I seed-loaded stent for the treatment of portal vein tumor thrombosis have been shown. However, the dosimetry characteristics of the seed-loaded stents remain unclear and there is no fast dose calculation technique available for seed-loaded stent. This paper aims to explore a fast and accurate analytical dose calculation method which takes into account the effect of stent and tissue inhomogeneity. **Methods:** An accurate model of the seed-

loaded stent was established using 3D modeling software and subsequently used in the MC simulation to calculate the dose distribution around stent. The dose perturbation caused by the presence of the stent was analyzed and dose perturbation kernels (DPKs) were derived. Then, the dose calculation method from AAPM TG43 was adapted and integrated with the DPK and inhomogeneity correction factor (ICF) correction to calculate dose distributions analytically. To validate the proposed method, extensive comparisons were performed with other methods in water phantom and voxelized CT phantoms of three patients. **Results:** The stent has a large effect on the dose distributions and reduces the dose up to 47% for single seed stent and 13% for 16-seed stent. In water phantom, dose distributions from MC simulations and the proposed method showed a good agreement with the relative error less than 3.3%. In voxelized CT phantoms, the relative errors of TG43 method can be up to 33% while those of the proposed method were less than 5%. For a dose matrix with 256×256×46 grid for 16 seeds-loaded stent, it only takes 17 seconds compared with 25 hours for MC simulation. **Conclusion:** With its good agreement with MC calculation and computational efficiency, the proposed method is adequate for dose evaluation and optimization in seed-loaded stent implantation treatment planning.

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WE-E-TRACK 6-03

Defining Dose Thresholds to the Optic Disc and Macula to Predict Vision Complications: A COMS Retrospective Study

C Oare*, B Gerbi, K Dusenbery, D Koozekanani, S Sun, C Ferreira, University of Minnesota, Minneapolis, MN

Purpose: Despite the long history of COMS (Collaborative Ocular Melanoma Study) eye plaques, limited consensus for dose thresholds exists for the optic disc and macula. A retrospective study was conducted to identify a correlation between complications and doses to optic structures for ocular melanoma patients. The primary objective of this work was to define dose-side effect relationships and identify dose thresholds for radiation retinopathy, vision decline, blindness, and cataract formation. **Methods:** A cohort of 52 patients treated with COMS plaques at our institution between 2004-2019, with adequate follow up data, were used in a single institution retrospective study. Patient demographics, dosimetry, and clinical outcomes were recorded. A univariate analysis and relative risk estimates were performed to analyze dose-side effect relationships for the optic disc, lens, and macula. A logistic regression was performed to identify dose percentiles for retinopathy, vision decline, blindness, and cataracts. **Results:** Doses to the optic disc and macula had significant relationships ($p \leq 0.05$) with incidence of radiation retinopathy, moderate vision decline (5 or more Snellen lines), and onset of blindness. Moreover, significant ($p \leq 0.05$) relative risk estimates exist for radiation retinopathy, vision decline, and blindness for doses greater than 35 Gy to the optic disc, and 55 Gy to the macula. Finally, a macula dose of 55 Gy will result in less than 13% of patients experiencing radiation retinopathy or vision loss. Optic disc doses of 24 and 33 Gy would result in less than 13% of patients experiencing retinopathy or vision loss, respectively. **Conclusion:** Based on the results of the study, it can be concluded that dose-effect relationships exist for the optic disc and macula with respect to radiation retinopathy, vision loss, blindness. To improve a patient's quality of life beyond treatment, further work is needed to expand upon dose reduction strategies for patients undergoing eye plaque brachytherapy.

WE-E-TRACK 6-04

Determination of the RBE of the Xofigo Axxent Source for Cervical Cancer Applications

A Walter*, K Nickel, R Kimple, L DeWerd, University of Wisconsin-Madison, Madison, WI

Purpose: To measure the relative biological effectiveness (RBE) of the Xofigo Axxent electronic brachytherapy source on HeLa cells with the bare S7600 Axxent source (BS) and source-in-titanium applicator (SIA). **Methods:** HeLa cells were suspended in 1 mL of cell culture media and placed into 1.5 mL microcentrifuge tubes. Cells were irradiated at a radial distance of 2 cm from the source and given doses ranging from 0-6 Gy for both the BS and SIA. The source was pulled back a distance of 3 mm for the SIA irradiations. Doses from the BS and SIA were determined using an Exradin A26 ionization chamber, TLD-100 microcubes, and the TOPAS Monte Carlo code. Irradiations were also performed in Co-60 using 15 mL tubes placed in a virtual water phantom at an SAD of 75 cm and depth of 1.5 cm to obtain reference dose points. Following irradiation, each suspension was seeded in a 6-well plate, with an equal concentration of cells per well. The plates were incubated for 1 week, stained with crystal violet, and the surviving colonies were counted. Survival curves were plotted and fit using a linear-quadratic equation. **Results:** The RBE values determined for the BS were found to be 1.19, 1.22, and 1.27 for the surviving fractions of 1%, 10%, and 37% respectively. The RBE determined for the SIA was found to be 1.17, 1.19, and 1.22 for the surviving fractions of 1%, 10%, and 37% respectively. **Conclusion:** The RBE of the Axxent source is approximately 20% greater than that of Co-60, which is representative of current external beam treatments. Additionally, filtration of the Axxent source from the applicator causes a reduction in dose rate and small reduction in the RBE of the Axxent source.

The sources and applicator used for this work were provided by Xofigo, a subsidiary of iCAD Inc.

WE-E-TRACK 6-05

Commissioning and Clinical Implementation of a Hybrid TG-43 Technique for Shielded Brachytherapy Applicators

E Choi, C Collins*, M Rivard, Brown University/Rhode Island Hospital, Providence, RI

Purpose: Commercial brachytherapy (BT) treatment planning systems (TPSs) rely on TG-43 calculations that treat all materials as water and fail to accurately accommodate shielded BT sources and applicators. While this works fine for conventional applications such as HDR prostate and GYN BT, treatment of superficial lesions and those in proximity to organs-at-risk will benefit from more sophisticated treatment delivery techniques. This includes the cylindrical AccuBoost breast and Leipzig skin BT applicators. Model-based dose calculation algorithms and applicator models for these devices are not readily available. Innovations using the TG-43 formalism make it possible for BT TPSs to approximate dose distributions in the vicinity of these shielded, high-Z applicators. **Methods:** This study created models for the AccuBoost and Leipzig BT applicators to function within a conventional BT TPS. Published Monte Carlo-derived dose distributions in Cartesian coordinates (along-away) were transformed into polar coordinates (radius and polar angle) for derivation of TG-43 dosimetry parameters through customized values of the dose-rate constant, radial dose function (RDF), 2D anisotropy function, active length, and calibrated source strength. RDF and 2D anisotropy function values were chosen to have sub-millimeter/sub-degree changes, permitting characterization of high-gradient dose regions (i.e., >40%/millimeter). These models were formatted as XML+MD5 files for TPS entry (Oncentra v.4.6.0, Elekta), then used to generate treatment plans for source/applicator commissioning. DICOM RD files were prepared and exported for direct comparison with published Monte Carlo-derived dose distributions. **Results:** Though the TPS limited the 3D dose calculation grid to voxels of at least 1.0 mm, high accuracy of absolute dose was observed with agreement typically within 0.2% and greater than 99.9% of the dose points being within 2%. Voxels outside 2% agreement were located in the high-gradient dose regions where volume averaging was not insignificant. **Conclusion:** This innovative method produced dosimetry results that were deemed acceptable for clinical use.

WE-E-TRACK 6-06

Adaptive Workflow for a Patient-Specific 3D-Printed Template with Direction Modulated Brachytherapy (DMBT) Tandem Applicator for HDR Cervical Interstitial Brachytherapy

J Sohn*, M Polizzi, E Fields, W Song, Virginia Commonwealth University, Richmond, VA

Purpose: We propose a novel workflow that utilizes a couple of emerging and certainly innovative technologies of direction modulated brachytherapy (DMBT)

tandem applicator along with rapid 3D printing for a patient-specific template augmented for a further improvement in dosimetric conformity and needle positioning accuracy. **Methods:** The proposed workflow consists of three steps to produce an optimized needle configuration with a DMBT tandem. The first step is composed of pre-and each inter-fraction MR imaging to acquire high-risk CTV and delineate the vaginal space for creating the patient-specific template. The second step includes the two-step inverse optimization to (1) determine optimal needle numbers and needle paths and (2) perform volume optimization with DMBT tandem and needle paths. The final step includes rapid 3D printing and QA. This process is iterative and the 3D printed template may be re-optimized using the patient's latest imaging. **Results:** We present a highly adaptable workflow and utilized it for a retrospective patient case. A conformal plan with optimal needle tracks and DMBT tandem was created, and a 3D vaginal template with the tandem and needle tracks was printed. A dosimetric planning comparison is provided to illustrate the additional plan quality improvement achievable with the proposed novel two-step optimization and 3D printing workflow. Radiation Oncology departments with access to MRI and 3D printers can implement this process to reduce inter and intra-fractional needle placement uncertainties. **Conclusion:** The proposed workflow attempts to optimally utilize the rich anatomic information present in the daily MRI by the power and the flexibility of needle positioning offered through 3D printed tracks in combination with the DMBT tandem that allows superior intensity modulation over conventional tandems. The combination of the two technologies, and powered by the daily MRI, offers a truly adaptive workflow that generates the most conformal plans for each fraction.

The senior author has Varian funding for the project.

WE-E-TRACK 6-07

Commissioning of the Ring Applicator and Quantifying Dwell Position Offset with Varian Bravos HDR Remote Afterloader

A Baghwala¹*, R Boopathy², (1) Radiation Physicist II, Houston Methodist Hospital, The Woodlands, TX, (2) Assistant Professor Division of Radiation Physics, OHSU School of Medicine, Portland, OR

Purpose: Despite new technology offered through BRAVOS afterloader from Varian, the issue of discrepancy between planned source dwell position and delivered in the ring applicator still exists. Purpose of our study was to propose a methodology and quantify the offset due to snaking effect in ring applicators before clinical use. **Methods:** 6 Ring applicators (2 sets of 30/45/60 degrees) from Eckert & Ziegler were used. The applicators are taped to Gafchromic™ EBT3 film and irradiated to 400MUs with a 16MeV electron beam on a linac with the x-ray marker BV 320mm inserted. A treatment plan was created and delivered with 10 equally spaced 1cm source dwell positions at a nominal dwell time of 2 seconds. Combination of applicator image and dwell positions in same radiograph helps in confirming the structural integrity and identifying the tip of the lumen better to compare the expected vs delivered source dwell positions. Films were scanned using Epson Expression 11000XL scanner and the analysis was done in Eclipse Brachytherapy 2D Entry. **Results:** Arc length was measured from the center of each delivered source/dwell position to the center of the ring. Subtracting with 0.25 (source center from distance tip) gives the path length. Subtracting the path length with applicator length (150 cm) will give us the shift in delivered source position. Results show an average of 5mm (σ +/-0.6mm) difference between planned and delivered source dwell position. The results also revealed that the offset increases with the angle of arc length. Applying a consistent correction offset of 4mm to all planned dwell position improved the positional accuracy to less than 1mm (σ +/-0.07mm). **Conclusion:** The data from this study warrants the need for robust applicator commissioning. Quantifying one consistent offset for ring applicators instead of multiple offset values will improve efficiency/workflow without compromising the safety/delivery accuracy.

SAM Therapy Educational Course

TRACK 7

The Evolving Paradigm of Patient Specific QA

WE-EF-TRACK 7-00

The Evolving Paradigm of Patient Specific QA

Moderator: Seng-Boh Lim, Memorial Sloan-Kettering Cancer Center, New York, NY

WE-EF-TRACK 7-01

J. Adamson, Duke University Medical Center:

Moving from a Simple "Type A" Independent Dose Calculation to a "Type B" (Convolution/Superposition) Or "Type C" (Monte Carlo) Based Independent Dose Calculation Algorithm

WE-EF-TRACK 7-02

S. Lim, Memorial Sloan-Kettering Cancer Center:

Moving from Measurement Based Pre-Treatment QA to Trajectory File Based QA

WE-EF-TRACK 7-03

B. McCurdy, CancerCare Manitoba:

Advances in In-Vivo Real-Time PSQA and Dose Reconstruction

WE-EF-TRACK 7-04

C. Bojchko, University of California San Diego:

Patient Specific QA for New Technologies and Online Adaptive Radiotherapy

WE-EF-TRACK 7-05

J. Andreozzi, Moffitt Cancer Center:

Scintillator Based Patient Specific QA in the Age of MRgRT

This educational program will cover the changing landscape of patient-specific QA, including experience of implementing new solutions into the clinic, as well as developing new technologies. Specific topics include: 1) Moving from measurement-based pre-treatment QA with Gamma Analysis in phantom to DVH based metrics 2) Moving from measurement-based pre-treatment QA to trajectory file-based QA 3) Moving from a simple 'Type A' independent dose calculation to a 'Type B' (convolution/superposition) or 'Type C' (Monte Carlo) based independent dose calculation algorithm 4) Patient-specific QA for new technologies and online adaptive radiotherapy 5) Scintillator based patient-specific QA in the age of magnetic guided radiation therapy (MRgRT).

Learning Objectives:

1. The benefits of DVH based analysis in patient-specific QA.
2. The strengths and weaknesses of log file-based QA
3. The latest advancement in the area of independent dose calculation algorithm.
4. The challenges of patient-specific QA in the era of online adaptive radiation therapy
5. New measurement-based methods in the presence of magnetic field

Dr. Adamson reports ownership of ClearSight RT LLC which is unrelated to this presentation, and grant support from Radialogica LLC.

Dr. Bojecho reports grant support from Varian Medical Systems.

Vendor Showcase

TRACK 8

Radformation

WE-SHOWCASE-TRACK 8-16-1

From CT to Treatment in 30 Minutes

E Van Wie^{1*}, (1) Radformation, New York, NY

Professional Symposium

TRACK 1

21 Years into the 21st Century: Modern Scientific Publishing in Medical Physics and the JACMP

WE-F-TRACK 1-00

21 Years Into the 21st Century: Modern Scientific Publishing in Medical Physics and the JACMP

Moderators: John Boone, UC Davis Medical Center, Sacramento, CA and Michael Mills, James Graham Brown Cancer Center, Louisville, KY

WE-F-TRACK 1-01

M. Mills, James Graham Brown Cancer Center:

JACMP: Emerging Trends in Open Access Publishing

WE-F-TRACK 1-02

A. Cetnar, The Ohio State University - James Cancer Hospital:

JACMP: Taxonomy and Video Abstracts

WE-F-TRACK 1-03

J. Boone, UC Davis Medical Center:

Medical Physics: The New Editorial Model

WE-F-TRACK 1-04

S. Benedict, UC Davis Cancer Center:

Medical Physics: Article Categories

WE-F-TRACK 1-05

Q&A

Scientific publishing is evolving rapidly and AAPM's journals are adapting to stay at the forefront of serving our patients and our profession. Authors and readers alike will benefit from hearing from the Editors-in-Chief and Deputy Editors of Medical Physics and the Journal of Applied Clinical Medical Physics (JACMP) about their latest work to modernize their publications as platforms for dissemination of important new medical physics knowledge.

As an open access journal, JACMP provides worldwide reach to authors and attracts readers who lack access to publications using a traditional subscription model. Open access publishing is affected by a number of trends of which both prospective authors should be aware. Authors will also benefit from learning about the taxonomy used by JACMP for its content. As modern publishing evolves from a print-only medium to a digital multimedia environment, JACMP has launched new initiatives such as video abstracts which will also be highlighted in this session.

A new editorial model for Medical Physics that provides a more horizontal construct for leadership and participation in the review process has been employed that includes 15 Deputy Editors. The Deputy Editors provide expertise that encompasses the entire taxonomy of topics published in Medical Physics, and the global reach of Deputy Editors includes Asia, North America and Europe. This presentation will expand on the benefits of this new editorial model, and the dynamic challenges it must address for the premier journal of our profession.

The Medical Physics journal publishes Research Articles and Technical Notes on a range of medical physics topics, which include Therapeutic interventions; Diagnostic Imaging (Ionizing and Non-Ionizing); Quantitative Imaging and Image Processing, Emerging Imaging and therapy Modalities; and Computational and Experimental Dosimetry. These article types and categories will be briefly presented in terms of the continuing expansion of our field.

Learning Objectives:

1. Explain the new editorial approach and review process for Medical Physics.
2. Discuss trends impacting open access publishing in the field of medical physics.
3. Apply correct journal taxonomy to manuscripts submitted for publication to Medical Physics and JACMP.
4. Describe the benefits of video abstracts.

M Mills receives compensation for serving as Editor-In-Chief of JACMP. A Cetnar has nothing to disclose. J Boone receives compensation for serving as Editor-In-Chief of Medical Physics. S Benedict receives compensation for serving as Therapy Editor of Medical Physics.

Ultrasound Scientific Session

TRACK 3

Quantitative Ultrasound and Emergent Imaging Technology (II)

WE-F-TRACK 3-01

High-Resolution Ultrasound Imaging Through Self-Supervised Learning

X Dai*, Y Lei, T Wang, M Axente, J Roper, D Xu, J Lin, J Bradley, T Liu, X Yang, Emory University, Atlanta, GA

Purpose: This study aims to develop a deep learning-based algorithm to reconstruct high-resolution 3D ultrasound (US) images from sparsely acquired 2D images without any extra atlas database. **Methods:** In conventional US imaging, in-plane spatial resolution is generally much higher than through-plane resolution. Under the assumption of isotropy of US imaging in biological tissue, the mapping from through-plane low-resolution to high-resolution images should be identical as that

for in-plane images. Therefore, the mapping can be learned using paired low-resolution and high-resolution in-plane images. Given the acquired 2D images which have high in-plane spatial resolution, the in-plane low-resolution images can be obtained through down-sampling. In this study, a model which is constructed by two independent cycle consistent generative adversarial networks (cycleGANs) is built to learn the mapping from low-resolution to high-resolution images and generate high-resolution through-plane images from original sparsely distributed 2D images. We performed a leave-one-out cross validation method to evaluate the proposed high-resolution method using a 3D breast database with 70 patients. Metrics including mean absolute error (MAE), peak signal-to-noise ratio (PSNR), and visual information fidelity (VIF) were computed to quantify the performance of our proposed method. **Results:** In the experimental studies with a spatial resolution enhancement factor of 3, for all patients, a mean and standard deviation of MAE value of 0.90 ± 0.15 , a PSNR value of 37.88 ± 0.88 dB, and a VIF value of 0.69 ± 0.01 were achieved by our proposed method, significantly outperforming the commonly used bicubic interpolation. **Conclusion:** In this study, a novel self-supervised learning framework has been investigated for reconstructing high-resolution 3D US images from sparsely acquired 2D images. Without using any extra atlas images, our proposed method achieves significant improvement on through-plane resolution. Its self-supervision capability could accelerate high-resolution US imaging, and this imaging tool will have a high impact on US-guided intervention.

WE-F-TRACK 3-02

A Novel Wireless Mechanical Oscillator for Power Doppler-Based Needle Tip Identification in Brachytherapy

N Orlando^{1,2*}, J Snir³, K Barker², D Hoover^{3,4}, A Fenster^{1,2,4}, (1) Department of Medical Biophysics, Western University, London, ON, CA, (2) Robarts Research Institute, Western University, London, ON, CA, (3) London Regional Cancer Program, London, ON, CA, (4) Department of Oncology, Western University, London, ON, CA

Purpose: Brachytherapy (BT) is a form of radiation therapy, which relies on the insertion of needles to deliver radiation. Accurate needle tip identification is key for safe and effective BT, as errors can result in radiation delivery that deviates from the planned dose. Typically, standard brightness (B)-mode ultrasound (US) imaging is used for needle tip identification; however, artifacts can limit the visibility of needles. We propose a novel wireless mechanical oscillator for needle perturbation in a power Doppler (PD)-based needle tip identification method to overcome these limitations and improve needle tip identification accuracy in BT. **Methods:** Our wireless mechanical oscillator is capable of oscillating at a fixed and controllable frequency and amplitude, featuring a cylindrical end-piece designed specifically for use with BT needle mandrins. We evaluated our method using a tissue equivalent phantom with a standard clinical needle implant pattern and an implant pattern designed to maximize needle shadowing. Needle tips were identified with B-mode US alone, with B-mode and PD US together, and with PD US alone by reducing B-mode gain to 0, simulating the clinical case of non-visible needles. Performance was quantified using a tip error metric computed using the clinical standard of ideal reference needles. **Results:** Absolute mean $\pm$ standard deviation tip error was 0.33 ± 0.27 mm (B-mode), 0.39 ± 0.23 mm (B-mode+PD), and 0.77 ± 0.89 mm (BlindedPD) for the standard clinical needle implant and 0.80 ± 1.70 mm (B-mode), 0.34 ± 0.47 mm (B-mode+PD), and 0.43 ± 0.62 mm (BlindedPD) for the shadowed needle implant. **Conclusion:** Our PD US method provided equivalent tip identification accuracy for the standard clinical needle implant and superior accuracy for shadowed needles, including the ability to visualize needles that could not be seen using B-mode US alone due to shadowing. Our wireless oscillator is easy to use and requires no modification to the clinical workflow, providing promise for the clinical utility of our method in BT procedures.

This work was supported by the Natural Sciences and Engineering Research Council of Canada (NSERC), the Canadian Institutes of Health Research (CIHR), and the Ontario Institute for Cancer Research (OICR). N Orlando was supported in part by the Queen Elizabeth II Graduate Scholarship in Science and Technology.

WE-F-TRACK 3-03

A Novel End-To-End Deep Learning Based Beamformer for Flexible Array Ultrasound Transducer

X Huang*, M Lediju Bell, K Ding, Johns Hopkins University, Baltimore, MD

Purpose: The flexible array transducer is a promising tool for image-guided tumor tracking in radiotherapy. Comparing with the traditional ultrasound transducer, it is user-independent and can avoid contact pressure that may cause an anatomical change. However, due to its flexible geometry, the conventional delay-and-sum (DAS) beamformer may apply incorrect time delay to the radio-frequency (RF) data and produce B-mode images with low quality and considerable distortion. Here, we propose a novel end-to-end deep learning approach that may alternate the conventional DAS beamformer when the transducer geometry is unknown. **Methods:** Our deep neural networks (DNNs) are designed based on the popular U-Net, Pix2Pix GAN, and Cycle GAN structures. They are trained with Field II simulation and real scanning RF data from transducers with different geometries and the corresponding properly beamformed images. The DNNs are expected to learn the proper time delays for each channel and generate the undistorted high-quality B-mode images directly from RF data. They are then tested with RF data from a transducer with random unknown geometry, and the results are evaluated using full-width-at-half-maximum (FWHM), contrast-to-noise ratio (CNR), and aspect ratio of the round-shaped cyst. **Results:** We compare the DNN results to the standard DAS results using different data. For point targets, the average FWHM of DNN results is 2.1 mm lower than that of DAS results. For anechoic cyst targets, the DNN results improve the CNR by 0.79 dB and aspect ratio by 0.23. For abdominal phantom and in vivo scans, the DNN approach improves the CNR by 1.80 dB and 2.18 dB, respectively. The U-Net structure has the overall best performance. **Conclusion:** The evaluations show that when the transducer geometry is unknown, the proposed DNNs can effectively reduce the distortion and improve the lateral resolution and contrast of the B-mode images compare with the conventional DAS beamformer.

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WE-F-TRACK 3-04

X-Ray Acoustic Computed Tomography of Linac Photon Beams: Medium Dependent Experiments

G Gonzalez^{1*}, S Wang², P Pandey², K Prather¹, J Caron¹, S Ahmad¹, L Xiang², Y Chen², (1) University of Oklahoma Health Sciences Center, Oklahoma City, OK (2) University of California, Irvine, CA

Purpose: The goal of radiotherapy is to deliver dose to the tumor while sparing organs at risk (OAR). Some cases can become challenging due to organ motion or breathing. X-ray induced acoustic tomography (XACT) offers an alternative for image guided radiotherapy (IGRT) without using additional imaging dose. This study compared XACT images from different medium. **Methods:** A low frequency single element transducer was used to acquire XACT signals on water and oil from a 4 cm square photon field. The transducer was fixed in front of the radiation field at an appropriate distance and depth of measurement. Beam delivery was performed using a Varian TrueBeam with 10X-FFF energy and 4 microsecond linac pulse to trigger acquisition. The XA signals passed through a shielded preamplifier, then collected in an oscilloscope with high averaging. Data collection was performed at every 6° collimator angle, for a total of 60 signals. MATLAB was used for signal processing and image reconstruction using simple backprojection. 2D images were obtained for the field in both water and oil medium. The frequency spectrum was compared for both medium and background signals. **Results:** The reconstructed square field was clearly visible in oil, while the image from water was considerably weaker presenting artifacts. Each signal contains rarefactions where the middle corresponds to the boundaries of the field. The signal intensity reached 1mVpp in water, while in oil it was 15mVpp. The frequency spectrum in oil presented larger response at lower frequencies, having at 50kHz a ratio of 11 to the water medium. **Conclusion:** This XACT system was able to reconstruct the image of the photon field. Signal strength is highly dependent on the acoustic properties and radiation absorption of the medium. Further analysis of frequency response could give insight on using dual mode ultrasound to improve image quality.

WE-F-TRACK 3-05

4D MR Reconstruction Based on Real-Time 3D Ultrasound for Image-Guided Radiotherapy

S Jupitz*, J Holmes, B Bednarz, University of Wisconsin, Madison, WI

Purpose: Utilizing a hands-free, MR-compatible 3D ultrasound transducer, we developed a method to bin multiphase 2D MR slices into MR volumes in time—effectively reconstructing a 4D MR dataset. The resulting dataset provides spatial and temporal information for image-guided radiotherapy applications. **Methods:** Simultaneous multiphase 2D sagittal MR slices and 3D ultrasound images of the liver were acquired from four healthy volunteers as well as end-exhale 3D MR images on a clinical 3T MRI. The imaging session was repeated for two volunteers. Liver vessels in ultrasound were tracked throughout the session. Ultrasound motion datapoints were temporally matched to 2D MR images. Deviations from the mean greater than 1.5 standard deviations were rejected as outliers. The remaining datapoints were sorted into 14 equidistant amplitude bins. Likewise, datapoints were sorted according to MR slice position. Thus, subsets of images were created for each amplitude-slice position bin pair. The image with the median position within each subset was selected to represent the group. Selected images were reorganized to create static MR volumes at each amplitude. This technique and resulting reconstructions were evaluated quantitatively and qualitatively. End-exhale 3D MR provided ground-truth for comparison to the binned 3D MR at end-exhale as measured in diaphragm position discrepancy. Difference maps were assessed for volunteers with two imaging sessions. **Results:** The average diaphragm position discrepancy was 2.14 ± 0.92 mm. Qualitative analysis showed reasonable out-of-plane diaphragm continuity. Difference maps demonstrated good agreement, especially for the liver. Ultrasound tracking revealed extreme baseline drift in one dataset. The change in breathing motion was effectively captured with this system indicating this workflow can determine whether an imaging session recorded consistent, representative respiratory signals. **Conclusion:** Simultaneous ultrasound was effectively used to reconstruct multiphase 2D MR images into 4D datasets. This technique could be applied to clinical datasets to assess target motion and deformation from respiration.

This work was supported by NIH grant R01CA190298 as well as the University of Wisconsin-Madison Graduate School Pandemic Affected Research Continuation Initiative (PARCI).

WE-F-TRACK 3-06

An Unsupervised Ultrasound Liver Motion Tracking Using Deep Convolutional Neural Network

Y Lei*, X Dai, S Momin, J Roper, E Schreibmann, P Patel, J Bradley, T Liu, X Yang, Emory University, Atlanta, GA

Purpose: The capability of providing video-rate volumetric images with zero ionizing radiation dose makes ultrasound (US) imaging suitable for intra-fraction motion tracking in radiotherapy. This study aims to develop a deep learning-based algorithm for liver motion tracking using US images. **Methods:** We propose an unsupervised deformable image registration network for US liver motion tracking. A Markov-like network is used to extract features from sequential US frames (one tracked frame is followed by untracked frames) to estimate a set of deformation vector fields (DVF) through the registration of the tracked frame and the untracked frames. This network is optimized in an unsupervised manner, namely, without the need for ground truth DVFs. The positions of the landmarks in the untracked frames are finally determined by transforming the landmarks in the tracked frame according to the estimated DVFs. The performance of the proposed method was evaluated on the testing dataset by calculating the tracking error between the predicted and ground truth landmarks on each frame. **Results:** The proposed method was assessed on the MICCAI CLUST 2015 dataset, which contains 63 2D and 22 3D US image sequences respectively from 42 and 18 subjects collected using 7 US scanners with 8 types of transducers. The experimental results show that our proposed method achieved a mean tracking error of 0.70 ± 0.38 mm on the 2D sequences and 1.71 ± 0.84 mm on the 3D sequences. **Conclusion:** We have proposed and investigated a new deep learning-based algorithm for liver motion tracking using US images and demonstrated its feasibility for real-time motion tracking. This imaging tool has strong potential for US-guided motion tracking during liver radiotherapy.

WE-F-TRACK 3-07

Motion Management in Abdomen with An Integrated Ultrasound and Abdominal Pressure Plate Device

W Li*, Y Huang, X Chen, Y Yang, University of Science and Technology of China, Hefei, Anhui, CN

Purpose: Radiotherapy to tumors in the abdomen is challenging because of the significant organ movement and deformation caused by respiration. We developed a motion management strategy using ultrasound (US) integrated with an abdominal compression plate (ACP), where the US was to monitor motion after the abdomen was compressed with the ACP. **Methods:** Twenty-one volunteers were immobilized using an in-house developed device that integrated an US probe with an ACP. For each subject, abdominal US data was successively collected with and without an ACP. Experiments were repeated three experiments to verify the reproducibility. A template matching algorithm based on normalized cross correlation was implemented for motion tracking. And the failure rates of the liver, pancreas and stomach tracking were analyzed. On the basis of the obtained organ locations in the US, linear correlation models between liver and pancreas/stomach were established to predict the motion of pancreas/stomach using liver as a surrogate. **Results:** The liver motion amplitudes for individual subjects with ACP compression were significantly reduced with ACP compression in both superior-inferior (SI) ($p < 0.0001$) and anterior-posterior (AP) ($p < 0.0001$) directions. The failure rates in liver, pancreas and stomach tracking were reduced from 1 %, 2.54% and 2.79 % to 0.03%, 0.73% and 1.07% after ACP compression respectively. The correlation coefficients between liver and pancreas/stomach were 0.98/0.97 without ACP and 0.96/0.94 with ACP in SI and were 0.68/0.68 and 0.43/0.42, respectively, in AP. **Conclusion:** While slightly reducing the linear correlation between organs, ACP effectively inhibits the abdominal motion and improves the reliability of organ tracking in US. The US and ACP combined device can work as an effective motion management strategy in abdominal radiotherapy.

SAM Therapy Scientific Symposium TRACK 5

Brachytherapy Technology Horizon

WE-F-TRACK 5-00

Brachytherapy Technology Horizon

Moderator: Bin Cai, UT Southwestern Medical Center, Dallas, TX

WE-F-TRACK 5-01

X. Jia, The University of Texas Southwestern Medical Ctr:
Deep Learning and Artificial Intelligence for Brachytherapy

WE-F-TRACK 5-02

W. Song, Virginia Commonwealth University:
Intensity Modulated and Anisotropic Brachytherapy Sources

WE-F-TRACK 5-03

Å. Carlsson Tedgren, Linköping University:
Advancements in Brachytherapy Plan Optimization

WE-F-TRACK 5-04

J. Robar, Nova Scotia Health Authority: Personalized Brachytherapy with Integration of 3D Printing Technologies

Brachytherapy is undergoing a dramatic transformation. There are tremendous advancements and automation across imaging, applicators, treatment planning, and QA. The future of brachytherapy will likely be shaped by these emerging technologies and this symposia is to take an in-depth look at what is on its horizon. First, we'll review the application of deep learning (DL) in brachytherapy. Undoubtedly, DL based approaches has become an effective and handy tools in tackling some of the most time-consuming tasks. Specifically, its role in image processing/registration, applicator/needle digitization, organ segmentation, and plan evaluation and toxicity prediction will be looked at. Second, we'll review the rise of cleverly designed shielded applicators that enable intensity modulated brachytherapy (IMBT). IMBT utilizes high-density shields to provide at least one additional degree of freedom in the dose delivery process (e.g., directional beam) and can result in marked improvement in plan quality. Third, we'll review a number of recent advances that improves the brachytherapy treatment planning in different aspects. For example, improved solution algorithms for new treatment technologies (IMBT), catheter placement and total number of catheters to improve treatment plan quality, robust optimization to mitigate the effects from uncertainty, and machine learning to automate the parameter tuning in optimization models. Finally, we'll review the state of the art in 3D printing and clinical applications as this technology has progressed in an impressive pace in the last few years. There are vast range of materials and printers to consider, and at least, one commercial solution available. It is poised to revolutionize the field and hence is a timely topic.

Learning Objectives:

1. Review of Deep Learning, Automation in Brachytherapy, including QA
2. Review of Intensity Modulated Brachytherapy and Plan Optimization
3. Review of 3D Printing in Brachytherapy and QA

Varian funding support.

Multi-Disciplinary Scientific Session

TRACK 6

Image-Guided Treatment Response Modeling and Assessment

WE-F-TRACK 6-01

A Bilateral Neural Network for Loco-Regional Recurrence Prediction in Head and Neck Squamous Cell Cancer

L Chen*, K Wang, C Shen, D Sher, J Wang, UT Southwestern Medical Center, Dallas, TX

Purpose: Accurate estimation of the risk of loco-regional recurrence (LRR) for patients with head and neck squamous cell cancer (HNSCC) prior to treatment is essential to physicians for making personalized treatment management. In this work, we proposed a bilateral neural network to predict LRR through using pre-treatment CT, PET and clinical data. **Methods:** In the proposed scheme, there are two classification convolutional neural network (cCNN) modules: the first cCNN module (cCNN_1) uses region of interest (ROI) containing tumor and its surrounding tissues as input while the second cCNN module (cCNN_2) uses tumor itself as input. cCNN_1 pays attention on the information of the tumor position and tumor contrast compared to surrounding tissues, while cCNN_2 concerns more on the tumor inside and tumor boundary. The features learned by these two networks are complementary, which are then fused with clinical data for the final LRR prediction. Additionally, due to the highly-imbalanced characteristics of the dataset (with vs. without LRR), we proposed a new loss function termed as focal-dual loss which focuses more on the samples hard to classify and also tries to decrease the probabilities that a data point is assigned to an incorrect class to train the classification model. This study included the images and clinical variables of 316 patients with HNSCC received radiation treatment, among which 60 patients experienced LRR. Five folder cross validation was used to evaluate the performance of the proposed scheme. **Results:** The sensitivity, specificity, accuracy, and area under the receiver operating characteristic curve (AUC) values obtained by the proposed scheme were 0.77, 0.75, 0.75 and 0.80 respectively, outperforming conventional cCNNs only using ROI or tumor itself, and radiomics approaches. **Conclusion:** The proposed method combining clinical feature and CT & PET imaging features learned from two cCNN modules provides a more accurate way to predict HNSCC LRR.

WE-F-TRACK 6-02

An Automated Multi-Objective Model with Hyperparameter Optimization for Treatment Outcome Prediction in Metastatic Melanoma

Z Zhou¹, M Zhou², Z Wang³, X Chen^{2*}, (1) University of Central Missouri, Warrensburg, MISSOURI, (2) Xi'an Jiaotong University, Xi'an, Shaanxi, China, (3) Peking University Cancer Hospital & Institute, Beijing, CN

Purpose: Accurately predicting immunotherapy response is of great importance to improve treatment effectiveness in metastatic melanoma. We aim to develop a new automated multi-objective model with hyperparameter optimization (AutoMO-HO) for treatment outcome prediction. **Methods:** Totally 50 patients who exam contrast-enhanced computed tomography (CECT) are used in this study. The images with both pre- and one-cycle post-treatment are obtained for extracting delta-radiomic features by calculating the difference between pre-treatment and one-cycle post-treatment radiomic features. To obtain more discriminative and complementary features, one-cycle post-treatment radiomic features are combined with delta-radiomic features to predict treatment response. To obtain the balanced sensitivity and specificity as well as higher confidence output, an automated multi-objective model (AutoMO) is applied in this study. However, there are several hyperparameters to be set manually before training, leading to the non-optimal model performance. As such, Bayesian optimization is introduced to automatically find the optimal model hyperparameter, and a new model termed as AutoMO-HO is developed based on AutoMO in this study. In AutoMO-HO, the training stage consists of two phases, they are Bayesian hyperparameter optimization through the Tree Parzen estimator algorithm and Pareto-optimal model set generation. In testing stage, the evidential reasoning (ER) strategy is used to fuse the output of each Pareto-optimal model to obtain more reliable results. Finally, the label with the maximal output confidence is taken as final output label. Accuracy, AUC, sensitivity and specificity are used for evaluation. **Results:** In this study, five-fold cross validation is performed. Accuracy, AUC, sensitivity and specificity of AutoMO-HO can achieve 0.88, 0.91, 0.96 and 0.88, respectively, while AutoMO without hyperparameter optimization is 0.84, 0.87, 0.82 and 0.85. **Conclusion:** A new automated multi-objective model with Bayesian hyperparameter optimization (AutoMO-HO) for predicting immunotherapy response is developed in metastatic melanoma. The experimental results demonstrated that AutoMO-HO can obtain better performance than AutoMO.

WE-F-TRACK 6-03

Changes in Post-Treatment Cardiac PET Avidity Predict Overall Survival in Lung Cancer Patients Treated with Chemoradiation

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Purpose: The relationship between thoracic radiation to the heart and overall survival (OS) has been recently demonstrated in non-small cell lung cancer (NSCLC) patients. Functional imaging, and in particular PET/CT, has been used to provide an early predictor for clinical outcomes. The purpose of this study was to use a

large, co-operative group clinical trial (ACRIN 6668) data-set to evaluate whether PET-CT changes in the heart are predictive of OS. **Methods:** We retrospectively evaluated data for 239 patients enrolled on the ACRIN 6668 study which evaluated the prognostic value of pre-treatment tumor PET-CT information for stage 2-3 NSCLC patients treated with chemo-radiation. 77 Eligible patients for the current analysis included patients for whom pre and post-treatment PET/CT data was acquired. The heart was contoured for each patient's PET-CT scan and the mean Standard Uptake Value (SUVmean) was calculated for each heart contour. The SUVmean cardiac change was calculated as post-treatment SUVmean minus pre-treatment SUVmean. The ability of cardiac SUVmean change to predict for OS was assessed using multivariable Cox modeling (controlling for cofactors of age and stage) and by comparing OS for patients with positive or negative cardiac SUVmean change. **Results:** The median OS was 469 days (range 144-1640 days). SUVmean cardiac change from pre to post-treatment was predictive of OS on multivariate analysis ($p=0.016$, Hazard ratio=0.81, 95% confidence interval=[0.68-0.96]). Median survival for patients with a negative SUVmean change (i.e. a post-treatment decrease in cardiac SUV) was 413 days while patients with a positive SUVmean change had a median OS of 585 days ($p<0.01$). **Conclusion:** The data demonstrates that cardiac SUV changes are significantly associated with OS in NSCLC patients. If validated in a prospective cohort, our data show the potential for cardiac metabolic changes to be an early predictor for clinical outcomes.

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WE-F-TRACK 6-04

Measuring Radiation-Induced Changes in Perfusion with Dynamic Contrast-Enhanced 4DCT and Non-Contrast Lung Density Analysis in a Novel Swine Model

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Purpose: To measure radiation-induced changes in perfusion using dynamic contrast-enhanced CT and correlate these changes to a non-contrast imaging biomarker in a novel swine population. **Methods:** Five Wisconsin Mini-Swine(WMSTM) received 60 Gy in 5 fractions following a clinical SBRT schedule using an MRI-guided LINAC. A dynamic contrast-enhanced 4DCT was taken prior to and 3 months post-RT. Vasculature was masked to enable the analysis of the non-vessel lung parenchyma and the vessels separately. All post-RT scans were deformably registered to the pre-RT scan using B-spline registration to allow voxel-wise comparisons. Analysis was performed in 10 Gy dose bins. Dose contours were copied onto the contralateral lung which received <5Gy for comparison. In each contour, measurements of the baseline HU measurement in the non-vessels and the max HU value of the vessels were obtained at pre and 3 months post-RT. **Results:** Statistically significant($p=0.001$) increases in the mean HU value in the non-vessel lung parenchyma were observed in the ipsilateral lung. The percent increases depended linearly with EQD2 dose ($R=0.989$). Additionally, statistically significant decreases ($p<0.05$) in the peak mean HU inside vessels were observed in each dose bin of the ipsilateral and contralateral lungs. The HU-increases outside the vessels strongly correlated ($R^2=0.85$) to HU-reductions inside the vessels suggesting radiation-induced vasculature leakage. The slope of the HU increase outside vessels vs. HU decrease inside vessels curve was -0.87. Additionally, there was an apparent threshold dose above which this leakage occurred of approximately 25Gy. **Conclusion:** This work suggests there is radiation-induced vessel leakage that occurs above a threshold dose of 25Gy. Additionally, this leakage can be predicted using a non-contrast biomarker (HU change outside vasculature). This presents the potential to develop non-contrast imaging bio-markers for radiation-induced change in perfusion to guide functional avoidance treatment planning.

Joseph M. Reinhardt is a shareholder in VIDA Diagnostics, Inc

John E. Bayouth is a co-owner of MR Guidance, LLC (MRG), which has performed business with vendors of technologies used in this publication

WE-F-TRACK 6-05

Quantitative Radiomics Approach to Assess Acute Radiation Dermatitis in Breast Cancer Patients

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Purpose: We applied a radiomics approach to skin surface images to objectively assess acute radiation dermatitis in patients undergoing radiotherapy for breast cancer. **Methods:** A prospective cohort study of 20 patients was conducted. Skin surface images in normal, polarised, and ultraviolet (UV) modes were acquired using a skin analysis device before starting radiotherapy ('before RT'), approximately 7 days after the first treatment ('RT D7'), on 'RT D14', and approximately 10 days after the radiotherapy ended ('after RT D10'). Eighteen types of radiomic feature ratios were calculated based on the values acquired 'before RT'. We measured skin doses in ipsilateral breasts using optically stimulated luminescent dosimeters on the first day of radiotherapy. Clinical evaluation of acute radiation dermatitis was performed using the Radiation Therapy Oncology Group scoring criteria on 'RT D14' and 'after RT D10'. Several statistical analysis methods were used in this study to test the performance of radiomic features as indicators of radiodermatitis evaluation. **Results:** As the skin was damaged by radiation, the energy for normal mode and sum variance for polarised and UV modes decreased significantly for ipsilateral breasts, whereas contralateral breasts exhibited a smaller decrease with statistical significance. The radiomic feature ratios at 'RT D7' had strong correlations to skin doses and those at 'RT D14' and 'after RT D10' with statistical significance. **Conclusion:** The energy for normal mode and sum variance for polarised and UV modes demonstrated the potential to evaluate and predict acute radiation, which assists in its appropriate management.

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WE-F-TRACK 6-06

Quantum Reinforcement and Deep Learning for Decision Support in Response-Adapted Lung Cancer Radiotherapy

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Purpose: To develop a robust decision support framework for personalized recommendations in response-adapted lung cancer Radiotherapy based on quantum computing and deep learning estimated tumor and normal tissue responses. **Methods:** Combined quantum reinforcement learning (QRL) and deep learning (DL) techniques were implemented for creating the proposed framework. DL models were used to create an Artificial Radiotherapy Environment, consisting of a transition function for governing the patients' state dynamics as a function of dose, and an RT outcome estimator of local control and grade 2 radiation induced pneumonitis for a given state. RT state features were selected from a multi-objective Bayesian network: IP10, GLSZM-ZSV, cxcrl-Rs2234671, Tumor-gEUD, and Lung-gEUD. Decisions were modeled as quantum states to reflect the associated uncertainty in treatment outcomes, and QRL was used for optimizing the decision-making process. Two new approaches were developed to improve the robustness of the framework: (1) a quantum controller circuit to evaluate the framework on a quantum processor and (2) a constrained DL approach to guide the RT outcome estimator learning of radiation response. We trained our framework on 67 non-small lung cancer patients treated with RT and then validated it on the independent data of 20 prospective patients. **Results:** The constrained DL modeling improved the learning of experimentally observed monotonic radiation response outcomes. The framework constructed with QRL and quantum controller using 5-qubit IBMQ quantum processor demonstrated improved performance over the ones trained in 3-qubit and 5-qubit quantum simulators. The RMSE values between the clinical dose decision and framework recommendations were 0.81 (0.78-0.84) Gy/frac, 0.93 (0.84-0.99) Gy/frac, and 0.86 (0.84-0.93) Gy/frac, respectively. The validation results showed reasonable performance in the institutional prospective dataset with an RMSE value of 0.96 (0.80-1.07) Gy/frac. **Conclusion:** This work demonstrates a robust QRL-DL framework for response-adapted lung cancer radiotherapy.

WE-F-TRACK 6-07

Computational Prediction of Symptomatic RP Based on Planning Computed Tomography Images Using Topologically Invariant Betti Numbers Prior to Stereotactic Body Radiation Therapy

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Purpose: Stereotactic body radiation therapy (SBRT) is commonly administrated for inoperable patients with early stage non-small-cell lung cancer. However, some patients, whose lung tissue is vulnerable to radiation, could experience radiation induced pneumonitis (RP). Therefore, we have developed a symptomatic RP prediction model by characterizing the vulnerability of lung regions based on planning computed tomography (pCT) images using topologically invariant Betti numbers (BNs) prior to SBRT. **Methods:** Two-hundred sixty-six patients who underwent SBRT were selected for this study. A total of 236 (20 with $\geq$ grade 2 symptomatic RP) and 30 cases (8 with $\geq$ grade 2 symptomatic RP) were used for training and independent test of the RP prediction models, respectively. A lung region on an axial plane of the pCT image, which included a centroid of a gross tumor volume, was analyzed for characterizing the vulnerability of lung regions. A total of 47,616 image features were obtained from 512 BN maps that may represent topologically invariant heterogeneous characteristics of lung cancer by applying histogram- and texture-based feature calculations. A BN-based signature was constructed using the least absolute shrinkage and selection operator logistic regression model. Support vector machine (SVM) models were utilized to build the RP prediction model using the BN-based signature in a leave-one-out cross validation. The performance of the BN-based model was compared to the SVM model trained with the conventional signature consisting of original pCT image features (pCT-based model) regarding a robustness index (RI) which defined higher total and lower difference of area under receiver operating characteristics curves (AUCs) in the validation and test. **Results:** The RIs for BN- and pCT-based models were 1.544 (AUC in the validation: 0.800, test: 0.778) and 1.191 (AUC in the validation: 0.750, test: 0.608), respectively. **Conclusion:** The BN-based model could be feasible for prediction of symptomatic RP prior to SBRT.

Vendor Showcase

TRACK 8

LAP Laser, LLC

WE-SHOWCASE-TRACK 8-17-1

Patient QA Solutions in Adaptive Radiotherapy with RadCalc

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Ultrasound Interactive ePoster Discussion

TRACK 1

Ultrasound

WE-lePD-TRACK 1-01

A Hybrid Method for Prostate Segmentation in Transrectal Ultrasound Based on Redefined Closed K-Segment Principal Curve and Improved Differential Evolution-Based Artificial Neural Network

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Purpose: Transrectal ultrasound (TRUS) is routinely used in ultrasound-guided brachytherapy of prostate cancer for treatment planning and verification. Manual delineation of the prostate volume is tedious, time-consuming, and subjective. The purpose of this work is to develop a prostate segmentation hybrid method using an improved redefined closed k-segment principal curve and improved differential evolution-based artificial neural network. **Methods:** TRUS scans of 55 brachytherapy patients obtained from the Vinno 70 Lab system were analyzed. We developed a novel method for TRUS prostate segmentation, using only a small number ($< 15\%$) of manually delineated points as the approximate initialization. This method contains three subnetworks: the first one uses an improved principal curve-based method to obtain the data sequences consisting of data points and projection indexes; the second one is using improved Memory-based and Cuckoo Search-based Differential Evolution (MCSDE) method for preliminarily searching for the optimal model of the machine learning method; lastly, we proposed a smooth mathematical model of prostate contour, which was denoted by the parameters of machine learning method. **Results:** Preliminary results show that the average Dice Similarity Coefficient (DSC), Jaccard Similarity Coefficient (Ω), and Accuracy (ACC) is 95.8%, 94.3%, and 95.4% on the testing dataset (15 patients). Meanwhile, the DSC, Ω , and ACC of the proposed method are as high as 93.3%, 91.9%, and 93%, respectively, at the influence of Gaussian noise (standard deviation of Gaussian function $\sigma=50$). Although σ increases from 10 to 50, the DSC, Ω , and ACC fluctuate about 2.5% at most, demonstrating the good robustness of our method. Compared with the competing models, our method has the best performance. **Conclusion:** Both qualitative and quantitative experimental results demonstrate that our hybrid method of prostate segmentation achieved superior performance compared with several state-of-the-art methods. Meanwhile, our method has good robustness to deal with the noise data effectively.

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WE-lePD-TRACK 1-02

Automatic Spatial Registration System for Whole-Breast Three-Dimensional Ultrasound

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Purpose: Mammography remains the standard in breast cancer screening, but challenges still exist in the 40-50% of women with radiographically dense breasts. Automated breast ultrasound (US) has enabled whole-breast imaging, improving reproducibility and reducing operator-dependence when compared to handheld US. However, these systems still require extensive user training and are costly. We propose an alternative spatially tracked system for automatic registration of three-dimensional (3D) US volumes for whole-breast imaging. This paper describes the system development and feasibility for accurate whole-breast 3DUS image reconstruction. **Methods:** The system is composed of an in-house 3DUS imaging subsystem, which can be adapted to any US transducer, and clinician-operated multi-jointed arm and counterbalanced stabilizer, which enables motion in six degrees-of-freedom. The system contains electromagnetic encoders in each of the six joints, which were integrated with custom software modules to track the 3D spatial coordinates of the 3DUS scanner and its image. Tracking accuracy was assessed for each joint independently by comparing the spatially-tracked position of a custom stylus to its optically-tracked position. Automatic spatial registration and 3DUS fusion was tested using string and breast phantoms. **Results:** The mean absolute tracking error (SD) was 0.87 ± 0.52 mm, 0.70 ± 0.46 mm, 0.53 ± 0.48 mm, 0.34 ± 0.32 mm, 0.43 ± 0.28 mm, and 0.78 ± 0.54 mm for joints 1-6 respectively. Quadratic-based lookup table corrections optimized the spatial tracking accuracy in translational joints 1-2 and rotational roll-joint. Compound motions utilizing all joints resulted in an absolute error of 1.08 ± 0.88 mm ($N=20$) within the intended clinical workspace for breast imaging. 3DUS images of the string and breast phantoms with simulated lesions demonstrated the utility for automatic registration and fusion of multiple 3DUS images using their spatially tracked positions. **Conclusion:** We demonstrated an accurate spatially tracked system for whole-breast 3DUS imaging. The system

shows potential as an alternative, cost-effective US-based approach with the capability for whole-breast imaging in women with radiographically dense breasts.

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WE-lePD-TRACK 1-03

Enhancement of Gene Therapy with MR-Guided Focused Ultrasound for Prostate Cancer

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Purpose: The purpose of this study was to investigate the effect of MR guided focused ultrasound (MRgFU) and AS-MDM treatment on investigate the inhibition of in vivo LNCaP tumor growth. **Methods:** LNCaP-orthotopic prostate tumor-bearing nude mice were divided into 3 groups (n=7/per group): group 1, MRgFU + AS-MDM2; group 2, AS-MDM2 only; and group 3, sham MRgFU. For group 1, tumors were allowed to grow to the size of 60 mm³ as determined by MRI and then treated with MRgFU for 6 treatments over three weeks (2 treatments per week). Each treatment was followed by an i.p. injection of AS-MDM (25 mg/kg) immediately after the MRgFU treatment. For group 2, only AS-MDM2 (25 mg/kg) was administered for 6 treatments (2 treatments per week). Group 3 received sham MRgFU treatment. Tumor growth was determined by measuring the tumor volume using weekly MRI. Mice were sacrificed when tumor reached about 600 mm³ or 7 weeks after treatment whichever occurred first. **Results:** Due to the limitation of the small animal model (25 g weight) 6 times of general anesthesia performed over three weeks caused high mortality (about 50%). Based on the 10 surviving animals that completed the treatment there were insignificant differences in tumor growth delay between the MRgFU+AS-MDM2 group and the AS-MDM2 only group. **Conclusion:** Possible reasons for the insignificant results of this study include (1) large variations in tumor growth due to limited number of mice, (2) insufficient tumor penetration of AS-MDM to enhance tumor growth inhibition, and (3) suboptimal conditions for AS-MDM2 as the greatest effect of AS-MDM2 was observed with androgen deprivation and/or RT. Further experiments are warranted to test MRgFU+AS-MDM2 under these other conditions.

WE-lePD-TRACK 1-04

Feasibility of An Automated Quality Assurance Tool for Ultrasound Shear Wave Elastography Images

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Purpose: Ultrasound shear wave elastography (SWE) can estimate tissue Young's modulus in selected regions of interest (ROI). It relies on careful acquisition and measurement by the operator. This study aims to develop an automated quality assurance (QA) tool for clinical liver SWE images on GE LOGIQ E9 scanners, which do not have an onboard quality indicator. **Methods:** Settings on LOGIQ E9 scanners which could affect SWE quality and quantitative measurement were investigated. A suite of custom MATLAB programs was built to check these settings and institutional protocol details, which were designed to increase the frame rate on this scanner model and minimize motion artifacts as reported before, from liver SWE DICOM images. Specifically, settings such as transducer model (C1-6D), acquisition parameters such as elastography gain, width of the SWE acquisition box, distance between the liver capsule and the SWE acquisition box, push and track beam output, percentage of color filling of the acquisition box, and diameter of the measurement ROI. The distance from rib shadowing artifact, if any, to the SWE box at transducer face level was also checked. Color-coded QA outputs were overlaid on the original SWE images off the scanners. Suggestions for optimization strategies were provided when certain conditions were met. **Results:** The QA tool has been established and retrospectively processed 1455 images acquired since 2018 by clinical sonographers. 1.1% of all parameters and settings was incorrect or could have been improved. The default elastography gain was found to be modified in 36 earlier images. This gain relates to cross correlation threshold for selecting pixels to be included in the quantitative measurement and therefore should be unchanged. **Conclusion:** Feasibility of an automated QA tool for liver SWE images that could provide users near real-time QA results and feedback was demonstrated. Artifact recognition will be included in the future.

WE-lePD-TRACK 1-05

MR to Ultrasound Image Registration with Segmentation-Based Learning for HDR Prostate Brachytherapy

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Purpose: To aid the organ contouring in high dose rate (HDR) prostate brachytherapy, a deep learning approach was first developed for automatic registration of magnetic resonance (MR) images to treatment planning ultrasound (US) images with severe needle artifacts. **Methods:** A new segmentation-based registration framework was proposed for this multi-modality image registration based on a weakly-supervised strategy, to address the problem of lacking training labels and difficulty of accurate registration from inferior image quality. Specifically, two prostate segmentation networks were trained to generate the weak supervision labels from the MR and US images automatically for the registration network training. To improve the registration accuracy, not only the image pair, but also the corresponding prostate probability maps from the segmentation were further fed to the registration network to predict the deformation matrix. A random augmentation method was designed to alleviate the map overfitting problem during the network training. MR data from 184 patient cases and US data from 167 cases were collected from our institution for the MR and US image segmentation networks, and the registration network learning. **Results:** The mean DSC, center-of-mass (COM) distance, Hausdorff distance (HD) and averaged symmetric surface distance (ASSD) results for the registration of prostate contours were 0.87±0.05, 1.70±0.89 mm, 7.21±2.07 mm, 1.61±0.64 mm, respectively. By providing the prostate probability map from the segmentation to the registration network, as well as applying the map augmentation method, the evaluation results of the four metrics were all improved, such as an increase of DSC from 0.83±0.08 to 0.86±0.06 and from 0.86±0.06 to 0.87±0.05, respectively. **Conclusion:** A novel segmentation-based registration framework was proposed to automatically register prostate MR images to treatment planning US images. It is potential to be applied in current HDR brachytherapy procedures to improve the treatment efficiency and quality.

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WE-lePD-TRACK 1-06

Quantitative Dimension Estimation and Clinical Utility of Three-Dimensional Ultrasound in Oral Cavity Tumors

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Purpose: Oral squamous cell carcinoma (OSCC) is the most common oral malignancy, representing approximately 90% of all oral cavity cancers. In consideration of the complexity of the oral cavity, imaging is integral to the classification of the primary tumor and disease extent. We propose the use of three-dimensional ultrasound (3DUS) as a point-of-care imaging modality for bedside oral cancer identification and tumor volume quantification. This work presents the clinical utility of 3DUS in a clinical case study of OSCC compared to healthy control. **Methods:** We developed a motorized handheld 3DUS device housing custom-printed 14L5 (10MHz) and 11L3 (7MHz) linear transducer-holders. Trans-submandibular 3DUS acquisition protocol involved 5.0cm linear translation at 4.0cm depth-of-field by a trained observer. 3DUS images of the patient and healthy control were acquired and anatomical structures were identified. In the pathological case, the maximum tumor dimensions in anteroposterior (AP) and craniocaudal (CC) planes and depth-of-invasion (DOI) were calculated and compared with preoperative CT and MRI measurements by quantifying the error in tumor volume estimation. **Results:** Bilateral visualization was attained of the oral floor anatomy with 10 MHz and 7 MHz transducers in both healthy and pathological cases. Preoperative CT and MRI images indicated tumor dimensions: 1.60x1.12mm (AP/CC) and 0.44cm DOI, and

1.45x1.90 cm (AP/CC) and 0.66cm DOI. 3DUS with the 10MHz transducer allowed for the approximation of tumor measurements at 1.44x1.37cm (AP/CC) with a 0.49cm DOI. The error in dimension estimation was 6%, 9%, and 11% for AP, CC, and DOI between the 3DUS and mean preoperative dimensions between CT and MRI images. **Conclusion:** We demonstrated clinical utility of 3DUS in OSCC in the oral floor. 3DUS acquisition enabled the complete visualization of the oral cavity anatomy, which supplants the demand for preoperative bedside imaging of the primary site and tumor dimension estimation.

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Imaging Interactive ePoster Discussion

TRACK 2

Imaging: Dose and Image Quality in CT and X-ray Imaging

WE-lePD-TRACK 2-01

Evaluation of the Feasibility of Universal Daily CT QC Protocols and Limits Across Different Manufacturers and Models

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Purpose: NYC requires daily CT QC to follow the requirements of the facility's accreditation body while NYS requires water CT #, noise, and uniformity to be performed using a common technique setting. With the goal to meet regulatory requirements, almost identical daily QC protocols were set up on all CT scanners at our institution. Initial limits were established by the QMP based on available data and ACR recommendations. This study retrospectively evaluates the feasibility of the approach and appropriateness of these limits. **Methods:** Manufacturers' CT phantoms were scanned axially and helically on 13 CT scanners (GE, Philips, Siemens, Toshiba). Center CT #, noise, and uniformity were recorded for both scan modes. According to the ACR, center CT # of water should be 0 ± 3 HU and must be 0 ± 5 HU; uniformity should be 0 ± 5 HU and must be 0 ± 7 HU, whereas limits for noise should be locally established by the QMP. Based on review of past daily CT QC, our institution established a universal limit of 0 ± 5 HU for center CT # and uniformity and 7HU for noise. **Results:** Center CT #, noise, and uniformity were analyzed for a median of 245 (183 – 360) days for each unit and mode. For all scanners and modes, the average values were -0.11 ± 1.02 HU, 1.02 ± 0.72 HU, and 4.56 ± 0.78 HU for center CT #, uniformity, and noise, respectively. Excluding the axial uniformity of the Philips unit, the difference between average and limit exceeded at least three standard deviations regardless of manufacturer, model, or scan mode. **Conclusion:** Universal daily CT QC is feasible on multiple makes and models. As part of quality improvement, decreasing the limits to 0 ± 3 HU for both center CT # and uniformity (excluding the Philips unit), and 6.5 for noise, is possible.

WE-lePD-TRACK 2-02

Local Diagnostic Reference Levels and Size-Specific Patient Dose in Computed Tomography at a Greek Hospital

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Purpose: Diagnostic Reference Levels (DRLs) are a useful tool for patient dose optimization, however, they do not consider patient size as the concept of the size-specific dose estimate (SSDE) introduced by AAPM. The aim of this study is to extract local DRLs for computed tomography (CT) exams at a Greek University Hospital in terms of volumetric CT Dose Index (CTDIvol) and dose-length product (DLP), as well as to evaluate the impact of patient size in patients' dose through SSDE. **Methods:** The retrospective data from a total of 164 CT studies of Abdomen-Pelvis (A-P), Brain, Chest-Abdomen-Pelvis (C-A-P), Pulmonary Angiography (CTPA) and Heart were extracted from a 16-slice and a 64-slice CT scanner and were utilized to extract DLP and CTDI values. This study estimated the effective dose (ED) from DLP measurements using the appropriate k-factors. SSDE was calculated from the product of CTDIvol with the appropriate conversion factor, based on the sum of the lateral and anteroposterior diameters. Results are compared with international DRL data. **Results:** Compared to national DRLs, the local DRLs regarding the CTDIvol values were lower by 14% for Abdomen-Pelvis, 43% for Brain and 25% for Chest-Abdomen-Pelvis exams. For CTPA and heart exams national DRLs were not suggested. In comparison to international DRLs, the picture of our results is mixed, having both higher and lower DRLs than other countries. However, for the CTPA exams, the local DRLs were higher by a range of 13%-30%. SSDE mean values were higher than CTDIvol mean values by a range of 19%-24%, nonetheless for the brain exams their mean values were the same. **Conclusion:** The metrics of CTDIvol, DLP and ED were useful to evaluate our institution's radiology practice regarding the radiation exposure. Since SSDE is a more accurate patient dose indicator, it would be advantageous to be established as a DRL.

WE-lePD-TRACK 2-03

Local Noise and Size of Region of Interest in CT Images

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Purpose: This study aims to investigate the correlation between the automated local image noise estimates with the manual noise estimates and effects of the size of region of interest (ROI) on the correlation. **Methods:** 76 clinical non-enhanced axial abdominal scans are randomly selected among the exams performed using the same protocol on the same scanner in the same quarter. One slice of each of 76 scans locating middle liver is used for comparison. For manual method, the standard deviation (SD) Hounsfield Unit (HU) within a 35-pixel radius circular ROI placed in a relatively uniformed area in the liver in ImageJ is used as the noise estimation. For local noise analysis, the SD HU of the $n \times n$ ROI ($n=3, 7, \dots, 67$) centering each body pixel is used as the noise estimate around that center pixel (Fig. 1). The local noise of each and every body pixel is calculated. This approach is automated in Matlab. The noise of the automated method is then compared with that of the manual method using linear regression approach. **Results:** The lower 25-percentile noise of all body pixel in general represents more uniform area shown in (Fig 2). Although the noises of the manual and automated method are well correlated, the slope and variance of residuals (Fig 3) increase with the square ROI size, suggesting that selecting the same smaller ROI for noise comparison across patients and scanners is required. **Conclusion:** The noise in clinic images estimated by a manual ROI is linearly correlated with the mean of lower 25 percentile of local noise obtained by an automatic method and a fixed small ROI is critical for intra- and inter-patient and scanner noise performance comparisons.

WE-lePD-TRACK 2-04

Global Noise Analysis of Single and Dual Source CT for Obese Patients

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Purpose: CT imaging of obese patients is limited by tradeoffs between image quality and radiation output (dose). For demanding clinical situations like abdominal imaging, sacrificing image quality may be required due to scanner output limits. In this study, we compared acquisition parameters and global noise (GN) between a standard, single-source (SS) CT protocol and a new dual-source (DS) protocol with enhanced radiation output capacity. **Methods:** A total of 97 matched SS and DS scans from 49 patients were compared. Matched pairs were included if the tube voltage, reconstruction kernel, and scanner platform were consistent between scans from each protocol, and changes in patient weight and water equivalent diameter were $<5\%$. GN was determined with a previously reported algorithm and its uncertainty within each protocol was established at the 95% prediction interval (PI, $\pm 1.96\sigma$). Prediction models were developed to estimate GN based on protocol type, CTDIvol, reconstruction kernel, and patient size parameters. **Results:** The median $\pm 1\sigma$ increase in CTDIvol for DS scans from the matched pairs was $52 \pm 32\%$

(21±11 mGy). The 95% PI for the uncertainty in GN was ±1.1 HU. GN in DS scans was reduced by more than 1.1 HU in 87% of matched pairs (Δ GN median±1 σ : -2.5±1.7 HU). The prediction models could estimate GN with RMS error of 0.9 and 0.6 HU, respectively, for SS and DS scans. Model predictions for three example patient sizes showed that DS scans provide the same GN at a lower dose under nearly all achievable scanning conditions when compared to SS scans. **Conclusion:** A DS protocol for obese patient CT imaging provides enhanced dose capabilities and reduces noise relative to our standard SS protocol which is limited in radiation output. The noise reduction benefits of the DS protocol improve with increasing patient size.

WE-lePD-TRACK 2-05

Deep Learning Vs. Iterative Reconstruction in CT Dose Reduction and Image Texture Preservation: A Live Animal Study

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Purpose: This study is to investigate the potential of deep learning (DL) in CT dose reduction while preserving image texture when compared to iterative reconstruction (IR) alone. **Methods:** The upper abdomen of a live 4-year-old sheep was scanned under anesthesia on a Siemens Force CT scanner at different exposures (120 kV, mA from 25 to 250). Images were reconstructed using both Filtered Back Projection (FBP) and Advanced Modeled Iterative Reconstruction (ADMIRE) at each exposure with various strengths (1-5). A recently-developed modularized deep neural network (MAP-NN) with 5 modules was used for CT imaging at each exposure level via progressive denoising. Region of Interest (ROI) were manually delineated over the liver for all image sets. Radiomic features were retrospectively extracted from the ROI with IBEX. The concordance correlation coefficient (CCC) was applied to quantify the agreement between any two sets of radiomic features. The structural similarity index measure (SSIM) was used to measure the similarity between any two images. **Results:** A total of 113 features are extracted. For FBP, the CCC shows consistent decrease with lower exposure. Both IR and DL have the potential to increase CCC of radiomic features with desired features (e.g., features at standard exposure with FBP), depending on DL/IR strength. DL has higher potential than IR to improve radiomic features at extremely lower exposure level. The relationship between exposure and IR/DL strength for dose reduction while preserving image quality is nonlinear. SSIM decreases with progressive increases of DL or IR strength. DL has higher potential than IR to preserve image structure at higher strength, independent of exposure. **Conclusion:** Using a multi-exposure experiment in a live animal model, deep learning shows better potential than iterative reconstruction to further reduce radiation dose while preserving structural similarity and feature correlation.

WE-lePD-TRACK 2-06

Radiation Dose Estimation for Small Animal Pencil Beam X-Ray Luminescence Computed Tomography Imaging

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Purpose: X-ray luminescence computed tomography (XLCT) is an emerging x-ray molecular imaging modality. Pencil beam XLCT imaging provides superior spatial resolution than other imaging geometries like sheet beam and cone beam geometries. However, the pencil beam XLCT suffers from long scan times resulting in concerns over dose, which discourages the use of pencil beam x-ray sources in XLCT imaging. **Methods:** To investigate the dose deposited by pencil beam x-ray in XLCT imaging, we used the Monte Carlo simulation platform, GATE (Geant4 Application for Tomographic Emission) to estimate the dose of one angular projection scan with three different x-ray sources. Furthermore, to investigate the dose deposited in a typical small animal XLCT imaging scan, dose estimations from six angular projections by the three different x-ray source energies were performed on a mouse trunk model composed of muscle, spine bone, and a tumor. **Results:** With the Sigray source from Sigray Inc, the bone marrow in the mouse leg model was estimated to have a radiation dose of 43 mGy for typical XLCT imaging with six angular projections, 100 micrometer scan step size, and 106 x-ray photons per linear scan. The spine bone, muscle tissues, and tumor structures of the mouse trunk model received a radiation dose of 38.49 mGy, 15.07 mGy, and 16.87 mGy respectively after six angular projections with 100 micrometer scan step size, and 106 x-ray photons per linear scan. **Conclusion:** A typical pencil beam based XLCT imaging has an acceptable dose deposited in small animals when six projections of measurements are used.

WE-lePD-TRACK 2-07

Assessing the Viability of Establishing Patient Dose Tables in Pediatric Radiography by Mining Quality Control Data

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Purpose: To determine the feasibility of establishing patient radiation dose look-up tables for pediatric X-ray exams by assessing the consistency of performance of 24 radiography X-ray tubes over a period of 5 years. **Methods:** Annual quality control testing records were collected over a period of 5 years between 2016 and 2020 for 24 Philips X-ray rooms. The data includes information about each tube's measured kVp, output/dose, and beam quality (HVL in mm Al). Measurements were taken using either a Raysafe X2 or RTI Piranha solid-state detector under narrow-beam in-air conditions at a source-to-detector distance of 58.5 cm. **Results:** The mean measured kVp had an average error of 0.52% over 5 years ranging from -3.8% to 2.48% deviation from the nominal. The mean kVp had an average percent standard deviation of 0.77% between machines over the same time period, where the largest percent standard deviation was 1.25% at 50 kVp in 2016 and 2018. The measured HVL averaged over 5 years was 2.38 mm Al at 60 kVp and 3.11 mm Al at 80 kVp with average percent standard deviations of 2.82% and 2.52% respectively. The maximum percent standard deviation was 3.32% at 60 kVp and occurred in 2016. The measured tube output in mR/mAs had an average percent standard deviation of 6.44% with a maximum deviation of 8.23% at 50 kVp in 2019. The mean standard deviation in average mR/mAs year-to-year was 1.18% with the maximum of 1.25% occurring at 50 kVp. Example look-up tables for pediatric chest and abdomen exams were generated based on average equipment performance data from the previous 5 years. **Conclusion:** Machine performance was found to be consistent between machines as well as year-to-year, suggesting that the creation of exam dose look-up tables is feasible.

Multi-Disciplinary Interactive ePoster Discussion TRACK 3

Multi-Disciplinary: Novel and Emerging Imaging Technologies in Radiation Therapy

WE-lePD-TRACK 3-01

An AI-Based Watchdog Application to Screen Out COVID-19 Patients in Radiation Therapy

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Purpose: To mitigate the risk of spreading COVID-19, it is crucial to detect people with the disease as early as possible. As Artificial intelligence (AI) assisted film reading technology is being increasingly adopted in radiology clinics, we developed a fully automated AI-based system to specifically screen out COVID-19 patients using routine CT simulation images. The tool will report the probability of a patient having contracted COVID-19. Particularly, it is not a diagnostic tool. **Methods:** The tool, which we call it AI-watchdog, consists of four components: monitoring of new CT simulation, prescreening of CT for the presence of lungs, detection of COVID-19 traits on CT images, and notification of positive patients to care team. As soon as a CT scan is found to be completed, it is copied to the AI-watchdog server. If lung is found on a CT scan, the AI component predicts the probability of COVID-19. The prediction is implemented using deep learning model convolutional neural network (CNN). It was trained and tested on three external datasets, and one internal dataset. To test generalizability, the model was tested on a cancer patient,

who had scans both before and after contraction of COVID-19. **Results:** The CT monitoring detects and make copies of new CT scans at a frequency of every 10 seconds. There is no discernable effect on the current clinic workflow. The prescreening takes 2 seconds. The prediction takes about 35 seconds, with accuracy of 96.7% and false positive 4.4%. The notification to care team includes 3 probabilities for COVID-19, common pneumonia, and normal. The model correctly identified all of the cancer patient scans. **Conclusion:** An AI-based watchdog application offers extra layer of barrier to prevent spreading of COVID-19 in clinics of radiation therapy. It has proven to be effective and practical.

WE-lePD-TRACK 3-02

Cherenkov Imaging of TSE Patients Using Cameras from Multiple Angles

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Purpose: Cherenkov signals generated in patients treated with TSET can be recorded in-real time to monitor the dose distribution on patient body surface during the treatment. However, the front facing camera alone cannot predict dose accurately near periphery areas (e.g. Vertex) of the patient due to large angle emission of Cherenkov light. **Methods:** The TSE patients were treated in Stanford technique at 500 cm source to patient midplane distance using dual-field 6-MeV electron beams at gantry angles 74° and 106° at 6 postures with HDTSE mode delivered by Varian Truebeam. Three ceiling mount time-gated intensified cameras were used to capture the Cherenkov signal generated from patient's skin surface. Scintillator luminescent detectors (SLD) and optically stimulated luminescent dosimeters (OSLDs) were taped to patient's chest, umbilicus, and side to record and verify the dose delivered to those locations. Prospective (geometrical) calibration was performed in 2-dimensions to correct the camera vignetting effect, lens correction and different pixel response. Dose was reconstructed from the Cherenkov imaging by normalizing intensity to the OSLD measurements at the same location. **Results:** Perspective calibration was studied in 2D for Cherenkov imaging for all cameras and applied to acquired patient data. The corrected Cherenkov intensity can be converted to dose by normalizing to OSLD or SLD measurements at one location. Using additional Cherenkov cameras improved agreement in vertex from 40-50% low to within 5% for most of the patients compared to the in vivo dose measurements. **Conclusion:** Cherenkov imaging provides valuable information about the dose distribution for TSET. Geometrical correction was established in 2D for Cherenkov imaging and dose distribution can be reconstructed from the Cherenkov imaging.

WE-lePD-TRACK 3-03

Developing Internal Liver Vascular Models for Radiation Therapy Assessment of Dose to Circulating Lymphocytes

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Purpose: Lymphopenia is a side effect of radiation therapy that correlates with treatment outcome. The purpose of this work was the development of a model of the internal vasculature of the liver for the adult female and male (AF/AM) in order to calculate dose to the blood. Computational phantoms currently have no intra-organ vasculature that allows the calculation of dosimetric quantities in the blood. **Methods:** Computer models of arterial, portal and hepatic vein trees were generated from optimization and physical principles using the Constrained-Constructive-Optimization (CCO) method. As liver has independent vasculature for each of the eight segments (following the Couinaud classification), AF/AM reference livers were divided into segments by matching the percentage of total liver volume to reported values of human liver. Next, the vessel generation algorithm was utilized to create independent vascular trees for all inflows (arterial and portal) and outflow (hepatic veins). Hemodynamic and geometric parameters of the main vessels were used as inputs. During the algorithm, pressure, blood flow, and radius of vessels in the tree are updated each time a new vessel is created and connected to the optimal bifurcation site. **Results:** A closed vascular tree of approximately 6000 straight cylindrical non-intersecting pipes representing the hepatic artery, portal and hepatic veins was created independently for the AF/AM reference liver. While a straightforward comparison is difficult because of the variation in vascular structures, the morphology of our models shows similar trends to those observed in actual human liver vasculature. **Conclusion:** We have developed computational models of human liver vasculature by implementing a vessel generation algorithm that creates binary trees representing realistic liver vasculature. Although the present model does not include the precapillary and capillary vasculature, or blood sinusoids, our model will be a useful tool to perform more realistic dose-calculations to circulating blood cells during radiation therapy.

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WE-lePD-TRACK 3-04

Preclinical Application of a Cross-Voxel Exchange Model for the Estimation of Tracer Transport and Drug Accumulation

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Purpose: Transport models, when paired with non-invasive imaging techniques, aim to predict drug accumulation in tumor tissue, greatly advantageous for tumor diagnosis and treatment outcome assessment. A Cross-Voxel Exchange Model (CVXM) describes the change in tracer concentration as the exchange of tracer between blood and tissue, in addition to the transport of tracer through the interstitium (diffusion and convection). At the level of a voxel, CVXM considers intra- and inter-voxel exchange of tracer. The goal is to derive kinetic properties of tumor tissue from DCE-MRI to predict drug perfusion and accumulation. **Methods:** Twenty TS-415 human cervical carcinoma xenografts were established inside the gastrocnemius muscle of adult female BALB/c nu/nu mice. The mice were subjected to DCE-MRI for a period of 15 minutes using a 1.5-T whole body scanner (Signa, General Electric) at a spatial resolution of 0.23x0.23x2.0mm³ and a time resolution of 14 seconds. The tracer injected was Gd-DTPA. Regions of interests were selected radially from the tumor center outwards. CVXM was fitted to the data to generate spatial maps for fluid flow velocities v , diffusion coefficients D and extravasation parameters Kext and v_e . **Results:** The fitting results allowed distinguishing higher velocity measures at the periphery averaging to 14.37±3.65µm/s. The other resulting parameters are D (mean 253.98±85.37µm²/sec), Kext (mean 0.02±0.01min⁻¹) and v_e (mean 0.09±0.02). These kinetic properties are in line with the observations in the MRI scans: strong contrast enhancements at the periphery (convection) and slower concentration uptakes towards the center of the tumor (slower acting diffusion). **Conclusion:** CVXM fittings show a heterogeneous distribution of transport parameters suggesting an uneven accumulation of drug in the tumor microenvironment. The results support the need to investigate CVXM as a non-invasive technique to determine kinetic properties of the tumor tissue leading to better predictions of tumor aggressiveness and more informed treatment decisions.

WE-lePD-TRACK 3-05

Understanding the Effect of CPAP On Respiration for Thoracic Radiation

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Purpose: To use implanted electromagnetic beacons to understand the effect of Continuous Positive Airway Pressure (CPAP) on the regularity and amplitude of respiration when undergoing thoracic radiation therapy. **Methods:** This was an exploratory objective of an MLC tracking trial in patients receiving stereotactic ablative body radiotherapy for lung tumours. Electromagnetic beacons (Calypso, Varian Medical Systems) were endobronchially inserted in the lungs of eight patients receiving stereotactic ablative body radiotherapy for lung tumours. Beacon motion traces were acquired for four treatment fractions per patient under regular

respiration (during radiation therapy). Subsequent to each fraction, CPAP was applied and the pressure was increased over 5-minutes from 5 cmH₂O to 15 cmH₂O, after which beacon motion traces were recorded for 3 to 8 minutes. 3D motion vectors were assessed; the difference in mean respiratory period and motion amplitude were compared between regular respiration and CPAP traces. **Results:** CPAP reduced the amplitude by (mean $\pm$ st.dev.) 1.0 $\pm$ 4.4 mm (range -4.6 to 11.5 mm); greater amplitude reduction was observed for larger breathing motions (Spearman rank ρ = 0.67). CPAP increased motion however in eighteen fractions in seven out of eight patients, although the increase in five of the fractions was under 1 mm. The mean respiratory period increased by 1.4 $\pm$ 2.1 seconds (range -6.7 to 2.3 seconds) when using CPAP. Five out of the eight patients had fewer breaths statistically considered outliers with the use of CPAP than with regular respiration, implying less erratic breathing. **Conclusion:** While on average CPAP reduced respiratory motion most patients experienced an amplitude increase in at least one fraction. Motion reduction was maximised for patients with larger respiratory motion, indicating a subset who may benefit from CPAP for motion margin reduction. CPAP resulted in an increase in respiratory period and a decrease in erratic breaths.

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WE-lePD-TRACK 3-06

Uptake and Retention of Radiosensitizing Nanoparticles in Key Cells of a Pancreatic Tumor Microenvironment

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Purpose: Pancreatic cancer is one of the deadliest types of cancer with a 5-year survival rate of only 10%. Nanotechnology offers a novel perspective to treat such deadly cancers through their incorporation into radiotherapy and chemotherapy. However, the interaction of nanoparticles (NPs) with cancer cells and with other major cell types within the pancreatic tumor microenvironment (TME) is yet to be understood. Therefore, our goal is to shed the light on the dynamics of NPs within a TME of pancreatic origin. **Methods:** In addition to cancer cells, normal fibroblasts (NFs) and cancer associated fibroblasts (CAFs) were examined in this experiment due to their important yet opposite roles of suppressing tumor growth and promoting tumor growth, respectively. We used gold nanoparticles as the model NP system due to their biocompatibility and physical & chemical proprieties, and their dynamics were studied both quantitatively and qualitatively in vitro and in vivo. **Results:** The in vitro studies revealed that both cancer cells and CAFs take up 50% more NPs compared to NFs. Most importantly, they all managed to retain 70-80% of NPs over a 24-hour time period. Uptake and retention of NPs within an in vivo environment was also consistent with in vitro results. **Conclusion:** The experiment concludes that NP-based targeting of both tumor cells and CAFs could be used as a smart novel strategy towards improving current anti-cancer therapies.

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WE-lePD-TRACK 3-07

Enabling Upright Radiotherapy Through the Deformation of Horizontally Acquired Images

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Purpose: Upright radiotherapy can lead to improved treatments for lung cancer and more widespread adoption of proton therapy. However, as CT/MRI scanners are orientated horizontally, treatment planning for upright radiotherapy relies on bespoke vertical imaging systems. The purpose of this work was to enable upright radiotherapy by mathematically deforming horizontally acquired images to represent upright anatomy. **Methods:** Horizontal (supine) and upright (seated) volunteer lung images were acquired on a 1T prototype open-bore MRI-Linac. Free-breathing MR slices were acquired with using gradient echo sequences (TE/TR=10/124 ms) in the coronal plane. Both volumes were corrected for gradient non-linearity distortion and rigidly aligned. MIM was then used to contour and register the horizontal images to the upright images using the contour-based hybrid deformable image registration (DIR) algorithm. From the DIR output, a novel deformation algorithm was applied to the horizontal images to generate new images that represented upright anatomy. The horizontal to upright transformation was quantified by (1) magnitude of deformation, (2) the structural similarity index measure (SSIM) and (3) the Dice similarity coefficient (DSC) of lung volumes. **Results:** The average deformation from horizontal to upright anatomy for lung voxels was 1.0 $\pm$ 6.2mm, 4.4 $\pm$ 9.9mm and -3.9 $\pm$ 7.2mm in the LR, SI and AP directions, respectively. Compared to acquired upright images, the generated upright images had a SSIM=0.36, while a repeated upright acquisition had a SSIM=0.46 and a horizontal acquisition had a SSIM=0.30. Lung volumes on the generated upright images had a DSC=0.86 with lung volumes on acquired upright images. This compared to a DSC=0.90 for repeated upright acquisitions and a DSC=0.76 with horizontal images. **Conclusion:** This work is the first to demonstrate the feasibility of using DIR to represent upright anatomy from horizontal images. This technique could be used to create upright treatment plans from horizontal CT/MR images and enable upright radiotherapy treatments.

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Therapy Interactive ePoster Discussion

TRACK 5

Therapy: SBRT/SRS Treatment Planning

WE-lePD-TRACK 5-01

Impact of Machine Uncertainties on Dosimetric Robustness of Modern-Day Stereotactic Radiosurgery Techniques

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Purpose: Multiple techniques are available for stereotactic radiosurgery (SRS) treatment of brain metastases on either a linac or Gamma Knife (GK) machine. Aside from the dosimetric differences, these techniques differ from each other in the magnitude of machine uncertainties inherently present and subsequently in the errors those uncertainties may cause. This study was designed to compare the dosimetric robustness of frameless GK, Elements, HyperArc and VMAT techniques as a function of various machine uncertainties. **Methods:** 16 patients previously treated for multiple brain metastases were retrospectively identified. 5 SRS plans were generated per patient: frameless GK, Elements, HyperArc, and two VMAT plans by independent planners. All plans met clinical planning standards. PTVs were defined as 1mm isotropic expansion of GTVs. We simulated dosimetric perturbations caused by machine uncertainties by applying a range of translational/rotational displacements to PTVs, up to maximum magnitudes of 0.1 mm/0.3 deg and 2 mm/0.5 deg for frameless GK and linac-based plans, respectively. The impact on V100% (V100), normalized D99% (nD99), and the Paddick Conformity Index (PCI) were assessed and compared for PTVs among plans. The one-way ANOVA and Tukey tests were performed for statistical analysis with a significance level of $p < 0.05$. **Results:** The mean V100, nD99, and PCI accounting for machine uncertainties were all significantly different ($p < 0.001$) among the plans. Overall, the GK Frameless plans demonstrated the best robustness across the metrics

(V100=0.877; nD99=1.180; PCI=0.962), which could be explained by the much tighter machine tolerance for GK. Among the Linac-based plans, Elements (V100=0.840; nD99=1.120; PCI=0.740) was more robust against machine uncertainties than the HyperArc/VMAT plans. Smaller variances were also observed within the other VMAT plans. **Conclusion:** Our results indicate that GK and Elements are more robust against machine uncertainties than the other common linac-based SRS treatment techniques.

WE-lePD-TRACK 5-02

Dosimetric Impact of Dose Calculation Algorithms on Lung SBRT Plan for COPD Patients

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Purpose: We evaluated the AAA and AcurosXB (AXB) algorithms for varying Hounsfield-Units (HUs) in lung SBRT patients to investigate the dosimetric impact of an extremely low-HU caused by Chronic Obstructive Pulmonary Disease (COPD) on target coverage and Organs-at-Risk. **Methods:** SBRT lung plans were generated using a Quasar Lung Phantom with a prescription of 12Gy $\times$ 5 to the PTVs located centrally and peripherally. The lung average-HU was varied from -990 to 650 to simulate the low-HU region caused by COPD. The AAA plans were calculated with varying HUs in lung using the Eclipse 15.6-AAA and normalized to cover 95% of the PTV with the prescription dose. With the same HU and monitor-unit, the AXB plans were generated. The dose-to-medium in AXB plans was compared with the dose-to-water in AAA plans with the dosimetric indices including D99% for GTV/PTV, maximum dose of spinal-cord and rib, and lung V20. PTV coverage in AAA plans was analyzed for varying HUs caused by COPD, and evaluated with respect to the PTV coverage in AXB plans. **Results:** The AAA overestimated the PTV coverage of > 10% for both targets in the HUs below -900. The central target was less impacted by varying HUs compared to the peripheral target. For an extremely low-HU of -990, the AAA overestimated the PTV D99% up to 45% compared to the AXB. However, the two dose algorithms were comparable in the normal lung HU range from -750 to -650. Spinal-cord and rib doses were relatively insensitive to the variation of HU for both targets. As the HU decreased below -900, the AAA overestimated the lung V20 and skin dose, and lung V20 was strongly dependent on varying HUs. **Conclusion:** For COPD SBRT cases, the AAA overestimates the PTV coverage over 10% in the HUs below -900 and the AXB is more accurate than the AAA.

WE-lePD-TRACK 5-03

Robust Optimization with Multiple-CT Sets to Account for Deep Inspiration Breath Hold Variation in Pancreatic Stereotactic Body Radiotherapy

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Purpose: Internal margin expansion on the gross tumor volume (GTV) is widely used to account for deep inspiration inter-breath hold (DIBH) variation in tumor positioning in pancreatic stereotactic body radiation (SBRT). This margin expansion promotes better target coverage potentially at the cost of increased dose to the gastrointestinal organs. We investigated whether robust optimization with multiple breath-hold CT sets can account for inter-breath hold variation with organ-at-risk (OAR) sparing in comparison with the margin-based compensation. **Methods:** Ten DIBH SBRT pancreatic cancer patients were selected for this retrospective study. At simulation, four DIBH CT sets were acquired. Conventional margin-based plans were generated using volumetric modulated arc therapy (PTV-VMAT), where planning tumor volume (PTV) definition included breath hold variation and a 2-mm setup margin. GTV-based plans (GTV-RO-VMAT) were generated using robust optimization (RO). OAR positioning uncertainties were considered using physician-approved contours on all DIBH CT sets with a 2-mm isotropic setup uncertainty. PTV-VMAT and GTV-RO-VMAT plans were considered matched when GTV coverage was equal at D99%. Target coverage and OAR doses were compared for both approaches. The metrics used for the dosimetric comparison were D99% and V130% for the GTV, and V20Gy and Dmax for duodenum, stomach and bowel. **Results:** RO reduced OAR V20Gy by an average of 10%, 20%, and 40% for duodenum, stomach, and bowel, respectively, at the cost of slightly increased Dmax (2-3%). Due to plan matching conditions, GTV D99% was consistent between plans. The V130% increased from an average of 0.4 to 6.4 cc in RO plans, however, these hot spots were well-contained within the GTV. **Conclusion:** We have demonstrated that RO techniques can be applied to pancreatic SBRT planning with DIBH to significantly reduce OAR dose while maintaining GTV coverage. RO tools are clinically available and may be applied to enhance the efficacy of photon-based SBRT planning.

WE-lePD-TRACK 5-04

Feasibility of Using RayStationTM TPS MonteCarlo Algorithm for Evaluating Ocular Dose Delivered Via Dedicated Proton Eyeline

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Purpose: Treatment of Ocular melanoma with protons, delivered via a dedicated fixed gantry beamline (aka Eyeline) has a long history and excellent clinical outcomes. However, creating a TPS that satisfies the requirements of modern multi-professional ocular treatment team, has been largely a matter of in-house program development at a few institutions. As new commercial treatment planning systems coming on the market, some of which are offering highly accurate computational methods, such as MonteCarlo, it stands to reason to attempt adapting such an algorithm to the purpose of calculating the dose for Eyeline treatment of Ocular Melanoma. **Methods:** A spectrum of PDDs and lateral profiles, corresponding to the options available in Eyeline was recreated in RaystationTM treatment planning system, using clinically commissioned pencil beam scanning option and MonteCarlo calculation algorithm. Semi-automated inverse optimization routine was constructed in RaystationTM optimization module to ensure target coverage. The size of the target was determined by the Range and Modulation of the option under investigation. 11 beams were inverse-optimized in this manner. In addition, a brass aperture was defined as an automatically created structure in Raystation. Block margins were kept to 2.5 mm as used clinically. **Results:** Distal and lateral dose fall-off distances were compared to the counterparts from eyeline measured PDD and profiles. It is shown that, given appropriate parameters of the optimization, layer-to-layer and spot-to-spot distance and appropriate overrides (brass block, air, negligible air gap between range shifter and the external planning structure), Distal Fall-off and Lateral 80-20% penumbra can be easily recreated in the vicinity (i.e., within 0.5mm on average, maximum deviation was 0.9mm for lateral penumbra) of the true measured clinical beam values: i.e., 3mm distal fall-off and 1.1-1.7 mm penumbra, respectively. **Conclusion:** It appears quite feasible to use Raystation MonteCarlo algorithm for Eyeline dose calculation and evaluation.

WE-lePD-TRACK 5-05

Uncertainties in the Lung Heterogeneity Correction and Local Control for Lung SBRT

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Purpose: In this study, local control for 93 prior lung SBRT patients was evaluated in regard to dosimetric uncertainties related to the lung heterogeneity correction. **Methods:** Under a retrospective IRB protocol, 93 patients who received SBRT for non-small-cell lung cancer from 2007-2014 with treatment plans originally calculated with AAA (n=86) or PBC (n=7) were retrospectively recalculated using Acuros. A subset (n=22) was also recalculated with Monte Carlo for comparison. Dosimetric uncertainty was quantified for PTV mean dose, D95%[Gy], D1%[Gy], mean lung dose, lung V12Gy[cc], biologically effective dose (BED), and tumor control probability (TCP). Correlation with local control was analyzed. **Results:** 8 of 93 patients experienced local failures within four years. For local failures and controls, BED from Acuros was 107.2 $\pm$ 14.8Gy and 127.7 $\pm$ 27.0Gy, respectively, for PTV mean dose (p=0.03) and 119.2 $\pm$ 10.8Gy and 151.8 $\pm$ 31.5Gy, respectively, for D1%[Gy] (p=0.005). After recalculation with Acuros, the original BED and TCP decreased by 5.2 $\pm$ 6.2% and 4.2 $\pm$ 4.9%, respectively, with 10% of all plans having their PTV mean dose decreasing by >12.2% and D95%[Gy] by >26.7%. For one particular failure, BED decreased from 109.9Gy to 72.5Gy and TCP from 0.55 to 0.26. Plan parameters that were correlated with dosimetric uncertainties in a multivariate analysis included beam energy (Δ D95%[Gy] of -5.8%, -7.2%, and -11.5% for 6X, 10X, and 15X, respectively), lung volume (Δ D95%[Gy] = -2.9% per 1000cc), PTV mean density (Δ D95%[Gy] = 1.9% per 100HU), and PTV size (Δ D95%[Gy] = -0.82% per 25cc). Consistent with previous studies, Acuros had better agreement with Monte Carlo (Δ D95%[Gy] = -1.7% $\pm$ 1.4%) than did AAA

($\Delta D95\%[Gy] = -9.7\% \pm 7.8\%$). **Conclusion:** For lung SBRT, dosimetric uncertainty from the heterogeneity correction can have a substantial effect on BED, potentially leading to increased probability of local failure. Future work will include expanding the patient cohort for further statistical analysis of local control.

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WE-lePD-TRACK 5-06

Feasibility of Using 6MV-FFF Halcyon RDS for Single-Isocenter/Two-Lesion Lung SBRT

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Purpose: To demonstrate the plan quality and delivery efficiency of volumetric modulated arc therapy (VMAT) with Halcyon ring delivery system (RDS) in the treatment of single-isocenter/two lesion lung stereotactic body radiation therapy (SBRT). **Methods:** Fourteen patients with two lung lesions previously treated with 6MV-FFF VMAT lung SBRT using a single-isocenter on C-arm TrueBeam Linac were selected. The mean isocenter to lesion distance was 6.0 ± 2.4 cm and combined target volume was 56.8 ± 37.4 cc. Prescribed dose was 50 Gy to each lesion in five fractions, treated every other day with 2-4 non-coplanar partial/full arcs using 6MV-FFF beam and AcurosXB dose calculation engine for tissue heterogeneity corrections. For comparison, TrueBeam single-isocenter plans were re-optimized for Halcyon RDS coplanar geometry. TrueBeam and Halcyon plans were normalized for identical combined target coverage. High and intermediate-dose spillage were compared. Normal lung metrics and dose to organs at risk (OARs) were evaluated. Plan complexity was assessed via total monitor units, beam-on time and modulation factor. **Results:** Halcyon plans were statistically similar to clinically delivered TrueBeam plans. No statistical differences in target conformity, dose heterogeneity and intermediate-dose spillage were observed (all, $p > 0.05$). Halcyon plans on average demonstrated statistically insignificant reduced maximum dose to most adjacent OAR. However, Halcyon yielded statistically significant lower maximal dose to the ribs ($p = 0.02$) and normal lung V10Gy ($p = 0.04$). Plan complexity slightly increased as seen in the average increase of monitor units, modulation factor and beam-on time by 533, 0.5 and 2.8 min, respectively. However, the estimated overall treatment time was reduced by 2.4 min. **Conclusion:** SBRT treatment of synchronous lung lesions via single-isocenter VMAT on Halcyon RDS is feasible and dosimetrically equivalent to clinically delivered TrueBeam plans. Halcyon provides excellent plan quality and shorter overall treatment time that may improve patient compliance, reduce intra-fraction movement and improve clinic efficiency.

This work is partially supported by Varian Medical Systems.

WE-lePD-TRACK 5-07

Impact of Deep Inspiration Breath-Hold Parameters and Gastrointestinal Gas Volume on Tumor Position Uncertainty in Stereotactic Body Radiation for Pancreatic Cancer

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Purpose: Deep-inspiration breath-hold (DIBH) is often employed to minimize respiratory-induced tumor motion in pancreatic stereotactic body radiotherapy (SBRT). Spirometer systems such as the Active Breathing Coordinator (ABC) assists patients during the DIBH process to promote reproducibility in tumor position, but how breath-hold parameters and daily variation in bowel gas patterns influence tumor position is unclear. We investigated the impact of these variables on tumor-position reproducibility (BH_Tumor_Var). **Methods:** Twenty pancreatic cancer patients who underwent DIBH SBRT were selected for the study. A total of 379 DIBH simulations and CBCT images and the corresponding ABC signals were analyzed. The tumor position was quantified as the centroid of three fiducial makers implanted into the tumor. The BH_Tumor_Var was defined as a relative shift of the centroid compared to that in the planning CT. The ABC signal of each image was characterized based on breath-hold duration (BH_Duration), percentage of inhaled volume relative to the total lung volume (Lung_Utilization), and rate of inhalation (Inhale_Rate). The total lung and gastrointestinal gas volumes (Lung_Volume and GI_Gas_Volume) were also quantified. Multivariate linear regression analysis was performed to understand a correlation between the five variables and BH_Tumor_Var. **Results:** The Inhale_Rate was the only statistically significant variable for BH_Tumor_Var in SI. As the Inhale_Rate increased, the tumor moved inferiorly. The BH_Duration, Lung_Utilization, and changes in GI_Gas_Volume were correlated with BH_Tumor_Var in AP. The tumor moved posteriorly when the BH_Duration, Lung_Utilization, and GI_Gas_Volume were shorter, higher, and increased, respectively. The BH_Tumor_Var in LR was correlated with all four variables except the GI_Gas_Volume. When the Inhale_Rate, Lung_Volume, and BH_Duration increased, the tumor moved to the left while higher Lung_Utilization caused the tumor motion to the right. **Conclusion:** Our study indicates that controlling DIBH parameters including visual-feedback of inhalation rate and minimizing variation of gastrointestinal gas volumes are important for reproducible tumor position.

Therapy Interactive ePoster Discussion TRACK 6

Therapy: Treatment Accuracy

WE-lePD-TRACK 6-01

The Role of Surface-Guided Radiation Therapy to Improve Patient Safety

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Purpose: Surface-guided radiation therapy (SGRT), while most useful for patient positioning and motion management during radiotherapy, can also improve safety and quality by acting as an "independent observer" during treatment. This study uses multiple incident learning systems (ILS) to identify the types and frequency of errors that could potentially be avoided by implementing SGRT in the clinic. **Methods:** Three ILS were retrospectively mined to categorize and quantify errors that could have been prevented with SGRT. As the availability of technology and types of errors may vary across practices, the ILS represented international (SAFRON, IAEA), regional (AvIC, Sweden), and institutional (UW-ILS, USA) databases. General incident categories relevant for SGRT were identified and used to search events. Each event was reviewed individually and assigned to the relevant Incident categories if considered preventable with SGRT. The severity of SGRT-related events was investigated. **Results:** A total of 179 events were identified to be preventable with SGRT, representing 21% of reports at the verification/treatment stage. The most common events were wrong isocenter (43%) and wrong immobilization (34%). The distribution of event types was similar between the UW-ILS and SAFRON databases. AvIC had a smaller absolute number of events and immobilization-related events, which was attributed to the widespread use of SGRT in Sweden. Severity data indicated that SGRT-preventable events tend to have high severity scores. **Conclusion:** To our knowledge this is the first study of its type to study the value of SGRT in error prevention. Errors related to the wrong isocenter and immobilization are the most common pathways that could be detected with SGRT. By reducing misalignments and monitoring the entire treatment process, SGRT has the potential to be a valuable tool for increasing patient safety in radiotherapy.

WE-lePD-TRACK 6-02

Leveraging a Multi-Institutional Consortium to Understand Dose Accuracy in the Near-Surface Region for Whole Breast Irradiation

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Purpose: To assess accuracy of dose calculations in the near-surface region for different treatment planning systems (TPSs), plan complexities, and energies to improve clinical decisions for patients receiving whole breast irradiation (WBI). This work is part of a multi-institutional effort investigating the correlation between mean dose calculated in the near-surface region and acute skin toxicities considering different WBI treatment techniques. **Methods:** A portable custom breast phantom was designed for measurements at eight institutions. Clinically-relevant treatment plans (open, field-in-field (FiF), FiF mixed-energy, and IMRT) were created in four TPSs on a patient dataset (50Gy/25fx) and peer reviewed by participants. After transfer to the phantom dataset, plans were re-calculated with fixed monitor units. For each irradiation, which included linear accelerators from different vendors, the phantom was aligned with predetermined moves or CBCT. Dose was measured with radiochromic film (1.5cmx13.0cm) placed at two depths (0.5 and 1.0cm) and three locations per depth within the phantom. Film was scanned (> 24h post-irradiation) and analyzed using FilmQA Pro. Dose differences were evaluated relative to the calculation. **Results:** Measured and calculated doses agreed well for all TPSs, complexities, and beam energies with 86.7%, 77.8%, 91.9%, and 80.6% of measurements within $\pm 5\%$ agreement for open field, FiF, FiF mixed-energy, and IMRT, respectively. Based on a Gaussian fit of the difference distribution, the mean percent difference was $-0.2 \pm 2.9\%$, with 95% of measurements within 6% agreement. Furthermore, Pearson correlation scores (# of measurements) of 0.8(60), 0.9(54), 0.7(36), and 0.8(72) showed good agreement for open fields, FiF, FiF mixed-energy, and IMRT, respectively. **Conclusion:** To the best of our knowledge, these results demonstrate that dose calculations from clinically relevant WBI plans for various complexities and beam energies are accurate in the near-surface region. This lays the foundation for future work to investigate the correlation between near-surface dose and patient skin toxicities.

The Michigan Radiation Oncology Quality consortium is financially supported by Blue Cross Blue Shield of Michigan and the Blue Care Network of Michigan as part of the BCBSM Value Partnerships Program.

WE-lePD-TRACK 6-03

Gradient-Dose Segmented Analysis (GDSA): An Improved Method for IMRT QA Analysis Compared to the Gamma Analysis

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Purpose: Presentation of a new method for IMRT QA comparisons that is more sensitive, more specific, and more clinically meaningful than gamma comparisons. **Methods:** Gradient-dose segmented analysis (GDSA) works by segmenting the dose map from calculations on an IMRT QA phantom into regions of high vs. low dose and dose gradient. The mean local dose difference in the high-dose low-gradient regions of the phantom comparison directly predicts the changes in the patient PTV mean dose using only information from the calculation on the detector array. GDSA is based on analysis of >180,000 comparisons in 40 IMRT and VMAT cases using various induced errors to determine appropriate dose/gradient thresholds and dose differences for phantom dose segmentation for GDSA. Here, GDSA is compared to five gamma criteria, using 3 different detector geometries (ArcCHECK, MapCHECK, Delta4), to evaluate the ability to predict $\pm 3\%$ changes in PTV mean dose (as a binary metric) using ROC analysis. An independent set of 25 clinical IMRT/VMAT cases were the validation dataset for evaluation of the GDSA dose/gradient thresholds and overall behavior. **Results:** ROC analyses comparing the sensitivity/specificity of GDSA to five different gamma criteria showed GDSA exhibited improved AUC compared to all five gamma criteria for all devices. AUC results from the validation dataset illustrate that sensitivity/specificity of GDSA is much improved compared to gamma with AUC values for GDSA vs gamma at 3%/3mm threshold 10%, global normalization were 0.893 vs. 0.79, 0.983 vs. 0.877, and 0.977 vs. 0.934 for ArcCHECK, MapCHECK, and Delta4, respectively. **Conclusion:** GDSA analysis of IMRT QA is capable of directly predicting changes in patient PTV mean dose using only information from the QA phantom measurement geometry, while also providing more clinically meaningful results and better identification of errors without any increases in clinical workload or effort.

WE-lePD-TRACK 6-04

Clinical Implementation of Deep Learning-Based VMAT Patient-Specific QA

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Purpose: To commission and implement an Autoencoder based Classification-Regression (ACLR) model for VMAT patient-specific quality assurance (PSQA) in a multi-institution scenario. **Methods:** 1836 VMAT plans from seven institutions were collected for the ACLR model commissioning and multi-institutional validation. We established three scenarios to validate the gamma passing rates (GPRs) prediction and classification accuracy with the ACLR model for different delivery equipment, QA devices, and treatment planning systems (TPS). The prediction performance of the ACLR model was evaluated using mean absolute error (MAE) and root mean square error (RMSE). The classification performance was evaluated using sensitivity and specificity. An independent end-to-end test (E2E) and routine QA of the ACLR model were performed to validate the clinical use of the model. **Results:** For multi-institution validations, the MAEs were 1.30-2.80% and 2.42-4.60% at 3%/3mm and 3%/2mm, respectively, and RMSEs were 1.55-2.98% and 2.83-4.95% at 3%/3mm and 3%/2mm, respectively, with different delivery equipment, QA devices, and TPS, while the sensitivity was 90% and specificity was 70.1% at 3%/2mm. For the E2E, the deviations between the predicted and measured results were within 3%, and the model passed the consistency check for clinical implementation. And the predicted results of the model were the same in daily QA, while the deviations between the repeated monthly measured GPRs were all within 2%. **Conclusion:** The performance of the ACLR model in multi-institution scenarios was validated on a large scale. Routine QA of the ACLR model was established and the model could be used for VMAT PSQA clinically.

WE-lePD-TRACK 6-05

Treatment Delivery Errors as a Component of IROC Phantom Dose Deviations

S Edward*, S Kry, C Peterson, R Howell, P Balter, J Pollard-Larkin, UT MD Anderson Cancer Center, Houston, TX

Purpose: To investigate whether delivery log files predict dose deviations in IROC phantom irradiations **Methods:** Delivery log files were collected for IROC lung and head and neck (H&N) phantom irradiations from January 2019 to present. We evaluated these files in Mobius FX to determine the difference between the TPS-calculated and machine-delivered doses (as a perturbation of the TPS). We evaluated whether this perturbed dose more accurately predicted the actual delivered dose (as measured by the TLD). A total of 41 lung phantoms (82 TLDs) and 36 H&N phantoms (216 TLDs) were evaluated. The maximum RMS errors in the MLCs, jaws, collimators, and gantry angles during treatment delivery were also analyzed for correlations with delivered dose. **Results:** Compared to the average dose

deviation (TPS vs TLD) of 2.4% (max = 11.6%) for the H&N phantom, and 2.6% (max = 9.6%) for the lung phantom, the dose perturbations predicted by log files were relatively small. The delivery errors ranged between -0.73% to 1.20% for the H&N phantom. Delivery played an even smaller role for the lung, which had a dose difference range of -0.75% to 0.17%. Minimal correlation between dose deviation and delivery error was found: $r = 0.13$ ($p = 0.04$) for H&N, $r = 0.03$ ($p = 0.7$) for lung. Delivered dose was not strongly correlated with MLC, jaw, collimator, and gantry angle maximum RMS errors ($-0.2 < r < 0.2$). **Conclusion:** Analysis of IROC H&N and lung phantom delivery log files showed minimal contributions to the total TLD dose deviations seen among these phantoms. This indicates that few delivery errors are present, or that these errors are not properly captured within the delivery log files. Future work will investigate the ability of this delivery data to catch/identify irradiation errors in IROC phantoms.

WE-lePD-TRACK 6-06

Streamlining Complex Multi-Isocentric VMAT Based Treatment Delivery Using a Newly Developed Software Tool

J Teruel*, P Galavis, K Osterman, S Taneja, B Cooper, N Gerber, C Hitchen, D Barbee, NYU Langone Health, New York, NY

Purpose: Multi-isocentric treatment delivery for CSI and TBI poses specific challenges for treatment delivery. We have developed a software tool to streamline all aspects of delivery for therapists and physicists at the machine, as well as to inform attending physicians of setup variability and image residuals at different locations. **Methods:** Our institution delivers VMAT-based CSI and TBI with up to 3 and 7 isocenters, respectively. A software tool was developed to assist with treatment delivery including initial patient setup, patient imaging, automatic calculation of the optimal global shift based on each isocenter's ideal shift, and automatic calculation of each isocenter's couch coordinates. Initial treatment couch coordinates are queried via the Eclipse scripting API. The global shift was calculated prioritizing the head isocenter for CSI treatments and the chest isocenter for TBI treatments by first maximizing residual tolerance at any other location prior to accepting any residual deviation at these locations. Maximum residuals tolerance was determined based on target margins, plan uncertainty and as per physician instructions. Delivery parameters are reported to a document uploaded to ARIA via API. **Results:** The developed tool was employed for 11 cases. The software tool replaced the need for plan shift comments or instructions for therapists. In particular, its use eliminated the need to provide isocenter shifts to therapists by directly providing final couch parameters for treatment, greatly reducing the risk of delivery errors. The software effectively informed the therapists if any expected tolerance was surpassed, triggering a patient setup evaluation. **Conclusion:** The described software tool is a core component to our multi-isocenter treatment programs and has streamlined delivery of these complex techniques that would otherwise require complicated instructions, including multiple shifts and on-the-fly calculations of optimal image alignment based on multiple imaging locations. This has substantially reduced the possibility of delivery errors.

Honorarium from Varian Medical Systems (Jose Teruel)

Honorarium from Sun Nuclear Corporation (David Barbee)

WE-lePD-TRACK 6-07

Double-Vertebral Segment SBRT Via Novel Ring-Mounted Halcyon Linac: Plan Quality, Delivery Efficiency and Accuracy

J Stephen*, A Webster, M Bernard, D Pokhrel, University of Kentucky, Lexington, KY

Purpose: Evaluate the plan quality, treatment efficiency and accuracy of single-isocenter volumetric modulated arc therapy (VMAT) of double-vertebral segment SBRT on Halcyon Linac. **Methods:** In-house multi-target end-to-end phantom tests and independent dose verification using MD Anderson's single-isocenter/multi-target thorax phantom (spine/lung targets) was completed. Six clinical patients with two-vertebral segments on thoracic and/or lumbar spine were replanned on Halcyon Linac with 6MV-FFF beam using a single-isocenter between vertebral segments. Three full arcs with 0° and ±10° collimator angles and advanced Acuros-based dose engine for heterogeneity corrections were used. Prescription was 35 Gy in 5 fractions to each vertebral-segment, synchronously. Halcyon VMAT-SBRT plans were retrospectively created on Truebeam Linac with a 6MV-FFF beam using identical planning geometry and optimization objectives. Plan quality metrics were evaluated per NRG-BR002 protocol. Treatment delivery parameters were recorded for both plans. **Results:** Phantom measurements showed acceptable spatial accuracy (<1mm within 5 cm from isocenter) of CBCT-guided Halcyon SBRT treatments. MD Anderson's phantom irradiation credentialing results met IROC requirements. Mean isocenter-to-tumor center distance was 3.3 ± 0.6 cm (range, 2.4–4.3 cm). Mean combined PTV was 57.3 ± 31.3 cc (range, 20.1–99.9 cc). Both Halcyon and Truebeam VMAT-SBRT plans met NRG-BR002 criteria and show similar CI, HI, GI, D2cm and maximal dose to adjacent OAR including dose to partial cord. Total monitor units, modulation, and overall treatment time were lower with Halcyon plans by 130 MU, 0.2, 3.8 min, on average, with similar beam-on time. Portal-dosimetry patient-specific QA on Halcyon showed a high pass rate of $99.6 \pm 0.3\%$ for 2%/2 mm clinical gamma passing criteria. **Conclusion:** SBRT treatment of double-vertebral segment via single-isocenter VMAT on Halcyon for select patients, is feasible and dosimetrically comparable to Truebeam Linac. Faster treatment delivery (≤ 10 min) of double-vertebral segment SBRT on Halcyon can improve patient tolerance due to back pains, potentially reduce intra-fraction motion errors, improve patient throughput and clinical efficiency.

This work is partially supported by VARIAN Medical Systems.

SAM Professional Symposium**TRACK 1*****"Picture a Scientist": Toward Gender Equity*****TH-A-TRACK 1-00****"Picture a Scientist": Toward Gender Equity**

Moderator: Kristi Hendrickson, University of Washington, Seattle, WA

TH-A-TRACK 1-01**K. Paradis (Younge), University of Michigan:****"Picture a Scientist": Toward Gender Equity and a More Diverse Medical Physics Workforce**

The gender imbalance in the field of medical physics continues. While increasing numbers of women enter the trainee ranks, we do not see these increases propagating up the ranks of the medical physics profession. This is sometimes called the "leaky pipeline" as it indicates that women are both interested and capable of succeeding in medical physics, but are often leaving the field before advancing to senior levels. There are unique obstacles and challenges faced by women physicists including gender bias, discrimination, and harassment that differ from those experienced by male colleagues. This session incorporates a three-part interactive experience. First, anonymous, real-world examples of challenges occurring in medical physics workplaces are shared from descriptions provided by women medical physicists (collected from an IRB-approved, national qualitative interview study). An anonymous audience response system will be used for attendees to share their reactions to the examples, tactics to address these issues, and advice on how to be an effective ally for our women colleagues. Then, we will break out into small groups to examine the examples and consider ways to respond in similar situations. Finally, participants will share the results of their small group discussions with the full group.

*Picture a Scientist is a 2020 documentary film that chronicles and examines gender bias and sexual discrimination experienced by women scientists. This film has been available for screening and discussion by AAPM membership.

Learning Objectives:

1. Understand the unique challenges and obstacles to a medical physics career that women face
2. Share and implement strategies to address the challenges
3. Be an active ally to early career women in medical physics

SAM Imaging Educational Course**TRACK 2*****Breast Imaging 2: Possible Future Directions*****TH-AB-TRACK 2-00****Breast Imaging 2: Possible Future Directions**

Moderator: Da Zhang, Harvard, Boston Children's Hospital, Sharon, MA

TH-AB-TRACK 2-01**J. Phillips, Beth Israel Deaconess Medical Center, Boston, Ma:****Future Directions for Contrast Enhanced Mammography****TH-AB-TRACK 2-02****R. Nosrati, Harvard Medical School:****A New Framework to Assess Localization Accuracy of Tomosynthesis Guided and Stereotactic Breast Biopsy Units****TH-AB-TRACK 2-03****P. Ghazi, MALCOVA LLC:****Dedicated Breast CT: Clinical Needs, Solutions, and Remaining Challenges****TH-AB-TRACK 2-04****K. Yang, Harvard Medical School, Massachusetts General Hospital:****Noise Power Spectrum Analysis of Breast Tissue Textures Presented in Mammographic Images****TH-AB-TRACK 2-05****Audience Q & A**

While mammography and tomosynthesis remain the mainstream for breast x-ray imaging, newer technologies are helping to fine-tune diagnosis and are playing an increasing role in follow-up studies and the early detection of cancer.

In this session, four speakers will describe and review recent technological advances in breast x-ray imaging, including dedicated breast CT, contrast enhanced spectral mammography, localization and biopsy via tomosynthesis/stereotactic guidance, and tissue texture analysis of breast x-ray images which showed correlation with breast density and cancer detection performance. The speakers will describe the clinical background, review the status quo and clinical adoption of current technologies, and report active investigations and ongoing/future directions of research in these topics.

Learning Objectives:

1. Understand the physics and principles behind contrast enhanced spectral mammography, stereotactic and tomosynthesis-based localization, breast CT, and tissue texture analysis.
2. Understand the clinical applications as well as current trend of investigations of the described emerging breast x-ray imaging technologies.

SAM Imaging Scientific Symposium**TRACK 3*****Going High or Going Low: MRI at Unconventional Field Strengths***

TH-AB-TRACK 3-00

Going High or Going Low: MRI at Unconventional Field Strengths

Moderator: Andrew Fagan, Mayo Clinic, Rochester, MN

TH-AB-TRACK 3-01

M. Rosen, MGH/Martinos Center for Biomedical Imaging:

How Low Can You Go: Contemporary Millitesla MRI

TH-AB-TRACK 3-02

A. Campbell-Washburn, National Heart, Lung and Blood Institute:

New Clinical Opportunities Using Lower Field MRI

TH-AB-TRACK 3-03

A. Fagan, Mayo Clinic:

7T MRI Part I: The Good, the Bad and the (Slightly) Ugly

TH-AB-TRACK 3-04

S. Mukundan, Brigham And Women's Hospital:

7T MRI Part II: Towards the Development of Clinical Operations

TH-AB-TRACK 3-05

Q&A & Panel Discussion

Since their introduction in the late 1970's MRI systems have evolved inexorably towards higher field strengths due to the greater spin polarization, and ultimately image signal-to-noise ratio, that higher fields allow. Commensurate with the increasing field strength was the need to overcome the many technical challenges associated primarily with the higher Larmor frequencies involved, from SAR-related safety considerations and imaging sequence modifications to new radiofrequency coil design concepts. This push to higher fields led to the recent arrival of 7 Tesla MRI systems into the clinical domain. The benefits and challenges of imaging at 7 Tesla will be reviewed, both from a technical/physics perspective and that of developing clinical applications of this technology. Future prospects for this technology will be reviewed.

At the same time, advances in MR technology and imaging techniques are facilitating renewed interest in imaging at lower field strengths. The benefits that pertain to imaging at low field will be reviewed, from 0.55 Tesla all the way down into the low milliTesla regime. New applications of a 0.55 Tesla MRI system equipped with contemporary hardware and software for cardiac imaging, lung imaging and MRI-guided cardiac interventions will be discussed, together with the capabilities of this system for routine diagnostic imaging. A high performance MRI system which operates at 6.5 mT will be described, highlighting the technical innovations which facilitate diagnostic impact. The use of modern image acquisition and reconstruction methods that can be used to maintain SNR and image quality at lower fields will also be discussed.

Learning Objectives:

1. Understand the underlying physics concepts describing the benefits of MR imaging at 7 T
2. Be familiar with the technical challenges of 7 T MRI and techniques to mitigate image artifacts
3. Demonstrate a knowledge of the safety aspects of 7T MRI
4. Understand the breath and limitations of current clinical applications of 7 T MRI
5. Apply knowledge of how modern image acquisition and reconstruction methods can be used to maintain SNR and image quality at lower fields
6. Understand the benefits of imaging at lower magnetic fields
7. Understand the opportunities and limitations of potential clinical applications for low field MRI

SAM Multi-Disciplinary Scientific Symposium

TRACK 4

Task-based Image Quality Optimization and Assessment in Diagnostic Imaging and Radiation Therapy

TH-AB-TRACK 4-00

Task-Based Image Quality Optimization and Assessment in Diagnostic Imaging and Radiation Therapy

Moderator: Hua Li, University of Illinois at Urbana-Champaign, Urbana, IL

TH-AB-TRACK 4-01

C. Abbey, University of California - Santa Barbara:

Training Model Observers for Predictive Performance in Novel Tasks

TH-AB-TRACK 4-02

M. Anastasio, Washington University in St. Louis:

Objective Image Quality Assessment by Use of Deep-Learning

TH-AB-TRACK 4-03

M. Kupinski, University of Arizona:

The Role of Dose in Task-Based Assessment of Image Quality — Diagnostic Imaging and Radiation Therapy

TH-AB-TRACK 4-04

Q&A

Medical imaging systems and image processing operations are widely employed to support clinical practice.

In medical imaging, images are often acquired for specific purposes and the use of objective measures of image quality (IQ) has been widely advocated for assessing imaging systems and image processing algorithms. Task-based measures of IQ quantify the ability of an observer to perform specific tasks.

Advanced deep-learning (DL) methods are being rapidly developed for a large number of medical imaging applications. The optimization of medical imaging systems and algorithms is still mainly guided by traditional physical measures of IQ, and the objective evaluation of IQ of medical imaging and image processing methods remains largely lacking. Traditional IQ measures such as spatial resolution, contrast, dice coefficient, and noise levels are widely employed; however, it is well-known, but perhaps under-appreciated, that their clinical value is limited because they are not always sensitive to the intended use of the image.

This symposium will address recent developments in task-based IQ assessment in diagnostic imaging and radiation therapy, and its applications for the assessment

and optimization of advanced DL-based image processing algorithms. In addition, the challenges of using task-based IQ measures to guide the optimization and assessment of imaging system and image processing will be discussed as well.

Learning Objectives:

1. Understand basic principles of task-based image quality assessment and its applications in optimizing medical imaging systems.
2. Understand the difference and relationship between objective/task-based and traditional physical metrics-based image quality assessment.
3. Understand the difference and relationship between model observers and human observers, understand the challenges in using task-based image quality assessment theory to guide clinical applications and assess DL-based imaging operators such as image reconstruction, image denoising, super-resolution, and others.

SAM Therapy Scientific Symposium

TRACK 5

Adaptive Radiation Therapy in the Era of Big Data with Artificial Intelligence

TH-AB-TRACK 5-00

Adaptive Radiation Therapy in the Era of Big Data with Artificial Intelligence

Moderator: X. Sharon Qi, UCLA School of Medicine, Los Angeles, CA

TH-AB-TRACK 5-01

X. Qi, UCLA School of Medicine:

Adaptive Radiotherapy – Current Clinical Challenges and Role of AI

TH-AB-TRACK 5-02

X. Gu, UT Southwestern Medical Center:

AI/DL Based Auto-Segmentation for ART: Technical Development and Clinical Translation

TH-AB-TRACK 5-03

Y. Wang, Massachusetts General Hospital:

Machine Learning-Based Auto Planning and Dose Prediction for ART

TH-AB-TRACK 5-04

S. Bentzen, University of Maryland:

Learning from Experience: AI and Data Mining for Outcome-Driven Radiation Treatment Planning

TH-AB-TRACK 5-05

X. Li, Medical College of Wisconsin:

Treatment Response Based on Delta Radiomics for ART

TH-AB-TRACK 5-06

I. El Naqa, Moffitt Cancer Center:

Automated Adaptive Decision Making with Deep Learning Neural Network (DLNN)

Adaptive radiation therapy (ART) aims to adjust the treatment plan based on anatomical and/or functional changes assessed based on frequent images acquired during the treatment course. Despite the appealing advantages, clinical implementations of ART, especially on-line ART, have been limited due to its complexity and lack of efficient and robust software tools in almost every step of the process. The rapid advances of artificial intelligence (AI) and AI-based technologies hold great promise for ART. This session will start with the current clinical ART workflows and its clinical challenges, then will discuss the latest AI-based development & clinical applications in each aspect of the ART workflow, including AI/deep learning (DL) based auto-segmentation; AI-based tools for auto-planning, secondary dose verification, plan check and patient-specific QA. Advanced development accounting for treatment response (as a feedback), including AI and data mining for outcome-driven radiation treatment planning; treatment response-based on delta radiomics for ART, and automated ART decision making frameworks built from deep learning neural networks will be also presented.

Learning Objectives:

1. Understand the current clinical practice and challenges of ART and the role of AI
2. Present the latest developments & clinical applications of AI-based automatic segmentation
3. Present AI-based tools for auto-planning, secondary dose verification, plan check and patient-specific QA
4. Learn AI and data mining for outcome-driven radiation treatment planning
5. Learn treatment response based on delta radiomics for ART
6. Present AI methods for automated ART decision making using deep learning neural networks for clinical implementation of ART

Received a grant from Varian Inc.

Therapy Scientific Session

TRACK 6

Machine QA

TH-A-TRACK 6-01

3D Isocenter Coincidence Measurement for MR-Linacs with Cherenkov Imaging

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Purpose: In this study, a novel Cherenkov imaging-based isocenter coincidence verification method is presented as a clinically deployable, integrated solution for MR-Linac daily quality assurance (QA). **Methods:** An acrylic water-filled cylindrical phantom was designed containing a custom-machined plastic conical structure fixed in place with 3D-printed spacers. Cherenkov emission from the cone was highly scattered, providing optical contrast from the surrounding water. The plastic is also visible with high contrast on low-field MR imaging, and the conical structure was etched with fiducials, which were visible on Cherenkov images. The phantom was aligned using co-registration of MR the images to baseline. A star shot plan was developed and delivered to the phantom to provide x-z isocenter localization

relative fiducials on the phantom, while a four-field thin sheet plan yielding a ring-shaped Cherenkov emission distribution, which encodes isocenter localization along the y-axis. A custom analysis program with a graphical user interface was developed in MATLAB to analyze results in near-real-time. **Results:** Strong linearity was observed ($R^2 > 0.99$) between longitudinal position and optical ring diameter using calibration data, and error was low (RMSE = 0.184 mm). A minimum circle radius of 0.34 mm was measured during the star shot analysis. Initial isocenter coincidence measurements in the lateral, longitudinal, and vertical directions were -0.61 mm, 0.55 mm, and -0.14 mm respectively, yielding a 3D coincidence of 0.83 mm, below the 2 mm tolerance defined in TG-142. Longitudinal analysis showed an average coincidence of $1.5 \text{ mm} \pm 0.4 \text{ mm}$ over 8 weeks of daily use. **Conclusion:** This method represents a robust and efficient method for MR-Linac isocenter coincidence verification which can be realistically deployed in the clinic for daily measurement. Longitudinal use data from our clinic suggests this method is a viable alternative to measurements which use reduced datasets or heavy time resources.

This work was funded by NIH grant R01EB023909. BP is the president and PB is the principal scientist at DoseOptics LLC, which supplied cameras and software for this study. DA receives consulting fees from DoseOptics outside the context of this work.

TH-A-TRACK 6-02

An Evaluation of the Use of EBT-XD Film for SRS/SBRT Commissioning of Unity MR-Linac System

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Purpose: The magnetic field (1.5-Tesla) of Unity MR-Linac (MRL) creates a high-intensity force on ferrous materials rendering many conventional QA detectors inoperable and poses significant challenges for clinical physicists. In this work, we evaluated using the Gafchromic EBT-XD film for the commissioning of SRS/SBRT in MRL. **Methods:** Unity MRL is a combined 1.5T MR scanner and 7X-FFF Linac. The output factors (St), for 1x1, 2x2, 3x3cm², were measured with film in solid water at d=5cm, SAD=143.5cm. The results were compared with (1) the measurements by a PTW CV diamond detector (CVD) in 3D water phantom; (2) the calculation by Monaco TPS (MTPS) in water using 1.0mm dose-grid and 0.5% statistical uncertainty. The inline and crossline profiles, measured by films and the CVD, were also compared. An SRS plan with three targets (1-3cm) was created in MTPS and was measured with EBT-XD film in a StereoPhan® phantom. The 3x3cm² field size was used for film calibration with doses ranging from 0 to 28.21Gy. Water was added to the phantom-film-phantom interface to reduce the electron-return-effect (ERE). Films were scanned on an Epson scanner following the One-scan dosimetry protocol. **Results:** The measured St between films and CVD were within 0.5% and 1.9% for 1x1 and 2x2cm² respectively. Compared with MTPS, films were within 1.5% and 2.4%. At the central B-field induced asymmetric region, films were within 0.6% on average from the CVD and the MTPS. 5-8% differences were observed between films and CVD in the 40%-5% gradient region in crossline profiles due to different magnitude of ERE. The differences in localization and dose were found to be 0.6mm and 3.3%. The $\gamma(3\%, 2\text{mm})$ and $\gamma(5\%, 2\text{mm})$ within 50% target dose region were 94.4% and 99.3%. **Conclusion:** EBT-XD film agreed well with the diamond detector and is a valuable QA tool for SRS/SBRT commissioning in MRL.

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TH-A-TRACK 6-03

Experimental Validation of a Prototype Ion Chamber for Accurate Magnetic Resonance Image Guided Radiation Therapy (MRgRT) Output Measurements in Solid Phantoms

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Purpose: Previous studies demonstrate large (up to 4 %) variations in response using commercial ionization chambers in solid phantoms. This work investigates an optimized chamber design for accurate output measurements in solid phantoms in MRgRT radiotherapy beams with a set-up most commonly used in conventional clinical workflow. **Methods:** We previously demonstrated an optimized chamber design that uses a thick (1.1 mm) brass wall to eliminate variations in response due to air gaps between the chamber and the solid phantom used for output measurements. Here, we construct and test a prototype chamber in a 7 MV Elekta Unity MR-linac with a 1.5 T magnetic field. The radiation field is collimated to 10x10 cm² at the surface of a plastic phantom with the chamber at a depth of 10 cm. The chamber is rotated about its axis through 360 degrees in the phantom with measurements made at each cardinal angle. This reveals any variation in chamber response by varying the thickness of the air gap between the chamber and the phantom. The chamber axis is perpendicular to the linac beam and tested both parallel and perpendicular to the magnetic field. **Results:** The prototype chamber exhibits less than 0.3 % variation in response when rotated about its axis in the plastic phantom, independent of whether its axis is parallel or perpendicular to the magnetic field. This is much smaller than the previously reported variations for thin-walled commercial Farmer-type chambers of similar designs. **Conclusion:** A prototype chamber has been constructed and validated for accurate output checks in a beam from an MR-linac. The chamber design enables the use of conventional equipment and procedures in MRgRT systems, minimizing disruption to current clinical workflow and allowing the use of solid phantoms unlike all other non-conventional solutions.

TH-A-TRACK 6-04

Assessment of the Dependence of the Field Shape On Calibration Coefficients in Dose Area Product (DAP) for Small Field Sizes

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Purpose: To assess the influence of the field shape on detectors calibrated in terms of Dose Area Product (DAP) for small field sizes. **Methods:** So far, primary references in radiotherapy are established in a 10x10cm² field in terms of absorbed dose to water at a point, in accordance with the recommendations of international protocols for more than two decades, despite the evolution of treatment modalities. The Laboratoire National Henri Becquerel (LNHB), the French Primary Standards Dosimetry Laboratory, proposed to use an innovative approach to obviate the difficulties of traceability for point measured quantities. Instead of considering a point measurement, an integrated measurement over an area larger than the irradiation field is adopted through another quantity: the Dose Area Product (DAP). To do so, a 30 mm core diameter graphite calorimeter and three plane-parallel ionization chambers of the same sensitive surface were built at LNHB. These chambers were calibrated in a 6 MV FFF field, at the maximum dose rate of 1400 UM/min with specifically designed circular collimators of 5, 7.5, 10, 13 and 15 mm diameter and jaws of 5, 7, 10, 13 and 15 mm side on the Varian TrueBeam accelerator of LNHB. **Results:** Preliminary results show the same behaviour for the three calibrated chambers with both circular collimators and jaws. All calibration coefficients are compatible with each other at a coverage factor of k=1, considering the same nominal field size. **Conclusion:** For small fields from 5 mm to 15 mm, calibration coefficients appear to be independent of the beam shape, allowing the primary standards carried out at the LNHB in DAP to be used independently by stereotactic cones or MLC/jaws. These promising results open the door of a paradigm shift in small field dosimetry.

TH-A-TRACK 6-05

Development of a Comprehensive Framework for the Validation of a Respiratory Gating System

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Purpose: To develop a framework to evaluate dosimetric and temporal characteristics of a respiratory gating system. **Methods:** We characterized the gating system dosimetrically and temporally using two distinct approaches. The gating system was composed of an automatic gating interface (Elekta Response™) and an in-house respiratory monitoring system featuring an optically tracked surface marker. Central-axis output and energy constancy were evaluated across 8 linear accelerators. Additionally, a representative set of 5 treatment plans were delivered both non-gated and gated on each accelerator to a 2D diode array (MapCHECK™). The surface marker was attached to a dynamic phantom (QUASAR™) which was programmed to replicate a typical DIBH breathing waveform. The passing rates between these modes of operation were evaluated using gamma analysis and a percent dose difference comparison. Modular and end-to-end approaches were used to quantify system latencies. The modular components studied included the optical tracking camera, sampling rate of the tracking software, signal travel time, and latency of the linear accelerator. The end-to-end approach involved measuring the displacement of a target moving at known velocities during the gating process. **Results:** Output and energy constancy were both within $\pm 0.5\%$ across each accelerator studied. The average difference in passing rates between non-gated and gated treatment deliveries were within $\pm 0.4\%$ using gamma analysis (2%, 1mm). Average passing rates were greater than 99% using a percent dose difference comparison (1%). No significant difference was observed $t(14)=0.21$, $p=.83$, between the modular ($M=1.55$ seconds, $SD=0.25$) and end-to-end ($M=1.49$ seconds, $SD=0.03$) approaches used to quantify beam-on latency. **Conclusion:** The proposed framework can be used to validate the operating characteristics of a respiratory gating system. The modular approach provides a convenient method to analyze the component latencies contributing to the end-to-end total.

This work was supported in part by a funding from the Karnival Krewe de Louisiane and from the Investor Collective

TH-A-TRACK 6-06

Geometric and Dosimetric Verification for a Novel Stereotactic Radiosurgery System: ZAP-X

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Purpose: The ZAP-X is a novel self-shielded radiosurgery system providing precise and highly conformal treatment to intracranial lesions. Here, we aimed to verify the delivery accuracy of ZAP-X and validate the dosimetric accuracy of its dedicated treatment planning system (TPS). **Methods:** ZAP-X is characterized by a 2.7-MV linac mounted on a gyroscope-like gantry rotating around isocenter. A tungsten wheel collimator provides circular collimation with diameters from 4 to 25 mm. A patient table enables translational head movement resulting in multi-iso treatment. Non-coplanar beams are optimized from the beam-geometry space with 2π steradians solid angles to achieve desired plan quality. Dose calculation uses ray-tracing algorithm with 0.5 or 1 mm grid sizes. Beam data were acquired using water tank, including the TPR converted from PDD, off-axis profiles and collimator output factors. For checking the iso-positioning accuracy, end-to-end test was performed using an anthropomorphic head phantom with skull structure. An integrated kV imaging system rotating around axial axis provides skull/phantom alignment. The head phantom accommodates the inserts holding thermo-/optically stimulated luminescence detectors (TLD/OSLD) and Gafchromic films to verify point absolute doses and relative dose profiles. Measurements of absolute dose profiles were performed using a polymethyl methacrylate cylindrical phantom with SRS MapCheck, i.e., a 2D diode array with 0.5 mm resolution. **Results:** Routinely performed end-to-end test for iso-positioning accuracy resulted in deviations of less than 0.7 mm. Absolute point dose measurements of TLD and OSLD agreed with TPS calculation within 3.5% with average difference $-1.9\pm1.8\%$ and $2.3\pm0.6\%$, respectively. Relative and absolute dose profiles measured by Gafchromic film and SRS MapCheck showed an average gamma pass rates of 98.06% and 98.93% when using 3%/1mm criteria. **Conclusion:** The positional and dosimetric accuracy of ZAP-X system were evaluated. The TPS and its commissioning beam data can be used to predict the delivered dose precisely and accurately.

TH-A-TRACK 6-07

Quality Assurance of CyberKnife Small Radiation Fields Using High-Resolution Scanning Diode Detector

K Yang^{1*}, S Sharma¹, A Ju¹, (1) East Carolina University, Greenville, NC

Purpose: The quality assurance of CyberKnife systems requires monthly beam profile measurement, to verify beam shape, symmetry, and field size. These QAs are typically performed using Gafchromic films due to their high spatial resolution. However, the film has drawbacks such as film polarization, nonuniformity, and relatively slow develop time. The objective of this study was to simplify these QA procedures and offer real-time reading using a custom-made 3D scanning tool. **Methods:** A diode was attached to an optical stage configured in linear XYZ translation, and stepper motor actuators with positioning accuracy $<5\mu\text{m}$ were mounted on all axis. The diode motion was programmed along crossline and inline directions in 0.2 mm step, in step-by-step scanning mode. The beam profiles were measured in air for diode and compared against films irradiated with no buildup material. All films were scanned in one image, and film-derived readings were corrected by a batch-specific calibration. **Results:** The beam profiles measured with diode showed less than 3% difference from the film references with a field size of 5 mm and 7.5 mm for fixed cone, and variable aperture Iris collimator. The dose map from film showed uniformity variations and noise, where diode profile showed fewer signal fluctuations. For diode, the FWHM calculated from the profile showed better agreement to nominal field size. In addition, the differences in FWHM between diode and film for the same field were typically within 0.2 mm range. **Conclusion:** We have designed a diode scanning tool with three-dimensional high-spatial resolution and accuracy, the program capability in acceleration/deceleration, and motion path. The scanning diode setup was shown to be within acceptable tolerance compared to film and enables potential monthly dosimetric verification for small radiation fields.

SAM Therapy Educational Course

TRACK 7

Hands-on Workshop: Simulated Error Training for the Physics Plan Review

TH-AB-TRACK 7-00

Hands-On Workshop: Simulated Error Training for the Physics Plan Review

Moderator: Perry Johnson, University of Florida Health Proton Therapy Institute, Jacksonville, FL

TH-AB-TRACK 7-01

L. Schubert, University of Colorado Denver:

Introduction to Simulated Error Training

TH-AB-TRACK 7-02

P. Johnson, University of Florida Health Proton Therapy Institute:

Overview of the WGPE Simulation Data Sets

TH-AB-TRACK 7-03

G. Kim, University of California, San Diego:

Hands On Session – Cases 1 & 2

TH-AB-TRACK 7-04

J. Fought, St. Jude Children's Research Hospital:

Hands On Session – Cases 3 & 4

Simulated error training is a method to practice error detection in situations where the occurrence of error is low. Such is the case for the physics plan review where a physicist may check several plans before encountering a significant problem. By simulating potentially hazardous errors, physicists can become familiar with how they manifest and learn from mistakes made during a simulated plan review. To this end, the WGPE has developed a series of mock treatment plans/charts which include a variety of errors based on the risk-assessment performed by TG-275. The mock datasets include documentation, screen shots, and digital content derived from common treatment planning and radiation oncology information systems. In this workshop, attendees are invited to review several mock datasets via remote interface. Moderators will guide a group discussion related to the types of errors encountered, the use of systematic search methods, and recommendations derived from TG-275's final report. The creation of the mock datasets will also be discussed along with personal experience implementing simulated error training as an educational tool for residents and trainees.

Learning Objectives:

1. Learn the rationale for simulated error training.
2. Learn how WGPE simulation data sets were created and their connection to TG-275 recommendations.
3. Learn how to implement the WGPE mock datasets as a training measure within a local setting.

This project is sponsored in part by an education grant provided by Elekta.

SAM Professional Symposium

TRACK 1

Standardization and Workflow Management in Larger Multi-site Radiation Oncology Practice: Challenges and Solutions

TH-B-TRACK 1-00

Standardization and Workflow Management in Larger Multi-Site Radiation Oncology Practice: Challenges and Solutions

Moderators: Raghavendiran Boopathy, OHSU School of Medicine, Portland, OR and Malcolm Heard, OHSU School of Medicine, Portland, OR

TH-B-TRACK 1-01

R. Boopathy, OHSU School of Medicine:

Introduction

TH-B-TRACK 1-02

J. Tanyi, Oregon Health & Science Univ:

Standardization and Consolidation Challenges in Multi-Site Radiation Oncology Practice

TH-B-TRACK 1-03

G. Niyazov, Memorial Sloan Kettering Cancer Center:

Team-Based Cross-Campus Planning Workflow

TH-B-TRACK 1-04

C. Nelson, UT MD Anderson Cancer Center:

Standardization of Physics QA Across Multi-Center Practice: Challenges and Lessons Learned

TH-B-TRACK 1-05

Audience Q&A

Despite the recent emphasis on and awareness of standardization and consolidation in health care, the complexity of radiation oncology processes poses a complex challenge and, potentially, a safety concern as it attempts to fully embrace this paradigm shift. Our leading professional bodies including the American Association of Physicists in Medicine (AAPM), the American College of Radiology (ACR), the American Society for Radiation Oncology (ASTRO), the National Comprehensive Cancer Network (NCCN), etc have outlined patterns of care and best practices reports, guidelines and recommendations. Nonetheless, departmental and institutional management and operations, including policies and standards for quality and patient safety, still may vary. This may be especially true in multi-center clinics where there is a likelihood of variation in personnel and physical resources, technology, workflow, and methodology. This session aims to discuss different experiences with standardization and consolidation with the understanding that there is currently not a prescription to achieving this end. Further, this session aims to look at the challenges that come with providing consistent high-level quality care when standardized and consolidation involves multiple sites. The session will conclude with a round table discussion to brainstorm for ideas and (new) approaches to facilitate the integration of clinics and their ultimate transition into one continuous program. The primary focus of the session discussions will be on technology integration and (physical and personnel) resource management and how they impact overall workflow.

Learning Objectives:

1. Highlight the benefits and challenges of standardization and consolidation across multi-site radiation oncology practices.
2. Learn about the impact of a teams-based enterprise-wide approach(es) to (treatment planning) workflow.
3. Communicate some best practices to staffing and standardization that are applicable to large satellite Radiation Oncology programs.

Multi-Disciplinary Scientific Session

TRACK 6

Real-time Tracking

TH-B-TRACK 6-01

Decompose KV Projections Using Neural Network for Improved Motion Tracking in Paraspinal SBRT

X He*, W Cai, F Li, Q Fan, P Zhang, L Cervino, X Li, T Li, Memorial Sloan Kettering Cancer Institute, New York, NY

Purpose: On-treatment kV X-ray images have been used in tracking patient motion during spine SBRT. One major challenge is the reduced contrast due to the passing of the X-ray through a large portion of the patient's body from the lateral direction. Besides, the overlap of the spine with the moving soft tissues could lead to significant errors in auto-registration. Our goal in this work is to automatically extract the spine component from the conventional 2D X-ray images to facilitate more robust and accurate motion management. **Methods:** A ResNet generative adversarial network (ResNetGAN) consisting of one generator and one discriminator was developed to learn the conditional mapping between the 2D kV X-ray images and the spine-only DRR. The trained model took a kV X-ray image as input and learned to generate the spine component of the X-ray image. The training dataset included 699 x-ray images from 11 patients and the corresponding matched spine-only DRRs. Another 185 x-ray images from additional three patients were used for validation. The resulted spine-only X-ray images and the original X-ray images

were registered to the DRRs, to compare the spine tracking accuracy. **Results:** The decomposed spine-only X-ray images matched with the calculated DRRs with submillimeter accuracy (165 out of 185), indicating that the model retains accurate information of the spine structure from the original X-ray images. For cases with low contrast in the original X-ray image, the decomposed spine-only X-ray significantly reduced the mean matching errors (in mm) to 0.62 and 0.16 from 0.69 and 0.21, and the maximum errors to 3.6 and 0.4 from 7.2 and 1.4, in x- and y- directions, respectively. **Conclusion:** We have developed a deep learning-based approach to suppress soft tissues in a kV projection image, leading to more accurate markerless spine tracking in paraspinal SBRT.

The presented study is partially supported by Varian Medical Systems Research Collaborations.

TH-B-TRACK 6-02

Near Real-Time Intra-Fractional 6D Monitoring of Prostate Position Using a Scheme Based on the Skeletonization of Implanted Marker Images

H Kuo^{1,2*}, D Lovelock¹, C Della-Biancia¹, A Damato¹, L Happersett¹, S Lin², M Zelefsky¹, L Cervino¹, S Berry¹, (1) Memorial Sloan Kettering Cancer Center, New York, CT, (2) Norwalk Hospital, Norwalk, CT

Purpose: Apply skeletonization scheme to extract gold marker locations from 2D planar images and to construct a 3D marker model for automatic 6D registration of prostate SBRT. Further, to evaluate the angle dependency of intrabeam 3D motion monitoring using 2D monoscopic X-ray images. **Methods:** 2D paired-planar images of markers were masked and skeletonized to represent the markers as 2D lines and to construct a 3D motion model of the marker position in space. An iterative closest point (ICP) algorithm was utilized to calculate the 6D transformation. The skeletonization scheme was compared at different kV/MV energies and techniques and the 6D registration was validated using a prostate phantom. 36 projected images were acquired over a full gantry rotation and were used to evaluate the range of angle directions and separations that preserves the approximation that two adjacent images can accurately localize the marker in 3D space. **Results:** Applying the skeletonization technique, kV and 2.5MV/6MV/16MV images showed similar Skeleton coordinates (differences <0.45mm). Using CBCT registration as ground truth for the 6D iso-center movement, the 2D to 3D registration accuracy of 2D-paired images from 11 observations had medians of 0.1mm, -0.2mm, -0.2mm, 0.1°, 0.0°, 0.0° in vertical, longitudinal, lateral, yaw, pitch, roll directions, respectively. The mean accuracy of 3D motion estimated from two consecutive kV images were <0.3mm for angle separation of 10° to 160°. The standard deviations (SD) were less than 0.3mm for angle separations of 20° to 120°. SD could be greater than 0.5mm if angle separation were 10°, 160° or 170°. **Conclusion:** The skeletonization scheme is feasible and has an accuracy in 6D comparable to 3D match using CBCT. The 2D line projection from the skeletonized 3D model can be used to estimate the 3D motion during treatment from 2D projected images with an angle separation as small as 10°.

TH-B-TRACK 6-03

Prediction of Lung Tumor Position Using 0.35 T Cine MRgRT in Retrospective Study

J Shin^{1,2*}, Z Ji¹, Y Huang¹, B Lewis¹, A Guta¹, S Oh³, J Kim², T Kim¹, (1) Washington University in St. Louis, Saint Louis, MO, (2) Yonsei University College of Medicine, Seoul, KR, (3) Allegheny General Hospital, Wexford, PA

Purpose: Real-time tumor tracking is a key feature of MRgRT. However, uncertainty in real-time tumor contours from low-resolution MRI acquired on a 0.35T MR-Linac system can lead to inaccurate beam delivery. To assist tumor tracking capabilities, we developed intra- and inter-fractional models for tumor position prediction from anatomical surrogates on 0.35T cine MR images. **Methods:** Cine MR images were acquired for real-time tumor tracking in standard MRgRT clinical workflow. Images included the tumor contour produced by the MRgRT system. Surrogates were automatically contoured using in-house software. Models were developed using linear regression and principal component analysis (PCA) to predict tumor motion from diaphragm and abdominal contours. A total of 1,069,071 images from 53 fractions for 10 lung cancer patients were analyzed retrospectively. The model for intra-fractional prediction was trained and tested on each fraction. The inter-fractional prediction was tested on each fraction using the patient's remaining fractions for training. The mean square error (MSE) and mean absolute error (MAE) between predicted and true position of the tumor were used for evaluation. **Results:** The regression model predicted the superior-inferior and anterior-posterior position of the tumor center from the diaphragm contour and abdomen contour with an average MSE of 0.57 and 0.54 mm² and an average MAE of 0.54 and 0.81mm, respectively. The PCA model predicted the tumor center with an average MSE of 1.15 and 1.70 mm² and an average MAE 0.80 and 1.24 mm, respectively. These errors were within two pixels on sagittal images. **Conclusion:** We developed models to predict tumor motion from anatomical surrogates using sagittal cine MRI. In this study, the models can predict the tumor motion using a linear model within a couple of millimeters. This study is on-going to apply the models to other treatment sites and additional imaging modalities, for example, on-board x-ray imaging.

TH-B-TRACK 6-04

Experimental and Monte Carlo Simulation Characterization of a kV Scattered X-Ray Imaging System for Real-Time Lung Tumor Motion Tracking

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Purpose: Motion management is a critical component of radiotherapy. We previously proposed a new scheme of using kV scattered x-ray photons for marker-less real-time tumor tracking in lung cancer radiotherapy. This study performs Monte Carlo (MC) studies and experimental validations to characterize its performance. **Methods:** A thin slice of x-ray beam (120kVp, 200mA, 50ms) was generated from the x-ray tube on a Varian TrueBeam linear accelerator and the slice was configured to pass the iso-center and the superior-inferior axis. The beam was directed to the moving target around the iso-center in a CIRS lung phantom. We measured the outgoing Compton-scattered photons using a photon-counting detector with a slat collimator. We set up a MC simulation system using an in-house developed GPU-based MC package for kV photon simulations and GATE/Geant4 for MV photon simulations modeling the image formation process. We also investigated effects of scattered photons from the therapeutic MV beam, the impact of scattered photons from bed and walls, and the imaging performance under several beam angles typically used in real treatments. We compared results in simulations and measurements. **Results:** The tumor can be clearly identified on the scattered images in both simulation and experimental results. MC simulations and experimental measurements agreed within 9 %. The scattered photons from the therapeutic MV beam generated a relatively uniform background signal without obvious anatomy structure; hence, they are expected not to affect tumor tracking. MC simulations showed that the amplitude of scattered photons from bed and walls was < 3% of the imaging signal of interest. The image quality does not significantly depend on kV beam angles. **Conclusion:** MC simulations and experimental measurements confirmed the feasibility of using scattered x-ray imaging as a method for real-time image lung tumor tracking.

TH-B-TRACK 6-05

Evaluation of Bronchoscopically Implanted Anchored Electromagnetic Transponders for Tumor Tracking and Monitoring During Deep Inspiration Breath Hold Lung Radiotherapy

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Purpose: To evaluate the efficacy of using implanted anchored electromagnetic transponders (EMTs) as surrogates for 1)tumor position and 2)repeatability of lung inflation during deep-inspiration breath-hold (DIBH) lung radiotherapy. **Methods:** 40 patients were treated with either SBRT or conventional lung radiotherapy on an IRB approved prospective protocol using coached DIBH. Three anchored EMTs were bronchoscopically implanted into small airways near or within the tumor.

DIBH treatment was gated with Calypso™ by tracking the EMT positions. Breath-hold cone-beam CTs (CBCTs) were acquired prior to every SBRT treatment or weekly for conventionally fractionated patients. Rigid registration between each CBCT and the patient's breath-hold planning-CT was performed to match to 1) spine 2) EMTs and 3) tumor. Differences in registration between EMT-centroid and spine were analyzed to determine surrogacy of EMTs for lung inflation. Differences in registration between EMT-centroid and tumor were analyzed to determine surrogacy of EMTs for tumor position. **Results:** 221 CBCTs were acquired and analyzed. The average differences between EMT-centroid and spine registration among all CBCTs were 0.05 ± 0.62 cm, -0.05 ± 0.41 cm, and 0.06 ± 0.23 cm in superior-inferior (SI), anterior-posterior (AP) and lateral directions, respectively. Only 58% of CBCTs had differences in registration <0.5 cm for EMT-centroid compared to spine, indicating that lung inflation is not reproducible from simulation to treatment. The average differences between the EMT-centroid and tumor registration among all CBCTs were 0.06 ± 0.18 cm, 0.06 ± 0.19 cm, and 0.02 ± 0.12 cm in SI, AP and lateral directions, respectively. 95% of CBCTs resulted in <0.5 cm change between EMT-centroid and tumor registration, indicating that EMT positions correspond well with tumor position during treatments. **Conclusion:** The implanted anchored EMTs are good surrogates for tumor position and are reliable for monitoring and maintaining tumor position during DIBH treatment when tracking with Calypso™. Large differences in registration between EMT-centroid and spine for most treatments suggest that the level of lung inflation achieved at simulation is often not reproduced.

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TH-B-TRACK 6-06

Markerless Tracking of Lung Tumor Motion Using Simulated Coded Aperture Scatter Imaging

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Purpose: Scattered treatment photons contain information about patient anatomy, yet the extremely low SNR makes scatter imaging challenging. A coded aperture is a device that encodes spatial information into the scatter image to boost the SNR and enable tracking of tumor motion without adding imaging dose to the patient. The purpose of this study was to explore the feasibility of Coded Aperture Scatter Imaging (CASI) for markerless tracking of lung tumors. **Methods:** 4DCT data of 10 lung cancer patients were imported into an MCNP6.2 radiotherapy model that included an MV treatment beam and an orthogonal kV on-board imager (kV-OBI). A MURA Coded Aperture was modeled between the patient and kV-OBI, and simulated scatter images were generated for different respiratory phases. These data were decoded, and the signals originating from the patients' tumors were analyzed. The tumor position was measured throughout the breathing cycle by a template matching algorithm and compared to the reference position. **Results:** Using the CASI method, tumor motion could be detected and tracked within the ITV throughout the patients' respiratory cycle, without adjusting the relative aperture position. Calculated deviations of the template correlation for large, mobile tumor positions were quantified with RMSE values in a range of [0.188-0.302] indicating the capability for accurate identification. Other anatomical features such as the diaphragm, chest wall, and ribs were visualized, along with their motion within the visible field edges. **Conclusion:** CASI is a novel markerless imaging technique which allows for visualizing tumor location and motion during radiation treatment. Results show that the CASI method is effective in visualizing a variety of tumor sizes' and different patient specific respiratory motion. Real-time, markerless tracking of lung tumor motion using scattered treatment photons during radiotherapy delivery might enable more accurate dose delivery while avoiding the risk of invasive fiducial marker placement.

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TH-B-TRACK 6-07

Markerless Motion Tracking with Simultaneous MV/kV Imaging in Spine SBRT Treatment

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Purpose: Motion tracking with simultaneous MV/kV imaging has the distinct advantage over single kV imaging system in detecting patient motions in three dimensions in real time. This is the first report evaluating the feasibility and accuracy of the technique in markerless motion tracking for spine SBRT treatments. **Methods:** First, a phantom study was carried out with clinical plans using 6FFF beam and 9 field-sliding window IMRT technique. Simultaneous MV/kV image pairs were acquired on a TrueBeam machine during beam delivery. High quality DRRs were generated using high-resolution CBCT (voxel size of 0.25mm) as the reference for matching. The MV images were taken continuously at 11.6 fps and kV images were acquired every 200MU, leading to 3~7 kV images per beam usually. Due to the limited size of the field aperture in each MV frame, an enhanced synthetic MV (ESMV) image were generated by combining multiple frames of MV images after corrections for scatters and frame intensity fluctuation. 3D translational shifts were then calculated by matching the ESMV/kV image pair to the reference DRRs based on mutual information. The motion tracking accuracy was evaluated by comparing the calculated shifts with the ground truth in the phantom study. Furthermore, the MV/kV motion tracking method was applied to two patient cases to investigate its feasibility in clinic (IRB approved). **Results:** The phantom study showed maximum error of 0.4mm in three directions for all beam angles for the proposed method. For the patient cases, the image acquisitions were seamless with no impact on the normal treatment workflow. The two patient data were retrospectively analyzed and revealed a motion up to 1.1mm and 0.6mm, respectively. **Conclusion:** The proposed simultaneous MV/KV markerless motion tracking technique is feasible of providing high quality orthogonal image pairs and sub-millimeter tracking accuracy in spine SBRT treatments.

The present study is partially supported by a research agreement with Varian Medical Systems.

Vendor Showcase

TRACK 8

LAP Laser, LLC

WE-SHOWCASE-TRACK 8-18-1

What's New from Sun Nuclear

R Schmidt*, (1) Lap of America Laser Applications, LLC

Imaging Interactive ePoster Discussion

TRACK 1

Imaging: Spectral and Dual Energy CT/CBCT

TH-IePD-TRACK 1-01

Estimation of Effective Atomic Number and Relative Electron Density Maps From Single-Energy CT Using Deep Learning

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Purpose: In general, parametric maps are derived from dual-energy (DE) computed tomography (CT) which has potential to differentiate materials. This study aims to develop deep generative models for parametric maps from single-energy (SE) CT with clinically used dual-layered DE-CT spectral results. **Methods:** Pix2Pix conditional generative adversarial network (cGAN) and U-net were used for generating parametric maps from conventional polychromatic CT data. For this purpose, effective atomic number (EAN) and 120 kVp CT data of 63 patients were collected from Philips iQon spectral results including: head & neck, chest and abdomen & pelvic images. For training the relationship between parametric maps and CT values, total sets of images were processed with cGAN and U-net, then U-net was applied to head & neck and chest images respectively. This study was implemented under IRB approval. The generated parametric maps were compared to the ones from the original spectral results. Image profiles of the input and output data were obtained and compared. The model performance was then assessed with R-square and pixel similarity values. **Results:** The preliminary results illustrated that U-net model outperformed the cGAN model. Compared to original parametric maps, the image profiles generated from U-net presented more smoothed lines. For head & neck and chest EAN maps, the R-square values were 0.9836 and 0.9793 and the pixel similarity values were 0.9010 and 0.8748, respectively. **Conclusion:** We developed deep learning model for generating parametric maps from conventional CT images and the accuracy of the model was evaluated. Further study consists of generating relative electron density with the commercial DE-CT spectral results and in-house calculated parametric maps.

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TH-IePD-TRACK 1-02

Are Spectral Modulator and Flying Focal Spot Beneficial for Fast-KV Switching Dual-Energy Cone-Beam CT ?

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Purpose: This work presents the first attempt to explore the potential of the combination of the spectral modulator and flying focal spot technology for material decomposition and scatter estimation in fast-kV switching cone-beam CT(CBCT)system. **Methods:** A preliminary simulation experiment was performed to investigate how the spectral modulator and flying focal spot affected the energy separation in fast-kV switching CBCT. Three sets of experimental configuration were used for kV switching: 80/140kV, 80/140kV+spectral modulator and 80/140kV+spectral modulator+flying focal spot. The Molybdenum and Copper are selected as the materials of the spectral modulator, and the thickness is set to 0.1mm and 0.4mm respectively taking the similar level of dose into consideration. The material decomposition simulation study was conducted in the three sets cases. The water phantom that we used has a size of 24 cm in diameter in which there are four rods of iodine solutions with concentrations at 5, 10, 15, 20 mg/cc. **Results:** In the simulation case with only spectral modulator applied, the energy separation can be improved from 23 keV to [23,27.5] keV for fast-kV switching CBCT system. In the other case combined with the flying focal spot technology, the spectral modulator can change the energy separation from [23,24] keV to [18,32] keV, while the energy separation of 24% pixels gets worse due to the flying focal spot. In material decomposition study, the CNR of water and iodine was improved in the case of using spectral modulator and flying focal spot. **Conclusion:** The preliminary simulation experiment shows that the spectral modulator and flying focal spot can improve energy separation for the most pixels, which is beneficial for material decomposition. In the future, the algorithms of material decomposition and scatter estimation will be developed for the fast-kV CBCT system combined with spectral modulator and flying focal spot.

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TH-IePD-TRACK 1-03

Assessment of Cardiac Gout Identification Using Dual EnergyCT and a Dynamic Phantom

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Purpose: There has been recent clinical interest in identifying gout (monosodium urate – MSU) vs. calcium deposits in the pericardium using dual energy CT (DECT). The unique challenge arises from the small size of the lesion in the fast-moving heart. This study was designed using a dynamic heart phantom and DECT with gated cardiac acquisition to evaluate identification of pericardial MSU in the presence of calcium lesions. **Methods:** The phantom was designed using two batches of kidney stones returned from clinical pathology; one was 80% MSU and 20% calcium oxalate (CaOx), while the other was 100% CaOx. The stones of size 1 - 4mm were loaded into separate 5mm polypropylene tubes, which were also filled with water and sealed. The tubes were inserted into a dynamic cardiac phantom with fillable epicardial coronary artery tracks. The phantom was controlled to execute the manufacturer's cardiac motion pattern at 60 bpm and 90 bpm. It was also kept stationary in combination with synthetic ECG. Imaging was performed on a Siemens Force CT using a DECT retrospectively-gated cardiac protocol at 70% R-to-R peak. For temporal and spatial resolution, the pitch defaulted to 0.19, the rotation time was 0.25 sec, and the tube voltages were 90 and 150(Sn) kVp. Images were reconstructed to thickness of 0.5 mm (Bv36) and processed on dual-energy software to distinguish calcium and MSU. The material decomposition parameters were chosen to maximize MSU identification without false positives from CaOx. **Results:** An average MSU volume (cc) of 0.18 and 0.17 was determined for the stationary and 60 bpm images, respectively. The MSU volume averaged 0.12 for 90 bpm images. The software correctly identified all MSU stones. **Conclusion:** This study shows potential for differentiating pericardial MSU and calcium in vivo, although our findings suggest that quantifying cardiac MSU is affected by lesion motion.

TH-IePD-TRACK 1-04

Dose Savings and Image Quality in Dual-Energy Non-Contrast CT Scans

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Purpose: To assess the benefit of dual-energy (DE) over single-energy (SE) CT and determine optimal tube potential settings across a range of phantom sizes in routine non-contrast abdominal scans. **Methods:** Images of the Catphan 500 phantom low-contrast module were acquired and reconstructed using standard clinical abdominal protocols on a dual-source (Siemens SOMATOM Force, Erlangen, Germany) CT scanner. Phantom scans were repeated with 30- and 35-cm rings added to evaluate SE and DE dose modulation. Six reference dose levels (0.75–24 mGy CTDIvol 32cm) were tested at four SE kVp settings (80, 100, 120, 140) and five DE kVp settings (70/Sn150, 80/140, 80/Sn150, 90/Sn150, 100/Sn150). Dose modulation (CAREDose4D) was enabled to adjust tube current based on phantom attenuation. The water-equivalent diameter (WED) of each phantom was determined. For each WED, size-specific dose estimates (SSDE) and image contrast-to-noise ratios (CNR) were computed. **Results:** Tube current modulation behaved similarly in SE and DE scans. SSDE was consistent across SE and DE kV settings for a given CTDIvol and WED. DE acquisitions (mixing ratio 0.6) consistently produced greater CNR than SE acquisitions. Dose efficiency (CNR per mGy SSDE) was consistently greater for DE acquisitions by 46–89%. Dose efficiency was also greater for lower kV settings when imaging smaller phantom diameters with either SE or paired DE acquisitions. As phantom diameter increased, low-kV settings became less dose-efficient and produced artifacts. **Conclusion:** For equivalent image CNR, non-contrast DE scans can reduce CT dose by approximately 40% over SE scans. For smaller patients, reduced kV settings should also be considered for greater dose efficiency and improved tissue contrast. These results indicate a dose-saving benefit to routine DE abdominal scanning. Contrast performance in vivo may differ from the low-contrast phantom data reported here. Additional benefits of low-contrast detectability in virtual monoenergetic images are being evaluated.

TH-IePD-TRACK 1-05

Evaluation of Intra-Manufacturer Dual-Energy CT Iodine Quantification Accuracy: Phantom Studies

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Purpose: To evaluate the accuracy of dual-energy CT iodine quantification comparing three different CT models from the a manufacturer, with two phantoms. **Methods:** Two phantoms (one with two setups to evaluate the effect of body habitus) with various nominal known iodine concentration inserts (0.5 to 15 mg/ml, and 2, 4 mg/ml mixed with blood mimicking materials) and other tissue/organ mimicking inserts were scanned with three Siemens CT scanners (Drive, X.cite and Force) that have two different dual-energy mechanisms, using clinically used abdomen/pelvis dual-energy scanning protocols, with different kilovolt peak or energy spectral separation settings and with automatic tube current modulation employed. Scans were performed with different insert locations in one phantom. Iodine concentration measurements were made on the images generated on Siemens Syngo.via. Linearity of measured iodine concentrations with nominal known insert concentrations were assessed. The absolute iodine concentration difference between aforementioned two were also analyzed. **Results:** Measured iodine concentrations from all three scanners are linear with nominal known iodine concentrations, with R^2 values between 0.9935 and 1. Among three scanners, absolute difference on iodine concentrations measured from two phantoms ranged from -1.8 to 0.7 mg/ml and from -1.5 to 1.6 mg/ml, respectively, with one scanner outperformed the other two on one phantom. Accuracy of iodine quantification was degraded prominently at 10 and 15 mg/ml on two scanners, especially when these inserts were placed in the periphery of the phantom that mimics large body habitus. One scanner showed notable degradation on contrast-enhance blood iodine concentration, while one scanner showed slight improvement, when those inserts were placed in the large-sized phantom. **Conclusion:** There's general agreement on iodine quantification accuracy with notable variations among three scanners from the same manufacturer. Locations of the iodine inserts and size of the phantom resulted in different accuracy of iodine quantification.

TH-IePD-TRACK 1-06

The Impact of Dual Energy CT (DECT) Protocol Settings on Image Quality

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Purpose: To explore the acquisition parameter space in dual source DECT by analyzing the impact of kVp, Quality Reference mAs ratio (QRM=QRM(A):QRM(B)), and pitch on image quality. **Methods:** A custom anthropomorphic phantom (25Hx32Wx25Lcm) was constructed with soft tissue-mimicking material and organ-mimicking inserts simulating liver, pancreas, kidney, aorta, bone, and water, with realistic organ-specific HU values, size and relative positions. Each organ contained a cylindrical insert that simulates iodinated organ tissue. The phantom was scanned on a Siemens dual-source scanner with single-energy (SE) and DECT. A standard DECT protocol (90/Sn150kVp, QRM=1.6, pitch=0.6) was considered as baseline and subsequent DECT scans were performed by individually varying pitch (0.9,1.2), QRM (1.2,2,4), and kVp (70/Sn150, 80/Sn150 and 100/Sn150). QRM(A) and QRM(B) were adjusted to maintain CTDIvol of 7.5mGy. CNR,SNR, and noise were calculated and compared using ANOVA for all organs on SECT and mixture DECT images. **Results:** Pitch of 0.6 on average improved the CNR by $11\pm4.2\%$ ($p<0.05$) compared to other investigated pitch values while changing pitch from 0.6 to 1.2 had no significant impact on the noise or organ HU ($p>0.7$). Among the four tested kVp pairs, although 70/Sn150kVp resulted in the highest average noise level ($p<0.05$), it improved the CNR by an average of $14.1\pm8.2\%$ ($p<0.05$) in iodinated tissues with no significant impact on CNR of non-iodinated organs. The changes in QRM from 1.2 to 4 did not introduce any visible artifacts or changes in noise performance ($p>0.5$). **Conclusion:** Increasing pitch from the vendor-recommended value (0.6) to higher ones did not significantly impact CT number accuracy and noise performance in mixture images, implying the possibility of using faster scans to mitigate motion artifacts. With similar CTDIv, CNR was higher in all DECT contrast-enhanced images compared to SECT, indicating the feasibility of reducing patient exposure with an optimized DECT protocol while maintaining/improving CNR.

TH-IePD-TRACK 1-07

Spectrally Neutral CT Calibration

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Purpose: To describe a computational approach to the design of spectrum-independent CT calibration materials and describe preliminary experimental verification. **Methods:** : The basis material paradigm asserts that the linear attenuation coefficient of an arbitrary material can be accurately represented by a combination of the linear attenuation coefficients of two dissimilar materials. The combined thicknesses, relative to the material being studied, of the basis materials do not, in general, equal unity. Adding a third material resolves that issue. Although the desired result is that the combined relative thicknesses of the three materials be equal to 1.0, volume changes when components are mixed may occur. Replacing the thickness condition of 1.0 with a constant, call it k, close to 1.0 allows for an iterative procedure that results in the desired match. If the constant is allowed to have values significantly greater or less than 1.0, and the material being characterized is water, the result is a mixture that has a constant ratio in attenuation coefficient relative to water, i.e., its CT number is independent of energy and is determined by the value of k. **Results:** Mixtures with nominal HU values greater than 0 were calculated using glycerol, water, and salt. Mixtures for negative CT numbers were calculated using isopropanol, water, and potassium nitrate. CT numbers up to 248 and down to -183 could be reached with the components studied. CT scans on a SIEMENS SOMATOM Force dual source CT scanner with samples calculated to produce HU values of 100 after correction for volume change were within 1 HU of the target value at kVs from 80 to 140. **Conclusion:** The method described produces material formulations that give CT numbers that are independent of x-ray spectrum. These formulations may be useful for the calibration of CT systems, especially in facilities with multiple scanners.

Imaging Interactive ePoster Discussion

TRACK 2

Imaging: MRI Quantitative Imaging and Applications

TH-IePD-TRACK 2-01

Comparison of Single Breathhold MDixon-Quant and Multi-Echo GRE Methods to Spin Echo Method for Estimating Liver Iron Concentration

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Purpose: Ferriscan is FDA approved to determine liver iron concentration (LIC). It is a spin echo (SE) free breathing technique which is time consuming and motion degraded. We evaluated two single breath-hold **Methods:** 3D mDixon-quant (mDixon) and multi-echo gradient echo (GRE). LIC from these breath-hold techniques were compared to LIC from Ferriscan. **Methods:** Patient population: 12 subjects (age 8-80 years; 8 males), referred to MRI to estimate LIC, were scanned on a 1.5T magnet. The data was retrospectively analyzed. Institutional review board approved waiver of individual consent. MRI parameters:GRE: 16 echoes were acquired at TE1/ΔTE 1.08ms/0.8ms, TR/14ms, Flip angle(FA)/20°, time of acquisition (TA) per breath-hold: 10.7sec. mDixon: 6 echoes were acquired at TE1/ΔTE 0.92ms/0.7ms, TR/5.3ms, FA/5°. TA per breath-hold: 11.6sec. Ferriscan: multi-acquisition to acquire 5 echoes (6ms, 9ms, 12ms, 15ms, and 18ms) for R₂ calculation, TR/1000ms, FA/90°. Each TA: 117sec. Data analysis: The GRE data was fitted to an exponential decay to yield T2* maps. mDixon T2* maps were derived from a six-point method. LIC(GRE) and LIC(mDixon) were calculated using the equation: $LIC[mg/g\ dry]=0.0254x(1000/T2^*[ms])+0.202$. Resonance Health calculated the Ferriscan LIC(SE). The agreement between LIC(SE) vs LIC(mDixon) and LIC(GRE) was determined using Bland-Altman analysis. **Results:** Ferriscan images from all subjects had significant motion, limiting the area of liver available for LIC evaluation. mDixon and GRE images had little motion. There was strong agreement (bias ± 2SD): $0.50 \pm 1.04\ mg/g\ dry$, $0.79 \pm 1.06\ mg/g\ dry$ and $0.29 \pm 0.48\ mg/g$ when comparing LIC(GRE) vs LIC(SE), LIC(mDixon) vs LIC(SE), and LIC(mDixon) vs LIC(GRE) respectively. **Conclusion:** LIC values from the mDixon and GRE techniques are consistent with the Ferriscan LIC. These results require validation in a larger cohort of patients. An advantage of mDixon, the entire liver is imaged, is the ability to provide both LIC and fat fraction.

TH-lePD-TRACK 2-02

Analysis of Diffusion and Microstructure Properties in Brain Tumors Using a Random Walk with Barriers Model

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Purpose: Restricted water diffusion in microstructural tissue depends upon measurement (diffusion) time. A random walk with barriers model (RWBM) could analyze these effects on diffusion to reveal microstructural properties from macroscopic imaging characteristics within heterogeneous brain tumors. Gross tumor volume in gadolinium enhanced T1-weighted images (GTV-Gd) usually contains tumor mixtures, which is difficult to be analyzed by a single diffusion measurement. This research aims to combine clustering analysis with the RWBM to demonstrate microstructural diffusion properties in brain tumors. **Methods:** Ten patients with brain tumors (eight with brain metastases, two with glioblastoma) had diffusion weighted (DW) images with three different diffusion times using pulsed diffusion gradients (PG) and sine-modulated oscillating gradients at $f=30\text{Hz}$ (OG30) and $f=50\text{Hz}$ (OG50) of a prototype sequence on a 3T scanner (MAGNETOM Skyra, Siemens Healthcare). DW images were acquired in 6 diffusion directions with 8 b-values (0 to 2600 s/mm²) for PG, 6 b-values (0 to 1300 s/mm²) for OG30, and 5 b-values (0 to 500 s/mm²) for OG50. The delineated GTV-Gd of each tumor, if $> 1\text{ cm}^3$, was classified to smaller volumes using DW image intensities and fuzzy c-means. The RWBM that characterizes time-dependent diffusion was applied to DW images in the GTV-Gd to yield five parameters: cell diameter, unrestricted diffusion coefficient (D_0) at a short time limit, bulk diffusion coefficient ($D(\infty)$) at a long time limit, membrane permeability and effective restriction. **Results:** Of 23 brain tumors, the mean tumor cell diameter of GTV-Gds was $(16.5 \pm 4.20)\text{ }\mu\text{m}$. D_0 was $(2.20 \pm 0.70)\text{ }\mu\text{m}^2/\text{ms}$. $D(\infty)$ decreased to $(0.60 \pm 0.14)\text{ }\mu\text{m}^2/\text{ms}$ due to microstructure restrictions with the mean effective restriction of 2.8 ± 1.2 and the mean membrane permeability of $(0.053 \pm 0.009)\text{ }\mu\text{m}/\text{ms}$. **Conclusion:** The RWBM yields parameters beyond analyzing DW images with a single diffusion time, and has potential to reveal microstructure of brain tumors with further investigation.

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TH-lePD-TRACK 2-03

Leveraging Prior Knowledge to Reconstruct Quantitative MR Relaxation Maps to Investigate Glioma Tumor Heterogeneity

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Purpose: The purpose of this study is to generate relaxation maps from clinical MRI scans by leveraging prior knowledge. Furthermore, we analyze joint histograms of these maps (T1 pre contrast, T1 post contrast, and T2 relaxation maps) to investigate glioma tumor heterogeneity. **Methods:** This study uses 50 patients from the multi-modal brain tumor image segmentation benchmark (BRATS) dataset, which contains four tumor sub-type volumes (necrotic core, edema, non-enhancing and enhancing tumor), T1-weighted (T1-w), T2-weighted (T2-w), and T1-weighted post contrast (T1c-w) MR images. By employing a novel method, we generate relaxation maps (T1, T2, and T1c maps) by leveraging relaxation values of white matter (WM) and cerebral spinal fluid (CSF) from the literature and calibrating the clinical MRIs based on these values. After the calibration process, the gray matter (GM) T1 and T1c relaxation values are automatically assessed for quality control purposes. 3D joint histograms (JH) were then created using all voxels within the tumor and each tumor sub-type. We calculated the JH center of masses, which are defined in the T1-T2-T1c relaxation coordinate system, for each sub-type, and quantitatively evaluated the relationship of these center of masses using a correlation matrix across all patients. **Results:** Relaxation maps for 6 patients were deemed unacceptable for further analysis using the developed GM quality control metric. Multiple subtypes' center of mass correlations across the remaining 44 patients showed correlation values of $r > 0.5$, corresponding to $p < 0.0005$. The highest correlation is 0.894 between the non-enhancing T1 and the non-enhancing T1c, meaning that the relationship between these relaxation values is most consistent across all patients. **Conclusion:** We present a novel method capable of estimating relaxation maps for any set of T1-w, T2-w, and T1c-w clinical MRIs, and believe these maps can be valuable tools capable of characterizing changes in tumor heterogeneity over time.

TH-lePD-TRACK 2-04

Obtaining New Contrasts From a Single T1-Weighted Image

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Purpose: While MRI contrast is predetermined by imaging protocol, it can be retrospectively tuned, which is potentially useful for normalizing MRI data for radiomics. In this study, we present a new paradigm to obtain various contrasts from a single T1-weighted image. **Methods:** We propose a contrast tuning framework that combines deep learning-based quantitative MRI with Bloch equations. Using deep neural networks, quantitative tissue relaxation parametric maps (T1 map, proton density map) and field map (B1 map) are predicted from a single T1 weighted image. Here, ground truth T1 map was obtained from four variable-flip-angle T1-weighted images and a B1 map (measured using the actual flip angle method); proton density map was calculated from T1-weighted image and T1 map. In these tasks, self-attention convolutional neural network is employed with unique shortcuts equipped and attention mechanism integrated. A total of 1,344 knee images from 56 subjects are utilized for training and testing in six-fold cross validation. Given estimated parametric maps, MR images with different contrasts are generated with the application of Bloch equations. While a wide spectrum of contrasts can be obtained with various imaging parameter values, the result is only validated at certain contrasts (corresponding to variable flip angles). **Results:** High accuracy has been achieved in quantitative parametric maps and qualitative images. From a T1-weighted image, quantitative T1 map, proton density map and B1 map are predicted with an averaged correlation coefficient of 0.95~0.99 and L1 error of 0.02~0.09; and T1-weighted images corresponding to alternative flip angles are obtained with an averaged correlation coefficient of 0.97~0.99 and L1 error of 0.04~0.09. **Conclusion:** A new data-driven strategy is proposed for retrospective MRI contrast tuning from a single T1-weighted image, which requires no additional data acquisition.

TH-lePD-TRACK 2-05

Deep Learning Based T2* Mapping From MRI Images with Different Types of Weighting

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Purpose: Quantitative T2* mapping is important for the diagnosis and management of knee diseases. However, derivation of T2* maps typically requires acquisition of multi-echo images, which is time consuming. In this study, we present a new paradigm to predict T2* map from two conventional images obtainable in a standard clinical MRI examination, thus eliminating additional data acquisition for quantitative MRI. **Methods:** A deep learning framework is proposed to predict T2* map from a T1-weighted image and a T2*-weighted image. The principle is validated on knee MRI. 1344 T2* maps from 56 subjects are retrospectively collected. Every T2* map was extracted from six T2*-weighted images acquired with different echo time (of 0.032, 4.4, 8.8, 13.2, 17.6, and 22ms); in addition, a T1-weighted image was acquired. To derive T2* map from two input images, a self-attention convolutional neural network is employed. Here, a hierarchical network architecture is adopted; global and local shortcuts are equipped to facilitate residual learning; and attention mechanism is integrated to make efficient use of non-local information. Separate deep neural networks are trained, each accepting T2*-weighted image obtained with a specific TE. In training, a mixed function of l1 loss and SSIM loss is employed, and network parameters are updated using the Adam algorithm. All 1344 images are utilized for training and testing in a six-fold cross-validation strategy. The prediction result is quantitatively evaluated within the region of interest (i.e. cartilage manually segmented). **Results:** Using well-trained models, T2* maps are predicted from single T2*-weighted images (acquired using TE of 4.4, 8.8, 13.2, or 17.6ms) along with corresponding T1-weighted image. High accuracy has been achieved. The averaged L1 error with ROI is 0.13. **Conclusion:** A data-driven strategy is proposed for T2* mapping from two conventional MR images, which has a potential to eliminate additional data acquisition for quantitative MRI.

TH-lePD-TRACK 2-06

Quantitative Inter-Vendor Evaluation of PDFF and R2* Mapping Accuracy for 3T MRI Scanners

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Purpose: To evaluate the relative accuracy of proton density fat fraction (PDFF) and R2* mapping sequences at 3T from different MRI scanner manufacturers and pulse sequence types using standardized quantitative MRI phantoms. **Methods:** PDFF and R2* phantoms manufactured by Calimetrix were scanned twice on GE PETMR, GE 750W, Siemens Vida, and Siemens Skyra platforms using the body coil and the two measurements averaged. For GE, IDEAL-IQ sequences were used to generate the fat fraction and R2* maps. For Siemens, product q-Dixon sequences were used, as well as a work-in-progress (WIP) q-Dixon sequence on the Skyra. MR parameters were kept at manufacturer's defaults. Data was processed by drawing a 1.6cm diameter circular ROI over the phantom vials DICOM images and recording the mean ROI value. **Results:** IDEAL-IQ sequences had lower overall average percent error for PDFF calculation, with 750W and PETMR measuring 2.6% and 7.1% error respectively. Skyra WIP q-Dixon had 4.8% error for PDFF; Vida product q-Dixon had 16.1% error. Siemens scanners were more accurate than GE for R2* measurements: Skyra WIP q-Dixon measured 7.1% error, and Vida product q-Dixon measured 7.8% error. In contrast, 750W and PETMR measured 14.6% and 17.7% error, respectively. Data from the Skyra product q-Dixon sequence was not accurate for both PDFF and R2* calculations, with an average error of 72% and 59%, respectively. Greater inaccuracy was observed at the high and low ends of the phantom's PDFF and R2* scale for both vendors, suggesting that extreme values are more challenging to reconstruct accurately. **Conclusion:** We observe modest accuracy for PDFF and R2* mapping sequences on both GE and Siemens MRI scanners, suggesting further refinement of reconstruction algorithms is possible. These results also highlight the importance of validation of quantitative results before integration into clinical workflows.

TH-lePD-TRACK 2-07

Automatic Stratification of Prostate Cancer Patients into Low- and High-Grade Groups Using a Support Vector Machine Model with Multiparametric Magnetic Resonance Image Features

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Purpose: To explore the potential of automatic stratification of prostate cancer patients into low- and high-grade groups using a support vector machine (SVM) model with multiparametric magnetic resonance (MR) image features. **Methods:** MR images [T1- and T2-weighted, diffusion weighted b=0, b=1,000, b=2,000, and apparent diffusion coefficient (ADC) images] of 101 cancer regions corresponding to histopathological images of prostate cancer were selected and divided into a training (n=72) and test datasets (n=29). Each dataset was divided into two groups, low-grade tumors [$\leq$ International Society of Urological Pathology (ISUP) Grade Group (GG) 2] and high-grade tumors (ISUP GG3 $\leq$). 1,620 MR image features were derived from original images and two-dimensional wavelet filtered images for the six types of images. Seven significant features were chosen by using a least absolute shrinkage and selection operator (LASSO) algorithm to build a SVM model. Parameters of the SVM model were optimized based on a leave-one-out cross validation test so that areas under receiver operating characteristic curves (AUCs) can be maximized. **Results:** The most significant feature was one of wavelet features from ADC images. The AUCs based on the SVM model of the automatic stratification of prostate cancer patients into two groups were 0.977 in the training dataset and 0.910 in the test dataset. **Conclusion:** This study suggested the potential of automatic stratification of prostate cancer patients into low- and high-grade groups using the SVM model with multiparametric MR image features.

Multi-Disciplinary Interactive ePoster Discussion

TRACK 3

Multi-Disciplinary: Imaging for Motion Management

TH-lePD-TRACK 3-01

Benchmark of 4DCT Derived Motion Models with a Biomechanical Lung Model for Tumor Position Estimation

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Purpose: Several CT-based surrogate motion models (SMM) have been proposed to simulate the complexities of respiratory motion and cycle-to-cycle variations in the thoracic anatomy. However, the lack of high-contrast internal imaging on most linear accelerators limits evaluating the model's intra- and inter-fractional reproducibility and validity. Biomechanical models can serve as a reference for SMM verification because they are able to capture the underlying biophysical principles of lung-tumor motion. We used a hybrid finite-element (FE) lung-tumor motion model to benchmark the accuracy of two SMMs for tumor position estimation. **Methods:** 4DCT, VisionRT surfaces (VRT), and ~30s lateral and anterior-posterior fluoroscopic images (FLs) were collected from five lung-SBRT patients and used to construct two previously published CT-based volumetric SMMs: SMM-1 used a correspondence between 4DCT and phase-averaged VRT surfaces to estimate deformation vector fields for new VRT surfaces. SMM-2 integrated lung-surface motion observed in a few FLs to update the external-internal correlation. A FE biomechanical lung-tumor model was constructed and used as reference to verify SMM estimated tumor position. Prior validation of our FE model for 4DCT found an accuracy of 3 mm for centroid position estimation. **Results:** Changes in the external-internal correlation were frequently encountered and degraded SMM-1 performance. The correlation between internal lung-surface and tumor centroid motion was more stable, and SMM-2 fidelity was preserved with updates. Model updates significantly improved agreement with the biomechanical model: the 99th percentile absolute difference in tumor SI/AP centroid motion was reduced from 7/5.2 mm (SMM-1) to 3.4/2.6 mm (SMM-2). Differences between the SMM-2 and biomechanical model were most notable for tumor positions beyond the original training data set (4DCT), where SMM-2 resorted to extrapolation. **Conclusion:** Biomechanical models offer a robust mechanism for verifying SMM-estimated internal lung-tumor motion. Updating the external-internal correlation preserves SMM fidelity and significantly improves agreements with the FE model.

TH-lePD-TRACK 3-02

Feasibility of Intrafraction Motion Management Using Triggered KV Imaging During Spine Stereotactic Radiosurgery

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Purpose: Precision is vital in spine stereotactic radiosurgery (SRS) due to highly-conformal dose distributions and ~1mm margins. However, most image-guided radiotherapy (IGRT) techniques only monitor interfraction motion or require specialized devices to monitor intrafraction motion. Feasibility of utilizing triggered kV imaging with the TrueBeam advanced imaging package to capture intrafraction motion during the treatment of spine SRS patients is investigated. **Methods:** Triggered kV imaging for intrafraction motion management was tested with an anthropomorphic phantom and simulated spine SRS treatments to the thoracic and lumbar spine. Various shifts/rotations were introduced in x, y, z, pitch, and yaw. Gantry-angle-based triggering was set to acquire kV images every 45s to investigate how well various shifts could be visualized. The anterior portion of the vertebra (to spinal canal) and spinous process were contoured as IGRT structures. Upon each triggered kV image acquisition, a 2D projection of the IGRT structures was automatically calculated and updated at arbitrary angles for display onto kV images (qualitative visual guide for patient alignment during treatment delivery). This method was implemented clinically for treatment of 32 patients (75 fractions). During treatment, patient positions were monitored with kV images triggered every 15 seconds. **Results:** Phantom images revealed that IGRT structure accuracy and therefore utility

of projected contour during triggered imaging improved with smaller CT slice thickness. Contouring vertebra superior and inferior to the treated site was necessary to detect clinically relevant phantom rotation. Clinical implementation verified robustness of patient immobilization. Based on visual inspection of projected IGRT contours on planar kV images, appreciable intrafraction motion was detected in only one fraction (1.3%). **Conclusion:** Feasibility of triggered imaging for spine SRS intrafraction motion management has been demonstrated in phantom experiments and implementation for patient treatments. This technique allows efficient, non-invasive monitoring of patient position using the TrueBeam OBI and patient anatomy as a direct visual guide.

TH-lePD-TRACK 3-03

Respiration-Induced Target Motion Prediction from a Surrogate Signal in Radiotherapy : A Bayesian Framework

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Purpose: To account for respiration-induced motion during lung radiotherapy, indirect tracking requires surrogate-based correlations and system latency compensation. The present study proposes a unified Bayesian framework addressing both issues. Its potential to predict both future target motion from a surrogate signal and the related uncertainty is investigated. **Methods:** The proposed framework is based on a Kalman filter theory specifically adapted for surrogate measurements. Since the quality of the surrogate motion affects the accuracy of the method, two external surrogates (abdominal and thoracic markers) and an internal surrogate (liver structure) are examined. Their motions and simultaneous lung target motion are collected for 9 volunteers during 4 minutes each, including drastic breathing changes to assess the robustness of the method. A comparison with an artificial non-linear neural network (NN) is performed. The first 15 seconds of each volunteer's sequence are used to train both methods, while testing is conducted under two scenarios: a worst-case scenario and a simplified one. **Results:** Latencies ranging from 200 to 800 ms are tested. Surrogate predictability and target-surrogate correlation are also independently assessed to provide interpretable insights into the accuracy of the method. While the prediction errors from the thoracic surrogate are reduced with the NN in some cases, the proposed framework outperforms the NN in most cases when using the liver and the abdomen surrogates. The Bayesian method is found more robust to increasing latencies and to breathing variations. The thoracic marker is found to be less reliable than the other surrogates to predict the target position, whereas the liver appears as a better surrogate. **Conclusion:** The proposed framework predicts both expected target positions and their associated confidence intervals. This study opens the way for real-time uncertainty prediction to further assist motion management decisions. Future work will investigate an adaptive version of the proposed framework.

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TH-lePD-TRACK 3-04

Verification of a System for Real-Time EPID Based Breath Hold Monitoring During DIBH Tangential Breast Cancer Radiotherapy

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Purpose: In breast cancer radiotherapy, the deep inspiration breath hold (DIBH) technique helps sparing heart, lungs, and liver from unintended radiation. The quality of DIBH is typically assessed via surrogates with varying degree of correlation with the patient's anatomy. This work reports on the accuracy validation of an anatomy based system for assessment of the quality of DIBH using live MV images collected with the electronic portal imaging device (EPID). **Methods:** The system consists of: (i) an image grabber application which acquires and saves MV EPID frames as TIFF files and (ii) image-processing software which analyses the files to assess the quality of the breath hold in real time. Measures used in the assessment are the lung depth (LD) and the distance from the breast surface to the posterior field edge (skin distance, SD) at user-defined locations. The system has been developed using 1000+ patient images. To assess the accuracy of the system, images of radiotherapy phantoms placed on and moved by a precision motion platform were analysed and the observed LD and SD were compared to those expected from the applied motion. A clinical surface imaging system was employed to confirm the phantom motions. **Results:** The accuracy of LD and SD measurements for a PMMA CTDI phantom were 0.31(1.09) mm (mean (standard deviation)) and -0.10(0.14) mm, respectively. The accuracy of LD and SD measurements on an E2E SBRT phantom were 0.01(0.18) mm and 0.05(0.08) mm, respectively. The latency of the system is <350 ms: the sum of the time delays of the image grabber (100ms) and the image processing (<250ms) applications. **Conclusion:** The system is able to assess the quality of DIBH breath hold via LD and SD with sub-millimetre accuracy. Processing times are short enough for real time clinical use. Clinical measurements with patients have begun.

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TH-lePD-TRACK 3-05

Inter-Fraction Motion Assessment and Accumulated Dose Quantification of Upper GI Organs During MR-Guided Ablative SBRT Treatment

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Purpose: To quantify inter-fraction deformations and dose accumulation of upper GI organs using T2w MRI in locally advanced pancreatic cancer (LAPC) patients undergoing MR-guided ablative SBRT on Elekta Unity MR-linac. **Methods:** Five LAPC patients underwent 5 fraction SBRT treatment using daily online plan adaptation to 50Gy. A pneumatic compression belt was used for all treatments to minimize GTV motion within 5mm. Between each consecutive fraction, pre-treatment MRI was deformably registered to next fraction using in-house Large Deformation Diffeomorphic Metric Mapping (LDDMM) deformable image registration (DIR) method and the total dose delivered to the duodenum-stomach, small and large bowels were accumulated. Inter-fraction deformations were quantified for all three directions using gradient magnitude of the Deformation Vector Fields (DVF). The DVFs were fit to a B-spline to provide elasticity to the converging/diverging vectors resulting from large organ deformations. Registrations were assessed using Dice and Hausdorff distance (HD) calculated between the deformed and physician's contours. DIR accumulated dose metrics (Dmax, D0.035cc, D2cc and D5cc) were compared against rigidly accumulated doses. **Results:** A total of 25 MRIs were analyzed. Average Dice and 95th percentile HD were: Stomach (0.93, 1.2mm), Small bowel (0.84, 4.4mm), and Large bowel (0.88, 3.2mm). Small bowels showed the largest motion with maximum (mean±SD) inter-fraction deformation of 21.9mm (9.1±2.5mm) compared to stomach: 15.8mm (6.8±2.8mm) and large bowels: 20.9mm (8.8±3.6mm). Mean difference between rigid and DIR accumulated dose for D0.035cc and D5cc was 1.52±0.59, 1.29±0.96 and 1.56±0.9 Gy and 0.36±0.26, 0.53±0.4 and 1.43±0.44 Gy for stomach, small bowel and large bowel respectively for all the fractions. **Conclusion:** LDDMM registration was able to account for large deformations of upper GI organs greater than 2cm. Rigidly accumulated doses to all organs at risk were significantly larger compared to DIR, suggesting use of deformable dose accumulation to spare major mobile structures for LAPC patients.

TH-lePD-TRACK 3-06

Tumor Motion Tracking in a KV Scattered X-Ray Imaging System for Lung Cancer Radiotherapy

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Purpose: Tumor motion tracking is a critical component of lung radiotherapy. We previously proposed a new scheme of using kV scattered x-ray photons for markerless tumor tracking and presented the imaging capability of using photon-counting detection. This study investigates suitability of this technique for tumor motion tracking accuracy via experiments and Monte Carlo (MC) simulations. **Methods:** A CIRS lung phantom was used in the measurements and modeled in the simulation. The targeted tumor with a diameter of 2 cm was aligned to the iso-center. The collimated primary X-rays from the on-board kV X-ray tube illuminated the tumor from 45° gantry direction. The scattered X-rays were detected using a photon-counting detector (XC_THOR, DirectConversion, Sweden) with a parallel multi-hole collimator. The kV x-ray technique was 120 kVp and 10 mAs (200 mA, 50 ms). Exposure time of 50ms was selected to achieve a 10 Hz imaging speed. An MV beam (6 MV, 600 MU/min, 2x2 cm²) was used to study the performance during treatment delivery. To investigate the tumor-tracking accuracy, we moved the couch in 1 mm steps in the superior-inferior (SI) direction in experiments. In MC simulation, the tumor was moved with a speed of 1 cm/s in the SI direction and along the kV beam direction. We tracked the tumor position in the scattering images using template-matching and compared with the ground truth. **Results:** The tumor can be visually identified clearly on the measured and the simulated kV scattered X-ray images. The measured tumor positions agreed well with the ground truth positions in the experiments and the simulation study, with the mean error of 0.65 mm and 1.0 mm for the experiments and simulation respectively. **Conclusion:** The experiments and simulations demonstrated sufficient motion tracking accuracy of real-time tumor tracking using the proposed technique.

TH-IePD-TRACK 3-07

Motion-Robust Free-Breathing Abdominal MR Fingerprinting (MRF) Using Radial Acquisition and Temporal Compressed Sensing Reconstruction: Initial Results

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Purpose: Quantitative relaxometry in the abdomen is challenging due to respiratory, pulsative, and peristaltic motion. We present an implementation of free-breathing MR-fingerprinting and study improvements in image quality as a function of reconstruction. **Methods:** The acquisition method relies on golden-angle radial sampling combined with a varying flip-angle (FA), gradient-spoiled, SSFP MRF sequence. The imaging sequence consists of an inversion pulse followed by the varying FA train and a delay time (Td) to allow for relaxation to thermal equilibrium. The exam parameters were: TR/TE/TI=8/4/15 ms, FOV=25x25cm², matrix=224x224, and slice thickness=5mm. Two FA schedules were studied. The number of spokes/dynamic was varied in the range [1 2 3 4 5 10] while Td was varied in the range [0.1 1 10] seconds. Golden-angle radial sparse parallel (GRASP) reconstruction was implemented with total variation (tTV) or principal-component analysis (tPCA) as regularization functions in the temporal domain. Template matching was used for T1/T2 calculations. The quantitative maps derived from tTV-GRASP, tPCA-GRASP, and conventional gridding were compared using a NIST phantom with known T1/T2 values. Image quality was also compared in a healthy volunteer during a free-breathing (FB) and (BH) acquisition. **Results:** The bias for T1 and T2 was comparable in parameters calculated with gridding and tTV-GRASP, and larger for tPCA-GRASP reconstructions. For T1, we find an approximately two-fold decrease in the standard deviation of maps calculated with tTV-GRASP compared to gridding and tPCA-GRASP. For T2, tTV-GRASP and tPCA-GRASP generated maps with comparable uncertainty but improved when compared to gridding. The improvements were dependent on the number of spokes/dynamic, Td, and FA schedule. **Conclusion:** GRASP provides a feasible method for free-breathing MR-fingerprinting with improved image quality and reduced sensitivity to artifacts from physiologic motion.

Therapy Interactive ePoster Discussion

TRACK 5

Therapy: Treatment Delivery

TH-IePD-TRACK 5-01

Evaluation of a Novel Film Dosimetry System: Orthochromic OC-1 Film

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Purpose: A new radiochromic film dosimetry system consists of the new OrthoChromicTM OC-1 film and a novel calibration procedure was evaluated. **Methods:** The novel calibration procedure involved both intensity-to-dose conversion and scanner spatial correction using OrthoChromic-Pro software. Two films, C1 and C2, were exposed simultaneously using the 6xFFF beam with a step-wedge pattern of five 15x3 cm² steps ranging from 590 to 3000 cGy. The films were placed in solid water at 5.0 cm depth and SAD 100 cm. C1 was used for calibration and C2 was used to validate the calibration curves. Another set of films were irradiated to characterize the film system. A film was exposed with an open 20x20 cm² field at SSD 100 cm and 5 cm depth. The beam profile measured with the film was compared to that measured with an IBA cc04 ion chamber in water. To assess the scatter response, the St were generated by exposing films with 1x1 to 10x10 cm² at SSD 100 cm and 10 cm depth. The film St were compared with the stereotactic diode, SD, and cc04 daisy-chained measurements in water. All films were scanned with the Epson 12000XL scanner with 150 dpi and analyzed using an in-house program. **Results:** The new calibration procedure was validated with an average dose difference of 1.6% between C1 and C2 and a $\gamma(2\%,2\text{mm})$ passing rate of 100%. For profile, the average absolute difference between cc04 and film within ~80% of field was 0.4% (i.e. profiles were not normalized to each other). The St between films and SD/cc04 were within -0.7% difference on average (range: 0.1% to -1.9%). **Conclusion:** Preliminary data has shown that the new film dosimetry system with OC-1 film can achieve highly accurate results. Further studies on energy and intensity modulated fields responses are in progress.

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TH-IePD-TRACK 5-02

Temporal Growth of the Polymerization in Irradiated EBT3 and EBT-XD Radiochromic Films

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Purpose: EBT-XD is the latest GafchromicTM film model designed to cover up to 40Gy dose level. Characterizing the temporal structure of its response to radiation is an important step toward better understanding its use and handling. The purpose of this work is to investigate the development of the response of EBT-XD and EBT3 films from several minutes up to 2 months after irradiation. **Methods:** Film samples (5x5cm² pieces) were placed between solid water phantoms and irradiated with a 6 MV beam. Two film samples for each model were irradiated in dose range 2 to 40 Gy for EBT-XD and 1 to 10 Gy for EBT3. The net optical density (OD) was measured using a flatbed scanner post-irradiation, every 5 to 10 min during the first 6 hours, every 20 to 40 min from 6-12 h, and every 24 hours for the first 5 days, and every 10 days beyond that until 2 months post-irradiation. **Results:** The largest growth rate in the net OD of the film was observed during the first 3 hours after irradiation, whose variation depended on the dose delivered and color channel used for analysis (e.g. ~3.3%/h in the 15-Gy-irradiated EBT-XD film). In the time interval of 3-12 hours, the increase in the film response was much slower (~0.7%/h). From 12-24 hours, the response continues to develop at a slower pace (~0.4%/h). From 24-120 hours a small change (~0.1%/h) was noted. Beyond the 5th day, ~0.02%/h in OD was observed. **Conclusion:** The polymerization of both film models was observed to be continuously developing post-irradiation manifested as an increase in the net OD. A two-exponential model could fit the net OD up to 5 days, beyond that a three-exponential model provided a better fit to the net OD.

TH-IePD-TRACK 5-03

Uncertainty in OSLD Dose Reading Due to CBCT-Based IGRT

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Purpose: Optically stimulated luminescence dosimeters (OSLDs) are commonly used for in-vivo dosimetry for in- and out-of-field applications. When kV-CBCT is part of IGRT, it is recommended to place OSLDs after imaging to avoid the dosimeter over-response to diagnostic x-rays. However, entering the treatment room between imaging and treatment is disruptive to IGRT clinical workflows. We quantify herein the OSLD dose reading overestimation due to kV-CBCT so that imaging and therapeutic doses can be separated. **Methods:** A thorax CBCT protocol (125kV, 270mAs) was utilized as the diagnostic beam quality. The OSLD kV dose linearity with number of scans was confirmed (predictive model). Standard and patient-specific setups were considered. For the standard setup (100cm SSD, 10x10cm² field size, d=1.5cm), in-field OSLDs were placed at the central axis, and out-of-field OSLDs were placed 10cm off-axis; outside the primary photon field but within the imaging field. Three irradiations were delivered with individual OSLD sets per irradiation: 200MU 6MV only, kV-CBCT only (12 scans), 200MU 6MV combined with kV-CBCT. For the patient-specific setup, a patient's head-and-neck VMAT plan was recalculated on a solid-water phantom with the OSLD placed under 1cm of bolus. The same three irradiation scenarios listed above were delivered for the patient plan, placing in-field OSLDs only. Only screened OSLDs were used and read according to manufacturer recommendations utilizing a 6MV calibration curve. **Results:** Per scan, and relative to MV only measurement, doses were overestimated on average by $1.9 \pm 0.02\%$ (in-field) and $60 \pm 0.3\%$ (out-of-field) when using the standard setup, and $4.6 \pm 0.05\%$ for the patient-specific setup. The added uncertainty was consistent with the prediction model. **Conclusion:** We quantified the OSLD readout uncertainty associated with the effect of kV-CBCT-based IGRT on MV readings. With overresponses as high as 60% for out-of-field applications, it is recommended to place OSLDs after IGRT or to create appropriate correction factors.

TH-IePD-TRACK 5-04

Design Considerations for CdTe-Based Radiation Detector for Correction-Less Dosimetry

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Purpose: An ideal dosimeter for radiation therapy applications is typically described as water-equivalent to eliminate inhomogeneity-related perturbations and simplify interpretation of measurements. In practice corrections are required, especially in small-field dosimetry where both field-size and detector-specific corrections are employed. Water-equivalency becomes even more challenging for solid state detectors, restricting choices to Si or diamond. We explore an alternative approach of purposely modifying electron spectra reaching the sensitive volume, studying high-efficiency Cadmium Telluride (CdTe) detector as a practical example. **Methods:** Since the stopping-power ratios are not affected by the field-size-dependency changes in photon fluence, modifying electron spectra within detector sensitive volume becomes feasible. MCNP5 package was used to model dosimeter designs combining a semiconductor layer with a metal back reflector. CdTe of 30 and 300 μ m thickness was simulated in combination with copper (Cu) and lead (Pb) of varying thickness, with surrounding PMMA layer. The absorbed dose was calculated based on electron fluences and stopping power. **Results:** Based on the electron spectrum generated at the surface of CdTe under 6MV photon beam representative of Varian TrueBeam linac, we were able to acquire fluences due to electrons backscattered in CdTe from the interface with a metal layer, thus adjusting the final dose in CdTe. Varying thickness of Cu and Pb and thus changing electron backscattered fluence and spectra from the interface we can adjust dose ratio of CdTe to water in a controlled manner. Further design modifications can be done using electron backscatter coefficients for the corresponding interfaces. **Conclusion:** Cadmium Telluride (CdTe) detectors offer high sensitivity, limited dark current, and mature manufacturing technology. High sensitivity of solid-state detectors is an attractive attribute for radiation dosimetry applications. Our approach of modifying electron spectra through addition of a back-reflector metal to CdTe detector opens a venue for design of a new class of correction-less dosimeters.

TH-IePD-TRACK 5-05

Validation of a Full Monte-Carlo Beam Source On the Varian TrueBeam for Primary Standards Establishment in Dose Area Product (DAP)

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Purpose: To implement a full Monte-Carlo beam source of the Varian TrueBeam head for primary standards establishment in terms of Dose Area Product (DAP) for 6 MV FFF small field sizes. **Methods:** In Monte-Carlo dosimetry, multiple phase-spaces files from different Monte Carlo codes are available and free access. However, for small fields, the lack of detailed information and the reported limited accuracy are even more critical. As part of the establishment of the primary standards for small fields in DAP at the Laboratoire National Henri Becquerel (LNHB), the French Primary Standards Dosimetry Laboratory, and in order to overcome limits of Varian phase-space files, a full 6 MV FFF TrueBeam head has been modelled as a beam simulation source with the EGSnrc code with data provided by the manufacturer. A substantiated set of 11 configurations with specifically designed circular collimators of 5, 7.5, 10, 13 and 15 mm diameter and jaws of 5, 7, 10, 13, 15 and 20 mm side were investigated with respectively DOSXYZnrc and DOSRZnrc user codes. All the measurements were performed with a PTW 60019 Microdiamond and the validation was carried out with a strict gamma-index criteria of 0.5%/0.5mm, in accordance with metrological expectations. **Results:** Experimental and Monte-Carlo calculations are found to be in excellent agreement. Commissioned Monte Carlo models reproduce the dose distributions of radiation field sizes from 5 mm to 20 mm with an average of 89 % and 98% of points passing the gamma-index criteria for respectively inline/crossline jaws profiles and percentage depth doses (PDD). For circular collimators, these values are 91 % and 96% for respectively radial profiles and PDD. **Conclusion:** This robust validation of the 6 MV FFF source parameters allows a more confident use in the calculation of the graphite to water conversion factor, and possibly other correction factors.

TH-IePD-TRACK 5-06

Dosimetric Effect of Tumor Treating Field Electrodes for Pediatric Infratentorial IMRT

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Purpose: The addition of tumor treating fields (TTF) designed to target the infratentorial brain to radiotherapy for diffuse intrinsic pontine glioma (DIPG) will be explored in a phase II clinical trial. This study explored the dosimetric effect of concurrent TTF on target and structures-at-risk in pediatric patients with newly diagnosed DIPG undergoing IMRT. **Methods:** Scalp dose with and without TTF electrodes was measured using a parallel-plate ion chamber in solid water for a 10x10 cm² open field. Buildup thicknesses varying from surface to 2 mm simulated different depths in the skin. Beam incident angles varied from 0-90 degrees. Maximum dose increase at 1 mm depth was measured for three six-field IMRT patient plans. A treatment planning study was conducted by transferring clinical targets from ten patients ages 2-14 years onto age-appropriate ATOM phantom reference CTs with no TTF electrodes present. IMRT plans were generated with Eclipse AAA. TTF electrodes were contoured on CTs of the phantom with electrodes reconstructed with a metal artifact reduction algorithm. These contours were transferred to the reference CT and changes in target coverage and scalp dose (5 mm from skin surface) due to TTF electrodes were evaluated. **Results:** Open field measurements demonstrated an increased sensitivity to electrode presence for shallower depths. The dose increase ranged from 150% at 2 mm to almost five-fold at the surface. Increases were greatest for enface beams. Maximum IMRT dose increase at 1 mm depth due to TTF electrode presence was approximately 150%. The treatment planning study showed minimal PTV coverage differences ($\Delta V_{95\%} < 1\%$) and slight volumetric scalp dose increase ($D_{1cc} + 2\%$) with TTF electrodes present. Surface point doses calculated agreed with measurement results. **Conclusion:** Presence of TTF electrodes during pediatric infratentorial IMRT delivery modestly increased the local scalp dose but had a minimal impact on volumetric dose metrics.

TH-IePD-TRACK 5-07

Monte Carlo Calculations of Photon Shielding Properties for Commercial Blocks

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Purpose: A commercial shielding company (Veritas Medical Solutions) provides shielding options for radiotherapy treatment rooms using modular, high-iron content interlinked blocks (VeriShield™). Monte Carlo simulations were used to calculate shielding properties and compare against density-scaled shielding parameters from the NCRP151 report. **Methods:** The simulation model utilized the Tool for Particle Simulation (TOPAS) code for radiation transport calculations where the realistic in-air model consisted of a photon source, concrete wall with varying thickness, and a detector. Vendor-provided (Varian Medical Systems) phase-space files for TrueBeam™ 6X, 6FFF, 10X, 10FFF, and 15X beams were used as sources for deriving primary tenth-value layer (TVL) values for ordinary concrete (V150) as well as concretes having higher iron content (V250 and V300). The concrete wall thickness varied between 0-100cm, with the source and wall exterior distance generally being 600cm. Circular-shaped detectors with 3cm radial bins and 0.05MeV-energy binning were placed 30cm from the wall exterior to score photon fluence. Subsequently, collisional kerma and primary shielding TVL values (1st TVL_i and equilibrium TVL_e) were calculated with consideration for detector volume-averaging. **Results:** Reproducing the NCRP151 geometry verified the primary TVL values for 6X in conventional concrete within 2%. Using the 6X phase-space files, TVL values of concrete were reduced by 2cm with (TVL_i, TVL_e) values in V150, V250, and V300 being (33.4cm, 31.2cm), (21.0cm, 19.4cm), and (16.7cm, 16.0cm) respectively. The 6FFF TVL values were (30.4cm, 30.4cm), (19.2cm, 18.9cm), and (15.3cm, 14.6cm) respectively. Similar trends were observed for 10X, 10FFF, and 15X beams. Sensitivity to angular volume-averaging was assessed towards providing reference values. Statistical uncertainties for TVL_e were typically < 1%. **Conclusion:** Radiological simulations to derive shielding parameters for VeriShield™ modular blocks indicated substantial differences between density-scaled results towards providing improved accuracy for vault shielding. Comprehensive simulations for remaining shielding parameters such as leakage and scatter are underway.

SAM Professional Symposium

TRACK 1

A Penny for Your Thoughts: Survey Science for Medical Physicists

TH-C-TRACK 1-00

A Penny for Your Thoughts: Survey Science for Medical Physicists

Moderators: Kristi Hendrickson, University of Washington, Seattle, WA and Jennifer Lynn Johnson, Kelsey-Seybold Clinic, Houston, TX

TH-C-TRACK 1-01

L. Schubert, University of Colorado Denver:

Start from Scratch: How to Craft Your Own Survey Questions, Methods, and Workflows

TH-C-TRACK 1-02

C. Buckley, Mayo Clinic Arizona:

Translation and Interpretation: Adapting Established and Validated Survey Questions

TH-C-TRACK 1-03

R. Ivie, American Institute of Physics:

Broaden the Horizons: Surveys at National and International Scale

TH-C-TRACK 1-04

Audience Q&A

Do you ever feel inundated from surveys, be it professional, political, or product-specific? Requests arrive via email, social media, and text message, and they're waiting for you in your break room and mailbox. Surveys are an established way to learn about others' feelings, perceptions, and preferences in addition to fact-finding. Medical physicists can use survey tools to identify and learn more about issues or trends in our working environment that we are responsible to resolve.

How can you prevent your survey from getting crushed by a respondent's survey fatigue, or simply forgotten about? What are the optimal guidelines for crafting thoughtful queries that get specific answers your team needs? How do you move forward with data analysis once the responses are back? This session shares best practices in survey usage, design, implementation, and analysis. Topics include: how should questions be crafted; how many questions is the right number; how do we deploy surveys internally within departments or externally to larger groups; what is an expected response rate; what do we do with the data once we have it?

We will share concrete examples of researchers who have written their own surveys, adapted outside surveys for their own use, and deployed large-scale surveys. We will help the audience to understand what works well and what is unexpectedly challenging when physicists seek to ask questions of other physicists. Who can use the data, who can see the data, what identifying information is present, and other subtle but important aspects will be included.

Learning Objectives:

1. Share best practices for creating your own survey questions, methodologies, and workflows
2. Offer a framework for attendees to approach the task of translating established and validated survey questions to their own site, department, team, hierarchy, infrastructure, IRB, etc.
3. Provide guidance to those hoping to send a survey outside of their own institution by describing the process of large-scale survey design

SAM Imaging Educational Course

TRACK 2

Effective Dose and Patient Dose

TH-C-TRACK 2-00

Effective Dose and Patient Dose

Moderator: Madan Rehani, Massachusetts General Hospital, Boston, MA

TH-C-TRACK 2-01

M. Rehani, Massachusetts General Hospital:

Effective Dose: What It Gives and What It Does Not

TH-C-TRACK 2-02

C. Martin, University of Glasgow:
Latest from ICRP On Effective Dose

TH-C-TRACK 2-03

E. Samei, Duke University:
What Is the Best Metric to Assess Patient Exposure

TH-C-TRACK 2-04

Audience Q & A

There have been recent changes and developments in ideas about Effective dose (E) that could lead to an expanded use of the quantity in coming years. Medical physicists should be aware of the expanding definition and potential future developments of E. Despite various limitations of the E which was originally developed by the International Commission on Radiological Protection (ICRP) for occupational radiation protection, E has remained a universally-accepted quantity. Recently, ICRP has allowed the use of E to represent patient dose with the ICRP reference phantoms. Consideration is now being given to encourage the development of patient specific dose. Some medical physicists have been apprehensive about the use of E for individual patients. The current position of ICRP has been clarified in an upcoming report, ICRP Publication 147, and recent journal publications have acknowledged the substantial progress in assessing patient-specific doses computed based on libraries of phantoms that allow matching of anatomy to individual patients. Organ and tissue doses derived from MC calculations can be combined with the ICRP tissue weighting factors to calculate a value for the patient specific effective dose, in which the organ doses represent estimates of those for the individual, rather than a reference phantom. This session details the justification and processes of these developments, and further puts the estimation of individual radiation exposure in terms of E in the context of the broader objective of assessing and ensuring the safety of radiation in medicine.

Learning Objectives:

1. To understand what effective dose means and what it does not.
2. To become familiar with the new developments in ICRP on effective dose.
3. To understand the difference between E as a radiation protection dose quantity and patient or size-specific effective dose relating to individual patients.
4. To understand the relationship between varying metrics of dose and methods to assess patient specific dose

Imaging Scientific Session

TRACK 3

MRI

TH-C-TRACK 3-01

BEST IN PHYSICS (IMAGING): Parallel Imaging Stream for Multi-Purpose Real-Time Adaptive MRI-Guided Prostate Radiotherapy

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Purpose: To develop a multiple-purpose 3D parallel imaging stream for real-time adaptive MRI-guided prostate radiotherapy. Here, we propose to modify the clinical 3D CINE to increase the temporal resolution, decrease the reconstruction latency and to include multiple image streams for improved motion detection. The CINE is modified in two ways: 1) the k-space data is sampled using a novel dual density Cartesian Acquisition with SPiral profile ordering (dd-CASPR), which enables viewsharing and multi-resolution image reconstruction; 2) the k-space data is rerouted and processed outside of the vendors software using an in-house built reconstruction framework (ReconSocket) to produce multiple image streams with low reconstruction latency. **Methods:** The dd-CASPR CINE was implemented on a 1.5T MR-linac system (Elekta Unity, Elekta AB, Stockholm, Sweden) and connected to the ReconSocket to generate the three image streams: 1) Real-time high frequency 1D navigators (projection) to detect extreme bulk motion at 3Hz, 2) Real-time low-resolution (10mm²) imaging to detect large target motion (e.g. gas bubbles) at 0.5Hz, 3) Real-time or retrospective high-resolution (2mm²) imaging for online replanning or dose accumulation at 0.15Hz. Images were reconstructed with parallel imaging using the BART toolbox on a GPU. The dd-CASPR CINE was evaluated using the Quasar motion phantom (Modus Medical Devices Inc., Canada) in which motion traces were compared. In addition, dd-CASPR was qualitatively evaluated in a volunteer scan, with deliberate onset of bulk motion. **Results:** The k-space data was streamed in real-time using the ReconSocket (latency=17ms, maximum jitter<10ms). The phantom experiments showed a high correlation between the input trajectory and the three different image streams. The in-vivo scans showed high quality images with reconstruction times 25ms, 0.8s and 7s for image streams 1-3, respectively. **Conclusion:** The proposed 3D dd-CASPR CINE produces multiple image streams with low reconstruction latency. Future work will focus on the integration of the multiple-image stream to facilitate start-and-stop real-time adaptive prostate radiotherapy.

TH-C-TRACK 3-02

Correcting Image Distortion for Brain Tumor Treatments on an MRI-Linac

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Purpose: MRI-Linac systems have been developed to enable real-time image guidance and adaptive tumor tracking during radiotherapy treatments. However, MR gradient design constraints in these hybrid systems can lead to geometric distortions and inaccurate tumor shapes and positions, risking unnecessary radiation dose to surrounding healthy tissues. Here, we quantitatively analyze and correct geometric distortions on a 1.0 T inline MRI-Linac to facilitate accurate brain tumor treatments. **Methods:** A grid phantom with 3718 markers was used to measure distortion on an MRI-Linac in a 20cm×20cm×30cm volume. Based on measured distorted and undistorted marker positions, a spherical harmonic (SH) model was established to characterize and to correct the geometric distortion. A head phantom was then scanned in an MRI-Linac and a CT system to provide test MR images and CT (undistorted) reference images. The contours of head phantom MR images before and after correction were compared with CT reference images. **Results:** The magnitude of geometric distortion in the MRI-Linac increased with distance from the isocenter. The maximum displacement within the head phantom volume (15cm×15cm×15cm) was 5.0±1.8 mm before correction. The slices containing the orbits, a key organ at risk, had maximum displacement 3.6±1.8 mm. Applying the SH correction model reduced distortion by a factor of 3. **Conclusion:** Geometric distortion on brain images was quantitatively analyzed and the SH method was used to effectively reduce image distortion. This work will improve the accuracy of brain tumor treatments on MRI-Linacs.

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TH-C-TRACK 3-03

Comparing Estimated Liver Elasticity Based on the Finite Element Method to MR-Elastography for Distribution of Liver Fibrosis

K Fujimoto*, T Shiinoki, Y Yuasa, Y Kawazoe, M Yamane, T Sera, H Tanaka, Yamaguchi University

Purpose: Finite element method (FEM) is a promising method to calculate the biomechanical parameters of human liver by performing the stress analysis. However, its performance to evaluate the distribution of liver fibrosis has not been demonstrated. In this study, to assess the performance to evaluate the distribution of liver fibrosis, the liver elasticity map estimated by using FEM (Elasticity-map) and the MRE were compared. **Methods:** Twelve patients who underwent magnetic resonance imaging (MRI) and MRE were enrolled in this study. The axial T1-weighted MR images were acquired at shallow inspiration and shallow expiration breath-hold. The displacement vector field (DVF) between T1-weighted MR images at the inspiration and expiration were calculated using deformable image registration. Elasticity-maps were calculated by using FEM and DVF. MRE images were acquired at shallow expiration breath-hold. The grade of liver fibrosis in all patients were determined by radiologist based on liver elasticity measured by MRE. The whole and regional liver (right lobe and left lobe of liver) elasticity estimated via Elasticity-map were evaluated for correlations with the elasticity obtained via MRE. **Results:** The whole liver elasticity estimated via Elasticity-map was significantly correlated with the elasticity measured via MRE ($r = 0.950$, $p < 0.001$) and the liver fibrosis grade ($r = 0.915$, $p < 0.001$). Furthermore, for the regional liver elasticity, the correlation coefficient between the Elasticity-map and MRE was 0.910 ($p < 0.001$) for right lobe of liver and 0.905 ($p < 0.001$) for left lobe of liver. **Conclusion:** The elasticity for whole liver, right lobe and left lobe of liver estimated via Elasticity-map were strongly correlated with the elasticity measured via MRE. Therefore, the Elasticity-map has a potential to be a feasible imaging biomarker to evaluate the distribution of liver fibrosis using only T1-weighted images without additional inspection and costs for special equipment for elastography.

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TH-C-TRACK 3-04

Visual Network in Primary Open Angle Glaucoma: A Functional Connectivity Analysis Based On Group Parcellation From Resting fMRI

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Purpose: Network analysis was an effective way to assess brain function. The atlases used in network analysis were generated from healthy population, which may fail to reflect functional characteristic of the diseased brain. The aim of this study was to carry out functional connectivity (FC) analysis on primary open-angle glaucoma (POAG) patients using a data-driven parcellation atlas generated from the exactly same cohort. **Methods:** Thirty-six POAG patients and 20 healthy controls underwent fMRI scans. K-means algorithm was used to generate a group-averaged parcellation atlas of visual cortices with 20 subregions using the FC profile from the 36 POAG patients. FC was then performed between POAG patients and healthy controls using this atlas among the 20 subregions in the visual cortices, and between the 20 subregions and the rest of the brain. **Results:** A visual atlas with 20 subregions was generated. FC analysis via this atlas revealed that several functional connections in POAG patients were significantly decreased compared with healthy controls, while some functional connections were significantly increased compared with healthy controls. The decreased FCs in POAG patients were mainly between the visual cortices and the right hemisphere, and the increased FCs were mainly located between the visual cortices and left hemisphere. Significant correlations between decreased FCs and increased FCs were found in POAG group. In addition, we found clinical measures such as the elevated intraocular pressure had negative correlations with the decreased FCs. **Conclusion:** POAG may lead to impaired visual networks and lateralization of functional changes in the brain. Functional compensation and inhibition loss mechanisms may occur in POAG brain. Furthermore, elevated intraocular pressure may decrease function connections between visual cortices and rest of the brain in POAG patients.

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TH-C-TRACK 3-05

Novel Sampling Strategy for Improved Reproducibility in Quantitative T1rho Mapping

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Purpose: T1rho (T1ρ) mapping emerges as a promising MRI tool for quantitative evaluation of low-frequency biochemical changes. However, the accuracy and reproducibility of T1ρ quantification can be compromised by B0 and B1 field inhomogeneities. The purpose of this study was to develop a novel sampling strategy of spin-locking time (TSL) for improved reproducibility in quantitative T1ρ mapping. **Methods:** Two TSL sampling strategies were compared in simulation and phantom experiments. The first one was the standard sampling strategy where TSLs were sampled in a linearly spaced fashion (TSL-LS). The second one was a novel TSL sampling strategy called precision- and reproducibility-guided TSL sampling (TSL-PRG) where all TSLs, except TSL=0 ms, were sampled near the expected T1ρ value of interest at a constant interval defined by the Mz oscillation frequency. Simulation was repeated 100 times with field offsets of B0 (±600 Hz) and B1 (±50%) in a 50×50 grid where the percentage of T1ρ errors < 5% (T1ρ_Error_5pt) and the coefficients of variation < 5% (T1ρ_CV_5pt) were compared. For phantom experiment, T1ρ standard deviations in the three tubes at 150 Hz B0 off-resonance were compared. **Results:** Simulation showed that TSL-PRG had slightly better T1ρ quantification accuracy within the range of B0 and B1 offsets (T1ρ_Error_5pt: 30.4% versus 32.8% for TSL-LS and TSL-PRG, respectively). Moreover, T1ρ variation was substantially reduced for TSL-PRG (T1ρ_CV_5pt: 44.3% versus 67.0% for TSL-LS and TSL-PRG, respectively). In the phantom experiment, standard deviation of T1ρ values at about 150 Hz B0 off-resonance was significantly reduced for TSL-PRG compared to TSL-LS in all three tubes (average of 5.40±0.75 ms versus 8.70±2.25 ms, $P < 0.001$). **Conclusion:** This study demonstrated that the proposed TSL-PRG sampling strategy produced more reproducible T1ρ quantification compared to standard linearly spaced TSL sampling (TSL-LS), suggesting the importance of TSL selection in quantitative T1ρ mapping.

TH-C-TRACK 3-06

A Simulated Comparison of Local SAR Near Orthopedic Implants at 7T Relative to Lower Field Strengths

P Jacobs*, A Fagan, Mayo Clinic, Rochester, MN

Purpose: Electromagnetic (EM) simulations were used to investigate the maximum local 1g-average SAR induced in tissue adjacent to orthopedic screws of varying lengths and field strengths. **Methods:** EM simulations were performed within a finite-difference time-domain solver package (Sim4Life, ZMT, Switzerland). Orthopedic screw models, ranging in length from 18-183mm, were placed within a tissue region assigned to dielectric properties of bone. A dipole antenna, rather than a birdcage-type RF coil model, was used to create uniform background electric fields in the vicinity of the implant for each field strength ranging from 3T to 7T. Simulations were performed using a dedicated GPU (Tesla V100, NVIDIA, USA), each taking approximately 4 hours to run. Two normalization methods were implemented on simulation input power. The first accounted for reflection losses and ensured identical transmit power (1W) into the tissue volume. The second ensured a uniform B1+ magnitude (0.45μT) was achieved at the implant location. Simulations were also performed on tissue with dielectric properties of muscle and white matter brain tissue at 7T. **Results:** After normalizing to a constant B1+ field, maximum 1g-average SAR as a function of screw length showed a peak near 4T; the value at 7T was 36% lower than the peak. The implant length at which the maximum SAR was measured for all field strengths occurred at approximately one-third of the wavelength at each field strength, as expected. The maximum SAR shifted to longer implant lengths in bone compared to muscle and white matter, reflecting the lower permittivity and conductivity in bone. **Conclusion:** For the specific set-up investigated, a low maximum SAR value was determined at 7T

compared to lower field strengths when normalized to the B1+ field. These results indicate RF-induced tissue may not necessarily be significantly higher at 7T compared to lower fields, as conventionally assumed.

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TH-C-TRACK 3-07

Reproducibility of Functional Modes Identified by Non-Negative Matrix Factorization of fMRI Depends On Pre-Processing Strategy

Z Ahmed*, D Kang, N Meyer, M In, Y Shu, Mayo Clinic, Rochester, MN

Purpose: Blind source separation algorithms such as independent component analysis (ICA) and non-negative matrix factorization (NMF) can identify functional modes from fMRI scans. Results from NMF are easier to interpret than ICA, but NMF assumes that input is non-negative whereas fMRI pre-processing typically produces values with zero-mean and unit-variance to identify functional signal change. Different studies have employed various strategies for dealing with these negative values. The goal of this work was to evaluate the impact of four strategies in a test-retest fMRI dataset. **Methods:** A public dataset was downloaded from openneuro.org/datasets/ds000114. It consisted of 10 subjects scanned twice, spaced 2-3 days apart, while performing five behavioral tasks. All images were spatially aligned and then temporally concatenated for each session and task, resulting in 10 pre-processed series (2 sessions x 5 tasks). The fMRI data was standardized to zero-mean and unit-variance for ICA while four strategies were applied to remove negativity for NMF by: (1) setting negatives to zero, (2) taking absolute value, (3) using spectra magnitude after Fourier Transform, or (4) using signal magnitude without any standardization. Sixteen spatial components were computed for each task using ICA and NMF and the agreement between the two scanning sessions was evaluated using the dice coefficient. **Results:** Among the 80 computed components (16 components x 5 tasks), the number of NMF components with a dice coefficient above 0.5 was 39 if the data was zeroed, 27 if absolute values were used, 11 if the spectra magnitude were used, and 29 if the signal magnitude was used. For ICA, 28 components had a dice score above 0.5. **Conclusion:** The strategy for removing negativity has substantial impact on NMF performance on fMRI detection. This study showed that the best approach was to set negative values to zero after data standardization.

SAM Multi-Disciplinary Scientific Symposium

TRACK 4

Novel Algorithms for High-quality Diagnostic and On-board Cone-Beam CT

TH-CD-TRACK 4-00

Novel Algorithms for High-Quality Diagnostic and On-Board Cone-Beam CT

Moderator: Alejandro Sisniega, Johns Hopkins University, Baltimore, MD

TH-CD-TRACK 4-01

A. Sisniega, Johns Hopkins University:

Deep-Learning-Enabled Autofocus for Image-Domain Deformable Motion Compensation in Cone Beam CT

TH-CD-TRACK 4-02

M. Kachelriess, DKFZ Heidelberg, FS05:

Data-Driven Methods for Image Reconstruction and Artifact Correction in Cone Beam CT

TH-CD-TRACK 4-03

A. Wang, Stanford University:

New Technologies for Dual Energy Cone Beam CT

TH-CD-TRACK 4-04

Y. Rong, Mayo Clinic Arizona:

Translating Recent Advances in CBCT to Clinical Practice

TH-CD-TRACK 4-05

Q&A

Cone-beam computed tomography (CBCT) offers an imaging platform suitable for design of highly adaptable, compact, and portable systems that can be tailored to specific applications. These unique characteristics resulted in increased presence of CBCT as an on-board imaging tool (for image-guided radiotherapy, interventional radiology, and surgery) and as a diagnostic imaging tool in specific applications (e.g., breast or extremities imaging). The intrinsic advantages of CBCT are however accompanied by a number of image-quality degradation factors that include non-idealities of the imaging chain, increased scatter, and long acquisition time that results in patient motion. Those limitations have conventionally deemed CBCT as a modality suitable only to imaging of high-contrast features within a static setting. Novel developments in algorithmic approaches for CBCT acquisition, reconstruction and image processing are closing the gap, pushing CBCT towards image quality compatible with applications involving visualization of low-contrast structures in dynamic environments.

This symposium will illustrate novel developments for enhancement of CBCT image quality via data-driven methods for image reconstruction and artifact correction, new technologies for dual energy CBCT, and image-based methods for compensation of deformable motion. Finally, the symposium will cover translation of novel algorithmic approaches to clinical environments, with example application in adaptive radiotherapy.

Data-driven methods for image reconstruction and artifact correction:

Various data-driven solutions for CBCT image reconstruction and artifact reduction have been proposed. For example, methods for sparse image reconstructions, methods to reduce noise, approaches to reduce artifacts such as ring artifacts or metal artifacts, algorithms to estimate scatter in real time, or methods that estimate and compensate for patient motion. The lecture will highlight such approaches and indicate how they compare with classical algorithms.

New Technologies for Dual Energy CBCT:

Dual energy CT has become an important tool for diagnostic CT. This talk will discuss applications, approaches, and challenges for dual energy CBCT. Applications include material-specific, quantitative imaging and improved image quality; approaches include kV switching, dual layer detectors, photon counting detectors, and filters; and challenges include patient scatter and system design to optimize dual energy acquisitions.

Image-Domain Deformable Motion Compensation in CBCT:

The moderately slow acquisition time of CBCT systems often results in artifacts from involuntary patient motion. Motion in CBCT ranges from rigid displacement of bony structures to complex deformation of soft-tissues, hampering methods for tracking motion with external markers that would, furthermore, fit poorly into the workflow of image-guided procedures. This lecture will present image-based approaches to motion compensation via combination of autofocus techniques with learning-based methods.

Translating Recent Advances in CBCT to Clinical Practice:

CBCT is the most common and readily available onboard imaging system for patient positioning, target localization, inter-fractional monitoring, and even adaptive radiotherapy (ART). It has been a challenge using CBCT for ART, especially for online ART, due to inferior image quality compared to CT. Previous studies have proved that CBCT is helpful in ART for reducing the volume of irradiated healthy tissue and the dose delivered to OAR, if image quality can be improved through various post-processing algorithms or with the help of image registration algorithms. This presentation focuses on the recent advancement in CBCT image quality improvement or co-registration methods with the planning CT, for clinical application towards offline and online ART.

Learning Objectives:

1. Learn about challenges to image quality in state-of-the-art CBCT systems.
2. Understand approaches for image quality improvement via novel algorithms for multi-energy CBCT image processing, data-driven image reconstruction and artifacts correction, and image-based motion compensation.
3. Learn the potential for clinical application of novel approaches to high-quality CBCT.

Research support from Siemens Healthineers

Therapy Scientific Session

TRACK 5

Novel treatment delivery and verification techniques I

TH-C-TRACK 5-01

A Novel 4D Dose Accumulation Algorithm for Predicting Dose to Organs at Risk During Motion Synchronized Treatments On Radixact

W Ferris*, W Culberson, J Bayouth, L DeWerd, University of Wisconsin, Madison, WI

Purpose: The Radixact is a helical tomotherapy device capable of delivering treatments with intrafraction target tracking, called Synchrony. The system dynamically collimates the radiation to follow target motion during treatment, addressing the effects of the motion by maintaining target dose. However, the radiation is directed without considering location changes of organs at risk (OARs). This study computed the dose to OARs from these motion-synchronized treatments. **Methods:** We developed a calculation algorithm that uses the patient's 4DCT scan as input and accounts for tissue deformations. The patient's respiratory amplitude throughout treatment was simulated by extracting the amplitude data from the 4DCT scan or assuming a generalized motion pattern (e.g. $\sin^2$). Based on the amplitude at each projection in the helical treatment, MLC leaf shifts and jaw corrections were applied, mimicking the corrections Synchrony applies during treatment. The helical treatment was broken up into 153 sub-projections per gantry rotation, which were each assigned to a corresponding discrete phase of the 4DCT. Dose was calculated on each phase separately and accumulated using a 4D deformable registration algorithm. **Results:** The algorithm has been used to predict dose for 4 lung SBRT treatments (10 Gy x 5 fx). Target motion ranged from 14 to 30 mm and PTV volume ranges from 12 to 130 cc. Dose to OARs was observed to deviate by up to 4 Gy between the planned dose and the synchrony-accumulated dose. In future work, the number of cases will be expanded, including upper abdominal cases. **Conclusion:** An algorithm for calculating dose from motion-synchronized treatments on Radixact has been developed and demonstrated. The algorithm can be used to predict dose to OARs and identify plans likely to deviate from planned dose when target tracking is activated.

TH-C-TRACK 5-02

A Trajectory Optimization Method for Non-Coplanar Volumetric Modulated Arc Therapy of Locally Advanced Unresectable Pancreatic Cancer

G Wang*, R Yang, Peking University Third Hospital, Beijing, CN

Purpose: To develop a trajectory optimization algorithm for non-coplanar volumetric modulated radiation therapy and investigate the potential of organ at risk sparing in locally advanced pancreatic cancer patients using non-coplanar VMAT. **Methods:** A cost map which represents the ray-OARs voxel intersections at each source position was generated using ray-tracing algorithm. A graph search algorithm was used to determine the least-cost path from the cost map. Lastly, full arcs or partial arcs were selected based on the least-cost path to generate the non-coplanar VMAT (ncVMAT) trajectories. Clinical coplanar VMAT (coVMAT) plans for 11 patients diagnosed with locally advanced unresectable pancreatic cancer (LAPC) were replanned using non-coplanar VMAT trajectories. The conformity index (CI), homogeneity index (HI), PTV coverage and dose to the organ at risk (OARs) were compared. **Results:** With ncVMAT, the mean coverage of PTV50Gy, PTV54Gy, PTV60Gy, and PTV70Gy increased significantly. The mean conformity index of PTV45Gy, PTV54Gy, and PTV70Gy was also improved in the ncVMAT plans. Compared with coVMAT plans, the ncVMAT plans resulted in significantly lower doses to the spinal cord, bilateral kidneys, stomach, and duodenum. The maximum dose to the spinal cord decreased by 6.11%. The mean dose to the left and right kidney decreased by an average of 5.52% and 11.71%. The Dmax, Dmean, and D15% of the stomach reduced by an average of 7.45%, 15.82%, and 16.79%, separately. The D15% and Dmean of duodenum decreased by 6.38% and 5.64%, respectively. **Conclusion:** A trajectory optimization algorithm was developed for non-coplanar VMAT. Compared with conventional coplanar VMAT, non-coplanar VMAT resulted in improved coverage and conformity to the targets. The sparing of OARs was significantly improved in non-coplanar VMAT compared with coVMAT plans for locally advanced pancreatic cancer.

TH-C-TRACK 5-03

Influence of the Freedom in Table Rotation on Plan Quality, Deliverability, Delivery Time and Robustness of Dynamic Trajectory Treatment Plans

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Purpose: To investigate the influence of freedom in dynamic table rotation on treatment plan quality, deliverability, delivery time and robustness against systematic machine position errors for dynamic trajectory radiotherapy (DTRT). DTRT extends VMAT by dynamic table and collimator rotation during beam-on, described by gantry-table and gantry-collimator paths. **Methods:** The gantry-table path is determined by minimizing overlap of OARs with PTV under following restrictions in freedom for table rotation: Maximum gradient of the path (G) between two adjacent control points ($\Delta\text{table-angle} / \Delta\text{gantry-angle}$) and maximum change of G (ΔG). A VMAT plan and 4 DTRT plans are created for different table rotation constraints $G \& \Delta G = 0.5 \& 0.125$ (DTRT1), $1 \& 0.125$ (DTRT2), $3 \& 0.125$ (DTRT3), $3 \& 3$ (DTRT4), including 2-4 arcs/trajectories, for a nasopharynx, a glioblastoma and a bilateral oropharynx case. For all plans, plan quality is evaluated by target coverage and OAR sparing, delivery time is estimated and dosimetric robustness against systematic errors between $\pm 1^\circ$ in gantry, table, collimator rotation and combination thereof is evaluated. Delivery accuracy is verified with film measurements on a TrueBeam for the nasopharynx plans. All dose calculations are Monte Carlo based. **Results:** With increasing $G \& \Delta G$, slight (oropharynx) and substantial (nasopharynx, glioblastoma) improvements in OAR sparing are observed compared to VMAT, while maintaining similar target coverage. However, delivery time was prolonged by up to 5 min. The VMAT (oropharynx), DTRT2 (nasopharynx) and DTRT1 (glioblastoma) plan revealed the best dosimetric robustness. Film measurements agreed by $>98\%$ (2%/2 mm Gamma-passing-rate) with the dose calculation for VMAT and DTRT1&2. DTRT3&4 resulted in interlocks due to gantry exceeding its tolerance limits. **Conclusion:** With increasing freedom in table rotation, plan quality can be improved for DTRT on cost of increased delivery time for the investigated cases. No clear dependency of dosimetric robustness on table rotation constraints was visible. This work was partially supported by Varian Medical Systems.

This work was partially supported by Varian Medical Systems.

TH-C-TRACK 5-04

Management of Larger Number of Multiple Brain Metastases Patients Via Restricted Single Isocenter Stereotactic Radiosurgery (RESIST) Method

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Purpose: SRS of multiple brain metastases (m-bm) via single-isocenter/multi-target (SIMT) VMAT can improve treatment efficiency and patient comfort. Due to setup uncertainty, aligning m-bm on a single CBCT is nearly impossible and has shown clinically unacceptable loss of target coverage. Moreover, due to island-blocking, m-bm sharing the same MLC pair, low and intermediate dose spill causes an increase in normal tissue dose. This study proposes RESIST method to improve plan quality by reducing normal brain dose from island-blocking. **Methods:** Six patients with 6-16 tumors were selected for this study with an average PTV size of 10.8 ± 2.1 cc (8.2-13.4cc). Patients were planned with SIMT-VMAT using 10MV-FFF beam with a 20Gy prescription to each lesion. Isocenter was placed at the geometric center of targets. Patients were replanned using RESIST, utilizing a single isocenter with lesions treated in two groups based on brain hemisphere location with corresponding non-coplanar partial-arc geometry. This technique uses DCA-based dose and aperture shape controller before VMAT optimization. It allows for two separate plans to be created for each group while sharing the same isocenter and all tumors to be treated in same session. **Results:** Compared to SIMT-VMAT, RESIST plans provide slightly better coverage and conformity with less low and intermediate dose spread. RESIST showed reductions in normal brain V8Gy and mean brain dose by an average of 64cc and 1Gy with slight increase in V12Gy by 3cc. Overall monitor units decreased by 2278MU and decrease modulation factor by 1.1, on average. Slight decreases in beam on time were evident; however, treatment time would increase with an additional CBCT and couch-kicks. **Conclusion:** For superior target coverage and reduced dose to normal brain, RESIST planning for m-bm is a simplified approach that provides patient comfort and compliance for those complex patients who may not tolerate traditional longer treatment times with multiple isocenters.

TH-C-TRACK 5-05

Treatment of Gynecological Cancer Patients with Extended Fields (EF) Via Dual-Isocenter Volumetric Modulated Arc Therapy (VMAT) Using 6MV-FFF O-Ring Halcyon Linac

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Purpose: A 28 cm² field size at isocenter is a limitation of the ring-mounted Halcyon V2.0 linac. Field size is extendable up to 36 cm utilizing dual-isocenter and junctional auto-feathering option during VMAT optimization. We report initial clinical experience using Halcyon dual-isocenter VMAT, enabling EF treatment for complex gynecological cancers. **Methods:** Five complex gynecologic cancer patients received EF irradiation with dual-isocenter VMAT, delivering 45 Gy in 25 fractions incorporating a 55 Gy simultaneous integrated boost (SIB) to PET-avid lymph nodes (LNs) on Halcyon. Two plans were generated with 4 full arcs (2 per isocenter) and three plans with 8 full arcs (4 per isocenter); different collimator settings were used for auto-feathering. Dual-isocenters were separated by 8 cm in the craniocaudal (CC) direction and the daily CBCT imaging isocenter was placed between them. We report plan quality, quality assurance results, treatment delivery accuracy and efficiency. **Results:** Mean EF size in CC direction was 32.1 ± 2.3 cm (29.2–34.3 cm). Mean target conformity and heterogeneity indexes were 1.05 ± 0.08 (1.01–1.09) and 0.15 ± 0.09 (0.06–0.25) respectively. 95% of the PTV45Gy and PTV55Gy received 46.8 ± 2.2 Gy (45.2–51.3 Gy) and 56.8 ± 2.3 Gy (55.6–60.2 Gy), respectively. Plans were evaluated per RTOG-0418 and all OAR were within protocol compliance. Pre-treatment portal dosimetry QA result was 99.7%, on average for 3%/3mm gamma passing criteria. The 3D radial couch correction and repositioning from daily CBCT imaging was 0.84 ± 0.57 cm (0.41–1.93 cm). Mean beam-on time was 2.6 ± 0.2 min (2.4–2.9 min). Mean patient couch time including CBCT was 9.5 ± 0.3 min (9.2–9.9 min). **Conclusion:** Treating extended fields in complex gynecologic cancers patients on Halcyon utilizing dual-isocenter VMAT auto-feathering is safe, feasible and efficient. Rapid treatment delivery on Halcyon may increase patient comfort, reduce intrafraction motion errors, and improve clinic efficiency and patient throughput. Follow up for disease-control and toxicity in a larger patient cohort is warranted.

TH-C-TRACK 5-06

BEST IN PHYSICS (THERAPY): Color Imaging of Cherenkov Emission in Vivo During Radiotherapy Exhibits Correlation with Superficial Tissue Composition and Blood Oxygenation Changes

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Purpose: Cherenkov imaging has demonstrated its usefulness for real-time beam monitoring and incident detection during X-ray and electron radiotherapy, as well as for remote in vivo surface dosimetry. In the latter case, sub-surface tissue composition introduces approximately 30% error in Cherenkov-dose accuracy. Here we introduce the concept of in vivo color Cherenkov imaging and investigate the potential for correcting these tissue composition factors, and explore novel application of seamless intra-fraction tissue oxygenation monitoring. **Methods:** A three-channel intensified CMOS camera was developed using a relay lens system and a pair of dichroic beamsplitters, which split the incoming image onto three separate intensifiers. The intensified RGB channels were read by three CMOS sensors and combined into 24-bit color image following spatial and color calibration. Tissue-mimicking phantoms containing water, 1% intralipid and blood were imaged during 6MV beam irradiation to study the effects of tissue composition (0-3.5% blood) and blood oxygenation (adjusted by glucose oxidase-catalase assay) on the color of Cherenkov emission. Lastly, in vivo color Cherenkov images of three right-sided breast patients were acquired for one treatment fraction each. **Results:** Oxygenated blood in tissue phantom images produced a visibly redder hue compared with deoxygenated blood due to decreased attenuation of red wavelengths by oxyhemoglobin. Variation in blood concentration produced a blue-red linear correlation ($R^2=0.96$) between blood concentration and the derived luminance parameter x in the CIE xyY color space. Cherenkov emission from patients exhibited pink color, and the luminance parameter was found to be in correlation with fibroglandular vs. fat tissue composition. **Conclusion:** This first investigation of color Cherenkov imaging has uncovered the natural relationship between tissue composition and the color of emitted Cherenkov light. The correlation between blood content and color will be translated into future work for in vivo sensing of variations in fat and fibroglandular content in irradiated tissues.

Hardware and software support for this project was provided by DoseOptics LLC, a company manufacturing Cherenkov imaging cameras. B. Pogue is the President and P. Bruza is an employee of DoseOptics LLC. D. Alexander reports receiving consulting fees from DoseOptics LLC. Intellectual Property application is pending.

TH-C-TRACK 5-07

Commissioning of a Novel Compensator-Based IMRT System for Low- and Middle-Income Countries

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Purpose: In support of a novel compensator-based IMRT system for low- and middle-income countries, here we demonstrate commissioning in Eclipse. This allows for high-quality treatment plans created locally and integration into the Radiation Planning Assistant, a web-based automatic planning system being developed

specifically for resource-limited clinics. **Methods:** The treatment device consists of a cobalt-60 unit and nine field compensator arrays. Each compensator has a 3D-printed plastic mold filled with reusable 2mm tungsten beads and is mounted around the patient. We commissioned it in Eclipse (v15.6). Validation tests described in MPPG 5.a were performed using Monte Carlo (MC) data calculated by GATE. Dose distributions and penumbra were evaluated for basic dose validation using a gamma index for the compensator-based IMRT dose validation. Compensator-based IMRT plans with the cobalt-60 unit were generated for five head-and-neck cancer cases, and PTV coverage and OAR sparing were compared to linac-based plans reviewed by physicians. **Results:** The PDDs and profiles of the Eclipse deviated from the MC simulation by $0.65 \pm 0.41\%$ (Range: 0.1-2.1) and $1.02 \pm 0.99\%$ (Range: 0.1-3.6) and are within 2% tolerance. The 80%-20% penumbra deviation is $0.5 \pm 0.3\text{mm}$ and is within 3mm tolerance. The gamma passing rate with criteria of 2%/2mm was 92.6%. Cobalt-60 compensator IMRT plans were found to deliver comparable quality treatment plans with standard linac-based VMAT plans based on DVHs. PTV D95% and D5% were within 1% for both plans evaluated. Each PTV received a conformal dose based on homogeneity index (0.04 vs. 0.05 for linac-based VMAT plan and compensator-based IMRT plan). Doses for critical structures were slightly higher in the cobalt-60 compensator plans but within clinical tolerances. **Conclusion:** The compensator-based cobalt-60 device can generate highly conformal IMRT plans on par with clinical linac-based technologies at a reduced cost. We are developing a test device with clinical partners in low- and middle-income countries.

Funded by National Cancer Institute

Multi-Disciplinary Scientific Session

TRACK 6

Novel and Emerging Technologies in Radiation Therapy

TH-C-TRACK 6-01

A Deconvolution-Based Image Fusion Technique for Heterogenous Multi-Layer Portal Imagers

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Purpose: Multi-layer (ML) portal imagers contain multiple imaging layers whose signals are combined to increase detector performance. For heterogeneous configurations with different properties in each layer, the overall performance depends on the different layer responses and how the signals are combined. This work introduces a joint Richardson-Lucy (RL) deconvolution to fuse the sub-images from the layers. **Methods:** Synthetic images of a bar-pattern and the Shepp-Logan phantom were generated for a ML detector containing one GOS layer and one with a high-efficiency plastic scintillator. The individual images are obtained by convolving the phantom geometry with Monte Carlo estimated point spread functions of each layer, scaling according to the quantum efficiency and applying Poisson noise. The final image was obtained by using a joint RL deconvolution of the two sub-images, both with and without a regularization based on total variation (TV). The spatial resolution, contrast-to-noise ratio (CNR) and signal-to-noise ratio (SNR) of the images were evaluated and compared with a simple summation of the signals. **Results:** The MTF evaluation of the bar-pattern showed an enhanced 10%-contrast level (f_{10}) for the fusion image ($f_{10}=0.9\text{ lp/mm}$) compared to the summation image ($f_{10}=0.7\text{ lp/mm}$). In this case, the TV regularization suppressed the noise, however slightly reduced the gain in spatial resolution. For the central high-density region of the head phantom, the fusion resulted in 2.7 times higher CNR if regularization is used. Similar improvements were observed for the SNR. **Conclusion:** A novel method for image fusion of ML detectors was presented that improves the performance of heterogeneous ML imagers compared to conventional image summation. Future work will investigate the optimal regularization and deconvolution parameters.

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TH-C-TRACK 6-02

Exploiting Pair Production for Nanoparticle Imaging and for In-Vivo Beam Monitoring in Megavoltage Radiotherapy

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Purpose: We propose a new imaging technique for detection of high-Z nanoparticles and for in-vivo beam monitoring in megavoltage X-ray beams by means of coincidence counting of annihilation photons following pair-production in nanoparticles and/or in tissue. **Methods:** The proposed megavoltage X-ray induced positron emission (MVIPE) imaging technique is studied by radiation transport computations using MCNP6/CEPXS-ONEDANT for two water phantoms: 35cm slab and similarly sized cylinder with 5cm gold nanoparticle (AuNP)-filled region in the center. MVIPE is compared to standard X-ray fluorescence computed tomography (XFCT). MVIPE adopts megavoltage X-ray sources and relies on the detection of 511keV annihilation photon pairs. XFCT uses kilovoltage sources and imaging is characterized by analysis of $K\alpha_{12}$ Au-characteristic lines. Four levels of AuNP concentration were studied: 0%, 0.1%, 1% and 10% by-weight. **Results:** Annihilation photons in the MVIPE technique originate along the X-ray beam path in the AuNP-loaded region and in water. Their flux is proportional to the MV source energy and linearly increases with AuNP concentration, while XFCT signal reaches saturation due to self-absorption within AuNP. The MVIPE technique using 15MV pencil beam and 10wt% AuNP detects about 4.5×10^3 511keV-photons/cm² at 90° w/r to the incident beam per 10° source-photons/cm²; 0.5×10^3 of these come from AuNP, while the rest come from water/tissue. The latter alone can be utilized for monitoring MV beams in tissue. In contrast, the XFCT technique using 150kVp detects about 100 $K\alpha_1$ -photons/cm² per 10° source-photons/cm² without collimation. **Conclusion:** In MVIPE, the number of detected annihilation photons secondary to pair-production is significantly greater than the $K\alpha_1$ -photons from fluorescence. Furthermore, coincidence counting in MVIPE enables collimation-free imaging, while collimation is a major limiting factor in XFCT. Additionally, MVIPE offers the potential of direct visualization of the x-ray beam in tissues during radiotherapy treatment, which may pave the way for in-vivo beam and dose monitoring.

TH-C-TRACK 6-03

Hypersensitivity of Cancer Cells to Photon and Proton Radiation Caused by Disruption of Cell Cycle Arrest by An ATR Inhibitor

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Purpose: Under the presence of DNA damage, the DNA repair protein ATR has essential roles in arresting cells in the G2 phase of the cell cycle so that DNA repair can take place. ATR is also a key protein in homologous recombination double strand break repair. Here we investigate the consequences of disrupting cell cycle arrest at G2 through ATR inhibition on cancer cell lines irradiated with photons and protons. **Methods:** H1299 and PANC-1 cancer cell lines were treated with DMSO (control), or an ATR inhibitor (ATRi) (AZD6738, 0.1 or 1 μM) for 1 h prior to irradiation. Cells were exposed to 5 Gy of 6 MV x-rays or 9.9 keV/ μm protons (dose-weighted LET in water), and then fixed and stained for flow cytometry analysis to evaluate cell cycle distribution (propidium iodide), mitosis (phospho-histone 3) and DNA damage ($\gamma\text{-H2AX}$). Clonogenic cell survival assays were also performed. **Results:** The relative biological effectiveness (RBE) had a non-significant increase for protons+ATRi compared to protons alone. Irradiated cells showed a pronounced G2 arrest after both photons and protons. ATRi 1 μM was effective in abrogating G2 arrest in cells exposed to photons or protons. A higher amount of mitotic cells was observed with 1 μM ATRi+5 Gy compared to DMSO+5 Gy (P-value <0.01). This effect was higher for protons alone compared to photons alone (P-value <0.01). Persistent $\gamma\text{-H2AX}$ foci was higher for ATRi treated cells compared to DMSO indicating that ATRi also suppressed DNA double strand break repair. **Conclusion:** Both photons and protons induced robust cell cycle arrest at G2. Administration of ATRi disrupted the G2 arrest and increased the amount of cells in mitosis. Our results suggest that ATRi is effective at inducing sensitivity of

cancer cells to radiation in part because of the role of ATRi on cell cycling.

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TH-C-TRACK 6-04

Plastic and Lead-Doped Scintillators for Ultrahigh Dose-Rate Irradiations Delivered with An X-Ray Tube

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Purpose: To examine the capabilities of plastic scintillators of different compositions to accurately measure dose in high dose-rate dose irradiations delivered with an x-ray tube. **Methods:** The output of five 1-mm in diameter and 3-mm long scintillators of varying materials was measured in short sub-second irradiation intervals using a traditional x-ray tube equipped with an in-house rotating shutter-wheel system. The shutter system was developed to limit exposure time to the scintillator by a continuous x-ray beam. Two plastic scintillator dosimeters made of polystyrene (BCF-10 and BCF-60) and three lead-loaded scintillators (0.5%, 1.4%, and 5% concentrations) were used. Scintillator output was normalized with respect to TOPAS Monte Carlo (MC) dose calculations using 100 and 200 ms exposures at a constant tube current of 25 mA and tube voltage of 80 kVp. Output linearity was measured for irradiation times between 10-100 ms. Dose-rate independence was tested with varying tube-current between 3-37.5 mA for the 80 kVp beam. Energy dependence was tested for all the probes at tube voltages ranging between 80-120 kVp. **Results:** All probes showed excellent agreement with MC for the 80 kVp beam and irradiation intervals of 100 and 200 ms. When irradiation time was varied between 10-100ms, scintillator output was shown to be linear with exposure time ($R^2 > 0.999$) for all probes. Scintillator output was also shown to be dose-rate independent ($R^2 > 0.999$) for all tested probes. At tube voltages higher than 80 kVp, it was shown that scintillators output exhibited some quenching, resulting in a maximum difference between experimental and MC data of 5.8% across the studied 80-120 kVp energy range for the 5% lead-loaded scintillator. **Conclusion:** Plastic scintillators of different compositions have the capability to accurately measure high dose-rate irradiations delivered in short ~10 ms intervals by a conventional x-ray tube.

Conflict of Interest: FT-P is a founder of Medscint

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TH-C-TRACK 6-05

The Feasibility of Measuring In Vivo Tumor Microenvironment pH During Radiotherapy Using a Novel Cerenkov Emission Multispectral Optical Probe Based On Silicon Photomultipliers

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Purpose: Tumor pH has been suggested as an indicator of cancer aggressiveness. In this work, we explore the use of a novel Cerenkov Emission (CE) multispectral optical probe based on silicon photomultipliers (SiPM), known as CMSI, for interrogating tumor microenvironment pH in head and neck cancer tumors in a proof-of-principle in vivo demonstration consisting of two mice. **Methods:** One mouse was implanted, subcutaneously, on the flank, with the UMSCC47 cell line while the other with an artificial tumor consisting of Matrigel (buffered at a pH of 7.4). During CMSI measurements, the mice were injected with a pH contrast agent consisting of phenol red (PR) and irradiated with 9MV photons. The intensity of CE spectral components at 445nm and 560nm were measured using CMSI and a quantitative value for pH was obtained by taking the ratio of these value (known as the CMSI ratio) in conjunction with a calibration curve, which was prepared from a collection of buffered solutions with PR added at varying pH values (6.5 to 7.8). Calibration was further refined to take into account changes in the CMSI ratio through the mouse tissue by using the artificial tumor pH as a baseline (7.4). **Results:** The CMSI ratio measured for the UMSCC47 tumor on the mouse flank is 1.15 compared to the 1.66 measured on the mouse with the benign artificial tumor. By using the pH calibration curve, the corresponding pH value of the UMSCC47 tumor is 6.5 (compared to the baseline of 7.4 in the artificial tumor), showing the cancerous tumor having a more acidic microenvironment. **Conclusion:** This work shows the feasibility of using CMSI for CE spectroscopic interrogation for real-time in vivo pH measurements during radiotherapy by successfully demonstrating tumor pH measurements during a pilot mouse study.

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TH-C-TRACK 6-06

Three-Dimensional Tumor Spheroid Model as a Tool to Optimize the Nano-Bio Interface

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Purpose: Radiotherapy and chemotherapy are the gold standard for treating patients with cancer in the clinic, which are limited by normal tissue toxicity. The use of nanomaterials, such as gold nanoparticles (GNPs), to improve radiosensitivity and act as drug delivery systems can mitigate toxicity while increasing deposited tumor dose. To expedite a quicker clinical translation, three-dimensional tumor spheroid models that can better approximate the tumor environment compared to a two-dimensional monolayer model have been used. **Methods:** We tested the uptake of 15 nm GNPs and 50 nm GNPs on a monolayer and on spheroids of two cancer cell lines, CAL-27 and HeLa, to evaluate the differences between a 2D and 3D model in similar conditions. The anticancer drug docetaxel, which can act as a radiosensitizer, was also utilized, informing future potential of gold nanoparticle-mediated combined therapeutics. The optimal treatment dose was elucidated using a resazurin-based proliferation assay. The uptake of GNPs into the cells was measured using inductively-coupled plasma mass spectrometry and imaged using darkfield microscopy. **Results:** In the 2D monolayer model, the addition of docetaxel induced a small, non-significant increase of uptake of gold nanoparticles of between 13% and 24%. In the 3D spheroid model, docetaxel increased uptake by between 47% and 186%, with CAL-27 having a larger increase relative to HeLa. Further, the depth of penetration of 15 nm gold nanoparticles over 50 nm gold nanoparticles increased by 33% for CAL-27 spheroids and 17% for HeLa spheroids. **Conclusion:** These results highlight the necessity to optimize gold nanoparticle treatment conditions in a more realistic tumor-life environment. A 3D spheroid model can capture important details, such as different packing densities from different cancer cell lines, which are absent from a simple 2D monolayer model. Inclusion of a more complex model into future lab experiments is imperative for future clinical translation.

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TH-C-TRACK 6-07

Measurement of Secondary Neutron Spectrum with He4 Fast Neutron Detector

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Purpose: He4 fast neutron scintillation detector can measure a wide range of neutron energies up to 10MeV with low gamma-sensitivity. This provides faster and wider spectral analysis than the standard Bonner spheres method, which needs measurements at different activation foil thicknesses and is mostly within thermal range. **Methods:** He4 fast neutron detector (Arktis Detection Systems, USA) was used to measure neutron spectrum from 18MV x-ray linac and IBA double-scattering proton gantry. Measurements were done with 20cm solid water (located at isocenter) irradiated with 18MV X-ray beam and 20x20 cm² collimation; and 180MeV proton beam with 15.1cm SOBP and 20cm² field size. A pulse-shape-discrimination algorithm was used to separate gamma-ray and neutron signals based on charge-integration method that calculates the integral of slow and fast components of each pulse; and spectrum unfolding algorithm, separately validated with a spectrum (up to 10MeV) using time-of-flight techniques calibrated by a deuteron-beryllium neutron source. Detector was placed near gantry room wall at 220cm from isocenter at distances of 30cm and 60cm lateral to edge of solid water. Measurements were done for 15cm, 20cm, 25cm, and 30cm water thicknesses. **Results:** Secondary neutrons are dominant in thermal range for the 18MV X-ray beam. He4 detector could not be used to measure neutron spectrum due to counting inefficiency. For <1MeV neutrons, other detectors (such as organic liquid scintillator EJ-309 and Cs₂LiYCl₆:Ce³⁺ (CLYC) scintillator) with higher detection efficiency were used. Neutron spectra at >1MeV were successfully measured with He4 for 180MeV proton beam. Differences in measured exiting neutron spectra from various solid water thicknesses (15cm, 20cm, 25cm, and 30cm) and detector-solid water distance (30cm versus 60cm) were observed. **Conclusion:** The secondary neutron spectrum and dose from 18MV x-rays and 180MeV protons were measured directly with He4 fast neutron detector. Measured results are comparable to simulations conducted by previous researchers.

SAM Therapy Educational Course

TRACK 7

Not for the Faint of Heart: Functional Radiosurgery

TH-C-TRACK 7-00

Not for the Faint of Heart: Functional Radiosurgery

Moderator: David Schlesinger, University of Virginia Health Systems, Charlottesville, VA

TH-C-TRACK 7-01

S. Braunstein, University of California San Francisco:

Functional Radiosurgery: Clinical Indications and Rationale

TH-C-TRACK 7-02

G. Kim, University of California, San Diego:

Functional Radiosurgery Using the Linear Accelerator

TH-C-TRACK 7-03

D. Schlesinger, University of Virginia Health Systems:

Functional Radiosurgery with Gamma Knife

TH-C-TRACK 7-04

Q&A

Functional radiosurgery procedures involve targeting morphologically normal brain anatomy with extremely high (80 Gy-160 Gy maximum) doses in order to lesion or selectively inhibit neurological activity to treat disorders such as trigeminal neuralgia, essential tremor, and obsessive-compulsive disorder. The techniques and acceptable procedural uncertainties for functional radiosurgery can differ substantially as compared to the more common radiosurgical indications for solid tumors and vascular malformations.

This session will introduce the clinical indications and rationale for functional radiosurgery, as well as the technical considerations for safely performing these radiosurgery procedures across different radiosurgery delivery platforms. In particular, speakers will provide a historical overview of functional radiosurgery; modern clinical indications; imaging, visualization, and treatment planning techniques; and quality assurance on various delivery platforms to ensure the required level of accuracy and precision. The session will highlight the differences between more common radiosurgical techniques used for tumors and vascular malformations and those required for functional radiosurgery.

Learning Objectives:

1. To enumerate the clinical indications treated with functional radiosurgery and published clinical outcomes.
2. To understand the significant differences between functional radiosurgery and radiosurgery as practiced for tumors/vascular malformations in terms of targeted tissue, dose, and acceptable uncertainties
3. To gain an understanding of how functional radiosurgery can be delivered on different SRS platforms, including imaging/visualization techniques, beam configurations, and quality assurance.

SAM Partners in Solutions Therapy

TRACK 8

Therapy: Adaptive Therapy AI Software Planning Systems

TH-CD-TRACK 8-00

Therapy: Adaptive Therapy AI Software Planning Systems

Moderators: Deborah Schofield, AdventHealth Orlando, Orlando, FL and Norman Lee Brown, Memorial Hospital Gulfport, Gulfport, MS

TH-CD-TRACK 8-01

D. Schofield, AdventHealth Orlando:

The Ghost in the Shell: Understanding How AI Can Improve Adaptive Radiation Therapy Workflows

TH-CD-TRACK 8-02

N. Dwivedi, Mim Software, Inc.:

Automated Adaptive Assessment Utilizing On-Board Imaging and Monte Carlo Dose Calculation

TH-CD-TRACK 8-03

A. Schulte, TheraPanacea:

Background of ART-Plan™ Proposal for ART-Plan™

TH-CD-TRACK 8-04

N. Dogan, University of Miami:

Varian's Adaptive Planning Solution - Online or Offline

TH-CD-TRACK 8-05

L. Schulte, Siris Medical:

InsightRT: Automated Adaptive Therapy Decision Support and Insurance Authorization

TH-CD-TRACK 8-06

Q&A

A treatment plan captures a single moment in time and fails to account for changes in anatomy over the course of treatment. Changes in weight or variations in internal anatomy, for example, can result in displacement of the target volume and OAR's relative to the planned dose cloud. The process of adapting a plan can be labor intensive and time consuming for both physicians and staff. Artificial intelligence (AI) has the potential to improve the accuracy, precision, efficiency, and overall quality of radiation therapy for patients by automating adaptive radiotherapy workflows such as target segmentation, treatment planning and, in some cases, radiotherapy delivery. Vendors will present their AI adaptive solutions and describe how their products can improve the workflow and empower the clinician and staff to make informed decisions related to patient cancer treatments.

Learning Objectives:

1. Review the rationale behind adaptive planning.
2. Explore the different means of adapting a plan including off-line, on-line, and real-time adaptive planning.
3. Explain how AI can help streamline the adaptive treatment process.

Professional Symposium

TRACK 1

Medical Physics Workforce Update

TH-D-TRACK 1-00

Medical Physics Workforce Update

Moderators: Wayne Newhauser, Louisiana State University, Baton Rouge, LA and Erli Chen, Cheshire Medical Center, Keene, NH and Michael Mills, James Graham Brown Cancer Center, Louisville, KY

TH-D-TRACK 1-01

Introduction

TH-D-TRACK 1-02

W. Newhauser, Louisiana State University:

The Medical Physics Workforce in the United States

TH-D-TRACK 1-03

E. Chen, Cheshire Medical Center:

A Special Report of 2020 AAPM Therapy Physicist Workforce Survey Results

TH-D-TRACK 1-04

D. Gress, American College of Radiology:

A Special Report of 2020 AAPM Diagnostic Physicist Workforce Survey Results

TH-D-TRACK 1-05

Audience Q&A

Review of the Current State and Future Demand of Medical Physics Workforce

The medical physics profession is a recognized, established, and mature workforce that is undergoing growth and change driven by scientific, technical, and medical advances. In this session we will review the size and general characteristics of the medical physics profession related to the Diagnostic and Therapy Medical Physics workforce and the national radiation professional workforce. Presenting 2020 diagnostic and therapy workforce survey results and discuss the factors that are driving the current and future demands and supply issues in the supply pathways for Diagnostic and Therapy Physicists. Participants will understand the current state of knowledge of the medical physics workforce, including uncertainties and gaps in knowledge. Authors will discuss the key workforce-related challenges facing the medical physics profession in the next decade. Presentations will examine supply and demand of qualified medical physicists for Diagnostic and Therapy Physicists. Presenters will summarize major challenges and opportunities to meet the profession's future needs, strategies and tactics. This information may be used to maintain an adequate and well-trained Medical Physics Workforce.

Learning Objectives:

1. Describe the General Characteristics of the medical physics profession.
2. Understand current state of the Diagnostic and Therapy Physics Workforce
3. Review Current and Future Issues in the Supply Pathways for Diagnostic and Therapy Physicists.
4. Discuss Projections of Demand and Supply for Diagnostic and Therapy Physicists.
5. Summarize major challenges, opportunities, strategies and tactics to maintain an Adequate and Well-Trained Medical Physics Workforce.

The Diagnostic Workforce survey received financial support from an AIP Venture Partnership Fund grant to AAPM

SAM Imaging Educational Course

TRACK 2

Ct Protocol Development and Management

TH-D-TRACK 2-00

CT Protocol Development and Management

Moderator: Megan Jacobsen, UT MD Anderson Cancer Center, Houston, TX

TH-D-TRACK 2-01

K. Little, Ohio State Univ:

CT Protocol Management Pearls and Pitfalls

TH-D-TRACK 2-02

T. Szczykutowicz, University Wisconsin-Madison:

CT Contrast Parameters for the Medical Physicist

TH-D-TRACK 2-03

M. Jacobsen, UT MD Anderson Cancer Center:

CT Protocol Case Studies: Safety and Image Quality

TH-D-TRACK 2-04

Audience Q & A

CT protocol management is an essential element of a quality assurance (QA) program to ensure patient safety and acceptable image quality. Furthermore, the regulations for many states and accrediting bodies, such as the American College of Radiology (ACR) and the Joint Commission, require CT protocol review to be performed annually. One of the goals of CT protocol management and review is to ensure all the protocols are standardized across a fleet of CT systems at a facility. However, this task can be daunting when faced with CT systems of various models from different manufacturers with potentially different technologies. These differences can lead to inconsistent image quality and disparate radiation dose from opposing acquisition and reconstruction techniques. Additional nonuniformities can result in tissue and vascular enhancement due differences in scan delay timing for different phases, injection site, injection rates, contrast injector pressure limits, needle size, and iodine concentration during the injection of iodinated contrast agents. Finally, there are encounters of nonideal patients due to implanted devices, noncompliance leading to patient motion, and obesity where artifacts are induced and image quality compromised.

The overall objective of the presentation is to provide a better understanding and process for developing a robust CT protocol management program. There will be particular emphasis on CT acquisition and reconstruction standardization, optimization of CT injected contrast optimization, and strategies for CT imaging of the nonideal patient.

Learning Objectives:

1. Understand the need for routine CT protocol management and methods for protocol standardization
2. Understand the basics of intravenous contrast dynamics and corresponding tissue enhancement and the relationship to the CT system for optimization of the contrast protocol
3. Understand how to adjust protocols to account for common artifacts related to the patient condition such as metallic implants, motion, and obesity

Imaging Scientific Session

TRACK 3

AI in Imaging

TH-D-TRACK 3-01

An Effective Deep Learning Framework for Lung Tumor Segmentation in 4D-CT

S Momin*, Y Lei, Z Tian, T Wang, J Roper, A Kesarwala, K Higgins, J Bradley, T Liu, X Yang, Department of Radiation Oncology and Winship Cancer Institute, Emory University, Atlanta, GA

Purpose: To present a deep-learning-based framework for tumor auto-segmentation in 4D lung CT images. **Methods:** We proposed a motion convolutional neural network (CNN) to perform lung tumor segmentation in 4D CT images. The unique network design first extracts tumor motion information from with consecutive 4DCT phases for input into an integrated backbone network architecture. The extracted motion information is advanced into the subsequent global and local motion head network architectures, respectively, to estimate the corresponding deformation vector fields. A self-attention strategy was incorporated in the final mask head network to remove any noisy features that might degrade segmentation performance. The mask head architecture was supervised by segmentation loss, which considers both, dice loss and binary cross entropy loss. A 4D-CT dataset with 200 images from 20 lung cancer patients was used to evaluate our method using cross-validation and hold-out experiments. The accuracy of the auto-segmented contours were benchmarked against expert contours drawn by physicians using the center-of-mass-distance (CMD), Dice-similarity-coefficient (DSC), 95th percentile Hausdorff distance (HD95) and mean surface distance (MSD). The performance of our approach was statistically compared against the U-Net architecture via paired t-test. **Results:** For all patients, our method framework yielded good overall agreement with contours drawn by the physicians. Our method yielded a CMD, DSC, HD95, and MSD of $0.71 \pm 0.36\text{mm}$, 0.91 ± 0.07 , $1.82 \pm 0.47\text{mm}$, and $0.45 \pm 0.16\text{mm}$, respectively. Our method significantly outperformed the U-Net architecture in terms of segmentation accuracy for both cross-validation ($p < 0.05$) and hold-out ($p < 0.05$) experiments. **Conclusion:** We developed a novel motion CNN – based approach for fully automated 4D tumor delineation for lung cancer patients and demonstrated its feasibility and reliability. The promising results of this work provide impetus for its potential applications in both current clinical practice and for real-time, image-driven automatic contouring to advance adaptive lung radiotherapy.

TH-D-TRACK 3-02

Automated Tumor Localization and Segmentation Through Hybrid Neural Network in Head & Neck Cancer

A Qasem^{1*}, Z Zhou², (1) , Warrensburg, MO, (2) University of Central Missouri, Warrensburg, MO

Purpose: Accurate localizing and segmenting tumors is important for diagnosis and treatment in head & neck (H&N) cancer. We aim to develop a hybrid neural network (HNN) model, fully automatic tumor localization and segmentation through a combination of Faster-RCNN / U-Net in CT images. **Methods:** In this preliminary study, a total of 48 patients with PET and CT images in H&N obtained from The Cancer Imaging Archive are used. It consists of complete CT/PET and Gross Tumor Volume (GTV) contours, and 825 slices were selected for training, 176 for validation and 79 for testing. HNN consists of two phases in the training stage, they are Faster-RCNN for tumor localization and U-Net for tumor segmentation. Each phase is trained independently. Since PET can detect the tumor location very well and Faster-RCNN is an outstanding model for learning object localization PET is fed into Faster-RCNN to learn the tumor localization. Furthermore, since U-Net has been proved an excellent medical image segmentation model, it is used for tumor segmentation in CT images. The testing stage consists of two steps: (1) tumor localization and segmentation: the PET image is fed into the trained Faster-RCNN which produces a class label and bounding box coordinates, these coordinates are used to automatically crop the testing CT image to a fixed-sized window, which is then fed into the U-Net for segmentation; (2) OTSU threshold method is applied for post-processing. The segmentation results were evaluated using accuracy, sensitivity, and specificity. **Results:** The proposed model achieved the average values of 0.959, 0.930, and 0.962 for accuracy, sensitivity, and specificity respectively. **Conclusion:** In this preliminary study, a hybrid neural network (HNN) for tumor localization and segmentation was proposed in H&N cancer. The experimental results demonstrated that HNN can localize tumors very well and obtain promising segmentation performance as well.

TH-D-TRACK 3-03

Deep Siamese Network for False Positive Reduction in Brain Metastases Segmentation

Z Yang*, M Chen, R Timmerman, T Dan, Z Wardak, W Lu, X Gu, UT Southwestern Medical Center, Dallas, TX

Purpose: Stereotactic radiosurgery requires precise target delineation, which can be highly time-consuming especially in patients with multiple (>4) brain metastases (BMs). We have developed a CNN-based BMs segmentation platform (BM-Seg) for accurate and efficient BMs delineation, but it suffers from false-positives caused by varied levels of contrast in planning MRI acquisition. Built on BM-Seg, we developed false-positive reduction Siamese network (FPrSN) to improve BM-Seg performance. **Methods:** The proposed FPrSN consists of a parallel feature extractor and a following classifier. The feature extractor utilized a pair of truncated pre-trained ResNet50 with shared weights. In the training stage, paired BM candidate images are used as the input into the feature extractor to 1) increase the training set size by a factor of N/2 and 2) identify the inter-class difference. Then the classifier will randomly take the output from one branch of the feature extractors to classify the corresponding input as a false-positive segmentation or true BM. In the testing stage, only one branch of the feature extractor will be utilized with the classifier to classify a single candidate input as false-positive segmentation or true-positive segmentation. The performance of FPrSN and FPrSN-integrated BM-Seg are evaluated. **Results:** The performance of the FPrSN reaches accuracy of 0.94, area under the curve of 0.92, sensitivity of 0.88 and specificity of 0.97 in a testing dataset. The BM-Seg has initial sensitivity of 0.96 and specificity of 0.59 in BMs detection/segmentation, and get improved to 0.86 and 0.89 after integrating FPrSN, respectively. Due to the flexibility of FPrSN, sensitivity/specificity balance of the FPrSN-integrated BM-Seg can be further adjusted to satisfy clinic needs. **Conclusion:** This developed FPrSN can effectively reduce the false-positive rate of the BMs segmentation candidates. Integrating FPrSN in the BM-Seg can improve the performance, and this integrated platform could be a beneficial tool for clinical implementation.

This work is supported by NIH R01 CA235723 and the seed grant from the Department of Radiation Oncology of the UT Southwestern Medical Center.

TH-D-TRACK 3-04

Development of Artificial Intelligence (AI) Based Platform for Locally Advanced Rectal Cancer Prognosis

Y Zhang¹*, L Shi², X Sun², S Jabbar¹, N Yue¹, K Nie¹, (1) Rutgers Cancer Institute of New Jersey, New Brunswick, NJ, (2) Department of Radiation Oncology, Sir Run Run Shaw Hospital, Zhejiang Univ., Hangzhou, China

Purpose: To develop a deep learning based platform to explicitly quantify multi-regional phenotypes of local advanced rectal cancer (LARC) and assess the association with neoadjuvant chemoradiation (CRT) outcome. **Methods:** A total of 203 patients receiving neoadjuvant CRT followed by total mesorectal excision (TME) were enrolled. All underwent anatomical T1/T2, diffusion-weighted MRI (DWI) and dynamic contrast-enhanced (DCE) MRI before CRT. 133 cases were used for training with four-fold validation, while remaining 70 cases treated at a later time was used as a separate validation set. Pixel-based deep learning algorithm with convolutional neural network (CNN) was developed to automatically localize the rectum and further identify tumor based on all-frame DCE sequences. A radiomics sequencer was then constructed using deep-learning based approach to understand tumor phenotypes across multi-modality MRI images. The radiomics sequencer was further utilized to associate with treatment outcome as pathologic complete response (pCR), good response (GR), and T-downstaging. Results were evaluated using area under the receiver operating characteristics curves (AUC). **Results:** The segmentation result was evaluated by comparing with physicians' contours using Dice Similarity Coefficient (DSC) with a mean of DSC is 0.84 among all cases. A deep-learning based radiomics sequencer was built in predicting pCR, GR and T-downstaging, with AUCs of 0.82, 0.79, and 0.77 respectively. It is also found that DCE sequences especially the second phase of post-contrast T1 contributed the most predictive features followed by DWI sequence. **Conclusion:** Based on the radiomics features from multiparametric MR imaging, machine learning is a feasible way to localize and predict the pathological outcome. Especially, support vector machines can provide better performance than other simple models.

TH-D-TRACK 3-05

Unsupervised COVID-19 Pneumonia Lesion Segmentation in CT Scans Using Cycle Consistent Generative Adversarial Network

Y Liu*, C Fang*, J Wen, Y Yang, University of Science and Technology of China, Hefei, Anhui, P. R. China

* The authors contribute equally

Purpose: COVID-19 has become a global pandemic and is still posing severe health risk for publics. Accurate and efficient segmentation of pneumonia lesions in CT scans is vital for the disease diagnosis and treatment decision-making. To automatically segment the pneumonia lesion, we proposed a novel unsupervised approach using cycle-consistent generative adversarial network (cycle-GAN) which could liberate people from heavy burden of manual labeling and accelerate the diagnosis process. **Methods:** To restrain the CT volume of interest used for lesion segmentation, a previously proposed method was used to delineate the whole lung first. The cycle GAN was developed to generate synthetic healthy CT lung volume from infected patients. Then, the pneumonia lesions were extracted by subtracting the synthesis healthy CT from the infected CT one. Noise and isolated points were eliminated using maximum connected component, threshold segmentation and gaussian smoothing. **Results:** Two public datasets (Coronacases and Radiopedia) were used to evaluate our approach. The dice reached 0.709 on average in the Coronacases, and 0.653 in the Radiopedia. Both are comparable with existing supervised approaches and outperforms previous unsupervised methods. Meanwhile, our approach reaches a specificity of 0.682 and sensitivity of 0.811 in Coronacases, and 0.655 and 0.706 respectively, in Radiopedia. Comparing with existing label-free method, the proposed method outperforms in all index except specificity for Coronacases which varies due to lesion shape, size and position. **Conclusion:** The proposed unsupervised method could automatically delineate the pneumonia lesion in a highly efficient and accurate way. It can also be used as a baseline for further manual modification and a quality assurance tool in clinical diagnosis.

TH-D-TRACK 3-06

BEST IN PHYSICS (IMAGING): Validation of a Deep-Learning Model Observer in a Realistic Lung-Nodule Detection Task with Convolutional Neural Network-Based CT Denoising

H Gong*, N Huber, C Koo, T Johnson, A Inuoe, J Marsh, J Thorne, S Leng, J Fletcher, C McCollough, L Yu, Mayo Clinic, Rochester, MN

Purpose: Deep convolutional neural network (CNN)-based denoising techniques have been used in clinical CT to improve image quality and reduce radiation dose. Objective assessment of its performance is desired but has been challenging due to its highly non-linear characteristics. We previously developed a deep-learning model-observer (DLMO) that can be directly applied to patient CT images. The purpose of this work is to validate the DLMO in a lung-nodule detection task in real patient images denoised by a CNN-based technique. **Methods:** Fifty nodule-free patient chest CT exams were retrospectively collected. Nodule-present cases at lower radiation dose levels were synthesized using clinically-validated projection-domain lesion- / noise-insertion tools: 10 experimental conditions involving two nodule types (ground-glass-nodule, partial-solid-nodule), three nodule sizes (3.4–7.4 mm) and four radiation dose levels (10%, 25%, 50%, and 100% of routine-dose, quality-reference-mAs 50). A ResNet-based CNN-denoising model was trained using only nodule-free training cases at 25% routine-dose, and deployed to independent testing cases across four dose levels. DLMO used sliding window to analyze local features and generate pixel-wise test-statistics, and used an internal-noise-component to calibrate the performance to human level. DLMO was established with two **Methods:** DLMO-1 (baseline method) – DLMOs were separately trained and deployed at each dose level; DLMO-2 (more computationally-efficient method) – one DLMO was trained at one dose level and deployed to other levels. Two fellowship-trained thoracic radiologists and one general radiologist interpreted the same DLMO test cases. Area-under-curve of localization-ROC was used as figure-of-merit. **Results:** In preliminary study, DLMO-1 and DLMO-2 performance yielded strong correlation with the averaged human performance (Pearson's

correlation coefficient: DLMO-1 – 0.995, P-value 0.0049; DLMO-2 – 0.996, P-value 0.004). Mean performance difference was small (DLMO-1 – 1.02%; DLMO-2 – 1.47%). **Conclusion:** DLMO demonstrated the potential for efficient and objective image quality assessment in realistic lung-nodule detection tasks with CNN-based CT denoising.

Research reported in this work was supported by the National Institutes of Health under award number R01 EB017095. Research support is provided to Mayo Clinic from Siemens Healthcare GmbH, unrelated to this work.

TH-D-TRACK 3-07

Towards Understandable AI in Lung Nodule Detection: Using the Genetic Algorithm for Interpretable, Human-Understandable Optimization of Nodule Candidate Generation in Lung CT Imaging

M Wahhi-Anwar*, N Emaminejad, G Kim, M Brown, M McNitt-Gray, David Geffen School of Medicine at UCLA, Los Angeles, CA

Purpose: Despite the recent growth in artificial intelligence (AI), the “black-box” nature brings a need for “explainable” AI (XAI) to better understand system performance and limitations. We apply this concept to an AI-based system for lung nodule detection in CT, which can be decomposed into steps of (1) candidate generation (CG), and (2) false positive reduction. The specific purpose of this work is to demonstrate the utility of a Genetic Algorithm (GA) in improving performance of CG in an explainable, understandable fashion. **Methods:** 1010 CT lung image cases from the Lung Image Data Consortium were used. Nodules ≥ 4 mm were identified based on markings from at least 3 radiologists. A previously validated SN CAD system was modified to enable tunable parameters for nodule CG, encoded as chromosomes in the GA. A training subset was used for progressive training, and a validation subset to select the optimal chromosome. Performance on the test set was measured for SNs: (a) hand-tuned, no GA, (b) GA, basic operations (GA-basic), or (c) GA, complex operations (GA-complex). GA-complex incorporated additional operations, to try to promote GA concepts of diversity (maintaining sufficient genetic heterogeneity in each generation) and elitism (propagation of the higher performing chromosomes). Performance was measured via sensitivity and false positive (FP) rate. Parameter importance was measured through random forest variable importance. **Results:** Trained from scratch, GA-complex-optimized CG matched hand-tuned model performance with 90.8% sensitivity with false positive rate greatly reduced to 498 FP/case. Tuning parameters “minimum x-y circularity”, “HU-range”, and “maximum x-y area” demonstrated the highest impact. **Conclusion:** GA-optimization permitted construction of a CG model from scratch that met hand-tuned performance. Further training could enable better performance than human-optimized model parameters. The GA allows examination and deconstruction of chromosomes in the optimization process, to easily follow and manage the evolution of parameters over time.

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Therapy Scientific Session

TRACK 5

Novel treatment delivery and verification techniques II

TH-D-TRACK 5-01

A Direct Machine-Specific Parameters Incorporated Spot-Scanning Proton Arc (SPArc) Algorithm

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Purpose: To address the challenges of generating a deliverable and efficient spot-scanning proton arc (SPArc) plan for a proton therapy system. We developed a novel SPArc optimization algorithm (SPArcDMSP) by directly incorporating machine-specific parameters such as mechanical constraints and delivery sequence. **Methods:** A SPArc delivery sequence model (DSMarc) was built based on the machine-specific parameters of the prototype arc delivery system, IBA ProteusONE®, including mechanical constraint (maximum gantry speed, acceleration, and deceleration speed) and delivery sequence (energy and spot delivery sequence and time). SPArcDMSP resamples and adjusts each control point's delivery speed based on the DSMarc calculation through the iterative approach (Figure 1). In SPArcDMSP, clinical users could set the expected arc delivery time and gantry max acceleration as a mechanical constraint during the plan optimization. Four representative cases (brain, head neck, liver, and lung cancer patients) were selected to test SPArcDMSP. Two kinds of SPArc plans were generated using the same planning objective functions: (1) SPArcDMSP plan meeting the maximum allowable gantry acceleration speed (0.6 deg/s²); (2) SPArcDMSP-user-speed plan with a user pre-defined delivery time and constraint the acceleration speed < 0.1 deg/s². Additionally, the arc delivery sequence was simulated based on the DSMarc and was compared. **Results:** With a similar objective value, number of energy layers and spots, both SPArcDMSP and SPArcDMSP-user-speed plans could be delivered continuously within the ± 1 degree tolerance window. The SPArcDMSP-user-speed plan could minimize the gantry momentum change based on clinical users' input compared to the SPArcDMSP, which could help relieve mechanical challenges to accelerate or decelerate the hundreds of tons of gantry (Figure 2&3). **Conclusion:** For the first time, clinical users not only could generate a SPArc plan meeting the mechanical constraint of their proton system but also directly controlled the arc treatment speed and momentum changes of the gantry during the plan optimization. This work paved the roadmap for the clinical implementation of proton arc therapy in the treatment planning system.

The research project in part is supported by Ion Beam Application S.A. (Belgium).

Xuanfeng Ding, Xiaoqiang Li and Di Yan has a patent related to particle arc therapy and the patent has been licensed to IBA.

Xuanfeng Ding reported personal fee from IBA speaker Bureau

TH-D-TRACK 5-02

Comparison of Robust Optimization and PTV Margins for Mixed-Beam Radiotherapy

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Purpose: Mixed-beam radiotherapy (MBRT) using intensity modulated electron and photon beams collimated by the photon MLC has been shown to give superior plan quality in the treatment of tumours with a superficial component. Conventional PTV margins may not be effective at compensating for setup errors in MBRT. We previously developed and validated a framework for robust optimization of deliverable MBRT plans. The aim of the current work is to apply this framework to clinical patient cases and to compare with PTV-based plans. **Methods:** The framework uses a hybrid direct aperture optimization approach which combines simulated annealing and column generation algorithms. To make plans robust against setup errors, the optimization minimizes the expectation value of the objective function evaluated across all specified setup error scenarios under simultaneous consideration of both beam modalities. Robust-optimized and PTV-based MBRT plans were designed on three clinical cases consisting of two head and neck cases and one brain case. The PTV-based plans used a 5 mm isotropic PTV margin while the robust plans considered ± 5 mm systematic translational setup errors along the principal axes (SI, LR, AP). For each plan, dose distributions were calculated for 5 mm isocenter shifts along the principal axes and were compared in terms of target coverage and OAR sparing. **Results:** The robust optimized plans respected the institutional planning guidelines for all considered setup error scenarios while the PTV-based plans had clinically unacceptable hotspots in the CTV and exceeded

OAR tolerances for many of the scenarios. Robust optimized plans achieved an 18% increase in the CTV dose homogeneity for the head and neck cases and, on average, a 6% reduction in Dmean and 15% reduction in D2% for the OARs in all patient cases. **Conclusion:** Robust optimization is a promising alternative to conventional PTV margins to account for setup uncertainties in MBRT.

This work was supported by grant 200021_185366 of the Swiss National Science Foundation and Varian Medical Systems.

TH-D-TRACK 5-03

Development of An Irradiation System for MRI-Guided Proton Therapy

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Purpose: To investigate a proposed design for an MRI-guided proton therapy system that simultaneously achieves both the large field size (FS) for protons and large field of view (FOV) for MRI. The basic characteristics of the proton beam delivered in the proposed system were analyzed. **Methods:** An open MRI design with iron magnetic poles was adopted in order to achieve both the large FS for protons and FOV for MRI and reduce the MR fringe fields at the beam scanning magnets and spot position monitor. A hole was made in the center of the yoke to ensure an unobstructed proton beam path. The shape of the magnetic poles was optimized assuming axis symmetry to ensure the magnetic field was uniform. Then the magnetic field strength map was calculated considering the yoke with a hole. A helium chamber was installed between the spot position monitor and dose monitors to reduce the beam size. Geant4 simulation (ver. 10.05.p01) was used to model these configurations and analyze the beam size with and without the helium chamber. **Results:** The achieved uniformity of the magnetic field strength was less than 0.8 ppm on the FOV surface. The maximum magnetic field strength in the area upstream from the yoke was 2.8 mT. This indicated that the MR magnetic flux was successfully confined in the yoke and the fringe fields decayed rapidly. The beam sizes of a 70 MeV proton beam were 10.2 and 9.2 mm at the iso center without and with the helium chamber, respectively. **Conclusion:** The proposed design of the MRI-guided proton therapy system realized both the large FS for protons and FOV for MRI, while the magnetic fringe fields were sufficiently small and did not affect the beam monitoring. Beam size was shown to be small enough to be used in clinics.

Authors Yusuke Fujii and Taisuke Takayanagi are employee of Hitachi Ltd., Tokyo, Japan.

TH-D-TRACK 5-04

Evaluation of GRID and Spatially Fractionated Radiation Therapy: Dosimetry and Preclinical Trial

T Johnson*, A Bassil, C Kent, N Williams, G Palmer, Y Mowery, M Oldham, Duke University Medical Center, Durham, NC

Purpose: To develop and implement novel methods of extreme spatially fractionated radiation therapy (including GRID therapy) and subsequent evaluation in pre-clinical mice trials investigating the potential of novel radiation treatments that more effectively promote anti-cancer immunogenic response. **Methods:** Spatially fractionated GRIDs were designed and precision-milled from 3mm thick lead sheets compatible with mounting on a 225 kVp small animal irradiator (X-Rad). Three pencil-beam GRIDs created arrays of 1mm diameter beams, and three "bar" GRIDs created 1x20mm rectangular fields. GRIDs projected 20x20mm fields at isocenter and beamlets were spaced at 1, 1.25, and 1.5mm, respectively. Output factors, peak-to-valley ratios, and dose distributions were determined with Gafchromic film. The bar GRID with 1mm beamlet spacing (50:50 open:closed ratio) was selected for the pre-clinical trial. Soft-tissue sarcoma (p53/MCA) was transplanted into C57BL/6 mice's flanks. Four treatment arms were compared: unirradiated control (n=14), conventional radiation therapy (n=8), GRID therapy (n=12), and hemi-irradiation (n=10) where one-half of the beam was blocked. All irradiated mice received a single fraction of 15 Gy to irradiated regions. **Results:** Very high peak-to-valley ratios were achieved (bar GRIDs: 11.9±0.9, 13.6±0.4, 13.7±0.5; pencil-beam GRIDs: 18.7±0.6, 26.3±1.5, 31.0±3.3). Pencil-beam GRIDs spared twice the number of intra-tumor immune cells as bar GRIDs but left more of the tumor untreated (2-3% vs 14-17% area receiving 95% prescription, respectively). Penumbra was halved when GRIDs were 50% closer to treatment isocenter. The GRID selected for mouse trials was capable of sparing approximately 30% of intra-tumor CD8+ and CD4+ T cells. Preliminary results indicate mean times to tumor quintupling were: 9, 12, 12, and 24 days for unirradiated, GRID, hemi-irradiated, and conventional treatment groups, respectively. **Conclusion:** Peak-to-valley ratios with kV grids were substantially superior to MV grids. In data collected to date, GRID did not control tumors as effectively as conventional treatment at the same dose level.

TH-D-TRACK 5-05

Experimental Study of An Innovative Thermo-Brachytherapy Seed for Simultaneous Delivery of Radiation and Hyperthermia

S Taghizadehghahremanloo*, E Parsai, D Shvydka, A Shan, University of Toledo Medical Center, Toledo, OH

Purpose: To experimentally characterize ferromagnetic materials for use in an innovative Thermo-brachytherapy (TB) seed, serving as a heat and radiation source for concurrent administration of radiation therapy and hyperthermia. The combination of two modalities has been confirmed very effective in the treatment of prostate cancer without an increase in complications. Measuring the quantity of relative permeability as a function of temperature will be utilized through the power-versus-temperature method for calculation of temperature distribution around the seed in a prototype treatment planning system. **Methods:** We measure μ_r of small (<1cm-long, ~1mm diameter) cylindrical TB seed samples with the same geometrical size and shape as the one used for prostate Brachytherapy seed implant. The nonlinear effects, demagnetization, and magnetic saturation effects in a finite-sized cylinder reduce the value of μ_r compared to those measured with a standard toroid sample (effective length of a toroid is infinite). The inductance, parasitic resistance, and relative permeability of the sample were derived from impedance measurements for a solenoid that was a custom-wound for the sample. The results are compared to the theoretical values and simulations from COMSOL Multiphysics for consistency. **Results:** We measured μ_r for ferrite and Ni-Cu samples. The Curie temperature for the ferrite and Ni-Cu cores are ~55 °C and ~45 °C correspondingly. Our values are closest to theoretical results for a long solenoid corrected for the circular wire shape, mutual inductance between the coil loops, wire self-inductance, the gradient resistance inside the wire, and diameter-to-length ratio. **Conclusion:** The ferrite core has a much steeper permeability fall-off with temperature. Rapid Curie transition, decoupling of the magnetic material properties from electrical properties of the coating, and higher value for permeability make ferrite samples much better candidates for the TB seed. These results can help us to identify a ferrite composition with ideal power generation.

TH-D-TRACK 5-06

Effect of Incoming Particle Energy and Ionization Cluster Size On the G-Value of Hydrated Electrons

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Purpose: To accurately calculate absorbed dose to water in hydrated electron ($e^-(aq)$) dosimetry, the radiation chemical yield (i.e. G-value) of hydrated electrons, $G(e^-(aq))$, must be precisely determined. The relationship between the G-value, linear energy transfer (LET), and ionization cluster size were also investigated for better understanding of factors affecting the G-value using Geant4-DNA. **Methods:** Different scenarios were investigated using Geant4-DNA track structure Monte Carlo toolkit. In scenario A, a 10 x 4 x 2 cm³ phantom of liquid water was placed at SSD 70 cm and irradiated with monoenergetic electron beams of different incoming energies from 1 keV to 1 MeV. $G(e^-(aq))$ and the G-value of all generated species with respect to incoming energy were scored. In scenario B, the relation

between $G(e^{-}(aq))$ and cluster size was investigated by irradiating a $0.1 \times 0.1 \times 0.1 \text{ cm}^3$ water phantom with electron point sources of energies from 0.01 keV to 1 MeV, starting at the center of the volume. **Results:** The time-evolution of the G-value showed good agreement with previous simulations and experiments. It was observed that as the incoming electron energy decreases (i.e. higher LET), $G(e^{-}(aq))$ generally decreases, and its rate of change over time becomes steeper. For 1 MeV electron beams, the initial $G(e^{-}(aq))$ at 1 picosecond was calculated to be 4.1 ± 0.1 . Moreover, we observed a negative correlation between $G(e^{-}(aq))$ and cluster size as a function of incoming electron energy. The results indicated that cluster size had a pattern similar to that of LET, which has been shown to remain relatively constant as the incoming electron energy increases. **Conclusion:** As the incoming electron energy increases to tens of MeV, $G(e^{-}(aq))$ will remain relatively constant, since it decreases with increasing cluster size and LET. This is beneficial, as Geant4-DNA can only simulate electron interactions up to 1 MeV.

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TH-D-TRACK 5-07

Ionoacoustic Treatment Monitoring for Radiosensitizer-Enhanced Carbon Ion Therapy: A Feasibility Study

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Purpose: To evaluate the impact of radiosensitizers on ion-induced acoustic emissions (ionoacoustics) and assess the accuracy of ionoacoustic range verification for carbon ion therapy at clinical synchrotrons. **Methods:** Acoustic emission resulting from the absorption of optical photons produced from scintillation and absorbed in optical dyes or photosensitizers (similar to photodynamic therapy) were investigated. Alternatively, the local enhancement of the energy deposition in presence of gold nanoparticles (radiosensitizers) was also evaluated. FLUKA Monte Carlo simulations were performed to model a monoenergetic carbon ion beams (215 MeV/u) in water in presence of scintillation materials or gold nanoparticles. The resulting acoustic emission was simulated using k-Wave, considering the temporal microstructures of carbon ions delivered by synchrotrons. The interference pattern (beats) between the direct ionoacoustic signal from the Bragg peak (BP) and the pressure produced due to radiosensitizers at the location where the ion beam enters the targeted volume was analyzed in the frequency-domain. The BP position relative to the tumor entrance was estimated from the beat frequency accounting for the speed-of-sound in the target. **Results:** Assuming sufficiently high light yield in the typical range of scintillation decay time, the absorption of optical photons as used in photodynamic therapy can enhance the ionoacoustic emission, e.g., light emission at wavelength of 300 nm with a light yield of 10^3 photons/MeV increases the signal amplitude by 50%. Alternatively, low concentration (<1 wt.%) of high-density materials like gold nanoparticles significantly enhance the ionoacoustic emission (>100 % increase). The additional pressure generated at the entrance of the target in presence of radiosensitizers allows for accurate BP localization (<1% error relative to the 9.8 cm range) for ion beam at synchrotrons. **Conclusion:** This simulation study shows the potential benefit of radiosensitizers on ionoacoustics-based range verification to enable accurate BP localization for carbon ion beams delivered by synchrotron accelerators.

Therapy Scientific Session

TRACK 6

Deep learning in Treatment Planning

TH-D-TRACK 6-01

3D DenseNet for Improving Dose Prediction of Volumetric Modulated Arc Therapy in Prostate Cancer

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Purpose: Accurate prediction of radiotherapy dose distributions could guide the treatment planning process to rapidly and consistently achieve high-quality plans. In this study, we proposed a novel 3D DenseNet for predicting dose distributions of volumetric modulated arc therapy (VMAT) for prostate cancer patients. **Methods:** 75 prostate cancer patients were included. Each patient was planned with a prescription dose of 40 Gy in 5 fractions using Eclipse. We built a 3D DenseNet to generate patient-specific dose distributions from the concatenation of the planning CT image, the contours of the external body, planning target volume (PTV), and organs-at-risk (OARs). The proposed DenseNet contains five dense blocks that could alleviate the vanishing gradient problem, encourage feature reuse, and reduce the number of model parameters. The patient cohort was randomly split into 53 patients for training, 11 patients for validation, and 11 patients for independent testing. A 3D U-Net was also trained for dose prediction and model comparison. The model performance was evaluated by comparing the predicted dose with the clinical dose using Dice similarity coefficients (DSCs) of isodose volumes and relevant dosimetric metrics. **Results:** The average DSCs of isodose volumes, between 0% and 100% (step size: 1%) of the prescription dose, achieved by the 3D U-Net and 3D DenseNet are 0.939 and 0.948, respectively. The average absolute differences in Dmean of the PTV and OARs are under 1.43 Gy (UNet) and 0.89 Gy (DenseNet). The average absolute difference in PTV V40Gy is under 2% for both models. **Conclusion:** The proposed 3D DenseNet could accurately predict the dose distribution of VMAT for a prostate cancer patient in about 8 seconds. It achieved more accurate dose prediction than the state-of-the-art 3D U-Net. The 3D DenseNet is a promising tool for fast dose prediction and guiding plan optimization in an online adaptive workflow.

TH-D-TRACK 6-02

A Deep Learning-Based Framework for Dose Prediction of Pancreatic Stereotactic Body Radiation Therapy

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Purpose: Efficacy of deep learning (DL) methods in dose prediction of pancreatic cancer is still to be fully explored. This study aims to implement and demonstrate the feasibility of a dual pyramid networks (DPNs) integrated DL model for predicting dose distributions of pancreatic stereotactic body radiation therapy (SBRT). **Methods:** The proposed framework is composed of four parts: CT-only feature-pyramid-network (FPN), contour-only FPN, late-fusion-network (LFN) and an adversarial network. During each phase of the network, combination of mean absolute error, gradient difference error, histogram matching, and adversarial loss was used for supervision. The performance of proposed model was demonstrated for pancreatic cancer SBRT plans with doses prescribed between 33 and 50 Gy across as many as three planning target volumes (PTV) in five fractions. Five-fold cross validation was performed on 30 patients, and 20 patients were used as hold-out tests of trained model. Predicted plans were compared with clinical plans through clinically relevant dose-volume parameters and paired t-test. For the same cross validation and hold-out datasets, our results were compared with the ones predicted by traditional 3D U-Net network architecture. **Results:** The proposed framework was able to predict 87% and 91% of clinically relevant dose parameters for cross-validation sets and holdout sets, respectively, without any significant differences ($P > 0.05$). The proposed model was also able to predict the intentional hotspots as feature characteristics of SBRT plans. In comparison to 3D U-Net, proposed network architecture increased prediction accuracy of 21/26 and 14/22 dose volume parameters for cross-validation and hold-out test, respectively. **Conclusion:** We developed a new DPN-based method to predict dose distributions of pancreatic SBRT and demonstrated its feasibility of dose distribution prediction for both single and multiple PTVs. The predicted prior information could help save pancreatic SBRT planning times for multi-level experienced dosimetrists.

TH-D-TRACK 6-03

Attention-Gated U-Net Implementation in Total Marrow Irradiation Plan Dose Prediction

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Purpose: To implement an attention-gated deep convolutional neural network (cnn) to predict the optimal 3D dose distribution for Total marrow irradiation (TMI) plans. TMI using intensity modulated radiation therapy has many potential benefits compared to total body irradiation for patients receiving hematopoietic cell transplant. The iterative optimization process in TMI is challenging due to dose constraints on organ-at-risks (OARs) and multiple target volumes (PTVs). **Methods:** Twenty-Seven Tomotherapy TMI plans with 20Gy/10fx prescription doses were collected. We focused on the most difficult to be optimized: the thoracic region where lung limits must be met. In this area, there are 6 PTV structures and 3 OARs. We incorporated an attention mechanism in an U-Net network to develop an Attention-Gated U-Net model for predicting 3D dose distributions. The model was trained on twenty randomly selected plans and was validated on the other 7 datasets. **Results:** The Attentional-gated U-Net model predicted the PTV and OAR doses accurately. The training time on was around 150 minutes (2500 epochs) on NVIDIA Quadro P6000 GPU. The average absolute difference ($|D_{\text{predict}} - D_{\text{true}}| / D_{\text{true}}$) of mean dose, max dose and D95 for (1) PTV_Ribs were $2.79\% \pm 1.86\%$, $3.24\% \pm 2.75\%$, $6.95\% \pm 5.18\%$. (2) PTV_Bone was $0.86\% \pm 0.71\%$, $5.09\% \pm 2.79\%$, $7.79\% \pm 5.05\%$. The average absolute difference of mean and max dose was $3.65\% \pm 2.64\%$, and $5.97\% \pm 4.01\%$ for lungs, $6.67\% \pm 5.1\%$, and $12.59\% \pm 11.45\%$ for heart. **Conclusion:** The Attention-Gated U-Net network models predicted the optimal dose distribution for complex multiple targets TMI plans. Our future work includes training and testing the model on a larger patient cohort with different prescription levels as input into plan optimization. We will also implement the model on other body regions and build a whole-body model for TMI planning.

TH-D-TRACK 6-04

Transfer Learning for Fluence Map Prediction in Adrenal Stereotactic Body Radiation Therapy

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Purpose: To predict fluence maps and automatically generate clinically acceptable plans for adrenal stereotactic body radiation therapy (SBRT) and to establish a deep learning framework that requires less demanding data volume using transfer learning (TL). **Methods:** Due to the small number of adrenal SBRT cases, we developed a TL framework that leverages the treatment planning knowledge from a pancreas SBRT model. This framework consists of two convolutional neural networks (CNN), which sequentially predict individual beam doses (BD-CNN) and fluence maps (FM-CNN) for 9-beam intensity modulated radiation therapy (IMRT) plans. One hundred pancreas and 45 adrenal plans were retrospectively included. The two types of plans had different dose prescriptions and beam settings. A base model (including two CNNs) was trained with all 100 pancreas cases. The adrenal plan cohort were first split into training/validation/test sets with the ratio of 20/10/15 without TL. The base BD-CNN was re-trained using 5, 10, 15, and 20 adrenal cases with TL to produce candidate adrenal BD-CNN TL models. The base FM-CNN was directly used for adrenal cases without TL. The resulting AI plans were evaluated by PTV and OAR dosimetric endpoints to produce a percentage score with the clinical plans as the benchmark. **Results:** Transfer learning significantly reduced the necessary training data size and training time. The TL models trained with 5/10/15/20 cases achieved the validation plan quality scores of 81.5/89.9/88.2/86.7, respectively. In contrast, a model using only adrenal training cases (maximum of 20) without TL only scored 78.2. For the test set, the 5/10/15/20-case TL models achieved the scores of 75.3/79.4/82.0/82.3. **Conclusion:** It is feasible to use transfer learning to train the proposed fluence prediction framework, therefore reducing the required data size, training time and effort. Further, this strategy is capable of accommodating different beam geometries and dose prescriptions.

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TH-D-TRACK 6-05

How Fluence-Prediction Error Impact Final Plan Quality: Insights Into a Deep-Learning-Based (DL-Based) Head-And-Neck (H&N) IMRT Planning AI Agent

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Purpose: To collect insights into dosimetric responses from fluence-map-prediction errors of a DL-based AI agent in H&N IMRT planning. **Methods:** A reported AI agent could automatically predict fluence maps and generate plans that are comparable to clinical plans for prostate IMRT. However, when trained for a more complex site, H&N, the AI agent tended to generate plans with higher quality variations. Most DL models are “black-boxes”, whose inner mechanism is not intuitive nor interpretable, and the relationship between fluence-map-prediction error and AI plan quality lacks thorough understanding and investigations. We designed five analytical protocols to collect insights on how the DL-based AI agent model performance affects the final plan quality. The DL model was trained with H&N 216 cases and tested with 15 additional cases. The fluence-map-prediction error (prediction minus ground truth) was analyzed for its dosimetric effects using five error-decomposition methods, including three spatial decompositions: ground-truth fluence-intensity thresholds, predicted fluence-intensity thresholds, and ground-truth fluence-gradient thresholds; and two frequency domain decompositions: Fourier-space frequency bands and Fourier-space truncated low-frequency disks. The decomposed error components were analyzed for their impact on resulting plans’ dosimetric metrics. The PTV dosimetric metrics included heterogeneity index, conformity index, whole-body maximum dose D_{2cc} , PTV $V_{105\%}$, $V_{110\%}$, and $V_{115\%}$. OAR dosimetric metrics included $D_{0.1cc}$ of brainstem and cord+5mm, D_{mean} of parotid left/right, oral cavity, larynx, and pharynx, and D_{2cc} of mandible. **Results:** Majority of PTV metrics were significantly correlated with various error components. Among different decompositions, the Fourier-space low-frequency disks could maximally extract error components that reveal plan quality impacts: error within ~20% area of Fourier space captures most of the dosimetric differences between prediction and ground truth. **Conclusion:** The fluence-map-prediction error in Fourier-space low-frequency region is critical to AI plans’ quality, especially for PTV-related metrics. This insight will help improving network architecture and loss function design.

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TH-D-TRACK 6-06

Feasibility Study of Cross-Modality IMRT Auto-Planning Guided by a Deep Learning Model

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Purpose: To evaluate the ability of our machine learning model to predict 3D dose distributions that can be converted to deliverable plans using the same and alternate treatment techniques than the training set plans. **Methods:** We developed a triple-stage cascaded U-Net model for 3D dose prediction for head and neck treatments using the data provided for the 2020 AAPM OpenKBP planning challenge. These plans were created using nine field step-and-shoot IMRT, prescribed to deliver 70 Gy in 35 fractions to a high risk PTV, and potentially 63 Gy to an intermediate risk PTV and/or 56 Gy to a standard risk PTV. We used our model to predict dose for 11 patients and imported the data into Raystation. Using the fall-back planning function, we created two plans that attempted to mimic the predicted dose, one using the same 9 field IMRT beam arrangement as the training plans and one using VMAT with two full arcs, which is the standard at our institution. The plans were normalized to give 95% coverage at 70 Gy to the high risk PTV. These plans were compared against the predicted dose based on several main clinical objectives: coverage of the standard risk PTV, max dose to the cord, mean dose and D50 to the left and right parotids, and max dose in the plan. **Results:** There were no

statistically significant differences in the majority of the clinical goals ($p > 0.05$), with the exception of the max spinal cord (for IMRT and VMAT) and the max overall (IMRT) dose. Adding an optimization objective for these goals in addition to the mimic dose objective for select plans removed this difference. **Conclusion:** Our model generates 3D dose predictions that are clinically achievable both using the technique used in the training plans and an alternate technique.

TH-D-TRACK 6-07

A Deep-Learning-Enabled Remote Teaching Platform for Pancreas SBRT Treatment Planning: A Pilot Study in Pandemic Era

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Purpose: Covid-19 pandemic calls for more effective remote teaching method necessary for social distancing, while remaining equal quality of teaching. This study innovatively developed a tutoring program to help physics residents to learn pancreas stereotactic body radiation therapy (SBRT) treatment planning. **Methods:** The pilot tutoring program is a compact front-end visualization platform composed of three modules: 1) teaching case module, 2) interactive knowledge module (IKM) and 3) scoring module. The teaching case module included five teaching cases. The trainee began with the benchmark case independently to benchmark the baseline skill level, which was revisited after completion of training of the other four cases to evaluate clinical planning readiness. The IKM included a deep neural network based dose-volume prediction model previously developed for pancreas SBRT. The trainee can seek reference and guidance from the IKM during learning. The scoring module summarized the plan quality by weighing key dosimetric endpoints and their relative importance. Grade point average (GPA) was introduced to qualitatively appraise the plan quality into A, B and C (within 3%, 3-10% and >10% difference of clinical plan score, corresponding to 4, 3, 2 point respectively). Five trainees with minimum planning experience completed the teaching course. **Results:** Trainees achieved an average of 65.1% of total points (3.6 GPA) with 84 minutes planning time for the benchmark case pre-teaching, and improved to an average of 75.7% (4.2 GPA) using 48 minutes post-teaching. The clinical plan scored 72.7% of total points. All trainees improved their teaching plans' scores after taking the virtual tutoring program. Post-teaching, all trainees received the GPA of A (clinical quality level) on the benchmark case planning. **Conclusion:** The remote teaching program with knowledge modules provides encouraging learning outcomes in pancreas SBRT planning for less experienced planners. This deep-learning-enabled remote teaching platform could provide valuable addition to the traditional human resource heavy in-person teaching of SBRT planning.

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SAM Therapy Educational Course

TRACK 7

Gamma Knife QA Update with TG-178

TH-D-TRACK 7-00

Gamma Knife QA Update with TG-178

Moderator: Nels Knutson, Washington University School of Medicine, St Louis, MO

TH-D-TRACK 7-01

S. Goetsch, San Diego Medical Physics:

TG-178 Recommendations

TH-D-TRACK 7-02

P. Alvarez, UT MD Anderson Cancer Center - IROC:

Implementing TG -178 into a Robust Modern QA Program

TH-D-TRACK 7-03

N. Knutson, Washington University School of Medicine:

Gamma Knife QA from TG 21 to TG-178

TH-D-TRACK 7-04

Q&A

This talk describes the report prepared by AAPM Task Group 178, beginning in 2008. The completed report will soon be published in Medical Physics. The scope of the investigation is the calibration and quality assurance of the Leksell Gamma Knife, in use worldwide since the original commercial unit was installed in 1987. The report adopts the formalism described by Alfonso et al (2008) for radiation therapy devices non-compliant with the AAPM Task Group 51 dose in water protocol, to extend its use to the Leksell Gamma Knife. Results of Monte Carlo calibration and subsequent verification by the manufacturer in 2010 provided the missing factors for use of 9 commonly used ionization chambers for three different phantoms. The round-robin measurement conducted by TG 178 in 2011 to 2012 are also reviewed. Quality Assurance recommendations for the latest Leksell Gamma Knife Perfexion and Icon will be presented.

This talk aims to present the results of the external audits offered by the IROC Houston QA office Gamma Knife. The program provides a verification of the output of unit and an end-to-end test with an anthropomorphic head phantom. Most of the institutions that use the service are NCI clinical trial's participants. The output verification requires the irradiation of TLD located in an insert that is compatible with the QA phantom used for ion chamber measurements (cassette for the Elekta ABS ball or insert for the solid water phantom). The SRS head phantom is a plastic shell with an insert that includes a centrally located target, TLD located in the target and gafchromic film. The information presented includes the distribution of the results of the dose comparison at the point of measurement as well as demographic information from these irradiations: models of the unit, calibration protocol use for determination of the output and reference medium.

This talk aims to give an example of the evolution of a Gamma Knife QA program, in particular in the context of the recent publication of TG-178 and the development of the Gamma Knife Icon. An extremely well developed for Gamma Knife QA program developed for the Gamma Knife Perfexion model. This system was based on Tg-21 calibration in the Elekta ABS ball. Then in November 2017 we received the Gamma Knife Icon. With the publication of TG-178 (released for public comment in 2018) and IAEA TRS 483 in November 2017, was a great catalyst for the revamping of our QA program. We reviewed every section of our qa program, safety, mechanical, calibration, relative dosimetry, imaging, intrafraction motion management, end to end testing, and secondary dose calculation. Most of the changes there are two common trends, 1) the need to address new technology, and 2) the desire to simplify and improve consistency. This talk discusses some lessons learned during this process.

Learning Objectives:

1. Understand the reasoning behind the creation of TG-178 and the recommendations made within the task group.
2. Understand the importance of external dosimetry audits for Gamma Knife and a review of results seen by IROC
3. See a realworld example of how to adopt some of the changes mentioned in TG-178

Imaging Interactive ePoster Discussion

TRACK 1

Imaging: CT Image Quality, Protocol Optimization, and Dose Reduction

TH-IePD-TRACK 1-01

A Method to Estimate the Half-Value Layer in CT from Two Exposure Measurements

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Purpose: Among other factors, dose to a CTDI phantom depends on the beam quality. If contributions from scatter to exposure to center-hole is excluded one can, in theory, estimate the central beam half-value layer (HVL). The proposed study investigates the validity of this concept. **Methods:** Transmission factors (TFs) through 15.4-cm PMMA were determined using a spectrum generator of beams whose qualities were modified by adding various aluminum filter thicknesses. The resulted HVLs and TFs from the simulations, at four tube-potentials, were fitted together using a second-order polynomial. To find the HVL from CTDI measurements, scatter-to-primary ratio (SPR) at each kV was determined. This was achieved through measurements in phantom (with 32-cm CTDI phantom, center-hole) and in-air on five scanners with HVLs manually-measured with Al filters. SPR was estimated as: $[\text{Center hole exposure} / \text{Primary exposure} - 1]$, where primary was calculated as: $\text{TF} \times \text{in-air exposure}$, where TF was determined using the polynomial relationship. Finally, by estimating the TF from the mean SPR at each kV, HVL was calculated for every scanner and kV combinations and compared to those manually-measured. Also, manually-measured HVLs were compared to values measured automatically by a calibrated multi-sensor. **Results:** SPR per kV did not vary significantly among different scanners/bowtie filters with different HVLs. Collimation showed no effect on SPR. Average SPRs were: 8.1, 7.6, 7.1 and 6.9 for 80, 100, 120 and 140 kV, respectively. Error of calculated vs. manually-measured HVLs was, on average, 4.7% (± 0.34 mm) over all four kV stations. Measured HVLs from the multi-sensor were found to result in an average error of 10.8% (0.75 mm). **Conclusion:** The method of estimating the HVL from in-air and in-phantom measurements was found to be reasonably accurate and practical. It can be used during the periodic CT testing. Furthermore, it showed higher accuracy than using the multi-sensor.

TH-IePD-TRACK 1-02

Half-Value Layer at Different Positions in Wide-Beam CT Scanners

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Purpose: To demonstrate how Half-Value Layer (HVL) and Effective Energy (E_{eff}) are spatially-dependent in the x-y plane and the z axes of wide CT beams. **Methods:** HVL in the x-y plane and z-axis, bowtie profiles in the x-y plane, and dose profiles along the z-axis were measured for all tube voltages (80-135 kVp) and bowtie filters in two 320-slice Canon CT scanners. All measurements were conducted in service engineer mode with the X-ray tube parked at the 12 o'clock position in the gantry. For HVL and bowtie profile measurements, 6-cm³ and 0.6-cm³ ion chambers were used, respectively. The dose profile along the z-axis was measured using OSL dosimeters and corrected for energy dependence by using calculated correction factors. Using HVL measurements, the effective energies for both scanners were estimated using SpekCalc, an x-ray spectrum generator. **Results:** Along the x-y plane, Scanner 2 had higher HVLs (4.27-11.3 mm-Al) and E_{eff} (39-71.2 keV) than Scanner 1 (4.10-8.40 mm-Al, 38.4-63.7 keV). Along the z-axis, the anode side had a higher HVL (5.17-11.2 mm-Al) and lower dose (5.63 mGy, which was the minimum dose at 80 kVp) than the cathode side of the x-ray tube (3.92-10.2 mm-Al, 57.8 mGy, which represented the peak dose at 135 kVp). The bow-tie profile showed a symmetrical curve for all energies for both scanners. **Conclusion:** Differences were observed for HVL and effective energy as a function of position within wide X-ray beams. Furthermore, Scanner 2 had a harder beam spectrum than Scanner 1. These findings imply differences in image quality and dosimetry in wide CT beams scanned with a single volumetric rotation. Individuals assessing image contrast, CT number accuracy, OSL dosimetry, or Monte Carlo simulations must consider the spatial dependence of HVL and effective energy within X-Ray beams.

TH-IePD-TRACK 1-03

Spatial Resolution of Extremity CTA On a Wide Coverage Diagnostic CT Scanner: Effects of Focal Spot Size and Reconstruction Algorithm

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Purpose: To study the effects of focal spot size and reconstruction algorithm on the spatial resolution performance of a wide coverage CT scanner (GE Revolution Apex) in the context of upper/lower extremity CTA imaging. **Methods:** A GE QA phantom for CT was used and the MTF was measured by analyzing the HU profile across a 0.05 mm diameter tungsten wire. A clinical extremity CTA protocol was modified for the use of different focal spot sizes and reconstruction settings. Four reconstructions settings are included: standard kernel with ASIR-V 50%, detail kernel with ASIR-V 50%, detail kernel with ASIR-V 30%, and detail kernel with ASIR-V 0%. For each reconstruction setting, three focal spot sizes, XL, L and S, were used by varying tube current while CTDIvol stayed constant. 10% MTF spatial frequency was used to quantify spatial resolution performance. The partial volume effect was also evaluated based on the relative change in the peak HU. **Results:** The change of reconstruction kernel from standard to detail, when focal spot size L and ASIR-V 50% were used, resulted in an increase in 10% MTF frequency from 6.35 lp/cm to 7.47 lp/cm. The change of focal spot size resulted in a maximum of 0.15 lp/cm difference when the standard kernel was used, and a maximum of 0.5 lp/cm difference when the detail kernel was used. The change of ASIR-V percentage resulted in minimal difference when the detail kernel was used. For partial-volume effect, the increase in focal spot size from S to XL resulted in a maximum of 14% decrease in 10% MTF frequency. **Conclusion:** Our results indicate more pronounced effect of reconstruction kernel on spatial resolution compared to both focal spot size and ASIR-V percentage. MTF measurements with varying signal contrast are desired for more comprehensive characterization of spatial resolution performance of the system.

TH-IePD-TRACK 1-04

Optimal Tube Voltage Selection in Coronary Artery Angiogram with Dual-Source CT

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Purpose: As we experienced in clinical practice, the default automatic tube voltage selection may not be optimal in coronary artery CT angiogram (CCTA). This study was designed to find alternative tube voltages that are optimal in CCTA with dual-source CT. **Methods:** Iodine targets of different diameters and densities (3 -5mm and 0.5-10 mg/cc) were inserted on the coronary artery track of a medium size anthropomorphic heart phantom. Large and extra-large phantom sizes were made by adding additional fat layers. The heart module in the phantoms was set in 3-dimensional motion by an ECG signal of 60 beats-per-minute. A Siemens Force CT was used to acquire CCTA with prospective gating at 65-75% of R-to-R peaks. The vendor CarekV was used first (with a typical CNR strength 9 for CCTA). Additional kVp (+/- 10) were manually selected with the scanner's automatic mA compensation for CNR maintenance. Images were reconstructed to thickness of 0.5 mm (Bv40 and ADMIRE 3). Cylindrical voxels of interest were placed inside the targets of 5 mg/cc and 10 mg/cc and nearby blood pool to measure the contrast-to-noise ratio (CNR). The figure-of-merit (FOM) was defined by the CNR normalized to the square root of volume CT dose index to focus on the impact of kVp selection. **Results:** On average the FOM was found 12% higher when the kVp was lowered by one step (10 kVp) from the machine auto-selection for all three phantom sizes. Visual comparison of image quality of the coronary iodine targets at different kVp values was consistent with the FOM results. 150 kVp was not found optimal for the extra-large phantom. **Conclusion:** When the kVp was lowered by one step (10 kVp) from the machine auto-selection, the CCTA with dual-source CT delivers better performance.

TH-lePD-TRACK 1-05

A New Approach for Low Dose Low KVp Coronary Artery Calcium Scoring

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Purpose: Dose reduction using lower kVp in coronary artery calcium scoring (CACs) CT is appealing due to the population dose concern from the wide use. Previously we proposed a vendor-independent approach for low dose low kVp CACs without changing the Agatston thresholds. This study was to further validate the approach by incorporating a large phantom size. **Methods:** By applying consistent calcium contrast-to-noise ratio (CNR) to two anthropomorphic heart phantoms (medium and large) with 3-cm hydroxyapatite (HA) inserts, scanned using a dual-source CT, the relationship was derived between the volume CT dose index (CTDIvol) at lower tube voltages and the baseline CTDIvol at 120 kVp based on consistent CNR. The baseline CTDIvol was obtained using the specified noise thresholds from the images acquired at 120 kVp. To preserve the conventional Agatston thresholds, down-scaling with the found calcium HU enhancement factors was applied to images acquired at lower voltages with a dynamic heart module in 3D motion simulating heart movement at 60 beats per minute with 1.2-5 mm calcium inserts (50- 400 mg/cc) on the coronary tracks. Scores were evaluated on the scaled images by six readers and compared against the ground truth scores at high CTDIvol. **Results:** The CTDIvol at lower voltages was related to the baseline CTDIvol following a power form of the voltage (index 1.246), regardless of the phantom size. The baseline CTDIvol was found to be 1.5 and 4.5 mGy, for the medium and large phantoms, respectively. Correspondingly, the reduced CTDIvol at 100-70 kVp were 1.28-0.76 mGy, and 3.57-2.32 mGy. The downscaling factors were 0.88-0.63. The calcium scores at lower voltages were found within 12% of the ground-truths. **Conclusion:** A vendor- and patient size-independent approach was established to obtain the reduced dose and correct coronary calcium scores at lower tube voltages. Up to 50% dose reduction was achieved.

TH-lePD-TRACK 1-06

A Quantitative Approach to Patient Size Dependent Optimal Dose for Coronary Artery Calcium Scoring CT at the Standard KVp

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Purpose: Coronary artery calcium scoring (CACs) CT is performed at 120-kVp in current practice. The dose becomes a concern because of CACs' screening nature. The dose delivery is empirical and may not be optimal. We aimed to find the optimal dose for different patient sizes. The result can also serve as the baseline for further dose reduction for CACs at lower kVp. **Methods:** A surrogate noise (SN) was constructed as a function of the patient effective diameter and the volume CT dose index (CTDIvol). For the specified threshold noises at two known patient sizes covering the concerned range, the corresponding CTDIvol values were experimentally determined. To derive the CTDIvol as a function of the patient size within the range, we assumed that the threshold noise, thus SN, varies linearly with the patient size. The experiment was conducted using two thoracic phantoms (medium and large) with a moving heart module. They were scanned at 120 kVp using a prospectively gated CACs with a Siemens Force CT at different CTDIvol (0.76 -6 mGy). Its motion was driven by an ECG of 60 beats-per-minute. Noise measurement were made in the heart blood pool. A constant noise threshold (20 HU) and the recommended variable noise thresholds (20 HU for medium and 23 HU for large subjects) from Society of Cardiovascular CT (SCCT) were applied separately. The results from the model were further compared retrospectively with 33 clinical cases using dose modulation with a reference effective mAs of 80. **Results:** According to the model, the doses to achieve the constant noise threshold (20 HU) were found more consistent with the observed patient doses, but switching to the SCCT variable thresholds resulted in a dose reduction of up to 25%. **Conclusion:** A quantitative model was established to obtain the patient-size dependent optimal dose for CACs scans.

TH-lePD-TRACK 1-07

Graph-Based Low Dose Computed Tomography

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Purpose: Computed tomography (CT) has played a vital role in medical diagnosis, assessment, and therapy planning, etc. Concerns about the increase of X-ray radiation exposure attract more and more attention in clinical practice. To lower the X-ray radiation, low-dose CT (LDCT) has been widely adopted in certain scenarios, while it will induce the degradation of CT image quality. In this paper, we proposed a graph-based denoising neural network to reconstruct high-quality LDCT images. **Methods:** Inspired by the non-local mean method, although conventional convolutional neural network (CNN) has obtained outstanding success in LDCT, the convolutional operation only focuses on the local area of CT pixels, which is inefficient at processing structural information across large regions. Therefore, we introduce the graph convolutional network (GCN) and employ a well-designed spatial GCN module and a Channel GCN module. Given the dimension of the LDCT image is $M \times N$. First, we used six convolutional layers to obtain low-resolution features ($M/4 \times N/4$) piecewise in case of memory overflow. Then we fed these features into four spatial GCN modules to explore large-range dependencies. Finally, we employed skip connections to introduce those features with the same resolution from the first six convolutional layers into the piecewise reconstruction, like U-Net. We employ residual learning to ease the network training. We use mean square error (MSE) as the loss function to train the proposed graph-based neural network. **Results:** In this work, we evaluate our method on the dataset from the 2016 NIH-AAPM-Mayo Clinic Low-Dose CT Grand Challenge. We randomly select seven patients as the training dataset, one for validation and two for testing. For visual inspection, we can see that the proposed method can tell more fine structure. **Conclusion:** The experimental results demonstrate that the proposed method can improve the LDCT image quality and verify the significance of large-range dependencies on LDCT.

Imaging Interactive ePoster Discussion

TRACK 2

Imaging: MRI Physics and QC

TH-lePD-TRACK 2-01

Comparative Analysis of T2W Contrast for Clinical Prostate Protocols Using a Dedicated MR System Phantom

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Purpose: Sequence parameters for most clinical diagnostic imaging protocols have traditionally been designed based upon subjective image interpretive feedback by radiologists. Imaging parameters are typically selected based on compromises between contrast, resolution, scan time, SNR, and motion. While perceived differences in contrast exist for different sequences, the standardization and validation of intrinsic sequence contrast across scanner platforms and field strengths is not addressed by standard QA procedures. Hence longitudinal imaging on disparate scanners can be associated with observed contrast changes between timepoints, negating radiologists' diagnostic confidence. In this study, we used the ISMRM-NIST system phantom to evaluate contrast of a T2-weighted acquisition protocol used for prostate imaging across several scanners in our institution. **Methods:** The system phantom was scanned on three 1.5T and three 3T scanners with a thin slice T2-weighted Turbo Spin Echo PIRADS 2.1 compliant acquisition. Within field strengths, the scanners had identical hardware configurations and executed identical scan parameters. Since the phantom cannot be imaged with an endorectal coil, coronal slices were acquired using the body coil to maximize signal homogeneity. Signal intensity of the T2-sensitive compartments was measured for evaluation of relative contrast, as defined by change in signal intensity as a function of T2. Background water signal near the spheres was also recorded as a measure of overall B1- inhomogeneity, which was applied as a correction factor to minimize residual signal inhomogeneity in the image. **Results:** Contrast as defined by signal percent difference between two designated spheres averaged 28.0% at 1.5T and 45.0% at 3T. For T2 ranging from 46.42 to 581.3 msec, contrast curves varied within $\pm 1.3\%$ of average at 1.5T and $\pm 9.4\%$ at 3T. **Conclusion:** A standardized system phantom

and measurement protocol enables the measurement and validation of T2 contrast between scanners. Notable contrast deviations were detected at 3T, which warrant further investigation.

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TH-lePD-TRACK 2-02

A Dynamic Susceptibility Contrast MRI Digital Reference Object for Testing Software with Leakage Correction

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Purpose: Relative cerebral blood volume (rCBV), derived from dynamic susceptibility contrast (DSC)-MRI, is a perfusion biomarker commonly used for evaluating brain tumors. To account for the extravasation of contrast agents, post-processing leakage correction is often applied to improve rCBV accuracy. Digital reference objects (DRO) are ideal for testing the post-processing software, because the DRO's biophysical model can be matched to the one implemented in the software. Our aim is to develop and test such DSC-DROs. **Methods:** Three DROs were generated using the Weisskoff model, each composed of nine foreground lesion objects with combinations of different levels of rCBV and contrast leakage effects. Three types of background were implemented for these DROs: 1) a multi-compartment brain-like background, 2) a sphere background with a constant signal time curve, and 3) a sphere background with signal time curve identical to that of the brain-like DRO's white matter (WM). The DROs were then analyzed with an FDA-cleared software with and without leakage correction. Leakage correction was tested with and without brain segmentation. **Results:** Accuracy of leakage correction was able to be verified using the brain-like phantom and the sphere phantom with WM background. The sphere with constant background did not perform well with leakage correction with or without brain segmentation. The DROs were able to verify that for the particular software tested, leakage correction with brain segmentation achieved the lowest error. **Conclusion:** DSC-MRI DROs with biophysical model matched to that of the post-processing software can be well used for the software's validation, provided that the background signals are also properly simulated for generating the reference time curve required by the model. Care needs to be taken to consider the interaction of the design of the DRO with the software's implementation of brain segmentation to extract the reference time curve.

TH-lePD-TRACK 2-03

Development of a Quality-Assurance Program for MR-Derived PDFF Measurements

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Purpose: MRI accreditation does not currently require inspection for quantitative MR measures such as proton density fat fraction (PDFF). Here we report on the initiation of such a program for MRI-derived PDFF measurements at a large, regional health system. A low-cost phantom composed of varying fat percentages was constructed and scanned repeatedly over the course of 1 year. Based on repeatability coefficients reported in the literature, we hypothesized that an action limit of 2% absolute would capture normal variation. **Methods:** Six fat/water vials were constructed with fat concentrations 0, 5, 10, 15, 22.5, 30% wt/wt. The phantom was scanned using the LiverLab product from Siemens Healthineers on five 1.5 T MRI systems. Software levels varied among systems: XA11 (1 system), VE11E (2 systems), and two systems were upgraded from VE11C to VE11E during the study period. The default CSE-PDFF protocol was used, with changes only to slice orientation to accommodate phantom geometry. **Results:** All slopes were not significantly different from one ($p > 0.10$ for all systems), albeit with wide confidence intervals (mean [CI]: 1.1 [0.9, 1.4]). Excellent agreement was found for repeated scans, with no repeated measures differing from baseline by more than 1.7% absolute. Small biases (maximum -0.2% absolute) and a maximum coefficient of repeatability of 1.8% were found, in good agreement with values from the literature. **Conclusion:** Given the excellent repeatability of CSE-PDFF measures, a quality assurance program that assessed longitudinal changes of a low-cost, home-made phantom with an action limit of 2% absolute change per vial found no changes over the course of one year.

TH-lePD-TRACK 2-04

Dependence of B₀ Homogeneity on Field Strength and Phantom Size When Measured with Four Common Methods

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Purpose: A uniform static magnetic field (B₀) is imperative in clinical MRI applications to minimize artifacts such as geometric distortion, signal loss, banding artifacts, and failures in frequency-selective techniques. Several methods to measure B₀ homogeneity (ΔB_0) have been developed. The purpose of this work is to assess the variability in measurements from different methods and potential contributing factors. **Methods:** ΔB_0 was measured by multiple users and scanner types. Four methods to measure ΔB_0 were analyzed: Spectral Peak (N=41), Bandwidth (N=45), Phase Difference Map (N=55), and Field Map (N=35). This dataset was retrospectively analyzed: average values of ΔB_0 and coefficients of variation (CV) were calculated for each method. These data were further grouped by field strength and phantom size and tested for significant differences to determine any contributing factors of ΔB_0 variability. **Results:** Average ΔB_0 in ppm and CV of the four methods were as follows: Spectral Peak (0.93, CV=0.68), Bandwidth (1.23, CV=0.82), Phase Difference Map (1.52, CV=0.43), Field Map (1.45, CV=0.43). Spectral peak was reported as root-mean-square, while the other methods were reported as peak-to-peak. The Bandwidth difference method had the highest CV while the two mapping methods had the lowest. Measurements for each method grouped by field strength did not differ significantly. ΔB_0 was significantly lower ($p=0.0464$) when measured in smaller phantoms (16-29 cm) compared with larger phantoms (30-45 cm) using the Spectral Peak method but did not differ significantly when measured with the other methods. **Conclusion:** This analysis of ΔB_0 measurements using four common methods serves as an initial reference for those conducting their own homogeneity tests. These initial data are highly variable, which may be explained by differences in scanner technology and reflect variability in methods across physicists. However, as more data points are added to the study, these reference values can be reported more precisely.

TH-lePD-TRACK 2-05

MRI Isocenter Consistency at Various Gantry Angles Using MODUS MRID3D Phantom

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Purpose: MRI isocenter consistency at all gantry angles is critical to ensuring accurate treatment delivery in MRgRT. Previous reports from our group showed an imaging isocenter shift of <1mm utilizing a rectangular phantom at multiple imaging locations with ViewRay's analysis software. In this study, we verified MRI isocenter fidelity using an independent 3rd party 3D geometric phantom and corresponding software. **Methods:** This experiment used a QUASAR™ MRID^{3D} phantom on a 0.35T ViewRay MRidian MR-linac alongside the QUASAR™ MRID^{3D} geometric distortion analysis system. Images were acquired at twelve gantry angles from 0–330° at 30° increments. Acquisitions utilized a 3D TRUF1 sequence with TR: 3ms, TE: 1ms, flip angle: 60°, FoV: 450x450x360mm³, imaging matrix: 300x300x240. Images were acquired with the onboard Siemens distortion correction function turned on and off to measure the ability of the system to correct geometric distortion. All images were imported into the distortion analysis software to quantify MRI isocenter shift and geometric distortion of the phantom per gantry angle. The baseline for all measurements is G300 for Gantry home position in RT. **Results:** The mean and standard deviation deformation vectors for Distortion

On were 0.28 ± 0.37 mm, 0.27 ± 0.32 mm, and 0.28 ± 0.33 mm in the AP, RL, and SI directions respectively, and 2.91 ± 4.44 mm, 3.15 ± 4.75 mm, and 1.82 ± 2.49 mm for Distortion Off. The mean isocenter shift was 0.62 mm with a range of 0.21 mm to 0.96 mm for Distortion On, and 0.65 mm ranged from 0.24 mm to 0.93 mm for Distortion Off. **Conclusion:** The magnitude of MRI isocenter shift from 3D MRID^{3D} phantom showed similar trends as previous work with a 2D slab phantom for all gantry angles. This study showed that the gantry position can impact the image isocenter yet can be corrected by applying a deformation vector field at all measured gantry angles using the QUASARTM MRID^{3D} phantom and software.

TH-lePD-TRACK 2-06

Need for a Saturation RF Band to Mitigate Flow Artifact in 3D-MPRAGE On a Compact 3T Scanner

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Purpose: Due to the smaller dimensions (inner diameter 37 cm, length 40 cm) of the RF-transmit coil on compact 3T (C3T) scanner [1], the B1 field falls off rapidly outside the head. This offers great potential for safe brain imaging of patients with implanted devices without sacrificing any imaging capabilities [2]. For a whole-body scanner, during a fast gradient-echo scan, the arterial blood usually experiences repeated RF pulses before flowing into the brain, so the magnetization becomes saturated. However, because of the relatively shorter RF coverage of the C3T, inflowing blood saturates less than with a conventional, whole-body transmit RF coil does (Fig. 1). This results in strong hyperintense intravascular signal and intense flow-related ghost artifacts. In this study, we demonstrate that these specific 3D-MPRAGE flow artifacts can be mitigated by applying an additional inferior spatial pre-saturation. **Methods:** An RF spatially selective saturation band module was added to the standard 3D-MPRAGE sequence to saturate incoming arterial flow from the heart (Fig. 2) and consequently suppress the intensity of intracranial vasculature. 3D-MPRAGE imaging was performed on the C3T both without and with use of the inferior RF saturation band in two human subjects under an IRB-approved protocol. Image assessment was performed by two board-certified neuroradiologists. **Results:** With the additional saturation RF pulses, the acquisition time was increased 20 seconds compared to the standard acquisition (5:02 vs. 5:22). While prominent ghost artifacts were observed along the in-plane phase-encoding (i.e., inferior-to-posterior) direction in the standard sagittal scan, the pulsation effects were substantially reduced in the proposed scan due to RF pre-saturation of the incoming arterial signal (Fig. 3). **Conclusion:** 3D-MPRAGE imaging with an additional RF saturation band is an effective method to suppress vascular ghost artifacts caused by the arterial blood flow inherent to smaller bore MRI system such as C3T.

TH-lePD-TRACK 2-07

MRI Scanner 3D Field Homogeneity Evaluation for ACR Acceptance and Performance Testing

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Purpose: To quantify and to improve the measurement accuracy of the “MRI scanner field inhomogeneity evaluation” for the “ACR MRI system Acceptance Test” and the “ACR MRI System Annual Performance Test”. **Methods:** A spherical phantom of 25 cm diameter (which is MRI site facility dependent, normally larger) at the isocenter of the MRI scanner bore was scanned with a 3D dual gradient echo sequence with TE1=7 ms and TE2=21.7 ms (which are scanner field strength dependent). A slab of 25-slices (which is phantom size dependent) was scanned and a total of 2x25 amplitude plus 2x25 phase images were acquired. An appropriate 3D phase unwrapping method was developed and applied to the phase images. The field variation map was then generated according to the guidance in the “ACR 2015 Quality Control Manual Medical Physicist Section” (Chapter IV, Section A-4) and the “ACR Acceptance Testing and Quality Assurance Procedures for MRI Facilities” (Chapter III, Section E-1) based on these two unwrapped 3D-phase images. The field at the isocenter was set as the reference point. **Results:** The result shows that each 3D-phase image is unwrapped perfectly as one unit and the 3D-field map generated based on these 3D-phase images is smooth and intuitive. It can be viewed with any 3D viewing tools from any viewing angle. The result shows that for this particular scanner, up to a diameter of 25 cm, the crossing field variation is about 0.3 ppm in the transverse plane while the field variation reaches over 1.0 ppm in the longitudinal direction. **Conclusion:** A semi-automated tool to generate the MRI scanner field variation map (2D and 3D) is developed based on this method and it provides a convenient, efficient and accurate mean for physicists to perform the MRI scanner field inhomogeneity evaluation for the “ACR MRI system Acceptance Test” and the “MRI System Annual Performance Test”.

Multi-Disciplinary Interactive ePoster Discussion

TRACK 3

Multi-Disciplinary: Segmentation III

TH-lePD-TRACK 3-01

The Segmentation of Main Intestinal Organs Based on CT Images With Deep Learning Method

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Purpose: Fully analyze the features of three intestinal organs including colon, small intestine and duodenum in morphology, texture, and location distribution, and used some specific structures to capture these features, so as to achieve the simultaneous and high-precision segmentation of the three intestinal organs. **Methods:** Considering the changes in the shape and size of the intestinal organs, the model needs to adequately capture the information at different scales. HRNet can maintain high-resolution feature map throughout the whole network, and can fully integrate features of different scales, which is helpful for the segmentation of large-scale organs such as colon, small intestine and small-scale organs like duodenum. In addition, by introducing location information and reasonable data enhancement operations, we further improve the segmentation ability of the network for different organs and alleviate the problem of mutual interference between multiple organs. **Results:** Our data is collected from 80 patients with 3316 CT images which is divided into three parts: 2652 for training, 332 for validation and 332 for the final test. For the 332 test CT images, dice coefficients of colon, small intestine and duodenum were 0.8963, 0.8792 and 0.8510, respectively, which all achieved excellent results so far. In the duodenum, which is the most difficult to segment, our results were 5% higher than those of some previous networks. **Conclusion:** The rich multi-scale information provided by HRNet is very important for large morphological changes of intestinal organs. Meanwhile, the maintenance of high-resolution feature maps also provides detailed information for fine segmentation of small target regions, such as organ edges and duodenum. Location information is the key to further distinguish small intestine and duodenum. By simply introducing coordinate information, the segmentation accuracy of duodenum can be improved. Meanwhile, a reasonable data enhancement strategy is also conducive to alleviating the class imbalance problem.

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TH-lePD-TRACK 3-02

Investigation of Clinical Target Volume Segmentation for Whole Breast Irradiation Using 3D Convolutional Neural Network and the Shape Regularization Model Without the Prior Information of the Target-Side Breast

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Purpose: This study aims to implement three-dimensional convolutional neural networks (3D-CNN) for clinical target volume (CTV) segmentation for whole breast irradiation and investigate the performance with and without shape regularization model (SRM). **Methods:** The 3D-UNet CNN was adopted to conduct automatic segmentation of the CTV for breast cancer. The 3D-UNet was trained using the 192 patients' dataset which contains both left- and right-sided breast cancer patients.

SRM was introduced to the 3D-UNet model to improve the segmentation performance by modifying the segmentation shape. SRM was trained in advance using the dataset of the incomplete CTV shapes and the ground truth CTVs. In the 3D-UNet+SRM model, the customized loss function was employed. The segmentation accuracy was evaluated using the Dice similarity coefficient (DSC) and Hausdorff distance (HD). To compare the performance of the 3D-UNet alone and 3D-UNet+SRM, the Wilcoxon signed-rank test was applied to the DSC and HD scores of the test dataset. **Results:** For the 3D-UNet alone, the DSC and the HD for the test dataset were 0.852 and 57.80, respectively. The DSC and the HD for the 3D-UNet+SRM were 0.854 and 52.67. The Wilcoxon signed-rank test demonstrated that both DSC and HD of the 3D-UNet+SRM performed better than 3D-UNet alone. The incomplete structures like small holes, jagged edges, and imperfect shapes were eliminated or compensated with SRM. SRM also worked to modify the images which had islands predicted as part of the CTV on the opposite side of the breast. However, if the 3D-UNet predicted the opposite side of the breast as the target breast, the small island on the correct side was eliminated. **Conclusion:** SRM modified the incomplete CTV shapes during the training process of 3D-UNet segmentation. If we make the SRM more sophisticated with a larger dataset, 3D-UNet+SRM could be improved more.

TH-lePD-TRACK 3-03

Attention Guided Network for Vestibular Schwannoma Growth Prediction

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Purpose: Knowledge of tumor growth behavior is key to clinical decision-making in the management of vestibular schwannoma (VS) patients. In clinical practice, a series of images acquired at a fixed interval are used to determine VS tumor growth, which may lead to delayed treatment for fast-growing tumors or over-imaging for slow-growing tumors. In this work, we aim to build an attention guided convolutional neural network to predict VS tumor growth with the initial diagnosis imaging to aid in personalized management of VS. **Methods:** Contrast-enhanced T1-weighted MR images of 103 VS patients were used in this study. Each patient has two MRI scans with median interval of 7.2 months. VS on these images were segmented by a neurologist. Sixty-two out of 103 patients experienced VS growth based on the volumetric measurements on two consecutive scans. Our attention guided model used an UNet segmentation module and an AlexNet-style classification module to exploit the tumor growth related structures for VS growth prediction. The segmentation module is used to identify tumor and its surrounding area relevant to tumor growth prediction and generate attention maps from different layers, while the classification module takes the MR image and attention maps as input and predicts tumor growth. These two modules are trained simultaneously through optimizing the combination of dice coefficient loss from the segmentation part and binary cross-entropy loss from the prediction part. **Results:** Eighty patients' data were used for model training, 10 for validation, and 13 for testing. The growth prediction accuracy and AUC of the proposed model on testing data are 0.85 and 0.82, which are higher than those from a separate classification network without the attention mechanism. **Conclusion:** We developed an attention guided network for VS tumor growth prediction using the initial diagnosis MR image. Comparison with separate modules demonstrated its efficiency and effectiveness.

TH-lePD-TRACK 3-04

Segmentation by Individualized Registration (SIR) for CBCT-Based Adaptive Radiation Therapy

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Purpose: The purpose of this study is to develop a method that can automatically generate segmentations on cone-beam CTs (CBCT) for head and neck adaptive radiation therapy (ART). Due to the prevalence of artifacts and truncations on CBCTs, directly contouring on CBCTs is very challenging. Thus, we propose to utilize a learning based deformable image registration method and contour propagation to get updated contours on CBCTs, where expert-drawn contours on the planning CT (pCT) can serve as prior knowledge. **Methods:** Different from other deep learning based image registration methods which are generally trained on a large datasets and tested on separate data, our proposed method uses deep learning architectures as a function f taking only a pair of moving and fixed images (pCT and CBCT in our case) as input and yielding warped moving images as outputs. The function f is composed of two parts: deformation vector field (DVF) generation and spatial transformer. DVF generation can predict a DVF from a pair of pCT and CBCT, then spatial transformer can warp pCT to deformed pCT (dpCT) based on predicted DVF. The function f can be learned by minimizing the cost function of image similarity between fixed and warped images plus regularization terms. **Results:** When compared with manual segmentations by a radiation therapy expert, the average dice coefficients of organ-at-risks and target by Elastix, 3DSlicer B-spline deformable registration, 3DSlicer demon deformable registration, FAIM, Voxelmorph, 5-cascaded Voxelmorph, VTN, and 10-cascaded VTN are 0.84, 0.83, 0.80, 0.83, 0.80, 0.84, 0.81, and 0.84 respectively. However, the average dice coefficients of Voxelmorph, 5-cascaded Voxelmorph, VTN, and 10-cascaded VTN using our proposed training strategy can reach up to 0.83, 0.85, 0.85, and 0.86. **Conclusion:** Our proposed method outperforms both the state-of-the-art learning based and traditional based image registration methods.

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TH-lePD-TRACK 3-05

Intentional Deep Overfit Learning (IDOL): An Application to CT Segmentation-Based Daily Adaptive Radiotherapy

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Purpose: Adaptive radiotherapy (ART) workflows have been increasingly adopted to achieve dose escalation and tissue sparing under shifting anatomic conditions, but the necessity of recontouring and the associated time burden hinders a real-time, online ART workflow. In response to this challenge, approaches to auto-segmentation involving deformable image registration, atlas-based segmentation, and deep learning-based segmentation (DLS) have been developed. Despite the particular promise shown by DLS methods, implementing these approaches in a clinical setting remains a challenge, namely due to the difficulty of curating a data set of sufficient size and quality so as to achieve generalizability in a trained model. In this study, we utilize a patient's prior information to leverage the behavior of an intentionally overfit model with the application of the Intentional Deep Overfit Learning (IDOL) framework to the auto-segmentation task. **Methods:** The intentionally overfit model is trained in two stages. In the first, a conventional, general model is trained with a diverse set of patient data ($n = 80$ patients) consisting of CT images and clinical contours. Following this, the same model is tuned with a data set consisting solely of deformations derived from data from the single patient of interest. The conventional model and proposed overfit models were evaluated using the Dice similarity coefficient computed for 2 structures in 20 test patients. **Results:** Adopting the IDOL framework improved segmentation performance for each patient. Dice scores improved from 0.85 with the conventional model to 0.92 with the patient-specific overfit model on average and by up to 14% in the best case. **Conclusion:** The IDOL framework applied to the auto-segmentation task achieves improved performance compared to the conventional DLS approach, demonstrating the promise of leveraging patient-specific prior information in a task central to online ART workflows.

TH-lePD-TRACK 3-06

Deep Learning Neural Network for Automatic Delineation of Para-Aortic Clinical Target Volume

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Purpose: To develop and evaluate an artificial-intelligence-based approach that automatically delineates clinical target volume (CTV) of para-aortic lymph nodes (PAN) for locally advanced cervical cancer (LACC) treatment. **Methods:** We retrospectively collected 47 planning CTs of LACC patients. For each CT, the Aorta, Inferior Vena Cava (IVC) and PAN-CTV were manually delineated for the elective para-aortic region from the left renal vein to the aortic bifurcation. The PAN-CTV was prophylactically contoured (ignoring gross para-aortic nodes) according to the recent Small et al. (2020) consensus guidelines. For each CT set, the spine structures were automatically delineated using threshold- and morphology-based segmentation. Then, all CTs were registered to a common reference frame based on the delineated spine structures and cropped to dimensions of 15x15x20 cm³ centered along the spine. A U-Net-based segmentation network was developed and trained to automatically segment Spine, Aorta, IVC, and PAN-CTV in the registered and cropped region. The segmented structures were converted into DICOM RTStruct with the original CT association. **Results:** We used 32 CTs for training, 8 for validation, and 7 for testing. The Dice-coefficient (DSC), Hausdorff-distance (HD95), and average symmetric-surface-distance (ASD) were used as the evaluation metrics. For PAN-CTV segmentations, our model achieved mean (SD) values of 75.5 (7.3)%, 8.42 (4.52) mm, and 1.55 (0.61) mm for DSC, HD95, and ASD, respectively. All PAN-CTV segmentations were visually checked by an attending radiation oncologist specializing in gynecologic malignancies and deemed clinically acceptable. **Conclusion:** An Artificial Intelligence approach comprised of registration and DL network was developed and validated for automatic segmentation of Spine, Aorta, IVC, and PAN-CTV in LACC. Registration was used to improve the boundary definition of the virtual structure, PAN-CTV, for DL segmentation without human intervention. This approach could facilitate efficient workflow and allow for optimal dose delivery to the involved nodal basins in LACC.

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TH-lePD-TRACK 3-07

Auto-Segmentation of Important Centers of Growth in the Pediatric Skeleton to Consider During Radiation Therapy Based on Deep Learning

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Purpose: Routinely delineating important skeleton growth centers is imperative to track and reduce exposure dose for pediatric neoplasms population with radiotherapy. The goal of this study is to explore network structure dimension effect on the efficiency and accuracy of deep learning segmentation models for skeleton growth plates. We devised a novel deep learning network called 2.5D UNet, that extends the semantics of adjacent slices to automatically segment the important centers of growing skeleton. **Methods:** The craniofacial, shoulder and pelvis growth plates, with a total of 19 structures, were manually delineated by a radiation oncologist on axial, coronal, sagittal CT images from 75 child cancer patients. Based on the UNet network structure, three models of segmentation networks including 2D, 3D and proposed 2.5D are compared. 2.5D refers to the use of three consecutive image slices to predict the segmentation of the middle slice, which can increase the associated semantic information. Dice Similarity Coefficient (DSC) and Hausdorff Distance 95% (HD95) were used to evaluate the segmentation accuracy of each model, and time consumption was recorded respectively. **Results:** Comprehensive evaluation of the 19 ROIs output by the three models of 2D, 3D and proposed 2.5D, the average DSC is 0.7972, 0.7975, 0.7987, and the average HD95 is 4.25, 1.88, 1.90, respectively. The HD95 performance of the latter two is clearly better than the former, while the DSC performance is not significantly different. The efficiency of the 3D model (23.5 hours of training time, 180 seconds of average inference time per patient) is significantly lower than 2D (3.1 hours, 45 seconds) and 2.5D (3.3 hours, 48 seconds). **Conclusion:** The deep learning-based models to automatically delineate pediatric growth plates have a satisfactory segmentation performance. Considering segmentation accuracy and time efficiency, 2.5D is more appropriate for clinical application than 2D and 3D models.

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Therapy Interactive ePoster Discussion

TRACK 5

Therapy: Treatment Verification

TH-lePD-TRACK 5-01

Development and Deployment of a Real-Time Transit EPID Dose Verification System

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Purpose: We have previously demonstrated feasibility of real-time transit dosimetry using EPID image prediction. This work aims to develop and deploy a clinically-ready real-time transit EPID dose verification system for modern treatment platforms that enables verification per control point of the delivery. **Methods:** A real-time transit dose verification system based on EPID image frames was developed and deployed for TrueBeam Linear Accelerators and aS1200 detectors (Varian Medical Systems, USA). Predicted images are generated for every CP using the DICOM treatment plan file and CT dataset and a previously published dual-source fluence model. Measured EPID images are acquired and evaluated in real-time using C# and Matrox Imaging Library (MIL) X (Matrox, Canada). Online real-time processing incorporates: 1) integration of measured frames for the selected arc-segment and synchronization to predicted images; 2) gamma and signal evaluations between measured and predicted images, and 3) evaluation of machine treatment parameters including delivered monitor units, multileaf collimator positions, and gantry and collimator angles. Six VMAT and four SBRT treatment plans from several treatment sites were irradiated in air and on an anthropomorphic phantom (CIRS, USA) for model testing purposes. Arc-segment resolution was set to 3 CPs per evaluation. **Results:** Gamma evaluation (5%, 2mm) passing rates for test cases were 98.8% (6X) and 98.0% (6XFFF) for in-air measurements, and 99.0% (6X) and 99.1% (6XFFF) for transit measurements. Differences in dose delivered per arc-segment were comparable for all test cases and averaged less than 1%, with a few isolated instances with differences of up to 2.5%. CP processing times including gamma evaluation and properties comparison are below 100ms. Preliminary results of the ongoing patient study suggest strong inter-fraction consistency in signal per patient. **Conclusion:** A real-time transit dose verification system has proven to be sufficiently stable, efficient, and automated to be integrated in the clinical practice.

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TH-lePD-TRACK 5-02

Evidence-Based Evaluation of Skin Flash for Prone Breast Treatments by Using Edge Detection on Daily Exit Dosimetry Images

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Purpose: In breast radiotherapy, skin flash is important to account for setup uncertainties and intrafraction motion. On Halcyon linear accelerator, exit dosimetry images are automatically acquired at each fraction, providing a method to assess the breast position with respect to the radiation field borders. We developed a tool to analyze the interfractional skin flash and breast anatomy variation using breast edge detection technique based on the exit dosimetry images. **Methods:** 288 daily exit dosimetry images were acquired for 9 patients undergoing 16-fraction prone breast treatments, with two opposed fields. The field border and breast contour map were acquired for every portal image using Canny algorithm. Algorithm parameters were derived automatically from a gradient-based initial estimate, to eliminate manual tuning. Noise and couch edges were removed combining local information and pixel connectivity. The distance between field border and apex of breast edge was calculated as the flash amount per fraction, and compared with the value from plans. The breast shape and position variation were evaluated by the range of breast contour change in beam's-eye-view (BEV) at three positions, upper, middle and lower part of the breast. **Results:** The mean and standard deviation of difference between fractional flash and planned flash is -1.8 ± 3.8 mm (negative sign represents actual breast closer to field border than planned). The variation of breast contour during treatment course has a mean and std of 9.5 ± 2.8 mm. **Conclusion:** An edge detection tool was developed for breast radiotherapy based on Halcyon exit dosimetry images, which is used to investigate skin flash and breast position change with treatment progression. Breast contour in BEV for this patient cohort varies up to 17 mm through the treatment course. This tool could be useful to automate daily breast contour detection, and to facilitate ongoing physics chart check by alerting user in case of insufficient skin flash

TH-lePD-TRACK 5-03

A Cross Institution Study of Anatomy Changes in Head-And-Neck Cancer Patients Using In-Vivo Electronic Portal Imaging Device (EPID) Transit Images

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Purpose: To develop automated tools and perform a cross-institutional study to evaluate patient anatomy trends using in vivo EPID transit images collected after every fraction of treatment. The tools were designed to forego a complex commissioning process and were deployed at two cancer centers, that have disparities in access to cancer care, to analyze head-and-neck cancer patients. **Methods:** For 252 patients treated at academic institutions in South Africa (40 patients) and United States (212 patients), analysis of in vivo EPID images for head-and-neck patients was performed. Both centers treated patients on the Varian Halcyon which collects EPID images for every fraction of treatment. Deviations relative to the first fraction of treatment were assessed, for 5630 fractions, with a novel technique, Gradient-Dose Segmented Analysis (GDSA) which correlates with changes in the PTV mean dose. The change in PTV mean dose over a treatment course was compared between centers and individual patient trends were investigated. **Results:** The mean and variance of the change in PTV mean dose were statistically equivalent for the two centers, demonstrating comparable magnitude and frequency of anatomical changes. Both centers observed patient weight loss/tumor shrinkage in the later fractions of treatment. One center observed cyclic patterns in the patient's external body contour, for half of the patients analyzed. These trends were not observed at the other institution and may be driven by differences in the access to cancer care for the two centers. **Conclusion:** We have developed an automated framework that can measure in vivo patient anatomy trends and how they compare between different clinics. In aggregate, anatomy deviations of head-and-neck patients at two different centers analyzed were equivalent. Further study is required to understand the source of different anatomy change trends observed. This framework is portable and can be deployed to other centers to perform similar analyses.

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TH-lePD-TRACK 5-04

Impact of Delivery and Measurement Uncertainties On the Calculation of Pixel Sensitivity Maps for MV Portal Imagers

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Purpose: To describe possible error pathways during calculation of the Pixel Sensitivity Maps (PSMs) and to study the impact on the calculated PSMs with different approaches. **Methods:** PSMs are used to normalize EPID's diode-pixels response to the incident radiation. Multiple approaches were developed to calculate EPID-specific PSMs. A shifting-based method uses multiple laterally shifted measurements of same flood field signals to find pixel-gain through utilizing the relative gains of pixels irradiated identically. A shiftless-based method, uses regression surface models to capture low-frequency features of the incident fluence and ascertain the device PSM by removal of these features from signals superimposing high-frequency PSM features with low-frequency fluence features. Here, we use Monte-Carlo calculated EPID signals with introduced errors to study the impact on the reconstructed PSMs. Errors introduced include misalignment of signals used during the PSM calculation, change of output between irradiations and inconsistent background removal from signals used in PSM generation algorithms. **Results:** In an ideal setup the shifting-based and the shiftless methods reconstructed the original PSM with 99% and 95% accuracies, respectively. For the shifting method; a one pixel misalignment during the PSM calculation resulted in 90% and 100% of the pixels retained 99% and 90% of the original gain values, respectively. A 0.1% and 0.5% change in linac output between images resulted in 27% and 11% of the gains have 99% of the original values. A 0.1% and 0.5% pixel-to-pixel variance in the individual pixel response resulted in 1.7% and 32% of the pixel gains have 99% and 90% of the original values, respectively. Introduction of the aforementioned errors in the shiftless method worsen the agreement results by a max of ~4%. **Conclusion:** The simulation conducted here shows that both methods are accurate in impeccable clinical setups. The shiftless PSM method performs better in sub-standard calibration scenarios.

This work is supported in part by Varian Medical Systems

TH-lePD-TRACK 5-05

Evaluation of Dynamic Tumour Tracking (DTT) uncertainty for a Fiducial-Based Stereotactic Ablative Radiotherapy (SABR) Treatment of the Liver Using EPID Images

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Purpose: To assess dynamic tumour tracking (DTT) uncertainty for fiducial-based stereotactic ablative radiotherapy (SABR) treatments of the liver using electronic portal imaging device (EPID) images. **Methods:** EPID images were acquired during 3D conformal radiotherapy (3DCRT) DTT using a Vero4DRT linac (Brainlab-Mitsubishi), equipped with an amorphous-silicon (aSi) EPID (maximum frame-rate: 2Hz). Images acquired in phantom and patients were pre-processed using a Laplacian-of-Gaussian (LoG) function. Three gold fiducials were inserted into an acrylic phantom. The phantom was placed on a 1D, moving Brainlab platform with sine-wave motion synchronized to a respiratory platform. EPID images were acquired for 3DCRT noncoplanar DTT beams. The EPID images of nine liver SABR patients treated with 3DCRT noncoplanar beams were also analyzed. All patients had 3-4 implanted fiducials. A chain-code algorithm was applied to the EPID images to detect field-edges of the MLC-defined apertures. Fiducials were detected using a connected-neighbor algorithm. For each beam, the centre-of-mass (COM) of the fiducials relative to the field-edge was reported (at isocentre plane). **Results:** Of the 123 EPID phantom images, 100% of the fiducials were located. The Standard-Deviation (SD) of the relative position between the fiducial COM and field-edge in the X(h) and Y(v) direction, for all fields, was 0.2 and 0.3 mm,

respectively. Of the 1418 EPID patient images acquired, 81% had detectable fiducials. Fiducials may not be detected if the images were noisy, or the location was occluded by the MLC. The SD of the relative position between the fiducial COM and field-edge in X and Y direction was between 0.27-0.82 mm and 0.53-1.61mm, respectively. Sources of uncertainty in the relative positions include field-edge localization, fiducial-localization, and physical tracking uncertainty during treatment. **Conclusion:** DTT uncertainty can be evaluated on a field-by-field basis using EPID images. The DTT uncertainty was <0.82 and 1.61mm in X and Y direction, respectively.

TH-IePD-TRACK 5-06

Theoretical Derivation and Experimental Verification of Beam and Tissue Factors Affecting the Reemitted Cherenkov Signal During External Beam Irradiation

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Purpose: To explain why the Cherenkov intensity emitted from MV x-ray delivery in tissue does not appear to change with beam energy in radiotherapy. The study developed a quantitative theoretical understanding of the effects of beam and tissue properties on the reemitted light. **Methods:** The reemitted Cherenkov signal was theoretically derived with a diffusion theory solution and experimentally measured as a function of the dose build-up slope and tissue optical properties over a range of clinical x-ray energies. Diffuse liquid tissue phantoms of various scattering and absorption properties were irradiated with 6, 10, and 18 MV x-rays. Each phantom was imaged with an intensified camera, and irradiated top-down for entrance surface imaging, as well as from underneath for exit surface imaging. Phantoms were built with the thickness of the depth of d(max) per energy. The reemitted Cherenkov intensity was calculated as the mean pixel intensity of a ROI at each phantom's surface. **Results:** At low energy, a higher surface dose and steeper build-up slope result in a greater reemitted Cherenkov signal per unit dose, a ~46% increase at 6 versus 18 MV. However, at high energy, more Cherenkov photons are generated per unit dose, resulting in a ~44% increase in the reemitted signal at 18 versus 6 MV. These competing effects yield an approximately constant reemitted Cherenkov signal from the surface of the phantoms at different energies. With increasing scattering and absorption properties of the phantoms, the reemitted Cherenkov signal increased ~10% and decreased ~20%, respectively. **Conclusion:** The results of this study explain why the reemitted Cherenkov signal during radiotherapy is not strongly dependent on beam energy, despite greater Cherenkov photon generation at higher energies. Quantifying the effects of beam and tissue properties is a progressive step towards verifying surface dose delivery with Cherenkov imaging during radiotherapy treatments in real-time.

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TH-IePD-TRACK 5-07

In Situ Radiochromic Film Dosimetry Using Portable Spectrophotometer

S Jung*, J Cho, J Han, J Kim, J Park, C Choi, J Son, J Kim

Purpose: This study aims to report the performance of using a portable spectrophotometer as a reader for radiochromic film dosimetry. The study compares net optical density (net OD) of the EBT3 film between a typical flatbed scanner and a portable spectrophotometer. **Methods:** The EBT3 films were irradiated by using 6 MV photon beam with 0 to 3 Gy. The films were read by the flatbed scanner (EPSON 10000XL) and the portable spectrophotometer (YS-3020, 3nh, China) at 10 min, 1 hour, 3 hours, 6 hours, and 24 hours after the irradiation. The net OD in red channel was acquired for each scanner. EBT3 films with 0.7 Gy and 1.8 Gy irradiation were read by the scanners and the measured doses were compared with the delivered doses. **Results:** During 10 min to 6 hours, the curves from the portable spectrophotometer showed larger fluctuations in net OD than those from the flatbed scanner. Calibration curves at 6 hours and 24 hours after the irradiation were acquired. Dose responses of the portable spectrophotometer were 28.2% to 51.7% higher than those of the flatbed scanner. The EBT3 film with 0.7 Gy irradiation read by the portable scanner at 6 hours after the irradiation showed large discrepancy (17.7%) comparing with the actual delivered dose. The discrepancies were reduced to 6.6% at 24 hours after the irradiation. For the same delivered dose read by flatbed scanner, the difference between the actual and estimated were 5.8% and 4.5% for 6 hours and 24 hours, respectively. **Conclusion:** The portable spectrophotometer has advantages over the flatbed scanner that it is much lighter, easier to use than the flatbed scanner. However, the portable spectrophotometer limits the film dosimetry to the point measurement and the impact of post-irradiation was more significant in the portable spectrophotometer.

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SAM Professional Symposium

TRACK 1

Professional Letters of Reference: Advice for Those Who Need Them and Those Who Need To Write Them

TH-E-TRACK 1-00

Professional Letters of Reference: Advice for Those Who Need Them and Those Who Need To Write Them

Moderators: Chelsea Page-Robertson, Banner MD Anderson Cancer Center, Greeley, CO and Robin Miller, Northwest Medical Physics Center, Seattle, WA

TH-E-TRACK 1-01

An Introduction to Professional Letters of Reference

TH-E-TRACK 1-02

R. Miller, Northwest Medical Physics Center:

Reference Letter Etiquette: How to Ask for a Letter of Reference

TH-E-TRACK 1-03

T. Pawlicki, UC San Diego:

How to Write a Great Letter of Reference

TH-E-TRACK 1-04

Audience Q&A

There have been many professional sessions at previous Annual and Spring Clinical meetings related to finding a job. One topic that has not been previously covered in depth is the important job-related topic of professional letters of reference. The PROFS Committee aims to host a session covering a broad range of topics relating to letters of reference. This will be coupled with an AAPM Newsletter Series of articles.

A letter of reference or recommendation letter is used to provide information about the candidate from an outside source, providing details about skills and

qualifications, sometimes giving a personal perspective of the candidate, and listing achievements. As Medical Physicists, we find ourselves needing letters of reference for new jobs, promotions, and awards but also needing to write them for colleagues, students, and residents. The goal of this session is to detail the differences and similarities in types of professional letters, because whether you are asking someone to write a letter of reference for you or writing a letter for someone else, it can be helpful to understand the difference types and perspectives of professional reference letters.

Learning Objectives:

1. Define and describe the different types of reference letters
2. Understand how to develop professional references and ask for letters of reference
3. Understand how to write different letters of reference

Multi-Disciplinary Scientific Symposium

TRACK 2

Principal Investigator Scientific Highlights

TH-EF-TRACK 2-00

Principal Investigator Scientific Highlights

Moderators: Carri Glide-Hurst, University of Wisconsin, Madison, WI and Lei Ren, University of Maryland, Baltimore, MD

TH-EF-TRACK 2-01

C. Glide-Hurst, University of Wisconsin:

The Funding Landscape for AAPM Members

TH-EF-TRACK 2-02

G. Prax, Stanford University:

NIH R01, New Tools for Tracking Single Cells In Vivo

TH-EF-TRACK 2-03

H. Li, University of Illinois at Urbana-Champaign:

NIH R01, Multimodal Biomarkers for Oropharyngeal Cancer

TH-EF-TRACK 2-04

K. Lafata, Duke University:

DOD, Multi-Scale Characterization of Radiation Resistance in Head and Neck Squamous Cell Carcinoma

TH-EF-TRACK 2-05

L. Ren, University of Maryland:

NIH R01, Hybrid Virtual-MRI/CBCT: A New Paradigm for Image Guidance in Liver SBRT

TH-EF-TRACK 2-06

W. Yang, Keck School of Medicine of USC:

NIH R21, Novel Four Dimensional Magnetic Resonance Imaging to Monitor Pancreatic Tumor Infiltrating Blood Vessels and Tumor Response to Chemoradiation Therapy

TH-EF-TRACK 2-07

N. Wen, Henry Ford Health System:

ACS, Physiological Adaptive Radiotherapy for Prostate Cancer Treatment

TH-EF-TRACK 2-08

K. Brock, UT MD Anderson Cancer Center:

NIH R01s, Anatomical Modeling to Improve the Precision of Image Guided Liver Ablation and Optimization and Evaluation of Anatomical Models of Liver Radiation Response

TH-EF-TRACK 2-09

C. Hruska, Mayo Clinic:

NIH R01, Density MATTERS Clinical Trial

TH-EF-TRACK 2-010

E. Samei, Duke University:

NIH P41, Center for Virtual Imaging Trials

TH-EF-TRACK 2-011

H. Paganetti, Physics Division, Department of Radiation Oncology, Massachusetts General Hospital and Harvard Medical School:

NIH R01s, Fast Individualized Delivery Adaptation in Proton Therapy and Developing Whole-Body Computational Phantoms for Blood Dosimetry to Model the Impact of Radiation on the Immune System

TH-EF-TRACK 2-012

Panel Discussion

In this invited scientific session, current extramurally funded Principal Investigators (PIs) from different areas of Medical Physics will shed light on the high impact scientific research being conducted by AAPM members. During each session, PIs will briefly describe their unique funding mechanisms and pathway to funding to highlight the depth and breadth of funding opportunities available to Medical Physics researchers. AAPM members who are single PIs and those participating as the primary PI in a multiple PI grant will be highlighted. For each grant, the PI will describe the clinical motivation and scientific underpinnings of the work. Specific aims and key initial scientific results will be highlighted along with a look toward future research directions. Finally, each PI will conclude with brief parting comments regarding grant funding to share their perspective and words of wisdom with the AAPM audience.

Learning Objectives:

1. To understand the different grant mechanisms currently funding AAPM members.
2. To learn about the scientific initiatives that have extramural funding.
3. To learn about the grant development process including developing preliminary data and specific aims.

Research funded in part by NIH R01CA204189

Imaging Scientific Session

TRACK 3

Dual Energy and Spectral CT and CBCT

TH-E-TRACK 3-01

Characterization and Optimization of Quantitative Greyscale Images from Photon Counting CT

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Purpose: Radiologists don't like to review extra images. Leveraging the abilities of photon-counting CT (PCCT) by using energy weighting to form a single set of quantitative greyscale images leverages the unique advantages of PCCT while still providing familiar content and workflow for radiologists, i.e. not requiring them to read additional spectral or material-basis images. We evaluated methods to generate such quantitative greyscale CT images. **Methods:** We used CatSim-based simulations of a 120 kVp polychromatic x-ray tube with varying dose coupled with a model of an 8 energy bin PCCT detector within a standard CT gantry. An elliptical water phantom contained inserts (5 cm in diameter) of liver, intestine, and adipose tissue, along with a blood/iodine mixture. The long axis of the phantom was varied from 30 to 50 cm in 2 mm steps. CT acquisitions of the 100 phantoms with independent noise realizations were simulated, and images reconstructed with FBP after applying energy-bin-specific beam-hardening corrections. The 8 energy-bin images were combined using several approaches: uniform, spectral, inverse-spectral, and RMS-optimized weightings. For each case, bias, noise, and RMS error (RMSE) were measured and compared to equivalent results for a standard energy-integrating CT (EICT). **Results:** For the EICT images the Hounsfield Unit (HU) values for the 4 inserts varied as phantom size increased – liver: 52 to 38 HU, intestine: 22 to 12 HU, adipose: -244 to -240 HU, and blood/iodine: 260 to 200 HU. However, for the PCCT greyscale images, HU values for all inserts were essentially constant as a function of phantom size for all energy-weighting schemes. PCCT images had generally higher noise but lower RMSE compared to EICT. **Conclusion:** PCCT can be used to form quantitative greyscale CT images that are invariant to object size and with lower RMSE relative to EICT.

Supported by a research grant from GE Healthcare

TH-E-TRACK 3-02

Development of a Prototype C-Arm Photon Counting Detector CT System for Interventional Imaging: First in Vivo Animal Results

X Ji, S Periyasamy, M Feng, G Chen, P Laeseke, K Li*, University of Wisconsin, Madison, WI

Purpose: This work aimed at bringing the energy-discriminating solid-state photon counting detector (PCD) technology to C-arm interventional systems to enable intraoperative PCD-CT imaging with superior low-contrast detectability and material differentiation capability. This abstract reports the hardware development of a rotating-gantry C-arm PCD-CT system and results from preliminary in-vivo porcine studies. **Methods:** The prototype system was constructed with minimal modification to an existing interventional system using a custom-made mounting device. A CdTe-based PCD strip (51cm×0.6cm; 100μm pixels) equipped with 2 energy thresholds and state-of-the-art anti-charge sharing technology was installed and can be translated in/out of the beam for switching between flat panel detector (FPD)-based imaging mode and PCD-CT mode. Artifacts due to gantry wobbling was corrected using an in-house geometric calibration phantom; the inter-CdTe panel response nonuniformity was corrected by imaging materials of known radiological pathlengths. Low-contrast detectability, image uniformity, and artifact level were characterized using physical phantoms. To explore the potential benefits of C-arm PCD-CT, four swine were imaged by both PCD-CT and FPD-based CBCT at clinical dose (35 mGy) and reduced-dose (7.1 mGy) levels with 7s gantry rotation time. **Results:** Phantom studies demonstrated, at matched collimation (2.5 cm), spatial resolution (8 lp/mm), slice thickness (4.2 mm), and dose level, PCD-CT resolved the 0.3% (3HU) 3mm inserts in Catphan and demonstrated excellent uniformity without any perceivable geometric distortion or ring artifacts. For the in vivo studies, interventional radiologists indicated that porcine abdominal PCD-CT showed a much better delineation of visceral detail and improved soft tissue contrast compared with FPD-CBCT acquired at matched dose levels; the reduced-dose PCD-CT porcine chest images closely matched the quality of the full-dose CBCT images. **Conclusion:** A prototype PCD-CT system was developed to generate high-quality and artifact-free images. In-vivo porcine studies demonstrated potential benefits in soft tissue imaging and radiation dose efficiency when compared to FPD-CBCT.

TH-E-TRACK 3-03

Dual Energy CT Using Flying Focal Spot and An Anti-Scatter Grid as a Filter

S Hsieh¹*, N Bennett², A Wang², (1) Mayo Clinic, Rochester, MN, (2) Stanford University, Stanford, CA

Purpose: Spectral CT is not available on all diagnostic systems because of its hardware requirements. A recent proposal is to achieve spectral CT by combining flying focal spot (FFS) with a fixed grating acting as a filter. We demonstrate that standard anti-scatter grid fabrication technologies can be used for this task, and we estimate the spectral separation that could be achieved. **Methods:** We envision an anti-scatter grid placed 100 mm from the source and focused at a first FFS focal spot. With a 10:1 grid ratio, the second FFS focal spot should be displaced 10 mm from the first spot to maximize spectral separation. We estimated spectral separation assuming 20 cm of water object filtration, 1.5 mm grid thickness, 0.036 mm lamellae at 60 lp/cm, 120 kVp, and several candidate materials. We provide an experimental tabletop system demonstration. A short focal distance grid was not easily available, so the concept was emulated by placing the grid at focus 105 cm from the source and acquiring two sequential scans with a 9.2 cm shift in source position. **Results:** Our experimental demonstration showed separation of multiple materials in a dual energy phantom, including adipose tissue and bone. We estimate 13 keV, 6 keV, and 2 keV spectral separation using a grid with tin, lead, or tungsten lamellae, respectively, with differences due to material K-edge. For comparison, a dual source CT system operating at 100 and 140 kVp provides a spectral separation of 12 keV. **Conclusion:** Anti-scatter grids could be combined with FFS for spectral imaging. The grid would need to be modified to have a short focal distance. A respectable spectral separation can be achieved using tin lamellae.

TH-E-TRACK 3-04

Quantitative Dual-Energy Imaging of Bone Composition on Multisource Cone-Beam CT with Model-Based Artifact Corrections

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Purpose: The recently introduced extremity Cone-Beam CT (CBCT) with a three-source x-ray unit enables single-rotation dual energy (DE) imaging by operating the x-ray tubes at different potential. We investigate the application of this novel DE acquisition mode for Bone Mineral Density (BMD) measurements in extremities with and without metal implants. **Methods:** The extremity CBCT implements three x-ray tubes arranged longitudinally (parallel to the long axis of the extremity). For DE imaging, the peripheral sources were operated at 70 kVp (low energy, LE) and the central source was operated at 100 kVp (high-energy, HE) in an alternating sequence. This yielded 105 views for each of the peripheral LE sources and 210 views for the central HE source; the total CTDI scan dose was ~10 mGy. Experimental studies involved a 16 cm water cylinder with BMD inserts (50 mg/mL, 100 mg/mL, 200 mg/mL) surrounding a Titanium tibial nail. Image domain DE decomposition was applied to estimate BMD (Calcium concentration). LE reconstruction for the decomposition was generated by weighted Filtered-Backprojection of the two peripheral source datasets. Scatter was estimated using accelerated Monte Carlo simulations (107 photons/view+denoising) from preliminary HE and LE reconstructions. To mitigate metal artifacts, constrained three-material decomposition (Ca, Ti and water) was first performed to obtain a model of the implant. The model was used to interpolate projection values behind the metal. Standard two-material DE decomposition was then applied to estimate BMD. **Results:** With the

proposed artifact correction framework, the BMD estimation errors were comparable for phantoms with and without metal: ~30% for the 50 mg/mL insert, ~16% for the 100 mg/mL insert, and ~12% for the 200 mg/mL insert. **Conclusion:** Single-scan DE imaging has been implemented on the multisource extremity CBCT. Model-based artifact corrections enabled accurate BMD estimation even in the presence of metal hardware.

TH-E-TRACK 3-05

Spectrum Optimization for Photon-Counting CT with a Silicon Detector

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Purpose: This study investigated improvement in material decomposition noise and iodine CNR by optimizing the tube voltage and filtration for photon-counting CT with a silicon detector. Silicon photon-counting detectors, which have demonstrated feasibility for human scanning, have unique spectral properties that may benefit from new tube potential and filtration approaches. **Methods:** A simulation study was performed across a range of conditions to identify promising filter and tube voltage combinations. The simulations modeled x-ray transmission through a 20, 30, or 40 cm water background and 1-cm iodine task with concentration varied from 1 to 20 mg/ml. Tube voltages of 80, 100, and 120 kV were modeled. Copper, tin, cerium, dysprosium, ytterbium, and tungsten filters of varying thicknesses were modeled in addition to inherent tube filtration. All spectra were modeled with skin exposure matched to the 120 kV beam without additional filtration. A silicon detector with eight energy bins was modeled assuming realistic detection efficiency and Compton scatter effects. The energy-bin thresholds were previously optimized for each tube voltage assuming no additional filtration. The Cramer-Rao-Lower-Bound material decomposition noise variance was calculated for each case and used to estimate the CNR of the resulting optimal virtual monoenergetic images. Promising filter and tube voltage combinations, considering realistic tube power constraints, were further investigated through photon-counting CT simulations. **Results:** Filters of 0.1-mm copper, 0.065-mm ytterbium, and 0.02-mm tungsten were identified as promising candidates. The study also demonstrated the improved noise and CNR when using 80 or 100 kV compared to 120 kV. For example, 100 kV spectra with 0.1-mm copper filtration demonstrated 12% CNR improvement and ~8% basis material noise reduction for the 20-cm phantom, and 22% CNR improvement for the 40-cm phantom. **Conclusion:** Photon-counting CT with silicon detectors may benefit from lower tube-voltage selection and additional filtration, for example 0.1-mm copper.

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TH-E-TRACK 3-06

Characterization of a Bench-Top Photon-Counting CT System

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Purpose: To characterize a photon-counting CT (PCCT) bench-top system with a high-flux photon-counting cadmium zinc telluride (CZT) detector with 6 energy bins to determine the limiting spatial resolution and the smallest detectable concentration of multiple high-Z contrast agents imaged simultaneously. **Methods:** Spatial resolution in 2D-radiographic images was determined by imaging a QC3 phantom, a QCkV-1 phantom, a custom CT resolution phantom, and a 0.33 mm thick sheet of Cu to obtain a straight edge. For the phantom data, the modular transfer function (MTF) was determined using the Droege and Morin method (1982). For the Cu edge, an oversampled edge spread function was obtained and the MTF calculated. In CT images, limiting spatial resolution was determined using the custom resolution phantom to obtain an MTF. The smallest detectable concentrations of the contrast agents were imaged in 0.2 ml vials placed in a 10-cm cylindrical polyethylene phantom at 120 kVp and 2mA with 2 mm of Al filtration. K-edge images were obtained using K-edge subtraction to separate the contrast agents. The contrast-to-noise ratio was measured in CT images to determine the lowest discernible concentration, which was determined when the CNR fell below 4. **Results:** The limiting spatial frequency in radiographic images was 2.02 lp/mm, determined at 20% of the MTF of the Cu edge. The MTF data from the three phantoms supported this finding. The limiting spatial frequency was found to be 1.31 lp/mm in CT images. The two contrast agents were well separated using K-edge subtraction and the limiting concentrations discernible in CT images were 1.47 mg/ml for gold and 1.62 mg/ml for gadolinium. **Conclusion:** The presented bench-top PCCT system with a spatial resolution of 1.31 lp/mm and the capability to separate multiple contrast agents distinguishable at low concentrations in a single image is a promising tool for future clinical applications.

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TH-E-TRACK 3-07

Iodine Quantification in a Full Field-Of-View Photon-Counting-Detector CT

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Purpose: To assess the multi-energy performance of a full field-of-view (FOV) photon-counting-detector (PCD) CT system in terms of iodine quantification accuracy. **Methods:** Five water phantoms of lateral size 25, 30, 35, 40 and 45 cm mimicking small to large adult body sizes were scanned on a full-FOV PCD-CT system (SOMATOM Count Plus, Siemens Healthcare GmbH) using a multi-energy abdomen protocol (144 x 0.4 mm collimation, 120kV, 0.6 pitch, 0.5s rotation and CAREdose4D with 150 quality reference mAs). The CTDIvol was matched to that of a routine abdomen protocol used in our clinical practice. Vials with iodine solutions at concentrations of 2, 5, 10, 15 and 20 mg/mL were placed inside the water phantoms and scanned. Threshold images at 20-120 keV and 65-120 keV were reconstructed using filtered back projection with a Qr40 kernel at 3 mm slice thickness and 250 mm FOV. Iodine contrast-to-noise ratio (CNR) was measured from the low threshold images for all concentrations and phantom sizes. Iodine and water maps were generated using a prior knowledge aware material decomposition technique. The quantitative accuracy of iodine maps was calculated using root-mean-squared-error (RMSE) for each phantom size. **Results:** Material decomposition enabled successful delineation of iodine inserts from the background (water) in the five phantom sizes. The 25 cm phantom exhibited the highest iodine CNR (average CNR = 33.4), and the CNR reduced by 2.6%, 21.5%, 38.1%, 48.9% for 30, 35, 40 and 45 cm, respectively. The iodine RMSE were 0.19, 0.25, 0.6, 1.12 and 0.99 mg/mL for 25, 30, 35, 40 and 45 cm phantoms, respectively. The average percent error for measured concentrations (against reference values) was 7.04% for all phantom sizes. **Conclusion:** We evaluated the iodine quantification accuracy in a full-FOV PCD-CT system. Accurate iodine quantification was achieved using a material decomposition technique with built-in noise reduction.

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Multi-Disciplinary Scientific Session

TRACK 4

AI Applications in Image Guided Adaptive Radiation Therapy

TH-E-TRACK 4-01

Few Shot Meta Learner for Post-Operative Prostate CTV Style Adaptation

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Purpose: Clinical target volumes (CTV) are contoured for postoperative prostate radiotherapy to treat microscopic disease with the aid of consensus guidelines. Adjustments are made in accordance with a physician's preference/clinical judgement, resulting in large observer-variations (segmentation-styles) that are distinguishable. Although autosegmentation models can be adapted to specific physician styles, this would require sufficient data from each physician for their styles to be learnable. To address this limited sample size issue, we propose a deep learning (DL)-based few-shot meta-learning architecture for CTV style adaptation. **Methods:** We used existing prior information for a specific CTV contouring style in the form of previously segmented patients as support dataset. To train the meta-learning architecture, we simulated 6 different CTV styles (4-training, 2-testing) by varying the inclusion of bladder/rectum/seminal-vesicles. Initial CTV segmentations were performed via an in-house DL model. Then, our adaptive learning architecture was tasked to learn the changes required for new CTV styles. Three-hundred and seventy-three patients treated at a single institution were used. The network consisted of four parts: a shared encoder, style block, averaging layer, and decoder. The encoder extracted features from prior patients and the new patient, the style block evaluated the difference between the DL model CTV and the provided support style, and the average layer averaged features across all prior patients. These were concatenated with the new patient features and decoded to obtain the adapted CTV. **Results:** The performance of the in-house CTV segmentation model on the two unexposed styles for 50 independent test patients were 71.9% and 63.0% dice score coefficient (DSC). With just three prior patients as input to the few-shot network, the mean DSC across test patients' increased to 77.8% and 73.0% for each style, respectively. **Conclusion:** The trained few-shot meta-learner can effectively adapt to new CTV styles.

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TH-E-TRACK 4-02

Multi-Year Clinical Experience with In-House Developed AI Auto-Segmentation for Radiotherapy Planning

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Purpose: To report on the clinical utility of in-house developed deep learning auto-segmentation (DLA) in radiotherapy at our institution. **Methods:** Two DLAs were developed to segment OAR and target volumes in routine prostate and head and neck (HN) radiation therapy. Output contours included bladder, rectum, prostate and seminal vesicles, penile bulb, left/right parotid (LP, RP), left/right submandibular glands (LS, RS), brainstem, and mandible. Both use different network architectures, a DeepLabV3+ pre-trained on ImageNet for prostate, and a Nested-Block Self-Attention method for HN. Training data consisted of 50 and 48 internally curated datasets for prostate and HN respectively. Each DLA is auto-triggered through our software pipeline, Ensemble of Voxel-wise Attributers (EVA), which manages DICOM transfers to our clinically approved high-performance computing cluster, input pre-processing, and RTSTRUCT generation. Clinical utility was gauged retrospectively through periodic random audits of comparison between the DLA output and final contours using published metrics Added Path Length (APL) and Surface DSC (SDSC). **Results:** To date our DLA have segmented 1277 prostate patients (314/25% audited) since June 2019 and 337 HN patients (79/23% audited) since May 2020. Mean APL (cm) and SDSC for each structure were bladder [107.9cm, 0.96], rectum [51.2cm, 0.92], prostate and seminal vesicles [129.1cm, 0.82], penile bulb [7.1cm, 0.94], LP [39.4cm, 0.86], RP [42.9cm, 0.85], LS [22.1cm, 0.82], RS [20.2cm, 0.83], brainstem [28.1cm, 0.88], and mandible [25.1cm, 0.94]. Average total manual contouring for all structures was reduced from 2586 cm to 473 cm, representing an average reduction of 82% across both DLAs. **Conclusion:** We successfully integrated two unique DLAs into routine clinical practice. This resulted in >80% reduction in APL, which has been shown to correlate with time savings. High contour surface similarity (>0.80) was achieved in all structures. This supports potential for artificial intelligence to improve the clinical workflow for segmentation in radiotherapy.

Erin Gillespie is a co-founder of the website eContour.org. Joseph Deasy is a co-founder/shareholder of PAIGE.AI. Michael Zelefsky serves as a consultant for Boston Scientific. Sean Berry/Harini Veeraraghavan held research grants with Varian Medical Systems.

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TH-E-TRACK 4-03

Physician Evaluation of Deep Learning-Based Dose Predictions for Head and Neck Radiotherapy

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Purpose: To predict high-quality, realistic dose distributions for head and neck (HN) VMAT plans that can be used as comparative dose distributions for automated peer review of plan quality. **Methods:** We trained a 3D dilated dense-UNet to predict 3D dose distributions for HN patients. The architecture takes a 24-channel input including CT images, target prescriptions, and normal tissue and target contours. 118 VMAT HN plans with 2-3 dose levels were split into train, cross-validation, and test datasets (3:1:1 respectively). The model was trained on random patches using a custom weighted mean squared error loss function, which assigns a customized weight to each contour. The predicted dose distributions were assessed by a HN radiation oncologist using a 5-point scale (1:No edits needed, 2:Minor edits optional, 3:Minor edits needed, 4:Major edits needed, 5:Fails to meet clinical criteria). A score of 1-3 indicates clinical acceptability. **Results:** The model predicted dose distributions of comparable plan quality to the clinical plans. The mean dose difference between the predicted and clinical dose distributions of the testing dataset was -0.42 ± 0.28 Gy. On average, the predicted target DVH metrics of D1, D95, and D99 normalized to the prescription were within $-1.94 \pm 1.35\%$, $-0.12 \pm 3.55\%$, and $0.64 \pm 6.23\%$ of the clinical plans, respectively. The predicted organ at risk metrics of mean and max doses were within -0.35 ± 1.41 Gy and -0.83 ± 1.85 Gy of the clinical plans, respectively. The predicted dose distributions on the test set were scored as clinically acceptable for 18/21 patients (score 1:5 patients, score 2:9 patients, score 3:4 patients). The other 3 patients received a score of 4 for poor target coverage and target dose fall-off. **Conclusion:** The 3D dilated dense-UNet can predict high-quality, realistic dose distributions for HN VMAT plans. Physician assessment indicates that greater than 85% of predictions are clinically acceptable, supporting the use of dose prediction to QA plan quality.

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TH-E-TRACK 4-04

Scale-Adaptive Convolutional Neural Network for Deformable Image Registration of Lung 4DCT

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Purpose: Deformable image registration between phases in lung 4DCT is essential for understanding respiratory motion pattern and facilitates temporally conformal radiotherapy delivery. Multi-resolution hierarchical strategy is typically used in conventional optimization-based registration to capture varying magnitudes of

motions while avoiding undesirable local minima. A rough concept of the scale is captured in deep networks by the reception field of kernels, and it has been realized to be both desirable and challenging to capture convolutions of different scales simultaneously in registration networks. In this study, we propose a registration network that is conscious of and self-adaptive to motions of various scales to improve registration performance. **Methods:** Dilated inception modules (DIMs) are proposed to incorporate receptive fields of different sizes in a computationally efficient way. Scale adaptive modules (SAMs) are proposed to guide and adjust shallow features using convolutional kernels with spatially adaptive dilation rate learned from deep features. DIMs and SAMs are integrated into a registration network which takes a U-net structure. The network was trained in an unsupervised setting with 15 4DCT scans and then tested using 10 scans in the DIR-Lab dataset, each with 300 annotated landmark pairs. **Results:** The method achieved a (2.52 ± 2.96) mm target registration error on the anatomical landmarks, better than a conventional B-spline method in SimpleElastix and networks without DIM or SAM, with statistical significance. The adaptive dilation rate in SAM corresponded well to the spatially and temporally varying motion scale. The average registration time was 0.42 s for images with size $256 \times 256 \times 96$. **Conclusion:** The introduction and integration of DIMs and SAMs address the spatio-temporal scale heterogeneity problem in an efficient and self-adaptive way and can be an efficacious alternative to the conventional multi-resolution strategy.

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TH-E-TRACK 4-005

Small Convolutional Neural Networks for Efficient 3D Medical Image Segmentation

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Purpose: To substantially reduce the computational overhead of convolutional neural networks (CNNs) to allow broader applicability in resource-constrained environments while maintaining the performance of full-sized networks. **Methods:** We propose a novel architecture modification that keeps the number of feature maps constant at each level of a CNN. Usually, the number of feature maps doubles at each layer of downsampling. This network design strategy is motivated by the similarity between CNNs and multigrid methods for solving partial differential equations. We call a network that uses this proposed design strategy a “pocket network,” or PocketNet for short. We apply our PocketNet strategy to three popular segmentation architectures (U-Net, ResNet, and DenseNet) and compare each of the modified network's performance to its unmodified counterpart on the publicly available Neurofeedback Skull-stripped (NFBS) repository. The dataset consists of isotropic 3D T1-weighted MR images and ground-truth brain masks (i.e., segmentation) from 125 subjects. We used a five-fold cross-validation scheme to train and evaluate each architecture. Predictions for full-sized and pocket-sized networks are evaluated using the Sorensen-Dice Similarity Coefficient (Dice) and the Hausdorff distance. **Results:** The pocket networks averaged a 97% reduction in the number of parameters and achieve segmentation accuracy comparable to the full architectures. The difference in Dice accuracy between the pocket and full architectures for NFBS segmentations is less than 0.003 for U-Net, ResNet, and DenseNet. The mean and maximum Hausdorff distances between the pocket and full architectures differed by less than 0.004 mm and 1.070 mm, respectively. Brain segmentations were visually indistinguishable. **Conclusion:** PocketNets demonstrated faster training and inference times and lower memory requirements while retaining segmentation performance. These savings in time and memory provided by PocketNet can make powerful segmentation architectures more accessible in resource-constrained environments that do not have access to specialized computing hardware.

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TH-E-TRACK 4-006

CBCT-Based Prostate and Organs-At-Risk Segmentation Using Deep Attention Network

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Purpose: To propose a deep attention network for cone beam CT (CBCT)-based multi-organ segmentation for facilitating CBCT-guided adaptive radiotherapy. **Methods:** Both the CBCT and synthetic MRI (sMRI) were used for the segmentation of soft tissue including the prostate, bladder, rectum, and the bones including left femoral head (LFH) and right femoral head (RFH). The synthetic MRI was generated using trained CycleGAN networks. The CBCT and sMRI were separately processed using two U-Nets for feature extraction. The CBCT has better ability in bony structure segmentation while the sMRI has better ability for soft tissue discrimination. It is important to effectively combine the feature maps learnt from both the CBCT and sMRI. To achieve this, a novel deep attention strategy was proposed to integrate the CBCT feature maps with the sMRI feature maps prior to final segmentation by a convolutional neural network (CNN). The deep attention strategy enables the network to focus on salient features that highlight the target organs. The network was trained and tested using 50 patients' datasets using five-fold cross validation. For evaluation, the dice similarity coefficient (DSC) and mean surface distances (MSD) were calculated for the target organs. **Results:** Our results show that the proposed method was able to segment all five organs simultaneously with a high accuracy. The DSCs and MSDs are 0.90 ± 0.06 and 0.94 ± 0.91 mm, 0.94 ± 0.04 and 0.87 ± 0.76 mm, 0.90 ± 0.03 and 0.88 ± 0.73 mm, 0.96 ± 0.04 and 1.16 ± 1.34 mm, as well as 0.95 ± 0.03 and 1.28 ± 1.43 mm for prostate, bladder, rectum, LFH and RFH, respectively. This proposed sMRI-aided CBCT segmentation outperforms CBCT-only based segmentations with better agreement with physicians' manual contours. **Conclusion:** We have developed a CNN network with deep attention strategy to segment prostate and multiple OARs on CBCT images. The accurate sMRI-aided organ delineation warrants further development of a CBCT-guided adaptive radiotherapy workflow for prostate cancer.

TH-E-TRACK 4-007

Automatic CT-Only Multi-Organ Segmentation for CT-On-Rails-Based Prostate Adaptive Radiotherapy

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Purpose: Fast and accurate segmentation of target and organs-at-risk (OARs) is a key requirement in adaptive radiotherapy (ART). In the scenario of CT-on-rails-based ART, the absence of MRI may lead to inaccurate prostate delineation. We propose to develop a CT-only, synthetic MRI (sMRI)-aided dual networks for rapid and accurate multi-organ segmentation for prostate ART. **Methods:** We propose a dual network method that combines the bony structure information provided by CT and superior soft-tissue information provided by sMRI for effective pelvic multi-organ segmentation. Cycle-consistent adversarial networks (CycleGAN) were used to generate CT-based sMRI. Independent features are extracted from CT and sMRI via the dual networks separately: the first network is used to extract features from CT that represent bony structures, and the second network is used to explore features from sMRI that represent the soft tissues. Deep supervision is used to force the extracted CT and sMRI feature maps from hidden layers of each pyramid level to be semantically discriminative. Then, the comprehensive feature maps from corresponding pyramid levels of the first and second pyramid networks are combined and processed via several deconvolution layers and attention gates. The attention gates further highlighted the informative elements of combined feature maps to well differentiate different organ tissues. The proposed method was trained and evaluated using a cohort of 140 prostate patients. **Results:** The Dice similarity coefficients and mean surface distances between our results and ground truth were 0.95 ± 0.05 , 1.16 ± 0.70 mm; 0.88 ± 0.08 , 1.64 ± 1.26 mm; 0.90 ± 0.04 , 1.27 ± 0.48 mm; 0.95 ± 0.04 , 1.08 ± 1.29 mm; and 0.95 ± 0.04 , 1.11 ± 1.49 mm for bladder, prostate, rectum, left and right femoral heads, respectively. The mean center of mass distances was within 3 mm for all organs. **Conclusion:** We demonstrated the feasibility of sMRI-aided dual networks for multi-organ segmentation on pelvic CT images. This method could reduce clinical workload and facilitate the development of CT-on-rail-based adaptive radiotherapy.

Therapy Scientific Session TRACK 5 Radiation Dose Calculation Algorithms

TH-E-TRACK 5-01

A Feasibility Study for Deep Learning-Based Efficient Monte Carlo Dose Calculation Method in Radiotherapy

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Purpose: We have proposed a Deep Learning-based Monte Carlo (MC) dose denoising framework, which innovatively uses MC statistical uncertainty (SU) distribution in combination with patient CT images for denoising to obtain accurate dose maps. **Methods:** First, we generate a database of MC dose maps and corresponding SU distributions for low- to high-histories using MC calculation engine. Then train the model to map the low-histories dose to high-histories dose maps. At the core of this framework is a multi-channel denoising network, which has multi-channel inputs with CT images and low-histories MC dose and its corresponding SU distributions, and outputs a denoised dose map. Finally, we evaluated the feasibility of the model to obtain high precision dose map through evaluation metrics MAE, ME, GPR, DVH, and dose maps. **Results:** The dose maps show the denoised dose is more consistent with the structural distribution shown in CT images, especially in the high-dose non-uniform areas. DVH curves provide more accurate target coverage and OARs distribution information. The predicted results effectively eliminate singularities in the input dose, and match better with the ground truth in both OARs and PTVs areas. The global prediction performance is also confirmed in the gamma maps. In the quantitative comparison results, the predicted dose demonstrated substantially improved match to the ground truth, with average GPR 99.38%±3.9%, the global ME and MAE were -0.25%±0.12% and 0.83%±0.33% respectively. The accuracy improvement between the input dose and the predicted dose can be up to tenfold. **Conclusion:** By using the SU distribution with dose difference information and CT images with electron density and anatomical structure information, the model better learned the potential inaccuracies in the input, further improving the accuracy of dose denoising. Our framework more efficiently obtain accurate dose distribution, to help physicists make more informed decisions when planning and evaluating treatment plans.

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TH-E-TRACK 5-02

A Generative Adversarial Network (GAN) Based Monte Carlo Model for a Biology-Guided Radiation Therapy Machine

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Purpose: Phase space (PHSP) is commonly used as a source model in Monte Carlo (MC) simulation. It is cumbersome and inconvenient to use despite the outstanding simulation accuracy. This study aims to model a biology-guided radiotherapy (BGRT) machine by a convenient and storage-saving technique using: generative adversarial networks (GAN). **Methods:** The BGRT machine is a novel radiotherapy machine that irradiates target using a 64-leaf binary MLC system. Each leaf projects a 0.625 cm opening along the x direction at 85 cm SAD. Two pairs of Y jaws move simultaneously to open 1 cm or 2 cm at isocenter. A series of PHSP sources below the collimation system were generated using BEAMnrc at field sizes from 0.625 x 2 cm² to 10 x 2 cm². For each field size, a GAN model was trained from the collimated PHSP using a python tool, GAGA-PHSP, which implemented a Wasserstein GAN with three hidden layers in the generator and discriminator. The training PHSP was 0.5 GB – 9 GB in size, depending on the field size. To validate the model, dose distributions in water were simulated with PHSP and GAN sources and compared in terms of percentage depth dose (PDD) and relative planar dose. **Results:** The difference of %dd(10) simulated with PHSP and GAN sources is 0.42%±0.4%. The passing rate of 1%/1mm gamma analysis of the relative dose distribution in transverse, sagittal, and coronal planes are 92.9%±4%, 96.9%±3%, and 96.2%±3.7%, respectively. The GAN-based model can generate unlimited particles and is only 40 MB in size, saving file storage by a factor up to 200. The average training time is 5 hours. **Conclusion:** The GAN-based MC model can simulate the BGRT Linac accurately. The high adaptivity, storage-saving, and time-saving features make this model potentially useful in the simulation of a new Linac instead of PHSP.

TH-E-TRACK 5-03

Development of a Virtual Source Model for Monte Carlo Based Independent Dose Calculation for Varian Linac

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Purpose: Monte Carlo (MC) simulation is often based on phase-space files (PSF), as they are able to accurately represent particle characteristics. However, the large-sized PSF creates a bottleneck for speeding up calculations, as the reading of the PSF takes appreciable time. In addition, the number of independent particles is limited by the PSF, preventing further reduction of statistical uncertainty. The purpose of this study is to develop and validate a virtual source model (VSM) to address these limitations. **Methods:** Particles from existing phase space files for Varian 6X, 6FFF, 10X, and 10FFF beams were tallied and analyzed. The particle density distribution, energy spectrum, particle direction, and the correlations between characteristics were computed. Virtual phase space models were created for these energies. The VSM approach was validated with water tank simulations as well as clinical cases against the original PSF. Computation time and gamma index passing rate were recorded for each case. **Results:** The new VSM only takes 2MB of disk space for each beam energy, compared to 50GB for PSF. The VSM approach reduces the calculation time by 10 mins on our server. At 5% MC uncertainty, the VSM approach was 45% faster in water tank simulations and reduces the calculation time for most clinical cases from ~15 mins to ~1.5 mins. Gamma index passing rates remain high at 5% statistical uncertainty for both approaches, with 98.5% ± 1.0% for VSM and 99.1% ± 0.9% for PSF. **Conclusion:** We have created VSM model for Varian Linacs. The VSM approach significantly saves computation time without sacrificing accuracy for second dose calculation.

Dr. Castle and Dr. Feng are employees of Carina Medical LLC. Dr. Chen is a co-founder of Carina Medical LLC.

TH-E-TRACK 5-04

Microdosimetry of Radionuclides Using Geant4 Simulation Considering the Auger Electron Contribution

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Purpose: The recent surge in the interest and development of radiopharmaceutical therapy (RPT) agents requires the investigation of microdosimetry due to the short range of emissions and heterogeneous activity distributions. Monte Carlo microdosimetric simulations can accurately account for the heterogeneous activity distributions that can be present at the cellular level and will aid in the development of RPT treatment planning by accounting for the tumor microenvironment. This study created a model of an idealized tumor microenvironment using the Monte Carlo code, Geant4. **Methods:** Geant4 simulations were compared cellular S-values of I-131, Lu-177, Y-90, and Br-77 to MIRD by modeling the tumor environment as an agglomeration of spherical cells. The tumor microenvironment was simulated as a cluster of liquid spherical 5 µm radius cells in both hexagonal and lattice configurations. Both source and target cellular S-values were calculated using Geant4

and validated using MIRDcell. Because the radionuclides investigated have different decay characteristics, physics lists, decay relative functions, and electron scattering cross-sections were analyzed using Geant4. **Results:** The differences between Geant4 and MIRD were under 10% for I-131, Lu-177, and Y-90 and 15% for Br-77. The ARMflag function was required for accurately model Auger electron emissions. Appropriate physics lists were required for different radionuclides to account for the different emitting radiation energy ranges and interactions based on each decay scheme. Accordingly, Livermore physics was applied to Y-90, DNA_option8 was applied to I-131 and Lu-177, and DNA_option7 was applied to Br-77. Lastly, the electron scattering cross-sections, de-excitation, and fluorescence process were also considered. **Conclusion:** A Geant4 microdosimetric tumor model was validated by a standard MIRD configuration. This tumor model is flexible and extendible to more realistic tumor cell and source activity configurations. Appropriate microenvironment modeling and radionuclide physics considerations can provide more optimal RPT treatment plans and account for more realistic dosimetric phenomenon.

BB is a co-founder of Voximetry, Inc., a nuclear medicine dosimetry company in Madison, WI.

TH-E-TRACK 5-05

Clinical Transition from AAA to Acuro XB: A Comprehensive and Treatment Site-Specific Report for Dose Comparison, Plan Analysis and Strategic Deployment

X Zhong*, Y Park, Y Zhang, Z Trivedi, J Wu, L Chen, S Kazemifar, A Godley, S Jiang, M Lin, UT Southwestern Medical Center, Dallas, TX

Purpose: Accurate dose calculation algorithms, such as AcurosXB (AXB) and Monte Carlo (MC), have become the only dose calculation engine available for some new adaptive therapy systems and accelerated the need to transition to AXB/MC. In this work, we describe the approach to transition from AAA to AXB in a large institution. **Methods:** Phase I implementation is in physics side, a large scale AAA plan recalculation and reoptimization study was performed with AXB for total of 205 cases covering seven disease sites. Gamma analysis of 1%/1mm, 2%/2mm and 3%/3mm and dose-volume-histogram (DVH) metric comparisons were performed to quantify the systematic dosimetric impact for the individual disease sites and develop the proposal to handle the dosimetric impacts. In phase 2 implementation, the representative cases in each disease site were replanned with AXB. The treated AAA plan recalculated with AXB, representing prior practice in AXB-scale; the AXB plan recalculated with AAA, representing the future practice in AAA-scale; along with the statistics from phase 1 were presented to the physicians, planners, and therapists of each disease-oriented team (DOT). New guidelines and workflows were developed based on the feedback obtained. **Results:** Breast plans demonstrated the lowest gamma passing rate followed by bone and lung. Gamma failure was observed in lung, cortical bone, and heterogeneity interfaces. In the DVH and replan studies, higher D_{0.03cc} and heterogeneity on PTV was found for all sites, and no significant impact was found in OAR doses. Planning timeline was adjusted for case suffering from AXB planning such as larynx and lung SBRT. Metal object handling starts from the physician consulting phase and added into the physician consultation questionnaire to allow time to obtain the material information. **Conclusion:** Transition of dose calculation engine requires team effort. The site specific findings and the implementation process presented in this work can be used as reference for other institution.

TH-E-TRACK 5-06

Dosimetric Comparison of HDR Surface Applicator Brachytherapy Between TG-43 and TG-186 for Angiosarcoma of the Scalp and Face

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Purpose: Cutaneous angiosarcoma is a rare cancer that forms in the blood and lymph vessels, occurring most often in the scalp and face -- regions involving complex topography. Surface applicator brachytherapy (SABT) may be effective in conformally treating such lesions while minimizing dose to organs at risk (OARs). TG-186 presents a model-based calculation method accounting for tissue heterogeneities. This study compares the TG-43 and TG-186 methodologies, assessing the dosimetric impact of different density materials on dose to angiosarcoma targets and OARs. **Methods:** 12 previously treated angiosarcoma plans were evaluated. Radio-opaque wires marked the tumor volume as well as a 5cm clinical margin expansion at CT simulation. Treatment planning was performed using Oncentra Brachy 4.5.3. 100% isodose was prescribed to 3mm depth in the marked target, allowing up to 8mm depth locally where visible evidence of disease on the CT. Contoured OARs include brain as well as bilateral orbits, lenses, lacrimal glands, cochlea, and parotids. The 100% TG-43 isodose volume was designated as the planning target volume (PTV) for comparison. The plans were recalculated using Oncentra's Advanced Collapsed-cone Engine (ACE) algorithm based on TG-186's recommendations. HU-based densities were applied with unchanged dwell times. A paired Wilcoxon signed-rank test was utilized to compare dose differences between the two calculations. **Results:** Using ACE compared to TG-43, the PTV median V_{100%} decreased 19.8% (p=0.001) and the D_{90%} decreased 7.1% (p=0.001). All OARs except for the parotids also showed decreases in median dose (p<0.05): Brain D_{max} 5.8%, V_{80%} 91.3%, V_{50%} 22.3%; mean orbit 9.5%, lens 11.0%, lacrimal gland 9.4%, and cochlea 7.4%. **Conclusion:** 12 scalp angiosarcoma SABT plans were recalculated using TG-186 formalism. Calculations using the TG-43 model are shown to underestimate doses for targets and OARs for superficial targets. Future studies will be required to evaluate any clinical impact of transitioning to TG-186 calculations.

TH-E-TRACK 5-07

Generation of 3D Dosimetric Reference Datasets Using Model-Based Dose Calculations for COMS Eye Plaque Brachytherapy

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Purpose: To evaluate the dosimetry of brachytherapy treatments for intraocular tumours by comparing results from several Monte Carlo (MC) systems and one commercial model-based dose calculation algorithm (MBDCA). This work is done in order to develop test cases for adopters of MBDCA. The scenario investigated here is the treatment of ocular melanoma using a Collaborative Ocular Melanoma Study (COMS) 16 mm eye plaque loaded with 13 model 6711 ¹²⁵I seeds. **Methods:** The levels of agreement among four MC codes (EGSnrc_eyes_brachy, Geant4 ALGEBRA, Penelope2014, and MCNP6) were evaluated by comparing the results from various test cases. Some comparisons were also done using a research version of the OncentraBrachy/ACE commercial TPS. The test cases included: i) modelling the 13 seeds individually, ii) combined in water positioned within the plaque, iii) the full COMS plaque in a water phantom, and iv) the full plaque in a realistic eye phantom. Local and global dose comparisons were made. **Results:** For all test cases, local agreement of ALGEBRA, MCNP, and Penelope with eyes_brachy was within 2% and global agreement was within 1%. These values are consistent with the typical uncertainties expected in the low energy photon domain. The presence of the fully modelled plaque (including metal backing and plastic insert as opposed to just the seeds) increased variations between the codes. TPS doses show larger disagreements, with local dose differing by 2% on average and by as much as 8% at some points. **Conclusion:** This work indicates that these MC codes exhibited excellent agreement for these test cases both in water and within an eye phantom, justifying their use as a benchmark. Further work will investigate discrepancies between the TPS and MC calculations, towards clinical applications of advanced dose evaluations.

Funding: Natural Sciences and Engineering Research Council (NSERC) of Canada, Canada Research Chairs, Ontario Ministry of Research and Innovation. COI: none.

Therapy Scientific Session

TRACK 6

SRS/SBRT: Segmentation, Dose Calculation, and Treatment Planning

TH-E-TRACK 6-01

Cardiac Radio-Ablation Target Delineation On 3D Electro-Anatomical Maps

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Purpose: Cardiac radio-ablation (CRA) target delineation and planning process evaluation using a novel multimodal 3D cardiac segmentation software. **Methods:** In this 3D-EAM-based workflow, the EP delineates the target on the epicardial surface of the left ventricle in the treatment planning interface of the proprietary software MUSIC (Liryc, Bordeaux, Fr). To guide the EP in the delineation process, the electroanatomic mapping (EAM) and cardiac- and fibrosis-segmented contrast-enhanced cardiac CT are in the same frame-of-reference. The resulting target is transmurally expanded and exported as a Dicom RT structure set in registry with the cardiac CT. In the subsequent planning process, the cardiac CT is registered with the planning CT to guide the ITV- and PTV delineation followed by treatment plan optimization. This 3D-EAM-based target delineation workflow replaces the axial-view-based targeting approach for CRA which relies on target delineation on CT scans without EAM data and fibrosis data being in registry. To quantify the time- and cost savings of our 3D-EAM-based workflow, we compared it to a previously performed time-driven activity-based costing (TD-ABC) analysis of the axial-view-based workflow. **Results:** The 3D-based targeting approach takes advantage of the EP's expertise and knowledge of the VT site. It removes the need for (i) the mental abstraction of the target from 3D to 2D, (ii) the use of external software to manually transfer EAM mapping and fibrosis segmentation to the treatment planning system (TPS), (iii) numerous follow-up consultations (further reducing total coordination efforts). Thus, the targeting process is more time-efficient and subject to less variability. TD-ABC demonstrates saving in overall procedural costs of 10.62% (\$1234.07), 7.25% (\$767.96) and, 8.80% (\$909.45) for the so far evaluated cases. **Conclusion:** The proposed workflow improves the time-efficiency and reliability of the targeting and planning process, leading to a much more streamlined workflow for CRA.

TH-E-TRACK 6-02

Dose Kernel Decomposition for GammaKnife and GammaPod Treatment Planning

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Purpose: Pre-calculation of accurate dose deposition kernels for GammaKnife (GK) and GammaPod (GP) treatment planning can be very time consuming with an enormous number of candidate shots. To address this issue, we propose a novel decomposition model of dose deposition kernels for GK and GP treatment planning. **Methods:** The dose deposition kernel for an arbitrary shot position is modeled as the product of a shift-invariant kernel and spatially variant scale factor. The shift-invariant kernels, one for each collimator, are pre-calculated on the commissioning phantom using the Monte Carlo (MC) method. The spatially variant scale factor is defined as the ratio of the mean tissue maximum ratio (TMR) at the candidate shot to that at the commissioning shot. The scale factors or normalized mean TMRs can be calculated on the whole treatment volume through parallel-beam ray tracing followed by averaging over all source directions. The proposed dose calculation was compared with the MC calculation for various shot locations and spot sizes on the phantom and twelve clinical plans. **Results:** The kernel decomposition (KD) model achieved similar doses to those calculated by the MC method. The Gamma index passing rates in 2%, 2 mm criteria were greater than 99% for the phantom study and clinical plans. The calculation time of the KD approach was within sub-seconds, which was several orders of magnitude faster than the MC method. **Conclusion:** The scale factors can be calculated very efficiently when implemented in GPU and only needs to be calculated once before treatment planning. The proposed dose calculation can provide accurate dose and is very fast with kernel shifting and scaling. In addition, both dose and derivatives can be calculated in real-time without pre-calculating the dose deposition kernels, which is critical to efficient search of optimal plans for GK and GP.

This work was supported in part by NIH grants (R01 CA235723, R01 CA218402).

TH-E-TRACK 6-03

Evaluation of Dose Calculation Accuracy of Intermediate-To-Low Dose Spillage in Single-Isocenter Multitarget Radiosurgery Using 3D Dosimetry

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Purpose: Single-isocenter multitarget (SIMT) stereotactic radiosurgery using volumetric modulated arc therapy provides rapid delivery of high dose radiation to multiple brain metastases. However, the intermediate-to-low dose spillage to normal brain is unclear. Dosimetric accuracy of the dose spillage calculation in treatment planning system involves modeling of out-of-field doses, and multileaf collimator characteristics, which is challenging to be evaluated using 1D/2D dosimeters. This study presents a comprehensive 3D dose comparison between calculated and measured dose distribution with various treatment geometries. **Methods:** Ten SIMT plans for treating multiple brain lesions with different sizes and target separations were created using HyperArc treatment planning system (Varian Medical System). Hybrid plan verification of each plan was generated on computed tomography images of the 3D dosimeters, PRESAGE (Heuris Inc.). Hybrid plans were then delivered to the dosimeters using a TrueBeam (Varian Medical System) linear accelerator. Using fiducial registration, target and normal tissue structures were registered to the measured dose image. A comprehensive comparison of the 3D dose distribution between measurement and calculation was performed. **Results:** Most of the gamma failing points are at the intermediate dose regions (50-70%). Measured 3D dose distribution presents higher dose spillage to normal tissue up to 5-10% in regions connecting the two close targets. In addition, dose discrepancies were observed in off-axis low dose regions where out-of-field dose is underestimated by the treatment planning system. Using convectional portal dosimetry, observed errors in 3D measurement were washed out due to dose summation. **Conclusion:** A comprehensive view of the dose discrepancies between calculated and measured inter-target dose spillage in different SIMT treatment geometries was presented. The results provide insights into the planning strategy of SIMT treatments and lay the path for future comparison of Gamma Knife to linear accelerator radiosurgery using 3D dosimetry.

The dosimeters were provided by Heuris Inc. free of charge.

TH-E-TRACK 6-04

Which Metric Quantifies Dose Fall-Off for SRS Treatments of Brain Lesions?

T Nano^{1*}, D Capaldi², C Chuang², L Wang², S Soltys², L Boreta¹, S Braunstein¹, L Ma¹, (1) University of California, San Francisco, San Francisco, CA, (2) Stanford University, Stanford, CA

Purpose: Dose fall-off guidelines for stereotactic radiosurgery (SRS) are critical for ensuring normal brain sparing. Two well-known but distinctive metrics are (1) dose gradient index (DGI) and (2) Paddick gradient index (PGI). We investigated the intrinsic correlations and the pros and cons of these two indices for SRS

treatment of brain lesions. **Methods:** Two cohorts of patient cases with multiple brain lesions from two institutions were treated with CyberKnife (n=72, 202 lesions, volume $2.94 \pm 6.99 \text{ mL}$ [range: 0.01–41.0]) and GammaKnife (n=435, 1983 lesions, volume $1.23 \pm 2.77 \text{ mL}$ [range: 0.01–28.89]). Dose metrics were extracted to compute DGI, where 100 is considered “ideal” and decreases with worst dose fall-off, and PGI, which increases with worst dose fall-off. Simple linear regression of DGI and PGI was compared for both CyberKnife and GammaKnife cases, and Fisher Z-transformation was used to compare the correlation coefficients. **Results:** For CyberKnife cases, targets smaller than 1 mL have DGI and PGI values of 89.96 ± 17.67 and 8.51 ± 4.14 respectively, and targets larger than 1 mL have values of 58.18 ± 19.59 and 4.26 ± 2.41 respectively. GammaKnife cases, targets smaller than 1 mL have a DGI and PGI value of 108.61 ± 8.94 and 4.03 ± 2.50 respectively and targets larger than 1 mL have values of 78.73 ± 18.40 and 3.08 ± 0.85 respectively. DGI was found to positively correlated with PGI (slope=0.2626, $R^2=0.002$, $p=0.7039$) for CyberKnife cases and negatively correlated with PGI for GammaKnife cases (slope=-0.9163, $R^2=0.009$, $p<0.0001$). CyberKnife and GammaKnife plan relationship between DGI and PGI were found to be different ($p=0.0139$) indicating the DGI is flawed in quantifying dose fall-off. **Conclusion:** PGI was found to best describes dose fall-off, whereas, DGI was found to be only weakly correlated with PGI and indicates that it is not a direct measure of dose fall-off. Guidelines for dose fall-off in SRS treatments of multiple brain lesions should be provided exclusively using PGI, with special consideration for small lesions and complex targets.

TH-E-TRACK 6-05

Demonstrating the Practical Limitation of a Previously Validated Knowledge-Based Planning Model for SIB Lung SBRT of Large (> 5 Cm) Tumors

J Visak*, A Webster, M Kudrimoti, R McGarry, M Randall, D Pokhrel, University of Kentucky, Lexington, KY

Purpose: An in-house lung SBRT knowledge-based planning (KBP) model was previously validated and deployed on a limited basis to support non-coplanar VMAT treatments for centrally-located NSCLC tumors. Increasingly at our institution, lung SBRT is delivered via a simultaneous integrated boost (SIB) approach to escalate dose to hypoxic tumor cores in tumors >5cm. This study demonstrates a practical limitation of a robust KBP model via SIB-SBRT of large tumors. **Methods:** Clinically deployed KBP model trained with 86 patients was adapted by reconfiguring DVH-estimate objectives. Five complex validation cases (average tumor volume, 189.7cc) were selected and reoptimized with risk-adapted KBP model. Patients received 40-50Gy to PTV in 5 fractions using 3-6 non-coplanar 6X-FFF beams. An inner GTV (1.0 cm interior to ITV) was escalated to 60Gy for a larger hotspot. Plans were reoptimized with KBP and renormalized for identical PTV coverage including escalated GTV dose. Plans were compared relative to RTOG-0813 criteria including target metrics, OAR dosing and optimization time. **Results:** KBPs were statistically similar and RTOG-0813 compliant, but clinically inferior to clinical plans due to lessened DVH-estimation capability with larger PTVs. KBPs were produced on average in 1.8 ± 0.8 hours and 2.4 optimizations with minimal planner burden and modest manual intervention. For similar conformity and intermediate dose-spillage, KBPs demonstrated mean GTV dose escalation of 1.7Gy ($p=0.08$). OAR dosing differences were statistically insignificant with minimal plan complexity costs. KBPs on average spared maximum bronchial tree dose 0.04Gy ($p=0.87$) but increased maximum heart dose by 1.0Gy ($p=0.42$). **Conclusion:** KBP is a powerful tool that can assist with planning in lung SBRT. Complex cases with tumors >5 cm may not be ideal for using KBP model despite potential RTOG compliance. For non-urgent scenarios, use of the KBP model should be limited in complex cases. Further development of sub-model may support complex SIB-SBRT with larger tumors.

TH-E-TRACK 6-06

Radiobiological Analysis of Isotoxic SBRT Simultaneous Integrated Boost of Multi-Parametric MR Defined Dominant Intraprostatic Lesions

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Purpose: To assess the utility of isotoxically boosting dominant intraprostatic lesions (DIL) by analyzing TCP and rectal NTCP with and without a DIL simultaneous integrated boost (SIB) using an MR-only workflow. **Methods:** A dosimetric study was performed on 10 patients with biopsy-proven prostate cancer who underwent T2, mDixon, and diffusion-weighted imaging on a 3T MRI. A physician contoured the DIL based on T2 and apparent diffusion coefficient (ADC) images. For planning, synthetic CT images were generated from fat and water mDixon sequences using a k-means segmentation-based method for assigning HU values to predefined tissue types. SIB and non-boosted plans were created for each patient. The entire prostate (PTVprostate) was prescribed 36.25Gy in 5 fractions. The PTVDIL was isotoxically boosted to the maximum dose allowed while maintaining the non-boosted rectal NTCP. Prescribed dose and rectum constraints were based on the hypo-FLAME and 50-Gy SBRT DIL boost clinical trials. The linear-Poisson tumor control probability (TCP) radiobiological model and Lyman-Kutcher-Burman (LKB) model for normal-tissue complication probability (NTCP) were employed. Radiobiological calculations were done with BioSuite software (Uzan et al 2011) to determine individualized TCPs based on each patient's ADC map and clonogen density. **Results:** Clonogen density in the DIL ranged from $4.4\text{E}08$ – $1.2\text{E}09$ clonogens/cc. Mean dose and D95 to the SIBs were 54.5 ± 1.9 Gy and 49.4 ± 2.7 Gy. The average TCP was $99.6 \pm 0.6\%$ with the SIB compared to $96.0 \pm 0.9\%$ without the boost. The average NTCP for incidence of late rectal toxicity grade ≥ 2 was $4.4 \pm 2.9\%$ (1.0–9.2%). Rectal doses to 1cc were within the protocol guidelines: 36.3 ± 0.7 Gy and 36.6 ± 0.9 Gy between the standard and SIB plans respectively. **Conclusion:** We demonstrated the viability of MR-only SBRT treatment planning for prostate cancer with an isotoxically boosted DIL. DIL dose was escalated to an average D95 of 49.4Gy, while maintaining rectal toxicity limits and improving overall TCP by 4%.

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TH-E-TRACK 6-07

Using Deep Learning Framework (pytorch) for Circular Cone Treatment Planning of CyberKnife System

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Purpose: With robot-controlled linac positioning, the CyberKnife system significantly increases the freedom in radiation beam placement, but also imposes more challenges on treatment plan optimization. This study proposed to use the deep learning framework (pytorch) for circular cone treatment planning of CyberKnife system. **Methods:** The candidate beams were initialized by simulating the process of covering the entire target volume with equivalent beam taper. For each voxel, the total dose was the weighted summation of the dose contributed by all beams. This dose summation was topologized into a simple feed-forward neural network, and the optimization was converted into network training. The pytorch framework, capable of automatic differentiation and dynamic adjustment, was utilized to train the network. The least absolute shrinkage and selection operator (lasso) and group lasso regularization terms were utilized to reduce the number of beams and nodes, respectively. **Results:** This method was validated and further compared with commercially available MultiPlan system on one typical lung case. This method protected the organs at risk (OARs) more effectively: the generated plan was of lower maximum dose within OARs. The homogeneity index (ratio of maximum dose to prescription dose, Dp) was reduced from 1.15 to 1.10. The volume covered by 50% (100%) Dp (V50 and V100) was reduced from 78.3cc to 59.3cc (14.6cc to 13.2cc), indicating more rapid dose fall-off. The number of nodes (beams) was reduced from 69 to 46 (146 to 93). In addition, the optimization time was shortened from around 30min to 52.3s. **Conclusion:** In conclusion, the generated treatment plan achieved more satisfactory dose distribution with fewer nodes and beams. Besides, the optimization was 1–2 orders of magnitude faster than the MultiPlan system. Using the deep learning framework for treatment planning optimization has the potential to be used clinically.

The work was supported by the National Natural Science Foundation of China (11875320 and 81801799).

SAM Therapy Educational Course

TRACK 7

Advances in Tomotherapy Treatment and QA - New Guidelines from TG 306

TH-E-TRACK 7-00

Advances in Tomotherapy Treatment and QA - New Guidelines from TG 306

Moderators: Quan Chen, University of Kentucky, Lexington, KY and Yi Rong, Mayo Clinic Arizona, Phoenix, AZ

TH-E-TRACK 7-01

J. Smilowitz, University of Wisconsin:

Updates to Tomotherapy Delivery Platform Since TG 148

TH-E-TRACK 7-02

Q. Chen, University of Kentucky:

Update of Tolerance Limits, Frequencies and QA Methodology for Machine QA

TH-E-TRACK 7-03

X. Qi, UCLA School of Medicine:

Update on Imaging, Treatment Planning and Patient Specific QA

TH-E-TRACK 7-04

J. Burmeister, Wayne State University, Karmanos Cancer Center:

Risk Based Assessment for Tomotherapy Delivery (FMEA Analysis)

Since the publication of Task Group 148 in 2010, the TomoTherapy platform has undergone substantial development involving treatment delivery, image guidance, motion management, and treatment planning systems (TPS). New hardware and software capabilities have dramatically changed its delivery and planning characteristics. Task Group 306 was formed to provide guidance on the implementation, management, and quality assurance for these technological advances. In this session, the work of Task Group 306 will be presented. We will start by summarizing changes to the TomoTherapy platform since 2010. The impact on clinical delivery and QA practices, including QA tests and tolerance limits will then be presented. Finally, results of an FMEA analysis performed by the task group members on the TomoTherapy treatment planning and delivery processes will be presented.

Learning Objectives:

1. Review updates/changes to tomotherapy delivery and planning since TG 148.
2. Present updated tolerance limits, frequencies and QA tests for tomotherapy and Radixact delivery systems
3. Present updated imaging, treatment planning and patient specific QA
4. Learn about a performing risk bases assessment for tomotherapy (FMEA analysis) based on clinical practice.

SAM Partners in Solutions Imaging

TRACK 8

Imaging: Artificial intelligence in Medical Imaging

TH-EF-TRACK 8-00

Imaging: Artificial Intelligence in Medical Imaging

Moderators: Yu Liu, Medical College of Wisconsin, Milwaukee, WI and Troy Crawford, Medical College of Wisconsin, Milwaukee, WI

TH-EF-TRACK 8-01

Y. Liu, Medical College of Wisconsin:

Artificial intelligence in Medical Imaging

TH-EF-TRACK 8-02

T. Crawford:

Artificial intelligence in Medical Imaging

TH-EF-TRACK 8-03

B. Rasolzadeh, Arterys:

The Journey from Code to Clinic: The Challenges of Bringing AI into Medical Practice

TH-EF-TRACK 8-04

R. Jones, Imagen Technologies:

Deep-Learning Systems for Fracture Detection in Musculoskeletal Radiographs

TH-EF-TRACK 8-05

B. Nett, GE Healthcare Technologies:

Deep Learning Image Reconstruction for CT: Principles, Performance and Clinical Applications

TH-EF-TRACK 8-06

Q&A

In the last a few years, diagnostic medical imaging has emerged as one of the most promising areas of healthcare innovation that artificial intelligence (AI) has clinical applications, including image processing and interpretation. AI is in general a technology involving computer science application in creating systems for tasks typically require human intelligence, and the term AI is also used when a technology simulates cognitive functions such as learning and problem solving. Machine learning (ML) is a subfield of AI involving all approaches that allow computers to learn from data without explicit programming, it utilizes computational models and algorithms mimicking the architecture of biological neural network in brain, which is defined as artificial neural network (ANN). When ML is used in medical imaging, it is narrowed to recognize patterns in imaging by using many example images to find the pattern with computer algorithms. On the other hand, deep learning (DL) is part of ML representing a branch of existing AI techniques. DL uses neural networks with many layers, and convolutional neural networks (CNNs) are common DL architectures for images, especially for image classification tasks.

Current and potential AI applications in diagnostic radiology may include improving image quality, reducing radiation dose, decreasing MRI imaging time, enhancing technical protocols, faster and standardized detection and characterization of image findings, and workflow improvement. With an ever increasing amount of data

and the potential use of AI, higher efficiency in diagnostic radiology will be available in the near future.

Learning Objectives:

1. Understand the basics concepts of Artificial Intelligence (AI) and its relation with Machine Learning (ML) and Deep Learning (DL).
2. Understand primary components of DL architectures and neural network.
3. Understand selected AI applications in diagnostic radiology.

SAM Professional Symposium

TRACK 1

Expanding the Scope of the Medical Physics Profession: Developing and Examining Patient-Facing Roles and Responsibilities

TH-F-TRACK 1-00

Expanding the Scope of the Medical Physics Profession: Developing and Examining Patient-Facing Roles and Responsibilities

Moderator: Todd Atwood, UC San Diego, La Jolla, CA

TH-F-TRACK 1-01

T. Atwood, UC San Diego:

Developing Patient-Facing Roles for Medical Physicists to Advance Patient Care

TH-F-TRACK 1-02

J. Burmeister, Wayne State University, Karmanos Cancer Center:

The Physicist's Role in Patient-Centered Communication

TH-F-TRACK 1-03

S. Geneser, University of Washington:

Implementing Patient Physics Consults: The University of Washington Experience

TH-F-TRACK 1-04

D. Brown, University of California, San Diego:

A Clinical Communication Skills Development Program for Medical Physicists

TH-F-TRACK 1-05

Audience Q&A

Patients routinely view diagnostic imaging studies and radiation therapy as complicated medical procedures. Misconceptions about radiation and confusion surrounding complex imaging and treatment equipment frequently result in patients having questions or concerns about the technical aspects of their care. These circumstances can result in patient-related distress and anxiety, delay the imaging necessary for a diagnosis, and even negatively impact outcomes, particularly in the case of radiation therapy.

Medical physicists are uniquely qualified to address patient questions and concerns related to the technical aspects of their care. The current training paradigm for imaging and therapy medical physicists produces analytical patient-care professionals with a propensity for problem solving. This training also instills a comprehensive understanding of the science and technology that dominate the fields of diagnostic imaging and radiation therapy. Expanding this education and knowledge to also include formal training in patient communication will allow medical physicists to further expand their roles to directly impact patient care.

This symposium will discuss new roles for medical physicists, where they establish independent professional relationships with radiation oncology patients. Speakers from institutions of different sizes and practice environments will discuss how new patient-facing roles are utilized to improve the patient experience, increase the quality of patient care, and facilitate new responsibilities. In addition, the role of the Medical Physics 3.0 Committee in promoting and training medical physicists for new responsibilities in direct patient care will be discussed.

Learning Objectives:

1. To outline the benefits of medical physicists establishing independent professional relationships with patients
2. To explore how current patient-facing initiatives are increasing the quality of patient care and facilitating new patient care responsibilities
3. To understand the role of the Medical Physics 3.0 Committee in promoting and training medical physicists for new patient-facing responsibilities

Therapy Scientific Session

TRACK 3

Radiation Dose Evaluation and Verification

TH-F-TRACK 3-01

A Dosimetric Evaluation of ^{223}Ra -Dichloride Using a Realistic Model of the Metastatic Castrate-Resistant Prostate Cancer Microenvironment and Whole-Body Pharmacokinetics

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Purpose: The mechanism of action of ^{223}Ra -dichloride is not well understood despite the growing use of the radiopharmaceutical for the treatment of metastatic castrate-resistant prostate cancer in the bone. To support preclinical studies of ^{223}Ra -dichloride we have developed an advanced microdosimetric model that combines a detailed anatomical model of bony anatomy and a whole-body pharmacokinetic (PK) model of radium once injected into the blood. **Methods:** A high-resolution CT scan ($10\ \mu\text{m} \times 10\ \mu\text{m} \times 10\ \mu\text{m}$) was taken of a mouse femur ex vivo using the MiLabs U-CT scanner (MiLabs, Netherlands). Multiple geometric regions in a section of the bone were segmented in MATLAB (MathWorks Inc., Natick, MA) using an intensity-based analysis. The non-uniform activity distribution of ^{223}Ra -dichloride was calculated in the bony anatomy based on the whole-body PK model of radium provided in the ICRP Report 137 using advanced PK modeling with the SimBiology application in MATLAB. Microdosimetric simulations were performed in the model using the general-purpose Monte Carlo code Geant4 accounting for the decay products of ^{223}Ra and its daughters. **Results:** The results of the PK modeling of the bony anatomy indicate that the endosteum and the cortical surface components showed the most rapid uptake and clearance. The non-exchangeable cortical and trabecular volumes showed the slowest uptake and clearance. The mean dose to the trabecular, endosteum, and marrow components were in accordance with previously published values. **Conclusion:** This work offers a method to understand the dosimetric behavior of ^{223}Ra -dichloride via the combined use of PK modeling, a microCT based geometric model, and GEANT4 Monte Carlo simulations. The model may provide further insight into the behavior of ^{223}Ra -dichloride and lead to more effective determinations of normal tissue toxicity and

therapeutic efficacy of metastatic castrate-resistant prostate cancer treatment regimens.

BB is a co-founder of Voximetry, Inc., a nuclear medicine dosimetry company in Madison, WI.

JJ is a co-founder of Phantech Medical, a nuclear medicine phantom company in Madison, WI.

TH-F-TRACK 3-02

Fetal and Maternal Atomic Bomb Survivor Dosimetry Using Kneeling and Lying Survivor Postures of J45 Pregnant Female Phantoms

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Purpose: Cancer risk models stemming from atomic bomb survivor data still rely on the results of the DS86 and DS02 dosimetry systems published by the Radiation Effects Research Foundation (RERF). This study investigates the dosimetry impact on the pregnant female (PF) population of atomic bomb survivors by utilizing a newly created series of kneeling and lying pregnant female computational phantoms. **Methods:** Utilizing the original standing series of J45 PF phantoms, a new series of both kneeling and lying J45 PF phantoms has been constructed to include gestational ages of 8, 15, 25, and 38 weeks. Kneeling phantoms were constructed using scripting methods within the 3D modeling program Rhinoceros®. Lying phantoms were created by rotating the original standing J45 series of PF phantoms in 3D space. Organ doses for both posture phantoms were computed via Monte Carlo transport code PHITS using particle fluences provided by DS02 for both Hiroshima and Nagasaki exposures. Maternal and fetal organ doses from the J45 PF phantoms were compared to the corresponding maternal organ and uterine wall, respectively, in the non-pregnant stylized adult female phantom used in DS02. **Results:** For kneeling phantoms, organ doses from photon spectra were overestimated compared to DS02 by as much as 40% for certain fetal and maternal organs. For lying phantoms, fetal organ doses from photon spectra were underestimated up to 25% while maternal organ doses were overestimated up to 15% related to DS02 organ doses. Neutron contributions to organ doses exhibited from neutron contributions exhibit an increasing overestimation in DS02 as gestational age increases. **Conclusion:** This work highlights the degree to which fetal and maternal dosimetry varies from the DS02 report with the introduction of newly created computational phantoms for both the kneeling and lying survivor cohort of pregnant women.

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TH-F-TRACK 3-03

New Dosimetric Standard for Radiotherapy: Bounded Dose Volumetric Histogram Derived from Gamma Criteria

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Purpose: Radiotherapy quality assurance (QA) requires quantitative methods to compare two different dose distributions to determine whether the differences are significant or within tolerances. The gamma index (GI), which is based on predetermined dosimetric criteria, is the most popular method for comparing 1, 2, or 3D dose distributions. This study proposed a gamma-defined bounded dose-volume histogram (bDVH) QA technique, which is based on a dose difference (DD) and a distance-to-agreement (DTA) tolerance to better reveal the dosimetric cause of plan failure/error. **Methods:** An algorithm that translates gamma criteria to a bDVH was developed. Gamma criteria of 1%/1mm, 2%/2mm and 3%/3mm were evaluated in three clinically relevant SBRT treatment plans for NSCLC with a dose calculation grid of 2x2x2 mm. Log file-based dose reconstruction is performed for QA deliveries of the 3 patient treatment plans. A DVH is extracted for each delivery and compared to the bDVH based on the 3 gamma criteria. Delivery is within tolerance if 100% of the points on the delivered DVH fall within the bDVH. All deliveries were synthetically shifted by 2x the distance criterion or scaled by 2x the dose criterion and tested again. **Results:** The bDVH test passed for all deliveries that met the gamma criteria from log file-based dose reconstruction at the 2%/2mm and 3%/3mm levels at 100% of points falling within the bDVH bounds. The bDVH test also failed at less than 100% when the dose was shifted or scaled for all 1%/1mm, 2%/2mm, and 3%/3mm tests. **Conclusion:** A bounded DVH generated based on gamma criteria can now be used as a dosimetric test for patient-specific quality assurance. For the first time, a DVH can be dosimetrically evaluated to determine whether it is within machine tolerances or it presents a risk.

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TH-F-TRACK 3-04

In Vivo Relative Dosimetry with XACT Imaging: A Feasibility Study

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Purpose: One of the potential applications capitalizing on the thermoacoustic phenomenon is X-ray acoustic computed tomography (XACT), which utilizes acoustic waves generated by clinical X-ray beams. The field of XACT has garnered growing interest as a potential tool for in vivo dosimetry at no extra radiation. This study aims to investigate the feasibility of using XACT as a relative dosimetry tool without the use of markers. **Methods:** XA signals generated from an X-ray beam in water were collected by a single element ultrasound transducer (Fig. 1a). The XA signal was amplified by a 60-dB preamplifier and acquired on an oscilloscope with 1024 averaging. The beam was delivered by a Varian TrueBeam system at 10 or 6 MV flattening filter free (FFF) energy with 4-μs pulse duration. To capture the change in X-ray dose with increasing water depth, the height of the transducer was varied from 2.60 cm to 13.00 cm at multi-millimeter steps. The peak-to-peak amplitude of the beam edge XA signal was plotted against transducer depth, and the XACT image was reconstructed in MATLAB using the back-projection method after high-pass filtering. **Results:** XA signals from the near and far edges of the beam corresponding to the field size are identified (Fig. 1b). As the transducer depth is increased, the signal amplitude decreases, while the signal profile remains largely unchanged (Fig. 2a). Fig. 2b demonstrates the linear relationship between the near-edge XA signal amplitude and transducer depth, which is comparable to the percentage depth dose (PDD) curves of 10 and 6 MV beams. A reconstruction of the depth plane is shown in Fig. 3, with the edges of the beam visible at 4 cm apart. **Conclusion:** This study demonstrates the feasibility of relative dosimetry using XACT imaging. Further investigation for a robust method to quantify XA signals is warranted.

TH-F-TRACK 3-05

Model Based Reconstruction Algorithm for Transperineal XACT Imaging

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Purpose: Radiation dosimetry is a crucial process in radiation therapy to ensure that the correct dose is accurately delivered to the desired location. Despite wide use in clinical intervention, the delivered radiation dose can only be planned and verified via simulation with phantoms, while in vivo and in-line verification of the delivered dose is still absent in the clinic making precision radiotherapy challenging. We have proposed a new imaging modality called X-ray-induced Acoustic Computed tomography (XACT) for in vivo dosimetry during radiotherapy. Specifically, we are developing model-based image reconstruction algorithm for XACT XACT imaging using transperineal ultrasound matrix array. **Methods:** Model-based reconstruction algorithm was employed to address the limited-view problem in transperineal sensor geometry. The algorithm builds up an exact X-ray acoustic propagation model in time domains to predict the signal. The reconstruction is performed by minimizing the least square error between the measured signals and predicted signals. The model-based reconstruction results are evaluated by comparing against the ground-truth images. **Results:** Model-based reconstruction achieves more satisfactory results than back-projection algorithm with less streak-type artifacts and more low-frequency information. The time-reversal reconstruction has similar performance as model-based reconstruction qualitatively. However, quantitatively, the voxel value has larger difference between time-reversal reconstruction and the ground-truth images while the model-based reconstruction has more similar voxel values. The model-based reconstruction algorithm is less time-consuming than the time-reversal algorithm when reconstructing small objects. For large objects, the acoustic model needs to be calculated during each iteration due to memory issues, making it more time-consuming than the time-reversal reconstruction. **Conclusion:** Model-based reconstruction algorithm is effective in the reconstruction of transperineal XACT imaging, which can be valuable in prostate radiation therapy. The study also demonstrated the potential of using transperineal XACT imaging as an in-vivo dosimetry tool for radiation beam localization during the prostate treatment.

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TH-F-TRACK 3-06

Investigation into the Relationship Patient Setup Accuracy and In-Vivo Transit Dosimetry for Image-Guided Volumetrically Modulated Total Body Irradiation (TBI)

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Purpose: In-vivo dosimetry for conventional total body irradiation (TBI) utilize point detectors placed along the patient surface to confirm the delivered dose matches prescription dose. However, in the volumetrically modulated arc therapy (VMAT) approach to TBI, the electronic portal imager device (EPID) can be utilized to acquire a 2-dimensional transmission fluence plane. This work explores the relationship between patient setup accuracy with transit in-vivo dosimetry. **Methods:** A total of 192 fields were investigated. Each VMAT plan consisted of four isocenters: head, chest, abdomen, and pelvis. Prior to treatment, the patient was imaged at the head, pelvis, and chest. Optimal couch shifts were determined for each isocenter under image guidance. The optimal IGRT shifts were determined using an in-house application that minimized dose deviation using criteria established through plan uncertainty analysis performed in Eclipse. Translational couch residuals were recorded and defined as the difference in the global shift calculated and the optimal couch position with shifts. Transit dosimetry was measured per arc, and analyzed using SNC PerFRACTION with a gamma criteria of 10%/5mm, 5%/5mm, and 5%/7mm. **Results:** Based on plan uncertainty analysis, clinical threshold for couch residuals were set to 7 mm (5 mm for chest isocenter) as there would be minimal impact on target coverage and organ sparing at those levels. Transit dosimetry showed that the average pass rate across all fields was 99.6%, 97.0%, and 99.2% for 10%/5mm, 5%/5mm, and 5%/7mm gamma criteria, respectively. Pearson correlation tests showed that there was weak correlation between gamma criteria and couch residuals. At stringent 3%/5mm gamma criteria, moderate correlation was found between lateral couch residuals for the head and chest and the head and chest arc analysis. **Conclusion:** Transit dosimetry showed high pass rates using our couch residual tolerances, which confirmed the plan uncertainty analysis performed during treatment planning.

TH-F-TRACK 3-07

Small-Cavity Detector Perturbation and Quality Correction Factors in Magnetic Resonance Imaging-Guided Radiotherapy Small Photon Beams

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Purpose: Provide physical insights on detector dose-response to multiple megavoltage photon beam sizes coupled to magnetic fields and determine optimal orientations for measurements. **Methods:** Small-cavity detector (solid-state: PTW60012 and PTW60019, ionization chambers: PTW31010, PTW31021, and PTW31022) dose-responses in water are determined with Monte Carlo simulations (EGSnrc), using Elekta Unity 7 MV FFF phase-spaces of field widths between 0.25 and 10 cm, for four detector axis orientations: perpendicular to the B-field (1.5 T) with the beam parallel (orientation 1) or perpendicular (orientation 2 or 3 with FLorentz towards stem or tip); perpendicular to the beam and parallel to B-field (orientation 4). The B-field effect on perturbation factors is evaluated. The overall perturbation factor (P_{MC}) accounts for the extracavitary components, atomic composition, and density. For each orientation, the density (P_{ρ}) and volume averaging (P_{vol}) perturbation factors and quality correction factors ($k_{Q(B)}$) are calculated. **Results:** For solid-state detectors, P_{ρ} is unaffected by the B-field and field size. For chambers in B=1.5 T, it increases up to 1.564 ± 0.001 with decreasing field size over all orientations. For chambers, the B-field effect on P_{MC} is only significant for field widths <1 cm, increasing with decreasing field size in all orientations. For solid-state detectors, the B-field effect on P_{MC} exhibits different trends with orientation, indicating that the beam incident angle and geometry play a crucial role. A maximal effect of 1.221 ± 0.005 occurs in orientation 2. P_{vol} remains the most significant perturbation both with and without magnetic fields. In most cases, the B-field effect on P_{vol} is 1% or less, except for chambers PTW31021 and PTW31022 that over-respond and under-respond in orientations 2 and 3, respectively, at smaller fields. **Conclusion:** Solid-state detectors dose-response is strongly affected by the B-field in all orientations. In general, ionization chambers yield a $k_{Q(B)}$ closer to unity, especially in orientations 2 and 4.

Multi-Disciplinary Scientific Session

TRACK 4

Novel Strategies Using Existing Imaging Technology for Planning, Delivery and Toxicity Analyses

TH-F-TRACK 4-01

A Non-Parametric Analysis to Identify High-Risk Vs. Low-Risk Anatomic Regions Associated with Reduced Overall Survival in the RTOG 0617 Clinical Trial Data

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Purpose: Radiation dose to the cardio-pulmonary system has been identified as a new key detriment for risk of death in locally-advanced non-small cell lung cancer (LA-NSCLC). However, primarily one-dimensional dose metrics of pre-defined structures have typically been used for deducing this association. Here, we introduce an approach based on a metric called the unbalanced optimal mass transport (OMT) distance, which uses neither pre-selection of sub-regions nor normalization (total mass need not be preserved). The method ranked risk of death in the RTOG 0617 based on the distance. **Methods:** The 409 patients retrieved from The Cancer Imaging Archive with evaluable data were randomly split into training and validation subsets. Multi-step B spline image registration was used to deform CT scans and the associated dose distributions to a chosen patient. Pairwise unbalanced OMT distances of all patients were efficiently computed through a newly proposed

approximation method via vector-valued OMT, which measured the dose spread pattern and total dose amount differences. The distances were translated into risk scores by C-indices. Risk status in validation data was measured as distances to the training reference patients. Lastly, high-risk and low-risk regions were deduced according to the 50 riskiest (or least risky, respectively) reference patients. The locations of the sub-regions were qualitatively compared to those of eight deep learning auto-segmented cardio-pulmonary substructures. **Results:** The identified riskiest region overlapped with the base of the heart, including the ascending aorta, pulmonary artery and the superior part of the left atrium. In contrast, the 'safe' distributions mostly avoided all cardio-pulmonary substructures except for parts of the superior vena cava and right pulmonary artery. **Conclusion:** This OMT-based approach provides a tool to identify regions associated with survival without requiring pre-defined contours. For the RTOG 0617 dataset, base of the heart was identified as a critical region that influenced overall survival.

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TH-F-TRACK 4-02

Blood Dose Calculation in Radiotherapy

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Purpose: To understand how radiation modality and dose distribution impacts the dose delivered to the circulating lymphocytes, a dynamic computational model was developed to estimate the dose to the circulating blood for a variety of radiation treatment modalities. **Methods:** To describe the 4D propagation of blood particles through organs and radiation fields, an explicit Monte Carlo blood flow model was developed based on the whole-body hemodynamics. For head-and-neck patients, the large-scale architecture of cerebral arteries is extracted from MRA images. The propagation of the blood was described as laminar along the vasculatures and through radiation fields. A dynamic delivery model was implemented to simulate different radiation modalities such as proton pencil beam scanning (PBS) and passive scattering (PS) as well as photon therapy. The model included dynamic details of clinical delivery system, such as up-/down-ramping of the current as well as the dead time during steering between spot positions and switching between energy layers were all taken into account. **Results:** For a 2 Gy fraction, PBS reduced the average dose to the circulating blood by 1.9% compared to PS and by 30% compared to photon therapy. The fraction of blood receiving any dose was only 1.1% for PBS, compared to 7.3% and 18.1% for PS and photon therapy, respectively. The blood volume receiving at least 0.07 Gy, which is known to deplete the lymphocyte count by 2 percentage points, increased from 0.65% for PBS to 1.4% for PS and 2.1% for photon therapy. **Conclusion:** A computational model able to estimate the dose accumulated by the circulating blood for different radiation modalities and various dose rates and fractionation schemes was developed. Said model enables us to analyze a radiation modality's potential impact on the circulating lymphocyte count, which may be especially important given the growing interest in immunotherapy.

TH-F-TRACK 4-03

Enabling Few-View 3D Tomographic Image Reconstruction by Geometry-Informed Deep Learning

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Purpose: Traditionally, a tomographic image is formulated as an inverse problem for a given set of measured data from different angular views. Here, we propose a geometric-integrated deep learning model for tomographic imaging of 2D-3D reconstruction with ultra-sparse sampling across various patients. **Methods:** We propose a geometry-informed deep learning framework for 3D tomographic image reconstruction. Specifically, the proposed framework consists of three modules: a) a 2D projection generation network is developed to learn to generate novel-view projections from the given sparse views; b) a geometric back-projection operator transforms the 2D projections to 3D images, referred to as geometric preserving images, which geometrically relates the pixelated 2D input data to the corresponding ray lines in 3D space; and c) a 3D image refinement network learns to refine and generate the final 3D images. To evaluate the proposed approach, we experiment on a public CT dataset containing 1018 lung CT images, where 80% and 20% data are for training and testing, with the projection images are digitally produced from CT images using cone-beam CT geometry. **Results:** We deploy the trained model on the held-out testing set for few-view 3D image reconstruction, and compare the results with ground truth. For single-/two-/three-view reconstruction, average NRMSE / SSIM / PSNR values over all testing data are 0.368 / 0.734 / 20.770, 0.300 / 0.807 / 22.687 and 0.274 / 0.838 / 23.669, respectively. Beyond, we observe the proposed model could generate images closely to the target images although anatomic structures of different patients have a large variance, indicating the potential of the proposed model for volumetric imaging even with few views. **Conclusion:** We present a novel geometric-integrated deep learning model for volumetric imaging with ultra-sparse sampling across patients, which may present a useful solution in image-guided interventional procedures with simplified imaging system design.

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TH-F-TRACK 4-04

Head-And-Neck IMRT Auto-Planning Through Fluence Map Prediction Using Progressive Growing of Generative Adversarial Networks

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Purpose: To develop an IMRT auto-planning algorithm for head-and-neck cases, where optimal fluence maps were predicted from patient anatomy using Progressive Growing of Generative Adversarial Networks (PGG). **Methods:** PGG enhances the traditional architecture of Generative Adversarial Networks (GAN) with a multi-resolution design that is potentially more effective at capturing the complexity of the head-and-neck cases. In PGG, input was a series of 128x128 2D projections representing patient anatomy for 9 template beam angles. The networks were trained in 6 stages, where the output fluence map dimensions were 4x4, 8x8, ..., and 128x128 for 9 beams. Upon a new stage, up-sampling layers and convolutional layers were added to the generator. Generator output gradually changed from up-sampling of previous stage's output to the average of previous output and new convolutional layers' result. Ground truth (128x128) was downsampled by average filters to the same dimension with the output for loss calculation. Furthermore, an additional discriminator taking higher resolution input was added to the existing discriminator group with a gradually increasing weight. The discriminator group contains discriminators using (downsampled) ground truth, anatomic input, and generator prediction with different resolutions. Training, validation, and test patient numbers were 200, 16, and 15. For test patients, last stage's output was post-processed into predicted fluence maps and was converted into IMRT plans in TPS (PGG plans). **Results:** Preliminary results showed that PGG plans had overall acceptable plan quality for test patients, except that the maximum dose is high (whole-body D_{2cc}=123.0%). PGG plan D_{mean} in parotid L/R (22.2±2.5Gy/23.4±3.0Gy), oral cavity (24.6±7.0Gy), larynx (22.1±5.7Gy), pharynx (35.4±2.9Gy) showed no clinically relevant difference from benchmark plans (23.1±2.0Gy/23.9±2.3Gy/23.9±4.3Gy/22.7±4.8Gy/34.7±2.5Gy). PGG plan D_{0.1cc} in brain stem (17.5±3.8Gy) and cord+5mm (30.5±2.5Gy) was higher than in benchmark plans (15.5±2.7Gy/25.8±1.9Gy). **Conclusion:** The PGG-based head-and-neck IMRT planning algorithm has demonstrated encouraging plan quality. Further development and training are underway to improve performance.

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TH-F-TRACK 4-05

Improvements in Beam's Eye View Fiducial Tracking Using A Novel Multilayer Imager

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Purpose: Electronic portal image devices (EPIDs) have been investigated previously for beams-eye view (BEV) applications such as tumor tracking but are limited by low contrast-to-noise ratio (CNR) and detective quantum efficiency (DQE). We used a novel multilayer imager (MLI) with better performance, consisting of four stacked flat-panels, to measure improvements in fiducial tracking during liver stereotactic body radiation therapy (SBRT) procedures. **Methods:** The prototype MLI was installed on a clinical TrueBeam linac in place of the conventional DMI single-layer EPID. The panel was extended during volumetric modulated arc therapy SBRT treatments in order to passively acquire data. A frame grabber card was connected to the imaging system. Images were acquired for six patients receiving SBRT to liver metastases over two fractions each, one with the MLI using all 4 layers and one with the MLI using the top layer only, representing a standard EPID. The acquired frames were processed by a previously published tracking algorithm modified to identify implanted radiopaque fiducials. Truth data was determined using respiratory traces combined with partial manual tracking. Results for 4- and 1-layer mode were compared against truth data for tracking accuracy and efficiency. Tracking and noise improvements as a function of gantry angle were determined. **Results:** Tracking efficiency with 4-layers improved to 82.8% versus 58.4% for the 1-layer mode, a relative improvement of 41.7%. Fiducial tracking with 1-layer returned a root mean square error (RMSE) of 2.1mm compared to 4-layer RMSE of 1.5mm, a statistically significant ($p<0.001$) improvement of 0.6mm. The increase in successfully tracked frames correlated strongly with reduction in noise using 4-layer mode ($r=0.913$). **Conclusion:** Increases in MV photon detection efficiency by utilization of a multi-layer imager (MLI) results in improved fiducial tracking for liver SBRT treatments. Future clinical applications utilizing BEV imaging may be enhanced by including similar noise reduction strategies.

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TH-F-TRACK 4-06

Quantifying Radiation-Induced Lung Injury as a Function of Regional Radiation Therapy Dose Using Hyperpolarized-¹²⁹Xe MRI

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Purpose: To establish a relationship between regional changes in lung gas exchange properties and radiation therapy (RT) dose. **Methods:** Six patients receiving conventional (2Gy/fx) RT for lung cancer underwent breath-hold hyperpolarized (HP)-¹²⁹Xe gas exchange MRI. Subjects were imaged pre- and 3-months post-RT. Gas exchange MRI utilizes the solubility and chemical shift of ¹²⁹Xe to facilitate 3D imaging of xenon in the lung airspaces (ventilation), alveolar interstitial barrier, and red blood cells (RBC). Voxel-wise ¹²⁹Xe signal ratios of barrier/airspace and RBC/airspace were computed to quantify the interstitial barrier uptake and RBC gas transfer properties of the regional lung tissue, the latter representing an end-to-end measure of function since it is a surrogate for oxygen transfer from alveoli to the bloodstream. Sequential MRI were rigidly registered, and deformed back to the planning CT. We excluded voxels in areas of high dose gradient (≥ 3 Gy/mm) and those with low ventilated xenon signal (signal-to-noise ratio < 5), and binned voxels according to the RT dose received (5Gy bin-width). The per-voxel changes in ventilation, barrier uptake, and RBC gas transfer were computed via registered image subtraction. In addition, whole-lung average changes in HP-¹²⁹Xe-measured function were calculated for comparison against changes in the pulmonary function tests (PFTs), forced vital capacity (FVC) and diffusing capacity for carbon monoxide (DLCO). **Results:** A least squares fit of data between 5-50Gy showed that ventilation decreased linearly ($R^2=0.96$) by 0.25% Gy⁻¹ between pre- and 3-month post-RT imaging, barrier uptake increased linearly ($R^2=0.94$) by 0.74% Gy⁻¹ (as expected from alveolar interstitial thickening post-RT), and RBC transfer decreased by 0.48% Gy⁻¹ ($R^2=0.73$). Whole-lung increases in barrier signal correlated significantly with decreases in both Δ FVC ($R^2=0.88$, $P<0.01$) and Δ DLCO ($R^2=0.71$, $P<0.04$). **Conclusion:** Hyperpolarized-¹²⁹Xe MRI can detect RT-induced, dose-dependent changes in regional lung function and may prove useful in diagnosing, grading, and quantifying RILI.

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TH-F-TRACK 4-07

Predicting Locoregional Recurrence Through Multi-Modality and Multi-View Deep Learning for in Head & Neck Cancer

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Purpose: Locoregional recurrence (LRR) remains one of leading causes in head and neck (H&N) cancer treatment failure despite the advancement of multidisciplinary management. Accurately predicting LRR in early stage can make an optimal personalized treatment plan. We aim to develop a multi-modality and multi-view convolutional neural network (mMmV-CNN) for LRR prediction. **Methods:** Totally 206 patients are used in this study, which includes the image and clinical parameters with H&N cancer received radiation treatment from September 2005 to November 2015. In mMmV-CNN, a dimension reduction operator is developed by rotating the 3D CT and PET data along the vertical axis and the average projection along horizontal axis is calculated to obtain 2D image in different directions to make fully use of the information from 3D data. The multi-view strategy is then adopted to improve the performance of 2D CNN for extracting image feature by aggregating features extracted from different views. As such, the number of network parameters is greatly reduced, and the spatial 3D context information can be effectively used as well. Furthermore, we design a multi-modality deep neural network which can be trained in an end-to-end manner and jointly optimize the deep features of CT, PET and clinical parameters. **Results:** Five-fold cross-validation is performed in this study. The area under the curve (AUC), sensitivity, specificity and accuracy when combining CT, PET and clinical data can achieve 0.8052, 0.7347, 0.8089, 0.7913, respectively. **Conclusion:** In this study, a new mMmV-CNN model was developed for accurately identifying H&N cancer patients at high-risk for LRR after definitive radiation or chemoradiation therapy. Compared with the single-view method, the multi-view method fully utilizes image data from different angles to obtain more discriminative information. Furthermore, the experimental results demonstrated that mMmV-CNN can obtain better performance by combining CT, PET and clinical features.

Multi-Disciplinary Scientific Session

TRACK 5

Translational Technologies and Techniques

TH-F-TRACK 5-01

A Development of An In-House Software for Motion Monitoring During the Stereotactic Paraspinal Treatment

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Purpose: Stereotactic paraspinal treatment has become increasingly popular due to its favorable clinical outcome. An often-overlooked factor that compromises the effectiveness of such treatment is the patients' involuntary intrafraction motion. This work introduces an in-house software that quantifies such motion for accurate motion monitoring. **Methods:** The software requires a separate full-trajectory CBCT after patient setup to establish reference projections. Once treatment starts, the software reads in the intrafraction motion review (IMR) image acquired by TrueBeam and compares it against the corresponding reference projection to instantly determine the 2D shifts of the vertebrae being monitored. To evaluate its performance, an anthropomorphic phantom was shifted 5 mm in three orthogonal directions, separately and altogether, immediately after the full-trajectory CBCT but prior to treatment. For each scenario of shift, the same nine-angle fixed-gantry IMRT plan was delivered. At each angle, three IMR images were acquired sequentially every 200 MU. For each IMR image, the software-reported 2D shift was compared with ground truth. These tests were repeated with three degree of rotation, pitch, and roll, respectively. **Results:** The maximum standard deviation of the software-reported shifts for each set of three IMR images was 0.16 mm, with 93rd percentile below 0.03 mm. For translational shift, the maximum registration error was 0.44 mm, with 93rd percentile below 0.26 mm. Left unaccounted for, rotation and pitch degraded the registration accuracy mainly in the longitudinal direction while roll degraded it mainly in the lateral direction. The maximum registration errors under rotation, pitch, and roll were 2.97, 1.44, 2.72 mm, respectively. **Conclusion:** Our in-house software has high repeatability and accuracy in registering IMR images with the reference projections for motion monitoring. Rotation, pitch, and roll degrade registration accuracy and need to be accounted for in future work.

The presented study is partially supported by Varian Medical Systems Research Collaborations.

TH-F-TRACK 5-02

A Real-Time Surveillance System to Improve the Safety of MRI in Radiation Oncology

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Purpose: To improve safety around MRI simulators and MRI-LINACs using an in-house developed Real-time Location System. **Methods:** A Real-Time Location System (RTLS) was developed using Bluetooth Low Energy (BLE) technology. A grid of Bluetooth Low Energy (BLE) sensors was installed throughout the radiation oncology treatment facility at University of Texas Southwestern Medical Center (UTSW), and an artificial intelligence (AI)-based localization engine was developed that enabled us to locate staff, patients, and devices within the facility at a high accuracy (99.99%). Based on the RTLS infrastructure, a suite of applications was developed to locate, verify and time tagged subjects. At each machine console, an iPad installed with the electronic checklist application, called RO-Lifeguard, helps the care team to locate and verify patients and patient-specific accessories prior to treatment. To increase the safety of our MRI simulator and MRI-LINACs, a warning system was implemented in RO-Lifeguard to alert the care team when a tagged MR-unsafe device approaches MR Zone 3. The warning can be issued by audible alarm and/or by a warning message on the RO-Lifeguard screen. **Results:** The whole radiation therapy facility was divided into 450 zones, each of which is given an MR safety level. Each tagged subject (person and/or device) is also assigned an MR safety status. The RTLS engine was able to identify subjects' location with high accuracy in treatment areas. When the MR safety policy logic is activated in the RTLS system, the entire facility is under MR safety surveillance in real-time. **Conclusion:** MRI simulators and MR-LINACs are powerful tools that improve tissue delineation and offer a great solution to adaptive radiation therapy for moving targets, but they also require an increased commitment to safety practices. Our RTLS-based MR safety surveillance system provides an extra layer of protection in areas where MRI simulators and MR-LINACs are present.

TH-F-TRACK 5-03

An International Validation of Knowledge-Based Planning

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Purpose: To carry out a large international validation of how dose prediction quality translates to plan quality in a radiotherapy knowledge-based planning (KBP) process. **Methods:** We collected dose predictions for head-and-neck cancer radiotherapy from 21 different research groups internationally who participated in the OpenKBP Grand Challenge. Each research group used the same training dataset (n=200) and validation dataset (n=40) to develop their methods. These methods predicted dose on a testing dataset (n=100), and those 2100 unique dose predictions were input to a previously published plan optimization method to generate 2100 treatment plans. The predictions and plans were compared to the ground truth dose via: (1) error, the mean absolute voxel-by-voxel difference in dose; and (2) quality, the mean and maximum deviation across 23 dose-volume histogram (DVH) criteria. **Results:** The range in median prediction error among the top 20 methods was 2.3Gy to 12.0Gy, which was 6.8Gy wider than the range in median plan error of 2.1Gy to 5.0Gy. One method also achieved significantly lower prediction error (P<0.05; one-sided Wilcoxon test) than all the other methods, however, it generated plans with error that was not significantly lower than 28.6% of the other methods. Additionally, predicted dose was consistently lower quality than plan dose. Half (n=1050) of all predictions and plans had an average deviation that was 0.1Gy worse and 0.8Gy better than the ground truth dose, respectively. Similarly, half of all predictions had a maximum deviation that was 3.7Gy worse than the ground truth dose, which was 1.0Gy worse than half of all plans. **Conclusion:** Many dose prediction methods can achieve low error, however, optimization often improves upon the predictions and eliminates significant differences between prediction methods. Thus, it is critical that we improve the optimization stage in KBP to get better utility out of the existing high-quality dose prediction methods.

TH-F-TRACK 5-04

Development of a Novel Dual-Modality Balloon Implant for Simultaneous HDR Brachytherapy and Magnetic Nanoparticle Hyperthermia of Brain Tumor Resection Cavities

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Purpose: Glioblastoma multiforme is an aggressive brain tumor that generally recurs locally and has a dismal median survival < 18 months. In vivo studies have shown that thermal enhancement ratio (TER) of up to 5 can be achieved when hyperthermia and radiation are applied simultaneously, which is not currently possible. The goal of this project is to develop a novel thermobrachytherapy (TBT) balloon implant to accomplish this. Our phase I development has 3 specific aims: i) to

fabricate TBT balloons; ii) to evaluate the device for compatibility of hyperthermia and HDR and to characterize thermal and radiation dosimetry in laboratory studies; and iii) to characterize heating patterns in perfused brain tissue via in vivo animal studies and to evaluate reliability and heating performance. **Methods:** The TBT balloon has two layers: an inner layer filled with saline to expand the resection cavity into approximately spherical shape, and an outer layer filled with magnetic nanoparticle (MNP) solution that absorbs energy from an external magnetic field to generate heat. The balloon is connected to a shaft that houses 4 ports to fill inner and outer layers, insert an HDR source into the balloon center and a fiberoptic sensor into the outer balloon for temperature monitoring. Five TBT balloons were fabricated. Gafchromic film was used to determine HDR dwell position to achieve radiation dose distribution centered at the balloon center. Optically stimulated luminescent dosimeters were used to measure radiation dose. **Results:** The presence of MNP, a 133 kHz magnetic field, and 43-55°C heating did not affect radiation dose significantly. Thermal mapping demonstrated spherically symmetric heating in phantom and will continue in vivo. **Conclusion:** Novel dual-modality balloons have been fabricated and tested in the lab. Because of the positive lab results, we will continue with in vivo dosimetry in preparation for phase II clinical development.

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TH-F-TRACK 5-05

Prescription Tradeoff Decision Support for Pancreas SBRT: From Templates to Artificial Intelligence Models

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Purpose: Radiation oncologists typically employ template constraints for dose prescription and OAR sparing. However, dose prescription and OAR constraints often form competing constraint pairs which require tradeoff discussions between the planner and radiation oncologist. In this study, we designed a novel artificial intelligence (AI) tool to support radiation oncologists in making optimal tradeoff decisions. It that can estimate achievable tradeoff scenarios before treatment planning starts. **Methods:** This decision support tool has an effective user interface module that integrates with a treatment planning system to facilitate the prescription decision process. It is powered by a machine learning core which is trained to learn the balance between PTV coverage and OAR sparing from a dataset of 98 pancreatic SBRT cases retrospectively selected for this study. A 10-fold cross validation was performed comparing clinical plan's PTV coverage (V100%) with model predicted values. These comparisons were further stratified by attending physicians to gain additional insights into decision making process. Cases with large discrepancy were identified, analyzed, and re-planned to validate and assure the achievability (i.e. accuracy) of predictions. **Results:** The clinical plan PTV V100% was $87.7 \pm 14.5\%$ while the model predicted value was $90.5 \pm 9.6\%$ ($p=0.005$). Model agreement discrepancy was observed among attending physicians. Among all 98 cases, nine cases were identified with large variation from model predictions. For the re-planned cases, an average of 15.3% improvement was achieved over the original clinical plans, indicating the AI model's accuracy and benefits for clinical planning guidance. **Conclusion:** The AI decision support assistant provides dynamic and interactive tradeoff guidance. Its decision support prior to treatment planning can save valuable efforts from both the planning team and radiation oncologist. Moreover, this tool should also work effectively as a training tool providing valuable tradeoff insights for fresh resident who are new to the treatment site.

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TH-F-TRACK 5-06

Understanding Physician's Preference in Treatment Planning of Stereotactic Body Radiation Therapy of Prostate Cancer

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Purpose: Understanding physician's preference in treatment planning is necessary to develop automatic treatment planning approaches capable of generating plans favorable by the physician. The purpose of this study is to develop a deep learning based Virtual Physician Network (VPN) that can understand physician's preference on plan approval for prostate cancer Stereotactic Body Radiation Therapy (SBRT) plans and suggest where to improve, if a plan is rejected. **Methods:** The VPN takes one Planning Target Volume (PTV) and eight Organs at Risk (OAR) contours, as well as a dose distribution of a plan seeking approval as input. It outputs the probability of approving the plan, and a dose distribution indicating improvements to be applied to the input dose. VPN includes two sub-networks. VPN1 uses an attention-gated U-Net to predict where to improve in the input dose. VPN2 has the first half architecture of VPN1 followed by fully-connected layers to output a probability of having the given plan approved. Due to the lack of unapproved plans in our database, two networks are alternatively trained under an adversarial framework. The intermediate results generated by VPN1 in the training process are suboptimal dose, which can naturally serve as training data for VPN2. 44 prostate cancer patients who received 45 Gy in 5-fraction SBRT were used in 4-fold cross-validation. **Results:** VPN can differentiate high- and low- quality plans with Area under the curve 0.91 ± 0.02 . For unapproved plans, after applying VPN's suggestion on dose improvement, V50 of bladder, rectum, and penile bulb, max dose of urethra, V95 and D95 of PTV agreed with ground truth with differences of $0.59 \pm 0.78\%$, $0.8 \pm 1.01\%$, $0.04 \pm 0.03\%$, $0.75 \pm 0.46\text{Gy}$, $0.34 \pm 0.21\%$, and $0.75 \pm 1.27\text{Gy}$. **Conclusion:** VPN was developed to accurately predict a physician's preference in plan approval and can provide useful suggestions on how to improve the dose distribution.

TH-F-TRACK 5-07

Use of CBCT for Reducing QACT Frequency in Intensity Modulated Proton Therapy: Head and Neck

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Purpose: Anatomic variation has a significant dosimetric impact in intensity modulated proton therapy. Traditionally weekly or bi-weekly CT scans, called QACTs, are used to monitor the anatomic and the resultant dose change to determine if an adaptive plan is needed. Frequent CT scans result in unwanted patient dose and clinical workload. This work proposes a CBCT-based approach to reduce the frequency of QACTs by analyzing patient setup CBCTs. **Methods:** we retrospectively analyzed 20 head and neck patients, associated with 885 CBCTs, 105 QACTs, and 22 adaptive plans. For each CBCT, water equivalent thickness (WET) along the pencil beam path between the target surface and the patient body was calculated. For each treatment plan, the WET of the first day CBCT (1st CBCT) was used as the reference, and the mean WET change in each CBCT from the 1st CBCT was used as the surrogate of proton range deviation. Using CBCTs acquired prior to a QACT, a linear regression model was generated to predict the mean WET change on QACT day. The impact of range deviation on target dose was calculated as (1) the predicted mean WET change weighted by the monitor units of each fields and (2) the ratio of mean WET change in (1) over the effective diameter of the target. **Results:** When the estimated mean WET change was larger than 8 mm or 8% of the effective diameter of the target, a QACT was required, and otherwise if the change was less than 8 mm AND 8%, the QACT could be skipped. Overall, 40 QACTs could be reduced from 105 QACTs. At least one QACT could have been omitted in 18 out of the 20 cases. **Conclusion:** This study shows that QACT frequency can be greatly reduced by calculating range deviation in patient setup CBCTs.

Therapy Scientific Session

TRACK 6

SRS/SBRT: Treatment Delivery and Verification

TH-F-TRACK 6-01

Development and Evaluation of a Bi-Polar Gated Respiratory Motion Management Strategy for Lung SBRT

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Purpose: A novel bi-polar (BP) gated motion management strategy for lung SBRT is proposed. The feasibility is evaluated geometrically and dosimetrically. **Methods:** In BP strategy, patient holds the breath at the end of inhale and exhale under audio/video coaching with dose only been delivered during breath hold. 10 previous lung SBRT patients with 4DCT were selected, each having tumor motion larger than 1cm. BP was compared with other strategies: free breathing (FB), gating at end of inhale (GI), exhale (GE), and real-time tracking (RT). Lung volume and tumor position were used to determine the end-exhale and end-inhale phases. The GTV was contoured at each 10-phases of 4DCT to determine the ITV for each strategy. PTV is formed by 1mm expansion of ITV. IMRT/VMAT plans were generated to give 60Gy to >95% of PTV and lung dose was compared. Breathing patterns' stability and repeatability were investigated on 8 volunteers and recorded by RPM and analyzed. **Results:** The breath-hold of the BP strategy is found to be comfortable over 3 minutes, with repeatability in periodicity at 1.00 ± 0.03 and duty cycle at 65%. Using GI as reference, the ITV for FB, GE, RT, and BP are 1.83 ± 0.50 , 1.00 ± 0.25 , 0.63 ± 0.14 , 0.59 ± 0.26 , respectively. RT is larger than BP due to the limited number of phases in 4DCT that produce larger GTV volume in fast-moving phases. In IMRT plans from 3 patients, the lung V5Gy normalized to GI are 1.33, 1.17, 1.02 and 0.95, respectively, with similar trends in VMAT plans. **Conclusion:** The bi-polar strategy is feasible for lung SBRT. The breathing pattern can sustain a long treatment time with a high duty cycle. Compared to other strategies, it has the smaller ITV, therefore a smaller treated volume and lower normal tissue toxicities. It is less complex to implement than real time tracking.

TH-F-TRACK 6-02

Predicting the Efficacy of Spacer Placement for Pancreatic Cancer Using a Novel Patient-Specific Virtual Spacer

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Purpose: Stereotactic body radiation therapy (SBRT) is an effective way to achieve local-regional control for pancreatic cancer but is limited by proximal organs' at risk (OARs). Our previous clinical trial demonstrated that endoscopic ultrasound-guided injection of hydrogel spacer increases separation between target and duodenum. Yet, spacer's clinical benefits highly depend on patient anatomy. We developed a virtual spacer for pancreatic hydrogel spacer to generate augmented data to design a decision support system (DSS) and predict patient-specific dosimetric benefits. **Methods:** A finite element method oriented spacer simulation algorithm (FEMOSSA) is developed and defined using the material properties, boundary conditions, surface interfaces, and interactions based on patient contour. Finite element analysis software is used to solve the differential equation system generated by FEMOSSA. Simulation accuracy was evaluated by pre-post injection CTs from four cases and quantified by dice similarity coefficient (DSC), overlap volume histogram (OVH), and radial nearest neighbor distance (RNND). Four virtual spacer scenarios were simulated for 15 SBRT patients: no injection, injection in pancreas interface with duodenal sections D1-D2, D1-D2-D3, and D1-D2-D3-D4. 60 corresponding SBRT plans (33Gy/5 fractions) are used for designing a Bayesian DSS. **Results:** Average target and duodenal DSC increased, 0.86 to 0.89 and 0.44 to 0.63, respectively. Average mean and 5th percentile RNNDs differed by 0.5 and 2.1mm. OVH thresholds show average of less than 0.75mm difference. Results demonstrate a spacer-location-independent decrease in duodenal V20Gy and a highly spacer-location-dependent change in V33Gy. L1cc and L20cc (tumor expansion overlapping 1cc and 20cc duodenal volume) are good predictors of V20Gy and V33Gy. L1cc>14mm results in V33Gy=0. Bayesian DSS predicts spacer injection related improvement in BED with root mean squared error of 3.6Gys. **Conclusion:** A novel patient-specific virtual spacer is developed and validated for pancreas SBRT, and used to design DSS that indicates hydrogel efficacy, optimum location, and expected improvement.

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TH-F-TRACK 6-03

Stereotactic Arrhythmia Radioablation for Treating Refractory Ventricular Tachycardia

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Purpose: To report the feasibility of single fraction stereotactic arrhythmia radioablation (STAR) in patients with refractory ventricular tachycardia (VT) based on co-registration of electroanatomic (EA) map and planning images. **Methods:** A cardiac electrophysiologist acquires EA maps using the Carto3® system (Biosense Webster, Irvine, CA, USA) and defines the target region during an electrophysiology study. The EA map consists of 3D cardiac anatomy and electrical activity at any given mapped point. The EA target map is registered to the planning CT by aligning the left-ventricle (LV), right-ventricle (RV), left-atrium (LA), and available reference structures with defined contours on the MR/CT image based on iterative closest point algorithm. In-house software was developed for converting the EA image format to DICOM and registering reference structures. The simulation, planning, and treatment is performed with a standard STAR technique: a single fraction of 25 Gy using volumetric-modulated arc therapy or dynamic conformal arc therapy depending on the target shape. **Results:** Three patients with refractory VT were treated by defining the target based on registering EA maps and contours on MR/CT. Dice similarity indices between transferred reference map and contours are 0.83 ± 0.04 and 0.72 ± 0.04 for LV and LA/RV, respectively. The target area is confirmed based on the consensus of cardiac electrophysiologists and radiation oncologists and expanded to an internal target volume (ITV) and a planning target volume (PTV), which provides a 5-mm margin on the ITV. **Conclusion:** The quality of the transferred map on the image was sufficient to localize the treatment region accurately although we continue to refine the transferred target map. Longer follow-up is required to determine the true safety and efficacy of this therapy.

TH-F-TRACK 6-04

Single-Isocenter/Multi-Target (SIMT) SRS to Multiple Brain Lesions Via Dynamic Conformal Arcs (DCA)-Based VMAT with MLC Aperture Shape Controller

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Purpose: Applying SIMT-SRS to multiple brain lesions has improved treatment efficiency, patient compliance, patient comfort, and clinic workflow. When targeting multiple brain lesions using highly modulated SIMT-VMAT, the inherent risk of small field dosimetry errors can compromise the treatment quality. Herein, an MLC aperture-shape controlled method is proposed to mitigate the effects of small field dosimetry errors. **Methods:** Seven complex SRS patients, each with 2-8 brain lesions, were selected with an average target size of 13.04 ± 8.83 cc. Treatment plans were developed for each patient using single-isocenter HyperArc VMAT with Encompass device and a 10X-FFF beam, prescribing 20 Gy to each lesion. Alternatively, patients were re-planned using modified HyperArc VMAT, implementing a DCA-based dose and a high priority MLC aperture-shape controller prior to optimization. Identical HyperArc geometry, isocenter location, planning parameters, and objectives were used. **Results:** For similar target coverage, conformity, and dose gradient, DCA-VMAT increased average dose to gross tumor volume by 1.5 Gy. It showed slightly higher V8Gy and V12Gy values, increasing by 4.2 cc and 0.3 cc on average, respectively. However, critical organ (optic apparatus, brainstem, and hippocampi) maximal doses and mean brain doses were decreased with DCA-VMAT. Moreover, the DCA-VMAT plan decreased the total monitor units, beam modulation factor, and beam-on time by an average of 1846 MU, 0.92, and 1.37 min, respectively. Independent dose verification through Monte Carlo (MC) code

showed treatment planning system agreement for DCA-VMAT within 3%, decreased from the 6% of HyperArc VMAT. **Conclusion:** Modified HyperArc VMAT, utilizing DCA-based dose and MLC aperture-shape controller, provided similar plan quality with substantially decreased total monitor units, beam modulation, and beam-on time. This arguably suggests that the modifications further improve SIMT HyperArc treatments for multiple brain lesions. Independent MC calculation showed reduction of small field dosimetry errors and likely patient setup uncertainties via DCA-VMAT.

TH-F-TRACK 6-05

Comparison Between the HYPERSCINT RP200 Scintillation Detector and Other Small Field Detectors for 10MV FFF SRS Beam Modelling On a VersaHD Linear Accelerator

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Purpose: Plastic scintillation detectors (PSDs) have advantageous dosimetric properties, including small size and energy independence, which make them ideal candidates for small field dosimetry. This work aims to quantify the performance of the HYPERSCINT RP200 (Medscent inc., Quebec city, Canada) in the process of beam modelisation for a small field stereotactic radiosurgery (SRS) program. **Methods:** Dose profiles, percent depth doses and output factors were acquired as part of the commissioning of a 10MV FFF field on a VersaHD linear accelerator (Elekta). These measurements were carried out using a HYPERSCINT RP200, a Razor diode (IBA), GafChromic EBT3 films (Ashland), an Advanced Markus parallel plate ion chamber (PTW), an Exradin A26 cylindrical ion chamber and an Exradin A1SL cylindrical ion chamber (Standard Imaging). Comparisons between each detector were made using percentage differences and distance to agreement criteria. **Results:** Better than 0.3 gamma index (1%/1mm) was achieved between the HYPERSCINT RP200, the Razor diodes and the GafChromic EBT3 film in small field dose profiles. Lower than 0.5% dose difference was met between the PSD and the Advanced Markus in percent depth doses. Smaller than 2% agreement was achieved when comparing output factors from the RP200 and every other detectors. Average differences $\pm$ standard deviations from output factors measurements were as follow: Razor Diode: $(1.1 \pm 0.4)\%$; GafChromic films: $(0.8 \pm 1.1)\%$; A26 ion chamber: $(0.3 \pm 0.3)\%$; A1SL ion chamber: $(0.5 \pm 0.4)\%$. **Conclusion:** These results demonstrate the capability of the HYPERSCINT RP200 to perform accurate small field dosimetry and beam data acquisition. This versatility in profiles, PDDs and output factor acquisition would allow for significant time savings during the commissioning process of small radiation fields. In the author's institution, the 10MV FFF beam modelisation for SRS was entirely performed using data from this PSD detector.

TH-F-TRACK 6-06

Assessment of the Dosimetric Impact of Rogue Leaf Pair Openings in Modulated Stereotactic Radiosurgery Plans

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Purpose: Optimization of modulated stereotactic radiosurgery (SRS) plans sometimes results in (i) leaf pair openings that do not irradiate the planning target volume (PTV) and (ii) individual leaf ends far (>2 mm) outside the PTV. Our goal is to understand the dosimetric impact of these rogue leaf openings. **Methods:** Single fraction, modulated Varian Eclipse SRS plans from 10 patients were selected retrospectively. After projecting a 2 mm expansion of the PTV (PTV+2mm) onto the isocenter plane, leaf pair openings not irradiating the PTV+2mm were flagged as superfluous. Individual leaves were also flagged as nonconformist if there was a gap between the leaf end and the PTV+2mm projections. Two modified plans were generated with in-house software while respecting machine parameters: for both "clean" and "conformed" plans, superfluous leaf openings were pushed behind the jaws. Additionally, in "conformed" plans, the nonconformist leaf ends were pushed to the edge of the PTV+2mm to enforce conformality. The doses for the modified plans were recalculated, renormalized to the same PTV coverage, and DVH metrics were compared. **Results:** Clean and conformed plans reduced total plan fluence by up to 12.9 and 14.7%, respectively. Despite this reduction in fluence, small changes in the maximum and minimum doses to 0.002cc of the PTV were observed (largest changes of 1.68 and -1.95%, respectively). Although conformed plans exhibited large relative changes in normal brain V2Gy, V5Gy, and V12Gy (up to -24.4, -15.8, and -16.4%, respectively) the maximum absolute change in V12Gy was -0.39 cc. **Conclusion:** Based on analysis of 10 patient plans, modulated SRS plans contain rogue leaf openings that unnecessarily irradiate normal tissue. Post processing can remove these rogue leaf positions without compromising PTV coverage. The resulting modified plans reduce normal brain V2Gy, V5Gy, and V12Gy.

TH-F-TRACK 6-07

Modeling Dosimetric Impact of Treatment Delivery Errors in Robotic Radiosurgery: Experimental Validation and Applications

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Purpose: A dose perturbation model is proposed to assess dose to target(s) that models delivery errors for treatments using respiratory motion tracking on CyberKnife. **Methods:** Motion-tracking residual errors are modeled as a spatial convolution of the planned dose with a probability distribution functions derived from data recorded in system log files. Uncorrected rotations are modeled by rotating the target volume about the imaging isocentre. The model estimates the accumulated dose volume by giving equal weights to each perturbation. For experimental validations, ten different experimental conditions are considered, including in-phase and out-of-phase internal and external breathing motion patterns (periodic motion and patient waveforms), with and without uncorrected rotations and for homogenous and heterogeneous phantoms emulating liver anatomy. The measured dose (using EBT3 radiochromic films) is compared to the perturbed dose using the gamma pass rate (2%/2mm or stricter criteria), and the penumbral spread and systematic dose shift if a motion phantom is employed. Four case studies are provided that demonstrate how the model can be used to assess plan robustness to delivery errors. It can also be used to evaluate the suitability of a patient-specific planning target volume (PTV) margins. **Results:** The gamma analyses ($> 96\%$ using 2%/2mm criteria) and the comparison of penumbral widths (≤ 2 mm) and systematic dose shift (< 2 mm) have validated the performance of this model. Dose to targets considering delivery errors can be significantly improved if treatments are planned following certain guidelines. **Conclusion:** The dose perturbation model is validated under various experimental conditions. It can be used to assess retrospectively dose to targets, evaluate the suitability of patient-specific PTV margins, or per-patient sensitivity to errors before treatment. It could enable adaptive planning processes, patient-specific PTV margins, and inform tolerances of treatments to uncorrected rotations.

This work was supported by The Ontario Consortium for Adaptive Interventions in Radiation Oncology, and a research grant from Accuray, Inc., CA.

SAM Therapy Educational Course

TRACK 7

Using Automation to Improve Linac-based VMAT for Total Body Irradiation (TBI)

TH-F-TRACK 7-00

Using Automation to Improve Linac-Based VMAT for Total Body Irradiation (TBI)

Moderator: Ping Xia, The Cleveland Clinic Foundation, Cleveland, OH

TH-F-TRACK 7-01

B. Guo, Cleveland Clinic Foundation:

Cleveland Clinic VMAT TBI Experience: Automation in Treatment Volume Divisions for Adult and Pediatric Patients

TH-F-TRACK 7-02

N. Kovalchuk, Stanford University Cancer Center:

Stanford VMAT TBI Experience: Automation in Treatment Planning

TH-F-TRACK 7-03

X. Gu, UT Southwestern Medical Center:

UT South Western VMAT TBI Experience: Automation in Treatment Planning and Delivery

Modern total body irradiation (TBI) uses inverse planning with volumetric modulated arc therapy (VMAT) and multiple iso-centers to achieve uniform dose for the total body while protecting critical organs such as lungs and kidneys. The feasibility and early clinical experience have been published in the literature. The entire process was extremely time consuming and complex. With different treatment planning systems and delivery platforms, this education session can share experts' clinical experience and technology development to automate and streamline processes of treatment planning, patient setup, and treatment delivery. Knowledge gained from this session can be applied and generalized to other treatment sites involving large treatment volumes such as total bone marrow and total lymph node irradiation. In this session, three experts from different institutions will discuss their clinical experiences of clinical implementation of modern TBI.

The learning objectives are:

1. Learn how to use an automated workflow to combine the head-first and feet-first CT scans to a total body CT scan.
2. Learn how to use an automated workflow to divide planning target volumes and to create multiple iso-centers and dose junction regions.
3. Learn how to use automations to streamline the planning, patient setup, and delivery processes for adult and pediatric patients.
4. Learn the early clinical experiences and clinical outcomes of VMAT TBI.

Ping Xia received research grants from AVO and Philips outside of this research topic

Education General ePoster Viewing AAPM ePoster Library

PO-GePV-E-01

A Chatbot for the Staff in Radiotherapy Using Artificial Intelligence and Machine Learning

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Purpose: A chatbot personalized by artificial intelligence and machine learning is built to provide educational information for staff in radiotherapy. The topics covered include interdisciplinary knowledge in radiation oncology, medical physics, radiotherapy, and radiation safety. **Methods:** The Bot was constructed using the IBM Watson Assistant functionalities on the IBM cloud. The Bot had ability to communicate with radiation staff by answering some simple questions regarding radiotherapy, or by multiple choice to the user from a list of options. A layered structure approach was used in the workflow of the Bot to interact with the user. Through selecting different topics in the list and communication between the Bot and user, he/she will finally acquire the requested information. In addition, communications between the Bot and user will be used in machine training leading to a more accurate response to the user's question. **Results:** The user interface of the Bot is a front-end window operating on an internet of things such as smartphone, tablet or laptop. The user can type in a question or anything to communicate with the Bot. In the communication, if the Bot cannot identify what the user needs, the Bot will provide a list of options to the user as guidance. Through selecting different topics on the list, another list with more detailed breakdown of the topic will pop up, so that the user can obtain his/her request accurately and quickly. The Bot acquires knowledge through interactions with the users. **Conclusion:** It is concluded that a chatbot with characterization supported by machine learning can provide information regarding radiotherapy to the radiation staff more efficiency, and can be used in staff training. This Bot is particularly useful for quick search such as looking for physics data or a review of some radiation safety protocol conveniently.

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PO-GePV-E-02

Clinical Diagnostic Reference Levels in Interventional Neuroradiology in Qatar

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Purpose: The objectives of this study were to: 1) evaluate patient radiation exposure in neuroradiology interventional (INR) procedures and 2) establish Diagnostic Reference Levels (DRLs) for stroke (S), brain aneurysms (AN) and brain arteriovenous malformation (AVM) clinical indications. **Methods:** Adult patient data undergoing cerebral endovascular treatment for S, AN and AVM were retrospectively collected using specially designed collection forms. The methodology described in the International Commission on Radiological Protection (ICRP) Report 135 was followed to establish clinical DRLs in terms of Kerma-Area Product (KAP), frontal (CDf) and lateral Cumulative Dose (CDl), fluoroscopy time (T), total number of images (Ni) and total number of runs (Nr). Collection forms were prepared in excel format in order to assist data analysis. The forms aimed to include patient demographic information (gender, age, weight, height, etc) X-ray modality technical parameters (fluoroscopy time (T), number of images (N), number of runs (r) and operator information (contact details, years of experience). Only adult patients (16≥ years of age) would be included in the study. **Results:** This study included data from 209 adult patients. National DRLs in terms of KAP, CDf, CDl, T, Ni and Nr for S, AN and AVM were 78 Gy cm², 380 mGy, 118 mGy, 12 min, 440, 157 for stroke, 85 Gy cm², 610 mGy, 96 mGy, 20 min, 720, 20 for AN and 180 Gy cm², 1144 mGy, 540 mGy, 36 min, 1380 and 30, respectively. Comparison was restricted as most recent literature refers to the technique and not the clinical indication. **Conclusion:** This is the first study reporting national DRLs based on clinical indication for INR procedures. The established national clinical DRL values will further facilitate patient dose optimization and quality improvement process within Hamad Medical Corporation and the state of Qatar.

PO-GePV-E-03

Brachytherapy Quality: An Innovative Approach to Addressing Institution and Practitioner Proficiency

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Purpose: To establish a technology-enhanced simulation environment for brachytherapy that facilitates implant quality assessment through simulated end-to-end delivery. **Methods:** Implantable anthropomorphic phantoms were developed to facilitate simulated implants for permanent seed implant brachytherapy for prostate and breast procedures. Institutional implant quality is primarily evaluated by seed placement accuracy, which is the difference between the planned and delivered seed locations. Implant quality is, in part, a measure of the quality of the brachytherapy commissioning process and programmatic implementation, which is a core competency of Medical Physicists. Separately, Radiation Oncologists were also evaluated for two qualitative metrics commonly used to assess proficiency in medical simulation tasks; a global rating scale and a procedural checklist. **Results:** Simulated implants were performed by five learner and five expert multi-disciplinary teams. The mean and standard deviation of seed placement accuracy was 8.1 ± 3.5 mm and 6.1 ± 2.6 mm for the learner and expert cohorts, respectively. Based on the expert cohort, the passing tolerance for institutional credentialing was set at two standard deviations above the mean values, which was 8.1 ± 3.4 mm. The median (range) procedural time-to-complete was 64 minutes (60 to 77) and 43 minutes (41 to 50) for learners and experts, respectively. In all metrics, expert team proficiency was superior to learner team proficiency. **Conclusion:** Anthropomorphic prostate and breast phantoms have been developed and characterized for use in technology-enhanced simulation education programs. Baseline institutional credentialing criteria have been defined, which will be later refined for greater participation in a multi-institutional study. Benchmark learner and expert practitioner performance was assessed, which provided validation evidence that the simulated implants are representative of the clinical experience. Next steps include quantifying learner progression in implant quality to approach expert proficiency.

Authors Roumeliotis and Meyer are Directors of Okolo Health, a company with financial interest in utilizing brachytherapy software and hardware devices discussed in this work.

PO-GePV-E-04

Development of Diagnostic X-Ray Sources – A Multi-Vendor Overview

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Purpose: The speed of scientific and industrial development of new diagnostic and therapeutic X-ray sources remains tremendous. Studying the history of medical X-rays and the evolution of the technology of its sources has been sparking fascination and heuristic insight over more than a century. Still to date, and despite of their deficiencies, vacuum electronic X-ray tubes and high voltage generators remain the dominant and most affordable sources of medical

diagnostic X-rays. **Methods:** This presentation will illustrate the various branches of the development of the technology and will pay credit to some of those scientists, craftsmen, business leaders, and artisans, who have driven this field of innovation. **Results:** Taking a multi-vendor perspective, this paper will discuss in depth the paths taken for improvement of medical X-ray tubes for radiography, CT and angiography. Thanks to contributions from various industrial sources, it presents a brief, but comprehensive overview. **Conclusion:** The development of modern diagnostic X-ray sources will continue to stay dynamic. There is no reasonable alternative to braking radiation generated by vacuum electronic devices.

PO-GePV-E-05

Enhancing Medical Physics Teaching with Image Repositories and Shared Resources

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Purpose: Provide a process to increase the effectiveness of medical physics teaching and enhance the professional performance of medical physicists, both as classroom/conference presenters/discussion leaders and as creators of visual learning resources. Also, to promote the value of collaborative teaching and the sharing of resources to enhance the teaching and learning of medical physics around the world. This is based on the concept that effective mental knowledge structures, especially for clinical applications of physical principles including diagnostic radiology, are based on images and visual representations of physical interactions and applications. Of major significance, it is how radiology residents learn. **Methods:** A comprehensive model and method has been developed providing for the publication and sharing of images and related visuals that can be used by medical physics educators in their teaching activities. It is based on teaching image Repositories on the websites of individual institutions or organizations. An article providing information and guidance on the creation and operation of Repositories has been published in an international medical physics journal and is available in all countries. Included are details on creating appropriate images and visuals for teaching, optimizing for web search engines, copyright and academic recognition issues, and management of Repositories. **Results:** Repositories are being developed and populated with teaching images by individual institutions and organizations and are available as open resources to all educators. A major value is it provides individual medical physicists an opportunity to publish and receive recognition for their creative efforts and contribute to medical physics education on a global basis. **Conclusion:** Teaching medical physics, especially for clinical applications including diagnostic radiology, is most effective when using images and visuals with presentations and discussions by medical physicists sharing their knowledge and experience. The development of Repositories increases the availability of high-quality images and visuals for that purpose.

PO-GePV-E-06

A Simulation-Based Mastery Learning Program for Ultrasound-Guided Interstitial Prostate Brachytherapy Using a Low-Cost 3D Printed Phantom

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Purpose: There has been a decline in the utilization of brachytherapy even though it has been shown beneficial for a number of indications. One of the reasons for this decline is inadequate training due to an insufficient patient caseload. Simulation-based mastery learning (SBML) has been observed to improve procedural skills among medical trainees. In this work, we developed an SBML program for interstitial transrectal ultrasound (TRUS)-based prostate LDR/HDR brachytherapy using a low-cost prostate phantom developed in-house. **Methods:** The SBML program is hands-on consisting of all components of interstitial brachytherapy, including (a) preparation of training materials including fabrication of prostate phantoms and dummy radioactive seeds, (b) mock calibration of radioactive seeds, (c) instruction in brachytherapy procedures/safety, (d) operation of survey meter, TRUS, and brachytherapy applicator, (e) brachytherapy needle placement, (f) treatment planning and treatment delivery, (g) quality assurance and radiation survey, (h) review of potential hazards, and (i) post-implant dosimetry. Pre-session and post-session surveys were taken by trainees to assess confidence in each step of the procedure, with scoring ranging from 1 (extremely uncomfortable) to 5 (extremely comfortable). **Results:** Each prostate phantom and dummy seed set were fabricated in 1.5-hours by trainees, with material costs being approximately \$30 for each phantom. Trainees in our SBML laboratory included seven physician residents and five medical physics residents. After participation in the laboratory, confidence in overall workflow increased from 2.1 to 4.2 ($p=0.005$), patient preparation from 2.2 to 3.5 ($p=0.038$), equipment operation from 1.9 to 4.1 ($p=0.005$), and familiarity with treatment planning from 1.7 to 3.9 ($p=0.013$). The confidence levels increased in all evaluation categories. **Conclusion:** This SBML approach supplements patient caseload and is a valuable opportunity for trainees to increase their knowledge and confidence in prostate LDR/HDR brachytherapy procedures. Low-cost prostate phantoms and dummy seeds make SBML affordable, promoting increased utilization of brachytherapy.

PO-GePV-E-07

Introduction of Mind Mapping Technique to Enhance Medical Physics Graduate Studies

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Purpose: Challenges exist in medical physics education due to its interdisciplinary nature requiring clinical knowledges and critical thinking skills. Medical physics professionals are also involved in advancing technology and safety in healthcare through their scientific and clinical research. Trainees also participate in such research developments either through graduate dissertation projects or residency rotations. In this work, we introduced the mind mapping technique to assist graduate students to succeed in their medical physics studies. **Methods:** The mind mapping technique utilizes a colorful spider diagram to help structure information from various sources visually resembling the human thinking process. The idea of “mind mapping” was first introduced by Tony Buzan in the 1970s and has since become popular with many educators. It has been applied in the education of many fields including medicine and basic sciences that are related to medical physics. Mind mapping tools have been introduced to some students in our graduate program in two areas, study of didactic courses and innovation of research projects. **Results:** Example mind maps will be presented. Mind maps in didactic courses, e.g., anatomy/physiology and therapy physics, demonstrated that organizing information through mind mapping assisted students in memory retention and concept understanding. Mind mapping on how to motivate students in their research bridged gaps between professor and students to help students more effectively achieve their goals. Mind maps developed by students during literature search and brain storming helped find opportunities for research projects through connections that have yet to be investigated. **Conclusion:** Mind mapping is a useful tool to some students in their didactic course studies and research projects. Developing mind maps visually make study more efficient and enjoyable by stimulating both brain hemispheres. It can also assist students to organize thoughts and balance logical and creative thinking in an interdisciplinary field such as medical physics.

PO-GePV-E-08

To Investigate the Effectiveness of Supervised Multisensory Instruction and Flipped Instruction in Physics Resident Education

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Purpose: To investigate the effectiveness of supervised multisensory instruction and flipped instruction in physics resident education. **Methods:** The SRS/SBRT physics group is trialing a didactic overhaul to better tailor teaching techniques to individual physics residents. A multisensory instruction technique is being implemented to utilize seven defined styles of learning. For optimal implementation, a weekly curriculum has been developed where each resident is provided a combination of new clinical responsibility, new reading material, discussion time with faculty physicists, and a flipped instruction presentation. Each piece of curriculum provided to the physics resident is designed so that both the didactic resources and interpersonal resources provide unique perspective.

The stereotactic radiation therapy physics rotation is four months: two months in the first year and two months in the second year. In the first year, the rotation is focused on treatment planning. Each week of the rotation focuses on a particular site and an associated IGRT such as lung SBRT without gating, lung SBRT with gating, liver SBRT, etc.. In the second year, the rotation is focused on small field dosimetry and stereotactic beam commissioning. Each week of the rotation focuses on a specific aspect of small field commissioning such as output factor measurements, profile measurements, treatment planning system validation, etc.. **Results:** Currently, six residents have completed the new first year rotation and three residents have completed the second year rotation. The feedback for each rotation is variable depending on the residents' learning styles. Each has favored the tasks of the rotation that better suit their preferred learning styles. **Conclusion:** Compared to a comprehension focused environment, the performance of the physics residents on exams demonstrate greater grasp of content. An evaluation sheet of the curriculum is being developed for physics residents to provide feedback designed to adjust each week of curriculum.

PO-GePV-E-09

Dosimetrist Knowledge Transfer Using Intelligent Agents

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Purpose: Despite the recent advanced developments in radiation therapy planning, treatment planning for head-and-neck cancers remains challenging due to multiple dose prescriptions and numerous organs-at-risk close to targets, particularly for difficult patient cases. Additionally, studies have shown that plan quality strongly depends on planner experience and skills. It is desirable to transfer this knowledge from experienced planners to trainees at the lowest possible cost so trainee plans can better meet clinical practice and physician preference. The potential for using an intelligent agent to transfer this valuable clinical knowledge was investigated in this work. **Methods:** A clinically deployed AI dose predictor for head-and-neck cancers was retrained, using a transfer learning technique, on the historic treatment plans of experienced dosimetrists. This new model was then used to generate 3D dose distributions for comparison with 10 plans created by physics residents. These were then compared using qualitative and quantitative clinical metrics to evaluate the potential for training junior planners with this intelligent agent. **Results:** Preliminary results show that an AI 3D dose predictor can be used to guide trainees when learning to create treatment plans. The generated dose distribution mimics the collective experience of the clinical practice and highlights areas for improvement with no educational oversight. The highlighted changes help to drive the planner closer to the pareto optimal surface. The dose prediction also displays overall plan dose distribution styles, such as posterior sparing or reducing dose leakage. **Conclusion:** An AI dose predictor for head-and-neck cancers can be used to highlight plan quality issues in trainee plans. With additional training and infrastructure, this intelligent agent can be a powerful learning tool and used to guide for plan quality improvements of junior dosimetrist and has the potential to assist them in meeting clinical goals in reduced planning time.

PO-GePV-E-10

The Experience of Writing a Medical Physics Article to Young Readers/Reviewers

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Purpose: Public education is an important aspect of medical physics education. Within the scope of public education, one of the most challenging tasks is to explain medical physics related concepts to the younger population (pre-teens, early teens, etc). This work shares the experience of writing a medical physics article to young readers/reviewers. **Methods:** An educational article titled "Treating Cancer with X-Ray Radiation Therapy" was written, targeting a reader group with age 8 to 15. The article briefly discusses the concept of cancer, production of X-ray using a LINAC, biology related to radiation therapy, etc. A group of young reviewers were invited to review and give comments about the article. At the time of reviewing, all reviewers were younger than 15, and were guided by a science mentor. The young reviewers were asked to summarize the main point of the article, explain its importance, and list the interesting and confusing parts. The young reviewers were also given the opportunity to ask further questions. **Results:** With three rounds of editing, the young reviewers, science mentor, editors, and authors reached an agreement in the final version of the article. The young reviewers find the concepts challenging, but are able to understand them. Some find the concept "X-ray production" interesting. Figures requested by the young reviewers include: a schematic of a LINAC, and radiation damages of DNAs. The concept "bremsstrahlung" was agreed to be too difficult and omitted from the final version. **Conclusion:** The experience of writing a medical physics article to young readers/reviewers is successful. The young reviewers are enthusiastic and are able to comprehend some abstract concepts in medical physics. The language of the article, illustrations, guidance from a science mentor, and interactive editing are essential in this experience.

PO-GePV-E-11

Development of a Flipped-Classroom Style Medical Physics Course for Radiation Therapist Students

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Purpose: The goal of this work was to convert an in-person, lecture-centered introductory medical physics course for radiation therapist (RT) students to a remote flipped-classroom approach, and to evaluate this approach through peer-review. We aimed for students to engage in more active and collaborative learning while maintaining a safe learning environment during the ongoing global pandemic. **Methods:** Traditionally, our RT medical physics course was team-taught using in-person lectures. This year, a flipped-classroom approach was developed where these lectures, pre-recorded, were converted to pre-class work. The originally scheduled class times were used for discussion sessions, which were conducted via Zoom. These included a brief review and then employed several methods to engage the students in active learning such as cold-calling, polls, group problem-solving, "game show" style activities, and dedicated Q&A time. Student feedback was solicited throughout the course, and the teaching team held regular meetings to share best practices. The efficacy of this approach was evaluated by reviewing periodic student feedback and end-of-semester course evaluations. Additionally, a teaching peer-review program was developed and implemented to improve collaboration and teaching quality among the team. **Results:** Implementing this new approach introduced challenges, both technological (e.g., video and audio quality, unfamiliarity with available tools and platforms) and pedagogical (e.g., student resistance to cold-calling, non-compliance completing pre-class work). However, student feedback from post-lecture surveys and course evaluations indicated high levels of satisfaction with the pre-class videos and the active-learning style of the discussion and review sessions. Several opportunities for improvement were identified from student feedback and peer-review. Future work will include the use of eLearning modules, monitoring pre-classwork compliance, and other strategies to further improve student engagement and learning outcomes. **Conclusion:** The teaching team successfully implemented a remote, flipped-classroom approach for our RT medical physics course along with a robust faculty peer-review program.

PO-GePV-E-12

Development of Visual Learning of AAPM Task Group Reports

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Purpose: Studies show that 83% of human learning occurs visually, demonstrating the clear importance of visual learning strategies. Visual learning utilizes graphs, charts, maps, diagrams, and other forms of visual stimulation to effectively process and interpret information. AAPM provides all sorts of materials, references, TG reports, and free resources on its website. However, many medical physicists still find that most AAPM TG reports are lengthy and difficult to interpret. This work aims to improve medical physicist learning of TG reports through the development of visual learning techniques based on adult learning theories. **Methods:** We have used Python, HTML, CSS, and JavaScript to develop the learning module to aid physicists in understanding the AAPM TG

reports. The website was also built with web scraping to allow keyword search within publicly available AAPM TG reports. We have selected the TG-100 as an example and extracted the essence of it, converting the central ideas of risk analysis methodologies into visual aids. The resulting visual teaching modules contain infographic techniques and a web-based learning module. **Results:** A web-based learning module was developed. As an example, we have successfully transformed the complex TG-100 risk analysis methodology into infographics. The website displays infographics that visually represent risk analysis methodologies and also allows a physicist to search a word within any AAPM TG reports that are available within one click from the AAPM Publication webpage. Once found by search, the title and ID of the TG report will be displayed, and the report can be viewed by clicking on the title. **Conclusion:** AAPM TG reports can be converted and displayed using both the infographic form and computer-based learning modules for ease of learning. Our work capitalized on what medical physicists already know and incorporated user-friendly interfaces on selected AAPM TG reports.

PO-GePV-E-13

Incorporating Statistical Learning and Artificial Intelligence Education to Medical Physics Graduate Training

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Purpose: To evaluate the value of a web-based educational tool developed to introduce medical physics graduate trainees to novel topics in statistical learning and artificial intelligence (AI). **Methods:** A self-paced tutorial-course was containing modules introducing trainees to 1) Python programming, 2) DICOM file operations, 3) Statistical Learning concepts, and 4) Deep Learning concepts was created. The materials were developed such as that no prior programming and/or statistical learning/AI background were required. This course was developed using Python and Jupyter notebooks, with many notebooks covering materials in each of the four topics. Eight first-year medical physics PhD graduate trainees at our institution used this tutorial-course materials over the course of a twelve-week research-tutorial to enhance knowledge for their projects. Weekly-meetings were held to go over materials covered and evaluate research progress. After completion of the self-paced course and research-tutorial, all trainees were asked to complete a survey to assess the value of the educational tool. **Results:** All trainees completed the tutorial-course while taking a typical first-year academic course load. 100% of trainees thought the tutorial-course was helpful in developing a strong background in statistical learning and AI. 75% of trainees feel confident that the tutorial has prepared them well to implement AI techniques into their PhD research. Several trainees submitted abstracts on their research-tutorial work for the AAPM annual meeting. One trainee won the Young Investigator Symposium award at the 2020 SWAAPM meeting based on her research-tutorial results. **Conclusion:** Training in statistical learning and AI are currently not part of the medical physics graduate education. We developed a structured self-paced tutorial-course to introduce trainees to basic principals in these topics while they worked on individual research projects. The course was positively-received by the trainees and has potential in bridging the gap in knowledge for future medical physicists in these fields.

PO-GePV-E-14

Implementation of Ra-223 Radiopharmaceutical Program in Radiation Oncology at a Small Community Cancer Center

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Purpose: Ra-223 has recently been adopted as a targeted radiopharmaceutical therapy for castration resistant prostate cancer patients with bone metastasis. Ra-223 therapy increases life expectancy with a relatively simple procedure involving the injection of Ra-223 for up to 6 fractions every four weeks. This treatment could be a routine procedure in a nuclear medicine department, its implementation is challenging in a small radiation oncology center, due to lack of staff experience. The purpose of this work is to give a guidance and share the experience for an independent small radiation oncology center about to launch the program. **Methods:** Physicians, physicists and nurses worked as a team in the process led by a physicist who will serve as RSO. Communication among staffs is critical for a successful workflow. We followed the code of Maryland regulation specific to radioactive material license consistent with federal nuclear regulatory commission. There were three phases of program build-up. It is imperative to obtain the license in the first place for some agreement states submission of detailed documentation of whole radiation safety procedures is essential. Secondly, emergency plan and safety training for the staffs were performed. The last phase involved equipment acceptance and QA, source logistics and workflow development. **Results:** The documentation of radiation safety procedures, staff training and acceptance and initial calibration of dose calibrator and well chambers and workflow development using ARIA® EMR were completed for first treatment. At our institution, we have treated 10 fractions of Ra-223 injections without any incident for 5 months since the launch of the program. **Conclusion:** The implementation involves team effort of administrators, physicians, nurses and physicists. For a small cancer center, a physicist who also should serve as a RSO plays as a leading role to establish and oversee radiation safety program as well as workflow establishment.

PO-GePV-E-15

VICTORIA: Online Monte Carlo and DICOM Data Viewer

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Purpose: To build a web-based DICOM and Monte Carlo (MC) dose viewer 'Voxel Interactive Contour Tool for Online Radiation Intensity Analytics (VICTORIA)' for the visualization and comparison of dose distributions with underlying anatomy that is free, accessible, simple to use, secure, and requires no downloads. The web viewer should be useful for researchers using .egsphant and .3ddose file types from the EGSnrc toolkit, or researchers in low- and middle-income countries without access to treatment planning systems. **Methods:** VICTORIA was coded in Javascript using D3.js and Conerstone.js to assist in visualization and file parsing. All processing is done client-side to avoid uploading files with patient information over the web. VICTORIA currently accepts .egsphant, .3ddose, CT DICOM, and RT DICOM file types, and can calculate dose differences between dose files. Web Workers were employed to cache the density slices to improve speed. **Results:** VICTORIA visualizes the position, density, dose, material, and dose profiles where the user clicks. Clicks can be used to navigate the image, sliders control the window and level of patient anatomy, and dose contours can be added or removed. The user can export the visualization and dose profiles to .png and .csv, respectively. Multiple different density and dose distributions may be loaded and compared. Using Google Chrome, a 21.7 MB .egsphant file took 1.4 s to load, and navigating slices before caching took 9.2 ms, while after caching it took 3.1 ms. A batch of 171 DICOM CT files with a total size of 90.4 MB took 13.2 s to load, while navigating slices took 45.5 ms before and 14.1 ms after caching, respectively. **Conclusion:** VICTORIA is a convenient web-based MC and CT dose comparison and visualizer tool that is free and accessible.

PO-GePV-E-16

A Practical Way to Strengthen Radiation Safety Culture in Healthcare

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Purpose: The incorporation of radiation safety culture into healthcare settings can help prevent injuries and deaths and help reduce unnecessary or unintended radiation dose to patients and staff overall. While there is broad agreement on the importance of safety culture, it is challenging to provide appropriate educational materials to a global, multidisciplinary audience from a wide range of economic environments. The IAEA's training material entitled "Strengthening Radiation Safety Culture in Medicine" has been developed to fill the gap in this area. **Methods:** Analysis of events in the SAFRON incident learning systems indicated that over 60% of reported events could have contributing factors relating to poor safety culture. A Technical Meeting held in November 2018 concluded that training material was needed that could improve safety culture practices in medical applications using ionizing radiation. Course

material was subsequently developed, and in 2021 the IAEA launched the Radiation Safety: Trait Talks handbook for strengthening radiation safety culture in healthcare. **Results:** The entire program, funded through the US Extra Budgetary Funds, is freely available. The course is structured around 10 important principles – traits – that contribute to a strong safety culture. These are adopted from the nuclear and aviation industry, and applicable to the use of radiation in medicine. Each trait includes case studies, accompanying exercises, and digital presentations. The material can be provided as a two-day instructor lead workshop or as individual modules tailored to the needs of the medical institutions. The educational material includes a student handbook, digital material, and an instructor's handbook. **Conclusion:** The Radiation Safety Culture: Trait Talks Handbook is available for free download on the IAEA website, aiming to fill the gap in training material for strengthening radiation safety culture in medicine in a wide range of settings.

PO-GePV-E-17

Brain Functional Connectivity Reveals RTMS Effectiveness Pain Reduction in Non-Specific Chronic Low Back Pain Patients

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Purpose: Non-specific chronic low back pain (CLBP) is one of the most common worldwide painful syndrome among the adult population without clear medical evidence and specific diagnosis, while the cure of CLBP patients is resistant to diverse pharmacological and non-pharmacological prescriptions. Repetitive transcranial magnetic stimulation (rTMS) as a non-invasive technique in spite of its prospective treatment effectiveness has not been examined deeply for the recovery solution of CLBP, yet. This study aimed to find out pre and post rTMS stimulation in neuronal vesicle activities using resting-fMRI brain functional connectivity (FC) data analysis. **Methods:** The pain intensity of nine non-specific CLBP patients (age 50±10.39) was determined by the conventional method of visual analog scale (VAS) before and 2-4 days after 20 Hz rTMS via an eight-shaped coil over the motor cortex. Stimuli were provided in field intensity of 95% of the motor thresholds and trains of 40 pulses, followed by a 28sec rest period. The rs-fMRI data were collected using a Siemens 3T MRI scanner with the standard 64-channel head coil, while, anterior cingulate cortex (ACC) as a key pain region selected for seed-based connectivity (SBC) analysis with other pain regions in every session. **Results:** Pre and post rTMS stimulation execution results showed the mean pain intensity of patients as 8.17±1.20 and 3.85±0.95, respectively (recovery rate=53%). The functional connectivity beta values represent Fisher-transformed correlation coefficients between ACC and insula, supramarginal gyrus (SMG), and prefrontal cortex (PFC) in pre-stimulation session were 0.67, 0.44, and 0.53, while in post-stimulation session significantly decreased as 0.54, 0.28, and 0.45, respectively (p-FDR < 0.05). **Conclusion:** Significant decremental brain connectivity coefficients in conformity with conventional VAS evaluation of CLBP patients showed the recovery in pain-related regions as the criteria for the effectiveness of rTMS pain reduction in CLBP patients.

PO-GePV-E-18

Effective Resident Training in the Medical Physics 3.0 Era

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Purpose: The world of medical physics is changing, from prescriptive and technical, to dynamic and patient-facing. Residency education needs to change with it, to prepare future leaders to thrive in this new ecosystem. Our residency made improvements and enhancements to multiple modules to begin incorporating this training. **Methods:** Earlier this year, we incorporated new facets, to guide the transition from a technical emphasis to one that resonates with the Medical Physics 3.0 ethos: expanding our role in patient care, with a mindset of both safety and value. We added role-playing exercises, in which staff physicists play the part of worried patients or caregivers and the residents are transmitting their knowledge. Residents learn about our significant advances in nomenclature, automation, and development of standard operating procedures—all of which have contributed to meaningful simplification. During coaching for motion management simulations, residents are getting both communication training and decision-making training, in addition to routine technical aspects. The resident's self-evaluated confidence score is surveyed prior to and after each training module for both technical and communication competency; this is paired with the staff's evaluation of their performance and both are tracked. **Results:** Our residents who have received this enhanced education are still in training, but we believe that they will be poised to drive systems level change, both in their work environments, and in the radiation oncology community. **Conclusion:** Enhancing the full value of physicists towards human health, from industry to clinic to research, is important in resident education. We are embracing Medical Physics 3.0 in our residency program by enhancing our curriculum, offering new and different opportunities to demonstrate independent professional relationships with our patients, and finding innovative ways to partner with our physician colleagues to leverage all of our unique skills in the pursuit of better patient treatment.

PO-GePV-E-19

Design and Implementation of An Interactive Virtual Reality Interface for Experimental Medical Physics Education

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Purpose: Teaching of experimental medical physics plays an important role in fostering students' abilities in the operation, observation, analyzing and solving problem. Due to the restriction and influence of the COVID-19 pandemic, it is difficult for schools to set up experiments in the laboratory to teach medical physics in order to meet the learning needs of students in China. The aim of this study is to design and implement an interactive virtual reality interface for medical physics students during the pandemic. **Methods:** The necessity of interactive virtual reality technology in medical physics experiment was analyzed. The virtual experiment system was designed to provide students with multi-level open-learning resources. A medical physics interactive virtual system based on the NI ELVIS hardware platform, NI LabVIEW software, and Moodle e-learning platform were created to simulate doing experiment in reality. A comparative study on the theoretical and virtual experimental results was conducted utilizing medical physics experiments as examples. **Results:** In experimental medical physics teaching, interactive virtual reality technology was flexibly applied in the 3D-displays of modern medical equipment, laboratory instruments and machine. Students could access and control those physics equipment and instrument in the virtual space to complete an experiment by following a given instruction. **Conclusion:** Applying interactive virtual reality technology in medical physics teaching can greatly change the traditional teaching concepts. The virtual interface could break through the difficulties of teaching in the real space, attract the attention of students to stimulate their interests in learning, and thereby enhance them to understand the appropriate teaching content. Consequently, there are broad prospects for its development and application.

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PO-GePV-E-20

SSGAN: Unpaired MR-To-CT Image Synthesis with Spatial Sequence GAN

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Purpose: Existing work on unpaired MR-to-CT image synthesis performs well at the 2D level. However, there is still some limitations regarding the synthetic image quality in the three dimensions. For instance, there is a lack of continuity between the previous and subsequent layers of the 3D sCT image, which is manifested in uneven levels in tissues and bones. In addition, the HU value of the 3D synthetic CT image has minor variations, indicating losing structural details in coronal and sagittal planes. Moreover, there is lack of shape consistency with the original MR. In order to solve the above problems, an effective spatial sequence generative adverse network (ssgan) was proposed to achieve unpaired MR to CT image synthesis. **Methods:** The convolutional long short-term memory (Conv-LSTM) model was embedded in the generator to guarantee the 3D spatial continuity of synthesis images by using a mixture of the input 2D images in different spatial orders. Secondly, three projection discriminators were introduced to discriminate the authenticity of the three projected images along axial, coronal and sagittal planes. Moreover, a non-local module was incorporated into the discriminator to further enhance the strength of distinguishing the voxel intensity relationship in the adjacent layers. **Results:** The feasibility and effectiveness of our SSGAN was demonstrated using unpaired head and neck MR and CT images. The synthetic CT (sCT) generated using our SSGAN shows superior similarity to the real CT in both 3D continuity and intensity authenticity. In addition, the cross-modality registration experiment illustrates that our SSGAN generates better sCT with highest mutual information **Conclusion:** the novel SSGAN method can provide better quality 3D sCT images from unpaired MR data compared with others methods.

PO-GePV-E-21

Fork-Net: Synthesis-Based Unsupervised Multi-Modality Image Registration

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Purpose: Multi-modality image registration plays an important role in disease diagnosis and staging, tumor delineation, and prognostic analysis. However, it is often challenging due to the distinct image variations among different modalities. In this study, we proposed a synthesis-based unsupervised multi-modality registration network--Fork-Net. **Methods:** In Fork-Net, the generator, discriminator and registration network were integrated into a single model with a reused encoder. The moving image was first synthesized into the same mode as the fixed image, and then registered through registration network, turning the multi-modality registration task into a mono-modality registration problem. A new Synthesis-Registration-Consistency loss was thus developed to strengthen the consistency constraint between generator and registration network. The method was demonstrated using two brain datasets: one open access (BraTS, 2018) for T1-T2 MRI registration and one in-house dataset for MR-CT registration. The model performance was compared with the state-of-the-art image synthesis and registration methods. The quality of the image synthesis was evaluated using mean absolute error (MAE) and larger peak signal-to-noise ratio (PSNR). The registration accuracy was evaluated through the Dice of the target contour before and after registration, and the average gradient of deformation field (aGDF). **Results:** Regarding synthetic image quality, Fork-net has smaller MAE and larger PSNR. The target contours using registered images through ForkNet has the highest Dice, i.e., 0.74 ± 0.12 and 0.68 ± 0.14 for dataset 1 and 2, respectively. ForkNet has the smallest aGDF, indicating smoothest deformation. In addition, the architecture of Fork-Net is superior to other models with less adjustable parameters and less memory requirement. **Conclusion:** The proposed synthesis-based unsupervised multi-modality registration network is superior in terms of synthesis quality, registration accuracy, and smoothness of the deformation field as compared with current state-of-the-art methods. It may be implemented into clinical workflow, facilitating routine practice of multi-modality image registration.

PO-GePV-E-22

A Novel Leaf Sequencing Optimization Method with Deep Reinforcement Learning

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Purpose: Due to the machine limitations inherent in the design of multi-leaf collimator (MLC), the current leaf sequence algorithm has problems such as insufficient execution efficiency and delivery accuracy. This study designed a leaf sequencing method based on reinforcement learning, trying to solve the above problems. **Methods:** The training engine consists of MLC simulation environment, data generator and discrete action network structure. First, a simulated fluence map was obtained through the data generator, which, together with the current collimator position and the cumulative conformal intensity map, was entered into agent network. The predicted collimator movement and monitor units was applied to environment, which made a new collimator state and intensity map. The reward was calculated with the final target and was back-propagated to complete a whole cycle. The design of MLC simulation environment would consider machine parameters and constraints, which greatly reduced the action space of discrete action model. The Bezier curve and elastic transformation were innovatively used in the data generator to simulate the intensity distribution, making the training target closer to the actual fluence map. Using nasopharyngeal carcinoma IMRT as a test bench, we tested and compared the accuracy difference between the proposed and traditional leaf sequencing algorithms. **Results:** During test using fluence map extracted from 20 IMRT plans of nasopharyngeal carcinoma, with an intensity level of 64, the proposed method achieved a relative intensity difference of 1.25% with 15 segments, while the traditional algorithm requires 40 segments to achieve 1.58%. When the strength level increased to 128, the method achieved 0.78% with 15 segments, while the traditional algorithm required 50 segments to achieve 1.06%. **Conclusion:** Our novel leaf-sequencing method based on reinforcement learning has obvious advantages in execution efficiency and delivery accuracy. Experiments shows that this method has great potential in the field of MLC leaf-sequencing.

PO-GePV-E-23

Enriching a Fully Remote Medical Physics Lab Culture: A Self-Assessment

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Purpose: COVID-19 drastically changed work environments and subsequently the medical physics trainee experience. To better understand and address the effects of this change, we performed a self-assessment of our lab culture. **Methods:** In an anonymous survey, we asked each lab member to identify and rank up to 10 values of a healthy lab culture. The participants then scored how well the lab embodied their contributed values before and during COVID-19 (1-5 scale). The results of the survey were analyzed in a live group session. In another live session, we performed a literature review where groups of 4-5 members were tasked with reviewing and presenting 2 peer-reviewed lab culture manuscripts. In three additional live sessions, we finalized 10 values, reflected on how well the lab embodied each value and proposed action items to address any deficiencies. Most of the lab was present at each session and the discussions were facilitated in large (21 people) and small (3-4 people) groups. **Results:** Values identified as being necessary to sustain a healthy lab culture were attitude, accountability, teamwork/collaboration, communication, diversity/inclusion, emotional intelligence, integrity, training, well-being, and adaptability/crisis-

management. The most frequently occurring and highest-ranked value among responses were teamwork/collaboration and attitude, respectively. The satisfaction score averaged over all values before and during COVID-19 was 4.39 ± 0.77 and 4.17 ± 0.95 , respectively. Changes made to the lab as a result of our self-assessment included simulating trainee office environments with 'coding hour' zoom rooms, dedicating a monthly meeting to socializing, and creating a wellness channel on Microsoft Teams. **Conclusion:** We successfully assessed the culture of our lab before and during COVID-19 using an anonymous survey and five, 2-3-hour group sessions. We determined 10 values for our group to embody, identified weak points in our lab culture, and proactively proposed and enacted solutions to address them.

AAPM ePoster Library

Professional General ePoster Viewing

PO-GePV-P-01

IROC Houston Performance Impacted by Covid-19

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Purpose: To describe and the effects of the Covid-19 pandemic on anthropomorphic phantom mailings and credentialing status inquiries (CSI). **Methods:** IROC Houston provides many services and quality control programs to support institutions that participate in clinical trials, including OSLD output checks, credentialing, and anthropomorphic phantom end-to-end remote audits. The phantoms are used for credentialing for NCI sponsored clinical trials as well as standard QA purposes. The CSI forms are submitted by institutions wishing to be credentialed for multi-institutional clinical trials. Phantom shipment numbers from 2019 and 2020 were compared to determine the effect that the pandemic had on the number of NCI credentialing phantoms and QA phantoms mailed. CSI numbers were also compared to determine the impact seen in credentialing requests. **Results:** 657 phantoms were shipped in 2019 and 623 phantoms were shipped in 2020. Overall, QA phantoms decreased by 77 and NCI phantom requests increased by 43 creating an overall decrease of 34 (5%) phantoms. A total of 4075 and 3744 CSI requests were completed in 2019 and 2020 respectively producing a decrease of 331 (8%) submissions. **Conclusion:** Though the overall number of phantoms shipped was less in 2020 than it was in 2019, it was not substantially less and was within the fluctuation that we typically see year to year. The CSI numbers dropped in March, April and May and then increased in the summer. This is likely due to many institutions stopping clinical trials temporarily when Covid-19 was first spreading in the United States. As institutions implemented safety protocols, they were able to reopen clinical trials and we saw requests likewise increased.

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PO-GePV-P-02

Quality Assurance for Small-Field VMAT SRS And Conventional-Field IMRT Using the Exradin W1 Scintillator

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Purpose: The plastic scintillator detector Exradin W1 has been proved to provide good dosimetry for a small field owing to its good water equivalence and small sensitive volume. However, few studies have reported on its dosimetry for fields of conventional size. The purpose of this study is to assess the accuracy of the Exradin W1 in measuring conventional-field IMRT plans and small-field VMAT SRS plans, and then to assess its clinical value in dose verification and providing quality assurance for clinical radiotherapy plans for different field sizes. **Methods:** 47 cases (i.e., 29 cases with IMRT clinical plans for conventional fields and 18 cases with VMAT SRS plans for a PTV < 8cm³) in our center were included in this study. The Exradin W1 and Exradin A19 standard ionization chamber were used for IMRT plans, and the Exradin W1 and Exradin A16 microchamber were used for the VMAT SRS plans. Finally, the measurements were compared with those of the TPS and the average differences were calculated. **Results:** For the 29 IMRT plans, the mean difference between the Exradin W1 and TPS was $0.16 \pm 1.83\%$; while for Exradin A19 chamber radio was $0.97 \pm 1.77\%$, which was significantly higher than that of the Exradin A19 chamber ($p = 0.048$). For the 18 VMAT SRS plans, the mean difference between the Exradin W1 and TPS was $1.13 \pm 1.33\%$ and the ratio for Exradin A19 microchamber was $0.93 \pm 1.57\%$. The accuracy of the Exradin W1 was nearly the same as that of the Exradin A16 and no statistical difference was observed. **Conclusion:** The Exradin W1 has good accuracy in measuring doses for conventional-field IMRT plans and small-field VMAT SRS plans, which makes it a more comprehensive radiation detector than others available and will be convenient for use in a clinic.

PO-GePV-P-03

An Automated, Vendor-Neutral Method for CT Protocols Documentation

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Purpose: The large number of protocols on each computed tomography (CT) scanner makes protocol review a laborious task to complete manually, especially for institutions with multiple CT scanners. Vendor-specific naming conventions, variable protocol templates, and immutable documents make the constant review of protocols time-consuming and prone to errors. The purpose of this study was to develop an automated, vendor-neutral software solution to reformat the protocol export files obtained from CT scanners to facilitate the protocol review process. **Methods:** Instead of manually maintaining a protocol book with all acquisition and reconstruction parameters for dozens of protocols on each CT scanner, we proposed taking advantage of the exported protocol files (in either .csv or .xml formats). Since these files are not reader- and print-friendly, a Python script-based program was created to identify and store values for the following parameters: tube potential, tube current, automatic exposure control reference, pitch, minimum and maximum tube current (if applicable), tube rotation time, collimation, computed tomography dose index (CTDI), CTDI notification value, dose-length product (DLP), slice thickness, slice interval, displayed field of view (DFOV), reconstruction kernel, and iterative reconstruction algorithm setting. An additional Python program was written to identify changes between two sets of protocols from the same scanner exported at different points in time. **Results:** The consistent, vendor-neutral format accelerates the identification of relevant protocol parameters, making protocol documentation and reviewing much easier. The additional comparison program automatically identifies changes in protocols between two points in time, allowing the protocol management team to quickly review the changes for accuracy. **Conclusion:** The program we developed saves the protocol management team a significant amount of time that would be spent manually parsing the raw protocol outputs or entering protocol changes. The program also opens new possibilities for more comprehensive analyses of protocols across vendors and throughout time.

PO-GePV-P-04

An Automatic CT Quality Assurance Program

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Purpose: The purpose of this study was to develop a digital, self-regulating program to monitor CT quality assurance (QA) results. There are two main goals of the study. The first was to transition from physical to digital records. The second was to reduce the QA review time spent by medical physicists. **Methods:** The program consists of two elements. The first element is a set of digital forms used to record daily, weekly, and monthly quality assurance results for all CT scanners. The daily test includes CT number, noise, and uniformity. The weekly test includes SMPTE pattern check. The monthly test includes visual

inspections, laser alignment, and vendor-recommended quantitative tests such as spatial resolution, contrast scale, and slice thickness. These Excel forms are stored on a cloud drive and incorporate limits specified by vendors and the American College of Radiology (ACR). The second element is a Python-based program that searches the forms for out of tolerance results and blank entries, generates weekly summaries, and sends email alerts when they are found. Pandas library was used for all data manipulation, while Pywin32 library was used for sending out email alerts. This program automatically runs daily using Windows Task Scheduler on a desktop computer. The accuracy of the program was evaluated for 2 months. **Results:** A defined group of users received an email alert whenever there were failures, warnings, or missing records. The results of the manual review all agreed with the program results. In addition, the program detected missing entries that the human review did not capture. For our enterprise health system with 20 CT scanners, it saves 100 minutes of a medical physicist's time per week. **Conclusion:** We successfully implemented an automated method for CT QA documentation and review. This method not only avoids human errors, but also can increase productivity.

PO-GePV-P-05

Connecting Tableau to An Oncology Informatics System to Identify Treatment Planning Workflow Delays

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Purpose: Over one-third of patient treatment plans at our institution are approved by the physician the day of, or the day prior to the new treatment start date. With the mean number of new starts increasing to over 10 per day, it is important to quickly identify plans held up at any stage in the planning process. This work develops a Tableau dashboard to visualize the status of quality check list (QCL) tasks for upcoming new start appointments. **Methods:** The Mosaic SQL database was connected to Tableau, a commercial data visualization toolkit. The treatment planning workflow was separated into seven components: simulation, contouring, planning, quality assurance, physics check, document approval, and therapist check. Each of these elements were classified as pending, completed, or in progress, with further information such as staff responsible, elapsed time, and any notes from Mosaic shown where relevant. Upcoming patient new start appointments were extracted from the database, sorted by date, and stratified by clinic location. Filters were added to allow users to search for patients or drill-down to specific start dates, clinic locations, and machines. **Results:** A web-based platform was created through which users can get an overview of plan status for upcoming new start appointments. Elapsed time of QCL tasks can be used to identify when and where tasks are held up, and the responsible staff member initials make it easy to identify a point of contact for follow-up. Notes from the QCL in Mosaic are displayed with a hover-over action, allowing additional information to be relayed. Comments can be added directly to the Tableau dashboard to facilitate communication about plans between staff. **Conclusion:** Tableau allows for a clean web-based display of information and eliminates multi-click searches within Mosaic. Thoughtful data visualization and simple filters facilitate a proactive approach to workflow bottlenecks and challenges.

PO-GePV-P-06

Implementation, Utilization, and Time Analysis of An Optional Physics Pre-Planning Check

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Purpose: TG-100 gives examples of checks that should be performed before treatment-planning, not afterwards as is commonly done. We have implemented a "physics pre-plan check," an optional check, performed at dosimetrist request, of 36 items that affect dose-calculation or plan-optimization with the intent of catching errors that would require repeating those steps. In this study, we report on utilization and time required for this additional check. **Methods:** For 3487 patients between February 2017 and February 2021, we recorded if and when a pre-plan check was requested and whether the check was performed. We calculated hours spent between contour completion and plan completion with and without pre-plan check. **Results:** Pre-plan checks were requested for 663 (19.0%) patients, but requests steadily declined over time from 43.9% in 2017 to 6.7% in 2021. 5.4% of checks were requested after planning but prior to plan documentation/upload. The likelihood of a request had a high positive correlation (0.899) with days remaining to the first fraction. Of the requested checks, 575 (86.7%) were completed. The remaining requests were eventually canceled by the dosimetrist. The median time from request to check completion was 2.3 hours. For canceled requests, the median time from request to cancellation was 5.6 hours. On average, a pre-plan check added 3.6 hours overall to the planning process. Plans without checks were completed in 9 hours compared to 12.6 hours for plans with checks (p-value: 9.014E-7). **Conclusion:** Our institutional experience is that the wait for a physics pre-plan check generally exceeded the expected time savings from catching an error early, and our dosimetry staff has mostly opted to forgo it. To retain the benefits of pre-plan checks, we developed a tool to automate the checking of 14 items and developed a checklist for the remainder for the dosimetrist to use without requiring a physicist.

PO-GePV-P-07

Automatic Parameter Extraction from Scanners to Wiki for CT Protocol Management

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Purpose: Joint Commission requires that all CT protocols are reviewed by a site's CT protocol optimization team. All protocols contain several series, each with multiple parameters (including kVp, mAs, pitch, resolution, etc), which require documentation and tracking, both being time consuming and labor intensive. We aimed to export protocol files from our scanners, and export scan parameters to a readily accessible and tracked wiki page. **Methods:** CT vendors export protocols in different formats, including .PROTO format for GE scanners, and .xml format for Siemens scanners. We developed Matlab programs to extract scan parameters directly from these protocol files, which are arranged in structured tables and written in a wiki format that can be displayed on web pages. Whenever scan parameters are changed on the scanners, a technologist exports the protocols to update the wiki. All history of protocol changes is also saved on the wiki. **Results:** The wiki-based CT protocol reviewing system allows immediate access to CT scan protocol parameters without requiring physical presence at the scanners. All essential scan parameters are summarized in one table rather than under multiple tags on vendor specific interfaces. This allows much quicker protocol comparison if discrepancies are noted during our CT radiation dose review process. The wiki site also serves as a reference for physicists and technologists to build similar protocols on new scanners, which can be laborious if from different vendors. **Conclusion:** Extracting CT scan parameters automatically from the scanners and exporting them in a wiki allows easily accessible protocol documentation and monitoring.

PO-GePV-P-08

Scripting Occupational Badge Dosimetry Analysis for a Large Number of Clinics Using MATLAB

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Purpose: To create a MATLAB code to automate and streamline occupational badge dosimetry monitoring and reporting for the oversight of a large number of hospitals and clinics. **Methods:** As part of a physics group that is responsible for radiation safety for about 50 institutions, it is often difficult and time-consuming to review dosimeter badge reports on a large scale. For this work, we only looked at sites that are covered by our physics group that use Landauer dosimeters. Landauer provides a tabular spreadsheet of badge measurements that can be downloaded from their website. However, challenges arise from the number of sites covered, site-specific ALARA levels, site-specific effective dose equivalent (EDE) factors, missed or unreported badges, and fetal badges. Manual analysis for the total number of sites as well as specificity of each site makes it tedious to notify any individual of exceeding ALARA levels. A MATLAB code was generated to organize badge categories, summarize any exceeding badge tolerances, and generate PDF reports for analysis. **Results:** The MATLAB script generates a modified Excel spreadsheet that organizes badge categories into separate tabs as follows: total deep dose equivalent badges, fetal

badges, extremity badges, EDE conversion doses, area monitoring, and missed or unreturned badges. Each tab is sorted in descending order with conditional formatting of any exceeding ALARA values. A summary tab is generated as a quick reference that includes only badges exceeding ALARA levels, any fetal badge readings, and any missed or unreturned badges. A word notification document is generated for any individual that exceeds an ALARA level. **Conclusion:** MATLAB is a powerful tool that can generate an organized spreadsheet and efficiently analyze necessary occupational dosimeter badge measurements.

PO-GePV-P-09

Value of Peer Review: An Outreach Facility Perspective

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Purpose: Peer review for outreach facilities is a valuable safety and quality tool. We describe an approach utilized to overcome the challenge of converging an outreach community practice with an affiliated academic clinic for a consistent level of excellence. **Methods:** The RO outreach practice developed a robust program to provide peer learning that promotes CQI in treatment practices. Prospective and retrospective processes are implemented through two primary mechanism, case-oriented and practice-oriented peer review. **Results:** Case-Oriented Peer Review Prospective peer review is performed with a four-layer peer review system on all patients' treatment courses for curative and palliative cases. Three days a week the outreach facility team participate in an affiliated academic center's dedicated 90 minute time slots for disease site groups. Pretreatment chart rounds include planned cases where indications, volumes, dose/fractionation and beam arrangements are considered. Review is recorded in a patient specific chart round task within RIS. Academic input facilitates an open discussion with all department staff represented. These prospective peer reviews are cascaded to the outreach facility chart rounds in an additional once/week time slot. A Chart Rounds widget enables electronic attendance records and patient specified tasks. Case-oriented peer reviews are performed subsequently by dosimetry, physicist and therapists prior to treatment initiation. Multidisciplinary level review is performed at the academic center and outreach facility tumor boards. Practice-oriented Peer Review The outreach staff participate in numerous Standing Committees. This peer review also includes a prospective, concurrent or retrospective case for educational purposes, professional feedback and future learning. Staff teach and review their CEU's. **Conclusion:** This peer review implementation helps establish a closer working relationship with the academic center and the process improves patient care and consistency. Scheduling complications between the groups are addressed. A successful comprehensive peer review program has been implemented, which can be applied to networked facilities.

PO-GePV-P-10

Comparison of Estimated Radiation Dose Indices for Abdominal Pediatric Computed Tomography Examinations - Patient Chronological Age Versus Patient Water Equivalent Diameter

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Purpose: To compare estimated radiation dose indices for abdominal pediatric computed tomography examinations when analyzed against patient chronological age versus patient water equivalent diameter. **Methods:** For the calendar years 2019 and 2020, a radiation dose index monitoring (RDIM) system collected dose indices, water equivalent diameter and chronological age for 3,236 pediatric CT examinations of the abdomen and abdomen/pelvis performed on forty-four (44) CT scanners located at sixteen (16) acute care hospitals and twelve (12) ambulatory imaging facilities. The RDIM harvested Dose Linear Product (DLP); Volumetric Computed Tomography Dose Index (CTDIVOL); Effective Dose (ICRP-103); Water Equivalent Diameter (WED); and Chronological Age (years) for selected pediatric single-phase examinations of the abdomen and abdomen/pelvis. **Results:** Enterprise-wide comparison of estimated dose indices confirm the benefits associated with acquisition parameters guided by patient attenuation coefficients as determined by water equivalent diameter estimates from CT localizer radiographs. **Conclusion:** The project confirms the benefit of CT localizer radiographs in setting acquisition parameters based on estimates of patient attenuation of radiation across the scan range.

PO-GePV-P-11

Optimizing Quality Assurance with Total QA® Implementation

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Purpose: To implement an easy-to-use cloud based quality assurance (QA) service to replace an existing framework of utilizing in-house spreadsheets and code to perform and analyze linac, brachytherapy, and CT QA. **Methods:** Performing, recording, and storing QA are essential to radiation therapy. The tasks of recording, analyzing, and storing normally employ customized, in-house spreadsheets and code that have to be internally maintained. Maintaining such a system can become complicated and error prone. Total QA® (Image Owl, Inc.), a QA creation, record, storage, analysis, and presentation service, is designed to meet an institution's QA needs and AAPM task group recommendations. For comparison, in-house safety, dosimetry, and mechanical QA tools, code, and spreadsheets for linacs, brachytherapy, and CT scanners were recreated in TQA and reviewed to assess the accuracy and validity of cloud-based system. After a satisfactory evaluation, a year's worth of QA were entered into TQA to create baseline metrics for each machine. **Results:** All existing linac, brachytherapy, and CT QA were successfully constructed in TQA to meet AAPM task group recommendations. Test creation, revisions, and maintenance were easier in TQA. Data entry, local and global analysis, and trend creations were faster and simpler in TQA. TQA eliminated the risk of accidentally altering spreadsheet formulas and/or code. TQA data was easily accessible and readily presentable for QA documentation necessary for regulatory/accreditation procedures and visits. **Conclusion:** Utilizing TQA to handle a clinics QA essential needs is more efficient, beneficial, and robust than maintaining in-house spreadsheets and code. TQA provides an easy, single point of access to facilitate any clinic's desire to organize, record, store, and present QA data.

PO-GePV-P-12

An Integrated, Automated Framework for Review and Approval of Policies and Procedures

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Purpose: Develop and implement a scalable framework for housing policy and procedure documents, with automated version control, approval workflows, and review cycle notifications. **Methods:** After exploring several commercial options (all found to be lacking in one regard or another), we ventured to build our own solution which would allow full control and customization. Microsoft Sharepoint was utilized as the backend data storage for our document library and instantiated with several custom categories of metadata to classify each document and allow easy navigation and filtering. An automated approval workflow was developed in Microsoft Power Automate, which allowed programmatic customization and where the metadata was again utilized for logic-based decision-making. Both the document library and approval workflow were integrated with Microsoft Teams and Outlook for front-end user interactions and document browsing. **Results:** Power automate provides programmatic logic to fully customize and automate a sequential approval chain workflow, sending notifications through email and Teams. The Sharepoint document library provides built-in version tracking and customizable permissions groups to restrict editing while also granting widespread availability through Teams integration. **Conclusion:** Moving from a file-based-system stored on a network drive to a cloud-based, automated in-house solution for review cycles and approval workflows will simplify and streamline the management of departmental policy and procedure

documents. Improving the contemporaneity and availability of policies and procedures should improve compliance with the information therein, and have a positive impact on overall quality program.

PO-GePV-P-13

In-House Incident Learning System for Failure Modes and Effects Analysis Utilizing Microsoft Sharepoints and Powerapps

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Purpose: We propose a simple in-house internal incident learning system utilizing available Office 365 features; Sharepoints and Powerapps. The goal of the designed app is to periodically perform and update our Failure Modes and Effects Analysis FMEA prospective risk management for continuous quality improvement using collected data and failure information of processes and/or equipment that are encountered throughout our daily workflows. **Methods:** We have broken down treatment workflows at our radiation oncology department by modality and technique. We chose GYN HDR Brachytherapy as a starting point. A radiation oncologist, medical physicist, oncology nurse, therapist and practice director teamed up to discuss the GYN HDR process map, to review identified failure modes from old data, and to approve the preliminary ranking and risk priority numbers RPN derived from AAPM TG100. 6 failure types and 19 failure modes were mapped to the consequences compromising treatment. A Sharepoint list in Office 365 organization workspace was created and founded the database for the Powerapp. The Powerapp building step is mainly automated and allows customization and connection to multiple data sources including SQL and excel spreadsheets. Patient MRN is entered by the user, a date of the encounter is selected, and dropdown menus of preidentified failure modes, associated causes and consequences are built-in to choose from, in addition to a free text box to enter comments. **Results:** The app was published for use by the GYN HDR Brachytherapy care team and being evaluated. The next step is to expand the app to include all modalities and techniques in our department. **Conclusion:** We have created a simple incident learning system for our department internal use that is expected to help us collect and accumulate relevant, accurate and reliable data to perform consistent and continuous FMEA at our department and compare our results to published data and reports.

PO-GePV-P-14

Proton Radiosurgery: Small PBS Field Dosimetry Validation

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Purpose: We present on the validation of small-collimated field dose calculations and delivery to enable the transition of a clinical proton radiosurgery (PSRS) program from a passive scattering (PS) to pencil beam scanning (PBS) delivery technique. **Methods:** 250 PSRS PS fields treated in the previous two-years at our institution were used to establish base parameters including range, modulation, aperture diameter and field aspect ratio. Previous Monte Carlo simulations of the treatment planning system's (TPS) small field halo algorithm and limited measurements paved the way to a more comprehensive validation for PSRS. Micro-diamond detectors were used to measure central axis depth dose profiles of PBS fields satisfying parameter combinations described above. Output measurements were repeated in a solid water phantom to establish equivalence when applying a solid water to water ratio to account for differences in scatter within the mediums. The solid water setup is intended for use for subsequent patient specific QA. Film were calibrated for doses between 50-1,300 cGyRBE. An alignment template was incorporated into a phantom to ensure setup reproducibility. The phantom allows the micro-diamond detector and films measurements to be performed simultaneously. Dose profiles were assessed using a hybrid 2/3D gamma-analysis. **Results:** In-water dose measurement agreement was -1.7 ± 2.9 and 2.5 ± 5.1 %, at isocenter and 10 mm depth, respectively. Distal and proximal SOBP 90% depth agreement was 0.10 ± 0.66 and 0.74 ± 0.48 mm, respectively. PASS rates for isocentric profile gamma-analysis within 2% global dose, 1 mm distance to agreement and 10% threshold was 97.2 ± 4.8 %. **Conclusion:** Measurements performed for small PBS collimated fields with a variety of range shifters and delivery parameters confirms the TPS can properly model small fields used for proton PSRS.

PO-GePV-P-15

Pre-ESD Evaluation in Early Gastric Cancer

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Purpose: Comparative analysis of the pathological diagnosis of endoscopic gastric biopsy and endoscopic submucosal dissection in patients with early gastric cancer and precancerous lesions in our hospital. Evaluate the size, nature and depth of the lesions before ESD, to reduce the risk of additional surgery after ESD. **Methods:** From January 2016 to December 2019, the 75 patients with gastric mucosal HGIN, severe atypical hyperplasia and in situ carcinoma which were treated by ESD were include. The relationship between lesion area, expression of white light endoscopy, magnification endoscopy, narrowband imaging endoscopy and postoperative pathological classification and depth of lesion was analyzed. **Results:** There were 75 patients with a total of 79 lesions. The coincidence rate of preoperative and postoperative pathology was LGIN 100.00% (4/4), HGIN 97.83% (45/46), Adenocarcinoma 89.66% (26/29). There was no significant difference between preoperative and postoperative pathological evaluation ($X^2=0.458$, $P=0.795$). The type of lesion and the degree of malignancy were independent with the location and size of the lesion. The endoscopic features of high grade intraepithelial neoplasia and early gastric cancer were clear border, irregular microvessel and irregular surface microstructure. Lesions > 3 cm, surface ulcer, spontaneous bleeding may be the risk factors for deeper lesions. **Conclusion:** 1. Endoscopic submucosal dissection is not only a means for the treatment of early gastric cancer and precancerous lesions, but also an important method for clear pathological diagnosis. 2. Magnification endoscopy combined with NBI can find early detection of HGIN and early gastric cancer. Preoperative biopsy had guidance significance in diagnosis and treatment, which can reduce the missed diagnosis of early gastric cancer.

PO-GePV-P-16

Risk Factors of Early Gastric Cancer Identified by Screening Endoscopy

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Purpose: A customized screening program for gastric cancer would optimize the benefits of screening endoscopy. This study investigated the risk factors for gastric cancer detected during screening and factors affecting clinical outcomes. **Methods:** The patients in Hebei General Hospital from January 2014 to December 2020 were selected as the research objects. The clinical outcomes of early gastric cancer identified during screening (screening group) were compared with age, sex, Helicobacter pylori infection, Family history, environment and bad habits of diagnosis matched subjects who were diagnosed with early gastric cancer in the outpatient clinic (outpatient group). **Results:** Of 39530 subjects, 226 were diagnosed with early gastric cancer. The median age of the screening group was 61.2 years (interquartile range 56–71 years), and the male-to-female ratio was 2.1:1. When comparing with the control group, Helicobacter pylori infection (odds ratio [OR] 2.721, $P < 0.001$), carcinoembryonic antigen (OR 9.623, $P = 0.004$), family history of gastric cancer (OR 3.154, $P = 0.005$), and drinking (OR 4.311, $P < 0.001$) were independent positive risk factors, and the use of aspirin a negative risk factor for gastric cancer (OR 0.345, $P = 0.011$) in multivariate analysis. The 5-year survival rate was significantly higher in the screening group than in the outpatient group ($P < 0.001$). Conditional Logistic regression was used for single factor and multivariate analysis, and OR value and 95% confidence interval were calculated. **Conclusion:** The high incidence of gastric cancer is related to many factors, especially the eating habits of pickled vegetables, smoking and drinking. Comprehensive measures should be taken to reduce its incidence.

PO-GePV-P-17

Influencing Factors of Daily Activities in Patients with Early Gastric Cancer After Endoscopic Treatment

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Purpose: To analyze the influencing factors of daily activities in patients with early gastric cancer after endoscopic treatment. **Methods:** Retrospective analysis the clinical data of 226 patients with early gastric cancer who underwent endoscopic submucosal dissection or endoscopic mucosal resection was performed in the Endoscopy Center of Hebei General Hospital from January 2014 to December 2020. We affiliated the university, including age, gender, base Basic disease, tumor size, location, morphology, postoperative pathological type and depth of invasion, postoperative complications, etc. Collect patients' daily activity scores during admission and discharge. Analyze the influencing factors of daily activities in patients with early gastric cancer after endoscopic treatment. **Results:** Age and postoperative bleeding were negatively correlated with postoperative daily activity scores ($r=-0.215$, $P<0.001$; $r=-0.155$, $P=0.025$). Logistic regression analysis showed that age was an independent risk factor for the decrease in daily activity score after operation (OR=1.085, 95%CI 1.003~1.204, $P=0.028$). **Conclusion:** The perioperative nursing work and health education for the elderly patients with early gastric cancer who undergo endoscopic treatment should be strengthened, and the patients' postoperative daily life ability should be closely monitored.

PO-GePV-P-18

The Diagnostic Rate and Analysis of the Clinical Pathological Features of Early Gastric Cancer

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Purpose: To calculate the diagnosis rate of early gastric cancer, and to analyze the relationship between clinicopathological features and lymph node metastasis of early gastric cancer. **Methods:** The clinical and pathological data of 201 patients with early gastric cancer who were admitted to Hebei General Hospital were collected from Jan. 2018 to Dec. 2020, and the early diagnosis rate was calculated. They were divided into lymph node negative group (121 cases) and lymph node positive group (80 cases), and the related factors were compared and analyzed. **Results:** The early diagnosis rates in 2018-2020 were 12.36%, 14.14%, and 15.17%, respectively. The average early diagnosis rate in 3 years was 12.40%. Gender, depth of invasion, and degree of differentiation were significantly associated with lymph node metastasis ($P<0.05$). Gender and depth of invasion were independent risk factors for lymph node metastasis ($P<0.05$). The AUC for predicting lymph node metastasis by gender was 0.682 (95% CI: 0.508-0.856); the AUC for predicting lymph node metastasis by depth of invasion was 0.722 (95% CI: 0.573-0.871); when the two factors were combined, the AUC for predicting lymph node metastasis was 0.806 (95% CI: 0.696-0.917), both higher than the AUC of individual gender or depth of invasion. Female and submucosal infiltration were independent risk factors for lymph node metastasis, and their predictive value for lymph node metastasis was higher when combined than when alone. **Conclusion:** The early diagnosis rate of early gastric cancer is about 12% in our hospital in the past 3 years. Female patients, submucosal infiltration or low differentiation degree of early gastric cancer are prone to lymph node metastasis, which should be carefully treated under endoscopic treatment.

PO-GePV-P-19

Regional Workshops as a Method to Facilitate the Implementation of Task Group Recommendations: The Experience of AAPM Task Group 275

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Purpose: To share the experience of AAPM Task Group 275 (TG275) in developing and delivering regional workshops to encourage the implementation of TG recommendations. **Methods:** A one-day workshop curriculum was developed by nine task group members to familiarize participants with the work and recommendations of TG275: Strategies for Effective Physics Plan and Chart Review in Radiation Therapy. In addition to a didactic component, at least 40% of each session's allotted time was devoted to active learning activities including individual exercises, small group activities, and full group discussions. Morning sessions included an overview of the work of TG275 along with education on the data collection phases. Afternoon sessions included education about the analysis phase, key recommendations, and suggestions for clinical incorporation of TG275 recommendations. The target audience included medical physicists with responsibility for plan review and chart checks. Each workshop was sponsored by an AAPM Chapter and feedback was gathered through program and speaker evaluations. **Results:** To date, two workshops have been held. The first was in-person with seventeen attendees and facilitated by two TG275 members. The second was virtual with fifteen attendees and facilitated by four members. Minor adjustments were made to the curriculum after the first workshop based on attendee feedback. The curriculum proved adaptable to both in-person and virtual formats. Attendees were actively engaged throughout both workshops. CAMPEP credits were offered in both instances with SAMs credits offered for the second workshop. Average program evaluation scores were 4.75 and 4.82 out of 5 for the two workshops respectively. **Conclusion:** A workshop curriculum was developed with the goal of familiarizing participants with the work and recommendations of AAPM TG275. To date, two workshops have been held. Workshops are a potentially useful mechanism to encourage the implementation of AAPM Task Group recommendations.

PO-GePV-P-20

Syed Template for Interstitial HDR: A 3D Printed Alternative

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Purpose: To have a template that is easy to handle during the Gynecological interstitial procedure. **Methods:** The Syed templates that are commercially available are very bulky and difficult to handle during the procedure. Suturing the template to perineum of the patient is very difficult. In addition to this due to the bulk of the template, it is easily movable when the patient is in discomfort which can lead to a re-plan on the subsequent fractions of treatments. The commercially available Syed Implants are about 3 cm in thickness and the passage of needles through that to position inside the tumor is difficult. Thinking about an alternative, we decided to use the 3D printer that was available in the department. Multiple iterations of Syed Templates were created and evaluated, dummy plans were created and checked to make sure practically this template is able to be used for improving patient comfort as well as positioning accuracy. **Results:** Using the 3D printer, we are able to print and deliver treatment with the guidance of a thinner and robust template by developing an in-house CAD plan data-file. **Conclusion:** Innovative technologies such as 3D printing is able to give clinically relevant solutions for day to day use.

PO-GePV-P-21

A Decentralized and Autonomous Blockchain-Based Medical Physics Education Platform: Toward a New Paradigm

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Purpose: Create a decentralized blockchain-based education platform to meet the modern medical physics education needs nationally and globally. **Methods:** The platform uses on two enterprise-level open source packages: Open edX and Hyperleger Fabric. The Open edX platform is responsible for course creation and hosting while the Hyperledger-based blockchain manages the course system. The blockchain 1)grants course creation permission; 2)issues course completion certificate; and 3)validates the course certificate upon request. The chaincodes of the blockchain are implemented through Go programming language. **Results:** We have successfully built up such a decentralized and autonomous education platform locally. An initial group of users such as all full or emeritus AAPM members can be enrolled through email verification and issued a private/public keypair. Other users are open to register and are subject to a single enrollment vote from an enrolled user. Any users can obtain time-constrained permission to create or modify a course, which are also subject to vote. Whether to publish a new course or a course improvement is also determined by consensus. Participation in the vote grants users reward points which determine the voting power of the user along with a power abuse prevention mechanism. Inactivity in participation would lead to the removal of a user. Furthermore, any activity be can monitored by any user anytime through the ledgers. Our local implementation is ready to migrate to a cloud-based environment. **Conclusion:** This blockchain-based medical physics education platform does not have a centralized authority to control its access and contents. Anyone can suggest what to teach and what to learn. The platform enables a decision based on collective intelligence and helps to create medical physics courses from the needs of the users, by the talents and visions of the whole medical physics community. This also can provide free medical physics education globally.

PO-GePV-P-22

Recommendations for Treatment Planning Dose Indices for Single-Isocenter Single Target VMAT Brain Stereotactic Radiosurgery/Radiotherapy; A Retrospective Analysis

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Purpose: Dose indices are effective variables that may serve as predictors in determining the plan quality for Stereotactic Radiosurgery (SRS) and Stereotactic Radiotherapy (SRT) treatments of brain lesions. The purpose of this study is to provide recommendations for dose indices versus target volumes based upon an institutional retrospective analysis of Linac based VMAT brain SRS/SRT treatments utilizing the Edge Radiosurgery System (Varian Medical Systems Inc, Palo Alto, CA). This study compares the dose indices of VMAT brain SRS/SRT treatments of patients treated at The University of Toledo Medical Center against previously established institutional guidelines for dose indices of this kind of treatment based upon Dynamic Conformal-Arc (DCA) radiotherapy. **Methods:** One hundred and forty patients with approved treatment plans planned in RayStation V.8A (8.0.1.10) TPS treated with VMAT SRS/SRT on Varian EDGE radiosurgery system using 6FFF with single brain metasis were randomly selected. Plans were grouped into three categories based upon lesion volume; small (0.10-2.0 cc, n=80), medium (2-10 cc, n=35), large (10-40 cc, n=24). Gradient Indices (GI), RTOG conformity indices (CI(RTOG)), and Paddick Conformity Indices (CI), were computed for each plan using a script developed in house and calculated via a collapsed cone dose calculation algorithm V.4.1 with a 1mm dose grid resolution. Plan indices for each plan were tabulated. **Results:** For patients presenting with small brain lesions (n=80), medium lesions (n=36), and (n=24) large lesions the recommendations suggested for plan quality metrics are presented in Table 1 of supporting documents. **Conclusion:** This study provides recommendations based upon an institutional retrospective analysis for plan indices versus lesion volumes for VMAT brain SRS/SRT Linac based treatments; serving as a guide to assess brain VMAT SRS/SRT plan quality. We recommend plan quality metrics be calculated for patients, preferably using scripts to allow this to be accepted into the clinical workflow.

PO-GePV-P-23

Volumetric Study for Hearing Preservation Using Fractionated Radiosurgery

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Purpose: To determine clinically relevant SBRT volumetric dose tolerance limits of hearing preservation in 1 to 5 fractions for Cyberknife based on statistical analysis of outcomes data. **Methods:** Retrospectively analyzed sixty-nine acoustic neuroma Cyberknife cases treated in 3 or 5 fractions. A logistic-dose response model of data was performed using the DVH Evaluator software. The selected endpoint was 50 dB of hearing loss in one ear, which is part of the Gardner-Robertson criteria for serviceable hearing. A 5 fraction regimen was chosen for the aggregate analysis to minimize the uncertainty of the alpha/beta parameter. QUANTEC used an alpha/beta parameter of 3 Gy and we iteratively determined that the alpha/beta parameter with maximally likelihood fit was 4.2 Gy. Confidence bands were estimated using the profile likelihood method. **Results:** QUANTEC recommends keeping single fraction maximum dose to the cochlea below 14 Gy to maintain less than 25% risk of serviceable hearing loss, so we also chose 25% as the risk level for comparisons, which corresponded to cochlea Dmax=25.6Gy/5fx in this updated analysis. In regard to volume effects, the cochlea D0.05cc<=20Gy/5fx and the median cochlea dose D50%<=20Gy/5fx both corresponded to 25% risk. **Conclusion:** The 50dB threshold level for serviceable hearing can be used to estimate the rate of hearing risk analysis from multisession acoustic neuroma radiosurgery. The resultant risk levels from the clinical datasets were compared to the QUANTEC 25% risk expectation, providing optimism for clinic use. Potentially the differential dose tolerance of the various volumes of the hearing apparatus could be exploited to improve hearing preservation.

PO-GePV-P-24

Dosimetric Benefits of Target-Tracking-Optimization Based Partial-Arc VMAT Technique for Mono-Isocenter Multiple Pulmonary Metastases ABC-SBRT

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Purpose: Treating multiple pulmonary metastases with stereotactic body radiation therapy (SBRT) becomes more challenging and promising. The purpose of the present study was firstly to apply the target-tracking-optimization based partial-arcs VMAT technique to further optimize and improve the dose distributions of target volumes and organs at risk (OARs) for mono-isocenter multiple pulmonary metastases active breathing control (ABC)-SBRT. **Methods:** Eighteen patients with multiple pulmonary metastases underwent mono-isocenter highly conformal coplanar partial-arcs VMAT SBRT treatment in our institution were retrospectively identified. All patients received ABC-CT simulation, ABC-gated kilovoltage cone-beam CTs (kV-CBCTs) and ABC-gated radiotherapy. A single-isocenter was placed among the lesions and doses were 60Gy in 8 fractions. Plans were calculated in Monaco TPS with target-tracking-optimization based partial-arcs VMAT (TTO-VMAT) technique and normalized to at least 95% of the planning target volume (PTV) receiving 100% of the prescribed dose. For comparison, single-isocenter without TTO-VMAT (unTTO-VMAT) and conventional multiple-isocenters VMAT plans (MultIso-VMAT) were created respectively. Dosimetric evaluation of planning target volume (PTV) and OARs were analyzed and compared. **Results:** The present TTO-VMAT SBRT showed better dosimetric advantages compared to other two methods. For gradient indexes (GIs), R50, R100 and D2cm, TTO-VMAT plans were superior to all plans, the unTTO-VMAT was slightly inferior to the MultIso-VMAT plans. There was no clinically significant difference in conformity indices, heterogeneity index, lung-V30 and other OARs among all SBRT plans. The normal lung V5(p = 0.04), V10(p = 0.02), V20 (p = 0.03) in TTO-VMAT were lower than those in MultIso-VMAT and unTTO-VMAT plans. **Conclusion:** The proposed target-tracking-optimization based partial-arcs VMAT optimization algorithm successfully improved dosimetric performance for mono-isocenter multiple pulmonary metastases SBRT and reduced doses to the normal lung and some of other OARs. This TTO-VMAT has the capability of immediate clinical application.

PO-GePV-P-25

Dosimetric Evaluation of Aperture-Based and Beamlet-Based Optimization for Conventional IMRT/VMAT and SBRT

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Purpose: The quality of radiotherapy plans depends on many factors such as optimization methods, dose constraints and delivery techniques. This study investigates the plan quality for conventional IMRT/VMAT and SBRT utilizing aperture-based and beamlet-based treatment optimization algorithms. **Methods:** The Varian Eclipse and Prowess RT-Pro TPS were used in this study, which employed beamlet-based and/or direct-aperture optimization (DAO) for IMRT/VMAT planning. Thirty-six patients were re-planned using Prowess with matching beam configuration and dose constraints, who were previously planned using Eclipse and treated on Varian accelerators. Treatment sites included head and neck, lung, liver and prostate. Plan quality was evaluated using the homogeneity index (HI), conformity index (CI), Dmax(D1%), Dmin(D99%), and dose falloff outside the target (Glouter) for conventional IMRT/VMAT, and R50% and D2cm for SBRT additionally. Statistical significance was tested using a paired Student's t-test for normality distribution and a Wilcoxon signed rank tests for non-normality distribution. **Results:** For IMRT, the same results were found in CI and Glouter except for small differences in Dmax, Dmin and HI between beamlet-based optimization (Eclipse) and DAO (Prowess). For VMAT, HI, Dmax and Glouter were identical for Eclipse and Prowess (both using DAO) except for small variations of CI and Dmin. Higher dose volumes were found for prostate OAR with DAO. Identical SBRT plan quality (CI, HI, Dmax, Dmin, R50% and D2cm) was obtained using either IMRT or VMAT delivery techniques for the two TPS/optimization methods when Dp was assigned to 65-85% isodose lines. **Conclusion:** Our study showed that both TPS using DAO and/or beamlet-based optimization produced high-quality plans that met our clinical acceptance criteria. DAO is more efficient than beamlet-based optimization because of the avoidance of leaf sequencing but its random nature in field shape selection results in larger HI (Dmax) for conventional IMRT/VMAT, which is not a problem for SBRT.

PO-GePV-P-26

First Quality Assurance Implementation of Treatment Planning System for Pencil Beam Scanning Proton Radiotherapy

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Purpose: A quality assurance (QA) program for pencil beam scanning (PBS) proton treatment planning system (TPS) was developed and implemented along with monthly CT QA. **Methods:** A program consisted of a periodic test procedure that was designed for PBS TPS quality assurance (QA). The program includes verification of the CT calibration for proton dose calculation algorithm, CT images transferred to TPS, and software functions in TPS. The consistency of the dose calculation algorithm included dose comparison between standard SFO/MFO plan and forwardly re-calculated plans. The consistency of CT images of Catphan transferred to TPS was examined, i.e patient orientation, image geometry, tissue density, and contour dimensions. The aspects of software evaluated include proton layer energy, monitor unit, dose calculation, conversion between HU number to electron density, and proton Relative Stopping Power ratio (RSP). **Results:** The maximum Energy layer for standard SOBP created on randomly CT images from one of our 3 CT simulators kept the same as 117.167MeV, overall the MU was 39993±23 MU, within the selected range 40000±800 MU, the geometry range is (9.62±0.1) cm, within selected range (9.65±0.2)cm, WED is (10.08±0.07), within preselected (10.0±0.2)cm. The maximum dose of the SOBP plan is (96.8±0.6) which is within (97±2)%, and the mean dose is (95.9±0.2)%, within (96±1)%. **Conclusion:** Based on TG53, TG185, and TG224, we are the first to develop the TPS QA program for our PBS proton planning. The program is efficient and effective to check the integrity of the TPS. Test results for the dose calculation algorithm are satisfactory. The implementation of the PBS TPS QA programs verified the consistency of TPS performance which can be traced back to the commission and verification of initial TPS. Periodic testing would have to be followed to ensure the TPS consistently meets the desired treatment quality.

PO-GePV-P-27

Evaluation of Varian Identify SGRT System for Localization of Cranial Targets

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Purpose: Identify (Varian, Palo Alto, CA) surface-guided localization system is a new imaging option offered to monitor inter-fractional and intra-fractional motion of patients. This study evaluates accuracy of target localization for non-coplanar beam treatments using Identify. **Methods:** An anthropomorphic head phantom was scanned using GE Discovery RT, 0.625mm slice thickness, and a multi-arc non-coplanar plan was created in Eclipse TPS (Varian). The DICOM structures of head contour and plan isocenter were sent to Identify server for surface image processing. The head phantom was placed on 6D TrueBeam (Varian) treatment couch in 10 different positions varying 3 translational (<1.5cm) and 3 rotational (<3deg) degrees of freedom, emulating inter-fractional variations. For each phantom placement CBCT imaging was performed using 1mm slice thickness. Imaging followed by a 6-DOF 3D automatic registration to reference image. Registration accuracy was manually reviewed by checking positions of three fiducials. The calculated shift parameters were compared to those generated by Identify system based on comparison of optically rendered head surface and radiographic external contour. After applying CBCT shifts the couch angle was shifted to +90deg position and Identify registration parameter were recorded. **Results:** For 0-degree couch position the maximum magnitude of translational offset recorded was 0.8mm (avg 0.7mm), with largest component in vertical direction (0.5mm). Rotational errors were 0.4deg(yaw), 0.1deg(roll) and 0.2deg(pitch). When Identify was generating shifts based on comparison of optical surface to optical surface captured during the first treatment session, translational match improved to 0.4mm magnitude, 0.2deg(yaw), 0.1deg(roll) and 0.1deg(pitch). When Identify tracked positional shifts during couch angle changes (+90, +45deg), the translational offset magnitude was 0.6mm and rotational errors were under 0.6deg. **Conclusion:** Identify surface guided positioning system is capable of localizing cranial targets with a better than 1 mm precision for non-coplanar treatments.

PO-GePV-P-28

Multichannel Vaginal Cylinder with Intrauterine Tube Applicators for Treating Vaginal, Cervical and Endometrial Cancer

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Purpose: Our institution offered patients brachytherapy with the single (SCVC) or multichannel vaginal cylinder (MCVC) for vaginal cancer and with the tandem and ring (TR) for cervical and endometrial cancer treatment. Each modality has advantages in some aspects over others. A combination of multichannel vaginal cylinder with intrauterine tube (MCVC-IUT) has shown a superior capability of treating vaginal, cervical and endometrial cancer in one procedure over to two procedures using, respectively, TR and cylinder applicators. The current work presents dosimetric evaluation and clinical application of MCVC-IUT applicator set in treating vaginal, cervical and endometrial cancer. **Methods:** Three sizes of MCVC in diameter 25, 30 and 35 mm and three types of intrauterine tubes of 20mm/15 deg, 40mm/30deg and 60mm/45deg were offered to patients. Each was assessed with physical measurements and CT scanning. The dwell source position of the cuffed catheters and intrauterine tubes was verified with a source position simulator and Gafchromic film EBT3. An Elekta Oncentra version 4.5.3 was served as a planning system. Radiation was delivered by an Elekta Flexitron. Radiation dose was prescribed either to the depth or the planned treatment volume (PTV) and point A. IPSA was used for radiation dose optimization. **Results:** Radiation dose 700cGy for a MCVC-IUT system was prescribed to the point A and PTV at 5mm away from the surface of a 25mm-MCVC with treatment length of 50mm. A Smit Sleeve was used for guiding intrauterine tube and identifying cervix. MCVC was able to sculpture radiation dose to the PTV while sparing dose to nearby organs at risk such as rectum and bladder. A

final dose distribution featured a joint dose distribution of tandem and ring with a cylinder. **Conclusion:** MCVC IUT applicators integrated outstanding dosimetric features from TR and MCVC in an effective treatment of vaginal, cervical and endometrial cancers.

PO-GePV-P-29

Daily KV Image Protocol of Prostate Patient Setup with Minimal CBCT

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Purpose: This study's intent is to test if first 3 fractions of a new IGRT protocol consisting of 3 CBCTs followed by daily kV pair and weekly CBCT are able to represent of all remaining treatment fractions. In this study, we are focused on discussing the setup errors of CBCTs. We will discuss neither the setup errors of kV pairs or between kV pairs and CBCT. **Methods:** We would like to compare a new IGRT protocol includes 3 CBCT followed by daily kV pair and weekly CBCT with the current IGRT protocol includes daily CBCT. We randomly selected 10 prostate patients who have daily CBCT in the entire treatment course (total 350 set of CBCT). Our evaluation criterion is the fraction of CBCT position mismatches in the first 3 fractions. An additional CBCT is acquired if only 1 fraction mismatches among the first three. We define as 'new protocol eligible' (eGroup) the group of patients for whom the setups with daily CBCT images match within the pre-selected thresholds of 5, 10, and 15 mm those of the consecutive CBCTs for at least 50% of the fractions. A new protocol ineligible (iGroup) is defined as the group of patients other than eGroup. **Results:** The fractions of patients in the eGroup were 10%, 80% and 90%, when the setup thresholds selected were 5, 10 and 15mm respectively. The sensitivities of determining iGroup from the first 3 fractions of CBCT were 0.88, 1.0, and 1.0, and the specificities were 1.0, 0.75 and 1.0 with thresholds 5, 10 and 15 mm respectively. **Conclusion:** The method suggested here well categorized cases for our new IGRT protocol. More patients are eGroup eligible if larger setup threshold is used. CBCT frequency in the daily kV plus weekly CBCT protocol may need to increase for protocol ineligible patients.

PO-GePV-P-30

Comparison of Physical and EQD2 Plan Parameters Between Two Kinds of Hybrid Plans for Stage III NSCLC

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Purpose: To compare physical and EQD2 plan parameters between two kinds of hybrid plans for stage III NSCLC. **Methods:** Two kinds of hybrid plans named CFRT&SBRT and CFRT&SIB were retrospectively made for 20 patients with stage III NSCLC. Prescription dose of CFRT&SBRT plans was 30×2Gy for PTVLN (lymph node) and sequential 4×12.5Gy for PTVPT (primary tumor), while prescription dose of CFRT&SIB plans was 26×2Gy for PTVLN and sequential 4×2Gy for PTVLN combined with 4×12.5Gy for PTVPT. Physical dose of SBRT and SIB plans were respectively superposed with 30 and 26 fractions dose of CFRT to acquire physical plan parameters of CFRT&SBRT and CFRT&SIB plans. SBRT and SIB plan dose were then corrected by voxel-wise EQD2 dose using an in-house Matlab script. EQD2 corrected SBRT and SIB dose were respectively superposed with 30 and 26 fractions dose of CFRT to acquire EQD2 plan parameters of CFRT&SBRT and CFRT&SIB plans. The physical and EQD2 plan parameters were respectively compared between two kinds of hybrid plans. Non-transit electronic portal imaging device (EPID) dosimetry analysis was performed to rule out dosimetric errors during hybrid plans delivery. **Results:** Physical plan parameters of PTVLN D2, HI, and CI60Gy, PTV(PT+LN) CI60Gy, esophageal V50 and Dmax had more advantages in CFRT&SIB plans, while EQD2 plan parameters of PTVLN HI and spinal cord Dmax were more advantageous in CFRT&SIB plans. There were no statistical difference in other physical and EQD2 parameters for two kinds of hybrid plans. The delivery errors of hybrid plans components were acceptable using the criterion of 2%/2mm and passing rate more than 90%. **Conclusion:** All physical dose volume objectives were met for two kinds of hybrid plans. CFRT&SIB plans had more advantages over CFRT&SBRT plans on physical and EQD2 plan parameters.

PO-GePV-P-31

Using Edge Detectors for Field Output Factor Measurements

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Purpose: Sun Nuclear Corporation (SNC) Edge Detector sensors are often picked as detectors of choice for measurements of small fields output factors (OF) due to their small measurement volume and high sensitivity. However these detectors show increased response for the measurement of larger field output factors. Here we review the limitations of edge detectors for large field output measurements. **Methods:** 4 Edge Detectors were chosen to evaluate TrueBeam output factors for fields in the range of 1cmx1cm to 40cmx40cm. The measurements were done using PTW water tank at 5cm depth and 95cm SSD. Each detector was used to measure output factors for 4 TrueBeam energies, 6x, 6xFFF, 10x and 15x. The field output measurements were initially normalized to 10cmx10cm field output for each of the detectors. Similarly, CC13 mini ion chamber (PTW) was used to measure the outputs for fields 3cmx3cm and higher. All the measured data was compared to output factors measured using Standard Imaging A19 ion chamber. Statistical uncertainty was evaluated for Edge Detector sample group for each field size measurement and the average output factor compared to ion chamber measurement. **Results:** PTW CC13 and SI A19 chambers showed excellent agreement for field sizes 5cmx5cm and larger. Edge Detectors matched the ion chamber output factors in the range of 5cmx5cm-10cmx10cm for low X energies 6x and 6xFFF and between 5cmx5cm and 15cmx15cm for 10x and 15x energies. For larger fields, edge detectors overresponded, and relative output factors exceeded ion chamber measurements with effect more pronounced for larger fields. The largest deviation noted was 2.3%(6x), 2.2%(6xFFF), 1.5%(10x), 1.2%(15x) for 40cmx40cm field. **Conclusion:** While being a good tool for measurement of small field dosimetry, edge detector measurements for fields 15cmx15cm and larger should be treated with care. Normalization of OF should be done to output at 5x5-10x10cm field.

PO-GePV-P-32

Design and Implementation of Internal Shielding to Reduce Electron Back Scattering

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Purpose: The objective was to report implementation of internal shielding to reduce electron back scattering in mouth-lip treatment. **Methods:** A customized mouth guard was designed to reduce electron back scattering (EBS) to protect the normal organs beyond the irradiated area. The mouth stent was made by a commercially available BiteLok by Klarity with lead foil to shield the transmission radiation. EZ-bolus pellets were coated surrounding lead to attenuate the electron back scattering in the interface of lead and internal normal tissue. The delivered dose was verified by OSLD in vivo dosimetry confirmed the prescribed dose. **Results:** An incident electron beam of 6 MeV was used to treat the upper lip of 1.0-cm thickness. Additional 1.0-cm thick bolus was applied to the lip surface to ensure the skin dose achieved. The energy Em at 2.0-cm depth was 1.76 MeV. The minimum lead thickness for shielding Em was 0.9 mm. Based upon the characterization on the interface of lead and soft material, EBS was estimated around 67%. The depth upstream in tissue for EBS approximately was estimated to be 5.9-mm range for a 10% transmission of EBS. These obtained thickness values of lead and EZ-bolus provide fundamental parameters in the mouth guard/stent design. An OSLD in vivo dosimetry further verified 1.8% deviation in measured and prescribed doses in the device used. **Conclusion:** The commercial BiteLok and EZ-bolus can be utilized to make an internal shielding device as a mouth guard to shield electron irradiation and reduce EBS. The design can be implemented quickly and reliably in clinic electron beam setup. In vivo dosimetry further verifies the shielding effects met design goal on the

transmitted dose through lead and residual EBS.

PO-GePV-P-33

Comparison of Intracranial Stereotactic Radiotherapy (SRS/SRT) Plan Quality Using CyberKnife, Eclipse MLC and Cone Techniques

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Purpose: To compare intracranial SRS/SRT plan quality among CyberKnife (CK), Multi-leaf collimator (MLC), and Cone techniques. **Methods:** Three groups of patients treated with CK were selected; group one (14 patients) with gross target volume (GTV) of 0.4 – 4.7 cc was re-planned with Eclipse Cone & MLC, group two (12 patients) with GTV of 0.1-1.3 cc was re-planned with Cone, and group three (12 patients) with GTV of 0.9 - 74.4 cc was re-planned with MLC. Plan quality was evaluated with Paddick's conformity index (CI), Gradient index (GI: the ratio of isodose volume V50%/V100%) to represent the normal tissue dose. One isocenter with the coplanar beam and the same GTV dose coverage were used for plan comparison. **Results:** For CK/MLC/Cone comparison in group one, three Cone-plans with GTV (>2.5cc) could not meet the GTV coverage and were eliminated from the comparison. The median CI was 1.16 (1.06-1.39), 1.18 (1.03-1.55) and 1.46 (1.06-2.9), and the median GI was 3.67 (2.68-4.38), 4.68(4.25-5.33) and 2.51(2.26-4.72), respectively. MLC-plans were not attempted for tumor volume of < 0.4 cc due to the MLC limitation. CK and MLC-plans were comparable in CI and were superior to Cone-plans. Cone-plans showed the lowest GI, and MLC-plans had the highest GI among all three techniques. Further comparison on MLC/CK and Cone/CK plans showed a good agreement with the group one results using patients with more different tumor size ranges. **Conclusion:** The Cone-plans were highly affected by tumor shape and size, and showed the most inferior CI and superior GI than the other two methods. The MLC-plans had some limitations for small size targets and also received the highest normal tissue doses in low dose areas. The CK-plans did not have any tumor size restriction, and the plan quality was either comparable or in between the other two methods.

PO-GePV-P-34

IAEA Guidance on the Certification of Clinically Qualified Medical Physicists

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Purpose: The global shortage of medical physicists has been highlighted in publications and by professional organizations using different platforms, e.g. IDMP. The International Medical Physics Certification Board (IMPCB) was formed in 2010 and one of their major objectives is to support the practice of medical physics through a certification programme in accordance with the International Organisation for Medical Physics (IOMP) guidelines. In the past decade, the IAEA has also produced consensus guidance to promote the profession. **Methods:** The lack of recognition of clinically qualified medical physicists (CQMPs) as health professionals, implies that formal national certification has not been widely adopted. In 2018, the IAEA embarked on a consultation process with professionals, as well as regional and international organizations, in order to produce guidance to encourage the formation of national certification bodies. **Results:** In February 2021, the IAEA published the Training Course Series No. 71 Guidance on the Certification of CQMPs (TCS71). The publication, endorsed by the IOMP and IMPCB, was launched through an IAEA press release; and outreach through emails and social media in collaboration with the co-endorsing organizations. Consultations with a focus group of expert medical physicists from different regions, highlighted the need for further practical guidance. Although certification is implemented in some Member States, the framework is applied variably. Medical physics practice is often approved by a radiation regulator or a national professional society, and not a health authority; hospitals accept certification from foreign certification bodies and/or regular re-certification is not based on continuing education. **Conclusion:** The IAEA aims to enhance the recognition of CQMPs as health professionals, in alignment with ILO's International Standard Classification of Occupations. TCS71 provides guidance on the application of best practices in the education of medical physicists, alignment of CQMP competencies to international standards and establishment of national certification bodies to recognize CQMPs.

PO-GePV-P-35

Application of Intensity Modulated Arc Therapy (IMAT) with Multi-ISO Planning Method to Sparing of Organs at Risk

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Purpose: We propose a new type of treatment that employs multi-isocenter technique instead conventional mono-isocenter approach and drastically reduce dose to the underlying sensitive structures often present in these cases. **Methods:** Two sets of IMAT plans(group A with multi-ISO centers and group B with mono-isocenter)were generated on Pinnacle 9.10 planning system formulated for a set of three representative cases: bilateral breast cancer,brain metastasis from lung carcinoma with intracranial lesions<=3, a scalp sarcoma,respectively. The dose distribution of the tumor and organs at risk and the delivery efficiency of the two sets of plans were compared. **Results:** Both sets of plans in two groups satisfied all clinical requirements and for targets there were no clinically significant difference in HI, CI and other PTV dose coverage.Paired comparison results of dose distributions indicated that group A plans were significantly superior to group B plans in organs at risk. In the bilateral breast case,V30 of heartin group A was better than that of group B. Especially for the brain cases that lesion located near by the stem, multi-isocenter plan reduced the stem dose about 950 cGy compared to the mono -isocenter plans. However, there were not such trend in middle ears and spinal cord. In the scalp case reduced the max and mean doses delivered to the brain by 30Gy (from 54Gy to 24Gy) and 7Gy (from 15Gy to 8Gy), respectively. In terms of monitor units and treatment time group A were slightly higher than group B. **Conclusion:** Multi-ISO IMAT is a feasible and practical method to significantly protect some OARs for superficial lesions while compromise a little planning efficiency. The benefits of this technique and its applicability should be further investigated.

PO-GePV-P-36

CyberKnife 0-View (xSight Spine) Tracking for Lung Tumors: Is It Safe at Distances Greater Than 5cm?

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Purpose: To determine the safety of delivering radiation with the CyberKnife system to a lung lesion using a 0-view plan where the tumor lies more than 5 cm from the spine tracking center. **Methods:** The CIRS lung motion phantom has an anatomical spine built in for spine tracking on the CyberKnife system. The ball cube built into the CIRS phantom is within a 5cm radius of the spine. Therefore, an external ball cube was placed in a reproducible fashion outside of the phantom. This way the spine within the phantom was leveraged to deliver radiation to a ball cube that was 20cm away from the spine tracking align center. The Accuray Precision Planning System version 3.0.1 was used to generate a treatment plan to deliver dose to the ball cube while tracking on the spine (0-view lung tracking). The initial end-to-end delivery was done without any ball cube motion. A future delivery will incorporate motion. After the radiation was delivered the Accuray provided E2E film analysis software v4.0 was used to calculate a targeting error. TG-135 recommends a targeting error of less than 1.5mm for extreme end-to-end scenarios. **Results:** The initial analysis within the E2E software yielded a targeting error of 1.77mm. **Conclusion:** Though this is a bit higher than the recommendation given in TG-135, given the extreme nature of this delivery the result is quite impressive. Since the ball cube was at 20cm from the spine, the precision planning system was forced to eliminate many more beams than normal during the optimization (using VOLO). The reduction in beams caused the treatment plan used for the initial delivery to provide less dose coverage to the ball cube as normally would be by a plan which was allowed more beams.

PO-GePV-P-37

Retrospective MR Image Review of Stereotactic Radiosurgery for Up to 7 Metastases

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Purpose: Stereotactic radiosurgery (SRS) is an established treatment option for patients with a limited number of brain metastases. SRS uses a high dose of radiation to treat tumors. SRS has shown advantages in the increased local control compared with whole brain radiotherapy (WBRT). However, SRS might bring about a risk of radiation induced side effect to be observed months to years after SRS. The current research is to review SRS treated metastases on MR images followed up to 8 years after SRS. **Methods:** This is a retrospective imaging study on brain SRS for patients with up to 7 brain metastases. SRS treatment option with a single iso-center technique was offered to the patients with 2 or more brain metastases. The 1- mm magnetic resonance (MR) images with contrast was registered with 1mm-CT images. Identified target volumes were delineated on the registered MR and CT images. A Varian Eclipse version 15.1 treatment planning system was used for treatment planning. Daily image guidance was used for target localization and patient alignment. The treatment was delivered with volumetric modulated arc therapy with 6MV flattened filter free (6MV FFF) mode at a rate of 1400 MU per minute from a Varian TrueBeam linear accelerator. **Results:** Followed up MR images in the mean 5 years (ranging from 2 to 8 years) after SRS showed an evolution of metastases (treated volume (TV)). A good control of metastases was seen from months to years after SRS. Some patients had necrosis formed within the treated volume and then degraded with time. Severe side effects from SRS are not observed by reviewing the MR images. **Conclusion:** A review of the followed up MR images indicated that both single and multiple iso-center technique are able to achieve similar tumor control. A single iso-center SRS is suggested for multiple metastases.

PO-GePV-P-38

A New General Maximum Intensity Projection Technology Via the Hybrid of U-Net and Radial Basis Function Neural Network

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Purpose: To develop a new general neural network-based technology that can better and clearly display the projected images. **Methods:** We propose a new method based on the hybrid of convolutional neural network (CNN) and radial basis function neural network (RBFNN) to synthesize the projection image. We first adopted CNN to obtain feature or enhanced images to be projected; subsequently, the RBF neural network performed further synthesis processing for these data; finally, the projection images were obtained. For experimental data, in order to increase the robustness of the proposed algorithm, the following three different types of datasets were adopted: the vascular projection of the brain, the bronchial projection of the lung parenchyma, and the vascular projection of the liver. In addition, radiologist evaluation and five classic metrics of image definition were implemented for effective analysis. **Results:** Compared to the traditional MIP technology and other neural network structures, the proposed method achieved the best comprehensive subjective evaluation and the best image quality-based objective metrics evaluation. **Conclusion:** The use of a large number of different types of data and superior experimental results proved the versatility and robustness of the proposed method. The new neural-network-based technology can display higher quality projected images.

PO-GePV-P-39

Imaging QA On CyberKnife According to TG-135_ a Practical Approach

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Purpose: "Report of AAPM TG 135: Quality assurance for robotic radiosurgery" recommends monthly and annual QA of the imaging system specifically for the signal to noise ratio, contrast-to-noise ratio, relative modulation transfer function and spatial resolution of the amorphous silicon detector. These tests are easily performed and analyzed on any other treatment modalities with customized phantoms and analysis software. With the complexity of the image extraction process and the image characteristics of the CyberKnife® system, it is a very difficult task to generate a report comparable to any other modalities on this area of Quality Assurance. **Methods:** By using the QC-kV-1 Phantom, the slanting phantom support table with inserted fiducial markers and PIPSpro™ software from Standard Imaging, we were able to generate customized plan for the imaging QA procedure and was able to analyze the images on a regular basis to generate reports. **Results:** Monthly QA and annual QA reports, according to the industry standards are generated and evaluated for the suggested tests in TG-135. **Conclusion:** Using the Imager extraction interface of the IDMS system, we are able to export the images from CyberKnife® and appropriate software and hardware combination we are able to generate standardized reports according to TG-135 requirements for Imaging QA.

PO-GePV-P-40

Inter-Fractional Dosimetric Variation Due to Daily Patient Position Variations in the Accelerated Partial Breast Irradiation with High Dose Rate Brachytherapy (HDR) Using the Multichannel SAVI Applicator with the 3 Fractionation Schema

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Purpose: To determine the inter-fractional dosimetric variation of accelerated partial breast irradiation (APBI) using the SAVI applicator due to daily patient positioning variations, through the first and second day of treatment. **Methods:** A CT is taken for treatment planning purpose, the PTV_eval is created according to the ASTRO recommendations and a treatment plan is made with a prescription of 22.5Gy in 3 fraction to the PTV_eval. Short CT scans were taken before the first treatment on day 1 and second fraction treatment on day2 to ensure no change in the integrity of the applicator and its position with respect to the anatomy. For the purpose of this study. Plans were created on the CTs of the first and second fractions, the dwell times and the source activity of the treatment plan were used in the comparison plans to keep the activity-hour intact. **Results:** The deviation of the PTV_Eval coverage -D90% varied between 0.08% and 6%. The variation in dose to skin ranged from 0.4Gy - 2.1Gy for 2cc and from 1.1 - 2.8Gy for 0.03cc. For the CW/ribs, the deviation of dose for the 2cc ranged from 0.3 to 4.6Gy and for the 0.03cc was from 0.7Gy to 6.5Gy. **Conclusion:** The positioning of the ipsilateral arm plays an important role in the variation of the proximity of the critical structures to the applicator. This variation determines changes in the new PTV_EVAL. Greater variation in dose was noted for 0.03cc as compared to 2cc for both CW/ribs and Skin. Variation in the coverage of PTV-Eval can be as high as 6%. As result of our finding, the highest grade of reproducibility in the patient setup is recommended to minimize the dosimetric variations.

PO-GePV-P-41

Clinical Experience of Pre-Selection Criterion of Patients with Left Breast Cancer for Free Breathing Vs Deep Inspiration Breath-Hold Radiotherapy Technique

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Purpose: The benefits of deep-inspiration breath-hold (DIBH) radiation technique (RT) for left breast patients (LBPs) are well known, but there is little awareness about challenges and risks associated with this technique. We propose the upfront criterion, based on estimation of heart V20Gy (volume of the heart which receives 20 Gy), to select the LBPs for free breathing (FB) RT. **Methods:** 38 patients, who were referred for adjuvant radiotherapy after left breast conserving surgery were analyzed in this study. All patients were simulated in the supine position and immobilized with breast board. Minimal computer tomography (CT) scan including heart volume was acquired for all patients. Heart V20Gy was estimated using pre-selection heart V20Gy method (based of maximum heart distance included in the field), and FB CT was acquired for patient with V20Gy<5% and DIBH for patients with V20Gy>5%. Using this

selection, 23 patients were selected for FB RT and 15 patients for DIBH RT. The prescription doses were 42.56 Gy/16 Fx and 40.05 Gy/15 Fx. Patients were treated with tangential conformal fields with a multileaf collimator using 6 MV and 10 MV. To avoid hot spots field-in-field techniques was used. All plans were normalized for $V95\% \geq 95\%$ with maximum dose of 107%. **Results:** All 23 pre-selected for FB RT patients complied the criterion calculated $V20Gy < 5\%$ with mean $V20Gy = 1.11$ Gy. Mean heart doses were in the range of 0.94 – 3.67 Gy with an average of 1.89 ± 0.84 Gy. 87% of the FB RT patients had mean heart dose of < 3 Gy. **Conclusion:** This work demonstrates the assessment of pre-selection criterion, based on evaluation of heart $V20Gy$, with respect to selecting the treatment technique of FB or DIBH for LBPs in order to avoid unnecessary complexity from simulation to treatment and also reducing the patient burden on DIBH techniques.

PO-GePV-P-42

Dosimetric Evaluation of Sub-Millimeter Robotic Corrections for Multiple Metastases Single Isocenter Plans

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Purpose: To quantify the dosimetric impact of 6DOF corrections for multiple-metastasis, single-isocenter SRS cases. Clinical practices vary on whether or not corrections are applied with couch rotations. This study aims to determine the dosimetric impact of this choice. **Methods:** Using a 21 Varian IX with fine beam isocenter and ExacTrac couch, patient specific QA was performed for several multi-met patients using a Sun Nuclear SRS Mapcheck. Plans included non-coplanar IMRT and VMAT optimized using Eclipse V15.6 and Brainlab's Elements Multiple Mets. The measurements were done in two ways. The first had the SRS Mapcheck precisely set up at isocenter at couch 180 (Varian scale) using the light field and verified by multiple physicists. The patient plans were delivered with all collimator angles and couch kicks (True Composite). The second setup utilized the ExacTrac system to correct the Stereophan position at each couch angle. Residual ExacTrac setup was limited to 0.2mm and 0.2 degrees. Measured doses were compared with plans calculated using Acuros with a 1mm dose grid size. A 3%, 1mm, 10% gamma index was used. **Results:** The residual setup correction from the light field couch 180 setup with ExacTrac was always under 0.7mm and 0.7%. Average gamma passing rates for the light field setup was 80.4%, whereas average gamma passing rates with the ExacTrac couch corrections was 93.0%. Note that this Varian IX has a couch walkout with a radius of 0.6mm. **Conclusion:** We have demonstrated that for couches with non-negligible walkout, ExacTrac setup corrections produce demonstrably better agreement between planned and calculated doses. This work supports current clinical practice to verify and adjust the patient at each couch angle.

PO-GePV-P-43

Comparison of Intrafractional Motion Monitoring During Spine Stereotactic Body Radiation Therapy Using Multiple Immobilization Systems

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Purpose: Given the proximity to spinal cord or cauda equina, spinal SBRT requires highly conformal doses with steep dose gradients. As a result, precise positioning and immobilization of the patient must be achieved and maintained for the duration of simulation/treatment. Our goal was to evaluate differences in intrafractional monitoring via ExacTrac during spinal SBRT resulting from a change in immobilization systems used in our clinic. **Methods:** Eight patients with a total of 10 spinal metastatic lesions were prospectively enrolled in a trial designed to evaluate setup/motion error for various combinations of immobilization and image guidance. All patients underwent 3 fraction spinal SBRT and were simulated/treated with one of two SBRT systems – Elekta BlueBAG/BodyFix (n=7) or Klarity SBRT system (n=3). After initial alignment to external marks, residual 6D alignment corrections were applied using the ExacTrac positioning system and subsequently verified to be within $1\text{mm}/1^\circ$. The ExacTrac x-rays system was further utilized to assess intrafractional motion and corrections were applied as necessary such that an overall positioning accuracy of $1\text{mm}/1^\circ$ was maintained. **Results:** Records for 28 treatment fractions were evaluated. The intrafractional shifts necessary to maintain $1\text{mm}/1^\circ$ positioning accuracy were applied in 3/19 (15.7%) and 1/9 (11.1%) fractions with the BlueBAG/BodyFix and Klarity SBRT systems, respectively. During one fraction with the Klarity SBRT system, intrafractional shifts were applied twice, otherwise all shifts were applied once mid-treatment. All applied shifts were $< 2\text{mm}/2^\circ$ in each direction. The averaged translational magnitude of applied intrafractional shifts was 1.4mm and 2.0mm for the BlueBAG/BodyFix and Klarity SBRT systems, respectively. **Conclusion:** Both spinal SBRT immobilization solutions achieve comparable results across the limited number of treatment fractions evaluated. Future study with a larger cohort is warranted. Some intrafractional motion was observed with both systems and demonstrates the necessity of motion management during treatment delivery for spinal SBRT.

PO-GePV-P-44

Validation of the Varian Machine Performance Check Enhanced Couch Accuracy

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Purpose: To determine suitability of the Varian Machine Performance Check (MPC) enhanced couch utility for couch isocentricity testing. This tool uses a specific phantom that contains fiducials for imaging at various gantry, collimator, and couch angles. The intent was to validate the use of MPC as a tool to address isocenter deviation issues. **Methods:** A phantom with a single BB was placed at the isocenter of the gantry, collimator, and couch by means of the Winston Lutz technique. The appropriate translations to reach to the isocenter were determined using an open-source python package. The BB phantom was then analyzed at couch 0, 90, 270 with gantry pointing down. Once at the actual machine isocenter, the actual isocenter size was determined by a Winston-Lutz measurement and this data was compared back to the MPC enhanced couch results. With early tests, the final shift to actual isocenter was not done, and larger discrepancies between measurement and MPC were found. **Results:** It was found that applying table isocenter offsets to the BB phantom was necessary to replicate the MPC values. The gantry isocenter measurement using the Winston Lutz test was not sensitive to minor offsets from actual isocenter. When the magnitude of the shift vector was less than 0.3 mm, the MPC analysis of the couch isocenter was within an average of 0.13 ± 0.12 mm of the Winston Lutz measurement. **Conclusion:** The accuracy of the MPC enhanced couch results as it pertains to the max table offset parameter was confirmed to be within acceptable limits. The MPC utility is an accurate tool for isocenter measurement.

PO-GePV-P-45

Machine Off-Isocenter Winston-Lutz Test in Single-Iso Multi-Mets Intracranial Stereotactic Radiosurgery

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Purpose: Develop a clinically feasible off-isocenter Winston-Lutz test to evaluate the capability of LINAC machine performance in single-iso and multi-mets intracranial stereotactic radiosurgery (SRS). **Methods:** A Brainlab Winston-Lutz test phantom containing one metallic ball (5mm diameter) was attached to tabletop using the mounting device. The distance between machine isocenter and ball center was 8.0cm along the couch direction. Six fields were generated to acquire the MV images with various combinations of gantry angles (0, 90, 180 and 270 degrees) and couch angles (0, 90 and 270 degrees). The field size was defined by MLC position of 3cmx3cm. The off-isocenter Winston-Lutz test was performed on a Varian Edge with HD-MLC. The acquired test images were analyzed using Doselab Pro, Python Pylinac. The delta value was calculated as the difference between field center and ball center in X and Y directions on image. Total delta is Root Sum Square of X and Y delta values. The results were compared with manual measurement in Aria Offline Review. **Results:** The off-isocenter test can be completed within 10 minutes after regular Winston-Lutz test was performed. Image quality was acceptable despite the asymmetric radiation penumbra on field border and the increased field size uncertainty due to off-isocenter effect. Consistent results were observed. The maximum and

mean total delta were 0.84mm and 0.46mm, 0.85mm and 0.46mm, 0.89mm and 0.47mm analyzed using Doselab Pro, Python Pylinac and manual measurement, respectively. When compared among the three methods for each off-isocenter test image, the delta in X and Y directions agreed within ± 0.2 mm. The total delta agreed well within ± 0.2 mm for each image. **Conclusion:** A practical off-isocenter Winston-Lutz test was developed, and its accuracy was validated in this study. The test can be easily integrated into machine morning QA or serve as an independent check for physicist to implement periodically.

PO-GePV-P-46

Phantom Validation of a CT-Based Navigation System Adapted for Interstitial Liver Procedures

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Purpose: To adapt a commercial CT-based neurosurgical navigation system for use in interstitial liver procedures and validate its performance on a representational liver phantom. CT-based navigation technology has been shown to improve efficacy and safety in neurosurgical procedures and offers potential to do the same for interstitial liver procedures, which are used in radiation oncology practice for implanting targeting fiducials or brachytherapy source transfer needles to facilitate precise delivery of radiation to lesions. **Methods:** A mobile intraoperative CT scanner and partnered optical-tracking navigation system were used as the imaging and navigation platforms. The default neurosurgical navigation software was augmented to display segmented structures from radiation-therapy treatment-planning software, enabling visualization of target and avoidance structures during image-guided needle placement. Fiducial needles were tested for compatibility and reliability with the navigation system. A phantom was fabricated in house containing pseudo costal/vascular avoidance structures and lesions visible on CT images, and this phantom was used to test the radiation oncologist's ability to accurately and reproducibly place fiducials in a target region without intersecting avoidance structures. **Results:** The navigation software was successfully modified to receive and display physician-drawn segmented structures, and this feature was utilized in tests involving the pseudo liver phantom. A pre-loaded fiducial needle was found to be compatible with the optical-tracking system and its tip location was accurately and reproducibly tracked in the navigation environment. The pseudo liver phantom was accurately and reproducibly implanted with fiducials in pre-planned locations by the radiation oncologist. **Conclusion:** We adapted a CT-based navigation system for use in interstitial liver procedures and developed a pseudo liver phantom to test the accuracy and reproducibility of navigated fiducial placement at pre-planned target locations in a controlled, simulated setting. This system could facilitate safe practices for fiducial-marker and/or brachytherapy-needle placement and permit faster radiation treatment throughput.

PO-GePV-P-47

Clinical Evaluation of Portal Dosimetry Ion Chamber Array and Mobius 3D for FiF Breast IMRT QA

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Purpose: The purpose of this study is to compare the Gamma analysis between Portal Dosimetry QA (PDQA), Ion chamber based 2D-array (Matrixx) and Mobius 3D for Field-in-Field (FiF) left breast IMRT. Motivation to pursue this study is to find easier way reference for point dose calculation compatible in RadCalc for secondary check. We decided to perform the secondary check with Gamma analysis comparison between Mobius3D, PDQA and 2D array, to determine the optimal clinical QA procedure. **Methods:** We evaluated number of left sided breast FiF IMRT plans. QA plans were calculated for 2D array and Varian Portal Dosimetry software (using portal dose image prediction module) and were delivered on a Varian linac. Also Plans were exported to Mobius3D software for Gamma analysis. Gamma criteria (3%/3mm >95%) was utilized for all the data analysis: OmniPro software for Matrixx; Varian software for PDQA and Mobius 3D. **Results:** For composite plans, Mobius 3D, PDQA & 2D-array Gamma analysis, average 3%/3mm Gamma is more than the passing criteria of 95%. For Mobius 3D, average Gamma (3%/3mm) is 99.2%. For Portal dosimetry average Gamma is 100%. For 2D array, average Gamma is 95.64%. All three software's results were in the same trends for passing rate. Per field Gamma analysis were performed for PDQA and 2D array. Average Gamma for per field comparison for 2D array is 91.54 % and for PDQA is 94.08 %. **Conclusion:** Mobius 3D, 2D array and PDQA can all be used with satisfactory results for the secondary plan check for FiF breast IMRT QA. For the composite plans, PDQA results shown 100% passing rate. Mobius 3D results are comparable with Portal Dosimetry, and Ion chamber based 2D array is more sensitive to the plan as of resolution. From our analysis, PDQA and Mobius are equally optimal for this QA task.

PO-GePV-P-48

Effect of Gantry Angle on Portal Dosimetry Vs Ion Chamber Array of Linac Based IMRT QA

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Purpose: Comparison of QA techniques for accelerated partial breast irradiation (APBI) IMRT using EPID and ion chamber based 2D array and to verify the response electronic portal imaging device (EPID) and 2D array with respect to treatment gantry angle vs gantry zero. **Methods:** Currently, all static IMRT QA plans are delivered at gantry angle zero. For APBI, comparison of patient QA between 2D array (Matrixx) and Linac based Portal Dosimetry QA (PDQA) was performed. QA plans were created at gantry angle of zero degree as well as planned gantry angles for both PDQA as well as Matrixx. QA plans were delivered on the Varian linac. Gamma analysis for each field and for composite plan was compared between PDQA and 2D array at both gantry zero and planned gantry angles. Gamma analysis for PDQA was analyzed using Varian Portal Dosimetry software and for Matrixx using OmniPro IMRT QA software. **Results:** 2D Gamma analysis results were analyzed for Portal Dosimetry and for Matrixx. Portal Dosimetry at gantry zero (2%/2mm Gamma criteria) resulted in fields 1 to field 5 are 96.8%; 94%, 96.5 %, 93.4% and 92.2% of the pixels passing Gamma criteria, respectively. Composite plan 2%/2mm Gamma analysis is 88.6%. Gamma analysis at planned gantry angles are: 97.5%, 95.1%, 96.9%; 94.8% and 93.5%, respectively. Composite plan result is 92.5%. For Matrixx Gamma analysis for composite plan for gantry zero is 89.37%. Composite at planned gantry angles is 89.16%. **Conclusion:** Both portal imager and 2D ion chamber array can be used for APBI IMRT QA. QA results are better with portal imager as compared to 2D array. However, portal imager results show more variation with gantry angle vs gantry zero. Gamma analysis results at planned gantry angle for Portal Dosimetry are better than gantry zero for both field-by-field analysis and composite plan analysis.

PO-GePV-P-49

Report of Image Distortion Based on Five Consecutive Years of Annual QA On Six MRI Scanners Used for Radiation Therapy

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Purpose: To determine MRI image distortion based on five consecutive years of annual measurements on six MRI scanners used for radiation therapy. **Methods:** We used a commercial MRI image phantom MRID3D to perform annual image distortion checks and monitor the stability of the distortion over a period of five years from 2016 to 2020 for six 1.5T and 3T MRI scanners that provided images for the treatment planning of radiation therapy. The phantom was scanned on these MRI machines with 1 mm image slice thickness to acquire 3D images by following the imaging protocol of Gamma Knife and Linac-based SBRT/SRS. For each imaging scan the phantom was aligned to its geometric center using the MRI's lasers and the localizer. Using the MRID3D software we investigated: (1) the magnitude of the image distortion; (2) the dependence of distortion magnitude on the distance from the imaging isocenter; (3) the difference in distortion magnitude for 1.5T and 3T machines; and (4) the stability of distortion across the five-year period. **Results:** The image distortion

magnitude depended on the distance from the imaging isocenter. For most MRI scanners the distortion magnitude was under 1 mm when within 10 cm of the isocenter. The magnitude increased with increasing distance from the isocenter and could reach 2 mm at the distance of 15 cm. In general, a 3T machine had worse distortion than that of a 1.5T machine but the distortion could still be less than 1 mm when the distance from the isocenter was less than 10 cm. Slight fluctuation in distortion was observed during 5 years. **Conclusion:** MRI image distortion varies for machines and manufacturers. It depends on the distance from the imaging isocenter. Additional margin due to MRI image distortion needs to be accounted for during treatment planning, especially for Gamma Knife or SRS/SBRT.

PO-GePV-P-50

An Implementation Strategy for Introducing Automated Treatment Planning into the Clinic

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Purpose: The purpose of this study is to design an implementation strategy for introducing new automated treatment planning scripts into clinical use. Deployment of automated planning for clinical use requires thoughtful preparation that ensures safety and consideration of additional failure modes introduced by automation of treatment planning tasks. **Methods:** The Eclipse Scripting Application Programming Interface (ESAPI) was used to develop an automated treatment planning script to be used for radiotherapy patients. A workflow was mapped to assist in clinical implementation of this script. The steps in this workflow involved: assessing clinical need for automation, discussions with dosimetry staff regarding preservation of current planning workflow, iterative testing for bugs and performance, code review by another physicist or programmer, introduction of unexpected inputs to find potential hidden failure modes, and training for physics and dosimetry staff. After release for clinical use, we also considered feedback from dosimetry staff for new features and updates as well as conducted ongoing physics review to assess the need for script changes due to adjustments in treatment planning strategy. **Results:** By following a regimented implementation procedure, we were able to deploy an automated planning script which is customized to our site's particular planning workflow. This procedure helped discover several failure modes that may have been missed otherwise. Blocks of code were added to prevent these failure modes which included potential incorrect isocenter placement and transient bugs that could have caused program crashes with rare, unexpected inputs. **Conclusion:** Automated treatment planning has the ability to reduce time spent planning, standardize plan quality, and reduce the probability of failure modes that are more likely to occur with manual planning. The time spent evaluating a new automated planning script is considerable, and the steps involved should be carefully planned prior to implementation.

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PO-GePV-I-01

Dosimetric Evaluation of O-Arm Cone Beam Computed Tomography (CBCT) Scan Modes for Pediatric Head Scans

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Purpose: To evaluate the additional doses to pediatric patients while using various CBCT acquisition modes in an O-arm and compare the volume Computed Tomography Dose Index (CTDI_v) values to conventional CT scans. **Methods:** O-arms (O2 O-arm, Medtronic, Minneapolis, MN) are increasingly being used for neurosurgery and interventional procedures in clinical settings for pediatric patients. Due to the latent carcinogenic risks of radiation and increased life expectancy of pediatric patients, doses accrued during surgical procedures is a serious concern, in addition to doses from pre-op conventional CT scans. To analyze this further, CTDI_v measurements were performed for various pediatric patient sizes and O-arm acquisition modes including standard, low dose, HD3D and enhanced3D. The CTDI_v values obtained for all patient sizes were averaged and compared to a Stealth and reference pediatric head protocol used in clinical practice. **Results:** The CTDI_v values measured were 16.7, 7.3, 25.2 and 70.2 mGy for standard, low dose, HD3D and enhanced3D modes respectively. The doses from conventional CT stealth protocol and reference pediatric head protocol were 32.4 and 33.7 mGy, respectively. **Conclusion:** Based upon the measured CTDI_v values, all modes except enhanced3D will result in lower radiation risk to pediatric patients when compared to conventional CT scans. Low dose mode is recommended for image registration purposes, if image quality is not a concern, to minimize radiation burden. Enhanced3D mode should only be utilized if the benefits outweigh the risks for diagnostic purposes.

PO-GePV-I-02

Evaluation of the Displayed Average Glandular Dose of Hologic Digital Breast Tomosynthesis (DBT) Systems

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Purpose: To investigate how the displayed average glandular dose (AGD) of Hologic DBT systems compares to computed AGD, and how this comparison is modulated by thickness, and the assumptions in women breast glandularity. **Methods:** For 2, 4, 6, and 8 cm of BR-12, exposures were acquired using the Auto-Filter AEC mode on six Hologic DBT systems. Entrance exposures and HVL were measured using a RaySafe X2 solid state detector, and the displayed entrance skin dose (ESD), displayed AGD, and displayed exposure index (EI) were recorded. The AGD was first estimated under the assumption of 50% glandularity, 50% adipose tissue composition. It was then computed using the assumption of breast composition for the average woman aged 40-49 years old (100%, 65%, 35% and 14% glandularity for 2-, 4-, 6- and 8-cm breast thickness, respectively). The AGD computed using these assumptions was compared to the displayed AGD. **Results:** Under the assumption of 50% glandularity, the absolute percentage difference between the computed AGD and displayed AGD ranged from 0.3% (for 2 cm of BR-12 in 3D) to 20.5% (2 cm in 2D). The deviation is most apparent at the largest thickness in 3D, where the displayed AGD overestimates the computed AGD. Using the assumption of glandularity based on the average woman aged 40-49 years old, the absolute percentage difference between the computed AGD and displayed AGD ranged from 0.04% (2 cm in 2D) to 23.2% (8 cm in 2D). The deviation is most apparent at the largest thickness in both 2D and 3D mode, where the displayed AGD underestimates the computed AGD. **Conclusion:** This study suggests that the discrepancy between displayed AGD and computed AGD may be correlated with the breast thickness. More data is needed to support this conclusion. The displayed AGD may overestimate or underestimate AGD based on the breast composition.

PO-GePV-I-03

Contrast-Enhanced Dual-Energy Digital Breast Tomosynthesis (CE-DE-DBT) Imaging: Iodine Concentration Estimation

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Purpose: To demonstrate the feasibility of estimation of iodine-contrast concentration in contrast-enhanced dual-energy digital breast tomosynthesis (CE-DE-DBT) through accurate image reconstruction within transverse planes. **Methods:** A digital breast phantom within a transverse plane was considered, that includes three iodine-contrast-enhanced ROIs. A typical DBT scanning configuration was employed for the transverse plane, with source-to-rotation and source-to-detector distances as 36 cm and 72 cm, and a detector of 37.5 cm consisting of 512 bins. Dual-energy data were generated over two overlapping scanning arcs with 80- and 140-kVp spectra, using a non-linear-data model incorporating the polychromatic X-ray spectrum. Three different angular ranges, 14°, 20°, and 30°, were investigated with an angular interval of 1° per view. 80- and 140-kVp images were reconstructed by a new directional total variation (DTV) algorithm

and were decomposed into basis images of breast tissue and iodine contrast. From basis images, monochromatic images at 34 keV were composed and assessed for artifacts reduction, and iodine-contrast concentrations were estimated. **Results:** The study reveals that, for the three limited-angular ranges, the DTV algorithm yields monochromatic images with minimized leaking artifacts that are otherwise observed in images obtained with existing algorithms. Furthermore, the iodine-contrast concentrations were accurately estimated, as compared to the results obtained with the full-scan data of 360° by using existing algorithms. Additional detailed results will be presented at the conference. **Conclusion:** It has been demonstrated in this work that monochromatic images with reduced artifacts can be obtained within transverse planes in DBT by use of the new DTV algorithm, which can lead to accurate estimation of iodine-contrast concentration in CE-DE-DBT. Future study will investigate the impact of additional physical factors, such as scatter.

PO-GePV-I-04

Design of a Display Test Pattern to Assess Ambient Light Levels in a Radiologists' Reading Environment

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Purpose: During the COVID-19 pandemic, many radiologists transitioned to remote reading at home. One concern when reading in a non-controlled reading environment is the appropriateness of ambient light levels. The purpose of this study was to develop a visual test pattern-based evaluation method to assess ambient light appropriateness. **Methods:** A series of six test patterns were created by adding letters at different contrasts to a uniform background. Each pattern contained 5 to 10 letters, distributed in a uniform, random fashion, to create an element of search. The choice of letters was loosely based on a Snellen chart. The uniform background pixel values ranged from 0 to 98% at 6 different levels. In a controlled reading environment and by use of a medical-grade DICOM GSDF calibrated display, three readers identified the lowest detectable contrast letter at two ambient light level settings (30 and 200 lux). To assess the generalizability of the approach, at each background brightness level, four sets of patterns were read. **Results:** In the darkest background, readers were able to detect the lowest contrast letter in 11 out of 12 reads (91% correct) at low ambient light levels (30 lux), but at 200 lux they were never able to detect the lowest contrast letter. In the second darkest background, low contrast detectability was 100% at 30 lux and 50% at 200 lux. The low contrast detectability was not affected by ambient light level in brighter backgrounds. **Conclusion:** The results indicate that these display test patterns enable the identification of excessive ambient light levels that potentially interfere with radiologists' low-contrast detection performance, particularly in darker luminance regions of the images. This test pattern-based approach does not require external photometers and can be readily implemented into PACS.

PO-GePV-I-05

Design of Nested Halbach Rings for Portable MRI Scanners

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Purpose: Advantages of magnetic resonance imaging (MRI) in medicine is well known and crucial, but developing a portable MRI device is challenging. A magnet system used to create a static magnetic field (B_0) in a portable MRI scanner should be simple, and lightweight. A lightweight Halbach magnet system might be used to yield suitable B_0 for portable MRI scanners. In this study, a lightweight Halbach magnet system was designed and its performance was evaluated for portable MRI scanners. **Methods:** In this study, B_0 was formed with multiple layer dipole Halbach rings that consist of octagon shape NdFeB (neodymium-iron-boron) permanent magnets. Two Halbach rings were placed facing each other with the same magnetization direction. Aluminum was chosen as the holder material. Investigation of the obtained magnetic field was made with Comsol Multiphysics simulations. Homogeneity of the magnetic field was also calculated. **Results:** As a result of the simulations, 0.37 T magnetic field was obtained and its stability was achieved along 60 mm on the x-axis, 38 mm on the y-axis, and 32 mm on the z-axis. Gradient results were also obtained from the permanent magnets system. **Conclusion:** Nested Halbach rings were used for the magnet system of portable MRI scanners. We present the light and inexpensive magnet design, but more studies should be done for a better understanding of the system. Homogeneity of the system and also the field of view (FOV) should be improved with further studies.

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PO-GePV-I-06

An Easy-To-Use Tool for Annual MRI Performance Evaluation Using the ACR Large Phantom

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Purpose: The ACR MRI Accreditation Program uses approved ACR phantoms for performance evaluation and quality control. A guidance documents standardize analysis by specifying seven quantitative procedures for evaluating scanner performance. These are evaluated at least annually by a qualified physicist, while three are performed weekly by technologists. The purpose of this work was to develop a fast, interactive tool to quickly assess ACR QC scans of the Large Phantom for use during annual evaluation and troubleshooting. **Methods:** An interactive program for inspecting scans of the ACR large phantom was developed using Matlab. The tool accepts ACR T1 sagittal, T1 axial and PD+T2 axial series (or site axial series) and evaluates them using procedures as described in the ACR guidance document. For steps that rely on subjective input (windowing, ROI selection, etc), the most pessimistic (i.e. least beneficial toward action limits) choice is selected. Results are displayed as an interactive annotated stack. Measurements are overlaid (lines, ROIs, boxes, text) on each slice with color-coding to indicate action limit outcomes (green=pass, yellow=warning, red=fail). Clicking on a measurement adjusts the presentation to best visualize the measurement and opens relevant graphs for inspection (e.g. 1D profiles). Images and results can then be saved and exported for reporting. **Results:** The tool was developed using archived annual and weekly QC scans from six scanners. The tool was then installed on a laptop and used during annual performance evaluation (5 scanners) and acceptance testing (2 scanners). Images were transferred to the laptop using USB export for evaluation. Results were rapid and because they follow the manual procedures, were easy to inspect interpret and verify in real-time. **Conclusion:** An interactive tool was developed for evaluation of the ACR large phantom. The tool closely follows ACR guidance which aids rapid interpretation and understanding.

PO-GePV-I-07

Developing and Validating a Radiomics Classification Model for Glioma Grading Using Multiplanar Reconstructive Magnetic Resonance T1-Weighted Imaging: A Reliable Multi-Institutional Study

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Purpose: Evaluation of a multiple plane reconstruction (MPR) T1-weighted imaging radiomics method for grading gliomas **Methods:** A total of 102 glioma patients treated in Shandong Cancer Hospital and Institute were selected as the training cohort. 16 patients from the tumor imaging file (TCIA) and 28 patients from the Second Hospital of Shandong University were enrolled in the validation cohort. Radiomics features, including axial, coronal, and sagittal planes, were extracted from the MPR T1-WI images of each patient. Then the feature is removed by Spearman correlation test and recursive feature removal (RFE) method

to make the feature non-redundant and predictive. Then, a single and combined classification model of the MPR plane is established and evaluated by using a variety of machine learning algorithms, such as support vector machine (SVM), logistic regression (LR), and random forest (RF). Finally, the classification model is further verified by the external validation cohort. **Results:** The optimal feature numbers of SVM, LR, and RF are 16, 14, and 10 respectively. The AUCs of the combination model was the highest in the training cohort (SVM 0.86 ± 0.06 , LR 0.89 ± 0.05 , and RF 0.81 ± 0.07 , respectively) and the validation cohort (SVM 0.84, LR 0.79, and RF 0.78, respectively), which was significantly higher than that of other single MPR classification models. **Conclusion:** MPR radiomics can be used to grade gliomas, and the combined model composed of axial, coronal, and sagittal MPR images is more satisfactory than the single model.

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PO-GePV-I-08

Fast T1 Mapping Optimization: A Phantom Study

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Purpose: The universal pipeline for T1 mapping is to acquire series of images with different T1 weightings and fitting the signal intensity to T1 relaxation for each pixel location. Optimization of T1 relaxometry is mostly focused on the efficiency of T1 estimation while trading off the total length of the experiment and T1 mapping computation. This research investigates the number of TIs to optimize the precision of T1 measurements using the classical inversion recovery (IR) technique. **Methods:** The standardized ACR phantom was scanned using IR sequences with a constant TR = 6000 ms and a range of TIs. A Matlab script was developed for the T1 mapping calculation based on the Barral JK methodology. Parametric maps were generated for 11 series of images each with different numbers of TIs (ranging from 3-6), and different choices of TI. The T1 map generated with 15TIs was considered to be the reference T1 map. The MultiScale_Structural Similarity Index (MS_SSIM) was the chosen method for image analysis. The SSIM was computed between the reference map and each other experimental maps. One-Way ANOVA and Tukey comparison was used for statistical analysis. **Results:** The p-value is less than 0.05 and the mean differences between the MS_SSIM of the different TIs is statistically significant. Also, the difference between the mean of 3TI vs. 4TI and 3TI vs. 5TI is statistically significant. The high predicted R2 value equal to 91.5% indicates that the model generates precise predictions for new observations. The groups of T1 maps generated with 3TIs showed statistically significant similarity to the reference map comparing with maps generated with 15TIs. **Conclusion:** T1 mapping generated using 3TIs provide comparable results to the T1 mapping generated using 15TIs. This reduces scanner time for experiments.

PO-GePV-I-09

Commissioning of Commercial Quantitative R2* Mapping Software for Clinical Use

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Purpose: In acceptance and annual testing of MR systems, physicists are key in assuring the high performance in acquisition, hardware fidelity and image quality. With the increase in dependence on quantitative measurements for patient management, there is an urgent need for physicists to commission the post-processing software used to generate this information. Here we illustrate a commissioning process developed for an FDA-cleared R2* mapping software package, which is used for such tasks as quantifying liver iron concentration. **Methods:** For initial commissioning, we created a digital reference object (DRO) to represent ground truth measurements for comparison with software output. DROs were generated using in-house software, with and without additive noise (SNR=5-20 on first TE), and logarithmically spaced R2* values with signal modeling to replicate the output of a clinical multi-echo gradient-echo protocol. R2* calculations using both a two-parameter and three-parameter fit were used to compare to the ground truth DRO value. For assuring for consistency in a software upgrade, six patients with recent iron overload scans were evaluated. A three-parameter fit was applied to calculate the R2* map on both software platforms. According to our clinical standard operating procedure, three ROIs were made on each R2* map from both software platforms. Results from two software were compared to assess their agreement. **Results:** DRO analysis demonstrated that the software's accuracy is dependent on the native liver R2*, acquisition parameters and that a 3-parameter fit should be used especially if moderate or higher liver iron overload is suspected. In our patient data for a range of 38-815 Hz, there was perfect agreement of mean ROI measurements between both software platforms. **Conclusion:** Using both DRO and patient data, we tested available software for its performance in calculating MR R2* measurements. This methodology can be applied to other applications when commissioning software for clinical use.

PO-GePV-I-10

Differential Diagnosis of Hepatocellular Carcinoma and Cirrhotic Nodules Via Radiomics Models Based on Magnetic Resonance Images

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Purpose: This study is to distinguish hepatocellular carcinoma and cirrhotic nodules using MR-radiomics features extracted from four different phases MRI images. **Methods:** In this study, the five kind of magnetic resonance images of 23 patients with hepatocellular carcinoma (HCC) were collected. Among them, 12 patients with liver cirrhosis were used to obtain cirrhotic nodules(CN).The dataset was used to extract MR-radiomics features from regions of interest (ROI). Using, then, statistical methods to screen MR-radiomics features, which could distinguish HCC and CN. And the ability of radiomics features distinguished HCC and CN was estimated by receiver operating characteristic (ROC) curve. **Results:** A total of 106 radiomics features were extracted, of which almost features showed relatively statistical difference ($p < 0.05$). They were 86 features in delay phase of contrast MRI, 86 features in spir phase of T2WI, 86 features in T1WI and 88 features in T2WI. Among them, the area under the curves (AUC) of these features larger than 0.05 were 74 features in delay phase of contrast MRI, 68 features in spir phase of T2WI, 77 features in T1WI and 75 features in T2WI. **Conclusion:** Radiomics features extracted from MRI images have the potential to distinguish HCC and CN.

PO-GePV-I-11

3D Computational Approach to Morphological Brain Abnormalities in Autism

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Purpose: Autism Spectrum Disorder (ASD) is defined as an idiopathic lifelong neurodevelopmental disorder that develops in early childhood. Its prevalence is broad, yet its nature remains elusive and inconsistent. This research describes a detailed research project related to the construction of computational phantoms, including certain brain structures which can be used in radiological, therapeutic, and diagnostic capacities. **Methods:** A collection of three hybrid computational brain phantoms corresponding to two pediatric ASD patients and one control patient were constructed using high quality magnetic resonance imaging (MRI) sets. A total of 12 brain structures, four from each patient, were outlined from radiological images via the 3DSlicer segmentation software and were imported to the three-dimensional (3D) modeling software package Rhinoceros for additional geometric manipulation and analysis. Rhinoceros allowed

the conversion of the segmentation into a non-uniform rational basis spline (NURBS) or mesh surface-based phantom. **Results:** This set of new phantoms can be used in the future to study the optimization of image quality and radiation dose for patients on the spectrum who have unique neuro-morphology. Given the wide neurological, behavioral, and morphological variations in patients with autism, individualized treatment plans involving an accurate, representative computational model are valuable. **Conclusion:** The aim is to develop a repository of data collected from these phantoms into a comprehensive dosimetry database. This novel set of computational brain models present morphological details that are crucial during radiation dose evaluation for ASD patients enduring diagnostic imaging or radiotherapy.

PO-GePV-I-12

CT-Based Radiomic Signature and Clinic Information Predicts Brain Metastasis in Lung Adenocarcinoma

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Purpose: This study evaluates computed tomography (CT) radiomic features and clinical information for their capability to predict risk factors for brain metastasis (BM) in patients with lung adenocarcinoma. **Methods:** The analysis retrospectively included $n=20$ BM patients and $n=40$ lung adenocarcinoma without BM. And the phenotype of the primary tumor was quantified on pre-treatment CT-scans using 106 radiomic features. Univariate and multivariate analysis was performed to evaluate clinic information using the X2 test and logistics regression. Radiomics statistical comparison between subgroups was evaluated by concordance index (CI). **Results:** Univariate analysis of brain Metastasis showed that age ≤ 60 years old, T4, N2-3, other distant metastases and the pretreatment of $LYM < 1.1 \times 10^9/L$, $LYM\% < 20$, $LDH > 250 U/L$, $CEA > 6.5 ng/ml$, $NSE > 16.3 ng/ml$ were risk factors for brain metastases of lung adenocarcinoma ($P < 0.05$). Multivariate logistic regression analysis showed that age, N stage, other distant metastases, CEA, NSE were independent risk factors. Radiomics showed a significant difference in first order features between subgroups. **Conclusion:** The occurrence of brain metastasis is related to the survival of patients with lung adenocarcinoma. The combination of functional CT radiomics parameters and clinical information could represent a novel integrated approach to predict brain metastatic disease with more accuracy than clinical information alone.

PO-GePV-I-13

Evaluation of Coronary CTA On the Anomalies Origin of Coronary Artery: Location, Types, and Hemodynamic Variation

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Purpose: To evaluate the location, types, and hemodynamic variation of coronary artery with anomalies origin by 128-slice Siemens dual-source computed tomography. **Methods:** A total number of 14820 coronary CTA were carried out in Hospital. According to Greenberg classification method, CTA images with abnormal origin were screened out (excluding myocardial bridging). The epidemic characters of sex, clinical symptoms, ECG, coronary atherosclerosis with coronary artery with anomalous origin were analyzed retrospectively. The correlation was analyzed between the atherosclerosis incidence and the characters of anomalous right coronary artery originating from the left coronary sinus with interarterial course, including the variation trend of the aortic-pulmonary artery space, take off angle, proximal and distal diameter of right coronary artery during the cardiac cycle. The results were analyzed by chi-square and Fisher exact test. P value < 0.05 was considered to indicate a statistically significant. All statistical data were analysed with SPSS 26.0 software. **Results:** 244 patient (1.6464%) with anomalous origin of coronary were reviewed in 14,820 patient. 139 patient (57%) with anomalous origin of the right coronary artery were most common in 244 patient. Sex ($P=0.672$), clinical symptoms ($P=0.181$), ECG abnormalities ($P=0.171$) with abnormal origin of coronary artery had no statistically significant with normal origin. The atherosclerosis incidence with ipsilateral anomalous coronary artery origination is not significant higher than that with normal coronary arteries origination ($P=0.038$). The atherosclerosis incidence with right coronary artery originating from the left coronary sinus with interarterial course is not significant correlated with the variation rate of take off angle ($P=0.806$), the aortic-pulmonary space ($P=0.670$), the proximal diameter ($P=0.671$) and the distal diameter ($P=0.099$) during the cardiac cycle. **Conclusion:** Coronary CTA can accurate and noninvasive assess depiction of coronary artery origin anomalies of location, types; especially in delineating the ostial origin and proximal path of an anomalous coronary more advantages.

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PO-GePV-I-14

Residual Convolutional Neural Network for Noise Reduction in Megavoltage Computed Tomography (MVCT)

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Purpose: Clinical application of megavoltage Computed Tomography (MVCT) is limited to three-dimensional visualization and position of the patient. The noise and low soft tissue contrast of MVCT make it hard to provide an accurate image registration for image-guided radiotherapy (IGRT) and accurate delineation for adaptive radiotherapy (ART), resulting in the less clinical potential of MVCT. In this work, we developed a -based approach to improve MVCT image quality. **Methods:** Kilovoltage CT (KVCT) and MVCT images of 34 head and neck (H&N) cancer patients were retrospectively analyzed in this study. The method incorporated a deconvolution network and shortcut connections into a Convolutional Neural Network (CNN) model, to learn a mapping between MVCT images and paired planning KVCT images. Compared with a CNN, the method enhances the transfer and reuse of features using shortcut connections. The peak signal-to-noise ratio (PSNR), structural similarity (SSIM), and root mean squared error (RMSE) were used to quantify the correction accuracy of the proposed algorithm. Also, the method is compared to -based (BM3D + DFR) MVCT image enhancement method. **Results:** The PSNR, SSIM, and RMSE were improved in processed images relative to the original MVCT images. Specifically, the PSNR and SSIM of the processed images improved to 29.4dB and 0.93 compared with 24.6dB and 0.88 in the original MVCT images, and the RMSE decreased from 23.5HU to 13.6HU. The proposed method processed images with less noise and higher soft tissue contrast in comparison with -based method. **Conclusion:** The MVCT image quality could be improved in the proposed method. The processed MVCT increases the feasibility of depicting the anatomical delineation of tumors and organs at risk, and therefore enables the promising application of image-guided and adaptive helical TomoTherapy.

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PO-GePV-I-15

Evaluating Motion Correction in a Commercial Subtraction CT Registration Algorithm

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Purpose: The purpose of this study was to assess the accuracy of algorithms in registering pre- and post-contrast images with and without motion using a subtraction CT software. **Methods:** A pre-contrast image of a cylindrical CT phantom was acquired. Five 28-mm wide water rods were replaced with iodine rods and the phantom was displaced vertically by 30 mm, in 5-mm increments, and post contrast images were acquired. Subtraction images were evaluated using a Matlab code by generating line profiles for each rod. The line profile data was used to estimate the diameter of each rod under stationary and all shifted positions by performing a linear interpolation to find the location in the image where the CT number exceeded 50 HU, and the location where it dropped below 50 HU. The difference in these locations determined the diameter of rods in each subtracted image. The rod diameters in the shifted images were compared with the rod diameters in the stationary image. **Results:** The motion correction was successful and maintained similar rods shape, location, and CT number for all displacements up to 25 mm. For example, the measured vertical diameter of the 9 o'clock rod decreased by 4.1 mm (-8.5% -10.6%). However, moving the phantom by 30 mm, more than the rod's diameter size (28 mm), resulted in misregistration and inaccurate subtraction. **Conclusion:** The subtraction algorithm corrected for motion between the pre- and post-contrast scans for displacements less than the size of the iodine rod.

PO-GePV-I-16

Improvement of A Tissue Equivalent Material for Multi-Modality Imaging

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Purpose: Phantoms are vital to evaluate imaging techniques; nonetheless, presently available phantom construction materials are not adequate for studies involving both magnetic resonance imaging (MRI) and computed tomography (CT). The purpose of this study is utilizing effective ingredients to fabricate a sort of synthetic material, which is adapt to multimodality imaging experiments. **Methods:** Agarose, gadopentetate dimeglumine (Gd-DTPA), micro-pearl powder, sodium chloride (NaCl) and sodium fluoride (NaF) was added during the formation of tissue-equivalent material (TMM). Both CT and MR scanning were performed, the T1 and T2 relaxation time and CT numbers (HU) were obtained for various combinations of water-based agarose gels. **Results:** The concentration of agarose (0- 4.5%) had little effect on the CT number of TMM, but had a significant effect on the T1 and T2 relaxation times (especially the T2 value), specifically the increase in agarose concentration, relaxation time reduce. The concentration of Gd-DTPA had a great influence on the T1 relaxation time of the material. The content of micro-pearl powder hardly changed the T1 or T2 relaxation time of the agarose gels. However, agarose loaded with micro-pearl powder was stable (in the range of 0-2%; gradient concentrations appear after the range is exceeded), and different concentrations produced TMMs that showed different CT values across different human tissues. **Conclusion:** The water-based agarose gel of the present study has tissue equivalency in dose attenuation with X-rays and relaxation times with MRI. It is an powerful tool for designing multimodality imaging phantoms.

PO-GePV-I-17

Beam-Shaping Filter and Dosimetric Characterization of the Airo Mobile CT Scanner

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Purpose: For a conventional CT with a 32 cm body phantom, the ratio between the peripheral and the center CTDI measurement (the P/C ratio) is close to 2. For the Airo mobile CT, the P/C ratio is approximately 4. This study is to investigate what causes the ratio difference and how it impacts radiation dose and image quality. **Methods:** We measured the HVL at the iso-center to verify the x-ray beam quality. For the assessment of the beam-shaping filter, exposure was measured under service mode with a RaySafe R/F detector in 14 lateral positions starting from the iso-center. The measurements were made with 2cm intervals between them for 120kV, 192mAs using service mode. All measurements were normalized to the measurement at the iso-center. The center and peripheral CTDIs were measured with a 32-cm body phantom for 80kV, 100kV, and 120kV. The image uniformity between the center and edge was analyzed. For comparison the same experiment was repeated in a Siemens Confidence. **Results:** The x-ray quality is comparable (6.85 vs. 7.84mmAl). The higher P/C ratio in the Airo is due to the beam-shaping filter where a flat filter is applied, while a bowtie filter is used for the Siemens Confidence. For the Airo, the P/C ratios are 4.83, 4.03, 3.64 for 80kV, 100kV, and 120kV, respectively. The P/C ratio for the Confidence is 2.11 at 120kV. The dose distribution across the body phantom is more homogeneous in the Confidence than the Airo. The use of a bowtie filter can homogenize the photon flux at the detectors and reduce radiation to the patient in those peripheral regions of the body. **Conclusion:** The Airo with a flat filter delivers relatively more radiation to patient especially to the skin. It may be of a concern for a complicated orthopedic procedure.

PO-GePV-I-18

CT Image Standardization for Radiomic Feature Enhancement in Non-Small Cell Lung Cancer

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Purpose: This study is to develop an end-to-end fully automatic CT image standardization approach named RadiomicGAN to improve the stability of radiomics features in NSCLC. **Methods:** RadiomicGAN was developed based on transfer learning and Generative Adversarial Network (GAN), which can efficiently learn the standard image feature distributions from relatively small medical image data. The performance of RadiomicGAN was enhanced by leveraging the wide dynamic range of CT imaging. Specifically, the model adaptively identified the effective pixel range in the current training cycle and smartly optimized its loss in the next training cycle. CT images for training and testing were acquired in a Siemens Sensation Force using three different kernels (B157, B164, Br40) and different slice thickness (0.5, 1, 1.5 and 3mm). The training data included a total of 14,372 images, while each test data contained 387 images. The RadiomicGAN was also trained and tested using CT images of a chest phantom. Radiomic features were extracted from the lung nodule delineated manually with the IBEX. In total, 1,401 radiomics features were evaluated and compared before and after image normalization. **Results:** RadiomicGAN achieves significantly better image normalization performance than other state-of-art models. In average, using B164 as the standard image reconstruction kernel, the averaged absolute error of the 1,401 radiomics features before and after normalization reduced from 51.82% to 29.8% for images reconstructed using B157 and from 78.45% to 30.5% for images reconstructed using Br40. The existing CT image standardization models such as Choe et al, GANai and STAN-CT reduced the error to 44.7%, 42.1%, and 40.5% respectively for B157, and 56.1%, 54.5%, and 48.6% respectively for Br40. **Conclusion:** RadiomicGAN provides a better end-to-end solution for CT image standardization and normalization. Further evaluation with larger amount of patient data are warranted.

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PO-GePV-I-19

Evaluation of Spatial Resolution Variability in Whole Body Bone Scans

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Purpose: To assess the variability in spatial resolution observed in whole body bone scans completed on different scintillation camera imaging devices within the same facility as reported by clinical staff. **Methods:** Upon installation of new scintillation camera, clinical staff reported variability in spatial resolution among cameras of same model and detector specifications when performing whole body bone scans. Static intrinsic and extrinsic spatial resolution FWHM measurements obtained with a four-quadrant bar phantom during annual/acceptance testing of the units were compared. The clinical whole body bone scan was replicated with two capillary line sources containing ^{99m}Tc spaced 20 cm apart and perpendicular to the direction of table travel. Detector contouring was

active and automatically adjusted the position of each detector relative to the line sources. Distances from the line sources to the camera heads were measured for all acquisitions. Static acquisitions of the line sources was obtained at same distance as measured above. **Results:** Comparison of static intrinsic and extrinsic FWHM measurements among the cameras showed minor variation. Significant variation in FWHM measurements of line sources between imaging devices was observed. The cause of variation was determined to be variability in the default table height position settings which resulted in variation in collimator-to-source distances on each imaging device. Default table height was set by the service engineer at the time of installation of each unit and was not standardized across the camera systems. The default table height setting was standardized and measurements were repeated. **Conclusion:** Variability in default table height settings for whole body bone scans resulted in reduced and inconsistent spatial resolution in clinical studies among imaging devices. Recommend routine verification of table height settings during whole body bone scans and inter-comparison of QC data to ensure optimum image quality during clinical scans.

PO-GePV-I-20

TOPAS for Imaging: A Monte Carlo Simulation Tool for PET/SPECT/CBCT Imaging

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Purpose: To develop an extension of medical imaging devices for the TOPAS Monte Carlo (MC) system. **Methods:** MC methods have been extensively used to simulate imaging systems, but there have so far been no examples for the TOPAS MC. We designed easily modifiable classes for the detector geometries, sources, and scorers in PET, SPECT, and CBCT systems. The output file is scored in CASToR (Customizable and Advanced Software for Tomographic Reconstruction) format that can directly be processed for image reconstruction using CASToR. **Results:** We developed TOPAS for Imaging, an extension of TOPAS designed to generate the geometries of PET, SPECT, and CBCT and record their measurements of simulated radioactive sources for reconstruction in CASToR. First comparisons to experimental data for validating the PET system will be presented. **Conclusion:** TOPAS for Imaging promises to advance our understanding of imaging systems by enabling researchers to have more platforms for simulating their systems.

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PO-GePV-I-21

Sparse-View CT Reconstruction Via Generative Adversarial Network (GAN) Using Fully Convolutional DenseNet (FC-DenseNet)

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Purpose: Sparse-view CT benefits from reduction in dose of radiation to patient with fewer projection views than general. Conventional filtered-back projection (FBP) algorithm is limited due to the insufficient projection information required to sparse-view CT reconstruction. The iterative reconstruction complement the disadvantage, while it is mostly time-inefficient and challenging to dramatically reduce the number of projection views. Deep-learning-based sparse-view reconstruction could address the drawbacks. Though various trials made thus far, this work proposes a reconstruction with Generative Adversarial Network (GAN) based model that utilizes FC-DenseNet as a backbone network architecture. **Methods:** The input and output of the network were set to be sparse-view CT (45-view), and ground-truth CT (720-view), which were obtained from FBP algorithm. Patient cohort consisted of 74 scans of head-and-neck patients: 37, 28, and 9 scans used for training, validating, and testing the proposed network. We trained GAN using FC-DenseNet and used L1 loss, L2 loss, and gradient difference loss to obtain training results. The metrics for reconstruction accuracy of the trained network employed Mean Absolute Error (MAE), Peak Signal to Noise Ratio (PSNR), and Structural Similarity Index (SSIM). **Results:** The results indicated that Our proposed model can improve the sparse-view CT image quality with minimal streaking artifacts. The generated image for our proposed model was assessed by MAE, PSNR, and SSIM. The original sparse-view CT reconstruction image for conventional FBP algorithm results 53.03, 51.28, 0.97. The model with L1 loss results 16.88, 44.52, 0.86. The model with L2 loss results 24.41, 36.59, 0.92. The model with gradient difference loss results 17.10, 31.86, 0.59. **Conclusion:** Our proposed network adopted GAN network architecture for sparse-view CT image reconstruction. It demonstrated that the trained network could achieve much more accurate reconstructed image than conventional FBP.

PO-GePV-I-22

Study of Cariporide Effect on Tumor Intracellular PH (pHi) Using Chemical Exchange Saturation Transfer (CEST) MRI

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Purpose: pHi is a hallmark of tumor microenvironment response to therapies. NHE1 is an acid-extruding membrane transport protein in cells.¹ In CEST-MRI contrast is produced by exciting exchangeable tissue protons at their specific absorption frequency and observing the transfer of magnetization to bulk tissue water. Amine and amide concentration-independent detection (AACID) is an approach that uses the sensitivity of amine and amide proton to CETS contrast.² The AACID value inversely relates to pHi. One way to achieve tumor acidification as a therapeutic strategy is by blocking the NHE1 transporter. Cariporide is a potent inhibitor of NHE1. We have shown that cariporide can selectively acidify U87MG glioma in mice.³ Our goal was to determine whether it also selectively acidifies a rat C6 glioma tumor model following injection by mapping tumor pHi. **Methods:** Cells were injected into the brain of six rats. To evaluate the effect of cariporide on tumor pHi, rats received an IP injection of the drug two weeks after implantation while inside a 9.4T scanner to measure the change in pHi following injection. **Results:** Five minutes after injection, CEST-MRI was collected for three hours. For data analysis, we compared the first maximum change in AACID value post-injection with the pre-injection. After one hour, the average AACID value in the tumor significantly increased. The average AACID value in tumor post-injection was 5.4% higher compared to pre-injection. The average AACID value in contralateral tissue also increased in a similar way. **Conclusion:** We did not observe selective tumor acidification following injection as was observed in the previous study. The reason for this discrepancy is currently unknown but may be related to potential differences in tumor vasculature that may limit the ability of cariporide to infiltrate the tumor.⁴ Future work includes increasing cariporide dose and modifying our quantification method of the AACID measurement.

PO-GePV-I-23

Conformance Analysis Between Positive Beam Limitation Quality Control Methods in Digital Roentgen-Ray Imaging

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Purpose: Conformance between alternate methods that evaluate the positive beam limitation (PBL) of X-ray systems is determined using Bland-Altman analysis. **Methods:** The PBL of GE Precision, Fuji Velocity and Philips Easy-Diagnost systems was evaluated using the gold-standard and two alternate methods, light-field-width and DR-coin. First, the tube was set at a detent position where the PBL engaged. The gold-standard method was carried by placing a CR imaging plate at a known distance from the image receptor and exposing it. For the light-field-width method, the light field projected on the Bucky's surface was measured with a measuring tape. For the DR-coin method, coins were placed on the Bucky's surface at the edges of the light field and the DR

detector inside the Bucky was exposed. All field sizes were measured in PACS and magnification corrected. Measurement cohorts were collected for both wall and table Buckies for the aforementioned units. The two methods were compared for conformance with the gold-standard using Bland-Altman technique. This method relies on plotting difference of results obtained using different methods against the average of results obtained by the two methods. **Results:** The mean difference between the gold-standard and DR-coin method for PBL assessment is -0.50 and the standard deviation of differences is 0.76. The mean difference between the gold-standard and light-field-width method for PBL assessment is 0.0 and the standard deviation of differences is 0.81. **Conclusion:** Preliminary results suggest that the DR-coin to gold-standard methods demonstrate slightly lesser variation between the observations as compared to the light-field-width method. This analysis along with our practical experience in the field favors the use of the DR-coin based method if the CR-based method is not available. Therefore, the DR-coin method is a more promising alternative to the conventional CR-based PBL assessment, given the shrinking availability of CR-based imaging in clinics.

PO-GePV-I-24

Direct Measurement of the Surface Dose Using a Plane-Parallel Ionization Chamber for Interventional Radiology

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Purpose: To directly measure the entrance surface dose (ESD) of a phantom using a plane-parallel ionization chamber for interventional radiology (IVR) procedures. **Methods:** A PTW 23344 plane-parallel chamber was calibrated in terms of the dose at a 1-cm depth determined using a PTW 30013 Farmer-type chamber. The ESD for six IVR devices of three manufacturers in three institutions was measured using the PTW 23344 chamber. The front surface of the PTW23344 was set at an IVR reference point on the examination tabletop located 15 cm below the isocenter. The irradiation conditions of the IVR procedures were determined according to the automatic exposure control for the abdomen and head in the digital acquisition mode and fluoroscopy mode. The ESD is presented as a function of a half-value layer (HVL (mm Al)) for the given quality index (QI) values, which are defined as the ratio of the effective energy to the maximum energy of photons. The relationship between the ESD and incident air kerma, $K_{air,si}$, was also analyzed. $K_{air,si}$ was calculated using x-ray fluence spectra for the IVR irradiation conditions created by the SpekCalc program. **Results:** The calibration coefficient of the PTW 23344 chamber was mostly constant for x-ray beam qualities with tube voltages of 60–140 kVp and HVLs of 2.24–9.22 mm Al. The ESD increased with an increasing tube voltage for x-ray beams with the same HVL, and even for the same QI. In addition, the ESD achieved a good linear relationship with $K_{air,si}$, regardless of the x-ray beam quality. **Conclusion:** To reduce the ESD, reducing the tube voltage and using the x-ray beam quality with the QI close to unity was found to be effective. It is also possible to estimate the ESD using $K_{air,si}$ based on the irradiation conditions for IVR procedures.

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PO-GePV-I-25

Optimisation of Radiation Dose and Image Quality for Six Different Flat Panel Detectors Using a Semi-Anatomical Chest Phantom

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Purpose: To evaluate the impact of radiation dose on image quality in six different flat panel detectors (FPD) and optimize radiation dose by altering exposure factors in low and high kVp chest radiography. **Methods:** Radiographic images of a semi-anatomical chest phantom were acquired from 6 FPDs from 4 different digital radiography (DR) manufacturers. Dose-area-product (DAP) was measured for three different protocols (Standard, AEC and low radiation dose) utilising 70 and 110 kVp. To normalize measurements across all FPDs, a standard DAP value of 51.05 μGym^2 obtained at 70 kVp and 9.43 μGym^2 at 110 kVp was used. The lowest possible radiation dose to provide acceptable image quality was achieved by manually adjusting mAs for both tube potentials. While selecting AEC exposure parameters, density, dose level or exposure (speed) classes were altered to obtain optimal radiation dose and image quality. Contrast to noise ratio (CNR), signal to noise ratio (SNR), figure of merit (FOM), contrast detail of the lung, heart and diaphragm regions and spatial resolution were quantified using a workstation. **Results:** In comparison with standard protocol, radiation dose reduction of 21%, 21.6%, 59.5%, 60.7%, 62.2% and 67.6% was achieved in GE Definium8k, Siemens Fusion, Fujifilm FGX, Fujifilm FGXR, Philips Digital Diagnost and Siemens Aristos with low dose protocol at 70 kVp. However, at 110 kVp, radiation dose could be reduced by 15.7%, 34.8% for Philips Digital Diagnost and Siemens Aristos alone. **Conclusion:** This study demonstrates the feasibility of using a semi-anatomical chest phantom in the optimisation of radiation dose and image quality for different FPDs. The FOM was a good indicator in assessing image quality between different detectors.

PO-GePV-I-26

1D and 2D MTF Measurement for a Dual-Detector Image Receptor Using the Noise Response (NR) Method Compared with Using the Standard Edge Method

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Purpose: To compare the modulation transfer functions (MTF) of a new dual-detector image receptor measured using the Noise-Response (NR) method and the standard edge-response method. **Methods:** The NR method was used to measure both 1D and 2D MTF's for the two built-in detectors of the Canon Alphenix system (194-micron pixel FPD and 76-micron pixel HiDef). A cascaded-linear-systems analysis based relationship between the noise power spectrum (NPS) and the MTF of a generalized detector system was used. A two-dimensional NPS was calculated for each detector at 7 different exposures (4 to 100 uR) with the RQA5 x-ray spectrum. To obtain the 2D MTF, the 1D MTF at each angle was determined using the radial component of the NPS at the corresponding angle. Each 1D NPS was plotted versus the mean signal level for all spatial frequencies; a linear regression was fitted to this data to isolate the quantum-noise component, the shape of which partly depends on the system resolution. The spatial-frequency response of the resulting slopes was used to obtain the corresponding 1D MTF. The set of 1D MTF's over all angles provides the 2D MTF. The NR 1D MTF's along the rows for both detectors were compared to those obtained using the standard edge-response method. **Results:** The 1D MTF's obtained using the noise and edge methods for both detectors agree within experimental uncertainty and demonstrate the improved resolution of the HiDef detector over the standard FPD. The 2D MTF shows the magnitude of reduced resolution along the diagonals. **Conclusion:** The noise-response method is a simple technique which can be used to accurately measure the 2D MTF of digital x-ray imagers, without the need for precision edge or slit devices. The unique capability of the NR method to provide a two-dimensional MTF provides more detailed detector/system characterization, including any potential non-isotropy.

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AAPM ePoster Library Multi-Disciplinary General ePoster Viewing

PO-GePV-M-01

Clinical Evaluation of Deep Learning Based Auto-Contouring Software for Prostate Radiotherapy

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Purpose: Deep learning-based auto-contouring (DLC) may be used to reduce the time required to generate organs-at-risk (OAR). In this study, we investigate both the time and accuracy of the DLC OARs compared to manually drawn OARs for prostate radiotherapy. **Methods:** The OARs of ten prostate patients were contoured by the DLC software and four experts (EC) (Senior dosimetrists with an average of 15 years of experience). The time to manually draw the Penile Bulb, Femurs, Bladder and Rectum was recorded. Additionally, the experts adjusted the DLC contours (User Adj DLC) and recorded the time. The Dice Similarity Coefficient (DSC) and the Mean Distance to Agreement (MDA) were used for comparisons. **Results:** The use of DLC resulted in a statistically significant ($p < 0.05$) reduction in contouring time, from a mean time for EC contours of 28 minutes (SD=10min) to a mean time of 10 minutes (SD=2min) for the User Adj DLC. The mean DSC values for the DLCs were: Penile Bulb = 0.62, Femurs = 0.94, Bladder = 0.94, and Rectum = 0.83. The DSC values for the mean User Adj DLC were: Penile Bulb = 0.69, Left and Right Femur = 0.96, Bladder = 0.94, and Rectum = 0.88. The mean MDA values for the DLCs were: Penile Bulb = 2.43mm, Femurs = 0.85mm, Bladder = 0.92 and Rectum = 2.31 mm. The MDA values for the mean User Adj DLC were: Penile Bulb = 1.90 mm, Left and Right Femur = 0.77 mm, Bladder = 0.81 mm, and Rectum = 1.57 mm. **Conclusion:** The use of auto contouring provides a starting point for the creation of OARs allowing for a significant time reduction. User adjustment of the DLC based contours resulted in a significant increase ($p < 0.05$) in DSC values and decrease in MDA values for all the OARs.

PO-GePV-M-02

Comparison of Deep Learning Based Auto-Contouring Software to Expert Inter-Observer Variability in Prostate Radiotherapy

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Purpose: Deep learning-based auto-counter (DLC) may be used to reduce the time required to generate organs at risk (OAR). In this study, we investigate the accuracy of DLC contouring relative to inter-observer variability. **Methods:** Ten prostate patients were contoured by four experts, senior dosimetrists with an average of 15 years of experience (EC). The penile bulb, femurs, bladder and rectum were contoured by the DLC software and each expert. The Dice Similarity Coefficient (DSC) and Mean Distance to Agreement (MDA) was used to compare the DLC contours to multiple EC contours. The worst DSC value for the Deep learning vs Experts (DLC-EC) combinations and inter-expert (EC-EC) combinations for each patient were used for the statistical analysis. This ensured that lower performing outliers were included. **Results:** No statistically significant ($p < 0.05$) differences were measured for the worst DSC metric for each case. The single-worst DSC for the DLC-EC and EC-EC were: penile bulb (0.30 vs 0.31), Femurs (0.87 vs 0.87), bladder (0.74 vs 0.74), and rectum (0.76 vs 0.77). The average-worst DSC were: penile bulb (0.49 vs 0.49), Femurs (0.92 vs 0.92), bladder (0.91 vs 0.90), and rectum (0.81 vs 0.82). The mean DSC values for DLC-EC were: Penile Bulb = 0.62, Femur = 0.94, Bladder = 0.94, and Rectum = 0.83. The mean DSC values for EC-EC were: Penile Bulb = 0.60, Femur = 0.93, Bladder = 0.92, and Rectum = 0.86. The mean MDA values: Penile Bulb = 2.43mm, Femurs = 0.85mm, Bladder = 0.92mm, and Rectum = 2.31mm. By comparison, the mean MDA values for the EC-EC inter-comparison are: Penile Bulb = 2.39mm, Femurs = 1.15mm, Bladder = 1.02mm, and Rectum = 2.26mm. **Conclusion:** The use of DLC based auto-contouring provides contours that are comparable to expert inter-observer variability.

PO-GePV-M-03

A 2D-3D Hybrid Convolutional Neural Network for Lung Lobe Auto-Segmentation on Standard Slice Thickness CT of Patients Receiving Radiotherapy

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Purpose: Accurate segmentation of lung lobe on routine CT images of locally advanced stagelung cancer patients undergoing radiotherapy can help radiation oncologists to implement lobar-level treatment planning, dose assessment and efficacy prediction. We aim to establish a novel 2D-3D hybrid convolutional neural network (CNN) to provide reliable lung lobe auto-segmentation results in the clinical setting. **Methods:** We retrospectively collected and evaluated thorax CT scans of 95 locally advanced Non-small-cell lung cancer (NSCLC) patients treated at our institution from June 2019 to February 2020. The CT images were acquired with 5mm slice-thickness. Two CNNs were used for lung lobe segmentation, a 3D CNN for extracting 3D contextual information and a 2D CNN for extracting texture information. Contouring quality was evaluated using three quantitative metrics and visual evaluation was performed to assess the clinical acceptability. **Results:** For the 25 cases in the test group, Dice Similarity Coefficient (DSC) of all lung lobes contours exceeded 0.75, which met the pass criteria of the segmentation result. Our model achieved high performances with DSC as high as 0.9579, 0.9459, 0.9472, 0.9452 and 0.8968 for left upper lobe (LUL), left lower lobe (LLL), right upper lobe (RUL), right lower lobe (RLL) and right middle lobe (RML), respectively. Clinician's visual assessment showed that 118/125 lobe contours met the requirements for clinical use, only 7 contours need manual correction. **Conclusion:** Our 2D-3D hybrid CNN model achieved accurate automatic segmentation of lung lobes on conventional slice-thickness CT of locally advanced lung cancer patients, and has good clinical practicability.

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PO-GePV-M-04

Automatic Contouring QA Approach Using a Deep-Learning-Based Auto-Contouring System

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Purpose: To establish a similarity metric to automatically flag clinically unacceptable contours in the female pelvis. **Methods:** Six female pelvic structures (Uterus/Cervix, CTVn, Para-aortic lymph nodes (PAN), Bladder, Rectum, and Kidneys) were generated using manual and auto-contouring. **Methods:** clinically acceptable and unacceptable contours were generated manually, and two auto-contours per structure were generated with two independently developed deep learning-based auto-contouring systems. These structures were generated on 87 CT image sets from 4 hospitals. The clinical acceptability of one of the two auto-contours ("primary contour") was confirmed by radiation oncologists, and the second auto-contour was used for QA purposes. Eleven similarity metrics (Dice-Similarity Coefficient (DSC), Hausdorff distance, 95% Hausdorff distance, mean surface distance, and surface DSC with incremental thicknesses of 1-10 mm) were calculated between primary and QA contours. The metrics were tested individually and in various combinations as inputs for a support vector machine. Model inputs were used to determine the optimal decision boundary between clinically acceptable and unacceptable contours. Four different kernels (linear, radial basis function (RBF), sigmoid, and polynomial) were tested when multiple metrics were used as inputs for the support vector machine. **Results:** The highest accuracies were achieved using surface DSCs with a thickness of 1, 2, or 3 mm and they were 0.91/0.90/0.89/0.92/0.95/0.97 for

UteroCervix/CTVn/PAN/bladder/rectum/kidneys, respectively. The average accuracies across different structures were 0.90/0.88/0.70/0.88 for linear/RBF/sigmoid/polynomial kernels, respectively, and therefore, the linear kernel was the most accurate when multiple metrics were used as inputs for the support vector machine. However, multiple metrics did not improve the accuracy significantly compared to a surface DSC as an input. **Conclusion:** We distinguished clinically acceptable contours from clinically unacceptable contours with an accuracy higher than 0.9 for the relevant structures in the female pelvis; the most accurate similarity metric was surface DSC with a thickness of 1, 2, or 3 mm.

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PO-GePV-M-05

Radiation Induced Lung Damage Tissue Segmentation and Classification Framework

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Purpose: Radiation-induced lung damage (RILD) is a common side effect of radiotherapy. The ability to automatically segment, classify and quantify different types of parenchymal change is essential to uncover underlying patterns of RILD and their evolution over time. **Methods:** The following five labels were introduced to classify parenchymal morphological appearance: A) default lung parenchyma; B) non-textured ground glass opacity; C) textured ground glass opacity; D) solid density with air-bronchograms; E) solid density. We utilized a two-stage ground truth data generation approach, akin to active learning. Manual segmentations were used to train an initial auto-segmentation method. These results were manually refined and used to train the final auto-segmentation algorithm. The auto-segmentation algorithm was an ensemble of six 2D-Unets with different loss functions and numbers of input channels. **Results:** Our development dataset consisted of 40 cases, each with a pre-radiotherapy, 3, 6, 12, and 24 month follow-up CT scan (n=200 CTs). The method was assessed on a hold-out test dataset of 6 cases (n=30 CTs). The Dice Score Coefficient (DSC) achieved for each tissue class were: A) 0.99 and 0.98, B) 0.71 and 0.44, C) 0.56 and 0.26, D) 0.79 and 0.47, and E) 0.96 and 0.92, for development and test subsets respectively. The lowest values for the test subsets were caused by imaging artefacts or in regions where annotations were less certain. We also performed qualitative evaluation on the test dataset presenting manual and auto-segmentations to a blinded independent radiologist to rate them as 'clinically acceptable', 'minor disagreement' or 'major disagreement'. The auto-segmentation ratings were similar to the manual segmentations, both having slightly over 80% of cases rated as 'clinically acceptable'. **Conclusion:** The proposed framework for auto-segmentation of different lung tissue classes produces acceptable results in the majority of cases and will be useful for future studies of RILD.

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PO-GePV-M-06

Hybrid Approach of Auto-Segmentation Based on Pixel Level Edge Detection

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Purpose: To explore semi-automatic approach for segmenting regions of interest based on pixel level information. **Methods:** Due to recent advances in machine learning and big data technologies, in radiotherapy field there is great interest in utilizing these advances to study treatment response, automating the organ segmentation, tracking moving tumors, online daily adaption of treatment plans and more. But in all of these studies possibility of achieving reliable results is directly proportional to reduction in inter-observer variability. We have developed a hybrid auto segmentation tool. Unlike atlas, model and deep learning based auto-contouring approaches users can interact in the automation process if necessary. To make it understand better our tool is like having a driverless car but rider can take control whenever it is necessary. Our INTERactive CONTouring (Intcont) tool draws contour around an edged region of interest (ROI). It combines automatic edge finding with the operator oversight and occasional intervention if the contour line deviates from the expected path. This is a universal tool and can be applied to segment tumor volumes as well as organs at risk. Moreover, Intcont allows to change local threshold at the edge if expert wants to include or exclude the actual edge from ROI (tighter or looser boundary around ROI). **Results:** Intcont has successfully drawn reliable ROIs on all datasets we considered during development of the tool. **Conclusion:** Intcont offers a novel approach to image contouring that provides combination of automated contouring with user interaction. It avoids contouring artifacts often introduced by fully automated contouring system and at the same time it is less subjective than outright manual drawing or manual editing when contours created by automated methods have artifacts. Rigorous analysis of the "correctness" of contours from Intcont will be done soon.

PO-GePV-M-07

Performance of a Convolutional Neural Network-Based Algorithm for Automatic Segmentation of the Whole Heart in Non-Contrast Radiotherapy Planning CTs with Image Artifacts

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Purpose: To evaluate the performance of a convolution neural network-based algorithm for automatic segmentation of the whole heart from non-contrast radiotherapy planning CTs with image artifacts **Methods:** We obtained non-contrast planning CTs with manual contours of the whole heart of 100 breast cancer patients in the Radiotherapy Comparative Effectiveness (RadComp) clinical trial. We employed a convolutional neural network, U-net, for the automatic segmentation of the whole heart. A total of 66 patients without image artifacts were used for training and 34 patients with image artifacts were used for testing the performance of the segmentation method. At least a single CT slice contained image artifacts: left breast (n=4), right breast (n=6), both breasts (n=15), heart (n=2), and other regions (n=7). The results were then compared with manual delineations to evaluate the performance of our method in terms of geometric similarities using dice similarity coefficients (DSC). **Results:** The initial training set ran 400 epochs, which stabilized in approximately 46 hours. The DSCs for the image sets with artifacts in left breast, right breast, both breasts, heart, and other regions were 0.90 ± 0.03 , 0.90 ± 0.04 , 0.91 ± 0.04 , 0.91 ± 0.01 , and 0.79 ± 0.15 , respectively. The lowest DSCs were observed in large patients extending outside the field of view. **Conclusion:** When compared to DSC results from literature based on convolutional neural network, we found that image artifacts substantially reduce the performance of automatic segmentations. We

plan to evaluate the dosimetric impact of the image artifacts.

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PO-GePV-M-08

A 3D Multi-Modality Lung Tumor Segmentation Method Based on Deep Learning

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Purpose: To develop an automatic lung tumor segmentation method based on multi-modality images and deep learning. **Methods:** A 3D segmentation neural network using PET and CT multi-modality image inputs were constructed based on the U-Net. The network architecture consists of two parallel convolution arms for independent feature extraction from the simulation CT and the diagnostic PET at multiple resolution levels. The extracted features were concatenated and fed into a single deconvolution path. The resulting tumor probability map was post-processed through the conditional random field and thresholded to obtain the tumor segmentation. The performance of this 3D multi-modality network was compared with that of the previously constructed 2D multi-modality network and the 2D and 3D CT-only single-modality networks. Furthermore, a volume-stratified training/validation strategy was evaluated for segmentation performance improvement. The networks were trained/validated/tested (3:1:1 split ratio) on 290 pairs of diagnostic PETs and planning CTs (166 SBRT and 124 conventionally-fractionated) from lung cancer patients treated at our clinic, with manual physician contours as the ground truth. The segmentation performance of the networks was evaluated via the DICE similarity coefficient (DSC) and the Hausdorff Distance (HD). **Results:** The mean DSC and HD of the single-modality models were 0.75 ± 0.16 and 8.5 ± 5.8 mm for 2D, and 0.77 ± 0.12 and 7.6 ± 4.7 mm for 3D, respectively. The mean DSC and HD of the multi-modality networks were 0.78 ± 0.15 and 7.6 ± 4.7 mm for 2D, and 0.79 ± 0.10 and 5.8 ± 3.2 mm for 3D, respectively. The small GTVs benefited from separate training/validation with a threshold of 25mL identified as ideal. The best overall performance was obtained by the multi-modality volume-stratified 3D model, with the overall DSC and HD of 0.83 ± 0.09 and 5.9 ± 2.5 mm, respectively. **Conclusion:** The 3D multi-modality deep learning network outperformed its 2D counterpart and the single-modality networks. The volume-based stratification can further improve the segmentation results.

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Disclosures:

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PO-GePV-M-09

A High-Throughput, High-Availability Automated Planning Tool for Radiotherapy Clinics in Low-Resource Settings

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Purpose: To assess the throughput and reliability of a web-based, automated radiotherapy planning tool (Radiation Planning Assistant, RPA) designed to support low-resource clinics. **Methods:** The RPA architecture consists of several, multi-capacity computing modules that process patients in a serial manner. Baseline timings for each module were measured by processing 25 cervix and 25 head & neck (H&N) patient datasets through the entire workflow. Each module was modeled in a manufacturing discrete event simulator (ManPy), and programmed with their respective timing data, to evaluate serial and parallel workflows. Model accuracy was evaluated by comparing the simulator's completion times for single- and multi-patient queues to those of the real RPA system. Finally, many module downtime scenarios were simulated to determine their impact on baseline performance of the RPA's daily throughput. **Results:** An independent t-test indicated that mean processing times were not significantly different between the RPA and the model ($p > 0.05$). The model gave an expectation value for plans successfully processed in 24 hours: 483 cervix plans, 255 H&N contours, or 258 H&N plans when all systems are operating. Cervix plan generation remained within 5% of its baseline throughput until any given module (except the plan/dose QA module) went down for 3+ hours. H&N contour generation remained within 5% of its baseline when either of its two contouring module's downtime did not exceed 1 hour. H&N plan generation remained within 5% of its baseline until at least 2 of 5 available VMAT optimization modules' downtime exceeded 1 hour, or any other module downtime exceeded 3 hours. Plan calculation and report generation module downtimes had <5% effect on output through 4 hours downtime. **Conclusion:** The RPA architecture is robust to downtime of its individual modules, thus having the ability to reliably address resource shortage in radiotherapy clinics.

PO-GePV-M-10

Development of Gell: A GPU-Powered Ultra-Large-Scale Cell-Based Simulator

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Purpose: To understand macroscopic imaging feature formation as a result of underlying cancer cell biology and growth dynamics, we build a novel fast GPU-powered cell-based simulation platform-Gell suited for ultra-large-scale tumor growth modeling. **Methods:** Tumor development is a multifaced and multiscale problem. At the biochemical microenvironment level, we implemented a fast paralleled PDE solver for the secretion, diffusion, uptake, and decay of oxygen and Vascular Endothelial Growth Factor (VEGF) with computational complexity proportional to volume voxel number. At the single-cell level, we included modules to describe cell proliferation, growth, death, nutrient uptake, and angiogenic factor release. At the multicellular level, we described the mechanical interactions for cell-cell adhesive and repulsion with computational complexity proportional to the number of cells. We combined these modules to simulate HCC827 non-small cell lung carcinoma spheroids' growth and compared the results with state-of-the-art CPU simulator PhysiCell. **Results:** Gell simulation results of the approximately constant spheroid diameter growth speed and the crack-like necrotic core microstructure are consistent with PhysiCell simulation and in vitro experimental observation. However, the computational time for 18 days of in vitro spheroid growth was reduced from 76 hours using CPU-based PhysiCell (dual 6-core Intel Xeon X5690) to seven minutes using GPU-based Gell (Nvidia RTX 2080Ti). In a large multicellular simulation test for a 1-hour biological process of 16 million cells, the simulation was accelerated by over one thousand times compared with PhysiCell. **Conclusion:** In this work, we developed a fast GPU-based simulator, Gell, for avascular multicellular processes. The GPU implementation afforded 1000X acceleration over the CPU method. The acceleration is essential to overcoming the bottleneck in CPU simulation for ultra-large-scale, morphologically complex centimeter-level tumors and tumor vasculature that form medical image features.

PO-GePV-M-11

Radiomic Analysis of Limited-Sized Data Using Unsupervised Methods: Prediction of HPV Status in Head and Neck Cancer

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Purpose: Working with limited-sized data is a common challenge in medicine. We investigated a promising unsupervised approach to radiomics that combines unsupervised optimal mass transport (OMT) on a network of radiomic features. tSNE (t-stochastic neighborhood embedding) was then used to cluster and visualize the results in 2-D with the novel Wasserstein metric. We demonstrate this approach on the important problem of predicting HPV status in head-and-

neck squamous cell cancer (HNSCC) patients with contrast enhanced computed tomography (CT) images. **Methods:** Radiomic analysis was performed on 77 HNSCC patients with pre-treatment CT scans downloaded from The Cancer Imaging Archive (TCIA). Using the CERR radiomics toolbox, 67 radiomic features were extracted from primary tumors manually contoured on all evaluable artifact-free slices. First, a partial correlation network was constructed, where a link between two nodes was determined using a partial correlation coefficient after conditioning on all other features. To further remove potential false positive links, a lasso-type regularization was applied to the resultant network. On the largest connected network component, an OMT metric called the W_1 Wasserstein distance was computed for all pairs of samples, and then the resultant distance matrix was input into tSNE to map the samples to a low-dimensional space. **Results:** The cohort had 13 HPV positive tumors. The resultant Wasserstein distance matrix computed in the largest network component with 10 radiomic features was fed into tSNE, using the OMT-based distance as a cost function. In the mapped two-dimensional space, tumors with HPV positivity were better clustered with a Chi-squared p-value of 0.0008 than the tSNE result with the Euclidean distance. **Conclusion:** We demonstrated that radiomic features represented as a network can predict HPV positivity in head and neck cancer with good accuracy, employing tSNE coupled with an OMT based metric. The unsupervised result could further be refined using supervised methods.

PO-GePV-M-12

The IAEA Directory of Radiotherapy Centres (DIRAC): Current Figures and New Developments

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Purpose: IAEA has maintained DIRAC since 1959. Nowadays, it is the only high-quality source of information on worldwide radiotherapy (RT) infrastructure and is publicly accessible on the web at [//dirac.iaea.org](http://dirac.iaea.org). The web portal allows data input by contributors and visualization of hospital, national and regional data using modern business intelligence tools. **Methods:** DIRAC is continuously updated by individuals and national focal points, and the updates are verified by the IAEA data curation team. Part of the DIRAC information is publicly available on the website, but more detailed analyses are provided upon request for non-commercial use only. Each request for information is processed individually. A major update of DIRAC data structure has been deployed recently. In addition, the website has been updated to include a tool to quantify the bibliometric impact of DIRAC. **Results:** To date, DIRAC records include over 7900 RT centres equipped with over 17600 RT units. Of those, 14258 are external beam units, 3319 brachytherapy units (including electronic BT) and 105 light ion facilities (as of 2021.02.22). Over the last 20 years, 140 peer-reviewed publications directly referred to DIRAC, and over 2300 articles cited these papers. DIRAC structure was expanded to include details of collimation, imaging and motion management systems of RT treatment units, specific features and algorithms of treatment planning systems for both tele- and brachytherapy, conventional CT scanners and simulators. Additionally, data collection and representation structure for light ion therapy facilities has been revised. **Conclusion:** DIRAC supports academic and health care research, particularly in the analysis of national, regional and global availability of RT infrastructure. Developments in the DIRAC structure and its web-based interface are expanding with advances in RT technology, techniques, process and services.

PO-GePV-M-13

Treatment Management Selection of HPV-Positive Oropharynx Cancer Patients Using Artificial Intelligence on Electronic Medical Records and Radiomics

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Purpose: Human papillomavirus (HPV)-associated oropharyngeal squamous cell carcinoma (OPSCC) can be treated with definitive radiotherapy (RT), combined chemoradiotherapy (CRT), and/or transoral robotic surgery (TORS). TORS is a surgical approach that provides a comparable overall survival (OS) rate of approximately 90% at 2 years with the potential for less toxicity, lower cost, and shorter treatment regimen. However, approximately one-third of TORS patients require adjuvant CRT (with the highest combined treatment toxicity profile) due to extranodal extension (ENE) or positive surgical margins on the final pathology. In retrospect, the decision for these patients to undergo TORS is unfavorable. This study describes a treatment management selection procedure which utilizes artificial intelligence (AI), applied to clinical and imaging data, such that it identifies the patients contraindicated for TORS. **Methods:** Electronic medical records (EMR) for 55 subjects were retrospectively reviewed. Patient data included demographics, history, vital signs, and labs. Radiomics from diagnostic CTs were also utilized. After statistical pre-processing and remapping to equivalent numerical representations, three statistically significant textures were added to the analyses. Extracted data were used as an input into a deep neural network (DNN) that computed a probabilistic value for each patient, indicating whether TORS is a suitable disease management option. The DNN (based on Google's TensorFlow) incorporated a supervised learning technique used to mimic the physician's decision-making. A 5-fold cross-validation was utilized for evaluating the model's performance. **Results:** The DNN was successfully implemented with variable parameters, allowing flexibility and easy reconfigurability. An average AUC of 0.89 was achieved using significant EMR and imaging data. **Conclusion:** Treatment selection of HPV-associated OPSCC patients may be facilitated by AI techniques, applied to commonly tallied clinical variables, and combined with diagnostic imaging radiomics. The proposed methodology will help in discriminating patients who would ultimately require tri-modality therapy with the highest toxicity profile.

PO-GePV-M-14

Hyperpolarized Gas MRI Ventilation Synthesis From CT: Comparison of Conventional and Deep Learning Methods

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Purpose: Hyperpolarized gas MRI can visualize and quantify regional lung ventilation with exquisite detail. However, clinical uptake is limited to a few centres worldwide due to the requirement of highly specialized equipment and exogenous contrast. Alternative, non-contrast techniques have been proposed to model ventilation, including multi-inflation CT-based models of ventilation that have shown moderate spatial correlations with hyperpolarized gas MRI. Recent advances in deep learning (DL) using convolutional neural networks (CNNs) have shown great promise for numerous medical imaging applications, including image synthesis. Here, we propose a multi-channel 3D CNN approach for synthesizing hyperpolarized gas MRI scans using multi-inflation CT and a conventional density-based CT ventilation imaging (CTVI) model as inputs. We compare the accuracy of the CNN-based ventilation images with those generated by the conventional CTVI model. **Methods:** The dataset comprises paired inspiratory and expiratory CT and helium-3 hyperpolarized gas MRI for 47 patients with a range of pulmonary pathologies. A VNet CNN with a root mean square error loss function was used for image synthesis. We performed 6-fold cross-validation on the dataset and evaluated the correlation of the synthetic ventilation and hyperpolarized gas MRI scans at full resolution using Spearman's ρ on all lung voxels. The combined DL and CTVI model was compared to the conventional CTVI model. **Results:** Qualitatively, there are numerous examples of the CNN-generated synthetic ventilation images accurately replicating ventilation defects in the hyperpolarized gas MRI scans. Quantitatively, the CNN approach yielded statistically significant improvements in Spearman's correlation compared to the CTVI model ($p < 0.0001$) with mean $\pm$ SD ρ of 0.5 ± 0.2 vs 0.4 ± 0.2 . **Conclusion:** We propose a multi-channel CNN-based approach to generate synthetic ventilation images from routinely acquired multi-inflation CT and CTVI models. We show that a synergy between conventional model-based CTVI and DL yields statistically significant improvements in spatial correlation compared with conventional CTVI modelling.

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PO-GePV-M-15

Limitations of the Structural Similarity Index in Medical Image Synthesis Evaluation

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Purpose: Structural similarity (SSIM) is a popular image quality metrics used in the medical image synthesis community because of its higher correlation with human visual quality assessment. It has been widely adopted, but its limitations are often overlooked. SSIM is designed to work on a positive intensity scale, which is not the case in medical imaging. Intensity scales like the Hounsfield units contain negative numbers, and image normalization techniques can also introduce negative numbers. **Methods:** To quantify the impact of misusing the SSIM metric, we trained two uncorrelated image synthesis models to perform T2 MRI synthesis from T1 MRI. The synthesized images are strictly positive and their real SSIM can be evaluated. The real SSIM score was compared to a SSIM score computed on the same synthetic images, but after subtracting the mean tissue value from the ground truth to mimic Z-normalization. Next, the suitability of SSIM as a loss function is evaluated by training two synthesis models, once to synthesize strictly positive T2, and once to synthesize Z-normalized T2. The quality of models is evaluated using MAE and PSNR. **Results:** SSIM was reduced from 0.895 to 0.774 in model 1 and from 0.891 to 0.784 in model 2 when the tissue mean was subtracted from the synthetic images. This means that SSIM is underestimated on images with negative values, and it introduced an error that changed the relative ranking of models. This error can be clinically significant considering the small SSIM difference between state-of-the-art models. Secondly, training a synthesis model with SSIM as loss function on images with negative values drastically reduces image quality, with the mean relative error increasing from 8.65% to 22.78%. **Conclusion:** We show that it is inappropriate to use SSIM on images containing negative values. Finally, we propose reporting guidelines for easier comparisons between articles.

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PO-GePV-M-16

Predicting Successful Clinical Candidates for Marker-Less Lung Tumor Tracking with a Deep Learning Classification Model

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Purpose: Robotic radiosurgery allows for marker-less lung tumor tracking by detecting pattern variations in tumor density in 2D orthogonal x-ray images. The ability to detect and track a lung lesion depends on its size, density, and location, and has to be evaluated on a case-by-case basis. The current method for identifying which patient can be successfully treated with marker-less lung tumor tracking is a time-consuming multi-step process involving CT acquisition, generation of a simulation plan, creation of the patient breathing model, and execution of the simulation plan on the treatment delivery platform. The aim of the study is to develop a tool based on binary classification of trackable and non-trackable lung tumors for automatic selection of optimal tracking methods for patient undergoing robotic radiosurgery. **Methods:** We developed a deep learning classification model and tested 5 different network architectures (AlexNet, Densenet201, InceptionResnetV2, VGG19, and NasNetLarge) to classify lung cancer lesions from digitally reconstructed radiographs (DRR) generated from planning CTs. This study included 138 IRB-approved patients. There were 109 patients with 115 lesions that were trackable and 20 patients with 29 lesions that were non-trackable, for a total of 271 images. We kept 80% of the images for training, 10% for validation, and the remaining 10% for testing. **Results:** The training time on a single 1.6GHz Intel i5 processor and 16GB of RAM was 18 minutes for AlexNet, 103 minutes for DenseNet, 170 minutes for InceptionResnetV2, 203 minutes for VGG19, and 245 minutes for NasNetLarge. There were no false classifications. The binary classification accuracy reached 100% after training, both in the validation and the testing phase, for all 5 network architectures. **Conclusion:** For patients undergoing robotic lung radiosurgery, candidates for marker-less lung tumor tracking can be successfully identified by classifying with a deep learning model DRR images sourced from simulation CT scans.

PO-GePV-M-17

Prediction of WEPL Image in Proton Radiography Using Deep Learning

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Purpose: To predict the WEPLs from energy-resolved dose functions (ERDF) using a deep learning algorithm. **Methods:** A solid water wedge of dimension 30 cm x 10 cm x 30 cm in the path of the scanned proton beams is simulated using Geant4. The beam energy is varied between 100 MeV to 225 MeV in steps of 3 MeV and the exit dose is estimated from the simulation. This setup is used to obtain the calibration datasets (i.e, ERDF). This database is used for training a deep-learning model. The optimized model is used to predict the WEPL for box and staircase configurations (three steps stairs). **Results:** The preliminary results show that the predicted WEPLs are in good agreement with the actual WEPLs for certain WEPL ranges. From the staircase results, the average error in predicting WEPL for 120 mm is 4.5% and the average error for 170 mm and 200 mm is 1.5%. However, the predicted WEPL varies significantly from the actual values in certain ranges of WEPL, and the work to achieve better accuracy is in progress. The advantage of the deep learning technique over the conventional technique is that the time taken for the deep learning model to predict the WEPL is considerably small (~ 1 minute to generate a 200 x 600 WEPL map). **Conclusion:** Deep learning techniques offer a new avenue of analysis in the field of proton imaging especially in reducing the computation time. The technique needs to be fine-tuned with respect to the model parameters and features in order to predict the WEPLs within clinically acceptable limits.

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PO-GePV-M-18

Using CNNs to Extract Standard Structure Names While Learning Radiomic Features

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Purpose: Although there is an increased interest in Big Data for research and quality assurance in radiation oncology, the lack of use of standard nomenclature has hindered such progress. AAPM TG 263 has proposed a solution to standardize structure labels but clinical adoption for prospective or retrospective data has been sparse. Previous work in structure name standardization has included radiomic features, such as mean, kurtosis, skew, and uniformity, but the selection of features were done manually. Deep Learning has demonstrated the ability to learn such features from the data alone, and so we investigate how CNN based models can learn from imaging and dose datasets. **Methods:** We used 9,750 structures from 550 prostate patients to automatically label the following structures: Bladder, Rectum, Left Femur, Right Femur, Small Bowel, Large Bowel, PTV, and a group containing all other structures. A bounding box was centered on each structure in the dataset and a bitmap was extracted. The same bounding box was used to extract the imaging and dose data. Four CNN models were tested: structure only, structure plus image, structure plus dose, structure plus image plus dose. **Results:** Our results showed that the combination of structure, image, and dose data provided the best results with weighed F1-scores of 0.842, 0.791, 0.845, and 0.861 respectively. When tested on an external dataset of 50 patients from another institution, the F1-scores were 0.829, 0.801, 0.831, 0.857. **Conclusion:** Adding dose information to the structure bitmap and imaging data can improve the ability to correctly label delineated structures. Based on their position relative to the target, each OAR or target may follow a unique pattern of dose delivery which may be contributing to the improved results. Although very large annotated radiotherapy datasets are not yet available, CNN models can still learn radiomic features without being manually defined.

PO-GePV-M-19

CNN-Based Clinical-Dose to Low-Dose CT Simulator

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Purpose: To simulate low dose CT images from clinical dose acquisitions in order to create datasets for testing image analysis or processing algorithms. **Methods:** A dataset of 2020 paired 2D slices from 101 clinical dose (400 mA; 80 kVp) and low dose (200 mA; 80 kVp) 320 slice non-contrast brain CT scans, reconstructed using 3D iterative reconstruction, was obtained retrospectively. Rigid registration was performed between the clinical dose and low dose images to correct for patient movement. Two neural networks were separately trained on this data for, first, denoising and then, generating the necessary noise expected at lower dose. The denoising network has a U-net architecture with residual layers, which estimates the average of the distribution, denoising the images. The noise generating network is a generative adversarial network (GAN), of which the generator is a U-net with residual layers and the discriminator is a modified version of the VGG16 model. The generator has to output realistic noise patterns for low dose CT images, in which the spatial correlation and magnitude depend on the content, location, and intensity of the image. As the real input image for the discriminator, the noise pattern obtained by subtracting the denoised image from the low dose image was used. **Results:** The denoising network reduces the noise effectively, with a reduction in the pixel variance in homogeneous areas by a factor of nine. The generator of the GAN produces realistic noise patterns of low dose images, which have spatial, content and intensity dependent correlation and magnitude, and it is able to generate realistic noise even across the edge of the skull. **Conclusion:** Using the denoising and generator networks enables generating multiple low dose images from a single clinical dose image with realistic noise patterns, opening the possibility to create datasets for testing purposes.

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PO-GePV-M-20

Liver Cancer Risk Quantification Through An Artificial Neural Network Based On Personal Health Data

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Purpose: Liver cancer is the sixth most common type of cancer worldwide and is the third leading cause of cancer related mortality. Several types of cancer can form in the liver. Hepatocellular carcinoma (HCC) makes up 75%-85% of all primary liver cancers and it is a malignant disease with limited therapeutic options due to its aggressive progression. While the exact cause of liver cancer may not be known, this study models a reduction in liver cancer risk of an individual using an artificial neural network (ANN) in response to change habits/lifestyle. **Methods:** To address this challenge, an artificial neural network (ANN) was developed, trained, and tested using the health data captured in the National Health Interview Survey (NHIS) and Pancreatic, Lung, Colorectal, and Ovarian cancer (PLCO) datasets to predict liver cancer risk. The ANN was trained on 70% (training dataset) of the data using 10-fold cross-validation, while the remaining 30% was withheld for further testing (testing dataset). The testing population was also stratified into low, medium, and high risk. **Results:** The best performance of the model was observed with an area under the curve (AUC) of 0.80 and 0.81 for training and testing dataset, respectively. **Conclusion:** Our results indicate that our ANN can be used to predict liver cancer risk with changes with lifestyle. This model may provide a novel approach to identify patients at higher risk who may benefited from early diagnosis and intervention.

PO-GePV-M-21

Deep Transfer Learning for the Multi-Label Classification of COVID-19 Presentation on Thoracic CT Scans

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Purpose: To characterize COVID-19 by six presentations – ground glass opacity (GGO), consolidation, nodule, linear/reticular, crazy paving, and architectural distortion – on chest CT for COVID-19 patients. **Methods:** Transfer learning exhibits robust performance when labeled images are limited. A transfer learning approach was implemented through feature extraction using the VGG-19 deep convolutional neural network (CNN) pretrained on the ImageNet database. Features were extracted from the pooling layers of the network on CT sections from 41 patients (221 scans) serving as input. Each patient was confirmed to have COVID-19 via PCR test. Multi-label classification methods of Binary Relevance (BR), Classifier Chains (CC), and Label Powersets (LP) with support vector machines were utilized to predict the multi-label output of COVID-19 presentation to distinguish if a given COVID-19 presentation was present or not. ROC analysis was used to evaluate the classifier performance. **Results:** In the task of distinguishing COVID-19 on thoracic CT, the CC pipeline outperformed BR and LP classifications as determined by F1-score, Hamming loss, and relative AUC. Classification of GGO yielded an area under the ROC (AUC) of 0.95 ± 0.015, consolidation yielded an AUC of 0.60 ± 0.024, nodule yielded an AUC of 0.92 ± 0.024, consolidation linear/reticular an AUC of 0.64 ± 0.021, crazy paving yielded an AUC of 0.75 ± 0.025, and architectural distortion yielded an AUC of 0.76 ± 0.042. **Conclusion:** The proposed deep learning algorithm with VGG-19 network can identify and differentiate COVID-19 patterns on CT images, potentially improving prognostic evaluations and informing treatment decision-making for COVID-19 patients. Future methods include improvements to the deep network feature extraction, such as fine-tuning and ensemble networks, as well as an investigation of the clinical implications of COVID-19 CT patterns on an additional 916 cases recently acquired (1935 imaging studies).

PO-GePV-M-22

Deep Learning-based COVID-19 Pneumonia Classification Using Chest CT Images: Model Generalizability

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Purpose: Since the outbreak of the COVID-19 pandemic, worldwide efforts have focused on using artificial intelligence (AI) technologies on various medical data of COVID-19-positive patients to classify various aspects of the disease, with promising results. However, concerns have been raised over their generalizability, given heterogeneous factors in datasets. This study examines the severity of this problem by evaluating deep learning (DL) classification models trained to identify COVID-19-positive patients on 3D CT datasets from different countries. **Methods:** We collected one internal dataset at UT Southwestern (UTSW) (337 patients), and three external datasets from three different countries: 1) CC-CCII Dataset (China), 2) COVID-CTset (Iran), and 3) MosMedData (Russia). We divided all the data into 2 classes: 1) COVID-19-positive and 2) COVID-19-negative patients. We divided the data, by patients, into 72% training, 8% validation, and 20% testing. We trained nine identical DL-based classification models using various dataset combinations. **Results:** The models trained on a single dataset achieved accuracy/area under the receiver operating characteristics curve (AUC) values of 0.87/0.826 (UTSW), 0.97/0.988 (CC-CCII), and 0.86/0.873 (COVID-CTset) when evaluated on their own dataset. The models trained on multiple datasets and evaluated on a test set from one of the datasets used for training performed better. However, the performance dropped close to an AUC of 0.5 for all models when evaluated on an unseen dataset. Including MosMedData, which only contained positive labels, into the training did not help the performance on the other datasets. **Conclusion:** The models could identify COVID-19-positive patients if the testing data were in the same dataset as the training data. However, we observed poor performance when evaluating the model on an unseen dataset. Multiple factors contribute to these results, including patient demographics, pre-existing clinical conditions, and differences in image acquisition or reconstruction, causing a data shift among different study cohorts.

PO-GePV-M-24

Feasibility Study of Deep Learning Based Radiosensitivity Binary Classification Model Using Gene Expression Profiling

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Purpose: In radiotherapy, the prediction of radiosensitivity of tumors has been considered to be clinically relevant. In this prospect, the radiosensitivity prediction model was presented using deep learning in previous study, whose radiosensitivity were represented by survival fraction at 2 Gy (SF2). However, what clinically important is whether the cell is radiosensitive or radioresistant, rather than the SF2 value. This binary classification can support decision making of physician more efficiently and require fewer computational resources. With this perspective, we investigated the feasibility of the deep learning binary classification model of radiosensitive or radioresistant using gene expression profiling. **Methods:** The gene expression profiling of the National Cancer Institute-60 tumor cell line was acquired from the gene expression omnibus database. SF2 values as a radiosensitivity indicator were obtained from previous publications, which were converted to one-hot encoded radiosensitivity as 0 if SF2>0.2 (radiosensitive) and 1 if SF2>0.8 (radioresistant). In this dataset, 36 triplicated data of 12 cell lines are included. The radiosensitivity prediction model was based on convolutional neural network, which consisted of 5 convolutional layers. The six fold cross-validation approach were applied to train and validate the model. The accuracy of the model was defined as a percentage of correctly classified samples. **Results:** The classification accuracy was 97.22%. The only sample failed to classify correctly was third sample of MDAMB435 cell line, which were also failed to predict correctly in previous study. In terms of computing time, the binary model converged to the saturation point 10 times faster than the previous model for predicting the precise SF2. **Conclusion:** This study suggests that the utilization of deep learning is feasible to develop the classification model deciding whether the cell is radiosensitive or radioresistant. Since the amount of data in this study was limited, further studies with additional data would be required.

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PO-GePV-M-25

Intentional Deep Overfit Learning (IDOL): Leveraging Patient-Specific Overfitting in the Synthetic CT Generation Task

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Purpose: With the steadily increasing interest in applications of deep learning (DL)-based approaches in the radiotherapy workflow, recent years have seen continued demonstrations of the promise of state-of-the-art DL algorithms. However, there remain several unsolved challenges when considering the practical implementation of these approaches in a clinical setting. A primary challenge is the curation of a training data set of sufficient size and quality that enhances the generalizability of the trained model. In this study, we propose an alternative to the traditional combat against overfitting with the application of the Intentional Deep Overfit Learning (IDOL) framework to the synthetic CT (sCT) reconstruction task in the breast. **Methods:** Training in the IDOL framework consists of two phases: 1) conventional training with a diverse population of patient data (n = 52 patients, 100 pairs per patient) consisting of pairs of MRI and CT images for 100 epochs, and 2) overfit training with synthetically perturbed and augmented data sets (n = 50 deformed sets, 80 images per set) derived from simulation planning data of the single patient of interest for 20 additional epochs. Testing of the IDOL model was performed with setup MRI data from three treatment fractions of the patient of interest. **Results:** Compared to sCT reconstructions produced by the conventional, general model (phase 1 training only), outputs of the IDOL model are characterized by improved definition at tissue boundaries and reproduction of bony anatomy. The mean absolute error computed over 147 test images improved from 68 ± 16 HU with the general model to 22 ± 5 HU with the IDOL model. **Conclusion:** The IDOL framework, explored here in the sCT reconstruction task, represents a powerful and widely applicable departure from the conventional wisdom of combatting overfitting that is inspired by the patient-centric ethos of the era of precision medicine.

PO-GePV-M-26

A Multiple Modality Comparison of Radiomic Features for Prostate Cancer

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Purpose: Most radiomics studies performed to date were assessed in a mostly pre-treatment setting using either MRI and CT. Unlike MR and CT images that are used for diagnostic and treatment planning purposes, CBCT images are used for daily patient setup prior to radiation delivery and may provide day-to-day radiomic feature changes of the prostate. While the predictive value of multiparametric (mp)MRI in prostate cancer (pCa) treatment is well-known, the potential of imaging features extracted from CBCT is novel for patients with pCa. The purpose of this study was to determine if there is any correlation between the prostate radiomic features between MRI, CT, and CBCT and to determine the mechanism through which they correlate. **Methods:** Twenty patients receiving radiotherapy for pCa who were treated on a variety of dose fractionations as part of IRB-approved protocols were included. Forty-two radiomic features were extracted from the prostate on CT, first fraction CBCT, and mpMRI sequences, including Apparent Diffusion Coefficient, Early and Late Dynamic Contrast Enhanced, T2-weighted, and T2 Fat-Saturated MRI. First-order intensity and second order statistical radiomic features from four gray-level matrix classes were considered. The correlation between radiomic features and prostate volume was also calculated. The Spearman correlation (R) was used to investigate the associations between the radiomics features of the aforementioned imaging modalities. Statistical significance was $p < 0.05$. The threshold for well-correlated radiomic features was $R > 0.85$. **Results:** CBCT and CT radiomic features were correlated with features from 5 and 3 mpMRI, respectively. Most well-correlated radiomic features were also well-correlated with volume. After volume-normalization, these radiomic features were no longer well-correlated between different modalities. **Conclusion:** The majority of radiomic features that correlated between CT, CBCT, and mpMRI was due to the strong volume-dependence of these features. This work highlights the importance of volume-normalization when considering radiomic features between different modalities.

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PO-GePV-M-27

Group-Lasso Regularized Artificial Neural Network for Embedded Feature Selection in Radiomics Study On Pediatric Patients with Craniopharyngioma Treated with Proton Therapy

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Purpose: Conventional wrapper and filter-based feature selection methods in radiomics study analyze the correlation of features and endpoints sequentially or individually, often resulting in excluding predictive features when combining with others. In this study, we developed an embedded feature selection method for an artificial neural network (ANN) using Group-Lasso regularization to predict obesity of pediatric patients after proton therapy using magnetic resonance imaging (MRI) features. **Methods:** Eighty-four children diagnosed with craniopharyngioma (aged 1-20 years) and treated with surgery and proton therapy were included in this study. Consent forms with Institutional review board approval was obtained. The endpoints were defined as obesity in patients with the

normalized body-mass index (BMI) crossing 95% after 5-year follow-up. Imaging features were retrospectively calculated from pre-operative MRI images with T1-, T2-weighted, and FLAIR sequences using LIFEx (Orsay, France) software. Group-Lasso regularization was implemented in a 4-layer ANN developed with TensorFlow (Google, USA) for feature selection. The optimal combination of features was selected by thresholding their L2-norm of weights. Five out of 105 features were selected with the embedded method. Data augmentation was used to reduce bias of the unbalanced grouping and 5-fold cross-validation was used to reduce statistical bias of the ANN model. **Results:** The average accuracy of the testing dataset in the 5-fold cross validation was $78.3\% \pm 7.8\%$, $76.7\% \pm 9.7\%$, and $80.0\% \pm 7.7\%$ for T1, T2, and FLAIR images, respectively. The macro average area under the ROC curve (AUC) was 0.79, 0.84, and 0.92 for T1, T2, and FLAIR images, respectively. **Conclusion:** An embedded feature selection method was implemented with the Group-Lasso regularization for an ANN radiomics model. The dimension of feature space was reduced without compromising average accuracy and AUC. The prediction of obesity could be applied in a personalized treatment plan to improve the quality of post-treatment life.

PO-GePV-M-28

Technical Artifact Detection in CBCT Imaging by Binary Image Classification

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Purpose: To train a deep learning TensorFlow model on cone-beam computed tomography (CBCT) images with various technical artifacts using transfer learning and to label image quality afterward as acceptable (flawless) or non-acceptable (somehow destructed). **Methods:** A pre-trained 101-layer variant of the Residual Network (ResNet v2) was used to generate a deep neural network (DNN) model with custom CBCT images to classify image quality one of two categories. The first category was employed by 2081 image slices from various body regions without artifacts (except small distortions created by marker seeds). The second category included 400 images with three types of artifacts observed during clinical CBCT imaging: organ motion, ring artifacts, and unknown types. Reference images were not further sub-divided, e.g., by tube parameter or region of interest (ROI). **Results:** Training for the configuration above took about 45 minutes (on an i7-10750H). The ratio between training and verification dataset was 80:20, and the evaluated internal accuracy was better than 99%. Prediction of external images took less than 3s, and all technical artifacts are assigned, while undistorted images and those with marker seeds are labeled as flawless. The algorithm typically predicts a label score of 97% or higher for those images. **Conclusion:** Transfer learning was successfully applied to detect technical artifacts in CBCT imaging for an O-ring gantry-type linear accelerator. For other imaging modalities like CTs in treatment planning, CNN and algorithms can be easily adopted. Future workflows should include automated image processing to reduce potential errors for patient position matching between treatment plan geometry and daily cone-beam CT.

PO-GePV-M-29

Classification of Attention-Deficit/hyperactivity Disorder Using Multisite Resting-State Functional MRI: A Radiomics Features Analysis

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Purpose: The precise pathology of ADHD is still unknown and apart from behavioral symptoms investigated by psychiatric methods, there is no standard biological test to diagnose ADHD. In this study, we aimed to utilize resting-state functional MRI (rs-fMRI) to extract radiomics features. We hypothesized that radiomics features based on rs-fMRI could be used as a potential biomarker to distinguish ADHD patients from healthy controls. **Methods:** In this study, rs-fMRI and clinical measures from 84 subjects with ADHD and 84 healthy controls were selected from 3 sites. Using preprocessed rs-fMRI, we extracted radiomics features by the 116 regions of interest (ROIs) of automated anatomical labeling (AAL) atlas. There are 93 features of 6 radiomics feature types extracted in each ROI. Finally, a total of 32364 radiomics features were obtained including 10788 features of mean regional homogeneity (mReHo), 10788 features of mean amplitude of low-frequency fluctuation (mALFF) and 10788 features of z-transformed voxel-mirrored homotopic connectivity (zVMHC). Then the Mann-Whitney U test and Spearman correlation were used to dimension reduction. The sequential backward elimination support vector machine (SBE-SVM) algorithm was used to features selection. Last, the selected features were used to train the SVM model to estimate the classification accuracy. **Results:** After dimension reduction and features selection, there were 47 radiomics features retained. By training SVM within the training dataset based on radiomics features, we got the accuracy of 98.5% (AUC = 0.998) and the accuracy in the testing dataset is 88.2% (AUC = 0.955). The classification results were further verified by permutation testing and leave-one-site-out strategy. **Conclusion:** This study provides a novel strategy to make more fully utilize of rs-fMRI base on radiomics features, which not only could effectively predict ADHD, but also have the potential to identify biomarker in it.

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PO-GePV-M-30

Low-Dose CT Images Denoising Integrating Machine Learning and Optimization

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Purpose: Although deep learning (DL) methods have shown promising performance in low-dose CT denoising, concerns have been raised regarding image fidelity loss. To better preserve the image fidelity while maintaining the strong statistical learning ability of DL, we proposed a hybrid method to combine DL with regularized optimization and then tested its performance on low-dose lung CT images. **Methods:** Lung CT images of 42 patients were reconstructed using filtered back-projection from full-dose and corresponding low-dose projections, which were downloaded from TCIA. The patients were split into train, test, and validation sets comprising 34, 4, and 4 patients, respectively. A denoising U-Net was trained to predict full-dose images from low-dose images. Low-dose images in test sets were fed into the trained U-Net to produce baseline DL-denoised images. Recovered images using the hybrid method were obtained by minimizing an objective function that penalized the L2 distance with denoised images and the L2 discrepancy with low-dose projections. To objectively evaluate the reconstruction quality, we trained an airway segmentation network on a separate lung CT dataset. Automated airway segmentation was then performed on the full-dose, DL-denoised, and hybrid recovered images using the trained model. Using the airway segmentation on the full dose images as the ground truth, we calculated the Dice coefficient and detectability of the DL-denoised and hybrid recovered image airway masks. **Results:** On average, hybrid recovered images showed an improvement of 0.00435 in Dice coefficient and 0.0165 in detectability over DL-denoised images, indicating the superior performance of our proposed method over the naive DL method. **Conclusion:** This work demonstrated a novel hybrid-denoising method combining the classical regularized reconstruction method with DL for low-dose CT reconstruction. Using automated airway segmentation for objective assessment, we showed that the hybrid method is superior to DL alone in preserving the imaging details while suppressing the noise.

PO-GePV-M-31

Can Clinical Data Uncover Physician's Decision Making Criteria On Patient's Prescription in Radiotherapy?

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Purpose: This project's goal is to utilize machine learning techniques to detect anomalous outlier prescriptions by modeling the radiation oncologist's

prescription decision making protocol process. The data-driven approach allows us to model the process without complete knowledge of the particular protocol used and their details. **Methods:** 14 years of clinical data (01/01/2007 – 01/01/2021) were queried from MOSAIQ, which includes all patients treated in the radiation oncology department of our institution. All features related to patients' treatment information were extracted, including treatment intent, treatment techniques, treatment site, tumor stages, tumor markers, biomarkers etc. Prescription data included number of fractions and total dose. Patients were grouped by their diagnostic code for site-specific model development. Prostate group data (6482 patients) was preprocessed, missing data was imputed, models were trained and tested with a Random Forest (RF) supervised learning base model. We trained hyper-parameter (maximum depth) to avoid overfitting. Feature importance analysis was performed by fitting the data with RF regression models. In addition, statistical analysis and visualization tools such as joyplot and scatter plot were employed to assist discovering important features and to explore the data. **Results:** Initial stage analysis shows that the treatment techniques and treatment intent are among the most important features for the Prostate group and average percentage prediction error was 70.87% for the RF model in contrast with 96.36% error from a constant mean prediction. **Conclusion:** Preliminary analysis shows that the treatment techniques are the most important features for predicting number of fractions as well as predicting total dose. Tumor stages and morphology codes are the least important features. Improvement on feature selection, imputation method, and multiple hyper-parameters optimization can potentially improve the model prediction result.

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PO-GePV-M-32

Intrafractional Tumor Motion Prediction Using Surrogates on Pancreatic Cine MRgRT

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Purpose: X-ray image-guided radiotherapy (IGRT) has been a vital tool in radiation oncology to visualize and correct for inter- and intra- fractional variations during radiotherapy treatment. However, X-ray imaging lacks good soft tissue contrast, limiting direct tumor visualization. This work uses 2D cine MR images obtained on a 0.35T MRgRT system to create a pancreatic tumor position prediction model using surrogates visible on X-ray imaging. **Methods:** A retrospective dataset of cine MRI images was collected from pancreatic tumor patients treated with 0.35T MRgRT at our institution, containing a total of 56 treatment fractions and 53,759 images. In-house software was used to extract tumor position with respect to diaphragm position from cine MR images. Diaphragm positions were determined using template-based auto-segmentation and tumor positions were extracted from the MRgRT systems. Diaphragm contours were manually evaluated and adjusted. Models were developed using ridge regression and feature reduction using principal component analysis (PCA). The intra-fractional prediction model was trained on each fraction using 70% of images and tested with the remaining 30%. The inter-fractional model was trained with all fractions except 10% of image in the target fraction, and tested with the remaining 90% of images in the target fraction. Models were evaluated with mean square error (MSE) and mean absolute error (MAE). **Results:** The predicted pancreatic tumor position had an average MSE (mm²) of 0.34/0.98(X/Y), and average MAE (mm) of 0.42/0.68(X/Y), for the intra- and average MSE (mm²) of 0.90/2.1(X/Y), and average MAE (mm) of 0.66/1.00(X/Y) for the inter-fractional models, respectively. After PCA of contours the models predicted tumor position with an average MSE (mm²) of 0.37/1.1(X/Y) and average MAE (mm) of 0.45/0.74(X/Y), for the intra- and average MSE (mm²) of 2.0/6.2(X/Y), and average MAE (mm) of 0.99/1.8(X/Y) for the inter-fractional models, respectively. **Conclusion:** Intra- and inter-fraction models were developed to predict pancreatic tumor position from diaphragm position, a surrogate visible on both MR and X-ray imaging.

PO-GePV-M-33

Three-Dimensional Dose Prediction Methodology for Gamma Knife

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Purpose: To develop a machine learning-based, 3D dose prediction methodology for Gamma Knife treatment plans. **Methods:** We obtained 322 Gamma Knife (GK) treatment plans from our institution, under REB approval. To handle the sparsity, size, and granularity of GK data, several data augmentations were made. The contoured MRI and clinical dose distributions were isolated and cropped based on tumor location and scaled to a standard size through interpolation. An accompanying 3D tensor containing distance information was created for each instance of cropped data to account for tumor size. This augmented dataset for 272 patients was used to train a convolution neural network (CNN-GK) and a generative adversarial network (GAN-GK). As a baseline, the same CNN (CNN-Baseline) and GAN (GAN-Baseline) were trained on the same dataset without the GK-specific data augmentations described above. All models were used to predict the dose distribution of 50 out-of-sample patients. Predictions were compared to their clinical counterparts using the gamma criterion (4%/2mm) and conformity index. **Results:** The predictions from CNN-GK and GAN-GK were generally similar to the clinical dose distributions, with gamma passing rates of 83.1 ± 17.2% and 84.9 ± 15.3%, respectively. In contrast, the gamma passing rate of CNN-Baseline (36.2 ± 33.8%) and GAN-Baseline (40.3 ± 36.4%) were both significantly lower than their respective GK-specific models ($p < 0.001$). Additionally, for 41% of the tumours in the out-of-sample patients, both baseline models predicted dose with a gamma pass rate of 0.0%. The conformity index of CNN-GK and GAN-GK predictions differed, on average, from that of the clinical distributions by 0.092 ± 0.10 and 0.086 ± 0.11 respectively. **Conclusion:** Our methodology that uses deep learning models and GK-specific data augmentation can accurately predict 3D dose distributions for Gamma Knife treatment plans. Off-the-shelf deep learning models applied to standard GK data generally generate poor predictions.

PO-GePV-M-34

Novel Paired Intensity-Pattern Recognition Technique Based on Perfect Intensity-Geometry (PIG)

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Purpose: To present a novel paired intensity-pattern recognition technique using pointwise mutual information (PMI) and image-pair joint probability (PAB) for perfect intensity-geometry (PIG). This technique can facilitate easy pre-/post-treatment comparison for therapy monitoring. **Methods:** Mutual information (MI) has been used as an optimization metrics for the intensity-based registration, but it becomes cumbersome during the repetitive classifications. Instead of using MI, PMI-to-PAB map, which describes “the dependence of pair’s intensity pattern” vs. “sub-volume ratio (i.e. PAB)”, was calculated and optimized to approximate to a theoretical reference (PIG). The number of intensity-pattern (N_p) of a registered image pair [A, B] was resolved by repetitive classifications (k-means) in the pair’s 2D intensity histogram. The optimal N_p was determined in the recognition routine based on the expectation of “PAB/exp(-PMI)” approaching to a unity, e.g. $|1 - \text{Exp}\{\text{PAB}/\text{exp}(-\text{PMI})\}| < 0.001$. Simulated image pairs using a patient’s PTV added with regional intensity variation (to simulate “treatment response”) were tested using this novel approach against functional diffusion map (fDM) method. In addition, the sensitivity of detecting response with the MRI ADC map before and after was also tested. **Results:** The novel PIG metrics was robust and was much faster without supervised training sessions compared to the traditional MI (information gain) metrics. It resulted in the patterns of underlying intensity range representing for the same tissues, physio-pathological variants, and mismatches. In the simulation, the PIG method successfully recognized paired intensity-pattern clusters as well as the regional response. In the brain MRI ADC map test, the routine had improved sensitivity to ADC responses compared to optimal fDM (0.844 vs 0.669). Note the traditional fDM can only reach 0.398 with fixed threshold. **Conclusion:** Our novel technique using PMI and PAB for perfect intensity-geometry (PIG) offers an unsupervised learning for paired intensity-pattern recognition. This technique can facilitate easy pre- and post-treatment comparison for therapy monitoring.

PO-GePV-M-35

Machine Learning Models to Predict the Delivered Positions of Elekta Multileaf Collimator Leaves for Volumetric Modulated Arc Therapy

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Purpose: To evaluate the performance of random forest (RF) regression, support-vector regression (SVR), and extreme gradient boosting (XGBoost) regression models in predicting the delivered positions of individual multileaf collimator (MLC) leaves. **Methods:** In this study, 43 MLC log files containing 2380 control points for volumetric modulated arc therapy (VMAT) treatment plans delivered using an Elekta linear accelerator were obtained. A total of 8 planned parameters, namely gantry angle, collimator angle, X1 jaw position, X2 jaw position, leaf gap, leaf position, leaf velocity, and leaf acceleration were extracted. The RF, SVR, and XGBoost regression models were trained using 70% of the log files. The delivered leaf positions were used as the target variable for training, and a 6-fold cross validation method was used to validate and tune the model's hyperparameters. The tuned models were tested on the remaining 30% of the MLC log files. The performance of the models was evaluated using mean absolute error (MAE), root mean square error (RMSE), R^2 value, and fitted line plots showing the relationship between delivered and predicted leaf positions. **Results:** The RF model achieved a MAE of 0.427 mm, RMSE of 0.434 mm, and R^2 of 0.957. The SVR model achieved a MAE of 0.798 mm, RMSE of 0.809 mm, and R^2 of 0.849. The XGBoost model achieved a MAE of 0.214 mm, RMSE of 0.223 mm, and R^2 of 0.989. **Conclusion:** Based on these results, the XGBoost model performed better than the RF and SVR models in predicting the delivered positions of the individual leaves for VMAT plans. Although the RF model had higher prediction errors, it still outperforms the SVR model. This study indicates that XGBoost has the potential to accurately predict the delivered MLC positions for VMAT treatment plans.

PO-GePV-M-36

Probabilistic Feature Extraction, Dose Statistic Prediction and Dose Mimicking for Automated Radiation Therapy Treatment Planning

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Purpose: We propose a general framework for quantifying predictive uncertainties of dose-related quantities and leveraging this information in a dose mimicking problem in the context of automated radiation therapy treatment planning. **Methods:** A three-step pipeline, comprising feature extraction, dose statistic prediction and dose mimicking, is employed. In particular, the features are produced by a convolutional variational autoencoder and used as inputs in a previously developed nonparametric Bayesian statistical method, estimating the multivariate predictive distribution of a collection of predefined dose statistics. Specially developed objective functions are then used to construct a dose mimicking problem based on the produced distributions, creating deliverable treatment plans. **Results:** The numerical experiments are performed using a dataset of 94 retrospective treatment plans of prostate cancer patients. We show that the features extracted by the variational autoencoder captures geometric information of substantial relevance to the dose statistic prediction problem, that the estimated predictive distributions are reasonable and outperforms a benchmark method, and that the deliverable plans agree well with their clinical counterparts. **Conclusion:** We demonstrate that prediction of dose-related quantities may be extended to include uncertainty estimation and that such probabilistic information may be leveraged in a dose mimicking problem. The treatment plans produced by the proposed pipeline resemble their original counterparts well, illustrating the merits of a holistic approach to automated planning based on probabilistic modeling.

PO-GePV-M-37

The Neurosubtyping of Autism Spectrum Disorders Based on Structural MRI Data

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Purpose: Clinical and etiological heterogeneity remain major obstacles to biomarker identification in Autism spectrum disorders (ASD). Recently, neuroimaging raised new hope in our ability to identify the subtypes of ASD for further understand the biological mechanisms. The purpose of this study is to utilize a novel semi-supervised machine learning method termed HYDRA (heterogeneity through discriminative analysis) to identify neurosubtyping of ASD. **Methods:** In this study, brain structural MRI data and clinical measures from 261 subjects with ASD and 314 healthy controls were selected from 6 sites of the Autism Brain Image Data Exchange database (ABIDE). A novel semi-supervised clustering method was utilized to divide individuals with ASD into several subtypes by the regional volumetric measures of grey matter, white matter, and CSF. This method can identify true disease subtypes by removing the influence of confounding variations introduced by age, sex, site, and other factors. The ASD subtypes obtained in this study were contrasted by voxel-wise volumetric, clinical measures and whole brain functional connectivity. To assess the reproducibility of subtypes, we take a series of analyses, including permutation tests, split-sample tests and leave-one-site-out cross validation. **Results:** Two reproducible neuroanatomical subtypes were found. The ASD subtypes with difference voxel-wise volumetric patterns were revealed. Besides, the IQ and ADOS scores is significantly different, atypical brain functional connectivity patterns were observed in two subtypes. By using HYDRA to divide the ASD into 2 subtypes, the classification accuracies were significantly improved between ASD and healthy controls. **Conclusion:** The method revealed distinct neurosubtyping of ASD and build stable brain-behavior relationships which could facilitate better understanding neuroanatomical heterogeneity of this disease and potentially be used to improve clinical decision-making and optimize treatment in the future.

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PO-GePV-M-38

What's Behind Auto-Segmentation Models: An Interpretability Analysis of Calculation Logic of Segmentation Model for Brain Tumor

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Purpose: We proposed a workflow to exam the implicit calculation logic behind segmentation models (U-NET, XGBoost) to better understand the underlying algorithms. We attempted to fit two segmentation models respectively through decision trees in a predefined feature space. The highly similar decision paths generated from these two models will be extracted to discover the associations among features. **Methods:** 369 patients from the BraTS20 were used and regrouped into 3 subgroups: long-term survival (>15 months), mid-term survival (10-15 months), and short-term survival (<10 months). Each patient had four preoperative multimodal MR scans: T1, T2, Flair, T1ce. A total of 444 radiomics features (74 first order features, 224 features from Gray Level Cooccurrence Matrix and 146 features from Gray Level Run Length Matrix) on MR images of 4 sequences were extracted from 295 patients to train an ensemble learning segmentation model and a U-NET segmentation model respectively. The decision tree was used to fit above models. The decision paths for segmenting each subregion of glioma, including the non-enhancing tumor region (NET) and the edema area (ED) respectively for the long-term subgroup, were filtered out to extract the most common feature associations. **Results:** A total of 14,636 decision paths in the segmentation models between NET and ED (6,836 for NET and 7,800 for ED) were examined. There were three highly similar decision paths between two models, which include 8 features. The most popular feature associations between NET and ED were identified respectively, including AutoCorrelation_180_f, AutoCorrelation_90_f and median_t1c, and AutoCorrelation_180_f, Con-trast_180_t1c, AutoCorrelation_270_t1c, min_t2, etc. **Conclusion:** We demonstrated a novel workflow to analyze the underlying

workflow of feature based auto-segmentation models. This method allows us to interpret the decision logic of the black box model in the boundary segmenting process, and show a promising way to explore and improve the interpretability of machine learning model.

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PO-GePV-M-39

Outcome Prediction of Prostate Cancer Patients After Radiotherapy Using Machine Learning Models Developed with Extrapolation Data

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Purpose: Outcome prediction with a small number of cases has been challenging task because of biased estimation of statistical population. Although mixup is effective technique that estimates statistical population with virtual data constructed by interpolation of training data, it could be more effective by extrapolation of limited number of training data. The purpose of this study was to develop machine learning (ML) models for prediction of prostate cancer recurrence using extrapolation data. **Methods:** The ML models trained with and without extrapolation data were compared. A total of 100 patients with prostate cancer who were treated with radiotherapy were included in this study. Effective features were selected by elastic net from 48 candidate features that include 35 clinical features and 13 dose features. The dataset was divided into training and test dataset. Virtual data were constructed by linear interpolation or linear extrapolation of two data randomly selected from training dataset. In this study, two kind of regression models (multiple linear regression: MLR and artificial neural network: ANN) were used as ML models. The ML models trained with and without extrapolation data were evaluated with the area under the receiver operating characteristic curve (AUC) for the prediction of prostate cancer recurrence in test dataset. **Results:** The AUCs of both ML models were significantly improved by using extrapolation data (MLR: $p < 0.0001$, ANN: $p < 0.0001$). The AUC of MLR with and without extrapolation data was 0.730 and, 0.671, respectively. The AUC of ANN with and without extrapolation data was 0.807 and, 0.764, respectively. **Conclusion:** Our results showed that the performance of ML models for the prediction of prostate cancer recurrence was improved by using extrapolation data.

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PO-GePV-M-40

Radiomics Analysis Improves Machine Learning Models Assessing Chronic Obstructive Pulmonary Disease

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Purpose: Computed tomography (CT) imaging can be used to quantitatively investigate chronic obstructive pulmonary disease (COPD). However, existing quantitative CT (QCT) features only assess structural characteristics (e.g. volumes and counts). Texture-based radiomics analysis is an emerging quantitative technique that investigates spatial relationships between voxel intensities and may therefore provide increased sensitivity to early or subtle disease changes. We hypothesized that CT radiomics features, in combination with existing QCT features, will increase accuracy for classifying COPD. **Methods:** Spirometry and CT images were obtained from the population-based CanCOLD study. Well-established QCT features consisted of the percentage of low-attenuation-areas below -950HU and -856HU, 15th percentile of the CT-density-histogram, low-attenuation-cluster slope, CT-tissue-volume, total-airway-count, and hypothetical airway of 10mm lumen perimeter (VIDA Diagnostics). For radiomics analysis, CT images were processed using whole lung and airway segmentation, voxel resampling, and thresholding between -1000HU-to-0HU prior to extraction of grey-level-co-occurrence, grey-level-run-length, grey-level-size-zone, grey-level-distance-zone, neighborhood-grey-tone-difference, and neighboring-grey-level-dependence texture-based radiomics features. Highly correlated features were removed and a least absolute shrinkage and selection operator feature selection method was completed. 5 iterations of a nested 10-fold cross validation training was performed using a Gaussian support vector machine with hyperparameter optimization. The mean classification accuracy was used to assess performance. Multivariable linear regression models for forced-expiratory-volume-in-1-second/forced-vital-capacity (FEV₁/FVC) were constructed. **Results:** A total of 1174 participants were evaluated (No COPD: n=582; COPD: n=592). The model trained with radiomics and QCT features had a 74.4% accuracy in classifying COPD, and was significantly greater than the accuracy from the only QCT features trained model (72.5%) ($p < 0.0001$). The combined feature model also had a stronger correlation with FEV₁/FVC ($r^2 = 0.53$, $p < 0.0001$) than the only QCT features model ($r^2 = 0.47$, $p < 0.0001$). **Conclusion:** CT radiomics features provides additional and complementary information when used with QCT features for characterizing COPD. Future studies should investigate using radiomics analysis for longitudinal COPD assessment.

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PO-GePV-M-41

Automatic Lung Field Segmentation on Chest Radiographs of COVID-19 Patients to Improve Diagnostic Deep Learning

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Purpose: COVID-19 from SARS-CoV-2 targets the lungs, and thus, there is a pressing need for effective surveillance using chest radiographs (CXR). Our study aims to use a deep-learning based lung field segmentation of CXR exams to aid in the task of COVID-19 diagnosis. **Methods:** In the first stage, quadrilateral masks were generated from manually delineated anatomic-point-based regions within the CXR lung. Using an initial cohort of 631 COVID-19 positive patients, a U-Net was trained from the CXR images and masks, split into 64%/16%/20% for training/validation/testing to identify the lung field. In the second stage, the trained U-Net model was used to segment the lung region on CXRs from an additional cohort of 9,126 patients tested with RT-PCR. This dataset was split for training/validation/testing to train and evaluate a DenseNet121 in the task of differentiating COVID-19 positive and negative patients. Classification performance was evaluated using receiver operating characteristic (ROC) analysis with area under the ROC curve (AUC) as the figure of merit. Gradient-weighted class activation mappings (Grad-CAM) were generated for explainability. **Results:** On the test set, U-Net segmentation yielded a Dice coefficient of 0.96 (95% CI: 0.90, 0.98) and average Hausdorff surface distance of 14.3 mm (11.7 mm, 15.9 mm). On the second test set, the DenseNet in the positive/negative classification yielded AUC values of 0.77 (0.74, 0.80) and 0.76 (0.72, 0.76) with and without the use of the thoracic mask, respectively. While the two curves failed to show a statistically significant difference, ΔAUC of 0.01 (-0.01, 0.03) with $P = 0.16$, the Grad-CAM heatmaps showed reduced influence from outside the lung region when masks were applied to the images. **Conclusion:** Automatic lung field segmentation achieved a promising classification performance while limiting predictions from regions outside the lung. Demonstrated performance of the U-Net segmentation indicates its high potential for future COVID-19 diagnosis classifications.

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PO-GePV-M-42

A Machine Learning Classifier for Predicting Recurrence in Oropharyngeal Cancer

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Purpose: To develop a machine learning classifier integrating radiomic image features and clinical data to predict recurrence in oropharyngeal cancer. **Methods:** A dataset of patients (n=295) with oropharyngeal cancer (OPC) treated with chemoradiation at the London Regional Cancer Program was used for this study. Primary tumor and nodal volumes were contoured on pre-treatment planning CT images. The Quantitative Image Feature Engine was used to compute size, shape, first-order intensity, and second-order textural radiomic features from these volumes of interest. The dataset was split into independent training (n=206) and testing (n=89) datasets. Feature selection was applied to select the optimal features to predict recurrence and a support vector machine (SVM) classifier was built using the selected features on the training dataset. The SVM's performance was assessed in the testing dataset based on the metric of the AUC. A baseline model comprised of only clinical features was compared to radiomics. **Results:** A total of 63 patients (21%) developed a recurrence. Through feature selection, 20 predictive features were selected. This included one first-order intensity feature, seven textural, nine shape, and three clinical features (HPV status, smoking status, nodal stage), among both the primary tumor and nodal volumes. The SVM model achieved an AUC on the testing dataset of 0.66 [95% CI: 0.53-0.77], in comparison to an AUC of 0.49 [95% CI: 0.36-0.62] for the model comprised of clinical features alone (p=0.06). **Conclusion:** Radiomic features augmented traditional clinical features for predicting recurrence in our relatively small testing dataset. To further improve predictive performance, we plan to implement additional clinical data, as well as investigate other machine learning models. Once refined and validated, this model could assist in identifying patients at a higher risk of recurrence, who may benefit from more personalized treatment options.

PO-GePV-M-43

CERRx: The Extended CERR Platform Integrated with Open Source Image Informatics Ecosystem

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Purpose: CERR, a widely used computational platform for dosimetry and image analysis including a research friendly data structure, was developed in Matlab considering the ease of development and wide use by physicists and computer scientists. This work aims to make CERR independent of Matlab license and GNU Octave-compatible. Python has recently emerged as the preferred choice for building scientific computing tools due to its open source nature, user-friendly syntax and ability to serve as a "glue language". Commonly used image informatics tools such as ITK, VTK, 3DSlicer provide Python APIs to allow simplified access to their C++ library. This work enables CERR functions to be accessible in Python, thus enhancing its utility in image informatics. **Methods:** CERR was made compatible with GNU Octave to support batch computation as well as graphical user interface. Octave community-maintained packages for image and statistical analysis were used as replacements for Matlab's built-in functions and toolboxes. Oct2py package in Python was used to call CERR functions from Python. This allows CERR to be used in JupyterLab with Octave-kernel and exchanging the results of CERR computations with a variety of programming languages. **Results:** Octave-compatible CERR codebase including JupyterLab notebooks demonstrating deep learning image segmentation and feature computation are distributed as open source, GPL copyright software at <https://www.github.com/cerr/CERR>. This work has resulted in expanding CERR's utility for medical image informatics including (1) accessing CERR data structure from XNAT via JupyterLab notebooks, (2) Using CERR functionality in 3DSlicer extensions, (3) Containerizing CERR's pre and post processing for deep learning image segmentation pipeline in a license-free environment, (4) Using novel features for dosimetry, radiomics and outcomes prediction models in other open-source software. **Conclusion:** This work makes CERR a truly open-source software platform that can be easily plugged into other components of the image informatics ecosystem.

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PO-GePV-M-44

Classification of Anatomic Structures by CT-Based Radiomics for Head-Neck Radiotherapy

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Purpose: To show that CT images' radiomics features can classify anatomic structures in the head and neck. **Methods:** We obtained 26 sets of CT image data for patients treated at a single institution for head and neck cancers using the IMRT technique. We calculated 174 radiomics feature values by the SIBEX software, which complies with the Image Biomarker Standardization Initiatives (IBSI) protocol. The radiomics calculations were done using three processing parameters: FBN, FBS with 25 and 10 bin widths. For this study, we chose several representative anatomical structures, i.e., 27 gross tumor volumes (GTV), a total of 47 right and left parotids, 25 mandibles, one brainstem, two eyes, and two sub-mandibles, which were segmented by attending radiation oncologists for treatment planning. First, we applied the principal component analysis (PCA) method to the feature values. Using the first two principal components, we studied if we can classify the anatomical structures by the k-means clustering method. Furthermore, the quantitative differences of 174 radiomics features were visualized by plotting heatmaps and correlation plots of all feature values among the structures. **Results:** The heatmaps of 174 normalized feature values of mandible showed a clear difference from GTV and parotids. The difference between GTV and parotids were also observed. The correlation plots showed the differences in feature values among GTV, parotid, and mandible. The first two principal components' two-dimensional plot indicated the five anatomical structures in the figure's specific locations. The radiomics features calculated with FBS=10 showed the clearest classification of GTV, parotid, and mandible among three different feature calculation methods. The clustering method could successfully classify those three structures. **Conclusion:** We could classify head-neck tumors, parotids, and mandibles by radiomics features. The results imply that the radiomics can decipher the differences of biological functions, microenvironment, and cellular structures from CT images.

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Prediction of HPV Status in the Head and Neck Cancer Patient Using CT Derived Radiomic Features: The Impact of CT Scanner

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Purpose: In this study, we assessed the impact of scanner choice on the computed tomography (CT)-derived radiomic features to predict association of oropharyngeal cancer (OPC) with human papillomavirus (HPV). **Methods:** A total of 1,294 OPC patients with known HPV status were retrospectively included in this study. In this patient cohort, CT images were acquired with two different scanner types. We extract a total of 1,874 IBSI (International Biomarker Standard Initiative) compliant radiomic features from the primary gross tumor volume (pGTV) from each patient. Feature selection has been done by Maximum Relevance Minimum Redundancy Ensemble (mRMRE)3 technique. The dataset was subsequently stratified by CT scanner type (Toshiba scanner, GE scanner, and Mix) and splitted into train (adjusted for class imbalance) and test sets with the proportion of 75/25. Next, a random forest classifier was trained on the following configurations: (1)Toshiba-Toshiba, (2)GE-GE, (3)Toshiba-GE, (4)GE-Toshiba, (5)Mix-Mix, (6)Toshiba-Mix, (7)GE-Mix, (8)Mix-Toshiba and

(9)Mix-GE. **Results:** The highest and lowest mean AUC (area under the curve) values were 0.79 (p-value<0.001) and 0.70 (p-value<0.001) and obtained with Toshiba-Mix and Toshiba-GE respectively (Figure 1). For models trained on one scanner type, the highest and lowest result obtained when they were tested on Mix sample (i.e GE-Mix [0.75,p-value<0.001], Toshiba-Mix [0.79, p-value<0.001]) and other scanner types (i.e GE-Toshiba [0.73,p-value < 0.001], Toshiba-GE [0.70,p-value < 0.001]) respectively. We identified that there is bias in the results in favor of one scanner type (Toshiba). Models either trained or tested on the Toshiba scanner resulted in higher predictive value compared to other scanners in all the same configurations. **Conclusion:** In this study, scanner type affects the prediction accuracy of the HPV status using hand-engineered radiomics features. The optimal prediction accuracy was achieved when the training set included only one specific type of scanner which reflects a bias in radiomic features towards the scanner type.

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PO-GePV-M-46

Repeatability and Reproducibility of Radiomics in Magnetic Resonance Fingerprinting of the Brain

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Purpose: Magnetic Resonance Fingerprinting (MRF) emerged as a quantitative MRI modality allowing for simultaneous measurement of multiple tissue properties in a single, time-efficient acquisition and offers enormous potential to improve diagnosis and therapeutic efficacy. Owing largely to the novelty of this technique, the feasibility of radiomics for quantitative measurements obtained using MRF has been little studied. This study aimed to assess the repeatability and reproducibility of radiomics in MRF-based relaxation time mapping of the brain. **Methods:** The imaging cohort included five healthy volunteers, each with the brain scanned on four MRI systems (two at 3.0T and two at 1.5T, all from a single vendor) located at two sites. MRF data was obtained using 2D steady-state free precession with a spiral readout trajectory. T1 and T2 maps were determined through voxel-wise pattern matching between the signal evolution and simulated dictionary. After registration of MRF-derived T1 and T2 maps to an anatomical atlas, 40 radiomics metrics related to gray-level run length, co-occurrence, zone size, and neighborhood difference were evaluated for each of the resultant parametric maps within one VOI located in the parietal lobe featuring strong spatial heterogeneity. Radiomics feature test/retest repeatability and inter-site reproducibility were measured with within-subject variability (WSV) and coefficient-of-variation (CV), respectively. **Results:** The test/retest repeatability of the studied radiomics features varied with mean±SD WSVs of 7.7%±12.1% at 1.5T and 8.2%±12.7% at 3.0T on T1 maps while varying with mean±SD WSVs of 7.9%±11.2% at 1.5T and 10.9%±13.4% at 3.0T on T2 maps. As to the inter-site reproducibility, CV of the studied radiomics features varied with mean±SDs of 7.1%±9.2% at 1.5T and 9.1%±11.9% at 3.0T on T1 maps while varying with mean±SDs of 12.0%±13.1% at 1.5T and 15.8%±18.4% at 3.0T on T2 maps. **Conclusion:** Compared to weighted images MRF-derived parametric maps may allow for more repeatable and reproducible radiomics analysis.

PO-GePV-M-47

Robustness and Performance of Radiomic Features in Diagnosing Cystic Renal Masses

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Purpose: Asses the performance of a machine learning algorithm to classify benign from potentially malignant cystic renal masses (CRM) presented in computerized tomography (CT) images based on the radiomics analyses. Evaluate the classification performance of the 1st order features alone in comparison to the inclusion of the higher order features. Investigate the effect of inter-reader variability in delineating the region of interest (ROI) for the same classification task. **Methods:** 230 region of interests (ROIs) were independently delineated by two radiologists. Through a combination of random fluctuations, dilation, and erosion operations over the original ROIs, we generated four additional sets of synthetic ROIs, which aimed to represent the inter-reader variability in a realistic manner. We evaluated the degree of variability among the ROIs visually and quantitatively based on the dice coefficient measurements. We applied a 10-fold stratified cross-validation (CV) to train and test the performance of the random forest model for the classification of CRMs. Each fold included the selection of the robust features based on ICC calculated over the 2 original and 4 synthetic sets of ROIs; removal of the highly correlated features; grid-search to tune the model training parameters; and fitting and testing the model. **Results:** The mean area under curve (AUC), sensitivity, specificity, positive predictive value, and negative predictive value calculated over the six ROIs were 0.87, 0.82, 0.90, 0.85, and 0.93, respectively, for the ~20 robust (ICC > 0.85) and uncorrelated (Pearson< 0.85) features, where 5 of them were 1st-order features. Similar results were obtained using 1st-order features only. **Conclusion:** We determined the usage of the 1st-order features alone is sufficient for the classification of cystic renal masses and the inclusion of higher order features does not necessarily improve the performance. We also determined the degree of variability introduced by the delineation process in the prediction outcome.

PO-GePV-M-48

Understanding Robustness of Deep Neural Network: A Comprehensive Robustness Analysis for a Deep Learning-Based Lung-Nodule Classification Model of CT Images with Respect to Image Noise

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Purpose: It was found that robustness of deep neural networks (DNNs) is often a concern and the reason for DNNs being robust or not is still unclear. In this work, we performed comprehensive numerical studies to get some insights on robustness behaviors of DNNs using lung nodule classification based on CT images as a testbed. **Methods:** We trained DNNs for lung nodule classification based on CT images. We added to the input of the trained DNNs realistic CT noise at 100mAs generated under a real noise power-spectrum and monitor if the output is affected. We investigated the correlation between robustness and network architecture by systematically studying a class of 16 DNNs with different numbers of layers and widths. To understand the unrobustness behavior, we visualized the extracted low-dimensional representations of the input by DNNs with/without noise added, and characterized the robustness via the distance of the representations to the decision boundary determined by the DNN in this low-dimension representation space. **Results:** The networks were trained to achieve Area Under the Curve of 0.89-0.93 in independent testing data. CT noise was able to affect the prediction results of all the DNNs models with up to 21.4% predicted labels altered by noise. We did not observe a trend between robustness and network architectures. By visualizing the input data in the representation space, the unrobustness behavior was ascribed to the sometimes large perturbations to the sample representations by noise, as well as the unfavorable placement of the decision boundary formed in the training process. **Conclusion:** Robustness concern existed in the class of DNNs studied here for the classification problem of interest. The reasons for not being robust were identified, which may help developing training strategies to improve model robustness.

PO-GePV-M-49

The Impact of the COVID-19 Pandemic on Cancer Care in Radiation Oncology

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Purpose: Current studies evaluating the impact of COVID-19 on Radiation Oncology (RO) departments describe alterations in practice patterns, assess acute effects, and offer management strategies. However, there is a lack of data documenting practice alterations. We aimed to assess the early impact of the first two waves of COVID-19 on RO practice. **Methods:** Interrupted time series analyses were conducted using electronic health record data from a major RO department from January 2018 through February 2021 (164-weeks). Segmented regression models were used to estimate the immediate impact of our state's COVID-19 Wave1 and Wave2 on weekly number of CT simulations (CT) and physics initial chart reviews (PCR), and percentage of patients receiving treatments hypofractionated at javascript:void(0); 4 Gy per fraction (g4Gy). State-level COVID-19 data were used to define the start and end of Wave1 as well as the start of Wave2. **Results:** Both CT and PCR presented a sudden drop at the start of Wave1 relative to the same time period in previous years. CT dropped by 22% (95% CI: -49.7%,-3.4%) and PCR dropped by 17.9% (95% CI:-36.0%,-3.8%). In contrast, there was a sudden 11.6% increase in patients receiving treatments g4Gy. Immediately after Wave1, CT and PCR increased by 23% and 27%, respectively. Weekly rates of CT, PCR, and g4Gy at the start of Wave2 did not deviate significantly from expected pre-pandemic rates. Wave2 is ongoing but preliminary data suggest decreasing CT and PC. Short post-pandemic within and between wave follow-up limit statistical power to detect a slope change among these points. **Conclusion:** Despite RO facilities remaining fully operational during the pandemic, large fluctuations in patient volumes and workload were observed. Contributing factors included reduced cancer screenings and elective surgery, internal policies to alter treatment regimens, and patients declining treatment. Our findings may help inform preparations for future novel aberrations in workflow.

PO-GePV-M-50

Predicting Overall Survival of Patients with Glioma Via A Novel Framework Consisting of Two-Phase Feature Selection and Fused Regression Forest
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Purpose: Accurate survival prediction may assist clinic-decision-making for personalized treatment for glioma. We aimed to develop a novel framework, consisting of a two-phase feature selection and a fused regression forest model, to achieve robust survival prediction of patients with glioma while reducing the computational cost. **Methods:** Brats19 211 high-grade patients with glioma were included. The patients were stratified into three subgroups according to the reported survival intervals: long-term (>15 months), mid-term (10-15 months), and short-term (<10 months). Each patient had four preoperative multimodal MR scans (T1, T1ce, T2 and Flair). Each tumor was further divided into three subregions: non-enhancing tumor (NET), enhancing tumor (ET) and peritumoral edema (ED). Instead of extraction of all features from all four MRIs, we first extracted a few robust features (18 first-order and 22 geometric features) from each subregion on the MR sequence that is most relevant (or optimized out of 64 combinations) for each subregion using a clustering evaluation method. Additional 1632 high-order texture features were then extracted from their corresponding MRI sequences. Recursive feature elimination algorithm was performed to generated 3 different feature sets for three subgroups, respectively. Each feature set was used to train an independent random forest (RF) model for each patient subgroup. The final prediction model was formed by fused the RF models. **Results:** The corresponding relationship that is most conducive to feature extraction between each tumor subregion and MRI sequence were determined. The optimized imaging feature sets were determined for patient survival subgroups, respectively. The proposed framework achieved an Area Under the ROC Curve (AUC) of 0.722 with least mean square error using a 10-Fold cross-validation. **Conclusion:** We developed a novel framework for tumor subregional based treatment response prediction on multiparametric MRIs. The proposed framework effectively identified an optimized feature set and achieved superior survival prediction for patients with glioma.

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PO-GePV-M-51

A Multichannel Feature-Based Approach for Longitudinal Lung CT Registration in the Presence of Radiation Induced Lung Damage

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Purpose: Tracking and quantifying local tissue changes in the lungs is imperative in furthering the study of radiation-induced lung damage (RILD). Traditional intensity-based registration approaches fail in this task due to dramatic geometric changes between timepoints. This work aims to successfully register longitudinal pre- and post-radiotherapy (RT) lung CT scans with considerable radiological changes due to RILD. This is achieved by extracting and registering only consistent anatomical features (lung boundaries, main airways, vessels). **Methods:** Fifteen pre-RT and 12-month post-RT CT pairs from a chemoradiation trial for lung cancer were available, all with varying degrees of RILD, ranging from mild parenchymal change to extensive consolidation and collapse. For each CT, signed distance transforms from delineations of the lungs and main airways were generated and the Frangi vesselness was calculated. These feature maps were concatenated into multi-channel images and diffeomorphic multichannel registration was performed for each image pair. For the evaluation, the pre- and post- registration landmark distance was calculated for all patients at a mean (range) of 44 (38-65) manually identified landmark pairs. Traditional intensity-based registrations were also performed for all patient datasets and evaluated in the same way as a comparison. **Results:** The mean (standard deviation) distance for all datasets decreased from 15.95 (8.09) mm pre-registration to 4.56 (5.70) mm post-registration. Qualitative improvements in image alignment were observed for all patient datasets. In comparison, the post-registration error of the traditional intensity-based registrations was higher, at 7.90 (8.97) mm and registrations of patients with presence of atelectasis and extensive consolidation failed. **Conclusion:** We have demonstrated that our novel registration method can successfully align 12-month follow up scans from RILD patients in the presence of large anatomical changes such as consolidation and atelectasis, outperforming the classical registration approach both quantitatively and through detailed visual inspection.

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Explore Potential Predictive Value of Microbiota for the Efficacy of Breast Cancer Radiotherapy

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Purpose: A growing number of studies suggest that human microbiome plays a significant role in cancer development and treatment. However, little is known about how the microbiome influences radiotherapy efficacy. The purpose of this study is to mine existing data to explore if microbiome plays any role in affecting outcome of radiotherapy of breast cancer. **Methods:** The Cancer Genome Atlas (TCGA), sponsored by National Cancer Institute and the National Human Genome Research Institute, consists of comprehensive clinical and genomic sequencing data with more than 11,000 human tumors across 33 different cancer types. In this study, we focused on a subset of 780 subjects with invasive ductal carcinoma. A total of 390 (50%) subjects received radiotherapy. We mined the genomic sequencing data for the presence of microbiome within both radiotherapy and non-treated groups and compared their presence among the

surviving and deceased groups. Our goal was to identify microbiota that may influence the survival of patients after radiotherapy. **Results:** Among radiotherapy patients, a total of 95 microorganisms were found to be significantly different ($p < 0.05$, t-test) between living and deceased groups. Among these, the presence of 30 microbiota were elevated in the deceased group, whereas 65 were elevated in the living group. Using criteria from Benjamini-Hochberg procedure to decrease false positive rate, we found marine bacterium *Maribius* was statistically significantly correlated with better survival in the radiotherapy group, with mean copy numbers 3 times higher in the surviving patients (0.72 ± 1.12) than those in the deceased (0.24 ± 0.64). In contrast, *Maribius* concentration level was not statistically different between deceased and living in the non-radiotherapy group. *Maribius* is a relatively novel marine bacterium discovered in 2007 which belongs to the Rhodobacteraceae family. **Conclusion:** Microbiome characterization may present opportunities to discover novel prognostic or predictive markers for radiotherapy outcome.

PO-GePV-M-53

Multi-Modality (PET/CT) Radiomics Based Recurrence Prediction in Head and Neck Cancers Prior to Radiotherapy

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Purpose: Radiomics machine learning models are a way to analyze existing imaging data to derive clinically relevant information that can be applied to a customized treatment paradigm. This study investigates the use of multi-modality (PET/CT) radiomics based machine learning to predict recurrence in head and neck cancer patients. **Methods:** Head and neck contrast enhanced radiation CT simulation scans, PET scans, radiation clinical target volumes (CTV), and clinical outcomes of 201 patients were obtained from The Cancer Imaging Archive (TCIA). Pyradiomics was used to extract intensity, shape, and texture based radiomics features (110) from CT and PET scans separately for all patients. Of the 220 (PET+CT) features extracted, the top 7 performing features were obtained by sequential feature selection in order to form a model for recurrence prediction. Multi-modality and single modality manual feature selection models, based on individual feature AUROC values, were run on the dataset for comparison. **Results:** The most accurate model was a multi-modality (PET/CT) model resulting from a combination of 5 PET and 2 CT features selected by backward sequential feature selection with a logistic regression estimator and five-fold cross validation. This model achieved the largest AUROC value (0.7895) compared to multi-modality (0.7701) and single modality (0.7364-CT, 0.7355-PET) manual feature selection models. **Conclusion:** Multi-modality radiomics results in more accurate machine learning models for recurrence prediction in head and neck cancer than single modality models. Multi-modality radiomics based recurrence prediction has the potential to improve the clinical outcome of head and neck cancer patients by applying the recurrence risk to pre-treatment management and surveillance patterns.

PO-GePV-M-54

Early Treatment Response Prediction Via Longitudinal CBCT Analysis for Locally Advanced Rectal Cancer

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Purpose: Early prediction of tumor response is the key for plan adaption towards individualized treatment management. We aimed to investigate the value of longitudinal radiomics analysis of daily CBCT images for earlier prediction of neoadjuvant chemoradiation treatment (nCRT) response in patients with locally advanced rectal cancer (LARC). **Methods:** A subset of 86 LARC patients who underwent nCRT with 50.4 Gy in 28 fractions was analyzed retrospectively. Clinical response scores according to Tumor Regression Grade (TRG), assessed from post-operative pathology, were collected for all patients. Among 19 cases studied, 10 patients achieved pathological complete response (pCR), 9 patients achieved partial or no response (non-pCR). Daily CBCTs were imported to Varian Ethos emulator, where segmentation of the gross tumor volume (GTV) and organ-at-risks (OARs) were automatically generated. The delivered doses based on daily patient anatomy were calculated from the CBCT of the day on Varian Ethos emulator. For each fraction, we extracted 1304 imaging features from the GTV on CBCT using PyRadiomics package. Highly correlated imaging features were eliminated according to Spearman's rank coefficient. Longitudinal correlations between the selected imaging features, dosimetric parameters and the pathological response were established using linear regression models. **Results:** Among 252 CBCTs assessed, the variations of the GTV and PTV coverage ranged 2.8% and 7.2% respectively over the course of the treatment. Dosimetric variations showed no clear relationships between pCR vs non-pCR patients. Correlations were observed using a small subset of imaging features and early treatment response detection as early as the 8th fraction. **Conclusion:** Early prediction of treatment response is possible via systematical analyses of CBCT radiomics over the treatment course. Quantitative imaging features could be used as biomarkers to predict treatment response for LARC during the treatment course. Varian Ethos emulator offers an integrated platform to streamline the process.

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PO-GePV-M-55

Assessment of Magneto-Plasmonic Nanoparticles as a Theragnostic for MRI-Based Photothermal Therapy: A Numerical 3-D Modeling Study in Mice

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Purpose: to present a three-dimensional photothermal therapy, finite element method based model in mice. which is developed to estimate temperature elevations of tumor due to notable optical absorption of magneto-plasmonic Nano particles imbedded in the tumor bearing mice. **Methods:** In our modelling, mice (male BALB/c, 18g weight) Geometry was acquired with Magnetic Resonance Imaging (MRI) and used slicer software for segment nanoparticle distribution. the wavelength of irradiated laser was 808nm and the size of nanoparticles was 40-70 nm. **Results:** the temperature elevation for 5 min was about 51 Celsius degree which is very promising result for this dual modality nanoparticle. **Conclusion:** Given this knowledge, controlling these parameters in photothermal treatment playing a key role:1. 3D in vivo visualization help us to selective targeting,2. The intensity of irradiated laser,3. The importance of aspect ratio 4.Correlation between nanoparticles volume fraction Overall, we found a good agreement between the experimental data and numerical modelling, obviously these agreements open new doors for further research in photo thermal therapy and enhanced models.

PO-GePV-M-56

Mixed Reality Visualization of Radiation Dose in Interventional Radiology

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Purpose: For interventional radiology, dose management has persisted as a crucially important issue to reduce radiation risks to patients and medical staff. This study examined the concept and feasibility of a real-time dose visualization system using mixed reality technology for dose management in interventional radiology. **Methods:** An earlier reported Monte-Carlo-based real-time dose estimation system for interventional radiology was adopted for our system. We also developed a system of acquiring fluoroscopic conditions to input them into the Monte Carlo system. Then, we combined the Monte Carlo system with the wearable device for three-dimensional holographic visualization. The estimated doses were transferred sequentially to the device. The patient's dose distribution

was then projected on the patient body. The visualization system also has a mechanism to detect one's position in a room to estimate the user's exposure dose based on the positional information. The dose was visualized continuously in front of the user. **Results:** An end-to-end system test was performed using a human phantom. The acquisition system accurately recognized conditions that were necessary for real-time dose estimation. The dose hologram represented the patient dose. The user dose was changed correctly depending on conditions and positions. **Conclusion:** Mixed reality dose visualization is expected to improve exposure dose management for patients and physicians by exhibiting invisible radiation exposure in real space.

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PO-GePV-M-57

Use of Crowd Innovation to Toward Development of An Artificial Intelligence Based Solution for Pancreatic Tumor Management

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Purpose: Success rates in predicting pancreatic resectability remain poor, as success is dependent on tumor size, shape, and location relative to surrounding vasculature. Pre-surgery image analysis is highly surgeon dependent, and quantitative methods remain uncommon. In this work, we determine the efficacy of using a non-traditional approach known as crowd innovation, to rapidly develop a suite of automated pancreatic tumor segmentation algorithms, for use in a pancreatic cancer management solution. **Methods:** We conducted a 10-week, prize based, competition open to Topcoder, Inc. community contestants (total prize money of \$30,000). Contestants were provided a training dataset containing 120 CT scans and contour data for pancreatic tumors and surrounding vessels (superior mesenteric artery, celiac axis/common hepatic artery, and superior mesenteric vein/portal vein), and competed in building algorithms to auto-segment a validation dataset containing 60 CT scans with no contours. An F-score metric was used to rank the contestants' algorithms using a separate, unrevealed dataset of 62 patients. F-score calculations included both the tumor and vessel contours, but with vessels weighted 7-times lower than the tumor. Contestants were offered a \$3,000 prize bonus for algorithms exceeding an F-score of 0.7 to encourage quality algorithms beyond simply winning prize money. **Results:** 167 contestants registered for the competition, across 34 countries, and submitted a total of 337 algorithms. The winning solution (average F-score=0.72) implemented a 3D-UNet model in a 3D semantic segmentation approach, which was found to be accurate in tumor and vessel delineation. All other winning algorithms used 2D model approaches. Top 5 algorithm performance was strongly correlated by patient. In 82% of patient cases, all top 5 algorithms succeeded in achieving an F-score >0.6. **Conclusion:** The use of crowd innovation for healthcare challenges has seen limited use, but offers great potential for rapid development of high quality segmentation algorithms with reasonable financial commitments.

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PO-GePV-M-58

A Navigation System for Real-Time Monitoring PDT Delivery

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Purpose: Photodynamic therapy (PDT) is used intraoperatively to treat malignant pleural mesothelioma. To achieve efficiency of PDT, it is crucial to deliver light dose uniformly. Currently, 8 light detectors are placed inside the pleural cavity to monitor the light fluence. A navigation system is developed to provide real-time information of the light dose delivery during the PDT treatment and improve the uniformity of light fluence further. **Methods:** An infrared (IR) tracking camera is used to track passive reflective markers on various wands, including treatment, contour and reference wands. Each wand can be identified automatically via its unique configuration. The newly developed software allows user communication with the camera. It also contains a user interface which visualizes the real-time feedback. The data presented is stored and can be exported for determination of the cavity geometry and light fluence calculation. These functions are integrated into the newly developed software. The calculated light fluence is also compared with data collected via the 8 detectors. **Results:** For application on a chest phantom, it showed that the light distribution was more uniform with the real-time feedback provided by the navigation system, compared to the distribution with detectors only. Furthermore, the navigation system allows 2D light fluence collection throughout the whole phantom rather than just the selected points. Light fluence distribution can be calculated based on that for further analysis. **Conclusion:** The navigation system is compatible with the latest computer operation system so that it can utilize the up-to-date hardware to speed up data processing and enable the system to provide real-time feedback. The system has been used successfully on chest phantom for the light source tracking and real-time display of 2D light fluence. It is possible to apply the system onto pleural PDT patient treatment for the purpose of a more uniform light dose delivery.

PO-GePV-M-59

A Clinical Evaluation of An Open-Source Toolkit for TG-142 Based Monthly Imaging Quality Assurance Via Comparison with Commercial Software

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Purpose: To assess and evaluate the quantitative differences of imaging parameters for TG-142 monthly quality assurance (QA) of MV- EPID (electron portal imaging device) and kV-CBCT (cone beam computed tomography) image post-processing utilizing an open-source toolkit Pylinac (PL) in comparison with the commercially available software PIPSpro (PP). **Methods:** The QC3 MV imaging phantom (Standard Imaging) and Catphan 504 CT phantom (The Phantom Lab) were imaged upon multiple Varian linear accelerators as specified by the manufacturer. Modular transfer functions (MTF) were measured using PL and PP respectively. The Catphan phantom was imaged via half-fan CBCT and analyzed in PL and PP to evaluate Hounsfield Unit (HU) uniformity, HU linearity, and spatial resolution. The results of each software system were then compared against each other and against the baseline previously established at our institution. **Results:** For MV imaging the MTF value differences obtained by PL were found to be 14 % higher than those established in PP for all the frequencies considered. PL based kV-CBCT testing revealed HU uniformity values greater by ~ 9% for all 5 regions of interests. For HU linearity, a difference of < 2% was found, both PL/PI reporting HU greater than standard values. kV-CBCT MTF comparisons demonstrate that PL values differ by < 2% for frequencies < 0.3 lp/mm, deviating more than 5% beyond that. **Conclusion:** PL and PP measured imaging parameters demonstrated similar results. PL can allow for quick and reliable estimation of the TG-142 related image post-processing parameters. The presented results warrant further investigation into other systems and phantom combinations to fully incorporate the PL in a clinical environment. Nevertheless, the use of the software to establish and monitor baseline conditions as outlined by TG142 and manufacturers maybe a cost-effective solution for further automation in the field.

PO-GePV-M-61

Range Modulated Proton Radiography Using High Energy Therapeutic Proton Scanning Pencil Beams

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Purpose: To investigate visibility of range modulated proton radiography using scanning pencil beam proton from high energy MEVION proton therapy

machine. The image quality including uniformity, linearity, contrast and spatial resolutions of the proton radiographs obtained were evaluated quantitatively. **Methods:** High energy proton beams from the scanning pencil beam MEVION proton therapy machines were used to obtain radiographs of different phantoms. These phantoms include LEEDS which is used for quality assurance of diagnostic KV photon beams. The proton radiographs of pelvis and head phantoms were acquired to evaluate visibility of proton radiography on the scale of patients. A phosphorus scintillation detector was used to acquire the proton radiographs. The effects of different beam parameters such as dose, depth, modulation, and separation between the detector and the object being imaged on the image quality of proton radiographs were evaluated quantitatively. **Results:** The contrast and spatial resolutions of the proton radiographs were decreasing with increasing depth of the testing objects in phantom, increasing spot size, increasing separation between the spot sizes and increasing range modulation. Both resolutions of the proton radiographs were affected by the proton beam scattering in the medium where it was degraded with increasing depth in phantom and increasing range modulation. Proton radiographic images showed high contrast of the soft tissue features in the beam view, while the intensity of photon radiographic images showed mainly high contrast of bone anatomy. The proton radiographs were affected by head and pelvis phantom thickness irregularities where the water equivalent depths across the image were largely variable. **Conclusion:** Proton radiography has a potential for image guided proton therapy as an alternative to photon radiography using attachable diagnostic kV photon beam systems. The use of range modulation of pencil beam scanning proton therapy system provides an approach to account for variations in patient thickness.

PO-GePV-M-62

Image Artifacts from High-Z Materials in Proton Radiography Using High Energy 250 MeV Therapeutic Proton Beam

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Purpose: To investigate image artifacts induced by high-z materials in proton radiographs obtained by high energy beams from a proton therapy machine. **Methods:** A high energy 250-MeV beam from the MEVION proton therapy system was used to irradiate a phantom that included different materials with low, intermediate and high z-composition and different shapes. The image quality of the proton radiographs was evaluated with the LEEDS phantom that is used usually for diagnostic kV-photon radiographs. The MEVION proton therapy machine provided open field beams of 25 mm diameter and depth ranges 5-32 cm. The proton images of different phantoms were captured with high spatial resolution and sensitivity phosphorus plates that are usually used in computed radiography. The proton radiographic images were captured at different depths of solid water 1-34 cm and separation between the phantom and CR plates downstream the proton beam. **Results:** The intensities in the proton radiographs showed strong interference patterns around high-z materials. The interference patterns depended on the composition of the density, position, shape and depth of the high-z objects in phantom. The image intensity varied significantly between the instructive and destructive interference patterns up to nearly 10% which resulted from the interference of the primary and scattered protons in the beam. The interference patterns varied with depth in phantom and position of the high-z materials in the beam path. These scatter patterns occur upstream with the energetic protons and in the Bragg peak dose deposition region. These interference patterns affected the imaged quality where they degraded the spatial and contrast resolutions of the proton radiographs. **Conclusion:** High-z materials induced strong image artifacts in proton radiographs acquired from high energy therapeutic proton beams. The intensity patterns resulting the interference of energetic primary and scattered protons affects the image quality of proton radiographs particularly spatial and contrast resolution.

PO-GePV-M-63

The Effects of Compton Camera Data Acquisition Timing and Readout On Prompt Gamma Images Acquired at Clinical Proton Beam Dose Rates

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Purpose: The purpose of this study was to determine how the characteristics of the data-acquisition (DAQ) electronics of a Compton camera (CC) affect the quality of the recorded prompt gamma (PG) interaction data and the reconstructed PG images, during clinical proton beam delivery. **Methods:** We used the Monte-Carlo-plus-Detector-Effect (MCDE) model to simulate the delivery of a 150 MeV clinical proton pencil beam to a tissue-equivalent plastic phantom at a low dose rate of 1.0×10^3 p+/sec, representing a rate at which only one gamma is estimated to interact in the CC at a time, and clinical dose rates from 1.0×10^9 p+/s up to 1.0×10^{10} p+/s. We studied how the recorded PG interaction data, at each dose rate, changed as two characteristics of the DAQ electronics of a CC were changed: (1) the number of data readout channels; and (2) the active charge collection, readout, and reset time. **Results:** As the dose rate increased, the number of recorded PG single-, double-, and triple-scatter events recorded decreased by a factor of 45x for the current CC DAQ configuration. However, as the DAQ readout channels increased and the readout/reset timing decreased, the number of recorded events decreased by $<5\times$ at the highest clinical dose rate. The increased readout channels and reduced readout/reset timing also resulted in a higher percentage of the recorded double- and triple-scatters being “true” events (caused by a single incident gamma) and not “false” events (caused by multiple incident gammas). **Conclusion:** The increase in the number and the quality of recorded double- and triple-scatter events resulted in a significant improvement to the reconstructed PG images, especially at the highest clinical dose rates.

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PO-GePV-M-64

A Novel Lung-Avoidance Planning Strategy Based On 4DCT-Ventilation Imaging and CT Density Characteristics for Stage III Non-Small-Cell Lung Cancer Patients

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Purpose: Functional planning based merely on 4DCT-ventilation imaging has limitations. In this study, we proposed a radiotherapy planning strategy based on 4DCT-ventilation imaging and CT density characteristics. **Methods:** For twenty stage III non-small-cell lung cancer (NSCLC) patients, clinical plans and lung-avoidance plans were generated. Through deformable image registration (DIR) and quantitative image analysis, a 4DCT-ventilation map was calculated. High-, medium- and low-ventilation regions of the lung were defined based on the ventilation value. In addition, the total lung was also divided into high-, medium-, and low-density areas according to the threshold. The lung-avoidance plan aimed to reduce the dose to functional and high-density lungs while meeting standard target and critical structure constraints. Standard and dose-function metrics were compared between the clinical and lung-avoidance plans. **Results:** Lung avoidance plans led to significant reductions in high-function and high-density lung doses, without significantly increasing other organs-at-risk (OARs) doses, but at the expense of a significantly degraded homogeneity index (HI) and conformity index (CI) ($p<0.05$) of PTV and a slight increase in monitor units (MU) as well as the number of segments ($p>0.05$). Compared with the clinical plan, the mean lung dose (MLD) in the high-function area and high-density area was reduced by 0.59 Gy and 0.57 Gy, respectively. **Conclusion:** A lung-avoidance plan based on 4DCT-ventilation imaging and CT density characteristics is feasible and implementable, with potential clinical benefits. Clinical trials will be crucial to show the clinical relevance of this lung-avoidance planning strategy.

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PO-GePV-M-65

A First In-Phantom Quantitative Concept Validation for a Prospective Respiratory Signal-Guided 4D CT Protocol

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Purpose: To quantitatively validate in-phantom effectiveness of a respiratory signal-guided 4D CT protocol based on retrospective reconstruction in GE CT scanners outlined by Pan et al. (2017) with the purpose of reducing breathing artifacts. This protocol has the striking advantage of not requiring any off-line data processing, new software or hardware besides the scanner itself. **Methods:** A custom made slab-like wooden insert was fit inside a CIRS motion phantom. Five motion curves were generated and loaded into the phantom during CT examination on a GE Optima 580 scanner. The motion curves simulate an increasingly irregular breathing pattern, ranging from a sinusoidal (meant to establish a baseline) to a breath-hold like signal. CT examinations were performed with and without using Pan et al.'s manual prospective gating 4D CT protocol. The wooden insert AP and LL motions were compared in each scenario by the means of in-house R-based image processing code. **Results:** For each CT acquisition, total angular deviations in the wooden insert motion are calculated. Considering ordinary examinations, compared to the sinusoidal baseline acquisition, a regular breathing profile one would display +22.1% angular deviations, increasing to +107.5%, +182.1% and +256.9% as the loaded motion curves raise in their degree of irregularity. If the gating signal-guided protocol is applied, though, these percentage ratios in the irregular profiles drop to +11.8%, +54.1% and +43.32% respectively. **Conclusion:** In-phantom qualitative validation of this protocol had already been provided by Pan et al. (2017) and Werner et al. (2020) with further development of AI integration. Concerning breathing artifact reduction, the concept of the applied protocol has proven itself to be effective from a quantitative point of view. Further work will be needed to include in this validation C-C motion.

PO-GePV-M-66

Impact of 4DCT Artifacts on Local Control for Lung SBRT

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Purpose: SBRT is a standard treatment for lung metastasis. Depending on patient abilities to provide consistent breathing pattern, 4DCT scans are used to delineate ITVs, hence the image quality of 4DCT is of uppermost importance for treatment planning and can potentially affect treatment outcome. A retrospective evaluation of 4DCT image artifacts is conducted and is correlated to local recurrence. **Methods:** This study uses clinical research protocol approved data of 96 patients diagnosed with lung metastases, 4DCT data sets with 10 phases were used for treatment planning, especially for the definition of the internal target volume (ITV). 4DCT artifacts were classified based on the severity of artifacts following the criteria outline in a prior study: images were classified as having minimal artifacts if they had ≤ 4 individual double structure artifacts at diaphragm or heart, and no interpolation artifacts in image center. Images were classified as having greater artifacts if they contained >4 individual double structure artifacts at diaphragm or heart, or any interpolation artifacts in image center. The dominant motion directions were recorded. The size of the GTV and proximity to closest artifacts were measured. Using RPM data, motion amplitude, standard deviation of patient breathing period, standard deviation of normalized patient breathing signal peak-to-peak amplitude during 4D CT acquisition are considered for covariates analysis. **Results:** The local failure control was observed for 8.3% (8/96) of patients. The preliminary artifacts scoring results shows that 11 out of 42 have severe CT artifacts, 31/42 have less or equal than 3 double structure artifacts at diaphragm. No correlation was found between 4DCT artifacts and local failure. **Conclusion:** Moderate 4DCT artifacts occurred in a non-negligible number of patients, however this was not found to have a measureable impact on local control. Future analysis will entail inclusion of more subjects and a multivariate analysis.

PO-GePV-M-67

Fiducial Registration-Error Analysis of PCA Lung Model for DVF Assessment

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Purpose: Original PCA lung model formulated by Zhang (2007) was validated using centroid position of a target volume. We (1) extend the analysis to the lung region using fiducial registration-error (FRE) analysis, which is more sensitive to accuracy of the deformation algorithm used to construct the model, and a more granulated evaluation of the deformation vector field (DVF). We (2) carry out preliminary analysis of the model to estimate DVF for lung ROI as well as target ROI for subsequent CT session, which can be used for margin assessment from respiration changes. **Methods:** We used public DIR-LAB 4DCT datasets and perform B-spline based deformable image registration (DIR) to acquire the displacement vector fields. Then perform FRE analysis between the actual and predicted landmark positions. Next, using two 4DCT images of the same patient acquired 30 minutes apart, the PCA model constructed from initial 4DCT was used to estimate subsequent 4DCT. Residual analysis was performed over each phase of breathing cycle to quantify differences between estimated and actual motions. **Results:** Mean FRE value for the DIR-LAB 4DCT cases was 1.6mm, consistent with gold-standard image-registration results, validating high fidelity of the PCA algorithm. When predicting a second 4DCT session, the mean residual over voxels in the lung region was 2.4mm. The directional components in eigenmode one and two indicate the first component is dominated by superior-inferior motion near the diaphragm, the second component is less predictable and has comparable contribution in all directions. **Conclusion:** The FRE evaluation is consistent with the previous centroid position evaluation. The PCA model can be used with surrogate information acquired during treatment to reconstruct target and organ motion for the purpose of dose accumulation. Our patient case study may indicate that the first principal component explains common diaphragm-dominated motion and the second component describes cycle-specific information.

PO-GePV-M-68

Impact of Flattening-Filter-Free Field on Online Adaptive Radiotherapy Using Virtual Couch Shift (VCS) Technique

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Purpose: Virtual couch shift (VCS) technique shifts MLC apertures rather than treatment couch to correct patient setup errors. When adapted to flattening-filter-free (FFF) linacs such as the MR-linac, VCS may lead to very different fluence pattern due to the non-flattened radiation field. This study aims to evaluate this effect for various anatomic sites. **Methods:** Four most commonly treated sites on MR-linacs (metastatic node, pancreatic, prostate and rectal cancers) were selected for this simulation study. Reference IMRT plan (P0) was generated in Pinnacle using 6MV FFF beam on an Elekta Versa linac with Agility MLC head. Translational shifts of 5, 10, 15 and 20-mm were introduced by shifting isocenter in the opposite direction. A Pinnacle script was developed to generate shifted MLC apertures taking beam divergence into account. For shift-only plan, dose distribution was computed with the original MUs. Shift-only plan was subsequently copied and re-optimized with segment weight optimization (SWO) using the same objectives from P0. Various dosimetric indices for planning target volume (PTV) and organs-at-risk (OARs) were collected and compared between P0, shift-only and SWO. **Results:** Shift-only resulted in decreased dose to PTV in the shifted direction and mostly affected V_{rx} (PTV received prescription dose). Dose reduction increased with shift distance and target size, and appeared more pronounced for shifts in the superior-inferior (SI) directions. For prostate and rectal cases, V_{rx} could drop by ~20% for 10-mm shift and ~50-60% for 20-mm shift. The impact on OAR indices (D0.5cc and D5cc) was relatively small and didn't show a clear trend. SWO quickly converged within ~20 optimization iterations. All SWO plans achieved comparable plan qualities as P0 except for one extreme situation (rectal case with 20-mm SI shift). **Conclusion:** Shift-only method mostly affects target coverage. Simple SWO is effective in restoring plan qualities even with relatively large shift.

PO-GePV-M-69

Adaptive Radiation Therapy with Deformable Image Registration and 4D Dose Calculation

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Purpose: To develop a four-dimensional (4D) dose optimization and calculation algorithm on 4D-CT-number maps with tissue heterogeneity using displacement-vector-fields (DVF) from different deformable-image-registration algorithms (DIR) in order to consider for anatomical variations and respiratory motion to achieve adaptive radiation therapy. **Methods:** A simulation study was performed to remap CT-numbers and dose distributions using DVF that were obtained from DIR. The DVF represented a voxel-by-voxel motion of the tumor and surrounding organs-at-risk in the region of interest. These DVF were employed to modify the intensity maps from 3D-conformal and intensity-modulation optimized plans to perform 4D-optimization on remapped CT-numbers to account for heterogeneity to consider anatomical variations and respiratory motion. **Results:** The DVF obtained from deformable image registration correlated linearly with the motion amplitudes of the phantom. The dose calculation algorithm produced 4D-optimized plans from the stationary 3D-conformal or intensity-modulated plans using DVF's extracted from DIR algorithms. The 4D-dose profiles became narrower for the targets moving with larger motion amplitudes. It performed optimization by shrinking and remapping the dose distribution that are obtained from dose optimization on stationary images. The unfolding of the 4D-dose distributions as the patient move during dose delivery produced equivalent dose distributions optimized in 3D for the stationary targets and organs at risk. Heterogeneity correction was also considered by remapping the CT-number values in the CT images using the DVF's in order to reverse motion artifacts. **Conclusion:** This approach provides a novel algorithm to perform adaptive radiation therapy using 4D dose optimization and calculation with heterogeneity correction using remapped CT numbers which is a practical and convenient alternative to manage respiratory motion of cancer patients. The advantages of this approach include achieving equivalent conformal doses without any restriction on anatomical variations and patient breathing during dose delivery, however, it is limited by reproducibility of the DVF from DIR.

PO-GePV-M-70

Respiratory Motion Detection and Tracking of Mobile Phantoms with a Microwave Imaging System

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Purpose: To detect and classify the motion of tissue equivalent phantoms by measuring the variation in the microwave signal. A model of motion classification was developed that employs the variations in the microwave signal to obtain the 3D-motion parameters for all voxels included in the phantom volume in the microwave field of view. **Methods:** A microwave system with 10.25 GHz that was made from one transmitter and one receiver was used to detect phantom motion. Several tissue equivalent phantoms with a mobile platform that moved with different motion patterns were used to characterize the microwave signal variations. The microwave signal variations with phantom thickness, position and density were quantified. These variables were used to detect, classify and quantify the phantom motion parameters using variations in the microwave signal induced by various motion patterns that simulate patient respiratory motion patterns. **Results:** The position of the minimal intensity of the microwave signal detected by the receiver varied linearly with the position of the phantom in the microwave field of view which can be predicted within 1.0 mm from the minimal microwave intensity. The microwave intensity varied non-linearly with the density of the phantom objects equivalent to lung, breast muscle, adipose, liver tissues. Furthermore, the microwave intensity decreased non-linearly with the increase in phantom thickness. A motion classification algorithm was developed that used the microwave projections with the previous parameters to extract the 3D-motion trajectory of a mobile phantom, phantom thickness, composition of the phantom in the microwave field of view. **Conclusion:** This microwave motion detection technique has potential clinical applications that are of interest in the imaging and treatment of cancer patients with radiation therapy that are superior to just tracking an external or internal marker currently used for patient motion tracking or gating used for cancer patient treated with radiation therapy.

PO-GePV-M-71

Testing Different Materials for a Quality Assurance Phantom for Proton Radiography Using the MEVION Proton Therapy System

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Purpose: To test different materials and develop a quality assurance phantom to evaluate quantitatively the image quality of proton radiographs obtained from high energy scanning beams using the MEVION proton therapy system. **Methods:** Different materials and phantoms (LEEDS and CIRS) were used to quantify the different imaging quality parameters of proton radiographs that included muscle, breast, adipose, liver, lung low, lung medium, bone. The images quality parameters that were tested included linearity, uniformity contrast and spatial resolutions. The proton radiographs were obtained using a high energy proton beam of 230MeV from the MEVION pencil-beam scanning proton therapy system. Proton pencil beams of a diameter of 3.5mm in air with a spot spacing of 2.5mm were used to obtain the proton radiographs. Large open proton beams of 20x20cm² and range of 32cm in water with one energy layer modulation were used. Phosphorous scintillation plates that are usually used photon computed radiography were used to acquire the proton radiographs. **Results:** High-Z materials such as lead, copper and other metals in the LEEDs phantom used for photon radiography were not appropriate for testing the image quality of the proton radiographs. Particularly the spatial resolution objects used for photon imaging did not work with proton imaging because the proton beam scattered with large angle degrading image quality. The contrast resolution was tested variable density and size plugs similar to the LEEDS phantom. The linearity of the proton radiographs was tested with inserts that are made from different tissue-equivalent materials with densities ranging from air to bone. The uniformity of the proton radiographs was tested using a solid water block. **Conclusion:** This quality assurance phantom provides a useful tool to test and quantify the image quality of proton radiographs that can be used for image guidance with on-board imaging systems integrated with proton therapy machines.

PO-GePV-M-72

Four-Dimensional Dosimeter and Quality Assurance Phantom for Adaptive Radiation Therapy

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Purpose: To develop a 4D-dosimeter and quality assurance phantom to quantify the effects of respiratory motion on 3D-optimized dose distributions and provide dose verification of futuristic 4D-optimized treatment plans and dose calculation algorithms. **Methods:** Dose distributions were measured with a MapCheck2 phantom that was placed on a mobile platform that can move with different motion patterns with adjustable motion amplitude and frequency. This motion phantom was used to measure the dose distributions optimized and calculated in 3D including conformal, IMRT and VMAT plans. The dose distributions were measured with this phantom system static and moving with different motion amplitudes (5-35mm) to quantify motion artifacts on the dose distributions. **Results:** Respiratory motion can lead to large dose discrepancies that depends on the patient motion pattern. The increase in motion amplitudes results more blurring in the dose distribution with an increased dose deficit in the tumor and increased dose deposition in the surrounding normal tissue. For a conformal open field plan, the dose profiles changed around the 50% isodose line, with increasing dose spelling outside the field for increasing motion amplitudes 5-35mm. The central axis dose and the surrounding flat region remained unchanged. However, for an intensity modulated plan, the motion artifacts affected the dose distribution overall with underdosing in the treatment planning volume and overdosing outside the treatment field. **Conclusion:** This mobile

dosimeter and phantom system provided a powerful tool for quantifying the dose deviations induced by cyclic motion on 3D optimized dose distributions. 4D-optimized dose distribution provides an alternative technique for the management of respiratory motion to conventional techniques such as breath holding and beam gating. This 4D phantom will be a necessary dosimetric tool that can be used for the verification of dose distribution calculated by 4D-optimization and dose calculation algorithms to achieve the goals of adaptive radiation therapy.

PO-GePV-M-73

Evaluation of Longitudinal Variation and Impact of Dose Calculation Algorithms Within a CBCT-Based Online Adaptive Workflow

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Purpose: To compare and evaluate the impactful dosimetric differences of three common treatment planning system algorithms longitudinally within the Ethos adaptive treatment workflow (EATW) and create practical expectations of each system. **Methods:** Dosimetric comparisons were analyzed across ten sessions of a prostate patient in an EATW at a satellite clinic with rotating physicians. Adaptive plans were created from contours segmented using artificial intelligence on a day of treatment CBCT. Initial calculations utilized the AcurosXB algorithm while simultaneously calculating externally with the Mobius3D-Collapsed Cone Convolution (CCC) algorithm. Plans were then exported offline to Varian Eclipse and recalculated using the Anisotropic Analytical Algorithm (AAA) 15.6.06 with preset monitor units and fluence parameters. Comparisons were made between the three algorithms using selected statistics and dose-volume histograms for each session. Additionally, longitudinal data was analyzed to evaluate trends over the course of the EATW. **Results:** Comparisons of DVH data between different algorithms indicate consistent differences in PTV coverage, as well as dose volume statistics in rectum and bladder organs-at-risk (OAR). AAA consistently resulted in the highest evaluation of dose volumes in each volume of interest, while CCC was consistently the lowest. AcurosXB is the clinically utilized dose algorithm, and in our analysis indicated a closer relationship with AAA. Longitudinal evaluations indicated minimal deviations in PTV coverage and only slight variations in OARs. Monitor units and structure volumes fluctuated between sessions accordingly, potentially related to different physician contouring criteria. **Conclusion:** As adaptive planning and treatment delivery grow more common with advancing technology, institutions may need to integrate new algorithms accordingly. This work establishes metrics for interpreting differences between three common treatment planning dose calculation algorithms in the Ethos workflow. This work establishes expected variations longitudinally and between algorithms that can be utilized by clinics new to adaptive workflows to develop confidence while incorporating unfamiliar processes.

PO-GePV-M-74

A Simple Adaptive Planning Workflow for MR-Guided Radiotherapy of Prostate Cancer

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Purpose: Two workflows, adaptive planning (adapt-to-shape, ATS) or repositioning (adapt-to-position, ATP) are used in MR-guided adaptive radiation therapy (MRgART) on a 1.5T MR-Linac. For MRgART of prostate cancer, ATP usually take 30 minutes, shorter than that for ATS (50 min), and is used for most fractions. However, ATP may not be sufficient if there is a large air-cavity in the rectum. We propose a simple adaptive workflow, Pseudo-ATS+, that can account for an air cavity change with a shorter time compared to ATS. **Methods:** The Pseudo-ATS+ workflow includes: (1) segmenting the air cavity and external contours on daily air scan specifically designed to image air, (2) populating the obtained air and external contours, along with the reference contours for other structures to the daily MRI, and (3) optimizing an adaptive plan on daily images. To evaluate the performance of Pseudo-ATS+, we performed standard ATP, reconstructed doses of the ATP and Pseudo-ATS+ plans on daily images with fully updated contours and compared the obtained doses with the ATS plans for selected daily MRI sets with the presence of an air cavity in the rectum. Various DVH parameters, e.g., minimum dose (Dmin) for the PTV and volume coverage of 63Gy (V63Gy) for rectum and bladder, were compared. **Results:** For Pseudo-ATS+, the differences in DVH parameters between the reconstructed dose and ATS plans were smaller than those for reconstructed ATP. The PTV Dmin of reconstructed Pseudo-ATS+ was 3.3% different from ATS, while the difference was 9.8% for reconstructed ATP from ATS for a case tested. Rectum V63Gy was within the planning goal for reconstructed Pseudo-ATS+ while reconstructed ATP plan dose was not. The Pseudo-ATS+ workflow required additional 5 minutes compared to ATP. **Conclusion:** This proposed simple adaptive workflow, Pseudo-ATS+, can properly account for rectal air cavities changes without significantly extending patient-on-couch time during MRgART for prostate cancer.

PO-GePV-M-75

Importance of Incorporating Rotational Setup Differences for QACT Dosimetric Evaluations

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Purpose: To develop an accurate method to incorporate rotational setup differences for QACT dosimetric evaluations using Eclipse and Precision treatment planning systems. **Methods:** Recalculation of the original treatment plan on a CBCT or updated fan-beam CT (QACT) is often used to provide the dosimetric impact of anatomical changes and to decide whether or not an adaptive plan is necessary. However, some existing technologies do not account for rotational differences (roll, pitch, and yaw) between the QACT and the planning CT. The VelocityAI software was used to resample the QACT to the planning CT frame-of-reference, to simulate rotational corrections that are applied at the treatment machines (up to 3° for TrueBeam and 5° for CyberKnife). Effect of omission of rotations on QACT dose estimates was studied on three pelvic treatment cases for TrueBeam/Eclipse with para-aortic and inguinal nodal targets and on an intracranial lesion treatment for CyberKnife/Precision. The original treatment plans were recalculated on the QACTs with and without rotational differences, to demonstrate the dosimetric sensitivity. **Results:** All three pelvic cases showed clinically unacceptable (<91% of prescription dose) minimum D0.03cc doses when rotational differences were not accounted for (63.5%-88.0%). Target coverage was acceptable when rotations were applied (minimum D0.03cc = 91.8%-95.8%). The CyberKnife case showed a small increase in optic nerve doses and a 16% increase in mean right cochlea dose when rotational differences were not taken into account. Dosimetric differences for OARs were negligible when rotations were applied. **Conclusion:** All four clinical examples presented would have triggered adaptive replanning if rotational differences in the patient CTs were not taken into account. Application of rotational corrections applied at the treatment machines is an important parameter when assessing the dosimetric impact of anatomical changes, and is a crucial component to adaptive RT workflows.

PO-GePV-M-76

Initial Experience with Online CT-Based Adaptive Radiotherapy in the Abdomen

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Purpose: CT-Based online adaptive radiotherapy using an integrated ring-gantry platform recently gained FDA clearance. This study describes our institution's early experience with abdominal CT-guided stereotactic adaptive radiotherapy (CT-STAR) to help guide other CT-based adopters. **Methods:** Our institution specifically targeted abdominal treatment sites with minimal tumor motion. We analyzed anatomical movement for both the target and OAR from the initial planning scans, determining the need for planning ITVs or motion planning risk volumes (mPRV). Patients were either treated at end-exhale breath hold or free breathing, which was determined by both patient compliance and tumor motion needs. Dosimetric indices from planning images, anatomy-of-the-day, and adapted anatomy were collected. Treatment times for each disease site were also collected. **Results:** Our institution treated two para-aortic lymph node (PA-

LN) and one anterior retroperitoneal lymph node (R-LN). Both PA-LN patients were scanned at end-exhale breath hold for adaptive contouring but treated at shallow free-breathing. The average tumor motion was 0.2cm and the average adjacent OAR motion was 0.4cm. In the R-LN patient, the patient was scanned and treated at end-exhale breath-hold. For the R-LN patient, the bowel dose was exceeded in all predicted plans. For the first PA-LN patient, the predicted plan exceeded small bowel constraints in all five fractions and exceeded the duodenum constraint in two of the five fractions. In the second PA-LN patient, the predicted plan exceeded small bowel constraints in all five fractions. For all three patients, adaptive treatment plans were within bowel tolerance. The average treatment times were 61.6min $\pm$ 23.6min (mini: 31min. max: 114min). **Conclusion:** Abdominal adaptive treatments with minimal motion were successfully treated in an acceptable time period. mPRVs can additionally account for predictable intra-fraction OAR motion in the absence of respiratory gating.

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PO-GePV-M-77

Longitudinal Review of Adaptive Therapy Planning Time for the Elekta Unity

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Purpose: The adaptive planning workflow on the Elekta Unity MR-Linac is a multi-step process for each treatment. These processes differ depending on whether the adapt-to-position (ATP) or adapt-to-shape (ATS) workflow has been selected. In the ATP workflow, a rigid registration is performed between the daily MR and the reference image and a warm-start gradient descent optimization is used. In the ATS workflow, deformable registration, re-contouring, and full objective-specific plan optimization is used. The time required for each step is dependent on the user's experience and is likely to improve with time. **Methods:** The total time required to complete online adaptive treatment planning and plan review was tracked at three separate time intervals, 18 days starting in December 2019 (146 fractions), 50 treatment days starting in March 2020 (133 fractions), and 26 treatment days starting in January 2021 for 125 fractions for ATP and ATS workflows. The distribution of time for each step in the planning process was recorded during treatment. The longitudinal mean and standard deviation of the times associated with each step in the planning process were calculated and a two tailed t-test was used to test if the time reductions were statistically significant. **Results:** Although changes in time were not statistically significant ($p>0.05$), a reduction in the mean time for certain planning steps was observed. The mean total time for ATP decreased from 15.7 to 14 minutes and decreased from 34.7 to 25.0 minutes for ATS over the three tracked time intervals. The detailed analysis of ATS planning times demonstrated a 40% decrease in the average registration (1.8 minutes) and contouring (3.9 minutes). **Conclusion:** Experience with the online adaptive RT planning tools and process is likely to decrease the time required for each treatment.

PO-GePV-M-78

Dose Accumulation of Daily Adaptive Plans Using MiM and RayStation for Elekta Unity MR-Linac

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Purpose: This study aims to test the daily dose accumulation of online adaptive plans using commercially available MiM (MiM Software; Cleveland, OH) and RayStation (RaySearch; Stockholm, Sweden) for Elekta Unity MR-Linac. Dose accumulation is essential to validate delivered dose to tumor and surrounding organs, which can be a prerequisite for further re-planning and/or an additional fraction. **Methods:** This study utilized a CT and 5 daily MR datasets of a prostate cancer patient to develop a reference plan (37.5Gy in 5 fractions with 7-fields and a 7FFF Unity® beam) and simulate two plan adaptations using an offline MRL Monaco (v5.40.01). In the first and second simulations, (1) five adapt-to-position(s) (ATP), rigidly registering CT-to-MR(s), were performed and (2) three ATP(s) for Day1, Day3 and Day5 MR(s) and two adapt-to-shape(s) (ATS) for Day2 and Day4 MR(s), deformably registering CT-to-MR or MR-to-MR, were performed for composite doses. MiM (Ver 7.0.6) and RayStation (Ver 9A) were used for rigid and deformable image registration for composite doses. Composite doses were compared in mean dose and 3D gamma analysis using an in-house software. **Results:** We found a minimal difference of mean dose for ATP plans (MiM<0.1% and RayStation<0.3% for PTV (Prostate+SeminalVes+5mm margin)) and slightly higher for ATS plans (MiM<0.3% and RayStation<2.5% for PTV). The 3D gamma analysis resulted in 91.1% and 93.3% for ATP plans, and 90.6% and 89.9% for ATS plans across MiM and RayStation, respectively. **Conclusion:** This is the first that we demonstrated the dose accumulation of ATP and ATS using MiM and RayStation across rigid and deformable registration. In addition, generating a composite plan in adaptive radiation therapy process can be achievable to validate delivered cumulative doses. Further investigations will need to include an improvement of 3D dose accumulation and speed for real-time process whilst optimizing image registrations and adapting daily target organ contours.

PO-GePV-M-79

Comparison of the Efficacy of CT On Rails Versus In-Room CBCT for Online Daily Adaptive Proton Therapy

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Purpose: To compare the efficacy of CT on rails versus in-room CBCT for daily adaptive proton therapy. **Methods:** We analyze a cohort of head-and-neck patients with daily CBCT images to assess the influence of near-treatment-position imaging on the efficacy of daily adaptation. Virtual daily CT images were generated by deformable image registration of the planning CT to daily CBCT images. We identified the necessity of moving the patient after a CT scan to be the most significant difference in the adaptation workflow, leading to an increased treatment execution uncertainty σ . It is a combination of the isocenter-matching σ_i and random patient movements σ_m induced by the couch motion. The former should not exceed 1 mm. The latter depends on the treatment site and might be patient-specific. Accordingly, to mimic the adaptation workflow with CT, we introduced random offsets between Monte-Carlo based adaptation and delivery of the adapted plan. The offsets followed Gaussian distribution with σ corresponding to the total uncertainty in each axis posterior-anterior, left-right, and inferior-superior. We studied three different scenarios with $\sigma_m=1, 2$, and 3 mm. **Results:** Results showed no significant differences in accumulated DVHs and dose distributions for $\sigma=1.4$ mm and 2.3 mm. Offsets with $\sigma=3.2$ mm resulted in underdosage to high- and low-risk CTV, more hot spots of considerable volume, and noticeably higher doses to OARs. For high-risk CTV, the mean D_{98} was found to be 97.9%, 98.0%, 96.7% for $\sigma=1.4$ mm, 2.3 mm, 3.2 mm, respectively. **Conclusion:** If the total uncertainty due to imaging with CT on rails does not exceed 2 mm, the efficacy of daily adaptation would not be significantly affected. Given other advantages of fan-beam CT, such as lower whole-body dose and better image quality, CT on rails can be considered a good alternative to CBCT for adaptive proton therapy.

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PO-GePV-M-80

Sun Nuclear DoseCHECK Vs Varian Mobius: Commissioning and Characterisation for Ethos Adaptive Independent Dose Calculations

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Purpose: To evaluate the accuracy and performance of DoseCHECK and Mobius as independent plan verification software for the purpose of pre-treatment quality assurance of Varian Ethos adaptive plans. **Methods:** DoseCHECK and Mobius were commissioned in accordance with international best practice documents on treatment planning system (TPS) commissioning and quality assurance, including: AAPM TG53, TG119, MPPG5a and IAEA TECDOC1583. All plans were sent to DoseCHECK and Mobius for independent dose calculation, and compared to measurement on the treatment delivery system (TDS). For end-to-end validation, Ethos clinical patient plans (n=50) were generated in the Ethos TPS encompassing a range of clinical sites and techniques (IMRT and VMAT), then sent to DoseCHECK and Mobius for independent dose calculation. Mean dose to the target and overall gamma pass rate were extracted and compared to phantom measurement on the TDS, with plan complexity defined as MU/Gy. **Results:** All results for commissioning test plans were within the published tolerances for DoseCHECK and Mobius when compared to the TPS and measurement. Higher degrees of uncertainty were noted for extreme cases, such as small field sizes, build-up region (surface dose) and heterogeneity. All patient plans were validated through measurement and in Mobius and DoseCHECK 3D gamma criteria of 3%/3mm (10% threshold, global normalisation) >95% points passing and mean dose to target difference within 3% was configured to achieve comparable results. The average gamma pass rate for both systems was >95% for all plans tested and mean dose to target difference <1%. In general, plans of higher complexity or significant heterogeneity contained higher uncertainty. **Conclusion:** DoseCHECK and Mobius were commissioned and found to be within tolerance of all tests recommended in international best-practice guidelines. Both systems were used to independently verify plan dose calculation for pre-treatment QA of Ethos adaptive plans.

NSCC has reference site agreements with Varian and SNC.

PO-GePV-M-81

CBCT-Based Plan Adaptation for Chest Wall Intensity Modulated Proton Therapy

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Purpose: Cone-beam CT (CBCT) is an important imaging modality for precise positioning and monitoring gross anatomical changes. However, CBCT is not suitable for dose calculation in intensity-modulated proton therapy (IMPT), largely limited by the lack of accuracy in Hounsfield units. This work demonstrated a new workflow that utilizes CBCT for fractional dose verification as well as adaptation for chest wall IMPT. **Methods:** A small cohort of four chest wall patients (three ipsilateral and one bilateral) were included in this study. All had one simulation helical CT and at least one setup CBCT. Three had verification helical CTs at some points in their radiotherapy course. A synthetic CBCT was created by deforming the simulation helical CT to a setup CBCT in RayStation. Structures contoured on the simulation CT were also propagated to the synthetic CBCT through the same chain of deformable imaging registration. Dose to the targets and normal organs were reported and compared among the simulation helical CT, the synthetic CBCT, and verification helical CT. **Results:** V95% of PTV and V5Gy heart were consistent among planning CT, synthetic CBCT, and verification CT with less than 3% and 1% variation, respectively. Except for one who suffered from significant swelling of tumor stroma. In the latter case, the synthetic CBCT was able to predict the PTV re-contoured independently and the dose within 1% on a replanning helical CT triggered by the synthetic CBCT. **Conclusion:** This work supports the assertion that CBCT imaging can be used as an early warning system capable of detecting anatomy changes that warrants adaptive planning. The synthetic CBCT we developed in this study could be a powerful tool for fractional dose verification and quick plan adaptation in IMPT.

PO-GePV-M-82

Evaluation of AI-Driven Adaptive Workflow for High Quality Palliative Radiotherapy

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Purpose: Approximately one third of radiotherapy patients are treated with palliative intent. Varian's Ethos online adaption system utilises AI contouring on daily cone-beam CT (CBCT) to auto-generate a plan of the day. This work evaluated the Ethos adaptive workflow in palliative patients comparing clinician contours against Ethos AI segmentation, and corresponding adaptive plans against original (scheduled) plans. **Methods:** Ten palliative patients (27 fractions) were replanned with dynamic MLCs using Ethos TPS, then duplicated and sorted into two test arms. Adaptive treatment was simulated in a test environment using patient-specific CBCTs. In arm 1, clinicians edited all AI contours before an adaptive auto-plan was generated. In arm 2, no contour edits were made. The adaptive plans generated were compared using metrics assessed in original plans (PTV D95% and 100% conformity index). On a subset of six fractions, AI and clinician-modified structures were compared using the DICE similarity coefficient. **Results:** All adaptive plans passed the primary constraint PTV D95%>95%, while two scheduled fractions failed. Mean PTV D95% for adaptive and scheduled fractions were 102.4% and 104.0%, respectively, with no statistically significant difference found (t-test P=0.29). Mean conformity indices measured 0.97 and 0.85 for adaptive and scheduled plans, respectively, were found to be statistically different (t-test P=0.017). The DCEs for CTV, kidneys, liver, small bowel and stomach were 0.90±0.09, 0.930±0.05, 0.98±0.02, 0.928±0.04 and 0.70±0.11, respectively, indicating accurate AI segmentation of all contours except for the stomach. **Conclusion:** Ethos AI-driven adaption was shown to achieve better target coverage than scheduled plans, indicating the system could account for inter-fraction anatomy changes associated with palliative cohorts. Good agreement was shown between preliminary Ethos and clinician contours, except for the stomach, which was most likely impacted by the presence of gas causing image artefacts, impacting AI segmentation. Future work includes online adaption in diagnostic CT-enabled palliative planning.

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PO-GePV-M-83

Performance Characterization of a Real-Time Dose Reconstruction Algorithm for Prostate Stereotactic Radiotherapy Treatment

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Purpose: There is a growing awareness that the dose a patient receives in SBRT often differs from the planned dose due to tumor motion. To improve dose accuracy, dose reconstruction should ideally occur during treatment. To address this, a real-time dose reconstruction method, DoseTracker has been developed. In a critical step to its clinical deployment, we benchmark DoseTracker against a treatment planning system (TPS)-based dose reconstruction method for prostate SBRT treatments and hypothesize that the motion-induced dose-errors determined from these two methods should agree. **Methods:** Simulated real-time motion-inclusive dose reconstruction was performed using DoseTracker for nine prostate cancer patients who received IGRT with gating during five-fraction SBRT (7.25Gy/Fx) in the TROG-15.01 SPARK trial. Dose reconstruction was performed for both actual-gated (45 fractions) and simulated non-gated sessions (11 fractions). The motion-induced dose-errors were calculated as the difference between planned and motion-inclusive doses using DoseTracker and compared against the motion-induced dose-errors calculated using an iso-center shift method with a clinical TPS. The motion-induced dose-errors were computed for the clinical target volume (CTV), planning target volume (PTV), and two risk-organs, bladder and rectum. **Results:** The median computation time for each dose calculation of DoseTracker ranged from 110-320 ms for calculation volumes ranging from 287-691 cm³. The mean and standard deviation of the differences of the motion-induced dose-errors calculated using DoseTracker and the TPS were 0.02±0.01 Gy for CTV(ΔDmean), 0.02±0.01 Gy for PTV(ΔDmean), 0.04±0.07 Gy for CTV(ΔD100%), 0.03±0.06 Gy for PTV(ΔD95%), 0.02±0.02 Gy for bladder(ΔDmean) and 0.02±0.07 Gy for rectum(ΔDmean). The close agreement in motion-induced dose-errors supports the hypothesis that the performance of DoseTracker is comparable with the

TPS-based method for determining motion-induced dose deterioration. **Conclusion:** DoseTracker accurately determined motion-induced dose-errors in real-time for a lower pelvic tumor site. This study demonstrates that clinical implementation of real-time motion-inclusive dose reconstruction with DoseTracker is feasible.

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PO-GePV-M-84

A Dilated Convolution Neural Network to Predict Three-Dimensional Dose Distribution in IMRT Treatment of Cervical Carcinoma

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Purpose: To implement a method for precise three-dimensional(3D) dose prediction in intensity-modulated radiation therapy(IMRT) using a dilated convolution deep convolutional neural network(CNN). **Methods:** Seventy-seven cases of patients with cervical carcinoma were retrospectively selected. Sixty-six cases were chosen randomly as the training set and the remaining as the test set. A deep convolutional neural network model named Deep-ResUnet101 was trained to predict 3D dose distribution. First, the dose distributions and all contours were converted to 3D matrix and 3D binary matrix respectively using in-house developed python programs. Second, the contours and dose distributions at voxel-level were fed to the dilated neural network. The model was trained from scratch with weights randomly initialized. The trained deep convolutional network was used to predict 3D dose distributions, and the accuracy was evaluated against the corresponding clinical dose distribution. **Results:** Our results demonstrate that the dilated neural network could predict 3D dose distribution precisely. For 11 tested cases, there was no significant difference between the predicted dose volume histograms (DVH) parameter values and the clinical ones. The mean absolute max dose differences and the mean absolute mean dose differences for PTV and OARs are within 2.2% and 2.8%, and nearly all the dosimetric indices for PTVs and OARs are within 0.1%. The average dice similarity coefficient (DSC) of different isodose volumes was 0.94-1. **Conclusion:** This study developed a new dilated convolutional neural network for voxelized 3D dose prediction, which can precisely predict the 3D dose distribution for cervical carcinoma IMRT patients. The predicted 3D dose distribution could be used to improve the design of IMRT plan, ensure the plan quality and consistency, and guide automatic IMRT treatment planning.

PO-GePV-M-85

Plan Quality and Feasibility of Adaptive CBCT-Guided Prostate and Nodal Volume SBRT

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Purpose: To investigate the clinical feasibility of utilizing a commercial CBCT-guided adaptive radiotherapy system (Ethos) for adaptive SBRT to the prostate and elective pelvic nodes. **Methods:** A template of clinical goals was created in Ethos to deliver 40Gy to prostate (P), 36.25Gy to P and proximal seminal vesicles (P+psV), and 25Gy to the pelvic nodes in 5 fractions with 3 full VMAT arcs, automatically optimized with an Intelligent Optimization Engine (IOE). The template was applied to five high-risk prostate patients without modification to simulate adaptive sessions, since anatomy can change but changes to Ethos planning goals are not editable during on-couch adaptive sessions. Plans were evaluated against NRG-GU009 & GU-010 constraints. Secondary calculations with Mobius3D and 3D gamma comparisons (3%/2mm criteria, 10% threshold) were evaluated. **Results:** Prostate volumes ranged from 42cc to 135cc. Rectum and bladder volumes ranged from 35cc to 149cc and 206cc to 507cc, respectively. For all 5 patient plans, IOE produced a plan that met all strict OAR constraints. All other constraints were at least within acceptable variation. The mean V100%Rx was 96.1%, 95.9%, and 99.5% for PTV_4000, PTV_3625, and PTV_2500, respectively. The minimum coverage of all PTVs by prescription dose was 93.9%; PTV coverage was sacrificed when necessary to spare overlapping organs-at-risk. Mean rectum max dose was 3837cGy [range: 3658-3911] (<4120 constraint) and mean bladder max dose was 4183cGy [range: 4163-4228] (<4350 constraint). MU ratios ranged from 4.0 to 6.1 (mean 4.9). All 3D gamma values with Mobius 3D dose calculation were greater than 99.6%. **Conclusion:** Clinically acceptable SBRT plans to P+psV and pelvic nodes were generated and feasibility to successfully create adaptive plans was demonstrated. Patient-specific optimization of the template could improve plan quality with confidence in on-couch adaptive plan fidelity.

PO-GePV-M-86

Comparison of Online Adaptation and Offline Re-Planning for Head and Neck Intensity-Modulated Proton Therapy

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Purpose: To compare daily online plan adaptation with offline treatment re-planning for head and neck intensity-modulated proton therapy (IMPT). **Methods:** IMPT treatment plans are created for a retrospective cohort of eight head and neck cancer patients with daily cone-beam CT (CBCT) images (median=33 fractions) and a re-scan CT acquired during the course of treatment due to large anatomy changes. Treatment plans are created based on the nominal planning CT. Dose tracking of the IMPT treatment is performed for three different delivery scenarios: re-planning with a 3 mm planning margin (Re-PTV), re-planning without margins (Re-CTV), and online adaptation without margins (OA). Both re-planning scenarios are realized by delivering the initial plan until the re-scan CT was acquired, at which point a completely new plan is created that fulfills the same clinical goals. For OA, the initial CT plan is adapted at each fraction using an in-house developed online adaptation workflow based on Monte Carlo dose calculations on scatter-corrected CBCTs. Plan adaptation is achieved by tuning the weights of a subset of beamlets without changing their energies or positions. Planning contours are propagated from the CT to each daily CBCT using deformable image registration. The same propagated contours are employed for dose tracking and accumulation throughout the treatment. **Results:** In terms of target coverage, OA outperformed both re-planning scenarios. For the low-risk CTV, the mean D₉₅ for individual fractions was 91.6%, 94.3%, and 98.4% of the prescribed dose for Re-CTV, Re-PTV, and OA, respectively. For the high-risk CTV, 97.0%, 98.2%, and 98.3% were acquired for the same metrics. In terms of OAR sparing, OA kept the dose comparable to Re-CTV while reducing the dose compared to Re-PTV. **Conclusion:** This study demonstrates the superiority of online IMPT plan adaptation over offline treatment re-planning in the context of target coverage and OAR sparing.

PO-GePV-M-87

Comparison of Target Coverage Between Adapted and Non-Adapted Treatment Plans for Prostate Radiation Therapy

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Purpose: To evaluate target coverage differences between adapted and non-adapted treatment plans for patients receiving adaptive radiation therapy to their prostate and proximal seminal vesicles. **Methods:** The D95 of the target structure was recorded for each session for both the adapted and non-adapted plans. All treatments were delivered on a Varian Ethos. For each treatment session contours were generated to reflect the anatomy on that day, including the target structures. The daily change in volume from baseline for the PTV, rectum, and bladder was also recorded. **Results:** Of the first 204 sessions delivered to 6 different patients, the average PTV D95 of the adapted plans was 99.7% of the prescription dose while the scheduled plans had an average D95 of 88.4%. Additionally, 198 of the 204 sessions had superior target coverage with the adaptive plan over the scheduled plan. The 6 sessions in which the scheduled plan provided better coverage were all from the same patient. During these 6 sessions the bladder was, on average, 63cc smaller than at the time of simulation while the rectum was 33cc larger. Comparatively, the bladder was 24cc underfilled and the rectum 12cc larger, on average, during all other sessions for this patient.

During these 6 sessions the adapted plan respected all specified normal tissue limits while the non-adapted plan exceeded the rectum and/or bowel limits. **Conclusion:** While adapted plans offered better target coverage in approximately 97% of treatment sessions there were instances in which the system intentionally under-covered the target in order to respect normal tissue limits as a result of variable bladder and rectum filling. This underscores the need to maintain a reproducible setup despite having adaptive capabilities.

PO-GePV-M-88

Developing On-Table ART with Conventional LINACs - TrueBeam

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Purpose: X-ray guided conventional LINAC-based RT is widely available and capable for ART, but currently lacks the ability to perform on-table adaptive procedures including adaptive CT and contour generation, and adaptive planning. In this project, we proposed a procedure for on-table adaptive radiotherapy on conventional LINACs (e.g., TrueBeam). **Methods:** Proposed on-table ART procedure consisted of 1) adaptive CT generation based on daily CBCT, 2) contour generation and 3) adaptive planning. We utilized offline adaptive feature of Velocity to generate adaptive CT and contours deformed from the planning CT and contours based on daily CBCT. The adaptive CT was then pushed into the treatment planning system (TPS) database where a collection of in-house scripts copied the beam's geometric parameters from the original plan and optimized dose for the adaptive CT using commercial KBP optimization. The feasibility of the proposed procedure was evaluated in terms of procedure time after daily CBCT and quality of three plans (prediction dose and two adaptive doses). **Results:** CTs and structures of head & neck cases were successfully generated based on daily CBCTs within 12 minutes. In addition to the predicted dose and two adaptive dose calculations within 5 minutes, the adaptive plan can be ready for delivery within 20 minutes. **Conclusion:** We demonstrated the clinical feasibility of on-table ART with conventional LINAC using a combined commercial solution with in-house planning scripts. The proposed approach can inspire on-table ART with conventional LINACs in community clinics.

PO-GePV-M-89

A Dosimetric Comparison of Stereotactic Body Radiation Therapy Delivery Techniques for Lung Cancer: Adaptive Versus Conventional

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Purpose: To compare dosimetric effect of stereotactic body radiation therapy (SBRT) using adaptive and conventional techniques on the Ethos system. **Methods:** A comparison for a stage T3N0 lung cancer patient is presented. This patient was originally treated on a linac using SBRT with a prescription dose of 50Gy in 5 fractions covering at least 95% of PTV. Cone beam CT (CBCT) was performed to setup the patient each fraction. Retrospectively, an SBRT plan with the same dose prescription and similar target and normal tissue constraints was generated using the Ethos planning system (Varian, Palo Alto). Simulated dose deliveries were performed using the Ethos emulator based on the CBCTs of each clinical treatment. Two treatment delivery schemes were compared, Conventional: five regular treatments using the initial plan, Adaptive: five treatments adapted by Ethos to match the target on the clinical CBCTs. Delivered dose was deformed back to the planning CT to compare the distributions on target and normal tissue for these two schemes. **Results:** The PTV volume and shape changed during the 5 fractions SBRT treatment process. The volume started at 120.3cm³, which is close to the planning CT volume of 121.4cm³, then increased to 134.9cm³ at the second fraction, and dropped to 84.8cm³ at the fifth fraction. The delivered PTV coverage of prescription dose is 92.9% for adaptive delivery and 88.1% for conventional delivery. The lung receiving 12.5Gy or more is 121cm³ for both delivery schemes. The maximum dose to heart is 54.4Gy and 55.5Gy for adaptive and conventional delivery scheme respectively. **Conclusion:** The adaptive dose delivery managed to maintain planning target coverage, while keeping equivalent normal tissue sparing to the conventional dose delivery.

PO-GePV-M-90

Impact of Dynamic Intrafraction Air Cavity Changes During MR-Guided Adaptive Radiation Therapy for Abdominal Tumors

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Purpose: The intrafraction anatomic change, especially the air cavity variability in the abdominal region, can be severe which can affect the accuracy of MR-guided adaptive radiation therapy (MRgART). The purpose of this study is to accurately quantify the intrafraction abdominal air cavity change using a dedicated MRI sequence and to study its dosimetric effect on MRgART. **Methods:** Abdominal air pockets were imaged using a dedicated, free-breathing air scan MRI sequence (3D FLASH, 1 msec echo time, 9 second scan time). A total of 21 air scans were acquired during pre-treatment imaging study sessions for 5 pancreatic cancer patients, each with 4 or 5 air scans separated by 2-5 minutes. Changes in all air cavities compared to the first air scan for each patient were quantified using the Dice similarity coefficient (DSC). An adaptive plan was generated on the first air scan and was re-calculated on subsequent air scans with the same beam settings. Changes of the dose to a 0.03cc volume (D0.03cc) for stomach, duodenum, colon and small bowel, and PTV coverage by 100% prescription (V100%) from the first air scan were calculated. **Results:** The volume of the air cavities ranged from 10cc to 50cc in the treatment field. Average DSC dropped to < 0.2 after 5 minutes. For two patients, the D0.03cc was increased by 15% to 20% in 2 to 5 minutes, while for the other three patients, the D0.03cc increment was < 4.7%. The PTV V100% dropped by 17% for one patient. **Conclusion:** Intrafraction changes in the position and shape abdominal air cavities during 20-25 minutes may be substantial, and can result in considerable dose increases to critical organs and/or decreases in target coverage. These data suggest the need for developing real-time adaptive planning solutions to account for dynamic intrafraction air cavities changes.

PO-GePV-M-91

Utilization of the QUART Phantom to Evaluate Imaging Stability of a CBCT-Based Online Adaptive Workflow

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Purpose: To evaluate the quality of image sets within the Ethos adaptive treatment workflow (EATW) using a vendor-provided phantom. **Methods:** The QUART phantom (Varian GmbH, Switzerland) supplied with the Ethos linear accelerator is made of acrylic with a central module containing material inserts (air, Teflon, and polystyrene). The central module is sandwiched between two uniform background modules. Three image sets were used in the evaluation: planning CT (pCT), kV cone-beam-computed-tomography (CBCT) Head-iCBCT protocol scan taken during EATW, and synthetic CT (sCT) created using deformable registration of pCT and CBCT during EATW. The image quality tests of focus were CT Hounsfield Unit (HU) constancy, uniformity, and noise. HU constancy was tested by comparing average HU values in inserts on the center slice to pCT in corresponding regions of interest (ROIs). Uniformity was computed using maximum absolute HU difference of the mean HU in ROIs drawn in each cardinal direction and the center. Noise was taken as the maximum standard deviation from the uniformity ROIs, and SNR and CNR values were computed following the methodology described by the vendor. **Results:** In comparison with the pCT, the maximum HU difference was 46.9±44.4 HU for the CBCT and 0.48±17.4 HU for the sCT. The uniformity value was 4.3±18.4 HU for the pCT, 33.4±34.7 HU for the CBCT, and 4.5±18.6 HU for the sCT. The noise was 14.0 HU for the pCT, 31.2 HU for the CBCT, and 14.4 HU for the sCT. **Conclusion:** Across the tests performed, a consistent trend was observed that showed a recovery of the pCT image quality in the sCT. This comparability to pCT is important because the sCT is used for contouring, plan optimization, and dose calculation within the EATW. Future work includes an automated analysis quality assurance solution to incorporate a more comprehensive set of image quality tests.

PO-GePV-M-92

Evaluation of Online Planning Strategies for CBCT-Based Adaptive Radiotherapy for Locally Advanced Lung Cancer

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Purpose: For patients with locally-advanced lung cancer treated with fractionated RT (60-66 Gy), toxicities, such as radiation pneumonitis, are driven by volume-effect dose indices (e.g. V20). If, during ART, daily lung volumes are aligned to the reference CT, it is unlikely that a significant reduction in V20 or mean-lung-dose (MLD) will be realized since large discrepancies will require mismatch of large volumes of lung. Here, we investigated strategies beyond those used traditionally to minimize OAR dose to understand the potential benefit of daily ART. **Methods:** Retrospective data were evaluated from 10 locally-advanced, NSCLC patients previously treated on the Varian Halcyon. Adaptive workflow simulation was performed using an Ethos emulator on a virtual machine. Patients were originally planned using target and OAR criteria outlined in RTOG 1106. Separate plans were calculated for a set of pre-defined treatment geometries (IMRT or VMAT). Planning strategies compared in this study included use of standard RTOG 1106 optimization constraints (OPT_S) alone, and combined with ipsilateral lung constraints (OPT_IL) and ring constraints (OPT_RING). Plans were adapted for every fraction. **Results:** Target coverage was maintained within RTOG 1106 constraints. Reduction of $1.11 \pm 0.78\%$ ($p=0.03$) and $0.26 \pm 0.13\text{Gy}$ ($p=0.01$) was observed in V20 and MLD, respectively for OPT_IL relative to OPT_S. Corresponding reduction for OPT_RING was $0.45 \pm 0.54\%$ ($p=0.03$) and $0.14 \pm 0.19\text{Gy}$ ($p=0.05$) relative to OPT_S. One patient exceeded the V20 constraint for OPT_S but met constraints when using OPT_IL. Relative to OPT_S, significant but minor improvement was observed in heart V30 and V50 ($0.65 \pm 0.57\%$ ($p=0.01$) and $0.46 \pm 0.55\%$ ($p=0.03$), respectively) for OPT_RING and in esophagus D2cc ($-0.31 \pm 0.25\text{Gy}$, $p=0.05$) for OPT_IL. **Conclusion:** Results suggest that a planning strategy incorporating ipsilateral lung constraints may offer the best option for sparing normal lung tissue doses and glean the most benefit from daily ART for patients with locally advanced lung cancers.

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PO-GePV-M-93

Necessity of Plan Adaptation in Stereotactic MR-Guided Online Adaptive Radiotherapy

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Purpose: Stereotactic MR-guided online adaptive radiotherapy (SMART) is a rapidly emerging technique in which new treatment plans are generated based on a patient's anatomy at each fraction. However, this process can also be time and resource intensive. The goal of this study is to determine which disease sites are most likely to meet criteria for the use of adaptive techniques. **Methods:** A total of 230 patients were treated with 5-fraction (or fewer) SMART treatments on MR-guided linear accelerator at our institution (1145 total fractions) between October 2019 and February 2021. At each fraction, the patient's anatomy was re-contoured by a physician and a treatment plan was re-calculated on the current anatomy to predict the dose that would have been delivered without adaptation. This prediction was compared to pre-specified prescription metrics, and adaptation was determined to be necessary if any critical organ at risk (OAR) metrics were exceeded, or if the target coverage was unacceptable. Patients were categorized by disease site (kidney/adrenal, liver, lymphoma, oligometastatic nodes, pancreas, prostate, spine and thoracic), and the timing of each adaptive fraction was recorded. **Results:** A total of 1038 of 1145 fractions (90.7%) showed indication for adaptation, with 734 due to OAR metrics being exceeded (64.1%) and 304 due to suboptimal target coverage (26.6%). The sites with the highest proportion of fractions meeting adaptive criteria were prostate (85/85, 100%) and pancreas (310/323, 96.0%), whereas the sites that least frequently met the criteria were kidney/adrenal (71/94, 75.5%) and oligometastatic nodes (345/395, 87.3%). The median time for plan adaptation was 31 minutes (range 11-68). **Conclusion:** There was a clinical benefit to plan adaptation for the majority of patients and fractions. However, adaptive need varied by disease site, with pancreas and prostate SBRT patients most likely to meet adaptive criteria.

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AI-Based Online Adaptive Radiation Therapy for Lung Cancer Treatment

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Purpose: Retrospective studies were performed to evaluate the accuracy of Varian Ethos AI-based automatically mapped structures and dosimetric consequences of daily adaptive radiation therapy (ART) for lung cancer treatment. **Methods:** Ten locally advanced non-small cell lung cancer patients (prescribed by 2Gyx30) with 297 fractions of treatment were selected for this retrospective study on an Ethos research emulator. All patient data were copied and simulated twice: AutomaticART – automatic contours were utilized without modification with the fully automatic workflow and SupervisedART – automatic contours were modified manually by physicians. All daily structure sets and doses were exported for dose accumulation and further analysis. Dosimetric results of scheduled plan, AutomaticART, and SupervisedART were compared as ratios of relative reference plan results and grouped as lower constraints for target coverage and upper constraint for sensitive structures and hot spots of targets. **Results:** 290 (of 297) fractions were successfully analyzed. Comparing target volumes between AutomaticART and SupervisedART, Dice similarity coefficient was 0.93 ± 0.05 , mean contour distance was 1.5 ± 1.2 mm, and Hausdorff distance was 4.0 ± 2.3 mm. AutomaticART and SupervisedART significantly improved minimum doses of planning treatment volumes by 4.85 ± 3.03 Gy and 4.46 ± 8.99 Gy, respectively, with a maximum increase of 22.62 Gy for one patient. For analysis over daily results of 290 fractions of treatment, average target dose coverages were improved from 0.96 ± 0.04 (scheduled plan) to 1.00 ± 0.02 and 1.02 ± 0.04 and average upper dose/volume constraints were lowered from 1.01 ± 0.11 to 0.94 ± 0.10 and 0.93 ± 0.12 for AutomaticART and SupervisedART, respectively. For cumulative dose comparison over 10 patients, target dose coverages were improved from 0.98 ± 0.02 (scheduled plan) to 1.01 ± 0.03 and 1.01 ± 0.02 , and upper dose/volume constraints were lowered from 0.99 ± 0.03 to 0.96 ± 0.02 and 0.96 ± 0.02 for AutomaticART and SupervisedART, respectively. **Conclusion:** Accuracy of Ethos AI-based automatic contouring is generally acceptable. The online adaptive radiation therapy improves target coverage and spares sensitive structures.

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PO-GePV-M-95

Intentional Deep Overfit Learning (IDOL): An Application to CBCT-Based Auto-Contouring

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Purpose: In cone-beam CT (CBCT)-guided adaptive radiotherapy (ART), deep learning (DL)-based automatic organ-at-risk (OAR) delineation can be an important time saving that enables an online ART workflow. However, there remain several barriers to the effective implementation of these approaches in a clinical setting, including severe image artifacts in CBCT images that hinder the delineation of critical structures and the difficulty of curating a training data set consisting of consistent manual annotations. We propose here a patient-specific auto-contouring approach utilizing the Intentional Deep Overfit Learning

(IDOL) framework that demonstrates a significant improvement in contouring accuracy using only a few training data sets. **Methods:** IDOL-based auto-contouring is an application of intentional overfitting that leverages patient-specific prior information. Training in the IDOL framework involves two stages: 1) conventional training of a generalized segmentation model with a training set of 30 CBCTs and manual contours, and 2) using the planning CT and contours of a 31st patient as prior information, the generalized model is trained further to achieve the IDOL model that is applied to the CBCT of the 31st patient. For testing, step 2 was repeated for patients 31-39 using the same generalized model as a starting point. **Results:** The proposed method was evaluated on a cohort of 9 head and neck cancer patients. The generalized model (step 1 training only) achieved Dice similarity coefficients (DSC) of 0.80 and 0.64 for the larynx and esophagus, respectively. By adopting the personalized IDOL model in each test case, the DSC improved to 0.89 (+0.09) and 0.74 (+0.10) for the same structures, showing a significant improvement in contouring accuracy. **Conclusion:** In this study, we have successfully applied the IDOL framework to the CBCT auto-contouring task. Our approach is widely applicable to common tasks in radiotherapy and can be utilized for CBCT or MRI-guided ART.

PO-GePV-M-96

Development of a Quality Assurance Process for Dose Accumulation in a CT-Guided Online Adaptive Radiotherapy System

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Purpose: The Varian ETHOS system provides the ability to adapt treatment based on daily changes in patient anatomy and also allows for dose accumulation. In this study, we develop a simple approach for routine quality assurance (QA) of dose accumulation. **Methods:** An electron density phantom was chosen for QA due to its wide availability in clinics. The inner section of the electron density phantom consists of nine cylindrical inserts. On a CT scan of the phantom one GTV (insert at 6 o'clock) and seven OARs (other inserts) were contoured in the treatment planning system (TPS). Two adapted sessions were then performed in which the GTV was re-contoured first at 7 o'clock and second at 5 o'clock. The accumulated dose was exported from the ETHOS TPS after delivery. Deformable vector fields were also exported to manually accumulate dose for comparison. Two approaches were considered to evaluate the dose accumulation. First, the mean dose in each structure was manually retrieved from each session, and then summed for comparison with the TPS-reported accumulated mean dose. In addition, manually-obtained and TPS-generated accumulates dose volumes were compared by Gamma analysis (2%/2mm). **Results:** The mean dose differences in the GTV was 0.2% and in the OAR1-7 were 38.3%, 16.0%, 0.8%, 0.7%, 0.3%, 1.0%, and 1.5%, respectively between the manually-summed dose and ETHOS-generated accumulated dose. Gamma passing rate between accumulated dose volumes was 99.4%. **Conclusion:** This study demonstrated strong agreement between manually obtained accumulated dose and the ETHOS dose accumulation in terms of structure mean dose values for the target and most OARs) and 3D dose distributions. Observed disagreement in the OAR1 and OAR2 is under further investigation and likely stems from the fact that these transformations (e.g. permutations in position) are not physically realistic.

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PO-GePV-M-97

Identification of Morphological Changes in Head-And-Neck Patients Through In-Beam PET Monitoring in Proton Therapy

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Purpose: In-beam Positron Emission Tomography (PET) can be used for in-vivo non-invasive treatment monitoring in proton therapy. Although this modality has been frequently applied, no straightforward method exists to translate the PET images into easy to interpret information for clinical personnel. The purpose of this work is to propose an innovative statistical method for analyzing in-beam PET monitoring images to locate possible morphological changes occurring over the course of treatment, and to quantify and easily visualize them. **Methods:** We investigated the impact of anatomical changes on the in-beam PET signal of a patient with Squamous Cell Carcinoma, that presented sinonasal cavity emptying during the course of treatment. For this purpose, we artificially created a set of CT scans with various volume changes of the cavity region and performed a vast amount of Monte Carlo simulations of a proton therapy treatment to model PET signal, using the characteristics of a dual head in-beam PET system. We performed voxel-wise two-tailed statistical tests, resembling the Voxel-Based Morphometry (VBM) method commonly used in neuroimaging data analyses, to locate regions with significant morphological changes. **Results:** The VBM resembling method could be applied to the simulated in-beam PET images, despite the image artifacts and limited statistics. Three-dimensional probability maps were obtained, that allowed to locate and quantify interfractional morphological changes. The characteristic color patterns resulting from the method is appropriate to trigger alarms if morphological changes occur along the course of treatment, resulting in an under- and/or overdose exposure. **Conclusion:** Our method is promising to apply to in-beam PET monitoring data. Using simulated PET monitoring images, we showed that the method allowed to correctly identify the regions that changed, and to quantify and visualize them in a straightforward manner. This new approach is promising and possibly brings us a step closer to dose reconstruction.

PO-GePV-M-98

Clinical Assessment of Patient Variations Between Treatment and Simulation in Breast Deep-Inhalation Breath Hold (DIBH) Using Optical Surface Imaging and On-Board 2D Radiography

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Purpose: To guide surface-guided radiotherapy (SGRT) setup for breast DIBH patients and determine if the simulation CT surface reference should be used or treatment surface reference is needed through a retrospective assessment of 28 patients' daily setup data of both SGRT and 2DkV in 25-fraction DIBH treatments. **Methods:** Twenty-eight left-sided breast cancer patients treated with DIBH in 25 fractions were studied retrospectively. These patients had locally advanced breast cancer and no previous RT. Volumetric-modulated arc therapy (VMAT) technique was applied, targeting both the breast and local lymph nodes, and both daily AlignRT surface imaging and 2DkV were prescribed to ensure the accuracy. AlignRT was first applied to match the body contours of free-breathing and DIBH CTs, and then 2DkV was used to align the anterior bones at DIBH for final adjustment and then capture a new on-site surface reference for DIBH gating. Therefore, setup differences between the 2 imaging modalities were recorded and the vertical shifts were compared for all patients and all fractions. It was hypothesized that patients who had no previous RT experience may need time to adapt and achieve a consistent DIBH, and therefore, they were likely to go through changes before settling down with consistency. To test this, the average of vertical shifts in first 5 fractions was compared to that of the remaining 20 fractions. **Results:** Among 28 patients, 14 patients' vertical setup shifts support the hypothesis: 8 patients have smaller mean shifts in the first 5 fractions than remaining fractions by at least 50% while 4 patients' data show the opposite. The remaining 14 patients' data show no significant difference. **Conclusion:** This study has demonstrated that 50% patients experience substantial difference between the first 5 fractions and remaining 20 fractions. This suggests that SGRT reference surface may need to update during DIBH treatment.

PO-GePV-M-99

Comparison of Three Setup Methods of Skin Marker Alignment Using Diamond-Shaped Light Field for Prone Breast EBRT

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Purpose: For the positioning of prone breast patients treated with tangential fields, a diamond-shaped light field(DSLF) is used daily at our institution to align with a corresponding skin marker besides IGRT. This study compares three different DSLF setups used in several hospitals to help standardize/optimize the clinical workflow. **Methods:** Sixty-one prone breast patients with daily kV tangential IGRT following DSLF were retrospectively categorized into three groups according to their DSLF setups in different hospitals: 1) field size(FS)=10x10cm² at central axis with direct lateral gantry(=90 or 270°), 2)Same as 1) except FS= 4x4cm², and 3) FS=4x4-6x8cm² at off-axis with tangential gantry as treatment beam (so that DSLF is approximately at the breast center). All the kV-based couch shifts in vertical and longitudinal directions were analyzed. The DSLF-only dose distribution was simulated by reversely shift the couch (to exclude IGRT) from the CT plan, which was assumed equivalent to the delivered dose when both DSLF and IGRT are used for patient alignment. The simulated DSLF-only dose to the target and OAR structures throughout the treatment course were calculated. **Results:** The mean daily couch shifts following DSLF No.1, 2, 3 are 0.64, 0.68, 0.8cm with standard deviation of 0.4, 0.3, 0.6cm respectively. The average target dose ratios of DSLF-only to DSLF-and-IGRT are 0.99, 0.99, 1 for three setups. Most planning objectives are still achieved for DSLF-only scenarios. The DSLF-only dose values to heart, chest wall and lung increases from setup No.1 to 2 to 3. Generally, Setup No.1 appears to spare OARs the best, whereas setup No.3 results in greater target coverage, and setup No.2 is in between No.1 and No.3. **Conclusion:** In general, the three DSLF setups are similar but yield different tradeoffs between target coverage and OAR dose sparing when daily IGRT is lacking. Our finding may help select DSLF setup.

PO-GePV-M-100

Evaluating Impact of Target Location on Intrafraction Target Motion for Lung SBRT

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Purpose: To assess the impact of anatomical target location on intrafraction target motion for SBRT lung treatments. **Methods:** 4DCBCTs from 89 SBRT lung patients treated between 2018-2019 were analyzed retrospectively. Patients were grouped based on target location (50 upper lobe (UL), 8 middle lobe (ML), 31 lower lobe (LL)). GTV defined at phase-0 of planning CT was propagated to phase images of pre-delivery (with couch correction applied) and post-delivery 4DCBCTs using deformable image registration (Elekta/Admire). Target centroid was used to calculate motion amplitude and mean target position (averaged over 10-phases). Differences in amplitude and mean target position between pre-delivery and post-delivery 4DCBCT was used to assess intrafraction motion variation (IFV). Results were compared to institutional PTV margins of 5mm (anteroposterior, mediolateral) and 6mm (craniocaudal) and statistical significance evaluated with ANOVA. **Results:** Absolute intrafraction target motion amplitude differences in mediolateral and anteroposterior directions was similar between lobes. In the craniocaudal direction, significant differences were seen ($p < 0.005$) between lobes, with mean $\pm$ sd [range] of 1.15 $\pm$ 1.21 [0.0-6.0]mm, 0.75 $\pm$ 0.63 [0.0-2.6]mm, and 0.57 $\pm$ 0.52 [0.0-2.7]mm for LL, ML and UL. Absolute mean target position differences were statistically significant in mediolateral ($p = 0.03$) and craniocaudal ($p < 0.005$) directions, while anteroposterior was not ($p = 0.09$). In the craniocaudal direction mean $\pm$ sd [range] differences were 2.00 $\pm$ 1.78 [0.01-13.13]mm, 1.56 $\pm$ 1.32 [0.06-4.79]mm, and 1.35 $\pm$ 1.44 [0.01-8.77]mm for LL, ML, and UL. Comparison of mean target position differences to PTV margins found 0% of treatments had ≥ 5 mm difference in the mediolateral direction for all lobes, whereas 5.26%(LL), 3.13%(ML) and 0.42%(UL) had ≥ 5 mm difference in anteroposterior direction and 1.50%(LL), 0.0%(ML) and 1.26%(UL) had ≥ 6 mm difference in craniocaudal direction. **Conclusion:** Absolute IFV difference in target motion amplitude and mean target position were largest in the craniocaudal direction for LL lesions, with some lesions surpassing existing PTV margins. Further investigation is needed to determine the dosimetric impact of the IFV differences observed.

PO-GePV-M-101

Respiratory Traces Analysis in Left-Breast Irradiation with Deep Inspiration Breath Hold

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Purpose: The use of deep inspiration breath-hold (DIBH) for respiratory gating in radiotherapy targeting left-sided breast cancer has led to significant reduction of radiation dose to the heart and lungs. The aim of this study was to assess the DIBH respiratory traces. **Methods:** 45 left breast cancer patients with a mean age of 60.6 (SD 11.1) years were enrolled in this study. The patient was coached and asked to perform at least three breath-hold (BH). In total, 146 DIBH traces were acquired and collected during CT simulation using Varian Real-time Position Management (RPM) system. The DIBH respiratory traces stability and reproducibility within the gating window were assessed. The correlation between the DIBH stability/reproducibility and other parameters such as age, isocenter drift, and DIBH amplitude were analyzed. **Results:** The mean of DIBH stability and DIBH reproducibility were 0.4385 cm (SD 0.2233) and 0.3462 cm (SD 0.2905) respectively. 65% of patients had stability less than 0.5 cm and 80.0% of patients had reproducibility less than 0.5 cm. The difference between free-breathing and DIBH amplitude showed an average of 2.10 cm (SD 0.75). The optimal gating window was 0.70 cm (SD 0.52). The DIBH reproducibility showed significant correlation with isocenter drift ($r = 0.47$, $p = 0.0012$). Patients older than 60 years showed significant increased variability in DIBH stability ($F = 4.87$, $P = .0007$). **Conclusion:** We analyzed DIBH respiratory traces for left breast cancer patients, the optimal gating window was 0.7 cm. The range of 0 to 0.5 cm is advisable for DIBH reproducibility and stability. Additional coaching may be needed for patients older than 60 years. The finding of this work could be used as a metric tool to identify the potential candidate for the DIBH technique.

PO-GePV-M-102

Integrating Temporal Physical and Biological Dose Distribution Analysis in Imaged-Guided Hypofractionated Radiosurgery of High-Risk Brain Lesions

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Purpose: High-risk brain lesions are large, located in eloquent brain region, and/or with aggressive histology. Online CBCT imaging guidance has primarily focused on daily patient setup without detailed analysis of dose accumulation effects along the full course of treatment. In this study, we aimed to develop a time series analysis approach by integrating temporal analysis with imaging guidance prior to the beam delivery for each treatment fraction in hypofractionated Gamma Knife radiosurgery of high-risk brain lesions. **Methods:** As a proof-of-concept, eight lesions (mean volume=17.4 $\pm$ 15.8 mL) from five patients treated to 30Gy in five fractions with online CBCT-guided hypofractionated SRS were retrospectively analyzed. CBCT scans were acquired at each fraction prior to delivery. 3D dose distributions were projected onto daily CBCT and compared to the planned treatment. The accumulated and biological equivalent dose over previously delivered treatment fractions were also computed to form a time series across all fractions. Comparison of 3D dose distribution against the original dose distributions for each treatment fraction was analyzed. **Results:** Examining the full treatment, a mean target dose deviation of -0.21Gy (-0.10Gy, -0.35Gy) was obtained averaging all cases/time points and represented an almost negligible deviation of <1.5% in target dosing. Further, a mean target EQD2 deviation of -0.42Gy (-0.2Gy, -0.7Gy) was also obtained, representing <3% deviation from the initial prescription. The largest dose deviation occurred with two lesions within 3cm. No systematic trend in the accumulated dose patterns was detected. **Conclusion:** Integrating temporal dose accumulation time series

analysis has been demonstrated for the first time to be best of our knowledge. Initial analysis on a small number of cases pointed to precision and robustness of CBCT image-guided hypofractionated SRS of high-risk intracranial targets. On-going analysis over a large number of cases treated across different modalities will lend more support to encouraging results already seen.

PO-GePV-M-103

Development of a Portable Body Surface Monitoring System Using Depth Camera

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Purpose: Currently commercially available surface guided radiation therapy (SGRT) device is usually fixed on ceil or mounted on gantry. This has limited the lay-out and usability. On the other hand, a portable SGRT system allows SGRT to be performed from any camera angle and position. The purpose of this study is to develop a potable body surface monitoring system and to evaluate the system for clinical implementation. **Methods:** We developed a body surface monitoring system that was consisted of Kinect (Azure Kinect DK, Microsoft), which has RGB and depth camera, and software programmed by Python. Coordinate systems of our system and treatment room were aligned by the calibration using RGB images of chessboard. Depth map of the phantom captured by Kinect were converted into 3D point cloud. Finally, the phantom movement was calculated by the rigid registration using point-to-point iterative closest point (p2p-ICP) and point-to-plane ICP (p2pl-ICP) algorithms. The phantom movement were derived from the transformation matrix by two ICP algorithms. Positional detectability was evaluated using general head and neck phantom. The phantom was moved in the range of 10 to 20 mm in the x, y, and z directions, respectively, using the treatment couch (Imaging Couch, Brainlab). The detection accuracy was compared between two ICP algorithms. **Results:** As results of detectability evaluation, translational error of p2p-ICP was not satisfied the tolerance of 2 mm defined by AAPM TG-147 in 2 conditions. On the other hand, translational error of p2pl-ICP was satisfied the tolerance in all conditions. The largest error of p2p-ICP and p2pl-ICP were 2.8 mm in z direction and 1.4 mm in x direction, respectively. **Conclusion:** We developed the portable body surface monitoring system and evaluated the basic performance. These results show our system with p2pl-ICP algorithm has the potential for clinically implementation.

PO-GePV-M-104

Management of Urgent Sim-And-Treat Palliative Cases Via Halcyon KV-ICBCT Images

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Purpose: Halcyon O-ring Linac is equipped with large field of view (FOV) kV-CBCT imager for image-guided RT. 3D reconstruction potentially enables patient's CT simulation on Halcyon. We report the feasibility of using Halcyon's kV-iCBCT imaging for simulation in extenuation circumstances, e.g. CT simulator downtime. **Methods:** Halcyon's kV-iCBCT reconstruction algorithm was used to image RANDO body phantom for different palliative treatment sites and beam arrangements (whole brain RT, single-field c-spine, AP/PA thorax, 4-fields pelvis and 2-fields thigh). Pelvis iCBCT protocol was used regardless of treatment site due the available maximal FOV of 25cm in superior/inferior (S/I) and 50cm in right/left lateral (R/L) directions. Three-point BB's were placed on each phantom to replicate standard clinic workflow for isocenter-localization in treatment planning system (TPS). Palliative 3D-CRT treatment plans were created without heterogeneity correction in Eclipse TPS via AAA algorithm for a TrueBeam Linac for 6/18MV beams. Overall phantom set up and scanning time was <5min. Total MU/fraction and qualifiers of minimum/maximum point doses and V100 were used for plan evaluation. **Results:** In palliative care setting, all five RT plans for various treatment sites produced expected outcomes for total MU with 3D-CRT planning regardless of homogenous calculation. AP/PA thorax/mediastinum plan demonstrated slight increase in expected MU due to tissue-to-air heterogeneity within lung volumes that was not being accounted for in dose calculation, although it was clinically acceptable in the palliative setting. **Conclusion:** Halcyon's kV-iCBCT was tested for simulation of palliative patients. FOV enables full viewing to permit the use of treatment fields. Dose calculation without heterogeneity correction is reasonable in most or all palliative cases, with caution needed for beams passing through the lung. Generating an electron density curve for Halcyon kV-iCBCT is ongoing. Stitching multiple scans together (if needed) for >25 cm S/I anatomy is possible, and work in this area is underway.

PO-GePV-M-105

A Non-Image Based Mechanism to Detect Significant Patient Motion for a Frame-Based Radiosurgery Treatment Platform

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Purpose: Motion management is important while delivering high-dose radiation therapy. Image-based systems are often considered the gold standard; however some frame-based radiosurgery platforms lack the ability to perform imaging. This work assesses the ability of time-of-flight (ToF) distance sensors to monitor and detect gross patient motion in the absence of an image-based patient position verification system. **Methods:** Two low-cost distance sensors were assessed – VL53L1X and the TF MiniPlus. Both sensors operate on the time-of-flight (ToF) principle by emitting a laser beam and measuring its travel time to measure distance. Accuracy and precision were measured from 20cm to 90cm in 10cm increments. Fifteen readings were obtained at each position. The sensors were controlled by a microcontroller (Arduino Uno) and the readings were transferred to a computer through a serial port for further analysis. Additionally, both sensors were utilized in conjunction with a head phantom to assess stability over a 200s period and the ability to detect arbitrary movement induced by the operator. **Results:** The VL53L1X sensor produced more accurate results over the range of measurements performed while the TF MiniPlus produced more precise results. Over the measured range of distances, the TF MiniPlus sensor accuracy varied from 3% to 6% and the precision varied from 0.03% to 0.21%. The VL53L1X sensor accuracy varied by $\pm 2\%$ and the precision varied from 0.57% to 0.16%. When utilized with the head phantom, the sensors were able to detect an arbitrary induced movement of 5mm. **Conclusion:** These devices provide a mean to assess gross (> 4 mm) intrafraction patient motion in situations where image guidance or clear visual line of sight challenges may exist. This work serves as a proof of concept for industrial-grade sensors and/or multiple sensors which can be utilized to further reduce uncertainty and improve motion detection.

PO-GePV-M-106

A Case-Study Investigation of the Feasibility of PTV Expansion Reduction for Dose Escalation for Anal Cancer

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Purpose: Several trials have investigated the impact of dose escalation on anal carcinoma and demonstrated increased colostomy-free survival. However, the high incident of treatment cessation due to acute toxicity may limit disease-free survival gains. ACCORD 03 and RTOG 0529 recommend PTV expansions of 3.0 and 3.5 cm, respectively. This study aims to assess the impact and feasibility of reducing these expansions. **Methods:** We assessed various PTV expansions (3.5, 3.0, 2.5, 2.0, and 1.5 cm) for a T3N1 locally advanced anal cancer patient treated with 54 Gy in 30 fractions. Daily CBCTs were registered to the planning CT using the treatment shift, and GTVs were contoured by a physician on each CBCT. GTV volumes ranging outside the PTVs were assessed and analyzed. **Results:** Daily GTVs remained well-within the 3.5 to 2.0 cm PTV expansions. GTVs were outside the 1.5 cm expansion PTV for 7 treatment days, with a maximum non-overlap volume of 1.1 cc and < 0.1 cc volume outside on two days. The patient's GTV volume on CBCT decreased markedly over the course of treatment, from 134.6 to 55.0 cc -- a reduction of 40.9%. **Conclusion:** In this case study, the daily GTVs remain well within the 2.0 cm expansion PTV with minimal volumes outside the 1.5 cm PTV expansion, indicating the feasibility to improve OAR sparing while retaining coverage. Given the 3.5 and 3.0 cm

PTV volumes are 1166.8 and 930.1 cc, respectively, compared with the initial planning GTV volume of 109.534.6 cc, the 2.0 cm PTV expansion resulting in 548.1 cc is reasonable. The GTV exhibited a substantial volume decrease, indicating it may be possible to monitor and safely further reduce the PTV expansion. This study demonstrates the feasibility of reducing PTV expansions for the purpose of anal cancer dose escalation.

PO-GePV-M-107

Target Motion Tracking with LLE-Based Features Using Surface Imaging Model

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Purpose: We propose to identify the target location inside human body based on features acquired from simple surface imaging with 4DCT and LLE (Locally Linear Embedding). **Methods:** Our model is based on two 10-phase lung 4DCTs taken from the same patient within half-hour interval. We labeled targets in 4DCT, and then assigned 10 target positions initially corresponding to each phase of 4DCT. The first 4DCT is for training, and second is for testing. Firstly, a bounding box containing the anterior breast region of patient is contoured. Secondly, the outer skin surface of chest region is transformed into a mask, and the mask is extended into a single column vector with a high dimension. LLE (Locally Linear Embedding) is applied to generate features in lower 9-dimension space. Test images are also transformed into column vectors, embedded into lower dimension, and the corresponding distances are calculated between training and testing images based on features. **Results:** LLE is able to preserve the spatial relations of 4DCT from different phases to the lower dimension. LLE-based on features showed high correlation (0.9, p-value 10⁻⁴ in SI Direction) with target position. We also investigated location of target on the second 4DCT with LLE-based features. We achieved best result when the number of neighbors in LLE was 2, and the number of selected features was 2. There were approximately 2 position deviations (0.7cm) between prediction and the actual positions. **Conclusion:** LLE can estimate the location of target inside body with simple skin surface imaging during treatment. This could be used with surface tracking methods such as Align RT and C-Rad to reduce tracking uncertainties.

PO-GePV-M-108

Fusion of Multi-Scale Feature Maps Improves Anisotropic Structure Detection from CT Images

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Purpose: Image guidance has been widely used in radiation therapy. Correctly identifying the bounding box that contains a region of interest (ROI), is an essential step in classical and deep learning image analysis. Unfortunately, not many anatomical structures have isotropic shapes. It is challenging to select feature maps from a single scale to detect an anisotropic structure. In this study, we improved our detection network architecture presented in AAPM2020 to achieve a better overall detection performance from CT images. **Methods:** We improved the network architecture by fusing the feature maps from multiple scales and letting data to decide how to combine them. To detect anisotropic shapes, this fusion was done independently in left-right, anterior-posterior and superior-inferior directions. We applied our network to two applications: 1) T12 detection for checking gross misalignments and 2) the prostate, bladder and rectum detection for adaptive therapy. The model of T12 detection was trained on thirty-five abdominal CT scans from the same imaging protocol and tested on fourteen CT scans. The model of the prostate, bladder and rectum detections was trained on thirty training CT data and tested on six testing datasets. In both applications, no test images were used during training. **Results:** Compared with the previous approach, our latest one reduces the box size uncertainty by 1/3 to 1/2; all mean box size errors are reduced, especially on T12 (11.07 mm to 5.33mm). The compromise of the center accuracy is negligible (<1mm). **Conclusion:** Our latest approach improves the reliability of accurately detecting structures with anisotropic shape in multiple sites for various applications. This can help to avoid major positioning errors in IGRT. Also the improved network demonstrated the ability to simultaneously detect multiple objects of a wide range of scales.

PO-GePV-M-109

X-Ray Imaging Practices in Radiation Therapy in the United States

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Purpose: To obtain a snapshot of X-ray imaging practices for Radiation Therapy (RT) in the United States. **Methods:** The ICRP TG-116 “Radiologic aspects of imaging in Radiation Therapy” developed a comprehensive survey to understand x-ray imaging practices in RT. Requests were sent to 72 physicists at US treatment centers. Responses were collected between August and October 2020. **Results:** Responses were received from 30 centers (57% were academic institutions) covering 143 linacs. The monthly patient loads (median:117 ranged between [20 – 380] patients per center. Reported imaging modalities for treatment planning include CT, MRI, PET, PET/CT, SPECT, angiograms, ultrasound, and optical systems. 92% of linacs are equipped with kV imaging, 70% of which are <10 years old. 80% of centers use IGRT for >75% patients, with planar kV-imaging and CBCT as standard modalities. All centers reported employing kV-CBCT routinely except for palliative patients. MV-MV pairs remain heavily used in public (tax-payer funded, not-for profit) centers and academic centers. The most frequently treated sites are lung, prostate and breast. The main imaging frequency is “daily” (>90%) except for breast (50% daily, 47% weekly, 3% T.I.W). Most cited reasons for optimizing imaging protocols were patient habitus and adjusting FOV to improve anatomical visualization. Reported imaging dose optimization methods included pediatric protocols for radiologic imaging, and employing Surface Guided RT (breast). Academic RT centers tend to spend more effort on imaging protocols optimization, facilitated by the easier access to local imaging physicists for consultation. Most centers perform monthly imaging QA, citing AAPM TG-142 as reference. However, only 63.3% of centers include imaging dose measurement as part of their QA protocols. **Conclusion:** IGRT has become the standard of care in the US. Most academic centers optimize their imaging protocols, but image quality is the main motivator rather than dose optimization, except for pediatric patients.

PO-GePV-M-110

Dosimetric Impact in Radiation Therapy When Using Dual-Energy CT Reconstructed Virtual Monoenergetic Images In Combination With Clinical Monoenergetic CT-Density Curves

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Purpose: The Dual Energy Monoenergetic Algorithm (DEMA) (Siemens AG) allows for the creation of virtual monoenergetic CT images (VMIs) from dual-energy CT (DECT) scans. VMI datasets produced for 40 keV through 150 keV were evaluated to determine the dosimetric impact associated with using a predetermined VMI dataset for radiation therapy treatment planning in concert with a standard 120 kVp CT-density curve. **Methods:** A Siemens SOMATOM Confidence RT Pro was used to scan an anthropomorphic phantom and a calibrated Gammex with materials ranging in density from 0.300 g cm⁻³ to 1.824 g cm⁻³. Scans were taken using a tube potential of 120 kVp. Additionally, two consecutive scans were collected for DECT with tube potentials of 80 and 140 kVp. 40 – 150 keV VMIs were created from the DECT scans. Mass density vs Hounsfield unit (HU) curves were analyzed for all Gammex scans. A VMAT treatment plan was created on the anthropomorphic phantom data sets and doses were computed and compared for the 80keV VMI and 120kVp scans. **Results:** The 80 keV VMIs agreed best with the standard 120 kVp CT-density curves. For mass densities ≤ 1.340 g cm⁻³, mean HU values differed by less than 5 HU. However, for mass densities ≥ 1.560 g cm⁻³, mean HU values differed by less than 35 HU. VMAT treatment plans computed on an anthropomorphic phantom demonstrated clinically equivalent dosimetry when computed on 80keV and 120kVp scans using a common CT-density curve. **Conclusion:** Dosimetric differences were found to be negligible when using an 80 keV VMI dataset for radiation therapy treatment planning with a standard 120 kVp CT-density curve.

It is feasible to implement DECT for clinical radiation therapy dose calculation while using a single common CT-density curve within a treatment planning system for all CT protocols.

PO-GePV-M-111

Liver Tumor Motion During SBRT with Different Motion Management and IGRT Strategies: A Systematic Review On Inter, Intra-Fraction and CT-Sim Motion

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Purpose: There is large variability among clinics how ITV and PTV margins are defined for liver SBRT. In this work we evaluated via systematic review the inter, intra and CT-Sim/4DCT the liver motion during stereotactic body radiation treatment (SBRT) with various immobilization devices and image-guided radiation therapy (IGRT) techniques. **Methods:** PRISMA guidelines were followed for systematic review and meta-analysis. Studies reporting liver motion during SBRT, and using motion management techniques and IGRT were searched in PubMed and Embase databases. From an initial 533 studies published before July 2020 were identified; 43 met our selection criteria. The 43 studies were categorized into 29 free-breathing (FB; n=690 patients), 10 abdominal compression (AC; n=257 patients) and 4 breath-hold (BH; n=74 patients) studies. For each motion category, we recorded amplitude mean, standard deviation, range, systematic and random errors in superior-inferior (SI), left-right (LR) and anterior-posterior (AP) directions. Pooled weighted bubble-plots were generated on mean motion with 95% confidence interval in each direction and one-way ANOVA tests were performed on SI-LR, SI-AP and LR-AP directions. **Results:** For FB treatments, the pooled weighted mean and standard deviation in inter-fraction, intra-fraction and CT-Sim/4DCT motion (SI,LR,AP) were (4.8±4.0mm, 1.6±1.4mm, 2.3±2.6mm), (8.8±4.4mm, 4.6±2.6mm, 3.3±2.2mm) and (9.2±3.6mm, 1.9±1.5mm, 4.7±2.3mm), respectively. There was significantly higher intra-fraction motion in comparison to inter-fraction motion (p<0.01). For BH treatments, inter-fraction, intra-fraction and CT-Sim/4DCT motion were (1.0±7.0mm, 0.0±5.0mm, 1.0±3.0mm), (0.2±6.4mm, 0.4±4.1mm, 0.1±4.9mm) and (-0.1±1.2mm, 0.1±1.2mm, 0.1±1.3mm), respectively. For AC treatments, inter-fraction, intra-fraction and CT-Sim/4DCT motion were (-0.9mm, 0.1mm, 0.2mm), (1.0±1.2mm, 0.5±0.8mm, 0.7±0.7mm) and (6.7±4.0mm, 2.2±3.6mm, 3.1±2.7mm), respectively. **Conclusion:** The average inter- and intra-fraction liver motion during FB, BH and AC SBRT varies based on the motion management method. For FB treatment, significant motion necessitates use of an internal target volume and could be reduced with BH and AC. Further analysis will be presented on optimizing margin protocols.

PO-GePV-M-112

Evaluation of Larynx Displacements in Pre-Treatment CBCT Scans for a Population of Laryngeal Cancer Patients

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Purpose: The purpose of this work was to evaluate the daily alignment accuracy of pre-treatment cone-beam computed tomography (CBCT) scans for laryngeal cancer patients. **Methods:** CBCTs from the 1st, 11th, and 21st treatment fractions were exported to the MIM software package for 20 laryngeal cancer patients treated with VMAT. Each CBCT was rigidly registered to the treatment planning CT separately with respect to bony and soft tissue alignment. The displacement of the larynx was defined as the difference between the bony and soft tissue alignments, recorded for each of three orthogonal directions (right/left, anterior/posterior, and superior/inferior). In addition, the magnitude of the soft tissue-based registration shift from MIM was compared to the clinically recorded shifts from the patient's treatment to provide quality assurance (QA) of the therapist generated registration shifts. **Results:** The average larynx displacements in the R/L, A/P, and S/I directions were 0.0+/-1.0, -0.4+/-1.4, and 0.6+/-2.5 mm, respectively, with a maximum of 2.6, 5.5, and 8.9 mm. For each direction, 95% of all displacements were within 2.3, 3.0, and 5.8 mm. 4 out of the 20 patients recurred locally; however, there is no appearance of correlation of local recurrence with large larynx displacements (greater than 5 mm). When aligned to the larynx via soft tissue-based registration, the average difference between the registration performed in this work and the on-treatment applied registration shift based on soft tissue alignment was 1.1+/-1.2 mm, indicating excellent agreement with shifts generated during the patient's treatment. **Conclusion:** Despite being advised not to swallow during CBCT acquisition and during treatment, large S/I larynx displacements were observed, indicating the potential for swallowing to occur. In addition, the magnitude of the observed larynx displacements should be considered in the planning margin to account for potential swallowing during treatment.

PO-GePV-M-113

Dosimetric Validation of Surface Guided Patient Set-Up for Breast Radiotherapy

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Purpose: To validate the dosimetric robustness of surface guided radiation therapy (SGRT) for breast radiotherapy using cone-beam CT (CBCT) dose calculations. **Methods:** A retrospective sample of 15 breast radiotherapy patients (40-50 Gy in 15-25 fractions) was selected for this analysis. All patients were set up using SGRT (VisionRT, London, UK) with weekly CBCT verification (chest wall alignment, 3DoF shifts, >75 CBCT datasets). The online matched CBCT images were imported into the treatment planning system (Raystation 8A) to compute CBCT fraction doses using a previously validated framework. Fraction doses with SGRT alone were simulated by repeating all CBCT dose calculations without any couch shifts. The differences between the doses calculated after CBCT and simulated SGRT alignments were evaluated to validate the dosimetric feasibility of using SGRT alone for setup. Observed trends between dose differences and CBCT shifts were assessed to determine minimum thresholds for CBCT shifts and SGRT monitoring. **Results:** The majority (>70%) of CBCT shifts were <5 mm. Minimal changes were observed between the CBCT and SGRT-aligned dose distributions (breast $\Delta D95 < 1\%$ for > 60% of fractions, < 3% for > 80% of fractions). **Conclusion:** The differences between the CBCT and SGRT-aligned doses increased with the magnitude of CBCT shifts and were minimal for small shifts (breast $\Delta D95 < 1\%$ for shifts <3 mm, <3% for shifts <5 mm), which supports the use of 3 mm thresholds for CBCT matching and SGRT monitoring. The use of SGRT alone was also found to be dosimetrically feasible for breast radiotherapy based on minimal degradation of delivered dose with SGRT alone (breast $D95 > 95\%$ or $\Delta D95 < 3\%$ for > 95% of fractions).

PO-GePV-M-114

Contrast Enhancement in Primary Oropharyngeal Tumors Through the Use of Split-Filter Dual Energy CT

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Purpose: To investigate the ability of split-filter dual-energy computed tomography (DECT) to enhance oropharyngeal tumor contrast and contrast-to-noise ratio (CNR) for radiation therapy treatment planning. **Methods:** Split-filter DECT contrast enhanced images were acquired for twenty patients with oropharyngeal cancer. Virtual monoenergetic images (VMIs) were generated at energies of 40 keV and 50 keV and compared to virtual 120 kVp-equivalent images. Additional images were reconstructed with an iterative reconstruction technique (SAFIRE strength 3). Average contrast and contrast-to-noise ratios were calculated using ROIs placed in the gross-tumor volume (GTV) and the geniohyoid muscle. **Results:** The VMIs increased the average GTV contrast from 4.44±10.0 HU for the 120 kVp-equivalent image to 42.0±32.9 HU (p=1x10⁻⁶) and 25.6±22.3 (p=1x10⁻⁶) for the 40 keV and 50 keV images, respectively. The 40 keV and 50 keV VMIs also demonstrated an increase in noise by 93% (p=4x10⁻¹²) and 46% (p=4x10⁻¹¹), respectively, relative to the 120 kVp-equivalent image. The average CNR values increased from 0.54±1.16 for the 120 kVp-equivalent image to 2.51±1.88 (p=1x10⁻⁶) for the 40 keV image and 1.99±1.64 (p=3x10⁻⁷) for the 50 keV image. The iterative reconstruction algorithm decreased noise in the VMIs by 19% (p=2x10⁻⁷) and 23% (p=3x10⁻¹⁰), and improved

CNR by 26% ($p=5 \times 10^{-4}$) and 31% ($p=8 \times 10^{-5}$), in the 40 and 50 keV images, respectively. **Conclusion:** Primary oropharyngeal tumor contrast and CNR were significantly improved using VMIs reconstructed from the split-filter DECT technique, and the use of iterative reconstruction further improved the CNR by decreasing the noise in the image. This gain in tumor contrast has the potential to enable more accurate tumor vs normal tissue delineation for radiation therapy treatment planning.

PO-GePV-M-115

An Evaluation of Translational Set-Up Errors to Define An Institutional Protocol for Set-Up Margins in the Treatment of Cancer Cervix Using Image Guided Radiotherapy (IGRT)

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Purpose: The primary aim of this study is to analyze the setup errors and thus defining an institutional protocol for the Clinical Target Volume(CTV) to Planning Target Volume(PTV) margins with three different methods in the patients of cancer cervix treated with Image Guided Radiotherapy(IGRT). **Methods:** A total of 37 patients with cancer cervix were selected for this retrospective study. Daily set-up errors were obtained by superimposing the planning CT as a reference image with the on-board Cone beam CT(CBCT) for each fraction delivered. The translational shifts in the Medio-Lateral(ML), Cranio-Caudal(CC), and Antero-Posterior(AP) directions given by CBCT were used for this study. Rotational shifts were found to be less than 0.1° and hence were not included. Further, these errors were analyzed to calculate systematic and random errors in each direction and the adequate CTV-PTV margins were estimated using Van Herk's, Stroom's, and ICRU 62 formulas. **Results:** Systematic errors were 0.13cm, 0.27cm and 0.14cm while random errors were 0.15cm, 0.43cm and 0.15cm in ML, CC and AP directions respectively. CTV to PTV margins calculated using Van- Herk's formula were 4.44mm in ML, 9.85mm in CC, and 4.48mm in AP directions. According to Stroom's formula, CTV-PTV margins were 3.77mm in ML, 8.48mm in CC, and 3.80mm in AP directions. Whereas based upon ICRU 62 formula CTV-PTV margins were 2.42mm in ML, 5.74mm in CC, and 2.43mm in AP directions. **Conclusion:** In order to maintain accuracy in the delivery of radiation, regular image verification is necessary. Although rigid immobilization is achieved, a significant variation in the day-to-day setup of patients still occurs. All the three formulas are satisfactory techniques for determining CTV to PTV margins. The suggested margins, based on the center's own experience, are calculated by the use of the Van Herk Formula.

PO-GePV-M-116

Head-To-Head Performance Comparison of Two Deep Learning Segmentation Algorithms for Radiotherapy Planning: A Study in Prostate

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Purpose: Deep learning (DL) based automated segmentation methods have demonstrated the potential to improve the radiotherapy (RT) treatment planning process by offering readily available, while consistent, anatomical contours. There is, however, a lack of knowledge of how reliable and valid these algorithms are for clinical use. This study aimed to evaluate performances of two marketed DL segmentation algorithms in a head-to-head study design. **Methods:** The algorithms being assessed included ProtégéAI (MIM Software Inc., Cleveland, OH) and AccuContour (Manteia Tech, Xiamen, China). Imaging data used consisted of CT simulation scans of 100 prostate cancer patients. The segmentation accuracy of each algorithm for prostate, bladder, femoral heads, rectum, and seminal vesicles was compared to the respective physician contour using metrics of Dice coefficient (DICE) and absolute volumetric difference (AVD) with differences being identified. **Results:** AccuContour significantly outperformed ProtégéAI in delineating bladder and right femoral head with mean DICE of 0.88 Vs. 0.74 (two-sided paired t-test; $p<0.001$) and 0.87 Vs. 0.82 ($p<0.001$) and mean AVD of 0.14 Vs. 0.25 ($p=0.007$) and 0.18 Vs. 0.36 ($p<0.001$); whereas the later excelled in delineating prostate and rectum with mean DICE of 0.83 Vs. 0.67 ($p<0.001$) and 0.83 Vs. 0.78 ($p<0.001$) and mean AVD of 0.17 Vs. 0.36 ($p<0.001$) and 0.13 Vs. 0.18 ($p=0.0149$). For seminal vesicle, ProtégéAI showed superior accuracy in terms of DICE (mean: 0.65 vs. 0.49; $p<0.001$) while no significant difference according to AVD (mean: 0.44 vs 0.54; $p=0.348$) in comparison to AccuContour. Regarding left femoral head, no significant difference was seen between the two algorithms regarding DICE (mean: 0.91 vs 0.90; $p=0.598$) or AVD (mean: 0.13 vs 0.10; $p=0.218$). **Conclusion:** Accuracy of DL-based segmentation varied based on algorithm and with anatomical location under evaluation. The findings from the current study indicate that caution should be exerted when implementing these algorithms into clinical workflow.

PO-GePV-M-117

Data-Driven Generation of CBCT-To-CT HU Mapping for Adaptive Radiotherapy in H&N Cancer

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Purpose: This study aims to develop a data-driven method to derive a patient-specific CBCT to CT HU mapping function and evaluate the method for dose calculation on CBCT in H&N cancer radiotherapy. **Methods:** Eleven patients receiving VMAT for oropharyngeal cancer in a dose de-escalation clinical trial were enrolled in this study. Daily CBCT and a mid-course CT for re-assessment were acquired. The study derived a CBCT-to-CT mapping from one pair of CT and its closest CBCT, and applied to CBCT in the other set for independent dose calculation evaluation. The mapping was created by spatially-constraint fuzzy c-mean clustering (FC) on (CBCT, CT) HU vectors in the regions of air, soft tissue and bone that were thresholded on CBCT. To generate pseudo-CT, the probabilities of a CBCT voxel belonging to subtypes were computed by clustering with the CBCT centroids, and a CT value was created from the CT centroids weighting with the probabilities. The methods of using a HU-density curve (HC) from Catphan phantom imaging and air-water-bone density assignment (DA) were also evaluated for CBCT-based dose calculations. Dose differences between the three methods and CT calculations were assessed. **Results:** Mean absolute differences of CT numbers in entire CBCT body of the patients are 110, 136 and 147 between FC, HC and DA method generated pseudo-CT and planning CTs, respectively. DVH differences between the three CBCT-based methods and CT calculated doses were within 3% for the PTV and related OARs. The percentage differences in parotid mean doses were $0.5 \pm 0.4\%$, $0.7 \pm 0.5\%$, $0.9 \pm 0.4\%$ for FC, HC, and DA methods, respectively. **Conclusion:** The fuzzy clustering on planning CT and 1st-day CBCT could generate a patient-specific CBCT-to-CT mapping for accurate dose calculation on patient CBCT in the subsequent fractions. The method may be used for CBCT-based daily dose assessment and adaptive replanning.

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PO-GePV-M-118

Quantitative Analysis of the Accuracy of CT to CBCT Deformable Image Registration for Contour Mapping for Prostate Cancer

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Purpose: Deformable Image Registration (DIR) can automatically propagate planning CT (pCT) contours to daily cone-beam CT (CBCT) images to account for anatomical changes such as bladder and rectal filling. The accuracy of deformed contours depends on several factors, including DIR process, algorithm and image modality/quality. This study hypothesizes that auto-propagated prostate contours through the commercially available DIR applied to daily CBCT are sufficiently robust and accurate. **Methods:** 28 prostate cancer patients enrolled on an IRB-approved protocol, treated with VMAT and receiving daily CBCT images were selected. Following rigid registration shifts, the pCT was deformably registered to the daily CBCTs using an intensity-based DIR using a B-spline

algorithm to generate deformed (auto) prostate contours. Gold standard (reference) contours on CBCT images were manually drawn by an experienced radiation oncologist. DIR algorithm performance was evaluated by comparing the auto and manually-drawn prostate contours on daily CBCT images (1,010 fractions) via dice similarity (DSI), mean distance-to-agreement (MDA), change in center of mass position (Δ CM) and Jacobian. 42 radiomic texture features were extracted from both contours and there correlation was determined through Pearson's correlation (PCC) and Lin's concordance correlation (LCC). **Results:** The average DSI, MDA and Δ CM between the auto- and manually-drawn contours were found to be 0.90 ± 0.04 , 1.81 ± 0.47 mm and 2.17 ± 1.26 mm respectively. The average minimum and maximum of the Jacobian determinant were 0.77 ± 0.18 and 1.30 ± 0.23 respectively. 94.8% of all auto contours were within tolerance provided by TG-132. 34 radiomic texture features were robust to small changes in contours as seen through high PCC and LCC (>0.80), indicating that small contour differences do not significantly change the radiomic texture feature data. **Conclusion:** The use of small ROI improved the DIR accuracy for prostate contour propagation to be within tolerance recommendations of TG-132 and appears to be suitable for radiomic analysis and/or dose accumulation.

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PO-GePV-M-119

Improvement of MVCT Image Quality for Adaptive Helical Tomotherapy Using CycleGAN-Based Image Synthesis with Small Datasets

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Purpose: Megavoltage computed tomography (MVCT) offers an opportunity for adaptive helical tomotherapy. However, high noise and a reduced contrast in the MVCT images resulted from a decrease in the imaging dose to patients limits the usability of MVCT. Therefore, we propose an algorithm to improve the image quality of MVCT with limited training data set. **Methods:** The proposed algorithm generates kilovoltage CT (kVCT)-like images from MVCT images using a deep learning-based image synthesis network. A total of 566 and 514 slices of CT images were prepared from 6 training and 5 test cases, respectively. Data augmentation using an affine transformation was used to produce the training data in order to overcome the lack of data diversity in the network training. The mean absolute error (MAE), root mean square error (RMSE), peak signal-to-noise ratio (PSNR), and structural similarity (SSIM) were used to quantify the correction accuracy of the images generated by the proposed algorithm. The proposed method was validated by comparing images generated with non-augmented datasets. **Results:** The average MAE, RMSE, PSNR, and SSIM values were 13.12 HU, 39.44 HU, 35.99 dB, and 0.96 in the proposed method, whereas the result with non-augmented data showed inferior outcomes (15.65 HU, 45.82 HU, 34.66 dB, 0.95). The voxel intensity of the image obtained by the proposed method also indicated similar distributions to those of the kVCT image. **Conclusion:** The proposed algorithm generates synthetic kVCT images from MVCT images using a deep learning network with small patient datasets. The image quality achieved by the proposed method was correspondingly improved to the level of a kVCT image while maintaining the anatomical structures of an MVCT image. With a further evaluation for clinical implementation, the proposed method can ensure an accurate treatment planning in adaptive radiation therapy.

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PO-GePV-M-120

Potential for Using Extracellular Volume Fraction Derived from Dual-Source DECT for Pancreatic Tumor Delineation

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Purpose: Low extracellular volume fraction (ECVf) correlates with poor vasculature and possible necrosis in pancreatic adenocarcinoma. In this work we investigate the potential of using contrast enhanced dual-source dual-energy CT (DECT) to map out ECVf spatial distribution for pancreatic tumor delineation. **Methods:** Contrast enhanced DECT acquired using a dual-source DECT scanner (SOMATOM Drive, Siemens) at 80 and 140 kVp during standard radiotherapy (RT) simulation for one pancreatic cancer patient was analyzed. An ECVf map was generated from the low and high energy CT series acquired simultaneously during the late arterial contrast phase in the following steps: (1) calibrating DECT iodine contrast using a phantom with known iodine densities; (2) constructing the patient-specific iodine map based on the iodine calibration; and (3) computing the ECVf map using the patient-specific iodine map and the patient's hematocrit level measured on the day of simulation. A region of interest (ROI) of low ECVf was delineated using an ECVf threshold of 0 to 7% from the obtained ECVf map. The volume of the ECVf ROI was compared with the clinical gross tumor volume (GTV) using the Dice similarity coefficient (DSC), and the overlap of the corresponding contour pair of the GTV and ECVf ROI on each image slice was measured using distance to agreement (MDA) and Hausdorff distance (HD). **Results:** The low ECVf ROI reasonably correlated to the clinical GTV. The DSC, MDA and HD between the ECVf ROI and the clinical GTV were 0.6 ± 0.1 , 2.0 ± 1 mm, and 5.0 ± 0.8 mm, respectively. **Conclusion:** This work demonstrated the potential of using the extracellular volume fraction map, derived from contrast enhanced dual-source DECT, to help define the pancreatic GTV for RT planning. Further studies with more patient data are required to determine the sensitivity, specificity, and limitations of this ECVf technique.

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Deep-Learning Based Auto-Classification of Anatomical Regions for IGRT Applications: Implementation and Explainability

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Purpose: Anatomical classification from single x-ray projections could help automate the selection of scan and reconstruction parameters in cone-beam CT (CBCT) for image-guided radiotherapy (IGRT). Here, deep learning (DL) methods are used for classification and the results are interpreted with the aid of visual explainability tools. **Methods:** Projection data from 1055 patients with multiple CBCT scans were used to train and test a DL model. Each scan was manually classified into five categories using projections at 45° intervals: abdomen, head, neck, pelvis and thorax. The dataset was limited to no more than 2-10 scans per patient, such that anatomical classes were balanced. Data were distributed for training (75%), testing (12.5%), and evaluation (12.5%) with no overlap between patients among the 3 sets. A VGG-16 neural network (NN) with pre-trained weights (imageNET dataset) was used; NN weights were adjusted by re-training the model with projections of a given angle (0°, 90°, 180°, and 270°). The classification performance was quantified in terms of precision and accuracy. Gradient-weighted Class Activation Maps (Grad-CAM) were reviewed to identify strengths and limitations of the proposed DL model. **Results:** The highest performance was achieved using the anterior-posterior projection (90°), with 92% accuracy. Precision was high for head, neck and pelvis ($>97\%$) but low for abdomen and thorax (87% and 76%, respectively) classes. Thorax projections were commonly misclassified as abdomen, and vice versa. Manual review of Grad-CAM results suggested that abdomen/thorax misclassifications were due to either large or insufficient presence of lung tissue. For all other classes, Grad-CAM showed a clear emphasis on boney anatomical features (e.g., iliac crest, mandible). **Conclusion:** A DL model was proposed and validated for the auto-classification of anatomical regions from single x-ray projection images. By using Grad-CAM it was possible to identify possible sources of class confusion in order to improve future models.

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Comparison of KV-CBCT Dose Measurement Techniques for Annual Linac QA

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Purpose: Modern linear accelerators (LINACs) are equipped with kV-imaging systems for 2D-images and 3D-cone-beam CT (CBCT). The delivered single kV-CBCT-dose is small compared to the treatment dose but can add up to a cumulative dose of 1-3% for multi-fraction courses according to TG180. Thus, TG142 recommends an annual assessment of the imaging dose. The purpose of this project was to review the current practice, compare approaches suggested in recent literature and standardize annual kV-CBCT dose measurements. **Methods:** Current institutional practice is to measure the CBCT dose with the 10cm-long RaySafe X2 pencil-ion-chamber inserted in the center position of a CIRS lung phantom. We introduced the PMMA-CTDI phantom in a satellite-location for all measurements. Dose was measured at 3/6/9/12 o'clock and center positions of a 32cm-wide CTDI phantom to determine max dose heterogeneity for a half-trajectory imaging protocol and at the 12 o'clock and center position of either the head or abdomen phantom for 11 protocols ranging from 80kVp/100mAs to 140kVp/1688mAs using the Raysafe X2, the Standard Imaging A101 pencil-ion-chamber, and the A12 Farmer-ion-chamber. Measurements were repeated with decreased kV-tube-collimator opening as suggested by Varian. **Results:** The dose discrepancy due to chamber position was largest between the 6 and 3 o'clock positions (10%) and the average was close to the peripheral measurement at 12 o'clock. All three detectors showed a linear dependency. Reducing CBCT beam collimation increased measurement complexity with only minimally improved agreement between measured and displayed CTDIw values. The standard deviation for dose measurements at different machines reduced from 22%/12% for the lung phantom to 7%/3% (max/mean) for the CTDI phantom. **Conclusion:** Dose measurements at 12 o'clock and center position of CTDI-phantom have been defined as our new standard using un-modified treatment imaging protocols; conversion factors for center-only or lung-phantom measurements have been provided.

PO-GePV-M-123

Abdominal 4DCT Image Quality/Visibility Between Iterative Model Reconstruction (IMR) and Iterative Reconstruction (iDose4)

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Purpose: IMR (Philips Healthcare) is widely used in diagnostic imaging. Use of IMR is not well documented in abdominal 4DCT for radiotherapy, where pixel counts are low. This study compared image quality and visibility between IMR and iDose4. **Methods:** A Gammex Model 467 tissue phantom was used for CT calibration. Contrast and resolution were evaluated using Philips QA CT phantom. Ten patients had 4DCT abdominal scans for treatment planning, with 160 mAs used for each phase image compared to 305 mAs used for conventional 3D imaging. Images were reconstructed using IMR and iDose4. The objective measurements included noise (pixel HU histogram), line profile crossing target (boundary gradient). Two independent observers provided subjective assessment on image quality and visibility for contouring target and organs. **Results:** For the tissue phantom, the mass density ranged 0 to 1.69 and HU ranged -1000 to 1170; the HU mean difference was -0.8 HU (STD 2.6 HU) between IMR(1) and iDose4(3). For the Philips QA phantom with multi plugs, HU ranged from 0 to 910 and, the HU mean difference was -0.9 HU (STD 0.8 HU) between IMR(1) and iDose4(3). There was no measurable difference in contrast and resolution using the Philips QA phantom, but IMR had less objective noise in rods and background compared to iDose4. For 4DCT in patients, there was no observable difference in target boundary gradient between IMR and iDose4. IMR had objective noise reduction (pixel histogram HU range) as much as 21% in targets and 29% in background tissue compared to that of iDose4. IMR had better image quality and visibility for contouring target and organ in all cases. **Conclusion:** For motion management in treatment planning, abdominal 4DCT has low pixel counts. We found IMR had better image quality and visibility when compared to that of iDose4.

PO-GePV-M-124

Comparison of CBCT Image Enhancement Using Patch-Based U-Net and GAN

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Purpose: Cone-beam computed tomography (CBCT) is an important imaging method in current practice of adaptive radiation therapy (ART). However, the CBCT images suffer from the relatively poor image quality. In this study, we developed a pure image domain deep convolutional neural network (DCNN) based method to improve the CBCT image quality. **Methods:** Paired CBCT and standard CT from 50 head neck cancer patients were split into three sets: training (32), validation (8) and testing (10). Optimization in pre-processing and data quality control were applied to guarantee the quality of our training set. A 2D U-Net was used as our backbone network structure to perform CBCT enhancement. Experiments were conducted to validate the patch-based training strategy and investigate the optimal training patch size. In addition, combined L1 and conditional generative adversarial network (GAN) loss was applied to explore whether it could yield visually sharper images. The trained networks were finally applied to an external testing set to check their deployment performance. **Results:** The U-Net trained with a patch size of 128 × 128 achieved the best performance with the lowest mean absolute error of 34.78 HU (57.91 HU for input images) and the highest structural similarity of 0.8455 (0.7546 for input images). After fixing to the optimal patch size, the network trained with a combined L1 and conditional GAN loss was able to maintain a higher resolution, especially in the soft tissue regions. **Conclusion:** The proposed DCNN-based method can generate enhanced CBCT with significant quality improvement compared with input images. This potentially enables further CBCT applications in the ART workflow.

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PO-GePV-M-125

Characterization of Imaging Performance of a Novel Helical KVCT System for Use in Image-Guided and Adaptive Radiotherapy

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Purpose: To evaluate the imaging performance of a novel helical kVCT system, the Radixact ClearRT, for use in image-guided radiotherapy and dose calculation in adaptive radiotherapy and to compare this performance to existing systems used for these purposes. **Methods:** A commonly used CT image quality phantom was scanned across clinically-relevant acquisition modes for image quality characteristics including noise, uniformity, contrast, spatial resolution, and mean CT number and CT number linearity. These metrics were analyzed using ImageJ software and compared to those measured for the Radixact MVCT, TomoTherapy MVCT, Varian TrueBeam OBI CBCT, and the Siemens SOMATOM Definition Edge. Individual metric performance was compared relative to the image noise for each acquisition. **Results:** Similar noise levels were seen for ClearRT and Siemens, while noise for the other systems was 2-5 times higher on average. Uniformity was highest for Siemens. On average, ClearRT performed best for contrast, followed closely by Siemens and Varian. Spatial resolution was best for Varian, while ClearRT showed slightly worse resolution than Siemens and TomoTherapy. For CT number vs. electron density calibration curves for analogous protocols, ClearRT showed a similar linear fit to the Siemens system commonly used for dose calculation in planning images. Mean CT numbers were also similar for ClearRT and Siemens, especially for inserts mimicking fat and soft tissue. Scan times for Siemens were less than 20s and approximately 30-40s for ClearRT. Times for the other systems were 2-10 times higher on average. **Conclusion:** ClearRT performed well for metrics most important to IGRT – mainly contrast. For image characteristics most important to dose calculation - mean CT number and CT number linearity -

similar values were measured for ClearRT and Siemens. Thus, ClearRT appears capable of providing accurate images for adaptive treatments, which will be further evaluated with measurements on dose calculation accuracy.

PO-GePV-M-126

Estimation of Conversion Factors for CTDI_w Calculation in MVCBCT Acquisitions

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Purpose: MVCBCT dosimetric performance using standardized dose indices can be used to optimize acquisition techniques in radiation therapy imaging. We propose a method to estimate CTDI_w for MVCBCT acquisitions directly from the delivered MU. This parameter will enable comparison with other x-ray based volumetric imaging (e.g. kVCBCT). **Methods:** An extended PMMA body phantom was modeled with a diameter and length of 32 cm and 30 cm, respectively. A 360-degree MVCBCT acquisition was simulated using the Varian MV imaging beam (2.5 MV) and field size equal to 26×26 cm² at isocenter. Correction factors for the large field size relative to the CTDI ion chamber length were calculated. Additional correction factors to account for dose distribution differences between MV and kV beams were calculated. The CTDI_w was calculated from the simulated dose distribution using the standard formula. A conversion coefficient to estimate CTDI_w from MU was derived. **Results:** Three-dimensional dose distributions were generated. Axial and radial dose profiles were produced to facilitate derivation of appropriate correction factors for center and peripheral CTDI locations. The correction factor for the wide field size was 2.45 for center and periphery. Correction factors for peripheral CTDI locations that fall outside the irradiated field ranged from 1.14 to 1.16. The conversion coefficient to estimate CTDI_w directly from delivered MU was equal to 0.212 cGy/MU. **Conclusion:** A method to directly estimate CTDI_w from delivered MU after an MVCBCT acquisition was proposed. The method relies on Monte Carlo simulations to produce correction factors for large field size and ion chamber location relative to the field size. The correction factors facilitate the derivation of conversion coefficients that allow direct calculation of CTDI_w from MU. The proposed method can be generalized for various field sizes and angles to provide a fast and reliable procedure for dose optimization of MVCBCT imaging techniques

PO-GePV-M-127

DeepDoseNet: A Deep Learning-Based Approach for 3D Dose Prediction

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Purpose: To develop an artificial intelligence—deep learning-based method for fully automated, accurate, and rapid prediction of optimized head-and-neck 3D dose distributions. **Methods:** The DeepDoseNet 3D dose prediction model is based on ResNet and Dilated DenseNet. The 340 head-and-neck datasets from the 2020 AAPM OpenKBP challenge were utilized, with 200 for training, 40 for validation, and 100 for testing. Structures include 56Gy, 63Gy, 70Gy PTVs, and brainstem, spinal cord, right parotid, left parotid, larynx, esophagus, and mandible OARs. Mean squared error (MSE) loss, mean absolute error (MAE) loss, and MAE plus domain-knowledge (DK)-based loss functions were investigated. Each model's performance was compared using a dose score, $\bar{S}(D)$, (mean absolute difference between ground truth and predicted 3D dose distributions) and a DVH score, $\bar{S}(DVH)$ (mean absolute difference between ground truth and predicted dose-volume metrics). **Results:** DeepDoseNet requires ~0.5 sec to predict the 3D dose distribution on NVIDIA Quadro RTX 8000 GPU with 48GB memory. The MAE+DK model had the lowest prediction error ($P < 0.0001$, Wilcoxon test) on validation and test datasets (validation: $\bar{S}(D)=2.3\text{Gy}$, $\bar{S}(DVH)=1.9\text{Gy}$; test: $\bar{S}(D)=2.0\text{Gy}$, $\bar{S}(DVH)=1.6\text{Gy}$) followed by the MAE model (validation: $\bar{S}(D)=3.6\text{Gy}$, $\bar{S}(DVH)=2.4\text{Gy}$; test: $\bar{S}(D)=3.5\text{Gy}$, $\bar{S}(DVH)=2.3\text{Gy}$). The MSE model had the highest prediction error (validation: $\bar{S}(D)=3.7\text{Gy}$, $\bar{S}(DVH)=3.2\text{Gy}$; test: $\bar{S}(D)=3.6\text{Gy}$, $\bar{S}(DVH)=3.0\text{Gy}$). **Conclusion:** The addition of the domain-knowledge reduced $\bar{S}(D)$ by ~60% and $\bar{S}(DVH)$ by ~70%. DeepDoseNet's optimized dose can be used to improve the efficiency of dose optimization algorithms by providing a good estimate of the final dose distribution. Because of DeepDoseNet's rapid and fully automated execution, it can be used in automated workflows to estimate the dosimetric relevance of delineated organs-at-risk (OARs).

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A Case Study of the Effects of Bladder Filling for Treating Prostate and Nodes Using Online Adaptive Radiotherapy On Ethos

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Purpose: In theory, image-guided online adaptive therapy (IGART) is a good choice for treating the prostate and the pelvic nodes by allowing the treatment plan to be adapted to account for daily changes in the OARs (bladder, rectum and bowel). However, present forms of the IGART process, including patient setup, imaging, contouring, reoptimizing, QA and re-imaging, have been shown to require 40-60minutes and bladder filling during this time could invalidate the potential dosimetric benefits offered by plan adaptation. The goal of this work is to evaluate the bladder-filling during IGART and its spatial effect on adjacent OARs. **Methods:** A single patient underwent twenty-five-CBCTs as part of 5-fraction-SBRT to the prostate and pelvic nodes. The first three-fractions of treatment were delivered without adaptation, and the last two were adapted based on daily CBCT. Additional CBCTs were acquired prior to each VMAT beam. The bladder and rectum were manually contoured on each CBCT. The relationship among the bladder volumes, the CBCT scan time, and the rectum anterior wall positions were studied. **Results:** Two out of three non-adapted fractions were delivered within a reasonable period (<15minutes). However, the patient's bladder filled quickly in all fractions (see supporting Fig.3), for example, from 200cc to 760cc in 50minutes. The patient had to discharge urine in 1 extended adaptive fraction (~70minutes). The position of the anterior rectal wall corresponded well with the bladder volume (Fig.4). The rectum anterior wall was pushed 1.9cm further posterior when the bladder was filled from 500cc to 900cc. **Conclusion:** In this study, bladder-filling was seen to affect the positioning of the rectum in a clinically meaningful way. For IGART to be effective, it is paramount to maximally reduce the total plan adaptation time, e.g. <20minutes, and carefully manage patients' liquid intake prior to treatment fractions to minimize intra-fraction bladder filling effects.

PO-GePV-M-129

Estimating Inconsistency in Deformable Image Registration-Based Dose Accumulation

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Purpose: To assess uncertainty in deformable image registration (DIR) based dose accumulation, we estimate the inverse consistency error (ICE) based on a DVH overlay method as applied to daily dose accumulation for prostate cancer cases. **Methods:** Dose accumulations of five fractions for ten prostate cancer cases treated with daily MVCT IGRT were performed by applying two commonly used DIR algorithms: a normalized correlation coefficient technique (NCC-DIR) and a free-form intensity-based registration (IB-DIR). Regions of interest (ROIs) included the PTV (prostate or prostate bed), bladder, and rectum. ICE was estimated using a DVH overlay technique, where a DIR applied between the plan CT and daily MVCT was used to forward-transfer the plan ROIs to the MVCT and inverse-transfer the daily dose to the plan CT. For the five summed daily doses, ICE was calculated as the weighted sum of the separation between the overlaid DVHs from the plan CT and daily MVCT for each fraction. **Results:** ICE for the PTV V(100%Rx) is similar for both algorithms, $1.6 \pm 1.6\%$ for IB-DIR and $1.2 \pm 0.6\%$ for NCC-DIR. For the bladder and rectum V(70%Rx), ICE is on average greater for IB-DIR (3.6% and 3.3%, respectively) than for NCC-

DIR (1.9% for both). Among all cases, the maximum value of ICE for PTV V(100%Rx) is 5.6% for IB-DIR and 2.1% for NCC-DIR; the maximum ICE for bladder V(70%Rx) was 6.3% for IB-DIR and 2.6% for NCC-DIR; and for rectum V(70%Rx) the maximum ICE was 6.3% for IB-DIR and 3.9% for NCC-DIR. **Conclusion:** We demonstrated that the inverse consistency error, a contributor to the overall uncertainty in DIR-based dose accumulation, can be estimated using the DVH overlay technique. The data obtained presently for prostate cancer cases indicate that caution should be exercised in using the dose accumulation for clinical decision making if ICE exceeds pre-defined thresholds.

PO-GePV-M-130

Accuracy of Pelvic CT-MR Deformable Image Registration: Comparison of Open-Source and Commercial Software

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Purpose: We evaluated the accuracy of deformable image registration (DIR) for the pelvic region between magnetic resonance (MR) and computed tomography (CT) images using two different software programs. **Methods:** This study used CT, T1-weighted (T1w) MR, and T2-weighted (T2w) MR images of the pelvic region of five male patients. DIRs were performed using Elastix, an open-source software, and Velocity AI, a commercial software. With Elastix, we used six patterns of cost functions with regularization weights $\lambda = 0.01, 0.1, 1, 10, 100$, and 200. For Velocity AI, we used a MR Corrected Deformable algorithm. The DIR accuracy was evaluated based on the Dice similarity coefficient (DSC) for prostate, bladder, rectum, and left and right femoral heads. **Results:** The mean DSCs from all structures in T1w images were 0.84 ± 0.08 (rigid, Elastix), 0.88 ± 0.05 (DIR, $\lambda = 0.1$), 0.86 ± 0.09 (DIR, $\lambda = 10$), 0.85 ± 0.08 (DIR, $\lambda = 200$), and 0.86 ± 0.07 (DIR, Velocity AI), respectively. For the T2w images, the mean DSCs were 0.85 ± 0.08 (rigid, Elastix), 0.85 ± 0.09 (DIR, $\lambda = 0.1$), 0.86 ± 0.09 (DIR, $\lambda = 10$), 0.85 ± 0.08 (DIR, $\lambda = 200$), and 0.85 ± 0.08 (DIR, Velocity AI), respectively. DIR accuracy in elastix was as good as Velocity AI when $\lambda = 0.1$ (10) between CT-T1w (CT-T2w) images. The λ at which the mean DSC was highest differed for each patient. **Conclusion:** The results suggested that DIR accuracy comparable to that of commercial DIR systems could be achieved by verifying the penalty weight for DIR between CT and MR images for each sequence and each facility.

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PO-GePV-M-131

Estimation of Uncertainty in Deformable Image Registration Based Daily Dose Accumulation for Head and Neck Cancer

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Purpose: Deformable image registration (DIR) based dose accumulation is essential for adaptive radiation therapy (ART) and can be subjective to errors. We accumulate daily doses based on DIR and estimate their uncertainties during RT for head and neck cancer (HNC). **Methods:** A commercial ART tool (PreciseART, Accuray, Inc.) was used to reconstruct daily doses and accumulate them via DIR for ten HNC patients treated with helical tomotherapy. The accuracy of dose accumulation was assessed through a consistency check that comprises: i) warping the structures defined in the planning kVCT to the daily MVCT, ii) mapping the dose calculated in the MVCT to the kVCT, iii) calculating and overlaying DVHs from both images. At a given volume level, the absolute difference between the doses computed in the DVH overlays was considered the leading uncertainty component of the assessed metric. By relating the evaluated dose/volume level in DVHs of both single fractions and cumulative distributions, daily uncertainties were propagated to estimate confidence intervals in the accumulated distribution. **Results:** Accumulated doses differed from planned values, e.g., the changes in PTV-D95%, larynx-D33%, esophagus-D50%, and ipsilateral parotid gland-D50% ranged from 97 to 101%, 90-111%, 63-106%, and 76-165%, respectively. For 87% of the regions-of-interest and metrics studied, uncertainties were less than 2% of the planned value, including all points of the PTV (D95%) and larynx (D33% and mean dose). For D50% in the esophagus and parotids, 24% of the points presented uncertainties from 3.5% to 5.5% of the accumulated dose; a maximum uncertainty of 20% was observed for the esophagus. **Conclusion:** The DIR-related dose reconstruction errors were estimated for HNC patients. Relative uncertainties can be greater than 5% for structures undergoing significant deformations during radiotherapy (around 20% volume change), e.g., the esophagus and parotids. In ART, both daily dose accumulation and its uncertainty should be considered.

PO-GePV-M-132

Improving Deformable Image Registration with Three Metrics for Postoperative Radiotherapy of Breast Cancer Patients

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Purpose: Deformable image registration (DIR) is increasingly used for target volume definition in radiotherapy. However, this kind of method is challenged by the situation of postoperative breast cancer patients due to the large deformations and non-correspondence caused by tumor resection and clip insertion. To deal with this situation, fiducials, region of interests (ROIs) and voxel intensities were jointly used for higher registration accuracy. **Methods:** Three metrics were combined to form the objective function of DIR algorithm. Fiducial-based metric minimizes the distance between 2 point sets with known correspondence. ROI-based metric utilized Kappa Statistic (KS) to maximize the overlap of corresponding ROIs. Intensity-based metric maximized mutual information (MI) between image intensities. Two sets of CT images before and after breast surgery were used for image registration. One set was the diagnostic CT image acquired before surgery, and another set was the planning CT image acquired after surgery for breast cancer radiotherapy. A total of 26 sets of CT images from 13 patients were collected retrospectively for the test. The registration accuracy of the proposed method was evaluated in terms of target registration error (TRE) and through visual assessment by the physician. **Results:** The difference of mean TRE between the proposed joint-metric method and the conventional intensity-based method was statistically significant for soft tissue (2.06 vs. 7.82, $P < 0.05$), rigid structure (3.02 vs. 3.34, $P < 0.05$) and boundary (3.70 vs. 6.93, $P < 0.05$). For visual assessment, the proposed method achieved better matching for soft tissue and boundary. **Conclusion:** The registration accuracy of the proposed method was higher than the conventional method based on the preliminary results. This method provides a feasible way for target volume definition in postoperative radiotherapy of breast cancer patients.

PO-GePV-M-133

Evaluation of Deformable Image Registration Accuracy for Liver Patients Using Contrast and Non-Contrast CT Images

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Purpose: Deformable image registration (DIR) facilitates cumulative dose estimation in re-irradiation of liver tumours. We quantify the accuracy of DIR in liver CT scans taken at multiple time points through use of contemporaneous intravenous contrast CTs. **Methods:** We selected patients who had undergone repeat courses of radiation therapy to liver tumours, who had contrast CTs (conCTs) acquired contemporaneously to non-contrast treatment planning CTs (non-ConCTs) for each course. Using VelocityAI (Varian Medical Systems, USA), rigid image registration (RIR) and DIR were performed between the two time-points based on either non-conCTs alone or conCTs alone. Deformable multipass (DMP), extended DMP (EDMP) and structure guided (SG) were evaluated. SG was based on liver contours or five random landmarks. Landmarks were identified on the conCTs and transferred to the contemporaneous nonConCTs via RIR and used to evaluate the accuracy of the RIR and DIR based on Target Registration Error (TRE). Landmarks used in SG registration were excluded from

TRE analysis of SG registration. **Results:** Mean RIR TRE was 5.8 mm for conCT and 6.0 mm for non-conCT registrations. The mean TRE was 3.9/6.0 mm for conCT/non-conCT, 4.3/5.9 mm for conCT/non-conCT for DMP and EDMP respectively. The mean TRE was 7.7/7.3 mm for conCT/non-conCT and 3.4/3.9 mm for conCT/non-conCT for SG based on liver contours or five landmarks respectively. **Conclusion:** SG DIR based on five landmarks achieved the lowest TRE, and SG DIR based on the liver contour the highest TRE. DIR based on contrast-CTs was more accurate than for non-contrast-CTs for DMP and EDMP DIR algorithms. For some landmarks, DIR increased the TRE compared to the RIR. Use of landmarks to guide the DIR improves accuracy of both contrast and non-contrast CT DIR.

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Evaluation of Various Deformable Image Registration Algorithms for Abdomen Images with Biomechanically Modelled Deformations

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Purpose: Deformable image registration (DIR) is increasingly used in radiotherapy applications and provides basis for previous treatment evaluation and dose accumulation. We examine the accuracy of DIR algorithms available on commercial planning platforms and evaluate their clinical applicability for the abdominal site. **Methods:** Following the guideline of AAPM TG-132, we implemented a patient-specific QA process to evaluate the accuracy of DIR for the abdomen site using digital phantoms created via ImSimQA (Shrewsbury UK). To simulate the impact of respiratory motion, we deformed the planning CT image at the end of exhalation to the end of inhalation using the biomechanical model embedded in ImSimQA. Consequently, the respiratory deformations were transferred onto the GTV in pancreas in the abdominal region, and we obtained the deformation vector field (DVF) as ground truth. Subsequently, we exported the deformed CT to Eclipse and MiM, performed DIR, and derived DVF_Eclipse and DVF_MiM for evaluation. Quantitative differences between DVFs, such as mean, max, and the distribution were tabulated. Geometrically, dice coefficient between the DVF-propagated and ground truth tumor was calculated. We generated a comprehensive report to illustrate the performance for the patient of interest as QA record. **Results:** The biomechanical model-derived DVF covers an FOV from the mid lung to the inferiormost pancreas boundary, with displacements ranges from 0 to 6.8mm. Data analysis comparing DIRs using DVFs shows the mean $\pm$ standard deviation 3D discrepancy is 1.8 ± 0.8 mm for Eclipse, and -1.2 ± 0.9 mm for MIM, respectively. Discrepancies near the diaphragm are larger than the mid lung or the pancreas. MIM tends to underestimate while Eclipse tends to overestimate the effect of deformation. **Conclusion:** DIR performed by Eclipse and MIM are evaluated, with a realistic biomechanical model derived DVF serving as ground truth. Further investigations on the impact of DIRs on dose distribution for radiation therapy is warranted.

PO-GePV-M-135

Evaluation of Synthetic CT Images Generated by Deformable Image Registration of Outdated Planning CT to CBCT of Treatment Day

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Purpose: Synthetic CT (sCT) generated by deformable image registration (DIR) of planning CT to CBCT was proposed for the delivered dose calculation in adaptive radiation therapy (ART). We aim to evaluate the registration accuracy when the acquisition times of the two images are different (patient anatomy has changed). **Methods:** A patient who had replanning CT in his abdomen due to significant weight loss during radiation treatment was chosen. Two sCT images were generated by deformable image registration (DIR) in Velocity using (1) initial planning CT ($\square$ sCT1) and (2) replanning CT ($\square$ sCT2), where planning CT images were deformed to CBCT that was taken on the same day of replanning CT. The DIR was performed using rigid and deformable multi pass option in the Velocity software system. After DIR, deformation vector and anatomical alignment were visually inspected. The deformed images were further analyzed for HU difference. **Results:** The magnitude of the difference between the two planning CT images was not fully realized between sCT1 and sCT2 due to the registration of initial CT to CBCT of a later day. A visual evaluation was satisfying except that the two sCT images showed a slight disagreement on external body contour and a more visible disagreement in the location of air in the internal organ. The two sCTs showed acceptable agreement in HU values as shown by the standard deviation of the HU difference of 54.5 HU centered around 0, across the entire voxels of the images. **Conclusion:** Using initial planning CT for sCT generation is acceptable when a patient's anatomy changes significantly in the abdomen. However, one should be aware of the possibility of losing the accuracy in the external outline and details of internal organs. Further dosimetry study is needed.

PO-GePV-M-136

Feasibility of Diagnostic Dose Fast-Helical Free Breathing CT Scan Protocol for Ventilation Measurement

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Purpose: Free-Breathing Fast Helical CT (FBFHCT) has enabled physical modeling of the lungs, such as ventilation mapping and elastography, to guide radiation treatment planning. However, the nature of FBFHCT provides information useful for applications outside of radiation oncology, such as downstream modeling to guide surgery. To realize these applications, the dose must be reduced to the diagnostic range. The aim of this project is to assess the feasibility of diagnostic dose FBFHCT for ventilation measurements by quantifying the difference in measured ventilation between CT protocols matching radiation oncology and diagnostic dose levels. **Methods:** In this study, noise was injected into six 40mAs FBFHCT scans to simulate 20mAs images, correlating to total doses of 3.6mSv. The first image was arbitrarily selected to be the reference, and the other five were chosen based on simultaneous respiratory bellows signals that sufficiently sampled the range of breathing amplitudes, simulating prospective scanning. The images were registered to the reference image with the pTVreg algorithm. To assess the effect of the dose reduction on registration accuracy, the Jacobians and their fit to the breathing amplitudes (dJ/dA) were calculated for each voxel and compared between doses. **Results:** Among the patient datasets, the mean absolute difference in the voxel-specific Jacobians between 40mAs and 20mAs scan sets was 0.031. 86.8% of voxels had an absolute Jacobian difference of 0.05 or less. The mean difference in dJ/dA between doses was 0.04 Jacobian per breath, and 80.3% of voxels were within 0.05. **Conclusion:** We have measured the change in the Jacobians when reducing the image protocol dose to be acceptable, thus demonstrating that CTs in the diagnostic dose range can be used to correctly measure ventilation. This work opens the door to motion modeling, biomechanical modeling, and regional ventilation mapping outside of the radiotherapy domain.

PO-GePV-M-137

Usefulness of 4DCT-Based Elasticity Map for Function Preserving Liver Radiotherapy Treatment Planning

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Purpose: Radiation-induced liver disease is the problem for liver radiotherapy. We have developed the 4DCT-based elasticity map and consider integrating the elasticity map into treatment planning. The purpose of this study is to evaluate the relationship between dosimetric metrics of functional liver based on the elasticity map and serum biomarkers of liver function. **Methods:** Twenty patients who underwent radiotherapy for liver were enrolled in this study. All patients were performed 4DCT scan for whole liver. Deformable image registration was performed using CT images at end-exhalation and end-inhalation and

deformable vector field (DVF) was acquired. Liver elasticity was calculated using DVF and finite element method. Functional liver regions were defined four categories using calculated liver elasticity (1) 5.00 kPa (2) 4.00 kPa (3) 2.93 kPa (4) whole liver. For all treatment planning, the prescribed dose was recalculated equivalent dose in 2 Gy fraction (EQD2). The relationships between dosimetric metrics (Mean, V5, V10, V20, V30Gy) for each functional liver, variation of aspartate aminotransferase to platelet ratio index (APRI) and fibrosis-4 index (FIB-4) were evaluated. **Results:** Functional liver regions could be classified based on the elasticity map. The mean functional liver volumes were 634.46, 274.77, and 120.00 cc for 5.00, 4.00 and 2.93 kPa, respectively. The means of the variation of APRI and FIB-4 were 0.49 and 2.85. For the anatomical liver region, all dosimetric parameters for functional liver were low or moderately correlated with APRI (R: 0.45-0.55) and FIB-4 (R: 0.36-0.54). For the functional liver region ≤ 2.93 kPa, all dosimetric parameters for functional liver were highly or moderately correlated with APRI (R: 0.66-0.81) and FIB-4 (R: 0.52-0.71). **Conclusion:** We have developed the loadmap to integrate the elasticity map based on the 4D-CT for radiotherapy treatment planning. That may be helpful for functional liver radiotherapy in clinical practice and deserves further analysis.

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PO-GePV-M-138

4DCT Based Ventilation Imaging Correlation to Pulmonary Function Test for SBRT Lung Patients

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Purpose: Advances in early diagnosis in lung cancer have increased the interest in preserving post-radiotherapy lung function, especially with commonly used lung SBRT/SABR radiation treatment; therefore, lung function map is extra important for radiotherapy treatment planning. During treatment simulation for lung cancer radiation therapy, a CT scan with different phases/4DCT can be obtained for the evaluation of lung function. This project was to validate and utilize the new ventilation calculation prototype that Siemens provides in order to implement lung function sparing radiotherapy treatment planning. **Methods:** Patients with lung disease treated with SBRT (total 50-55Gy) at a single institution between 2011-2019 were retrospectively identified from a prospectively collected database. Pretreatment and post-treatment 10-phase 4DCT scans were included that are compatible with the Siemens E-Xamine software (Siemens Healthineers, Germany). We imported secondary data from these 4D scans to create a CT ventilation and aeration map using the E-Xamine software. The ventilation map calculated based on Hounsfield Unit changes with deformable registration was used in the E-Xamine. Pre-treatment CT with isodose levels, ventilation levels and tumor contours (GTV, ITV and PTV) and delivered doses (20Gy and 5Gy) were registered to post-treatment CT. The coefficient of ventilation (variance of the whole lung map) and lung V20 and V5 were correlated with pre and post pulmonary function test (PFT) results. **Results:** All 6 patients have pre/post PFTs within 5 months from the pre-treatment and post-treatment 4DCT scans. Relative changes in FEV1/FVC for V5Gy trended negatively with slope of -0.9 and r square value of 0.77. Relative changes in FVC for V20Gy trended positively with slope of 3.3 and r square = 0.61. **Conclusion:** E-Xamine calculated ventilation maps have a medium acceptable level respective to the PFT value changes.

PO-GePV-M-139

Multimodality Image Registration for Application in Brachytherapy Based On Automated Organ Segmentation

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Purpose: Treatment planning for modern brachytherapy requires imaging acquisitions such as computed tomography (CT) and magnetic resonance imaging (MRI). CT remains the first choice for treatment planning. MRI provides detailed information about treatment target delineation and soft tissue anatomy. Multimodality image registration is usually needed but often needs manual intervention. A novel automated multimodality image registration method based on automated organ segmentation is proposed and validated in this study, with the expectation to improve efficiency of treatment planning workflow for brachytherapy. **Methods:** CT and MRI images acquired on the same day from fourteen patients with locally advanced cervical cancer (stages IB3-III) treated with brachytherapy were used. A U-Net model was first trained to automatically segment the bladder in both CT and MRI. Data from eight patients were used for training, and data from the other six patients were used for validation. The Dice scores were used to evaluate segmentation accuracy. Rigid image registration was performed based on automated alignment of the contours. The average L2 differences between the translation vectors from the automatic registration and those from human were calculated to evaluate quality of registration. The proposed method was compared with conventional registration method based on mutual information provided by Advanced Normalization Tools (ANTs). **Results:** The Dice metrics for bladder segmentation in MRI and CT datasets were 0.89±0.04 and 0.80±0.06 respectively. Compared to ANTs using MRI and CT images directly (23.5±22.2mm, 3.5-52.3mm.), the U-Net-based registration significantly reduced the L2 distance error (5.9±4.0mm, 2.0-10.9mm) in the six testing patients against the ground truth. **Conclusion:** In this pilot study, a new multimodality image registration method based on automated organ segmentation is proposed and tested. Initial results showed a relatively high accuracy and robustness of this method with potential for future clinical applications.

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PO-GePV-M-140

A Study of 2D/3D with 3D/3D Registration Using KV/kV, KV/2.5X/6X

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Purpose: To validate the 2D/3D registrations by comparing the registration accuracy, image quality, and dose with the 3D/3D registrations. **Methods:** Cranial (CIRS STEEV) and body (BrainLab Pelvis) phantoms were used as an imaging target. Each phantom was placed near the treatment positions with deliberate small random translations and rotations. The 6D transformations were measured independently by seven techniques: 3D/3D match with CBCT, 2D/3D match with kV/kV on TrueBeam (kV/kV-T), kV/kV with ExacTrac (kV/kV-E), and kV/MV with four MV MU settings (2.5X-Hi, 2.5X-Lo, 6X-Hi, 6X-Lo). The 3D/3D registrations were used as the ground truth to evaluate the accuracy of the planar image group using 2D/3D registration. The difference between 3D/3D and 2D/3D was calculated and assessed with the Wilcoxon Rank-Sum Test (p-value<0.05). A Varian Las Vegas phantom was used to compare the image quality of different kV or MV to evaluate the impact of image quality at registration. 2 OSLD chips were placed at the surface, mid-plane, and exit of the phantoms to measure imaging doses. **Results:** From 23 measurements (cranial-11, body-12), the mean residual errors of each 2D/3D method in each translational and rotational dimension were small ($\leq 0.3\text{mm}/0.3^\circ$). kV/kV-E and kV-2.5X-Lo had the smallest standard deviations (SD) of 0.3mm/0.3° in all dimensions; kV/kV-T, kV-2.5X-Hi, and kV-6X-Hi showed SD of 0.5-0.7mm/0.3-0.5° at Lateral/Roll direction; kV-6X-Lo had the highest SD of 1mm/0.8° at Lateral/Roll direction. Compared to 3D/3D, 2D/3D registrations showed no significant difference (p-values: 0.13-1.00). 2.5X showed comparable contrast to kV. 6X contrast was worse than kV/2.5X. The maximum surface/mid-plane/exit doses (mGy) were 23.0/16.9/8.8, 24.8/28.9/19.7, 0.22/0.16/0.02, and 1.50/0.23/0.05 for 2.5X, 6X, 80/100kV-E, and 80/100kV-T, respectively. **Conclusion:** All 2D/3D registration methods in this study showed no significant difference to 3D/3D. 2.5X has comparable image quality to kV and lower doses than 6X. The MV imaging doses were 10 times higher than kV.

PO-GePV-M-141

Using Previously Registered CBCT Image to Facilitate Online HCT to CBCT Image Registration in Lung Stereotactic Body Radiation Therapy

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Purpose: In our conventional image registration workflow, 4D-CBCT was directly registered to reference HCT using a dual registration approach in the Elekta XVI software. In this study, we proposed a new HCT-CBCT auto-registration strategy using previously registered CBCT as the reference image. **Methods:** From a previous CBCT session, the average 4D-CBCT was selected as CBCT0 and the registration vector of HCT-CBCT0 from clinician's registration was recorded as F0. In the new CBCT session, auto-registration was first performed between the new average 4D-CBCT (CBCT1) and CBCT0, and the registration vector of CBCT0-CBCT1 was recorded as F1. During this auto-registration step, the reference point in CBCT0 was shifted from its image center by F0, while the reference point in CBCT1 was kept at its original position. This operation would eliminate the uncertainty associated with rotation vectors. Combining F0 and F1 mathematically, the final registration matrix of HCT-CBCT1 was derived and 6 registration parameters were solved. The final registration parameters were compared with clinician's results (ground-truth) to evaluate its accuracy using a test dataset consisted of 32 challenging registration cases. **Results:** The uncertainty of the new auto-registration method was -0.1 ± 0.5 , -0.0 ± 1.0 and -0.2 ± 0.7 mm in 3 translational directions (lateral, longitudinal and vertical, respectively) and 0.0 ± 1.5 , 0.5 ± 1.7 and 0.5 ± 0.8 degree for rotations. Two patients (6.25%) had translational uncertainty >2 mm (max=2.5mm) and both occurred in the longitudinal direction. Meanwhile, the uncertainty of the conventional HCT-CBCT1 auto-registration was -0.4 ± 2.6 , -0.2 ± 7.4 , -1.4 ± 3.6 mm for translations and -0.3 ± 1.2 , 0.1 ± 1.5 and 0.0 ± 1.1 degree for rotations with eleven patients (34.4%) had translation uncertainty >2 mm (max=30mm) in at least one direction. Accuracy in translation was significantly improved with the new method while rotation accuracy stayed in the same order. **Conclusion:** By utilizing prior registration knowledge, the proposed auto-registration method provides a quick and reliable starting solution for online HCT-CBCT registration.

PO-GePV-M-142

Comparing Dose Summation Techniques for External Beam and LDR Brachytherapy for Combined Modality Prostate Cancer Treatment

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Purpose: To compare composite biological effective dose (BED) of prostate cancer patients from external beam radiation therapy (EBRT) and low dose rate (LDR) brachytherapy contributions through image-based rigid registration, fiducial-based rigid registration, b-spline deformable image registration (DIR), and dose volume histogram (DVH) parameter summing techniques. **Methods:** Ninety one (91) patients received intensity-modulated EBRT before ¹⁰³Pd-based LDR. Post-implant LDR plan CTs were registered to EBRT planning CTs via image-based rigid registration, fiducial-based rigid registration, and DIR. EBRT and LDR physical doses were converted to BED and summed via the 3 image registration techniques and scalar summation of select DVH parameters without registration. Results were compared to summed dose of 44 "control" patients whose post-implant and EBRT planning CTs were inherently registered via DICOM because they received LDR before EBRT. Prostate, urethra, bladder, and rectum dosimetric parameters were compared for statistically significant differences using ANOVA ($\alpha=0.05$). **Results:** Among the three image registration techniques, prostate D90 was higher for fiducial-based rigid registration (122.6 ± 19.6 GyBED) compared with image-based rigid and DIR (114.6 ± 18.4 GyBED, 116.1 ± 16.5 GyBED respectively), and all three were lower than the control group (130.1 ± 19.7 GyBED). Scalar summation of prostate D90 was significantly higher than all 3 image registration techniques and the control group (168.3 ± 17.2 GyBED), potentially due to interobserver variation in EBRT and LDR prostate contour. Composite urethra D10, rectum D2cc, and rectum D0.1cc from image-based rigid registration were significantly higher than the other registration techniques (3-7% for urethra, 16-20% for rectum D2cc, 26-30% for rectum D0.1cc.) **Conclusion:** Fiducial-based rigid registration and deformable registration yielded combined BED distributions that were closest (within 5-10%) to BED distributions from inherently-registered treatment plans. Scalar summation of DVH parameters may overestimate combined dosimetry, particularly with discrepancies in contour delineation between external beam and brachytherapy treatment plans.

PO-GePV-M-143

Evaluation of the Displacement of the Target Position Caused by Residual Setup Errors in Stereotactic Radiosurgery with Single-Isocenter Volumetric Modulated Arc Therapy for Multiple Brain Metastases

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Purpose: To evaluate the displacement of the target position caused by residual setup error (RSE) in stereotactic radiosurgery (SRS) with single-isocenter volumetric modulated arc therapy (SI-VMAT) for multiple brain metastases. **Methods:** A total number of 417 gross tumor volumes (GTVs) from 72 consecutive patients who underwent SRS with SI-VMAT between April 2017 and June 2019 were included. All patients were immobilized with a frameless mask. All treatment plans consisted of 3 or 5 arcs including non-coplanar arcs. Patient positioning was performed with ExacTrac X-ray system (ETX) (Brainlab) and a six-degree-of-freedom (6-DOF) robotic couch. Verification and correction of the patient position was performed until within tolerance values (0.7 mm/ 1.5°) for each arc. The RSEs were regarded as the values on the ETX obtained just before irradiation. The planned center positions of each GTV from treatment isocenter for all patients were calculated. Also, the distance between the planned center position of each GTV and treatment isocenter (d) was calculated. Then, the shifted center positions of each GTV caused by RSEs were calculated using in-house program. Finally, the distance between the planned center position and the shifted center position for all GTVs (Δd) was calculated, and the correlation of Δd and d was analyzed. **Results:** A total number of 7,270 shifted center positions were analyzed. All 1,185 RSEs were within the tolerance. The median of d was 49.5 mm (range: 4.9 – 95.0 mm). The frequency of Δd within 1 mm was 97.0%. No correlation was observed between Δd and d ($r=0.11$). The maximum value of Δd was 1.93 mm, which was observed in a tumor with d of 73.3 mm. **Conclusion:** The patient positioning with a frameless mask, ETX, and 6-DOF robotic couch is highly accurate in SRS with SI-VMAT for multiple brain metastases even when targets are far from treatment isocenter.

PO-GePV-M-144

A Quantitative Analysis of Lung Registration Error in a Clinically Used Deformable Image Registration Implementation Using Manually Identifiable Features in Fast Helical Free-Breathing CT Scans

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Purpose: Deformable image registration (DIR) of fast helical free-breathing CT (FBFHC) scans produces deformation vector fields (DVF) which describe the motion within the lung at the voxel level. The accuracy of these results are difficult to quantify and remain unverified. In this study, we aim to measure the error of a clinically used lung DIR implementation using identifiable landmark feature tracking. **Methods:** We developed a feature tracking algorithm of manually identified and selected structures within the deformably registered reference and target lung images. A set of 25 patient datasets (source-target image pair and associated DVFs) were retrospectively employed for this study. DVFs were generated using dense displacement sample (deeds) DIR. 50 manually identifiable structures were selected on each reference scan and the corresponding structures were then identified on the associated target scan. The algorithm accepts the input of a precise point (single voxel) on the identified feature selected by a user in the reference image. After displaying the corresponding deformed voxel location in accordance with the DVF, the algorithm accepts a true voxel location, which was selected based on the corresponding feature on the target image. The Euclidean distance and percent error between the deformed voxel location and the true voxel location was then calculated and reported by the

algorithm for each set of feature points. **Results:** The average distance/registration error measured across all features and datasets was 0.83mm (min: 0.28mm, max: 1.65mm, std: 0.37mm). The average percent error was observed to be 13.18% (min: 4.74%, max: 30.84%, std: 7.76%). **Conclusion:** This work employed a feature tracking tool to quantify the image registration accuracy of a clinically used DIR lung registration algorithm for FHFBC scans.

PO-GePV-M-145

A Method to Consider Dose Distribution and Dose Gradient in Evaluating Image Registration for Dose Accumulation

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Purpose: Deformable dose-accumulation accuracy depends both on local spatial dose gradients of each treatment fraction and the accuracy of the underlying image registration. Conventional deformable registration evaluation methods focus only on the geometrical target registration error (TRE) without considering the dose-distribution characteristics. In this study, we propose a general procedure to incorporate dose distribution features in the registration evaluation process. It can support the selection of registration methods for patient-specific treatment planning and the estimation of dose accumulation uncertainty from the chosen registration methods. **Methods:** Our method generates a TRE tolerance map based on the dose distributions and the user-selected dose-error tolerance level in 4 steps: 1) Region of interest (ROI) are selected to excludes low-dose areas. 2) In a 15 mm radius neighborhood around each voxel in the ROI, the neighborhood voxels with doses that deviate from the center voxel dose by more than the dose-error tolerance (5% of the prescribed dose) are labeled. 3) For each voxel, the TRE tolerance vector is created that points to the nearest labelled voxel in its neighbor and has a magnitude equal to the distance between center and nearest labelled voxels. 4) The TRE tolerance map was composed by the TRE tolerance vectors for all voxels in the ROI. **Results:** We applied the proposed method to generate the dose-based TRE tolerance map for a SBRT case, and evaluated a Demons image registration method based on landmarks and the tolerance maps. The average landmark TRE was 2.7 mm and >90% of landmark pairs passed the TRE tolerance test. **Conclusion:** A procedure was developed to generate the dose-based TRE tolerance map, which could assist the selection of image registration method for subject-specific cases or the production of dose accumulation uncertainty from the registration methods.

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PO-GePV-M-146

High-Resolution Spine Digital Reconstructed Radiograph (DRR) for Motion Tracking in Paraspinal Treatment

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Purpose: High quality DRR is essential to image-based motion tracking systems. In this work, we propose a method of creating high-resolution digital reconstructed radiograph (DRR) to facilitate image registration for a recently developed MV-KV imaging-based motion tracking system for paraspinal SBRT treatments. **Methods:** High-resolution spine-only DRRs were computed by incorporating planning CT information into high-resolution CBCT reconstructions. Varian's proprietary software iTools Reconstruction can reconstruct with an isotropic voxel size as small as 0.25mm, which is comparable to kV image resolution. A major challenge is to segment bones from CBCT because of its high noise level and numerous artifacts. Our strategy is to create a binary bone mask using planning CT images as they have much better image quality. The mask defines the volumes occupied by bones in CT, and in CBCT as well, provided that the patient is set up well against planning CT. Major steps of creating bone mask include: (1) Identify CBCT volume in CT; (2) Apply a soft threshold to extract cortical bones; and (3) Connect mask edges using convex hull. Bone volumes are segmented by multiply CBCT with the mask and are then projected to create bone-only DRRs from desired gantry angles using a detector pixel-driven approach. **Results:** The quality of DRRs indicated that high noise level and artifacts associated with high-resolution CBCT did not impact DRR quality. Compared to DRR created from CT sim, the high spatial resolution CBCT created DRRs sharper and show more anatomic details. High-resolution DRR profiles represent actual KV image profiles well. DRR-to-KV registration uncertainties are generally around 0.25mm. **Conclusion:** The proposed method is able to produce DRRs with a resolution and image quality comparable to kV images, which makes it possible for achieving sub-millimeter tracking accuracy for paraspinal SBRT.

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PO-GePV-M-147

Automated Detection of IGRT Shift Errors Due to Mis-Registration

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Purpose: Critical errors can result from incorrect shifts being applied during Image-Guided Radiation Therapy (IGRT). Incorrect shifts can result from mis-registering cone-beam CT (CBCT) images to reference images, or failing to move the linac couch by the intended shift. Once couch coordinates are captured, tolerance tables can be effective at identifying IGRT shift errors due to mis-registration. However, if the initial couch coordinates are captured incorrectly during verification simulation, a systematic error in the treatment could result. We demonstrate here an automated method to detect IGRT shift errors due to mis-registrations. **Methods:** Daily CBCT images and spatial registration object files containing actual IGRT shifts for six patients were transferred to MIM. Residual shifts were determined by the difference between actual IGRT shifts and an automated rigid registration performed by MIM ART Assist. To determine an action level for residual shift errors, treatment plans were recalculated on daily CBCT images with isocenter shifts of +/- 0.5, 1.0, 2.0, and 5.0 cm applied along the three cardinal directions using MIM SureCalc ART. Automated detection of residual shift errors was then implemented clinically in MIM ART Assist. **Results:** For shift errors of +/- 0.5 cm, no difference in target coverage was detected. For shift errors of +/- 1.0 cm, target coverage fell to 94% on average (range 92-97%). Beyond +/- 1.0 cm, target coverage fell below 80%. Based on these results, a residual shift error action level of +/- 1.0 cm was employed. During clinical use over 323 fractions, 8 fractions (3 chest, 1 abdomen, 4 pelvis patients) demonstrated residual shift errors exceeding +/- 1.0 cm that required further investigation. **Conclusion:** We have demonstrated an automated method to detect IGRT shift errors due to mis-registration. A residual shift error action level of +/- 1.0 cm may be appropriate to trigger further investigation.

PO-GePV-M-148

Intentional Deep Overfit Learning (IDOL) : An Application to Deep Learning Based MRI Super-Resolution

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Purpose: Deep learning (DL)-based MRI super-resolution (SR) reconstruction has recently been receiving attention due to the significant improvement in spatial resolution compared to conventional SR techniques. However, there remain several barriers to the effective implementation of these approaches in a clinical setting. Low-resolution (LR) MRIs captured in the clinic contain complex tissues obfuscated by noise that are difficult for a simple DL framework to reconstruct. Moreover, training a robust SR network requires abundant, perfectly matched pairs of LR and high-resolution (HR) images that are often clinically

unavailable. We propose here a patient-specific SR approach utilizing the Intentional Deep Overfit Learning (IDOL) framework that demonstrates a significant improvement in SR MRI quality using only a few training data sets. **Methods:** IDOL-based SR is an intentional application of overfitting that leverages patient-specific prior information. Training in the IDOL framework involves two stages: 1) conventional training of a generalized SR model with a training set of LR and HR MRIs of 4 human volunteers, and 2) using the inhale respiratory phase of a 5th human subject as prior information, the generalized model is trained further to achieve the IDOL model that is applied to the exhale respiratory phase of the 5th subject. For testing, step2 was repeated for subjects 5-8 using the same generalized model as a starting point. **Results:** The proposed method was evaluated on a cohort of 4 volunteers. The generalized model (step1 training only) achieved a normalized mean absolute error (MAE) of 446.5 and peak signal-to-noise ratio (PSNR) of 31.33. By adopting the personalized IDOL model in each test case, the MAE and PSNR improved to 321.5 and 34.72, respectively. **Conclusion:** In this study, we have successfully applied the IDOL framework to the MRI super-resolution task. Our approach is widely applicable to common tasks in radiotherapy and can be utilized for MRI-guided ART.

PO-GePV-M-149

Left Ventricle Avoidance in Non-Small Cell Lung Cancer Radiotherapy: A Planning Comparison Study

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Purpose: To provide a dosimetric comparison among NSCLC treatment plans to determine whether minimizing dose to the left ventricle (LV) will significantly increase the dose to other organs at risk, while maintaining target coverage. **Methods:** Fifteen Stage III non-small cell lung cancer (NSCLC) patients who have received a standard 60Gy in 30 fraction volumetric-modulated arc therapy (VMAT) treatment and with a V20Gy = 20-30% were retrospectively replanned. The LV and left anterior descending artery (LAD) were additionally contoured on the end inhale and end exhale CT scans extracted from the clinical 4D-CT dataset and registered to the untaged average CT dataset that contained clinically defined targets and normal structures. The clinical treatment plans were retrospectively optimized with the goal of minimizing the mean(LV), and mean(heart) dose while maintaining the dose constraint of V20GyTotal Lung ≤ 30% and V95%PTV ≥ 95%. Dose-volume histograms and paired T-tests were used to compare the heart and lung dosimetric parameters, including mean heart, LV, and LAD doses, V5GyHeart, mean lung, V5GyLung and V20GyLung between the original and optimized treatment plans. **Results:** Significant reduction ($p \leq 0.044$) were observed in mean(LV), mean(heart), mean(LAD), max(LAD) dose and V5GyHeart in the re-optimized treatment plans compared to the clinically delivered plans. V20GyLung ≤ 30% and V95%PTV ≥ 95% were maintained and insignificant differences were determined for parameters of mean(Lung), V5GyLung and V20GyLung, mean(esophagus) and mean(cord). **Conclusion:** This study demonstrated that minimizing left ventricle dose in NSCLC treatment plans were achievable and dosimetrically advantageous in the heart while maintain dose constraints to normal lung and maximizing tumor control. Radiation dose reduction to cardiac substructures such as the left ventricle and LAD may decrease the risk of radiation-induced cardiac toxicity in this patient population.

PO-GePV-M-150

Comparison of a Commercial Calculation-Based and Offline Measurement-Based Systems for Online Adaptive Radiotherapy

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Purpose: Measurement-based patient-specific quality assurance (PSQA) is not possible for online adaptive radiotherapy (oART). The Varian Ethos CBCT-guided adaptive system utilises MobiusAdapt for online PSQA. This work compares MobiusAdapt and measurement-based PSQA for original and oART plans. **Methods:** Fifty-two clinical IMRT plans (17 original and 35 adapted) were generated with Ethos. A variety of treatment sites were selected to ensure the plan complexity range reflected that of typical clinical practice. MobiusAdapt calculations were performed pre-treatment for original, and pre-beam-on for oART plans. Post-treatment, phantom-based dose measurements were performed for all plans using Delta4+ for 2D dose verification and a homogenous CIRS phantom for point dose verification. Different analysis criteria were evaluated to aid the comparison. For MobiusAdapt, gamma analysis was performed with 95% pass-rate for 3%/2mm and 3%/3mm (department criteria). For Delta4+, gamma analysis was performed with 95% pass-rate for 2%/2mm, and 3%/2mm (department criteria). CIRS point dose deviations (DD) were evaluated with department criteria 3% and compared to MobiusAdapt target structure meanDD, and Delta4+ medianDD. Results were compared for each plan, and the pass-rates relative to plan complexity metric MU/Gy evaluated. **Results:** All plans passed department PSQA criteria. Average gamma pass-rates were 98.3% and 99.8% for MobiusAdapt 3%/2mm and 3%/3mm criteria, and 92.4% and 99.9% for Delta4+ 2%/2mm and 3%/2mm criteria. MobiusAdapt 3%/2mm pass-rates were on average 1% higher (range -2% to 3%) compared to Delta4+ 3%/2mm. Both CIRS point dose and Delta4+ medianDD measured on average 1% higher than Ethos, indicating agreement between measurement methods. Conversely, MobiusAdapt target meanDD was on average 1.4% lower than measurement. Greater plan complexity (MU/Gy) resulted in lower pass-rates for MobiusAdapt and Delta4+, and greater medianDD for Delta4+. **Conclusion:** Online calculation-based QA and post-treatment measurement-based QA was performed on Ethos IMRT plans of varying complexity. MobiusAdapt gamma pass-rates were comparable to Delta4+.

PO-GePV-M-151

Evaluation of Dosimetric Impact of Organ Variation on Daily Basis Using MIM Image Registration Software

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Purpose: Evaluating the dosimetric impacts of the variations of organs during day to day delivery is a challenge. The aim of this study is to demonstrate a simple and efficient method to evaluate rectum and bladder dose volume histogram (DVH) on daily basis using MIM image registration software for prostate cancer radiotherapy. **Methods:** A prostate cancer patient with 76 Gy in 38 fractions was selected for this pilot study. The rectum and bladder contours were drawn on the daily CBCT images transferred in MIM. Each daily CBCT and original CT image data set was rigid-image-registered and the dose was transferred to the CBCT in MIM. The DVHs of bladder and rectum for each CBCT and the average of all fractions of DVH with one standard deviation (SD) generated in MIM were analyzed and graphed for the bladder and rectum. **Results:** Figure 1 displays the volume variations of bladder and rectum on daily CBCT. Figure 2 and 3 shows the DVH of bladder and rectum on CBCT. The variation of DVH on daily basis can be seen. Figure 4 and 5 compared the DVH of bladder and rectum between initial planning and the daily averaged DVH on CBCT. The dose parameters such as Maximum Dose and Vx are noticeable different. Table 1 lists the averaged dose parameters of maximum dose and Vx, e.g., V76, V70, V60, V50, and V40 of the bladder and rectum on CBCTs, and the corresponding dose parameters for the original planning CT. **Conclusion:** This study demonstrated the dosimetric impacts due to the changes of bladder and rectum as detected on daily CBCT during the course of prostate cancer radiotherapy. The comparison of dosimetric parameters and DVH between original treatment plan and day to day delivery on CBCT shows a noticeable difference due to the organ changes.

PO-GePV-M-152

Deep Learning Markerless Spine Tracking Using Intrafraction Motion Management (IMR) Images: A Feasibility Study for Paraspinal SBRT Motion Management

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Purpose: To investigate the feasibility and accuracy of a deep-learning based spine tracking algorithm using on-treatment kV imaging. **Methods:** Varian's on-treatment IMR allows visual inspection of patient movement during treatment, it however does not provide quantitative information about the motion. In this

study, we trained a deep convolutional neural network to quantify the translational movement indicated by an IMR image. The proposed neural network registers an IMR to the corresponding spine-only DRR which is a pre-processed reference image representing the initial spine position established during patient setup. The neural network modifies Resnet101 to allow the input of a pair of images and regression for the relative translation (ΔX , ΔY) between them. The training and testing images were collected from nine patients treated for C-, T-, and L-spines, among which 390 images from eight patients were used for training and 41 images from the last patient were used for testing. To generate the full training data, we augmented the image set by randomly translating the IMR images to simulate patient movement. Each translated IMR is paired with the corresponding spine-only DRR to form an input sample, and the applied artificial translation is the training label. **Results:** The trained neural network were evaluated on the test dataset with simulated spine movements up to 5mm. The mean detection error was below 0.7mm. The performance was compared to mutual information (MI) based matching implemented by Matlab. We found that our model is significantly more accurate and robust at predicting spine movements larger than 2mm, in which case MI-based method were undermined by inherent soft tissue interference. **Conclusion:** The proposed deep convolutional neural network is capable of detecting spine motions with sub-millimeter accuracy for movements up to 5mm. Our results suggest that the proposed deep-learning approach is feasible for spine tracking during a SBRT treatment.

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PO-GePV-M-153

Analysis of a Surface Imaging System Using a Six Degree of Freedom Couch

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Purpose: To compare surface imaging (SI) logs with MV images and linear accelerator trajectory logs to validate SI reported offsets on an anthropomorphic phantom for commissioning and routine quality assurance of a SI system used for stereotactic radiosurgery (SRS). **Methods:** An anthropomorphic Styrofoam phantom (Floracraft, Ludington, MI) with a radiopaque tungsten carbide ball bearing (BB) placed midline was tracked with SI with a region-of-interest (ROI) representative of a typical SRS ROI. Couch motion was programmed in developer mode and delivered on an Edge (Varian Medical Systems, Inc., Palo Alto, CA) linac. IDENTIFY (Varian Medical Systems, Inc., Palo Alto, CA) logs were synchronized with linac trajectory logs. Ten random couch positions were selected at couch rotations of 0, 45, 90, 270, 315° with MV images taken to account for couch walkout. The IDENTIFY residual error (ϵ), the difference between SI reported offset and MV or trajectory log position, was calculated. For lateral and longitudinal directions, ϵ was calculated from MV images. For vertical, and rotational ϵ , linac trajectory logs were used. **Results:** The median [range] of magnitude of translational ϵ was (0.17 [0.05, 0.27], 0.39 [0.21, 0.47], 0.43 [0.30, 0.53], 0.43 [0.33, 0.60], 0.46 [0.41, 0.63]) mm. The mean $\pm$ standard deviation (std) roll ϵ was (-0.01 $\pm$ 0.01, 0.28 $\pm$ 0.01, 0.55 $\pm$ 0.02, -0.30 $\pm$ 0.02, -0.26 $\pm$ 0.01) °; mean $\pm$ std pitch ϵ was (0.03 $\pm$ 0.01, 0.06 $\pm$ 0.01, 0.21 $\pm$ 0.01, 0.44 $\pm$ 0.02, 0.22 $\pm$ 0.01) °; mean $\pm$ std yaw ϵ was (0.02 $\pm$ 0.01, 0.08 $\pm$ 0.02, -0.02 $\pm$ 0.03, -0.08 $\pm$ 0.02, -0.02 $\pm$ 0.01) °, for couch rotations of 0, 45, 90, 270, 315°, respectively. **Conclusion:** SI system and linear accelerator trajectory log analysis can be used to assess SI system performance with automated couch motion to validate accuracy for systems used for SRS.

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PO-GePV-M-154

Impact of Various Noise Reduction Techniques on Markerless Tumor Tracking in Dual Energy Imaging

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Purpose: This study aims to quantitatively evaluate the performance of various noise reduction algorithms on markerless tumor tracking using dual-energy (DE) imaging. **Methods:** DE images were obtained using the fast-kV switching capabilities of the on-board imager (OBI) of a commercial linear accelerator with alternating acquisitions at 60 and 120 kVp. Weighted log-subtraction was then used on paired 60/120 kVp images to generate soft tissue images. Both the anti-correlated noise reduction (ACNR) technique, and 2D-median filter were applied to subsequent DE images. Additionally, the adaptive histogram equalization (AHE) algorithm was also evaluated for comparison. The tumor motion on DE-unfiltered and DE-filtered image sequences was tracked using a template-based matching algorithm. Ground truth positions from the DE-unfiltered and DE-filtered images were estimated using Bayesian inference. **Results:** A total of 1145 DE-unfiltered and DE-filtered image frames were evaluated for 5 patients. The DE-filtered images showed slight variation in tracking results vs. DE-unfiltered images, with reduced tracking error > 0.1 mm in 30.2% (346/1145), 13.8% (159/1145), 12.5% (144/1145) of image frames using median filter, ACNR, and AHE methods respectively. However, DE-filtered images only showed improved results for tracking error > 1.0 mm in 1.57% (18/1145), 1.57% (18/1145), 2.18% (25/1145) image frames, with median filter, ACNR, and AHE techniques respectively. **Conclusion:** These noise reduction techniques do not significantly affect tumor tracking on DE images. Thus, DE images are suitable to track the tumor without using any additional post-processing. This is advantageous since the addition of noise reduction techniques increases image processing times and affects system latency.

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PO-GePV-M-155

Feasibility Study of An InBore Real-Time Surface Monitoring System

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Purpose: To evaluate the performance of the InBore real-time monitoring system (Vision RT Ltd., London, UK) on the Halcyon and Ethos-type linacs. **Methods:** The study was designed with 14 test sets to evaluate the InBore camera system safety and accuracy. The Catphan phantom was used to test the InBore camera system's impact on beam delivery for typical clinical cases using different IGRT techniques. The InBore camera system's physical stability was tested for the potential collision risk between the treatment couch and treatment accessories as well as access for Halcyon service. The monitoring accuracy was measured with the VisionRT Cube phantom, static head phantom, thorax phantom, and the breathing thorax phantom. **Results:** There was no dosimetric difference between the pre-and post-InBore camera system installation for multiple plans and several disease sites. The InBore camera system does not obscure the beam delivery pathway and no impact was observed on the image quality on the Halcyon IGRT techniques. The result of the monitoring accuracy for static phantoms (> 20 min monitoring) was < 0.4 mm in any direction and < 0.1 degree in any rotation axis. The result of geometric accuracy using the known translational shifts (1, 3, 5, 10, and 20 mm) in each direction, the average deviation was 0.08 mm, and maximum deviation was 0.5 mm. The monitoring range of the InBore camera system was ± 10 cm Vert, ± 7 cm Lat, ± 5 cm Lng direction. **Conclusion:** The InBore camera system on the Halcyon linac is appropriate for clinical use and active patient monitoring. It is likely that this system will be applicable for cases such as SRS, SBRT, DIBH, and adaptive cases. This monitoring technique may be useful for the adaptive therapy workflow to monitor the patient while waiting for the plan to adapt to the patient's anatomy of the day.

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PO-GePV-M-156

An iOS Surface Audiovisual Biofeedback Smartphone Application for Respiratory Monitoring in Radiation Oncology

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Purpose: Radiation dose delivered to a target located near the upper abdomen or thorax is significantly affected by respiratory motion, necessitating large margins and limiting dose escalation. Currently, normal tissue sparing can be improved by using motion monitoring systems. The amalgamation of these systems with audiovisual biofeedback can further improve the quality and delivery of radiation as the patient is able to see their breathing pattern and correct for variations leading to consistent and more predictable target motion. Alternative to current solutions, we have developed and evaluated the feasibility of an audiovisual feedback system that leverages depth information from the depth sensor onboard iOS devices to provide visual coaching to improve patient compliance and reduce treatment delivery time. **Methods:** The iOS application, coined iOS Surface Audiovisual Biofeedback (iSAVB), was developed in Swift (Apple Inc.) and implemented on an iPhone® with the TrueDepth camera system. Validation was performed using the QUASAR™ Respiratory Motion Phantom (Modus Medical Devices); previously recorded motion traces during free-breathing and breath-hold treatments were programmed into the motion platform and played while recording depth information using iSAVB. iSAVB measurement of displacement was compared to the input signal trace using linear regressions and Bland-Altman analysis. **Results:** Free-breathing and breath-hold traces show excellent agreement between the input signal trace from the motion phantom and iSAVB. iSAVB and QUASAR™ motion phantom traces were significantly related (free-breathing: $r=0.999$, $p<0.0001$; bias= 0.00 ± 0.04 cm, 95%-confidence-interval= -0.07 cm, 0.07 cm; breath-hold: $r=0.991$, $p<0.0001$; bias= 0.00 ± 0.08 cm, 95%-confidence-interval= -0.16 cm, 0.16 cm). **Conclusion:** Feasibility of an iOS application to provide depth information for real-time respiratory motion monitoring is demonstrated. With the ubiquity of smartphone devices, this work has the potential to provide audiovisual biofeedback to patients in a low-cost platform that will improve the efficacy of radiation therapy.

PO-GePV-M-157

Towards Real-Time Markerless Prostate IGRT During VMAT Treatment

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Purpose: Image-guided radiation therapy (IGRT) and real-time IGRT for prostate cancer typically relies on markers to determine the target location in kilovoltage (kV) images. However, the placement of markers is an additional procedure that introduces costs and side effects. We present a novel deep learning method for markerless prostate IGRT. **Methods:** A real-time tracking system using a conditional generative adversarial network (cGAN) was developed. A base model was trained on an imaging database of prostate cancer patients in the TROG 15.01 SPARK trial and was fine-tuned individually on unseen patients. The base model was trained using 17,000 kV images from 34 fractions of 17 patients. A patient-specific model was then trained for five different patients using 3,600 digitally reconstructed radiographs (DRR) per patient generated over 360 degrees. The DRRs were produced using the planning CTs and clinical target volume contours. The system was evaluated on 1,000 kV images/patient from two fractions. The implanted markers were used as the ground truth and were masked out for training and testing. The centroid geometric error was defined as the cGAN segmentation versus the ground truth and the Dice similarity coefficient (DSC) was calculated to gauge the similarity of the segmentations. **Results:** On patient images, the mean error was 0.2 ± 1.9 mm and -0.2 ± 1.5 mm in the lateral/anterior-posterior and superior-inferior directions, respectively. The [2.5th, 97.5th] percentiles of the error were [-3.4, 3.9]mm in the lateral/anterior-posterior direction, and [-2.6, 2.2]mm in the superior-inferior direction. The mean DSC was 0.92 ± 0.03 . **Conclusion:** The first algorithm for markerless prostate segmentation during an arc treatment was developed and evaluated on patient kV images. The results demonstrate that real-time markerless prostate IGRT is achievable to a high degree of accuracy using deep learning. This solution could enable more patients access to real-time IGRT without the costs, time and side effects for marker placement.

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PO-GePV-M-158

Enhancing Cherenkov Photon Yield by Gold Implants

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Purpose: To study how much Cherenkov photons are generated with and without a gold implant/marker in a water phantom for utilizing the Cherenkov photons in radiation therapy. **Methods:** The Geant4-based Monte Carlo code, GAMOS, was employed for simulating the number of Cherenkov photons produced with and without a gold marker implanted in a homogeneous water phantom under photon beam irradiation. A $10 \times 10 \times 10$ cm³ phantom was modeled. A thin gold plate with a 10 mm diameter and 1.0 mm thickness was placed at the center of the phantom. 6, 10, and 15 MV photon beams with a 5×5 cm² field size directed towards the phantom. The integrated dose and the number of Cherenkov photons produced within 5 mm above and below the gold plate, designated as ROI1 and ROI2, were calculated. Those quantities with the gold were compared with those without the gold. **Results:** The integrated dose in the ROI1 increased by 28.0, 27.2, and 24.4% for 6, 10, and 15 MV beams, respectively. The dose difference in the ROI2 increased as the photon energy increased. The differences in the numbers of the Cherenkov photons in ROI1 between with and without the gold were 23.2, 11.8, and -19.2% for the 6, 10, and 15 MV beam, respectively, whereas the difference was -0.06, 7.71, and -6.04% in the ROI2. **Conclusion:** Monte Carlo simulations showed the increase of the dose in the ROI1; thus, resulting in an enhancement of the Cherenkov photon yield by a gold marker. The difference in the Cherenkov photon yield between with and without the gold marker in the ROI1 was significant for 6 and 10 MV beams. A potential application of this phenomenon includes a non-invasive real-time identification of the marker location implanted in a tumor for motion tracking.

PO-GePV-M-159

Towards Faster and More Accurate Target Localization on Intrafraction X-Ray Fluoroscopy for Lung Radiotherapy Using Modified YoloV3 Detector

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Purpose: Recently, there has been renewed interest in using object detection algorithms to localize the treatment target on X-ray Fluoroscopy to overcome the time-consuming shortage of template matching. However, the major obstacle of deep-learning-based object detection is that the accuracy drops severely when the target is blocked by vertebral bodies. In this study, we set out to verify the usefulness of a modified YoloV3 framework combined with training data augmentation strategies to mitigate this problem. **Methods:** To compare with the previous study applying the Faster-RCNN framework, we repeated the experiments with the dynamic thorax phantom (CIRS), in which a tumor surrogate was programmed to move independently from the chest. Besides, we verified the reproducibility of the proposed method using datasets provided by The Markerless Lung Target Tracking Challenge (MATCH). The datasets consist of CT and X-ray imaging files of a 3D-printed anthropomorphic phantom and files recording 3D trajectories of the complex phantom motion. In both experiments,

the programmed 3D positions were mapped to 2D X-ray images to serve as the ground truth. **Results:** For the CIRS phantom, the mislocalization problem was addressed when the surrogate size was 18mm or 27mm. But when it decreased to 12mm, the localization failure rate(LFR) dropped to 16% compared with the result of 41% for Faster-RCNN. In MATCH datasets, for target1, targets in all frames were successfully detected and errors of 86% results were less than 1mm with an average error of (0.58±1.7mm, -0.35±2.4mm). For target2, the LFR was 7% and errors of 72% results were less than 1mm with an average error of (0.74±1.9mm, 1.32±1.47mm). The overall end-to-end processing time of an image is less than 60ms(~16.7FPS). **Conclusion:** The proposed method is obviously improved, considering the trade-off between accuracy and localization latency. It nearly addresses the mislocalization problem while further approaching real-time operation.

PO-GePV-M-160

Quantitative Interpretation of Data Acquired During Intrafraction Motion Review Imaging

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Purpose: The current clinical interface for Varian's Intrafraction Motion Review is limited, providing only qualitative data for review at the treatment console. This study provides ways of extracting and interpreting data from combined log files for quantitative evaluation offline. **Methods:** Combined log files acquired during patient treatment were exported for analysis. We developed a parsing code in C# to scan the combined file looking for unique identifiers pertaining to the data of interest. Data was saved as a comma separated .txt file that could easily be opened with spreadsheet software for further analysis. We used a time-based system to correlate treatment arcs time to specific patient treatments. **Results:** We were able to extract clinically relevant parameters from the log files including: date and time, gantry angle, expected marker position, found marker position, pixel size, and detection result. Detection result is the written classification of the color-coded results seen on the console during treatment. The stored data does not record found versus expected seed position difference, but only the X and Y positions and the detection result (ie. within or outside tolerance). Such numerical detection positional differences need to be calculated externally. **Conclusion:** The current data available for clinical decision-making is limited when using IMR. However, data is recorded in the combined log files, which can be utilized to better understand motion during patient treatment. Understanding data recorded during treatment within the combined log files can be helpful for retrospectively reviewing motion during treatment to continue to improve patient care.

PO-GePV-M-161

Clinical Performance of a New Commercial Surface Imaging System for Frameless Stereotactic Radiosurgery

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Purpose: To assess the performance of a new commercial surface imaging (SI) system by analyzing intrafraction motion from the initial cohort of patients treated with frameless stereotactic radiosurgery. Aggregate data was used to analyze system performance at non-zero couch angles and when camera pods are blocked by gantry motion. **Methods:** Patients were immobilized with an open face thermoplastic Encompass™ (Qfix, Avondale, PA) mask. The SI region of interest (ROI) was contoured in the treatment planning system (TPS) and exported to the SI system. Patients were aligned by kV followed with CBCT then the SI reference was captured. After treatment, linear accelerator trajectory logs were synchronized with SI logs to correlate SI reported offsets with couch and gantry positions. **Results:** 58 fractions from 25 patients were monitored on an Edge linear accelerator (Varian Medical Systems, Palo Alto, CA) with IDENTIFY (Varian Medical Systems, Palo Alto, CA). The median magnitude of translational SI reported offsets at the end of treatment was 0.31 mm. The median magnitude before beam-on at non-zero couch angles was 0.57 mm. SI reported offsets were shown to increase when camera pods are blocked by the gantry with larger increases seen at non-zero couch angles. **Conclusion:** SI and linear accelerator trajectory logs were used to analyze intrafraction motion during frameless stereotactic radiosurgery and evaluate SI system performance. Performance is comparable to other commercially available surface imaging systems where offsets are shown to increase at non-zero couch angles and during camera pod blockage.

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PO-GePV-M-162

Comparing Patient Setup Techniques Between Optical Surface Guidance and Conventional Tattoo/Laser Alignment Using Cherenkov Image Consistency Metrics

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Purpose: For breast radiation therapy treatments, the standard of care is traditionally to mark the patient with small tattoos which are used for alignment each day, though in over 1,700 oncology centers worldwide, optical surface guidance (SG) systems are used for patient positioning prior to receiving radiation. In this study, Cherenkov imaging metrics were analyzed in a clinic that utilizes SGRT for pre-treatment patient alignment (site 1) and compared to the metrics from a clinic that practices conventional tattoo/alignment (site 2). This was done to determine whether setup techniques were more consistent using one technique over the other. **Methods:** Free-breathing breast treatments were recorded with Cherenkov imaging each day for n = 15 patients, (fx = 134 fractions) at site 1, and for n = 5 patients, (fx = 54 fractions) at site 2. A two-tailed t-test of unequal variance was carried out to determine significance for comparison of DICE analytics (% similarity) and MDC (Mean Distance to Conformity). **Results:** MDC and DICE t-test results reveal that the null hypothesis could not be rejected, indicating that the setup techniques yield no significant difference (DICE, p = 0.432; MDC, p = 0.094). The p-value describing MDC comparison was smaller, however, possibly indicating that this may be the analytical metric more sensitive to changes in beam shape. **Conclusion:** While SGRT may be a less invasive technique via the negation of tattoos and possibly a faster setup technique, ultimately, the Cherenkov data acquired in our two clinics showed no significant difference between the surface-guidance setup technique and the conventional skin marks/laser technique. In the two clinics studied, the quality or accuracy of patient setup was not compromised by using one setup technique over the other. Cherenkov imaging for clinical real-time monitoring was shown to provide a useful tool for this analysis.

Authors Prof. Brian Pogue, Prof. M Jermyn, Prof. P. Bruza, and Dr. Lesley Jarvis M.D. have competing interests with DoseOptics LLC, a company that manufactures Cherenkov cameras used to monitor radiation therapy.

PO-GePV-M-163

Clinical Detectability of MR Suitable Prostate Fiducial Markers Utilizing the TrueBeam Advanced Imaging Package During VMAT Prostate Treatment in Phantom

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Purpose: We study the detection of MR suitable fiducial markers (FM) in an anthropomorphic phantom. We report the clinical detection rate, angular dependence, and optimized detection algorithm choice for fiducial marker tracking during VMAT prostate treatment on the Varian TrueBeam. **Methods:** A Rando phantom, holding a beefsteak prostate phantom seeded with 1, Platinum-coil (0.35x10mm) and 2, Platinum-coil (0.35x20mm) FM's, was used. The phantom received a (VMAT) treatment utilizing on-board kV imaging and our clinical FM SOP. In each image, the Varian software found and located each

marker within a 3mm radius of its planned location. Treatment was repeated 22 times using all available marker detection settings (MDS), collecting 913 images. **Results:** The “GoldSeed_1_3x3_3” MDS was optimal for identifying the platinum-coil (0.35x10mm) marker, finding it in 100% of the projection images. The platinum-coil (0.35x20mm) marker was much more robust to MDS selection, where eight different settings each identified the marker in 100% of the projection images. Little projection-angle dependence was observed, with an association only appearing when the worst performing MDS's were selected. The mean and standard deviation of the distance from the recognized marker position to the expected position in the projection images was found to be 1.25mm and 0.2mm respectively for the (0.35x20mm) marker and 1.5mm and 1.2mm respectively for the (0.35x10mm) marker using optimized MDS's. **Conclusion:** The TrueBeam Advanced Imaging Package was used during a VMAT treatment of a prostate phantom housed in an anthropomorphic pelvis phantom. The clinical detectability of two MR suitable FM's was assessed. The longer platinum markers show improved detectability. Currently, marker detection accuracy is being studied and will be compared to prior measurements of markers from the same and different vendors.

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PO-GePV-M-164

Auto-Segmentation of Organs at Risk for CT Thoracic Cases: A Single Institution's Experience

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Purpose: The purpose of this experiment is to examine one institution's experience with a 3D Convolutional Neural Network (CNN) based automated contouring method. **Methods:** A CNN trained on 658 expert-segmented images from multiple institutions was used to provide baseline contours for 129 patients undergoing thoracic radiation therapy. Five structures were included: heart, left/right lungs, esophagus, and spinal cord. Six dosimetrists edited those baseline contours to match their institutional standards. The Dice score (DSC) and Hausdorff distance (HD, mean/maximum) were calculated for each edited contour compared to the CNN contour. For each patient, the dosimetrists provided a self-reported scale of the edits required for each structure, choosing from “Minimal Edits”, “Moderate Edits”, “Not Usable”, and “N/A”. The “N/A” option was chosen if the structure was not required for the treatment plan or an automatically generated segmentation was not considered for use. **Results:** The following average statistics were calculated for each structure (DSC, mean HD, maximum HD): Heart (0.92, 2.17, 14.89), Left Lung (0.98, 0.58, 12.61), Right Lung (0.98, 0.56, 13.16), Esophagus (0.76, 10.11, 44.79), Spinal Cord (0.86, 5.50, 35.71). Of the 292 structures for which the amount of edits were reported, 83% required minimal edits, 11% required moderate edits, and 6% were redrawn. **Conclusion:** The CNN contours required minimal edits to yield clinically acceptable results, as demonstrated by the DSC, HD, and self-reported editing amounts. The esophagus and spinal cord yielded lower DSC and higher HD, (mean and standard deviation) due to variability in the superior/inferior extent of the structure that was contoured when using the CNN versus manual editing. In the future, we plan to investigate solutions to stylistic differences to improve the CNN performance with structures with variable extents.

Aaron Nelson is a part-owner of MIM Software Inc.

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PO-GePV-M-165

Deep Learning-Based Auto-Segmentation of Glioblastoma in Brain Cancer Radiotherapy

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Purpose: Contouring is a potential source of uncertainties in radiotherapy treatment planning that could affect treatment outcomes. It is important to reduce contouring uncertainties in terms of inter-and intra-observer variation. In this study, we review the most recent researches and introduce an automatic brain tumor segmentation method. Our method is able to predict multiple brain tumor regions using the Brats 2019 dataset; We evaluate and report the dosimetric impact of automatic segmentation for treatment planning. **Methods:** Two contouring methods are compared. We use manual contours as the gold standard against our deep-learning-based auto-contour. In each method, Gross Tumor Volume (GTV) was generated on 15 patients with glioblastoma cancer. Workflows were compared according to contour similarity, and dosimetric evaluation. Different treatment plans were compared using gamma index analyses were performed with 1%/1 mm, 3%/3 mm. **Results:** The GTV segmentation network achieved a DSC of 0.88 ± 0.04 on the test set. The proposed network achieved a HD of 1.27 ± 0.39 mm for the GTV. The automatically generated contours met the dose coverage constraints using gamma analyses in all of the cases for GTV. Gamma index measurements for 1%/1 mm and 3%/3 mm were achieved at 97.92% and 99.375% respectively. **Conclusion:** The proposed deep-learning-based auto-contour is fast, reliable, and achieved an acceptable agreement between the ground truth and predicted contours in relation to similarity and dosimetry. This method can provide a useful tool for automatic segmentation of the brain tumor and improve the reproducibility of contouring in the radiotherapy workflow.

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Clinical Evaluation of Deep Learning Based Auto Segmentation (DLAS) of Organs at Risk in the Head and Neck Region

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Purpose: To clinically evaluate the accuracy of DLAS of organs in computed tomography (CT) images in head and neck region utilizing deep-learning based and commercially available software. **Methods:** 9 organs at risk (brainstem, esophagus, eyes, larynx, mandible, optic nerves, oral cavity, parotids and spinal cord) in 16 head-and-neck planning CT images were firstly segmented by skilled clinicians as the ground truth. For comparison, auto-segmentation was conducted by AccuContour (Version 3.0, Manteia) using both the vendor's default and customized models. The customized model was generated by Manteia AccuLearning AI model training platform using 20 training cases following the vendor's minimum recommendations. Segmentation accuracy of both the default and customized model was assessed using Dice similarity coefficient (DSC) and mean distance metrics. **Results:** The average time to auto segment all 9 structures was 1 min/patient. DLAS with default model performed well, with an average DSC and mean distance of 0.90 and 0.46mm for mandible, 0.86 and 0.8mm for brainstem, 0.84 and 0.7mm for eyes, 0.77 and 2.0mm for esophagus, 0.77 and 3.3 mm for spinal cord, 0.77 and 3.3 mm for oral cavity, 0.74 and 7.2mm for parotids, 0.58 and 0.6mm for optic nerves, and 0.41 and 2.9mm for larynx. Significant improvement ($P=0.034$) of DLAS accuracy using a customized model was observed for all OARs except for optic nerves, with DSC and mean distance increased by 0.10 and 0.96mm respectively. The largest difference was found for larynx (DSC=0.85, distance mean=1.2mm). **Conclusion:** AccuContour DLAS of computed tomography images proved to be a useful tool in the rapid delineation of normal tissues. AccuLearning training platform can build customized models to further improve DLAS accuracy for an individual clinic with a small training cohort.

PO-GePV-M-167

PTV Auto Contouring for Prostate and Nodes Utilizing Deep Learning Artificial Intelligence

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Purpose: Create a 3D deep learning model capable of generating a Planning Target Volume (PTV) on the prostate and nodes volume with CT input images for radiotherapy treatment planning. **Methods:** We anonymized and exported 265 CT images and PTV_4500 volumes of prostate cancer patients for this project. PTVs with 45 Gy were contoured by 7 MDs and reviewed to be clinically acceptable. Two groups were formed from 212 cases for training and 53 for testing. We developed a Pixel-Deconvolutional Network (PDN) model by tweaking the weighting parameters to minimize the dice loss function using the training cases. The performance of the auto-segmentation was analyzed through the similarity DICE and Average Hausdorff Distance (AHD) between manual (gold-standard) and auto segmented PTV_4500 (predicted). Given the restrictions of computing resources, all images required resizing to lower resolution initially as input for the network. **Results:** The mean DICE/AHD scores with standard deviation (SD) for the testing set were (0.716±0.076) and (2.061±1.749 cm) for the CT image modality. If excluding one testing case with AHD > 10 cm, the mean and SD were (0.723±0.062) and (1.885±1.209 cm). The computation time for training was about 222 minutes. **Conclusion:** A deep-learning network was developed to contour the PTV for radiotherapy of prostate with nodes to 45 Gy with simulation CT images. The predicted contours have a reasonable dice score and AHD value compared with the manual contouring. We believe increased contouring accuracy is possible through more curating of the training data.

PO-GePV-M-168

Synthetic Lung MRI for Auto-Segmentation Using Unet Training: Supervised Cross-Domain Adaptation from CT to MRI

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Purpose: Magnetic resonance imaging (MRI) has become an emerging tool in radiation therapy. It improves treatment accuracy over conventional CT-based radiotherapy due to superior soft-tissue contrast. While dose calculation is usually performed with CT images, MRI contrast can be critical for accurate tumor and organ delineation and segmentation. Therefore, we developed a supervised cross-domain adaptation based deep learning method to generate synthetic MR from CT datasets, facilitating tumor and organ auto-segmentation. **Methods:** Current deep learning approach achieves reasonably accurate tumor segmentation from limited MRI data by leveraging large CT datasets. This method helps to preserve tumors on synthesized MRIs produced from CT images. We present a supervised cross-domain adaptation-based deep learning approach to generate synthetic MR from registered CT and T2-weighted MRI datasets. This approach was implemented following the steps: (1) performed image registration of CT and MRI on 15 lung cancer patients (IRB approved); (2) paired CT and MR data from 15 cancer patients was used for Unet training to generate training model for adaption; (3) a CT image from the 16th patient served as the test image for the training model, to generate synthetic MRI. The test CT image also served as the reference for image quality evaluation. Mutual information was calculated for the synthetic MRI and the test CT, and it was used as an evaluator of the trained model. This method was subsequently evaluated by comparing supervised cross-domain adaptation model with unsupervised cross-domain adaptation. **Results:** Mutual information of the synthetic MRI and reference CT calculated from supervised cross-domain adaptation is 1.41; with unsupervised cross-domain adaptation it is 1.18, lower than the former. **Conclusion:** The novel supervised cross-domain adaptation based deep learning approach showed promising results, achieving more accurate image features on synthetic MRIs, compared with unsupervised cross-domain adaptation based deep learning approach.

PO-GePV-M-169

Prior Information Guided Auto-Contouring of Breast Tumor Bed for Post-Operative Breast Cancer Radiotherapy

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Purpose: Contouring of breast tumor bed in planning CT is important to post-operative radiotherapy of patients after breast conserving surgery (BCS). However, this contouring is highly uncertain due to specimen volume, seroma size, clarity, surgical clips, inter-observer variability, and other factors. To alleviate this issue, prior information was introduced in a 3D U-Net model to predict the contour of breast tumor bed automatically. **Methods:** The pre-operative CT was first aligned to the planning CT via intensity-based B-splines registration. Then the contour of breast tumor expansion in pre-operative CT was transformed to planning CT accordingly. This transformed contour indicates the approximate area of breast tumor bed in planning CT and would be valuable prior information in searching for more accurate contour. Next, the transformed contour with CT images were fed into a 3D U-Net model to predict the fine contour of breast tumor bed. 110 sets of CT images (each including pre-operative CT and planning CT) and the corresponding label images were used for model training with five-fold cross-validation. The dice similarity coefficient (DSC) and Hausdorff distance (HD) were used to evaluate the accuracy of 3D U-Net model. **Results:** The average DSC and HD of prediction model were 0.685±0.065 and 20.816±14.565 for breast tumor bed without the input of prior information, while the average values were 0.795±0.038 and 13.896±8.737 with the input of prior information. The differences were statistically significant (0.685 vs. 0.795, $P = 0.0014 < 0.05$; 20.816 vs. 13.896, $P = 0.002 < 0.05$). **Conclusion:** The introduction of prior information improved the prediction accuracy of breast tumor bed contour. This method provides an automatic way to contour breast tumor bed more accurately in CT image and will be useful in other image modalities.

PO-GePV-M-170

The Physics of Cellular Self-Assembly to Investigate Cancer Migration in Interface with Healthy Tissues

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Purpose: A more precise understanding of cancerous biological systems may help devise novel treatments in oncology. To that end, the underlying physics of the dynamics of multicellular systems may provide valuable insight into the process of oncogenesis and consequently develop remedies to halt or decelerate cancer progression. One avenue of exploration is the physics of cellular self-assembly that may shed more light onto the biology of tumor invasion into healthy tissues. **Methods:** A model based on the physics of tissue engineering, which investigates tissue migration microscopically and is known as "Zipper CAMs", was modified to enable its application to the cellular interactions at the interface between cancerous and healthy tissues. The kinetics of cell adhesion molecules in the surfaces of individual cells were investigated to predict and control the final tissue location. Statistical mechanics was used to calculate the force involved in this cellular self-assembly. **Results:** The model successfully predicted the speed of cancerous tissue invasion into healthy tissues. The data show that the migration procedure must abide by some physical principles including tissue surface tension (ATST), and that the biomechanical properties of both tissues types play a critical role in both the invasion and its speed. The ATST represent the strength of adhesion molecules that drive the invasion and provide the energy and force for migration. Such physical parameters may therefore provide important information to oncologists on how to prevent or decelerate the process. **Conclusion:** The physics of cellular self-assembly has been used over the past 10 years in tissue engineering to fabricate healthy human tissues and organs. However, the application of physics in the interaction between cancerous and healthy tissues is barely explored. Here, we introduce a mathematical and microscopic model based on the physical principles of the dynamics of multicellular systems to comprehend and control cancer invasion.

PO-GePV-M-171

Apparent Diffusion Coefficient Calculation On a 0.35T MR-Linac System

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Purpose: Diagnostic radiology techniques continue to be utilized for improving patient outcome in radiotherapy, including diffusion weighted imaging (DWI) for tumor staging and monitoring. MR-guided radiotherapy (MRgRT) has become a highly desirable treatment modality and provides the capability for daily DWI. This study assesses apparent diffusion coefficients (ADC) from EPI and TSE sequences with a NIST traceable phantom on 0.35T MR-linac. **Methods:** DWI acquisitions of a NIST traceable diffusion phantom with 13 vials of aqueous polyvinylpyrrolidone (PVP), concentrations from 0% to 50%, were performed on a 0.35T MR-linac. DWI-EPI acquisitions were performed with b-values of 0, 200, 500, and 800 s/mm², diffusion prepared (DP)-TSE acquisitions utilized b-values of 0, 300, 500, and 800 s/mm². Acquisitions were performed with pre-scan normalization on and off, along with noise fields. Additionally, signal averages (NSA) of 1, 4, 16, and 32 were acquired. ADC values were calculated using nonlinear least-squares fitting of all b-values for original and denoised images. Two denoising methods were used including quadratic background subtraction and noise field subtraction. **Results:** EPI acquisitions with noise field subtraction had the lowest ADC error of -7.3% and 4.4% from reference for 0% and 40% PVP, NSA=1. ADC values from initial EPI-DWI acquisitions had an ADC error of -14.2% and -9.8% from reference for 0% and 40% PVP, NSA=1. Initial TSE images had an error of -6.9% and -57.6% from reference for 0% and 40% PVP with NSA=1. ADC values were significantly greater than reference value for 50% PVP on all scans and postprocessing methods but is outside the physiological range in humans. **Conclusion:** EPI and TSE had similar ADC accuracy but were not equivalent to reference values. However, TSE acquisitions had reduced geometric distortion and greater consistency across NSA values. Further correction is required for accurate ADC calculation on low field MRgRT systems.

PO-GePV-M-172

Automated Analysis of QA Images Acquired with Low-Field MRgRT

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Purpose: Test the accuracy and reproducibility of Magphan(R) phantom, designed for quality assurance (QA) needs of MRI-guided radiotherapy. It allows automated measurements for image analysis and acquisition of important imaging parameters. **Methods:** QA was performed weekly. Magphan(R) RT phantom in a low-field MR-LINAC system was scanned over a period of 10 months. At least 35 measurements for each imaging parameter were acquired. Three MRI sequences were used for QA: TRUFI, T1, and T2. DICOM files were used to calculate the following imaging parameters: geometric distortion, uniformity, resolution, signal-to-noise ratio (SNR), and laser alignment. Point spread function and edge spread function are also calculated for resolution analysis. All data was collected and analyzed for accuracy and reproducibility. Mean, median, standard deviation, maximum and minimum values were used in combination with plots to check for data consistency over the 10-month period. **Results:** Data showed small standard deviation and high consistency for each tested imaging parameter with each MRI sequence. Highest variability in data was seen with the TRUFI sequence, which was expected because of low resolution and short scan time (25sec). Point spread function, edge spread function, signal uniformity and SNR measurements remained consistent over 10 months. Laser alignment traditional offsets and angular deviation remained consistent over 10 months, except for one day where phantom was misaligned. Distortion analysis, performed for two different phantom fiducial diameters per imaging sequence, remained consistent for both diameters. **Conclusion:** The Magphan(R) RT phantom showed to be both reliable, reproducible, and easy to use. Imaging parameters that are not available with the current accepted low-field LINAC QA phantom, such as laser alignment and geometric distortion, were found to be both important and relevant. These results show promise for a more streamlined, time-saving, and less error-prone low-field MR-LINAC QA workflow when compared to current practices.

PO-GePV-M-173

Investigation of Unity MR-Linac Adaptive Planning On Prostate SBRT Plan Quality

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Purpose: The Elekta Unity MR-Linac utilizes 1.5 T magnetic resonance (MR) images and software tools to create an adaptive plan prior to each patient treatment. This study aims to investigate the impact of adaptive planning on SBRT prostate treatments. **Methods:** In this preliminary retrospective planning study, two adaptive planning methods were investigated for a single prostate patient following the PACE-B SBRT protocol (3,625 cGy delivered in 5 Fractions). The first method mimicked the time constraints of online adaptive planning and allowed only a single optimization attempt. The second method mimicked an offline planning approach and allowed for multiple optimization runs in an attempt to maximize the treatment plan quality. Both adaptive planning methods were then compared to a Versa HD VMAT plan calculated on the daily MRI. All dose calculations on MRI used bulk density overrides, and all fractions were added using a DVH addition technique to determine final DVH metrics. **Results:** All organ-at-risk DVH metrics met the PACE protocol requirements for all planning methods evaluated, but PTV coverage was determined to be 76% in the VMAT plan which was not modified for anatomical changes. Prostate and proximal seminal vesicle coverage was 95% and 92% respectively for the VMAT plan but was 99.9% for both adaptive planning methods with PTV coverage normalized to 95% for both adaptive planning approaches. It was found that the offline adaptive plan approach yielded additional OAR sparing for the rectum and bladder compared to the online plan approach. **Conclusion:** PTV coverage was significantly improved with an adaptive planning strategy, as were the prostate and proximal seminal vesicle coverage. Both the adaptive and non-adaptive VMAT plan met all OAR constraints of the PACE-B protocol, but it was found that the time constraints related to online adaptive planning may lead to sub-optimal OAR sparing.

PO-GePV-M-174

The Dosimetric Impact of Magnetic Field on Lung Cancer Treatment When Using MR-Linac

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Purpose: To compare dose distribution of stereotactic body radiation therapy (SBRT) treatment for lung cancer with or without the presence of magnetic field on 3D and IMRT plans using MR-Linac. **Methods:** A comparison for a stage T3N0 lung cancer patient, which was originally treated on a conventional linac using SBRT with a prescription dose of 50Gy in 5 fractions covering at least 95% of PTV, is presented in this study. Retrospectively, 3D and IMRT based SBRT plans with the same PTV dose prescription were generated using the Monaco planning system for Unity MR-Linac (Elekta, Stockholm, Sweden) without magnetic field correction. Those plans were re-calculated using magnetic field corrected beam data to represent the presence of magnetic field during treatment. Dose distributions with and without magnetic field in exactly same beams were compared on 3D and IMRT plans. Thereafter, the IMRT plan was re-optimized using magnetic field corrected beam data with the same target and normal tissue dose constraints. Dose distribution of IMRT plans optimized with and without magnetic field were also compared. **Results:** The dose on lung side of lung-target interface was reduced with the presence of magnetic field for both 3D and IMRT plans. Therefore, the target coverage was compromised by the presence of magnetic field if the planning system ignored the effects of the field. After re-optimization, the IMRT plan can compensate for the reduced coverage of the target while keep sparing of the normal tissue. **Conclusion:** The presence of magnetic field changes the dose distribution of lung treatment and the lung target interface. Re-optimization is required for IMRT plans with magnetic field corrected beam data.

PO-GePV-M-175

Dosimetric Evaluation for MR-Linac Online Adaptive Planning Approaches: A Case Study of Recurrent Adrenocortical Carcinoma

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Purpose: Elekta Unity MR-Linac provides two online plan adaptation approaches: daily plan re-optimization with isocenter adaptation (i.e., adapt-to-position, ATP) and daily re-planning considering anatomical changes (adapt-to-shape, ATS). We aimed to analyze the dosimetric differences between these two approaches for a patient treated with ATS in order to shed a light for optimizing online plan adaptation strategy. **Methods:** A 77 year-old female, whom was diagnosed with recurrent adrenocortical carcinoma, was treated with hypofractionated radiotherapy for a metastatic pelvic sidewall tumor that was located proximal to the large bowel. The patient was treated with the Elekta Unity system, with a prescription of 60 Gy delivered in 10 fractions. The ATS workflow was applied to each fraction for online plan adaptation, including re-contouring of the GTV and normal tissues (large bowel, rectum, bladder, and small bowel). Retrospectively, the ATP workflow was applied by re-planning based on the isocenter shift on each daily MR scan. Accumulated dose for each of the two plan adaptation methods was calculated by summation of the absolute DVH of each structure. **Results:** The tumor volume did not show significant changes during treatment, but the proximity of large bowel to tumor varied from day to day. The clinical goal of GTV D90%>60Gy was achieved using both methods (60.5Gy vs. 60.1Gy, ATP vs. ATS). However, there were significantly higher doses using the ATP method for large bowel, bladder, rectum, and small bowel ($p=0.0001$, 0.0001 , 0.02 , 0.0002 , respectively), with the large bowel violated the $D_{max}<40\text{Gy}$ constraint using the ATP approach (46.1Gy vs. 39.4Gy, ATP vs. ATS). Additionally, the maximum doses of large bowel had a larger variance on the ATP workflow, as compared to the ATS (stdev=3.6Gy vs. 0.9Gy). **Conclusion:** ATS approach may offer higher precision than ATP approach in treating tumors proximal to radiosensitive normal tissues, without sacrificing target coverage.

PO-GePV-M-176

Urethra Inter-Fractional Geometric Variations of Prostate Cancer Patients: A Study Using an On-Board MRI

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Purpose: Urethral injury plays a significant role in genitourinary (GU) toxicities for prostate cancer patients receiving radiation therapy (RT). However, urethra sparing has been challenging due to non-trivial urethra delineation with conventional RT technologies. MRI provides strong soft tissue contrast and potentially high urethra visibility. Moreover, MR-guided online adaptive RT (MRgART) based on patient's daily MRI shows the potential of improved organ-at-risk (OAR) sparing. In this study, we used MR-guided radiation therapy (MRgRT) on-board MRI to retrospectively analyze inter-fractional urethra localization and geometric variation for a cohort of prostate cancer patients and evaluated the value of MRgART for urethra sparing. **Methods:** Eleven prostate cancer patients were scanned on a 0.35T MRgRT system using a urethra-specific T2-weighted 3D HASTE sequence at two different timepoints between simulation and the final fraction. The average time between urethral imaging pair was 19.8 days. A resident radiation oncologist retrospectively contoured the prostatic urethra for all patient images. Additionally, the oncologist qualitatively scored the prostatic urethra visibility. For each patient, the urethra imaging pairs were rigidly registered based on the prostate gland and urethra inter-fractional geometric variation was evaluated using 95th percentile Hausdorff distance (HD95), mean-distance-to-agreement (MDA), and DICE coefficient. **Results:** Qualitatively, 3D HASTE provided good conspicuity and the urethra could be identified clearly for the majority of the patient scans. The average length and diameter of the prostatic urethra was $3.54\pm0.52\text{cm}$ and $0.78\pm0.12\text{cm}$. The average HD95, MDA, and DICE between urethra imaging pairs was $0.26\pm0.10\text{cm}$, $0.11\pm0.04\text{cm}$, and 0.58 ± 0.14 . The majority of urethral inter-fractional shifts were observed in the anterior-posterior and superior-inferior direction. **Conclusion:** Urethral inter-fractional geometric variation is significant, with respect to the urethra diameter, when shifts are introduced in the anterior-posterior direction. MRgART workflow with optimal on-board imaging may improve urethra sparing throughout treatment.

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Feasibility of Adapt-To-Position Treatment of Prostate Patients Undergoing Ultra-Hypofractionated MR-Guided Adaptive Radiotherapy

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Purpose: To investigate the feasibility and dosimetric outcome of performing adapt-to-position (ATP) workflow on prostate cases undergoing ultra-hypofractionated radiotherapy with boost to dominant intraprostatic lesion (DIL). **Methods:** 10 prostate patients were treated on Unity MR-Linac using a five-fraction regimen with 40Gy to prostate and 45Gy simultaneous-integrated-boost to DIL. All patients were treated clinically using adapt-to-shape (ATS) workflow with full re-plan and contour adjustments. Retrospectively, an ATP plan was generated on the original planning CT (pCT) to account for daily prostate shifts based on soft tissue match. The ATP plan was also re-calculated on the daily MR (dMR) with the adjusted contours-of-the-day to see the realistic impact of performing ATP. Dosimetric data to OAR's (bladder, rectum, bowel) and targets (PTV_Prost, PTV_DIL) were compared between the treated ATS (ATS_txt), ATP (ATP), and the ATP plan recalculated on dMR (ATP_recalc). Volume and positional/center of mass (COM) changes for targets were compared between pCT and dMR's. **Results:** PTV_Prost and PTV_DIL coverage decreased more than 10% on ATP_recalc compared to the ATS_txt in 20% and 28% of fractions respectively. This coverage drop was not seen on any of the ATP plans. The average volume change of the PTV_DIL contour on dMR compared to pCT for the course of treatment ranged from 0.5-10 cc indicating during treatment changes. The difference in relative distance between the PTV_Prost COM and the PTV_DIL COM between the pCT and dMR's ranged from 0.3-1.42 cm. Certain OAR constraints were violated on ATP_recalc but were met on ATP. **Conclusion:** The decrease in target coverage and change in relative COM highlights that ATP planning is not an optimal approach for MR-guided adaptive prostate radiotherapy when boosting the DIL since ATP does not reflect daily variation in position and deformation of soft tissue anatomy and may result in OAR dosimetric parameters being exceeded.

PO-GePV-M-178

Dosimetric Effects of Air Cavities for MRI-Guided Online Adaptive Radiation Therapy (MRgART) of Prostate Bed After Radical Prostatectomy

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Purpose: To evaluate the dosimetric impact of air cavity electron density correction for MR-guided Online Adaptive Radiation Therapy (MRgART) of prostate bed patients after radical prostatectomy. **Methods:** Five prostate bed patients treated on a 0.35T MR Linac following institutional bladder and rectum filling protocol were retrospectively included to simulate the MRgART workflow. For two fractions of each patient, workflow1 'Recalc': the initial clinical plan was calculated on the daily setup MRI with electron density directly obtained from simulation CT deformably registered to daily MRI; and workflow2 'Override': the initial clinical plan was calculated with one additional step, air cavities were manually contoured with bulky air density assignment based on daily MRI anatomy. Dosimetric metrics for targets and OAR were compared between the calculated dose with and without air cavity electron density correction and its impact on clinical decision for the necessity of adaptive planning was evaluated for each fraction. **Results:** The average air cavity change between simulation CT and daily MRI was $8.15\pm9.16\text{cc}$. Clinical dosimetric endpoints between the 'Recalc' and 'Override' workflow showed little clinical dosimetric differences, with the maximum deviations being 3.47% and 2.82% for the rectum (V27.5Gy) and bladder (V32.5Gy). Furthermore, the dosimetric difference did not lead to change in clinical decision for adaptive planning except for only one fraction, where the dose calculation with air cavity density correction indicated that the bladder D_{max} dose exceeded the constraint while the calculated dose without correction showed acceptable dose, although the absolute difference was only 0.4Gy. **Conclusion:** With strict bladder and rectum filling protocol, air cavity changes between simulation and daily treatment did not result in large dosimetric changes on our 0.35T MRgRT system. However, caution may need to be paid for potential large air cavity variations or under higher magnetic fields.

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Assessment of Dose Accuracy of Bulk Synthetic CT in MR-Guided Online Adaptive Radiotherapy for Nasopharyngeal Carcinoma

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Purpose: This study aims to evaluate the dose accuracy of bulk relative electron density (rED) assignment approach to generate synthetic CT (sCT) for application in 1.5T MR-Linac, to assess the reliability of this approach in the case of online adaptive MR-guided radiotherapy for Nasopharyngeal Carcinoma (NPC) patients. **Methods:** The original CT and MRI T2 were acquired for 10 NPC patients. All region of interests (ROIs) were delineated on CT in Monaco system (v5.40.02). Each patient's MRI/CT image pair was aligned using a rigid registration algorithm and ROIs were copied to MRI from CT. The sCT was generated in MRI using bulk rED assignment approach, based on the rED values of the ROIs. Three sCT images were generated: The sCT(ICRU) uses the rED values recommended by ICRU46, the sCT(tailor) uses the patient-specific mean rED values, and the sCT(Homogeneity) was created using water equivalent value for entire patient data set. The same treatment plan was calculated on three sCTs and original CT. Dose calculation accuracy was evaluated by gamma analysis, point dose comparison and dose volume histogram (DVH) parameters. **Results:** For both criteria, 1%/1mm and 2%/2mm gamma passing rate results of plans on sCT(tailor) were always higher than plans on sCT(ICRU) and sCT(Homogeneity) when compared with original CT based plans ($p = 0.005$). The difference on point dose in PTVnx for each case calculated on sCT(tailor) ($-0.14\% \pm 1.44\%$) was smaller if compared to those calculated on sCT(ICRU) ($-8.77\% \pm 2.33\%$) and sCT(Homogeneity) ($1.65\% \pm 2.57\%$). The DVH differences for plan based on sCT(tailor) were much smaller than sCT(ICRU) and sCT(Homogeneity). **Conclusion:** The sCT generated with bulk rED assignment approach guarantees an acceptable level of dose accuracy for NPC patients with 1.5T MR-Linac, making this approach suitable to online adaptive MR-guided radiotherapy. The accuracy can be improved by using patient-specific rED values.

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PO-GePV-M-180

Superparamagnetic Iron Oxide Enhanced MRI for Elekta Unity MR-Linac: A Pre-Clinical Phantom Study

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Purpose: This study aims to sufficiently visualize Super-Paramagnetic-Iron-Oxide enhanced MR images for Elekta Unity MR-Linac (Elekta; Stockholm, Sweden). Once IV of Ferumoxytol (Feraheme, AMAG Pharmaceuticals, Waltham, MA) injected as an alternative contrast media, it is prolonged hepatic blood pool phase and delayed intracellular uptake in liver Kupffer cells over a month. Then, it alters MR signals due to susceptibility effect. Hence, we performed a pre-clinical phantom study to investigate the sensitivity of Feraheme on MR images of Unity. **Methods:** Varying concentrations of Feraheme and Gadobutrol (Gd, Gadavist, Bayer HealthCare) were diluted in water into clinically relevant concentrations and poured into cylindrical containers: (1) 10 μ g/mL, (2) 15 μ g/mL and (3) 30 μ g/mL of Feraheme, (4) 0.1mmol/kg Gd, and (5) distilled water only and an identical set with the same concentrations with 1% w/w agarose. The varying concentrations were imaged using CT and MRI (Siemens and Unity) with T1-Weighted, T2, Proton Density (PD) and T2* sequences. All MR images were visually inspected to evaluate signal enhancements. **Results:** Feraheme successfully enhanced MR images but not for CT. Feraheme especially enhanced T1-Weighted (TR<8.0ms/TE<5ms) images brighter than Gd. Feraheme negatively enhanced T2 (TR/TE=1000.0/100.0ms) images but it is insufficient to differentiate the three Feraheme concentrations of contrast agents. Feraheme also negatively enhanced PD (TR/TE=1400.0/30.0ms) images, similar to the T2 images but it slightly differentiate them. In addition, Feraheme enhanced T2* (TR/TE=1074.7/5.8ms) images of Siemens but negatively-minimally enhanced T2* (TR/TE=1457.7/18.4ms) images of Unity. The 1% agarose resulted in signal reduction on only T2 images. **Conclusion:** This study utilized a clinical range of concentrations of Feraheme to enhance MR images with positive or negative signal changes on T1, T2, PD and T2* images whilst compared to the ground truth of diagnostic images. It is evident that Feraheme can be applicable for Unity and clinical trials for investigations liver functional imaging.

PO-GePV-M-181

Evaluation of Cranial FSRT Plan Quality on a Magnetic Resonance Image Guided Linear Accelerator

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Purpose: To investigate treatment plan quality for intracranial post-operative tumor beds treated with fractionated stereotactic radiotherapy (FSRT) for delivery on a Magnetic Resonance Image Guided Linear Accelerator (MR-Linac). **Methods:** Ten intracranial FSRT patients were replanned on a Monte Carlo-based commercial treatment planning system (TPS), MRIdian v5.4.0.97 (ViewRay, Mountain View CA). The MR-linac system combines a 0.345T magnet and a 6MV flattening filter free (FFF) treatment beam. Plans were generated using a step-and-shoot IMRT technique, generated with a double-stacked MLC with an equivalent leaf width of 4.15mm. Eleven beams were utilized, spaced 20-35 degrees apart. Target volumes were prescribed to 24-30Gy in 3 fractions (8-10Gy per fraction) and normalized so that 95% of the target volume received prescription dose. Plans were optimized to achieve AAPM TG 101 organ at risk tolerances and sharp dose fall off. Dose fall off parameters were evaluated including conformality index (CI), gradient index (GI), volume of Normal Brain (Brain-Target volume) receiving 12Gy (NormBrainV12Gy), 18Gy (NormBrainV18Gy), and 24Gy (NormBrainV24Gy) as well as maximum dose to the target volume (MaxTargetD0.03cc). Deliverability of five of the ten cases was verified with ion chamber and film measurements. **Results:** The average target volume was 26.20cc (range: 10.57cc-59.36cc), with several planned with a simultaneous integrated boost (SIB). The average MaxTargetD0.03cc normalized to the SIB dose was 120% (range: 112%-126%). The corresponding CI and GI were 1.15 and 3.69, respectively. Average volumes to NormBrainV12Gy, NormBrainV18Gy, NormBrainV24Gy were 65.87cc, 22.09cc, and 5.38cc, respectively. Average delivery time was 12.2 minutes. Average ion chamber point dose difference from planned values was -1.92% and average gamma pass rate utilizing 3%/1mm criteria was 97.94%. **Conclusion:** This study demonstrates the feasibility of planning intracranial FSRT on an MR-Linac. Delivery was verified and dose distributions were deemed clinically acceptable and accomplished without sacrificing maximum target dose.

PO-GePV-M-182

Synthetic CT Generation from MRI for Stereotactic Radiosurgery

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Purpose: To evaluate the accuracy of synthetic CT from MRI using a conditional generative adversarial network (cGAN) for stereotactic radiosurgery (SRS). **Methods:** A method to generate synthetic CT was developed using an image-to-image translation cGAN with a loss function including a least-square adversarial term, mean-absolute-error (MAE) regularization and mutual information. Training datasets included 25 pairs of CT-sim images and diagnostic T1-weighted MRI with gadolinium contrast acquired for SRS planning. Before training, non-uniformity corrections and image normalization using z-score were applied to MR images. Synthetic CT images were compared with true CT images for 5 cases (brain metastases, arteriovenous malformation and vestibular schwannoma) in terms of image quality and dosimetric accuracy. To evaluate the dosimetric accuracy, clinical plans were recalculated on synthetic CT images with the fixed MUs in an Eclipse treatment planning system. Dose volume histograms (DVHs) and 3D gamma analyses (1%/1mm) were compared between

synthetic CT (with and without density corrections) and true CT datasets. **Results:** For the test datasets, CT number MAEs for air, soft tissues and bone were 456, 53 and 320 respectively. The peak signal-to-noise ratios (PSNR) were in the range of 23.2-25.2 and the structural index similarity measures (SSIM) were in the range of 0.905-0.92. The 3D gamma analyses ($\gamma < 1$) for true CT data against synthetic CT data with and without density corrections were 98%-100% and 87%-95%, respectively. Compared to DVHs for the true CT data, D99.5% for synthetic CT data with and without density corrections was $-0.4\% \pm 0.7\%$ and $1.2\% \pm 1.9\%$, respectively, while D2% for the data with and without corrections was $-0.2\% \pm 0.6\%$ and $2.2\% \pm 0.5\%$. **Conclusion:** Our study demonstrated the potential of using synthetic CT from MRI for SRS treatment planning. Our future work will validate our method on more cases and also investigate the accuracy of synthetic CT for SRS position verification.

PO-GePV-M-183

CINE Viewer in Eclipse for Tumor Motion Evaluation in MRgRT

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Purpose: 4D MRI has been limitedly utilized in clinic unlike 4D CT for tumor motion evaluation. In contrast, 2D CINE MRI has been actively used in MRgRT through MRI vendors' display tools, resulting in limited access from Radiation Oncology practice. In the study, we investigated the feasibility of utilizing CINE MRI through a treatment planning system. **Methods:** We developed CINE viewer in Eclipse using Varian Eclipse Scripting Application Programming Interface (ESAPI). It consisted of 1) single/orthogonal viewer, 2) display controller,s and 3) measurement tools for tumor motion evaluation. **Results:** In Eclipse, CINE viewer loaded 1) CINE MRIs from 1.5T Philips Ingenia MRI which were DICOM imported to Eclipse database and 2) CINE MRIs from ViewRay 0.35T MR-LINAC which were saved to a local database from ViewRay TPS as a movie file. CINE viewer handled two display modes (single plane or orthogonal planes) depending on acquisition planes. Multiple measuring markers were placed on tumor at various respiratory phases, which provided tumor motion ranges in x-, y-axis of CINE images among the respiratory phases. **Conclusion:** We demonstrated the clinical feasibility of tumor motion evaluation using CINE viewer in Eclipse for MRI simulation and MRgRT review. The proposed approach streamlines tumor motion evaluation using CINE MRI using Eclipse in MRgRT procedures.

PO-GePV-M-184

Evaluate the Jointly Model-Based Reconstruction Method to Correct MRI Images Geometrical Distortion for Precise MRI-Based Brain Stereotactic Radiosurgery Planning: Using Pari QC Head Phantom

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Purpose: The main problem of using MRI images for radiotherapy planning is its geometrical distortion. The primary source of sequence-dependent MRI image geometrical distortion changes in magnetic susceptibilities. The conventional B0 field-mapping method is not accurate. This work aims to improve the distortion correction method and evaluate the alternative approach using the jointly model-based reconstruction. **Methods:** MRI and CT images of customized QC head phantom "Pari head phantom," by Spintecx.LLC, which includes air cavities and internal tissue mimicking structures, were acquired. We used multi echo-GRE MRI pulse sequence to generate magnitude images at different TEs and derive field map. The k-space data obtained using a 3D T1-MPRAGE sequence. The Cartesian k-space trajectory modeled as a regularized least-square cost function to estimate the undistorted field-map and corrected image. To generate the field map, the quadratic regularizer parameter was set to 2^{-3} , and for the image, the non-quadratic edge preserving regularizer parameter was set to 2^3 by cross-validation. We minimized this cost function by using the CG-NUFFT algorithm in MATLAB. Convergence criterion met by an iterative process. Finally, both corrected images using a field map and our proposed method compared with MRI image register with CT images. **Results:** We estimate NRMSE error of the corrected and uncorrected MRI images by registering to CT images. Maximum improvement in NRMSE error was 47% and 19.48% for jointly model-based reconstruction and correction by distorted field-map. The most-distorted areas were around air cavities. **Conclusion:** MRI image's geometrical distortion close to air cavities is considerably large. Using the "Pari QC head phantom" lets us simulate the MRI image's geometrical distortion close to air-cavities. And Our method shows better results than the conventional B0 field map method to correct MRI images more effectively and faster. This method will improve and facilitate using MRI images for precise SRS planning.

PO-GePV-M-185

Application of Super-Resolution Reconstruction to Prostate 0.35T MRI Scans

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Purpose: Super-resolution (SR) is a technique to combine multiple low-resolution images acquired with sub-voxel frame shifts into a single high-resolution image by inverting a linear image acquisition model, which may be used to address resolution limitations arising from subject motion and hardware. In this study we evaluate the performance of SR on 0.35T magnetic resonance imaging (MRI) prostate scans towards an application of low-field MRI-guided focal therapy. **Methods:** Balanced steady-state free precession (bSSFP) multi-slice images of an American College of Radiology (ACR) MRI accreditation phantom and one prostate patient were acquired on a 0.35T MRI-guided radiotherapy machine. All images had a pixel spacing of 1.5mm x 1.5mm. Couch translations of 0.5mm and 1.0mm were applied in both in-plane directions, for a total of five acquisitions per object. Registrations were performed using Elastix. We used a maximum a posteriori (MAP) based SR algorithm to reconstruct the reference object by a factor of 2 in each dimension. We assumed uniform Gaussian blurring in the SR forward model from high-resolution to low-resolution and a Gaussian noise prior. The resulting images were compared to bilinear upsampling and an upsampled average of the registered images. Example edges in the patient image were chosen to extract line profiles, which were then fit to a sigmoid function. The sigmoid widths quantified spatial resolution of the object. **Results:** In the ACR phantom high-contrast spatial resolution slice, the SR image showed improvement in the visibility of the hole-array pairs. Similarly, greater detail was present in the prostate patient SR reconstruction. Sigmoid curve fitting demonstrated smaller widths for SR over single image and average image interpolation. **Conclusion:** MAP-based SR is a feasible method to improve 0.35T MR prostate image resolution. Additional improvements, such as incorporating a measured imager point-spread function, may be beneficial.

PO-GePV-M-186

Automated Detection of Freiburg Flap Applicator Catheters for Surface Brachytherapy on PETRA MR Images

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Purpose: MR-only treatment planning for skin High Dose Rate surface applicator brachytherapy (SABT) relies on accurate depiction of applicator channels on MR images. This work presents automatic detection of empty Freiburg Flap (FF) applicator catheters on optimized Pointwise Encoding Time Reduction with Radial Acquisition (PETRA) images, where catheters exhibit positive contrast. **Methods:** An FF applicator (12 catheters with 24 spheres of 10mm diameter) was sandwiched between two plexiglass blocks and placed on a MagPhan TMR008 phantom in a 3T Siemens Vida. Coronal 0.8mm isotropic resolution 3D PETRA images (TR/TE/TI 3.3/0.07/100 ms, FOV 306x306mm², BW 407Hz/px) were acquired using UltraFlexLarge18 and Spine32 coils. A ROI encompassing a visible catheter and its surrounding beads on axial slices was drawn in MATLAB. After manual definition of the center column of the first sphere, the center and width of the following spheres was derived using known dimensions. Normalized signal intensity column profiles and their numerical

gradients were computed. Profile mean and standard deviation (PM, SD) and gradient mean (GM) over the 13 columns width of each sphere was extracted. Locations of maximum PM found within 2 pixels of minima of SD and of GM were catheter position candidates for a sphere. A catheter was reconstructed by selecting the candidate position of a sphere within 1 pixel from its preceding one. **Results:** Since FF catheters appeared as bright lines with dark borders inside bright circles, the characteristics of signal intensity profiles perpendicular to the catheters revealed accurately catheter position. The presented algorithm detected successfully all 12 catheters inside their 24 surrounding spheres. Six catheters were reconstructed as completely straight lines, five presented 0.8mm and one 1.6mm maximum deviation from a straight line. **Conclusion:** Automated detection of empty catheters in FF applicators for skin SBT was feasible, based on their signal intensity properties on PETRA MRI.

PO-GePV-M-187

MRI-Based Prostate SBRT Treatment Planning Using Synthetic CT

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Purpose: To evaluate the feasibility of using a commercial MRI-derived synthetic CT (sCT) protocol for CyberKnife- and VMAT-based treatment planning for prostate SBRT. **Methods:** Same-day CT and MRI simulations in the treatment position were performed on ten patients who were prospectively enrolled into this IRB-approved study and treated with SBRT to the prostate (40Gy in 5 fractions). Dixon in-phase and out-of-phase MRIs were acquired on a 3T scanner (MAGNETOM Vida, Siemens Healthcare) using a 3D T1-weighted gradient echo sequence (TE1/TE2/TR=1.23/2.46/4ms, readout bandwidth=1090 Hz/pixel, 1x1x1.5mm³, scan time=5mins) to generate sCT datasets with a bulk-assignment algorithm based on in-phase, out-of-phase, fat-only, and water-only MRIs. CT and sCT datasets were co-registered and HU values compared using mean absolute error (MAE). CyberKnife and VMAT treatment plans were generated on each patient CT and recomputed onto corresponding sCTs using the same plan parameters and CT HU calibration curve. Dose distributions were compared using 3%/3mm local gamma criteria and dose-volume-histograms. **Results:** Relative electron density values computed from CT datasets and assigned to sCT datasets were 1.04, 0.95, 1.1, and 1.7g/cc for soft tissue, adipose, inner bone, and cortical bone, respectively. The proposed sCT protocol produced total MAE of 94±20HU. As expected, disagreement could be seen in instances of variable bowel/rectal gas filling, fiducials or calcifications appearing in the CT but not the sCT, and/or bone distribution differences, a common challenge in mapping sCT from MRI. Gamma criteria values were 98.9±0.9% (97.1-100%) and 97.7±1.3% (95.3-99.3%) for VMAT and CyberKnife plans, respectively. Dose delivered to 95% PTV volume in sCT plans was 100.5±0.8% (99.8-102.5%) and 97.0±6.0% (81.9-103.0%) of the prescription dose in the CT plans for VMAT and CyberKnife plans, respectively. **Conclusion:** MRI-derived sCT using the proposed approach shows excellent dosimetry agreement with conventional CT simulation, demonstrating the feasibility of using MRI-derived sCT for prostate SBRT treatment planning.

PO-GePV-M-188

Geometric Accuracy of Spiral FLAIR for Treatment Planning in Stereotactic Radiosurgery: A Preliminary Study

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Purpose: Recently a spiral FLAIR MRI technique has been developed for treatment planning in stereotactic radiosurgery of trigeminal neuralgia, providing good delineation of the nerve root, adequate suppression of CSF, improved contrast, and insensitivity to field disturbance due to the metal frame. This study aims to determine the geometric accuracy of spiral FLAIR for treatment planning. **Methods:** A CIRS MR distortion and image fusion head phantom 603A was used. CT images were acquired as reference. The phantom was scanned on a 3T research MRI scanner with a metal frame using SPGR, FIESTA, and the proposed spiral FLAIR (only available on the 3T research scanner), and on a 1.5T clinical MRI scanner without the metal frame using SPGR and conventional FLAIR. To measure the geometric accuracy, all MRI images were registered to CT images in Leksell GammaPlan. The coordinates of 13 points were measured from CT and all MRI images, 4 corner markers and 1 center marker at the middle slice, 4 corners at a superior and an inferior slice, respectively. The point-to-point deviation in each coordinate direction and corresponding distance between CT and each MRI data set were computed. Mean error in each direction, and mean and standard deviation of distance, are derived. **Results:** Mean errors in (X, Y) (mm) at 3T are (0.19, 0.32), (0.37, 0.40), (0.68, 0.56) for SPGR, FIESTA, spiral FLAIR, respectively, (0.60, 0.24) and (0.56, 0.18) at 1.5T for SPGR and conventional FLAIR, respectively. Mean and standard deviation of distance at 3T are (0.40±0.26), (0.59±0.25), (0.98±0.56), for SPGR, FIESTA, spiral FLAIR, respectively, (0.66±0.38) and (0.61±0.25) at 1.5T for SPGR and conventional FLAIR, respectively. The error in Z is 0 in all measurements. **Conclusion:** This study demonstrated that spiral FLAIR achieves good geometric accuracy, providing a potential tool for stereotactic radiosurgery treatment planning of trigeminal neuralgia.

Barrow Neurological Institute has a master research agreement with Philips Healthcare.

PO-GePV-M-189

Feasibility of Lung Stereotactic Body Radiation Therapy (SBRT) Treatment with MR-Image Guided Linear Accelerator (MRI-Linac)

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Purpose: MR image guided radiation therapy (MRigRT) offers real time imaging of tumor motion to monitor lung lesions and gate treatments for high dose schemes like lung stereotactic body radiation therapy (SBRT). The feasibility of performing lung SBRT on MR guided linear accelerator was investigated and compared to conventional C-arm linac-based treatment. **Methods:** Six patients had their clinically optimized treatment plans completed on one system and replanned on the other. Target definitions included GTV for MR-linac gated plans (3mm expansion for PTV) and ITV (5mm expanded PTV) for C-arm linac non-gated plans. Organs at risk (OAR) were defined as per RTOG 0813. Target doses were 50-55Gy delivered with 6 MV FFF X-rays using either VMAT (C-arm linac) or step-and-shoot IMRT (MR-linac). Plans were normalized to give identical PTV coverage (V_{95%}). Target dose, dose conformity index (CI), dose gradient index (GI) and OAR volume dose constraints were compared between both modalities. The CI was compared as a function of target motion amplitude as measured by 4DCT. **Results:** Compared to C-arm linac, plans delivered on the MR-linac gave an average higher median OAR dose of 0.4±0.6Gy and a higher maximum OAR dose of 1.0±0.8% with no difference in PTV and GTV/ITV coverage. For targets with motion amplitude ≤5mm, MR-linac based plans had a worse CI (1.31±0.17 vs. 1.08±0.07, p=0.046) compared to C-arm linacs. For larger motion amplitude (>10mm), the CI was similar (1.12±0.09 vs. 1.13±0.07, p=0.913). MR-linac had a lesser GI (7.4±2.1 vs 4.9±1.2, p=0.084) and comparable GI (4.5±0.9 and 4.3±0.6, p=0.818) compared to C-arm linac for targets with motion amplitude ≤5 mm and ≥10 mm, respectively. **Conclusion:** MR-linac based IMRT plans gave similar dosimetric results compared to C-arm linac VMAT plans for lung SBRT for targets that had motion amplitude >10mm and slightly higher doses for targets with motion amplitude <5mm.

PO-GePV-M-190

Dosimetric Evaluation of Compressed Sensing Respiratory Gating in MRI-Guided Radiotherapy

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Purpose: A commercial 0.35T magnetic resonance imaging (MRI)-guided radiotherapy vendor recently introduced upgraded real-time imaging frame rate based on compressed-sensing techniques. Furthermore, changes to motion tracking algorithms were introduced. Compressed sensing can be used to increase

image frame rate, but may also compromise image quality. In order to assess the impact of this upgrade on respiratory gating accuracy, we evaluated gated dose distributions pre- and post-upgrade using a motion phantom and radiochromic film. **Methods:** Five motion traces were used to drive an MRI-compatible motion phantom, including two artificial waveforms with 4 and 6 second periods, and two free-breathing and one breath-hold breathing traces acquired from previously treated patients. A treatment plan was developed to deliver a 3 cm diameter spherical dose distribution typical of a stereotactic body radiotherapy plan. Gating was performed using 4 frames-per-second (fps) imaging pre-upgrade, and 8 fps post-upgrading using two different motion tracking algorithms, namely “small mobile targets” (SMT) and “large deformable targets” (LDT). Radiochromic film was placed in a moving insert within the phantom to measure dose. Triple-channel film dosimetry was performed and the planned and delivered dose distributions were compared using gamma with 3%/3mm criteria. One-tailed Wilcoxon signed-rank tests compared gamma indices between 4 fps and two algorithms at 8 fps. **Results:** Gamma pass rates ranged from 63–93% over the 15 combinations of breathing trace and gating conditions examined. At a significance level of 0.05, median gamma indices for 4 fps and 8 fps SMT each exceeded that for 8 fps LDT. Median gamma indices between 4 fps and 8 fps SMT were not statistically distinguishable. **Conclusion:** Respiratory gating at 8 fps provides dose distributions similar to that at 4 fps. The particular algorithm selected for motion tracking in the real-time images measurably influences gated dose distributions and should be chosen deliberately.

PO-GePV-M-191

Evaluating a Daily QA RT to MRI Isocentricity Test for an MR-Guided Linear Accelerator

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Purpose: To evaluate a proposed daily isocentricity test for MRI and RT coincidence for an MR-guided linear accelerator **Methods:** An MR-compatible A28 PTW chamber, inserted into a holder bored along the central axis of a cylindrical water phantom, was aligned under MR image guidance such that the center of its sensitive volume was located at the MRI isocenter of our MR-guided linear accelerator. A 0.60 cm x 0.83 cm field (comparable to the detector's dimensions) was irradiated at each of the four cardinal angles, and the charge measured in each case was normalized to the charge measured for a 10.04 x 9.96 cm field (to eliminate sensitivity of the readings to temperature, pressure, and the machine's output). A series of known shifts to the phantom position along each of the cardinal axes was then introduced, and the same sequence of measurements was repeated in order to establish a relationship between the detector's response and its offset from MRI isocenter. The relationship in each direction was fitted by a second order polynomial function, from which a global three-dimensional fitting function was derived. It is proposed that daily measurement of the five fields above with the detector aligned to the MRI isocenter, coupled with this three-dimensional fitting function, can be used to reconstruct, and thereby assess, the offset of radiation isocenter from MRI isocenter as part of a daily quality assurance program. **Results:** The detector's response is proven to be highly sensitive to displacements between radiation isocenter and MRI isocenter on the desired millimeter scale, and is well described by a three-dimensional second order polynomial function. **Conclusion:** The method shows promise as a daily QA test of the congruence between radiation and imaging isocenters, although it cannot quantify it absolutely, being sensitive also to daily variations in the field-bounding leaf positions.

PO-GePV-M-192

Dosimetric Evaluation of Irradiation Geometry and Potential Air-Gaps in An Acrylic Miniphantom Used for External Audit of Absolute Dose Calibration for a Hybrid 1.5T MR-Linac (MRL) System

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Purpose: To investigate the impact of partial lateral scatter (LS), back scatter (BS) and presence of airgaps on OSLD measurements in a standard IROC acrylic miniphantom on the Unity MRL. **Methods:** The following irradiation geometries were investigated using OSLDs, A26MR/A12MR ion-chamber (IC) and Monaco Monte Carlo system: (a) IC/OSLD in IROC miniphantom (partial LS, partial BS) (b) IC/OSLD in a miniphantom placed on a solid water (SW) stack at a depth of 1.5cm (partial LS, full BS) (c) IC/OSLD placed at a depth of 1.5cm inside a 3cm slab of SW/buildup (full LS, partial BS) (d) IC/OSLD centered inside a 3 cm slab of SW/buildup at a depth of 1.5cm placed on top of a SW stack (full LS, full BS). Average of two irradiated OSLDs with and without water was used at each setup. An airgap of 1mm and 2mm, mimicking presence of potential airgap around the OSLDs in the IROC geometry was also simulated. The calibration condition of the machine was 1cGy/MU at SAD=143.5cm, d=5cm, G90, and 10x10cm². **Results:** The expected dose at d=1.5cm was 110cGy. The Monaco calculation (0.5% uncertainty and 1.0mm voxel-size) for the 4 setups were 108.2, 108.1, 109.4 and 110.0 cGy. The corresponding IC measurements were 109.0±0.03, 109.5±0.06, 110.2±0.02 and 109.8±0.03 cGy. Without water, OSLDs measurements were ~10% higher than the expected. With added water to minimize airgaps, the measurements were significantly improved to within 2.2%. The dosimetric impacts of 1mm and 2mm airgaps were also verified with Monaco to be 13.3% and 27.9% higher respectively due to the electron-return-effect. **Conclusion:** IROC MP geometry with partial LS and BS did not have any impact at the measurement depth. A minimal amount of air around or within the OSLDs can cause measurement discrepancies of 10% or higher when placed in a high b-field Unity MRL.

PO-GePV-M-193

Assessment of Inter- and Intrafraction Organs Motion with Online MRI for Cervical Cancer Patients Treated On a 1.5T MR-Linac

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Purpose: Internal bladder and rectum motion occur during radiotherapy for cervix cancer. IMRT planning studies have shown that the quality of radiation delivery is influenced by these anatomical changes. The study is to evaluate bladder and rectum inter- and intrafraction motion using MRI during the full course of MR-Linac radiotherapy fractions, in preparation of MR-guided online adaptive radiotherapy. **Methods:** Five stage II A - III B cervix cancer patients were treated with 25×1.8 Gy fractions on a 1.5T MR-Linac. Each fraction, pre-treatment MRI scans were obtained at the start of every treatment session, and post-treatment MRI scans were obtained at the end of every treatment session. The bladder and rectum were delineated on each MRI by the same radiation oncologist. Inter-fraction and intrafraction variability were evaluated with the Dice similarity coefficient (DSC), the Hausdorff distance (HD), mean distance to agreement (MDA), volume (V) and dose volume parameters. **Results:** Large inter-fraction motion was found for bladder and rectum due to their filling variation (Bladder: DSC: 0.68±0.17, HD: 30.58±15.32, MDA: 7.77±4.65, V: -23.02%±13.13%, Dmean: 2.23%±7.10; Rectum: DSC: 0.69±0.04, HD: 22.98±4.02, MDA: 3.45±0.63, V: 13.57%±18.09%, Dmean: 0.4%±0.7%). Median in-room treatment time pre fraction was 35 min which was much longer than conventional treatment time. The intrafraction motion for bladder was obvious that was explained by the bladder filling variation due to prolonged in-room treatment time per fraction (DSC: 0.88±0.07, HD: 14.66±8.48, MDA: 1.89±1.15, V: 13.06%±15.66%, Dmean: -0.89%±2.45%). The intrafraction rectum motion was much smaller. **Conclusion:** Considerable inter-fraction bladder and rectum motion is observed and intrafraction bladder motion which caused by longer treatment time is also obvious in cervical cancer patients on 1.5T MR-Linac. Intrafraction MR-guided online adaptive radiotherapy may potentially further correct this motion.

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PO-GePV-M-194

Optimizing the Accuracy and Training Speed of An Artificial Neural Network for Three-Dimensional Tumour Trajectory Prediction in Linac-MR Image-Guided Radiation Therapy

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Purpose: To predict 3D tumor trajectories during respiratory motion using an artificial neural network (ANN) for applications to real-time IGRT using a hybrid linac-MR, and to improve the accuracy and training speed of the ANN through hyperparameter optimization, adaptive learning, data augmentation, ensemble learning and GPU acceleration. **Methods:** External surrogate estimates of 3D internal thoracoabdominal tumor trajectories during normal respiration are obtained from Suh et al. (Phys. Med. Biol. 53 3623 (2008)). Long short-term memory recurrent neural networks (LSTM-RNNs) with a range of structures and hyperparameter selections are trained on a 30s subset of data obtained at the start of each fraction, and their accuracy in predicting the tumor position at a forecasting delay of 200ms is evaluated for the following 450s of motion. **Results:** Preliminary results indicate that shallow, narrow LSTM-RNNs can perform as well as or better than the larger LSTM-RNNs employed previously in this work, significantly reducing computational expense and training time. Across all fractions, the mean absolute error at a 200ms forecasting time is found to be 0.40 mm, increasing by a factor of 1.5 between 0-30s post-training and 420-450s post-training. Early experiments with ensemble learning indicates that improvements up to 30% are attainable when the output of 5 randomly initialized models are averaged, and since the individual networks are small, large ensembles may be trained simultaneously on a single GPU. **Conclusion:** Sub-mm mean absolute error (30s training, 450s testing, 240ms input intervals, 200ms forecasting interval) is achievable with small LSTM-RNNs when applied to a dataset of 3D thoracoabdominal tumor trajectories during normal respiratory motion. This suggests that computationally inexpensive, patient- and fraction-specific ANNs may be practically trained on-demand prior to each fraction of IGRT on a linac-MR.

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PO-GePV-M-195

Implementation of a 3D Printing Program in a Radiation Oncology Clinic: A Case for Simple Custom Breast Bolus Fabrication

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Purpose: Customized bolus is an important tool in modulating radiotherapy treatment planning in post-mastectomy (+/- reconstruction) patients. Three-dimensional printing of boluses can streamline custom bolus fabrication in radiotherapy. We present guidelines for implementing a 3D printing program specifically tailored for the production of custom breast boluses in the context of a radiation oncology clinic. **Methods:** We considered 10 different printers having a large enough footprint to produce breast boluses in one piece, and a fast-printing speed to allow timely clinical production of breast boluses. We selected the one with the fastest printing speed (Ultimaker S5) to conduct our evaluation. We compared cost, fabrication time, dosimetric properties and geometric accuracy of wax bolus versus 3D printed bolus. The absorbed dose was measured with OLSDs for 20 different bolus thicknesses and materials (wax, thermoplastic, PLA and TPU), and with a field size of 10 cm by 10 cm, 200 MUs at 6 MV, and 100 SAD. **Results:** The conventional wax bolus fabrication method used at our institution takes about 5.5 hours. The 3D printed bolus fabrication process can save about 5 hours of work for each bolus made since about 30 minutes is required to supervise the 3D printing process. Moreover, custom bolus production cost can be cut down significantly once the 3D printer is purchased, since 1 kg of wax costs about \$60, and 1 kg of PLA costs about \$20. A layer height of 0.4 mm enables both sufficient thickness precision and efficient printing speed. Using 3D printing allow to decrease the bolus thickness uncertainty associated with the manual bolus fabrication process. There are no significant dosimetric differences between the wax boluses and the 3D printed boluses. **Conclusion:** Our results show that 3D printing is an efficient, cost-effective and accurate method to produce breast boluses.

PO-GePV-M-196

X-Ray Imaging Properties of Tissue Equivalent Materials

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Purpose: Tissue equivalent materials have been widely used in medical imaging phantom. For radiologic image simulation, imaging characteristics and physical properties of phantom should be similar to real tissue they replicate. The mechanical properties of materials have been reported. However, medical imaging properties of materials, that are essential for phantom production, are not provided by the manufacturer. This work would be studied in detail the X-ray imaging properties of a selected number of tissue equivalent materials. **Methods:** A variety of tissue equivalent materials were evaluated included several formulations of polylactic acid (PLA), acrylonitrile butadiene styrene (ABS), light curing resin (LCR), epoxide resin, polyethylene (PE), silica gel and hydrogel (n = 9). All of these materials were fabricated using instructions of manufacturer recommended. X-ray attenuation coefficient (μ) was tested on a digital radiography (CXDI-55G, Canon, Japan) with dosimeter at tube voltage from 40-120 kV and 100 mA tube current. CT image data were acquired on a CT scan (Lightspeed, GE, USA) at 120 kV and 100 mA. **Results:** These tissue equivalent materials had different μ (Fig.1). The CT numbers at 120 kV of these materials were also different, ranging from -42-630 HU (Table1). CT number of ABS was negative value, indicating that X-ray attenuation of ABS was weaker than that of water. Other materials had positive CT numbers and stronger X-ray attenuation than water. Several factors affect X-ray attenuation. Some are related to the x-ray beam and the others to properties of the material. The properties of material include the density (ρ) and the effective atomic number (Zeff). In general, substance with higher Zeff or ρ will absorb more X-rays, resulting in more X-ray attenuation. **Conclusion:** These results would make us more familiar with tissue equivalent materials, as well as providing clear guidance in selecting materials suitable as real tissue.

PO-GePV-M-197

Evaluation of 3D Filaments for Creating Patient-Specific Radiotherapeutic Phantoms

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Purpose: To evaluate the radiodensity of modified 3D filaments for creating patient specific anthropomorphic head and neck radiotherapeutic phantoms. **Methods:** A DICOM file of a head and neck phantom was downloaded from the NIH cancer imaging archive. Using Vesalius 3 software the file was segmented to bone (226-2071 HU) and skin (-718 to -177 HU) converted to STL files and then to g-code using the 3D slicer Cura. 3D filaments with various potentially varying radio densities were prepared by melting polylactic acid (PLA) pellets at 180°C and mixing with the mineral powders listed in the Table (Filabot EX2, Barre VT, 05641). The resulting filaments were 3D printed with either the Ender3 or CR10 models (Creality). The resulting 3D prints were imaged on a CT scanner (Siemens) with scan parameters of 140 kV and 300mA. **Results:** The measured HU values ranged from -185 for PLA without additive to 2800 for PLA containing 60% steel. PLA containing 20% Bronze (20% in-fill) had an HU result of 840 while 5% bismuth trioxide (85% in-fill) was higher at 1100. PLA containing calcium silicate was 355 (85% in fill) while human skull measured at 1500 HU. **Conclusion:** The HU results for the two controls, PLA filament

without additives and human skull, are consistent with prior reported values. The 20% Bronze and 5% bismuth trioxide fall within the HU range of bone. The HU result for the PLA filament is consistent with that of skin. Additional fine adjustments of the HU values can be made by varying the % of mineral content and the % in-fill density in the Cura slicer before 3D printing.

PO-GePV-M-198

Development and Performance Evaluation of a Portable Respiratory Self-Training System Using Patch Type Magnetic Sensor

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Purpose: Respiratory training to patients has been recommended to deliver planned radiation accurately to tumors that are moved by respiration in the thorax and abdomen. However, it is difficult to provide respiratory training to all patients because it requires additional space, time, and professional human resources. We developed portable self-respiratory training system that is based on patch type magnetic sensor; and evaluated the system for basic performance and clinical applicability. **Methods:** The system, which is used with a sensor attached to the chest and a magnet to the back, is designed to monitor the patient's respiration by measuring changes of magnetic field strength according to movements of the chest surface. It consists of a micro-electro-mechanical-system (MEMS) based patch type magnetic sensor capable of wireless communication, magnet for measurement, and mobile application. To evaluate the performance of the system, we measured the level of systemic noise, the precision of a sensor for various breathing patterns, signals change for varying distances, and effects of the presence or absence of intermediate substances between the sensor and the magnet. Various respiration patterns were produced by the QUASAR respiratory motion phantom, and the data were analyzed by fitting method and peak value method. **Results:** The sensor had a noise ratio <0.54% of the signal; average errors of signal amplitude and period for breathing patterns were 78.87 μ m and 72 ms, respectively. The signal could be measured consistently when the sensor-magnet distance was 15-20 cm. The signal difference was 1.89% for the presence or absence of a material, indicating that its influence on the measurement signal is relatively small. **Conclusion:** The potential of the respiratory self-training system was confirmed by performance tests and clinical usability evaluation. In the future, we believe that the system could provide precise respiratory training to various patients through clinical trials.

PO-GePV-M-199

Developing and Testing a Real-Time Collision Avoidance System for Couch and Gantry of a Linear Accelerator (LINAC)

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Purpose: Linear accelerators are equipped with safety mechanisms to avoid collision within the patient zone. However, they lack the safety features to detect collision between gantry and couch with the surrounding environment inside the treatment room. In this study, we present a low-cost solution for collision avoidance using commercially available sensors. **Methods:** Two types of sensors were considered for this study. A low-cost ultrasonic distance sensor (HC-SR04) that provides a non-contact measurement for distances between 2 cm and 400 cm. A thin film pressure sensor (ZD10-100) for contact-based force-sensing measurement, which is rated at 500 grams. The pressure sensor was covered with a 5 mm layer of foam tape to transmit applied pressure to the force-sensitive resistor embedded inside the sensor. For testing purposes, the sensors were mounted at the base of the couch. The distance and force measurements were monitored in real-time (every 250 ms) by a micro-controller (Arduino Uno). The micro-controller was programmed to activate a trigger when the distance dropped below 3 cm or when a force was detected simulating a hard stop. **Results:** The ultrasound sensor underestimated the distance over the measured range (3-50 cm) with an accuracy of 3-9 mm. The pressure sensor demonstrated high sensitivity when a small force was applied. The micro-controller successfully activated the trigger signal when an object was detected within 3 cm range or when an object came in contact with the pressure sensor. **Conclusion:** We have demonstrated the feasibility of constructing a collision avoidance system using distance and force sensitive sensors. These sensors can be arranged in different configurations along the base of the couch and various locations along the gantry to achieve optimal performance. An industrial grade ultrasonic sensor is recommended for better stability and accuracy.

PO-GePV-M-200

The Feasibility Study of In-Vivo Detection of Retained Gadolinium in Brain and Bone Using Magnetic Resonance Imaging (MRI) and K X-Ray Fluorescence (K-XRF) Systems

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Purpose: Gadolinium-based Contrast Agents (GBCAs) have been used in MRI for a long time. Gd is cleared from the body via glomerular filtration within hours. However, latest researches indicated that Gd could accumulate in organs such as brain and bone as a long-term deposit. A non-invasive technique to measure retained Gd in the body would allow investigation of the potential negative health effects. Gd retention was only thought to be a concern for those with impaired renal function. Nevertheless, new studies have been pointing to the retention of Gd in the brain of individuals with healthy renal function, raises new questions and concerns regarding the safety of the widely used GBCAs. **Methods:** In a small animal study, Gd concentration in the brain was measured by a 7T animal MRI at different time intervals after contrast injection (1 hour, 24 hours & 7 days). The mice bones were extracted and Gd deposit was assessed by the 109Cd γ -ray induced K-XRF system. The correlation between Gd concentration in bone and brain will be investigated. **Results:** The minimum detection limit of Gd in a bare bone phantom using K-XRF was calculated based on μ g Gd per gram phantom and found to be 0.8 ppm. The average brain Gd concentration using MRI relaxivity (R1) mode was measured to be 0.00022 Mol/L for left and right caudoputamen, an hour post-contrast injection. The Gd concentration of 24-hour and 7-day post-contrast injection for both regions of interest was 0.00021 Mol/L, indicating no substantial reduction of Gd retention in brain. **Conclusion:** MRI and XRF systems can be used to quantify Gd retention in the brain and bone. There was no significant decrease of Gd deposit in brain at different time intervals post-contrast injection. Accurate assessment of Gd deposit is crucial for investigating potential toxicity.

PO-GePV-M-201

An Indoor Real-Time Location System (RTLS) to Improve Patient Safety and Clinical Workflow of Radiation Therapy

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Purpose: To present the infrastructure and technologies of an indoor RTLS system and its applications in a radiation therapy facility to improve patient safety and clinical workflow. **Methods:** Bluetooth Low Energy (BLE) technology was utilized as the foundation of the RTLS system. As sensors 384 Raspberry Pis were installed throughout our 136,000 sf facility which is divided into 363 zones. BLE beacons carried by subjects (patients, staff and devices) transmit BLE signals that are detected by nearby sensors. Variations in received signal strength due to distance and building structure form a unique pattern that is classified by an artificial intelligence based algorithm to tell a beacon's location. Based on the location engine infrastructure, a suite of RTLS applications were developed to improve clinical operations. **Results:** Through iterations of R&D, the RTLS system achieved a 99.9% location accuracy at zone level after it was trained. To meet clinical needs for better practice, eight applications were developed: 1) the beacon registration and asset management tool; 2) the Live Map that indicates live locations of subjects on floor maps; 3) the location search engine; 4) the electronic checklist that locates and verifies patients and accessories; 5) the MR safety surveillance system; 6) the attendance logging; 7) the timing tool for improvement of workflow efficiency and patient experience; 8) a set of data analytics and reporting tools. Since deployed in 2018, RTLS has made it more efficient to locate and verify patients and devices, and reduced wrong or

missing treatment device errors from 63 in 2018 to 23 in 2020. **Conclusion:** BLE RTLS system is proven to be an effective tool at low cost for care teams to boost patient safety, experience and operation efficiency in radiation therapy clinics, and could be easily adapted to other hospital settings.

PO-GePV-M-202

Total-Body Potassium in Rats Determined by Compact in Vivo Neutron Activation Analysis Assembly

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Purpose: This study aimed to evaluate the feasibility of a compact neutron generator-based in-vivo measurement system to quantify total body potassium in small animals using rat carcasses. **Methods:** A cohort of thirty-nine rats (n= 20 males and 19 females, average weight 255±15 and 163±7 gram), which were part of a manganese exposure study, were sacrificed and inserted in polyethylene bottles. These rats were then placed and irradiated in a carefully designed irradiation cave built beside the neutron generator with an optimized thermal neutron flux and radiation dose ratio. The irradiation time was 10 minutes, followed by a 10-minute decay and 2-hour measurement using the high purity germanium detector. **Results:** The average K concentration in male and female rats was significantly different (3546±473 vs. 2258±/-281µg/g). There was a significant correlation between potassium concentration and weight in both male and female groups (for male r(20) =.48, P=0.03, for female r(19)=.59, P=.007). We evaluated the impact of manganese exposure on potassium, and potassium concentration was not affected by manganese exposure. **Conclusion:** This work shows that in-vivo neutron activation analysis can be a potential way to explore in-vivo potassium storage and metabolism.

PO-GePV-M-203

Correlation Analysis Between the Kyoto Gastritis Classification and Early Gastric Cancer

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Purpose: To explore the value of Endoscopic performance score for risk of gastric cancer. **Methods:** 578 patients with chronic atrophic gastritis were selected as the study subjects. We selected endoscopic appearance and background mucosal appearance, biopsy pathology, and Helicobacter Pylori examination, to analyze the relationship between Endoscopic performance score and gastric carcinoma. **Results:** In 578 patients with chronic atrophic gastritis, 156 cases LGIN and 86 cases HGIN were detected. The detection rate of HGIN in male patients was 22.3% (57/256) and that of female patients was 9.0 % (29/322). The difference was statistically significant (P<0.001). There are Kyoto gastritis score less than 3 points 213 cases, score 4-6 points 198 cases, score 7-8 points 167 cases. The scoring grade was significantly correlated with the occurrence and degree of intraepithelial neoplasia (R=0.260, P < 0.001). The total score of patients with gastric cancer in the fundus of the stomach and cardia was significantly increased (P<0.01). In univariate analysis, the risk factor for gastric cancer is intestinal metaplasia, with an OR value of 4.811(95% CI 2.905-7.967, P<0.01). **Conclusion:** Endoscopic performance score has a good predictive value for the risk of gastric cancer.

PO-GePV-M-204

Detection of Cell Apoptosis with Gaussian Markov Random Field Algorithm

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Purpose: Cell apoptosis plays critical roles in cancer research and treatments. Flow cytometry (FCM) in combination with cell staining is commonly used for detecting cell apoptosis, which is of higher cost and time consuming. We developed a new machine learning based stain-free method for the detection and study of cell apoptosis. **Methods:** Suspension-cultured K562 cells were stained with Annexin V and nuclear dyes after apoptosis induction and then sorted by fluorescence-activated cell sorting (FACS) to obtain the cell samples in different stages. Texture features of diffraction images of the cells acquired with our polarization diffraction imaging flow cytometry (p-DIFC) system were extracted with a Gaussian Markov Random Field (GMRF) algorithm. The data were separated into training and test groups. Four classifiers were chosen to explore the model and to obtain the best performance on the data. **Results:** The new method could achieve a high precision of 92% in the classification of cells in different status during apoptosis on independent test datasets of the cell line. The chosen classifiers performed well in the classification. And the results also showed that the detection precision of the model on the independent test datasets was stable. **Conclusion:** The new method constituted by a GMRF algorithm for feature extraction and a SVM for classification has advantages in the apoptosis measurements with alive cells to keep their structure and function integrity for further experimental studies.

PO-GePV-M-205

Carbon Nanotubes Role in KV X-Rays: Enhancement or Diminution?

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Purpose: To experimentally ascertain the impact of carbon nanotube (CNT) on deposited energy in solid water by keV x-rays. In our previous work, we have found that adding CNTs to water enhances the measured dose deposited by x-ray beams in the MeV range. **Methods:** The radiation dose from 80,100,120,140 kVp x-rays was measured using an ion chamber at 1 cm depth in a solid water phantom. In an area of 5 cm x 5 cm, a layer of carbon nanotubes in two forms (foil and powder) was placed on the surface of the phantom, at 99 cm SSD. The pinpoint-type ion chamber is at 100 cm SAD, a centimeter below the carbon nanotubes. The measured dose enhancement ratio (DER) is the ratio of the dose to solid water + CNTs / dose to solid water only. **Results:** With CNTs in tape form, the measured DERs are 1.089, 1.083, 1.020 and 1.016 for 80 kVp, 100 kVp, 120 kVp and 140 kVp, respectively. When the tape CNTs is replaced by 650 mg of powder form CNTs, the measured DER is 1.089, 1.083, 1.051 and 1.048 for 80 kVp, 100 kVp, 120 kVp and 140 kVp, respectively. **Conclusion:** For all measured energies, the dose was enhanced when carbon nanotubes were added to the beam path. The dose increase is as high as 9% for 80 kVp and as low as 2% for 140 kVp.

PO-GePV-M-206

Virtual Collision Checks with a Laser Scanning and Depth Sensing Camera

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Purpose: Radiotherapy with non-coplanar beams typically require patients to attend a mock treatment ('collision check') wherein the motions of a radiotherapy plan are performed with the patient immobilized, but without radiation delivery. Treatment staff observe clearance throughout the simulation. Collision checks require LINAC time, staff time, and patient travel. Alternatively, plans and support structures can be conservatively designed to avoid the need for collision checks, but this can limit available beam orientations and patient setups (e.g., arms up) and result in sub-optimal plans. In this work we demonstrate that virtual simulation of collision checks are possible using a low-cost hand-held laser scanning and depth sensing (LiDAR) camera. **Methods:** An inexpensive (\$500 USD) camera was built using a Raspberry Pi, an Intel RealSense L515 LiDAR camera module, and a 3D-printed housing. Models of a General Electric CT

scanner and Varian TrueBeam LINAC, including couches, were captured. Models were spatially registered, and the LINAC model was rigged for virtual motion. An anthropomorphic phantom placed on the CT couch was LiDAR scanned, registered to the base CT model, and a 3D mesh model was extracted. It was then virtually transferred to the LINAC. LINAC rotational motion was simulated via a Minkowski product and intersection tests were performed to identify intentionally-created collision scenarios. **Results:** Spatial errors in the LiDAR-scanned surfaces were dependent on the material, ranging from the LINAC exterior (± 0.7 mm) to glossy carbon fibre (± 2.1 mm). In the absence of other positional uncertainties (e.g., support structures), couch-LINAC collisions due to couch translation could be predicted virtually within approximately 2.2 mm. **Conclusion:** Virtual collision checks have the potential to reduce resource usage for plans with non-coplanar beams, atypical patient setups, and bulky support structures. By allowing radiotherapy planners to simulate collision checks as part of the planning stage, additional treatment options are possible.

PO-GePV-M-207

Introduction of Automation in Radiation Treatment Plan Assignment and Daily Monitoring

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Purpose: To report on the development of ROSS (Radiation Oncology Smart Scheduling), an automated tool to assist with assignment of radiation therapy plans to planners and daily monitoring for a busy medical physics department with treatments administered over 8 locations and centralized treatment planning (71 planners, 31 linacs, 12 simulators). **Methods:** Plan scheduling assigns plans associated to completed simulations at the end of each business day and monitors the assigned plans for unexpected conflicts (start date change, planner absence, etc). ROSS uses constraint programming to generate assignments with no violations (staff credentialing, time-off, clinical conflicts) and optimizes the schedule for equitable distribution of workload among planners. ROSS accesses databases with assigned plans and new simulations to extract relevant information (required credentialing, technique, dates, etc.) and staff schedules. ROSS monitors the databases daily to flag violations (e.g., changes in start dates, new staff conflicts). Constraints and objectives were continuously adjusted and analyzed during the limited clinical rollout since fall 2020 to determine sufficient and optimal choices. Flagged violations were analyzed for a one-month period and divided into high-impact (likely clinical impact) and low-impact (clerical). **Results:** Adjustments to ROSS included constraints: limits on the number of plans due daily and in total, ramp-up/down near vacation periods, sufficient availability during planning/fusions, number of new assignments, multi-sites assigned to one planner. Development objectives inserted included: minimizing cross-campus planning, number of fusions assigned, reaching a dynamic target goal in number of plans (defined by clinical duties such as MR linac, same day SRS program, brachytherapy). ROSS' daily checks flagged 62 (14 high-priority and 48 low-priority) violations. ROSS produces assignments with a clinically feasible schedule that contains 0 violations in scheduling, with average runtimes of 244 seconds. **Conclusion:** ROSS demonstrates that automation can meaningfully assist in scheduling and provides early detection of emergent plan assignment issues.

PO-GePV-M-208

A Supplementary Management System for Personalized Ultra-Fractionated Stereotactic Adaptive Radiotherapy (PULSAR)

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Purpose: Personalized ultra-fractionated stereotactic adaptive radiotherapy (PULSAR) is a promising radiation therapy paradigm taking advantage of the clinically proven effectiveness of stereotactic ablative radiotherapy (SABR) and the opportunity to adapt more effectively with long intervals (weeks to months) between individual SABR treatments (pulses). However, the long time between consecutive pulses creates a management challenge to the care team. A specialized whiteboard application was developed to oversee the process of PULSAR treatments to ensure patient treatment is completed correctly. **Methods:** The web-based whiteboard application was designed to track the whole process of treatment planning and delivery throughout the entire course of PULSAR. When the prescription and timing is defined, tasks for the entire course will be automatically scheduled. Responsible parties will be reminded of upcoming work via email and page based on the due dates. The patient's progress is visually presented to optimize effectiveness and ease of use. The whiteboard application is connected to databases of clinical information systems for automatic data acquisition. **Results:** A whiteboard application was developed in the ecosystem of our in-house applications. It is drastically different from any in-house or commercial system in a) visually presenting the progress of the entire patient treatment and within each pulse; b) automatically laying out schedules of the entire course; c) automatically updating some of the statuses and sending reminders to responsible parties. Functions for access control and audits are also embedded in this application to ensure safety and a smooth workflow. **Conclusion:** The PULSAR whiteboard application is a feasible means to organize and track PULSAR treatments. It helps the care team carry out non-conventional adaptive radiation therapy treatments on schedule. Profession-sensitive whiteboards and analytics tools can also be added for higher efficiency and better workflows.

PO-GePV-M-209

Analysis of the Dose Enhancement Factor in the Presence of Gold Nanoparticles in Clinical Radiotherapy Beams

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Purpose: Radiotherapy aims to realize tumor control using ionizing radiation sources, preserving adjacent healthy tissues. Recent studies aim to deliver the dose in a more conformed way to tumor tissues. Among new strategies to enhance dose in tumor volume, the use of high-Z materials combined with radiotherapy emerged as a proposal for this. Thus, the combination of radiotherapy with gold nanoparticles as an alternative to maximize the efficiency of the treatment, aiming to reduce the toxicity. Therefore, this work analyzed the increase in dose deposition resulting from the effects of the interaction between gold nanoparticles and radiation from different clinical radiotherapy beams, using Monte Carlo Simulation with PENELOPE code. **Methods:** A homogeneous model of gold incorporation in water was used to quantify the interaction effect of gold with ionizing radiation (Ir-192; photon beam - 6 MV; electron beam - 6 MeV). The Dose Enhancement Factor (DEF) was calculated for all simulated conditions. **Results:** The results showed that DEF increases according to the increase in gold concentration, being also dependent on the spectrum of primary radiation used. The DEF values using the highest concentration studied for the 192Ir, 6 MV, e 6 MeV were 2.82 ± 0.09 , 1.41 ± 0.08 and 1.0297 ± 0.0002 , respectively. Thus, the work showed that the use of gold nanoparticles for electron beams does not result in a significant DEF value. **Conclusion:** The use of gold nanoparticles can provide a more effective tumor control, minimizing toxicity in adjacent healthy tissues, increasing the dose around the gold nanoparticles. This is a preliminary study with a simplified model for a previous analysis of DEFs in different types of radiation. Studies with more realistic cell models are needed to estimate an increase in local dose at different energies and beams.

PO-GePV-M-210

Measuring Distribution of An I-125 Labeled Elastin-Like Polypeptide (ELP) Nanoparticle Within Mice Tumors for Consideration as a Novel Technique of Delivering Brachytherapy

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Purpose: An elastin-like polypeptide (ELP) nanoparticle labeled with I-131 has been demonstrated to deliver curative radiation doses to xenografted tumors

in mice. The ELP conjugate is soluble below body temperature and undergoes a thermally triggered phase transition to become an insoluble coacervate at body temperature, establishing a stable radioactive intratumoral depot. This work is to quantify activity distribution for the purpose of calculating radiation dose distribution. **Methods:** Human pancreatic adenocarcinoma cells (BxPc3-luc2) were subcutaneously injected in 5 male nu/nu mice. When tumors grew to about 500 mm³, each mouse was injected with 300 uCi of ELP labeled with the X-ray emitter I-125 (instead of beta emitter I-131) for SPECT imaging. With ELP stabilized within tumor at 24 hours, the mice were sacrificed. The excised tumor was fixed in agarose in a plastic holder that was designed to allow for biopsy at two known locations. The tumor was then imaged with micro-SPECT/CT. After SPECT/CT imaging, small biopsy punches were taken. The absolute radioactivity of each biopsy sample was measured in a gamma counter and was used to convert the SPECT image intensity to absolute distribution of I-125 within the tumor. **Results:** SPECT images show that ELP distribution within the tumor is inhomogeneous and an-isotropic. The intensity decays quickly with distance from the injection depot but not monotonously. Local hot spots were observed outside of the injection depot. For quantitative analysis, inconsistencies between SPECT intensity and radioactivity measured at the two biopsy points were observed, preventing reliable conversion of SPECT intensity to radioactivity. **Conclusion:** If ELP intratumoral delivery is engineered for improved homogeneity within tumors, especially within larger tumors, ELPs have the potential to become a novel technique for delivering brachytherapy to the whole tumor volume. Inhomogeneous patterns of ELP diffusion in inhomogeneous tissue poses challenges to quantify its distribution.

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Generating Voxel-Based Dose from Activity Distribution for Radionuclide Therapy by Deep Learning Method

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Purpose: To provide a potential approach to automatically generate a voxel-based dose for radionuclide therapy from activity distribution by Generative Adversarial Nets (GANs) for visualizing process, guiding treatment strategies or calculating cumulative dose etc.. **Methods:** The activity-density superposition image is input into the Generator (G) as the referred image to generate the pseudo dose image, then it is identified by the Discriminator (D) via mixing with the true dose image (i.e. GT). If the pseudo dose image is identified, the information is fed back and the G will re-generate a pseudo dose image, until the D falls to obtain the 'false for true' dose image. As a feasibility study, dose distributions based on CT-based Zubal phantom with various source organs and activity distributions were trained and tested. The dose distribution was calculated by a voxel-based Monte Carlo dose algorithm. The activity source was assumed to be 1, 2, 3, 4 or 7 blocks of subsources, and the 3-subsource model was as test dataset and the others were as training dataset. The activity distribution at the subsource scale was assumed to be homogeneous or heterogeneous (i.e. Gaussian diffusion with sigma 0.0, 0.3 or 0.6). The differences were compared by whole body-based or organ-based, and evaluated by Gamma analysis. **Results:** The dose of different activity heterogeneous degrees could be well predicted by its same or quasi-heterogeneous degree model. The whole body-based conversion is suitable for activity homogeneous distribution, whereas the organ-based conversion is suitable for activity highly heterogeneous distribution (e.g. sigma 0.6). Although the 1-source model has been trained via very few dataset, it shows the best balance in precision and training efficiency. There are offsets in the mean dose prediction, but they show high Gamma pass ratio (> 93%). **Conclusion:** Deep learning method is explored to automatic generation of radionuclide dose.

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Prediction of Dose Discrepancies Using Differences in Plan Normalization Values of Head-And-Neck VMAT Plans

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Purpose: This study evaluated whether plan normalization value (PNV) differences could be used as predictors of potential discrepancies between planned and delivered doses. **Methods:** Forty-five patients who underwent volumetric-modulated arc therapy (VMAT) for head-and-neck cancer at our institution were included. The absolute dose and dose distribution were verified using ionization chamber and a GafChromic film and Delta4 PT, respectively. The VMAT plan to change the PNV used computed-tomography images and a simulated head-and-neck radiotherapy structure-set provided by the American Association of Physicists in Medicine Task Group (TG) 119. Seven treatment plans with varying PNVs were selected: 95%, 97%, 99%, 100%, 101%, 103%, and 105% PNV. For dose distribution verification, we used Delta4 PT in clinical cases. **Results:** PNV ranged from 98% to 102% in 38 (84.4%) of the 45 cases of head-and-neck VMAT. Further, PNVs ranged from 98% to 102% with 14 measurement locations exceeding $\pm 2.0\%$, and five measurement locations exceeded $\pm 3\%$. All GafChromic film and Delta4 PT gamma analysis (GA) results were above 90%, regardless of the PNV range. There was no apparent difference between the results of PNV and dose verification for 45 cases of head-and-neck VMAT. The 2 mm/2% result for GA in TG119 showed a trend of decreasing GA pass rates as the PNV moved away from 100%. **Conclusion:** The linear accelerator beam did not hold in the 95–105% PNV range, and the delivery was accurate with little effect on the accuracy of the planned and delivered doses.

PO-GePV-M-213

Effect of VMAT Improvement Before and After Referring to Radiation Therapy Planning QA Software

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Purpose: Software that evaluates the quality of treatment plans (PlanIQ™) has become commercially available in recent years. It includes a feasibility assessment tool that provides the ideal dose volume histogram (DVH) for each organ at risk, based on the ideal dose falloff from the prescribed dose at the target boundary. It is important to investigate whether the PlanIQ™ assessment tool (Feasibility DVH™) can assist treatment planners who are not experienced in treatment planning. Therefore, the present study aimed to evaluate the usefulness of this tool for improving the quality of treatment plans. **Methods:** This study included five participants with prostate cancer. Volumetric-modulated arc therapy (VMAT) planning data obtained before and after using the Feasibility DVH™ were scored with PlanIQ™. The impact of the feasibility assessment tool on the quality of the VMAT plans was evaluated for different patients and treatment planners. The treatment planners include two graduate students, two undergraduate students, and one clinical planner. **Results:** Four out of five cases showed improved treatment planning by referring to the Feasibility DVH™. In addition, four of five treatment planners showed improvement in treatment

planning by referring to the Feasibility DVH™. **Conclusion:** The present findings suggest that referring to the Feasibility DVH™ tool may improve the quality of treatment plans for different types of planners and patients. However, planners at any level of experience should be trained to check the dose distribution in addition to checking the DVH.

Yuji Nakaguchi is employees of TOYO MEDIC CO., LTD.

PO-GePV-M-214

Geometric and Dosimetric Impact of 3D Generative Adversarial network-Based Metal Artifact Reduction Algorithm On VMAT and IMPT for the Head and Neck Region

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Purpose: We investigated the geometric and dosimetric impact of 3D generative adversarial network (GAN)-based metal artifact reduction (MAR) algorithms on VMAT and IMPT for the head and neck region, based on artifact-free CT volumes with dental fillings. **Methods:** Two different CT volumes, one with four (m4) and the other with eight (m8) pseudo-dental fillings with a CT value of 4,000 HU, were created as ground truths for each case from thirteen metal-free CT volumes of the head and neck region (Reference). CT volumes with metal artifacts were then generated from the Reference CT volumes (Artifacts). On the Artifacts CT volumes, metal artifacts were manually corrected for using the water density override method with a value of 1.0 g/cm³ (Water). By contrast, the CT volumes with reduced metal artifacts using 3D GAN were also generated (GAN-MAR). The structural similarity (SSIM) index within the target were calculated between the Reference CT volumes and other volumes. After creating VMAT and IMPT plans on the Reference CT volumes, the original plans were recalculated for the remaining CT volumes. **Results:** The calculation time required to generate a single GAN-MAR CT volume was approximately 30 seconds. The median PTV volume was 133.6 ml (range, 98.7–169.1 ml), and median overlap ratio of the artifact corrected volume to the PTV volume was 7.5% (range, 0.0–18.5%) in the m4 group and 14.3% (range, 0.0–22.8%) in the m8 group. The SSIM showed that the GAN-MAR CT volumes were more similar to the Reference CT volumes than Water and Artifacts. The median dosimetric differences from the original plans were within 3% for VMAT and IMPT. **Conclusion:** GAN-MAR yielded identical CT volumes with the Reference CT volumes. The observed dosimetric difference compared to the original plan were clinically acceptable.

PO-GePV-M-215

Physical and Dosimetric Characteristic Properties of BeO OSL for Clinical Dosimetric Measurements

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Purpose: The study aims to figure out the physical and the dosimetric characteristics of the Beryllium Oxide (BeO) optically stimulated luminescence dosimeters (OSLDs) system and to investigate the ability to use OSLDs for machine quality assurance measurements by using necessary factors as mentioned in AAPM 191 for dose calculation. **Methods:** OSLs were irradiated with 6 MV, 10 MV, and 15 MV photon beams using Elekta Versa HD linear accelerator. The measurements were taken by placing BeO OSLs at the maximum dose depth and ion chamber at the reference depth for each energy at the source surface distance (SSD) at 100 cm. For the angular/directional dependence test, the setup for the cylindrical phantom was the same as the slab phantom, but a SAD of 100 cm was set to the isocenter of the phantom and center of the OSL dosimeter. The tests performed were reproducibility/sensitivity, dose-response linearity, energy dependence, angular dependence, field size dependence, and SSD dependence. As a result of the tests, correction factors were obtained, and the OSL measurement accuracy was increased by using these factors. **Results:** Reproducibility/sensitivity and angular dependence are smaller than %2, dose-response linearity and energy dependence are smaller than %1. After making SSD corrections as recommended in IAEA Human Health Reports No:8, the dependency of SSD is %3 for SSD 80cm and smaller than %2 for other SSDs. Field size dependence is larger than %2. k(L), for linearity, k(Q) for beam quality, k(Θ) for angular dependence, k(SSD) for SSD correction, k(Field) for field correction are obtained for all energies. **Conclusion:** It is possible to increase the accuracy of measurements with OSLDs when all factors are obtained and added to the measurement results. Using BeO OSLD in radiotherapy measurements is promising with further studies like percentage depth dose measurements.

BeO dosimeters used in the study were provided by RADKOR Personal Dosimeter Measurement and Assessment Laboratory company.

PO-GePV-M-216

Head and Neck Multi-Organ Segmentation on Dual-Energy CT Using Dual Pyramid Convolutional Neural Networks

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Purpose: Organ delineation is crucial to diagnosis and therapy, while is also labor-intensive and observer-dependent. Dual energy CT (DECT) provides additional image contrast than conventional single energy CT (SECT), which may facilitate automatic organ segmentation. This work aims to develop an automatic multi-organ segmentation approach using deep learning for head-and-neck region on DECT. **Methods:** We proposed a Mask scoring regional convolutional neural network (R-CNN) where comprehensive features are firstly learnt from two independent pyramid networks and are then combined via deep attention strategy to highlight the informative ones extracted from both two channels of low and high energy CT. To perform multi-organ segmentation and avoid misclassification, a mask scoring subnetwork was integrated into the Mask R-CNN framework to build the correlation between the class of potential detected organ's volume of interest (VOI) and the shape of that organ's segmentation within that VOI. We trained and tested our model on DECT images from 66 head-and-neck cancer patients with manual contours of 19 organs as training target and ground truth. **Results:** For large- and mid-sized organs such as brain and parotid, the proposed method successfully achieved average Dice similarity coefficient (DSC) larger than 0.8. For small-sized organs with very low contrast such as chiasm, cochlea, lens and optic nerves, the DSCs ranged between 0.5 and 0.8. With the propose method, using DECT images outperforms using SECT in all 19 organs with statistical significance in DSC ($p < 0.05$). Meanwhile, by using the DECT, the proposed method is also significantly superior to a recently developed CNN-based method in most of organs. **Conclusion:** Quantitative results demonstrated the feasibility of the proposed method, the superiority of using DECT to SECT, and the advantage of the proposed R-CNN over CNN. The proposed segmentation tool has the potential to facilitate the current radiation therapy workflow in treatment planning.

PO-GePV-M-217

Foreign Bodies Detection in Pediatrics with Low Dose CT

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Purpose: To develop a reasonable image quality CT protocol to detect the foreign bodies (FBs) in the respiratory tract of pediatrics with both a minimum patient radiation dose and an equivalent radiation risk to a conventional radiograph chest exam. **Methods:** CT Data were collected on a Siemens CT system (SOMATOM Flash). The imaging techniques of the child CT chest protocol were optimized, and vendor dose optimization features were utilized to further lower the dose and maintain good image quality. Twenty CT exams using the modified CT protocol were performed with different FBs distributions in a five years old size anthropomorphic phantom (LUNGBODY) shown in Figure(1). The FBs included small size radiopaque bodies such as fish bones, peanuts, LEGO

toys, etc. The exams were then presented randomly to the radiologist team to interpret the images and report the results. The CT effective and organ doses were calculated by Radimetrics™ software using Monte Carlo (MC) simulations. For dose comparison purposes, we used the routine chest radiograph exam to image the LUNGBODY phantom. The chest radiograph's organ dose is calculated using MC PCXC2.0 software for five years of phantom's lung, heart, and liver. **Results:** The effective dose of the original protocol was 3 mSv, while the dose of the modified protocol was only 0.26 mSv. The reduction in dose by one order of magnitude was mainly caused by the kVp and mAs reduction. The reported effective doses of the LUNGBODY for the modified CT using MC simulations and the chest radiograph using MC PCXC2.0 software and ICRP 103 weighting factors are listed in table(1). **Conclusion:** Low-dose CT protocols can be very reliable for FB detection with high image quality and low patient dose, especially in pediatric patients where the aspiration of FBs is a frequent and severe cause of respiratory problems in children.

PO-GePV-M-218

Magnetic Particle Spectrometer for Diagnosis of Magnetic Nanoparticles

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Purpose: Magnetic particle imaging (MPI) is an emerging medical imaging technique that only relies on superparamagnetic iron oxide nanoparticles (SPIONs) as tracer agents. Magnetic particle spectrometer and relaxometer are used as diagnostic tools for evaluation and selection of the best magnetic nanoparticles for MPI usage. In this study, a magnetic particle spectrometer (MPS) is designed at 9.9 kHz to measure the 5th/3rd harmonic ratio, full width at half maximum (FWHM) of the point spread function, and relaxation time of the nanoparticles at 15 mT excitation field. **Methods:** Hardware components of the MPS were simulated in COMSOL Multiphysics 5.4. Drive and receive coils were enclosed in cylindrical copper shielding to eliminate the environmental noise disturbance. The direct feedthrough effect of the fundamental harmonics was minimized with the optimized gradiometric design of the receive coil. A highly homogenous excitation field of 15 mT was targeted at the sample position (midpoint of the setup). The setup of the MPS was implemented based on simulated parameters to achieve optimum results. **Results:** Commercially available SPIONs (Vivotrax, Perimag) were first tested with home-built MPS. Mn_{0.75}Fe_{2.25}O₄ coated with citric acid and Mn_{0.75}Fe_{2.25}O₄ coated with polyacrylic acid were characterized at 9.9 kHz with 15 mT (peak amplitude) excitation field, and then compared with the commercial nanoparticles. Mn_{0.75}Fe_{2.25}O₄ has a 0.49 (5th/3rd) ratio, 7.60 mT FWHM, and 4.09 μs relaxation time. These results are in good agreement with Perimag. Similarly, Mn_{0.75}Fe_{2.25}O₄ has a 0.46 (5th/3rd) ratio, 8.95 mT FWHM, and 4.259 μs relaxation time. The effect of different coatings on the same iron core was analyzed as well. **Conclusion:** The home-made MPS provides an opportunity to diagnose magnetic nanoparticles and helps to mark the best tracer agent for MPI scanning. The expected resolution of MPI scanners for specific SPIONs can be estimated from FWHM (mT) of MPS.

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PO-GePV-M-219

An Multi-Stage Explainable System for Lobe Injury Diagnosis

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Purpose: To develop an accurate and explainable diagnosis system for lobe injury of Magnetic Resonance Imaging(MRI) image. **Methods:** A multi-stage explainable system(MSES) was constructed to make an accurate and explainable diagnosis for lobe injury on MRI images. MSES adopted a two-stage strategy. At the first stage, a 2.5D multi-task and multi-depth group UNet(MMGU) was constructed, which concatenated upper current and lower slices as inputs to supply context information and merged two segmentation branches, a positive branch pre-trained on positive slices with multi-depth paths, features at different levels of decoder fused to enhance feature's semantic information, a negative branch, structure same as the positive part, pre-trained on slices with positive and negative sample ratio of 1:4. Multi-task including region and boundary segmentation was proposed to enhance the lobe injury boundary effect. At the second stage, during the training step, the segmentation result of MMGU was used to form ROI patches to fine-tune a resnet-18 model as a false patch discrimination model(FPDM). During the case prediction stage, a similar process was performed to the training step to predict ROI patch probabilities, then the maximum was selected as the case probability. The Dice's coefficient(DSC) was used to evaluate the accuracy of MMGU, while the receiver operating characteristic(ROC) curves were used to evaluate the performance of the MSES. **Results:** On the testing data set, MMGU achieved superior segmentation performance with a mean DSC of 0.7865. the MSES gained superior diagnosis performance with an AUC of 0.98. **Conclusion:** The quantitative and visualization evaluation results demonstrates the effectiveness of the MSES, which also has a good diagnosis explanation, for it affords the lobe jury probability of interest patches on MRI images, which befits clinicians to make further analysis and decision.

PO-GePV-M-220

Develop a Rotational Robust Optimized Spot-Scanning Proton Arc (SPArc) Algorithm

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Purpose: Develop a new rotational robust optimization SPArc algorithm (SPArcrot) to mitigate the dosimetric impact from patient's rotational setup error. **Methods:** The SPArcrot incorporates a multi-CT robust optimization framework. Compared to the standard SPArc optimization algorithm, the SPArcrot takes into account two more CT image sets by rotating +5°, and -5° of the planning CT. Five cases from different disease sites such as brain, lung, head neck (HN), prostate, and liver cancer were selected to evaluate the effectiveness of SPArcrot. Both SPArc and SPArcrot plans were generated using the same translational robust optimization parameters. Then, both SPArc and SPArcrot plans were recalculated using a series of pseudo-CT introducing different rotational setup errors (±1°, ±2°, ±3°, and ±5°). Dosimetric metrics such as D98%, D95% of CTV, and 3D gamma analysis were used to assess the dose distribution changes inside the target and patient bodies. **Results:** The magnitudes of dosimetric perturbation in the targets due to rotational errors were significantly reduced by the SPArcrot in all the cases (Figure 1, 2). The max dose uncertainties to brainstem, optic nerves, spinal cord, and esophagus were reduced using SPArcrot for brain and lung cases, respectively(Figure 1). The uncertainties of the mean dose to the OARs such as the liver and oral cavity, parotid is comparable between the two planning techniques. The gamma pass-rate (3%/3mm) was significantly improved for CTV of all the disease sites in SPArcrot plan group (Table 1). **Conclusion:** SPArcrot could effectively mitigate the max dose perturbations in the adjunct OARs which is critical to the series organ. This new algorithm could be implemented through the multi-CT robust optimization framework using the existing commercial TPS.

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Xuanfeng Ding, Xiaoqiang Li and Di Yan have a patent related to the particle arc therapy and the patent has been licensed to IBA.

Xuanfeng Ding reported personal fee from IBA speaker bureau

PO-GePV-M-221

Inverse Shielding for Super High-Frequency (SHF) Microwaves to Ensure Contemporary PET Imaging Quality

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Purpose: Prevalent PET imaging reconstructs 2γ photon pairs from electron-positron annihilations. Due to QED, ~60% of annihilations occur directly and 40% occur through positronium (Ps) process. The ratio of triplet Ps (3S1, ortho-Ps, decaying to 3γ) to singlet Ps (1S0, para-Ps, decaying to 2γ) is 3:1 in vacuum. Although 3S1 to 1S0 transitions are forbidden, most ortho-Ps convert to para-Ps through "Pick-off process", hence 2γ events predominated. The energy difference between 3S1 and 1S0 (hyperfine splitting or HFS) is 203.4GH at ground state and 25.4GHz at first excited state. With magnetic fields, excited energy levels are split (Zeeman Effect), hence transitions between 23S1 to 21S0 become possible. Human body provided such conditions through biological tuning. Absorbing 25.4GHz microwave, many 23S1 converts to 21S0 and decays to 3γ rather than 2γ . This downgrades imaging quality because contemporary PET imaging rejects 3γ events. Such loss of para-Ps becomes crucial in PET-guided radiotherapy since minimal tracer was administered. Hence shielding against 25.4GHz source is necessary. **Methods:** Typical PET facility shielding is to protect area outside imaging room, this microwave shielding is inverse shielding because it tends to isolate patient against external microwave radiations. Microwaves could penetrate through 2-3 feet concrete walls, but will be stopped at thin metal layer, e.g., aluminum of 1-2mm thickness. For this reason, a metal layer should be always put into PET facility shielding in all directions of the vault, since pure concrete shielding is insufficient. **Results:** Microwaves of HFS energy 25.4GHz (or other higher excited states) drop PET imaging quality significantly. **Conclusion:** In PET-guided radiotherapy, PET imaging at a fractional basis serves for target localization rather than diagnosis, thus minimal injected dose is administered to reduce exposures. Losing para-Ps is unaffordable because image quality might downgrade to unacceptable level. Hence shielding microwaves of HFS energy levels are necessary.

PO-GePV-M-222

Direct Proton Dosimetry for Fractionated Radiotherapy: Dose Invariance Features and Mono-Endpoints

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Purpose: Proton radiotherapy has been booming since the new millennium. Contemporary proton dosimetry is to use Relative Biological Effectiveness (RBE=1.1) factor to convert reference photon dose to perspective proton dose regardless treatment strategies. This shortcut simplifies proton dosimetry into an add-on to the existing photon dosimetry, causing dose uncertainties in fractionated strategies. As a direct-ionization modality, protons essentially differ from X-rays/ γ -rays in cell-killing mechanism. In this study, we intended to establish direct proton dosimetry for fractionated treatment strategies. **Methods:** Unlike X-rays/ γ -rays, the cellular survival percentage for proton is a straight line in logarithm scale against dose, hence escalation doses remain a straight line. Different fractional strategies are mathematically equivalent if they have the same escalated doses, regardless of number of fractions or fractional dose in each strategy. This dose invariance feature is essential for fractionation themes in proton dosimetry. In X-rays/ γ -rays dosimetry, biological effective dose (BED) is used to compare 2 conventional fractionation (CF) themes, but the endpoints are different due to approximation with LQ modeling. In proton dosimetry, the endpoint is identical (mono-endpoint), and no approximation is needed. **Results:** We evaluated dose uncertainties and deviation of endpoint of using fixed RBE=1.1 in all fractionated themes, referencing to direct proton dosimetry. For CF the dose uncertainties are minimal. For hyper-fractionation, the real RBE>1.1, hence using fixed RBE will overdose. For hypo-fractionation, SBRT and SRS, the real RBE<1.1, hence using fixed RBE will be under-dosed, and the uncertainties increase with decrease of number of fractions for same endpoint. **Conclusion:** We established an independent proton dosimetry with high precision based on radiobiological nature, in which dose invariance and mono-endpoint are 2 essential features. We evaluate dose uncertainties for all fractionated themes using fixed RBE=1.1. For its accuracy, proton dosimetry could be used as the primary reference in radiotherapy, rather than vice versa.

PO-GePV-M-223

Respiratory Motion Management from MD Consultation to Radiotherapy

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Purpose: Respiratory motion management is a key component in thoracic and abdominal radiotherapy. In this project, we developed a respiratory motion management program from MD consultation to treatment including respiratory trace acquisition, respiratory motion evaluation and visual-guided radiotherapy. **Methods:** MRI/CT-compatible respiratory sensor and corresponding software were developed in our departmental respiratory motion management program. Respiratory trace can be acquired at MD consultation by clinical staff, analyzed and reported by Physics team, and visually utilized during CT/MRI simulation and treatment (visual guidance). **Results:** Hardware includes an air pressure sensor (Venier Barometer), placed on patient abdomen using a Velcro belt. The Barometer measures the pressure changes from patient abdominal displacement against the velcro belt. An USB device (Venier Go!Link) interfaces with the air pressure sensor for data-collection via developed software. Software has two components. 1) Real-time data collection and display, which can be switched between wave mode and bar mode. Wave mode shows the breath signal as a continuous wave, while the bar mode displays a horizontal bar moving up down corresponding to the patient's breath cycle. The real-time data collection part also includes a breath-holding timer function to guide the patient during breath-holding period. 2) Breathing pattern analyzer. Analysis for breath includes selections of breath time range and automatic breath period detection. Parameters such as Global Amplitude Variation (GAV), Period Variation (PV) are calculated. Local Amplitude Variation (LAV) is calculated using an adjustable window, suggesting a range of the gating window. Breath-holding analysis also includes time range selection. An adjustable window is also laid on the breath-holding analysis plot to calculate AV. **Conclusion:** We successfully developed the respiratory motion management from MD consultation to Radiotherapy. It is an efficient clinical solution for patient respiratory trace acquisition, patient training, breathing pattern analysis and corresponding respiratory motion management determination for radiotherapy.

PO-GePV-M-224

Dosimetric Properties of An HDR Ir-192 Source

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Purpose: We simulated the dose rate distributions around domestically produced HDR Ir-192 source by means of the Monte Carlo technique using the MCNP4c code. **Methods:** Based on the dose rate calculation equation in the TG43 protocol, we established the Ir-192 source model with the MCNP4c code. We simulated and calculated the spatial dose distribution of this model, and obtained and analyzed the radial dose function and anisotropy function of the Ir-192 source. **Results:** The simulated anisotropy function was in line with and within the allowed error range of published results. At short radial distance, the simulated radial dose function was more in line with published results; at long radial distance, the simulated radial dose function was more in line with some published results, but showed bigger discrepancy with other published results. **Conclusion:** The dosimetric properties of a domestically produced HDR Ir-192 source can be better studied and simulated with the Monte Carlo technique based on the MCNP4c code. The study results lay a theoretical foundation for further exploration of factors that impact the dose distribution of after-loading radiotherapy.

PO-GePV-M-225

An Incremental-Based Model for Organ at Risk in Multi-Scene Cervical Cancer Radiotherapy

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Purpose: Accurate delineation of organs at risk(OARs) is the key to protect the organs during radiotherapy. There exists multiple scenes in cervical cancer radiotherapy, such as after-loading and external irradiation, which have huge differences in the image domain and organ form. Using multiple independent models delineate organ at risk for each scene is inefficient and fails to use information between scenes. Therefore, an incremental-based segmentation model(IBSM) for OARs in multi-scene radiotherapy for cervical cancer was put forward. **Methods:** IBSM adopted a three-branch structure, the after-loading branch pre-trained on the positive slices of after-loading Computed Tomography (CT) images with object organs of rectum, bladder, and sigmoid, the external irradiation branch, structure same as the after-loading branch, pre-trained on the positive slices of irradiation CT images with object organs of rectum and bladder, the fusion branch merging the after-loading and irradiation branch as feature extract module trained on full data including after-loading and external irradiation CT images with full object organs above. During training, IBSM accepted after-loading and external irradiation data as inputs alternately. Dynamic weighted multi-task loss function among multiple organs and samples was proposed to guide the model to focus more on the hard samples and organs. The Dice's coefficient(DSC) was used to evaluate the accuracy of IBSM. **Results:** On the testing data set, IBSM achieved superior performance on most organs, with the mean DSC of 76.5(±6.18)% on the rectum, 89.6(±1.92)% on the bladder, 63.9(±1.3)% on sigmoid of after-loading CT images, 47.2(±30.769)% on the rectum, 84.5(±6.47)% on the bladder of external irradiation CT images. The IBSM gained consistent good delineation result on most organs. **Conclusion:** This work indicates the potential and effectiveness of ensemble and incremental-based model in multiple scene radiotherapy with a distinct data source and task.

PO-GePV-M-226

Determination of the Phantom Surface Dose Using a Diode Detector for Diagnostic X-Ray Beams

R Tanabe*

Purpose: To determine the surface dose for diagnostic kilovoltage x-ray beams from the air kerma and a half-value layer (HVL, mm Al) measured using a diode detector. **Methods:** A kV/dose multi-sensor (AGMS-DM+ detector, Radcal Co. USA) was used as a diode detector, and calibrated in terms of the air kerma measured with a PTW 30013 Farmer-type ionization chamber. The air kerma was measured for 20 x-ray beams with tube voltages of 50-140 kVp and HVL of 2.2-9.7 mm Al for a quality index (QI) of 0.4, 0.5 and 0.6, which is specified by the ratio of the effective energy to maximum energy corresponding to the tube voltage. The air kerma was converted into the phantom surface dose using the backscatter factor and the ratio of mass-energy absorption coefficient for water to air. Finally, the air kerma measured by the AGMS-DM+ detector was expressed as a ratio of the surface dose for 10 × 10 and 20 × 20 cm² fields. The ratio of both was represented as a function of HVL for the given QI values and verified by comparing it with that calculated using the Monte Carlo method. **Results:** The air kerma calibration factor for the AGMS-DM+ detector ranged from 0.986 to 1.016 (0.9% in k = 1), and was almost independent of the x-ray fluence spectra for the given QI values. The ratio of the surface dose to the air kerma determined by the PTW 30013 chamber and the AGMS-DM+ detector was less than 1.8% for the values calculated using the Monte Carlo method, showed a good correlation with HVL for the given QI values. **Conclusion:** It is possible to determine the surface dose from the air kerma and HVL measured by a diode detector for the given QI values for diagnostic kilovoltage x-ray beams.

The diode detector, AGMS-DM+ detector used in this study were provided by Toyo Medic Co. in Japan.

PO-GePV-M-227

A New Intracavitary Applicator Design: Dosimetric Characterization of Shielded Multichannel Applicator for High Dose Rate 60Co Brachytherapy Gynecological Cancer Treatment

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Purpose: The present study aims to construct a flexible multichannel applicator with several channels placed along the periphery of a cylinder and a rigid cylinder with interchangeable shields and evaluate dosimetric properties of this novel intracavitary applicator using MCNPX Monte Carlo code in terms of modulating between full dose to the target and minimum dose to the mucosa and non-involved parts of the cervix wall and rectum using HDR 60Co radioisotope. **Methods:** In this report, TG43 report and MC was used as reference tools, and then the comparison of dose distribution using both MC of both the new applicator was calculated shielded and unshielded cylinder applicators. **Results:** MCNPX Monte Carlo calculation shows that shielded multichannel applicator designs reduce the dose by 66% and 59%, but 66% and 63%, and 79 and 76% between 1.8 cm and 3 cm for 5channels, 3 channels, and 3 channels applicators with a thickness of 1.35 cm, 1.5cm and 2.1 cm from the applicator surface in desired regions. Furthermore, the dose rate in front of the shield and shielding ability is closely independent of the size of the shield thickness when calculated from the applicator surface, and dose rate in more distant volumes will to some extent be not influenced by shield thickness. **Conclusion:** Shielded multichannel applicator considerably spares OARs, but it needs to significantly compromise the treatment plan in cancerous regions. Therefore, further optimization of shielding design that compromises the treatment plan in parts that need the full dose.

PO-GePV-M-228

Quantification of Spinal Bone Marrow Fat Fraction Using Three-Material Decomposition Technique on Dual-Energy CT: A Feasibility Study

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Purpose: Two-material decomposition is insufficient to quantify the fat fraction of spinal bone marrow, which is comprised of a mixture of bone minerals, water, and yellow marrow (fat). To develop a three-material decomposition-based bone marrow fat fraction (FF3MD) quantification technique using dual-energy CT, and investigate its feasibility to characterize osteoporotic and pathologic spinal fractures in adults. **Methods:** Bone marrow edema phantoms containing trabecular bone minerals, water, and fat were constructed using fat fractions and bone mineral density values matching those expected in healthy and edematous bone, and scanned on a commercial dual-energy CT. Fat quantified by FF3MD were compared to MRI fat fraction FF_(MRI) and conventional two-material-decomposition-based fat fraction FF_(2MD). Subsequently, FF(3MD) was applied in a retrospective case study including seven adults (four male and three female; average age: 60.4 y.o.; range: 23-89 y.o.) with 10 spinal fractures (six acute osteoporotic, two hemangiomas, and two metastases) imaged at the time of admission. Findings were compared to MRI or pathology. **Results:** FF_(3MD) demonstrated an excellent correlation with FF_(MRI) (r=0.97, R2=0.96) in the phantom, significantly more accurate than FF_(2MD) when confounding bone minerals are present (50 mg/cm3: r=1.02, R2=0.95 vs. r=0.65, R2=0.79 (p<0.01); 100 mg/cm3: r=0.81, R2=0.47 vs. r=0.21, R2=0.21 (p<0.05)). The fat fractions of the 6/6 acute osteoporotic fractures were significantly lower than the healthy anatomical counterparts (p=0.0377). The fat fractions of the 2/2 vertebral hemangiomas and the 2/2 metastatic lesions were increased and decreased, respectively, when compared to healthy anatomical counterparts. These findings correlated well with T1-/T2-weighted MRI or pathology findings. **Conclusion:** FF_(3MD) accurately quantified bone marrow fat fraction, comparing with FFMRI, in the phantom. Three-material decomposition-based findings correlated well with MRI or pathology in the case study to characterize osteoporotic and pathologic fractures.

B Li, V Andreu-Arasa, C LeBedis, S Anderson received a research grant from GE Healthcare on dual-energy CT clinical applications.

PO-GePV-M-229

The First Investigation of Spot-Scanning Proton Arc (SPArc) Delivery Time and Accuracy with Different Delivery Tolerance Window Settings

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Purpose: The tolerance window has been playing a critical role in the arc treatment delivery in the photon treatment. However, there is no study has been done on the proton arc delivery module yet. This is the first investigation of Spot-scanning proton arc (SPArc) treatment delivery time and accuracy with various delivery tolerance window settings. **Methods:** SPArc plans were generated for four representative disease sites (lung, brain, head neck, liver cancer) with an angle sampling frequency of 2.5 degrees. An in-house dynamic arc controller was used as a simulator mimicking the arc treatment delivery with various tolerance windows (± 0.25 , ± 0.5 , ± 1 , and ± 1.25 degrees). The controller generates machine mechanical and irradiation information (logfilesimulation) with one degree of sampling frequency during the arc delivery simulation, such as gantry speed, acceleration and deceleration, spot position, and delivery sequence similar to the actual machine logfiles. The logfilesimulation was then imported to the treatment planning system (TPS) to reconstruct the delivered dose distribution and compared to the initial SPArc plan. A three-dimensional gamma index with criteria 3mm/3% was used to assess delivery accuracy quantitatively. Total treatment delivery time and relative lost time (dynamic arc delivery time – fix beam delivery time) / fix beam delivery time $\times 100$ were reported. **Results:** The 3D gamma pass ratio was great than 97% for all the cases. The total delivery time for dynamic arc delivery increased with the decreasing of delivery tolerance windows length. The average delivery time and the relative lost time (%) were 1024 ± 516 s (235% ± 31 %), 527 ± 262 s (73% ± 13 %), 375 ± 185 s (23% ± 7 %), 322 ± 149 s (7% ± 1 %), 311 ± 141 s (5% ± 2 %) for tolerance windows as ± 0.25 , ± 0.5 , ± 1 , and ± 1.25 degrees respectively. **Conclusion:** This is the first investigation of SPArc delivery time and accuracy with different delivery tolerance window settings. The simulated result indicated that the SPArc plan with 2.5degrees sampling frequency can be delivered efficiently and accurately with ± 1 or ± 1.25 degrees tolerance window.

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PO-GePV-M-230

MRI Characterization of 3D Printed Anatomic Models Using a Dual-Echo Ultrashort TE (UTE) Sequence

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Purpose: Three-dimensional printing (3DP) has been used increasingly in medicine over the past several years for various applications such as surgical planning, trainee education, and patient education. 3DP allows for the creation of comprehensive patient-specific models or models with complex geometries that otherwise may not be achievable. Although many different printing materials are available for 3DP, most materials are hard plastics or based on powders which generally have very short transverse relaxation time (< 1 ms), therefore are not readily visualized using conventional MRI techniques. The purpose of this study was to use an UTE MRI sequence to visualize a 3DP anatomic model with complex geometry and evaluate the volume accuracy of the MRI acquisition method. **Methods:** A patient-specific 3DP renal mass model was created from volumetric medical imaging data using binder jetting (CJP 660 Pro, 3D Systems). The printed model was immersed in saline water and scanned at 3T using a dual-echo ultrashort TE sequence with isotropic (0.6mm)³ spatial resolution and TR/TE1/TE2/FA=7.8ms/50 μ s/2.2ms/7°. 3D MR datasets from two echoes were subtracted to improve contrast before model segmentation. MR images were evaluated for image quality and the accuracy of the MR technique was evaluated. **Results:** The first echo with TE=50 μ s had positive MR signal from the 3DP model, while the second echo had negligible MR signal with TE=2.2ms. High positive contrast between 3DP model and background was achieved after image subtraction, leading to easy 3DP model segmentation. Renal mass volumes were 82523.12mm², 79247.75mm³, 75906.68mm³ for the 3D printed design from the patient-specific imaging data, the MRI of the model, and the CT of the model respectively, with DICE correlation coefficients of 0.98 for the MRI and 0.96 for the CT. **Conclusion:** 3D dual-echo UTE MRI can be utilized to generate positive MR contrast images from 3DP anatomic models to accurately determine geometrical volume.

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PO-GePV-M-231

Feasibility of Federated Learning for Head and Neck (HN) Tumor Segmentation in PET/CT

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Purpose: Generalization of deep-learning-based medical image segmentation strongly relies on the number and the diversity of training cases; however, it is practically challenging to aggregate data from different institutions due to various technical, legal, privacy, and data ownership barriers. This study aims to investigate the feasibility of employing federated learning to collaboratively train an auto-segmentation model for HN tumor segmentation in PET/CT from multi-institutional cases without data sharing. **Methods:** We used 200 training cases from the HECKTOR challenge in MICCAI 2020. These cases were collected from four institutions, including 55 from CHGJ, 18 from CHMR, 55 from CHUM and 72 cases from CHUS. We randomly selected 20% cases from each institution to form a testing dataset of 40 cases, and the rest of them were used for training. The segmentation was based on scale-attention-network (SA-Net) that we recently developed and achieved the 4th place in the HECKTOR challenge. We implemented FedAvg where the weights from each local model were aggregated in the server to form a global model that was re-distributed to each client for further training. The performance of the FedAvg model was compared with the baseline models that were trained exclusively on the local data within each institution. **Results:** The average DSC on the testing dataset was 0.716, 0.434, 0.680 and 0.738 when applying the baseline model from CHGJ, CHMR, CHUM and CHUS respectively. The FedAvg model yielded an average DSC of 0.752, which achieved 97% performance of the pooled model (DSC: 0.775) that was trained by pooling all the data across institutions. **Conclusion:** Our experiments show that the segmentation performance of the baseline models was restricted by the limited local data. Despite the significant imbalance in numbers of cases per institution, federated learning has potential to train a better model across collaborating institutions without sharing their data.

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PO-GePV-M-232

Clinical Evaluation Study of Image Guided Radiotherapy (IGRT) for Prostate Cancer Comparing Kv Imaging of Fiducial Markers and Cone Beam Computed Tomography (CBCT) Using SpaceOAR-View as a Matching Structure

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Purpose: To study the effectiveness of Space OAR-View gel as volume to match to during the Cone Beam CT acquisition as part of image guidance and comparing the matching to fiducial markers for patients with prostate cancer. **Methods:** From September 2020 we selected 12 patients with the following criteria: Age < 80 years; P.S. 0-1; Stage T1a – T2b; Gleason Score 3+3 with PSA ≤15 ng/ml. The patients selected did not undergo prior radiation therapy or surgery to the pelvis area. Space OAR-View by Boston Scientific was placed between anterior rectal wall and posterior prostate and 4 fiducial markers were implanted using a trans perineal approach. Each patient was immobilized in a supine position and simulated with 2.5mm CT scan with empty bladder. Contours of critical structures included the Space-OAR-View gel. Image guided radiation therapy using both cone beam CT (CBCT) and KV/KV fiducial markers match were used daily to assess the correlation and comparison between the Space-OAR-View contoured structure with respect to the Markers. Gel volume was also evaluated for the duration of the treatments. **Results:** Review of our preliminary data suggests that the use of Space OAR-View gel as structure to match to after cone beam CT is a valid and accurate methodology that helps the users during the fusion; it presents benefits over matching to other structures such as anterior rectal wall due to day to day variations and changes; it correlated well with a comparison to the fiducial markers match. We have also noticed a small variation in the Space OAR-View volume and more data is needed to validate our observations. **Conclusion:** Our study procedural imaging prostate patients with CBCT and matching to the gel findings may benefit patients with prostate fossa whom don't get fiducial markers and CBCT can be used with the gel.

PO-GePV-M-233

A Knowledge-Based Automatic Lung IMRT Planning Method for Partial Heart Sparing

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Purpose: Recent findings suggest that sparing of cardiac sub-volumes might be critical to maintain normal organ function. However, a lack of established dose constraints and planning experience hinder the clinical adaptation of cardiac substructure sparing in IMRT planning. This study was to establish clinically achievable cardiac substructure dose constraints based on prior plans and investigate their correlation with patient anatomy. An automatic lung IMRT planning method was developed to incorporate patient specific normal tissue dose constraints. **Methods:** Seventeen locally-advanced lung cancer patients who were treated with fixed-field IMRT were included in this study. The cardiac-substructures and the right heart sub-volume were contoured. Knowledge models were trained to predict the OARs and cardiac-substructure dose metrics utilized a step-wise regression method to quickly build models with good generalizability using only limited dataset. The predicted dose metrics were compared with the actual dose in clinical plans and a leave-one-out cross-validation method was used to evaluate the prediction accuracy of the dose metrics. The optimization objectives for automatic planning were determined based on model prediction. The creation of the objectives and plan optimization were all handled by a computer script interfacing with the treatment planning system with no human intervention. The dose metrics obtained by the automatic plans are compared with the clinical plans by paired signed-rank tests. **Results:** Knowledge models showed good predictive accuracy: the determination coefficients (R^2) for the dose metrics range between 0.3-0.9. Eight of the twelve predicted dose metrics had $R^2 > 0.5$. In particular, R^2 was 0.81 for right sub-volume of heart. Automatic plans improved all heart dose metrics and esophagus V60Gy and were comparable for all other dose metrics. **Conclusion:** Knowledge models can be utilized to determine patient-specific dose constraints for heart sub-volume and other OARs. Automatic planning can achieve comparable or better plan quality than clinical plans.

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PO-GePV-M-234

Real-Time Volumetric Image Guidance Via Deep Learning

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Purpose: Volumetric image guidance (VIG) requires X-ray computed tomography (CT) image and the corresponding multi-target segmentation. However, a large number of projection views from different angular positions are needed for CT reconstruction. The notorious multi-target segmentation then relies on the preceding CT. The time-consuming processes, including the multiple projection views acquisition, reconstruction, and segmentation, make the real-time VIG infeasible. **Methods:** Here, we develop a Deep Learning (DL) method to generate fully high-fidelity volumetric CT images and the multi-target segmentation simultaneously in real-time from a single-view X-ray projection. **Results:** We demonstrate the feasibility of the approach with a dynamic thorax phantom and more than twenty patients. **Conclusion:** Real-time VIG via deep learning allows on-the-fly visualization and dramatically reduces the imaging dose for patients, which could be useful in image-guided interventional procedures. It might help simplify the hardware of tomographic imaging systems.

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PO-GePV-M-235

A Site Agnostic 3D Dose Distribution Prediction Model for Radiation Therapy with Deep Neural Networks

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Purpose: Radiotherapy data is hard to collect. Normally we can only get a couple hundred patient's data to train the model on. Current dose prediction models are site specific, and need to be trained on the limited data available, this can reduce the overall potential performance. We propose a site agnostic model that can leverage data from any treatment site thus increasing total data available to train the model with. **Methods:** The input to our proposed model is the patient's 3D anatomical information and physician imposed tradeoff preferences to predict the 3D-dose prediction. Our supervised convolutional neural network, is based on 3D-U-net model. Data preprocessing includes mapping dose to voxels based on their distance ranks to PTV. The dataset contains 70 prostate cancer IMRT plans (source data) and 38 head and neck cancer VMAT plans (target data). We tested the generalizability of the model to different treatment sites via transfer learning technique. We have tested the performance of model with quantitative dose-volume-histogram (DVH) metrics for planning target volume (PTV) and organs-at-risk (OARs). We also tested the dependence of the model performance on the training data size to ensure the efficiency gain. **Results:** We implemented a deep neural network for volumetric dose prediction that is independent of the number of OARs and the treatment site. We tested the performance benefit by leveraging data from 2 different sites: prostate and head and neck. The model trained on one treatment site, efficiently performed high quality predictions on the second site even with small training sizes. **Conclusion:** Our proposed model is independent of the number of input OARs and the treatment site. Our model is able to make accurate predictions with limited training data

PO-GePV-M-236

Optimization of a FLASH Proton Beam Using a Track-Repeating Algorithm

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Purpose: We optimize characteristics of a 86.4 MeV proton beam for FLASH experiments using the Fast Dose Calculator (FDC), a track-repeating Monte Carlo algorithm. **Methods:** A phase space file located at 383 mm upstream of the isocenter is generated through tuning parameters to match FDC results with measured integrated dose distribution in water and MCNPX simulated lateral dose files in air. To spread out the Bragg Peak, widen the beam and reduce the penumbra, we insert a Ridge Filter, a high-density material Scatterer and a Collimator in the beam path and optimize their shapes, sizes and positions. The FDC calculations are validated by comparing Geant4 simulations. In addition, an algorithm to automatically choose the optimal dimension and position of the beam shaping elements is developed and tested using the same beam. **Results:** The optimized beam had a 10.5 mm SOBP (from 80% to 80% of the maximum dose), 13.5 and 12.5 mm lateral width with dose above 80% and 90% respectively and a 2.5 mm penumbra from 80% to 20% in the lateral profile. The FDC calculations had passing rates above 99% using 3mm/3% as gamma-index criterion comparing with Geant4 simulations. The automatic algorithm can choose the proper radius of high-density Scatterer within 20 minutes. **Conclusion:** The 86.4 MeV FLASH proton beam can be optimized by inserting proper high-density material Scatterer, Ridge Filter and Collimator. The track-repeating Monte Carlo algorithm FDC can accurately predict dose distribution when beam shaping elements are included. The automatic algorithm works efficiently in choosing dimensions for high-density Scatterer.

PO-GePV-M-237

Interpretable Artificial Intelligence-Based Extracapsular Extension Prediction in Head and Neck Cancer Analysis

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Purpose: In medical image-based computer-aided diagnosis, various artificial intelligence (AI)-based models have been proposed and demonstrated high performance. However, there is still a gap between the human's understanding and how these models learned, which influence how we understand and trust these AI algorithms. Therefore, the interpretability of AI model is essential. Our objective is to automatically perform Extracapsular Extension (ECE) detection with an interpretable deep learning technique. ECE is a decisive indication for treatment planning of patients with head and neck squamous cell carcinoma (HNSCC). Current detection practice often requires the visual identification of professionals and ultimately pathologic confirmation. We investigate AI techniques to assist with human identification. **Methods:** In this research, we propose a deep learning method to detect ECE from 3D computed tomography (CT) scans. The proposed model has multi-input that captures information with different scales. The deep convolutional neural network (DCNN) structure has been implemented. To check the explainability of the model, 3D possibility map is utilized to highlight the possible ECE region. Then, the results can be compared with the locations of lymph node region. **Results:** The experimental results have demonstrated that our model can identify ECE and non-ECE patients. Specifically, we have achieved the ECE detection with 96.92% accuracy, and 98.84% AUC. In addition, ECE probability maps are generated for patients, and ECE high potential regions are shown, which is able to enhance the interpretability of the model. **Conclusion:** The research demonstrates the capability to use artificial intelligence for ECE detection. By introducing the possibility map, we can identify where ECE has possibly occurred. Therefore, this will contribute to the explainability of the proposed model. The outcome of this study is expected to promote the implementation of interpretable artificial intelligence-assist ECE detection.

PO-GePV-M-238

Using Deep Learning Auto-Planning for Evaluating the Dosimetric Impact of Deep Learning Auto-Segmentation Without Human Intervention Over Multiple Dose Escalation Schemes for Lung SBRT Treatment Planning

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Purpose: This work presents an integrated solution for automatic segmentation and planning for lung SBRT. Running auto-planning with multiple dose escalation schemes allowed the dosimetric impact of auto-segmentation to be evaluated. **Methods:** The DL planning model was trained in RayStation 10B, using 99 lung SBRT plans delivering 50 Gy in five fractions. The DL segmentation model originated from an official RaySearch model for heart, spinal cord, esophagus, and both lungs. This was extended with a sub-model for chest wall, trained in RayStation 10B using the same lung SBRT patients. The planning model was previously validated, allowing it to be used as a virtual planner. It generates three types of auto-plans, following different dose escalation schemes (110%, 115%, and 120%) to an ITV-4mm heater in the center of the lesion. For 15 test patients, auto-planning was run based on manual and DL ROIs, separately. The former was evaluated on manual ROIs only, while the latter was evaluated on both manual and DL ROIs. PTV, CTV, and heater were always manually delineated. **Results:** When evaluating on manual ROIs, auto-planning based on manual and DL ROIs resulted in similar lung V20 dose, heart mean dose, and spinal cord maximum dose for all test patients and all dose escalation schemes. Studying target coverage and chest wall sparing, however, only 12/15 patients had adequate coverage in the DL ROI-based plans. The remaining three patients had lesions abutting the chest wall, for which the lesion was subtracted from the chest wall in the manual contours. This behavior was not captured by the DL segmentation model. **Conclusion:** Generally, auto-planning based on DL ROIs instead of manual ROIs did not have significant dosimetric effects, except for some patients with lesions abutting the chest wall. Further, the choice of dose escalation scheme had no effect in these dosimetric analyses.

Mats Holmstrom and Elin Samuelsson are employees of RaySearch Laboratories, Stockholm, Sweden.

PO-GePV-M-239

Validation of a Track-Repeating Monte Carlo Method for Isotopes Calculation

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Purpose: A small amount of positron-emitting isotopes, such as 10C, 11C, 14O, 15O, 13N, 30P and 38K, are produced in a patient body during carbon treatment. This phenomenon can be used to estimate the dose range through PET scan. When the scan is performed after the treatment, most of these generated isotopes will not be detected due to their fast decay, except for 11C. We implemented positron-emitting isotope production in Fast Dose Calculator (FDC), a track-repeating Monte Carlo method. The purpose of this work is to validate the accuracy of FDC isotopes simulation for dose range estimates. **Methods:** During the carbon trajectories simulation by FDC, we save stopped isotopes mentioned above. The same calculation is performed by Geant4 with physics list QSGP_BIC_AIHHP. The FDC isotopes simulation is validated by comparing the ranges of isotope distribution from FDC and Geant4 after blur, decay and body washout processed. We used a cohort of ten patients treated with carbon therapy with the following cases: 4 Thorax, 4 Pelvis and 2 Head & Neck. In addition, six databases with different tissues including water, muscle, adipose, breast, brain and cartilage were generated for FDC. **Results:** The mean range

difference between FDC and Geant4 of 1008 range estimates are -0.4539, -0.09474, -0.180, -0.2257, -0.3993 and -0.3859 mm with FDC database of water, muscle, adipose, breast, brain and cartilage respectively. The standard deviation of the range difference for the 1008 channels is around 4 for different databases. The speed of FDC is thousands of times faster than Geant4. **Conclusion:** FDC isotope distribution calculation is validated. Using muscle database, FDC can give more accurate average range compared with databases using other tissues.

PO-GePV-M-240

Dual-Energy Subtraction Efficiency: Development of An Objective Quality Metric for Tissue-Subtraction Radiography

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Purpose: To develop a figure of merit for the objective evaluation and comparison of tissue-selective images generated from dual-energy (DE) radiography systems. **Methods:** The requirements for a successful metric are identified and a metric is proposed based on the spectral transfer of noise, contrast, and resolution properties from the captured (or primary) DE image data to the generated (or secondary) tissue-subtracted images. It combines the relative noise and resolution properties of the tissue-subtracted images with respect to the primary data with the measured quantity and quality of tissue cancellation. An experimental method and an imaging phantom are presented for the capturing all the necessary primary data. Lastly, an analytical method for computing this metric is detailed and verified in a variety of scenarios through first-order X-ray imaging simulations. **Results:** The proposed Dual-Energy Subtraction Efficiency (DSE) metric adequately describes a multitude of image properties for all simulated image scenarios, indicating its ability to accurately and objectively evaluate image quality. It is even possible to identify the particulars of the image deficiencies by studying the DSE spectral shape. **Conclusion:** The DSE and its measurement method described can become a tool to characterize dual-energy radiographic image quality objectively and quantitatively, allowing for improved system comparison, development, and optimization. We conclude that, alongside quality metrics dedicated to conventional radiography such as detective quantum efficiency, it is now possible to extend objective X-ray system evaluation to include dual-energy subtraction radiography.

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A Multi-Modal Deep Learning-Based Decision Support System for Individualized Radiotherapy of Non-Small Cell Lung Cancer

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Purpose: There have been rapidly growing applications of artificial intelligence in precision radiotherapy, particularly involving deep learning, which often outperform physicians by utilizing big data in the clinic. The goal of this study is to develop a multi-modal deep learning-based decision support system to help physicians in individualizing radiotherapy for their non-small cell lung cancer patients. **Methods:** With IRB approval, data of 200 non-small cell lung cancer patients was obtained, which included treatment planning data, images, and clinical notes. Factors extracted from treatment planning system were focused on clinical, demographic, and dose-volume indexes. Radiomics features were extracted by a convolutional neural network from imaging data, while the clinical notes were analyzed using natural language processing to cluster phrases into predictive features. The toxicities of each patient were ranked on a RTOG scale of 0-5 on radiation pneumonitis. Initial optimization was performed on individual sub-models, the results of which were normalized and input into a deep neural network to make a final prediction. For each model, the data was split into training and testing data, while a variety of Python tools such as Tensorflow and PyDicom were used to optimize the number of hidden layers and weights for the final predictions. Missing data was imputed using open source packages. **Results:** The accuracy of the individual models was in the 75% range depending on the parameters. Our study indicated that additions beyond three hidden layers had diminishing returns. The final model ended up predicting patient's radiation pneumonitis with an accuracy of 83%. **Conclusion:** Our work indicates that there is statistical validity in adopting a multi-modality-based decision support system for radiation therapy, particularly in predicting radiation-induced toxicity. More work will be needed on determining predictive model calibration and creating a data infrastructure for streamlined integration of structured and unstructured data in radiotherapy.

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Decreasing NTCP Toxicity in Lung Cancer RT by Reducing PTV Margins Using a Combined Deep Learning Auto-Segmentation and Automated Treatment Planning Approach

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Purpose: Large planning target volumes (PTVs) in locally-advanced non-small cell lung cancer (LA-NSCLC) frequently result in high doses to nearby normal tissues that can consequently lead to severe complications. We reduced current PTV margins and studied the impact on normal tissue complication probabilities (NTCPs). Tumors and considered normal tissues were derived from deep learning (DL) auto-segmentation and treatment was simulated using automated planning with the ambition of using an overall unbiased approach. **Methods:** The cord, esophagus, heart, lungs and tumors (GTV) of 8 randomly selected patients with LA-NSCLC previously treated with definitive concurrent chemo-radiation to 60 Gy in 30 fractions were segmented using a multiple residual network DL auto-segmentation algorithm. The CTV was created via an added 7mm isotropic margin. Another 2mm and 5mm isotropic margin were used to generate a study-specific small PTV (PTVS) and a clinical practice large PTV (PTVL), respectively. One PTVS plan and one PTVL plan was generated for all patients using automated hierarchical optimization applying all currently used dose-volume criteria. Plans were compared by means of NTCP using published models (Esophagus: Esophagitis; Lung: Pneumonitis) and by clinically relevant dose metrics: max dose (cord), mean dose (heart) and D95% (PTV). **Results:** For all patients, PTV coverage was similar in the PTVS and in the PTVL plans (D95%=60Gy in all plans) while dose to all normal tissues was systematically lower in the PTVS plans: The risk of esophagitis was on average reduced from 60% to 57%, the risk of pneumonitis from 8% to 6%, cord max dose from 24Gy to 17Gy and heart mean dose from 9Gy to 8Gy. **Conclusion:** This modestly reduced PTV margin study has demonstrated considerable dose sparing for the key normal tissues in lung cancer RT by means of an unbiased automated segmentation and automated planning approach.

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Multi-Scale Deep Learning CT Liver and Spleen Segmentation Network for Radiation-Induced Toxicity Outcomes Analysis

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Purpose: Dose toxicity to immune-related organs-at-risk during radiation therapy (RT) may contribute to lymphopenia and radiation-induced liver disease. Our goal is to provide an open-source tool to automatically segment the liver and spleen from clinical contours to enable maximal tissue sparing and perform clinical outcomes analysis. To facilitate this, we built and validated a Deep Learning Segmentation (DLS) framework for consistent and robust auto-

segmentations. **Methods:** The DLS framework leveraged contextual information from 182 contrast and non-contrast enhanced Computed Tomography (CT) scans with well-curated clinical contours of patients previously treated with thoracic/abdominal RT. The DLS model was trained on a nested deep neural network architecture, Unet++, for multi-label segmentation (learning rate: 0.1; batch size: 8 images for 70 epochs) using 90% of the CT scans. The multi-scale, deep-supervision enabled network learned representations from dense skip-connections on entire CT body scans. The remaining 10% of the scans were used for quantitative evaluation of the final model vs. reference contours using Dice Similarity Coefficients (DSC), 95th percentile of Hausdorff Distances (HD95) and the mean dose. Statistical comparison was performed using the Wilcoxon signed-rank test and Spearman's correlation coefficient tests. **Results:** The DLS model reduced segmentation time per patient from about half an hour of manual segmentation to 40 seconds. The liver and spleen achieved accuracy of (median DSC=(0.94(0.93-0.95), 0.92(0.89-0.93)) and HD95=(7.5mm(5.3mm-8.4mm), 3.7mm(3.3mm-6.2mm)). No statistically significant difference was observed between the calculated mean dose for the auto-generated and the expert contours for both structures (p-value>=0.4). **Conclusion:** The model was robust against variability in image characteristics and large image background, including the presence/absence of contrast. The segmentation accuracy is considered adequate for extracting dose-volume histograms to use in outcomes analyses. The abdominal DLS model is distributed as a part of CERR's Model Implementations library.

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Performance Evaluation and Validation of Knowledge-Based Treatment Planning of VMAT for Post-Mastectomy Loco-Regional Radiotherapy Involving Internal Mammary Chain

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Purpose: Clinical implementation of knowledge-based treatment planning for Volumetric Modulated Arc Radiotherapy for post-mastectomy loco-regional radiotherapy. **Methods:** A knowledge-based treatment planning (KBP) model was built (Rapid-Plan v13.5.35, Eclipse v13.5, Varian Medical System) using 73 left-sided patients who had undergone irradiation of the ipsilateral chest wall with internal mammary nodes (IMN) and supra-clavicular fossa (SCF) with prescribed dose of 26 Gy/5 fractions. All the clinical were generated using 6MV photon beams with two tangential partial arcs with isocentre located in the centre of PTV chest wall (Eclipse v13.5). For the validation, automatic plans(KBP) were generated for 13 patient datasets in a single optimization step without any manual intervention. Various dose volume parameters were compared and analysed using paired t-test and Wilcoxon signed rank test. P<0.05 was considered significant. **Results:** Mean $\pm$ Standard deviation (SD) between clinical plans and KBP for various DVH parameters have been analysed and comparisons are given as :D 95[%] of PTV for CW, IMN and SCF were 93.2 $\pm$ 2.4vs 93.2 $\pm$ 3.1 (p=0.95), 89.48 $\pm$ 2.94 vs 91.21 $\pm$ 2.49 (p=0.17) and 94.95 $\pm$ 1.4 vs 97.27 $\pm$ 1.7 (p=0.00) respectively. For OARs, mean dose (Gy) $\pm$ SD to heart, contralateral breast, ipsilateral lung was 2.92 $\pm$ 0.54 vs 2.79 $\pm$ 0.55 (p=0.06), 2.14 $\pm$ 0.34 vs 2.27 $\pm$ 0.24 (p=0.162) and 6.67 $\pm$ 0.71vs 7.48 $\pm$ 0.39 (p<0.001) respectively. For contralateral lung, mean dose (Gy) with (IQR) was 2.46 (2.21,2.51) vs 2.2 (1.97,2.28) with (p =0.012). Maximum dose to left anterior descending artery (LAD) was 15.1 $\pm$ 3.1 vs 13.16 $\pm$ 2.43 (p=0.001). Homogeneity index (HI) and Conformation Number (CN) of total PTV for clinical and KBP were 0.12 $\pm$ 0.02 vs 0.12 $\pm$ 0.02 (p=0.39) and 0.93 $\pm$ 0.02vs 0.93 $\pm$ 0.03 (p=0.47). Mean $\pm$ SD of MU for clinical and rapid plan were 1994 $\pm$ 275.3 vs 1704 $\pm$ 315.3 (p=0.002). **Conclusion:** The KBP were either better or comparable to the clinically acceptable plans generated by experienced physicist for various Dosimetric parameters analysed. Hence KBP can be used for clinical implementation.

Varian Medical Systems,Palo Alto, California, United States for providing licence to use Knowledge base planning (Rapid Plan) module

PO-GePV-M-245

Benchmarking of Three Spatial Fractionated Radiation Therapy Techniques for Multiple Treatment Sites

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Purpose: To benchmark three techniques for Spatial Fractionated Radiation Therapy (SFRT) and evaluate their dosimetric performance in multiple treatment sites. **Methods:** Five patients (multiple sites) who underwent conventional single-field GridBlock therapy were replanned using GridBlock, MLC-based, and Lattice-based SFRT (15Gy/1fraction, 6 MV). The GridBlock consisted of a hexagonal pattern of divergent holes (hole diameter=1.43cm, center-to-center spacing=2.11cm). The MLC-based grid was generated by converting a static beam to a step and shoot beam delivering grid pattern (square aperture=1cm, open-to-closed ratio=0.375/0.625). The Lattice-based grid was generated by placing cylinders with 1.5cm diameter and 3cm center-to-center spacing within the GTV to mimic conventional GridBlock therapy. The quality of the plans was evaluated using different dosimetric parameters (Valley-to-Peak Ratio (VPR=D10valley/D90peak), mean GTV dose, equivalent uniform doses (EUD) for both GTV and Normal Tissue, delivery time). **Results:** The VPR is the major postulated contributor of the bystander and abscopal effects in SFRT. The lattice technique achieved a better mean VPR (54.03 $\pm$ 15%) when compared to those of the Grid-Block (19.09 $\pm$ 4%) and MLC (19.40 $\pm$ 16%) techniques. For dose coverage of the GTV, the lattice technique also achieved a better dose coverage (D=9Gy $\pm$ 3Gy) compared to Grid-Block (D=5.8Gy $\pm$ 1Gy) and MLC (D=4Gy $\pm$ 1Gy). The EUD predicts the effectiveness of tumor control and normal tissue toxicity and the lattice technique predicted better tumor control and normal tissue toxicity (mean EUD_tumor =7.97Gy, mean EUD_normal=5.87Gy) than the GridBlock (mean EUD_tumor=1.08Gy, mean EUD_normal=6.91Gy) and MLC (mean EUD_tumor=1.67Gy, mean EUD_normal=5.45Gy). The beam-delivery time was clinical acceptable for all the techniques (mean T_lattice=8.1min, mean T_GridBlock=2.83min, mean T_MLC=7.3min). **Conclusion:** When available, the lattice technique should be preferred above the GridBlock and MLC techniques because it produces better quality plans. It achieves better VPRs and coverage of the GTV. It gives better predictions of the tumor control and normal tissue toxicity, and its delivery times are clinical acceptable.

PO-GePV-M-246

A Motion Phantom Study on RefleXion X1: The Dosimetric Impacts of Stereotactic Radiation Therapy Delivery Technique and Motion

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Purpose: RefleXionTM X1 is a novel therapy machine designed to support both biology-guided radiotherapy (BgRT) and SBRT. To determine the dosimetric effects of motion during RefleXion SBRT treatment delivery, a lung phantom film dosimetry study was performed under multiple motion conditions using the first clinical installation of RefleXion X1. **Methods:** CIRS Dynamic Thorax Motion Phantom with a 3.0cm diameter spherical tumor film insert was used to program respiratory waveforms with varying peak-to-peak amplitude (A) and period (T). 4DCT scans were acquired for the following 3 dynamic conditions: A=0.6cm with T=5s, A=1.2cm with T=4s, and A=2cm with T=6s to simulate human respiratory motion of lung tumours. Phase-based binning was performed, and GTVs were contoured on each respiratory-phase. ITVs were then generated, and PTVs were created with 2mm and 5mm expansions. To determine the effects of different motion and delivery parameters, treatment plans were created for both PTVs using 1 and 2cm jaw sizes with a prescription of $\geq$ 95% coverage of PTV with 5.0Gy. Plans were delivered to a phantom under planned dynamic conditions. Two-dimensional dose was measured using Gafchromic EBT-3 film along the sagittal plane of the target center. The film measurements were used to calculate the 2D cumulative DVH, percentage of ITV receiving the prescription dose, and the minimum dose received by 95% of the ITV. **Results:** The larger motion amplitude resulted in more under-dosing of the ITV, and this effect was greater for 1cm-jaw plans. The 2cm-jaw plans provided adequate coverage to the ITV for smaller amplitudes of motion (i.e., 0.6cm/1.2cm) with ITV-to-PTV margins of 2mm or 5mm. **Conclusion:** The interplay of RefleXion SBRT delivery and target motion was measured for several clinically-relevant lung cases. Clinically acceptable ITV coverage was obtained for smaller amplitudes of motion with 2cm-jaws whereas significant under-dosing was

observed for larger motion amplitudes with 1cm-jaws.

PO-GePV-M-247

Tattooless Patient Setup with 3D Optical Surface Imaging For Paraspinal Radiation Therapy

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Purpose: Permanent skin tattoos have been part of standard care for paraspinal patients undergoing EBRT. In this study, we prospectively evaluated tattooless setup with 3D surface imaging and compared it with the skin tattoo-based approach. **Methods:** Patients receiving paraspinal radiation therapy underwent a standard tattoo-based setup, alternating daily with a tattooless setup by surface imaging. Following the initial setup, the patient position was verified by daily CBCT imaging, with matching on the bone or surgical hardware to represent ground truth. Translational and rotational shifts determined by CBCT imaging were recorded, as were setup time duration and total in-room time duration. Statistical analysis was performed using the non-parametric Wilcoxon Rank-Sum test and F-test for Equality of Two Variances. **Results:** We evaluated 57 individual treatments from 14 paraspinal patients, 22 treated under surface imaging, and 35 under a traditional tattoo-based setup. The distributions of entire in-room time duration and set up time duration are right-skewed in both cohorts and are not normally distributed. The median entire in-room time duration and set up time duration in the surface imaging cohort are indifferent from those of the traditional skin tattoo cohort ($p = 0.417$ and 0.597 , respectively). The variance of position shift in LNG direction in the 3D surface imaging cohort is significantly different from that of the traditional skin tattoo cohort ($p = 0.0069$). There is insufficient evidence ($p > 0.05$) to indicate a difference in variance or mean between the two cohorts in other directions. **Conclusion:** Our preliminary results suggest that a tattooless setup approach to paraspinal radiotherapy with surface imaging may be sufficiently accurate for initial patient setup. Further analyses among large cohorts and including delivered dosimetry will determine whether standard tattoo-based approaches can be replaced by 3D surface imaging.

PO-GePV-M-248

A Medical Image Challenge Infrastructure

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Purpose: To develop an infrastructure for benchmarking the performance of task-specific software tools in medical imaging challenge competitions. **Methods:** The open-source challenge platform, CodaLab, has become a popular tool for benchmarking scientific competitions. We customized an instance of CodaLab for use in medical imaging applications, by adding capabilities to accept challenge proposals, provide guidelines, and distribute data from public image repositories such as The Cancer Imaging Archive, or from a private server. Additional developments include allowing docker submissions, a documentation site, and a helpdesk to assist users. **Results:** The National Cancer Institute's Medical Imaging Challenge Infrastructure (MedICI - <https://github.com/QTIM-Lab/MedICI>) can be deployed locally or in a public cloud. It allows for challenge design submissions. Organizers are able to use scripts and codes from public libraries in R, Python, Javascript, etc to evaluate the results based on previously defined metrics. To date, over 20 medical imaging challenges, with applications in cancer have been conducted through MedICI, either independently or in conjunction with scientific conferences. These have included challenges in benchmarking tools for image-based detection, diagnosis, classification of cancers, and prediction of response to radiotherapy. Many of the tools developed in response to challenges on MedICI have deployed innovative approaches to machine learning and deep learning in medical imaging. **Conclusion:** Challenges provide a robust way to promote open data science by assessing the performance of software tools in addressing practical technical or clinical tasks in medical imaging. They facilitate a head-to-head assessment of innovative tools for diagnosis and treatment of disease using standardized datasets and high-quality ground truth data. As a resource, MedICI provides a comprehensive platform for the implementation of challenges in medical imaging. It will continue to evolve and be a valuable resource to the community by serving challenges to scientific organizations, including the AAPM.

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Automatic Segmentation of Head and Neck Tumors During Treatment: Can We Remove the Mask Yet?

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Purpose: To investigate the feasibility of automatic segmentation of head and neck (H&N) tumors in kV images using simulated projections. Success will lead to the ability to track tumor motion during H&N radiotherapy with the goal of removing the need for the thermoplastic mask for radiotherapy. **Methods:** CT scans of two patients were used from the HNSCC database from the Cancer Imaging Archive. For each patient, 3600 evenly spaced digitally reconstructed radiographs (DRRs) were generated (0.1° projection angle difference) from the CT scan. For each DRR the ground truth tumor location was forward projected from the planning CT and corresponding structures file. A Cycle Generation Adversarial Network (cGAN) was trained to detect the tumour location in each DRR. The testing data consisted of 360 DRRs from a rigidly transformed copy of the CT volume, with the projection angles of the DRRs being taken from a typical CBCT scan for H&N patients. The CT scan was rotated by 3° and then translated by 3mm in the AP, LR and SI directions. The estimated tumor location from the cGAN was compared with the ground truth for each patient by measuring the centroid error and the DICE similarity coefficient (DSC). **Results:** The mean($\pm$ Standard deviation) DSC coefficients for patients 1 and 2 were 0.89 ± 0.02 , 0.90 ± 0.02 , and for all patients the DSC was 0.89 ± 0.02 . The mean($\pm$ Standard deviation) absolute centroid error for patients 1 and 2 were 1.43 ± 0.76 mm and 1.54 ± 0.88 mm, and for all patients was 1.49 ± 0.82 mm. **Conclusion:** The results in this study demonstrate potential for tracking the tumor location for head and neck cancers in silico. Further development of this technology will pave the way for the removal of the need for thermoplastic masks for treating head and neck cancers during radiotherapy.

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PO-GePV-M-250

Efficient and Stable Evaluation of Image Registration Using Deep Learning

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Purpose: The alignment of the region of interest is of primary concern in radiotherapy. Although quantitative assessment can be obtained by manually contouring the same structures on both fixed and moving images, this process is time-consuming and observer-dependent. Here, we established an efficient and stable evaluation tool using deep learning to evaluate the accuracy of image registration quantitatively and objectively. **Methods:** Ten patients with head and neck tumors were enrolled. Two sets of CT images for each patient were acquired in different time, and then were registered with rigid and deformable methods, respectively. Eight OARs were segmented by oncologists and the deep learning method in each image, named as Manual-fixed, Manual-moving, Auto-fixed and Auto-moving respectively. Then, both Manual-moving and Auto-moving were introduced to the transformation matrices to get Manual-moving-trans and Auto-moving-trans. The similarities between Manual-fixed and Manual-moving-trans were quantified by the geometric index, including Dice similarity coefficient (DSC) and mean distance to agreement (MDA), which was set as the ground truth (GT) in this study. The same analysis was applied to

Auto-fixed and Auto-moving-trans, obtaining the value of DSC and MDA. Finally, it was compared with the GT to verify its reliability. The paired t-test was performed for statistical analysis. A p-value less than 0.05 was statistically significant. **Results:** For the rigid registration, Δ DSC for all OARs were smaller than 0.012%, and all Δ MDA less than 0.2mm. There were no significant differences between GT and evaluation tool. In the deformable case, Δ DSC for all OARs were within 0.02%, and Δ MDA were always less than 0.09mm. There were no significant except for the spinal cord and the mandible. **Conclusion:** The proposed deep learning-based evaluation tool could substitute the conventional evaluation method with the quantitative measurement, reflecting the accuracy of registration stably and efficiently in the interested region.

PO-GePV-M-251

A Generalizable Target Contour Validation Method Using Radiomics Based Regression Analysis

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Purpose: To analyze the changing trend of radiomics features of the targeted volumes, especially those with fuzzy edges, on the boundary area in CT images of nasopharyngeal carcinoma patients, and to explore this approach as a method to evaluate the quality of automatic or manual delineation of GTVs and organ at risks (OARs) with low contrast boundaries. **Methods:** Contours of GTVs and OARs in CT images of nasopharyngeal carcinoma patients were expanded and retracted by 1 mm, 2 mm and 3 mm respectively. Next, extract the radiomics features of the original image, the expanded images and the shrunk images. Then, the difference of adjacent features is calculated to get the feature differences. Regression model is built on features extracted using decision tree algorithm. The error of the model in the test set is calculated and analyzed. Performance was assessed on a separate test dataset. **Results:** We conducted sufficient experiments on 2 datasets of nasopharyngeal cancer patients. One set of the data was used as the ground truth for model training, and the other data set was used for model testing. We used the mean square error as the evaluation index of the model, and the error is 0.02. We compared the results of the model prediction to doctor's judgment, and found that our model can distinguish and gives evaluation on these situations very well: incomplete delineation of the area, delineation area is too large, and delineation is of good quality. **Conclusion:** We found that there are changing trend of radiomics features in the boundary area of targeted volumes with fuzzy edges in CT images of nasopharyngeal carcinoma patients. By exploring this characteristic change using regression models, we can evaluate the quality of the target area delineation automatically to accelerate quality assurance procedure in clinical practice.

PO-GePV-M-252

Monte Carlo Simulation of Varian Linac Using PRIMO

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Purpose: PRIMO is a new Monte-Carlo program and has a user-friendly graphical interface. A commissioned PRIMO program can be used in quality assurance. **Methods:** A VitalBeam with 6X and 6X FFF equipped with the 120 Millennium MLC was simulated by PRIMO. We adjusted four parameters(initial energy, energy full width at half maximum(FWHM), focal spot FWHM, and beam divergence) to match the measurements in the water-tank. PTW BEAMSCANTM water-tank and Semiflex 3D 0.07cm³ ion-chamber were used in the measurement. Percentage depth dose(PDD) and off-axis ratio(OAR) were evaluated with gamma passing rate. PDDs were matched at different widths of standard square fields(3cm, 4cm, 6cm, 8cm, 10cm, 15cm, 20cm, 30cm and 40cm) in central axis. OARs were matched at five depths(1.5cm, 5cm, 10cm, 20cm and 30cm) in cross-field direction. Dose leaf gap (DLG) was measured by an electronic portal imaging device(EPID) using a sweeping gap method with SDD=105cm. **Results:** Gamma passing rate of PDD and OAR showed a high level of agreement between Monte Carlo calculations and measurements. For the criterion of 2%/2mm, 1%/2mm, and 1%/1mm, the gamma passing rate(GIPR) of 6X PDD were 99.33%~100%, 99%~100%, 99%~100%, respectively. The GIPR of 6X FFF PDD were 99.33%~100%, 98.99%~99.66%, 97.64~98.99, respectively. The GIPR of 6X OAR were 96.4%~100%, 90.99%~100%, 85.12%~98.62%, respectively. The GIPR of 6X FFF OAR were 95.15%~100%, 89.32%~100%, and 87.02%~99.74%, respectively. The calculated DLG matched well with the measured DLG(6X: 1.36mm vs. 1.45mm; 6X FFF: 1.05mm vs 1.35mm, calculation vs measurement). The transmission factors were also similar(6X: 1.25% vs 1.39%; 6X FFF: 0.8% vs 1.15%, calculation vs measurement). **Conclusion:** The calculated PDD, OAR, and DLG were all in good agreement with measurements. As an independent and accurate Monte Carlo tool, PRIMO can be used in beam configuration, quality assurance, and treatment plan verification.

PO-GePV-M-253

Translating Predictive Models Into Treatment Planning Guidelines for Use in Clinical Practice: The Example of the Cardiopulmonary Dose Model for Overall Survival in RTOG 0617

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Purpose: New knowledge on normal tissue response requires updated NTCP models. Based on accumulated experience translating such models into treatment planning optimization goals, this study derived evidence-based dose-volume criteria from the recently published RTOG 0617 overall survival (OS) model. **Methods:** The published model included Atria D45%, Mean Lung Dose (MLD), Pericardium MOH55%, and Ventricles MOH55%. This study focused on reducing Pericardium MOH55% since MLD is already constrained internally (MLD<20Gy), Atria and Ventricles are included in the Pericardium, and the strongest association with OS was observed with Pericardium dose. Objective function input in commercial treatment planning systems is limited to Dx and Vx. Hence, MOH55% was most strongly represented by a combination of D15 and pericardium mean dose (PMD, Spearman's rho: 0.93, 0.99). These two variables were also among the best OS predictors in RTOG 0617. Dose-volume goals were D15<6Gy and PMD<5Gy, which represented low risk in the OS model. Nine of 105 randomly selected internal patients previously treated to 60Gy in 30 fractions with concurrent chemo-radiation and durvalumab were re-planned incorporating D15 and PMD goals using standard planning practice. Pericardium was generated from our open-source deep-learning algorithm. **Results:** Compared to the original plans, D15 was on average reduced from 43Gy to 29Gy; PMD was reduced from 20Gy to 13Gy; the normalized relative reduction ranged between 6-72% and 7-64%, respectively. Only 2/9 patients met D15<6Gy and PMD<5Gy; the majority (7/9 patients) were reduced to an intermediate-moderate risk (D15<49Gy and PMD<20Gy). Pericardium sparing was less in patients with a larger (>8%) Pericardium-PTV overlap; median relative reduction: 18% vs. 58% for D15 and 20% vs. 48% for PMD. **Conclusion:** This study has demonstrated a practical treatment planning-based approach to translate a published NTCP model that included multiple and new organs and novel dose parameterizations to form dose-volume criteria for use in treatment planning.

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Image Quality Assessment of Low CT Dose for Foreign Bodies Detection

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Purpose: To develop a suitable image quality CT protocol to detect the foreign bodies (FBs) in the respiratory tract of pediatrics with both a minimum patient radiation dose and an equivalent radiation risk to a conventional radiograph chest exam. **Methods:** CT Data were collected on a Siemens Flash. The

modification to the protocol included setting the reference kVp to 80, which lowered the CTDI value by 58%, the reference mAs to 9 mAs which reduced the CTDI by an additional 90%, using X-CARE and Adaptive Dose Shield, SAFIRE was set to order 4. Twenty CT exams using the modified CT protocol were performed with different FBs distributions in a five years old size anthropomorphic phantom (LUNGBODY) shown in Figure (1). The FBs included small size radiopaque bodies such as fish bones, peanuts, LEGO toys, etc. The exams were then presented randomly to the radiologist team to interpret the images and report the results. The GE QC phantom (Figure a) was used to measure the modulation transfer function (MTF). The ACR CT phantom (Figure 2) used to measure the (CNR). Image quality metric (Q) is calculated to characterize and to compare the different CT protocols in terms of dose, spatial resolution, and noise using: $Q = c \cdot (p(10\%)^4) / (\sigma \cdot D)$. **Results:** The CNR is reduced by 20% due to the decrease in mAs. The images were reconstructed with kernel I30, the limiting spatial resolution at 10% MTF was improved. The Q-factor was increased, indicating improvement in image quality, due to the decrease in the CTDI and the increase in the MTF. The results are summarized in table(1) **Conclusion:** Low-dose CT protocols can be very reliable for FB detection with high image quality and low patient dose, especially in pediatric patients where the aspiration of FBs is a frequent and severe cause of respiratory problems in children.

PO-GePV-M-255

Ventilation Measurements from Free-Breathing CT

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Purpose: To develop methods for measuring free-breathing ventilation using free-breathing CT scans **Methods:** A multi-scan free-breathing CT protocol was used to scan ten patients. The CT scans were acquired with a simultaneous breathing surrogate (abdominal pneumatic bellows). To reduce calculation time, the image registration was conducted on regions-of-interest (ROIs) from the upper right lungs. The ROIs were first registered using a published registration technique (pTV), which was followed up with a secondary registration that used a two-term objective function; a ventilation-adjusted Hounsfield Unit difference and a conservation-of-mass term labeled $\Delta\Gamma$ that denoted the difference between the deformation Jacobian and the tissue density ratio. The ventilations were calculated voxel-by-voxel as the slope of a first-order fit of the Jacobian as a function of the breathing amplitude. **Results:** The ventilation patterns showed different patterns and magnitudes. The mean ventilations calculated from the DVFs of the pTV and secondary registration were nearly identical, but the standard deviations of the voxel-to-voxel differences were approximately 0.1. The mean of the 90th percentile values of $\Delta\Gamma$ were reduced from 0.153 to 0.079, implying first that the secondary registration improved the conservation-of-mass criterion by almost 50% and that on average the correspondence between the Jacobian and density ratios were less than 0.1. This improvement occurred in spite of the average of the 90th percentile changes in the DVF magnitudes being only 0.58 mm. **Conclusion:** This work introduces the use of multiple free-breathing CT scans for free-breathing ventilation measurements. The approach has some benefits over the traditional use of 4DCT or breath-hold scans, including not having sorting artifacts. The benefits over breath-hold scans include the relatively small motion induced by quiet respiration versus deep-inspiration breath hold and the potential for characterizing dynamic breathing processes that disappear during breath hold.

PO-GePV-M-256

Investigation of Light-Transparent Bolus for Potential Breast SGRT Applications

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Purpose: Standard light-opaque bolus is suboptimal for radiotherapy (RT) patient setup because it obscures surface skin marks. ClearSight bolus is a transparent polymer gel bolus designed for enhanced patient setup with surface visibility. We evaluated the clinical characteristics of ClearSight bolus to determine its suitability for breast SGRT applications. **Methods:** Room-light transparency was evaluated by placing bolus (30 cm x 30 cm x 3 mm) on an AlignRT calibration plate. The visibility, deformation, and sharpness of the black blobs were examined. Radiation attenuation was measured by placing bolus on a 15-cm stack of solid water, with ion chamber at d5 for 6x, 6xFFF, 15x, d3 for 6E, 9E, and d4 for 12E, 16E, and 20E. The readings were compared with those obtained from an equivalent thickness of solid water. Bolus-to-skin conformity was evaluated by placing bolus over surfaces with various curvatures and examining the air gap between the two. Effects on 3D optical surface imaging tracking were assessed by placing bolus on various surfaces and measuring 6D shifts using AlignRT Advance. **Results:** ClearSight bolus had good room-light transparency. The black blobs and crosshair on the calibration board were easily visible through the bolus. The mean attenuation ratio between bolus and solid water was 1.0010 ± 0.0006 for photon energies and 1.0499 ± 0.0682 for electron energies. Bolus-to-skin conformity was good for flat and cylindrical surfaces, but inadequate for hemispherical surfaces. ClearSight bolus was not entirely transparent to the structured light from AlignRT Advance: it induced a modest translational shift (range -1.0 to 1.5mm) and rotational shift (range -1.5° to 1.5°) depending on surface topography. **Conclusion:** ClearSight bolus is suitable for breast SGRT applications. When using AlignRT Advance, capturing a new reference surface is necessary after bolus placement and, for some patients, larger than usual bolus may be required.

PO-GePV-M-257

A Deep Learning-Based Physician-Specific OAR Segmentation Framework for Radiotherapy Treatment Planning

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Purpose: To assist the organ contouring in current treatment planning procedures, we developed a deep learning-based framework to automatize the whole contouring process contributing to the clinical practice. **Methods:** This framework was designed for the auto-segmentation of organs at risk (OARs) in head and neck, and pelvis sites, which consisted of a pre-processing module, a deep learning-based segmentation module, and a post-processing module. In the pre-processing module, the original image data was extracted from the clinical DICOM Image file, and then processed for the segmentation network learning. A U-Net like network was designed in the segmentation module, in which three input branches for three consecutive image slices were designed to consider the contour consistency among adjacent image slices, and a deep supervision strategy was integrated to speed up the convergence of the network. The segmentation results would be converted to a DCIOM Structure file in the post-processing module for treatment planning. Moreover, this framework is being developed toward physician-specific to comply with the contour style of each radiation oncologist separately. **Results:** The organ-averaged Dice similarity coefficient (DSC) performance of this auto-segmentation framework is 0.71 and 0.74 for 16 head and neck and 11 pelvis OARs, respectively. Once a clinical DICOM Image data is input to the framework, a corresponding DICOM Structure file would be automatically generated containing the OAR segmentation results. The overall operation takes about 1-2 min per case without the need for additional manual interference. **Conclusion:** A deep learning-based framework was developed for the auto-segmentation of head and neck, and pelvis OARs with promising contouring accuracy and operation time. This framework is ready to be applied in our clinical practice to improve the efficiency of current treatment planning procedures.

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PO-GePV-M-258

Decision On SBRT, regular Or Less Fractions RT with a Simple TCP and EQD2 Calculation

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Purpose: The determination to use SBRT or fractionated RT for none-small lung cancer (NSCLC) are typically determined by tumor size and location without patient specific TCP and EQD2 calculations. A method is proposed here to quickly assess the patient cancer and tissue cell distributions, TCP and EQD2. **Methods:** Physician-defined CTV (or GTV) is decomposed into small ROIs, that are defined by intersections of radiopacity on an average of 4DCT scans and SUV of FDG PET images for an apparent cancer cell density. Nearby OARs are prioritized according to the patient conditions. Clinical SBRT plans with GTV < 22 cc and IMRT plans with CTVs > 500 cc are retrospectively analyzed. A unified multi-activation (UMA) model for cell survival of squamous cell lung carcinoma (SW1573, n=12, Do=1.1-Gy), adenocarcinoma of Lung (HX144, n=4.0, Do=2.2-Gy), (normal epithelial cell lines, n = 9.0, Do=1.1-Gy) and tumor cell density of 10⁵/gram are used for TCP and EQD2 calculations. **Results:** Plans of standard SBRT in five 10-Gy fractions have approximately 100%TCP for most density levels of the cancer cells. A low density volume at the periphery of an adenocarcinoma with a GTV of 10 cc has the lowest 98%TCP and a high but acceptable EQD2 of 90-Gy to a 25-cc of the surrounding lung. TCP and EQD2 to the treatment in options of: (1) 1.8-Gyx37, (2) 3-Gyx22, or (3) 13.2-Gyx5 of a 750-cc squamous cell lung cancer, 0.1-cc spinal cord, 25-cc lungs, 5-cc heart, and 2-cc esophagus are (1) 33%, 36, 65, 40, 58-Gy, (2) 98%, 37, 99, 43, 75-Gy, and (3) 100%, 46, 227,73, 248-Gy, respectively, favor of the option (2). **Conclusion:** The new method can quickly calculate TCP and EQD2 with apparent cell density and dose distribution which may help clinicians make accurate treatment decisions.

PO-GePV-M-259

Penalized Entropy Loss Function with High Quality Uncertainty Estimation for Pneumonia Diagnosis in Pediatric Patients

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Purpose: In the uncertainty-informed decision system, models may make wrong predictions with high certainty which leads to omission of false predictions. Therefore, we propose to set up a model which could make correct prediction with low uncertainty and wrong prediction with high uncertainty. **Methods:** From accuracy and certainty perspective, we categorized the predictions into four kinds, correct and certain (CC), correct and uncertain (CU), wrong and certain (WC), wrong and uncertain (WU). The desirable model is that almost all predictions belong to CC or WU and fewer WC cases. Therefore, we integrated uncertainty estimation into training phase and proposed penalized entropy (PE) loss. PE has two terms, accuracy related and certainty related terms. Accuracy related term is cross entropy. Certainty related term is penalizing WC predictions and is the multiplication of negative likelihood (NLL) and certainty, where certainty is calculated by Monte Carlo dropout. High NLL and high certainty will result in high certainty-related loss which means wrong and certain decision. A hyper-parameter is used for relative weighting of certainty-related term with respect to accuracy-related term. **Results:** Validating on the Chest X-ray dataset, we achieved higher quality of uncertainty estimation that accuracy vs. uncertainty (AvU) is 0.89 which outperformed state-of-the-art losses. Furthermore, PE can produce more accurate predictions with accuracy of 0.90 and AUC of 0.96. **Conclusion:** In this study, we proposed a penalized entropy loss function in a prediction model for pneumonia diagnosis in pediatric patients. This model achieves better uncertainty estimation that makes correct predictions with high certainty, wrong predictions with high uncertainty, especially avoiding wrong but certain results. Compared to state-of-the-art loss functions, this loss can not only give higher quality uncertainty estimation but also more accurate results. Therefore, model trained with this novel loss function can be used in clinical for precise and comprehensive diagnosis.

PO-GePV-M-260

LiteUnet : A Lightweight 2DUnet Combine GRU for Cardiac Substructures Segmentation

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Purpose: To propose a lightweight and universality two-dimensional convolution neural network for whole heart segmentation of CT data on few-shot learning. And explore what kind of deep learning model architecture is more suitable for CT data that is three-dimensional and less training data. **Methods:** Every case in our CT data is 512*512*n (n for CT slice, depending on the actual case). Firstly, we use 2D UNet to roughly cut out the region of the heart, then we can obtain data that is 192*192*48 and contains the entire heart. Secondly, we use 2D UNet combine bidirectional-GRU and attention mechanism on 3D heart CT to segment out 14 cardiac substructures. **Results:** Our model has few parameters and is easy to train, but performance is equal to or even exceed some models of three-dimensional convolution. **Conclusion:** This work indicates the advantage of the model of two-dimensional convolution combine some time-series processing methods like RNN when we use three-dimensional CT dataset on few-shot learning.

PO-GePV-M-261

Predicting Lymph Node Metastasis Through Automated and Reliable Multi-Objective Model in Head & Neck Cancer

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Purpose: Accurate diagnosis of lymph node metastasis (LNM) is critical in treatment management for patients with head & neck (H&N) cancer. We aim to develop an automated and reliable multi-objective model (ARMO) for reliable LNM prediction. **Methods:** Totally 129 surgical H&N cancer patients with pathology-confirmed LNs status are used and there are 130 malignant nodes and 413 benign nodes. Two hundred and fifty-six hand-crafted feature are extracted from PET and CT images, respectively. To obtain balanced outcome between sensitivity and specificity, a multi-objective model is developed to obtain Pareto-optimal model set in the training stage. In addition, to obtain calibrated confidence, which aims at maximizing the output probability with correct prediction label and minimizing the probability with incorrect prediction label, a weight that measures the relative importance of each Pareto-optimal model is calculated in training stage. Meanwhile, the individual reliability which measures how reliable of each Pareto-optimal model output is estimated in testing stage. Furthermore, an analytic evidential reasoning rule is used to combine the output of each Pareto-optimal model. Finally, the overall reliability of the model output is obtained by averaging the individual reliability to estimate the model uncertainty, which can be used to alter clinicians on model outputs with high uncertainty. Accuracy (ACC), AUC, sensitivity (SEN) and specificity (SPE) are used to measure the model performance, whilst expected calibration error (ECE) and maximum calibration error (MCE) are used to measure the confidence of calibration, where both ECE and MCE of a perfectly calibrated model are 0. **Results:** The mean $\pm$ standard deviation of AUC, ACC, SEN and SPE for ARMO are 0.97 $\pm$ 0.00, 0.93 $\pm$ 0.00, 0.88 $\pm$ 0.01, 0.94 $\pm$ 0.00, respectively. Meanwhile, the ECE is 0.0085 $\pm$ 0.0018 and MCE is 0.4483 $\pm$ 0.0336. **Conclusion:** We developed a reliable LNM prediction model, which can not only obtain promising performance, but also better calibrated confidence and uncertainty estimation.

PO-GePV-M-262

Development of Markerless Pancreatic Tumor Tracking Realized by Deep Learning

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Purpose: To realize markerless pancreatic tumor tracking by using a deep learning model to detect the contour of the clinical target volume (CTV) on the orthogonal X-ray images acquired by the on-board imagers of the gimbaled linac system. **Methods:** The input of the model was either the digitally reconstructed radiograph (DRR) image or the X-ray image acquired by the on-board imagers. The output was the image with the contour of the CTV. After the four-dimensional computer tomography was acquired, a data augmentation procedure contains interpolation between respiratory phases and tilt of gantry and ring angle was conducted to generated sufficient images to train the deep learning model. For one patient, a total of 2500 DRR images were generated for one X-ray tube angle. Two-thousand images were used to fine-tune the pre-trained model and the remaining images were used to evaluate the accuracy of the model. The contour of CTV was extracted from CTV-only DRR and then set as the ground truth of contour prediction. The dice similarity coefficient (DSC) and the centroid difference of contour at the isocenter level were used to evaluate the accuracy of prediction. **Results:** We initially performed markerless pancreatic tumor contour prediction on the DRR images of one patient from orthogonal X-ray tube angles, 45° and 135°. The mean DSC between prediction and ground truth were 0.91 and 0.82 for 45° and 135°, respectively. The mean centroid differences were 2.10 mm and 5.25 mm, respectively. Running on a single NVIDIA GV100 graphics processing unit, the mean calculation time on an image was 172 ms. **Conclusion:** This work demonstrates the success of the current workflow and adaption of the deep learning model on markerless tumor tracking for pancreatic cancer. The prediction accuracy and calculation time also indicate a bright future in clinical practice.

PO-GePV-M-263

Using New Feature Extraction Framework to Predict Heart Toxicity in Breast Cancer

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Purpose: Radiation-induced cardiac toxicity such as acute coronary event (ACE) is a key issue of breast radiotherapy. As it has been mostly discussed in mean heart dose, however, there are few studies about cardiac substructures and ACE after radiotherapy. This work proposes a new heart toxicity prediction model by a new feature extraction and processing framework with cardiac sub-structure and dose distribution. **Methods:** The patient cohort consists of 29 toxicity and 16 non-toxicity cases after breast radiotherapy. A recent study demonstrated that the feature extraction from segmenting target volume followed by feature selection produced better prediction accuracy in survival prediction after lung SBRT than radiomics and conventional deep neural network (convolution/fully-connected layers). We applied the framework for toxicity prediction. It compared three types of prediction models: a) radiomics, b) conventional deep neural network, and c) feature extraction and processing model with CT image and dose information as input, and heart segmentation as output. The output of the auto-segmentation model in (c) was varied from entire heart to 6 sub-structures and 3 sub-structures (left-ventricle, right-ventricle, and left-anterior descending artery). The relationship between the features and prediction was visualized by a method, called Grad-CAM. The network training and testing was conducted with 4-fold cross-validation. **Results:** The conventional deep neural network (50-60% accuracy) performed worst. The proposed model led to higher prediction accuracy (over 78%) than radiomics (69%). Impressively, the predictive model with 6 and 3 sub-structures (81%) outperformed that with entire heart structure (78%). The model with 3 sub-structures showed greater specificity and sensitivity than that with 6-substructures, though both having the same accuracy. When visualized by Grad-CAM, the hot-spots were generated around left-ventricle region in most cases. **Conclusion:** Our proposed prediction model characterized by 1) new feature extraction/processing process and 2) cardiac sub-structure dose/structure information enhanced the prediction accuracy.

PO-GePV-M-264

Local Temperature as a Surrogate of Inflammatory State Induced by Skin Irritant

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Purpose: To visualize changes in microcirculation of irritated skin through Eulerian amplification of high-resolution monochrome videos correlated to known detected temperature differences by a thermal camera as a control. **Methods:** MATLAB code was written to implement Eulerian amplification, a technique developed by the M.I.T. Media Lab, to exaggerate changes in skin color to make them more discernable in video sequences. In an investigator self-monitoring feasibility trial, a monochrome machine vision camera (model Edmund Optics EO-2323) with detector resolution of 1920 x 1200 pixels and frame rate of 180 fps was used to collect high-resolution video of a region of interest (ROI) undergoing circulatory changes induced by a cold compress. Heart rate was extracted from subtle color changes, then the video was amplified by a factor of 100 with a bandwidth of 0.1 Hz encompassing the heart rate. A thermal camera (model FLIR A325sc) with detector resolution of 320 x 240 pixels was used to collect static thermal images to monitor temperature changes over time to act as a standard against which the visualization of changes in blood flow by the monochrome camera could be correlated. **Results:** Following removal of a cold compress, blood flow increased to the region of interest as a result of vasodilation and was noted by the increase in amplitude of non-visible skin color changes in the magnified video as time increased. These increases in amplitude were correlated to increases in temperature observed by the thermal camera. **Conclusion:** The thermal camera temperature differences were mapped to blood flow changes and demonstrated the monochrome camera's ability to detect non-visible changes in microcirculation based on amplification of subtle skin color changes. This technique was evaluated for application in radiotherapy to assess skin changes throughout treatment and detect erythema before clinical manifestation in patients at risk for skin reactions.

AAPM ePoster Library

Therapy General ePoster Viewing

PO-GePV-T-01

A Nomogram to Estimate Required Ci-S in Prostate HDR Brachytherapy

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Purpose: To improve patient safety by providing an independent mathematical model to estimate Curie seconds in prostate HDR brachytherapy. **Methods:** HDR prostate brachytherapy, particularly for boost treatment with a single fraction of 15Gy, has become a regular procedure in recent years. For a given source activity, the time needed to deliver the prescribed dose is correlated with the target volume (TV). We calculate the Curie seconds (Ci-s) for each patient plan by multiplying the Iridium-192 source activity in Curies and the total treatment time in seconds. We create a nomogram (Ci-s vs. TV) based on a training dataset of 90 HDR prostate cases with a treatment volume of 15cc to 70cc. Three different curve-fitting models are used in this study (power, polynomial, and logarithmic). We test the models in a validation dataset of 11 additional patients from a different institution and different HDR afterloader vendor. **Results:** The coefficient of determination (R²) for the models are 90.45% (power), 90.29% (Polynomial), and 89.68% (logarithmic). The average absolute percentage errors for 11 patients are 4.4% (power), 4.6% (polynomial), and 9.2% (logarithmic). The power curve fit shows the smallest estimation error on 11 patients with a maximum error of 12.4% for 6.84CC and a minimum error of 0.5% for 14.8CC TV. The power curve model for the estimation of Ci-s is

$704.39x(TV)^{0.5342}$. The polynomial model parameters are $A2=-0.8245$, $A1=133.66$, $A0=1115.7$, and the logarithmic model is $2433.2x\ln(TV)-3839$. **Conclusion:** This method can be used clinically as a sanity check by physicists. The power curve fitting shows the best performance for typical prostate volumes (25-40cc). More extensive clinical tests will finalize the determination of the model for clinical use. Future work will incorporate more variables into the training data to improve the predictive power. We are also going to increase more training data for TV less than 10cc.

PO-GePV-T-02

Independent Determination of Absorbed Dose to Water Depth-Dose Curves for Ru-106 Ophthalmic Plaques Traceable to the Primary Standards of External Beam Radiotherapy

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Purpose: To propose an approach for independent determination of the absorbed dose to water ($D(water)$) depth-dose curves traceable to the external beam primary standards of absorbed dose to water for the dosimetry of ophthalmic ^{106}Ru plaques used in brachytherapy. **Methods:** A microSilicon diode was cross-calibrated against a reference air ionization chamber in terms of the $D(water)$ in a 6 MeV external electron beam. The diode and a high precision setup equipment was used to determine the $D(water)$ depth-dose curves for ^{106}Ru CCB-type plaques manufactured in 2018 and 2019. Detector blueprints were available and a Monte Carlo (MC) code system PENELOPE-2018 was used to correct for the absorbed-dose energy dependence of the detector between calibration and measurement beam qualities (from 2 to 10 mm distance from the plaque surface). The results were compared with the $D(water)$ depth-dose curves calculated with MC in the absence of the detector and with those determined by the vendor with a plastic scintillator with other traceability. Further analysis was made to determine possible reasons of the discrepancies. **Results:** Differences between experimentally determined $D(water)$ depth-dose curves using the microSilicon diode and those provided by the vendor were from -5 % at the reference depth of 2 mm to nearly 20 % at 10 mm depth. The MC-calculated $D(water)$ depth-dose curves in the absence of the detector differed systematically from the vendor values but agreed within their provided uncertainties with the values obtained using the microSilicon detector. **Conclusion:** We show that a detector calibration in external beam allows for accurate and independent dosimetry of ^{106}Ru CCB plaques with lower uncertainties than following an existing method of plaque calibration.

PO-GePV-T-03

HDR Ring Applicator Commissioning: Eliminating the Need for Film-Processors to Generate Autoradiographs

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Purpose: Commissioning of HDR applicators requires dwell position verification. This is particularly challenging for ring applicators, which are known to have unreliable offsets of the distal dwell due to the sharp curvature of the ring channel. Historically, this verification was performed via autoradiograph: first by imaging the applicator with an inserted dummy source of nominal dwell positions and then by delivering an HDR plan at the same expected dwell positions to radiographic film. This required the use of film processors, which have since become largely extinct in radiation oncology departments. Therefore, this work aims to investigate a novel and accessible method to generate an autoradiograph with a linac in order to commission a ring applicator. **Methods:** Varian 450 plastic and titanium ring applicators were affixed to Gafchromic EBT3 radiochromic film and secured using a clamping device. In order to determine the optimal energy/MU settings, it was irradiated with 6, 12, and 18 MeV electrons at MU=100, 500, 1000 and 1500 using a 10x10 cone and SSD=106cm. The applicator/film geometry was then carefully transferred for HDR plan delivery with 10s per dwell for plastic and 15s for titanium, respectively. Films were qualitatively assessed for resolution and quantitatively analyzed for contrast. **Results:** Plastic and titanium applicator image resolution improved with increasing energy and contrast increased with MU. For the titanium applicator contrast/MU plateaued above 1000MU and showed no further improvement (1000MU: 0.07, 1500MU: 0.07). **Conclusion:** The method presented demonstrates a novel and efficient approach to generate an autoradiograph, thereby enabling accurate dwell position verification during applicator commissioning. It was evident that appropriate energy/MU selection will vary with applicator material/thickness and is the subject of future investigation.

PO-GePV-T-04

Quantifying Risk Associated with Physics Errors in HDR Prostate Brachytherapy Using Automated Simulations

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Purpose: To quantitatively evaluate clinical significance of potential HDR prostate brachytherapy (HDRPB) physics errors through automated simulations. **Methods:** Three idealized HDRPB geometries with rectum, urethra and different sized prostate contours were generated in BrachyVision. High and low dwell-time modulated plans using clinical constraints were created per geometry. Based on FMEA analysis, four high-impact errors applicable to HDRPB were selected for quantitative analysis. Plans were exported, and error simulations and dose calculations were performed using MATLAB and Python routines: single ($N=1,722$) and systematic ($N=126$) catheter shifts (cranio-caudal; -1cm:1cm); single catheter digitization errors (tip and connector needle-tips displaced independently in random directions; 0.1cm:0.5cm; $N=44,318$); and swaps (two catheters swapped during digitization or connection; $N=528$). The deviations due to each error on Prostate_D_90%, Urethra_D_20% and Rectum_D_1cc were analyzed using two thresholds considered for clinical significance: 5-20% (possible clinical impact [PCI]) and >20% (likely clinical impact [LCI]). **Results:** Single catheter shifts resulted in a PCI for <6% of simulations for small geometry with large shifts (≥ 0.8 cm), and no LCI. Systematic catheter shifts had the largest clinical impact on Prostate_D_90% with 100% of simulations resulting in a PCI for shifts ≥ 0.5 cm and LCI for shifts ≥ 0.8 cm. Systematic shifts had no clinical impact on Urethra_D_20%; for Rectum_D_1cc PCI occurred in 25% of shifts ≥ 0.5 cm with no LCI. Digitization errors resulted in PCI for all structures but LCIs only for Urethra_D_20cc. As prostate size and modulation decreased, impact on prostate and rectum increased while urethra decreased. Swaps had PCI 14%, 21% and 9% of the time for Prostate_D_90%, Urethra_D_20% and Rectum_D_1cc, respectively. **Conclusion:** Prostate_D_90% and Rectum_D_1cc had the highest chance of PCI from systematic catheter shifts, while Urethra_D_20% was from swaps. LCI was uncommon for all combinations, besides systematic catheter shift's impact on Prostate_D_90%. Digitization errors most consistently caused PCI for all scenarios.

PO-GePV-T-05

Error Detection in HDR Prostate Brachytherapy with Plan-Specific Optimization of In-Vivo Dosimetry (IVD) Parameters

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Purpose: To demonstrate that plan-specific optimization of IVD detector position increases the specificity and sensitivity of error detection. **Methods:** 19 HDR prostate plans were used to generate virtual simulation of treatment delivery with virtual errors introduced in the plan and virtual IVD monitoring using 4 detectors with realistic characteristics (noise, etc). An algorithm was created to optimize detector position and signal threshold based on plan. Three schemes were investigated: using the algorithm (optimized), using a standard positioning [unoptimized], and positioned at random [random]. A comparison of theoretical use of different detectors (Plastic scintillators, MOSFET, and semiconductor diodes) was performed. **Results:** For plastic scintillator detectors, sensitivity and specificity for the optimized scheme (average $\pm$ SD) were (0.88 ± 0.06) , and (0.98 ± 0.01) respectively, in comparison with (0.18 ± 0.06) , $p < 0.01$, T-test), and (0.97 ± 0.07) , $p > 0.05$) for the unoptimized and $((0.39 \pm 0.00)$, $p < 0.01$, T-test), and (0.94 ± 0.01) , $p > 0.05$) for the random. For the optimized scheme, compared

to plastic scintillator detectors, sensitivity (specificity) were 0.88 ± 0.06 $p > 0.05$ (0.98 ± 0.01 $p < 0.01$), 0.88 ± 0.07 $p > 0.05$ (0.95 ± 0.02 , $p < 0.01$), and 0.87 ± 0.08 $p > 0.05$ (0.86 ± 0.06 , $p < 0.01$) respectively for plastic scintillators, semiconductor diodes, and mosfet. **Conclusion:** We demonstrated that our algorithm improves sensitivity and specificity using plan-specific IVD optimization when compared with unoptimized and random groups. Permitting the tailoring of IVD at the end of the planning process before treatment, minimizing disruption from false positives and maintaining an acceptable true positive detection rate. Our simulations show that high sensitivity and specificity can be achieved with a range of detector types, although practical and logistical considerations were not considered in this analysis.

PO-GePV-T-06

Performance Evaluation of a Novel Quality Assurance Device for High Dose-Rate Brachytherapy

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Purpose: A novel system composed of a monocrystalline scintillator and optical camera has been developed for Quality Assurance (QA) on high dose-rate (HDR) brachytherapy afterloaders. HDR QA requires adherence to US-NRC regulations, and includes high definition archival image documentation of the source positioning. HDR QA is required for (1) daily use of the system, (2) periodic source exchanges, (3) validation of applicators, and new developments call for (4) patient specific QA. The new proto-type commercial system uses radioluminescence imaging to determine position and dwell-time accuracy of the source and can provide patient specific QA for complex deliveries. **Methods:** The scintillator and optical camera system (OriQA, Luca Medical Systems, Palo Alto, CA) with high spatial and temporal resolution was tested on a Varian GammaMediX HDR afterloader (Palo Alto, CA). The evaluations included position and timer accuracy as well as cumulative activity measurement over a range of doses typically prescribed for gyn brachytherapy. The accuracy and standard deviation were then evaluated for dwell time and position, and cumulative activity for various configurations. **Results:** The system readily acquires and analyzes images of the source position in accordance with standard practice for HDR daily QA. Several dwell positions over a 10 cm range with varying dwell times between 0.3 - 20 seconds were employed. The position error of the QA device was identified as $-0.01 (\pm 0.04)$ centimeters, the time error was $-0.08 (\pm 0.11)$ seconds, and the error in cumulative activity was $< 1\%$. **Conclusion:** A new, novel, and practical HDR QA system has been devised that includes a monocrystalline scintillator and optical camera system. The radioluminescence technology of the device is highly accurate and precise, and with its low-cost camera based optical imaging system provides QA documentation in accordance with NRC regulations. The potential for applying this technology to patient specific QA is also investigated.

UC Davis has a research agreement with OriQA, Luca Medical Systems

PO-GePV-T-07

Commissioning of Lumen Markers for Needle Visualization on MRI Scans in HDR-Gyn Brachytherapy

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Purpose: Use of interstitial needles with pose a challenge during MRI-imaging due to patient/organ movement, lack of signal and due to diameter of sheathe being comparable to imaging slice thickness. Due to challenging catheter reconstructions on MRI, despite inferior soft-tissue contrast, CT scans are still used in addition to MRI for manual reconstruction of needles in image-guided brachytherapy. This workflow does not account for change in plastic needle position due to missing stylet during MRI and treatment. Flexi-needles are often in a different position due to changing tissue, organ forces and proximity to ovoids. To address these challenges, MRI Lumen markers were commissioned for clinical use to improve delineation of needles on MRI-scan for treatment planning in HDR gynecological-intracavitary implants with needles. **Methods:** MRI lumen markers™ (Theragenics Inc.) were inserted in the plastic needle sheaths in different media (water, soft-bolus material, in-house poured silicone and on solid-water slabs with soft bolus on top) to assess accuracy of visualization of tip of the needle and delineation of interstitial needles, alongside an empty flexi-needle. Images acquired using clinical protocols on 3T MRI scanner and 0.35T MR-Linac TRUFI (Viewray Inc) scans were compared to CT images of the needles with metal stylets for verification of the tip. **Results:** Use of soft-bolus (102 HU) and in-house poured silicone phantom (194 HU) were verified as suitable media for needle visualization study using CT and 0.35T MR linac. Needle marker tips on the TRUFI Images were localized to within ± 1 mm of the CT-scans. Needle without the marker was visualized within ± 1.3 mm. Results on the 3T MR scanner were inconclusive due to age/resue of markers warranting additional investigation. **Conclusion:** Needle markers are visible on 0.35MRI across multiple sequences, and image orientations facilitating treatment planning dosimetry directly on MR-Linac through accurate source path localization and reconstruction.

PO-GePV-T-08

Determination of Changes in Dosimetric Parameters with Co-60 and Ir-192 Source in High Dose Rate Brachytherapy of Cervical Carcinoma

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Purpose: To determine the differences in dosimetric parameters of cervical cancer patients treated with interstitial and intracavitary high dose rate brachytherapy using Co-60 and Ir-192 sources. **Methods:** CT-based treatment plans of 20 patients with cervical carcinoma treated with brachytherapy were retrospectively studied. Patients with interstitial implants were planned for 24 Gy in 4 fractions with inverse planning simulated annealing using Co-60 and Ir-192 source keeping same dose objectives. Patients of intracavitary brachytherapy (ICBT) were planned for 21 Gy in 3 fractions with both sources keeping the same dwell positions. The dosimetric parameters Co-60 and Ir-192-based plans were compared in terms of HR-CTV coverage, OAR doses, homogeneity index (HI), and conformity index (CI). **Results:** In interstitial brachytherapy, mean V100 values for Co-60 and Ir-192-based plans were $97.03 \pm 1.99\%$ and $96.42 \pm 2.25\%$ respectively. Average values of D2cc in the bladder for Co-60 and Ir-192-based plans were 4.30 ± 0.57 Gy and 4.27 ± 0.54 Gy respectively while in the rectum were 3.96 ± 0.43 Gy and 4.08 ± 0.49 Gy respectively. There were no significant differences between V100 and D2cc values from both sources. Ir-192 based plans were more homogeneous (HI= 0.68 ± 0.09) than the Co-60 plans (HI= 0.59 ± 0.10), however, the conformity was similar with both the sources. In ICBT plans, mean D90 values for Co-60 and Ir-192 were 4.86 ± 1.29 Gy and 4.94 ± 1.43 Gy respectively. Mean D2cc values in the bladder for Co-60 and Ir-192-based plans were 6.19 ± 2.72 Gy and 5.93 ± 2.44 Gy respectively while in the rectum were 4.17 ± 1.73 Gy and 4.29 ± 1.94 Gy respectively. We did not find any significant difference in the dosimetric parameters of Co-60 and Ir-192-based ICBT plans. **Conclusion:** There are no significant differences in dosimetric parameters and no clinical advantage of using Co-60 over Ir-192-based brachytherapy. However, when it comes to cost-management, Co-60 has a longer half-life, which reduces the cost of frequent source replacement and disposal.

PO-GePV-T-09

MRI-Based Organ Segmentation of Pelvis Using Deep Learning for Brachytherapy Treatments

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Purpose: In this study, we investigate the accuracy of deep learning-based contouring algorithm compared to manual contouring practice and how this

algorithm can help in our clinic to standardize consistent contouring. **Methods:** Over 500 T2- weighted MRI slices from 9 different cancer patients were selected for this study. The MRI slices and contour structures were exported for pre-processing to create masks. The mask is acquired from the MRI slice by detecting edges of OAR contour and by assigning pixel value of 1 inside the contour's boundary and 0 outside. For training, U-net based deep learning approach was implemented by using Pytorch open source machine learning framework. In training phase, the algorithm should make one complete pass through dataset to extract millions of trainable parameters. 30 epochs were used in our study and dice similarity coefficient was employed to quantify accuracy of prediction with respect to expected result which is also called ground truth **Results:** We have carried out this pilot study by only using 9 patients MRI data and only single GPU (graphics processing unit) computing power. The dice similarity coefficient for the bladder and rectum segmentation were around 69 %. Although accuracy of predicted location of said OARs are excellent, the algorithm is coming short on predicting whole contour in respective slice. We are expecting the similarity between predicted contours and ground truth contours to be over 95 % with adequate computing power and patient data which are essential in deep learning algorithms **Conclusion:** Deep learning approach has benefits for radiation oncology doctors by improving their time efficiency in labor-intensive contouring process. This, in turn, will lead to increased patient comfort and quality patient care. However more patient datasets are needed for training step to increase contouring accuracy to accepted level.

PO-GePV-T-10

Dose Optimization of Vaginal Cylinder: A New Planning Method for Patients with Endometrial Adenocarcinoma

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Purpose: To explore the relationship between applicator surface dose and 5 mm depth dose and to optimize both locations simultaneously for three most used cylinder sizes (2.5, 3.0 and 3.5 cm in diameter) in treating patients with endometrial adenocarcinoma. **Methods:** Total 216 plans were created (9 dose levels x 4 plans/level x 3 Cylinder size x 2 patients/size): if the prescription is on the surface, dose will be optimized on the surface alone or both. In the later plan, surface points will be given a higher priority (100 ~ 150%) than points at 5 mm lines (60% ~ 70%). The same strategy is applied to dose prescribed at 5 mm depth. We use VEGO TG-43 Volume Optimization (Varian Eclipse) for planning and single channel to treat patients. **Results:** A dose table between surface and 5 mm depth and its fifth order polynomial mapping function were established for each applicator size, so any prescribed dose at one site can find the prescription dose on the other for optimization at both locations. For dose prescribed at 5 mm, the maximum surface dose can be as high as 30% and 40% on the surface, and 5% and 12% high at 5 mm w/o surface dose optimization. For dose prescribed on the surface, the maximum doses are similar (10% ~ 15%) w/o 5 mm dose optimization, but the addition of 5 mm dose optimization can slightly improve the minimum dose (~5%) on both locations. **Conclusion:** 1. Proposed three dose tables and their mapping functions between surface dose and their corresponding 5mm depth doses for three different applicator sizes; 2. Proposed a new method to optimize doses at both locations simultaneously; 3. Reduce the maximum surface dose from 30 - 40% to 10% or so if changing prescribed dose from 5 mm to applicator surface.

PO-GePV-T-11

Uncertainties in MR-To-MR Image Registration for HDR Cervical Brachytherapy

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Purpose: To evaluate the dosimetric effects of uncertainties in shared DICOM coordinates registration between MR sequences used clinically in planning the HDR brachytherapy component for cervical cancer patients. **Methods:** Thirteen locally advanced cervical cancer patients treated with HDR intracavitary/interstitial boost were included in this study. The HDR treatment was performed using Varian titanium tandem and ovoid applicators with optional free-hand interstitial needles. The implants were stabilized with Alatus balloons. The imaging protocol consisted of a planning CT for applicator digitization, a T1 weighted isotropic spoiled GRE sequence for registration with planning CT, and T2 weighted sequences for target and organ at risk contouring. A custom workflow was created in MIM Maestro to break the inherent registration between the MR sequences and re-register the T1- and T2-weighted series using intensity-based rigid registration. The physician contours were transferred from T2 images to CT using the new registration. Clinically relevant equi-effective dose metrics for the clinical CTV_{HR}, CTV_{IR} (D₉₀, D₉₈) and bladder, rectum, sigmoid, bowel and vagina (D_{2.0} [cm]³) were then compared with metrics calculated on the new contours. **Results:** A total of 60 fractions were used in this study. The mean percent difference (±standard deviation) between clinical and reregistered contours were -0.60% ±1.49, 0.56% ±1.06, -0.49% ±1.71, -0.53% ±1.13 for CTV_{HR} D₉₀, CTV_{IR} D₉₀, CTV_{HR} D₉₈, CTV_{IR} D₉₈, respectively and -0.05% ±2.13, 0.61% ±2.85, 1.67% ±1.93, 1.76% ±2.04, and 1.75% ±2.10 for bladder, bowel, rectum, sigmoid, and vagina D_{2.0} [cm]³, respectively. **Conclusion:** As expected for the target, due to the stabilizing presence of the vaginal balloons there is minimal difference between the two registrations. However, organs at risk have more variability due to potential movement between scans. Further analysis will allow us to identify these specific cases and adjust the clinical flow to account for potential anatomical differences between MR scans.

PO-GePV-T-12

Dosimetric Evaluation of Rectal Hydrogel Spacer in Prostate HDR Treatments

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Purpose: An injectable rectal hydrogel spacer (RHS) is used in both external beam and HDR treatments of the prostate to create spatial separation between the target and the rectum. In a re-planning study, we evaluated the dosimetric impact of the presence of RHS on prostate HDR plan quality. **Methods:** Eighteen patients with RHS who recently underwent prostate HDR treatment, were re-planned with modified objectives in order to simulate the absence of the spacer. Patients with three types of treatments were included in this study: monotherapy (9 patients; 27Gy in 2 fractions), salvage therapy (7 patients; 24Gy in 2 fractions), and boost (2 patients; 15Gy in one fraction, following external beam treatment of 46Gy in 23 fractions). The clinical plans were dosimetrically compared to the post-treatment re-plans, which were optimized with modified goals aiming at sparing the urethra, bladder, and RHS (as a surrogate for rectum). The trends in target coverage as well as organ-at-risk sparing were statistically analyzed. **Results:** The analysis of 33 plan pairs showed that the presence of RHS significantly reduced the rectal volume receiving >75% of the prescription dose: from mean V75%=0.63cc without RHS to 0.02cc with RHS. Without RHS, the mean prostate coverage was reduced by 1.5% and the urethral volume receiving >125% of the prescription dose on average increased by 0.1cc (from 0.02cc to 0.12cc). No change in the bladder dose was observed. The plan heterogeneity index (V150%/V100%) was slightly worsened for the plans without RHS: from 0.35(5) to 0.36(4). **Conclusion:** This dosimetric study clearly demonstrated the benefits of injectable RHS in prostate HDR treatments. Excellent rectal sparing was achieved as well as improved target coverage and marginally better urethral sparing. Injectable RHS was shown to be indispensable in both salvage and boost HDR treatments, where organ-at-risk sparing is of critical importance.

PO-GePV-T-13

Commissioning of High Dose Rate Brachytherapy Surface Applicator System

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Purpose: A high-dose-rate brachytherapy surface applicator set has been commercially available for several years with sparse description in the literature. A thorough verification of the available dose distributions and dose output from all applicators as well as determining the optimal clinical settings for patient use

was necessary for clinical implementation. **Methods:** Measurements of dose distribution and absolute dose at depth under the applicators were made. Absolute dose was measured using a commercial diamond detector and radiochromic film. The diamond detector was calibrated for Ir-192 using a known dose in water. This calibration was used to measure the dose given in water at 1-5 mm depths for each surface applicator. Radiochromic film was used in plastic water at 3 mm depth to measure dose distributions and absolute dose of different dwell positions. **Results:** All applicator/insert combinations have a pronounced dip in the center of the dose distribution due to the metal channel entering the applicator perpendicular to the patient surface. There are two logical dwell position options, and for all applicator/insert combinations the 129 cm dwell position led to a smaller dip in the center of the distribution (generally 20-25% vs. 20-44% for 128.5 cm dwell position). Using the 129 cm dwell position, dose in water was measured for each applicator/insert combination and tabulated for use in patient calculation. Absolute dose measurements from the radiochromic film agreed with the diamond detector measurements with a mean difference of 4.5%. **Conclusion:** The skin applicator set is clinically useful though must be used with a specific dwell position to reduce the asymmetry of the dose distribution. Further recommendations for the use of the applicator set should be followed closely to assure optimal patient treatment, as there are multiple possibilities for asymmetric dose distribution in the center and periphery of the applicator.

PO-GePV-T-14

A Retrospective Dose Analysis of High-Dose Rate (HDR) Tandem and Ovoid and Interstitial Needle Cervical Brachytherapy Cases with a Model-Based Dose Calculation Algorithm (MBDCA)

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Purpose: To investigate and compare calculated dose in cervical cancer patients using Varian's model-based dose calculation algorithm Acuros 13.5, and standard Task Group 43 (TG-43) formalism. **Methods:** A set of locally advanced cervical cancer patients treated with MR based HDR brachytherapy using tandem and ovoid applicators and free-hand interstitial needles were selected. These implants were stabilized using Alatus vaginal balloons with a 5% IsoVue to 95% saline fill ratio. The current clinical flow utilizes manual digitization of all applicators followed by planning and dose calculation using the TG-43 formalism. For this study, every patient fraction was replanned using the Fleture-Suit Delclos-style solid applicator (SA) library in Brachy Vision that matched the set used clinically. The Ti needles were copied from the clinical plan. The dose-to-medium-in-medium ($D_{m,m}$) was calculated using Acuros and several key equi-effective dose (EQD2) dosimetric parameters were compared between the two formalisms. We report D_{90} , D_{98} for CTV(HR) and CTV(IR), and D_{2cm^3} for organs at risk (OAR) including bladder, rectum, sigmoid, bowel, and vagina. Mean and absolute differences from the two plans were investigated with linear regression, and Bland Altman analysis. **Results:** In this subset of 15 patients each with 5 fractions, where prescription dose is 550 cGy, the Acuros plans resulted in an average dose reduction to CTV(HR) by $4.5\% \pm 0.37$ (D_{98}) and $4.4\% \pm 0.22\%$ (D_{90}), and to CTV(IR) by $4.1\% \pm 0.19$ (D_{98}) and $4.0\% \pm 0.17$ (D_{90}). Normal structures also showed a dose reduction to D_{2cm^3} regions of bladder as $5.0\% \pm 0.31$, rectum $8.1\% \pm 0.39$, sigmoid $5.1\% \pm 0.43$, bowel $6.2\% \pm 0.30$, and vagina $4.8\% \pm 0.28$. **Conclusion:** Acuros estimates lower doses to target and normal structures. However, considering that the cervical brachytherapy consensus guidelines are based on TG43 data, it is unclear if using MBDCA calculations will be a clinical improvement.

PO-GePV-T-15

HDR Brachytherapy as An Alternative to SBRT for Treating Selective Lung Cancer Patients

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Purpose: High-dose-rate (HDR) brachytherapy is attractive modality for several reasons, including steep dose-fall off, less susceptibility to target/organ motion, better sparing of normal tissue and organs at risk (OARs), and reduced chance of geometric miss. Aim of this study is to investigate the potential of HDR brachytherapy as an alternative to stereotactic body radiation therapy (SBRT) for lung cancer treatment. **Methods:** Randomly selected 27 patients who had prior SBRT near the chest wall were evaluated by simulating interstitial HDR plans. The HDR plans were created in Oncentra TPS (version 4.5, Ir-192 source), utilizing 5-10 needles inside of the target volume and placed in-between ribs. Clinically used SBRT prescriptions ranged 50-55Gy in 4-5 fractions, while the HDR plans had 30Gy in one fraction. **Results:** The HDR plans showed comparable dosimetric coverage of the target volume and marked reduction in the OARs' doses, especially for the chest wall and ribs. Average values of D_{1cc} , D_{5cc} , and D_{10cc} to the chest wall for SBRT were 50.0Gy, 44.0Gy, and 39.7Gy. Respective HDR doses were 16.5Gy, 11.5Gy, and 9.3Gy. Average maximum dose to the ribs from SBRT was 53.9Gy vs 24.6Gy for HDR. The mean lung dose from SBRT was 6.4Gy, while 1.5Gy was for HDR. For target coverage, average D_{90} for SBRT was 104.2% of Rx and for HDR was 107.0% of Rx. Average gradient and conformity indices for SBRT were 5.0 and 1.1, and those for HDR were 3.1 and 0.8, respectively. **Conclusion:** This study indicates that single fraction HDR brachytherapy can be a plausible and attractive modality for treating selective lung cancer patients who have tumors near the chest wall. The HDR brachytherapy can have similar target coverage, while having considerable reduction in dose to the OARs.

PO-GePV-T-16

Comparison of Dose Distribution in Different Types of Cylindrical Applicators Using High Dose Rate Cobalt-60 Source Brachytherapy in Gynecological Cancer

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Purpose: The aim of this study is to conduct an assessment of dosimetric properties of single channel, multichannel and shielded cylinder applicators through dose-volume-histograms. **Methods:** This study utilizes a uterus phantom with nine different volumes of bladder and rectum to perform dose measurement around the cylindrical applicators. GAFCHROMIC EBT3 model film was used to measure dose distributions in water phantom around the bladder, rectum, applicator and CTV. In order to properly compare the performance of the shielded applicator with multichannel and single channel applicators, we consider the both cases of a one sided CTV and two CTVs placed on both sides of the vagina. **Results:** Dose-volume histograms analysis shows no statistically significant difference in tumoral region for the two types of CTV between cylindrical applicators. The received dose in a volume of at least 2 cm³ (D_{2cc}) for two-sided CTVs for single channel and multichannel is 2Gy, 0.7Gy in bladder and 2.52Gy, 0.39Gy in rectum. As for the one-sided CTVs, the results are the same for single channel, but it is 0.39Gy, 0.7Gy in bladder and 0.69Gy, 2.2Gy rectum for multichannel applicators. When CTV assumed one side of vagina, the multichannel applicator demonstrates a significant dose reduction in both bladder and rectum compared with single channel, and shielded applicator exhibits a better dose reduction in rectum compared with multichannel applicator. However, in the case of two-sided CTV coverage, the multichannel applicator demonstrates a noticeable dose reduction in bladder and rectum. **Conclusion:** The results indicate that employing multichannel applicator in vaginal high dose rate cobalt-60 source brachytherapy can reduce the received dose to organs at risk without any significant loss of received dose in CTV, which is contributed to its asymmetric or symmetric dose distribution that depend on which channels are activated.

PO-GePV-T-17

Evaluation of Post-Implant Dosimetry in LDR Permanent Interstitial Brachytherapy for Prostate Cancer

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Purpose: To present the dosimetry analysis of low dose rate (LDR) prostate brachytherapy patients treated over a three-year period. To compare dosimetry on the implant day (D_0) and post-implant dosimetry at the nominal optimal dosimetry time (D_{30}) for patients implanted with 125I. **Methods:** A total of 118

patients diagnosed with prostate cancer were treated with LDR brachytherapy using I-125 seeds (activity 0.371mCi at implant) between 2018-2020. 67% patients received 145 Gy while 33% received 108/110 Gy boost prescription. Transrectal Ultrasound (TRUS)-based intraoperative planning was used to obtain real time images for dosimetry on D0. Prostate (target) and organs-at-risk (OAR) volumes were contoured based on MR-TRUS registration or TRUS-only. The planning was performed while the patient was under anesthesia in real-time using Oncentra Prostate treatment planning system based on inverse-planning to optimize and analyze target and OAR dose metrics. After plan approval, patient specific needles were built using QUICKLINK® seed loader (BARD). Day 30 post-implant dosimetry was performed using MR-CT or CT-only for contouring, and CT-based planning. Target and OAR dosimetric parameters were reported. **Results:** D0 target dosimetry was as followed: D90 111.7±3.7%, V100 96.1±2.8%, V150 48.6±5.7%. The corresponding D30 target dosimetry was D90 108.9±7.3%, V100 94.0±3.3%, V150 49.7±9.4%. The urethra D0 V125 0.03±0.08cm3 and V150 0.00±0.01cm3 shifted to D30 V125 0.14±0.18cm3 and V150 0.02±0.04cm3. Rectum D0 metrics were V100 0.05±0.11cm3 and V69 1.43±0.62cm3. Rectum D30 values settled at V100 0.23±0.43 cm3 and V69 1.25±1.30 cm3. **Conclusion:** D30 dosimetric parameters revealed high consistency in dosimetry compared to D0. Post-implant dosimetry complied with the American Brachytherapy Society recommendations. The intraoperative technique takes advantage of the real-time images to allow for accurate needle placement, seed localization, and dose distribution optimization, all of which may contribute to dosimetric stability from D0 to D30.

PO-GePV-T-18

Dosimetric Differences of Line-Source Versus Point-Source Approximation for Prostate Seed Implant

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Purpose: Line-source consideration is appropriate for dose calculations in low-dose-rate prostate seed implant (LDR-PSI) cases when stranded sources are used, as the orientation of the seed is known. Some RTOG protocols, however, recommend use of a point-source approximation. The purpose of this study is to evaluate the differences in target and organs-at-risk (OAR) dosimetry of point-source as compared to line-source approximations on intra-operative planning of ultrasound-guided LDR-PSI. **Methods:** Out of 30 randomly selected PSI patients, 15 utilized I-125 (Theragenics AgX100), 15 had Pd-103 (Theragenics TheraSeed-200), and 15 patients who were treated with I-125 were replanned using Cs-131 (IsoRay CS-1 model). Dosimetric computation was performed on TRUS images using MIM Symphony TPS for both point-source and line-source approximations to evaluate dosimetric differences in OARs and target. **Results:** In most cases, point-source underestimated dose to OARs. For rectum, mean dose difference between line-source and point-source for D0.1cc D1cc, and D2cc were -1.9% (+/-1.6), -3.1% (+/-1.3), and -3.3% (+/-1.9), respectively. Similarly, D10 of urethra was underestimated by point-source, with mean difference of -3.4% (+/-1.9). For prostate, mean difference for D90 was -0.3% (+/-1.7), but for V150 and V200, point-source dosimetry overestimated in comparison to line-source by 2.3% and 9.0%, respectively. I-125, Pd-103, and Cs-131 isotopes exhibited mixed results on OARs. Rectum D1cc and D2cc for Pd-103 and Cs-131 were approximately 1% lower for point-source compared to I-125, while point-source overestimated for Cs-131 target parameters. The largest noted difference was found in V100 for urethra, with a mean difference for I-125 of -23.1% (+/-79.9%), and for Pd-103 -4.5 (+/-5.6). **Conclusion:** Overall, point-source approximation, compared to line-source, underestimated dose. Dose difference between I-125, Pd-103, and Cs-131 showed variation, depending on OAR parameters considered. When orientation of the implanted source is known, it is reasonable to use line-source approximation to achieve accurate dosimetry for LDR-PSI.

PO-GePV-T-19

MRI-Based Virtual Volume Study for Permanent Prostate Brachytherapy (PPB)

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Purpose: Traditionally, an ultrasound-based volume study is performed during an in-person visit for patients who choose to receive PPB. This study is to investigate the feasibility of using diagnostic staging MR images for pre-planning of PPB in replacement of ultrasound-based volume study to minimize patients' in-person visits. **Methods:** Diagnostic staging MR images of fifteen previous PPB patients were retrospectively retrieved. High resolution T2 images (0.5mmx0.5 mm in plane, 3mm thickness) were imported into brachytherapy treatment planning system. Prostate volumes were delineated and compared with volumes from both ultrasound-based pre-treatment volume study and intra-operative planning study. Treatment plans using I-125 seeds were generated by the same physician/physics team. The prescription dose, seeds activity and dosimetric criteria of MRI-based plans were kept the same as their corresponding ultrasound-based plans. The number of seeds required in MRI-based plans were compared with ultrasound-based plans. The patterns of needles and seeds were visually compared too. **Results:** The prostate volumes of 15 patients from MRI-based volume study range from 21.5cc to 76.2cc with average of 36.1cc ± 15.1cc (1SD). The volume difference between MRI-based pre-plan and ultrasound-based pre-plan is 2.0cc ± 1.5cc (1SD) and not statistical significant (p value=0.77). The similar small difference 2.9cc ± 2.2cc (1SD), (p value=0.88) was observed between MRI-based pre-plan and ultrasound-based intra-operative plan. The number of seeds used in MRI pre-plan is very close to the number used in ultrasound-based plans with average difference 3±2 (1SD), (p value=0.81). The implant pattern, particularly the seeds pattern in each needle can be very different due to probe caused gland deformation and imaging slicing orientation between MRI imaging and ultrasound study. **Conclusion:** MRI-based virtual volume study can generate similar plan as ultrasound-based volume study and be used for ordering loaded seeds. However, using MRI-based plan to order pre-loaded needles need to be further studied.

PO-GePV-T-20

Evaluation of Dosimetric Parameters of Two Brachytherapy Ophthalmic Sources Using Monte Carlo Simulation

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Purpose: Dosimetry in brachytherapy is of paramount importance to ensure equality between the prescribed and delivered dose to the patient. Due to the high dose gradient in regions close to the source, experimental dosimetry is highly complex. Therefore, the American Association of Physicists in Medicine (AAPM) recommends using Monte Carlo simulation as an alternative. In this work, the relative depth dose, the radial dose function and the anisotropy function (at the reference distance for dosimetry in brachytherapy, 1 cm) were analyzed, for two sources commonly used in ophthalmic brachytherapy, using Monte Carlo simulation with PENELOPE code. **Methods:** The models used were IR06-103Pd and OncoSeed 6711 for ¹⁰³Pd and ¹²⁵I, respectively. Both were modeled and positioned in the center of a 30 cm cubic phantom, filled with water and divided into 101 pixels in the three directions of the Cartesian axis. The 8 and 5 main emission lines were inserted for the ¹⁰³Pd and ¹²⁵I sources, respectively. **Results:** Comparing the relative depth dose from the two sources, the source of ¹²⁵I has 5.4 p.p less than ¹⁰³Pd source, due to the distribution of the radioactive material being different in both sources. Analyzing the radial dose function, the major difference was 0.03 at 1.2 cm, being greater for ¹²⁵I. In the anisotropy function, both showed the same behavior, but with higher values for the ¹⁰³Pd source, with the major difference 0.13 at 135°. **Conclusion:** For the dosimetric parameters analyzed, the sources of low dose rate showed similar behavior in dosimetric parameters and difference in relative depth dose, due to the distribution of radioactive material by the source geometry. Thus, the PENELOPE package presents as a promising tool for dosimetric calculations for brachytherapy sources.

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PO-GePV-T-21

A Comparison of Surgical Bed Biologically Effective Dose from Permanently Implanted Collagen Tile Embedded Cs-131 Brachytherapy Vs Stereotactic Radiosurgery in Patients with Resected Brain Tumors

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Purpose: To compare, in patients with brain tumors, surgical bed biologically effective dose (BED) from GammaTile (GT), a permanently implanted collagen tile embedded with Cs-131 seeds (GT Medical Technologies, Tempe, AZ) versus BED with commonly used stereotactic radiosurgery (SRS) protocols. **Methods:** In four patients who underwent GT implants (60Gy prescribed at 5mm depth from the operative bed surface), absolute dose was calculated on post-operative thin-slice CT using a commercial treatment planning software (MIM Symphony, MIM Software Inc, Beachwood, OH) and AAPM TG-43U1 formalism. Voxel dose was converted to BEDGT using a validated linear/quadratic model ($\alpha/\beta=10$) that accounted for half-life (9.7days), cellular repair half-life (1.5hours), and tumor doubling time (5.4days). Absolute dose and BED covering 90% (DGT90% and BEDGT90%) and 95% (DGT95% and BEDGT95%) of the target volumes (tumor bed [TB]+0-5mm) were calculated. Prescription doses to target volumes (TB+2-3mm) commonly used for single- and fractionated-SRS -- 12, 14, 15, 17, 18, or 20Gy/1 fraction; 27Gy/3 fractions; and 30Gy/5 fractions -- were converted to BED1fx, BED3fx, and BED5fx, respectively, using the external beam linear/quadratic model ($\alpha/\beta=10$). BEDGT90% and BEDGT95% were compared to BED1fx, BED3fx, and BED5fx. **Results:** BED for SRS was 26.4–60Gy (12-20/1fx), 51.3Gy (27 Gy/3fx), and 48Gy (30 Gy/5fx). For TB+margins of 0, 1, 2, 3, 4, and 5 mm, mean BEDGT90% was 145.5±56.9Gy, 116.3±40.2Gy, 92.9±29.9Gy, 78.5±25.1Gy, 65.6±19.9Gy, and 55.2±17.1Gy, respectively. Mean BEDGT95% for TB+0, 1, 2, 3, 4, and 5mm was 126.5±51.8Gy, 100.6±35.2Gy, 80.6±26.0Gy, 67.9±21.7Gy, 56.7±17.1Gy, and 47.8±15.0Gy, respectively. **Conclusion:** This preliminary work indicates that GT delivers equivalent or higher BED than commonly used SRS protocols. Investigations with a larger sample size are underway, and may be informative for interpreting efficacy and safety results using this novel brachytherapy device, which is FDA-cleared to deliver radiation therapy in patients with newly-diagnosed malignant intracranial neoplasms and recurrent intracranial neoplasms.

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PO-GePV-T-22

A Comprehensive Dosimetric Evaluation of Circular Collaborative Ocular Melanoma Study (COMS) Plaques and Eye Physics Plaques On a Standard Eye Model

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Purpose: To evaluate dosimetric differences between circular COMS and Eye Physics (EP) plaques for uveal melanomas to provide clinical guidance in the selection of eye plaque for various tumor sizes and heights. **Methods:** Tumors with basal diameters (D_{base}) ranging from 6mm to 18mm, and heights ranging from short (3mm) to tall (10mm), were simulated on the equator of a standard eye model in the Plaque Simulator planning system. Treatment plans with a COMS plaque selected based on the COMS convention and one or two EP plaques with similar diameter were calculated for each tumor. 85Gy was prescribed to tumor apex for tall tumors and prescribed to cover to at least tumor apex and 95% of the tumor base plus 2mm margin for short tumors. **Results:** For short tumors, most EP plaques produce lower tumor base D_{max} (D_{max_TB}) and V85Gy of entire eye, and achieve similar coverage at shorter prescription height, than their corresponding COMS plaque, with an average reduction of 66Gy, 1.1%, and 0.7mm, respectively; this difference is more pronounced for D_{base} ≥ 12mm. For tall tumors with D_{base} > 14mm, EP and COMS plaques show similar D_{max_TB}, but, with D_{base} ≤ 14mm, EP plaques reduce D_{max_TB} by an average of 161Gy. However, for both short and tall tumors, EP plaques increase outside-of-margin D_{max} (D_{max_OOM}) by an average of 29Gy and 103Gy, respectively, although D_{max_OOM} is on average 40% lower than D_{max_TB}. **Conclusion:** EP plaques generate lower max dose at tumor base, indicating potentially reduced ocular toxicity, compared to similar-sized COMS plaques, especially for tall tumors with small to medium base size and short tumors with medium to large base size. EP plaques do produce increased dose-spill outside of tumor, which should be carefully evaluated when tumor is located close to critical structures such as optic nerve and macula.

PO-GePV-T-23

Dosimetric Impact of Atlas Generated Contours for Post-Prostate Seed Implant Evaluation

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Purpose: The purpose of this work is to generate an atlas for the automatic segmentation of contours for post-prostate seed implant evaluation and to investigate the dosimetric impact of using these contours for post-plan analysis. **Methods:** A prostate atlas was generated in the MIM software package using expert (radiation oncologist) defined contours for the bladder, rectum, and prostate for 40 patients. A workflow was developed that implemented several pre-processing steps to improve the contouring accuracy. Processing steps included the automated detection of metal (seeds) and air and masking these image values in order to allow the atlas registration framework to focus on the patient anatomy and not on the metal/air that could bias the result. Post-plans from 10 additional patients were exported from the clinically implemented VariSeed TPS to MIM for dosimetric analysis. The RTplan from VariSeed was used to calculate a new dose distribution in MIM using the same seed locations and source strengths. Relevant dosimetric parameters were calculated using both the expert and atlas defined contours. **Results:** An atlas and workflow was successfully generated in MIM for the automatic segmentation of the prostate, rectum, and bladder. Minimal dosimetric differences were observed between VariSeed and MIM when using expert defined contours, consistent with previously published results. When atlas generated contours were used, there were larger differences in dosimetric parameters for the prostate and rectum. The average difference in prostate volume, V₁₀₀, and D₉₀ was -0.3+/-4.3 cm³, -0.7+/-1.4%, and -1.3+/-6.6%, respectively. The average difference in V₅₀ for the rectum was 0.6+/-2.6 cm³. **Conclusion:** Despite large differences in dosimetric parameters observed when using atlas defined contours for post-plan analysis, the magnitude of these differences are generally small with respect to clinical tolerances. In addition, the improved efficiency of using automatically generated contours has the potential to significantly improve clinical workflow.

PO-GePV-T-24

Dosimetric Analysis of Real-Time Intra-Operative Planning and Post Implant Planning for the Prostate Permanent Brachytherapy Implants

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Purpose: To compare and analyze various dose-volume-parameters of real-time intra-operative plans using ultrasound-images with those of CT-based postimplant plans for prostate-brachytherapy. **Methods:** 15 patients were selected for this study who received permanent brachytherapy implants using Iodine-125. Five patients received monotherapy of 145Gy and 10 received 110Gy as a boost combined with external-beam. 5mm ultrasound axial images were captured into the VariSeed treatment-planning-system. Prostate, urethra, and rectum were contoured, a treatment-plan was generated to cover 90-97 % of prostate with prescription-dose at the same time not to exceed tolerance doses to urethra and rectum. Seeds were implanted using real-time ultrasound images utilizing Mick applicator. Post implant CT based planning was performed after 4-6 weeks to evaluate the dose coverage to the prostate and organs at risk

volumes(OARs) doses. Prostate volumes and dose volume parameters such as V100, V150, V200, D90 for prostate and V100, D0.1cc, D1cc and D2cc for the rectum were compared and analyzed. **Results:** The mean intra-op prostate volume was 40.3cc(SD14.5), mean post implant prostate volume was 37.8cc(SD14.8)($p<0.06$). The intra-op means for prostate V100, V150, V200, D90 were 93.8(SD2.6), 57.0(SD5.1), 30.8(SD6), and 108.5(SD5.8), while the post implant means were 92.3(SD5.2), 64.6(SD13.6), 38.1 (SD10.1), and 110.1(SD18.3), with their P-values being 0.103, 0.026, 0.011, and 0.343, respectively. Rectum mean V100, D0.1cc, D1cc, D2cc for Intra-op plans were 0.4(SD0.3), 110.4(SD14.6), 88.1(SD9.0), 78.4(SD8.5) and post-implant plans were 0.3(SD0.4), 104.5(SD29.6), 75.8(16.1), 63.8(SD12.2) with p-values of 0.173, 0.251, 0.008, 0.000 respectively. **Conclusion:** Small differences exists between intra-op and post implant dosimetric results, however this study suggests that intra-operative real-time dosimetry system provides a close match to the actual delivered doses and a good predictor of quality of the implant. These data support the use of real-time intra-op system to modify and adapt the implantation of seeds during the surgery to achieve more favorable dosimetric results.

PO-GePV-T-25

Evaluation of Dose Distribution from An Ophthalmic Brachytherapy Through the TOPAS Code

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Purpose: Ophthalmic applicators used in brachytherapy are placed in contact or close to the region to be treated, administering a high dose gradient in a few millimeters, making the determination of dosimetric parameters become a challenge. Thus, the TG-43 protocol proposed by the American Association of Physicists in Medicine (AAPM) recommends the use of Monte Carlo simulation to determine the dosimetric parameters of this type of source. Therefore, the objective of this work was to analyze the relative dose distributions through the TOPAS simulation code from a source $^{90}\text{Sr}/^{90}\text{Y}$ used in brachytherapy. The obtained results were compared with the PENELOPE code. **Methods:** The applicator simulated in this work was the model SIA.20, which contains a source of $^{90}\text{Sr}/^{90}\text{Y}$, with 9 mm in diameter used in the treatment of ophthalmic brachytherapy. A spherical phantom with 30 mm in diameter, representing the patient's eye, was placed below the applicator. For the evaluation of the TOPAS code, a beam profile was obtained through a dose distribution at 1 mm depth of the phantom, as recommended by dosimetry protocol. The shape of the beam profile is similar to a Gaussian curve, so, the values of average width at half maximum (FWHM) were compared with the PENELOPE code, as the same condition of simulations. **Results:** The FWHM values were 0.219 and 0.362 for PENELOPE and TOPAS, respectively. Comparing the values of dose relative at 2.5 mm from the center of the source, a region with a high dose gradient, a difference of 5.5% was found, due to the contribution of the electron edge effects. **Conclusion:** The results obtained in this work show that the TOPAS code presents as a promising tool for dosimetry in brachytherapy.

PO-GePV-T-26

Evaluation of MIM Symphony Treatment Planning System for Post Prostate Seed Implant (PPSI) Planning

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Purpose: To evaluate MIM Symphony treatment planning system for PPSI planning. **Methods:** PPSI planning procedure was evaluated and compared between MIM Symphony and Varian VariSeed on five patients, including image import, contour, seed placement/identification, dose calculation, plan evaluation, and plan printout documentation. **Results:** MIM Symphony is an add-on function of MIM software, and it is compatible with all other MIM functions. We believe this module could potentially improve workflow for PPSI planning, especially for patients boosted with external beam radiation. It is also possible to generate cumulative EBRT and brachytherapy equivalent dose and contour sharing. In MIM, the imported PPSI CT images can be fused to MRI, PSI ultrasound, and other external beam planning CTs. Contouring can be efficiently performed based on various contrasts provided by different image modalities and within one module. MIM Symphony seed auto-identification function is comparable with the widely used VariSeed planning system. The seed auto-identification parameters can be constructed in MIM Symphony and adjusted for individual PPSI plans, including seed intensity threshold, PTV/CTV margins for seed search, and seed size upper and lower limits. The dose calculation algorithms for MIM Symphony and other commercially available LDR treatment planning systems are identical, which is based on AAPM TG-43. Dose constraints for target and organs-at-risk can be incorporated to standardize plan evaluation. The plan printout document of MIM Symphony can be customized to fit each clinics' individual requirements, and templates can be generated. **Conclusion:** Our initial tests have shown that MIM Symphony for PPSI is comparable to other LDR treatment planning systems, and has the extra advantages of ease of contouring, contour sharing and equivalent dose accumulation. We believe it has the potential for more streamline workflow and therefore improving PPSI process efficiency.

PO-GePV-T-27

Automating Brachytherapy Treatment Plan Document Generation by Using Eclipse Scripting API (v15.6)

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Purpose: To simplify generating brachytherapy treatment plan documents in Eclipse (Varian) and to streamline the process of independent dose check, an Eclipse Script API (ESAPI) "uPRINT Brachy" has been developed to automate the preparation of treatment plan documents. **Methods:** ESAPI (v15.6) has been used to retrieve brachytherapy treatment plan parameters from Eclipse. An independent dose check was built in based on simple inverse-square law algorithm. Eclipse viewing plane parameters are not accessible in current ESAPI version; therefore, an ARIA SQL query was implemented for retrieving these parameters in order to replicate the display in the Eclipse application. Additionally, an XML parser and "VarianSolidApplicator" class were implemented in order to import the solid applicator definition from Varian applicator files (.acrx) and render its 3D geometry in isodose display. The planning document is generated in HTML format and can be printed or saved as PDF via web browsers. **Results:** Scripting plan printing simplifies the brachytherapy treatment plan document preparation, and the built-in independent dose check further streamlines our workflow. The dose check automatically applies all reference points with physical locations, and users can select/unselect them for reporting. Since brachytherapy plans are often displayed in arbitrary rotated views, aligning applicators using pre-set viewing plane parameters from Eclipse (retrieving from ARIA database) is highly recommended for automation and minimizing user intervention. The solid applicator rendering feature retains the same visual quality compared to an Eclipse screenshot when there is no CT/anatomy present. **Conclusion:** ESAPI version 15.6 enables users to access brachytherapy plan treatment parameters. Our work demonstrates an implementation of automating the brachytherapy plan document preparation process by using ESAPI. This work not only can improve efficiency and reduce errors, but can also provide matching graphic quality of documents currently prepared using Eclipse printing/screenshot functions.

PO-GePV-T-28

Evaluation of Eclipse Smart Segmentation for Liver Yttrium-90 Selective Internal Radiation Therapy

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Purpose: In resin Yttrium-90 (Y-90) selective internal radiation therapy (SIRT) of metastatic liver cancer, liver volume sizes are needed for determining Y-90 activity with the body-surface-area method, which are obtained from liver contours delineated by physicians. The aim is to evaluate Eclipse Smart Segmentation, a knowledge-based auto-segmentation module, for potential application in liver delineation for resin Y-90 SIRT. **Methods:** CT images of 10 patients treated with resin Y-90 were studied. Auto-segmentation of liver was performed with the Smart Segmentation in an Eclipse treatment planning system

(Version 15.6). For each case, an expert case in the built-in library, which had the highest similarity to the studied case as indicated by the system, was selected for auto-segmentation. Liver contours generated with the Smart Segmentation were compared with physician manually-delineated contours. The latter were taken as the standard. To evaluate the auto-segmentation accuracy, Dice similarity coefficient (DSC) was calculated and liver volume sizes were compared. **Results:** The average DSC was 0.77 (standard deviation: 0.10; range: 0.56-0.9). The average relative volume difference was 12% (standard deviation: 9%; range: 0.4%-22.6%). **Conclusion:** The Smart Segmentation can be used to generate initial liver contours, which need to be reviewed and edited by physicians. Application of the Smart Segmentation will save contouring time and improve the efficiency of the workflow in resin Y-90 SIRT.

PO-GePV-T-29

Validation of ACE: Advanced Collapsed Cone Engine Heterogeneity-Corrected Brachytherapy Dose Calculation with Gafchromic EBT3 Film

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Purpose: To validate Elekta's Oncentra ACE TG-186 heterogeneity-corrected calculation of dose with heterogeneous materials using Gafchromic EBT3 film dosimetry. **Methods:** A phantom was created to deliver a uniform dose to a 5 cm by 10 cm area of EBT3 Gafchromic film using a Nucletron MicroSelectron HDR afterloader with an Ir-192 source. The phantom composed of bolus, five plastic catheters attached to a solid water support, a variable density layer (e.g. bolus, cork, bone), and EBT3 film attached to a solid water base. The phantom was imaged using a GE Lightspeed CT simulator with a slice thickness of 1.25 mm. Layers were contoured within Varian's Eclipse and assigned program-defined material with uniform density values. Materials of water and air were assigned to bolus/solid water and cork respectively. Plans were created using Oncentra Brachy TPS. Doses of 200 cGy were delivered to a normalization point located at the center of film. EBT3 film was calibrated for absolute dosimetry using a 6MV linear accelerator beam. **Results:** Film irradiated through a layer of cork produced an average dose of 194 cGy at the normalization point. Doses for each trial were less than the ACE-calculated 200 cGy in Oncentra. **Conclusion:** Model-based dose calculation algorithms (MBDCAs) are of great importance in cases where heterogeneities are present, and its accuracy is paramount for research. There is a need for experimental validation of these engines to assure meaningful results, as well as guidance for best practice when assigning material definitions. Gafchromic EBT3 film is a great practical option since it offers minimal additional heterogeneities into the dose validation calculation. This work indicates the possibility of using film as a means for dosimetric validation of ACE calculated plans.

PO-GePV-T-30

Investigation of Impact of Systematic Errors in Applicator Reconstruction On Doses of Organs at Risk and Target Volumes in Intracavitary Brachytherapy

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Purpose: To investigate the impact of systematic errors in applicator reconstruction on DVH parameters of different organs at risk (OARs) and target volumes (TV) in intracavitary brachytherapy (ICBT) using GEC ESTRO EMBRACE protocol. **Methods:** Retrospectively, 10 patient plans of ICBT were taken in this study in which Nucletron CT and MR imaging compatible ring applicator of 3cm ring diameter and 6 cm tandem length (Nucletron, Veenendaal, The Netherlands) were used. Planning and contouring of OARs (Bladder, Rectum, Sigmoid) and TVs (HRCTV) were done on Oncentra TPS (Nucletron) with MRI images and applicator geometry was reconstructed manually and considered as zero position. Further, systematic errors were introduced in the spatial coordinates of applicator geometry by translating the applicator in -5 mm to +5 mm in a step size of 1 mm from the zero position in x-, y- and z-axis respectively by using an in-house program. The impact of systematic errors was quantified using DVH parameters of different OARs and TVs respectively. **Results:** The mean value of relative percentage variation of dosimetric parameters for 0.1cc, 1cc, 2cc, and 5cc volumes of the bladder, rectum, sigmoid colon and D90%, D98% and V90% for HRCTV in x-, y- and z- directions were calculated and shown in Table 1. **Conclusion:** The reconstruction of applicator geometry is very sensitive to the systematic errors that significantly affect the doses of OARs and TVs. This study reveals the quantitative impact of systematic errors on the OARs and TVs doses.

PO-GePV-T-31

Focal-Boost Dose Escalation in High-Dose-Rate Brachytherapy

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Purpose: High-Dose-Rate Brachytherapy (HDR-BT) is one treatment options for localized prostate cancer. Magnetic resonance imaging (MRI) allows for delineation of the dominant lesion (DL) and thus offers the potential of a focal boost. The question remains of how much dose can safely be prescribed to the dominant lesion without increasing toxicity during whole gland treatments. **Methods:** The retrospective planning study includes 50 patients who previously underwent HDR-BT to the whole gland. All patients had the DL contoured. Plans were re-optimized with our in-house planning-system allowing for constrained optimization. Plans were constrained to our current clinical metrics with the objective to maximize the dose to the DL. The dose escalation for the focal boost was analyzed with respect to lesion size and location. **Results:** Volumes for the gland and DL ranged from 6.0cc-47.2cc and 0.2cc-16.8cc, respectively. D90 of the whole gland was 107.8%±1.9%. Based on a Wilcoxon rank-sum test, key metrics for urethra and rectum did not differ significantly (p>0.05) when compared to clinical approved plans. D90 for the DL was (141.6±23.5)%. The maximum dose escalation was 220% for the smallest and 110% for the largest lesion. As per study design dose metrics for organs at risk were always met. Dose escalation for the DL is anti-proportional to its size and plateauing to D90=135% of the prescribed dose for lesions greater than 4cc. Boosting the DL in the posterior portion of the gland is limited by urethra and rectum with a median D90=129.3%. This contrasts with the anterior portion with a D90=145.4%. Those findings are independent of the laterality. **Conclusion:** Focal boosts are limited by the size of the dominant lesion and location within the gland. The results are used as a baseline for achievable dose escalation and can be incorporated in the treatment planning for MRI guided HDR-BT.

PO-GePV-T-32

Recommendations for Previous Treatment Assessment of Non-CT Based IVRT Brachytherapy

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Purpose: To review dosimetry and associated toxicity in patients treated with CT-guided IVRT in an effort of predicting prior radiation dose volume constraints in patients who have received non-CT guided IVRT, and present with recurrence requiring re-irradiation. **Methods:** 29 patients with endometrial cancer treated at our institution with upfront surgery +/- RT +/- chemo subsequently presented with an isolated vaginal recurrence and were salvaged with brachytherapy +/- EBRT. All patients received CT-guided IVRT (prescriptions ranging from 4 x 5 - 5 x 5 (Gy x fx)) and 18/29 received EBRT (prescription 1.8 Gy x 25fx). Bowel, rectum, bladder, and rectosigmoid) were contoured at the time of the IVRT. D2cc was retrospectively analyzed for each OAR as physical dose (Gy) from the IVRT course only, using DVHs. Toxicity was collected for each patient prospectively at routine clinical follow-up and graded according to Common Terminology Criteria for Adverse Events v5.0. **Results:** The median (range) D2cc for bowel was 1.86 (4.2 - 0.94 Gy), for rectosigmoid was 2.1 (4.75 - 0.4 Gy), for bladder was 3.4 (5.3 - 2.0 Gy), and for rectum was 3.4 (5.0 - 1.7 Gy). One patient treated with EBRT and IVRT experienced a grade 3 sacral insufficiency fracture (3.4%). Grade 1 and 2 toxicity was recorded in 14 (48.3 %) and 13 (44.8%) patients respectively. D2cc for bladder, rectum, bowel and rectosigmoid was less than prescription dose in 90%, 86%, 100%, and 100% of the patients respectively. **Conclusion:** Overall salvage IVRT was well tolerated.

14% of patients received OAR D2cc greater than prescription with only 2 with grade 2 toxicities. These preliminary results suggest that using D2cc equal to the prescription dose is clinically acceptable for all OAR as a prior radiation dose volume constraint in patients who have received non-CT IVRT, and requiring re-irradiation.

PO-GePV-T-33

Role of Planning Ring Structure in Volume Optimization for High-Dose-Rate Prostate Brachytherapy

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Purpose: Manual HDR prostate planning can be time-consuming and user-dependent. Anatomy-based inverse optimization is an effective way to achieve a HDR prostate plan with desirable dose distribution. This study evaluated the optimum planning ring structure for HDR prostate using BrachyVision. **Methods:** Five patients treated with single fraction HDR prostate brachytherapy were selected. Five HDR plans were created for each patient by varying the planning ring structure during volume optimization. The planning ring structures were 4 cm wide with 0, 1, 2, 3, and 4 mm gap from the high-risk CTV, respectively. Other planning parameters were kept constant, including the step size, max dwell time, source dwell positions, and dose constraints. Each plan was evaluated for target coverage (V100, V125, V150), urethra, rectum, bladder sparing (D0.1cc), and conformity index (CI=ratio of prescription isodose volume to target volume receiving prescription dose). One-way ANOVA tests were performed. **Results:** All plans achieved over 97% target coverage. V100 significantly increased with the planning ring structures with increasing gap from the CTV ($p < 0.001$). The mean V100 values were 97.2%, 98.0%, 98.3%, 98.5% and 98.6% for ring structures with 0, 1, 2, 3, and 4 mm gap. No significant differences were found in V125, V150, D0.1cc rectum, D0.1cc bladder, and D0.1cc urethra. The CI significantly increased with planning ring structures with increasing gap from the CTV ($p = 0.024$), indicating less conformal plans. The mean CI were 1.32, 1.41, 1.46, 1.52, and 1.54 for ring structures with 0, 1, 2, 3, and 4 mm gap. **Conclusion:** When using planning ring structures with increasing gap from the CTV, the target coverage significantly increased at the cost of losing conformity. Since the ring structure with 0 mm gap achieved the most conformal plans with target coverage over 97%, we recommend using it as a starting point for volume optimization.

PO-GePV-T-34

Interfractional Dosimetric Variations of the Target and OARs for HDR GYN Brachytherapy

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Purpose: To investigate the interfractional dose variations of the target and organs at risk (OAR) with respect to the prescription and its potential effect. **Methods:** We retrospectively analyzed 29 cervical cancer patients treated with intracavitary high dose rate (HDR) brachytherapy using tandem and ovoids from January 2020 to November 2020. Prescriptions ranged from 5 Gy to 7 Gy per fraction for 2 to 5 fractions. Treatment planning were conducted in the Varian Eclipse brachytherapy treatment planning system (TPS). The HRCTV dosimetric parameter of interest was the dose that covers 90 % of the HRCTV (D90). The dosimetric parameter for OAR evaluation was the absorbed dose by 2 cc of the volume of interest (D2cc). Each fraction's dosimetric parameter was normalized to the prescription and first fraction value to determine the proportional, and thus, percentage difference. For prescription normalization, OARs were normalized to the recommended OAR tolerance values from the International Commission on Radiation Units report 89 (ICRU 89) of 80% of the prescription. Statistical analysis was performed in Python 3.7 with the utilization of the Statistics package. **Results:** The mean variations for the HRCTV were 23.9 ± 17.3 % for prescription, respectively. Mean dose variations from the dose constraints were 19.1 ± 19.9 , 16.6 ± 17.3 , and 26.4 ± 17.5 % for rectal, bladder, and sigmoid, respectively. The dose deviation from the first fraction followed a similar pattern. A shift to high dose regions indicated the dose delivered to the target and critical structures was higher than the prescription. **Conclusion:** A pattern of dose variation including a shift to higher dose in both the HRCTV and OARs was observed. Such deviations may have potential effect on dose prescription and plan evaluation, which will be further investigated.

PO-GePV-T-35

Calculation of Stopping-Power Ratios for Electron Beams Used in Total Skin Electron Therapy

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Purpose: The electron beams for total skin electron therapy (TSET) are always used at extended distances and degraded by a scatter plate. Dose measurements according to AAPM TG-51 dosimetry protocol require the beam quality conversion factor, k_Q , which is derived from a standard beam ($SSD=100$ cm with 10×10 cm²). Since k_Q is directly related to the water/air stopping-power ratios (SPRs) at electron reference depth, d_{ref} , it is important to know if the SPRs of a clinical degraded beam differ from that of a standard beam. This study calculates SPRs for these beams in order to determine if k_Q is still applicable. **Methods:** The calculation of SPRs in this study is carried out in two steps, viz. the generation of the clinical incident beams followed by their use as incident beams to compute the SPRs. The Monte Carlo code BEAMnrc is used to simulate clinical 6 MeV HDTEe beams from Varian accelerators. **Results:** R_{50} in water decreases approximately 1 mm for each additional 100 cm extended distance ranging from 2.24 cm at $SSD=100$ to 1.72 cm at $SSD=700$ cm. For beams passing through a scatter plate, R_{50} is 1.76 cm (1.14) at $SSD=300$ and 1.48 cm (0.85 cm) at $SSD=600$ cm with an Acrylic plate thickness of 3 mm (9 mm) respectively. Although significant differences in calculated SPRs as a function of depth occur at depth beyond d_{max} between a clinical beam and a standard beam, less than 1% variation is found at d_{ref} for all beams. **Conclusion:** The SPRs as a function of depth for a standard electron beam provided by AAPM TG-51 protocol may not be accurate to convert a measured depth-ionization curve into a depth-dose curve accurately. However, k_Q in the protocol, is still applicable in the dose determination at d_{ref} for a beam used in TSET.

PO-GePV-T-36

Variability of Bremsstrahlung Among Modern Detectors in Total Skin Electron Treatment

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Purpose: Bremsstrahlung component in electron beam can be a limiting factor for the treatment especially for total skin electron treatment (TSET). Measurements are challenging due to the magnitude of bremsstrahlung at 3-5 meter distances. Additionally, suitability of electrometers in pico-Coulomb readings and detectors that are sensitive enough in electron and photon beams are investigated in this study. **Methods:** Multiple detectors (0.6 cm³ Farmer chamber, Markus parallel plate ion chamber, Electron diode, Edge detector, microSilicon, microDiamond and EBT3 film) were used to measure bremsstrahlung component from a Varian TrueBeam® 6MeV high-dose-rate (2500MU/min) beam. Measurements were made in solid water and in small water tank. PTW Unidos reference class electrometer was used for data collection. Analysis is made in terms of signal-to-noise ratio. For EBT film proper scaling for exposure in electron and bremsstrahlung and other film precautions were used. **Results:** Previously published data on Saturn machine using Markus chamber showed that component of bremsstrahlung increases with increasing distance up to 5% (Med Phys 21:1733-1738, 1994). However, this could be due to signal-to-noise ratio of some detectors and electrometer combination. It is confirmed that Markus parallel plate and microDiamond could give erroneous results due to saturation of the signal. Our data with various detectors which provide large signal indicated that bremsstrahlung increase is relatively modest (0.6-2%) with increasing distance. EBT3 film and other detectors provided nearly identical bremsstrahlung magnitude in TSET. A fitted exponential function ($Br = 0.67e^{0.0017x}$) from mean values of measurement data is provided. **Conclusion:** Care should be warranted for bremsstrahlung measurement as it is overwhelmed with poor signal with clinical electrometers. Farmer chamber, Edge, EFD, microSilicon and EBT films are best suited for TSET. The bremsstrahlung increase with distance is modest and a functional form is used to estimate the magnitude.

PO-GePV-T-37

Verification of Electron Beam Energy Stability

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Purpose: Monthly energy verification quality assurance (QA) evaluates the stability of the energy spectrum delivered by a Linac. Modeling the QA setup may allow physicists to determine whether failures during QA are due to changes in Linac operation or measurement setup. **Methods:** Electron energy verification records measured with an IBA Matrixx detector array on an Elekta Versa Linac were collected from the years 2018-2020. Electron energies included 6MeV, 9MeV, 12MeV, and 15MeV. Matrixx assesses the electron energy using eight ion chambers shielded by different thicknesses of high-Z inserts to simulate different effective depths. The signal measured in each detector is normalized by an unshielded chamber on the central axis to obtain an effective percent depth dose (PDD_e) curve. Measurements were compared to the reference measurements acquired during commissioning with a $\pm 2\%$ tolerance applied to each chamber. The detector array and high-Z inserts were simulated in Pinnacle Treatment Planning System (TPS). The dose to each chamber was calculated with the reference conditions used during energy verification QA: 100 monitor units and a 20x20cm² field. For each electron energy, the PDD_e was compared between the Pinnacle simulation and those obtained during each energy verification measurement. **Results:** From 2018-2020, 42 measurements were acquired with one failure for both 6MeV and 15MeV beams; 9MeV and 12MeV beams resulted in no failures. Differences between measurements and reference ranged from 0.0%-2.9% (mean: 0.37%), 0.0%-1.8% (mean: 0.19%), 0.0%-1.2% (mean: 0.29%), and 0.0%-2.1% (mean: 0.62%) across chambers for 6MeV, 9MeV, 12MeV, and 15MeV beams, respectively. Simulated doses differed from reference measurements on average by 3.16%, 2.65%, 1.53%, and 1.04% for 6MeV, 9MeV, 12MeV, and 15MeV beams, respectively. **Conclusion:** Simulating the energy verification measurement setup and adjusting the energy in the TPS may allow for an assessment of beam stability relative to commissioning and changes in the energy spectrum.

PO-GePV-T-38

An Investigation of Dosimetric Effects From Handlebars During Total Skin Electron Therapy

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Purpose: Total skin electron therapy (TSET) is a radiotherapy technique to treat Mycosis Fungoides; a condition where tumors form on the skin. Spoiled, low-energy electron beams (3-7 MeV) can be utilized to treat very superficial lesions on the skin. The Stanford method consists of six patient-oriented dual-fields that uniformly deposit a prescribed dose of 36 Gy to the skin. Commonly, skin areas that do not react to the radiation require a radiation boost. To aid in patient comfort and treatment position reproducibility, handlebars may be used; however, poorly characterized handlebars may attenuate low-energy electrons. The purpose of this study is to assess the dosimetric effects from holding handlebars during TSET. **Methods:** All measurements were performed with a newly commissioned high-dose rate (2500 MU/min) 6 MeV energy (Varian TrueBeam) that is spoiled by 10 mm Lucite. At our center, the Stanford six dual-field technique (gantry angles of 288° and 252°) is employed at an SSD of 424.5 cm on a total body irradiation stand by Radiation Products Design, Inc. Both, parallel-plate chamber (P11) and OSLD measurements were acquired at an off-axis (OAX) position typical for hand placement. Three configurations were investigated: no attenuation, handlebar, and handlebar with bolus (1cm, Superflab) to simulate additional self-shielding from fingers; all measurements were performed with an SAD of 424.5 cm to assess relative attenuation effects. **Results:** The average dose from both measurement modalities (P11 and OSLD) decreased by 79.8 $\pm$ 6.3% and 82.2 $\pm$ 5.9% when comparing the handlebar and handlebar/bolus to the standard (i.e. no buildup), respectively, indicating a large attenuation by the handlebars. **Conclusion:** We recommend investigating low-energy attenuation by handlebars when commissioning TSET to guide personalized clinical decisions, such as handlebar usage, radiation boosting of the palms, and, if possible, educating the patient to place opened-hands before the handlebars.

PO-GePV-T-39

Real-Time in Vivo Dosimetry for Electron Beam Therapy Using Amorphous Silicon Based Flexible Thin Film Combined with Scintillating Powder

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Purpose: The present study evaluated the clinical feasibility of the amorphous silicon (a-Si) based flexible thin film solar cell coated with scintillating powder for in vivo dosimetry for electron beam therapy. **Methods:** Scintillating powder was coated on a thin film solar cell using an optical adhesive to enhance the sensitivity of in vivo dosimetry system. Calibration factors of this system was acquired by dividing the dose measured at a reference depth for 6-20 MeV electron beam energy using the Markus chamber which was used as reference chamber and the signal measured using solar cell under the same condition. Energy dependence, source-to-surface distance (SSD) dependence, angular dependence and dose linearity of this system were evaluated. Also, surface doses measured using solar cell in vivo dosimetry system, Markus chamber, and thermoluminescent dosimeter (TLD) were compared to evaluate the feasibility of thin film solar cell for in vivo dosimetry for electron beam therapy. **Results:** The signals generated through interaction between solar cell and electron beam increased linearly with absorbed dose and were independent to electron energy, SSD and beam angle. Calibration factors for each electron beam energy were verified through comparison between the dose calculated using solar cell and that measured using Markus chamber at R90. The dose difference between the solar cell and the Markus chamber was less than 5%, while the dose measured using TLD showed a difference of about 7%. **Conclusion:** Clinical feasibility of the a-Si flexible thin film solar cell based real-time in vivo dosimetry system for electron beam therapy was evaluated by comparing the surface dose measured using this system with those measured using Markus chamber and TLD. The results in this study suggest that this system has the potential for in vivo dosimetry of electron beam therapy.

PO-GePV-T-40

Evaluating a Commercial Electron Monte Carlo Algorithm Using Oblique Incidence and Irregular Surfaces Present On Custom Wax Phantoms

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Purpose: To evaluate obliquity and irregular surfaces as addressed by a commercial electron Monte Carlo dose calculation algorithm using custom made phantoms. **Methods:** GAFChromic EBT-3 film was exposed using a Varian Linac and an edge-on orientation between acrylic slabs in order to validate obliquity factors as calculated by RayStation's eMC algorithm along the central axis of the beam for 6, 9, 12, 16, and 20 MeV electron beams. Calculated and measured obliquity factors are tabulated in increments of 10 percent of each beam's practical range. Calculated and measured shifts in dmax, d80, and d50 were also recorded. Next, phantoms approximating the challenging patient geometries of a nose and forearm were created out of beeswax (~100 HU). Additionally, an acrylic phantom approximating the curvature of a chest wall was used. Again, GAFChromic EBT-3 film was exposed using the edge-on orientation – this time to compare dose distributions in an axial slice along the beam's central axis. Gamma analysis was performed using RIT's Complete Patient QA tool. **Results:** The majority of obliquity factors matched calculated values within 5 percent, and oblique incidence of 45 degrees was sufficient to result in dmax shifting upstream by over 5 mm in 6 MeV and 9 MeV beams, while shifting downstream approximately 20 MeV beams. An average of 88.72 percent of pixels passed gamma analysis (criteria between 3%/3mm and 5%/5mm). **Conclusion:** This work demonstrates that custom-made phantoms can be constructed out of inexpensive materials in order to determine the performance of a TPS in challenging clinically relevant situations. Access to complex phantoms allows the user to go beyond the minimum requirements for a commercial system outlined in MPPG 5a. RayStation's eMC algorithm handled the complexity introduced by oblique patient geometries effectively. Future work could benefit from added complexity by introducing heterogeneities within the same wax phantoms.

PO-GePV-T-41

Development of a Software Tool for FLASH Dose and Dose Rate Optimization for IntraOp Mobetron

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Purpose: To develop a method to select optimal combinations of PRF, PW and N based on current pre-measured machine characteristics in order to achieve the requested dose and the dose rate with IntraOp Mobetron for FLASH dose delivery. **Methods:** We have developed a mixed integer optimization based computer program in MATLAB to evaluate the various irradiation geometries and find the optimum combination of operating parameters in order to best provide a prescribed dose and dose rate (DR). At FLASH DRs, this machine can only be programmed to deliver an integer number of pulses (N) with a set pulse repetition frequency (PRF) and pulse width (PW). Many different irradiation geometries are possible utilizing 3 different cones and various cone sizes. The cone and cone size used to shape the beam and SSD determines the dose per pulse (Dpp). The optimization code minimizes a function of dose and DRs for a given beam energy, cone size and irradiation geometry. When the exact dose and DR cannot be achieved, optimization proposes PRF, PW, N, and SSD resulting in the closest dose and the DRs. **Results:** The developed optimization allows for optimal combinations of PRF, PW and N based on current pre-measured machine characteristics in order to achieve the requested dose and DR; hence eliminating the creation, use and updates of big spreadsheets to cover all PRF, PW, N, cone size, and geometry combinations and increases efficiency when planning for irradiation of large number of samples. **Conclusion:** We have developed a code that selects the optimum combination of PRF, PW, and N parameters in order to achieve the requested dose and the dose rate, thus increasing efficiency when planning for the irradiation of a large number of samples.

PO-GePV-T-42

eFLASH Dosimetry On a Conventional Linac Using Pulse-Gated Delivery

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Purpose: To build on previous experiments and improve reproducibility of electron FLASH delivery on a conventional linear accelerator, a pulse-gating circuit was constructed and tested with several dosimeters. **Methods:** FLASH mode was achieved by operating a Varian iX linear accelerator with 15MV photon operating parameters, with the target and flattening filter retracted and a 12MeV electron foil in the beam's path. Pulse gating was achieved by an Arduino-based circuit interfaced with the linac gating module and a scintillator-photodiode sensor that was in the beam's path to count pulses. Dosimeters were placed at a source-to-surface distance of 47-47.5cm, with 0-5mm of build up and 2-3cm backscatter. Measurements were performed in open fields and on cut-outs ranging from 1x2cm² to 3x3cm². Doses were measured using a 0.01cc volume ion chamber, optically stimulated luminescence dosimeters (OSLDs), Gafchromic MD film and a novel plastic scintillation detector and Hyperscint spectral analysis. Film profiles and PDD were used to confirm dosimetry measurements were performed in a region receiving >90% of maximum dose. **Results:** A dose rate of 225 ± 11 Gy/s was achieved in FLASH mode. The pulse-gating method delivered the specified pulse count in 89% of irradiations and was within 2 extra pulses for 11% of the beams, as confirmed by scintillator. Corrected ion chamber, OSLD and scintillator doses showed excellent agreement for a range of 0-60Gy, while film showed saturation effects past 30Gy. The dose per pulse varied from 1.1Gy/pulse (7 pulses delivered) to 1.3Gy/pulse (40 pulses delivered). Oscilloscope and scintillator readings indicated consistently lower amplitude of the first pulse compared with subsequent pulses. **Conclusion:** A prototype pulse-gating method for eFLASH deliveries on a conventional linear accelerator was implemented. Further efforts will aim to reduce circuitry delays resulting in extra pulse delivery and compensation for pulse height for more consistent dose delivery.

PO-GePV-T-43

Analog FLASH Pulse Integration Using An External Diode Circuit

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Purpose: Dosimetric control of electron FLASH radiotherapy systems demand pulse level control, as each pulse delivers a non-trivial fraction of the overall dose. In our delivery system, an initial beam current ramp results in a variable dose per pulse, limiting the accuracy of dose determination through digital pulse counting. Here we describe our initial attempts at describing this effect through analog dose integration using an external diode. **Methods:** We created a 10MeV FLASH beam that delivers roughly 200Gy/s at the mylar window of the machine. We placed an external diode into the beam and connected a custom low-noise amplifier circuit which delivers a time-dependent analog signal to an oscilloscope and a digital pulse counting signal to an Arduino Uno. We used the Arduino to interrupt the pulse forming network of the linac after a preset number of pulses and attempted to relate pulse height as measured on the oscilloscope to delivered dose. We used radiochromic film to measure absolute dose. **Results:** Our diode circuit saturated at the 6th pulse in all trials. We, therefore, delivered pulse trains of 2, 3, 4, and 5 pulses and plotted integrated pulse height (V) vs dose (Gy). The goodness of fit of the linear curve was $R^2=0.9965$. Removal of the 5 pulse data point improved the fit to $R^2=1$. This is likely due to the 5th pulse nearing diode saturation. **Conclusion:** Preliminary tests indicate a simple proportional relationship between pulse height when not near diode saturation. Tuning the diode amplifier circuit and performing analog integrating of pulse heights should provide better dosimetric beam control than our current digital pulse counting system.

PO-GePV-T-44

Bremsstrahlung Dose of the Electron Beam at Extended Distances Used in Total Skin Electron Therapy

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Purpose: Electron beam from a Linear accelerator is commonly used in total skin electron Therapy (TSET) for advance cutaneous T-cell lymphoma. The high percentage bremsstrahlung dose up to 5% was reported for a 6 MeV electron beam at extended distance of 500 cm from source by Das et al. (Med Phys 21, p.1733, 1994). The significant increase in bremsstrahlung dose is a clinical concern as the electron beam for TSET is always used at extended distances. This study re-evaluates the reported high bremsstrahlung dose by both measurements and Monte Carlo simulations. **Methods:** We measured bremsstrahlung doses of 6 MeV HDTSe beams from Varian accelerators by using different detectors at extended distances. The Monte Carlo simulations were done by using the EGSnrc Monte Carlo codes, BEAMnrc/DOSXYZnrc, to simulate incident beams and to calculate dose distribution in the measurement phantoms. The simulated beams with jaws opening to 40 x 40 cm² were benchmarked with experiments in water phantom. **Results:** The result of previous reported high bremsstrahlung dose up to 5% at extended distances can only be reproduced by measurement using a parallel plate ion chamber. The results from Monte Carlo simulations show an insignificant (1%) of percentage bremsstrahlung even at SSD=500 cm. The measurements by using cylindrical chambers and other detectors confirmed the results from Monte Carlo simulations. **Conclusion:** Previously reported significant high percentage bremsstrahlung doses for 6 MeV electron beam at extended distances were inaccurate because a parallel plate ion chamber was used in the measurements. The reason that an incorrect high bremsstrahlung dose is measured by a parallel plate chamber needs further investigation. Although the bremsstrahlung tail increases with the increasing of extended distances, the percentage bremsstrahlung doses of a 6 MeV electron beam are 0.5% to 1% for SSD from 100 to 500 cm respectively.

PO-GePV-T-45

Creation and Validation of An Interactive Electron Cutout Library in a Clinical Environment

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Purpose: Electron treatments often require custom blocks for each field. Blocks may be purchased from a fabricator, often used once, and returned to the vendor or discarded. Previously, we designed a software tool to catalog and match existing electron cutouts to desired cutouts to encourage cost and time saving through reutilization. The current objective is to test the feasibility and functionality of the tool in a clinical environment by surveying three dosimetrists. **Methods:** A library of 140 electron plans were created from all-electron treatments between 10/01/2019-10/01/2020. A sample of 10 electron plans not included in the library was chosen. Electron blocks from the sample were exported and compared using the software. The software's closest 3 matches were imported back into the treatment plan for each patient. Each dosimetrist had the flexibility to adjust the SSD, collimator angle, and isocenter to get the best coverage and conformity. The dosimetrists were asked to rank the coverage, conformity, and most suitable plan (if any). **Results:** In both the conformity and coverage tests, the dosimetrists agreed that a library-selected block conformed and gave adequate coverage for 7 out of 10 of the chosen patients. The average rank of the chosen block assigned by the dosimetrists agreed with the algorithm. The D90 for each cutout was compared against the clinical cutout, with the accepted cutouts per plan varying by less than 3.7%. The D90 varied more than 3.7% from clinical for the 3 rejected cutouts. **Conclusion:** All dosimetrists agreed that the tool was easy to use. On average the dosimetrists agreed that the system works well in a clinical environment and that it would significantly increase the speed of treating patients. Future work includes translation into an API script and growth of the electron cutout library.

PO-GePV-T-46

A Thorough Validation of Electron Monte-Carlo Dose Calculation Algorithm and Generation of Correction Factors for Independent Monitor Units Hand Calculations in Clinic

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Purpose: The purpose of this study is to thoroughly validate electron Monte-Carlo (eMC) calculation algorithm from simple to complex custom-made phantoms that simulates real clinical cases and generate correction factors for independent hand calculations. **Methods:** For five TrueBeam accelerators, beam modeling was performed for seven electron energies (6, 9, 12, 15, 18, 20, and 22 MeV) utilizing Eclipse treatment planning system. First part of the study validated the eMC dose calculation algorithm in the planning system by following AAPM Medical Physics Practice Guidelines (MPPG) 5.a. tests 8.1, 8.2, and 8.3. The data collection includes point measurements with ion chamber and profile checks with ion chamber array. To test eMC calculation accuracy for challenging geometries, such as obliquity, irregular surface and heterogeneity corrections, various tests were set up to simulate real clinical scenarios utilizing homemade phantoms. The collected data for each scenario for these cases was compared with water tank measurements. Three correction factors were calculated, gantry obliquity factor (GOF), irregular surface factor (ISF), and heterogeneity correction factor (HCF) respectively. These factors were plugged in our MU hand calculation and compared with TPS MUs. **Results:** : The results showed a good agreement between ion chamber measurements and Eclipse eMC calculations. The point measurements of all MPPG 5.a. tests were under 5%, and profiles and PDDs were 97±2% (3% dose difference and 3mm DTA tolerance) for all electron energies. Three correction factors calculated as gantry obliquity (GOF, range 0.89-0.99), irregular surface (ISF, range 0.89-.95), and heterogeneity correction factor (HCF, range 0.88-0.97 respectively. After plugging these correction factors, an adequate agreement was found (<5%) for all clinical cases. **Conclusion:** Our methodology provides a unique method to compare the hand calculation MUs with planning system. Treatment planning system with eMC model accuracy calculates the dose for complex cases with obliquity, irregularity and heterogeneity involved.

PO-GePV-T-47

A Monte Carlo Study of Patient Dose Distributions Between Stanford Six-Field and Rotational Techniques in Total Skin Electron Therapy

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Purpose: Total skin electron therapy (TSET) has proven to be one of the most effective treatments for advanced-stage cutaneous T-cell lymphoma. Many techniques have been developed to deliver a uniform dose to skin. Two most used techniques are the Stanford six-field and rotational techniques. This study compares patient skin dose distributions as function of depth between these two techniques. **Methods:** The Monte Carlo code BEAM/DOSXYZnrc was used to simulate a 6 MeV HDTSe beam from a Varian Linac. The simulated beams were benchmarked and validated against experiments. The simulated dual-field beams passed a 3 mm thick of Acrylic scattering plate of 50 cm x 200 cm before incident on the patient at extended distance. The dual-field beams were then used to calculate the patient dose distributions on a CT-based patient images. The calculations assumed the same patient standing on a platform and the patient's different postures were ignored for the Stanford six-field technique in the dose calculation for direct comparison of the two techniques. The resultant patient doses were analyzed as a function of skin depth-dose coverage and evaluated using dose-volume-histograms (DVH). **Results:** Although small dose differences were observed in some local areas, no clinically significant differences were found in the patient 3D dose distributions between the Stanford six-field and rotational techniques. When dose coverage DVH curves were compared, they are visually the same between these two irradiation techniques for mean dose to skin depth of 0-5 mm, 5-10 mm and 10-15 mm from the skin surface, respectively. **Conclusion:** Both the Stanford six-field and rotational techniques provide similar dose distributions to patient skin with only small differences in some local areas. The comparison did not consider different patient postures in the Stanford technique, which could affect the comparison of the doses to the limbs.

PO-GePV-T-48

A Clinical Workflow for Creating 3D Printed Bolus with Isocenter and Orientation Labels for Placement

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Purpose: To facilitate the clinical use of 3D printed bolus (3DPB) with accurate placement during setup based on projected isocenter and bolus orientation and with a new clinical workflow to allow a coherent link between radiotherapy planners, machine-shop engineers, and therapists. **Methods:** An Eclipse plugin program to export 3DP_Bolus.stl file with 3DPB structure and ISO-points.txt file containing 3 key-point coordinates, including the projected isocenter and two points that are 1cm away from the isocenter in the X1 and Y2 directions (Varian jaw definition) to the external surface of the 3DPB. A special ARIA carepath with patient treatment information and bolus labeling instruction was created to pass the 3DPB task from planners, to machine-shop, and to therapists. Based on the ISO-points.txt file, three 2mm-diameter hemispheres were created and merged with the 3DPB-bolus.stl file before printing. Machine-shop staff labeled it with a crosshair at the isocenter, writing 4 orientation labels and patient ID, and shipping it to a specified treatment room. 3DPB was placed after the initial patient setup by aligning the room lasers with the crosshair. Seven patients were used in this clinical 3DPB test, including 4 breast electronic-boost, 2 nasal, and 1 extremity cases. **Results:** The 3DPB labeling is accurate for 7 patients' cases by visually checking the isocenter locations and bolus orientation against the plans. Bolus placement with laser alignment is taken seconds and taped down to secure its position. It provides a conformal shape to the skin without distorting the local anatomy. The new clinical workflow ensures necessary handshakes between clinical staff. **Conclusion:** This study has demonstrated that

the new tool for printing and labeling 3DPB is accurate, facilitating bolus placement and that the new clinical workflow helps to ensure a reliable custom bolus production for treatment. This procedure is ready to be implemented in our hospitals.

PO-GePV-T-49

Quality Assurance and Acceptance of Eyeshields for Use in Electron Radiotherapy

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Purpose: Eyeshields offer protection to the organs underneath by virtue of the beam energy attenuation through the materials. This is especially important for radiotherapy of the eyelid with a need to protect the cornea and lens. An important specification from the manufacturer is the transmission. As a quality assurance (QA) measure and part of acceptance, this needs to be confirmed onsite. We describe our clinical filmless approach for a quick evaluation of the eyeshield prior to use on patients. **Methods:** Our work was based on the Tungsten eyeshield (RPD Inc., Albertville, TN) which comes in 2 thicknesses, 2mm/3mm and 5 sizes. We used the 3mm, S-size with a 0.5 and 1mm Al caps. In the first method, we used an IC profiler to determine the transmission with and without the Al caps. The RPD transmission is specified at a depth of 3mm. The IC profiler has an inherent depth of 9mm; therefore, we used several additional water slabs to measure transmission at several depths and by linear extrapolation determined the transmission at 3mm. In the second method, we used Exradin1DH 97241 diode, inherent effective depth of 0.8mm, with additional slab on the diode to measure transmission at 3mm depth. **Results:** The RPD specified transmission at 3mm, 9MeV beam was 3.3%, 2.9%, and 2.8% for the eyeshield with no cap, 0.5mm Al cap and 1mm Al cap respectively. The corresponding IC profiler values (inherent depth of 9mm) were 10.9%, 11.3%, and 11.9%. However, after applying the linear extrapolation to 3mm depth, the corresponding transmission were 1.0%, 1.3%, and 2.1%. The corresponding measured transmission using Exradin1DH were 3.2%, 3.1%, and 3.3%. **Conclusion:** We have described a simple technique of acceptance and QA of an eyeshield prior to patient use. This was done for a 9MeV energy but can be repeated for 6MeV.

PO-GePV-T-50

A Practical Approach for Commissioning a Total Skin Electron Treatment Program

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Purpose: To efficiently commission and clinically implement a program for total skin electron (TSE) treatments using an affordable surrogate phantom and dual-field angle optimization jig designed from readily available materials and optically stimulated luminescent dosimeters (OSLDs). **Methods:** The TSE treatment modality for the modified Stanford technique was commissioned for a Varian Novalis Tx linear accelerator. This dual-field technique relies on the delivery of two equal exposures, one from each of the two angled components of the dual field, to improve dose uniformity over areas large enough to be representative of typical patient size. The optimal value for the angle between the two angled fields was experimentally determined from OSLD measurements made at 30 cm intervals along the vertical profile of a simple jig constructed using solid water slabs affixed to a supporting foam board. A TG-51 calibration was completed and then extended SSD and B-factor values were characterized and validated via an end-to-end test. Measurements made with a parallel plate ion chamber were used to determine the TSE beam quality and the inverse square factor at extended SSD. The setup for the determination of the B-factor involved the placement of 18 OSLDs on two 6-gallon buckets filled with water and stacked on top of each other, meant to approximate a patient's head and torso, respectively. The OSLDs were placed at different heights – meant to represent the abdomen, chest, and head positions. **Results:** Optimal gantry angles of 70.5° and 109.5° were established for the Stanford dual-field technique using the angle optimization jig. A value of 3.06 was established for the B-factor using the practical surrogate phantom, which is within the recommended range of 2.5 – 3.1 specified in TG-30. **Conclusion:** The TSE modality was efficiently commissioned using the surrogate phantom and angle optimization jig constructed from inexpensive, readily available materials.

PO-GePV-T-51

Development and Validation of ESAPI Script Based Electron Cutout Digitizer

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Purpose: To develop an Eclipse script based digitizer for electron cutout using photos of plastic plate templates with cutout outlines, and evaluate the accuracy of the method using simulated images. **Methods:** Executable script was developed and tested with Eclipse Scripting API (ESAPI). Multiple testing cases were generated on transparent plastic plate templates of all available sizes. Experimental outlines of cutouts were contoured on the plates, and photos were taken with the camera central axis pointing to the plate at a relatively vertical angle and at least 50cm away from the surface of the plate. The images were stored electronically in .jpg format with standard pixel size and default compression settings. The testing photos were read by the script and displayed in a Graphical User Interface (GUI). The four corners of the template were manually picked to establish a projective transformation mapping the pixels in the photo to the corresponding positions in the Beam Eye View (BEV) of electron beams. The cutout contours were then semi-manually (automatically if the photo was taken with white background) digitized in the 2D domain of BEVs for the generation of electron cutouts. **Results:** 10 testing photos have been generated and digitized to generate electron cutouts importable to treatment plans in Eclipse. The projective transformations have been successfully established for all testing photos. Cutout outlines were digitized with acceptable accuracy (max difference <2mm) for testing cases. **Conclusion:** The script-based digitization provided a free and reliable alternative solution for the generation of electron cutout from plastic templates. The accuracy of the method has been verified on limited set of testing cases. For following work, the script will be revised to make the digitization process more automated, and clinically used electron cutouts with more complicated shapes will be re-digitized to further validate the accuracy of the method.

PO-GePV-T-52

Comparison of Electron Beam Characteristics According to Applicator Type in Small Field

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Purpose: This study focused on comparison of electron beam characteristics according to applicator type in small field. **Methods:** Measurement were carried out in small field (2,3,4,5 cm) of 6 MeV electron beam of Elekta Infinity Linear accelerators. The two type of applicator were general applicator with circular Cerrobend block and stainless steel of tubular applicator by applied vendor(Elekta).The beam characteristics of electron were as follows that beam quality (R_{50}) from PDD curve depending on SSD (95,100,105 cm), output factor(SSD 100, ref. field :10 cm x 10 cm) and beam profile. All data were measured by micro diamond detector Type 60019 (PTW Freiburg, Germany) in 3D water phantom (BeamScan, PTW Freiburg, Germany).The gamma evaluation(2% DD/ 2 mm DTA) was used to compare beam profile in two type applicators. **Results:** The beam quality($R_{50,mm}$) of circular Cerrobend block were 23.58 ± 0.27 (2 cm), 25.03 ± 0.10 (3 cm), 25.39 ± 0.24 (4 cm), 25.35 ± 0.07 (5 cm). Those of stainless steel of tubular applicator were 23.13 ± 0.10 (2 cm), 25.03 ± 0.16 (3 cm), 25.36 ± 0.10 (4 cm), 25.44 ± 0.18 (5 cm). Output factor of circular Cerrobend block were 0.685 (2 cm), 0.789 (3 cm), 0.843 (4 cm) and 0.867 (5 cm). Those of stainless steel of tubular applicator were 0.452 (2 cm), 0.650 (3 cm), 0.763 (4 cm) and 0.791 (5 cm). The results of comparison on beam profile using gamma analysis were passed (100%) in all field size. **Conclusion:** The beam characteristics of two applicator types were compared in 6 MeV small field. Beam quality was overall similar, and the beam profile was consistent within 2% /2 mm. On the other hand, Output factor could confirm that the circular Cerrobend block

applicator was larger than the stainless steel of tubular applicator

PO-GePV-T-53

A Comparative Study of Different Collimation Methods for Quantification of Required Skin Margin During Electron Treatment

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Purpose: To quantify and compare the effectiveness of electron treatment with cutout and skin blocks for high and low energy electrons in terms of required physical margin and dose uniformity. **Methods:** Monte Carlo simulations were performed in a cube water tank phantom (30 cm sides). RayStation 8A, modeling electrons in Elekta Versa HD and Eclipse v 15.5, modelling electrons in Varian Truebeam STx were used for dose calculation. Each TPS performed 12 different simulations including three different circular cutouts of 3 cm, 5 cm and 7 cm in diameter and their equivalent skin blocks for two different electron energies of 6 MeV and 15 MeV. For all setups, an SSD of 100 cm and a 10 cm x 10 cm cone was used. Doses were calculated by delivering predefined 100 MUs for each setup and uncertainty of 0.5%. Dose was prescribed to 2 cm for 6 MeV and to 4 cm for 15 MeV electrons. The calculated isodose distributions were compared based on the physical margin required for each method, and dose uniformity within the prescribed isodoseline. **Results:** Both TPS systems produced comparable dose distributions for all 12 setups. Our results showed that a margin of 5 mm on the skin surface provide enough coverage at 50% of prescribed dose for both 6 MeV and 15 MeV electrons. Larger margins should be used for treatment of deeper targets. **Conclusion:** Although skin collimators can reduce the required margin on the skin level, no significant difference was observed for treating deep lesions. Electron cutouts still provides sufficient dose coverage laterally to the targets. Smaller cutouts and lower energy electrons may use an additional 1 mm margin as compared to higher energies and larger cutouts. Skin collimator creates hot spots that are more pronounced for high energy electrons and smaller diameters.

PO-GePV-T-54

Dosimetric Effects of a Commercial Breast Board Patient Support System

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Purpose: In prone breast treatments a carbon fiber support device resides under the contralateral breast. Tangent beams are designed to encompass the treated breast and these often pass through the board at a shallow angle, resulting in significant attenuation. Our planners account for this attenuation by added field-in-field dose to the deep part of the breast, through the board. Concern was raised about how accurate the treatment delivery is when the inherent uncertainties of patient position are accounted for. Further, transmission measurements are usually carried out perpendicular to the board; a non-clinical situation. The goal of this study is to evaluate the dosimetric effect of board and the robustness of the plan to positional uncertainty. **Methods:** Twenty two breast patients treated on a commercial prone breast board between 2017 and 2020 were selected for this retrospective study. To evaluate the board's attenuation we compared the plans with the board removed from the dose calculation. To quantify the robustness of this technique we moved the beam isocenter wrt the patient and board. **Results:** When the breast board is removed from a plan which was designed to account for the board attenuation, the average point dose increases by 7.48%, with a maximum of 22%. Comparing results with a mixed ANOVA and least square means analysis, our robustness evaluation indicates that anterior shifts at every magnitude (1mm through 5mm) make a significant difference in all dose statistics (D95, max, 95% prescription coverage and homogeneity index) investigated. Right/left shifts resulted in an insignificant change in dose statistics. **Conclusion:** Prone breast boards can add significant dosimetric uncertainties into the treatment delivery process. Accounting for plan robustness in the design of the plan is highly recommended. Prone breast board design, with supports moved away from the beam-path, is warranted.

PO-GePV-T-55

Initial Characterization and Validation of a Commercial Head and Neck Immobilization System for MR-Guided Radiotherapy

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Purpose: Head-and-neck (H&N) cancers are challenging to treat with non-coil bridge MR-guided radiotherapy (MRgRT) linacs due to non-integration/compatibility of the receiver coils into common radiotherapy H&N immobilization systems. We developed, characterized, and clinically implemented the first commercial H&N immobilization system for a non-coil bridge MRgRT linac. **Methods:** The H&N immobilization system includes an indexable custom baseplate that affixes to a head cradle, an open-face thermoplastic mask, and head-mold cushion. The baseplate is composed of clear acrylic and white Delrin and is constructed as the negative imprint of the H&N receiver coil. We quantified the baseplate's radiographic integrity, dosimetric attenuation, dosimetric impact of daily IGRT localization positioning shifts relative to the baseplate, and image quality effects due to its presence in a 0.35T magnetic field. **Results:** Mean baseplate CT # was -54.7±212.5 HU. For a single posterior beam, attenuation measured was 4.1% versus TPS calculated of 2.8%--difference of 1.3%. For a 13-field IMRT plan delivered to a 3D-IMRT QA array positioned anterior to the baseplate, the 3D gamma 3%/2mm passing rate was 98.8%. The dosimetric impact of IGRT localization was negligible (<0.1%) for couch shifts of 1, 2, 5, 10, and 50mm in AP/LR/CC directions as measured by shifting the baseplate ROI and comparing 3D gamma 3%/2mm passing rate between shifted and non-shifted calculated dose distributions. Image quality metrics in presence of baseplate and without baseplate (n=6 months) was 0.95 versus 0.89 (max geometric fidelity diameter 350mm, p=0.05), 37.5 versus 35.9 (SNR, p=0.35), 224 versus 435 (uniformity diameter 300mm, p<0.01), 2.12 versus 1.91 (high-contrast spatial resolution FWHM PSF, p=0.01), respectively. **Conclusion:** The H&N immobilization system presented here provides a clinically viable immobilization solution for H&N patients on an MRgRT linac. The immobilization system does not appear to induce significant dosimetric perturbations and does not significantly impact the MR image quality.

Dr. Mittauer reports personal fees from ViewRay Inc, other from MR Guidance LLC, grants from ViewRay Inc.

PO-GePV-T-56

Accuracy and Precision of Surface Tracking on Bra-Fitted Breast Radiation Treatment

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Purpose: Radiotherapy for patients with large and pendulous breasts is often associated with higher lung and heart doses. One way to localize the pendulous breast away from the chest wall is to lift it into a tightly fitted bra. Surface detection-based localization is often utilized for real-time tracking for breast radiotherapy to ensure accurate treatment delivery. However, the breast and adjacent surface topography is manipulated by way of fitting the bra for large breast patients. This study is to investigate the accuracy and precision of Chabner XRT (CiVCO, Orange city, IA, USA) bra with AlignRT® (Vision RT, London, UK) surface tracking. **Methods:** From 12/2020-2/2021, 58 breast patients received radiation treatment with surface tracking. Nineteen patients were treated with Chabner XRT bra and localized with AlignRT®. MV verification films were taken daily to compare with VisionRT surface tracking deltas in the patient SI, AP and LR directions. **Results:** About 30% of breast patients utilized Chabner XRT bra during their radiation treatment in our institution. 85% (16 out of 19) of these bra-fitted patients showed good surface tracking (within 1mm in all 3 directions) with AlignRT®. However, for smaller breast depths (differences between upper chest and bust circumferences, e.g., C cup) together with a relatively larger lower chest band (e.g. 40 inches), the breast tissue of

these patients (3 out of 19) tends to be equally distributed to their entire breast in the treatment position, resulting in a lower bra precision to reflect the breast surface. The bra surface could mislead the surface tracking. Because of this, these three patients required daily film verification. **Conclusion:** Bra with surface tracking is accurate and precise for normal breast sizes. When the patient has a small breast depth and larger lower chest band circumference, surface tracking could be compromised.

PO-GePV-T-57

Clinical Implementation of An Abdominal Compression Belt on a Double Scattering Proton Machine

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Purpose: To evaluate the dosimetric effects of the Qfix SBRT Abdominal Compression Belt in proton radiotherapy and to validate its water equivalent thickness (WET) modelled by the Varian Eclipse treatment planning system (TPS). **Methods:** Physical measurements of WET were performed using EBT3 film sandwiched between solid water slabs wrapped around by the compression belt in a way as it would be used clinically. A CT scan was acquired with the same setup for TPS WET calculation. The percent depth dose (PDD) was measured by delivering proton beam with range of 12cm on a Mevion S250 proton machine through a slit, so that the film was parallel to the beam direction and perpendicular to the long side of slit. Different parts of the belt were put in the beam path for each measurement. The relative range shift due to the presence of device was evaluated and compared to the WET modeled in the TPS. **Results:** The relative range shift measured by film showed that the belt and velcro had almost 0mm WET, which is consistent with the reading of 0.1mm for belt only and 0.6mm for belt with velcro in TPS. The WETs of the paddle center and 5cm superior/inferior to the paddle center were estimated to be 4mm and 7.5mm on the film measurement, respectively, comparable to 5.3mm and 8mm, respectively, from TPS. The differences between TPS and measurements were within 1.5mm. The buckle of the belt showed high WET in TPS, with 10.5mm WET. **Conclusion:** The TPS model of the compression system agrees well with measurement results, proving feasibility to use abdominal compression belt in proton therapy, provided that the system is included in dose calculation in TPS. The buckle should be avoided in the beam path due to its high WET and poor reproducibility in patient daily setup.

PO-GePV-T-58

Comparison of Three Cushion Types for Gamma Knife Icon Mask Treatment

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Purpose: For mask treatment on Gamma Knife Icon, our clinic uses Moldcare cushion (QFix, Avondale, PA, USA) to mold underneath the patient's head and upper neck for comfort and reproducibility inside the mask adaptor before a thermoplastic mask is molded around their face for immobilization. In order to treat an anterior lesion, the mold is made very thin to avoid collision. Occasionally the lesion is so anterior that, instead of the moldable cushion, a 3-mm bolus, as readily available in all radiation oncology department, or a grip shelf liner, a common consumer product, may be used to further reduce the distance between the skull and the mask adaptor. This study investigates the relative dosimetric effect from the two non-conventional material compared to the Moldcare cushion. **Methods:** Two plans were delivered to an anthropomorphic head phantom with an ionization chamber inside. Plan-A had one shot of all 8 sectors of 16-mm collimation centered at the ion chamber. Plan-B has 3 posterior sectors at 16-mm and the rest at Block position. The phantom was placed inside Moldcare, or on top of the 3-mm bolus, or the grip shelf liner. **Results:** Compared to Moldcare, both plans delivers > 1% less dose when the phantom is on top of the 3-mm bolus. One layer of liner received the same dose as the Moldcare from Plan-A, but 0.8% more dose from Plan-B. The 3-layer liner configuration received the same dose as the Moldcare from Plan-B. **Conclusion:** The 3-mm bolus, while more accessible and easy to use, has more than 1% dose difference from the standard of care. The grip shelf liner is very easy to use and has less attenuation than the Moldcare. While all measured dose are lower than the planned dose, the dose from the liner is the closest to planned.

PO-GePV-T-59

Risk Mitigation Strategies for Custom Software in the Clinic

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Purpose: One of the evolving roles of MedPhys 3.0 practice encompasses in-house software development and software development oversight. While in-house clinical software development is not new in medical physics, its widespread adoption and available vendor-provided tools are growing rapidly. Many physicists who are untrained in the best practices of software development have begun creating scripts for clinical use. There is an increasing need on guidance for both developers and medical physicists to code wisely in the clinic. **Methods:** We surveyed both medical physics and non-medical physics literature and regulatory documents for guidance which could be applicable for custom clinical software. We then began practicing these techniques internally for clinical software development. The review was integrated with our institutional experience to form a practical guide to educate medical physicists in software best practices and to begin implementing these techniques in the clinic. **Results:** We found little literature for custom software development guidelines targeted at medical physicists. However, we found several risk mitigation techniques including unit testing, code review, source control, end user testing, and commissioning from non-medical physics literature and created institutional guidelines for evaluating software for risk and implementing these strategies. The mitigation technique requirements were stratified by risk level of the software: low risk, moderate risk and high risk. **Conclusion:** Our findings indicate there is a lack of guidance for developing in-house software for medical physicists. We integrated a literature review and institutional experience to form guidelines to help facilitate a safe software development practice for the next generation of medical physicists.

PO-GePV-T-60

Creation of a Database to Utilize Patient Outcomes for Radiation Plan Improvement

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Purpose: To deliver truly personalized cancer care, we need to bridge the gap between patients and clinical decision-makers. During the treatment of a patient, there are countless decisions made by physicians, physicists, and dosimetrists that impact the plan and delivery of therapy. Clinicians record treatment outcome information obtained during interactions with patients, but often there is no integrated way to systematically monitor and fully utilize this data. Our initiative strives to create an environment in which patient outcome information is automatically pulled into a global database in real-time, for immediate evaluation of the treatment plan delivery and its impact. By analyzing and trending this data, it will be possible to directly correlate a multitude of planning techniques and parameters with outcomes and recognize areas that have potential for improvement. **Methods:** Patient on-treatment and follow-up information, including side effects and quality of life assessment, is exported from the EMR, then automatically imported into a patient database via in-house scripting. This outcome data is coupled with treatment planning data, such as beam dose, spot distribution, etc. that has been exported directly from the TPS. The data can then be analyzed for trends and correlations between planning parameters and patient outcome information. **Results:** Patient outcome information is readily available to physicians, physicists, and dosimetrists in a format that can be used to evaluate and improve therapy delivery for individual patients, as well as planning techniques for the entire patient population. **Conclusion:** This initiative has provided a useful tool for optimizing planning techniques, based on real-time analysis of patient outcome information. Future directions for the work involve inclusion of a more comprehensive set of patient complication, recurrence, and survival information, as well as the integration of analysis techniques and tools including machine learning to further illuminate correlations between plan

parameters and their therapeutic impact.

PO-GePV-T-61

Recommendations for a Medical Physics Pre-Simulation Breath Hold Consultation: Medical Physics 3.0 in Practice

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Purpose: To take initiative within the clinic by preparing and delivering patient education materials regarding breath hold treatments prior to simulation. **Methods:** Several PowerPoint presentations were prepared to educate breath hold patients prior to simulation. These presentations were designed to be relevant to L Breast, general DIBH, and general EEBH patients. These presentations take less than 10 minutes to deliver. Several key pieces of information are communicated to the patient. First, we explain that this process is important for reproducible positioning during treatment. We next explain that they will be asked to hold their breath during some of the images taken at simulation and during each treatment. To alleviate patient concern, it is important to communicate that a breath hold of roughly 15 seconds will be beneficial, and that patients will have time to catch their breath. Shifting focus to the start of treatment, we emphasize that the treatment machine will automatically turn off when the patient stops holding their breath, and patients will have time to catch their breath. Usually, around 2.5 minutes of total breath hold time is sufficient to complete an entire treatment. **Results:** In general, patients are receptive to this education. By educating the patients in a consultation room, there are no other distractions or demands of the patient; they can focus on the instructions. Physicians have noted that simulations take less time when the MP has consulted the patient. Patients are generally more comfortable and confident when educated about breath hold treatments. As a cautionary note, patients almost always ask about potential side effects. **Conclusion:** Patient education prior to breath hold simulations/treatments is an excellent opportunity for medical physicists to ensure that patient care is effective and efficient.

PO-GePV-T-62

Comparison Between Two Treatment Planning Systems On IMRT Planning: UIH Versus Monaco

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Purpose: To investigate the performances of IMRT plans on various cancers between two commercial treatment planning systems (TPS): UIH and Monaco, and evaluate the feasibility of develop photon beams calculation model of Elekta infinity linear accelerators (LINACs) with UIH TPS. **Methods:** Twenty patients, include nasopharynx, breast, lung and rectal cancer were enrolled in this study. Two IMRT plans were generated using both UIH and Monaco TPS for each patient. The direction, number and energy of beams were the same in the plans from both systems for each treatment site. Same clinical objectives were provided for each plan. The TPS were compared in terms of homogeneity and conformity, mean and maximum doses about PTV or relevant organs-at-risk (OARs). **Results:** For the nasopharynx cancer, homogeneity and conformity were comparable for both TPSs. UIH provided a significantly lower maximum and mean dose for OARs, particularly the Optic nerve (-19.2%), chiasm (-12.02%) and parotid (-4.8%), but the spinal cord (5.98%) and larynx (6.01%) were higher. For the breast cancer, UIH had a better conformity and lower maximum dose (-3.21%) in PTV, while the mean dose in lung (6.87%) and heart (8.5 cGy) were higher. For the lung cancer, UIH plans had a lower mean dose of lung (-3.29%), however the maximum dose of spinal cord (1.72%) was larger. There was no significant difference between two TPSs for rectal cancer. Meanwhile, Monaco require longer optimization time and more monitor units. **Conclusion:** For various cancers, a comparable quality IMRT plans was found for both TPSs. Further, comparisons between measurements and calculations for two TPS should be done in absolute dose terms.

PO-GePV-T-63

Skin Toxicity Assessment Using Optical-Wavelength Video Processing

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Purpose: To evaluate skin toxicity through daily changes in blood flow to the skin from optical-wavelength video processing. **Methods:** A conventional digital camera was used to record daily videos for two weeks of skin damage caused by an incidental sunburn in an investigator self-monitoring feasibility study. The camera and investigator's leg were positioned reproducibly to maintain imaging conditions. A fingertip pulse oximeter was utilized to provide a ground-truth heart rate. Videos were post-processed to identify cardiac signals using a binned pixel technique. For each frame, the original videos were down-blurred and progressively resampled to obtain a final 1D signal with improved signal-to-noise ratio (SNR). Heart rate was then identified by Fourier transforming this signal. The original 1920x1080 pixels video was downsampled to a coarse grid of 4x4 pixels for increased SNR. Each bin was Fourier transformed, and the frequency component amplitude corresponding to heartrate was presented in a false color image power map. Variations in the amplitudes of these power maps were evaluated. **Results:** Subtle oscillations of skin tone during the cardiac cycle were able to be extracted from 10-second videos. The technique evaluated changes in the magnitude observed in binned regions of the power map for both undamaged and damaged skin, theoretically representing differences in blood moving into and out of areas of the skin. Regions of higher magnitude in the power map were correlated with skin damage. The first day, the average measure of amplitude was 37.6% higher in the damaged skin, whereas at the end of the two weeks it was only 8.6% higher. **Conclusion:** This feasibility study evaluated this novel signal processing method's capability to track changes in power map magnitude over time. These power map changes should be correlated with blood flow and may be usable as prognostic indicators for skin toxicity during radiotherapy.

PO-GePV-T-64

Impact on Patient Reported Quality of Life From Reducing Prostate Margins in Patients Who Underwent Radiotherapy: A Prospective Clinical Trial

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Purpose: To investigate the impact of reducing planning target volume (PTV) margins on patient reported quality-of-life (QOL) for patients with localized prostate cancer after radiation therapy (RT). **Methods:** A cohort of 20 patients were included in a single institution IRB-approved prospective study, out of which 9 were planned with reduced margins (5 mm uniformly except 4 mm at prostate/rectum boundary), while 11 were control patients with standard margins (10/6 mm). At time points pre-treatment, post-treatment, and at 2, 6, 12, 18, 24, 36, 48, and 60 months follow-ups, EPIC-26 based patient reported QOL data was collected. Validated EPIC-26 methodology was used to baseline correct and standardize all post-RT QOL scores to five separate EPIC QOL domains on a [0-100] scale. Correlations between QOL scores and dosimetric parameters were investigated, and the overall QOL differences between the two groups (QOL(M-R)-QOL(control)) were calculated for each time point. **Results:** Age, Gleason score, and PSA level differences between the two groups were statistically insignificant ($p>0.2$). Median QOL follow-up length for the 20 patients was 48 months at the time of reporting. At 48 months post-RT, standardized and baseline-corrected mean QOL(M-R)-QOL(control) were improved for "Urinary Incontinence", "Urinary Irritative/Obstructive", "Bowel", and "Sexual" EPIC domains by 3.5, 14.8, 10.2, and 16.1, respectively. QOL differences between groups were clinically significant for the last 3 domains, while statistically significant for only "Urinary Irritative/Obstructive" domain ($p=0.03$). PTV/rectum ($p=0.03$) and PTV/bladder ($p=0.14$) intersection volumes were larger for the control group (7.2 ± 5.8 , 18.2 ± 8.1 cc) compared with the margin-reduced group (2.6 ± 1.8 , 12.5 ± 8.3 cc), while the dose to these intersection volumes did not reach statistical significance ($p>0.1$) between the groups. PTV/rectum intersection volume showed a moderate correlation ($R=-0.6$, $p=0.01$) to "Bowel" EPIC domain. **Conclusion:** Margin-reduced group exhibits clinically meaningful improvement in Urinary, Bowel, and Sexual QOL domains. More patients are needed to draw unequivocal conclusions.

PO-GePV-T-65

NTCP Modeling of Patient Reported Dysphagia Using the EAT10 Scoring System 12 Months After Radiotherapy for Oropharyngeal Cancer

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Purpose: To determine the values of two normal tissue complication probability (NTCP) models for pharyngeal constrictors regarding the endpoint of patient reported severity of dysphagia based on the EAT10 scoring system 12 months after radiotherapy for oropharynx cancer. Additionally, to identify risk thresholds. **Methods:** 151 patients with HPV-associated oropharyngeal squamous cell carcinoma were treated with 60Gy IMRT. Dysphagia was assessed based on the EAT10 score as the top 5% percentile of net difference from baseline at 12 months post-RT. The DVHs of the superior (SPC) and combined pharyngeal constrictors were exported for each patient. The clinical data was fitted by the Lyman-Kutcher-Burman (LKB) and Relative Seriality (RS) NTCP models. **Results:** Dysphagia was observed in 5.3% of the patients. The values of the D50, m and n parameters of the LKB model were 69.1Gy, 0.11 and 1.0 for SPC and 65.4Gy, 0.10 and 0.22 for the combined constrictors. Similarly, the values of the D50, γ and s parameters of the RS model were 74.6Gy, 2.1 and 0.0001 for SPC and 65.6Gy, 6.7 and 1.0 for the combined constrictors. The AUC values of the correlations ranged between 0.75-0.77. Statistically significant Odds Ratios (ORs) of 5.5 (95%CI: 1.3-23.6) and 6.5 (1.5-28.3) were found for the SPC for the two NTCP models for a gEUD threshold of 58Gy. The respective ORs for the combined constrictors were 9.2 (1.5-57.5) and 11.6 (1.8-76.2) for the LKB and RS models for a gEUD threshold of 58Gy and 63Gy, respectively. **Conclusion:** The LKB and RS NTCP models could fit the clinical data and determine the dose-response curve of SPC and combined constrictors regarding dysphagia based on the EAT10 score at 12 months post-RT. gEUD < 58Gy to SPC or combined constrictors was found to significantly reduce by 5-12 times the risk for patient reported dysphagia.

PO-GePV-T-66

Investigating the Time and Organ Dependency of NTCP Modeling of Patient Reported Sticky Saliva After HPV Associated Oropharyngeal Cancer

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Purpose: This study aims at analyzing the dependency of normal tissue complication probability (NTCP) modeling on follow-up time and organ dose regarding the endpoint of patient reported (PRO) sticky saliva after radiotherapy. **Methods:** A dose of 60Gy treated with IMRT was prescribed to 151 patients with HPV-associated oropharyngeal squamous cell carcinoma. Sticky saliva was assessed based on the PRO-CTCAE score and it was defined as a ≥ 2 point increase from baseline at 6, 12 and 24 months post-RT. The dose volume histograms (DVH) of the contralateral glands (parotid and submandibular glands combined), contralateral parotid and salivary glands (ipsilateral and contralateral parotid and submandibular glands combined) of each patient were exported. The dose and response data were fitted by the Lyman-Kutcher-Burman (LKB) and Relative Seriality (RS) NTCP models. **Results:** Sticky saliva was observed in 31%, 20% and 15% of the patients at 6, 12 and 24 months, respectively. The AUC values were highest (0.7) for the contralateral glands at 12mo post-RT. For the contralateral glands, the values of the D50, m and n parameters of the LKB model were 44.1Gy, 0.43 and 0.65, whereas the values of the D50, γ and s parameters of the RS model were 48.7Gy, 0.74 and 0.92 at 12mo post-RT. Statistically significant Odds Ratios (ORs) of 6.0 (1.3-28.6) and 7.8 (1.7-35.1) were found for the contralateral glands for biological doses (gEUD) < 43Gy and 45Gy for the LKB and RS models, respectively. **Conclusion:** The dose-response curves of PRO sticky saliva at 6, 12 and 24 months post-RT could be determined by the LKB and RS NTCP models for different combinations of salivary glands. Keeping the gEUD to the contralateral glands < 43Gy was found to significantly reduce by 6-8 times the risk for patient reported sticky saliva.

PO-GePV-T-67

NTCP Modeling and Dose-Volume Correlations of Significant Blood Count Drops 3 Months After Prostate Radiation Therapy

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Purpose: The aim of this study was to model the correlation of the dose to different parts of the pelvis with a significant drop in hematocrit 3 months after prostate cancer radiotherapy. **Methods:** 221 patients with complete treatment and follow-up data were included in this analysis. Patients received either 78Gy to primary prostate or 70Gy post-prostatectomy and 54Gy to the pelvic lymph nodes. All patients had complete blood counts (CBCs) collected at baseline (pre-RT) and 3 months post-RT. The net difference of each CBC metric post-RT vs. baseline was calculated and the level of the 15 percentiles defined the thresholds of response in each case. The doses to 8 different pelvic structures were derived and fitted 20 CBC metrics using the Relative Seriality (RS) NTCP model and the biologically equivalent uniform dose (BEUD). **Results:** The structure os coxae upper (above acetabulum) bilateral (OCUB) showed the highest correlation for a significant level of hematocrit drop. The values for the D50, γ and s parameters of the RS model were 26.9 Gy (25.9-27.9), 1.3 (1.2-2.2) and 0.12 (0.10-0.83), respectively. The AUC of BEUD was 0.73 and patients with BEUD to UCUB ≥ 27 Gy had 8.2 times higher rate of significant hematocrit drop vs. < 27Gy. The bone volume of the whole pelvis (BVWP) and the os coxae total bilateral (OCTB) also showed significant correlations for the same endpoint (AUC values of 0.72 and 0.69, respectively). In those cases, similar BEUD thresholds were found (27Gy and 28Gy, respectively) but with lower odds ratios (2.8 and 2.9, respectively). **Conclusion:** The dose-response curve of os coxae upper bilateral for a significant hematocrit drop could be determined by fitting the clinical data with the RS NTCP model. A threshold of BEUD < 27Gy was found to significantly reduce the risk for this endpoint.

PO-GePV-T-68

A Framework for Exploring the Correlation Between Radiation-Induced Toxicities and Dosimetric Quantities

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Purpose: To develop a framework to automate the analysis of large cohorts of patients regarding the correlation of toxicities and patterns of failure with various dosimetric and geometric parameters. **Methods:** A software package has been built to process data from patients with radiation-induced toxicities or recurrence after proton and photon radiation therapy at our institution. The package is written in C++ and allows for flexibility in processing large cohorts of patients and exploring a large parameter space. It takes as input CT, RTPLAN, RTDOSE, REG and RTSTRUCT dicom files from the TPS, as well as Monte Carlo-generated LET and dose distributions. Correlation metrics include: DVH metrics, Dice Coefficient, Jaccard Index, Hausdorff and centroid distances and volume of overlap between the toxicity structure and iso-dose and iso-LET levels. Toxicity structures and recurrence volumes were contoured by physicians. **Results:** The software is used to analyze a cohort of brain patients. The metrics were computed within the TPS and compared with our calculated values to check for correctness. The code is found to be robust and can be used to analyze a large cohort of patients efficiently. For proton patients, it also facilitates the evaluation of biological dose models, allowing us to explore the relationship between their predictions with physician-contoured radiation-induced injury regions. **Conclusion:** Understanding the correlation between radiation-induced toxicities and dosimetric quantities within the framework of various biologic dose models is of great interest. As a first step, we built and tested an in-house framework for automating the calculation of relevant metrics for a large cohort of patients.

PO-GePV-T-69

4D Treatment Planning for Lung Proton Therapy: Treatment Design and Evaluation

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Purpose: Intensity-modulated proton therapy for lung tumors with large tumor movement is challenging. Often a 5mm tumor movement cut-off is used for proton patient selection. We propose a robust and easily implementable treatment planning strategy for lung tumors with motion larger than 5mm, and a 4DCT robust evaluation strategy. **Methods:** We created an internal target volume (ITV) based treatment planning strategy. The plans were robustly optimized on the 4DCT-based average-CT. Clinical plan acceptability was judged on robust evaluation, computing voxel-wise minimum/maximum (VWmin/max) doses over 28 error-scenarios on the average-CT. Plans were calculated for sixteen lung patients with 4-26mm tumor motion in RayStation (RaySearch) with Monte Carlo dose-engine for a Mevion S250i Hyperscan system (Mevion Medical Systems). We developed a 4D robust evaluation (4DRobAvg). The 28 scenario-doses were computed on all eight 4DCT-phases. For each scenario, the doses on the individual phases were deformed to the reference phase (50%expiration) and combined to an average scenario-dose. From these 28 phase-averaged scenario-doses, VWmin/max were computed. The 4DRobAvg was compared to two other 4D evaluation strategies: re-computing nominal dose on each 4DCT-phase (4DNom), and computing robust VWmin/max doses on each phase (4DRobInd). **Results:** A clinically acceptable plan was feasible for all patients with adequate tumor coverage and no organs-at-risk over-dosage. The 4DNom and 4DRobInd evaluations under- or overestimated the dosimetric effect of tumor movement, respectively; 4DNom on average showed 1Gy higher target dose, and 4DRobInd showed target under-dosage for five patients, not observed in 4DRobAvg. The accuracy of dose deformation in 4DRobAvg was acceptable and dose differences were generally less than 1Gy. **Conclusion:** The proposed ITV-based planning strategy on the average-CT was found clinically feasible even for large tumor movement. The proposed 4D robust evaluation, 4DRobAvg, was shown to give an easily interpretable understanding of the effect of respiratory motion on the dose distribution.

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PO-GePV-T-70

Dose Enhancement Effects of High-Z Nanoparticles in Proton Therapy

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Purpose: To study using Monte Carlo simulation techniques the dose enhancement due to gold foils in energetic proton beam irradiation. The presence of high-Z nanoparticles has been recently studied for their ability to increase the number of secondary electrons thus enhancing dose to the target and shifting the Bragg peak location towards proximal depth. This Monte Carlo study compares the dose of a custom water phantom with gold foils to that of a control water phantom. **Methods:** Two water phantoms of 30cmx30cmx40cm were created using the Topas 3.6.1 toolkit. The control phantom consisted of pure water and the custom phantom consisted of a 0.4 mm wide gold foil in water. The location of the foil was in the proximal to distal 80% Bragg Peak width of a 226 MeV proton beam with spot size of 2.7 mm diameter. Dose was scored in 0.1 mm resolution. The dose enhancement factor (DEF) as well as the shift in the Bragg Peak were quantized. Modular physics list which include g4em-standard_opt3, g4h-phy_QGSP_BIC_HP, g4decay, g4ion-binarycascade, g4h-elastic_HP and g4stopping were used. 5 million histories for DEF, and 0.2 million for the Bragg Peak shift were used. **Results:** The introduction of gold foils in water phantom shifted the Bragg peak upstream by about 6 mm, and narrowed the peak from 8.1 to 3.1 mm. Our DEF findings showed an increase in enhancement as high as 1.46 situated around the Bragg Peak. **Conclusion:** In this study, we simulated gold nano-particles placed in a foil with a proton beam. Our results showed a dose enhancement effect as well as a Bragg Peak shift. This preliminary study validates our simulation environment for future investigations. Future work will include simulation with many beam energies using different densities of gold nano-foils and other foils of different materials.

PO-GePV-T-71

Radiobiological Evaluation of Physical Metrics in IMPT and VMAT Planning for Prostate Cancer Patient with Metal Hip Prostheses

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Purpose: The aims of this study are to 1) evaluate dosimetric deviations between treatment plans of optimized on the original CT and re-calculated on the metal-artifact-reduced CT for prostate cancer patient with metal-hip-prostheses, and 2) investigate radiobiological deviation of physical metrics. **Methods:** Optimized and re-calculated plans of 16 cases were for intensity-modulated-proton-therapy (IMPT) using two-field pencil-beam-scanning proton beams and for volumetric-modulated-arc-therapy (VMAT) using two-photon-arcs. The prescribed dose to clinical-target-volume (CTV) over 44 fractions in optimized plan of each patient was 72 Gy for IMPT and 79.2 Gy for VMAT. Two IMPT plans sets were evaluated, LAT plan with two lateral-fields and HipAvd plan with one-lateral and the other posterior-oblique avoiding hip-prostheses. Obtained IMPT/VMAT plans were evaluated with previously developed Radio-Therapy-Plan-Evaluation-Tool (RTPET). The 1.1 constant radiobiological effectiveness for IMPT target was applied in RTPET. The radiobiological evaluation was conducted with tumor-control-probability (TCP), and normal-tissue-complication-probability (NTCP) with physical metrics e.g. Cumulative DVHs, Conformity-Index (CI), Dose-Homogeneity-Index (DHI). **Results:** The presence of metal artifacts barely affected VMAT plans with all metrics within +/- 2.0% differences, while it significantly affected IMPT plans in TCP/NTCP values that could result in tumor control deficiency. Dividing plans with single and bilateral hip-prosthesis, the TCP_CTV for LAT plans varied <10% for single, but > 80% for bilateral. The larger TCP_CTV deviation of bilateral corresponds to > 40 Gy deviation of minimum dose in CTV. The values of NTCP for bladder and rectum for LAT varied < 0.5% for single, but more > 2% for bilateral. The variation of the mean and maximum dose to bladder or rectum were < 2 Gy. **Conclusion:** The presence of metal artifacts without any correction in planning could compromise radiobiological outcomes as well as surrounding normal tissue sparing power of IMPT.

PO-GePV-T-72

Developing a Workflow Using Supercomputer Accelerated Monte Carlo Simulation Based On Machine Log-Files for Patient Specific Quality Assurance of the Mevion S250i Proton Therapy System

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Purpose: The measurement based proton specific quality assurance (PSQA), a time-consuming process, plays an important role for proton therapy. The purpose of this work has been to develop a unique workflow of running machine log-files based Monte Carlo simulation on supercomputer for PSQA of the Mevion S250i proton therapy system. The proposed method not only takes the delivery errors into account but also saves personnel and machine beam-on time. **Methods:** The Mevion beamline was modeled in detail using TOPAS Monte Carlo code. The machine log file was acquired during delivery and converted by our in-house batch file converter to create the input file for running machine log-files based Monte Carlo simulation on supercomputer. The 3D dose was then reconstructed in a solid water phantom. Similar to measurement based PSQA, the 2D Gamma analysis was conducted in three different depths that include isocenter, proximal and distal fall-off and compared with the corresponding doses calculated from the treatment planning system. **Results:** Simulation time for 10⁸ protons on 5x5x5 cm³ uniform cubic plan was reduced from almost 2 days using 10 cores on the local server to 1.9 hours using 20 compute nodes on supercomputer. The 2D dose profile comparison at three different depths showed good agreement between simulation and treatment planning system. The Gamma index passing rates for both one beam 5x5x5 cm³ uniform cubic plan and two beam IROC pelvis plan were passed more than 95% satisfying 4%/4mm

criteria. **Conclusion:** A unique workflow of how to utilize supercomputer for log-file based Monte Carlo simulation has been developed, which can improve both efficiency and authenticity of utilizing Monte Carlo simulation for PSQA of proton therapy within the clinically acceptable level.

PO-GePV-T-73

Analysis of the Difference in Calculation of Physical Dose for Each Patient Site of SECT and DECT Using TOPAS

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Purpose: To calculate the carbon ion dose based on single-energy CT (SECT) and dual-energy CT (DECT) through Monte Carlo (MC) simulation, and analyze the dose calculation results by patient site. **Methods:** We used TOPAS code (version 3.4) for MC simulation and modeling. For carbon ion beam scanning, a 3-D active beam scanning system was modeled. Both SECT images and DECT images were acquired with Philips IQon spectral CT which is a dual-layer detector type. 120 kVp CT images for SECT and effective atomic number (EAN) and electron density (ED) images for DECT were obtained. Then, material information for each pixel was imported into TOPAS for simulation. SECT images were imported through the stoichiometric method embedded in the code, and DECT images were imported through the in-house material converter. After dose calculation, in order to compare the dose calculation results using the two images, integral depth dose (IDD) profiles were obtained for the monoenergetic carbon ion beam irradiated from the lateral direction (case 1) and the perpendicular direction (case 2), respectively. In addition, the dose difference was analyzed after beam scanning, using the modeled beam scanning system (case 3). **Results:** Before evaluating the dose calculation results for various patient sites, one site (abdomen) of the patient CT with an IRB-approved was evaluated as a preliminary result. In Cases 1 and 2, the Bragg peak position of monoenergetic ion in DECT was $-1.6 / +1.7$ mm from the SECT, and the dose difference at the peak was $-3.8 / +3.7\%$ respectively. In Case 3, the dose difference for each pixel was up to 47% from the SECT at Bragg peak. **Conclusion:** The difference in dose calculation for SECT and DECT image was confirmed through preliminary calculation. Dose calculations and analyzes will be performed for various patient sites.

PO-GePV-T-74

A Fast Deterministic Method for Calculating Doses and Fluence Spectra for Proton Beam Therapy

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Purpose: Treatment plan optimization in proton beam therapy involves multiple calculations of spatial distributions of physical dose and relative biological effectiveness (RBE). We develop a novel fast and accurate method for calculation of doses and fluence spectra. Fluence spectra provide physical data that is sufficient for RBE calculations using almost any of the current models. **Methods:** Our method is deterministic, it does not involve any random sampling. It is based on exactly the same physics as the most rigorous Monte Carlo algorithms, which ensures high accuracy of our method. Monte Carlo algorithms rely on multiple scattering models to generate random particle trajectories. Our algorithm uses the same models, but differently, to find the solution using an iterative procedure. By eliminating random sampling, we achieve high computing speed. Our algorithm accounts for energy loss straggling using Vavilov distribution, inelastic nuclear reactions and particle trajectory curvature. **Results:** We have tested our algorithm for 40 to 220 MeV protons propagating in water. The CPU time was 1.6-3.8 sec in Matlab on a quad-core 4.2 GHz processor. In each run our code calculates doses and fluence spectra at ~40 depths. Our method agreed with MCNPX 2.7.0 Monte Carlo code. In the plateau region the root mean square difference in doses (Gy/particle) between our code and Monte Carlo was 0.3%. We did not find the difference exceeding 2% in any of the tests. The maximum distance to agreement near the Bragg peak was 0.8 mm. The maximum difference in the energies of fluence spectra maximum was 0.6 MeV and the maximum discrepancy in full widths at half maximum was 0.9 MeV. These spectra comparisons included depths on both sides of the Bragg peak. **Conclusion:** Our new method achieves high calculation speed with uncompromised accuracy. There is a potential for substantial further improvement of its performance.

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PO-GePV-T-75

Development of a Machine Learning Model to Predict Dosimetric Impact of Proton Spot Delivery Errors

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Purpose: A machine learning model was trained on historic data of spot position errors and MU delivery errors, taken from log files from our Mevion S250i Hyperscan proton therapy system. The output of the model will serve as input into a workflow to predict the spot position errors before treatment and the objective of this work is to assess the dosimetric impact of those errors as part of pre-treatment quality assurance. **Methods:** In previous works, a MATLAB script was developed to extract spot position errors and MU delivery errors from a large collection of log files. This script generated data for approximately 4×10^6 spots from 108 patients. Planned spot positions and Monitor Units (MUs), delivered spot positions and MUs, gantry angle and snout extension were used as input for a supervised machine learning algorithm. The MATLAB Statistics and Machine Learning toolbox was used to train a shallow feedforward neural network with 5 neurons. Seventy percent of the data was used for training, 15% for validation and 15% for testing. **Results:** The mean squared error of the model fit on the testing data was 0.301 and $R=0.99985$. **Conclusion:** The model was trained well, as evidenced by the R value close to unity. Output from this model will serve as input for the next step in a workflow to input the predicted spot position errors and MU errors back into the RaySearch RayStation Treatment Planning System and recalculate the dose on the patient CT. This recalculated dose, based on the predicted beam delivery, will then be compared to the dose calculated with the optimal spot positions and MUs. These comparisons will be reviewed as part of a pre-treatment quality assurance program.

PO-GePV-T-76

Open Source TPS for Bragg Peak Proton FLASH Treatment Plans

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Purpose: FLASH irradiation implies delivering the treatment dose at high dose-rates (typically > 40 Gy/s) to elicit sparing in irradiated healthy tissue. To fulfill the FLASH time constraints and achieve high dose-rates in proton PBS irradiation, a working approach is scanning the pencil-beam in a monoenergetic delivery, whilst ensuring the target coverage within the generated Spread-Out-Bragg-Peak (SOBP). **Methods:** We present a way to generate a proton FLASH treatment plan unraveling the full potential of proton Bragg-peak for high dose-rate FLASH irradiations, alleviating the limitations in the current shoot-through, transmission-FLASH approach that negates the advantage of the Bragg-peak. **Results:** The solution consists in passing the proton pencil-beam at the highest cyclotron energy through a range shifter setting the beam maximum range and through a spatially non-uniform ridge filter (the "hedgehog"). The ridge filter base height controls the distal modulation while the height of its spikes defines the proximal modulation depth at each position, giving a conformal dose distribution over the non-uniform CTV surfaces. The spike shapes determine the SOBP plateau flatness. The ridge filter and range shifter are patient and field specific, responsible in generating the SOBP covering the target volume. Its design is based on the weight of each beamlet in the field, by optimizing a standard IMPT plan using an analytical algorithm in MIROpt (<http://openmiropt.org/>), alongside the MCsquare dose engine (<http://www.openmcsquare.org/>). A

modified MIROpt version optimizes the spot scanning trajectory and computes the percentile dose-rate at each CT voxel. The shape of the ridge filter generated by MIROPT and MCsquare is validated by comparing with the dose distribution computed using MCNPX Monte Carlo. **Conclusion:** An open-source TPS considering the time constraints of FLASH irradiation was designed, based on a Conformal Energy Ridge Filter approach that converts a monoenergetic 230MeV proton beam delivery into an SOBP.

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PO-GePV-T-77

Virtual Particle Monte Carlo (VPMC), a New Concept to Avoid Simulating Secondary Particles in Proton Therapy Dose Calculation

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Purpose: In proton therapy dose calculation, conventional Monte Carlo (MC) simulation may not take full advantage of GPUs because the random secondary-particle-generating processes result in unbalanced workload for different threads of GPUs. Therefore, we propose a novel concept of virtual particle (VP) in MC (VPMC) to avoid simulating secondary particles with clinically acceptable accuracy in proton therapy dose calculation. **Methods:** We interpreted the track histories from a conventional MC code, MCsquare, in a new way (Figure 1). The dose of one primary proton and its secondaries, named realistic particles, can be regarded as if it is deposited by multiple VPs, which results in the same dose distribution as realistic particles. Each VP corresponds to one realistic particle. However, VPs start at the same starting position of the primary proton. After converting the histories of realistic particles into histories of VPs, we can induce a probability model to describe the behaviors of VPs. The model includes a continuing-slowing-down-approximation (CSDA) model and a large angle event model corresponding to nuclear interactions for VPs. In this study, to simplify the calculation, we ignored neutrons and gamma rays, locally deposited the dose of electrons, heavy ions, and nuclear fragments, and converted the tracks of deuterons into tracks of protons. Finally the dose distribution from realistic particles was calculated by simulating only VPs in VPMC, which was then carefully benchmarked with the one calculated by MCsquare. **Results:** The curves of integrated-depth dose, percentage-depth dose, iso-dose contours, and lateral-dose profiles all matched very well (Figure2). The 3D Gamma passing rate with a threshold of 1%/1mm/2%/2mm between VPMC and MCsquare is 97.8%/99.4% in water phantoms and 96.5%/99.4% in inhomogeneous phantoms, respectively. The calculation is highly efficient (1.3×10^7 particles per second with 6 Nvidia T4 GPUs). **Conclusion:** VPMC achieved high accuracy and efficiency in proton therapy dose calculation.

PO-GePV-T-78

Experimental Validation of Clinical Proton Dose Calculation Algorithms for Breast Prostheses and Tissue Expanders

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Purpose: The clinical proton dose calculation algorithms using pencil-beam convolution superposition (PCS) and Monte Carlo (MC) methods in Eclipse™ TPS were validated in a phantom with the metallic port of a Natrelle-133 tissue expander (Allergan, PLC) placed between a silicone-filled prosthesis (Mentor, LLC) and solid water. **Methods:** A proton spot-scanning field was generated to deliver a homogeneous dose to a target volume in the phantom. The metallic port was contoured based on geometrical dimensions provided by the manufacturer. The Hounsfield units (HUs) were assigned based on the corresponding stopping powers from lectures (silicone: 0.94, Titanium: 3.17, magnet: 5.50, solid water: 1.02) for PCS (PCS 15.6) algorithm. The materials used for MC (AcurosPT 13.7) calculation were selected (silicone: wax, Titanium: Aluminum, magnet: Brass) to have the closest stopping powers. The 2D planar doses at multiple depths close to the distal-dose fall off were measured using a MatriXX-PT detector (IBA-dosimetry, Germany) and compared with the calculations. **Results:** The measured 2D planar doses at distal target edge (36.5 mm below the prosthesis) matched both PCS (35.4 mm) and MC (37.4 mm) calculations within 1 mm discrepancy in depth. The depth-doses collected from 2D planar doses behind the breast prosthesis were found to over-range in measurements by 1.0 mm and 0.0 mm versus PCS and MC calculations, respectively. However, a 5-8 mm over-range outside the target volume was observed when protons passed through the metallic port. **Conclusion:** The assigned HU (-124) for silicone-filled breast prostheses with the institutional CT curve was experimentally validated for both PCS and MC algorithms. Limited by the lateral scattering in PCS algorithm and material selections for MC method, the disturbed doses caused by the high-Z metallic port cannot be predicted precisely. A two-field beam arrangement is suggested to average out and confine the uncertain doses inside the tissue expander.

PO-GePV-T-79

Development of a Fast and Memory-Efficient Monte Carlo Code for Proton Dose Calculation

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Purpose: Monte Carlo (MC) code's running times are significantly reduced with the development of general-purpose graphic processing unit (GPGPU) based algorithms. However, some calculations, like dose influence matrix (D_{ij}) calculation or temporal dose calculation, cannot take full advantage of the GPGPU due to the limited memory size and the expensive computational cost of memory transfer. We developed a memory-efficient framework to fully utilize GPGPU within our novel GPU-based MC code for proton therapy, "moqui". **Materials and Methods:** To increase the efficiency of the code, we have adopted the hash table, one of the key-value pair data structures, to the scoring system of our MC code. We put the voxel index in the key and accumulated the voxel's dose deposition in the value. There are two advantages of using the hash table. (1) Only voxels that interact with particles occupy the memory (2) A hash function of a key determines the address of a pair, thus reducing the time for searching over the data structure to check a pair's existence and the existing pair's address. To demonstrate the efficiency of the developed code, we calculated the D_{ij} matrix and compared with an array scoring system, which is a conventional data structure used for scoring. **Results:** For simulating 40 million particles, the hash table scoring system took about 115.9 seconds, while the array scoring system took 151.4 seconds. With the hash table scoring system, we can score every voxel that interacts with particles. In contrast, the array scoring system was only able to score selected voxels using twice of the memory used for the hash table scoring system. **Conclusion:** The MC code with the hash table scoring system runs faster and has better memory-efficiency than the conventional scoring system. It is incorporated in the "MOnte carlo for QUICK proton dose calculation" (moqui).

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PO-GePV-T-80

Three-Dimensional Dose and LETd Prediction in Proton Therapy Using Artificial Neural Networks

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Purpose: The dose weighted linear energy transfer (LETd) is one of the parameters defining the response of the tissue in proton therapy. Estimating LETd requires treatment planning information and time-consuming Monte Carlo (MC) simulations. We expand our reported artificial neural networks (ANN) three-dimensional dose prediction model to include MC dose and LETd. **Methods:** 27 patient subset from an institutional craniopharyngioma treatment protocol comprised the study. All patients received a prescribed dose of 54 CGE. MC dose and LETd for model training were calculated using the Tool for Particle Therapy Simulation. Predicted beam path masks were created by expanding existing clinical target volume contours and raytracing along the beam directions.

Using MATLAB's neural network toolbox, voxel features within the predicted beam path were used to train feed forward ANN models that predicted dose and LETd. ANNs were trained with several network training and activation functions with varying number of deep layers and neurons. The final model is the one with lowest root mean square error (RMSE). **Results:** The final trained model used 3 layers and 25 neurons with RMSE of predicted dose and LETd 5.8 CGE and 0.5 MeV/mm/(g/cm³), respectively. The percentage of voxels within $\pm 5\%$ of the MC calculated values were 70.8% and 50.2% within the CTV, 44.9% and 38.1% 0-5mm from the CTV and 13.5% and 28.9% 5-15mm from the CTV in dose and LETd, respectively. Model agreement trended worse near sharp dose gradients, suggesting spatial uncertainty in the predicted values. **Conclusion:** We implemented an ANN algorithm to predict dose and LETd. The benefits of upfront dose prediction are well documented in the knowledge-based planning community. Supplementing the treatment planning process with an LETd prediction allows for careful consideration of treatment geometries to limit the potential biologic impact of high LETd regions in normal tissues.

PO-GePV-T-81

Systematic Analysis of the Neutron Equivalent Dose in a Mevion S250 Proton Therapy Machine Using Geant4 Monte Carlo Simulation

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Purpose: To study the contribution of different components in a Mevion S250 proton therapy system's vault, including the room, proton therapy machine, and the patient itself, in the neutron equivalent dose. **Methods:** Geant4 Monte-Carlo was used to simulate the room geometry and the proton therapy machine. A $40 \times 40 \times 40$ cm³ water phantom was placed at the isocenter. The measured proton depth dose and lateral profiles were used to benchmark the simulation. Also, the experimental results obtained from a previous study of the neutron spectrum was used for benchmarking. Three different proton energies corresponding to the 1st, 9th, and 18th steps of the range modulation wheel were considered. We utilized the Root data analysis framework to collect the information on individual neutron interactions at any point within the simulation geometry. Such information was used to find the contribution of each object to the total neutron equivalent dose absorbed in the water phantom. The impact of the brass aperture opening on the neutron dose to the patient was also studied. **Results:** Most of the neutron dose (>65%) absorbed to the water phantom representing a patient comes from the internal neutrons generated inside the water phantom. Among the sources of the external neutron dose, the major contributor is the brass aperture followed by the synchrocyclotron body. A small fraction of dose comes from the floor and the ceiling, however, the impact of the walls to the dose was small. A quasi-exponential behavior in the reduction of the neutron dose as a function of the longitudinal distance from the treatment nozzle was noted. A quasi-Gaussian distribution of neutron dose in lateral direction with respect to the nozzle was noted. **Conclusion:** Our results shed light on the origin and energy spectrum of the secondary neutrons in a Mevion S250 proton therapy vault.

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Determination of Plan Deliverability and Treatment Time Estimation for Proton Pencil Beam Therapy Using a Machine Emulator Script

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Purpose: Availability of field delivery time estimate during the treatment planning allows to better assess if a given plan is tolerable for a patient delivery time-wise and can provide an input for patient scheduling in the treatment rooms. It is also desirable to determine the deliverability of plan prior to patient specific QA or the actual treatment. The aim of this study is to evaluate a tool which can determine the plan deliverability and the treatment time of proton pencil beam therapy. **Methods:** Treatment Time Calculator (TTC, Varian) algorithm has been developed and tested to estimate treatment delivery time and deliverability by considering several room-specific parameters such as minimum spot duration, layer switching time, beam currents, and beam line transmission. Fifteen deliverable (46 total fields) and 10 undeliverable treatment plans (47 total fields) of various anatomic sites were selected and tested by TTC tool. The actual delivery time of each deliverable plan were obtained for multiple fractions (average=7) from oncology information system and compared to TTC estimated time. **Results:** The TTC average estimated treatment time predicted to be lower ($-22.7 \pm 5.6\%$) than the actual delivery time as TTC reports the ideal treatment delivery and doesn't consider daily fluctuations of the system. By applying a form factor of 23%, average difference can be reduced to 6%. Based on our daily QA records, daily treatment time can vary up to 15%. For the same plan, both calculated and measured delivery time can vary up to 30% in different treatment rooms. Additionally, prediction of treatment plan deliverability by TTC shows high sensitivity (85.0%) and specificity (99.7%). **Conclusion:** Using TTC, undeliverable plans can be filtered out at the time of treatment planning and prior to delivery, which can eliminate last-minute replanning. Treatment time determination enables proper selection of treatment rooms depending on the plan configuration.

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Characterizing the Entrance Dose Induced by a Dynamic Collimation System for Pencil Beam Scanning Proton Therapy Using a Commercial Multilayer Ionization Chamber

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Purpose: To characterize changes in entrance dose resulting from low-energy scatter induced from the Dynamic Collimation System (DCS), an energy layer-specific collimator designed for low-energy pencil beam scanning proton therapy using a commercially available multilayer ionization chamber (MLIC). **Methods:** Uncollimated integral depth dose (IDD) curves were acquired with an IBA Stingray parallel plate ionization chamber scanned in a water tank and the IBA Giraffe multilayer ionization chamber (MLIC). IDD for beamlets collimated by various collimator orientations were then measured with the Giraffe MLIC. Results were compared to identical simulations performed using a benchmarked Monte Carlo model of the DCS. Simulated IDDs consisted of tallying the dose to water in a 0.1mm mesh with a diameter identical to the 12-cm diameters of Giraffe MLIC and Stingray chambers. Simulated and measured collimated IDDs were compared using 1D overall 0.5%/0.5mm gamma tests. **Results:** For the uncollimated scenarios, the MLIC measurements exhibited a significant decrease in dose at the Bragg peak when compared to the uncollimated Stingray measurements that is most likely due to rise a rapid change in the mass stopping power ratios of water to Giraffe-equivalent material near the end of range. To account for this, uncollimated Stingray and MLIC measurements were then used to derive energy- and depth-dependent correction functions that were then applied to the collimated MLIC measurements. For the pristine collimated beams, 0.5%/0.5mm gamma pass rates increased from 17.5% to 89.6% with the use of the correction. For the range shifted collimated beams, pass rates increased from 39.0% to 95.4%. **Conclusion:** We have demonstrated a need and implemented a method to correct IDDs acquired with an MLIC for the purpose of characterizing changes in entrance dose resulting from collimation.

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Log File-Based Patient Specific Quality Assurance: 5 Years of Experience in Pencil Beam Scanning Pediatric Proton Therapy

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Purpose: To report the trend analysis of an in-house log file-based patient specific quality assurance (QA) method in pencil beam scanning pediatric proton therapy over 5 years from 2016 to 2020. **Methods:** In-house log file-based software was implemented as a part of clinical patient specific QA process to complement 2D array absolute dose measurements. The software extracts beam specific information from treatment delivery log and automatically compares spot MU, lateral position of each spot against the intended values from treatment plan to determine any discrepancies in beam delivery. This software monitors spot width for various energies and gantry angles recorded by spot position monitor against the established baseline at the time of acceptance. Since 2016, the software has analyzed 286 pediatric patients, 476 plans, 1182 fields, and more than 22M proton spots. **Results:** The trend analysis showed that proton system has been very stable in delivering patient QA fields within the proton energies of 69.4-221.3 MeV and MU range between 0.003 and 1.473 MU per spot. The planned mean and standard deviation (STD) of energy and spot MU were found to be 114.5 ± 26.4 MeV, and 0.0108 ± 0.0096 MU, respectively. The mean and STD of spot MU and position differences between delivered and planned spots were $1.49 \times 10^{-7} \pm 2.35 \times 10^{-4}$ MU, and $0.0289/-0.0073 \pm 0.0497/0.0447$ mm in X/Y axes for random, $0.0126/0.1343 \pm 0.1874/0.1758$ mm in X/Y axes for systematic positional differences. The mean and STD of spot width comparison were determined as $0.0927/0.0981 \pm 0.1368/0.1734$ mm in X/Y axes between the acceptance and delivered values. **Conclusion:** We developed an in-house tool to extract crucial information about the stability of the proton pencil beam scanning delivery and monitor the system performance since 2016. Each patient's plan has been verified pretreatment to ensure the delivery accuracy per machine delivery tolerance.

PO-GePV-T-85

Feasibility of Delivering Exchanged PBS Treatment Plans in Two Gantries of Multiroom Proton Therapy System

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Purpose: Inspired by the beam-matching concept in medical linear accelerators, the feasibility of delivering exchanged pencil beam scanning treatment plans without replanning and recalculations in gantries of IBA ProteusPLUS proton system through analyzing the commissioning data and quality assurance (QA) results, is studied in this work. **Methods:** Proton beam characteristics including integral depth dose (IDD) and spot profiles for 33 energy levels (70-226.09MeV) of two gantries (GTR2/GTR4 in our hospital) were measured and established beam models. 36 cases of nasopharyngeal-carcinoma, lung cancer and prostate cancer were retrospectively selected to generate patient-specific QA plans based on the beam model of gantries. Each QA plan was respectively delivered in both gantries, meanwhile, planar dose measurements were done, and were compared and statistically analyzed. **Results:** Through comparison between two gantries' models, their IDD were analyzed. It is found that the spot profiles at each energy level had slight differences. Independent of delivering the same QA plans in GTR2 or GTR4, the average passing rates of measured doses had not changed much, reaching a high level of 95% when using a 3mm/3% criterion. While the paired sample t-test shows that these differences were not statistically significant, the Pearson correlation analysis proved a certain correlation between the passing rates of two gantries. Few abnormal cases could be seen only when the stringent criterion of 2mm/3% was used, but when using the less rigorous 3mm/3% criterion which was widely used in clinical practice, there was no exception. **Conclusion:** The beam dosimetric characteristics in different rooms showed high consistency and the tolerance was kept within a reasonable range while the difference of the same treatment plan delivered on two gantries met the requirement of clinical treatment. It was recognized that the same treatment plan could be delivered directly in different treatment rooms without affecting the treatment efficacy.

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PO-GePV-T-86

Quantitative & Statistical Analyses of Monthly OBI QA on Varian ProBeams

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Purpose: Routine quality assurance of onboard imaging systems (OBI) for proton therapy is a standard practice performed monthly according to the recommendations given in the American Association of Medical Physics Task Group Report No. 224 (AAPM TG #224). Methods utilized to carry out such tests are nearly identical to those in standard photon radiotherapy (AAPM TG #142) for both planar-kV imaging and CBCT imaging, where all measurements taken in subsequent months are with respect to a pre-determined baseline measurement. This study examines the variability of obtained image quality parameters across four proBeam treatment rooms and varying imaging techniques. Furthermore, this study examines the effects of possible artifacts and routine imager recalibration specific to ProBeam systems. **Methods:** ProBeam OBI monthly quality assurance statistics were extracted from myQA and placed into another spreadsheet for statistical analysis. Descriptive statistics were obtained for each imaging parameter, from which a distribution of imaging parameters were obtained. A two way ANOVA test was also performed to compare the distribution of imaging parameters across all four rooms to determine if room dependence existed with respect to image quality. **Results:** Of the tests undertaken, image quality statistics across the four treatment rooms were shown to be fairly uniform per specific room, but varied slightly when compared to each other. This was expected due to variability across rooms as well as initial commissioning data. **Conclusion:** OBI systems for proton therapy are fundamental to ensure proper patient setup and alignment for treatment, and routine quality assurance tests are fundamental for proper safety. Furthermore, the utilization of a succinct SOP ensures that imaging parameters obtained will be as uniform as possible, despite that parameters will most likely be different across rooms.

PO-GePV-T-87

Radiation Therapy Interference with Cardiovascular Implantable Electronic Devices; Comparison of Proton Beam Therapy Versus Intensity-Modulated Radiation Therapy

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Purpose: To determine whether Proton Beam Therapy (PBT) is associated with less radiation dose than Intensity-Modulated Radiation Therapy (IMRT) to cardiac implantable electronic devices (CIEDs). **Methods:** We investigated retrospectively the radiation dose to the CIEDs for 33 patients underwent PBT at our institution between 2010-2020, with targets located close to the CIED, including thoracic, H&N, and esophagus. For each patient, quantitative comparison of dose to the CIEDs between PBT and IMRT was performed. **Results:** The CIED distance from edge of field (50% isodose line) ranged 1.5 cm to 22 cm for PBTs. However, due to spread out of dose in IMRTs, edge of fields was closer to the CIEDs, where the distances ranged 1.4 cm to 18 cm. Our results indicate that CIED dose for both PBT and IMRT of all 33 patients were <2Gy, except one patient that IMRT exceed 2 Gy. For this case, IMRT was delivered as initial plan in first 5 fractions, then switched to PBT. We found that the CIED dose in PBT is negligible compare to IMRT. For PBTs of 33 patients, max and mean doses to the CIED were 52.6 and 4.2 cGy, respectively. However, IMRT delivered overall greater dose to the CIED, in which max and mean doses were 216.9 and 73.4 cGy, respectively. **Conclusion:** Our study revealed that the clinical treatments with PBT delivered lower dose to the CIEDs (with potentially less induced malfunctions) than those with IMRT. Regardless of treatment modality, the CIED dose depends directly on the distance of the device to the radiation

field. However, in PBT, field arrangement is crucial for avoiding irradiation of the device with the distal edge of the Bragg-Peak.

PO-GePV-T-88

Output Estimation by Analytical and Machine Learning Models in a Double Scattering Proton System

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Purpose: This study aims to design an empirical method using the analytical and machine learning (ML) models to estimate proton output in a double scattering proton system. **Methods:** Three analytical models using polynomial, linear fitting, and logarithm-polynomial equations were generated on a training datasets consisting 1544 clinical measurements to estimate proton output for each option. Meanwhile, three ML models using Gaussian process regression GPR with exponential kernel, squared exponential kernel, and rational quadratic kernel were also created for all options combined. The accuracy of each model was validated against additional 241 clinical measurements as testing datasets. Two most robust models were selected, and the minimum number of samples needed for either model to achieve sufficient accuracy ($\pm 3\%$) was determined by evaluating the mean average percentage error (MAPE) with increasing sample number. The differences between the estimated outputs using the two models were also compared for 1000 proton beams with randomly generated range, and modulation for each option. **Results:** The polynomial model and the ML GPR model with exponential kernel yielded the most accurate estimations with less than 3% deviation from the measured outputs. At least 20 samples of each option were needed to build the polynomial model with less than 1% MAPE, whereas at least a total of 400 samples were needed for all beam options to build the ML GPR model with exponential kernel to achieve comparable accuracy. The two independent models agreed with less than 2% deviation using the testing datasets. **Conclusion:** The polynomial model and the ML GPR model with exponential kernel were built for proton output estimation with less than 3% deviations from the measurements. They can be used as an independent output prediction tool for a double scattering proton beam, and a secondary output check tool for cross check between themselves.

PO-GePV-T-89

Clinical Evaluation of Optically Stimulated Luminescence Dosimeters Dose Accuracy in Intensity Modulated Proton Therapy

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Purpose: To evaluate the accuracy of measured dose using optically-stimulated-luminescence-dosimeters (OSLD) for intensity-modulated-proton-therapy (IMPT) plans with and without range shifter, with pencil-beam (PBA) or Monte-Carlo (MC) algorithms, and at various beam angles. **Methods:** OSLDs were calibrated in solid water under two conditions: 1) without range shifter at 1.2 cm depth using a uniform planar dose, and 2) with a range shifter in the middle of a uniform cubic dose at superficial depth (5cm range and 5cm modulation). In both cases, OSLD dose was compared to PBA and MC calculated dose over a dose range of 0.5-10 Gy, RBE. Once calibrated, OSLDs were used to compare PBA and MC calculated doses for uniform fields and clinical breast plans. Angular dependence of OSLD was investigated in a spherical phantom from 0 to 330°. **Results:** A polynomial function yielded the best fit ($R^2=0.9997$) using both calibration conditions. A 2.7% difference between the calibrations with and without range shifter was noted—however, this difference was within OSLD response uncertainties of 2.9%. For uniform fields, the differences of OSLD and calculated doses were 2.5% for MC and 3.5% for PBA. For clinical breast plans, the differences were 5% for MC and 8% for PBA. OSLD response exhibit variations to beam angles within 2%, with the largest discrepancies at 90 and 270 degree where it is parallel to beams. Therefore, the total uncertainties of OSLD dose are 7% for PBA plans, and 10% for MC plans. **Conclusion:** The OSLD calibration conditions with or without range shifter do not significantly differ. A single calibration curve may be used for plans with or without range shifter and for MC or PBA algorithms. OSLDs should be placed en-face to the beam and avoid tangential beams. The overall uncertainties of OSLD are 7% for PBA plans, and 10% for MC plans.

PO-GePV-T-90

Influence of Beam Quality Correction Factors on Measurement Depth in Scanning Proton Beams Using Monte Carlo Calculation

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Purpose: In radiotherapy, determination and management of delivery dose by measuring the absolute dose are important. The reference dose for clinical proton beams is based on ionization chamber dosimetry. However, there are few data on the beam quality correction factors ($k(Q)$) required for calculating the absolute dose for the proton beam. Furthermore, the perturbation correction factor ($P(Q)$) of ionization chambers is assumed to be unity for proton beams in the IAEA TRS-398 CoP. Hence, the aim of this study is to calculate $k(Q)$ and $P(Q)$ for ionization chambers in scanning proton beams using the Monte Carlo simulation (MC). **Methods:** We used PTSIM a particle therapy simulation framework based on MC code Geant4 version 10.05.p01. We reproduced a spot scanning nozzle (PROBEAT-III: Hitachi,Ltd) and calculated the $f(Q)$ and $P(Q)$ for the cylindrical ionization chamber (30013: PTW). f is determined by the ratio of the absorbed dose of water $D(w)$ to dose of cavity $D(air)$ in the chamber. We investigated four different monoenergetic beams and one modulated beam (SOBP) with a range of 12 cm. The former was measured at 2 cm, the latter was measured at various depths with different dose gradient. **Results:** For the maximum difference in monoenergy, the $f(Q)$ between this work and Baumann et al (2020) was 0.4%. The $f(Q)$ of the modulated proton beam at the SOBP center was 1.1249. At proximal region the $f(Q)$ increased up to about 2.0%. Perturbation correction factor $P(cel)$ was about 0.4% different than unity and was independent in the energy and measurement depth. **Conclusion:** We calculated $f(Q)$ and $P(cel)$ for cylindrical ionization chamber in monoenergetic and modulated proton beams. We will investigate other individual components of $P(Q)$ using cylindrical ionization chamber.

PO-GePV-T-91

Adaptation of A commercial Daily QA Device for Daily Spot Positioning QA

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Purpose: For Pencil Beam Scanning systems, AAPM TG-224 recommends measuring the absolute and relative spot positioning accuracy within 2mm and 1 mm respectively. We have adapted a the IBA MyQA Daily commercial daily QA device to measure spot position with the required accuracy. **Methods:** We created a pattern of 5 spots directed at ion chambers on the MyQA Daily Device. The measurement database is queried after spot delivery, and a Gaussian fit is performed to the measured dose distributions to determine the spot locations. The measured spot positions are then compared to the expected and a pass/fail message is presented to the operator. The software used in the analysis was created with C# using open source packages available on GitHub. To test sensitivity, intentional offsets from the expected distributions were delivered to see if the system would reject out of tolerance spots. **Results:** The analysis code run over the data collected by the MyQA daily device was shown pass spot distributions when expected to and to flag mismatches appropriately. **Conclusion:** Compared to dedicated spot measurement systems such as scintillator or aSi panels, use of the MyQA Daily has the advantage of being relatively inexpensive and simpler for operators to use. The myQA Daily device is also used for the daily machine output check with the same setup, speeding up the daily QA measurement compared to using different detectors for output and spot positioning.

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Evaluation of the PRESAGE Dosimeters for Spot- Scanning Proton Therapy

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Purpose: To evaluate the energy and dose rate dependence of the PRESAGE response for the application of the 3D dosimeters to Proton therapy dosimetry. **Methods:** Measurements were made using 4 PRESAGE dosimeters of 10 cm diameter and 10 cm length. Each phantom was setup at a 226-cm SSD and irradiated using 4 single-spot proton pencil beams with intended maximum doses around 1 Gy, 3 Gy, 5 Gy and 10 Gy using Varian ProBeam system. Proton pencil beams with 2 energies (100 MeV and 70 MeV) and 2 dose rates (700 nA and 350 nA Cyclotron currents) were used. The irradiated phantoms were scanned using a newly designed broad-beam optical CT scanner that utilizes fiber optic taper for collimation of transmitted lights. The scanning time for each phantom is about 10 minutes with 1 mm resolution. Dose response curves were generated from each phantom along the central axes of the pencil beams at 3 different depths and analyzed for the energy dependence and the dose rate dependence. Three-dimensional comparisons were also made between the dose distributions from the measurements and the treatment planning system. **Results:** The measured Bragg peak depths for the 100 MeV and 70 MeV beams are 70 mm and 38.0 mm, compared with 75.3 mm and 40.3 mm in water, respectively. The measured water-equivalent thickness is 1.07 which is comparable with the physical density of PRESAGE (1.07 gm/cm³). No energy and dose rate dependence was observed for this PRESAGE formula within the accuracy of the experimental approach. The measured central depth dose curve for each energy agrees well with the planning system for small depths but is in general 10-35% lower than the plan in the Bragg peak region. **Conclusion:** Preliminary results indicate that the Presage dosimeters can be used for proton therapy dosimetry with adequate corrections for the physical density of the dosimeters and the dose quenching effect.

PO-GePV-T-93

Can An Ionization-Chamber Array Be Used for QA of Proton SBRT Beams?

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Purpose: SRS and SBRT are beginning to be used in proton therapy. Currently, there are no criteria for patient specific quality assurance for proton SBRT. This presents certain challenges due to the small size of the fields and high dose fall off. Film provides high spatial resolution and measures absolute dose but can be cumbersome to use. In this study, we look at clinically available detectors and their ability to address issues. **Methods:** Two patients were selected: one with a spinal lesion and one with a brain lesion. The evaluation PTVs were both under 60 cc. Each field was measured with a MatriXX PT (32x32 array, 7.6mm spacing). Additionally, the MatriXX was shifted by half the detector distance in the x and y directions. These two measurements were combined with Matlab code and interpolated to create a measurement with higher spatial resolution. A high resolution relative dose measurement was taken with a scintillation detector (Lynx, IBA). The dose was scaled to the maximum dose of the plane. For volume averaging effects, the planned dose was calculated at a 0.5mm resolution. The dose to each detector was calculated with this fine resolution plane and compared to the MatriXX measurements. **Results:** The three measurement methods showed similar passing rates across all criteria. The MatriXX alone had a 2%/2mm passing rate of 99.74%±0.62% across all fields, the MatriXX with interpolation had a passing rate of 98.16%±0.62%, and the Lynx a passing rate of 97.8%±1.34%. The volume averaging showed an average of -40.22±34.02 mGy across all fields. This corresponds to a 1.67%±1.25% dose difference. **Conclusion:** The MatriXX with interpolation will provide a satisfactory method for measuring small fields with more heterogeneities. This has the advantage over the Lynx by measuring absolute dose. Volume averaging also provides a promising solution to account for heterogeneities.

PO-GePV-T-94

Novel Collimator for High-Throughput Proton FLASH-RT Murine Hemithorax Irradiations

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Purpose: To develop and validate a novel collimator with for high-throughput hemithoracic irradiations of mice using 250 MeV protons delivered at FLASH-RT dose rates. **Methods:** Varian ProBeam 250 MeV FLASH-RT protons have a range in water of ~38 cm. We designed a brass collimator of 7.62 cm width, corresponding to ~44 cm in water equivalent thickness. Six 13 mm apertures (approximate length of the mouse lung) were precision-machined to allow concurrent ipsilateral lung irradiation of six mice while sparing contralateral lung, head and neck, and abdominal organs. Radiochromic EBT-XD Gafchromic film was used to measure radiation dose profiles at various depths of solid water below the apertures to assess radiation penumbra and field size. A radiation survey of the neutron activation of the block was also performed to ensure that the dose exposure to technical staff would not exceed federal dose limits. **Results:** A “halo effect” caused by scatter within the collimator was observed, as the dose at the surface nearest to the collimator edge exceeded the central axis (CAX) by ~50%. The homogeneity of the radiation profile improved with depth in solid water, but at the cost of a wider penumbra. At 20 mm depth, the radiation field edges delivered only 80% of the central axis dose (CAX), but showed a sharp profile edge, with a 80-20% penumbra measuring at ~0.4 mm. The four outer apertures had full width half-maxima (FWHM) of 13.5 mm while the two nearest to the isocenter had FWHM of 13.3 mm despite best attempts to match beam divergence. According to the radiation survey, radiation dose to technical staff would stay below federal limits if the workload stays below 240 Gy/hr. **Conclusion:** We designed and constructed a brass collimator able to perform 6 concurrent hemi-thoracic proton irradiations while sparing other organs.

PO-GePV-T-95

Design, Fabrication, and Characterization of Range Modulators for SOBP Proton FLASH Beams

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Purpose: FLASH radiotherapy, beam delivery at ultra-high dose rates (>40 Gy/s), is an emerging technique to better spare healthy tissues compared to radiotherapy at conventional dose rates (~0.5 Gy/s). The purpose of this work is to design, fabricate, and characterize range modulators to create proton SOBP FLASH radiation fields using a proton therapy synchrocyclotron. **Methods:** The design of the range modulator was composed of arrays of holes with diameter a and spacing between them b on a block of PMMA plastic with thickness t. The parameter t gives the desired modulation depth. The a and b parameters were optimized at each modulation depth to provide the most flat depth dose profile. 3D printing or mechanical machining was used to fabricate the range modulators. A clinical Mevion HYPERSCAN synchrocyclotron was fine tuned to generate ultra-high dose rate proton pencil beams. The range modulator was placed before a boron carbide absorber in the beam path and irradiated with proton pencil beams with nominal energy of 230 MeV. The integral depth dose (IDD) was measured with a PTW Bragg Peak® ionization chamber. The lateral beam profile was measured with radiochromic film. The average dose rate was measured using a PPC05 plane-parallel ionization chamber. **Results:** By selecting proper combination of hole sizes and modulator thickness, SOBP proton fields with >98% depth dose flatness can be generated from a single pencil beam. The optimal hole size depends on the desired modulation depth. By scanning four spots with optimal distancing, lateral flatness >98% over a 1×1 cm² area can be obtained. **Conclusion:** SOBP proton beams can be generated from pencil beams using range modulators composed of arrays of holes on a flat plastic slab. A flat SOBP profile with at least 1 cm³ volume can be generated using such range modulators.

Townsend Zwart, Miles Wagner, and Tucker Evans are employees of Mevion Medical Systems.

PO-GePV-T-96

Dosimetry and Dose Rate Evaluation of Transmission Proton Pencil Beam Scanning FLASH Treatment Planning for Hypofractionated Lung Cases: A Comparison Study with Intensity Modulated Proton Therapy

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Purpose: To date, the plan quality of the transmission FLASH plans is not well defined. This study aims to evaluate transmission proton pencil beam scanning (PBS) FLASH radiotherapy (RT) planning for lung cancer patients. The dosimetry characteristics of the FLASH plans were quantified in comparison to conventional intensity-modulated proton therapy (IMPT)-stereotactic body radiation therapy (SBRT) plans. **Methods:** Proton transmission plans of 34Gy in 1 fraction for a cohort of 7 consecutive lung cancer patients were optimized with 5 fields using multiple-field-optimization (MFO) and an in-house developed tool. 400 minimal MU/spot with a spot peak dose rate (SPDR) of 670Gy/s was used for planning. The 3D average dose rate (ADR) and dose-averaged dose rate (DADR) distributions were calculated. RTOG0915 dose metrics were used to compare the FLASH and clinical lung IMPT plans. **Results:** The FLASH plans resulted in worse CTV Dmax compared to the IMPT plans, as 114.7±4.6% vs 107.7±2.1% (p<0.01). The lung V7.0Gy and V7.4Gy were 717.5±378.0cc and 667.8±359.2cc compared to 488.2±245.5cc and 464.5±231.2cc between FLASH and IMPT plans (p<0.01). The FLASH plans yielded significantly higher (both p<0.05) D5cc (13.9±7.2Gy vs. 6.7±11.5Gy) and Dmax (20.9±13.6Gy vs. 17.1±13.1Gy) for esophagus. Similar trends were observed in the other OARs but did not achieve statistical significance due to the variation of target location and small cohort size. The average ADR V40Gy/s of the OARs is 79.0±3.5%, lower than that using the DADR which is 97.0±0.9%. **Conclusion:** Transmission plans result in inferior plan quality to conventional IMPT due to the nature of transmission beam and less optimization flexibility. A 400 minimal MU/spot can ensure 80-100% OAR FLASH dose rate coverage for both ADR and DADR. The potential biological benefit of transmission FLASH despite its inferior dosimetry requires investigation. New planning strategies and delivery techniques are needed to improve delivery quality of FLASH RT.

PO-GePV-T-97

Optimizing Design And commissioning of a Synchrotron Proton Beamline for FLASH Preclinical experiments

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Purpose: The purpose of this work was to optimize beam shaping devices to produce small volumes (up to ~8 cm³) of uniform dose in a spread-out Bragg peak to achieve ultrahigh dose rates appropriate for FLASH experiments. We modified the characteristics of our ocular beamline to spill the accelerated protons in times as short as 2 milliseconds. **Methods:** Monte Carlo simulations were employed to design custom beam modifiers. These included a scattering foil and a conical flattening filter to maximize the flux of protons into the region of interest, ridge filters, range compensators, and apertures. Shapes, sizes and positions of the last three components were optimized to provide various field sizes and SOBPs in preparation of experiments to be performed. The stability and reliability of the beamline was ascertained, and dose distributions computed with Monte Carlo simulations for each configuration have been validated experimentally. **Results:** With the modified devices we produced circular field sizes of 10, 15 and 20 mm diameter and SOBP modulation widths of 10, 15 and 20 mm. System tests revealed excellent stability and flatness of lateral dose profiles at the center of the SOBP was within ± 3%. Assessment of systematic uncertainties, such as impact of misalignments and positioning uncertainties was performed using simulations and the results used to make appropriate adjustments. **Conclusion:** As a result of this project, the modified beamline is now capable of delivering proton flash beams for in vitro and in vivo experiments.

PO-GePV-T-98

Achievable Flash Dose-Rate by Cyclotron Based Proton Therapy Systems

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Purpose: The current beam configuration use the transition mode of proton Flash beam to irradiate treatment targets. This practice is dictated by the limited capability to the current proton therapy systems to deliver Flash dose-rate over clinically sufficient beam range span. In this study, we investigate the dependency of achievable dose-rate to the main beam parameters for the cyclotron based proton therapy systems. **Methods:** The achievable dose-rate versus the beam energy, the beamline transmission efficiency and the cyclotron's beam current were computed. The beam energy is varied between 70 MeV to 250 MeV (1 MeV step-size), the beamline efficiency is varied from one-tenth to ten folds of the typical beamline transmission at any given proton beam energy. The calculations are repeated for a set of cyclotron beam currents ranging from 100 nA to 1000 nA (100 nA step-size). The maximum achievable dose-rate by the various designs of the commonly used clinical proton therapy systems is indicated thereafter. **Results:** Based on our simulation, the maximum achievable dose-rate is 1080 Gy/sec at the beam of 250 MeV where the machine's beamline transmission efficiency is about 100% and with cyclotron current of 1 µA. For other beam energies, the dose-rate value strongly depends on the beamline transmission efficiency, which varies in average between 10% (high energy) to 0.05% (low energy). In this case, the maximum achievable integral dose-rate is about 110 Gy/sec for 249 MeV and the lowest achievable Flash dose-rate (40 Gy/sec) occurs at the beam energies between 102 MeV and 226 MeV depending on every machine specific beamline transmission efficiency. **Conclusion:** We developed an open source calculator for achievable proton Flash dose-rate with input parameters that can be made specific to any cyclotron based proton therapy facility.

PO-GePV-T-99

The Software Controller of Isocentric Rotation Chair Positioner for Treating Patients with Head/Neck Cancers in Fixed Carbon-Ion Beamline

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Purpose: To utilize an isocentric rotating chair as a positioner for a clinical application, a software controller was built by implementing the mathematical manipulations of specific coordinate transformation to the IEC fixed (If) reference coordinate for each movement mechanism of chair. **Methods:** A software controller to execute any required displacement under a treatment workflow was evaluated. The three major motion mechanisms in the chair positioner are a 360° rotating platform, a Hexapod, and a 3D translation module. To execute a 6 DoF displacement by the controller over the chair, the mathematical manipulations was developed in a matrix formula in the If coordinate through a proper transformation on each mechanism. The trend of deviations on the displacements between ideal and misaligned mechanism for each parameter was investigated and measured by using an optical tracking system. The tolerance of each parameter was investigated to achieve the accuracy of +/- 1.0 mm in translation and +/- 1.0° in rotation on any required displacement executed by the chair positioner. **Results:** Developed mathematical manipulations in the matrix formula with measured deviations of parameters were integrated into a visual user interface. When a large deviation of 6 DoF displacement occurred due to misalignment of any mechanism that were reproducible in the Validation-and-Verification (VnV) procedure, the reproducible deviations were internally corrected in the Hexapod. A commissioning was conducted to meet clinical requirements after the VnV procedure. 17 patients was successfully treated by using this chair positioner. **Conclusion:** The chair positioner using the software control with implemented mathematical manipulations was embedded to conduct a clinical trial under the workflow with an existing image-guided treatment system. The deformation of tilted posture on the Hexapod and/or the 3D platform will be further investigated as one of concern for using over 5° tilt fields.

PO-GePV-T-100

3D Thoracic Motion Phantom Study for Interplay Effect of Pencil Beam Proton Scanning

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Purpose: Interplay effect which causes dose degradation from intended delivery depends on a number of factors including proton beam spot size in a pencil beam scanning (PBS) proton therapy. This study accesses interplay effect for different range of motion amplitude with a three-dimensional thoracic phantom using EBT3 films. **Methods:** The study involved Mevion S250i with Hyperscan PBS beam delivery system with spot size relevant to this research from 17.4mm ($\approx 33\text{MeV}$) to 10.4mm ($\approx 120\text{MeV}$). Four-dimensional computed tomographic scan of QUASAR respiratory motion phantom was obtained for 6-, 10-, and 14-mm motion amplitude. Three respiratory gated and nine free-breathing (FB) plans with volumetric repainting (VR) number of one, three and five for three motion amplitudes were created using appropriate scans. The phantom used 15 breaths per minute to mimic adult breathing pattern and gated treatment plan used 20%-40% gating window for planning and delivery. Monte-carlo optimization and dose calculation algorithm were used. Dose distribution of the irradiated EBT3 film was compared with planning using three gamma analysis criteria: 3%/2mm (Task Group 218), 3%/3mm (clinical), and 7%/5mm (IROC lung) with passing criteria of 90%, 95%, and 85%, respectively. **Results:** Gated treatment for three motion amplitudes results in more than 95% of gamma passing rate for all gamma criteria studied. The 6-mm and 10-mm FB motion amplitude meets gamma passing criteria but 14-mm did not with 82.2% and 90.5% for 3%/2mm and 3%/3mm, respectively. As the number of VR increases, gamma index passing percentage improves as a general trend for all FB motion amplitudes. **Conclusion:** With large proton beam spot size, PBS technique can be implemented for moving thoracic tumors with an acceptable accuracy up to 10 mm motion amplitude. Respiratory gated and VR appeared to decrease interplay effect and can thus increase tumor motion treatable amplitude range beyond 10 mm.

PO-GePV-T-101

Experimental Assessment of Conformal 4D-Optimization in Lung Cancer Particle Therapy

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Purpose: For this study, we compared treatment quality and projected treatment outcomes for two approaches to treating moving tumors with ion beams: synchronization of delivery to the motion of a moving target, and delivery to a motion-encompassing internal tumor volume (ITV) with rescanning. **Methods:** We created 4D optimized treatment plans for proton and carbon ion beams. Two 4DCTs of NSCLC patients were used to generate several treatment plan types: motion-synchronized deliveries, ITV treatments with rescanning, static deliveries and uncompensated deliveries. Each plan was delivered with the same dose delivery system. Dose delivery data was measured with a 2D ionization chamber array detector that was mounted to a linear stage, moving with periodic respiratory-like motion. Likewise, beam monitoring data from log files was used to reconstruct the delivered dose distributions in water and in patient 4DCTs, to compare to measurements and assess delivery quality. Conformity, uniformity, tumor coverage, and extent of hotspots within the target volume were assessed, and projected clinical outcomes (normal tissue complication probability (NTCP) and tumor control probability (TCP)). **Results:** The motion compensated and static treatments showed acceptable dosimetric performance, and dose coverage was above 95% for all these treatments. The greatest conformity (62 %) was achieved with the motion-synchronized approach, while the lowest was observed for ITV rescanned treatments (49 %). However, higher overdose and lower homogeneity were observed for the motion-synchronized delivery approach due to radiation leakage during beam gating. Despite this, projected therapeutic outcomes favored the motion-synchronized treatment approach, with the highest equivalent uniform dose and the lowest NTCP. **Conclusion:** Our findings strongly suggest the feasibility to implement a motion-synchronized treatment strategy, using an M-DDS that provides superior organ sparing compared to ITV based rescanning.

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PO-GePV-T-102

Practical Implementation and Dosimetric Implications of a Novel Dose Painting Delivery Strategy: Minimal Energy Modification (MEM) During Proton Therapy of Thoracic and Abdominal Tumors

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Purpose: Dose painting in proton treatment is often implemented by splitting a field into two or three, which increases number of beams, resulting in prolonged treatment duration. This study examines feasibility and dosimetric implications of a novel dose painting delivery strategy with reference to non-dose painting (NDP) based planning for proton therapy treatment plans of mobile tumors. **Methods:** A practical method to implement dose painting is to split layers of proton beam in a field rather than split the field. To make this field deliverable, the energies of the split layers are minimally altered from the NDP plan (a difference of ~ 0.1 MeV per energy layer and delivery of a fraction $\sim 50\%$ of the MU's per painting iteration). Above alterations were made after an optimized NDP plan was approved clinically for treatment. Dose was then recalculated to yield a minimal energy modification dose painting (MEM-DP) based plan. Dosimetric comparison of the NDP and MEM-DP based plans was performed retrospectively on 10 patient cases to ensure there were no significant differences clinically. **Results:** Target (CTV) D95 for the NDP and MEM-DP plans differed from each other by $0.1 \pm 0.1\%$, indicating good agreement. No significant differences in normal tissue sparing were observed. Total average savings in beam delivery time with MEM DP approach was ~ 4 minutes, with a min and max of 55 seconds and 14 minutes per treatment session respectively. **Conclusion:** It is feasible to deliver MEM-DP based treatments and achieve savings in beam delivery time within our existing clinical environment without any significant changes to NDP based plan quality.

PO-GePV-T-103

Dosimetric Comparison of Intensity Modulated Radiation Therapy (IMRT), 3D Conformal Proton Therapy and Intensity Modulated Proton Therapy (IMPT) for Metastatic Brain Cancer

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Purpose: To compare IMRT against proton treatment (conformal or IMPT) plans for metastatic brain cancer. **Methods:** Ten IMRT patients with brain cancer were retrospectively chosen, all of which received doses in the range of 20 to 40 Gy and in 3 to 5 fractions by Varian TrueBeam STx. Three proton plans (conformal: double scattering, static-IMPT: single collimation, and dynamic-IMPT: layer-by-layer collimation) were generated for the same patient using Mevion S250 (conformal) and S250i (IMPT) to compare. Each plan had respective treatment planning systems: Brainlab iPlan for IMRT, Varian Eclipse for proton conformal, and RaySearch Raystation for IMPT. Dosimetric comparisons were carried out through dose-volume histogram analysis of the target and the organs at risk. **Results:** Average target dose conformity indices are 2.42, 2.79, 2.25 and 2.30 and homogeneity indices are 1.61, 5.08, 8.05, and 9.07 for IMRT, proton conformal, static-IMPT and dynamic-IMPT, respectively. Average maximum target doses differ with respect to prescription dose by 1.19%,

5.2%, 8.19%, and 9.26% for IMRT, proton conformal, static-IMPT and dynamic-IMPT, respectively. Normal tissues are in general spared more on average with proton compared to photon treatments. The following percentages represent average maximum dose relative to prescription dose to the brainstem and optic chiasm, respectively. IMRT: 37.61% and 14.86%, Proton Conformal: 22.18% and 10.89%, IMPT Static: 28.12% and 12.93%, IMPT Dynamic: 27.27% and 14.08%. **Conclusion:** Target dose homogeneity was better with IMRT compared to proton, whereas overall dose conformity was found somewhat better with IMPT than IMRT. Normal tissue doses were better for both conformal proton and IMPT compared to IMRT when the tissues are not in close proximity to the target.

PO-GePV-T-104

Dosimetric Comparison of Volumetric Modulated Arc Therapy (VMAT), 3D Conformal Proton Therapy and Intensity Modulated Proton Therapy (IMPT) for Lung Cancer

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Purpose: To compare VMAT against proton treatment (conformal and IMPT) plans for lung cancer. **Methods:** Ten patients with lung cancer were retrospectively chosen, each of which received doses of 55 Gy in 5 fractions using 6X photon beam from Varian TrueBeam STx machine with VMAT technique. The treatment plans were done using Varian Eclipse treatment planning system (TPS). Two proton treatment plans (conformal: double scattering, and static-IMPT) were subsequently generated for each patient using Mevion S250 and Mevion S250i for conformal and IMPT, respectively. Varian Eclipse TPS was used for proton conformal, and RaySearch Raystation was used for IMPT. Dosimetric comparisons were carried out through dose-volume histogram analysis of the target volume and the organs at risk. **Results:** Average target dose conformity indices are 2.90, 2.70, and 2.0 and homogeneity indices are 3.2, 6.1, and 11.4 for VMAT, proton conformal and static-IMPT, respectively. Average maximum target doses differ with respect to prescription dose by 6.8%, 9.1%, and 13.3% for VMAT, proton conformal, and static-IMPT, respectively. Normal tissues are in general spared more on average with proton treatments when compared to photon treatments. The following percentages represent average maximum dose relative to prescription dose to the spinal cord, uninvolved lung, and heart, respectively. VMAT: 15.8%, 20.6%, and 15.5%; Proton Conformal: 3.3%, 5.2%, and 15.1%; and IMPT: 6.1%, 5.3%, and 12.3%. **Conclusion:** Target dose homogeneity was better with VMAT compared to proton treatments, whereas overall dose conformity was found to be somewhat better with IMPT than proton conformal and VMAT. Normal tissue doses were better for both conformal proton and IMPT compared to VMAT, with IMPT being the best.

PO-GePV-T-105

Dose-Averaged LET Evaluation in Head and Neck Adenoid Cystic Carcinoma Patients for Carbon-Ion Radiotherapy

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Purpose: In carbon ion radiation therapy (CIRT), the dose-averaged LET (LET_d) is one of the important factors in determining clinical outcomes. In order to investigate the local control of adenoid cystic carcinoma (ACC) and the optic neuropathy of optic nerve (ON)s in CIRT, the effects of the LET_d were evaluated clinically. We propose novel tumor control probability (TCP) and normal tissue complication probability (NTCP) models with D90-value of RBE-weighted dose and LET_d as clinical parameters. **Methods:** Total 99 patients with head and neck ACC who received CIRT at the National Institute of Radiological Sciences were investigated. The physical dose and LET_d distributions for each patient were recalculated using in-house treatment planning system. For ACC tumor, minimum-value of dose-averaged LET and D90-value of RBE-weighted dose as an indicator of evaluation were estimated. For ONs, maximum-value of LET_d and maximum-value of RBE-weighted dose were evaluated. **Results:** For each prescribed RBE-weighted dose of 57.6 and 64 Gy (RBE), no statistical significance for local control by RBE-weighted dose was found, but for 64 Gy (RBE), we observed the LET_d may be a marginally significant. For ONs, it was obvious that the optic neuropathy had a greater effect from RBE-weighted dose rather than dose-averaged LET. The proposed models could be applied to both TCP and NTCP prediction, and was found that the clinical outcomes from low to high LET_d region were predictable. **Conclusion:** We analysed clinically the physical dose and LET_d distributions for 99 patients with head and neck ACC. Our results indicated that LET_d may be a determining factor for local control for ACC tumors. For ON, the RBE-weighted dose was related to optic neuropathy rather than LET_d. The proposed approach may be helpful to predict whether LET_d will affect clinical outcomes and be useful when constraining not only RBE-weighted dose but also LET_d.

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PO-GePV-T-106

Dose to Cardiac Implantable Electronic Devices in Patients Receiving Proton Beam Therapy

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Purpose: To investigate the dose to Cardiac Implantable Electronic Devices (CIEDs) for patients receiving proton beam therapy (PBT). We also characterized major factors contributing to delivered dose to the CIED in PBT. **Methods:** 118 adult patients with the CIED who received PBT- either pencil beam scanning or double-scattering- for brain, Head&Neck, thorax, abdomen, pelvis had their treatment plans and delivery information retrospectively analyzed. Eclipse treatment planning system (TPS) was utilized for proton planning in this study. Patients without the CIED included in their simulation computed tomography scan were excluded from studies investigating the distance of the CIED from 50% and 5% isodose lines. **Results:** Patient information including age, gender, anatomical site treated, OSLD dose, TPS estimated dose to the CIED, proton beam delivery technique, and fractionation scheme information is tabulated. Outdata show that the CIEDs receive negligible to no dose when the device is at medium/significant distance from the treatment area, with its relative location to the treatment volume being the most influential factor for evaluating the dose (both estimated and measured). **Conclusion:** For PBT, OSLD and TPS assessed dose to the CIEDs does not have a level of accuracy to draw any significant conclusions. It is generally seen that planning an appropriate treatment with effort taken to avoid the CIEDs will result in doses that do not exceed or approach the 2 Gy limit. Thus, it is our proposal that patients with a CIED receiving proton therapy be managed in proportion to their dependence on their cardiac device and in line with recommendations from TG-203.

PO-GePV-T-107

Dosimetric Comparison for Unfavorable Early Stage Lung Cancer: Intensity Modulated Proton Therapy Versus Volumetric Arc Photon Therapy

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Purpose: Stereotactic body radiotherapy is now the standard of care for management of early stage lung cancer; however, for exceptionally large or ultra-central tumors a less aggressive hypo-fractionated approach (60Gy, 15 fractions) is available. We sought to identify patients previously treated with hypo-fractionated volumetric arc photon radiotherapy (VMAT) that may benefit from proton therapy and identify what organ-at-risk (OAR) dose reductions could be statistically significant. **Methods:** Ten non-small cell lung cancer patients with either large (PTV largest diameter > 5cm) and/or ultra-central (7 patients) treatment volumes treated with hypo-fractionated (60Gy, 15 fractions) VMAT were re-planned using 3 to 5 field pencil beam scanning (PBS) proton therapy using the same fractionation. The iGTV was robustly optimized (multi-field) using 3.5% range uncertainty and 5mm isocenter shifts. The PTV was nominally optimized and field-specific targets were generated for spot placement. Planning was performed on the average CT and forward calculated on the exhale and inhale phase CTs to check coverage stability. Statistical analysis was completed using a 2-sided paired t-test for each OAR. **Results:** The spinal cord (D5cc),

esophagus (D0.03cc, D5cc), trachea (D10cc), bronchus (D10cc), and Lungs-iGTV (mean) dose differences were all found to be statistically significant ($p < 0.05$) showing improvement with protons. All patients also had reduced mean doses to heart, great vessels and ipsilateral lung. Five patients had not met the chest wall constraint for the clinical plan due to PTV overlap and the proton plan was able to significantly ($p = 0.03$) reduce the dose (D50.00cc) to the chest wall for these patients (average reduction 3.3Gy). The proton plans were able to meet conformity index and gradient measure metrics and coverage remained stable on phase CTs. **Conclusion:** PBS proton therapy could improve dose metrics for patients whose anatomy is unfavorable to meet hypo-fractionated planning constraints using VMAT, particularly those near the chest wall.

PO-GePV-T-108

Effective Treatment Plan Integrity Verification for Passive Scattering and Pencil Beam Scanning Proton Therapy

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Purpose: The majority of the automatic error detection studies are focused on photon radiation therapy. Few have been reported in the proton community. The purpose of this study was to develop and implement a framework for automating aspects of the initial physics plan check for passive scattering and pencil beam scanning (PBS) proton radiotherapy, with the overall goal of improving workflow efficiency and increasing error detection. **Methods:** A RayStation script, Auto-IniCheck, was developed based on a set of comprehensive checklists designed to standardize the initial physics plan check and incorporate recommendations from AAPM's TG-275 report. The checklists are categorized by treatment site and proton modality. Specific items are characterized into beam set independent and dependent groups. The former includes items related to patient assessment and data management; the latter contains items related to patient modeling, plan design, optimization, and evaluation. Each item that could be automated was included in Auto-IniCheck. Items that could not be automated are highlighted for manual review. The results are summarized in a PDF report displayed to the user. **Results:** 75%-80% of checklist items could be automated. The script takes less than 15 seconds to run and drastically decreases the amount of time needed to check each item. Additionally, Auto-IniCheck enables the detection of specific errors that would not have been readily apparent through manual reviews, such as validating the isocenter coordinates truncated to one decimal place or that PTV margins match institutional protocols. **Conclusion:** A new script, Auto-IniCheck, was developed to automate several aspects of the initial physics check for passive scattering and PBS proton plans. By improving the efficiency of the review and the visibility of specific errors, the system enhances the plan check. It allows the reviewer more time to focus on critical areas such as plan quality and clinical decision making.

PO-GePV-T-109

Dosimetric Comparison of Pencil Beam Scanning Proton Plans with Different Aperture Modes

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Purpose: The Mevion S250i HyperScan proton system employs a mini-MLC (Adaptive Aperture (AA)) to trim lateral spots for sharpened penumbra, which can be achieved through a static single layer trimming (SA) or dynamic layer-by-layer trimming (MLA). This study investigates potential dosimetric benefits that maybe offered by MLA in pancreatic cancer treatment. **Methods:** Six clinical hypofractionated pancreatic treatment plans were retrospectively re-planned using these two different spot trimming techniques. Raystation TPS (V.10a) was used for planning, which models the AA either as a single static collimator in SA or multiple collimators of varying sizes in MLA. Treatment plans, which consists of 2 to 3 beams, were optimized to meet the clinical goals at the minimum and further optimized for additional improvements possible. Robust optimization incorporating a 3.5% range uncertainty and 5 mm position uncertainty in all directions were performed. Isodose distributions and DVHs for targets and OARs were compared. Conformity index (CI) of the 50% isodose line and a dose gradient index (V20%-V80%) were calculated and compared. **Results:** While both SA and MLA plans meet the clinical goals for similar CTV coverage, on average, plans with MLA achieved better sparing of GI OARs (e.g., 20% reduction of V15Gy for small bowel). No statistically significant difference was observed for the CI. The SA plans yielded slightly better (4%) gradient index. **Conclusion:** For pancreatic cancer treatments using the Mevion HyperScan proton system, the benefits of MLA were demonstrated in reduced bowel dose. Benefits in other anatomical sites will need further evaluation and are likely site dependent.

PO-GePV-T-110

Systematic Study of Penumbra Reduction with Adaptive Aperture for Hyperscan Spot Scanning Proton Beams

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Purpose: To quantify lateral penumbra reduction of field with adaptive aperture (AA) compared to the same field without AA for the Mevion HyperScan proton system under different energies, air gaps, field sizes, and depths. **Methods:** RayStation Monte Carlo proton TPS was used to optimize various single energy layer plans without AA to thin target of size of 3x3, 10x10, and 18x18 cm at ISO. The same plans were re-calculated with AA that conforms to the known field size. EBT3 film and a PTW Octavius 1500XDR detector were used to measure lateral penumbra at the depth of Bragg peak for energies of 28, 120, 180, and 227 MeV. For 10x10 field of 120 MeV, plans with varied air gaps of 1.4, 5, 10, 15, 20 cm were created and measured. For 10x10 fields of 120 and 227 MeV, we also measured penumbra at surface and other 3 intermediate depths. **Results:** Penumbra reduction decreases greatly from 78% to -4% as energy increases from 28 to 227 MeV. A change of 55% to 47% is observed for air gap of 1.4 to 20 cm for 120 MeV. There is no clear field size dependence except for 227 MeV where the change of 3.5% to -11% for field of 3 to 18 cm indicates no benefit with AA for the field larger than 6 cm. The smallest penumbra of 1.2mm is observed at surface of 227 MeV field with AA where the spot size is the smallest too. Consistent with energy change, penumbra reduction decreases linearly as depth (or range) increases. There exists a turning point of 29.5cm (216MeV) after which the penumbra with AA becomes worse than without AA. **Conclusion:** The significant penumbra reduction is observed for low energy or shallow depth. A mild advantage with AA is demonstrated by reducing air gap.

PO-GePV-T-111

Reducing Beam Delivery Time for Proton PBS Treatment with Breath-Hold Motion Management Technique

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Purpose: Breath-hold is an effective motion mitigation strategy to reduce tumor motion interplay effect with active scanning pencil beam proton therapy. This work explores the possibility of reducing beam delivery time by decreasing the number of energy layers or proton spots. **Methods:** Ten plans were re-created in a commercial TPS for 5 esophageal and liver cancer patients, with the same parameters as the clinical plans (Plan_C), except for the energy layer spacing (ELS) or the spot spacing (SS). In Plan_C, ELS and SS were usually set to values smaller than the system default. For each Plan_C, a Plan_ELS was optimized with the energy layer spacing changed to the default, i.e., fewer number of energy layers, while keeping all other parameters unchanged. Similarly, a Plan_SS with setting the spot spacing to default. All 15 plans were delivered on a clinical PBS proton machine and the beam delivery time was recorded for each field. **Results:** Compared with the Plan_C, both Plan_ELS and Plan_SS achieved the same target dose. The Plan_ELS also had near identical OAR doses, while Plan_SS had slightly higher doses to selected OARs. Recorded beam delivery times were 55.5±11.9 s, 43.6±9.0 s, and 53.7±10.7 s for Plan_C, Plan_ELS, and Plan_SS, respectively. Number of energy layers were reduced by 34.9±4.4% in Plan_ELS; as a result, the beam delivery times were 21.4±4.0%, or 12.0±3.7 s, less than Plan_C. The Plan_SS had 26.5±4.8% fewer number of spots, while the delivery times were reduced by 2.9±2.6% (1.8±1.7 s) compared to Plan_C.

Conclusion: Although the spot spacing setting only had a marginal improvement on beam delivery time; using less energy layers can significantly reduce the beam delivery time without compromising plan quality. Future investigation will focus on achieving the optimal trade-off between plan quality and beam delivery time.

PO-GePV-T-112

A Novel Worst-Case Robust Optimization Method That Combines Geometric and Target Volume Definition Uncertainties

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Purpose: A fixed margin expansion (GTV-to-CTV) holds some limitations when the uncertainty of target volume definition is considered explicitly in the treatment planning process. For instance, worst-case scenarios can become unnecessarily improbable when a fixed GTV-to-CTV expansion is combined with extreme geometric uncertainty scenarios. This work proposes a statistically sound worst-case approach that combines target volume uncertainties with geometric uncertainties. **Methods:** The method consists of two steps: (1) scenario definition and (2) robust optimization. (1) The entire space of uncertainties (so-called scenario space) is considered by sampling n error scenarios according to their uncertainty distributions. This includes a dimension with tumor infiltration errors to account for the uncertainty in target volume definition. Infiltration errors are modeled by isotropically dilating the GTV, followed by correcting for anatomical barriers. The scenario space is subsequently partitioned in k -equiprobable clusters with a k -means clustering algorithm. Finally, the optimization scenarios are defined as the k cluster centers. (2) During optimization, the 10% worst scenarios are discarded in order to guarantee a 90% confidence interval in dosimetric space. Minimax is performed with the 90% best scenarios, re-evaluated at each iteration. The proposed method ($n = 1000$, $k = 60$) is tested for a lung tumor patient (60 Gy prescription), treated with IMPT. The results are benchmarked against regular CTV-based worst-case optimization. The evaluation procedure consists of recomputing the dose for 250 scenarios after sampling tumor infiltration errors, setup errors, range errors and breathing motion. **Results:** The proposed method achieves a reduced esophagus dose (-7.9 Gy of D2) whilst maintaining similar levels of robustness: target worst-case D95 of 57.4 Gy and 57.6 Gy for regular method and proposed method, respectively. **Conclusion:** A novel worst-case approach is proposed that allows for improved control over the confidence interval with improved OAR sparing.

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Monte Carlo Modeling of a Proton Portal Imaging System Using in Vivo Generated Neutrons

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Purpose: To present a proton portal imaging (PPI) system which uses in situ generated spallation neutrons to produce images of patient anatomy; and to provide an accurate Monte Carlo model which enables the optimization of the detector placement and geometry, as well as to provide spectral data on the spallation neutrons and photons in order to ensure optimal detection efficiency. **Methods:** The proposed PPI system will make use of scintillation cameras (modeled as LiF-ZnS) to capture the in vivo generated neutrons. An MCNP 6.1 model was constructed to optimize the detector placement, geometry, and to ensure adequate sensitivity to neutrons while also displaying low sensitivity to gamma radiation. This model also serves to generate modulation curves, which can be compared to measurements to determine the limiting spatial resolution for the system. A line pair phantom was constructed with calcium bars of thicknesses and spacings from 2-8 mm to determine the limiting spatial resolution. The Monte Carlo model was compared to measurements obtained in a clinical double-scattering proton system with a typical treatment field. **Results:** Modulation plots are presented for the limiting bar thicknesses and spacings. The bar pattern is indiscernible at about 4-5 mm, which is the limiting spatial resolution of the system. Photon spatial intensity maps are also presented to verify that neutron interactions in the scintillator dominate photon interactions, as these are the interactions of interest in this study. Neutron and photon spectra plots were also produced in the phantom to study the sensitivity of detector placement on neutron capture efficiency. **Conclusion:** The limiting spatial resolution for the proposed PPI system is between 4-5 mm. The interaction of neutrons in the scintillator outweighs the contribution of photons to the acquired signal, which is essential for PPI using in vivo generated exit neutrons.

PO-GePV-T-114

Initial Experience with a Head and Neck RapidPlan Model for Pencil Beam Scanning Proton Therapy

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Purpose: Varian Eclipse 16.1 includes the capability to create a proton RapidPlan (RP) model. We tested the feasibility of using RP for generating clinically appropriate pencil beam scanning head and neck treatment plans. **Methods:** Twenty-six left, right, and bilateral head and neck cases treated at our center were used to create the RP proton model. Five of the cases that were used for training and two cases not included in the training set were re-planned with RP for model validation. The RP model was trained without optimization and ring structures. Additionally, the RP cases were run only once using the same beam arrangement as the clinical plan. The re-planned and dosimetrist-planned cases were blinded and evaluated by an academic radiation oncologist specializing in head and neck cancer and proton therapy. **Results:** While the 7 RP model plans passed our institutional plan and dose criteria, the radiation oncologist preferred the dosimetrist planned case over the RP in all 7 cases. Five of the 7 RP plans were determined to be clinically appropriate and not meaningfully different than the clinical plan. Two RP plans were deemed to require minor edits; no RP plans were judged inappropriate or to require major edits. Dose homogeneity and conformity were identified as the primary distinctions between the planning methods. **Conclusion:** While the RP cases were not chosen as the preferred final plan, the model has demonstrated significant potential as a consistent starting point for the dosimetry team. The RP model also offers substantial time savings as it does not require generation of optimization structures. The relatively minor issues noted by the radiation oncologist were parameters that can be added to the model or addressed by a dosimetrist on editing the RP. Additional investigation is warranted into the clinical role of this tool.

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Should We Be Considering Biological Robustness in Proton Therapy?

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Purpose: Treatment planning strategies that minimize end-of-range uncertainties associated with high linear energy transfer (LET) in proton therapy are needed. We replanned clinical cases to assess changes in LET and compromises to dose. **Methods:** Four patients were chosen to represent various brain tumor shapes, locations, and proximity to the brainstem. Treatment plans were developed for each patient by changing one of the following factors from the clinical plan and re-optimizing: optimization method (i.e., single field versus multi field), spot restrictions at end range, beam orientation, number of beams, and

scenario-based robust optimization parameters. Including the original clinical plans, a total of 28 unique treatment plans were used to calculate dose and dose-weighted LET (LETd) using TOPAS Monte Carlo code. Dose distributions with a constant RBE of 1.1 were assessed to judge clinical feasibility of the plan. Volume of LETd exceeding 6 keV/micron in regions receiving greater than 80% of the prescription dose, defined as a high-risk volume, mean LETd, and dose-times-LETd were used to assess changes in potential biological implications. **Results:** Clinically acceptable plans were achieved for all strategies in 3 of 4 patients. Planning strategies inconsistently affected the high-risk volume (range = 0.3 to 8.1cc) and dose-times-LETd distribution in normal brain. Mean LETd in brainstem was higher in the modified plans by an average of 1.1 and 1.9 keV/micron when adding a third beam or decreasing the intersection angle between beams, respectively. The maximum change in the brainstem among all other techniques was 0.3 keV/micron. **Conclusion:** Forward-based strategies to mitigate LETd hotspots are patient dependent, but generally possible in a way that does not compromise dose distribution quality. Beam geometry is a greater predictor for undesirable LETd placement than optimization parameters or restricted spot placement. We recommend the use of two parallel-opposed fields in the treatment of cranial targets.

PO-GePV-T-116

Assessment of A Knowledge-Based Planning for Robustly-Optimized Intensity Modulated Proton Therapy for Head and Neck Cancer

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Purpose: Knowledge-based planning (KBP) tools, which incorporates prior treatment planning experience, has the potential to improve the quality and consistency of treatment plans. In this work, we evaluated the performance of a proton-specific KBP model in the generation of robustly optimized IMPT (RO-IMPT) plans for treatment of head and neck (HN) patients. **Methods:** Sixty head and neck cancer patients previously treated with volumetric modulated arc therapy (VMAT) were selected and replanned with RO-IMPT. Plans were robustly optimized using ± 3 mm setup uncertainty (in cardinal directions) along with $\pm 3\%$ proton range uncertainty. Fifty IMPT plans were selected to build the HN proton KBP model library. The remaining ten patients were used for model validation. The model quality was assessed using model generated plots such as DVH plots, regression and residual plots. The comparison of the dosimetric indexes for target and OARs between KBP generated and expert plans were performed by two-sided paired t-test, $p < 0.05$ indicating significance. **Results:** Compared with expert plans, the nominal KBP plans delivered significantly lower doses to almost all OARs ($-4.9\text{Gy} \pm 2.74\text{Gy}$ for larynx Dmean, $-6.16\text{Gy} \pm 3.87\text{Gy}$ for constrictor Dmean, $-6.84\text{Gy} \pm 4.65\text{Gy}$ for spinal cord Dmax, $-7.06\text{Gy} \pm 4.66\text{Gy}$ for left cochlea Dmean) except the mandible and the oral cavity which showed no statistically different significance. When including all uncertainty scenarios, the target coverage was better in the KBP plans ($4.59\% \pm 7.58\%$ for CTV70 V100, $4.07\% \pm 5.84\%$ for CTV56 V100). **Conclusion:** This work demonstrated that the proton-specific KBP model was able generate plans with similar target coverage and much better OAR sparing as compared to the ones generated by the expert for HN patients. The HN KBP plans were also more robust and homogenous. Use of proton KBP model for generation of IMPT plans has a potential to improve treatment planning efficiency.

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Knowledge-Based Planning for Robustly-Optimized Intensity Modulated Proton Therapy of Prostate Cancer

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Purpose: To assess the performance of a Proton-specific Knowledge Based Planning (KBP) model in creation of robustly optimized IMPT plans for treatment of prostate cancer patients. **Methods:** Forty localized prostate cancer patients previously treated with volumetric modulated arc therapy (VMAT) were included. All the patients were replanned with robustly optimized Intensity Modulated Proton Therapy (IMPT). Thirty prostate patients were included in creation of proton KBP model. The remaining ten patients were used for model validation. KBP model quality and accuracy was evaluated based on the model-provided organ at risk (OAR) regression plots and metrics. To compare the robustness of the IMPT plans, we simulated the $\pm 3\%$ proton range uncertainty with 5mm translational error in six directions, resulting in twelve uncertainty scenarios for each plan. The expert and KBP generated IMPT plans were compared using the clinical dose volume constraints for clinical target volume (CTV), bladder, and rectum. **Results:** The resulting R^2 (0.87 ± 0.07) between dosimetric and geometric features, as well as χ^2 (1.17 ± 0.07) between the original and estimated data, showed good quality of the model. All the KBP plans were clinically acceptable. Looking only at the nominal plans, KBP plans delivered marginally higher dose to the rectum ($0.42 \pm 0.56\%$ for rectum V65Gy) than the expert plan, but otherwise there was no statistically significant difference in dose-volume indices for CTV and bladder. When all uncertainty scenarios are included, the KBP plans showed much greater CTV V100 ($11.64 \pm 20.09\%$) and lower HI (-1.81 ± 1.91) than the expert plan did. **Conclusion:** This study demonstrated that the robustly optimized IMPT plans created by a proton KBP model are of high quality and are at least comparable to the expert plans. Furthermore, the KBP model are able to generate more robust and homogenous plans compared to the expert plans.

PO-GePV-T-118

Assess the Interplay Effect Based On the Cyclotron Accelerator Proton Therapy System Machine-Specific Delivery Sequence Model

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Purpose: Proton Pencil Beam scanning delivery sequence is critical to the evaluation of the motion interplay effect. We proposed an experimental approach to build a precise machine-specific model for standard, volumetric, and layer repainting delivery based on a cyclotron accelerator system. Then we assessed the interplay effect using a 4D mobile lung target phantom compared to a generic delivery sequence model (WPE) using the same type of cyclotron accelerator proton system published previously. **Methods:** Test fields and clinical treatment plans were used to drive each beam delivery parameters that impacted BDT: energy layer switching time, spot switching time, and spot drill time. To quantitatively evaluate the interplay effect, a series of digital thoracic 4DCT image sets were used. The interplay effect was assessed based on the 4D dynamic dose accumulation method. Target motion was simulated based on the periodic respiratory motion with an amplitude 5 mm and breathing cycle 4s. Two-field Single Field Uniform Dose plans were generated using PA and lateral field. Standard delivery, volumetric and layer repainting delivery were simulated based on the machine-specific delivery sequence model and WPE model. The total BDT is also calculated. D99 (Dose received by 99% of target volume) of the target is used to estimate the delivery accuracy. **Results:** The results show that the interplay effect is very different between the two models even though the BDT is similar. More specifically, the WPE model leads to a very different result from the machine-specific model from this study in layer repainting technique. The delivery sequence comparison with the actual log files shows that the WPE model has a difference delivery sequence, resulting in the difference in the interplay effect evaluation. **Conclusion:** A precise machine-specific delivery sequence is highly recommended to ensure an accurate estimation of mobile target treatment's interplay effect.

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PO-GePV-T-119

Determination of Virtual Source Points for Mevion S250i Pencil Beam Scanning Proton System

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Purpose: The Mevion S250i proton system utilizes for energy selection a technique of binary insertion of Lexan plates of various thicknesses, resulting in an energy dependence of virtual source distance (VSD). We present our study on the VSD determination. **Methods:** Two methods were used to determine the VSDs: 1) EBT3 film and XRV3000 scintillation camera were used to measure spot size in air of different energy beams at points of -20, -10, 0, 10, 16.5 cm distance from ISO along the beam axis; 2) A Markus chamber was used to measure dose of single spot at the same points. Corrections to the measured doses were made to factors introduced by chamber size and beam shape. A linear fit to spot sigma vs position yields both VSD and beam divergence, and an inverse square law fit to dose also yields the VSD. **Results:** For the 11 energies selected from 227 to 28.7 MeV, the spot sigma at ISO and beam divergence increase linearly from 4.0 to 25.6 mm and 0.15 to 3.17 degree, respectively. The VSD measured with the two methods agree within 5%, demonstrating a rapid decrease from 154 to 52 cm when the energy changes from 227 to 184 MeV, followed by only a slight further decrease from 52 to 47 cm when the energy is reduced to below 184 MeV. Incidentally, this seemingly transitioning point of 184 MeV coincides with insertion in the beam path of the two most downstream and thickest plates of 3.324 and 6.641 cm WET. **Conclusion:** The unique design of the energy selection system of the Mevion S250i has resulted in an unique energy dependence of the VSD, exhibited by a rapid decrease in the higher energy range of 227-184 MeV, and remains virtually unchanged in the lower energies of 184 to 28.7 MeV.

PO-GePV-T-120

Effects of the Activated Dental Metal Induced by Proton Beam Irradiation for Head and Neck Cancer: A Phantom Study

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Purpose: In proton therapy (PT) for patients with head and neck cancer, dental metals are known to largely perturb the dose distribution compared to that in photon therapy. The dental metals are basically removed before PT but removing them are often difficult. If the dental metals are irradiated by the proton beam, these can be activated by protons or secondary neutrons. However, there are no reports on effects of the activated dental metals. We aimed to fundamentally investigate the effects of the activated dental metals using Monte Carlo simulation and measurement. **Methods:** The beam data on 150 MeV proton beam with 60 mm spread-Out Bragg peak (SOBP) size was used. Ti alloy and 12% Au-Ag-Pd alloy that are used as standard dental material in Japan were placed in the SOBP center in water phantom. Radionuclides induced by the proton beam irradiation were identified using Particle and Heavy Ion Transport code System (PHITS) and DCHAIN. Also, gamma-rays emitted from the activated dental metals were counted by high purity germanium (HP-Ge) detector. In addition, dose enhancement caused by the activated dental metals was calculated by PHITS. **Results:** The spectra of the activated dental metals obtained in HP-Ge detector had the same energy peaks as the radionuclides estimated by PHITS. In addition, the dose around Ti alloy and Au-Ag-Pd alloy increased slightly as compared with the prescription dose. **Conclusion:** The radionuclides induced from the dental metals were mostly dominated by nuclides with short half-lives. The dose enhancement caused by the activated dental metals was significantly lower compared with the prescription dose.

PO-GePV-T-121

Tuning, Calibration and Testing of a Dynamic Collimation System Controller for Pencil Beam Scanning Proton Therapy

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Purpose: The purpose of this work was to tune and calibrate the Dynamic Collimation System (DCS) controller and measure the delivery time of an example treatment plan for a number of different jerk values where jerk is defined as the rate of change of acceleration. The control system needs to be tuned for the given electro-mechanical setup and the desired motion parameters in order to reduce oscillations in the motion. **Methods:** A DCS prototype was built consisting of two pairs of orthogonal nickel trimmers independently translated by Parker T1S linear motors. Test motions were used to identify the motor response and tune the control system. Trimmer motion was observed for different jerk values before and after tuning the controller. The trimmer sequence from a sample test plan consisting of 244 trimmer positions and 3 ms dwell times was simulated with variable jerk values to determine the impact of this parameter on treatment time. Trimmer position accuracy was also tested for different DCS orientations of 90 and 180 degrees. **Results:** Controller tuning and calibration provided stable and accurate trimmer motion. A maximum overshoot of 0.7 mm was observed prior to tuning as compared to 0.04 mm after tuning. Treatment time for the sample test plan with a jerk of 4×10^5 mm/s³ was 21.6% (98 s) faster than with a jerk of 5×10^4 mm/s³ (125 s). Trimmer position profiles were nearly identical for different orientations of the DCS with the positional accuracy found to be within ± 0.1 mm. **Conclusion:** The DCS controller can be integrated within a commercial Ion Beam Applications (IBA) Pencil Beam Scanning (PBS) system and used to accurately and automatically position the trimmers for treatment delivery, improving dose conformity with minimal impact to treatment time.

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PO-GePV-T-122

Dosimetric Investigation of Inverse-Optimized Proton GRID and Lattice Treatment Plans with a Clinical Pencil Beam Scanning System

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Purpose: We report inverse-optimized GRID and Lattice pencil beam scanning (PBS) proton treatment techniques and their dosimetric characteristics. **Methods:** GRID and Lattice plans were created in a water phantom in a commercial treatment planning system with a clinical PBS proton system. An in-house script generated evenly spaced cylinders and spheres within a 10 cm cube centered at 10 cm from surface; the cylinder axes and the sphere lattice were positioned along the vertical direction. The diameter of the cylinders and spheres was 3 mm; patterns with 4 different center-to-center spacings were tested: 1.5 cm, 2.0 cm, 2.5 cm, and 3.0 cm. Two plans were created for each center-to-center spacing, a GRID plan and a Lattice plan, using the cylinders and the spheres as targets, respectively, with a single beam from gantry angle 0. All plans were optimized to deliver a uniform dose (15 Gy in 1 fraction) to the target while minimizing dose outside of the targets. **Results:** At the depth of 10 cm (center of cube), the peak-to-valley ratios (PVRs) are 1.54, 3.47, 9.52, 26.79 for the GRID plans and 1.52, 3.26, 9.15, 22.39 for the Lattice plans for the center-to-center distances of 1.5 cm, 2.0 cm, 2.5 cm, and 3.0 cm, respectively. The ranges of PVR at different depths for the GRID plans are: 1.27-1.71, 2.48-4.32, 5.39-13.54, and 17.65-34.88, respectively. For the Lattice plans, the ranges are: 1.23-1.67, 2.33-3.90, 5.13-9.15, and 15.00-29.41, respectively. **Conclusion:** Proton GRID and Lattice plans were feasible with multiple center-to-center distances. The achievable PVRs are mainly determined by the center-to-center distances for both GRID and Lattice plans. For the same center-to-center distance, GRID plans have slightly higher PVRs than the Lattice plans. The PVRs decrease with depth for both GRID and Lattice plans.

PO-GePV-T-123

Evaluation of Distorted Lateral Beam Profiles in MRI-Guided Proton Therapy

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Purpose: In order to realize MRI-guided proton beam therapy system, it is important to evaluate the characteristics of distorted dose and relative biological effectiveness (RBE) distributions in the MRI scanner. The deflected proton beams in the perpendicular magnetic field result in asymmetrically distorted lateral

beam profiles. The purpose of this study is to clarify these profiles by analyzing dose, mean energy and dose-averaged linear energy transfer (LETd) distributions under the various magnetic fields. **Methods:** Monte Carlo simulations using Geant4.10.2.p02 were performed for proton beam analysis in magnetic fields ranging from 0 to 3 T. Initial proton energies are set from 70 to 220 MeV and the proton source is assumed to have no energy spread and angular divergence. A water phantom was installed in the MRI scanner where the magnetic field was applied perpendicular to the beam incident angle. The three-dimensional dose distributions in the phantom were calculated. The lateral dose profiles were fitted using the primary Gaussian, secondary Left, and Right tail functions. Then the ratio of the coefficient of Left to Right function (γ_R to γ_L) was evaluated to quantify the magnitude of asymmetric distortion. **Results:** The distortions of the lateral beam profile were observed near Bragg peak region, leading to non-negligible asymmetrical changes in the dose distributions. The asymmetric ratio of γ_R to γ_L , which indicates the magnitude of distortion, increased as the strength of the magnetic field increased. The ratio became 2.4 under the 3T magnetic field. In addition, it was shown that the mean energy and LETd distributions also have non-negligible distortions which will affect RBE distributions. **Conclusion:** We quantitatively evaluated the distortion of the dose distribution in the magnetic field using the fitting functions. Non-negligible dose deformations at the Bragg peak area were identified and distortion of LETd also affected RBE.

PO-GePV-T-124

Feasibility of Treating Large Mobile Targets Using a Breath-Hold Technique with a New Adaptive Collimation Leaf Sequencing Algorithm for Intensity Modulated Proton Therapy

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Purpose: Fast delivery of an intensity modulated proton therapy (IMPT) plan utilizing an adaptive aperture (AA) enables treatment of patients under breath-hold (BH). Field delivery time has several influencers, including dose-per-pulse, and number of energy-layers and spots. In the past the AA has prohibitively increased time such that only small targets could be treated with BH. The goal of this study is to determine the feasibility of treating larger targets on BH with a new commercially available AA positioning algorithm for a single-room proton therapy system. **Methods:** Eight patients, ranging in target volume from 2.3 cc to 153 cc were planned with an AA to meet clinical standards. Based upon the dose per spot, number of energy layers and spots, and leaf sequence of the AA, delivery times were calculated for each field both with and without the new leaf positioning algorithm. Delivery time was also calculated without the AA for comparison. Assuming a maximum unassisted patient BH capability of 15 seconds, the delivery times were compared with target volume to predict the threshold target size such that fields could be delivered in a single BH. **Results:** Delivery time is related to the target volume being treated, largely due to the strong influence on treatment time from AA motion, accounting for about 50% of the total time of delivery per field. Without using the AA, fields for target volumes over 100 cc could be treated in a single BH. However, the new leaf positioning algorithm significantly reduces delivery time with the AA, such that target sizes up to 60 cc could likely be treated in a single BH per field. **Conclusion:** The once prohibitive time requirement of AA motion has greatly been reduced with a new leaf positioning algorithm, making a BH IMPT delivery technique feasible for larger targets.

The authors of this abstract are customers of or employed by Mevion Medical Systems

PO-GePV-T-125

Monte Carlo Evaluation of Neutron Dose Equivalent Topatients Treated with Vertex Beams in IMPT and PSPT

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Purpose: Patients treated with protons are subjected to secondary neutrons originating from the beam delivery system and inside the patient. Treatment plans featuring vertex beams, i.e., proton beams oriented in the superior-inferior direction, raised concerns of increased neutron exposure to the patients. A study was performed to assess the neutron dose equivalent to patients treated with such beam configurations. **Methods:** A Monte Carlo model of the complete treatment room, including all relevant beam line equipment, was produced for IMPT and PSPT treatments, each equipped with a simplified representation of an adult and a 10 year old pediatric patient. Proton beams of about 15 cm range in water were simulated to the heads of the phantoms in the vertex beam orientation and from a lateral direction. The neutron fluences in the phantom's arms, legs, torso, as well as head and neck were scored as a function of neutron energy and converted to neutron dose equivalent. **Results:** The IMPT treatments yielded moderately reduced neutron dose equivalent values for Vertex Beams, mostly to the neck of the phantoms. Using PSPT, there was a noticeable systematic reduction in neutron dose equivalent in the Vertex Beam configuration. **Conclusion:** We found that the neutron dose equivalent to the phantoms was more dependent on the distance to the beam line/nozzle than on beam direction, and that the application of Vertex Beams does not pose additional hazards to a proton therapy patient due to neutron exposure.

PO-GePV-T-126

Novel Dynamic Apertures for Reducing Spot Sizes in Proton Therapy

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Purpose: To improve nearby normal tissue protection in proton therapy, spot sizes must be controlled and reduced if possible, which can be achieved by a novel concept of dynamic pinhole aperture (DPA). In this work, we investigate the clinical viability of the DPA based on its ability to reduce spot sizes while retaining a reasonable dose rate. **Methods:** The DPA is a small device mounted to a motor assembly, capable of moving along the beam axis and two lateral directions. The DPA has a cylindrical opening with a range shifter attached that follows and trims each spot. The DPA alters the dose distribution by changing its positions instead of changing its shapes. MCsquare, an open-source Monte Carlo code, has been enhanced to support DPAs and used to calculate the dose distributions with varying parameters of DPAs to find the optimal parameters for clinical use in shallow tumors. The resulting spot sizes are compared among the DPA, an optimal thickness range shifter, and the extended range shifter (ERS). The beam transmission ratio is characterized for the DPA, on- and off-axis, to indicate the reduction of the dose rate. **Results:** Compared to ERS, spot sizes are reduced 50% when the DPA is placed 5 cm from the patient at depths less than 50 mm. For a DPA of 3 mm radius positioned 5 cm from the patient, the spot size is reduced by 30% compared to an optimal range shifter placed at the same position. The beam transmission ratio is between 10-50% for a 3 mm radius DPA. **Conclusion:** The device is simple, compact, and effective, especially for shallow tumors. Since the DPA adds treatment time for each spot, its use should be limited to a subset of spots of the treatment plan where the reduction in spot sizes is most beneficial.

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PO-GePV-T-127

Considerations in Lung SABR IMPT Using Two Different PBS Systems

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Purpose: Stereotactic Ablative Body Therapy (SABR) is an effective treatment for early-stage lung cancer. To effectively implement Lung SABR IMPT, effective spot size, plan dosimetric quality, and delivery time needed for practical and effective clinical treatment were evaluated for two different proton

systems. **Methods:** A compact proton therapy delivery system (Proteus-one, P1) was configured to deliver large spot IMPT similar to Proteus-plus with universal nozzle (P+). Total of 9 patients with 10 tumors were selected for this study. Seven targets were prescribed 60Gy in 10 fractions; three targets were 48Gy in 4 fractions. Raystation 8A were used to generate treatment plans for each nozzle type with exact same optimization objectives. P1 plans used 4cm range-shifter placed at 45cm from isocenter to create large in-air beam spots, which were compared to the P+ in-air spots without range-shifter. P+ plans used 7cm range-shifter placed 4cm away from patients, to minimize spot size degradation. Target coverage was optimized to prescription dose covers 100-percent GTV with 90-percent covers PTV. Dose to the OARs were compared to QUANTEC metrics. Three volumetric dose-repainting was performed. Plan beam-on times were obtained to evaluate treatment delivery efficiency. **Results:** The in-air beam spots were comparable between P+ without and P1 with range shifter. All target coverage are acceptable. OAR doses were comparable and within the QUANTEC limits. The beam-on time of P+ plans were 3 to 6 time of those of P1 plans. When combined with patient setup and image guidance time, patient discomfort could lead to higher uncertainty in P+ treatment. **Conclusion:** P1 system range-shifter configuration can effectively simulate a large spot P+ system. For the two proton systems, plan qualities were dosimetrically comparable; volumetric dose-repainting were applicable to reduce interplay effect. However, P1 delivery was significantly shorter than that of P+.

PO-GePV-T-128

Dynamically Collimated Spatially Fractionated (GRID) Pencil Beam Scanning Proton Therapy

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Purpose: The dosimetric advantages of protons, which include a pronounced Bragg peak and low entrance dose, are desirable traits when considering GRID therapies. The objective of this work was to investigate the applicability of using a dynamic collimation system (DCS), an energy layer specific collimator, to improve PBS GRID treatments by allowing variable grid ratios and to demonstrate the viability of this technique between different cancer cell lines. **Methods:** GRID dose distributions were simulated using Geant4 with an experimentally benchmarked IBA UN beam line model and physics list. Individual 4 cm spread-out-Bragg peak beamlets were calculated that were collimated with equivalent square fields ranging from 0.25 cm to 1.0 cm. Treatment planning dose distributions were generated in MATLAB by optimizing the spot spacing and collimation width for various cancer cell types. A linear quadratic model was used to assess the cell survival between malignant and nonmalignant tissues when irradiated under full-field and GRID irradiation conditions. **Results:** As expected, the effectiveness of GRID treatments were largely impacted by changes in spot spacing or collimation, as these variables directly influence the resulting dose distribution. However, it is not necessarily the case that maximizing the difference between adjacent low- and high-dose regions resulted in a superior dose-volume effect. In fact, no combination of spot spacing and collimation width was universally optimal but rather depended on the cell radiosensitivity and the desired cell kill fraction. **Conclusion:** An optimal GRID dose distribution reflects the malignant and non-malignant cell radiosensitivities at the desired dose level. Given the capacity of the DCS to perform energy-layer specific collimation, it appears that this technology may help tailor treatments towards the specific radiobiology of a patient's tumor in order to maximize the therapeutic benefits from GRID treatments.

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PO-GePV-T-129

Impact of Noise Level on the Analysis of Proton Exposed Targets

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Purpose: Accurate online determining of the elemental composition of irradiated tissue in proton therapy plays an important role in treatment verification, since it leads to accurate estimation of the depth of Bragg peak, the proton range and the stopping power map of irradiated tissue. A hopeful approach for elemental analysis in proton therapy is employing the proton induce characteristic prompt gamma-rays. In this work, the effect of gamma spectrum noise level on estimating elemental analysis of irradiated target and proton range are investigated. **Methods:** In the present study, the Geant4 toolkit has been used to simulate the prompt gamma-ray spectra of phantom in different noise levels. Analysis done using correlation between the mass concentration of elements and net counts of selected peaks in proton induced prompt gamma spectra of irradiated phantoms. To investigate the effect of noise levels on the accuracy and precision of the results, different noise levels are simulated, and the errors of elemental mass are calculated. Also the error propagation applied to estimate error of analytical calculated proton range. **Results:** The results show that both the accuracy and the precision of estimated elemental weight percent decrease by increasing noise levels. The quantification results show that the elemental weight percent errors in high noise level lead to deviations less than 1% in mass rang (in g/cm²) of protons. **Conclusion:** In conclusion, the applied analysis method can be used to estimate weight percent of irradiated tissue from its recorded prompt gamma spectra even in the presence of high noise levels. However, the increment in the standard deviation of the results is unavoidable.

PO-GePV-T-130

A Preliminary Study of DECT- and Kernel-Based Dose Calculation Algorithm for Carbon Beam Therapy

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Purpose: This study aims to develop a kernel-based dose calculation algorithm for carbon beam therapy using DECT image and to verify the plausibility of the dose calculation algorithm. **Methods:** At first, the carbon beam kernels of 100-430 MeV/u are assessed using Geant4 Monte Carlo toolkit. The pencil beam dose calculation algorithm of matRAD software is revised for using the calculated beam kernel and dual energy computed tomographic (DECT) image. The DECT and single energy CT (SECT) images of abdomen case are obtained by Philips IQon Spectral double-layer CT and given algorithm. And then, the images are converted to relative stopping power ratio (RSP) maps for the dose calculation. In order to evaluate the influence of DECT based dose calculation method, the doses on DECT and SECT images are compared with each other. **Results:** The RSP maps obtained by DECT and SECT images basically show similar tendency, however, the SECT image overestimates RSP up to 5% compared to that of DECT even in homogeneous media. Moreover, the physical dose calculation results show slight range disagreements due to the RSP differences. The carbon range in the target on SECT image is shorter than the range of DECT up to 2 mm in the same treatment plan. **Conclusion:** The plausibility of the beam kernel-based dose calculation algorithm has been verified in this study. We figured out that DECT can be used to correct the range uncertainty due to the CT conversion of conventional SECT about 2 mm. This work does not consider the biological effectiveness of the carbon beam, in further, the influence of DECT on biologically effective dose will be quantitatively evaluated with more patient cases in the other therapeutic regions.

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PO-GePV-T-131

Assessment of Metal Artifact Reduction in Dual-Layer Dual-Energy Computed Tomography for Carbon Ion Radiotherapy

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Purpose: This study aims to assess a dosimetric impact metal artifact reduction in carbon ion radiotherapy with dual-layer dual-energy computed tomography (DL-DECT) in terms of relative stopping power ratio (RSP). **Methods:** DL-DECT images of 2 patients implemented metallic structure were analyzed. The scan were performed with a commercial DL-DECT system (IQon Spectral CT, Philips Healthcare). One patient had metal hip implants and the other had a dental implant. Each patient was scanned once and the images were reconstructed with metal artifact reduction (O-MAR) and without metal artifact reduction. When the images were reconstructed, the spectral based images (SBLs) were also acquired to extract effective atomic number (Zeff) and relative electron density (RED) map. By using the Zeff and RED, RSP for each image (i.e., with O-MAR and without O-MAR) were calculated. **Results:** Metal artifacts in 120 kVp conventional images were significantly reduced when O-MAR was applied to each image set. In addition, artifacts in Zeff and RED map were also reduced. In both cases of metal hip replacement and dental implant, the difference in RSP ranged from -1.5 to 1.5. **Conclusion:** The metal artifacts significantly affected the accuracy of Zeff, RED, RSP, and dose calculation for carbon ion radiotherapy. Adequate metal artifact reduction algorithm is required to be applied in case the patient had metallic implant.

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PO-GePV-T-132

Preliminary Investigation of the Dosimetric Impact of Common Dental Restorative Materials On Proton Beams

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Purpose: To quantify the dosimetric impact of 70 MeV proton beams in the vicinity of materials commonly used in current dental practice. **Methods:** Two-millimeter-thick metallic, Titanium (Acnis), Cobalt-Chromium (MESA), and Zirconium (Nacera), and non-metallic, PEEK (Juvora), Pekkton (Cendres+Metaus), and PMMA, samples were studied. A uniform monoenergetic 70 MeV scanned proton field was delivered by an IBA ProteusPLUS PBS proton therapy system (Ion Beam Applications, Louvain-la-Neuve, Belgium). Samples were placed 60 mm apart edge-to-edge in a 3D printed tray on top of a 10x25x25 cm³ block of solid water with an additional 1, 2, or 3 cm of solid water added to the top surface. Relative profiles were measured using Gafchromic EBT3 film placed immediately proximal or distal to the sample tray and normal to the beam direction. Pixel values in regions of interest (ROIs) (1x1 cm²) centered at and in the absence of the materials were averaged and compared. **Results:** Backscatter measured directly at the entrance surface was negligible compared to measurements made in absence of the samples, with relative pixel value differences (RPVDs) of -1.2%±1.4% to 0.9%±1.5%. The dose effects at the distal surface were greater for metals than for nonmetals, with RPVDs for non-metals ranging from -1.7%±1.5% to 0.9%±1.3% and -25.7%±1.5% to 61.9%±2.2% for metals. **Conclusion:** Metallic dental restorations can cause downstream dose perturbations with the extent dependent on the material and position of the material relative to the Bragg peak. Future work will include validating these results with ion chamber measurements and with additional energies with and without the presence of a range shifter and comparing measurements to the WET of samples to examine dose perturbations from secondary interactions.

Credit Panthera Dental Lab as they prepared all the samples to Evan Rosen at no cost.

PO-GePV-T-133

Wireless Device for Adaptive Planning Based Upon Thermoacoustic Range Estimates - Design and Benchtop Validation

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Purpose: Transitioning thermoacoustic range verification from benchtop experiments with cumbersome research equipment to clinical practice is the purpose of our work. A wireless device for thermoacoustic range verification with correlation to online ultrasound images has been developed for initial application to adaptive planning (Fig. 1). **Methods:** A 7 x 5.5 x 3.5 cm detector (NucSafe) requiring 5V power supply replaces the 0.5 m long prompt gamma detector that required -2 kV (Fig. 2a). The new detector provides both analog and digital signals to quantify proton pulse shape and serve as a trigger. Oscilloscopes are replaced by a wireless data acquisition board (DAQ) with 1 MSPS ADC and vertical resolution of 38 uV. 6 acoustic channels are amplified. 2 channels for compact gamma detector inputs (analog and digital) are not amplified. Data is acquired continuously for up to 30 minutes and stored on a secure data card for retrieval via a serial port after the exam. Two different types of transducers provide broadband response in a compact form factor (Fig. 2c). **Results:** A custom receive chain provides improved 10-18 dB greater sensitivity over the relevant frequency band (10-100 kHz, Fig 2b). Transducers positioned to either side of the ultrasound imaging array are up to 8 dB more sensitive to low frequencies (Fig 2c). The new DAQ provides 10 dB greater amplification. A wireless ultrasound imaging array (Clarius P4-1) provides greater imaging depth, 30 cm vs 7 cm, and smaller footprint (Fig. 2d). **Conclusion:** This improved device provides greater ease of use, positioning flexibility and receive sensitivity which could enable range verification during pulsed delivery of a 2 Gy fraction. Although validation on phantoms in proton therapy vaults is required, nonintervention patient studies could quickly follow because "listening" for thermoacoustic signals generated during the normal course of therapy is entirely passive.

NIH-NCI SBIR #R43CA243764. Dr. Patch founded Acoustic Range Estimates and holds US patent #10,758,127. Mr. Narayanaswamy is founder of Swamy Enterprises.

PO-GePV-T-134

Sensitivity of Target Coverage and Patient Specific IMPT QA to Spot Positioning Errors

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Purpose: The clinical efficacy of intensity modulated proton therapy (IMPT) requires accurate delivery of the spot pattern. As a result, patient specific quality assurance (QA) is required for each new plan. However, sensitivity of commonly used patient QA systems to spot positional errors, and the conditions where position delivery errors can be detected in patient QA have not previously been examined. This work evaluates the sensitivity of current patient-specific IMPT QA procedures to minor spot positioning errors. **Methods:** Dosimetric effects of delivery uncertainties in proton spot position was evaluated for ten head-and-neck patients. Each spot position was perturbed randomly assuming a uniform distribution with interval-width of 0.75, 1, 1.5, 2.0, 2.5, and 3.0 mm at isocenter. Dose distributions and dose-volume histograms of original and perturbed plans were recalculated and compared using CTV coverage metrics. Additionally, absolute dose and 2D-planar dose measurements using 2D array ion chamber (MatriXX-PT) were performed. The impact of spot position uncertainties on the patient QA results was examined using a gamma criteria of ±3%/3mm and the gamma passing rate greater than 90% considered as criteria for validation of a clinical plan. **Results:** Comparison of TPS calculated dose between original and perturbed plans suggest the effect of random spot position uncertainties on target dose parameters such as D2%, D95%, D98% was not significant (less than 2%) until 1.5mm internal-width. Additionally, the patient-specific IMPT QA all perturbed plans up to 1.5mm tolerance was acceptable within clinical criteria (±3%/3mm). **Conclusion:** Our results show that the target coverage was not sensitive to random spot position uncertainty up to 1.5 mm. Additionally, this uncertainty did not affect patient IMPT QA results.

PO-GePV-T-135

Impact of Spot Position Uncertainties on Intensity Modulated Proton Therapy Plans for Head and Neck Cancers

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Purpose: Intensity modulated proton therapy (IMPT) utilizes complex patterns of spot placement to generate highly conformal dose distributions and rapid fall-offs. Therefore, a high confidence level in the positional accuracy of each spot is demanded. However, the dosimetric impact of uncertainties associated with delivered spot positions, particularly in regions that involve several complex anatomical structures such as head-and-neck has not been investigated. **Methods:** Fifteen patients with head and neck cancers were randomly selected for this study. Patients were treated with $\geq 95\%$ of the largest CTV receiving the prescription dose (Rx: 59.4 Gy) according to a single integrated boost regimen. For each patient plan, random uncertainties in spot position were simulated by perturbing the spot position according to uniform (or Gaussian) distribution with interval width equal to the vendor-set tolerances (1.2 mm/1.45 mm in X/Y direction at monitoring chamber for Varian ProBeam). Impact of spot position uncertainties on target coverage was studied using the minimum, maximum (dose to 0.01cm³), and dose to 95% of CTV volume (D95%). **Results:** Distributions for target coverage metrics were normal with standard deviations ranging between 0.5% to 5% of Rx dose (180cGy/fx). D95% exhibited the lowest sensitivity, and was within 2% of the clinical plan for all patients. Larger deviations in Dmin (<10%) and Dmax (<6%) were observed suggesting potential for target under-dosage, or unwanted hot spots. No significant correlation between coverage variation and target volume was found. **Conclusion:** While in comparison to MLC based intensity modulated radiation therapy (IMRT), IMPT is more sensitive to positional uncertainties, spot positional uncertainties within the vendor supplied 1.2 mm tolerance do not pose significant clinical impact.

PO-GePV-T-136

Impact of Esophageal and Biliary Stents on Proton Dose Calculation

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Purpose: Hepatobiliary and esophageal malignancy patients who are commonly implanted with stents are often treated with radiotherapy. To date, the impact of the newer stents on proton calculated dose is unknown. This study aims to analyze and compare the radiation dosimetric impact of newer esophageal and biliary stents to determine if dose adjustments in areas with stent presence are needed. **Methods:** Five esophageal and seven biliary stents were secured inside a water phantom and CT scans were acquired. The water equivalent thickness (WET) values of each sample were obtained using a multilayer ionization chamber and 2D-planar dose measurements of a proton field were performed using an ion chamber array. Dosimetric impact and WET of stents on treatment planning system (TPS) calculated dose was compared with measurements. **Results:** WET values for all esophageal stents were less than 1mm (0.35 - 0.85 mm) and values for all biliary stents were less than 0.5mm (0.15 - 0.45 mm). For planar dose measurements in water, no difference (<1%) was observed for both esophageal and biliary stents, thereby implying that the perturbation effect of the stents was within the measurement uncertainty and not detectable. For biliary stents TPS indicated changes in WET values as large as 2 mm. This inconsistency between measured and TPS calculated WET values are due to CT artifacts and inaccuracy of SPR estimation. **Conclusion:** Our study's results show that the presence of biliary and esophageal stents has minimal impact on proton dose. However, inaccuracies associated with SPR estimation suggest that these devices can perturb calculated dose.

PO-GePV-T-137

Non-Radioactive Elements for Prompt-Gamma Production Enhancement in Proton Therapy

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Purpose: To develop a new Prompt Gamma Spectroscopic (PGS) technique that enhances the production of PG rays emitted from ¹²C* and ¹⁶O* de-excitations, by injecting a non-radioactive element in the tumor area. Positive results will lead in accurate monitoring of proton range and target-tissue characteristics during the treatment. **Methods:** The Tool for Particle Simulations (TOPAS) Monte Carlo (MC) package was used to simulate 10⁷ proton histories of 75 MeV and 100 MeV incident in a co-centric cylindrical geometry. The outer cylinder was filled with water, and the inner cylinder was filled with water plus 0.01%-25% fractions in weight of the elements ¹⁹F, ¹⁷O, ¹²C, ¹H, and ¹²⁷I, respectively. We identified the characteristic peaks in the PG spectra, and then we assessed the PG enhancement of the 4.44 MeV peak (P1) resulting from ¹²C* de-excitations, and the 6.13 MeV peak (P2) resulting from ¹⁶O* de-excitations. **Results:** For the 75 MeV incident proton beam the maximum P1 and P2 enhancement was 20.1% and 3.71% when 0.01% of ¹²⁷I and 1% ¹⁹F enriched the inner cylinder placed at surface, with respect the outer cylinder. For 100 MeV incident proton beam the maximum P1 enhancement was 8.88% when 0.01% ¹⁹F was added to the inner cylinder. When the inner cylinder was placed at depth of 2 cm for the 75 MeV incident proton beam, and at 5 cm for the 100 MeV incident proton beam, with respect to the outer cylinder, the PG enhancement dropped to half. The Bragg peak was not affected by the enrichment of the inner cylinder with the studied elements. **Conclusion:** It was found that the enhancement of PG production yield is possible by enriching the inner cylinder with the used non-radioactive elements. This is an ongoing study and simulations with more tissue-like materials such as brain-tissue are in progress.

PO-GePV-T-138

The Benefit of Novel Beam Arrangements Made Feasible by Range Uncertainty Reductions in Proton Therapy for Brain and Skull Base Tumors

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Purpose: In proton therapy, conforming the dose distribution to an organ at risk (OAR) using the sharp dose fall-off at the distal beam edge may improve OAR sparing. Currently, the lateral beam edge is often used instead to reduce the effects of uncertainties in the in vivo proton range. However, range uncertainty reductions may allow for greater use of the distal beam edge. We quantified the benefits of such novel beam arrangements in a set of nine cases with targets abutting the brainstem. **Methods:** For each case, six treatment plans robust to 2 mm set-up as well as 0-5% range uncertainty were created for the traditional clinical beam arrangement and a novel beam arrangement making greater use of the distal beam edge to conform the dose distribution to the brainstem. Metrics including the brainstem normal tissue complication probability (NTCP) with the endpoint of necrosis were determined for all plans and set-up and range uncertainty scenarios. **Results:** For the traditional beam arrangement, reducing the range uncertainty from the current level of approximately 4% to a potentially achievable level of 1% reduced the brainstem NTCP by up to 0.6 percentage points when averaging over all scenarios and by up to 0.9 percentage points in the worst-case scenario. A switch to the novel beam arrangement at 1% range uncertainty increased these values by a factor of about 2, to 2.0 percentage points and 2.9 percentage points. The novel beam arrangement achieved a lower brainstem NTCP in all cases at a range uncertainty of 2% when averaging over scenarios and at a level of 1% in the worst-case scenario. **Conclusion:** The benefits of novel beam arrangements may be of the same magnitude or even exceed the direct benefits of range uncertainty reductions. The benefits of range uncertainty reductions may therefore previously have been underestimated.

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PO-GePV-T-139

Incident Learning During the Early COVID-19 Pandemic

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Purpose: To assess the impact of the early COVID-19 pandemic on incident learning through evaluation of RO-ILS: Radiation Oncology Incident Learning System. The Radiation Oncology Healthcare Advisory Council hypothesized that the pandemic would impact participant engagement in reporting to the patient safety organization and the characteristics of reported events. **Methods:** The RO-ILS database was queried for events reported pre-COVID (March 1, 2019 to July 31, 2019) and during early COVID (March 1, 2020 to July 31, 2020). Events were segregated based on practices in early COVID hotspot states as identified by the CDC (WA, CA, AZ, TX, FL, NY, NJ, CT, MA, PA, MD) and their reporting status (Top 5). Descriptive statistics were used to describe trends in reporting and differences in data elements pre-COVID and during-COVID. **Results:** There was a 16% absolute reduction in event reporting during-COVID (n=1255) as compared to pre-COVID (n=1759). Practices located in COVID-hotspots had a 33% absolute reduction in reporting, while those not in hotspots had a 23% reduction. However, initial analysis did not identify drastic change in event classification. Amongst the Top 5 reporting practices, there was a 48% absolute reduction in incident reporting; of note, three of these practices did not report any events during-COVID. During-COVID, errors more often occurred and were discovered during treatment planning, regardless of hotspot status. Practices rated more events as moderate-severe during-COVID (25%) than pre COVID (18%). No appreciable difference was observed in the types of treatment techniques nor magnitude of dosimetric deviations pre-COVID and during-COVID. **Conclusion:** Reporting to RO-ILS declined during the early COVID-19 pandemic, especially in hotspot areas. Three of the five top reporting practices that ceased reporting have since reported events after the analysis timeframe, suggesting the decline may be temporary. Stability in event classification suggests that practices continued to report a variety of events.

The Radiation Oncology Healthcare Advisory Council (RO-HAC), managed by Clarity PSO, provides subject-matter expertise, analyzes, interprets and reports on RO-ILS data. RO-HAC members receive an honorarium from program sponsors ASTRO or AAPM.

PO-GePV-T-140

Radiotherapy Prescription Quality

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Purpose: In 2020, 16% of the incident reports submitted to our Incident Learning program were related to Radiotherapy prescriptions. As these are a critical part of Radiotherapy treatments, a large amount of resources and time are invested within our department to check for and correct these issues before treatment starts. We decided to investigate in further detail the error rates, and their causes to better understand the quality of Radiotherapy prescriptions and how to improve it. **Methods:** We used three different methods to measure and investigate Radiotherapy prescription errors: first, we reviewed past incident reports to find all reports related to Radiotherapy prescriptions; second, we organised a two-week survey to identify and quantify the type and number of issues with Radiotherapy prescriptions; third, a chart audit of 100 targeted patients was done to identify any unreported errors. Patient selection was done purposefully to ensure that all Radiation Oncologists, treatment sites, techniques and modalities were represented within the audit. **Results:** Whereas the incident reports represented 2.3% of Radiotherapy prescriptions treated in our department in 2020, the survey identified 13 radiotherapy prescriptions having errors out of 135 investigated (9.6%), and the audit identified 20 radiotherapy prescriptions having errors out of 100 investigated (20%). We categorized the errors for the survey and audit investigations, noticing that most of the errors that could have led to treatment errors where associated with treatment sequencing and treatment site identification. **Conclusion:** The important number of reports submitted to our Incident Learning program alerted the team to investigate prescription errors, but it was clearly underreporting the importance of these issues despite being a highly functional program. We hope to review and improve our processes with a Failure Modes and Effect Analysis, making sure to repeat our survey and audit experiments to measure the outcomes of any process changes.

PO-GePV-T-141

Retrospective Review and Analysis of Treatment Planning Errors

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Purpose: Our department performs regular process review as part of the quality improvement efforts. The goal of this study was (1) to retrospectively analyze treatment planning- and chart setup-related errors in a controlled sub-sample of patients and (2) to identify potential sources of these errors to reduce and avoid future occurrences. **Methods:** To ensure consistency and accuracy of data statistics, a complete set of treatment plans, which were checked by a single physicist over a 7-month period, was included in the study. The plans, in which no errors were found, consist the control sample, while the rest of the plans, where errors were found by the physicist or other staff involved in the patient care, define the variance sample. Errors related to treatment planning and chart setup were analyzed. Analysis metrics included hypothesized correlations of various timelines along the treatment planning process with error rate. Significant difference between the control and variance groups was investigated. **Results:** 207 plans in the control group and 115 reported errors in the variance group were analyzed. No significant difference was observed between the control and variance groups in errors occurring at various time points. The variance group had a slightly larger fraction of plans that violated our planning timeline (0.42 vs. 0.39) and a larger fraction of plans that had a shorter timeline between patient simulation and the start of treatment. Planning efficiency and planner variable were found not to be factors in clinical errors. **Conclusion:** The results did not find any correlation between the control sample and the variance sample with respect to the treatment planning timelines. As continuous review of adverse events helps improve clinical processes and prevents future similar errors, further examination is needed to investigate other root causes of the errors.

PO-GePV-T-142

Automated Plan Checking Software Demonstrates Continuous and Sustained Improvements in Safety and Quality: A 3-Year Longitudinal Analysis

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Purpose: Perform a longitudinal analysis of the performance of our automated plan check tool (PCT) by retrospectively evaluating the number of errors identified in plans delivered to patients in three, month-long, data collection periods taken between 2017 and 2020. **Methods:** 11 automated checkers were retrospectively run on 1169 external beam radiotherapy treatment plans identified as meeting the following criteria: planning target volume (PTV) based multi-field photon plans receiving a status of treatment approved in either March 2017, March 2018, or March 2020. The number of “passes” and “flags” were recorded. “Flags” were sub-categorized into “false flags”, “false flags due to naming conventions”, or “real flags”. 2x2 contingency tables using a two-tailed Fischer’s exact test were utilized to determine whether there were nonrandom associations between the output of PCT and whether the check was manual or automated at the original time of treatment approval. **Results:** A statistically significant decrease in flags between the pre- and post-automation datasets was observed for four contour-based checkers, namely adjacent structures overlap, empty structures and missing slices, overlap between body and couch, and laterality, as well as a checker that determined whether the plan’s global maximum dose was within the planning target volume. Analysis of the origin of false flags resulted in recommendations for algorithmic or configuration modifications to circumvent such flags in the future. **Conclusion:** Periodic and longitudinal review of the performance of automated software is essential for monitoring and understanding its impact on error rates as well as for optimization of the tool

to adapt to regular changes of clinical practice. PCT has demonstrated continuous contributions to the safe and effective delivery of external beam radiotherapy to our patient population, an impact that extends beyond its initial implementation and deployment.

The plan check tool project is part of a co-development agreement between MSKCC, University of Michigan, and Varian Medical Systems. At the time of data collection Sean Berry held a grant from Varian Medical Systems unrelated to this work.

PO-GePV-T-143

The Impact of Using Daily CT for 2D EPID-Based In-Vivo Dosimetry

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Purpose: The difference of anatomical structure and positioning between planning and treatment may lead to bias in EPID based in-vivo dosimetry calculation. The purpose of this study is to use daily CT instead of planning CT as reference for EPID-based in-vivo dosimetry calculation and to analyses the necessity of using daily CT for EPID-based in-vivo dosimetry calculation in terms of patient quality assurance. **Methods:** Twenty patients were enrolled in this study. These plans include 8 sites (Cervical, Nasopharyngeal, oral cavity, rectum, prostate, bladder, lung, Esophagus). All treatments were delivered with (CT-linac 506c, UIH, Shanghai) using 6MV photon beams, this machine was equipped in-orbit diagnosis level fan-beam CT and amorphous silicon EPID (XRD1642, Varex Imaging Corporation, Utah). A Monte Carlo algorithm was developed to calculate the in-vivo transit dose distribution. A pre-treatment measurement was performed to access system accuracy by delivery on homogeneous phantom (RW3 slab, PTW, Freiburg). During treatment, each patient was received a CT scanning before delivery by in-orbit CT every one week. Meanwhile, transmit EPID image was acquired for each field when delivery. The reference of transmit EPID image was calculated based on both planning and daily CTs. The gamma passing rate were calculated. **Results:** The average (standard deviation) of gamma pass rate in RW3 slab phantom was $99.6\% \pm 1.0\%$ (3mm3%), $98.9\% \pm 2.1\%$ (2mm3%), and $97.2\% \pm 3.9\%$ (2mm2%). For patient measurement, the average (standard deviation) of gamma pass rate was $87.8\% \pm 14.0\%$ (3mm3%), $82.2\% \pm 16.9\%$ (2mm3%) and $74.2\% \pm 18.9\%$ (2mm2%) by using planning CT as reference, while $93.6\% \pm 8.2\%$ (3mm3%) $89.7\% \pm 11.0\%$ (2mm3%) and $82.8\% \pm 14.7\%$ (2mm2%) by using daily CT as reference. There were significant differences between planning CT and daily CT results. All p-values (paired t-test) were less than 0.01. **Conclusion:** Due to changes in internal anatomical structure, planning CT may introduce bias in reference image calculation for EPID-based in-vivo dosimetry verification. Using daily CT is recommend for EPID-based in-vivo dosimetry.

PO-GePV-T-144

Radiotherapy Dose Accumulation Routine (RADAR) - a Fast Dose Accumulation Tool for Planning with Previous Treatments

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Purpose: To pre-clinically evaluate a new treatment planning tool for re-irradiation risk evaluation which accumulates equivalent dose in 2-Gy fractions (EQD2) or physical dose from previous treatments with either rigid or deformable registration using a novel method to account for uncertainty in registration. **Methods:** Radiotherapy Dose Accumulation Routine (RADAR), developed as an Eclipse plug-in, matches previous and current plans using the existing rigid or deformable registrations in Eclipse and accumulates either physical dose or EQD2 with structure-specific α/β . Physical dose can be converted into EQD2 either with structures in previous plans, or with structures in the current plan after physical dose is transferred. To account for registration uncertainty, dose given to a voxel inside organs-at-risk is conservatively estimated as the local maximum within an ellipsoid centered on the voxel. The ellipsoid's 3 principal radii are user-configurable depending on the mobility of the structure and accuracy of registration. To further remove ambiguity, RADAR provides a structure matching option for the uncertainty ellipsoid to only include voxels with paired structure label between the previous and current plans. Overlay ratios of matched structures are calculated to inform possible issues of contouring and registration. **Results:** In a sanity check, physical dose accumulation via RADAR generates the same dose distribution as the Eclipse plan sum with rigid registration. RADAR finished within 7 and 40 seconds, for a 2.5mm and 1.25mm grid resolution, respectively, for a typical brain metastases case. Structure matching prevents non-relevant dose from being unreasonably assigned to organs-at-risk. Site-specific quality assurance checks have been performed to ensure its applicability. **Conclusion:** We are developing an EQD2 dose accumulation tool capable of handling both rigid and deformable registrations, additionally with local search options to account for registration uncertainty, providing an efficient and integrated workflow clinically needed to assess the risk for patients with previous treatments.

PO-GePV-T-145

Validation of Radiochromic.com Gamma Analysis Software

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Purpose: Agnew and McGarry presented a suite of images to evaluate software for gamma analysis, and verified with it a comprehensive set of commercial programs [Agnew CE, McGarry CK. A tool to include gamma analysis software into a quality assurance program. Radiotherapy and Oncology. 2016 Mar 1;118(3):568-73]. The purpose of our study was to validate the implementation of gamma analysis in Radiochromic.com (Radiochromic SL, Benifaió, Spain) by using these tests. Two different versions of Radiochromic.com were evaluated: currently available version 3.3 and a pre-release version. **Methods:** The Agnew and McGarry's (AG) tests consist of four synthetic geometric images and two clinical images (IMRT and VMAT plans). The synthetic images have known gamma results assuming that the evaluated image is not interpolated. However, real dose distributions are continuous, and more accurate gamma values are obtained by interpolating the evaluated images. The results of the clinical AG tests depend on the interpolation used. Both Radiochromic.com versions apply bicubic interpolation, although the resolution of the interpolated images may differ. Also, the pre-release version has the option of not interpolating, employs parallel computation, and can calculate 1D to 3D gammas. **Results:** In the clinical tests, Radiochromic.com v3.3 had a mean passing rate and standard deviation of $99.0 \pm 1.4\%$, while in the pre-release version it was $98.9 \pm 1.5\%$. For comparison, from the AG paper, mean passing rates for FilmQAPro, PTW Verisoft v6.1, and Sun Nuclear SNC v6.5 programs were $96.2 \pm 6.0\%$, $98.8 \pm 1.9\%$, and $98.5 \pm 2.1\%$, respectively. In the synthetic tests, the pre-release version obtained exact results and v3.3 had errors of $-0.1 \pm 0.5\%$ as a consequence of interpolation. **Conclusion:** The accuracy of the implementation of the gamma analysis was validated in both versions of Radiochromic.com.

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PO-GePV-T-146

Radiation Oncology Community Site Integration Experience - Tools and Checklist

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Purpose: To share our experience, findings, tools, and checklists from a successful integration of radiation oncology information system (ROIS) of community site into a centralized ROIS and to illustrate the role of medical physicists and IT support. **Methods:** An integration steering committee led by a qualified

medical physicist (QMP) oversees the whole integration for safe patient care. This integration procedure can be categorized into six major sub-processes: staff readiness, treatment delivery machine readiness, patient data integration, machine reconfiguration, clinical process integration, and post-integration verifications. Based on their natures, these processes can progress before or after the integration, in parallel or in sequence. Staff readiness and machine readiness are a pre-integration process without system disruptions. During patient data integration and machine reconfiguration, no patient treatments can be performed due to system disruption/configurations. Clinical process integration and post-integration verification will happen right after patient data integration and machine reconfiguration. **Results:** These integration processes produced successful integration of a new member practice site into our academic radiation oncology system (already integrated with five clinics) including a Proton site with a centralized radiation oncology information system. The successful planning and detailed technical preparedness resulted in no patient treatment delays at the existing sites. However, due to the availability of the vendor's assistance for machine reconfiguration to our centralized database, the patient treatments at the newly integrated community practice were postponed for one day. Although we integrated a community practice with its specific infrastructure, the whole integration procedure suggested in this report is infrastructure- and vendor-neutral and can be easily adopted by other institutions. **Conclusion:** Our experience shows that careful planning, effective communication, various combination of expertise, and coordination of functional groups (including vendors) would lead to efficient and seamless integration of a community radiation oncology practice to an existing ROIS.

PO-GePV-T-147

Automated Verification of Treatment Parameter Transfer Between TPS and OIS Databases Via Direct Value Comparison

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Purpose: Transferring treatment plans from a treatment planning system (TPS) to an oncology information system (OIS) for treatment delivery can introduce data errors, especially when manual manipulation is required. The accuracy of data transfer must be validated before commencing a treatment. The current practice of manual checks does not review all field parameters (e.g., MLC leaf positions). The purpose of this work was to automate this critical component of chart checks between Varian's Eclipse TPS and Elekta's Mosaic OIS. **Methods:** Eclipse-Mosaic Transfer Validation (EMTV) was developed using the Eclipse Scripting API to read field parameters directly from Eclipse. It queries values from the Mosaic database for comparison. EMTV checks the prescription dose, fractionation, and field parameters including individual control points. To test and validate the EMTV, two physicists performed chart checks with and without using EMTV. Errors detected by both processes were recorded to determine the validity of EMTV. Times used in chart check with and without EMTV were recorded to determine time savings. **Results:** A total of 53 patient plans were evaluated. Manual checks revealed one error, while the EMTV flagged two plans with errors. A prescription mismatch for a patient transferred from a satellite was discovered by both; the fraction number was adjusted in Mosaic but not in Eclipse. Another error, only detected by EMTV, occurred in individual control points; caused by rounding during the data transfer. The manual check averaged 2.5 ± 1.8 minutes while EMTV averaged 6.2 ± 5.3 seconds. **Conclusion:** An automated tool for treatment parameter transfer verification has been developed and validated. EMTV was shown capable of catching mismatches in the plan parameters. It was 16 times faster than the current methods while being more comprehensive. The EMTV has a potential to streamlining the clinical workflow while improving patient safety.

PO-GePV-T-148

Application of FMEA to Review and Analyze Software Error Messages

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Purpose: Error messages from vendors can often be confusing and lacking in clarity. Our department has implemented an internal "Problem Ticket" website where staff can enter problems with clinical equipment, including error messages encountered on clinical systems. We propose a systematic review of these errors so they can be fully understood in a multidisciplinary manner through including working with radiation oncology staff, vendors, and departmental/hospital IT. The overall goal of this process is to provide staff with appropriate and timely feedback to help mitigate any negative consequences caused by error messages. The purpose of this study is to report on high impact error messages observed within our department. **Methods:** TG-100 describes a method to prospectively find/rank failure modes (and associated failure pathways) using process mapping, FMEA, and fault tree analysis. In this study, we perform a systematic review of error messages using FMEA to retrospectively analyze and rank error messages reported at our institution. Reports are categorized and prioritized using FMEA/RPN formalism. **Results:** In total, 307 error messages were reviewed from the departmental "Problem Ticket" website and assigned values for Occurrence(O), Severity(S), and Detectability(D) to calculate RPN. Of the 307 messages, 24 were identified with RPN>200 and further analyzed/categorized. Two high-frequency categories were Network/Services and Data Import to the TPS. Both issues caused increases in response time to resolve the error. For both categories, the corrective action/troubleshooting step was not clear from the error message content, so reviewing and communicating the recommended steps to staff has the potential to improve the effectiveness and efficiency in real-time response to both categories. **Conclusion:** Development of a systematic and department-wide method to analyze error messages using FMEA can help to fully understand error messages and ensure that appropriate staff are notified, thereby improving both effectiveness and efficiency in troubleshooting error messages in the clinical setting.

PO-GePV-T-149

A Custom-Made Cassette for In-Water Film Dose Measurements and Quality Assurance of Cs-137 Blood Irradiators

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Purpose: To develop a cassette capable of holding custom-sized radiochromic film in a reproducible manner for in-water dose measurements and quality assurance (QA) of Cs-137 blood irradiators. **Methods:** The currently developed film cassette was made of two thin (0.8mm) Lexan sheets cut to 19.4cm x 12.3cm. It contained a rectangular hole at the center (9.7cm above the bottom of the canister) and circular holes at six off-center locations (2.5cm and 5cm lateral to the center, and 8 cm above and below the center). The holes facilitated uniform exposure within the designated regions of measurement without any interference from the cassette material. To measure and verify the dose output and dose distribution of Cs-137 blood irradiators, a GAFChromic EBT-3 film was sandwiched in the cassette and placed diametrically vertical inside the water-filled canister. The film was then irradiated for the specified set time to deliver the desired dose (~8 Gy) based on the manufacturer-quoted dose rate. The film was scanned using Epson Expression 1000XL scanner one day after the irradiation. The measured optical density of the film was converted to the dose using the film calibration curve derived from prior measurements with an ionization chamber-calibrated (reference) Cs-137 irradiator. The film calibration curve was verified on the date of measurement by exposing two 2cm x 2cm pieces of the same film to a known dose (~8 Gy) using the reference Cs-137 irradiator. Another 2cm x 2cm piece of unexposed film provided the background reading. **Results:** The measured doses at the center and off-center locations using the film cassette agreed with the manufacturer-stated doses within $\pm 5\%$ over three periodic QA sessions. **Conclusion:** The currently developed film cassette was found efficient and reliable for in-water dose measurements and periodic QA of Cs-137 blood irradiators.

PO-GePV-T-150

Development of An In-House Off Axis Mechanical Isocentricity Test

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Purpose: To develop and analyze an off-axis isocentricity test for the evaluation of a LINAC's multiple metastasis single isocenter treatments and determine

the machine's off-axis limitations. **Methods:** A QA plan was developed for use with an existing phantom in our department. The existing QA phantom has a series of 0.75 cm diameter BBs positioned throughout itself. The two BBs used for evaluation in this test were located at the beam's isocenter and at a combined 3 cm lateral, 2 cm longitudinal, and 2 cm vertical displacement. The radial displacement of this BB from the isocenter is 4.12 cm. Images were acquired with various couch, gantry, and collimator rotations using the EPID. An in-house script was developed to analyze these images and calculate the absolute 3D error of the machine's isocenter off axis. **Results:** The error in the test machine's mechanical isocenter was observed to be 0.74 mm. At the off axis BB the absolute 3D error was observed to be 1.49 mm. A visual inspection was used to verify the accuracy of the script's BB identification routine. The location of the BB as denoted by the script agreed with our visual inspection. **Conclusion:** This proof of concept tool can successfully identify and localize two BBs in three dimensional space using a series of MV images. These tests confirmed that small mechanical errors are magnified away from the machine's isocenter. This work paves the way for future simultaneous evaluation of several points off the beam axis. This future work will allow one to determine an acceptable cutoff for single isocenter multi-metastasis treatments.

PO-GePV-T-151

Validation of Automated Segmentation Algorithms for Clinical Radiation Therapy Treatment Planning

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Purpose: To introduce a systematic method of clinical validation of automated-segmentation algorithms including patient- and organ-specific variations. **Methods:** Patient- and organ- specific variations were quantified for two algorithms (A1-2). Eighty computed-tomography scans including a range of treatment sites and scanning protocols from head to pelvis with 34 different organs-at-risk were included. For each patient, Algorithm-generated contours were compared to humans using nine volume, surface, and statistical measures. Validation of A1 and A2 were performed based on accuracy, consistency, and potential problems in specific-patients and specific-OARs prior to clinical introduction. **Results:** Sample means were compared to the population means to identify outliers flagging OARs including femurs, esophagus, larynx, and pharynx. The range of volume and surface metrics in femurs was 0.27-0.52 and 17.8-66.4mm in A1 implying the algorithm was failing. However, these differences represented the difference between A1 contouring only the femoral head, and clinical contours including lateral and inferior aspects of the femur; this is an example of institutional vs. algorithmic definition, not contouring error. This was one of several examples of variations in the "ground truth" dataset and the auto-contour models. Per-patient variations revealed issues in image artifacts, large slice spacing (>5mm) combined with small structures, and issues predicting bladder in scans extending significantly beyond pelvis. Per-organ analysis revealed volume- and boundary- metrics combined to correlate well to high-quality auto-segmentations. However, statistical metrics such as variation of information identified different issues like large boundary error in few slices. **Conclusion:** Automated-segmentation will reduce human workload and standardize clinical practice but has the potential to introduce clinical errors; manual review/editing of erroneous auto-contours can negate efficiency benefits. Per-patient and per-organ validation of automated segmentation will detect problematic contours and enable seamless clinical integration.

PO-GePV-T-152

Application of Automatic Plan Checking and Report Printing to Improve Clinical Workflow Accuracy and Efficiency

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Purpose: The pre-treatment plan check by a physicist is a key component to ensuring patient safety and treatment plan quality. The report is also a critical part for the radiation therapy. The purpose of a computer assisted plan check and report is to ensure plan quality, accurate patient setup instruction, and correct documentation of patient and plan information. **Methods:** To implement an assisted plan check a script was written using the ESAPI and C#. The plan check list and report follow TG275 and TG315 guidelines. The plan check includes prescription, structure, collision, constraints and parameters checks. The "print" button generates the report and saves it in patient's folder. The report includes the coversheet with the ACR recommended checklist and patient setup instruction, the prescription table, plan summary, 2D-view, constraints table, parameters table and DVH. The setup instruction includes the import information about the plan such as DIBH, Bolus etc, shifts, SSD and table vertical position. This information is helpful to therapists to setup and treat the patient correctly. **Results:** The script has been tested by physicists for 6-month, and was then implemented in clinic for 5-month. The script saved at least 30-minute for each plan check and printing. The script created report includes all the recommended items in the TG315 report. The most errors recorded by the script are the "reference-point-dose-limit" and "tolerance-table". Since the implementation of this assisted plan check script, our near miss rate has dropped from the value prior to its implementation. **Conclusion:** Automating some processes of the plan check and report reduces the errors, voids manual input mistakes, and improve plan check efficiency. This standardized treatment plan report can be used as a consistency check for planners, a pretreatment check by therapists, a quick review for the physicist's initial, weekly and final checks, and billing verification.

PO-GePV-T-153

Towards Implementation of a Knowledge-Based Decision Support System for Treatment Plan Auditing Through Automation

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Purpose: To identify the needs and opportunities in standardizing and improving our external beam radiotherapy (EBRT) treatment planning procedure across all 7 campuses of our institution, a knowledge-based anomaly detection algorithm to provide decision support for our manual retrospective plan auditing process was developed and implemented. **Methods:** 710 lung SBRT (Stereotactic Body Radiation Therapy) plans that were delivered from 2015 to 2020 have been acquired automatically from our clinical treatment planning systems. A total of 44 plan parameters were extracted from each plan and pre-processed. A knowledge-based anomaly detection algorithm, namely "isolation forest" (iForest), was then applied. Each plan was recursively isolated by randomly selecting a plan parameter and a split value, represented by an isolation tree (iTree). Subsequently, an iForest is assembled from a collection of developed iTrees and an anomaly score is determined from the path length between the root node and a terminating node on the iTree. Anomalous plans are more susceptible to such isolation mechanism, thereby having higher anomaly scores. To evaluate the feasibility of the proposed method, top 20 plans ranked with the highest anomaly scores together with the extracted plan parameters were validated manually by two plan auditors. **Results:** The two auditors certified that 12 out of 20 plans with the highest iForest anomaly scores have similar "alarming" qualities to further investigate and improve our planning procedures and staff training. A manual audit of one chart is estimated to take 45 minutes, while with the guidance of iForest it is estimated to save approximately 7.5 minutes per chart, a time savings of 16.7%. As our internal audit process typically consists of review of 250 charts annually, the total time savings are substantial. **Conclusion:** The iForest method effectively detects anomalous plans with high sensitivity and provides decision support to our cross-campus manual plan auditing procedure.

PO-GePV-T-154

A Methodology and Open Source Tool for Radiotherapy Dose Transformation QA

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Purpose: Owing to improved palliative treatment options and increased survivorship, a growing number of patients require multiple distinct radiotherapy treatments. This presents unique planning challenges, and it is now commonplace to use image registration and biologically-effective dose (BED) methods for

lifetime organ-at-risk dose estimation. However, quality assurance (QA) options are limited, and often involve spot-checks. In this work, we present a simple and universal method and software implementation for QA of dose transformation methods like BED. **Methods:** The equivalent dose in 2 Gy fractions (EQD2) paradigm is based on the linear-quadratic cell survival model. How dose is transformed in clinical scenarios depends on the fractionation and α/β , which is tissue- and outcome-specific. Inverting the EQD2 model provides a way to estimate α/β across the entire body, voxel-by-voxel, given the fractionation, EQD2 dose, and original dose. Histograms of α/β -volume can be extracted for individual regions-of-interest and analyzed automatically via percentile extraction, or maps can be visually inspected to survey the entire body quantitatively. Model inversion is sensitive to human transcription and selection errors involving the original fractionation, ensuring appropriate courses were used, α/β settings, selection of appropriate regions-of-interest, and also that the anticipated EQD2 model was selected. A free and open source implementation of EQD2 model inversion is provided. Simulated errors were tested to confirm detectability. Results of being used clinically in a supplementary role to spot-check QA for 21 plans are provided. **Results:** Errors resulting from incorrect region-of-interest selection or overlap handling, use of incorrect EQD2 models, incorrect fractionation and course selection can all be detected. Clinically, instances of incorrect α/β were detected for two plans with EQD2 model inversion that were not detected using spot-checks (invalid region-of-interest selection and invalid α/β). **Conclusion:** Model inversion is an effective approach for independent QA of BED-based dose transformations.

PO-GePV-T-155

Challenges On Implementing An Hybrid AI-And-Rules Based Plan Check Tool in Clinical Practice - a Pilot Study

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Purpose: Artificial Intelligence (AI) has become a focus of research in radiation oncology in recent years, demonstrating potential for improvements in clinical safety in multiple studies. However, there are hurdles to the implementation of AI applications in ways that can realize those improvements clinically. In this work, we report a pilot study on implementing an in-house developed hybrid rules-and-AI based tool designed to assist medical physicists in initial plan checks. The goals of the study are to identify the key challenges and evaluate these aspects of system implementation into clinical workflow. **Methods:** The web-based plan check tool takes a hybrid method approach, consisting of a rules-based component and a Bayesian network (BN)-based AI model. In the pilot study, three volunteering clinical medical physicists use the tool to assist their routine plan check process. Potential errors that are flagged by the tool are verified by the physicists and report as true positive or false positive results. Feedback is collected from the volunteers on their concerns using the tool. **Results:** The simple logic system has shown excellent performance in the pilot study, while the BN-based AI tool successfully flags different potential logical errors in clinical plans. However, the AI tool also reported multiple false positive flags, and volunteers have reported that these repeated false positive errors combined with the lack of understanding on the AI model reduced their confidence in relying on the tool. **Conclusion:** The BN-based AI tool was found to be useful to physicists in identifying errors in treatment plans during routine checks. We also found that interpretability of results presented a challenge to users, and understanding how to manage trade-offs between reliability, efficiency, and efficacy of the model needs to be explored.

PO-GePV-T-156

Development of An Automated Pre-Treatment Plan Evaluation

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Purpose: To ensure high-quality care, radiation oncology teams conduct pre-treatment plan evaluations which review the machine treatment parameters, patient set-up instructions, prescriptions, and radiation dose distributions. The current process is performed manually, which consumes a great deal of time and introduces the risk of human errors. Automating pre-treatment plan evaluations can improve clinical efficiency and reduce errors. **Methods:** Using Python, a web-based user-interface was created to automatically compare the data from the DICOM export and the Mosaic SQL database. Various machine parameters, prescription details, and patient information are taken into consideration when cross-comparing between the two systems. The results are displayed side-by-side for each field being evaluated, and any discrepancies are highlighted for easy recognition. Additionally, the DVH is calculated and structures are assessed to determine if, and by how much, all constraints are being met. These scripts are publicly available through PyMedPhys. **Results:** A data-transfer check has been developed to ensure a plan's DICOM files and Mosaic's database information are consistent between general patient information and beam specific information. The comparison can be performed in a matter of seconds, making it significantly less time consuming than a manual comparison. Additionally, plan quality can be determined through DVH analysis and compared to an archive of patients. **Conclusion:** The data transfer portion of a pre-treatment check can be automated and used to drastically reduce the amount of time required to complete the task while minimizing opportunity for human error. Implementing this into clinical workflow has the potential to increase efficiency while also increasing error detection sensitivity and improving plan quality.

CPRIT Pre-Doctoral Trainee (RP 170345)

PO-GePV-T-157

An End-To-End Test to Evaluate Off-Axis Targeting Accuracy for Single-Isocenter Multitarget Stereotactic Radiosurgery

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Purpose: Single-isocenter multitarget radiosurgery (SIMS-RS) is a treatment option for multiple brain metastases. Precise alignment of rotational axes of the linear accelerator is crucial and must be checked frequently. The drawback of current quality assurance (QA) procedures is that targeting accuracy is only evaluated at isocenter. This study introduces a hidden target end-to-end (E2E) test allowing to check targeting accuracy for off-axis targets. **Methods:** The E2E follows our current clinical workflow. Targets are three BBs placed inside the Rando head phantom. The procedure is similar to the Winston and Lutz isocentricity test (WL-test). A 10cmx10cm field size allows that the implanted BBs were visible in the recorded portal images. For evaluation, an in-house tool for image analysis and 3D reconstruction of target positions was developed. The algorithm allows reconstruction of multiple BB positions in a non-homogeneous phantom. In addition, a multi-view 3D reconstruction is used to automatically identify the BBs and to calculate its 3D positions. The automated analysis consists of a classic WL-test for the central BB as well as a comprehensive 3D analysis of all BBs. **Results:** The median distance for the center BB to the CAX was 0.46mm. The reconstructed size of the isocenter was 0.36mm for collimator/gantry combinations and 1.0mm for couch rotations. This is in agreement with an independent WL-test. For the off-axis BBs with an displacement of 5.6cm and 7.0cm the reconstructed center size was 0.49mm and 0.72mm, respectively. Couch rotation led to an uncertainty 1.05mm for the proximal and 1.26mm for the distal BB. A 3D point registration showed that the total target accuracy was 0.43mm. **Conclusion:** The proposed E2E test in combination with the developed processing algorithm allows a streamlined assessment of SIMS-RT targeting accuracy and can be used as an additional QA step for those complex treatments.

PO-GePV-T-158

How AI Can Support Personalized Risk Mitigation During Automated Treatment Planning

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Purpose: To automatically determine individual risk priority numbers for patients during the radiation therapy process due to potential human and technical errors and to demonstrate how to mitigate some of these risks through artificial intelligence (AI). **Methods:** APIs (Application Programming Interfaces) for

treatment planning systems are used to expand their scope of functionality and be used to interface AI algorithms. In this work, a C# environment was used to implement an Expert System (own activity), supervised Machine Learning (ML.NET, Microsoft), and Eclipse Scripting (ESAPI, Varian Medical Systems). Ratings for the risk factors severity S, occurrence O, and detectability D were created for several error scenarios and grouped into three phases of an automated planning workflow (load patient with associated demographic information; checking prescription and consistency of PTV/OAR structures; calculating plan setup, dose, and quality metrics). The program calculates risk priority numbers ($RPN = S \times O \times D$) during each planning step and compares these with values mitigated by the AI algorithms. **Results:** Several AI mitigations were successfully integrated, among them binary classification, anomaly detection, and recommendations. In more than 80% of automated planning workflows with initially critical (RPN between 30 and 125) or unacceptable risk levels (RPN above 125), the AI-supported treatment planning found mitigations to reduce values to an acceptable level (RPN below 30). Where algorithms could perform no mitigations, root cause analysis identified either missing branches in the Expert System or missing data. **Conclusion:** This work demonstrates the potential for significant risk reduction by embedded AI support during treatment planning. With prior trained machine learning models and expert systems, the gain in safety comes basically without additional costs; and the modular structure of ESAPI also allows the implementation of other mitigations programmed by third-party institutions.

PO-GePV-T-159

Failure Modes and Effects Analysis of Low Dose Bilateral Whole Lung Irradiation Treatment for COVID-19 Positive Patients Requiring Respiratory Ventilation

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Purpose: To identify high priority risks in a new radiation treatment workflow for alleviating severe COVID-19 pneumonia using a multi-phase failure modes and effects analysis (FMEA). **Methods:** A comprehensive FMEA survey of 133 possible causes of failure was developed for a clinical trial workflow treating mechanically ventilated COVID-19+ patients with low-dose whole lung radiation therapy (Phase I). The occurrence, severity, and detection risk of each possible cause of failure was scored by 3 physicists using an adapted scoring guide based on TG-100 recommendations. High-risk potential failure modes were identified using the average risk priority number (RPN) and severity scores. The responses were used to create a second targeted survey containing high-priority risks throughout the treatment workflow (Phase II). The second survey was completed by 13 participants in Radiation Oncology including therapists, physicians, and physicists. Scores from the Phase II survey were evaluated to identify steps requiring possible intervention and examine risk perception patterns. **Results:** Based on the results of the initial survey, 30 high-priority failure modes were selected for the Phase II survey. Strong internal consistency was shown in both surveys using Cronbach's alpha ($\alpha > 0.82$). The 10 failures with the largest RPN values concerned COVID-19 transmission ($N = 6$), wrong treatment ($N = 3$), and patient injury ($N = 1$). The average RPN was larger for COVID-related failures ($RPN = 108$) than other failure types ($RPN = 67$) due to the perceived difficulty of failure detection. **Conclusion:** The high ranking of failures concerning nosocomial transmission of COVID-19 suggest that these steps may require the most quality management intervention for the treatment of critically ill COVID-19+ patients. The results also suggest that a multi-phase FMEA survey may provide useful information for assessing risks in radiation therapy procedures, supporting efforts to adapt FMEAs to clinical procedures in the literature.

PO-GePV-T-160

Failure Modes and Effects Analysis (FMEA) for Multi-Site, Multi-Institutional Deep Inspiration Breath Hold (DIBH) Treatments

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Purpose: Deep inspiration breath hold (DIBH) is a motion management technique used for various treatments, notably left breast irradiation. We performed a multi-institutional TG100 recommended risk analysis of DIBH for lung, abdomen and breast irradiations to identify where quality management can be improved. **Methods:** A detailed process map was outlined for DIBH treatments of both institutions and fault tree analysis (FTA) of all potential failure modes (FM) was performed. A team comprised of two radiation oncologists, four physicists and one radiation therapist ranked occurrence (O), severity (S) and detectability (D) on a 1 to 10 scale. Ranking was subjective based on personal expertise treating DIBH within each center. FMs with the both the highest risk priority number (RPN) and highest severity scores were considered top priority and fault tree analysis was examined to identify ways to improve our workflow. **Results:** The process of DIBH treatment was divided into four sub-processes: motion study, DIBH image acquisition, treatment planning, and treatment delivery, each with multiple steps. We identified 22 FM's for the motion study sub-process, 52 for DIBH image acquisition, 8 for treatment planning and 20 for treatment delivery, for a total of 102. The RPN scores ranged from 12-245, 36-120, 18-150 and 24-144 for each sub-process respectively. The severity scores ranged 3-5 for each sub-process. Many quality assurance and control steps have been suggested for clinical implementation based on this analysis. **Conclusion:** FMEA analysis revealed portions of inconsistent workflow and areas needing quality management. The analysis will be discussed to improve the workflow and quality of patient care.

PO-GePV-T-161

Comparing Leakage Doses Between Three Gamma Knife Icon Machines

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Purpose: With Icon's capability to treat a patient fixated with a mask and the ability of temporally suspending the treatment as soon as the patient moves out of a pre-set tolerance, it is vital to verify the leakage since the sources are parked at the Block position for a maximum of 5 times per shot, each time could be as long as 30 seconds during treatment suspension. **Methods:** Three methods were employed to measure leakage dose on three Icon machines. All methods calculate the leakage percentages using the ion chamber readings with all sectors On and the readings with some sectors Off. Method-A turns off some combinations of sectors. Method-B starts with turning Sector-1 on, and adds one consecutive sector at a time. Method-C turns on only one sector at a time. **Results:** The results from machines 1, 2, and 3 for Method-(A, B, C) are (0.08%, 0.02%, 0.10%), (0.18%, 0.13%, 0.15%) and (0.14%, 0.09%, 0.14%), respectively. **Conclusion:** Since GammaPlan requires a minimum shot time of 8.8 seconds for a shot containing 16-mm collimation (5.5 sec for 4-mm or 8-mm shots), a specification of 0.2% leakage at the Block position translates to 3.4-5.5% extra dose to the patient in the worst case. Our results show that we have met this spec.

PO-GePV-T-162

Failure Mode and Effect Analysis for Volumetric Modulated Arc Therapy for Total Body Irradiation

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Purpose: To apply failure mode and effects analysis (FMEA) to the Volumetric-Modulated Arc Therapy (VMAT) technique for Total Body Irradiation (TBI) following TG-100 guideline. **Methods:** A process map for VMAT TBI was developed following the AAPM TG-100 guidelines. This process map gives a visual representation of the flow of steps in the treatment process from the initial simulation, treatment planning, and treatment delivery. From the process map, all potential failure modes in the process were identified. Evaluation of the occurrence (O), detectability (D), and severity of impact (S) of each failure mode according to scoring criteria (1-10) by radiation oncology staff members was carried out. Subsequent calculation of the risk priority number ($RPN = O \times D \times S$)

S) and identification of effective quality assurance methods for the highest risks proceeded evaluation. **Results:** A total of 26 failure modes were derived. Five failure modes (top 20%), identified as being high-risk were (1) CBCT patient positioning check failure (2) the treatment planning system reports incorrect MUs after re-optimization (3) incorrect patient set-up SSDs recorded in the record-and-verify system (4) patient movement during treatment and (5) organs-at-risk are contoured incorrectly, producing an inaccurate dose to critical structures. Further quality assurance and quality control measures were proposed to improve effectiveness of the procedure. **Conclusion:** Quality assurance and quality control interventions were implemented based on the FMEA results. Further checks of the treatment planning system reported dose, patient contouring, patient positioning and of the record-and-verify system were proposed to improve the effectiveness of the treatment procedure.

PO-GePV-T-163

Efficiency Analysis of the Automatic Plan Printing Application

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Purpose: To evaluate the efficiency and effectiveness of a script based automatic treatment plan report printing application. **Methods:** An Eclipse Scripting API (ESAPI) based C# application has been developed in two hospitals where they were printing the treatment plan document manually. The script was designed to convert the treatment planning data, such as plan parameters, image information, dose volume histogram, and dose distributions, into the department-specific plan reports. The report is saved as a single PDF file after automatic attachment of independent MU calculation form etc. Total time spent during the plan printing was measured by parsing the time stamp of an individual PDF file or manually counting the elapsed time by the planner before and after the automation. Other qualitative feedback was also collected from the planners. **Results:** In the A hospital, the mean manual plan printing time was 24.1±20.7 min (median 16.3 min) per plan, while 0.4±0.4 min (median 0.3 min) per plan was spent after the automation, which gives 19.7 min time saving per plan. Similarly, B hospital could save 3.8 min per plan after the automation, where the mean manual printing time was 5.2 min per plan. Considering the average number of treatment plans of 2,470 and 14,690 cases at each hospital, their time savings could be converted into 0.46 and 0.48 FTE, respectively. In addition to the quantitative time saving, the planners could reduce their stress with being free from mistakes and avoid reprinting. Automation improved the workflow to print out the treatment plan, resulting in more consistency, and allowing to provide more information in the report. **Conclusion:** It was found that significant time and effort are spent on the plan report printing. The plan finalization time could be reduced, and the planning workflow could be improved using the automatic plan printing tool.

PO-GePV-T-164

Sources of Systemic and Random Errors in Bleached Optically Stimulated Luminescence Dosimeter (OSLD) In-Vivo Measurements: A Retrospective Clinical Study

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Purpose: To investigate sources of systemic and random errors in in-vivo OSLD measurements during regular clinical use, by comparing adjacently placed pairs of OSLDs in total body irradiation (TBI) and total skin electron therapy (TSET) under the assumption of equal dose delivered to both OSLDs. **Methods:** Directly adjacent pairs of NanoDot OSLDs were placed on patients with a medical tape prior to TBI and TSET, with or without 1.5cm of bolus respectively, at each anatomical site monitored. Each OSLD was read 5 times (standard error of mean $\sigma=0.2\%$). Resulting raw readings were divided by vendor-provided sensitivity (mean 0.85 ± 0.02) to obtain corrected readings (CR). Used OSLDs were bleached between each use until 1500cGy was accumulated. OSLD database was queried January 2019 to November 2020 to obtain 622 paired samples possibly used for 200cGy prescription TBI and TSET. Pairs of 101 TBI and 110 TSET measurements were verified to be present within clinical records. Relative difference in corrected counts ($\Delta CR_r = \text{pair-difference}/\text{mean}$) was fitted linearly versus sensitivity, bleaching history, cumulative dose, and age differences between paired OSLDs. **Results:** OSLDs were bleached 8 times (0~28), accumulated 942cGy (60~3015), and were used for 238days (0~819) on average before termination. ΔCR_r was normally distributed ($\sigma=\pm5.9\%$). ΔCR_r for TBI and TSET subsets deviated $\sigma=\pm4.7\%$ and $\pm6.3\%$ respectively, despite similar predictor distributions. OSLD with higher sensitivity yielded -0.68% less per +0.01 than the other ($p=4e-17$). Partial multiple linear models taking sensitivity into account found little contributions from bleaching ($p=0.89$) or cumulative dose ($p=0.047$). TBI and TSET samples yielded $\text{RMSE}=\pm4.8\%$ after sensitivity correction. **Conclusion:** Our findings suggest vendor-recommended reading correction via sensitivity division results in a small but significant systemic error. OSLD sensitivities should be considered during reading-to-dose calibration and clinical measurements. Bleaching and cumulative dose likely minimally affect measurements. Random errors were higher for TSET than TBI.

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PO-GePV-T-165

The Use of Statistical Process Control Methods to Evaluate Patient-Specific Delivery QA Failure as a Function of Plan Complexity

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Purpose: To evaluate the suggestions that QA passing rates decrease as a function of plan complexity and the subsequent need for site-specific tolerance limits. **Methods:** Patient-specific ionchamber array-based delivery QA analysis for 250 previously treated, non-SBRT VMAT patients analyzed at 3%, 3mm and a threshold of 10% was repeated with gamma criteria of 3%, 2mm and a threshold of 10%. Cases were divided into H&N (n=76), Thorax (40), Abdomen (61), and Pelvis (73) with passing rates plotted as a function of an approximate modulation scaling factor (MSFapprox.). Statistical Process Control (SPC) methods were used to compare the action limit of each subset to the universal action limit (ALuniv) of 90%. **Results:** The site-specific action limits differ from the ALuniv by 0.9, -1.1, -2.3 and -4.4% for the Abdomen, H&N, Pelvis and Thorax respectively with MSFapprox. and passing rates being significant between the groups at the 1% and 5% levels, respectively. The passing rates did not reach the same level of significance due to the lower number of thorax cases available for evaluation. The correlation of passing rate as a function of MSFapprox. was "low" for all sites with pelvis having the highest r value at -0.35. The tolerance limits (TL) for the above body sites were 92.8, 91.0, 89.3 and 88.7%, respectively. **Conclusion:** Both site-specific TL and AL for our system show some variation. The site order presented above demonstrated difference from ALuniv as well as general increasing MSFapprox.. It is interesting that the Pelvis (GI, GU, GYN) site demonstrated higher plan complexity and lower AL than H&N at our institution which is contrary to what is generally thought. Additionally, correlation with plan complexity being "low" for all suggests that the need for site-specific action levels may be unwarranted.

PO-GePV-T-166

Variability of Control Limits for Mean Percent Dose Differences Across Different Anatomical Contours

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Purpose: To investigate the use of Statistical Process Control (SPC) to monitor use of a web-based auto-contouring and planning tool. **Methods:** The Radiation Planning Assistant (RPA) is such a tool that provides plans calculated on standard linacs, which must then be recalculated by the user in their own treatment planning system for their own linac. Thirty-three 4-field box cervix plans (8 structures) and 34 VMAT head and neck plans (30 structures) were generated by the RPA. These plans were downloaded and imported into Raystation v10B where they were recalculated using fixed MUs. The final plans were uploaded

back to the RPA where the differences in mean dose for each structure was calculated. Utilizing SPC, control charts were generated with mean and control limits at the 3 standard deviation. Mean percent dose differences that fell outside of our control limits were removed from the calculation of the mean and standard deviation. **Results:** Head and neck mean percent dose difference and control limits varied per structure. For example, PTV1 (2.28%, [1.00%, 3.56%]), PTV2 (2.19%, [5.59%, -1.21%]), PTV3 (1.67%, [-0.67%, 4.01%]), and spinal cord (0.59%, [1.09, 0.09%]), and similarly, for cervix plans, L5 contour (-2.59%, [-3.43%, -1.74%]) and right femoral head (-1.79%, [-0.77%, -2.81%]). This indicates that SPC will give more individualized results per contour than can be achieved with a fixed threshold (e.g., 5%). Variability in SPC thresholds may be partially attributed to differences in dose calculations near the surface, particularly for targets near the surface for head and neck VMAT cases, and to differing body contours between Eclipse and Raystation. **Conclusion:** Specific control limits for individual contours based on percent mean dose difference can be determined and differ across anatomical sites. More study is needed to understand the underpinning of these differences and if they are effective in indicating plan errors.

PO-GePV-T-167

An Investigation of a Statistical Process Control Based Quality Assurance for Optical Stimulated Luminescence Detector (OSLD) System

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Purpose: Optically stimulated luminescence detectors (OSLD) are widely used for in-vivo dosimetry. The current vendor recommended quality assurance (QA) process is susceptible to errors, such as calibration curve changes. In this study, we investigated a statistical process control (SPC) based quality assurance process to ensure improved consistency and accuracy within an institution's OSLD program. **Methods:** All the OSLD used were nanoDot™ type OSLD (Landauer, Glenwood, IL) and read out in a MicroStarr® II system (Landauer, Glenwood, IL) calibrated with a specific set of calibration OSLDs, each exposed to doses ranging from 5 cGy to 200 cGy. A control OSLD was exposed with 100 cGy. All OSLDs were exposed in a Varian 6EX. The control OSLD was read out daily after performing the vendor suggested QA. The daily variation, $V(t)$, on day t was calculated by taking the ratio between the readout dose at t and $t-1$. The upper and lower control limits, UCL and LCL, were defined as 3 standard deviations, 3σ , which were established with the initial 26 data points. **Results:** The σ of $V(t)$ was found to be 0.8% translating to a UCL and LCL of $\pm 2.5\%$. Using these limits, an additional 71 data measurements were taken. Three instances were found to breach the control limits. Two of the three failures were attributed to either calibration curve changes or chip defects that were not captured by the vendor QA. One was a false positive. The accuracy and precision of the system was estimated to be 98.5% and 66.7% respectively. **Conclusion:** A quantitative QA process was established to enhance the quality control of an OSLD program and can potentially be scaled to a large hospital network. More statistics will be acquired to establish the long-term accuracy and precision of the system.

PO-GePV-T-168

4D Monte Carlo Assessment of the Impact of Inter- and Intra-Fractional Respiratory Motion Variation On Delivered Dose

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Purpose: To quantify the impact of respiratory motion variations on the delivered dose, using a previously validated 4D Monte Carlo-based dose reconstruction tool, for a cohort of non-small cell lung cancer patients undergoing curative radiation therapy treatments. **Methods:** Surface motion measurements recorded during treatment delivery using the RADPOS 4D dosimetry system were used for input in the 4D dose reconstruction tool, 4Ddefdosxyzncr, along with linac log files and the deformation vectors between the exhale and inhale phases of the 4D CT. Deformation vectors were calculated in Velocity AI (V3.2.1), and the reconstructed doses were compared to the clinical VMAT plans calculated on the average CT scan set using Monaco (V5.51.02). **Results:** We present here the results for two patients, for whom we have collected and analyzed the data over five treatment fractions. Each of the patients had the tumor located in the upper right lobe. No motion control device was used during planning CT scans and treatments. The average peak-to-peak breathing amplitude, r , was 5.23 ± 1.35 mm and 4.74 ± 0.21 mm for Patients 1 and 2, respectively. The 4D reconstructed doses show less than 0.5% inter-fraction variation for the CTV D98% for both Patient 1 and 2 and are within 3.0% of the CTV V66Gy planned in Monaco. These results are consistent with relatively small breathing motion amplitude measured for these patients. However, percent change of up to 22.3% in the mean dose to the esophagus was found for Patient 2 when comparing the planned dose to the reconstructed doses. **Conclusion:** This preliminary study demonstrated the use of a 4D reconstruction tool for a small patient sample to evaluate the effects of respiratory motion on delivered dose. Future studies will include larger number of lung patients and variety of tumor locations.

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PO-GePV-T-169

Dosimetric Impact on VMAT Plan Delivery From Tumor Motion Variations Due to Breathing Irregularities

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Purpose: To simulate treatment delivery with tumor motion by moving the isocenter based on tumor motion waveforms and evaluate the dosimetric impact from the tumor motion variations that differ from the internal tumor volume (ITV) used in volumetric-modulated arc therapy (VMAT) planning. **Methods:** A motion-incorporated VMAT planning was applied to simulate plan delivery with actual and simulated tumor motion variations based on 4-dimensional computed tomography (4DCT) acquired at simulation. First, the clinical ITV-based VMAT plans of 9 lung cancer patients with single lesions were segmented into composite plans with multi-beam fields at the control points. Second, tumor motion waveforms were applied to the isocenter. Tumor motion waveforms were created by scaling down the diaphragm motion extracted from the 4DCT images to the tumor motion. The single 4DCT breathing cycle was replicated and concatenated to form a long motion waveform to cover the duration of the 2-4 arc(s) in the clinical plans. Additional tumor motions were simulated by scaling up the tumor waveforms with extra 0.5, 1.0, and 2.0cm motions. Four motion-incorporated plans were created per patient and compared to the clinical VMAT plan, which covered planning tumor volume (PTV) at D95%=100%. The native-motion-incorporated plans were first compared with the clinical VMAT plan as a control, followed by 3 simulated-motion-incorporated plans using D95%=90% as a breach for ITV and PTV. **Results:** Using the original 4DCT motion, the mean D95% from all 9 patients is found to be 0.99 ± 0.02 . The D95%=90% breach point increases with ITV and PTV, and requires consistent motion variations of 0.5-1.0cm for small tumors (ITV=5-30cc) and ~2.0cm for large tumors (ITV=80-400cc). **Conclusion:** This study has demonstrated that large tumor motion variations deviating from simulation can reduce the ITV/PTV coverage substantially. Under clinical conditions, it seems necessary to apply patient breathing monitoring for threshold-based beam gating during treatment delivery.

PO-GePV-T-170

A Practical Workflow to Streamline Treatment Planning for Reirradiation with Considering Biological Doses

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Purpose: With recent substantial increase of radiation retreatments, solutions to include biological effects of the initial irradiation (II) in the re-irradiation (RI) are demanded. In this study, we develop a practical workflow to consistently consider biological dose distributions of the II in the treatment planning of reirradiation. **Methods:** The workflow was developed using MIM software, including its extensions, to (1) host patient retreatment plan data, tumor and organ-specific radiobiological model parameters extracted from pooled clinical data using common dose volume constraints with a linear-quadratic-linear (LQ-L)

model, and (2) perform biologically effective dose (BED) calculation, deformable image registration (DIR), dose summation and plan review. To plan a RI, the workflow includes (1) converting physical dose distributions to BED distributions using the selected model parameters for both II and RI, (2) creating the composite BED distribution by DIR of both planning images with considering tissue repair between the II and RI, (3) creating planning organ at risk volume (PRV) to consider the uncertainty in DIR, (4) converting the composite BED distribution into the physical dose distribution of the prescribed RI fractionation for plan evaluation, and (5) adjusting the RI plan if target coverage and doses to organs at risk are unacceptable and repeating the workflow until the composite plan achieves the prescription goal. **Results:** The workflow has been successfully used for 6 patients. It helps to adjust dose fractionation, in addition to beam and field arrangements, to achieve the prescription for the irradiation. Up to 18% dose difference was found between calculations with the current workflow and the conventional EQD2 method. **Conclusion:** The newly developed workflow has been successfully used in the clinic and will facilitate a prospective trial that is being designed to investigate efficacy of retreatment for multiple tumor sites.

PO-GePV-T-171

Daily Portal Exit Dosimetry Images as A Way to Monitor Interfractional Dosimetric Variation: A Feasibility Study

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Purpose: Exit dosimetry images are automatically acquired on the Halcyon™ linear accelerator for all treatment beams and could potentially be used to monitor daily treatment. We investigate the correlation between image analysis metrics calculated within the Portal Dosimetry Scripting API (PDSAPI) (version 15.1) and dosimetric changes throughout treatment. **Methods:** Daily exit dosimetry images were compared to first-fraction exit images for 8 patients undergoing 16-fraction prone breast treatments. Dose-difference images, gamma analysis (2%/2mm and 3%/3mm, with 10% threshold) images, and gamma indices were calculated within the PDSAPI. For dose-difference images, a 10-70% pixel intensity region of interest (ROI) was selected and percent variations from the first-fraction pixel values were calculated; for daily 10-70% ROI images, an average dose-difference value was calculated. A 10-70% pixel threshold helped to focus on fluences traversed through breast tissue by eliminating out-of-field and out-of-breast dose. Dose-volume differences between the planned and daily delivered PTV parameters (dV90, dV95, dV100, dV103, dV105, and dV107) and Breast Evaluation PTV size were calculated on synthetic-CTs derived from daily-CBCT images. Pearson correlation coefficients with 0.05 significance levels were calculated between image analysis metrics and clinically-relevant DVH metrics. **Results:** No gamma analysis metric produced significant correlations with any dose-volume parameters. Daily average dose-differences within ROIs and differences between -1-3% and +1-3% pixel deviations from dose-difference image metrics provided strong correlations ($r \geq 0.8$) with deviation in DVH parameters (dV90, dV95, dV103, dV105, and dV107) normalized to planning values, as well as with breast volume changes. **Conclusion:** Dose-difference tests between first and subsequent daily fraction exit portal dosimetry images could serve as a quick metric to identify overdose/underdose developments and changes in breast tissue volumes. These metrics can be easily implemented into clinical workflow using PDSAPI scripting capabilities for automated monitoring and alert for manual evaluation when large deviations are detected.

PO-GePV-T-172

Initial Experience with Varian Portal Dosimetry Patient-Specific QA

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Purpose: This work details our initial experience transitioning from ArcCheck to Varian Portal Dosimetry (PD) as our primary method for intensity-modulated radiation therapy quality assurance (IMRT QA). **Methods:** A total of 347 IMRT plans (866 beams) were measured in the initial months using Varian PD across four TrueBeams. Gamma analysis results using a 10%, 5%, and 1% threshold, with a dose difference/distance to agreement (DD/DTA) criteria of 3%/3mm and 2%/2mm were recorded for both composite and individual fields as well as mean and max calibrated unit (CU) differences. Development of departmental IMRT QA policies included a detailed troubleshooting flow chart with goal- and acceptable-level passing rates. PD plans failing to meet the action limit tolerance of $\Gamma \geq 95\%$ or $\Gamma \geq 90-95\%$ with all individual fields passing $\Gamma \geq 95\%$ at 3%/3mm (IMRT) or 2%/2mm (SBRT/SRS) would be deemed acceptable if ArcCheck passed $\Gamma \geq 90\%$ with 3%/3mm. **Results:** The number of plans with a gamma passing rate $\Gamma \geq 90\%$ and $\Gamma \geq 95\%$ was 97.1% and 91.6% using 3%/3mm and 80.1% and 66.0% using 2%/2mm (10% threshold). Min, mean(stdev), and max average CU differences were 0.5%, 2%(1.3%), and 6.9%, respectively. For PD plans with $\Gamma < 95\%$ at 2%/2mm, the mean DD/DTA needed for $\Gamma \geq 95\%$ was 2.8%/2.8mm. Out of the 17 plans that failed to meet the PD action limit and were measured on the ArcCheck, 2 failed on the ArcCheck and required re-planning. Qualitatively, some of the observed benefits of PD IMRT QA include a notably reduced physics time burden, ability to perform composite and field-by-field analysis, and ability to overlay MLC motions on gamma map. Common workflow challenges encountered include increased imaging panel calibration drift for new linacs, large fields not fitting on the EPID without a collimator rotation, and incorrect panel distances selected. **Conclusion:** Varian PD has been an effective, practical, and reliable means of IMRT QA.

PO-GePV-T-173

Long-Term Monitoring of aSi-1200 EPID Performance

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Purpose: Electronic portal imaging devices have established their usefulness as two-dimensional dosimetry systems in radiotherapy. However, their performance monitoring as a routinely used clinical dosimeter has not been thoroughly examined in literature. The purpose of this research is to monitor stability and reproducibility of the dosimetric response of the EPID over an extended time frame. **Methods:** Weekly measurements of the entire two-dimensional aSi1200 EPID (Varian Medical Systems) dose response against an ion chamber positioned on central axis and an ion chamber array (MatriXX, IBA Dosimetry) were carried out. Weekly measurements of calibration fields (flood and dark) and defective pixel counts were run in parallel. The measurements were performed on two clinically used Varian TrueBeam linear accelerators. Both imaging modes (continuous and integrated), and all available beam energies (6X, 6FFF, 10X, 10FFF and 23X) were tested. Measurements took place over 18 months between September 2019 and February 2021, except when temporarily suspended from March - May 2020 due to the initial shutdown caused by the covid-19 pandemic. **Results:** The entire EPID response to the large open 'flood' field and its ratios to the ion chamber reading and the ion chamber array demonstrated good constancy with 80% of all ratios falling within 1% from their mean values over the period of observation, and 100% of measurements within 2%. Measured dark and flood fields demonstrated good constancy with the individual pixel signal variations below 2.0% over the entire 18 months period for both imaging modes and all energies tested. The defective pixel count increased by 4% and 8% depending on the acquisition mode. **Conclusion:** The long-term dosimetric behavior of the aSi1200 EPID was monitored and shown to be stable and reproducible for routine clinical dosimetry.

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PO-GePV-T-174

Machine Learning Based Prediction for Patient-Specific Quality Assurance: Understand the Plan Complexity for VMAT

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Purpose: To evaluate complexity metrics of volumetric modulated arc therapy (VMAT) plans and predict the portal dosimetry based patient-specific quality assurance (QA) results with machine learning. **Methods:** A hundred thirty consecutive VMAT plans treated for various treatment sites from Sep 2020-Feb 2021 were included in the study. Each beam was delivered on TrueBeam Linacs with portal dosimetry and the results were analyzed using 2%/2mm global dose/distance-to-agreement (DTA) gamma passing rate (GPR). Total of 313 beams, including 60 beams exceeding the GRP tolerance, were used to build the machine learning model. Forty-six features of both the plan complexity and machine performance were collected for model building. Random forest algorithm was applied in the feature selection process. With selected features, conventional logistic regression and machine-learning based method, were used to create the classification model that can predict potential failure. In addition, five-fold cross-validation was employed to evaluate the model performance. **Results:** The area under the curve (AUC) and sensitivities of the machine-learning based model and logistic regression model were 0.88 and 85%, 0.76 and 78%, respectively. MLC modulation complexity, plan irregularity and jaw tracking are the most important features to predict the potential QA failure. **Conclusion:** Portal dosimetry based QA failure might be potentially predicted by machine-learning based model for VMAT plans. This information may be integrated into the plan optimization to generate more robust plan prior to the physical QA check.

PO-GePV-T-175

Quantitative Comparison of 2D and 3D Modes of a Commercial EPID-Based Hybrid Patient-Specific QA System for Spine SBRT Treatments

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Purpose: To evaluate different dose verification modes of an EPID-based hybrid dose verification system for SBRT treatments of cervical, thoracic and lumbar spine regions. **Methods:** An EPID-based hybrid system (PERFRACTION) was commissioned at our institution with 2D (Fraction-Zero Absolute Dose, FZAD) and two different 3D dosimetry modes using EPID or log-files. The EPID 3D mode extracts the MLC positions from the measured EPID images while the log-file mode extracts MLC positions from the log-files. Both 3D dosimetry modes reconstruct volumetric dose on the patient specific CT images while FZAD mode generate a 2D dose map in the water-equivalent phantom comparing with measured EPID images. Twenty nine spine SBRT plans with the prescription dose of 18 Gy in one fraction were delivered on EPID panel in both dosimetry and cine modes on a Varian Edge machine. The treatment plans, measured EPID images and log-files were transferred to PERFRACTION system for calculations and 2D/3D global gamma analysis at 2%/2mm, 3%/2mm and 3%/3mm levels with 10% low dose threshold. The relative point dose differences (% Δ D) were compared against the TPS calculated dose at the high dose low gradient regions. **Results:** The gamma passing rates from all modes were in a good agreement with the predicted dose. The average passing rates at 2%/2mm were 96.3% $\pm$ 2.1%, 96.9 $\pm$ 4.0% and 98.7 $\pm$ 1.1% for FZAD, EPID and log-file modes respectively. For 3%/2mm and 3%/3mm, the average passing rates were observed >98% for all modes. The average % Δ D for all modes were <0.4%. **Conclusion:** EPID-based dose verification provides a high resolution for SBRT QA. Gamma passing rates from all three modes were comparable while 2D-FZAD and 3D-EPID modes were more sensitive than the log-file based mode at a stricter gamma metric.

PO-GePV-T-176

Comparison of Pixel Sensitivity Map Methods for EPID Dosimetry Measurement

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Purpose: To intercompare EPID imagers PSMs created with a vendor-developed shift-based calibration method with a method that does not require shifting the imager. **Methods:** The pixel sensitivity map (PSM) is used to convert the raw EPID signal output to a relative dosimetry response through correcting for pixel-specific response variations. In this study, we compared two different PSM calculation methods. The first method calculates the PSM via different image sets of large-overlapping-field irradiations acquired with the EPID. The second one is estimates the underlying fluence by fitting EPID images with 2D polynomials and then extracting PSM by division with raw images. Two EPID imaging systems aS1000 and aS1200 were utilized for this comparison. Raw signals from different field sizes of 2x2, 5x5, 10x10, and 15x15 cm² and flood field at source to detector distance (SDD) of 150 and 180 cm were taken. The inline and crossline profiles of calculated PSMs were compared and a 2D relative difference map was calculated to estimate the discrepancy between the two methods. **Results:** Both methods show good agreement in the center of the imager (relative difference up to 4%), however, discrepancies become more visible at the border with a maximum of 8%. Central crossline profiles of corrected images show good agreement, while central inline profiles show major deviations. **Conclusion:** The proposed PSM calculation method can be used in replacement for the current shifting method within some degree of agreement.

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PO-GePV-T-177

A New Generation of X-Ray Irradiator for Preclinical Research: Dosimetric Characterization

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Purpose: Recent trend of preclinical research on highly conformal radiotherapy, motion management, and FLASH irradiation compels the need for a new generation of advanced preclinical irradiation system. Here, we study the feasibility of a novel x-ray irradiation cabinet system employing medical imaging x-ray sources with rotating anode technology, as an enabling technology to enhance preclinical research capabilities. **Methods:** The proposed system employs two commercially available high-capacity 150kVp fluoroscopy x-ray sources with rotating anode technology. The tubes have independent gantry rotational and radial motions. Simulation was performed for parallel-opposed and angled beam arrangements with Geant4 (v.10.04) Monte-Carlo platform. Simulated dosimetric properties were characterized for output, uniformity and field size as a function of SSD, beam current, exposure time, and filters. Additionally, dose, dose-rate, and cross-plane profiles were measured from a single kV x-ray fluoroscopy source using radiochromic films to validate the Monte-Carlo models. **Results:** The irradiator can deliver FLASH and conventional dose-rate irradiations with the fields of 25-200 mm in length. Doses over 50 Gy can be delivered to a 20-mm thick water equivalent medium at FLASH dose-rates of 40-240 Gy/s in a parallel-opposed arrangement at SSDs <50mm. A uniform depth dose-rate within $\pm$ 5% deviation is achieved over 8-12 mm in phantom central region. At extended SSD, conventional dose-rates (< 0.1Gy/s) can be achieved with the depth-dose uniformity improved to a 27-mm thickness in a 40-mm thick water phantom. Beam angling reduces high surface doses up to 39% and provides a first order capability of conformal irradiation for the parallel-opposed beam arrangement. **Conclusion:** It is feasible to irradiate small animals and cell cultures with a wide range of doses and dose-rates, including FLASH and conventional irradiation with kV x-rays from rotating anode x-ray sources in the parallel-opposed and angled beam arrangements. The system will greatly enhance basic and translational researches in the laboratory setting.

PO-GePV-T-178

An Improved Statistical Model for Gamma Index Pass Rates in Intensity Modulated Radiation Therapy Patient-Specific Quality Assurance

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Purpose: The beta distribution was recently proposed as an appropriate statistical distribution for modelling gamma index passing rates (GIPRs) computed during patient-specific quality assurance (PSQA). As such, it has been used to infer control limits, which improve the robustness of PSQA by defining thresholds beyond which abnormal GIPRs are automatically identified. This study presents a more appropriate statistical distribution for GIPRs which produces control limits with better theoretical justification. **Methods:** Our presented distribution is a modified beta distribution, derived from the beta distribution with an

additional nonzero probability at 100% GIPR. To represent this probability, the distribution includes a separate parameter equal to the fraction of fitting data points at 100% GIPR. This parameter accounts for the cluster of plans that fall well within the gamma index's pass thresholds and achieve exactly 100% GIPR. The beta distribution assumes that these plans are infinitesimally unlikely to exist and models them incorrectly. The data points between 0% and 100% GIPR are then used to fit a separate beta distribution. The modified beta distribution was compared to the beta distribution according to the control limits produced by each distribution. These control limits were generated assuming a false positive rate of 1/370 from two years of individual plan field GIPR data from PSQA of volumetric modulated arc therapy. **Results:** The control limits derived from the patient GIPR data were 92.8% and 93.6% for the beta distribution and modified beta distribution, respectively. This difference is comparable to the differences between outliers in the data and it can impact abnormal plan detection. **Conclusion:** The modified beta distribution produces control limits which are meaningfully different from the control limits produced by the original beta distribution. Given its stronger theoretical justification, the modified beta distribution should be used over the beta distribution to improve the robustness of PSQA.

PO-GePV-T-179

EBT3 Film Calibration Using Deep Learning with Keras Functional API

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Purpose: Ashland Inc. EBT3 film is a widely used dosimetry quality assurance tool. The film usually is recalibrated each quarter due to the 'aging effect', and calibration uncertainties always exist between individual films even in the same lot. Instead of performing recalibration, a new method is proposed using an adaptive power function with a deep neural network (DNN) model. **Methods:** EBT3 film was calibrated several times at different dates. Following "one red-channel after three-channel method", the dose-dependent optical density (OD) of the red channel is fitted to the delivered dose using a power function with a constant fitting parameter. Instead of recalibrations, film-dose can be calculated using all the fitting parameters of the first calibration and the fresh measured background OD. A bias τ is added to the input OD to improve the fitting results and is trained using a designed DNN through the Keras functional Application Program Interface (API). The given dose is compared with the film-doses calculated using the presented DNN API method, using all new recalibrated fitting parameters, using all first fitting parameters, and using the adaptive method without τ . **Results:** The results demonstrate that the difference between the film-dose with the delivered dose using the adaptive DNN API method is within 3.5%. Without using τ , the differences may reach 7% for a delivered dose lower than 50 cGy. If using the fitting parameters of first calibration, the difference is even greater than 50% around a given 50 cGy. **Conclusion:** With the DNN model, the film-dose can be calculated using the first calibrated form and using the new measured background OD to adapt the constant fitting parameter rather than recalibrating the film. To precisely calculate the film-dose, this method can be adapted especially for each individual film.

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PO-GePV-T-180

Longitudinal Shift Limitation for Long-Field Delta4 Measurements

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Purpose: This work aims to determine the limits to longitudinal shifts in measuring patient IMRT QA plans using the Delta4+ device. This is applicable to extended field measurements on radiotherapy plans longer than 40 cm, such as for total body irradiation plans on a Tomotherapy treatment delivery system. **Methods:** An IMRT plan was created on an anonymized CT scan of a patient, where a uniform dose distribution was optimized in a target, 60 cm long. This was to ensure that the dose distribution was longer than the length of the phantom. An IMRT QA plan was created for with a Delta4+ phantom within the RayStation treatment planning system. Three separate measurements were then performed with the phantom centered at three different locations with respect to the plan, shifted -8.0 cm, 0.0 cm and +8.0 cm. A comparison of dose accumulation in a horizontal line of the plan was performed to determine the effect of the amount of phantom material near the measurements. **Results:** Figure 1 shows the setup to measure a line profile with different shifts of the Delta4+ phantom. It was determined that the center line profile of the dose distribution measured with the phantom centered is equivalent to the measurement performed when the phantom was shifted 8 cm superiorly. However, the measurement for the same line was less when the phantom was shifted 8 cm inferiorly. This shows that the amount of phantom material preceding a line in the dose distribution affects the accumulated measured dose. **Conclusion:** Delta4+ phantom position affects dosimetry in the IMRT QA process in plans whose dose distribution is longer than the Delta4+ phantom, limiting the amount one may shift the phantom. There is an amount of needed phantom material, above which measurements remain equivalent. Additional measurements are needed to establish this amount.

PO-GePV-T-181

Initial Experience in Comparing IQM and SRS MapCHECK for Single ISO Multiple Mets Brain SRS/SRT Plan Quality Assurance

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Purpose: To compare two commercial systems, Sun Nuclear SRS MapCHECK and iRT Systems Integral Quality Monitor(IQM), on quality assurance(QA) of single isocenter(ISO) multiple metastases(Mets) stereotactic radiosurgery(SRS) and stereotactic radiation therapy(SRT) plans. **Methods:** Single ISO multiple Mets SRS/SRT plans with multiple arcs were created in Brainlab Multiple Mets Elements(MME). The SRS MapCHECK is a 2D diode array of 77mmx77mm. It was designed to be inserted into StereoPHAN secured on a designated base mounted on a Linac couch. The SRS MapCHECK/StereoPHAN system was CT scanned and imported into Brainlab MME as a patient-specific QA phantom. Gamma pass rates per Met was analyzed. IQM is a single large-area 550cc ionization chamber mounted onto the Linac head. It has a sensitive area of 40cmx40cm projected at the isocenter, which only needs one plan delivery regardless of Mets number. With IQM, the measured signal per segment and cumulative signal per arc are compared to the calculated. **Results:** The patient-specific QA on three clinical plans totaling 13 Mets and 24 arcs were performed with both systems and all passed QA. With the SRS MapCHECK, the Gamma passing rates @3%/1mm criteria per Met were 90%- 100% (mean 96.9%±3.4%). With IQM, the difference between calculated and measured signal per Arc was mostly within 3%, except for the smallest segments, in which the deviations increased up to 4-7%, but well within the acceptable tolerance. The time to complete the measurements was on average 120 minutes for SRS MapCHECK and 15 minutes for IQM. The difference in time was due to the fact that with a limited measurement size, the SRS MapCHECK had to be irradiated multiple times to cover all of the Mets. **Conclusion:** There was good agreement between the QA results on both systems. The QA time was dramatically decreased with IQM by approximately a factor of 10.

PO-GePV-T-182

Clinical Implementation of Varian TrueBeam Portal Dosimetry: Comparison to Sun Nuclear MapCHECK2 for Patient-Specific Quality Assurance

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Purpose: Current patient-specific quality assurance (QA) methodologies adopted by the medical physics community are well established, but lack in guidance on the utilization of portal dosimetry (PD), as it continues to grow in popularity. The focus of this study was to clinically implement PD, compare the patient-specific QA analysis features to those employed by SunNuclear (SNC) MapCHECK2, and design a more robust QA program which utilizes PD to improve the efficiency of faculty resources and accuracy of complex dose distribution analyses. **Methods:** Implementation of Varian Portal Dosimetry requires 4 sets of

data: (I) diagonal profiles at dmax for largest field size; (II) Varian 5 bar fluence pattern; (III) dose normalization of the flat panel detector (FPD); (IV) measured output factors of the FPD. These data characterize the dosimetric effects of the MLC and configure the dose deposition kernel used to compute the fluence deposited on the FPD. Patient-specific QA was measured and analyzed using both PD and MapCHECK2 methods. The treatment fields assessed were comprised of clinically relevant sites associated with each beam energy (6MV, 10MV, and 16MV). SNC Patient software was used for gamma analysis of the MapCHECK2 measurements and ARIA Portal Dosimetry was used for gamma analysis of the PD measurements. **Results:** The average pass-rates across the treatment fields using a dose difference (DD) of 3% and distance-to-agreement (DTA) of 3mm was 95.2% and 97.7% for PD and MapCHECK2 respectively. For a DD of 2% and DTA of 2mm, the rates were 90.7% and 95.2%. All fields which did not pass using PD gamma analysis, passed when using SNC gamma analysis. **Conclusion:** Portal dosimetry shows promise as an improved method of patient-specific QA as it requires less faculty resources, contains increased sensitivity within its employment of gamma analysis, and has an intrinsic improvement in measurement spatial resolution.

PO-GePV-T-183

Extraction of a Large Dataset Populating the Gamma Criteria Parameter Space for EPID-Based Patient-Specific QA

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Purpose: To develop a large dataset that explores the parameter space of gamma criterion using our institutional patient-specific quality assurance (PS-QA) data from the past five years of using EPID-based QA. This dataset is to be used to train and test a machine learning model that will be used to identify plans that may require additional quality review. **Methods:** A Portal Dosimetry Application Programming Interface (PDAPI, Varian Medical Systems, Palo Alto, CA, USA) application has been developed to process and extract PS-QA results applying various gamma evaluation criteria. All 27 combinations of gamma criteria involving dose differences of 1, 2, and 3%, distance-to-agreement of 1, 2, and 3 mm, and evaluation thresholds of 10, 20, and 50% of dose maximum were applied to EPID-QA images acquired at our institution since 2015. We collected average, maximum, percentage of points with gamma >1, and >1.2 gamma values for all image-criteria combinations. Average and variance value for each QA session were also calculated for comparison with values presented in the literature. **Results:** Our dataset consists of 94000 analyses with >3300 images involving >700 patients. Average percentage of points with gamma >1 for 3%/3mm/10% threshold was 3.8%, which agrees with published results of 4.1% in smaller series from other institutions. The higher the threshold for allowable points, the worse the gamma results were for all evaluations. Trending was performed demonstrating an improvement in gamma results over time from 202/735 images to 100/581 images with area of gamma >1 <90% using TG-218 criteria of 3%/2mm/20% Threshold. **Conclusion:** We have automated a gamma analysis data extraction tool and will continue to build our dataset to use in future studies evaluating contributing factors for poor gamma passing rates using EPID-based QA. Our current dataset is amongst the largest extracted datasets of patient specific QA results in literature.

PO-GePV-T-184

Development of a User-Friendly Correction Method for the Lateral Response Artifact From Flatbed Scanners in Radiochromic Film Dosimetry

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Purpose: In radiochromic film dosimetry, it is common practice to scan films with a flatbed scanner. A known issue affecting the accuracy is the lateral response artifact (LRA). The aim of this work is to correct the LRA with a user-friendly graphical user interface (GUI). **Methods:** This work is based on a LRA correction using a one-dimensional linear correction function. This approach was enhanced by increasing the resolution of correction function. For this purpose, two sheets of film, one blank and one irradiated uniformly with 2 Gy, were used, which enables correcting films by means of a user-friendly GUI. The films were scanned in the center of the scanner and subsequently shifted laterally in both directions. After averaging the resulting values longitudinally, the correction functions were obtained. For validation, dose measurements with films inside a solid water phantom of two VMAT plans with a prescribed dose of 1 Gy and 2 Gy were performed. The irradiated films were scanned once in the center of the scanner and shifted 5 cm and 10 cm laterally in both directions. Then, the enhanced LRA correction was applied. The corrected and uncorrected dose values were evaluated in FilmQA Pro™ (Ashland) using a gamma evaluation (3% global/ 2 mm, 20% dose threshold). **Results:** The gamma passing rate for the uncorrected films was above 99% for the red color channel in the center of the scanner but decreased to values between 70% and 91% and between 22% and 44% when scanned 5 cm and 10 cm away from the center, respectively. For the corrected films, the passing rates improved for all situations with the lowest passing rate being 97%. **Conclusion:** An enhanced correction method for LRA with a GUI was developed, validated for VMAT plans and lead to substantially reduced LRA compared to uncorrected films.

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PO-GePV-T-185

Stereotactic Patient QA Evaluation Between MapCheck SRS 2D Array, Electronic Portal Image Dose (EPID) and EBT3 Gafchromic Film Dosimetry

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Purpose: Comparing Stereotactic plans patient QA between Sun Nuclear's SRS Mapcheck (SRSMC), Varian's Electronic Portal Dosimetry (EPID) suite and Gafchromic film (EBT-3). **Methods:** For this evaluation the three different QA process mentioned above (EPID, SRSMC, EBT-3 films) were used. Two different EPID resolutions were used, an aSi-500 and an aSi-1000. Patients were selected based on their lesion size and site in order to review a spectrum of sites and sizes. Lesion sizes were specifically chosen to test the robustness of the QA processes. All plans included in the study were re-planned (Eclipse v15.6) as multiple field IMRT fields in order to have a direct comparison between the different QA methods. Plans were delivered on two different Varian-iX machines, each one having a different Eclipse model with a different CT curve. Data normalization was performed based on the Eclipse models and CT curves in order to exclude variations related with these factors. QA passing rates were set to 95% (2% - 2mm). Total number of calculation points, passing and failing points were recorded and evaluated for each one of the techniques used. **Results:** Time of setup, ease of use, accuracy, resolution, dependability, and dependencies were examined when comparing the 4 (SRSMC, EBT-3, aSi1000, aSi500) different QA devices. EPID aSi-1000 had the highest resolution while the lowest was for SRSMC. Gamma analysis showed that for 93.8% of the cases SRSMC had higher passing rates than EPID results, though the gamma passing rate standard deviation among the different measurements did not exceed 2% (average difference was 0.87%). Comparison between aSi-500 and aSi-1000 showed that passing rates for aSi-1000 was higher for 84.4% of the cases. **Conclusion:** Comparison between the different patient QA processes showed that aSi1000 will be an appropriate system for SRS QA.

PO-GePV-T-186

CNN-Based Classification of IMRT Errors from Sparsely Sampled Measurements

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Purpose: Error detection and classification in intensity-modulated radiation therapy (IMRT) is a difficult problem exacerbated by the lack of sufficiently resolving measurement equipment. The commonly used Sun Nuclear MapCheck is a diode-based measurement device which could fulfill the purpose of measuring IMRT fields for error classification. However, its low resolution (7mm detector pitch) prevents the creation of a continuous distance-to-agreement (DTA) map from the measurement and the corresponding calculated fluence distribution, and the data sparsity hinders detection of small deviations from

expected field fluence. In this work, a convolutional neural network (CNN)-based method for error detection and classification is presented. **Methods:** Diode array measurements are simulated by retrospective undersampling of perturbed fluence maps, which are used as input training data for the CNN with the original calculated maps. Dropout is employed in the network architecture prior to classification to improve validation accuracy. The proposed CNN is evaluated on its ability to classify eight separate types of spatial error, including systematic leaf shift, source size, and single-leaf under- and over-travel. **Results:** While the precision and recall are poor for distinguishing source size errors from statistical noise, the network achieves better than 85% accuracy with an F1 score of 0.855, Matthews correlation coefficient of 0.827, Cohen kappa of 0.826. The highest recall and precision for individual features are 98.3% and 95.3% for systematic X bank shift and systematic Y shift, respectively. **Conclusion:** Diode array devices present singular utility in initial characterization of errors in IMRT. Though the sampling space is sparse, careful choice of interpolation modality provides a suitable training set for an NN – the modified Akima (makima) method was used in this work as it avoids large undulations which are not expected in the interior of most IMRT fields. Further refinement of learning parameters may increase training speed and performance.

PO-GePV-T-187

Validation of Rapid Arc Gating Technique for Lung and Liver Stereotactic Body Radiation Therapy (SBRT) with In-House 4D-Phantom

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Purpose: In recent years, volumetric arc therapy combined with 4DCT for moving target, became more popular. It can be either performed by gating or tracking the tumour. The purpose of this dosimetric study to validate the rapid arc gating technique for lung and liver stereotactic body radiation therapy (SBRT) with an in-house made 4D wax phantom. **Methods:** 4DCT scan of the phantom with CC13 ion chamber inserted into the groove given, acquired using real-time position management system (RPM) (Varian Medical System, Palo Alto, CA) for respiratory motion monitoring. Appropriate phases were selected for delineation purpose, and the plan created in eclipse V13.5. The respiratory pattern of phantom recorded at the time of the 4DCT scan serves as a reference pattern during dose delivery. We took a gated cone-beam computed tomography (CBCT) scan for the setup verification before dose delivery. Quality assurance performed on in-house made 4D wax phantom having CC13 ion chamber inserted. Dosimetric validation of rapid arc gating technique has been performed for gated rapid arc plans created with 6MV, 6MV-FFF, 10MV, 10MV-FFF energies. **Results:** Dosimetric evaluation for gated rapid arc plans created with 6MV, 6MV-FFF, 10MV, 10MV-FFF energies showed $\leq 3\%$ variation for all the cases. Plans created for 6MV showed 0.50% and 1.00% variation, plans created for 6MV-FFF, 10MV, 10MV-FFF energies respectively showed the variation of 1.59%, 2.60%, 3.00%, additionally we have created conformal arc plan which showed variation within tolerance limit between the treatment planning system (TPS) dose and measured dose with ion chamber. **Conclusion:** This dosimetric study showed that the Point dose validation of a rapid arc gating technique carried out successfully using indigenous 4D wax phantom is well within tolerance limits compared to TPS calculated dose. So, Gated RapidArc can clinically be used to treat SBRT lung and liver.

PO-GePV-T-188

A Convolutional neural Network Model for Predicting Gamma Passing Rate of 3D Array Detector-Based VMAT QA

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Purpose: The predicting models of gamma passing rate (GPR) via artificial intelligence have been proposed for improving the efficiency of the patient-specific quality assurance (QA) workloads. The current study aims to develop a convolution neural network (CNN) model for predicting GPR, and to propose a novel omitted workflow of prediction-based VMAT QA. **Methods:** Two hundred and twenty-five VMAT plans measured GPR by ArcCHECK were used in this study. The dose and DUP distribution on the detector element plane were used as input data for the predicting model. The DUP distribution was generated by accumulated field edges weighted by a segmental monitor unit. Our CNN model comprised nineteen layers and was trained for 200 epochs. 10-fold cross-validation was applied to verify the predicting performance. The GPR values were predicted for thirty-five test cases, and the difference between the measured and predicted GPR values was evaluated. The omission rate of the measurement was calculated for cases with GPR 2SD of prediction error higher or lower from 90% set as the action limit at 3%/3-mm and 3%/2-mm tolerances. **Results:** Mean absolute error (MAE) between measured GPR and predicted GPR in the CNN model using the dose distribution was 2.0% for 3%/3-mm, 3.1% for 3%/2-mm, and 6.6% for 2%/2-mm tolerances. By adding the DUP distribution to the CNN model, the MAE was improved to 1.8%, 3.0%, 6.0% for 3%/3-mm, 3%/2-mm, and 2%/2-mm tolerances, respectively. The effort of the measurements for patient-specific QA could reduce by 83% and 23% for 3%/3-mm and 3%/2-mm tolerances, respectively. **Conclusion:** The CNN model with the dose and DUP distributions on the detector enabled the prediction of GPR value with high accuracy. The proposed workflow with the CNN model could potentially help to omit the patient-specific QA workloads.

PO-GePV-T-189

Machine- and Site-Specific Analyses of TomoTherapy Patient-Specific QA Results Planned with Accuray Planning Station and RaySearch RayStation

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Purpose: To analyze treatment planning system calculation accuracy relative to patient-specific quality assurance (PSQA) measurements among two TomoTherapy treatment delivery systems and two treatment planning systems and evaluate any dependencies observed based on machine, planning system, or treatment site. **Methods:** Data were extracted from PSQA reports created from Delta4+ measurements on two TomoTherapy HDA units over more than two years, during which time treatment planning was transitioned from Accuray Planning Station (PS) to RaySearch RayStation (RS). Data points included the treatment site, the median of the planned-to-measured dose deviations over all detectors, and the proportions of detectors passing gamma calculation with a range of dose and distance to agreement criteria. These data were analyzed based on sorting by treatment machine, originating treatment planning system, and by treatment site. **Results:** Median dose deviation histograms considering all treatment sites revealed Gaussian probability distributions with mean values of -0.5% for PS and -0.3% for RS on Machine A, and -0.2% for PS and +0.3% for RS on Machine B. Welch's t-tests demonstrated that all these distributions were unique, indicating that despite their independent calibrations all delivery and treatment planning system combinations produced good agreement between dose prediction and delivery. Gamma pass rates pooling all treatment sites were 98-100% for 3%/3mm evaluation criteria, dropping to approximately 91-96% for 3%/2mm and 60-66% for 2%/2mm. **Conclusion:** Median dose deviation and gamma analysis both suggest similar dose calculation and delivery performance for both the Accuray and RaySearch treatment planning systems on two independent delivery systems. Because no clear dependence was observed based on generalized treatment site, future work is focused on investigating correlations with other plan characteristics such as target volume, machine parameters, and plan complexity.

PO-GePV-T-190

Quasi-3D Dosimetry System with Solid Polyurethane-Based Dosimeter

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Purpose: To evaluate a quasi-3D dosimetry system using a solid polyurethane-based dosimeter when conducting patient-specific quality assurances (QAs) in radiation therapy. **Methods:** The patient-specific QA is the procedure to verify the coincidence between measurement and calculated dose distribution for intensity-modulated radiotherapy (IMRT) and volumetric modulated arc therapy (VMAT) plans by using gamma analysis. A VMAT plan for a pancreas patient was analyzed with a criterion of both 2%/2 mm and 3%/3 mm, a threshold value of 10%, and a tolerance level of 90%. For the plan, the 2D gamma analysis was fulfilled with a MapCHECK2, and the 3D gamma analysis was performed using in-house solid polyurethane-based dosimeters and a quasi-3D phantom. The phantom was produced by drilling 25 holes of 9.4 mm in diameter and 56 mm in depth at even intervals, and it was made to be used by inserting a solid polyurethane-based dosimeter of the same size as the hole. A total of 9 solid polyurethane-based dosimeters were positioned according to the dose distribution of the plan. The 3D gamma analysis was performed on a region of interest of 5.5 mm in diameter and 35 mm in depth using a free platform for biomedical research, (3D) Slicer. **Results:** For the 3D gamma analysis, the average passing rates with 2%/2 mm and 3%/3 mm criteria for relative dose distribution were $91.6\% \pm 1.4\%$ and $99.4\% \pm 0.7\%$, respectively. For the 2D gamma analysis showed passing rates of 97.5%, 99.3% with 2%/2 mm and 3%/3 mm criteria, respectively. **Conclusion:** The quasi-3D dosimetry system was evaluated by performing patient-specific QAs with the solid polyurethane-based dosimeters and quasi-3D phantom. As the 3D gamma index for all solid polyurethane-based dosimeters showed more than 90% for 2%/2 mm criteria, this result means there was a possibility that the quasi-3D dosimetry system was clinically acceptable.

PO-GePV-T-191

Utilization of a Plan Quality Prediction Software to Investigate Patient Specific QA Passing Rates

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Purpose: To investigate the use of a Plan Quality Prediction Software in improving patient specific quality assurance (PSQA) passing rates. **Methods:** Five prostate patients, whose clinical boost plans were generated using Pinnacle³ v16.2.1 (Philips, Amsterdam, Netherlands) and had failed institutional PSQA tolerances (95% passing at 3%/2mm, global gamma criteria), were investigated. PlanIQ (Sun Nuclear Corp., Melbourne, FL) was utilized to retroactively investigate the difference between the clinical DVH for the Bladder, Rectum, and Penile Bulb and the feasibility DVH. These differences were compared with five patients whose clinical QA passed institutional criteria. Three re-plans were created in Pinnacle³ v16.4.2 for each failing plan: 1.) pushing the DVHs to represent the DVHs of the clinical plan, 2.) using PlanIQ to set planning objectives, and 3.) using PlanIQ but restricting the minimum segment area. PSQA was measured for each re-plan and the relationship between plan passing rates was analyzed. **Results:** Comparing the failing and passing clinical plans, no significance was found in the difference between the clinical DVH and feasibility DVH for the Bladder or Rectum. However, at a feasibility score of 0.1, the difference between the clinical DVH and feasibility DVH for the Penile Bulb was found to be significant ($p=0.032$) with the failing plans exhibiting an average DVH difference of -2.23 ± 0.88 Gy and the passing plans exhibiting an average DVH difference of -0.57 ± 0.44 Gy. After re-planning these cases in v16.4.2, improved PSQA passing rates for failing plans were found when using PlanIQ, with average improvement of $1.1 \pm 2.5\%$. Combining PlanIQ with a restricted minimum segment area showed further improvement in PSQA passing rates, with average improvement of $2.6 \pm 1.7\%$. **Conclusion:** PSQA passing rates are influenced by many factors, but PlanIQ was shown to be a useful tool in improving PSQA passing rates for failing plans especially when combined with restricted segment areas.

PO-GePV-T-192

Design of a Respiratory Motion Quality Assurance System with Portal Dosimetry and Customizable Tumor Phantom

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Purpose: To design a high throughput respiratory motion QA system using portal dosimetry (PD) and customizable tumor with 3D printing. **Methods:** The design is divided into the stage control and tumor printing part. A closed-loop stages was assembled and the tumor was mounted on a rod attached to the stage and driven in the Sup-inf direction. The stage can be programmed to move with variable amplitude, period or according to patient's breathing curve. 3D printing was used to build customizable tumor to provide realistic representation of the treated tumor. Different PLA fill patterns and percentages were optimized to obtain a HU of close to zeros and the shortest printing time. The set-up was tested with an actual treatment plan comparing the static and moving phantom. Spherical tumors with 2 cm and 4 cm diameter were used for this testing. Gamma passing rate was used to quantify the dose agreement between static and moving phantom. **Results:** A linear pattern with 80% in-fill yields the shortest printing time and gives a HU closest to 0. A maximum HU of about 100 is achievable with PLA material. A moving phantom does blurs out the portal dose and causes a disagreement between the dose distribution between the static and moving phantom. A smaller tumor gives a gamma passing rate of 97% while a larger tumor gives a smaller passing rate of 94% with a 2%/2mm criteria. **Conclusion:** Appreciable dose difference between moving and static phantom with 4 cm and 2 cm water equivalent tumor phantom can be detected with portal dosimetry. Further work is underway to test the system with phase gating and determine the sensitivity of detecting breathing curve discrepancy and dose degradation due to motion interplay.

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PO-GePV-T-193

Automatic Plan Quality Assurance Using Knowledge-Based DVH Prediction for Head and Neck Cancer Patients

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Purpose: To investigate the use of knowledge-based planning to identify poor quality VMAT plans for head and neck cancer (HNC) patients. **Methods:** We used a commercial knowledge-based planning system (RapidPlan, Varian Medical Systems) to create two dose-volume histogram (DVH) prediction models using data from 25 HNC patients. One was trained by clinical plans (manual plans: MP); the other used automatically created plans (AP). 7 normal tissues and 2 automatically-generated standardized planning structures were included. DVHs were then predicted for an evaluation cohort of 25 patients (EC1) and compared with achieved DVHs of MP and AP plans. Next we predicted DVHs for 25 other patients (EC2) on which we intentionally generated plans with suboptimal normal tissue sparing while satisfying dose-volume limits of standard practice. These plans were also reviewed by one radiation oncologist when blinded to DVH predictions. Oncologist reviews on plan quality were analyzed together with the predicted DVHs. **Results:** Predicted DVH ranges (upper – lower predictions) were consistently wider for the MP model than the AP model for all structures. The average range of mean dose predictions among all structures was 9.9 Gy (MP model) and 3.4 Gy (AP model) for EC1 patients. RapidPlan models identified 7 MP plans as outliers according to mean dose or D1% on one of normal tissues while none of AP plans was flagged. For EC2 patients, 22 suboptimal plans were identified by prediction. While re-generated AP plans validated that these plans could be improved and pass RapidPlan QA, 19 out of 45 structures with predicted poor sparing were also identified by oncologist review for additional planning due to inadequate sparing. **Conclusion:** Our study shows that RapidPlan could provide accurate DVH prediction for HNC cancer patients. Agreement between prediction and oncologist review also shows promising use of knowledge-based DVH prediction for detecting poor quality plans.

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PO-GePV-T-194

Estimation of Dosimetric Leaf Gap Correction Factor as Part of Commissioning a Log File Based QA System

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Purpose: During the commissioning of the Mobius3D (M3D) QA software for TrueBeam (Varian Medical Systems Inc. Palo Alto, CA), an independent, log file-based QA system, it was observed that for more complex plans (IMRT/RapidArc) while performing dose comparison, the target volumes consistently came out as hot. This problem demanded a need for the determination of a Correction Factor (CF) for Dosimetric Leaf Gap (DLG). Even though DLG is a dosimetric parameter hardcoded in the M3D system to account for partial transmission through the rounded leaf ends on the MLC, provisions are incorporated within the software for its optimization by applying additional corrections. A Linear regression method recommended by the vendor was followed to overcome this scenario. This paper explains the steps performed to accomplish this task as part of M3D commissioning. **Methods:** The process for determining the DLG CF involved a spreadsheet for the calculation and a verification phantom (VP) provided by the vendor, subtracting MLC transmission contribution from ionization chamber measurements of multiple sweeping gap plans, plotting the resulting differences against the nominal gaps, and using linear regression to finding the optimal DLG CF. MFX, the value from the VP, is compared with the measured values to find the percentage deviation. Seven different plans, IMRT/RapidArc for various sites, were used in this process. **Results:** Dosimetric Leaf Gap (DLG) Correction Factor (CF) was determined for the energy 6MV and used these values to configure the M3D system. After the correction, the plan comparison for complex plans showed better results. **Conclusion:** This method proved to be very easy and reliable, which could be used as an essential tool for fine-tuning the Dosimetric Leaf Gap (DLG) Correction Factor (CF) for M3D commissioning. This method's approach and principles can be applied to any independent QA verification system's commissioning.

PO-GePV-T-195

Sensitivity of OSLD with Fully Filled Deep Electron/hole Traps in Kilovoltage Energy Range

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Purpose: To evaluate the sensitivity of optically stimulated luminescence dosimeters (OSLDs) in kilovoltage energy range with fully filled deep electron/hole traps. **Methods:** The experimental group was consisted of 24 OSLDs which were pre-irradiated greater than 5 kGy with Co-60 gamma ray source for fully filling deep electron/hole traps (OSLD(Full)), while the non-pre-irradiated 24 OSLDs were arranged as a control group (OSLD(Empty)). The bleaching device was equipped with 520 nm long-pass filter attached to light-emitting diode source, and the signal was read out with linear mode. For OSLD(Full) group, bleaching-and-readout cycle was repeated until the remained signal reached around background level. According to the in-phantom method in AAPM TG-61 reference dosimetry protocol, absorbed doses to water at 2 cm for 75, 80, 85, 90, 95, 100, and 105 kVp were measured. Due to the over-response to low energy X-ray of OSLDs, three dosimeters were calibrated with respect to the energy. For each energy, an exposure with 200 mA and 40 ms was defined as unit dose. To find an optimal bleaching time, bleaching-and-readout after fifty times of unit dose exposures was repeated until the signal reached low stable level. To investigate the sensitivity, exposures-readout-bleaching-readout cycle was repeated. The normalized sensitivity was defined as the signal counts divided by initial value. **Results:** For both OSLD(Empty) and OSLD(Full) group, 3.5 days were found as optimal for bleaching time with filtered bleaching device. In case of no cumulated dose, the average normalized sensitivities were improved as 11.7%, 13.6%, 14.8%, 7.9%, 15.8%, 20.1%, and 16.7% for 75, 80, 85, 90, 95, 100, and 105 kVp, respectively. Variations with cumulated dose were less than 7.8% for both OSLD(Empty) and OSLD(Full) group regardless of bleaching condition. **Conclusion:** Filling deep electron/hole traps in OSLDs can possibly increase the sensitivity in the kilovoltage energy range.

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PO-GePV-T-196

Measuring Superficial Total Doses Delivered by Helical Tomotherapy using EPR Dosimetry

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Purpose: This phantom study investigates the applicability of Electron Paramagnetic Resonance (EPR) dosimetry for measuring superficial total doses delivered by helical tomotherapy treatments. **Methods:** For this purpose, eight EPR dosimeters were attached to distinct spots next to the surface of an anthropomorphic head phantom: i) within a superficial simultaneously integrated boost (SIB) volume, ii) near to organs at risk including the parotids and the lenses, iii) at the thyroid lying out-of-field. Each dosimeter consisted of two pellets: one lithium formate monohydrate (LFM) pellet and one L-alanine (ALA) pellet. A helical tomotherapy plan representing a realistic head and neck treatment was applied daily over a period of about 6 weeks (32 fractions). Planned total doses to the dosimeters ranged from about 2 Gy (lenses, thyroid) to 64 Gy (SIB volume). EPR read out was performed by means of a previously proposed EPR dosimetry system after all fractions were applied. The results were compared to thermoluminescence dosimeter (TLD) measurements and to planned total doses derived from the treatment planning system (TPS). **Results:** By taking uncertainties into account, EPR measurements were in agreement with TLD and the TPS. Exceptions occurred at the thyroid (out-of-field) and lenses (extreme sparing). Remarkably, each LFM and ALA pellet placed inside the SIB volume provided dose values that were within plus/minus 3% of the respective planned dose. LFM and ALA provided comparable results. However, LFM is preferable at lower doses due to lower measuring uncertainties. **Conclusion:** This study suggests that EPR dosimetry is a valuable tool for measuring superficial total doses in-vivo being applicable in a wide dose range and in various irradiation conditions.

PO-GePV-T-197

Evaluating the Accuracy of Different Methods for Estimating the Dose to An IMD (implanted Medical Device) of a TBI Patient, Treated at An Extended Distance with External Blocks

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Purpose: We treat TBI patients at an extended distance (440 SAD), with Cerrobend blocks approx. 5 cm from the patient. A CT scan in the treatment planning position is not possible (the MUs are determined by a hand calculation), making estimating the dose to an IMD challenging. We compare the accuracy of different methods for estimating the dose to the IMD. **Methods:** We estimated the dose using three different **Methods:** Eclipse calculations using AAA and Acuros algorithms, and ion chamber and OSLD measurements in solid water. The results were analyzed and compared to OSLD measurements from treatment. Four-7.5 cm thick blocks of different sizes were fabricated. Patient separation (16 cm) and depth of the IMD from the anterior surface (2.5 cm) were estimated from a CT scan. A 16*25*25 cm solid water phantom using an A12 ion chamber was used to perform measurements at IMD depth. NanoDot OSLDs, under 0.5 cm bolus, were used for measurements on the surface. A virtual water phantom of 16*25*25 cm, with a block 5 cm above the phantom, was used for the calculations. Measurements and calculations were performed to estimate the dose to the IMD, under the center of the block, for a single fraction in a 6 MV AP/PA treatment. A patient was treated using one of the blocks, and OSLD measurements were performed on the patient's skin during the treatment. **Results:** The OSLD and ion chamber measurements in solid water agree well: ion chamber 34 cGy, OSLD 35 cGy, Acuros 38 cGy, AAA 65 cGy. They also agree with OSLD measurement during treatment (34 cGy). **Conclusion:** Ion chamber and OSLD measurements in solid water predicted the results of the patient

measurements well. Acuros was accurate enough to provide a rough estimate, while AAA overestimated the dose to IMD significantly.

PO-GePV-T-198

Dosimetric Evaluation of Prostate VMAT Plans Optimized by the Photon and Progressive Resolution Optimization Algorithm

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Purpose: This study evaluated the dosimetry of prostate VMAT plans optimized by the photon optimization (PO) algorithm, which is introduced to replace the progressive resolution optimization (PRO) algorithm in the Varian Eclipse treatment planning system. **Methods:** The dosimetric evaluation was carried out by studying ten prostate patients with VMAT plans created by the 6 MV photon beams and dual-arc technique using the Eclipse treatment planning system. The prescription of the dose was equal to 78 Gy/39 fractions. For each patient, two plans optimized by the PO and PRO algorithm, as per the same dose-volume criteria for the target and organs-at-risk, were created. Dose-volume histograms (DVHs) for the planning target volume and critical organs, namely, rectum, bladder, left and right femur, were determined. From the DVHs, dose-volume index such as conformity (CI) and homogeneity index (HI) were calculated, together with radiobiological variables such as tumour control probability (TCP) and normal tissue complication probability (NTCP). **Results:** Comparing the dosimetry of VMAT plans optimized by the PO and PRO algorithm, it is found that the average CI values of plans optimized by the PO algorithm (0.912) was slightly higher than that by the PRO (0.908). On the other hand, the average rectal NTCP of plans using the PO algorithm was higher than that using the PRO (0.054 vs. 0.052). For the average HI and TCP of the prostate, both algorithms produced the same values of 0.054 and 0.955 from their VMAT plans, respectively. **Conclusion:** It is concluded that prostate VMAT plan optimized by the PO algorithm had a slightly higher average CI and rectal NTCP value compared to the PRO, with other plan dosimetry remaining almost the same. Considering other advantages of the PO algorithm, namely, less multileaf collimator aperture complexity and faster optimization speed, the PO algorithm is a good successor for the PRO.

PO-GePV-T-199

ADiar: Architecture of Dose-Image Agent Retrieval—an Architecture to Automatically Retrieve Similar Radiotherapy Plans

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Purpose: The design of treatment planning in radiation therapy (RT) represents the key to dosimetrists' experiences. This paper proposes an automated architecture using dose image as the agent to retrieve a similar treatment plan from a database and employing the optimization parameters from the retrieved plan to initialize the searching plan. **Methods:** From May 2018 to June 2019, 612 IMRT treatment plans of lung cancer patients were retrospectively selected. Architecture of Dose-image Agent Retrieval system consists of two parts separated by a firewall: one is the in-hospital workstation, and the other is the search engine in cloud. After the architecture is developed we did two studies. The first retrospective study is conducted to validate the percentage of fully automated plans generated by this architecture is qualified for clinical practice. 96 patients were involved in this study. The second comparison study is performed to validate whether this architecture (α Diar) may improve the quality of plan designing. Two dosimetrists are involved and design only one trial with and without α Diar. 26 patients were involved in this study. **Results:** The first study shows that 54.17% (52/96) of plans are qualified according to our department guideline. The second study shows that with the help of α Diar, the quality of treatment planning is improved for junior dosimetrists. **Conclusion:** With the help of α Diar, novel treatment planning systems may be developed, and a large part of treatment plans could be designed automatically, and those automated plans not qualified for clinical practice may also help improve the quality for plan designing.

PO-GePV-T-200

Deep Learning Dose Distribution Prediction Models for Cervical Cancer Volumetric Modulated Arc Therapy

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Purpose: To develop and test 2D and 3D-Unet dose prediction models to predict the three-dimensional dose distribution of volumetric modulated arc therapy (VMAT) for cervical cancer. **Methods:** 8 independent channels of contoured structures were input to the model, and the dose distribution was used as the output of the model. 2D and 3D-Unet prediction models were trained and validated on the training set (n=86) and validation set (n=11), respectively, and then tested using the test set (n=20). The study included 117 cases of cervical cancer patients treated with VMAT, with a prescribed dose of 50.4 Gy in 28 fractions. The results between clinical dose distribution and predicted dose distribution were compared in the following aspects: (a) the mean absolute error (MAE) within the body (b) the Dice similarity coefficients (DSCs) under different isodose volumes (c) the dose parameters include the mean dose (Dmean) and V40Gy (the volume of PTV receiving 40Gy dose) and PTV conformity index (CI) and homogeneity index (HI). (d) dose-volume histogram (DVH) **Results:** Both 2D and 3D-Unet dose prediction models can accurately predict the dose distribution of VMAT plans for cervical cancer. The overall average MAE of the 2D and the 3D-Unet model are $2.92 \pm 4.08\%$ and $2.43 \pm 3.17\%$ of the prescribed doses, respectively. The average DSCs are all greater than 0.9 in different isodose volumes, most of the dose parameters including Dmean, V40Gy, CI and HI are equivalent to those of clinical plans, and there is no statistically significant difference. The DVH of the two models is similar to the clinical plans. **Conclusion:** We have developed two dose prediction models, 2D-Unet and 3D-Unet, which can accurately predict the three-dimensional dose distribution of VMAT for cervical cancer. In terms of MAE and DSCs, the 3D-Unet model is better than the 2D-Unet model.

PO-GePV-T-201

Automated and Clinical-Criteria-Driven Planning for Locally Advanced Non-Small-Cell Lung Cancer Using the Expedited Constrained Hierarchical Optimization (ECHO) System

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Purpose: To develop and implement a fully automatic, clinical-criteria-driven treatment planning solution for locally advanced non-small-cell lung cancer (LA-NSCLC). **Methods:** 21 patients with LA-NSCLC of various tumor locations and sizes with prescription of 60Gy in 30 fractions were retrospectively included in this study. Clinical plans were created manually using static IMRT. Automated plans with the same beam arrangements were designed using in-house-developed automated planning system: ECHO. ECHO uses hierarchical constrained optimization technique and is integrated into Eclipse using API scripting. Department clinical criteria were separated into 2 categories: limits and guidelines. Limits were strictly enforced by ECHO as constraints, while guidelines were optimized as much as reasonably achievable in objective functions. Dosimetric metrics were evaluated for plans in terms of tumor coverage and normal tissue sparing. **Results:** Planning Target Volume (PTV) sizes ranged from 179cc to 1371cc. Planning time for ECHO averaged 59 minutes (range: 17min-107min). Both clinical and ECHO plans met all clinical criteria. Compared with clinical plans, ECHO plans were more consistent in target coverage and normal tissue sparing. Percentage dose to 95% of PTV for ECHO plans ranged 96.7%-100% (median 100%), compared to clinical plans with 87.1%-101.3% (median 100%). Median volume of total lungs excluding GTV receiving >5Gy and >20Gy for ECHO was 59.7% (46.9%-78.9%) and 31.8% (19.5%-36.9%), and for the clinical plans was 60.3% (42.7%-73.4%) and 33.3% (16.9%-36.3%) respectively. Doses to other organs at risk (OARs), including cord, esophagus and heart did not vary significantly between ECHO and clinical plans. **Conclusion:** We developed a flexible automated treatment planning system for LA-NSCLC, driven by modifiable constraints and planning limits/guidelines, and integrated with Eclipse. The automated ECHO plans met established

clinical criteria with better consistency for tumor coverage and OARs sparing. Use of ECHO saves time and effort while resulting in improved or non-inferior plans.

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PO-GePV-T-202

Monitor Unit Efficient VMAT Treatment Planning Facilitated by the MU Objective Tool

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Purpose: Identify an MU-efficient treatment planning methodology that maintains the dosimetric quality of the treatment plan. **Methods:** Seventeen hypofractionated (7-18Gy/fx) and twenty-five conventional fractionated (1.8-3Gy/fx) volumetric modulated arc therapy (VMAT) treatment plans were selected utilizing the Varian Eclipse™ version AcurosXB_15.6. The treatment sites included H&N, lung, liver, and prostate. A base plan was created without utilizing the MU objective tool. Then, an MU-efficient plan was created by utilizing the MU objective tool with a strength setting of 70, a lower objective of zero, and an upper objective (MaxMU) of different values. Dosimetric quality and MU-efficient metrics were used to compare the base plan and the MU-efficient plan. The MaxMU parameter was identified for each treatment site that best reduced the total MU, but at the same time gave a similar to or better plan than the base plan by as determined by fulfilment of the dose constraint metrics. **Results:** For the hypofractionated lung and liver plans, a MaxMU of 1.25 times the prescription dose per fraction (1.25x) resulted in the total MU reduction by 24% and 25% respectively. For the hypo-fractionated prostate, a MaxMU of 2x resulted in the total MU reduction by 13%. For the conventional fractionated bilateral H&N and prostate with lymph nodes plans, a MaxMU of 600MU resulted in the total MU reduction by 19% and 15% respectively. For the conventional fractionated unilateral H&N and prostate minus lymph nodes plans, a MaxMU of 400MU resulted in the total MU reduction by 13% and 18% respectively. MU-efficient plans with the above-mentioned MaxMUs resulted in similar plans compared to the base plans, and in some cases, gave better dosimetric results for OARs. **Conclusion:** The MaxMU for four studied sites were identified that led to the optimum MU reduction without sacrificing target coverage or any other studied plan quality metrics.

PO-GePV-T-203

Evaluation of Performance of Static MLC Leaf-Sequencing Algorithms in IMRT

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Purpose: The aim of this study is to compare the performance of three multileaf collimator (MLC) leaf-sequencing algorithms, proposed by Engel et al, Bortfeld et al, and Kalinowski et al using megavoltage external radiation beams with various beam angles for lumbar vertebrae intensity modulated radiation therapy (IMRT). **Methods:** Different IMRT plans for lumbar vertebrae were created using the ARTS treatment planning system (TPS). The fluence maps for all photon beams were determined by the TPS after plan optimization. Three different MLC leaf-sequencing algorithms based on Engel et al, Bortfeld et al and Kalinowski et al were programmed to calculate the final beam segments and their monitor units. For comparison purpose, the matrices of fluence maps were kept constant in different plans. The number of beam segments and total number of monitor units were calculated for the three algorithms. **Results:** Although values of the number of beam segments and total number of monitor units calculated by the three algorithms varied with the lumbar vertebrae plans, it is found that the algorithm by Engel et al performed better with the lowest number of beam segments. Moreover, the algorithms by Kalinowski et al and Engel et al had larger total number of monitor units than that of Bortfeld et al. **Conclusion:** Although the performance of leaf-sequencing algorithms varied with different IMRT plans having different fluence maps, the best MLC field segmentation can be identified through comparisons among different algorithms for the calculated number of beam segments and monitor units. Our results suggested that ideally, a treatment planning system for IMRT should implement multiple algorithms for the MLC field segmentation.

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PO-GePV-T-204

Sparing Lung Tissue with Virtual Block Method in VMAT Planning for Locally Advanced Non-Small Cell Lung Cancer

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Purpose: Despite that VMAT technique has been widely used in the radiotherapy of non-small cell lung cancer, it may increase the low irradiation dose volume to lungs. This study aims to develop a new virtual block method for sparing the normal lung tissue in VMAT planning. **Methods:** A retrospectively VMAT planning study was performed and compared without and with virtual block method for 17 patients, named as non-block plan and block plan respectively. The identical initial optimization parameter for the non-block plans were designed by the mdaccAutoPlan system. The block plan adopted all optimization parameters in non-block plan for each patient, in addition of the maximum dose constraint for the virtual block. The two types of plans were compared with dosimetric indexes of targets and normal structures, and the results were statistically analyzed with Wilcoxon signed rank test. **Results:** Both block and non-block plans achieved sufficient dose coverage in target volume and similar conformity of target volume. For the structures, block plans have advantages in the following dosimetric metrics: 1) for total lungs, lower V5, V10 and mean dose, NTCP were observed ($p < 0.01$, respectively); 2) for the contralateral lung, significant reductions in V5 and V10 ($P < 0.001$, respectively); 3) decrease in the mean dose and V40 of heart; 4) decrease in esophagus V40; 4) significant reductions in mean dose, V5 and V10 of normal tissue. The quality scores of block plans were significantly higher than that of non-block plans (Mean: 82.51 ± 7.07 vs. 80.74 ± 7.22 , $P < 0.001$). **Conclusion:** The VMAT plans with the virtual block can significantly reduce the low irradiation dose volume of the lungs and the doses to other normal structures whereas keep similar conformity and homogeneity of targets. The block method might reduce the risk of radiation-related toxicity for patients.

PO-GePV-T-205

MLC Leaf Position Optimization for Direct Aperture Optimization Treatment Planning

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Purpose: Direct-aperture optimization (DAO) is an efficient optimization method that combines both dose optimization and leaf sequencing in the same process. Due to the random selection of aperture shapes it is often time-consuming to find optimal aperture shapes in DAO. This study investigates MLC leaf position optimization for DAO treatment planning. **Methods:** Fifteen patients were unarchived from our clinical database, who were previously planned using Eclipse TPS and treated on Clinac iX accelerators. These patients were re-planned using Prowess RT-Pro (Prowess, Concord, CA) using the same beam angles, dose constraints and optimization parameters. Manual MLC leaf position adjustment was performed for these Prowess plans to remove hot spots inside the target and cold spots near the target border. Plan quality was evaluated using the homogeneity index (HI), conformity index (CI), Dmax (D1%), and Dmin (D99%) for IMRT/VMAT, and R50% and D2cm for SBRT, and dose-volume and Dmean for organs at risk (OAR). **Results:** Treatment plan quality was

significantly improved after adjusting MLC leaf positions following very simple rules for Prowess plans generated using the DAO algorithm. The removal of hot spots inside the target and cold spots near the target border resulted in much improved CI, HI and OAR mean doses as well as OAR dose-volumes. However, the initial selection of aperture shape and monitor unit optimization played an important role in the critical structure sparing as shown in the bladder and rectum mean doses for prostate plans. **Conclusion:** Minor adjustment of aperture shapes and/or removal of invalid segments for DAO generated plans could significantly improve the plan quality to generate identical superior treatment plans using Eclipse and Prowess employing either DAO and/or beamlet-based approach. The method developed in this work can be further programmed as a post-optimization tool to improve DAO treatment planning efficiency.

PO-GePV-T-206

Using a Novel Leaf Sequencer and Segment Shape Optimization Algorithm to Reduce Treatment Session Times on the Elekta Unity

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Purpose: A current limitation of MR-guided radiotherapy is extended treatment session times. This study evaluates a new Elekta developed research optimizer and leaf sequencing algorithm with respect to plan quality and the potential to reduce optimization as well as treatment delivery times for Elekta Unity patients. **Methods:** Three prostate, three oligometastases, two pancreatic, and two liver patients previously treated on the MR-linac were retrospectively evaluated. Reference plans for all ten patients were generated using the clinical Hyperion optimizer and leaf sequencer. These plans were then re-optimized using a novel Optimal Fluence Levels (OFL) leaf sequencing algorithm and Pseudo Gradient Descent optimization algorithm (OFL+PGD). Beam angles and IMRT objectives were held constant for clinical and research algorithm optimizations. All plans were normalized for equivalent PTV coverage. Optimization time, treatment delivery time, number of plan segments, total MU, and gamma passing rate were all evaluated. A paired t-test was used to determine statistical significance. Relative changes in plan quality were evaluated for the PTV and critical OAR's. **Results:** The PGD and OFL algorithms had a reduction in optimization time of $51.4 \pm 5.0\%$ and a reduction in treatment time of $10.6 \pm 7.5\%$ as compared to the Hyperion plans. The OFL+PGD plans used $5.6 \pm 9.1\%$ fewer MU, $13.0 \pm 12.4\%$ fewer plan segments, and had an average gamma passing rate of 99.1% compared to 99.2% for the Hyperion algorithm. Reductions in optimization time ($p=0.002$), delivery time ($p=0.005$), and number of plan segments ($p=0.009$) are statistically significant. The plan quality between the Hyperion and OFL+PGD algorithms was observed to be equivalent in terms of OAR sparing and PTV hotspot. **Conclusion:** The OFL+PGD generated plans optimized faster and had shorter delivery times while maintaining equivalent plan quality. These algorithms have the potential to decrease overall treatment session times for MR-guided radiotherapy.

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PO-GePV-T-207

Volumetric Dose Reconstruction for Iso-Dose Tuning

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Purpose: To develop a model to generate volumetric dose distribution from two isodose surfaces (iso-surfaces), and to interactively tune dose distribution by iso-surface dragging. **Methods:** We model volumetric dose distribution as analytical extension of two iso-surfaces with the extension variables as distances to iso-surfaces. We built a 3D lookup table (LUT) which are generated based on clinical dose distributions. Two LUT tables store the mean and standard deviation of voxel dose values of clinical doses and binned as distance to 100% iso-surface, reference iso-surface and reference dose level. The process of interactive tuning starts from a given base plan. A user drags iso-surface for a desired carving. Our method responds with tuned dose. The derivation of tuned dose follows two steps. Dose is extended from the two user-desired iso-surfaces (eg. 100% and 50%) to the whole patient volume by table lookup, using distances to two iso-surfaces and reference dose level as keys. Then we fine tune the extended dose by a correction strategy utilizing the information of base plan. **Results:** We validated this method on coplanar VMAT doses of post-operative prostate plans. The LUT was populated by dose distributions of 30 clinical plans. We optimized two plans with different rectum sparing for an independent case to mimic the process of dose tuning. The plan with less rectum sparing is set as base plan. The 50% iso-surface of the more-sparing plan is defined as the desired iso-surface input. The dose output by our method (expansion and correction) agrees with the more-sparing plan obtained by optimization, in terms of gamma (90%), DVH and profiles. The overall dose reconstruction time is within two seconds. **Conclusion:** We developed a distance-to-isosurface based volumetric dose reconstruction method, and applied it to interactive tuning with iso-surface dragging.

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PO-GePV-T-208

A Multi-Staged Treatment Planning Framework for Large-Volume Ventricular Tachycardia Stereotactic Ablative Radiotherapy

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Purpose: Stereotactic ablative radiotherapy (SABR) is emerging as a non-invasive ventricular tachycardia (VT) ablation regimen and has attained initial success of reducing refractory VT. However, a single-fraction biological potent dose (25 Gy) ablating a large planning target volume (PTV, median 98.9cc, range 60.9–298.8cc in ENCORE-VT trial) has high risk of radiation toxicity. We propose consideration of an optimized, multi-stage treatment planning strategy to mitigate radiation risk to adjacent organs at risk (OARs). **Methods:** The rationale for multi-staged SABR-VT is to divide an otherwise very large single-fraction treatment volume into sub-volumes allowing adjacent normal tissues to experience more tolerable fractionated radiation thereby minimizing the overall OARs biological effective dose (BED) and toxicity. The N-staged planning problem is stated: given a set of sub-targets and their respective prescription doses, and treatment sequence, find treatment plans such that each plan delivers the prescription dose p to target T in a single treatment session, and the total dose of all plans satisfies the OAR dose constraints. We formulate this problem as a constrained optimization. The objective of the constrained optimization is to ensure PTV coverage at prescription dose and minimize the weighted sum of OARs' BED. We use alpha beta ratio as 2 to estimate OAR BED reduction. **Results:** One of the VT cases we treated has PTV of 188.2cc. In a 1-stage treatment plan, the maximum doses to stomach and esophagus are 20.46Gy and 21.57Gy, respectively. Under a 3-stage treatment strategy, the biologically-equivalent 1-stage doses to stomach and esophagus are estimated as 14.85Gy and 15.63Gy constituting 28 percent reduction. **Conclusion:** Clinical investigation is required to confirm our modelled predictions regarding toxicity reduction with multi-stage SABR-VT and characterize the success of the approach in controlling arrhythmia.

PO-GePV-T-209

An Evaluation of Complex Plan Quality Using RayStation's Collapsed Cone Convolution (CCC) Algorithm as a Function of Differing Hardware Given a Time Constraint

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Purpose: The introduction of graphical processing unit (GPU) acceleration for RayStation's collapsed cone convolution (CCC) algorithm saw tremendous increases in the speed of dose calculations, but these speed increases and their effect on plan quality has not been extensively documented. The complexity of the optimization problem provides no guarantee of reaching acceptable plan results in reasonable timeframes. This study investigates the impact of hardware

on radiation treatment plan quality, pertaining to time burdens and optimization setup. **Methods:** Several complicated treatment plans were developed in RayStation v8A SP1 with optimization criteria that produce clinically acceptable plans without additional user input. These criteria have been optimized using different sets of initializations and optimization steps (1x240, 2x120...). These were evaluated for speed of computation and plan quality under GPU acceleration. CPU optimized plans were created under time constraints matching the accelerated plans, and results were compared to evaluate the impact of acceleration on plan quality. **Results:** CCC calculations entail a five-fold increased runtime in the CPU-accelerated environment over GPU-accelerated. Total runtime increases quadratically for long, continuous optimization runs, with decreased differential returns on time invested. The 1x240 optimization resulted in a plan with the most volume exceeding Rx (58.7%), while the 4x60 optimization resulted in the least excess (6.92%). **Conclusion:** The combination of quadratically increasing runtime for long optimizations and the superior metrics of multi-initialized plans suggests that this strategy is best, but multi-initialized plans give more of their runtime to CCC. Given the same amount of runtime, GPU-accelerated optimizations can perform more iterations than CPU optimization by completing CCC evaluations efficiently – producing higher quality treatment plans. This suggests that clinics using RayStation's CCC algorithm with tight timeframes can produce better plans with improved hardware. Further study intends to investigate the effect of varying GPU hardware on plan quality and optimization times.

PO-GePV-T-210

Automated Deliverable Plan Generation of Intensity Modulated Radiation Treatment Using Dose Distribution Predicted from Deep Learning

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Purpose: Intensity-modulated radiotherapy(IMRT) is widely used to achieve higher target conformity and normal tissue sparing. This report developed automated treatment planning strategies for nasopharyngeal carcinoma(NPC) and rectal cancer, including 3D dose prediction based on deep learning and development of automatic optimization algorithm based on dose distribution. **Methods:** IMRT plans for 117 NPC patients and 200 patients with rectal cancer were enrolled in this study. A deep learning model based on U-Net was trained to predict dose distribution of new patients. Model input was composed of patients' CT images and clinically delineated PTV contour, and model output was three dimensional dose distribution of patients. An optimized algorithm composing MSE and DVH constraints was developed to achieve dose distribution-based plan optimization. Planning results were illustrated by comparing DVH curves and quality evaluation parameters of PTVs and OARs. **Results:** Dose distribution similar to the clinical executed plan can be obtained by deep learning method, and cross-sectional dose map comparison results show no significant difference between clinical and predicted dose distribution. Automatically generated plans using MSE-based optimization target meet the clinical evaluation criteria, meanwhile the maximum brainstem dose is reduced by 5.2 Gy. V30 of left and right parotid glands are declined by 10.5% and 9.2% respectively. The overall planning time of a single NPC patient was less than 5 minutes. **Conclusion:** This research developed a novel automated IMRT planning process. Report demonstrated this method's ability to predict IMRT dose distribution precisely, providing more information accounting for an individual patient's treatment planning. Furthermore, the use of innovative objective function composed of MSE & DVH objectives allowed a standardized planning process that showed clinical gains over existing manual IMRT planning.

PO-GePV-T-211

Dosimetric Differences of Plans Using Flattened 6MV Versus Flattening-Filter-Free 10MV Beams for the Treatment of Prostate with and Without Seminal Vesicles

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Purpose: To investigate the dosimetric differences in treatment plans for prostate with and without seminal vesicle using volumetric modulated arc therapy (VMAT) plans with flattened and flattening-filter-free (FFF) beams. **Methods:** Nine patients treated with 1.8-2.5Gy/tx and 6MV-flattened beam (6x) were identified. Each patient's VMAT plans were recalculated for 10MV FFF beams (10FFF). All of the optimization parameters were set to be identical for flattened and FFF beam plans. The maximum dose rate for 6x (600 MU/min) and for 10FFF (2400 MU/min) were allowed during optimization; however, the treatment planning system automatically optimized at a lower dose rate for FFF plans. Plans optimized with 10FFF beams were compared with the plans optimized and calculated with 6x beams. Various metrics of these two plans were evaluated in this study. Studied metrics included mean dose to rectum, bladder, bowel, body, target; maximum dose to femoral heads, and penile bulb; conformity index; target dose inhomogeneity; and Total MU. **Results:** All generated plans, regardless of the beam modality, met clinically accepted objectives. Both 6x and 10FFF beams provided similar target coverage and dose conformity. In comparison, 10FFF beam provided better solution in sparing rectum, bladder, bowel, femoral heads and penile bulb than 6x beam. Mean dose to the body decreased with 10FFF by 5.2%. Average maximum dose to the femoral heads also decreased with 10FFF by 4.2%. 10FFF increased target dose inhomogeneity. In most cases, failure after external beam radiation show that the area responsible for the local recurrence is the intraprostatic dominant nodule. Finally, total MU was lower in plans using 10FFF. **Conclusion:** Both treatment modalities provided clinically acceptable treatment plans. The results of this study indicated that 10FFF plans resulted in better plan quality metrics, with relatively shorter delivery times. FFF beam's target dose inhomogeneity may be used for a focal dose escalation.

PO-GePV-T-212

Evaluation of Elekta Unity IMRT Beam Complexity Using a Novel Leaf Sequencer and Shape Optimizer in the Monaco Treatment Planning System

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Purpose: During online adaptive planning with the Elekta Unity MR-Linac, new intensity modulated radiotherapy (IMRT) treatment plans are created for each patient immediately prior to treatment. To reduce both offline and online planning time, Elekta has developed a new Optimal Fluence Levels leaf sequencer and pseudo-gradient descent shape optimizer (OFL+PGD) in a research version of the Monaco treatment planning system. **Methods:** Unity MR-Linac Step-and-shoot IMRT beam complexity metrics were calculated for plans generated using Elekta's research OFL+PGD algorithms and Elekta's current optimizer and leaf sequencer (Hyperion). A total of 10 patients were included in this preliminary study (3 prostate, 2 pancreas, 3 oligometas, and 2 liver). Beam complexity metrics include plan averaged beam area normalized to PTV area (NPA), plan averaged beam irregularity (PI), plan normalized MU (PMU), and plan averaged beam modulation (PM). Total number of segments, and number of MLC subsegments (i.e. separated open areas within a segment) was also quantified. Statistical significance between the OFL+PGD and Hyperion algorithms results was assessed via student t-test and correlations through Spearman's rank test. **Results:** Differences in IMRT beam complexity metrics between OFL+PGD and Hyperion algorithms were determined as follows: $\Delta NPA = -0.004 \pm 0.025$, $\Delta PI = -0.2 \pm 0.5$, $\Delta PMU = -37.0 \pm 59.4$ MU, and $\Delta PM = -0.1 \pm 0.1$. OFL+PGD algorithms also produced fewer segments and subsegments (-10.1 ± 9.7 and -0.2 ± 0.2 respectively) compared to the Hyperion algorithm. The pairwise reduction in PM, number of segments, and number of subsegments was statistically significant ($p=0.004$, $p=0.009$, $p=0.02$ respectively). Additionally, PMU was found to be correlated to PM for both the OFL+PGD and Hyperion algorithms. **Conclusion:** For the 10 plans evaluated, the new OFL+PGD algorithm produced plans with reduced PM, number of segments, and number of subsegments compared with Hyperion. Differences in NPA, PI and PMU were not found to be statistically different between the two algorithms. For both Hyperion and OFL+PGD algorithms, PMU was correlated with PM.

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PO-GePV-T-213

Sub-Segmentation of Laryngopharynx Whole-Mean Dose Constraints for Reducing Intra-Organ Dose Variability

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Purpose: QUANTEC recommends whole-mean radiotherapy dose constraints for various organs-at-risk, but mean dose constraints allow dose to concentrate in arbitrary regions of the constrained volume. Here we demonstrate a radiotherapy treatment planning method that uses sub-segmentation to improve spatial control of modulated dose distributions. **Methods:** An open-source binary plug-in script for Varian Eclipse™, “Plan n Check,” was developed for structure sub-segmentation, applying constraints, and automating Volume Modulated Arc Therapy (VMAT) plan optimization. Plan n Check was used on 6 head-and-neck VMAT patients (70Gy, 35 fractions) to divide laryngopharynx structures into 8 sub-structures of equal volume using nested segmentation. An institutional whole-mean dose constraint of 45Gy (adapted from QUANTEC) was imposed on each sub-structure in an effort to mitigate the concentration of dose to sub-regions of the laryngopharynx. Plans were optimized with Plan n Check using the Analytical Anisotropic Algorithm (AAA). In addition, a standard plan which did not apply sub-segment dose constraints was created for each patient as a control. The number of sub-structures violating the adopted dose constraint was compared for plans with and without sub-structure constraints. The standard deviation of mean dose within laryngopharynx sub-structures was also calculated for all plans to determine whether sub-structure optimization can reduce intra-structure dose variability. **Results:** The number of laryngopharynx sub-structures with dose violations was significantly reduced (mean: 2.2 to 0.33) when plans imposed sub-structure whole-mean constraints (Student’s $t(5) = 3.38$, $p < 0.02$). Imposing sub-structure constraints significantly reduced the standard deviation (8.2Gy to 4.1Gy) of mean laryngopharynx sub-structure doses ($t(5) = 2.58$, $p < 0.025$). This technique did not significantly impact other organ or target volume dose distributions. **Conclusion:** Mean dose constraints may in some cases insufficiently constrain organ doses. Imposing additional sub-segment constraints is a practical method for exerting greater spatial control and reducing intra-organ dose variability.

PO-GePV-T-214

Dosimetric Advantages of Split-Field Fixed-Jaw Intensity-Modulated Radiation Therapy On Lung Protection for T-Shaped Esophageal Cancer

H Chen*, Z Xu

Purpose: To evaluate the dosimetric characteristics of split-field fixed-jaw (SF-FJ) intensity modulated radiation therapy (IMRT) technique used in treating T-shaped esophageal cancer and the lung sparing, and assess the correlation between the lung tissue dose volumes reduced by this technique and the target vertical axis length. **Methods:** SF-FJ IMRT plans for Thirty T-shaped esophageal cancer patients were compared to jaw tracking (JT) plans and static jaw (SJ) plans. The dose distribution in target volumes, the protection of organs at risk (OARs), MU as well as the correlation between the lung irradiation volumes reduced by the SF-FJ plan and the length of the target vertical axis were evaluated. Wilcoxon signed-rank test and Pearson rank test ($p < 0.05$) were used as the verification methods. **Results:** The HI and CI of PTV-G in the JT and SJ plans were superior to those of the SF-FJ plan. Compared with the JT and SJ plan, the SF-FJ plan significantly reduced the V5, V10, V13, V15, V20 for lungs and the maximum dose for the spinal cord, but slightly increased the V30 of lungs. The SF-FJ plan delivered higher MU than the SJ plan, and lower MU than the JT plan. The correlation coefficients between $\Delta V5$ JT-FJ, $\Delta V10$ JT-FJ, $\Delta V13$ JT-FJ, $\Delta V15$ JT-FJ and the length of the target vertical axis were 0.718, 0.725, 0.717 and 0.599, respectively, and $\Delta V30$ JT-FJ was negatively correlated with the length of the target vertical axis, with correlation coefficient value 0.631. The correlation coefficients between $\Delta V10$ SJ-FJ, $\Delta V13$ SJ-FJ, $\Delta V15$ SJ-FJ and the length of the target vertical axis were 0.747, 0.743 and 0.776, respectively. **Conclusion:** SF-FJ IMRT planning could maximize the protection of lung tissues, and as the length of the target vertical axis was longer, the low-dose to lungs was further reduced.

shanghai jiaotong university(YG2019ZDA25)

PO-GePV-T-215

Evaluation of Planning for Upper Thoracic Esophageal Carcinoma Based on Auto-Planning

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Purpose: To evaluate the progressive optimization algorithm-based Auto-Planning module of Pinnacle3 9.10 treatment planning system (TPS) whether can effectively reduce the difference in the plan quality caused by the lack of experience of the dosimetrist compared with the manual plan for the upper thoracic esophageal carcinoma. **Methods:** Ten patients with upper thoracic esophageal carcinoma were selected, set the same initial conditions, and Manual-IMRT and Auto-IMRT were designed separately. Change the lung dose limit conditions to generate a new plan. The results obtained are normalized based on the results of the initial plan and compare the effects of different initial conditions on automatic and manual plans. **Results:** Compared with the Manual-IMRT plans, the V5, V20 of Auto-IMRT plans were no statistical difference ($P = 0.704$, 0.085), and the V30 significantly reduced by 0.23% ($P = 0.003$). The mean dose for the lung decreased 0.1 Gy ($P = 0.045$) by using Auto-Planning. Compared with Manual-IMRT plans, Auto-IMRT plans were more convergent with the change of lung dose restriction conditions. **Conclusion:** Auto-Planning module can effectively reduce the difference in the plan quality caused by the lack of experience of the dosimetrist and improve overall plan quality and consistency.

PO-GePV-T-216

Dosimetric Impact in GNP-Enhanced Radiotherapy Using the 10 MV FF and FFF Photon Beams: A Monte Carlo Depth Dose Study

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Purpose: This study investigated the effect of dose enhancement and beam attenuation on the depth dose, when gold nanoparticles (GNP) were added to the soft tissue, using the flattening filter-free (FFF) photon beams in GNP-enhanced radiotherapy. **Methods:** A cubic water phantom irradiated with the 10 MV FF and FFF photon beams (field size = 10×10 cm²) was used. GNPs with concentrations ranging from 3 to 40 mg/mL were added to water. The depth doses of the phantom were determined using Monte Carlo simulation under the macroscopic approach. The depth dose enhancement ratio (DDER), defined as the ratio of the dose with GNP addition at a depth to the dose without GNP addition at the same depth, was calculated for each GNP concentration. **Results:** When GNPs were added, the DDERs were found larger than one in the depth range of 4-6 and 4-8 cm for the 10 MV FF and FFF photon beams, respectively. However, when the depth was larger than 6 (FF beam) and 8 cm (FFF beam), the enhancement ratios were less than one. This shows that the presence of GNPs did not increase the dose beyond 6 and 8 cm, and it is because the GNP addition not only enhances the dose, but attenuates the beam along the depth. Comparing the FF and FFF photon beams with GNP concentration increased from 3 to 40 mg/mL, the DDERs were found increased from 1 to 1.004 (FF beam) and 1 to 1.015 (FFF beam). This is because the FFF beam contained more low-energy photons than the FF. **Conclusion:** When GNPs were added to the phantom, the dose enhancement decreased with depths deeper than 6 and 8 cm for the FF and FFF photon beams, respectively. In addition, the dose enhancement was found decreasing at a higher rate when the GNP concentration increased.

PO-GePV-T-217

A Study of Surface Dosimetry for Whole Breast Irradiation Geometry Using EBT3 Gafchromic Film and Monte Carlo Simulation

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Purpose: This in-phantom study was performed to validate a film technique for measurement of surface dose in isocentric, tangential whole breast irradiation by comparison with Monte Carlo (MC) simulations. **Methods:** Solid water phantoms of 20 cm and 30 cm separation were irradiated at 100 cm SAD with lateral half-blocked 6 MV and 15 MV x-ray beams, on a Varian TrueBeam linac. Two centimetres beam clearance on the anterior and inferior phantom edges was used to simulate whole breast tangential field geometry, with a field length of 20 cm and asymmetric field width of 15 cm. Measurements were performed using EBT3 Gafchromic™ film on the phantom surface under 0.0 g/cm², 0.03 g/cm² and 0.0995 g/cm² buildup materials to simulate potential breast support devices on the entrance, exit and inferior surfaces of the phantom. Dose was normalized to the mid separation dose. Film calibrations were performed at SAD 100 cm, 10x10 cm² field size at depth of 10 cm for both energies. Surface dose was calculated in depth increments of 0.025 g/cm² from the surface to a depth of 0.1 g/cm² using EGSnrc using the following parameter settings: ECUT=0.521 MeV, PCUT=0.010 MeV, BCA EXACT with a skin depth of 3, ESA PRESTA-II. **Results:** Film measurements agreed with MC within 5% at the beam entrance and exit surfaces for both beam energies and all buildup materials. Agreement was within 10% at the inferior phantom surfaces receiving side scatter from the phantom. **Conclusion:** Investigations of this EBT3 film technique to measure surface dose are ongoing. Significant corrections appear not be required when measuring entrance or exit surface dose but further work is needed to improve agreement with MC simulation at the inferior phantom surface, where dose is primarily due to side scatter from the irradiated volume.

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PO-GePV-T-218

Dosimetric Considerations of Moldable Silicones for Radiotherapy Boluses and Phantoms

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Purpose: Due to their favourable characteristics, moldable silicones (MS) have gained popularity in medicine and recently, in radiotherapy. We investigate the dosimetric properties of MS in high-energy radiotherapy photon and electron beams and determine their suitability as water substitutes for constructing radiotherapy boluses and phantoms. **Methods:** Two MS types were used ($\rho=1.04$ and 1.07 g/cc). Various dosimetric properties were characterized (e.g., Z_{eff} , relative electron density, mass-energy absorption coefficients). For each type, two MS slabs were molded with a base of 16×16 cm², and 1.5 cm and 5.0 cm thicknesses. Two additional 5.0 cm thick slabs were molded to fit a Markus ionization chamber (IC) flush to the surface. For each MS type, the doses at 1.5 cm and 5.0 cm depths were measured with 5.0 cm Solid Water (SW) backscatter material (MS-SW), and with a full MS setup (MS-MS), then compared with doses at the same depths in a full SW setup (SW-SW). Relative doses were reported as MS-SW/SW-SW and MS-MS/SW-SW. Measurements were repeated using film. Experimental results were verified using Monaco Treatment Planning System (TPS) dose calculations and DOSXYZnrc Monte Carlo (MC) simulations. **Results:** Dose ratios varied according to MS and beam types. For photon beams, a bolus setup caused ~5% dose reduction (MS-SW/SW-SW). Using MS alone resulted in ~2% dose reduction (MS-MS/SW-SW). For electrons, up to 2% dose increase was observed for MS-SW and MS-MS relative to a full SW setup (SW-SW). **Conclusion:** MS compositions differ from water/SW's and vary between manufacturers. Accurate dosimetry using these materials requires consideration of these differences. Compared with dose in SW, an interface of MS-SW can cause relatively high differences. Using MS alone for phantom construction provides a more tissue-equivalent setup. However, it is important to characterize a particular MS's properties for a given beam quality prior to clinical use.

PO-GePV-T-219

Feasibility of 18-MV Grid Therapy from the View of Radiation Protection

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Purpose: In Spatially Fractionated Grid Radiation Therapy (SFGRT), the grid block in the collimation system modifies the neutron production, photon scattering, and electron contamination in and out of the radiation field. The main purpose of this study is to investigate the photoneutron production and the secondary cancer risk in SFGRT with 18-MV photons. **Methods:** Using the MCNPX code, three types of grid made of brass, cerrobend, and lead, located at the tray of an 18-MV medical linear accelerator, were simulated. Plans with different field sizes were generated to compare the neutron and electron contaminations. A revised female MIRD phantom with an 8-cm spherical tumor inside the liver was simulated to study the lifetime risks of fatal cancer due to neutrons and scattered photons using a 20-Gy SFGRT plan and a 40-Gy CFRT plan with a 9×9 cm² field size. **Results:** The neutron fluences per electron-history in SFGRT plans with brass, cerrobend and lead blocks are averagely 55%, 31% and 31% less than that of the CFRT plan, respectively. In both SFGRT and CFRT plans with a 9×9 cm² field size, the contribution of the primary and scattered photons to the fatal cancer risk is 2 times or more than the photoneutrons. The total risks from photons in SFGRT with brass, cerrobend, and lead blocks are 1.733, 1.374, and 1.260%, respectively. These values are significantly lower than the total photon-risk in CFRT (5.827%). **Conclusion:** In the brass, cerrobend, and lead grids, the attenuation of photoneutrons outweighs its photoneutron production in SFGRT. The total cancer risks in the SFGRT plan were less than the risk in the CFRT plan. The grids also decrease the amount of electron contamination.

PO-GePV-T-220

Impact of Depth-Induced Spectrum Change on the Dose Enhancement of Gold Nanoparticle Aided Radiotherapy at Megavoltage Energy

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Purpose: The magnitude of dose enhancement ratio (DER) induced by GNPs is sensitive to the beam quality, such as particle type and energy spectrum - traversing matter changes the energy spectrum. Monte Carlo simulations were used to assess the impact of depth-induced spectrum change on the micro-dosimetry induced by GNPs irradiated using megavoltage (MV) beams. **Methods:** Simulations were carried out using TOPAS-nBio. The micro-dosimetry calculation was split into three distinct steps. Step one: a simulated 10648cm³ cubic water phantom was irradiated by 6 MV x-rays, and phase space data (PSD) of the particles passing through a 5cm² plane at different depths (0.25cm to 10cm) of phantom were recorded. Step 2: the recorded PSD was used as a source of GNP irradiation. For increasing interaction probability, the PSD acquired in the first step was shrunk to be nanoscopic (16nm²). Ejected secondary particles were recorded as spherical PSD around the GNP. Step 3: the spherical PSD recorded in the step two was used to calculate the dose distribution in nanoscale - utilizing track structure-based physics models within TOPAS-nBio. The dose was scored in bins of 1nm for from 0nm-100 nm from the GNP center, 10 nm from 100nm-1µm and 100 nm from 1µ-100µm. **Results:** Simulations were validated using known characteristics of 6 MV clinical beams in water and found to be accurate. The fraction of low energy particles (<500 keV) in the incident beam increased from 6.56% to 8.44% as depth increased from 0.25cm to 10 cm. GNP-mediated DER was very localized around the NP regardless of depth, however, a depth dependency is still apparent. **Conclusion:** The incident photon spectrum does show a depth dependency, which should be accounted for when considering clinical GNP aided radiation therapy. The optimal energy for GNP-mediated DER needs to be further refined before clinical use.

PO-GePV-T-221

Validation of the Monte Carlo Model of a Novel Biology-Guided Radiotherapy (BgRT) Linac

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Purpose: RefleXion is a novel radiotherapy machine that tracks tumor using real-time positron emission signals and delivers radiation through a binary multi-leaf collimation (MLC) system. A Monte Carlo (MC) model of the RefleXion Linac is validated against commissioning measurement in this study. **Methods:** The RefleXion Linac consists of a 6 MV FFF photon beam, a 64-leaf MLC system, and two pairs of y-jaws locating above and below the MLCs. The leaf side focuses on the source and each leaf provides a 0.625 cm opening along the x direction at 85 cm SAD, giving a total of 40 cm opening when all MLCs are retracted. The y-jaw pairs move simultaneously to open 1 or 2 cm at 85 cm SAD. The beam commissioning measurements were performed in a Blue Helix Phantom using a diode and a W2 scintillation detector for scanning and output factor acquisition, respectively. The MC model provided by the vendor includes phase space files above MLCs and detailed structures of MLC leaves and lower y-jaws. Beam characteristics of field sizes from 1.25 x 1 cm² to 40 x 2 cm², including percentage depth dose (PDD), beam profiles, and output factors were simulated using MC codes, BEAMnrc and GATE, and compared with measurement. The relative simulation uncertainty is approximately 0.6%. **Results:** The mean and maximum differences of %dd(10) between measurement and simulation are 0.31% and 0.82%, respectively. The inline FWHM shows a deviation of 0.47 mm on average and 0.58 mm at most. The crossline off-axis ratio of the central 80% of field size differs by 0.88% on average (std = 0.92%). The mean output factor disagreement is 0.93% (std = 0.72%). **Conclusion:** A complete and accurate MC model of RefleXion Linac has been validated. This model will help substantially in the upcoming dosimetric, imaging and quality assurance studies.

PO-GePV-T-222

Focal Spot Size Effect of the RefleXion X1 Radiotherapy Machine: A Monte Carlo Simulation Study

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Purpose: The RefleXion™ X1 radiotherapy machine is a novel combination of positron emission tomography (PET) and computed tomography (CT) subsystems in the same gantry with a 6MV linac, with the intent of delivering biology-guided radiotherapy treatment (BgRT). Small treatment beam penumbra is desirable for small clinical targets. This study evaluated the focal spot size influence on treatment beam penumbra using a Monte Carlo (MC) technique. **Methods:** A reference field of 10 cm x 2 cm at source to axis distance (SAD) of 85 cm was modeled and simulated using the BEAMnrc MC package. The electron beam incident on the X-ray target was empirically modeled with a Gaussian function and the FWHM of the beam for the linac head model was varied between 0.4 and 2.0 mm in steps of 0.2 mm. Dose distributions were calculated in a water phantom of 60 x 30 x 22 cm³ using the DOSXYZnrc MC code. The effect of X-ray beam spot size on percentage depth dose (PDD), penumbra width, and field width based on profiles at depth of 15 mm and profiles in the air at SAD=85 cm was evaluated. **Results:** The variation in X-ray beam spot size did not have a significant effect on the PDD, with an average difference < 0.34% for PDD at depth of 100 mm. Penumbra width increases as X-ray beam spot size increases (R=1, P<0.01), while the field width is not affected by the variation of X-ray beam spot size. **Conclusion:** The beam spot size has a sizeable effect on penumbra width. This study shows that a small X-ray beam spot is preferred for small treatment fields due to the resulting narrow penumbra width.

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PO-GePV-T-223

Evaluating Backscattered Radiation into the Dose Monitor Chamber in the RefleXion X1 Using Monte Carlo Simulation

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Purpose: The RefleXion™ X1 is a novel ring gantry system with a unique structural design made to accommodate both a linear accelerator (LINAC) and positron emission tomography (PET) detectors for biology-guided radiotherapy (BgRT). A dose monitor chamber (DMC) for measuring beam output from the LINAC is located between a fixed primary collimator and a movable secondary multi-leaf collimator (MLC). Backscattered radiation from the secondary collimator may contribute to the total charges collected in the DMC, may result in an artifact in the measurement. The present investigation utilized Monte Carlo (MC) simulation to characterize the impact of backscattered radiation to the DMC measurements. **Methods:** The EGSnrc Monte Carlo package was used to model the X1 LINAC output of a 6 MV photon beam in flattening filter-free mode with 521 keV global electron cutoff energy and spin-on for reasonable backscattering calculation. The backscatter effect from downstream LINAC components was quantified by calculating the contribution of forward-directed and backscattered radiation to the energy deposited in the DMC. Scenarios involving 2, 4, 8, and 16 open MLC leaves with 1, 2, and 3 cm jaw settings were computed. **Results:** A maximum backscatter contribution of 0.12%±0.1% was obtained for a simulated field of 10 cm (16 MLC leaves) with a 2 cm jaw opening. For a simulated field of 1.25 cm (2 MLC leaves), a maximum backscattering contribution of 0.27%±0.1% was recorded. On average, backscattering contribution decreased with an increase in the field size. Minimal variation in backscattering was observed for different jaw openings. **Conclusion:** For the X1 LINAC photon beam, the effect of backscattered radiation is minimal even with very small field sizes and is not correlated with jaw size. Therefore, correcting for field-size dependence of DMC output to account for backscatter is not necessary for the RefleXion X1 design.

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On the Determination of Source Parameter Probability Distribution Functions During the Monte Carlo Modeling of a CyberKnife M6 Unit

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Purpose: As prescribed by the IAEA-AAPM nonstandard beam dosimetry formalism, specialized radiotherapy treatment units which cannot achieve reference conditions established by conventional dosimetry protocols require detector quality correction factors. Monte Carlo is recognized as an essential method for the evaluation of these correction factors, but requires an accurate modeling and source parametrization of the radiotherapy device of interest. The purpose of this study is to propose a new maximum likelihood estimation method for the fitting of the source parameters of a CyberKnife M6 stereotactic radiotherapy unit. **Methods:** Based on the approach of Francescon et al 2008, a formalism is developed to estimate the likelihood distributions of the energy and spot size of the electron beam incident on the target. A rigorous uncertainty budget, which includes detailed consideration of positioning uncertainties, is estimated. The EGSnrc code system is used to model: 1) the CyberKnife M6 unit and 2) four detectors recommended by the IAEA-AAPM protocol for relative dosimetry of small photon fields. Absorbed dose in the detectors' sensitive volume are simulated for two setups, output factors and tissue-phantom ratios, using phase-space files generated with the CyberKnife model as the beam source. Corresponding measurements are performed on the CyberKnife M6 unit. **Results:** The method is used to estimate the probability distribution functions of the electron energy and spot size. From these distributions, the parameters, and their respective uncertainties, are extracted using the solutions of maximum likelihood and the standard deviation of the obtained distributions. Results illustrate the agreement, using a coverage factor of 95%, between measured and simulated output factors and tissue-phantom ratios for most considered detectors and field sizes. **Conclusion:** The proposed method allows an accurate beam parameterization of a CyberKnife M6 unit and a rigorous evaluation of uncertainties on the

determined energy and spot size.

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KQ Calculations of the ZAP-X Radiosurgical Device

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Purpose: The purpose of this work is to characterize the ZAP-X (ZAP Surgical Systems, Inc., San Carlos, CA) radiosurgical device beam quality using EGSnrc Monte Carlo k(Q) calculations of several ionization chambers. The TG-51 and TRS-398 clinical dosimetry protocols require k(Q) factors for calibrating external radiotherapy beams. **Methods:** ZAP-X produces a 3 MV photon beam with varying narrow conical beam collimation settings. For the k(Q) calculations, the beam was collimated to a diameter of 25 mm at 45 cm SSD (source to surface distance). This is a small field set-up compared to reference calibration conditions of 10 x 10 cm² at 100 cm SSD, that were used for the reference Co-60 beam. No machine-specific reference field has yet been defined for the ZAP-X linac, so as a first step only an overall k(Q) factor was calculated, with the chamber measurement point set at 10 cm depth in a 30 x 30 x 30 cm³ tank of water. For the Co-60 beam, the chamber depth was 5 cm as usual. Three ionization chambers were considered: the PTW 31010, Exradin A28 and Exradin A1SL. The Monte Carlo models for these chambers were previously produced from detailed manufacturer specifications and validated. **Results:** The overall k(Q) factors for the PTW 31010, Exradin A28 and Exradin A1SL under the 3 MV ZAP-X beam were calculated. Machine-specific reference k(Q) factors were then investigated based on additional simulations for various set-up conditions. **Conclusion:** Beam quality of the ZAP-X radiosurgical device was investigated using EGSnrc Monte Carlo simulations. Despite the small field size and low beam energy, overall k(Q) factors seemed relatively insensitive to varying ionization chamber models, and machine-specific reference conditions are proposed for the new ZAP-X linac.

PO-GePV-T-226

Characterization of Phosphorescent Strontium Aluminate as a Real-Time Dosimeter

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Purpose: A novel dosimeter capable of displaying dose in real time would enable one to verify the distribution of dose on the surface of a patient during treatment. The purpose of this study was to characterize the response of phosphorescent strontium aluminate (SrAl₂(2)O(4):Eu²⁺Dy³⁺) to high energy x-rays and to subsequently determine how the signal decays with respect to time. The wide range of application and low cost makes it a promising and attractive alternative to relative dosimetry. **Methods:** Glass cuvettes were filled with phosphorescent strontium aluminate powder and individually irradiated by a 6 MV beam at 100 cm SSD with a 10x10cm(2) field size at 1.5 cm depth. Post-irradiated cuvettes were placed in a black box and remotely photographed using a Nikon DSLR camera 1.5 minutes after irradiation. The resultant images were analyzed in MATLAB. Using a dose rate of 600 MU/min, we irradiated samples with 25, 50, 100, 200 400, 600, 800, and 1000 MU's to test linearity. Signal decay over time was measured by irradiating a cuvette at 600 MU and obtaining images in 5-minute intervals. **Results:** Strontium Aluminate exhibited a linear relationship to an increase in dose, within a dynamic range of 400-1000 MU with an R-square of .9922. For the signal decay over time test, the signal decreased roughly by a factor of 5 within the first 5 minutes and gradually decayed for the remaining intervals. The decay in signal was best fit by a rational function to the first degree. **Conclusion:** Phosphorescent Strontium Aluminate showed a favorable response in its linearity trials with changing dose and could potentially be used as a real time dosimeter during treatment or for routine QA procedures. Future experiments will focus on solidifying its functionality as an alternative to the current and more expensive relative dosimeters in use.

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When to Use Flattening Filter Free(FFF) Mode for Elekta Versa HD Delivered with VMAT

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Purpose: To compare and evaluate the performance of Elekta Versa HD with and without flattening filter two operating mode(FF and FFF)delivery with volumetric modulated arc therapy. **Methods:** Treatment plans were graded four(Low,Moderate,Moderate-High and High complexity)according to the complexity.This includes 1 low complexity(brain metastasis 20Gy single dose),2 moderate complexity(Lung and Liver 12 Gy fraction dose),1 Moderate-High complexity(prostate 2 Gy fration dose) and 1 high complexity (head and neck 1.8 Gy fration dose) cases.All cases were designed two VMAT plans,one with flattening filter (group A) and the other with flattening filter free (group B).All plans were done on Elekta Versa HD with Monaco treatment planning system. All plans were generated with 6 MV X-rays.Plans were evaluated based on the ability to meet the dose volume histogram, radiation conformity index, estimated radiation delivery time,dose homogeneity index(HI) and monitor units(MU) needed to deliver the prescribed dose. **Results:** Plans of group A achieved the best HI (1.06 Vs.1.07) at the low complexity cases while plans of group B were slightly better at the high complexity cases (1.15 Vs.1.13). Treatment time of group B reduction more pronounced with increasing fraction dose for moderate target complexity (Low complexity: 535 Vs.262s,Moderate complexity:247s Vs.132s, and 341s Vs.137s),especially for the liver case with paradigms delivered time can be decreased 60%.The more MU for group B than group A as plan complexity increased (Low complexity:3325 Vs.3263, Moderate complexity:2001 Vs.2888, Moderate-High complexity:765 Vs. 920,High complexity:637 Vs.1333).No significant changes were observed between the group B and group A plans in terms of the healthy tissue mean dose.Both plans respected the planning objective for all organs at risk. **Conclusion:** The study concludes that VMAT plans with flattening filter free can significant decrease the delivering time at the moderate complexity cases with hypofractionation dose,while a slight compromise in the MU increased should be noted.

PO-GePV-T-228

Evaluation of a Scripted Planning Technique in Lung SBRT

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Purpose: The efficacy of SBRT in thoracic lesions depends on the effective dose deposited in the lesions. To reduce plan variability between multiple planners, a semi-automated process was validated for consistency of CI and R50 while increasing ITV mean dose and centralizing the hot spot to the ITV. **Methods:** Twenty-two lesions were investigated for patients who received a five fraction SBRT regimen via VMAT. A two-step process included a python code where user selects a set of initial parameters which automates the creation of tuning ROIs, machine settings, and loads the optimizer with a consistent set of objectives. The tuning ROIs include sets of rings at certain distances from PTV, as well as overlap ROIs with nearby organs at risk. The process focuses primarily on dose falloff and PTV coverage and then, meeting nearby organ at risk dose tolerances. The clinically delivered plan, which was planned by experienced staff, was used for comparison. **Results:** For the clinical plans, the CI was 1.04±0.05; R50 was 4.07±0.597; and ITV Mean relative to prescription was 117%±6.4%. For the scripted plans, the CI was 1.01±0.03; R50 was 3.61±0.351; and ITV Mean relative to prescription was 123.8%±2.8%. The scripting process was successful in meeting 335/336 OAR metrics resulting in 21 of 22 successful plans. For the clinical plans, the hot spot was located in the ITV 77.3%, while this occurred

in 95.3% of scripted plans. **Conclusion:** Through programmatic planning, standard deviations can be reduced compared to plans from a multiuser environment. In addition, the CI improved with a decrease of R50 without violation of critical organ dose tolerances in majority of cases. This script allowed for more planned dose deposition to PTV while also maintaining a safe dose falloff and organ tolerance preservation. This technique's clinical implementation can improve consistency with minimal user interaction.

PO-GePV-T-229

Retrospective Study of a Novel Automated Software for the Evaluation of Hypofractionated Treatment Plans Using Conventional Constraints

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Purpose: Assess the use of a novel automated software in the treatment planning system to evaluate hypofractionated plans using conventional constraints. **Methods:** Eighteen Eclipse(v15.5) hypofractionated treatment plans of varying sites and dose per fraction regimens were retrospectively reviewed using Radformation's plan check software, ClearCheck(v1.67). The hypofractionated doses were compared to conventional 2Gy constraints using Equivalent-Dose(EQD2) formula, $EQD2=TD[(d+\alpha/\beta)/(2+\alpha/\beta)]$, where the $\alpha/\beta=2$. Within each treatment plan, EQD2 doses from ClearCheck were evaluated for soundness, accuracy, and consistency. **Results:** Variation in values of EQD2 doses determined by ClearCheck, compared to calculated values for the 18 plans, ranged between ± 0.1 cGy. Software-reported values demonstrated accuracy in the calculation itself and consistency amongst plans. In one plan, hypofractionated doses were within the defined objectives in ClearCheck. Subsequent conversion to the EQD2 however, indicates values just slightly in excess of the tolerances in ClearCheck for two critical structures. In theory, EQD2 values should be greater than the doses from the hypofractionated treatment plan. Invalid EQD2 values were observed for structures that were receiving doses less than 2Gy per fraction. **Conclusion:** ClearCheck is a convenient tool in the evaluation of hypofractionated plans and the conversion to conventional 2Gy-fraction constraints. Along with the user's ability to modify the α/β ratio, there is an instantaneous assessment of EQD2. With control in the manner reported, a planner is also able to ensure critical structure constraints are expressed in terms of maximum dose, mean dose, or dose for a specified volume, all essential for the conversion of EQD2 doses. While the EQD2 values are accurately calculated within the software, opportunity for erroneously reported values does, however, exist. In these instances, such as with normal tissues objectives receiving less than 2Gy/fraction, a warning should be provided, alerting the user of the priority in verifying the validity of results.

PO-GePV-T-230

Individualized Planning Target Volume for Cervical Cancer Patients Treated with Intensity-Modulated Radiotherapy Based On Bladder Filling Change

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Purpose: To establish an individualized planning target volume (PTV) based on bladder filling. To investigate the value of individualized PTV in clinical application, through comparing individualized PTV and RTOG standard PTV in tumor coverage and dosimetry. **Methods:** Twenty patients received radical radiotherapy for cervical cancer were selected in this study. Each patient scanned the CT image during full bladder (FB) and empty bladder (EB), respectively. IMRT based on an individualized PTV(PTVi) covering the CTV (Clinical Target Volume) position variations from FB to EB status instead of a uniform CTV-PTV margin of 1.5 and 2.0 cm (PTVs-1.5 and PTVs-2.0) at FB position were given to each patient. The target volume coverage was evaluated for PTVi, PTVs-1.5 and PTVs-2.0. The dose volume histogram was used to evaluate the treatment planning and assess the dosimetric parameter of OARs. **Results:** Compared with PTVi, the target volume was increased by 32.9% and 58.6% for PTVs-1.5 and PTVs-2.0, and the CTV to PTV was increased by 71.8% and 127.8%, respectively. PTVi had better target coverage than PTVs-1.5 and PTVs-2.0. For the OARs, the pelvic bone received V25Gy was $73.5\% \pm 4.89\%$, $79.5\% \pm 4.34\%$ and $82.3\% \pm 4.12\%$ for PTVi, PTVs-1.5 and PTVs-2.0, respectively ($p < 0.05$); V40Gy was $20.7\% \pm 2.90\%$, $31.6\% \pm 4.07\%$, $40.1\% \pm 4.93\%$, respectively ($p < 0.05$). Small bowel received V45Gy was $131.8cc \pm 69.23cc$, $145.7cc \pm 76.61cc$, $157.0cc \pm 82.94cc$, respectively ($p < 0.05$). For rectum and bladder, PTVi showed significantly less radiation dose, compared with PTVs-1.5 and PTVs-2.0. **Conclusion:** Compared with PTVs-1.5 and PTVs-2.0, individualized PTV results in better target coverage and organ-at-risk sparing. An individualized PTV is desired choice of the cervical cancer for radiotherapy.

PO-GePV-T-231

An Evaluation of An EQD2 Conversion Via Commercially-Available Software for Hypofractionated Treatment Plan Tolerances: A Retrospective Study

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Purpose: Assess the use of an automated software to evaluate hypofractionated plans using conventional constraints. **Methods:** Eighteen hypofractionated treatment plans were retrospectively reviewed using MIM's EQD2 Variation 1 of the LQ model. The hypofractionated doses were compared to conventional 2Gy constraints using Thames and Withers' 2Gy Equivalent Dose (EQD2) formula. Within each treatment plan, EQD2 doses were evaluated within MIM for accuracy and consistency. **Results:** The 18 patients plans excel-generated hand calculations and the MIM-converted EQD2 doses demonstrated less than 5% difference for structures receiving more than 2Gy per fraction. In theory, EQD2 values should be greater than the doses from the hypofractionated treatment plan. Variations were observed with critical structures receiving fractional doses < 2 Gy. There is also disparity ($> 1\%$) between the hand calculated and derived EQD2 maximum and mean values for critical structures with a small volume. The MIM EQD2 is calculated per voxel and its handling of voxels within contours contributes to the discrepancy. A small change to the volume contributes to the discrepancy due to the software's preferences set to use "nearest neighbor" interpolation for datasets. **Conclusion:** MIM is a robust tool in evaluating hypofractionated treatment plans by converting to EQD2 doses to critical structures using traditional 2Gy/fxn constraints. Assessment of dose accumulated from hypofractionated and conventional plans with reference to one dataset is achievable, and the software makes for an easy workflow and rapid evaluation. While the EQD2 values are accurately calculated within the software, the user must exercise caution in cases when the fractional dose to a structure is ≤ 2 Gy. Acquiring a more comprehensive understanding of the way in which the EQD2 is applied to the dose cloud would be of additional benefit and begs further investigation into the software's handling of the EQD2 conversion and whether order of conversion matters.

PO-GePV-T-232

Evaluation of Bladder Dosimetric Property for Localized Prostate Cancer Using Deformable Registration of Simulation CT

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Purpose: To evaluate the dosimetric consequence caused by the inter-fractional variations in bladder during the prostate radiation therapy using deformable registration of simulation CT to Daily Cone-Beam CT (CBCT). **Methods:** Four patients who underwent volumetric modulated arc therapy (VMAT) for localized prostate cancer under RTOG 0126 protocol were chosen in this study, with total prescription dose of 7920 cGy delivered in 44 fractions in two stages. Patients were under daily CBCT guidance through the treatment course. To simulate the effect on consistent bladder under-filling, nine daily CBCT instances with smaller bladder volumes were selected for each patient, with seven selected for initial plan, and two selected for boost plan. For each instance, the simulation CT was deformably registered to daily CBCT image. During the registration, the shape of prostate and target region was preserved to stabilize the original planning ISO-center. The deformed simulation CT was then used to generate fractional dose distribution using original beam configuration. Finally, dose distributions from all nine instances were then deformed to the original simulation CT and summed up to generate the final accumulated dose distribution

through the radiation course. **Results:** The average reduction of the bladder volume among the four patients was $39.9\% \pm 12\%$ when compared with simulation CT. An increase was observed for bladder dose-volume parameters: for bladder V65Gy, the increase was $17.2\% \pm 11.6\%$; for bladder V75Gy, the increase was $15.8\% \pm 13\%$. **Conclusion:** A workflow has been developed to evaluate the accumulative dose through treatment course accounting for inter-fractional organ shape and position variations. Inconsistent bladder filling leads to increased bladder doses. It can be used to evaluate the dose-volume parameters variation due to inter-fractional bladder volume change and has the potential to guide future analysis of dose-volume relationship for bladder toxicity.

PO-GePV-T-233

Dosimetric Comparison of VMAT Prostate Plans Designed with 10MV Vs 18MV Photon Beams

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Purpose: To compare prostate volumetrically modulated arc therapy (VMAT) treatments planned with 10MV vs 18MV photon beams to determine the practicality of 18 MV VMAT treatments. Recent studies indicate significantly lower than previously reported neutron doses associated with high-energy photon beams. **Methods:** Clinically delivered prostate patient treatments planned using a 10MV photon VMAT technique were selected and re-planned with 18MV photon beams. Both sets of plans were made to meet the same clinical goals and follow the same optimization constraints. The resultant plans were then compared across various criteria such as modulation factor, 50% isodose volume, and average dose to OARs. All treatments were planned in RayStation v8 utilizing clinically accepted beam model data. **Results:** All selected plans had the same prescription, and were treated on the same treatment machine. On average, 18MV photon plans were delivered with a ~17% lower modulation factor than the corresponding 10MV plans. The average 50% isodose volume of 18MV plans was found to be ~10 % larger than that of the 10MV treatments and average doses to the bladder and rectum were also larger on average. It was consistently observed however that in plans designed to treat larger patients 18MV photon plans were more likely to outperform their 10MV counterparts. **Conclusion:** On average treatments planned with 10 MV photons proved to be superior in 50% isodose volume and OAR averages doses. For larger patients 18MV prostate treatments are likely to produce dosimetrically preferable plans. Criteria used to distinguish that patient population will be presented. Since high-energy intensity-modulated treatments for prostate therapy may not be as detrimental as previously estimated, their place in clinical radiation therapy has to be firmly established.

PO-GePV-T-234

Evaluation of Two Different Dose Calculations Algorithms Between Two Treatment Planning Systems

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Purpose: To determine if there is any significant difference between two different dose calculation algorithms of two different treatment planning systems. **Methods:** Five Head and Neck VMAT plans generated in RayStation (version 9A) using collapsed cone algorithm were exported to and recalculated in Monaco (version 5.11.02) treatment planning system using Monte Carlo dose calculation algorithm. The five plans consisted of two arcs and each were recalculated on a Delta 4 QA phantom (Delta 4) with a voxel size of 3 mm. These calculated plans were compared to measurements using Delta 4. RayStation and Monaco algorithms were commissioned for an Elekta Versa HD Linac with Agility MLC and for 6 MV and 6 MV FFF photon beams. Point-dose measurements using a CC13 ion chamber in a large water tank were also taken to compare RayStation and Monaco calculated values. **Results:** Using 1% for the dose difference and 2mm for the DTA, the gamma analysis for the composite fields using Monaco was 94.0%, while for the convoluted cone was 90.3%. The two-tailed P value equals 0.037, which is statistically significant. Increasing up to 3% in dose difference and 3 mm in DTA, there was no statistically significant difference between the two calculations compared to measurements. For the point-dose measurements, RayStation showed less than 0.3% for different points, while Monaco showed 0.6% for 6 MV. For 6 MV FFF, Raystation showed 0.05% and Monaco 0.20% agreement for 17 different fields. **Conclusion:** The beam models in Monaco and RayStation are comparable when compared to point dose measurements. Monaco may generate a better dose agreement with measurement when using a tighter QA passing criteria than RayStation. However, they do not show a statistically significant difference in agreement between calculation and measurement when using the common passing criteria of 3% and 3 mm.

PO-GePV-T-235

Can a Seven-Day Subset of CBCTs Be Representative of All CBCTs When Reconstructing the Dose Delivered Over the Entire Course of Treatment for Prostate Bed Radiotherapy?

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Purpose: To determine if, for prostate bed treatment, generating synthetic CTs (sCT) from one daily cone-beam CT (CBCT) per week represents the average dose delivered to the patient over the full course of treatment. **Methods:** The planning CT plus 34 daily CBCT scans of a post-operative prostate bed patient were anonymized and sCTs were generated. The planning and organ at risk (OAR) volumes were deformed to the new sCT and checked for consistency. The treatment plan was recalculated on each of sCT. Single fraction Dose Volume Histogram (DVH) data was recorded for AnoRectum, Bladder, Body, Bowel, Femoral Heads, and Planning Target Volumes including prostate bed and pelvic lymph nodes. DVH data for all 34 fractions sCTs were averaged to get the treatment average daily dose to targets and OARs. Average DVHs were calculated for subsets of the sCT DVH data including for the first 7 days, the last 7 days, and every fifth day starting at day 1, 2, 3, and 4. ANOVA tests used to determine if the subgroup averages differed significantly from the average over all 34 fractions. **Results:** According to the ANOVA tests, which compared the subgroup averages to the total daily averages, the subset are not statistically different than the total average. For each structure, the p value was greater than 0.64. **Conclusion:** The recalculated DVH data for any of the checked 7 day subsets of the 34 sCTs were not significantly different than the average of the entire 34 fractions, so a once weekly subset appears to be an acceptable alternative to using the whole data set.

PO-GePV-T-236

A Comparison Study of AI Contouring and the Effect of Organ Tolerances and Reducing Dose Toxicity

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Purpose: To compare the use of standard contouring and AI contouring systems for use in Eclipse (Varian Medical Systems, Palo Alto, CA) treatment planning systems. **Methods:** Using the structures created by dosimetry and Limbus (Limbus AI, London, UK) and Mvision (Mvision AI, Helsinki, Finland) AI Systems, a comparison was performed to determine the dose toxicity difference in three head and neck patients chosen at random. Using the approved treatment plan, each set was applied to the plan. A dose volume histogram was generated for the brain, brainstem, mandible, oral cavity, and left and right parotids. The percent difference of the minimum, mean, and maximum dose (Gray) from the treatment structures were calculated and compared to determine if the dose toxicity would decrease using AI contouring. **Results:** The brain contour difference for the treatment plan, Limbus, and Mvision was under 5% difference for the minimum and maximum dose, however the mean for both systems were over 20%. The brainstem contour and mandible were under 5% for Mvision while Limbus was 10%. The mandible contour using Limbus was under 5% for the mean and max dose but the minimum was almost 10% different. The oral cavity contour using Limbus was under 5% (min) and 10-45% different (mean and max) while Mvision produced under 5% difference (min and max) but 10% difference for the mean. The parotid contours for both systems yielded a mean difference of over 10%, a min difference of under 5%, but the Limbus had a

max difference equal to 25% and MVision was under 5%. **Conclusion:** The use of AI contouring systems can reduce treatment planning time for patients and reduce dose toxicity. The MVision produced more contours that were within 5% of the treatment planning structure set while Limbus produced large differences of 10- 45%.

PO-GePV-T-237

Overlapping Volume Histogram (OVH) as a Tool for OAR Delineation Outliers Detection and DVH Prediction

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Purpose: In contrast to conventional usage of OVH to determine the lowest feasible dose to an organ-at-risk (OAR), this study examines the feasibility of using OVH in detecting outliers incurred in the delineation of OARs in treatment plans, and determining the optimal OVH (amongst a large cohort of patient OVH) in predicting dosimetric DVH properties. **Methods:** An OVH characterizing the spatial relationship between the planning-target-volume (PTV) and an OAR was implemented by iteratively expanding the PTV, and determining the volume of overlap between the OAR and the expanded PTV. The OVH was determined for a cohort of 71 head-and-neck cancer cases. In evaluating the use of OVH as a tool for detecting outliers in OAR delineation, the OVH was applied to the entire data set and mean and median OVH from the OARs to PTV-1, -2, and -3 were evaluated along with the 1-standard-deviation of the OVH of all patient plans. Properties derived from the mean and median OVH profile were evaluated for accuracy in DVH prediction. **Results:** A smaller spread in the OVH curves, and a closer spacing between the mean and median OVH curves are observed for the serial organs (e.g. spinal cord) vs the parallel OARS (e.g. right parotids). For parallel organs, the OVH is confounded by the laterality of the OAR with respect to the PTV. The median OVH, which has a steeper gradient and lies close to the center of the OVH distribution, is relatively insensitive to outliers. Larger differences in the predicted DVH was observed between using mean and median OVH features for parallel organs. **Conclusion:** New insights on the sensitivity and feasibility of using OVH as a tool in detecting outliers of target contours is presented. Prediction of DVH for serial OAR is sensitive to features extracted from the mean and median OVH.

PO-GePV-T-238

To Assess the Impact of Bowel Gas Override on Pelvis Treatment Plans: Case Studies

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Purpose: To assess the impact of bowel gas override on pelvis treatment plans as a contributor to overall treatment delivery uncertainty. **Methods:** Two VMAT pelvis plans for cervical carcinoma, on cases with noticeable bowel gas (BG) volumes (P1, 11.3% and P2, 27.3% gas in bowel bag) were optimized with mass density override (DO) of 0, 0.5 and 1 g/cc to get clinically acceptable plans. Each optimization was re-calculated with remaining 2 BG-DO overrides to assess the robustness of the optimization on varying bowel-gas conditions in-vivo. PTV coverage metrics of V100%, Dmax(0.03cc) and Dmin(0.03cc) and OAR metrics of Bowel Dmax(0.03cc) and V40Gy and Sigmoid V45Gy were chosen for comparing the studies. **Results:** Impact of BG-DO on PTV (V100) was low - in the worst case scenario of no BGO at optimization and recalculated with DO of 1 g/cc, the coverage dropped 2.2% on P1 and 3.8% on P2. Plans optimized with BG-DO of 1 g/cc are least variant (~0.5%) to changes in bowel gas. Changes in bowel gas with DO of 0.5 g/cc resulted in [-4.6%- 1.3%] difference in MinD0.03cc within the PTV. Impact of DO had a maximum change of 2% on D0.03cc for bowel. On P2, all 3 optimizations resulted in 4-6% variation on V40Gy for bowel as compared to <1% on P1. Impact of BGO on sigmoid V45 was substantial on both plans resulting in >7% variation. However, Sigmoid V44Gy was less sensitive to changes in bowel gas. **Conclusion:** Impact of bowel gas in overall treatment delivery uncertainty is of the order of 2-4% on PTV coverage. It is <2% for bowel itself, and OARs overlapping with PTV and in-plane with bowel should be evaluated on less sensitive metrics for robust planning.

PO-GePV-T-239

API Driven Plan Scorecard Designer for the Evaluation of External Beam Planning Quality

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Purpose: Treatment plan quality for external beam radiation therapy contains several quantifiable metrics that may assist the clinician in evaluating dosimetric plan quality. Metric points along the dose-volume histogram (DVH) curves and other calculated metrics can provide assessment to the normal tissue organ sparing and target coverage. A single quantifiable metric, the plan score, can be used to determine plan quality from multiple dose quality parameters and has shown efficacy in plan quality events such as planning competitions and knowledge-based DVH estimation algorithm refinement. **Methods:** An integrated planning scorecard tool, the Eclipse Plan Scorecard (EPS) allows for the generation of custom plan scorecard templates and the import of templates from commercial plan scoring software applications. EPS allows for users to generate evaluation structures automatically and graphically displays plan scoring metrics to the user. To evaluate the scoring functionality, the score for several plans were evaluated against a commercial plan evaluation tool with scorecard functionality. **Results:** The plan scores for 8 patients were extracted by generating scorecard templates within the EPS and comparing them to existing scorecards from ProKnow. The percentage difference across the 8 plans ranged from -3.16% to 2.42% with a mean percentage difference of -0.15% across all plans. Sampling differences between the DVH extracted with ESAPI and the DICOM dose file used by ProKnow leads to minor discrepancies in the reported plan score. **Conclusion:** Integrating a planning scorecard into the treatment planning system allows for the immediate extraction of dosimetric quality parameters and an overall score, thereby facilitating the immediate review of plan quality and improvement across planning iterations.

Mr. Schmidt reports teaching honoraria outside of the submitted work and personal consulting fees with Varian Medical Systems associated with the submitted work.

PO-GePV-T-240

The Use of Daily Transit Dosimetry to Monitor Anatomy Changes and Setup Errors in Prone Breast Radiotherapy

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Purpose: To investigate the use of daily transit dosimetry to monitor anatomy changes and setup errors in prone breast radiotherapy. **Methods:** Five breast cancer patients treated with 3D tangential fields were retrospectively studied. All patients were treated prone and received 42.4-Gy in 16 fractions to the whole breast. Transit dosimetry and kVCBCT were acquired at each fraction. Using the field border structure on planning CT, PTV_CBCT was generated for each fraction same way as clinical planning PTV. Plans were recalculated on CBCT scans and the mean doses to PTV_CBCT were recorded. A reference PTV_CBCT was selected from the first three fractions, and absolute mean dose difference (AMDD) was calculated for each fraction. Transit doses were compared to the reference fraction dose using mean absolute error (MAE) in percentage within a region of interest defined by the planning PTV with 3 cm expansion to account for the daily breast motion. The AMDD in PTV_CBCT was evaluated as a function of MAE in transit dosimetry. **Results:** The median AMDD from all fractions ranges from 0.8% to 6.2% for five cases. While one case overall has larger AMDD ranging from 1.1% to 9.6%, other cases have

AMDD within 3%. The median MAE for all cases ranges from 3.4% to 11.3%. The same case mentioned above also has the largest MAE with a range of 6.6% - 16.2%, when other cases have MAE within 11%. In general, AMDD increases as MAE increases with a R2 value of 0.6102 using a linear regression fitting on all cases. **Conclusion:** Our study demonstrated a potential of using daily transit dosimetry to track anatomy changes and setup errors in prone breast radiotherapy. Current results suggest a threshold of 10% in MAE indicates setup position check or adaptive planning. Future work will include more patients including supine breast cases.

PO-GePV-T-241

Evaluating Bladder and Rectum Doses for Cervical Cancer Based on Dose Guided Radiation Therapy Using Kilovoltage Cone-Beam Computed Tomography

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Purpose: The bladder and rectum filling change causes uncertainty in radiation doses for cervical cancer patients. To estimate the bladder and rectum doses for cervical cancer patients with dose guided radiation therapy (DGRT) using weekly kilovoltage cone-beam computed tomography (KV-CBCT). **Methods:** Six cervical cancer patients were included in this study. All patients were required to have a full bladder for the planning CT and weekly KV-CBCT scan. In our hospital, weekly three-dimensional images of patients in treatment position are acquired for image-guided radiation therapy with KV-CBCT. In each KV-CBCT study, the bladder and rectum volumes were delineated and matched to the planning CT scan to appraise the volume change. These images can be used for calculating the actual dose delivered to the patient during treatment. The histogram matching algorithm was used for calibrating the KV-CBCT images. The treatment plans were recalculated on the KV-CBCT scans. Dose volume histograms were used to analyze the dose distribution in the bladder and rectum during each KV-CBCT scan. **Results:** The mean volume of the bladder delineated on weekly CBCT image was 390.5 ± 154.9 cc (range 130.1-541.9cc), and the mean rectum volume was 30.9 ± 7.1 cc (range 25.7-43.3cc). There was a large variation in rectum and bladder volume based on the weekly KV-CBCT image. The bladder and rectum received V40Gy were $61.3 \pm 8.2\%$ (range 47.8-72.9%) and $33.2 \pm 8.8\%$ (range 19.7-41.9%), respectively; and V45Gy of the bladder and rectum were $47.8 \pm 6.2\%$ (range 41.5-57.8%) and $4.9 \pm 2.6\%$ (range 0.93-7.8%), respectively. **Conclusion:** Considering the large changes of bladder and rectum volumes in the inter-fractional treatment, it is strongly suggested to perform in-room bladder volume measurements, using a 3D ultrasound scanner. DGRT can be used to determine the actual dose in the bladder and rectum using weekly KV-CBCT scan.

PO-GePV-T-242

A Comparison of Modulated Planning Techniques for Whole Brain Hippocampal-Avoidance Radiotherapy

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Purpose: Whole brain radiotherapy is a widely utilized palliative option for patients with brain metastases. Despite improving overall survival, associated reductions in cognitive function have a significant impact on quality of life. Hippocampal avoidance whole brain radiotherapy can delay the onset of neurocognitive decline. A planning technique published in 2015 demonstrated a class solution by partitioning the brain into sub-volumes and utilizing a VMAT split arc technique. The current study investigates an application of this technique, with simplified sub-volumes, for five clinical cases, with a comparison to alternative modulated techniques. **Methods:** Five patients who previously received hippocampal avoidance whole brain radiotherapy were selected for this retrospective planning study. The hippocampal avoidance region (HA) was a 5mm expansion of the hippocampi. The planning target volume (PTV) was the whole brain minus the HA region. The PTV was partitioned into three volumes: superior; mid; and inferior to the HA. Raystation was used to generate plans using the split VMAT technique, VMAT coplanar arcs, VMAT non-coplanar arcs and IMRT. The optimization prioritized dose homogeneity on the superior and inferior volumes. Dose falloff adjacent to the hippocampi was prioritized in the PTV mid-volume. Dose metrics and DVH data were used to assess plan quality. VMAT plans were perturbed in orthogonal directions to assess robustness. **Results:** PTV partitioning and standardized optimization parameters were able to meet dosimetric goals. VMAT delivered a more homogenous dose to the brain than IMRT. The VMAT split field was able to provide superior hippocampal avoidance and homogeneity. This technique resulted in the most robust plans under simulated setup variations. **Conclusion:** A simple split field VMAT technique consistently produces plans with superior homogeneity and hippocampal sparing. This technique does not require couch rotation and is relatively robust, which results in a simple plan delivery to patients who are often medically compromised.

PO-GePV-T-243

Dosimetric Comparison of the New Pinnacle Evolution Versus Pinnacle Version 16.2

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Purpose: To investigate the optimization performance of two different versions, v16.2 and v16.4.2, of the Pinnacle Treatment Planning System by evaluating treatment plans and analyzing their dosimetric difference. **Methods:** For 25 patients, two VMAT plans were generated with the same beam setup and optimization goals for the targets and OARs: one with Pinnacle v16.2 and one with Pinnacle v16.4.2. The patients were planned for the following sites: brain, lung, head and neck (HN), pelvis, and prostate with 5 patients per site. The optimization time was limited to one hour for each plan. After the plans were completed, both systems were adjusted to have the same V100 value for comparison. **Results:** Pinnacle v16.2 provided a better conformity index ($CI = TV_{PIV}/TV$) with a mean value of 1.1003 when considering all treatment sites compared to v16.4.2, which had a mean of 1.1335. The homogeneity index ($HI = I_{max}/I$) was closer to 1 for v16.4.2 with a mean value of 1.1213. The gradient index ($GI = V50\%/V100\%$) was superior for lung, HN, and prostate patients for v16.4.2 plans, however when all sites were evaluated the GI was better for v16.2 plans with a mean value of 4.3677 compared to 4.4158 for v16.4.2. When analyzing individual sites, the CI was closer to 1 in four out of five sites when the plan was created using v16.2. Pinnacle v16.4.2 allowed for a greater number of optimizations within the one hour time limit, but it was noted that a higher amount of dose constraints were not met for the v16.4.2 plans, specifically for prostate, brain, and lung patients. **Conclusion:** For multiple treatment sites, VMAT treatment plans optimized within 1 hour using Pinnacle TPS v16.2 and v16.2.4 consisting of the same setup were found to be reasonably comparable. Further investigations involving more data points and evaluating other plan quality metrics are in progress.

PO-GePV-T-244

Investigating the Differences Between PlanIQ and IPA Planning Tools for Plan Optimization in Head & Neck Cancers

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Purpose: To investigate treatment plan guidance for head and neck cancers via the feasibility tool PlanIQ (Sun Nuclear) and an open-source Ideal Planning Assistant (IPA) application. **Methods:** Three patients with oropharyngeal squamous cell carcinoma were used to generate two sets of DVH guidance values for simultaneous integrated boost (SIB) treatment plans. In each case, 60Gy and 54Gy were prescribed to the high and standard risk PTVs. The PlanIQ feasibility tool generated values for the feasibility of 0.0, 0.1, and 0.5 for each OAR. To obtain similar feasibility values in the IPA application each OAR structure was pushed to the 90% value, providing a comparable 0.1 feasibility. Both tools assumed a 100% coverage of the PTVs with the prescription doses. **Results:** At the regions of maximum doses and dose cutoffs (OAR - target overlap regions) both tools showed good agreement in predicting high difficulty of feasibility.

However, lower doses for PlanIQ demonstrated the potential for greater sparing. For example, for the left and right parotids, planIQ indicates an average D50% 10.2Gy vs. 23.1Gy of IPA and 10.2Gy vs. 21.1Gy, respectively. Also, for the constrictors, PlanIQ indicates an average mean dose 26.3Gy vs. 45.2Gy of IPA. Those deviations mainly stem from the fact that PlanIQ accounts only for the target volume and OAR correlation, while PlanIQ considers the target volume in addition to dose sparing on all other defined OARs. **Conclusion:** The comparison of PlanIQ and IPA showed good agreement in the high doses but significant differences in the middle and lower dose region. The results reflect their fundamental approaches since planIQ calculates its feasibility doses without considering any tradeoffs between the OARs, whereas IPA accounts for OAR to OAR dose tradeoffs to achieve a balanced sparing between organs.

PO-GePV-T-245

Preliminary Patient Specific Dose Constraints of Rectum for Prostate Cancer Treatment

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Purpose: External beam radiation therapy has gained popularity in treating prostate tumors. This study aims to re-evaluate the rectum dose constraints from conventional prostate treatment plans. **Methods:** RTOG 0924 protocol has been used as the guideline for the treatment plan with dose prescription, rectum contour and dose constraints. There were 41 patients treated during the period from May 2016 to Feb 2018 following RTGO 0924 protocol. Assume the proctitis is related to the relative high dose area in the rectum wall during the prostate treatment. Three volumes were used for the dose constraints re-evaluation: PTV prostate, Rectum and Rectum wall (R-wall-5mm-3cm) defined as 5mm outer wall of rectum structure within 3cm from the PTV prostate. RTOG 0924 recommended the rectum dose constraints as relative volume dose constraints such as V70<15%. As individual prostate patient has different prostate volume (also different overlap length with rectum in SI direction) and different total rectum volume, patient-specific rectum dose constraints were made here and used for re-evaluation. For all 41 prostate patients, the volumes of PTV prostate, R-wall-5mm-3cm and Rectum are 146.10cc, 37.75cc and 99.41cc respectively. The ratio of R-wall-5mm-3cm and PTV prostate volume and ratio of rectum and PTV prostate volume were used for patient specific rectum dose constraints re-evaluation. **Results:** The ratio of R-wall-5mm-3cm and PTV prostate volume is 0.29±0.09. And the ratio of rectum and PTV prostate volume is 0.80±0.50. The second ratio is in a very large random range. Therefore, it's hard to predict the side effect based on the relative volume dose constraints. **Conclusion:** This study showed that, for prostate radiation therapy, 1) patient-specific dose constraints are vital for individual patients and 2) the organ at risk for the dose constraints need re-evaluation.

PO-GePV-T-246

Evaluation of An Angle Optimization Algorithm in Breast Cancer Treatment Planning

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Purpose: To evaluation a commercial automatic beam angle optimization (BAO) algorithm in breast cancer radiotherapy. **Methods:** Twenty-one patients who previously treated with radiotherapy and breast conserving surgery were enrolled in this study, including 10 patients with left breast cancer and 11 with right breast cancer. Among these patients, 11 were treated with prescription dose 48Gy, 10 were treated with prescription dose 50Gy. For each patient, two IMRT plans were developed with different set of beams angles. One set of beams angles was generated by an automatic beam angle optimization and the other was manually set by physicists. The algorithm was developed by United Image (Shanghai, China) which was based on a new Multi-stage optimization approach. The Method introduces Branch-and-Prune to generate a feasible solution in the first phase. A simplex method is applied to find a local optimal solution from the Phase I starting point in the second phase. Both two plans were generated on UIH TPS (United Imaging, Shanghai, China) and use same number of beams, prescription and optimization constraint objective. The maximum dose (Dmax), homogeneity and conformity Index (HI, CI) of PTV and mean doses of relevant organs-at-risk (OARs) were compared. **Results:** There were no significant different in PTV Dmax, HI and CI for two plans (p>0.05). In terms of OARs, BAO plans achieved less value in ipsilateral lung V5 (35.5%±5.6%) compared to manual beam angle plans (36.9%±4.3%, P=0.034). There were no significant different in ipsilateral lung V20 and heart mean dose (p>0.05). No significant different were observed for all contralateral lung and contralateral breast dose volume indices. **Conclusion:** The UIH automatic beam angle optimization algorithm can achieve comparable quality plans compared to manual beam angles plans.

PO-GePV-T-247

Planning Strategies for Whole Head-And-Neck Radiation with Robustness Against Shoulder Motion Using Volumetric-Modulated Arc Therapy

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Purpose: To investigate planning strategies for head-and-neck radiation using volumetric-modulated arc therapy with robustness against interfractional shoulder motion and to evaluate each approach for reduction of shoulder dose and planning efficiency. **Methods:** Ten patients were randomly selected for planning in RayStation v10A. Clinical plans served as base plans and had no mechanism for robustness against interfractional shoulder motion. Plan variations included the following: 'shoulder-protect': using a feature in the TPS that closes the MLC leaves over a selected structure to avoid entrance dose, and 'shoulder-PRV': adding shoulder-avoid structures to the optimizer. Two patients were also planned using 'skip-arcs': partial arcs which skip the segments around the shoulders, 'shifted-iso': shifting the isocenter and utilizing beam divergence to avoid beams passing through the shoulders, and 'couch-kicks': using couch rotations to avoid the shoulders. Plans were evaluated for target coverage, shoulder sparing, OAR doses and planning efficiency. **Results:** V95 target coverage for all variations was within 2% of the base plan. All variations achieved lower mean shoulder dose relative to the base plan: 73%(shoulder-protect), 71%(shoulder-PRV), 80%(skip-arcs), 94%(shifted-iso), 59%(couch-kicks); however, the dose to 1% of the shoulder structures (D1 shoulder) increased relative to the base plan: 113%(shoulder-protect), 107%(shoulder-PRV), 109%(skip-arcs), 103%(shifted-iso), 104%(couch-kicks). The complexity metric (ratio of MU's over prescription dose) remained similar, except for an increase for the 'shoulder-PRV' approach(109%) and a reduction with the 'couch-kick' approach(88%). Dose homogeneity increased for the 'shoulder-protect'(108%) and 'shoulder-PRV'(106%) variations and decreased for the 'skip-arcs'(90%), 'shifted-iso'(94%), and 'couch-kick'(99%) variations. The 'shoulder-PRV' variation required the least amount of manual adjustments and planning time. **Conclusion:** Adding shoulder-avoid structures directly to the optimizer was found to be the best compromise between increased plan robustness against interfractional shoulder motion and increased planning complexity. More cases will be studied in the future to confirm the statistical significance of the findings.

PO-GePV-T-248

Dosimetric Study for Spine SRT Plans with Carbon Fiber Stabilization Devices

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Purpose: Presence of metal stabilization devices causes several issues in radiotherapy including imaging artifacts, possible dosimetric discrepancy and beam energy limitation. Carbon-Fiber Reinforced Polyetherthetketone (CFR-PEEK) material has been a potential alternative fixation with its radiolucency, and strength comparable to commonly used titanium devices. Previous studies have only focused on the dosimetric comparison for the screws themselves which are only part of the fixations. In this study, we present the dosimetric comparisons between the plans and the measurements on radiochromic films using a bone phantom with the entire CFR-PEEK fixations. **Methods:** CT scans were obtained with a specially designed bone phantom with CFR-PEEK fixations under

water phantoms representing neck and abdomen. The bone phantom was designed to resemble the shape and the HU values of vertebrae and spinal cord inside. A bone density phantom was scanned with the bone phantom for calibration. The bone phantom consists of two pieces, between which a film was placed for dosimetry measurement. Three stereotactic radiation plans were generated with mock targets near the fixations (one for neck and two for abdomen) and delivered to the bone phantom and the fixations. Coronal dose profiles were acquired with radiochromic films to evaluate the dose differences. **Results:** CFR-PEEK fixations did not determine any HU artifacts on CT images resulting in better image quality and easier delineation. The coronal profiles compared between the plans and measurements showed a reasonable agreement. **Conclusion:** This work evaluated dosimetric profiles on plans and measurements on CFR-PEEK fixations with a bone phantom. It demonstrated that the fixations provide better image quality and reduced dose uncertainty in target near the fixations. Further comparison with metal fixations is warranted.

PO-GePV-T-249

The Feasibility Study of Using Light Detection and Ranging (LiDAR) Application for Total Body Irradiation (TBI)

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Purpose: This study aims to evaluate the feasibility of obtaining TBI patient specific measurements from available technology, specifically, LiDAR, on Apple iOS hardware and software application. The off-axis distance, reference plane distance, and patient field separations were measured with different devices for various anatomical sites according to our institutions TBI policy to determine monitor units (MU) and compare the differences. **Methods:** A direct comparison between standard measurement devices, caliper or measuring tape, and the Apple iPhone® iOS application was performed. To test the capability of LiDAR application and the effect of room light on application's performance, a 30cm and 15.2cm lengths were evaluated using a measuring tape and a ruler in horizontal and vertical setups. Vertical measurements were used to determine the distance limit in one-meter increments. To remove inherent motion of the device being hand-held, and the effect of room light on the software's capability, a 3D printed stand, a linear motion slide (Accuride International, Santa Fe Springs, CA), and a digital lux meter (Dr. Meter, Union City, CA) were utilized in house. **Results:** Considering the LiDAR application limitations (+/-1 cm), positioning, and motion effect, LiDAR measurements showed reasonable agreement with standard measuring tools. (+/-1 cm). Although stationary measurement was accurate with +/-1 cm, the LiDAR application was unable to report sub-centimeter and unable to define a starting point for the 15.2cm ruler at 4 meters. By calculating MU using the off-axis distance, reference plane distance, and patient field separations measured from the volunteers, agreement within 5% difference were found (2133 MU using standard tape measure vs. 2245 MU using LiDAR application). However, there are always random uncertainties with measuring and positioning volunteers/patients. **Conclusion:** The feasibility of the iOS application accuracy as compared to the standard tape measure has been demonstrated. Comparable measurements were obtained but great care must be taken when utilizing LiDAR.

PO-GePV-T-250

Extending the Body Outline Out into the Air to Improve the Superficial Dose Evaluation Accuracy for Breast Cancer Intensity Modulated Radiotherapy

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Purpose: The superficial dose for breast cancer patients are usually underestimated by the irradiation treatment planning system (TPS) due to ignore the dose contribution which come from out of the body outline. This research aims to explore a reasonable body contouring method to improve the superficial dose evaluation accuracy for breast cancer intensity modulated radiotherapy (IMRT). **Methods:** Twelve modified radical mastectomy cases were involved. Automatic contour the outline of body with threshold value -350HU, then add 1cm margin along the planning target volume (PTV) anterior edge. 7 fields IMRT plans were designed to meet the clinical requirement. Duplicate the plan and recontouring the body with the threshold value of -350HU, then keep the fields and fluence as the same as the previous one and recalculate the new plan. Analyze the dose coverage difference of the PTV and OARs between the two groups. For one typical modified radical mastectomy case, the automatic contouring threshold value was set as -350HU, -650HU, -650HU plus 1cm margin, respectively. The superficial dose change with different distance by incident and exiting directions was analyzed. Plans were designed by Varian Eclipse TPS version 13.5. **Results:** The dose difference under the skin is much smaller than in the air. The trend is more obvious in exiting direction than in incident direction. The 1cm margin group have a much better target dose coverage than -350HU group. The PTV dose parameter V95% (99.2%±1.14 vs 96.8%±3.26), V100% (94.8%±3.53 vs. 88.7%±5.87), D98% (46.7±0.84 vs 39.6±6.54, Gy), CI (0.78±0.06 vs 0.68±0.03), HI (0.12±0.02 vs. 0.26±0.13), all P<0.05. All the OARs irradiation dose has no difference. **Conclusion:** Add 1cm margin to the body outline could represent the real irradiation environment in TPS, as thus to improve the superficial dose evaluation accuracy for breast IMRT. Especially for the case which PTV was contoured align the body surface.

PO-GePV-T-251

Conformal Radiotherapy with Sequential Boost Versus Intensity-Modulated Radiation Therapy with a Simultaneously Integrated Boost

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Purpose: The purpose of this study is to compare the dosimetric and radiobiological differences between conformal radiotherapy (CRT) to the whole breast with sequential boost to the primary tumor and intensity-modulated radiation therapy (IMRT) to the whole breast with simultaneously integrated boost to the primary tumor. **Methods:** This study included 5 primary left-sided breast cancer patients. In total 5 CRT plans and 5 IMRT plans were created. In the CRT plans the whole breast receives a dose of 50Gy in 25 fractions with a sequential boost to the primary tumour of 10Gy in 5 fractions. In the IMRT plans the whole breast receives 50Gy and the primary tumor 60Gy in 25 fractions. Each plan was evaluated in terms of PTV coverage, primary tumour conformity index, maximum plan dose, ipsilateral lung dose (mean, V20), heart dose (mean, V5), contralateral breast doses and spinal cord dose. The doses of the respective plans were compared between each other and against the clinical goals. Furthermore, the respective tumor control and normal tissues complication probabilities (TCP and NTCP) were calculated. **Results:** IMRT created more conformal plans in all cases, (conformity index: average CRT = 0.33, IMRT = 0.76). CRT plans delivered lower mean heart doses in all cases (average CRT = 2.2Gy, IMRT = 2.8Gy). CRT plans delivered higher mean ipsilateral lung doses in 4 of 5 cases (average CRT = 6.8Gy, IMRT = 6.3Gy) and V20Gy in 4 of 5 cases (average CRT = 11.0%, IMRT = 10.1%). The NTCP values of ipsilateral lung and heart were lower for IMRT than CRT (0.02% and 0.0% vs. 0.04% and 0.23%, respectively). **Conclusion:** IMRT produced plans of superior dose conformity, lower NTCP values and shorter treatment duration at the expense of slightly higher maximum doses to contralateral breast and spinal cord.

PO-GePV-T-252

Dosimetric Impact of Multi-Leaf Collimator Positional Errors on Standard-Fractionated and Hypofractionated Radiotherapy Using Flattened and Flattening Filter-Free Beams

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Purpose: This study investigated the dosimetric impact of multi-leaf collimator (MLC) positional errors and determined the desired MLC positional accuracy for flattened (FF) and flattening filter-free (FFF) beams in standard-fractionated and hypofractionated radiotherapy. **Methods:** Eighty patients (20 each for treatment site) with prostate cancer, lung cancer, brain metastases, or spinal metastases were evaluated. Standard-fractionated and hypofractionated (2 and 7.25, 2 and 13, 4.2 and 10, and 3 and 16 Gy per fraction for prostate, lung, brain metastases and spinal metastases, respectively) volumetric-modulated arc therapy (VMAT) plans were generated using FF and FFF beams. Systematic open/closed MLC positional errors (± 0.25 , 0.5 , 1.0 , and 2.0 mm in opposite directions) were introduced in the VMAT plans. The dosimetric impact for the targets and organs-at-risk (OARs) were evaluated based on the generalized equivalent uniform dose (gEUD). **Results:** The gEUD deviations with MLC errors of the target volumes were strongly dependent on treatment site (2.46 ± 0.38 , 4.47 ± 0.88 , 1.74 ± 0.46 and 6.61 ± 1.25 %/mm for prostate, lung, brain, and spine metastases, respectively). The beam modality and dose per fraction had less impact on the gEUD of the target volumes. Among OARs, the gEUD deviations with MLC errors were strongly dependent on the dose per fraction (average 1.28 ± 0.29 and 1.29 ± 0.30 %/mm in prostate and spinal metastases) except for lung and brain metastases (average 0.06 ± 0.03 , and 0.14 ± 0.17 Gy/mm, respectively). Maintaining the target to within 2% corresponded to within 0.63, 0.42, 0.81, and 0.27 mm for prostate, lung, spinal metastases, and brain metastases, respectively. **Conclusion:** Since the dose errors due to MLC positional errors for VMAT may depend on the treatment sites, beam modalities (FF and FFF beams), and doses per fraction, these should be considered for quality assurance.

PO-GePV-T-253

Treatment of Large-Size Prostate Cancer Patients On An O-Ring Linac with 6MV FFF: Plan Quality Comparison with a C-Arm Linac Using 10MV Or 15MV Photons

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Purpose: To evaluate the dosimetric performance of the Halcyon Linac equipped with 6MV FFF photon beam in planning for large-size prostate cancer patients, compared to TrueBeam with Millennium MLC using 10MV or 15MV photons. **Methods:** Eight large-size prostate cancer patients ($BMI = 27.7 \pm 2.2$ kg/m² and pelvis lateral separation = 36.3 ± 1.1 cm) were retrospectively included in the treatment planning study. The dose prescription (79.2Gy to prostate and proximal seminal vesicles in 44 fractions) and clinical planning objectives were defined as in RTOG 0126. Halcyon and TrueBeam plans were optimized using 3 full arcs. For each patient, three plans were developed: Halcyon plan with 6MV FFF (Hal), TrueBeam plan with 10MV (TB-10X), and TrueBeam plan with 15MV (TB-15X). The plans were normalized so that prescription isodose surface covers 98% of the PTV volume. **Results:** The dosimetric parameters of all plans are meeting RTOG 0126 objectives. The Dmax of all plans are below 107% of prescription dose, and no difference of Dmax was observed among Hal, TB-10X and TB-15X. The V75Gy, V70Gy, V65Gy, and V60Gy of rectum on Hal are: $11.5\% \pm 3.6\%$, $17.8\% \pm 5.5\%$, $23.9\% \pm 7.0\%$, and $30.8\% \pm 8.4\%$, respectively; on TB-10X are: $11.1\% \pm 3.7\%$, $17.1\% \pm 5.6\%$, $23.1\% \pm 7.3\%$, and $29.9\% \pm 8.9\%$, respectively; on TB-15X are: $11.2\% \pm 3.7\%$, $17.3\% \pm 5.6\%$, $23.5\% \pm 7.1\%$, and $30.6\% \pm 8.6\%$, respectively. The V80Gy, V75Gy, V70Gy, V65Gy of bladder on Hal are: $4.3\% \pm 3.6\%$, $8.4\% \pm 6.7\%$, $11.5\% \pm 8.9\%$, and $14.6\% \pm 11.0\%$, respectively; on TB-10X are: $4.2\% \pm 3.6\%$, $8.2\% \pm 6.7\%$, $11.2\% \pm 9.0\%$, and $14.2\% \pm 11.1\%$, respectively; and on TB-15X are: $4.2\% \pm 3.5\%$, $8.1\% \pm 6.8\%$, $11.2\% \pm 9.1\%$, and $14.2\% \pm 11.3\%$, respectively. The mean penile bulb doses are: $19.9\text{Gy} \pm 8.5\text{Gy}$, $16.2\text{Gy} \pm 6.7\text{Gy}$, and $16.3\text{Gy} \pm 7.0\text{Gy}$, respectively for Hal, TB-10X, and TB-15X. **Conclusion:** This retrospective treatment planning study, based on the dose constraints derived from RTOG 0126 protocol, suggested that the essential equivalence between Halcyon plan using 6MV FFF and TrueBeam plans using 10MV/15MV for the arc treatment of large-size prostate cancer patients.

PO-GePV-T-254

The Future Is Here: An Inter-Comparative Study of AI and Human-Generated Plans Using Dosimetric Indices as a Guide to Plan Efficacy

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Purpose: The purpose of this study is to assess the efficacy of Artificial Intelligence (AI)-generated treatment plans using Varian Ethos system. **Methods:** Six large field pelvic/abdomen sites, treated with Varian Ethos v.2.0, were chosen for the present study. Ethos created 2 VMAT and 3 IMRT AI-driven plans for these cases, based on priority goals in the RT intent and the user chose the most optimal plan. For all cases, IMRT plans were dosimetrically superior to VMAT plans and were selected for treatment. Typically, these cases would have been treated using VMAT in Eclipse. Ethos VMAT plans were exported to Eclipse v.16.1 and optimized by an experienced dosimetrist, using the same treatment geometry. All plans were calculated for dose using AcurosXB and were normalized, such that, 95% of the PTV received 100% of the prescription dose. **Results:** Global maximum doses for Ethos vs. Eclipse plans were 114.6% vs. 114.1% , respectively. PTV indices (D98/D50/D2) for Ethos vs. Eclipse VMAT plans were 98.3% vs. 98.4% , 104.2% vs. 105.0% , and 108.3% vs. 109.4% , respectively. Conformity and Homogeneity indices for Ethos and Eclipse plans were 1.06 vs. 1.08 , and 0.10 vs. 0.10 , respectively. On average, total plan monitor units were 985.7 and 946.2 for Ethos and Eclipse plans, respectively. Based on RTOG criterion, specific to the organ in question, both sets of plans were deemed to be clinically acceptable. **Conclusion:** Varian AI-driven Ethos system creates highly-modulated IMRT plans that are comparable or superior to conventional IMRT plans generated by dosimetrists. VMAT plans from Ethos are not as good as their IMRT plans but comparable to typical VMAT plans created in Eclipse. The inferiority of Ethos VMAT plans seems to be systemic. It appears AI VMAT plans are not running sufficient iterations due to time constraints and, therefore, plans are being trapped in local minima.

PO-GePV-T-255

Are Auto-Generated Organ-At-Risk Contours Good Enough in Prostate Radiotherapy?

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Purpose: Precision manual contouring is a time- and labor-intensive process for radiotherapy that is subject to inter- and intra-observer variabilities. Furthermore, contours drawn prior to treatment do not necessarily represent the real patient anatomy over an extended treatment course. This study investigates the impact of auto-generated organ-at-risk (OAR) contours on treatment outcome. **Methods:** Prostate, seminal vesicles (SV), bladder, and rectum were manually delineated on planning CT and daily CBCT images for 20 prostate patients. OARs were automatically generated with atlas-based segmentation algorithms on planning CTs. Treatment plans were created via a novel automated planning application accessing a knowledge-based planning solution for both auto-generated OARs (aOAR-plan) and manual OARs (mOAR-plan). Both low-risk patients (LRP, CTV=Prostate) and intermediate-risk patients (IRP, CTV=Prostate+SV) were studied. Calculated doses were transferred from planning CT to daily CBCTs based on clinical shifts, then contour-based deformable registrations were used to obtain the cumulative doses. Differences in cumulative dose for the bladder and rectum were used as an indicator of the agreement between auto-plans and manual-plans. Furthermore, 5 patients were selected to test the difference between obtaining cumulative dose from transferred doses versus doses recalculated on CBCT. Maximum dose, V55, V60, V65, V70 and V75 were used for the dosimetric evaluation. **Results:** For both LRP and IRP groups, good dosimetric agreement between the aOAR-plans and mOAR-plans was observed. The dose differences were within 2.5% for bladder and within 4% for rectum, respectively. No significant dose differences between the transferred and recalculated doses were observed for both the bladder and rectum. **Conclusion:** Manually drawn precision contours are used to represent patient anatomy that often fluctuate by multiple millimeters on a daily basis, thereby making the time and effort questionable. Automated contouring tools offer improvement in contour consistency, and provide acceptable delivered doses compared with precision contours.

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PO-GePV-T-256

Dosimetric Effects Due to Inter-Observer Variability of Organ Contouring When Utilizing a Knowledge-Based Planning System for Prostate Cancer
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Purpose: Radiotherapy is a widely accepted standard of care for early stage prostate cancer, and it is believed that the plan quality and treatment outcome are associated with contour accuracy of both the target and organs-at-risk (OAR). The purposes of this study are to: (1) assess geometric and dosimetric uncertainties due to inter-observer contour variabilities and (2) evaluate the effectiveness of geometric indicators to predict target dosimetry in prostate radiotherapy. **Methods:** Twenty prostate patients were selected for this retrospective study. Five experienced clinicians created unique structure sets containing prostate, seminal vesicles, bladder, and rectum for each patient. A fully automated script and knowledge-based planning routine was utilized to create standardized and unbiased plans that could be used to evaluate changes in isodose distributions due to inter-observer variability in structure segmentation. Plans were created on a “gold-standard” structure set, as well as on each of the user defined structure sets. The correlation between geometric and dosimetric indices was evaluated using R2 statistics for plans generated with different combination of structure sets. **Results:** Inter-observer variability of contours during structure segmentation was very low for clearly defined organs such as the bladder but increased for organs without well-defined borders (prostate, seminal vesicles and rectum). For plans generated with the user defined structure sets, strong / moderate correlations (correlation coefficient $0.4 < |r| < 1$) were observed between the geometric indicators for target structure agreement and target coverage for both low-risk and intermediate risk patient groups, while OAR indicators showed no correlation ($|r| < 0.2$) to final dosimetry. **Conclusion:** Level of target delineation accuracy is crucial in order to maintain adequate dosimetric coverage regardless of the associated inter-observer uncertainties in OAR contours that had limited impact upon final dosimetry.

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PO-GePV-T-257

A Single Institution Evaluation of a Machine Learning Based Decision Support Software for Head and Neck Radiotherapy

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Purpose: To evaluate the performance of a machine learning (ML) based decision support software (InsightRT, Siris Medical, CA) for head and neck (H&N) radiation therapy. **Methods:** Twenty-two sets of H&N patients' data were retrospectively evaluated in this study. All patients were treated with volumetric modulated arc therapy and planned using the Monaco treatment planning system (Elekta Inc.). Three simultaneous integrated boost (SIB) treatment schemes were prescribed including PTV60/54-59.3/53.9 (11 patients), PTV69.96/59.4-63/54-59.3/53.9 (5 patients) and PTV66/59.4-63/54-59.3/53.9 (6 patients). The H&N prediction model of InsightRT was built upon a total of 813 patients with the aforementioned three Rx schemes. D95 for all PTVs (PTV69.96, PTV66, PTV60, PTV<59.3) and organ at risk (OAR) constraints including cord max (CM), mandible max (MM), left parotid mean (LPM) and right parotid mean (RPM) were compared between planned and predicted data through a paired t-test. Software running time was also evaluated to assess the efficiency of the clinical workflow. **Results:** Statistically, there was no significant difference between planned value and predicted value for D95 of PTVs (pPTV69.96=0.45, pPTV66=0.3, pPTV60=0.07, pPTV<59.3=0.38) and OAR constraints (pCM=0.49, pMM=0.09, pLPM=0.46, pRPM=0.28). Prediction could be performed in less than 5 seconds for each individual case, and structures could be automatically associated provided the structure labeling followed pre-defined rules, thus demonstrating a robust algorithm. In addition, matched patient data used for prediction were provided to assist dosimetric tradeoff decisions during the planning process. **Conclusion:** InsightRT could provide plan result prediction in good agreement with clinically approved plans for H&N radiotherapy in a rapidly produced and automated fashion. Predicted plan results and matched patient data could assist with dosimetric tradeoff decisions during the planning process and help prevent unnecessary plan modifications or replanning cycles.

PO-GePV-T-258

An Unsupervised Method for Error Detection of Treatment Plan in Radiotherapy

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Purpose: An unsupervised machine learning method, autoencoder (AE), was applied to error detection of treatment plans in radiotherapy. **Methods:** Treatment plans of more than 500 breast cancer patients between 2010 and 2020 were collected from MOSAIQ system and analyzed retrospectively. Those plans were hybrid intensity-modulated radiotherapy (IMRT) plans with two tangential conformal fields and two tangential intensity-modulated fields. The most important 9 parameters were extracted from plan for each patient. A classical autoencoder (AE) method was used to encode the original features to fewer features and then reconstruct the original features from the encoded features. The average mean-square error (MSE) between the reconstructed features and the original features for all training data was calculated and set as baseline for error detection. For a new patient, the 9 features were extracted from treatment plan and fed into trained AE model. The reconstruction MSE was then calculated by AE and compared to the baseline value for the difference. If the difference is more than a threshold value, this treatment plan is considered as an outlier and need further checking. In addition, errors in various levels were purposely introduced to each parameter in testing dataset. The resulting error detection rates were used to evaluate the sensitivity of AE model. **Results:** AE model detected all errors with 100% detection rate in testing dataset. For sensitivity analysis, 1 parameter was not affected as its value always fixed to two constants. 4 parameters were less sensitive to errors and another 4 parameters were more sensitive to errors. For all tested parameters, the false positive rate is about 0.01%. **Conclusion:** AE method provides a novel way for automatic treatment plans checking, which is more efficient and robust.

PO-GePV-T-259

The Effect of Dimensionality-Reduction Algorithms on Parameter Checking of Treatment Plan in Radiotherapy

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Purpose: Dimensionality reduction is important for clustering method in detecting error of radiotherapy treatment plan. The traditional method is principal component analysis (PCA) which is linear representation of high-dimensional features and might be failed for data with non-linear relationship. In this study, the non-linear components contained in high-dimensional feature space were extracted by an autoencoder (AE) method for more effective clustering capability. **Methods:** Treatment plans of more than 500 breast cancer patients between 2010 and 2020 were analyzed retrospectively. Those plans were hybrid intensity-modulated radiotherapy (IMRT) plans with two tangential conformal fields and two tangential intensity-modulated fields. The most important nine parameters were extracted from plan for each patient. First, the input feature vectors in nine dimensions were reduced to three components by PCA and AE methods, respectively. Then the feature vectors in three dimensions were assigned to different groups using K-means clustering algorithm. Next, the outliers were detected by a distance threshold value. Finally, the effects of two dimensionality-reduction methods on the clustering performance were evaluated by silhouette coefficient index. **Results:** The performance of error detection by AE-based method is better than that by PCA-based method according to the values of precision rate and recall rate. The performance of clustering resulted by AE-based method is higher than that of PCA-based method according to the silhouette coefficient index. **Conclusion:** AE-based feature vectors show higher clustering capability than those of PCA-based feature vectors. It provides an effective

dimensionality-reduction method for error detection of treatment plan in radiotherapy.

PO-GePV-T-260

Patient Treatment Transferability Between Unmatched Medical Linacs

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Purpose: Linacs with unmatched beam characteristics create operational challenges. The feasibility of transferring patients between unmatched Linacs was investigated for a limited number of treatment fractions. **Methods:** Twenty 4-field 3DCRT and 20 VMAT prostate plans were evaluated on unmatched Elekta Infinity and Versa Linacs using Pinnacle TPS. With MUs held constant, the mean dose to four OARs (rectum, right femur, left femur, and bladder), PTV, and prescription point were evaluated. Differences in mean dose were assessed with paired t-tests ($p < 0.05$). Relative differences in mean dose were calculated. The number of interchangeable fractions allowable before surpassing a 5% difference in dose to the prescription point was determined as recommended by ICRU-50. VMAT plans were delivered on both Linacs for QA with ArcCheck. Measured dose distributions were compared with distributions calculated on the Infinity. Gamma analysis passing rates were compared between machines using 2.0%/2.0mm criteria. **Results:** The mean dose to the PTV and prescription point was greater for all 3DCRT plans calculated on the Versa and reduced for 17 and 16 out of 20 VMAT plans, respectively. The averaged mean dose difference for the rectum, right femur, left femur, bladder, and PTV were 0.9%, 1.3%, 5.5%, 0.6%, and 0.9% among 3DCRT plans, and 2.9%, 0.9%, 1.4%, 2.8%, and 0.6% among VMAT plans, respectively. Mean dose significantly differed between Linacs for all OARs, PTV, and prescription point. The relative difference to the prescription point ranged from 0.75%-1.35% (mean: 0.96%) among 3DCRT plans and 0.06%-1.40% (mean: 0.43%) among VMAT plans resulting in an average of 5 (range: 4-7) and 11 (range: 4-90) allowable interchangeable fractions between Linacs, respectively. Among VMAT plans, gamma passing rates differed by 0.0%-3.0% (mean: 0.18%). **Conclusion:** Despite small differences in calculated dose, transferring patients between unmatched machines for a limited number of fractions (<5) is possible to improve clinical flexibility.

PO-GePV-T-261

Development of a Progressive Plan Quality Evaluation Solution

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Purpose: Plan quality assessments have become a ubiquitous component of plan evaluation in radiotherapy. The typical methodology assesses dose-volume histogram (DVH) constraints and applies a binary pass/fail evaluation. In this work, a framework was developed to continuously accumulate plan quality data and provide additional context during plan assessment by comparing the current plan results to the past patient population. **Methods:** At our institution, a third party commercial software is utilized to maintain a database of plan quality constraint templates, categorized by treatment site, technique, prescription dose and fractionation. This software does not have the capability of storing plan evaluations in their database, but is capable of exporting the individual patient reports in JavaScript Object Notation (JSON) format. A reactive web application was developed to parse this data, calculate statistics, compile a patient database, present the results for comparison and evaluation, and print plan quality reports for inclusion in the patient chart. **Results:** Initial development focused on external beam radiotherapy of the breast, which has well established plan constraints and a large population of treated patients. The data for 111 patients have been collected so far. Plan quality reports can now display the current plan performance overlaid on a boxplot of the past patient population. Each subsequent patient can then be added to the population database. **Conclusion:** A framework for continuous accumulation of plan quality data and feedback to the planner has been implemented. Further study is needed, but this framework is expected to lead to improvements in overall plan quality for two reasons: (1) evaluation that looks beyond pass/fail DVH constraints with the context of past patients to see where a plan may improve; and (2) the objectives currently in use can be evaluated against what has been historically achievable and adjusted accordingly to improve their efficacy.

PO-GePV-T-262

A Retrospective Study to Establish Recommendations for Plan Quality Metrics in Linac Lung SBRT

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Purpose: The quality of planned dose distributions in lung SBRT was evaluated by characterizing dose distributions for conformity, homogeneity, and gradient parameters using an in-house script. The goal of this study was to establish size-dependent recommendation for achievable values of these plan quality metrics. **Methods:** Sixty-nine patients (73 lesions) treated with lung SBRT during the period 2017- 2020 were retrospectively reviewed for this study. Included patients had been treated on the Varian Edge and have a prescription of 50-60 Gy in 5 fractions. All cases were classified based on the lesion size into four groups, very small ($V < 5cc$), small ($5 < V < 10cc$), medium ($10 < V < 30cc$), and large ($V > 30cc$). Plans were generated in RayStation treatment planning system using the collapsed cone algorithm with a computation grid size of 2 mm. Plans were retrospectively evaluated using dosimetric indices from respective RTOG and ICRU protocols. **Results:** Our recommendations for the acceptable values for the plan quality metrics, HI, GI, CI and PCI is shown in table 1 of the supporting documents. These are based on clinically approved plans, after removing statistical outliers that we would not have approved if the metrics had been calculated. Additionally, we observed that a sharper dose fall-off and a more homogeneous plan were found using 6FFF compared to 10FFF energy. **Conclusion:** Using various dosimetric indices to characterize dose distributions in lung SBRT is a powerful tool to assess plan quality. The plan recommendations are now being used in our clinics. We recommend that these values be calculated for all plans, utilizing a script or program so as to improve clinical workflow. Our retrospective analysis identified 9 out of 73 plans that would have been rejected. Also, we recommend using the 6FFF beam for lung SBRT treatment planning based on our institutional data evaluation.

PO-GePV-T-263

Evaluation of Treatment Plan Uncertainties for VMAT TBI

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Purpose: To investigate the effect of patient positioning in Volumetric Modulated Arc Therapy (VMAT) for Total Body Irradiation (TBI) given the use of multiple isocenters, by simulating offsets in patient positioning and evaluating changes to planned dose distributions. **Methods:** VMAT treatment plans for seven TBI patients treated as part of a prospective stage II clinical trial were evaluated. Plan uncertainties were calculated by introducing 5mm and 10mm translational shifts to the plans' isocenters in the lateral (x), vertical (y), and longitudinal (z) directions. Dose distributions were then re-calculated in the treatment planning system (Eclipse), in order to evaluate dosimetric robustness to one global imaging shift at treatment. Differences in target volume (PTV) coverage and doses to organs at risk were evaluated based on four parameters: lung mean dose, PTV-V100%, PTV-D98%, and kidney mean doses. **Results:** Lung mean dose increased an average of 8.2cGy, 4.4cGy, and 3.3cGy when shifted 5mm in the x,y,z directions (respectively) across seven patients; 33.2cGy, 18.5cGy, 18.3cGy for 10mm shifts in x,y,z. Target coverage V100% decreased an average of 0.3%, 0.03%, 0.1% for 5mm shifts, and 1.1%, 0.8%, 0.4% for 10mm shifts in x,y,z. D98% decreased 0.9%, 0.3%, 0.3% when shifted 5mm; 3.5%, 2.1%, 1.0% when shifted 10mm in x,y,z. Mean dose to the left kidney increased 6.6cGy, 9.7cGy, 2.8cGy for 5mm, and 28.1cGy, 32.7cGy, 18.0cGy for 10mm shifts in x,y,z. Right kidney mean dose increased 11.9cGy, 8.9cGy, 3.1cGy for 5mm, and 36.5, 30.5, 19.8cGy for 10mm. **Conclusion:** Though small in relation to total dose, the largest increase in mean lung dose and decrease in coverage was seen with lateral shifts as compared to vertical or longitudinal shifts. These results support the use of an approach with preferential alignment to the chest region (lung-sparing), as long as residual imaging alignment outside the chest is kept below 10mm.

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PO-GePV-T-264

Three-Year Experience with Active Breathing Coordinator Device Used in Reducing Radiation Dose to the Heart in Left-Sided Whole Breast Cancer Patients

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Purpose: Radiation dose to the heart during breast radiotherapy (RT) has been associated with increased risk of ischemic heart diseases. In our clinic about three years ago, we started deep inspiration breath hold (DIBH) technique using Active Breathing Coordinator (ABC) device for reducing mean heart dose (MHD) in suitable left-sided breast cancer (LBC) patients. **Methods:** Out of 120 LBC patients treated with 3D/field-in-field plans during the last three years, 80 were found suitable candidates for using ABC. These patients underwent training with the ABC device (Elekta Oncology, Stockholm, Sweden), during which lung capacity, optimal breath hold duration and level, patient comfort and compliance were assessed. Out of these 80 patients, 12 patients declined using ABC; 33 patients did not require DIBH/ABC use due to a negligible volume of heart within the treatment fields as determined via a free-breathing (FB) CT scan and remaining 35 patients underwent CT simulation with both FB and ABC for comparison of dosimetry. The median prescription dose was 42.56Gy plus 10Gy boost. All 35 patients were treated using ABC on Elekta Infinity via automated response gating interface. **Results:** Out of 80 eligible patients, 35 patients received RT using ABC, corresponding to 44% procedural success. Use of ABC in LBC patient's RT course reduced MHD by 20% or greater when compared with MHD from FB based planning. The median values for absolute and relative reduction in MHD were 1.4 Gy and 65%, respectively. **Conclusion:** ABC device during RT should be considered to reduce MHD and hence the risk of ischemic heart disease in left-sided breast patients. However, high proportion of the LBC patients were ineligible due to poor tolerability of the device. DIBH with the ABC device remains preferred approach for cardiac sparing. Data for additional patients treated with ABC device will be added to this analysis

PO-GePV-T-265

Investigating the Level of Dose Trade-Offs Between Left Lung and Heart in Lung Cancer Cases to Identify Optimal Sparing

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Purpose: To identify the current state of dose trade-offs between ipsilateral lung and heart as a function of PTV size and target distance from heart. This study will investigate whether one organ should be systematically spared over the other, with the potential to investigate different levels of dose trade-offs between ipsilateral lung and heart. **Methods:** The dosimetric data of 118 consecutive patients, who were treated for lung cancer with IMRT/VMAT or SBRT were retrospectively collected. The dose volume histograms of heart and left lung were obtained from each patient and used to calculate the corresponding normal tissue control probability (NTCP) values. The Relative Seriality model was used to calculate NTCP values. The mean lung dose (MLD), mean heart dose (MHD), lung V20, and heart V5 were also obtained from each patient treatment plan. The minimum distance from target to heart was measured directly on the planning CT. **Results:** The median PTV volume was 93cc and the mean minimum distance between the target and heart was 3.5cm, respectively. The patients with PTV > 93cc had an average lung NTCP value of 11.6%, which is much larger than the average heart NTCP of 3.0% (p=0.001). For PTV < 93cc the corresponding NTCP values are almost equivalent (0.2% vs 0.1%, respectively). Regarding the target to heart distance for both cut-offs with the average lung and heart NTCP values being 9.6% vs 2.5%, respectively for the 3cm cut-off and 8.2% vs 2.1%, respectively for the 5cm cut-off. **Conclusion:** Our results show that there is an unbalanced level of sparing between left lung and heart, which varies based on the minimum target to heart distance and PTV size. Using NTCP metrics for heart and lung would help in finding better levels of sparing.

PO-GePV-T-266

Automated and Clinical-Criteria-Driven IMRT Planning for Prostate Using Expedited Constrained Hierarchical Optimization (ECHO)

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Purpose: To report our initial clinical experience with automated IMRT planning for prostate radiotherapy plans using ECHO. ECHO is a home-grown independent optimization engine based on hierarchical constrained optimization, integrated with Eclipse® system through application program interface. ECHO does not use Eclipse optimization engine and the optimal fluence map generated by ECHO is imported into Eclipse for leaf sequencing and final dose calculation. **Methods:** ECHO was released to clinical use at our institution starting from Dec.2019 for prostate post-brachy SBRT (25Gy in 5fractions) and moderate hypofractionation prostate radiotherapy (MODHYPO) (70.2Gy in 26fractions) without nodal involvement. Prior to clinical release, training was performed on randomly selected 10 SBRT and 8 MODHYPO cases to fine-tune the ECHO parameters, and validated retrospectively on 50 cases (21 SBRT, 29 MODHYPO). We performed a comparison study with the manually generated clinical plans for all dose metrics used for prostate SBRT and MODHYPO patients in our clinic. Over a 14 months period of ECHO clinical implementation, ECHO has generated 152 clinical plans (130 SBRT and 22 MODHYPO) using 9 coplanar beams IMRT. **Results:** In pre-clinical validation, ECHO plans met all the clinical criteria and had high quality with less variability compared with manually generated clinical plans. After clinical implementation, the average time to produce an ECHO plan was 28 minutes (range 16-39). All ECHO plans were produced by one single run. They all met or were better than the institutional clinical criteria. All ECHO plans were delivered after passing institutional quality assurance process. **Conclusion:** ECHO required only few patients for training and produced high-quality plan within short time consistently for post-brachytherapy and moderate hypofractionation prostate radiotherapy. It streamlined the clinical planning workflow, improved the plan quality on average and consistency, and shortened planning time by one day from simulation to treatment in our clinic.

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PO-GePV-T-267

Auxiliary Structures-Assisted Plan Optimization for Advanced Left Breast Cancer Radiotherapy

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Purpose: To evaluate the IMRT and VMAT treatment plan quality of auxiliary structures (ASs)-assisted plan optimization in advanced left breast cancer radiotherapy when using the ASs to spare the regions with aorta, pulmonary artery, superior vena cava and contralateral tissue of upper chest and neck, and the regions with cardia and fundus of the stomach, left liver lobe and splenic flexure where there are no intact delineated structures. **Methods:** A total of 20 left-sided advanced breast cancer patients' planning CT data were selected. The ASs were created separately in the regions of the aorta, pulmonary artery, superior vena cava and contralateral tissue of upper chest and neck, and in the regions of cardia and fundus of the stomach, left liver lobe and splenic flexure. Using these two ASs during optimization, the two kinds of plans, IMRT and VMAT were created for each study case. Dosimetric parameters for the target and OARs were compared for the two kinds of plans. **Results:** With the use of ASs, both the IMRT and VMAT plans were clinically acceptable and deliverable. The ASs significantly realized the dose sparing for the regions and bring a better conformity index from 0.70 to 0.79 (p < 0.05) and homogeneity index from 1.12 to 1.08 (p < 0.05) in VMAT plans. Besides, the volume receiving at least 20 Gy (V20) for the heart (4.5% vs 5.4%, p < 0.05), the V40 for the left lung (7.9% vs

9.2%, $p < 0.05$), and the V40 for the axillary-lateral thoracic vessel juncture (ALTJ) region (27.22% vs 41.04%, $p < 0.05$) were all lower in VMAT plans. **Conclusion:** The use of the defined auxiliary structures in VMAT was more effective than that in the IMRT in improving the target dose distribution as well as sparing OARs for advanced left breast cancer radiotherapy.

PO-GePV-T-268

Evaluation of Initial Planning Techniques in Ethos for Adaptive Head and Neck Patients

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Purpose: Initial planning techniques and plan quality with the novel Ethos CBCT-guided ring-gantry adaptive radiotherapy system are evaluated for head and neck(H&N) patients. **Methods:** Clinical data(planning CT/structures) are used to replan previously treated H&N patients(n=9). The Ethos Intelligent Optimization Engine generates 5 plans(7/9/12 fixed-field IMRT and 2/3 full-arc VMAT) based on clinical goal prioritization (rather than cost function parameter inputs). Four priority levels exist (P4-P1) with fine-tuning provided by ordering clinical goals within each level. Planning target volume(PTV) goals and critical organ-at-risk(OAR) Dmax are assigned higher priority(P1-P2), while other OAR goals are assigned lower priority(P3-P4). Within this paradigm, different planning techniques are assessed: utilization of physician goals; PTV coverage preference ignoring overlapping OARs; clinical goals applied to OARs cropped from PTVs. Generic helper structures for isodose shaping and hot-spot control are evaluated. Dosimetric comparison of Ethos(AcurossXB) and clinical(Collapsed-cone-convolution superposition(n=8); Monte Carlo(n=1)) plans is based on NRG guidelines. **Results:** The first two planning techniques result in under-covered PTVs, unacceptable PTV heterogeneity, and high dose to OARs intersecting PTVs. The third technique, with the use of helper structures, provides the most comparable results to clinical plans. These Ethos versus clinical plan PTV average D99%, D95%, and Dmax are within the NRG guidelines: 103.6% vs. 100.5%, 102.2 vs. 100.9%, and 108.8% vs. 105.5%, respectively. Average global hot-spot is 110.5% in ethos(occasionally outside of PTVs) and 105.6% in clinical plans (potentially due to dose calculation algorithm differences). Most OAR doses are within NRG guidelines for both plans(with lower doses in Ethos) except for larynx, pharynx and submandibular glands (with higher doses in Ethos) due to PTV proximity. **Conclusion:** Ethos generates clinically comparable initial H&N plans with better PTV coverage and OAR sparing. However, care must be taken to precisely evaluate the magnitude and location of hot-spots in the produced plans to ensure clinical acceptability.

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PO-GePV-T-269

Analyzing Performance of Atlas-Based and Deep Learning-Based Auto-Segmentation in Breast Cancer Patients

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Purpose: Auto-segmentation has become more widespread for contouring, saving time and decreasing inter-observer variability. In this study we analyzed the performance of both an atlas- and a deep learning-based auto-segmentation of patients undergoing breast radiation. **Methods:** The atlas-based segmentation was built using 12 patients from our clinic in MIM v6.9.3. The model for the deep learning-based auto-segmentation (MIM Contour ProtégéAI) was provided by the vendor. CT scans of 12 unique patients were used for testing, and contours were manually delineated for four different structures (heart, left lung, right lung, and left breast) to be used as reference structures. The 12 test patients were then auto-segmented using the atlas and the deep learning-based methods. Three geometrical metrics were then used to quantitatively evaluate the auto-contours: the Dice similarity coefficient (DSC), the mean absolute distance (MAD), and the Hausdorff distance (HD). **Results:** The mean DSC for the heart and the lungs were greater than 0.90 for both the atlas and deep learning techniques. For the breast, which was only available via atlas-based segmentation, the DSC value was 0.83 ± 0.04 (mean $\pm$ standard deviation). The mean MAD for the lungs, heart, and breast were all less than 0.4cm for both techniques. The HD for atlas and deep learning techniques were 2.4 ± 0.8 cm and 3.1 ± 0.6 cm for the lungs, and 1.8 ± 0.8 cm and 1.4 ± 0.4 cm for the heart, respectively. The atlas resulted in HD of 3.6 ± 0.7 cm for the breast. The largest discrepancies seen were in the lung and breast where the auto-contours tapered off at the inferior and superior edges. **Conclusion:** Based on these results, we conclude that both auto-segmentation methods were sufficient for clinical use for heart and lungs, and the atlas-based segmentation will be implemented in our clinical workflow this year. The breast auto-segmentation will be used to facilitate automated beam setup for tangent fields.

PO-GePV-T-270

Gamma Analysis to Validate Dose Calculation Algorithms in a Lung Phantom: AAA Vs AcurosXB

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Purpose: This study investigated the accuracy of two dose calculation algorithms (AAA version 15.6.05 and AcurosXB (AXB) version 15.6.05) in lung, fat, and bone regions of a lung phantom using 2D Gamma analysis. **Methods:** A commercial thorax phantom (CIRS) was marked and CT simulated with 1.5 mm slice thickness then planned in Eclipse v15.6 (Varian) using an oblique 3x3 cm field traversing soft tissue, lung, and spine regions. Dose was calculated using AAA and AXB algorithms at 1.0 mm resolution for four different beam energies with the same number of monitor units: 6X, 6X-FFF, 10X and 10X-FFF. Each field was separately delivered to the phantom with Gafchromic EBT3 film placed in the axial plane through the beam isocenter. Film alignment was performed using built-in phantom pins as registration marks. Gamma analysis was performed in SNC Patient 8.2 (Sun Nuclear Corp.) software using distance-to-agreement (DTA) to compare AAA and AXB calculated doses against one another and against measured dose. **Results:** Considerable dosimetric differences occurred between AAA and AXB especially at tissue interfaces and in the beam penumbra within the lung. About 10% and 30% of analyzed points had dose differences larger than 3% for 6X and 10X, respectively. The differences were a few percentages smaller for non-flattened beams. In comparison of calculation with measurement, the 3%/2mm-DTA passing rate for AAA was 3.7% higher than AXB for the 6X beam (3.9% for 6X-FFF), but -15.9% lower for 10X (-24.9% for 10X-FFF). **Conclusion:** Dose calculation algorithm accuracy was assessed in a heterogeneous thorax phantom using 2D gamma analysis and Gafchromic film. Large differences in the passing rate, especially at 10X(FFF), suggest to verify that clinical plan evaluation metrics such as coverage and dose constraints are not compromised by the choice of dose calculation algorithm.

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Clinical Implementation of Automated Treatment Planning for Rectum Intensity-Modulated Radiotherapy on UIH Using Voxel-Based Dose Prediction and Clinical Optimization Strategies

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Purpose: This study aims to demonstrate the feasibility of clinical implementation of automated treatment planning (ATP) for rectum cancer combining a deep learning based three-dimensional (3D) dose prediction and clinical optimization strategies on UIH (TPS; United Image, Shanghai, China) for rectum cancer. **Methods:** 160 previous treated rectum cancer cases were enrolled in this study including 140 cases for training, 10 for validation and 10 for testing. Using data of CT images, planning target volumes and contours delineation the organs at risk as input and 3D dose distribution as output, a DL model was developed for predicting a 3D dose distribution for new patients. Based on these 3D dose prediction automated IMRT plans were then generated through two optimization objective functions: MSE and MSE plus more clinical optimization strategies, which include complex clinical dose target metrics, such as high priority of

target, HI and CN improving, OARs protection, target hot point control and low dose area adaptive control. To evaluate the performance of the proposed ATP model, the DVH parameters of OARs and PTV and the 3D dose distributions of plan were compared with manual plans. **Results:** By combining clinical optimization strategies, the automatically generated treatment plan can achieve better homogeneous PTV coverage and dose sparing for OARs except mean dose for femoral-head. Compared with manual plan, no statistically significant differences were found on HI, CN and global maximum dose. The manual plan outperforms slightly better in other dosimetric indexes, but our proposed clinical optimization strategies still can obtain clinically acceptable plan. **Conclusion:** With the help of these clinical intelligent optimization strategies, the proposed new automated treatment planning solution can be used to improve the quality and consistency of the treatment plans for intensity modulated rectal cancer radiation therapy.

PO-GePV-T-272

Implementation of An Automated Workflow for Planning Tangential Breast Radiotherapy

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Purpose: Automated planning for breast radiotherapy can improve efficiency and consistency for one of the most commonly treated disease sites. While several in-house solutions have been proposed, few have been translated to other clinical practices. This work aims to implement a previously developed auto-planning workflow for tangential breast radiation in an independent clinic. **Methods:** Each step of planning is automated, including the beam geometry. The contours are generated via atlas-based auto-segmentation in MIM. The posterior border of the tangents and thus the beam geometry is determined with a support vector machine in MATLAB. Eclipse scripts create the plan in the TPS, and EZFluence optimizes the dose distribution. Breast plans were generated for 12 patients, 4 to tune the beam generation parameters and 8 for independent testing. Auto-plan quality was compared to the clinical plan using DVH metrics. Beam geometry similarity and treatment coverage were assessed with the 50% and 90% isodose overlap, respectively, using the Dice similarity coefficients (DSC) and mean absolute distance (MAD). **Results:** All DVH parameters were comparable to the clinical plans. On average, the heart mean dose changed by 0 ± 13 cGy, the lung V20Gy by $+1.4 \pm 3.8\%$, and D95% of the tumor bed by $+1.0 \pm 1.2\%$ of prescription. The average maximum dose was $+0.5 \pm 1.0\%$ and the homogeneity (V105%/V95%) was 0.02 ± 0.04 . We confirmed similarity of beam geometry and coverage with DSC averaging 0.91(50%: ± 0.06 , 90%: ± 0.07) and MAD averaging 0.27cm(50%: ± 0.23 cm, 90%: ± 0.21 cm). **Conclusion:** We successfully integrated a suite of commercial and in-house tools for breast auto-planning at our institution. The methodology for determining beam geometry, first developed at an outside institution, was implemented in our workflow, showing the generalized applicability of this approach. Automated plans were comparable to clinical plans. Next, we will implement this automated workflow in our clinic and assess the efficiency gains.

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PO-GePV-T-273

Adequacy of Target Volume Drawn Based On "slow" CT Scan for SBRT Lung Treatment in Helical TomoTherapy

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Purpose: In this project, we investigated adequacy of "slow" CT for Stereotactic Body Radiation Therapy (SBRT) utilizing helical TomoTherapy treatment technique and confirmation of dose delivered to the moving target volume. **Methods:** Presently various CT simulation techniques such as slow CT, fast CT, four-dimensional CT(4DCT) and breath-hold(BH) are employed for treatment planning and delivery of radiation therapy treatment to account for tumor motion in chest and abdominal area. Due to limitations of Hi Art TomoTherapy systems, only slow scan CT scan technique can be employed to account for tumor motion. Hi Art TomoTherapy systems are equipped with MVCT that not only provide means to localize tumor volume, but can also be used for adaptive planning to confirm dose delivered to the target volume. Four patients with lung lesions were prescribed SBRT treatment and were gone through CT simulations employing "Slow CT Scan" technique in conjunction with planning CT on Siemens Biograph PET/CT. CT images were imported in Pinnacle treatment planning system. CT/Slow CT images were used to draw GTV/PTV and other critical structures volumes. All patients have gone through MVCT imaging prior to each treatment delivery. Shifts were employed based on MVCT/Planning CT fusion prior to the treatment delivery. Latter MVCT acquired for localization prior to each treatment delivery was utilized to compute dose utilizing the TomoTherapy Planned Adaptive application. **Results:** TomoTherapy Planned Adaptive application enabled calculation of actual dose delivered to the patient during each treatment fraction utilizing the pretreatment megavoltage computed tomography (MVCT). Dose coverage to PTV, GTV, and D2cm computed for each fraction were compared to the values obtained per approved treatment plan. **Conclusion:** Results of our investigations show that tumor volumes drawn based on slow CT scan appropriately account for tumor motion and enables to provide adequate coverage to the target volumes during high dose treatment.

PO-GePV-T-274

Interpolation of Dose Planes for Use in the Optimization of Multi-Leaf Collimator Dosimetric Leaf Gap and Transmission

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Purpose: To develop computationally efficient and fast process of plan generation for use in the optimization of multi-leaf collimator dosimetric leaf gap and transmission. **Methods:** Eighteen sliding window intensity modulated radiotherapy treatment plans with varying number of fields, collimator angles, and monitor units were selected. The planar doses corresponding to a set of transmission and dosimetric leaf gaps were calculated using the treatment planning system. Each pixel value in these series of planar doses was fit with an independent linear function for transmission and dosimetric leaf gap. The interpolated dose planes within the bounding values of DLG and transmission were compared to the calculated dose planes from the treatment planning system. Mean absolute difference (MAE), mean squared difference (MSE), and structure similarity metric (SSM) were used to compare the interpolated dose planes to the TPS generated and exported dose planes. **Results:** The interpolation method produced 116,688 dose planes in 2.5×10^5 seconds, or 2.16s per dose plane, as opposed to 13s per dose plane when calculating in the treatment planning system (without accounting for read-write operations required to export the dose plane). The use of multivariate linear interpolation to calculate dose planes produces nearly identical results to the dose planes calculated in the treatment planning system with MAE of 0.0207 ± 0.0286 cGy, MSE of 0.0484 ± 0.1768 cGy², and SSM of 0.9999 ± 0.0001 . **Conclusion:** Using multivariate linear interpolation provides an accurate and time efficient process for generating dose planes required for optimization of dosimetric leaf gap and transmission.

PO-GePV-T-275

Accounting for Dose Accumulation Uncertainty in Spine Reirradiation

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Purpose: To estimate dosimetric uncertainty in registration-based dose accumulation for spine reirradiation and propose a novel method to account for this uncertainty. **Methods:** Dose accumulation was performed for 3 SBRT spine reirradiation patients using 3 methods. First, a rigid vertebral registration between primary and secondary scans was used. Second, the rigid registration was independently perturbed 2-3 mm in the superior, inferior, anterior, posterior, left and right directions to estimate the effect of registration uncertainty on results. Third, an in-house API script (Radiotherapy Dose Accumulation Routine, RADAR) was used. RADAR accounts for registration uncertainty by associating with each dose voxel in the secondary scan the maximum dose within a small search

space of user specified dimensions centered on that voxel. The RADAR search dimensions were set equal to the perturbations from method 2 for comparison purposes. Results of the 3 methods were compared using maximum spinal cord (patients 1 and 2) or cauda (patient 3) dose (Dmax). RADAR was run for 2 use cases: “unrestricted”, where the entire search space was considered for each secondary voxel and “restricted”, where the search space for each voxel was limited to the portion within the cord/cauda contours. **Results:** Dmax for each patient respectively was 3790/2463/3548 cGy using method 1. Dmax increased to 4062/2555/3787 cGy (SI perturbation), 4904/3808/3773 cGy (LR perturbation), and 4955/3538/3735 cGy (AP perturbation) using method 2. Using RADAR, Dmax was 4163/2595/3802 cGy (SI), 4878/3723/3751 cGy (LR), and 4878/3477/3777 cGy (AP) for the unrestricted use case and 3947/2496/3704 cGy (SI), 4846/2496/3633 cGy (LR), and 4575/2496/3777 cGy (AP) for the restricted use case. Typical computation times for RADAR were ~2-4 minutes. **Conclusion:** RADAR provides an automated solution for incorporating registration uncertainty into dose accumulation for spine reirradiation. RADAR can be applied to other disease sites with adjustments to search radii, which are structure specific.

PO-GePV-T-276

Determining the Optimal Assigned Density for Various High-Density Prostheses Calculated with AcurosXB and AAA

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Purpose: To investigate Acuros®XB and AAA algorithms with the presence of high-density prostheses commonly found in the body and to determine the optimal material and density assigned in treatment planning system (TPS). **Methods:** Prosthetic devices (hip, knee, and dental implants) were utilized to develop patient-like phantoms. Prostheses were contoured in the TPS and divided into regions with assigned appropriate materials and densities. An iterative process determined the optimal assigned materials/densities for each phantom in the TPS. All treatment plans were computed using both algorithms to compare with the measured data. For the first hip prosthesis phantom two plans were generated, a single field plan and a clinical VMAT plan. The measured data were acquired using Sun Nuclear (SNC) ArcCHECK® with prothesis inside the central cavity. Measurement data for a second hip prosthesis phantom were acquired utilizing SNC MapCHECK® along with two pieces of Gafchromic film at two additional depths. The doses to the knee and dental phantoms were measured with Gafchromic film. **Results:** The two hip prostheses phantoms yielded same optimal assigned materials, despite their different sizes. For the femoral head and acetabular components, optimal assigned material/density were Stainless Steel (7.24 g/cm³) and Aluminum (2.6113 g/cm³), respectively. The femoral stem was assigned Titanium Alloy. When the femoral stem was contoured to separate the “neck” and “body” regions the optimal assigned densities of the “neck” was 4.31g/cm³ and the “body” was 4.4001g/cm³. The optimal material/density for the “metal” of the knee prosthesis was Titanium Alloy (4.6144 g/cm³) and its “plastic liner” was PMMA (1.19 g/cm³). The dental fillings were assigned Stainless Steel (8.0 g/cm³). **Conclusion:** This work indicates the advantage of contouring a prosthesis into regions to reflect the actual materials present and by assigning the appropriate densities to each region. The two algorithms yielded comparable pass rate results.

PO-GePV-T-277

Implementing Recommendations of TG 329

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Purpose: To implement recommendations of TG 329 for dose-to-tissue corrections **Methods:** Traditionally, our department reported dose as dose-to-water. We have undertaken the task of changing the reporting to dose-to-tissue, following the guidance of TG 329. Our department is using Varian Eclipse treatment planning system for external beam therapy plans, with AAA algorithm for photon beams and eMC for electron beams. For both algorithms, TG 329 recommends applying 0.99 correction factor to reference calibration dose in water. We have applied this correction factor and then recalculated the models. Verification of this change was then performed. This included both phantom plans as well as selected patient plans. We have compared Monitor Units, isodose distributions, dose profiles, and dose-volume histograms of plans done before the change and after the dose-to-tissue corrections were applied. **Results:** For photon beams, plan recalculation after adjusting reference calibration dose leads to different Monitor Units but the same isodose distribution when static MLC is used. In case of VMAT plans (that have not been re-optimized) MUs stay the same, but there is a change of isodose distribution. For both such scenarios, we verified that the 0.99 factor is correctly reflected in the isodose distribution and MUs. For electron beams, due to the stochastic nature of Monte Carlo algorithm, in order to verify the expected 1% change, we had to reduce statistical uncertainty below the level used clinically. **Conclusion:** A change to dose-to-tissue reporting was successfully implemented in our clinic.

PO-GePV-T-278

A Hybrid Full-Body Image Set Generation for TBI Using Both CT and 3D Optical Imaging

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Purpose: To implement TBI-VMAT, we developed a technology that can obtain 3D whole-body information using both CT and 3D optical surface imaging and bring it to a treatment planning system (TPS). **Methods:** This process consists of 5 steps. Step1: Take a CT scan, only the upper body (i.e., head to pelvis). Step2: Take a surface image for the lower body using a handheld 3D scanner while keeping the same position as CT scanning. At this time, reference markers are placed on the body surface and that part of the body is overlapped in both image sets for proper alignment later. Step3: Convert the surface image to the “.stl” format to get the lower body outline. Step4: Expand the field-of-view of the CT image and assign a density equivalent to air. Merge both the expanded CT and the lower body outline utilizing the overlapped area and markers. Then it is saved as a new CT DICOM set. Step5: Import the new DICOM set to the TPS (Eclipse) and assign water density inside the lower body. Finally, organ-at-risk structures in the upper body including the pelvis are delineated. The feasibility of the proposed method was tested using a Rando phantom and a cost-effective Structure Sensor 3D scanner (Occipital, Inc., USA). **Results:** Image registration accuracy between CT and the 3D surface image was within 3 mm on average. The time required for surface imaging was about 5 minutes. The process of image alignment and registration took about 30 minutes. A treatment plan with homogeneous dose distribution in the whole body while effectively reducing the dose to normal organs was successfully obtained without compensators/shields. **Conclusion:** It was demonstrated that a whole-body image set for modern TBI treatment planning could be obtained in a single setup by using both CT and 3D surface imaging.

PO-GePV-T-279

PenMRT: A Multi-Scale Dose Calculation Engine for Microbeam Radiation Therapy

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Purpose: Microbeam radiation therapy (MRT) is a treatment modality based on high flux, synchrotron generated x-rays. The negligible beam divergence together with the high x-ray source coherence gives the possibility of spatial fractionation of the radiation beam in a micrometric array to push the dose-volume effect towards its theoretical limits. So far, the most advanced MRT treatment planning system (hybrid dose calculation algorithm) is limited to a macroscopic rendering of the dose and do not account for the special features of MRT. For this reason, a pure Monte Carlo dose calculation engine with micrometric resolution is currently been developed for MRT (penMRT). **Methods:** PenMRT, is based on the PENELOPE (2018) Monte Carlo code, modified to take into account the voxelised geometry of the patients (CT-scans) and offering an adaptive micrometric dose calculation grid. To further improve the speed, parallelization using MPI was added together with an optional source replication protocol. The performance of penMRT is compared to its original version (penmain) for a 2x2 cm² incident broad beam and for a 50m x 2cm single microbeam, in terms of dose distribution assessment in a water equivalent phantom.

The so-called peak to valley dose ratio (PVDR) obtained in a 2x2 cm² MRT field using penMRT were also compared to the literature. **Results:** A good agreement was observed between penmain and penMRT with a maximum relative difference of $\pm 0.05\%$ and $\pm 0.2\%$ in the irradiation field and out-off field, respectively. PVDR values obtained using a spectrum with an average energy of 99 keV are in coherence with the ones found in literature. **Conclusion:** This study shows the performances and reliability of penMRT to provide micrometric dose maps for MRT treatments using synchrotron generated x-ray microbeams arrays. The code is now in experimental validation phase using Gafchromic films and microdiamonds detectors.

PO-GePV-T-280

Evaluation of Beam Data Correspondence for Five TrueBeam Linear Accelerators

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Purpose: To evaluate consistency in beam-data for a group of TrueBeam linacs. **Methods:** Five TrueBeam linacs were installed between 2010 and 2018 at our institution. For each linac, beam-data was measured for input into TPS, including scanning data (PDD, crossline profile, diagonal profile) acquired using one of two scanning chamber types (0.04cc volume and 0.13cc volume) and output factors (0.13cc chamber volume). From the full dataset, an average dataset ("consensus") was generated for analysis, and variation of each machine dataset was evaluated relative to consensus data. For scanning data, field sizes larger than 4x4cm² were included for analysis, ranging from 6x6cm² to 40x40cm² (6 field sizes in total). For output-factor data, field sizes 3x3cm² and larger were included for analysis. The data was evaluated as follows: (1) output factors – relative-st.dev. is reported, (2) PDD – average dose difference magnitude and overall max dose difference for depths beyond dmax are reported, (3) crossline/diagonal profiles – average dose difference magnitude and overall max dose difference within the inner field (defined as inner 80% of field). **Results:** For output-factor comparison, the average relative-st.dev. was 0.15% and largest relative-st.dev. was 0.37%. For PDD measurements, range in average dose difference magnitude was [0.18%, 0.26%] and range in overall max dose difference was [0.25%, 0.83%]. For crossline profiles (field sizes of 30x30cm² and smaller), the range in average dose difference magnitude was [0.06%, 0.19%], and range in overall max dose difference was [0.52%, 0.65%]. For 40x40cm² field size, overall max dose differences were slightly larger, with range of [0.64%, 1.17%]. For diagonal profiles, range in average dose difference magnitude was [0.19%, 0.37%] and range in overall max dose difference was [0.55%, 1.29%]. **Conclusion:** For field sizes ranging from 6x6cm² to 30x30cm², our results indicate a level of agreement well within 1% for all measurements. For largest field size (40x40cm²), our results indicate a level of agreement within 1.5% for all measurements. Overall, our results suggest that average beam-data set may be used to model the broad-beam-data for TrueBeam linacs.

PO-GePV-T-281

Solving Beam Orientation Optimization (BOO) of IMRT Treatment Plan Using Tree Embedded Reinforcement Learning for Prostate Cancer

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Purpose: To propose a method to solve BOO problem in a matter of seconds without dose-influence matrices information available. We compare the proposed method with the state of the art column generation method (CG) to show its capability in solving BOO problem. **Methods:** A tree search embedded reinforcement learning method (RL) is proposed to solve beam orientation optimization problem (BOO) in IMRT treatment planning for patients with prostate cancer. This method uses a pre-trained deep neural learning (DNN) model introduced in [1] and improves on it by adding a tree-search algorithm during the training of the model. During clinical use, after the model is trained, it can suggest a set of beams for treatment planning in less than one second with only patient anatomical information as input. The same set of patients previously used to train and validate the DNN model is used here and a separate independent set of data is used for testing. **Results:** Although the model is still in training stages, because of the complex and memory intensive training steps, initial results are promising. In about 50% of the cases, RL successfully surpassed CG, and in the rest, although it may not surpass CG, the model will be able to find a solution as good as or better than the DNN model. In all cases, the beam selection process takes 1 seconds to propose five beams for each patient. **Conclusion:** A fast method for solving BOO problem that uses only anatomical features of the patient and can perform better compared to DNN model and CG.

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PO-GePV-T-282

Clinical Impact of the Dose Calculation Engine for Vestibular Schwannoma Treatments on Cyberknife

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Purpose: Historically for Cyberknife, long computation time was a barrier to using Monte Carlo (MC) instead of Ray Tracing (RT) for repeat and final dose calculations during plan optimization. With current computing power, MC can be used during optimization for more accurate dose calculations. In this study, the impact of changing the primary engine from RT to MC was evaluated for radiosurgery treatment planning of vestibular schwannoma with varying target sizes. **Methods:** Retrospective analysis was performed in Precision v2.0.1.1 for 23 vestibular schwannoma patients treated at our institution to an Rx of 18Gy in 3 fractions. The mean GTV volume was 2.6 +/- 3.7cc. From the MC and RT calculations for each patient, we extracted the minimum dose to the GTV, the percent PTV volume receiving the full Rx, and the maximum dose to the PRV of the ipsilateral cochlea. Results were evaluated against 3 GTV sizes: >5cc, 1.5-5cc, and <1.5cc. **Results:** For the minimum dose to the GTV, MC vs RT results showed minimal sensitivity to target sizes >5cc. For the second (1.5-5cc) and third (<1.5cc) bins, MC was higher by an average of 2% and 3.5%, respectively. This implies that with MC-based optimization that maintains the nominal Rx, future patients will, on average, receive a lower actual dose to the GTV by the amounts above. Dose to the neighboring OARs will be lower (an aside: for the PRV of the ipsilateral cochlea, MC results were higher by 1%, 2.5% and 5% for the above bins, respectively). The change in the percent PTV receiving full Rx changed by -1.0 +/- 3.3%, implying the need for a small scaling in either direction with MC-based optimization. **Conclusion:** Switching from RT- to MC-based optimization is not expected to affect the continuity of historical clinical outcomes for vestibular schwannoma patients, including small targets.

PO-GePV-T-283

Improved Dose Prediction Performance with Data Clustering and Transfer Learning On H&N Cancer Patients

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Purpose: We propose a deep learning (DL)-based clustering and Transfer Learning technique for improved model performance and test it on dose prediction of head and neck (H&N) cancer patients. **Methods:** We used H&N cancer patient data available from the OpenKBP Challenge, consisting of 200 training, 40 validation, and 100 testing. This includes CTs, planning target volumes (PTV) and organs-at-risk (OAR) masks, 3D dose distributions. The dose distribution was generated from a 9 beam equidistant IMRT setup with 6 MV fields. As baseline, we trained a U-net style architecture model on all of the training data to take the patient anatomy as input to predict the dose distribution. For the clustering method, we first utilized an encoder-decoder style model and trained it to take the anatomy as input to predict the dose distribution, and then used the encoder to transform all the data into the latent space. A TSNE calculation is made on the data in the latent space, and clusters of data are determined. Models of the same U-net style architecture as the baseline are then trained to predict the dose distribution in one of two ways: 1) training from scratch on each data cluster or 2) Transfer Learning (TL) from the baseline model to each data cluster.

Results: Compared to the ground truth data, the Cluster+TL method performed the best in regard to minimizing the error in structure mean and max doses, dose coverage, and isodose similarities between its prediction and the ground truth dose data. Training from scratch on each cluster without TL performed similarly to the baseline model likely due to tradeoff between data quantity and precision. **Conclusion:** We show that our method of Clustering+TL can further improve the result from baseline. Clustering without TL may require more data before it can outperform the baseline model.

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PO-GePV-T-284

Feasibility Study of a Novel and Efficient Treatment Planning Technique for Total Body Irradiation and Total Marrow Irradiation On Conventional Linacs

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Purpose: We evaluated the feasibility of using a novel treatment planning technique for total body irradiation (TBI) using modern VMAT fields on conventional linacs with the aim of improving planning efficiency and minimizing dose variation in field junction regions. **Methods:** Two clinical TBI cases were used in this study. The head-in upper body CT simulation and feet-in lower-extremity CT simulation for each case were registered and concatenated to form a whole-body simulation from the top of skull to toes. A treatment plan was optimized to treat the whole-body target volume for 13.2 Gy total in 8 fractions, in which nine arc fields were used to treat the upper body and three wide aperture arc fields were used to treat lower extremities from the mid-thigh level. The optimized plan was separated into an upper-body and a lower-extremity plan. An in-house software program converted the lower-extremity plan into a new plan with the feet-in position by re-arranging control points and rotating field apertures at every control point. The converted lower extremity plan was re-imported into the TPS for recalculation and validation and the total dose was verified by an independent calculation algorithm. **Results:** In both the whole-body TBI plans the mean lung dose was < 8 Gy with the upper-body and lower-extremity targets receiving the prescribed dose to at least 85% volume. With the feet-in patient position, the re-calculated lower-extremity plans showed identical dose distribution to the original plans. Both the upper-body and lower-extremity plans showed a linear dose gradient in the mid-thigh junction region with an average drop-off of 30 cGy/mm. **Conclusion:** Using this planning technique, VMAT TBI plans can be generated efficiently with homogeneous dose in the junction region while minimizing dose variation from setup uncertainties. Our clinical experience showed that this takes similar time and resources as conventional TBI.

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Generalizability Study On a Deep Learning-Based Dose Conversion Model

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Purpose: Recently a deep-learning model was proposed to boost the accuracy of AAA doses to the level of AcurosXB doses (AXB). A robustness study was performed to evaluate the model on plans across different sites, planning techniques, and machine models to examine its generalizability. **Methods:** In total 66 AAA plans of 5 sites from the same institution were recalculated using the AXB dose engine, and the resulting AAA-AXB dose pairs were used as the testing cohort. A prior deep-learning model trained and validated using VMAT lung plans on Elekta LINACs (VersaHD) was deployed on the testing cohort to derive deep-learning-boosted doses (DLB-AAA) from original AAA doses. The DLB-AAA doses were evaluated using gamma passing rates with the calculated AXB doses as reference. The testing cases were grouped into VMAT-only, same machine model-only, and VMAT & same machine model-only to evaluate the effects of planning techniques and machine models on the deep-learning model accuracy. **Results:** When comparing the gamma passing rates for all testing cases, the DLB-AAA doses do not show better match to AXB doses than the original AAA doses, indicating overall limited model generalizability. However, when restricting the evaluation plans to VMAT-only, the DLB-AAA doses are significantly closer to AXB doses than the original AAA doses across all sites except for head-and-neck (HN), potentially due to the substantial difference in modulation factors between HN and lung VMAT plans. To improve the generalizability of the model, we propose to include plans from all sites and planning techniques for training, with potential fine-tuning through transfer learning warranted. **Conclusion:** Our DL-based AAA-AXB dose conversion model shows robustness towards machine models and treatment sites (except for HN), while lacks robustness towards different planning techniques (VMAT vs. IMRT vs. 3D). Transfer learning or a more generalized model is needed to further improve its robustness.

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PO-GePV-T-286

Dosimetric Evaluation of Dose Accumulation in MIM and Pinnacle System

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Purpose: Imaging solutions such as MIM Software are commonly utilized to perform dose accumulation for previous treatment. The purpose of this study was to evaluate the dosimetry of dose accumulation performed using algorithms in MIM and Pinnacle treatment planning system (TPS). **Methods:** Six SBRT NSCLC patients with a previous SBRT lung treatment were retrospectively analyzed in this study. The new treatment planning CT was used as the primary dataset. Rigid image registration was performed in Pinnacle TPS. In the first technique, a Pinnacle host script was used to transfer and recompute the previous treatment beams to the new CT; and then summed with the dose from the current treatment beams. The image registration, original RT dose files, original CT datasets, and the new RT structures were exported to MIM software. In the second technique, MIM utilized the imported image registration from Pinnacle to transfer the original old dose from the old CT to the new CT followed by a second dose accumulation generated in MIM. To remove sampling grid bias, both MIM and Pinnacle generated dose accumulations were evaluated in MIM. Dosimetric analysis was performed based on clinical OAR dose limitations. **Results:** The OAR doses in the composite plans were similar between the two software. For the Dmax(0.1cc) constraints for spinal cord, heart, esophagus, etc, the mean and standard deviation of the absolute difference between the two methods were 0.24 ± 0.28 Gy (range: 0 to 1.31 Gy). A significant statistical difference was not observed ($p > 0.5$). The difference between the two methods for the volumetric dose constraints were within 0.36%. Neither was statistically significant. **Conclusion:** Minimal dose differences were observed between the dose accumulation generated from Pinnacle recalculation method and MIM direct summation. These differences are not clinically nor statistically significant. Either methods of dose accumulation can be clinically implemented.

PO-GePV-T-287

Knowledge-Based Planning Assisted Automatic Prostate Cancer Radiation Treatment Planning

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Purpose: For modern radiation therapy (RT), a human planner needs to adjust planning objectives and constraints in the treatment planning system (TPS) to define the optimization problem for plan generation. Such a process is often tedious and time-consuming, while the plan quality can be affected by many factors, such as the planner's experience and available planning time. To address this issue, we developed a RapidPlan assisted automatic scheme to improve

treatment planning efficiency and quality. **Methods:** The proposed scheme has two stages: Firstly, RapidPlan is employed to define the planning objectives and constraints in TPS to generate an intermediate plan. Secondly, the intermediate plan was evaluated by a set of planning criteria consisting of the safety limits and pre-defined physician's objectives. If the intermediate plan fails to meet all the planning criteria, a set of rules summarized corresponding to the criteria are applied to adjust the objectives and constraints to improve the plan quality. This step can be repeatedly executed until a clinically acceptable plan is obtained. In this study, we considered the prostate-cancer-SBRT (4500cGy in 5 fractions) and prostate-Post-Operative-VMAT (7020cGy in 39 fractions) as two testbeds. 10 patient cases were employed for each type to evaluate the performance of the proposed scheme. **Results:** For 7 of the 10 prostate SBRT, plans generated directly by RapidPlan achieved similar or better plan quality compared to clinical plans. The other three needed to proceed with the second stage, and clinically acceptable plans were successfully generated. Similar results were observed for Post-Operative VMAT planning with 6 of 10 going through the second stage. **Conclusion:** The proposed Rapidplan assisted automatic TPS is effective to generate high-quality plans for prostate SBRT and Post-Operative VMAT. It can be useful for other planning tasks, and has the potential to serve as a tool for training new planners.

PO-GePV-T-288

Dose Evaluation of Deep Inspiration Breath-Hold On the Left Anterior Descending Coronary Artery for Left-Sided Breast Irradiation

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Purpose: To evaluate the radiation dose of the left anterior descending coronary artery (LADA) for the patients treated with deep inspiration breath-hold (DIBH) technique. In routine clinical treatment planning for the left breast cancer patients, the dose-volume histogram (DVH) analysis of the LADA is not commonly evaluated because its dose constraint information is not enough to analyze clinical toxicities in terms of a critical volume with total dose in number of fractions. **Methods:** Ten left breast patients with various tumor sizes and positions previously treated with field-in-field tangential beam and DIBH were randomly selected. The LADAs were manually contoured. DIBH based on 4DCT (4-dimensional computer tomography) was used for each patient to minimize the heart irradiation. For treatment planning, the prescription dose was 42.6 Gy in 16 fractions. Organs at risk (OAR) included LADA, heart, lumpectomy cavity and lungs. **Results:** The volume of LADA from this study was measured to be 0.25 ± 0.07 cm³. There was not a difference to the sizes, but the positions were changed by the patients' anatomic structure. The dose to LADA were affected by these. The mean, minimum and maximum dose of LADA were 5.37 ± 1.94 , 2.03 ± 0.87 , and 13.15 ± 8.20 , while the mean, minimum and maximum dose of heart were 1.31 ± 0.36 , 0.13 ± 0.04 , and 18.16 ± 11.99 , respectively. The mean dose ratio of LADA to heart was 4.1. **Conclusion:** Our evaluation of the LADA dose demonstrated lower mean dose and standard deviation than other institutions. More clinical evaluations are required from this study in terms of the discrepancy of LADA delineation among the planners and inclusion of the dose effect from electron boost.

PO-GePV-T-289

Discrepancies in Out-Of-Field Dose Prediction Between Treatment Planning Systems

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Purpose: Physicians routinely rely on dosimetric parameters for organs outside of the treatment field to make clinical decisions regarding plan quality. The purpose of this study is to quantify variance in predicted out-of-field dose by the TPS for relevant OAR constraints in a simple treatment plan. **Methods:** This investigation compared dosimetric data from a simple breast tangent plan prescribed to 4005 cGy in 15 fractions with open jaw-based rectangular fields calculated using different beam models and calculation algorithms within several software systems. The plan parameters such as energy, field size, gantry angle, collimator angle, MU, and isocenter were kept consistent throughout all iterations and the same CT and structure set was utilized. The plan was evaluated on a Varian Truebeam, Varian Edge, Varian Trilogy and Elekta Synergy with an Agility head (Elekta-SA) modeled in various treatment planning systems: Eclipse using AAA, Xio using CCC, and Monaco using Monte Carlo. **Results:** The mean heart dose was recorded for each iteration of this plan as well as the Lung V20. The mean heart doses ranged from 44.7 cGy (Trilogy, Eclipse AAA) to 98.3 cGy (Elekta-SA, Eclipse AAA), the standard deviation was 19.58 cGy. This research also found a 29% difference in the calculated mean heart dose for the same plan on the same machine that had been modeled in two different TPS. Total Lung V20 did not change significantly with distinct beam model, ranging from 12.2% to 12.5%. **Conclusion:** The research reinforces that out-of-field dose cannot be quantified with consistent accuracy across treatment planning systems. The relationship between out-of-field dose predicted by a particular beam model and absolute dose being delivered to that area is not well characterized and necessitates further investigation. This calls into question the strength with which physicians should weigh dosimetric values outside of the treatment field.

PO-GePV-T-290

Investigation of the Dosimetric Effects of An SBRT Positioning System

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Purpose: Investigate the potential dosimetric impact of the Bionix SBRT system on the patient. **Methods:** The Bionix RT-4500 Omni-V SBRT system was investigated for the potential dosimetric influence on patients. Structures not unaccounted for during planning may result in inaccurate skin dose and dose coverage. Skin dose measurements acquired using Varian 21iX and 6MV beam. The Edge diode-detector was placed at 100cm SAD, adjacent to each component, for varying field sizes (10x10, 6x6, 4x4, 2x2cm²). The components included the compression belt (with and without insert), chest compression plate (thickest and thinnest part) of the Turner Arch, and the body support vacuum cushion (2, 5, 10cm thicknesses). For attenuation effects, the diode detector was placed at 100cm SAD and 10cm depth in solid water (90cm SSD) for the same field sizes. The SecureFit Bar (center and lock tab), side-rail belt-attachment, Turner Arch, and the base of the SBRT system were placed on top of the phantom. **Results:** With the Eclipse TPS, anything outside the body contour is not considered dosimetrically. Components of the immobilization system must be scrutinized and appropriately considered. Oversight of a device can result in increased surface dose or reduced coverage of the target due to attenuation. Skin dose increases as the field size increases. For attenuation effects, there is a slight decrease of the radiation dose as the field size decreases. **Conclusion:** Measurements traversing various components of the Bionix SBRT system have been reported for user reference within the radiation oncology community. Apart from the T-Vac Cushion on which the patient lies, most components can be knowingly avoided within the treatment planning process. While minimal attenuation of the beam through parts of the device may be observed, unintended skin dose may be significant.

PO-GePV-T-291

The Impact of Treatment Planning System Errors on Patient Dose Delivery

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Purpose: To evaluate the impact of different beam modeling errors within the treatment planning system (TPS) on patient care. **Methods:** The Imaging Radiation Oncology Core (IROC) recently conducted a study to understand suboptimal TPS beam modeling that was associated with IROC phantom failures. The resultant dataset describes TPS parameter selection in the radiotherapy community. The current study evaluated the impact of this spread in TPS parameters on clinical treatment plans. The following parameters were modified in RayStation: multi-leaf collimator offset (MLC(O)), transmission (MLC(T)), gain (MLC(G)), curvature (MLC(C)), source size (SS), leaf tip width (LTW) and tongue and groove (T&G). Each parameter was independently modified to match

the community data at the 2.5, 25, 75 and 97.5 percentile levels. The impact of these modifications was evaluated on fifteen patient cases, including prostate, lung and brain plans, generating a total of four hundred and twenty perturbations. The difference in the mean dose delivered to the clinical target volume (CTV) and parallel organs at risk (OAR), and the maximum dose to the serial OAR, were evaluated with respect to the dose delivered using 50-percentile parameter values. **Results:** Variation in the MLC(O) and MLC(T) parameters resulted in the greatest dose difference, relative to the 50-percentile level, for all anatomical sites. For the MLC(O) parameter, the greatest impact to the mean CTV and OAR dose was 5.5% and 14.6%, respectively, at the 97.5-percentile. For MLC(T), the greatest CTV and OAR dose perturbations were -2.8% and -26.7%, respectively, at the 2.5-percentile. **Conclusion:** MLC(O) and MLC(T) beam modeling parameters play a substantial role in patient dose calculations. Extreme values for these parameters (e.g., 2.5 and 97.5 percentile values), which have previously been associated with failing the IROC phantom, negatively impact the quality of patient care. Careful attention should be given to these beam modeling parameters.

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PO-GePV-T-292

Dose Prediction Using Three-Dimensional Convolutional Neural Network for Nasopharyngeal Carcinoma with Tomotherapy

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Purpose: To accurately predict 3D dose distribution based on the patient-specific gap between organs at risk (OARs) and PTVs, and small OARs for nasopharyngeal carcinoma (NPC) with Tomotherapy. **Methods:** 3D matrix for a deep learning model of dose prediction was trained to learn more about the relationship between dose distribution and patient anatomical information by increasing the height of training matrix. And 3D HD U-net was used as model. 127 NPC patients treated with Tomotherapy were enrolled in this study, including 84 for training and 43 for testing. The multi-channel 3D input matrix including CT and ROIs, dose array was used as the output matrix. A comparative experiment was done between the training matrix height of 48 (Model I) and 16 (Model II). Mean absolute error (MAE) was taken as the loss function. The models' performance was evaluated by the comparison and statistical analysis. **Results:** The corresponding errors were calculated by comparing the predicted dose with ground truth dose. The deviations among the mean and maximum dose of planning target volumes (PTVs) and OARs were within 4.37%. Error for max dose of optic nerves-L in Model I were $4.05 \pm 3.49\%$, comparing with $7.11 \pm 5.10\%$ in Model II ($p=0.002$). The gamma passing rates of body and PTV for 3%/3mm criteria were $76.5 \pm 10.7\%$ and $83.8 \pm 5.4\%$ in Model I, comparing with $43.3 \pm 5.4\%$ and $73.6 \pm 6.6\%$ in Model II ($p<0.001$). The prediction error of D95 for PTV was $0.48 \pm 0.33\%$ in model I, comparing with $2.72 \pm 1.28\%$ in Model II ($p<0.001$). **Conclusion:** It is important for deep learning dose prediction model to learn the distance of OARs and PTV. We found that large height of training matrix could improve the accuracy of dose prediction. Our dose prediction methods provide a milestone to build automatic Tomotherapy planning.

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PO-GePV-T-293

ROI Contour Based Deep Learning AAA Dose Improvement Method

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Purpose: Deep learning methods have been explored to accelerate speed and boost accuracy of dose calculation, which of great significance in improving radiotherapy efficiency, by converting fast AAA dose calculation results to much slower but more accurate AXB dose calculation results. In this study we aim to develop a new deep learning scheme, by incorporating ROI contours into the training structure, to improve the performance of dose conversion from AAA dose to AXB dose. **Methods:** We developed a 3D HD U-NET to convert AAA dose to AXB dose. In this deep learning scheme ROI contour binary masks were able to be incorporated as input data in the network structure, as well as the CT and AAA dose. Corresponding AXB dose were calculated using AXB algorithm as the ground-truth. 95 VMAT female pelvic cancer cases were selected for the research, where 76 cases were for training, and 19 cases were for testing. MSE is the loss function. Decay learning rate ranged from 10-4 to 10-6. The training process is stopped when the loss and learning rate stopped to decrease. The comparison study used only CT and AAA dose as input data. **Results:** We calculated the Gamma passing rate (1 mm / 1%, 10% threshold) to evaluate our method using corresponding AXB dose calculation as comparison reference. The Gamma passing rate of original AAA dose and converted AXB dose are $73.64 \pm 3.46\%$ and $95.57 \pm 1.06\%$, respectively. Time consumed for the conversion is 3.06 ± 1.78 s. Compared to deep learning schemes using only CT and AAA dose as input data, Gamma passing rate of our newly devised method increased by 1.1% (TTest, $p=0.004$). **Conclusion:** Deep learning method performs well in converting AAA dose to AXB dose in VMAT female pelvic cancer cases. Adding ROI contour as training data can improve the accuracy of dose conversion.

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The Effect of Contrast Agents On Dose Calculations in Volumetric Radiotherapy for Critical Structures

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Purpose: Accuracy of radiotherapy dose calculation requires high quality CT imaging while tissue delineation may require the use of Contrast Agents (CA). It is widely accepted practice to acquire two sets of CT images in radiotherapy CT simulators, one with contrast and another with no contrast. This study aims to evaluate the effect of CA on the dose calculations in critical structures and planning target volumes (PTV). **Methods:** Two hundred and twenty-six patients' Volumetric Modulated Arc Therapy (VMAT) plans of different cancer sites (Brain, Head and Neck (H&N), Chest, Abdomen, and Pelvis) were evaluated in this study. Selection criteria included all VMAT patients that had planning CT with contrast (CCT) and non-contrast CT (NCCT) at the same time. Approved treatment plans were recalculated using CCT without any further changes, then compared to NCCT. The variation in Hounsfield Units (HU) and dose distributions for critical structures and PTV's were analyzed using absolute HU values, relative dose values, D2%, D98%, and 3D-gamma analysis. **Results:** HU variation due to CA were statistically significant ($p\text{-value}<0.05$) for the majority of critical structures and PTVs. However, this was not clinically significant as the variation in HU values was within 40 HU. Variation in PTV D2% and D98% values was insignificant for all sites except in Brain and Nasopharynx. Relative dose maximum difference was $\leq 2\%$ for the majority of critical structures. Minimum, maximum, mean, and median values for D2% and D98% were within 1.5%. Gamma analysis results revealed that 202 and 18 patients passed with 2%/2mm, 3%/3mm, criteria respectively, other 6 patients failed. **Conclusion:** CCT could be acquired for VMAT radiotherapy planning purposes instead of NCCT, since there is no clinically significant difference between dose calculations based on either image set.

PO-GePV-T-295

A Simple Method for Accurate Multi-Target Winston-Lutz Testing

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Purpose: Treatment modalities such as Varian's HyperArc allow convenient and efficient treatment of multiple stereotactic targets in a single isocenter. Confirming geometric accuracy of the mechanical, radiation, and imaging systems for off-axis targets is crucial for these treatments, but centering an MLC field on an off-axis target poses a significant challenge due to geometric offsets in the source-target direction and fixed MLC leaf width. While methods have been proposed to perform Winston-Lutz testing of off-axis targets relative to baseline measurements, this work aims to provide a simple solution for multi-target Winston-Lutz testing that is absolute, efficient, and reproducible. **Methods:** The VisionRT Cube phantom was used, which includes five 7.5mm ceramic spheres, including 4 targets offset in three dimensions from a central target. Total distances vary from 2.3cm to 4.1cm. Eight fields were planned which included gantry, collimator, and table rotations. Within each field, 1.5cm MLC openings were centered on three targets including the central target. Exact MLC locations for centering on each target in the leaf motion direction, as well as known target offsets in the interleaf direction, were calculated from simple beam divergence projections. These known offsets were then subtracted from the Winston-Lutz results for each field. This test was repeated on five occasions to confirm reproducibility. **Results:** Initial results showed a maximum isocenter offset of 1.35mm. After applying offset corrections, all targets for all fields showed accurate results with the largest offset being 1.03mm. The average offsets ranged from 0.32mm to 0.54mm for the various targets, while standard deviations ranged from 0.15mm to 0.21mm. **Conclusion:** This proposed multi-target Winston-Lutz method provides an accurate and reproducible means to characterize off-center targeting performance. The process can be utilized during commissioning of a single isocenter multi-target SRS program and is simple enough to perform routinely.

PO-GePV-T-296

Physics Quality Assurance: A Pilot Survey of Global Needs

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Purpose: To assess physics QA and quality management in low- and middle-income countries (LMIC). An exploratory preliminary study was conducted to investigate the relationships between physics QA needs and the economic factors in the radiotherapy centers surveyed. **Methods:** A survey was conducted amongst 13 centers in 6 countries. 3 LMIC and 10 upper middle-income countries (UMIC). Questions covered education and training, recognition of medical physics as a profession, the nature and type of external beam equipment, patient workload on Linac/Co-60, HDR and LDR brachytherapy and, physics QA needs and practices. **Results:** The survey response rate was 92%. In countries with lower gross domestic product per capita (GDPpc) there were a higher number of patients treated per year and higher linac workloads. In terms of staff workload on average there were 360 patients/year/physicist and 210 patients/year/radiation oncologist (or therapist). This workload is comparable to other surveys in Eastern Europe and other regions. In lower GDPpc countries there were more QA needs, including: film/array detectors, independent dose calculation software, calibration of ion chambers, diodes, TLDs, phantoms for verification, TPS test phantoms, imaging test phantoms and film dosimeters education and training. The study also revealed centers follow AAPM (75%), IAEA (66.7%) and other international guidelines (20%) for their QA practices. **Conclusion:** Our study shows the relation between QA needs and RT centers in low GDP regions. Though centers follow international QA guidelines, they lack recommended devices and infrastructure to execute these protocols. These results support the need to further investigate current physics QA practices in these centers and develop collaborations to build comprehensive local QA programs.

PO-GePV-T-297

Accelerator Beam Data Variability with Detectors

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Purpose: Accurate beam parameters (percent depth dose-PDD and profiles) are essential for treatment planning and dose calculations that are acquired during machine commissioning. However, these are dependent on the selection of a detector. The suitability of modern detectors for a wide range of field sizes and beam energies are investigated. **Methods:** Beam parameters are measured using IBA blue phantom scanning water tank on a Varian TrueBeam machine with 6, 10, 18MV, 6FFF and 10FFF energies for field sizes ranging from 1x1 cm² to 30x30 cm². Modern detectors such as IBA CC-01, CC-04, Scanditronix SFD, PFD, microDiamond (PTW-60017), microSilicon (PTW-60023) and Standard imaging-W2 plastic scintillator detector (PSD) were used with recommended bias (300V and 0V). IBA Stealth chamber and Scanditronix RFD were used as the reference chambers for small to intermediate and large fields, respectively. IBA CC-04 was treated as a reference detector for 5x5cm² to 30x30cm² and W2 for 1x1cm² for PDD and profile analysis and comparisons. **Results:** The PDD and profiles varies significantly among the detectors, field size and beam energies. The average variation in PDD beyond buildup region, among detector was $\pm 1\%$, $\pm 2\%$ and $\pm 5\%$ for 1x1cm², 10x10 cm² and 30x30 cm², respectively but nearly independent ($\pm 2\%$) on beam energy. The variation in profiles at 10 cm depth was $\pm 5\%$, $\pm 4.5\%$ and $\pm 3\%$ for 1x1cm², 10x10cm² and 30x30cm² for 6MV and slightly larger ($\pm 1\%$) for 18MV but mainly in penumbral region. Differences are pronounced with beam energy and field sizes due to μ_{en}/ρ variation associated with photon spectrum. **Conclusion:** Selection of detector is critical for acquiring accurate PDD and profile for various energies and field sizes. It is concluded that PSD W2 and small volume ion chamber are preferred for field sizes from 1x1-5x5cm² and 3x3-30x30cm² respectively.

PO-GePV-T-298

Deep Learning for Patient-Specific Quality Assurance : Predicting Gamma Passing Rates for IMRT Based Log File

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Purpose: To propose a deep learning-based approach to predict IMRT QA gamma passing rates using log files. **Methods:** 112 IMRT plans for thoracic tumors, including 584 fields, were planned and delivered with portal dosimetry on TrueBeam LINAC. Then the machine recorded log files data were collected. Gamma passing rates for 3%/3 mm, 2%/3mm, 3%/2mm, 2%/2mm local dose/distance-to-agreement (DTA) were analyzed and recorded. The CNN based learning model was trained using log files as inputs and gamma passing rates of four different criteria as output. Fivefold cross-validation was utilized to tune the model and prevent "overfitting" and MSE was used for loss function. Model performance for both validation set and test set were assessed using MAE, RMSE, Spearman rank correlation coefficients and Determination coefficient were analyzed between measured and predicted GPR values. **Results:** In test set, the MAE of prediction model were 0.402, 0.511, 1.724, 2.530, the RMSE were 0.800, 0.993, 2.580, 3.083, the Sr were 0.643, 0.684, 0.821, 0.824 (P<0.001) and the R2 were 0.4110, 0.4666, 0.6677, 0.6769 for 3%/3mm, 3%/2 mm, 2%/3 mm, 2%/2 mm, respectively. The accuracy of the prediction model decreased with stricter gamma criteria while the Sr and R2 between measured and predicted GPR values increased. **Conclusion:** The CNN prediction model based on log files could accurately IMRT QA passing rates for different gamma criteria. It could reduce QA workload and improve efficiency in pre-treatment and on-treatment dose verification. Our results suggest that CNN prediction model based on log files may be a promising tool for gamma evaluation of IMRT QA.

PO-GePV-T-299

Toward Ending the Manual Annual: Lessons Learned From the Expansion of An Automated Output Quality Assurance Technique to Outside Institutions

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Purpose: To adapt a single-setup in-house linac QA automation method to outside institutions. **Methods:** Annual output quality assurance was automated in previous work for our Varian TrueBeam through the SAGE tool, TrueBeam Developer Mode, and the PTW UNIDOSwebline electrometer with in-house software. We adapted this work to 2 outside institutions based on their 2020 annual output QA procedures. Crucially, we hypothesized that outside institutions would see a strong hysteresis effect depending on the length of the beam off time between irradiations. Extensive scripts were created to address the tradeoff between short scripts and hysteresis across a hypothesized 140 measurements over 30 minutes. **Results:** The expansion of our automation method to an outside institution highlighted the implicit clinic-specific assumptions that were made in the original Developer Mode scripts. Adaptation required adjustment of each beam to the energies, test parameters, and measurement repetition used in each clinic. Also, our in-house analysis program needed to be adjusted to accept the output file format from a Standard Imaging SuperMAX electrometer. Furthermore, the MLC model needed to be switched from HD to standard. It is essential that clinics using our technique are able to create their own scripts in a user-friendly manner. Our early emphasis on minimizing the time between irradiations while optimizing the inherent tradeoff with hysteresis was misplaced. The contributing clinics required only 53% of the measurements of our own clinic. Minimizing the duration of beam-off pauses yielded a small gain for the outside institutions. A 5-second beam off time after each irradiation was measured to be sufficient to reduce hysteresis to within 0.5% over 100 irradiations. **Conclusion:** Widespread adaptation of the work will help to determine the appropriateness of digital electrometers and their analysis software for automation. Generalization is necessary to enable customization of this work to any cancer center.

PO-GePV-T-300

An Innovative Water-Equivalent Solid Device for Beam Scanning to Replace Computerized 3D Water Scanning Systems

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Purpose: To develop a solid phantom that is capable of replacing the computerized water scanning system in its entirety. This phantom is designed such that it is easy to install at the head of the gantry and within a few minutes of data collection, the user can extract profiles and percent depth dose (PDD) data quickly through the software associated with this device. **Methods:** Using Monte Carlo package MCNP5 we modeled our LINAC (Varian Edge 6FFF), and confirmed that simulated PDD ($10 \times 10 \text{ cm}^2$) and profiles at different depths agree favorably with measured data acquired in a 3D water tank ($40 \times 40 \times 40 \text{ cm}^3$). We then created a virtual solid device placed at 60cm source to surface distance (SSD) on modeled Linac, which is near closest distance to gantry head. The device modeled in MCNP consists of two new phantoms that interlock and are modeled using published data for solid water material and a detector grid with much smaller volume and special geometrical shape to accommodate acquisition of data for PDD and profiles. Scoring points were placed in virtual phantom representing arrays of diodes for future prototype for the measurements of PDD and profiles. New data are acquired through simulation and compared with data from the 3D water tank to demonstrate the feasibility of this new device. **Results:** Simulated PDD and profiles in new designed solid water phantoms agree favorably with data from a 3D water tank (<1% error). The effective mass of the prototype phantom for profiles is 9% of 3D water tank while the phantom for PDD is only 1%, indicating much easier handling and setup. **Conclusion:** Monte Carlo simulations show the feasibility of our new solid phantoms to replace 3D scanning water tank from the points of saving time in data acquisition, accurate dosimetry evaluation and convenience of equipment setup.

PO-GePV-T-301

Energy Dependence And dosimetric Stability of EPIDs Used for Linac QA

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Purpose: Electronic portal imaging devices (EPIDs) have been widely used for linac QA. The purpose of this study is to determine the energy dependence and long-term stability of EPID pixel response. **Methods:** Six Varian Truebeam linacs (3 equipped aS1200 and 3 equipped aS1000 EPID panel) with different ages (1 to 10 years) were included in this study. Since the vendor-provided flood-field calibration method cannot separate beam response and pixel gain, we generated a pixel sensitivity map (PSM) independent of the beam profile to quantify the pixel response. PSM was derived by applying a recursive algorithm from five EPID images acquired in different imager locations. All the pixel responses were normalized to the central pixel. PSM measurements were performed for 6 MV and 10 MV consecutively from 8-25 weeks for different imagers. The pixel response of EPID was compared for 6MV and 10 MV. The time-dependent response of EPID pixels was evaluated to determine long-term stability. **Results:** Central axis response changes for the six machines were as follows: 0.24% (Tr5-aS1000), 0.31% (Tr7-aS1000), 0.23% (Tr14-aS1000), 0.23% (Tr15-aS1200), 0.24% (CHNW TB1-aS1200), 0.18% (BJSP TB2-aS1200). The mean absolute changes of pixel response for the entire panel during the study period were: 0.48% (Tr5-aS1000), 0.76% (Tr7-aS1000), 0.45% (Tr14-aS1000), 0.98% (Tr15-aS1200), 0.59% (CHNW TB1-aS1200), 0.51% (BJSP TB2-aS1200). The mean difference of EPID response for 6MV and 10 MV was: 1.10% (Tr5-aS1000), 0.89% (Tr7-aS1000), 0.83% (Tr14-aS1000), 0.65% (Tr15-aS1200), 0.66% (CHNW TB1-aS1200), 0.57% (BJSP TB2-aS1200). **Conclusion:** The PSM corrected images enable the evaluation of pixel response independent beam profile. EPID Pixel response has a small dependence on energy. Overall, the EPID pixel response was found to be very stable over a relatively long period.

PO-GePV-T-302

API Assisted Treatment Planning System Validation for Medical Linear Accelerators

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Purpose: The characterization of the dosimetric properties of a medical linear accelerator (linac) in radiation therapy is a critical physics task prior to the linac's clinical use. Vendor provided application programming interface (API) access can assist in the efficient collection of validation data. **Methods:** Over the past year, two linear accelerators, TB1 and TB2, were commissioned with the aid of custom software assistance. The beam data validation was assisted by a software application that accesses calculated treatment planning system (TPS) percent-depth-dose (PDD) and profile data for electron and photon beam qualities. Data was also collected and analyzed for stereotactic conical collimators (SRS CC) and electronic portal imaging device images. **Results:** Gamma pass rates of scanned beam commissioning data were above 94% using a 1% dose-difference (DD) and a 1mm distance-to-agreement (DTA) for PDD and 2% DD and 2mm DTA for profiles for photon beams. Electron pass rates for PDD and profile data were above 98% for TB1 and 92% for TB2 using 2%/2mm and 3%/3mm for DD/DTA for PDD and profiles, respectively. SRS CC output factors for all cone sizes were within 2% for all cone sizes and tissue maximum ratio point data were within 3% for TB1. Validation of image prediction algorithms were performed with a 2D gamma image comparison with 2%/2mm DD/DTA and an average pass rate of 99.42% for over 42 acquired images. **Conclusion:** API driven analysis tools can assist commissioning physicists in the efficient and precise collection of beam scan data, calculated dose volumes, and dosimetric images. Thousands of data points and dozens of images can be analyzed while simultaneously measuring other energies and fluence modes. From the experience of commissioning two linacs throughout the past year, TPS validation scripts assist in gathering the dosimetric data needed to confidently release a linac into clinical service.

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PO-GePV-T-303

Independent Evaluation of MPC Using Winston-Lutz Test

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Purpose: The Varian Machine Performance Check (MPC) system is a QA tool used to quickly test the functions and performance of the Varian TrueBeam and Halcyon. This study was to evaluate MPC with geometry test independently using the Winston-Lutz test. **Methods:** MPC was performed on the Varian TrueBeam using an IsoCal phantom sitting on the six-degree-freedom treatment couch. Beam isocenter size/location, Isocenter coincidence, collimator rotation offset, gantry position accuracy, couch accuracy, MLC leaf position (inner leaves), and Jaw position are checked by MPC. The Winston-Lutz test (WL) was conducted with a 10x10x10cm³ plastic cube. A small metal ball (BB) sits in the center of the cube. The cube was set on the couch with the BB located at the isocenter. The WL results were analyzed using the Pylinac Winston-Lutz program. **Results:** MPC performed six tests with the IsoCal phantom aligned with the isocenter, and three tests with the IsoCal phantom off the isocenter. Correspondingly, nine WL tests were performed. The average isocenter diameter measured by WL was 0.33 mm, and 0.35 mm by MPC with the IsoCal phantom aligned with the isocenter, and the difference was 0.02 mm. With the phantom off isocenter, the isocenter diameter was 0.32 mm in average for both WL and MPC. **Conclusion:** MPC tests of the isocenter and couch are highly accurate. The accuracy of tests are not affected by phantom setup uncertainty and thus reliable. It is comparable with WL test in accuracy for isocenter test and also a convenient and efficient tool for SRS/SBRT QA.

PO-GePV-T-304

Quantification of Peripheral Dose in the New Ring-Gantry RefleXion X1 Radiotherapy Machine

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Purpose: In radiation treatment delivery, scattering and secondary particles exist far from the treatment target region, contributing unaccounted doses to the patient. This study aims to quantify the peripheral dose in the RefleXion™ X1 radiotherapy machine. **Methods:** Measurements were carried out with an Exradin-A14SL microchamber inside a solid-water phantom in the RefleXion X1 machine under the conditions of a flattening filter-free 6MV photon beam. Peripheral dose was measured at 5 points (2, 4, 6, 8, and 10 cm) from the edge of the fields in the IEC-X direction at 85 cm SSD for 10,000 MU. The fraction of dose was calculated using the CAX measurement as a reference. In addition, the dose was measured at 1.5, 5, and 10 cm depths in the solid-water phantom for beam segments defined with a 2 cm fixed jaw opening in the IEC-Y direction and 2.5, 5, 7.5, and 10 cm multi-leaf collimator (MLC) opening in the IEC-X direction. **Results:** Peripheral dose for the RefleXion X1 machine was 0.39%, 0.55%, 0.69%, and 0.79% of the CAX dose, respectively, for the corresponding 2.5, 5, 7.5, and 10 cm MLC openings at a 1.5 cm depth in the phantom at a 2 cm extended distance from the field edge. The peripheral dose increased with depth from 75 cGy at 1.5 cm to 146.24 cGy at 10 cm depth for 10 cm MLC x 2 cm jaw field size. Also, radiation dose decreased with larger extended distances from the field edge: At 10 cm off-axis from the field edge, the ion chamber measurement was 0.11%±0.02%, 0.17%±0.07%, 0.20%±0.06%, and 0.23%±0.08% of the CAX measurement for the corresponding 2.5, 5, 7.5 and 10 cm MLC openings, respectively. **Conclusion:** This study demonstrates that the RefleXion X1 machine's peripheral dose is low, even in scenarios where the scattering effect is pronounced.

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PO-GePV-T-305

Efficient Beam Modeling of a Linear Accelerator with the Reference Data Set to Enable a Faster Clinical Go-Live

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Purpose: An ultraefficient program that optimizes linac commissioning and beam modeling times to enable a faster clinical go-live using golden beam data (GBD) is proposed. The aim of this paper is to address this issue and quantify the dose variation using commissioning data. **Methods:** Two Elekta Infinity linacs with agility heads were installed and matched with the reference data set in our institution. Then, a customer acceptance test (CAT) and model validations were performed on measured percent depth doses (PDDs) and profiles for open-field point-doses in a 3D water phantom. Gamma analysis was used to assess the agreement for these two machines. Before clinical use, intensity modulated radiation therapy (IMRT) and volumetric modulated arc therapy (VMAT) plans were used to test the validity of the two beam models. **Results:** For comparing Monaco-calculated PDDs and profiles with measured data, gamma index values greater than 93.3% above 1 were observed for both beam models. A total of 96.8% and 97.2% open field point doses of TPS calculations were within the 2% tolerance for linac 1 and linac 2, respectively. Electron beam TPS calculations of R50 and R90 in a homogenous water phantom yielded results that were within 3 mm at all evaluated points. Output factors derived from TPS-calculated dose values for field sizes greater than 6 × 6 cm² agreed to within 3% for all energies, cones, and cutout sizes tested. Comparisons between TPS calculated plans and Delta4 measurements and gamma analysis at 3% and 3 mm yielded passing rates of greater than 90% for all plans measured. Point-dose agreement was within 3% for all plans. **Conclusion:** The proposed method has the potential to provide treatment flexibility and to simplify the linac commissioning procedure, save time and manpower and increase the accuracy of the commissioning and annual QA processes.

PO-GePV-T-306

Streamline the Annual Relative Dosimetry QA Check with MapCheck

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Purpose: The study goal is to streamline the annual relative dosimetry QA checks to increase efficiency while maintain precision. The performance of MapCheck was evaluated against ion chamber (IC) and razor diode for various measurements. **Methods:** The scope of annual Linac QA requires the checking of a large number of beam parameters. The possible use of a MapCheck diode array to perform multiple annual tests was investigated in order to determine if time required to perform the tests could be reduced, and that results of the tests were consistent with those obtained using a single ionization chamber and a razor diode. **Results:** The relative standard deviation from repetitive MapCheck point dose measurements across all beams was 0.08% on average with the maximum of 0.79%. In photon checks, MapCheck provided the same measurement uncertainty as IC for dose linearity, dose rate linearity, output factor for ≥3x3cm field, and enhanced dynamic wedge output. MapCheck measured output factor for small field was <2% different from a razor diode for 3x3cm down to 1x1cm square field. The beam flatness measured by MapCheck yielded >2% error comparing to IC. For electron beams, cone factors measured by MapCheck had upto 1.42% uncertainty. Based on such observations, four QA plans were designed to cover all annual relative dosimetry checks: plan 1: photon beam measured by MapCheck; plan 2: photon beam measured by IC; plan 3: electron beam measured by IC; and plan 4: electron beam measured by MapCheck. Comparing to doing QAs in service mode with IC, it can potentially reduce the total QA time by half. **Conclusion:** MapCheck demonstrated good quality as an IC substitute for relative dosimetry QA task except for absolute beam flatness and electron cone factor measurements. A streamlined QA scheme was

developed by combining MapCheck and IC to achieve high efficiency.

PO-GePV-T-307

Validation of Non-Isocentric Delivery and Analysis of VMAT QA Plans Using ArcCHECK

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Purpose: To validate non-isocentric delivery and analysis of VMAT QA plans using ArcCHECK. **Methods:** ArcCHECK (Sun Nuclear Corp., Melbourne, FL) is routinely used in rotational delivery for patient pre-treatment QA. A CavityPlug insert with ion chamber is used for isocenter dose measurement. Plans with high dose targets not at isocenter typically require one delivery with isocentric setup for gamma analysis and another with ArcCHECK shifted to position the ion chamber at a high dose region for dose measurement only. The shifted non-isocentric ArcCHECK measurements, particularly in transverse directions, are not standard and not recommended by the manufacture for gamma analysis. In this study, we demonstrated and validated non-isocentric delivery and analysis of VMAT QA plans using ArcCHECK. Ten clinical VMAT plans with the isocenter outside high dose targets were used for the study. All plans were generated in Varian Eclipse TPS with HD120 and Millennium 120 MLCs using 6FFF, 6X, 10FFF and 10X photon energies. For each clinical plan, two QA plans were created, one without and one with shifts from the ArcCHECK isocenter. QA plans delivered without shifts were used for gamma analysis as baseline. QA plans delivered with shifts were used for both absolute dose measurement and gamma analysis. The average gamma passing rates for non-isocentric delivery were compared with the corresponding results of isocentric delivery. **Results:** With 10% threshold and local 3%/3mm passing criteria, the average gamma passing rate for non-isocentric and isocentric deliveries were $(98.7 \pm 1.4)\%$ vs. $(97.0 \pm 2.4)\%$, respectively. With 3%/2mm passing criteria, they were $(95.2 \pm 4.0)\%$ vs. $(90.2 \pm 5.7)\%$, respectively. The absolute dose measurements of the non-isocentric deliveries were all within $\pm 3\%$ of the calculated dose. **Conclusion:** The non-isocentric delivery and analysis of VMAT QA plans using ArcCHECK were demonstrated and validated for patient pre-treatment QA.

PO-GePV-T-308

On the Performance of LINAC Annual Quality Assurance Without the Need for a 3D Water Tank

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Purpose: 3D water tanks are utilized to acquire data as part of the annual quality assurance on linear accelerators. A 1D water tank alone or in combination with a modern array detector may offer an attractive alternative in terms of cost and expeditious results. In this work, we compared aspects of our annual QA acquired with and without the use of a 3D water tank. **Methods:** The 3D water tank used was Standard Imaging's DoseView 3D and the alternate equipment was Sun Nuclear's 1D Scanner as well as the Ion Chamber (IC) profiler. With the IC profiler's inherent depth of 0.9cm, additional water slabs were used as needed to acquire profiles for photons (6MV, 6FFF, 10MV, 10FFF, 15MV), and electron energies (6e, 9e, 12e, 15e, 18e) for different field sizes and different depths. Furthermore, by irradiating the 1D tank from the side, that is a gantry of 90 (or 270) with the tank atop the couch at 100cm SSD to the side, we also acquired profiles using the 1D tank alone. These results were compared with those acquired using the 3D water tank. **Results:** We observed excellent agreement between the 3D tank versus the IC profiler with a measured discrepancy within 1% for all photons/electrons profile flatness and symmetry and within 2% for all electron flatness. The calculated Beam Profile Constancy for the 6MV, 10MV and 15MV beams between the 3D and 1D water tank were excellent, observing 0.49%, 0.44% and 0.79% respectively for the 10x10cm field and correspondingly 0.38%, 0.54% and 0.62% for the 20x20cm field. **Conclusion:** A 1D scanner and a detector array produce comparable profile results vis-a-vis a 3D scanner and therefore offer an attractive alternative. The limitation is with the maximum field size attainable as well as the range of depths achievable using a 1D scanner.

PO-GePV-T-309

Small Field Dosimetry of a Novel Biology-Guided Radiotherapy System

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Purpose: A biology-guided radiotherapy (BgRT) system - Reflexion™ X1 was developed and equipped with PET imaging. This study reports the small field measurement results of the first clinical BgRT machine at Stanford. **Methods:** The X1 linac produces 6MV flattening filter free beam and consists of a binary multi-leaf collimator (MLC) system with sixty-four 6.25mm-thick leaves and y-jaws switching between 1cm and 2cm. All clinical beams are delivered with the narrow longitudinal beam geometry which resulted in a lack of charged particle equilibrium. Therefore, it is critical to ensure the accuracy of small field dosimetry for the system. We performed profile scans and relative output factor measurements of field sizes from 1.25x1cm² to 40x2cm² using a diode detector in a 3D water tank during the commissioning process. Subsequently, we repeated the tests using diode, film, and tissue-equivalent W2 scintillator in plastic water focusing on small fields down to 0.625x1cm². **Results:** For the fields with 2cm Jaw, the mean differences of relative output factors measured using diode and scintillator were 0.57% and a maximum of 2.79% for the 0.625x2cm² field. The film results agreed better with the scintillator, with mean and max differences of 0.2% and 0.7%. For the fields with 1cm Jaw, the mean differences of relative output factors measured using diode and scintillator were 1.35% and a maximum of 3.15% for the 0.625x1cm² field. The film results maintained good agreements to the scintillator with mean and max differences of 0.44% and 0.99%. The FWHM differences calculated from the longitudinal and transverse profiles measured with diode and film were within 0.5mm across the depth from 1.5cm to 20cm. **Conclusion:** This evaluation represents the first small field dosimetry of the clinical BgRT system serving as a future reference.

PO-GePV-T-310

A Novel Method to Verify Deformable Dose Accumulation for Patients Treated with Helical Tomotherapy

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Purpose: To develop an independent dose mapping program for verification of adaptive radiotherapy (ART) for patients treated with helical tomotherapy. **Methods:** An in-house developed ART platform (Hertz) was adapted to verify dose accumulations performed in a commercial ART system (PreciseART, Accuracy Inc.). Specifically, in PreciseART, deformable image registrations were performed from daily MV-CTs to a planning kV-CT. The resultant displacement vector fields (DVF) were used to map doses from the MV-CTs to the kV-CT. All the CTs, doses and DVFs were exported to Hertz. In Hertz, HU-to-mass density tables for kV and MV-CTs were used to convert HU and dose to mass density and energy. DVFs were used to generate their pseudo-inverse DVFs. An energy-and-mass congruent mapping (EMCM) method was implemented to reconstruct the deformed doses on the planning CT using the pseudo-inverse DVFs. The doses reconstructed by the two methods were compared to verify the accuracy of dose accumulation in PreciseART. The doses reconstructed by EMCM and PreciseART were converted to energy using the mass densities of the daily MV-CT and planning kV-CT, respectively. Energy distributions derived from the planned, delivered, and reconstructed doses were compared with each other to detect potential DVF errors, mass changes or imaging artifacts in each organ. The program was tested with the data acquired from three tomotherapy patients. **Results:** The dose and energy reconstructed by PreciseART and Hertz were compared within six representative structures. The dose difference averaged over all the structures is $2.7 \pm 1.7\%$ and energy difference is $5.4 \pm 0.8\%$. Compared to the energy on the daily images, the reconstructed energy on the planning images changed by $4.1 \pm 0.4\%$ for Hertz and $8.4 \pm 0.0\%$ for PreciseART. **Conclusion:** An independent dose reconstruction program has been developed to verify dose accumulations in PreciseART. This program may also help calculate energy distributions within each organ for root cause analysis.

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PO-GePV-T-311

Evaluation of Out-Of-Field Dose Calculation Accuracy of Monaco Treatment Planning System for a 1.5T MR-Linac

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Purpose: To evaluate the accuracy of the out-of-field dose calculation for a 1.5T MR-Linac using a Monte Carlo fast dose algorithm in the Monaco treatment planning system. **Methods:** Using optically stimulated luminescence dosimeters, 25 checkpoints were measured on surfaces (20 checkpoints) and internal organs (5 checkpoints) of the anthropomorphic phantom for square radiation fields, with similar prescribed doses for the head, thoracic, and pelvic sites. Each measurement was repeated five times, and the average was compared with the calculated dose values of the Monaco treatment planning system. The inverse of the function that describes the out-of-field dose with respect to the isocentric distance from the radiation field was fitted. In addition, the relationship between the dose evaluation error of the treatment planning system and the variation in the isocentric distance from the radiation field is described. **Results:** Among the 25 checkpoints, the thyroid gland received the highest dose. However, the measured dose did not exceed 1.45% of the prescribed dose. The dose measured both on the surface and internal checkpoints of the anthropomorphic phantom for plans with different sites, followed an exponential function of dose reduction over a distance of 10–90 cm from the isocenter of the plan. Compared with the measured data, the results of the Monte Carlo dose calculation algorithm in the Monaco treatment planning system underestimated the out-of-field checkpoints dose by 0–35% in an isodose area between 2% and 5% and overestimated the out-of-field checkpoint dose by 0–270% in an isodose area between 0% and 2%. **Conclusion:** In an isodose area below 5%, the out-of-field dose evaluated through the fast Monte Carlo algorithm in the Monaco treatment planning system deviates from the measured dose, and the deviation increases as the distance from the target volume increases.

PO-GePV-T-312

Cross Dose Engine Transformation Based Fast Dose Calculation for Online MR-Guided Adaptive Radiotherapy

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Purpose: To propose a novel magnetic field dose calculation method based on transformation from pencil beam to monte carlo distribution. **Methods:** The novel magnetic field dose calculation algorithm is established by a pencil beam dose engine and a magnetic field influence correction network. The correction network is constructed by a Res-UNet framework, including a residual module and an encoding-decoding path. Beam dose calculated by pencil beam engine without considering magnetic field and by Monte Carlo with magnetic field are generated and enhanced respectively to train the network. 100 clinical treated static beam IMRT plans were collected, including brain, head and neck, lung, and abdominal cancer cases. 80% of them are used to train the model and the other 20% to test. By comparing the predicted dose distribution with the original input PB dose and the corresponding MC dose, the feasibility and effectiveness of the method is evaluated. **Results:** Results of both lung and HN cases show that the predicted dose distribution from the cross-dose engine transformation is very close to MC results, and the profiles are basically consistent, compared to PB dose VS. MC. The average voxel absolute dose error could decrease from 4.2% (PB to MC) to 0.9% (predicted value to MC), and the Gamma pass rate increase from 48.8% to 99.2%. The PB calculation and transformation process can be completed in ms. **Conclusion:** We have successfully proposed a fast and precise magnetic field dose calculation algorithm, with the capability to perform accurate and real-time dose calculation during optimization process for MR-guided adaptive radiotherapy.

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PO-GePV-T-313

Second Dosimetry Check with Deep DoseNet

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Purpose: We propose a second dosimetry check framework based on data collected from model-based calculations or measured beam doses, from which a convolution neural network is trained to perform dose calculation with high accuracy and speed. **Methods:** We first estimate the dose from primary photon transport using the cross-axis profile of field size 40 cm and the PDD for 100 cm SSD and 10 cm beam size corrected by Mayneord factor using radiological distances calculated from raytracing at the 5 mm voxel size. The network, named as Deep DoseNet (DDN), takes the estimated 3D doses and corresponding CT scans as the input and outputs up sampled final doses at 2.5 mm. DDN uses Residual in Residual Dense Block as the backbone where local features learned at each layer are sent to all subsequent layers to promote feature reuse, which is widely used and the state of art in the image super resolution field. For the training, we generated 2000 beams at 6 MV in Eclipse TPS on 20 patient CT scans. The beam isocenter, gantry angle, collimator angle, jaw position and MLC position are all randomly generated. The doses are calculated with the Acuros algorithm at 2.5x2.5x2.5 mm. The mean square error of the DDN converges after training for ~20 epochs using the ADAM optimization algorithm with an initial learning rate of 10^{-4} . **Results:** For 11 randomly selected vmat arc fields on patient CTs which the network hasn't seen before, the average 3D gamma passing rate for 1%1mm is 74.8% (58.6%-88.3%), 2%2mm is 96.5% (93.9%-99.1%) and 3%3mm is 99.5% (98.5%-99.8%). The average calculation time is ~30 seconds with one Nvidia V100 GPU card. **Conclusion:** DDN successfully generalizes from limited training dataset. It can serve as an independent, fast and accurate second dosimetry check tool for VMAT treatments.

PO-GePV-T-314

Validation of ClearCalc Photon Secondary Check Dose Calculation Algorithm

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Purpose: ClearCalc (RADformation, NY) is an independent dose calculation software and has increasingly gained popularity for secondary dose check. This work aims to validate its photon dose calculation algorithm. **Methods:** We first created a 10x10 cm² open-field plan irradiating a 30x30x30 cm³ solid water phantom (SW) in Eclipse treatment planning system (TPS, v15.6), with and without a Styrofoam insert. Both the central-axis and off-axis point doses at dmax, 5cm and 10cm depths were measured with pinpoint ion chamber and compared with those from TPS and ClearCalc, respectively. Subsequently, we replaced the above open-field plan with 20 clinical treatment plans, then measured the isocenter doses with ion chamber, and compared the measurements with the corresponding point doses in ClearCalc. Lastly, we randomly selected 91 clinical plans (25 brain, 10 head&neck, 13 breast, 10 lung, 12 abdomen, 11 pelvis, 10 prostate) and retrospectively compared the calculation point doses between TPS and ClearCalc. **Results:** In the SW, the average central-axis point dose difference between ClearCalc and measurement for the open-field plans was $-0.99\% \pm 0.89\%$ (without the insert) and $-1.21\% \pm 1.34\%$ (with the insert), respectively. The ranges span from -2.41% to 0.34% and from -4.01% to 0.04%, respectively. As to the 20 clinical plans, the average isocenter dose difference between ClearCalc and measurement was $-1.27\% \pm 7.22\%$ with the range spanning from -13.8% to 18.6%. Additionally, 13 out of 20 plans had agreement within 5%. For 91 clinical plans, the average difference between TPS and ClearCalc was $-1.16\% \pm 0.43\%$ and the difference range is (-1.46%, 8.76%). Specifically for each individual treatment sites, the difference was $-0.22\% \pm 0.39\%$ (SRS cone plans), $-0.05\% \pm 1.26\%$ (brain), $0.52\% \pm 1.95\%$ (head&neck),

0.49%±2.59% (breast), 1.90%±1.64% (lung), -0.12%±1.17% (abdomen), 1.14%±1.17% (pelvis), and -0.10%±2.35% (prostate). **Conclusion:** ClearCalc photon dose calculations corroborate relatively well with both the treatment planning system and measurement in most cases.

PO-GePV-T-315

Comparison of Secondary MU Calculation Softwares for Clinical Use: RadCalc and ClearCalc

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Purpose: To compare two MU calculation softwares in their calculation accuracy and automation workflow for secondary MU check of AAA and AcurosXB calculated plans. **Methods:** ClearCalc is an API-based software that integrates to the treatment planning system (TPS). RadCalc is a standalone software equipped with two modes of heterogeneity correction: one based on path length and one performs the correction based on the structure contours (RadCalc-ROI). We developed an API script to eliminate the manual plan export/import and carry over the SSDs and effective-path-length information to RadCalc. The arrival of DICOM files to RadCalc triggers its own automatic calculation and reporting workflow. A total of 30 plans covering various sites were analyzed to test the MU calculation accuracy and workflow. Both AAA and AXB plans were created for each case to evaluate the accuracy of the secondary MU calculation for different algorithms. Planner's user experience survey was performed to score the software/workflow design of the two software. **Results:** The results are similar between AAA and AXB plans. With the finite-sized pencil beam (FSPB) algorithm and patient's CT information, ClearCalc exhibited a better calculation accuracy in general. The frequency distribution of calculation failure (>5%) demonstrated that the RadCalc-ROI module improves the calculation accuracy to a comparable level as ClearCalc. For the lung site, all methods failed with very large difference (40-80%) which evidences that these softwares are not best suited for lung SBRT plans. ClearCalc provides an intuitive user interface and seamless process to switch calculation point while RadCalc offers automatic report printing and the calculation failure warning/stop-sign to prevent users from blindly passing a failed plan. **Conclusion:** Both systems offer accurate secondary MU calculation. The integration with TPS via API minimizes the manual steps and ensures a safe workflow.

PO-GePV-T-316

Modulation Area Index (MAI): A Plan-Specific Metric for Quantifying VMAT Modulation Complexity

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Purpose: The constraint on a minimum segment area in VMAT optimization is utilized to restrain small MLC apertures due to limited accuracy on TPS's small-field output factor modeling. Since this constraint is applied to a total MLC aperture area per control point (CP), it does not effectively prevent small island apertures, e.g., island aperture area<1cm². We quantify the amount of these apertures in clinical plans as the first step for investigating their dosimetric and delivery impact. **Methods:** 21 head-and-neck, 21 pelvis-node, and 21 prostate-bed clinical VMAT plans are selected, which are optimized with Elekta Agility MLC in Pinnacle³. Each DICOM RT-plan is exported from TPS for analysis. A CP may have multiple unconnected, island apertures. A table is created for every CP to record the MLC parameters of each island aperture: index of leaf pair(s) forming the aperture, leaf positions, and aperture area. The process is performed for all CPs. Finally, Modulation Area Index, MAI(r), is defined as a cumulative histogram of these island-aperture areas, r, calculated from all CPs and normalized by the total MLC opening area. The MAI(r) quantifies as a percentage of the island aperture for r<0.1cm², r<0.8cm², and r<0.6cm². **Results:** Average MAI(r<1cm²) is 3.4%, 1.3%, and 0.3% with maximal 8.22%, 2.67% and 1.6% for the head-and-neck, prostate-node, and prostate-bed plans; for r<0.8cm², average MAI is reduced to 2.3%, 0.8%, and 0.2%; when r is further decreased to <0.6cm², MAI is still non-zero as 0.6%, 0.1%, and 0.1%. Statistical significance, p<0.05, is observed with the three disease sites. **Conclusion:** We propose a plan-specific metric quantifying VMAT modulation complexity: optimization for head-and-neck produces the most island apertures, while prostate-bed produces the least. Future study is to investigate the impact of these apertures on the passing rates of patient-specific QA and TPS's dose calculation accuracy on target coverage.

PO-GePV-T-317

The Impact of Flattening Filter Free on Remaining Volume at Risk in Lung Cases for VMAT and IMRT Techniques

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Purpose: Flattening filter free beam is a modern modality used in new radiotherapy machines, especially in SBRT cases to consuming treatment time. The remaining volume at risk (RVR) is known as undefined normal tissue, and it assists in evaluating very late effects like carcinogenesis. The present study is aimed to compare between the effect of flattening and unflattening 6MV beams on (RVR) in the lung cases that treated by conventional dose for volumetric modulated arc therapy (VMAT) and Intensity modulated radiation therapy IMRT. **Methods:** Eleven lung cancer cases with prescribed dose 60Gy in 30 fractions were selected retrospectively for this study. Four plans were generated for each case (VMAT FF, VMAT FFF, IMRT FF and IMRT FFF plan). Plans are developed using Eclipse treatment planning system and Truebeam linear accelerator HD MLC. Two arcs for VMAT plan and five coplanar fields are selected according to lesion volume and location for IMRT plans. RVR mean dose, V15Gy, V10Gy and V5Gy are introduced in this study as RVR evaluation parameters. **Results:** The study revealed that the variance percentage comparison between FFF and FF for the RVR evaluation parameters are (2.95, 1.1, 1.95 and 2.6), (1.95, 0.9, 1.6 and 2.15) for VMAT and IMRT respectively. However, for the comparison between VMAT and IMRT in RVR evaluation parameters study shows that variance percentage is (1.74, 2.06, -7.8 and -4.8), (2.86, 2.28, -7.2 and -4.4) for FF and FFF respectively. **Conclusion:** The study present that using FFF is clinically benefit than FF for all RVR evaluation parameters in addition to consuming time. On the other hand VMAT is preferred in RVR mean dose and V15Gy while oppositely in the V10Gy and 5Gy.

PO-GePV-T-318

Field Junction Technique for Total Body Irradiation Using Tomotherapy with TomoDirect Mode

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Purpose: The present study aims to verify the dose distribution of the field junction region using a volumetric gradient matching technique (VGMT) with TomoDirect (TD) mode. **Methods:** Two planning CT image sets were acquired with the head first position (HFP) and feet first position (FFP) using two solid water phantoms mimicked the thighs. Planning target volume was divided into upper, lower and gradient volumes. The junction region was comprised of one 2.5 cm and four 2 cm thick gradient volumes to create descending slope dose distribution. TD plans were generated with 12 and 4 ports for HFP and FFP plans, respectively. Gafchromic film placed at the central coronal plane and measured dose profile was obtained. Dose verification, especially in the junction area, was performed by shifting deliberately the position of the FFP plan by 0.5, 1.0, and 1.5 cm in the superior and inferior direction. Absolute dose measurements were performed at 3 points for respectively HFP and FFP plans using ionization chamber at the matching plane. **Results:** The TD plans produced a linear gradient in junction region. An optimum alignment of both HFP and FFP leads to a relatively constant dose profile. Relative dose deviations via film dosimetry of 6.5%, 11.1%, 17.2% and 11.7%, 15.1%, 16.9% were observed for 0.5, 1.0, and 1.5 cm setup errors in the superior and inferior direction, respectively. The absolute measured dose showed a good correlation with the calculated dose in the gradient junction except for 1 out of 6 measured points. The average percent difference (±SD) in five points was +1.9 ± 2.1%. **Conclusion:** VGMT with TD enables to create a linear dose gradient across the junction region and reduce the impact of longitudinal setup error on dose distribution.

PO-GePV-T-319

Commissioning and Clinical Implementation of Pseudo 4Pi On An MR-Guided Radiotherapy (MRgRT) Linac

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Purpose: The maximum field size on a commercial MR linac is 27x24cm. In the event that a treatment volume exceeds the field dimensions, a multi-isocenter technique can be employed to elongate the total treatment coverage area. We commissioned and clinically implemented a Pseudo 4Pi workflow on an MRgRT linac. **Methods:** An eight-coplanar beam, dual isocenter step-and-shoot IMRT plan with 6cm displacement of isocenters across the IEC-Y axis was created. The plan was delivered to a 3D IMRT diode array phantom. In order to treat each isocenter, the treatment couch moves between the isocenters during the treatment workflow. A second plan was developed to perform an end-to-end validation in treatment workflow on a cylindrical water phantom with an A1SL ion chamber in the region of the field junction. The chamber dose was compared between the calculated TPS (i.e., mean dose of A1SL active volume ROI) and the measured dose. Additionally, treatment interrupts with IGRT shifts were performed to validate the couch localization accuracy in the Pseudo 4Pi workflow. **Results:** The gamma analysis using 3%/2mm criteria resulted in a passing result of 98%. There was no observed overlap or separation at the dosimetric junction on the diode array measured dose profiles. For the ionization chamber absolute dose measurement, the difference between the measured (10.21±0.06Gy) and calculated (10.10±0.20Gy) dose was 1.04%. The treatment time was twice as long due to the doubling of treatment beams for the two isocenter approach. The couch localized within ±0.1mm during interrupts irrespective of positioned isocenter at respective delivery. **Conclusion:** There was no overlap visible in the measured profiles and the difference between the expected and measured dose to the ion chamber was less than 5%. The accuracy of Pseudo 4Pi planning and execution make it a clinically acceptable technique when target volumes exceed the field size dimensions.

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PO-GePV-T-320

Dosimetric Analysis of ElastoGel Bolus

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Purpose: ElastoGel Bolus has been gaining increasing popularity. It has a density of 1.2g/cm³, instead of the traditional 1.0g/cm³. This difference in density will have an effect on treatment beams traversing the bolus, especially electron beams. This study aims to quantify the difference for typical electron beam energies. **Methods:** Two methods were used to quantify the dose difference caused by the higher density of ElastoGel. First, solid water phantoms were scanned with ElastoGel and Superflab boluses with thicknesses of 0.3, 0.5, and 1.0cm. Treatment plans with a 10x10cm² field, 100MUs and 100cm SSD to the bolus surface were calculated for 6, 9, 12, 16, and 20MeV electron beams. A dose profile was taken from these beams along the central axis and 2cm off-axis. Second, 9 plans, previously treated using ElastoGel were retroactively analyzed. The bolus depths included 0.3, 0.5, and 1.0cm. The electron energies were 6, 6 and 9, 9, 12, and 16MeV. For each plan, the patient was scanned with no bolus and the bolus was then added in the treatment planning software. To simulate Superflab, the bolus was assigned a Hounsfield Unit of 0. For the purposes of this study, a PTV was created to match the 90% isodose volume for the plan. The bolus was then changed to simulate ElastoGel, with 300HU. PTV coverage was compared using mean dose, D_{max}, V5, V20, V80, V90, V95, and V100. **Results:** The largest difference was seen for 6MeV with 1cm of bolus. For the solid water plans, past the buildup within the bolus, this combination had a dose difference of 0.3%. For patients, V95 and V100 changed by up to 4.2%. All other metrics changed by less than 1%. **Conclusion:** ElastoGel Bolus presents little dosimetric influence and is not a risk if the density is assigned incorrectly.

PO-GePV-T-321

Validation of Spatial Fractionated GRID Treatment Planning Using Volumetric Modulated Arc Therapy (VMAT)

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Purpose: To validate a novel and practical volumetric modulated arc therapy (VMAT) planning method for Biology Spatial Fractionated (SF) therapy (GRID). **Methods:** Spherical contours measuring 1.5 cm in diameter are distributed in a bulky tumor ranging in sizes from 500 to 3000 cm³. Each contour must be at least 3 cm apart, center to center and at least 1 cm from any organ at risk (OAR). The spheres are then grouped together and planned as one target. Three concentric ring structures (0.5, 1 and 3 cm) are created to optimize the dose to the center of the spheres and minimize the dose surrounding the spheres. The result is alternating regions of high and low dose to the tumor and minimal dose to the OAR's. The number of spheres varied from 12 to 30 depending on the size of the tumor. High dose rate filter-free (FFF) beams are used to deliver the entire treatment, typically in less than 20 minutes. **Results:** The validation is performed on 5 previously treated bulky liver or stomach patients using anonymized data. The spheres were placed by expanding the ROI size to 3 cm arranging them to not overlap, until the tumor was completely filled. Once the tumor was filled, the sphere size was reduced to 1.5cm diameter. The planning was optimized using similar objectives as other VMAT multi-target SRS planning techniques and then delivered on a Varian TrueBeam STX. Dose measurements were collected using a standard multidetector planar array. The treatments varied from 2-8 arcs. All five treatment plans passed 95% gamma at 3%/3mm 10% threshold. **Conclusion:** We have validated a novel approach to planning GRID therapy patients. VMAT can be used to treat bulky tumors in a SF approach both as a single treatment or as a boost, without sacrificing dose to surrounding tissue.

PO-GePV-T-322

Treatment Planning and Verification for Radiodynamic Therapy Using 45 MV Beams

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Purpose: Radio-dynamic therapy (RDT) uses a combination of high-energy photon beams and photo/radio-sensitizers to kill cancer cells. RDT treatment plans were generated following our institutional protocol for a phase I study on RDT. In this work, we present the experimental and theoretical studies that were conducted to insure accurate dose delivery for RDT. **Methods:** A LA45 RaceTrack Microtron was commissioned for RDT. LA45 uses scanning electron beams to produce uniform photon dose distributions and a purging magnet to remove the contaminant electrons. Purging efficiency was tested as it could influence machine calibration. The Eclipse system was used for treatment planning. Monte Carlo simulation was used to model 45 MV beams to serve as a second check for RDT. Matrixx 2d array was used treatment plan QA check. Initial trials of portal imaging with the 45MV beam using humanoid phantom showed blurry images, which was attributed to scattered electrons generated by the patient. A 5-mm copper build-up plate was therefore used to insure optimal image contrast and film exposure. **Results:** Purging magnet was found to be functional as was seen by the comparison between percentage depth curves of a 10cm x10cm field with and without the purging magnet. The machine was calibrated following TG51. A comparison between the calculated and measured dose distributions was made resulting in 95% of pixels analyzed passing our acceptance criteria. This was achieved with our phantom studies as well as with patients treated in the phase I trial. Portal images using 45 MV photons showed better image quality compared to lower energies with the copper buildup. **Conclusion:** Treatment plans for RDT were designed to facilitate post treatment PET imaging, which were used to dose delivery verification and treatment assessment, and were successfully delivered to patients.

PO-GePV-T-323

Dosimetric Impact of Materials Override of Hydrogel Spacer in Stereotactic Body Proton Therapy (SBPT) for Locally Advanced Pancreas Cancer (LAPC)

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Purpose: To quantitatively assess the dosimetric impact of three materials override methods for injectable iodinated hydrogel spacer utilized in stereotactic body proton therapy (SBPT) for pancreatic cancer. **Methods:** TraceIT is an absorbable, iodinated polyethylene glycol (PEG) hydrogel for tissue marking and spacing. A patient diagnosed with locally advanced pancreas cancer (LAPC) underwent endoscopic hydrogel injection and CT simulation with breath-hold. The CT with patient and spacer contours was imported to Raystation 9B for proton pencil beam scanning (PBS) planning. The patient was prescribed with 3300 cGy (RBE) in 5 fractions to the GTV. Two posterior beams were used, one of which has a distal end close to the spacer. The ground truth of the spacer composition was calculated based on previous publication and imported to the Raystation. SBPT plans with the same beam but different material override methods for spacer were recalculated with Monte Carlo algorithm: 1) HC plan assigning spacer with ground truth material; 2) W plan with spacer assigned with water; and 3) CT plan with CT Hounsfield Units of spacer used in planning. The target and organs-at-risk doses and Gamma index of 2 mm / 2% were used to quantify the dosimetric differences. **Results:** All three plans achieve the same target coverage with D98 around 3316 cGy (RBE). The mean doses to the duodenum are 1364 cGy (RBE), 1359 cGy (RBE), and 1364 cGy (RBE) for plan 1), 2) and 3), respectively. For stomach, the mean dose is 256 cGy (RBE), 249 cGy (RBE) and 255 cGy (RBE), respectively. The Gamma index analysis confirms the plan differences for plans 1) and 2), and plans 1) and 3) are minimal. **Conclusion:** With TraceIT as a spacer for proton dose planning, we found no significant dosimetric difference existing among the plans using three override methods

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PO-GePV-T-324

Quantifying the Dosimetric effect of Tumor Treating Fields (TTF) Transducer Arrays: Is a Correction Needed?

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Purpose: Current studies report the protocol for the use of TTF in treatment of glioblastoma as CT simulation using a foam pad as a surrogate for the high-density TTF transducer array followed by the delivery of external beam radiotherapy with the array in place. This work assesses the need for a dosimetric correction factor using these conditions. **Methods:** Measurements were taken on an Elekta VersaHD accelerator with and without the transducer array for depths of 0–20 cm using solid water slabs for the following energies: 6 MV, 10 MV, 18 MV, 6 MV FFF, and 10 MV FFF. Surface measurements were taken with OSLDs, measurements up to 1 cm were taken with a parallel plate chamber, and measurements 1 cm and greater were taken with a cylindrical chamber. For each depth and energy, 100 MU was delivered with a 10x10 cm² open field at 100 cm SAD, and ratios of the readings with the array to readings without the array were determined. **Results:** Ratios ranged from 0.96–2.35 for 6 MV, 0.97–3.34 for 10 MV, 0.97–4.08 for 18 MV, 0.96–2.43 for 6 MV FFF, and 0.96–2.87 for 10 MV FFF. The deviations between readings with and without the transducer array were greatest for shallow depths for each energy. The ratios decreased with increasing depth throughout the build-up region before stabilizing near d_{max} for each energy. **Conclusion:** Utilizing lower energy beams for radiation treatments with TTF transducer arrays will reduce the discrepancies in the build-up region between dose calculated without the array and dose delivered with the array in place. It is recommended to apply a correction factor for targets located at shallow depths, as differences for measurements taken with and without the transducer arrays were most pronounced for depths shallower than d_{max} for each energy.

PO-GePV-T-325

Reducing Delivery Time of 3D DIBH Breast Treatments Using a Sliding Window Technique Over Step and Shoot Field in Field

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Purpose: A concern over the time taken to deliver DIBH left breast tangent treatments warranted us to investigate methods to reduce delivery times by switching from a field in field delivery to a sliding window technique. **Methods:** DIBH left breast patients treated with static tangents using a step-and-shoot, field in field, forward planned technique were retrospectively replanned and compared using the sliding window tangents. For an accurate comparison of the difference in delivery times, the replanned treatments were created to the same clinical quality as the original plans. Beam energy, gantry, collimator and blocks were not changed when the plans were redesigned using sliding window. Both the original and new plans for each patient were then delivered on the machine with no gating and the delivery times were compared. **Results:** From the delivery time statistics gathered, initial results suggest that a sliding window delivery may reduce delivery times by 10% or more over field in field for breast treatments due to not having to stop the beam while the MLCs move. It was found though that the sliding window plans have slightly more 105% dose in the breast as well, but that may be simply an issue of plan quality which could be improved by replanning again, rather than a significant difference between the two different delivery techniques. **Conclusion:** Given that sliding window delivery takes less time than step and shoot field in field, this indicates that sliding window may be the preferred method specifically for DIBH left breast patients. Many patients have difficulty holding their breath for an extended period of time, so any reduction in delivery time will greatly increase comfort for these patients. This should also come with no detriment to the patient as it seems plans of nearly identical quality can be produced with either method.

PO-GePV-T-326

An Investigation into the Dosimetric Properties of a Three-Dimensional (3D) Printing Material for Potential Use in Radiation Therapy

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Purpose: To measure and evaluate the dosimetric properties of a three-dimensional (3D) printing material, polylactic acid (PLA), for use in radiation therapy. This study evaluates the effect of several different infill percentages as well as the effect of infill pattern on the dose delivered to a phantom to establish reasonable guidelines for 3D printed bolus usage. **Methods:** Six slabs of PLA, each with dimensions 5x5x1 cm³, were designed and printed. The cubic infill pattern was used for five slabs with infill percentages of 25%, 50%, 67%, 75%, and 100%. One slab with 50% infill and the tri-hexagonal pattern was printed. Preliminary dose calculations were made in RayStation, using two dose specification points, one at the surface between the slab and the backscatter material, and one at a depth of 1.5 cm into the backscatter material. The dose at each point was recorded from delivering 100 monitor units (MUs) to each slab material and density, with each photon energy, and for both 5x5 cm² and 10x10 cm² field sizes. **Results:** The results present the ratios of calculated dose under the PLA bolus material to the dose with either no bolus (for data at 1.5 cm depth) or a water bolus (for data at the surface). The results show that the dose was consistent at 1.5 cm depth for multiple slabs with 10 MV energy, while 100% and 75% infill slabs were consistently similar to the water bolus. **Conclusion:** PLA is an adequate replacement for traditional bolus material, and 3D printed, patient specific bolus materials could significantly improve patient outcomes. Additional studying of how the measured dose and the spatial dose distribution are affected by the choice of infill, as well as the effect on electron dose distribution, are ongoing.

PO-GePV-T-327

Volumetric Axillary Boost (VMAB): Novel Planning for Nodal Coverage in Breast/Chestwall Patients

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Purpose: To demonstrate the superior quality of the dose distribution given by a novel axillary boost planning technique for nodal coverage during breast/chestwall radiotherapy. **Methods:** Six breast cancer patients previously planned with the traditional monoisocentric posterior axillary boost (PAB) technique were replanned with a volumetric axillary boost (VMAB) to complement the current 3D tangent and supraclavicular (SCLV) fields. The at-risk nodal volumes had been delineated during the planning process as well. The dose distributions were then compared for major differences with the key endpoints being the nodal region dose volume histograms (DVH), ipsilateral lung DVHs, and maximum dose hotspot within the plan. The original PAB plans were designed balancing the tradeoff of nodal dose coverage and overall hotspot while the VMAB plans optimized both. The ideal hotspot was considered < 110% of the whole breast/cw dose and the ideal nodal coverage was considered D95 = 95-100% of 45-50 Gy depending on the original patient prescription. **Results:** Due to the limited sample size, no statistically significant differences were observed between the traditional 3D PAB technique and VMAB. Two statistically significant differences ($p < 0.05$) were observed for lymph node dose coverage for level 1 V95% and level 2 V100%, where the VMAB increased coverage by a mean value of 14.9% and 30.3%, respectively. Level 2 V95%, was not significant ($p = 0.053$) with a mean increase of 20.7%, but may become significant with a higher sample size. Differences in maximum dose, level 3 lymph node dose coverage, and lung doses were not found to be close to significance. **Conclusion:** This novel planning technique, VMAB, results in significantly improved dose coverage of nodal regions while avoiding increase of undesirable hotspots and/or lung dose. Though implemented clinically, future work will confirm these findings with a larger sample size to discern true statistical certainty.

PO-GePV-T-328

Development of a Collision Prediction Tool Between Gantry and Table Using Blender

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Purpose: Non-coplanar treatment techniques such as dynamic trajectory radiotherapy (DTRT) have the potential to improve treatment plan quality in comparison to conventional volumetric modulated arc therapy. However, collision avoidance must be considered during treatment plan creation. In this work, we developed a detailed collision-prediction tool for gantry and table-top or table-base, considering possible six degree of freedom (DoF) set-up table correction. **Methods:** A virtual linac model was created using blender, a free and open-source 3D computer graphics software toolset, that can be accessed via an integrated python API. DTRT plans described with XML files can be simulated on the virtual linac and tested for possible collisions between gantry and table for a given table position. If an additional tolerance table is provided, all combinations of maximal setup shifts given in the tolerance table are tested for potential collision interlocks. Collision prediction was verified experimentally by setting the gantry to a certain angle and then translating or rotating the table until a collision interlock occurred. The resulting table angles and positions were compared to the predicted values. This was repeated with additional 3° pitch and roll rotations in order to verify the model for setup correction. **Results:** For collision due to table angle, the mean difference between predicted and measured table angles leading to collision interlock of the gantry with table-top or table-base were 2.1° and 4.2°, respectively, with 70% of the values within 2° difference. For collision due to table translations with and without setup correction, the mean difference between predicted and measured table positions between gantry and table-top was 0.7 cm with 97% of the values within 2 cm difference. **Conclusion:** Using blender, a collision prediction framework between gantry and table capable of considering patient shifts with a 6 DoF table was successfully developed and verified with measurements.

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PO-GePV-T-329

Preliminary Treatment Planning System Commissioning Results for the First Clinical Biology-Guided Radiotherapy Machine

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Purpose: To report the preliminary treatment planning system (TPS) commissioning results for the first clinical installation of Reflexion X1- biology-guided radiotherapy (BgRT) system. Reflexion TPS generates IMRT/SBRT plans using the gradient-descent-optimization and Collapsed-Cone-Convolution algorithm (2.1mm grid) for 6MV-FFF beam delivered axially via 50 firing positions with couch advancing every 2.1mm. **Methods:** CT images of multiple phantoms were acquired and transferred to Reflexion TPS to test the accuracy of data acquisition, anatomical modeling, plan evaluation and dose calculation. Dosimetric parameters for various open static fields and SADs were evaluated in water and heterogeneous slabs phantoms. TG-119 and representative clinical IMRT/SBRT cases were planned and verified with ion chamber, film and ArcCheck (gamma:3%/2mm). End-to-end (E2E) testing was performed using anthropomorphic head and lung phantoms. **Results:** The maximum difference between Reflexion TPS and known CT numbers was 10.3±6.0HU, maximum difference in the structure volume was 20.2±4.8cc(Eclipse) and 2.93±0.6cc(MIM); maximum difference in the structure dose was 5.0±1.2Gy(Eclipse) and 1.7±0.4Gy(MIM). For static fields, PDD10 matched between TPS and measurement within 0.8%, crossline profiles matched within 1.6% in 80% from CAX for all fields≥2.5cm, and inline profiles agreed within 0.5mm FWHM. Output factors matched scintillator measurements within 1% for all fields. Dose in heterogeneous slab phantoms agreed within 3.5%. Lung phantom verification with film passed with gamma≥98.6%. The ion chamber measurements for TG-119 plans were within 2.9%; and ArcCheck QA passed>90% for all plans. E2E film analysis showed 0.8mm total targeting error for isocentric and 1.1mm for off-axis treatments. Iterations to the beam model are in progress to further improve the passing rate for smaller targets≤3cm. **Conclusion:** In this current preliminary report, the test results agreed within the tolerances specified by TG-53, MPPG 5.a, TG-119, TG244, and TG-148. A subset of the commissioning tests has been identified as a baseline data for an ongoing QA program.

PO-GePV-T-330

Capacity Assessment of African Clinics to Conduct Hypofractionated Radiotherapy Clinical Trials for Localized Prostate Cancer

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Purpose: Recent developing work has demonstrated the efficacy and benefits of adopting hypofractionated radiotherapy (HFRT) for prostate cancer in low- and middle-income countries. Adopting HFRT in limited-resource countries, such as those in sub-Saharan Africa, can reduce treatment costs and increase treatment access where cancer burden is great. Before HFRT can be fully implemented, clinical trials on African populations are necessary. Clinics in these regions are significantly under-represented in radiotherapy clinical trial participation. In this work, we assess the capacity of African cancer centers to participate in radiotherapy clinical trials in terms of capability and quality assurance. **Methods:** 14 African radiotherapy clinics with linear accelerators were invited to participate in a survey inquiring about past clinical trial experience, available resources, and relevant infrastructure. Survey items included experience in specific clinical trial phases, facility accreditation and documentation, specialist personnel on-site, and quality assurance methods. These reports were tabulated and compared to determine missing critical items required to participate in cancer clinical trials. **Results:** Among surveyed cancer centers; 36%, 93%, 93%, and

79% of clinics have reported capability to participate in clinical trials of Phase I, II, III, and IV, respectively. Specialist on-site personnel varied greatly from 57% to 93%. Specialists included roles such as research coordinators, pathologists, and biostatisticians. All surveyed clinics reported to have computed tomography imaging. **Conclusion:** While most of the surveyed clinics have the resources and accreditation for clinical trial participation, preliminary analysis shows that most sites are missing a number of specialist personnel which may be necessary for HFRT clinical trials. Additionally, only a handful of our surveyed clinics report to have Phase-I clinical trial capability. This could be due to lack of quality assurance necessary for Phase-I, lack of investment, or cultural stigma. Further investigation is required to determine the influence on the lack of Phase-I participation.

PO-GePV-T-331

Photoelectric Dose-Enhancement Using FDA-Approved Materials

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Purpose: Photoelectric dose enhancement using metallic biomaterials continues gaining popularity among methods of radiotherapy (RT) improvement. High-Z materials such as gold, platinum, or gadolinium are commonly attached to targeting proteins or formed into colloid nanoparticles which eject photoelectrons while under low-energy x-rays. When these biomaterials are introduced to the tumor microenvironment, typical external beam RT can deliver increased local dose by this method. Already existing FDA-approved metallic biomaterials such as specific chemotherapy drugs or MRI contrast agents could be implemented with multiple functions by providing RT dose enhancement in addition to their primary function. In this work, we estimate the potential dose enhancement provided by FDA-approved materials. **Methods:** Dose enhancement estimations were determined by a Monte Carlo-simulated fluence from a 6MV clinical LINAC directed to a homogeneous metallic biomaterial distribution at 20cm depth (10cm x 10cm field) in an infinite slab tissue-equivalent phantom. Photoelectron dose was calculated corresponding to the fluence energy spectrum, concentration of metallic biomaterial, and the metallic photoelectric absorption coefficients. This was repeated for gadolinium and platinum nanoparticles; gadolinium-based contrast agents; and platinum-based chemotherapy drugs. **Results:** To achieve a clinically significant dose-enhancement factor (DEF) of 1.2, the metallic biomaterial concentrations were estimated as 6mg/ml for Gd and 4.9mg/ml for Pt. Using FDA-approved intravenous delivery, this translates to 1%ID/ml of Omniscan or 0.5%ID/ml of Magnevist contrast agents; or 2%ID/ml of Carboplatin or 0.5%ID/ml of Cisplatin chemotherapy drugs. If 100%ID were to localize to the cancer site, the DEF would be 20,38,11, and 40 for Omniscan, Magnevist, Carboplatin, and Cisplatin, respectively. **Conclusion:** Preliminary estimations showed that FDA-approved biomaterials could potentially deliver clinically significant DEF at reasonably achievable concentrations. In future work, we seek to investigate local delivery or protein-targeting of these biomaterials to improve localization and increase the DEF.

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Comparing Ideal and Realistic Surrogate Motions with the Radixact Synchrony System

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Purpose: To evaluate the effect of the surrogate motion on the target motion modelling for Radixact Synchrony system. **Methods:** A modified version of Delta4 from Scandidos was used to test Accuray Synchrony motion tracking. A 3D printed cube with density close to that of tissue containing a cylindrical high-density material was used to simulate the tumor inside the lung. High density material was delineated as target and planned with Lung with Respiratory motion tracking option in Accuray Precision planning software. A Delta4 HexaMotion platform was used to simulate the patient breathing motion. Four motion traces provided for AAPM MATCH Challenge were used to simulate respiration. All motions were delivered twice (1) with surrogate LEDs attached to the phantom in which case the surrogate motion is the same as target motion (2) with surrogate LEDs on the cardboard that was placed on top of the Delta4 such that the surrogate motion is closely related with the target motion but not exactly the same. Accuray delivery raw files were analyzed to compare the modelled motion traces and HexaMotion driving traces. **Results:** Average shift of the modelled target traces in x, y, z directions were -0.22 mm, -0.57 mm, -0.56 mm for cases (1) and -0.40 mm, -0.62 mm, -0.58 mm for cases (2). These corresponds to setup uncertainty. The tracking accuracy can be estimated with standard deviation of the difference between driving traces and modelled traces. The average value of the standard deviation in x, y, z directions were 0.34 mm, 0.30 mm, 0.31 mm for cases (1) and 0.63 mm, 0.46 mm, 0.63 mm for cases (2). **Conclusion:** This test showed that Synchrony system can accurately model the target motion with submillimeter precision even if the surrogate motion is not the same as target motion.

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End-To-End Testing of Hybrid VMAT-Based Total Body Irradiation Technique: Simulation, Planning and Delivery Verifications

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Purpose: To conduct an end-to-end verification for VMAT based total body irradiation. **Methods:** Total body irradiation (TBI) is traditionally delivered using anterior and posterior (AP/PA) beams. Recently a VMAT based hybrid planning technique was developed at our institution to improve the target coverage and patient comfort while sparing the critical organs. To validate this technique, an end-to-end test was conducted using an anthropomorphic phantom, including simulation, planning and treatment delivery following the clinical procedures of VMAT-TBI. The CT images were acquired at 5mm slice thickness and whole-body CT was generated. Planning was done in Pinnacle TPS prescribing 13.2 Gy in 8 fractions. The goals were to achieve >85% of PTV receiving the prescription dose while reducing the mean dose to lungs <80% and mean dose to kidneys <50% of the prescription dose. Hotspots were limited between 110-120% of the prescription dose. Treatment was delivered on rando phantom in HFS orientation for upper body and in FFS orientation for lower body under image guidance. Measurements were performed with MOSFET dosimeter at four locations (central and lateral thorax, abdomen and pelvis) and two transverse EBT3 films at the lung and pelvis at the junction region between the head first and feet first treatment. **Results:** The 88.5% of PTV received prescription dose. Mean lung dose and mean kidney dose were 10.4Gy and 6.3Gy meeting the planning constraints. A good agreement was achieved between the measured and calculated dose. The mean dose difference for both films at lungs and pelvis were 5.1 ± 14.9 cGy and 11.8 ± 6.6 cGy respectively. The average dose difference for MOSFET dosimeters was 4.1%. **Conclusion:** The hybrid VMAT based TBI planning technique was capable of delivering uniform dose to the whole body and spare critical structures such as lungs and kidneys. End-to-End test proved that the workflow of VMAT-TBI achieves accurate dose delivery.

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Pre-Treatment 4DCT Fiducial Marker Combination Selection for Robotic Lung SBRT

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Purpose: This study aims to make recommendations for fiducial marker selection that optimizes lung tumor tracking in SBRT patients. It is rare that all fiducials are tracked during treatment delivery and the selection is currently based on proximity to the GTV. **Methods:** 10 patients were retrospectively reviewed. Each patient had 4 fiducials placed via bronchoscopy per manufacturer recommendations. The gross tumor volume (GTV) and each fiducial were contoured in each of the 10 phases from a simulation 4DCT. The center of mass (COM) was calculated for each fiducial and the GTV in every phase using a Matlab program with post-processing to eliminate noise. The centroid of every fiducial combination was found and compared to determine which centroid

combination was closest to the GTV's COM and which centroid's motion was most like the GTV's. **Results:** 8 of the 10 patients had the centroid tracked that was not the centroid closest to the GTV's COM and 1 patient had the centroid that matched the tumors motion the best tracked. The centroid that was closest to the GTV's COM was closer than the tracked centroid on average by 8.5 mm and the centroid that kept the most consistent distance was more consistent on average by 0.3 mm. **Conclusion:** The fiducials that are tracked during robotic SBRT delivery per usual clinical practice are not always the fiducials that are closest to the GTV's COM, or that best correlate with GTV COM motion during breathing. An analysis of GTV and fiducial motion on 4DCT using the technique described here, along with machine data during clinical simulation on how well different fiducial combinations are tracked, can improve tumor tracking during SBRT delivery or inform the treating physician on the expansions needed to create the PTV that would minimize the risk of marginal tumor recurrence.

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Prediction of Fiducial Marker Positions in Robotic SBRT Using 4D-CTs at Different Treatment Dates

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Purpose: To predict the marker positions using deformation vectors from the registration of 4D-CTs of patients who may have had multiple tumors treated over different dates and compare the predicted positions with the reference positions **Methods:** Two sets of 4D-CTs were acquired from a patient receiving robotic SBRT for separate lung tumors in 2017 and 2019. Deformation vector fields (DVF) were acquired by B-spline registration between the expiration phases in 2017 and 2019, and the expiration and the inspiration phase in 2017. Two hypothetical markers were inserted in the 4D-CTs in 2017 to the positions acquired from the registration of 4D-CTs in 2019 and the 4D-CTs of 2017 were deformed using the DVFs acquired above. A scaling factor was applied to compensate the difference between the breathing amplitudes in 2017 and 2019 and it was determined iteratively. The center of masses of the predicted markers from 2017 and the reference markers from 2019 were identified and the Euclidean distances between two points were calculated and compared. **Results:** The 4D-CTs were reconstructed using the DVFs and the positions of hypothetical markers were acquired. The average distance between the center of masses of prediction markers and reference markers was 7.8 mm (Marker A: 7.5 mm, Marker B: 8.0 mm). After a scaling factor was applied, the distance was reduced to 2.0 mm (Marker A: 1.1 mm, Marker B: 2.9 mm). **Conclusion:** The data would suggest that the position discrepancy can be no more than 3.0 mm. However, more research is needed to validate this approach with more cases. Several uncertainties can be associated with these results such as registration accuracy, image quality of 4D-CTs, and DVFs variations with and without tumors and markers. This data can be very helpful when planning the fiducial marker placement for a clinical or research application.

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The 4D Composite Dose Study of the Breath Motion Effect During the Lung Stereotactic Body Radiotherapy (SBRT)

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Purpose: Tumor motion due to the breathing cycle affects the dosimetry of the lung treatment especially for SBRT cases. We retrospectively studied the correlation between the dose coverage and the tumor motion including rigid motion and deformable motion for SBRT lung treatment. **Methods:** Six patients are enrolled into this study. Simulation 4D-CT is acquired and divided to 10 individual phases. The gross tumor volume (GTV) is contoured on each phase of the breathing cycle. The original treatment plan is generated on the average phase. This plan is copied to each individual phase CT and recalculated with same beams orientation and with a weighting factor of 10% prescription dose. The dose on each phase is transformed back to the average phase based on the deformable registration. A 4D dose is then determined by combining the weighted dose from all 10 phases. The displacement of the GTV centroid is used as the index of rigid motion; and the GTV volume change is used as the index of the deformable motion. **Results:** The 4D composite dose is compared with the clinical plan. With rigid motion from 0.22 cm to 1.89 cm, the 4D PTV dose coverage dropped by up to 20% from the clinical plan. The PTV coverage loss gets higher while the tumor motion is higher, on the order of ~10% for 1 cm rigid motion and R2 coefficient is 0.93. When the tumor has similar rigid motion, the tumor with higher deformable motion gets worse dose coverage, with ~3% PTV coverage loss for 20% GTV volume change. **Conclusion:** For SBRT lung cases, breath motion causes worse dose coverage and lower maximum dose. Rigid motion plays a dominant factor for motion effect on dose coverage. Deformable motion can change the dose coverage too at a smaller scale.

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Effect of Target Positioning Error on Tumor Control Probability in Stereotactic Radiosurgery for Metastatic Brain Tumors Using the CyberKnife M6

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Purpose: The purpose of this study was to evaluate the effect of target positioning error (TPE) on dose distribution and tumor control probability (TCP) in stereotactic radiosurgery (SRS) plans for metastatic brain tumors of different sizes using the CyberKnife M6. **Methods:** The reference plans in SRS were created using the circular cone of the CyberKnife M6 for each spherical planning target volume (PTV) with diameters (Φ) of 5, 10, 15, and 20 mm, contoured on CT images of a head phantom. The dose of 20 Gy per single fraction was prescribed to the 70% isodose lines, covering 95% of the PTV. Subsequently, TPE plans were created by shifting the beam center by 0.1–2.0 mm in three dimensions relative to the reference plans, using the same monitor units and beam arrangements. Dosimetric parameters such as V20Gy of the PTV, conformity index (CI) and generalized equivalent uniform dose (gEUD)-based TCP estimates were evaluated for each plan. Shifts corresponding to the TCP reduction rate of 3% were also evaluated for each PTV size. **Results:** The V20Gy, CI and TCP values of the reference plans for Φ 5-mm PTV were 95.1%, 0.90 and 81.4%, respectively. The corresponding values for TPE plans with 0.5-mm, 1.0-mm, and 2.0-mm shifts were 79.0%, 0.62 and 79.6%, 62.9%, 0.40 and 72.3%, and 34.2%, 0.12 and 19.5%, respectively, all of which decreased as TPE values increased. The TCP values for TPE plans with 2-mm shifts were 19.5% for Φ 5-mm PTV, 68.9% for Φ 10-mm PTV, 74.3% for Φ 15-mm PTV, and 79.0% for Φ 20-mm PTV. Shifts corresponding to the TCP reduction rate of 3% were 0.6, 1.1, 1.2, and 1.6 mm, respectively. **Conclusion:** The TPE had a significant effect on target dose coverage and TCP in SRS for metastatic brain tumors using the CyberKnife M6, in particular, for small PTV.

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Predicting the Effect of Indirect Cell Kill for Radiosurgery of Multiple Brain Metastases Via Single-Isocenter/Multi-Target Volumetric Modulated Arc Therapy (SMT-VMAT)

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Purpose: Patient setup errors are of major concern for stereotactic radiosurgery (SRS) of multiple brain metastases (m-bm) when using SIMT-VMAT treatments. However, recent clinical outcome studies show high rates of tumor local control for SIMT-VMAT. This could be from contributions of both direct cell kill (DCK) and indirect cell kill (ICK) via devascularization for a single-dose of 15Gy or more. This study quantifies the role of ICK in dosimetric errors as a function of spatial patient setup uncertainty. **Methods:** Nine patients with 2-16 metastases (61 total) were planned using SIMT-VMAT with HyperArc planning geometry using a 10MV-FFF beam with a 20Gy prescription to each lesion. Isocenter was placed at the geometric center of all tumors with an average distance to isocenter of 5.4 cm (2.2-8.9cm). Plans were recalculated with clinically realistic patient setup errors [± 2 mm, $\pm 2^\circ$] in all 6-directions. Boolean structures were generated to calculate the effect of DCK as 20Gy isodose volume (IDV) and ICK as 15Gy minus 20Gy IDV. Contributions of DCK and ICK to PTV coverage were analyzed along with brain toxicity. Induced uncertainty and minimum dose covering the entire PTV were analyzed to determine the maximum acceptable patient setup errors to utilize ICK for SIMT-VMAT. **Results:** Patient setup errors of 1.3mm/1.3° in all 6-directions must be maintained to achieve PTV coverage of 15Gy for ICK with SIMT-VMAT. Setup errors of 2mm/2° showed clinically unacceptable loss of PTV coverage of 29.4 $\pm$ 14.6% even accounting for ICK—suggesting repositioning the patient. However, no significant effect on normal brain toxicity was observed. **Conclusion:** SRS of m-bm using SIMT-VMAT treatments have shown positive outcomes even with patient setup errors. These clinical outcomes, while largely due to DCK, are also due to ICK. Mechanisms, such as devascularization, should be explored to provide better understanding of the radiobiological response of SRS of m-bm using SIMT-VMAT.

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Treating Vestibular Schwannomas Abutting the Brainstem with SRS/fSRS Alone

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Purpose: To present our experience in treating large vestibular schwannomas abutting the brainstem with stereotactic radiosurgery (SRS) or fractionated stereotactic radiosurgery (fSRS) alone without surgery. **Methods:** At our institution, we have increasingly performed Gamma Knife (GK) SRS/fSRS alone for patients clinically diagnosed as having a Koos Grade 3 or 4 acoustic neuroma. In this study, we retrospectively reviewed the treatment plans of 23 patients treated with this approach from 3/2014-1/2021, and reported the dosimetric indexes and treatment outcomes. All patients were prescribed a dose of 12Gyx1 or 6Gyx5 fractions, planned using the Leksell GammaPlan (LGP), and treated with the GK Perfexion (before 5/2018) or ICON system (after 6/2018) of our institution. Dosimetric indexes collected included gradient index (GI), Paddick Conformity Index (CI), and maximal dose and V10/23Gy (per TG101 dose constraints) of the brainstem. To measure changes in tumor volume, post-treatment imaging was imported into LGP and the index lesion was individually contoured. **Results:** All treatments demonstrated excellent local control with limited toxicity. All except 2 treated targets touched the brainstem. The mean $\pm$ standard deviation for the CI is 1.23 $\pm$ 0.08 and for the GI is 3.00 $\pm$ 0.24. For the brainstem, the mean maximal dose and V10Gy are respectively 12.20 $\pm$ 3.77 Gy and 0.097 $\pm$ 0.097 cm³ for 12Gyx1 fractionation. The mean maximal dose and V23Gy are respectively 26.53 $\pm$ 1.078 Gy and 0.096 $\pm$ 0.130 cm³ for 6Gyx5 fractionation. The majority of the follow-up scans showed increased tumor volumes at 6 months and decreased/stabilized volume in the 18-month plan. **Conclusion:** Large vestibular schwannomas can be safely treated with SRS/fSRS alone. The brainstem constraints in TG101 for single and five fractions can be met. This treatment approach is proven effective based on the follow-up scans. Further data is also needed to appropriately compare hypofractionated vs single-fractionated SRS in this population.

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Evaluation of Three Dosimeters on Output Factor Measurement of the New Aktina SRS Cone System

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Purpose: To evaluate the performance of three dosimeters on output factor(OF) measurement of the newest Aktina SRS Cone system attached to Elekta VersaHD LINAC using 6FFF beam. **Methods:** The Aktina SRS cone system is equipped with cone sizes of 5mm, 7mm, 9mm, 11mm, 13mm and 15mm. OF for each cone defined as the cone output at 10 cm depth divided by the output of open field of size 10cm at 90cm SSD was measured. A daisy chain correction factor was applied to calculate the final OF value using field size of 3cm. Dosimeters evaluated in this study include EBT3 Gafchromic film, EDGE detector, and Exradin A16 Micro chamber. EDGE & A16 measurements were performed in water tank, while EBT3 film was conducted in solid water. EDGE & A16 OF were also corrected based on published small field output correction factors for comparison purpose. In addition, IROC SRS OF phantom was utilized to independently verify the OF for cone size of 15mm. **Results:** The OFs measured with EBT3 were 0.502, 0.609, 0.664, 0.720, 0.761 and 0.783 respectively. OFs measured by EDGE were 0.550, 0.645, 0.705, 0.746, 0.773 and 0.795 respectively which were consistently higher (mean diff= 2.9%) than EBT3 measurements. A16 measurements were consistently lower than EBT3 measurements (mean diff= -2.4%) which were 0.455, 0.574, 0.651, 0.703, 0.743 and 0.770 respectively. EBT3 OFs, corrected EDGE OFs and corrected A16 OFs on average were within $\pm 1\%$ of each other. IROC results showed a satisfactory independent check of OF for cone size of 15mm (TLD/Inst=0.96 at 1.5cm & TLD/Inst= 0.97 at 7.5cm). **Conclusion:** Overall, EDGE over-responded to small field measurement and A16 Chamber under-responded to small field measurement due to volume averaging. OFs measured by EBT3 were found to be in good agreement with corrected EDGE and A16 OFs.

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A Multi-Clinic Validation of An End-To-End Procedure for MLC-Based Stereotactic Radiosurgery with a Novel Phantom

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Purpose: An end-to-end procedure for a new anthropomorphic phantom was developed for MLC-based SRS commissioning on a linear accelerator. This procedure was developed and tested by Northwest Medical Physics Center at three independent clinical sites with Varian Edge and TrueBeam accelerators, following local standards for MLC-based SRS planning. **Methods:** The end-to-end procedure uses a novel anthropomorphic SRS phantom recently developed by Varian Medical Systems. This phantom was used to verify isocenter coincidence, targeting accuracy, treatment modulation and absolute dosimetry for dose delivery to a single target and single-isocenter, multitarget geometries. Testing was performed at three clinical sites in an all-Varian environment consisting of ARIA, Eclipse treatment planning, and TrueBeam/Edge machines with Millennium and HD-MLCs. CT simulation and treatment delivery was performed using Qfix Encompass and Brainlab mask systems. Treatment plans were developed for RapidArc and dynamic conformal arcs at 6FFF and 10FFF. Consistent planning criteria was applied across all three clinical sites. A nominal dose of 16 Gy was prescribed for each plan using a sites' standard beam geometry for SRS cases. **Results:** Pinpoint ion chamber readings showed agreement with the planned dose to within 3% for treatment delivery to a 2-cm target. DoseLab software was used to perform relative gamma analysis of the film dose planes compared to extracted data from the treatment planning system. Most single and multitarget treatment plans showed gamma passing rates >90% for 3% and 1 mm after auto-registration shifts ≤ 1 mm in any direction. **Conclusion:** Validation of the end-to-end procedure at three independent clinical sites indicates that it is acceptable for the commissioning of MLC-based SRS treatments. Data is now being acquired at additional clinical sites for the purposes of establishing acceptance criteria for end-to-end measurements performed on TrueBeam and Edge machines.

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PO-GePV-T-342

Consistency in Response Between Two Copies of a High-Resolution Diode Array for SRS Dosimetry

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Purpose: To evaluate variability in response between two copies of SunNuclear's SRS MapCHECK diode array. **Methods:** Measurements suitable for device commissioning were collected with each copy of the SRS MapCHECK. The irradiations tested the devices' dependencies on beam angle, pulse repetition rate (PRR), dose per pulse (DPP), total dose, and field size for flattened and unflattened 6MV and 10MV beams. Angular dependence was tested with both axial and vertex beams. DPP was varied by changing SDD. Additionally, three treatment plans (lung SBRT, spine SBRT, 3D conformal SRS) for each energy were delivered to each device. Measurements were collected on one Varian TrueBeam on the same day to reduce variability of linac behavior. Differences in dependence on each variable were compared between the two devices. **Results:** Differences in dependence on beam angle were less than 1% for angles >10° from parallel with the array for axial and vertex orientations. Changing PRR and DPP resulted in differences between devices of less than 0.5% across the available range of PRR, less than 0.8% across the range of DPP, and within 0.4% for all but the highest DPP. Dose linearity varied by <0.2% between the two devices from 10 to 3500 cGy. Variability in field size dependence was <0.5% for all field sizes except for the 10FFF 1cmx1cm field, where the greatest difference was 0.9%. Finally, passing rates from each device were at least 94.8% and within 3% of each other using a 1%/1mm gamma comparison (lung SBRT and 3D SRS, all energies) and >89% and within 4.3% of each other with 2%/2mm criteria for the spine SBRT plans. **Conclusion:** The two diode arrays responded similarly across a wide range of tests. Only small differences in dependence to various conditions were observed. Gamma passing rates matched closely with more stringent criteria than used clinically.

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Evaluation of a New Diode Detector for Beam Data Measurements

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Purpose: The purpose of this study is to evaluate a new semiconductor detector for photon beam dosimetry. Especially consideration was given to the narrow field dosimetry. **Methods:** The PTW MicrosiliconX diode is being evaluated for its application in measurements for beam data collection. An ELEKTA VersaHD linac and the PTW BEAMSCAN water tank were used for the measurements. We used 6MV, 10MV, 18MV, 6MV FFF and 10MV FFF energies for all measurements and field sizes ranging from 0.5x0.5cm² to 30x30cm². We measured profiles at 5,10, 20cm depths, percent depth dose curves, and output factors. The same measurements were collected for fields with the motorized wedge in place. Finally, all measurements were repeated using the SNC Edge detector and the PTW Photon Diode 61016P detector. The dose per pulse dependency was also checked for all detectors. **Results:** The dose per pulse dependency was acceptable for all detectors. The MicrosiliconX and 61016P detector had the lowest and the SNC edge had the highest dose per pulse dependency. The SNC Edge is also the most aged detector amongst them. The dose per pulse was 0.29%, 0.31%, 1.15% for the MicroSiliconX, 61016P and SNC Edge detectors. There was less than a 2% difference in all output factors. PDDs and profiles had small differences between them. The SNC Edge detector showed higher PDD values past 15cm depths but that the difference was within 1-2%. The profile penumbras were very similar. The PTW detectors had less noise in the measurements of PDDs and profiles. **Conclusion:** The new PTW MicrosiliconX detector was compared against established detectors for photon beam measurements. The detector performed similarly to the other ones and had less noise in the PDD and profile scans which in turn requires less data smoothing and less time for processing while obtaining reliable measurements.

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A Comparison of Field Output Correction Factors for a Diode Detector Among Various Measurement Setups and Among Various Energies for Stereotactic Cones

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Purpose: Diode detectors over-respond in small fields and their over-response needs to be corrected for field output factors (FOFs). The goal of this study is to compare field output correction factors (FOCFs) for the Sun Nuclear Edge detector among various measurement setups and among four different energies for stereotactic cones. **Methods:** FOCFs for Varian stereotactic cones (4, 5, 7.5, 10, 12.5, 15 and 17.5 mm in diameter) (Varian Medical Systems, Palo Alto, CA) referenced to a field size of 10 × 10 cm² were measured on Varian TrueBeam STx using a plastic scintillation detector (W2, Standard Imaging Inc., Middleton, WI) and the Sun Nuclear Edge detector (Sun Nuclear Corp., Melbourne, FL). Measurements were performed for four different energies (6MV, 6MV flattening filter free (FFF), 10MV and 10MV FFF) at depths of 1.5 cm, 5 cm and 10 cm in source-to-skin distance (SSD) and source-to-axis distance (SAD) setups. The daisy-chain method was used only for the Edge. FOCFs were determined by taking a ratio of FOFs for the W2 to those for the Edge. **Results:** First, absolute differences in FOCFs among three depths range from 0.002 to 0.011 (6MV), from 0.001 to 0.008 (6MV FFF), from 0 to 0.004 (10MV) and from 0.01 to 0.013 (10MV FFF) for the SSD setup and from 0.002 to 0.004 (6MV), from 0.002 to 0.006 (6MV FFF), from 0.001 to 0.003 (10MV) and from 0.012 to 0.02 (10MV FFF) for the SAD setup. Second, FOCFs for the SAD setup are usually larger than for the SSD setup and differences are largest for the 4 mm cone. Third, FOCFs for 6MV FFF are largest in both setups. **Conclusion:** FOCFs for the Edge detector can be different depending on the measurement setup and energy. Therefore, FOCFs for each setup and energy should be considered for accurate FOFs.

PO-GePV-T-345

A Method for Estimating the Geometrical Accuracy for Single-ISO Multiple-Target Treatment

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Purpose: To present a method for estimating the geometrical targeting accuracy in single-ISO multiple-target (SIMT) linac-based stereotactic radiosurgery. **Methods:** Seven BBs (4 mm diameter) were inserted into a Rando head phantom, with CT image acquired. A treatment plan was created to irradiate the BBs with SIMT. Plan contained 69 static beams, 1.5x1.5cm field size (MLC-shaped), with various couch, gantry and collimator angles. Each beam irradiated one BB only. MLC aperture was designed to align BB and field centers as closely as possible. Distances from ISO to BB centers ranged from 0 to 7.5 cm. Before irradiation, the phantom was aligned to the planned position with CBCT guidance so that the final positional mismatch was ≤0.1 mm (translation) and ≤0.2 degree (rotation). During irradiation, EPID MV imager was used to capture exit field images of the BBs and 1.5x1.5cm fields. Distance between the BB centers on the exit MV image against DRR was measured, revealing the inaccuracy intrinsic to target alignment, and the uncertainty of the MV imager and the couch. Distance of the field centers on the MV image versus the DRR was measured, revealing the inaccuracy due to the MV imager, MLC leaf positioning, gantry sag, and collimator rotation. Distances between BB and field centers on the MV image and the corresponding values on the DRR were measured and compared to distinguish various inaccuracy sources. **Results:** For our Varian Edge System, the distance of the BB center on MV image against DRR was 0.4±0.2mm (max=0.7 mm). The distance of the field center on MV image against DRR was 0.3±0.1mm (max=0.6 mm). The difference of the BB center relative to field center on MV image against DRR was 0.5±0.2mm (max=1.0 mm). **Conclusion:** Our method confirmed that our Varian Edge System can deliver SIMT treatment with submillimeter geometrical targeting accuracy.

PO-GePV-T-346

Optimal Frequency of Winston-Lutz Test for Stereotactic Radiosurgery Treatment: A Five-Year Clinical Experience

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Purpose: To determine optimal frequency of Winston-Lutz(WL) test in our clinic that assures the integrity of linac radiation isocentricity for stereotactic radiosurgery(SRS) treatment while effectively detecting abnormal deviations at minimum cost of clinical machine time. **Methods:** An Elekta Versa HD linac equipped with Agility MLC and Aktina cone system is used for MLC- or cone-based SRS. WL test is done with Elekta BB phantom and MV imager and analyzed with in-house software. After our SRS program commissioning in 2015, WL test is implemented in three phases with different frequencies: phase I(2015-2017) -on every day that had SRS treatment, phase II(2018) -bimonthly/monthly, and phase III(2019 to date) -on weekly basis if that week has SRS treatment, or after an isocentricity-affecting event, i.e. beam tuning, etc. Individual and average deviations of BB-cone/MLC field centers in MV images are calculated and compared. 2D distribution of those deviations is evaluated by beam-eye-view(BEV) plots. **Results:** The isocentricity of our SRS system is overall stable across all three phases of WL test frequencies. Average BB-cone field center deviations are $0.71 \pm 0.35 \text{ mm}$ ($n=54$), $0.75 \pm 0.35 \text{ mm}$ ($n=14$) and $0.73 \pm 0.40 \text{ mm}$ ($n=35$) for phase I,II and III, respectively. For MLC-based case, average deviations are $0.88 \pm 0.31 \text{ mm}$ ($n=23$), $0.83 \pm 0.31 \text{ mm}$ ($n=15$) and $0.69 \pm 0.37 \text{ mm}$ ($n=23$), respectively. The trend of average MLC-based field center deviation has relatively large fluctuation ($0.49\text{--}1.33 \text{ mm}$), compared to that of cone-based case ($0.61\text{--}1.03 \text{ mm}$). Weekly WL did help to catch abnormal average deviations (tolerance $< 1.2 \text{ mm}$) due to beam tuning and drifting. BEV plots indicate that MLC-based field centers (mean coordinates $(-0.41 \text{ mm}, -0.27 \text{ mm})$), are relatively more off isocenter than cone-based ones $(-0.08 \text{ mm}, -0.38 \text{ mm})$. **Conclusion:** Given an SRS system with historical stability in isocentricity and that daily WL could be machine time-demanding and monthly WL could miss opportunity identifying possible issue, it appears that weekly WL QA may promptly assure the integrity of SRS system isocentricity while being temporally practical in our clinic.

PO-GePV-T-347

Dosimetrical Verification of Output of Small Cones in a Radiosurgery Device

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Purpose: Dosimetry for small fields presented a challenge for determining output factors for small-sized cones, so the accuracy delivered by small-sized cones could be affected if there are some errors in output factors. This work designed a method to verify the dose accuracy for small cones used in a Cyberknife. **Methods:** A "a large uniform dose area" plan using 5 mm cone was generated for a prostate (60 mm lesion size) and a brain case with a large lesion (35 mm in size), and then recalculated with a phantom. A point dose (placed at a uniform dose area) was measured with a pinpoint ion chamber (0.015cc) for each phantom plan. The method to measure the point dose actually has been practiced in our routine clinical QA for plans that have a large uniform dose area, which has been approved correct. To compare, plans for these two cases are also generated with large cones (35 mm for the brain case and 60 mm for the prostate case), and their point doses for phantom plans were also measured. **Results:** The average of three repeated measurements were taken as the measured dose. For plans using 5 mm cone, the measured dose is larger than the plan dose by 2.71% for the brain case while the prostate case shows that the measured dose is smaller than the plan dose by 3.8%. For the plan using large cones (35 mm above), the differences between the measured and plan doses are 1.41% and 0.8%, respectively for the brain and prostate case. **Conclusion:** With acknowledging the method used for a point dose measurement is correct for a large uniform dose area, this work showed that the output for small cones in our Cyberknife are correct within about 3%.

PO-GePV-T-348

A Simple Method for Quantifying and Reducing Lateral Response Artifact of Flatbed Scanner for Film Dosimetry

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Purpose: To quantify radiochromic film flatbed scanner's lateral response artifact in a technically simple manner and to determine the region of the flatbed scanner where the lateral response is minimized, reducing or eliminating the need for correction factors. **Methods:** The scanning area ($30 \times 45 \text{ cm}^2$) of the expression 12000XL photo scanner is virtually divided into a matrix of $5 \times 5 \text{ cm}^2$ squares. A single piece of $3 \times 3 \text{ cm}^2$ EBT3 film is irradiated, then scanned separately at the center of each "square" location 72 hrs later using a ROI of $5 \times 5 \text{ cm}^2$. The center $2 \times 2 \text{ cm}^2$ ROI of each image is analyzed with red channel in RIT, and the average pixel value is converted into dose. Film calibration is generated from $3 \times 3 \text{ cm}^2$ film pieces from the same sheet, irradiated to known dose quantity, and scanned at the center of the flatbed. The regions with similar pixel values and low ROI standard deviation is determined to be the area on the scanner for which correction factors for clinical dosimetry is not needed. **Results:** Dose response of the single piece of $3 \times 3 \text{ cm}^2$ film is not observed from excessive scanning. The pixel value of all analyzed ROI ranges from 17317 to 19628, which corresponds to a dose range of 6.3 Gy to 8.5 Gy. The regions where lateral response artifact is minimized is approximately 10 cm from the scanner glass edge in each direction except on the right side ($\sim 5 \text{ cm}$ from the edge). In this region the pixel value ranges from 18557 to 18645 (dose range between 7.13 Gy to 7.21 Gy), and the standard deviation is < 131 pixels. **Conclusion:** This work describes a straightforward method to identify regions of a flatbed scanner where the lateral response artifact is minimal, which can improve the outcome quantitative film dosimetry.

PO-GePV-T-349

Thresholding Gamma Analysis On SRS Anthropomorphic Phantoms

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Purpose: To calculate gamma results for SRS anthropomorphic phantoms using different thresholding methods and compare to the standard IROC region of interest method. **Methods:** The SRS anthropomorphic phantoms have a 1.9 cm target that houses two TLD and radiochromic film in the sagittal and coronal planes. It is mailed to institutions that create a plan treating the center of the target to 30 Gy with target coverage per institutional protocol. They irradiate and return the phantom and submit the plan in DICOM format. TLD are read and a 5%/3mm gamma analysis is done on the film. A newly commissioned in-house software addition includes the ability to use dose thresholding techniques to define the area used in the gamma analysis. Prior versions only allowed the area to be defined by as a rectangle relative to the target and was chosen to be a 2.9 cm square centered on the target. 28 recent SRS phantom results were recalculated using 10%, 20% and 30% of maximum dose thresholds as well as using a 10 Gy limit. Results were compared to the original regions of interest. **Results:** Analyzed phantoms had a pass rate of 86% using all methods except the 10 Gy threshold measurement. This method created one more failure making a pass rate of 82%. The average changes in the gamma results were 0% for each percent thresholds and 1% for the 10 Gy threshold with maximum changes of 2%, 3%, 5% and 14% respectively. **Conclusion:** The new ability of the software allows more flexibility in setting the area on the film use for gamma analysis. It will be further tested results of other phantoms before our methods are changed.

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PO-GePV-T-350

An End-To-End, Low-Cost, Universal Winston-Lutz Type Quality Assurance Procedure for Hypofractionated Image-Guided Stereotactic Radiosurgery

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Purpose: Image-guided stereotactic radiosurgery (SRS) is rapidly disseminating worldwide. A robust low-cost and universally accessible QA method is critically needed for its safe implementation and global collaborations. In this study, we developed a low-cost, universal, end-to-end Winston-Lutz method to address such a need. **Methods:** After surveying the state-of-the-art SRS phantoms and QA strategies on the market, a simple two-piece detachable plastic film slab was created and specifically implemented for isocentric and off-center Winston-Lutz measurements. The phantom is constructed similar to the OSLD slabs shipped from IROC for Gamma Knife units. To conduct off-center Winston-Lutz testing, the user was instructed to cut a 0.3-mm solder wire into small bits (~0.5 mm). These fiducial bits were then randomly taped onto a radiochromic film (~250 cm²). The film was then needle-pricked on its back side at central positions of the taped fiducial bits. The film was then sandwich pressed between the two slabs. Treatment plan was created to deliver a near-spherical dose toward individual fiducials via online imaging guidance. The film and RTDose file were collected and analyzed centrally, a process similar to IROC workflow. **Results:** The initial measurements with the method yielded excellent results, showing submillimeter matching in mechanical-vs-radiological isocenters and peripheral isodose lines. Most excitingly, the technique for CBCT-guided trigeminal neuralgia treatments yielded <0.5 mm isocenter deviation that cemented clinical workflow. Additional benefits of our technique is its low cost and ease of implementation with the consumable cost estimated to be \$10.20. Since final dose distribution associated with different phantom constructs was embedded into the DICOM RTDose file, no special restrictions were placed for the film slabs if made locally. **Conclusion:** A low-cost, high-functional, end-to-end test procedure is developed and demonstrated for the first time. Implementations of the technique across modalities including carbon machines are underway with results forthcoming.

PO-GePV-T-351

Evaluation of Delivery Quality Assurance of Cyberknife Stereotactic Radiosurgery

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Purpose: The purpose of this study is to establish and assess a practical delivery quality assurance method for stereotactic radiosurgery with Cyberknife by analyzing the geometric and dosimetric accuracies obtained using a PTW31016 PinPoint ionization chamber and EBT3 films. **Methods:** 150 cases of delivery quality assurance plans were performed on Cyberknife for assessing point dose and planar dose distribution respectively using a PTW31016 PinPoint ionization chamber and Gafchromic EBT3 films. The measured chamber doses were compared with the planned mean doses in the sensitive volume of the chamber, and the measured planar doses were compared with the calculated dose distribution using gamma index analysis. The gamma passing rates were evaluated using criteria of 3%/1 mm and 2%/2 mm. **Results:** For point dose comparison, the averaged dose differences ($\pm$ standard deviations) were $1.6 \pm 0.73\%$ for all the cases. For planar dose distribution, the mean gamma passing rate for 3%/1 mm, and 2%/2 mm evaluation criteria were $94.26\% \pm 1.89\%$, and $93.86\% \pm 2.16\%$, respectively. The gamma passing rates were higher than 90% for all the delivery quality assurance plans with the criteria of 3%/1 mm and 2%/2 mm. **Conclusion:** PTW31016 PinPoint ionization chamber and EBT3 film can be used for routine Cyberknife delivery quality assurance. The point dose difference should be within 3%. The gamma passing rate should be higher than 90% for the criteria of 3%/1 mm and 2%/2 mm.

PO-GePV-T-352

Comparison of SRS QA Tools On IROC Head Phantom Plan

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Purpose: Intracranial stereotactic radiosurgery (SRS) requires high positional and dosimetric accuracy. This study evaluated three different SRS QA tools for IROC SRS phantom test. **Methods:** Portal dosimetry (PD), SunNuclear SRS MapCHECK, and IROC film/TLD were used to evaluate the agreement between delivered and planned dose for IROC SRS phantom test plan QA. Gamma Index for each of the three SRS QA methods was computed for a combination of criteria from 1-3mm distance-to-agreement (DTA), and 1%-5% dose difference (DD) for all detector points measured > 10% threshold. A VMAT plan consisting of 6-field non-coplanar beams of 6MV FFF was delivered to a 1.9 cm diameter target using TrueBeam V2.7 and 120 Millennium MLC. Plan was calculated by Eclipse AAA (Version 15.6.03) in 1mm calculation grid. **Results:** PD, MapCheck and film/TLD all are good tools for SRS QA. For tightest tolerance of 1%/1 mm, the Gamma passing rate were 95.4%/92.5% for PD/MapCheck, respectively. For clinical use 2%/2mm tolerance, the passing rates were 99.6%/98.2%/94.7% for PD/Mapcheck/film, respectively. We found that PD had slightly higher passing rate than MapCHECK QA in all combination of DTA and DD criteria, with the largest difference at the tightest bound of 1mm/1%. Gamma Index difference was reduced to <1.5% at 2mm/2%, and <1% at 2mm/3%. **Conclusion:** Portal dosimetry does not require the use of a specialized phantom, provides sub-millimeter detector resolution, and is convenient to use. However, due to the fixed geometry of the portal imager with the gantry, the PD QA does not simulate couch walk, and hence does not represent the true delivered dose distribution. Before an institution commissions the protocol for patient-specific QA using PD, it is recommended to use multiple SRS QA tools and determine a machine specific gamma passing criteria adequate for an accurate SRS treatment delivery.

PO-GePV-T-353

Implementing a QA Phantom to Test Discrepancies in Stereotactic Space Definition for Gamma Knife ICON

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Purpose: To develop a QA phantom that can quantitatively evaluate stereotactic spaces defined among MRI, CT, and Cone Beam CT (CBCT). Variations in stereotactic space definition have been an issue when acquiring imaging with different modalities for gamma knife SRS patients. These discrepancies dispute the systematic integrity and are especially significant since variation in object localization may result in lack of prescribed dose to the target with additional radiation being absorbed by parenchyma. **Methods:** A Leksell Gamma Knife dosimetry phantom was placed in the immobilized stereotactic frame so it can be locked in position when being imaged across multiple modalities. A total of 12 CT and 4 MRI spot markers were placed inside and around the phantom. These markers were used as the means of comparison among different modalities. The phantom was imaged in CT and MRI with dedicated localizer box and a CBCT image was also taken using on board imager from a Gamma Knife ICON machine. All coordinates for the markers were determined in Gamma plan with different stereotactic space definition accordingly. The coordinates from CT and MRI were compared with that of CBCT and a translational shift was calculated by a taking vector magnitude between the two stereotactic space coordinates. **Results:** Minimum, median, and maximum total translational shifts of 0.65, 1.24, and 1.85mm, respectively were obtained for CT compared to CBCT. Interior markers were more susceptible to shifts compared to exterior. The MRI to CBCT comparison resulted in similar total translational shifts, with minimum, median, and maximum values of 0.64, 1.17, and 1.41mm. The 3 markers used for this comparison were on the phantom exterior. **Conclusion:** With the magnitude of most of the shifts being close to 1mm, this experiment suggests that further investigation may need for systematic error when co-registering images from different modalities.

PO-GePV-T-354

Evaluation of a New Winston Lutz Tool for the Treatment of Multiple Brain Metastases for Mono-Isocentric Beams

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Purpose: To evaluate the use of a new Winston Lutz tool for the treatment of multiple brain metastases for mono-isocentric beams. **Methods:** The Winston Lutz phantom has 5 built in BBs positioned at different distances from isocenter including off axis positions. The phantom was imaged before delivering the plan to ensure setup position. In total, twenty-one Winston Lutz tests were performed. Each test included eight different gantry, couch, and collimator configurations. The actual position of the phantom and each BB was also calculated for all the measurements. **Results:** The average position of the phantom ranged from 0.08-0.19 mm in the x,y, z direction and 0.1 degrees for the pitch, roll and yaw positions. Target 4 (5.8 cm from isocenter) and Target 5 (7 cm from isocenter) had the largest number of Winston Lutz test with a maximum 2D deviation between the BBs and the field center of 0.75 mm and 1 mm for a configuration with the gantry, couch and collimator at 0 degrees. The same targets showed a larger percent of times with a Winston Lutz > 1 mm. Overall target 5 showed 3% of the time a value ≥ 1.0 . **Conclusion:** Winston Lutz tests were performed for a total of 168 beam arrangements for a phantom aligned within our clinical tolerances < 0.5 mm and < 0.5°. Overall targets 4 and 5, the targets further from isocenter, had the largest number of Winston Lutz > 1 mm among all shifts and all gantry combinations. Longitudinal shifts seemed to affect the overall results the most. Nevertheless, a Winston Lutz < 0.75 mm seems to be achievable for most circumstances. Our result might be an indication that careful consideration should be taken for those targets further away from isocenter even when tight setup constraints are used during treatment.

PO-GePV-T-355

Simple and Reliable Calibration Fields for OrthoChromic Film Dosimetry Protocol

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Purpose: To demonstrate various calibration fields for the newly introduced OrthoChromic Film Dosimetry Protocol to optimize dose accuracy in order to deliver stable film dosimetry results over a wide dose range. **Methods:** The OrthoChromic OC-1 film was tested using Eclipse v15.6 with Truebeam and MultiPlan v5.3 with Cyberknife M6, respectively. The films were scanned in reflection mode on an Epson flatbed scanner with 48bit at 75dpi. A glass plate was used to flatten films on the scanner. OrthoChromic protocol requires a calibration film exposed to a calibration plan with multiple dose levels in scan direction and uniform dose in lateral direction, and a QA film exposed to patient specific QA plan. When the two films are scanned together in parallel in scan direction, the possible dose errors caused by lateral effect from scanner sensors and film coating variations are corrected. OrthoChrome Pro software was used to register films with corresponding plans, calculate delivered dose from film, and compare delivered dose to planned dose. Quality of QA film compared with planned dose were determined by gamma analysis at 2% and 2mm. **Results:** (1) Truebeam jaw can generate a step wedge calibration field that provides reliable dosimetric results, with over 95% gamma passing rate. (2) Cyberknife MLC can generate calibration field comparable to the Truebeam 8-level step wedge, with over 95% gamma passing rate. (3) Using overlapping circular fields generated by Cyberknife Cone, an equally reliable calibration field can be generated, with over 95% gamma passing rate. (4) An actual patient treatment case QA was performed with the overlapping cone calibration. The reliability of this dosimetric method was confirmed with over 95% gamma passing rate. **Conclusion:** There are many ways to design reliable calibration fields to obtain accurate dosimetric results using the newly introduced OrthoChromic Film Dosimetry Protocol.

PO-GePV-T-356

The Effect of Collimator Setting on Small Photon Field Output in a Commercial Treatment Planning System

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Purpose: More than one third of SBRT/SRS fields treated in our clinic were less than 5 mm in size. Due to limitation of linear accelerator collimator jaw positioning accuracy (Y: +/- 2 mm, X: +/- 1 mm, manufacture specification) the effect of the collimator setting on small photon field output needs to be studied. **Methods:** A commercially available treatment planning system (Eclipse TPS, Version 15.5.11, Varian Medical System) was used in this study. The outputs of small photon fields of collimator setting in 10 x 10 mm and 20 x 20 mm were first calculated to validate the accuracy of small field dose calculation accuracy of our treatment planning system. A number of treatment plans were then created with a small photon MLC field of 5 x 5 mm in size with various X and Y jaw collimator settings. The calculated outputs of the small photon fields were compared, as well as to that of 100 x 100 mm reference field. **Results:** The outputs for 10 x 10 mm and 20 x 20 mm photon field shown comparable results with publishing results in the literature, validating the dose calculation accuracy of our dose calculation algorithm. The outputs of the same field size of 5 x 5 mm with various collimator setting from 5 x 5 mm to 20 x 20 mm shown a difference up to 24.9%. For the same collimator setting 5 x 20 mm, by interchanging X, Y jaw settings the output also shown 4% difference. **Conclusion:** There was significant difference among plans with the same MLC field size but different collimator jaw settings, which would be verified by measurement with plastic scintillation detector. Recently we have adopted new planning guideline to have fixed jaws of 20 x 20 mm for all small field conformal arc plans,

PO-GePV-T-357

Comparison of Commercial Pre-Treatment QA Devices for Stereotactic Radiotherapy Plans

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Purpose: The goal of this research is to compare commercial quality assurance (QA) devices to the EBTXD film dosimetry method for pre-treatment evaluation of stereotactic body radiation therapy (SBRT) & Intracranial stereotactic radiosurgery (SRS) treatment plans. **Methods:** SBRT & SRS plans were created using IMRT or VMAT technique with varying target volumes (smallest volume of 11.98cc and largest of 40.01 cc) were created using Eclipse 15.6 treatment planning system. Plans were delivered using a Varian Edge LINAC and measured on QA devices at beam energies of 6 MV and 10 MV with and without flattening filter. The QA plans measured using EBTXD film on a 15cm solid water phantom and a A1SL ion chamber, SNC ArcCHECK (0.64mm² detector area with 10mm spacing) and Varian DMI EPID. Results analyzed at 3%/3mm, 2%/2mm, 2%/1mm, and 1%/1mm for each QA method. **Results:** EBTXD films were scanned using an Epson10000XL flatbed scanner and analyzed using RIT software and had average gamma analysis passing rates of 97.42%, 93.42%, 84.52%, and 76.07% for 3%/3mm, 2%/2mm, 2%/1mm, and 1%/1mm gamma criteria, respectively. The Varian DMI EPID produced an average gamma analysis passing rates of 99.46%, 99.90%, 99.29%, and 97.92% for 3%/3mm, 2%/2mm, 2%/1mm, and 1%/1mm gamma criteria, respectively. The SNC ArcCHECK produced an average gamma analysis passing rates of 96.82%, 90.84%, 78.18%, and 70.80% for 3%/3mm, 2%/2mm, 2%/1mm, and 1%/1mm gamma criteria, respectively. **Conclusion:** Passing rates decreased for all devices as gamma criteria becomes stricter. EBTXD film and ArcCHECK show agreement within 3.5% for 3%/3mm and 2%/2mm gamma criteria and begin to diverge from agreement as criteria becomes more stringent. In contrast, the EPID system resulted in high pass rates regardless of target size, modulation factor, and gamma criteria. To further study this topic, an SNC SRS MapCheck and IBA myQA SRS will also be used in the comparison.

PO-GePV-T-358

On the Optimal Number of Dose-Limiting Shells in the SBRT Auto-Planning Design for Peripheral Lung Cancer

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Purpose: The number of dose limiting shells in the optimization process is one of the key factors determining the quality of stereotactic body radiotherapy (SBRT) auto-planning. This study attempted to derive the optimal number of shells by evaluating the auto-plans designed with different number of shells for peripheral lung cancer patients treated with SBRT. **Methods:** Identical treatment technique, optimization process, constraints, and dose calculation algorithm were retrospectively applied to fifty peripheral lung cancer patients who underwent SBRT in our center. For each of the patients, auto treatment plans were optimized based on 2 shells, 3 shells, 4 shells, 5 shells, 6 shells, 7 shells, 8 shells, respectively. The optimal number of shells for the SBRT auto-planning was derived through the evaluations and comparisons of various dosimetric parameters of planning target volume (PTV) and organs at risk (OARs), monitor units (MU) and optimization time of the plans. **Results:** The conformity index (CI) and the gradient index (GI) of PTV, the maximum dose outside the 2cm of PTV (D2cm), Dmax of Spinal Cord (SCmax), the percentage of volume of total lung excluding ITV receiving 20Gy (V20) and 10Gy (V10), and the mean lung dose (MLD) were improved when the number of shell increased, but the improvement became not significant as the number of shell reached 6. The monitor units (MUs) varied little among different plans where no statistical differences were found. However, as the number of shell increased, the auto plan optimization time increased significantly. **Conclusion:** It appears that for peripheral lung SBRT auto planning using 6 shells can yield satisfactory plan quality with acceptable beam MUs and optimization time.

PO-GePV-T-359

Volume-Staged Stereotactic Radiosurgery (VS-SRS) Treatment of Large Arteriovenous Malformation AVM: Volumetric Modulated Arc Therapy Approach

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Purpose: Volume-staged SRS downsizes treated volume per stage which allows tolerating higher SRS dose to target's sub-volumes and reducing risk of brain necrosis. However, meeting SRS treatment plans quality matrix in VS-SRS plans is challenging given the angioarchitecture of staged-volumes and total volume, delivery technique and modality limitations. We present a VMAT VS-SRS planning approach for large AVM treated with VS-SRS to three sub-volumes of nidus. **Methods:** A patient with large AVM (16.5cc) in the right parietal lobe was simulated with Brainlab SRS-mask and 1 mm CT slice thickness were obtained and imported into Varian Eclipse TPSV15.6 and registered to MRA images data-sets and Angio-CBCT scans for contouring AMV nidus which was divided to feeding nidus (8 cc), draining-superior (4.25 cc) and draining-inferior (4.0cc) for geometric optimization purposes. 20 Gy was prescribed to the entire nidus volume and portions of total dose were delivered to sub volumes with interval of 2-4 weeks. VMAT plans were generated for each volume/iso center using non-coplanar arcs delivered on HDMLC Novalis-Tx. Collimator and arc angles were configured to minimize dose spillage between staged volumes. Planning objectives including generalized uniform equivalent dose gEUD were assigned to planning volumes and OARs. Normal tissue objective was enabled, structure resolution was set to 1.25mm, and AAA calculation grid was set to 1mm. Individual and composite plans isodoses and DVHs were examined. All individual sub-volume plans passed IMRT QA. **Results:** V95% was >95% and D95% was >95% and minimum dose was >93.5% for all three sub volumes and total volume in composite plan. CI was 0.63, PTV was 1.57, R50 was 4.36 while GI was 2.77, V12 was 52.9cc (3.9%). **Conclusion:** VMAT VS-SRS meets SRS plans quality parameters when compared with published literature covering VS-SRS delivery with different treatment modalities and techniques such as Cyberknife and GammaKnife, DCA and IMRT.

PO-GePV-T-360

Clinical Evaluation of a Novel Automatic Inverse Planning System in Gamma Knife Lightning

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Purpose: To evaluate the dosimetric qualities of plans generated using a novel automatic inverse planning system for Gamma Knife stereotactic radiosurgery (GKSRS). **Methods:** Patients treated with GKSRS for pituitary adenoma (PA), vestibular schwannoma (VS), and post-operative brain metastases (pBM) at a single institute were randomly selected. The original clinical plans were manually generated and reviewed prior to treatment. Each case was re-planned by two independent planners using the inverse planning (IP) system in Gamma Knife Lightning. The planning goal was to achieve at least 99% prescription dose coverage to targets, while minimizing beam-on time and dose to organs at risk (OARs), if any. The beam-on time, scaled by plan dose rate, and dosimetric metrics were compared among three plans. Statistical analyses were performed using the ANOVA and Tukey tests. **Results:** Thirty GKSRS patients were identified with 10 in each diagnosis type. At least 99% target coverages were achieved in the IPs, comparable to those in the clinical ones. There were no significant differences in the Paddick Conformity Index. Whereas the Gradient Indices were equivalent among the plans for PA and VS, the GIs in the IPs for pBM were significantly lower ($p < 0.01$) due to the relatively large targets. Similar dosimetric metrics of OARs were also observed between the IPs and clinical plans. The average beam-on times in the IPs were reduced by 25 minutes for PA ($p < 0.01$) and by 10 minutes for VS ($p = 0.60$), but unchanged for pBM. However, the reductions in the IPs for PA and VS were offset by the increased numbers of shots used to control the dose to OARs, resulting in comparable total treatment durations. **Conclusion:** Inverse planning in GK Lightning allows planners to generate GKSRS plans of equivalent plan qualities compared to manual planning. The automation accelerates GKSRS planning and therefore improves the clinical workflow.

PO-GePV-T-361

Dose Gradient Improvement for SBRT Lung Plans During Optimization

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Purpose: In this work, as a preliminary study and proof of concept, patient specific normal tissue objective (NTO) based on PTV volumes is applied during the optimization process to improve dose gradient R50% (ratio of the 50% isodose volume/PTV volume). **Methods:** Assuming PTV as a sphere, the radius can be derived from simple geometric formula. The desired 50% isodose volume can be obtained from the PTV volume times the R50% constraint from RTOG 0813 protocol, thus the radius of 50% isodose volume can be derived. The difference of the radius of these two spheres (the distance from PTV edge to 50% isodose line) is an indication of dose falloff. This distance (DNTO) is calculated according to PTV volumes and incorporated in the normal tissue objective (NTO) module of Eclipse TPS's photon optimizer. The setting is set to achieve the distance from PTV edge to the 50% isodose line be equal to DNTO. Retrospectively 5 lung SBRT patients are re-planned utilizing the NTO module, while maintaining other OAR constraints. The R50%, CI (Conformity index) and V20 (percent lung volume receiving 20 Gy) for each plan are then compared with previously treated clinical plans where NTO objectives were not used. **Results:** CI remains unchanged as expected. R50% is reduced for all plans, ranging from 1.2% to 7.7%, with an average of 4.4% when patient specific NTO are used. V20 are also reduced, ranging from 2.8-6.8%, with an average of 5.4% utilizing NTO. Further reduction is possible if more aggressive NTO setting is employed, which is under current investigation. **Conclusion:** R50% can be effectively reduced using patient specific NTO settings based on PTV sizes. V20 can also be improved utilizing NTO. The correlation of R50% reduction and potential better clinical outcomes is also manifested in the improved V20 values.

PO-GePV-T-362

Investigation of Treatment Planning Strategies for Multi-Isocenter Intracranial stereotactic Radiosurgery

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Purpose: To assess the impact of utilizing avoidance structures versus avoidance sectors on dosimetric performance, planning efficiency and patient-specific quality assurance (PSQA) of multi-isocenter frameless stereotactic radiosurgery (SRS). **Methods:** Five patients each with 4 to 6 lesions were included in this study. Three VMAT plans were generated for each patient, to treat one target and avoid the remaining lesions. While the treatment geometry was identical between the three plans, the first VMAT plan employed avoidance sectors (SRS-AvS) to minimize the dose to the untargeted lesions. The second and third plans designated the untargeted lesions as avoidance structures and prohibited "Entrance and Exit" of the beam through them for the second plan (SRS-NEE) and only "Entrance" for the third plan (SRS-NE). The three plans were evaluated using commonly employed dose-volumes and plan quality indices (Paddick Conformity Index [CIPaddik], Homogeneity Index [HI], normal brain [V12Gy], maximum dose to OARs and untargeted lesions) and total number of monitor units (MUs). PSQA was performed for the SRS-AvS and SRS-NEE plans and evaluated for gamma passing rate of 3%/3mm. **Results:** All three plans achieved similar target coverage, CIPaddik, HI, V12Gy and sparing of OARs and untargeted lesions. On average, the SRS-AvS plans resulted in 44% and 17% reduction in the total MUs when compared to SRS-NEE and SRS-NE plans, respectively. The greater the overlap between the targeted lesion and avoidance structures, the larger the reduction in total MUs offered by SRS-AvS plans. Regarding PSQA, two of the SRS-NEE plans that resulted in highest increase in total MUs did not satisfy the clinical criteria of 95% passing rate. **Conclusion:** Although the use of avoidance structures improves planning efficiency by eliminating the need of identifying avoidance sectors in the VMAT arcs it could potentially result in substantial increase in MUs and ultimately beam-on time.

PO-GePV-T-363

Validation and Clinical Implementation of VMAT Liver SBRT on 6MV-FFF O-Ring Halcyon Linac

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Purpose: Evaluate the plan quality, delivery efficiency, and accuracy of VMAT liver SBRT on 6MV-FFF O-ring Halcyon. **Methods:** Eight liver cancer patients received VMAT-SBRT, delivering 30-50 Gy in 5 fractions, prescribed to 80% isodose line on Halcyon. Mean PTV volume was 167.4±70.88 cc (range, 71.7-304.1cc). VMAT liver SBRT plans were generated with full RTOG-1112 compliance, advanced AcurosXB algorithm for heterogeneity corrections with Halcyon couch and SBRT board insert. Patients were treated every other day using 2-4 partial/full arc geometry with 6X-FFF beams including fast 15-second kV-iCBCT imaging for set up corrections. For comparison, clinical Halcyon plans were retrospectively reoptimized on TrueBeam Linac with identical geometry and optimization parameters. Plan quality, organs-at-risk (OAR) and treatment delivery parameters were compared after identical target normalization. For clinical Halcyon plans, quality assurance and couch corrections results are reported. **Results:** Halcyon and TrueBeam plans showed similar conformity index, total monitor units, modulation factors and maximal dose fall-off at 2cm away from the PTV. Mean PTV dose was 106.0±2.4% and 105.9±1.8% of the prescribed dose for Halcyon and TrueBeam, respectively. RTOG volumetric doses for OAR were similar: 8.7Gy vs 8.3 Gy (0.5cc of duodenum); 13.1Gy vs 12.9Gy (0.5cc of stomach); 9.6 Gy vs 9.9 Gy (0.5cc of small bowel) and 4.4 Gy vs 4.1 Gy (30cc of heart) for Halcyon vs TrueBeam plans. Halcyon pre-treatment portal-dosimetry QA result was 98.7%, on average for 2%/2mm gamma passing criteria. The 3D radial couch correction was 0.65±0.47cm (0.41–0.95cm). Mean beam-on time was 3.4±1.2min (2.0–5.7min). Overall treatment time on Halcyon was <15 min. **Conclusion:** For select patients, SBRT of liver cancer on Halcyon is safe, feasible, efficient and dosimetrically similar to TrueBeam Linac. Faster overall liver SBRT treatment on Halcyon may increase patient compliance, reduce intra-fraction movement and improve clinic workflow. Clinical follow up of these patients is underway.

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PO-GePV-T-364

Retrospective Characteristic Analysis of New Exponential Dosimetric Variable, Dose Dropping Speed (DDS) in the Treatment Plans of Arteriovenous Malformations (AVM) to Evaluate Dose Fall-Off

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Purpose: The purpose of this study is to analyze the trend of Exponential dosimetric variable Dose Dropping Speed (DDS) in the treatment planning of Arteriovenous Malformations (AVM) to evaluate dose fall-off and normal tissue sparing. **Methods:** A total of 30 patients of Arteriovenous Malformations (AVM) of volume ranging 0.03 cc-9.775 cc, Prescription Dose-ranging 14 Gy-25 Gy were included in this study. Treatment Plans were done using Iris or Fixed collimators depending upon the volume of the target. The DDS included two decay co-efficient: one which is in the vicinity of Gross Tumor Volume (GTV) named greater decay coefficient (b1) and another at a distance from GTV named lesser decay coefficient (b2). For performance evaluation of DDS, various equally situated ring-like shells were generated around the GTVs, and the mean dose in each shell was noted. Further, curve fitting was done using the exponential decay function of second-order and two different decay coefficients were obtained. **Results:** Prescription line for all the cases were in the range of 60%-80%. The Conformity Index (CI), New Conformity Index (nCI), Homogeneity Index (HI) and equivalent diameter of GTVs were 1.23±0.20, 1.29±0.20, 1.43±0.12 and 1.49±0.63. The average value of Decay co-efficient b1 and b2 were 0.33±0.10 and 0.21±0.12. The maximum and minimum difference between b1 and b2 was 0.3455 and -0.307. **Conclusion:** It was observed that 90% of the cases were having a larger value of b1 as compared to b2 which concluded that dose fall off near the GTVs was more and compared to dose fall away from the GTVs. A higher b2 value in 10% of the cases was observed where nCI was more. DDS shall be included in the Treatment Plan evaluation process. Furthermore, Institutional decay coefficient values can be standardized depending upon the tumor volume.

PO-GePV-T-365

Evaluation of Model-Based Dose Algorithms Using HU as a Density Surrogate for FFF-SBRT of Island Lung Lesions

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Purpose: To investigate dosimetric parameters of 4 algorithms for lung cancer patients treated with flattening filter free stereotactic body radiotherapy using HU as a surrogate for PTV density. **Methods:** We studied 15 lung cancer patients treated with SBRT using 6xFFF. Dose regimen was 12 Gy/fx in 4 fractions. Treatment plans were created using the pencil beam convolution in iPlan with IMRT and recalculated using identical parameters with iPlan's Monte Carlo Algorithm (MC), Analytical Anisotropic Algorithm (AAA), AcurosXB (AXB), and a collapsed cone convolution algorithm (Sun Nuclear Dose Check (SDC)). These lung-island tumors had ITV volumes of 12.44 ± 13.90 cm³. Data sets were indexed by average PTV density and several dosimetric indices for the target and normal lung tissue were evaluated. % difference relative to MC calculation is reported. **Results:** Average tumor densities were -594.3 ± 146.7 (-828.9 to -189.9 HU). The ΔD95 was -8.35 ± 3.22% (-13.00% to -2.99%), 1.16 ± 2.85% (6.47% to -1.93%), and -1.52 ± 1.55% (-4.58% to 1.44%) for SDC, AAA, and AXB respectively across all densities. Agreement with MC was moderately (0.576 ≤ r ≤ 0.713) correlated with density with a p < 0.05. A threshold for clinical agreement between the algorithms was noted around -450 HU. Below -600 HU ΔD95 was -11.39 ± 1.63%, 3.25 ± 3.29%, and -1.41 ± 1.71% for SDC, AAA, and AXB respectively. Of note, AAA and AXB were closer to MC than SDC but AXB and SDC both predicted under coverage relative to MC with AAA predicting over coverage. Lung V20 and V5 showed a tight spread with max difference within 2.0% of MC. **Conclusion:** Preliminary results suggest that target

coverage agreement has moderate correlation with target density and a threshold for clinical agreement might be established. Lung V20 and V5 showed a much tighter spread of agreement.

PO-GePV-T-366

Evaluation of Parameters Affecting Gamma Passing Rate in Patient-Specific QA's for Multiple Brain Lesions IMRS Treatments

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Purpose: Using single isocenter for treatment of multiple brain lesions with Intensity Modulated Radiosurgery (IMRS) have gained acceptance in recent years. One of the challenges of this technique is conducting a patient-specific quality assurance (QA), involving accurate gamma passing rate (GPR) calculations for small and wide spread-out targets. We evaluated effects of parameters such as dose grid and energy on GPR using our clinical IMRS plans. **Methods:** 10 patients with total of 40 volumetric modulated arc therapy (VMAT) plans were created in Raystation(V.8A) treatment planning system (TPS) for the Varian Edge Linac using 6 and 10 Flattening Filter-Free (FFF) beams and planned dose grids of 1mm and 2mm resulting in 4 plans with 6 to 10 targets per patient. All parameters and objectives except dose grid and energy were kept the same in all plans. Next, patient specific QA's were measured evaluating GPR with 10% threshold, 3%/3mm objective, and an acceptance criterion of 95%. Modulation factors (MF) and confidence intervals were calculated. Two modes of measurements, standard density (SD) and high density (HD) were used. **Results:** 1mm planned dose grid has higher GPR's than 2mm dose grid. The GPR's of 6FFF plans were higher than those of 10FFF. GPR showed no noticeable difference between HD and SD measurements. No correlation between MF and GPR was observed. The SD pass rates fall within the confidence interval of HD. **Conclusion:** Calculated dose grid should be less than or equal to 1/3 of distance to agreement, thus 1mm planned dose grid is recommended to reduce artifacts in gamma calculation. GPR of SD and HD measurement modes is almost the same which indicates SD mode is clinically preferable for performing patient specific QAs. According to our results using 6FFF with 1mm planned dose grid is more accurate and reliable for dose calculation of IMRS plans.

PO-GePV-T-367

Gamma Knife Thalamotomy – Imaging Verification of the Isocenter Location

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Purpose: As functional treatment of essential tremor (ET) becomes a popular protocol for Gamma knife radiosurgery, the geometrical location of the isocenter was determined based on the anatomical formula to identify the Ventral intermediate (thalamic) nucleus (VIM). Registered MRI allows for lesion positioning/localization, as well as guidance for GK isocenter placement, verification of this location from the post-MR was accomplished. **Methods:** Distortion or mis-registration of the MR introduces the uncertainties of target localization. The standard formula for determining the core of VIM was challenging. A Series of treated ET images were studied to identify the actual treatment location, 130 Gy prescription of dmax with 4 mm collimator was prescribed and the necrosis is observed to identify the margin accuracy. Image fusion analysis was performed to analyze the geometrical accuracy, the distance was measured with the necrosis ring with image fusion to show the 11 mm lateral to the midline of third ventricle was accurately correlated. MR fusion was performed. **Results:** With typical intracranial image fusion of pre/post treatment datasets of same patient, a case study has shown that post treatment MR verification of treated isocenter. Framed based GK radiosurgery has recommended that the lateral distance for the core should be half of the 3rd ventricle width plus 11 mm lateral shift, as this patient with 14 mm lateral shift to correlate the observable lesion was confirmed. Image fusion using frame has demonstrated that the target has been achieved with T1 necrosis with ring, which addressed radiation dose of ET thalamotomy was successfully delivered with proper OAR avoidance. **Conclusion:** Frame based GK thalamotomy with current formula to identify the core of Vim is accurate enough to generate necrosis with proper margin, which caused the noticeable clinical outcome for ET. The core of Vim was properly identified and delivered dose as planned.

PO-GePV-T-368

To Determine the Impact of Rotational Uncertainties On Dosimetric Coverage of Distant Targets Using VMAT Planning and a TPS Tool to Measure Plan Uncertainty

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Purpose: To determine the impact of rotational uncertainties on dosimetric coverage of distant targets using VMAT planning and a TPS tool to measure plan uncertainty. **Methods:** VMAT plans containing a spherical 8 mm GTV with uniform 1 mm PTV expansion at 5 cm from isocenter were generated in a phantom. Similar plans were also generated for a target at 6 cm to assess differential impact based on distance. All plans were normalized such that V100% > 98% for each PTV; CI ranged from 1.23-1.32. As the TPS tool for assessing plan uncertainty uses translational shifts only, shifts were calculated to represent the location of the target after rotations of 0.4 and 0.5 degrees. PTV and GTV dosimetric coverage was then analyzed to determine the impact of each rotation in pitch, roll, and yaw individually. The impact of positive versus negative roll was also evaluated to determine sensitivity to rotational direction at each distance. **Results:** GTV coverage remained at 100% for all tests. In no case did the d=5 target coverage fall below 93%, whereas in 2/6 cases the d=6 target did. Worst-case coverage for the d=5 target was 93.0% versus 88.3% for the d=6 target. Dosimetric impact was found to be sensitive to rotational direction for the d=6 target, with coverage suffering more with roll in one direction versus the other. **Conclusion:** Rotational deviation reduced PTV coverage at both distances evaluated. The effect was larger for greater degrees of rotation and was more variable at greater distance. Despite decreased PTV coverage, the 1 mm GTV to PTV margin ensured that GTV coverage remained at 100%. This data demonstrates that target coverage may be sensitive to uncorrected rotation. The potential impact of rotational deviations should be considered when evaluating targets far from isocenter with a specific GTV to PTV margin.

PO-GePV-T-369

Feasibility of Skull Base Stereotactic Radiotherapy on Elekta MR-Linac

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Purpose: Reirradiation of skull base lesions is often challenging due to the high dose gradient required to spare previously-treated nearby structures. This study investigates the feasibility of stereotactic radiotherapy for skull base lesions on Elekta MR-Linac. **Methods:** Two patients with locally-recurrent skull base lesions were treated with stereotactic radiotherapy in 2020. IMRT plans(mIMRT) were re-generated in Monaco for MR-Linac and compared to clinical RayStation VMAT plans(rVMAT) for Varian TrueBeam sTx. Simulation MR images were registered with planning CT images, and then manually shifted 2mm and 4mm individually in each of the lateral, anterior-posterior, and superior-inferior(SI) directions to simulate and test the online adapt-to-position(ATP) daily planning workflow. The quality of daily adaptive plans(dIMRT) was evaluated in terms of target coverage and dose to organ-at-risks(OARs). **Results:** Patient-1 had a smaller CTV1(1.4cm³) and a more irregular CTV2(17.4cm³) compared with Patient-2 larger CTV1(21.6cm³) and CTV2(34.3cm³). The mean registration error in Monaco was 0.3mm(range: 0.0 – 0.7mm). mIMRT plans achieved comparable target coverage and uniformity to rVMAT plans. Conformity and dose gradient were superior for rVMAT plans. All clinical goals were met in mIMRT plans for non-adjacent OARs, however mean doses of these OARs were inferior to rVMAT. Volumes enclosed by 50% prescription isodose line were larger in mIMRT plans. dIMRT plans achieved same clinical goals that used in corresponding mIMRT plans on simulated 2mm shift images but not on all 4mm images. This reflects the features of current MR-Linac with coplanar beams, sagittal MLCs, and 7mm MLC width. All ATP plans were concluded within 3-minutes. Brainstem and cochlea goals were not met for Patient-1 with 4mm lateral shift and required an Adapt-to-Shape plan(>20-minutes). **Conclusion:** MR-Linac has the potential for stereotactic radiotherapy to treat aggressive

tumors in the skull base region. Daily adaptive planning can track lesion changes while maintaining nearby critical structure dose constraints.

PO-GePV-T-370

Impact of MLC Uncertainty in Single Isocenter Multiple-Lesions SRS VMAT Plans

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Purpose: To evaluate the dosimetric impact of multiple-leaf collimator (MLC) positional uncertainties (including systematic and random errors) in single-isocenter multiple lesions stereotactic radiosurgery (SRS) with volumetric modulated arc therapy (VMAT). And we also explore the impact on dose distributions due to MLC positional uncertainties in different lesion volumes. **Methods:** Five single-isocenter multi-lesion SRS VMAT plans with planning target volumes (PTVs) of varying size (<0.2 cc to >10 cc) are selected to introduce simulated MLC leaf position errors in RayStation. There are two types of systematic errors, MLC banks shift in the same direction and in opposite directions are simulated for magnitudes of 0.5 and 1 mm. For random errors, MLC leaf position offsets follow normal distributions and uniform distributions are simulated for magnitude of 0.5 and 1 mm standard deviations and bounds. The resultant dose distributions re-calculated in RayStation using the same Monitor Units (MUs) as in original plans for all control points are compared with original dose distributions of PTVs. **Results:** With both systematic and random MLC optional errors, small volume lesions (<0.2 cc), the percentage dose differences between modified plans and the original ones at D98 and D95 are much higher than those of lesions in relative larger volume (~ 0.5 to 1 cc). For hot spots in plans, the percentage dose differences at D2 randomly fluctuate with lesion volumes which are smaller than 3 cc. For the lesion with large volume (>10 cc), the percentage dose differences at both D95 and D2 are just change slightly with systematic and random MLC errors. **Conclusion:** Data demonstrates that both systematic and random MLC uncertainties have an effect on D2, D95, and D98 of single isocenter multiple-lesion SRS VMAT plans. The MLC positional uncertainties' impact on dose is dependent on lesion volumes--more pronounced effect for lesions with small volumes.

PO-GePV-T-371

Use Monte-Carlo Simulation to Determine the Coverage-Based PTV Margin

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Purpose: To determine the PTV margins required to achieve a desired reliability (e.g., 95% of the time) for covering a specified percentage (e.g., at least 95%) of the CTV. **Methods:** Monte Carlo simulation was performed to simulate the CTV motion, assuming changes in CTV position follow an independent normal distribution, $N(0, \mu)$ in each of the three motion axes. We also limited the CTV to be perfectly spherical. Applying circular symmetry, the three-dimensional random motion was simplified to a one-dimensional random motion in the radius direction following the Chi-square distribution ($k=3$). For each simulated motion, the percentage of CTV covered by PTV was calculated. The simulation was run for 1 million times for each combination of CTV radius and PTV margin, from which the histogram of percent CTV coverage (by PTV) was obtained. **Results:** The reliability function for percent CTV coverage was plotted for CTVs of radius from 1μ to 30μ and PTV margins (M) from 0 to 3μ . We found that the PTV margin required to achieve a desired reliability for covering a specified percentage of the CTV is significantly affected by the CTV dimension. If the clinical goal is at least 95% CTV coverage 95% of the time, a smaller CTV will require a larger PTV margin to achieve this goal, e.g., $M=1.97\mu$ for $r=5\mu$, and $M=1.19\mu$ for $r=15\mu$. If $M=0$, only 59.1% of a CTV with $r=5\mu$ can be reliably covered by PTV 95% of the time, while it is 86% for $r=15\mu$. An exponential equation was derived to relate the CTV radius to the required PTV margin, $M = 2.72 \cdot \exp(-r/19.78) - 0.11$ for achieving this clinical goal. **Conclusion:** This study demonstrated a size dependence for coverage-based PTV margin. A useful equation was developed for quick assessment of the required PTV margin for a given CTV radius.

PO-GePV-T-372

Assessment of Dosimetric Robustness Against Manually-Introduced Tumor Positional Shifts Among Lung SBRT Patients

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Purpose: The lung tumor lesions were occasionally displaced from planned positions in reference to the bony anatomy during the treatment course. Due to the uncertainties in the dosimetric effects of tumor positional shifts on the quality of lung SBRT plan, the patient treatment was oftentimes interrupted or postponed for re-planning. The purpose of this work is to assess the dosimetric consequences of the "would-be" tumor positional shifts. **Methods:** We selected one lung SBRT patient and "manually" shifted the original target contours and the isocenter along six different orthogonal directions up to 6 cm. Both original and shifted target densities were overridden with either lung density or soft-tissue density to account for the changes. For each shifted plan, the dose was re-calculated with the same beam arrangement and MU as the original plan. The dose-volume histogram (DVH) of the internal/planning target volumes (ITV/PTV) and organs-at-risk (OAR) was assessed, including ITV V98%[%], PTV V95%[%], Ipsilateral Lung V10.5Gy[cc], Heart D0.035cc[Gy], Esophagus D0.035cc[Gy], and Cord D0.035cc[Gy]. **Results:** For lateral shift up to 3cm, PTV V95% decreased by 3% and the max percent increase of cord D0.035cc was 50%. For inferior shift up to 4cm, PTV V95% changed only by 0.4% while the max percent increase of lung V10.5Gy was 15%. For superior shift up to 2cm, PTV V95% decreased by 4% (the most percent change among all directions) while DVH percent changes of all OAR were less than 6%. For anterior-posterior shifts up to 6 cm, PTV V95% decreased by 2% whereas the max percent increase of heart D0.035cc was 40%. For all directional shifts, ITV V98% was 100%. **Conclusion:** Target coverage was robust against all directional tumor displacements. The DVH percent change of OAR were sensitive to both shift direction and magnitude, particularly lateral shift.

PO-GePV-T-373

Dosimetric Impact of Anatomical Changes Between Photon and Proton Stereotactic Body Radiation Therapy for Lung Cancer

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Purpose: To investigate the impact of tumor displacement (TD) on target coverage in photon and proton stereotactic body radiation therapy (SBRT) treatments for lung cancer patients under daily IGRT. **Methods:** For patients treated with photon SBRT for lung cancer in 2017-2019 in our institution, 16 fractions from 16 patients with daily iso-center shifts >1.5 cm in any directions were identified according to kV-cone-beam CT (kV-CBCT) guidance. TDs of these fractions were identified by the difference between the iso-center shifts of bony alignment and tumor alignment. Eight fractions from eight patients had TDs in any directions >3 mm. For each of these eight patients, one photon plan with VMAT technique and one proton plan with IMPT were generated retrospectively. Photon and proton plans were optimized to achieve $>95\%$ of the PTV and $>98\%$ of the ITV receiving the prescription dose, respectively. For dose recalculation with TD, we shifted the tumor contours according to the corresponding TD on the patient specific planning CT. HU override was applied to shifted ITV (0.7 g/cm³) and original ITV (0.26 g/cm³). We evaluated dosimetric impact of TD with the percentage volume of GTV and ITV receiving the prescription dose (V100%). **Results:** The range of TDs was 3.58–12.18 mm, where the translational shifts was 0.24–4.45 mm in ML direction, 0.11–11.69 mm in AP direction and 0.9–10.57 mm in SI direction, respectively. For photon SBRT, all fractions had V100% of the GTV $\geq 99\%$ and V100% of the ITV $\geq 98\%$. For proton SBRT, V100% of the GTV and ITV had a range of 12.60%–96.21% and 34.17%–83.75%, respectively. **Conclusion:** Adequate target coverage was achieved in photon SBRT plans. TD had greater impact in proton SBRT plans, leading to target underdosage and adaptive planning was needed to provide adequate tumor dose coverage.

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PO-GePV-T-374

The Dosimetric Effect of Using a Titanium Material Assignment Versus Limited Hounsfield Units for SBRT Spine Patients with Hardware

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Purpose: To evaluate the dosimetric differences observed in spine SBRT patients when assigning a material to contoured spine hardware as compared to utilizing the Hounsfield units (HU) obtained during computed tomography simulation. **Methods:** Forty-five spine SBRT patients with titanium hardware were retrospectively analyzed. The clinical plan was re-calculated with AcurosXB (version 15.6.05) set to report dose-to-medium with the spine hardware set to 3071 HU. A duplicate of the clinical plan was calculated with the material of the hardware contour set to titanium (density of 4.42 g/cm³ and 8610 HU) via the physical materials table available. A planning target volume for evaluation (PTVeval) was created by subtracting the hardware from the clinical PTV. Dosimetric statistics for PTVeval, spinal cord, cauda equina, and esophagus were computed and analyzed. The PTVeval was used for comparison purposes as dose within the rods is not clinically significant. **Results:** The compared PTVs displayed average percentage differences for maximum dose, mean dose, minimum dose, median dose, and dose to 95% of the volume (D95) of 1.95%, -0.18%, -0.44%, -0.16%, and -0.86%, respectively. Whereby a positive percent difference means that the plans with titanium assigned show higher dose. The average percent difference for the spinal cord, cauda equina, and the esophagus are 0.04%, -0.14%, and -0.22%. When the hardware encompassed more than 10% of the PTV volume, the average percentage differences for maximum dose, mean dose, minimum dose, median dose, and D95 were 2.90%, -0.34%, -0.74%, -0.30%, and -1.41%, respectively. **Conclusion:** The largest observed dosimetric difference for the D95 was -3.5%, representing a decrease in coverage when assigning titanium material versus 3071 HU. Additionally, no dosimetric differences for the organ at risk metrics exceeded 2.5% when assigning titanium material versus 3071 HU.

PO-GePV-T-375

Quantitative Assessment of Inter-Fractional Anatomy Change On Dose Delivery of Prostate SBRT

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Purpose: Target coverage and OAR sparing deficiency may be more prone to inter-fractional anatomy change due to the fewer fraction nature of hypofractionation. This study quantifies inter-fractional anatomy variations and evaluates their dosimetric impact during prostate SBRT using image deformation technique. **Methods:** Ten patients diagnosed with favorable intermediate risk prostate cancer were retrospectively included in this study. Total dose was prescribed to 36.25Gy in 5 fractions. Following guidelines from RTOG 0938, a CTV to PTV expansion of 5mm in all directions except for 4mm posteriorly were used. Patients were instructed to maintain a full bladder and empty rectum prior to CT simulation and each treatment. Localization was achieved through aligning to 3 fiducial markers in the prostate on daily CBCT. To evaluate the dosimetric impact from inter-fractional anatomy change on actual delivery, the body and OARs (bladder, rectum, penile bulb, femurs) were deformed to daily CBCT before performing forward dose calculations on each deformed CT. Inverse vector-field deformation was applied before creating a 5-fraction composite back onto the planning CT. **Results:** A 4.32%±32.32%(4.58cc±81.28cc) inter-fraction volume variation was observed for bladder, and -4.06%±4.86%(-2.02cc±2.41cc) for rectum. Discrepancy was noted between fiducial marker/CTV-focused versus pelvic bone-focused rigid registrations. The translation misalignment ranged from 0.05mm-9.38mm, and rotational angles from 0.0°-3.71°. CTV coverage from the composite dose remained comparable, with D95% at 100.73%±1.29% of the prescription dose(Rx) for the original plan and 100.31%±1.48%Rx for the delivery. The dosimetric parameters of OARs, including D0.03cc, D3cc, D90%, D80%, and D50% of rectum, and D0.03cc, D90%, and D50% of bladder remained comparable to those in treatment plan. **Conclusion:** In spite of the inter-fractional anatomical change (including, but not limited to, bladder filling, rectal emptying and bony anatomy change), the CTV coverage, the rectum and bladder sparing of the overall course of delivery remained similar to the planned condition.

PO-GePV-T-376

Analysis of Plan Quality for Intensity-Modulated Proton Therapy Vs. Photon Stereotactic Radiosurgery for Brain Metastases

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Purpose: The purpose of this work was to quantify the difference in plan quality between intensity- modulated proton therapy (IMPT) and photon-based stereotactic radiosurgery (SRS) treatment planning for multiple brain metastases. **Methods:** Twenty-two brain lesions from six patients, were planned using scanning beam proton and compared against photon-based SRS plans. Proton plans were optimized and evaluated using the Monte Carlo algorithm and photon plans were optimized using a collapsed cone convolution algorithm. Dose coverage to each lesion was set to 99% of the target volume receiving the prescription dose for all plans. Conformity index (CI), gradient index (GI), total V4.5Gy, mean brain dose and V12Gy for each lesion, were used to compare plan quality between the proton and photon plans using dose-volume histograms (DVHs) generated from each plan. **Results:** Both average CI and GI across all lesions were generally lower than the proton plans than those from the photon plans, but both were comparable, with a mean CI of 2.29 ± 1.47 and 2.48 ± 2.48 for protons and photons, respectively, with p>0.05. Mean GI was 4.58 ± 3.13 and 5.18 ± 2.60 for protons and photons respectively, with p>0.05. Mean V4.5Gy was 59.3 ± 35.7 cm³ and 203.8 ± 121.9 cm³ for proton plans and photon plans, respectively, with p<0.05. Mean brain dose was 36.7 ± 25.8 cm³ and 135.2 ± 47.5 cm³ for protons and photons, respectively, with p<0.05. **Conclusion:** We quantified the plan quality for both IMPT treatment plans and photon SRS plans for brain metastases. Proton therapy for brain lesions offers a good alternative to photon-based SRS treatment, with a lower average CI and GI, V4.5Gy and mean brain dose, and can significantly increase preservation of normal brain tissue.

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PO-GePV-T-377

Robustness of AI Treatment Planning to Account for Inter-Fractional Changes in Targets and OARs

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Purpose: To evaluate robustness of an AI model developed from a different institution to create adaptive plans for our pancreas patients to account for inter-fractional changes in the targets and organs-at-risk during a course of radiation therapy for pancreatic cancer. **Methods:** Fifteen patient CTs, including planning and daily CTs, were contoured using an image analysis software. Major organs-at-risk contoured include the duodenum, stomach, small bowel and colon, along with a GTV and PTV. Treatment plans for each patient CT were planned using an artificial intelligence model developed for pancreas patients. These fifteen patient CTs were not used for the model development. Prescription dose was 33 Gy and 25 Gy to the GTV and PTV, respectively. DVHs were generated and evaluated for each GTV, PTV and each major organ-at-risk. Dosimetric constraints for each organ-at-risk include dose at 0.03 cc < 33 Gy, 0.5 cc < 30 Gy and 5 cc < 25 Gy. Target volumes were evaluated for dose at 95% volume and percent volume at 95% dose. **Results:** Overall, the AI model was successful to create adaptive plans with adequate dose coverage to all targets. Most OAR constraints were also met. Average PTV percent volume at 95% dose for the AI model was 96.29 ± 5.20%. Average dose at 95% volume PTVs was 32.21 ± 0.68 Gy. **Conclusion:** This study demonstrates that the AI model developed from another institution is robust and can be applied to our patient images for daily adaptive planning.

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Selection of Single-Isocenter for Multiple Targets Based On Simulated Annealing Algorithm

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Purpose: Treating multiple targets with a single isocenter improves efficiency but requires margins to account for rotation which is related with target-to-isocenter distance. A method to select the isocenter position for multiple targets that minimizes the total volume of expanded target is presented. **Methods:** A statistical model was developed considering both the conventional setup error and the additional rotational uncertainty. Both translational and rotational uncertainties are assumed to follow the three-dimensional independent normal distribution with a zero mean, and standard deviations of σ_s and σ_R respectively. The relationship between margin of each target point and the distance d (from isocenter to target point), setup uncertainties, and significance level was established. In order to find the optimal iso-center position quickly, simulated annealing optimization algorithm was used. This method was implemented in the Pinnacle3 treatment planning system and compared with isocenters at the center-of-mass (COM) and center-of-surface-area (CSA). Spherical tumors and ten multiple brain metastases targets were chosen for evaluation. **Results:** For multiple brain metastases targets, isocenter placement at the optimal location, COM, and CSA, reduced the total margin volume versus an isocenter at geometric center up to 18.9%, 17.5%, and 18.6%, respectively. CSA provide useful approximations to select the optimal isocenter for multi-target. Simulated annealing algorithm can find the optimal isocenter for multiple targets within five minutes. The result has nothing to do with the position of the initial point used in simulated annealing algorithm. This method is not only suitable for two targets, but also can be used for dozens of targets. **Conclusion:** Optimal treatment isocenter selection for multiple targets of large size differences can significantly reduce total margin volume. This method could be used for clinical margin determination and might be useful for the protection of risk organs.

PO-GePV-T-379

Dosimetric Comparison of ZAP-X and CyberKnife Stereotactic Radiosurgery for Single Brain Metastasis

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Purpose: To evaluate the dosimetric characteristics of ZAP-X and CyberKnife (CK) G4 stereotactic radiosurgery (SRS) for single brain metastasis. **Methods:** Ten patients with single brain metastasis treated with CK were re-planned in the ZAP-X planning system using the same prescribed dose. The prescribed dose of planning target volume (PTV) was 20-24Gy for 1-3 fractions. The PTV volumes ranged from 0.68-11.52 cm³. The same target and organs at risk (OARs) constraints were used, and the prescription dose of PTV was normalized to 70% for both ZAP-X and CK. The dosimetric parameters and total MUs, treatment time (TT) were evaluated for PTV and normal structures. A paired t-test was performed to show the significance of difference. **Results:** For coverage, CK is significantly higher than ZAP-X (98.89±0.89% vs 97.69±1.54%, $p<0.01$), but ZAP-X shows a higher conformity index (0.80 ±0.05 vs 0.77±0.08), and a lower Paddick gradient index (2.97±0.26 vs 3.04±0.24) than CK. Because all the prescribed doses of PTV are normalized to 70%, the plans have the same HI. The total MUs of ZAP-X are significantly lower than CK (12159.9 ±5423.0 vs 25940.1 ±2752.3, $p<0.01$) and shows shorter TT than ZAP-X (25.67 ±6.67 vs 37.89 ±4.94 min) ($p<0.01$). For V2-V21 and Dmean of brain tissues, there are no significant differences between two modalities, but ZAP-X have a little lower than CK. But CK may have a certain advantage for the target close to OARs, such as lens, due to no function of avoiding certain organs in ZAP-X. **Conclusion:** For SRS treating single brain metastasis, ZAP-X can provide a high quality plan equivalent to or even better than CK, especially reducing the treatment time. ZAP-X may become a new SRS platform for the treatment of brain metastasis.

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PO-GePV-T-380

Direct Dosimetric Comparison of LINAC Vs GK FSRT for Large Brain Tumors

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Purpose: To directly compare the plan quality of Gamma Knife (GK)- vs linear accelerator (LINAC)-based delivery techniques for fractionated stereotactic radiotherapy (fSRT) of large brain metastases. **Methods:** For 15 patients with >10cc clinical target volumes (CTVs), comparative plans were run for 9Gy×3 fractions utilizing Eclipse vs Leksell GammaPlan (LGP) treatment planning system (TPS). Each GK plan was first developed using LGP's automatic planning initially, followed by manual adjustment/refinement. The same MRI and structures (CTV+OARs) were then DICOM-transferred to Eclipse TPS, and used to run Volumetric Modulated Arc Therapy (VMAT) plans and Dynamic Conformal Arc (DCA) plans for a Truebeam with high-definition MLCs, using a single isocenter and 3 non-coplanar arcs. No PTV margins were added and no heterogeneity correction was used for either TPS. GK plans were prescribed to the 50% isodose line, and Eclipse VMAT and DCA plans allowed the maximal dose up to 170% and ~125%, respectively. Gradient index (GI), selectivity, Paddick Conformity Index (CI), V20GyRind, and V4GyRind of all three techniques were calculated and compared. **Results:** Median CTV volume was 13.9cc (range 10.3cc-39.1cc). Mean±standard deviation of Paddick CI were 1.18±0.05, 1.11±0.03, and 1.37±0.16 for GK, VMAT and DCA plans, respectively. They were respectively 2.64±0.17, 2.47±0.17, and 2.76±0.40 for GI; 13.9±5.8, 9.8±3.29 and 17.80±6.6 for V20GyRind; and 280.1±88.4, 305.4±84.9, and 368.53± 86.8 for V4GyRind. The differences were all statistically significant with $p<0.01$ for most cases. **Conclusion:** A direct comparison using the same MRI scan and contours was performed to evaluate the plan quality of various FSRT delivery techniques for CTV>10cc. LINAC VMAT provided the best dosimetric outcome in regards to CI, GI, and V20GyRind. GK outcomes were similar while DCA outcomes were significantly worse. Even though GK has a smaller collimator penumbra, VMAT outperformed GK in this study due to enhanced penumbra sharpening and better beam optimization.

PO-GePV-T-381

Dosimetric Evaluation of Lung Stereotactic Body Radiation Therapy (SBRT) Treatment Plans

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Purpose: To retrospectively analyze SBRT treatment plans. **Methods:** Twenty-Five Lung SBRT treatment plans with the prescription dose of 5000 cGy delivered over 5 fractions were retrospectively analyzed. Abdominal compression was used to control the respiratory motion at 4D CT simulation and during treatments. An internal target volume (ITV) was defined based on findings of 4D CT scans and PET/CT scans. ITV was expanded isotropically by 3-5 mm to create planning target volume(PTV). All treatment plans were developed using Pinnacle treatment planning system using 1-2 VMAT arcs. Various dosimetric parameters of target volumes and normal structures were calculated and compared with RTOG 0813 protocol constraints. **Results:** The average percentage PTV volume that received the prescription dose (Rx) was (96±1) %. The percentage PTV volume that received 95% of Rx ranged between 99% to 100%. The average conformity (R100%) and homogeneity (HI5%/95%) indices were (1.01 ±0.04) and (1.3±0.1) respectively. The average R50% for all plans was (5.8±1.3). The average maximum dose to any point at 2 cm from the PTV in any direction was (53±9)% of prescription dose. The average geometry index (GI=R50%/R100%) was (5.9±1.6) and the average Paddick conformation number was(0.97±0.07). Dosimetric constraints for all normal structures were met,

except for rib in fifteen treatment plans. **Conclusion:** The conformity index and the maximum dose 2 cm away from the PTV for all plans were well within the RTOG 0813 protocol criteria. However, the criterion of R50% was difficult to meet in some plans. This might be the result of irregular shape of some tumors and the proximity of PTV to OARs. Rib were either overlapping with PTVs or very close to them in plans that failed to meet the dosimetric constraints of this structure. In conclusion, there is potential improvement for periphery dose gradient and intermediate-dose spillage.

PO-GePV-T-382

Thoracic Spine SBRT: Is Free Breathing CT Sufficient for VMAT Planning?

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Purpose: Successful stereotactic body radiation therapy (SBRT) requires both accurate patient positioning and adequate motion management. Spine SBRT, although generally considered as treating a rigid target with minimum motion, could nonetheless be affected by breathing rhythms due to changes of tissue intersected in the beam path. This study focuses on dosimetric variation of VMAT SBRT treating spines approximate to diaphragm. **Methods:** Spine SBRT plans were generated on images of three previously treated patients who had both 4D-CT and free breathing CT performed in the same session. Two spinal vertebrae parallel to the diaphragm apex were identified as target. A treatment plan of 16Gy in 1 fraction was generated on the free breathing CT, following TG101 constraints for spinal cord and esophagus. The treatment plan was then projected onto each phase of the 4D-CT, and the calculated doses were accumulated through deformable dose composition and considered as the 'treated' dose on the free breathing CT. Differences between planned dose and 'treated' dose were analyzed. **Results:** PTV volumes were between 100.9cc (T6-7) and 126.5cc (T10-11). Diaphragm motion in the superior-inferior direction were between 4.8mm and 5.9mm. Compared with the planned dose, 'treated' dose differed -3.3~0.2%, -2.5~0.4% for PTV V100% and CTV V100%, -0.12~0.51Gy for cord D0.03cc, -0.24~0.17Gy for esophagus D0.03cc, and -4.8~9.7cc for lung V5Gy. Global Dmax varied less than 0.15Gy and stayed within PTV. Locally, the largest dose differences were observed in lung and liver where doses increased up to 2.74Gy. Noticeable continuous dose variations approximately coincided with the superior and inferior field boundaries, as well as domes of diaphragm. **Conclusion:** Statistic parameters revealed discernable difference between the planned and 'treated' dose, although small. Treatment planning should strive to leave rooms for such dosimetric variations. In addition, extent of dose perturbation effect on different planning techniques should be included into analysis.

PO-GePV-T-383

Variance in DVHs Based on Machine Repeatability

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Purpose: Dose volume histogram (DVH) data presented during the treatment planning process generates discrete values for structures. Patient movement, daily alignment variation, and deviations in machine delivery parameters generate uncertainty in the expected dose distribution within the patient. Quantifying how these variables impact plan delivery can help to produce more realistic DVHs. In this study, we assess the differences in DVHs generated from multiple patient deliveries. **Methods:** Log files produced from three Elekta Versa HD linacs were collected from ten unique head and neck patients. Each patient underwent at least 30 fractions. Each log file was used to generate a new radiation plan (RP) file that mirrored dose rates, gantry and collimator angles, and MLC locations. New RP files were used in conjunction with the original patient CT and structures to recalculate dose distributions and DVHs for each fraction of a patient's delivery using the Monaco treatment planning system. Dose distributions were exported and interpreted using the original structures in MATLAB. **Results:** Normal tissue volumes were found to have an average DVH variation of 25 cGy, and target volumes had 60.5 cGy. Deviations occurred most often for structures receiving higher doses to a lower normalized volume. There was no correlation between total structure volume and DVH variance. No specific region (i.e. PTV or OAR) demonstrated significant deviation due to machine delivery from the planned dose. **Conclusion:** DVHs generated from treatment delivery log files on the Elekta Versa HD linear accelerator maintain tight tolerances for delivery reproducibility. All deviations were within 1% of their expected value. The DVHs generated in this study provide valuable insight into how delivered dose correlates to a planned dose based on machine reproducibility. We plan to incorporate daily CBCTs for dose calculation to study how alignment variability and machine delivery impact treatment quality.

PO-GePV-T-384

Comparison of Monte Carlo and Pencil Beam Algorithms in Brain Dose Calculations for Stereotactic Radiosurgery of Multiple Brain Metastases

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Purpose: In stereotactic radiosurgery, dose to brain tissue is a major factor for necrosis. When multiple brain tumors are present, it is more challenging for accurate brain dose calculations because the brain dose might be the sum of scattered low doses from individual tumor treatments. This retrospective study compared brain doses calculated with Pencil Beam(PB) and Monte Carlo(MC) algorithms for SRS patients with multiple brain metastases. **Methods:** Fifteen previously treated patients with number of brain tumors ranging from 7 to 14 were selected. Tumor volume varied from 0.1 to several cc. Treatment planning system was Brainlab iPlan. Each tumor was planned individually with 3 to 4 conformal dynamic arcs, with the isocenter located at the PTV center. Prescription dose was up to 24Gy, dependent on PTV volume. In this study, all these plans originally calculated with PB algorithm were re-calculated with MC algorithm. All arc parameters were kept the same, so that the sole difference in calculations was the algorithm. The data collected for comparison was the 12Gy volume (V12) to brain tissue, which is a common element in evaluating brain radionecrosis. **Results:** Differences in brain V12 were discovered. MC usually presented higher V12 than PB. However, the difference varied from case to case. V12 was almost identical for a few cases. For some other cases, the difference reached near 4%. The percentage difference in V12 was not associated with either total PTV volume or number of PTVs. The cause of V12 variation would require further investigations. Even at 4%, the change in V12 might not be significant enough clinically to justify the adoption the MC algorithm. **Conclusion:** Even though noticeable difference in brain V12 was discovered between the algorithms, the difference may not be significant enough to revise dose calculation algorithms. PB algorithm is accurate enough for most cases.

PO-GePV-T-385

Algorithm for Forward SBRT Planning with Constant Dose Rate Flattening Filter Free Dynamic Conformal Arcs

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Purpose: Dynamic Conformal Arcs (DCA) are an efficient planning alternative to VMAT. Recent published techniques include variable dose rate or additional inverse planning. Pointed Flattening Filter Free (FFF) beams is more challenging. This algorithm presents a concise forward planning constant dose rate DCA procedure for rapid SBRT FFF planning. **Methods:** Qualifying PTV contours are smooth concave shapes without close proximity or overlap with critical structures. Basically the technique is suitable if it is determined that a conformal dose distribution will meet all constraints. A partial arc with isocenter near the PTV center is first selected to conform to the target shape as much as possible. The dose is normalized to 100% at isocenter and the 95% isodose is converted into a structure S95. A Boost structure is then defined as Boost = PTV - S95. The "FIT and SHIELD" feature in ECLIPSE treatment planning system can automatically create additional arcs to simultaneously treat the Boost but avoid additional dose to S95. Often two arcs are sufficient. The relative weights of the initial arc conforming to the entire PTV and the Boost arcs are adjusted to provide the desired distribution. If necessary a new Boost structure is created and the process repeated. Note that normalization that may create a desired area of high dose in the PTV will also increase the 50% isodose volume and consequently the Gradient Index. **Results:** The algorithm was successfully tested using phantom plans with various PTV configurations. Three peripheral lungs

and two adrenal cases were planned which were comparable to acceptable protocol SBRT VMAT plans. **Conclusion:** An algorithm for rapid forward SBRT DCA planning was developed and tested. Unlike previous techniques only DCA are used at constant dose rate. Applications include use for high throughput without IMRT patient specific QA and adaptive therapy with automated script planning.

PO-GePV-T-386

The Influence of Iso-Center Location, Beam Type and Lesion Size on Plan Specific QA in Lung SBRT

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Purpose: Lung SBRT is used in clinics routinely to treat early stage and oligometastatic tumors. This work investigates how treatment iso-center (Tx-iso) location with respect to PTV centroid, beam type (6MV vs 6MV FFF) and lesion size effect plan specific QA. This work aims to provide a framework to dosimetry team when planning for these cases in order to achieve a clinically deliverable plan. **Methods:** An anthropomorphic phantom was scanned along the thorax region. An ITV (Internal Treatment Volume) was defined in posterior right lung having diameter of 0.5 cm and density of 1 g/cm³. PTV (Planning Target Volume) was constructed by expanding the ITV by 0.5 cm. Four scenarios were explored with regards to iso-center location and beam type as below: 1) Tx-iso at PTV centroid using 6 MV FFF beam; 2) Tx-iso at PTV centroid using 6 MV flat beam; 3) Tx-iso away from PTV centroid anteriorly by 10 cm using 6 MV FFF beam; and 4) Tx-iso away from PTV centroid anteriorly by 10 cm using 6 MV flat beam. In all cases VMAT plans were developed using Pinnacle Treatment Planning System to deliver 48 Gy in 4 fraction to at least 95% of the PTV while meeting OARs constraints following RTOG 0915 protocol. The plans were then transferred to SNC ArcCHECK phantom for plan specific QA and delivered using a TruBeam ver. 2.7 machine. **Results:** The gamma analysis pass rate for 3% dose, 3mm distance criteria and 10% threshold were obtained as 100%, 100%, 88.0% and 95.4% for scenarios 1 to 4 above respectively. **Conclusion:** Our results indicate the best QA results are obtained when the tx-iso is positioned at the PTV centroid. Having the Tx-iso moved away from the PTV center results in a drop of pass rate specially for 6 MV FFF beam.

PO-GePV-T-387

A Dynamic Conformal Arc Technique Based on a High Dose Rate Flattening Filter Free Beams for Peripheral Lung Cancer

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Purpose: To compare two kinds of dynamic conformal arc techniques in the high dose rate flattening filter free (FFF) beams and evaluate the dosimetric differences. **Methods:** 20 patients with early peripheral non-small cell lung cancer were selected, and two dynamic conformal arc plans were designed in the Eclipse 10.0 treatment planning system (TPS), one of them based on tumor-center (T-DCA) and the other based on iso-center (Iso-DCA). All plans were created using the TrueBeam linear accelerator, based on 6 MV FFF photons with a dose rate of 1400 MU/min. All patients received the prescribed dose of 4800 cGy in four fractions (1200cGy/fraction). Target coverage and organ at risk limits were planned and designed according to the radiation therapy oncology group (RTOG) criteria. **Results:** There was no significant difference in the target coverage between the T-DCA and Iso-DCA plans. Both conformal index (conformity index, CI) and uniformity index (homogeneity index, HI) had no significant differences, but D2cm and R50% showed significant differences. For the monitor units (MU), Iso-DCA increased by 21% compared with T-DCA. For the organs at risk (OARs), the maximum dose of spinal cord and esophagus with Iso-DCA increased by 112.5% and 146.8% over T-DCA, respectively. There was no significant difference in the other OARs. **Conclusion:** The dose fall-off of Iso-DCA is better than T-DCA, but T-DCA was consistently superior in sparing dose to spinal cord and esophagus, and T-DCA has fewer monitor units.

PO-GePV-T-388

Dosimetric Study of Stereotactic Body Radiotherapy Using Flattening Filter Free Beams for Peripheral Lung Cancer

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Purpose: To evaluate the dosimetric differences between two stereotactic body radiotherapy (SBRT) techniques for the early stage peripheral non-small cell lung cancer (NSCLC). **Methods:** Fifteen patients with early peripheral NSCLC were selected and underwent 4D-CT scan. The target volumes and organs at risk (OARs) were delineated according to RTOG protocols. SBRT plans based on dynamic conformal arc (DCA) and rapidarc (RA) were created respectively in the Eclipse 15.0 treatment planning system. All plans used Truebeam accelerator with 6MV photons using flattening filter free beams with a dose rate of 1400MU/min. All patients received a prescription dose of 48 Gy given in 4 fractions. The target dose coverage and organs at risk limits were considered according to the RTOG constraints. **Results:** There was no significant difference between DCA and RA plans for the dose coverage of target area and OAR exposure dose ($P > 0.05$). All plans passed the RTOG criteria. The conformal index (CI) of DCA and RA plans were 1.45 ± 0.16 and 1.06 ± 0.02 , the difference was statistically significant. And there was no significant difference in the uniformity index (HI) ($P > 0.05$). Both DCA and RA plans showed significant differences for the D2cm, R50%, monitor units (MU) and planning time ($P < 0.05$). **Conclusion:** There was no difference in sparing dose to OARs in all plans, but RA was significantly better than DCA for the conformity index. DCA plans had significantly less monitor units and shorter planning times over that of RA.

PO-GePV-T-389

Dosimetric Impact of Peripheral MLC Leaves for Stereotactic Radiosurgery

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Purpose: To investigate the impact of multi-leaf collimator (MLC) characteristics on patient-specific quality assurance (PSQA) measurement agreement for stereotactic radiosurgery (SRS) treatment plans **Methods:** 328 multi-target 10MV flattening-filter-free HyperArc (Varian Medical Systems, Palo Alto, CA) plans using AcurosXB₁₅₅₁₁ were measured with the SRS MapCHECK (Sun Nuclear Corporation, Melbourne, FL). For each plan, measurements were done for the smallest and largest targets with the phantom positioned such that the target was located at the central diode. Peripheral dosimetric leaf gap (pDLG) and central dosimetric leaf gap (cDLG) were measured via ion chamber. Plans were recalculated to simulate different pDLG and cDLG values using the treatment planning system (TPS) scripting interface. The difference between the center diode dose and the TPS point dose, δ , was obtained for clinical and recalculated plans. For each measurement, the fraction of MU delivered to the center diode through apertures defined by the peripheral leaves, f , was calculated. For original and recalculated plans, δ was compared with f . **Results:** The measured cDLG and pDLG were 0.48mm and 0.29mm, respectively. For the clinical plans, 67% of the measurements had f zero, for which the mean δ was -0.8%. The mean δ for the recalculation was also -0.8%, as expected. 8.8% of plans had $f > 0.5$, for which the mean δ was -4.6% for clinical plans. The mean δ for the recalculation was -3.8%. The largest difference between the clinical and recalculated δ was 1.9% with f of 0.8. **Conclusion:** By modeling the pDLG as a separate value, dose agreement between TPS and measurement was modestly improved. However, a different pDLG accounted for only a portion of the difference for targets treated through peripheral leaves. Work is ongoing to identify other contributing sources to the dosimetric difference.

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PO-GePV-T-390

Organ-At-Risk Motion Analysis for Pancreatic Tumor Stereotactic Radiotherapy

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Purpose: Using daily diagnostic CT to quantify inter-fractional anatomical changes and motions in the pancreas tumors and nearby organs-at-risk (OARs) in a course of radiation therapy for pancreatic cancer. **Methods:** Thirteen daily CT scans were contoured and evaluated for interfractional motion of target volumes and surrounding OARs organs-at-risk. The GTV, PTV, duodenum, stomach, small bowel and colon were manually contoured on each of thirteen CT scans. Using an imaging analysis software program, contours from each daily CT scan were compared with those from the corresponding planning CTs with the Dice Similarity coefficient (DSC), mean slice-wise Hausdorff distance (MSHD), and average centroid motion. **Results:** Median DSCs and Hausdorff coefficients for each target and OAR were plotted and compared. Mean DSCs and Hausdorff coefficients for each target and OAR were also compared in a table with standard deviations. Comparing DSC between the planning CT and daily CT for all thirteen CTs, the largest DSCs were observed for the PTV and stomach, with average values of 0.63 ± 0.07 and 0.65 ± 0.08 , respectively. Median values were 0.64 and 0.66, respectively. For the small bowel and colon, the Hausdorff distance was greatest with average values of 13.5 ± 5.1 mm and 11.7 ± 6.1 mm, respectively. Average change in contour centroid was seen for the small bowel and stomach, with average values of 76.3 mm and 88.6 mm, respectively. **Conclusion:** This study demonstrates a method to quickly and effectively evaluate target and organ-at-risk motion and that these structures can shift greatly during a patient's radiation therapy treatment course. This study also shows the great importance in implementing better adaptive radiotherapy measures to treat pancreatic cancer more effectively.

PO-GePV-T-391

Intra-Fraction Motion of Brain SRS/SRT with Multiple Targets: an Evaluation Based On a Robotic Radiosurgery System

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Purpose: Concurrent multiple target SRS/SRT is increasingly used for brain metastasis management. However, the accuracy of dose delivery is sensitive to patient motion for plan with considerable inter-target distance. This study aimed to evaluate the intra-fraction motion during SRS/SRT by analyzing patient motion tracking data and may provide useful clinical data. **Methods:** 40 treatment fractions from 12 patients were included who were treated for 8-15 targets in a single plan with Cyberknife. Patient intra-fraction motion, defined as position offset from plan position, was calculated through registering KVs to plan DRRs. Orthogonal KV pairs were acquired every ~20 seconds during treatment. These motion data were retrieved in treatment log files and processed to include 5016 on-treatment time points. Patient's translation in anterior-posterior, superior-inferior, left-right directions as well as rotation (i.e., yaw, pitch, roll) were analyzed. **Results:** The distance of translation has a range of [0, 3.9]mm (mean: 0.84 ± 0.43 mm). For each direction, these values are [-3.3, 2]mm (mean: -0.03 ± 0.57 mm) in S-I, [-2.3, 2.7]mm (mean: -0.04 ± 0.62 mm) in L-R, [-1.8, 2.1]mm (mean: 0.08 ± 0.41 mm) in A-P. Patient rotation has a range of [-2.8, 2.1]° with a mean -0.1 ± 0.7 ° for yaw, [-1.8, 3.1]° (mean: -0.19 ± 0.57 °) for pitch, and [-2.0, 1.9]° (mean: 0.0 ± 0.45 °) for roll. Results also show 71.8% of time patients move <1mm distance, and 97.9% (<2mm). For patient rotation, 47.4% of time is for <0.5° yaw and 83.8% (<1°). These values are 51.5% (<0.5°) and 91.3% (<1°) for pitch, and 74.4% (<0.5°) and 97.5% (<1°) for roll. **Conclusion:** Most of intra-fraction motion is <2mm (97.9% of time). However, patient's yaw rotation seems significant with only 83.8% of time <1°. With 1.8mm offset caused by 1° rotation for a 10cm inter-target distance, this means ~16% of time, targets away from the isocenter may have ~2mm offset versus plan position (~1 mm for targets 5cm away from isocenter).

PO-GePV-T-392

Brain Stereotactic Radiosurgery Using MR-Guided Online Adaptive Planning to Daily Setup Variation: An End-To-End Test

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Purpose: An end-to-end test to evaluate the normal brain dose for stereotactic brain radiosurgery (SRS) treated with an MR-Linac using the adapt-to-position (ATP) and adapt-to-shape (ATS) workflows to account for daily setup variations. **Methods:** The RTsafe Prime brain phantom was used with a delineation insert that includes two pre-defined structures mimicking gadolinium contrast-enhanced lesions. Daily adaptive plans were generated using 6 presets (3 mm and 3 degrees) and 6 random setup variations (> 10 mm). Two adaptive plans per daily MR image were generated using the ATP and ATS workflows. In the ATS workflow, GTV volumes on daily MRIs were recorded and PTVs were recreated with a 1mm margin. Target doses were measured with two ionization chambers inserted into the phantom. The adaptive plans were compared with the reference plan using the target coverage, conformity index, gradient index (GI), V12Gy, R50, optimization time, and total monitor units (MUs). **Results:** For small daily setup variations (≤ 3 mm), GI and V12Gy of combined PTVs in the ATS workflow were 8% and 14% lower than those from the ATP workflow. The GTV volume change was < 1% after contour deformation in the ATS workflow. For large setup variations (> 10 mm), both GI and V12Gy from the ATS workflow were 17% lower than those from the ATP workflow but the GTVs increased by 3% for the large lesion (5 cm³) and 5% for the small lesion (1.4 cm³). The differences between the measured and planned target doses were within 1% for both treatment workflows. **Conclusion:** Brain SRS on MRL is feasible but it is important to reduce daily setup variation. Normal brain dose could be lower with the ATS workflow at the cost of a longer plan optimization time (5-10 min) comparing to the ATP workflow (< 2min).

PO-GePV-T-393

Optimizing Isocenter Placement Based On Rotational Uncertainty in Single-Isocenter SRS

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Purpose: Single-isocenter stereotactic radio surgery (SRS) to multiple targets allows patients to be treated significantly faster than multi-isocenter SRS but is more susceptible to rotational uncertainty. Currently, in single-isocenter SRS, the isocenter is placed at the geometric center of all the lesions. Large targets and those closer to the isocenter are less influenced by rotational uncertainty. The aim of this work is to optimize the placement of the isocenter to minimize the effect that a rotational error would have on target coverages. **Methods:** Targets from three previous patients with multiple brain metastases with 2, 4 and 5 targets were used to optimize isocenter placement in single-isocenter SRS. The targets were converted to equivalent spheres with a second sphere being calculated for each target representing dose coverage. The intersecting volume of each target sphere with its respective dose sphere was calculated after a 1° rotational was applied with respect to isocenter placement during optimization iteration. The optimization function sought to minimize the difference in coverage between targets while maximizing the coverage of each individual target using the weighted Tchebycheff technique. **Results:** The average variation in target coverage across targets with the optimized isocenter was 1.04%, 10.71% and 9.07%, while placing the isocenter at the geometric center resulted in 1.38%, 16.49% and 12.86% for 2, 4 and 5 target patients respectively. For all cases, the average difference in coverage across targets was reduced with the optimized isocenter. **Conclusion:** Small targets and those furthest from the isocenter are more influenced by rotational uncertainty in single-isocenter SRS. Optimization may be used to place the isocenter in such a way that target coverage is less perturbed by rotational uncertainty. Absolute coverage can be adjusted by dose normalization but minimizing variation in coverage reduces the extent to which particular targets suffer greater losses than others.

PO-GePV-T-394

Dosimetric Study of the Electron Return Effect On Breast Treatment Planning On a 1.5 T MR-LINAC

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Purpose: The advent of MR-LINAC makes it feasible to adapt treatment plans based on the anatomy-of-the-day, especially for low x-ray contrast tumor sites including abdomen and breast. However, under the presence of a magnetic field (especially for a 1.5T model), the known 'electron return effect' (ERE) at high-gradient tissue boundaries may make the planning more challenging. In this study we investigated such an effect on the dosimetric endpoints of breast planning. **Methods:** A pilot study was performed on pre-operative breast SBRT cases, with a prescription of 40Gy in 5 fractions. We planned and compared the dosimetric results based on four techniques: 1. A non-parallel-opposed beam configuration featuring 6 beams from 45° to 330° (counter-clockwise), with the magnetic field turned 'on'; 2. The same beam configuration as 1, but with the magnetic field turned 'off'; 3. A parallel-opposed beam configuration featuring 12 beams ranging from 45° to 150° (counter-clockwise), with the magnetic field turned 'on'; and 4. The same beam configuration as 3, but with the magnetic field turned 'off'. The plans were optimized under the same objectives and constraints. **Results:** Results showed a clear presence of ERE, in the form of high dose regions extending into the chest wall and lungs for plans with the magnetic field 'on'. However, with the introduction of proper tuning structures, all four planning techniques achieved adequate target coverage and minimized chest wall and lung doses. Although non-parallel-opposed beam plans show similar dosimetric endpoints to parallel-opposed ones, they need to be further compared regarding to their relative robustness against intra-fractional motion/deformation under ERE. **Conclusion:** ERE in the presence of magnetic fields is non-negligible in MR-LINAC plans. Inverse treatment planning with curated tuning structures can mitigate its dosimetric impact, while potential intra-fractional patient motion should be taken into account in the selection of a robust plan.

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PO-GePV-T-395

Dosimetric Quality Comparisons Between MRI-Linac Vs Conventional VMAT for Prostate SBRT

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Purpose: Preparing a clinic for the introduction of magnetic resonance (MR)-guided radiotherapy requires determination of how current clinical practices compare or will be adapted to this new treatment planning paradigm. The goal of this work was to determine if MR-linac treatment plans (IMRT) could be created that match our current clinical practice (VMAT) plan quality for prostate SIB SBRT cases. **Methods:** Ten prostate cancer patients, who recently received SBRT VMAT treatment with a prescription to 45 Gy in 5 fractions on a conventional linac in our clinic were selected. Clinically acquired CT simulations with associated target and OAR contours were used for MR-linac treatment planning simulation. MR-linac treatment plans were created using a predefined template of 13 beams arranged to limit dose to the femurs, rectum and bladder. MR-linac and clinical plans were compared and differences in target and OAR doses assessed. **Results:** Compared to clinically delivered plans, the treatment plans created for the MR-linac system were able to meet the clinical objectives for SBRT prostate plans with 45 of 48 or 94% of the DVH based objectives met. In all cases, the objectives that did not meet were approved for treatment and structures that overlapped with the PTV volume, such as the bladder wall. In particular, the most common failure was of the V1830ccGy < 18 cc objective with the average failure of 1.8 cc. **Conclusion:** This study demonstrates that clinically acceptable plans for prostate SBRT cases can be created using the MR-linac treatment planning system. However, creation of clinically optimal plans will take additional time and effort and may require adjustment of clinical expectations. For example, prostate SIB SBRT plans have additional treatment planning challenges and are currently being added to the testing data. All these considerations should also take into account the benefits of MR guided imaging.

PO-GePV-T-396

Investigation of Dosimetric Accuracy for CyberKnife Intracranial SRT/SRS Treatment Planning Using a Consistent CT Calibration Model

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Purpose: To investigate the dosimetric impact of the density model established based on the head configuration and body configuration on CyberKnife intracranial SRT/SRS planning. **Methods:** Two phantoms of different diameters representing body and head size, with 11 electron density inserts, were used to acquire the CT calibration curves for CyberKnife treatment planning. Fifteen brain SRT/SRS plans with small (<0.5cc), intermediate (>0.5cc, <5cc) or large (>5cc) tumors located in different brain lobes (frontal, parietal, temporal, occipital and cerebellum) were retrospectively recalculated using the body CT calibration curve, whereas the initial calculations used the head CT calibration curve. The dosimetric changes were evaluated by calculating the differences in mean dose to target, conformity index and target coverage. **Results:** Comparing the body and head CT calibration curves, the CT number mainly differs in the bone region and this difference increases with bone density. The CT number of cortical bone insert is lower by 160HU in body compared to that in head. When correlating CT number to electron and physical density for dose calculation, the head skull (860±400HU) has 4.3% increase in electron density and 4.9% increase in physical density when using the body CT calibration curve instead of head. In the fifteen SRT/SRS plans, when using the body CT calibration curve, on average the mean dose to target is decreased by 0.15%, CI decreased by 0.01 and target coverage decreased by 0.1%. The maximum difference among the fifteen plans was found to be 0.3% decrease in mean dose to target, 0.02 decrease in CI and 0.7% decrease in coverage. **Conclusion:** Our results indicated that applying a consistent CT calibration curve for CyberKnife intracranial treatment planning would introduce a less than 1% difference in the target dose regardless of tumor size and location, which should be considered as clinically acceptable.

PO-GePV-T-397

New Dosimetric Guidelines for Acuros Through Comparative Evaluation of SBRT Lung Treatment Plans Against AAA

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Purpose: To propose equivalent guidelines for lung SBRT when using Acuros XB to RTOG guidelines, suited for convolution algorithms such as AAA, considering the difference between the algorithms and its biological impact. **Methods:** A retrospective analysis was performed on 30 lung patients previously treated with SBRT by generating two sets of plans. The first set compared the original AAA plans, based on dynamic-conformal arcs (DCAs), to new plans first recalculated and then renormalized for PTV coverage using AXB. The second set was calculated in AXB based on DCAs and additional static beams, and then recalculated and renormalized in AAA. The recalculated and renormalized plans were compared to the original plans based on V100 and V90 PTV coverage as well as V105, conformality index, D2cm, Rx/Dmax, R50, and Dmin. These metrics were analyzed both nominally and based on variations according to RTOG guidelines. **Results:** AAA plans recalculated in AXB had a significant reduction in most dosimetric metrics with extreme changes in V100 (4%) and the conformality index (7.5%). To achieve equal PTV coverage, AXB required an average of 1.8% more MU. Based on the relative differences between AAA and AXB in the above criteria, new guidelines for AXB, equivalent to RTOG guidelines, have been developed. Minor violations of the new guidelines by AXB plans of the 1st set were found to have significantly increased from those of the RTOG guidelines, with no change in major violations, by 19 and 34% for recalculated and renormalized plans, respectively. The violations became comparable to those of AAA plans against the RTOG guidelines. A similar trend was found for the second set (original) by 225%. The number of violations of the two sets against the new guidelines were similar to each other. **Conclusion:** The new guidelines are recommended for planning with AXB.

PO-GePV-T-398

Evaluation of Dosimetric Uncertainties in Small Fields of the Varian EDGE Radiosurgery System Using Monte Carlo Simulations

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Purpose: The measured dosimetry uncertainties in some clinical cases treated with diminutive MLC fields ($\sim 0.5 \times 0.5 \text{ cm}^2$) of 10MV FFF arc beams in stereotactic radiosurgery have suggested that such variability mostly emerged from tiny shift of the MLC. This research aims to evaluate uncertainties due to slight MLC drift in very small fields at different clinical settings and to explore their clinical significance. **Methods:** A Monte Carlo (MC) dose model was developed for both 6FFF and 10FFF beams of Varian EDGE radiosurgery system using the EGSnrc MC program. The phase space files provided by the vendor were used for simulations. The simulations were performed for 10×10 – $1 \times 1 \text{ cm}^2$ fields w/o MLC at (a) SSD=100cm, d=dmax and (b) SSD=95cm, d=5cm settings in water phantom until the dose uncertainty $< 5\%$ was achieved. The dose profiles and depth dose curves were compared to treatment planning system (Eclipse). For the MLC field of $0.5 \times 0.5 \text{ cm}^2$ with fixed jaws setting at 95cm SSD, d=5cm, the dosimetric parameters were analyzed with 1mm MLC shift. Major input parameters varied during simulations were the number of incident particles, directional Bremsstrahlung splitting, range rejection, particle splitting, and voxel size. **Results:** Preliminary results showed acceptable agreements ($\leq 4\%$) between MC simulated and TPS calculated dose profiles along with $\leq 4\%$ dose uncertainty for reference fields (10×10 & 5×5) cm^2 at set-ups (a) & (b) in water phantom. Similarly, good dose agreements of $\leq 4\%$ for 6FFF and $\leq 3\%$ for 10FFF between MC versus TPS were found for (5×5 – 1×1) cm^2 fields at 95cm SSD, d=5cm. However, for the MLC field of $0.5 \times 0.5 \text{ cm}^2$, early results showed relatively higher dosimetric uncertainty ($\sim 5\%$). **Conclusion:** A MC dose model was developed for the Varian EDGE radiosurgery system and was successfully validated ($\leq 4\%$) against the TPS. Evaluation of the dosimetric uncertainties in the smallest field is in progress.

PO-GePV-T-399

Automatic Spine SBRT Treatment Planning Using Pinnacle Scripting

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Purpose: To evaluate automatic spinal stereotactic body radiation therapy (SBRT) treatment planning using Pinnacle scripting. **Methods:** Pinnacle scripts were developed and implemented for generating spine SBRT plans in Pinnacle v16.2 environment. The script interface allows plan generation using different fractionations for different target locations and organs-at-risk such as spinal cord or cauda equina. In combining with Pinnacle Auto-Planning, the scripts allows the user to enter the prescription, select the beam energy and arrangement. It automatically optimizes the plan and generate an evaluation scorecard. We tested the script for patients with thoracic or lumbar spine tumors. The treatment plan was created with the prescription of 16 Gy or 18 Gy in 1 fraction, 24 Gy in 2 fractions, or 30 Gy in 4 fractions following the guidelines from RTOG 0631 or our internal protocols. For the treatment of the thoracic spine, 3 VMAT arcs with the energy of 6 MV FFF or 10 MV FFF were used. For treatment of lumbar spine, 3 VMAT arcs with the energy of 10 MV FFF were used. Each plan was evaluated with protocol recommended DVH constraints and the Paddick conformity index (CI) and gradient index (GI). **Results:** We used these Pinnacle scripts to generate 10 spine SBRT plans. All except one plan was successfully created on the first pass and met the DVH criteria for the targets and organs-at-risk. The CI was on average 1.03 ± 0.04 and the GI was on average 3.67 ± 0.39 . All plans passed patient-specific QA ($> 95\%$) with the threshold of 3%/2 mm. **Conclusion:** Plans created with our Pinnacle scripts were all clinically acceptable. Automatic planning for spine SBRT using scripting reduces human errors, improves efficiency, and standardizes plan quality.

PO-GePV-T-400

The Influence of Metal Artifacts in Ray Tracing and Monte Carlo Calculation Algorithms for Thoracic Spine Lesions Treated with CyberKnife

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Purpose: Metal implants in CT systems can cause severe streak artifacts that affect the accuracy of dose calculation. This work investigates the effect of metal implants in Ray Tracing (RT) and Monte Carlo (MC) calculation algorithms for thoracic spine lesions in CyberKnife radiotherapy. **Methods:** 22 cases of postoperative spinal implant patients who had previously received Cyberknife treatment were used in this study. Four different patient plans using RT and MC algorithms with and without artifact correction have been applied. CT artifacts of metal implants were manually outlined based on image-based Hounsfield unit threshold segmentation method, and assigning an average value HU value of soft tissue for dose calculation. Dosimetric parameters of planning target volume (PTV), spinal cord, esophagus, and lungs were compared. **Results:** There were no significant differences in two plans with or without artifact correction using RT and MC algorithms ($P > 0.05$). For RT algorithm, compared the plan without artifact corrections, the average Gradient index (GI) was improved (3.04 vs 2.60). The differences of the target and organs were generally less than 4% between two plans. For MC algorithm, the average GI of two plans with or without artifact corrections were 3.13 and 2.69, respectively. The differences of the target and organs were reduced, less than 1% between two plans. **Conclusion:** Ignoring the difference between RT and MC algorithms, there were no significant differences between plans with or without correcting manually for the artifacts of metal implants in each algorithm. However, the gradient index was slightly improved, which may help to improve the quality of the plan.

PO-GePV-T-401

Enhanced Dose Conformity by Manually Sectional Adjustment of High-Definition Multileaf Collimators: A Dosimetric Comparison Between Two Linac-Based Stereotactic Radiosurgery/Radiotherapy BrainLab iPlan Planning Techniques

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Purpose: The purpose of this study is to use the available BrainLab iPlan RT Dose 4.5.5 treatment planning workstation's draw object tool to enhance conformity of dose distribution, by manually fine adjustment of high-definition multileaf collimators (HD-MLCs) for Linac-based Stereotactic radiosurgery/radiotherapy planning techniques and to compare between selected techniques, with and without using this HD-MLCs fine tuning tool. **Methods:** Seventeen irregular complex tumor shaped patients with different intracranial indications (Brain Mets, Schwannoma, Recurrence Nasopharynx, Glioma, and Chordoma) were recruited for this study. CT datasets were acquired using stereotactic head frame, thermoplastic mask, nose bridge and mouth bite, for contouring and treatment planning purpose. Four plans, with and without the manual sectional adjustments of HD-MLCs were created for comparison purpose using conformal beams (CBs) and dynamic conformal arcs (DCA) techniques, two plans for each. In CB technique, the median number of non-coplanar beams were 12 (range from 11 to 13) but in DCA technique, the median number of non-coplanar arcs were 6 (range from 5 to 7). Minimum dose coverage criteria for GTV+2mm was fixed to 95% of the prescribed dose, for all plans. Dose-Volume histogram (DVH) was used to dosimetrically analyze maximum dose, mean dose, minimum dose, Conformity index and surrounding normal structures doses. **Results:** The CBs technique, without the adjustment of HD-MLCs showed mean CI=1.77 but with fine adjustment of MLCs, CI=1.19 for all Patients. DCAs technique showed mean CI=1.82 without sectional adjustment of MLCs but with precise manually sectional adjustments in MLCs, CI=1.20. D100% of GTV+2mm was 96-98% of prescribed dose in all plans. Significant dose reduction in surrounding normal Organs at risk was observed with manual MLCs adjustments. **Conclusion:** The BrainLab iPlan's draw object tool for manually sectional adjustment of MLCs can achieve conformal dose distribution in Irregular complex shaped tumors, without compromising the tumor dose coverage.

PO-GePV-T-402

Clinical Implementation of PDT Dose and Reactive Oxygen Species Explicit Dosimetry (ROSED) for Pleural Photodynamic Therapy (pPDT)

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Purpose: To model the amount of total reacted reactive oxygen species([ROS]_{rx}) and PDT dose based on explicit measurements of light fluence rate, photosensitizer concentration and tissue blood flow, on various locations in pleural cavity of patient receiving Photofrin-mediated photodynamic therapy for malignant pleural mesothelioma. **Methods:** PDT was performed on patient's lung cavity after complete surgical debulking of malignant pleural mesothelioma tissues. Light fluence rates were measured on 8 sites inside the lung cavity using 8 isotropic detectors sutured on the pleural wall. 4 out of the 8 isotropic detectors were connected to 4 spectrometers through bifurcated fibers to simultaneously acquired Photofrin fluorescence excited by the 630nm treatment light. The measured fluorescence spectra were fitted using an SVD algorithm to separate the contribution of Photofrin from other spectral components including ambient room light and tissue autofluorescence. The fluorescence SVD amplitudes were then corrected for tissue optical properties effect and quantified for absolute in vivo Photofrin concentrations. Tissue blood flow was measured on 2 sites where Photofrin fluorescence measurements were performed, using diffuse correlation spectroscopy (DCS). 2 DCS contact probes were sutured adjacent to the 2 isotropic detectors and tissue blood flow measurements were performed using a separate 785nm long coherent laser concurrently. PDT dose was then computed based on measured light fluence rate and Photofrin concentration, and [ROS]_{rx} was computed using ROSED model based on the measured light fluence rate, Photofrin concentration and tissue blood flow. These two dosimetric quantities were then compared among 10 pPDT patients. **Results:** The average PDT dose delivered to 40 sites on 10 patients is 614±202.6μMJcm⁻² and the average [ROS]_{rx} delivered to 18 sites on the same 10 patients is 0.59±0.25mM. **Conclusion:** Large intra- and inter-patient variations in PDT dose and [ROS]_{rx} were observed although a constant 60Jcm⁻² light fluence was prescribed.

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PO-GePV-T-403

Impact of Dose Calculation Algorithm On Airway Dosimetry: Comparison of AAA and Acuros

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Purpose: Minimizing post-radiotherapy (RT) respiratory function loss entails better understanding of RT-induced airway injury. Previously, we showed the importance of including individual airways as organs at risk (OARs) to reduce lung stereotactic ablative radiotherapy (SABR) toxicity after treatment. Estimating and fulfilling airway dose constraints during RT plans require accurate dose calculations to these structures. Therefore, here we extend the investigations on common dose calculation algorithms to include the study of airway dosimetric accuracy. **Methods:** We retrospectively evaluated the anisotropic analytical algorithm (AAA) and Acuros algorithm by using a high-resolution breath-hold CT (BHCT), simulation 4DCT, and prescribed RT plan from four lung SABR patients. A virtual bronchoscopy software was used to autosegment 856 airways on the BHCT images. Planning target volumes (PTVs) and standard OARs were contoured on the average image of the 4DCT phases (AVG). Conformal RT (CRT) and intensity-modulated RT (IMRT) plans were recreated on the BHCT (with OARs warped to this image and breath-hold PTV contoured on this image) and on the AVG (with airways warped to this image), for a total of four plan types/patient. Dose calculations were performed using AAA and Acuros, and the differences in mean and maximum dose (Dmax) were calculated. **Results:** Dose differences in PTVs/OARs were comparable to those previously reported. Median differences in airway Dmax ranged from -0.03 to +0.22Gy with respect to airway diameter without clear dependence, and from -0.52 to +0.21Gy with respect to Dmax for all plan types. AAA underestimations (negative values) were typically observed in BHCT plans, overestimations in AVG plans. Translating these results to airway injury risk, these dose differences corresponded to variations in airway collapse ≤±0.7%. **Conclusion:** These results suggest that airway dosimetric differences between AAA and Acuros are not clinically relevant and that using either algorithm is appropriate when including patient airways as individual OARs.

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PO-GePV-T-404

Dosimetric Comparison of Gamma Knife® Icon and Linear Accelerator-Based FSRT Plans for the Re-Irradiation of Recurrent Glioblastomas

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Purpose: To perform a dosimetric comparison between clinically deliverable (treatment time <20 minutes) Gamma Knife® Icon (GK) and linac-based FSRT plans on the basis of normal brain dose sparing for recurrent glioblastomas (GBM) with target volumes <14 cc. **Methods:** Twelve patients with recurrent GBM who were treated using re-irradiation via linac-based FSRT with a dose of 35 Gy in 10 fractions or 60 Gy in 5 fractions were selected for this study. New clinically deliverable plans were created in the Leksell GammaPlan® V11 TPS with identical image sets, structures and prescription doses to the original linac-based plans for two of the twelve patients. Planning objectives for both modalities included target coverage of PTV V100 ≥95%, and maximum dose to cranial organs-at-risk (optic nerves, chiasm, brainstem) D0.03cc <15 Gy. Plan comparison was performed in MIM. Dosimetric indices for comparison were target coverage (TC), Conformity Index (CI), Gradient Index (GI), normal brain (Brain-PTV) dose (V4, V12, V20, and mean dose) and treatment time. Statistical significance was measured using a two-tailed, paired t-test with p<0.05. **Results:** For all patients replanned at this time, comparable target coverage and CI were seen between the linac and GK plans. GK plans showed lower GI and lower normal brain in all quantities evaluated. As expected, treatment times for GK plans were longer than linac treatment times. **Conclusion:** GK treatment plans resulted in lower volume of normal brain dose and mean brain dose compared to linac-based FSRT, with improvements on all dosimetric indices while maintaining clinical deliverability.

PO-GePV-T-405

Comparison of Ray Tracing and Monte Carlo Algorithms for Spine Lesions Treated with CyberKnife

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Purpose: This study attempts to evaluate Ray Tracing (RT) and Monte Carlo (MC) algorithm for CyberKnife treatments of spine lesions and determine whether MC algorithm is necessary for all spine treatment and analyze which sites of spine lesion for which RT algorithm is comparable to MC algorithm. **Methods:** The CyberKnife is used for stereotactic body radiotherapy for lesions in the cervical spine (30), thoracic spine (50), lumbar spine (30) and sacral spine (15). The PTV volume ranged from 51.7 cc to 319.8cc. Patients were treated in 3 fractions with a dose of 24-30Gy. Treatment plans were generated using 1-4 IRIS aperture collimators and an average 148 (range 104-227) non-coplanar non-isocentric beams. Dose was calculated using RT and MC algorithms for patients planned with the same beam angles and monitor units. DVH of the target and selected critical structures are evaluated, including V100 of PTV, D0.1cc, D1cc and D5cc of the esophagus, spinal cord and bowels, and Dmean and V16 of kidneys. **Results:** V100 of PTV with MC was 94.80%, 88.47%, 92.52% and 93.41% respectively in cervical, thoracic, lumbar and sacral spine. For thoracic spine, RT algorithm significantly overestimates the percentage volume of target covered by the prescribed dose, and overestimates doses to organs at risk in most cases, including lung, spinal cord and esophagus. For cervical, lumbar and sacral spine, the differences of the target coverage of prescription dose were generally less than 3% between the RT and MC. The differences of doses to organs

at risk varied with lesion sites and surrounding organs. **Conclusion:** In the thoracic spine lesions with beams through air cavities, RT algorithm should be limited and verified with MC algorithm, but the RT algorithm is adequate for treatment of cervical, lumbar and sacral spine lesions without or small amount of beams passing through the lungs.

PO-GePV-T-406

Treatment Planning of Intra-Cranial Lesions with Zap Surgical & Comparison with CyberKnife

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Purpose: Stereotactic Radiosurgery (SRS) has been utilized as a minimally invasive option to treat intracranial lesions. Zap Surgical, a Gyroscopic radiosurgery system with frameless and image-guided patient positioning and tracking, and sharp dose fall off, is capable to treat various sizes of intracranial lesions. We present treatment planning studies for intracranial lesions of varying sizes and locations. The planning studies will encompass both Forward and Inverse planning aspects of the Zap TPS. These plans will be compared with other frameless SRS system CyberKnife. **Methods:** CT studies of 1.25 mm axial slices were registered with post-contrast T1-weighted MRI of 2 mm axial slices. Additionally, T2-weighted Fiesta series were registered for Trigeminal Neuralgia. The prescription doses for the targets and OAR dose constraints followed standard clinical practice and guidance from published literature. Targets were varying in sizes, complexities and proximities to OARs. Planning studies incorporated single and multiple lesions. Planning studies compared both forward and inverse planning results of Zap TPS. These studies were repeated in Accuray Precision TPS. Zap planning results were compared with Accuray planning in terms of target coverage, conformality, beam on time and OAR dose constraints. **Results:** All plans in Zap forward and inverse planning and CyberKnife Precision achieved the planning goals of 95% target coverages and met all published OAR dose constraints. Zap inverse forward planning yields shorter beam on time, but with a little less conformality as compared to the inverse plans. Zap treatment plans, both forward and inverse plans, were comparable with CyberKnife plans. **Conclusion:** Zap Surgical's treatment planning system is capable of producing highly conformal treatment plans with sharp dose fall off for intracranial lesions. Both forward and inverse planning aspects result in clinically equivalent plans, with a little better conformality and OAR sparing with inverse planning. The plans are comparable with CyberKnife plans.

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PO-GePV-T-407

Dosimetric Case Study of Lung SBRT with Two Adjacent Targets: One ISO Vs Two ISOS

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Purpose: LINAC-based stereotactic body radiotherapy (SBRT) has gained popularity in treating early stage lung cancer. This study aims to compare dosimetric outcomes of different SBRT **Methods:** one-ISO plan (one-I) and two-ISO plan (two-I). **Methods:** There are two early stage lung tumors within the same lung lobe at the same level. PTV volumes are 10.0cc and 17.4cc. The distance between these two tumors is 3.5 cm. 48Gy in 4fx were prescribed to both two tumors. Partial arcs VMAT plans were implemented for both the one-I and two-I at Varian Truebeam machine with 6X FFF beam. VALOR dose constraints were used for plan evaluation. **Results:** For both SBRT plans, the PTV coverages were the same as 95% PTV volumes covered by 100% prescribed dose. For all organ-at-risk listed in the trial, the Dmax of one-I is 4.6% lower than that of two-I. The volume dose of one-I is 3.46% lower than that of two-I. CI of one-I is 2.73% lower than that of two-I and the D2cm of one-I is 3.3% lower than that of two-I. R50 of one-I is 16.67% lower than that of two-I. MU of one-I is 46.58% lower than that of two-I. **Conclusion:** This study showed that one-isocenter plans, compared with two-isocenter plans, provided better normal lung sparing and incurred much less MUs for shorter beam-on time and less radiation leakage to patient.

PO-GePV-T-408

Can a Single Beam Model Be Used for SBRT Planning On Multiple Linear Accelerators?

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Purpose: To verify the feasibility of using a single beam model for SBRT treatments across multiple linear accelerators through patient-specific QA measurements of representative SBRT plans. **Methods:** Ten previously treated SBRT plans for various sites (lung, brain, spine, pancreas, adrenal gland, iliac lymph node) were created using a single unified beam model(AAA v.15.6) and 6FFF mode in Eclipse TPS. Input beam data (PDD,Crossline Profiles,Output Factors) used for the model were based on institutional consensus data, which used average measured data from multiple TrueBeam machines for each data type. The measured DLG and MLC transmission agreed within 0.1 mm and 0.01%(absolute), respectively, for the two machines analyzed. To test the beam model, plans were mapped to an acrylic phantom for point dose measurement using micro-ionization chamber(PTW PinPoint) and planar dose measurement(Gafchromic EBT3 film) on two linacs, with both dose measurements performed at isocenter. The same plan was delivered on each machine, and the plan dose was compared to measured dose using point dose difference, Gamma analysis, and line profile analysis. **Results:** For point dose measurements, the average±st.dev. percent difference was 0.36±1.60% for Machine1 and 2.07%±1.47% for Machine2. All point dose measurements were within 4% of planned dose values. For film measurements, Gamma(3%,2mm) passing rates for both machines were in excess of 98.5% for all plans. Additionally, line profile analysis was performed to compare planar dose measurements to calculated plan dose using unified beam model. Line profile analysis indicated agreement in high dose region, penumbra, and low-dose tail for each machine relative to the planned dose. **Conclusion:** Our results for point and planar dose indicate an acceptable level of agreement between measured dose and plan dose for each machine. We conclude that using a single unified beam model for SBRT plan delivery is a feasible option for clinics with multiple TrueBeam machines.

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A Trajectory Sampling Metric for the Optimization of Dose Distributions in Radiosurgery

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Purpose: C-arm linac-based radiotherapy has seen a recent interest in deliveries with dynamic rotations, principally 4 π methods using simultaneous rotations of couch and gantry to reduce doses to organs-at-risk (OARs) and increase conformity. While many methods use heuristics to generate trajectories that avoid OARs combined with arbitrary trajectory restrictions to prevent oversampling, a metric has not yet been developed to succinctly quantify the sampling of the deliverable space with trajectories. **Methods:** Evenly spaced sampling points were distributed across a 4 π sphere centred on isocentre. Six trajectories which increased sampling of the 4 π space were applied, and the minimum arc distance between each sampling point and the nearest point of the trajectory was measured. The distribution of all arc distances was then condensed by taking the mean arc distance to represent the 4 π sampling of each trajectory. An open-source treatment planning system, matRad, was used to generate 192 treatment plans to establish the relationship between isodose volumes and trajectory mean arc distance for eight target volumes in four positions and for the six trajectories. **Results:** A monotonic relationship between mean arc distance and trajectories

that intuitively increased sampling was found. The impact of isodose sparing increases with increased sampling at all levels above 2.0 % of prescription dose. The impact of sparing is exponential, with isodoses between 2.5% and 50% of prescription being reduced most significantly. Reduction of absolute isodose volume with increased 4π sampling is highly dependent on target volume. **Conclusion:** Mean arc distance is a representative means of quantifying 4π sampling. Trajectories that increase sampling are useful in decreasing the volume of isodoses relevant for sparing normal tissues. By quantifying this feature, candidate dynamic trajectories can be efficiently compared for sampling. This metric may also have broader applications in radiotherapy when seeking to increase sampling by adding non-axial trajectories.

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PO-GePV-T-410

Predictive Model of Lung Tumor Intrafraction Motion Variability

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Purpose: The aim of the work was to create a predictive model of the intrafraction motion variability in the superior/inferior (SI) direction applicable in margin delineation of internal target volume. **Methods:** Log file analysis from online respiratory tumor tracking was performed in 145 patients. We used univariate analysis to assess the suitability of the variables recorded during data collection. Data from 100 patients were selected for model creation, and the remaining data were used for model cross validation. Categorical variables included sex (male / female), origin of lesion (metastasis / primary), lungs (left / right), combination of sex and origin. Continuous variables included mean motion along the SI axis (mm), GTV size (mm³), and localization of the target in the lung tissue: individual partial distances (mm) and relative target location (0–1) in the SI, AP, and LL directions. Backward elimination was used in the creation of the model (using $p < 0.05$ as the criterion of statistical significance). **Results:** A model was created in the following form: Intrafraction variability SI (mm) = $-0.82 + 0.003 \cdot \text{Ant} + 0.38 \cdot \text{TPSI} + 0.006 \cdot \text{Md}$ where Ant is the perpendicular distance from the center of the lesion to the outermost edge of the lung in the Anterior direction (mm), TPSI is the mean Target Motion in the SI direction (mm), and Md is the length of the perpendicular line from the center of the target to the vertebra. The mean prediction error was 0.55 ± 0.48 mm. A prediction error exceeding 1 mm was recorded for five targets (11%). Predicted values highly correlates with real observed values ($R=0.89$, $p<0.001$). **Conclusion:** The developed model exhibits a good ability to estimate intrafraction variability. We propose patient specific margin expansion in the SI direction based on the model output after the impact of the ITV margin expansion on dose distribution will be further tested.

PO-GePV-T-411

On the Interplay Effect in Lung SBRT Treated with CyberKnife Static Tracking System

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Purpose: To assess the amplitude of the interplay effect existing between different planned MU distributions and respiratory patterns in CyberKnife system when treating moving targets together with a static tracking technique. **Methods:** Measurements were performed using a semi-anthropomorphic dynamic thorax phantom. Small- and Large-Respiratory motions (SRM and LRM, respectively) different in amplitude and frequency were simulated. Bi-dimensional dose distributions were measured by means of GAFCHROMIC EBT3 films placed orthogonally in axial and coronal planes. A fraction dose of 7.5 Gy was prescribed to the PTV, generated by adding 3 mm isotropically to the resulting ITV. The interplay effect was evaluated for plans designed with the minimum number of MU per beam of 6, 30 and 60. Each plan was delivered three times starting the treatment at different phases on the breathing cycle. The coverage and the dose conformity of the GTV were evaluated for the different treatment configurations. **Results:** Generally, SRM plans showed higher GTV coverage and less dependency on beam-on phases than LRM plans. For SRM plans the GTV coverage ranged from 95.2% to 99.7% while for LRM plans from 85.9% to 99.8%. Maximum GTV coverage was found for 60MU plans in SRM and for 6MU plans for LRM. SRM plans showed also less conformity variations on beam-on phase changes than LRM. **Conclusion:** This study confirmed the presence of the interplay effect between target motion and beam irradiation time when static tracking technique is employed in CyberKnife. The magnitude of the effect was considerable for large respiration motion where the GTV coverage fell below the 90%, while it remained relatively independent from MU for small respiration motion. To maximize GTV coverage, small number of MU per beam for large motion amplitude and large number of MU per beam for small motion amplitude should be used.

PO-GePV-T-412

How Low Can We Go? Modified HyperArc VMAT Geometry for Recurrent Head and Neck SRT

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Purpose: HyperArc VMAT is used to treat brain lesions and gaining globe interest. To expand this modality, we demonstrated feasibility of the modified HyperArc module to treat locally recurrent/advanced or previously-irradiated head&neck cancers with stereotactic radiotherapy (SRT). **Methods:** RANDO head phantom was imaged, planned, and irradiated (35Gy in 5 fractions, 6MV-FFF beam) with eight HyperArc VMAT plans (4 right and 4 left neck tumors) at different anatomical locations (C1-C4) using the original HyperArc fixed-geometry. Mean tumor volume was 21.7cc (13.0–32.3cc). Average distance to isocenter from the central marker of the Qfix Encompass device down to the neck tumors was 25.8cm (21.8–28.0cm) for left and 24.3cm (21.0–27.1cm) for right-sided neck tumors and about 9cm from the lateral markers providing patient's protection zone. The HyperArc fixed-geometry was modified by adding new partial arcs with 20° couch spacing and selectively assigning 5-unilateral arcs for these neck tumors down to C4-level avoiding entrance through the brain. Original plans were re-planned using modified HyperArc geometry. Plan quality and feasibility variables were compared. **Results:** For similar target coverage, conformity, dose fall-off and heterogeneity, using modified HyperArc plans, mean/maximum doses to all critical organs (cord and skin) were reduced by 5% to 20%/10% to 30%. The 50% dose-spillage and brain dose was decreased by 2Gy/3Gy. Both plans showed similar total monitor units, modulation factor, beam-on time, and were delivered in less than 15min (including set up and CBCT imaging) with no collision issue. Six-degrees-of-freedom, couch corrections average was <1.0 mm and 1.5°. **Conclusion:** Modified HyperArc VMAT geometry provided highly conformal dose distribution, rapid dose fall-off, excellent sparing of critical organs including brain. These highly accurate, fast, and efficient SRT treatments could be delivered for locally advanced/recurrent head&neck tumors to as low as C4-level. Our modified technique avoids collision issue with precise patient set and could become an attractive head&neck SRT.

PO-GePV-T-413

Single Fraction SRS Planning Strategies for Dual Metastases Have Different Doses

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Purpose: Stereotactic radiosurgery SRS is indicated technique for multiple brain metastases treatment. SRS metastases doses are depending basically on metastases volume size. Variations in metastases doses and short distance between metastases make a plan is more complicated than normal. **Methods:** Three retrospective cases are selected with 2 nearly metastases PTV18 and PTV22 that have prescribed doses 18Gy and 22Gy respectively in single fraction. Three planning strategies are applied in this study using Truebeam machine and Eclipse planning system with same optimization constrains. SIB method (SIB):

Single plan with 22Gy to PTV22 and 18Gy to PTV18 in same plan. Sequential method (SEQ): Two plans are generated, 18Gy for all PTVs then plan boost 4Gy to PTV22 using base dose plan option. Separated method (SEP): Two plans are generated one with 18Gy for PTV18 then creating plan with 22Gy for PTV22 using base dose plan option. **Results:** Study Dose statistics comparison between (SIB), (SEQ) and (SEP) show that the average of PTV18 and PTV22 are (0.86, 0.89 and 0.88), (0.89, 0.9 and 0.86) for PI, (1.09, 1.11 AND 1.15), (0.94, 0.93 and 0.94) for CI, (0.88, 0.88 and 0.9) (0.93, 0.94 and 0.93) for QI, (1.15, 1.19 and 1.24), (1.13, 1.14 and 1.15) for HI. Furthermore the GI average are (2.48, 2.42 AND 2.72) for (SIB), (SEQ) and (SEP) respectively, while the average volumes of 12Gy for (Brain – PTV) are (8.9, 8.4 and 10cc) for (SIB), (SEQ) and (SEP) respectively, while (SIB) and (SEP) failed to achieve the criteria in one case. There is no noticeable difference between methods in other OARs. Otherwise the MU result data were (7634±740, 7444±724 and 11368±156) for (SIB), (SEQ) and (SEP) respectively. **Conclusion:** The study proved that Sequential is the superior strategy in most of evaluation parameters while the separated strategy is least one.

PO-GePV-T-414

Treatment of Brain Metastases with ZAP-X and CyberKnife: A Dosimetric Comparison

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Purpose: The ZAP-X is a novel first-of-its-kind self-shielded radiosurgery system dedicated to cranial/head and neck lesions. CyberKnife was designed for both intracranial and extracranial purposes with thousands of treatments delivered to date. The objective of this study is to compare each platform's dosimetric characteristics when treating brain metastases. **Methods:** Eleven patients with 28 brain metastases (volume range 0.06 – 4.89 cm³) previously treated using CyberKnife were selected to generate comparative ZAP-X plans using its dedicated treatment planning system. For ZAP-X planning, sphere packing was used to place isocenter positions, and beam weights were optimized using linear and quadratic programming. CyberKnife plans were optimized using an improved sequential algorithm (VOLO). For both systems, ray-tracing algorithm was used to generate beam sets and calculate final doses with high resolution. For all plans, the prescription isodose percentage was chosen to optimize target-volume coverage between 97-100%. TG101 dose constraints were utilized for all organs at risk. Conformity, homogeneity and gradient indices (CI, HI and GI) were calculated to evaluate plan quality. **Results:** Comparing identical prescriptions (15-30 Gy), the median prescribed isodose lines achieved using ZAP-X were lower than those achieved using CyberKnife, median: 75% (range: 55%-88%) vs 83% (range: 80%-86%), respectively. The median CI (1.44 vs. 1.38, p=0.26) and target-volume coverage (99.79% vs. 99.81%, p=0.29) did not differ significantly between the ZAP-X and CyberKnife plans. Significant differences in the HI and GI were noted between the plans. CyberKnife produced plans with a higher median HI compared to ZAP-X (1.32 vs. 1.19, p<0.0001) while ZAP-X produced plans with a lower median GI compared to CyberKnife (3.20 vs. 5.83, p<0.0001). **Conclusion:** ZAP-X and CyberKnife produced dosimetrically comparable plans that were nearly identical in target coverage and conformity. CyberKnife produced more homogenous plans compared to ZAP-X. ZAP-X produced plans with sharper peripheral dose falloff compared to CyberKnife.

PO-GePV-T-415

Dosimetric Evaluation of the Galaxy Rotating Gamma System for SRS/SRT

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Purpose: The Galaxy® RTi system, a gamma stereotactic radiosurgery (SRS) system with real-time image guidance technology, has been developed for high-precision SRS or frameless, fractionated stereotactic radiotherapy (SRT). This work investigated the treatment plan quality of Galaxy by comparing both the machine setup parameters and plan dosimetry parameters with those of the Leksell Gamma Knife® Perfexion™ (GK), which is widely used in the clinic for SRS/SRT. **Methods:** The GK system (Elekta AB, Sweden) uses 192 static cobalt-60 sources to deliver noncoplanar beams while the latest rotating Gamma system Galaxy (Akesis, Concord, CA) uses 30 cobalt-60 sources to deliver non-coplanar, non-overlapping arcs simultaneously. Ten brain cancer patients were unarchived from our clinical database, which were previously treated on GK in our institution. The lesion volume for these cases vary from 0.1 cm³ to 15.4 cm³. Galaxy plans were generated using the Prowess TPS (Prowess, Concord, CA) using the same dose constraints and optimization parameters. Plan quality metrics of Galaxy plans such as target coverage (%volume receiving the prescription dose), D99, conformity index (CI), cone size, shots number, beam on time, and DVH curve, were compared with those of GK plans. **Results:** Superior treatment plans were generated for the Galaxy system that met our clinical acceptance criteria. For the 10 patients investigated, the mean CI and dose coverage for Galaxy was 1.77±0.22 and 99.24±0.33%, compared to 1.94±0.48 and 99.19±0.67% for GK, respectively. The beam-on time for Galaxy was 17.42±5.79 minutes compared to 21.34±7.35 minutes for GK (both assuming dose rates at the initial installation). **Conclusion:** The Galaxy® RTi system can provide dose distributions with similar quality to that of GK with less beam-on time. The system also capable of real-time image guidance at treatment position to ensure accurate dose delivery for SRS/SRT.

PO-GePV-T-416

Northwestern Memorial Hospital's Experiences in Gamma Knife ICON Unit in First 15 Months

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Purpose: To Share useful clinical experiences with Gamma Knife and non-Gamma Knife users alike. **Methods:** Clinical data from models B, 4C, PERFEXION and ICON Gamma Knife units at Northwestern Memorial Hospital were used to extract the number of patients and fractions treated with each model. For ICON unit, the data were also separated into frame-based and frameless (mask)-based cases to illustrate the current practice. **Results:** Northwestern Memorial installed the Model B Gamma Knife unit in 1998, which was then successively upgraded to 4C, PERFEXION and to ICON in 2019. Approximately 4,000 patients were treated in the past 20 years. The data show an appreciable increase of caseloads/year by +40.0% and treatment fractions/year by as much as +116.2% in the first 15 months of ICON uses, as compared to the most recent model PERFEXION unit. These increases are ascribed to the capability of ICON units to provide mask-based fractionated treatments. Currently, slightly more patients are treated with mask-base system (56.4%) with most patients being treated with 1, 3 or 5 fractions, averaging 2.0 fractions per patient. The patient disease site distribution was also tabulated. On the other hand, due to the gating nature of the mask-based treatments, the treatment time per treatment could increase as much as +75% and highly depends on patient's conditions. Discussions were also presented on new hardware and treatment planning software features associated with ICON upgrade. **Conclusion:** Introduction of optional frameless treatment with on-board CBCT and Infrared gating system, ICON unit enables many more intracranial patients to be treated and thus leads to significant higher patient caseload albeit at the expense of possible longer treatment time.

PO-GePV-T-417

Appropriate Modality for Salvage Radiation Therapy for Local Failure After Primary Lung SBRT

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Purpose: To evaluate the appropriate modality for treating locally recurrent lung cancer patients following SBRT, using single fraction stereotactic body radiation therapy (SBRT), single fraction proton therapy, and single fraction interstitial high-dose-rate (HDR) brachytherapy for retreatment. **Methods:** Six patients with recurrence (local failure) after SBRT were planned and evaluated using three modalities for retreatment: linac-based SBRT (VMAT using Pinnacle, ver-16.2), proton therapy (Pinnacle, ver-16.2), and HDR brachytherapy (Oncentra, ver-4.5, Ir-192 isotope); all treatments were prescribed to 30Gy

in one fraction. All the plans for retreatment were exported to MIM Maestro (ver-7.0) for generating dose composites of each trial with the initial CyberKnife SBRT dose of 60Gy. Various parameters of target coverage and dose to the organs at risk (OARs) were compared. **Results:** Interstitial HDR and proton plans showed reduction in OARs' doses, specifically for heart, spinal cord, and trachea, while achieving comparable dosimetric coverage to the target volume. The average cumulative mean lung dose for proton therapy and HDR brachytherapy reduced by 2.0% and 5.2%, respectively. Both proton therapy and HDR brachytherapy also presented large decrease in average cumulative mean heart dose, 38.4% and 41.1%, respectively, in comparison to SBRT. For proton therapy and HDR brachytherapy, the maximum (0.035cc) spinal cord dose reduced by 38.4% and 36.9%; trachea (4cc) dose reduced by 31.5% and 68.7%, respectively. **Conclusion:** This study reveals the benefit of using proton or interstitial HDR brachytherapy to reduce dose to OARs, which potentially can reduce side effects after salvage radiotherapy. Although single fraction proton therapy is uncommon, the findings of the current study may provide the clinicians with options of treating recurrent lung cancer patients with single fraction HDR brachytherapy or equivalent fractionated proton therapy.

PO-GePV-T-418

An Innovative Modification of a Commercial Mask Used in Gamma Knife Icon Stereotactic Radiosurgery

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Purpose: The Leksell Gamma Knife (GK) ICON added an onboard cone-beam computed tomography (CBCT) and an infrared intrafraction motion management (IFMM) system which allow a thermoplastic mask to be used for immobilization of a patient instead of an invasive frame for a single or multiple fraction treatment. During the treatment, the IFMM system pauses the GK treatment when the motion vector reaches to a set tolerance value and it requires a new CBCT scan to realign the system to patient position for an accurate dose delivery when it stays out of the tolerance, which extends the treatment time and patient discomfort. In our clinical practice, we introduced a mouth piece with the commercial mask and we report the results comparing the effect with and without the mouth piece on the number of pausing treatment due to patient movement during the treatment. **Methods:** When making a mask, we ask patients to open their mouth when the mask is warm and we press the mask to generate a mouth bite on a wooden piece which provides an additional anchor point. The hypothesis is that this mouth piece helps with reducing patient movement within a mask. 30 patients each with and without a mouth piece of treatment time about 1 hour were randomly selected for this study. The number of CBCT scan required to complete the GK treatment was recorded and used for the analysis. A student T-test was used for the comparison. **Results:** The average number of CBCT scan required to complete a treatment including the initial one is 1.7 with and 2.1 without the mouth piece. **Conclusion:** The added mouth piece has demonstrated to reduce the number of CBCT scan during a treatment to less treatment interruptions and ultimately reduced treatment time.

PO-GePV-T-419

Measuring Cone Factors for Stereotactic Radiosurgery Applicators on a Varian Edge

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Purpose: Measure cone factors (CF) and commission a set of Stereotactic Radiosurgery (SRS) Cones for a Varian Edge machine with field sizes ranging from 17.5-4 mm diameter. **Methods:** Cone Factors were measured using three detectors: GafChromic EBT3 Dose+ film, a Standard Imaging DIV diode, and an Exradin A16 microionization chamber. Measurements were repeated twice and averaged. Absolute dose output was determined by 'daisy-chaining' outputs in a 5x5cm² field using a calibrated Exradin A1SL ionization chamber, and verified with absolute film dosimetry. CFs were measured at 100 SAD, 5cm depth. Ionization chamber measurements were completed using both polarities set to 300 V. EBT3 film was calibrated in 30 cGy intervals from 0-240 cGy and scanned on an Epson 11000XL flatbed scanner. Calibration curves were determined using a third-order polynomial fit to red-channel film measurements. The 17.5mm diameter cone was chosen as the machine-specific reference field. No small-field correction factors were applied. CFs were entered into BrainLab iPlan Treatment Planning System (v4.5.1), and compared against Varian Representative Beam Data (VRBD). End-to-end testing was performed using film embedded in a cranial phantom. **Results:** Measured CFs were consistently below the VRBD, but within 5% for all field sizes. Film and diode measured CFs were within 2% for all field sizes. Microionization chamber measurements were far below film and diode measurements and were not included in calculating the final CF. End-to-end test results demonstrated <3% differences between predicted maximum dose. **Conclusion:** Our measured CFs were consistently lower than the VRBD. This was expected as the VRBD was acquired with a SNC Edge shielded diode. Microchamber measurements were not sufficient to accurately quantify CFs for field sizes < 10mm diameter. Film and unshielded diode readings demonstrated agreement to within 2% for all field sizes. End-to-end testing demonstrated acceptable agreement between calculated and measured plans.

PO-GePV-T-420

Dosimetric Comparison of Coplanar and Noncoplanar Beam Arrangements for Radiotherapy of Patients with Lung Cancer: A Meta-Analysis

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Purpose: Radiotherapy plays an important role in the treatment of lung cancer, and both coplanar beam arrangements (CBA) and noncoplanar beam arrangements (NCBA) are adopted in clinic practice. The aim of this study is to answer the question whether NCBA are dosimetrically superior to CBA. **Methods:** Search of publications were performed in PubMed, Web of Science, and the Cochrane Library till March 2020. The searching terms were as following: ((non-coplanar) or ("non coplanar") or ("4pi") or ("4π")) AND (("lung cancer") or ("lung tumor") or ("lung carcinoma")) AND ((radiotherapy) or ("radiation therapy")). The included studies and extracted data were manually screened. All forest and funnel plots were carried out with RevMan software, and the Egger's regression asymmetry tests were conducted with STATA software. **Results:** Nine studies were included and evaluated in the meta-analysis and treatment plans were designed with both CBA and NCBA. For the planning target volumes (PTV), D98%, D2%, the conformity index (CI), and the gradient index (GI) had no statistically significant difference. For organs-at-risk (OAR), V20 of the whole lung and the maximum dose of the spinal cord were significantly reduced in NCBA plans compared with CBA ones. But V10, V5, and mean dose of the whole lung, the maximum dose of the heart, and the maximum dose of the esophagus exhibited no significant difference when the two types of beam arrangements were compared. **Conclusion:** After combining multicenter results, NCBA plans have significant advantages in reducing V20 of the whole lung and max dose of spinal cord.

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PO-GePV-T-421

End to End Verification of BrainLAB's Element Planning System for Intracranial SRS Treatments of Arteriovenous Malformations On Varian TrueBeam

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Purpose: The objective of this work is to perform end-to-end verification test of Brainlab Elements cranial vascular planning module designed for arteriovenous malformation (AVM) treatment. **Methods:** A dosimetric end-to-end test was performed using a customized phantom setup. High resolution angiogram, CT, and MRI scan of the phantom was obtained and imported into the Brainlab Elements cranial vascular planning module. A clinical treatment plan with 5 arcs was generated and delivered to the phantom on a Varian TrueBeam STx Linac having HD-MLCs using 6-FFF photons. The delivered dose was verified using

a ADCL calibrated ionization chamber placed in the phantom. **Results:** The Angio, MRI and the CT scans obtained with the custom-made phantom were fused to generate a treatment plan with SRS Rx dose of 15 Gy. The total dose measured by the ionization chamber and the planned dose agreed within 1%. **Conclusion:** An end-to-end dose verification of the Brainlab Elements cranial vascular planning module for intracranial SRS treatments of AVM was successfully performed using a custom-made phantom with <1% measurement uncertainty.

PO-GePV-T-422

Characterization of Single-Dose Radiotherapy (SDRT) Performance in a New High-Speed Ring Gantry-Based LINAC System

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Purpose: The RefleXionTM X1 is a novel ring gantry-based radiotherapy machine that utilizes rapid rotation and a new pneumatic spring-based binary multi-leaf collimator that transitions at 100Hz. These features may provide better dosimetric conformality, which is especially pertinent in very high-dose regimens such as single-dose radiotherapy (SDRT). SDRT itself is attractive—particularly for multiple target (oligometastatic) scenarios—as a means of reducing interfraction error, treatment time, cost, and patient inconvenience. As such, this investigation was carried out to assess the SDRT performance of the RefleXion X1 in multiple target scenarios using ArcCHECK (AC) analysis. **Methods:** Three multi-target lung cancer cases, each with 2-lesions ≤ 5 cm were planned with a free-breathing single-dose (28 Gy) regimen protocol (TROG 13.01 SAFRON-II). The treatment plans were carried out using a research version of the RefleXion treatment planning system which utilizes a Collapsed Cone Convolution Superposition algorithm, using a single-isocenter technique. Afterward, the SDRT plans were delivered in stereotactic treatment mode to an AC dosimetry device. Sun Nuclear patient-software was used to compare the AC measured dose with planned dose using the 2%/2mm gamma criteria with 10% low-dose threshold, as demonstrated by the absolute gamma passing rate (GP) > 90%. **Results:** All three patient plans met dosimetric coverage and organ-at-risk avoidance goals. With the treatment machine set at 850MU/Min, the average delivery duration was 46.19 mins (range: 37.47-59.98 mins). The delivery duration was associated with the lesion volume size with a correlation coefficient of 0.97 (P=0.11). Likewise, the 2-lesion plan deliveries met the GP for delivery accuracy with an average of 98.13% (range: 95.5-100%). **Conclusion:** RefleXion plans for SDRT in the multiple target setting met dosimetric and physics quality criteria. These results demonstrate treatment planning robustness and treatment delivery feasibility for relevant clinical scenarios such as SDRT or oligometastatic disease using the RefleXion X1.

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PO-GePV-T-423

Feasibility Study of Precision-CyberKnife, iPlan-Novalis, Raystation-VersaHD and Monaco-VersaHD to Satisfy the UNC Treatment Planning Protocol of Stereotactic Radiosurgery for Multiple Brain Lesions

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Purpose: The purpose of this study is to provide an end-to-end validation of the capability of different radiotherapy modalities to treat challenging multi-lesion stereotactic radiosurgery (SRS) cases. **Methods:** The study uses the CT simulation scan of a real patient, who was treated for five brain lesions ranging from 7-20 mm in diameter within 6 cm of isocenter. 20 Gy in a single fraction was prescribed to all lesions following the clinical protocol of UNC. Four treatment plans were produced using the following combinations of treatment planning software (TPS) and delivery machine: 1) Precision-CyberKnife (Accuray), 2) iPlan-Novalis (BrainLab), 3) Raystation-VersaHD (RaySearch Laboratories-Elekta), 4) Monaco-VersaHD (Elekta). Plans were delivered to a 3D-printed phantom emulating the specific patient's anatomy while including a gel-dosimetry insert to derive the dose distribution. **Results:** The conformity index values were 0.76, 0.79, 0.80 and 0.69, respectively (clinical goal: ≥ 0.70). All the plans failed to satisfy the constraint of V12Gy < 10cc for brain minus PTVs (Precision: 22.8cc, iPlan: 18.0cc, Raystation: 32.5cc, Monaco: 22.5cc). All the plans performed satisfactory regarding the constraint of D0.035cc < 15Gy for brainstem (Precision: 15.9cc, iPlan: 15.5cc, Raystation: 14.9cc, Monaco: 13.3cc). CyberKnife and VersaHD achieved clinically acceptable results for all the targets regarding 3D global gamma analysis (3%/2mm). More specifically, the 7mm, 9mm, 12mm, 15mm and 20mm lesions had passing rates of 100%, 99.5%, 99.3%, 98.6%, 95.9%, respectively for CyberKnife, 69.5%, 95.5%, 66.8%, 88.1%, 79.9% for Novalis, 100%, 100%, 100%, 99.8%, 94.9%, for VersaHD (Raystation), and 94.3%, 97.0%, 100%, 100%, 92.7% for VersaHD (Monaco). **Conclusion:** All the plans were comparable in terms of quality, satisfying the requirements of the clinical protocol with some deviations within clinical acceptance. CyberKnife and VersaHD accurately delivered the corresponding radiosurgery plans for targets within 6cm of isocenter. Clinically acceptable gamma pass rates were achieved for all the lesion sizes.

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Effect of Dose Rate on Cardiac Synchronized Volumetric Modulated Arc Therapy

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Purpose: To investigate the effects of different dose rates on the plan fidelity for a cardiac synchronized volumetric modulated arc therapy (CSVMT) treatment. **Methods:** A QUASAR body phantom with an acrylic film cassette placed in the centre receptacle and a wood insert placed in the peripheral receptacle was CT scanned and imported into the Eclipse treatment planning system. The film insert was contoured as the heart, the wood insert contoured as lung, and additional structures were added for the esophagus and great vessels. The planning target volume (PTV) was created near the centre of the heart. Plans were optimized to deliver 25 Gy to 100% of the PTV in two full arcs, using a 10MV FFF beam with dose rates of 1600, 2000, and 2400 MU/min. CSVMT plans were then adapted from each original treatment plan and synchronized to an artificial heart rate increasing from 50 to 100 beats per minute. A Varian TrueBeam linear accelerator was used to deliver the original and CSVMT plans, irradiating separate strips of Gafchromic EBT-XD film positioned horizontally at the isocentre within the QUASAR phantom. The dose distribution of each CSVMT plan was compared with its original plan by performing a gamma analysis on the corresponding films using FilmQA Pro. **Results:** Gamma comparison (2%/2mm criteria, global normalization to dose maximum, 10% low dose threshold) between the CSVMT and original plans returned passing rates of 98.69%, 99.13%, and 86.22% for the 1600, 2000, and 2400 MU/min dose rates, respectively. The dose profile across the region of interest showed that the 2400 MU/min CSVMT plan delivery was under-dosed compared to the original plan. **Conclusion:** The CSVMT technique on a TrueBeam linear accelerator can provide accurate treatment plan delivery up to a dose rate of 2000 MU/min. Further investigation as to delivery robustness is warranted.

Authors Steven Thomas, Kirpal Kohli, and Justin Poon have filed a patent application for CSVMT.

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API Based Automated Treatment Planning of Lung Stereotactic Radiation Therapy

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Purpose: To develop an API based automated treatment planning tool for lung stereotactic radiation therapy. **Methods:** The automated treatment planning tool is a combination of two components: a patient anatomy dependent beam setup (including gantry, collimator and couch optimization) and an a priori model-based optimization setup. The initial component of the software creates two VMAT fields. The gantry angles, collimator angles, couch angles and isocenter placement depend on the patients' planning CT data and physician orders. Deciding factors include, but are not limited to, the request of gated delivery, number of targets, and presence and location of implanted cardiac devices. The second component of this software creates preliminary optimization goals depending on the size and location of both the targets and organs at risk. The priorities of the optimization objectives are determined from both previous plans generated in an a priori fashion and from proximity based analytical functions of targets to organs at risk. The physician orders for target coverage and organ at risk dose constraints are evaluated on the optimized plan. If any constraints are not met, the tool performed re-optimization automatically with appropriate updates to objective values and priorities. **Results:** The tool has successfully planned 20 lung SBRT patients including the following: both single and multiple targets in one isocenter, gated treatments, patients with implanted cardiac devices, and patients with prior thoracic radiation history. Two separate radiation oncologists reviewed the clinical acceptability of the plans generated by the treatment planning tool. **Conclusion:** Compared to a manual treatment planning workflow, an automated treatment planning tool demonstrated similar quality treatment plans. The goal of this tool is to decrease overall planning time for lung SBRT and to consolidate the process into a set of rules. Future goals for this tool include the improvement of plan consistency between patients.

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Dosimetric Evaluation of Different Rotating Gamma Ray System Designs

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Purpose: Rotating gamma ray systems have been developed for stereotactic radiosurgery (SRS) and fractionated stereotactic radiotherapy (SRT). Different designs are available commercially that differ in the source number, source dimensions, and source distribution/arrangement. In this study, we investigated two designs with regard to their capabilities and dosimetric parameters for SRS/SRT. **Methods:** The CybeRay system (OUR United RT Group, Xian, China) consists of a ring gantry containing 13 Co-60 sources focusing at the isocenter with 7 changeable collimators. The treatment head can also swing 35° in the superior direction. Galaxy RTi (Akesis, Concord, CA) contains 30 Co-60 sources, which can rotate as a group creating 30 non-overlapping arcs. It includes 4 changeable collimators. Both systems are equipped with inline cone beam CT for real-time image guidance, which share the same isocenter with the treatment beams. Ten previous patients treated on the GammaKnife were unarchived from our clinical database. Treatment plans were generated for the two machines using the Prowess TPS (Prowess, Concord, CA). Plans were evaluated based on the beam-on time, conformity index (CI), isodose distribution, and dose-volume histograms. **Results:** Comparable plans were achieved in most cases. The mean beam-on time for the studied cases was 25 minutes for CybeRay as compared to 17 minute for Galaxy. The mean CI was in the order of 1.9 for CybeRay as compared to 1.7 for Galaxy. Galaxy showed faster dose fall off in the axial view compared to CybeRay as can be seen by the spread of the lower isodose lines. On the other hand, the ring configuration of CybeRay allows treatment of extra-cranial body sites. **Conclusion:** Both systems were capable of producing superior clinical treatment plans to meet our clinical acceptance criteria although Galaxy is specific for intracranial SRS/SRT while CybeRay is also capable of stereotactic body radiotherapy (SBRT).

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Monte Carlo (GATE/Geant4) Estimation of Clinician Hand Dose from Radionuclide Injection

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Purpose: To determine the average absorbed hand dose to the administering clinician from radionuclide injection of Ra-223 dichloride (Xofigo) and Lu-177 dotatate (Lutathera) over a 1-minute injection. **Methods:** BD 10 mL and 60 mL syringes, radioactive sources, and 2 mm thick tungsten syringe shields were modeled utilizing the GATE/Geant4 environment language. Various simulations were conducted to approximate injections with and without syringe shields: the traditional case simulating a prototypical injection, and the extreme case in which a clinician clenches the syringe with their hand centered directly over the source. Each simulation transported 50 million particles using the emstandard_opt4 physics list. Alpha, beta, and gamma emission probabilities and radioactive materials were user-defined within GATE based on the literature and prescribing information. **Results:** For Lutathera, average absorbed dose to the hand was found to be on the order of 700 µGy/min and 20 µGy/min for the extreme and traditional cases respectively. For Xofigo, in the non-shielded syringe, average absorbed dose to the hand was found to be on the order of 3 µGy/min and 0.01 µGy/min for the extreme and traditional cases. Shielded Xofigo syringes produced average estimates of doses to the hand on the order of 0.05 µGy/min. In the extreme scenario, for the same clinician injecting one patient per day for 250 workdays, a total annual dose to the hand would be approximately 700 µGy for the unshielded syringe and 10 µGy for the shielded syringe. **Conclusion:** Syringe shields were found to reduce dose from Xofigo by a factor of 52 in the extreme case and 3.1 in the traditional case. We estimate that reaching the annual occupational exposure limit of 500 mSv would take approximately 12 hours of time clenching an unshielded Lutathera syringe- which is never recommended. Clinicians should take care while delivering radionuclides to observe ALARA principles.

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The Capabilities and Characteristics of Helical Tomotherapy and Volumetric-Modulated Arc Therapy in Sparing the Hippocampus During Prophylactic Cranial Irradiation

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Purpose: To investigate the capabilities and characteristics of helical tomotherapy (HT) and volumetric-modulated arc therapy (VMAT) in sparing the Bilateral hippocampal tissues (BHT) during prophylactic cranial irradiation (PCI). **Methods:** HT and VMAT treatment plans were generated with a prescription dose of 30Gy/10 fractions in 16 patients treated with PCI. The dosimetry parameters of BHT, organs at risk (OARs), and planning target volume (PTV) were determined when the average dose of BHT was reduced by about 4Gy to serve as an observation point. Statistical plots were used to present dosimetry changing trends when sparing the BHT with both HT and VMAT technologies. **Results:** HT plans spared BHT V7.3Gy by up to 40%, with the average Dmean-BHT = 8.42±0.41Gy, while VMAT plans failed to spare BHT V7.3Gy by such an amount. HT authored significantly lower doses to OARs, including Dmean-BHT (HT 16.37Gy vs. VMAT 17.81Gy, P=0.03), Dmean-inner ear (P=0.02), Dmean-parotid glands (P=0.02), Dmax-lens (P=0.02), and Dmax-brainstem (P=0.02), except for Dmax-optic nerves (P=0.87), compared to VMAT. Also, compared to VMAT, HT plans provided better target coverage, with lower average D2% (P=0.02), higher average D98% (P=0.02), and better conformal index (0.87 vs. 0.84, P=0.02) and homogeneity index (0.15 vs. 0.21, P=0.05). With smaller BHT doses, the dose distribution to PTV occurred in three dosimetry regions for both HT and VMAT plans; the plateau region (>20.0Gy for HT vs >16.0Gy for VMAT), gradient region (20.0-12.0Gy vs 16.0-11.0Gy), and falling region (<12.0Gy vs <11.0Gy). The average delivery duration for HT was determined at almost 7.7 times longer than VMAT. **Conclusion:** Compared to VMAT, HT was better at sparing the BHT and OARs and had better target coverage but much longer treatment duration. More BHT dose decreases would yield worse target dose coverage, which would present a 'three regions' stair-stepping characteristic.

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A Comparative Study of Radiation Dosimetry on Acute Hematotoxicity Caused by Fixed Field Intensity Modulated Radiotherapy and Helical Tomotherapy for Cervical Cancer

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Purpose: To compare the difference of acute hematotoxicity caused by fixed field intensity modulated radiotherapy (ff-IMRT) and helical tomotherapy (HT) in patients with cervical cancer, and to analyze the radiation dosimetry index which caused the difference. **Methods:** The study included 232 patients were pathological diagnosis with cervical cancer, and the patients were randomly assigned into ff-IMRT group and HT group to complete radiotherapy plan and treatment. The dose volume of pelvic bone irradiation in the two groups was evaluated, the blood cells count of acute leukopenia and neutropenia during radiotherapy were recorded, and the numbers of the patients receiving recombinant human granulocyte colony stimulating factor (rhGM-CSF) were recorded. Independent sample T-test was performed to test the difference of acute hematotoxicity and pelvic bone dose volume between ff-IMRT and HT groups. **Results:** (1) Clinical characteristic: there was no significant difference in patient age ($P=0.35$), pathological type ($P=0.88$), TNM stage ($P=0.14$) and radiation prescription dose ($P=0.22$) between ff-IMRT and HT groups. (2) The incidence of acute hematotoxicity of RTOG: 118 (100%) and 109 (95.6%) ($P=0.03$) of $\geq$ grade 1 hematotoxicity in ff-IMRT and HT groups, respectively (3) There were significant differences in dose-volume parameter of pelvic bone marrow at V5Gy~45Gy ($P<0.05$). **Conclusion:** Compared with ff-IMRT, HT significantly reduced the incident of $\geq$ grade 1 acute hematotoxicity for cervical cancer with concurrent chemotherapy, and the dose volume parameters in V5Gy~45Gy HT group were lower than those in ff-IMRT group. It is demonstrated that HT technique has advantages in reduce the dose of pelvis bone marrow and in avoiding acute hematological toxicity caused by radiation.

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PO-GePV-T-430

Investigation on the Use of Comet Assay for the Measurement of High Dose Radiation

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Purpose: To standardize the comet assay as a biodosimetric tool for estimating high levels of radiation dose. **Methods:** 10 ml blood from two healthy individuals were collected using venepuncture method in EDTA containers under aseptic conditions. The blood samples were irradiated for known doses of 15, 25 and 35 Gy in Co⁶⁰ teletherapy unit. After irradiation, the samples were gently overlaid on histopaque 1077 and were centrifuged. The buffy coat of lymphocytes were mixed with 1% low melting point agarose. The mixture was gently smeared over a frosted slide and allowed to dry for 15 minutes. The slides were immersed into lysis solution for 12 hours with pH adjusted to 9.5. After lysis, the slides were washed with neutral buffer and were subjected to neutral electrophoresis using TAE buffer for 30 minutes at 25 V. Fluorescence microscope was used to capture images which were stained using propidium iodide (2.5 μ g/ml). Comet assay parameters such as tail DNA % and OTM (Olive Tail Moment) were the intensity values were measured using Fiji software. **Results:** The arbitrary units of tail DNA % were 38.36 ± 3.40 , 53.42 ± 1.95 , 60.34 ± 4.24 and OTM values were 11.41 ± 0.8 , 17.14 ± 0.62 , 29.35 ± 2.05 for 15, 25 and 35 Gy respectively. For each dose point the control values were subtracted in order to avoid inter individual effects. The dose (Gy) vs tail DNA % and OTM showed a correlation with R^2 value of 0.9563 and 0.9583. **Conclusion:** Comet assay allows the estimation of DNA damage in single cell level and can be used as a potential biodosimeter to estimate high levels of radiation dose above 15 Gy thereby overcoming the demerit of gold standard dicentric chromosomal assay which can only measure doses up to 6 Gy.

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Effect of PEGylated Gold Nanoparticles in Intraoperative Radiotherapy(IORT) Dose Enhancement in the MCF-7 Human Breast Cancer Cell Line Using Gafchromic™ EBT3 Films

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Purpose: To perform a precise Gafchromic EBT3 film dosimetry to evaluate the dose enhancement factor (DEF) of The PEGylated gold nanoparticles in breast cancer MCF-7 cell line under irradiation of 50 kV X-rays Intraoperative radiotherapy (IORT). **Methods:** The PEG-GNPs were synthesized and characterized, ranging from 40 to 60nm in diameter. The different concentrations cytotoxicity effects of the PEG-GNPs on MCF-7 cells were evaluated using MTT assay. The untreated and treated MCF-7 Cells with 1200 μ l of 25, 75, and 150 μ g/mL concentrations PEG-GNPs were placed on the surface EBT3 film and irradiated with a calibrated system IORT with the 5 cm spherical applicator at a single dose of 4Gy. The standard EBT3 films were selected for physical DEF calculation after dosimetric evaluation. Experimental DEFs were calculated using the optical density (OD) data obtained from the selected densitometer type. The films were exposed to 6MV and (50 KV) X-rays IORT using 2Gy/min up to 7Gy radiation doses. The films were scanned in the red channel to obtain the OD-dose curves. **Results:** DEF is measured by the EBT3 film for MCF-7 cells with 25, 75, and 150 μ g/mL concentrations of PEG-GNP exposed to X-rays in IORT is 1.008, 1.038, and 1.066, respectively. In the buildup region, no significant effect has been seen on the absorbed dose measured for cells treated with PEG-GNP by the film. The Rx of films presents significant energy response at (50 KV) IORT relative to that for 6-MV x-rays varied from 0.85–0.79 for doses 50–200 cGy. **Conclusion:** PEG-GNPs demonstrated acceptable dose enhancement efficacy in IORT with low-energy X-rays (50kV), which is also the optimal beam energy for dose enhancement (DE). Gafchromic™ EBT3 film dosimetry revealed significant physical DE signals when the PEG-GNP is used as a radiosensitizer with MCF-7 human breast cancer cell line.

PO-GePV-T-432

Concurrent Radiosensitization and Chemoradiotherapy for Brain Tumors

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Purpose: Glioblastoma remains the most malignant and most common primary brain tumor in adults, with a median overall survival of only 15 months. This bleak outcome is partly due to the high chemoresistance and high radioresistance of glioblastoma. Current standard of care includes surgical resection, chemoradiotherapy and temozolomide (TMZ) as adjuvant therapy. The purpose of this work is improve treatment outcomes for highly radioresistant and

chemoresistant cancers employing nano-particle mediated radiosensitization and chemoradiotherapy. **Methods:** We are using nanoparticle (NP) spectroscopy with a broad range of novel biocompatible nanoparticles including carbon quantum dots (CQD) and graphene quantum dots (GQD) based on our recently developed assay involving fluorescence intensity modulation of biocompatible CdSe/ZnS quantum dots (QDs), to assess reactive oxygen species (ROS) generation during chemotherapy and radiotherapy. Our strategy here is to use the NPs for concurrent measurement of ROS and radiosensitization, while applying chemoradiotherapy. Using a standard laboratory Cell Irradiator (Faxitron), we irradiate glioblastoma cancer cell lines (T98G and U87 cells) treated with QDs, CQDs and GQDs as well as chemotherapeutic drugs such as Temozolomide (TMZ). We quantify their QD/CQD/GQD fluorescence intensity and their migration following irradiation with clinically relevant doses (5 Gy, 10 Gy, 20 Gy, 50 Gy). Outcomes are characterized using cell proliferation and migration assays as well as clonogenic assays. **Results:** Our proof of concept results with U87 and T98G cells treated with TMZ, 5 Gy and 20 Gy irradiation show significant changes in fluorescent intensity between the conditions. Effects of radiosensitization on these conditions are among our ongoing work and the results will be presented. **Conclusion:** We are applying our nanoparticle-based assays for simultaneous measurement of ROS and radiosensitization of cancer cells to chemoradiotherapy. The proof of concept results suggest possible improvement of radio- and chemotherapeutic outcomes for brain cancers through NP-mediated radiosensitization and chemoradiation.

PO-GePV-T-433

The Plant Hemoglobin Can Be Developed as a Radio-Enhancer in Solid Tumor Radiation Treatment

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Purpose: Intratumoral hypoxia is a major regulator of radio-resistance in the cancer treatment due to reduced blood supply by the immature blood vessels and necrotic tumor core. An African indigenous *Justicia* plant (family Acanthaceae), has been shown to be evolutionized to contain hemoglobin which shows similarity with human α and β Hemoglobin whereas, plant usually have rudimentary hemoglobin called hemoglobin γ . Here, we used this plant hemoglobin extract to increase the radiosensitivity of the cancer treatment during radiotherapy. **Methods:** Lyophilized *Justicia* plant extract was analyzed for its hemoglobin content. ELISA was performed using Human Hemoglobin Kit (Sigma-Aldrich) to quantify the Hemoglobin concentration (human type). Western blot assay was performed to confirm α and β hemoglobin chains using Human Hemoglobin α and β antibody (abcam, 1:150 ratio), and a study as remedy for anemia common in cancer patients undergoing radiotherapy was performed in a mouse model. To evaluate the radio-sensitizing effect in cancer treatment, Panc-02 tumors were generated in C57BL/6 mice and treated with Plant hemoglobin (0.1mg/Kg) intratumorally just before radiation treatment (10 Gy) using the small animal radiation research platform (SARRP). Tumor measurement and survival analysis were performed. **Results:** Well characterized plant hemoglobin was injected intratumorally prior to radiation treatment in pancreatic adenocarcinoma mouse model comparing with a cohort only with same dose of radiation treatment. Control group was treated only with PBS. Significant reduction of tumor volume was observed in the plant Hemoglobin+Radiation treated group compare to the cohort with only radiation treatment ($p < 0.0001$). Significant longer survival duration was also observed in this combination treated group ($p < 0.0001$) confirming significant radio-enhancer effect of this plant hemoglobin for cancer treatment. **Conclusion:** The Plant hemoglobin can be developed as a radio-enhancer in solid tumor radiation treatment.

PO-GePV-T-434

Investigate Depth-Dependent Dose Enhancement of PSMA-Targeted Gold Nanoparticles Assisted Radiosensitization at Megavoltage Energy

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Purpose: Gold nanoparticles (GNPs) are promising targeted radiosensitizers due to their capability of specific tumor uptake and locally deposited high dose. This study evaluated the dose enhancement of PSMA-targeted GNPs in prostate cancer radiotherapy at megavoltage energy. Moreover, we investigated the impact of depth-induced spectrum change on radiosensitization, to explore the potential of beam modification in clinical GNP-assisted radiotherapy. **Methods:** 15-nm PEGylated GNPs were conjugated to anti-PSMA antibodies using 1-Ethyl-3-(3-dimethylaminopropyl) carbodiimide and N-hydroxy sulfo succinimide. LNCaP prostate cancer cells were pretreated with 250 μ g/mL PSMA-targeted GNPs. Cells were irradiated with 160kV x-rays on RadSource small-animal irradiator, to verify the dose enhancement with GNPs at kV energy and offer reference to MV irradiation group. To evaluate GNP-aided dose enhancement at MV energy, LNCaP cells were treated using a 6MV flattening-filtered photon beam from Varian Trilogy LINAC with by 20cm $\times$ 20cm field size. Cells located in culture plates were sandwiched in Elasto-gel bolus (1.0cm thick) and covered by solid water with different thickness to mimic tumors at different depths (1.5cm, 5cm, 10cm, and 20cm). A 5cm-thickness solid water was placed under the plate for backscatter along with OSLDs for dose confirmation. Radiation dose enhancement by GNPs was assessed via clonogenic assay, and quantification of DNA double-strand break via γ -H2AX foci counting. **Results:** Under 2Gy-irradiation, clonogenic assay revealed statistically significant GNP-induced dose enhancement with kV beam (150%) and MV beam (140%) at 2.5cm depth. Additionally, γ -H2AX showed statistically significantly increased foci counts under 2Gy-kV (125%) and 2Gy-MV-2.5cm (120%) irradiation, respectively. Clonogenic assay of LNCaP cells at varying depth from MV irradiation showed increased radiosensitization at 20cm depth. **Conclusion:** PSMA-targeted GNPs are effective radiosensitizers for prostate cancer cells irradiated with both kV and MV beams. Furthermore, beam quality plays a critical role in optimizing GNP-induced radiosensitization on prostate cancer cells.

PO-GePV-T-435

A Treatment Planning System for Small Animal Radiation with the Gamma Knife

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Purpose: To develop a treatment planning system (TPS) based on Monte Carlo (MC) dose algorithm for small animal radiation with the Gamma Knife (GK) Perfexion machine. **Methods:** High precision with small collimation focus makes the GK an excellent modality for small animal radiation research. It is ideal to follow the clinical procedure as received during patient treatments to achieve a high level of precision. However, the patient TPS (Leksell GammaPlan®) requires a manufacturer defined fiducial box for the GK coordinate geometry which is too large compared with the size of a small animal. The image spatial resolution of typical clinical CT or MR is too coarse for small animals. To overcome these difficulties, we developed an in-house TPS which uses high-resolution micro-CT or micro-MR images and obviates a fiducial box with markers. Instead, we built an acrylic box with a reference point that correlates to the GK focal point. The acrylic box is mounted to the GK couch. A cradle with a marker that has a known relative position to the box is scanned with a mouse using micro-CT /MR. Our previously developed MC algorithm for the GK is used to calculate accurate dose distributions based on the high-resolution images. **Results:** A process of irradiating mice based on micro-CT/MR is established using our in-house TPS. A graphical user interface is developed to allow a planner to easily 1) create irradiation plans; 2) determine the treatment sites guided by the image; 3) place multiple shots with the option of different collimator sizes; 4) calculate dose using MC method which is capable of taking heterogeneity into account and 5) generate shot coordinates and irradiation time for machine delivery. **Conclusion:** Our image guided TPS greatly facilitates the procedure for small animal radiation with the Gamma Knife.

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Analysis of Preclinical IMRT Plan Quality Versus Number of Beam Angles

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Purpose: To evaluate 3D printed compensator-based preclinical IMRT plan quality as a function of number of angles for efficient hypoxia dose-painting in a murine tumor model. **Methods:** Twenty mice with hind-limb tumors and previously acquired electron paramagnetic resonance oxygen images were randomly selected for this study. A preclinical inverse treatment planning system was used to generate IMRT plans targeting hypoxic tumor regions, ranging from one to seven per animal, using 2, 3, 5, 7, 9, and 15 angles. Each plan was normalized such that 95% of the hypoxic volume received 35.5 Gy. Plan quality was assessed in terms of conformity index (CI), difference between effective radii of the prescription and 50% isodose volumes ($\Delta 50$), homogeneity index (HI), and mean dose (Dmean) to surrounding non-hypoxic tissue. Quality metrics were evaluated for significance ($p < 0.05$) using two-tailed paired t-tests versus plans with 3 angles. **Results:** Preclinical IMRT dose distributions were significantly ($p = 0.01$) less conformal with 2 angles than 3 angles, with mean CI values of 10.65 and 1.93, respectively. However, there was no significant difference ($p \geq 0.06$) for the 5 (CI=1.69) and 7-angle (CI=1.58) scenarios. CI was 1.56 with 9 ($p = 0.04$) and 1.59 ($p = 0.05$) with 15 angles. Dose fall-off ($\Delta 50$) measured 2.5, 3.1, 2.8, 2.7, 2.7, and 2.7 mm for 2, 3, 5, 7, 9, and 15 angles, respectively. From 3 to 15 angles, the mean HI varied from 120.0-118.8% versus 115.0% for the 2-beam case. Use of 3 or more angles significantly reduced non-target tissue Dmean by 6.2%, 8.2%, 8.6%, 9.1%, and 8.1% with increasing angle number relative to a mean dose of 22.1 Gy for the 2-angle plans. **Conclusion:** The results indicate that using 3 angles for compensator-based preclinical IMRT does not compromise plan quality relative to treatments with 5 or more angles while providing improved experimental workflow and treatment delivery efficiency.

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Precise Pulse Delivery Control Using Monitor Units in Electron FLASH-RT

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Purpose: In electron FLASH-RT, precise delivery of the correct number of pulses is critical to accurate dose administration in preclinical radiobiological studies. This work investigates the use of LINAC monitor ion chambers to most precisely control FLASH pulse delivery. **Methods:** A Varian Clinac 21EX was modified and tuned to deliver 16 MeV electrons at FLASH dose rates ($> 100 \text{ Gy/s}$, $> 0.55 \text{ Gy per pulse}$). Instead of stopping the beam with a custom circuit asserting an external interlock, we use monitor units (MUs) in the primary monitor ion chamber (MU1) in a range of 1-60, which corresponds to a dose range of 2-110 Gy. A calibrated plastic scintillation detector and EBT-XD Gafchromic films were used for online and passive dosimetry, respectively. The plastic scintillation detector also served as a direct pulse counter. **Results:** The circuit board interpreting the monitor ion chamber signal is saturated by the ultra-high instantaneous FLASH dose rate ($\sim 10^6 \text{ Gy/s}$), which significantly impacts the relationship between MUs counted and radiation dose delivery. However, this saturation can be exploited to convert the monitor ion chamber system into a real-time internal pulse counter. The number of pulses delivered exhibits a highly reproducible, perfectly linear relationship with programmed MU1 (3.19 pulses/MU, $R^2 = 1$). In contrast to our custom external circuit, which delivers with a ± 1 pulse uncertainty, no pulse delivery uncertainty was observed when using MU1. Given the stable dose rate of our FLASH system ($SD = \pm 1.1\%$), the dose delivered is linear with MU1 over a large dose range (2-110 Gy) for both scintillator and film measurements. **Conclusion:** This study demonstrates that the primary monitor ion chamber can act as a simple, stable and reliable control for precise FLASH pulse delivery.

PO-GePV-T-438

Dosimetry and Quality Assurance for the Small Animal Radiation Research Platform (SARRP)

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Purpose: Preclinical studies in radiation oncology have a major impact on the design of human clinical trials. Accordingly, similar standards are required to ensure reliable and reproducible workflow and dose delivery for preclinical irradiations on murine subjects. Here we present findings from our four year experience on daily, weekly, monthly, and annual quality assurance (QA) of an orthovoltage preclinical small animal radiation research platform (SARRP, Xstrahl). **Methods:** Irradiations on the SARRP can be performed both in vitro and in vivo, with the latter being implemented as image-guided radiation therapy (IGRT) using the integrated cone beam computed tomography system (CBCT). Thus, QA tasks are performed three-fold: (i) Radiation dosimetry is measured using two independent systems using ion chamber and EBT3 radiochromic films. (ii) Geometric and mechanical accuracy tests are performed to evaluate alignment and targeting of a region of interest on the couch with CBCT, radiation beam, and the setup lasers. (iii) Imaging quality parameters are assessed in terms of spatial resolution. **Results:** Dosimetric output constancy measurements showed monthly variations within 2.5% over a course of four years. A cross-check to a clinical 6MV beam showed a difference of 1.1% to the SARRP beam. Dosimetric characteristics of the different collimators - including percentage depth dose (PDD), beam profile, symmetry, and flatness - were found consistent with results obtained at commissioning. Geometric and mechanical accuracy are confirmed on a weekly basis. Imaging QA revealed a slow decline in spatial resolution over time, which was corrected by a CBCT calibration resulting in improved image quality compared to commissioning. **Conclusion:** The results showed the successful implementation and importance of rigorous QA for preclinical irradiators. Solutions for challenges associated with implementation of clinical protocols on preclinical irradiators, which often operate on different energies and differ in their physical setup, are highlighted.

PO-GePV-T-439

Evaluation of A New Method to Measure Colony Growth In-Vitro

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Purpose: A novel method for accurate measurement of cell colony growth in complex in-vitro culture conditions is presented, suitable for radiation cell survival and growth assays involving fluorescing cells. **Methods:** An Algorithm using CellProfiler was created to measure mCherry-labelled 4T1 murine mammary carcinoma cell colonies grown in 3D organotypic culture on rat brain slices. The accuracy of the algorithm was investigated by application to realistic simulated image sets. 10,000 mCherry-labelled 4T1 were plated on 400 μm rat brain slices on top of transwell-inserts within 6 well plates. The cells were imaged once a day for 5 days filtered for the mCherry signal. Representative simulated images corresponding to days 1-5 were created using MATLAB with similar colony intensity, distribution, size distribution, variable background, colony overlap, and background noise. A CellProfiler pipeline was developed to measure the integrated intensity of the mCherry signal. Parameters in the pipeline were varied to determine an optimal pipeline which gave values closest to ground truth values for the simulations. The optimized CellProfiler pipeline was then applied to the original 4T1 tumor cell images to determine colony growth over five days. **Results:** Key parameters to be optimized in the pipeline include colony diameter, minimum intensity, background subtraction method and size. The optimized CellProfiler pipeline performed extremely well on 10 realistic simulated data sets, with an accuracy of $99.23\% \pm 0.75\%$. Further analysis on images

with increasing colonies, background, and noise showed the pipeline was accurate for variable features. Analysis of the 4T1 real cell image data showed an increase in integrated intensity following day 2 approximately consistent with the 13.6 ± 1.5 hour doubling time for 4T1 cells. **Conclusion:** The new algorithm has strong potential to bring much needed improvement in rigor and systematic consistency to the counting of fluorescing cell colonies in radiation survival and cell growth assays.

PO-GePV-T-440

Radiation Cell Survival Fraction After a FLASH Radiation Pulse

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Purpose: Ultra-high dose-rate (FLASH) irradiation has produced remarkable normal tissue sparing while keeping the tumor cell control rate similar to Conventional (CONV) irradiation in recent in-vitro/in-vivo experiments. This study mathematically explores the cell survival fraction (SF) and radiation dose relationship based on the radiation-induced oxygen depletion hypothesis. A straightforward single ultra-short radiation pulse FLASH experimental setup is utilized for this proof-of-concept investigation. **Methods:** When an ultra-short radiation pulse hits a cell, it causes a sudden drop in intracellular oxygen tension. Oxygen diffusion will subsequently restore the concentration over a time called characteristic time τ . Radiation can cause direct and indirect damages to a cell. The probability that the damages are 'fixed' by oxygen molecules to form DNA lesions in the cell is assumed to decrease exponentially in time with a decaying constant λ . Using quasi-static approximation, we estimate the cell's Oxygen Enhancement Ratio reduction during its intracellular oxygen tension recovery. Combining SF at the high dose rate limit from Curtis' Lethal and Potentially Lethal (LPL) Model, we then derive the radiation cell SF following a single FLASH radiation pulse. **Results:** We have derived the cell radiation SF following a single FLASH pulse. The result indicates that single-pulse FLASH irradiation's sparing effect depends on τ , λ , the cell's LPL model parameters and OER parameters, and its initial intracellular oxygen tension p_0 . A large amount of radiation dose is usually needed to make the FLASH sparing effect observable. The effect is prominent for a cell in aerobic condition and diminishes when a cell is in a hypoxic state. Numerical analysis is performed for the C3H10T1/2 cell to demonstrate our results. **Conclusion:** The radiation-induced oxygen depletion hypothesis is a leading theory to explain the FLASH RT effect. Our results lend potential quantitative validations. More studies are needed to understand the FLASH effect clearly.

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PO-GePV-T-441

A Study of Indirect Apoptotic Death by Vascular Damage On Cell Survival After Single-Fraction Radiotherapy by Using An Improved Cellular Automata Model

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Purpose: To investigate the effects of indirect apoptotic cell death due to vascular damage by a single large dose with an improved cellular automata (CA) model. **Methods:** The tumor growth process was simulated by considering the oxygen and nutrients supplied from the blood vessels. The cell damage process was simulated by considering the direct radiation damage with the LQ-model and the indirect apoptotic death by the vessel damage. The radiation caused increased permeation of plasma protein through the blood vessel or the breakdown of the vasculature. The amount of oxygen in cancer cells lead to a change in radiation sensitivity of cancer cells and tumor growth rate after irradiation. The indirect apoptotic death was evaluated with a single dose of 6 to 24 Gy on 100 day after the irradiation. The tumor control probability (TCP) was evaluated by varying the probability of apoptotic death, P^{apop} , the threshold of the oxygen level for indirect apoptotic death, O^{thre} , and the average oxygen level, O^{avg} in cancer cells. **Results:** The apoptotic death increased with a higher O^{thre} and a higher P^{apop} . Consequently, the TCP increased. TCP as a function of O^{avg} exhibited the minimum at a certain O^{avg} . The apoptosis increased as O^{avg} decreased to a very low value, leading to increased TCP. On the other hand, the radiation damage and proliferation of the tumor increased for higher O^{avg} . Specially, the direct radiation damage became dominant for cell killing for higher O^{avg} . **Conclusion:** We showed that the indirect apoptotic death of cancer cells, which was caused by the reduction of the oxygen level due to the radiation-induced vascular damage, enhanced the radiation cell kill ability by explicitly modeling the damage of blood vessels by radiation in a CA simulation.

PO-GePV-T-442

EQD2 of HDR Brachytherapy for Vaginal Cancer Based on a Novel Model

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Purpose: The α/β ratio of the linear-quadratic (LQ) model is obtained by fitting the cell-survival curves (CSC) in the low-dose domain. Applying the same α/β ratio might be inappropriate to predict the biological effects for treatments with large fractional dose. Recently a unified multi-activation (UMA) model has been proposed to fit CSC over the entire dose range. Based on the UMA model, we calculate EQD2 and compare with LQ-EQD2. **Methods:** Fifteen HDR cases treated with multi-channel vaginal cylinders are selected for this study. D90 (minimum dose received by 90% or less of the volume) of the high-risk CTV (HRCTV) with and without applicator subtraction are collected from the treatment plans for each fraction. EQD2 of the HRCTV and HRCTV-applicator D90 are calculated based on the LQ model (assuming $\alpha/\beta=10$ Gy) and the UMA model with fitting parameters corresponding to two cell-lines, D0=2.71Gy, $n=2.8$ (NHK3025 with 40Gy pre-irradiation to medium, "UMA-1"), and D0=1.11Gy, $n=1.5$ (RL95-2, "UMA-2"). **Results:** For 10 cases with 7Gy/fraction prescription, the UMA-1 and UMA-2 EQD2 cumulative D90 doses are different from the LQ EQD2 on average by 2.47% and -21.25% for HRCTV, and by 3.06% and -18.76% for HRCTV-applicator, respectively. For 5 cases with 6Gy/fraction, the differences are 3.39% and -16.82% for HRCTV, and 3.71% and -14.77% for HRCTV-applicator, respectively. We also note that excluding the HDR applicator or not from HRCTV could be significant for the evaluation of biological effects of radiation treatment. For example, the LQ EQD2 cumulative doses of HRCTV-applicator are different from those of HRCTV by -11.26% on average (-8.36% to -16.67%). **Conclusion:** Using a single α/β ratio obtained from low-dose domain of CSC for HDR EQD2 calculation could be overly simplistic. The novel UMA-model fits CSC over entire dose range and provides a new way to calculate EQD2, which may be more accurate.

PO-GePV-T-443

Effect of Total and Hypoxic Tumor Volumes on the TCP in Hypofractionated Radiotherapy

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Purpose: Clinical studies in the hypofractionated stereotactic body radiotherapy (SBRT) have shown a reduction in the probability of local tumor control with increasing initial tumor volume. A plausible mechanism for these volume effects would be useful for the analysis of treatment outcomes and the design of dose escalation protocols. **Methods:** In our earlier work, we obtained and tested an analytical dependence of the TCP (tumor control probability) on the total and hypoxic tumor volumes using conventional radiotherapy model. We used a two-level cell population tumor model for hypoxic tumors and the TCP equation derived from the Poisson probability distribution. Compared to radiotherapy with conventional fractionation schedules, simulations of hypofractionated radiotherapy may require different models for cell survival and the Oxygen Enhancement Ratio (OER). Our TCP simulations in hypofractionated radiotherapy

are based on the Liner-Quadratic (LQ) model and the Universal Survival Curve (USC) developed for the high doses used in SBRT. The predicted trends in local control as a function of the initial tumor volume were evaluated in SBRT for non-small cell lung cancer. **Results:** Both LQ and USC based models cannot describe the TCP reduction for larger tumor volumes if the tumor is completely oxygenated. The TCP calculations are in agreement with the clinical data if the subpopulation of radio-resistant hypoxic cells is considered with the volume that increases as initial tumor volume increases. **Conclusion:** First, the extent of hypoxia is likely a primary reason of the TCP reduction with increasing the initial tumor volume in SBRT for non-small cell lung cancer. Second, the LQ model can be an acceptable approximation for the TCP calculations in hypofractionated radiotherapy if the tumor response is defined primarily by the hypoxic fraction. The larger value of OER in the hypofractionated radiotherapy extends the applicability of the LQ model to larger doses.

PO-GePV-T-444

A Biology-Driven Deep-Learning Based Cell Survival Model for Proton Radiation

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Purpose: It is challenging to construct a generic biophysical model with good generality to accurately predict cell survivals for proton radiation. This study proposes a Biology-guided Cell Survival deep neural network (BgCSDNN) to directly derive cell surviving fraction (SF) based on different types of DNA damages computed by microscopic Monte Carlo (MC) simulations. **Methods:** GPU-based MC computation platform gMicroMC was used to simulate the physical and chemical stages of proton radiations and calculate different types of DNA damages following setup of a proton irradiation experiment on lung cancer H460 cells. Spectrum of DNA damage was calculated to account for the dose inhomogeneities among cells. A biology-driven neural network that predicts SF as a function of the DNA damage spectrum was constructed with the network structure reflecting major pathways of DNA damage repair. We performed an end-to-end training of the network using MC simulated DNA damage data with ten different LETs and tested the validity of the SF prediction model for two different LETs. Weights of pathways related to different DNA damage types in the trained network were analyzed. **Results:** At 6 Gy, when LET increases from 0.9 to 19.0 keV/μm, double-strand breaks (DSBs) increases from 334±26.7 to 574±101.7 (mean ± standard deviation). The trained network achieved a mean relative error of 0.1 and 0.11 against experimental ground truth data on training data and test data, indicating generalizability of the model, which performed better than 0.23 for a LET-based linear-quadratic model. In the trained network, weights for DSB related pathways were ~0.4, substantially larger than the one for single strand break (~0.01), indicating the importance of DSBs in cell kills. **Conclusion:** The novel BgCSDNN can accurately predict cell survival for proton radiation and provides biological insights to the cell survival process.

PO-GePV-T-445

Relative Response of Murine Lung Tumors from Conventional Radiotherapy (CRT) Versus Pulsed Low-Dose-Rate (PLDR) Radiotherapy

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Purpose: This work investigates the radiation response of conventional radiotherapy (CRT) and pulsed low-dose-rate (PLDR) radiotherapy using in vivo small cell lung cancer murine tumor models. **Methods:** Lung cell line KP1 (murine small cell lung cancer) was prepared for inoculation of ~20 gram C57BL/6 mice (The Jackson Laboratory). Exponentially growing cells were harvested and suspended in serum-free medium, and 5x10⁵ cells were injected subcutaneously into the right hind leg of 6 mice. Tumors were allowed to develop to a diameter of 5-7 mm before being separated into CRT and PLDR groups. A Varian Clinac iX machine was used to deliver 20Gy (4Gy/fraction) to the tumor using a SSD set-up, while the mice were under general anesthesia by way of isoflurane vapor with oxygen used as a carrier gas. The novelty of using isoflurane is that it safely allows for repeated and prolonged administration, which lends itself well to fractionated treatments and extended intra-fraction PLDR treatments. The dose rate for the CRT group was 500 cGy/min, while the effective dose rates (EDR) for PLDR was 8.3 cGy/min (a train of radiation pulses, each 0.25Gy, delivered in 3 minute intervals). After each fraction, mice were revived and tumors were measured with a caliper. **Results:** Radiation responses of tumors in the CRT and PLDR groups were statistically indistinguishable, while both showing tumor control. **Conclusion:** PLDR effect is comparable to that of CRT against KP1 murine tumors. This result adds to the body of research showing PLDR's clinical efficacy, due its equivalent tumor control and reduced normal tissue toxicity with decreased EDRs. Further studies are underway to replicate a standard treatment protocol for lung cancer using stereotactic body radiotherapy. PLDR may be developed into a clinically viable alternative for treating large tumor masses and/or recurrent cancers with decreased normal tissue tolerances.

PO-GePV-T-446

Optimal Treatment Scheduling in the Presence of Stress-Induced Phenotypic Switching of Cancer Cells

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Purpose: Intratumoral phenotypic diversity and switching enables a population of cancer cells to cope with the fast-changing external stress without the need of slow genetic modifications. There is a growing evidence that stress-induced phenotypic switching of cancer cells is responsible for therapy resistance and tumor recurrence in oncology. Our goal is to find the optimal therapy scheduling in the presence of phenotypic switching. **Methods:** We consider a mathematical model with reversible switching between sensitive/growing and tolerating/quiescent phenotypes in an oscillating environmental stress which represents the cycles of therapy and rest period. The sensitive/growing cells grow at a constant growth rate. The tolerating/quiescent cells don't grow and are tolerant of therapy. The sensitive cells switch to the tolerating cells only in a stressful environment and the tolerating cells switch to the sensitive cells only in a stress-free environment. Stress represents both extrinsic therapy and intrinsic cellular competition over limited resources. **Results:** In the limit of infinitely fast switching rates, the cancer population size is determined by its exponential growth during stress-free periods. The optimal therapy scheduling is a continuous therapy without a rest period. In the limit of infinitely large stress-induced cell-death rate, the cancer population size is determined by its exponential decrease due to reverse switching of tolerating cells to sensitive cells during stress-free periods. The optimal strategy is the repetition of a long rest followed by a very short yet extremely effective therapy (such as FLASH). In the intermediate values of switching/death rates, there exists an optimal inter-therapy rest period which is long enough to facilitate the switching of tolerating cells to sensitive cells, but short enough to limit the proliferation of the sensitive cells. **Conclusion:** The treatment scheduling needs to take into account the phenotypic switching and the optimal rest period lest therapy cause cancer growth.

PO-GePV-T-447

DNA-Scale Dose Deposition Surrounding Heavy-Metal Nanoparticles Irradiated with a Monoenergetic Kilovoltage-Energy Beam

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Purpose: In some studies of radiotherapy enhancement with gold nanoparticles, cell kill per Gy exceeds that predicted by models considering only gross excess energy deposition due to nanoparticle interactions. It has been proposed that the fine structure of energy deposition around the nanoparticles leads to an explanation of excess cell kill, marking this fine structure as a surrogate for dose enhancement. Here we investigate the energy deposition around nanoparticles other than gold to identify possible dose enhancement opportunities. **Methods:** Simulations were performed using the Geant4 Monte Carlo toolkit and Geant4-DNA extension. Simulation geometry consisted of a focused, mono-energetic 40keV photon beam impinging perpendicularly on a 10μm cube of water containing a single 20nm nanoparticle of varying composition. Experiments consisted of 10⁸ primary events and were repeated for statistical purposes. For

each nano-material radial dose deposition was scored in concentric shells nanoparticle to a distance of 300nm. **Results:** Interactions in the gold nanoparticle produced the largest number of secondary particles per ionization (21.0 + 0.29), followed by bismuth (13.7+ 0.49), iodine (13.2 + 0.30), and iron oxide (9.60 + 0.22). Normalized radial dose functions for gold and bismuth were qualitatively similar, displaying greater dose near the nanoparticle compared to iodine and Fe2O3. However, iodine produced a dose an order of magnitude higher than gold at a distance of 300nm, indicating a higher fractional dose due to low energy photons. **Conclusion:** The qualitative similarity of the normalized radial dose functions indicates that the efficacy of nanoparticle enhanced radiotherapy may be most sensitive to bioconcentration and limited by potential cytotoxicity in normal tissues. The observed similarity between gold and bismuth radial dose functions suggests equivalent dynamics of biological effect, while the extended range of dose from iodine could allow for an effectiveness less dependent upon biological distribution of the nanoparticles within the cell.

PO-GePV-T-448

Alpha Particle Microdosimetry Calculations Aided by Shallow Neural Network Regression Fitting

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Purpose: To train a neural network to accurately predict alpha particle microdosimetric parameters, namely, the average and average square single-hit specific energy, $\langle z_1 \rangle$ and $\langle z_1^2 \rangle$ respectively. **Methods:** A two-layer feed forward neural network consisting of 4 inputs and 2 outputs was adapted from MATLAB and trained on sufficiently large previously published microdosimetric data. The data spans the common range of alpha-particle energies (3.97 – 8.78 MeV), nuclear radii (2 – 10 microns), cell radii (3 – 18 microns), and several source-target configurations. The mean square error (MSE) was used to measure the performance of the function fitting neural network. The random data partition for percent training/validation/testing and the number of nodes in the hidden layer were chosen to minimize MSE while ensuring the network could generalize well. The output layer transfer function was chosen to restrict outputs to physically meaningful positive values. The Levenberg-Marquardt training algorithm was implemented. A neural network of 8 hidden nodes and a 70%/15%/15% data partition was trained on 672 data points. The network can be exported and shared for tests on independent data sets and calculations of $\langle z_1 \rangle$ and $\langle z_1^2 \rangle$ in Gy and Gy² respectively. **Results:** The network predictions and target values showed good agreement. The best training performance of MSE = 8.323 x 10⁻⁵ was observed at epoch 153, and training was stopped after satisfying 6 validation checks. The network performed with a final test MSE = 1.306 x 10⁻⁴. The overall target to output correlation was R = 0.99981. The mean percent error was -1.33% for $\langle z_1 \rangle$ and 3.05% for $\langle z_1^2 \rangle$. The largest percent errors were observed for very small values. **Conclusion:** This trained neural network can produce microdosimetric parameters used for the study of alpha-particle emitters. Further directions include expanding to more realistic configurations and testing the network on independent data sets.

AAPM ePoster Library

Ultrasound General ePoster Viewing

PO-GePV-U-01

The Accuracy and Radiomics Feature Effects of Multiple U-Net Based Automatic Segmentation Models for Transvaginal Ultrasound Images of Cervical Cancer

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Purpose: Ultrasound (US) imaging has been recognized and widely used as a screening and diagnostic imaging modality for cervical cancer all over the world. The purpose of this study is to investigate the U-net based automatic segmentation models for cervical cancer on US images and investigated the effects of automatic segmentations on radiomics features. **Methods:** A total of 1102 transvaginal US images from 796 cervical cancer patients were collected and randomly divided into training (800), validation (100) and test sets (202), respectively, in this study. Four U-net models (U-net, U-net with Resnet, context encoder network (CE-net), and Attention U-net) were adapted to segment the target volumes of cervical cancer automatically on these US images. Radiomics features were extracted and evaluated from both manually and automatically segmented target volumes. **Results:** The mean Dice similarity coefficient (DSC) of U-net, Attention U-net, CE-net, and U-net with Resnet were 0.88, 0.89, 0.88, and 0.90, respectively. The average Pearson coefficients for the evaluation of the reliability of US image based radiomics were 0.94, 0.96, 0.94 and 0.95 for U-net, U-net with Resnet, Attention U-net, and CE-net, respectively, in their comparison with manually segmentation. The reproducibility of the radiomics parameters evaluated by intraclass correlation coefficients (ICC) showed robustness of automatic segmentation with an average ICC coefficient of 0.99. **Conclusion:** multiple U-net based automatic segmentations were accurate and feasible enough to delineate the target volumes on US images for patients with cervical cancer. Radiomics features extracted from automatic segmented target volumes showed high reliability and reproducibility for further radiomics investigations.

PO-GePV-U-02

The Use of Statistics-Based Analyses of Random Spherical Void Phantoms as a Potential Means for Quantitatively Assessing the Performance of Ultrasound Imaging Systems

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Purpose: To explore the potential of using a relatively high-density random spherical void (RSV) phantom along with statistically-based, quantitative metrics as an approach for assessing the performance of ultrasound imaging systems. **Methods:** Ten computer-simulated RSV phantoms with a RSV volume fraction of 0.15 were generated. To determine the number of image planes necessary to provide robust measurements, a series of 1 to 150 “ideal image planes” were analyzed and the observed circle cross-section radii histograms were compared with the expected theoretical histogram. Simulated ultrasound image planes were produced by adding speckle and degradation of imaging system performance was modeled by averaging 1 to 9 neighboring planes to represent increasing elevation plane thicknesses. Quantification of the performance of the imaging system was determined by measuring the: 1) mean number of observed circle cross sections per frame; 2) mean fractional area of circle cross-sections per frame; and 3) agreement of observed circle cross section radii histograms with the theoretical distribution (Chi-square statistic). **Results:** Results suggest that analyses of over 100 “ideal image planes” (>3000 circle cross-sections) provide excellent agreement between the observed and expected theoretical histogram distributions (mean chi-square < 0.004). For 150 image plane analyses, the degraded RSV phantom images show decreasing: mean number of circle cross sections detected per frame (31.5±0.3, 28.4±0.3, 28.2±0.3, 26.3±0.3 and 25.3±0.3); mean fractional area of circle cross-section per frame (0.157±0.002, 0.133±0.001, 0.133±0.001, 0.111±0.001 and 0.108±0.001); and agreement with theoretical histogram distribution (chi-square: 0.070±0.004, 0.140±0.005, 0.149±0.007, 0.379±0.011 and 0.518±0.010) for 1, 3, 5, 7 and 9 plane averages, respectively. **Conclusion:** This simulation suggests that using a relatively high-density a RSV phantom, with an appropriate number of image frames analyzed, may represent an approach for providing quantitative, statistically-based assessment of ultrasound system performance in a reproducible fashion.

with an Intel Core™ i7-6700 CPU@3.40 GHz PC. **Conclusion:** The CNN model can quickly predict target center on x-ray radiograph with accuracy equal or slightly better than the current system detection. With further development using large datasets, the deep learning may be used to improve target prediction for real-time markless lung tumor tracking.